

Group schemes

SGA 3

Unofficial English draft

Seminar directed by M. Demazure and A. Grothendieck

About this draft

This volume collects the foreword, introduction and Exposés I–XXVI, including the A and B parts, of *Schémas en groupes* (SGA 3). The French source is the recomposition edited by P. Gille and P. Polo. Statement numbers and references retain the printed French numbering. Clear typographical slips corrected in the English are marked in the LaTeX source; apparent mathematical slips are retained and recorded in the per-exposé READMEs. Independent scholarly proofreading remains outstanding. The translation contribution is licensed under the MIT License; the French original remains copyright of its authors and publishers.

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Foreword and introduction

Foreword

We present here a slightly revised re-edition of the original Seminar, whose aim and content are indicated in the Introduction. The revision consisted essentially in the correction of misprints, the addition (in footnotes) of a few supplementary remarks or references, the present division into three volumes each provided with a detailed table of contents and an index of notations, and the addition of a terminological index at the end of volume 3. Moreover, exposé VI_B of J.-E. Bertin has been partially rewritten by him, notably paragraphs 5 and 10, so that certain references to this exposé differ from the references to the original exposé. The reader will find a list of the exposés of the Seminar at the beginning of the present volume.

Since the publication of the first edition of the present Seminar, the whole of the *Éléments de Géométrie Algébrique*, Chap. IV, has appeared, which renders certain passages of the Seminar unnecessary; we have sometimes indicated in footnotes the relevant references to EGA IV which make it possible to bypass such passages.

For another exposé on algebraic groups making systematic use of the language of schemes, we mention the book of M. Demazure and P. Gabriel, *Groupes algébriques* (North-Holland & Masson et Cie). Unlike the present Seminar, this book assumes no knowledge of algebraic geometry, but contains all the necessary preliminaries of scheme theory, and its reading can therefore serve as an introduction to the study of our Seminar. (It contains, moreover, themes not covered in the Seminar, such as the structure theory à la Dieudonné of commutative affine algebraic groups, in *loc. cit.* Chap. V.)

Bures-sur-Yvette, March 1970

Introduction

1. The aim of the present seminar is twofold.

On the one hand, we aim to give convenient foundations for the theory of group schemes in general. Exposés I to IV will give, in this respect, the indispensable preliminary exercises in schematic and categorical syntax. In order to obtain a language which “sticks” without effort to the geometric intuition, and to avoid circumlocutions which are unbearable in the long run, we always identify a prescheme X over another S with the functor $(\text{Sch})^\circ/S \rightarrow (\text{Ens})$ which it represents^(*), and it is necessary to give numerous definitions in such a way that they apply to arbitrary functors, representable or not. Moreover, almost all the functors which we shall have to use will be “sheaves” (for the “faithfully flat quasi-compact topology”); exposé IV, which treats groups only incidentally, gives a sketch of the language of “localization” and of sheaves, which likewise proves very convenient in questions of representability of functors. This exposé will above all provide us, for questions of passage to the quotient, with the most convenient framework for the sequel.

Exposé V gives some general results on the existence of quotients, taken up again in exposé VI_A in the case of the quotient of an algebraic group over a field (or, more generally,

^(*)Such a point of view seems to have been envisaged for the first time eight or nine years ago, in connection with the theory of formal groups, by P. Cartier, who unfortunately did not take the trouble to make it precise and to systematize it as it deserved.

over an artinian ring) by a subgroup^(**). This latter exposé and exposé VI_B, which follows it, also contain various elementary results special to algebraic groups over a field, commonly used in the sequel.

Exposé VII studies certain facts connected with the characteristic of the base field and develops in particular, with the appropriate generality, the correspondence between radicial group schemes of height 1 and restricted Lie p -algebras.

Finally, exposé XVIII contains the generalization, in the theory of schemes, of Weil's theorem on the “birational” definition of algebraic groups.

On the other hand, we propose to generalize, to groups over an arbitrary base prescheme, the Borel–Chevalley structure theory of affine algebraic groups. It has moreover become apparent, on the occasion of the writing up of the notes of the seminar, that the affine hypothesis was unnecessary for many results of the theory. The most complete results are of course obtained in the cases of “semi-simple group schemes”, or more generally “reductive” ones, with which we shall concern ourselves exclusively from exposé XIX on. Chevalley himself had already given the construction of the groups “of Tôhoku” over the ring of integers, a construction which will be taken up again in the present seminar. The principal uniqueness theorem⁽²⁾ gives a simple characterization of the “twisted” variants of these Tôhoku groups, over a base prescheme S : these are the groups which are affine and smooth over S , whose geometric fibers are connected semi-simple groups in the usual sense^(***) ⁽²⁾.

2. As in the case of the known theory over an algebraically closed field, a crucial role is played by the subtori of the group schemes under consideration. Thus the preliminary study of tori, and more generally of “group schemes of multiplicative type”, (both from the intrinsic point of view and from the point of view of the subgroups of multiplicative type of a given group), takes a rather large place in this Seminar (exposés VIII to XII). Their remarkable rigidity (greater even, in certain respects, than that of abelian schemes, or of semi-simple group schemes) makes them very effective working tools for the study of certain more general groups.

3. From exposé XII on (to the exclusion of exposé XVIII, already mentioned) we shall commonly use the theory of affine algebraic groups over an algebraically closed field, which the reader will find in the Chevalley Seminar 1956, more particularly in exposés IV to IX of this Seminar. We shall also use, but to a lesser extent, the later exposés of the Chevalley Seminar, devoted to the structure of semi-simple algebraic groups. Indeed, we shall take up Chevalley's theory directly in the framework of schemes: one will see that in this way (even over a base field) the exposé gains in simplicity and in precision.

4. The principal object of the present Seminar is of course to develop techniques which apply to the study of group schemes over an arbitrary base, i.e. essentially to the study of families of algebraic groups. As such, the infinitesimal properties of such families, and in particular the case of an artinian base scheme, play an important role. These properties intervene even for the study of algebraic groups over a field k , in the case where the latter is not perfect, so as

^(**)For a more thorough study of passage to the quotient, notably by reductive groups, see the important study of D. Mumford, *Geometric Invariant Theory*, Ergebnisse der Mathematik, Bd 34, Springer 1965. Let us observe that on one important point, the terminology of this book does not agree with ours, for over a field of characteristic $p > 0$, the groups which Mumford calls “reductive”⁽¹⁾, are the smooth groups of multiplicative type in the sense of the Seminar (cf. Exp. IX). One may doubtless consider that the sense in which Mumford takes the word “reductive”, which loses its meaning over a base which is not a field, was adopted by him provisionally and as a sort of makeshift (and this is also more or less what Mumford explains, for other reasons, from the second paragraph of his preface!). ⁽¹⁾ **N.D.E.** see the remarks added at the end of this Introduction.

⁽²⁾**N.D.E.** This refers to corollary XXIII.5.6, which follows easily from the uniqueness theorem for split reductive groups (cf. XXIII, th. 4.1 and cor. 5.2), given that every semi-simple S -group is a “twisted form” (for the étale topology) of a group “of Tôhoku” (cf. XXII, cor. 2.3).

^(***)This is the essential result of the thesis of M. Demazure (*Schémas en groupes réductifs*, Bull. Soc. Math. France **93** (1965), 369–413).

to be able in particular to apply the technique of descent in the non-Galois case. Among the new results obtained in this case, let us mention the existence of maximal tori and of Cartan subgroups in an arbitrary smooth algebraic group, the rationality of the variety of maximal tori, and various related results (Exp. XIV ⁽³⁾), or the correspondence between the “forms” of a semi-simple group and the principal homogeneous bundles under a suitable semi-simple algebraic group (in general not connected) (Exp. XXIV, 1.16–1.20). In general, one may say that the methods required in order to work over a base field which is not perfect are essentially those used for arbitrary base preschemes, and thereby fall outside the framework of classical algebraic geometry.

5. It did not seem useful to indicate at the head of the written exposés the date or the dates of the corresponding oral exposés of the Seminar. Let us be content to say that the order of the mimeographed exposés (from I to XXVI) corresponds well to the order of the oral exposés. Moreover, the writing of the definitive text is sometimes distinctly later than that of the oral exposé, and often differs from it rather substantially, the written text being generally more detailed and more complete (such as Exp. IV and VII_B), or even appreciably more general (such as Exp. XII or VII_B) than the oral exposé. Other written exposés correspond to no oral exposé (VI_B, VII_A, XV, XVI, XVII, and the essential part of XXVI), and were written and inserted into the mimeographed Seminar, either to provide convenient references for various other exposés (this is notably the case of VI_B), or because they constitute a natural extension of the notions and techniques already developed. One will note as a consequence that the reading of exposés VII_A, VII_B, XV, XVI, XVII is not necessary for the study of the rest of the Seminar, and in particular for the part of this Seminar devoted to reductive group schemes.

6. From the theory of schemes, we shall use above all the general language of schemes, set out in EGA I, the notions of flat morphism, étale morphism, smooth morphism, set out in SGA 1, I to V, and finally the theory of faithfully flat descent of SGA 1, VIII. We have as far as possible avoided formulating unnecessary noetherian hypotheses, which has on the other hand obliged us to replace the usual hypothesis “of finite type” by the hypothesis “of finite presentation”. For the notion of morphism of finite presentation, the reader will consult EGA IV, 1.4 and 1.6. The results of SGA 1, I to IV, stated most often in the noetherian context, will be developed in the general case in EGA IV^(****), where standard methods will likewise be developed in detail for reducing certain types of statements (involving hypotheses of finite presentation) to the noetherian case (EGA IV, paragraphs 8, 9, 11). The reader who does not wish to admit these results of EGA IV may simplify certain statements or their proof by supposing the base prescheme to be locally noetherian. He does, however, expose himself to difficulties in the cases where the proof proceeds by descent from $\hat{A}$ to A , where $\hat{A}$ is the completion of a noetherian local ring A , for this method leads one to introduce the ring (in general non-noetherian) $\hat{A} \otimes_A \hat{A}$.

7. References will be made according to the usual decimal system: the reference 5.7.11 refers to the proposition (or lemma, definition, etc.) of that name in the same exposé; in the reference XVII 7.8 the Roman numeral indicates the number of the exposé. We shall use the following sigla for our standard references:

⁽³⁾**N.D.E.** in particular, theorems 1.1 and 6.1.

^(****)Since the writing of this introduction, the four parts (§§ 1 to 21) of EGA IV have appeared.

- Bible = Séminaire Chevalley “Groupes de Lie algébriques”, 1956/58.
- EGA X , $x.y.z$ = J. Dieudonné and A. Grothendieck, *Éléments de Géométrie Algébrique*, Chap. X , statement $x.y.z$ (or subsection $x.y$, etc.).
- SGA n , X $y.z$ = Séminaire de Géométrie Algébrique du Bois-Marie, year n , statement $y.z$ of exposé X .
- TDTE = A. Grothendieck, *Techniques de descente et Théorèmes d’existence en Géométrie Algébrique*, exposés given in the Bourbaki Seminar between 1959 and 1962.

(1)

⁽¹⁾**N.D.E.** Concerning the quotient by a reductive group, the situation has evolved considerably since the writing of Note (**) by A. Grothendieck. Indeed, for an affine algebraic group over an arbitrary field k , the “good” notion, introduced by Mumford, is that of a “geometrically reductive” group. (One says, on the other hand, that G is “linearly reductive” if every rational representation of G is completely reducible, but, as pointed out in Note (**), this condition is too restrictive if $\text{car}(k) > 0$). According to the results of M. Nagata and W. J. Haboush ([Na64], [NM64], [Ha75]), the geometrically reductive k -groups are exactly the reductive k -groups in the sense of the present Seminar. For all of this, see the later editions of Mumford’s book ([MF82]). Moreover, the extension to the case of an arbitrary base of the notion of “geometric reductivity”, and of its consequences for passage to the quotient, has been carried out by C. S. Seshadri ([Se77]), and additions, due to M. Raynaud, are to be found in the article [CTS79] of J.-L. Colliot-Thélène and J.-J. Sansuc. Finally, for more recent developments concerning passage to the quotient by an algebraic group or, more generally, by a groupoid (cf. Exposé V of the present Seminar), let us mention the articles [Ko97] and [KM97] of J. Kollár and of S. Keel and S. Mori.

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Exposé I. Algebraic structures. Cohomology of groups

by *M. Demazure*

⁽⁰⁾This exposé consists of two parts; the first collects a certain number of general definitions and sets down notations which will often be used in the sequel, the second treats the cohomology of groups and leads to Theorem 5.3.3 (vanishing of the cohomology of diagonalizable groups).

We choose once and for all a Universe.⁽¹⁾ All the definitions introduced and all the constructions carried out will be relative to this Universe. We shall systematically allow ourselves the following abuse of language: in order to define a functor $f: \mathcal{C} \rightarrow \mathcal{C}'$, we shall content ourselves with defining the object $f(S)$ of $\mathcal{C}'$ for every object S of $\mathcal{C}$, whenever there will be no ambiguity as to the manner of defining $f(h)$ for an arrow h of $\mathcal{C}$. In practice, we shall say: let $f: \mathcal{C} \rightarrow \mathcal{C}'$ be the functor defined by $f(S) = \dots$.

1. Generalities

1.1. — Let $\mathcal{C}$ be a category. One will denote by $\widehat{\mathcal{C}}$ the category $\text{Hom}(\mathcal{C}^\circ, (\text{Ens}))$ of contravariant functors from $\mathcal{C}$ to the category (Ens) of sets.⁽²⁾ There exists a canonical functor $\mathbf{h}: \mathcal{C} \rightarrow \widehat{\mathcal{C}}$ which associates with every $X \in \text{Ob}(\mathcal{C})$ the functor $\mathbf{h}_X$ such that

$$\mathbf{h}_X(S) = \text{Hom}(S, X).$$

For every functor $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$, one defines (cf. for example EGA 0_{III}, 8.1.4) a bijection⁽³⁾

$$\text{Hom}(\mathbf{h}_X, \mathbf{F}) \xrightarrow{\sim} \mathbf{F}(X).$$

In particular, for every pair X, Y of objects of $\mathcal{C}$, the canonical map below is bijective:

$$\text{Hom}_{\mathcal{C}}(X, Y) \xrightarrow{\sim} \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{h}_X, \mathbf{h}_Y);$$

i.e. the functor $\mathbf{h}$ is *fully faithful*. It therefore defines an isomorphism of $\mathcal{C}$ onto a *full* subcategory of $\widehat{\mathcal{C}}$, and an equivalence of $\mathcal{C}$ with the *full* subcategory of $\widehat{\mathcal{C}}$ formed of the *representable* functors (i.e. isomorphic to a functor of the form $\mathbf{h}_X$). In the sequel, we shall often identify X and $\mathbf{h}_X$. The following numbers are intended to show that this identification can be made without danger.

Remark 1.1.1. ⁽⁴⁾One sometimes needs the following variant. Let $\mathcal{D}$ be a *full* subcategory of $\mathcal{C}$ and let $X, Y \in \text{Ob}(\mathcal{D})$; denote by $\mathbf{h}'_X$ and $\mathbf{h}'_Y$ the restrictions to $\mathcal{D}$ of $\mathbf{h}_X$ and $\mathbf{h}_Y$. Then one has

$$\text{Hom}_{\mathcal{C}}(X, Y) = \text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_{\widehat{\mathcal{D}}}(\mathbf{h}'_X, \mathbf{h}'_Y)$$

and hence: to give a morphism $X \rightarrow Y$ “is the same thing” as to give, in a manner functorial in T , a map $\varphi(T): \text{Hom}(T, X) \rightarrow \text{Hom}(T, Y)$, for every $T \in \text{Ob}(\mathcal{D})$.

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** Cf. SGA 4, Exp. I, §0 and the Appendix; see also the discussion in [DG70], p. xxvi.

⁽²⁾**N.D.E.** One calls it the category of presheaves on $\mathcal{C}$, cf. IV 4.3.1.

⁽³⁾**N.D.E.** This result is often called the “Yoneda lemma”; we shall use this terminology in other N.D.E.

⁽⁴⁾**N.D.E.** This remark has been added.

1.2. — ⁽⁵⁾One will say that $\mathbf{F}$ is a *subobject* (or a *subfunctor*) of $\mathbf{G}$ if $\mathbf{F}(S)$ is a subset of $\mathbf{G}(S)$ for each S .

In $\widehat{\mathcal{C}}$ “arbitrary” projective limits exist and are computed by:⁽⁶⁾

$$(\varprojlim_i \mathbf{F}_i)(S) = \varprojlim_i \mathbf{F}_i(S).$$

In particular, fibered products are defined by:⁽⁷⁾

$$(\mathbf{F} \times_{\mathbf{G}} \mathbf{F}')(S) = \mathbf{F}(S) \times_{\mathbf{G}(S)} \mathbf{F}'(S).$$

We shall choose as final object of $\widehat{\mathcal{C}}$ the functor $\mathbf{e}$ such that $\mathbf{e}(S) = \{\emptyset\}$.⁽⁸⁾ Every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$ possesses a unique morphism into $\mathbf{e}$, and one sets

$$\mathbf{F} \times \mathbf{F}' = \mathbf{F} \times_{\mathbf{e}} \mathbf{F}'.$$

The functor $\mathbf{h}$ commutes with projective limits; in particular, for $X \times X'$ to exist ($X, X' \in \text{Ob}(\mathcal{C})$), resp. for $\mathcal{C}$ to admit a final object e , it is necessary and sufficient that $\mathbf{h}_X \times \mathbf{h}_{X'}$ be representable, resp. that $\mathbf{e}$ be representable, and one has

$$\mathbf{h}_X \times \mathbf{h}_{X'} \simeq \mathbf{h}_{X \times X'} \quad \text{and} \quad \mathbf{h}_e \simeq \mathbf{e}.$$

A monomorphism of $\widehat{\mathcal{C}}$ is nothing other than a morphism $\mathbf{F} \rightarrow \mathbf{G}$ such that, for $S \in \text{Ob}(\mathcal{C})$, the corresponding map of sets $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is injective.⁽⁹⁾

The functor Γ . For every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$ one sets

$$\Gamma(\mathbf{F}) = \text{Hom}(\mathbf{e}, \mathbf{F});$$

an element of $\Gamma(\mathbf{F})$ is thus a family $(\gamma_S)_{S \in \text{Ob}(\mathcal{C})}$, $\gamma_S \in \mathbf{F}(S)$, such that for every arrow $f: S' \rightarrow S''$ of $\mathcal{C}$, one has $\mathbf{F}(f)(\gamma_{S''}) = \gamma_{S'}$.

One sets $\Gamma(X) = \Gamma(\mathbf{h}_X)$ for $X \in \text{Ob}(\mathcal{C})$. If $\mathcal{C}$ has a final object e , one thus has an isomorphism $\Gamma(X) \simeq \text{Hom}(e, X)$.

1.3. — Let $S \in \text{Ob}(\mathcal{C})$. One denotes by $\mathcal{C}/S$ the category of objects of $\mathcal{C}$ over S , i.e. the category whose objects are the arrows $f: T \rightarrow S$ of $\mathcal{C}$, the set $\text{Hom}(f, f')$ being the subset of $\text{Hom}(T, T')$ formed of the u such that $f = f' \circ u$. If $\mathcal{C}$ possesses a final object e , then $\mathcal{C}/e$ is isomorphic to $\mathcal{C}$. The category $\mathcal{C}/S$ possesses a final object: the identity arrow $S \rightarrow S$.

If $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, then one can form the category $(\mathcal{C}/S)/f$, which one denotes, by abuse of language, $(\mathcal{C}/S)/T$, and one has a canonical isomorphism

$$\mathcal{C}/T \simeq (\mathcal{C}/S)/T.$$

This construction also applies to the category $\widehat{\mathcal{C}}$; one defines in particular the category $\widehat{\mathcal{C}}/\mathbf{h}_S$. On the other hand, one can form the category $\widehat{\mathcal{C}}/S$.

If $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, then $\Gamma(f)$ is identified with the set $\Gamma(T/S)$ of sections of T over S , that is, of the arrows $S \rightarrow T$ which are right inverses of f . Let us observe that $\mathbf{h}_f: \mathbf{h}_T \rightarrow \mathbf{h}_S$ is then an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$ and that one has:

$$\Gamma(\mathbf{h}_f) \simeq \Gamma(\mathbf{h}_T/\mathbf{h}_S) \simeq \Gamma(T/S) \simeq \Gamma(f).$$

⁽⁵⁾**N.D.E.** The order has been modified, so as to introduce fibered products before monomorphisms, cf. N.D.E. (9).

⁽⁶⁾**N.D.E.** Likewise, “arbitrary” inductive limits exist and are computed “argument by argument”, i.e. $(\varinjlim_i \mathbf{F}_i)(S) = \varinjlim_i \mathbf{F}_i(S)$; but in general the functor $\mathbf{h}$ does not commute with inductive limits.

⁽⁷⁾**N.D.E.** In particular, the kernel of a pair of morphisms $u, v: \mathbf{F} \rightrightarrows \mathbf{G}$ is the subfunctor $\text{Ker}(u, v)$ of $\mathbf{F}$ defined by $\text{Ker}(u, v)(S) = \{x \in \mathbf{F}(S) \mid u(x) = v(x)\}$.

⁽⁸⁾**N.D.E.** $\{\emptyset\}$ (the set of subsets of the empty set) denotes the one-element set.

⁽⁹⁾**N.D.E.** If $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is injective for every S , it is clear that $\mathbf{F} \rightarrow \mathbf{G}$ is a monomorphism; the converse is seen by considering the diagram $\mathbf{F} \times_{\mathbf{G}} \mathbf{F} \rightrightarrows \mathbf{F} \rightarrow \mathbf{G}$. One thus obtains that: “ $\mathbf{F} \rightarrow \mathbf{G}$ is a monomorphism if and only if the diagonal morphism $\mathbf{F} \rightarrow \mathbf{F} \times_{\mathbf{G}} \mathbf{F}$ is an isomorphism” (cf. EGA I, 5.3.8). Likewise, it is clear that if $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is surjective for every S , then $\mathbf{F} \rightarrow \mathbf{G}$ is an *epimorphism*, and the converse is seen by considering the amalgamated sum $\mathbf{G} \amalg_{\mathbf{F}} \mathbf{G}$, cf. the proof of Lemma 4.4.4 in Exp. IV.

1.4. — One now proposes to define an equivalence of the categories $\widehat{\mathcal{C}/S}$ and $\widehat{\mathcal{C}}/\mathbf{h}_S$, that is, to prove that “to give a functor on the category of objects of $\mathcal{C}$ over S , is “the same thing” as to give a functor on $\mathcal{C}$ equipped with a morphism into $\mathbf{h}_S$ ”.

(i) *Construction of $\alpha_S: \widehat{\mathcal{C}}/\mathbf{h}_S \rightarrow \widehat{\mathcal{C}/S}$.*

Let first $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ be an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$. One must define a functor $\alpha_S(\mathbf{H})$ on $\mathcal{C}/S$. Let therefore first $f: T \rightarrow S$ be an object of $\mathcal{C}/S$; let us define $\alpha_S(\mathbf{H})(f)$ to be the inverse image of $f \in \mathbf{h}_S(T)$ under the map $\mathbf{H}(T): \mathbf{F}(T) \rightarrow \mathbf{h}_S(T)$.⁽¹⁰⁾

Let next $u: f \rightarrow f'$ be an arrow of $\mathcal{C}/S$; then $\mathbf{F}(u): \mathbf{F}(T') \rightarrow \mathbf{F}(T)$ induces a map from $\alpha_S(\mathbf{H})(f')$ to $\alpha_S(\mathbf{H})(f)$, which one denotes $\alpha_S(\mathbf{H})(u)$. One checks at once that the maps

$$f \mapsto \alpha_S(\mathbf{H})(f) \quad \text{and} \quad u \mapsto \alpha_S(\mathbf{H})(u)$$

do define a functor on $\mathcal{C}/S$, hence an object $\alpha_S(\mathbf{H})$ of $\widehat{\mathcal{C}/S}$.

Let finally $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ and $\mathbf{H}': \mathbf{F}' \rightarrow \mathbf{h}_S$ be two objects of $\widehat{\mathcal{C}}/\mathbf{h}_S$ and $\mathbf{U}: \mathbf{F} \rightarrow \mathbf{F}'$ a morphism of $\widehat{\mathcal{C}}/\mathbf{h}_S$:

$$\begin{array}{ccc} \mathbf{F} & \xrightarrow{\mathbf{U}} & \mathbf{F}' \\ & \searrow \mathbf{H} & \swarrow \mathbf{H}' \\ & \mathbf{h}_S & \end{array}$$

Then for every $f: T \rightarrow S$, the map $\mathbf{U}(T): \mathbf{F}(T) \rightarrow \mathbf{F}'(T)$ induces a map

$$\alpha_S(\mathbf{U})(f): \alpha_S(\mathbf{H})(f) \rightarrow \alpha_S(\mathbf{H}')(f),$$

which defines a morphism of functors

$$\alpha_S(\mathbf{U}): \alpha_S(\mathbf{H}) \rightarrow \alpha_S(\mathbf{H}').$$

One checks easily that the maps

$$\mathbf{H} \mapsto \alpha_S(\mathbf{H}) \quad \text{and} \quad \mathbf{U} \mapsto \alpha_S(\mathbf{U})$$

do define a functor $\alpha_S: \widehat{\mathcal{C}}/\mathbf{h}_S \rightarrow \widehat{\mathcal{C}/S}$.

(ii) **Proposition 1.4.1.** *The functor $\alpha_S: \widehat{\mathcal{C}}/\mathbf{h}_S \rightarrow \widehat{\mathcal{C}/S}$ is an equivalence of categories.*

Let us merely indicate the principle of the construction of a quasi-inverse functor $\beta_S: \widehat{\mathcal{C}/S} \rightarrow \widehat{\mathcal{C}}/\mathbf{h}_S$. Let $\mathbf{G}$ be a functor on $\mathcal{C}/S$; for every object T of $\mathcal{C}$, one sets

$$\beta_S(\mathbf{G})(T) = \text{sum of the sets } \mathbf{G}(f) \text{ for } f \in \text{Hom}(T, S) = \mathbf{h}_S(T),$$

which defines a functor $\beta_S(\mathbf{G})$ on $\mathcal{C}$, equipped with an obvious projection onto $\mathbf{h}_S$.

1.5. — *The equivalence α_S commutes with the functors Γ .* In other words, if $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ is an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$ and $\alpha_S(\mathbf{H})$ the corresponding object of $\widehat{\mathcal{C}/S}$, one has

$$\Gamma(\alpha_S(\mathbf{H})) \simeq \Gamma(\mathbf{H}) \simeq \Gamma(\mathbf{F}/\mathbf{h}_S).$$

The equivalence α_S commutes with the functors $\mathbf{h}$, i.e. if $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, $\mathbf{h}_f: \mathbf{h}_T \rightarrow \mathbf{h}_S$ is an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$ whose transform by α_S is none other than $\mathbf{h}_{\mathcal{C}/S}(f)$, where

$$\mathbf{h}_{\mathcal{C}/S}: \mathcal{C}/S \rightarrow \widehat{\mathcal{C}/S}$$

is the canonical functor⁽¹¹⁾. Consequently:

⁽¹⁰⁾**N.D.E.** For example, if $\mathbf{F} = \mathbf{h}_X$, then $\mathbf{H}$ corresponds to a morphism $h: X \rightarrow S$ and $\mathbf{H}(T): \text{Hom}_{\mathcal{C}}(T, X) \rightarrow \text{Hom}_{\mathcal{C}}(T, S)$ is the map $g \mapsto h \circ g$, whence $\alpha_S(\mathbf{h}_X) = \text{Hom}_{\mathcal{C}/S}(-, X)$.

⁽¹¹⁾**N.D.E.** Cf. N.D.E. (10).

Proposition 1.5.1. *Let $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ be an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$. For $\alpha_S(\mathbf{H}): (\mathcal{C}/S)^\circ \rightarrow (\text{Ens})$ to be representable, it is necessary and sufficient that $\mathbf{F}: \mathcal{C}^\circ \rightarrow (\text{Ens})$ be representable; if $\mathbf{F} \simeq \mathbf{h}_T$, then $\alpha_S(\mathbf{H})$ is representable by the object $T \rightarrow S$ of $\mathcal{C}/S$.*

The equivalence α_S is transitive in S : if $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, one has a commutative diagram of equivalences

$$\begin{array}{ccccc} (\widehat{\mathcal{C}}/\mathbf{h}_S)/\mathbf{h}_T & \xrightarrow{\alpha_S/\mathbf{h}_T} & \widehat{(\mathcal{C}/S)}/\mathbf{h}_T & \xrightarrow{\alpha_f} & \widehat{(\mathcal{C}/S)}/T, \\ & \searrow \simeq & & & \swarrow \simeq \\ & & \widehat{\mathcal{C}}/\mathbf{h}_T & \xrightarrow{\alpha_T} & \widehat{\mathcal{C}}/T \end{array}$$

where $\alpha_S/\mathbf{h}_T$ denotes (provisionally) the *restriction* (cf. 1.6) of the functor α_S to the objects over $\mathbf{h}_T$.

1.6. Base change in a functor. — For every $S \in \text{Ob}(\mathcal{C})$, one has a canonical functor

$$i_S: \mathcal{C}/S \rightarrow \mathcal{C}$$

defined by $i_S(f) = T$ if f is the arrow $T \rightarrow S$. If $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, one denotes $i_{T/S} = i_f$ the functor:

$$i_{T/S}: (\mathcal{C}/S)/T \rightarrow \mathcal{C}/S,$$

and one has the commutative diagram:

$$\begin{array}{ccccc} (\mathcal{C}/S)/T & \xrightarrow{i_{T/S}} & \mathcal{C}/S & \xrightarrow{i_S} & \mathcal{C} . \\ & \searrow \simeq & & & \swarrow i_T \\ & & \mathcal{C}/T & & \end{array}$$

That is, identifying $(\mathcal{C}/S)/T$ with $\mathcal{C}/T$ as we shall do henceforth,

$$i_S \circ i_{T/S} = i_T.$$

In the same way, if one identifies $\mathcal{C}$ and $\mathcal{C}/e$, when $\mathcal{C}$ possesses a final object e , then $i_{S/e}: \mathcal{C}/S \rightarrow \mathcal{C}/e$ is identified with i_S .

For $X \in \text{Ob}(\mathcal{C})$ (resp. $Y \in \text{Ob}(\mathcal{C}/S)$), let $p_S(X)$ (resp. $p_{T/S}(Y)$) be the object of $\mathcal{C}/S$ (resp. of $\mathcal{C}/T$) when it exists, defined by $X \times S$ (resp. $Y \times_S T$) equipped with its second projection:

$$\begin{array}{ccc} X \times S & & Y \times_S T \\ p_S(X) \downarrow & \text{resp.} & \downarrow p_{T/S}(Y) \\ S & & T \end{array}$$

The (partially defined) functor p_S (resp. $p_{T/S}$) is called the *base change functor*. It is, by definition of the product (resp. of the fibered product), the functor right adjoint to the functor i_S (resp. $i_{T/S}$)⁽¹²⁾. One likewise denotes

$$p_S(X) = X_S \quad \text{and} \quad p_{T/S}(Y) = Y_T.$$

The functor i_S defines a functor (*restriction*)

$$i_S^*: \widehat{\mathcal{C}} \rightarrow \widehat{\mathcal{C}}/S;$$

one denotes $\mathbf{F}_S = i_S^*(\mathbf{F}) = \mathbf{F} \circ i_S$. One evidently has

$$i_{T/S}^* \circ i_S^* = i_T^*,$$

⁽¹²⁾**N.D.E.** I.e., if $g: U \rightarrow S$ (resp. $h: V \rightarrow T$) is an object of $\mathcal{C}/S$ (resp. $\mathcal{C}/T$), then $i_S(g) = U$ and $i_{T/S}(h)$ is the object $f \circ h: V \rightarrow S$ of $\mathcal{C}/S$, and one has $\text{Hom}_{\mathcal{C}/S}(U, X \times S) \simeq \text{Hom}_{\mathcal{C}}(U, X)$, resp. $\text{Hom}_{\mathcal{C}/T}(V, X \times_S T) \simeq \text{Hom}_{\mathcal{C}/S}(V, X)$.

that is, for every functor $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$,

$$(\mathbf{F}_S)_T = \mathbf{F}_T.$$

The notation calls for a justification, which is as follows:

Proposition 1.6.1. *For the functor $(\mathbf{h}_X)_S: (\mathcal{C}/S)^\circ \longrightarrow (\text{Ens})$ to be representable, it is necessary and sufficient that the product $X \times S$ exist. One then has*

$$(\mathbf{h}_X)_S \simeq \mathbf{h}_{X_S}.$$

This shows that $\mathbf{F}_S$ has two interpretations: *restriction* of the functor $\mathbf{F}$ to $\mathcal{C}/S$, functor obtained by *base change* $\mathbf{e} \leftarrow S$. This leads to the following notation:

$$\begin{array}{ccccc} \mathbf{F} & & \mathbf{F}_S & & \mathbf{F}_T \\ & \swarrow & & \searrow & \\ \mathbf{e} & \longleftarrow & S & \longleftarrow & T \end{array}$$

which accounts well for the two preceding interpretations.

Let us observe that one has

$$\Gamma(\mathbf{F}_S) \simeq \text{Hom}(\mathbf{h}_S, \mathbf{F}) \simeq \mathbf{F}(S),$$

in particular

$$\Gamma(X_S) \simeq \text{Hom}(S, X).$$

1.7.0. — ⁽¹³⁾Let $\mathbf{E}$ be an object of $\widehat{\mathcal{C}}$. Let us consider the category $\mathcal{C}/_{\mathbf{E}}$ of objects of $\mathcal{C}$ over $\mathbf{E}$: its objects are the pairs (V, ρ) consisting of an object V of $\mathcal{C}$ and of a $\widehat{\mathcal{C}}$ -morphism $\rho: \mathbf{h}_V \longrightarrow \mathbf{E}$, i.e. $\rho \in \mathbf{E}(V)$; a morphism from (V, ρ) to (V', ρ') is the datum of a $\mathcal{C}$ -morphism $f: V \longrightarrow V'$ such that $\rho' \circ f = \rho$ (i.e. $\mathbf{E}(f)(\rho') = \rho$). Let us denote by $\mathbf{L}$ the functor

$$\varinjlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{h}_V,$$

i.e. for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{L}(S) = \varinjlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{h}_V(S)$ is the set of equivalence classes of triples (V, ρ, v) , where $v: S \longrightarrow V$ is a $\mathcal{C}$ -morphism, and where one identifies (V, ρ, v) with $(V', \rho', f \circ v)$ for every $\mathcal{C}$ -morphism $f: V \longrightarrow V'$ such that $\rho' \circ f = \rho$.

Then the map $\varphi_{\mathbf{E}}(S)$ which, to the class of (V, ρ, v) , associates the element $\rho \circ v$ of $\mathbf{E}(S)$ is well-defined, and defines a morphism of functors

$$\varphi_{\mathbf{E}}: \varinjlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{h}_V \longrightarrow \mathbf{E}.$$

Lemma. $\varphi_{\mathbf{E}}$ is an isomorphism.

Indeed, let $S \in \text{Ob}(\mathcal{C})$. Every $x \in \mathbf{E}(S)$ is the image under $\varphi_{\mathbf{E}}(S)$ of the triple (S, x, id_S) ; this shows that $\varphi_{\mathbf{E}}(S)$ is surjective. On the other hand, let $\ell_1 = (V_1, \rho_1, v_1)$ and $\ell_2 = (V_2, \rho_2, v_2)$ be two elements of $\mathbf{L}(S)$ having the same image in $\mathbf{E}(S)$; set $\gamma = \rho_1 \circ v_1 = \rho_2 \circ v_2$. Then ℓ_1 and ℓ_2 are both equal, in $\mathbf{L}(S)$, to the class of the triple (S, γ, id_S) . This shows that $\varphi_{\mathbf{E}}(S)$ is injective.

Corollary. For every object $\mathbf{F}$ of $\widehat{\mathcal{C}}$, one has

$$\text{Hom}(\mathbf{E}, \mathbf{F}) = \varprojlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{F}(V).$$

⁽¹³⁾**N.D.E.** The lemma below has been added (cf. SGA 4, I.3.4); it is used in the proof of 1.7.1 and will be useful several times in the sequel.

1.7. — Hom, Isom, etc., objects. Let $\mathbf{F}$ and $\mathbf{G}$ be two objects of $\widehat{\mathcal{C}}$. We are going to define another object of $\widehat{\mathcal{C}}$ in the following way:

$$\begin{aligned}\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})(S) &= \text{Hom}_{\widehat{\mathcal{C}}/S}(\mathbf{F}_S, \mathbf{G}_S) \\ &\simeq \text{Hom}_{\widehat{\mathcal{C}}/\mathbf{h}_S}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G} \times \mathbf{h}_S) \simeq \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G}).\end{aligned}$$

The object $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ defined above possesses the following properties:

- (i) $\underline{\text{Hom}}(e, \mathbf{G}) \simeq \mathbf{G}$.⁽¹⁴⁾
- (ii) *The formation of $\underline{\text{Hom}}$ commutes with extension of the base:*

$$\underline{\text{Hom}}(\mathbf{F}_S, \mathbf{G}_S) \simeq \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})_S.$$

- (iii) $(\mathbf{F}, \mathbf{G}) \mapsto \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ is a bifunctor, contravariant in $\mathbf{F}$ and covariant in $\mathbf{G}$.

These three properties are obvious from the definitions.

We are going to show that, for every object $\mathbf{E}$ of $\widehat{\mathcal{C}}$, one has

$$\underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

Let $\varphi: \mathbf{E} \times \mathbf{F} \rightarrow \mathbf{G}$; we must associate with it a morphism from $\mathbf{E}$ to $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$. Let therefore $S' \rightarrow S$ be an arrow of $\mathcal{C}$. One has maps

$$\mathbf{E}(S) \times \mathbf{F}(S') \rightarrow \mathbf{E}(S') \times \mathbf{F}(S') \xrightarrow{\varphi(S')} \mathbf{G}(S').$$

Every element e of $\mathbf{E}(S)$ therefore defines, for every $S' \rightarrow S$, a map $\mathbf{F}(S') \rightarrow \mathbf{G}(S')$ functorial in S' , i.e. an element $\theta_\varphi(e)$ of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})(S)$. One has therefore obtained a map

$$\varphi \mapsto \theta_\varphi, \quad \text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \rightarrow \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})),$$

which is “functorial in $\mathbf{E}$ ”.

Proposition 1.7.1. ⁽¹⁵⁾ *Let $\mathbf{E}, \mathbf{F}, \mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$.*

- (a) *The map $\varphi \mapsto \theta_\varphi$ is a bijection:*

$$\text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \xrightarrow{\sim} \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})).$$

- (b) *Moreover, one has an isomorphism of functors:*

$$\underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

(a) Let us consider the two sides as functors in $\mathbf{E}$. The stated result is true if $\mathbf{E} = \mathbf{h}_X$; indeed, in this case it is nothing other than the definition of the functor $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$. On the other hand, the two sides, as functors in $\mathbf{E}$, transform inductive limits into projective limits. Finally, by lemma 1.7.0, every object $\mathbf{E}$ of $\widehat{\mathcal{C}}$ is isomorphic to the inductive limit of the $\mathbf{h}_X$, where X runs through the category $\mathcal{C}/\mathbf{E}$. This proves (a).

Let us sketch a direct proof of (a). To every $\theta \in \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G}))$, one associates the element φ_θ of $\text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G})$ defined as follows. For every $S \in \text{Ob}(\mathcal{C})$, one has a map

$$\theta(S): \mathbf{E}(S) \rightarrow \text{Hom}(\mathbf{F} \times S, \mathbf{G}),$$

functorial in S . If $(e, f) \in \mathbf{E}(S) \times \mathbf{F}(S)$, then f is a morphism $S \rightarrow \mathbf{F}$, hence $f \times \text{id}_S$ is a morphism $S \rightarrow \mathbf{F} \times S$; on the other hand, $\theta(S)(e)$ is a morphism $\mathbf{F} \times S \rightarrow \mathbf{G}$, hence by composition one obtains a morphism

$$\theta(S)(e) \circ (f \times \text{id}_S): S \rightarrow \mathbf{G},$$

i.e. an element $\varphi_\theta(S)(e, f)$ of $\mathbf{G}(S)$. One checks easily that the correspondence $S \mapsto \varphi_\theta(S)$ is functorial in S , hence defines a morphism φ_θ from $\mathbf{E} \times \mathbf{F}$ to $\mathbf{G}$. One leaves it to the reader to check that the maps $\theta \mapsto \varphi_\theta$ and $\varphi \mapsto \theta_\varphi$ are bijections inverse to one another.

⁽¹⁴⁾**N.D.E.** And, if $\mathbf{E}$ is a third object of $\widehat{\mathcal{C}}$, one has $\text{Hom}(\mathbf{E}, \mathbf{F} \times \mathbf{G}) \simeq \text{Hom}(\mathbf{E}, \mathbf{F}) \times \text{Hom}(\mathbf{E}, \mathbf{G})$.

⁽¹⁵⁾**N.D.E.** One has added point (b), which will be useful in II.1 and II.3.11. On the other hand, one has sketched a second, more direct, proof of point (a).

Let us prove (b). If $S \in \text{Ob}(\mathcal{C})$, one has, by 1.7 (ii) and (a) applied to $\mathcal{C}/S$:

$$\begin{aligned} \underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G}))(S) &\simeq \text{Hom}_S(\mathbf{E}_S, \underline{\text{Hom}}_S(\mathbf{F}_S, \mathbf{G}_S)) \\ &\simeq \text{Hom}_S(\mathbf{E}_S \times_S \mathbf{F}_S, \mathbf{G}_S) \\ &\simeq \text{Hom}(\mathbf{E} \times \mathbf{F} \times S, \mathbf{G}) \\ &\simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G})(S) \end{aligned}$$

and these isomorphisms are functorial in S .

Corollary 1.7.2. *One has:*

$$\begin{aligned} \underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) &\simeq \underline{\text{Hom}}(\mathbf{F}, \underline{\text{Hom}}(\mathbf{E}, \mathbf{G})), \\ \underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) &\simeq \underline{\text{Hom}}(\mathbf{F}, \underline{\text{Hom}}(\mathbf{E}, \mathbf{G})). \end{aligned}$$

In particular, setting $\mathbf{E} = e$, and taking into account $\underline{\text{Hom}}(e, \mathbf{G}) \simeq \mathbf{G}$, one has

$$\Gamma(\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \text{Hom}(\mathbf{F}, \mathbf{G}).$$

Let us note that the composition of the $\underline{\text{Hom}}$ furnishes functorial morphisms

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{H}) \longrightarrow \underline{\text{Hom}}(\mathbf{F}, \mathbf{H}).$$

If $\mathbf{F}$ and $\mathbf{G}$ are two objects of $\widehat{\mathcal{C}}$, one denotes by $\text{Isom}(\mathbf{F}, \mathbf{G})$ the subset of $\text{Hom}(\mathbf{F}, \mathbf{G})$ consisting of the isomorphisms from $\mathbf{F}$ onto $\mathbf{G}$. One then defines a subobject $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$ of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ by:

$$\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})(S) = \text{Isom}(\mathbf{F}_S, \mathbf{G}_S).$$

One then has isomorphisms

$$\begin{aligned} \Gamma(\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})) &\simeq \text{Isom}(\mathbf{F}, \mathbf{G}), \\ \underline{\text{Isom}}(\mathbf{F}, \mathbf{G}) &\simeq \underline{\text{Isom}}(\mathbf{G}, \mathbf{F}). \end{aligned}$$

In the particular case where $\mathbf{F} = \mathbf{G}$, one sets

$$\begin{aligned} \underline{\text{End}}(\mathbf{F}) &= \underline{\text{Hom}}(\mathbf{F}, \mathbf{F}), \quad \text{End}(\mathbf{F}) = \text{Hom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\text{End}}(\mathbf{F})), \\ \underline{\text{Aut}}(\mathbf{F}) &= \underline{\text{Isom}}(\mathbf{F}, \mathbf{F}), \quad \text{Aut}(\mathbf{F}) = \text{Isom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\text{Aut}}(\mathbf{F})). \end{aligned}$$

The formation of the objects $\underline{\text{Hom}}$, $\underline{\text{Isom}}$, $\underline{\text{Aut}}$, $\underline{\text{End}}$ commutes with base changes.

Remark 1.7.3. ⁽¹⁶⁾ Let us observe that one can construct an object isomorphic to $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$ in the following way: one has a morphism

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\text{End}}(\mathbf{F});$$

interchanging $\mathbf{F}$ and $\mathbf{G}$, one deduces from it a morphism

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G}).$$

On the other hand, the identity morphism of $\mathbf{F}$ is an element of $\underline{\text{End}}(\mathbf{F})$ and therefore defines a morphism $e \longrightarrow \underline{\text{End}}(\mathbf{F})$. Doing the same with $\mathbf{G}$ and taking the product, one finds a morphism

$$e \longrightarrow \underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G}).$$

It is then immediate that the fiber product of e and of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F})$ over $\underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G})$ is isomorphic to $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$.

All these definitions apply in particular to the case where $\mathbf{F} = \mathbf{h}_X$, $\mathbf{G} = \mathbf{h}_Y$. In the case where $\underline{\text{Hom}}(\mathbf{h}_X, \mathbf{h}_Y)$ is representable by an object of $\mathcal{C}$, one denotes this object by $\text{Hom}(X, Y)$. It possesses the following property: if $Z \times X$ exists, then

$$\text{Hom}(Z, \text{Hom}(X, Y)) \simeq \text{Hom}(Z \times X, Y).$$

This property characterizes it when products exist in $\mathcal{C}$.

⁽¹⁶⁾**N.D.E.** One has added the numbering 1.7.3, for later references.

One likewise defines (whenever they are willing to exist) objects

$$\text{Isom}(X, Y), \quad \text{End}(X), \quad \text{Aut}(X);$$

let us simply observe that, by the construction given above, $\text{Isom}(X, Y)$ exists whenever fiber products exist in $\mathcal{C}$ and $\text{Hom}(X, Y)$, $\text{Hom}(Y, X)$, $\text{End}(X)$ and $\text{End}(Y)$ exist.

Everything that precedes applies equally to categories of the form $\mathcal{C}/S$. The corresponding objects will be denoted as explicitly as possible by appropriate symbols: for example, if T and T' are two objects of $\mathcal{C}$ over S , one will denote by $\text{Hom}_S(T, T')$ the object $\text{Hom}_{\mathcal{C}/S}(T/S, T'/S)$.

1.8. — Constant objects. Let $\mathcal{C}$ be a category in which direct sums and fiber products exist and in which direct sums commute with base changes (for example the category of schemes⁽¹⁷⁾). For every set E and for every object S of $\mathcal{C}$, let us set

$$(1.8.1) \quad E_S = \begin{cases} \text{the direct sum of a family } (S_i)_{i \in E} \\ \text{of objects of } \mathcal{C} \text{ all isomorphic to } S. \end{cases}$$

This object is characterized by the formula:

$$(1.8.2) \quad \text{Hom}_{\mathcal{C}}(E_S, T) = \text{Hom}(E, \text{Hom}_{\mathcal{C}}(S, T)),$$

for every $T \in \text{Ob}(\mathcal{C})$, where the second Hom is taken in the category of sets.

The object E_S is equipped with a canonical projection onto S , in such a way that $E \mapsto E_S$ is in fact a functor from (Ens) to $\mathcal{C}/S$.

⁽¹⁸⁾ If $S' \rightarrow S$ is an arrow of $\mathcal{C}$, one has, since direct sums commute with base changes,

$$E_{S'} = (E_S)_{S'}.$$

In particular, if $\mathcal{C}$ possesses a final object e , one has

$$E_S = (E_e)_S.$$

The functor $E \mapsto E_S/S$, from (Ens) to $\mathcal{C}/S$, commutes with finite products. It suffices, for this, to see that

$$(\times) \quad E_S \times_S F_S = (E \times F)_S.$$

Now, by the results of 1.7 applied to $\mathcal{C}/S$, one has, for every $T \in \text{Ob}(\mathcal{C}/S)$, natural isomorphisms (all unspecified Hom being taken in $\mathcal{C}/S$):

$$\begin{aligned} \text{Hom}((E \times F)_S, T) &\simeq \text{Hom}_{(\text{Ens})}(E \times F, \text{Hom}(S, T)) \\ &\simeq \text{Hom}_{(\text{Ens})}(E, \text{Hom}_{(\text{Ens})}(F, \text{Hom}(S, T))) \simeq \text{Hom}_{(\text{Ens})}(E, \text{Hom}(F_S, T)) \end{aligned}$$

and

$$\begin{aligned} \text{Hom}(E_S \times_S F_S, T) &\simeq \text{Hom}(E_S, \text{Hom}(F_S, T)) \\ &\simeq \text{Hom}_{(\text{Ens})}(E, \text{Hom}(S, \text{Hom}(F_S, T))). \end{aligned}$$

Now, $\text{Hom}(S, \text{Hom}(F_S, T)) = \text{Hom}(F_S, T)$, whence $(\times)$.

Suppose that, $\emptyset$ denoting an initial object of $\mathcal{C}$, the diagram

$$(*) \quad \begin{array}{ccc} \emptyset & \longrightarrow & S \\ \downarrow & & \downarrow \\ S & \longrightarrow & S \amalg S \end{array}$$

is cartesian. (This is the case for the category of schemes.)⁽¹⁹⁾ Then the functor $E \mapsto E_S$ commutes with finite projective limits.

⁽¹⁷⁾**N.D.E.** One has replaced the old terminology “preschemes/schemes” by the current terminology “schemes/separated schemes”.

⁽¹⁸⁾**N.D.E.** In the three paragraphs which follow, one has modified the order of the sentences and added a few clarifications concerning the role of hypothesis $(*)$ below.

⁽¹⁹⁾**N.D.E.** $(*)$ is *not* satisfied if $\mathcal{C}$ is the category whose arrows are $A \rightarrow B$ and id_A, id_B ; in this case $B \amalg B = B$ and $B \times_B B = B \not\cong B$.

Indeed, taking $(\times)$ into account, it suffices to see that $E \mapsto E_S$ commutes with fiber products. Let $u: E \rightarrow G$ and $v: F \rightarrow G$ be two maps of sets. Since in $\mathcal{C}$ direct sums commute with base changes, one has

$$(1) \quad F_S \times_{G_S} E_S \simeq \coprod_{f \in F} S_f \times_{G_S} E_S \simeq \coprod_{\substack{f \in F \\ x \in E}} S_f \times_{G_S} S_x.$$

If $v(f) \neq u(x)$, there exists in $\mathcal{C}/S$ a morphism

$$S_f \times_{G_S} S_x \longrightarrow S_f \times_{S_{v(f)} \coprod S_{u(x)}} S_x;$$

now, by hypothesis $(*)$, the right-hand term is $\emptyset$. Consequently,

$$(2) \quad S_f \times_{G_S} S_x \simeq \emptyset \quad \text{if } v(f) \neq u(x).$$

On the other hand, if $v(f) = u(x)$, there exists in $\mathcal{C}/S$ a morphism

$$S \longrightarrow S_f \times_{G_S} S_x;$$

since $S \xrightarrow{\text{id}} S$ is a final object of $\mathcal{C}/S$, it follows that

$$(3) \quad S_f \times_{G_S} S_x \simeq S \quad \text{if } v(f) = u(x).$$

Combining (1), (2) and (3), one obtains an isomorphism functorial in S

$$F_S \times_{G_S} E_S \simeq \coprod_{F \times_G E} S = (F \times_G E)_S.$$

An object of the form E_S will be called a *constant object*. Let us observe that one has a morphism functorial in E :

$$E \longrightarrow \Gamma(E_S/S)$$

which associates with each $i \in E$ the section of E_S over S defined by the isomorphism of S onto S_i . Suppose condition $(*)$ satisfied for every object S of $\mathcal{C}$; then the morphism $E \rightarrow \Gamma(E_S/S)$ is a monomorphism for every $S \neq \emptyset$.

If $\mathcal{C}$ is the category of schemes, then $\Gamma(E_S/S)$ is identified with the locally constant maps from the topological space S to the set E , the preceding map associating with each element of E the corresponding constant map. Let us observe that it follows from what one has just said that E_S may also be defined as representing the functor which, to every S' over S , associates the set of locally constant functions from the topological space S' to the set E .⁽²⁰⁾

2. Algebraic structures

Given a species of algebraic structure in the category of sets, we propose to extend it to the category $\mathcal{C}$. Let us first treat an example: the case of groups.

2.1. — Group structures. We keep the notations of the preceding paragraph.

Definition 2.1.1. Let $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$. One calls structure of $\widehat{\mathcal{C}}$ -group on $\mathbf{G}$ the datum, for every $S \in \text{Ob}(\mathcal{C})$, of a group structure on the set $\mathbf{G}(S)$, in such a way that for every arrow $f: S' \rightarrow S''$ of $\mathcal{C}$, the map $\mathbf{G}(f): \mathbf{G}(S'') \rightarrow \mathbf{G}(S')$ is a group homomorphism. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, one calls morphism of $\widehat{\mathcal{C}}$ -groups from $\mathbf{G}$ to $\mathbf{H}$ every morphism $u \in \text{Hom}(\mathbf{G}, \mathbf{H})$ such that for every $S \in \text{Ob}(\mathcal{C})$ the map of sets $u(S): \mathbf{G}(S) \rightarrow \mathbf{H}(S)$ is a group homomorphism.

One denotes by $\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H})$ the set of morphisms of $\widehat{\mathcal{C}}$ -groups from $\mathbf{G}$ to $\mathbf{H}$ and by $(\widehat{\mathcal{C}}\text{-Gr.})$ the category of $\widehat{\mathcal{C}}$ -groups.

⁽²⁰⁾**N.D.E.** Let us also note that the diagonal morphism $E_S \rightarrow E_S \times_S E_S = (E \times E)_S$ is a closed immersion, i.e. E_S is separated over S .

Examples. Let $\mathbf{E} \in \text{Ob}(\widehat{\mathcal{C}})$; the object $\underline{\text{Aut}}(\mathbf{E})$ is equipped in an evident way with a structure of $\widehat{\mathcal{C}}$ -group. The final object e possesses a unique structure of $\widehat{\mathcal{C}}$ -group which makes it a final object of $(\widehat{\mathcal{C}}\text{-Gr.})$.

For every $S \in \text{Ob}(\mathcal{C})$, let $e_{\mathbf{G}}(S)$ be the unit element of $\mathbf{G}(S)$. The family of the $e_{\mathbf{G}}(S)$ defines an element $e_{\mathbf{G}} \in \Gamma(\mathbf{G}) = \text{Hom}(e, \mathbf{G})$ which is a morphism of $\widehat{\mathcal{C}}$ -groups $e \rightarrow \mathbf{G}$ and which one calls the *unit section* of $\mathbf{G}$. Let us observe that giving a structure of $\widehat{\mathcal{C}}$ -group on $\mathbf{G}$ amounts to giving a law of composition on $\mathbf{G}$, that is, a $\widehat{\mathcal{C}}$ -morphism

$$\pi_{\mathbf{G}}: \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$$

such that for every $S \in \text{Ob}(\mathcal{C})$, $\pi_{\mathbf{G}}(S)$ equips $\mathbf{G}(S)$ with a group structure.

In the same way, $f: \mathbf{G} \rightarrow \mathbf{H}$ is a morphism of $\widehat{\mathcal{C}}$ -groups if and only if the following diagram is commutative:

$$\begin{array}{ccc} \mathbf{G} \times \mathbf{G} & \xrightarrow{\pi_{\mathbf{G}}} & \mathbf{G} \\ (f,f) \downarrow & & \downarrow f \\ \mathbf{H} \times \mathbf{H} & \xrightarrow{\pi_{\mathbf{H}}} & \mathbf{H}. \end{array}$$

A subobject $\mathbf{H}$ of $\mathbf{G}$ such that, for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{H}(S)$ is a subgroup of $\mathbf{G}(S)$ evidently possesses a structure of $\widehat{\mathcal{C}}$ -group induced by that of $\mathbf{G}$: it is the only one for which the monomorphism $\mathbf{H} \rightarrow \mathbf{G}$ is a morphism of $\widehat{\mathcal{C}}$ -groups. The $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ equipped with this structure is called a sub- $\widehat{\mathcal{C}}$ -group of $\mathbf{G}$.

If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, the product $\mathbf{G} \times \mathbf{H}$ is equipped with an evident structure of $\widehat{\mathcal{C}}$ -group: for every $S \in \text{Ob}(\mathcal{C})$, one equips $\mathbf{G}(S) \times \mathbf{H}(S)$ with the product group structure of the group structures given on $\mathbf{G}(S)$ and $\mathbf{H}(S)$. The $\widehat{\mathcal{C}}$ -group $\mathbf{G} \times \mathbf{H}$ equipped with this structure will be called the product $\widehat{\mathcal{C}}$ -group of $\mathbf{G}$ and of $\mathbf{H}$ (it is, moreover, the product in the category of $\widehat{\mathcal{C}}$ -groups).

If $\mathbf{G}$ is a $\widehat{\mathcal{C}}$ -group, then for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{G}_S$ is a $\widehat{\mathcal{C}}_S$ -group. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, one will define the object $\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H})$ of $\widehat{\mathcal{C}}$ by:

$$\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H})(S) = \text{Hom}_{\widehat{\mathcal{C}}_S\text{-Gr.}}(\mathbf{G}_S, \mathbf{H}_S).$$

(Note: $\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}$ is not in general a $\widehat{\mathcal{C}}$ -group, nor a fortiori the object Hom in the category $(\widehat{\mathcal{C}}\text{-Gr.})$.)

One defines in the same way the objects

$$\text{Isom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H}), \quad \text{End}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}), \quad \text{Aut}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}).$$

Definition 2.1.2. Let $G \in \text{Ob}(\mathcal{C})$. One calls a structure of $\widehat{\mathcal{C}}$ -group on $\mathbf{h}_G \in \text{Ob}(\widehat{\mathcal{C}})$ a structure of $\mathcal{C}$ -group on G . One calls an element $u \in \text{Hom}(G, H) \simeq \text{Hom}(\mathbf{h}_G, \mathbf{h}_H)$ which defines a morphism of $\widehat{\mathcal{C}}$ -group from $\mathbf{h}_G$ to $\mathbf{h}_H$ a morphism from the $\mathcal{C}$ -group G to the $\mathcal{C}$ -group H .

One denotes by $(\mathcal{C}\text{-Gr.})$ the category of $\mathcal{C}$ -groups. Let us note that there exists in (Cat) a cartesian square

$$\begin{array}{ccc} (\mathcal{C}\text{-Gr.}) & \longrightarrow & (\widehat{\mathcal{C}}\text{-Gr.}) \\ \downarrow & & \downarrow \\ \mathcal{C} & \xrightarrow{\mathbf{h}} & \widehat{\mathcal{C}}. \end{array}$$

All the preceding definitions and constructions are therefore transported at once to $(\mathcal{C}\text{-Gr.})$ whenever the functors which they involve (products, Hom objects, etc.) are representable. They also apply to the categories $\mathcal{C}_S$. In this case, we shall write $\text{Hom}_{\mathcal{C}_S\text{-Gr.}}$ for $\text{Hom}_{\widehat{\mathcal{C}}_S\text{-Gr.}}$, etc.

2.2. — More generally, if (T) is a species of structure on n base sets defined by finite projective limits (for example, by commutativities of diagrams constructed with cartesian products: structures of monoid, of group, of set with operators, of module over a ring, of Lie algebra over a ring, etc.), the preceding construction makes it possible to define the notion of “structure of species (T) on n objects $\mathbf{F}_1, \dots, \mathbf{F}_n$ of $\widehat{\mathcal{C}}$ ”: such a structure will be the giving, for each S of $\mathcal{C}$, of a structure of species (T) on the sets $\mathbf{F}_1(S), \dots, \mathbf{F}_n(S)$ in such a way that for every arrow $S' \rightarrow S''$ of $\mathcal{C}$, the family of maps $(\mathbf{F}_i(S'')) \rightarrow (\mathbf{F}_i(S'))$ is a poly-homomorphism for the species of structure (T) . One defines in a similar manner the morphisms of the species of structure (T) , whence a category $(\widehat{\mathcal{C}} \times \widehat{\mathcal{C}} \times \dots \times \widehat{\mathcal{C}})^{(T)}$. The fully faithful functor $(\mathbf{h} \times \mathbf{h} \times \dots \times \mathbf{h})$ then makes it possible to define by inverse image the category $(\mathcal{C} \times \mathcal{C} \times \dots \times \mathcal{C})^{(T)}$, then, as it commutes with projective limits, to transport thereto all the functorial properties, notions and notations introduced in $\widehat{\mathcal{C}}$. Suppose now that fibered products exist in $\mathcal{C}$, and let (T) be a species of algebraic structure defined by the *giving* of certain morphisms between cartesian products satisfying *axioms* consisting of certain commutativities of diagrams constructed by means of the preceding arrows. A structure of species (T) on a family of objects of $\mathcal{C}$ will therefore be defined by certain morphisms between cartesian products satisfying certain commutation conditions. It follows that if $\mathcal{C}$ and $\mathcal{C}'$ are two categories possessing products and $f: \mathcal{C} \rightarrow \mathcal{C}'$ is a functor *commuting with products*, then for every family of objects (F_i) of $\mathcal{C}$ equipped with a structure of species (T) , the family $(f(F_i))$ of objects of $\mathcal{C}'$ will thereby be equipped also with a structure of species (T) . Every $\mathcal{C}$ -group will be transformed into a $\mathcal{C}'$ -group, every pair ($\mathcal{C}$ -ring, $\mathcal{C}$ -module over this $\mathcal{C}$ -ring) into an analogous pair in $\mathcal{C}'$, etc.

Let in particular $\mathcal{C}$ be a category satisfying the conditions of 1.8⁽²¹⁾; the functor $E \mapsto E_S$ defined in *loc. cit.* commutes with finite projective limits; it therefore transforms group into S -group (i.e. $\mathcal{C}_S$ -group), ring into S -ring, etc.

Remark. It is well to remark that the preceding construction procedure, applied to the category $\widehat{\mathcal{C}}$, does indeed give back the notions which one has already defined there; in other words, it amounts to the same to give, on an object of $\widehat{\mathcal{C}}$, a structure of species (T) when one considers this object as a functor on $\mathcal{C}$, or to give a structure of species (T) on the representable functor on $\widehat{\mathcal{C}}$ defined by this object.⁽²²⁾

We shall now treat two further particular cases of the preceding construction, the case of structures with groups of operators and the case of modules.

2.3. Structures with groups of operators. —

Definition 2.3.1. Let $\mathbf{E} \in \text{Ob}(\widehat{\mathcal{C}})$ and $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}}\text{-Gr.})$. A structure of object with $\widehat{\mathcal{C}}$ -group of operators $\mathbf{G}$ (or of $\mathbf{G}$ -object) on $\mathbf{E}$ is the giving, on $\mathbf{E}(S)$, for every $S \in \text{Ob}(\mathcal{C})$, of a structure of set with group of operators $\mathbf{G}(S)$ in such a way that, for every arrow $S' \rightarrow S''$ of $\mathcal{C}$, the map of sets $\mathbf{E}(S'') \rightarrow \mathbf{E}(S')$ is compatible with the homomorphism of operators $\mathbf{G}(S'') \rightarrow \mathbf{G}(S')$.

As usual, it amounts to the same to give a morphism

$$\mu: \mathbf{G} \times \mathbf{E} \rightarrow \mathbf{E}$$

which for every S equips $\mathbf{E}(S)$ with a structure of set with operators $\mathbf{G}(S)$. But $\text{Hom}(\mathbf{G} \times \mathbf{E}, \mathbf{E}) \simeq \text{Hom}(\mathbf{G}, \text{End}(\mathbf{E}))$, hence μ defines a morphism $\mathbf{G} \rightarrow \text{End}(\mathbf{E})$ and it is immediate that the latter maps $\mathbf{G}$ into $\text{Aut}(\mathbf{E})$ and that it is a morphism of $\widehat{\mathcal{C}}$ -groups. Consequently: *to give on $\mathbf{E}$ a structure of object with $\widehat{\mathcal{C}}$ -group of operators $\mathbf{G}$ is equivalent to giving a morphism*

⁽²¹⁾**N.D.E.** Including the condition $(*)$.

⁽²²⁾**N.D.E.** For example, for the group structures: let $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$; if the functor $\widehat{\mathcal{C}} \rightarrow (\text{Ens}), \mathbf{F} \mapsto \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F}, \mathbf{G})$ is equipped with a group structure, the same is true of its restriction to $\mathcal{C}$, $X \mapsto \mathbf{G}(X)$. Conversely, if $\mathbf{G}$ is a $\widehat{\mathcal{C}}$ -group, then the “multiplication” morphism $\pi_{\mathbf{G}}: \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$ induces, for every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$, a group structure on $\text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F}, \mathbf{G})$, functorial in $\mathbf{F}$.

of $\widehat{\mathcal{C}}$ -groups

$$\rho: \mathbf{G} \longrightarrow \text{Aut}(\mathbf{E}).$$

In particular, every element $g \in \mathbf{G}(S)$ defines an automorphism $\rho(g)$ of the functor $\mathbf{E}_S$, that is, an automorphism of $\mathbf{E} \times \mathbf{h}_S$ commuting with the projection $\mathbf{E} \times \mathbf{h}_S \longrightarrow \mathbf{h}_S$, and in particular an automorphism of the set $\mathbf{E}(S')$ for every $S' \longrightarrow S$.

Definition 2.3.2. One denotes by $\mathbf{E}^{\mathbf{G}}$ the subobject of $\mathbf{E}$ defined as follows:

$$\mathbf{E}^{\mathbf{G}}(S) = \{x \in \mathbf{E}(S) \mid x_{S'} \text{ invariant under } \mathbf{G}(S') \text{ for every } S' \longrightarrow S\},$$

where $x_{S'}$ denotes the image of x by $\mathbf{E}(S) \longrightarrow \mathbf{E}(S')$.

Then $\mathbf{E}^{\mathbf{G}}$ (“subobject of the invariants of $\mathbf{G}$ ”) is the largest subobject of $\mathbf{E}$ on which $\mathbf{G}$ operates trivially.

Definition 2.3.3. Let $\mathbf{F}$ be a subobject of $\mathbf{E}$. One denotes by $\text{Norm}_{\mathbf{G}} \mathbf{F}$ and $\text{Centr}_{\mathbf{G}} \mathbf{F}$ the sub- $\widehat{\mathcal{C}}$ -groups of $\mathbf{G}$ defined by

$$\begin{aligned} (\text{Norm}_{\mathbf{G}} \mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}_S = \mathbf{F}_S\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}(S') = \mathbf{F}(S'), \text{ for every } S' \longrightarrow S\}, \\ (\text{Centr}_{\mathbf{G}} \mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}_S} = \text{identity}\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}(S')} = \text{identity}, \text{ for every } S' \longrightarrow S\}, \end{aligned}$$

where the vertical bar after $\rho(g)$ denotes restriction.⁽²³⁾

Scholium 2.3.3.1. ⁽²⁴⁾ In particular, let $x \in \Gamma(\mathbf{E})$, i.e. (cf. 1.2) a collection of elements $x_S \in \mathbf{E}(S)$, $S \in \text{Ob}(\mathcal{C})$, such that for every arrow $f: S' \longrightarrow S$ one has $\mathbf{E}(f)(x_S) = x_{S'}$ (if $\mathcal{C}$ possesses a final object S_0 one has $\Gamma(\mathbf{E}) = \mathbf{E}(S_0)$). Then x defines a subfunctor of $\mathbf{E}$, which one will denote by x , and one has $\text{Norm}_{\mathbf{G}} x = \text{Centr}_{\mathbf{G}} x$. One will denote by $\text{Stab}_{\mathbf{G}}(x)$, and one will call stabilizer of x , this functor; for every $S \in \text{Ob}(\mathcal{C})$ one therefore has:

$$\text{Stab}_{\mathbf{G}}(x)(S) = \{g \in \mathbf{G}(S) \mid \rho(g)x_S = x_S\}.$$

Suppose that fibered products exist in $\mathcal{C}$; if $\mathbf{G} = \mathbf{h}_G$ (resp. $\mathbf{E} = \mathbf{h}_E$), where G is a $\mathcal{C}$ -group (resp. $E \in \text{Ob}(\mathcal{C})$), and if $\mathcal{C}$ possesses a final object S_0 , so that x is a morphism $S_0 \longrightarrow E$, then $\text{Stab}_{\mathbf{G}}(x)$ is representable by the fibered product $G \times_E S_0$, where $G \longrightarrow E$ is the composite of $\text{id}_G \times x: G = G \times S_0 \longrightarrow G \times E$ and of $\mu: G \times E \longrightarrow E$.

Remark 2.3.3.2. ⁽²⁴⁾ The formation of $\mathbf{E}^{\mathbf{G}}$, $\text{Norm}_{\mathbf{G}} \mathbf{F}$ and $\text{Centr}_{\mathbf{G}} \mathbf{F}$ commutes with base change, i.e. for every $S \in \text{Ob}(\mathcal{C})$ one has

$$(\mathbf{E}^{\mathbf{G}})_S = (\mathbf{E}_S)^{\mathbf{G}_S}, \quad (\text{Norm}_{\mathbf{G}} \mathbf{F})_S \simeq \text{Norm}_{\mathbf{G}_S} \mathbf{F}_S, \quad (\text{Centr}_{\mathbf{G}} \mathbf{F})_S \simeq \text{Centr}_{\mathbf{G}_S} \mathbf{F}_S.$$

Definition 2.3.4. If G is a $\mathcal{C}$ -group and $\mathbf{E}$ an object of $\widehat{\mathcal{C}}$ (resp. E an object of $\mathcal{C}$), a structure of G -object on $\mathbf{E}$ (resp. on E) is a structure of $\mathbf{h}_G$ -object on $\mathbf{E}$ (resp. $\mathbf{h}_E$).

In view of this definition, all the notions and notations defined above carry over to $\mathcal{C}$, when they involve only representable functors: for example if $\text{Norm}_{\mathbf{h}_G}(\mathbf{h}_F)$ is representable, then there exists one and only one subobject of G which represents it and which is then a sub- $\mathcal{C}$ -group of G ; one denotes it by $\text{Norm}_G(F)$, etc.

Definition 2.3.5. a) One says that the $\widehat{\mathcal{C}}$ -group $\mathbf{G}$ operates on the $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ if $\mathbf{H}$ is equipped with a structure of $\mathbf{G}$ -object such that, for every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{H}(S)$ defined by g is a group automorphism.

⁽²³⁾**N.D.E.** One has corrected the original, replacing the inclusion $\rho(g)\mathbf{F}_S \subset \mathbf{F}_S$ by an equality, in order to ensure that $\text{Norm}_{\mathbf{G}}(\mathbf{F})$ is indeed a group (see VI_B 6.4 for conditions under which the “transporter” coincides with the “strict transporter”).

⁽²⁴⁾**N.D.E.** One has added scholium 2.3.3.1 and remark 2.3.3.2.

It amounts to the same to say that for every $g \in \mathbf{G}(S)$, the automorphism $\rho(g)$ of $\mathbf{H}_S$ is an automorphism of $\widehat{\mathcal{C}}_S$ -groups, or again that the morphism of $\widehat{\mathcal{C}}$ -groups $\mathbf{G} \rightarrow \text{Aut}(\mathbf{H})$ maps $\mathbf{G}$ into $\text{Aut}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{H})$.

b) In the above situation, there exists on the product $\mathbf{H} \times \mathbf{G}$ a unique structure of $\widehat{\mathcal{C}}$ -group such that, for every S , $(\mathbf{H} \times \mathbf{G})(S)$ is the semi-direct product of the groups $\mathbf{H}(S)$ and $\mathbf{G}(S)$ relative to the given operation of $\mathbf{G}(S)$ on $\mathbf{H}(S)$. One will denote this $\widehat{\mathcal{C}}$ -group by $\mathbf{H} \cdot \mathbf{G}$ and call it the *semi-direct product of $\mathbf{H}$ by $\mathbf{G}$* . One therefore has by definition

$$(\mathbf{H} \cdot \mathbf{G})(S) = \mathbf{H}(S) \cdot \mathbf{G}(S).$$

Let $\mathbf{G}$ be a $\widehat{\mathcal{C}}$ -group. For every arrow $S' \rightarrow S$ of $\mathcal{C}$ and every $g \in \mathbf{G}(S)$, let $\text{Int}(g)$ be the automorphism of $\mathbf{G}(S')$ defined by $\text{Int}(g)h = ghg^{-1}$. This definition extends to that of a morphism of $\widehat{\mathcal{C}}$ -groups

$$\text{Int}: \mathbf{G} \rightarrow \text{Aut}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}) \subset \text{Aut}(\mathbf{G}).$$

Definition 2.3.3 therefore applies, and one has sub- $\widehat{\mathcal{C}}$ -groups of $\mathbf{G}$

$$\text{Norm}_{\mathbf{G}}(\mathbf{E}) \quad \text{and} \quad \text{Centr}_{\mathbf{G}}(\mathbf{E})$$

for every subobject $\mathbf{E}$ of $\mathbf{G}$.

Definition 2.3.6. One calls *center* of $\mathbf{G}$, and one denotes by $\text{Centr}(\mathbf{G})$, the sub- $\widehat{\mathcal{C}}$ -group $\text{Centr}_{\mathbf{G}}(\mathbf{G})$ of $\mathbf{G}$. One says that $\mathbf{G}$ is commutative if $\text{Centr}(\mathbf{G}) = \mathbf{G}$ or, what amounts to the same, if $\mathbf{G}(S)$ is commutative for every S .

One says that the sub- $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ of $\mathbf{G}$ is *invariant* in $\mathbf{G}$ if $\text{Norm}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, what amounts to the same, if $\mathbf{H}(S)$ is invariant in $\mathbf{G}(S)$ for every S .⁽²⁵⁾

Definition 2.3.6.1. ⁽²⁶⁾ Let $f: \mathbf{G} \rightarrow \mathbf{G}'$ be a morphism of $\widehat{\mathcal{C}}$ -groups. One calls *kernel* of f , and one denotes by $\text{Ker } f$, the sub- $\widehat{\mathcal{C}}$ -group of $\mathbf{G}$ defined by

$$(\text{Ker } f)(S) = \{x \in \mathbf{G}(S) \mid f(S)(x) = 1\} = \text{Ker } f(S)$$

for every $S \in \text{Ob}(\mathcal{C})$; it is an *invariant* sub- $\widehat{\mathcal{C}}$ -group.

Let us note that if $\mathbf{G} = \mathbf{h}_G$ and $\mathbf{G}' = \mathbf{h}_{G'}$, if $\mathcal{C}$ possesses a final object S_0 and if fibered products exist in $\mathcal{C}$, then $\text{Ker}(f)$ is *representable* by $S_0 \times_{G'} G$.

Definition 2.3.6.2. ⁽²⁶⁾ Let $\mathbf{E} \in \widehat{\mathcal{C}}$ and let $\mathbf{G}$ be a $\widehat{\mathcal{C}}$ -group operating on $\mathbf{E}$. One says that the operation of $\mathbf{G}$ on $\mathbf{E}$ is *faithful* if the kernel of the morphism $\mathbf{G} \rightarrow \text{Aut}(\mathbf{E})$ is trivial, i.e. if for every $S \in \text{Ob}(\mathcal{C})$ and $g \in \mathbf{G}(S)$, the condition $gs' \cdot x = x$ for every $S' \rightarrow S$ and $x \in \mathbf{E}(S')$, implies $g = 1$.

Many definitions and propositions of the elementary theory of groups transpose easily. Let us simply point out the following one, which will be useful to us:

Proposition 2.3.7. Let $f: \mathbf{W} \rightarrow \mathbf{G}$ be a morphism of $\widehat{\mathcal{C}}$ -groups. Set $\mathbf{H}(S) = \text{Ker } f(S)$. Let $u: \mathbf{G} \rightarrow \mathbf{W}$ be a morphism of $\widehat{\mathcal{C}}$ -groups which is a section of f (and which is then necessarily a monomorphism). Then $\mathbf{W}$ is identified with the semi-direct product of $\mathbf{H}$ by $\mathbf{G}$ for the operation of $\mathbf{G}$ on $\mathbf{H}$ defined by $(g, h) \mapsto \text{Int}(u(g))h$ for $g \in \mathbf{G}(S)$, $h \in \mathbf{H}(S)$, $S \in \text{Ob}(\mathcal{C})$.

The set of these definitions and propositions carries over as usual to $\mathcal{C}$. One defines in particular the semi-direct product of two $\mathcal{C}$ -groups H and G (G operating on H) when the cartesian product $H \times G$ exists, and one has the analogue of proposition 2.3.7 in the following form:

⁽²⁵⁾**N.D.E.** Moreover, one says that $\mathbf{H}$ is *central* in $\mathbf{G}$ if $\text{Centr}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, what amounts to the same, if $\mathbf{H}(S)$ is central in $\mathbf{G}(S)$ for every S .

⁽²⁶⁾**N.D.E.** One has added definitions 2.3.6.1 and 2.3.6.2.

Proposition 2.3.8. *Let $H \xrightarrow{i} W \xrightarrow{f} G$ be a sequence of morphisms of $\mathcal{C}$ -groups such that for every $S \in \text{Ob}(\mathcal{C})$, $(H(S), i(S))$ is a kernel of $f(S): W(S) \rightarrow G(S)$. Let $u: G \rightarrow W$ be a morphism of $\mathcal{C}$ -groups which is a section of f . Then W is identified with the semi-direct product of H by G for the operation of G on H such that if $S \in \text{Ob}(\mathcal{C})$, if $g \in G(S)$ and $h \in H(S)$, one has $\text{Int}(u(g))i(h) = i(g^h)$.*

3. The category of $\mathbf{O}$ -modules, the category of $\mathbf{G}$ - $\mathbf{O}$ -modules

Definition 3.1. Let $\mathbf{O}$ and $\mathbf{F}$ be two objects of $\widehat{\mathcal{C}}$. One says that $\mathbf{F}$ is a $\widehat{\mathcal{C}}$ -module over the $\widehat{\mathcal{C}}$ -ring $\mathbf{O}$, or for short an $\mathbf{O}$ -module, if, for every $S \in \text{Ob}(\mathcal{C})$, one has endowed $\mathbf{O}(S)$ with a ring structure and $\mathbf{F}(S)$ with a module structure over this ring, in such a way that for every arrow $S' \rightarrow S''$ of $\mathcal{C}$, $\mathbf{O}(S'') \rightarrow \mathbf{O}(S')$ is a ring homomorphism and $\mathbf{F}(S'') \rightarrow \mathbf{F}(S')$ a homomorphism of abelian groups compatible with this ring homomorphism.

If $\mathbf{O}$ is fixed, one defines in the usual way a morphism of the $\mathbf{O}$ -modules $\mathbf{F}$ and $\mathbf{F}'$, whence the commutative group $\text{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and the category of $\mathbf{O}$ -modules, denoted $(\mathbf{O}\text{-Mod.})$.

Lemma 3.1.1. ⁽²⁷⁾ *$(\mathbf{O}\text{-Mod.})$ is endowed with the structure of an abelian category, defined “argument by argument”. Moreover, $(\mathbf{O}\text{-Mod.})$ satisfies axiom (AB 5) (cf. [Gr57, 1.5]), i.e. arbitrary direct sums exist and if $\mathbf{M}$ is an $\mathbf{O}$ -module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i \in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then*

$$\bigcup_{i \in I} (\mathbf{F}_i \cap \mathbf{N}) = \left(\bigcup_{i \in I} \mathbf{F}_i \right) \cap \mathbf{N}.$$

Indeed, let $f: \mathbf{F} \rightarrow \mathbf{F}'$ be a morphism of $\mathbf{O}$ -modules. One defines the $\mathbf{O}$ -modules $\text{Ker } f$ (resp. $\text{Im } f$ and $\text{Coker } f$) by setting, for every $S \in \text{Ob}(\mathcal{C})$, $(\text{Ker } f)(S) = \text{Ker } f(S)$ (resp. $\dots$). Then $\text{Ker } f$ (resp. $\text{Coker } f$) is a kernel (resp. a cokernel) of f , and one has an isomorphism of $\mathbf{O}$ -modules $\mathbf{F}/\text{Ker } f \xrightarrow{\sim} \text{Im } f$. This proves that $(\mathbf{O}\text{-Mod.})$ is an abelian category.

Arbitrary direct sums exist and are defined “argument by argument”. Finally, if $\mathbf{M}$ is an $\mathbf{O}$ -module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i \in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then the inclusion

$$\bigcup_{i \in I} (\mathbf{F}_i \cap \mathbf{N}) \subset \left(\bigcup_{i \in I} \mathbf{F}_i \right) \cap \mathbf{N}$$

is an equality: indeed, if $S \in \text{Ob}(\mathcal{C})$ and $x \in \mathbf{N}(S) \cap \bigcup_i \mathbf{F}_i(S)$, there exists $i \in I$ such that $x \in \mathbf{N}(S) \cap \mathbf{F}_i(S)$.

Proposition 3.1.2. ⁽²⁷⁾ *One assumes that the category $\mathcal{C}$ is small, i.e. that $\text{Ob}(\mathcal{C})$ is a set. Then $(\mathbf{O}\text{-Mod.})$ has a generator, and hence enough injective objects.*

PROOF. For every object S of $\mathcal{C}$, one defines the $\mathbf{O}$ -module $\mathbf{O}_S$ as follows. For every object S' , $\mathbf{O}_S(S')$ is a direct sum of copies of $\mathbf{O}(S')$ indexed by $\text{Hom}(S', S)$, i.e.

$$\mathbf{O}_S(S') = \bigoplus_{g: S' \rightarrow S} \mathbf{O}(S') g.$$

For every morphism $f: S'' \rightarrow S'$, denoting by f^* the ring homomorphism $\mathbf{O}(S') \rightarrow \mathbf{O}(S'')$, the morphism $\mathbf{O}_S(f): \mathbf{O}_S(S') \rightarrow \mathbf{O}_S(S'')$ sends each element ag of $\mathbf{O}_S(S')$ (where $a \in \mathbf{O}(S')$ and $g: S' \rightarrow S$) to the element $f^*(a)g \circ f$ of $\mathbf{O}_S(S'')$. One easily checks that this does indeed define a functor in $\mathbf{O}$ -modules.

Since, by hypothesis, $\text{Ob}(\mathcal{C})$ is a set, one may consider the direct sum $\mathbf{G} = \bigoplus_{S \in \text{Ob}(\mathcal{C})} \mathbf{O}_S$.

⁽²⁷⁾**N.D.E.** Statements 3.1.1 and 3.1.2 have been added in order to show that the category $(\mathbf{O}\text{-Mod.})$ is abelian and satisfies axiom (AB 5), and moreover has enough injective objects if the category $\mathcal{C}$ is small. An error in 3.1.2, pointed out in 2016 by Linyuan Liu, is corrected here.

Lemma 3.1.2.1. *Let $\mathbf{F}$ be an $\mathbf{O}$ -module. For every $S \in \text{Ob}(\mathcal{C})$, every element x of $\mathbf{F}(S)$ defines a unique morphism of $\mathbf{O}$ -modules $x: \mathbf{O}_S \rightarrow \mathbf{F}$ such that $x(1 \text{id}_S) = x$.*

PROOF. Let ϕ be a morphism of $\mathbf{O}$ -modules $\mathbf{O}_S \rightarrow \mathbf{F}$ such that $\phi(1 \text{id}_S) = x$. Let $g: S' \rightarrow S$ and $a \in \mathbf{O}(S')$. The $\mathbf{O}(S')$ -linearity and the commutativity of the diagram

$$\begin{array}{ccc} \mathbf{O}_S(S) & \xrightarrow{\phi} & \mathbf{F}(S) \\ \mathbf{O}_S(g) \downarrow & & \downarrow \mathbf{F}(g) \\ \mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \end{array}$$

imply that $\phi(ag) = a\phi(1g) = a\mathbf{F}(g)(\phi(1 \text{id}_S)) = a\mathbf{F}(g)(x)$. Thus ϕ , if it exists, is determined by the preceding equality. Conversely, if one defines ϕ in this way, then for every $f: S'' \rightarrow S'$ one has

$$\phi \circ \mathbf{O}_S(f)(ag) = \phi(f^*(a)g \circ f) = f^*(a)\mathbf{F}(g \circ f)(x) = \mathbf{F}(f)(a\mathbf{F}(g)(x)) = \mathbf{F}(f) \circ \phi(ag),$$

i.e. the diagram

$$\begin{array}{ccc} \mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \\ \mathbf{O}_S(f) \downarrow & & \downarrow \mathbf{F}(f) \\ \mathbf{O}_S(S'') & \xrightarrow{\phi} & \mathbf{F}(S'') \end{array}$$

is commutative. This proves the lemma. $\square$

Now let $\mathbf{F}$ be an $\mathbf{O}$ -module. For every $S \in \text{Ob}(\mathcal{C})$, let $F_0(S)$ be a system of generators of the $\mathbf{O}(S)$ -module $\mathbf{F}(S)$, and let $\mathbf{L}_S$ be the direct sum, indexed by $F_0(S)$, of copies of $\mathbf{O}_S$. By the lemma, one obtains a morphism of $\mathbf{O}$ -modules $\mathbf{L}_S \rightarrow \mathbf{F}$ which is such that the morphism $\mathbf{L}_S(S) \rightarrow \mathbf{F}(S)$ is surjective, hence an epimorphism of the category of $\mathbf{O}(S)$ -modules.

Set $I = \bigsqcup_{S \in \text{Ob}(\mathcal{C})} F_0(S)$. Then one obtains a morphism of $\mathbf{O}$ -modules

$$\mathbf{G}^{\oplus I} \rightarrow \bigoplus_{S \in \text{Ob}(\mathcal{C})} \mathbf{L}_S \rightarrow \mathbf{F}$$

which is an epimorphism “argument by argument”, hence an epimorphism of $(\mathbf{O}\text{-Mod.})$. This shows that $\mathbf{G}$ is a generator of $(\mathbf{O}\text{-Mod.})$ (cf. [Gr57, 1.9.1]). Since $(\mathbf{O}\text{-Mod.})$ satisfies (AB 5), it then follows from [Gr57, 1.10.1] that $(\mathbf{O}\text{-Mod.})$ has enough injective objects. $\square$

Remark 3.1.3. If $\mathbf{O}_0$ is the $\widehat{\mathcal{C}}$ -ring defined by $\mathbf{O}_0(S) = \mathbb{Z}$ (which must not be confused with the functor associated with the constant object $\mathbb{Z}$), then the category of $\mathbf{O}_0$ -modules is isomorphic to the category of commutative $\widehat{\mathcal{C}}$ -groups.

Definition 3.1.4. Let us observe that, if $\mathbf{F}$ is an $\mathbf{O}$ -module, then for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{F}_S$ is an $\mathbf{O}_S$ -module. One may therefore define an abelian $\widehat{\mathcal{C}}$ -group $\text{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$ by

$$\text{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')(S) = \text{Hom}_{\mathbf{O}_S}(\mathbf{F}_S, \mathbf{F}'_S).$$

One will likewise define objects

$$\text{Isom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}'), \quad \text{End}_{\mathbf{O}}(\mathbf{F}) \quad \text{and} \quad \text{Aut}_{\mathbf{O}}(\mathbf{F}),$$

the last being a $\widehat{\mathcal{C}}$ -group for the structure induced by the composition of automorphisms.

Definition 3.2. Let $\mathbf{O}$ be a $\widehat{\mathcal{C}}$ -ring, $\mathbf{F}$ an $\mathbf{O}$ -module, and $\mathbf{G}$ a $\widehat{\mathcal{C}}$ -group. A structure of $\mathbf{G}$ -object such that, for every $S \in \text{Ob}(\mathcal{C})$ and every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{F}(S)$ defined by g is an automorphism of its structure of $\mathbf{O}(S)$ -module, is called a *structure of $\mathbf{G}$ - $\mathbf{O}$ -module on $\mathbf{F}$* .

It amounts to the same to say that the morphism of $\widehat{\mathcal{C}}$ -groups

$$\rho: \mathbf{G} \rightarrow \text{Aut}(\mathbf{F})$$

defined in 2.3 sends $\mathbf{G}$ into the $\widehat{\mathcal{C}}$ -subgroup $\text{Aut}_{\mathbf{O}}(\mathbf{F})$ of $\text{Aut}(\mathbf{F})$. To give a *structure of $\mathbf{G}$ - $\mathbf{O}$ -module* on the $\mathbf{O}$ -module $\mathbf{F}$ is thus to give a *morphism of $\widehat{\mathcal{C}}$ -groups*

$$\rho: \mathbf{G} \longrightarrow \text{Aut}_{\mathbf{O}}(\mathbf{F}).$$

One defines in a natural way the abelian group $\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, hence the additive category of $\mathbf{G}$ - $\mathbf{O}$ -modules, denoted $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$.

Remark 3.2.1. The reader will observe that $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ may equally be defined as the category of $\mathbf{O}[\mathbf{G}]$ -modules, where $\mathbf{O}[\mathbf{G}]$ is the algebra of the $\widehat{\mathcal{C}}$ -group $\mathbf{G}$ over the $\widehat{\mathcal{C}}$ -ring $\mathbf{O}$, whose definition is clear.⁽²⁸⁾ Consequently, by 3.1.1 and 3.1.2, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ is an abelian category satisfying axiom (AB 5); moreover, if $\mathcal{C}$ is *small*, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects.

All the constructions of this paragraph specialize at once to the case where $\mathbf{G}$ (or $\mathbf{O}$, or both) are representable by objects of $\mathcal{C}$ which are thereby endowed with the corresponding algebraic structures.

We have treated succinctly the case of the principal algebraic structures encountered in the rest of this seminar. For the others (the structure of $\mathbf{O}$ -Lie algebra, for example), we believe that the reader will have had enough examples in this paragraph to be able, in each particular case, to put into operation for himself the general mechanism sketched in 2.2.

We are now going to apply what we have just done to the category of schemes, denoted (Sch) , and more generally to the categories $(\text{Sch})/S$ (which one will also denote (Sch/S)).

4. Algebraic structures in the category of schemes

We shall permit ourselves, whenever there is no ambiguity, the following abuses of language: one will denote by T the object $T \xrightarrow{f} S$ of $\mathcal{C}/S$, the datum of f ("structural morphism of T ") being understood, and one will identify $\mathcal{C}$ with a subcategory of $\widehat{\mathcal{C}}$. In view of the compatibilities stated in the preceding paragraphs, these identifications may be made without danger.

On the other hand, we shall simplify the terminology on the following model: a (Sch) -group will also be called a *group scheme*, a $(\text{Sch})/S$ -group a *group scheme over S* , or an *S -group scheme*, or an *S -group*, or an *A -group* when S is the spectrum of the ring A .

4.1. Constant schemes. — The category of schemes satisfies the conditions required in 1.8. One may therefore define constant objects there; for every set E , one has a scheme $E_{\mathbb{Z}}$ and, for every scheme S , an S -scheme $E_S = (E_{\mathbb{Z}})_S$. Let us recall (cf. *loc. cit.*) that for every S -scheme T , $\text{Hom}_S(T, E_S)$ is the set of locally constant maps from the topological space T to E .

The functor $E \mapsto E_S$ commutes with finite projective limits. In particular, if G is a group, G_S is an *S -group scheme*; if O is a ring, O_S is an *S -ring scheme*, etc.

4.2. Affine S -groups. — Let us recall a certain number of things about affine S -schemes (EGA II, § 1). One says that the S -scheme T is *affine over S* if the inverse image of every affine open of S is affine. The $\mathcal{O}_S$ -algebra $f_*(\mathcal{O}_T)$, which one denotes by $\mathcal{A}(T)$, is then *quasi-coherent* (f denotes the structural morphism of T). Conversely, to every quasi-coherent $\mathcal{O}_S$ -algebra $\mathcal{A}$, one may associate an S -scheme affine over S , denoted $\text{Spec}(\mathcal{A})$. These functors $T \mapsto \mathcal{A}(T)$ and $\mathcal{A} \mapsto \text{Spec}(\mathcal{A})$ are equivalences quasi-inverse to one another between the category of S -schemes affine over S and the category opposite to that of quasi-coherent $\mathcal{O}_S$ -algebras.

⁽²⁸⁾**N.D.E.** The following sentence has been added; this will be used in Section 5.

It follows that to give an algebraic structure on an S -scheme T affine over S is equivalent to giving the corresponding structure on $\mathcal{A}(T)$ in the category opposite to that of quasi-coherent $\mathcal{O}_S$ -algebras. In particular, if G is an affine S -group over S , $\mathcal{A}(G)$ is endowed with the structure of an augmented $\mathcal{O}_S$ -bialgebra, that is, one has two *morphisms of $\mathcal{O}_S$ -algebras*

$$\Delta: \mathcal{A}(G) \longrightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \quad \text{and} \quad \varepsilon: \mathcal{A}(G) \longrightarrow \mathcal{O}_S$$

corresponding to the morphisms of S -schemes

$$\pi: G \times G \longrightarrow G \quad \text{and} \quad e_G: S \longrightarrow G.$$

The maps Δ and ε satisfy the following conditions (which express that G is an S -monoid):⁽²⁹⁾

(HA 1) Δ is coassociative: the following diagram is commutative

$$\begin{array}{ccccc} & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) & & \\ & \Delta \nearrow & & \searrow \text{Id} \otimes \Delta & \\ \mathcal{A}(G) & & & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ & \Delta \searrow & & \nearrow \Delta \otimes \text{Id} & \\ & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) & & \end{array}$$

(HA 2): Δ is compatible with ε : the following two composites are the identity

$$\begin{aligned} \mathcal{A}(G) &\xrightarrow{\Delta} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\text{Id} \otimes \varepsilon} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathcal{A}(G), \\ \mathcal{A}(G) &\xrightarrow{\Delta} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\varepsilon \otimes \text{Id}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\sim} \mathcal{A}(G). \end{aligned}$$

Let us take the opportunity to remark once again that it follows from the definition of an S -group structure that, in order to give such a structure on an S -scheme G affine over S , it is not necessary to check anything at all on $\mathcal{A}(G)$, but simply to endow each $G(S')$, for S' over S , with a group structure functorial in S' . This remark applies *mutatis mutandis* to morphisms.

4.3. The groups $\mathbb{G}_a$ and $\mathbb{G}_m$. The ring $\mathbb{O}$

4.3.1. — Let $\mathbf{G}_a$ be the functor from $(\text{Sch})^\circ$ to (Ens) defined by

$$\mathbf{G}_a(S) = \Gamma(S, \mathcal{O}_S),$$

endowed with the structure of $(\widehat{\text{Sch}})$ -group defined by the additive group structure of the ring $\Gamma(S, \mathcal{O}_S)$. It is representable by an affine scheme which one will denote by $\mathbb{G}_a$, and which is therefore a *group scheme*

$$\mathbb{G}_a = \text{Spec } \mathbb{Z}[T].$$

Indeed, $\mathbf{G}_a(S) = \text{Hom}(S, \mathbb{G}_a) = \text{Hom}_{\text{Alg.}}(\mathbb{Z}[T], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S)$.

For every scheme S , one therefore has an affine S -group over S , denoted $\mathbb{G}_{a,S}$, which corresponds to the $\mathcal{O}_S$ -bialgebra $\mathcal{O}_S[T]$, with the diagonal map and the augmentation defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \quad \varepsilon(T) = 0.$$

4.3.2. — Let $\mathbf{G}_m$ be the functor from $(\text{Sch})^\circ$ to (Ens) defined by

$$\mathbf{G}_m(S) = \Gamma(S, \mathcal{O}_S)^\times,$$

⁽²⁹⁾**N.D.E.** And, of course, the inversion morphism $G \longrightarrow G$ induces a morphism of $\mathcal{O}_S$ -algebras $\tau: \mathcal{A}(G) \longrightarrow \mathcal{A}(G)$ which, together with Δ and ε , makes $\mathcal{A}(G)$ into a Hopf $\mathcal{O}_S$ -algebra.

where $\Gamma(S, \mathcal{O}_S)^\times$ denotes the multiplicative group of invertible elements of the ring $\Gamma(S, \mathcal{O}_S)$, endowed with its natural structure of $(\widehat{\mathbf{Sch}})$ -group. It is representable by an affine scheme denoted $\mathbb{G}_m$

$$\mathbb{G}_m = \mathrm{Spec} \mathbb{Z}[T, T^{-1}] = \mathrm{Spec} \mathbb{Z}[\mathbb{Z}],$$

where $\mathbb{Z}[\mathbb{Z}]$ denotes the algebra of the group $\mathbb{Z}$ over the ring $\mathbb{Z}$. Indeed

$$\mathbb{G}_m(S) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[T, T^{-1}], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S)^\times.$$

For every scheme S , one therefore has an S -group affine over S , denoted $\mathbb{G}_{m,S}$, which corresponds to the $\mathcal{O}_S$ -algebra $\mathcal{O}_S[\mathbb{Z}]$, with the diagonal map and the augmentation defined by:

$$\Delta(x) = x \otimes x \quad \text{and} \quad \varepsilon(x) = 1, \quad \text{for } x \in \mathbb{Z}.$$

4.3.3. — Let $\mathbf{O}$ be the functor $\mathbf{G}_a$ endowed with its structure of $(\widehat{\mathbf{Sch}})$ -ring. It is represented by the scheme $\mathrm{Spec} \mathbb{Z}[T]$, which one will denote by $\mathbb{O}$ when one considers it as endowed with its structure of *ring scheme*.

For every scheme S , $\mathbb{O}_S = S \times_{\mathrm{Spec} \mathbb{Z}} \mathrm{Spec} \mathbb{Z}[T] = \mathrm{Spec}(\mathcal{O}_S[T])$ is therefore a ring S -scheme, affine over S . (*Nota:* in EGA II 1.7.13, $\mathbb{O}_S$ is denoted by $S[T]$).

4.3.3.1. — ⁽³⁰⁾ For every object $\mathbf{X}$ of $(\widehat{\mathbf{Sch}})$, $\mathbf{O}(\mathbf{X}) = \mathrm{Hom}(\mathbf{X}, \mathbf{O})$ is endowed with a ring structure, functorial in $\mathbf{X}$. In particular, if X' is a scheme and if one is given morphisms $x: X' \rightarrow \mathbf{X}$ and $f: \mathbf{X} \rightarrow \mathbf{O}$ (i.e., $f \in \mathbf{O}(\mathbf{X})$), then $f(x) = f \circ x$ is an element of $\mathbf{O}(X') = \Gamma(X', \mathcal{O}_{X'})$.

Definition. — Let $\pi: \mathbf{M} \rightarrow \mathbf{X}$ be a morphism of $(\widehat{\mathbf{Sch}})$, and let $\mathbf{O}_{\mathbf{X}} = \mathbf{O} \times \mathbf{X}$. One says that $\mathbf{M}$ is an $\mathbf{O}_{\mathbf{X}}$ -module if one is given, for every $\mathbf{X}$ -scheme X' , a structure of $\mathbf{O}(X')$ -module on $\mathrm{Hom}_{\mathbf{X}}(X', \mathbf{M})$, functorial in X' .

This amounts to giving a law of abelian $\mathbf{X}$ -group $\mu: \mathbf{M} \times_{\mathbf{X}} \mathbf{M} \rightarrow \mathbf{M}$ and an “external law”

$$\mathbf{O} \times \mathbf{M} = \mathbf{O}_{\mathbf{X}} \times_{\mathbf{X}} \mathbf{M} \rightarrow \mathbf{M}, \quad (f, m) \mapsto f \cdot m$$

which is an $\mathbf{X}$ -morphism (i.e. $\pi(f \cdot m) = \pi(m)$) and which, for every $x \in \mathbf{X}(S)$, endows $\mathbf{M}(x) = \{m \in \mathbf{M}(S) \mid \pi(m) = x\}$ with a structure of $\mathbf{O}(S)$ -module.

In this case, for every $\mathbf{Y} \in \mathrm{Ob}(\widehat{\mathbf{Sch}})_{/\mathbf{X}}$ (not necessarily representable), $\mathrm{Hom}_{\mathbf{X}}(\mathbf{Y}, \mathbf{M}) = \Gamma(\mathbf{M}_{\mathbf{Y}}/\mathbf{Y})$ is an $\mathbf{O}(\mathbf{Y})$ -module, functorially in $\mathbf{Y}$.

4.4. Diagonalizable groups.

4.4.1. — The construction of $\mathbb{G}_m$ generalizes as follows: let M be a commutative group and $M_{\mathbb{Z}}$ the group scheme associated with it by the procedure of 4.1. Consider the functor defined by

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{groups}}(M, \mathbb{G}_m(S)) \simeq \mathrm{Hom}_{S\text{-Gr.}}(M_S, \mathbb{G}_{m,S}).$$

It is a commutative $(\widehat{\mathbf{Sch}})$ -group which is representable by a *group scheme* which one will denote by $\mathbf{D}(M)$; one will therefore have by definition:

$$\mathbf{D}(M) \simeq \mathrm{Hom}_{(\mathrm{Sch})\text{-Gr.}}(M_{\mathbb{Z}}, \mathbb{G}_m).$$

Indeed, set

$$\mathbf{D}(M) = \mathrm{Spec} \mathbb{Z}[M],$$

where $\mathbb{Z}[M]$ is the algebra of the group M over the ring $\mathbb{Z}$; one has

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[M], \Gamma(S, \mathcal{O}_S)) \simeq \mathrm{Hom}_{\mathrm{groups}}(M, \Gamma(S, \mathcal{O}_S)^\times)$$

by the very definition of the algebra $\mathbb{Z}[M]$.

4.4.2. — For every scheme S one therefore has an S -group scheme affine over S

$$\mathbf{D}_S(M) = \mathbf{D}(M)_S = \mathrm{Hom}_{(\mathrm{Sch})\text{-Gr.}}(M_{\mathbb{Z}}, \mathbb{G}_m)_S = \mathrm{Hom}_{S\text{-Gr.}}(M_S, \mathbb{G}_{m,S}).$$

⁽³⁰⁾**N.D.E.** This paragraph has been added; it will be useful later (cf. II 1.3).

It is associated with the $\mathcal{O}_S$ -bialgebra $\mathcal{O}_S[M]$, equipped with the diagonal map and the augmentation defined by

$$\Delta(x) = x \otimes x \quad \text{and} \quad \varepsilon(x) = 1, \quad \text{for } x \in M.$$

4.4.3. — If $f: M \rightarrow N$ is a homomorphism of commutative groups, one defines in an evident way a morphism of S -groups

$$\mathbf{D}_S(f): \mathbf{D}_S(N) \rightarrow \mathbf{D}_S(M),$$

whence a *functor*

$$\mathbf{D}_S: M \mapsto \mathbf{D}_S(M)$$

from the category of abelian groups to the category of affine S -groups over S , which one can also define as the composite of the functor $M \mapsto M_S$ and of the functor $M_S \mapsto \text{Hom}_{S\text{-Gr.}}(M_S, \mathbb{G}_{m,S})$. This functor *commutes with extensions of the base*.

An S -group isomorphic to a group of the form $\mathbf{D}_S(M)$ is said to be *diagonalizable*. Let us note that the elements of M are interpreted as certain characters of $\mathbf{D}_S(M)$, that is, certain elements of $\text{Hom}_{S\text{-Gr.}}(\mathbf{D}_S(M), \mathbb{G}_{m,S})$. (*Confer* VIII 1).

4.4.4. — Let us give some examples of diagonalizable groups. First, one has

$$\mathbf{D}(\mathbb{Z}) = \mathbb{G}_m \quad \text{and} \quad \mathbf{D}(\mathbb{Z}^r) = (\mathbb{G}_m)^r.$$

One sets

$$\mu_n = \mathbf{D}(\mathbb{Z}/n\mathbb{Z}),$$

and one calls it the group of n -th roots of unity. Indeed, one has

$$\mu_n(S) = \text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, \Gamma(S, \mathcal{O}_S)^\times) = \{f \in \Gamma(S, \mathcal{O}_S) \mid f^n = 1\}.$$

The S -group $\mu_{n,S}$ corresponds to the $\mathcal{O}_S$ -algebra $\mathcal{O}_S[T]/(T^n - 1)$. Suppose in particular that S is the spectrum of a field k of characteristic $p = n$. Setting $T - 1 = s$, one finds

$$k[T]/(T^p - 1) = k[s]/(s^p),$$

which shows that the underlying topological space of $\mu_{p,k}$ is reduced to a point, the local ring of this point being the artinian k -algebra $k[s]/(s^p)$. (In the same vein, let us point out that the S -schemes $\mathbb{G}_{a,S}$, $\mathbb{G}_{m,S}$, $\mathbb{O}_S$ are *smooth* over S , that $\mathbf{D}_S(M)$ is flat over S and that it is formally smooth (resp. *smooth*) over S if and only if no residual characteristic of S divides the torsion of M (resp. and if moreover M is of finite type), cf. Exp. VIII, 2.1).

4.5. Other examples of groups. — The preceding procedure applies to the “classical groups” (linear groups $\mathbf{GL}_n$, symplectic groups $\mathbf{Sp}_n$, orthogonal groups $\mathbf{O}_n$, etc.). One will define for example $\mathbf{GL}_n$ as representing the functor $\mathbf{GL}_n$ such that:

$$\mathbf{GL}_n(S) = \mathbf{GL}(n, \Gamma(S, \mathcal{O}_S)) = \text{Aut}_{\mathcal{O}_S}(\mathcal{O}_S^n).$$

One can construct it for example as the open subscheme of $\text{Spec } \mathbb{Z}[T_{ij}]$ ($1 \leq i, j \leq n$) defined by the function $f = \det((T_{ij})_{i,j=1}^n)$, that is, $\text{Spec } \mathbb{Z}[T_{ij}, f^{-1}]$.

4.6. Module-functors in the category of schemes. — We propose to associate with every $\mathcal{O}_S$ -module on the scheme S , an $\mathbf{O}_S$ -module (where $\mathbf{O}_S$ denotes the ring-functor introduced in 4.3.3). This can be done in two different ways. To be precise:

Definition 4.6.1. — Let S be a scheme. For every $\mathcal{O}_S$ -module $\mathcal{F}$ one denotes by $\mathbf{V}(\mathcal{F})$ and $\mathbf{W}(\mathcal{F})$ the contravariant functors on $(\text{Sch})_S$ defined by:

$$\begin{aligned} \mathbf{V}(\mathcal{F})(S') &= \text{Hom}_{\mathcal{O}_{S'}}(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}, \mathcal{O}_{S'}), \\ \mathbf{W}(\mathcal{F})(S') &= \Gamma(S', \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \end{aligned}$$

(where $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ denotes the inverse image over S' of the $\mathcal{O}_S$ -module $\mathcal{F}$).

Then $\mathbf{V}(\mathcal{F})$ and $\mathbf{W}(\mathcal{F})$ are endowed in an evident way with a structure of $\mathbf{O}_S$ -modules (one recalls that $\mathbf{O}_S(S') = \Gamma(S', \mathcal{O}_{S'}) = \mathbf{W}(\mathcal{O}_S)(S')$), in such a way that one has in fact defined

two functors $\mathbf{V}$ and $\mathbf{W}$ from the category of $\mathcal{O}_S$ -modules to the category of $\mathbf{O}_S$ -modules, $\mathbf{V}$ being contravariant and $\mathbf{W}$ covariant.

We restrict ourselves in the rest of this paragraph to the case in which the $\mathcal{O}_S$ -modules in question are quasi-coherent, that is, we consider $\mathbf{V}$ and $\mathbf{W}$ as functors from the category $(\mathcal{O}_S\text{-Mod.q.c.})$ of quasi-coherent $\mathcal{O}_S$ -modules to the category of $\mathbf{O}_S$ -modules

$$\begin{aligned}\mathbf{V}: (\mathcal{O}_S\text{-Mod.q.c.})^\circ &\longrightarrow (\mathbf{O}_S\text{-Mod.}), \\ \mathbf{W}: (\mathcal{O}_S\text{-Mod.q.c.}) &\longrightarrow (\mathbf{O}_S\text{-Mod.}).\end{aligned}$$

Remark 4.6.1.1. — ⁽³¹⁾ The reader will notice that, in the sequel, all the propositions which involve only the functor $\mathbf{W}$ are valid for arbitrary $\mathcal{O}_S$ -modules, not necessarily quasi-coherent.

Proposition 4.6.2. —

- (i) *The functors $\mathbf{V}$ and $\mathbf{W}$ commute with extension of the base: if S' is over S and if $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_S$ -module, one has*

$$\mathbf{V}(\mathcal{F} \otimes \mathcal{O}_{S'}) \simeq \mathbf{V}(\mathcal{F})_{S'} \quad \text{and} \quad \mathbf{W}(\mathcal{F} \otimes \mathcal{O}_{S'}) \simeq \mathbf{W}(\mathcal{F})_{S'}.$$

- (ii) *The functors $\mathbf{V}$ and $\mathbf{W}$ are fully faithful: the canonical maps*

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') &\longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F})) \\ \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') &\longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}'))\end{aligned}$$

are bijective.

- (iii) *The functors $\mathbf{V}$ and $\mathbf{W}$ are additive:*

$$\mathbf{V}(\mathcal{F} \oplus \mathcal{F}') \simeq \mathbf{V}(\mathcal{F}) \times_S \mathbf{V}(\mathcal{F}') \quad \text{and} \quad \mathbf{W}(\mathcal{F} \oplus \mathcal{F}') \simeq \mathbf{W}(\mathcal{F}) \times_S \mathbf{W}(\mathcal{F}').$$

Parts (i) and (iii) are evident from the definitions. For (ii), one takes for S' opens of S . We leave the proof to the reader (for $\mathbf{V}$, use EGA II, 1.7.14).

Let us recall (cf. 3.1.4) that if $\mathcal{F}, \mathcal{F}'$ are $\mathcal{O}_S$ -modules, $\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}')$ denotes the S -functor (in abelian groups) which to every $S' \rightarrow S$ associates $\mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{F}_{S'}, \mathcal{F}'_{S'})$.

Proposition 4.6.3. — ⁽³²⁾ *One has canonical morphisms in $(\mathbf{O}_S\text{-Mod.})$:*

$$\begin{array}{ccc}\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}')) & & \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F})) \\ & \nwarrow \quad \nearrow & \\ & \mathbf{W}(\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}')) & \end{array}$$

This follows immediately from 4.6.2 (i) and (ii).

Notation 4.6.3.1. — Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$ -module. One knows (EGA II, 1.7.8) that the S -functor $\mathbf{V}(\mathcal{F})$ is representable by an S -scheme affine over S , which one denotes by $\mathbf{V}(\mathcal{F})$ and which one calls the vector fibration⁽³³⁾ defined by $\mathcal{F}$:

$$\mathbf{V}(\mathcal{F}) = \mathrm{Spec}(\mathcal{S}(\mathcal{F})),$$

where $\mathcal{S}(\mathcal{F})$ denotes the symmetric algebra of the $\mathcal{O}_S$ -module $\mathcal{F}$.⁽³⁴⁾

⁽³¹⁾**N.D.E.** The numbering 4.6.1.1 has been added, for later references.

⁽³²⁾**N.D.E.** The isomorphism $\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}')) \simeq \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F}))$ has been added.

⁽³³⁾**N.D.E.** “vector bundle” has been replaced by “vector fibration”; current usage being to call “vector bundle of rank r ” a vector fibration which is locally trivial of rank r , i.e., whose sheaf of sections is locally isomorphic to $\mathcal{O}_S^{\oplus r}$.

⁽³⁴⁾**N.D.E.** Let us point out here the articles [Ni04] (resp. [Ni02]) which show that if S is noetherian and $\mathcal{F}$ is a coherent $\mathcal{O}_S$ -module, then $\mathbf{W}(\mathcal{F})$ (resp. the S -group which to every $T \rightarrow S$ associates $\mathrm{Aut}_{\mathcal{O}_T}(\mathcal{F} \otimes \mathcal{O}_T)$) is representable if and only if $\mathcal{F}$ is locally free.

Proposition 4.6.4. — *Let $\mathcal{F}$ and $\mathcal{F}'$ be two quasi-coherent $\mathcal{O}_S$ -modules, and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_S$ -algebra. One has a functorial isomorphism:*

$$\mathrm{Hom}_S(\mathrm{Spec}(\mathcal{A}), \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}'), \mathbf{W}(\mathcal{F}))) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}', \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}).$$

Indeed, if one denotes $X = \mathrm{Spec}(\mathcal{A})$, the first member is canonically isomorphic to $\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}'), \mathbf{W}(\mathcal{F}))(X)$, that is, by definition, to

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{F}')_X, \mathbf{W}(\mathcal{F})_X) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{F}' \otimes \mathcal{O}_X), \mathbf{W}(\mathcal{F} \otimes \mathcal{O}_X))$$

(cf. 4.6.2 (i)), which by 4.6.2 (ii) may also be written

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}' \otimes \mathcal{O}_X, \mathcal{F} \otimes \mathcal{O}_X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}', \pi_*(\pi^*(\mathcal{F}))),$$

where one denotes by $\pi: X \rightarrow S$ the structural morphism. But, by EGA II, 1.4.7, one has $\pi_*(\pi^*(\mathcal{F})) \simeq \mathcal{F} \otimes \mathcal{A}$, which completes the proof.

Corollary 4.6.4.1. — *One has a canonical isomorphism*

$$\mathbf{W}(\mathcal{F} \otimes \mathcal{A}) \simeq \mathrm{Hom}_S(\mathrm{Spec}(\mathcal{A}), \mathbf{W}(\mathcal{F})).$$

Indeed,⁽³⁵⁾ let $f: S' \rightarrow S$ be an S -scheme and $X' = X \times_S S'$; one has a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & X \\ \pi' \downarrow & & \downarrow \pi \\ S' & \xrightarrow{f} & S \end{array}$$

and by EGA II, 1.5.2, X' is affine over S' and $\pi'_*(\mathcal{O}_{X'}) = f^*(\mathcal{A})$. One therefore has

$$\mathrm{Hom}_S(\mathrm{Spec}(\mathcal{A}), \mathbf{W}(\mathcal{F}))(S') = \mathrm{Hom}_{S'}(\mathrm{Spec}(f^*(\mathcal{A})), \mathbf{W}(f^*(\mathcal{F})))$$

and by 4.6.4 applied to $f^*(\mathcal{F})$, $\mathcal{F}' = \mathcal{O}_{S'}$ and $f^*(\mathcal{A})$, this equals

$$\Gamma(S', f^*(\mathcal{F}) \otimes f^*(\mathcal{A})) = \Gamma(S', f^*(\mathcal{F} \otimes \mathcal{A})) = \mathbf{W}(\mathcal{F} \otimes \mathcal{A})(S').$$

Proposition 4.6.5. — *If $\mathcal{F}$ and $\mathcal{F}'$ are two locally free $\mathcal{O}_S$ -modules of finite type, the morphisms of 4.6.3 are isomorphisms.*

Indeed, for every $S' \rightarrow S$, one has

$$\begin{aligned} \mathbf{W}(\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}'))(S') &= \Gamma(S', \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') \otimes \mathcal{O}_{S'}) \\ &= \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{F} \otimes \mathcal{O}_{S'}, \mathcal{F}' \otimes \mathcal{O}_{S'}). \end{aligned}$$

But the second member is indeed isomorphic to

$$\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}'))(S')$$

and to

$$\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F}))(S'),$$

by 4.6.2 (i) and (ii).

Corollary 4.6.5.1. — *Let $\mathcal{F}$ be a locally free $\mathcal{O}_S$ -module of finite type. Set*

$$\mathcal{F}^\vee = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{O}_S).$$

One has canonical isomorphisms:

$$\begin{aligned} \mathbf{W}(\mathcal{F}^\vee) &\simeq \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{O}_S) \simeq \mathbf{V}(\mathcal{F}), \\ \mathbf{V}(\mathcal{F}^\vee) &\simeq \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}), \mathbf{O}_S) \simeq \mathbf{W}(\mathcal{F}). \end{aligned}$$

Finally, one has the following proposition:

⁽³⁵⁾**N.D.E.** The original has been spelled out in what follows.

Proposition 4.6.6. — *Let $f: \mathcal{F} \rightarrow \mathcal{F}'$ be a morphism of locally free $\mathcal{O}_S$ -modules of finite rank. In order that $\mathbf{W}(f): \mathbf{W}(\mathcal{F}) \rightarrow \mathbf{W}(\mathcal{F}')$ be a monomorphism, it is necessary and sufficient that f identify $\mathcal{F}$ locally with a direct factor of $\mathcal{F}'$.*

The direct statement is essentially contained in EGA 0_I, 5.5.5.⁽³⁶⁾ Conversely, if $\mathcal{F}$ is locally a direct factor of $\mathcal{F}'$ then, for every $\pi: S' \rightarrow S$, $\pi^*\mathcal{F}$ is a submodule of $\pi^*\mathcal{F}'$ (since it is locally a direct factor), hence $\mathbf{W}(\mathcal{F})(S') = \Gamma(S', \pi^*\mathcal{F})$ is a submodule of $\mathbf{W}(\mathcal{F}')(S') = \Gamma(S', \pi^*\mathcal{F}')$.

4.7. The category of G - $\mathcal{O}_S$ -modules. — Let G be an S -group and $\mathcal{F}$ an $\mathcal{O}_S$ -module; $\mathbf{W}(\mathcal{F})$ is endowed with a structure of $\mathbf{O}_S$ -module.

Definition 4.7.1. — One calls a structure of G - $\mathcal{O}_S$ -module on $\mathcal{F}$ a structure of h_G - $\mathbf{O}_S$ -module on $\mathbf{W}(\mathcal{F})$ (cf. 3.2). A morphism of G - $\mathcal{O}_S$ -modules is by definition a morphism of the associated h_G - $\mathbf{O}_S$ -modules. One thus obtains the category $(G\text{-}\mathcal{O}_S\text{-Mod.})$, and one denotes by $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ the full subcategory formed by the G - $\mathcal{O}_S$ -modules which are quasi-coherent as $\mathcal{O}_S$ -modules.

To give a structure of G - $\mathcal{O}_S$ -module on $\mathcal{F}$ is therefore, by 3.2, to give a morphism of $(\widehat{\mathbf{Sch}})_{/S\text{-groups}}$

$$\rho: h_G \rightarrow \text{Aut}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F})).$$

Remark 4.7.1.0. — ⁽³⁷⁾ Since one has, by 4.6.2, an *anti-isomorphism* of S -group functors

$$(\dagger) \quad \text{Aut}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F})) \simeq \text{Aut}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F})),$$

one sees that it is equivalent to give a structure of h_G - $\mathbf{O}_S$ -module on $\mathbf{W}(\mathcal{F})$ or on $\mathbf{V}(\mathcal{F})$. Indeed, let $\rho: h_G \rightarrow \text{Aut}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}))$ be as above. For every $T \rightarrow S$ and $g \in G(T)$, denote by $\rho^*(g)$ the image of $\rho(g)$ under the anti-isomorphism $(\dagger)$; one therefore has $\rho^*(gh) = \rho^*(h) \circ \rho^*(g)$, i.e. ρ^* defines a structure of h_G - $\mathbf{O}_S$ -module “on the right” on $\mathbf{V}(\mathcal{F})$. Setting $\rho^\vee(g) = \rho^*(g^{-1})$, one indeed obtains a structure of h_G - $\mathbf{O}_S$ -module on $\mathbf{V}(\mathcal{F})$, the datum of which is equivalent to that of ρ .

Remark 4.7.1.1. — One may say that the categories which one has just defined have been constructed by the cartesian squares:

$$\begin{array}{ccccc} (G\text{-}\mathcal{O}_S\text{-Mod.q.c.}) & \hookrightarrow & (G\text{-}\mathcal{O}_S\text{-Mod.}) & \hookrightarrow & (h_G\text{-}\mathbf{O}_S\text{-Mod.}) \\ \downarrow & & \downarrow & & \downarrow \text{forgetful} \\ (\mathcal{O}_S\text{-Mod.q.c.}) & \hookrightarrow & (\mathcal{O}_S\text{-Mod.}) & \xrightarrow{\mathbf{W}} & (\mathbf{O}_S\text{-Mod.}) \end{array}$$

⁽³⁸⁾ The categories $(\mathcal{O}_S\text{-Mod.})$ and $(\mathbf{O}_S\text{-Mod.})$ are abelian, but one must take care that in general the functor $\mathbf{W}$ is not exact, neither on the left nor on the right.

Remark 4.7.1.2. — ⁽³⁹⁾ Let $\mathcal{F}$ be a G - $\mathcal{O}_S$ -module. The subsheaf of invariants $\mathcal{F}^G$ is defined as follows: for every open U of S ,

$$\mathcal{F}^G(U) = \mathbf{W}(\mathcal{F})^G(U) = \{x \in \mathcal{F}(U) \mid g \cdot x_{S'} = x_{S'} \text{ for every } S' \xrightarrow{f} U, g \in G(S')\},$$

where $x_{S'}$ denotes the image of x in $\Gamma(S', f^*(\mathcal{F})) = \Gamma(U, f_*f^*(\mathcal{F}))$.

One must take care that the natural morphism $\mathbf{W}(\mathcal{F}^G) \rightarrow \mathbf{W}(\mathcal{F})^G$ is not an isomorphism in general. For example, if $S = \text{Spec}(\mathbb{Z})$ and G is the constant group $\mathbb{Z}/2\mathbb{Z} = \{1, \tau\}$ acting on $\mathcal{F} = \mathcal{O}_S$ by $\tau \cdot 1 = -1$, one has $\mathcal{F}^G = 0$ but, if R is an $\mathbb{F}_2$ -algebra, $\mathbf{W}(\mathcal{F})^G(\text{Spec}(R)) = R$.

⁽³⁶⁾**N.D.E.** The following sentence has been spelled out.

⁽³⁷⁾**N.D.E.** This remark has been added, taken from [DG], II, §2, 1.1.

⁽³⁸⁾**N.D.E.** The original has been corrected by removing the assertion that the category $(G\text{-}\mathcal{O}_S\text{-Mod.})$ is abelian; see 4.7.2.1 below.

⁽³⁹⁾**N.D.E.** This remark has been added.

4.7.2. — One assumes henceforth, until the end of no. 4.7, that G is *affine* over S .⁽⁴⁰⁾ Then, by virtue of 4.6.4, the datum of a morphism of S -functors

$$\rho: h_G \longrightarrow \underline{\text{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{F}))$$

is equivalent to that of a morphism of $\mathcal{O}_S$ -modules

$$\mu: \mathcal{F} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

To say that ρ is a morphism of $(\widehat{\text{Sch}})_{/S}$ -groups is then equivalent to saying that μ satisfies the following axioms:

(CM 1) *the following diagram is commutative*

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\mu} & \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ \mu \downarrow & & \downarrow \text{id} \otimes \Delta \\ \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) & \xrightarrow{\mu \otimes \text{id}} & \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G). \end{array}$$

(CM 2) *the composite below is the identity*

$$\mathcal{F} \xrightarrow{\mu} \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\text{id} \otimes \varepsilon} \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathcal{F}.$$

These axioms (CM 1) and (CM 2) are those of the structure of a (right) comodule⁽⁴¹⁾ over the bialgebra $\mathcal{A}(G)$.

Set $\mathcal{A} = \mathcal{A}(G)$. If $\mathcal{F}$ and $\mathcal{F}'$ are $\mathcal{A}$ -comodules, a morphism of comodules $f: \mathcal{F} \longrightarrow \mathcal{F}'$ is a morphism of $\mathcal{O}_S$ -modules such that the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{f} & \mathcal{F}' \\ \mu_{\mathcal{F}} \downarrow & & \downarrow \mu_{\mathcal{F}'} \\ \mathcal{F} \otimes \mathcal{A} & \xrightarrow{f \otimes \text{id}} & \mathcal{F}' \otimes \mathcal{A}. \end{array}$$

One thus obtains the category $(\mathcal{A}\text{-Comod.})$, and one will denote by $(\mathcal{A}\text{-Comod.q.c.})$ the full subcategory consisting of the $\mathcal{A}$ -comodules which are quasi-coherent as $\mathcal{O}_S$ -modules. One has thus obtained:

Proposition 4.7.2. — *Let G be an S -group affine over S . One has equivalences of categories:*

$$(G\text{-}\mathcal{O}_S\text{-Mod.}) \simeq (\mathcal{A}(G)\text{-Comod.})$$

$$(G\text{-}\mathcal{O}_S\text{-Mod.q.c.}) \simeq (\mathcal{A}(G)\text{-Comod.q.c.}).$$

⁽⁴²⁾ *If moreover $S = \text{Spec}(\Lambda)$ is affine and if one denotes $\Lambda[G] = \Gamma(S, \mathcal{A}(G))$, one has an equivalence of categories*

$$(\mathcal{A}(G)\text{-Comod.q.c.}) \simeq (\Lambda[G]\text{-Comod.}).$$

⁽⁴³⁾ Suppose moreover that $\mathcal{A} = \mathcal{A}(G)$ is a *flat* $\mathcal{O}_S$ -module. Let $\mathcal{E}$ be an $\mathcal{A}$ -comodule and $\mathcal{F}$ a sub- $\mathcal{O}_S$ -module of $\mathcal{E}$. Since $\mathcal{A}$ is flat over $\mathcal{O}_S$, one can identify $\mathcal{F} \otimes \mathcal{A}$ (resp. $\mathcal{F} \otimes \mathcal{A} \otimes \mathcal{A}$) with a sub- $\mathcal{O}_S$ -module of $\mathcal{E} \otimes \mathcal{A}$ (resp. $\mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A}$). Suppose that $\mu_{\mathcal{E}}$ maps $\mathcal{F}$ into $\mathcal{F} \otimes \mathcal{A}$; then its restriction $\mu_{\mathcal{F}}: \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}$ equips $\mathcal{F}$ with a structure of $\mathcal{A}$ -comodule; one says that $\mathcal{F}$ is a *subcomodule* of $\mathcal{E}$. By passage to the quotient, $\mu_{\mathcal{E}}$ defines a morphism of $\mathcal{O}_S$ -modules $\mathcal{E}/\mathcal{F} \longrightarrow \mathcal{E}/\mathcal{F} \otimes \mathcal{A}$, which equips $\mathcal{E}/\mathcal{F}$ with a structure of $\mathcal{A}$ -comodule. If $f: \mathcal{E} \longrightarrow \mathcal{E}'$ is a morphism of $\mathcal{A}$ -comodules, $\text{Ker } f$ (resp. $\text{Im } f$) is a sub- $\mathcal{A}$ -comodule of $\mathcal{E}$ (resp. $\mathcal{E}'$), and f induces an isomorphism of $\mathcal{A}$ -comodules: $\mathcal{E}/\text{Ker } f \xrightarrow{\sim} \text{Im } f$. Moreover, if $\mathcal{E}$ and $\mathcal{E}'$ are

⁽⁴⁰⁾**N.D.E.** Cf. VI_B, §§ 11.1–11.6 for the extension of the results of 4.7.2 to the case where G is not necessarily affine, but where G and $\mathcal{F}$ are assumed *flat* over S .

⁽⁴¹⁾**N.D.E.** Left $G\text{-}\mathcal{O}_S$ -modules correspond naturally to right $\mathcal{A}(G)$ -comodules.

⁽⁴²⁾**N.D.E.** The following sentence has been added.

⁽⁴³⁾**N.D.E.** The following paragraph has been added, taken from [Se68, § 1.3].

quasi-coherent $\mathcal{O}_S$ -modules, the same is true of $\text{Ker } f$ and $\text{Im } f$. Consequently, $(\mathcal{A}\text{-Comod.})$ and $(\mathcal{A}\text{-Comod.q.c.})$ are *abelian* categories.

Corollary 4.7.2.1. — *One assumes that G is affine and flat over S . Then the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ (resp. $(G\text{-}\mathcal{O}_S\text{-Mod.})$), equivalent to the category of quasi-coherent $\mathcal{A}(G)$ -comodules over $\mathcal{O}_S$ (resp. of $\mathcal{A}(G)$ -comodules), is abelian.*

4.7.3. — Suppose now that G is a *diagonalizable* group, that is, that $\mathcal{A}(G)$ is the algebra of a commutative group M over the sheaf of rings $\mathcal{O}_S$. If $\mathcal{F}$ is an $\mathcal{O}_S$ -module, one has

$$\mathcal{F} \otimes \mathcal{A}(G) = \coprod_{m \in M} \mathcal{F} \otimes m\mathcal{O}_S.$$

To give a morphism of $\mathcal{O}_S$ -modules

$$\mu: \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}(G)$$

is therefore equivalent to giving $\mathcal{O}_S$ -endomorphisms $(\mu_m)_{m \in M}$ of $\mathcal{F}$, such that for every section x of $\mathcal{F}$ over an open of S , $(\mu_m(x))$ is a section of the direct sum $\coprod_{m \in M} \mathcal{F}$ (this means that over every sufficiently small open, only finitely many of the restrictions of the $\mu_m(x)$ are nonzero).

In order that μ , defined by

$$\mu(x) = \sum_{m \in M} \mu_m(x) \otimes m,$$

satisfy (CM 1) (resp. (CM 2)), it is necessary and sufficient that one have

$$\mu_m \circ \mu_{m'} = \delta_{mm'} \mu_m, \quad \left(\text{resp. } \sum_{m \in M} \mu_m = \text{Id}_{\mathcal{F}} \right),$$

which means that the μ_m are pairwise orthogonal projectors summing to the identity. One has thus proved:

Proposition 4.7.3. — *If $G = \mathbf{D}_S(M)$ is a diagonalizable S -group, the category of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules (resp. of $G\text{-}\mathcal{O}_S$ -modules) is equivalent to the category of quasi-coherent $\mathcal{O}_S$ -modules (resp. of $\mathcal{O}_S$ -modules) graded of type M .*

Remark. — If $\mathcal{F}$ is endowed with a structure of $\mathcal{O}_S$ -algebra preserved by the operations of G , then the grading of $\mathcal{F}$ is an algebra grading. More precisely:

Corollary 4.7.3.1. — *The functor $\mathcal{A} \mapsto \text{Spec } \mathcal{A}$ induces an equivalence between the category of quasi-coherent $\mathcal{O}_S$ -algebras graded of type M and the opposite category to that of the S -schemes affine over S with S -group of operators $G = \mathbf{D}_S(M)$.*

Proposition 4.7.4. — *Let G be a diagonalizable S -group. If $0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$ is an exact sequence of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules which splits as a sequence of $\mathcal{O}_S$ -modules, then it likewise splits as a sequence of $G\text{-}\mathcal{O}_S$ -modules.*

Indeed, if $G = \mathbf{D}_S(M)$, each of the $\mathcal{F}_i$ is graded by the $(\mathcal{F}_i)_m$ and for each $m \in M$ the sequence

$$0 \longrightarrow (\mathcal{F}_1)_m \longrightarrow (\mathcal{F}_2)_m \longrightarrow (\mathcal{F}_3)_m \longrightarrow 0$$

of $\mathcal{O}_S$ -modules is split. The preceding proposition then yields the result.

5. Cohomology of groups

5.1. The standard complex. — ⁽⁴⁴⁾Let $\mathcal{C}$ be a category, $\mathbf{G}$ a $\widehat{\mathcal{C}}$ -group, $\mathbf{O}$ a $\widehat{\mathcal{C}}$ -ring and $\mathbf{F}$ a $\mathbf{G}$ - $\mathbf{O}$ -module. One sets, for $n \geq 0$,

$$\underline{C}^n(\mathbf{G}, \mathbf{F}) = \underline{\text{Hom}}(\mathbf{G}^n, \mathbf{F}) \quad \text{and} \quad C^n(\mathbf{G}, \mathbf{F}) = \text{Hom}(\mathbf{G}^n, \mathbf{F}),$$

where $\mathbf{G}^0$ is the final object e . Then $\underline{C}^n(\mathbf{G}, \mathbf{F})$ (resp. $C^n(\mathbf{G}, \mathbf{F})$) is endowed in an evident way with a structure of $\mathbf{O}$ -module (resp. of $\Gamma(\mathbf{O})$ -module), and one has

$$C^n(\mathbf{G}, \mathbf{F}) \simeq \Gamma(\underline{C}^n(\mathbf{G}, \mathbf{F})) \quad \text{and} \quad \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S).$$

To give an element of $C^n(\mathbf{G}, \mathbf{F})$ is to give, for each $S \in \text{Ob}(\mathcal{C})$, an n -cochain of $\mathbf{G}(S)$ in $\mathbf{F}(S)$, functorially in S . The boundary operator

$$\partial: C^n(\mathbf{G}(S), \mathbf{F}(S)) \longrightarrow C^{n+1}(\mathbf{G}(S), \mathbf{F}(S)),$$

which, let us recall, is given by the formula

$$\begin{aligned} \partial f(g_1, \dots, g_{n+1}) &= g_1 f(g_2, \dots, g_{n+1}) + \sum_{i=1}^n (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &\quad + (-1)^{n+1} f(g_1, \dots, g_n), \end{aligned}$$

is functorial in S and therefore defines a homomorphism

$$\partial: C^n(\mathbf{G}, \mathbf{F}) \longrightarrow C^{n+1}(\mathbf{G}, \mathbf{F})$$

such that $\partial \circ \partial = 0$. One has thus defined a complex of abelian groups (and even of $\Gamma(\mathbf{O})$ -modules), denoted $C^*(\mathbf{G}, \mathbf{F})$. One defines in the same way the complex of $\mathbf{O}$ -modules $\underline{C}^*(\mathbf{G}, \mathbf{F})$, and one has

$$C^*(\mathbf{G}, \mathbf{F}) = \Gamma(\underline{C}^*(\mathbf{G}, \mathbf{F})).$$

One denotes by $H^n(\mathbf{G}, \mathbf{F})$ (resp. $\underline{H}^n(\mathbf{G}, \mathbf{F})$) the homology groups (resp. the homology $\widehat{\mathcal{C}}$ -groups) of the complex $C^*(\mathbf{G}, \mathbf{F})$ (resp. $\underline{C}^*(\mathbf{G}, \mathbf{F})$).

One has in particular

$$\underline{H}^0(\mathbf{G}, \mathbf{F}) = \mathbf{F}^{\mathbf{G}} \quad \text{and} \quad H^0(\mathbf{G}, \mathbf{F}) = \Gamma(\mathbf{F}^{\mathbf{G}}).$$

Remark 5.1.1. — ⁽⁴⁵⁾The “set-theoretic” description of ∂ given above is convenient for checking that $\partial \circ \partial = 0$. One can also define ∂ in terms of the multiplication $m: \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}$ and of the action $\mu: \mathbf{G} \times \mathbf{F} \longrightarrow \mathbf{F}$ as follows: for every $f \in C^n(\mathbf{G}, \mathbf{F})$,

$$\partial f = \mu \circ (\text{id}_{\mathbf{G}} \times f) + \sum_{i=1}^n (-1)^i f \circ (\text{id}_{\mathbf{G}^{i-1}} \times m \times \text{id}_{\mathbf{G}^{n-i}}) + (-1)^{n+1} f \circ \text{pr}_{[1,n]},$$

where $\text{pr}_{[1,n]}$ denotes the projection of $\mathbf{G}^{n+1} = \mathbf{G}^n \times \mathbf{G}$ onto $\mathbf{G}^n$. Likewise, for every $S \in \text{Ob}(\mathcal{C})$ and $f \in \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S)$, one has

$$\partial f = \mu_S \circ (\text{id}_{\mathbf{G}_S} \times f) + \sum_{i=1}^n (-1)^i f \circ (\text{id}_{\mathbf{G}_S^{i-1}} \times m_S \times \text{id}_{\mathbf{G}_S^{n-i}}) + (-1)^{n+1} f \circ \text{pr}_{[1,n]},$$

where m_S and μ_S are deduced from m and μ by base change.

5.2. — ⁽⁴⁶⁾One recalls (cf. § 3) that $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ is endowed with the structure of an abelian category, defined “argument by argument”; thus,

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

is an exact sequence of $\mathbf{G}$ - $\mathbf{O}$ -modules if, and only if, the sequence of abelian groups

$$0 \longrightarrow \mathbf{F}'(S) \longrightarrow \mathbf{F}(S) \longrightarrow \mathbf{F}''(S) \longrightarrow 0$$

is exact, for every $S \in \text{Ob}(\mathcal{C})$.

⁽⁴⁴⁾**N.D.E.** This complex is often called the “Hochschild complex”; see for example § II.3 of [DG70].

⁽⁴⁵⁾**N.D.E.** This remark has been added.

⁽⁴⁶⁾**N.D.E.** The following reminder has been added.

⁽⁴⁷⁾ Suppose $\mathcal{C}$ is small; then, by 3.2.1, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects, so that the derived functors of the left exact functors H^0 and $\underline{H}^0$ are defined. We now propose to show that the functors H^n (resp. $\underline{H}^n$) are indeed the derived functors of H^0 (resp. $\underline{H}^0$).

Definition 5.2.0. — ⁽⁴⁸⁾For every $\mathbf{O}$ -module $\mathbf{P}$, one denotes by $E(\mathbf{P})$ the object $\underline{\text{Hom}}(\mathbf{G}, \mathbf{P})$ of $\widehat{\mathcal{C}}$ endowed with the structure of $\mathbf{G}\text{-}\mathbf{O}$ -module defined as follows: for every $S \in \text{Ob}(\mathcal{C})$ one has $\underline{\text{Hom}}(\mathbf{G}, \mathbf{P})(S) = \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$, and one lets $g \in \mathbf{G}(S)$ and $a \in \mathbf{O}(S)$ operate on $\phi \in \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ by the formulas

$$(g \cdot \phi)(h) = \phi(hg) \quad \text{and} \quad (a \cdot \phi)(h) = a\phi(h),$$

for every $h \in \mathbf{G}(S')$, $S' \rightarrow S$. Moreover, for every $\phi \in \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ one sets

$$\varepsilon(\phi) = \phi(1) \in \mathbf{P}(S)$$

(where 1 denotes the unit element of $\mathbf{G}(S)$).

This defines a functor $E: (\mathbf{O}\text{-Mod.}) \rightarrow (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ and a natural transformation $\varepsilon: E \rightarrow \text{Id}$, where Id denotes the identity functor of $(\mathbf{O}\text{-Mod.})$.

Remark 5.2.0.1. — ⁽⁴⁸⁾In what follows, let us denote by $\mathbf{G}_1$ and $\mathbf{G}_2$ two copies of $\mathbf{G}$. Then the morphism

$$\mathbf{G}_1 \times E(\mathbf{P}) \rightarrow E(\mathbf{P}), \quad (g_1, \phi) \mapsto (g_2 \mapsto \phi(g_2 g_1))$$

corresponds, via the isomorphisms

$$\begin{aligned} \text{Hom}(\mathbf{G}_1 \times E(\mathbf{P}), E(\mathbf{P})) &\simeq \text{Hom}(E(\mathbf{P}), \underline{\text{Hom}}(\mathbf{G}_1, \underline{\text{Hom}}(\mathbf{G}_2, \mathbf{P}))) \\ &\simeq \text{Hom}(E(\mathbf{P}), \underline{\text{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})) \end{aligned}$$

to the morphism $\phi \mapsto ((g_2, g_1) \mapsto \phi(g_2 g_1))$, i.e. to the morphism

$$\underline{\text{Hom}}(\mathbf{G}, \mathbf{P}) \rightarrow \underline{\text{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})$$

induced by the multiplication $\mu_{\mathbf{G}}: \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$, $(g_2, g_1) \mapsto g_2 g_1$.

Lemma 5.2.0.2. — ⁽⁴⁸⁾

- (i) *The functor E is right adjoint to the forgetful functor $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}) \rightarrow (\mathbf{O}\text{-Mod.})$; more precisely, $\varepsilon: E \rightarrow \text{Id}$ induces, for every*

$$\mathbf{M} \in (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}) \quad \text{and} \quad \mathbf{P} \in \text{Ob}(\mathbf{O}\text{-Mod.}),$$

a bijection

$$\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}\text{-Mod.}}(\mathbf{M}, \mathbf{P})$$

functorial in $\mathbf{M}$ and $\mathbf{P}$.

- (ii) *Consequently, if $\mathbf{I}$ is an injective object of $(\mathbf{O}\text{-Mod.})$ then $E(\mathbf{I})$ is an injective object of $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$.*

PROOF. To every $\mathbf{O}$ -morphism $f: \mathbf{M} \rightarrow \mathbf{P}$, one associates the element ϕ_f of

$$\text{Hom}_{\mathbf{O}}(\mathbf{M}, E(\mathbf{P}))$$

defined as follows. For every $S \in \text{Ob}(\mathcal{C})$ and $m \in \mathbf{M}(S)$, $\phi_f(m)$ is the element of $\text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ defined by: for every $g \in \mathbf{G}(S')$, $S' \rightarrow S$,

$$\phi_f(m)(g) = f(gm) \in \mathbf{P}(S').$$

Then, for every $h \in \mathbf{G}(S)$, one has $\phi_f(hm) = h \cdot f(m)$, i.e. ϕ_f is an element of

$$\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})).$$

⁽⁴⁷⁾**N.D.E.** The following sentence has been added.

⁽⁴⁸⁾**N.D.E.** The original has been modified, in order to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

If

$$\phi \in \text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P}))$$

and if one denotes, for every $m \in \mathbf{M}(S)$, $f(m) = \phi(m)(1)$, then

$$\phi_f(m)(g) = f(gm) = \phi(gm)(1) = (g \cdot \phi(m))(1) = \phi(m)(g),$$

i.e. $\phi_f = \phi$. Conversely, it is clear that $\phi_f(m)(1) = f(m)$. This proves (i), and (ii) follows from it at once. $\square$

Definition 5.2.0.3. — Let $\mathbf{M}$ be a $\mathbf{G}\text{-}\mathbf{O}$ -module; the identity map of $\mathbf{M}$ (considered as an $\mathbf{O}$ -module) corresponds by adjunction to the morphism of $\mathbf{G}\text{-}\mathbf{O}$ -modules

$$\mu_{\mathbf{M}}: \mathbf{M} \longrightarrow E(\mathbf{M})$$

such that for every $S \in \text{Ob}(\mathcal{C})$ and $m \in \mathbf{M}(S)$, $\mu_{\mathbf{M}}(m)$ is the morphism $\mathbf{G}_S \longrightarrow \mathbf{M}_S$ defined by: for every $S' \longrightarrow S$ and $g \in \mathbf{G}(S')$, $\mu_{\mathbf{M}}(m)(g) = g \cdot m_{S'} \in \mathbf{M}(S')$.

Note that $\mu_{\mathbf{M}}$ is a *monomorphism*. Indeed, $\varepsilon_{\mathbf{M}}: E(\mathbf{M}) \longrightarrow \mathbf{M}$ is a morphism of $\mathbf{O}$ -modules such that $\varepsilon_{\mathbf{M}} \circ \mu_{\mathbf{M}} = \text{id}_{\mathbf{M}}$; consequently $\mathbf{M}$ is a direct factor, as an $\mathbf{O}$ -module, of $E(\mathbf{M})$.

Proposition 5.2.1. — *One assumes that $\mathcal{C}$ is small, that finite products exist in it, and that $\mathbf{G}$ is representable. Then the functors $H^n(\mathbf{G}, _)$ (resp. $\underline{H}^n(\mathbf{G}, _)$) are the derived functors of the left exact functor $H^0(\mathbf{G}, _)$ (resp. $\underline{H}^0(\mathbf{G}, _)$) on the category of $\mathbf{G}\text{-}\mathbf{O}$ -modules.*

By virtue of the well-known general results⁽⁴⁹⁾, it suffices to check that the $H^n(\mathbf{G}, _)$ (resp. $\underline{H}^n(\mathbf{G}, _)$) form an effaceable cohomological functor in degrees > 0 .

Let

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

be an exact sequence of $\mathbf{G}\text{-}\mathbf{O}$ -modules, and let $S \in \text{Ob}(\mathcal{C})$. By hypothesis, $\mathbf{G}$ is representable by an object $G \in \text{Ob}(\mathcal{C})$, and finite products exist in $\mathcal{C}$; in particular $\mathcal{C}$ has a final element e . Thus each $G^n \times \mathbf{h}_S$ is representable by $G^n \times S$ (with $G^0 = e$), and the sequence

$$0 \longrightarrow \mathbf{F}'(G^n \times S) \longrightarrow \mathbf{F}(G^n \times S) \longrightarrow \mathbf{F}''(G^n \times S) \longrightarrow 0$$

is exact. This shows that the sequence of $\mathbf{O}$ -modules

$$0 \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}') \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}) \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}'') \longrightarrow 0$$

is exact. It follows that $\underline{C}^*(\mathbf{G}, _)$, considered as a functor on $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ with values in the category of complexes of $(\mathbf{O}\text{-Mod.})$, is exact. This shows that the $\underline{H}^n(\mathbf{G}, _)$ do indeed form a cohomological functor. As the functor Γ is exact, the same holds for the $H^n(\mathbf{G}, _)$. It will now suffice for us to prove:

Lemma 5.2.2. *For every $\mathbf{P} \in \text{Ob}(\mathbf{O}\text{-Mod.})$, one has:*

$$H^n(\mathbf{G}, \text{Hom}(\mathbf{G}, \mathbf{P})) = 0 \quad \text{and} \quad \underline{H}^n(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \quad \text{for } n > 0.$$

It suffices for us to prove that $C^*(\mathbf{G}, \text{Hom}(\mathbf{G}, \mathbf{P}))$ and $\underline{C}^*(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$ are homotopically trivial in degrees > 0 . It even suffices to do so for the second, the corresponding result for the first being deduced from it by base change.⁽⁵⁰⁾ Now, one defines for every $n \geq 0$ a morphism

$$\sigma: \underline{C}^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P})) \longrightarrow \underline{C}^n(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$$

as follows. Let $f \in \underline{C}^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$; for every $S \in \text{Ob}(\mathcal{C})$ and $g_1, \dots, g_n \in \mathbf{G}(S)$, $\sigma(f)(g_1, \dots, g_n)$ is the element of $\text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ defined by: for every $S' \longrightarrow S$ and $x \in \mathbf{G}(S')$,

$$\sigma(f)(g_1, \dots, g_n)(x) = f(x, g_1, \dots, g_n)(e) \in \mathbf{P}(S')$$

⁽⁴⁹⁾**N.D.E.** Cf. [Gr57], 2.2.1 and 2.3. Moreover, the original has been spelled out in what follows.

⁽⁵⁰⁾**N.D.E.** The original has been expanded in what follows.

(where e denotes the unit element of $\mathbf{G}(S')$). Then σ is a homotopy operator in degrees > 0 . Indeed, for every $g_1, \dots, g_{n+1} \in \mathbf{G}(S)$ and $x \in \mathbf{G}(S')$ one has, on the one hand:

$$\begin{aligned} \partial\sigma(f)(g_1, \dots, g_{n+1})(x) &= f(xg_1, g_2, \dots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^n (-1)^i f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1})(e) \\ &\quad + (-1)^{n+1} f(x, g_1, \dots, g_n)(e), \end{aligned}$$

and on the other hand:

$$\begin{aligned} \sigma(\partial f)(g_1, \dots, g_{n+1})(x) &= (xf(g_1, g_2, \dots, g_{n+1}))(e) - f(xg_1, g_2, \dots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^n (-1)^{i+1} f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &\quad + (-1)^{n+2} f(x, g_1, \dots, g_n)(e), \end{aligned}$$

whence

$$(\partial\sigma(f) + \sigma(\partial f))(g_1, \dots, g_{n+1})(x) = f(g_1, \dots, g_{n+1})(x),$$

i.e. $\partial\sigma + \sigma\partial$ is the identity map of $\underline{C}^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$, for every $n \geq 0$.

Remark 5.2.3. ⁽⁵¹⁾ The hypothesis “ $\mathcal{C}$ small” is used only to ensure the existence of the derived functors $R^n H^0$ and $R^n \underline{H}^0$. In all cases, what precedes shows that the functors $H^n(\mathbf{G},)$ resp. $\underline{H}^n(\mathbf{G},)$ form a cohomological functor, effaceable in degrees > 0 , hence they are the (right) satellite functors of the left exact functor $H^0(\mathbf{G},)$ resp. $\underline{H}^0(\mathbf{G},)$, in the sense of [Gr57], 2.2; if $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects (which is the case if $\mathcal{C}$ is small), they coincide with the derived functors (*loc. cit.* 2.3).

5.3. Cohomology of the $G\text{-}\mathcal{O}_S$ -modules. — Let S be a scheme, G an S -group and $\mathcal{F}$ a quasi-coherent $G\text{-}\mathcal{O}_S$ -module. One defines the cohomology groups of G with values in $\mathcal{F}$ by

$$H^n(G, \mathcal{F}) = H^n(\mathbf{h}_G, \mathbf{W}(\mathcal{F})).$$

(for the notations, cf. 4.6).

Suppose G is *affine* over S . Then, in view of proposition 4.6.4, this cohomology is computed in the following way: $H^n(G, \mathcal{F})$ is the n -th homology group of the complex $C^*(G, \mathcal{F})$ whose n -th term is:

$$C^n(G, \mathcal{F}) = \Gamma\left(S, \mathcal{F} \otimes \underbrace{\mathcal{A}(G) \otimes \dots \otimes \mathcal{A}(G)}_{n \text{ times}}\right).$$

If f resp. a_i is a section of $\mathcal{F}$ resp. of $\mathcal{A}(G)$ over an open of S , one has

$$\begin{aligned} \partial(f \otimes a_1 \otimes \dots \otimes a_n) &= \mu_{\mathcal{F}}(f) \otimes a_1 \otimes a_2 \otimes \dots \otimes a_n \\ &\quad + \sum_{i=1}^n (-1)^i f \otimes a_1 \otimes \dots \otimes \Delta a_i \otimes \dots \otimes a_n \\ &\quad + (-1)^{n+1} f \otimes a_1 \otimes a_2 \otimes \dots \otimes a_n \otimes 1, \end{aligned}$$

where $\Delta: \mathcal{A}(G) \rightarrow \mathcal{A}(G) \otimes \mathcal{A}(G)$ and $\mu_{\mathcal{F}}: \mathcal{F} \rightarrow \mathcal{F} \otimes \mathcal{A}(G)$ describe the coalgebra structure of $\mathcal{A}(G)$ and the comodule structure of $\mathcal{F}$. Let us note in passing that the cohomology of G with values in $\mathcal{F}$ therefore depends only on the comodule structure of $\mathcal{F}$, and in particular only on the S -monoid structure of G .

One has in particular

$$H^0(G, \mathcal{F}) = \Gamma(S, \mathcal{F}^G),$$

⁽⁵¹⁾**N.D.E.** This remark has been added.

where $\mathcal{F}^G$, the *sheaf of invariants* of $\mathcal{F}$, is defined as the sheaf whose sections over the open U of S are the sections of $\mathcal{F}$ over U whose inverse image in every S' over U is invariant under $G(S')$ (cf. 4.7.1.2).

Theorem 5.3.1. *Let S be an affine scheme, G an S -group affine and flat over S . The functors $H^n(G, _)$ are the derived functors of $H^0(G, _)$ on the category of quasi-coherent G - $\mathcal{O}_S$ -modules.*

PROOF. ⁽⁵²⁾ As G is affine and flat over S , then, by 4.7.2.1, the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ is equivalent to the category $(\mathcal{A}(G)\text{-Comod.q.c.})$ of quasi-coherent $\mathcal{A}(G)$ -comodules on $\mathcal{O}_S$, and is therefore abelian. On the other hand, $\mathcal{A}(G)$ being a flat $\mathcal{O}_S$ -module, each functor $\mathcal{F} \mapsto \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G)^{\otimes n}$ is exact; as moreover S is affine, one obtains that $C^*(G, _)$ is an exact functor on $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

Let us denote by Δ resp. η the comultiplication resp. the augmentation of $\mathcal{A}(G)$. For every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{P}$, one denotes $\text{Ind}(\mathcal{P}) = \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ equipped with the $\mathcal{A}(G)$ -comodule structure defined by

$$\text{id}_{\mathcal{P}} \otimes \Delta: \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G);$$

this defines a functor $\text{Ind}: (\mathcal{O}_S\text{-Mod.q.c.}) \longrightarrow (G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

It follows from 4.6.4.1, 5.2.0 and 5.2.0.1 that one has an isomorphism of G - $\mathcal{O}_S$ -modules:

$$(*) \quad \mathbf{W}(\text{Ind}(\mathcal{P})) \simeq E(\mathbf{W}(\mathcal{P})) = \text{Hom}(G, \mathbf{W}(\mathcal{P})).$$

Via this identification, the morphism $\varepsilon: E(\mathbf{W}(\mathcal{P})) \longrightarrow \mathbf{W}(\mathcal{P})$ corresponds to the morphism of $\mathcal{O}_S$ -modules $\text{id}_{\mathcal{P}} \otimes \eta: \text{Ind}(\mathcal{P}) \longrightarrow \mathcal{P}$.

One has already used that the functor $\mathbf{W}: (\mathcal{O}_S\text{-Mod.}) \longrightarrow (\mathcal{O}_S\text{-Mod.})$ is fully faithful; the same holds, by definition 4.7.1, of its restriction to $(G\text{-}\mathcal{O}_S\text{-Mod.})$, i.e. if $\mathcal{M}, \mathcal{M}'$ are G - $\mathcal{O}_S$ -modules, one has a functorial isomorphism

$$\text{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathcal{M}, \mathcal{M}') \simeq \text{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathbf{W}(\mathcal{M}), \mathbf{W}(\mathcal{M}')).$$

Consequently, one deduces from lemma 5.2.0.2 the

Corollary 5.3.1.1. *(i) The functor Ind is right adjoint to the forgetful functor*

$$(G\text{-}\mathcal{O}_S\text{-Mod.q.c.}) \longrightarrow (\mathcal{O}_S\text{-Mod.q.c.}).$$

More precisely, the map

$$\text{id}_{\mathcal{P}} \otimes \eta: \text{Ind}(\mathcal{P}) \longrightarrow \mathcal{P}$$

induces, for every object $\mathcal{M}$ of $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$, a bijection

$$\text{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathcal{M}, \text{Ind}(\mathcal{P})) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_S}(\mathcal{M}, \mathcal{P}).$$

(ii) Consequently, if $\mathcal{I}$ is an injective object of $(\mathcal{O}_S\text{-Mod.q.c.})$ then $\text{Ind}(\mathcal{I})$ is an injective object of $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

Let $\mathcal{F}$ be a G - $\mathcal{O}_S$ -module and $\mu_{\mathcal{F}}: \mathcal{F} \longrightarrow \text{Ind}(\mathcal{F})$ the map defining the $\mathcal{A}(G)$ -comodule structure. It follows from 5.2.0.3 (or else from the axioms (CM 1) and (CM 2) of 4.7.2) that $\mu_{\mathcal{F}}$ is a morphism of G - $\mathcal{O}_S$ -modules, and that $(\text{id}_{\mathcal{F}} \otimes \eta) \circ \mu_{\mathcal{F}} = \text{id}_{\mathcal{F}}$, hence that $\mathcal{F}$ is a direct factor of $\text{Ind}(\mathcal{F})$ as an $\mathcal{O}_S$ -module; in particular, $\mu_{\mathcal{F}}$ is a monomorphism. As one has, by (*) and 5.2.2,

$$H^n(G, \mathbf{W}(\text{Ind}(\mathcal{F}))) \simeq H^n(G, \text{Hom}_S(G, \mathbf{W}(\mathcal{F}))) = 0 \quad \text{for } n > 0,$$

one therefore obtains that $H^n(G, _)$ is effaceable for $n > 0$.

Finally, S being affine, $(\mathcal{O}_S\text{-Mod.q.c.})$ has enough injective objects. Let therefore $\mathcal{F} \hookrightarrow \mathcal{I}$ be a monomorphism of $\mathcal{O}_S$ -modules, where $\mathcal{I}$ is an injective object of $(\mathcal{O}_S\text{-Mod.q.c.})$; then, $\mathcal{A}(G)$ being flat over $\mathcal{O}_S$, $\text{Ind}(\mathcal{F})$ is a sub- G - $\mathcal{O}_S$ -module of $\text{Ind}(\mathcal{I})$, whence:

⁽⁵²⁾**N.D.E.** The original has been modified, in order to show that the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ is abelian and has enough injective objects. One may compare with [Ja03], Part I, 3.3–3.4, 3.9, 4.2 and 4.14–4.16 (where one will take care that “ k -group scheme” means “affine k -group scheme”, cf. *loc. cit.*, 2.1).

Corollary 5.3.1.2. *The abelian category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ has enough injective objects.*

Taking into account [Gr57], 2.2.1 and 2.3 (already used in the proof of 5.2.1), this completes the proof of theorem 5.3.1. $\square$

Remark 5.3.1.3. One can also prove 5.3.1.1 by the following computation. To every morphism of $G\text{-}\mathcal{O}_S$ -modules $\phi: \mathcal{M} \rightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ one associates the $\mathcal{O}_S$ -morphism $(\text{id}_{\mathcal{P}} \otimes \eta) \circ \phi: \mathcal{M} \rightarrow \mathcal{P}$. Conversely, to every $\mathcal{O}_S$ -morphism $f: \mathcal{M} \rightarrow \mathcal{P}$ one associates the morphism of $G\text{-}\mathcal{O}_S$ -modules $(f \otimes \text{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}}: \mathcal{M} \rightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$. One sees at once that

$$(\text{id}_{\mathcal{P}} \otimes \eta) \circ (f \otimes \text{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}} = (f \otimes \text{id}_{\mathcal{O}_S}) \circ (\text{id}_{\mathcal{P}} \otimes \eta) \circ \mu_{\mathcal{M}} = f.$$

On the other hand, for every ϕ the diagram below is commutative:

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{\phi} & \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ \mu_{\mathcal{M}} \downarrow & & \downarrow \text{id}_{\mathcal{P}} \otimes \Delta \\ \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{A}(G) & \xrightarrow{\phi \otimes \text{id}_{\mathcal{A}(G)}} & \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G). \end{array}$$

It follows that

$$\begin{aligned} ((\text{id}_{\mathcal{P}} \otimes \eta) \circ \phi) \otimes \text{id}_{\mathcal{A}(G)} \circ \mu_{\mathcal{M}} &= (\text{id}_{\mathcal{P}} \otimes \eta \otimes \text{id}_{\mathcal{A}(G)}) \circ (\phi \otimes \text{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}} \\ &= (\text{id}_{\mathcal{P}} \otimes \eta \otimes \text{id}_{\mathcal{A}(G)}) \circ (\text{id}_{\mathcal{P}} \otimes \Delta) \circ \phi = \phi. \end{aligned}$$

This proves 5.3.1.1 (i) (and (ii) follows).

Let $\mathcal{F}$ be a $G\text{-}\mathcal{O}_S$ -module; one has seen above that axiom (CM 2) of 4.7.2 shows that, considered as an $\mathcal{O}_S$ -module, $\mathcal{F}$ is a direct factor of $E(\mathcal{F})$. This entails:

Proposition 5.3.2. *Let S be an affine scheme and G an affine and flat S -group; suppose that every exact sequence $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$ of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules, which splits as a sequence of $\mathcal{O}_S$ -modules, likewise splits as a sequence of $G\text{-}\mathcal{O}_S$ -modules.*

Then the functors $H^n(G, _)$ are zero for $n > 0$ (or, what amounts to the same, the functor $H^0(G, _)$ is exact).

Indeed, by the hypothesis, the sequence of $G\text{-}\mathcal{O}_S$ -modules

$$0 \rightarrow \mathcal{F} \rightarrow E(\mathcal{F}) \rightarrow E(\mathcal{F})/\mathcal{F} \rightarrow 0$$

is split; $\mathcal{F}$ is therefore a direct factor, as a $G\text{-}\mathcal{O}_S$ -module, in $E(\mathcal{F})$, but the cohomology of the latter is zero.

One deduces immediately from this and from proposition 4.7.4:

Theorem 5.3.3. *Let S be an affine scheme and G a diagonalizable S -group. For every quasi-coherent $G\text{-}\mathcal{O}_S$ -module $\mathcal{F}$, one has $H^n(G, \mathcal{F}) = 0$, for $n > 0$.*

Remark. Proposition 5.3.2 remains valid when G is not necessarily flat over S ; the proof then appeals to relative cohomology.⁽⁵³⁾

6. G -equivariant objects and modules

⁽⁵⁴⁾ Let $\mathcal{C}$ be a category having a final object and in which fiber products exist. Let $\mathbf{G}$ be a $\hat{\mathcal{C}}$ -group, $\pi: \mathbf{M} \rightarrow \mathbf{X}$ a morphism of $\hat{\mathcal{C}}$, and $\lambda = \lambda_{\mathbf{X}}: \mathbf{G} \times \mathbf{X} \rightarrow \mathbf{X}$ an action of $\mathbf{G}$ on $\mathbf{X}$. In what follows, one will denote by $\mathbf{Y} \times_f \mathbf{M}$ the fiber product of $\pi: \mathbf{M} \rightarrow \mathbf{X}$ and of an $\mathbf{X}$ -functor $f: \mathbf{Y} \rightarrow \mathbf{X}$.

⁽⁵³⁾**N.D.E.** The editors have not sought to develop this remark.

⁽⁵⁴⁾**N.D.E.** This section has been added.

For every $U \in \text{Ob}(\mathcal{C})$ and $x \in \mathbf{X}(U)$, one will denote $\mathbf{M}_x = U \times_x \mathbf{M}$, i.e. for every $\varphi: U' \rightarrow U$, one has

$$\mathbf{M}_x(U') = \{m \in \mathbf{M}(U') \mid \pi(m) = x_{U'} = \varphi^*(x)\}.$$

Finally, if $g \in \mathbf{G}(U)$ one will also denote by gx the element $\lambda(g, x)$ of $\mathbf{X}(U)$.

Definition 6.1. a) One says that $\mathbf{M}$ is a G -equivariant $\mathbf{X}$ -object if one has given oneself an action $\Lambda: \mathbf{G} \times \mathbf{M} \rightarrow \mathbf{M}$ of $\mathbf{G}$ on $\mathbf{M}$ lifting λ , i.e. such that the square below is commutative:

$$\begin{array}{ccc} \mathbf{G} \times \mathbf{M} & \xrightarrow{\Lambda} & \mathbf{M} \\ \text{id}_{\mathbf{G}} \times \pi \downarrow & & \downarrow \pi \\ \mathbf{G} \times \mathbf{X} & \xrightarrow{\lambda} & \mathbf{X}. \end{array}$$

This amounts to giving, for every morphism $(g, x): U \rightarrow \mathbf{G} \times \mathbf{X}$, maps

$$\Lambda_x^U(g): \mathbf{M}_x(U) \rightarrow \mathbf{M}_{gx}(U), \quad m \mapsto g \cdot m$$

satisfying $1 \cdot m = m$ and $g \cdot (h \cdot m) = (gh) \cdot m$ and functorial in the $(\mathbf{G} \times \mathbf{X})$ -object U . This amounts again to giving morphisms of U -objects:

$$\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$$

satisfying $\Lambda_x(1) = \text{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$.

b) Let now $\mathbf{O}$ be a $\widehat{\mathcal{C}}$ -ring and let $\mathbf{O}_{\mathbf{X}} = \mathbf{O} \times \mathbf{X}$. Under the conditions of (a), one says that $\mathbf{M}$ is a G -equivariant $\mathbf{O}_{\mathbf{X}}$ -module if it is an $\mathbf{O}_{\mathbf{X}}$ -module (cf. definition 4.3.3.1, valid for every ring-valued functor on a category $\mathcal{C}$) and if the action Λ is compatible with the $\mathbf{O}_{\mathbf{X}}$ -module structure of $\mathbf{M}$, i.e. if for every $(g, x) \in \mathbf{G}(U) \times \mathbf{X}(U)$ ($U \in \text{Ob}(\mathcal{C})$), the map $\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$, $m \mapsto g \cdot m$ is a morphism of $\mathbf{O}_U$ -modules.

Remark 6.2. (i) In (a) above, the conditions $\Lambda_x(1) = \text{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ evidently entail that each $\Lambda_x(g)$ is an *isomorphism*, with inverse $\Lambda_{gx}(g^{-1})$. Conversely, if one supposes that each $\Lambda_x(g)$ is an isomorphism, the condition $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ applied to $h = 1$ gives $\Lambda_x(1) = \text{id}$.

(ii) Let $\mathbf{M}$ be an $\mathbf{O}_{\mathbf{X}}$ -module. First, one sees that giving a morphism $\Lambda: \mathbf{G} \times \mathbf{M} \rightarrow \mathbf{M}$ making the diagram of 6.1 commutative and such that each morphism $\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$, $m \mapsto g \cdot m$, is an isomorphism of $\mathbf{O}_U$ -modules, amounts to giving an *isomorphism* of $\mathbf{O}_{\mathbf{G} \times \mathbf{X}}$ -modules:

$$\begin{aligned} \theta: \mathbf{G} \times \mathbf{M} &= (\mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}}} \mathbf{M} \xrightarrow{\sim} (\mathbf{G} \times \mathbf{X}) \times_{\lambda} \mathbf{M} \\ (g, x, m) &\mapsto (g, x, g \cdot m). \end{aligned}$$

As one has supposed that each $\Lambda_x(g)$ is an isomorphism, the equality $\Lambda_x(1) = \text{id}$ will be a consequence of the equality $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ applied to $h = 1$. One therefore obtains that Λ is an *action* of $\mathbf{G}$ on $\mathbf{M}$ (i.e. $g \cdot (h \cdot m) = (gh) \cdot m$) if and only if the diagram of $(\mathbf{G} \times \mathbf{G} \times \mathbf{X})$ -isomorphisms below is commutative (where one denotes by μ the multiplication of $\mathbf{G}$ and by $f^*(\theta)$ the isomorphism deduced from θ by a base change $f: \mathbf{G} \times \mathbf{G} \times \mathbf{X} \rightarrow \mathbf{G} \times \mathbf{X}$):

$$\begin{array}{ccc} (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ \text{pr}_{2,3}} \mathbf{M} & \xrightarrow[\sim]{\text{pr}_{2,3}^*(\theta)} & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ \text{pr}_{2,3}} \mathbf{M} \\ \\ (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ (\mu \times \text{id}_{\mathbf{X}})} \mathbf{M} & & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ (\text{id}_{\mathbf{G}} \times \lambda)} \mathbf{M} \\ (\mu \times \text{id}_{\mathbf{X}})^*(\theta) \downarrow \sim & & (\text{id}_{\mathbf{G}} \times \lambda)^*(\theta) \downarrow \sim \\ (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ (\mu \times \text{id}_{\mathbf{X}})} \mathbf{M} & & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ (\text{id}_{\mathbf{G}} \times \lambda)} \mathbf{M}. \end{array}$$

One therefore sees that giving a structure of G -equivariant $\mathbf{O}_{\mathbf{X}}$ -module on $\mathbf{M}$ amounts to giving an isomorphism θ of $\mathbf{O}_{\mathbf{G} \times \mathbf{X}}$ -modules as above, such that the diagram above is commutative.

(iii) Everything that precedes extends to the case where $\mathbf{G}$ is only a $\widehat{\mathcal{C}}$ -monoid: in this case, giving an action $\Lambda: \mathbf{G} \times \mathbf{M} \rightarrow \mathbf{M}$ lifting λ and such that each $\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$ is a morphism of $\mathbf{O}_U$ -modules, amounts to giving a *morphism* θ of $\mathbf{O}_{\mathbf{G} \times \mathbf{X}}$ -modules as in (ii), such that the diagram above (without the signs $\sim$ under the arrows) is commutative, and such that $p_{\mathbf{M}} \circ \theta \circ (\varepsilon_{\mathbf{G}} \times \text{id}_{\mathbf{M}}) = \text{id}_{\mathbf{M}}$, where $\varepsilon_{\mathbf{G}}$ denotes the unit section of $\mathbf{G}$ and $p_{\mathbf{M}}$ the projection onto $\mathbf{M}$.

6.3. G -equivariant morphisms. — Let $\mathbf{Y}$ be a second object of $\widehat{\mathcal{C}}$, equipped with an action $\lambda_{\mathbf{Y}}: \mathbf{G} \times \mathbf{Y} \rightarrow \mathbf{Y}$ of $\mathbf{G}$, and let $\mathbf{N}$ be a second G -equivariant $\mathbf{O}_{\mathbf{X}}$ -module. One says that a $\widehat{\mathcal{C}}$ -morphism $f: \mathbf{Y} \rightarrow \mathbf{X}$ resp. a morphism of $\mathbf{O}_{\mathbf{X}}$ -modules $\phi: \mathbf{M} \rightarrow \mathbf{N}$ is *G -equivariant* if it commutes with the action of $\mathbf{G}$, i.e. if one has, set-theoretically, $f(g \cdot y) = g \cdot f(y)$ resp. $\phi(g \cdot m) = g \cdot \phi(m)$, which amounts to saying that $f \circ \lambda_{\mathbf{Y}} = \lambda_{\mathbf{X}} \circ (\text{id}_{\mathbf{G}} \times f)$ resp. $\phi \circ \Lambda_{\mathbf{M}} = \Lambda_{\mathbf{N}} \circ (\text{id}_{\mathbf{G}} \times \phi)$.

One then obtains at once the following lemma:

Lemma 6.3.1. *Let $f: \mathbf{Y} \rightarrow \mathbf{X}$ be a G -equivariant morphism and $\mathbf{M}$ a G -equivariant $\mathbf{O}_{\mathbf{X}}$ -module. Then the inverse image $f^*(\mathbf{M}) = \mathbf{Y} \times_f \mathbf{M}$ is a G -equivariant $\mathbf{O}_{\mathbf{Y}}$ -module.*

On the other hand, $\mathbf{G}$ acts on $\text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{Y}, \mathbf{X})$. Indeed, let $T \in \text{Ob}(\mathcal{C})$, $g \in \mathbf{G}(T)$ and φ a T -morphism $\mathbf{Y}_T \rightarrow \mathbf{X}_T$; then g defines automorphisms $\lambda_{\mathbf{Y}}(g)$ and $\lambda_{\mathbf{X}}(g)$ of $\mathbf{Y}_T$ and $\mathbf{X}_T$, and one will denote by $g \cdot \varphi$ (or also $g\varphi g^{-1}$) the morphism $\lambda_{\mathbf{X}}(g) \circ \varphi \circ \lambda_{\mathbf{Y}}(g^{-1})$. This defines an action of $\mathbf{G}(T)$ on $\text{Hom}_T(\mathbf{Y}_T, \mathbf{X}_T)$, functorial in T .

Definition 6.3.2. If $\varphi: \mathbf{Y} \rightarrow \mathbf{X}$ is an arbitrary $\widehat{\mathcal{C}}$ -morphism, one may therefore consider its stabilizer $\text{Stab}_{\mathbf{G}}(\varphi)$ (cf. 2.3.3.1): for every $T \in \text{Ob}(\mathcal{C})$, $\text{Stab}_{\mathbf{G}}(\varphi)(T)$ is the subgroup of $\mathbf{G}(T)$ formed of the g such that $g \circ \varphi_T = \varphi_T \circ g$, i.e. such that the diagram

$$\begin{array}{ccc} \mathbf{Y}_T & \xrightarrow{\varphi_T} & \mathbf{X}_T \\ g \downarrow & & \downarrow g \\ \mathbf{Y}_T & \xrightarrow{\varphi_T} & \mathbf{X}_T \end{array}$$

commutes. Then the morphism $\varphi: \mathbf{Y} \rightarrow \mathbf{X}$ is equivariant under $\text{Stab}_{\mathbf{G}}(\varphi)$.

6.4. Global sections. — Let $\mathbf{M}$ be a G -equivariant $\mathbf{O}_{\mathbf{X}}$ -module. Denote by S_0 the final object of $\mathcal{C}$ and (cf. Exp. II, 1.1) by $\prod_{\mathbf{X}/S_0} \mathbf{M}$ the “functor of sections of $\mathbf{M}$ over $\mathbf{X}$ ”: it is the functor which to every $T \in \text{Ob}(\mathcal{C})$ associates

$$\text{Hom}_{\mathbf{X}}(\mathbf{X}_T, \mathbf{M}) = \text{Hom}_{\mathbf{X}_T}(\mathbf{X}_T, \mathbf{M}_T) = \Gamma(\mathbf{M}_T/\mathbf{X}_T).$$

Let us recall on the other hand that every morphism $g: \mathbf{Z} \rightarrow \mathbf{Y}$ of $\widehat{\mathcal{C}}$ -objects over $\mathbf{X}$ induces a morphism of abelian groups $\mathbf{M}(g): \mathbf{M}(\mathbf{Y}) \rightarrow \mathbf{M}(\mathbf{Z})$, which is compatible with the ring morphism $g^*: \mathbf{O}(\mathbf{Y}) \rightarrow \mathbf{O}(\mathbf{Z})$. In particular, when $\mathbf{Z} = \mathbf{Y}$ (g then being an $\mathbf{X}$ -endomorphism of $\mathbf{Y}$), one obtains a morphism of abelian groups

$$\mathbf{M}(g): \mathbf{M}(\mathbf{Y}) \rightarrow \mathbf{M}(\mathbf{Y})$$

which is not in general a morphism of $\mathbf{O}(\mathbf{Y})$ -modules, but which satisfies, for every $m \in \mathbf{M}(\mathbf{Y})$ and $\alpha \in \mathbf{O}(\mathbf{Y})$:

$$\mathbf{M}(g)(\alpha \cdot m) = g^*(\alpha) \cdot \mathbf{M}(g)(m).$$

This being said, one will simply write, in what follows, g^* instead of $\mathbf{M}(g)$.

Let $T \in \text{Ob}(\mathcal{C})$ and let $\mathbf{X}_T = \mathbf{X} \times T$ and $\text{pr}_{\mathbf{X}}$ be the projection $\mathbf{X}_T \rightarrow \mathbf{X}$. For every $\alpha \in \mathbf{O}(\mathbf{X}_T)$ and $g \in \mathbf{G}(T)$, set $g(\alpha) = (g^{-1})^*(\alpha)$: this is the element of $\mathbf{O}(\mathbf{X}_T)$ defined

set-theoretically by: for every $S \in \text{Ob}(\mathcal{C})$ and $x \in \mathbf{X}(S)$, $t \in T(S)$,

$$g(\alpha)(x, t) = \alpha(g^{-1}x, t).$$

One thus obtains an action (on the left) of $\mathbf{G}(T)$ by ring automorphisms on $\mathbf{O}(\mathbf{X}_T)$, functorial in T , and such that $g(\alpha) = \alpha$ if α is the image in $\mathbf{O}(\mathbf{X}_T)$ of an element of $\mathbf{O}(T)$.

Let us now denote by ϕ the identity morphism of $\mathbf{X}_T$ (cf. 6.3 and the generalization below to a morphism $\phi: \mathbf{Y} \rightarrow \mathbf{X}$) and denote $\text{Hom}_{\mathbf{X}}(\mathbf{X}_T, \mathbf{M})$ by $\mathbf{M}(\phi g^{-1})$ resp. $\mathbf{M}(\phi)$, according as $\mathbf{X}_T$ is regarded as an $\mathbf{X}$ -object via $\text{pr}_{\mathbf{X}} \circ \lambda(g^{-1})_T$, resp. $\text{pr}_{\mathbf{X}}$.

Then $\lambda(g^{-1})_T$ is an $\mathbf{X}$ -morphism between these two $\mathbf{X}$ -objects, hence, by what precedes, one obtains a morphism of abelian groups

$$(g^{-1})^*: \mathbf{M}(\phi) \rightarrow \mathbf{M}(\phi g^{-1}), \quad m \mapsto m \circ \lambda(g^{-1})_T$$

which satisfies $(g^{-1})^*(\alpha \cdot m) = (g\alpha) \cdot (g^{-1})^*(m)$ for every $m \in \mathbf{M}(\phi)$ and $\alpha \in \mathbf{O}(\mathbf{X}_T)$. (If m is a section of $\mathbf{M}_T$ over $\mathbf{X}_T$, then $(g^{-1})^*(m)$ is the section defined set-theoretically by $(x, t) \mapsto m(g^{-1}x, t)$.) In particular, $(g^{-1})^*$ is a morphism of $\mathbf{O}(T)$ -modules.

By the functoriality of the morphisms of $\mathbf{O}(\mathbf{X}_T)$ -modules $\Lambda_x(g)$, one obtains a commutative diagram:

$$\begin{array}{ccc} \mathbf{M}(\phi) & \xrightarrow{(g^{-1})^*} & \mathbf{M}(\phi g^{-1}) \\ \Lambda_{\phi}(g) \downarrow & & \downarrow \Lambda_{\phi g^{-1}}(g) \\ \mathbf{M}(g\phi) & \xrightarrow{(g^{-1})^*} & \mathbf{M}(g\phi g^{-1}) \end{array}$$

and $g\phi g^{-1} = \phi$, since ϕ is the identity map. Writing $A(g) = (g^{-1})^* \circ \Lambda_{\phi}(g)$, one therefore obtains a morphism of abelian groups

$$A(g): \mathbf{M}(\phi) \rightarrow \mathbf{M}(\phi)$$

which is “compatible with the action of $\mathbf{G}(T)$ on $\mathbf{O}(\mathbf{X}_T)$ ”, i.e. which satisfies

$$A(g)(\alpha \cdot m) = (g\alpha) \cdot A(g)(m).$$

Finally, if h is a second element of $\mathbf{G}(T)$, it follows from the functoriality of $\mathbf{M}$ and of the morphisms $\Lambda_x(g)$ that one has a commutative diagram

$$\begin{array}{ccccc} & & \mathbf{M}(\phi) & & \\ & \swarrow \Lambda_{\phi}(g) & & \searrow \Lambda_{\phi}(hg) & \\ \mathbf{M}(g\phi) & \xrightarrow{\Lambda_{\phi}(h)} & \mathbf{M}(hg\phi) & & \\ (g^{-1})^* \downarrow & & \downarrow (g^{-1})^* & \searrow ((hg)^{-1})^* & \\ \mathbf{M}(\phi) & \xrightarrow{\Lambda_{\phi}(h)} & \mathbf{M}(h\phi) & \xrightarrow{(h^{-1})^*} & \mathbf{M}(\phi) \end{array}$$

whence $A(h) \circ A(g) = A(hg)$. Consequently, one has obtained the following proposition.

Proposition 6.4.1. *For every T , the $\mathbf{O}(\mathbf{X}_T)$ -module $(\prod_{\mathbf{X}/S_0} \mathbf{M})(T) = \Gamma(\mathbf{M}_T/\mathbf{X}_T)$ is endowed, in a manner functorial in T , with an action of $\mathbf{G}(T)$ “compatible with the action of $\mathbf{G}(T)$ on $\mathbf{O}(\mathbf{X}_T)$ ”, i.e. which satisfies*

$$A(g)(\alpha \cdot m) = g(\alpha) \cdot A(g)(m).$$

As $g(\alpha) = \alpha$ for $\alpha \in \mathbf{O}(T)$, this shows, in particular, that $\prod_{\mathbf{X}/S_0} \mathbf{M}$ is a $\mathbf{G}\text{-}\mathbf{O}_{S_0}$ -module.

More generally let, as in 6.3, $\mathbf{Y}$ be a second G -equivariant $\hat{\mathcal{C}}$ -object, $\phi: \mathbf{Y} \rightarrow \mathbf{X}$ a $\hat{\mathcal{C}}$ -morphism and $\mathbf{H} = \text{Stab}_{\mathbf{G}}(\phi)$. Then the fiber product $\mathbf{M}_{\phi} = \mathbf{Y} \times_{\phi} \mathbf{M}$ is an $\mathbf{H}$ -equivariant $\mathbf{O}_{\mathbf{Y}}$ -module. One therefore obtains the:

Corollary 6.4.2. *The functor $\prod_{\mathbf{Y}/S_0} \mathbf{M}_{\phi}$ is a $\text{Stab}_{\mathbf{G}}(\phi)\text{-}\mathbf{O}_{S_0}$ -module.*

6.5. G -equivariant $\mathcal{O}_X$ -modules. — Let us apply what precedes in the following situation. Let S be a scheme, G an S -group scheme acting on an S -scheme X , and $\mathcal{F}$ an $\mathcal{O}_X$ -module (not necessarily quasi-coherent).

Definition 6.5.1. One says that $\mathcal{F}$ is a G -equivariant $\mathcal{O}_X$ -module if the $\mathbf{O}_X$ -module $\mathbf{M} = \mathbf{W}(\mathcal{F})$ is G -equivariant, i.e. if one has given oneself, for every morphism $(g, x): U \rightarrow G \times_S X$, isomorphisms of $\mathbf{O}_U$ -modules

$$\Lambda_x(g): \mathbf{W}(x^*(\mathcal{F})) \xrightarrow{\sim} \mathbf{W}((gx)^*(\mathcal{F})),$$

functorial in U and satisfying $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$. As the functor $\mathbf{W}$ is fully faithful (cf. 4.6.1.1 and 4.6.2), one therefore obtains isomorphisms of $\mathcal{O}_U$ -modules

$$\Lambda_x(g): x^*(\mathcal{F}) \xrightarrow{\sim} (gx)^*(\mathcal{F}),$$

where one recalls that gx denotes the morphism $\lambda \circ (g, x): U \rightarrow X$. In particular, applying this to the identity morphism of $G \times_S X$, one obtains an isomorphism of $\mathcal{O}_{G \times_S X}$ -modules

$$(\star) \quad \theta: \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

such that the diagram $(\star\star)$ of morphisms of sheaves on $G \times_S G \times_S X$ below is commutative:

$$\begin{array}{ccc} \mathrm{pr}_{2,3}^* \circ \mathrm{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{\mathrm{pr}_{2,3}^*(\theta)} & \mathrm{pr}_{2,3}^* \circ \lambda^*(\mathcal{F}) & (\mathrm{id}_G \times \lambda)^* \circ \mathrm{pr}_X^*(\mathcal{F}) \\ & & (\star\star) & \downarrow (\mathrm{id}_G \times \lambda)^*(\theta) \\ (\mu \times \mathrm{id}_X)^* \circ \mathrm{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{(\mu \times \mathrm{id}_X)^*(\theta)} & (\mu \times \mathrm{id}_X)^* \circ \lambda^*(\mathcal{F}) & (\mathrm{id}_G \times \lambda)^* \circ \lambda^*(\mathcal{F}) \end{array}$$

Moreover, if $\mathcal{E}$ is a second G -equivariant $\mathcal{O}_X$ -module, one says that a morphism of $\mathcal{O}_X$ -modules $\phi: \mathcal{E} \rightarrow \mathcal{F}$ is G -equivariant if the morphism $\mathbf{W}(\phi): \mathbf{W}(\mathcal{E}) \rightarrow \mathbf{W}(\mathcal{F})$ is. With the notations above, this amounts to saying that $\theta_{\mathcal{F}} \circ \mathrm{pr}_X^*(\phi) = \lambda^*(\phi) \circ \theta_{\mathcal{E}}$.

Remark 6.5.2. If $f: Y \rightarrow X$ is a G -equivariant morphism, it follows from 6.3.1 that $f^*(\mathcal{F})$ is a G -equivariant $\mathcal{O}_Y$ -module.

Remark 6.5.3. Suppose moreover that $\mathcal{F}$ is *quasi-coherent*. Then it follows from what precedes that the datum of a structure of G -equivariant $\mathbf{O}_X$ -module on $\mathbf{M} = \mathbf{W}(\mathcal{F})$ amounts to the datum of such a structure on $\mathbf{V}(\mathcal{F})$, which still amounts to giving oneself an action of G on the vector fibration $\mathbf{V}(\mathcal{F})$, compatible with the action on X and “linear” on the fibers.

Indeed, denote by ϕ the isomorphism $\lambda^*(\mathcal{F}) \xrightarrow{\sim} \mathrm{pr}_X^*(\mathcal{F})$ inverse to θ . For every morphism $(g, x): U \rightarrow G \times_S X$, one has an isomorphism of $\mathcal{O}_U$ -modules $\phi_x(g) = \Lambda_x(g)^{-1} = \Lambda_{gx}(g^{-1})$ from $(gx)^*(\mathcal{F})$ to $x^*(\mathcal{F})$; it induces an isomorphism of $\mathbf{O}_U$ -modules

$$\mathbf{V}((gx)^*(\mathcal{F})) \xrightarrow{\sim} \mathbf{V}(x^*(\mathcal{F}))$$

which one will denote by ${}^t\phi_x(g)$. One then has

$${}^t\phi_{gx}(h) \circ {}^t\phi_x(g) = {}^t(\Lambda_{gx}(h) \circ \phi_x(g))^{-1} = {}^t\Lambda_x(hg)^{-1} = {}^t\phi_x(hg).$$

As, for every X -scheme $f: Y \rightarrow X$, one has $\mathbf{V}(\mathcal{F}) \times_f Y \simeq \mathbf{V}(f^*(\mathcal{F}))$ (cf. 4.6.2), one therefore obtains that the isomorphism

$$\begin{aligned} {}^t\phi: G \times_S \mathbf{V}(\mathcal{F}) &= (G \times_S X) \times_{\mathrm{pr}_X} \mathbf{V}(\mathcal{F}) = \mathbf{V}(\mathrm{pr}_X^*(\mathcal{F})) \\ &\xrightarrow{\sim} \mathbf{V}(\lambda^*(\mathcal{F})) = (G \times_S X) \times_{\lambda} \mathbf{V}(\mathcal{F}) \end{aligned}$$

endows $\mathbf{V}(\mathcal{F})$ with a structure of G -equivariant $\mathbf{O}_X$ -module. Finally, identifying each functor $\mathbf{V}(-)$ with the vector fibration which represents it, and composing ${}^t\phi$ with the projection onto $\mathbf{V}(\mathcal{F})$, one obtains an action of G on the vector fibration $\mathbf{V}(\mathcal{F})$, compatible with the action on X and “linear” on the fibers.

One thus recovers the definition given, for example, in [GIT, Chap. 1, § 3], except that in *loc. cit.* Mumford considers an $\mathcal{O}_X$ -module $\mathcal{E}$ locally free of finite rank, and defines an

action of G on $\mathbf{V}(\mathcal{E}) = \mathbf{W}(\mathcal{E}^\vee)$. Indeed, the preceding diagram $(\star\star)$, with θ replaced by ϕ and the direction of the arrows reversed, is exactly the one found in *loc. cit.*, Def. 1.6, and the isomorphism ${}^t\phi$ above coincides with the isomorphism Φ of *loc. cit.*, p. 31.

Remark 6.5.4. Let us consider in particular the case where $X = S$, endowed with the trivial action of G . In this case, a G -equivariant $\mathcal{O}_S$ -module $\mathcal{F}$ is the same thing as a G - $\mathbf{O}_S$ -module (cf. 4.7.1). Moreover, if one denotes by f the morphism $G \rightarrow S$ (here equal both to pr_X and to λ), then the isomorphism

$$(\star) \quad \theta: f^*(\mathcal{F}) \xrightarrow{\sim} f^*(\mathcal{F})$$

is an element of

$$\begin{aligned} \text{Hom}_{\mathcal{O}_G}(f^*(\mathcal{F}), f^*(\mathcal{F})) &= \text{Hom}_{\mathbf{O}_G}(\mathbf{W}(f^*(\mathcal{F})), \mathbf{W}(f^*(\mathcal{F}))) \\ &= \text{End}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}))(G) \end{aligned}$$

which is none other than the morphism $\rho: G \rightarrow \text{End}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}))$ defining the operation of G on $\mathbf{W}(\mathcal{F})$. Moreover, θ corresponds by adjunction to the morphism of $\mathcal{O}_S$ -modules $f_*(\theta) \circ \tau: \mathcal{F} \rightarrow f_*f^*(\mathcal{F})$, where $\tau: \mathcal{F} \rightarrow f_*f^*(\mathcal{F})$ is the “unit” morphism of the adjunction. (This will be used in VI_B, 11.10.bis.)

6.6. The functors $\prod_{X/S} \mathbf{W}(\mathcal{F})$ and $\mathbf{W}(p_*(\mathcal{F}))$. — Let S be a scheme, G an S -group scheme acting on S -schemes X and Y via the morphisms $\lambda: G \times_S X \rightarrow X$ and $\mu: G \times_S Y \rightarrow Y$, and let $p: X \rightarrow Y$ be a G -equivariant morphism. Suppose that the morphisms $p: X \rightarrow Y$ and $\nu_Y: Y \rightarrow S$ are *quasi-compact* and *quasi-separated*, and that $\pi: G \rightarrow S$ is *flat*.

Then the projection $\text{pr}_Y: G \times_S Y \rightarrow Y$ is flat, as is μ (since μ is the composite of pr_Y and of the automorphism $(g, y) \mapsto (g, gy)$). Finally, for a variable S -scheme $f: T \rightarrow S$, one will denote by $p^T: X_T \rightarrow T$ and $f_X: X_T \rightarrow X$ the morphisms deduced from p and f by base change.

Let $\mathcal{F}$ be a *quasi-coherent* and G -equivariant $\mathcal{O}_X$ -module, and let θ be the isomorphism $\text{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$ of 6.5.1 $(\star)$. As p is quasi-compact and quasi-separated, $p_*(\mathcal{F})$ is quasi-coherent (cf. EGA I, 9.2.1).

Lemma 6.6.1. *The quasi-coherent $\mathcal{O}_Y$ -module $p_*(\mathcal{F})$ is G -equivariant.*

Indeed, one has the two cartesian squares below:

$$\begin{array}{ccccc} G \times_S X & \xrightarrow{\text{pr}_X} & X & \xleftarrow{\lambda} & G \times_S X \\ q \downarrow & & \downarrow p & & \downarrow q \\ G \times_S Y & \xrightarrow{\text{pr}_Y} & Y & \xleftarrow{\mu} & G \times_S Y. \end{array}$$

As p is quasi-compact and quasi-separated and pr_Y and μ are flat, it follows from EGA III, 1.4.15 (supplemented by EGA IV₁, 1.7.21) that one has $q^*\text{pr}_X^*(\mathcal{F}) = \text{pr}_Y^*p_*(\mathcal{F})$ and $q^*\lambda^*(\mathcal{F}) = \mu^*p_*(\mathcal{F})$. Consequently, θ induces an isomorphism

$$\theta_Y: \text{pr}_Y^*p_*(\mathcal{F}) \xrightarrow{\sim} \mu^*p_*(\mathcal{F}),$$

and one likewise obtains that θ_Y satisfies the “cocycle condition” 6.5.1 $(\star\star)$ (since the morphisms $G \times_S G \times_S Y \rightarrow G \times_S Y$ which occur are flat). This proves the lemma.

In particular, take $Y = S$ endowed with the trivial action of G . Taking remark 6.5.4 into account, one then obtains that $p_*(\mathcal{F})$ is a G - $\mathbf{O}_S$ -module. If moreover G is affine over S and if one writes $\mathcal{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ is therefore an $\mathcal{A}(G)$ -comodule, by 4.7.2.

On the other hand, by 6.4.1, the functor $\prod_{X/S} \mathbf{W}(\mathcal{F})$, which to every $f: T \rightarrow S$ associates

$$\mathbf{W}(f_X^*(\mathcal{F}))(X_T) = \Gamma(X_T, f_X^*(\mathcal{F})) = \Gamma(T, p_*^T f_X^*(\mathcal{F}))$$

is a $G\text{-}\mathbf{O}_S$ -module. Moreover, one has a canonical morphism $\tau: \mathbf{W}(p_*(\mathcal{F})) \rightarrow \prod_{X/S} \mathbf{W}(\mathcal{F})$, which is given for every $f: T \rightarrow S$ by the canonical morphism

$$\Gamma(T, f^*p_*(\mathcal{F})) \rightarrow \Gamma(T, p_*^T f_X^*(\mathcal{F}))$$

and which is an isomorphism when one restricts it to the full subcategory of schemes T flat over S , and one checks without difficulty that τ is a morphism of $G\text{-}\mathbf{O}_S$ -modules. Consequently, one has obtained the following proposition (for point (ii), compare with [GIT], p. 32).

Proposition 6.6.2. *Let S be a scheme, G an S -group scheme acting on an S -scheme X , and $\mathcal{F}$ a quasi-coherent and G -equivariant $\mathcal{O}_X$ -module. Suppose that $\pi: G \rightarrow S$ is flat and that $p: X \rightarrow S$ is quasi-compact and quasi-separated.*

- (i) *Then $p_*(\mathcal{F})$ is a quasi-coherent $G\text{-}\mathbf{O}_S$ -module. On the other hand, the canonical morphism $\mathbf{W}(p_*(\mathcal{F})) \rightarrow \prod_{X/S} \mathbf{W}(\mathcal{F})$ is a morphism of $G\text{-}\mathbf{O}_S$ -modules, and these two functors coincide on the category of flat S -schemes.*
- (ii) *If moreover G is affine over S and if one writes $\mathcal{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ is endowed with a structure of $\mathcal{A}(G)$ -comodule.⁽⁵⁵⁾*

6.7. Stabilizers. — Let S be a scheme, G an S -group scheme acting on an S -scheme X , and let $\mathcal{F}$ be a quasi-coherent G -equivariant $\mathcal{O}_X$ -module. Let Y be a second S -scheme endowed with an action of G (possibly trivial), let $\phi: Y \rightarrow X$ be an S -morphism, not necessarily G -equivariant, and let $\text{Stab}_G(\phi)$ be the stabilizer in G of ϕ (cf. 6.3.2).

Let us note at once (see Exp. VI_B, § 6) that $\text{Stab}_G(\phi)$ is representable by a closed group subscheme H of G if X is separated over S and if Y is essentially free over S (cf. *loc. cit.*, Def. 6.2.1). Indeed, consider the morphism $r: G \times_S Y \rightarrow X \times_S X$ given set-theoretically by $r(g, y) = (\phi(y), g\phi(g^{-1}y))$, and let $P = G \times_S Y$ and P' be the inverse image by r of the diagonal $\Delta_{X/S}$. Then one has (cf. *loc. cit.*, 6.2.4 (a))

$$\text{Stab}_G(\phi) = \prod_{P/G} P'$$

and hence, by *loc. cit.*, $\text{Stab}_G(\phi)$ is representable by a closed group subscheme H of G if X is separated over S and if Y is essentially free over S ; this second condition being automatically satisfied if S is the spectrum of a field, or else if $Y = S$. Under these hypotheses, $\phi^*(\mathcal{F})$ is then a quasi-coherent H -equivariant $\mathcal{O}_Y$ -module (cf. 6.5.2).

Therefore, if moreover $\pi: G \rightarrow S$ and $p: Y \rightarrow S$ are quasi-compact and quasi-separated over S , and π is flat, then $p_*\phi^*(\mathcal{F})$ is an $H\text{-}\mathbf{O}_S$ -module, by 6.6.2. In particular, one obtains the:

Corollary 6.7.1. *Let S be a scheme, G an S -group scheme acting on an S -scheme X , and $\mathcal{F}$ a quasi-coherent G -equivariant $\mathcal{O}_X$ -module. Suppose that $\pi: G \rightarrow S$ is flat and that $X \rightarrow S$ is quasi-compact and separated. Let $\tau: S \rightarrow X$ be a section of X over S .*

- (i) *The stabilizer $H = \text{Stab}_G(\tau)$ is a closed group subscheme of G , and $\tau^*(\mathcal{F})$ is a quasi-coherent $H\text{-}\mathbf{O}_S$ -module.*
- (ii) *If moreover H is affine over S (for example, if G is) and if one writes $\mathcal{A}(H) = \pi_*(\mathcal{O}_H)$, then $\tau^*(\mathcal{F})$ is an $\mathcal{A}(H)$ -comodule.*

6.8. G -equivariant sheaves on G . — To conclude, let us indicate two results (6.8.1 and 6.8.6 below) which will be used in exposés II and III (cf. in particular III, 4.25).

Proposition 6.8.1. *Let S be a scheme, $\pi: G \rightarrow S$ an S -group scheme, and $\varepsilon: S \rightarrow G$ the unit section. One considers the action by left translations of G on itself. Then the functors $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ and $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ induce equivalences, quasi-inverse to one another, between the category of quasi-coherent $\mathcal{O}_S$ -modules and that of quasi-coherent G -equivariant $\mathcal{O}_G$ -modules.*

⁽⁵⁵⁾**N.D.E.** The same holds if G is flat, quasi-compact and quasi-separated over S and if $p_*(\mathcal{F})$ is a flat $\mathcal{O}_S$ -module, cf. Exp. VI_B, 11.6.1 (ii).

PROOF. Denote by μ the multiplication of G and by pr_2 the second projection $G \times_S G \rightarrow G$. As $\pi \circ \mu = \pi \circ \text{pr}_2$, for every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$ one has a canonical isomorphism $\mu^* \pi^*(\mathcal{E}) = \text{pr}_2^* \pi^*(\mathcal{E})$, and one easily checks that this isomorphism satisfies the “cocycle condition” 6.5.1(★), i.e. $\pi^*(\mathcal{E})$ is a G -equivariant $\mathcal{O}_G$ -module. As $\varepsilon^* \pi^*(\mathcal{E}) = \mathcal{E}$, the functor $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ is fully faithful; it therefore remains to see that for every G -equivariant quasi-coherent $\mathcal{O}_G$ -module $\mathcal{F}$, one has $\mathcal{F} \simeq \pi^* \varepsilon^*(\mathcal{F})$.

By hypothesis, one has an isomorphism $\theta: \text{pr}_2^*(\mathcal{F}) \xrightarrow{\sim} \mu^*(\mathcal{F})$; taking the inverse image of θ by the morphism $\tau: G \rightarrow G \times_S G$ with components $(\text{id}_G, \varepsilon \circ \pi)$, one obtains an isomorphism $\mathcal{F} \xrightarrow{\sim} \pi^* \varepsilon^*(\mathcal{F})$.

Remarks 6.8.2. (a) Let us consider the action of $G \times_S G$ on G defined by $(g_1, g_2) \cdot g = g_1 g g_2^{-1}$; then the stabilizer of the unit section $\varepsilon: S \rightarrow G$ is the diagonal subgroup H of $G \times_S G$. Consequently, if $\mathcal{F}$ is a quasi-coherent $(G \times_S G)$ -equivariant $\mathcal{O}_G$ -module, then, by 6.5.2, $\mathcal{E} = \varepsilon^*(\mathcal{F})$ is endowed with the structure of an H - $\mathcal{O}_S$ -module.

(b) One can show that $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ is an equivalence of categories, between the category of quasi-coherent $(G \times_S G)$ -equivariant $\mathcal{O}_G$ -modules and that of quasi-coherent H - $\mathcal{O}_S$ -modules. This is a particular case of more general “descent” results (cf. Exp. IV, §2 and SGA 1, VIII); see for example [Th87], 1.2–1.3.

Remark 6.8.3. One keeps the notations of 6.8.1. For every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$, let us denote by $\pi_*^G \pi^*(\mathcal{E})$ the sub- $\mathcal{O}_S$ -module of $\pi_* \pi^*(\mathcal{E})$ whose sections over every open V of S are the $\gamma \in \Gamma(\pi^{-1}(V), \pi^*(\mathcal{E}))$ such that $g \cdot \gamma_{S'} = \gamma_{S'}$ for every $S' \rightarrow V$ and every $g \in G(S')$. Then the natural morphism $\mathcal{E} \rightarrow \pi_*^G \pi^*(\mathcal{E})$ is an isomorphism: this is immediate if $\pi_* \pi^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_S} \pi_*(\mathcal{O}_G)$ (for example if $G \rightarrow S$ is affine, or if $G \rightarrow S$ is quasi-compact and quasi-separated and $\mathcal{E}$ is flat), and this is verified without difficulty in the general case.

Remark 6.8.4. Let S be a scheme, H an S -group scheme operating on an S -scheme X , and $\mathcal{F}$ an $\mathcal{O}_X$ -module. One assumes that H is flat over S , and one denotes by $\mathbf{W}_P(\mathcal{F})$ the restriction of the functor $\mathbf{W}(\mathcal{F})$ to the full subcategory consisting of the flat S -schemes. Since, by 6.5.1, endowing $\mathcal{F}$ with an H -equivariant module structure is equivalent to giving an isomorphism $\theta: \text{pr}_X^*(\mathcal{F}) \rightarrow \lambda^*(\mathcal{F})$ of sheaves on $G \times_S X$, satisfying the “cocycle condition” (★), one sees that, in order to give an H -equivariant module structure on $\mathcal{F}$, it suffices to give such a structure on $\mathbf{W}_P(\mathcal{F})$.

Recall 6.8.5. Let X be an S -scheme and Y a sub- S -scheme of X . One denotes by $\mathcal{N}_{Y/X}$ the conormal sheaf of the immersion $i: Y \hookrightarrow X$ (cf. EGA IV₄, 16.1.2). If $S' \rightarrow S$ is a flat morphism and if one denotes by $i': Y' \hookrightarrow X'$ the immersion deduced from i by base change, then, by loc. cit., 16.2.2 (iii), one has $\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'} = \mathcal{N}_{Y/X}$.

Proposition 6.8.6. Let S be a scheme, X an S -group scheme, and Y a group sub- S -scheme of X . One assumes that Y is flat over S . Then the conormal sheaf $\mathcal{N}_{Y/X}$ is a $(Y \times_S Y)$ -equivariant $\mathcal{O}_Y$ -module.

Indeed, $H = Y \times_S Y$ is flat over S , so, by remark 6.8.4, it suffices to endow $\mathbf{W}_P(\mathcal{N}_{Y/X})$ with the structure of an H -equivariant module. Let S' be a flat S -scheme, let $Y' \hookrightarrow X'$ be the immersion obtained by base change, and let $\mathcal{N}' = \mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'}$. By 6.8.5, one has $\mathcal{N}' = \mathcal{N}_{Y'/X'}$.

Every $h \in H(S')$ induces an automorphism of Y' , and one therefore obtains, for every $y \in Y'(S')$, isomorphisms

$$\Gamma(S', y^*(\mathcal{N}')) \xrightarrow{\sim} \Gamma(S', y^* h^*(\mathcal{N}'))$$

which endow $\mathbf{W}_P(\mathcal{N})$ with the structure of an H -equivariant module (cf. 6.1).

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⁽⁵⁶⁾**N.D.E.** : additional references cited in this Exposé.

Exposé II. Tangent bundles. Lie algebras

by M. Demazure

⁽⁰⁾We propose in this exposé to construct the analogue, in the theory of schemes, of the *tangent bundles* and *Lie algebras* of the classical theory. It will however be useful not to restrict oneself to schemes properly so called, but also to take an interest in certain functors on the category of schemes which are not necessarily *representable* (for example functors Hom , Norm , etc.). As was announced in the previous exposé (cf. I 1.1), we shall identify a scheme with the functor associated with it.

On the other hand, the constructions set out below go beyond the framework of the theory of schemes. They are equally valid, for example, in the theory of *analytic spaces* with nilpotent elements, modulo a few modifications of detail.

Before beginning this construction, we must set down a few general definitions which complete those of I 1.7.

1. The functors $\text{Hom}_{Z/S}(X, Y)$

Let us take up again the notations of I 1.1. One identifies the category $\mathcal{C}$ with a full subcategory of $\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}^\circ, (\text{Ens}))$ (in particular one omits the underlinings⁽¹⁾ which allowed us to distinguish graphically an object of $\widehat{\mathcal{C}}$ from an object of $\mathcal{C}$).

Let us consider the following situation: four objects of $\widehat{\mathcal{C}}$, denoted S, X, Y, Z , the first being in fact an object of $\mathcal{C}$, X and Y over Z , Z over S :

$$\begin{array}{ccc} X & & Y \\ & \searrow p_X & \swarrow p_Y \\ & Z & \\ & \downarrow & \\ & S. & \end{array}$$

Definition 1.1. One defines an object of $\widehat{\mathcal{C}}_S$, denoted $\text{Hom}_{Z/S}(X, Y)$, by:

$$(1) \quad \text{Hom}_{Z/S}(X, Y)(S') = \text{Hom}_{Z_{S'}}(X_{S'}, Y_{S'}) = \text{Hom}_Z(X \times_S S', Y),$$

for every object S' of $\mathcal{C}_S$. One sees at once that $\text{Hom}_{Z/S}(X, Y)$ is nothing other than the subobject of $\text{Hom}_S(X, Y)$ formed of the morphisms compatible with p_X and p_Y , that is, the kernel of the pair of morphisms

$$\text{Hom}_S(X, Y) \rightrightarrows \text{Hom}_S(X, Z)$$

defined, the first by composition with p_Y , the second as being the constant morphism whose “image” is p_X .

⁽⁰⁾**N.D.E.** Version of 14/10/2024

⁽¹⁾**N.D.E.** Rendered in this edition by bold characters **F**, **G**, **V**, **W**, cf. Exposé I. However, for functors such as Norm and Centr (cf. 5.2) the underlinings have been kept as in the original, and one has added them for the functor Lie, in order to distinguish the functor Lie(G/S) from the $\Gamma(S, \mathcal{O}_S)$ -Lie algebra $\text{Lie}(G/S) = \underline{\text{Lie}}(G/S)(S)$, used for example in Exposé VII_A.

⁽²⁾On the other hand, one sees as in I 1.7 that, for every object T of $\widehat{\mathcal{C}}$ over S , one has a natural bijection:

$$(2) \quad \text{Hom}_S(T, \text{Hom}_{Z/S}(X, Y)) \simeq \text{Hom}_Z(X \times_S T, Y).$$

Moreover, by I 1.7.1, if E, F are objects of $\widehat{\mathcal{C}}$ over Z , one has:

$$\text{Hom}_Z(E, \text{Hom}_Z(F, Y)) \simeq \text{Hom}_Z(E \times_Z F, Y) \simeq \text{Hom}_Z(F, \text{Hom}_Z(E, Y)).$$

Applying this to $E = X$ and $F = Z \times_S T$, one obtains natural bijections, for every object T of $\widehat{\mathcal{C}}_S$:

$$(3) \quad \text{Hom}_S(T, \text{Hom}_{Z/S}(X, Y)) \simeq \text{Hom}_Z(X \times_S T, Y) \simeq \begin{cases} \text{Hom}_Z(Z \times_S T, \text{Hom}_Z(X, Y)) \\ \text{Hom}_Z(X, \text{Hom}_Z(Z \times_S T, Y)) \end{cases}.$$

Moreover, these bijections are functorial in T , so one obtains isomorphisms of S -functors:

$$(4) \quad \begin{array}{ccc} \text{Hom}_S(T, \text{Hom}_{Z/S}(X, Y)) & \xrightarrow{\sim} & \text{Hom}_{Z/S}(X, \text{Hom}_Z(Z \times_S T, Y)) \\ & \searrow \simeq \quad \swarrow \simeq & \\ & \text{Hom}_{Z/S}(X \times_S T, Y) & \end{array}$$

Let us point out two particular cases of the definition. If $Z = S$, one has:

$$\text{Hom}_{S/S}(X, Y) = \text{Hom}_S(X, Y).$$

On the other hand, when $X = Z$, one sets

$$(5) \quad \prod_{Z/S} Y = \text{Hom}_{Z/S}(Z, Y),$$

one therefore has by definition

$$\left(\prod_{Z/S} Y \right)(S') = \text{Hom}_Z(Z \times_S S', Y) \simeq \Gamma(Y_{S'}/Z_{S'}).$$

The functor $\prod_{Z/S}: \widehat{\mathcal{C}}_Z \rightarrow \widehat{\mathcal{C}}_S$ is right adjoint to the functor of base change from S to Z : for every S -functor U one has

$$\text{Hom}_S\left(U, \prod_{Z/S} Y\right) = \text{Hom}_Z(U \times_S Z, Y).$$

(If $\mathcal{C} = (\text{Sch})$ and if Z is an S -scheme, the functor $\prod_{Z/S}$ is called “*Weil restriction of scalars*”).⁽³⁾ Let us also note that one has an isomorphism:

$$(6) \quad \text{Hom}_{Z/S}(X, Y) \simeq \text{Hom}_{X/S}(X, Y \times_Z X) = \prod_{X/S} (Y \times_Z X),$$

which gives in particular, for $Z = S$, an isomorphism:

$$(7) \quad \text{Hom}_S(X, Y) \simeq \prod_{X/S} Y_X.$$

Remark 1.2. The functor $Y \mapsto \text{Hom}_{Z/S}(X, Y)$ commutes with the product in the following sense: one has a functorial isomorphism

$$(*) \quad \text{Hom}_{Z/S}(X, Y \times_Z Y') \simeq \text{Hom}_{Z/S}(X, Y) \times_S \text{Hom}_{Z/S}(X, Y').$$

It follows that if Y is a Z -group, resp. a Z -ring, etc., then $\text{Hom}_{Z/S}(X, Y)$ is an S -group, resp. an S -ring, etc.

⁽²⁾**N.D.E.** Points (2), (3), (4) have been spelled out; in particular, (4) will be used in 3.11.

⁽³⁾**N.D.E.** The two preceding sentences have been added.

Remark 1.3. ⁽⁴⁾Moreover, let $\pi: M \rightarrow Y$ be a Y -functor in $\mathbf{O}_Y$ -modules (cf. I, 4.3.3.1). Set $H = \mathrm{Hom}_{Z/S}(X, Y)$. Then $\mathrm{Hom}_{Z/S}(X, M)$ is endowed with a natural structure of $\mathbf{O}_H$ -module; more precisely, for every H' over H , $\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$ is endowed with a natural structure of $\mathbf{O}(H' \times_S X)$ -module.

Indeed, let us denote by $m: M \times_Y M \rightarrow M$ and $\lambda: \mathbf{O}_Y \times_Y M \rightarrow M$ the morphisms defining the structures of Y -group (abelian) and of $\mathbf{O}_Y$ -module. Let H' be an S -scheme over $H = \mathrm{Hom}_{Z/S}(X, Y)$, that is, one is given a Z -morphism $f: X \times_S H' \rightarrow Y$, which therefore makes $X \times_S H'$ a Y -object. Then

$$\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$$

is the set of Z -morphisms $\phi: X \times_S H' \rightarrow M$ such that $\pi \circ \phi = f$, that is, of Y -morphisms $X \times_S H' \rightarrow M$.

If ϕ, ψ are two such morphisms, one defines $\phi + \psi$ as the composite Y -morphism

$$X \times_S H' \xrightarrow{\phi \times \psi} M \times_Y M \xrightarrow{m} M$$

and one checks that this endows $\mathrm{Hom}_{Z/S}(X, M)$ with the structure of an abelian group over $H = \mathrm{Hom}_{Z/S}(X, Y)$.⁽⁵⁾

Likewise, if a is an element of $\mathbf{O}(X \times_S H')$, i.e. an S -morphism $a: X \times_S H' \rightarrow \mathbf{O}_S$, one defines $a\phi$ as the composite $\lambda \circ (a \times \phi)$, where $a \times \phi$ denotes the Y -morphism from $X \times_S H'$ to $\mathbf{O}_Y \times_Y M \simeq \mathbf{O}_S \times_S M$ with components a and ϕ ; one checks that this endows $\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$ with the structure of an $\mathbf{O}(X \times_S H')$ -module, functorial in the H -object H' .

2. The schemes $I_S(\mathcal{M})$

Definition 2.1. Let S be a scheme and $\mathcal{M}$ a quasi-coherent $\mathcal{O}_S$ -module. One denotes by $\mathcal{D}_{\mathcal{O}_S}(\mathcal{M})$ the quasi-coherent $\mathcal{O}_S$ -algebra $\mathcal{O}_S \oplus \mathcal{M}$ (where $\mathcal{M}$ is regarded as a square-zero ideal). One denotes by $I_S(\mathcal{M})$ the S -scheme $\mathrm{Spec} \mathcal{D}_{\mathcal{O}_S}(\mathcal{M})$.⁽⁶⁾ In particular one denotes $\mathcal{D}_{\mathcal{O}_S} = \mathcal{D}_{\mathcal{O}_S}(\mathcal{O}_S)$, $I_S = I_S(\mathcal{O}_S)$ and one calls them respectively the *algebra of dual numbers over S* and the *scheme of dual numbers over S* .

Then $\mathcal{M} \mapsto I_S(\mathcal{M})$ is a contravariant functor from the category of quasi-coherent $\mathcal{O}_S$ -modules to that of S -schemes. In particular the morphisms $0 \rightarrow \mathcal{M}$ and $\mathcal{M} \rightarrow 0$ define respectively the structural morphism $\rho: I_S(\mathcal{M}) \rightarrow I_S(0) = S$ and a section $\varepsilon_{\mathcal{M}}$ of the latter, which one calls the *zero section*.⁽⁷⁾

2.1.1. — ⁽⁸⁾As $\mathcal{M} \rightarrow I_S(\mathcal{M})$ is a contravariant functor, every $a \in \mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ defines an S -endomorphism a^* of $I_S(\mathcal{M})$, and one has $1^* = \mathrm{id}$, $(ab)^* = b^* \circ a^*$, $0^* = \varepsilon_{\mathcal{M}} \circ \rho$ and $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$. Consequently, the S -scheme $I_S(\mathcal{M})$ is endowed with a right action of the multiplicative monoid of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$, which commutes with the S -morphisms $I_S(\mathcal{M}) \rightarrow I_S(\mathcal{M}')$ coming from morphisms $\mathcal{M}' \rightarrow \mathcal{M}$; in particular, the operators a^* preserve the zero section of $I_S(\mathcal{M})$.

⁽⁴⁾**N.D.E.** This remark has been added; it is useful for Corollary 3.11.1.

⁽⁵⁾**N.D.E.** Of course, if M is a Y -group (not necessarily abelian) and if one sets $\phi \cdot \psi = m \circ (\phi \times \psi)$, this makes $\mathrm{Hom}_{Z/S}(X, M)$ a group over $\mathrm{Hom}_{Z/S}(X, Y)$.

⁽⁶⁾**N.D.E.** Note that $I_S(\mathcal{M})$ has the same underlying space as S .

⁽⁷⁾**N.D.E.** Compare with 3.1.1.

⁽⁸⁾**N.D.E.** The original has been set out in detail in what follows, and the numbering 2.1.1 to 2.1.3 has been added.

For every $a \in \text{End}_{\mathcal{O}_S}(\mathcal{M})$ and $f: S' \rightarrow S$ and $m \in I_S(\mathcal{M})(S')$, one will denote $m \cdot a = a^*(m)$; then $m \cdot 1 = m$, $(m \cdot a) \cdot b = m \cdot (ab)$, $m \cdot 0 = \varepsilon_{\mathcal{M}}(\rho(m))$ and, if $m = \varepsilon_{\mathcal{M}}(f)$, then $m \cdot a = m$.⁽⁹⁾

Remark 2.1.2. The formation of the $I_S(\mathcal{M})$ commutes with extension of the base: one has canonical isomorphisms

$$I_S(\mathcal{M})_{S'} \simeq I_{S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}).$$

For simplicity, one will denote $I_{S'}(\mathcal{M}) = I_S(\mathcal{M})_{S'}$; more generally, if X is an S -functor (not necessarily representable), one will denote $I_X(\mathcal{M}) = I_S(\mathcal{M}) \times_S X$.

2.1.3. — ⁽¹⁰⁾By what precedes, the multiplicative monoid of $\mathbf{O}(S')$ operates on the S' -scheme $I_{S'}(\mathcal{M})$, functorially in $\mathcal{M}$, i.e. the S -scheme $I_S(\mathcal{M})$ is endowed with the structure of an object with monoid of operators $\mathbf{O}_S$, this structure being functorial in $\mathcal{M}$.

One therefore has a morphism of S -schemes

$$\lambda: I_S(\mathcal{M}) \times_S \mathbf{O}_S \rightarrow I_S(\mathcal{M}),$$

satisfying obvious conditions. For every S -functor X , one obtains by base change a morphism of X -functors:

$$\lambda_X: I_X(\mathcal{M}) \times_S \mathbf{O}_S \rightarrow I_X(\mathcal{M})$$

which makes the S -functor $I_X(\mathcal{M})$ an object with monoid of operators $\mathbf{O}(X)$: every element a of $\mathbf{O}(X) = \text{Hom}_S(X, \mathbf{O}_S)$ defines an X -endomorphism

$$a^* = \lambda_X \circ (\text{id}_{I_X(\mathcal{M})} \times (a \circ \text{pr}_X))$$

of $I_X(\mathcal{M})$; explicitly, if $x \in X(S')$ and $m \in I_S(\mathcal{M})(S') = I_{S'}(\mathcal{M})(S')$, then $a(x) = a \circ x$ belongs to $\mathbf{O}(S')$ and one has:

$$(m, x) \cdot a = (m \cdot a(x), x).$$

This operation is functorial in $\mathcal{M}$ and preserves the zero section $\varepsilon_{\mathcal{M}}: X \rightarrow I_X(\mathcal{M})$, i.e. $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$ for every $a \in \mathbf{O}(X)$.

Moreover, this operation is “functorial in X ” in the following sense: if $\pi: Y \rightarrow X$ is a morphism of S -functors and $u: \mathbf{O}(X) \rightarrow \mathbf{O}(Y)$ the corresponding morphism of rings (i.e. $u(a) = a \circ \pi \in \mathbf{O}(Y)$ for every $a \in \mathbf{O}(X) = \text{Hom}_S(X, \mathbf{O}_S)$), then the following diagram is commutative:

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{a^*} & I_X(\mathcal{M}) \\ \pi \downarrow & & \downarrow \pi \\ I_Y(\mathcal{M}) & \xrightarrow{u(a)^*} & I_Y(\mathcal{M}). \end{array}$$

2.2. — Let now $\mathcal{M}$ and $\mathcal{N}$ be two quasi-coherent $\mathcal{O}_S$ -modules. The commutative diagram

$$\begin{array}{ccc} & \mathcal{M} \oplus \mathcal{N} & \\ \swarrow & & \searrow \\ \mathcal{M} & & \mathcal{N} \\ \searrow & & \swarrow \\ & 0 & \end{array}$$

⁽⁹⁾**N.D.E.** In the sequel, one will be concerned mainly with the case where $a \in \mathbf{O}(S)$, acting by homotheties by $\mathcal{M}$. For example, if $\mathcal{M}$ is a free $\mathcal{O}_S$ -module of rank r , then, for every $S' \rightarrow S$, $\text{Hom}_S(S', I_S(\mathcal{M}))$ is identified with the set of r -tuples $(\varepsilon_1, \dots, \varepsilon_r) \in \mathbf{O}(S')$ such that $\varepsilon_i \varepsilon_j = 0$ for all i, j , and one has $(\varepsilon_1, \dots, \varepsilon_r) \cdot a = (a\varepsilon_1, \dots, a\varepsilon_r)$ for every $a \in \mathbf{O}(S)$.

⁽¹⁰⁾**N.D.E.** This paragraph has been added; it will be useful in 3.4.2 and in 4.6.2.

defines a commutative diagram of S -schemes

$$(*) \quad \begin{array}{ccccc} & & \mathbf{I}_S(\mathcal{M} \oplus \mathcal{N}) & & \\ & \nearrow & \uparrow \varepsilon_{\mathcal{M} \oplus \mathcal{N}} & \nwarrow & \\ \mathbf{I}_S(\mathcal{M}) & & & & \mathbf{I}_S(\mathcal{N}) \\ & \nwarrow \varepsilon_{\mathcal{M}} & \downarrow & \nearrow \varepsilon_{\mathcal{N}} & \\ & & S & & \end{array}$$

Proposition 2.2. *For every S -scheme X , the diagram of functors over S obtained by applying the functor $\mathrm{Hom}_S(-, X)$ to the diagram $(*)$ is cartesian:*

$$\begin{array}{ccccc} & & \mathrm{Hom}_S(\mathbf{I}_S(\mathcal{M} \oplus \mathcal{N}), X) & & \\ & \swarrow & & \searrow & \\ \mathrm{Hom}_S(\mathbf{I}_S(\mathcal{M}), X) & & & & \mathrm{Hom}_S(\mathbf{I}_S(\mathcal{N}), X) \\ & \searrow & \downarrow & \swarrow & \\ & & \mathrm{Hom}_S(S, X) = X & & \end{array}$$

One must check that for every $S' \rightarrow S$, the diagram of sets obtained by taking the value of the functors on S' is cartesian. As the formation of $\mathbf{I}_S(\mathcal{P})$ commutes with extension of the base in the sense made explicit above, it suffices to do this for $S' = S$, hence to check that the following diagram of sets is cartesian:

$$\begin{array}{ccccc} & & X(\mathbf{I}_S(\mathcal{M} \oplus \mathcal{N})) & & \\ & \swarrow & \downarrow X(\varepsilon_{\mathcal{M} \oplus \mathcal{N}}) & \searrow & \\ X(\mathbf{I}_S(\mathcal{M})) & & & & X(\mathbf{I}_S(\mathcal{N})) \\ & \searrow X(\varepsilon_{\mathcal{M}}) & \downarrow & \swarrow X(\varepsilon_{\mathcal{N}}) & \\ & & X(S) & & \end{array}$$

Now, if $x \in X(S)$, it follows from SGA 1, III 5.1⁽¹¹⁾, that $X(\varepsilon_{\mathcal{M}})^{-1}(x)$ is isomorphic, functorially in $\mathcal{M}$, to

$$\mathrm{Hom}_{\mathcal{O}_S}(x^*(\Omega_{X/S}^1), \mathcal{M}),$$

where $\Omega_{X/S}^1$ denotes the sheaf of relative differentials of X with respect to S . Now this latter functor (in $\mathcal{M}$) evidently transforms a direct sum of $\mathcal{O}_S$ -modules into the product of the corresponding sets, whence the result.

Corollary 2.2.1. *Let X be an S -scheme and $\mathcal{M}$ a free $\mathcal{O}_S$ -module of finite type. The functor $\mathrm{Hom}_S(\mathbf{I}_S(\mathcal{M}), X)$ is isomorphic (as a functor over X) to a finite product (over X) of copies of $\mathrm{Hom}_S(\mathbf{I}_S, X)$.*

Nota 2.2.2. It follows from the proof of the proposition that $\mathrm{Hom}_S(\mathbf{I}_S, X)$ is isomorphic as an X -functor to $\mathbf{V}(\Omega_{X/S}^1)$ (I 4.6.1), hence representable by the vector fibration⁽¹²⁾ $\mathbf{V}(\Omega_{X/S}^1)$.

3. The tangent bundle, the condition (E)

In this section, unless the contrary is indicated, the letters $\mathcal{M}$, $\mathcal{M}'$, $\mathcal{N}$, etc., will always denote free $\mathcal{O}_S$ -modules of finite type (that is, isomorphic to a finite direct sum of copies of $\mathcal{O}_S$).

⁽¹¹⁾**N.D.E.** See also the addition 0.1.8 in Exp. III.

⁽¹²⁾**N.D.E.** Cf. N.D.E. (33) of Exposé I.

We shall systematically use the identifications justified in Exposé I; thus we shall say “functor over S ” to designate indiscriminately a functor equipped with a morphism to S ($= \mathbf{h}_S$) or a functor on the category of objects over S . One will likewise say “group functor over S ”, etc.

Definition 3.1. Let S be a scheme and $\mathcal{M}$ a free $\mathcal{O}_S$ -module of finite type. Let X be a functor over S . One calls the tangent bundle to X over S relative to the $\mathcal{O}_S$ -module $\mathcal{M}$, and one denotes by $T_{X/S}(\mathcal{M})$, the S -functor

$$T_{X/S}(\mathcal{M}) = \text{Hom}_S(I_S(\mathcal{M}), X).$$

In particular, one calls the tangent bundle to X over S , and one denotes by $T_{X/S}$, the functor

$$T_{X/S} = T_{X/S}(\mathcal{O}_S) = \text{Hom}_S(I_S, X).$$

(13)

Then $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from the category of free $\mathcal{O}_S$ -modules of finite type to the category of S -functors. In particular the morphisms⁽¹⁴⁾ $\mathcal{M} \rightarrow 0$ and $0 \rightarrow \mathcal{M}$ define respectively an S -morphism

$$\pi_{\mathcal{M}}: T_{X/S}(\mathcal{M}) \rightarrow T_{X/S}(0) \simeq X$$

and a section τ_0 of the latter, called the zero section (or null section).

Remark 3.1.1. ⁽¹⁵⁾ One will note that the projection $\pi_{\mathcal{M}}: T_{X/S}(\mathcal{M}) \rightarrow X$ is induced by the zero section $\varepsilon_{\mathcal{M}}: S \rightarrow I_S(\mathcal{M})$, while the null section $\tau_0: X \rightarrow T_{X/S}(\mathcal{M})$ is induced by the structural morphism $\rho: I_S(\mathcal{M}) \rightarrow S$; i.e. for every point $t \in T_{X/S}(\mathcal{M})(S')$ (resp. $x \in X(S')$), corresponding to an S -morphism $f: S' \times_S I_S(\mathcal{M}) \rightarrow X$ (resp. $g: S' \rightarrow X$), one has $\pi(t) = f \circ (\text{id}_{S'} \times \varepsilon_{\mathcal{M}})$ (resp. $\tau_0(x) = g \circ (\text{id}_{S'} \times \rho)$).

It follows from 3.1 that $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from the category of free $\mathcal{O}_S$ -modules of finite type to that of functors over X . In particular $\mathbf{O}(S)$ is a monoid of operators of the X -functor $T_{X/S}(\mathcal{M})$ which respects “the functoriality in $\mathcal{M}$ ”.

Scholium 3.1.2. ⁽¹⁵⁾ What precedes means, in particular, the following things. For every S -morphism $\phi: X' \rightarrow X$, set

$$\Sigma(X', \mathcal{M}) = \text{Hom}_X(X', T_{X/S}(\mathcal{M})).$$

One has an action of the multiplicative monoid $\text{End}_{\mathcal{O}_S}(\mathcal{M})$ on $\Sigma(X', \mathcal{M})$, denoted $(\lambda, x) \mapsto \lambda * x$, such that $\lambda * (\mu * x) = (\lambda\mu) * x$, $1 * x = x$, and $0 * x = \tau_0 \circ \phi$, where τ_0 is the null section $X \rightarrow T_{X/S}(\mathcal{M})$. One likewise has an action of $\text{End}_{\mathcal{O}_S}(\mathcal{M} \oplus \mathcal{M})$ on $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$.⁽¹⁶⁾

Moreover, let $m: \mathcal{M} \oplus \mathcal{M} \rightarrow \mathcal{M}$ (resp. $\delta: \mathcal{M} \rightarrow \mathcal{M} \oplus \mathcal{M}$) be the addition (resp. the diagonal map) of $\mathcal{M}$, and denote by $m_{X'}: \Sigma(X', \mathcal{M} \oplus \mathcal{M}) \rightarrow \Sigma(X', \mathcal{M})$ and $\delta_{X'}: \Sigma(X', \mathcal{M}) \rightarrow \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ the morphisms induced by m and δ . For $\lambda, \mu \in \mathbf{O}(S)$, denote by ℓ_{λ} (resp. $\ell_{\lambda, \mu}$) multiplication by λ in $\mathcal{M}$ (resp. by (λ, μ) in $\mathcal{M} \oplus \mathcal{M}$). Since $m \circ \ell_{\lambda, \lambda} = \ell_{\lambda} \circ m$ and $m \circ \ell_{\lambda, \mu} \circ \delta = \ell_{\lambda + \mu}$, one has, for every $z \in \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ and $x \in \Sigma(X', \mathcal{M})$:

$$(\dagger) \quad \lambda * m(z) = m((\lambda, \lambda) * z), \quad m((\lambda, \mu) * \delta(x)) = (\lambda + \mu) * x.$$

Definition 3.2. Let $u \in X(S) = \text{Hom}_S(S, X) = \Gamma(X/S)$. One calls the tangent space to X over S at the point u relative to $\mathcal{M}$, and one denotes by $L_{X/S}^u(\mathcal{M})$, the S -functor obtained

⁽¹³⁾**N.D.E.** When $S = \text{Spec}(k)$ and X is a k -scheme, one has $T_{X/S}(S') = \text{Hom}_S(S' \otimes_k k[\varepsilon], X) = X(S' \otimes_k k[\varepsilon])$; one thus recovers one of the usual definitions of the tangent bundle.

⁽¹⁴⁾**N.D.E.** “ $0 \rightarrow \mathcal{M}$ and $\mathcal{M} \rightarrow 0$ ” has been corrected to: “ $\mathcal{M} \rightarrow 0$ and $0 \rightarrow \mathcal{M}$ ”.

⁽¹⁵⁾**N.D.E.** Paragraphs 3.1.1 and 3.1.2 have been added.

⁽¹⁶⁾**N.D.E.** In fact, one is interested only in the action of $\mathbf{O}(S)$ (resp. $\mathbf{O}(S) \times \mathbf{O}(S)$) on $\Sigma(X', \mathcal{M})$ (resp. $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$), cf. below and the proof of 3.6.

from the X -functor $T_{X/S}(\mathcal{M})$ by inverse image under the morphism $u: S \rightarrow X$:

$$\begin{array}{ccc} L_{X/S}^u(\mathcal{M}) & \longrightarrow & T_{X/S}(\mathcal{M}) \\ \downarrow & & \downarrow \pi \\ S & \xrightarrow{u} & X. \end{array}$$

In particular $L_{X/S}^u(\mathcal{O}_S)$ is denoted by $L_{X/S}^u$. It is the tangent space to X over S at the point u .

Remark 3.2.1. ⁽¹⁷⁾ It follows from 3.1.1 that, for every $t: S' \rightarrow S$, $L_{X/S}^u(\mathcal{M})(S')$ is the set of S -morphisms $f: I_{S'}(\mathcal{M}) \rightarrow X$ such that $f \circ \varepsilon_{\mathcal{M}} = u \circ t$, where $\varepsilon_{\mathcal{M}}: S' \rightarrow I_{S'}(\mathcal{M})$ is the zero section.

Let us immediately record the

Proposition 3.3. *If X is representable by an S -scheme denoted $\tilde{X}$, then $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ are representable. In particular $T_{X/S}$ and $L_{X/S}^u$ are representable by vector fibrations over $\tilde{X}$ and over S which are respectively $\mathbf{V}(\Omega_{\tilde{X}/S}^1)$ and $\mathbf{V}(u^*(\Omega_{\tilde{X}/S}^1))$.*

It obviously suffices to prove the proposition for $T_{X/S}(\mathcal{M})$, the analogous results for $L_{X/S}^u(\mathcal{M})$ being deduced from it by inverse image. By Corollary 2.2.1, it even suffices to do so for $T_{X/S}$, and in this case the proposition is nothing other than the remark indicated in 2.2.2.

Remark 3.3.1. From this proposition there follows a particularly simple description of the vector fibration representing $L_{X/S}^u$: the image of the section u of X over S is locally closed⁽¹⁸⁾, hence defined by a quasi-coherent ideal $\mathfrak{m}$ of a scheme induced on an open of X . The quotient $\mathfrak{m}/\mathfrak{m}^2$ may be regarded as a quasi-coherent module over S . It is this module which defines the vector fibration sought.

For example, let X be an algebraic scheme over a field k and u a point of X rational over k . Let $\mathfrak{m}$ be the maximal ideal of the local ring $\mathcal{O}_{X,u}$ and let t be the k -vector space dual of $\mathfrak{m}/\mathfrak{m}^2$; it is the Zariski tangent space of $\mathcal{O}_{X,u}$ at the point u . Then, with the notations of I 4.6.5.1, one has:

$$L_{X/k}^u = \mathbf{V}(\mathfrak{m}/\mathfrak{m}^2) = \mathbf{W}(t).$$

This parenthesis being closed, let us return to the general situation. Let us first remark that $L_{X/S}^u(\mathcal{M})$ is a covariant functor from the category of free $\mathcal{O}_S$ -modules of finite type to that of functors over S . In particular $\mathbf{O}(S)$ is a set of operators of the S -functor $L_{X/S}^u(\mathcal{M})$ which respects the functoriality in $\mathcal{M}$.⁽¹⁹⁾

Proposition 3.4. *The formation of $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ commutes with extension of the base: for every S -scheme S' , one has isomorphisms functorial in $\mathcal{M}$*

$$\begin{aligned} T_{X_{S'}/S'}(\mathcal{M} \otimes \mathcal{O}_{S'}) &\xrightarrow{\sim} T_{X/S}(\mathcal{M})_{S'}, \\ L_{X_{S'}/S'}^{u'}(\mathcal{M} \otimes \mathcal{O}_{S'}) &\xrightarrow{\sim} L_{X/S}^u(\mathcal{M})_{S'}, \quad \text{where } u' = u_{S'}. \end{aligned}$$

This follows immediately from the fact that the Hom commute with extension of the base.

Corollary 3.4.1. *The X -functor $T_{X/S}(\mathcal{M})$ (resp. the S -functor $L_{X/S}^u(\mathcal{M})$) is endowed naturally with a structure of object with operators $\mathbf{O}_X$ (resp. $\mathbf{O}_S$), this structure being functorial in $\mathcal{M}$.*

⁽¹⁷⁾**N.D.E.** This remark has been added.

⁽¹⁸⁾**N.D.E.** Cf. EGA I, 5.3.11.

⁽¹⁹⁾**N.D.E.** Cf. 3.1.2.

Let us first show this for $L_{X/S}^u(\mathcal{M})$. For each S' over S , $\mathbf{O}(S')$ operates on $\mathcal{M} \otimes \mathcal{O}_{S'}$, hence on $L_{X_{S'}/S'}^u(\mathcal{M} \otimes \mathcal{O}_{S'}) = L_{X/S}^u(\mathcal{M})_{S'}$; now one checks that this operation is functorial in S' . It therefore endows $L_{X/S}^u(\mathcal{M})$, as announced, with a structure of functor with operators $\mathbf{O}_S$.

For $T_{X/S}(\mathcal{M})$ it is a little more complicated. For each X' over X , set $T_{X/S}(\mathcal{M})_{X'} = T_{X/S}(\mathcal{M}) \times_X X'$; one must endow $T_{X/S}(\mathcal{M})_{X'}(X') = \text{Hom}_X(X', T_{X/S}(\mathcal{M}))$ with a structure of set with monoid of operators $\mathbf{O}(X')$, in a manner functorial in X' . For this one constructs the following diagram, where $X_{X'}$ denotes $X \times_S X'$ and f' the section of $X_{X'}$ over X' defined by $f: X' \rightarrow X$.

$$\begin{array}{ccc}
 & T_{X_{X'}/X'}(\mathcal{M}) & \\
 \swarrow & & \searrow \\
 T_{X/S}(\mathcal{M}) & & T_{X/S}(\mathcal{M})_{X'} \\
 & & \\
 & X_{X'} & \\
 \swarrow & \xleftarrow{f'} & \searrow \\
 X & \xleftarrow{f} & X' \\
 \searrow & & \swarrow \\
 & S &
 \end{array}$$

⁽²⁰⁾ This diagram, together with 3.2.1, shows that $T_{X/S}(\mathcal{M})_{X'}(X')$ is identified with

$$L_{X_{X'}/X'}^{f'}(\mathcal{M})(X') = \{X'\text{-morphisms } \psi: I_{X'}(\mathcal{M}) \rightarrow X_{X'} \text{ such that } \psi \circ \varepsilon_{\mathcal{M}} = f'\},$$

on which every $a \in \mathbf{O}(X')$ operates via its action on $I_{X'}(\mathcal{M})$, i.e. with the notations of 2.1.1, one has: $a\psi = \psi \circ a^*$, i.e. for every $X'' \rightarrow X'$ and $x \in I_{X'}(\mathcal{M})(X'')$, $(a\psi)(x) = \psi(x \cdot a)$. One then checks easily that this construction is functorial in X' .

The isomorphisms of Proposition 3.4 are then by construction isomorphisms for the structures of $\mathbf{O}_{X_{S'}}$ -objects, resp. $\mathbf{O}_{S'}$ -objects.

Remark 3.4.2. ⁽²¹⁾ The operation of $\mathbf{O}_X$ on $T_{X/S}(\mathcal{M})$ may be seen, more simply, as follows. For every $f: X' \rightarrow X$, one has

$$\text{Hom}_X(X', T_{X/S}(\mathcal{M})) = \{\phi \in \text{Hom}_S(I_{X'}(\mathcal{M}), X) \mid \phi \circ \varepsilon_{\mathcal{M}} = f\},$$

and one has seen in 2.1.3 that $I_{X'}(\mathcal{M})$, considered as an S -functor, is endowed with an operation of the monoid $\mathbf{O}(X')$ which preserves the zero section $\varepsilon_{\mathcal{M}}: X' \rightarrow I_{X'}(\mathcal{M})$. Consequently, if one denotes by a^* the X' -endomorphism of $I_{X'}(\mathcal{M})$ defined by $a \in \mathbf{O}(X')$, one has: $a\phi = \phi \circ a^*$, i.e. for every $S' \rightarrow S$ and $(m, x') \in \text{Hom}_S(S', I_S(\mathcal{M}) \times_S X')$,

$$(a\phi)(m, x') = \phi(m \cdot a(x'), x')$$

(note that $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$, whence $(a\phi) \circ \varepsilon_{\mathcal{M}} = \phi \circ \varepsilon_{\mathcal{M}} = f$).

Likewise, the operation of $\mathbf{O}_S$ on $L_{X/S}^u(\mathcal{M})$ may be described as follows. For every $t: S' \rightarrow S$, $L_{X/S}^u(\mathcal{M})(S')$ is the set of S -morphisms $\phi: I_{S'}(\mathcal{M}) \rightarrow X$ such that $\phi \circ \varepsilon_{\mathcal{M}} = u \circ t$; for such a ϕ and $a \in \mathbf{O}(S')$, one has: $a\phi = \phi \circ a^*$.

Remark 3.4.3. ⁽²¹⁾ The observations of 3.1.2 hold equally for the operation of $\mathbf{O}_S$ on $L_{X/S}^u(\mathcal{M})$ and for that of $\mathbf{O}_X$ on $T_{X/S}(\mathcal{M})$.

Definition 3.5. Let S be a scheme and X an S -functor. One says that X satisfies the condition (E) relative to S if, for every S' over S and all free $\mathcal{O}_{S'}$ -modules of finite type $\mathcal{M}$

⁽²⁰⁾**N.D.E.** The original has been set out in detail in what follows.

⁽²¹⁾**N.D.E.** Remarks 3.4.2 and 3.4.3 have been added.

and $\mathcal{N}$, the diagram of sets

$$\begin{array}{ccc} & X(\mathrm{I}_{S'}(\mathcal{M} \oplus \mathcal{N})) & \\ \swarrow & & \searrow \\ X(\mathrm{I}_{S'}(\mathcal{M})) & & X(\mathrm{I}_{S'}(\mathcal{N})) \\ \searrow & & \swarrow \\ & X(S') & \end{array}$$

obtained by applying X to the diagram $(*)$ defined in 2.1, is cartesian.⁽²²⁾

3.5.1. — It comes to the same to say that the functor $\mathcal{M} \mapsto \mathrm{T}_{X/S}(\mathcal{M})$ takes direct sums of free $\mathcal{O}_S$ -modules of finite type to products of X -functors;⁽²³⁾ in this case, the same holds for the functor $\mathcal{M} \mapsto \mathrm{L}_{X/S}^u(\mathcal{M}) = S \times_X \mathrm{T}_{X/S}(\mathcal{M})$, for every $u \in \Gamma(X/S)$.

Proposition 2.2 shows that *every representable functor satisfies the condition (E)*.

Abbreviation: instead of saying “ X satisfies the condition (E) with respect to S ”, one will sometimes say “ X/S satisfies the condition (E)”.

If X/S satisfies the condition (E), the functor $\mathcal{M} \mapsto \mathrm{T}_{X/S}(\mathcal{M})$ commutes with products, hence takes groups to groups. In particular $\mathrm{T}_{X/S}(\mathcal{M})$ is a commutative X -group. For the same reason, $\mathrm{L}_{X/S}^u(\mathcal{M})$ is a commutative S -group.

Proposition 3.6. *If X/S satisfies (E), the structure of abelian group on $\mathrm{T}_{X/S}(\mathcal{M})$ (resp. $\mathrm{L}_{X/S}^u(\mathcal{M})$) and the operation of $\mathbf{O}_X$ (resp. $\mathbf{O}_S$) endow $\mathrm{T}_{X/S}(\mathcal{M})$ (resp. $\mathrm{L}_{X/S}^u(\mathcal{M})$) with a structure of $\mathbf{O}_X$ -module (resp. $\mathbf{O}_S$ -module).*

The operation of $\mathbf{O}_X$ (resp. $\mathbf{O}_S$) is functorial in $\mathcal{M}$; it therefore respects the structure of abelian group which is deduced by functoriality from that of $\mathcal{M}$.⁽²⁴⁾ Indeed, let us take up again the notations of 3.1.2. The structure of (abelian) X -group on $\mathrm{T}_{X/S}(\mathcal{M})$ is defined by the composite:

$$\mathrm{T}_{X/S}(\mathcal{M}) \times_X \mathrm{T}_{X/S}(\mathcal{M}) \simeq \mathrm{T}_{X/S}(\mathcal{M} \oplus \mathcal{M}) \xrightarrow{m} \mathrm{T}_{X/S}(\mathcal{M}),$$

and on the other hand the morphism

$$\mathrm{T}_{X/S}(\mathcal{M}) \xrightarrow{\delta} \mathrm{T}_{X/S}(\mathcal{M} \oplus \mathcal{M}) \simeq \mathrm{T}_{X/S}(\mathcal{M}) \times_X \mathrm{T}_{X/S}(\mathcal{M})$$

is the diagonal morphism. Taking Remark 3.4.3 into account, one deduces from the equalities $(\dagger)$ of 3.1.2 that

$$\lambda(x + y) = \lambda x + \lambda y, \quad (\lambda + \mu)x = \lambda x + \mu x,$$

for every $f: X' \rightarrow X$, $x, y \in \mathrm{Hom}_X(X', \mathrm{T}_{X/S}(\mathcal{M}))$ and $\lambda, \mu \in \mathbf{O}(X')$.

Remark 3.6.1. If X is representable, in which case, on the one hand it satisfies (E), and on the other hand $\mathrm{T}_{X/S}$ and $\mathrm{L}_{X/S}^u$ are representable by vector fibrations, the preceding laws are the same as those which are deduced from the structures of vector fibration (cf. I 4.6).⁽²⁵⁾

Proposition 3.4. bis. *If X/S satisfies (E), then $X_{S'}/S'$ satisfies (E) and the isomorphisms of 3.4 respect the structures of $\mathbf{O}_{X_{S'}}$ -modules, resp. of $\mathbf{O}_{S'}$ -modules.*

Without comment.

⁽²²⁾**N.D.E.** For examples of S -functors and S -groups not satisfying (E), see 6.2 and 6.3.

⁽²³⁾**N.D.E.** What follows has been added.

⁽²⁴⁾**N.D.E.** What follows has been set out in detail.

⁽²⁵⁾**N.D.E.** That is, for every S -morphism $f: X' \rightarrow X$, the action of $\mathbf{O}(X')$ on $\mathrm{Hom}_X(X', \mathrm{T}_{X/S}(\mathcal{M}))$ corresponds, via the identification $\mathrm{Hom}_X(X', \mathrm{T}_{X/S}(\mathcal{M})) \simeq \mathrm{Hom}_{\mathcal{O}_{X'}}(f^*(\Omega_{X/S}^1), \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{X'})$, to the natural action of $\mathbf{O}(X')$ on the term on the right; this follows from the proof of SGA 1, III 5.1 (see also the addition 0.1.8 in Exposé III).

Proposition 3.7. *The functors $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ are functorial in X , i.e. if $f: X \rightarrow X'$ is an S -morphism, one has commutative diagrams:*

$$\begin{array}{ccc} T_{X/S}(\mathcal{M}) & \xrightarrow{T(f)} & T_{X'/S}(\mathcal{M}) \\ \downarrow & & \downarrow \\ X & \xrightarrow{\quad} & X', \end{array} \quad \begin{array}{ccc} L_{X/S}^u(\mathcal{M}) & \xrightarrow{L(f)} & L_{X'/S}^{fou}(\mathcal{M}) \\ & \searrow & \swarrow \\ & S. & \end{array}$$

⁽²⁶⁾ Moreover, if f is a monomorphism, the same holds of $T(f)$ and $L(f)$.

The existence of $T(f)$ and $L(f)$, as well as the last assertion, are deduced immediately from the definitions. The commutativity of the diagrams then results from the functoriality of these morphisms with respect to $\mathcal{M}$ and from the fact that $T_{X/S}(0) = X$.

Remark 3.7.1. ⁽²⁷⁾ Suppose X and X' representable, and let r be the rank of the free $\mathcal{O}_S$ -module $\mathcal{M}$. Then, by 2.2.2, $T_{X/S}(\mathcal{M})$ is isomorphic to the product over X of r copies of $V(\Omega_{X/S}^1)$, and likewise for $T_{X'/S}(\mathcal{M})$. Consequently, the square above is cartesian when f is an open immersion, more generally when $f^*(\Omega_{X'/S}^1) = \Omega_{X/S}^1$, for example if f is étale; under these conditions, one has an isomorphism of S -functors

$$L_{X/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{X'/S}^{fou}(\mathcal{M}).$$

In general, the cartesian square of 3.7 defines a morphism of X -functors:

$$\begin{array}{ccc} T_{X/S}(\mathcal{M}) & \xrightarrow{\quad} & T_{X'/S}(\mathcal{M}) \times_{X'} X \\ & \searrow & \swarrow \\ & X. & \end{array}$$

Proposition 3.7. bis. *Let $f: X \rightarrow X'$ be an S -morphism; if X and X' satisfy (E) with respect to S , then*

$$T_{X/S}(\mathcal{M}) \xrightarrow{T(f)} T_{X'/S}(\mathcal{M})_X \quad \text{resp.} \quad L_{X/S}^u(\mathcal{M}) \xrightarrow{L(f)} L_{X'/S}^{fou}(\mathcal{M})$$

is a morphism of $\mathcal{O}_X$ -modules (resp. of $\mathcal{O}_S$ -modules).

This follows from Proposition 3.7 by functoriality in $\mathcal{M}$.

Proposition 3.8. *Let X and Y be two functors over S . One has isomorphisms, functorial in $\mathcal{M}$:*

$$(3.8.1) \quad T_{X/S}(\mathcal{M}) \times_S T_{Y/S}(\mathcal{M}) \xrightarrow{\sim} T_{(X \times_S Y)/S}(\mathcal{M}),$$

$$(3.8.2) \quad L_{X/S}^u(\mathcal{M}) \times_S L_{Y/S}^v(\mathcal{M}) \xrightarrow{\sim} L_{(X \times_S Y)/S}^{(u,v)}(\mathcal{M}).$$

⁽²⁸⁾ The first isomorphism follows from 1.2 (*), and the second is deduced from it by the base change $(u, v): S \rightarrow X \times_S Y$.

Remark 3.8.0. Let us observe that (3.8.1) may also be interpreted as an isomorphism of $X \times_S Y$ -functors

$$(T_{X/S}(\mathcal{M}) \times_X (X \times_S Y)) \times_{X \times_S Y} (T_{Y/S}(\mathcal{M}) \times_Y (X \times_S Y)) \xrightarrow{\sim} T_{(X \times_S Y)/S}(\mathcal{M}).$$

⁽²⁶⁾**N.D.E.** The sentence which follows has been added.

⁽²⁷⁾**N.D.E.** The hypothesis that X and X' are representable has been added, and what follows has been set out in detail.

⁽²⁸⁾**N.D.E.** What follows has been spelled out.

Corollary 3.8.1. *If X/S is equipped with an algebraic structure defined by finite cartesian products, then $T_{X/S}(\mathcal{M})$ is equipped with a structure of the same species, and the projection $T_{X/S}(\mathcal{M}) \rightarrow X$ is a morphism of this species of structure.*

Proposition 3.8. bis. *If X/S and Y/S satisfy (E), then $(X \times_S Y)/S$ satisfies (E), and (3.8.1) (resp. (3.8.2)) is an isomorphism of $\mathcal{O}_{X \times_S Y}$ -modules (resp. of $\mathcal{O}_S$ -modules).*

PROOF. ⁽²⁹⁾ Suppose that X/S and Y/S satisfy (E). Then, by 3.5.1 and (3.8.1), the same holds for $(X \times_S Y)/S$. Let us show that (3.8.1) is an isomorphism of $\mathcal{O}_{X \times_S Y}$ -modules.

Let $(x, y): Z \rightarrow X \times_S Y$ be an S -morphism; taking 3.4.2 into account, it suffices to see that the map

$$\begin{aligned} & \{\phi \in \text{Hom}_S(I_Z(\mathcal{M}), X) \mid \phi \circ \varepsilon_{\mathcal{M}} = x\} \times \{\psi \in \text{Hom}_S(I_Z(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = y\} \\ & \longrightarrow \{\theta \in \text{Hom}_S(I_Z(\mathcal{M}), X \times_S Y) \mid \theta \circ \varepsilon_{\mathcal{M}} = (x, y)\} \end{aligned}$$

which to (ϕ, ψ) associates $\phi \times \psi$ is a morphism of $\mathcal{O}(Z)$ -modules. But this is clear, for if $a \in \mathcal{O}(Z)$, then $a \cdot (\phi, \psi) = (\phi \circ a^*, \psi \circ a^*)$ is sent to

$$(\phi \circ a^*) \times (\psi \circ a^*) = (\phi \times \psi) \circ a^* = a \cdot (\phi \times \psi).$$

Likewise, using 3.2.1, one shows that (3.8.2) is an isomorphism of $\mathcal{O}_S$ -modules. $\square$

3.9.0. — ⁽³⁰⁾ If X is an S -group and if $e: S \rightarrow X$ denotes its unit section, one denotes:

$$\text{Lie}(X/S, \mathcal{M}) = L_{X/S}^e(\mathcal{M}),$$

i.e. $\text{Lie}(X/S, \mathcal{M})$ is defined by the cartesian square:

$$\begin{array}{ccc} \text{Lie}(X/S, \mathcal{M}) & \xrightarrow{i} & T_{X/S}(\mathcal{M}) \\ \downarrow & & \downarrow p \\ S & \xrightarrow{e} & X. \end{array}$$

By Corollary 3.8.1, the projection $p: T_{X/S}(\mathcal{M}) \rightarrow X$ is a morphism of S -groups, and it follows that $\text{Lie}(X/S, \mathcal{M})$ is equipped with a structure of S -group, and is isomorphic via i to the kernel of p .

If, moreover, X/S satisfies condition (E), one will see in Proposition 3.9 that the structure of S -group on $\text{Lie}(X/S, \mathcal{M})$, induced by that of X , *coincides with the structure of abelian group induced by the functoriality in $\mathcal{M}$* (cf. 3.5.1). In fact, this result is valid under the weaker hypothesis that X be an S -functor in monoids or, more generally, an S -functor in H -sets (cf. the definition below).

Definitions 3.9.0.1. a) Let us introduce the following terminology:⁽³¹⁾ an H -set is a set X equipped with a composition law with two-sided unit, denoted e_X or simply e . If $f: X \rightarrow Y$ is a morphism of H -sets, its kernel $\text{Ker } f$ is $f^{-1}(e_Y)$; it is a sub- H -set of X .

b) An H -object in a category $\mathcal{C}$ is defined in the usual way: it is thus an object X of $\mathcal{C}$, equipped with a morphism $X \times X \rightarrow X$ such that there exists a section of X (over the final object) possessing the properties of a two-sided unit. Every $\mathcal{C}$ -monoid, in particular every $\mathcal{C}$ -group, is thus a $\mathcal{C}$ - H -object. In particular, an H -object of the category of functors over the scheme S will be called an S - H -functor.

c) If X is an S - H -functor (for example, an S -group), and if $e: S \rightarrow X$ denotes the unit section of X , one denotes:

$$\text{Lie}(X/S, \mathcal{M}) = L_{X/S}^e(\mathcal{M}) \quad \text{and} \quad \text{Lie}(X/S) = \text{Lie}(X/S, \mathcal{O}_S).$$

⁽²⁹⁾**N.D.E.** This proof has been added.

⁽³⁰⁾**N.D.E.** This paragraph has been added, in order to explain the introduction of the notion of S - H -functor.

⁽³¹⁾**N.D.E.** This is inspired by the notion of H -space in topology.

Let us then make explicit the following particular case of 3.8.1.⁽³²⁾

Corollary 3.9.0.2. *If X is an S - H -functor (resp. an S -group), then $T_{X/S}(\mathcal{M})$ and $\text{Lie}(X/S, \mathcal{M})$ are also S - H -functors (resp. S -groups), and one has morphisms of S - H -functors (resp. of S -groups):*

$$\text{Lie}(X/S, \mathcal{M}) \xrightarrow{i} T_{X/S}(\mathcal{M}) \xrightleftharpoons[s]{p} X,$$

where i is an isomorphism of $\text{Lie}(X/S)(\mathcal{M})$ onto $\text{Ker } p$ and s is a section of p .

Proposition 3.9. *Let X be an S - H -functor satisfying (E) with respect to S . The structure of S - H -functor on $\text{Lie}(X/S, \mathcal{M})$ coming from that of X coincides with the structure of S -group defined in 3.5.1.*

It follows from what has been said above that $\text{Lie}(X/S, \mathcal{M})$ is an H -object in the category of $\mathcal{O}_S$ -modules. The proposition will then follow from the following lemma:⁽³³⁾

Lemma 3.10. *Let $\mathcal{C}$ be a category. Let G be an H -object in the category of $\mathcal{C}$ - H -objects; G is thus a $\mathcal{C}$ - H -object (whose composition law we shall denote by $f: G \times G \rightarrow G$) equipped with a morphism of $\mathcal{C}$ - H -objects $h: G \times G \rightarrow G$.⁽³⁴⁾ Then $f = h$ and f is commutative.*

By taking the values of the functors on a variable argument, one reduces in the usual way to checking the lemma when $\mathcal{C}$ is the category of sets. One thus has a set G and two maps $f, h: G \times G \rightarrow G$ such that

$$h(f(x, y), f(z, t)) = f(h(x, z), h(y, t)).$$

On the other hand one has two elements of G , namely e and u , with $f(e, x) = f(x, e) = x$, $h(u, x) = h(x, u) = x$. One sees first that

$$h(f(u, y), f(x, u)) = f(x, y) = h(f(x, u), f(u, y)).$$

In particular, for $y = e$, resp. $x = e$, one obtains, respectively,

$$x = f(x, e) = h(f(u, e), f(x, u)) = h(u, f(x, u)) = f(x, u),$$

$$y = f(e, y) = h(f(e, u), f(u, y)) = h(u, f(u, y)) = f(u, y),$$

whence, substituting back into the original equality,

$$h(y, x) = f(x, y) = h(x, y).$$

This proves the lemma, and also Proposition 3.9. One then deduces from 3.9 the following corollaries.

Corollary 3.9.1. *If X is an S - H -functor satisfying (E) with respect to S , every element of $X(I_S(\mathcal{M}))$ which projects onto the unit element of $X(S)$ is invertible.*

Corollary 3.9.2. *If X is an S -monoid satisfying (E) with respect to S , an element of $X(I_S(\mathcal{M}))$ is invertible if and only if its image in $X(S)$ is invertible.*

Corollary 3.9.3. *If X is an S -group satisfying (E) with respect to S , the two S -group laws on $\text{Lie}(X/S, \mathcal{M})$ coincide.*

Corollary 3.9.4. ⁽³⁵⁾ *Let G be an S -group satisfying (E) with respect to S . For $n \in \mathbf{Z}$, let $n_G: G \rightarrow G$ be the morphism of S -functors defined by $g \mapsto g^n$. Then the derived morphism $L(n_G): \text{Lie}(G/S) \rightarrow \text{Lie}(G/S)$ is the “multiplication by n ”, i.e. the map which to every $x \in \text{Lie}(G/S)(S')$ associates nx .*

⁽³²⁾**N.D.E.** Corollary 3.9.0.2 has been added; it will be useful in 4.1.

⁽³³⁾**N.D.E.** Inspired by the standard proof that the fundamental group of an H -space is abelian.

⁽³⁴⁾**N.D.E.** $G \times G$ is equipped with the law $(G \times G) \times (G \times G) \rightarrow G \times G, ((x, z), (y, t)) \mapsto (f(x, y), f(z, t))$.

⁽³⁵⁾**N.D.E.** This corollary has been added; it amplifies Remark 4.1.1.2 further on.

Let us first observe that n_G is not in general a morphism of groups, but that it preserves the unit section $e: S \rightarrow G$, so the derived morphism $L(n_G)$ does indeed send $\text{Lie}(G/S) = L_{X/S}^e$ into itself. If one denotes by i the inclusion $\text{Lie}(G/S) \hookrightarrow T_{G/S}$, then $L(n_G)$ is defined by the equality $i(L(n_G)(x)) = i(x)^n$, for every $S' \rightarrow S$ and every $x \in \text{Lie}(G/S)(S')$. Now, by 3.9, the two group laws on $\text{Lie}(G/S)$ (the one coming from condition (E) and the one coming from the law of G) coincide, i.e. one has $i(x)^n = i(nx)$, whence $L(n_G)(x) = nx$.

Before drawing further consequences from Proposition 3.9, let us prove another functoriality result:

Proposition 3.11. *In the situation of Section 1, one has an isomorphism, functorial in $\mathcal{M}$,*

$$T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

Indeed, one has by definition (cf. 3.1):

$$T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) = \text{Hom}_S(I_S(\mathcal{M}), \text{Hom}_{Z/S}(X, Y)).$$

By isomorphism (4) of 1.1, applied to $T = I_S(\mathcal{M})$, one has:

$$\text{Hom}_S(I_S(\mathcal{M}), \text{Hom}_{Z/S}(X, Y)) \simeq \text{Hom}_{Z/S}(X, \text{Hom}_Z(Z \times_S I_S(\mathcal{M}), Y)).$$

Taking account of the isomorphism $Z \times_S I_S(\mathcal{M}) \simeq I_Z(\mathcal{M})$, this gives

$$\begin{aligned} T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) &\simeq \text{Hom}_{Z/S}(X, \text{Hom}_Z(I_Z(\mathcal{M}), Y)) \\ &= \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})). \end{aligned}$$

Corollary 3.11.1. *If Y/Z satisfies (E), then $\text{Hom}_{Z/S}(X, Y)/S$ satisfies (E), and the isomorphism of 3.11 respects the structures of $\mathcal{O}$ -modules over $\text{Hom}_{Z/S}(X, Y)$.⁽³⁶⁾*

PROOF. Let $\mathcal{M}, \mathcal{N}$ be two free $\mathcal{O}_S$ -modules of finite type. If Y/Z satisfies (E), then

$$T_{Y/Z}(\mathcal{M} \oplus \mathcal{N}) \simeq T_{Y/Z}(\mathcal{M}) \times_Y T_{Y/Z}(\mathcal{N}).$$

The right-hand term is a subfunctor of $T_{Y/Z}(\mathcal{M}) \times_S T_{Y/Z}(\mathcal{N})$, and via the isomorphism of 1.2 (*), one obtains an isomorphism

$$\begin{aligned} \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M} \oplus \mathcal{N})) \\ \simeq \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \times_{\text{Hom}_{Z/S}(X,Y)} \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{N})). \end{aligned}$$

Combined with 3.11, this yields:

$$\begin{aligned} T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M} \oplus \mathcal{N}) \\ \simeq T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) \times_{\text{Hom}_{Z/S}(X,Y)} T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{N}), \end{aligned}$$

so $\text{Hom}_{Z/S}(X, Y)$ satisfies (E) with respect to S . This proves the first assertion of the corollary.

Let us turn to the second. Write $H = \text{Hom}_{Z/S}(X, Y)$, and let us take an S -morphism $\Delta: H' \rightarrow \text{Hom}_{Z/S}(X, Y)$, i.e. a Z -morphism $\delta: H' \times_S X \rightarrow Y$, which thus makes $H' \times_S X$ into a Y -object.

⁽³⁶⁾**N.D.E.** Setting $H = \text{Hom}_{Z/S}(X, Y)$, this implies, in particular, that $T_{H/S}(\mathcal{M})$ is equipped with a natural structure of module over $\prod_{X \times_S H/H} \mathcal{O}_H$. This result seemed a little surprising to the editors; for this reason its proof has been spelled out.

On the one hand, one has a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, \mathrm{T}_{Y/Z}(\mathcal{M}))) & \hookrightarrow & \mathrm{Hom}_S(H', \mathrm{Hom}_{Z/S}(X, \mathrm{T}_{Y/Z}(\mathcal{M}))) \\
\downarrow & & \downarrow \\
\mathrm{Hom}_Y(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M})) & \hookrightarrow & \mathrm{Hom}_Z(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M})) \\
\downarrow & & \downarrow \\
\{\psi \in \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\} & \hookrightarrow & \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y).
\end{array}$$

By 1.3, the action of $\alpha \in \mathcal{O}(H' \times_S X)$ on $\Psi \in \mathrm{Hom}_Y(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M}))$ is given by: for every $U \rightarrow S$ and every $(h, x) \in \mathrm{Hom}_S(U, H' \times_S X)$ (with U then lying over Y via $\delta \circ (h, x)$), one has:

$$(\alpha\Psi)(h, x) = \alpha(h, x) \Psi(h, x),$$

where $\alpha(h, x) \in \mathcal{O}(U)$ acts on $\Psi(h, x) \in \mathrm{T}_{Y/Z}(\mathcal{M})(U)$ via the structure of $\mathcal{O}_Y$ -module on $\mathrm{T}_{Y/Z}(\mathcal{M})$. By 3.4.2, the latter is given, via the identification

$$\begin{aligned}
& \mathrm{Hom}_Y(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M})) \\
&= \{\psi \in \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\},
\end{aligned}$$

by: for every $(m, h, x) \in \mathrm{Hom}_S(U, \mathrm{I}_S(\mathcal{M}) \times_S H' \times_S X)$,

$$(1) \quad (\alpha\psi)(m, h, x) = \psi(m \cdot \alpha(h, x), h, x).$$

On the other hand, let us consider the tangent space $\mathrm{T}_{H/S}(\mathcal{M}) = \mathrm{Hom}_S(\mathrm{I}_S(\mathcal{M}), H)$; one has a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Hom}_H(H', \mathrm{T}_{H/S}(\mathcal{M})) & \hookrightarrow & \mathrm{Hom}_S(H', \mathrm{T}_{H/S}(\mathcal{M})) \\
\downarrow & & \downarrow \\
\{\Phi \in \mathrm{Hom}_S(\mathrm{I}_{H'}(\mathcal{M}), H) \mid \Phi \circ \varepsilon_{\mathcal{M}} = \Delta\} & \hookrightarrow & \mathrm{Hom}_S(\mathrm{I}_{H'}(\mathcal{M}), H) \\
(*) \downarrow & & \downarrow \\
\{\phi \in \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y) \mid \phi \circ \varepsilon_{\mathcal{M}} = \delta\} & \hookrightarrow & \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y),
\end{array}$$

where the bijection $(*)$ is given as follows (cf. 1.1 (2) and I 1.7.1): for every $U \rightarrow S$ and every $(m, h, x) \in \mathrm{Hom}_S(U, \mathrm{I}_S(\mathcal{M}) \times_S H' \times_S X)$ (so that U lies over Z via $U \xrightarrow{x} X \rightarrow Z$), one has $\Phi(m, h) \in \mathrm{Hom}_Z(X \times_S U, Y)$ and

$$(\dagger) \quad \phi(m, h, x) = \Phi(m, h) \circ (x \times \mathrm{id}_U) \in \mathrm{Hom}_Z(U, Y).$$

By 3.4.2 (where one replaces X by $\mathrm{Hom}_{Z/S}(X, Y)$ and X' by H'), the action of $a \in \mathcal{O}(H')$ on $\Phi \in \mathrm{Hom}_S(\mathrm{I}_{H'}(\mathcal{M}), H)$ is given by: for every $U \rightarrow S$ and every $(m, h) \in \mathrm{Hom}_S(U, \mathrm{I}_S(\mathcal{M}) \times_S H')$,

$$(a\Phi)(m, h) = \Phi(m \cdot a(h), h).$$

Consequently, if ϕ (resp. $a\phi$) is the element of $\mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y)$ associated with Φ (resp. $a\Phi$), one has, by $(\dagger)$,

$$\begin{aligned}
(2) \quad (a\phi)(m, h, x) &= \Phi(m \cdot a(h), h) \circ (x \times \mathrm{id}_U) \\
&= \phi(m \cdot a(h), h, x).
\end{aligned}$$

Together with (1), this shows that the isomorphism $\mathrm{T}_{H/S}(\mathcal{M}) \xrightarrow{\sim} \mathrm{Hom}_{Z/S}(X, \mathrm{T}_{Y/Z}(\mathcal{M}))$ of 3.11.1 is an isomorphism of $\mathcal{O}(H)$ -modules; moreover, for every $H' \rightarrow H$, the structure of $\mathcal{O}(H')$ -module on $\mathrm{Hom}_H(H', \mathrm{T}_{H/S}(\mathcal{M}))$ extends, in a manner functorial in H' , to a structure of $\mathcal{O}(H' \times_S X)$ -module. $\square$

In particular, for $Z = S$, one obtains the following corollary.

Corollary 3.11.2. *One has an isomorphism, functorial in $\mathcal{M}$,*

$$T_{\mathrm{Hom}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \mathrm{Hom}_S(X, T_{Y/S}(\mathcal{M})).$$

Moreover, if Y/S satisfies (E), then $\mathrm{Hom}_S(X, Y)/S$ satisfies (E), and the preceding isomorphism respects the structures of $\mathcal{O}$ -modules over $\mathrm{Hom}_S(X, Y)$.

⁽³⁷⁾ Let $u: X \rightarrow Y$ be an S -morphism; one identifies it with the constant morphism $u: S \rightarrow \mathrm{Hom}_S(X, Y)$ such that $u(f) = u$ for every $f: S' \rightarrow S$. One then sees at once that the fibered product of u and of $\mathrm{Hom}_S(X, T_{Y/S}(\mathcal{M})) \rightarrow \mathrm{Hom}_S(X, Y)$ is identified with $\mathrm{Hom}_{Y/S}(X, T_{Y/S}(\mathcal{M}))$, where X lies over Y via u . Consequently, from the definition of $L_{\mathrm{Hom}_S(X,Y)/S}^u(\mathcal{M})$ and from the preceding corollary, one deduces the following:

Corollary 3.11.3. *Let $u: X \rightarrow Y$ be an S -morphism. One has an isomorphism functorial in $\mathcal{M}$ (where, in the right-hand term, X lies over Y via u):*

$$L_{\mathrm{Hom}_S(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \mathrm{Hom}_{Y/S}(X, T_{Y/S}(\mathcal{M})).$$

⁽³⁸⁾ It is an isomorphism of $\mathbf{O}_S$ -modules if Y/S satisfies (E).

In particular, for $Y = X$, $\mathrm{End}_S(X)$ is an S -functor in monoids, hence *a fortiori* an S -H-functor; recalling that $\mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M})$ denotes $L_{\mathrm{End}_S(X)/S}^e(\mathcal{M})$, where e is the unit section (cf. 3.9.0.1), one obtains:

Corollary 3.11.4. *One has an isomorphism functorial in $\mathcal{M}$*

$$\mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M});$$

⁽³⁸⁾ it is an isomorphism of $\mathbf{O}_S$ -modules if Y/S satisfies (E).

Remark 3.11.5. ⁽³⁹⁾ Suppose that X/S satisfies (E). Then

$$\prod_{X/S} T_{X/S}(\mathcal{M}) = \mathrm{Hom}_{X/S}(X, T_{X/S}(\mathcal{M}))$$

is endowed with a structure of $(\prod_{X/S} \mathbf{O}_X)$ -module, i.e. for every $S' \rightarrow S$,

$$\begin{aligned} & \mathrm{Hom}_{X/S}(X, T_{X/S}(\mathcal{M}))(S') \\ &= \{ \psi \in \mathrm{Hom}_X(I_{S'}(\mathcal{M}) \times_S X, X) \mid \psi \circ (\varepsilon_{\mathcal{M}} \times \mathrm{id}_X) = \mathrm{pr}_X \} \end{aligned}$$

is endowed with a structure of $\mathbf{O}(X \times_S S')$ -module, functorial in S' . This follows, at one's choice, from 3.6 and from the properties of the functor $\prod_{X/S}$ (cf. 1.2), or else from the proof of 3.11.1.

We shall now interpret geometrically the definition of the tangent bundle. ⁽⁴⁰⁾ Let U be an S -functor; by I 1.7.2, one has isomorphisms functorial in $\mathcal{M}$

$$\begin{aligned} T_{X/S}(\mathcal{M})(U) &= \mathrm{Hom}_S(U, \mathrm{Hom}_S(I_S(\mathcal{M}), X)) \\ &\simeq \mathrm{Hom}_S(I_S(\mathcal{M}), \mathrm{Hom}_S(U, X)) \\ &\simeq \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}). \end{aligned}$$

⁽³⁷⁾ **N.D.E.** The following sentences have been added.

⁽³⁸⁾ **N.D.E.** The following sentence has been added.

⁽³⁹⁾ **N.D.E.** This remark has been added.

⁽⁴⁰⁾ **N.D.E.** The original has been simplified in what follows.

In particular, the morphism $\mathcal{M} \rightarrow 0$ gives a commutative diagram:

$$\begin{array}{ccc} \mathrm{Hom}_S(U, T_{X/S}(\mathcal{M})) & \xrightarrow{\sim} & \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \\ \pi_{\mathcal{M}} \downarrow & & \downarrow \\ \mathrm{Hom}_S(U, X) & \xrightarrow{\text{identity}} & \mathrm{Hom}_S(U, X), \end{array}$$

where the second vertical arrow is obtained by the base change $\varepsilon_{\mathcal{M}}: S \rightarrow I_S(\mathcal{M})$.⁽⁴¹⁾

Consequently:

Proposition 3.12. *Let $h_0: U \rightarrow X$ be an S -morphism. Then $\mathrm{Hom}_X(U, T_{X/S}(\mathcal{M}))$ is identified with the set of $I_S(\mathcal{M})$ -morphisms from $U_{I_S(\mathcal{M})}$ into $X_{I_S(\mathcal{M})}$ which restrict to h_0 on U (viewed as a subobject of $U \times_S I_S(\mathcal{M})$ via $\mathrm{id}_U \times_S \varepsilon_{\mathcal{M}}$).*

⁽⁴²⁾ In particular, for $U = X$ and $h_0 = \mathrm{id}_X$, one obtains:

Corollary 3.12.1. *The set $\Gamma(T_{X/S}(\mathcal{M})/X)$ is identified with the set of $I_S(\mathcal{M})$ -endomorphisms φ of $X_{I_S(\mathcal{M})}$ which induce the identity on X , i.e. such that the diagram below is commutative.⁽⁴³⁾*

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{\varphi} & I_X(\mathcal{M}) \\ \varepsilon_{\mathcal{M}} \uparrow & & \uparrow \varepsilon_{\mathcal{M}} \\ X & \xrightarrow{\mathrm{id}} & X \end{array}$$

⁽⁴⁴⁾ On the other hand, by 3.11.2, $\Gamma(T_{X/S}(\mathcal{M})/X) \simeq \mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M})(S)$; if moreover X/S satisfies (E), then $\mathrm{End}_S(X)/S$ satisfies (E), so $\mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M})$ is an $\mathbf{O}_S$ -module, by 3.6 (and in fact a $(\prod_{X/S} \mathbf{O}_X)$ -module, by 3.11.5). Applying 3.9, one deduces from this:

Proposition 3.13. *If X/S satisfies (E), the abelian group $\Gamma(T_{X/S}(\mathcal{M})/X)$ is identified with the set of $I_S(\mathcal{M})$ -endomorphisms of $X_{I_S(\mathcal{M})}$ which induce the identity on X . Consequently, every $I_S(\mathcal{M})$ -endomorphism of $X_{I_S(\mathcal{M})}$ which induces the identity on X is an automorphism.*

Corollary 3.13.1. *Let $u: X \rightarrow Y$ be an S -isomorphism, with Y/S satisfying (E). Every $I_S(\mathcal{M})$ -morphism from $X_{I_S(\mathcal{M})}$ into $Y_{I_S(\mathcal{M})}$ which extends u is an isomorphism.*

Corollary 3.13.2. *If Y/S satisfies (E), then the monomorphism*

$$\mathrm{Isom}_S(X, Y) \rightarrow \mathrm{Hom}_S(X, Y)$$

induces, for every $u \in \mathrm{Isom}_S(X, Y)$, an isomorphism

$$L_{\mathrm{Isom}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\mathrm{Hom}_S(X, Y)/S}^u(\mathcal{M}).$$

PROOF. ⁽⁴⁵⁾ One must see that $L_{\mathrm{Isom}_S(X, Y)/S}^u(\mathcal{M})(S') \rightarrow L_{\mathrm{Hom}_S(X, Y)/S}^u(\mathcal{M})(S')$ is a bijection, for every $S' \rightarrow S$. By base change (cf. 3.4), it suffices to do this for $S' = S$. In this case, $L_{\mathrm{Hom}_S(X, Y)/S}^u(\mathcal{M})(S)$ (resp. $L_{\mathrm{Isom}_S(X, Y)/S}^u(\mathcal{M})(S)$) is the set of $I_S(\mathcal{M})$ -morphisms (resp. automorphisms) $X_{I_S(\mathcal{M})} \rightarrow Y_{I_S(\mathcal{M})}$ which extend u , and one applies the preceding corollary. $\square$

⁽⁴¹⁾**N.D.E.** Via the isomorphism $\mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \simeq \mathrm{Hom}_S(U \times_S I_S(\mathcal{M}), X)$, this is equivalent to Remark 3.1.1. Likewise, 3.12 is equivalent to the first part of Remark 3.4.2.

⁽⁴²⁾**N.D.E.** The original has been spelled out in what follows.

⁽⁴³⁾**N.D.E.** When $\mathcal{M} = \mathcal{O}_X$, these are called “infinitesimal endomorphisms” of X ; see 6.2 at the end of this exposé.

⁽⁴⁴⁾**N.D.E.** The original has been spelled out in what follows.

⁽⁴⁵⁾**N.D.E.** The following proof has been added.

Corollary 3.13.3. *If X/S satisfies (E), the monomorphism $\text{Aut}_S(X) \rightarrow \text{End}_S(X)$ induces, for every $u \in \text{Aut}_S(X)$, an isomorphism $L_{\text{Aut}_S(X)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\text{End}_S(X)/S}^u(\mathcal{M})$. In particular, one has*

$$\text{Lie}(\text{Aut}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \text{Lie}(\text{End}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}),$$

⁽⁴⁶⁾ so that $\text{Lie}(\text{Aut}_S(X)/S, \mathcal{M})$ is endowed with a structure of $(\prod_{X/S} \mathbf{O}_X)$ -module.

3.14. — Suppose, to conclude, that X is representable.⁽⁴⁷⁾ In this case, one has seen in 2.2.2 that the X -functor $T_{X/S}$ is representable by $\mathbf{V}(\Omega_{X/S}^1)$, whence the bijections:

$$\Gamma(T_{X/S}/X) \simeq \text{Hom}_X(\Omega_{X/S}^1, \mathcal{O}_X) \simeq \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X).$$

This can also be deduced from what precedes, as follows. By 3.13, $\Gamma(T_{X/S}/X)$ is identified with the set of infinitesimal endomorphisms of X (i.e. the I_S -endomorphisms of X_{I_S} inducing the identity on X). Now X and X_{I_S} have the same underlying topological space, the corresponding sheaves of rings being $\mathcal{O}_X$ and $\mathcal{D}\mathcal{O}_X = \mathcal{O}_X \oplus \mathcal{M}$, where $\mathcal{M} = \mathcal{O}_X$ is regarded as a square-zero ideal. Denoting by $\pi: \mathcal{D}\mathcal{O}_X \rightarrow \mathcal{O}_X$ the morphism of $\mathcal{O}_X$ -algebras which vanishes on $\mathcal{M}$, one deduces from this that giving an infinitesimal endomorphism of X is equivalent to giving a morphism of $\mathcal{O}_S$ -algebras $\varphi: \mathcal{O}_X \rightarrow \mathcal{D}\mathcal{O}_X$ such that $\pi \circ \varphi = \text{id}_{\mathcal{O}_X}$, which is equivalent to giving an $\mathcal{O}_S$ -derivation of the sheaf of rings $\mathcal{O}_X$.

Moreover, one easily sees that if $D, D' \in \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X)$ and if one denotes by φ_D the infinitesimal endomorphism corresponding to D , then

$$\varphi_{D+D'} = \varphi_D \circ \varphi_{D'}.$$

This shows that the identification

$$\{\text{infinitesimal endomorphisms of } X\} \simeq \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X)$$

is an isomorphism of abelian groups. Taking 3.13 (and 3.11.5) into account, one has therefore constructed an isomorphism of abelian groups (and even of $\mathbf{O}(X)$ -modules)

$$\Gamma(T_{X/S}/X) \xrightarrow{\sim} \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X)$$

which recovers the classical interpretation of tangent vector fields in terms of derivations of the structure sheaf.⁽⁴⁸⁾ Let us moreover observe that $\Gamma(T_{X/S}/X)$ is equal to $H^0(X, \mathfrak{g}_{X/S})$, where $\mathfrak{g}_{X/S}$ is the dual of $\Omega_{X/S}^1$.

4. Tangent space to a group – Lie algebras

4.1. — Let G be a group functor over S . By 3.9.0.2, $T_{G/S}(\mathcal{M})$ and $\text{Lie}(G/S, \mathcal{M})$ are endowed with structures of groups over S , and one has group morphisms

$$\text{Lie}(G/S, \mathcal{M}) \xrightarrow{i} T_{G/S}(\mathcal{M}) \xrightarrow{\overset{s}{\varphi}} G;$$

by definition i is an isomorphism of $\text{Lie}(G/S)(\mathcal{M})$ onto the kernel of p , and s is a section of p . It then follows from I 2.3.7 that this sequence of morphisms allows one to identify $T_{G/S}(\mathcal{M})$ with the semidirect product of G by $\text{Lie}(G/S, \mathcal{M})$.

Definition 4.1.A. ⁽⁴⁹⁾ The corresponding operation of G on $\text{Lie}(G/S, \mathcal{M})$ is denoted

$$\text{Ad}: G \rightarrow \text{Aut}_{S\text{-gr.}}(\text{Lie}(G/S, \mathcal{M}))$$

⁽⁴⁶⁾**N.D.E.** The following has been added.

⁽⁴⁷⁾**N.D.E.** The reminder of 2.2.2 has been added, and the sequel has been spelled out.

⁽⁴⁸⁾**N.D.E.** Moreover, this isomorphism is functorial in S : if $S' \rightarrow S$, setting $X' = X_{S'}$, one has $\text{Lie}(\text{Aut}_S(X)/S)(S') \simeq \Gamma(T_{X'/S'}/X') \simeq \text{Dér}_{\mathcal{O}_{S'}}(\mathcal{O}_{X'})$.

⁽⁴⁹⁾**N.D.E.** The numbering 4.1.A and 4.1.B has been added in order to set off these definitions.

and called the *adjoint representation* (relative to $\mathcal{M}$) of G ; one therefore has by definition, for $x \in G(S')$ and $X \in \text{Lie}(G/S, \mathcal{M})(S')$:

$$\text{Ad}(x)X = i^{-1}(s(x) i(X) s(x)^{-1}).$$

If G is commutative, then so is $T_{G/S}(\mathcal{M})$, and $\text{Ad}(x)X = X$.

Definition 4.1.B. ⁽⁴⁹⁾ If G and H are two group functors over S and if $f: G \rightarrow H$ is a group morphism, one deduces from it by functoriality a morphism of exact sequences compatible with the canonical sections:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \text{Lie}(G/S, \mathcal{M}) & \longrightarrow & T_{G/S}(\mathcal{M}) & \longrightarrow & G \longrightarrow 1 \\ & & \downarrow L(f) & & \downarrow T(f) & & \downarrow f \\ 1 & \longrightarrow & \text{Lie}(H/S, \mathcal{M}) & \longrightarrow & T_{H/S}(\mathcal{M}) & \longrightarrow & H \longrightarrow 1 \end{array}$$

$L(f)$, which one will also denote $\text{Lie}(f)$, is the *derived morphism* of f .⁽⁵⁰⁾

Remark 4.1.C. ⁽⁵¹⁾ If G/S and H/S satisfy (E), then $L(f)$ respects the structures of $\mathbf{O}_S$ -modules deduced from the “functoriality in $\mathcal{M}$ ” (cf. 3.6).

Proposition 4.1.1. *Let $g \in G(S)$. Then $\text{Ad}(g): \text{Lie}(G/S, \mathcal{M}) \rightarrow \text{Lie}(G/S, \mathcal{M})$ is the derived morphism of $\text{Int}(g): G \rightarrow G$.*

Indeed $\text{Ad}(g)X = i^{-1}(\text{Int}(g) i(X))$, which is none other than $L(\text{Int}(g))(X)$ by the very definition of the derived morphism.

Suppose that G/S satisfies (E). Then, by proposition 3.9, the group structure of $\text{Lie}(G/S, \mathcal{M})$ defined as above is none other than the structure induced by its structure of $\mathbf{O}_S$ -module (defined by means of (E)). One then deduces from the preceding proposition and from the functoriality of the operation of $\mathbf{O}_S$ (cf. 3.6) the corollary:

Corollary 4.1.1.1. *Suppose that G/S satisfies (E). Then Ad sends G into the subgroup*

$$\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\text{Lie}(G/S, \mathcal{M}))$$

of $\text{Aut}_{S\text{-gr.}}(\text{Lie}(G/S, \mathcal{M}))$, i.e. for every $g \in G(S')$, $\text{Ad}(g)$ respects the structure of $\mathbf{O}_{S'}$ -module of $\text{Lie}(G'/S', \mathcal{M})$ (where $G' = G \times_S S'$). In other words, Ad is a linear representation (cf. I, 3.2) of G in the $\mathbf{O}_S$ -module $\text{Lie}(G/S, \mathcal{M})$.

Remarks 4.1.1.2. a) In order that G/S satisfy (E), it is necessary and sufficient that, for every pair $(\mathcal{M}, \mathcal{N})$ of free $\mathbf{O}_S$ -modules of finite type, the diagram

$$\begin{array}{ccc} & \text{Lie}(G/S, \mathcal{M} \oplus \mathcal{N}) & \\ \swarrow & & \searrow \\ \text{Lie}(G/S, \mathcal{M}) & & \text{Lie}(G/S, \mathcal{N}) \\ \searrow & & \swarrow \\ & \text{Lie}(G/S, 0) = S & \end{array},$$

obtained by applying the functor $\text{Lie}(G/S, _)$ to the diagram (*) of 2.1, be cartesian.

b) Suppose that G/S satisfies (E). Then the derived morphism of the group law $\pi: G \times_S G \rightarrow G$ is none other than the addition law in $\text{Lie}(G/S, \mathcal{M})$. (N.B. π is not a group morphism, but $\pi(e, e) = e$, so the derived morphism $L(\pi)$ sends $T_{(G \times_S G)/S}^{(e, e)}(\mathcal{M}) = \text{Lie}(G/S, \mathcal{M}) \times_S \text{Lie}(G/S, \mathcal{M})$ into $\text{Lie}(G/S, \mathcal{M})$, cf. 3.7 and 3.8.) For every $n \in \mathbf{Z}$, one shows likewise that if

⁽⁵⁰⁾**N.D.E.** If f is a monomorphism, then so are $L(f)$ and $T(f)$, cf. 3.7.

⁽⁵¹⁾**N.D.E.** This remark has been added.

one denotes by $n_G: G \rightarrow G$ the morphism of S -functors defined by $g \mapsto g^n$, then the derived morphism $L(n_G)$ is multiplication by n on $\text{Lie}(G/S)$, cf. 3.9.4.

4.1.2.0. — ⁽⁵²⁾ Let us now consider the S -functor $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))$; for every $S' \rightarrow S$, one has $T_{G/S}(\mathcal{M})_{S'} \simeq T_{G_{S'}/S'}(\mathcal{M})$ (cf. 3.4) and therefore:

$$\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))(S') \simeq \text{Hom}_{G_{S'}}(G_{S'}, T_{G_{S'}/S'}(\mathcal{M})) = \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}).$$

Let us first note that one has an isomorphism, functorial in S' ,

$$(*) \quad \text{Hom}_{S'}(G_{S'}, \text{Lie}(G_{S'}/S', \mathcal{M})) \xrightarrow{\sim} \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$$

which associates to every $f: G_{S'} \rightarrow \text{Lie}(G_{S'}/S', \mathcal{M})$ the section $s_f: G_{S'} \rightarrow T_{G_{S'}/S'}(\mathcal{M})$ such that, for every $S'' \rightarrow S'$ and $g \in G(S'')$:

$$s_f(g) = i(f(g)) s(g).$$

Let h be an automorphism of the functor $G_{S'}$ over S' , not necessarily respecting the group structure. To every section τ of $T_{G_{S'}/S'}(\mathcal{M})$, one may associate $h(\tau)$ defined by transport of structure: it is for instance the only section of $T_{G_{S'}/S'}(\mathcal{M})$ making the diagram

$$\begin{array}{ccc} G_{S'} & \xrightarrow{\tau} & T_{G_{S'}/S'}(\mathcal{M}) \\ \downarrow h & & \downarrow T(h) \\ G_{S'} & \xrightarrow{h(\tau)} & T_{G_{S'}/S'}(\mathcal{M}) \end{array}$$

commutative.

In particular, let us take for h the right translation t_x by an element x of $G(S')$, i.e. $h(g) = t_x(g) = g \cdot x$, for every $g \in G(S'')$, $S'' \rightarrow S'$. Then one immediately has

$$t_x(s_f) = s_{t_x(f)}$$

where $t_x(f)$ denotes the morphism from $G_{S'}$ into $\text{Lie}(G_{S'}/S', \mathcal{M})$ defined by

$$t_x(f)(g) = f(g \cdot x^{-1})$$

for every $g \in G(S'')$, $S'' \rightarrow S'$.

It follows that if one lets G operate by right translations on

$$\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M})) \quad \text{and} \quad \text{Hom}_S(G, \text{Lie}(G/S, \mathcal{M}))$$

in the following way: for every $S' \rightarrow S$, $x \in G(S')$, $\sigma \in \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and $f \in \text{Hom}_{S'}(G_{S'}, \text{Lie}(G_{S'}/S', \mathcal{M}))$,

$$(\sigma \cdot x)(g) = \sigma(g \cdot x^{-1}) \cdot s(x) \quad \text{and} \quad (f \cdot x)(g) = f(g \cdot x^{-1}),$$

for every $g \in G(S'')$, $S'' \rightarrow S'$, then the isomorphism $(*)$ above respects the operations of G .

In particular, via this isomorphism, the elements of $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ correspond to the *constant* morphisms from $G_{S'}$ into $\text{Lie}(G_{S'}/S', \mathcal{M})$ (i.e. factoring through the projection $G_{S'} \rightarrow S'$) or else to the elements of $\text{Lie}(G_{S'}/S', \mathcal{M})(S') = \text{Lie}(G/S, \mathcal{M})(S')$.

Terminology. — The elements of $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ will be called “sections of $T_{G_{S'}/S'}(\mathcal{M})$ invariant under right translation”.

One then obtains:

⁽⁵²⁾**N.D.E.** The original has been spelled out in what follows, in order to exhibit the action of G on the functors $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))$ and $\text{Hom}_S(G, \text{Lie}(G/S, \mathcal{M}))$.

Proposition 4.1.2. ^{*} *The map $\text{Lie}(G/S, \mathcal{M})(S) \longrightarrow \Gamma(T_{G/S}(\mathcal{M})/G)$ which associates to $X \in \text{Lie}(G/S, \mathcal{M})(S)$ the section $x \mapsto X \cdot x$ is a bijection of $\text{Lie}(G/S, \mathcal{M})(S)$ onto the part of $\Gamma(T_{G/S}(\mathcal{M})/G)$ consisting of the sections invariant under right translation.*

Likewise, one lets G act on the right on $\text{End}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$ as follows: for every $S' \longrightarrow S$, $x \in G(S')$ and $u \in \text{End}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})(S') = \text{End}_{I_{S'}(\mathcal{M})}(G_{I_{S'}(\mathcal{M})})$,

$$(u \cdot x)(g) = u(g \cdot x^{-1}) \cdot x,$$

for every $g \in G(S'')$, $S'' \longrightarrow I_{S'}(\mathcal{M})$. Then the morphism of 3.12.1

$$\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M})) \longrightarrow \text{End}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$$

respects the operations of G and therefore induces, for every $S' \longrightarrow S$, a bijection between $\Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and the set of $I_{S'}(\mathcal{M})$ -endomorphisms u of $G_{I_{S'}(\mathcal{M})}$ which induce the identity on G and which “commute with the right translations”, i.e. which satisfy $u_{S''} \cdot x = u_{S''}$ for every $S'' \longrightarrow S'$ and $x \in G(S'')$. One therefore obtains:

Proposition 4.1.3. ^(*) *There exists a bijection, functorial in G , between the set*

$$\text{Lie}(G/S, \mathcal{M})(S)$$

and the set of $I_S(\mathcal{M})$ -endomorphisms of $G_{I_S(\mathcal{M})}$ inducing the identity on G and commuting with the right translations of G (in the sense indicated above).

Now taking 3.13 into account:

Theorem 4.1.4. ^(*) *Let G be an S -group functor; suppose that G/S satisfies (E). Then the group $\text{Lie}(G/S, \mathcal{M})(S)$ is identified, functorially in G , with the group of $I_S(\mathcal{M})$ -automorphisms of $G_{I_S(\mathcal{M})}$ inducing the identity on G and commuting with the right translations of G (in the sense indicated above).*

One thus recovers (in the case $\mathcal{M} = \mathbf{O}_S$) one of the classical definitions of the Lie algebra of a group.

4.2.0. — ⁽⁵⁴⁾ Before going further, let us establish new corollaries to 3.11. Let X, Y be over Z , and Z over S , as in section 1. As one saw in 3.11, the isomorphisms 1.1 (4):

$$(1) \quad \begin{array}{ccc} \text{Hom}_S(I_S(\mathcal{M}), \text{Hom}_{Z/S}(X, Y)) & \xrightarrow{\sim} & \text{Hom}_{Z/S}(X, \text{Hom}_Z(I_Z(\mathcal{M}), Y)) \\ & \searrow \simeq \quad \swarrow \simeq & \\ & \text{Hom}_{Z/S}(X \times_S I_S(\mathcal{M}), Y) & \end{array}$$

induce the isomorphism θ below

$$(2) \quad \begin{array}{ccc} T_{\text{Hom}_{Z/S}(X, Y)}(\mathcal{M}) & \xrightarrow[\sim]{\theta} & \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \\ & \searrow \simeq \quad \swarrow \simeq & \\ & \text{Hom}_{Z/S}(X \times_S I_S(\mathcal{M}), Y). & \end{array}$$

By 1.3, if Y is a Z -group, the same is true of $\text{Hom}_Z(V, Y)$ for every $V \longrightarrow Z$ (in particular for $V = I_Z(\mathcal{M})$); explicitly, if $Z'' \longrightarrow Z' \longrightarrow Z$ and $\varphi, \psi \in \text{Hom}_Z(V_{Z'}, Y)$, then $\varphi \cdot \psi$ is defined by $(\varphi \cdot \psi)(v) = \varphi(v)\psi(v)$, for every $v \in V_{Z'}(Z'')$.

Suppose now that X and Y are Z -groups, and let us set the following definition.

^{*}The statements 4.1.2, 4.1.3 and 4.1.4 are obtained more simply by observing that the automorphisms of G invariant under right translations are the left translations. ⁽⁵³⁾

⁽⁵³⁾**N.D.E.** See for instance the proof of propositions 4.6 and 2.5 of [DG70], § II.4.

⁽⁵⁴⁾**N.D.E.** One has added paragraph 4.2.0, whose results are used implicitly in 4.2 and 4.7 of the original.

Definition 4.2.0.1. Let $\text{Hom}_{(Z/S)\text{-gr.}}(X, Y)$ be the subfunctor of $\text{Hom}_{Z/S}(X, Y)$ defined by: for every $S' \rightarrow S$,

$$(3) \quad \text{Hom}_{(Z/S)\text{-gr.}}(X, Y)(S') = \text{Hom}_{Z_{S'}\text{-gr.}}(X_{S'}, Y_{S'}).$$

This definition applies equally when one replaces Y by the Z -group $T_{Y/Z}(\mathcal{M})$.

One then sees easily that $T_{\text{Hom}_{(Z/S)\text{-gr.}}(X, Y)}(\mathcal{M})(S')$ corresponds, in the preceding isomorphisms (2), to the $Z_{S'}$ -morphisms

$$\varphi: X_{S'} \times_{S'} I_{S'}(\mathcal{M}) \rightarrow Y_{S'}$$

which are “multiplicative in X ”, i.e. which satisfy $\varphi(x_1 x_2, m) = \varphi(x_1, m) \varphi(x_2, m)$, and that these correspond to the morphisms of $Z_{S'}$ -groups $X_{S'} \rightarrow T_{Y/Z}(\mathcal{M})_{S'}$. One has thus obtained:

Proposition 4.2.0.2. *Let X, Y be Z -groups, with Z over S . One has isomorphisms of S -functors, functorial in $\mathcal{M}$:*

$$T_{\text{Hom}_{(Z/S)\text{-gr.}}(X, Y)}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{(Z/S)\text{-gr.}}(X, T_{Y/Z}(\mathcal{M})).$$

In particular, for $Z = S$, one obtains the following corollary. Before stating it, let us remark that if Y is a commutative S -group, the same is true of $T_{Y/S}(\mathcal{M})$, then of $H = \text{Hom}_{S\text{-gr.}}(X, Y)$ and $\text{Hom}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M}))$, and finally of $T_{H/S}(\mathcal{M})$.

Corollary 4.2.0.3. *Let X, Y be S -groups. One has isomorphisms of S -functors, functorial in $\mathcal{M}$:*

$$T_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M})).$$

If Y is commutative, these are moreover isomorphisms of abelian S -groups.

Definition 4.2.0.4. ⁽⁵⁵⁾ If Y is an $\mathbf{O}_S$ -module, the functor $T_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}(Y/S, \mathcal{M})$) is equipped with an $\mathbf{O}_S$ -module structure deduced from that of Y . Equipped with this structure, one will denote it by $T'_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}'(Y/S, \mathcal{M})$).

Consequently, if X, Y are $\mathbf{O}_S$ -modules, then $T'_{Y/S}(\mathcal{M}) = \text{Hom}_S(I_S(\mathcal{M}), Y)$ and $H = \text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, Y)$, and then $\text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, T'_{Y/S}(\mathcal{M}))$ and $T'_{H/S}(\mathcal{M})$, are equipped with an $\mathbf{O}_S$ -module structure, and one obtains the following:

Corollary 4.2.0.5. *If X, Y are $\mathbf{O}_S$ -modules, one has isomorphisms of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$T'_{\text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, Y)/S}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, T'_{Y/S}(\mathcal{M})).$$

Definition 4.2.A. ⁽⁵⁶⁾ Let X, L be S -groups, with X operating on L by group automorphisms (cf. I 2.3.5). One defines the subfunctor $Z_S^1(X, L)$ of $\text{Hom}_S(X, L)$ as follows: for every $S' \rightarrow S$,

$$Z_S^1(X, L)(S') = \left\{ \varphi \in \text{Hom}_{S'}(X_{S'}, L_{S'}) \left| \begin{array}{l} \varphi(x_1 x_2) = \varphi(x_1) (x_1 \cdot \varphi(x_2)), \\ \text{for every } x_1, x_2 \in X(S''), S'' \rightarrow S' \end{array} \right. \right\}.$$

One calls it the “functor of crossed homomorphisms from X to L ”.

Remark 4.2.B. ⁽⁵⁶⁾ a) If L' is a second S -group on which X operates by group automorphisms, one has

$$Z_S^1(X, L \times_S L') \simeq Z_S^1(X, L) \times_S Z_S^1(X, L').$$

b) If L is a $G\text{-}\mathbf{O}_S$ -module, $Z_S^1(X, L)$ coincides with the kernel of the differential

$$\partial: \text{Hom}_S(X, L) \rightarrow \text{Hom}_S(X^2, L)$$

⁽⁵⁵⁾**N.D.E.** One has inserted here this definition, which in the original appeared in 4.3.

⁽⁵⁶⁾**N.D.E.** One has expanded the original in what follows; in particular, one has added definition 4.2.A and remark 4.2.B.

defined in I 5.1; in particular, $Z_S^1(X, L)$ is in this case an $\mathbf{O}_S$ -module.

4.2. — Let $u: X \rightarrow Y$ be a morphism of S -groups; then $L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M})$ is described as follows. First, one saw in 3.11.3 that one has an isomorphism of S -functors, functorial in $\mathcal{M}$:

$$(\dagger) \quad L_{\text{Hom}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{Y/S}(X, T_{Y/S}(\mathcal{M})).$$

On the other hand, as Y is an S -group; one then has

$$T_{Y/S}(\mathcal{M}) = \text{Lie}(Y/S, \mathcal{M}) \cdot Y = \text{Lie}(Y/S, \mathcal{M})_Y;$$

there results an isomorphism of S -functors, functorial in $\mathcal{M}$:

$$L_{\text{Hom}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_S(X, \text{Lie}(Y/S, \mathcal{M})).$$

For every $S' \rightarrow S$, let us denote by $u': X' \rightarrow Y'$ the morphism deduced from u by base change. Let us consider the S -functor defined as follows: ⁽⁵⁷⁾ for every $S' \rightarrow S$,

$$\begin{aligned} \text{Hom}_{(Y/S)\text{-gr.}}(X, \text{Lie}(Y/S, \mathcal{M}) \cdot Y)(S') &= \text{Hom}_{Y'\text{-gr.}}(X', (\text{Lie}(Y/S, \mathcal{M}) \cdot Y)_{S'}) \\ &= \text{Hom}_{Y'\text{-gr.}}(X', \text{Lie}(Y'/S', \mathcal{M}) \cdot Y'). \end{aligned}$$

Then one sees easily that the isomorphism $(\dagger)$ induces an isomorphism

$$(\dagger') \quad L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{(Y/S)\text{-gr.}}(X, \text{Lie}(Y/S, \mathcal{M}) \cdot Y).$$

On the other hand, the morphism

$$X \xrightarrow{u} Y \xrightarrow{\text{Ad}} \text{Aut}_{S\text{-gr.}}(\text{Lie}(Y/S, \mathcal{M}))$$

defines an operation of X on $L = \text{Lie}(Y/S, \mathcal{M})$ by group automorphisms.

If $\Phi \in \text{Hom}_{Y/S}(X, \text{Lie}(Y/S, \mathcal{M}) \cdot Y)$ then, for every $S'' \rightarrow S' \rightarrow S$ and $x \in X(S'')$, one can write in a unique way

$$\Phi(S')(x) = \varphi(S')(x) \cdot u'(x), \quad \text{where} \quad \varphi(S')(x) \in \text{Lie}(Y'/S', \mathcal{M})(S'');$$

this determines an element φ of $\text{Hom}_S(X, \text{Lie}(Y/S, \mathcal{M}))$. Then $\Phi(S')$ is a morphism of groups if, and only if, for every $x_1, x_2 \in X(S'')$ one has:

$$\begin{aligned} \varphi(S')(x_1 x_2) &= \varphi(S')(x_1) (u(x_1) \varphi(S')(x_2) u(x_1)^{-1}) \\ &= \varphi(S')(x_1) (x_1 \cdot \varphi(S')(x_2)), \end{aligned}$$

i.e. if and only if $\varphi \in Z_S^1(X, \text{Lie}(Y/S, \mathcal{M}))$. One has thus obtained the following:

Proposition 4.2. *Let $u: X \rightarrow Y$ be a morphism of S -groups. One has an isomorphism of S -functors, functorial in $\mathcal{M}$:*

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(Y/S, \mathcal{M})).$$

⁽⁵⁸⁾ Suppose moreover that Y/S satisfies (E). Then it follows from 4.2.0.3, exactly as in the proof of 3.11.1, that $\text{Hom}_{S\text{-gr.}}(X, Y)/S$ satisfies (E). Thus one has (cf. 3.5.1):

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M} \oplus \mathcal{N}) \simeq L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \times_S L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{N}).$$

(This also follows from 4.2 and 4.2.B a.) Consequently, $L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M})$ is equipped, as is $Z_S^1(X, \text{Lie}(Y/S, \mathcal{M}))$ (cf. 4.2.B b)), with an $\mathbf{O}_S$ -module structure, deduced from the functoriality in $\mathcal{M}$. One deduces from this that the isomorphism of 4.2 is, in this case, an isomorphism of $\mathbf{O}_S$ -modules:

Proposition 4.2. bis. ⁽⁵⁸⁾ *Let $u: X \rightarrow Y$ be a morphism of S -groups; suppose that Y/S satisfies (E). One has an isomorphism of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(Y/S, \mathcal{M})).$$

⁽⁵⁷⁾ **N.D.E.** This is definition 4.2.0.1 applied to $Z = Y$ and to the Y -groups $u: X \rightarrow Y$ and $T_{Y/S}(\mathcal{M})$.

⁽⁵⁸⁾ **N.D.E.** One has added the sentences which follow and proposition 4.2 bis, implicit in the original.

Moreover, when Y/S satisfies (E), one deduces from 3.13.1, exactly as in the proof of 3.13.2, that for every $u \in \text{Isom}_{S\text{-gr.}}(X, Y)$ one has an isomorphism functorial in $\mathcal{M}$:

$$(*) \quad L_{\text{Isom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}).$$

One deduces from this the following two corollaries.

Corollary 4.2.1. *Let $u: X \rightarrow Y$ be a morphism of S -groups; if Y/S satisfies (E), one has an isomorphism of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$L_{\text{Isom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(Y/S, \mathcal{M})).$$

Corollary 4.2.2. *Let X be an S -group; if X/S satisfies (E), one has an isomorphism of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$\text{Lie}(\text{Aut}_{S\text{-gr.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(X/S, \mathcal{M})).$$

⁽⁵⁹⁾ Furthermore, if Y is commutative, the adjoint action of Y on $L = \text{Lie}(X/S, \mathcal{M})$ is trivial, whence $Z_S^1(X, L) = \text{Hom}_{S\text{-gr.}}(X, L)$. Thus:

Corollary 4.2.3. *Let Y be a commutative S -group; one has an isomorphism of S -functors, functorial in $\mathcal{M}$:*

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{S\text{-gr.}}(X, \text{Lie}(Y/S, \mathcal{M})).$$

4.3. — Let us now consider the case in which X and Y are $\mathbf{O}_S$ -modules. Let us recall (cf. 4.2.0.4) that one denotes by $T'_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}'(Y/S, \mathcal{M})$) the functor $T_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}(Y/S, \mathcal{M})$) equipped with the $\mathbf{O}_S$ -module structure deduced from that of Y .

When Y/S satisfies (E), one will always denote by $\text{Lie}(Y/S, \mathcal{M})$ the functor $\text{Lie}(Y/S, \mathcal{M})$ equipped with the $\mathbf{O}_S$ -module structure defined for every functor satisfying (E). In this case, we know (cf. 3.9) that the abelian group structures of $\text{Lie}(Y/S, \mathcal{M})$ and $\text{Lie}'(Y/S, \mathcal{M})$ coincide, but the same is not a priori true of the module structures (see a counter-example in paragraph 6.3). For every $S' \rightarrow S$ and $a \in \mathbf{O}(S')$, one will denote by $a \cdot' m$ (resp. $a \cdot m$) the action of a on $m \in \text{Lie}'(Y/S, \mathcal{M})(S')$ (resp. on $m \in \text{Lie}(Y/S, \mathcal{M})(S')$), and likewise for the action of a on $T'_{Y/S}(\mathcal{M})$ and $T_{Y/S}(\mathcal{M})$.

One has $T'_{Y/S}(\mathcal{M}) \simeq \text{Lie}'(Y/S, \mathcal{M}) \oplus Y$ as $\mathbf{O}_S$ -modules; consequently, one obtains, exactly as for propositions 4.2 and 4.2 bis, the following:

Proposition 4.3. *Let $u: X \rightarrow Y$ be a morphism of $\mathbf{O}_S$ -modules. One has an isomorphism of S -functors, functorial in $\mathcal{M}$:*

$$(*) \quad L_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, \text{Lie}'(Y/S, \mathcal{M})).$$

⁽⁶⁰⁾ If Y/S satisfies (E), then $\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, Y)/S$ satisfies (E) and $(*)$ is an isomorphism of $\mathbf{O}_S$ -modules when one equips the two members with the $\mathbf{O}_S$ -module structure deduced from the functoriality in $\mathcal{M}$. ⁽⁶¹⁾

Remark 4.3. bis. ⁽⁶²⁾ Let $u: X \rightarrow Y$ be a morphism of $\mathbf{O}_S$ -modules; let us denote by τ_u the map which to every morphism of $\mathbf{O}_S$ -modules $\varphi: X \rightarrow \text{Lie}'(Y/S, \mathcal{M})$ associates the morphism

$$u \oplus \varphi: X \rightarrow T'_{Y/S}(\mathcal{M}) = Y \oplus \text{Lie}'(Y/S, \mathcal{M}).$$

⁽⁵⁹⁾**N.D.E.** One has added the sentence which follows.

⁽⁶⁰⁾**N.D.E.** One has added the sentence which follows.

⁽⁶¹⁾**N.D.E.** Note that on the right-hand term, it is the structure defined by $(a\varphi)(x) = a \cdot \varphi(x)$, where $a \cdot \varphi(x)$ denotes the action of a on $\varphi(x) \in \text{Lie}(Y/S, \mathcal{M})$. This differs from the action $(a \cdot' \varphi)(x) = a \cdot' \varphi(x) = \varphi(ax)$, but coincides with the latter if $\text{Lie}(Y/S, \mathcal{M}) = \text{Lie}'(Y/S, \mathcal{M})$.

⁽⁶²⁾**N.D.E.** One has added this remark, which will be useful in 4.5.1.

Then the isomorphism of 4.3 fits into the following commutative diagram, functorial in $\mathcal{M}$:

$$\begin{array}{ccc} L_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}^u(\mathcal{M}) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, \text{Lie}'(Y/S, \mathcal{M})) \\ \downarrow & & \downarrow \tau_u \\ T_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})). \end{array}$$

Moreover, when Y/S satisfies (E), one deduces from 3.13.1, exactly as in the proof of 3.13.2, that for every $u \in \text{Isom}_{\mathbf{O}_S\text{-mod.}}(X, Y)$, one has

$$(*) \quad L_{\text{Isom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}^u(\mathcal{M}) = L_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}^u(\mathcal{M}).$$

Corollary 4.3.1. *Let X be an $\mathbf{O}_S$ -module satisfying (E) with respect to S . One has a functorial isomorphism in $\mathcal{M}$:*

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, \text{Lie}'(X/S, \mathcal{M}))$$

which respects the $\mathbf{O}_S$ -module structures deduced from the functoriality in $\mathcal{M}$. ⁽⁶³⁾ In particular, $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S$ satisfies (E).

PROOF. The first assertion follows from $(*)$ and 4.3; let us prove the second. Since X/S satisfies (E), one has isomorphisms of $\mathbf{O}_S$ -modules $\text{Lie}'(X/S, \mathcal{M} \oplus \mathcal{N}) \simeq \text{Lie}'(X/S, \mathcal{M}) \times_S \text{Lie}'(X/S, \mathcal{N})$, and hence:

$$\begin{aligned} \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{M} \oplus \mathcal{N}) &\simeq \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \\ &\quad \times_S \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{N}). \end{aligned}$$

In view of 4.1.1.2 a), this proves that $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S$ satisfies (E). $\square$

4.3.2. — Before continuing in this direction, let us examine more closely the relations between Y , $\text{Lie}(Y/S)$ and $\text{Lie}'(Y/S)$. Let us first remark that

$$(1) \quad \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) = \text{Lie}'(\mathbf{O}_S/S, \mathcal{M}) = W(\mathcal{M})$$

(where $W(\mathcal{M})$ is defined in I 4.6) and that one therefore has a canonical isomorphism

$$(2) \quad d: \mathbf{O}_S \xrightarrow{\sim} \text{Lie}(\mathbf{O}_S/S).$$

Now let F be an $\mathbf{O}_S$ -module. For every $S_2 \rightarrow S_1 \rightarrow S$ ⁽⁶⁴⁾, one has a dihomomorphism

$$(3) \quad \begin{cases} F(S_1) \rightarrow F(S_2) \\ \mathbf{O}(S_1) \rightarrow \mathbf{O}(S_2), \end{cases}$$

whence a morphism of $\mathbf{O}(S_2)$ -modules

$$F(S_1) \otimes_{\mathbf{O}(S_1)} \mathbf{O}(S_2) \rightarrow F(S_2).$$

In particular, setting $S_1 = S'$ and $S_2 = I_{S'}(\mathcal{M})$, one deduces from this morphisms of $\mathbf{O}(S')$ -modules, functorial in $\mathcal{M}$,

$$F(S') \otimes_{\mathbf{O}(S')} T_{\mathbf{O}_S/S}(\mathcal{M})(S') \rightarrow T'_{F/S}(\mathcal{M})(S');$$

letting S' vary, one obtains morphisms of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$,

$$(4) \quad F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) \rightarrow T'_{F/S}(\mathcal{M}).$$

These morphisms are functorial in $\mathcal{M}$, hence compatible with the projections of the tangent bundles onto their bases; they therefore define morphisms of $\mathbf{O}_S$ -modules

$$(5) \quad F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \rightarrow \text{Lie}'(F/S, \mathcal{M})$$

⁽⁶³⁾**N.D.E.** Cf. N.D.E. (61). On the other hand, one has added the sentence which follows, as well as its proof.

⁽⁶⁴⁾**N.D.E.** One has changed $S' \rightarrow S$ into $S_2 \rightarrow S_1 \rightarrow S$, for one must let S_2 and S_1 vary (cf. below). On the other hand, one has expanded the original in what follows.

such that one has the commutative diagram:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M}) & \longrightarrow & F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) & \longrightarrow & F \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \mathrm{Lie}'(F/S, \mathcal{M}) & \longrightarrow & T'_{F/S}(\mathcal{M}) & \longrightarrow & F \longrightarrow 0.
 \end{array}$$

One may regard the morphisms (5) as morphisms of abelian S -groups

$$(6) \quad F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M}) \longrightarrow \mathrm{Lie}(F/S, \mathcal{M});$$

by tensoring F with the isomorphism $d: \mathbf{O}_S \xrightarrow{\sim} \mathrm{Lie}(\mathbf{O}_S/S)$, one deduces from this (for $\mathcal{M} = \mathbf{O}_S$) a morphism of abelian S -groups

$$(7) \quad F \xrightarrow{\sim} F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S) \longrightarrow \mathrm{Lie}(F/S)$$

also denoted $d: F \longrightarrow \mathrm{Lie}(F/S)$.

Remark 4.3.3. ⁽⁶⁵⁾ When F/S satisfies (E), the morphisms (6) and (7) are not necessarily morphisms of $\mathbf{O}_S$ -modules, when one equips the two members with the module structures deduced from that of $\mathcal{M}$ by means of condition (E).

For example, let k be a field of characteristic $p > 0$, $S = \mathrm{Spec}(k)$, and F the $\mathbf{O}_S$ -module which to every S -scheme T associates $F(T) = \Gamma(T, \mathbf{O}_T)$ equipped with the $\mathbf{O}(T)$ -module structure obtained by letting the scalars operate via the p -th power, i.e. $r \cdot f = r^p f$, for $r \in \mathbf{O}(T)$, $f \in F(T)$. As an S -group functor, F is isomorphic to $\mathbf{G}_{a,S}$. Thus F satisfies (E) and $\mathrm{Lie}(F/S)$ is identified with $\mathrm{Lie}(\mathbf{G}_{a,S}/S) \simeq \mathbf{O}_S$. Then the canonical morphism $d: F \longrightarrow \mathrm{Lie}(F/S)$ is, for every $T \longrightarrow S$, the identity map $F(T) \longrightarrow \mathbf{O}(T)$: it does respect the abelian group structure but not the $\mathbf{O}_S$ -module structure.

Remark 4.3.4. ⁽⁶⁶⁾ One can make the morphisms (4) and (5) explicit as follows. The morphism $\Theta: F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}) = \mathrm{Hom}_S(I_S(\mathcal{M}), F)$ is defined by: for every $S' \longrightarrow S$, $\alpha \in \mathbf{O}(I_{S'}(\mathcal{M}))$, and $f: S' \longrightarrow F$,

$$\Theta(f \otimes \alpha) = \alpha \cdot (\tau_0 \circ f) = \alpha \cdot (f \circ \rho),$$

where τ_0 is the zero section $F \longrightarrow T'_{F/S}(\mathcal{M})$ and ρ the structural morphism $I_{S'}(\mathcal{M}) \longrightarrow S'$. Then Θ induces a morphism $\theta: F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M}) \longrightarrow \mathrm{Lie}'(F/S, \mathcal{M})$; this results from the “functoriality in $\mathcal{M}$ ” already mentioned after (4), and can be seen explicitly as follows. On the one hand, one has

$$\mathrm{Lie}'(F/S, \mathcal{M})(S') = \{\varphi \in \mathrm{Hom}_S(I_{S'}(\mathcal{M}), F) \mid \varphi \circ \varepsilon_{\mathcal{M}} = e\},$$

where e denotes the unit section $S' \longrightarrow F$. On the other hand, $\mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M})(S') = \Gamma(S', \mathcal{M})$ is the kernel of the augmentation $\eta: \mathbf{O}(I_{S'}(\mathcal{M})) \longrightarrow \mathbf{O}(S')$, and it is therefore a matter of seeing that if $f \in F(S')$ and $\alpha \in \Gamma(S', \mathcal{M})$, then $\alpha \cdot (f \circ \rho) \circ \varepsilon_{\mathcal{M}} = e$.

Let us consider the dihomomorphism (3), in the case where $S_2 \longrightarrow S_1$ is the S -morphism $\varepsilon_{\mathcal{M}}: S' \longrightarrow I_{S'}(\mathcal{M})$; one then has a commutative diagram

$$\begin{array}{ccc}
 F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S') \\
 \alpha \downarrow & & \downarrow \eta(\alpha) \\
 F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S').
 \end{array}$$

⁽⁶⁵⁾**N.D.E.** One found in the original the assertion that when F/S satisfies (E), the morphisms (6) (and hence (7)) are morphisms of $\mathbf{O}_S$ -modules, an assertion contradicted by a counter-example given in 6.3. One has deleted this assertion, inserted here the aforementioned example, and modified definition 4.4 accordingly. This is of no consequence for what follows.

⁽⁶⁶⁾**N.D.E.** One has added this remark, which will be useful in 4.6.2.

For every $\varphi: I_{S'}(\mathcal{M}) \rightarrow F$, one therefore has $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = \eta(\alpha) \cdot (\varphi \circ \varepsilon_{\mathcal{M}})$, whence $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = e$ if $\eta(\alpha) = 0$.

In particular, taking $\mathcal{M} = t\mathbf{O}_S$, one has $\text{Lie}(\mathbf{O}_S/S) = t\mathbf{O}_S$ and the morphism $F \rightarrow \text{Lie}'(F/S)$ is given by $f \mapsto t \cdot (f \circ \rho)$.

Remark 4.3.5. ⁽⁶⁷⁾ Let F be an $\mathbf{O}_S$ -module, set $E = \text{End}_{\mathbf{O}_S\text{-mod.}}(F)$ and denote by d_F and d_E the morphisms of $\mathbf{O}_S$ -modules given by 4.3.2 (5):

$$\begin{aligned} d_F: F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) &\rightarrow \text{Lie}'(F/S, \mathcal{M}), \\ d_E: E \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) &\rightarrow \text{Lie}'(E/S, \mathcal{M}). \end{aligned}$$

One deduces from 4.3.4 the following commutative diagram of morphisms of $\mathbf{O}_S$ -modules:

$$\begin{array}{ccc} E \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) & \xrightarrow{d_E} & \text{Lie}'(E/S, \mathcal{M}) \\ & & \uparrow \simeq (*) \\ \text{Hom}_{\mathbf{O}_S}(F, F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \text{Hom}_{\mathbf{O}_S}(F, \text{Lie}'(F/S, \mathcal{M})) \end{array}$$

where the right-hand vertical arrow is the isomorphism $(*)$ of 4.3. Thus (*loc. cit.*), if F/S satisfies (E), then E/S satisfies (E) and $(*)$ is also an isomorphism of $\mathbf{O}_S$ -modules when one equips the right-hand terms with the $\mathbf{O}_S$ -module structure deduced from (E).

Remark 4.4.0. ⁽⁶⁸⁾ In 4.3.2, the morphisms (4) are isomorphisms if and only if the morphisms (5) are. Moreover, if these conditions are satisfied, then F/S satisfies (E). Indeed, it suffices to check that $\text{Lie}'(F/S, \mathcal{M} \oplus \mathcal{N}) \simeq \text{Lie}'(F/S, \mathcal{M}) \times_S \text{Lie}'(F/S, \mathcal{N})$. Now one has the commutative diagram below, in which by hypothesis the horizontal arrows are isomorphisms:

$$\begin{array}{ccc} F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M} \oplus \mathcal{N}) & \xrightarrow{\sim} & \text{Lie}'(F/S, \mathcal{M} \oplus \mathcal{N}) \\ \downarrow \wr & & \downarrow \\ F \otimes_{\mathbf{O}_S} (\text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \times_S \text{Lie}(\mathbf{O}_S/S, \mathcal{N})) & \xrightarrow{\sim} & \text{Lie}'(F/S, \mathcal{M}) \times_S \text{Lie}'(F/S, \mathcal{N}); \end{array}$$

the second vertical arrow is therefore also an isomorphism, i.e. F/S satisfies (E).

Definition 4.4. One says that F is a *good* $\mathbf{O}_S$ -module if the morphisms

$$F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) \rightarrow T_{F/S}(\mathcal{M})$$

or, equivalently,

$$F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \rightarrow \text{Lie}(F/S, \mathcal{M}),$$

are isomorphisms of abelian S -groups (so that F/S satisfies (E)) and if moreover they respect the $\mathbf{O}_S$ -module structures deduced from condition (E).

Corollary 4.4.1. ⁽⁶⁹⁾ Let F be an $\mathbf{O}_S$ -module. Consider the following conditions:

- (i) F is a good $\mathbf{O}_S$ -module.
- (ii) F/S satisfies (E) and $d: F \rightarrow \text{Lie}(F/S)$ is an isomorphism of $\mathbf{O}_S$ -modules.
- (iii) $\text{Lie}(F/S, \mathcal{M}) = \text{Lie}'(F/S, \mathcal{M})$.

Then one has (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii).

PROOF. (i) $\Rightarrow$ (ii) follows from the definition. To prove (ii) $\Rightarrow$ (i), one must show that the morphisms of abelian S -groups (functorial in $\mathcal{M}$)

$$F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \text{Lie}(F/S, \mathcal{M})$$

⁽⁶⁷⁾N.D.E. One has added this remark, which will be useful in 4.5 and 4.7.

⁽⁶⁸⁾N.D.E. In view of the modifications in definition 4.4 (cf. N.D.E. (65)), one has inserted here this remark, which in the original appeared in the proof of 4.4.1.

⁽⁶⁹⁾N.D.E. The original showed that (i) implies (ii) and (iii); one has added the implication (ii) $\Rightarrow$ (i).

are isomorphisms of $\mathbf{O}_S$ -modules. Since F/S satisfies (E), the two members transform finite direct sums of copies of $\mathbf{O}_S$ into finite products of abelian S -groups. This reduces us to the case $\mathcal{M} = \mathbf{O}_S$, which follows from the hypothesis.

Finally (i) $\Rightarrow$ (iii) follows from the definition and from the fact that the isomorphisms

$$F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \text{Lie}'(F/S, \mathcal{M})$$

of 4.3.2 (5) are morphisms of $\mathbf{O}_S$ -modules. $\square$

Examples 4.4.2. For every quasi-coherent $\mathbf{O}_S$ -Module $\mathcal{E}$, the $\mathbf{O}_S$ -modules $V(\mathcal{E})$ and $W(\mathcal{E})$ defined in I 4.6 are good.

(70) Indeed, for every $f: S' \rightarrow S$, the morphisms

$$\begin{aligned} V(\mathcal{E})(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathcal{M})) &\rightarrow T_{V(\mathcal{E})/S}(\mathcal{M})(S') \\ W(\mathcal{E})(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathcal{M})) &\rightarrow T_{W(\mathcal{E})/S}(\mathcal{M})(S') \end{aligned}$$

correspond, respectively, to the morphisms:

$$\begin{aligned} \text{Hom}_{\mathbf{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), \mathbf{O}_{S'}) \otimes_{\mathbf{O}(S')} \Gamma(S', \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})) \\ \rightarrow \text{Hom}_{\mathbf{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})) \\ \Gamma(S', f^*(\mathcal{E})) \otimes_{\mathbf{O}(S')} \Gamma(S', \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})) \\ \rightarrow \Gamma(S', f^*(\mathcal{E}) \otimes_{\mathbf{O}_{S'}} \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})); \end{aligned}$$

these are isomorphisms since $\mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})$ is isomorphic, as an $\mathbf{O}_{S'}$ -module, to a finite direct sum of copies of $\mathbf{O}_{S'}$.

Proposition 4.5. *Let F be a good $\mathbf{O}_S$ -module. Then:*

(i) $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(F)/S$ satisfies (E) and one has a functorial isomorphism

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_S\text{-mod.}}(F, \text{Lie}(F/S, \mathcal{M}))$$

which respects the $\mathbf{O}_S$ -module structures deduced from condition (E). In particular, one has an isomorphism of $\mathbf{O}_S$ -modules

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(F)/S) \xrightarrow{\sim} \text{End}_{\mathbf{O}_S\text{-mod.}}(F).$$

(ii) (71) Moreover, $\text{End}_{\mathbf{O}_S\text{-mod.}}(F)$ is a good $\mathbf{O}_S$ -module.

Indeed, by 4.4.1, F/S satisfies (E) and

$$(1) \quad \text{Lie}(F/S, \mathcal{M}) = \text{Lie}'(F/S, \mathcal{M}) \simeq F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}).$$

Assertion (i) then follows from 4.3.1. Set $E = \text{End}_{\mathbf{O}_S\text{-mod.}}(F)$. By (1) and remark 4.3.5, one has the following commutative diagram of morphisms of abelian S -groups:

$$\begin{array}{ccc} \text{End}_{\mathbf{O}_S\text{-mod.}}(F) \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) & \xrightarrow{d_E} & \text{Lie}(\text{End}_{\mathbf{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \\ & & \uparrow \simeq (*) \\ \text{Hom}_{\mathbf{O}_S}(F, F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \text{Hom}_{\mathbf{O}_S}(F, \text{Lie}(F/S, \mathcal{M})) \end{array}$$

where d_F and $(*)$ are isomorphisms of $\mathbf{O}_S$ -modules; consequently, the same is true of d_E . This proves (ii).

(70) **N.D.E.** One has added what follows.

(71) **N.D.E.** One has added point (ii), an immediate consequence of what precedes.

Scholium 4.5.1. ⁽⁷²⁾ Set (cf. 2.1) $\mathbf{O}_{I_S} = \mathbf{O}_S \oplus t\mathbf{O}_S$ (with $t^2 = 0$), and let F be a good $\mathbf{O}_S$ -module. Then, for every $S' \rightarrow S$, the morphism

$$F(S') \oplus tF(S') = F(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}) \rightarrow F(I_{S'}) = F(S') \oplus \text{Lie}(F/S)(S')$$

(which is the identity on $F(S')$) induces an isomorphism of $\mathbf{O}(S')$ -modules $tF(S') \simeq \text{Lie}(F/S)(S')$.

Letting S' vary, one obtains an isomorphism which one may denote $\text{Lie}(\mathbf{F}/S) \simeq t\mathbf{F}$. For every $S' \rightarrow S$, one therefore has, by 4.5, a commutative diagram:

$$\begin{array}{ccc} \text{End}_{\mathbf{O}_{S'}\text{-mod.}}(\mathbf{F}_{S'}) & \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_{S'}\text{-mod.}}(\mathbf{F}_{S'}, t\mathbf{F}_{S'}) & \xrightarrow{\sim} \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S)(S') \\ \downarrow & & \downarrow \\ \text{Aut}_{\mathbf{O}_{I_{S'}}\text{-mod.}}(\mathbf{F}_{I_{S'}}) & & T_{\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S}(S') \end{array}$$

and one deduces from 4.3 bis that every $X \in \text{End}_{\mathbf{O}_{S'}\text{-mod.}}(\mathbf{F}_{S'})$ corresponds to the element $\text{id} + tX$ of $\text{Aut}_{\mathbf{O}_{I_{S'}}\text{-mod.}}(\mathbf{F}_{I_{S'}})$.

Definition 4.6. One says that the group S -functor $\mathbf{G}$ is *good* if $\mathbf{G}/S$ satisfies condition (E) and if $\text{Lie}(\mathbf{G}/S)$ is a good $\mathbf{O}_S$ -module. ⁽⁷³⁾ Note that if $\mathbf{F}$ is a good $\mathbf{O}_S$ -module, it is a good S -group; indeed $\mathbf{F}/S$ satisfies (E) and $\text{Lie}(\mathbf{F}/S) \simeq \mathbf{F}$ is a good $\mathbf{O}_S$ -module.

Example 4.6.1. If $\mathbf{G}$ is representable, it is good. Indeed, $\mathbf{G}/S$ satisfies (E) and $\text{Lie}(\mathbf{G}/S)$ is of the form $\mathbf{V}(\mathcal{E})$, hence good, by 4.4.2.

Lemma 4.6.2. ⁽⁷⁴⁾ Let $\mathbf{G}$ be a group S -functor such that $\mathbf{G}/S$ satisfies (E), and let $\mathbf{F} = \text{Lie}(\mathbf{G}/S)$. Then $\mathbf{F}$ satisfies (E) and the morphism of abelian groups $d: \mathbf{F} \rightarrow \text{Lie}(\mathbf{F}/S)$ respects the $\mathbf{O}_S$ -module structures.

Consequently, $\mathbf{G}$ is good if and only if $\mathbf{F} \rightarrow \text{Lie}(\mathbf{F}/S)$ is bijective.

PROOF. Suppose that $\mathbf{G}/S$ satisfies (E). Let $\mathcal{M}, \mathcal{N}$ be free $\mathcal{O}_S$ -modules of finite rank. Write $\mathbf{F}(\mathcal{N}) = \text{Lie}(\mathbf{G}/S, \mathcal{N})$ and let e be the unit section of $\mathbf{G}$.

For every $S' \rightarrow S$, one has $\mathbf{F}(\mathcal{N})(S') = \{g \in \text{Hom}_S(I_{S'}(\mathcal{N}), \mathbf{G}) \mid g \circ \varepsilon_{\mathcal{N}} = e\}$ and $\text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M})(S') = \text{Hom}_S(I_{S'}(\mathcal{M}), \mathbf{F}(\mathcal{N}))$ is identified with

$$\left\{ \Phi \in \text{Hom}_S(I_{S'}(\mathcal{N}) \times_{S'} I_{S'}(\mathcal{M}), \mathbf{G}) \left| \begin{array}{l} \Phi \circ (\varepsilon_{\mathcal{N}} \times \text{id}_{I_{S'}(\mathcal{M})}) = e \in \mathbf{G}(I_{S'}(\mathcal{M})) \\ \Phi \circ (\text{id}_{I_{S'}(\mathcal{N})} \times \varepsilon_{\mathcal{M}}) = e \in \mathbf{G}(I_{S'}(\mathcal{N})) \end{array} \right. \right\}.$$

This shows that

$$(1) \quad \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}) \simeq \text{Lie}(\mathbf{F}(\mathcal{M})/S, \mathcal{N}).$$

As $\mathbf{G}/S$ satisfies (E), one deduces from this:

$$\begin{aligned} \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) &\simeq \text{Lie}(\mathbf{F}(\mathcal{M}_1 \oplus \mathcal{M}_2)/S, \mathcal{N}) \\ &\simeq \text{Lie}((\mathbf{F}(\mathcal{M}_1) \times_S \mathbf{F}(\mathcal{M}_2))/S, \mathcal{N}). \end{aligned}$$

By 3.8, the right-hand term is isomorphic to

$$\text{Lie}(\mathbf{F}(\mathcal{M}_1)/S, \mathcal{N}) \times_S \text{Lie}(\mathbf{F}(\mathcal{M}_2)/S, \mathcal{N}) \simeq \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1) \times_S \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_2).$$

It follows that

$$(2) \quad \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) \simeq \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1) \times_S \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_2),$$

hence $\mathbf{F}(\mathcal{N})/S$ satisfies (E), by remark 4.1.1.2 a).

⁽⁷²⁾**N.D.E.** One has added this scholium, implicit in the original, which will be useful in 4.7.1. (Here and in what follows, one denotes by t, t' , etc., variables of square zero, the letter ε being reserved for the unit section of groups.)

⁽⁷³⁾**N.D.E.** One has added the following sentence.

⁽⁷⁴⁾**N.D.E.** One has added this lemma.

Let us now show that the morphism of abelian groups $d: \mathbf{F}(\mathcal{N}) \rightarrow \text{Lie}(\mathbf{F}(\mathcal{N})/S)$ respects the $\mathbf{O}_S$ -module structures. Let us consider the free $\mathcal{O}_S$ -module $\mathcal{M} = t\mathcal{O}_S$, so that

$$\mathbf{O}(\text{I}_{S'}(\mathcal{M})) = \mathbf{O}(S')[t]/(t^2) = \mathbf{O}(S') \oplus t\mathbf{O}(S')$$

and $\text{Lie}(\mathbf{O}_S/S) \simeq t\mathbf{O}_S$. One will denote by ρ_t the structural morphism $\text{I}_S(\mathcal{M}) \rightarrow S$, which corresponds to the injection $u_t: \mathcal{O}_S \hookrightarrow \mathcal{D}_{\mathcal{O}_S}(\mathcal{M}) = \mathcal{O}_S \oplus t\mathcal{O}_S$. Recall (cf. 3.4.2) that, for every S -functor X , $\mathbf{F}(\mathcal{N})(X) = \text{Lie}(\mathbf{G}/S, \mathcal{N})(X)$ is the set of S -morphisms $\phi: \text{I}_X(\mathcal{N}) \rightarrow \mathbf{G}$ such that $\phi \circ \varepsilon_{\mathcal{N}} = e$, and that the operation of $a \in \mathbf{O}(X)$ is given by $a \cdot \phi = \phi \circ a^*$, where a^* is the endomorphism of $\text{I}_X(\mathcal{N})$ associated with a , cf. 2.1.3.

Consequently, by 4.3.4, the morphism $d: \mathbf{F}(\mathcal{N}) \xrightarrow{\sim} \mathbf{F}(\mathcal{N}) \otimes_{\mathbf{O}_S} t\mathbf{O}_S \rightarrow \text{Lie}(\mathbf{F}(\mathcal{N})/S)$ is given by: for every $S' \rightarrow S$ and $f \in \text{Hom}_S(S', \mathbf{F}(\mathcal{N}))$,

$$f \mapsto f \otimes t \mapsto t \cdot (f \circ \rho_t) = f \circ \rho_t \circ t^*,$$

where, in the last term,

$$f \circ \rho_t \in \mathbf{F}(\mathcal{N})(\text{I}_{S'}(\mathcal{M}))$$

is regarded as an S -morphism $\phi: \text{I}_{S'}(\mathcal{M})(\mathcal{N}) \rightarrow \mathbf{G}$ (such that $\phi \circ \varepsilon_{\mathcal{N}} = e$). It remains to see that d is a morphism of $\mathbf{O}_S$ -modules, i.e. that $d(a \cdot f) = u_t(a) \cdot d(f)$ for every $a \in \mathbf{O}(S')$.

Regarding $f \in \mathbf{F}(\mathcal{N})(S')$ as a morphism $\text{I}_{S'}(\mathcal{N}) \rightarrow \mathbf{G}$, one likewise has $a \cdot f = f \circ a^*$. On the other hand, by the functoriality in X of the operation of $\mathbf{O}(X)$ on $\text{I}_X(\mathcal{N})$ (cf. 2.1.3), one has the commutative diagram below:

$$\begin{array}{ccc} \text{I}_{S'}(\mathcal{N}) & \xrightarrow{a^*} & \text{I}_{S'}(\mathcal{N}) \\ \uparrow \rho_t & & \uparrow \rho_t \\ \text{I}_{\text{I}_{S'}(\mathcal{M})}(\mathcal{N}) & \xrightarrow{u_t(a)^*} & \text{I}_{\text{I}_{S'}(\mathcal{M})}(\mathcal{N}) \end{array}$$

One therefore has

$$\begin{aligned} d(a \cdot f) &= f \circ a^* \circ \rho_t \circ t^* = f \circ \rho_t \circ u_t(a)^* \circ t^* \\ &= f \circ \rho_t \circ t^* \circ u_t(a)^* = u_t(a) \cdot d(f) \end{aligned}$$

(the penultimate equality resulting from the fact that $\mathbf{O}$ is commutative). This completes the proof of lemma 4.6.2. $\square$

Theorem 4.7. *If $\mathbf{F}$ is a good $\mathbf{O}_S$ -module, the S -group $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$ is good.*

(75) Indeed, by 4.5, $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S$ satisfies (E) and

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S) \simeq \text{End}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$$

is a good $\mathbf{O}_S$ -module.

4.7.1. — (76) Let now $\mathbf{G}$ be an S -group and $\mathbf{F}$ a good $\mathbf{O}_S$ -module. Suppose given a *linear representation* of $\mathbf{G}$ in $\mathbf{F}$, that is (I 4.7.1), a morphism of S -groups

$$\rho: \mathbf{G} \rightarrow \text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F}).$$

If $\mathbf{G}/S$ satisfies (E), one deduces from this, by 4.1.C and 4.5, a morphism of $\mathbf{O}_S$ -modules, denoted ρ' or $d\rho$:

$$\text{Lie}(\mathbf{G}/S) \rightarrow \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S) \simeq \text{End}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F}).$$

(77) Moreover, setting $\mathcal{O}_{\text{I}_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$ (with $t^2 = 0$), one deduces from 4.5.1 that, if $S' \rightarrow S$ and $X \in \text{Lie}(\mathbf{G}/S)(S') \subset \mathbf{G}(\text{I}_{S'})$, then one has in $\text{Aut}_{\text{I}_{S'}\text{-mod.}}(\mathbf{F}_{\text{I}_{S'}})$ the following equality:

$$(*) \quad \rho(X) = \text{id} + t\rho'(X),$$

(75) **N.D.E.** One has simplified the original, using the additions made in 4.3.1 and 4.5.

(76) **N.D.E.** One has added the numbering 4.7.1, 4.7.2, 4.7.3.

(77) **N.D.E.** One has added the following sentence.

i.e. for every $S'' \rightarrow I_{S'}$ and $f \in \mathbf{F}(S'')$, one has in $\mathbf{F}(S'')$ the equality $\rho(X)(f) = f + t\rho'(X)(f)$.

Definition 4.7.2. Let $\mathbf{G}$ be a *good* S -group. Then $\mathrm{Lie}(\mathbf{G}/S)$ is a good $\mathbf{O}_S$ -module, and one has a morphism of S -groups $\mathrm{Ad}: \mathbf{G} \rightarrow \mathrm{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathrm{Lie}(\mathbf{G}/S))$. One deduces from this, by 4.7.1, a morphism of $\mathbf{O}_S$ -modules

$$\mathrm{ad}: \mathrm{Lie}(\mathbf{G}/S) \rightarrow \mathrm{End}_{\mathbf{O}_S\text{-mod.}}(\mathrm{Lie}(\mathbf{G}/S)),$$

or, what amounts to the same thing, an $\mathbf{O}_S$ -bilinear morphism:

$$\mathrm{Lie}(\mathbf{G}/S) \times_S \mathrm{Lie}(\mathbf{G}/S) \rightarrow \mathrm{Lie}(\mathbf{G}/S), \quad (x, y) \mapsto [x, y] = \mathrm{ad}(x) \cdot y$$

(where x and y denote two arbitrary elements of $\mathrm{Lie}(\mathbf{G}/S)(S') = \mathrm{Lie}(\mathbf{G}_{S'}/S')(S')$). If $\mathbf{G}$ is commutative, then $[x, y] = 0$.

4.7.3. — One may give an equivalent definition of the bracket, as follows: let us first observe that it suffices to do so for $x, y \in \mathrm{Lie}(\mathbf{G}/S)(S)$. Let us next observe that there is a canonical isomorphism $I_S \times_S I_S \simeq I_{I_S}$; to avoid confusion, let us denote by I and I' two copies of I_S and set $\mathcal{O}_I = \mathcal{O}_S[t]$, $\mathcal{O}_{I'} = \mathcal{O}_S[t']$, where $t^2 = 0 = t'^2$. One then has a commutative diagram

$$\begin{array}{ccc} I \times I' & \longrightarrow & I' \\ \downarrow & & \downarrow \\ I & \longrightarrow & S, \end{array}$$

the two arrows issuing from $I \times I'$ identifying the latter with the scheme of dual numbers on I or on I' . There results a commutative diagram of groups (where one writes $L = \mathrm{Lie}(\mathbf{G}/S)$):

$$(1) \quad \begin{array}{ccccccc} & & 1 & & 1 & & \\ & & \downarrow & & \downarrow & & \\ & & L(I) & \longrightarrow & L(S) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & L(I') & \longrightarrow & \mathbf{G}(I \times I') & \longrightarrow & \mathbf{G}(I') \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & L(S) & \longrightarrow & \mathbf{G}(I) & \longrightarrow & \mathbf{G}(S) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 1 & & 1 & & 1 \end{array}$$

The ninth piece of the puzzle is none other than $\mathrm{Lie}(L/S)(S)$. If $\mathbf{G}$ is *good*, this is $L(S)$, and one therefore has the following commutative diagram, in which the rows and the columns are exact sequences of groups, the five $L(\)$ are commutative,⁽⁷⁸⁾ and in which, taking into account the identification $L(I) = L(S) \oplus tL(S)$ (resp. $L(I') = L(S) \oplus t'L(S)$), the injection $L(S) \hookrightarrow L(I)$ (resp. $L(S) \hookrightarrow L(I')$) is given by $u \mapsto tu$ (resp. $u \mapsto t'u$):

$$(2) \quad \begin{array}{ccccc} z \in L(S) & \xrightarrow{t} & L(I) & \longrightarrow & L(S) \ni x \\ t' \downarrow & & \downarrow & & \downarrow \\ L(I') & \longrightarrow & \mathbf{G}(I \times I') & \longrightarrow & \mathbf{G}(I') \\ \downarrow & & \downarrow & & \downarrow \\ y \in L(S) & \longrightarrow & \mathbf{G}(I) & \longrightarrow & \mathbf{G}(S). \end{array}$$

Now in such a diagram, if one takes two elements x and y as marked, and if one lifts x resp. y arbitrarily to an element $\tilde{x} \in L(I)$ resp. $y' \in L(I')$, the commutator $\tilde{x}y'\tilde{x}^{-1}y'^{-1}$ in

⁽⁷⁸⁾**N.D.E.** One has expanded the original in what follows.

$\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ does not depend on the chosen lifts and is the image of an element z as marked. The reader will verify that one has $z = [x, y]$.⁽⁷⁹⁾

Indeed, if one still denotes by x the image of x under the canonical section $L(S) \rightarrow L(\mathbf{I})$ (and likewise for y), then $\tilde{x} = xu$ and $y' = yv$, with $u, v \in L(S) = L(\mathbf{I}) \cap L(\mathbf{I}')$, and since $L(\mathbf{I})$ and $L(\mathbf{I}')$ are commutative one has

$$\tilde{x} y' \tilde{x}^{-1} y'^{-1} = x u y v u^{-1} x^{-1} v^{-1} y^{-1} = x u y u^{-1} v x^{-1} v^{-1} y^{-1} = x y x^{-1} y^{-1}.$$

Moreover, this element is sent to the unit element of $\mathbf{G}(\mathbf{I})$ and of $\mathbf{G}(\mathbf{I}')$, hence comes from a (unique) z as indicated. Finally, regarding y (resp. x) as an element of $L_{\mathbf{I}'}(\mathbf{I}')$ (resp. of $L(S) \subset \mathbf{G}(\mathbf{I}')$), one has by 4.7.1 (*):

$$xyx^{-1} = \text{Ad}(x)(y) = (\text{id} + t' \text{ad}(x))(y) = y + t'[x, y],$$

hence the element $xyx^{-1}y^{-1}$ of $L(\mathbf{I}')$ is the image of the element $z = [x, y]$ of $L(S)$.

In this construction the following two properties appear:

- (i) the bracket is “functorial in $\mathbf{G}$ ”: precisely, $\mathbf{G} \mapsto \text{Lie}(\mathbf{G}/S)$ is a functor from the category of good S -groups to the category of good $\mathbf{O}_S$ -modules equipped with an $\mathbf{O}_S$ -bilinear composition law.
- (ii) One has $[x, y] + [y, x] = 0$: indeed the diagram is symmetric with respect to the main diagonal.⁽⁸⁰⁾

Proposition 4.8. *Let $\mathbf{F}$ be a good $\mathbf{O}_S$ -module. Via the identification*

$$\text{Lie}(\text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})/S) = \text{End}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})$$

one has

$$\text{Ad}(g) \cdot Y = g \circ Y \circ g^{-1} \quad \text{and} \quad [X, Y] = X \circ Y - Y \circ X,$$

for every $S' \rightarrow S$, $g \in \text{Aut}_{\mathbf{O}_{S'\text{-mod}}}(\mathbf{F}_{S'})$, and

$$X, Y \in \text{Lie}(\text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})/S)(S') = \text{End}_{\mathbf{O}_{S'\text{-mod}}}(\mathbf{F}_{S'}).$$

PROOF. ⁽⁸¹⁾ By base change, one reduces to $S' = S$, which makes it possible to lighten the notation. Set $\mathbf{I} = \mathbf{I}_S$ and $\mathcal{O}_{\mathbf{I}} = \mathcal{O}_S[t]$ (with $t^2 = 0$). Recall (cf. 4.5.1) that the inclusion $i: \text{End}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F}) \hookrightarrow \text{Aut}_{\mathbf{O}_{\mathbf{I}\text{-mod}}}(\mathbf{F}_{\mathbf{I}})$ sends Y to $\text{id} + tY$. Then, by definition of $\text{Ad}(g)$ (cf. 4.1.A), one has:

$$\text{id} + t \text{Ad}(g)(Y) = g \circ (\text{id} + tY) \circ g^{-1} = \text{id} + t(g \circ Y \circ g^{-1}),$$

whence $\text{Ad}(g)(Y) = g \circ Y \circ g^{-1}$.

Let $\mathbf{I}'$ be a second copy of $\mathbf{I}_S$, and set $\mathcal{O}_{\mathbf{I}'} = \mathcal{O}_S[t']$ (with $t'^2 = 0$). Let us apply the results of 4.7.3 to $\mathbf{G} = \text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})$ and $L = \text{Lie}(\mathbf{G}/S) = \text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})$. One identifies X with its image under the canonical section $L(S) \hookrightarrow L(\mathbf{I})$; its image in $\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ is then $\text{id} + t'X$, whose inverse is $\text{id} - t'X$. Likewise, Y lifts to $\text{id} + tY$, whose inverse is $\text{id} - tY$. Then the commutator

$$(\text{id} + t'X) \circ (\text{id} + tY) \circ (\text{id} - t'X) \circ (\text{id} - tY) = \text{id} + tt'(X \circ Y - Y \circ X)$$

is the image in $\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ of the element $Z = X \circ Y - Y \circ X$ of $L(S)$ (indeed, Z is sent to $tZ \in L(\mathbf{I})$, then to $\text{id} + t'tZ \in \mathbf{G}(\mathbf{I} \times \mathbf{I}')$). By 4.7.3, this shows that $[X, Y] = X \circ Y - Y \circ X$. $\square$

⁽⁷⁹⁾**N.D.E.** The original indicated in a footnote: “(*) The writer acknowledges, at Gabriel’s request, that the exercise is not immediate; this is moreover the reason why it is not in the text”. One has given the details, taking into account the addition made in 4.7.1; see also [DG70], § II.4, 4.2.

⁽⁸⁰⁾**N.D.E.** I.e., x and y play symmetric roles: the image in $\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ of $[y, x]$ equals the commutator $y'\tilde{x}y'^{-1}\tilde{x}^{-1}$, which is the inverse of $\tilde{x}y'\tilde{x}^{-1}y'^{-1} = [x, y]$. On the other hand (cf. 4.10 below), one does not know whether one necessarily has $[x, x] = 0$: if one regards x as an element of $\tilde{x} \in L(\mathbf{I})$ (resp. $x' \in L(\mathbf{I}')$) it is not a priori clear that $\tilde{x}$ and x' commute...

⁽⁸¹⁾**N.D.E.** One has expanded and simplified the original in what follows.

Corollary 4.8.1. *Let $\mathbf{G}$ be a good S -group and $x, y, z \in \text{Lie}(\mathbf{G}/S)(S')$. One has:*

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

⁽⁸²⁾ Indeed, since $\mathbf{G}$ is good, $\text{Lie}(\mathbf{G}/S)$ is a good $\mathbf{O}_S$ -module and hence, by 4.7,

$$\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\text{Lie}(\mathbf{G}/S))$$

is a good S -group. Then the morphism of S -groups

$$\text{Ad}: \mathbf{G} \longrightarrow \text{Aut}_{\mathbf{O}_S\text{-mod.}}(\text{Lie}(\mathbf{G}/S))$$

given by the functoriality 4.7.3 (i): $\text{ad}[x, y] = [\text{ad } x, \text{ad } y]$. Combined with 4.8, this gives:

$$\text{ad}[x, y] = [\text{ad } x, \text{ad } y] = \text{ad } x \circ \text{ad } y - \text{ad } y \circ \text{ad } x,$$

which, applied to an element z , gives the Jacobi relation.

Corollary 4.8.2. *Let $\mathbf{G}$ be a good S -group operating linearly on a good $\mathbf{O}_S$ -module $\mathbf{F}$ (i.e. let $\mathbf{F}$ be a $\mathbf{G}$ - $\mathbf{O}_S$ -module, $\mathbf{G}$ and $\mathbf{F}$ good). Then the linear map $\rho': \text{Lie}(\mathbf{G}/S) \longrightarrow \text{End}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$ is a representation, that is, one has*

$$\rho'([x, y]) = \rho'(x) \circ \rho'(y) - \rho'(y) \circ \rho'(x).$$

Scholium 4.9. *To every good S -group (for example a representable one), one has associated a good $\mathbf{O}_S$ -module $\text{Lie}(\mathbf{G}/S)$ equipped functorially with a bilinear map satisfying*

$$[x, y] + [y, x] = 0, \quad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

We shall call $\text{Lie}(\mathbf{G}/S)$ equipped with this structure the “Lie algebra” of $\mathbf{G}$ over S (the quotation marks being justified by the fact that one does not know whether $\text{Lie}(\mathbf{G}/S)$ is strictly speaking an $\mathbf{O}_S$ -Lie algebra⁽⁸³⁾). To every linear representation of $\mathbf{G}$ in a good $\mathbf{O}_S$ -module $\mathbf{F}$ is associated a representation of its “Lie algebra”. In particular, to the adjoint representation of $\mathbf{G}$ is associated the adjoint representation of its “Lie algebra”.

Definition 4.10. A group functor $\mathbf{G}$ over S is said to be *very good* if it is good and if $\text{Lie}(\mathbf{G}/S)$ is an $\mathbf{O}_S$ -Lie algebra (that is, if one has identically $[x, x] = 0$).

Examples 4.10.1. The following S -groups are very good: $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$ for every good $\mathbf{O}_S$ -module $\mathbf{F}$ (cf. 4.7 and 4.8), every representable group (see below), every good S -group admitting a monomorphism into a very good S -group, for example every good group subfunctor of a representable group, or every good S -group admitting a faithful linear representation in a good $\mathbf{O}_S$ -module, for example every good S -group such that Ad is a monomorphism. . .

4.11. — Suppose now that G is a group scheme over S . By 4.1.4, $\text{Lie}(G/S)(S)$ is identified with the group of right-invariant infinitesimal automorphisms of G/S , that is, by 3.14, with the group of derivations of $\mathcal{O}_G$ over $\mathcal{O}_S$ invariant under right translation. Moreover this identification respects the module structure and is an⁽⁸⁴⁾ anti-isomorphism of Lie algebras, as one sees by arguing as in 4.7.3: let us set $\mathcal{O}_I = \mathcal{O}_S[t]$ and $\mathcal{O}_{I'} = \mathcal{O}_S[t']$, and let $x \in L(I)$ and $y \in L(I')$. The left translation λ_x (resp. λ_y) is an S -automorphism of $G_{I \times I'}$ which induces the identity on $G_{I'}$ (resp. G_I) and which corresponds to an $\mathcal{O}_S$ -automorphism

$$u = \text{id} + td_x \quad \text{resp.} \quad v = \text{id} + t'd_y$$

of $\mathcal{O}_{G_{I \times I'}} = \mathcal{O}_G[t, t']/(t^2, t'^2)$, where d_x, d_y are $\mathcal{O}_S$ -derivations of $\mathcal{O}_G$ invariant under right translation. As the correspondence between S -automorphisms of $G_{I \times I'}$ and $\mathcal{O}_S$ -automorphisms of $\mathcal{O}_{G_{I \times I'}}$ is contravariant, $\lambda_x \lambda_y \lambda_x^{-1} \lambda_y^{-1}$ corresponds to $v^{-1} u^{-1} v u = \text{id} + tt'(d_y d_x - d_x d_y)$. One deduces from this, by 4.7.3, that the map $x \mapsto -d_x$ is an isomorphism of Lie algebras (for more details, see [DG70], § II.4, 4.4 and 4.6). The foregoing is valid

⁽⁸²⁾N.D.E. One has expanded the original in what follows.

⁽⁸³⁾N.D.E. For one does not know whether $[x, x] = 0$; see 4.10 below.

⁽⁸⁴⁾N.D.E. One has corrected a sign error in the original, cf. [DG70], § II.4, 4.4 and 4.6.

for $\mathrm{Lie}(G/S)(S') = \mathrm{Lie}(G_{S'}/S')(S')$ for every $S' \rightarrow S$. One then recovers the classical definition.⁽⁸⁵⁾

Scholium 4.11.1. *Via the isomorphism $x \mapsto -d_x$, $\mathrm{Lie}(G/S)$ is identified with the functor which, to every S' over S , associates the $\mathcal{O}(S')$ -Lie algebra of the derivations of $G_{S'}$ with respect to S' invariant under right translation.*

As one already knows, by 4.6.1, that every representable group is good, it follows from the foregoing:

Corollary 4.11.2. *Every representable group is very good.*

Let $\varepsilon: S \rightarrow G$ be the unit section of G . Set $\omega_{G/S}^1 = \varepsilon^*(\Omega_{G/S}^1)$ and recall (cf. 3.3) that $\mathrm{Lie}(G/S)$ is representable by the vector bundle $\mathbb{V}(\omega_{G/S}^1)$.

Scholium 4.11.3. *One has therefore associated functorially, to every S -group scheme G , a vector bundle $\mathbb{L}\mathrm{ie}(G/S) = \mathbb{V}(\omega_{G/S}^1)$ over S , which represents the functor $\mathrm{Lie}(G/S)$, and hence is equipped with a structure of S -scheme in $\mathcal{O}_S$ -Lie algebras. Moreover (cf. 3.4 and 3.8), this construction commutes with base change and with finite products.*

Remarks 4.11.4. ⁽⁸⁶⁾ Let π denote the morphism $G \rightarrow S$.

a) The $\mathcal{O}_G$ -module $\Omega_{G/S}^1$ is evidently $(G \times_S G)$ -equivariant (cf. I, § 6) and hence, by I 6.8.1, one has $\Omega_{G/S}^1 \simeq \pi^*(\omega_{G/S}^1)$. It follows for example that $\Omega_{G/S}^1$ is locally free (resp. locally free of finite type) if $\omega_{G/S}^1$ is, which is in particular the case if S is the spectrum of a field (resp. if S is the spectrum of a field and G is locally of finite type over S).

b) Moreover, by I 6.8.2, $\omega_{G/S}^1$ is equipped with a canonical structure of G - $\mathcal{O}_S$ -module, which induces on $\mathbb{V}(\omega_{G/S}^1) = \mathbb{L}\mathrm{ie}(G/S)$ the adjoint operation.⁽⁸⁷⁾

c) On the other hand (cf. EGA I, 5.3.11 and 5.4.6), ε is an immersion, and is a closed immersion if G is separated over S . Thus $\omega_{G/S}^1$ is identified with $\mathcal{I}/\mathcal{I}^2$, where $\mathcal{I}$ is the quasi-coherent ideal defining $\varepsilon(S)$ in an open U of G in which $\varepsilon(S)$ is closed (if $G \rightarrow S$ is separated, one may take $U = G$, and if $G = \mathrm{Spec} \mathcal{A}(G)$ is affine over S , $\mathcal{I}$ is none other than the augmentation ideal of $\mathcal{A}(G)$, i.e. the kernel of $\varepsilon^\sharp: \mathcal{A}(G) \rightarrow \mathcal{O}_S$, cf. I 4.2).⁽⁸⁸⁾

Remark 4.11.5. One can deduce from the isomorphism $\Omega_{G/S}^1 \simeq \pi^*(\omega_{G/S}^1)$ that the $\mathcal{O}_S$ -module $\omega_{G/S}^1$ is identified with the sheaf $\pi_*^G(\Omega_{G/S}^1)$ of “differentials of G with respect to S invariant on the right”, that is, the sheaf whose sections over an open U of S are the sections of $\Omega_{G/S}^1$ over $\pi^{-1}(U)$, invariant under right translation (cf. I, 6.8.3, compare also with VII_A, 2.4).⁽⁸⁹⁾

⁽⁸⁵⁾**N.D.E.** One has set out the original in more detail in what follows; in particular, one has added, for later references, the numbering 4.11.1 to 4.11.8.

⁽⁸⁶⁾**N.D.E.** One has set out what follows in more detail, taking into account the additions made in Exp. I, § 6.8.

⁽⁸⁷⁾**N.D.E.** For this reason, the linear operation of G on $\omega_{G/S}^1$ could be called the “pre-adjoint” operation; in fact, by abuse of language, one will still speak of “the adjoint operation” on $\omega_{G/S}^1$. Let us point out here a somewhat more general construction, which will be used in Exp. III (cf. in particular III 0.8). Suppose given, for every $Y \rightarrow S$, an $\mathcal{O}_Y$ -module $\mathcal{N}(Y)$, and for every S -morphism $\varphi: Z \rightarrow Y$, a morphism of $\mathcal{O}_Z$ -modules $\mathcal{N}(\varphi): \varphi^*\mathcal{N}(Y) \rightarrow \mathcal{N}(Z)$, functorial in φ (this is the case, for example, if $\mathcal{J}$ is a quasi-coherent ideal of $\mathcal{O}_S$ and if one sets $\mathcal{N}(Y) = \mathcal{J}\mathcal{O}_Y$, cf. Exp. III); then the operation of G on $\omega_{G/S}^1$ induces a (generalized) “adjoint operation” of G on the S -functor in abelian groups which, to every $f: Y \rightarrow S$, associates $\mathrm{Hom}_{\mathcal{O}_Y}(f^*(\omega_{G/S}^1), \mathcal{N}(Y))$.

⁽⁸⁸⁾**N.D.E.** One has added the preceding description of $\omega_{G/S}^1$, which will be useful later (for example, in VII_A, 5.5).

⁽⁸⁹⁾**N.D.E.** This description of $\omega_{G/S}^1$ in terms of invariant differentials will not be used in the sequel.

Notation 4.11.6. One denotes by $\mathcal{L}ie(G/S)$ the sheaf of sections of the vector bundle $\mathbb{L}ie(G/S) \rightarrow S$; it is the $\mathcal{O}_S$ -module $(\omega_{G/S}^1)^\vee = \mathcal{H}om_{\mathcal{O}_S}(\omega_{G/S}^1, \mathcal{O}_S)$ dual to $\omega_{G/S}^1$ (cf. EGA II 1.7.9). It is equipped with a structure of $\mathcal{O}_S$ -Lie algebra.

As this construction does not commute with base change (in general), the Lie algebra structure on this module does not make it possible to reconstruct the structure of S -scheme in $\mathcal{O}_S$ -Lie algebras on $\mathbb{L}ie(G/S)$.⁽⁹⁰⁾

However one has:

Lemma 4.11.7. *One supposes $\omega_{G/S}^1$ locally free of finite type (which occurs in particular if G is smooth over S (cf. EGA IV₄, 17.2.4), or if S is the spectrum of a field and G is locally of finite type over S). Then $\mathcal{L}ie(G/S)^\vee \simeq (\omega_{G/S}^1)^{\vee\vee} \simeq \omega_{G/S}^1$ and hence*

$$\mathbb{L}ie(G/S) = \mathbb{V}(\omega_{G/S}^1) = \mathbb{V}(\mathcal{L}ie(G/S)^\vee) = \mathbb{W}(\mathcal{L}ie(G/S))$$

(the last equality resulting from I 4.6.5.1).

Finally, let $G \rightarrow H$ be a monomorphism of group functors. Then $\mathcal{L}ie(G/S) \rightarrow \mathcal{L}ie(H/S)$ is likewise a monomorphism (cf. 3.7). As every vector monomorphism of vector bundles is a closed immersion⁽⁹¹⁾ one obtains:

Corollary 4.11.8. *Let $G \hookrightarrow H$ be a monomorphism of S -group schemes.*

- (i) $\mathbb{L}ie(G/S) \rightarrow \mathbb{L}ie(H/S)$ is a closed immersion, and hence $\omega_{H/S}^1 \rightarrow \omega_{G/S}^1$ is an epimorphism.
- (ii) If $\omega_{G/S}^1$ is locally free of finite type, then the corresponding morphism $\mathcal{L}ie(G/S) \rightarrow \mathcal{L}ie(H/S)$ is an isomorphism of $\mathcal{L}ie(G/S)$ onto a submodule of $\mathcal{L}ie(H/S)$ which is locally a direct factor.

5. Computation of some Lie algebras

5.1. Examples of Lie algebras: the diagonalizable groups. — Let $G = D_S(M)$ be a diagonalizable group over S (I 4.4). As the formation of $\mathcal{L}ie(G/S)$ commutes with base change, it suffices to carry out the construction for $G = D(M)$. One then has:

$$G(I_S) = \mathrm{Hom}_{\mathrm{gr.}}(M, \Gamma(I_S, \mathcal{O}_{I_S})^\times) = \mathrm{Hom}_{\mathrm{gr.}}(M, \Gamma(S, D\mathcal{O}_S)^\times).$$

Now one has a split exact sequence

$$1 \rightarrow \Gamma(S, \mathcal{O}_S) \rightarrow \Gamma(S, D\mathcal{O}_S)^\times \rightarrow \Gamma(S, \mathcal{O}_S)^\times \rightarrow 1,$$

which gives that $\mathcal{L}ie(G)(S)$ is identified with $\mathrm{Hom}_{\mathrm{gr.}}(M, \mathcal{O}(S))$ equipped with its evident structure of $\mathcal{O}(S)$ -module. One therefore obtains, after base change:

Proposition 5.1. *One has isomorphisms*

$$\mathrm{Hom}_{S\text{-gr.}}(M_S, \mathcal{O}_S) \xrightarrow{\sim} \mathcal{L}ie(D_S(M)/S) \quad \text{and} \quad \mathcal{H}om_{\mathrm{gr.}}(\widetilde{M}_S, \mathcal{O}_S) \xrightarrow{\sim} \mathcal{L}ie(D_S(M)/S),$$

(where, in the second isomorphism, $\widetilde{M}_S$ denotes the constant sheaf of groups on S defined by M , and $\mathcal{H}om_{\mathrm{gr.}}$ the sheaf of homomorphisms of sheaves of groups).

⁽⁹⁰⁾**N.D.E.** For $\mathcal{L}ie(G/S) \simeq (\omega_{G/S}^1)^\vee$ does not necessarily determine $\omega_{G/S}^1$. For example, if $S = \mathbf{A}_k^1$ (k a field) and if G is the S -group whose fiber at $s = 0$ is $\mathbf{G}_{a,k}$ and whose other fibers are the trivial group, then $\mathcal{L}ie(G/S) = 0$ but $\mathcal{L}ie(G_s/k)(k) = k$.

⁽⁹¹⁾**N.D.E.** Let $f: \mathcal{M} \rightarrow \mathcal{N}$ be a morphism of $\mathcal{O}_S$ -modules and $\mathcal{P} = \mathrm{Coker}(f)$. If $\mathbb{V}(\mathcal{N}) \rightarrow \mathbb{V}(\mathcal{M})$ is a monomorphism, the surjective morphism $\mathrm{Sym}(\mathcal{N}) \rightarrow \mathrm{Sym}(\mathcal{P})$ factors through $\mathcal{O}_S$, hence $\mathcal{P} = 0$.

Corollary 5.1.1. *If M is free of finite type (or, as we shall say later, if $D_S(M)$ is a split torus) then (see I, 4.6.5 for the definition of $\mathbb{W}$)*

$$\begin{aligned}\mathbb{W}(\mathrm{Lie}(D_S(M)/S)) &\xrightarrow{\sim} \mathrm{Lie}(D_S(M)/S), \\ M^\vee \otimes_{\mathbf{Z}} \mathcal{O}_S &\xrightarrow{\sim} \mathrm{Lie}(D_S(M)/S),\end{aligned}$$

where $M^\vee$ denotes the dual of the abelian group M . In particular

$$\mathcal{O}_S \xrightarrow{\sim} \mathrm{Lie}(\mathbf{G}_{m,S}/S) \quad \text{and} \quad \mathcal{O}_S \xrightarrow{\sim} \mathrm{Lie}(\mathbf{G}_{m,S}/S).$$

5.2. Normalizers and centralizers. — Let us first prove a few lemmas. Recall (cf. I 3.1.1) that a sequence $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ of $\mathcal{O}_S$ -modules is said to be *exact* if for every $S' \rightarrow S$ the sequence $0 \rightarrow F'(S') \rightarrow F(S') \rightarrow F''(S') \rightarrow 0$ of $\mathcal{O}(S')$ -modules is exact.

Likewise, a sequence $1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1$ of S -groups is said to be exact if for every $S' \rightarrow S$ the sequence of groups $1 \rightarrow G'(S') \rightarrow G(S') \rightarrow G''(S') \rightarrow 1$ is exact.

Lemma 5.2.1. ⁽⁹²⁾ *Let $1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1$ be an exact sequence of S -groups.*

(i) *The sequences $1 \rightarrow T_{G'/S}(\mathcal{M}) \rightarrow T_{G/S}(\mathcal{M}) \rightarrow T_{G''/S}(\mathcal{M}) \rightarrow 1$ and*

$$1 \rightarrow \mathrm{Lie}(G'/S, \mathcal{M}) \rightarrow \mathrm{Lie}(G/S, \mathcal{M}) \rightarrow \mathrm{Lie}(G''/S, \mathcal{M}) \rightarrow 1$$

are then exact.

(ii) *Let $1 \rightarrow H' \rightarrow H \rightarrow H'' \rightarrow 1$ be a second sequence of groups; it is exact if and only if the sequence below is exact:*

$$1 \rightarrow G' \times_S H' \rightarrow G \times_S H \rightarrow G'' \times_S H'' \rightarrow 1.$$

(iii) *If two of the S -groups G', G, G'' satisfy (E), the third also satisfies (E).*

(iv) *If $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ is an exact sequence of $\mathcal{O}_S$ -modules and if two of the modules F', F, F'' are good, the third is also good.*

(v) *If two of the S -groups G', G, G'' are good, the third is also good.*

PROOF. ⁽⁹³⁾ The first part of (i) is immediate, and the second part follows from it. Likewise, (ii) is immediate. Let us prove (iii). For brevity, let us write $L(\mathcal{M}) = \mathrm{Lie}(G, \mathcal{M})$, $L'(\mathcal{M}) = \mathrm{Lie}(G', \mathcal{M})$, etc. Then one has the following commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & L'(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & L(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & L''(\mathcal{M} \oplus \mathcal{N}) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & L'(\mathcal{M}) \times_S L'(\mathcal{N}) & \longrightarrow & L(\mathcal{M}) \times_S L(\mathcal{N}) & \longrightarrow & L''(\mathcal{M}) \times_S L''(\mathcal{N}) \longrightarrow 1 \end{array}$$

in which the first (resp. the second) row is exact by (i) (resp. by (i) and (ii)). Assertion (iii) follows from this.

Let us prove (iv). One has a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & F' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F'' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & T_{F'/S}(\mathcal{M}) & \longrightarrow & T_{F/S}(\mathcal{M}) & \longrightarrow & T_{F''/S}(\mathcal{M}) \longrightarrow 0 \end{array}$$

with exact rows (the first, because $T_{\mathcal{O}_S/S}(\mathcal{M})$ is a free $\mathcal{O}_S$ -module, hence flat, the second by (i)). It follows that if two of the modules F', F, F'' are good, the third is also good. Finally, (v) follows from (iii) and (iv).

Lemma 5.2.2. *Let G be an S -group, E, H two G -objects, and F a G - $\mathcal{O}_S$ -module.*

⁽⁹²⁾**N.D.E.** One has set out the statement of the lemma (and its proof) in more detail; this will be useful in 5.3.1.

⁽⁹³⁾**N.D.E.** One has added the proof of points (iii) and (v).

- (i) The canonical homomorphism $E^G \times_S H^G \longrightarrow (E \times_S H)^G$ is an isomorphism.
- (ii) If F is a good $\mathcal{O}_S$ -module, then so is F^G .

PROOF. ⁽⁹⁴⁾ The first assertion is immediate; let us prove the second. For every $S' \longrightarrow S$, one has a commutative diagram

$$\begin{array}{ccc} F^G(I_{S'}(\mathcal{M})) & \xhookrightarrow{\quad} & F(I_{S'}(\mathcal{M})) \\ \uparrow \varphi & & \simeq \uparrow \varphi_1 \\ F^G(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) & \xhookrightarrow{\quad} & F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) \end{array}$$

and one must prove that φ is bijective; now, it is evidently injective; let us show that it is surjective. Let $(t_0, \dots, t_n)$ be a basis of the free $\mathcal{O}(S')$ -module $\mathcal{O}(I_{S'}(\mathcal{M}))$ and let

$$u = \sum_{i=0}^n f_i \otimes t_i$$

be an element of $F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M}))$ such that $\varphi_1(u)$ belongs to $F^G(I_{S'}(\mathcal{M}))$. Let us show that the f_i belong to $F^G(S')$.

Let $S'' \longrightarrow S'$; one may regard S'' as lying over $I_{S'}(\mathcal{M})$ via the zero section $\varepsilon_{\mathcal{M}}$. Then, for every $g \in G(S'')$, one has:

$$u_{S''} = g \cdot u_{S''} = \sum_{i=0}^n g \cdot (f_i)_{S''} \otimes (t_i)_{S''}.$$

As the $(t_i)_{S''}$ form a basis of $\mathcal{O}(I_{S''}(\mathcal{M})) \simeq \mathcal{O}(I_{S'}(\mathcal{M})) \otimes_{\mathcal{O}(S')} \mathcal{O}(S'')$ over $\mathcal{O}(S'')$, it follows that $g \cdot (f_i)_{S''} = (f_i)_{S''}$, whence $f_i \in F^G(S')$ for every i .

Notations. — ⁽⁹⁵⁾ If E is an S -group and F a sub- S -group of E , one denotes by E/F the S -functor which to every $S' \longrightarrow S$ associates the set $E(S')/F(S')$ of classes $x = x F(S')$, $x \in E(S')$. If E is a commutative S -group then E/F is a commutative S -group.

Let now G be an S -group and K a sub- S -group of G ; set $E = \text{Lie}(G/S, \mathcal{M})$ and $F = \text{Lie}(K/S, \mathcal{M})$. The adjoint action of K on E stabilizes F , hence induces an action of K on the S -functor E/F . Then, for every $S' \longrightarrow S$, one has:

$$(E/F)^K(S') = \left\{ x \in E(S')/F(S') \left| \begin{array}{l} \text{for every } f: S'' \longrightarrow S' \text{ and } k \in K(S''), \\ f^*(x^{-1}) \text{ Ad}(k)(f^*(x)) \in F(S'') \end{array} \right. \right\},$$

where $f^*(x)$ denotes the image of x in $E(S'')$.

Theorem 5.2.3. *Let G be an S -group, and K a sub- S -group of G . Let us denote (I 2.3.3)*

$$N = \text{Norm}_G(K), \quad Z = \text{Centr}_G(K).$$

Let us make K operate on $\text{Lie}(G/S, \mathcal{M})$ by means of the adjoint representation of G .

- (i) *If the group law of $\text{Lie}(G/S, \mathcal{M})$ is commutative^(*)⁽⁹⁶⁾, then*

$$\text{Lie}(N/S, \mathcal{M})/\text{Lie}(K/S, \mathcal{M}) = (\text{Lie}(G/S, \mathcal{M})/\text{Lie}(K/S, \mathcal{M}))^K.$$

- (ii) *If the group law of $\text{Lie}(G/S, \mathcal{M})$ is commutative $(*)$, then*

$$\text{Lie}(Z/S, \mathcal{M}) = \text{Lie}(G/S, \mathcal{M})^K.$$

- (iii) *If G satisfies (E) (resp. if G and K satisfy (E)), then Z satisfies (E) (resp. N satisfies (E)).*

⁽⁹⁴⁾**N.D.E.** One has set out the original in more detail in what follows.

⁽⁹⁵⁾**N.D.E.** One has set out these notations in more detail.

^(*)Condition automatically satisfied if G satisfies (E) (cf. 3.9), for example if G is representable.

⁽⁹⁶⁾**N.D.E.** For an example where the S -group $\text{Lie}(G/S)$ is not commutative, see 6.3.

- (iv) Suppose G is good; then Z is good; if moreover G is very good, then Z is very good.
- (v) Suppose G and K are good; then N is good; if moreover G is very good, then N is very good.

To prove (i) and (ii)⁽⁹⁷⁾ we shall use the following lemma, which results from the diagram of exact sequences considered in 4.1.B (with G and H interchanged).

Lemma 5.2.3.0. *Let G be an S -group, H a sub- S -group of G , and $\mathcal{M}$ a free $\mathcal{O}_S$ -module of finite type. Then $T_{H/S}(\mathcal{M})$ and $\text{Lie}(G/S, \mathcal{M})$ are sub- S -groups of $T_{G/S}(\mathcal{M})$ and one has:*

$$T_{H/S}(\mathcal{M}) \cap \text{Lie}(G/S, \mathcal{M}) = \text{Lie}(H/S, \mathcal{M}),$$

where one has set $T_{H/S}(\mathcal{M}) \cap \text{Lie}(G/S, \mathcal{M}) \stackrel{\text{def}}{=} T_{H/S}(\mathcal{M}) \times_{T_{G/S}(\mathcal{M})} \text{Lie}(G/S, \mathcal{M})$.

As the functors considered in (i) and (ii) commute with base change, it suffices to show the equalities of S -points.

Set $H = N$ (resp. $= Z$) and let $\alpha = \pm 1$. By the preceding lemma and the definition of N and Z (cf. I 2.3.3), one has: $\text{Lie}(H/S, \mathcal{M})(S) =$

$$\left\{ \begin{array}{l} X \in \text{Lie}(G/S, \mathcal{M})(S) \subset G(I_S(\mathcal{M})) \mid \\ \text{for every } f: S' \longrightarrow I_S(\mathcal{M}) \text{ and } u \in K(S'), \\ f^*(X^\alpha) \cdot u \cdot f^*(X^{-\alpha}) \cdot u^{-1} \in K(S') \\ \text{resp. } f^*(X) \cdot u \cdot f^*(X^{-1}) \cdot u^{-1} = 1 \end{array} \right\} \quad (*),$$

where $f^*(X)$ denotes the image of X in $G(S')$.

To simplify the writing, let us set

$$\mathfrak{g} = \text{Lie}(G/S, \mathcal{M}), \quad \mathfrak{k} = \text{Lie}(K/S, \mathcal{M}), \quad \mathfrak{n} = \text{Lie}(N/S, \mathcal{M}), \quad \mathfrak{z} = \text{Lie}(Z/S, \mathcal{M}).$$

If $X \in \text{Lie}(H/S, \mathcal{M})(S)$, the equalities $(*)$ are valid for every $f: S' \longrightarrow S$, since $f = \rho \circ \varepsilon_{\mathcal{M}} \circ f$ factors through $I_S(\mathcal{M})$ (where $\rho: I_S(\mathcal{M}) \longrightarrow S$ is the structural morphism and $\varepsilon_{\mathcal{M}}: S \longrightarrow I_S(\mathcal{M})$ the zero section). One deduces from this that

$$\mathfrak{n}(S)/\mathfrak{k}(S) \subset (\mathfrak{g}/\mathfrak{k})^K(S) \quad \text{and} \quad \mathfrak{z}(S) \subset \mathfrak{g}^K(S).$$

To prove the reverse inclusions, let us henceforth suppose $\mathfrak{g} = \text{Lie}(G/S, \mathcal{M})$ commutative; then $\mathfrak{g}^K$ and $(\mathfrak{g}/\mathfrak{k})^K$ are commutative S -groups. Let $X \in \mathfrak{g}(S)$ and X its image in $(\mathfrak{g}/\mathfrak{k})(S)$, suppose that $X \in (\mathfrak{g}/\mathfrak{k})^K(S)$ (resp. $X \in \mathfrak{g}^K(S)$) and let us show that $X \in \mathfrak{n}(S)$ (resp. $X \in \mathfrak{z}(S)$).

Let $f: S' \longrightarrow I_S(\mathcal{M})$; let us show that the preceding condition $(*)$ is satisfied for every $u \in K(S')$. Consider the cartesian square

$$\begin{array}{ccc} I_{S'}(\mathcal{M}) & \xrightarrow{p} & I_S(\mathcal{M}) \\ \rho' \downarrow & & \downarrow \rho \\ S' & \xrightarrow{\rho \circ f} & S \end{array}$$

and let h be the section of ρ' defined by f . It suffices to show that, for every $v \in K(I_{S'}(\mathcal{M}))$, one has

$$\left. \begin{array}{l} p^*(X^\alpha) \cdot v \cdot p^*(X^{-\alpha}) \cdot v^{-1} \in K(I_{S'}(\mathcal{M})) \\ \text{resp. } p^*(X) \cdot v \cdot p^*(X^{-1}) \cdot v^{-1} = 1 \end{array} \right\} \quad (**).$$

Indeed, taking $v = \rho'^*(u)$ and applying h^* to $(**)$, one obtains $(*)$, since $\rho' \circ h = \text{id}_{S'}$ and $p \circ h = f$.

Let us now show $(**)$; to simplify, one will write X in place of $p^*(X)$. Every $v \in K(I_{S'}(\mathcal{M}))$ may be written uniquely $Y \cdot k$ where $Y \in \text{Lie}(K/S, \mathcal{M})(S')$ and $k \in K(S')$. The expression $X^\alpha \cdot u \cdot X^{-\alpha} \cdot u^{-1}$ then becomes $X^\alpha \cdot Y \cdot (k \cdot X^{-\alpha} \cdot k^{-1}) \cdot Y^{-1}$ which, as $\text{Lie}(G/S, \mathcal{M})$ is

⁽⁹⁷⁾**N.D.E.** One has set out the proof in more detail, adding in particular Lemma 5.2.3.0, implicit in the original.

commutative, is written $X^\alpha \text{Ad}(k)(X^{-\alpha})$. Now the latter lies a priori in $\text{Lie}(G/S, \mathcal{M})(S')$; taking Lemma 5.2.3.0 into account, the condition $(**)$ for X therefore becomes: for every $k \in K(S')$,

$$\begin{cases} \text{Ad}(k)(X) = X & \text{if } H = Z; \\ X^\alpha \text{Ad}(k)(X^{-\alpha}) \in \text{Lie}(K/S, \mathcal{M})(S') & \text{if } H = N. \end{cases}$$

When $H = Z$, this condition is indeed a consequence of the hypothesis $X \in \mathfrak{g}^K(S)$. When $H = N$, the condition may also be written:

$$(*)' \quad \text{Ad}(k)(X) = X \quad \text{and} \quad \text{Ad}(k)(X^{-1}) = X^{-1},$$

now the second condition of $(*)'$ is a consequence of the first, since the operation of K on $\mathfrak{g}/\mathfrak{k}$ respects the group structure of the latter. Thus, when $H = N$, the condition $(**)$ for X is indeed a consequence of the hypothesis $X \in (\mathfrak{g}/\mathfrak{k})^K(S)$. This proves (i) and (ii).

To prove (iii)–(v), let us write $\mathfrak{g}(\mathcal{M}) = \text{Lie}(G/S, \mathcal{M})$ and let us define likewise $\mathfrak{k}(\mathcal{M})$, $\mathfrak{z}(\mathcal{M})$ and $\mathfrak{n}(\mathcal{M})$. If G/S satisfies (E), then $\mathfrak{g}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{g}(\mathcal{M}) \times_S \mathfrak{g}(\mathcal{N})$ and hence, by 5.2.2 (i),

$$\mathfrak{g}(\mathcal{M} \oplus \mathcal{N})^K \simeq \mathfrak{g}(\mathcal{M})^K \times_S \mathfrak{g}(\mathcal{N})^K$$

whence $\mathfrak{z}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{z}(\mathcal{M}) \times_S \mathfrak{z}(\mathcal{N})$, hence Z satisfies (E). If moreover K/S satisfies (E), one obtains successively isomorphisms:

$$\begin{aligned} (\mathfrak{g}/\mathfrak{k})(\mathcal{M} \oplus \mathcal{N}) &\simeq (\mathfrak{g}/\mathfrak{k})(\mathcal{M}) \times_S (\mathfrak{g}/\mathfrak{k})(\mathcal{N}) \\ (\mathfrak{g}/\mathfrak{k})^K(\mathcal{M} \oplus \mathcal{N}) &\simeq (\mathfrak{g}/\mathfrak{k})(\mathcal{M})^K \times_S (\mathfrak{g}/\mathfrak{k})(\mathcal{N})^K, \end{aligned}$$

then a commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{k}(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & \mathfrak{n}(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & (\mathfrak{n}/\mathfrak{k})(\mathcal{M} \oplus \mathcal{N}) \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow & & \downarrow \wr \\ 0 & \longrightarrow & \mathfrak{k}(\mathcal{M}) \times_S \mathfrak{k}(\mathcal{N}) & \longrightarrow & \mathfrak{n}(\mathcal{M}) \times_S \mathfrak{n}(\mathcal{N}) & \longrightarrow & (\mathfrak{n}/\mathfrak{k})(\mathcal{M}) \times_S (\mathfrak{n}/\mathfrak{k})(\mathcal{N}) \longrightarrow 0 \end{array}$$

whence it follows that $\mathfrak{n}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{n}(\mathcal{M}) \times_S \mathfrak{n}(\mathcal{N})$, hence N satisfies (E).

Henceforth, let us write $\mathfrak{g} = \mathfrak{g}(\mathcal{O}_S) = \text{Lie}(G/S)$ and let us define likewise $\mathfrak{k}$, $\mathfrak{z}$, $\mathfrak{n}$. If G is *good*, $\mathfrak{g}$ is a *good* $\mathcal{O}_S$ -module, hence, by 5.2.2 (ii), so is $\mathfrak{z} \simeq \mathfrak{g}^K$, hence Z (which satisfies (E) by (iii)) is *good*. If moreover K is *good*, then $\mathfrak{k}$ is a *good* $\mathcal{O}_S$ -module, hence, by 5.2.1 (iv) and 5.2.2 (ii), the same is true of $\mathfrak{g}/\mathfrak{k}$ and $(\mathfrak{g}/\mathfrak{k})^K$. In view of the exact sequence

$$0 \longrightarrow \mathfrak{k} \longrightarrow \mathfrak{n} \longrightarrow (\mathfrak{g}/\mathfrak{k})^K \longrightarrow 0,$$

one obtains, by 5.2.1 (iv) again, that $\mathfrak{n}$ is *good*. Finally, if in addition to the preceding conditions G is *very good*, that is, if one has identically $[x, x] = 0$ for every $x \in \mathfrak{g}$, it is clear that Z and N are *very good*. This proves (iii), (iv) and (v).

Corollary 5.2.3.1. *One has $\text{Lie}(\text{Centr}(G)/S) = \text{Lie}(G/S)^G$ if the group law of $\text{Lie}(G/S)$ is commutative.*

Corollary 5.2.3.2. *If the group law of $\text{Lie}(G/S)$ is commutative and if K is an invariant subgroup of G , then*

$$(\text{Lie}(G/S)/\text{Lie}(K/S))^K = \text{Lie}(G/S)/\text{Lie}(K/S).$$

5.3. Linear representations. — Let G be a *good* S -group operating linearly on a *good* $\mathcal{O}_S$ -module F via

$$\rho: G \longrightarrow \text{Aut}_{\mathcal{O}_S\text{-mod.}}(F).$$

One has defined (4.7.1 and 4.8.2) a corresponding linear representation

$$\rho': \text{Lie}(G/S) \longrightarrow \text{End}_{\mathcal{O}_S\text{-mod.}}(F).$$

The sub- S -groups $\text{Norm}_G(E)$ and $\text{Centr}_G(E)$ are defined for every subset E of F , for example for every sub- $\mathcal{O}_S$ -module E of F .

Definition 5.3.0. One will set, analogously: for every $S' \rightarrow S$,

$$\text{Norm}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid \rho'(X)E_{S'} \subset E_{S'}\};$$

$$\text{Centr}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid \rho'(X)E_{S'} = 0\}.$$

(Let us note that this construction is carried out for every linear representation of an $\mathcal{O}_S$ -“Lie algebra” (in the sense of 4.9) and that the two subobjects constructed are sub- $\mathcal{O}_S$ -modules stable under the bracket).

Theorem 5.3.1. *Let G be a good S -group operating linearly on a good $\mathcal{O}_S$ -module F and let E be a sub- $\mathcal{O}_S$ -module of F .*

- (i) *One has $\text{Lie}(\text{Centr}_G(E)/S) = \text{Centr}_{\text{Lie}(G/S)}(E)$ and $\text{Centr}_G(E)$ is a good S -group; it is very good if G is.*
- (ii) *Suppose that E is a good $\mathcal{O}_S$ -module. Then⁽⁹⁸⁾ $\text{Lie}(\text{Norm}_G(E)/S) = \text{Norm}_{\text{Lie}(G/S)}(E)$ and $\text{Norm}_G(E)$ is a good S -group; it is very good if G is.*

The proof is left to the reader.

5.3.2. — Let G be a good S -group; the foregoing applies in particular to the case where one takes for ρ the adjoint representation of G . Let E be a *good*⁽⁹⁹⁾ submodule of $\text{Lie}(G/S)$; one therefore associates with it two subgroups of G , its centralizer and its normalizer. By 5.3.1, their Lie algebras are respectively the centralizer and the normalizer of E in $\text{Lie}(G/S)$, computed as usual by means of the bracket:

$$\text{Centr}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid [X, E_{S'}] = 0\}.$$

$$\text{Norm}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid [X, E_{S'}] \subset E_{S'}\}.$$

5.3.3. — Let K be a sub- S -group of G , then $\text{Lie}(K/S)$ is a sub- $\mathcal{O}_S$ -module of $\text{Lie}(G/S)$; suppose that $\text{Lie}(K/S)$ is a *good* $\mathcal{O}_S$ -module⁽⁹⁹⁾ (which is the case if K is a *good* S -group). One evidently has

$$\text{Centr}_G(K) \subset \text{Centr}_G(\text{Lie}(K/S))$$

$$\text{Norm}_G(K) \subset \text{Norm}_G(\text{Lie}(K/S)),$$

whence, by 5.3.1,

$$\text{Lie}(\text{Centr}_G(K)/S) \subset \text{Centr}_{\text{Lie}(G/S)}(\text{Lie}(K/S))$$

$$\text{Lie}(\text{Norm}_G(K)/S) \subset \text{Norm}_{\text{Lie}(G/S)}(\text{Lie}(K/S)),$$

but none of these four inclusions is a priori an identity; we shall see many examples of this later.

It follows in particular from these inclusions that if K is an *invariant subgroup* of G , then $\text{Lie}(K/S)$ is an *ideal* of $\text{Lie}(G/S)$.

6. Miscellaneous remarks

6.1. — One can define the bracket of two infinitesimal automorphisms for an S -functor X which is not necessarily a group. It suffices to apply the results of this exposé to the group $\text{Aut}_S(X)$. In order to arrive at a pleasant formalism, one is led to suppose X *good*, that is, to suppose that the $\mathcal{O}_X$ -module $T_{X/S}$ is *good* (if X is an S -group, this definition evidently coincides with definition 4.6).

⁽⁹⁸⁾**N.D.E.** One has corrected the original, which stated the following equality without hypotheses on E ; see 6.5.1 for counterexamples.

⁽⁹⁹⁾**N.D.E.** One has added this hypothesis; cf. 6.5.1.

6.2. — There exist functors possessing infinitesimal endomorphisms which are not automorphisms, hence *a fortiori* not satisfying condition (E).⁽¹⁰⁰⁾ For every pointed set (E, x_0) , let $\text{MA}(E)$ be the free abelian monoid generated by E and let $\text{MA}_P(E, x_0)$ be the abelian monoid obtained by quotienting $\text{MA}(E)$ by the equivalence relation generated by the relation $m \sim x_0 + m$. Then $(E, x_0) \mapsto \text{MA}_P(E, x_0)$ is the left adjoint of the forgetful functor from the category of abelian monoids to that of pointed sets; one will say that $\text{MA}_P(E, x_0)$ is the “free abelian monoid on the pointed set (E, x_0) ”.

Let us then take for X the functor which to every scheme S associates the free abelian monoid on the set $\text{O}(S)$, pointed by the zero element. Each morphism $f: S \rightarrow I_{\mathbf{Z}} = \text{Spec}(\mathbf{Z}[\varepsilon])$ corresponds to an element u_f of square zero of $\text{O}(S)$, hence defines an endomorphism of $X(S)$ by $x \mapsto x + u_f$ (sum in $\text{MA}_P(\text{O}(S), 0)$). One thus obtains an endomorphism ϕ of $X_{I_{\mathbf{Z}}} = X \times_{\mathbf{Z}} I_{\mathbf{Z}}$, defined as follows. For every $f \in I_{\mathbf{Z}}(S)$ and $x \in X(S)$,

$$\phi((x, f)) = (x + u_f, f).$$

If $f_0: S \rightarrow I_{\mathbf{Z}}$ is the composite of the structural morphism $S \rightarrow \text{Spec}(\mathbf{Z})$ and of the zero section of $I_{\mathbf{Z}}$, the corresponding element is $u_{f_0} = 0$, and hence $\phi((x, f_0)) = (x, f_0)$, since $x + 0 = x$ in $\text{MA}_P(\text{O}(S), 0)$. This shows that ϕ induces the identity on X ; it is therefore an infinitesimal endomorphism of X which is evidently not an automorphism.

6.3. — There exist modules which are not *good*. On the one hand, one can slightly modify the preceding counterexample:⁽¹⁰¹⁾ let us take for $F(S)$ the free $\text{O}(S)$ -module with basis the elements of $\text{O}(S)$; then F does not satisfy condition (E) with respect to $\text{Spec}(\mathbf{Z})$; moreover, let G be the $\mathbf{Z}$ -group defined by $G(S) = \text{GL}_n(R(S))$, where $n \geq 2$ and $R(S)$ is the $\text{O}(S)$ -algebra of the abelian group $\text{O}(S)$; then the $\mathbf{Z}$ -group $\text{Lie}(G/\mathbf{Z})$ is not commutative.

⁽¹⁰²⁾ On the other hand, one can give the following counterexamples. Let $S = \text{Spec}(k)$, k a field of characteristic $p > 0$.

a) Let F be the $\mathcal{O}_S$ -module which to every S -scheme T associates $F(T) = \Gamma(T, \mathcal{O}_T)$ equipped with the structure of $\text{O}(T)$ -module obtained by making the scalars operate via the p -th power, that is, $r \cdot f = r^p f$, for $r \in \text{O}(T)$, $f \in F(T)$. As an S -group functor, F is isomorphic to $\mathbf{G}_{a,S}$. Hence F/S satisfies (E) and $\text{Lie}(F/S)$ is identified with $\text{Lie}(\mathbf{G}_{a,S}/S) \simeq \mathcal{O}_S$. Then, the canonical morphism $F \rightarrow \text{Lie}(F/S)$ is, for every $T \rightarrow S$, the identity map $F(T) \rightarrow \text{O}(T)$: it does respect the structure of abelian group but not the structure of module. Hence F is not *good* (cf. 4.4.1).

b) Let $\alpha_{p,k}$ be the k -group functor which to every S -scheme T associates

$$\alpha_{p,k}(T) = \{x \in \text{O}(T) \mid x^p = 0\},$$

it is represented by $\text{Spec}(k[X]/(X^p))$ hence it is a *very good* S -group; it is also equipped with a structure of $\mathcal{O}_S$ -module, but it is not a *good* $\mathcal{O}_S$ -module, for the canonical morphism $\alpha_{p,k} \rightarrow \text{Lie}(\alpha_{p,k}/k) = \mathbf{G}_{a,k}$ is not bijective.

6.4. — Let G be a group functor over S . One has by definition the following implications

$$(G/S \text{ satisfies (E)}) \iff (G \text{ is good}) \iff (G \text{ is very good}).$$

⁽¹⁰³⁾ It would be interesting to prove or to give counterexamples for the implications in the reverse direction.

⁽¹⁰⁰⁾ **N.D.E.** One has set out the original in more detail in what follows. In particular, the correct definition of the “free abelian monoid on a pointed set” was pointed out to us by O. Gabber.

⁽¹⁰¹⁾ **N.D.E.** One has added what follows.

⁽¹⁰²⁾ **N.D.E.** In what follows, one has set out example a) in more detail and added example b).

⁽¹⁰³⁾ **N.D.E.** And G is very good if it is representable (4.11). For criteria of representability, see for example [MO67]. On the other hand, for the automorphisms of an affine algebraic group (over an algebraically closed field of characteristic 0), let us cite [HM69].

6.5. — ⁽¹⁰⁴⁾Let $\mathbf{Nil}$ be the $\mathbf{Z}$ -group functor defined as follows: for every scheme S , $\mathbf{Nil}(S)$ is the ideal of $\mathcal{O}(S)$ consisting of the nilpotent elements, i.e.

$$\mathbf{Nil}(S) = \{x \in \mathcal{O}(S) \mid \text{there exists } n \in \mathbf{N} \text{ such that } x^n = 0\}.$$

($\mathbf{Nil}$ is *very good* but is not representable). Let $\mathbf{Nil}^2$, $\mathcal{O}_{\text{red}}$ and F be the $\mathbf{Z}$ -group functors which to every scheme S associate, respectively, the ideal $\mathbf{Nil}(S)^2$ and

$$\mathcal{O}_{\text{red}}(S) = \mathcal{O}(S)/\mathbf{Nil}(S), \quad F(S) = \mathcal{O}(S)/\mathbf{Nil}(S)^2.$$

One sees easily that $\text{Lie}(\mathcal{O}_{\text{red}}/\mathbf{Z}) = 0$, hence the $\mathcal{O}_{\mathbf{Z}}$ -module $\mathcal{O}_{\text{red}}$ is not *good* (although it is a *good* $\mathbf{Z}$ -group). If $\mathcal{M}, \mathcal{N}$ are free $\mathbf{Z}$ -modules of finite rank, one has

$$\mathbf{Nil}^2(I_S(\mathcal{M} \oplus \mathcal{N})) \simeq \mathbf{Nil}^2(S) \oplus \mathbf{Nil}(S) \otimes_{\mathbf{Z}} \mathcal{M} \oplus \mathbf{Nil}(S) \otimes_{\mathbf{Z}} \mathcal{N}$$

and hence

$$F(I_S(\mathcal{M} \oplus \mathcal{N})) \simeq F(S) \oplus \mathcal{O}_{\text{red}}(S) \otimes_{\mathbf{Z}} \mathcal{M} \oplus \mathcal{O}_{\text{red}}(S) \otimes_{\mathbf{Z}} \mathcal{N}.$$

One deduces from this, on the one hand, that the $\mathbf{Z}$ -group functor F satisfies condition (E) and, on the other hand, that $\text{Lie}(F/\mathbf{Z}) = \mathcal{O}_{\text{red}}$; as the latter is not a *good* $\mathcal{O}_{\mathbf{Z}}$ -module, this shows that F is an example of a $\mathbf{Z}$ -group which satisfies (E) but which is not *good*.

6.5.1. — Let us also give the counterexamples indicated in 5.3.1–5.3.3. Let S be a scheme, F the *good* $\mathcal{O}_S$ -module $\mathcal{O}_S^{\oplus 2}$ equipped with the natural action of the *good* S -group $G = \text{GL}_{2,S}$, and E the sub- $\mathcal{O}_S$ -module of F consisting of the pairs (x_1, x_2) such that x_2 is nilpotent. Set $N = \text{Norm}_G(E)$. Then $\text{Lie}(N/S) = \text{Lie}(G/S)$ whereas, for every $S' \rightarrow S$, one has

$$\text{Norm}_{\text{Lie}(G/S)}(E)(S') = \left\{ \begin{pmatrix} a & b \\ x & c \end{pmatrix} \mid a, b, c, x \in \mathcal{O}(S'), x \text{ nilpotent} \right\}$$

hence $\text{Lie}(\text{Norm}_G(E)/S) \neq \text{Norm}_{\text{Lie}(G/S)}(E)$.

By considering the semidirect product $G_1 = F \cdot G$, one obtains an analogous counterexample in which E is a sub- $\mathcal{O}_S$ -module of $\text{Lie}(G_1/S)$; moreover, with the notations introduced above, $E = \text{Lie}(K/S)$ where K is the subgroup $\mathcal{O}_S \oplus \mathbf{Nil}^2$ of F (that is, for every $S' \rightarrow S$, $K(S')$ consists of the pairs (x_1, x_2) such that $x_2 \in \mathbf{Nil}(S')^2$).

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⁽¹⁰⁴⁾**N.D.E.** One has added this paragraph.

⁽¹⁰⁵⁾**N.D.E.** Additional references cited in this Exposé.

Exposé III. Infinitesimal extensions

by M. Demazure

⁽⁰⁾In this exposé, one places oneself in the following general situation. One has a scheme S and a coherent nilpotent ideal $\mathcal{I}$ on S . One denotes by S_n the closed subscheme of S defined by the ideal $\mathcal{I}^{n+1}$ ($n \geq 0$). In particular S_0 is defined by $\mathcal{I}$. As $\mathcal{I}$ is nilpotent, S_n equals S for n large enough, and the S_i have the same underlying topological space. A typical example of this situation is the following: S is the spectrum of a local artinian ring A , $\mathcal{I}$ is the ideal defined by the radical of A , hence S_0 is the spectrum of the residue field of A .

In the preceding situation, one is given a certain number of data above S_0 , and one seeks above S data which lift them, that is to say, which give them back by base change from S to S_0 . This is done step by step, by means of the S_n . At each step, one proposes to define the obstructions encountered and to classify, when they exist, the solutions obtained.

The passage from S_n to S_{n+1} may be generalized as follows: one has a scheme S , two ideals $\mathcal{I}$ and $\mathcal{J}$ with $\mathcal{I} \supset \mathcal{J}$, and $\mathcal{I} \cdot \mathcal{J} = 0$ (in the preceding case, S , $\mathcal{I}$ and $\mathcal{J}$ are respectively S_{n+1} , $\mathcal{I}/\mathcal{I}^{n+2}$, $\mathcal{I}^{n+1}/\mathcal{I}^{n+2}$). One denotes by S_0 (resp. $S_{\mathcal{J}}$) the closed subscheme of S defined by $\mathcal{I}$ (resp. $\mathcal{J}$), and one poses a problem of extension from $S_{\mathcal{J}}$ to S .

In SGA 1 III were treated the problems of extension of morphisms of schemes and of extension of schemes. We shall pose here the problems of extension of morphisms of groups, of extension of group structures and of extension of subgroups.

We have gathered in an n° 0 the results of SGA 1 III which will be useful to us, so as to put them in the form most practical for our purpose, and so as to spare the reader from having constantly to refer to SGA 1 III.⁽¹⁾ The n° 1 gathers computations of cohomology of groups, useful in the sequel, and which have nothing to do with the theory of schemes. Numbers 2 and 3 treat respectively of *the extension of morphisms of groups and of the extension of group structures*. In n° 4, we have briefly recalled the proof of a result stated in TDTE IV concerning the extension of subschemes, and applied this result to the problem of extension of subgroups. *For the sequel of the Seminar, only the result of n° 2, concerning the extension of morphisms of groups, will be indispensable.*⁽²⁾

The idea of reducing the problems of infinitesimal extensions to the usual cohomology computations in group extensions was suggested by J. Giraud during the oral exposé (whose computations were distinctly more complicated and less transparent). Unfortunately, it seems that this method applies well only to the first two problems studied, and we have been unable to avoid rather painful computations in the case of extensions of subgroups.

To simplify the language, we shall call Y -functor, resp. Y -scheme, etc., a functor, resp. scheme, etc., equipped with a morphism into the functor Y , thus extending the definitions of Exposé I (which concerned only the case of a representable Y).

0. Reminders from SGA 1 III and various remarks

Let us first state a general definition.

⁽⁰⁾**N.D.E.** Version of 14/10/2024

⁽¹⁾**N.D.E.** The reader may prefer to begin by reading section 1, which is easier, and which can serve as motivation and as a guide for the results obtained in section 0.

⁽²⁾**N.D.E.** However, 3.10 is used in XXIV, 1.13. On the other hand, 4.37 is used in the proof of IX, 3.2 bis and 3.6 bis, but these also follow from results of Exp. X, which do not use 4.37.

Definition 0.1. Let $\mathcal{C}$ be a category, X an object of $\widehat{\mathcal{C}}$, G a $\widehat{\mathcal{C}}$ -group operating on X . One says that X is formally principal homogeneous⁽³⁾ under G if the following equivalent conditions are satisfied:

- (i) for each object S of $\mathcal{C}$, the set $X(S)$ is empty or principal homogeneous under $G(S)$;
- (ii) the morphism of functors $G \times X \longrightarrow X \times X$ defined set-theoretically by $(g, x) \mapsto (gx, x)$ is an isomorphism.

This done, we are going to put the results of SGA 1 III, § 5⁽⁴⁾ in the form which will be the most useful to us. We shall employ the following general notations throughout this number. One has a scheme S and on S two quasi-coherent ideals $\mathcal{I}$ and $\mathcal{J}$ such that

$$\mathcal{I} \supset \mathcal{J} \quad \text{and} \quad \mathcal{I} \cdot \mathcal{J} = 0.$$

One will therefore have in particular $\mathcal{J}^2 = 0$. One will denote by S_0 (resp. $S_{\mathcal{J}}$) the closed subscheme of S defined by the ideal $\mathcal{I}$ (resp. $\mathcal{J}$). For every S -functor X , one will systematically denote by X_0 and $X_{\mathcal{J}}$ the functors obtained by base change from S to S_0 and $S_{\mathcal{J}}$. The same notations for a morphism.

Definition 0.1.1. ⁽⁵⁾Let X be an S -functor. Let us define a functor X^+ above S by the formula

$$\mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_{\mathcal{J}}}(Y_{\mathcal{J}}, X_{\mathcal{J}}) = \mathrm{Hom}_S(Y_{\mathcal{J}}, X)$$

for a variable S -scheme Y . In the notations of Exp. II, 1, one has

$$X^+ \simeq \coprod_{S_{\mathcal{J}}/S} X_{\mathcal{J}}.$$

The identity morphism of $X_{\mathcal{J}}$ defines by construction an S -morphism

$$p_X: X \longrightarrow X^+.$$

⁽⁶⁾Explicitly, for every S -scheme Y , the map

$$p_X(Y): \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_S(Y_{\mathcal{J}}, X)$$

is the map induced by the morphism $Y_{\mathcal{J}} \longrightarrow Y$.

Remark 0.1.2. Let us now remark that if X is an S -functor in groups, $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -functor in groups; then the defining formula of X^+ equips it with a structure of S -functor in groups, and the description of p_X above shows that $p_X: X \longrightarrow X^+$ is a morphism of S -functors in groups.

Remark 0.1.3. On the other hand, for every S -functor in groups Y , one has

$$\mathrm{Hom}_{S\text{-gr.}}(Y, X^+) = \mathrm{Hom}_{S_{\mathcal{J}}\text{-gr.}}(Y_{\mathcal{J}}, X_{\mathcal{J}}).$$

Indeed, let $f \in \mathrm{Hom}_S(Y, X^+)$, corresponding to $f_{\mathcal{J}} \in \mathrm{Hom}_{S_{\mathcal{J}}}(Y_{\mathcal{J}}, X_{\mathcal{J}})$; the condition for $f \in \mathrm{Hom}_{S\text{-gr.}}(Y, X^+)$ is that, for every $T \longrightarrow S$ and $y, y' \in Y(T)$, one has $f(y \cdot y') = f(y) \cdot f(y')$, and this is equivalent to

$$f_{\mathcal{J}}(y_{\mathcal{J}}) \cdot f_{\mathcal{J}}(y'_{\mathcal{J}}) = f_{\mathcal{J}}((y \cdot y')_{\mathcal{J}});$$

as $(y \cdot y')_{\mathcal{J}} = y_{\mathcal{J}} \cdot y'_{\mathcal{J}}$ (since $Y(T) \longrightarrow Y(T_{\mathcal{J}}) = Y_{\mathcal{J}}(T_{\mathcal{J}})$ is a morphism of groups), this is the condition for $f_{\mathcal{J}}$ to be a morphism of groups. Applying this to $Y = X$, one finds again that p_X , which corresponds to $\mathrm{id}_{X_{\mathcal{J}}}$, is a morphism of S -functors in groups.

⁽³⁾**N.D.E.** One also says “pseudo-torsor”, cf. EGA IV₄, 16.5.15. On the other hand, the more general notion of formally homogeneous object (not necessarily principal homogeneous) is defined in Exp. IV, 6.7.1.

⁽⁴⁾**N.D.E.** See also EGA IV₄, 16.5.14–18.

⁽⁵⁾**N.D.E.** One has added the numbering 0.1.1, ..., 0.1.13, so as to bring out the definitions and results which are found there.

⁽⁶⁾**N.D.E.** One has added the following sentence, and spelled out the two remarks which follow.

Let us now return to the general case, but suppose that X is an S -scheme. An S -morphism of a variable S -scheme Y into X^+ being by definition an $S_{\mathcal{J}}$ -morphism $g_{\mathcal{J}}$ of $Y_{\mathcal{J}}$ into $X_{\mathcal{J}}$, one is going to define an X^+ -functor in abelian groups L_X as follows.

Scholium 0.1.4. ⁽⁷⁾If $\pi: Y \rightarrow S$ is a morphism of schemes and $\mathcal{M}$ an $\mathcal{O}_S$ -module, one denotes by $\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_Y$ the inverse image $\pi^*(\mathcal{M})$. If $\mathcal{J}$ is an ideal of $\mathcal{O}_S$, one denotes by $\mathcal{J}\mathcal{O}_Y$ the ideal of $\mathcal{O}_Y$, image of the morphism

$$\pi^*(\mathcal{J}) = \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_Y \rightarrow \mathcal{O}_Y.$$

Let us note that, for every morphism of S -schemes $f: Z \rightarrow Y$, one has an epimorphism of $\mathcal{O}_Z$ -modules

$$(0.1.4) \quad f^*(\mathcal{J}\mathcal{O}_Y) \rightarrow \mathcal{J}\mathcal{O}_Z,$$

as follows from the commutative diagram below:

$$\begin{array}{ccc} \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_Y \otimes_{\mathcal{O}_Y} \mathcal{O}_Z & & \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_Z \\ \downarrow & & \downarrow \\ f^*(\mathcal{J}\mathcal{O}_Y) = (\mathcal{J}\mathcal{O}_Y) \otimes_{\mathcal{O}_Y} \mathcal{O}_Z & \longrightarrow & \mathcal{J}\mathcal{O}_Z. \end{array}$$

Definition 0.1.5. ⁽⁸⁾Let X be an S -scheme. For every X^+ -scheme Y , given by a morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$, one sets

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y),$$

where Ω_{X_0/S_0}^1 denotes the module of relative differentials of X_0 with respect to S_0 (cf. SGA 1, I.1 or EGA IV₄, 16.3), and where one regards $\mathcal{J}\mathcal{O}_Y$ as an $\mathcal{O}_{Y_0}$ -module thanks to the fact that it is annihilated by $\mathcal{I}$.

Then, L_X is an X^+ -functor in abelian groups.⁽⁹⁾ Indeed, for every X^+ -morphism $f: Z \rightarrow Y$, the functor f_0^* and the morphism $f_0^*(\mathcal{J}\mathcal{O}_Y) \rightarrow \mathcal{J}\mathcal{O}_Z$ of (0.1.4) induce a natural morphism of abelian groups $L_X(f)$:

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) &\longrightarrow \mathrm{Hom}_{\mathcal{O}_{Z_0}}(f_0^*g_0^*(\Omega_{X_0/S_0}^1), f_0^*(\mathcal{J}\mathcal{O}_Y)) \\ &\longrightarrow \mathrm{Hom}_{\mathcal{O}_{Z_0}}(f_0^*g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Z). \end{aligned}$$

Finally, let us remark that $L_X(f)$ is described in a simple way locally, as follows. Let us first note that Y and $Y_{\mathcal{J}}$ have the same underlying topological space, and that the same holds for V and $V_{\mathcal{J}}$, if V is an open of Y . Let then $U = \mathrm{Spec}(A)$ be an affine open of X above an affine open $\mathrm{Spec}(\Lambda)$ of S , $V = \mathrm{Spec}(B)$ an affine open of Y such that $g_{\mathcal{J}}(V_{\mathcal{J}}) \subset U$, and $W = \mathrm{Spec}(C)$ an affine open of $f^{-1}(V)$. Let J and I be the ideals of Λ corresponding to $\mathcal{J}$ and $\mathcal{I}$. Then f (resp. $g_{\mathcal{J}}$) induces a morphism of Λ -algebras $\theta: B \rightarrow C$ (resp. $\phi: A \rightarrow B/IB$), and one evidently has $\theta(JB) \subset JC$. On the other hand, $m \in \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathcal{J}\mathcal{O}_Y)$ induces an element D of

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_{V_0}}(g_0^*(\Omega_{U/S}^1), \mathcal{J}\mathcal{O}_V) &= \mathrm{Hom}_{B/IB}(\Omega_{A/\Lambda}^1 \otimes_A B/IB, JB) \\ &= \mathrm{Der}_{\Lambda}(A, JB), \end{aligned}$$

⁽⁷⁾**N.D.E.** One has added this scholium.

⁽⁸⁾**N.D.E.** The introduction of the functor L_X leads to an extension “functorial in Y ” (and in X) of SGA 1, III, Prop. 5.1; see below 0.1.8 and 0.1.9, as well as 0.1.10 and 0.2. Moreover, when X is an S -functor in groups, L_X gives rise, under certain hypotheses, to an S -functor in groups L'_X and to an exact sequence $1 \rightarrow L'_X \rightarrow X \rightarrow X^+$ (cf. 0.4, 0.9 and 0.11), which play an essential role in this exposé (cf. Theorems 2.1, 3.5 and 4.21).

⁽⁹⁾**N.D.E.** One has added what follows.

and the image of $L_X(f)(m)$ in

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_{W_0}}(f_0^* g_0^*(\Omega_{U/S}^1), \mathcal{J}\mathcal{O}_Z) &= \mathrm{Hom}_{C/IC}(\Omega_{A/\Lambda}^1 \otimes_A C/IC, JC) \\ &= \mathrm{Der}_\Lambda(A, JC) \end{aligned}$$

is none other than $\theta \circ D$.

Remark 0.1.6. ⁽¹⁰⁾Let $f: X \rightarrow W$ be an S -morphism. It induces an S -morphism $f^+: X^+ \rightarrow W^+$ defined as follows: if g is an element of $\mathrm{Hom}_S(Y, X^+)$, corresponding to an S -morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X$, then $f^+(g)$ is the element $f \circ g_{\mathcal{J}}$ of $\mathrm{Hom}_S(Y, W^+)$. It is clear that the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{f} & W \\ p_X \downarrow & & \downarrow p_W \\ X^+ & \xrightarrow{f^+} & W^+. \end{array}$$

Reminders 0.1.7. ⁽¹¹⁾In this paragraph, given an S -scheme X , one “recalls” certain functorial properties of the module of differentials $\Omega_{X/S}^1$ and of the first infinitesimal neighborhood of the diagonal, $\Delta_{X/S}^{(1)}$, properties which follow easily from EGA IV₄, §§ 16.1–16.4, but which do not appear there explicitly.

a) Let us begin by recalling the following facts (cf. EGA II, §§ 1.2–1.5). Let $g: Y \rightarrow X$ be a morphism of schemes, $\pi: X' \rightarrow X$ an *affine* X -scheme, $\mathcal{B}$ the quasi-coherent $\mathcal{O}_X$ -algebra $\pi_*(\mathcal{O}_{X'})$; then the Y -scheme $Y \times_X X'$ is affine and corresponds to the quasi-coherent $\mathcal{O}_Y$ -algebra $g^*(\mathcal{B})$, and one has a commutative diagram of bijections:

$$\begin{array}{ccc} \mathrm{Hom}_X(Y, X') & \xrightarrow{\sim} & \mathrm{Hom}_Y(Y, Y \times_X X') \\ \wr \downarrow & & \downarrow \wr \\ \mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(\mathcal{B}, g_*(\mathcal{O}_Y)) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^*(\mathcal{B}), \mathcal{O}_Y). \end{array}$$

Moreover, these bijections are functorial in the pair (X, X') , i.e., if W' is an affine W -scheme, corresponding to the quasi-coherent $\mathcal{O}_W$ -algebra $\mathcal{A}$, if one has a commutative diagram of morphisms of schemes

$$\begin{array}{ccccc} & & X' & \xrightarrow{f'} & W' \\ & \nearrow g' & & & \\ Y & \xrightarrow{g} & X & \xrightarrow{f} & W \end{array}$$

and if one denotes by $\phi: \mathcal{A} \rightarrow f_*(\mathcal{B})$ and $\phi^\sharp: f^*(\mathcal{A}) \rightarrow \mathcal{B}$ (resp. $\theta: \mathcal{B} \rightarrow g_*(\mathcal{O}_Y)$ and $\theta^\sharp: g^*(\mathcal{B}) \rightarrow \mathcal{O}_Y$) the algebra morphisms associated with f' (resp. with a variable X -morphism $g': Y \rightarrow X'$), then one has the following commutative diagram (where Y is

⁽¹⁰⁾**N.D.E.** One has spelled out this remark.

⁽¹¹⁾**N.D.E.** One has added this paragraph (see also [DG70], § I.4, n^{os} 1–2).

viewed as a W -scheme via $f \circ g$):

$$\begin{array}{ccc}
 \mathrm{Hom}_X(Y, X') & \xrightarrow{g' \mapsto f' \circ g'} & \mathrm{Hom}_W(Y, W') \\
 \downarrow \wr & & \searrow \sim \\
 & & \mathrm{Hom}_{\mathcal{O}_W\text{-alg.}}(\mathcal{A}, f_* g_*(\mathcal{O}_Y)) \\
 \mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(\mathcal{B}, g_*(\mathcal{O}_Y)) & \xrightarrow{\theta \mapsto \theta \circ \phi^\#} & \mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(f^*(\mathcal{A}), g_*(\mathcal{O}_Y)) \\
 \downarrow \wr & & \downarrow \wr \\
 \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^*(\mathcal{B}), \mathcal{O}_Y) & \xrightarrow{\theta^\# \mapsto \theta^\# \circ g^*(\phi^\#)} & \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^* f^*(\mathcal{A}), \mathcal{O}_Y).
 \end{array}$$

$\theta \mapsto f_*(\theta) \circ \phi$ (arrow from $\mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(f^*(\mathcal{A}), g_*(\mathcal{O}_Y))$ to $\mathrm{Hom}_{\mathcal{O}_W\text{-alg.}}(\mathcal{A}, f_* g_*(\mathcal{O}_Y))$)

b) Let now X be an S -scheme. Let $\Omega_{X/S}^1$ be the module of differentials of X over S , and $\Delta_{X/S}^{(1)}$ the first infinitesimal neighborhood of the diagonal immersion $\delta_X: X \rightarrow X \times_S X$; it is a subscheme of $X \times_S X$, of which the diagonal $\Delta_{X/S}$ is a closed subscheme. One denotes by pr_X^i ($i = 1, 2$) the two projections of $X \times_S X$, and by π_X the restriction of pr_X^1 to $\Delta_{X/S}^{(1)}$.

On the one hand, every morphism $f: X \rightarrow W$ of S -schemes induces an S -morphism $\Delta^{(1)} f: \Delta_{X/S}^{(1)} \rightarrow \Delta_{W/S}^{(1)}$ such that the following diagram is commutative:

$$\begin{array}{ccccccc}
 X & \xrightarrow{\delta_X} & \Delta_{X/S}^{(1)} & \longrightarrow & X \times_S X & \xrightarrow{\mathrm{pr}_X^i} & X \\
 f \downarrow & & \Delta^{(1)} f \downarrow & & f \times f \downarrow & & \downarrow f \\
 W & \xrightarrow{\delta_W} & \Delta_{W/S}^{(1)} & \longrightarrow & W \times_S W & \xrightarrow{\mathrm{pr}_W^i} & W.
 \end{array}$$

On the other hand, $\Delta_{X/S}^{(1)}$ is, via the projection π_X , an affine X -scheme, spectrum of the augmented quasi-coherent $\mathcal{O}_X$ -algebra

$$\mathcal{P}_{X/S}^1 = \mathcal{O}_X \oplus \Omega_{X/S}^1,$$

where $\Omega_{X/S}^1$ is an ideal of square zero; the augmentation is the morphism of $\mathcal{O}_X$ -algebras $\varepsilon_X: \mathcal{P}_{X/S}^1 \rightarrow \mathcal{O}_X$ which vanishes on $\Omega_{X/S}^1$ and which corresponds to the closed immersion $\delta_X: X \hookrightarrow \Delta_{X/S}^{(1)}$. Then, every morphism of S -schemes $f: X \rightarrow W$ induces a morphism of augmented $\mathcal{O}_X$ -algebras

$$f^*(\mathcal{P}_{W/S}^1) = \mathcal{O}_X \oplus f^*(\Omega_{W/S}^1) \rightarrow \mathcal{P}_{X/S}^1 = \mathcal{O}_X \oplus \Omega_{X/S}^1,$$

i.e. , in an equivalent way, a morphism of $\mathcal{O}_X$ -modules

$$f_{X/W/S}: f^*(\Omega_{W/S}^1) \rightarrow \Omega_{X/S}^1,$$

cf. EGA IV₄ (16.4.3.6) (and (16.4.18.2) for the notation $f_{X/W/S}$).

As $\pi_X: \Delta_{X/S}^{(1)} \rightarrow X$ is affine, then, by a), $\Delta^{(1)} f$ is entirely determined by $f_{X/W/S}$ and, for every X -scheme $g: Y \rightarrow X$, the set

$$\mathrm{Hom}_X(Y, \Delta_{X/S}^{(1)}) \simeq \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(\mathcal{O}_Y \oplus g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$$

is identified with a subset of $\mathrm{Hom}_{\mathcal{O}_Y}(g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$, namely the subset

$$\mathrm{Hom}_{\mathcal{O}_Y}^\square(g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$$

formed of the $\mathcal{O}_Y$ -morphisms $\psi: g^*(\Omega_{X/S}^1) \rightarrow \mathcal{O}_Y$ such that $\text{Im}(\psi)$ is an ideal of $\mathcal{O}_Y$ of square zero.⁽¹²⁾

Consequently, applying a) to the diagram

$$\begin{array}{ccccc} & & \Delta_{X/S}^{(1)} & \xrightarrow{\Delta^{(1)}f} & \Delta_{W/S}^{(1)} \\ & \nearrow g' & & & \\ Y & \xrightarrow{g} & X & \xrightarrow{f} & W \end{array}$$

and taking into account the fact that $\Delta^{(1)}f$ is the restriction to $\Delta_{X/S}^{(1)}$ of $f \times f$, one obtains the following commutative diagram, functorial in the X -scheme $Y \xrightarrow{g} X$:

$$(0.1.7 \text{ (*)}) \quad \begin{array}{ccc} \text{Hom}_X(Y, X \times_S X) & \xrightarrow{(g, g') \mapsto (f \circ g, f \circ g')} & \text{Hom}_W(Y, \Delta_{W/S}^{(1)}) \\ \uparrow & & \uparrow \\ \text{Hom}_X(Y, \Delta_{X/S}^{(1)}) & \xrightarrow{g' \mapsto (\Delta^{(1)}f) \circ g'} & \text{Hom}_W(Y, \Delta_{W/S}^{(1)}) \\ \uparrow \simeq & & \uparrow \simeq \\ \text{Hom}_{\mathcal{O}_Y}^\square(g^*(\Omega_{X/S}^1), \mathcal{O}_Y) & \xrightarrow{\psi \mapsto \psi \circ g^*(f_{X/W/S})} & \text{Hom}_{\mathcal{O}_Y}^\square(g^*f^*(\Omega_{W/S}^1), \mathcal{O}_Y) \end{array}$$

Remark 0.1.7.1. Let us end this paragraph with the following remark, which will be useful later (cf. 0.1.10). If one denotes by $L_X^\square$ the X -functor which, to every X -scheme $g: Y \rightarrow X$, associates $\text{Hom}_{\mathcal{O}_Y}^\square(g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$, and by $L_f^\square: L_X^\square \rightarrow L_W^\square$ the morphism of functors defined above (which, to every $\psi \in L_X^\square(Y)$, associates $\psi \circ g^*(f_{X/W/S})$), what precedes shows that one has a commutative diagram of functors:

$$\begin{array}{ccccc} X \times_X L_X^\square & \xleftarrow{\sim} & \Delta_{X/S}^{(1)} & \hookrightarrow & X \times_S X \\ f \times L_f^\square \downarrow & & \Delta^{(1)}f \downarrow & & \downarrow f \times f \\ W \times_W L_W^\square & \xleftarrow{\sim} & \Delta_{W/S}^{(1)} & \hookrightarrow & W \times_S W. \end{array}$$

Theorem 0.1.8. (SGA 1, III 5.1)⁽¹³⁾ Let Y, X be two S -schemes, $\mathfrak{J}$ a quasi-coherent ideal of $\mathcal{O}_Y$ of square zero, $Y_{\mathfrak{J}}$ the closed subscheme of Y defined by $\mathfrak{J}$, and $g_{\mathfrak{J}}: Y_{\mathfrak{J}} \rightarrow X$ an S -morphism.

a) The set $P(g_{\mathfrak{J}})$ of S -morphisms $g: Y \rightarrow X$ which extend $g_{\mathfrak{J}}$ is either empty, or principal homogeneous under the abelian group

$$\text{Hom}_{\mathcal{O}_{Y_{\mathfrak{J}}}}(g_{\mathfrak{J}}^*(\Omega_{X/S}^1), \mathfrak{J}).$$

b) If $i: Y_0 \hookrightarrow Y_{\mathfrak{J}}$ is the closed immersion defined by a quasi-coherent ideal $\mathfrak{I} \supset \mathfrak{J}$ such that $\mathfrak{I}\mathfrak{J} = 0$, and if $g_0 = g_{\mathfrak{J}} \circ i$, the preceding abelian group is isomorphic to

$$\text{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathfrak{I}).$$

⁽¹²⁾**N.D.E.** In the proof of SGA 1, III 5.1, one must accordingly correct the sentence “Now the algebra homomorphisms $\mathcal{A} \rightarrow \mathcal{O}_Y$ correspond . . .” (the sequel of *loc. cit.* being unchanged).

⁽¹³⁾**N.D.E.** One has reproduced here the statement and the proof of SGA 1, III 5.1, which one needs in order to prove Proposition 0.2 below (see already Corollary 0.1.9 which follows). On the other hand, $\mathfrak{J}$ (resp. $\mathfrak{I}$) is another calligraphy of the letter J (resp. I); returning to the earlier notations ($\mathcal{J} \subset \mathcal{I}$ ideals of $\mathcal{O}_S$ such that $\mathcal{I}\mathcal{J} = 0$), one will apply this to $\mathfrak{J} = \mathcal{J}\mathcal{O}_Y$ (resp. $\mathfrak{I} = \mathcal{I}\mathcal{O}_Y$).

PROOF. (b) follows at once from (a). Indeed, $\mathfrak{J}$, being annihilated by $\mathfrak{J}$, may be regarded as an $\mathcal{O}_{Y_0}$ -module, whence, by adjunction,

$$\mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J}) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(i^*g_3^*(\Omega_{X/S}^1), \mathfrak{J}).$$

To prove (a), one may suppose $P(g_3) \neq \emptyset$, i.e. that there exists an S -morphism $g: Y \rightarrow X$ extending g_3 . Let us denote by j the immersion $Y_3 \hookrightarrow Y$. Then $P(g_3)$ is the set of S -morphisms $g': Y \rightarrow X$ such that $g' \circ j = g_3$. The datum of such a g' is equivalent to the datum of an S -morphism

$$h: Y \rightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ h = g$ and $h_3 = \delta \circ g_3$, where $h_3 = h \circ j$ and δ is the diagonal immersion $X \hookrightarrow X \times_S X$:

$$\begin{array}{ccc} X \times_S X & \xleftarrow{h_3 = \delta \circ g_3} & Y_3 \\ \mathrm{pr}_1 \downarrow & \swarrow h & \downarrow j \\ X & \xleftarrow{g} & Y. \end{array}$$

As h_3 factors through δ and Y is in the first infinitesimal neighborhood of the immersion $j: Y_3 \rightarrow Y$ (since $\mathfrak{J}^2 = 0$), then, by functoriality (cf. EGA IV₄, 16.2.2 (i)), the sought-for h factor, in a unique way, through $\Delta_{X/S}^{(1)}$ (cf. 0.1.7). Let us set

$$Y' = \Delta_{X/S}^{(1)} \times_X Y \quad \text{and} \quad Y'_3 = Y' \times_Y Y_3 = \Delta_{X/S}^{(1)} \times_X Y_3.$$

Then the sought-for h are in bijection with the sections u of $Y' \rightarrow Y$ which extend the section $u_3 = (\delta \circ g_3, \mathrm{id})$ of $Y'_3 \rightarrow Y_3$. On the other hand, Y' (resp. Y'_3) is a scheme *affine over* Y (resp. Y_3), corresponding to the quasi-coherent algebra

$$\mathcal{A} = \mathcal{O}_Y \oplus g^*(\Omega_{X/S}^1), \quad \text{resp.} \quad \mathcal{A}_3 = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_3} = \mathcal{O}_{Y_3} \oplus g_3^*(\Omega_{X/S}^1).$$

Let us denote by $\varepsilon: \mathcal{A} \rightarrow \mathcal{O}_Y$ the canonical augmentation of $\mathcal{A}$ (i.e. the morphism of $\mathcal{O}_Y$ -algebras $\mathcal{A} \rightarrow \mathcal{O}_Y$ which vanishes on $g^*(\Omega_{X/S}^1)$), and let us define likewise $\varepsilon_3: \mathcal{A}_3 \rightarrow \mathcal{O}_{Y_3}$. Then

$$\Gamma(Y'/Y) \simeq \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(\mathcal{A}, \mathcal{O}_Y), \quad \Gamma(Y'_3/Y_3) \simeq \mathrm{Hom}_{\mathcal{O}_{Y_3}\text{-alg.}}(\mathcal{A}_3, \mathcal{O}_{Y_3}),$$

and, via these isomorphisms, the section $u = (\delta \circ g, \mathrm{id})$ (resp. u_3) corresponds to ε (resp. ε_3).

Consequently, $P(g_3)$ is in bijection with the set of morphisms of algebras $\mathcal{A} \rightarrow \mathcal{O}_Y$ which reduce according to ε_3 , and via this bijection, g corresponds to ε .

Set $\mathcal{M} = g^*(\Omega_{X/S}^1)$. Then $\mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(\mathcal{A}, \mathcal{O}_Y)$ is identified with the set of $\mathcal{O}_Y$ -morphisms $\psi: \mathcal{M} \rightarrow \mathcal{O}_Y$ such that $\mathrm{Im}(\psi)$ is an ideal of square zero, and one is interested in those which induce the zero morphism $\mathcal{M} \rightarrow \mathcal{O}_{Y_3} = \mathcal{O}_Y/\mathfrak{J}$, i.e. which map $\mathcal{M}$ into $\mathfrak{J}$. Conversely, since $\mathfrak{J}^2 = 0$, every $\mathcal{O}_Y$ -morphism $\phi: \mathcal{M} \rightarrow \mathfrak{J}$ comes from a (unique) morphism of algebras $\mathcal{A} \rightarrow \mathcal{O}_Y$, reducing according to ε_3 . Finally, one has

$$\mathrm{Hom}_{\mathcal{O}_Y}(g^*(\Omega_{X/S}^1), \mathfrak{J}) = \mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J})$$

since $\mathfrak{J}^2 = 0$ (cf. the proof of (b) already seen). One therefore obtains a bijection

$$P(g_3) \simeq \mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J})$$

by which g corresponds to the zero morphism.

For every $m \in \mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J})$, let us denote by $m \cdot g$ the element of $P(g_3)$ associated with g and m by the preceding bijection. One has already seen that $0 \cdot g = g$; it remains to see that

$$(0.1.8 \text{ (*)}) \quad m' \cdot (m \cdot g) = (m + m') \cdot g.$$

This is verified locally.⁽¹⁴⁾ Indeed, the two preceding morphisms $Y \rightarrow X$ induce the same continuous map as g between the underlying topological spaces; it therefore suffices to check that for every affine open $U = \operatorname{Spec}(A)$ of X lying over an affine open $\operatorname{Spec}(\Lambda)$ of S , and every affine open $V = \operatorname{Spec}(B)$ of $g^{-1}(U)$, they induce the same morphism of Λ -algebras $A \rightarrow B$.

Let $J = \Gamma(V, \mathfrak{J})$ and let ϕ, ψ and η be the morphisms $A \rightarrow B$ induced by $g, m \cdot g$ and $m' \cdot (m \cdot g)$ respectively; they coincide modulo J . One can write in a unique way $\psi = \phi + D$ (resp. $\eta = \psi + D'$), where D (resp. D') is an element of

$$\operatorname{Dér}_\phi(A, J) = \{\delta \in \operatorname{Hom}_\Lambda(A, J) \mid \delta(ab) = \phi(a)\delta(b) + \phi(b)\delta(a)\}$$

(resp. $\operatorname{Dér}_\psi(A, J)$). But $\operatorname{Dér}_\phi(A, J) = \operatorname{Dér}_\psi(A, J)$ since $J^2 = 0$, and both are identified with

$$\operatorname{Hom}_{B/J}(\Omega_{A/\Lambda}^1 \otimes_A B/J, J),$$

and via this identification D corresponds to m and D' to m' . Then $\eta = \phi + D + D'$ and $D + D'$ corresponds to $m + m'$, whence the equality (*). $\square$

Corollary 0.1.9. ⁽¹⁵⁾ *Let X be an S -scheme; let us take up again the notations of 0.1.5. Then X is equipped with a (left) operation of the abelian X^+ -group L_X , which makes X into a formally principal homogeneous object under L_X over X^+ , i.e. one has an isomorphism of X^+ -functors:*

$$L_X \times_{X^+} X \xrightarrow{\sim} X \times_{X^+} X$$

(defined set-theoretically by $(m, x) \mapsto (x, m \cdot x)$).

PROOF. Let i_0 be the immersion $X_0 \hookrightarrow X$. Let us first note that, since $X_0 = X \times_S S_0$, one has $i_0^*(\Omega_{X/S}^1) \simeq \Omega_{X_0/S_0}^1$ (cf. EGA IV, 16.4.5).

Let Y be an X^+ -scheme, given by an S -morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X$, and let $g_0: Y_0 \rightarrow X_0$ be the morphism obtained by base change. By 0.1.8, if $\operatorname{Hom}_{X^+}(Y, X)$ is non-empty, it is a principal homogeneous set under the group

$$\operatorname{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathcal{J}\mathcal{O}_Y),$$

which is identified with $\operatorname{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) = L_X(Y)$. One therefore has a bijection

$$L_X(Y) \times \operatorname{Hom}_{X^+}(Y, X) \xrightarrow{\sim} \operatorname{Hom}_{X^+}(Y, X \times_{X^+} X)$$

given by $(m, g) \mapsto (g, m \cdot g)$. Let us show that this is “functorial in Y ”.

Let $f: Z \rightarrow Y$ be a morphism of S -schemes. The point is to show that the diagram below is commutative:

$$\begin{array}{ccc} L_X(Y) \times \operatorname{Hom}_{X^+}(Y, X) & \xrightarrow{\sim} & \operatorname{Hom}_{X^+}(Y, X \times_{X^+} X) \\ \downarrow L_X(f) \times f & & \downarrow f \times f \\ L_X(Z) \times \operatorname{Hom}_{X^+}(Z, X) & \xrightarrow{\sim} & \operatorname{Hom}_{X^+}(Z, X \times_{X^+} X). \end{array}$$

If $\operatorname{Hom}_{X^+}(Y, X) = \emptyset$, there is nothing to show. It therefore suffices to see that, for every S -morphism $g: Y \rightarrow X$ extending $g_{\mathcal{J}}$ and every $m \in L_X(Y)$, one has:

$$(0.1.9 \text{ (*)}) \quad (m \cdot g) \circ f = L_X(f)(m) \cdot (g \circ f).$$

These two S -morphisms $Z \rightarrow X$ coincide on $Z_{\mathcal{J}}$ with $g_{\mathcal{J}} \circ f_{\mathcal{J}}$; in particular, they induce the same continuous map as $g \circ f$ between the underlying topological spaces. Consequently, it suffices to see that, if $z \in Z$, $y = f(z)$, $x = g(y)$, and if A, B, C denote respectively the local rings $\mathcal{O}_{X,x}, \mathcal{O}_{Y,y}, \mathcal{O}_{Z,z}$, then the morphisms $A \rightarrow C$ induced by $(m \cdot g) \circ f$ and

⁽¹⁴⁾**N.D.E.** What precedes is reproduced almost word for word from SGA 1, III 5.1; what follows has been spelled out.

⁽¹⁵⁾**N.D.E.** One has added this corollary of SGA 1, III 5.1, which proves point (i) of proposition 0.2 further on.

$L_X(f)(m) \cdot (g \circ f)$ coincide. Let us denote by s the image of x in S , $\Lambda = \mathcal{O}_{S,s}$, and let J and I be the ideals of Λ corresponding to $\mathcal{J}$ and $\mathcal{I}$, and let $\phi, \psi: A \rightarrow B$ and $\theta: B \rightarrow C$ be the morphisms of Λ -algebras induced by $g, m \cdot g$ and f . Then m induces an element D of

$$\mathrm{Hom}_{B/IB}(\Omega_{A/\Lambda}^1 \otimes_A B/IB, JB) = \mathrm{Dér}_\Lambda(A, JB)$$

and one has $\psi = \phi + D$; hence $(m \cdot g) \circ f$ corresponds to $\theta \circ \psi = \theta \circ \phi + \theta \circ D$. Now, one has seen in 0.1.5 that $\theta \circ D$ is the image of $L_X(f)(m)$ in

$$\mathrm{Hom}_{C/IC}(\Omega_{A/\Lambda}^1 \otimes_A C/IC, JC) = \mathrm{Dér}_\Lambda(A, JC);$$

consequently, $\theta \circ \phi + \theta \circ D$ is the image of $L_X(f)(m) \cdot (g \circ f)$. This proves the equality (0.1.9 (*)). $\square$

Corollary 0.1.10. ⁽¹⁶⁾a) L_X depends functorially on X : for every S -morphism $f: X \rightarrow W$, there exists an S -morphism $L_f: L_X \rightarrow L_W$ which is a morphism of abelian groups “over f^+ ”, i.e. the diagram

$$\begin{array}{ccc} L_X & \xrightarrow{L_f} & L_W \\ \downarrow & & \downarrow \\ X^+ & \xrightarrow{f^+} & W^+ \end{array}$$

is commutative and, for every $Y \rightarrow X^+$,

$$L_f(Y): \mathrm{Hom}_{X^+}(Y, L_X) \rightarrow \mathrm{Hom}_{W^+}(Y, L_W)$$

(where Y is over W^+ via f^+) is a morphism of abelian groups.

b) Moreover, the following diagram is commutative:

$$\begin{array}{ccc} L_X \times_{X^+} X & \xrightarrow{\sim} & X \times_{X^+} X \\ \downarrow L_f \times f & & \downarrow f \times f \\ L_W \times_{W^+} W & \xrightarrow{\sim} & W \times_{W^+} W. \end{array}$$

PROOF. a) L_f is induced by the morphism of $\mathcal{O}_{X_0}$ -modules $f_{X_0/W_0/S_0}: f_0^*(\Omega_{W_0/S_0}^1) \rightarrow \Omega_{X_0/S_0}^1$ (cf. 0.1.7 b)): for every X^+ -scheme Y , given by an S -morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X$, one has a commutative diagram, functorial in Y :

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) & \xrightarrow{L_f(Y)} & \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*f_0^*(\Omega_{W_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) \\ \downarrow & & \downarrow \\ \{g_{\mathcal{J}}\} & \xrightarrow{\quad} & \{f_{\mathcal{J}} \circ g_{\mathcal{J}}\} \end{array}$$

where $L_f(Y)$ is the map $\psi \mapsto \psi \circ g_0^*(f_{X_0/W_0/S_0})$, which is indeed a morphism of abelian groups. ⁽¹⁷⁾

Let us prove (b). Let Y be an X^+ -scheme; if $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$ there is nothing to show. Let therefore $g \in \mathrm{Hom}_{X^+}(Y, X)$; one must see that for every $m \in L_X(Y)$, one has:

$$(0.1.10 \text{ (*)}) \quad f \circ (m \cdot g) = L_f(Y)(m) \cdot (f \circ g).$$

Now, g being fixed, $\mathrm{Hom}_X(Y, X \times_{X^+} X)$ is a subset of $\mathrm{Hom}_X(Y, \Delta_{X/S}^{(1)})$ and

$$L_X(Y) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) = \mathrm{Hom}_{\mathcal{O}_Y}(g^*(\Omega_{X/S}^1), \mathcal{J}\mathcal{O}_Y)$$

⁽¹⁶⁾**N.D.E.** One has spelled out the “functoriality in X ” of L_X , in particular point b) below.

⁽¹⁷⁾**N.D.E.** One will note that L_f depends only on $f_0: X_0 \rightarrow W_0$.

a subset of $L_X^\square(Y)$ (cf. 0.1.7); finally $L_f(Y)$ is the restriction to $L_X(Y)$ of the map $L_f^\square(Y)$. Moreover, the bijection

$$L_X(Y) \xrightarrow{\sim} \text{Hom}_X(Y, X \times_{X^+} X), \quad m \mapsto (g, m \cdot g)$$

is (the inverse of) the restriction to $L_X(Y) \subset L_X^\square(Y)$ of the bijection $\text{Hom}_X(Y, \Delta_{X/S}^{(1)}) \xrightarrow{\sim} \{g\} \times L_X^\square(Y)$ considered in 0.1.7.1. Consequently, the equality (0.1.10 (*)) follows from (0.1.7 (*)); indeed, if one denotes by g' the X -morphism $Y \rightarrow \Delta_{X/S}^{(1)}$ defined by $(g, m \cdot g)$, then the element of $L_W(Y)$ corresponding to $(f \circ g, f \circ g')$ is $L_f^\square(Y)(m) = L_f(Y)(m)$, i.e. one indeed has

$$L_f(m) \cdot (f \circ g) = f \circ (m \cdot g).$$

□

Lemma 0.1.11. *Let X, X' be two S -schemes. One has a commutative diagram:*

$$\begin{array}{ccc} L_X \times_S L_{X'} & \xrightarrow{\sim} & L_{X \times_S X'} \\ \downarrow & & \downarrow \\ X^+ \times_S X'^+ & \xrightarrow{\sim} & (X \times_S X')^+. \end{array}$$

PROOF. ⁽¹⁸⁾First, for every S -scheme Y , $\text{Hom}_S(Y, X^+ \times_S X'^+)$ equals $\text{Hom}_S(Y, X^+) \times \text{Hom}_S(Y, X'^+)$ and the latter is isomorphic to

$$\text{Hom}_S(Y_{\mathcal{J}}, X) \times \text{Hom}_S(Y_{\mathcal{J}}, X') = \text{Hom}_S(Y, (X \times_S X')^+);$$

this proves that $X^+ \times_S X'^+ \simeq (X \times_S X')^+$.

Next, let Y be a scheme over $X^+ \times_S X'^+$ via a morphism $h: Y_{\mathcal{J}} \rightarrow X \times_S X'$; set $f = p \circ h$ and $g = q \circ h$, where one has denoted by p, q the projections of $X \times_S X'$ to X and X' . Since

$$\Omega_{(X_0 \times_{S_0} X'_0)/S_0}^1 \simeq p_0^*(\Omega_{X_0/S_0}^1) \oplus q_0^*(\Omega_{X'_0/S_0}^1)$$

(cf. EGA IV₄, 16.4.23), one obtains a natural isomorphism:

$$\begin{aligned} & \text{Hom}_{\mathcal{O}_{Y_0}}(f_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) \times \text{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X'_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) \\ & \simeq \text{Hom}_{\mathcal{O}_{Y_0}}(h_0^*(\Omega_{(X_0 \times_{S_0} X'_0)/S_0}^1), \mathcal{J}\mathcal{O}_Y), \end{aligned}$$

i.e. $L_X(Y) \times L_{X'}(Y) \simeq L_{X \times_S X'}(Y)$. □

Remark 0.1.12. ⁽¹⁹⁾Let $\mathcal{C}$ be a category stable under fibered products, S an object of $\mathcal{C}$, T_1, T_2 two objects over S and, for $i = 1, 2$, L_i and X_i two objects over T_i :

$$\begin{array}{ccccc} L_1 & & X_1 & & L_2 & & X_2 \\ & \searrow & \swarrow & & \searrow & \swarrow & \\ & T_1 & & & T_2 & & \\ & & \searrow & & \swarrow & & \\ & & S & & & & \end{array}$$

Then one has a natural isomorphism:

$$(L_1 \times_{T_1} X_1) \times_S (L_2 \times_{T_2} X_2) \simeq (L_1 \times_S L_2) \times_{T_1 \times_S T_2} (X_1 \times_S X_2).$$

Consequently, one deduces from the preceding lemma the:

⁽¹⁸⁾**N.D.E.** One has spelled out the proof.

⁽¹⁹⁾**N.D.E.** One has added this remark, as well as the corollary which follows.

Corollary 0.1.13. *Let X_1, X_2 be two S -schemes. One has a commutative diagram of isomorphisms:*

$$\begin{array}{ccc} L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & & \\ \downarrow (0.1.11) \simeq & \searrow \simeq & \\ (L_{X_1} \times_S L_{X_2}) \times_{(X_1^+ \times_S X_2^+)} (X_1 \times_S X_2) & \xrightarrow[(0.1.12)]{\simeq} & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2). \end{array}$$

We can now state:

Proposition 0.2. *For every S -scheme X , one can define a (left) operation of the abelian X^+ -group L_X on the X^+ -object X , such that:*

- (i) *this operation makes X into a formally principal homogeneous object under L_X over X^+ , i.e. the morphism*

$$L_X \times_{X^+} X \longrightarrow X \times_{X^+} X$$

is an isomorphism of X^+ -functors;

- (ii) *this operation is functorial in the S -scheme X , i.e. for every S -morphism $f: X \longrightarrow W$, the following diagram is commutative:*

$$\begin{array}{ccc} L_X \times_{X^+} X & \longrightarrow & X \\ L_f \times f \downarrow & & \downarrow f \\ L_W \times_{W^+} W & \longrightarrow & W; \end{array}$$

- (iii) *this operation “commutes with fibered product”, i.e. for all S -schemes X_1 and X_2 , the following diagram is commutative:*

$$\begin{array}{ccc} L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ \downarrow \simeq & \searrow \simeq & \uparrow \\ (L_{X_1} \times_S L_{X_2}) \times_{(X_1^+ \times_S X_2^+)} (X_1 \times_S X_2) & \xrightarrow{\simeq} & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2). \end{array}$$

PROOF. ⁽²⁰⁾(i) and (ii) follow respectively from corollaries 0.1.9 and 0.1.10. To prove (iii), set $P(X) = L_X \times_{X^+} X$, for every S -scheme X . Then, by (ii) applied to the projections $p_i: X_1 \times_S X_2 \longrightarrow X_i$, one obtains commutative squares

$$\begin{array}{ccc} P(X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ L_{p_i} \times p_i \downarrow & & \downarrow p_i \\ P(X_i) & \longrightarrow & X_i \end{array}$$

for $i = 1, 2$, and hence a commutative square:

$$\begin{array}{ccc} P(X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ \downarrow & & \\ P(X_1) \times_S P(X_2) & \longrightarrow & X_1 \times_S X_2. \end{array}$$

Combining this with corollary 0.1.13, one obtains that the vertical arrow is an isomorphism, and that one has the commutative diagram announced in (iii). $\square$

⁽²⁰⁾**N.D.E.** One has modified and spelled out the original, taking into account the additions made previously; these incorporate, spelling it out, the content of page 89 of the original.

Remark 0.3. Suppose the X^+ -scheme Y *flat* over S (cf. SGA 1, IV). One can then write

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}).$$

Remark 0.4. Let us denote by $\pi_0: X_0 \rightarrow S_0$ the structural morphism and suppose that there exists an $\mathcal{O}_{S_0}$ -module ω_{X_0/S_0}^1 such that $\Omega_{X_0/S_0}^1 \simeq \pi_0^*(\omega_{X_0/S_0}^1)$ (the case will present itself in particular when X_0 is an S_0 -group, cf. II, 4.11). If one defines a functor L'_X over S by the formula

$$(0.4.1) \quad \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{J}\mathcal{O}_Y),$$

one then has $\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X)$ for every X^+ -scheme Y , i.e.

$$L_X = L'_X \times_S X^+.$$

⁽²¹⁾Then, since $L_X \times_{X^+} X = L'_X \times_S X$, the operation of L_X on X induces an operation of L'_X on X , and this operation respects the morphism $p_X: X \rightarrow X^+$; indeed, if Y is an S -scheme, $h: Y \rightarrow X$ an S -morphism and m an element of $L'_X(Y)$, then h and $m \cdot h$ have the same restriction to $Y_{\mathcal{J}}$, i.e. $p_X(m \cdot h) = p_X(h)$.

Remark 0.5. Let us keep the hypotheses and notations of 0.4 and suppose moreover that Y is an S -scheme *flat* over S . One then has

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{S_0}(Y_0, L_{0X}),$$

where the S_0 -functor in abelian groups L_{0X} is defined by the following identity (with respect to the variable S_0 -scheme T):

$$(0.5.1) \quad \mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T).$$

In the notations of II, 1, one has therefore shown that the functors L'_X and $\prod_{S_0/S} L_{0X}$ have the same restriction to the full subcategory of $(\mathbf{Sch})_S$ whose objects are the S -schemes Y *flat* over S .

Remark 0.6. Let us keep the hypotheses and notations of 0.5⁽²²⁾ and suppose moreover that there exists a section ε_0 of $\pi_0: X_0 \rightarrow S_0$; one then has $\omega_{X_0/S_0}^1 \simeq \varepsilon_0^*(\Omega_{X_0/S_0}^1)$. First, one has (independently of the preceding hypothesis):

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma(T, \mathcal{H}\mathrm{om}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T)).$$

Suppose now that ω_{X_0/S_0}^1 admits a finite presentation (cf. EGA 0_I, 5.2.5), which will in particular be the case if X_0 is *locally of finite presentation* over S_0 (cf. EGA IV₄, 16.4.22). Then, if T is *flat* over S_0 , it follows from EGA 0_I, 6.7.6 that

$$\mathcal{H}\mathrm{om}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T) \simeq \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T,$$

whence

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma(T, \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T).$$

Introducing the notation $\mathbf{W}(\)$ of I, 4.6.1, one has therefore proved that for every S_0 -scheme T *flat* over S_0 , one has

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{S_0}(T, \mathbf{W}(\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}))).$$

In summary, if ω_{X_0/S_0}^1 *admits a finite presentation*, and if one restricts oneself to the category of S -schemes *flat* over S , one has

$$(0.6.1) \quad L'_X = \prod_{S_0/S} \mathbf{W}(\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})),$$

and $\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})$ is a *quasi-coherent* $\mathcal{O}_{S_0}$ -module, by EGA I, 9.1.1.

⁽²¹⁾**N.D.E.** One has spelled out what follows.

⁽²²⁾**N.D.E.** One has added the hypothesis which follows, implicit in the original.

Let us remark finally that if ω_{X_0/S_0}^1 is moreover locally free (of finite rank), for example if X_0 is *smooth* over S_0 (in which case it is automatically locally of finite presentation over S_0), one has

$$(0.6.2) \quad \mathrm{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \simeq \underline{\mathrm{Lie}}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J},$$

where one denotes by abuse of language (X_0 not necessarily being an S_0 -group) $\underline{\mathrm{Lie}}(X_0/S_0)$ the dual of the $\mathcal{O}_{S_0}$ -module ω_{X_0/S_0}^1 .⁽²³⁾

Proposition 0.2 (and its proof) has two important corollaries.⁽²⁴⁾

Corollary 0.7. *Let X be an S -scheme.*

- a) *Every S -endomorphism of X inducing the identity on $X_{\mathcal{J}}$ is an automorphism.*
- b) *One has an exact sequence of groups:*

$$0 \longrightarrow \mathrm{Hom}_{\mathcal{O}_{X_0}}(\Omega_{X_0/S_0}^1, \mathcal{J}\mathcal{O}_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathcal{J}}}(X_{\mathcal{J}}).$$

- c) *Moreover, if one lets $\mathrm{Aut}_S(X)$ operate on the first group by transport of structure, one has, for all $u \in \mathrm{Aut}_S(X)$ and $m \in \mathrm{Hom}_{\mathcal{O}_{X_0}}(\Omega_{X_0/S_0}^1, \mathcal{J}\mathcal{O}_X)$:*

$$i(um) = u i(m) u^{-1}.$$

PROOF. By 0.2 (i), $\mathrm{Hom}_{X^+}(X, X)$ is a principal homogeneous set under $\mathrm{Hom}_{X^+}(X, L_X)$, for it is certainly non-empty; indeed, it contains a marked point: the identity automorphism of X .⁽²⁵⁾ Consequently, the map $m \mapsto m \cdot \mathrm{id}_X$ induces a *bijection*

$$\mathrm{Hom}_{\mathcal{O}_{X_0}}(\Omega_{X_0/S_0}^1, \mathcal{J}\mathcal{O}_X) = L_X(X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, X).$$

Let $m \in L_X(X)$ and let $f = m' \cdot \mathrm{id}_X$ be an element of $\mathrm{Hom}_{X^+}(X, X)$. Applying 0.2 (ii) to f , one obtains that:

$$f \circ (m \cdot \mathrm{id}_X) = L_f(X)(m) \cdot f = L_f(X)(m) \cdot (m' \cdot \mathrm{id}_X).$$

On the other hand, as f is an X^+ -endomorphism of X , one has $f_{\mathcal{J}} = \mathrm{id}_{X_{\mathcal{J}}}$ and hence $f_0 = \mathrm{id}_{X_0}$; as L_f depends only on f_0 (cf. N.D.E. (17) in 0.1.10), one therefore has $L_f(X)(m) = m$. Consequently, the equality above is rewritten:

$$(m' \cdot \mathrm{id}_X) \circ (m \cdot \mathrm{id}_X) = m \cdot (m' \cdot \mathrm{id}_X) = (m + m') \cdot \mathrm{id}_X.$$

This shows that the bijection $m \mapsto m \cdot \mathrm{id}_X$ transforms the group law of $\mathrm{Hom}_{X^+}(X, L_X)$ into the composition law of the X^+ -endomorphisms of X .

It follows first that every element of $\mathrm{Hom}_{X^+}(X, X)$ is invertible, which is the first assertion of the statement, and then that one has an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathcal{J}}}(X_{\mathcal{J}}),$$

which is the second.

Let us remark now that the morphism i defined above is functorial in X for isomorphisms, for it is defined in structural terms from the operation of L_X on X over X^+ , itself functorial in X by assertion (ii) of proposition 0.2.⁽²⁶⁾ Therefore every automorphism u of X over S induces by transport of structure isomorphisms

$$h: \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, L_X)$$

⁽²³⁾**N.D.E.** This is justified by II, 3.3 and 4.11. Indeed, the functor $L_{X_0/S_0}^{\varepsilon_0}$, “tangent space to X_0 over S_0 at the point ε_0 ”, is represented by the vector fibration $\mathbf{V}(\varepsilon_0^*(\Omega_{X_0/S_0}^1)) = \mathbf{V}(\omega_{X_0/S_0}^1)$ over S_0 , whose sheaf of sections is the dual module $\mathrm{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{O}_{S_0})$; the latter is $\underline{\mathrm{Lie}}(X_0/S_0)$ when X_0 is an S_0 -group and ε_0 the unit section.

⁽²⁴⁾**N.D.E.** One has added to 0.7 the corollary 0.7.bis, which was used implicitly in the proof of 0.8. Let us point out here that the numbers 0.8 to 0.12, as well as 0.17, which play an important technical role in the sequel of this exposé, are consequences of 0.7 and 0.7.bis.

⁽²⁵⁾**N.D.E.** One has set out what follows in detail.

⁽²⁶⁾**N.D.E.** One has set out what follows in detail.

and $f: \text{Aut}_S(X) \xrightarrow{\sim} \text{Aut}_S(X)$ such that the following diagram is commutative:

$$\begin{array}{ccc} \text{Hom}_{X^+}(X, L_X) & \xrightarrow{i} & \text{Aut}_S(X) \\ h \downarrow & & \downarrow f \\ \text{Hom}_{X^+}(X, L_X) & \xrightarrow{i} & \text{Aut}_S(X) \end{array}$$

i.e. such that $f \circ i = i \circ h$. On the other hand, f is given by the commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{a} & X \\ u \downarrow & & \downarrow u \\ X & \xrightarrow{f(a)} & X, \end{array}$$

that is, $f(a) = u \circ a \circ u^{-1}$, for every $a \in \text{Aut}_S(X)$. Writing $i(h(m)) = f(i(m))$, one finds the desired formula. $\square$

Corollary 0.7.bis. *Let X be an S -scheme such that $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -monoid. Then L_X is equipped with a structure of S -monoid, one has a split exact sequence of S -monoids:*

$$1 \longrightarrow L'_X \xrightarrow{i} L_X \xrightleftharpoons[p]{p} X^+ \longrightarrow 1$$

and the monoid law induced on L'_X coincides with its structure of abelian group. In particular, if $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -group, then L_X is an S -group and is the semidirect product of X^+ and of L'_X .

PROOF. Indeed, as $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -monoid, then $X^+ = \prod_{S_{\mathcal{J}}/S} X_{\mathcal{J}}$ is an S -monoid (indeed, one has $X^+(Y) = X_{\mathcal{J}}(Y_{\mathcal{J}})$ for every $Y \rightarrow S$). For every S -scheme Y , let us denote by $\tilde{Y}_{\mathcal{J}}$ the affine $Y_{\mathcal{J}}$ -scheme corresponding to the quasi-coherent $\mathcal{O}_{Y_{\mathcal{J}}}$ -algebra $\mathcal{O}_{Y_{\mathcal{J}}} \oplus \mathcal{J}\mathcal{O}_Y$ (i.e. the graded algebra associated with the filtration $\mathcal{O}_Y \supset \mathcal{J}\mathcal{O}_Y$). Then $L_X(Y)$ is identified with $X_{\mathcal{J}}(\tilde{Y}_{\mathcal{J}})$ and $L'_X(Y)$ with the kernel of the morphism $p: X_{\mathcal{J}}(\tilde{Y}_{\mathcal{J}}) \rightarrow X_{\mathcal{J}}(Y_{\mathcal{J}})$ induced by the “zero section” $Y_{\mathcal{J}} \rightarrow \tilde{Y}_{\mathcal{J}}$ (i.e. by the morphism of $\mathcal{O}_{Y_{\mathcal{J}}}$ -algebras $\mathcal{O}_{\tilde{Y}_{\mathcal{J}}} \rightarrow \mathcal{O}_{Y_{\mathcal{J}}}$ vanishing on the ideal $\mathcal{J}\mathcal{O}_Y$). One therefore has, for every $Y \rightarrow S$, a split exact sequence of monoids, functorial in Y :

$$1 \longrightarrow L'_X(Y) \xrightarrow{i} L_X(Y) \xrightleftharpoons[p]{p} X^+(Y) \longrightarrow 1.$$

It remains to see that the monoid law induced on L'_X coincides with its structure of abelian group. Let us denote by μ the monoid law of L_X and by e its unit section; one must show that for all $m, m' \in L'_X(Y)$, one has

$$\mu(m \cdot e, m' \cdot e) = (m + m') \cdot e.$$

This can be seen in one or the other of the following ways. On the one hand, one can take up again the proof of the equality (0.1.10 (*)) upon replacing the morphism $f: X \rightarrow W$ which figures there by the morphism $\mu: L_X \times_S L_X \rightarrow L_X$. Identifying $X^+(Y) = X_{\mathcal{J}}(Y_{\mathcal{J}})$ with its image by s in $L_X(Y) = X_{\mathcal{J}}(\tilde{Y}_{\mathcal{J}})$ one obtains that, for all $g, g' \in X_{\mathcal{J}}(Y_{\mathcal{J}})$ and $m, m' \in L'_X(Y)$, one has

$$(\star) \quad \mu(m \cdot g, m' \cdot g') = L_{\mu}^{(g, g')}(m, m') \cdot \mu(g, g'),$$

where $L_{\mu}^{(g, g')}$ denotes the derived morphism of μ at the point (g, g') (i.e. $\tilde{Y}_{\mathcal{J}}$ lies over $L_X \times_S L_X$ via (g, g')). In particular, one has $\mu(m \cdot e, m' \cdot e) = L_{\mu}^{(e, e)}(m, m') \cdot e$; but $L_{\mu}^{(e, e)}(m, m') = L_{\ell_e}(m') + L_{r_e}(m)$, where ℓ_e (resp. r_e) denotes the left translation (resp. the right translation) by e , which is the identity map of $X_{\mathcal{J}}$, whence $L_{\mu}^{(e, e)}(m, m') = m + m'$.

Or else, one can proceed as follows (cf. the proof of [DG70], § II.4, Th. 3.5). By lemma 0.1.11, the formation of X^+ and of L_X “commutes with the product” and the same therefore holds for L'_X ; it follows that the morphism $\mu': L'_X \times L'_X \rightarrow L'_X$, induced by μ , is

a homomorphism for the structure of abelian groups. One then deduces from lemma 3.10 of Exp. II that μ' coincides with the abelian group law. $\square$

0.8. — ⁽²⁷⁾ Suppose now that X is an S -scheme such that $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -group. Suppose that there exists an S -morphism

$$P: X \times_S X \longrightarrow X$$

such that the morphism obtained by base change

$$P_{\mathcal{J}}: X_{\mathcal{J}} \times_{S_{\mathcal{J}}} X_{\mathcal{J}} \longrightarrow X_{\mathcal{J}}$$

is the group law of $X_{\mathcal{J}}$. (An important particular case of the preceding situation will be the case where X is an S -group and where one takes for P its group law.) One deduces from this a morphism

$$L_P: L_X \times_S L_X \simeq L_{X \times_S X} \longrightarrow L_X$$

which, in fact, does not depend on P , for it is computed by means of the group law $P_{\mathcal{J}}$ of $X_{\mathcal{J}}$ as we are going to see now.⁽²⁸⁾ Indeed, by (ii) and (iii) of 0.2, for every $Y \longrightarrow S$ and $x, x' \in X(Y)$, $m, m' \in L'_X(Y)$, one has

$$P(m \cdot x, m' \cdot x') = P((m, m') \cdot (x, x')) = L_P^{(x, x')}(m, m') \cdot \mu(g, g')$$

where g (resp. g') is the image of x (resp. x') in $X^+(Y)$. Moreover (cf. the proof of 0.10), $L_P^{(x, x')}$ equals $L_{\mu}^{(g, g')}$ and, by 0.7.bis ($\star$), the latter is the element of $L'_X(Y)$ defined by the following equality in $L_X(Y)$:

$$L_{\mu}^{(g, g')}(m, m') \cdot \mu(g, g') = \mu(m \cdot g, m' \cdot g'),$$

that is, if one denotes by $\times$ (instead of μ) the group law of L_X and by Ad “the adjoint operation” of X^+ on L'_X (which factors through X_0 and which is induced by the adjoint operation of X_0 on ω_{X_0/S_0}^1), one obtains that

$$L_{\mu}^{(g, g')}(m, m') \times g \times g' = m \times g \times m' \times g' = (m \times \text{Ad}(g)(m')) \times g \times g'$$

whence finally $L_P^{(x, x')}(m, m') = m \times \text{Ad}(g)(m')$. One therefore obtains the following:

Proposition 0.8. *Let $P: X \times_S X \longrightarrow X$ be an S -morphism such that $P_{\mathcal{J}}$ equips $X_{\mathcal{J}}$ with a structure of $S_{\mathcal{J}}$ -group. Let us denote by $\times$ the group law of L'_X and by $(m, x) \mapsto m \cdot x$ the morphism $L'_X \times_S X \longrightarrow X$ defining the action of L'_X on X , and let $\text{Ad}: X^+ \longrightarrow \text{Aut}_{S\text{-gr.}}(L'_X)$ be “the adjoint operation” of X^+ on L'_X (which is induced by the adjoint operation of X_0 on ω_{X_0/S_0}^1). Then, for every $S' \longrightarrow S$ and $x, x' \in X(S')$, $m, m' \in L'_X(S')$, one has:*

$$(0.8.1) \quad P(m \cdot x, m' \cdot x') = (m \times \text{Ad } p_X(x)(m')) \cdot P(x, x').$$

If X is an S -group, one will denote by $*$ its law, by e its unit section, and by i the S -morphism defined by:

$$i(m) = m \cdot e,$$

for every $S' \longrightarrow S$ and $m \in L'_X(S')$.

Corollary 0.9. *Let X be an S -group. Then X^+ is equipped naturally with a structure of S -group, and p_X is a morphism of S -groups. Moreover, the S -morphism*

$$i: L'_X \longrightarrow X, \quad m \mapsto m \cdot e$$

is an isomorphism of S -groups of L'_X onto $\text{Ker}(p_X)$, and one has, for all $S' \longrightarrow S$, $x' \in X(S')$, $m \in L'_X(S')$:

$$(0.9.1) \quad m \cdot x' = (m \cdot e) * x' = i(m) * x'.$$

⁽²⁷⁾**N.D.E.** One has added here the number 0.8 in order to mark the return to the original.

⁽²⁸⁾**N.D.E.** One has set out the original in detail in what follows.

The first two assertions have already been proved in 0.1.2. As X is formally principal homogeneous over X^+ under $L_X = L'_X \times_S X^+$, the morphism i is indeed an isomorphism of S -functors of L'_X onto the kernel of p_X . The fact that i is a morphism of groups and the formula (0.9.1) result from the formula (0.8.1) applied respectively to $x = x' = e$, and to $x = e, m' = 1$.

Corollary 0.10. *Let X be an S -group. With the preceding notations, for every $S' \rightarrow S$ and all $x \in X(S')$ and $m' \in L'_X(S')$, one has*

$$(0.10.1) \quad x * i(m') * x^{-1} = i(\text{Ad } p_X(x)(m')).$$

This results from the equality $i(m') * x^{-1} = m' \cdot x^{-1}$ and from (0.8.1) applied to $m = 1$ and $x' = x^{-1}$.

When X is an S -group, we have therefore determined explicitly the kernel of $X \rightarrow X^+$ and the operation of the inner automorphisms of X on this kernel. We are now going to see that one can do likewise for certain S -functors in groups, not necessarily representable. One case will be useful to us, that of the functors Aut (I 1.7). Let us state at once:

Proposition 0.11. *Let E be an S -scheme. Set $X = \text{Aut}_S(E)$. The kernel of the morphism of S -functors in groups*

$$p_X: X \rightarrow X^+$$

is identified canonically with the commutative S -functor in groups L'_X defined by

$$\text{Hom}_S(Y, L'_X) = \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{J}\mathcal{O}_{E \times_S Y}),$$

where Y denotes a variable S -scheme.

Indeed, if Y is a variable S -scheme, one has $\text{Hom}_S(Y, X) = \text{Aut}_Y(E \times_S Y)$, and

$$\text{Hom}_S(Y, X^+) = \text{Hom}_S(Y_{\mathcal{J}}, X) = \text{Aut}_{Y_{\mathcal{J}}}(E \times_S Y_{\mathcal{J}}) = \text{Aut}_{Y_{\mathcal{J}}}((E \times_S Y) \times_Y Y_{\mathcal{J}}).$$

By applying 0.7 b) to the Y -scheme $E \times_S Y$, one obtains an isomorphism of groups:

$$\text{Hom}_S(Y, L'_X) \simeq \text{Ker}(\text{Hom}_S(Y, X) \rightarrow \text{Hom}_S(Y, X^+)),$$

an isomorphism which one easily verifies to be functorial in the S -scheme Y . One therefore obtains an isomorphism of S -groups

$$L'_X \simeq \text{Ker}(X \rightarrow X^+),$$

which completes the proof of proposition 0.11.

Corollary 0.12. ⁽²⁹⁾ *One keeps the notations of 0.11: E is an S -scheme and $X = \text{Aut}_S(E)$. One has a natural operation f of X on L'_X defined in the following manner. For every S -scheme Y , one has*

$$\text{Hom}_S(Y, X) = \text{Aut}_Y(E \times_S Y)$$

and

$$\text{Hom}_S(Y, L'_X) = \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0/Y_0}, \mathcal{J}\mathcal{O}_{E \times_S Y})$$

(N.B. $\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0} \simeq \Omega_{E_0 \times_{S_0} Y_0/Y_0}^1$, cf. EGA IV₄, 16.4.5); the first group operates on the second by transport of structure and this operation is indeed functorial in Y . One then has the formula:

$$(0.12.1) \quad x i(m) x^{-1} = i(f(x)m),$$

for every $Y \rightarrow S$ and all $x \in \text{Hom}_S(Y, X)$, $m \in \text{Hom}_S(Y, L'_X)$.

Indeed, this results from 0.7 c) applied to the Y -scheme $E \times_S Y$.

⁽²⁹⁾**N.D.E.** One has changed “Remark” into “Corollary”.

Recall 0.13. The direct image of a quasi-coherent module by a morphism of finite presentation is quasi-coherent. Under the same conditions, the formation of the direct image commutes with *flat* base change: in the situation

$$\begin{array}{ccc} T & \xleftarrow{g'} & T' = T \times_S S' \\ f \downarrow & & \downarrow f' \\ S & \xleftarrow{g} & S', \end{array}$$

if one supposes f (and hence f') of finite presentation and g (and hence g') *flat*, one has for every quasi-coherent $\mathcal{O}_T$ -module $\mathcal{F}$

$$f_*(\mathcal{F}) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} = f'_*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

or, in a more aesthetic manner

$$g^*(f_*(\mathcal{F})) = f'_*(g'^*(\mathcal{F})).$$

These two facts are more generally valid for a morphism f quasi-compact and quasi-separated, cf. EGA I, 9.2.1 and EGA III₁, 1.4.15 in the quasi-compact separated case (taking into account EGA III₂, Err_{III} 25) and EGA IV₁, 1.7.4 and 1.7.21.

Remark 0.14. ⁽³⁰⁾ Let us take up again the notations of 0.11: let E be an S -scheme, $X = \text{Aut}_S(E)$ and L'_X the commutative S -functor in groups defined by:

$$\begin{aligned} L'_X(Y) &= \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{JO}_{E \times_S Y}) \\ &= \text{Hom}_{\mathcal{O}_{E \times_S Y}}(\Omega_{E/S}^1 \otimes_{\mathcal{O}_S} \mathcal{O}_Y, \mathcal{JO}_{E \times_S Y}) \\ &= \Gamma(E \times_S Y, \mathcal{H}om_{\mathcal{O}_{E \times_S Y}}(\Omega_{E/S}^1 \otimes_{\mathcal{O}_S} \mathcal{O}_Y, \mathcal{JO}_{E \times_S Y})). \end{aligned}$$

Suppose Y is flat over S ; then one has isomorphisms:

$$\mathcal{JO}_{E \times_S Y} \xleftarrow{\sim} (\mathcal{JO}_E) \otimes_{\mathcal{O}_S} \mathcal{O}_Y \xrightarrow{\sim} (\mathcal{JO}_E) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}.$$

Suppose moreover that E is of finite presentation over S ; then $\Omega_{E/S}^1$ is an $\mathcal{O}_E$ -module of finite presentation (cf. EGA IV₄, 16.4.22) and hence, by EGA 0_I, 6.7.6, one has:

$$\mathcal{H}om_{\mathcal{O}_{E \times_S Y}}(\Omega_{E/S}^1 \otimes_{\mathcal{O}_S} \mathcal{O}_Y, (\mathcal{JO}_E) \otimes_{\mathcal{O}_S} \mathcal{O}_Y) \simeq \mathcal{H}om_{\mathcal{O}_E}(\Omega_{E/S}^1, \mathcal{JO}_E) \otimes_{\mathcal{O}_S} \mathcal{O}_Y.$$

Let us denote by $\pi: E \rightarrow S$ and $g: Y \rightarrow S$ the structural morphisms; applying 0.13 to the diagram

$$\begin{array}{ccc} E & \xleftarrow{g'} & E \times_S Y \\ \pi \downarrow & & \downarrow \pi' \\ S & \xleftarrow{g} & Y, \end{array}$$

and to the $\mathcal{O}_E$ -module $\mathcal{F} = \mathcal{H}om_{\mathcal{O}_E}(\Omega_{E/S}^1, \mathcal{JO}_E)$, one obtains

$$\Gamma(E \times_S Y, g'^*\mathcal{F}) = \Gamma(Y, \pi'_*g'^*\mathcal{F}) = \Gamma(Y, g^*\pi_*\mathcal{F}) = W(\pi_*\mathcal{F})(Y).$$

One has therefore shown that, if E is of finite presentation over S , one has

$$(0.14.1) \quad L'_X = W(\pi_*\mathcal{H}om_{\mathcal{O}_E}(\Omega_{E/S}^1, \mathcal{JO}_E))$$

on the category of S -schemes *flat* over S . Let us note moreover that the module of which one takes the W is *quasi-coherent*, by EGA I, 9.1.1 and 9.2.1.

⁽³¹⁾Let us denote by L_0 the S_0 -functor

$$W(\pi_{0*}\mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, \mathcal{JO}_{E_0})).$$

⁽³⁰⁾**N.D.E.** One has set out the first part of this remark in detail, and one has added a second part.

⁽³¹⁾**N.D.E.** One has added what follows.

Then, returning to the definition of $L'_X(Y)$ and taking into account the isomorphism

$$J\mathcal{O}_{E \times_S Y} \simeq (J\mathcal{O}_E) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0},$$

one obtains, by reasoning as above, that

$$L'_X(Y) = L_0(Y_0) = L_0(Y \times_S S_0) = \left(\prod_{S_0/S} L_0 \right)(Y).$$

Thus, on the category of S -schemes *flat* over S , one has:

$$L'_X = \prod_{S_0/S} W(\pi_{0*} \mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J\mathcal{O}_{E_0})).$$

It is not evident that the action of X on L'_X defined in 0.12 comes from an action of $X_0 = \text{Aut}_{S_0}(E_0)$ on L_0 ; this is however the case when, moreover, E is *flat* over S .

Indeed, one has in this case canonical isomorphisms:

$$\begin{aligned} J\mathcal{O}_E &\simeq J \otimes_{\mathcal{O}_S} \mathcal{O}_E \simeq J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0}. \\ L_0 &\simeq W(\pi_{0*} \mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0})), \end{aligned}$$

$$(0.14.2) \quad L'_X = \prod_{S_0/S} W(\pi_{0*} \mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0})).$$

Then, for every S_0 -scheme T , one has

$$L_0(T) \simeq \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} T}}(\Omega_{E_0 \times_{S_0} T/T}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0 \times_{S_0} T})$$

and $\text{Hom}_{S_0}(T, X_0) = \text{Aut}_T(E_0 \times_{S_0} T)$ acts by transport of structure on $L_0(T)$, in a manner functorial in T ; finally, for every S -scheme Y flat over S , the action by transport of structure of $\text{Hom}_S(Y, X) = \text{Aut}_Y(E \times_S Y)$ on $L'_X(Y) = L_0(Y_0)$ factors through $\text{Aut}_{Y_0}(E_0 \times_{S_0} Y_0)$.

Let us finally extract from SGA 1 III the following two propositions.

Proposition 0.15. (SGA 1 III, 6.8)⁽³²⁾ *For every S_J -scheme Y smooth over S_J and affine, there exists an S -scheme X smooth over S such that $X \times_S S_J \simeq Y$, and such an X is unique up to (non-unique) isomorphism.*

Proposition 0.16. (SGA 1 III, 5.5)⁽³³⁾ *Let X be an S -scheme smooth over S . For every affine S -scheme Y , the canonical map*

$$p_X(Y): \text{Hom}_S(Y, X) \longrightarrow \text{Hom}_S(Y, X^+) = \text{Hom}_{S_J}(Y_J, X_J)$$

is surjective.

Corollary 0.17. *Let E be an S -scheme smooth over S and affine; let us write $X = \text{Aut}_S(E)$. For every affine S -scheme Y , the canonical map*

$$\text{Aut}_Y(E \times_S Y) = \text{Hom}_S(Y, X) \longrightarrow \text{Hom}_S(Y, X^+) = \text{Aut}_{Y_J}(E_J \times_{S_J} Y_J)$$

is surjective.

PROOF. Indeed, $Y \times_S E$ is affine over Y , itself affine, hence affine. Applying 0.16, one deduces from this that every S_J -morphism $Y_J \times_{S_J} E_J \longrightarrow E_J$ extends to an S -morphism $Y \times_S E \longrightarrow E$.⁽³⁴⁾ In other terms, every Y_J -endomorphism of $Y_J \times_{S_J} E_J$ lifts to a Y -endomorphism of $Y \times_S E$. Then, 0.7 a) shows that every Y_J -*automorphism* of $Y_J \times_{S_J} E_J$ lifts to a Y -*automorphism* of $Y \times_S E$, which is the announced property. $\square$

⁽³²⁾**N.D.E.** This “global” lifting result uses the “local” lifting result of *loc. cit.*, 4.1, which is stated for S locally noetherian; see EGA IV₄ 18.1.1 for the general case.

⁽³³⁾**N.D.E.** This is an immediate consequence of the definition of (formally) “smooth” adopted in EGA IV₄, 17.3.1 (and 17.1.1).

⁽³⁴⁾**N.D.E.** One has slightly modified the original, so as to be exactly under the hypotheses of 0.7.

1. Extensions and cohomology

1.1. — Let $\mathcal{C}$ be a category stable under fiber products.⁽³⁵⁾ Let S be an object of $\mathcal{C}$, G an S -group (*representable*) and F a commutative group S -functor on which G operates. One has defined in I, 5.1 the cohomology groups $H^n(G, F)$. One recalls that these are the homology groups of a complex denoted $C^*(G, F)$ where, writing $(G/S)^n = G \times_S \cdots \times_S G$ (n factors),

$$C^n(G, F) = \text{Hom}_S((G/S)^n, F).$$

As G and hence⁽³⁵⁾ the $(G/S)^n$ are representable, one also has

$$C^n(G, F) = F((G/S)^n);$$

from this, and from the definition of the boundary operator, one sees that the complex $C^*(G, F)$ depends only on the restriction of F to the full subcategory of $\mathcal{C}_S$ whose objects are the cartesian powers of G over S . Consequently, one has the

Lemma 1.1.1. *Let $\mathcal{C}$ be a category stable under fiber products⁽³⁵⁾, S an object of $\mathcal{C}$, G a representable S -group. Let us denote by $\mathcal{C}(G)$ the full subcategory of $\mathcal{C}_S$ whose objects are the cartesian powers of G over S . Let F and F' be two commutative group S -functors on which G operates. If F and F' have the same restriction to $\mathcal{C}(G)$, one has a canonical isomorphism*

$$H^*(G, F) \xrightarrow{\sim} H^*(G, F').$$

1.1.2. Cohomology and restriction of scalars. — ⁽³⁶⁾ Let us state another comparison result. Let now $T \rightarrow S$ be a morphism of $\mathcal{C}$. If F is a commutative group T -functor, then the functor obtained by “restriction of scalars” (cf. Exp. II, 1)

$$F_1 = \prod_{T/S} F$$

is a commutative group S -functor and one has a morphism of group S -functors

$$u: \prod_{T/S} \text{Aut}_{T\text{-gr.}}(F) \longrightarrow \text{Aut}_{S\text{-gr.}}(F_1).$$

⁽³⁷⁾ Let now G be a group S -functor and let

$$G_T \longrightarrow \text{Aut}_{T\text{-gr.}}(F)$$

be an operation of G_T on F . By definition of the functor $\prod_{T/S}$, one deduces from this a morphism of group S -functors

$$G \longrightarrow \prod_{T/S} \text{Aut}_{T\text{-gr.}}(F),$$

whence, by composition with u , an operation of G on $F_1 = \prod_{T/S} F$.⁽³⁸⁾

⁽³⁵⁾**N.D.E.** One has added the hypothesis that $\mathcal{C}$ be stable under fiber products (which is the case in the applications where $\mathcal{C}$ is the category of schemes (Sch) or that of the functors $(\text{Sch})^\circ \rightarrow (\text{Ens})$). If one omits this hypothesis, one must suppose in the sequel that the fiber products $G \times_S \cdots \times_S G$ are representable.

⁽³⁶⁾**N.D.E.** One has added the title of this paragraph.

⁽³⁷⁾**N.D.E.** Indeed, let $S' \rightarrow S$ and $\alpha \in \text{Aut}_{(T \times_S S')\text{-gr.}}(F_{T \times_S S'})$; for every $U \rightarrow T \times_S S'$, let us denote by $\alpha_U \in \text{Aut}_{\text{gr.}}(F_{T \times_S S'}(U))$ the element defined by α ; then $u(S')(\alpha)$ is the element β of $\text{Aut}_{S\text{-gr.}}(F_1)(S') = \text{Aut}_{S'\text{-gr.}}(\prod_{T \times_S S'/S'} F_{T \times_S S'})$ such that $\beta_{S''} = \alpha_{T \times_S S''}$ for every $S'' \rightarrow S'$.

⁽³⁸⁾**N.D.E.** Explicitly, for every $S' \rightarrow S$, the operation of $G(S')$ on $F_1(S') = F(T \times_S S')$ is given by the operation of $G_T(T \times_S S') = G(T \times_S S')$ on $F(T \times_S S')$ and the morphism of groups $G(S') \rightarrow G(T \times_S S')$ corresponding to the projection $T \times_S S' \rightarrow S'$. One may also say that the operation of G_T on F is given by a morphism $G_T \times_T F \rightarrow F$; applying the functor $\prod_{T/S}$ and denoting by G_1 the S -group $\prod_{T/S} G_T$, one obtains, taking into account II 1.2, a morphism $G_1 \times_S F_1 \rightarrow F_1$ which defines an operation of G_1 on F . The operation of G is then obtained via the natural morphism $G \rightarrow G_1$, which corresponds by adjunction to id_G .

Lemma 1.1.2. *Under the preceding conditions, one has a canonical isomorphism*

$$H^*\left(G, \prod_{T/S} F\right) \simeq H^*(G_T, F).$$

PROOF. Indeed, by the definition of the cohomology, the standard complexes are canonically isomorphic. $\square$

1.2. Lifting of morphisms of groups. — ⁽³⁹⁾ Following the general principles, one sets down the following definition:

Definition 1.2.1. Let $1 \rightarrow M \xrightarrow{u} E \xrightarrow{v} G$ be a sequence of morphisms of $\widehat{\mathcal{C}}$ -groups. One says that it is *exact* if the following equivalent conditions are verified:

(i) for every $S \in \text{Ob } \mathcal{C}$, the sequence of ordinary groups below is exact:

$$1 \rightarrow M(S) \xrightarrow{u(S)} E(S) \xrightarrow{v(S)} G(S)$$

(ii) for every object H of $\widehat{\mathcal{C}}$, the sequence of ordinary groups below is exact:

$$1 \rightarrow \text{Hom}(H, M) \xrightarrow{u(H)} \text{Hom}(H, E) \xrightarrow{v(H)} \text{Hom}(H, G)$$

Taking in particular $H = G$ in (ii), one sees that the set of sections of v (not respecting *a priori* the group structures) is empty or principal homogeneous under $\text{Hom}(G, M)$. Let us suppose it non-empty; let therefore

$$s: G \rightarrow E$$

be a section of v . Then for every $S \in \text{Ob } \mathcal{C}$ and every $x \in G(S)$, the element $s(x)$ of $E(S)$ defines an inner automorphism of E_S which normalizes M_S (more correctly the image of M_S under u_S), hence an automorphism of M_S .

Scholium 1.2.1.1. ⁽⁴⁰⁾ *If M is commutative, one sees “set-theoretically” that this automorphism does not depend on the chosen section, but only on x , and that it depends multiplicatively on it. In summary, to every exact sequence*

$$(E) \quad 1 \rightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

such that M is commutative and that v possesses a section, there is associated a morphism of $\widehat{\mathcal{C}}$ -groups

$$G \rightarrow \text{Aut}_{\widehat{\mathcal{C}}\text{-gr.}}(M)$$

which one calls the operation of G on M defined by the extension (E) .

Definition 1.2.1.2. One has seen in I, 2.3.7 that v possesses a section which is a morphism of $\widehat{\mathcal{C}}$ -groups if and only if the extension (E) is isomorphic (“as an extension”) to the semidirect product of M by G relative to the preceding operation. Such a section of v will be called a *section of the extension (E)* , or simply a *section of (E)* .

If s is a section of (E) and if $m \in \Gamma(M) \simeq \text{Ker}(\Gamma(E) \rightarrow \Gamma(G))$ (for the definition of Γ , see I, 1.2), then the morphism $G \rightarrow E$ defined by⁽⁴¹⁾

$$x \mapsto u(m) s(x) u(m)^{-1}$$

is likewise a section of (E) , said to be *deduced from s by the inner automorphism defined by m (or by $u(m)$)*.

⁽³⁹⁾ **N.D.E.** One has added this title.

⁽⁴⁰⁾ **N.D.E.** One has added the numbering 1.2.1.1 and 1.2.1.2 in order to bring out the notions which are introduced there.

⁽⁴¹⁾ **N.D.E.** In the right-hand term, one has corrected x to $s(x)$.

Lemma 1.2.2. *Let $(E): 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$ be an exact sequence of $\widehat{\mathcal{C}}$ -groups such that M is commutative and that v possesses a section. Let us make G operate on M in the manner defined by (E) .*

- (i) *The extension (E) canonically defines a class $c(E) \in H^2(G, M)$ whose vanishing is necessary and sufficient for the existence of a section of (E) .*
- (ii) *If $c(E) = 0$, the set of sections of (E) is principal homogeneous under the group $Z^1(G, M)$, and the set of sections of (E) modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$ is principal homogeneous under the group $H^1(G, M)$.*
- (iii) ⁽⁴²⁾ *Let s be a section of (E) ; the set of the conjugates of s by the inner automorphisms defined by $\Gamma(M)$ is in bijection with $\Gamma(M)/H^0(G, M)$.*

PROOF. The proof is carried out exactly as in the case of ordinary groups, the fact that one starts from a section of v ensuring the functoriality of the set-theoretic computations. Let us briefly indicate the principal steps of the proof.

a) To every section s of v one associates the morphism

$$Ds: G \times G \longrightarrow M,$$

defined set-theoretically by

$$u(Ds(x, y)) = s(xy) s(y)^{-1} s(x)^{-1}.$$

One shows that Ds is a 2-cocycle by the following computation.⁽⁴³⁾ By the definition of the differential of the standard complex (I, 5.1), one has:

$$(\partial^2 Ds)(x, y, z) = (s(x) Ds(y, z) s(x)^{-1}) \cdot Ds(x, y)^{-1} \cdot Ds(xy, z)^{-1} \cdot Ds(x, yz);$$

it suffices to substitute the definition of Ds into this formula in order to find (without using any commutativity) $Ds \in Z^2(G, M)$.

b) If s and s' are two sections of v , there exists $f: G \longrightarrow M$ such that $s(x) = f(x)s'(x)$. One then has

$$\begin{aligned} Ds'(x, y) &= f^{-1}(xy) Ds(x, y) s(x) f(y) s(x)^{-1} f(x), \\ &= Ds(x, y) \cdot f^{-1}(xy) \cdot (s(x) f(y) s(x)^{-1}) \cdot f(x), \end{aligned}$$

that is,

$$Ds' = Ds \cdot \partial^1 f.$$

⁽⁴⁴⁾ This shows that the class of Ds in $H^2(G, M)$ does not depend on the chosen section s of v ; it is the class $c(E)$ of the extension (E) .

c) Let s and s' be two sections of v and let $m \in \Gamma(M)$. Then, the equality $s(x) = m^{-1}s'(x)m$ (for every $x \in G(S)$, $S \in \text{Ob } \mathcal{C}$) is equivalent to

$$s(x) = m^{-1}s'(x)m s'(x)^{-1}s'(x), \quad \text{i.e.} \quad s = \partial^0 m \cdot s'.$$

In particular, the stabilizer of s in $\Gamma(M)$ is the subgroup of those $m \in \Gamma(M)$ such that $\partial^0 m = e_M$, i.e. the subgroup $H^0(G, M)$. This already proves (iii).

d) The reasoning is now the usual one:⁽⁴⁵⁾ Let s_0 be an arbitrary section of v ; there exists a section s , necessarily of the form $s = f \cdot s_0$, which is a morphism of groups, i.e. which satisfies $Ds = 0$, if and only if $(Ds_0)^{-1} = \partial^1 f$, i.e. , if and only if the class $c(E)$ is zero. This proves (i).

In this case, the set of sections of (E) is formed of the sections $s' = h \cdot s$, where $h: G \longrightarrow M$ satisfies $\partial^1 h = 0$, i.e. $h \in Z^1(G, M)$. Moreover, by point c), two such sections $h_1 \cdot s$ and $h_2 \cdot s$

⁽⁴²⁾**N.D.E.** One has added this point, the counterpart of 1.2.4 (iii). Let us note on the other hand that the assertion is valid for *every* section of v , cf. the proof.

⁽⁴³⁾**N.D.E.** One has corrected the original, so as to make it compatible with I, 5.1.

⁽⁴⁴⁾**N.D.E.** One has added the sentence which follows.

⁽⁴⁵⁾**N.D.E.** One has set out in detail what follows.

are conjugate under $\Gamma(M)$ if and only if h_1 and h_2 have the same image in $H^1(G, M)$. This proves (ii). $\square$

Let still

$$(E) \quad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathcal{C}}$ -groups with M commutative. Let

$$f: H \longrightarrow G$$

be a morphism of $\widehat{\mathcal{C}}$ -groups. Consider $E_f = H \times_G E$; it is a $\widehat{\mathcal{C}}$ -group and the projection $v_f: E_f \longrightarrow H$ is a morphism of $\widehat{\mathcal{C}}$ -groups. Likewise for $e_f: E_f \longrightarrow E$. On the other hand, if one sends M into E by u and into H by the unit morphism, one defines a morphism of $\widehat{\mathcal{C}}$ -groups $u_f: M \longrightarrow E_f$. One has therefore constructed a commutative diagram of $\widehat{\mathcal{C}}$ -groups:

$$\begin{array}{ccccccc} (E) & 1 & \longrightarrow & M & \xrightarrow{u} & E & \xrightarrow{v} & G \\ & & & \uparrow \text{id} & & \uparrow e_f & & \uparrow f \\ (E_f) & 1 & \longrightarrow & M & \xrightarrow{u_f} & E_f & \xrightarrow{v_f} & H. \end{array}$$

One has immediately:

Lemma 1.2.3.

- (i) *The sequence (E_f) is exact.*
- (ii) *The map $s \mapsto e_f \circ s = f'$ realizes a bijective correspondence between the morphisms*

$$s: H \longrightarrow E_f$$

such that $v_f \circ s = \text{id}$ (that is, the sections of v_f) and the morphisms

$$f': H \longrightarrow E$$

such that $v \circ f' = f$ (that is, the morphisms f' “lifting” f).

- (iii) *In the preceding correspondence, sections of (E_f) and morphisms of groups f' lifting f correspond.*

Applying lemma 1.2.2 to the extension (E_f) and taking 1.2.3 into account, one obtains the following proposition (which formally contains 1.2.2):

Proposition 1.2.4. *Let $(E): 1 \longrightarrow M \longrightarrow E \xrightarrow{v} G$ be an exact sequence of $\widehat{\mathcal{C}}$ -groups with M commutative. Let*

$$f: H \longrightarrow G$$

be a morphism of $\widehat{\mathcal{C}}$ -groups; suppose that it lifts to a morphism (not necessarily of groups) $f': H \longrightarrow E$. Let us make H operate on M by the composite morphism (multiplicative and independent of the choice of f'),

$$H \xrightarrow{f'} E \xrightarrow{\text{int}} \text{Aut}_{\widehat{\mathcal{C}}\text{-gr.}}(M).$$

- (i) *The morphism f defines canonically a class $c(f) \in H^2(H, M)$ whose vanishing is necessary and sufficient for the existence of a morphism of $\widehat{\mathcal{C}}$ -groups*

$$f': H \longrightarrow E$$

lifting f .

- (ii) *If $c(f) = 0$, the set of morphisms of $\widehat{\mathcal{C}}$ -groups f' lifting f , modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$ (i.e. by the elements m of $\Gamma(E)$ such that $v(m) = e$) is a principal homogeneous space under $H^1(H, M)$.*
- (iii) *If $f': H \longrightarrow E$ is a morphism of groups lifting f , the set of the transforms of f' by the inner automorphisms defined by the elements of $\Gamma(M)$ is isomorphic to $\Gamma(M)/\Gamma(M^H) = \Gamma(M)/H^0(H, M)$.*

1.3. Extensions of group laws. — Consider the following situation: one has a morphism of $\widehat{\mathcal{C}}$

$$(\dagger) \quad p: X \longrightarrow Y$$

and a commutative $\widehat{\mathcal{C}}$ -group M operating on X , such that X is formally principal homogeneous over Y under M_Y .

If $g: Y \longrightarrow Z$ is an arbitrary morphism of $\widehat{\mathcal{C}}$, then $g \circ p: X \longrightarrow Z$ is invariant under M : for each $S \in \text{Ob}(\mathcal{C})$, $(g \circ p)(S)$ is invariant under the action of $M(S)$ operating on $X(S)$. Conversely, we shall suppose the following condition verified for $n = 1, 2, 3, 4$.

$(+)_n$: Every morphism from X^n into M , invariant under the action of M^n operating on X^n , factors uniquely through $p^n: X^n \longrightarrow Y^n$ (where the powers n denote cartesian powers).

Lemma 1.3.1.

- (i) If h is a morphism from Y into M , the automorphism u_h of X defined set-theoretically by $x \mapsto h(p(x)) \cdot x$ preserves the fibers of p and commutes with the operations of M on X ,⁽⁴⁶⁾ i.e. for all $S \in \text{Ob}(\mathcal{C})$ and $x \in X(S)$, $m \in M(S)$, one has

$$p(h(p(x)) \cdot x) = p(x), \quad m \cdot h(p(x)) \cdot x = h(p(m \cdot x)) \cdot m \cdot x.$$

- (ii) This construction realizes a bijective correspondence between morphisms from Y into M and automorphisms of X preserving the fibers of p and commuting with the operations of M .

The first part is clear, since $p(m \cdot x) = p(x)$ and since M is commutative. Conversely, an automorphism u of X preserving the fibers of p is written set-theoretically $x \mapsto g(x)x$, where g is some morphism from X into M . If u commutes with the operations of M , the morphism g is invariant under $M^{(47)}$ and one concludes by condition $(+)_1$.

We now suppose that there are given in addition a group law on Y and an operation of Y on M , that is, a morphism of $\widehat{\mathcal{C}}$ -groups:

$$(\ddagger) \quad f: Y \longrightarrow \text{Aut}_{\widehat{\mathcal{C}\text{-gr.}}}(M).$$

Definition 1.3.2. A composition law on X

$$P: X \times X \longrightarrow X$$

is said to be *admissible* if it satisfies the following two conditions:

- (i) P lifts the group law of Y , i.e. the diagram

$$\begin{array}{ccc} X \times X & \xrightarrow{P} & X \\ (p,p) \downarrow & & \downarrow p \\ Y \times Y & \longrightarrow & Y \end{array}$$

is commutative.

- (ii) For every $S \in \text{Ob}(\mathcal{C})$ and all $x, y \in X(S)$, $m, n \in M(S)$, one has the following relation

$$(++) \quad P(m \cdot x, n \cdot y) = m \cdot f(p(x))(n) \cdot P(x, y).$$

Proposition 1.3.3. In order that a group law $*$ on X be admissible, it is necessary and sufficient that the following four conditions be satisfied:

- (i) $p: X \longrightarrow Y$ is a morphism of groups.
 (ii) The morphism $i: M \longrightarrow X$ defined by $i(m) = m \cdot e_X$ is an isomorphism of groups from M onto $\text{Ker}(p)$, that is: one has set-theoretically $(m \cdot e_X) * (n \cdot e_X) = (mn) \cdot e_X$.

⁽⁴⁶⁾**N.D.E.** Here and in the sequel, one has replaced “commutes with p and with the operations of M ” by “preserves the fibers of p and commutes with the operations of M ”. On the other hand, one has added the equalities which follow.

⁽⁴⁷⁾**N.D.E.** Indeed, the equality $mg(x)x = mu(x) = u(mx) = g(mx)mx$ implies $g(mx) = g(x)$.

- (iii) One has $m \cdot x = (m \cdot e_X) * x = i(m) * x$ for each $m \in M(S)$, $x \in X(S)$.
- (iv) The inner automorphisms of X operate on $\text{Ker}(p)$ according to the set-theoretic formula:

$$x * i(m) * x^{-1} = i(f(p(x))m).$$

The proof is immediate.

Lemma 1.3.4. ⁽⁴⁸⁾ Let h be a morphism $Y \rightarrow M$ and u_h the automorphism $x \mapsto h(p(x)) \cdot x$ of X (cf. 1.3.1). Let P be an admissible composition law (resp. a group law) on X and let P' be the composition law on X deduced from P by means of u_h , i.e., $P'(x, y) = u_h^{-1}(P(u_h(x), u_h(y)))$. Then:

- (i) P' is an admissible composition law (resp. group law).
- (ii) For all $x, y \in X(S)$ ($S \in \text{Ob}(\mathcal{C})$), set $v = p(x)$ and $w = p(y)$; then

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

PROOF. One has $u_h^{-1} = u_{h^\vee}$, where $h^\vee: Y \rightarrow M$ is defined by $h^\vee(y) = h(y)^{-1}$. By 1.3.2 (i) and (ii), one has $P(h(v) \cdot x, h(w) \cdot y) = h(v) \cdot f(v)(h(w)) \cdot P(x, y)$ and $p(P(x, y)) = vw$, whence

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

It is then immediate that P' satisfies conditions (i) and (ii) of 1.3.2. $\square$

Definition 1.3.5. Two admissible composition laws deduced from one another by the procedure of 1.3.4 are said to be *equivalent*. ⁽⁴⁹⁾

Proposition 1.3.6. Suppose that there exists an admissible composition law on X . Then:

- (i) There exists a class $c \in H^3(Y, M)$ (determined canonically), whose vanishing is necessary and sufficient for the existence of an associative admissible composition law on X .
- (ii) If $c = 0$, the set of associative admissible composition laws (resp. of equivalence classes of associative admissible composition laws) on X is a principal homogeneous space under $Z^2(G, M)$ (resp. $H^2(G, M)$).

The proof is carried out in several steps.

a) Let P be an admissible composition law on X . As P lifts the composition law of Y , which is associative, there exists a unique morphism $a: X^3 \rightarrow M$ such that

$$(*) \quad P(x, P(y, z)) = a(x, y, z) P(P(x, y), z).$$

By applying conditions 1.3.2 (i) and (ii), one sees at once that a is invariant under the action of M^3 on X^3 , ⁽⁵⁰⁾ whence, by applying hypothesis $(+)_3$, the following result:

- (1) There exists a unique morphism $DP: Y^3 \rightarrow M$ such that

$$P(x, P(y, z)) = DP(p(x), p(y), p(z)) P(P(x, y), z),$$

and P is associative if and only if $DP = 0$.

b) Let us compute step by step $P(P(P(x, y), z), t)$ by means of the preceding formula. Setting $p(x) = u$, $p(y) = v$, $p(z) = w$, $p(t) = h$, one obtains ⁽⁵¹⁾ the following pentagonal

⁽⁴⁸⁾**N.D.E.** One has set out the statement as well as its proof in detail, with a view to point (e) of the proof of 1.3.6.

⁽⁴⁹⁾**N.D.E.** This is indeed an equivalence relation, since $u_h^{-1} = u_{h^\vee}$.

⁽⁵⁰⁾**N.D.E.** For $x, y, z \in X(S)$, set $u = p(x)$, $v = p(y)$ and $w = p(z)$. Then $P(mx, P(m'y, m''z)) = P(mx, m'f(v)(m'')P(y, z)) = mf(u)(m')f(uv)(m'')P(x, P(y, z))$. On the other hand, as $p(P(x, y)) = uv$, one also has $P(P(mx, m'y), m''z) = P(mf(u)(m')P(x, y), m''z) = mf(u)(m')f(uv)(m'')P(P(x, y), z)$, and the comparison of these equalities with $(*)$ gives $a(mx, m'y, m''z) = a(x, y, z)$.

⁽⁵¹⁾**N.D.E.** One has added the diagram which follows.

diagram, where an arrow $a \xrightarrow{m} b$ means that $b = m \cdot a$:

$$\begin{array}{ccc}
 & P(x, P(y, P(z, t))) & \\
 DP(u, v, wh) \swarrow & & \searrow f(u)(DP(v, w, h)) \\
 P(P(x, y), P(z, t)) & & P(x, P(P(y, z), t)) \\
 DP(uv, w, h) \downarrow & & \downarrow DP(u, vw, h) \\
 P(P(P(x, y), z), t) & \xleftarrow{DP(u, v, w)} & P(P(x, P(y, z)), t)
 \end{array}$$

hence one finds

$$\begin{aligned}
 & DP(u, v, w) \cdot DP(u, vw, h) \cdot f(u)DP(v, w, h) \\
 & \cdot DP(u, v, wh)^{-1} \cdot DP(uv, w, h)^{-1} = e_M,
 \end{aligned}$$

i.e., $\partial^3 DP(u, v, w, h) = e_M$. As on the other hand the left-hand side of the preceding formula can be written, with the aid of P and of a , as the expression in (x, y, z, t) of some morphism $X^4 \rightarrow M$, it follows from the uniqueness hypothesis in $(+)_4$ that $\partial^3 DP$ and e_M , which factor the same morphism, are equal, hence

(2) DP is a cocycle, i.e. one has $DP \in Z^3(Y, M)$.

c) If P and P' are two admissible composition laws on X , there exists a unique morphism

$$b: X^2 \rightarrow M$$

such that $P'(x, y) = b(x, y)P(x, y)$. Applying 1.3.2 (ii) to P and to P' , one sees that b is invariant under M^2 , whence, by $(+)_2$:

(3) For every pair of admissible composition laws (P, P') , there exists a unique

$$d(P, P'): Y^2 \rightarrow M$$

such that

$$P'(x, y) = d(P, P')(p(x), p(y)) P(x, y),$$

and the set of admissible composition laws thus becomes a principal homogeneous space under $\text{Hom}(Y^2, M) = C^2(Y, M)$.

d) Under the preceding conditions, one has the formula:

$$(4) \quad DP' - DP = \partial^2 d(P, P').$$

e) P and P' are equivalent if and only if there exists a morphism $h \in C^1(Y, M) = \text{Hom}(Y, M)$ such that $d(P, P') = \partial^1 h$; this follows from the definition of equivalence and from 1.3.4 (ii).

f) It remains only to conclude: one seeks a P' which is associative, i.e. such that $DP' = e_M$. Now DP is a cocycle whose class in $H^3(Y, M)$ does not depend on the admissible composition law P chosen (by (3) and (4)). This class is the obstruction c sought. One will be able to choose a P' satisfying the conditions if and only if $c = 0$; indeed, choosing an arbitrary P , one will have to solve, by (1):

$$0 = DP' = DP + \partial^2 d(P, P'),$$

which is possible by (3) and (4) if and only if $c = 0$. The set of associative P' is a principal homogeneous space under $Z^2(Y, M)$, again by (3) and (4). The set of associative P' up to equivalence is a principal homogeneous space under $H^2(Y, M)$ by (e).

2. Infinitesimal extensions of a morphism of group schemes

Let us resume the notations of n° 0. Let Y and X be two group S -functors. Let M be the kernel of the morphism of groups $p_X: X \rightarrow X^+$. One therefore has an exact sequence of group S -functors

$$1 \rightarrow M \rightarrow X \xrightarrow{p_X} X^+.$$

By the definition of X^+ , one has isomorphisms

$$\begin{aligned}\mathrm{Hom}_S(Y, X^+) &\xrightarrow{\sim} \mathrm{Hom}_{S_J}(Y_J, X_J) \\ \mathrm{Hom}_{S\text{-gr.}}(Y, X^+) &\xrightarrow{\sim} \mathrm{Hom}_{S_J\text{-gr.}}(Y_J, X_J),\end{aligned}$$

and the morphism

$$\mathrm{Hom}_S(Y, p_X): \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)$$

associates with an S -morphism $f: Y \longrightarrow X$, the S -morphism $f^+: Y \longrightarrow X^+$ corresponding, via the preceding isomorphisms, to the S_J -morphism $f_J: Y_J \longrightarrow X_J$ obtained by base change from f . If M is *commutative*, one can apply proposition 1.2.4 to this situation.

2.0. — ⁽⁵²⁾ In the sequel, we shall be interested in the following case: Y is *flat* over S , and X is a group S -functor of one of the following two kinds:

- (a) X is an S -group scheme,
- (b) $X = \mathrm{Aut}_S(E)$ where E is an S -scheme, *of finite presentation over S* .

Let us denote by $(\mathbf{Plats})_S$ the category of S -schemes *flat* over S . In case (a) (resp. (b)), the group S -functor $M = \mathrm{Ker}(X \longrightarrow X^+)$, its restriction L to $(\mathbf{Plats})_S$, and the operations of the inner automorphisms of X on M , have been computed in 0.9, 0.5, and 0.10 (resp. 0.11, 0.14, and 0.12). That is, in case (a), let L_0 be the commutative group S_0 -functor defined by: for every S_0 -scheme T_0 ,

$$\mathrm{Hom}_{S_0}(T_0, L_0) = \mathrm{Hom}_{\mathcal{O}_{T_0}}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}),$$

on which X_0 operates via its adjoint representation in ω_{X_0/S_0}^1 ; then $L = \prod_{S_0/S} L_0$, i.e. for every S -scheme T , one has $L(T) = L_0(T \times_S S_0)$.

In case (b), let us denote by π the structural morphism $E \longrightarrow S$; then L is the abelian group functor on $(\mathbf{Plats})_S$ defined by

$$\mathrm{Hom}_S(T, L) = \Gamma(T, \pi_*(\mathcal{H}\mathrm{om}_{\mathcal{O}_E}(\Omega_{E/S}^1, J \mathcal{O}_E)) \otimes_{\mathcal{O}_S} \mathcal{O}_T),$$

on which X , considered as a functor on $(\mathbf{Plats})_S$, operates as one has seen in 0.12.

Then, one has an exact sequence of group functors on $(\mathbf{Plats})_S$:

$$(E) \quad 1 \longrightarrow L \longrightarrow X \longrightarrow X^+.$$

On the other hand, Y being supposed *flat* over S , the groups $H^i(Y, M)$ depend, by 1.1.1, only on the restriction L of M to $(\mathbf{Plats})_S$. As $L = \prod_{S_0/S} L_0$ in case (a), then by 1.1.2, one has in this case isomorphisms $H^i(Y, L) \simeq H^i(Y_0, L_0)$.

Then, taking account of what precedes, one deduces from proposition 1.2.4 the⁽⁵³⁾

Theorem 2.1. *Let S be a scheme, I and J two quasi-coherent ideals such that $I \supset J$ and $I \cdot J = 0$, defining the closed subschemes S_0 and S_J , and let:*

- X a group S -functor of type (a) or (b), and L_0, L as above;
- Y an S -group scheme *flat* over S and $f_J: Y_J \longrightarrow X_J$ a morphism of S_J -groups.

Then:

- (i) *In order that f_J lift to a morphism of S -groups $Y \longrightarrow X$, it is necessary and sufficient that the following two conditions be satisfied:*
 - (i₁) *f_J lifts to a morphism of S -functors $Y \longrightarrow X$ (by 1.2.4, this defines an operation of Y on L , which does not depend on the chosen lifting; moreover, in case (a), the operation of Y_0 on L_0 thus obtained comes from the morphism $f_0: Y_0 \longrightarrow X_0$ and from the “adjoint action” of X_0 on L_0);*

⁽⁵²⁾**N.D.E.** One has added the numbering 2.0, for later references.

⁽⁵³⁾**N.D.E.** One has placed higher up the definitions which in the original figured in the statement of theorem 2.1 and formed the essential part of page 113 of the original.

- (i₂) A certain obstruction $c(f_J)$, defined canonically by f_J , is zero, where $c(f_J)$ is a class in $H^2(Y, L) (\simeq H^2(Y_0, L_0)$ in case (a)).
- (ii) If the conditions of (i) are satisfied, the set E of morphisms of group S -functors $Y \rightarrow X$ extending f_J is a principal homogeneous space under $Z^1(Y, L) (\simeq Z^1(Y_0, L_0)$ in case (a)), and E modulo the action of the inner automorphisms of X defined by the sections of X over S inducing the unit section of X_J over S_J , is a principal homogeneous space under $H^1(Y, L) (\simeq H^1(Y_0, L_0)$ in case (a)).
- (iii) If $f: Y \rightarrow X$ is a morphism of group S -functors extending f_J , the set of the morphisms $Y \rightarrow X$ transforms of f by the inner automorphisms defined by the sections of X over S inducing the unit section of X_J over S_J , is isomorphic to $\Gamma(L)/H^0(Y, L) (\simeq \Gamma(L_0)/H^0(Y_0, L_0)$ in case (a)).

Remark 2.1.1. ⁽⁵⁴⁾ If $f, f': Y \rightarrow X$ are morphisms of group S -functors extending f_J , one therefore obtains a cocycle $d(f, f') \in Z^1(Y, L) (\simeq Z^1(Y_0, L_0)$ in case (a)), such that

$$(*) \quad f' = d(f, f') \cdot f.$$

⁽⁵⁵⁾ One will denote by $d(f, f')$ the image of $d(f, f')$ in $H^1(Y, L) (\simeq H^1(Y_0, L_0)$ in case (a)).

Remark 2.2. One keeps the preceding notations; in particular, Y is flat over S . In case (b), L is, by (0.14.1), the restriction to $(\mathbf{Plats})_S$ of the functor

$$\mathbf{W}(\pi_*(\mathcal{H}\text{om}_{\mathcal{O}_E}(\Omega_{E/S}^1, J\mathcal{O}_E))),$$

where $\pi: E \rightarrow S$ is the structural morphism. In case (a), suppose moreover that X is locally of finite presentation over S ; then by (0.6.1), L is the restriction to $(\mathbf{Plats})_S$ of the functor

$$\coprod_{S_0/S} \mathbf{W}(\mathcal{H}\text{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, J)).$$

In both cases, the module of which one takes the $\mathbf{W}$ is quasi-coherent, by EGA I, 9.1.1. Suppose moreover that Y is affine over S ⁽⁵⁶⁾. Then, by I, 5.3, one obtains:

- (a) $H^i(Y, L) = H^i(Y_0, L_0) = H^i(Y_0, \mathcal{H}\text{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, J)),$
- (b) $H^i(Y, L) = H^i(Y, \pi_*(\mathcal{H}\text{om}_{\mathcal{O}_E}(\Omega_{E/S}^1, J\mathcal{O}_E))).$

Remark 2.3. 1) By 0.16 and 0.17, condition (i₁) is automatically satisfied when Y is an affine scheme and

$$(*) \quad \begin{cases} \text{in case (a), } X \text{ is smooth over } S; \\ \text{in case (b), } E \text{ is smooth and affine over } S. \end{cases}$$

2) Moreover, under these conditions (Y being still supposed flat over S , cf. 2.0), one can write in case (a), by 2.2 a) and (0.6.2),

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} J),$$

⁽⁵⁷⁾ and in case (b), by (0.14.2), 1.1.2 and I, 5.3,

$$H^i(Y, L) = H^i(Y_0, \pi_{0*} \mathcal{H}\text{om}_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0})).$$

⁽⁵⁴⁾**N.D.E.** One has added this remark, analogous to 4.5.1, in order to introduce the notations $d(f, f')$ and $d(f, f')$, used in 4.38; consequently, one has also added in 2.1 (ii) above, the part concerning E itself.

⁽⁵⁵⁾**N.D.E.** One has conformed to the sign conventions of the original. One could have chosen to write $f' = d(f', f) \cdot f$, but then, in order to have in 4.27 the equality $d^1(d(i, i')) = d(Y, i'(Y))$ when i, i' are two immersions $Y \hookrightarrow X$, one would have had to change the sign of the class $d(Y, Y')$ introduced in 4.5.1, and that would have led to changes of sign in the formulas of 4.8, 4.14, 4.17. One has preferred to keep the signs given in the original (all correct!).

⁽⁵⁶⁾**N.D.E.** One has added this hypothesis, as well as the reference to I, 5.3.

⁽⁵⁷⁾**N.D.E.** One has added what follows, cf. the N.D.E. (31) in 0.14.

Let us now state a certain number of corollaries concerning the case where Y is a diagonalizable S -group (I, 4.4); one then knows (*loc. cit.* 5.3.3) that if S is affine, $H^n(Y, \mathcal{F}) = 0$ for $n > 0$ and every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. First a particular case:

Corollary 2.4. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let Y be a diagonalizable S -group and let:*

- (a) *X an S -group locally of finite presentation over S ,*
- (b) *$X = \text{Aut}_S(E)$ where E is an S -scheme locally of finite presentation.*

Let $f: Y \rightarrow X$ be a morphism of S -groups such that the morphism $f_0: Y_0 \rightarrow X_0$ obtained by base change is the unit morphism. Then f is the unit morphism.

Indeed, the question is local on S and (in (b)) on E . One may therefore suppose S affine and (in (b)) E of finite presentation over S . Introducing now the closed subschemes S_n of S defined by the powers of the ideal defining S_0 , one is reduced to the case where S_0 is defined by an ideal of square zero, and in this case the stated assertion follows from the theorem, via 2.2.

In the case where one does not necessarily suppose that f_0 is the unit morphism, one has:

Corollary 2.5. *Let S and S_0 be as in 2.4. Suppose moreover S affine. Let Y be a diagonalizable S -group, X a group S -functor and $f_0: Y_0 \rightarrow X_0$ a morphism of group S_0 -functors.*

- (i) ⁽⁵⁸⁾
- (ii) *Suppose that one has one of the following two properties:*
 - (a) *X is an S -group smooth over S ;*
 - (b) *$X = \text{Aut}_S(E)$ where E is smooth and affine over S .*

Then f_0 extends to a morphism of S -groups $Y \rightarrow X$; two such extensions are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0 .

PROOF. Let us introduce the S_n as above.⁽⁵⁹⁾ For (ii), let us first note that a scheme smooth over S is necessarily locally of finite presentation over S ; hence, in case (b), E , being smooth and affine over S , is necessarily of finite presentation over S , i.e. one is indeed under hypothesis (b) of 2.0.

Then, under the hypotheses of (ii), condition (i_1) of 2.1 is automatically satisfied by 0.16 and 0.17; moreover every section of X_{S_n} over S_n lifts to a section of $X_{S_{n+1}}$ over S_{n+1} , by the definition of “smooth over S ” in case (a), and by 0.17 in case (b). Consequently, if f and f' are two liftings of f_0 , one may suppose step by step that $f_n = f'_n$, by lifting the inner automorphism whose existence is asserted by part (ii) of the theorem, which completes the proof.

Arguing in the same way, one obtains, taking Remark 2.3 into account:

Corollary 2.6. *Let S be a scheme, $\mathcal{I}$ a nilpotent ideal defining the closed subscheme S_0 , Y an S -group flat over S and affine, X an S -group smooth over S .*

- (i) *If, for every $n \geq 0$, one has $H^2(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$, every morphism of S_0 -groups $f_0: Y_0 \rightarrow X_0$ lifts to a morphism of S -groups $f: Y \rightarrow X$.*

⁽⁵⁸⁾**N.D.E.** The original stated: “Suppose that one has one of the following two properties: (a) X is an S -group locally of finite presentation; (b) $X = \text{Aut}_S(E)$ where E is of finite presentation over S . Then f_0 extends to a morphism of S -groups $Y \rightarrow X$ if and only if it extends to a morphism of S -functors $Y \rightarrow X$.” and indicated: “(i) follows step by step from part (ii) of the theorem.”. This proof does not seem sufficient: if for example $I^3 = 0$ and $J = I^2$, and if $f: Y \rightarrow X$ is a morphism of S -functors lifting f_0 , then f_0 lifts to a morphism of S_J -groups $g_J: Y_J \rightarrow X_J$; next, does g_J lift to a morphism of S -functors $g: Y \rightarrow X$? In any event, this assertion (i) is not used in the sequel, where X is everywhere supposed smooth over S .

⁽⁵⁹⁾**N.D.E.** In what follows, one has slightly modified the original and set it out in detail.

- (ii) If, for every $n \geq 0$, one has $H^1(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$, two such liftings are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0 .

Now one has the following lemma:

Lemma 2.7. *Let S be an affine scheme, G an affine S -group, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_S$ -module, $\mathcal{L}$ a locally free $\mathcal{O}_S$ -module. Suppose that one has an operation of G on $\mathcal{F}$ in the sense of Exposé I, which defines an operation of G on $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L}$ ⁽⁶⁰⁾. Denote by Λ the ring of S , and by L the Λ -module defining $\mathcal{L}$ (which is therefore a projective module). One has a canonical isomorphism*

$$H^*(G, \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L}) \simeq H^*(G, \mathcal{F}) \otimes_{\Lambda} L.$$

PROOF. ⁽⁶¹⁾ Indeed, denote by $\mathcal{A}$ the $\mathcal{O}_S$ -algebra $\mathcal{A}(G)$ and consider the complex $\mathcal{C}$ of quasi-coherent $\mathcal{O}_S$ -modules:

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow \cdots$$

By I, 5.3, $H^*(G, \mathcal{F})$ (resp. $H^*(G, \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L})$) is the cohomology of the complex $\Gamma(S, \mathcal{C})$ (resp. $\Gamma(S, \mathcal{C} \otimes_{\mathcal{O}_S} \mathcal{L})$). Now, as S is affine, one has (cf. EGA I, 1.3.12)

$$\Gamma(S, \mathcal{C} \otimes_{\mathcal{O}_S} \mathcal{L}) \simeq \Gamma(S, \mathcal{C}) \otimes_{\Lambda} L.$$

As L is a projective Λ -module (hence flat), one also has $H^*(\Gamma(S, \mathcal{C}) \otimes_{\Lambda} L) \simeq H^*(\Gamma(S, \mathcal{C})) \otimes_{\Lambda} L$, whence the announced result.

Using the lemma, one transforms 2.6 into:

Corollary 2.8. *Let S be an affine scheme, $\mathcal{I}$ a nilpotent ideal on S defining the closed subscheme S_0 . Suppose that the $\mathcal{I}^{n+1}/\mathcal{I}^{n+2}$ are locally free over S_0 . Let Y be an S -group flat over S and affine, X an S -group smooth over S , and $f_0: Y_0 \rightarrow X_0$ a morphism of S -groups.*

- (i) *If $H^2(Y_0, \text{Lie}(X_0/S_0)) = 0$, f_0 lifts to a morphism of S -groups $Y \rightarrow X$.*
- (ii) *If $H^1(Y_0, \text{Lie}(X_0/S_0)) = 0$, two such liftings are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0 .*

In particular, setting $Y = X$:

Corollary 2.9. *Let S and S_0 be as above. Let X be an S -group smooth over S and affine.*

- (i) *If $H^1(X_0, \text{Lie}(X_0/S_0)) = 0$, every endomorphism of X over S inducing the identity on X_0 is the inner automorphism defined by a section of X over S inducing the unit section of X_0 over S_0 .*
- (ii) *If*

$$H^2(X_0, \text{Lie}(X_0/S_0)) = 0,$$

every S_0 -automorphism of X_0 extends to an S -automorphism of X .⁽⁶²⁾

Remark 2.10. The assertions concerning the H^1 have converses by the theorem. Let us point out the following as an example: if $S = \mathbb{I}_{S_0}$ is the scheme of dual numbers over S_0 (II, 2.1) and if X is a flat S -group such that every automorphism of X over S inducing the identity on S_0 is the inner automorphism defined by a section of X over S inducing the unit section of X_0 over S_0 , then $H^1(X_0, \text{Lie}(X_0/S_0)) = 0$.⁽⁶³⁾

⁽⁶⁰⁾N.D.E. Where $\mathcal{L}$ is equipped with the trivial action of G .

⁽⁶¹⁾N.D.E. One has set out in detail (and simplified) the proof of the original (the latter moreover invoked the isomorphisms $\mathcal{H}^n(\mathcal{C} \otimes_{\mathcal{O}_S} \mathcal{L}) \simeq \mathcal{H}^n(\mathcal{C}) \otimes_{\mathcal{O}_S} \mathcal{L}$ and $H^n(\Gamma(-)) \simeq \Gamma(\mathcal{H}^n(-))$, where the $\mathcal{H}^n(\mathcal{C})$ denote the cohomology sheaves of the complex $\mathcal{C}$).

⁽⁶²⁾N.D.E. Indeed, let f_0 be an S_0 -automorphism of X_0 and let g_0 be its inverse. By 2.8 (i), f_0 (resp. g_0) lifts to an S -endomorphism f (resp. g) of X . Then $g \circ f$ and $f \circ g$ are endomorphisms of X inducing the identity on X_0 ; they are therefore, by 0.7, S -automorphisms of X , and the same is therefore true of f and g .

⁽⁶³⁾N.D.E. This is used in XXIV, 1.13.

Corollary 2.11. *Let $S, \mathcal{I}$ and $\mathcal{J}$ be as in 2.1. Let Y be an S -group scheme flat over S , X an S -group scheme, $f: Y \rightarrow X$ a morphism of S -groups. The set of morphisms from Y to X deduced from f by conjugation by elements $x \in X(S)$ inducing the unit of $X(S_{\mathcal{J}})$ is isomorphic to the quotient*

$$E = \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) / \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})^{\text{ad}(Y_0)},$$

where the second group is formed of the $\mathcal{O}_{S_0}$ -morphisms $\omega_{X_0/S_0}^1 \rightarrow \mathcal{J}$ which, by every base change $S' \rightarrow S_0$, give morphisms $\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{S'} \rightarrow \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{S'}$ invariant under the action of $Y_0(S')$ on the first factor.

PROOF. By 2.1 (iii), one knows that the set sought is isomorphic to $\Gamma(L_0)/H^0(Y_0, L_0)$. Now $\Gamma(L_0) = \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})$ and $H^0(Y_0, L_0)$ is evidently none other than $\Gamma(L_0)^{\text{ad}(Y_0)}$ in the sense of the preceding statement.

Corollary 2.12. *Under the conditions of 2.11, suppose moreover ω_{X_0/S_0}^1 locally free of finite rank. Then*

$$E \simeq \Gamma(S_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) / H^0(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

PROOF. ⁽⁶⁴⁾ Indeed, if ω_{X_0/S_0}^1 is locally free of finite rank, one has $\text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \simeq \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$.

Corollary 2.13. *Suppose moreover Y_0 diagonalizable. Then*

$$E \simeq \Gamma(S_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) / \Gamma(S_0, \text{Lie}(X_0/S_0)^{\text{ad}(Y_0)} \otimes_{\mathcal{O}_{S_0}} \mathcal{J})$$

where $\text{Lie}(X_0/S_0)^{\text{ad}(Y_0)}$ may be constructed as the factor of the decomposition of I , 4.7.3, corresponding to the zero character of Y_0 .

PROOF. Indeed, if $Y_0 \simeq D_{S_0}(M)$, one has by *loc. cit.* a decomposition into a direct sum:

$$\text{Lie}(X_0/S_0) = \text{Lie}(X_0/S_0)_0 \oplus \bigoplus_{\substack{m \in M \\ m \neq 0}} \text{Lie}(X_0/S_0)_m.$$

On tensoring by $\mathcal{J}$, one finds an analogous decomposition for $\text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$, whence the relation

$$H^0(Y_0, \text{Lie}(X_0/S_0) \otimes \mathcal{J}) \simeq \Gamma(S_0, \text{Lie}(X_0/S_0)_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

Corollary 2.14. *Suppose moreover S_0 affine. Then*

$$E \simeq \Gamma\left(S_0, [\text{Lie}(X_0/S_0) / \text{Lie}(X_0/S_0)^{\text{ad}(Y_0)}] \otimes_{\mathcal{O}_{S_0}} \mathcal{J}\right).$$

3. Infinitesimal extensions of a group scheme

Still in the notations of no. 0 ($S, \mathcal{I}, \mathcal{J}$, etc.), let us give ourselves an S -scheme X and suppose $X_{\mathcal{J}}$ endowed with a group structure. We propose to find the S -group structures on X inducing on $X_{\mathcal{J}}$ the given structure.

From now on, we suppose X flat over S . Let $\mathcal{C}$ be the category of S -schemes flat over S . One therefore has $X \in \text{Ob } \mathcal{C}$. We shall denote by Y , resp. M , the functor on $\mathcal{C}$ defined by X^+ , resp. L'_X . The canonical morphism $p_X: X \rightarrow X^+$ defines a morphism of $\widehat{\mathcal{C}}$

$$p: X \rightarrow Y$$

⁽⁶⁴⁾**N.D.E.** One has added the sentence which follows.

and the operation of L'_X on X in $(\widehat{\text{Sch}})_{/S}$ defines an operation of M on X in $\widehat{\mathcal{C}}$. One checks at once that X does indeed thus become formally principal homogeneous under M_Y over Y (cf. 0.2 (i) and 0.4).

The operation of X^+ on L'_X defined in 0.8 (denoted Ad in *loc. cit.*) defines an operation, denoted f , of Y on M . One knows, on the other hand (0.5), that

$$\text{Hom}_{\widehat{\mathcal{C}}}(Z, M) \simeq \text{Hom}_{S_0}(Z_0, L_0), \quad Z \in \text{Ob } \mathcal{C},$$

where L_0 is the functor defined in 0.5.

Lemma 3.1.

- (i) *The condition $(+)_n$ of 1.3 is satisfied for every positive integer n .*
- (ii) *If one lets the S_0 -group X_0 act on the S_0 -functor L_0 via its adjoint representation, one has a canonical isomorphism*

$$H^*(X_0, L_0) \simeq H^*(Y, M),$$

(the first cohomology is computed in $(\text{Sch})_{/S_0}$, the second in $\mathcal{C}$).

PROOF. The two parts of the lemma result from the relation:

$$\begin{aligned} \text{Hom}_{\widehat{\mathcal{C}}}(Y, M) &\simeq \text{Hom}_{(\widehat{\text{Sch}})_{/S_0}}(X^+ \times_S S_0, L_0) \\ &\simeq \text{Hom}_{S_0}(X_0, L_0) \\ &\simeq \text{Hom}_{\widehat{\mathcal{C}}}(X, M), \end{aligned}$$

which comes at once from the definition of M as a $\prod_{S_0/S}$. This relation being more generally satisfied on replacing X, Y by X^n, Y^n , one deduces from it that every morphism $X^n \rightarrow M$ factors uniquely through Y^n , which entails $(+)_n$. One also deduces from it the relation $C^*(Y, M) = C^*(X_0, L_0)$, which entails (ii).

We may therefore apply the constructions of 1.3. In particular:

Lemma 3.2. *Let $P: X \times_S X \rightarrow X$ be a morphism. For P to induce the group law of $X_{\mathcal{J}}$, it is necessary and sufficient that P be an admissible composition law (cf. 1.3.2) on X .*

PROOF. Indeed, for P to induce the group law of $X_{\mathcal{J}}$, it is necessary and sufficient that P lift the group law of X^+ , or again that of Y . One therefore only has to show that every morphism P lifting the group law of $X_{\mathcal{J}}$ satisfies the identity $(++)$ of 1.3.2 (ii), which is exactly what one saw in 0.8.

Proposition 3.3. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let X be an S -scheme flat, and quasi-compact or locally of finite presentation over S . Let $P: X \times_S X \rightarrow X$ be a composition law on X . For P to be a group law, it is necessary and sufficient that the following two conditions be satisfied:*

- (i) *P is associative.*
- (ii) *P induces on $X_0 = X \times_X S_0$ a group law.*

PROOF. These conditions are evidently necessary. Let us show that they are sufficient. Suppose first that $X \rightarrow S$ has a section. As $X(S')$ is then nonempty for each $S' \rightarrow S$, it suffices⁽⁶⁵⁾ to show that, for every $x \in X(S')$, the translations *on the left and on the right* by x are isomorphisms of $X_{S'}$.⁽⁶⁶⁾

⁽⁶⁵⁾**N.D.E.** One has corrected the original, by removing the inadequate reference to an exercise of Bourbaki on semigroups (cf. [BAI], § I.2, Exercises 9 to 13) and by indicating the role of the translations on the left and on the right; see the following N.D.E.

⁽⁶⁶⁾**N.D.E.** Let E be a nonempty set equipped with an associative composition law, such that every left translation ℓ_x is bijective; let us fix $x_0 \in E$. There exists a unique $e \in E$ such that $x_0 \cdot e = x_0$; then $x_0 \cdot e \cdot x = x_0 \cdot x$ entails $e \cdot x = x$, for every $x \in E$. On the other hand, for every x there exists a unique x' such that $x \cdot x' = e$. Suppose moreover that there exists $b \in E$ such that the right translation r_b is injective. Then,

One may evidently suppose $S' = S$; the translation t in question induces on X_0 a translation t_0 of X_0 , which is therefore an automorphism since X_0 is a group. One concludes by flatness (SGA 1 III 4.2).⁽⁶⁷⁾

No longer assuming now that X has a section over S , suppose that there exists an $S' \rightarrow S$ such that $X_{S'}$ has a section over S' . Then $X_{S'}$ is an S' -group by what one has just seen; let us consider its unit section e' . The inverse image of e' under $\text{pr}_i: S' \times_S S' \rightarrow S'$ ($i = 1, 2$) is the unit section of $X_{S''}$ for the group law inverse image of $P_{S'}$ under pr_i . But as P is “defined over S ”, these two group laws coincide, hence also their unit section. One therefore has $\text{pr}_1^*(e') = \text{pr}_2^*(e')$.

If $S' \rightarrow S$ is a descent morphism (cf. Exp. IV no. 2), there will exist a section of X giving e' by extension of the base, and one will be done. As X_X has a section over X (the diagonal section), one sees that it now suffices to prove that $X \rightarrow S$ is a descent morphism. Now it is flat and surjective, and quasi-compact or locally of finite presentation, hence covering for (fpqc), hence a descent morphism (Exp. IV, no. 6).

Remark. In fact the hypothesis that $X \rightarrow S$ be quasi-compact or locally of finite presentation is superfluous, by virtue of the following result, which the reader will prove as an exercise on Exposé IV:

Under the conditions of the text on S and S_0 , if $X \rightarrow S$ is a flat morphism and $X_0 \rightarrow S_0$ a morphism covering for (fpqc), then $X \rightarrow S$ is a descent morphism.

Lemma 3.4. *For two admissible composition laws on X to be equivalent (cf. 1.3.5), it is necessary and sufficient that they be deduced from one another by an automorphism of X over S inducing the identity on $X_{\mathcal{J}}$.*

PROOF. Indeed, the morphisms constructed in 1.3.1 are exactly those of the preceding statement (by 0.7).⁽⁶⁸⁾

Taking all the preceding results into account, Proposition 1.3.6 gives:

Theorem 3.5. *Let S be a scheme, $\mathcal{I}$ and $\mathcal{J}$ two ideals on S such that $\mathcal{I} \supset \mathcal{J}$, $\mathcal{I} \cdot \mathcal{J} = 0$, and S_0 and $S_{\mathcal{J}}$ the closed subschemes of S which they define. Let X be an S -scheme flat over S (and locally of finite presentation or quasi-compact over S), and X_0 and $X_{\mathcal{J}}$ the schemes obtained by base change. Suppose $X_{\mathcal{J}}$ endowed with a structure of $S_{\mathcal{J}}$ -group, and denote by L_0 the S_0 -functor in abelian groups defined by the formula*

$$\text{Hom}_{S_0}(T, L_0) = \text{Hom}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T)$$

on which X_0 acts via its adjoint representation.

- (i) *For there to exist an S -group structure on X inducing the given structure on $X_{\mathcal{J}}$, it is necessary and sufficient that the following conditions be satisfied:*

for every x , the equality $x \cdot e \cdot b = x \cdot b$ gives $x \cdot e = x$ (i.e. e is a neutral element), and $x \cdot x' \cdot x = x = x \cdot e$ entails $x' \cdot x = e$, i.e. x' is the inverse of x on the left and on the right, hence E is a group.

Note that the hypothesis “ r_b injective” is necessary: on every set E one may define a composition law by $x \cdot y = y$, for all $x, y \in E$; then every left translation is the identity (whence the associativity of the law), but for every y one has $r_y(E) = \{y\}$, hence E is not a group if $|E| > 1$.

⁽⁶⁷⁾**N.D.E.** Since X and X_0 have the same underlying topological space and t_0 is an automorphism, t is a homeomorphism, hence an *affine* morphism, cf. Exp. VI_B, 2.9.1 or EGA IV₄, 18.12.7.1. It therefore suffices to see that if J is a nilpotent ideal of a ring Λ , and $\phi: A \rightarrow B$ a morphism of Λ -algebras, with B flat over Λ , such that $\phi \otimes_{\Lambda} (\Lambda/J)$ is bijective, then ϕ is bijective. By the “nilpotent Nakayama lemma”, ϕ is surjective; moreover, B being flat over Λ , one also has $\text{Ker}(\phi) \otimes_{\Lambda} (\Lambda/J) = 0$, whence $\text{Ker}(\phi) = 0$, hence ϕ is bijective.

⁽⁶⁸⁾**N.D.E.** Indeed, by the proof of 0.7, the S -endomorphisms of X inducing the identity on $X_{\mathcal{J}}$ are the automorphisms $m \cdot \text{id}_X$, for m running through $M(X) = \text{Hom}_S(X, M)$ (for every $S' \rightarrow S$ and $x \in X(S')$, one has $(m \cdot \text{id}_X)(x) = m(x) \cdot x$). Now, by the proof of 3.1, each $m: X \rightarrow M$ factors uniquely as a morphism h from $Y = X^+$ to M , and therefore $m \cdot \text{id}_X$ is the automorphism u_h introduced in 1.3.1. The lemma then follows from the definition of the equivalence, cf. 1.3.4 and 1.3.5.

- (i₁) *There exists a morphism of S -schemes $X \times_S X \rightarrow X$ inducing the group law of $X_{\mathcal{J}}$.*
- (i₂) *A certain obstruction class belonging to $H^3(X_0, L_0)$ (defined canonically by the datum of X and of the group law of $X_{\mathcal{J}}$) is zero.*
- (ii) *If the conditions of (i) are satisfied, the set E of group laws on X inducing the given law of $X_{\mathcal{J}}$ is a principal homogeneous set under $Z^2(X_0, L_0)$, and E modulo the S -automorphisms of X inducing the identity on $X_{\mathcal{J}}$, is a principal homogeneous set under $H^2(X_0, L_0)$.*

PROOF. ⁽⁶⁹⁾ Indeed, every morphism of S -schemes $f: X \times_S X \rightarrow X$ inducing the group law of $X_{\mathcal{J}}$ is, by 3.2, an *admissible* composition law on X ; then, by 1.3.6 (i), the existence of an *associative* admissible composition law $P: X \times_S X \rightarrow X$ is equivalent to the vanishing of a certain class $c(f) \in H^3(X_0, L_0)$, and in this case, by 3.3, P is a group law. This proves (i), and (ii) then follows from 3.3 and 1.3.6 (ii).

Remark 3.5.1. ⁽⁷⁰⁾ If μ, μ' are group laws on X inducing the given law of $X_{\mathcal{J}}$, one therefore obtains a cocycle $\delta(\mu, \mu') \in Z^2(X_0, L_0)$, the sign convention chosen being that $\mu' = \delta(\mu, \mu') \cdot \mu$, i.e. for every $S' \rightarrow S$ and $x, y \in X(S')$,

$$\mu'(x, y) = \delta(\mu, \mu')(x_0, y_0) \cdot \mu(x, y).$$

(71)

One will denote by $\delta(\mu, \mu')$ the image of $\delta(\mu, \mu')$ in $H^2(X_0, L_0)$. Finally, if X equipped with the group law μ (resp. μ') is denoted simply by X (resp. X'), one will write $\delta(X, X')$ instead of $\delta(\mu, \mu')$, and likewise for $\delta(X, X')$.

Remark 3.6. Let X_J be an S_J -scheme *smooth* over S_J and *affine*. By 0.15, there exists, up to isomorphism, a unique S -scheme X , *smooth* over S , and reducing to X_J . If X_J is equipped with a structure of S_J -group, it follows from 0.16 that condition (i₁) is automatically satisfied. Moreover, by 0.6 the definition of L_0 simplifies and one obtains:

Corollary 3.7. *Let $S, \mathcal{I}$ and $\mathcal{J}$ be as in 3.1. Let X_J be an S_J -group smooth over S_J and affine.*

- (i) *The set of S -groups smooth over S and reducing to X_J , up to isomorphism (inducing the identity on X_J), is empty or a principal homogeneous set under the group*

$$H^2(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

- (ii) *There exists an S -group smooth over S reducing to X_J if and only if a certain obstruction in*

$$H^3(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J})$$

is zero.

One deduces from this, as usual, the following corollaries:

Corollary 3.8. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal $\mathcal{I}$. Let X_0 be an S_0 -group smooth over S and affine.*

- (i) *If $H^2(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$ for every $n \geq 0$, two S -groups smooth over S and reducing to X_0 are isomorphic (by an isomorphism inducing the identity on X_0).*

⁽⁶⁹⁾N.D.E. One has added what follows.

⁽⁷⁰⁾N.D.E. One has added this remark, analogous to 4.5.1, in order to introduce the notation $\delta(\mu, \mu')$ (or $\delta(X, X')$), used in 4.38; consequently, one has also added in 3.5 (ii) above the part concerning E itself.

⁽⁷¹⁾N.D.E. One has conformed to the sign conventions of the original, so as to have in 4.38 (5) the equality $\partial^1 d(X, X') = \delta(X, X')$ (see also N.D.E. (54)).

- (ii) If $H^3(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$ for every $n \geq 0$, there exists an S -group smooth over S , reducing to X_0 .

Corollary 3.9. *Let S be an affine scheme and S_0 a closed subscheme defined by a nilpotent ideal $\mathcal{I}$. Suppose the $\mathcal{I}^{n+1}/\mathcal{I}^{n+2}$ are locally free over S_0 . Let X_0 be an S_0 -group smooth and affine over S_0 .*

- (i) *If $H^2(X_0, \mathcal{L}ie(X_0/S_0)) = 0$, two S -groups smooth over S and reducing to X_0 are isomorphic.*
 (ii) *If $H^3(X_0, \mathcal{L}ie(X_0/S_0)) = 0$, there exists an S -group smooth over S reducing to X_0 .*

Corollary 3.10. *Let S_0 be a scheme and $S = \mathbf{I}_{S_0}$ the scheme of dual numbers over S_0 . Let X_0 be an S_0 -group smooth over S_0 . In order that every S -group Y , smooth over S , such that Y_0 is S_0 -isomorphic to X_0 , be S -isomorphic to $X = X_0 \times_{S_0} S$, it is necessary and sufficient that $H^2(X_0, \mathcal{L}ie(X_0/S_0)) = 0$.⁽⁷²⁾*

Indeed, by virtue of 3.5, the set of classes, up to an isomorphism of S -groups “inducing the identity on X_0 ”, of such groups Y , is in bijection with $H^2(X_0, \mathcal{L}ie(X_0/S_0))$, hence the set of classes, up to an arbitrary isomorphism of S -groups, is in bijection with

$$H^2(X_0, \mathcal{L}ie(X_0/S_0))/\Gamma_0,$$

where

$$\Gamma_0 = \text{Aut}_{S_0\text{-gr.}}(X_0)$$

(which operates in an evident way on the H^2). The conclusion follows at once from this.⁽⁷³⁾

4. Infinitesimal extensions of closed subgroups

Let us first state a result valid in an arbitrary *abelian category*.

Lemma 4.1. *Let $0 \rightarrow A' \xrightarrow{i} A \xrightarrow{p} A'' \rightarrow 0$ be an exact sequence, $\phi: A' \rightarrow Q$ a morphism and $\pi: A'' \rightarrow P$ an epimorphism with kernel C . Let E be the set (up to isomorphism) of the quadruples (B, f, g, h) such that the sequence*

$$0 \rightarrow Q \xrightarrow{f} B \xrightarrow{g} P \rightarrow 0$$

is exact and the diagram below is commutative:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' \longrightarrow 0 \\ & & \downarrow \phi & & \downarrow h & & \downarrow \pi \\ 0 & \longrightarrow & Q & \xrightarrow{f} & B & \xrightarrow{g} & P \longrightarrow 0. \end{array}$$

- (i) *In order that E be non-empty, it is necessary and sufficient that the image in $\text{Ext}^1(C, Q)$ of the element A of $\text{Ext}^1(A'', A')$ be zero.*
 (ii) *Under these conditions, E is a principal homogeneous set under the abelian group $\text{Hom}(C, Q)$.*

⁽⁷²⁾**N.D.E.** This is used in XXIV, 1.13.

⁽⁷³⁾**N.D.E.** Indeed, $\text{Aut}_{S_0\text{-gr.}}(X_0)$ operates by group automorphisms on the abelian group $H^2(X_0, \mathcal{L}ie(X_0/S_0))$, hence the orbit of 0 is the singleton $\{0\}$; consequently the quotient set is a singleton if and only if $H^2(X_0, \mathcal{L}ie(X_0/S_0)) = \{0\}$.

Let us introduce the amalgamated sum $B' = A \amalg^{A'} Q$. One then has a commutative diagram in which the rows are exact:⁽⁷⁴⁾

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' \longrightarrow 0 \\ & & \downarrow \phi & & \downarrow & & \downarrow \\ 0 & \longrightarrow & Q & \xrightarrow{j} & B' & \longrightarrow & A'' \longrightarrow 0, \end{array}$$

and it is clear that the category of solutions of the problem posed is canonically isomorphic to the category of solutions of the corresponding problem for the sequence

$$0 \longrightarrow Q \longrightarrow B' \longrightarrow A'' \longrightarrow 0$$

and the morphisms id_Q and $\pi: A'' \rightarrow P$.⁽⁷⁵⁾ In this case, the set E is in bijection with the set of subobjects N of B' such that $B' \rightarrow A''$ induces an isomorphism of N with the kernel C of $A'' \rightarrow P$, that is, the set of morphisms $\varepsilon: C \rightarrow B'$ lifting the canonical morphism $C \rightarrow A''$. The abelian group $G = \text{Hom}(C, Q)$ acts on E by $g \cdot \varepsilon = g + \varepsilon$ (addition in $\text{Hom}(C, B')$), and if $E \neq \emptyset$ this makes E into a principal homogeneous set under G .

One deduces from this:

Proposition 4.2. ⁽⁷⁶⁾ Let S be a scheme, S_J the closed subscheme defined by a quasi-coherent ideal $\mathcal{J}$ of square zero, X an S -scheme, $\mathcal{F}$ an $\mathcal{O}_X$ -module, $X_J = X \times_S S_J$, $\mathcal{F}_J = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_J}$, and $\mathcal{G}_J = \mathcal{F}_J / \mathcal{H}_J$ a quotient module of $\mathcal{F}_J$. Suppose given a morphism of $\mathcal{O}_{X_J}$ -modules

$$f: \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J \longrightarrow \mathcal{Q}.$$

Let $\mathcal{E}$ be the sheaf of sets on X defined as follows: for each open U of X , $\mathcal{E}(U)$ is the set of quotient modules $\mathcal{G}$ of $\mathcal{F}|_U$, such that $\mathcal{G}/\mathcal{J}\mathcal{G} = \mathcal{G}_J|_U$ and such that there exists an isomorphism

$$h: \mathcal{J}\mathcal{G} \xrightarrow{\sim} \mathcal{Q}|_U$$

making the following diagram commutative:

$$\begin{array}{ccc} \mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{G}_J|_U) & \xrightarrow{f|_U} & \mathcal{Q}|_U \\ & \searrow \text{can.} \nearrow h & \\ & \mathcal{J}\mathcal{G} & \end{array}$$

(h is then unique, since $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{G}_J|_U) \rightarrow \mathcal{J}\mathcal{G}$ is an epimorphism). Then $\mathcal{E}$ is a formally principal homogeneous sheaf under the sheaf of commutative groups

$$\mathcal{A} = \text{Hom}_{\mathcal{O}_X}(\mathcal{H}_J, \mathcal{Q}) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{H}_J, \mathcal{Q}).$$

PROOF. If $\mathcal{E}(U) = \emptyset$ there is nothing to prove; one may therefore suppose that $\mathcal{E}(U)$ contains an element $\tilde{\mathcal{G}}$. Then, in the diagram below, h is an isomorphism and all the arrows are epimorphisms:

$$\begin{array}{ccccc} \mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{F}_J|_U) & \longrightarrow & \mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{G}_J|_U) & \xrightarrow{f|_U} & \mathcal{Q}|_U \\ \text{can.} \downarrow & & \text{can.} \downarrow & \nearrow h & \\ \mathcal{J}\mathcal{F}|_U & \longrightarrow & \mathcal{J}\tilde{\mathcal{G}} & & \end{array}$$

⁽⁷⁴⁾**N.D.E.** One has $\text{Coker}(j) = B' \amalg^Q 0 = A \amalg^{A'} 0 = A''$, and one sees that $\text{Ker}(j) \simeq \text{Ker}(i) = 0$ by arguing “as if C were a category of modules”; for a proof solely in terms of arrows, see for example [Fr64], Th. 2.5.4 (*).

⁽⁷⁵⁾**N.D.E.** In what follows, one has replaced A by B' , and set out the end of the argument.

⁽⁷⁶⁾**N.D.E.** One has rewritten the statement so as to lie exactly in the setting of the application which is made of it in 4.3; on the other hand, one has set out the proof, following the indications given by M. Demazure.

Thus, the morphism $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{F}_J|_U) \rightarrow \mathcal{Q}|_U$ induces an epimorphism (necessarily unique) $\phi: \mathcal{J}\mathcal{F}|_U \rightarrow \mathcal{Q}|_U$, and if $\mathcal{G}$ is an $\mathcal{O}_U$ -module such that $\mathcal{G}/\mathcal{J}\mathcal{G} = \mathcal{G}_J|_U$ and one has a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{J}\mathcal{F}|_U & \longrightarrow & \mathcal{F}|_U & \longrightarrow & \mathcal{F}_J|_U \longrightarrow 0 \\ & & \phi \downarrow & & \downarrow & & \pi \downarrow \\ 0 & \longrightarrow & \mathcal{Q}|_U & \longrightarrow & \mathcal{G} & \xrightarrow{p_J} & \mathcal{G}_J|_U \longrightarrow 0 \end{array}$$

(where p_J is the projection $\mathcal{G} \rightarrow \mathcal{G}/\mathcal{J}\mathcal{G} = \mathcal{G}_J|_U$, so that $\mathcal{Q}|_U = \text{Ker}(p_J) = \mathcal{J}\mathcal{G}$), then one may identify $\mathcal{G}$ with a quotient module of $\mathcal{F}|_U$. Consequently, by 4.1 (ii), the set $\mathcal{E}(U)$ is a principal homogeneous set under the abelian group

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{H}_J, \mathcal{Q})(U) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{H}_J, \mathcal{Q})(U).$$

□

Proposition 4.3. (TDTE IV 5.1) *Let S be a scheme, S_J the closed subscheme defined by a quasi-coherent ideal $\mathcal{J}$ of square zero, X an S -scheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, $X_J = X \times_S S_J$, $\mathcal{F}_J = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_J}$. Let $\mathcal{G}_J = \mathcal{F}_J/\mathcal{H}_J$ be a quasi-coherent quotient module of $\mathcal{F}_J$, flat over S_J .*

For every open U of X , let $\mathcal{E}(U)$ be the set of quasi-coherent quotient modules $\mathcal{G}^{(77)}$ of $\mathcal{F}|_U$, flat over S and such that $\mathcal{G}/\mathcal{J}\mathcal{G} \simeq \mathcal{G}_J|_U$. Then the $\mathcal{E}(U)$ form a sheaf of sets $\mathcal{E}$ on X , which is formally principal homogeneous under the sheaf of commutative groups

$$\mathcal{A} = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{H}_J, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J).$$

PROOF. Let us denote by $\pi: X \rightarrow S$ the structural morphism. Let U be an open of X and $\mathcal{G}$ an $\mathcal{O}_U$ -module flat over S and such that $\mathcal{G}/\mathcal{J}\mathcal{G} \simeq \mathcal{G}_J|_U$. Then, for every $x \in U$, $\mathcal{G}_x$ is a module flat over the local ring $\mathcal{O}_{S,s}$ (where $s = \pi(x)$), and therefore the morphism

$$\mathcal{J}_s \otimes_{\mathcal{O}_{S,s}} (\mathcal{G}/\mathcal{J}\mathcal{G})_x = \mathcal{J}_s \otimes_{\mathcal{O}_{S,s}} \mathcal{G}_x \rightarrow (\mathcal{J}\mathcal{G})_x$$

is bijective; one therefore has an exact sequence

$$0 \rightarrow \mathcal{J} \otimes_{\mathcal{O}_S} (\mathcal{G}_J|_U) \rightarrow \mathcal{G} \rightarrow \mathcal{G}_J|_U \rightarrow 0$$

and since $\mathcal{J} \otimes_{\mathcal{O}_S} (\mathcal{G}_J|_U)$ and $\mathcal{G}_J|_U$ are quasi-coherent $\mathcal{O}_U$ -modules, so is $\mathcal{G}$ (cf. EGA III, 1.4.17).

Conversely, since one has supposed $\mathcal{G}_J$ flat over S_J , if $\mathcal{G}$ is a quasi-coherent $\mathcal{O}_U$ -module such that $\mathcal{G}/\mathcal{J}\mathcal{G} \simeq \mathcal{G}_J$ and the morphism $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J \rightarrow \mathcal{J}\mathcal{G}$ is bijective, then $\mathcal{G}$ is flat over S , by the “fundamental criterion of flatness” (cf. SGA 1 IV, 5.5⁽⁷⁸⁾).

Consequently, the set $\mathcal{E}(U)$ considered here coincides with the set considered in 4.2, taking for f the identity morphism of $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J$, and the conclusion therefore follows from 4.2. □

⁽⁷⁹⁾One keeps the preceding notations. Let Y_J be a closed subscheme of X_J , defined by a quasi-coherent ideal $\mathcal{I}_{Y_J}$. One supposes Y_J flat over S_J . Then, applying 4.3 to $\mathcal{F} = \mathcal{O}_X$ and $\mathcal{G}_J = \mathcal{O}_{Y_J} = \mathcal{O}_{X_J}/\mathcal{I}_{Y_J}$, one obtains the following corollary.

Corollary 4.3.1. *Let $S, S_J, \mathcal{J}, X, X_J, Y_J$ and $\mathcal{I}_{Y_J}$ be as above; one supposes Y_J flat over S_J . Let us denote by $\mathcal{A}_J$ the sheaf of commutative groups*

$$\text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J})$$

⁽⁷⁷⁾N.D.E. In order to lighten the statement, one has added here the hypothesis that $\mathcal{G}$ be quasi-coherent, and deferred to the proof the remark that this hypothesis is automatically verified; one has set out the proof accordingly.

⁽⁷⁸⁾N.D.E. See also [BAC], § III.5, th. 1.

⁽⁷⁹⁾N.D.E. One has set out what follows and added corollary 4.3.1. On the other hand, one recalls that “pseudo-torsor” is synonymous with “formally principal homogeneous”.

on X_J and $\mathcal{A} = i_*(\mathcal{A}_J)$, where i is the immersion $X_J \hookrightarrow X$.

For every open U of X , let $\mathcal{E}(U)$ be the set of closed subschemes Y of U , flat over S , such that $Y \times_S S_J = Y_J \cap U$. Then $\mathcal{E}$ is an $\mathcal{A}$ -pseudo-torsor.

If moreover a Y exists locally (i.e., if every $x \in X$ has an open neighborhood U such that $\mathcal{E}(U) \neq \emptyset$), then $\mathcal{E}$ is an $\mathcal{A}$ -torsor. Now one knows (see for example EGA IV₄, 16.5.15) that the $\mathcal{A}$ -torsors on X are parametrized by the group $H^1(X, \mathcal{A}) = H^1(X_J, \mathcal{A}_J)$, and that $\mathcal{E}$ possesses a global section (i.e., $\mathcal{E}(X) \neq \emptyset$) if and only if the cohomology class corresponding to $\mathcal{E}$ is zero. One therefore obtains the:

Corollary 4.4. *Let $S, S_J, \mathcal{J}, X, X_J, Y_J$ and $\mathcal{I}_{Y_J}$ be as above; one supposes Y_J flat over S_J . Let E be the set of closed subschemes Y of X , flat over S , such that $Y \times_S S_J = Y_J$.*

(i) *The set E is empty or a principal homogeneous set under the abelian group*

$$H^0(X, \mathcal{A}) = H^0(X_J, \mathcal{A}_J) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J}).$$

(ii) *In order that E be non-empty, it is necessary and sufficient that the following two conditions be satisfied:*

- (a) *There exists locally on X a solution of the problem.*
- (b) *A certain obstruction is zero, which lies in*

$$H^1(X_J, \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J})).$$

Complement 4.4.1. ⁽⁸⁰⁾Let us keep the notations of 4.4 and suppose that E contains an element Y . Let us denote by $\mathcal{I}_Y$ the ideal of $\mathcal{O}_X$ defining Y , and by $\mathcal{I}_{Y_J}$ its image in $\mathcal{O}_{X_J}$. Then, as one has seen in the proof of 4.2, one has a commutative diagram

$$\begin{array}{ccc} \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{X_J} & \longrightarrow & \mathcal{J} \mathcal{O}_X \\ \downarrow & \nearrow & \downarrow \\ \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J} & \xrightarrow{\sim} & \mathcal{J} \mathcal{O}_Y \end{array}$$

hence an epimorphism of $\mathcal{O}_X$ -modules $\phi: \mathcal{J} \mathcal{O}_X \rightarrow \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J}$; let us denote by $\mathcal{K}$ its kernel. Then, for every element Y' of E , the morphism $\mathcal{O}_X \rightarrow \mathcal{O}_{Y'}$ factors through $\mathcal{O}_X/\mathcal{K}$ (which is the amalgamated sum B' of the proof of lemma 4.1) and, denoting by $\mathcal{I}_{Y'}$ the ideal of Y' in $\mathcal{O}_X$, one has a commutative diagram:

$$\begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ & & \mathcal{I}_{Y'}/\mathcal{K} & \xrightarrow{\sim} & \mathcal{I}_{Y_J} & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & (\mathcal{J} \mathcal{O}_X)/\mathcal{K} & \longrightarrow & \mathcal{O}_X/\mathcal{K} & \longrightarrow & \mathcal{O}_{X_J} \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J} & \longrightarrow & \mathcal{O}_{Y'} & \longrightarrow & \mathcal{O}_{Y_J} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

Thus, replacing X by the closed subscheme defined by $\mathcal{K}$, one reduces to $\mathcal{K} = 0$. Then, giving Y' is equivalent to giving the sub- $\mathcal{O}_X$ -module $\mathcal{I}_{Y'}$ of $\mathcal{O}_X$, mapping bijectively onto $\mathcal{I}_{Y_J}$ by the projection $p: \mathcal{O}_X \rightarrow \mathcal{O}_{X_J}$; let us denote by $f': \mathcal{I}_{Y_J} \xrightarrow{\sim} \mathcal{I}_{Y'}$ (resp. $f: \mathcal{I}_{Y_J} \xrightarrow{\sim} \mathcal{I}$) the inverse isomorphism. Then $f' - f$ is an element of

$$\text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J}) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \mathcal{O}_Y)$$

⁽⁸⁰⁾**N.D.E.** One has added this complement, useful for proving point (ii) of proposition 4.8.

which one will denote by $d(Y', Y)$. (Note that $d(Y, Y') = -d(Y', Y)$.)

For our fixed Y and variable Y' , let us consider the morphism

$$\mathcal{I}_{Y'} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y = \mathcal{O}_X / \mathcal{I}_Y ;$$

since the composite with $\mathcal{O}_Y \longrightarrow \mathcal{O}_{Y_{\mathcal{J}}}$ is zero, one knows that it takes values in $\mathcal{J}\mathcal{O}_Y = \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}$. More precisely, if V is an open of X , x' a section of $\mathcal{I}_{Y'}$ over V and $x_{\mathcal{J}}$ its image in $\Gamma(V, \mathcal{I}_{Y_{\mathcal{J}}})$, then

$$x' = f'(x_{\mathcal{J}}) = f(x_{\mathcal{J}}) + (f' - f)(x_{\mathcal{J}}) = f(x_{\mathcal{J}}) + d(Y', Y)(x_{\mathcal{J}}).$$

Consequently: the morphism $\mathcal{I}_{Y'} \longrightarrow \mathcal{J}\mathcal{O}_Y$ is given by $d(Y', Y)$.

4.5.0. — ⁽⁸¹⁾ Let us keep the notations of 4.3.1 and 4.4 and carry out a certain number of transformations: $\mathcal{I}_{Y_{\mathcal{J}}} / \mathcal{I}_{Y_{\mathcal{J}}}^2$ is a quasi-coherent $\mathcal{O}_{X_{\mathcal{J}}}$ -module annihilated by $\mathcal{I}_{Y_{\mathcal{J}}}$, hence is the direct image of a quasi-coherent $\mathcal{O}_{Y_{\mathcal{J}}}$ -module denoted $\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$, and called the *conormal sheaf* to $Y_{\mathcal{J}}$ in $X_{\mathcal{J}}$. ⁽⁸²⁾ As $\mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}$ is annihilated by $\mathcal{I}_{Y_{\mathcal{J}}}$, the sheaf of commutative groups $\mathcal{A}_{\mathcal{J}}$ of 4.3.1 is identified with

$$\mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{I}_{Y_{\mathcal{J}}} / \mathcal{I}_{Y_{\mathcal{J}}}^2, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}) = \mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}),$$

whence, for every $i \geq 0$,

$$\mathcal{H}^i(X_{\mathcal{J}}, \mathcal{A}_{\mathcal{J}}) = \mathcal{H}^i(Y_{\mathcal{J}}, \mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}})) .$$

⁽⁸³⁾ One may then drop the hypothesis “ Y closed”, as follows. Let us first note that every open $U_{\mathcal{J}}$ of $X_{\mathcal{J}}$ arises by base change from the open subscheme U of X having the same underlying topological space as $U_{\mathcal{J}}$. Let now $Y_{\mathcal{J}}$ be a closed subscheme of $U_{\mathcal{J}}$, flat over $S_{\mathcal{J}}$, and $\mathcal{I}_{Y_{\mathcal{J}}}$ the quasi-coherent ideal of $\mathcal{O}_{U_{\mathcal{J}}}$ defining $Y_{\mathcal{J}}$. If $Y_{\mathcal{J}}$ lifts to a subscheme Y of X , then Y , having the same underlying topological space as $Y_{\mathcal{J}}$, is a closed subscheme of U ; consequently, the obstruction to lifting $Y_{\mathcal{J}}$ to a subscheme, flat over S , of X or of U is “the same”, and it lies in

$$\mathcal{H}^1(Y_{\mathcal{J}}, \mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}})) .$$

Finally, let us return to the notations of no. 0: let $\mathcal{I}$ be a quasi-coherent ideal of $\mathcal{O}_S$ such that $\mathcal{J} \subset \mathcal{I}$ and $\mathcal{I}\mathcal{J} = 0$, and let S_0 be the closed subscheme of $S_{\mathcal{J}}$ defined by $\mathcal{I}$. For every S -scheme Z , one denotes $Z_{\mathcal{J}} = Z \times_S S_{\mathcal{J}}$ and $Z_0 = Z \times_S S_0$. Then, as $\mathcal{J}$ is annihilated by $\mathcal{I}$, one has, with the notations of 4.4,

$$\mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}} = \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}$$

$$\mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}) = \mathcal{H}\text{om}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}),$$

etc. One therefore obtains:

Proposition 4.5. *Let S be a scheme, S_0 and $S_{\mathcal{J}}$ the closed subschemes defined by the quasi-coherent ideals $\mathcal{I}$ and $\mathcal{J}$, such that $\mathcal{I} \supset \mathcal{J}$ and $\mathcal{I} \cdot \mathcal{J} = 0$. Let X be an S -scheme and $Y_{\mathcal{J}}$ a subscheme of $X_{\mathcal{J}}$, flat over $S_{\mathcal{J}}$. Let $\mathcal{A}_0$ be the $\mathcal{O}_{Y_0}$ -module defined by*

$$\mathcal{A}_0 = \mathcal{H}\text{om}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}).$$

- (i) *In order that there exist a subscheme Y of X , reducing to $Y_{\mathcal{J}}$, flat over S , it is necessary and sufficient that the following conditions be satisfied:*
 - (a) *Such a Y exists locally on X .*

⁽⁸¹⁾**N.D.E.** One has added the numbering 4.5.0 in order to mark the return to the original.

⁽⁸²⁾**N.D.E.** One has corrected the following sentence.

⁽⁸³⁾**N.D.E.** One has set out what follows in more detail.

- (b) *A certain obstruction in $R^1\Gamma(Y_0, \mathcal{A}_0)$ is zero.*⁽⁸⁴⁾
 (ii) *Under these conditions, the set of the Y satisfying the required conditions is a principal homogeneous space under the commutative group $\Gamma(Y_0, \mathcal{A}_0)$.*

Remark 4.5.1.⁽⁸⁵⁾ It follows from 4.5(ii) that there is given, for every pair (Y, Y') of subschemes⁽⁸⁶⁾ of X , flat over S and reducing to $Y_{\mathcal{J}}$, a “deviation”

$$d(Y', Y) \in \Gamma(Y_0, \mathcal{A}_0) = \text{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0});$$

the sign convention adopted in 4.4.1 being that $d(Y', Y)$ corresponds to the morphism of $\mathcal{O}_X$ -modules

$$\mathcal{I}_{Y'} \hookrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y$$

(which takes values in $\mathcal{J}\mathcal{O}_Y \simeq \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}$ and factors through $\mathcal{I}_{Y'_{\mathcal{J}}} = \mathcal{I}_{Y_{\mathcal{J}}}$ and then through $\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$).

Remark 4.6.⁽⁸⁷⁾ If X is flat over S and if $Y_{\mathcal{J}}$ is *locally a complete intersection* in $X_{\mathcal{J}}$, then condition (a) is always satisfied, and every Y flat over S lifting $Y_{\mathcal{J}}$ is then locally a complete intersection in X . If moreover Y_0 is *affine*, condition (b) is likewise satisfied.

Definition 4.6.1. (cf. SGA 6, VII 1.1) Let B be a commutative ring, $f: E \rightarrow B$ a B -linear morphism, where E is a free B -module of finite rank d , and I the ideal $f(E)$ (if one chooses a basis of E , f is given by a d -tuple $(f_1, \dots, f_d)$ of elements of B , and I is the ideal generated by the f_i). The *Koszul complex* $K_{\bullet}(f)$ is the graded B -module $\bigwedge_B^{\bullet} E$, equipped with the differential (of degree -1)

$$x_1 \wedge \dots \wedge x_i \longmapsto \sum_{j=1}^i (-1)^{j-1} f(x_j) x_1 \wedge \dots \wedge \widehat{x_j} \wedge \dots \wedge x_i.$$

One therefore has an augmented chain complex (B/I being in degree -1):

$$\dots \longrightarrow \bigwedge^2 E \longrightarrow E \xrightarrow{f} B \longrightarrow B/I \longrightarrow 0$$

which by definition is exact in degree 0, since $I = f(E)$. One says that f is *regular* if $K_{\bullet}(f)$ is acyclic in degrees > 0 , i.e. if the augmented complex above is a *resolution* of $C = B/I$.

In this case, the proof of SGA 6, VII 1.2.b) shows that the C -modules I^n/I^{n+1} ($n \in \mathbf{N}$) are free, I/I^2 being of rank d .

Definition 4.6.2. (cf. SGA 6, VII 1.4) Let X be a scheme, Y a subscheme, U an open of X such that Y is a closed subscheme of U , defined by the quasi-coherent ideal $\mathcal{I}_Y$.

One says that Y is *locally a complete intersection* in X if $Y \hookrightarrow X$ is a *regular immersion* in the sense of SGA 6, VII 1.4, i.e. if for every $y \in Y$ there exist an affine open neighbourhood V of y in U , a finite free $\mathcal{O}_V$ -module $\mathcal{E}$, and a regular morphism $f: \mathcal{E} \rightarrow \mathcal{O}_V$ with image $\mathcal{I}_Y|_V$, i.e. such that $K_{\bullet}(f)$ is a resolution of $\mathcal{O}_{Y \cap V}$.

This implies that the immersion $Y \hookrightarrow X$ is *locally of finite presentation*, and, by 4.6.1, that the conormal sheaf $\mathcal{N}_{Y/X} = \mathcal{I}_Y/\mathcal{I}_Y^2$ is a *finite locally free* $\mathcal{O}_Y$ -module.

⁽⁸⁴⁾**N.D.E.** Here, one has denoted by $R^1\Gamma(Y_0, \mathcal{A}_0)$ the “coherent” cohomology group $H^1(Y_0, \mathcal{A}_0)$ of the $\mathcal{O}_{Y_0}$ -module $\mathcal{A}_0$, in order to distinguish it from the “Hochschild” cohomology groups $H^i(Y_0, \mathcal{M}_0)$ (Y_0 an S_0 -group, $\mathcal{M}_0$ an $\mathcal{O}_{S_0}$ -module) which will be considered starting from 4.16.

⁽⁸⁵⁾**N.D.E.** One has placed this remark here; it replaces Remark 4.7 of the original.

⁽⁸⁶⁾**N.D.E.** One has corrected “closed subschemes” to “subschemes”.

⁽⁸⁷⁾**N.D.E.** One has kept, for the record, Remark 4.6 of the original, in which the definition of “locally a complete intersection” does not appear. One has added afterwards the “right” definition, taken from SGA 6, VII 1.4 (which *replaces* that of EGA IV₄, 16.9.2), and the proof of the three results stated in the remark.

Lemma 4.6.3. ⁽⁸⁸⁾ *Let A be a ring, J an ideal of A of square zero, $\bar{A} = A/J$, B a flat A -algebra, E a free B -module of finite rank, $f: E \rightarrow B$ a morphism of B -modules. One assumes that the morphism $g: \bar{E} = E \otimes_A \bar{A} \rightarrow \bar{B} = B \otimes_A \bar{A}$ induced by f is regular and that $\bar{B}/g(\bar{E})$ is flat over $\bar{A}$.*

Then f is regular and $B/f(E)$ is flat over A .

PROOF. Set $C = B/f(E)$ and $\bar{C} = C \otimes_A \bar{A} = \bar{B}/g(\bar{E})$. First, the $\bigwedge_B^i(E)$ are free B -modules, hence flat A -modules, since B is flat over A . As $\bigwedge_B^\bullet E \otimes_A \bar{A} \simeq \bigwedge_{\bar{A}}^\bullet \bar{E}$, one therefore obtains an exact sequence of complexes

$$0 \rightarrow J \otimes_A \bigwedge_B^\bullet E \rightarrow \bigwedge_B^\bullet E \rightarrow \bigwedge_{\bar{A}}^\bullet \bar{E} \rightarrow 0.$$

Moreover, as $J^2 = 0$, one has $J \otimes_A M = J \otimes_{\bar{A}} \bar{A} \otimes_A M$ for every A -module M . Denoting by $--\rightarrow$ the augmentation arrows, and by d the rank of E , one therefore obtains the following double complex, in which the rows are exact:

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & J \otimes_A \bigwedge_B^d E & \longrightarrow & \bigwedge_B^d E & \longrightarrow & \bigwedge_{\bar{A}}^d \bar{E} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \vdots & & \vdots & & \vdots \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & J \otimes_A E & \longrightarrow & E & \longrightarrow & \bar{E} \longrightarrow 0 \\ & & \downarrow \text{id} \otimes g & & \downarrow f & & \downarrow g \\ 0 & \longrightarrow & J \otimes_A B & \longrightarrow & B & \longrightarrow & \bar{B} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & J \otimes_A \bar{C} & \longrightarrow & C & \longrightarrow & \bar{C} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}.$$

Moreover, the right-hand and left-hand columns are exact, since $K_\bullet(g)$ is a resolution of $\bar{C}$ and since the latter is flat over $\bar{A}$. Thus, considering the long exact sequence in homology associated with the exact sequence of non-augmented complexes, one obtains that $K_\bullet(f)$ is acyclic in degrees > 0 , and that in degree 0 one has an exact sequence

$$0 \rightarrow J \otimes_A \bar{C} \rightarrow C \rightarrow \bar{C} \rightarrow 0.$$

Hence C is flat over A , by the “fundamental criterion of flatness” (cf. [BAC], §III.5, th. 1). $\square$

Lemma 4.6.4. ⁽⁸⁸⁾ *Let A be a commutative ring, J a nilpotent ideal, $N \subset M$ A -modules such that M/N is flat over A . If $x_1, \dots, x_n$ are elements of N whose images generate the image $\bar{N}$ of N in M/JM , then they generate N .*

Indeed, denote by N' the submodule of N generated by the x_i , and $Q = N/N'$. Then the morphism $N' \otimes (A/J) \rightarrow \bar{N}$ is surjective. On the other hand, as M/N is flat over A , the morphism $N \otimes (A/J) \rightarrow \bar{N}$ is bijective. One therefore obtains that $Q \otimes (A/J) = 0$, whence $Q = 0$ by the “nilpotent Nakayama lemma” (one has $Q = JQ = J^2Q = \dots = 0$).

One may now prove the following:

Proposition 4.6.5. ⁽⁸⁸⁾ *Let $S, \mathcal{I}, \mathcal{J}$ and $X, Y_{\mathcal{J}}$ be as in 4.5. Assume moreover X flat over S and $Y_{\mathcal{J}}$ locally a complete intersection in $X_{\mathcal{J}}$.*

⁽⁸⁸⁾**N.D.E.** In order to prove the results stated in Remark 4.6, one has added Lemmas 4.6.3, 4.6.4 and Proposition 4.6.5, as well as Remark 4.6.6.

a) Then, condition (a) of 4.5 (i) is satisfied; moreover, every Y flat over S lifting $Y_{\mathcal{J}}$ is locally a complete intersection in X .

b) If moreover Y_0 is affine, condition (b) of loc. cit. is likewise satisfied.

PROOF. The first assertion of (a) follows from Lemma 4.6.3; the second then results from Lemma 4.6.4. On the other hand, the hypothesis implies (cf. 4.6.2) that $\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$ is a finite locally free $\mathcal{O}_{Y_{\mathcal{J}}}$ -module, hence the $\mathcal{O}_{Y_0}$ -module

$$\mathcal{A}_0 = \mathcal{H}om_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0})$$

is quasi-coherent (cf. EGA I, 1.3.12), whence $R^1\Gamma(Y_0, \mathcal{A}_0) = 0$ if Y_0 is affine. $\square$

Remark 4.6.6. ⁽⁸⁸⁾ Let us end this paragraph with the following example, which shows that, under the hypotheses of Lemma 4.6.3, if (g_1, g_2) is a regular sequence generating the ideal $I = g(E)$, it does not necessarily lift to a regular sequence in B .

Let k be a field, $A = k[X, Y]$, denote by $k\varepsilon$ the A -module $A/(X, Y)$ (i.e. $P \cdot \varepsilon = P(0, 0)\varepsilon$ for every $P \in A$), and let $A = \bar{A} \oplus k\varepsilon$, where $J = k\varepsilon$ is an ideal of square zero. One has $A/J = \bar{A}$.

The algebra $B = A \otimes_k k[Z, T]$ is free over A , hence flat; one has $\bar{B} = k[X, Y, Z, T]$. Set $g_1 = XZ - YT$ and $g_2 = XZ - 1$. As the polynomial g_1 is irreducible, $\bar{B}/(g_1)$ is an integral domain, and hence (g_1, g_2) is a regular sequence in $\bar{B}$, generating the ideal $I = (XZ - 1, YT - 1)$. Thus

$$\bar{C} = \bar{B}/I = k[X, Y, X^{-1}, Y^{-1}] = \bar{A}[X^{-1}, Y^{-1}]$$

is a flat $\bar{A}$ -algebra (and also a flat A -algebra). But every lifting in B of g_1 is of the form $XY - ZT + \lambda\varepsilon$, where $\lambda \in k[Z, T]$, hence annihilates ε .

4.7. — One has deleted here Remark 4.7, placed in 4.5.1.

Remark 4.8.0. ⁽⁸⁹⁾ Let S be a scheme, S' a closed subscheme, X an S -scheme, Y a sub- S -scheme of X , and $X' = X \times_S S'$, $Y' = Y \times_S S'$. Then one has a *surjective* morphism of $\mathcal{O}_{Y'}$ -modules

$$\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\text{surj.}} \mathcal{N}_{Y'/X'}.$$

Indeed, after replacing X by a suitable open, one may assume that Y is closed, defined by an ideal $\mathcal{I}_Y$ of $\mathcal{O}_X$; then the image of $\mathcal{I}_Y$ in $\mathcal{O}_{X'}$ is the ideal $\mathcal{I}_{Y'}$ defining Y' , and one has a surjective morphism of $\mathcal{O}_{Y'}$ -modules

$$\pi: (\mathcal{I}_Y/\mathcal{I}_Y^2) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\text{surj.}} \mathcal{I}_{Y'}/\mathcal{I}_{Y'}^2.$$

Assume moreover that $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}_Y$ is *flat* over $\mathcal{O}_S$; then the natural morphism

$$\mathcal{I}_Y \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \longrightarrow \mathcal{I}_{Y'}$$

is bijective (cf. EGA IV₂, 2.1.8). One then has the following commutative diagram with exact rows:

$$\begin{array}{ccccccc} \mathcal{I}_Y^2 \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} & \longrightarrow & \mathcal{I}_Y \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} & \longrightarrow & (\mathcal{I}_Y/\mathcal{I}_Y^2) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} & \longrightarrow & 0 \\ \downarrow \text{surj.} & & \downarrow \wr & & \downarrow \pi \text{ surj.} & & \\ 0 \longrightarrow & \mathcal{I}_{Y'}^2 & \longrightarrow & \mathcal{I}_{Y'} & \longrightarrow & \mathcal{I}_{Y'}/\mathcal{I}_{Y'}^2 & \longrightarrow 0 \end{array}$$

⁽⁸⁹⁾**N.D.E.** One has inserted this remark here, used in the proposition which follows; it appeared in 4.10 of the original.

whence one deduces, by the snake lemma:⁽⁹⁰⁾

$$(4.8.0) \quad \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\sim} \mathcal{N}_{Y'/X'} \quad \text{if } Y \text{ is flat over } S.$$

Proposition 4.8. *Let $S, S_0, S_{\mathcal{J}}$ and $\mathcal{I}, \mathcal{J}$ be as in 4.5.⁽⁹¹⁾ Let X be an S -scheme, Y a subscheme of X , and i the immersion $Y \hookrightarrow X$.*

- (i) *For every S -morphism $f: T \rightarrow X$ such that $f_{\mathcal{J}}: T_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$ factors through $Y_{\mathcal{J}}$, one may define an obstruction*

$$(*) \quad c(X, Y, f) \in \text{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T)$$

whose vanishing is equivalent to the existence of a factorization of f through Y .

- (ii) *Let Y' be a second subscheme of X . Assume that $Y'_{\mathcal{J}} = Y_{\mathcal{J}}$ and that Y, Y' are flat over S . One then has isomorphisms (cf. 4.8.0)*

$$\mathcal{J}\mathcal{O}_Y \simeq \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0} \simeq \mathcal{J}\mathcal{O}_{Y'} \quad \text{and} \quad \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_{\mathcal{J}}} \xrightarrow{\sim} \mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$$

whence an isomorphism

$$u: \text{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J}\mathcal{O}_Y) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}, \mathcal{J}\mathcal{O}_{Y'}).$$

Denoting by $i': Y' \rightarrow X$ the canonical immersion and by $d(Y, Y')$ the deviation of 4.5.1, one has:⁽⁹²⁾

$$(**) \quad c(X, Y, i') = u(d(Y, Y')).$$

- (iii) *The canonical morphism $\mathcal{N}_{Y/X} \xrightarrow{D} i^*(\Omega_{X/S}^1)$ (cf. SGA 1 II, formula 4.3)⁽⁹³⁾ induces a morphism*

$$D_0: \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} \rightarrow \Omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_{Y_0}$$

and thus, for every S -morphism $f: T \rightarrow X$ as in (i), a morphism

$$v_{f_0}: \text{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_T) \rightarrow \text{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T),$$

$$a \mapsto a \circ f_0^*(D_0)$$

where above the first group is $\text{Hom}_{X^+}(T, L_X)$, cf. 0.1.5. For $a \in \text{Hom}_{X^+}(T, L_X)$, one has

$$(***) \quad c(X, Y, a \cdot f) - c(X, Y, f) = v_{f_0}(a),$$

where one has denoted by $a \cdot f$ the composite morphism $T \xrightarrow{a \times f} L_X \times_{X^+} X \rightarrow X$.

We shall prove part (i) of the proposition, leaving to the reader the task of (not) verifying the assertions (ii) and (iii); this verification is carried out by reduction to the affine case, then by comparison of the explicit definitions.⁽⁹⁴⁾

Let us therefore prove (i). The morphism $f: T \rightarrow X$ defines a morphism of sheaves of rings $\varphi: \mathcal{O}_X \rightarrow f_*(\mathcal{O}_T)$.⁽⁹⁵⁾ Let U be an open subscheme of X in which Y is closed; as T (resp. $Y_{\mathcal{J}}$) has the same underlying space as $T_{\mathcal{J}}$ (resp. Y), the underlying continuous map

⁽⁹⁰⁾**N.D.E.** In the original, this was indicated in Remark 4.10, under the additional hypothesis that Y' be locally a complete intersection in X' . This hypothesis also appeared, consequently, in the statements 4.12–4.14; it seems in fact superfluous, and one has removed it from the aforementioned statements.

⁽⁹¹⁾**N.D.E.** One has removed the hypothesis that $\mathcal{I}$ be nilpotent, which appears superfluous (cf. the proof).

⁽⁹²⁾**N.D.E.** See also 4.27 further on.

⁽⁹³⁾**N.D.E.** See also EGA IV₄, 16.4.21. Recall that if U is an affine open of X such that $Y \cap U$ is defined by the ideal I of $A = \mathcal{O}_X(U)$, if one denotes by d the differential $A \rightarrow \Gamma(U, \Omega_{X/S}^1)$, and if $x \in I$, then $D(x + I^2)$ is the element $d(x) \otimes 1$ of $\Gamma(U, \Omega_{X/S}^1) \otimes_A (A/I)$.

⁽⁹⁴⁾**N.D.E.** One has carried out these verifications further down.

⁽⁹⁵⁾**N.D.E.** On the one hand, one has removed the hypothesis that $\mathcal{I}$ be nilpotent, i.e. that X_0 have the same underlying topological space as X ; on the other hand, one has set out the following sentence in more detail.

of f sends T into U , and as U is an open of X , φ induces a morphism of sheaves of rings $\mathcal{O}_U = \mathcal{O}_X|_U \rightarrow f_*(\mathcal{O}_T)$, i.e. f factors through U .

Thus, one may restrict oneself to the case where Y is closed, hence defined by a sheaf of ideals $\mathcal{I}_Y$. In order that f factor through Y , it is necessary and sufficient that the composite map $\mathcal{I}_Y \rightarrow \mathcal{O}_X \rightarrow f_*(\mathcal{O}_T)$ be zero. As $f_{\mathcal{J}}$ factors through $Y_{\mathcal{J}}$, the composite map $\mathcal{I}_{Y_{\mathcal{J}}} \rightarrow \mathcal{O}_{X_{\mathcal{J}}} \rightarrow f_*(\mathcal{O}_{T_{\mathcal{J}}})$ is zero. Considering the commutative diagram in which the first row is exact

$$\begin{array}{ccccccc} 0 & \longrightarrow & f_*(\mathcal{J}\mathcal{O}_T) & \longrightarrow & f_*(\mathcal{O}_T) & \longrightarrow & f_*(\mathcal{O}_{T_{\mathcal{J}}}) \\ & & \nwarrow \varphi & & \uparrow \varphi & & \uparrow \varphi_{\mathcal{J}} \\ & & & & \mathcal{O}_X & \longrightarrow & \mathcal{O}_{X_{\mathcal{J}}} \\ & & & & \uparrow & & \uparrow \\ & & & & \mathcal{I}_Y & \longrightarrow & \mathcal{I}_{Y_{\mathcal{J}}} \end{array}$$

one deduces from this that φ sends $\mathcal{I}_Y$ into $f_*(\mathcal{J}\mathcal{O}_T)$.⁽⁹⁶⁾ Since $\mathcal{J}^2 = 0$, it follows that $f_*(\mathcal{J}\mathcal{O}_T)$, viewed as an $\mathcal{O}_X$ -module via φ , is annihilated by $\mathcal{I}_Y$; consequently, φ induces a morphism of $\mathcal{O}_X$ -modules

$$h: i_*(\mathcal{N}_{Y/X}) = \mathcal{I}_Y/\mathcal{I}_Y^2 \rightarrow f_*(\mathcal{J}\mathcal{O}_T).$$

On the other hand, one has cartesian squares

$$\begin{array}{ccccc} T_0 & \xrightarrow{f_0} & X_0 & \xleftarrow{i_0} & Y_0 \\ \tau_{T_0} \downarrow & & \tau_{X_0} \downarrow & & \downarrow \tau_{Y_0} \\ T & \xrightarrow{f} & X & \xleftarrow{i} & Y \end{array}$$

where i_{T_0} etc. are the closed immersions deduced by base change from $S_0 \hookrightarrow S$. As $\mathcal{J}\mathcal{O}_T$ is a quasi-coherent $\mathcal{O}_T$ -module annihilated by $\mathcal{I}$, one has an isomorphism

$$\mathcal{J}\mathcal{O}_T \simeq (\tau_{T_0})_* \tau_{T_0}^*(\mathcal{J}\mathcal{O}_T),$$

whence $f_*(\mathcal{J}\mathcal{O}_T) \simeq (\tau_{X_0})_*(f_0)_* \tau_{T_0}^*(\mathcal{J}\mathcal{O}_T)$. Thus h corresponds, by adjunction, to a morphism of $\mathcal{O}_{T_0}$ -modules

$$h_0: f_0^* \tau_{X_0}^* i_*(\mathcal{N}_{Y/X}) \rightarrow i_{T_0}^*(\mathcal{J}\mathcal{O}_T).$$

Now, $\tau_{X_0}^* i_*(\mathcal{N}_{Y/X}) \simeq (i_0)_* \tau_{Y_0}^*(\mathcal{N}_{Y/X}) = \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}$. Thus, reverting to the abuse of notation $i_{T_0}^*(\mathcal{J}\mathcal{O}_T) = \mathcal{J}\mathcal{O}_T$ constantly used, h_0 is identified with a morphism of $\mathcal{O}_{T_0}$ -modules

$$h_0: f_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}) \rightarrow \mathcal{J}\mathcal{O}_T$$

which is the desired obstruction $c(X, Y, f)$. This proves (i).

When f is the immersion $i': Y \hookrightarrow X$, one sees that $c(X, Y, i')$ arises from the morphism $\mathcal{I}_Y \hookrightarrow \mathcal{O}_X \rightarrow \mathcal{O}_{Y'}$, hence corresponds, by 4.4.1 and 4.5.1, to the class $d(Y, Y')$. This proves (ii).

Let us prove (iii). First, $D: \mathcal{N}_{Y/X} \rightarrow i^*(\Omega_{X/S}^1)$ induces a morphism

$$D_0: \tau_{Y_0}^*(\mathcal{N}_{Y/X}) \rightarrow \tau_{Y_0}^* i^*(\Omega_{X/S}^1) = i_0^* \tau_{X_0}^*(\Omega_{X/S}^1)$$

and, as $X_0 = X \times_S S_0$, one has $\tau_{X_0}^*(\Omega_{X/S}^1) \simeq \Omega_{X_0/S_0}^1$ (cf. EGA IV₄, 16.4.5). One therefore obtains the announced morphism

$$D_0: \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} \rightarrow \Omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_{Y_0}.$$

Finally, one will verify the equality $(***)$ after the remark below.

Remark 4.9. The obstruction $c(X, Y, f)$ is computed locally on T . Let $U = \text{Spec}(C)$ be an affine open of T lying over an affine open $\text{Spec}(A)$ of X , itself lying over an affine open $\text{Spec}(\Lambda)$ of S , let $\mathcal{J} \subset \mathcal{I} \subset \Lambda$ (resp. $\mathcal{I}_Y \subset A$) be the ideals corresponding to $\mathcal{J} \subset \mathcal{I}$ (resp.

⁽⁹⁶⁾**N.D.E.** One has set out what follows in more detail.

to $\mathcal{I}_Y$), let $B = A/\mathcal{I}_Y$ and let $\varphi: A \rightarrow C$ be the morphism of Λ -algebras corresponding to $f: T \rightarrow X$; as $f(T_{\mathcal{J}}) \subset Y_{\mathcal{J}}$ one has $\varphi(\mathcal{I}_Y) \subset \mathcal{J}C$ and hence φ induces a morphism of Λ -algebras $B \rightarrow C/\mathcal{J}C \rightarrow C_0 = C/\mathcal{I}C$. Then the obstruction $c = c(X, Y, f)$ is computed by the following commutative diagram

$$\begin{array}{ccccc} \mathcal{I}_Y & \longrightarrow & A & \xrightarrow{\varphi} & C \\ \downarrow & & & & \uparrow \\ \mathcal{I}_Y/\mathcal{I}_Y^2 & \longrightarrow & \mathcal{I}_Y/\mathcal{I}_Y^2 \otimes_B C_0 & \xrightarrow{c} & \mathcal{J}C \end{array}$$

i.e. it is defined, above the open U , as the unique element of

$$\mathrm{Hom}_{C_0}(\mathcal{I}_Y/\mathcal{I}_Y^2 \otimes_B C_0, \mathcal{J}C) = \mathrm{Hom}_{B_0}(\mathcal{I}_Y/\mathcal{I}_Y^2 \otimes_B B_0, \mathcal{J}C)$$

such that, with the obvious notations, one has $c(x \otimes_B 1) = \varphi(x)$, for every $x \in \mathcal{I}_Y$.

⁽⁹⁷⁾One can now complete the proof of 4.8 (iii). The equality $(***)$ is checked locally on T , so one is reduced to the affine situation described above. Denote by $d_{A/\Lambda}$ the differential $A \rightarrow \Omega_{A/\Lambda}^1$. Then a corresponds, above U , to an element a_U of

$$\mathrm{Hom}_{C_0}(\Omega_{A_0/\Lambda_0}^1 \otimes_{A_0} C_0, \mathcal{J}C) \simeq \mathrm{Hom}_{B_0}(\Omega_{A/\Lambda}^1 \otimes_A B_0, \mathcal{J}C) \simeq \mathrm{Hom}_A(\Omega_{A/\Lambda}^1, \mathcal{J}C).$$

Then, on the one hand, $v_{f_0}(a)$ corresponds above U to the element $a_U \circ D_0$, where D_0 is the morphism of B -modules⁽⁹⁸⁾

$$I_Y/I_Y^2 \rightarrow \Omega_{A/\Lambda}^1 \otimes_A B_0, \quad x + I_Y^2 \mapsto d_{A/\Lambda}(x) \otimes 1.$$

On the other hand (cf. the proof of 0.1.8 and 0.1.9), the morphism of Λ -algebras $\phi': A \rightarrow C$ corresponding to $a \cdot f$ differs from ϕ by the Λ -derivation $A \rightarrow \mathcal{J}C$ associated with a_U , i.e. one has:

$$\phi' = \phi + a_U \circ d_{A/\Lambda} = \phi + a_U \circ (d_{A/\Lambda} \otimes 1).$$

Consequently, writing $c' = c(X, Y, a \cdot f)$, one has for every $x \in I_Y$, writing x for its image in I_Y/I_Y^2 :

$$(c' - c)(x \otimes 1) = a_U(d_{A/\Lambda}(x) \otimes 1) = (a_U \circ D_0)(x) = v_{f_0}(a)(x).$$

This shows that $c' - c = v_{f_0}(a)$.

4.10. — One has deleted remark 4.10 of the original, rendered obsolete by the addition of remark 4.8.0.

4.11. — We now propose to study the following situation. Let $S, S_{\mathcal{J}}$ and S_0 be as in 4.8; one has three S -schemes X, X', T , a subscheme Y of X (resp. Y' of X'), and morphisms $f: T \rightarrow X'$ and $g: X' \rightarrow X$.

$$\begin{array}{ccccc} & & Y' & & Y \\ & & \downarrow i' & & \downarrow i \\ T & \xrightarrow{f} & X' & \xrightarrow{g} & X \end{array}$$

One supposes that by reduction modulo $\mathcal{J}$, this diagram completes to a commutative diagram

$$\begin{array}{ccccc} & & Y'_{\mathcal{J}} & \dashrightarrow & Y_{\mathcal{J}} \\ & & \downarrow i'_{\mathcal{J}} & & \downarrow i_{\mathcal{J}} \\ T_{\mathcal{J}} & \xrightarrow{f_{\mathcal{J}}} & X'_{\mathcal{J}} & \xrightarrow{g_{\mathcal{J}}} & X_{\mathcal{J}} \end{array}$$

⁽⁹⁷⁾**N.D.E.** One has added what follows.

⁽⁹⁸⁾**N.D.E.** Cf. N.D.E. (93).

One therefore has, by 4.8, obstructions:

$$\begin{aligned} c(X, Y, g \circ i') &\in \operatorname{Hom}_{\mathcal{O}_{Y'_0}}(i'^*_0 g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_{Y'}), \\ c(X', Y', f) &\in \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J}\mathcal{O}_T), \\ c(X, Y, g \circ f) &\in \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T), \end{aligned}$$

whose relations one seeks to calculate.⁽⁹⁹⁾

Lemma 4.12. *Suppose Y' flat over S , so that $\mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y'_0} = \mathcal{J}\mathcal{O}_{Y'}$.*

(i) *One has a natural morphism*

$$b_{f_0}: \operatorname{Hom}_{\mathcal{O}_{Y'_0}}(i'^*_0 g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_{Y'}) \longrightarrow \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T).$$

(ii) *One also has a natural morphism, functorial in T* ⁽¹⁰⁰⁾

$$a_{g_0}(f_0): \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J}\mathcal{O}_T) \longrightarrow \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T).$$

PROOF. ⁽¹⁰¹⁾Let us first remark that, X, X', Y, Y' being fixed, giving a T as above is equivalent to giving a morphism $(f, f_{\mathcal{J}}): T \longrightarrow X' \times_{X'_+} Y'_+$. Set $Z = X' \times_{X'_+} Y'_+$ and denote by M and M' the Z -functors defined by: for every $f: T \longrightarrow Z$,

$$\begin{aligned} M(T) &= \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T), \\ M'(T) &= \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J}\mathcal{O}_T). \end{aligned}$$

⁽¹⁰²⁾One has in any case a commutative diagram:

$$\begin{array}{ccc} f_0^*(\mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y'_0}) & & \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0} \\ \downarrow & & \downarrow \\ f_0^*(\mathcal{J}\mathcal{O}_{Y'}) & \dashrightarrow & \mathcal{J}\mathcal{O}_T \end{array}$$

and as Y' is flat over S , the left-hand arrow is an isomorphism, so one obtains a morphism of $\mathcal{O}_{T_0}$ -modules $f_0^*(\mathcal{J}\mathcal{O}_{Y'}) \longrightarrow \mathcal{J}\mathcal{O}_T$. The latter induces a morphism of abelian groups

$$\operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), f_0^*(\mathcal{J}\mathcal{O}_{Y'})) \longrightarrow M(T)$$

and, composing with the morphism

$$M(Y') \longrightarrow \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), f_0^*(\mathcal{J}\mathcal{O}_{Y'})),$$

induced by f_0^* , one obtains the morphism $b_{f_0}: M(Y') \longrightarrow M(T)$.

Likewise, one has in any case a diagram

$$\begin{array}{ccc} g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}) & \dashrightarrow & \mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0} \\ \downarrow & & \downarrow \\ g_0^*(\mathcal{N}_{Y_0/X_0}) & \longrightarrow & \mathcal{N}_{Y'_0/X'_0} \end{array}$$

⁽⁹⁹⁾**N.D.E.** From 4.17 onwards, one will apply this to the case where X is an S -group, $g: X \times_S X \longrightarrow X$ the multiplication, Y a subscheme of X such that $Y_{\mathcal{J}}$ is a subgroup of $X_{\mathcal{J}}$, $Y' = Y \times_S Y$, and to the two morphisms $Y^3 \longrightarrow X^2$ which send (y_1, y_2, y_3) to $(y_1 y_2, y_3)$, resp. $(y_1, y_2 y_3)$. In this case, the comparison of the obstructions above will show that the obstruction to Y being a subgroup of X lies in a certain cohomology group (of Hochschild) $H^2(Y_0, N_0)$.

⁽¹⁰⁰⁾**N.D.E.** One has deleted the hypothesis “ Y'_0 locally a complete intersection in X'_0 ”, superfluous by 4.8.0; on the other hand, one has added that $a_{g_0}(f_0)$ is “functorial in T ”, this playing a crucial role in the proof of 4.17.

⁽¹⁰¹⁾**N.D.E.** One has spelled out the proof, in order to bring out the “functoriality in T ” of a_{g_0} .

⁽¹⁰²⁾**N.D.E.** The situation will simplify from 4.16 onwards: one will restrict oneself to schemes *flat* over S , Y will be a flat S -group and $Y' = Y \times_S Y$, and one will then obtain S_0 -functors N_0 and N'_0 .

and as Y' is flat over S , the second vertical arrow is an isomorphism, by 4.8.0. One therefore obtains an $\mathcal{O}_{Y'_0}$ -morphism

$$i_0^* g_0^* (\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}) \longrightarrow \mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}$$

which induces a morphism $a_{g_0}(\text{id}_{Y'}) : M'(Y') \longrightarrow M(Y')$ and, for every $f : T \longrightarrow Z$, a morphism $a_{g_0}(f) : M'(T) \longrightarrow M(T)$ such that one has a commutative diagram

$$\begin{array}{ccc} M'(Y') & \xrightarrow{a_{g_0}(\text{id}_{Y'})} & M(Y') \\ b'_{f_0} \downarrow & & \downarrow b_{f_0} \\ M'(T) & \xrightarrow{a_{g_0}(f_0)} & M(T) \end{array}$$

(where b'_{f_0} is defined as b_{f_0}).

Q.E.D.

Remark 4.12.1. ⁽¹⁰³⁾Let us denote by M_0 and M'_0 the Y'_0 -functors defined by: for every $f : T_0 \longrightarrow Y'_0$,

$$M_0(T) = \text{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^* (\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}),$$

$$M'_0(T) = \text{Hom}_{\mathcal{O}_{T_0}}(f_0^* (\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}).$$

Let us remark at once that $Z_0 = Y'_0$ and that on the category of Z -schemes T which are *flat* over S , M and M' coincide, respectively, with the functors $\prod_{S_0/S} M_0$ and $\prod_{S_0/S} M'_0$. In this case, b_{f_0} is simply the morphism

$$f_0^* : M_0(Y'_0) \longrightarrow M(T_0)$$

induced by f_0 , and for every morphism $u : U \longrightarrow T$, setting $h = f \circ u$, one has a commutative diagram

$$\begin{array}{ccc} M'_0(T_0) & \xrightarrow{a_{g_0}(f_0)} & M_0(T_0) \\ u_0^* \downarrow & & \downarrow u_0^* \\ M'_0(U_0) & \xrightarrow{a_{g_0}(h_0)} & M_0(U_0) \end{array}$$

i.e. a_{g_0} becomes a morphism of functors $\prod_{S_0/S} M'_0 \longrightarrow \prod_{S_0/S} M_0$.

Proposition 4.13. *Suppose Y' flat over S . One then has the formula:*

$$c(X, Y, g \circ f) = a_{g_0}(c(X', Y', f)) + b_{f_0}(c(X, Y, g \circ i')).$$

As the definition of the various obstructions and of the morphisms a_{g_0} and b_{f_0} is local, one sees easily that it suffices to check the given formula when the various schemes in question are affine. Let us therefore write $S = \text{Spec}(\Lambda)$, $S_J = \text{Spec}(\Lambda/J)$, $S_0 = \text{Spec}(\Lambda/I)$, $T = \text{Spec}(C)$, $X = \text{Spec}(A)$, $Y = \text{Spec}(A/I_Y) = \text{Spec}(B)$, $X' = \text{Spec}(A')$, $Y' = \text{Spec}(A'/I_{Y'}) = \text{Spec}(B')$.

One therefore has a diagram of rings and ideals⁽¹⁰⁴⁾

$$\begin{array}{ccccc} & & B' & & B \\ & & \uparrow \pi' & & \uparrow \pi \\ C & \xleftarrow{f} & A' & \xleftarrow{g} & A \\ & & \uparrow & & \uparrow \\ & & I_{Y'} & & I_Y. \end{array}$$

⁽¹⁰³⁾**N.D.E.** One has added this remark, used in the proof of 4.17.

⁽¹⁰⁴⁾**N.D.E.** One has kept the notations of the original, writing $f : A' \longrightarrow C$ and $g : A \longrightarrow A'$ for the ring morphisms corresponding to $f : T \longrightarrow X'$ and $g : X' \longrightarrow X$. This explains the formula $c(X, Y, g \circ f)(x \otimes 1) = f(g(x))$, for $x \in I_Y$.

Let us study the various terms of the formula to be proved. In what follows, if $x \in I_Y$ (resp. $u \in I_{Y'}$), one denotes by x (resp. u) its image in I_Y/I_Y^2 (resp. $I_{Y'}/I_{Y'}^2$); on the other hand, if m belongs to a Λ -module M , one denotes by m_0 its image in $M_0 = M/IM$.

One has seen that $c = c(X, Y, g \circ f)$ is the unique C_0 -morphism $I_Y/I_Y^2 \otimes_B C_0 \rightarrow JC$ such that $c(x \otimes 1) = f(g(x))$, for every $x \in I_Y$.

Let us fix $x \in I_Y$; one has $g(x) \in I_{Y'} + JA'$ since $g_J(Y_J') \subset Y_J$. Let us write $g(x) = x' + \sum \lambda_i a'_i$, with $x' \in I_{Y'}$, $\lambda_i \in J$, $a'_i \in A'$. One therefore has

$$(1) \quad c(X, Y, g \circ f)(x \otimes 1) = f(g(x)) = f(x') + \sum \lambda_i f(a'_i).$$

Let us now consider $a_{g_0}(c(X', Y', f))$. By the definitions set down, it is defined by the diagram

$$\begin{array}{ccccc} & & I_{Y'} & \xrightarrow{f} & C \\ & & \downarrow & & \\ I_{Y'_0}/I_{Y'_0}^2 \otimes_{B'_0} C_0 & \xleftarrow{\sim} & I_{Y'}/I_{Y'}^2 \otimes_{B'} C_0 & \xrightarrow{c(X', Y', f)} & JC \\ & \uparrow g_0 & \uparrow & \nearrow a_{g_0}(c(X', Y', f)) & \\ I_{Y_0}/I_{Y_0}^2 \otimes_{B_0} C_0 & \xleftarrow{\sim} & I_Y/I_Y^2 \otimes_B C_0 & & \end{array}$$

One therefore has $a_{g_0}(c(X', Y', f))(x \otimes 1) = f(u)$, where u is an element of $I_{Y'}$ whose image u in $I_{Y'_0}/I_{Y'_0}^2$ satisfies $u_0 \otimes 1 = g_0(x_0) \otimes 1 = g_0(x_0) \otimes 1$. One may therefore take $u = x'$, and one finds

$$(2) \quad a_{g_0}(c(X', Y', f))(x \otimes 1) = f(x').$$

Let us finally consider $b_{f_0}(c(X, Y, g \circ i'))$. By hypothesis, the morphism of Λ_0 -algebras $f_0: A'_0 \rightarrow C_0$ factors through B'_0 and hence, as $J \otimes_{\Lambda_0} B'_0 \xrightarrow{\sim} JB'$ (B' being flat over Λ), one obtains a morphism of B'_0 -modules $\psi: JB' \rightarrow JC$ such that one has a commutative diagram:

$$\begin{array}{ccccc} J \otimes_{\Lambda_0} A'_0 & \xrightarrow{\text{id} \otimes \pi'} & J \otimes_{\Lambda_0} B'_0 & \xrightarrow{\text{id} \otimes f_0} & J \otimes_{\Lambda_0} C_0 \\ \downarrow & & \downarrow \wr & & \downarrow \\ JA' & \xrightarrow{\pi'} & JB' & \xrightarrow{\psi} & JC. \end{array}$$

Denote by $\phi: JB' \otimes_{B'_0} C_0 \rightarrow JC$ the morphism of C_0 -modules deduced from ψ ; then one has, for every $a' \in A'$, $\lambda \in J$,

$$(\dagger) \quad \phi(\lambda \pi'(a') \otimes 1) = \lambda f(a').$$

Then, $b_{f_0}(c(X, Y, g \circ i'))$ is defined by the commutative diagram:

$$\begin{array}{ccc} I_Y & \xrightarrow{\pi' \circ g} & B' \\ \downarrow & & \downarrow \\ I_Y/I_Y^2 \otimes_B B'_0 & \xrightarrow{c(X, Y, g \circ i')} & JB' \\ \downarrow & & \downarrow \\ I_Y/I_Y^2 \otimes_B C_0 & \xrightarrow{\quad} & JB' \otimes_{B'_0} C_0 \\ & \searrow b_{f_0}(c(X, Y, g \circ i')) & \downarrow \phi \\ & & JC. \end{array}$$

One therefore has at once

$$(3) \quad b_{f_0}(c(X, Y, g \circ i'))(x \otimes 1) = \phi\left(\sum \lambda_i \pi'(a'_i) \otimes 1\right) = \sum \lambda_i f(a'_i),$$

the last equality following from (†) above. The comparison of the three explicit results (1), (2), (3) gives the desired formula.

Corollary 4.14. *Let Y, Y' be two subschemes of X , flat, reducing to $Y_{\mathcal{J}}$; suppose Y_0 locally a complete intersection in X_0 . If $f: T \rightarrow X$ is an S -morphism such that $f_{\mathcal{J}}$ factors through $Y_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$, one has the formula*

$$c(X, Y, f) - c(X, Y', f) = b_{f_0}(d(Y, Y')).$$

Indeed, applying the preceding formula to the diagram

$$\begin{array}{ccccc} & & Y' & & Y \\ & & \downarrow i' & & \downarrow i \\ T & \xrightarrow{f} & X & \xrightarrow{\text{id}} & X \end{array}$$

one finds $c(X, Y, f) - c(X, Y', f) = b_{f_0}(c(X, Y, i'))$. Moreover, by 4.8 (ii), one has $c(X, Y, i') = d(Y, Y')$.

Proposition 4.15. *Let X be an S -group smooth over S and Y an S -subgroup flat and locally of finite presentation over S . Then Y is locally a complete intersection (cf. 4.6.2) in X .*

PROOF. ⁽¹⁰⁵⁾One will show that the immersion $Y \rightarrow X$ is regular in the sense of EGA IV₄, 16.9.2, which implies that it is so also in the sense of 4.6.2, by EGA IV₄, 19.5.1 (moreover, by *loc. cit.*, the two definitions are equivalent if S is locally noetherian). Thus, in what follows, one takes “regular immersion” in the sense of EGA IV₄, 16.9.2. As X and Y are flat and locally of finite presentation over S , then, by EGA IV₄, 19.2.4, it suffices to show that, for every $s \in S$, $Y_s \rightarrow X_s$ is a regular immersion. By EGA IV₄, 19.1.5 (ii), one is reduced to checking the assertion on the geometric fibers of S , hence when S is the spectrum of an algebraically closed field k .

Then, by VI_A, 3.2, the quotient X/Y exists and is smooth, the morphism $\pi: X \rightarrow X/Y$ is flat, and one has a cartesian square

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ \downarrow & & \downarrow \pi \\ e & \xrightarrow{i} & X/Y \end{array}$$

(where e is the image in X/Y of the unit point of X). Thus, by flat base change (cf. EGA IV₄, 19.1.5 (ii)), it suffices to see that i is a regular immersion, which is immediate since the noetherian local ring $\mathcal{O}_{X/Y, e}$ is smooth, hence its maximal ideal is generated by a regular sequence. $\square$

⁽¹⁰⁵⁾**N.D.E.** One has added to the statement the hypothesis that Y be locally of finite presentation over S , and one has given the proof which follows, more direct than the one sketched in the original. For completeness, let us also spell out the latter. As in the proof given above, one first reduces to the case where $S = \text{Spec}(k)$, k being an algebraically closed field. By EGA IV₄, 16.9.10 and 19.3.2, it suffices to see that, for every $y \in Y$, the completion of the local ring $\mathcal{O}_{Y, y}$ is the quotient of a complete noetherian local ring by a regular sequence. By *loc. cit.*, 19.3.3, the set of $y \in Y$ satisfying this property is an open subset U of Y ; as Y is of finite type over k , it suffices to show that U contains every closed point. As Y is a k -group it suffices, by a translation argument, to show that the property holds for the completion of $\mathcal{O}_{Y, e}$, i.e. for the “formal group” $\hat{Y}$ corresponding to Y (cf. Exp. VII_B). Now, as X is smooth, the affine algebra $A(\hat{X})$ is an algebra of formal power series $k[[X_1, \dots, X_n]]$, and one concludes by means of the structure theorem of Dieudonné, which shows that $A(\hat{Y})$ is isomorphic to a quotient $k[[X_1, \dots, X_{r+s}]]/(X_1^{p^{n_1}}, \dots, X_r^{p^{n_r}})$, where p is the characteristic exponent of k and $r + s \leq n$, cf. VII_B, Remark 5.5.2 (b).

4.16. — ⁽¹⁰⁶⁾Let X be an S -group *smooth* over S ; denote by $\mu: X \times_S X \rightarrow X$ its group law. Let us be given an $S_{\mathcal{J}}$ -subgroup $Y_{\mathcal{J}}$ of $X_{\mathcal{J}}$, *flat and locally of finite presentation* over $S_{\mathcal{J}}$. By 4.15, $Y_{\mathcal{J}}$ is locally a complete intersection in X .

Thus, by 4.6.5, every *flat*⁽¹⁰⁷⁾ S -scheme Y lifting $Y_{\mathcal{J}}$ is locally a complete intersection in X . For such a Y one has, by 4.8.0,

$$(4.16.1) \quad \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} = \mathcal{N}_{Y_0/X_0} = \mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}.$$

On the other hand, let us denote by $\varepsilon_0: S_0 \rightarrow Y_0$ the unit section of Y_0 and by $\mathbf{n}_{Y_0/X_0}$ the quasi-coherent $\mathcal{O}_{S_0}$ -module:

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathcal{N}_{Y_0/X_0}).$$

As Y_0 and X_0 are S_0 -groups, one sees easily that $\mathcal{N}_{Y_0/X_0}$ is invariant under the translations (say on the left) of Y_0 , hence⁽¹⁰⁸⁾ is the inverse image under $Y_0 \rightarrow S_0$ of $\mathbf{n}_{Y_0/X_0}$, i.e. one has

$$(4.16.2) \quad \mathcal{N}_{Y_0/X_0} = \mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}.$$

Taking (4.16.1) and (4.16.2) into account, one deduces on the one hand from 4.5 that the set of sub- S -schemes Y of X , *flat* over S , lifting $Y_{\mathcal{J}}$, is empty or a principal homogeneous space under

$$(4.16.3) \quad \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}),$$

and one deduces on the other hand from 4.8 (i) that, for every such Y and every S -morphism $f: T \rightarrow X$ such that $f_{\mathcal{J}}: T_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$ factors through $Y_{\mathcal{J}}$, the obstruction $c(X, Y, f)$ to f factoring through Y is an element of

$$\mathrm{Hom}_{\mathcal{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0});$$

if moreover T is *flat* over S , this last group equals

$$\mathrm{Hom}_{\mathcal{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}).$$

This leads one to introduce the group functor N_0 below:

Definition 4.16.1. Let N_0 be the commutative group S_0 -functor defined by: for every $Z \in \mathrm{Ob}(\mathrm{Sch})/S_0$,

$$(*) \quad \mathrm{Hom}_{S_0}(Z, N_0) = \mathrm{Hom}_{\mathcal{O}_Z}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_Z, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_Z).$$

Then the set of sub- S -schemes Y of X , *flat* over S , lifting $Y_{\mathcal{J}}$, is empty or a principal homogeneous space under

$$\mathrm{Hom}_{S_0}(Y_0, N_0) = C^1(Y_0, N_0).$$

For each such Y , let us consider the following diagram:

$$\begin{array}{ccc} Y \times_S Y & & Y \\ \downarrow (i,i) & & \downarrow i \\ X \times_S X & \xrightarrow{\mu} & X \end{array}$$

and let us denote by $DY = c(X, Y, \mu \circ (i, i))$ the obstruction to $\mu \circ (i, i)$ factoring through Y , i.e. to Y being stable under the group law of X ; by what precedes, DY is an element of

$$N_0(Y_0 \times_{S_0} Y_0) = C^2(Y_0, N_0).$$

⁽¹⁰⁶⁾**N.D.E.** One has reorganized 4.16 by gathering in it, on the one hand, the hypotheses stated at the end of 4.15 and, on the other hand, the definition of the obstruction DY .

⁽¹⁰⁷⁾**N.D.E.** One has corrected the original by adding “flat”.

⁽¹⁰⁸⁾**N.D.E.** See 4.25 below.

Lemma 4.17. ⁽¹⁰⁹⁾ *Let X be an S -group smooth over S , and $Y_{\mathcal{J}}$ a sub- $S_{\mathcal{J}}$ -group of $X_{\mathcal{J}}$, flat and locally of finite presentation over $S_{\mathcal{J}}$. For each subscheme Y of X , flat over S and lifting $Y_{\mathcal{J}}$, let us consider the obstruction defined in 4.16.1:*

$$DY \in \operatorname{Hom}_{S_0}(Y_0 \times_{S_0} Y_0, N_0) = C^2(Y_0, N_0)$$

(i) *In order that Y be a sub- S -group of X , it is necessary and sufficient that $DY = 0$.*

(ii) *If one lets Y_0 act on N_0 by functoriality, starting from the inner automorphisms of Y_0 , then $DY \in Z^2(Y_0, N_0)$.*

(iii) *If Y and Y' are two subschemes of X , flat over S , lifting $Y_{\mathcal{J}}$ (so that the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ is defined, cf. 4.5.1), one has $DY' - DY = \partial^1 d(Y, Y')$.* ⁽¹¹⁰⁾

Let us prove these various assertions in succession.

4.18. — *Proof of 4.17 (i).* By definition, one has $DY = 0$ if and only if Y is stable under the group law of X . Thus $DY = 0$ if Y is a subgroup of X . Conversely, if $DY = 0$, Y is equipped with the induced law μ^Y , which is associative and reduces modulo $\mathcal{J}$ according to the group law on $X_{\mathcal{J}}$; since Y is flat and locally of finite presentation over S , it follows from 3.3 that μ^Y is a group law.

4.19. — *Proof of 4.17 (ii).* This is done by comparing the two values of $u = c(X, Y, \mu^2 \circ (i, i, i))$ computed in the following two diagrams (D_j) , $j = 1, 2$:

$$(D_j) \quad \begin{array}{ccccc} Y \times_S Y \times_S Y & & Y \times_S Y & & Y \\ \downarrow (i, i, i) & & \downarrow (i, i) & & \downarrow i \\ X \times_S X \times_S X & \xrightarrow{f_j} & X \times_S X & \xrightarrow{\mu} & X \end{array}$$

where $f_1 = (1, \pi)$, $f_2 = (\pi, 1)$, and where one denotes by μ^2 the morphism

$$\mu \circ f_1 = \mu \circ f_2: X \times_S X \times_S X \longrightarrow X.$$

⁽¹¹¹⁾ Set $\mu^Y = \mu \circ (i, i)$, $f_j^Y = f_j \circ (i, i, i)$ and $\mu^{2,Y} = \mu^2 \circ (i, i, i)$. For $j = 1, 2$, let us denote by a_j and b_j the morphisms

$$a_j = a_{\mu_0}((f_j^Y)_0) \quad \text{and} \quad b_j = b_{(f_j^Y)_0},$$

associated with the pair of morphisms (f_j^Y, μ) by Lemma 4.12; one therefore has:

$$(\dagger) \quad \begin{cases} \operatorname{Hom}_{\mathcal{O}_{Y_0^3}}((f_j^Y)_0^*(\mathcal{N}_{Y_0 \times Y_0/X_0 \times X_0}), \mathcal{J} \mathcal{O}_{Y_0^3}) \xrightarrow{a_j} \operatorname{Hom}_{\mathcal{O}_{Y_0^3}}((\mu^{2,Y})_0^*(\mathcal{N}_{Y_0/X_0}), \mathcal{J} \mathcal{O}_{Y_0^3}) \\ \operatorname{Hom}_{\mathcal{O}_{Y_0^2}}((\mu^Y)_0^*(\mathcal{N}_{Y_0/X_0}), \mathcal{J} \mathcal{O}_{Y_0^2}) \xrightarrow{b_j} \operatorname{Hom}_{\mathcal{O}_{Y_0^3}}((\mu^{2,Y})_0^*(\mathcal{N}_{Y_0/X_0}), \mathcal{J} \mathcal{O}_{Y_0^3}). \end{cases}$$

Since $\mathcal{N}_{Y_0 \times Y_0/X_0 \times X_0} \simeq \operatorname{pr}_1^* \mathcal{N}_{Y_0/X_0} \oplus \operatorname{pr}_2^* \mathcal{N}_{Y_0/X_0}$ (since X_0 and Y_0 are flat over S_0), and $\mathcal{N}_{Y_0/X_0} \simeq \mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}$, one has:

$$(f_j^Y)_0^*(\mathcal{N}_{Y_0 \times Y_0/X_0 \times X_0}) \simeq (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathcal{O}_{Y_0^3}$$

and, likewise,

$$(\mu^{2,Y})_0^*(\mathcal{N}_{Y_0/X_0}) \simeq \mathbf{n}_{Y_0/X_0} \otimes \mathcal{O}_{Y_0^3} \quad \text{and} \quad (\mu^Y)_0^*(\mathcal{N}_{Y_0/X_0}) \simeq \mathbf{n}_{Y_0/X_0} \otimes \mathcal{O}_{Y_0^2}.$$

Moreover, since Y_0^2 and Y_0^3 are flat over S_0 , $(\dagger)$ is rewritten in the following form:

$$(\ddagger) \quad \begin{cases} a_j: \operatorname{Hom}_{S_0}(Y_0^3, N_0 \oplus N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0) \\ b_j: \operatorname{Hom}_{S_0}(Y_0^2, N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0). \end{cases}$$

⁽¹⁰⁹⁾**N.D.E.** One has modified 4.17 and 4.18, taking into account the additions made in 4.16.

⁽¹¹⁰⁾**N.D.E.** In the original, one finds $DY' - DY = -\partial d(Y, Y')$, but ∂ there is the opposite of the differential ∂^1 defined in I, 5.1.

⁽¹¹¹⁾**N.D.E.** One has slightly modified the notations, and spelled out the beginning of the argument.

Applying 4.13 twice to $c(X, Y, \mu^{2,Y}) = u$, one obtains:

$$a_1(c(X^2, Y^2, f_1)) + b_1(c(X, Y, \mu^Y)) = u = a_2(c(X^2, Y^2, f_2)) + b_2(c(X, Y, \mu^Y)).$$

Now, $c(X, Y, \mu^Y) = DY$ and, since $f_1 = (1, \mu)$ and $f_2 = (\mu, 1)$, one has, with evident notations:

$$c(X^2, Y^2, f_1) = (0, DY) \quad \text{and} \quad c(X^2, Y^2, f_2) = (DY, 0).$$

Thus one obtains:

$$u = a_1((0, DY)) + b_1(DY) = a_2((DY, 0)) + b_2(DY).$$

The first thing that one notices is that b_j is none other than $\text{Hom}_{S_0}((f_j^Y)_0, N_0)$, that is, the morphism deduced from $(f_j^Y)_0$ by functoriality.

The identity above therefore becomes:

$$a_1((0, DY)) - \text{Hom}((\mu, 1), N_0)(DY) + \text{Hom}((1, \mu), N_0)(DY) - a_2((DY, 0)) = 0.$$

One recognizes the two middle terms: they are the parts “ $DY(xy, z)$ ” and “ $DY(x, yz)$ ” of the formula of the 2-coboundary. It therefore only remains to identify the two other terms.

We must first compute the map a_j . Now it arises, by inverse image under $(f_j^Y)_0$, from the morphism of $\mathcal{O}_{Y_0^2}$ -modules

$$P: \mathbf{n}_{Y_0/X_0} \otimes \mathcal{O}_{Y_0^2} \longrightarrow (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathcal{O}_{Y_0^2}$$

induced by the product in Y_0 . Now this morphism is described in the following way: let us consider the vector bundle $\mathbf{V} = \mathbf{V}(\mathbf{n}_{Y_0/X_0})$; P gives by duality a morphism

$$\mathbf{V}(P): \mathbf{V} \times_{S_0} \mathbf{V} \times_{S_0} Y_0 \times_{S_0} Y_0 \longrightarrow \mathbf{V} \times_{S_0} Y_0 \times_{S_0} Y_0$$

which is expressed set-theoretically by

$$\mathbf{V}(P)(u, v, a, b) = (u + \text{Ad}(a)v, ab, b).$$

⁽¹¹²⁾ This is proved exactly like the corresponding fact for Lie algebras, that is, for the module ω_{Y_0/S_0}^1 . One first observes that $\mathbf{V}$ is equipped, by functoriality in Y_0 , with a group structure in the category of vector bundles over S_0 ; by virtue of the lemma already used for Lie algebras (Exposé II, 3.10), this structure coincides with the group structure underlying its structure of $\mathcal{O}_S$ -module. One then sees that $\mathbf{V}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}) = \mathbf{V}(\mathcal{N}_{Y_0/X_0})$ is likewise equipped with a structure of S_0 -group which is none other than the semidirect product of that of $\mathbf{V}$ by that of Y_0 . It only remains to identify the operations of Y_0 on $\mathbf{V}$ in order to establish the desired formula.

Let us now compute the two remaining terms. Let us first consider $a_1((0, DY))$. One computes it by the diagram (where $\mathbf{n}$ denotes $\mathbf{n}_{Y_0/X_0}$):

$$\begin{array}{ccc} \mathbf{n} \otimes \mathcal{O}_{Y_0^2} & \xrightarrow{P} & (\mathbf{n} + \mathbf{n}) \otimes \mathcal{O}_{Y_0^2} \\ \downarrow (f_1^Y)_0^* & & \downarrow (f_1^Y)_0^* \\ \mathbf{n} \otimes \mathcal{O}_{Y_0^3} & \searrow a_1((0, DY)) & (\mathbf{n} + \mathbf{n}) \otimes \mathcal{O}_{Y_0^3} \\ & & \downarrow (0, DY) \\ & & \mathcal{J} \otimes \mathcal{O}_{Y_0^3}. \end{array}$$

Considering now the vector bundles defined by these various modules as so many schemes over S_0 , and taking the points with values in anything whatsoever, one has, denoting by

⁽¹¹²⁾**N.D.E.** One has replaced a, b by ab, b , in order to make it plain that $\mathbf{V}(P)$ arises by inverse image on Y_0^2 from the multiplication morphism $\mathbf{V}_{Y_0} \times_{S_0} \mathbf{V}_{Y_0} \longrightarrow \mathbf{V}_{Y_0}$.

(u, x, y, z) a point of $\mathbf{V}(\mathcal{J}) \times Y_0^3$:

$$\begin{array}{ccc}
 (\mathrm{Ad}(x) DY_{y,z}(u), x, yz) & \longleftarrow & (0 + DY_{y,z}(u), x, yz) \\
 \uparrow & & \uparrow \\
 & & (0 + DY_{y,z}(u), x, y, z) \\
 & & \uparrow \\
 (\mathrm{Ad}(x) DY_{y,z}(u), x, y, z) & \longleftarrow & (u, x, y, z)
 \end{array}$$

One has therefore obtained $a_1((0, DY))(x, y, z) = \mathrm{Ad}(x) DY(y, z)$, which is indeed the first term of the coboundary. One would likewise have $a_2((DY, 0))(x, y, z) = DY(x, y)$, whence⁽¹¹³⁾

$$0 = \mathrm{Ad}(x) DY(y, z) - DY(xy, z) + DY(x, yz) - DY(x, y) = (\partial^2 DY)(x, y, z).$$

4.20. — *Proof of 4.17 (iii).*⁽¹¹⁴⁾ This is done by comparing the two values of $v = c(X, Y, \mu \circ (i', i'))$ computed in the following two diagrams

$$\begin{array}{ccc}
 (*) & & \\
 & Y' & Y \\
 & \downarrow i' & \downarrow i \\
 Y' \times_S Y & \xrightarrow{\mu \circ (i', i')} & X \\
 \\
 (\dagger) & & \\
 & Y \times_S Y & Y \\
 & \downarrow (i, i) & \downarrow i \\
 Y' \times_S Y' & \xrightarrow{(i', i')} X \times_S X \xrightarrow{\mu} & X
 \end{array}$$

Let us denote $f = \mu \circ (i', i')$; then $(*)$ gives

$$(1) \quad v = DY' + f_0^*(c(X, Y, i')).$$

Now $Y'_0 = Y_0$ and f_0 is the multiplication $Y_0^2 \rightarrow Y_0$; one deduces from this that

$$(2) \quad f_0^*(c(X, Y, i'))(x_0, y_0) = c(X, Y, i')(x_0 y_0).$$

Set $c = c(X, Y, i')$; via the identification $N'_0 \simeq N_0 \oplus N_0$, $c(X \times_S X, Y \times_S Y, (i', i'))$ is identified with the pair (c, c) . Then, denoting $h = (i', i')$, $(\dagger)$ gives

$$(3) \quad v = h_0^*(DY) + a_{\mu_0}(c, c).$$

Now h_0 is the identity map of Y_0^2 , whence $h_0^*(DY) = DY$. Finally, by the computation of a_{μ_0} carried out previously, one has, for every $S' \rightarrow S$ and $x_0, y_0 \in Y_0(S'_0)$,

$$(4) \quad a_{\mu_0}(c, c)(x_0, y_0) = c(x_0) + \mathrm{Ad}(x_0)(c(y_0)).$$

One therefore obtains:

$$\begin{aligned}
 (DY' - DY)(x_0, y_0) &= \mathrm{Ad}(x_0)(c(X, Y, i')(y_0)) - c(X, Y, i')(x_0 y_0) + c(X, Y, i')(x_0) \\
 &= (\partial^1 c(X, Y, i'))(x_0, y_0).
 \end{aligned}$$

Since $c(X, Y, i') = d(Y, Y')$ (cf. 4.8 (ii)), this shows that $DY' - DY = \partial^1 d(Y, Y')$.

Theorem 4.21. *Let S be a scheme, $\mathcal{I}$ and $\mathcal{J}$ two ideals⁽¹¹⁵⁾ on S such that $\mathcal{I} \supset \mathcal{J}$ and $\mathcal{I} \cdot \mathcal{J} = 0$. Let X be an S -group smooth over S , and $Y_{\mathcal{J}}$ a sub- $S_{\mathcal{J}}$ -group of $X_{\mathcal{J}}$, flat and*

⁽¹¹³⁾**N.D.E.** One has changed the signs, in order to make them compatible with I 5.1.

⁽¹¹⁴⁾**N.D.E.** One has spelled out the original in what follows.

⁽¹¹⁵⁾**N.D.E.** One has removed the hypothesis that $\mathcal{I}$ be nilpotent, which seems superfluous.

locally of finite presentation over $S_{\mathcal{J}}$. Let us consider the commutative S_0 -group functor N_0 defined by

$$\mathrm{Hom}_{S_0}(T, N_0) = \mathrm{Hom}_{\mathcal{O}_T}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T), \quad T \in \mathrm{Ob}(\mathrm{Sch})_{/S_0},$$

on which Y_0 acts by means of the inner automorphisms of X_0 .

(i) In order that there exist a sub- S -group of X , flat over S , which reduces to $Y_{\mathcal{J}}$, it is necessary and sufficient that the following two conditions be satisfied:

(i₁) There exists a subscheme Y of X , flat over S , lifting $Y_{\mathcal{J}}$ (a condition automatically satisfied if Y_0 is affine, cf. 4.6.5).

(i₂) A certain canonical obstruction, an element of $H^2(Y_0, N_0)$, is zero.

(ii) If the conditions of (i) are satisfied, the set of sub- S -groups Y of X , flat over S and reducing to $Y_{\mathcal{J}}$, is a principal homogeneous set under the group $Z^1(Y_0, N_0)$.⁽¹¹⁶⁾

Indeed, condition (i₁) is necessary. Let us assume it satisfied, and let Y be flat over S and lifting $Y_{\mathcal{J}}$. We must seek a Y' flat over S , also lifting $Y_{\mathcal{J}}$, such that $DY' = 0$,⁽¹¹⁷⁾ cf. 4.17 (i). By 4.17 (iii), this amounts to seeking a $d(Y', Y) \in C^1(Y_0, N_0)$ such that $DY = \partial^1 d(Y', Y)$.⁽¹¹⁸⁾

Let $c \in H^2(Y_0, N_0)$ be the image class of DY , which is a cocycle by 4.17 (ii). It does not depend on the choice of Y by 4.17 (iii), and its vanishing is necessary and sufficient for the existence of a $d(Y', Y)$ satisfying the preceding equation. This proves (i).

If one has now chosen Y such that $DY = 0$, the equation to be solved reads $\partial^1 d(Y', Y) = 0$, which proves (ii).

Remark 4.22. One keeps the notations of 4.21. By 4.15, Y_0 is locally a complete intersection in X_0 , so $\mathcal{N}_{Y_0/X_0}$ is a finite locally free $\mathcal{O}_{Y_0}$ -module, and consequently $\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathcal{N}_{Y_0/X_0})$ is a finite locally free $\mathcal{O}_{S_0}$ -module. Thus, denoting $\mathbf{n}_{Y_0/X_0}^\vee = \mathcal{H}\mathrm{om}_{\mathcal{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{O}_{Y_0})$, one has

$$\mathrm{Hom}_{\mathcal{O}_T}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T) \simeq \mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T$$

for every $T \rightarrow S_0$.⁽¹¹⁹⁾ Consequently, the S_0 -functor N_0 is isomorphic to the functor

$$\mathbf{W}(\mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) \simeq \mathbf{W}(\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{J})).$$

There result the isomorphisms:⁽¹²⁰⁾

$$H^2(Y_0, N_0) \simeq H^2(Y_0, \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{J})) \simeq H^2(Y_0, \mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J}),$$

$$Z^1(Y_0, N_0) \simeq Z^1(Y_0, \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{J})) \simeq Z^1(Y_0, \mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

4.23. — Still under the hypotheses of 4.21, we shall now study how the set of the Y lifting $Y_{\mathcal{J}}$ behaves with respect to conjugation by sections of X . If x is a section of X over S inducing the unit section of $X_{\mathcal{J}}$, the inner automorphism $\mathrm{Int}(x)$ defined by x transforms flat subgroups of X lifting $Y_{\mathcal{J}}$ into flat subgroups of X lifting $Y_{\mathcal{J}}$. Now, under the conditions of 4.21 (ii), the set of these subgroups is principal homogeneous under $Z^1(Y_0, N_0)$; we shall see that there then exists a subgroup Γ of $B^1(Y_0, N_0)$ ⁽¹²¹⁾ such that two subgroups of X , flat over S and lifting $Y_{\mathcal{J}}$, are conjugate (by $x \in X(S)$ inducing the unit element of $X(S_{\mathcal{J}})$) if and only if their “difference” in $Z^1(Y_0, N_0)$ is an element of Γ . In the most favorable cases, we shall

⁽¹¹⁶⁾**N.D.E.** The question whether the preceding set, modulo conjugation by the $x \in X(S)$ inducing the unit element of $X(S_{\mathcal{J}})$, is principal homogeneous under $H^1(Y_0, N_0)$, occupies nos. 4.23 to 4.36.

⁽¹¹⁷⁾**N.D.E.** One has corrected $\partial DY'$ to DY' .

⁽¹¹⁸⁾**N.D.E.** Cf. N.D.E. (110).

⁽¹¹⁹⁾**N.D.E.** One has spelled out what precedes; this shows that the isomorphism which follows is valid without a flatness hypothesis. On the other hand, since 4.16, one has restricted oneself to the S -schemes $f: T \rightarrow S$ flat over S , in order to ensure that the group $\mathrm{Hom}_{\mathcal{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0})$, in which the obstruction $c(X, Y, f)$ lies, coincides with $N_0(T_0)$ (cf. the end of 4.16).

⁽¹²⁰⁾**N.D.E.** with the notations of I 5.3, assuming Y_0 affine over S_0 .

⁽¹²¹⁾**N.D.E.** One has replaced $Z^1(Y_0, N_0)$ by $B^1(Y_0, N_0)$, since the proof shows that Γ is a subgroup of $B^1(Y_0, N_0)$, cf. 4.27–4.29.

show that Γ equals $B^1(Y_0, N_0)$, so that the set of flat subgroups of X lifting $Y_{\mathcal{J}}$, modulo conjugation by the $x \in X(S)$ inducing the unit element of $X(S_{\mathcal{J}})$, is empty or principal homogeneous under $H^1(Y_0, N_0)$ (cf. 4.29 and 4.36).

4.24. — One keeps the notations of 4.21. Let Y be a flat subgroup of X , reducing to $Y_{\mathcal{J}}$. Let us recall that we introduced in 0.5 the functor L_{0X} (resp. L_{0Y}) defined by the following identity with respect to the variable S_0 -scheme T :

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T)$$

(resp. likewise upon replacing X by Y), as well as the functor $L'_X = \prod_{S_0/S} L_{0X}$.

Now one has:

Lemma 4.25. *There exists a canonical exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules*

$$(+) \quad \mathbf{n}_{Y_0/X_0} \xrightarrow{d} \omega_{X_0/S_0}^1 \longrightarrow \omega_{Y_0/S_0}^1 \longrightarrow 0$$

having the following properties:

- (i) By inverse image on Y_0 , d gives the morphism D_0 of 4.8 (iii).
- (ii) If X_0 and Y_0 are smooth over S_0 , then d is injective. Since the two ω^1 are then locally free of finite type, the same holds for $\mathbf{n}_{Y_0/X_0}$, and the sequence is locally split.

PROOF. ⁽¹²²⁾ Let π_0 denote the morphism $Y_0 \rightarrow S_0$. By SGA 1 II, formula (4.3) (see also EGA IV₄, 16.4.21), one has a canonical exact sequence of $\mathcal{O}_{Y_0}$ -modules

$$(\dagger) \quad \mathcal{N}_{Y_0/X_0} \xrightarrow{D_0} \Omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_{Y_0} \longrightarrow \Omega_{Y_0/S_0}^1 \longrightarrow 0.$$

Since this sequence is formed of $(Y_0 \times_S Y_0)$ -equivariant modules and morphisms, its inverse image $(+)$ by ε_0^* is an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules, and $(\dagger)$ is the inverse image of $(+)$ by π_0^* (cf. Exp. I, § 6.8). This proves (i).

Suppose moreover that X_0 and Y_0 are smooth over S_0 . Then, by SGA 1 II 4.10 (see also EGA IV₄, 17.2.3 (i) and 17.2.5), D is injective and the sequence $(\dagger)$ is formed of $\mathcal{O}_{Y_0}$ -modules locally free of finite type (hence is locally split). By the equivalence of categories I, 6.8.1, d is likewise injective, and thus the sequence $(+)$ has the properties indicated. $\square$

4.26. — ⁽¹²³⁾ For every S_0 -scheme $f: T \rightarrow S_0$, $(+)$ gives an exact sequence of $Y_0(T)$ - $\mathcal{O}(T)$ -modules

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_{\mathcal{O}_T}(f^*(\omega_{Y_0/S_0}^1), f^*(\mathcal{J})) &\longrightarrow \mathrm{Hom}_{\mathcal{O}_T}(f^*(\omega_{X_0/S_0}^1), f^*(\mathcal{J})) \\ &\longrightarrow \mathrm{Hom}_{\mathcal{O}_T}(f^*(\mathbf{n}_{Y_0/X_0}), f^*(\mathcal{J})), \end{aligned}$$

one therefore has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$(4.26.1) \quad 0 \longrightarrow L_{0Y} \longrightarrow L_{0X} \xrightarrow{d} N_0.$$

One deduces from this an exact sequence of complexes of abelian groups:

$$0 \longrightarrow C^*(Y_0, L_{0Y}) \longrightarrow C^*(Y_0, L_{0X}) \xrightarrow{d^*} C^*(Y_0, N_0),$$

and in particular, a commutative diagram with exact rows

$$\begin{array}{ccccc} 0 \longrightarrow C^0(Y_0, L_{0Y}) & \longrightarrow & C^0(Y_0, L_{0X}) & \xrightarrow{d^0} & C^0(Y_0, N_0) \\ & \downarrow \partial & \downarrow \partial & & \downarrow \partial \\ 0 \longrightarrow C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \xrightarrow{d^1} & C^1(Y_0, N_0). \end{array}$$

Let us note that $C^0(Y_0, L_{0Y})$ (resp. $C^0(Y_0, L_{0X})$) is none other than $\mathrm{Hom}_{S_0}(S_0, L_{0Y}) = \mathrm{Hom}_S(S, L'_Y)$ (resp. $\cdots$), i.e. (cf. 0.9) the group of sections of Y (resp. of X) over S inducing

⁽¹²²⁾**N.D.E.** One has spelled out the proof, taking into account the additions made in Exp. I, § 6.8.

⁽¹²³⁾**N.D.E.** One has spelled out the original, in order to make it plain that one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules.

the unit section of $X_{\mathcal{J}}$. Let us also note that d^1 is none other than the morphism $v_{i_{Y_0}}$ of 4.8 (iii), where $i_{Y_0}: Y_0 \rightarrow X_0$ is the canonical immersion.⁽¹²⁴⁾

Lemma 4.27. *Under the conditions of 4.21 for $S, \mathcal{I}, \mathcal{J}$ and X , let Y be a subgroup of X , flat over S and lifting $Y_{\mathcal{J}}$. Let $i: Y \hookrightarrow X$ denote the canonical immersion.⁽¹²⁵⁾*

(i) *Let $i': Y \rightarrow X$ be a morphism of S -schemes lifting i_0 (so that i' is also an immersion), let $Y' = i'(Y)$ and let $d(i, i')$ be the element of $C^1(Y_0, L_{0X})$ such that $i' = d(i, i') \cdot i$ (cf. 1.2.4). Then the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by the formula:*

$$d(Y, Y') = d^1(d(i, i')).$$

(ii) *Let $x \in C^0(Y_0, L_{0X})$ be a section of X over S inducing the unit section of $X_{\mathcal{J}}$ over $S_{\mathcal{J}}$. Then the deviation $d(Y, \text{Int}(x)Y) \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by the formula:*

$$-d(Y, \text{Int}(x)Y) = d^1 \partial x = \partial d^0 x.$$

Indeed, Y' is the image of Y by the composite morphism:⁽¹²⁶⁾

$$Y \xrightarrow{(d(i, i'), i)} L'_X \times_S X \rightarrow X,$$

which is denoted $d(i, i') \cdot i$ in 4.8 (iii); by *loc. cit.* and the equality $v_{i_0} = d^1$, one therefore has:

$$c(X, Y', d(i, i') \cdot i) - c(X, Y', i) = v_{i_0}(d(i, i')) = d^1(d(i, i')).$$

But $d(i, i') \cdot i = i'$ factors through Y' by definition, so the first term is zero; moreover, by 4.8 (ii), one has $c(X, Y', i) = d(Y', Y) = -d(Y, Y')$. Thus $d(Y, Y') = d^1(d(i, i'))$, which proves (i).

Now let x be as in (ii). By the formula

$$xyx^{-1} = xyx^{-1}y^{-1}y = (x - \text{Ad}(y)x)y = (-\partial x)(y) \cdot y,$$

one sees that Y' is the image of Y by the immersion $i' = (-\partial x) \cdot i_Y$. Thus, by (i), one obtains

$$-d(Y, \text{Int}(x)Y) = d^1 \partial x = \partial d^0 x.$$

Corollary 4.28. *In order that two subgroups Y and Y' of G , flat over S and lifting $Y_{\mathcal{J}}$, be conjugate by a section of X over S inducing the unit section of $X_{\mathcal{J}}$, it is necessary and sufficient that $d(Y, Y') \in \partial d^0 C^0(Y_0, L_{0X}) \subset \partial C^0(Y_0, N_0) = B^1(Y_0, N_0)$.*

Corollary 4.29. *If d^0 is surjective, Y and Y' as above are conjugate by a section of X over S inducing the unit section of $X_{\mathcal{J}}$ if and only if $d(Y, Y') \in B^1(Y_0, N_0)$.*

Corollary 4.30. *Let Y be as in 4.27; the set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathcal{J}}$ is isomorphic to:*

$$d^1 \partial C^0(Y_0, L_{0X}) = C^0(Y_0, L_{0X}) / \text{Ker } d^1 \partial.$$

Let us now note that $C^0(Y_0/L_{0X})/\text{Ker } d^1 \partial$ is computed solely by means of the left-hand square of the commutative diagram of 4.26. It follows in particular that one can also compute it in any diagram of the same type having the same left-hand square. Let us consider in particular the functor L_{0X}/L_{0Y} over S_0 defined by

$$\text{Hom}_{S_0}(T, L_{0X}/L_{0Y}) = \text{Hom}_{S_0}(T, L_{0X}) / \text{Hom}_{S_0}(T, L_{0Y}).$$

⁽¹²⁴⁾**N.D.E.** This follows from the definition of $d: \mathbf{n}_{Y_0/X_0} \rightarrow \Omega^1_{X_0/S_0}$ (cf. 4.25) and from that of $v_{i_{Y_0}}$ (cf. 4.8).

⁽¹²⁵⁾**N.D.E.** One has added point (i) below, which will be useful in 4.35.1 and then in 4.38 (4) and (5).

⁽¹²⁶⁾**N.D.E.** One recalls that $L'_X = \prod_{S_0/S} L_{0X}$.

One has a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^0(Y_0, L_{0Y}) & \longrightarrow & C^0(Y_0, L_{0X}) & \longrightarrow & C^0(Y_0, L_{0X}/L_{0Y}) \longrightarrow 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \longrightarrow & C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \longrightarrow & C^1(Y_0, L_{0X}/L_{0Y}) \longrightarrow 0, \end{array}$$

whence, by the preceding remark:

Corollary 4.31. *Let Y be as in 4.27; the set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathcal{J}}$ is isomorphic to*

$$E = \partial C^0(Y_0, L_{0X}/L_{0Y}) = C^0(Y_0, L_{0X}/L_{0Y})/H^0(Y_0, L_{0X}/L_{0Y}).$$

Corollary 4.32. *Suppose moreover that S_0 is affine and that ω_{Y_0/S_0}^1 ⁽¹²⁷⁾ is finite locally free. If one denotes $\mathcal{F}_0 = (\mathcal{L}ie(X_0/S_0)/\mathcal{L}ie(Y_0/S_0)) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$, one has $E = \Gamma(S_0, \mathcal{F}_0)/H^0(Y_0, \mathcal{F}_0)$.*

⁽¹²⁸⁾ Indeed, since ω_{Y_0/S_0}^1 is finite locally free, as is ω_{X_0/S_0}^1 (since X is assumed smooth over S), one has, by 0.6:

$$L_{0Y} = \mathbf{W}(\mathcal{L}ie(Y_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) \quad \text{and} \quad L_{0X} = \mathbf{W}(\mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

On the other hand, by 4.25, one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$0 \longrightarrow \mathcal{K} \longrightarrow \omega_{X_0/S_0}^1 \xrightarrow{\phi} \omega_{Y_0/S_0}^1 \longrightarrow 0$$

(where $\mathcal{K} = \text{Ker}(\phi)$). Since ω_{Y_0/S_0}^1 and ω_{X_0/S_0}^1 are finite locally free, one has a locally split exact sequence:

$$0 \longrightarrow \mathcal{L}ie(Y_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \longrightarrow \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \longrightarrow \mathcal{F}_0 \longrightarrow 0$$

It follows that one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$0 \longrightarrow L_{0Y} \longrightarrow L_{0X} \longrightarrow \mathbf{W}(\mathcal{F}_0).$$

By the argument which we used to prove 4.31, we may compute E as the image of the composite map

$$C^0(Y_0, L_{0X}) \xrightarrow{\pi} C^0(Y_0, \mathbf{W}(\mathcal{F}_0)) \xrightarrow{\partial} C^1(Y_0, \mathbf{W}(\mathcal{F}_0)).$$

Now the map π above is the map $\Gamma(S_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) \longrightarrow \Gamma(S_0, \mathcal{F}_0)$. Thus, S_0 being affine, π is surjective, and one indeed finds the result announced.

Corollary 4.33. *Let S , $\mathcal{I}$, $\mathcal{J}$ and X be as in 4.21, and let Y be a diagonalizable subgroup of X . Suppose that ω_{Y_0/S_0}^1 is finite locally free and that S_0 is affine. ⁽¹²⁹⁾ The set of subgroups of X , conjugates of Y by a section of X over S inducing the unit section of $X_{\mathcal{J}}$, is isomorphic to*

$$E = \Gamma\left(S_0, \mathcal{L}ie(X_0/S_0)/\mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)}\right) \otimes_{\Gamma(S_0, \mathcal{O}_{S_0})} \Gamma(S_0, \mathcal{J})$$

⁽¹³⁰⁾ that is, to $L_{0X}(Y_0)/H^0(Y_0, L_{0X})$.

Indeed, one writes, by I 4.7.3 (cf. 2.13):

$$\mathcal{L}ie(X_0/S_0) = \mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)} \oplus \mathcal{R}.$$

⁽¹²⁷⁾ **N.D.E.** One has corrected $\mathcal{L}ie(Y_0/S_0)$ to ω_{Y_0/S_0}^1 .

⁽¹²⁸⁾ **N.D.E.** One has spelled out the original in what follows.

⁽¹²⁹⁾ **N.D.E.** One has added the hypothesis on ω_{Y_0/S_0}^1 and replaced the hypothesis “ S affine” by “ S_0 affine”.

⁽¹³⁰⁾ **N.D.E.** One has added what follows, cf. 4.34.

Since Y_0 is commutative, one has $\mathcal{L}ie(Y_0/S_0) \subset \mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)}$, hence

$$\begin{aligned} \mathcal{F}_0 &= (\mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)} / \mathcal{L}ie(Y_0/S_0)) \otimes \mathcal{J} \oplus \mathcal{R} \otimes \mathcal{J}, \\ \mathcal{F}_0^{\text{ad}(Y_0)} &= (\mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)} / \mathcal{L}ie(Y_0/S_0)) \otimes \mathcal{J}. \end{aligned}$$

By 4.32, one therefore has $E \simeq \Gamma(S_0, \mathcal{R} \otimes \mathcal{J})$. Returning to the definition of $\mathcal{R}$, one is done.

Corollary 4.34. *Let $S, \mathcal{I}, \mathcal{J}$ and X be as in 4.21, and let Y be a diagonalizable subgroup of X . Suppose that ω_{Y_0/S_0}^1 is finite locally free and that S_0 is affine.⁽¹³¹⁾ If $x \in X(S)$ induces the unit section of $X_{\mathcal{J}}$ and normalizes Y , then it centralizes Y .*

This follows immediately from the comparison of the preceding corollary and of 2.14. Indeed, 4.33 shows that the elements of $C^0(Y_0, L_{0X})$ which globally preserve Y are the elements of $H^0(Y_0, L_{0X})$, and one has seen in 2.14 that they are precisely those which act trivially on the canonical immersion $Y \rightarrow X$.

4.35. — Let us return to the general situation of 4.21 and assume $Y_{\mathcal{J}}$ smooth over $S_{\mathcal{J}}$. Then, by 4.25 (ii), one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$(*) \quad 0 \rightarrow \mathcal{L}ie(Y_0/S_0) \rightarrow \mathcal{L}ie(X_0/S_0) \rightarrow \mathbf{n}_{Y_0/X_0}^{\vee} \rightarrow 0$$

and these are finite locally free $\mathcal{O}_{S_0}$ -modules.

On the other hand, by SGA 1, II 4.10, every subscheme Y of X lifting $Y_{\mathcal{J}}$ and *flat* over S will be *smooth* over S .⁽¹³²⁾ Suppose moreover that S_0 and $Y_{\mathcal{J}}$ are *affine*. Then, since $\mathbf{n}_{Y_0/X_0}^{\vee}$ is a locally free $\mathcal{O}_{S_0}$ -module, the sequence $(*)$ remains exact when one applies $\otimes_{\mathcal{O}_{S_0}} \mathcal{J}$ to it, and then takes the inverse image on Y_0^n , and since the Y_0^n are affine, one therefore obtains an exact sequence of complexes of abelian groups:

$$0 \rightarrow C^*(Y_0, L_{0Y}) \rightarrow C^*(Y_0, L_{0X}) \xrightarrow{d^*} C^*(Y_0, N_0) \rightarrow 0$$

and in particular, a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \rightarrow & C^1(Y_0, L_{0Y}) & \rightarrow & C^1(Y_0, L_{0X}) & \xrightarrow{d^1} & C^1(Y_0, N_0) \rightarrow 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \rightarrow & C^2(Y_0, L_{0Y}) & \rightarrow & C^2(Y_0, L_{0X}) & \xrightarrow{d^2} & C^2(Y_0, N_0) \rightarrow 0. \end{array}$$

Now let Y, Y' be two subgroups of X lifting $Y_{\mathcal{J}}$ and *flat*, hence *smooth*, over S . Since $Y_{\mathcal{J}}$ is affine, then, by 0.15, Y and Y' are isomorphic as schemes extending $Y_{\mathcal{J}}$, i.e. there exists an isomorphism of S -schemes

$$f: Y \xrightarrow{\sim} Y'$$

inducing the identity on $Y_{\mathcal{J}}$. On the one hand, by 1.2.4, f defines an element a of $C^1(Y_0, L_{0X})$ such that $f(y) = a(y_0)y$, for every $y \in Y(S')$, $S' \rightarrow S$, and by 4.27 (i), one has

$$d^1(a) = d(Y, Y').$$

Moreover, as Y, Y' are subgroups of X , the above element belongs to $Z^1(Y_0, N_0)$ (cf. 4.21). Then ∂a is an element of $Z^2(Y_0, L_{0Y})$ whose image ∂a in $H^2(Y_0, L_{0Y})$ depends only on the class $d(Y, Y') \in H^1(Y_0, N_0)$; this being the definition of the boundary map $\partial^1: H^1(Y_0, N_0) \rightarrow H^2(Y_0, L_{0Y})$, one therefore has:

$$\partial^1(d(Y, Y')) = \partial a.$$

On the other hand, let us transport by f the group structure of Y' and let Y_1 be the group obtained (which therefore has Y as underlying scheme), that is, the group law μ_1 of Y_1 is defined by: for every $S' \rightarrow S$ and $x, y \in Y(S')$,

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

⁽¹³¹⁾ **N.D.E.** Cf. N.D.E. (129).

⁽¹³²⁾ **N.D.E.** One has added what follows and Proposition 4.35.1, implicit in the original, cf. 4.38 (5).

By 3.5.1, Y_1 defines a cocycle $\delta(Y, Y_1) \in Z^2(Y_0, \mathcal{L}ie(Y_0/S_0))$ such that, for every $S' \rightarrow S$ and $x, y \in Y(S')$, one has

$$\delta(Y, Y_1)(x_0, y_0) xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$

Let us set $b = \delta(Y, Y_1)$. For every $S' \rightarrow S$ and $x, y \in Y(S')$, one has $(b(x_0, y_0) xy)_0 = x_0 y_0$, and hence one obtains that $f(b(x_0, y_0) xy)$ equals, on the one hand, $a(x_0 y_0) b(x_0 y_0) xy$ and, on the other hand,

$$f(x)f(y) = a(x_0) x a(y_0) y = a(x_0) \text{Ad}(x_0)(a(y_0)) xy.$$

Comparing the two expressions, one obtains that $b(x_0, y_0)$ equals

$$\begin{aligned} a(x_0 y_0)^{-1} a(x_0) \text{Ad}(x_0)(a(y_0)) &= \text{Ad}(x_0)(a(y_0)) - a(x_0 y_0) + a(x_0) \\ &= (\partial a)(x_0, y_0), \end{aligned}$$

i.e. $\delta(Y, Y_1) = \partial a$. One has therefore obtained the

Proposition 4.35.1. ⁽¹³²⁾ *Under the hypotheses of 4.21, suppose in addition that S_0 is affine and that $Y_{\mathcal{J}}$ is smooth over $S_{\mathcal{J}}$ and affine. Let Y, Y' be two subgroups of X lifting $Y_{\mathcal{J}}$ and flat (hence smooth) over S , let $f: Y \xrightarrow{\sim} Y'$ be an isomorphism of S -schemes inducing the identity on $Y_{\mathcal{J}}$, and denote by Y_1 the group obtained by transporting by f the group structure of Y' . Then one has*

$$\partial^1 d(Y, Y') = \delta(Y, Y_1).$$

Proposition 4.36. *Under the hypotheses of 4.21, suppose in addition that $Y_{\mathcal{J}}$ is smooth over $S_{\mathcal{J}}$ and that S_0 is affine. The set of S -subgroups Y of X flat (or smooth) over S , reducing to $Y_{\mathcal{J}}$, modulo conjugation by sections of X over S inducing the unit section of $X_{\mathcal{J}}$, is either empty or a principal homogeneous set under the group*

$$H^1(Y_0, [\mathcal{L}ie(X_0/S_0)/\mathcal{L}ie(Y_0/S_0)] \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

It is enough for us to check that Corollary 4.29 applies, that is, that

$$d^0: \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \rightarrow \text{Hom}_{\mathcal{O}_{S_0}}(\mathfrak{n}_{Y_0/X_0}, \mathcal{J})$$

is surjective. Now this follows from the fact that the sequence (+) of 4.25 (ii) is split, S_0 being affine.⁽¹³³⁾

Let us finally state a corollary common to 4.21 and 4.36, which will, in fact, be the only form in which we shall use in the sequel the general results of this number.⁽¹³⁴⁾

Corollary 4.37. *Let S be a scheme and S_0 the closed subscheme defined by a nilpotent ideal $\mathcal{I}$. Let X be an S -group smooth over S , and Y_0 an S_0 -subgroup of X_0 , flat over S_0 .*

(i) *If S_0 is affine, Y_0 is smooth over S_0 , and if*

$$H^1(Y_0, [\mathcal{L}ie(X_0/S_0)/\mathcal{L}ie(Y_0/S_0)] \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$$

for every $n \geq 0$, two S -subgroups of X , flat (or smooth) over S , reducing to Y_0 , are conjugate by a section of X over S inducing the unit section of X_0 .

(ii) *If Y_0 is affine and of finite presentation and if⁽¹³⁵⁾*

$$H^2(Y_0, \mathfrak{n}_{Y_0/X_0}^{\vee} \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$$

for every $n \geq 0$, there exists an S -subgroup of X , flat over S , reducing to Y_0 .

⁽¹³³⁾ **N.D.E.** This also follows from the proof of 4.32.

⁽¹³⁴⁾ **N.D.E.** For example, 4.37 is used in Exposé IX to prove the statements 3.2 bis and 3.6 bis.

⁽¹³⁵⁾ **N.D.E.** One has replaced $\text{Hom}_{\mathcal{O}_{S_0}}(\mathfrak{n}_{Y_0/X_0}, \mathcal{I}^{n+1}/\mathcal{I}^{n+2})$ by $\mathfrak{n}_{Y_0/X_0}^{\vee} \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}$, in accordance with Remark 4.22.

4.38. — It remains for us to relate the three constructions that we have made in this exposé. In order to avoid inessential complications, we shall place ourselves in the following situation: S_0 is the spectrum of a field k , S is the spectrum of the dual numbers $D(k)$, G is an S -group smooth over S , and K an S -subgroup, *smooth* over S and *affine*.⁽¹³⁶⁾ Let us denote $\mathfrak{g}_0 = \mathcal{L}ie G_0$ (which here equals $\Gamma(S_0, \mathcal{L}ie G_0) = \text{Lie}(G_0/S_0)(S_0)$) and $\mathfrak{k}_0 = \mathcal{L}ie K_0$. One has an exact sequence of k -vector spaces⁽¹³⁷⁾:

$$0 \longrightarrow \mathfrak{k}_0 \xrightarrow{i} \mathfrak{g}_0 \xrightarrow{d} \mathfrak{n}_{K_0/G_0}^\vee \longrightarrow 0,$$

giving rise to an exact cohomology sequence:

$$\begin{aligned} 0 \longrightarrow H^0(K_0, \mathfrak{k}_0) \xrightarrow{i^0} H^0(K_0, \mathfrak{g}_0) \xrightarrow{d^0} H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) \\ \xrightarrow{\partial^0} H^1(K_0, \mathfrak{k}_0) \xrightarrow{i^1} H^1(K_0, \mathfrak{g}_0) \xrightarrow{d^1} H^1(K_0, \mathfrak{n}_{K_0/G_0}^\vee) \xrightarrow{\partial^1} H^2(K_0, \mathfrak{k}_0). \end{aligned}$$

Now these various groups all have a geometric meaning.

- a) $H^0(K_0, \mathfrak{k}_0) = \mathcal{L}ie \text{ Centr}(K_0)$ ⁽¹³⁸⁾ by II 5.2.3.
- b) $H^0(K_0, \mathfrak{g}_0) = \mathcal{L}ie \text{ Centr}_{G_0}(K_0)$ ⁽¹³⁸⁾ (*idem*).
- c) $H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) = \mathcal{L}ie \text{ Norm}_{G_0}(K_0)/\mathfrak{k}_0$ ⁽¹³⁸⁾ (*idem*).
- d) $H^1(K_0, \mathfrak{k}_0) = \mathcal{L}ie(\text{Aut}_{S_0\text{-gr.}}(K_0))/\text{Im}(\mathfrak{k}_0)$, where $\text{Im}(\mathfrak{k}_0)$ denotes the image of $\mathfrak{k}_0$ under the morphism $\mathcal{L}ie(\text{Int}_0)$ deduced from $\text{Int}_0: K_0 \longrightarrow \text{Aut}_{S_0\text{-gr.}}(K_0)$. Indeed, it follows from 2.1 (ii), applied to $Y = X = K$ and $f_0 = \text{id}_{K_0}$, that $Z^1(K_0, \mathfrak{k}_0)$ is the group of infinitesimal automorphisms of the S_0 -group K_0 , and that $H^1(K_0, \mathfrak{k}_0)$ is obtained by quotienting by the inner infinitesimal automorphisms, i.e. by the image of $\mathfrak{k}_0$. Moreover, by II 4.2.2, one also has $Z^1(K_0, \mathfrak{k}_0) = \mathcal{L}ie(\text{Aut}_{S_0\text{-gr.}}(K_0))$.⁽¹³⁹⁾
- e) $H^1(K_0, \mathfrak{g}_0)$ is, by 2.1 (ii), the group of deviations between homomorphisms $K \longrightarrow G$ extending the canonical immersion $i_0: K_0 \longrightarrow G_0$, modulo the deviations obtained by the action of the inner automorphisms of G defined by elements of $G(S)$ giving the unit of $G(S_0)$ (that is, elements of $\mathfrak{g}_0$).
- f) $H^1(K_0, \mathfrak{n}_{K_0/G_0}^\vee)$ is, by 4.36, the group of deviations between subgroups K' of G extending K_0 and flat over S (hence *smooth* over S , cf. SGA 1, II 4.10), modulo the deviations obtained by the action of the inner automorphisms of G constructed as previously.
- g) $H^2(K_0, \mathfrak{k}_0)$ is, by 3.5 (ii), the group of deviations between group structures on K extending that of K_0 , modulo the S -automorphisms of K inducing the identity on K_0 .

We now propose to show how one can make explicit the six morphisms of the preceding exact sequence in the geometric interpretation which we have just given.

- 1) i^0 and d^0 are none other than the morphisms obtained by passage to the Lie algebra (then by passage to the quotient, for d^0), starting from the canonical monomorphisms:

$$\text{Centr}(K_0) \longrightarrow \text{Centr}_{G_0}(K_0) \longrightarrow \text{Norm}_{G_0}(K_0).$$

This is indeed what follows immediately from the definition of the identifications (a), (b), and (c).

- 2) One constructs ∂^0 as follows. Let $x \in \mathcal{L}ie \text{ Norm}_{G_0}(K_0)/\mathfrak{k}_0$. Let us lift it to an $x \in \mathcal{L}ie \text{ Norm}_{G_0}(K_0) \subset \text{Norm}_G(K)(S)$. Then $\text{Int}(x)$ defines an automorphism of K inducing the

⁽¹³⁶⁾**N.D.E.** One has slightly modified the original in what follows. In particular, one has replaced X by G and Y by K , and one has denoted by $\mathfrak{g}_0$ and $\mathfrak{k}_0$ their Lie algebras. On the other hand, one has written explicitly $H^i(K_0, \)$ in place of the abbreviation $H^i(\)$ of the original.

⁽¹³⁷⁾**N.D.E.** Equipped with the adjoint action of K_0 .

⁽¹³⁸⁾**N.D.E.** As the formation of centralizers and normalizers commutes with base change (cf. I 2.3.3.1), one has written $\text{Centr}(K_0)$ in place of $\text{Centr}(K)_0$ in the original, and likewise $\text{Centr}_{G_0}(K_0)$ and $\text{Norm}_{G_0}(K_0)$ in place of $\text{Centr}_G(K)_0$ and $\text{Norm}_G(K)_0$.

⁽¹³⁹⁾**N.D.E.** And this is the Lie algebra $\text{Dér}_k(\mathfrak{k}_0)$ of the derivations of $\mathfrak{k}_0$, so $H^1(K_0, \mathfrak{k}_0)$ is the quotient of $\text{Dér}_k(\mathfrak{k}_0)$ by the inner derivations (i.e. by the image of $\text{ad}: \mathfrak{k}_0 \longrightarrow \text{Dér}_k(\mathfrak{k}_0)$).

identity on K_0 , hence an element of $\mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0)$. Denote by $\text{Int}(x)$ the image of this element in $\mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0)/\text{Im}(\mathfrak{k}_0)$. Then one has:

$$(*) \quad \partial^0(x) = -\text{Int}(x) = \text{Int}(x^{-1}).$$

Indeed, let us compute the element of $\mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0)$ defined by $\text{Int}(x)$. By definition it will correspond to an element a of $Z^1(K_0, \mathfrak{k}_0)$ such that

$$xyx^{-1} = a(y_0)y, \quad \text{for every } y \in K(S'), \quad S' \longrightarrow S.$$

But this may also be written $a(y_0) = xyx^{-1}y^{-1} = x - \text{Ad}(y)x = -\partial(x)(y_0)$, whence $a = -\partial(x)$.

⁽¹⁴⁰⁾ On the other hand, the image of $x \in \mathcal{L}\text{ie Norm}_{G_0}(K_0) \subset \mathfrak{g}_0$ under ∂ is an element of $Z^1(K_0, \mathfrak{k}_0)$, whose image $\partial(x)$ in $H^1(K_0, \mathfrak{k}_0)$ depends only on x , and by definition of the boundary map ∂^0 , one has

$$\partial^0(x) = \partial(x);$$

combined with the equality $a = -\partial(x)$, this proves $(*)$.

3) ⁽¹⁴¹⁾ Denote by $i: K \longrightarrow G$ the canonical immersion. Let u be an element of $H^1(K_0, \mathfrak{k}_0)$, the image of an

$$u \in \mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0) \subset \text{Aut}_{S\text{-gr.}}(K).$$

Then one has:

$$(**) \quad i^1(u) = d(i, i \circ u),$$

where $d(i, i \circ u)$ is the class defined in 2.1.1.

Indeed, u is the image of an element $v \in Z^1(K_0, \mathfrak{k}_0)$ such that $u(y) = v(y_0)y$, and $i^1(u)$ is the image in $H^1(K_0, \mathfrak{g}_0)$ of the cocycle $i \circ v \in Z^1(K_0, \mathfrak{g}_0)$.

Now, as i is a morphism of groups, the equality $u(y) = v(y_0)y$ implies $iu(y) = iv(y_0)i(y)$. It follows that $i \circ v = d(i, i \circ u)$, whence $(**)$.

4) Let $i': K \longrightarrow G$ be a morphism of groups lifting i_0 , let $d(i, i')$ be the class defined in 2.1.1, and let $d(K, i'(K)) \in C^1(K_0, \mathfrak{n}_{K_0/X_0})$ be the deviation defined in 4.5.1; by 4.21, $d(K, i'(K))$ belongs to $Z^1(K_0, \mathfrak{n}_{K_0/X_0})$. Denote by $d(K, i'(K))$ its image in $H^1(K_0, \mathfrak{n}_{K_0/X_0})$. Then, by 4.27 (i), one has:

$$(\dagger) \quad d^1(d(i, i')) = d(K, i'(K)).$$

5) Finally, let K' be a subgroup of G lifting K_0 and flat, hence *smooth*, over S . One has supposed that K_0 is *affine*. Then one knows that K and K' are isomorphic as schemes extending K_0 (cf. 0.15), hence that there exists an isomorphism of S -schemes

$$f: K \xrightarrow{\sim} K'$$

inducing the identity on K_0 . Let us transport by f the group structure of K' and let K_1 be the group obtained (which therefore has K as underlying scheme), that is, the group law μ_1 of K_1 is defined by: for every $S' \longrightarrow S$ and $x, y \in K(S')$,

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

⁽¹⁴²⁾ By 3.5.1, K_1 defines a cocycle $\delta(K, K_1) \in Z^2(K_0, \mathfrak{k}_0)$ such that, for every $S' \longrightarrow S$ and $x, y \in K(S')$, one has

$$\delta(K, K_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$

Then, by 4.35.1, one has:

$$(\ddagger) \quad \partial^1 d(K, K') = \delta(K, K_1).$$

⁽¹⁴⁰⁾**N.D.E.** One has added the following sentence.

⁽¹⁴¹⁾**N.D.E.** One has set out the original in more detail in what follows, and in $(**)$ one has corrected $u \circ i$ to $i \circ u$.

⁽¹⁴²⁾**N.D.E.** One has modified the original in what follows, taking into account the additions made in 3.5.1 and 4.35.1.

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⁽¹⁴³⁾**N.D.E.** : additional references cited in this Exposé.

Exposé IV. Topologies and sheaves

by M. Demazure^(*)

⁽⁰⁾This exposé is intended to acquaint the reader with the essentials of the language of topologies and of sheaves (without cohomology), particularly convenient in questions of passage to the quotient (among others).

The first three paragraphs develop the language of passage to the quotient. The fourth, which is the central part, is the exposition of the theory of sheaves, oriented principally towards the application to questions of quotients; the fifth is an application to passage to the quotient in groups and to principal homogeneous bundles. The last paragraph concerns more specifically the category of schemes, and defines various useful topologies on this category.

The reader will find it useful to refer to [AS], [MA], [D], and SGA 4; to [D] especially concerning the applications of topologies to descent theory, and to SGA 4 for questions of universes (particularly ill-treated in this exposé).

1. Universal effective epimorphisms

In the sequel of this exposé, one assumes fixed a category $\mathcal{C}$.

Definition 1.1. A morphism $u: T \rightarrow S$ is called an *epimorphism* if, for every object X , the corresponding map

$$X(S) = \text{Hom}(S, X) \rightarrow X(T) = \text{Hom}(T, X)$$

is injective⁽¹⁾. One says that u is a *universal epimorphism* if for every morphism $S' \rightarrow S$, the fiber product $T' = T \times_S S'$ exists, and $u': T' \rightarrow S'$ is an epimorphism.

Definition 1.2. A diagram

$$A \xrightarrow{u} B \begin{array}{c} \xrightarrow{v_1} \\ \xrightarrow{v_2} \end{array} C$$

of maps of sets is said to be *exact* if u is injective and if its image consists of the elements b of B such that $v_1(b) = v_2(b)$. A diagram of the same type in $\mathcal{C}$ is said to be *exact* if for every object X of $\mathcal{C}$, the corresponding diagram of sets

$$A(X) \longrightarrow B(X) \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} C(X)$$

is exact; one then also says that u makes A a *kernel* of the pair of arrows (v_1, v_2) . Dually, a diagram

$$C \begin{array}{c} \xrightarrow{v_1} \\ \xrightarrow{v_2} \end{array} B \xrightarrow{u} A$$

in $\mathcal{C}$ is said to be *exact* if it is exact as a diagram in the opposite category $\mathcal{C}^\circ$, i.e. if for every object X of $\mathcal{C}$, the corresponding diagram of sets

$$X(A) \longrightarrow X(B) \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} X(C)$$

^(*)This text develops the substance of two oral exposés by A. Grothendieck, completing the latter on several important points, which had been passed over in silence or barely touched upon.

⁽⁰⁾**N.D.E.** : Version of 13/10/2024

⁽¹⁾**N.D.E.** : that is, if u is right-cancellable.

is exact.⁽²⁾ One also says that u makes A a *cokernel* of the pair of arrows (v_1, v_2) .

Definition 1.3. A morphism $u: T \rightarrow S$ is called an *effective epimorphism* if the fiber square $T \times_S T$ exists, and if the diagram

$$T \times_S T \begin{array}{c} \xrightarrow{\text{pr}_1} \\ \xrightarrow{\text{pr}_2} \end{array} T \xrightarrow{u} S$$

is exact, i.e. if u makes S a cokernel of $(\text{pr}_1, \text{pr}_2)$. One says that u is a *universal effective epimorphism* if for every morphism $S' \rightarrow S$, the fiber product $T' = T \times_S S'$ exists, and the morphism $u': T' \rightarrow S'$ is an effective epimorphism.

One evidently has the implications:

$$\begin{array}{ccc} \text{universal effective} & & \text{effective} \\ \text{epimorphism} & \longrightarrow & \text{epimorphism} \\ \downarrow & & \downarrow \\ \text{universal} & & \\ \text{epimorphism} & \longrightarrow & \text{epimorphism}, \end{array}$$

but in general no other implication is valid.⁽³⁾

Definition 1.4.0. ⁽⁴⁾One “recalls” that a morphism $u: T \rightarrow S$ is said to be *quarrable* if for every morphism $S' \rightarrow S$, the fiber product $T \times_S S'$ exists.

Lemma 1.4. Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then

- a) u, v epimorphisms $\Rightarrow uv$ epimorphism $\Rightarrow u$ epimorphism,
- b) u, v universal epimorphisms $\Rightarrow uv$ universal epimorphism and u quarrable $\Rightarrow u$ universal epimorphism.

Lemma 1.4 is trivial from the definitions. One concludes from it:

Corollary 1.5. Let $u: X \rightarrow Y$ and $u': X' \rightarrow Y'$ be universal epimorphisms, such that $Y \times Y'$ exists, then $X \times X'$ exists and $u \times u': X \times X' \rightarrow Y \times Y'$ is a universal epimorphism.

Let us also note:

Definition 1.6.0. ⁽⁵⁾One says that an object S of $\mathcal{C}$ is *quarrable* if its product by every object of $\mathcal{C}$ exists. (If $\mathcal{C}$ has a final object e , this amounts to saying that the morphism $S \rightarrow e$ is quarrable, cf. 1.4.0.)

Lemma 1.6. Let $u: X \rightarrow Y$ be a morphism in $\mathcal{C}_{/S}$; for it to be an epimorphism (resp. a universal epimorphism, resp. an effective epimorphism, resp. a universal effective epimorphism), it suffices that the corresponding morphism in $\mathcal{C}$ be one, and this is also necessary if one assumes that S is a quarrable object of $\mathcal{C}$.

PROOF. Immediate, left to the reader. One uses the hypothesis “ S quarrable” in order to interpret the $\mathcal{C}$ -morphisms from an object Y of $\mathcal{C}_{/S}$ into an object Z of $\mathcal{C}$, as being the $\mathcal{C}_{/S}$ -morphisms from Y into $Z \times S$. $\square$

⁽²⁾**N.D.E.** : This implies, in particular, that u is an epimorphism.

⁽³⁾**N.D.E.** : For example, if $\mathcal{C} = (\text{Sch})$ is the category of schemes, one easily sees that every universal epimorphism is surjective. Let $T = \coprod_{p \text{ prime}} \text{Spec}(\mathbf{F}_p)$ and $S = \text{Spec}(\mathbf{Z})$; then the morphism $u: T \rightarrow S$ is an epimorphism which is *not universal*. On the other hand, one sees that $T \times_S T$ is identified with T , so that the two projections $T \times_S T \rightrightarrows T$ coincide; as id_T does not descend to a morphism $S \rightarrow T$, this shows that u is *not* an effective epimorphism.

⁽⁴⁾**N.D.E.** : One has added the numbering 1.4.0, for later references.

⁽⁵⁾**N.D.E.** : One has added the numbering 1.6.0, for later references.

Lemma 1.7. *With the notations of 1.4: u, v effective epimorphisms and v universal epimorphism $\Rightarrow uv$ effective epimorphism.*

To see this, one considers the diagram

$$\begin{array}{ccccc}
 S & \xleftarrow{u} & T & \xleftarrow{\quad} & T \times_S T \\
 & & \uparrow v & & \uparrow v \times_S v \\
 & & U & \xleftarrow{\quad} & U \times_S U \\
 & & \uparrow \uparrow & \nearrow & \\
 & & U \times_T U & &
 \end{array}$$

One notes that by hypothesis, row 1 and column 1 are exact, and that by virtue of 1.5 and 1.6, $v \times_S v$ is an epimorphism (v being a *universal* epimorphism). The conclusion follows from this by an evident diagram-chasing: if an element of $X(U)$ has the same images in $X(U \times_S U)$, it *a fortiori* has the same images in $X(U \times_T U)$, hence comes from an element of $X(T)$ since column 1 is exact. As row 1 is exact, it suffices to check that the element under consideration has the same images in $X(T \times_S T)$, and as $v \times_S v$ is an epimorphism, it suffices to check that the images in $X(U \times_S U)$ are the same, which is indeed the case.

Proposition 1.8. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then u, v universal effective epimorphisms $\Rightarrow uv$ universal effective epimorphism and u quarrable $\Rightarrow u$ universal effective epimorphism.*

The first implication follows at once from 1.7. For the second, one looks at the diagram (of “bisimplicial” type):

$$\begin{array}{ccccc}
 S & \xleftarrow{u} & T & \xleftarrow{\quad} & T \times_S T \\
 \uparrow uv & & \uparrow & & \uparrow \\
 U & \xleftarrow{\quad} & U \times_S T & \xleftarrow{\quad} & U \times_S T \times_S T \\
 \uparrow \uparrow & & \uparrow \uparrow & & \uparrow \uparrow \\
 U \times_S U & \leftarrow & U \times_S U \times_S T & \not\leftarrow & U \times_S U \times_S T \times_S T
 \end{array}$$

Columns 1, 2, 3 are exact by virtue of the hypothesis “ uv universal effective epimorphism”, row 2 is exact, for $U \times_S T \rightarrow U$ is an effective epimorphism (for it has a section over U), and the same holds for row 3 (same reason). An evident diagram-chasing then shows that row 1 is exact, i.e. u is an effective epimorphism. As the hypotheses made are invariant under an arbitrary base change $S' \rightarrow S$, it follows that u is even a universal effective epimorphism.

Corollary 1.9. *Let $u: X \rightarrow Y$ and $u': X' \rightarrow Y'$ be universal effective epimorphisms, such that $Y \times Y'$ exists; then $X \times X'$ exists and $u \times u': X \times X' \rightarrow Y \times Y'$ is a universal effective epimorphism.*

PROOF. As for 1.5, by the diagram

$$\begin{array}{ccccc}
 & & Y' & \xleftarrow{u'} & X' \\
 & & \uparrow & & \uparrow \\
 X & \xleftarrow{\quad} & X \times Y' & \xleftarrow{\quad} & X \times X' \\
 \downarrow u & & \downarrow & \nearrow u \times u' & \\
 Y & \xleftarrow{\quad} & Y \times Y' & &
 \end{array}$$

□

Corollary 1.10. *Consider a quarrable morphism $u: T \rightarrow S$, and a base change morphism $S' \rightarrow S$, which is a universal effective epimorphism. For u to be a universal effective epimorphism, it is necessary and sufficient that $u': T' = T \times_S S' \rightarrow S'$ be so:*

$$\begin{array}{ccc} T & \xleftarrow{v'} & T' \\ u \downarrow & & \downarrow u' \\ S & \xleftarrow{v} & S' \end{array} .$$

Only the “it suffices” requires a proof. Now if u' is a universal effective epimorphism, the same holds for vu' thanks to 1.8, and since $vu' = uv'$, one concludes by 1.8 that u is a universal effective epimorphism.

Remark 1.11. The same reasoning shows that in 1.10 one may replace “universal effective epimorphism” by “universal epimorphism” or “universal and effective epimorphism”, or simply by “epimorphism” (and in this last case, the hypothesis “ u quarrable” is evidently useless).

In the proof of 1.8 we have used the following result, which deserves to be spelled out:

Proposition 1.12. *Let $u: T \rightarrow S$ be a morphism which admits a section. Then u is an epimorphism, and if $T \times_S T$ exists, it is an effective epimorphism, and a universal effective epimorphism if moreover u is quarrable.*

The first assertion is contained in 1.4 a), and the third will follow at once from the second, which it will therefore suffice to establish. In fact one has a stronger conclusion: for every functor $F: \mathcal{C}^\circ \rightarrow (\text{Ens})$ (not necessarily representable), the diagram of sets

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact. This may be considered as a particular case of the formalism of Čech cohomology (in dimension 0!) which we content ourselves with recalling here. Let us simply assume that $T \times_S T$ exists; one then sets

$$H^0(T/S, F) = \text{Ker}(F(T) \rightrightarrows F(T \times_S T)).$$

One may regard $H^0(T/S, F)$ in an evident way as a contravariant functor in the argument T varying in $\mathcal{C}_S$, every S -morphism $T' \rightarrow T$ defining a map

$$(+)\quad H^0(T/S, F) \longrightarrow H^0(T'/S, F).$$

Let us fix T and T' in $\mathcal{C}_S$. A well-known computation shows that if there exists an S -morphism from T' into T , the corresponding map $(+)$ is in fact independent of the choice of this morphism⁽⁶⁾, so that $H^0(T/S, F)$ may be regarded as a functor on the category associated with the set $\text{Ob } \mathcal{C}_S$ preordered by the relation of “domination” (T' dominates T if there exists an S -morphism from T' into T). In particular, if T and T' are isomorphic in this latter category, i.e. if each dominates the other, then $(+)$ is an isomorphism of sets. This applies in particular to the case where T' is the final object of $\mathcal{C}_S$, i.e. essentially S itself; in any case T dominates $T' = S$, and the converse is true precisely if T/S has a section. This establishes 1.12 in the strengthened form announced.

Remark 1.13. For various applications, the notions introduced in the present exposé, and the results stated, are to be developed more generally relative to a *family* of morphisms

⁽⁶⁾**N.D.E.** : This is the following argument, communicated by M. Demazure. Let $f, g: T' \rightarrow T$, and let $\varphi: T' \rightarrow T \times_S T$ be the morphism with components f and g , whence $p_1 \circ \varphi = f$ and $p_2 \circ \varphi = g$. Then $F(\varphi): F(T \times_S T) \rightarrow F(T')$ satisfies $F(\varphi) \circ F(p_1) = F(f)$ and $F(\varphi) \circ F(p_2) = F(g)$. Now, for every $x \in H^0(T/S, F)$, one has $F(p_1)(x) = F(p_2)(x)$. Therefore, applying $F(\varphi)$ to both sides, one obtains $F(f)(x) = F(g)(x)$, which shows that f and g induce the same morphism.

$u_i: T_i \rightarrow S$ with the same target (instead of a single morphism $u: T \rightarrow S$). Thus, such a family will be said to be *epimorphic* if for every object X of $\mathcal{C}$, the corresponding map

$$X(S) \rightarrow \prod_i X(T_i)$$

is injective, and one likewise introduces the notion of effective epimorphic family and the “universal” variants of these notions. We shall admit if need be, in the sequel, that the results of the present exposé extend to this more general situation.

Remark 1.14. Every morphism which is at the same time a monomorphism and an effective epimorphism is an isomorphism. Indeed, in the notations of 1.3, the fact that $T \rightarrow S$ is a monomorphism implies that the two morphisms

$$\text{pr}_1, \text{pr}_2: T \times_S T \rightrightarrows T$$

are equal (and are isomorphisms). Now a diagram

$$C \begin{array}{c} \xrightarrow{v} \\ \xrightarrow{v} \end{array} B \xrightarrow{u} A$$

is exact only if u is an isomorphism, as follows immediately from the definition.⁽⁷⁾

2. Descent morphisms

Let us recall the following definitions:

Definition 2.1. Let $f: S' \rightarrow S$ be a morphism such that $S'' = S' \times_S S'$ exists, and let $u': X' \rightarrow S'$ be an object over S' . One calls *gluing datum* on X'/S' , relative to f , an S'' -isomorphism

$$c: X_1'' \xrightarrow{\sim} X_2''$$

where X_i'' ($i = 1, 2$) denotes the inverse image (assumed to exist) of X'/S' by the projection $\text{pr}_i: S'' \rightarrow S'$. One says that the gluing datum c is a *descent datum* if it satisfies the “cocycle condition”

$$\text{pr}_{3,1}^*(c) = \text{pr}_{3,2}^*(c) \text{pr}_{2,1}^*(c)$$

where $\text{pr}_{i,j}$ ($1 \leq j < i \leq 3$) are the canonical projections of $S''' = S' \times_S S' \times_S S'$ into S'' (N.B. one now assumes that S''' also exists), where $\text{pr}_{i,j}^*(c)$ is the inverse image of c , considered as an S''' -morphism from X_j''' into X_i''' , and where for every integer k between 1 and 3, X_k''' denotes the inverse image (assumed to exist) of X'/S' by the projection of index k , $q_k: S''' \rightarrow S'$.

In the second part of the definition, one has therefore used identifications and abuses of notation in current use⁽⁸⁾, which experience proves to be harmless, but which it is obviously advisable to avoid in a rigorous exposé of the theory of descent, (which must precisely justify to a certain extent these current abuses of language). Such a formal exposé ([D]) has been written by Giraud, (with a view to justifying and making precise SGA 1, VII, which has never been written). For a detailed exposé of the results of faithfully flat descent, of which constant use will be made in the present Seminar, one may consult SGA 1, VIII.

Let $f: S' \rightarrow S$ still be a morphism such that $S'' = S' \times_S S'$ exists, and let X be an object over S such that $X' = X \times_S S'$ and $X'' = X \times_S S''$ exist; then the inverse images of X' by pr_i ($i = 1, 2$) exist and are canonically isomorphic, and consequently X'/S' is endowed with a canonical gluing datum relative to f . When S''' and $X''' = X \times_S S'''$ exist, it is even a descent datum. If Y is another object over S , satisfying the same conditions as X/S , then for

⁽⁷⁾**N.D.E.** : One will note that a monomorphism which is an epimorphism is not necessarily an isomorphism. For example, in $\mathcal{C} = (\text{Sch})$, the morphism $\text{Spec}(\mathbf{F}_p) \coprod \text{Spec}(\mathbf{Z}[1/p]) \rightarrow \text{Spec}(\mathbf{Z})$ is a monomorphism and a surjective epimorphism, but is not an isomorphism.

⁽⁸⁾**N.D.E.** : for example, one has identified, on the one hand, $\text{pr}_{2,1}^*(X_1'') = \text{pr}_{3,1}^*(X_1'')$ and, on the other hand, $\text{pr}_{2,1}^*(X_2'') = \text{pr}_{3,2}^*(X_2'')$.

every S -morphism $X \rightarrow Y$, the corresponding S' -morphism $X' \rightarrow Y'$ is “compatible with the canonical gluing data” on X', Y' . If in particular $S' \rightarrow S$ is a *quarrable* morphism, then

$$X \mapsto X' = X \times_S S'$$

is a functor from the category $\mathcal{C}_S$ into the “category of objects over S' endowed with a descent datum relative to f ”—a category whose definition is left to the reader, and which is a full subcategory of the “category of objects over S' endowed with a gluing datum relative to f ”.

This being said:

Definition 2.2. One says that a morphism $f: S' \rightarrow S$ is a *descent morphism* (resp. an *effective descent morphism*) if f is *quarrable* (i.e. for every $X \rightarrow S$, the fiber product $X \times_S S'$ exists), and if the preceding functor $X \mapsto X' = X \times_S S'$ from the category $\mathcal{C}_S$ of objects over S , into the category of objects over S' endowed with a descent datum relative to f , is fully faithful (resp. an equivalence of categories).

One will note that the first of these two notions can be expressed with the help of only the notion of gluing datum (hence without involving the triple fiber product S'''), f being a descent morphism if f is *quarrable* and $X \mapsto X'$ is a fully faithful functor from the category $\mathcal{C}_S$ into the category of objects over S' endowed with a gluing datum relative to f . When one makes this definition explicit, one finds that this means that for two objects X, Y over S , the following diagram of sets

$$(x) \quad \text{Hom}_S(X, Y) \longrightarrow \text{Hom}_{S'}(X', Y') \xrightleftharpoons[p_2]{p_1} \text{Hom}_{S''}(X'', Y'')$$

is exact, where $p_i(u')$ denotes the inverse image of $u' \in \text{Hom}_{S'}(X', Y')$ by the projection $\text{pr}_i: S'' \rightarrow S'$, for $i = 1, 2$; indeed, the kernel of the pair (p_1, p_2) is by definition nothing other than the set of S' -morphisms $X' \rightarrow Y'$ compatible with the canonical gluing data.

Let us note that, by definition of the inverse images Y', Y'' , one has canonical bijections

$$\text{Hom}_{S'}(X', Y') \simeq \text{Hom}_S(X', Y) \quad \text{and} \quad \text{Hom}_{S''}(X'', Y'') \simeq \text{Hom}_S(X'', Y),$$

so that the exactness of the diagram (x) is equivalent to that of

$$(xx) \quad \text{Hom}_S(X, Y) \longrightarrow \text{Hom}_S(X', Y) \rightrightarrows \text{Hom}_S(X'', Y),$$

obtained by applying $\text{Hom}_S(-, Y)$ to the diagram in $\mathcal{C}_S$:

$$(xxx) \quad X'' \rightrightarrows X' \longrightarrow X$$

which is deduced from

$$S'' \rightrightarrows S' \longrightarrow S$$

by the change of base $X \rightarrow S$. This proves, taking 1.6 into account, the first part of the following proposition:

Proposition 2.3. *Let $f: S' \rightarrow S$ be a morphism. If it is a universal effective epimorphism, it is a descent morphism, and the converse is true if S is a quarrable object of $\mathcal{C}$ (i.e. its product by every object of $\mathcal{C}$ exists).*

It remains to prove that if f is a descent morphism, it is a universal effective epimorphism, that is, that for every morphism $X \rightarrow S$, the diagram (xxx) is exact, i.e. for every object Z of $\mathcal{C}$, the transform of this diagram by $\text{Hom}(-, Z)$ is an exact diagram of sets. Now by hypothesis $Z \times S$ exists; let Y be the object of $\mathcal{C}_S$ which it defines; then the transform of diagram (xxx) by $\text{Hom}(-, Z)$ is isomorphic to the transform by $\text{Hom}_S(-, Y)$, now the latter is exact by the hypothesis on f .

One may therefore apply to universal effective epimorphisms the results on descent morphisms, such as the following:

Proposition 2.4. *Let $f: S' \rightarrow S$ be a descent morphism (for example a universal effective epimorphism). Then:*

a) *For every S -morphism $u: X \rightarrow Y$, u is an isomorphism (resp. a monomorphism) if and only if $u': X' \rightarrow Y'$ is.*

b) *Let X, Y be two subobjects of S , and X' and Y' the subobjects of S' , inverse images of the preceding ones. In order that X be majorized by Y (resp. be equal to Y), it is necessary and sufficient that the same hold for X', Y' .*

For (a), it follows from the definition that if u' is an isomorphism in the category of objects with gluing datum, then u is an isomorphism; now one sees at once that every isomorphism of objects over S' , compatible with gluing data, is an isomorphism of objects with gluing datum, i.e. its inverse is likewise compatible with the gluing data. For b), one is reduced to proving that if X' is majorized by Y' , i.e. if there exists an S' -morphism $X' \rightarrow Y'$, then the same holds for X, Y over S . Now since $Y' \rightarrow S'$, hence also $Y'' \rightarrow S''$, is a monomorphism, one sees that $X' \rightarrow Y'$ is automatically compatible with the gluing data, hence comes from an S -morphism $X \rightarrow Y$. Let us note that the proof holds more generally when one has two objects X, Y over S , with $Y \rightarrow S$ a monomorphism, and one asks whether the morphism $X \rightarrow S$ factors through Y : it suffices that $X' \rightarrow S'$ factor through Y' .

Corollary 2.5. *Let $f: S' \rightarrow S$ be a universal effective epimorphism and $g: S \rightarrow T$ a morphism such that $S \times_T S$ exists. Suppose that $S'' = S' \times_S S'$ is also a fiber product of S' by itself over T , i.e. $S' \times_S S' \xrightarrow{\sim} S' \times_T S'$. Then $g: S \rightarrow T$ is a monomorphism (and conversely, of course).*

Indeed, let us consider the cartesian diagram

$$\begin{array}{ccc} S' \times_S S' & \xrightarrow{\sim} & S' \times_T S' \\ \downarrow & & \downarrow f \\ S & \longrightarrow & S \times_T S, \end{array}$$

where the second horizontal arrow is the diagonal morphism. By virtue of 1.9 the second vertical arrow f is a universal effective epimorphism, by hypothesis the first horizontal arrow is an isomorphism, hence by virtue of 2.4 a) or b), at one's choice⁽⁹⁾, the same holds of $S \rightarrow S \times_T S$, which means precisely that $g: S \rightarrow T$ is a monomorphism.

Remark 2.6. The notions introduced in 2.1, in terms of morphisms between certain projective limits, are made explicit in an evident way in terms of the contravariant functors defined by the objects S, S', X' envisaged: subject to the existence of the fiber products envisaged in 2.1, there is a one-to-one correspondence between the gluing data (resp. descent data) on X'/S' relative to $S' \rightarrow S$, and the gluing data (resp. descent data) for the corresponding objects in $\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}^\circ, (\text{Ens}))$. This makes it possible, if one so desires, to extend these notions to the case where one makes no existence hypothesis on fiber products in $\mathcal{C}$.

Remark 2.7. The notions introduced in this number generalize to the case of families of morphisms. They can on the other hand be presented in a more intrinsic manner with the help of the notion of sieve (4.1). For these questions, the reader is referred to [D].

3. Universal effective equivalence relations

3.1. Equivalence relations: definitions

⁽⁹⁾**N.D.E.** : applied to $f: S' \times_T S' \rightarrow S \times_T S = Y$.

Definition 3.1.1. A *representable subfunctor* R of the functor $X \times X$, such that for every $S \in \text{Ob}(\mathcal{C})$, $R(S)$ is the graph of an equivalence relation in $X(S)$, is called a $\mathcal{C}$ -*equivalence relation* in $X \in \text{Ob}(\mathcal{C})$.

This definition applies in particular to the category $\widehat{\mathcal{C}}$. If one considers X as an object of $\widehat{\mathcal{C}}$, one sees then that a $\widehat{\mathcal{C}}$ -equivalence relation in X is nothing other than a subfunctor R of $X \times X$ such that $R(S)$ is the graph of an equivalence relation in $X(S)$ for every object S of $\mathcal{C}$.⁽¹⁰⁾

If R is a $\mathcal{C}$ -equivalence relation in X , one denotes by p_i the morphism from R into X induced by the projection $\text{pr}_i: X \times X \rightarrow X$. One therefore has a diagram

$$p_1, p_2: R \rightrightarrows X.$$

Definition 3.1.2. A morphism $u: X \rightarrow Z$ is said to be *compatible* with R if $up_1 = up_2$. An object of $\mathcal{C}$, cokernel of the pair (p_1, p_2) , is also called a *quotient-object* of X by R and denoted X/R .

One therefore has an exact diagram

$$R \xrightarrow{p_1, p_2} X \xrightarrow{p} X/R$$

and X/R represents the *covariant* functor

$$\text{Hom}_{\mathcal{C}}(X/R, Z) = \{\text{morphisms from } X \text{ into } Z \text{ compatible with } R\}.$$

As the quotient-objects have been chosen in $\mathcal{C}$, the quotient X/R is unique (when it exists).

These definitions generalize at once to the case of a $\widehat{\mathcal{C}}$ -equivalence relation in X , but one will observe that the identification of the objects of $\mathcal{C}$ with their images in $\widehat{\mathcal{C}}$ *does not commute with the formation of quotients*, that is, that the quotient X/R of X by R in $\mathcal{C}$ is not *a priori* a quotient of X by R in $\widehat{\mathcal{C}}$. One will therefore take care not to identify $\mathcal{C}$ rashly with its image in $\widehat{\mathcal{C}}$ in questions involving passages to the quotient.⁽¹¹⁾

In the sequel, *we shall say simply equivalence relation for $\widehat{\mathcal{C}}$ -equivalence relation*; we shall specify, when the case arises, whether it is a matter of $\mathcal{C}$ -equivalence relations.⁽¹²⁾

Definition 3.1.3. If X is an object of $\mathcal{C}$ over S , an equivalence relation R in X such that the structural morphism $X \rightarrow S$ is compatible with R is called an equivalence relation in X over S .

The canonical monomorphism $R \rightarrow X \times X$ then factors through the monomorphism

$$X \times_S X \rightarrow X \times X$$

⁽¹⁰⁾**N.D.E.** : The condition is obviously necessary. Conversely, if for every $S \in \text{Ob}(\mathcal{C})$, $R(S)$ is the graph of an equivalence relation, then this equivalence relation extends to $R(F)$ for every $F \in \text{Ob}(\widehat{\mathcal{C}})$, by declaring that two morphisms $\varphi, \psi: F \rightarrow R$ are equivalent if, for every $S \in \text{Ob}(\mathcal{C})$ and $x \in F(S)$, $\varphi(x)$ and $\psi(x)$ are equivalent in $X(S)$.

⁽¹¹⁾**N.D.E.** : Let us illustrate this remark by giving a glimpse of the sequel of this Exposé. Let G be a $\mathcal{C}$ -group, H a $\mathcal{C}$ -subgroup, R the equivalence relation in G defined by $G \times H \rightarrow G \times G$, $(g, h) \mapsto (g, gh)$ (cf. 3.2). The functor Q defined by $Q(S) = G(S)/H(S)$ is a quotient in $\widehat{\mathcal{C}}$ (by 4.4.9 applied to the coarsest topology, cf. 4.4.2), but it is not in general the quotient that one wishes. For example, for $\mathcal{C} = (\text{Sch})$, one has an exact sequence of (affine) group schemes:

$$1 \rightarrow \mu_2 \rightarrow \mathbf{G}_m \xrightarrow{p} \mathbf{G}_m \rightarrow 1$$

which identifies $\mathbf{G}_m$ with the quotient $\mathbf{G}_m/\mu_2$. Moreover, as p is a finite and locally free morphism, $\mathbf{G}_m$ is then the *quotient-sheaf* of $\mathbf{G}_m$ by μ_2 in the larger category of *sheaves for the (fppf) topology*, cf. 4.6.6 (ii) and 6.3.2. By contrast, the quotient Q in $\widehat{\mathcal{C}}$ is not isomorphic to $\mathbf{G}_m$ since, for example, $Q(\mathbf{Z}) = \{1\}$ whereas $\mathbf{G}_m(\mathbf{Z}) = \{\pm 1\}$. Hence Q is not an (fppf) sheaf, and a fortiori Q is not representable.

⁽¹²⁾**N.D.E.** : Take care that, even for a $\widehat{\mathcal{C}}$ -equivalence relation in X , one is interested in the existence of a quotient *in $\mathcal{C}$* .

and defines an equivalence relation in the object $X \rightarrow S$ of $\mathcal{C}/S$. When the quotient X/R exists, it is endowed with a canonical morphism into S and the corresponding object of $\mathcal{C}/S$ is a quotient of $X \in \text{Ob}(\mathcal{C}/S)$ by the preceding equivalence relation. Conversely, if S is a *quarrrable* object of $\mathcal{C}$ and if $Y \rightarrow S$ is a quotient of X by this equivalence relation (in $\mathcal{C}/S$), then Y is a quotient of X by R in $\mathcal{C}$. In any case, we shall never have to consider quotients in $\mathcal{C}/S$ which are not already quotients in $\mathcal{C}$.

Definition 3.1.4. If X (resp. X') is an object of $\mathcal{C}$ endowed with an equivalence relation R (resp. R'), a morphism

$$u: X \rightarrow X'$$

is said to be *compatible with R and R'* if the following equivalent conditions are satisfied:

- (i) for every $S \in \text{Ob}(\mathcal{C})$, two points of $X(S)$ congruent modulo $R(S)$ are transformed by u into two points of $X'(S)$ congruent modulo $R'(S)$;
- (ii) there exists a morphism $R \rightarrow R'$ (necessarily unique) making the following diagram commutative

$$\begin{array}{ccc} R & \longrightarrow & R' \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow{u \times u} & X' \times X' \end{array}$$

By the universal property of X/R , there then exists (when the quotients X/R and X'/R' exist) a unique morphism v making the following diagram commutative

$$\begin{array}{ccc} X & \xrightarrow{p} & X/R \\ \downarrow u & & \downarrow v \\ X' & \xrightarrow{p'} & X'/R' \end{array}$$

Definition 3.1.5. A subobject (= a representable subfunctor) Y of X is said to be *stable* under the equivalence relation R if the following equivalent conditions are satisfied:

- (i) For every $S \in \text{Ob}(\mathcal{C})$, the subset $Y(S)$ of $X(S)$ is stable under $R(S)$.
- (ii) The two subobjects of R , inverse images of Y by p_1 and p_2 , are identical.

An important particular case of a stable subobject of X is the following: the quotient X/R exists and Y is the inverse image on X of a subobject of X/R .

Definition 3.1.6. Let R be an equivalence relation in X and $X' \rightarrow X$ a morphism. The equivalence relation R' in X' obtained by the cartesian diagram

$$\begin{array}{ccc} R' & \longrightarrow & R \\ \downarrow & & \downarrow \\ X' \times X' & \longrightarrow & X \times X \end{array}$$

is called the equivalence relation in X' , *inverse image* of the equivalence relation R in X . In particular, if X' is a subobject of X , one will say that it is the equivalence relation *induced* in X' by R ; one will denote it by $R_{X'}$.

The morphism $X' \rightarrow X$ is compatible with R' and R ; one therefore has, when the quotients exist, a morphism $X'/R' \rightarrow X/R$ (3.1.4). If X' is a subobject of X , we shall see later that in certain cases one can prove that $X'/R' \rightarrow X/R$ is a monomorphism, hence identifies X'/R' with a subobject of X/R . When this is the case, the inverse image of this subobject in X will be a subobject of X majorizing X' and stable under R : the *saturation* of X' for the equivalence relation R .

Proposition 3.1.7. *If the subobject Y of X is stable under R , one has two cartesian squares, for $i = 1, 2$:*

$$\begin{array}{ccc} R_Y & \longrightarrow & R \\ p_i \downarrow & & \downarrow p_i \\ Y & \longrightarrow & X. \end{array}$$

PROOF. Immediate. □

3.2. Equivalence relation defined by a group operating freely

Definition 3.2.1. Let X be an object of $\mathcal{C}$ and H a $\mathcal{C}$ -group operating on X (that is, equipped with a morphism of $\hat{\mathcal{C}}$ -groups

$$H \longrightarrow \text{Aut}(X)).$$

One says that H operates *freely* on X if the following equivalent conditions are satisfied:

- (i) For every $S \in \text{Ob}(\mathcal{C})$, the group $H(S)$ operates freely on $X(S)$;
- (ii) The morphism of functors

$$H \times X \longrightarrow X \times X$$

defined set-theoretically by $(h, x) \mapsto (hx, x)$ is a monomorphism.

(In the preceding definition, one has supposed that the group H operated “on the left” on X . One evidently has an analogous notion in the case where the group operates “on the right”, that is, when one has been given a morphism of groups from the group H° opposite to H into $\text{Aut}(X)$.)

If H operates freely on X , the image of $H \times X$ by the morphism of (ii) is an equivalence relation in X , called the *equivalence relation defined by the action of H on X* . When the quotient of X by this equivalence relation exists, one denotes it by $H \backslash X$ (X/H when H operates on the right). It represents the following covariant functor: if Z is an object of $\mathcal{C}$, one has

$$\text{Hom}(H \backslash X, Z) = \{\text{morphisms from } X \text{ into } Z \text{ invariant under } H\}$$

where the morphism $f: X \longrightarrow Z$ is said to be invariant under H if for every $S \in \text{Ob}(\mathcal{C})$, the corresponding morphism $X(S) \longrightarrow Z(S)$ is invariant under the group $H(S)$.

Lemma 3.2.2. *Under the conditions of 3.2.1, let Y be a subobject of X . The following conditions are equivalent:*

- (i) Y is stable under the equivalence relation defined by H (3.1.5);
- (ii) For every $S \in \text{Ob}(\mathcal{C})$, the subset $Y(S)$ of $X(S)$ is stable under $H(S)$;
- (iii) There exists a morphism f , necessarily unique, making the following diagram commutative

$$\begin{array}{ccc} H \times Y & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ H \times X & \longrightarrow & X. \end{array}$$

Under these conditions, f defines a morphism of $\hat{\mathcal{C}}$ -groups

$$H \longrightarrow \text{Aut}(Y)$$

and the equivalence relation defined in Y by this operation of H is none other than the equivalence relation induced in Y by the equivalence relation defined in X by the action of H .

PROOF. The proof is immediate. One evidently has an analogous statement for a “right operation”. The operation of H on Y defined above will be called the operation *induced* in Y by the given operation of H on X . □

Let us now consider the following situation: H and G are two $\mathcal{C}$ -groups and one has been given a morphism of groups

$$u: H \longrightarrow G.$$

Then H operates on G by translations (one sets, set-theoretically, $hg = u(h)g$) and operates there *freely* if and only if u is a *monomorphism*. The quotient of G by this operation of H is denoted, when it exists, by $H \backslash G$. One likewise defines a right operation of H on G and a quotient G/H . These quotients are functorial with respect to the groups in question; precisely, one has the following lemma, stated for the right quotients:

Lemma 3.2.3. *Let $u: H \longrightarrow G$ and $u': H' \longrightarrow G'$ be two monomorphisms of $\mathcal{C}$ -groups. Suppose given a morphism of $\mathcal{C}$ -groups*

$$f: G \longrightarrow G'.$$

The following conditions are equivalent:

- (i) *f is compatible with the equivalence relations defined in G and G' by H and H' .*
- (ii) *For every $S \in \text{Ob}(\mathcal{C})$, one has $fu(H(S)) \subset u'(H'(S))$.*
- (iii) *There exists a morphism $g: H \longrightarrow H'$, necessarily unique and multiplicative, such that the following diagram is commutative*

$$\begin{array}{ccc} H & \xrightarrow{g} & H' \\ u \downarrow & & \downarrow u' \\ G & \xrightarrow{f} & G' \end{array}$$

Under these conditions, if the quotients G/H and G'/H' exist, there exists a unique morphism f making the following diagram commutative

$$\begin{array}{ccc} G & \xrightarrow{f} & G' \\ p \downarrow & & \downarrow p' \\ G/H & \xrightarrow{f} & G'/H' \end{array}$$

PROOF. The first part is proved by reduction to the set-theoretic case. The second follows immediately from (i). $\square$

One could carry over into the present situation the notions introduced above for general equivalence relations. Let us merely point out the following lemma, whose proof is immediate by reduction to the set-theoretic case:

Lemma 3.2.4. *Let $u: H \longrightarrow G$ be a monomorphism of $\mathcal{C}$ -groups and G' a $\mathcal{C}$ -subgroup of G . For the subobject G' of G to be stable under the equivalence relation defined by H , it is necessary and sufficient that u factor through the canonical monomorphism $G' \longrightarrow G$, and under this condition the operation of H on G' induced by the operation of H on G defined by u is none other than the operation deduced from the monomorphism $H \longrightarrow G'$ through which u factors.*

3.3. Universal effective equivalence relations

Definition 3.3.1. Let $f: X \longrightarrow Y$ be a morphism. One calls the *equivalence relation defined by f in X* , and one denotes by $\mathcal{R}(f)$, the $\hat{\mathcal{C}}$ -equivalence relation in X which is the image of the canonical monomorphism

$$X \times_Y X \longrightarrow X \times X.$$

Definition 3.3.2. Let R be an equivalence relation in X . One says that R is *effective* if

- (i) R is representable (i.e. is a $\mathcal{C}$ -equivalence relation);
- (ii) the quotient $Y = X/R$ exists in $\mathcal{C}^{(13)}$;
- (iii) the diagram

$$R \begin{array}{c} \xrightarrow{p_1} \\ \xrightarrow{p_2} \end{array} X \xrightarrow{p} Y$$

makes R the fiber square of X over Y , that is, R is the equivalence relation defined by p .

Scholium 3.3.2.1. ⁽¹⁴⁾ *If R is an effective equivalence relation in X , then p is an effective epimorphism (1.3). If $f: X \rightarrow Y$ is an effective epimorphism, then $\mathcal{R}(f)$ is an effective equivalence relation in X , one quotient of which is Y . There is therefore a one-to-one “Galois” correspondence between effective equivalence relations in X and effective quotients of X (i.e. equivalence classes of effective epimorphisms with source X).*

Definition 3.3.3. One says that the equivalence relation R in X is *universal effective* if the quotient $Y = X/R$ exists, and if, for every $Y' \rightarrow Y$, the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist and R' is a fiber square of X' over Y' . It amounts to the same to say that R is *effective* and that $p: X \rightarrow X/R$ is a *universal effective epimorphism*.

Scholium 3.3.3.1. ⁽¹⁴⁾ *There is therefore, as above, a one-to-one correspondence between universal effective equivalence relations in X and universal effective quotients of X .*

Remark 3.3.3.2. ⁽¹⁴⁾ Suppose that $\mathcal{C}$ is the category of S -schemes, and denote by $\mathbf{A}^1$ the affine space of dimension 1 over S . Let $R \subset X \times_S X$ be a universal effective equivalence relation and $p: X \rightarrow Y$ the quotient. Then, for every open U of Y , $\mathcal{O}(U) = \text{Hom}_S(U, \mathbf{A}_S^1)$ is the set of elements ϕ of $\mathcal{O}(p^{-1}(U)) = \text{Hom}_S(p^{-1}(U), \mathbf{A}_S^1)$ such that $\phi \circ \text{pr}_1 = \phi \circ \text{pr}_2$. In particular, if R is given by the action of a group H operating freely on the right on X (cf. 3.2.1), then $\mathcal{O}(U)$ is the set of $\phi \in \mathcal{O}(p^{-1}(U))$ such that $\phi(xh) = \phi(x)$, for every $S' \rightarrow S$ and $x \in X(S')$, $h \in H(S')$.

Proposition 3.3.4. *Let R be a universal effective equivalence relation in X . Let $f: X \rightarrow Z$ be a morphism compatible with R , hence factoring through $g: X/R \rightarrow Z$. The following conditions are equivalent:*

- (i) g is a monomorphism;
- (ii) R is the equivalence relation defined by f .

PROOF. Indeed, (i) implies (ii) trivially, and the converse is none other than 2.5. □

Definition 3.3.5. Let H be a $\mathcal{C}$ -group operating freely on X . One says that H operates in an *effective* manner, or that the operation of H on X is *effective* (resp. *universal effective*), if the equivalence relation defined in X by the action of H is effective (resp. universal effective).

3.4. (M)-effectivity. — In practice, it is most often difficult to characterize the universal effective epimorphisms. One nevertheless often has at one’s disposal a certain number of morphisms of this type, for example in the theory of schemes, the faithfully flat quasi-compact morphisms. This leads to the developments below.

3.4.1. — Let us first state a certain number of conditions on a family (M) of morphisms of $\mathcal{C}$:

- (a) (M) is stable under base extension, i.e. every $u: T \rightarrow S$ which is an element of (M) is quarrable (cf. 1.4.0) and for every $S' \rightarrow S$, $u': T \times_S S' \rightarrow S'$ is an element of (M) .
- (b) The composite of two elements of (M) lies in (M) .

⁽¹³⁾N.D.E. : One has added “in $\mathcal{C}$ ”.

⁽¹⁴⁾N.D.E. One has added the numbering 3.3.2.1 (resp. 3.3.3.1) for later references. On the other hand, one has added remark 3.3.3.2.

- (c) *Every isomorphism is an element of (M) .*
 (d) *Every element of (M) is an effective epimorphism.*

Let us note that (a) and (b) imply:

(a') *The cartesian product of two elements of (M) lies in (M) : Let $u: X \rightarrow Y$ and $u': X' \rightarrow Y'$ be two S -morphisms which are elements of (M) . If $Y \times_S Y'$ exists, then $X \times_S X'$ exists and $u \times_S u'$ is an element of (M) .*

This follows from the diagram

$$\begin{array}{ccccc}
 & & Y' & \xleftarrow{u'} & X' \\
 & & \uparrow & & \uparrow \\
 X & \xleftarrow{\quad} & X \times_S Y' & \xleftarrow{\quad} & X \times_S X' \\
 \downarrow u & & \downarrow & \swarrow u \times_S u' & \\
 Y & \xleftarrow{\quad} & Y \times_S Y' & &
 \end{array}$$

Likewise (a) and (d) imply:

(d') *Every element of (M) is a universal effective epimorphism.*

3.4.2. — The family (M_0) of universal effective epimorphisms satisfies conditions (a) to (d) of 3.4.1. Indeed, (a), (c) and (d) are satisfied by definition, and (b) follows from 1.8. In what follows, we shall assume given a family (M) of morphisms of $\mathcal{C}$ satisfying these conditions: our results will therefore apply to the family (M_0) in particular.

Definition 3.4.3. One says that the equivalence relation R in X is of *type (M)* if it is representable and if $p_1 \in (M)$ (which by (b) and (c) implies that $p_2 \in (M)$).

One says that R is *(M) -effective* if it is effective and if the canonical morphism $X \rightarrow X/R$ is an element of (M) .

One says that the quotient Y of X is *(M) -effective* if the canonical morphism $X \rightarrow Y$ is an element of (M) .

From this definition there follow the following consequences:⁽¹⁵⁾

Proposition 3.4.3.1.

- (i) *An (M) -effective equivalence relation is of type (M) and universal effective.*
- (ii) *An (M) -effective quotient is universal effective (cf. 3.3.3).*
- (iii) *The maps $R \mapsto X/R$ and $p \mapsto \mathcal{R}(p)$ realize a one-to-one correspondence between (M) -effective equivalence relations in X and (M) -effective quotients of X .*
- (iv) *(M_0) -effective is equivalent to universal effective.*

PROOF. Let us prove point (i). Since R is (M) -effective, one has a cartesian square

$$\begin{array}{ccc}
 R & \xrightarrow{p_2} & X \\
 p_1 \downarrow & & \downarrow p \\
 X & \xrightarrow{p} & X/R,
 \end{array}$$

and $p \in (M)$. Then, by 3.4.1 (a), p_1 and p_2 belong to (M) , hence R is of type (M) .

Set $Y = X/R$, and let $Y' \rightarrow Y$ be an arbitrary morphism. By 3.4.1 (a), the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist, and the morphisms $X' \rightarrow Y'$ and $p'_i: R' \rightarrow X'$ belong to (M) . Finally, since $R = X \times_Y X$, one obtains, by associativity of the fiber product:

$$R' = X \times_Y X \times_Y Y' = X' \times_{Y'} X'.$$

⁽¹⁵⁾**N.D.E.** : One has added the numbering 3.4.3.1, for later references, and one has set out in detail the proof of point (i).

This shows that R' is (M) -effective; hence, in particular, R is universal effective. This proves (i), and also (iv). Points (ii) and (iii) follow from this, taking definition 3.3.2 into account. $\square$

3.4.4. — Let H be an S -group whose structural morphism is an element of (M) . Then if H operates *freely* on the S -object X , it defines on it an equivalence relation of type (M) . Indeed, by (a) the fiber product $H \times_S X$ exists and $\text{pr}_2: H \times_S X \rightarrow X$ is an element of (M) . One will say that the operation of H is (M) -*effective* if the equivalence relation defined in X by this operation is (M) -effective.

Proposition 3.4.5 ((M)-effectivity and base change). *Let R be an (M) -effective equivalence relation in X over S . Set $Y = X/R$. Let $S' \rightarrow S$ be a base change such that $Y' = Y \times_S S'$ exists. Then $X' = X \times_S S'$ exists, $R' = R \times_S S'$ is an (M) -effective equivalence relation in X' over S' , and $X'/R' \simeq (X/R)'$.*

PROOF. Indeed, the canonical morphisms $X \rightarrow Y$ and $R \rightarrow Y$ are elements of (M) , hence by (a') X' and R' are representable. By associativity of the product, R' is the equivalence relation defined in X' by the canonical morphism $X' \rightarrow Y'$, which is an element of (M) , whence the conclusion. $\square$

Proposition 3.4.6 ((M)-effectivity and cartesian products). *Let R (resp. R') be an (M) -effective equivalence relation in X (resp. X') over S . If $(X/R) \times_S (X'/R')$ exists, then $X \times_S X'$ exists, $R \times_S R'$ is an (M) -effective equivalence relation in $X \times_S X'$ over S , and*

$$(X \times_S X')/(R \times_S R') \simeq (X/R) \times_S (X'/R').$$

PROOF. Set $Y = X/R$, $Y' = X'/R'$. By (a'), the fiber product $X \times_S X'$ exists and the canonical morphism

$$q: X \times_S X' \rightarrow Y \times_S Y'$$

is an element of (M) . Now the formula

$$(X \times X') \times_{(Y \times Y')} (X \times X') \simeq (X \times_Y X) \times (X' \times_{Y'} X')$$

(all unindexed products are taken over S) shows that $R \times_S R'$ is the equivalence relation defined by q in $X \times_S X'$, which completes the proof. $\square$

3.4.7. — ⁽¹⁶⁾Suppose that $\mathcal{C}$ possesses a final object e and let $f: G \rightarrow G'$ be a morphism of $\mathcal{C}$ -groups, such that $f \in (M)$. Then, by 3.4.1 (a), the kernel $\text{Ker}(f)$ is representable by $e \times_{G'} G$, and the morphism $\text{Ker}(f) \rightarrow e$ belongs to (M) .

On the other hand, the equivalence relation defined by f is the same as that defined by the action of $\text{Ker}(f)$ (say, on the right) on G , i.e. it is the image of the morphism $G \times \text{Ker}(f) \rightarrow G \times G$, defined set-theoretically by $(g, h) \mapsto (g, gh)$. Consequently, one deduces from 3.3.2.1 the following corollary:

Corollary 3.4.7.1. *Suppose that $\mathcal{C}$ possesses a final object e and let $f: G \rightarrow G'$ be a morphism of $\mathcal{C}$ -groups, such that $f \in (M)$. Then the action of $\text{Ker}(f)$ on G is (M) -effective and G' is the quotient $G/\text{Ker}(f)$.*

3.5. Construction of quotients by descent. — It frequently happens that one does not know how to construct a quotient directly, but that one knows how to do so after a suitable base change. The present number gives a useful criterion in this situation.

One has seen in paragraph 2.1 the definition of a descent datum on an object X' over S' relative to a morphism $S' \rightarrow S$.

⁽¹⁶⁾N.D.E. : One has added this paragraph.

Definition 3.5.1. One says that a descent datum on X' relative to $S' \rightarrow S$ is *effective*, if X' equipped with this descent datum is isomorphic to the inverse image on S' of an object X over S , equipped with its canonical descent datum.

If $S' \rightarrow S$ is a descent morphism (2.2), then the X of the definition is unique up to unique isomorphism. To say that $S' \rightarrow S$ is an effective descent morphism (2.2) is to say that it is a descent morphism and that every descent datum relative to this morphism is effective.

Let us now consider an equivalence relation R in an object X over S . Let X' (resp. X'' , resp. X''') be the inverse images of X on S' , $S'' = S' \times_S S'$ and $S''' = S' \times_S S' \times_S S'$, and let R' , R'' , R''' be the equivalence relations deduced from R by inverse image. Suppose that the equivalence relation R' in X' is (M) -effective, and consider the quotient $Y' = X'/R'$, which is an object over S' . Its two inverse images on S'' are isomorphic to X''/R'' by 3.4.5. The S' -object Y' is therefore equipped with a canonical gluing datum. Using likewise the uniqueness of X'''/R''' , one sees that it is even a descent datum. (Remark: one has implicitly supposed in this proof that all the fiber products written existed, which is the case in particular if $S' \rightarrow S$ is quarrable, for example a descent morphism.)

Proposition 3.5.2. *Let R be an equivalence relation in the object X over S . Let $S' \rightarrow S$ be a universal effective epimorphism. Suppose that every S -morphism whose inverse image on S' is in (M) is itself in (M) . The following conditions are equivalent:*

- (i) R is (M) -effective in X ;
- (ii) R' is (M) -effective in X' and the canonical descent datum on X'/R' is effective.

If this is so, the “descended” object of X'/R' is canonically isomorphic to X/R .

The fact that (i) entails (ii) is nothing other than the translation into the language of descent of 3.4.4. If one shows the converse, the last assertion of the proposition will be a consequence of the fact that a universal effective epimorphism is a descent morphism (2.3), hence that the “descended object” is unique (up to unique isomorphism).

Let us therefore prove (ii) $\Rightarrow$ (i). Let Y' be the quotient X'/R' and Y the descended object. As the canonical morphism $p': X' \rightarrow X'/R' = Y'$ is compatible by construction with the descent data (its inverse images on S'' coincide with the canonical morphism $X'' \rightarrow X''/R''$), it arises by inverse image on S' from an S -morphism $p: X \rightarrow Y$. As p' is an element of (M) , it follows from the hypothesis made on the morphism $S' \rightarrow S$ that p is likewise an element of (M) . As p' is compatible with the equivalence relation R' , p is compatible with R , again because a universal effective epimorphism is a descent morphism. One therefore has a morphism

$$R \longrightarrow X \times_Y X.$$

To prove that R is (M) -effective and that Y is isomorphic to X/R , it suffices to prove that this morphism is an isomorphism. Now it becomes one by extension of the base from S to S' , for R' is effective; it is therefore an isomorphism for the same reason as previously (2.4).

Let us remark that the hypothesis of the text is verified if one takes for (M) the family (M_0) of universal effective epimorphisms and if $\mathcal{C}$ possesses fiber products (1.10). One deduces from this the

Corollary 3.5.3. *Suppose that $\mathcal{C}$ possesses fiber products (over S would suffice). Let R be an equivalence relation in X over S and $S' \rightarrow S$ a universal effective epimorphism. The following conditions are equivalent:*

- (i) R is universally effective in X ,
- (ii) R' is universally effective in X' and the canonical descent datum on X'/R' is effective.

If this is so, the “descended” object of X'/R' is canonically isomorphic to X/R .

4. Topologies and sheaves

The notion of sieve, and the presentation of the notion of topology (4.2.1) adopted here, (more intrinsic and more convenient in many respects than that by covering families of [MA]), are due to J. Giraud [AS].

4.1. Sieves

Definition 4.1.1. One calls a sieve of the category $\mathcal{C}$ a subfunctor C of the final functor $\mathbf{e}: \mathcal{C}^\circ \longrightarrow (\mathbf{Ens})$.

With every sieve C of $\mathcal{C}$ one associates the set $E(C)$ of the objects X of $\mathcal{C}$ such that $C(X) \neq \emptyset$, that is, such that the structural morphism $X \longrightarrow \mathbf{e}$ factors through C . One therefore has the equivalences

$$(+) \quad \begin{cases} X \in E(C) & \Longleftrightarrow C(X) = \mathbf{e}(X) = \{\emptyset\}. \\ X \notin E(C) & \Longleftrightarrow C(X) = \emptyset. \end{cases}$$

The set $E = E(C)$ enjoys the following property:

(++) If $X \in E$ and if $\text{Hom}(Y, X) \neq \emptyset$, then $Y \in E$.

(Let us remark that if one equips the set $\text{Ob}(\mathcal{C})$ with its *natural preorder structure*, (Y dominating X if there exists an arrow from Y to X), the sets E satisfying (++) are the subsets of $\text{Ob}(\mathcal{C})$ which contain every majorant⁽¹⁷⁾ of one of their elements.)

Conversely, if E is a subset of $\text{Ob}(\mathcal{C})$ enjoying the property (++) , then E may be written in a unique manner in the form $E(C)$ and C is defined by the formulas (+). There is therefore a one-to-one correspondence between the sieves of $\mathcal{C}$ and the subsets of $\text{Ob}(\mathcal{C})$ satisfying the condition (++) . By *abuse of language*, we shall sometimes say that the set $E(C)$ is a sieve of $\mathcal{C}$.

⁽¹⁸⁾Let C and C' be two sieves of $\mathcal{C}$; as they are two subfunctors of the final functor $\mathbf{e}$, it amounts to the same to say that C dominates C' (for the domination relation in $\text{Ob}(\widehat{\mathcal{C}})$), or that C is a *subfunctor* of C' , or again that $E(C) \subset E(C')$; one will then say that C is *finer* than C' . One sees that one is dealing in this case with an order structure on the set of sieves of $\mathcal{C}$. Moreover, one has $E(C) \cap E(C') = E(C \times C')$ and hence the set of sieves of $\mathcal{C}$ is *filtered* for the order relation “being finer”.

Every subset E of $\text{Ob}(\widehat{\mathcal{C}})$, for example a subset of $\text{Ob}(\mathcal{C})$, defines a sieve $C(E)$: the set of the $X \in \text{Ob}(\mathcal{C})$ such that $F(X) \neq \emptyset$ for at least one $F \in E$ satisfies the condition (++) and defines the sieve sought.

This sieve may also be defined as the *image* of the family of morphisms $\{F \longrightarrow \mathbf{e}, F \in E\}$ in the sense of the following definition:

Definition 4.1.2. Let $\{F_i \longrightarrow F\}$ be a family of morphisms of $\widehat{\mathcal{C}}$ with the same target F . One calls the *image* of this family the subfunctor of F defined by

$$S \longmapsto \bigcup_i \text{Im } F_i(S) \subset F(S).$$

Proposition 4.1.3. *The formation of the image commutes with extension of the base: in the preceding notations, let us denote by I the image of the family $\{F_i \longrightarrow F\}$; for every morphism $G \longrightarrow F$ of $\widehat{\mathcal{C}}$, the image of the family of morphisms $\{F_i \times_F G \longrightarrow G\}$ is the subfunctor $I \times_F G$ of G .*

⁽¹⁷⁾**N.D.E.** : Here, “majorant” is taken in the sense of the preorder relation mentioned above, that is, Y majorizes X if there exists an arrow $Y \longrightarrow X$. On the other hand, if X, Y are two subobjects of an object Z , one says (cf. 2.4) that Y majorizes X if $X \subset Y$. To avoid any ambiguity between these two terminologies, one has in the sequel replaced “majorant” by “dominating” in the first case, and by “containing”, in the second.

⁽¹⁸⁾**N.D.E.** : One has spelled out the sentence which follows.

Definition 4.1.4.0. ⁽¹⁹⁾Let C be a sieve of $\mathcal{C}$. If E is a subset of $\text{Ob}(\mathcal{C})$ such that $C(E) = C$, one says that E is a *base* of C . Every sieve C possesses a base, for example the set $E(C)$.

We propose to describe the set $\text{Hom}(C, F)$, where C is a sieve of $\mathcal{C}$ and F an object of $\widehat{\mathcal{C}}$, by means of a base $\{S_i\}$ of C . For each pair (i, j) , one has a diagram in $\widehat{\mathcal{C}}$:

$$\begin{array}{ccc} S_i \times S_j & \longrightarrow & S_i \\ \downarrow & & \downarrow \\ S_j & \longrightarrow & \mathbf{e}, \end{array}$$

whence a diagram of sets

$$\Gamma(F) = \text{Hom}(\mathbf{e}, F) \xrightarrow{\sigma} \prod_i \text{Hom}(S_i, F) \xrightleftharpoons[\tau_2]{\tau_1} \prod_{i,j} \text{Hom}(S_i \times S_j, F)$$

such that $\tau_1\sigma = \tau_2\sigma$. One therefore has a morphism

$$\text{Hom}(\mathbf{e}, F) \longrightarrow \text{Ker} \left(\prod_i \text{Hom}(S_i, F) \Rightarrow \prod_{i,j} \text{Hom}(S_i \times S_j, F) \right).$$

One verifies immediately:

Proposition 4.1.4. *One has an isomorphism, functorial in F ,*

$$\text{Hom}(C, F) \xrightarrow{\sim} \text{Ker} \left(\prod_i \text{Hom}(S_i, F) \Rightarrow \prod_{i,j} \text{Hom}(S_i \times S_j, F) \right),$$

such that the diagram

$$\begin{array}{ccc} \text{Hom}(\mathbf{e}, F) & \longrightarrow & \text{Ker} \left(\prod_i \text{Hom}(S_i, F) \Rightarrow \prod_{i,j} \text{Hom}(S_i \times S_j, F) \right) \\ & & \uparrow \wr \\ \text{Hom}(\mathbf{e}, F) & \longrightarrow & \text{Hom}(C, F), \end{array}$$

where the last line is induced by the canonical morphism $C \longrightarrow \mathbf{e}$, is commutative.

Corollary 4.1.5. *Suppose that the fiber products $S_i \times S_j$ are representable, for example that the S_i be quarrable. One then has, for every $F \in \text{Ob}(\widehat{\mathcal{C}})$, an isomorphism*

$$\text{Hom}(C, F) \xrightarrow{\sim} \text{Ker} \left[\prod_i F(S_i) \Rightarrow \prod_{i,j} F(S_i \times S_j) \right].$$

Remark 4.1.6. Let R be a sieve of $\mathcal{C}$; let us denote by $\mathcal{R}$ the full subcategory of $\mathcal{C}$ whose set of objects is $E(R)$ and by

$$i_R: \mathcal{R} \longrightarrow \mathcal{C}$$

the inclusion functor. One has an isomorphism, functorial in $F \in \text{Ob}(\widehat{\mathcal{C}})$,

$$\text{Hom}(R, F) \xrightarrow{\sim} \Gamma(F \circ i_R)$$

such that the diagram

$$\begin{array}{ccc} \text{Hom}(\mathbf{e}, F) & \longrightarrow & \text{Hom}(R, F) \\ \downarrow \wr & & \downarrow \wr \\ \Gamma(F) & \longrightarrow & \Gamma(F \circ i_R), \end{array}$$

⁽¹⁹⁾**N.D.E.** : One has added the numbering 4.1.4.0, in order to bring this definition into evidence.

where the second line is induced by the functor i_R , is commutative.

Definition 4.1.7. Let $\mathcal{C}$ be a category. One calls a sieve of the object S of $\mathcal{C}$ a sieve of the category $\mathcal{C}_{/S}$.

A sieve of S is therefore a sub- $\widehat{\mathcal{C}}$ -object of S . There corresponds to it canonically a subset of $\text{Ob}(\mathcal{C}_{/S})$ containing the source of every arrow whose target it contains. By *abuse of language*, such a set will also be called a sieve of S .

4.2. Topologies: definitions

Definition 4.2.1. Let $\mathcal{C}$ be a category. One calls a topology on $\mathcal{C}$ the giving, for each S of $\mathcal{C}$, of a set $J(S)$ of sieves of S , called *covering* sieves or *refinements* of S , this giving satisfying the following axioms:

- (T 1) For every refinement R of S and every morphism $T \rightarrow S$, the sieve $R \times_S T$ of T is *covering* (“stability under *base change*”).
- (T 2) If R, C are two sieves of S , if R is covering and if for every $T \in \text{Ob}(\mathcal{C})$ and every morphism $T \rightarrow R$, the sieve $C \times_S T$ of T is covering, then C is a refinement of S .⁽²⁰⁾
- (T 3) If $C \supset R$ are two sieves of S and if R is covering, then C is covering.
- (T 4) For every S , S is a refinement of S .

One may reformulate these axioms in the following manner. Suppose given a topology $S \mapsto J(S)$ on $\mathcal{C}$ and, for every object F of $\widehat{\mathcal{C}}$, let us denote by $J(F)$ the set of subfunctors R of F such that for every morphism $T \rightarrow F$ of $\widehat{\mathcal{C}}$, where T is *representable*, $R \times_F T$, which is a sieve of T , is *covering*. By virtue of (T 1), this notation is indeed compatible with the preceding one. One will likewise say that $R \in J(F)$ is a *refinement* of F . One verifies immediately that the preceding axioms entail the following properties:

- (T' 0) If $F \supset G$ are two objects of $\widehat{\mathcal{C}}$, and if for every $S \in \text{Ob}(\mathcal{C})$ and every morphism $S \rightarrow F$, $G \times_F S \in J(S)$, then $G \in J(F)$.
- (T' 1) If $G \in J(F)$, and if $H \rightarrow F$ is a morphism of $\widehat{\mathcal{C}}$, then $G \times_F H \in J(H)$.
- (T' 2) If $F \supset G \supset H$ are three objects of $\widehat{\mathcal{C}}$, if $G \in J(F)$ and $H \in J(G)$, then $H \in J(F)$.
- (T' 3) If $F \supset G \supset H$ are three objects of $\widehat{\mathcal{C}}$ and if $H \in J(F)$, then $G \in J(F)$.
- (T' 4) For every $F \in \text{Ob}(\widehat{\mathcal{C}})$, $F \in J(F)$.

Conversely, if one gives oneself for every $F \in \text{Ob}(\widehat{\mathcal{C}})$ a set $J(F)$ of subobjects of F satisfying the properties (T' 0) to (T' 4), the map $S \mapsto J(S)$ defines a topology on $\mathcal{C}$ and the two preceding constructions are inverse to one another.

From (T' 1), (T' 2) and (T' 3)⁽²¹⁾ there results the following property:

- (T' 5) If G and H are two subobjects of F and if $G, H \in J(F)$, then $G \cap H \in J(F)$.

The set $J(F)$, ordered by the relation $\supset$, is therefore *filtered*; this remark will be of use to us later.

4.2.2. — One will say that the topology defined by J is finer than the topology defined by J' if for every $S \in \text{Ob}(\mathcal{C})$, $J(S) \supset J'(S)$ (it amounts to the same to say that for every $F \in \text{Ob}(\widehat{\mathcal{C}})$, $J(F) \supset J'(F)$).

Every set of topologies on $\mathcal{C}$ possesses a greatest *lower* bound: let I be a set of indices, and for each $i \in I$, let $S \mapsto J_i(S)$ be a topology on $\mathcal{C}$. Set $J(S) = \bigcap_{i \in I} J_i(S)$; it is immediate

⁽²⁰⁾**N.D.E.** : that is: if C is covering “locally with respect to the covering sieve R ”, then C is covering.

⁽²¹⁾**N.D.E.** : (T' 1) and (T' 2) suffice: $G \cap H = G \times_F H$ belongs to $J(H)$, by (T' 1), hence to $J(F)$, by (T' 2).

that one has thus defined a topology on $\mathcal{C}$, and that it is indeed the greatest *lower* bound of the given set.

In particular, let us give ourselves, for each $S \in \text{Ob}(\mathcal{C})$, a set $E(S)$ of sieves of S . One calls the topology *generated* by these sets the *least fine* topology for which the elements of $E(S)$ are refinements of S for every S .

Definition 4.2.3. Let $\{F_i \rightarrow F\}$ be a family of morphisms of $\widehat{\mathcal{C}}$. Let $G \subset F$ be the image (4.1.2) of this family. The family is said to be *covering* if $G \in J(F)$. A morphism is said to be *covering* if the family reduced to this morphism is covering.

This definition applies in particular to an inclusion: a sieve C of S is covering if and only if the canonical morphism $C \rightarrow S$ is covering.

The axioms (T' 0) to (T' 5) entail for covering families the following properties:

(C 0) Let $\{F_i \rightarrow F\}$ be a family of morphisms of $\widehat{\mathcal{C}}$. If for every representable base change $S \rightarrow F$, the family $\{F_i \times_F S \rightarrow S\}$ is covering, then the original family is covering too.

(C 1) For every covering family $\{F_i \rightarrow F\}$ and every morphism $G \rightarrow F$, the family $\{F_i \times_F G \rightarrow G\}$ is covering (“stability under base change”).

(C 2) If $\{F_i \rightarrow F\}$ is a covering family and if, for each i , $\{F_{ij} \rightarrow F_i\}$ is a covering family, then the composite family $\{F_{ij} \rightarrow F\}$ is covering (“stability under composition”).

(C 3) If $\{G_j \rightarrow F\}$ is a covering family, and if $\{F_i \rightarrow F\}$ is a family of morphisms with target F such that for each j there exists an i such that $G_j \rightarrow F$ factors through $F_i \rightarrow F$, then $\{F_i \rightarrow F\}$ is covering (“saturation”).

(C 4) Every family consisting of a single isomorphism is covering.

Note that (C 2) and (C 3) also imply:

(C 5) If $\{F_i \rightarrow F\}$ is a family of morphisms with target F such that there exists a covering family $\{G_j \rightarrow F\}$ such that for every j the family $\{F_i \times_F G_j \rightarrow G_j\}$ is covering, then the family $\{F_i \rightarrow F\}$ is covering (“a locally covering family is covering”).

4.2.4. — Conversely, let $\mathcal{C}$ be a category possessing fibered products and, for each $S \in \text{Ob } \mathcal{C}$, let us give ourselves a set of families of morphisms of $\mathcal{C}$ with target S , called covering families, this datum satisfying axioms (C 1) to (C 4) (and hence also (C 5), which is a consequence of them). For every $S \in \text{Ob } \mathcal{C}$, let $J(S)$ be the set of sieves of S possessing a covering basis (or, what amounts to the same by (C 3), all of whose bases are covering). Then $S \mapsto J(S)$ defines a topology on $\mathcal{C}$. The two preceding constructions are inverse to one another.

In fact, in applications it is not very practical to consider all covering families, for one sometimes possesses fairly simple descriptions of a “sufficient” number of these families. This leads to the following definitions.

Definition 4.2.5.0. ⁽²²⁾ Let $\mathcal{C}$ be a category. Suppose that, for each $S \in \text{Ob } \mathcal{C}$, a set $P(S)$ of families of morphisms of $\mathcal{C}$ with target S is given. One calls the *topology generated by P* the least fine topology for which the given families are covering.

Definition 4.2.5. Let $\mathcal{C}$ be a category. One calls a *pretopology* on $\mathcal{C}$ the datum, for each $S \in \text{Ob } \mathcal{C}$, of a set $R(S)$ of families of morphisms $\{S_i \rightarrow S\}$ with target S , called covering for the pretopology in question, satisfying the following axioms:

(P 1) For every family $\{S_i \rightarrow S\} \in R(S)$ and every morphism $T \rightarrow S$, the fibered products $S_i \times_S T$ exist and $\{S_i \times_S T \rightarrow T\} \in R(T)$.

(P 2) If $\{S_i \rightarrow S\} \in R(S)$ and if for each i , $\{T_{ij} \rightarrow S_i\} \in R(S_i)$, then the composite family $\{T_{ij} \rightarrow S\}$ belongs to $R(S)$.

(P 3) Every family consisting of a single isomorphism is covering.

⁽²²⁾**N.D.E.** This definition (placed in the original after 4.2.5) has been put here, since it will be used in 6.2.1 in a slightly more general setting than that of 4.2.5.

Proposition 4.2.6. *For every S , let $J(S)$ be the set of sieves of S covering for the topology generated by the pretopology R . Let $J_R(S)$ be the subset of $J(S)$ consisting of the sieves defined by families in $R(S)$. Then $J_R(S)$ is cofinal in $J(S)$: every refinement of S contains a sieve defined by a family in $R(S)$.*

PROOF. For every S , let $J'(S)$ be the set of sieves of S containing a sieve in $J_R(S)$. One evidently has $J'(S) \subset J(S)$. To show that $J(S) = J'(S)$, it suffices to show that the $J'(S)$ form a topology on $\mathcal{C}$, that is, satisfy axioms (T 1) to (T 4). Now (T 1), (T 3), (T 4) are evidently satisfied. It remains to verify (T 2).⁽²³⁾

Thus let U be an element of $J'(S)$ and C a sieve of S ; one assumes that for every $T \rightarrow U$, the sieve $C \times_S T$ is in $J'(T)$, and one must prove that $C \in J'(S)$. By definition of J' , U contains a refinement U' defined by a family $\{S_i \rightarrow S\} \in R(S)$. Since one has verified (T 3), it suffices to prove that $U' \cap C \in J'(S)$; one may therefore assume that $U = U'$. By hypothesis, for every i , $C \times_S S_i \in J'(S_i)$; there therefore exists, for each i , a covering family $\{T_{ij} \rightarrow S_i\} \in R(S_i)$ such that $T_{ij} \rightarrow S_i$ factors through $C \times_S S_i \rightarrow S_i$. The morphism $T_{ij} \rightarrow S$ therefore factors through $C \rightarrow S$, which shows that C contains the sieve defined by the composite family $\{T_{ij} \rightarrow S\}$, and one has finished by (P 2). $\square$

Axioms (P 1) to (P 3) are those of [MA]. In view of the practical interest of pretopologies, we shall interpret each important result with the aid of a pretopology defining the given topology.

Remark 4.2.7. One may introduce a slightly more general notion: for each S one gives a set of covering families satisfying (P 1), (P 3) and Proposition 4.2.6. This occurs in particular when the given families satisfy (P 1), (P 3) and (C 5). The reader may consult [D].

Definition 4.2.8. Let $\mathcal{C}$ be endowed with a topology, and let S be an object of $\mathcal{C}$. Let $P(S')$ be a relation involving an argument $S' \in \text{Ob } \mathcal{C}/_S$. One assumes that $\text{Hom}(S'', S') \neq \emptyset$ implies $P(S') \Rightarrow P(S'')$. One says that P is *true locally on S* for the topology considered if the following equivalent conditions are satisfied:

- (i) The set of $S' \rightarrow S$ such that $P(S')$ is true is a refinement of S .
- (ii) There exists a refinement of S such that $P(S')$ is true for every S' of this refinement.
- (iii) (If the given topology is defined by a pretopology.) There exists a covering family for this pretopology such that $P(S')$ is true for every S' of this family.

Example 4.2.9. Let $f : X \rightarrow Y$ be an S -morphism. One will say that f is locally an isomorphism if there exists a covering family $\{S_i \rightarrow S\}$ such that, for every i , $f \times_S S_i$ is an isomorphism. It amounts to the same to require that there exist a refinement R of S such that, for every $T \rightarrow R$, $X(T) \rightarrow Y(T)$ is an isomorphism.

One will see many other examples of “local” language in what follows.

4.3. Presheaves, sheaves, sheaf associated to a presheaf

Definition 4.3.1. Let $\mathcal{C}$ be a category. One calls a *presheaf of sets* on $\mathcal{C}$ any contravariant functor from $\mathcal{C}$ to the category of sets. The category $\hat{\mathcal{C}} = \text{Hom}(\mathcal{C}^{\circ}, (\text{Ens}))$ is called the *category of presheaves* on $\mathcal{C}$. If $\mathcal{C}$ is endowed with a topology, one says that the presheaf P is *separated* (resp. is a *sheaf*) if for every $S \in \text{Ob } \mathcal{C}$ and every $R \in J(S)$, the canonical map

$$(+) \quad P(S) = \text{Hom}(S, P) \rightarrow \text{Hom}(R, P)$$

⁽²³⁾**N.D.E.** In what follows, misprints in the original have been corrected.

is injective (resp. bijective). One calls the full subcategory of $\widehat{\mathcal{C}}$ whose objects are the sheaves the *category of sheaves*, and denotes it by $\widetilde{\mathcal{C}}$.⁽²⁴⁾

Proposition 4.3.2. *Let P be a separated presheaf (resp. a sheaf). For every functor $H \in \text{Ob } \widehat{\mathcal{C}}$ and every $R \in J(H)$, the canonical map*

$$(+)\quad \text{Hom}(H, P) \longrightarrow \text{Hom}(R, P)$$

is injective (resp. bijective).

PROOF. Indeed, let P be a separated presheaf, H a presheaf, $R \in J(H)$, and $u, v : H \rightarrow P$ such that $uj = vj$. For every $f : S \rightarrow H$, $S \in \text{Ob } \mathcal{C}$, $R \times_H S$ is a refinement of S and $ufj_S = vfj_S$:

$$\begin{array}{ccccc} & & R & \xrightarrow{j} & H & \xrightarrow{u,v} & P \\ & \nearrow f_R & & & \nearrow f & & \\ R \times_H S & \xrightarrow{j_S} & S & & & & \end{array}$$

Since P is separated, one deduces $uf = vf$. Since this is true for every representable S , one has $u = v$.

Suppose now that P is a sheaf. Let $g : R \rightarrow P$; let us show that it factors through H . For every $f : S \rightarrow H$, $S \in \text{Ob } \mathcal{C}$, $g \circ f_R : R \times_H S \rightarrow P$ factors uniquely through S , hence defines a morphism $h : S \rightarrow P$ which is evidently functorial in f , by uniqueness:

$$\begin{array}{ccccc} R \times_H S & \xrightarrow{f_R} & R & \xrightarrow{g} & P \\ \downarrow j_S & & \downarrow j & \nearrow h & \\ S & \xrightarrow{f} & H & & \end{array}$$

One has therefore defined, for every S , a map from $H(S)$ to $P(S)$, functorial in S , hence a morphism from H to P which does satisfy the required conditions. $\square$

Corollary 4.3.2.1.⁽²⁵⁾ *Let R, F be two sheaves. If R is a refinement of F , then $R = F$.*

PROOF. Indeed, suppose that R is a refinement of F and denote by j the inclusion $R \hookrightarrow F$. By 4.3.2, one has $\text{Hom}(F, R) = \text{End}(R)$, hence there exists $\pi : F \rightarrow R$ such that $\pi \circ j = \text{id}_R$. Similarly, one has $\text{End}(F) = \text{Hom}(R, F)$, and the equality $j \circ \pi \circ j = j$ implies $j \circ \pi = \text{id}_F$, so j is an isomorphism. $\square$

Proposition 4.3.3. ([AS], 1.3). *Let $\mathcal{C}$ be a category. Let P be a presheaf on $\mathcal{C}$; for every $S \in \text{Ob } \mathcal{C}$, denote by $J(S)$ the set of sieves R of S such that for every $T \rightarrow S$, the map*

$$(+)\quad \text{Hom}(T, P) \longrightarrow \text{Hom}(R \times_S T, P)$$

is injective (resp. bijective). Then the $J(S)$ define a topology on $\mathcal{C}$, i.e. satisfy axioms (T 1) to (T 4).

Corollary 4.3.4. *For every $S \in \text{Ob } \mathcal{C}$, let $K(S)$ be a family of sieves satisfying (T 1). Let P be a presheaf on $\mathcal{C}$. For it to be separated (resp. a sheaf) for the topology generated by the $K(S)$, it is necessary and sufficient that, for every $S \in \text{Ob } \mathcal{C}$ and every $R \in K(S)$, the canonical map*

$$(+)\quad \text{Hom}(S, P) \longrightarrow \text{Hom}(R, P)$$

⁽²⁴⁾**N.D.E.** One will note that if Q is a subpresheaf of a separated presheaf P , then Q is separated. Indeed, for every sieve R of S , the composite map $Q(S) \hookrightarrow P(S) \hookrightarrow P(R)$ is injective and factors through $Q(S) \rightarrow Q(R)$.

⁽²⁵⁾**N.D.E.** This corollary has been added.

be injective (resp. bijective).

Corollary 4.3.5. *For each $S \in \text{Ob } \mathcal{C}$, let $R(S)$ be a set of families of morphisms of $\mathcal{C}$ with target S , satisfying (P 1) (for example defining a pretopology). Let P be a presheaf on $\mathcal{C}$. For P to be separated (resp. a sheaf) for the topology generated by R , it is necessary and sufficient that for every $S \in \text{Ob } \mathcal{C}$ and every family $\{S_i \rightarrow S\} \in R(S)$ the map*

$$P(S) \rightarrow \prod_i P(S_i)$$

be injective (resp. the diagram

$$P(S) \rightarrow \prod_i P(S_i) \rightrightarrows \prod_{i,j} P(S_i \times_S S_j)$$

be exact).

Definition 4.3.6. Let $\mathcal{C}$ be a category. One calls the *canonical topology* on $\mathcal{C}$ the finest topology for which all representable functors are sheaves.

Corollary 4.3.7. *For a sieve R of S to be a refinement for the canonical topology, it is necessary and sufficient that, for every morphism $T \rightarrow S$ of $\mathcal{C}$ and every $X \in \text{Ob } \mathcal{C}$, the canonical map*

$$\text{Hom}(T, X) \rightarrow \text{Hom}(R \times_S T, X)$$

be bijective.

Definition 4.3.8. A covering sieve for the canonical topology will be called a *universal effective epimorphic sieve*.

Corollary 4.3.9. *A universal effective epimorphic family defines a universal effective epimorphic sieve. Conversely, every base-changeable family defining a universal effective epimorphic sieve is universal effective epimorphic.*

Let us return to the case where $\mathcal{C}$ is endowed with an arbitrary topology and pass to the construction of the sheaf associated to a presheaf P . Let S be an object of $\mathcal{C}$. If $R \supset R'$ are two refinements of S , one has a diagram

$$\begin{array}{ccc} \text{Hom}(S, P) & \longrightarrow & \text{Hom}(R, P) \\ & \searrow & \downarrow \\ & & \text{Hom}(R', P) \end{array}$$

The ordered set $J(S)$ is filtered, as one has already remarked. Since S is an element of $J(S)$, one has an evident morphism

$$\text{Hom}(S, P) \rightarrow \varinjlim_{R \in J(S)} \text{Hom}(R, P).$$

Definition 4.3.10.0. ⁽²⁶⁾ Set $\check{H}^0(S, P) = \varinjlim_{R \in J(S)} \text{Hom}(R, P)$. One verifies that $\check{H}^0(S, P)$ depends functorially on S , hence defines a functor LP by

$$(++)\quad \text{Hom}(S, LP) = \check{H}^0(S, P) = \varinjlim_{R \in J(S)} \text{Hom}(R, P).$$

⁽²⁶⁾**N.D.E.** The numbering 4.3.10.0 has been added for later references. Moreover, it follows from the definition that if $Q \rightarrow P$ is a monomorphism, the same is true of $LP \rightarrow LQ$; thus L “preserves monomorphisms” (see also 4.3.16 for a more general result: L “commutes with finite projective limits”).

By construction one has morphisms

$$\begin{aligned}\ell_P : P &\longrightarrow LP \\ z_R : \text{Hom}(R, P) &\longrightarrow \text{Hom}(S, LP).\end{aligned}$$

Lemma 4.3.10. (i) For every refinement R of S and every $u : R \longrightarrow P$, the diagram

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ \uparrow u & & \uparrow z_R(u) \\ R & \xrightarrow{i_R} & S \end{array}$$

is commutative.

(ii) For every morphism $S \xrightarrow{v} LP$, there exists a refinement R of S and a morphism $u : R \longrightarrow P$ with $v = z_R(u)$.

(iii) Let Q be a functor and $u, v : Q \longrightarrow P$ such that $\ell_P u = \ell_P v$. Then the kernel of the pair (u, v) is a refinement of Q .

(iv) Let $u : R \longrightarrow P$ and $u' : R' \longrightarrow P$; for $z_R(u) = z_{R'}(u')$, it is necessary and sufficient that there exist a refinement $R'' \subset R \cap R'$ of S such that u and u' agree on R'' .

PROOF. (i): One must verify that $z_R(u)i_R = \ell_P u$. For this, it suffices to verify that the composites of these two morphisms with every morphism $T \xrightarrow{g} R$, where T is representable, are equal. Now consider $f = i_R g$ and the fibered product $R' = R \times_S T$:⁽²⁷⁾

$$\begin{array}{ccccc} P & \xrightarrow{\ell_P} & LP & & \\ \uparrow u & & \uparrow z_R(u) & & \\ R & \xrightarrow{i_R} & S & & \\ \uparrow p & \swarrow g & \uparrow f & & \\ R' & \xrightarrow{\sim} & T & & \end{array}$$

By definition of ℓ_P , $\ell_P u g = z_R(u) f$ (this is the particular case of what one is trying to prove in which i_R is an isomorphism), and $z_R(u) f = z_R(u) i_R g$.

(ii) and (iv) merely express the definition of $\text{Hom}(S, LP)$ as an inductive limit.

(iii): If K denotes the kernel of the pair (u, v) , for each morphism $f : S \longrightarrow Q$, where S is representable, $K \times_Q S$ is a subfunctor of the kernel of the pair of arrows $u f, v f : S \longrightarrow P$. By (T' 0), one is therefore reduced to proving the assertion in the case where $Q = S$ is representable. But in this case it follows from (ii) and (iv) that K contains a refinement of S , hence is a refinement of S . $\square$

Finally one verifies that $P \mapsto LP$ defines a functor

$$L : \widehat{\mathcal{C}} \longrightarrow \widehat{\mathcal{C}}$$

and $P \mapsto \ell_P$ a morphism of functors

$$\ell : \text{Id}_{\widehat{\mathcal{C}}} \longrightarrow L.$$

Let us now state the essential result:

Proposition 4.3.11. (i) If P is any presheaf, LP is separated and $\ell_P : P \longrightarrow LP$ is covering (4.2.3).

(ii) If P is a sheaf, $P \longrightarrow LP$ is an isomorphism.

⁽²⁷⁾**N.D.E.** Denote by p the projection $R' \longrightarrow R$. Since i_R is a monomorphism, one has $p = g i_{R'}$. Let g' be the section of $i_{R'}$ defined by g ; then $p g' = g$, hence $p g' i_{R'} = g i_{R'} = p$; this implies $g' i_{R'} = \text{id}_{R'}$, and thus $i_{R'} : R' \longrightarrow T$ is an isomorphism, with inverse g' .

(iii) For every presheaf P and every separated presheaf (resp. sheaf) F , the map

$$\text{Hom}(\ell_P, F) : \text{Hom}(LP, F) \longrightarrow \text{Hom}(P, F)$$

is injective (resp. bijective).

(iv) If P is separated, $\ell_P : P \longrightarrow LP$ is a covering monomorphism (hence P is a refinement of LP), and LP is a sheaf.

PROOF. (i) First, $P \longrightarrow LP$ is covering; indeed, for every morphism $S \xrightarrow{v} LP$, there exists, by 4.3.10 (i) and (ii), a refinement R of S such that one has a commutative diagram:

$$\begin{array}{ccccc} & & P & \xrightarrow{\ell_P} & LP \\ & \nearrow & \uparrow v' & & \uparrow v \\ & P \times_{LP} S & \longrightarrow & S \\ & \nearrow & \uparrow i_R & & \\ R & & & & \end{array}$$

hence $P \times_{LP} S \longrightarrow S$ is covering. It follows, by (C 0), that $P \longrightarrow LP$ is covering.

If, on the other hand, two morphisms $S \xrightarrow{v_1, v_2} LP$ induce the same morphism from a refinement R of S , let us show that they are equal. There exist refinements R_i , $i = 1, 2$, and morphisms $u_i : R_i \longrightarrow P$ such that $z_{R_i}(u_i) = v_i$. Taking R sufficiently small, one may assume that $R_1 = R_2 = R$. It then follows from the commutative diagram of 4.3.10 (i) that $\ell_P u_1 = \ell_P u_2$. By *loc. cit.* (iii), u_1 and u_2 therefore agree on a refinement of R , hence a refinement of S , which implies $z_R(u_1) = z_R(u_2)$, by *loc. cit.* (iv).

(ii) is clear, for if P is a sheaf, $\text{Hom}(S, P) \longrightarrow \text{Hom}(R, P)$ is already an isomorphism for every refinement R of S .

(iii) Let u, v be two morphisms $LP \longrightarrow F$ such that $u\ell_P = v\ell_P$. To show that $u = v$, it suffices to see that $uf = vf$ for every $f : S \longrightarrow LP$, where S is representable. Now there exists a refinement R of S and a morphism $g : R \longrightarrow P$ with $f = z_R(g)$. Then uf and vf agree on R with $u\ell_P g = v\ell_P g$, hence agree on R . If F is separated, one therefore has $uf = vf$. Suppose now that F is a sheaf; one then has the commutative diagram

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ \downarrow & \nearrow & \downarrow \\ F & \xrightarrow{\sim} & LF \end{array}$$

which shows that $\text{Hom}(\ell_P, F)$ is surjective.

(iv) Let us show that if P is separated, $P \longrightarrow LP$ is a monomorphism. For this, it suffices to see that for every pair of morphisms $u, v : S \longrightarrow P$ (where S is representable) such that $\ell_P u = \ell_P v$, one has $u = v$. Now *loc. cit.* (iii) shows that u and v agree on a refinement of S , hence agree, since P is separated. This shows that $P \longrightarrow LP$ is a monomorphism; since it is covering by (i), one obtains that P is a refinement of LP .

Finally let us show that LP is a sheaf. Since one already knows by (i) that it is a separated presheaf, it suffices to see that for every $S \in \text{Ob } \mathcal{C}$, every refinement R of S , and every morphism $h : R \longrightarrow LP$, there exists a morphism $u : S \longrightarrow LP$ with $u i_R = h$. Now $R' = P \times_{LP} R$ is a refinement of R , since P is a refinement of LP , hence R' is a refinement of S . Set $u = z_{R'}(h')$:

$$\begin{array}{ccccc} P & \xrightarrow{\ell_P} & LP & & \\ \uparrow h' & & \uparrow h & \nwarrow u & \\ R' & \xrightarrow{j} & R & \xrightarrow{i_R} & S \end{array}$$

One has $ui_{R'} = \ell_P h' = hj$, hence $ui_R j = hj$. Since R' is a refinement of R and LP is separated, 4.3.2 shows that $ui_R = h$. $\square$

Corollary 4.3.12. ⁽²⁸⁾ *Let F be a sheaf and R a sub- $\widehat{\mathcal{C}}$ -object of F . Then R is a separated presheaf, $\ell_R : R \rightarrow LR$ is a covering monomorphism, and one has a commutative diagram*

$$\begin{array}{ccc} R^{\mathcal{C}} & \xrightarrow{i} & F \\ \ell_R \searrow & & \nearrow j \\ & LR & \end{array}$$

Consequently, R is a refinement of F if and only if j is an isomorphism.

PROOF. One has already noted that R is separated and that j is a monomorphism, cf. N.D.E. (24) and (26). By 4.3.11 (iv), ℓ_R is a covering monomorphism. Thus, if j is an isomorphism, R is a refinement of F . Conversely, if i is covering, j is too, hence is an isomorphism by 4.3.2.1. $\square$

Remark 4.3.13. If $J'(S)$ is a cofinal subset of $J(S)$, one has

$$\mathrm{Hom}(S, LP) = \varinjlim_{R \in J'(S)} \mathrm{Hom}(R, P).$$

In particular, let $S \mapsto R(S)$ be a pretopology generating the given topology. The functor L may be described with the aid of the covering families that are elements of $R(S)$. By making the formula above explicit, one recovers the construction of [MA].

Denote by i the inclusion functor $\widetilde{\mathcal{C}} \rightarrow \widehat{\mathcal{C}}$. The following theorem follows from Proposition 4.3.11:

Theorem 4.3.14. *There exists a unique functor $a : \widehat{\mathcal{C}} \rightarrow \widetilde{\mathcal{C}}$ such that the following diagram is commutative*

$$\begin{array}{ccc} \widehat{\mathcal{C}} & \xrightarrow{L} & \widehat{\mathcal{C}} \\ a \downarrow & & \downarrow L \\ \widetilde{\mathcal{C}} & \xrightarrow{i} & \widetilde{\mathcal{C}} \end{array}$$

i.e. for every presheaf P , $L(L(P))$ is a sheaf. The functors i, a are adjoint to one another: for every presheaf P and every sheaf F , one has an isomorphism functorial in P, F

$$\mathrm{Hom}_{\widehat{\mathcal{C}}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathcal{C}}}(a(P), F),$$

that is,

$$\mathrm{Hom}(P, F) \simeq \mathrm{Hom}(a(P), F).$$

Definition 4.3.15. The sheaf $a(P)$ is said to be *associated* to the presheaf P .

Remark 4.3.16. Since the functor L is constructed with the aid of projective limits and filtered inductive limits, it commutes with finite projective limits.⁽²⁹⁾ Moreover, if one identifies $L(P \times P)$ with $LP \times LP$, the morphism $\ell_{P \times P}$ is identified with $\ell_P \times \ell_P$. It follows, for example, that if P is a presheaf of groups, LP is also canonically endowed with the structure of a presheaf of groups and the canonical morphism $P \rightarrow LP$ is a morphism of groups. The same holds for the functor a , which shows that if P is a presheaf of groups and F a sheaf of groups, one has an isomorphism

$$\mathrm{Hom}_{\widehat{\mathcal{C}}\text{-gr.}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathcal{C}}\text{-gr.}}(a(P), F).$$

⁽²⁸⁾N.D.E. The statement of the corollary and its proof have been given in detail.

⁽²⁹⁾N.D.E. In particular, if K is the kernel of a pair of morphisms of presheaves $u, v : Q \rightrightarrows P$, then LK is the kernel of $Lu, Lv : LQ \rightrightarrows LP$ (this will be used in 4.4.5).

See [D] for more details.

4.3.17. — If $\mathcal{V}$ is any category, one calls a *presheaf on $\mathcal{C}$ with values in $\mathcal{V}$* a contravariant functor from $\mathcal{C}$ to $\mathcal{V}$. To define sheaves with values in $\mathcal{V}$, we must first recall the definition of the *projective limit* of a functor. If $\mathcal{R}, \mathcal{V}$ are two categories, and

$$F : \mathcal{R}^\circ \longrightarrow \mathcal{V}$$

is a contravariant functor from $\mathcal{R}$ to $\mathcal{V}$, one denotes by $\varprojlim F$ the object of $\widehat{\mathcal{V}}$ defined as follows:

$$(\varprojlim F)(X) = \text{Hom}_{\widehat{\mathcal{V}}}(X, \varprojlim F) = \varprojlim_{U \in \text{Ob } \mathcal{R}} \text{Hom}_{\mathcal{V}}(F(U), X) = \text{Hom}(c_X, F),$$

where X is a variable object of $\mathcal{V}$, where c_X denotes the contravariant functor from $\mathcal{R}$ to $\mathcal{V}$ which sends each object of $\mathcal{R}$ to X and each arrow of $\mathcal{R}$ to id_X , and where the last Hom is taken in the category $\text{Hom}(\mathcal{R}^\circ, \mathcal{V})$. If $\mathcal{R}$ possesses a final object $e_{\mathcal{R}}$, one has $\varprojlim F = F(e_{\mathcal{R}})$. If $\mathcal{V}$ is the category of sets, the functor $\varprojlim$ is identified with the functor Γ .

If S is an object of $\mathcal{C}$ and R a sieve of S , denote by $\mathcal{R}$ the full subcategory of $\mathcal{C}_{/S}$ whose set of objects is $E(R)$, and by $i_R : \mathcal{R} \longrightarrow \mathcal{C}_{/S}$ the canonical functor. If P is a presheaf on $\mathcal{C}$ with values in $\mathcal{V}$, it defines by restriction a functor $P_S : (\mathcal{C}_{/S})^\circ \longrightarrow \mathcal{V}$. The functor i_R induces a morphism of $\widehat{\mathcal{V}}$:

$$P(S) = P_S(S) = \varprojlim P_S \longrightarrow \varprojlim (P_S \circ i_R).$$

One denotes by $\varprojlim_R P$ the object $\varprojlim (P_S \circ i_R)$ of $\widehat{\mathcal{V}}$. By 4.1.6, Definition 4.3.1 generalizes to the

Definition 4.3.18. The presheaf P on $\mathcal{C}$ with values in $\mathcal{V}$ is said to be *separated* (resp. a *sheaf*) if, for every $S \in \text{Ob } \mathcal{C}$ and every $R \in J(S)$, the canonical morphism of $\widehat{\mathcal{V}}$

$$P(S) \longrightarrow \varprojlim_R P$$

is a monomorphism (resp. an isomorphism).

In the case where $\mathcal{V}$ is the category (Gr.) of groups (or any other category of sets endowed with algebraic structures defined by finite projective limits), one can see (cf. [D]) that there is an equivalence between the following notions: a presheaf on $\mathcal{C}$ with values in (Gr.) whose underlying presheaf of sets is a sheaf, and a group in the category of sheaves of sets. Taking these identifications into account, we shall always consider sheaves with values in a category of sets endowed with algebraic structures defined by finite projective limits as sheaves of sets endowed, in the category $\widetilde{\mathcal{C}}$, with the corresponding algebraic structure.

4.4. Exactness properties of the category of sheaves

Theorem 4.4.1. (i) *Arbitrary projective limits exist in $\widetilde{\mathcal{C}}$; “they are calculated in $\widehat{\mathcal{C}}$ ”, i.e. the inclusion functor $i : \widetilde{\mathcal{C}} \longrightarrow \widehat{\mathcal{C}}$ commutes with projective limits: if (X_α) is a projective system of sheaves, the presheaf*

$$\varprojlim i(X_\alpha) : S \longmapsto \varprojlim X_\alpha(S)$$

is a sheaf, and one has $i(\varprojlim X_\alpha) = \varprojlim i(X_\alpha)$.

(ii) *Arbitrary inductive limits exist in $\widetilde{\mathcal{C}}$: if (X_α) is an inductive system of sheaves, one has*

$$\varinjlim X_\alpha = a(\varinjlim i(X_\alpha)),$$

where $\varinjlim i(X_\alpha)$ is the inductive-limit presheaf of the $i(X_\alpha)$:

$$\varinjlim i(X_\alpha) : S \longmapsto \varinjlim X_\alpha(S).$$

(iii) The functor $a : \widehat{\mathcal{C}} \longrightarrow \widetilde{\mathcal{C}}$ commutes with arbitrary inductive limits and finite projective limits.

PROOF. Assertions (i) and (ii) follow formally from the adjunction formula (4.3.14), and⁽³⁰⁾ the first assertion of (iii) follows from (ii). Finally, the second assertion of (iii) has already been pointed out in 4.3.16. $\square$

Scholium 4.4.2. *This theorem allows one to use the following method to prove in $\widetilde{\mathcal{C}}$ an assertion concerning simultaneously arbitrary inductive limits and finite projective limits (for example: “every epimorphism is universal effective”, cf. below). One begins by proving the corresponding assertion in the category of sets, then extends it “argument by argument” to the category of presheaves. Next, one uses the preceding theorem to pass from the category of presheaves to the category of sheaves. One will see many examples of this method in what follows (4.4.3, 4.4.6, 4.4.9, etc.).*

Finally let us remark that the assertions concerning the category of presheaves are formally corollaries of the assertions concerning the category of sheaves. Indeed, it suffices to take as topology the least fine topology (“chaotic”), that is, the topology defined by $J(S) = \{S\}$ for every $S \in \text{Ob } \mathcal{C}$; every functor is indeed a sheaf for this topology.

Proposition 4.4.3. *Let $\mathcal{F} = \{F_i \longrightarrow F\}$ be a family of morphisms of sheaves. The following conditions are equivalent:*

- (i) $\mathcal{F}$ is an epimorphic family.
- (ii) $\mathcal{F}$ is a universal effective epimorphic family (1.13).
- (iii) $\mathcal{F}$ is covering (4.2.3).
- (iv) The image sheaf of $\mathcal{F}$ (that is, the sheaf associated to the image presheaf of $\mathcal{F}$ (4.1.2)) is F .

The equivalence of (iii) and (iv) follows from 4.3.12. The other equivalences will follow from the following lemmas.

Lemma 4.4.4. *Let $f : X \longrightarrow Y$ be a monomorphism of sheaves which is an epimorphism. Then f is an isomorphism.*

PROOF. The lemma is first clear in the category of sets. Let us next prove it in the category of presheaves. Consider the presheaf

$$V : S \longmapsto Y(S) \amalg^{X(S)} Y(S);$$

it is the amalgamated sum of Y and Y under X in the category of presheaves.⁽³¹⁾ Denote by i_1, i_2 the two “coordinate” morphisms $Y \longrightarrow V$. If $X \longrightarrow Y$ is an epimorphism in the category of presheaves, then $i_1 = i_2$. In this case, for each S , the map $X(S) \longrightarrow Y(S)$ is surjective; since it is also injective, it is a bijection, and thus $X \longrightarrow Y$ is an isomorphism.

Finally let us work in the category $\widetilde{\mathcal{C}}$ of sheaves. By 4.4.1 (ii), the amalgamated sum in $\widetilde{\mathcal{C}}$ of the sheaves Y, Y under X is the sheaf Z associated to the presheaf V . Consider the diagram of morphisms:

$$X \xrightarrow{f} Y \xrightarrow[i_2]{i_1} V \xrightarrow{\tau} Z = a(V).$$

One has $i_1 f = i_2 f$, hence $\tau i_1 f = \tau i_2 f$, and thus $\tau i_1 = \tau i_2$, since f is an epimorphism in $\widetilde{\mathcal{C}}$. By point (iii) of the lemma below, the presheaf V is separated, hence τ is a monomorphism (4.3.11 (iv)). Thus $i_1 = i_2$, and one has seen above that this implies that $f : X \longrightarrow Y$ is an isomorphism. This proves Lemma 4.4.4, once one has verified that V is separated. $\square$

⁽³⁰⁾N.D.E. The original has been modified here.

⁽³¹⁾N.D.E. The rest of the proof has been slightly modified.

(32) Let $R_\emptyset$ be the presheaf which associates the empty set to every $S \in \text{Ob } \mathcal{C}$; it is an initial object of $\widehat{\mathcal{C}}$.

Lemma 4.4.5. (i) Suppose that $R_\emptyset \notin J(S)$, for every $S \in \text{Ob } \mathcal{C}$. If (X_i) is a family of separated presheaves, the direct-sum presheaf $\coprod_i X_i$ is separated. (33)

(ii) Consider an equivalence relation in the category of presheaves:

$$X \xrightarrow[u]{u} Y$$

and let $w : Y \rightarrow Z$ be the quotient. If X is a sheaf and Y is separated, then Z is separated.

(iii) Consider an amalgamated sum in the category of presheaves, where u, u' are monomorphisms:

$$\begin{array}{ccc} & Y & \\ u \nearrow & & \searrow \\ X & & V \\ u' \searrow & & \nearrow \\ & Y' & \end{array} .$$

If Y, Y' are separated and X is a sheaf, then V is separated.

PROOF. (i) Set $X = \coprod_i X_i$. Let $S \in \text{Ob } \mathcal{C}$ and $\tau : R \hookrightarrow S$ be a refinement of S , and let x_1, x_2 be two elements of $X(S)$ such that $x_1 \circ \tau = x_2 \circ \tau$; there exist indices i, j such that $x_1 \in X_i(S)$, $x_2 \in X_j(S)$. Since $R \neq R_\emptyset$, there exists a morphism $\phi : T \rightarrow R$, with $T \in \text{Ob } \mathcal{C}$. Then $x_1 \circ \tau \circ \phi = x_2 \circ \tau \circ \phi$, and since $X(T)$ is the disjoint union of the $X_k(T)$, this implies $i = j$. Then, since X_i is separated and is a subobject of X , the map $X_i(S) \rightarrow X_i(R) \rightarrow X(R)$ is injective, hence $x_1 = x_2$. This proves that X is separated.

Let us prove (iii). Consider the morphisms $i : Y \rightarrow V$ and $j : Y' \rightarrow V$, and let K be the kernel of:

$$Y \times Y' \xrightarrow[p]{q} V ,$$

where $p = i \circ \text{pr}_1$, $q = j \circ \text{pr}_2$. Let $S \in \text{Ob } (\mathcal{C})$. Since u, u' are monomorphisms, one may identify $X(S)$ with its image in $Y(S)$ (resp. $Y'(S)$); denote by $Z(S)$ (resp. $Z'(S)$) the complement. Then $V(S)$ is identified with the disjoint union of $Z(S), Z'(S), X(S)$, and one easily sees that the maps $i(S) : Y(S) \rightarrow V(S)$, $j(S) : Y'(S) \rightarrow V(S)$ are injective, and that the map

$$(u \times u')(S) : X(S) \rightarrow K(S)$$

is bijective. Consequently i, j are monomorphisms, and $u \times u' : X \rightarrow K$ is an isomorphism.

Set $U = Y \amalg Y'$, and let $\tau : R \hookrightarrow S$ be a refinement of S ; one has a commutative diagram:

$$\begin{array}{ccc} U(S) & \xrightarrow{U(\tau)} & U(R) \\ \downarrow & & \downarrow \\ V(S) & \xrightarrow{V(\tau)} & V(R) \end{array} .$$

(32) **N.D.E.** Point (i) of Lemma 4.4.5 has been corrected, and the proofs of the three points have been given in detail.

(33) **N.D.E.** In general, the direct sum of two sheaves F, G is not a sheaf. Indeed, let $S_1, S_2 \in \text{Ob } \mathcal{C}$; one assumes that the direct sum $S = S_1 \amalg S_2$ exists in $\mathcal{C}$ and that the fibered product $S_1 \times_S S_2$ is an initial object $\emptyset$ of $\mathcal{C}$ (cf. I, 1.8). Let R be the sieve of S with basis $\{S_1, S_2\}$; then $(F \amalg G)(R)$ is the disjoint union of $F(S) \amalg G(S)$ and of $F(S_i) \times G(S_j)$ for $i \neq j$, hence $F \amalg G$ is not a sheaf in general. On the other hand, if $\mathcal{C}$ is the category with a single object S and id_S as its only morphism, endowed with the topology defined by $J(S) = \{R_\emptyset, S\}$, then the only separated presheaves are $R_\emptyset$ and $X = h_S$ (which is a sheaf), and $X \amalg X$ is not separated.

Let $v_1, v_2 \in V(S)$ have the same image in $V(R)$. By definition of V , v_1, v_2 lift to elements y_1, y_2 of $U(S)$; denote by z_1, z_2 their images in $U(R)$. Then z_1, z_2 have the same image in $V(R)$.

Since $i : Y \rightarrow V$, $j : Y' \rightarrow V$ are monomorphisms, the maps $Y(R) \rightarrow V(R)$, $Y'(R) \rightarrow V(R)$ are injective. Thus, since Y, Y' are separated, if y_1, y_2 both belong to $Y(S)$ or to $Y'(S)$, then $y_1 = y_2$. Otherwise one may assume that $y_1 \in Y(S)$, $y_2 \in Y'(S)$, hence $z_1 \in Y(R)$, $z_2 \in Y'(R)$. But then, since $i \circ z_1 = j \circ z_2$, the morphism $z_1 \boxtimes z_2 : R \rightarrow Y \times Y'$ factors through $K = X$. Since moreover $X(R) = X(S)$ (because X is a sheaf), there exists $x \in X(S)$ such that $u(x) \circ \tau = z_1 = y_1 \circ \tau$ and $u'(x) \circ \tau = z_2 = y_2 \circ \tau$, whence, since Y, Y' are separated, $u(x) = y_1$, $u'(x) = y_2$, and thus $v_1 = v_2$. This proves that V is separated.

Let us prove (ii). First remark that the morphism of presheaves

$$X \xrightarrow{u \boxtimes v} K,$$

where K denotes the kernel of the pair of morphisms $w \circ \text{pr}_i : Y \times Y \rightarrow Z$, is an isomorphism. Indeed, for every $T \in \text{Ob } \mathcal{C}$, $X(T)$ is an equivalence relation in $Y(T)$, so that the diagram

$$X(T) \xrightarrow{u \boxtimes v} Y(T) \times Y(T) \xrightarrow[w \circ \text{pr}_2]{w \circ \text{pr}_1} Z(T)$$

is exact in the category of sets.

Now let $S \in \text{Ob } \mathcal{C}$ and $g_1, g_2 : S \rightarrow Z$ be two morphisms which agree on a refinement $\tau : R \hookrightarrow S$ of S . Since $S \in \text{Ob } \mathcal{C}$, one has $Z(S) = Y(S)/X(S)$, by construction of Z , and hence there exist morphisms $f_1, f_2 : S \rightarrow Y$ such that $wf_i = g_i$. Then $wf_1\tau = wf_2\tau$, hence, by the preceding argument, there exists a morphism $\phi : R \rightarrow X$ such that $u\phi = f_1\tau$, $v\phi = f_2\tau$. Since X is a sheaf, there exists $\psi : S \rightarrow X$ such that $\phi = \psi\tau$, and one therefore has in $Y(R)$ the equalities:

$$u\psi\tau = f_1\tau, \quad v\psi\tau = f_2\tau.$$

Since $Y(S) \rightarrow Y(R)$ is injective (Y being separated), this implies $u\psi = f_1$, $v\psi = f_2$, whence $g_1 = g_2$. This proves that Z is separated. $\square$

Lemma 4.4.6. ⁽³⁴⁾ (i) Let $\{F_i \xrightarrow{f_i} F\}$ be a family of morphisms of presheaves, and let $G \subset F$ be the image presheaf. Then the following diagram in $\hat{\mathcal{C}}$ is exact:

$$(\coprod_i F_i) \times_G (\coprod_j F_j) \xrightarrow[\text{pr}_2]{\text{pr}_1} \coprod_i F_i \longrightarrow G$$

that is, for every presheaf H , the following diagram of sets is exact:

$$(*) \quad \text{Hom}(G, H) \longrightarrow \prod_i \text{Hom}(F_i, H) \xrightarrow[p]{q} \prod_{i,j} \text{Hom}(F_i \times_G F_j, H).$$

(ii) Every covering family of morphisms of sheaves is universal effective epimorphic.

PROOF. (i) Let H be a presheaf. The map f^* which associates to a morphism $\phi : G \rightarrow H$ the family of morphisms $\phi \circ f_i : F_i \rightarrow H$ is injective, for, for every $S \in \text{Ob } \mathcal{C}$, $\phi(S)$ is determined by the $(\phi \circ f_i)(S)$, since the family $f_i(S) : F_i(S) \rightarrow G(S)$ is surjective. It is clear that the image of f^* is contained in $\text{Ker}(p, q)$. Conversely, let $\phi_i : F_i \rightarrow H$ be a family of morphisms such that, for every i, j , the diagram below is commutative:

$$\begin{array}{ccc} F_i \times_G F_j & \longrightarrow & F_j \\ \downarrow & & \downarrow \phi_j \\ F_i & \xrightarrow{\phi_i} & H \end{array}$$

⁽³⁴⁾ **N.D.E.** The statement of the lemma and its proof have been given in detail.

Then, for every S , the map $\coprod_i F_i(S) \rightarrow H(S)$ factors uniquely through a map $\phi(S) : G(S) \rightarrow H(S)$, and this defines a morphism $\phi : G \rightarrow H$ such that $f^*(\phi) = (\phi_i)$. This proves the exactness of sequence (*), and point (i).

Let us prove (ii). Since the notion of covering family is stable under base extension, it suffices to show that every covering family is effective epimorphic. Thus let $\{F_i \rightarrow F\}$ be a covering family of morphisms of sheaves, and let G be the image presheaf of this family. Since the family is covering, the monomorphism $G \hookrightarrow F$ is too, hence, by 4.3.12, one has $a(G) = F$.

On the other hand, since $G \hookrightarrow F$ is a monomorphism, the fibered products $F_i \times_G F_j$, $F_i \times_F F_j$ are the same. Thus, by (i), the following diagram of sets is exact, for every presheaf H :

$$(**) \quad \text{Hom}(G, H) \longrightarrow \prod_i \text{Hom}(F_i, H) \xrightarrow[p]{q} \prod_{i,j} \text{Hom}(F_i \times_F F_j, H).$$

If H is moreover a sheaf, one has

$$\text{Hom}(G, H) = \text{Hom}(a(G), H) = \text{Hom}(F, H),$$

and then (**) shows that $\{F_i \rightarrow F\}$ is indeed an effective epimorphic family in the category $\tilde{\mathcal{C}}$ of sheaves. $\square$

Lemma 4.4.7. *Every family of morphisms of sheaves $\{F_i \rightarrow F\}$ factors into a covering family $\{F_i \rightarrow G\}$ and a monomorphism $G \rightarrow F$.*

PROOF. Indeed, it suffices to take for G the image sheaf of the given family. $\square$

PROOF OF PROPOSITION 4.4.3. One has seen in 4.4.6 that (iii) implies (ii), and evidently (ii) implies (i). Finally let $\{F_i \rightarrow F\}$ be an epimorphic family; by Lemma 4.4.7, it factors into a covering family followed by a monomorphism. But the latter, being dominated ⁽³⁵⁾ by an epimorphic family, is an epimorphism, hence an isomorphism by 4.4.4. $\square$

Remark 4.4.8. ⁽³⁶⁾ Since the image presheaf of the family $\mathcal{F}$ is separated, the construction of the associated sheaf shows that the conditions of Proposition 4.4.3 are also equivalent to the following:

- (v) For every $S \in \text{Ob } \mathcal{C}$, every $f \in F(S)$ is locally in the image of $\mathcal{F}$, that is:
- (vi) For every $S \in \text{Ob } \mathcal{C}$ and every $f \in F(S)$, the set of $S' \rightarrow S$ such that the image of f in $F(S')$ is in the image of one of the $F_i(S')$ is a refinement of S .
- (vii) (If the topology is defined by a pretopology R .) For every $S \in \text{Ob } \mathcal{C}$ and every $f \in F(S)$, there exists a family $\{S_j \rightarrow S\} \in R(S)$ such that for every j the image f_j of f in $F(S_j)$ is in the image of one of the $F_i(S_j)$.

Remark 4.4.8.bis. If the sheaf F is representable, the preceding conditions are also equivalent to:

- (viii) The set of $T \rightarrow F$ ($T \in \text{Ob } \mathcal{C}$) such that there exist an i and a commutative diagram

$$\begin{array}{ccc} F_i & \longrightarrow & F \\ \uparrow & \nearrow & \\ T & & \end{array}$$

is a refinement of F .

Indeed, if (viii) is satisfied, the image presheaf of the F_i is dominated by ⁽³⁷⁾ a refinement of F , which implies that the family is covering. Conversely, one applies (vi) to $\text{id}_F \in F(F)$.

⁽³⁵⁾**N.D.E.** We recall (cf. N.D.E. (17)) that a morphism $g : G \rightarrow H$ is said to be dominated by a family of morphisms $F_i \rightarrow H$ if this family factors through g .

⁽³⁶⁾**N.D.E.** To agree with later references, the numbering of the original, which had two nos. 4.4.7, has been corrected.

⁽³⁷⁾**N.D.E.** “majorizes” has been replaced by “is dominated by”, cf. N.D.E. (17).

This condition is expressed in figurative language as follows: locally on F , there exists an i such that $F_i \rightarrow F$ possesses a section. In particular a morphism $G \rightarrow F$, where G is a sheaf and F a representable sheaf, will be covering if and only if it possesses a section locally (on F).

(38) The following lemmas will be useful in VIB, 8.1 and 8.2.

Lemma 4.4.8.1. *Let $Q \subset P$ be presheaves of groups on $\mathcal{C}$, with Q normal in P . Suppose that P is separated and that Q is a sheaf. Then the presheaf of groups P/Q is separated. Consequently one has*

$$a(P/Q) = L(P/Q) = \varinjlim_{R \in J(S)} (P/Q)(R).$$

PROOF. It suffices to show the first assertion, for the second follows from it by 4.3.11 (iv) and (ii). Let $S \in \text{Ob } \mathcal{C}$, let R be a sieve of S , and let $y \in P(S)/Q(S)$ have the identity as its image in $(P/Q)(R)$. One must show that $y = 1$. Now, by hypothesis, the morphism $P(S) \rightarrow P(R)$ (resp. $Q(S) \rightarrow Q(R)$) is injective (resp. an isomorphism), and in the commutative diagram below the upper row is exact:

$$\begin{array}{ccccccc} 1 & \longrightarrow & Q(S) & \longrightarrow & P(S) & \longrightarrow & P(S)/Q(S) \longrightarrow 1 \\ & & \simeq \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & Q(R) & \longrightarrow & P(R) & \longrightarrow & (P/Q)(R) \end{array}$$

The result will follow if one shows that the lower row is exact. Let $f : R \rightarrow P$ have the identity as its image in $(P/Q)(R)$, and let $\phi : T \rightarrow R$, with $T \in \text{Ob } \mathcal{C}$. Then $f \circ \phi$ is the identity of $(P/Q)(T) = P(T)/Q(T)$, i.e. $f \circ \phi \in Q(T)$. Thus $\phi \mapsto f \circ \phi$ is a functorial map $R(T) \rightarrow Q(T)$, hence defines a morphism of functors $\pi : R \rightarrow Q$ such that $\pi(\text{id}_R) = f$, whence $f \in Q(R)$. This proves the exactness of the lower row, and the lemma is proved. $\square$

Lemma 4.4.8.2. *Let $H \subset G$ be presheaves of groups on $\mathcal{C}$.*

- (i) *If H is normal in G , then $L(H)$ is normal in $L(G)$.*
- (ii) *If H is central in G , then $L(H)$ is central in $L(G)$.*

PROOF. Let $S \in \text{Ob } \mathcal{C}$; one must show (cf. I 2.3.6) that $L(H)(S)$ is normal (resp. central) in $L(G)(S)$. Let $h \in L(H)(S)$, $g \in L(G)(S)$; there exist a sieve R of S and elements $h' \in H(R)$, $g' \in G(R)$ such that $h = z_R(h')$, $g = z_R(g')$ (notation of 4.3.10). Since z_R is a morphism of groups, $ghg^{-1}h^{-1} = z_R(g'h'g'^{-1}h'^{-1})$. In case (i), one has $g'h'g'^{-1}h'^{-1} \in H(R)$, whence $ghg^{-1}h^{-1} \in LH(S)$; in case (ii), $g'h'g'^{-1}h'^{-1} = 1$, hence $ghg^{-1}h^{-1} = 1$. $\square$

Proposition 4.4.9. *Every equivalence relation in $\tilde{\mathcal{C}}$ is universal effective (3.3.3): let R be a $\tilde{\mathcal{C}}$ -equivalence relation in the sheaf X ; then the sheaf associated to the separated presheaf*

$$i(X)/i(R) : S \mapsto X(S)/R(S)$$

is a universal effective quotient of X by R .

PROOF. Let X/R be the quotient sheaf of X by R , which exists by 4.4.1 (ii): $X/R = a(i(X)/i(R))$. We must show that $X \rightarrow X/R$ is a universal effective epimorphism, and that the morphism $f : R \rightarrow X \times_{X/R} X$ is an isomorphism. The first assertion has already been proved (4.4.3). As for f , it comes, by application of the functor a , from the morphism $i(R) \rightarrow i(X) \times_{i(X)/i(R)} i(X)$ or, since $i(X)/i(R)$ is separated (4.4.5 (ii)), so that $i(X)/i(R) \rightarrow i(X/R)$ is a monomorphism, from the canonical morphism $i(R) \rightarrow i(X) \times_{i(X)/i(R)} i(X)$.

One is therefore reduced to proving the same assertion in the category of presheaves. But $i(X)/i(R)$ is the presheaf $S \mapsto X(S)/R(S)$, and one is reduced to proving the analogous assertion in the category of sets, where it is immediate. $\square$

(38) **N.D.E.** Lemmas 4.4.8.1 and 4.4.8.2 have been added.

Proposition 4.4.10. *Under the conditions of 4.4.9, let Y be a subsheaf of X . Denote by R_Y the equivalence relation induced in Y by R . Then the canonical morphism (3.1.6)*

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism: it identifies Y/R_Y with a subsheaf of X/R , which is the image sheaf of the composite morphism

$$Y \longrightarrow X \longrightarrow X/R.$$

PROOF. The morphism of presheaves

$$i(Y)/i(R_Y) = i(Y)/i(R)_{i(Y)} \longrightarrow i(X)/i(R)$$

is a monomorphism. Since the functor a is left exact (4.3.16), it transforms monomorphisms into monomorphisms, and thus

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. The last assertion follows from the commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y/R_Y & \longrightarrow & X/R \end{array},$$

and the fact that $Y \longrightarrow Y/R_Y$ is covering. $\square$

By virtue of this proposition, we shall always identify Y/R_Y with a subsheaf of X/R .

Proposition 4.4.11. *Let R be a $\tilde{C}$ -equivalence relation in the sheaf X . For every subsheaf Y of X stable under R , denote by Y' the quotient Y/R_Y , considered as a subsheaf of $X' = X/R$. Then $Y = Y' \times_{X'} X$, and the maps $Y \mapsto Y/R_Y$ and $Y' \mapsto Y' \times_{X'} X$ establish a bijective correspondence between the set of subsheaves Y of X stable under R and the set of subsheaves Y' of X' .*

PROOF. If Y' is a subsheaf of X' , then $Y' \times_{X'} X$ is a subsheaf of X stable under R . ⁽³⁹⁾ If Y' is obtained by passage to the quotient from a subsheaf Y of X , then Y is a subobject of $Y' \times_{X'} X$. It therefore suffices to show that if one has two subsheaves Y, Y_1 of X , stable under R , with Y_1 containing Y , and if the quotients $Y/R_Y, Y_1/R_{Y_1}$ are identical, then $Y = Y_1$. One is evidently reduced to proving the same assertion in the case where $Y_1 = X$. Then denoting by P (resp. Q) the presheaf $i(X)/i(R)$ (resp. $i(Y)/i(R_Y)$), the diagram

$$\begin{array}{ccc} X & \longrightarrow & P \\ \uparrow & & \uparrow \\ Y & \longrightarrow & Q \end{array}$$

is cartesian. Since one has a commutative diagram

$$\begin{array}{ccc} P & \hookrightarrow & a(P) \\ \uparrow & & \uparrow \\ Q & \hookrightarrow & a(Q) \end{array}$$

and since $Q \hookrightarrow a(Q)$ is covering (4.3.11), the monomorphism $Q \hookrightarrow P$ is covering, hence Q is a refinement of P . By base change, Y is a refinement of X . Since X, Y are sheaves, this implies (4.3.12) $Y = X$. $\square$

⁽³⁹⁾**N.D.E.** And one has $(Y' \times_{X'} X)/R = Y'$.

4.4.12. — In particular, if Y is a subsheaf of X , and if $Y' = Y/R_Y$, then the preceding correspondence defines a subsheaf $\bar{Y}$ of X , stable under R , containing Y , and minimal for these properties, which one calls the *saturation* of Y for the equivalence relation R .

4.5. The case of a topology less fine than the canonical topology

By 4.3.6 and 4.3.8, the following conditions are equivalent for a topology $\mathcal{T}$ on $\mathcal{C}$:

- (i) $\mathcal{T}$ is less fine than the canonical topology of $\mathcal{C}$.
- (ii) Every representable presheaf is a sheaf for $\mathcal{T}$.
- (iii) Every covering sieve for $\mathcal{T}$ is universal effective epimorphic.

If $\mathcal{T}$ is defined by a pretopology $S \mapsto R(S)$, these conditions are also equivalent to

- (iv) Every family belonging to $R(S)$ is universal effective epimorphic.

In the case where these conditions hold, the canonical functor $\mathcal{C} \rightarrow \hat{\mathcal{C}}$ factors through a functor $j_{\mathcal{C}} = j : \mathcal{C} \rightarrow \tilde{\mathcal{C}}$ (one will also write $j(S) = \tilde{S}$).⁽⁴⁰⁾

Proposition 4.5.1. *The functor $j : \mathcal{C} \rightarrow \tilde{\mathcal{C}}$ is fully faithful and commutes with arbitrary projective limits. In particular, it is left exact, and therefore preserves algebraic structures defined by finite projective limits.*

PROOF. This follows immediately by considering the commutative diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\quad} & \hat{\mathcal{C}} \\ & \searrow j & \nearrow i \\ & \tilde{\mathcal{C}} & \end{array}$$

and from 4.4.1 (i). □

Before exhibiting other properties of the functor j , we must define the topology induced on a category $\mathcal{C}_{/S}$. No longer necessarily assuming that the given topology is less fine than the canonical topology, one does this as follows: if C is a sieve of T in $\mathcal{C}$ and if one has a morphism $T \rightarrow S$, then C naturally defines a sieve of T in $\mathcal{C}_{/S}$, denoted $J(T)$ (for the definition of a sieve of T depends only on the category $\mathcal{C}_{/T} = (\mathcal{C}_{/S})_{/T}$). If, for example, C is defined by the family $\{T_i \rightarrow T\}$, then its image in $\mathcal{C}_{/S}$ is defined by the same family considered as a family of morphisms of $\mathcal{C}_{/S}$. This said, the map $T \mapsto J(T)$ defines a topology on $\mathcal{C}_{/S}$, called the topology induced by the given topology. With the definitions of [AS], 2.3, this is the least fine topology on $\mathcal{C}_{/S}$ for which the canonical functor

$$i_S : \mathcal{C}_{/S} \rightarrow \mathcal{C}$$

is a comorphism.⁽⁴¹⁾ One will remark that the identifications

$$(\mathcal{C}_{/S})_{/T} = \mathcal{C}_{/T}$$

respect the topologies by definition.

Proposition 4.5.2. *Let S be a representable sheaf on $\mathcal{C}$ and $F \rightarrow S$ a morphism of $\hat{\mathcal{C}}$. For $S' \mapsto \text{Hom}_S(S', F)$ to be a separated presheaf (resp. a sheaf) on $\mathcal{C}_{/S}$, it is necessary and sufficient that F be a separated presheaf (resp. a sheaf) on $\mathcal{C}$.*

PROOF. For every functor P , one has (I 1.4.1)

$$\text{Hom}(P, F) = \coprod_{f \in \text{Hom}(P, S)} \text{Hom}_f(P, F).$$

⁽⁴⁰⁾**N.D.E.** And one will also write $h_S = \hat{S}$, cf. the first commutative diagram of 4.5.4.

⁽⁴¹⁾**N.D.E.** That is, such that for every object $T \rightarrow S$ of $\mathcal{C}_{/S}$, every covering sieve R' of $i_S(T \rightarrow S) = T$, considered as a sieve of $(T \rightarrow S) \in \text{Ob } \mathcal{C}_{/S}$, is covering.

$$\mathrm{Hom}_{\hat{\mathcal{C}}}(S', F) \longrightarrow \mathrm{Hom}_{\hat{\mathcal{C}}}(C', F)$$
$$\mathrm{Hom}_f(S', F) \longrightarrow \mathrm{Hom}_{f \circ \eta}(C', F).$$
☐
$$\tilde{\mathcal{C}}_{/\tilde{S}} \longrightarrow \widetilde{\mathcal{C}}_{/S}.$$
$$\begin{array}{ccccccc}
\mathcal{C}_{/S} & \xrightarrow{(jc)_{/S}} & \widetilde{\mathcal{C}}_{/\widetilde{S}} & \xrightarrow{(ic)_{/\widetilde{S}}} & \widehat{\mathcal{C}}_{/\widehat{S}} & \xrightarrow{(ac)_{/\widehat{S}}} & \widetilde{\mathcal{C}}_{/\widetilde{S}} \\
\parallel & & \downarrow \simeq & & \downarrow \simeq & & \downarrow \simeq \\
\mathcal{C}_{/S} & \xrightarrow{j_{\mathcal{C}_{/S}}} & \widetilde{\mathcal{C}}_{/S} & \xrightarrow{i_{\mathcal{C}_{/S}}} & \widehat{\mathcal{C}}_{/S} & \xrightarrow{a_{\mathcal{C}_{/S}}} & \widetilde{\mathcal{C}}_{/S} \\
(\widetilde{\mathcal{C}}_{/\widetilde{S}})_{/\widetilde{T}} & \longrightarrow & (\widetilde{\mathcal{C}}_{/S})_{/\widetilde{T}} & & & & \\
\downarrow \simeq & & & \searrow & & & \\
\widetilde{\mathcal{C}}_{/\widetilde{T}} & \longrightarrow & \widetilde{\mathcal{C}}_{/T} & \xrightarrow{\simeq} & \widetilde{(\mathcal{C}_{/S})_{/T}} & &
\end{array}$$
$$\begin{array}{ccc} \widehat{\mathcal{C}}_{/\widehat{S}} & \xrightarrow{L'} & \widehat{\mathcal{C}}_{/\widehat{S}}, \\ \downarrow & & \downarrow \\ \widehat{\mathcal{C}}_{/S} & \xrightarrow{L''} & \widehat{\mathcal{C}}_{/S} \end{array}$$

Scholium 4.5.5. *The various assertions of this number show that in the case where the given topology is less fine than the canonical topology, one may identify $\mathcal{C}$ with a full subcategory of $\widehat{\mathcal{C}}$, itself a full subcategory of $\widehat{\widehat{\mathcal{C}}}$, and that in this identification one may indulge in the customary abuses of language concerning $\mathcal{C} \rightarrow \widehat{\mathcal{C}}$, justified by the preceding commutativities. Let us remark explicitly that the first diagram of 4.5.4 shows that one may use the functor α without special precautions.*

⁽⁴²⁾**N.D.E.** The original has been given in detail in what follows.

We shall see in the next number that the identification of $\mathcal{C}$ with a full subcategory of $\tilde{\mathcal{C}}$ (contrary to what happened for $\widehat{\mathcal{C}}$) commutes with the formation of certain inductive limits, and we shall then say how to use this fact.

From now on, unless expressly stated otherwise, we shall assume that the given topology is less fine than the canonical topology, and we shall systematically make the identifications set out above.

Proposition 4.5.6. *Let F, G be two sheaves over S , and $f : F \rightarrow G$ an S -morphism. The following conditions on f are equivalent:*

- (i) f is a monomorphism (resp. an epimorphism, resp. an isomorphism) in $\tilde{\mathcal{C}}$.
- (ii) f is a monomorphism (resp. an epimorphism, resp. an isomorphism) in $\tilde{\mathcal{C}}_S = \widetilde{\mathcal{C}_S}$.

PROOF. For monomorphisms and isomorphisms this is evident (it is a question of presheaves). For epimorphisms, this follows from the description of epimorphisms as covering morphisms (4.4.3) and the fact that, by definition of the induced topology, these are the same in $\tilde{\mathcal{C}}$ and $\tilde{\mathcal{C}}_S$. $\square$

Proposition 4.5.7. *Let $f : F \rightarrow G$ be a morphism of sheaves. The following conditions are equivalent:*

- (i) f is a monomorphism (resp. an epimorphism, resp. an isomorphism).
- (ii) For each $S \in \text{Ob } \mathcal{C}$, $f_S : F_S \rightarrow G_S$ is locally a monomorphism (resp. an epimorphism, resp. an isomorphism), that is:
- (iii) For each $S \in \text{Ob } \mathcal{C}$, the set of $T \rightarrow S$ such that $F_T \rightarrow G_T$ is a monomorphism (resp. an epimorphism, resp. an isomorphism) is a refinement of S .

If the given topology is defined by a pretopology R , these conditions are also equivalent to the following:

- (iv) For each $S \in \text{Ob } \mathcal{C}$, there exists a covering family $\{S_i \rightarrow S\} \in R(S)$ such that for every i , $F_{S_i} \rightarrow G_{S_i}$ is a monomorphism (resp. an epimorphism, resp. an isomorphism).

If the category $\mathcal{C}$ possesses a final object e , one may restrict oneself to taking $S = e$ in conditions (ii), (iii), (iv).

PROOF. One evidently has (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv). To prove the equivalence of (i) and (ii), as well as the supplement concerning the final object, one must show that (ii) implies (i) and that the notions considered are stable under base extension. Let us first prove the latter point. For monomorphisms and isomorphisms, this is evident (it is a question of presheaves). For epimorphisms, it follows from the fact that every epimorphism of sheaves is universal (4.4.3).

Finally let us show that (ii) implies (i). First suppose that $f_S : F_S \rightarrow G_S$ is locally a monomorphism (resp. an isomorphism). There then exists a covering sieve C of S such that for every $T \rightarrow C$, f_T is a monomorphism (resp. an isomorphism). Since a projective limit of monomorphisms (resp. isomorphisms) is one, $\text{Hom}(C, f)$ will be a monomorphism (resp. an isomorphism) (cf. 4.1.4). The commutative diagram

$$\begin{array}{ccc} \text{Hom}(C, F) & \xrightarrow{\text{Hom}(C, f)} & \text{Hom}(C, G) \\ \simeq \uparrow & & \uparrow \simeq \\ F(S) & \xrightarrow{f(S)} & G(S) \end{array}$$

then shows that $f(S)$ is injective (resp. bijective). Finally suppose that $f : F \rightarrow G$ is locally an epimorphism, and let $H \subset G$ be its image. For each $S \rightarrow G$, $H \times_G S$ is the image of $f \times_G S : F \times_G S \rightarrow S$. To show that f is an epimorphism, one must show that H is a refinement of G , that is, $H \times_G S$ is a refinement of S for each S . But since this is so after every base change $T \rightarrow S$ from a refinement of S (since $f \times_G S$ is locally covering), $H \times_G S$ is indeed a refinement of S (Axiom (T 2)). $\square$

Corollary 4.5.8. *Let F, G be two sheaves over S , and $f : F \rightarrow G$ an S -morphism. For f to be a monomorphism, resp. an epimorphism, resp. an isomorphism, it is necessary and sufficient that it be so locally on S .*

Remark 4.5.9. The proof of the proposition shows that it remains valid, for the part concerning monomorphisms (resp. isomorphisms), when one assumes only that F is a separated presheaf (resp. a sheaf) and G is any presheaf (resp. a separated presheaf).

Let us temporarily return to the case of an arbitrary topology and give a definition.

Definition 4.5.10. Let $G \rightarrow F$ be a morphism of $\widehat{\mathcal{C}}$. One says that G is a *relative sheaf over F* if for each F -functor H and each refinement R of H , the canonical map

$$(+) \quad \text{Hom}_F(H, G) \xrightarrow{\sim} \text{Hom}_F(R, G)$$

is bijective.

Proposition 4.5.2 generalizes at once:

Proposition 4.5.11. *If F is a sheaf, G is a relative sheaf over F if and only if it is a sheaf.*

Lemma 4.5.12. *In the situation $X \rightarrow T \rightarrow S$ (where X, T, S are three objects of $\widehat{\mathcal{C}}$), if X is a relative sheaf over T , then $U = \prod_{T/S} X$ is a relative sheaf over S .*

PROOF. Indeed, for every S -functor Y , one has

$$\text{Hom}_S(Y, U) = \text{Hom}_T(T \times_S Y, X).$$

If C is a sieve of Y , then $T \times_S C$ is a sieve of $T \times_S Y$; one concludes at once. $\square$

Corollary 4.5.13. *The presheaves $\underline{\text{Hom}}_{T/S}(X, Y)$, $\underline{\text{Isom}}_S(X, Y)$, etc., are sheaves when the arguments which occur in them are also sheaves.*

PROOF. Indeed, all these presheaves are constructed with the aid of fibered products and presheaves $\prod$ (I 1.7 and II 1). It therefore suffices to verify the result for a presheaf $\prod_{T/S} X$; in this case, the assertion follows from 4.5.11 and 4.5.12. $\square$

4.6. Description of the quotient of a sheaf by an equivalence relation

Recall that we assume that the given topology $\mathcal{T}$ is less fine than the canonical topology.

Proposition 4.6.1. *Let $R \xrightarrow{p_1, p_2} X$ be a $\widetilde{\mathcal{C}}$ -equivalence relation in the sheaf X . Let $F \in \text{Ob } \widehat{\mathcal{C}}$ be defined as follows: for each S of $\mathcal{C}$,*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{whose quotient by } R_Z \text{ is } S, \text{ that is, such that} \\ \text{the diagram } R_Z \rightrightarrows Z \rightarrow S \text{ is exact} \end{array} \right\}.$$

(43) *Then, for every sheaf Y , $\text{Hom}(Y, F)$ is identified with the set:*

$$\left\{ \begin{array}{l} \text{sub-}Y\text{-sheaves of } X \times Y \text{ stable under } R \times Y \\ \text{and whose quotient is } Y \end{array} \right\}.$$

In particular, the subsheaf R of $X \times X$ corresponds to an element p of $\text{Hom}(X, F)$, and the diagram

$$R \xrightarrow[p_2]{p_1} X \xrightarrow{p} F$$

is exact, hence identifies F with the quotient sheaf X/R .

(43) **N.D.E.** $R \times S$ denotes the equivalence relation in $X \times S$ defined by $R \times S_{\text{diagonal}} \subset X \times X \times S \times S$, and R_Z is the equivalence relation which it induces in Z (cf. 3.1.6).

PROOF. Indeed, set $Q = X/R$. For every sheaf Y and every morphism $f \in \text{Hom}(Y, Q)$ corresponding to a section $s : Y \hookrightarrow Q \times Y$, consider the diagram

$$(*) \quad \begin{array}{ccccc} R \times Y & \rightrightarrows & X \times Y & \longrightarrow & Q \times Y \\ & & \uparrow & & \uparrow s \\ & & Z & \longrightarrow & Y \end{array}$$

Here the square is cartesian. It is immediate by 4.4.11 that Z is a sub- Y -sheaf of $X \times Y$, stable under $R \times Y$, whose quotient is Y , and that conversely, every Z of this type comes from a unique section of $Q \times Y$ over Y . Taking Y representable first, one deduces an isomorphism $Q \simeq F$. Taking arbitrary Y next, one deduces the announced form of $\text{Hom}(Y, F)$. Finally considering the canonical morphism $X \rightarrow Q$, one sees at once that it corresponds to the sub- X -sheaf R of $X \times X$, which completes the proof. $\square$

Corollary 4.6.2. *Let G be any subfunctor of F such that $\text{Hom}(X, G) \subset \text{Hom}(X, F)$ contains R . Then the canonical morphism $p : X \rightarrow F$ factors through G . Since p is covering (4.4.9 and 4.4.3), it follows that G is a refinement of F . In particular, every subsheaf G of F satisfying the preceding condition is equal to F (4.3.12).*

4.6.3. — We shall now concern ourselves with the case where X, R are representable. Let us first introduce some terminology. Besides conditions (a) to (d) introduced in 3.4.1, we shall use other conditions on a family (M) of morphisms of $\mathcal{C}$, which we state below, recalling conditions (a) to (c) already given, for completeness.

- (a) (M) is stable under base extension.
- (b) The composite of two elements of (M) is in (M).
- (c) Every isomorphism is an element of (M).
- (d $\mathcal{T}$) Every element of (M) is covering. ⁽⁴⁴⁾
- (e $\mathcal{T}$) Let $f : X \rightarrow Y$ be a morphism of $\mathcal{C}$. If there exists a refinement R of Y such that for every $Y' \rightarrow R$, $X \times_Y Y' \rightarrow Y'$ is an element of (M), then f is an element of (M).

Recall that (a) and (b) imply

- (a') The cartesian product of two elements of (M) is an element of (M).

On the other hand, (a) and (d $\mathcal{T}$) imply, by 4.3.9:

- (d') Every element of (M) is a universal effective epimorphism.

4.6.4. — The preceding conditions are satisfied by the family of covering morphisms, denoted (M $\mathcal{T}$), when $\mathcal{C}$ possesses fibered products. Indeed (cf. 4.2.3), (a) follows from (C 1), (b) from (C 2), (c) from (C 4), (d $\mathcal{T}$) from the definition, and (e $\mathcal{T}$) from (C 5). The results that we shall establish for a family satisfying these conditions will apply in particular to the family (M $\mathcal{T}$). In particular, one may take for $\mathcal{T}$ the canonical topology, and for (M) the family of universal effective epimorphisms.

Lemma 4.6.5. *Let (M) be a family of morphisms satisfying the preceding properties (a) to (e $\mathcal{T}$). Let R be a $\mathcal{C}$ -equivalence relation in $X \in \text{Ob } \mathcal{C}$, of type (M). Let $\tilde{X}$ be the sheaf defined by X , $\tilde{R}$ the $\tilde{\mathcal{C}}$ -equivalence relation in $\tilde{X}$ defined by R , and $\tilde{X}/\tilde{R}$ the quotient sheaf. For R to be (M)-effective, it is necessary and sufficient that $\tilde{X}/\tilde{R}$ be representable. If this is so, $\tilde{X}/\tilde{R}$ is represented by the quotient X/R .*

PROOF. First suppose that R is (M)-effective and write $Y = X/R$. The canonical morphism $p : X \rightarrow Y$ is an element of (M), hence covering by (d $\mathcal{T}$). The corresponding morphism

$$\tilde{p} : \tilde{X} \rightarrow \tilde{X}/\tilde{R}$$

is therefore a universal effective epimorphism of $\tilde{\mathcal{C}}$ (4.4.3), hence identifies $\tilde{X}/\tilde{R}$ with the quotient of $\tilde{X}$ by the equivalence relation R' defined in $\tilde{X}$ by $\tilde{p}$. Since the canonical functor

⁽⁴⁴⁾**N.D.E.** Recall that $\mathcal{T}$ denotes the given topology on $\mathcal{C}$, less fine than the canonical topology.

$\mathcal{C} \longrightarrow \tilde{\mathcal{C}}$ commutes with fibered products, R' is precisely $\tilde{R}$, since R is the equivalence relation defined by p .

Conversely, suppose that $\tilde{X}/\tilde{R}$ is represented by an object Y of $\mathcal{C}$. Let $p : X \longrightarrow Y$ be the morphism deduced from the canonical morphism $\tilde{X} \longrightarrow \tilde{X}/\tilde{R}$; it is a covering morphism by 4.4.3. It is clear, as before, that R is the equivalence relation defined by p . It remains only to show that $p \in (M)$. Now the cartesian square

$$\begin{array}{ccccc} R & \xrightarrow{\sim} & X \times_Y X & \longrightarrow & X \\ & \searrow p_2 & \downarrow \text{pr}_2 & & \downarrow p \\ & & X & \xrightarrow{p} & Y \end{array}$$

shows that p becomes p_2 , which is an element of (M) , after base change by the covering morphism p . One concludes by $(e_{\mathcal{T}})$. $\square$

Corollary 4.6.5.1. ⁽⁴⁵⁾ *Let (M) satisfy the preceding properties (a) to $(e_{\mathcal{T}})$, and let $f : G \longrightarrow G'$ be a morphism of $\mathcal{C}$ -groups such that $f \in (M)$. Suppose that $\text{Ker}(f)$ is representable (which is the case if $\mathcal{C}$ possesses a final object e). Then the equivalence relation in G defined by $H = \text{Ker}(f)$ is (M) -effective, and G' represents the quotient sheaf $\tilde{G}/\tilde{H}$ for the topology $\mathcal{T}$.*

PROOF. This follows from 4.6.5 and 3.4.7.1. $\square$

We are now in a position to state the principal result of this number.

Theorem 4.6.6. *Let (M) be a family of morphisms satisfying axioms (a) to $(e_{\mathcal{T}})$ of 4.6.3. Let R be a $\mathcal{C}$ -equivalence relation of type (M) (cf. 3.4.3) in the object X of $\mathcal{C}$. Consider the functor $F \in \text{Ob } \hat{\mathcal{C}}$ defined as follows:*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}\mathcal{S}\text{-sheaves } Z \text{ of } X_S \text{ stable under } R \times S \\ \text{whose quotient by } R_Z \text{ is } S \end{array} \right\}.$$

Let F_0 be the subfunctor of F defined as follows: $F_0(S)$ consists of the representable $Z \in F(S)$, that is:

$$F_0(S) = \left\{ \begin{array}{l} \text{sub-}\mathcal{C}_{/S}\text{-objects } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{such that } R_Z \text{ is } (M)\text{-effective with quotient } S \\ \text{(i.e. such that } Z \longrightarrow S \text{ belongs to } (M) \text{ and } R_Z \simeq Z \times_S Z) \end{array} \right\}.$$

Then:

(i) The morphism $p \in \text{Hom}(X, F) = F(X)$ defined by the subobject R of $X \times X$ identifies F with the quotient sheaf of X by R .

(ii) The following conditions are equivalent:

- a) F is representable.
- b) F_0 is representable.
- c) R is (M) -effective.

Under these conditions, $F = F_0 = X/R$.

(iii) Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S_i \longrightarrow S\}$ and every family $\{T_i \longrightarrow S_i\}$ of morphisms of (N) , every descent datum on the T_i relative to $\{S_i \longrightarrow S\}$ is effective. Suppose that X is base-changeable (cf. 1.6.0) and that the morphism $R \longrightarrow X \times X$ is an element of (N) . Then $F_0 = F$.

PROOF. (i) has already been proved (4.6.1).

(ii) One has seen the equivalence of a) and c), as well as the equality $F = X/R$. It remains to prove that b) or c) implies $F_0 = F$. First remark, as indeed asserted in the statement, that F_0 is a subfunctor of F ; indeed, for every $S \in \text{Ob } \mathcal{C}$ and every $Z \in F_0(S)$, the morphism

⁽⁴⁵⁾N.D.E. This corollary has been added.

$Z \rightarrow S$ is base-changeable, hence $Z \times_S S'$ is an element of $F_0(S')$ for every $S' \rightarrow S$. Since $R \in F(X)$ belongs to $F_0(X)$, 4.6.2 shows that b) implies $F_0 = F$.

Now suppose that c) holds and let Q be an object of $\mathcal{C}$ representing X/R . Then the morphism $X \rightarrow Q$ is an element of (M), and for every $S \in \text{Ob } \mathcal{C}$ and every $Z \in F(S)$, diagram (*) of 4.6.1 shows that $Z = S \times_{(Q \times_S)} (X \times S)$ is representable, and $Z \rightarrow S$ belongs to (M), hence $Z \in F_0(S)$.

(iii) Let $f \in \text{Hom}(S, F)$ correspond to $Z \in F(S)$. One must show that f factors through F_0 , that is, that Z is representable. This is first clear if f factors through X , by virtue of:

Lemma 4.6.7. *Let $x_0 \in X(S)$. The image of x_0 in $F(S)$ corresponds to the subsheaf Z of X_S defined by the two cartesian squares*

$$\begin{array}{ccccc} X_S & \xrightarrow{\text{id}_{X_S} \times x_0} & X_S \times_S X_S & \longrightarrow & X \times X \\ \uparrow & & \uparrow & & \uparrow \\ Z & \longrightarrow & R_S & \longrightarrow & R \end{array}$$

PROOF. This lemma follows at once from the description of the morphism $X \rightarrow F$. $\square$

Let us return to the proof of the theorem. If f factors through X , then Z is representable, and, since $R \rightarrow X \times X$ is an element of (N), so is $Z \rightarrow X_S$. In general f does not necessarily factor through X ; but, since $X \rightarrow F$ is covering (4.4.3), there exists by 4.4.8 (vii) a covering family $\{S_i \rightarrow S\}$, and for each i a morphism $S_i \rightarrow X$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & F \\ \uparrow & & \uparrow f \\ S_i & \longrightarrow & S \end{array}$$

commutative. By the preceding argument, the morphism $f_i : S_i \rightarrow F$ defined by the preceding diagram belongs to $\text{Hom}(S_i, F_0)$ and corresponds to the subsheaf $Z \times_S S_i$ of X_{S_i} . The morphism $Z \times_S S_i \rightarrow X_{S_i}$ is an element of (N), and the family $X_{S_i} \rightarrow X_S$ is covering. It therefore remains only to establish:

Proposition 4.6.8. *Let $\{S_i \rightarrow S\}$ be a covering family, and Z a sheaf over S . Suppose that for each i , the S_i -functor $Z \times_S S_i$ is represented by an object T_i . Then the family of the T_i is endowed with a canonical descent datum relative to $S_i \rightarrow S$. For Z to be representable, it is necessary and sufficient that this datum be effective; if it is so, the “descended” object represents Z .*

PROOF. First remark that by 4.4.3, $S_i \rightarrow S$ is a universal effective epimorphic family in $\tilde{\mathcal{C}}$, hence a descent family in $\tilde{\mathcal{C}}$ (2.3). If Z is represented by the object T , then $T \times_S S_i$ (considered as a sheaf) is isomorphic to $Z \times_S S_i$, hence the descent datum on the T_i is effective, and the descended object (unique) is isomorphic to Z . Conversely, suppose that the canonical descent datum on the T_i is effective, and let T be the descended object. Since the family $\{S_i \rightarrow S\}$ is a descent family in $\tilde{\mathcal{C}}$, there exists an S -morphism $T \rightarrow Z$ which by base extension to each S_i gives back the canonical morphism $T_i \rightarrow Z \times_S S_i$. This morphism is locally an isomorphism; since T, Z are sheaves, it follows from 4.5.8 that it is an isomorphism. $\square$

$\square$

Corollary 4.6.9. *Let R be an (M) -effective equivalence relation in X . For every sheaf F , the map*

$$\text{Hom}(X/R, F) \longrightarrow \text{Hom}(X, F)$$

identifies the first set with the subset of the second consisting of morphisms compatible with R .

Corollary 4.6.10. *Let $\mathcal{T}'$ be a topology less fine than $\mathcal{T}$, for which the morphisms of (M) are covering. Under the conditions of 4.6.6 (iii), X/R is also the quotient sheaf of X by R in every topology intermediate between $\mathcal{T}'$ and the canonical topology.*

Remark 4.6.11. If, in the statement of 4.6.6 (iii), one assumes moreover that, under the hypotheses of the text, if one denotes by T the descended object, the morphism $T \rightarrow S$ is an element of (N) , then the inclusion morphisms $Z \hookrightarrow X_S$ are also elements of (N) , as follows at once from the construction of Z by descent.

Remark 4.6.12. The implications $c) \Rightarrow b) \Rightarrow a)$ and $c) \Rightarrow [F_0 = F = X/R]$ have been established without recourse to the “sufficient” part of Lemma 4.6.5, which is the only place where condition $(e_{\mathcal{T}})$ is used. They therefore remain valid if (M) satisfies only conditions (a) to $(d_{\mathcal{T}})$. An example of such a family (M) is that of base-changeable covering morphisms (compare with 4.6.4). In the case of the canonical topology, these are precisely the universal effective epimorphisms. One therefore has:

Corollary 4.6.13. *Let R be a universal effective equivalence relation in X . Then the object X/R of $\mathcal{C}$ is the quotient sheaf of X by R for the canonical topology. It represents the following functor: $(X/R)(S)$ is the set of sub- $\mathcal{C}_{/S}$ -objects Z of X_S stable under $R \times S$ and such that the induced equivalence relation is universal effective with quotient S .*

Similarly, for an arbitrary topology:

Corollary 4.6.14. *Let (M) be the family of base-changeable covering morphisms. If R is an (M) -effective equivalence relation in X , then the object X/R of $\mathcal{C}$ is the quotient sheaf of X by R , and represents the functor F_0 of 4.6.6.*

Scholium 4.6.15. *We may now add the following precisions to 4.5.5. Whereas in questions involving exclusively projective limits (fibered products, algebraic structures, etc.), one may, by the results of Exposé I and 4.5.5, identify $\mathcal{C}$ indifferently with a full subcategory of $\tilde{\mathcal{C}}$ or $\hat{\mathcal{C}}$, the same is not true in questions mixing projective and inductive limits. In all questions involving both projective limits and inductive limits, in particular passages to the quotient (example: group structure on the quotient of a group by an invariant subgroup), we shall consider the given category as embedded in the category of sheaves; thus if R is a $\mathcal{C}$ -equivalence relation in the object X of $\mathcal{C}$, X/R will denote the quotient sheaf of X by R (previously denoted by $j(X)/j(R)$), hence, in the case where this sheaf is representable, the object which it represents. The preceding results show that in the most important cases a quotient in $\mathcal{C}$ will also be a quotient in the category of sheaves; in any event we forbid ourselves the use of the notation X/R for a quotient in $\mathcal{C}$ which does not coincide with the quotient in $\tilde{\mathcal{C}}$ (for example, one which is not universal), thus modifying the definitions of no. 3.*

To study a problem of the above type, one therefore first works in the category of sheaves, where all the customary results are valid (cf. no. 4.4), then specializes the results obtained to the initial category, using the results of this number and, when available, criteria for effectiveness of descent. We shall see examples of this method in the following numbers.

4.7. Use of effectiveness criteria: isomorphism theorem

In this number we give an example of the use of effectiveness criteria. The initial data are a topology $\mathcal{T}$ on $\mathcal{C}$ (still less fine than the canonical topology), a family (M) of morphisms of $\mathcal{C}$ satisfying axioms (a) to $(e_{\mathcal{T}})$ of 4.6.3, and a family (N) of morphisms of $\mathcal{C}$ which may satisfy the following axioms:

- (a) (N) is stable under base extension.

(f_T) “Morphisms of (N) descend under the given topology”; that is: for every $S \in \text{Ob } \mathcal{C}$, every covering family $\{S_i \rightarrow S\}$ and every family $\{T_i \rightarrow S_i\}$ of morphisms of (N), every descent datum on the T_i relative to $\{S_i \rightarrow S\}$ is effective, and if one denotes by T the descended object, the morphism $T \rightarrow S$ is an element of (N).

Since every element of (M) is covering (condition 4.6.3 (d_T)), (f_T) implies the following axiom: ⁽⁴⁶⁾

(f_M) If $Y \rightarrow X$ is an element of (N) and $X \rightarrow X_1$ an element of (M), every descent datum on Y relative to $X \rightarrow X_1$ is effective; if one denotes by Y_1 the descended object, $Y_1 \rightarrow X_1$ is an element of (N).

Let us immediately point out an example of this situation, which will be treated later: $\mathcal{C}$ is the category of schemes, $\mathcal{T}$ the faithfully flat quasi-compact topology; (M) the family of faithfully flat quasi-compact morphisms, (N) the family of closed immersions or that of quasi-compact immersions. ⁽⁴⁷⁾

Recall the principal result of 4.6.6 (taking account of 4.6.11):

Proposition 4.7.1. *If X is a base-changeable object of $\mathcal{C}$, R an equivalence relation of type (M) in X such that $R \rightarrow X \times X$ is an element of (N), with (N) satisfying (a) and (f_T), then the quotient sheaf X/R is defined by*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-objects } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{such that } Z \rightarrow X_S \text{ belongs to (N), } Z \rightarrow S \text{ is covering} \\ \text{(or an element of (M)), and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

Moreover, one has:

Proposition 4.7.2. *Let $X \in \text{Ob } \mathcal{C}$, and R an (M)-effective equivalence relation in X . Let (N) be a family of morphisms satisfying (a) and (f_M).*

For every subobject Y of X stable under R and such that $Y \rightarrow X$ belongs to (N), the equivalence relation induced in Y by R is (M)-effective, and the quotient $Y/R_Y = Y'$ is a subobject of $X' = X/R$ such that $Y' \rightarrow X'$ belongs to (N).

The map $Y \mapsto Y'$ is a bijection between the set of subobjects Y of X stable under R , such that $Y \rightarrow X$ belongs to (N), and the set of subobjects Y' of X' such that $Y' \rightarrow X'$ belongs to (N). The inverse map is $Y' \mapsto Y' \times_{X'} X$.

PROOF. Since R is (M)-effective, the morphism $X \rightarrow X'$ belongs to (M). Let Y' be a subobject of X' such that the canonical morphism $Y' \rightarrow X'$ belongs to (N). Then the subobject $Y = Y' \times_{X'} X$ of X is stable under R , and the morphism $Y \rightarrow X$ (resp. $Y \rightarrow Y'$) belongs to (N) (resp. (M)), since (N) and (M) are stable under base change. Denote by R_Y the equivalence relation induced in Y by R . By 4.4.11, the quotient sheaf Y/R_Y is represented by Y' , and hence, by 4.6.5, R_Y is (M)-effective.

Conversely, let us show that every subobject Y of X stable under R , such that the structural morphism $Y \rightarrow X$ belongs to (N), is obtained in this way. Indeed, if Y is stable under R , its two images in $R = X \times_{X'} X$ are identical, and Y is endowed with a descent datum relative to $X \rightarrow X'$; the desired result follows, since the family (N) satisfies axiom (f_M). $\square$

Corollary 4.7.3. *Let $X \in \text{Ob } \mathcal{C}$, and R an (M)-effective equivalence relation in X ; suppose moreover that $R \rightarrow X \times X$ belongs to (N), where (N) satisfies (a) and (f_T). Then, for every Y as in 4.7.2, $R_Y \rightarrow Y \times Y$ also belongs to (N), and hence, by 4.7.1, one has:*

$$(Y/R_Y)(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-objects } Z \text{ of } Y_S \text{ stable under } R_Y \times S, \\ \text{such that } Z \rightarrow Y_S \text{ belongs to (N) (then } Z \rightarrow X_S \text{ does too),} \\ Z \rightarrow S \text{ is covering, and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

⁽⁴⁶⁾N.D.E. Axiom (f_M) (which appeared before Proposition 4.7.2) has been placed here.

⁽⁴⁷⁾N.D.E. Cf. § 6.4, see also VIA, 5.3.1.

5. Passage to the quotient and algebraic structures

5.1. Principal homogeneous bundles

Definition 5.1.0. ⁽⁴⁸⁾ Recall (III 0.1) that an object X with a group of operators H (on the right) is said to be *formally principal homogeneous* under H ⁽⁴⁹⁾ if the canonical morphism (of functors)

$$X \times H \longrightarrow X \times X$$

defined by $(x, h) \mapsto (x, xh)$ is an isomorphism. It amounts to the same to say (cf. *loc. cit.*) that for every $S \in \text{Ob } \mathcal{C}$, $X(S)$ is formally principal homogeneous under $H(S)$, that is, empty or principal homogeneous under $H(S)$. In particular, if one lets H act on itself by translations (on the right), H becomes formally principal homogeneous under itself.

Definition 5.1.1. The object X with a group of operators H is said to be *trivial* if it is isomorphic (as an object with a group of operators H) to H on which H acts by translations.

Proposition 5.1.2. *Let X be formally principal homogeneous under H . One has an isomorphism*

$$\Gamma(X) \xrightarrow{\sim} \text{Isom}_{H\text{-obj.}}(H, X)$$

of principal homogeneous sets under $\Gamma(H)$.

PROOF. To every section x of X one associates the morphism from H to X defined set-theoretically by $h \mapsto xh$. The stated assertion is immediate, by reduction to the set-theoretic case. $\square$

Corollary 5.1.3. *One has an isomorphism of objects with operators H*

$$X \xrightarrow{\sim} \underline{\text{Isom}}_{H\text{-obj.}}(H, X).$$

Corollary 5.1.4. *For an object with a group of operators to be trivial, it is necessary and sufficient that it be formally principal homogeneous and that it possess a section.*

Definition 5.1.5. Let $\mathcal{C}$ be a category endowed with a topology. One says that the S -object X with an S -group of operators H is a *principal homogeneous bundle* under H if it is locally trivial, that is, if the following equivalent conditions hold: ⁽⁵⁰⁾

- (i) The set of $T \longrightarrow S$ such that (the functor) $X \times_S T$ is trivial under $H \times_S T$ is a refinement of S .
- (ii) There exists a covering family (for a pretopology defining the given topology) $\{S_i \longrightarrow S\}$ such that for each i , the S_i -functor $X \times_S S_i$ with an S_i -group-functor of operators $H \times_S S_i$ is trivial (= possesses a section over S_i).

Proposition 5.1.6. *Let $\mathcal{C}$ be a category endowed with a topology $\mathcal{T}$. Let (M) be a family of morphisms of $\mathcal{C}$ satisfying axioms (a) to (e $\mathcal{T}$) of 4.6.3. Let H be an S -group such that the structural morphism $H \longrightarrow S$ is an element of (M) , and P an S -object with an S -group of operators H . The following conditions are equivalent:*

- (i) P is a principal homogeneous bundle under H (Definition 5.1.5).
- (ii) P is formally principal homogeneous under H and the structural morphism $P \longrightarrow S$ is an element of (M) .

⁽⁴⁸⁾N.D.E. The numbering 5.1.0 has been introduced for later references.

⁽⁴⁹⁾N.D.E. One also says “pseudo-torsor”, cf. EGA IV₄, 16.5.15. Moreover, the more general notion of formally homogeneous object (not necessarily principal homogeneous) is defined in the addition 6.7.1 at the end of this exposé.

⁽⁵⁰⁾N.D.E. In this case one also says that X is an H -torsor.

- (iii) *There exists a morphism $S' \rightarrow S$ which is an element of (M) , such that by base extension from S to S' , P becomes trivial, that is, $P \times_S S'$ is trivial under $H \times_S S'$.*
- (iv) *H acts freely on P , in an (M) -effective manner, and the quotient P/H is isomorphic to S .*

PROOF. First remark that (ii) and (iv) are equivalent, taking account of the fact that in either case $P \rightarrow S$ is an element of (M) , hence base-changeable, which ensures the representability of the fibered products $H \times_S P$, $P \times_S P$. It is clear that (ii) implies (iii), for one may take P itself as S' , the hypothesis that P is formally principal homogeneous implying that $P \times_S P$ is trivial under $H \times_S P$ (5.1.4), for it has a section (the diagonal section). It is clear that (iii) implies (i), for $\{S' \rightarrow S\}$ is a covering family by axiom $(d_{\mathcal{T}})$. It therefore remains to show that (i) implies (ii). The morphism of sheaves $P \times_S H \rightarrow P \times_S P$ is locally an isomorphism, hence an isomorphism (4.5.8); thus P is formally principal homogeneous. The structural morphism $P \rightarrow S$ is locally isomorphic to the structural morphism $H \rightarrow S$, which is an element of (M) . It is therefore itself an element of (M) by $(e_{\mathcal{T}})$. $\square$

The equivalence between (i) and (iv) generalizes:

Proposition 5.1.7. *Under the same hypotheses on $\mathcal{C}$ and (M) , let H be an S -group, and X an S -object on which H acts (on the right). Suppose that the structural morphism $H \rightarrow S$ is an element of (M) . The following conditions are equivalent:*

- (i) *H acts freely on X , in an (M) -effective manner.*
- (ii) *There exists an S -morphism $p : X \rightarrow Y$ compatible with the equivalence relation defined in X by the action of H , such that the action of $H \times_S Y$ on X over Y deduced from it makes X a principal homogeneous bundle under H_Y over Y .*

Under these conditions p identifies Y with the quotient X/H .

PROOF. If $p : X \rightarrow Y$ is a morphism compatible with the action of H , then the action of $H \times_S Y$ on X over Y deduced from it defines in X the same equivalence relation as the action of H , by virtue of the formula

$$H_Y \times_Y X \xrightarrow{\sim} H \times_S X.$$

The proposition follows from this remark and the preceding equivalence (iv) $\Leftrightarrow$ (i). $\square$

Corollary 5.1.7.1. ⁽⁵¹⁾ *Let $\mathcal{C}$ be a category possessing a final object, stable under fibered products and endowed with a topology $\mathcal{T}$ less fine than the canonical topology. Let $f : G \rightarrow H$ be a morphism of $\mathcal{C}$ -groups, and $K = \text{Ker}(f)$. Suppose that f is covering for the topology $\mathcal{T}$.*

Then H represents the quotient sheaf G/K , and f is a K_H -torsor. (N.B. One will also say: “ G is a K -torsor over H ”.)

PROOF. Indeed, since f is covering, it is a universal effective epimorphism (4.4.3), hence by 3.3.3.1, H is the quotient of G by the equivalence relation $\mathcal{R}(f) = G \times_H G$. On the other hand, the morphism $G \times K \rightarrow G \times_H G$, $(g, k) \mapsto (g, gk)$, is an isomorphism of objects with a group of operators $K_G = G \times_H K_H$ (its inverse being given by $(g, g') \mapsto (g, g^{-1}g')$). Thus, on the one hand, $\mathcal{R}(f)$ is the equivalence relation defined by K ; on the other, since the morphism $f : G \rightarrow H$ is covering, f is a K_H -torsor by 5.1.6 (ii) (or directly by Definition 5.1.5 (ii)). $\square$

We may now specify Theorem 4.6.6 in the case of passage to the quotient by a group of operators:

Proposition 5.1.8. *Under the hypotheses of 5.1.7, denote by F_0 the functor over S defined as follows: for each $S' \rightarrow S$, $F_0(S')$ is the set of representable sub- S' -functors Z of $X \times_S S'$,*

⁽⁵¹⁾**N.D.E.** This particular case has been added as a corollary, and will be used several times in the following exposés.

stable under $H \times_S S'$ and principal homogeneous bundles under this S' -group for the induced action (3.2.2).

(i) The following conditions are equivalent:

a) The action of H on X is (M) -effective and free. ⁽⁵²⁾

b) F_0 is representable.

Under these conditions $F_0 = X/H$.

(ii) Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S'_i \rightarrow S'\}$ and every family $\{T_i \rightarrow S'_i\}$ of morphisms of (N) , every descent datum on the T_i relative to $\{S'_i \rightarrow S'\}$ is effective. Suppose that the morphism $X \times_S H \rightarrow X \times_S X$ is an element of (N) , and that X is base-changeable. Then the element p of $\text{Hom}(X, F_0)$ corresponding to the subobject $X \times_S H$ of $X \times_S X$ identifies F_0 with the quotient sheaf X/H .

5.2. Group structures and passage to the quotient

In this number we concern ourselves with the algebraic structures that one can put on the quotient G/H of a group by a subgroup. We shall first work in the category of sheaves on $\mathcal{C}$ for an arbitrary topology. Taking the canonical topology and using 4.5.12, we shall obtain results for passage to the universal effective quotient in $\mathcal{C}$.

Proposition 5.2.1. *Let $u : H \rightarrow G$ be a monomorphism of sheaves of groups. There exists on the quotient sheaf G/H a unique structure of object with a group of operators G such that the canonical morphism*

$$p : G \rightarrow G/H$$

is a morphism of objects with a group of operators G . This structure is functorial in the pair (G, H) : if one has a commutative diagram

$$\begin{array}{ccc} H & \longrightarrow & G \\ \downarrow & & \downarrow f \\ H' & \longrightarrow & G' \end{array}$$

the morphism $\bar{f} : G/H \rightarrow G'/H'$ (3.2.3) is compatible with the morphism f on the groups of operators.

PROOF. Indeed, the sheaf G/H is the sheaf associated to the presheaf

$$i(G)/i(H) : S \mapsto G(S)/H(S);$$

since the functor a is left exact, it transforms objects with groups of operators into objects with a group of operators. Since the presheaf $i(G)/i(H)$ is endowed with a structure of object with a group of operators $i(G)$, $G/H = a(i(G)/i(H))$ is endowed with a structure of object with operators $a(i(G)) = G$. This structure evidently enjoys all the stated properties. $\square$

Corollary 5.2.2. *Let $u : H \rightarrow G$ be a monomorphism of $\mathcal{C}$ -groups. Suppose that the action of H on G is universal effective. There exists on the quotient object $G/H \in \text{Ob } \mathcal{C}$ a unique structure of object with a group of operators G such that $p : G \rightarrow G/H$ is a morphism of objects with operators. This structure is functorial in the pair (H, G) (H acting in a universal effective manner on G) in the preceding sense.*

Proposition 5.2.3. *Let $u : H \rightarrow G$ be a monomorphism of sheaves of groups identifying H with an invariant subsheaf of groups of G . There exists on the quotient sheaf G/H a unique structure of sheaf of groups such that the canonical morphism $p : G \rightarrow G/H$ is a morphism of groups. This structure is functorial in the pair (H, G) (H invariant).*

PROOF. The proof is similar to that of 5.2.1. $\square$

⁽⁵²⁾N.D.E. “and free” has been added.

Corollary 5.2.4. *Let $u : H \rightarrow G$ be a monomorphism of $\mathcal{C}$ -groups identifying H with an invariant subgroup of G . Suppose that the action of H on G is universal effective. There exists on the quotient object $G/H \in \text{Ob } \mathcal{C}$ a unique group structure such that the canonical morphism $G \rightarrow G/H$ is a morphism of groups. This structure is functorial in the pair (H, G) (H invariant, H acting in a universal effective manner).*

One can characterize the group structure of G/H in a more expressive manner:

Proposition 5.2.5. *Under the conditions of 5.2.4, let K be a $\mathcal{C}$ -group, and $f : G \rightarrow K$ a morphism. The following conditions are equivalent:*

- (i) *f is a morphism of groups compatible with the equivalence relation defined by H .*
- (ii) *f is a morphism of groups inducing the trivial morphism $H \rightarrow K$.*
- (iii) *f factors through a morphism of groups $G/H \rightarrow K$.*

In particular, one has an isomorphism functorial in the group K

$$\text{Hom}_{\mathcal{C}\text{-gr.}}(G/H, K) \xrightarrow{\sim} \{f \in \text{Hom}_{\mathcal{C}\text{-gr.}}(G, K) \mid f \circ u = e\}.$$

PROOF. The equivalence of (i) and (ii) is proved set-theoretically. One evidently has (iii) $\Rightarrow$ (ii). The equivalence of (iii) and (ii) follows from the formula

$$\text{Hom}(G/H, K) \simeq \text{Hom}(i(G)/i(H), K)$$

and the definition of the group structure of G/H . $\square$

Remark 5.2.6. In the preceding situation, if the kernel of f is exactly H , the morphism $G/H \rightarrow K$ through which f factors is a monomorphism. This follows at once from 3.3.4.

In the case of sheaves of groups, one can specify 4.4.11 by the

Proposition 5.2.7. *Let G be a sheaf of groups and H an invariant subsheaf of groups. For every subsheaf of groups K of G containing H , let K' be the quotient group K/H considered as a subgroup of $G' = G/H$.*

One has $K = K' \times_{G'} G$, and the maps $K \mapsto K/H$, $K' \mapsto K' \times_{G'} G$ establish a bijection between the set of subsheaves of groups of G containing H and the set of subsheaves of groups of G' . Under this correspondence, invariant subsheaves of groups of G, G' correspond.

PROOF. The first part follows easily from 4.4.11 and 3.2.4. It remains to see that K is invariant in G if and only if K' is invariant in G' . If K is invariant in G , then the presheaf $i(K)/i(H)$ is invariant in $i(G)/i(H)$. The same is true of the associated sheaves, by virtue of the customary argument. Conversely, if K' is invariant in G' , then the fibered product $K \times_{G'} G$ is invariant in G , as one sees immediately. $\square$

If now L is any subsheaf of groups of G , let $\bar{L}$ be the saturation of L for the equivalence relation defined by H ; one will also write $\bar{L} = L \cdot H$.

Proposition 5.2.8. *Under the preceding conditions, $L \cdot H$ is a subsheaf of groups of G containing H , and the image of L in G/H is identified with*

$$(L \cdot H)/H \simeq L/(H \cap L).$$

PROOF. Indeed, denote by L' the image sheaf of L in G/H . It is a subsheaf of groups of G/H corresponding to $L \cdot H$ under the correspondence of the preceding proposition. Since the morphism $L \rightarrow L'$ is covering, hence a universal effective epimorphism of sheaves, it follows from 4.4.9 that L' is identified with the quotient of L by the kernel of $L \rightarrow L'$, which is evidently precisely $H \cap L$. $\square$

Finally consider the following situation: one has a sheaf of groups G , a subsheaf of groups K , and a subsheaf of groups H of K , invariant in K . First let us define an action (on the right) of the sheaf of groups $H \backslash K (= K/H)$ on G/H . The group K acts by right translations on G . Since H is invariant in K , this action is compatible with the equivalence relation

defined by the action of H , and therefore defines an action of K on G/H , that is, a morphism from the group K° opposite to K to $\underline{\text{Aut}}(G/H)$. Since the latter is a sheaf (4.5.13) and this morphism is trivial on H , it factors through K/H and defines the desired action. Since the right and left actions of G on itself commute, the actions of $G, K/H$ on G/H commute.

Proposition 5.2.9. *Under the preceding conditions, K/H acts freely (on the right) on G/H , and one has a canonical isomorphism of sheaves with a group of operators G*

$$(G/H)/(K/H) \simeq G/K.$$

When K is invariant in G , in which case K/H is invariant in G/H (5.2.7), this isomorphism respects the group structures of the two members.

PROOF. One has an isomorphism of presheaves

$$(i(G)/i(H))/(i(K)/i(H)) \xleftarrow{\sim} i(G)/i(K),$$

which respects the structures of objects with a group of operators $i(G)$. The announced result is obtained by applying the functor a to this relation. $\square$

Corollary 5.2.10. *Let G be a $\mathcal{C}$ -group, K a sub- $\mathcal{C}$ -group of G , and H an invariant sub- $\mathcal{C}$ -group of K . Let (M) be a family of morphisms of $\mathcal{C}$ satisfying axioms (a) to $(e_{\mathcal{T}})$. Suppose that the action of H on G (resp. K) on the right is (M) -effective. Then K/H acts naturally and freely on the right on G/H ; this action commutes with that of G . The following conditions are equivalent:*

- (i) *The action of K on G is (M) -effective.*
- (ii) *The action of K/H on G/H is (M) -effective.*

Under these conditions one has an isomorphism of objects⁽⁵³⁾ with a group of operators G :

$$(G/H)/(K/H) \simeq G/K.$$

5.3. Use of effectiveness criteria: Noether's theorem

⁽⁵⁴⁾ Let $\mathcal{C}, \mathcal{T}$, and (M) be as usual. Let (N) be a family of morphisms satisfying axioms (a) and (f_M) of 4.7. Putting together 5.2.7 and 4.7.2, one obtains:

Proposition 5.3.1. *Let G be a $\mathcal{C}$ -group. Let H be an invariant sub- $\mathcal{C}$ -group of G , acting in an (M) -effective manner on G .*

For every sub- $\mathcal{C}$ -group K of G containing H and such that the morphism $K \rightarrow G$ belongs to (N) , H acts on K in an (M) -effective manner, and the quotient $K/H = K'$ is a sub- $\mathcal{C}$ -group of $G/H = G'$, such that the morphism $K' \rightarrow G'$ belongs to (N) .

The map $K \mapsto K'$ is a bijection between the set of sub- $\mathcal{C}$ -groups K of G containing H and such that $K \rightarrow G$ belongs to (N) , and the set of sub- $\mathcal{C}$ -groups K' of G' such that $K' \rightarrow G'$ belongs to (N) . The inverse map is $K' \mapsto K \times_{G'} G$. Under this correspondence, invariant subgroups of G, G' correspond.

Corollary 5.3.2. *If $H \rightarrow G$ is an element of (N) , then $\mathcal{C}$ possesses a final object e , and the unit section $e \rightarrow G/H$ is an element of (N) .⁽⁵⁵⁾*

6. Topologies in the category of schemes

6.1. The Zariski topology

⁽⁵³⁾N.D.E. Of $\mathcal{C}$.

⁽⁵⁴⁾N.D.E. In fact, the isomorphisms established in 5.2.8 to 5.2.10 would deserve to be called “Noether's isomorphism theorems”.

⁽⁵⁵⁾N.D.E. This follows from the proposition applied to $K = H$.

This is the topology generated by the following pretopology: a family of morphisms $\{S_i \rightarrow S\}$ is covering if each morphism is an open immersion and the union of the images of the S_i is all of S . It is denoted (Zar).

Definition 6.1.1. A sheaf for the Zariski topology is also called a *functor of local nature*: it is a contravariant functor from (Sch) to (Ens) such that for every scheme S and every covering of S by open sets S_i , one has an exact diagram:

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \cap S_j).$$

In particular, a functor of local nature transforms direct sums into products. Since every representable functor is a sheaf, this topology is less fine than the canonical topology. As regards terminology, whenever we say “local”, “locally”, without further specification, this will refer to the Zariski topology, hence have the customary meaning.

6.2. A procedure for constructing topologies

Proposition 6.2.1. Let $\mathcal{C}$ be a category, $\mathcal{C}'$ a full subcategory, P a set of families of morphisms of $\mathcal{C}$ with the same target, stable under base change and composition (i.e. satisfying axioms (P 1) and (P 2) of 4.2.5), and P' a set of families of morphisms of $\mathcal{C}'$ containing the families consisting of a single identity isomorphism. Endow $\mathcal{C}$ with the topology generated by P, P' (cf. 4.2.5.0), and suppose that the following three conditions hold:

(a) If $\{S_i \rightarrow S\} \in P'$ (thus $S_i, S \in \text{Ob } \mathcal{C}'$), and if $T \rightarrow S$ is a morphism of $\mathcal{C}'$, then the fibered products $S_i \times_S T$ (in $\mathcal{C}$) exist, and the family $\{S_i \times_S T \rightarrow T\}$ belongs to P' (thus $S_i \times_S T \in \text{Ob } \mathcal{C}'$). (Remark: this condition implies that P' is stable under base change in $\mathcal{C}'$, but is not equivalent to it, for it assumes moreover that the inclusion functor from $\mathcal{C}'$ to $\mathcal{C}$ commutes with certain fibered products.)

(b) For every $S \in \text{Ob } \mathcal{C}$, there exists $\{S_i \rightarrow S\} \in P$, with $S_i \in \text{Ob } \mathcal{C}'$ for each i .

(c) In the following situation:

$$\begin{array}{ccc} & S_{ijk} & \\ & \downarrow (P') & \\ S_i & \xleftarrow{(P)} & S_{ij} \\ (P') \downarrow & & \\ S & & \end{array}$$

where $S, S_i, S_{ij}, S_{ijk} \in \text{Ob } \mathcal{C}'$; $\{S_i \rightarrow S\} \in P'$; $\{S_{ij} \rightarrow S_i\} \in P$ for each i ; $\{S_{ijk} \rightarrow S_{ij}\} \in P'$ for each ij , there exists a family $\{T_n \rightarrow S\} \in P'$ and, for each n , a multi-index (ijk) and a commutative diagram

$$\begin{array}{ccc} S_{ijk} & \xleftarrow{\quad} & T_n \\ & \searrow & \swarrow \\ & & S \end{array}$$

Then, for a sieve R of $S \in \text{Ob } \mathcal{C}$ to be covering, it is necessary and sufficient that there exist a composite family

$$\begin{array}{ccc} R & \xleftarrow{\quad} & S_{ij} \\ \downarrow & & \downarrow (P') \\ S & \xleftarrow{(P)} & S_i \end{array}$$

where $S_i, S_{ij} \in \text{Ob } \mathcal{C}'$, $\{S_i \rightarrow S\} \in P$, $\{S_{ij} \rightarrow S_i\} \in P'$ for each i , and that the resulting morphisms $S_{ij} \rightarrow S$ factor through R (in other words, that the sieve generated by this composite family be contained in R).

PROOF. Since families that are elements of P, P' are covering, a composite family of such families will also be covering (C 2); hence a sieve of the indicated form will be covering, since it contains a covering sieve.

Conversely, it suffices to see that the sieves of this form do form a topology, i.e. it suffices to verify axioms (T 1) to (T 4) of 4.2.1.

Axiom (T 4). Let $S \in \text{Ob } \mathcal{C}$. By (b), there exists a family $\{S_i \rightarrow S\} \in P$, with $S_i \in \text{Ob } \mathcal{C}'$. The families $\{S_i \xrightarrow{\text{id}_{S_i}} S_i\}$ are elements of P' by hypothesis. The sieve S of S is therefore of the desired form:

$$\begin{array}{ccc} S_i & \longrightarrow & S \\ (P') \downarrow \text{id} & & \downarrow \text{id}_S \\ S_i & \xrightarrow{(P)} & S \end{array} .$$

Axiom (T 3). Evident.

Axiom (T 2). Let R be a sieve of S of the desired form, and let C be a sieve⁽⁵⁶⁾ of S such that, for every $T \in \text{Ob } \mathcal{C}$ and every morphism $T \rightarrow S$ factoring through R , the sieve $C \times_S T$ of T is of the desired form. Then, since $S_{ij} \rightarrow S$ factors through R , the sieve $C_{ij} = C \times_S S_{ij}$ of S_{ij} :

$$\begin{array}{ccccc} & & S_{ij} & \longleftarrow & C_{ij} \\ & & \downarrow (P') & & \downarrow \\ & & S_i & & \\ & \swarrow & \downarrow (P) & & \downarrow \\ R & \longrightarrow & S & \longleftarrow & C \end{array}$$

is of the desired form; thus, for every ij , one has a diagram of the form:

$$\begin{array}{ccc} S_{ijkl} & & \\ (P') \downarrow & \searrow & \\ S_{ijk} & & \\ (P) \downarrow & & \\ S_{ij} & \longleftarrow & C_{ij} \end{array} .$$

One has therefore proved that there exists a composite family

$$S_{ijkl} \xrightarrow{(P')} S_{ijk} \xrightarrow{(P)} S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S$$

belonging to $P \circ P' \circ P \circ P'$, factoring through C , and in which all objects other than S are in $\mathcal{C}'$. Applying condition (c) to each family $\{S_{ijkl} \rightarrow S_i\}$, one deduces that for each i there exists a family $\{T_{in} \rightarrow S_i\} \in P'$ such that $T_{in} \rightarrow S$ factors through one of the S_{ijkl} , hence

⁽⁵⁶⁾N.D.E. The original has been corrected here.

through C :

$$\begin{array}{ccc} T_{in} & \longrightarrow & S_{ijkl} \\ (P') \downarrow & & \downarrow \\ S_i & & \\ (P) \downarrow & & \downarrow \\ S & \longleftarrow & C \end{array}$$

The sieve C of S is therefore of the desired form, which completes the verification.

Axiom (T 1). Let R be a sieve of S of the given form, and $T \rightarrow S$ a morphism of $\mathcal{C}$. Let us show that the sieve $R \times_S T$ of T is of the desired form.

$$\begin{array}{ccccc} & S_{ij} & \longleftarrow & U_{ikj} & \\ & \downarrow (P') & & \downarrow (P') & \\ & S_i & \longleftarrow & T_i & \longleftarrow U_{ik} \\ & \downarrow (P) & & \downarrow (P) & \swarrow (P) \\ R & \longrightarrow & S & \longleftarrow & T \end{array}$$

Let $T_i = S_i \times_S T$. The family $\{T_i \rightarrow T\}$ belongs to P (by (P 1)). Applying (b), one constructs $\{U_{ik} \rightarrow T_i\} \in P$, with $U_{ik} \in \text{Ob } \mathcal{C}'$. By hypothesis (condition (P 2) on P), one has $\{U_{ik} \rightarrow T\} \in P$. By (a), $U_{ik} \times_{S_i} S_{ij} = U_{ikj}$ is an object of $\mathcal{C}'$, and for every ik , $\{U_{ikj} \rightarrow U_{ik}\} \in P'$. Then the commutative diagram below

$$\begin{array}{ccc} R & \longleftarrow & U_{ikj} \\ \downarrow & & \downarrow (P') \\ S & \longleftarrow & T \longleftarrow U_{ik} \end{array}$$

shows that the morphisms $U_{ikj} \rightarrow T$ factor through the sieve $T \times_S R$ of T , which is therefore of the desired form, and this completes the proof. $\square$

Corollary 6.2.2. *If $S \in \text{Ob } \mathcal{C}'$ and R is a sieve of S , R is covering if and only if there exists a family $\{T_i \rightarrow S\} \in P'$ factoring through R .*

PROOF. Indeed, such a sieve is covering. On the other hand, it suffices to apply (c), taking the family $\{S_i \rightarrow S\}$ to consist of the identity isomorphism of S , to deduce from the proposition that a covering sieve is of the indicated form. $\square$

Corollary 6.2.3. *For a presheaf F on $\mathcal{C}$ to be separated (resp. a sheaf), it is necessary and sufficient that the morphism*

$$F(S) \longrightarrow \prod_i F(S_i)$$

be injective (resp. that the diagram

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times_S S_j)$$

be exact) in the following two cases:

- (i) $\{S_i \rightarrow S\} \in P$,
- (ii) $S, S_i \in \text{Ob } \mathcal{C}'$; $\{S_i \rightarrow S\} \in P'$.

PROOF. Indeed, the conditions are necessary, for the families in question are covering. If C is the sieve of S which is the image of a family of morphisms $\{S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S\}$, an easy diagram chase shows that the conditions of the corollary imply that $\text{Hom}(S, F) \xrightarrow{g} \text{Hom}(C, F)$

is injective (resp. bijective). But every refinement R of S contains a sieve C of the above type, and one has a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(S, F) & \xrightarrow{f} & \mathrm{Hom}(R, F) \\ & \searrow g \quad \swarrow h & \\ & \mathrm{Hom}(C, F) & \end{array}$$

One knows that g is injective, hence f is too. Thus F is separated. But R is a refinement of C , hence h is also injective. If g is bijective, then f is too, hence F is a sheaf. $\square$

Remark 6.2.4. The preceding corollary does not follow from 4.3.5, since P' is not stable under base extension.

Remark 6.2.5. Condition (c) holds in particular in the case where

- (i) P' is stable under composition.
- (ii) If $\{S_j \rightarrow S\}$ is a family of morphisms of $\mathcal{C}'$, an element of P , there exists a subfamily of it which is an element of P' .

6.3. Application to the category of schemes

Take for $\mathcal{C}$ the category of schemes, for $\mathcal{C}'$ the full subcategory consisting of affine schemes, and for P the set of surjective families of open immersions. One will consider several sets P' :

P'_1 : finite surjective families consisting of flat morphisms.

P'_2 : finite surjective families consisting of flat morphisms of finite presentation which are quasi-finite. ⁽⁵⁷⁾

P'_3 : finite surjective families consisting of étale morphisms.

P'_4 : finite surjective families consisting of finite étale morphisms.

For each of these sets P'_i , except P'_4 , the conditions of Proposition 6.2.1 hold ((c) by 6.2.5, for, since an affine scheme is quasi-compact, every family of morphisms of $\mathcal{C}'$ which is an element of P contains a finite subfamily which is also in P , hence in P'_i for $i = 1, 2, 3$). The topology $\mathcal{T}_i$ generated by P, P'_i is denoted and called as follows: ⁽⁵⁷⁾

$\mathcal{T}_1 = (\mathrm{fpqc}) =$ faithfully flat quasi-compact topology.

$\mathcal{T}_2 = (\mathrm{fppf}) =$ faithfully flat (locally) finite presentation topology.

$\mathcal{T}_3 = (\mathrm{ét}) =$ étale topology.

$\mathcal{T}_4 = (\mathrm{étf}) =$ finite étale topology.

Since $P'_1 \supset P'_2 \supset P'_3 \supset P'_4$, one has

$$(\mathrm{fpqc}) \geq (\mathrm{fppf}) \geq (\mathrm{ét}) \geq (\mathrm{étf}) \geq (\mathrm{Zar}).$$

Proposition 6.3.1. (i) For the sieve R of S to be covering for $\mathcal{T}_i$, $1 \leq i \leq 3$, it is necessary and sufficient that there exist a covering (S_p) of S by affine open sets and, for each p , a family $\{S_{pq} \rightarrow S_p\}$ which is an element of P'_i , the S_{pq} being affine, such that each morphism $\{S_{pq} \rightarrow S\}$ factors through R . ⁽⁵⁸⁾

⁽⁵⁷⁾**N.D.E.** Let P'_2 be the set of finite surjective families consisting of flat morphisms of $\mathcal{C}'$ of finite presentation. By Proposition 6.3.1 below, the topology $\mathcal{T}_2$ generated by P, P'_2 coincides with that generated by P, P'_2 . This follows from the results of EGA IV₄, § 17.16 on quasi-sections; see the proof of 6.3.1. Let us cite here the following particular case of EGA IV₄, 17.16.2: let S be affine and $f : X \rightarrow S$ a surjective flat morphism locally of finite presentation; then there exists a faithfully flat morphism $S' \rightarrow S$ of finite presentation which is quasi-finite, with S' affine, and an S -morphism $S' \rightarrow X$.

⁽⁵⁸⁾**N.D.E.** By hypothesis, each family $\{S_{pq} \rightarrow S\} \in P'_i$ is finite, hence $S'_p = \coprod_q S_{pq}$ is affine, and the family can therefore be replaced by the morphism $S'_p \rightarrow S_p$, which still belongs to P'_i .

(ii) For a presheaf F on (Sch) to be a sheaf for (fpqc) (resp. (fppf) , (ét) , (étf)), it is necessary and sufficient that

- a) F be a sheaf for (Zar) , i.e. a functor of local nature.
- b) For every faithfully flat (resp. faithfully flat of finite presentation and quasi-finite, resp. surjective étale, resp. surjective finite étale) morphism $T \rightarrow S$, where T, S are affine, one have an exact diagram:

$$F(S) \rightarrow F(T) \rightrightarrows F(T \times_S T).$$

- (iii) The topologies $\mathcal{T}_i$, $i = 1, 2, 3, 4$, are less fine than the canonical topology.
- (iv) Every surjective family consisting of flat and open (resp. flat and locally of finite presentation, resp. étale, resp. finite étale) morphisms is covering for (fpqc) (resp. (fppf) , resp. (ét) , resp. (étf)).
- (v) Every finite surjective family consisting of flat quasi-compact morphisms is covering for (fpqc) . In particular, every faithfully flat quasi-compact morphism is covering for (fpqc) .

PROOF. (i) follows from 6.2.1, (ii) from 6.2.3, taking account of the fact that a sheaf for the Zariski topology transforms direct sums into products. Since every representable functor is a sheaf for (Zar) and satisfies condition (b) of (ii) by SGA 1, VIII 5.3, $\mathcal{T}_1$ is less fine than the canonical topology, which proves (iii).

Let us prove (iv). Let $\{S_i \rightarrow S\}$ be a family of morphisms as in the statement. Considering a covering of S by affine open sets, one is immediately reduced to the case where S is affine.⁽⁵⁹⁾ First let us treat the case where the morphisms $S_i \rightarrow S$ are flat and open (resp. étale). Let S_{ij} be a covering of S_i by affine open sets. Since the morphisms in question are open, the images T_{ij} of the S_{ij} form an open covering of S . Since S is affine, hence quasi-compact, it is covered by a finite number of open sets T_{ij} , with (i, j) ranging over a finite set F . Then $S' = \coprod_F S_{ij}$ is affine, and the morphism $S' \rightarrow S$ belongs to P'_1 , resp. P'_3 , hence is covering. Since it factors through the given family, the latter is covering.

In the case (étf) , each S_i is finite over S , hence affine; in the preceding argument, one may then take the covering $\{S_i\}$ of S , and one obtains that $S' \rightarrow S$ belongs to P'_4 .

Now consider the case where the morphisms $f_i : S_i \rightarrow S$ are flat and locally of finite presentation. For every $s \in S$, there exists, by (the proof of) EGA IV₄, 17.16.2, an affine subscheme $X(s)$ of some S_i , such that $s \in f_i(X(s))$ and the morphism $g_i : X(s) \rightarrow S$, the restriction of f_i , is flat, of finite presentation, and quasi-finite. Then $g_i(X(s))$ is an open neighborhood $U(s)$ of s (EGA IV₂, 2.4.6), and, since S is affine, it is covered by a finite number of such open sets $U(s_j)$, $j = 1, \dots, n$. Consequently $X' = \coprod_j X(s_j)$ is affine, and the morphism $X' \rightarrow S$ is surjective, flat, of finite presentation and quasi-finite, hence belongs to P'_2 .⁽⁶⁰⁾ This completes the proof of (iv).

Let us prove (v). Let $\{S_i \rightarrow S\}$ be a finite faithfully flat quasi-compact family. Let T_j be a covering of S by affine open sets. The $S_{ij} = T_j \times_S S_i$ are quasi-compact, and therefore possess finite affine open coverings T_{ijk} . Each morphism $T_{ijk} \rightarrow T_j$ is flat, and the family $\{T_{ijk} \rightarrow T_j\}$ is finite and surjective, hence covering for $\mathcal{T}_1$. The family $\{T_{ijk} \rightarrow S\}$ is therefore also covering, by composition. It factors through the given family, which is therefore also covering:

$$\begin{array}{ccccc} S_i & \longleftarrow & S_{ij} & \longleftarrow & T_{ijk} \\ \downarrow & & \downarrow & \swarrow & \\ S & \longleftarrow & T_j & & \end{array}$$

□

⁽⁵⁹⁾N.D.E. The rest has been simplified, taking advantage of the fact that one now assumes S affine.

⁽⁶⁰⁾N.D.E. This shows that, if one denotes by P'_2 the set of finite surjective families of flat morphisms of $\mathcal{C}'$ of finite presentation, the topology generated by P, P'_2 equals $\mathcal{T}_2$. On the other hand, with the notation at the beginning of the proof of (iv), if one takes a covering of S_i by affine open sets of finite presentation over S , one obtains that $S' \rightarrow S$ belongs to P'_2 .

Corollary 6.3.2. *Let (M_i) be the following family of morphisms:*

- (M_1) : *faithfully flat quasi-compact morphisms.*
- (M_2) : *faithfully flat morphisms locally of finite presentation.*
- (M_3) : *surjective étale morphisms.*
- (M_4) : *surjective finite étale morphisms.* ⁽⁶¹⁾ *The family (M_i) satisfies axioms (a), (b), (c), $(d_{\mathcal{T}_i})$, and $(e_{\mathcal{T}_i})$ of 4.6.3.*

PROOF. Indeed, for (a), (b), (c), this is classical (EGA and SGA 1, passim).⁽⁶²⁾ By 6.3.1 (iv) and (v), (M_i) satisfies $(d_{\mathcal{T}_i})$. It remains to see that (M_i) satisfies $(e_{\mathcal{T}_i})$; for this it suffices to see that (M_i) satisfies $(e_{\mathcal{T}_1})$, which implies the others. This follows from SGA 1, VIII (nos. 4 and 5). $\square$

Corollary 6.3.3. *If X is a scheme and R an equivalence relation in X of type (M_i) , R is (M_i) -effective if and only if the quotient sheaf of X by R for $\mathcal{T}_i$ is representable, and in this case it is represented by the quotient X/R .*

PROOF. Indeed, this is 4.6.5. $\square$

6.4. Effectiveness conditions

We now seek families (N) of morphisms satisfying axiom $(f_{\mathcal{T}})$ of 4.7. First remark that $(f_{\mathcal{T}_1})$ implies $(f_{\mathcal{T}_i})$, so that we may restrict ourselves to the case of the topology $(fpqc)$.

Lemma 6.4.1. *The following families of morphisms satisfy axiom $(f_{\mathcal{T}_1})$ of 4.7, that is, “descend under $(fpqc)$ ”:*

- (N) : *open immersions.*
- (N') : *closed immersions.*
- (N'') : *quasi-compact immersions.*

PROOF. By 6.3.1 (ii), it suffices to verify that the given families descend under the Zariski topology and under a faithfully flat quasi-compact morphism. The first assertion is clear; let us verify the second. For (N) , this is SGA 1, VIII 4.4; for (N') , it is *loc. cit.*, 1.9. For (N'') , one argues as in *loc. cit.*, 5.5, with the aid of the two preceding results. $\square$

Corollary 6.4.2. *The same result is valid for quasi-compact open immersions.*

These results allow one to apply to the present situation the general results of 4.7.1, 4.7.2, 5.1.8, 5.3.1, etc. Let us state one as an example, the first.

Corollary 6.4.3. $(= 4.7.1 + 4.6.10)$. *Let X be a scheme and R an equivalence relation in X . Suppose that $R \rightarrow X$ is faithfully flat and quasi-compact, and that $R \rightarrow X \times X$ is a closed immersion (resp. open, resp. quasi-compact, resp. quasi-compact open). Then the quotient sheaf X/R is the same for the topology $(fpqc)$ and for the canonical topology, and for each scheme S , one has*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{closed (resp. open, resp. retrocompact,} \\ \text{resp. retrocompact open) subschemes } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{such that } Z \rightarrow S \text{ is faithfully flat quasi-compact} \\ \text{and the diagram } R_Z \rightrightarrows Z \rightarrow S \text{ is exact} \end{array} \right\}. \quad (63)$$

⁽⁶¹⁾**N.D.E.** The original has been corrected by adding for (M_3) , (M_4) the surjectivity hypothesis, which is automatically satisfied in the other cases.

⁽⁶²⁾**N.D.E.** Cf. EGA I, 6.6.4 for “quasi-compact”, EGA II, 6.1.5 for “finite”, EGA IV₁, 1.6.2 for “locally of finite presentation”, EGA IV₂, 2.2.13 for “faithfully flat”, and EGA IV₄, 17.3.3 for “étale”.

⁽⁶³⁾**N.D.E.** Recall that a subscheme Z of a scheme T is said to be retrocompact if the immersion $Z \hookrightarrow T$ is quasi-compact, cf. EGA 0_{III}, 9.1.1.

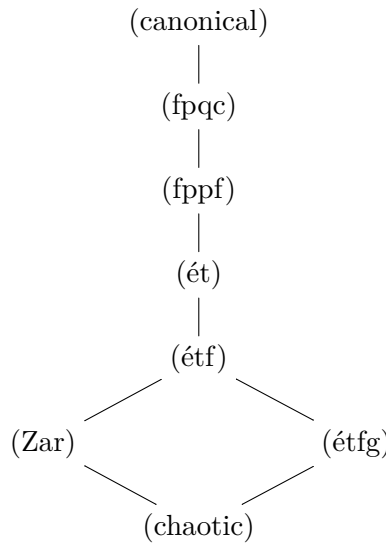
6.5. Principal homogeneous bundles

Let us simply point out the terminology:

Topology	Principal homogeneous bundles
(fpqc)	(without further qualification)
(ét)	quasi-isotrivial
(étf)	locally isotrivial
(Zar)	locally trivial.

6.6. Other topologies

One sometimes uses other topologies on the category of schemes. Let us point out one: the *global finite étale topology* (étfg), generated by the pretopology whose covering families are surjective families consisting of finite étale morphisms. It is not finer than the Zariski topology. The corresponding principal homogeneous bundles are called “isotrivial”.



(64)

6.7. Homogeneous spaces

⁽⁶⁵⁾ Let G be an S -group scheme, X an S -scheme with a group of operators G (on the left), and

$$\Phi : G \times_S X \longrightarrow X \times_S X$$

the morphism of S -schemes defined set-theoretically by $(g, x) \mapsto (gx, x)$. Recall (cf. 5.1.0 and III.0.1) that X is said to be a formally principal homogeneous space under G if the following equivalent conditions are satisfied:

- (i) For every $T \rightarrow S$, the set $X(T)$ is empty or principal homogeneous under $G(T)$.
- (ii) Φ is an isomorphism of S -functors.
- (iii) Φ is an isomorphism of S -schemes.

(The equivalence (i) $\Leftrightarrow$ (ii) is clear, and one has (ii) $\Leftrightarrow$ (iii), since $\mathcal{C} = (\text{Sch})_S$ is a full subcategory of $\widehat{\mathcal{C}}$.)

The definition of a formally homogeneous space (not necessarily principal homogeneous) is obtained by requiring that Φ be an epimorphism in the category of sheaves for an appropriate topology $\mathcal{T}$. Indeed, the condition that Φ be an epimorphism of S -functors is equivalent

⁽⁶⁴⁾**N.D.E.** Recall (cf. 4.4.2) that the chaotic topology is the least fine topology, defined by $J(S) = \{S\}$ for every $S \in \text{Ob } \mathcal{C}$.

⁽⁶⁵⁾**N.D.E.** The following numbers have been added.

to saying that for every $T \rightarrow S$, the set $X(T)$ is empty or homogeneous (not necessarily principal homogeneous) under $G(T)$, but this condition is too restrictive, as the following simple example shows. Let $S = \text{Spec } \mathbf{R}$, $G = \mathbf{G}_{m, \mathbf{R}}$, and $X = \mathbf{G}_{m, \mathbf{R}}$, on which G acts by $t \cdot x = t^2 x$. Then the morphism Φ is étale, finite, and surjective, hence an epimorphism in the category of sheaves for the topology (étf) (a fortiori, an epimorphism of S -schemes); on the other hand, the points $1, -1$ of $X(\mathbf{R})$ are not conjugate by an element of $G(\mathbf{R})$, so that the morphism $G(\mathbf{R}) \times X(\mathbf{R}) \rightarrow X(\mathbf{R}) \times X(\mathbf{R})$ is not surjective. ⁽⁶⁶⁾ One is therefore led to give the following definition:

Definition 6.7.1. Let G be an S -group, X an S -scheme with a group of operators G , and $\mathcal{T}$ a topology on $(\text{Sch})_S$ less fine than the canonical topology. One says that X is a *formally homogeneous space* under G (relative to the topology $\mathcal{T}$) if the following equivalent conditions are satisfied:

- (i) The morphism $\Phi : G \times_S X \rightarrow X \times_S X$ is an epimorphism in the category of sheaves for the topology $\mathcal{T}$.
- (ii) For every $T \rightarrow S$ and $x, y \in X(T)$, there exists a morphism $T' \rightarrow T$ covering for the topology $\mathcal{T}$, and $g \in G(T')$, such that $y_{T'} = g \cdot x_{T'}$.

Remark 6.7.2. Condition (i) implies in particular that Φ is a universal effective epimorphism in $(\text{Sch})_S$ (cf. 4.4.3). This implies, as one easily sees, that Φ is surjective (cf. 1.3, N.D.E. (3)).

Proposition and definition 6.7.3. ⁽⁶⁷⁾ Let G be an S -group, X an S -scheme with a group of operators G , and $\mathcal{T}$ a topology on $(\text{Sch})_S$ less fine than the canonical topology. The following conditions are equivalent:

- (i) X satisfies the following two hypotheses:
 - (1) The morphism $\Phi : G \times_S X \rightarrow X \times_S X$ is covering, i.e. X is a formally homogeneous G -space.
 - (2) The morphism $X \rightarrow S$ is also covering, that is, locally for the topology $\mathcal{T}$, it possesses a section (cf. RefIV.4.4.8bis.bis).
- (ii) “Locally on S for the topology $\mathcal{T}$ ”, X is isomorphic, as a scheme with a group of operators G , to the quotient sheaf (for $\mathcal{T}$) of G by a subgroup scheme H , that is, there exists a covering family $\{S_i \rightarrow S\}$ such that each $X \times_S S_i$ represents the quotient sheaf of $G \times_S S_i$ by some subgroup scheme H_i .

Under these conditions, one says that X is a *homogeneous G -space* (relative to the topology $\mathcal{T}$).

PROOF. Suppose that (ii) holds. Set $G_i = G \times_S S_i$, $X_i = X \times_S S_i$. Then X_i possesses a section over S_i , namely the composite of the unit section $\varepsilon_i : S_i \rightarrow G_i$ and the projection $\pi_i : G_i \rightarrow X_i = G_i/H_i$. Thus $X \rightarrow S$ is covering.

⁽⁶⁶⁾N.D.E. Evidently, this difficulty comes from the fact that if $\mathcal{C}'$ is a full subcategory of $\widehat{\mathcal{C}}$ containing $\mathcal{C}$, for example the category $\widetilde{\mathcal{C}}_{\mathcal{T}}$ of sheaves on $\mathcal{C}$ for a topology $\mathcal{T}$ less fine than the canonical topology, and if $f : X \rightarrow Y$ is a morphism in $\mathcal{C}$, then the implications:

$$\begin{aligned} f \text{ epimorphism of } \widehat{\mathcal{C}} &\Rightarrow f \text{ epimorphism of } \mathcal{C}' \\ &\Rightarrow f \text{ epimorphism of } \mathcal{C} \end{aligned}$$

are in general strict.

⁽⁶⁷⁾N.D.E. Cf. [Ray70], Def. VI.1.1.

On the other hand, π_i is covering, hence $\pi_i \times \pi_i$ is too (cf. 4.2.3 (C 1) and (C 2)), and one has a commutative diagram:

$$\begin{array}{ccc} G_i \times_{S_i} X_i & \xrightarrow{\Phi_i} & G_i \times_{S_i} X_i \\ \text{id} \times \pi_i \uparrow & & \uparrow \pi_i \times \pi_i \\ G_i \times_{S_i} G_i & \xrightarrow[\sim]{\Psi_i} & G_i \times_{S_i} G_i \end{array}$$

where Φ_i is deduced from Φ by the base change $S_i \rightarrow S$, and Ψ_i is the isomorphism defined set-theoretically by $(g, g') \mapsto (gg', g)$. Then $(\pi_i \times \pi_i) \circ \Psi_i$ is covering, hence Φ_i is too (4.2.3 (C 3)). This shows that Φ is “locally covering”, hence covering (4.2.3 (C 5)). This proves (ii) $\Rightarrow$ (i).

Conversely, suppose that (i) holds, and suppose moreover that the structural morphism $X \rightarrow S$ possesses a section σ . By EGA I, 5.3.13, σ is an immersion. Define $H = G \times_X S$ by the diagram below, both of whose squares are cartesian:

$$\begin{array}{ccccc} H & \longrightarrow & G & \xrightarrow{\text{id}_G \boxtimes \sigma} & G \times_S X \\ \downarrow & & \downarrow \pi & & \downarrow \Phi \\ S & \xrightarrow{\sigma} & X & \xrightarrow[\text{id}_X \boxtimes \sigma]{} & X \times_S X \end{array}$$

where $\pi, \text{id}_G \boxtimes \sigma$ and $\text{id}_X \boxtimes \sigma$ denote the morphisms defined set-theoretically, for $T \rightarrow S$ and $g \in G(T)$, $x \in X(T)$, by:

$$\begin{aligned} \pi(g) &= g \cdot \sigma_T, & (\text{id}_G \boxtimes \sigma)(g) &= (g, \sigma_T), \\ (\text{id}_X \boxtimes \sigma)(x) &= (x, \sigma_T). \end{aligned}$$

Then π is covering, and H is a subgroup scheme of G , representing the stabilizer $\text{Stab}_G(\sigma)$ of σ (cf. I, 2.3.3), that is, for every $T \rightarrow S$, one has:

$$H(T) = \{g \in G(T) \mid g \cdot \sigma_T = \sigma_T\}.$$

Denote by G/H the *presheaf* $T \mapsto G(T)/H(T)$, and by $a(G/H)$ the associated sheaf for the topology $\mathcal{T}$. By the preceding argument, one obtains a commutative diagram of morphisms of presheaves with a group of operators G :

$$\begin{array}{ccc} G & \xrightarrow{\pi} & X \\ \downarrow & \nearrow \bar{\pi} & \\ G/H & & \end{array}$$

where $\bar{\pi}$ is a monomorphism. Since π is covering, $\bar{\pi}$ is too, and hence, by 4.3.12, $\bar{\pi}$ induces an isomorphism $a(G/H) \xrightarrow{\sim} X$. One has therefore proved that: *if X is a homogeneous G -space such that $X \rightarrow S$ admits a section σ , then X represents the quotient sheaf G/H , where $H = G \times_X S$ is the stabilizer of σ .*

In the general case, there exists by hypothesis a covering family $\{S_i \rightarrow S\}$ such that each morphism $X_i = X \times_S S_i \rightarrow S_i$ possesses a section σ_i . Set $G_i = G \times_S S_i$; then the morphism $\Phi_i : G_i \times_{S_i} X_i \rightarrow X_i \times_{S_i} X_i$ deduced from Φ by the base change $S_i \rightarrow S$ is still covering. Thus, by the preceding argument, $X_i \simeq G_i/H_i$, where H_i is the stabilizer in G_i of σ_i . This completes the proof of the implication (i) $\Rightarrow$ (ii). $\square$

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⁽⁶⁸⁾ **N.D.E.** This reference has been added.

Exposé V. Construction of quotient schemes

by P. Gabriel

The object of this exposé is to prove the theorems stated in TDTE III.⁽¹⁾ If X and T are two objects of a category $\mathcal{C}$, we write $X(T)$ instead of $\text{Hom}_{\mathcal{C}}(T, X)$. Similarly, if $\varphi : Y \rightarrow X$ (resp. T) is an arrow (resp. an object) of $\mathcal{C}$, $\varphi(T)$ denotes the map $g \mapsto \varphi \circ g$ from $Y(T)$ to $X(T)$:

$$\begin{array}{ccc} & T & \\ g \swarrow & & \searrow \varphi \circ g \\ Y & \xrightarrow{\varphi} & X, \end{array}$$

and $T(\varphi)$ denotes the map $g \mapsto g \circ \varphi$ from $T(X)$ to $T(Y)$:

$$\begin{array}{ccc} Y & \xrightarrow{\varphi} & X \\ & \searrow g \circ \varphi & \swarrow g \\ & T. & \end{array}$$

Finally, if P is a scheme, one denotes by $\underline{P}$ the underlying set of P .

Exceptionally, in the present exposé we do not follow the convention stated in IV 4.6.15 concerning the notation for quotients (loc. cit., top of page 227 of the original), for we wish to give here a construction of quotients that also applies to “pre-equivalence relations”⁽²⁾ that are not equivalence relations.

1. $\mathcal{C}$ -groupoids

a) $\mathcal{C}$ is a category in which products and fibered products exist. Let us first recall that a diagram

$$X_1 \xrightleftharpoons[d_0]{d_1} X_0 \xrightarrow{p} Y$$

of $\mathcal{C}$ is said to be exact if $pd_0 = pd_1$ and if, for every $T \in \mathcal{C}$, $T(p)$ is a bijection from $T(Y)$ onto the subset of $T(X_0)$ consisting of the arrows $f : X_0 \rightarrow T$ such that $fd_0 = fd_1$. One also says that (Y, p) is the cokernel of (d_0, d_1) , and one writes

$$(Y, p) = \text{Coker}(d_0, d_1).$$

b) Let, for example, $\mathcal{C}$ be the category (Esp.An) of ringed spaces. In this case there is always a cokernel (Y, p) , of which one can give the following description: the underlying topological space of Y is obtained from X_0 by identifying the points $d_0(x)$ and $d_1(x)$ and giving Y the

⁽¹⁾**N.D.E.** Namely, Theorems 5.1, 5.3, 6.1, 6.2 and 7.2 of TDTE III. The first two (resp. the next two) correspond to Theorem 4.1 (resp. Theorems 7.1 and 8.1) of this exposé. Theorem 7.2 of TDTE III is proved in Exp. VIA, 3.2 and 3.3.

⁽²⁾**N.D.E.** That is, to groupoids with base X , cf. the terminology at the end of section 1. When $\mathcal{C}$ is the category of schemes, the quotient $p : X \rightarrow Y$ of a groupoid X_* with base X exists under certain hypotheses (cf. 4.1, 6.1, 7.1); if, moreover, X_* is an equivalence relation, p is, under the same hypotheses, faithfully flat and quasi-compact, hence a universal epimorphism, cf. loc. cit.

quotient topology. The canonical map $\pi : X_0 \rightarrow Y$ and d_0, d_1 then induce a pair of arrows of sheaves of rings on Y :

$$\pi_*(\mathcal{O}_0) \rightrightarrows^{\delta_0}_{\delta_1} \pi_*(d_{0*}\mathcal{O}_1) = \pi_*(d_{1*}\mathcal{O}_1),$$

where $\mathcal{O}_i$ is the structure sheaf of X_i . For the sheaf of rings on Y one chooses the subsheaf of $\pi_*(\mathcal{O}_0)$ whose sections s satisfy $\delta_0(s) = \delta_1(s)$. The arrow p is defined in the obvious way.

⁽³⁾Let $X_1 \rightrightarrows X_0$, with arrows d_1, d_0 , be a diagram in (Esp.An), and let (Y, p) be its cokernel. An open subset U of X_0 is said to be saturated if $d_0^{-1}(U) = d_1^{-1}(U)$, which is equivalent to saying that $U = p^{-1}(p(U))$. In this case, since Y has the quotient topology, $p(U)$ is an open subset of Y .

Lemma 1.1. *Let U be a saturated open subset of X and $V = p(U)$. If one denotes by U_1 the open subset $d_0^{-1}(U) = d_1^{-1}(U)$ of X_1 , and by $\tilde{d}_0, \tilde{d}_1$ and $\tilde{p}$ the restrictions of d_0, d_1 to U_1 , and of p to U , then $(V, \tilde{p})$ is a cokernel in (Esp.An) of:⁽⁴⁾*

$$U_1 \rightrightarrows^{\tilde{d}_1}_{\tilde{d}_0} U \xrightarrow{\tilde{p}} V.$$

The verification is easy.

Lemma 1.2. *Let $X_1 \rightrightarrows X_0$, with arrows d_1, d_0 , be a diagram in (Sch) and let (Y, p) be its cokernel in (Esp.An).*

- (i) *If Y is a scheme and p a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in (Sch).*
- (ii) *Suppose every point of X_0 has a saturated open neighborhood U such that, denoting by $\tilde{d}_0$ and $\tilde{d}_1$ the restrictions of d_0 and d_1 to $d_0^{-1}(U) = d_1^{-1}(U)$, and by (Q, q) the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp.An), Q is a scheme and q a morphism of schemes. Then (Y, p) is a cokernel of (d_0, d_1) in (Sch).*

(i) is proved in §4.c; since the proof is short, let us repeat it here. Let $f : X_0 \rightarrow Z$ be a morphism of schemes such that $fd_0 = fd_1$. By hypothesis there is one and only one morphism of ringed spaces $r : Y \rightarrow Z$ such that $f = rp$. One must show that, for every $y \in Y$, the homomorphism $\mathcal{O}_{r(y)} \rightarrow \mathcal{O}_y$ induced by r is local. This follows from the fact that p is surjective, hence y has the form $p(x)$, and that the homomorphism $\mathcal{O}_{f(x)} \rightarrow \mathcal{O}_x$ induced by f is local.

(ii) follows from (i) and the preceding lemma.

c) In this exposé we study the existence of $\text{Coker}(d_0, d_1)$ when the pair of arrows (d_0, d_1) is placed in a richer context; precisely, denote by $X_2 = X_1 \times_{d_1, d_0} X_1$ the fibered product of the diagram

$$(*) \quad \begin{array}{ccc} & X_1 & \\ & \downarrow d_1 & \\ X_1 & \xrightarrow{d_0} & X_0, \end{array}$$

⁽³⁾**N.D.E.** Lemmas 1.1 and 1.2, used several times in sections 5 to 9, have been added.

⁽⁴⁾**N.D.E.** This is not the case in the category of schemes. Let, for example, $S = \text{Spec}(\mathbf{C})$, $X_0 = \mathbb{A}_S^2 = \text{Spec}(\mathbf{C}[x_1, x_2])$, $d_1 : \mathbf{G}_{m, S} \times_S \mathbb{A}_S^2 \rightarrow \mathbb{A}_S^2$ the action of $\mathbf{G}_{m, S}$ by homotheties on $\mathbb{A}_S^2$, d_0 the projection onto the second factor, and $U = \mathbb{A}_S^2 - \{m\}$, where m is the point $(0, 0)$. Then the projective space $\mathbb{P}_S^1$ is the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp.An) and in (Sch), and the cokernel Y of (d_0, d_1) in (Esp.An) is the union of $\mathbb{P}_S^1$ and the point $y_0 = \{p(m)\}$; the only open subset containing y_0 is Y , and one has $\Gamma(Y, \mathcal{O}_Y) = \mathbf{C}$. If $f : \mathbb{A}_S^2 \rightarrow T$ is a morphism of S -schemes such that $fd_0 = fd_1$, and if $V = \text{Spec}(A)$ is an affine open subset of T containing the point $t_0 = f(y_0)$, then $f^{-1}(V) = \mathbb{A}^2$ and the ring homomorphism $A \rightarrow \mathbf{C}[x_1, x_2]$ factors through $\mathbf{C}$; this shows that $S = \text{Spec}(\mathbf{C})$ is the cokernel of (d_0, d_1) in the category (Sch/ S).

and by d'_0 and d'_2 the two canonical projections of X_2 onto X_1 ; one therefore has, by definition, a cartesian square

$$(0) \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_0} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_0} & X_0. \end{array}$$

Moreover, give oneself a third arrow $d'_1 : X_2 \rightarrow X_1$; we say that $(d_0, d_1 : X_1 \rightrightarrows X_0, d'_1)$ is a $\mathcal{C}$ -groupoid if for every object T of $\mathcal{C}$, $X_1(T)$ is the set of arrows of a groupoid $X_*(T)$ whose set of objects is $X_0(T)$, whose source map is $d_1(T)$, whose target map is $d_0(T)$, and whose composition map is $d'_1(T)$ (as usual, one identifies $(X_1 \times_{d_1, d_0} X_1)(T)$ with $X_1(T) \times_{d_1(T), d_0(T)} X_1(T)$; one also recalls that a groupoid is a category all of whose arrows are invertible).⁽⁵⁾

If φ is an arrow of the groupoid $X_*(T)$, the map $f \mapsto \varphi \circ f$ is a bijection from the set of arrows f whose target coincides with the source of φ onto the set of arrows having the same target as φ . One easily sees that this fact can be expressed by saying that the square

$$(1) \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ d'_0 \downarrow & & \downarrow d_0 \\ X_1 & \xrightarrow{d_0} & X_0 \end{array}$$

is cartesian.

Similarly, the map $g \mapsto g \circ \varphi$ is a bijection from the set of arrows g of $X_*(T)$ whose source is the target of φ onto the set of arrows having the same source as φ . One can again express this fact by saying that the square

$$(2) \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_1} & X_0 \end{array}$$

is cartesian.

On the other hand, let $s : X_0 \rightarrow X_1$ be the unique arrow of $\mathcal{C}$ such that, for every T , $s(T) : X_0(T) \rightarrow X_1(T)$ assigns to every object of $X_*(T)$ its identity arrow.⁽⁶⁾ The arrow s satisfies the equalities

$$(3) \quad d_1 s = \text{id}_{X_0},$$

and

$$(3 \text{ bis}) \quad d_0 s = \text{id}_{X_0}.$$

Finally, associativity of the composition maps $d'_1(T)$ is expressed by commutativity of the diagram

$$(4) \quad \begin{array}{ccc} X_1 \times_{d_1, d_0} X_1 \times_{d_1, d_0} X_1 & \xrightarrow{d'_1 \times X_1} & X_1 \times_{d_1, d_0} X_1 \\ X_1 \times d'_1 \downarrow & & \downarrow d'_1 \\ X_1 \times_{d_1, d_0} X_1 & \xrightarrow{d'_1} & X_1. \end{array}$$

⁽⁵⁾**N.D.E.** Thus, in this case, $X_2(T)$ is the set of pairs (f_2, f_1) of composable arrows, that is, such that $d_0(f_1) = d_1(f_2)$, and d'_0, d'_1, d'_2 send (f_2, f_1) to $f_2, f_2 \circ f_1, f_1$ respectively.

⁽⁶⁾**N.D.E.** $T \mapsto s(T)$ defines an element of $\text{Hom}(h_{X_0}, h_{X_1})$, and the latter equals $\text{Hom}(X_0, X_1)$ by Yoneda's lemma.

Conversely, conditions (1), (2) and (4) and the existence of an arrow s satisfying (3) imply that $(X_1 \rightrightarrows X_0, d'_1)$, with arrows d_1, d_0 , is a $\mathcal{C}$ -groupoid. Condition (3) is harmless; it simply ensures that the map $d_1(T) : X_1(T) \rightarrow X_0(T)$ is surjective for every $T \in \mathcal{C}$. In the remainder of this exposé we chiefly use the cartesian squares (0), (1) and (2), which we summarize in the diagram

$$(0,1,2) \quad \begin{array}{ccccc} X_2 & \xrightleftharpoons[d'_0]{d'_1} & X_1 & \xrightarrow{d_0} & X_0 \\ d'_2 \downarrow & & \downarrow d_1 & & \\ X_1 & \xrightleftharpoons[d_0]{d_1} & X_0 & & \end{array} .$$

In this diagram the two squares on the left (that is, squares (0) and (2)) are cartesian; the first row is exact and X_2 is identified with the fibered product $X_1 \times_{d_0, d_0} X_1$.

We use associativity only indirectly, for example to ensure the existence of an arrow s satisfying (3) and (3 bis), or to ensure the existence of an arrow

$$(\dagger) \quad \sigma : X_1 \rightarrow X_1 \quad \text{such that} \quad d_0 \sigma = d_1 \text{ and } d_1 \sigma = d_0$$

(one chooses σ so that $\sigma(T) : X_1(T) \rightarrow X_1(T)$ sends each arrow of $X_*(T)$ to its inverse arrow).⁽⁷⁾

By abuse of language, we shall sometimes call a diagram

$$X_2 \xrightleftharpoons[d_0]{d'_0, d'_1, d'_2} X_1 \xrightleftharpoons[d_0]{d_0, d_1} X_0$$

a $\mathcal{C}$ -groupoid if (0), (1) and (2) are cartesian, (4) is commutative and there exists s satisfying (3). The object X_2 may thus be “a” fibered product of $(*)$ without being “the” fibered product of $(*)$.⁽⁸⁾

Terminology. Instead of the $\mathcal{C}$ -groupoid X_* , we shall also speak of the groupoid X_* with base X_0 , or of the pre-equivalence relation X_* in X_0 .

2. Examples of $\mathcal{C}$ -groupoids

a) Let X be an object of $\mathcal{C}$ and G a $\mathcal{C}$ -group acting on X on the left. We denote by $d_0 : G \times X \rightarrow X$ the arrow defining the action of G on X , by $d_1 : G \times X \rightarrow X$ the projection of the product onto the second factor, by $\mu : G \times G \rightarrow G$ the arrow defining the $\mathcal{C}$ -group structure of G , and finally by $\text{pr}_{2,3}$ the projection of $G \times G \times X = G \times (G \times X)$ onto the second factor. Then

$$G \times G \times X \xrightarrow[\begin{smallmatrix} \xrightarrow{\mu \times X} \\ \xrightarrow{G \times d_0} \end{smallmatrix}]{\text{pr}_{2,3}} G \times X \xrightleftharpoons[d_0]{d_1} X$$

is a $\mathcal{C}$ -groupoid.

b) Let $d_0, d_1 : X_1 \rightarrow X_0$ be an equivalence pair, that is, if $d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times X_0$ is the arrow with components d_0 and d_1 , we suppose that $(d_0 \boxtimes d_1)(T)$ is, for every object T of $\mathcal{C}$, a bijection from $X_1(T)$ onto the graph of an equivalence relation on $X_0(T)$. The set $X_1(T)$ is consequently identified with the set of pairs (x, y) of elements of $X_0(T)$ such that $x \sim y$; similarly, the set $X_2(T) = (X_1 \times_{d_1, d_0} X_1)(T)$ is identified with the set of triples (x, y, z) of elements of $X_0(T)$ such that $x \sim y$ and $y \sim z$. There is therefore one and only one arrow $d'_1 : X_2 \rightarrow X_1$ making squares (1) and (2) commutative: $d'_1(T)$ must send $(x, y, z) \in X_2(T)$ to $(x, z) \in X_1(T)$. For this choice of d'_1 , $(d_0, d_1 : X_1 \rightrightarrows X_0, d'_1)$ is a $\mathcal{C}$ -groupoid.

⁽⁷⁾**N.D.E.** It follows from Yoneda’s lemma that σ is an involutive automorphism of X_1 ; this will be used, for example, in 3.e) and in Theorem 4.1.

⁽⁸⁾**N.D.E.** See example 2.a) below.

Conversely, consider a $\mathcal{C}$ -groupoid X_* such that $d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times X_0$ is a monomorphism. Then (d_0, d_1) is an equivalence pair, and X_* can be reconstructed from (d_0, d_1) as explained a few lines above.⁽⁹⁾

c) If $p : X \longrightarrow Y$ is any arrow of $\mathcal{C}$, and if pr_1 and pr_2 are the two projections of $X \times_{p,p} X$ onto X , then $(\text{pr}_1, \text{pr}_2) : X \times_{p,p} X \rightrightarrows X$ is an equivalence pair. One says that p is an effective epimorphism if the diagram

$$X \times_{p,p} X \begin{array}{c} \xrightarrow{\text{pr}_1} \\ \xrightarrow{\text{pr}_2} \end{array} X \xrightarrow{p} Y$$

is exact, that is, if $(Y, p) = \text{Coker}(\text{pr}_1, \text{pr}_2)$.

Let, for example, S be a noetherian scheme and $\mathcal{C}$ the category of schemes finite over S . Let us show that an epimorphism of $\mathcal{C}$ is not necessarily effective: one chooses $S = \text{Spec } k[T^3, T^5]$, where k is a commutative field, $Y = S$ and $X = \text{Spec } k[T]$. If i is the inclusion of $B = k[T^3, T^5]$ into $A = k[T]$, p is chosen to be $\text{Spec } i$. In this case $X \times_{p,p} X$ is identified with $\text{Spec}(A \otimes_B A)$ and $\text{Coker}(\text{pr}_1, \text{pr}_2)$ with $\text{Spec } B'$, where B' is the subring of A consisting of the a such that $a \otimes_B 1 = 1 \otimes_B a$. Now

$$T^7 \otimes_B 1 = (T^2 T^5) \otimes_B 1 = T^2 \otimes_B T^5 = T^2 \otimes_B (T^3 T^2) = T^5 \otimes_B T^2 = 1 \otimes_B T^7.$$

Thus T^7 belongs to B' , does not belong to B , and $\text{Spec } B'$ is distinct from $\text{Spec } B$, whence the counterexample.⁽¹⁰⁾

3. Some sorites on $\mathcal{C}$ -groupoids

Here, in no particular order, are some remarks used below:

a) Let

$$X_2 \begin{array}{c} \xrightarrow{d'_0, d'_1, d'_2} \\ \xrightarrow{d_0, d_1} \end{array} X_1 \xrightarrow{d_0, d_1} X_0$$

be a $\mathcal{C}$ -groupoid and $f_0 : Y_0 \longrightarrow X_0$ an arrow of $\mathcal{C}$. We shall define a $\mathcal{C}$ -groupoid with base Y_0

$$Y_2 \begin{array}{c} \xrightarrow{e'_0, e'_1, e'_2} \\ \xrightarrow{e_0, e_1} \end{array} Y_1 \xrightarrow{e_0, e_1} Y_0$$

which will be said to be induced by X_* and f_0 . One will also say that Y_* is the inverse image of X_* by the base change f_0 .

For Y_1 we choose the fibered product of the diagram

$$\begin{array}{ccc} Y_1 & \xrightarrow{f_1} & X_1 \\ \downarrow & & \downarrow d_0 \boxtimes d_1 \\ Y_0 \times Y_0 & \xrightarrow{f_0 \times f_0} & X_0 \times X_0, \end{array}$$

and for e_0 and e_1 the arrows obtained by composing the canonical arrow $Y_1 \longrightarrow Y_0 \times Y_0$ with the first and second projections of $Y_0 \times Y_0$. The morphism $Y_1 \longrightarrow Y_0 \times Y_0$ is then $e_0 \boxtimes e_1$, and one has $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, where f_1 denotes the projection of Y_1 onto X_1 .

One puts $Y_2 = Y_1 \times_{e_0, e_1} Y_1$, cf. 1.c). One can say that the pair (e_0, e_1) is defined so that, for every $T \in \mathcal{C}$, and for every pair (y, x) of elements of $Y_0(T)$, there is a certain one-to-one correspondence $\psi \longmapsto {}_y\psi_x$ between the arrows ψ of $X_*(T)$ with source $f_0(x)$ and target $f_0(y)$, and the arrows ${}_y\psi_x$ of $Y_*(T)$ with source x and target y . One therefore determines $e'_1 : Y_2 \longrightarrow Y_1$ by defining, for every $T \in \mathcal{C}$, the composition of arrows of $Y_*(T)$ by the formula

$$z\psi_y \circ {}_y\varphi_x = z(\psi \circ \varphi)_x.$$

⁽⁹⁾**N.D.E.** In particular, if G is a $\mathcal{C}$ -group acting on the left on an object X of $\mathcal{C}$ and if X_* is the $\mathcal{C}$ -groupoid defined in a), then (d_0, d_1) is an equivalence pair if and only if G acts freely on X , cf. Exp. III, 3.2.1.

⁽¹⁰⁾**N.D.E.** The same argument applies to $B = k[T^3, T^4]$ and $T^5 \otimes_B 1$; more generally, to $B = k[T^n, T^{n+r}]$ and the element $T^{n+2r} \otimes_B 1$, provided n does not divide $2r$.

It is clear that this definition makes each $Y_*(T)$ a groupoid.

b) Knowing the $\mathcal{C}$ -groupoid X_* and the base change $f_0 : Y_0 \rightarrow X_0$, one can reconstruct the pair $(e_0, e_1) : Y_1 \rightrightarrows Y_0$ in another way:⁽¹¹⁾ construct $Y_0 \times_{X_0} X_1$, pr_1 and pr_2 so that the square

$$\begin{array}{ccc} Y_0 \times_{X_0} X_1 & \xrightarrow{\text{pr}_2} & X_1 \\ \text{pr}_1 \downarrow & & \downarrow d_0 \\ Y_0 & \xrightarrow{f_0} & X_0 \end{array}$$

is cartesian. One then verifies without difficulty, by reduction to the set-theoretic case, that one has the cartesian square:

$$\begin{array}{ccc} Y_1 & \xrightarrow{e_0 \boxtimes f_1} & Y_0 \times_{X_0} X_1 \\ e_1 \downarrow & & \downarrow d_1 \circ \text{pr}_2 \\ Y_0 & \xrightarrow{f_0} & X_0, \end{array}$$

where f_1 denotes the canonical projection of $Y_1 = (Y_0 \times Y_0) \times_{(X_0 \times X_0)} X_1$ onto X_1 .

c) We shall give two examples of inverse images of a $\mathcal{C}$ -groupoid. Take $Y_0 = X_1$, $f_0 = d_0$. For every object T of $\mathcal{C}$, $Y_1(T)$ is then identified with the set of diagrams of the form

$$\begin{array}{ccc} b & \xrightarrow{\varphi} & d \\ f \uparrow & & \uparrow g \\ a & & c \end{array}$$

of $X_*(T)$. The source of such a diagram is the arrow f , the target is the arrow g . These diagrams compose in the obvious way.

Now put $Y'_0 = X_1$, $f'_0 = d_1$ (we add primes⁽¹²⁾ to avoid any confusion with the preceding example). In this case, for every $T \in \mathcal{C}$, $Y'_1(T)$ is identified with the set of diagrams of the form

$$\begin{array}{ccc} b & & d \\ f \uparrow & & \uparrow g \\ a & \xrightarrow{\psi} & c \end{array}$$

of the groupoid $X_*(T)$. The source of such a diagram is f , the target is g ; the composition of these diagrams is obvious.

This being said, it is clear that the identity map of $Y_0(T)$ and the map

$$\begin{array}{ccc} b & \xrightarrow{\varphi} & d \\ f \uparrow & & \uparrow g \\ a & & c \end{array} \quad \mapsto \quad \begin{array}{ccc} b & & d \\ f \uparrow & & \uparrow g \\ a & \xrightarrow{g^{-1}\varphi f} & c \end{array}$$

from $Y_1(T)$ onto $Y'_1(T)$ define an isomorphism of the groupoid $Y_*(T)$ onto $Y'_*(T)$. Moreover, this isomorphism depends functorially on T , so that the $\mathcal{C}$ -groupoids Y_* and Y'_* are isomorphic.⁽¹³⁾

d)

⁽¹¹⁾**N.D.E.** This second viewpoint will be used in 3.f) and in the proof of 6.1.

⁽¹²⁾**N.D.E.** “Accents” in the original.

⁽¹³⁾**N.D.E.** This will play a crucial role in the proof of Lemma 6.1.

Proposition 3.1. *We retain the notation of a), and suppose that f_0 is an effective and universal epimorphism. Then $\text{Coker}(d_0, d_1)$ exists if and only if $\text{Coker}(e_0, e_1)$ exists.⁽¹⁴⁾ Moreover, in this case f_0 induces an isomorphism*

$$\text{Coker}(d_0, d_1) \xrightarrow{\sim} \text{Coker}(e_0, e_1).$$

Let us first recall that an epimorphism $f_0 : Y_0 \rightarrow X_0$ is said to be universal if, for every cartesian square

$$\begin{array}{ccc} Y' & \longrightarrow & Y_0 \\ f' \downarrow & & \downarrow f_0 \\ X' & \longrightarrow & X_0, \end{array}$$

f' is an epimorphism. This being said, denote by $C(d_0, d_1)$ the covariant functor from $\mathcal{C}$ to sets which assigns to every $T \in \mathcal{C}$ the kernel of the pair $T(d_0), T(d_1) : T(X_0) \rightrightarrows T(X_1)$. Define $C(e_0, e_1)$ similarly. For every $T \in \mathcal{C}$, one therefore has a commutative diagram

$$\begin{array}{ccccc} C(d_0, d_1)(T) & \longrightarrow & T(X_0) & \xrightleftharpoons[T(d_0)]{T(d_1)} & T(X_1) \\ T(f) \downarrow & & T(f_0) \downarrow & & \downarrow T(f_1) \\ C(e_0, e_1)(T) & \longrightarrow & T(Y_0) & \xrightleftharpoons[T(e_0)]{T(e_1)} & T(Y_1), \end{array}$$

where $T(f)$ is the injection induced by the injection $T(f_0)$. If we show that $T(f)$ is a surjection for every T , we shall have a functorial isomorphism $f : C(d_0, d_1) \xrightarrow{\sim} C(e_0, e_1)$, so that the representability of one of these functors is equivalent to that of the other; this will prove our proposition.

To prove surjectivity of $T(f)$, consider the diagram

$$\begin{array}{ccccc} & & Y_1 & \xrightarrow{f_1} & X_1 \\ & \nearrow \Delta & \downarrow e_1 & \downarrow e_0 & \downarrow d_1 \\ Y_0 \times_{X_0} Y_0 & \xrightarrow{\text{pr}_2} & Y_0 & \xrightarrow{f_0} & X_0, \\ & \searrow \text{pr}_1 & & & \downarrow d_0 \end{array}$$

where Δ is the section of $Y_1 \rightarrow Y_0 \times Y_0$ defined by the morphism $s \circ f_0 \circ \text{pr}_1 : Y_0 \times Y_0 \rightarrow X_1$, the arrow $s : X_0 \rightarrow X_1$ satisfying equalities (3) and (3 bis) of paragraph 1.

If the arrow $g : Y_0 \rightarrow T$ satisfies $g \circ e_0 = g \circ e_1$, one has $g \circ e_0 \circ \Delta = g \circ e_1 \circ \Delta$, hence $g \circ \text{pr}_1 = g \circ \text{pr}_2$. Since f_0 is an effective epimorphism, g is the composite of f_0 with an arrow $h : X_0 \rightarrow T$, that is, one has $g = T(f_0)(h)$. It remains to show that h belongs to $C(d_0, d_1)(T)$, that is, satisfies $hd_0 = hd_1$; now one has

$$hd_0f_1 = hf_0e_0 = ge_0 = ge_1 = hf_0e_1 = hd_1f_1,$$

whence the desired equality, since f_1 is an epimorphism (for f_0 is a universal epimorphism).

e) Now consider a scheme S and choose $\mathcal{C} = (\text{Sch}/S)$. Giving a $\mathcal{C}$ -groupoid

$$X_2 \xrightleftharpoons[d'_0]{d'_1} X_1 \xrightleftharpoons[d_0]{d_1} X_0$$

allows one to define an equivalence relation on the underlying set $\underline{X}_0$ of the scheme X_0 : if $x, y \in \underline{X}_0$, one writes $x \sim y$ when there exists $z \in \underline{X}_1$ such that $x = d_1z$ and $y = d_0z$. Reflexivity and symmetry of this relation are obvious;⁽¹⁵⁾ let us prove transitivity: if $x \sim y$

⁽¹⁴⁾**N.D.E.** The original has been modified to bring out the isomorphism below.

⁽¹⁵⁾**N.D.E.** Reflexivity follows from the existence of $s : X_0 \rightarrow X_1$, which is a section of d_0 and d_1 ; symmetry follows from the existence of the involution σ of X_1 which “interchanges d_0 and d_1 ”, that is, satisfies $d_0\sigma = d_1$ and $d_1\sigma = d_0$, cf. §1, (3), (3 bis) and (†).

and $y \sim z$, there exist $u, v \in \underline{X}_1$ such that $x = d_1 u$, $y = d_0 u$, $y = d_1 v$, $z = d_0 v$. It follows that (v, u) belongs to the set-theoretic fibered product $\underline{X}_1 \times_{d_1, d_0} \underline{X}_1$. Since the canonical map

$$\underline{X}_1 \times_{d_1, d_0} \underline{X}_1 \longrightarrow \underline{X}_1 \times_{d_1, d_0} \underline{X}_1$$

from the underlying set of the fibered product to the fibered product of the underlying sets is surjective, (v, u) is the image of some $w \in \underline{X}_2$. One then has $x = d_1 d'_1 w$ and $z = d_0 d'_1 w$, whence $x \sim z$.

f) Retain the notation of a) and b), $\mathcal{C}$ still being (Sch/S) . If x, y are points of Y_0 , we shall see that $x \sim y$ if and only if $f_0(x) \sim f_0(y)$ (the inverse image of the equivalence relation defined by a groupoid is the equivalence relation defined by the inverse image of the groupoid).

Indeed, suppose $x \sim y$. There thus exists $z \in \underline{Y}_1$ such that $x = e_1(z)$ and $y = e_0(z)$. Since $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, one then has $f_0(x) = d_1 f_1(z)$ and $f_0(y) = d_0 f_1(z)$, whence $f_0(x) \sim f_0(y)$.

Conversely, suppose $f_0(x) \sim f_0(y)$, and let $z \in \underline{X}_1$ satisfy $f_0(y) = d_1(z)$ and $f_0(x) = d_0(z)$. Using the construction and notation of b), there is then a point t of $Y_0 \times_{X_0} X_1$ such that $\text{pr}_1(t) = x$ and $\text{pr}_2(t) = z$. Similarly, since $f_0(y) = d_1 \text{pr}_2(t)$, there is an $s \in \underline{Y}_1$ such that $y = e_1(s)$ and $(e_0 \boxtimes f_1)(s) = t$. One then has $e_0(s) = \text{pr}_1(e_0 \boxtimes f_1)(s) = \text{pr}_1(t) = x$. Whence $x \sim y$.

4. Passage to the quotient by a finite flat groupoid (proof of a special case)

Theorem 4.1. *Consider a (Sch/S) -groupoid:*

$$X_2 \begin{array}{c} \xrightarrow{d'_1} \\ \xrightarrow{d'_0} \end{array} X_1 \xrightarrow{d_0} X_0$$

Suppose the following conditions hold.⁽¹⁶⁾

- a) d_1 is finite locally free,
- b) for every $x \in X_0$, the set $d_0 d_1^{-1}(x)$ is contained in an affine open subset of X_0 .⁽¹⁷⁾

Then:

- (i) *There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S) ; moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*
- (ii) *p is integral and open, and Y is affine if X_0 is affine.*⁽¹⁸⁾
- (iii) *The morphism $X_1 \longrightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*
- (iv) *If (d_0, d_1) is an equivalence pair, $X_1 \longrightarrow X_0 \times_Y X_0$ is an isomorphism⁽¹⁹⁾ and $p : X_0 \longrightarrow Y$ is finite locally free.⁽²⁰⁾ Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology and, for every base change $Y' \longrightarrow Y$, Y' is the cokernel of the groupoid $X_* \times_Y Y'$ obtained from X_* by the base change $X_0 \times_Y Y' \longrightarrow X_0$.*

In particular, for every base change $S' \longrightarrow S$, $Y' = Y \times_S S'$ is the cokernel of the S' -groupoid $X'_ = X_* \times_S S'$. Thus, in this case, “formation of the quotient commutes with base change”.*

⁽¹⁶⁾**N.D.E.** Since $d_0 = d_1 \sigma$, where σ is an involutive automorphism of X_1 , these two conditions are symmetric in d_1 and d_0 ; moreover one has $d_0 d_1^{-1}(x) = d_1 d_0^{-1}(x)$.

⁽¹⁷⁾**N.D.E.** Hypothesis b) cannot be omitted. Indeed, H. Hironaka gave an example of an action of the finite group $G = \mathbf{Z}/2\mathbf{Z}$ on a proper $\mathbf{C}$ -variety X_0 , such that the quotient X_0/G is an algebraic space which is not a scheme ([Hi62], see also [Mum65], Chap. 4, §3).

⁽¹⁸⁾**N.D.E.** The assertion that p is open has been added, using the analogous proof given in 6.1.

⁽¹⁹⁾**N.D.E.** Note that, in this case, $X_1 \longrightarrow X_0 \times_S X_0$ is therefore an immersion (EGA I, 5.3.10); see also VIB, 9.2.1. On the other hand, for the existence of the quotient (in the category of schemes or that of algebraic spaces) under the weaker hypothesis that d_0 and d_1 are finite (but not necessarily flat), see [An73], §1.1, [Fe03], [Ko08]. . .

⁽²⁰⁾**N.D.E.** The following consequences have been made explicit; see [Ray67a], th. 1 (iii) and the proof given below, at the end of section 5.

It follows obviously from (i) that the underlying topological space of Y is the quotient of the underlying topological space of X_0 by the equivalence relation defined by the (Sch/S) -groupoid X_* .

We shall first prove this theorem when X_0 is affine and d_1 is locally free of constant rank n . We shall then see how one can reduce to this special case.

In the case in which we have placed ourselves, X_0, X_1 and X_2 are affine. We can therefore suppose that one has

$$X_i = \text{Spec } A_i, \quad d_j = \text{Spec } \delta_j \quad \text{and} \quad d'_k = \text{Spec } \delta'_k,$$

the A_i being commutative rings and the δ_j, δ'_k ring homomorphisms. One can then replace diagram $(0,1,2)$ by the following

$$((0,1,2)^*) \quad \begin{array}{ccccc} & & \delta'_0 & & \\ & & \swarrow & \searrow & \\ A_2 & \xleftarrow{\delta'_0} & A_1 & \xleftarrow{\delta_0} & A_0 \\ \delta'_2 \uparrow & & \delta'_1 \uparrow & & \delta_1 \uparrow \\ & & \delta_0 & & \\ A_1 & \xleftarrow{\delta_0} & A_0 & & \\ & & \delta_1 & & \end{array}$$

where the two squares on the left are cocartesian.

Denote by B the subring of A_0 consisting of the $a \in A_0$ such that $\delta_0(a) = \delta_1(a)$.

a) Let us show that A_0 is integral over B . If a belongs to A_0 , let

$$P_{\delta_1}(T, \delta_0(a)) = T^n - \sigma_1 T^{n-1} + \cdots + (-1)^n \sigma_n$$

be the characteristic polynomial of $\delta_0(a)$ when one regards A_1 as an algebra over A_0 by means of the homomorphism δ_1 (cf. Bourbaki, Alg. VIII, §12 and Alg. comm. II, §5, exercise 9). Since the squares on the left of $(0,1,2)^*$ are cocartesian, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_0 \delta_0(a))$$

and

$$\delta_1(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_1 \delta_0(a)).$$

Since $\delta'_0 \delta_0 = \delta'_1 \delta_0$, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = \delta_1(P_{\delta_1}(T, \delta_0(a))),$$

that is, $\delta_0(\sigma_i) = \delta_1(\sigma_i)$ for every i . On the other hand, Hamilton–Cayley tells us that one has

$$\delta_0(a)^n - \delta_1(\sigma_1) \delta_0(a)^{n-1} + \cdots + (-1)^n \delta_1(\sigma_n) = 0.$$

Since $\delta_1(\sigma_i) = \delta_0(\sigma_i)$, one also has

$$\delta_0(a)^n - \delta_0(\sigma_1) \delta_0(a)^{n-1} + \cdots + (-1)^n \delta_0(\sigma_n) = 0,$$

whence

$$a^n - \sigma_1 a^{n-1} + \cdots + (-1)^n \sigma_n = 0,$$

for there exists a homomorphism $\tau : A_1 \rightarrow A_0$ such that $\tau \delta_0 = \text{id}_{A_0}$, so δ_0 is injective. It follows that A_0 is integral over B .

b) Now consider two prime ideals x and y of A_0 . We shall show that the equality $x \cap B = y \cap B$ implies the existence of a prime ideal z of A_1 such that $x = d_0(z)$ and $y = d_1(z)$.

Indeed, if the assertion were not true, x would be distinct from $\delta_0^{-1}(t)$ for every prime ideal t of A_1 such that $\delta_1^{-1}(t) = y$. For such a t one would have $\delta_0^{-1}(t) \cap B = \delta_1^{-1}(t) \cap B = y \cap B = x \cap B$, whence, by Cohen–Seidenberg (cf. Bourbaki, Alg. comm. V, §2, cor. 1 of prop. 1), it would follow that x would not be contained in any $\delta_0^{-1}(t)$.⁽²¹⁾ Now there are at most n prime ideals t of A_1 such that $\delta_1^{-1}(t) = y$ (cf. loc. cit., prop. 3), so, by the “Prime ideal avoidance lemma” (loc. cit., II, §1, prop. 3), there would be an $a \in x$ belonging to none of the $\delta_0^{-1}(t)$.

⁽²¹⁾**N.D.E.** The original has been expanded in what follows; in particular, Lemma 4.1.1, taken from [DG70], III, §2.4, Lemma 4.3, has been added.

Consequently, $\delta_0(a)$ would belong to none of these ideals t , and therefore, by the lemma below, the norm $N_{\delta_1}(\delta_0(a))$ would not belong to y (one calculates this norm by regarding A_1 as an algebra over A_0 by means of the homomorphism δ_1 ; one has $N_{\delta_1}(\delta_0(a)) = \sigma_n$ with the notation of a)). Now, since $(-1)^{n-1}\sigma_n = a^n + \sum_{i=1}^{n-1} (-1)^i \sigma_i a^{n-i}$, this norm belongs to $B \cap x = B \cap y$, whence the contradiction.

Lemma 4.1.1. *Let $A \rightarrow A'$ be a morphism of commutative rings making A' a projective A -module of rank n . Let $\mathfrak{p} \in \operatorname{Spec}(A)$, let $\mathfrak{q}_1, \dots, \mathfrak{q}_r$ be the elements of $\operatorname{Spec}(A')$ above $\mathfrak{p}$, and let $a \in A'$. Then a belongs to $\mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_r$ if and only if its norm $N(a)$ belongs to $\mathfrak{p}$.*

Indeed, replacing A and A' by the localizations $A_{\mathfrak{p}}$ and $A'_{\mathfrak{p}}$, one reduces to the case where $(A, \mathfrak{p})$ is local and A' semilocal, with $\operatorname{Spec}(A') = \{\mathfrak{q}_1, \dots, \mathfrak{q}_r\}$. In this case, A' is a free A -module of rank n (cf. Bourbaki, Alg. comm. II, §3.2, cor. 2 of prop. 5), and $N(a)$ is the determinant of the endomorphism $\ell_a : a' \mapsto aa'$ of A' ; one therefore has the following equivalences:

$$N(a) \notin \mathfrak{p} \iff N(a) \text{ invertible} \iff \ell_a \text{ invertible} \iff a \notin \mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_r.$$

c) Proof of (i):

Put $Y = \operatorname{Spec} B$ and $p = \operatorname{Spec} i$, where i is the inclusion of B into A_0 . By a), the morphism $p : X_0 \rightarrow Y$ is surjective. We shall first show that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces: indeed, it follows from b) that the underlying set of $\operatorname{Spec} B$ is obtained from the underlying set of X_0 by identifying the points x and y such that there exists $z \in X_1$ with $d_1 z = y$, $d_0 z = x$. Moreover, since i is integral, $p = \operatorname{Spec} i$ is closed, so that Y has the quotient topology of that of X_0 . It follows that p is open. Indeed, let U' be any open subset of X_0 ; since d_1 is surjective and finite locally free, hence faithfully flat and of finite presentation, and therefore open, the saturation $U = d_1(d_0^{-1}(U'))$ of U' for the equivalence relation defined by X_* is open. Then $p(U') = p(U)$ is open, since Y has the quotient topology.

Finally, it follows from the choice of B and from the fact that p, d_0 and d_1 are affine that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_{X_0}) \rightrightarrows_{p_*(\delta_0)}^{p_*(\delta_1)} p_*(d_{0*}(\mathcal{O}_{X_1})) = p_*(d_{1*}(\mathcal{O}_{X_1}))$$

is exact.

It remains to show that (Y, p) is also the cokernel of (d_0, d_1) in the category of schemes (more generally in that of spaces ringed in local rings). Let, then, $q : X_0 \rightarrow Z$ be a morphism of schemes such that $qd_0 = qd_1$. By what precedes, there is one and only one morphism of ringed spaces $r : Y \rightarrow Z$ such that $q = rp$. One must show that, for every $y \in Y$, the homomorphism $\mathcal{O}_{r(y)} \rightarrow \mathcal{O}_y$ induced by r is local. This follows from the fact that p is surjective, hence y has the form $p(x)$, and that the homomorphism $\mathcal{O}_{q(x)} \rightarrow \mathcal{O}_x$ induced by q is local.

d) Proof of (ii): follows from a) and c).

e) Proof of (iii):

Recall that $\underline{P}$ denotes the underlying set of a scheme P , and $\underline{d} : \underline{P} \rightarrow \underline{Q}$ the map induced by a morphism $d : P \rightarrow Q$.

Lemma 4.1.2. ⁽²²⁾ *Let $(A, \mathfrak{m})$ be a local ring, k its residue field, and K an extension of the field k . Then there exists a local flat A -algebra B such that $B/\mathfrak{m}B$ is k -isomorphic to K ; moreover, one can choose B finite and free over A if K has finite degree over k .*

This is proved in EGA 0_{III}, 10.3.1, where it is further shown that one can choose B noetherian if A is. For the reader's convenience, let us indicate the proof.

⁽²²⁾**N.D.E.** This lemma, used several times in this exposé and in subsequent exposés (VIA, VIB), has been inserted. It appeared as Lemma VIB, 4.5.1 in the original edition of SGAD (1965).

Put $A' = A[T]$, where T is an indeterminate. If $K = k(T)$, let $\mathfrak{p} = \mathfrak{m}A'$ and $B = A'_{\mathfrak{p}}$. Then $B/\mathfrak{m}B \simeq k[T]_{(0)} = k(T)$, and B is flat over A' , which is a free A -module, so B is flat over A .

If $K = k(t) = k[t]$, where t is algebraic over k , put $B = A'/(F)$, where $F \in A'$ is a monic polynomial whose image in $k[T]$ is the minimal polynomial f of t over k . Then B is a free A -module of finite rank $\deg(F) = \deg(f)$. In particular, B is integral over A , so every maximal ideal of B contains $\mathfrak{m}$. Since $B/\mathfrak{m}B \simeq k[T]/(f) \simeq K$, it follows that B is local with maximal ideal $\mathfrak{m}B$. This already shows that if $[K : k] < \infty$, one can choose B finite and free over A .

In the general case, let $(t_i)_{i \in I}$ be a system of generators of K over k , and give I a well-order (that is, a total order $\leq$ such that every nonempty subset of I has a least element). For every $i \in I$, denote by k_i (resp. $k_{<i}$) the subfield of K generated by the t_j for $j \leq i$ (resp. $j < i$). By adding an element if necessary, one can suppose that I has a greatest element ξ , so that $K = k_{\xi}$. Consider the subset J of I consisting of those indices i for which there exists an inductive system $(A_j)_{j \leq i}$ of local flat A -algebras such that $A_j/\mathfrak{m}A_j \simeq k_j$ and A_j is flat over A_{ℓ} for every $\ell < j$. Suppose $I - J$ nonempty; let i be its least element and let $A' = \varinjlim_{j < i} A_j$. Since tensor product commutes with inductive limits, A' is flat over A and over each A_j for $j < i$, and one has $A'/\mathfrak{m}A' \simeq A' \otimes_A (A/\mathfrak{m}) \simeq k_{<i}$. Moreover, A' is local with maximal ideal $\mathfrak{m}A'$. Indeed, if $x = f_j(x_j)$ is not invertible, then x_j is not invertible, hence belongs to the maximal ideal $\mathfrak{m}A_j$ of A_j , whence $x \in \mathfrak{m}A'$. It then follows from the monogenic case treated above that there exists a local flat A' -algebra A_i such that $A_i/\mathfrak{m}A_i \simeq k_{<i}(t_i) = k_i$; then A_i is flat over each A_j for $j < i$, and hence $i \in J$, contrary to the hypothesis. This contradiction shows that $J = I$, and therefore A_{ξ} answers the question. Lemma 4.1.2 is proved.

Let us now prove 4.1 (iii). Recall that $\underline{P}$ denotes the underlying set of a scheme P , and $\underline{d} : \underline{P} \rightarrow \underline{Q}$ the map induced by a morphism $d : P \rightarrow Q$. One can then express b) by saying that the map

$$d_0 \boxtimes d_1 : \underline{X}_1 \rightarrow \underline{X}_0 \times_Y \underline{X}_0$$

with components $\underline{d}_0$ and $\underline{d}_1$ is surjective; now this map factors as follows

$$\underline{X}_1 \xrightarrow{d_0 \boxtimes d_1} \underline{X}_0 \times_Y \underline{X}_0 \xrightarrow{q} \underline{X}_0 \times_Y \underline{X}_0,$$

q being the canonical map; the image of $d_0 \boxtimes d_1$ therefore contains all points v of $X_0 \times_Y X_0$ such that $\{v\} = q^{-1}(q(v))$. This last condition⁽²³⁾ holds, in particular, if v is rational over Y , that is, if the residue field $\kappa(v)$ of v is identified with the residue field $\kappa(w)$ of the image w of v in Y .

If $v \in X_0 \times_Y X_0$ is not rational over Y , let w again be its image in Y . By Lemma 4.1.2, there exists a local ring C with radical $\mathfrak{m}$ and a local flat homomorphism $f : \mathcal{O}_w \rightarrow C$ such that $C/\mathfrak{m}$ is isomorphic to $\kappa(v)$ as a $\kappa(w)$ -algebra. If one puts $Y' = \text{Spec } C$ and denotes by $\pi : Y' \rightarrow Y$ the morphism induced by f , it is clear that the canonical projection from $(X_0 \times_Y X_0) \times_Y Y'$ to $X_0 \times_Y X_0$ sends to v a point v' of $(X_0 \times_Y X_0) \times_Y Y'$ that is rational over Y' . Since

$$(X_0 \times_Y X_0) \times_Y Y' \simeq (X_0 \times_Y Y') \times_{Y'} (X_0 \times_Y Y'),$$

and since the hypotheses of Theorem 4.1 and the preceding results, in particular point b), remain valid after the base change $\pi : Y' \rightarrow Y$, v' is the image of an element $u' \in X_1 \times_Y Y'$ under the morphism obtained from $d_0 \boxtimes d_1$ by base change. If u is the image of u' in X_1 , one indeed has $v = (d_0 \boxtimes d_1)(u)$.

f) Proof of (iv):⁽²⁴⁾

⁽²³⁾**N.D.E.** Note the permutation of the pages in Lecture Notes 151: the actual order is 265–266–268–269–267–270–271.

⁽²⁴⁾**N.D.E.** The following lemma, taken from [DG70], I, §5, Prop. 1.5 (see also the proof of EGA IV₃, 8.11.5), has been added; it is used implicitly in the original and explicitly in [DG70], III, §2, 4.6. It is also useful in th. 7.1 below.

Lemma 4.1.3. *If a monomorphism of schemes $f : T \rightarrow Z$ is a finite morphism, it is a closed immersion.*

Indeed, by covering Z with affine open subsets Z_i and replacing f by the induced morphisms $f^{-1}(Z_i) \rightarrow Z_i$, one reduces (f being finite, hence affine) to the case $Z = \operatorname{Spec} B$ and $T = \operatorname{Spec} A$. Since f is a monomorphism, the diagonal morphism $T \rightarrow T \times_Z T$ is an isomorphism (EGA I, 5.3.8), i.e. $A \otimes_B A \rightarrow A$ is an isomorphism. Consequently, for every maximal ideal $\mathfrak{m}$ of B , one has an isomorphism

$$(A/\mathfrak{m}A) \otimes_k (A/\mathfrak{m}A) \simeq (A/\mathfrak{m}A),$$

where one has put $k = B/\mathfrak{m}$. Since A is finite over B , $A/\mathfrak{m}A$ is a finite-dimensional k -vector space of dimension d , and the isomorphism above implies that $d = 0$ or 1 , so that the morphism $k = B/\mathfrak{m} \rightarrow A/\mathfrak{m}A$ is surjective. Thus, by Nakayama's lemma (A being finite over B), the morphism $B_{\mathfrak{m}} \rightarrow A_{\mathfrak{m}}$ is surjective. It follows that the morphism of B -modules $B \rightarrow A$ is surjective (since its cokernel C satisfies $C_{\mathfrak{m}} = 0$ for every $\mathfrak{m}$, and is therefore zero). This proves the lemma.

Let us now prove (iv). By hypothesis $X_0 = \operatorname{Spec} A_0$, $X_1 = \operatorname{Spec} A_1$, and, for $i = 0, 1$, the morphism $\delta_i : A_0 \rightarrow A_1$ makes A_1 a finitely generated A_0 -module; thus, a fortiori, the morphism $A_0 \otimes_B A_0 \rightarrow A_1$ is finite.

One further supposes that $d = d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times_Y X_0$ is a monomorphism; thus, by the preceding lemma, the morphism $A_0 \otimes_B A_0 \rightarrow A_1$ is surjective.

We shall show that it is an isomorphism (in the course of the proof we shall show that $p : X_0 \rightarrow Y$ is finite locally free). It suffices to show that, for every prime ideal $\mathfrak{p}$ of B , the homomorphism $(A_0)_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} (A_0)_{\mathfrak{p}} \rightarrow (A_1)_{\mathfrak{p}}$ with components $\delta_{0\mathfrak{p}}$ and $\delta_{1\mathfrak{p}}$ is bijective. In other words, one may suppose B local. It then follows from b) that $(A_0)_{\mathfrak{p}}$ is semilocal; indeed, if $\mathfrak{m}$ is a maximal ideal of $(A_0)_{\mathfrak{p}}$, the other maximal ideals have the form $\delta_0^{-1}(\mathfrak{n})$, where $\mathfrak{n}$ ranges over the prime ideals of A_1 such that $\delta_1^{-1}(\mathfrak{n}) = \mathfrak{m}$; the assertion therefore follows from the fact that there are at most $n = [A_1 : A_0]$ such prime ideals $\mathfrak{n}$. By making a faithfully flat base change if necessary,⁽²⁵⁾ one can also suppose that the residue field of B is infinite, so that one can use the following lemma:

Lemma 4.2. *Let B be a local ring with infinite residue field, A a semilocal ring, and $i : B \rightarrow A$ a homomorphism sending the maximal ideal $\mathfrak{n}$ of B into the radical $\mathfrak{r}$ of A . Let M be a free A -module of rank n and N a B -submodule of M generating M as an A -module. Then N contains a basis of M over A .*

Indeed, recall that a sequence $m_1, \dots, m_n$ of elements of M is an A -basis of M if and only if the canonical images of $m_1, \dots, m_n$ in $M/\mathfrak{r}M$ form a basis of $M/\mathfrak{r}M$ over $A/\mathfrak{r}$. One can therefore replace M by $M/\mathfrak{r}M$, N by $N/(N \cap \mathfrak{r}M)$, A by $A/\mathfrak{r}$ and B by $B/\mathfrak{m}$. In this case the lemma is easy (if A is a product of fields $K_1 \times \dots \times K_r$, one can identify M with the module $K_1^n \times \dots \times K_r^n$; if x_j is an element of N whose j th component in $K_1^n \times \dots \times K_r^n$ is nonzero, show that a certain linear combination x of the x_j with coefficients in B has all its components nonzero; then replace M by M/Ax and proceed by induction on n).

We apply the preceding lemma in the following situation: $B = B$, $A = A_0$, i is the inclusion of B into A_0 , $M = A_1$ regarded as an A_0 -module by means of the homomorphism δ_1 , $N = \delta_0(A_0)$. Indeed, since $d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times_Y X_0$ is a closed immersion, the homomorphism $A_0 \otimes_B A_0 \rightarrow A_1$ with components δ_0 and δ_1 is surjective; this means precisely that $\delta_0(A_0)$ generates the A_0 -module A_1 .

Let, then, $a_1, \dots, a_n$ be elements of A_0 such that $\delta_0(a_1), \dots, \delta_0(a_n)$ form a basis of A_1 over A_0 . If we show that $a_1, \dots, a_n$ is a basis of A_0 over B , it will follow that the homomorphism $A_0 \otimes_B A_0 \rightarrow A_1$ sends the basis $(1 \otimes a_i)_{1 \leq i \leq n}$ to the basis $(\delta_0(a_i))_{1 \leq i \leq n}$, and is therefore bijective. Consequently, if $\varepsilon : \mathbf{Z}^n \rightarrow A_0$ is the homomorphism of abelian groups sending

⁽²⁵⁾**N.D.E.** Cf. Lemma 4.1.2.

the natural basis of $\mathbf{Z}^n$ to $a_1, \dots, a_n$, it suffices to prove that the map $B \otimes_{\mathbf{Z}} \mathbf{Z}^n \rightarrow A_0$ with components i and ε is bijective.

Now diagram $(0, 1, 2)^*$ considered at the beginning of this proof induces the following commutative diagram:

$$\begin{array}{ccccc} A_2 & \xleftarrow{\delta'_0} & A_1 & \xleftarrow{\delta_0} & A_0 \\ \uparrow u_2 & & \uparrow u_1 \simeq & & \uparrow u_0 \\ A_1 \otimes_{\mathbf{Z}} \mathbf{Z}^n & \xleftarrow[\delta_1 \otimes \mathbf{Z}^n]{\delta_0 \otimes \mathbf{Z}^n} & A_0 \otimes_{\mathbf{Z}} \mathbf{Z}^n & \xleftarrow{i \otimes \mathbf{Z}^n} & B \otimes_{\mathbf{Z}} \mathbf{Z}^n, \end{array}$$

where u_0, u_1 and u_2 have components respectively i and ε , δ_1 and $\delta_0 \varepsilon$, δ'_2 and $\delta'_0 \delta_0 \varepsilon$. We know that u_1 is an isomorphism. Since the two squares on the left of $(0, 1, 2)^*$ are cocartesian, u_2 is bijective. Now the two horizontal rows of our diagram are exact; thus u_0 is bijective.⁽²⁶⁾ This shows that A_0 is a locally free B -module of rank n , and, by the preceding reductions, this implies that $\delta_0 \otimes \delta_1 : A_0 \otimes_B A_0 \rightarrow A_1$ is an isomorphism. This completes the proof of Theorem 4.1 in the special case considered (X_0 affine and d_1 locally free of constant rank n).

5. Passage to the quotient by a finite flat groupoid (general case)

a) Let $U^{(n)}$ be the largest open subset of X_0 above which d_1 is finite locally free of rank n . One knows that X_0 is the direct sum of the $U^{(n)}$. It follows, on the other hand, from the two cartesian squares

$$\begin{array}{ccc} X_2 & \xrightarrow{d'_0} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_0} & X_0 \end{array} \quad \text{and} \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_1} & X_0 \end{array}$$

that the inverse images of $U^{(n)}$ under d_0 and d_1 both coincide with the largest open subset of X_1 above which d'_2 is locally free of rank n ;⁽²⁷⁾ one therefore has $d_0^{-1}(U^{(n)}) = d_1^{-1}(U^{(n)})$, so that the groupoid X_* is the direct sum of the groupoids $X_*^{(n)}$ induced by X_* on the open and closed subsets $U^{(n)}$. Consequently, as one easily sees, it suffices to prove Theorem 4.1 for each $X_*^{(n)}$: one is reduced to the case where d_1 is finite locally free of rank n .

b) We are now in a position to prove our theorem in the general case.

By a), one can suppose d_1 locally free of rank n . Let, then, (Y, p) be a cokernel of (d_0, d_1) in the category of all ringed spaces. The argument at the end of paragraph 4.c) shows that, to prove 4.1 (i), it suffices to prove that Y is a scheme and $p : X_0 \rightarrow Y$ a morphism of schemes. By Lemma 1.2, the question is local on Y : let $y \in Y$ and $x \in X_0$ satisfy $p(x) = y$; if x has an affine saturated open neighborhood U , $p(U)$ will be an affine open subset of Y by §4, and $p|_U$ will be a morphism of schemes. It therefore suffices to prove that every $x \in X_0$ has an affine saturated open neighborhood U . Here is how one proceeds (the proof is taken from SGA 1, VIII, cor. 7.6).

$$\begin{array}{ccccccccccc} d_1(d_0^{-1}(x)) \subset & U = (V_f)' & \subset & V_f & \subset & V' & \subset & V & \subset & X_0 \\ & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\ & \text{largest saturated} & & \text{distinguished affine} & & \text{largest saturated} & & \text{affine open subset} & & \\ & \text{open subset of } V_f & & \text{open subset of } V & & \text{open subset of } V & & & & \end{array}$$

⁽²⁶⁾N.D.E. What follows has been added.

⁽²⁷⁾N.D.E. Indeed, since d_0 (resp. d_1) is surjective, flat and finite, hence faithfully flat and affine, d'_2 has rank n above a neighborhood of a point x of X_1 if and only if d_1 has rank n above a neighborhood of $d_0(x)$ (resp. $d_1(x)$).

By condition b) of 4.1, there exists an affine open subset V of X_0 containing $d_1(d_0^{-1}(x))$; ⁽²⁸⁾ if $F = X_0 - V$, $d_1(d_0^{-1}(F))$ is closed because d_1 is integral, and $V' = X_0 - d_1(d_0^{-1}(F))$ is the largest saturated open subset contained in V . Since V' is a neighborhood of the finite set $d_1(d_0^{-1}(x))$, there exists a section f of the structure sheaf of V which vanishes on $V - V'$ and such that $d_1(d_0^{-1}(x))$ is contained in the open subset V_f of V consisting of the points where f does not vanish. We shall see that the largest saturated open subset $(V_f)'$ of V_f is affine, and thus answers the question.

Indeed, let $Z(f) = V' - V_f$. Then $d_0^{-1}(Z(f))$ is the set of points of $d_0^{-1}(V') = d_1^{-1}(V')$ where the image $d_0^*(f)$ of f under the map induced by d_0 vanishes. On the other hand, since d_1 induces a locally free morphism of rank n from $d_0^{-1}(V') = d_1^{-1}(V')$ onto V' , ⁽²⁹⁾ by Lemma 4.1.1, $d_1(d_0^{-1}(Z(f)))$ is the set of points where the norm N of $d_0^*(f)$ for the morphism d_1 vanishes. It follows that $(V_f)' = V' - d_1(d_0^{-1}(Z(f)))$ is the set of points of V_f where N does not vanish; consequently $(V_f)'$ is affine.

This proves 4.1 (i); assertions (ii), (iii), and the first part of (iv) are then clear. Finally, let us show the consequences noted at the end of point (iv) (cf. [Ray67a], th. 1 (iii)).

By hypothesis, the groupoid X_* comes from an equivalence relation $i : R \rightarrow X_0 \times X_0$ (i thus being an immersion, cf. N.D.E. (19)), and one has established that R is effective (cf. Exp. IV, 3.3.2) and that $p : X_0 \rightarrow Y = X_0/R$ is a surjective finite locally free morphism, hence, in particular, faithfully flat and of finite presentation. Consequently, denoting by (M) the family of faithfully flat morphisms locally of finite presentation, R is (M) -effective. Thus, by Exp. IV, 6.3.3, (Y, p) represents the quotient sheaf of X_0 by R for the fppf topology, and the assertions concerning base change follow from IV, 3.4.3.1.

Remark 5.1. ⁽³⁰⁾ Retain the hypotheses and notation of 4.1, and further suppose S locally noetherian and $\pi_0 : X_0 \rightarrow S$ quasi-projective. Let us then show that $\pi : Y \rightarrow S$ is quasi-projective.

The hypotheses above imply that $Y \rightarrow S$ is of finite type; see the proof of 6.1 (ii). Let $\mathcal{A}$ be an invertible $\mathcal{O}_{X_0}$ -module ample for π_0 . By EGA II, 6.1.12, $p_*(\mathcal{A})$ is an invertible $p_*(\mathcal{O}_{X_0})$ -module. There therefore exists a covering $(V_i)_{i \in I}$ of Y by affine open subsets such that $\mathcal{A}$ is trivial above each of the saturated affine open subsets $U_i = p^{-1}(V_i)$.

For each index i , denote by $A_{i,0} = \mathcal{O}_{X_0}(U_i)$, by $A_{i,1}$ the ring of the affine open subset $d_0^{-1}(U_i) = d_1^{-1}(U_i)$ of X_1 , by $\delta_{i,0}$ (resp. $\delta_{i,1}$) the morphism $A_{i,0} \rightarrow A_{i,1}$ induced by d_0 (resp. d_1), and by $B_i = \mathcal{O}_Y(V_i) = \{b \in A_{i,0} \mid \delta_{i,0}(b) = \delta_{i,1}(b)\}$.

Following EGA II, §6.5, consider the invertible $\mathcal{O}_{X_0}$ -module $N_{d_1}(d_0^*(\mathcal{A}))$, the norm, relative to the finite locally free morphism $d_1 : X_1 \rightarrow X_0$, of the invertible $\mathcal{O}_{X_1}$ -module $d_0^*(\mathcal{A})$. If $\mathcal{A}$ is given, relative to the covering $(U_i)_{i \in I}$, by transition functions $c_{ij} \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times$, then $N_{\delta_1}(d_0^*(\mathcal{A}))$ is given by the transition functions $N_{d_1}(\delta_0(c_{ij})) \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times$; since, by paragraph 4.a), these elements belong to $\mathcal{O}_Y(V_i \cap V_j)^\times$, they define an invertible $\mathcal{O}_Y$ -module $\mathcal{L}$ such that $p^*(\mathcal{L}) = N_{d_1}(d_0^*(\mathcal{A}))$. Note further that, for every $n \in \mathbf{N}^*$, one has $p^*(\mathcal{L}^n) = N_{d_1}(d_0^*(\mathcal{A}^n))$, cf. loc. cit., (6.5.2.1).

Let us show that $\mathcal{L}$ is ample for the morphism $\pi : Y \rightarrow S$. To this end, replacing S by an affine open subset, one can suppose S affine. Let, then, $y \in Y$, $x \in X_0$ such that $p(x) = y$, V an affine open subset of Y containing y , and $U = p^{-1}(V)$. Since $\mathcal{A}$ is π_0 -ample, there exist $n \in \mathbf{N}^*$ and a section $s \in \Gamma(X_0, \mathcal{A}^n)$ such that the open subset $(X_0)_s$ satisfies $x \in (X_0)_s \subset U$. With the preceding notation, s is given by sections $a_i \in A_{i,0} = \mathcal{O}_{X_0}(U_i)$ such that $a_i = c_{ij}a_j$ on $U_i \cap U_j$, and $(X_0)_s$ is the union of the open subsets $U'_i = \{\mathfrak{p} \in \text{Spec}(A_{i,0}) \mid a_i \notin \mathfrak{p}\}$.

For each index i , put $N(a_i) = N_{\delta_1}(\delta_0(a_i)) \in B_i$. By 4.1 (i) and Lemma 4.1.1, one has:

$$p(U'_i) = p d_1 d_0^{-1}(U'_i) = p d_1(\{\mathfrak{q} \in \text{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\})$$

⁽²⁸⁾ N.D.E. One has $d_1(d_0^{-1}(x)) = d_0(d_1^{-1}(x))$, cf. N.D.E. (16) in Theorem 4.1.

⁽²⁹⁾ N.D.E. The reference to Lemma 4.1.1 has been added, cf. [DG70], III, §5.2, p. 313.

⁽³⁰⁾ N.D.E. This paragraph has been added.

and $d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\}) = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid N_{\delta_1}(\delta_{i,0}(a_i)) \notin \mathfrak{p}\}$, whence

$$p(U'_i) = \{\mathfrak{p} \in \operatorname{Spec}(B_i) \mid N(a_i) \notin \mathfrak{p}\}.$$

It follows that $p((X_0)_s)$ equals $Y_{N(s)}$, where $N(s)$ denotes the section of $\mathcal{L}^n$ on Y defined by the sections $N(a_i) \in \mathcal{O}_Y(V_i)$. One therefore has

$$(*) \quad y \in p((X_0)_s) = Y_{N(s)} \subset p(U) = V.$$

This shows that $\mathcal{L}$ is ample for $\pi : Y \rightarrow S$, which completes the proof that $\pi : Y \rightarrow S$ is quasi-projective.

6. Passage to the quotient when a quasi-section exists

We shall now prove a technical lemma which will serve us in the proof of the two theorems we have in view. Let S be a scheme and

$$X_2 \xrightarrow[d'_0, d'_1, d'_2]{d_0, d_1} X_1 \xrightarrow[d_0, d_1]{d_0, d_1} X_0$$

a (Sch/ S)-groupoid. We shall call a quasi-section of the groupoid X_* any subscheme U of X_0 such that (1) and (2) hold:

(1) The restriction v of d_1 to $d_0^{-1}(U)$ is a finite, locally free and surjective morphism from $d_0^{-1}(U)$ onto X_0 .

(2) Every subset E of U consisting of points pairwise equivalent for the equivalence relation defined by X_* (§3.e)) is contained in an affine open subset of U .⁽³¹⁾

If U is a quasi-section of X_* , the (Sch/ S)-groupoid

$$U_2 \xrightarrow[u'_0, u'_1, u'_2]{u_0, u_1} U_1 \xrightarrow[u_0, u_1]{u_0, u_1} U$$

induced by X_* and the inclusion of U into X_0 (cf. §3.a)) satisfies the hypotheses of Theorem 4.1. Indeed, put $V = d_0^{-1}(U)$ and let u and v be the morphisms with source V induced respectively by d_0 and d_1 :

$$X_0 \xleftarrow{v} V \xrightarrow{u} U.$$

By paragraph 3.b), one has a cartesian square

$$\begin{array}{ccc} U_1 & \longrightarrow & V \\ u_1 \downarrow & & \downarrow v \\ U & \xrightarrow{\text{inclusion}} & X_0, \end{array}$$

so u_1 is surjective and finite locally free by (1). Together with (2), condition (1) therefore ensures that the groupoid U_* satisfies the hypotheses of Theorem 4.1. In particular, $\operatorname{Coker}(u_0, u_1)$ exists in (Sch/ S). Moreover, d_0 has a section, so u is an effective universal epimorphism (cf. III 1.12); it follows, by Proposition 3.1, that $\operatorname{Coker}(u_0, u_1)$ coincides with the cokernel $\operatorname{Coker}(v_0, v_1)$ of the groupoid V_* :

$$V_2 \xrightarrow[v'_0]{v'_2} V_1 \xrightarrow[v_0]{v_1} V,$$

the inverse image of U_* by the base change $u : V \rightarrow U$, that is, also the inverse image of X_* by the base change:

$$V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_0} X_0.$$

⁽³¹⁾**N.D.E.** If $x, y \in E$, there exists $z \in X_1$ such that $d_1(x) = z$ and $d_0(z) = y$, that is, z belongs to the set $v^{-1}(x)$, which is finite by (1). Thus E is contained in the finite set $d_0(v^{-1}(x)) = d_0 d_1^{-1}(x) \cap U$.

(32) By paragraph 3.c), V_* is isomorphic to the groupoid V'_* , the inverse image of X_* by the base change:

$$v : V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_1} X_0,$$

and hence V'_* has a cokernel in (Sch/S) . Now, being flat, surjective and finite, $v : V \rightarrow X_0$ is faithfully flat and quasi-compact, hence an effective universal epimorphism by III 6.3.2. Consequently, by Proposition 3.1, the groupoid X_* has a cokernel $\text{Coker}(d_0, d_1)$ in (Sch/S) . We have therefore proved the first assertion of point (i) of the following lemma:

Lemma 6.1. *Suppose the (Sch/S) -groupoid X_* has a quasi-section. Then:*

- (i) *There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S) ; moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*
- (i') *p is surjective, and is open (resp. universally closed) if d_0 is.*
- (ii) *Suppose S locally noetherian and X_0 locally of finite type (resp. of finite type) over S . Then p and $Y \rightarrow S$ are locally of finite presentation (resp. of finite presentation).*
- (iii) *The morphism $X_1 \rightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*
- (iv) *If (d_0, d_1) is an equivalence pair, $X_1 \rightarrow X_0 \times_Y X_0$ is an isomorphism. Moreover, if $d_0 : X_1 \rightarrow X_0$ is flat, p is faithfully flat.*

Before proving the second assertion of (i), we shall prove (i'), (ii) and (iii).

a) Proof of (i') and (ii):

We have just seen that (Y, p) is identified with $\text{Coker}(v_0, v_1)$ and $\text{Coker}(u_0, u_1)$. Let, then, q and r be the canonical epimorphisms from U and V to Y :

$$\begin{array}{ccccc} X_0 & \xleftarrow{v} & V & \xrightarrow{u} & U \\ & \searrow p & \downarrow r & \swarrow q & \\ & & Y & & \end{array}$$

By hypothesis, v is surjective and finite locally free, hence open. On the other hand, if $d_0 : X_1 \rightarrow X_0$ is open (resp. universally closed), then u , obtained from it by base change, is also so. Since, by Theorem 4.1, q is surjective, integral and open, r is surjective, and is open (resp. universally closed) if d_0 is. The same therefore holds for p , since v is surjective. This proves (i').

Now suppose S locally noetherian and X_0 locally of finite type over S , so that X_0 is locally noetherian.

Let us show that Y is locally of finite presentation over S . Let $S' = \text{Spec } R$ be an affine open subset of S , $Y' = \text{Spec } B$ an affine open subset of Y projecting into S' , and $U' = \text{Spec } A$ the inverse image of Y' in U . Since R is noetherian, it suffices to show that B is a finitely generated R -algebra; now, by paragraphs 4 and 5, B is contained in A , which is a finitely generated R -algebra; the assertion therefore follows from the facts that R is noetherian and A integral over B .

Finally, since $X_0 \rightarrow S$ is locally of finite type, p is also so (EGA I, 6.6.6), hence p is locally of finite presentation since Y is locally noetherian.

It remains to show the last assertion of (ii). Further suppose X_0 of finite type over S . Then, since p is surjective, Y is moreover quasi-compact over S , hence of finite type over S . Since S is locally noetherian, $X_0 \rightarrow S$ and $Y \rightarrow S$ are of finite presentation, and hence $p : X_0 \rightarrow Y$ is also so (EGA IV₁, 1.6.2 (v)).

b) Proof of (iii):

Since the groupoid V_* with base V is isomorphic both to the inverse image of U_* by the base change u and to the inverse image of X_* by the base change v , one has a double

(32) **N.D.E.** What follows has been slightly modified; in particular, in Lemma 6.1, the additional hypothesis that d_0 is flat has been moved to (iv), (ii) has been split into (i') and (ii), and the second assertion of (i') has been added.

cartesian square

$$\begin{array}{ccccc} X_1 & \xleftarrow{\quad} & V_1 & \xrightarrow{\quad} & U_1 \\ d_0 \boxtimes d_1 \downarrow & & v_0 \boxtimes v_1 \downarrow & & u_0 \boxtimes u_1 \downarrow \\ X_0 \times_Y X_0 & \xleftarrow[v \times v]{} & V \times_Y V & \xrightarrow[u \times u]{} & U \times_Y U. \end{array}$$

Since $u_0 \boxtimes u_1$ is surjective, so is $v_0 \boxtimes v_1$. Since $v \times v$ is surjective, so is the composite morphism $V_1 \longrightarrow X_0 \times_Y X_0$, and therefore $d_0 \boxtimes d_1$.

c) Proof of (i):

It remains to prove that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. We first show that Y is obtained from X_0 by identifying points x and y such that there exists $z \in X_1$ with $d_0(z) = x$ and $d_1(z) = y$. Indeed, p is surjective and one has $pd_0 = pd_1$; moreover, if $p(x) = p(y)$, there is a point z' of $X_0 \times_Y X_0$ whose first projection is x and second projection y . If z is a point of X_1 such that $(d_0 \boxtimes d_1)(z) = z'$, one indeed has $d_0(z) = x$ and $d_1(z) = y$.

On the other hand, if W is a saturated open subset of X_0 , $W \cap U$ is a saturated open subset of U ; by 4.1, $q(W \cap U)$ is an open subset of Y . Since $q(W \cap U)$ is none other than $p(W)$, one sees that Y has the quotient topology of that of X_0 .

It remains to show that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_{X_0}) \rightrightarrows p_*d_{0*}(\mathcal{O}_{X_1}) = p_*d_{1*}(\mathcal{O}_{X_1})$$

is exact.

Let, then, Y' be an open subset of Y , and put $U' = q^{-1}(Y')$, $X'_0 = p^{-1}(Y')$, etc.⁽³³⁾ Then U' is an open subset of U , saturated for the equivalence relation defined by the groupoid U_* , and it follows from Lemmas 1.1 and 1.2 that Y' is the cokernel, in (Sch/S) and in (Esp. An) , of the groupoid induced by U_* on U' . Similarly, X'_0 is an open subset of X_0 , saturated for the equivalence relation defined by X_* , and one has the following commutative diagram, whose two squares are cartesian:

$$\begin{array}{ccccc} X'_0 & \xleftarrow{\tilde{d}_1} & \tilde{V}' = d_0^{-1}(U') & \xrightarrow{\tilde{d}_0} & U' \\ \downarrow & & \downarrow & & \downarrow \\ X_0 & \xleftarrow{d_1} & V = d_0^{-1}(U) & \xrightarrow{d_0} & U. \end{array}$$

Then $\tilde{d}_1$ is surjective and finite locally free. On the other hand, let $x \in U'$. Since U is a quasi-section, the set $E := d_0 d_1^{-1}(x) \cap U$ is finite and contained in an affine open subset W of U . Then $E' = E \cap U'$ is a finite set contained in the quasi-affine open subset $W \cap U'$. Consequently, there exists an affine open subset W' of $W \cap U'$ containing E' . This shows that U' is a quasi-section of the groupoid X'_* induced by X_* on X'_0 . The first assertion of (i), applied to X'_* and U' , then shows that Y' is the cokernel in (Sch/S) of X'_* .

In particular, for every S -scheme T , one has the exact sequence

$$T(Y') \xrightarrow{T(p|_{X'_0})} T(X'_0) \xrightarrow[T(d_0|_{X'_1})]{T(d_1|_{X'_1})} T(X'_1).$$

Now, if T is the “affine line” $\mathbf{G}_{a,S}$ (I 4.3), this sequence is identified with the sequence:

$$\Gamma(Y', \mathcal{O}_Y) \rightarrow \Gamma(p^{-1}(Y'), \mathcal{O}_{X_0}) \xrightarrow[\delta_0]{\delta_1} \Gamma(d_0^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}) = \Gamma(d_1^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}),$$

which is therefore exact for every open subset Y' . This completes the proof of 6.1 (i).

⁽³³⁾**N.D.E.** What follows has been expanded, in particular the fact that U' is a quasi-section of the groupoid induced on X'_0 .

d) Proof of (iv):

If (d_0, d_1) is an equivalence pair, so is (u_0, u_1) . It follows that $u_0 \boxtimes u_1 : U_1 \rightarrow U \times_Y U$ is an isomorphism (Theorem 4.1), and therefore also $v_0 \boxtimes v_1$ (cf. the cartesian squares of b)); since $v \times v$ is faithfully flat and quasi-compact, $d_0 \boxtimes d_1$ is an isomorphism (SGA 1, VIII 5.4).

Moreover, if d_0 is flat, so is u . Now q is flat by Theorem 4.1, hence so is r . Since v is faithfully flat, p is flat, and therefore faithfully flat since it is surjective.

7. Quotient by a proper flat groupoid

Theorem 7.1. ⁽³⁴⁾ *Let S be a locally noetherian scheme and*

$$X_2 \begin{array}{c} \xrightarrow{d_1'} \\ \xrightarrow{d_0'} \end{array} X_1 \xrightarrow{d_0} X_0$$

a (Sch/S) -groupoid such that d_1 is proper and flat, X_0 is quasi-projective over S ,⁽³⁵⁾ and the morphism $d : X_1 \rightarrow X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite. Then:

- (i) *There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S) ; moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*
- (ii) *p is surjective, open, proper, of finite presentation, and $Y \rightarrow S$ is of finite presentation and separated.*⁽³⁶⁾
- (iii) *The morphism $X_1 \rightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*
- (iv) *If (d_0, d_1) is an equivalence pair, $X_1 \rightarrow X_0 \times_Y X_0$ is an isomorphism and p is faithfully flat.*⁽³⁷⁾ *Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology and, for every base change $Y' \rightarrow Y$, Y' is the cokernel of the groupoid $X_* \times_Y Y'$ obtained from X_* by the base change $X_0 \times_Y Y' \rightarrow X_0$.*

In particular, for every base change $S' \rightarrow S$, $Y' = Y \times_S S'$ is the cokernel of the S' -groupoid $X'_ = X_* \times_S S'$. Thus, in this case, “formation of the quotient commutes with base change”.*

Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Lemma 1.2 shows that, to prove (i), it suffices to show that every point z of X_0 has a saturated open neighborhood U_z such that, denoting by $\tilde{d}_0$ and $\tilde{d}_1$ the restrictions of d_0 and d_1 to $d_0^{-1}(U_z) = d_1^{-1}(U_z)$, and by (Q, q) the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp.An) , Q is a scheme and q a morphism of schemes.

By Lemma 6.1 (i), it therefore suffices to show that every point z of X_0 has a saturated open neighborhood U_z such that the groupoid induced on U_z by X_* has a quasi-section. One can even suppose that z is closed in the fiber of z over S (we shall say that z is closed relative to S).⁽³⁸⁾ The existence of U_z then follows from the lemmas below:

⁽³⁴⁾**N.D.E.** Let us mention here the article of S. Keel and S. Mori ([KM97]), in which the following theorem is established. Let X be an algebraic space of finite type over a locally noetherian base S and $j : R \rightarrow X \times_S X$ a flat groupoid whose stabilizer $j^{-1}(\Delta_X)$ is finite over X ; there then exists an algebraic space which is a geometric quotient of X by R and a uniform categorical quotient; moreover, if j is separated, this quotient is separated. In particular, if a flat S -group scheme G acts properly on X , with finite stabilizer (i.e. the morphism $G \times_S X \rightarrow X \times_S X$, $(g, x) \mapsto (x, g \cdot x)$ is proper and the stabilizer of the diagonal is finite over X), then there exists a geometric quotient $X \rightarrow X/G$. In the case of a reductive S -group scheme G , this is a result of J. Kollár ([Ko97]).

⁽³⁵⁾**N.D.E.** This hypothesis on X_0 is necessary, cf. N.D.E. (17) in Th. 4.1.

⁽³⁶⁾**N.D.E.** In TDTE III, Th. 6.1, it is stated that $Y \rightarrow S$ is quasi-projective if S is noetherian. The editors have not seen how to deduce this from the local existence of quasi-sections.

⁽³⁷⁾**N.D.E.** The following consequences have been made explicit; see [Ray67a], th. 1 (iii) and the end of the proof of the theorem. Let us also mention that another proof of th. 7.1 in the case of an equivalence relation, based on the existence of Hilbert schemes, is given in [AK80], Th. 2.9; see also [BLR90], §8.2, Th. 12; it is moreover shown there, in this case, that $Y \rightarrow S$ is quasi-projective.

⁽³⁸⁾**N.D.E.** Indeed, if one has constructed such an open neighborhood U_z for every point z closed relative to S , the union of these U_z covers X_0 , since each fiber over S of the closed complement is a noetherian scheme without closed points, hence empty.

Lemma 7.2. *Let T be an affine noetherian scheme, X, Y and Z be T -schemes of finite type, X being quasi-projective over T , and*

$$\begin{array}{ccc} Y & \xrightarrow{u} & X \\ v \downarrow & & \downarrow \\ Z & \dashrightarrow & T \end{array}$$

a diagram of (Sch/T) . Let, on the other hand, z be a point of $v(Y)$ closed relative to T and such that v is flat at the points of $v^{-1}(z)$. Then there exists a closed subscheme F of X such that $u(u^{-1}(F) \cap v^{-1}(z))$ is finite nonempty and the restriction of v to $u^{-1}(F)$ is flat at the points of $v^{-1}(z)$.

Let $T = \text{Spec } A$. One may suppose X of the form $\text{Proj } \mathcal{S}$, where $\mathcal{S}$ is the symmetric algebra of a finitely generated A -module E .

If $u(v^{-1}(z))$ is finite, one can choose $F = X$. Otherwise, denote by $y_1, \dots, y_n$ the points of the fiber $v^{-1}(z)$ associated to its structure sheaf $\mathcal{O}_{v^{-1}(z)}$ (the y_i are thus such that, if $\mathcal{O}_i$ denotes the local ring of $v^{-1}(z)$ at y_i , the maximal ideal of $\mathcal{O}_i$ consists of zero divisors). If t is the image of z in T , $u(v^{-1}(z))$ is an infinite constructible subset of the fiber of t in X . There therefore exists a point x closed in this fiber, belonging to $u(v^{-1}(z))$ and distinct from $u(y_1), \dots, u(y_n)$. Then $X - \{x\}$ is an open neighborhood of $u(y_1), \dots, u(y_n)$, and therefore contains an open neighborhood of the form $D_+(f)$, where f is a homogeneous element of degree d of $\mathcal{S}$ (the notation is that of EGA II, §2.3).

Consequently, the closed subscheme $X_1 = V_+(f)$ defined by f contains x and avoids the points $u(y_1), \dots, u(y_n)$. It follows obviously that the inverse image $Y_1 = u^{-1}(V_+(f))$ of this subscheme is distinct from Y and meets $v^{-1}(z)$. We shall even show that the restriction v_1 of v to Y_1 is flat at the points of $v^{-1}(z)$; if $u(v_1^{-1}(z))$ is finite, one will therefore only have to choose $F = X_1$; otherwise, one repeats the argument just developed, replacing Y by Y_1 , v by v_1 and u by the morphism u_1 induced on Y_1 by u ; in this way one obtains a decreasing sequence $X, X_1, \dots$ of closed subschemes of X ; since such a sequence terminates, $u(u^{-1}(X_n) \cap v^{-1}(z))$ will be finite nonempty for some n , and one chooses $F = X_n$.

It therefore remains to show that v_1 is flat at the points of $v^{-1}(z)$; let, then, y be a point of Y_1 above z , $\mathcal{O}_y$ the local ring of y in Y , $\overline{\mathcal{O}}_y$ the local ring of y in $v^{-1}(z)$, and $\mathcal{O}_{v(y)}$ the local ring of $v(y)$ in Z . If $g \in \mathcal{S}_1$ is such that $D_+(g)$ is a neighborhood of $u(y)$ in X , let φ be the image of f/g^d in $\mathcal{O}_y$, and $\overline{\varphi}$ its image in $\overline{\mathcal{O}}_y$. It follows from the construction of f that $\overline{\varphi}$ is not a zero divisor in $\overline{\mathcal{O}}_y$; since $\mathcal{O}_y$ is flat over $\mathcal{O}_z$, φ is not a zero divisor in $\mathcal{O}_y$ and $\mathcal{O}_y/\mathcal{O}_y\varphi$ is flat over $\mathcal{O}_z$ (SGA 1, IV 5.7). Now $\mathcal{O}_y/\mathcal{O}_y\varphi$ is none other than the local ring of y in Y_1 .

Lemma 7.3. *Retain the notation and hypotheses of 7.1. Every point z of X_0 closed relative to S then has a saturated open neighborhood U_z such that the groupoid induced by X_* on U_z has a quasi-section.*

The statement being local on S , one can suppose S affine noetherian and apply the preceding lemma to the diagram

$$\begin{array}{ccc} X_1 & \xrightarrow{d_0} & X_0 \\ d_1 \downarrow & & \downarrow \\ X_0 & \dashrightarrow & S \end{array}$$

of (Sch/S) . Let, then, F be a closed subscheme of X_0 such that $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite nonempty and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $d_1^{-1}(z)$. Denote by F_1 and F_2 the inverse images of F under d_0 and $d_0 d'_0 = d_0 d'_1$, and by $\tilde{d}_0, \tilde{d}_1$, etc., the morphisms

induced by d_0, d_1 , etc. One therefore has a commutative diagram

$$\begin{array}{ccccc}
 F_2 & \xrightarrow{\tilde{d}'_1} & F_1 & \xrightarrow{\tilde{d}_0} & F \\
 \tilde{d}'_2 \downarrow & & \tilde{d}_0 \downarrow & & \searrow \tilde{q} \\
 X_1 & \xrightarrow[d_0]{d_1} & X_0 & \dashrightarrow & S,
 \end{array}$$

where the two squares on the left are cartesian and the first row exact (cf. (0,1,2), §1), and where q and $\tilde{q}$ denote the structure morphisms.

Let us first show that there are only finitely many points of F_1 above z .⁽³⁹⁾ Indeed, let s be the image of z in S ; since F is of finite type over S , the fiber $\tilde{q}^{-1}(s)$ is a noetherian scheme. On the other hand, since $\tilde{d}_0$ is proper, $\tilde{d}_0(\tilde{d}_1^{-1}(z))$ is a closed subscheme of $\tilde{q}^{-1}(s)$ consisting of finitely many points. Consequently (cf. EGA I, 6.2.2), the points of this set are closed in $\tilde{q}^{-1}(s)$ and also (since F is closed in X_0) in the fiber $q^{-1}(s)$ of s in X_0 . Let y be one of these points; since the fiber $q^{-1}(s)$ is of finite type over $\kappa(s)$, it contains affine open neighborhoods $\text{Spec } B$ and $\text{Spec } C$ of y and z respectively, where B and C are finitely generated $\kappa(s)$ -algebras. Then y and z correspond to maximal ideals $\mathfrak{p} \subset B$ and $\mathfrak{q} \subset C$, the fields $B/\mathfrak{p}$ and $C/\mathfrak{q}$ have finite degree over $\kappa(s)$, and hence $(B/\mathfrak{p}) \otimes_{\kappa(s)} (C/\mathfrak{q})$ is a finite-dimensional $\kappa(s)$ -algebra whose maximal ideals correspond exactly to the points of $X_0 \times_S X_0$ whose second (resp. first) projection is z (resp. y). There are therefore only finitely many points u of $X_0 \times_S X_0$ whose second projection is z and whose first projection belongs to $\tilde{d}_0(\tilde{d}_1^{-1}(z))$. Finally, since $X_1 \rightarrow X_0 \times_S X_0$ has finite fibers, such a point u comes from finitely many points of X_1 , whence the desired assertion.

The morphism $\tilde{d}_1$ is therefore quasi-finite and flat at the points of F_1 above z . Since $\tilde{d}_1$ is of finite type, by SGA 1, IV 6.10 and EGA III, 4.4.10,⁽⁴⁰⁾ the set Φ of points of F_1 where $\tilde{d}_1$ is not both flat and quasi-finite is closed in F_1 , hence in X_1 (since F_1 is closed in X_1). Since d_1 is proper, $\tilde{d}_1(\Phi)$ is closed, and it does not contain z by what precedes. Put $W = \tilde{d}_1(F_1) - \tilde{d}_1(\Phi)$. Then the restriction of $\tilde{d}_1$ to $\tilde{d}_1^{-1}(W)$ is⁽⁴¹⁾ of finite presentation (in view of the noetherian hypotheses), flat, proper and quasi-finite, hence finite, locally free and open, by EGA III, 4.4.2 and EGA IV₂, 2.1.12 and 2.4.6. Consequently, $\tilde{d}_1(F_1)$ is a neighborhood of z , and W is the largest open subset of X_0 contained in $\tilde{d}_1(F_1)$ above which $\tilde{d}_1$ is both flat and quasi-finite.

We shall see in Lemma 7.4 that the inverse images of Φ under $\tilde{d}'_1$ and $\tilde{d}'_0$ are both identified with the set of points of F_2 where $\tilde{d}'_2$ is not both flat and quasi-finite. It follows that $\tilde{d}_0^{-1}(W) = \tilde{d}'_2(F_2) - \tilde{d}'_2(\tilde{d}'_0^{-1}\Phi)$ coincides with $\tilde{d}_1^{-1}(W) = \tilde{d}'_2(F_2) - \tilde{d}'_2(\tilde{d}'_1^{-1}\Phi)$, that is, W is saturated. Consequently, if one puts $W_1 = \tilde{d}_1^{-1}(W)$, the equality $\tilde{d}_0^{-1}(W) = \tilde{d}_1^{-1}(W)$ implies $\tilde{d}'_2^{-1}\tilde{d}_0^{-1}(W) = \tilde{d}'_2^{-1}\tilde{d}_1^{-1}(W)$, that is, $\tilde{d}'_0^{-1}(W_1) = \tilde{d}'_1^{-1}(W_1)$. Since $\tilde{d}_0$ is faithfully flat

⁽³⁹⁾**N.D.E.** Details have been added and the role of the properness hypothesis on d_0 and d_1 in Theorem 7.1 has been brought out. (One may compare with the statement and proof of Theorem 8.1, where this properness hypothesis is omitted.)

⁽⁴⁰⁾**N.D.E.** If $f : X \rightarrow Y$ is a morphism locally of finite type, the set of $x \in X$ isolated in their fiber $f^{-1}(f(x))$ is open in X : in EGA III, 4.4.10, this is deduced, for Y locally noetherian, from Zariski's "Main Theorem"; on the other hand, for arbitrary Y , it follows from Chevalley's semicontinuity theorem (EGA IV₃, 13.1.3 and 13.1.4). Consequently, f is quasi-finite at x if and only if f is of finite type at x and x is isolated in $f^{-1}(f(x))$; this will be used below, cf. N.D.E. (42).

⁽⁴¹⁾**N.D.E.** The original has been expanded in what follows.

and quasi-compact (for d_0 is, like d_1 , surjective, proper and flat), and the square

$$\begin{array}{ccc} F_2 & \xrightarrow{\tilde{d}'_1} & F_1 \\ \tilde{d}'_0 \downarrow & & \downarrow \tilde{d}_0 \\ F_1 & \xrightarrow{\tilde{d}_0} & F \end{array}$$

is cartesian, it follows that W_1 has the form $\tilde{d}_0^{-1}(U)$, where U is an open subset of F (SGA 1, VIII 4.4). This open subset U of F is a quasi-section for the groupoid with base W induced by X_* . One can therefore choose $U_z = W$.

It remains to state Lemma 7.4, whose proof is classical:

Lemma 7.4. *Consider a cartesian square of schemes*

$$\begin{array}{ccc} F_2 & \xrightarrow{v} & F_1 \\ d' \downarrow & & \downarrow d \\ X_1 & \xrightarrow{u} & X_0 \end{array}$$

and let x be a point of F_2 .

- (i) *If u is flat, d' is flat at x if and only if d is flat at $v(x)$.*
- (ii) *If d is locally of finite type, d' is quasi-finite at x if and only if d is quasi-finite at $v(x)$.⁽⁴²⁾*

We have therefore proved that there exists a covering of X_0 by saturated open subsets W such that the groupoid W_* induced by X_* on W has a quasi-section.⁽⁴³⁾

By Lemma 6.1 and the reductions stated after Theorem 7.1, this implies assertions (i) and (iii) of Theorem 7.1, and the facts that p is surjective and open, and that p and $Y \rightarrow S$ are locally of finite presentation. Moreover, since $X_0 \rightarrow S$ is quasi-projective, hence separated and of finite type, p is separated, and the proof of point (ii) of Lemma 6.1 shows that p and $Y \rightarrow S$ are of finite presentation.

To show that p is proper, it therefore remains to show that it is universally closed. Since the assertion is local on Y , one can work on a saturated open subset W such that the groupoid W_* induced by X_* on W has a quasi-section U (since X_0 is covered by such open subsets). Returning to the notation of 6.a), one has a commutative diagram

$$\begin{array}{ccccc} & & W & \xleftarrow{v} & V & \xrightarrow{u} & U \\ & \searrow & & & \downarrow r & & \swarrow q \\ & & & & Z, & & \end{array}$$

where Z is an open subset of Y , all arrows are surjective, and q is integral. Moreover, by hypothesis, $d_0 : X_1 \rightarrow X_0$ is proper, so u , obtained from it by base change, is also so. Consequently, r is universally closed, and hence so is p , since v is surjective.

Finally, p being surjective and universally closed, and X_0 quasi-projective, hence separated, the diagonal $\Delta_{Y/S}(Y)$ is closed in $Y \times_S Y$, being the image under $p \times p$ of the diagonal $\Delta_{X_0/S}(X_0)$. Thus Y is separated over S . This completes the proof of 7.1 (ii).

⁽⁴²⁾**N.D.E.** The conditions are sufficient, by base change (cf. EGA II, 6.2.4 (iii) and EGA IV₂, 2.1.4). Conversely, put $y = d'(x)$ and $z = u(y) = d(v(x))$, and suppose d' flat at x and u (hence also v) flat. Then $\mathcal{O}_{v(x)} \rightarrow \mathcal{O}_x$ is faithfully flat, as is $\mathcal{O}_z \rightarrow \mathcal{O}_y \rightarrow \mathcal{O}_x$. Consequently $\mathcal{O}_z \rightarrow \mathcal{O}_{v(x)}$ is faithfully flat (cf. EGA IV₂, 2.2.11 (iv)). Finally, suppose d locally of finite type and d' quasi-finite at x . Then $v(x)$ is isolated in its fiber $d^{-1}(z)$, since x is isolated in its fiber $d'^{-1}(y) = d^{-1}(z) \otimes_{\kappa(z)} \kappa(y)$. Thus, by Chevalley's semicontinuity theorem, there exists an open neighborhood of $v(x)$ every point of which is isolated in its fiber (EGA IV₃, 13.1.3 and 13.1.4), so d is quasi-finite at $v(x)$.

⁽⁴³⁾**N.D.E.** What follows has been modified, taking advantage of the additions made to Lemma 6.1.

The assertions to be proved in 7.1 (iv) are local on Y ; since X_0 is covered by the saturated open subsets U_z , it suffices to verify these assertions by replacing X and Y by U_z and $V = p(U_z)$. As already seen at the beginning of the proof of 7.1, it follows from Lemmas 1.1, 1.2 and 6.1 (i) that $(V_z, p|_{U_z})$ is the cokernel in (Sch) and in (Esp.An) of the groupoid induced by X_* on U_z . Now the hypothesis that $d = (d_0, d_1)$ is a monomorphism is preserved by the base change $U_z \rightarrow X_0$. Consequently, the first two assertions of 7.1 (iv) follow from 6.1 (iv).

Finally, let us show the consequences noted at the end of point (iv) (cf. [Ray67a], th. 1 (iii)). By hypothesis, the groupoid X_* comes from an equivalence relation $R \rightarrow X_0 \times X_0$, and one has established that R is effective (cf. Exp. IV, 3.3.2) and that $p : X_0 \rightarrow Y = X_0/R$ is faithfully flat and of finite presentation. Consequently, denoting by (M) the family of faithfully flat morphisms locally of finite presentation, R is (M) -effective. Thus, by Exp. IV, 6.3.3, (Y, p) represents the quotient sheaf of X_0 by R for the fppf topology, and the assertions concerning base change follow from IV, 3.4.3.1.

8. Passage to the quotient by a flat groupoid that is not necessarily proper

Theorem 8.1. ⁽⁴⁴⁾ *Let S be a noetherian scheme and*

$$X_2 \begin{array}{c} \xrightarrow{d_2'} \\ \xrightarrow{d_1'} \\ \xrightarrow{d_0'} \end{array} X_1 \xrightarrow{d_1} X_0$$

a (Sch/ S)-groupoid such that d_1 is flat and of finite type, X_0 is of finite type over S , and the morphism $X_1 \rightarrow X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite.

There then exists a dense saturated open subset W of X_0 satisfying the following properties:

- (i) *If $W_2 \xrightarrow{w_2'} W_1 \xrightarrow{w_1'} W$ is the groupoid induced by X_* on W , (w_0, w_1) has a cokernel (V, p) in (Sch/ S); moreover (V, p) is a cokernel of (w_0, w_1) in the category of all ringed spaces.*
- (ii) *p is surjective and open.*
- (ii') *p and $V \rightarrow S$ are of finite presentation.*
- (iii) *The morphism $W_1 \rightarrow W \times_V W$ with components w_0 and w_1 is surjective.*

⁽⁴⁴⁾**N.D.E.** There exists a largest open subset W of X_0 satisfying the conclusions of the theorem. Indeed, let W be an open subset as in the theorem and $W^\#$ a dense saturated open subset contained in W . Since p is open, $V^\# = p(W^\#)$ is an open subset of V , and $W^\# = p^{-1}(V^\#)$, since $W^\#$ is saturated. By Lemma 1.1, $V^\#$ is a cokernel for the groupoid induced on $W^\#$. Thus one can glue along their intersection $W^\#$ two open subsets W and W' satisfying the conclusions of the theorem, and conditions (i), (ii), (iii), (iv), as well as the fact that p and $V \rightarrow S$ are locally of finite presentation, are preserved. Conclusion (ii') follows, as in the proof of 6.1 (ii), from the hypothesis that X_0 is of finite type over noetherian S .

Furthermore, Lemmas 1.1 and 1.2 also show that the union of all saturated open subsets U of X_0 such that the open subset $p(U)$ of Y is a scheme and $p|_U : U \rightarrow p(U)$ is a morphism of schemes is the largest saturated open subset Ω of X_0 satisfying condition (i) of 8.1. Theorem 8.1 shows that Ω contains a dense open subset W , but it is not immediate that Ω satisfies properties (ii) to (iv).

On this subject, the reader may consult [Ray67a], [Ray67b] and appendix I of [An73], which give more precise results and study the question of representability of the quotient fppf S -sheaf $\widetilde{X/R}$ (where R denotes the groupoid X_* with base $X = X_0$), all this under weaker hypotheses (S an arbitrary scheme, X a scheme locally of finite type over S , and R an S -groupoid with base X such that d_0 (and hence d_1) is flat and of finite presentation). Let us mention in particular the following results. If $\widetilde{X/R}$ is represented by an S -scheme Y , then Y is also the cokernel in the category (Esp.An). The converse is false in general (cf. example 0.4 of [Mum65], Chap. 0, §3, cited in [Ray67a], Rem. 1), but is true if $d = (d_0, d_1)$ is an immersion. Under this hypothesis, the morphism $p : \Omega \rightarrow Z := \Omega/R_\Omega$ is faithfully flat and of finite presentation; if moreover S is locally noetherian, a point x of codimension 1 in X belongs to Ω if and only if the graph of the groupoid induced on $\text{Spec}(\mathcal{O}_{X,x})$ is closed. For all this, see [Ray67a], Prop. 1, [Ray67b], Prop. 1 and Theorems 2, 1 and 4, and [An73], Theorems 5 and 6 on pages 66–67 and Prop. 3.3.1 on page 49. (See also, in the case of an action of an algebraic group over an algebraically closed field k , the article [DR81].)

- (iv) If (d_0, d_1) is an equivalence pair, $W_1 \rightarrow W \times_V W$ is an isomorphism and p is faithfully flat.

We shall show that one can choose W so that the (Sch/S) -groupoid W_* induced by X_* has a quasi-section (cf. §7). Theorem 8.1 will then follow from Lemma 6.1.

Let us provisionally admit that, for every point $z \in X_0$ closed relative to S (cf. §7), there exists a saturated open subset W_z which has a quasi-section and meets all irreducible components of X_0 passing through z . Then the exterior $X_0 - \overline{W}_z$ of W_z in X_0 is saturated (for the saturation $d_1(d_0^{-1}(X_0 - \overline{W}_z))$ of this exterior is open and does not meet W_z). If this exterior is nonempty, one can choose in it a point z' closed relative to S and associate to z' an open subset $W_{z'}$ as above; one can moreover suppose $W_{z'}$ contained in $X_0 - \overline{W}_z$; then W_z and $W_{z'}$ are disjoint and the groupoid induced by X_* on $W_z \cup W_{z'}$ has a quasi-section. The process must terminate because X_0 has only finitely many irreducible components. It therefore remains to construct W_z .

For this one may suppose S affine; in this case, let y be a point of X_1 such that $d_1(y) = z$, X an affine open subset of X_0 containing $d_0(y)$, Y the inverse image of X in X_1 under d_0 , and finally $u : Y \rightarrow X$ and $v : Y \rightarrow X_0$ the morphisms induced by d_0 and d_1 . Since X is affine, hence quasi-projective, one can apply Lemma 7.2: there is therefore a subscheme F of X_0 such that $d_0^{-1}(F) \cap d_1^{-1}(z)$ is nonempty, $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite, and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $v^{-1}(z)$. This allows us to return to the notation of Lemma 7.3, denoting by F_1 and F_2 the inverse images of F in X_1 and X_2 , etc.

$$\begin{array}{ccccc} F_2 & \xrightarrow{\tilde{d}'_1} & F_1 & \xrightarrow{\tilde{d}_0} & F \\ \tilde{d}'_2 \downarrow & & \tilde{d}'_0 \downarrow & & \tilde{d}_1 \downarrow \\ X_1 & \xrightarrow[d_0]{} & X_0 & & \end{array} .$$

One then shows, as in 7.3, that $\tilde{d}_1$ is quasi-finite at the points of $\tilde{d}_1^{-1}(z)$, so it is natural to consider the open subset F'_1 of F_1 consisting of the points where $\tilde{d}_1$ is both flat and quasi-finite. By 7.4, the two inverse images of F'_1 under $\tilde{d}'_1$ and $\tilde{d}'_0$ consist of the points of F_2 where $\tilde{d}'_2$ is flat and quasi-finite, so these two inverse images coincide and F'_1 has the form $\tilde{d}_0^{-1}(F')$, where F' is an open subset of F (SGA 1, VIII 4.4). By replacing F by F' if necessary, one can therefore suppose $\tilde{d}_1$ quasi-finite and flat. In this case, denote by W_z the largest open subset of $\tilde{d}_1(F_1)$ above which $\tilde{d}_1$ is finite and flat.

This open subset W_z does not necessarily contain z , but it contains the generic points of the irreducible components of X_0 passing through z .⁽⁴⁵⁾ Since d_0 (resp. d_1) is faithfully flat and of finite presentation (hence open), it follows from SGA 1, VIII 5.7 that $d_0^{-1}(W_z)$ and $d_1^{-1}(W_z)$ both coincide with the largest open subset of $\tilde{d}'_2(F_2)$ above which $\tilde{d}'_2$ is finite and flat. Consequently, one sees, as in 7.3, that the two inverse images of $\tilde{d}_1^{-1}(W_z)$ under $\tilde{d}'_0$ and $\tilde{d}'_1$ coincide, hence that $\tilde{d}_1^{-1}(W_z)$ has the form $\tilde{d}_0^{-1}(U)$, where U is an open subset of F which is a quasi-section for the groupoid induced by X_* on W_z .

9. Elimination of the noetherian hypotheses in Theorem 7.1

a) Return to the notation and hypotheses of Lemma 6.1, and let $\pi : S' \rightarrow S$ be an arbitrary base change. Denote by $f' : X' \rightarrow Y'$ the morphism of S' -schemes obtained by the extension π of the base from a morphism of S -schemes $f : X \rightarrow Y$. With this convention, $p' : X'_0 \rightarrow Y'$ is surjective, as is the morphism $X'_1 \rightarrow X'_0 \times_{Y'} X'_0$ with components d'_0 and d'_1 . The underlying set of Y' is therefore identified with the quotient of the underlying set of

⁽⁴⁵⁾**N.D.E.** Indeed, let η be such a generic point. The hypotheses imply that $\mathcal{O}_{X_0, \eta}$ is an artinian local ring, and $\mathcal{O}_{F_1, \eta}$ a finitely generated $\mathcal{O}_{X_0, \eta}$ -module. Thus, by SGA 1, VIII 6.5, there exists an open neighborhood of η above which d_1 is finite.

X'_0 by the equivalence relation defined on X'_0 by the S' -groupoid X'_* . Moreover, $q' : U' \rightarrow Y'$ is integral surjective, so the topology of Y' is the quotient topology of that of U' , and thus also of that of X'_0 (cf. the proof in §6.c).

On the other hand, it is clear that U' is a quasi-section of the S' -groupoid X'_* , to which one can therefore apply Lemma 6.1. In particular, X'_* has a cokernel (Y_1, p_1) , and the underlying topological space of Y_1 is obtained from the underlying topological space of X'_0 by identifying the points equivalent for the relation defined by X'_* . It follows that the canonical morphism $Y_1 \rightarrow Y'$ is a homeomorphism; I even say that $Y_1 \rightarrow Y'$ is a universal homeomorphism: indeed, if S'' is over S' , let Y_2 be the cokernel of $(d_0 \times_{S'} S'', d_1 \times_{S'} S'')$. By what precedes, applied to the base changes $S'' \rightarrow S'$ and $S'' \rightarrow S$,

$$Y_2 \rightarrow Y_1 \times_{S'} S'' \quad \text{and} \quad Y_2 \rightarrow Y \times_S S'' \simeq Y' \times_{S'} S''$$

are homeomorphisms, so the same holds for $Y_1 \times_{S'} S'' \rightarrow Y' \times_{S'} S''$.

b) Analogous remarks obviously apply to the case where the groupoid X_* “locally” has quasi-sections (cf. the proof of Theorem 7.1).

⁽⁴⁶⁾For example, one has the following theorem:

Theorem 9.0. *Let S be an arbitrary scheme and $X_2 \xrightarrow{d'_j} X_1 \xrightarrow{d_i} X_0$ a (Sch/S) -groupoid such that: X_0 is of finite presentation and quasi-projective over S , d_1 is of finite presentation, proper and flat, and the morphism $d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times_S X_0$ is quasi-finite. Then:*

- (1) *Every point x of X_0 has an open neighborhood W which is saturated and such that the groupoid induced by X_* on W has a quasi-section.*
- (2) *Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Then Y is a scheme, p a morphism of schemes, and (Y, p) is a cokernel of (d_0, d_1) in (Sch/S) .*
- (3) *p is surjective, open and universally closed.*
- (4) *The morphism $d : X_1 \rightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*
- (5) *If (d_0, d_1) is an equivalence pair, then:*
 - (a) *$d : X_1 \rightarrow X_0 \times_Y X_0$ is an isomorphism and p is faithfully flat.*
 - (b) *p and $Y \rightarrow S$ are of finite presentation, and (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.*

PROOF. For (1), the question is local on S , so one can suppose $S = \text{Spec } B$ affine. There then exist a ring A finitely generated over $\mathbf{Z}$, a morphism $S \rightarrow T = \text{Spec } A$ and a (Sch/T) -groupoid Z_* such that X_* is identified with $Z_* \times_T S$ (cf. EGA IV₃, 8.8.3, applied to $S_0 = \text{Spec } \mathbf{Z}$ and $S_i = \text{Spec } A_i$, the A_i ranging over the finitely generated $\mathbf{Z}$ -subalgebras of B). Moreover, one can suppose that Z_* satisfies the conditions of Theorem 7.1 (cf. EGA IV₃, 8.10.5). Consequently, Z_* “locally” has quasi-sections.

The same therefore holds for X_* by a), and assertions (2), (3), (4) and (5) (a) follow from 6.1, as in the proof of 7.1.

c) Let us show that $Y \rightarrow S$ is of finite presentation.⁽⁴⁷⁾ By hypothesis, $(d_0^{X_*}, d_1^{X_*})$ is an equivalence pair, that is, $d^{X_*} : X_1 \rightarrow X_0 \times_S X_0$ is a monomorphism. By EGA IV₃, 8.10.5, one can suppose, enlarging A if necessary, that $d^{Z_*} : Z_1 \rightarrow Z_0 \times_T Z_0$ is a monomorphism. Since $T = \text{Spec } A$, with A noetherian, it then follows from Theorem 7.1 that the groupoid Z_* has a cokernel (Q, q) in (Sch/T) , that q and $Q \rightarrow T$ are of finite presentation, and moreover that $q : Z_0 \rightarrow Q$ is faithfully flat and d^{Z_*} induces an isomorphism $Z_1 \xrightarrow{\sim} Z_0 \times_Q Z_0$. Put $Q_S = Q \times_T S$.

Since $X_i \simeq Z_i \times_T S$, one therefore obtains an isomorphism:

$$d^{Z_*} \times_T S : X_1 \xrightarrow{\sim} X_0 \times_{Q_S} X_0.$$

⁽⁴⁶⁾**N.D.E.** What follows has been expanded to bring out Theorem 9.0 below.

⁽⁴⁷⁾**N.D.E.** The original states that this follows from Proposition 9.1 below. We have been unable to reconstruct this argument. The proof that follows was indicated to us by O. Gabber.

Denote its inverse by φ , and let π be the canonical morphism

$$X_0 \times_Y X_0 \longrightarrow X_0 \times_{Q_S} X_0.$$

Then $\varphi \circ \pi$ is the inverse of $d_0 \boxtimes d_1 : X_1 \xrightarrow{\sim} X_0 \times_Y X_0$. It follows that the equivalence relation defined by X_* , that is, the monomorphism

$$X_1 \xrightarrow{\sim} X_0 \times_Y X_0 \hookrightarrow X_0 \times_S X_0,$$

is identified with the equivalence relation $R(q_S)$ defined by the morphism $q_S : X_0 \longrightarrow Q_S$. Since the latter is faithfully flat and of finite presentation, hence an effective universal epimorphism, $R(q_S)$ has quotient Q_S (cf. IV 3.3.2). Thus $Y \simeq Q \times_T S$, so $Y \longrightarrow S$ and p are of finite presentation. Moreover, by IV 6.3.3, (Y, p) is also a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology. $\square$

Proposition 9.1. *Consider morphisms of schemes*

$$X_0 \xrightarrow{p} Y \xrightarrow{q} S$$

such that qp is of finite type (resp. of finite presentation) and p faithfully flat of finite presentation. Then q is of finite type (resp. of finite presentation).⁽⁴⁸⁾

Since p is surjective and qp quasi-compact, q is quasi-compact. One can therefore suppose S, Y and X_0 affine with rings A, B, C . One has $B = \varinjlim B_i$, where the B_i range over the finitely generated A -subalgebras of B . Since C is of finite presentation over B , there exist an index i_0 , a finitely presented B_{i_0} -algebra C_{i_0} and an isomorphism $C \simeq C_{i_0} \otimes_{B_{i_0}} B$; if one puts $C_i = C_{i_0} \otimes_{B_{i_0}} B_i$ for $i \geq i_0$, one therefore has $C \simeq C_i \otimes_{B_i} B$.

$$\begin{array}{ccc} B & \longrightarrow & C \\ \uparrow & & \uparrow \\ B_i & \longrightarrow & C_i \\ \uparrow & & \\ A & & \end{array}$$

Since C is faithfully flat over B , one deduces from EGA IV₃, 11.2.6 and 8.10.5 (vi) the existence of an $i_1 \geq i_0$ such that C_{i_1} is faithfully flat over B_{i_1} ; consequently C_i is faithfully flat over B_i for $i \geq i_1$. For $i \geq i_1$, the canonical map $C_i \longrightarrow C$ is then injective, for it is obtained from $B_i \longrightarrow B$ by faithfully flat extension of the base.

If C is of finite type over A , it follows that there exists an index $j \geq i_1$ such that $C_j = C$, whence $B_j = B$, since C_j is faithfully flat over B_j . Consequently B is of finite type over A .

Now suppose C of finite presentation over A . By what precedes, B is of finite type over A , hence of the form $\overline{B}/I$, where $\overline{B}$ is a polynomial algebra over A in finitely many indeterminates and I an ideal of $\overline{B}$. Then I is the union of its finitely generated subideals I_α ; whence $B = \varinjlim B_\alpha$ with $B_\alpha = \overline{B}/I_\alpha$. Proceeding as above, there exist an index α_0 , a finitely presented B_{α_0} -algebra C_{α_0} and an isomorphism $C \simeq C_{\alpha_0} \otimes_{B_{\alpha_0}} B$. For $\alpha \geq \alpha_0$, one again puts $C_\alpha = C_{\alpha_0} \otimes_{B_{\alpha_0}} B_\alpha$, so that $C \simeq C_\alpha \otimes_{B_\alpha} B$ for $\alpha \geq \alpha_0$. Again by EGA IV₃, 11.2.6 and 8.10.5 (vi), one concludes as above that C_α is faithfully flat over B_α for sufficiently large α . In this case, the kernel of the map $C_\alpha \longrightarrow C$ (resp. $C_\alpha \longrightarrow C_\beta$ for $\beta \geq \alpha$) is identified with $C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ (resp. $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha)$).

Since C_α and C are of finite presentation over A and $C_\alpha \longrightarrow C$ is surjective, $C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ is a finitely generated ideal⁽⁴⁹⁾ and is the union of the ideals $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha)$. One therefore has $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) = C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ for sufficiently large β , whence also $I_\beta = I$ (for C_α is faithfully flat over B_α); thus B is of finite presentation over A .

⁽⁴⁸⁾Cf. EGA IV₄, 17.7.5 for a more general result.

⁽⁴⁹⁾**N.D.E.** Cf. EGA IV₁, 1.4.4.

10. Supplement: quotients by a group scheme

Paragraphs 10.2–10.4 below, written following indications of M. Raynaud, aim to apply the preceding theorems to the case of an action of a group scheme. For the reader's convenience, we have begun by reproducing, in §10.1, statements 2.1 to 2.3 of Exp. XVI.

10.1. — Representability theorems for quotients. — Let us first “recall” the following result:

Theorem 10.1.1. *Let S be a scheme, X and Y two S -schemes, and $f : X \rightarrow Y$ an S -morphism. Suppose one is in one of the following two cases:*

- α) The morphism f is locally of finite presentation.*
- β) The scheme S is locally noetherian and X is locally of finite type over S .*

Then the following conditions are equivalent:

- (i) *There exist an S -scheme X' and a factorization of f :*

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S -morphism locally of finite presentation and f'' is a monomorphism.

- (ii) *The (first) projection*

$$p_1 : X \times_Y X \rightarrow X$$

is a flat morphism.

Moreover, if the preceding conditions hold, (X', f') is a quotient of X by the equivalence relation defined by f (for the fppf topology), so the factorization $f = f'' \circ f'$ of i) is unique up to isomorphism.

The case Y locally noetherian, X of finite type over Y , is treated in [Mur65], cor. 2 of th. 2. We shall see that one can reduce to this case.

Let us first make some remarks:

- a) The implication (i) $\Rightarrow$ (ii) is trivial. Indeed, the first projection

$$p'_1 : X \times_{X'} X \rightarrow X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism, since f'' is a monomorphism, and p'_1 is flat, since f' is flat; hence p_1 is flat.

b) The assertions of 10.1.1 are local on Y (hence local on S); they are also local on X , as follows easily from the fact that a flat morphism locally of finite presentation is open (EGA IV₃, 11.3.1).

c) Under the hypotheses of 10.1.1 α), in view of what precedes, we are reduced to the case where X and Y are affine and f of finite presentation. By replacing S by Y if necessary, one can suppose X and Y of finite presentation over S . One then reduces to the case S noetherian by EGA IV₃, 11.2.6.

d) Under the hypotheses of 10.1.1 β), one can suppose S, X, Y affine, S noetherian and X of finite type over S . Regard Y as a filtered projective limit of affine schemes Y_i of finite type over S . The schemes $X \times_{Y_i} X$ form a decreasing filtered family of closed subschemes of $X \times_S X$, whose projective limit is $X \times_Y X$. Since $X \times_S X$ is noetherian, one has $X \times_{Y_i} X = X \times_Y X$ for sufficiently large i , so that $f_i : X \xrightarrow{f} Y \rightarrow Y_i$ satisfies the hypotheses of 10.1.1 ii) if f does. Since the equivalence relation defined by f on X coincides with that defined by f_i , it is clear that it suffices to prove ii) $\Rightarrow$ i) for f_i , which reduces us to the case where Y is of finite type over S .

Application to group schemes. Let S be a scheme and G an S -group scheme locally of finite presentation over S , acting (on the left) on an S -scheme X . If $X \rightarrow S$ has a section ξ , recall that the stabilizer $\text{Stab}_G(\xi)$ is representable by a subgroup scheme of G (cf. I, 2.3.3).

Theorem 10.1.2. *Let S be a scheme and G an S -group scheme locally of finite presentation over S , acting on an S -scheme X .*

Suppose $X \rightarrow S$ has a section ξ whose stabilizer H in G is flat over S . If one of the hypotheses below holds:

- a) X is locally of finite type over S ,
- b) S is locally noetherian,

then the quotient fppf sheaf G/H is representable by an S -scheme locally of finite presentation over S , and the S -morphism

$$f : G \rightarrow X, \quad g \mapsto g \cdot \xi$$

factors as:

$$\begin{array}{ccc} G & & \\ p \downarrow & \searrow f & \\ G/H & \xrightarrow{i} & X, \end{array}$$

where p is the canonical projection, a faithfully flat morphism locally of finite presentation, and i is a monomorphism.

PROOF. The morphism f makes G an X -scheme. By definition of the stabilizer of ξ , the morphism

$$G \times_S H \rightarrow G \times_X G, \quad (g, h) \mapsto (g, gh)$$

is an isomorphism. Since H is flat over S , $G \times_S H$ is flat over G , hence the first projection $p_1 : G \times_S G \rightarrow G$ is a flat morphism. Furthermore, if X is locally of finite type over S , f is locally of finite presentation (EGA IV₁, 1.4.3 (v)); otherwise S is supposed locally noetherian. One then need only apply 10.1.1 to the morphism f . It remains to see that G/H is locally of finite presentation over S , but this follows immediately from 9.1. $\square$

Corollary 10.1.3. *Let S be a scheme and $u : G \rightarrow H$ a morphism of S -group schemes. Suppose G locally of finite presentation over S , and either H locally of finite type over S or S locally noetherian.*

Then, if $K = \text{Ker}(u)$ is flat over S , the quotient group G/K is representable by an S -group scheme locally of finite presentation over S , and u factors as:

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \searrow & & \nearrow i \\ & G/K & \end{array}$$

where p is the canonical projection and i a monomorphism.

Proof: one applies 10.1.2 with $X = H$ and ξ the unit section of H .

10.2. — Stabilizer of the diagonal. — Let S be a noetherian scheme, X an S -scheme of finite type, and G a flat S -group scheme of finite type acting on X on the left, i.e. one has an S -action $d_0 : G \times_S X \rightarrow X$. Denote by $d_1 : G \times_S X \rightarrow X$ the projection onto the second factor. Following §2.a), one has the groupoid

$$G \times_S G \times_S X \begin{array}{c} \xrightarrow{\text{pr}_{2,3}} \\ \xrightarrow[-\mu \times X]{G \times d_0} \end{array} G \times_S X \begin{array}{c} \xrightarrow{d_1} \\ \xrightarrow{d_0} \end{array} X,$$

whose cokernel, if it exists, is denoted $G \backslash X$.

Definition 10.2.1. Denote by $F \subset G \times_S X$ the stabilizer of the diagonal section, i.e. the X -scheme defined by the cartesian product

$$\begin{array}{ccc} F & \longrightarrow & X \\ \downarrow & & \downarrow \Delta \\ G \times_S X & \xrightarrow{(d_0, d_1)} & X \times_S X. \end{array}$$

Then F is a subgroup X -scheme of $G \times_S X$. Since $G \times_S X$ is of finite type over noetherian S , hence noetherian, F is of finite type over S and over X (EGA I, 6.3.5 and 6.3.6). Moreover, if $X \rightarrow S$ is separated, F is a closed subgroup X -scheme of $G \times_S X$.

Recall that one says G acts freely on X if the morphism

$$G \times_S X \xrightarrow{(d_0, d_1)} X \times_S X$$

is a monomorphism (cf. Exp. III, 3.2.1). This amounts to saying that F is the trivial group scheme with base X .

10.3. — Case where F is quasi-finite over X . — Since F is of finite type over X , it is quasi-finite over X if and only if the stabilizers of the geometric points of X are finite.

Theorem 10.3.1. ⁽⁵⁰⁾ *Under the hypotheses of 10.2, suppose F quasi-finite over X . Then there exists a dense G -saturated open subset U of X satisfying the following properties:*

- (i) *In (Sch/S) , the cokernel $V = G \backslash U$ exists; moreover, the scheme V is a quotient in the category of ringed spaces.*
- (ii) *$p : U \rightarrow V$ is surjective, open and of finite presentation.*
- (iii) *V is of finite presentation over S .*
- (iv) *The morphism $G \times_S U \rightarrow U \times_V U$, $(g, x) \mapsto (gx, x)$, is surjective.*
- (v) *Further suppose G acts freely on X . Then $U \rightarrow V$ is a (left) G -torsor locally trivial for the fppf topology. In particular, $U \rightarrow V$ is faithfully flat.⁽⁵¹⁾*

PROOF. One has supposed that the morphism $G \times_S X \rightarrow X \times_S X$, $(g, x) \mapsto (gx, x)$, is quasi-finite. Theorem 8.1 therefore applies to the groupoid defined by (X, G) . Thus there exists a dense saturated open subset $U \subset X$ such that the quotient $G \backslash U$ exists; it satisfies properties (i), (ii), (iii).

To establish (iv), recall that G acts freely on X if and only if (d_0, d_1) is an equivalence pair (III 3.2.1). In this case, Theorem 8.1 (iv) shows that the morphism $G \times_S U \rightarrow U \times_V U$ is an isomorphism and p is faithfully flat and of finite presentation. Thus U is a G -torsor with base V , locally trivial for the fppf topology. $\square$

10.4. — Case where F is flat over X . — Denote by

$$d = (d_0, d_1) : G \times_S X \rightarrow X \times_S X$$

the morphism $d(g, x) = (gx, x)$. Recall that the sheaf graph $\tilde{\Gamma}$ of the equivalence relation associated to (X, G) is the fppf S -subsheaf of $X \times_S X$ that is the image of (d_0, d_1) . It is the fppf sheaf associated to the graph functor:

$$T \mapsto \Gamma(T) = \{(x_0, x_1) \in X(T) \times X(T) \mid x_0 \in G(T)x_1\}.$$

Put $G_X = G \times_S X$. For every S -scheme T , one has a surjective map

$$G_X(T) \rightarrow \Gamma(T), \quad (g, x) \mapsto (gx, x),$$

⁽⁵⁰⁾**N.D.E.** Here too there exists a largest open subset U of X satisfying the conclusions of the theorem, cf. N.D.E. (44).

⁽⁵¹⁾**N.D.E.** If one further supposes that G is a reductive S -group scheme and the (free) action of G on X is linearizable, one then knows that $G \backslash X$ is representable and $X \rightarrow G \backslash X$ is a (left) G -torsor. This follows from results of Raynaud and Seshadri and appears in the article [CTS79] (proposition 6.11).

which induces a bijective map

$$\varphi(T) : G_X(T)/F(T) \xrightarrow{\sim} \Gamma(T);$$

indeed, if $(g, x), (g', x') \in G_X(T)$ satisfy $(gx, x) = (g'x', x')$, then $x' = x$ and $g^{-1}g'x = x$, hence $(g^{-1}g'x, x) \in F(T)$, and (g, x) and (g', x) have the same image in $G_X(T)/F_X(T)$.

By definition (cf. IV, 4.4.1 (ii) or the proof of 5.2.1), the quotient sheaf G_X/F is the fppf sheaf associated to the presheaf

$$T \mapsto G_X(T)/F(T) \simeq \Gamma(T).$$

One therefore has an isomorphism of sheaves $\varphi : G_X/F \rightarrow \tilde{\Gamma}$.

Theorem 10.4.1. ⁽⁵²⁾ *Under the hypotheses of 10.2, one has:*

- a) $\tilde{\Gamma}$ is representable if and only if F is flat over X .
- b) Suppose F flat over X . Then the morphisms induced by d_1 and d_0 :

$$G_X/F \begin{array}{c} \xrightarrow{\bar{d}_1} \\ \xrightarrow{\bar{d}_0} \end{array} X$$

are faithfully flat and of finite presentation.

Proof of a): Suppose the fppf sheaf G_X/F represented by an X -scheme Y . Then, by IV 6.3.3, $p : G_X \rightarrow Y$ is faithfully flat and locally of finite presentation, and the second square of the diagram below is cartesian:

$$\begin{array}{ccccc} F & \longrightarrow & F \times_X G_X & \longrightarrow & G_X \\ \downarrow & & \downarrow & & \downarrow p \\ X & \xrightarrow{e_X} & G_X & \longrightarrow & Y, \end{array}$$

the first square being obtained by base change by the unit section $e_X : X \rightarrow G_X$. Since p is faithfully flat and locally of finite presentation, the same holds for $F \rightarrow X$.

Conversely, suppose F flat over X . Put $X^2 = X \times_S X$. The morphism $d : G_X \rightarrow X^2$ allows one to form the fibered product:

$$\begin{array}{ccc} G_X \times_{X^2} G_X & \longrightarrow & G_X \\ \downarrow & & \downarrow \\ G_X & \longrightarrow & X^2. \end{array}$$

Then the morphism $G_X \times_{X^2} G_X \rightarrow X^2$ is an $F \times_X X^2$ -torsor over X^2 , and is therefore flat and of finite type (since F is). By Theorem 10.1.1, the morphism d factors uniquely:

$$G_X \xrightarrow{\psi} Y \xrightarrow{\tau} X \times_S X,$$

where ψ is faithfully flat (of finite type) and τ is a monomorphism of schemes.

Thus the morphism of sheaves $\psi : G_X \rightarrow Y$ is F -invariant, and one obtains a morphism of sheaves $\bar{\psi} : G_X/F \rightarrow Y$. Furthermore, since ψ is faithfully flat (of finite type), the monomorphism of sheaves τ factors through the image sheaf of d , that is, $\tilde{\Gamma}$. The isomorphism of sheaves $G_X/F \simeq \tilde{\Gamma}$ therefore factors through the monomorphism $Y \rightarrow \tilde{\Gamma}$. One concludes that Y represents G_X/F .

Proof of b): Suppose F flat over X . Then, by a) and its proof, G_X/F is representable, and the morphism $p : G_X \rightarrow G_X/F$ is faithfully flat and of finite presentation. On the other hand, the morphisms $d_i : G_X \rightarrow X$ ($i = 0, 1$) are faithfully flat and of finite presentation by hypothesis. Since $d_i = \bar{d}_i \circ p$, it follows from EGA IV₂, 2.2.13 (iii) and EGA IV₃, 11.3.16 that $\bar{d}_i$ is faithfully flat and of finite presentation.

⁽⁵²⁾**N.D.E.** This is point (2) of Theorem 3 of [Ray67b]. Another proof of th. 10.1.1 is sketched in this Note.

Theorem 10.4.2. ⁽⁵³⁾ *Under the hypotheses of 10.2, suppose F flat over X . Then there exists a dense saturated open subset U of X such that the quotient fppf sheaf $V = G \backslash U$ is an S -scheme of finite type and $U \rightarrow V$ is faithfully flat and of finite presentation.*

PROOF. Theorem 10.4.1 shows that $G_X/F \simeq \tilde{\Gamma}$ is representable. Then the fppf sheaf $G \backslash X$ is identified with the quotient sheaf of

$$G_X/F \xrightarrow[\bar{d}_0]{\bar{d}_1} X.$$

By what precedes, $\bar{d}_i : G_X/F \rightarrow X$ is faithfully flat and of finite presentation ($i = 0, 1$), and the morphism

$$G_X/F \xrightarrow{\sim} \tilde{\Gamma} \hookrightarrow X \times_S X$$

is a monomorphism, that is, $(\bar{d}_0, \bar{d}_1)$ is an equivalence pair. Consequently, Theorem 8.1 applies. There therefore exists a dense saturated open subset U of X such that the quotient fppf sheaf $V = G \backslash U$ is an S -scheme of finite type and $U \rightarrow V$ is faithfully flat and of finite presentation. $\square$

Taking account of the generic flatness theorem (EGA IV₂, 6.9.3), one obtains the

Corollary 10.4.3. *Under the hypotheses of 10.2, suppose X reduced. Then there exists a dense saturated open subset U of X such that the quotient fppf sheaf $G \backslash U$ is an S -scheme of finite type and $U \rightarrow G \backslash U$ is faithfully flat and of finite presentation.*

⁽⁵³⁾**N.D.E.** Here too there exists a largest open subset U of X satisfying the conclusions of the theorem; moreover, a point $x \in X$ of codimension 1 in X belongs to U if and only if the morphism $(G_X/F) \times_X \text{Spec}(\mathcal{O}_{X,x}) \rightarrow \text{Spec}(\mathcal{O}_{X,x}) \times_S \text{Spec}(\mathcal{O}_{X,x})$ is a closed immersion, cf. N.D.E. (44).

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⁽⁵⁴⁾N.D.E. Additional references cited in this exposé.

Exposé VIA. Generalities on algebraic groups

by P. Gabriel

⁽¹⁾Throughout this chapter, A will denote an artinian local ring with residue field k . A group scheme G over $\mathrm{Spec} A$ will simply be called an A -group. This A -group is said to be locally of finite type if the underlying scheme is locally of finite type over A ; it is said to be algebraic if the underlying scheme is of finite type over A .

0. Preliminary remarks

0.1. — First consider a group scheme G over any scheme S . We call the structure morphism $\mu : G \times_S G \rightarrow G$ multiplication, and the morphism $c : G \rightarrow G$ defined by the equalities $c(T)(x) = x^{-1}$ inversion (T being a scheme over S and x an element of $G(T)$). If U and V are subsets of the underlying set of G , we denote by $U \cdot V$ the image under the multiplication morphism of the subset of $G \times_S G$ consisting of the points whose first projection belongs to U and second to V . Similarly, the notations U^{-1} and $c(U)$ are equivalent.

Let pr_1 be the projection of $G \times_S G$ onto the first factor, and $\sigma : G \times_S G \rightarrow G \times_S G$ the morphism with components pr_1 and μ . For every S -scheme T , $\sigma(T)$ is the map $(x, y) \mapsto (x, xy)$; it follows that σ is an automorphism. The composite of this automorphism with the projection pr_2 of $G \times_S G$ onto the second factor is the multiplication morphism. When G is flat over S , pr_2 , and hence μ , are flat morphisms; when G is smooth over S , pr_2 , and hence μ , are smooth morphisms, etc.

0.2. — From now on we suppose that S is the spectrum of an artinian local ring A with residue field k . We denote by $(\mathrm{Sch}/k)_{\mathrm{red}}$ the category of reduced schemes over k . For every scheme X over A , the reduced scheme X_{red} is an object of $(\mathrm{Sch}/k)_{\mathrm{red}}$, and the functor $X \mapsto X_{\mathrm{red}}$ is right adjoint to the inclusion of $(\mathrm{Sch}/k)_{\mathrm{red}}$ into (Sch/A) . It follows that, for every A -group G , G_{red} is a group in the category $(\mathrm{Sch}/k)_{\mathrm{red}}$,⁽²⁾ that is, for every reduced k -scheme T , $G_{\mathrm{red}}(T)$ has a group structure functorial in T . One must beware that G_{red} is not necessarily a k -group, for “multiplication” is only a morphism from $(G_{\mathrm{red}} \times_k G_{\mathrm{red}})_{\mathrm{red}}$ to G_{red} .⁽³⁾

⁽⁴⁾However, if k is a perfect field, the inclusion of $(\mathrm{Sch}/k)_{\mathrm{red}}$ into (Sch/k) commutes with products, so that groups in the category $(\mathrm{Sch}/k)_{\mathrm{red}}$ are identified with group schemes over k whose underlying scheme is reduced. In this case, if G is a k -group, G_{red} is a subgroup scheme of G ; this subgroup is not in general invariant in G .

For example, if k is a field of characteristic 3, the constant group $(\mathbf{Z}/2\mathbf{Z})_k$ acts nontrivially on the diagonalizable group $D_k(\mathbf{Z}/3\mathbf{Z})$ (cf. Exp. I, 4.1 and 4.4); if G denotes the semidirect product of $D_k(\mathbf{Z}/3\mathbf{Z})$ by $(\mathbf{Z}/2\mathbf{Z})_k$ defined by this action, G_{red} is identified with $(\mathbf{Z}/2\mathbf{Z})_k$ and is not invariant in G .⁽⁵⁾

⁽¹⁾**N.D.E.** Version of 13 October 2024.

⁽²⁾**N.D.E.** The following sentences have been added.

⁽³⁾**N.D.E.** For examples of group schemes G over an imperfect field k such that G_{red} is not a k -group scheme, see 1.3.2 below.

⁽⁴⁾**N.D.E.** What follows in 0.2, as well as 0.3, has been slightly modified.

⁽⁵⁾**N.D.E.** For an analogous example with G connected, one can consider, for $\mathrm{char}(k) = p > 0$, the semidirect product of $\alpha_{p,k}$ and $\mathbf{G}_{m,k}$.

Let k be any field, $k^{p^{-\infty}}$ its perfect closure, and H a group in the category $(\text{Sch}/k)_{\text{red}}$. Then $(H \otimes_k k^{p^{-\infty}})_{\text{red}}$ is a group scheme over the perfect field $k^{p^{-\infty}}$. Since $H \otimes_k k^{p^{-\infty}}$ and $(H \otimes_k k^{p^{-\infty}})_{\text{red}}$ have the same underlying topological space, one sees that groups in $(\text{Sch}/k)_{\text{red}}$ share with k -groups certain topological properties invariant under extension of the base field: for example, it will follow from 0.3 and the remarks just made that every group in $(\text{Sch}/k)_{\text{red}}$ is separated.

We shall encounter groups in $(\text{Sch}/k)_{\text{red}}$ below whenever we deal with a nonempty locally closed subset U of an A -group G such that $U \cdot U = U$ and $U^{-1} = U$: indeed, the reduced subscheme of G defined by U is a group in $(\text{Sch}/k)_{\text{red}}$.

0.3. — An A -group G is always separated, for the unit section $e : \text{Spec } A \rightarrow G$ is a closed immersion. Indeed, let x be the unique point of $\text{Spec } A$ and η the structure morphism $G \rightarrow \text{Spec } A$. Since $\eta \circ e = \text{id}_{\text{Spec } A}$, for every affine open subset $U = \text{Spec } B$ of G containing $e(x)$ the morphism $B \rightarrow A$ has a section, hence is surjective. It follows that e is a closed immersion.⁽⁶⁾ Now the diagonal of $G \times_A G$ is identified with the functor from $(\text{Sch}/A)^\circ$ to (Ens) assigning to every scheme S over A the inverse image of the neutral element of $G(S)$ under the map $\varphi(S) : (x, y) \mapsto x \cdot y^{-1}$ from $G(S) \times G(S)$ to $G(S)$. One consequently has the cartesian square below, so the diagonal morphism, obtained from a closed immersion by base change, is itself a closed immersion:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\varphi} & G \\ \uparrow \text{diagonal morphism} & & \uparrow \text{unit section} \\ G & \longrightarrow & \text{Spec } A. \end{array}$$

0.4. — ⁽⁷⁾Let G be an A -scheme. We shall say that a point g of G is strictly rational over A if there exists an A -morphism $s : \text{Spec } A \rightarrow G$ sending the only point of $\text{Spec } A$ to g , i.e. if the morphism $A \rightarrow \mathcal{O}_{G,g}$ admits a retraction. Note that one then has $\kappa(g) = k$, and therefore g is a closed point of G (if B is the ring of an affine open neighborhood of g and $\mathfrak{p}$ the prime ideal of B corresponding to g , then $B/\mathfrak{p} \subset k$ is a finite A -algebra, hence an integral artinian ring, hence a field).

Henceforth suppose G an A -group; then such an $s : \text{Spec } A \rightarrow G$ defines an automorphism r_s of the scheme G over A , called right translation by s : for every morphism $\pi : S \rightarrow \text{Spec } A$, $r_s(\pi)$ is the automorphism of $G(S)$ defined by $x \mapsto x \cdot G(\pi)(s)$ for every $x \in G(S)$. Similarly, ℓ_s denotes left translation by s , that is, the automorphism of G defined by $\ell_s(\pi)(x) = G(\pi)(s) \cdot x$ for every $x \in G(S)$.

Since $G \otimes_A k$ and G have the same underlying topological space $\underline{G}$, $G \otimes_A k$ is a k -group, and $s \otimes_A k$ depends only on g and not on s , one sees that the automorphisms of $\underline{G}$ induced by r_s and ℓ_s (or by $r_{s \otimes k}$ and $\ell_{s \otimes k}$) depend only on g and not on s ; when P is a subset of $\underline{G}$, we can therefore write $r_g(P)$ or $P \cdot g$ (resp. $\ell_g(P)$ or $g \cdot P$) instead of $r_s(P)$ (resp. $\ell_s(P)$), consistently with 0.1.

Remark 0.4.1. If g is a strictly rational point of G and $A \rightarrow A'$ is a morphism of artinian local rings, then $G' = G \otimes_A A'$ has a unique point g' above g , and g' is strictly rational over A' ; moreover, if P' denotes the inverse image of P in G' , then $P' \cdot g'$ is the inverse image of $P \cdot g$ (cf. EGA I, 3.4.8).

⁽⁶⁾**N.D.E.** This argument holds for any zero-dimensional local ring A , and shows that if S is a discrete scheme, every S -group is separated, cf. VIB, 5.2; on the other hand (cf. VIB, 5.6.4), if S contains a nonisolated closed point s , the S -scheme G obtained by gluing two copies of S along the open subset $S - \{s\}$ is not separated over S , but has an S -group structure.

⁽⁷⁾**N.D.E.** The original has been expanded in what follows; in particular, Remark 0.4.1 has been added.

Proposition 0.5. ⁽⁸⁾ *Let G be an A -group and U, V two dense open subsets of G . Then $U \cdot V$ (i.e. the image of $U \times_A V$ under multiplication) equals the whole underlying space of G .*

Indeed, since G and $G \otimes_A k$ have the same underlying space, one can suppose, replacing A by k and G by $G \otimes_A k$ if necessary, that $A = k$. Let $g \in G$. Put $K = \kappa(g)$; then left translation ℓ_g is an automorphism of G_K . Since the projection $G_K \rightarrow G$ is open, U_K and V_K are dense open subsets of G_K , as is the image of V_K under λ_g . There therefore exists $v \in V_K$ such that $u = \ell_g(v)$ belongs to U_K . Let L be an extension of K containing $\kappa(v)$, and hence $\kappa(u)$, and let g_L and v_L be the L -points of G_L obtained from g and v . Then $g_L \cdot v_L = u'$ is a point of G_L above u , and therefore $g_L = u' \cdot v_L^{-1}$ lies above $U \cdot V$, whence $g \in U \cdot V$, which proves the proposition.

Corollary 0.5.1. ⁽⁹⁾ *If G is an irreducible A -group, G is quasi-compact.*

Indeed, let U be a nonempty affine open subset of G ; then U is dense in G , so, by 0.5, the morphism $\mu : U \times_A U \rightarrow G$ is surjective, hence G is quasi-compact since $U \times_A U$ is.

Corollary 0.5.2. ⁽¹⁰⁾ *Let G be an A -group scheme and H a subgroup A -scheme of G . Then H is closed.*

PROOF. Let k' be the perfect closure of the residue field k of A . Since the underlying topological spaces of G and H are unchanged by the base change $A \rightarrow k \rightarrow k'$, one can suppose $A = k$ a perfect field. One can then suppose G and H reduced, hence geometrically reduced.

Let $\overline{H}$ be the closure of H ; then $\mu^{-1}(\overline{H})$ is a closed subset of $G \times G$ containing $H \times H$. Since the morphism $H \rightarrow \text{Spec } k$ (resp. $\overline{H} \rightarrow \text{Spec } k$) is universally open, and H is dense in $\overline{H}$, $H \times H$ is dense in $H \times \overline{H}$, and $H \times \overline{H}$ is dense in $\overline{H} \times \overline{H}$, so $H \times H$ is dense in $\overline{H} \times \overline{H}$. Thus $\mu(\overline{H} \times \overline{H}) \subset \overline{H}$, and hence, since $\overline{H} \times \overline{H}$ is reduced, μ induces a morphism $\mu' : \overline{H} \times \overline{H} \rightarrow \overline{H}$.

Let, then, $g \in \overline{H}$ and put $K = \kappa(g)$. Since the projection $\overline{H}_K \rightarrow \overline{H}$ is open, H_K and $\ell_g(H_K)$ are two dense open subsets of $\overline{H}_K$, so there exist $u, v \in H_K$ such that $\ell(g)(v) = u$. One concludes, as in the proof of 0.5, that g belongs to $H \cdot H = H$, whence $\overline{H} = H$. $\square$

1. Local properties of an A -group locally of finite type

We shall first see that, if G is locally of finite type and flat over A , one can “make every closed point of G strictly rational” by a finite flat extension of the base.

1.1. — Unless expressly stated otherwise, we shall henceforth suppose G an A -group locally of finite type. When A is a field k , we shall obtain in Exposé VIIIB very precise results on the local rings of G .⁽¹¹⁾ Here we content ourselves with some elementary results:

Proposition 1.1.1. *Let x be a point of an A -group G locally of finite type and flat over A . Then the local ring $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and there exists a system $a_1, \dots, a_n$ of parameters of $\mathcal{O}_{G,x}$ such that $\mathcal{O}_{G,x}/(a_1, \dots, a_n)$ is a finite flat (hence finite free) A -module.*

We first suppose A equal to its residue field k ; it then suffices to prove that $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and one can restrict to the case where x is a closed point (cf. EGA 0_{IV}, 16.5.13). By Lemma 1.1.2 below, G contains a closed point y such that $\mathcal{O}_{G,y}$ is Cohen–Macaulay. By

⁽⁸⁾**N.D.E.** The original stated this result under the hypothesis that G is locally of finite type over A . Since it is useful to have it in the general case, and the proof is essentially the same, the result has been stated and proved in the general case. This will be used several times below.

⁽⁹⁾**N.D.E.** This corollary has been inserted here, cf. 2.4 and 2.6.2 below.

⁽¹⁰⁾**N.D.E.** This corollary, used implicitly in VIII, 6.7, has been added; see also VIB, 6.2.5.

⁽¹¹⁾**N.D.E.** For example, they are always complete intersections, cf. VIIB, 5.5.1. Moreover, if $\text{char}(k) = 0$, G is smooth (VIB, 1.6.1; see also VIIB, 3.3.1).

SGA 1, I §9, this amounts to saying that, for every finite extension K of the base field k and every point $\bar{y}$ of $\bar{G} = G \otimes_k K$ above y , $\mathcal{O}_{\bar{G}, \bar{y}}$ is Cohen–Macaulay. If the extension K has been chosen sufficiently large, i.e. if K contains a normal extension of k containing the residue fields $\kappa(y)$ and $\kappa(x)$, then $\bar{y}$ is (strictly) rational over K , as is every point $\bar{x}$ of $\bar{G}$ above x .⁽¹²⁾ Since the automorphism $r_{\bar{x}} \circ (r_{\bar{y}})^{-1}$ sends $\bar{y}$ to $\bar{x}$, it follows that $\mathcal{O}_{\bar{G}, \bar{x}}$, and hence $\mathcal{O}_{G, x}$ (SGA 1, I §9), are Cohen–Macaulay.

When A is again arbitrary, the preceding argument applies to $k \otimes_A G$, so $k \otimes_A \mathcal{O}_{G, x}$ is Cohen–Macaulay. If $a_1, \dots, a_n$ is a sequence of elements of $\mathcal{O}_{G, x}$ whose image in $k \otimes_A \mathcal{O}_{G, x}$ is a system of parameters, it follows from SGA 1, IV 5.7 or EGA 0_{IV}, 15.1.16 that $a_1, \dots, a_n$ is an $\mathcal{O}_{G, x}$ -regular sequence and $\mathcal{O}_{G, x}/(a_1, \dots, a_n)$ is finite flat (hence finite free) over A .

Lemma 1.1.2. *Every nonempty scheme X locally of finite type over an artinian ring A contains a closed point x whose local ring is Cohen–Macaulay.*

One can obviously suppose X affine with algebra B and argue by induction on $\dim X$ (the assertion is clear if X is discrete, all local rings then being artinian). Since B is of finite type over A , if $\dim B > 0$, B contains a noninvertible element a that is not a zero divisor.⁽¹³⁾ The closed subscheme $X' = \operatorname{Spec} B/(a)$ of X then has dimension strictly smaller than $\dim X$, and contains, by the induction hypothesis, a closed point x such that $\mathcal{O}_{X', x}$ is Cohen–Macaulay. Since $\mathcal{O}_{X', x} = \mathcal{O}_{X, x}/(a)$ and a is noninvertible and not a zero divisor in $\mathcal{O}_{X, x}$, $\mathcal{O}_{X, x}$ is Cohen–Macaulay (see also EGA IV₂, 6.11.3).

Proposition 1.2. *Let A be an artinian local ring, G an A -group locally of finite type and flat over A , and x a closed point of G . There then exists a local A -algebra A' , finite and free over A , such that every point of $G \otimes_A A'$ above x is strictly rational over A' .*

⁽¹⁴⁾Indeed, let k_1 be a normal extension of finite degree of k containing the residue field $\kappa(x)$ of x . By Lemma V 4.1.2, there exists a local A -algebra A_1 , finite and free over A , with residue field k_1 . In this case (cf. N.D.E. (11)), all points $g_1, \dots, g_n$ of $G \otimes_A A_1$ above $x \in G$ have residue field k_1 (i.e. $g_1, \dots, g_n$ are rational over A_1 in the sense of Exp. V, §4.e)).

Let, then, $B_1, \dots, B_n$ be the local rings of $g_1, \dots, g_n$. By 1.1.1, $B_1, \dots, B_n$ have quotients $B'_1, \dots, B'_n$ which are artinian and finite free over A_1 . Put $A' = B'_1 \otimes_{A_1} \dots \otimes_{A_1} B'_n$. Then A' is local, finite and free over A_1 , and, for each $i = 1, \dots, n$, one has surjective homomorphisms

$$B_i \otimes_{A_1} A' \twoheadrightarrow B'_i \otimes_{A_1} A' \twoheadrightarrow A',$$

the second being induced by the multiplication map $B'_i \otimes_{A_1} B'_i \rightarrow B'_i$. Consequently A' answers the question.

1.3. — Let e be the neutral element (or origin) of G , that is, the image of the only point of $\operatorname{Spec} A$ under the unit section $\operatorname{Spec} A \rightarrow G$. By the very definition, e is strictly rational over A .

Proposition 1.3.1. ⁽¹⁵⁾ *Let G be a group locally of finite type and flat over an artinian ring A , and let K (resp. $\bar{k}$) be the perfect closure (resp. an algebraic closure) of the residue field k of A .*

⁽¹²⁾N.D.E. Indeed, the hypothesis on K implies that, for every extension L of K , every k -morphism $\kappa(x) \rightarrow L$ (resp. $\kappa(y) \rightarrow L$) factors through K ; consequently all points of $G \otimes_k K$ above x or y have residue field K , i.e. are (strictly) rational over K .

⁽¹³⁾N.D.E. Indeed, B is a noetherian Jacobson ring (cf. [BAC], V §3.4). If every noninvertible element is a zero divisor, every prime ideal is an associated prime ideal of B , so, in particular, B has only finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$. Since B is a Jacobson ring, the intersection of the $\mathfrak{m}_i$ is the nilradical of B , and it follows that each $\mathfrak{m}_i$ is a minimal prime ideal of B , so $\dim B = 0$.

⁽¹⁴⁾N.D.E. The original has been expanded in what follows.

⁽¹⁵⁾N.D.E. Point (1), useful in the examples 1.3.2 below, has been added.

(1) For every closed point x of $\overline{G} = G \otimes_k \overline{k}$, the local rings $\mathcal{O}_{\overline{G},e}$ and $\mathcal{O}_{\overline{G},x}$ are isomorphic. In particular, the tangent spaces $T_e \overline{G}$ and $T_x \overline{G}$ have the same dimension.

(2) The following assertions are equivalent:

- (i) $G \otimes_A K$ is reduced.
- (i bis) $\mathcal{O}_{G,e} \otimes_A K$ is reduced.
- (ii) G is smooth over A .
- (ii bis) G is smooth over A at the origin.

PROOF. (1) Let x be a closed point of $\overline{G}$; there is exactly one $\overline{k}$ -morphism $s : \text{Spec } \overline{k} \rightarrow \overline{G}$ whose image is x ; right translation r_s then induces an isomorphism from $\mathcal{O}_{\overline{G},e} = \mathcal{O}_{G,e} \otimes_k \overline{k}$ onto $\mathcal{O}_{\overline{G},x}$, whence assertion (1).

Let us prove assertion (2). By means of SGA 1, II.2.1, one immediately reduces to the case where A is a field ($A = k$). The implications (i) $\Rightarrow$ (i bis), (ii) $\Rightarrow$ (ii bis), (ii) $\Rightarrow$ (i) and (ii bis) $\Rightarrow$ (i bis) are obvious, so it suffices to prove that (i bis) implies (ii).

Now it follows from (i bis) that $\mathcal{O}_{G,e} \otimes_k \overline{k}$ is reduced. Thus, by (1), $\mathcal{O}_{\overline{G},x}$ is reduced for every closed point x of $\overline{G}$, so $\overline{G}$ is reduced. Hence, since $\overline{G}$ is locally of finite type over $\overline{k}$, there exists at least one closed point y such that $\mathcal{O}_{\overline{G},y}$ is regular. Since, by (1), the local rings of the closed points of $\overline{G}$ are all isomorphic to $\mathcal{O}_{\overline{G},e}$, one sees that all these local rings are regular, so $\overline{G}$ is smooth over $\overline{k}$, and therefore G smooth over k . $\square$

⁽¹⁶⁾One can now give the examples below, pointed out by M. Raynaud, of group schemes G over an imperfect field k such that G_{red} is not a k -group scheme.

Examples 1.3.2. Let k be an imperfect field of characteristic $p > 0$, $t \in k - k^p$, $\overline{k}$ an algebraic closure of k , and $\alpha \in \overline{k}$ such that $\alpha^p = t$.

(1) Consider the additive group $\mathbf{G}_{a,k} = \text{Spec } k[X]$, and let G be the subgroup scheme, finite over k , defined by the additive polynomial $X^{p^2} - tX^p$. Then

$$G_{\text{red}} = \text{Spec } k[X]/X(X^{p(p-1)} - t)$$

is étale at the origin. If it were a k -group scheme, it would be smooth (by 1.3.1); now G_{red} is not geometrically reduced, so it is not a k -group scheme.

(2) Consider $\mathbf{G}_{a,k}^4 = \text{Spec } k[X, Y, U, V]$, and let G be the subgroup scheme defined by the ideal I generated by the additive polynomials $P = X^p - tY^p$ and $Q = U^p - tV^p$. Then G has dimension 2 and is irreducible, for $(G_{\overline{k}})_{\text{red}} \simeq \text{Spec } \overline{k}[Y, V]$ is so.

Let $A = k[X, Y, U, V]$ and $\mathfrak{m}$ its augmentation ideal. Denote by x, y, u, v the images of dX, dY, dU, dV in $\Omega_{A/k}^1 \otimes_A (A/\mathfrak{m})$, regarded as linear forms on the tangent space $k^4 = T_0 \mathbf{G}_{a,k}^4$. Let us show that the subspace $E = T_0 G_{\text{red}}$ equals k^4 . Otherwise there would exist a linear form $f = ax + by + a'u + b'v$, with $a, b, a', b' \in k$ not all zero, vanishing on E . Recall that formation of $\Omega_{A/k}^1$ (and therefore of tangent spaces) commutes with base change (cf. EGA IV₄, 16.4.5), and identify f with its image in $(\overline{k}^4)^*$. Since $(G_{\overline{k}})_{\text{red}} \subset (G_{\text{red}})_{\overline{k}}$, f vanishes on the subspace $T_0 (G_{\overline{k}})_{\text{red}}$ of $\overline{k}^4$, defined by the equations $g_1 = x - \alpha y$ and $g_2 = u - \alpha v$, and hence $f = \lambda g_1 + \mu g_2$, with $\lambda, \mu \in \overline{k}$. Now $\lambda g_1 + \mu g_2$ belongs to k^4 only if $\lambda = \mu = 0$! This contradiction shows that $E = k^4$, and therefore $T_0 (G_{\text{red}})_{\overline{k}} = \overline{k}^4$.

On the other hand, $R = XV - YU$ belongs to $\sqrt{I}$, for $R^p = (X^p - tY^p)V^p - Y^p(U^p - tV^p)$. Consequently, the tangent space F at the point $(\alpha, 1, \alpha, 1)$ of $(G_{\text{red}})_{\overline{k}}$ is contained in the hyperplane H of $\overline{k}^4$ with equation $\alpha dV + dX - dU - \alpha dY = 0$, hence has dimension ≤ 3 .⁽¹⁷⁾ Thus, by point (1) of 1.3.1, G_{red} is not a k -group scheme.

⁽¹⁶⁾**N.D.E.** The following sentence and examples 1.3.2 have been added.

⁽¹⁷⁾**N.D.E.** In fact, one sees without difficulty that $\sqrt{I}$ is generated by P, Q, R at every point $\neq 0$, so $F = H$.

2. Connected components of an A -group locally of finite type

2.1. — First consider any A -group G , and let G' be the connected component of the origin e of G . This connected component is obviously closed, so we can identify it with the reduced closed subscheme of G having G' as its underlying space.⁽¹⁸⁾

Proposition 2.1.1. *For every extension K of the residue field k of A , $G' \otimes_A K$ has as its underlying space the connected component of the origin in the K -group $G \otimes_A K$ (i.e. G' is geometrically connected).*

Indeed, let $(G \otimes_A K)'$ be the connected component of the origin in $G \otimes_A K$. Since the image of $(G \otimes_A K)'$ in G is connected and contains the neutral element of G , this image is contained in G' , so $(G \otimes_A K)'$ is contained in the inverse image $G' \otimes_A K$ of G' in $G \otimes_A K$. The proposition therefore follows from connectedness of $G' \otimes_A K$, proved in Lemma 2.1.2:

Lemma 2.1.2. *Let X and Y be two connected schemes over a field k . If X contains a rational point, $X \times_k Y$ is connected.*

We give below a direct proof of this result of EGA IV₂ (4.5.8 and 4.5.14).

First suppose Y nonempty, connected and affine with algebra B . In this case, $X \times_k Y$ is the spectrum of the quasi-coherent $\mathcal{O}_X$ -algebra $\mathcal{B} = \mathcal{O}_X \otimes_k B$. We want to show that every open, closed and nonempty subset U of $X \times_k Y$ coincides with $X \times_k Y$. Now U is affine over X and has as its affine $\mathcal{O}_X$ -algebra a direct factor of $\mathcal{B}$. It therefore follows from Lemma 2.1.3 below that the image of U in X is open and closed, i.e. coincides with the whole of X . This image contains, in particular, a rational point x of X , so U meets the inverse image of x in $X \times_k Y$. Since the latter is isomorphic to Y , hence connected, U contains this inverse image. The same result would hold for the complement of U in $X \times_k Y$ if U were distinct from $X \times_k Y$, which would be absurd.

If Y is now any k -scheme, what precedes shows that the fibers of the canonical projection $X \times_k Y \rightarrow Y$ are connected. If x is a rational point of X , these fibers all meet the subscheme $\{x\} \times_k Y$, which is itself connected, whence the proposition.

Lemma 2.1.3. *Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$ -algebra which is locally⁽¹⁹⁾ a direct factor of a free $\mathcal{O}_X$ -module. The image of $\text{Spec } \mathcal{A}$ in X is then open and closed.*

Let V be this image. It is clear that V is contained in the support of $\mathcal{A}$. Conversely, if x belongs to the support of $\mathcal{A}$, $\mathcal{A}_x$ is nonzero and is a free $\mathcal{O}_{X,x}$ -module, since, by Kaplansky, every projective module over a local ring is free. Consequently, the fiber of $\text{Spec } \mathcal{A}$ at x , affine with algebra $\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, is nonempty. Thus the image of $\text{Spec } \mathcal{A}$ coincides with the support of $\mathcal{A}$.

If s is the unit section of $\mathcal{A}$, the equality $s_x = 0$ implies that s , and hence $\mathcal{A}$, are zero in a neighborhood of x . Thus the support of $\mathcal{A}$ is closed. On the other hand, one can suppose $\mathcal{A}$ a direct factor of the direct sum L of a family $(\mathcal{O}_X^\alpha)$ of copies of $\mathcal{O}_X$.⁽²⁰⁾ For every β , denote by φ^β the restriction to $\mathcal{A}$ of the canonical projection of $L = \bigoplus_\alpha \mathcal{O}_X^\alpha$ onto $\mathcal{O}_X^\beta$. For $x \in X$, denote by $\mathcal{A}_x^*$ the dual module $\text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x})$. Since $\mathcal{A}_x$ is a direct factor of L_x , the restriction map

$$\text{Hom}_{\mathcal{O}_{X,x}}(L_x, \mathcal{O}_{X,x}) \longrightarrow \text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}) = \mathcal{A}_x^*$$

is surjective. If $\mathcal{A}_x \neq 0$, since $\mathcal{A}_x$ is free and nonzero, there exist $a \in \mathcal{A}_x$ and a linear form $\phi \in \mathcal{A}_x^*$ such that $\phi(a) = 1$. There therefore exists a family (ξ^α) of elements of $\mathcal{O}_{X,x}$ such that, for every $u \in \mathcal{A}_{X,x}$, one has $\phi(u) = \sum_\alpha \xi^\alpha \varphi^\alpha(u)$ (this sum being finite, since $\varphi^\alpha(u) = 0$ except

⁽¹⁸⁾**N.D.E.** One will see below (2.2) that G' is a group in the category $(\text{Sch}/k)_{\text{red}}$.

⁽¹⁹⁾**N.D.E.** The word “locally” has been added. Furthermore, the reference to Kaplansky is: *Projective modules*, Ann. of Maths. **68** (1958), 372–377; see also [BAC], §II.3, Ex. 3.

⁽²⁰⁾**N.D.E.** The original has been expanded in what follows.

for finitely many indices). Applying this to $u = a$, one obtains that there exist $\alpha_1, \dots, \alpha_n$ such that

$$1 = \phi(a) = \xi^{\alpha_1} \varphi^{\alpha_1}(a) + \dots + \xi^{\alpha_n} \varphi^{\alpha_n}(a).$$

There then exists an open neighborhood U of x such that a and the ξ^{α_i} come from sections $\tilde{a} \in \mathcal{A}(U)$ and $\tilde{\xi}^{\alpha_i} \in \mathcal{O}_X(U)$, and the equality $\sum_i \tilde{\xi}^{\alpha_i} \varphi^{\alpha_i}(\tilde{a}) = 1$ on U shows that $\tilde{a}_y \neq 0$ for every $y \in U$. (“The support of a projective module is open.”)

2.2. — The notation still being that of 2.1, it is clear that G' is a reduced k -scheme. Lemma 2.1.2 shows that $G' \times_k G'$ is connected, so $(G' \times_k G')_{\text{red}}$ is the reduced subscheme of $G \times_A G$ whose underlying space is the connected component of the origin. In particular, the multiplication morphism $\mu : G \times_A G \rightarrow G$ induces a morphism $\mu' : (G' \times_k G')_{\text{red}} \rightarrow G'$ making G' a group in $(\text{Sch}/k)_{\text{red}}$.

2.2.bis. — ⁽²¹⁾Recall (cf. Exp. V) that, if P is a scheme, $\underline{P}$ denotes its underlying topological space. One then defines an A -subfunctor G^0 of G by putting, for every A -scheme S ,

$$G^0(S) = \{u \in G(S) \mid \underline{u}(\underline{S}) \subset \underline{G'}\}.$$

Let $c : G \rightarrow G$ be the inversion morphism; since $c(G') = G'$, one has $c \circ u \in G^0(S)$ for every $u \in G^0(S)$. On the other hand, if $u, v \in G^0(S)$, $u \boxtimes v$ sends $\underline{S}$ into the subspace of $G \times_A G$ consisting of the points whose two projections belong to G' ; this subspace is identified with the underlying space of $G' \times_A G'$, connected by Lemma 2.1.2. Consequently, $\mu \circ (u \boxtimes v)$ sends $\underline{S}$ into G' . This shows that G^0 is an A -subgroup functor of G .

If the connected component of e is an open subset of G , the subfunctor G^0 is represented by the subscheme induced by G on this open subset, which is therefore a subgroup scheme of G ; it will also be denoted G^0 . In this case, with the notation of 2.1, one has $G' = (G^0)_{\text{red}}$, and the topological spaces G' and G^0 coincide.⁽²²⁾

2.3. — In accordance with our conventions in 1.1, we again suppose from now on that G is locally of finite type over A . Then G is locally noetherian, hence locally connected,⁽²³⁾ hence: every connected component of G is open.

We shall then denote by G^0 the subscheme induced by G on the connected component G' of the neutral element. By 2.2.bis, G^0 is a subgroup scheme of G which we shall call the neutral component of G ; for every A -scheme S one therefore has:

$$G^0(S) = \{u \in G(S) \mid \underline{u}(\underline{S}) \subset \underline{G^0} = \underline{G'}\}.$$

Let G^α be any connected component of G , and $\nu^\alpha : G^\alpha \times_A G^0 \rightarrow G$ the morphism defined by the equalities

$$\nu^\alpha(S)(g, \gamma) = g\gamma g^{-1}$$

for every $S \in (\text{Sch}/A)$, $g \in G^\alpha(S)$, $\gamma \in G^0(S)$.

If e is the origin of G , the restriction of ν^α to $G^\alpha \times_A \{e\}$ is the zero morphism; since $G^\alpha \times_A G^0$ is connected by 2.1.2, one sees that ν^α factors through G^0 . Thus, for every A -scheme S , $G^0(S)$ is an invariant subgroup of $G(S)$. One has therefore obtained the following proposition:⁽²⁴⁾

⁽²¹⁾**N.D.E.** This paragraph has been added to make the connection with VIB, §3.

⁽²²⁾**N.D.E.** In this exposé, the notation G^0 is reserved for the case where the connected component of e is open; in VIB, §3, this connected component will be denoted G^0 in every case. This is a slight abuse of notation, but is compatible with what precedes when the connected component is the underlying topological space of an open subgroup scheme G^0 of G .

⁽²³⁾**N.D.E.** Indeed, in a noetherian space, the connected components are finite in number, so each is open; see also EGA I, 6.1.9.

⁽²⁴⁾**N.D.E.** The numbering 2.3.1 has been added to bring out this statement. Note further that G^0 is even a characteristic subgroup of G , cf. 2.6.5 (ii).

Proposition 2.3.1. *Let G be an A -group locally of finite type. Then the neutral component G^0 is an open invariant subgroup scheme of G .*

Proposition 2.4. *Let G be an A -group locally of finite type.*

- (i) G^0 is irreducible and $G^0 \otimes_A k$ is geometrically irreducible over k .
- (ii) G^0 is quasi-compact, hence of finite type over A .

(25)

PROOF. (i) Since G^0 and $G^0 \otimes_A k$ have the same underlying topological space, it suffices to show the second assertion. Let $\bar{k}$ be an algebraic closure of k . By 2.2, $(G \otimes_A \bar{k})_{\text{red}}$ is a reduced $\bar{k}$ -group locally of finite type, hence smooth over $\bar{k}$ (1.3.1). A fortiori the local rings of $(G \otimes_A \bar{k})_{\text{red}}$ are integral domains, so, ⁽²⁶⁾ since $G \otimes_A \bar{k}$ is locally noetherian, the connected components of $G \otimes_A \bar{k}$ are irreducible (cf. EGA I, 6.1.10). In particular, the connected component $G^0 \otimes_A \bar{k}$ (cf. 2.1.1) is irreducible.

(ii) Let us now show that G^0 is of finite type over A . Since G^0 is locally of finite type over A , it suffices to prove G^0 quasi-compact. Since G^0 is irreducible, this follows from 0.5.1. $\square$

Corollary 2.4.1. *Every connected component of G is irreducible, ⁽²⁷⁾ of finite type over A , ⁽²⁸⁾ and of the same dimension as G^0 .*

Indeed, one can suppose A equal to its residue field k . Let, then, C be a connected component of G , x a closed point of C , $\kappa(x)$ the residue field of x , and k' a normal extension of k containing $\kappa(x)$ and of finite degree over k . The canonical projection $\pi : C \otimes_k k' \rightarrow C$ is open and closed; ⁽²⁹⁾ consequently, if C' is a connected component of $C \otimes_k k'$, the projection $C' \rightarrow C$ is surjective, so C' contains a point $y \in \pi^{-1}(x)$, and such a point is rational over k' (cf. the proof of 1.2), so C' is the image of $G^0 \otimes_k k'$ under translation r_y . Now $G^0 \otimes_k k'$ is of finite type over k' by 2.4, and $\pi^{-1}(x)$ is finite (of cardinality $\leq [k' : k]$), so $C \otimes_k k'$ is of finite type over k' , and hence C is of finite type over k .

On the other hand, since $G^0 \otimes_k k'$ is irreducible by 2.4, the same holds for C' , ⁽³⁰⁾ and therefore also for C , since the projection $C' \rightarrow C$ is surjective.

Finally, we have seen above that $C \otimes_k k'$ is a disjoint union of finitely many translates of $G^0 \otimes_k k'$. Since dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), it follows that C has the same dimension as G^0 . (Moreover, by EGA IV₂, 5.2.1, one has $\dim_g G = \dim G^0$ for every point $g \in G$.)

2.5. — This paragraph has been added. The results below appear in Exp. VIB, but could (or should) have appeared in VIA, and it is useful to have them already, in order to make Theorem 3.2 below more precise.

Lemma 2.5.1. *Let $(A, \mathfrak{m})$ be an artinian local ring, and $k = A/\mathfrak{m}$ its residue field.*

- (i) *If X is an A -scheme such that $X \otimes_A k$ is locally of finite type (resp. of finite type) over k , then the same holds for X over A .*
- (ii) *Let $u : X \rightarrow Y$ be a morphism of A -schemes. If $u \otimes_A k$ is an immersion (resp. a closed immersion), the same holds for u .*

⁽²⁵⁾**N.D.E.** In the statement, “ G^0 is geometrically irreducible” has been replaced by “ $G^0 \otimes_A k$ is geometrically irreducible over k ”, and the proof has been expanded.

⁽²⁶⁾**N.D.E.** The original has been expanded by adding the reference to EGA I, 6.1.10.

⁽²⁷⁾**N.D.E.** One must beware that a nonneutral connected component is not geometrically connected in general. For example, if $k = \mathbf{R}$, the group $\mu_{3,\mathbf{R}}$, represented by $\mathbf{R}[X]/(X^3 - 1)$, has two connected components: $\{e\} = \text{Spec } \mathbf{R}$ and $C = \text{Spec } \mathbf{R}[X]/(X^2 + X + 1)$, and $C \otimes_{\mathbf{R}} \mathbf{C}$ has two components.

⁽²⁸⁾**N.D.E.** The following assertion has been added, cf. VIB, 1.5.

⁽²⁹⁾**N.D.E.** The original has been expanded in what follows.

⁽³⁰⁾**N.D.E.** The original has been simplified here.

PROOF. (i) Suppose $X \otimes_A k$ locally of finite type over k . Let $U = \operatorname{Spec} B$ be an affine open subset of X . By hypothesis, there exist elements $x_1, \dots, x_n$ of B whose images generate $B/\mathfrak{m}B$ as a k -algebra, and it follows from the “nilpotent Nakayama lemma” that the x_i generate B as an A -algebra. This proves that X is locally of finite type over A . If moreover $X \otimes_A k$ is quasi-compact, the same holds for X (which has the same underlying topological space), and hence X is of finite type over A . This proves (i).

Let us prove (ii). Suppose $u \otimes_A k$ an immersion (resp. a closed immersion). Then u is a homeomorphism from X onto a locally closed (resp. closed) subset of Y and, for every $x \in X$, the ring morphism $\varphi_x : \mathcal{O}_{Y, u(x)} \rightarrow \mathcal{O}_{X, x}$ is such that $\varphi_x \otimes_A k$ is surjective. By the nilpotent Nakayama lemma, it follows that φ_x is surjective, so u is an immersion (resp. a closed immersion). $\square$

Proposition 2.5.2. *Let A be an artinian local ring with residue field k , and let $u : G \rightarrow H$ be a quasi-compact morphism between A -group schemes locally of finite type.*

(a) *The set $u(G)$ is a closed subset of H whose connected components are irreducible and all have the same dimension.*

(b) *One has $\dim G = \dim u(G) + \dim \operatorname{Ker}(u)$.*

(c) *If u is a monomorphism, it is a closed immersion.*

PROOF. By the preceding lemma, it suffices to prove the proposition in the case $A = k$. Moreover, since the properties under consideration are stable under fpqc descent, and dimension is invariant under extension of the base field, one can suppose k algebraically closed.

Let us prove (a). Denote by C the reduced subscheme of H whose underlying topological space is $\overline{u(G)}$. Since $u(G)$ is stable under the inversion morphism of H , so is C . On the other hand, $u : G \rightarrow C$ is quasi-compact and dominant, so, by EGA IV₂, 2.3.7, the same holds for $u \times_k \operatorname{id}_G$ and $\operatorname{id}_H \times_k u$, hence for their composite $u \times_k u : G \times_k G \rightarrow C \times_k C$. Consequently, multiplication in H sends $C \times_k C$ into C , so C is a subgroup scheme of H .

Thus, replacing H by C , one reduces to the case where u is dominant. Then $G(k)$ is dense in H , hence meets every connected component of H , and therefore acts transitively on the set of these connected components. It thus suffices to show that $u(G)$ contains H^0 . Replacing G by $u^{-1}(H^0)$, one can therefore suppose $H = H^0$; in this case, by 2.4, H is irreducible and of finite type over k , hence noetherian. On the other hand, u is locally of finite type (cf. EGA I, 6.6.6) and quasi-compact, hence of finite type. Consequently, by Chevalley’s constructibility theorem (cf. EGA IV₁, 1.8.5), $u(G)$ is a constructible (and dense) subset of $H = \overline{u(G)}$, and therefore contains a dense open subset U of H (cf. EGA 0_{III}, 9.2.2). Then, by 0.5, one has $H = U \cdot U \subset u(G)$, whence $u(G) = H$. Taking account of 2.4.1, this proves assertion (a).

Let us prove (b). First recall that the functor $\operatorname{Ker}(u)$ (cf. I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the neutral element of H . Since u is of finite type, $\operatorname{Ker}(u)$ is of finite type over k . On the other hand, replacing H by the reduced closed subscheme $u(G)$, one can suppose u surjective. Denote by u^0 the restriction of u to G^0 . Since G and $\operatorname{Ker}(u)$ are equidimensional, and $\operatorname{Ker}(u)^0 \subset \operatorname{Ker}(u^0)$, one reduces to the case where G , and hence also H , are irreducible.

Then, by EGA IV₃, 9.2.6.2 and 10.6.1 (ii), the set of $y \in H$ such that $\dim u^{-1}(y) = \dim G - \dim H$ contains a nonempty open subset V . Since u is surjective, $U = u^{-1}(V)$ is then a nonempty open subset of G , and therefore contains a closed point x of G , since G is a Jacobson scheme (cf. EGA IV₃, 10.4.8). Then right translation r_x is an isomorphism from $\operatorname{Ker}(u)$ onto $u^{-1}(u(x))$, whence:

$$\dim \operatorname{Ker}(u) = \dim u^{-1}(u(x)) = \dim G - \dim H.$$

Let us prove (c), following [DG70], I, §3.4. (Another proof is given in Exp. VIB, 1.4.2.) Suppose u a monomorphism. If C is a connected component of G , there exists a closed point $x \in G$ such that $C = r_x(G^0)$, and, denoting by u_C (resp. u^0) the restriction of u to C (resp.

G^0), one has $u_C = r_{u(x)} \circ u^0 \circ r_x^{-1}$; thus it suffices to show that u^0 is a closed immersion. One can therefore suppose $G = G^0$, so G is irreducible and of finite type over k .

Let ξ be the generic point of G ; then $\mathcal{O}_{G,\xi}$ is an artinian local ring; denote its maximal ideal by $\mathfrak{m}$. On the other hand, let $h = u(\xi)$, $\mathfrak{n}$ the maximal ideal of $\mathcal{O}_{H,h}$, and $A = \mathcal{O}_{G,\xi}/\mathfrak{n}\mathcal{O}_{G,\xi}$. Since u is a monomorphism, so is the morphism $u_h : \text{Spec}(A) \rightarrow \text{Spec}(\kappa(h))$ obtained by base change, so the multiplication morphism $A \otimes_{\kappa(h)} A \rightarrow A$ is an isomorphism (cf. EGA I, 5.3.8), whence $A = \kappa(h)$. By Nakayama's lemma (since $\mathfrak{n}\mathcal{O}_{G,\xi}$ is contained in $\mathfrak{m}$, hence nilpotent), it follows that the morphism $\mathcal{O}_{H,h} \rightarrow \mathcal{O}_{G,\xi}$ is surjective.

Let, then, V be an affine open subset of H containing h , U a nonempty affine open subset of G contained in $u^{-1}(V)$, $\varphi : \mathcal{O}_H(V) \rightarrow \mathcal{O}_G(U)$ the morphism of k -algebras induced by u , $\mathfrak{p}$ the prime ideal of $\mathcal{O}_G(U)$ corresponding to η , and $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$. Since G is of finite type over k , $\mathcal{O}_G(U)$ is generated as a k -algebra by finitely many elements $a_1, \dots, a_n$. By what precedes, there exist elements $b_1, \dots, b_n, s$ in $\mathcal{O}_H(V)$ such that $s \notin \mathfrak{q}$ and the equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ hold in $\mathcal{O}_G(U)_{\mathfrak{p}}$. There therefore exist elements $t_1, \dots, t_n$ of $\mathcal{O}_G(U) - \mathfrak{p}$ such that $t_i(a_i\varphi(s) - \varphi(b_i)) = 0$. Then, putting $t = t_1 \cdots t_n\varphi(s) \in \mathcal{O}_G(U) - \mathfrak{p}$, the equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ already hold in $\mathcal{O}_G(U)_t$, and, since $t \in \mathcal{O}_G(U) = k[a_1, \dots, a_n]$, there exists $b \in \mathcal{O}_H(V)$ such that $t/1 = \varphi(b)/\varphi(s)^r$ for some $r \in \mathbb{N}$. Consequently, φ induces a surjection from $\mathcal{O}_H(V)_{sb}$ onto $\mathcal{O}_G(U)_t$, and hence u is a local immersion at ξ .

The open subset W of G consisting of the points where u is a local immersion is therefore nonempty. Since G is a Jacobson scheme, W contains a closed point y , and, to show that $W = G$, it suffices to show that every closed point of G belongs to W . Now every closed point x is the image of y under translation $r_x \circ r_y^{-1}$, hence belongs to W , whence $W = G$. This proves that u is a local immersion.

Since G is irreducible, it follows that u is an immersion. Indeed, for every $x \in G$, let U_x and V_x be open subsets of G and H such that $x \in U_x$ and u induces a closed immersion from U_x into V_x . Since U_x is dense in G , $u(U_x)$ is dense in $u(G) \cap V_x$, and since $u(U_x)$ is closed in V_x , one therefore has $u(U_x) = V_x$. Since moreover u is injective, one has $U_x = u^{-1}(V_x)$, and it follows that u induces a closed immersion from G into the open subscheme of H covered by the V_x . Thus $u : G \rightarrow H$ is an immersion. But we have already seen that $u(G)$ is a closed subset of H , so u is a closed immersion. $\square$

Lemma 2.5.3. ⁽³¹⁾ *Let A be an artinian local ring, k its residue field, G a flat A -group, X an A -scheme with a left action $\mu : G \times_A X \rightarrow X$ of G and a section $s_0 : \text{Spec } A \rightarrow X$. (This is the case, for example, if G' is a second A -group and one is given a morphism of A -groups $G \rightarrow G'$.)*

Let φ be the morphism $\mu \circ (\text{id}_G \times s_0)$ from $G = G \times_A A$ to X . If φ is flat at one point g of G , then φ is flat.

PROOF. Since G is flat over A , by the fiberwise flatness criterion (EGA IV₃, 11.3.10.2), it suffices to show that $\varphi \otimes_A k$ is flat, so one can suppose $A = k$. In this case, giving s_0 amounts to giving a k -point $x_0 \in X(k)$, and φ is the morphism $h \mapsto hx_0$.

Let, then, $h \in G$; let us show that φ is flat at h . Let K be an extension of k containing a copy of $\kappa(g)$ and $\kappa(h)$; one has a cartesian square

$$\begin{array}{ccc} G_K & \xrightarrow{\varphi_K} & X_K \\ \downarrow & & \downarrow \\ G & \xrightarrow{\varphi} & X \end{array}$$

in which the two vertical arrows are faithfully flat. Thus, by V 7.4 (i), φ_K is flat at every point $g' \in G_K$ above g , and, to show that φ is flat at h , it suffices to show that φ_K is flat at

⁽³¹⁾**N.D.E.** No finiteness hypotheses are made in this statement; this will be useful below (cf. 6.2 and VIB, §12).

a point h' above h . One is therefore reduced to the case where g and h are rational. Let, then, $u = hg^{-1}$, and let ℓ_u (resp. μ_u) be the left translation of G (resp. X) defined by u ; since $\varphi \circ \ell_u = \mu_u \circ \varphi$, one obtains a commutative square

$$\begin{array}{ccc} \mathcal{O}_{G,h} & \xrightarrow{\sim} & \mathcal{O}_{G,g} \\ \uparrow & & \uparrow \\ \mathcal{O}_{X,hx_0} & \xrightarrow{\sim} & \mathcal{O}_{X,gx_0} \end{array}$$

in which the horizontal arrows are isomorphisms. Since the morphism $\mathcal{O}_{X,gx_0} \rightarrow \mathcal{O}_{G,g}$ is flat, the morphism $\mathcal{O}_{X,hx_0} \rightarrow \mathcal{O}_{G,h}$ is also so. $\square$

Proposition 2.5.4. *Let A be an artinian local ring, k its residue field, G an A -group locally of finite type, and $X \neq \emptyset$ an A -scheme locally of finite type with a left action of G . Suppose the morphism $\varphi : G \times_A X \rightarrow X \times_A X$, defined set-theoretically by $(g, x) \mapsto (gx, x)$, is surjective. Then:*

- (i) *The connected components of X are of finite type, irreducible and all of the same dimension.*
- (ii) *More precisely, let $\bar{k}$ be an algebraic closure of k and x a closed point of $X \otimes_A \bar{k}$; its stabilizer F is a closed subgroup scheme of $G \otimes_A \bar{k}$, and the dimension of the irreducible components of X is $\dim G - \dim F$.*

Taking account of Lemma 2.5.1, one can suppose $A = k$. First suppose k algebraically closed. Then G_{red} is a k -group locally of finite type, and therefore, replacing G by G_{red} and X by X_{red} , one can suppose G and X reduced.

Since $G \times_k X$ is locally of finite type over k , φ is locally of finite type (cf. EGA I, 6.6.6 (v)), hence locally of finite presentation since $X \times_k X$ is locally noetherian. Let x be a rational point of X ; then the morphism $\varphi_x : G \rightarrow X$ obtained from φ by base change is surjective and locally of finite presentation. If η is a maximal point of X , $\mathcal{O}_{X,\eta}$ is a field (since X is reduced), hence φ_x is flat at every point of G above η . Thus, by Lemma 2.5.3, φ_x is flat. Consequently, $\varphi_x : G \rightarrow X$ is faithfully flat and locally of finite presentation, hence open (cf. EGA IV₂, 2.4.6). Since G^0 is open in G , irreducible and quasi-compact (by 2.4), each orbit $G^0x = \varphi_x(G^0)$, as x ranges over the rational points of X , is an irreducible quasi-compact open subset of X , hence of finite type over k (since X is locally of finite type over k).

Since every nonempty open subset of X contains a rational point, it follows that X is covered by these open subsets. Moreover, two such open subsets are either disjoint or equal. Indeed, if $\varphi_x(G^0) \cap \varphi_y(G^0)$ is nonempty, it contains a rational point z , so there exist two rational points $g, h \in G^0$ such that $gx = z = hy$, whence $x = g^{-1}z$ and $y = h^{-1}z$, and hence $\varphi_x(G^0) = \varphi_z(G^0) = \varphi_y(G^0)$. It follows that the orbits $\varphi_x(G^0)$ are also closed, and are therefore both the connected components and the irreducible components of X .

Finally, let x, y be two rational points of X . Since φ_x is surjective, there exists a rational point $g \in G$ such that $y = gx$, and, since G^0 is an invariant subgroup of G , the orbit G^0y is the image of G^0 under translation ℓ_g of X , so G^0y and G^0x have the same dimension.

Moreover, by I, 2.3.3.1, the stabilizer of x is represented by the closed subscheme F of G defined by the cartesian square below:

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \varphi_x \\ \text{Spec } k & \longrightarrow & X. \end{array}$$

Then F is a k -group locally of finite type, $F \cap G^0$ is a k -group of finite type containing F^0 , and, by 2.4.1, F and $F \cap G^0$ are equidimensional, of the same dimension as F^0 . Let $C = \varphi_x(G^0)$ be the irreducible component of X containing x . Proceeding as in the proof of point (b) of 2.5.2, one obtains $\dim C = \dim G^0 - \dim F^0 = \dim G - \dim F$.

In the general case (i.e. for any field k), let $\bar{k}$ be an algebraic closure of k . Let C be a connected component of X and C' a connected component of $C \otimes_k \bar{k}$; then C' is a connected component of $X' = X \otimes_k \bar{k}$. The morphism $\pi : X' \rightarrow X$ is open (cf. EGA IV₂, 2.4.10), and, being integral, it is also closed; consequently $\pi(C') = C$. Since C' is irreducible and quasi-compact, C is irreducible and quasi-compact, hence of finite type over k (since X is locally of finite type over k).

Finally, since dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), $\dim C = \dim C'$, and, since all irreducible components of X' have the same dimension, the same holds for those of X .

2.6. — Supplements. — This paragraph, taken from [Per75] II, §§1–2, with supplements due to O. Gabber, has been added.⁽³²⁾ It shows that the preceding results hold for every group scheme G over a field k . (This will be used in sections 5, 6 and 7 of Exp. VIB.)

Fix a field k . Let us begin with the following lemma (loc. cit., II 2.1.1), which does not appear explicitly in EGA IV₂, §4.4 (although one can doubtless read it between the lines at the beginning of loc. cit., §4.4.1).

Lemma 2.6.0. *Let X be an irreducible k -scheme, K an extension of k , and X' an irreducible component of X_K . The projection $X' \rightarrow X$ is surjective.*

Indeed, let B be a transcendence basis of K over k and $L = k(B) \subset K$. By EGA IV₂, 4.3.2 and 4.4.1, X_L is irreducible and X' dominates X_L . The morphism $X' \rightarrow X_L$ is therefore integral and dominant, hence surjective. Since $X_L \rightarrow X$ is surjective (loc. cit., 4.4.1), $X' \rightarrow X$ is also so.

For the remainder of 2.6, fix a k -group scheme G and an action $\mu : G \times X \rightarrow X$ of G on a k -scheme X satisfying the following condition:

($\star$) the morphism $\Phi : G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$ is surjective

(this is the case, in particular, for G acting on itself by left translations). For brevity, one will then say: “Let X be a G -scheme satisfying ($\star$)”. Finally, denote by k' the perfect closure of k .

Proposition 2.6.1. *Let X be a G -scheme satisfying ($\star$).*

- (i) *X is geometrically pointwise irreducible over k , i.e. for every extension K of k , every $x \in X_K$ belongs to a unique irreducible component of X_K .*
- (ii) *Every local ring of $(X_{k'})_{\text{red}}$ is normal.*
- (iii) *Let η be a maximal point of $(X_{k'})_{\text{red}}$, $C = \overline{\{\eta\}}$, and L the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme and is geometrically irreducible over L (i.e. $C \otimes_L K$ is irreducible for every extension K of L).*
- (iv) *In particular, if x is a rational point of X , the irreducible component of X containing x is geometrically irreducible over k .*

PROOF. (i) Since hypothesis ($\star$) is preserved by every base change $k \rightarrow K$, it suffices to show that every $x \in X$ belongs to a unique irreducible component of X . Since the morphism $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is a universal homeomorphism, one can further suppose k perfect. One can then suppose G and X reduced. Let η be a maximal point of X and z any point of $Z = \overline{\{\eta\}}$. Since X is reduced, the local ring $\mathcal{O}_{X,\eta}$ equals $\kappa(\eta)$, and, since k is perfect, for every extension K of k , $\kappa(\eta) \otimes_k K$ is normal (cf. EGA IV₂, 6.14.2), so every point of X_K above η is normal.

Since Φ is surjective, there exists a point γ of $G \times X$ such that $\Phi(\gamma)$ has projections z and η . Let $K = \kappa(\gamma)$; there then exist rational points g and η' of G_K and X_K such that η' lies above η and $z' = g\eta'$ above z . By what precedes, η' is a normal point of X_K , hence so is

⁽³²⁾**N.D.E.** These results were communicated to us by O. Gabber, in particular 2.6.6, which plays an important role in section 5 of VIB.

z' . Since $\pi : X_K \rightarrow X$ is flat, it follows that $z = \pi(z')$ is a normal point of X (cf. EGA IV₂, 2.1.13). This proves (ii) and (i).

The first assertion of (iii) then follows from (ii). Next, since L is algebraically closed in $\kappa(\eta)$, C is geometrically irreducible over L , by EGA IV₂, 4.5.9.

Finally, if X has a rational point x , it follows from (iii) that $L = k$, and hence X is geometrically irreducible over k . This can also be seen directly as follows (cf. [Per75], II 2.1): let C be the irreducible component of X containing x , and let K be an extension of k ; X_K has a unique point e_K above e , and, by (i), e_K belongs to a unique irreducible component C' of X_K ; on the other hand, by 2.6.0, every irreducible component of C_K contains e_K , hence equals C' . $\square$

Notation. — Temporarily denote by C^0 the reduced closed subscheme of G whose underlying space is the unique irreducible component of G containing the neutral element e .

Corollary 2.6.2. *C^0 is geometrically irreducible over k and is stable set-theoretically under the group law, i.e. $(C_{k'}^0)_{\text{red}}$ is a subgroup of $(G_{k'})_{\text{red}}$. Consequently C^0 is quasi-compact.*

Indeed, by 2.6.1, C^0 is geometrically irreducible over k , so $C^0 \times C^0$ is irreducible, hence $\nu(C^0 \times C^0) \subset C^0$, where ν denotes the morphism $(g, h) \mapsto gh^{-1}$. Since $\text{Spec } k' \rightarrow \text{Spec } k$ is a universal homeomorphism, one has the same conclusion for $C_{k'}^0$, then for $H = (C_{k'}^0)_{\text{red}}$, and hence, since $H \times_{k'} H$ is reduced, ν induces a morphism $H \times_{k'} H \rightarrow H$, i.e. H is a subgroup of $(G_{k'})_{\text{red}}$. Consequently, by 0.5.1, H (and therefore also C^0) is quasi-compact.

Recall 2.6.3. Let Y be a scheme. Recall (EGA 0_{III}, 9.1.1) that a subset E of Y is said to be retrocompact if the inclusion $E \hookrightarrow Y$ is quasi-compact, and that, by EGA IV₁, 1.9.5 (v) and 1.10.1, if U is a retrocompact open subset of Y , the closure $\overline{U}$ of U is the union of the closures $\overline{\{y\}}$ of the points $y \in U$, and of course it suffices to let y range over the maximal points of U , i.e. the maximal points of Y contained in U .

Consequently, if Y is quasi-separated and η a maximal point of Y , the intersection of the closed neighborhoods of η equals $\overline{\{\eta\}}$: indeed, if $y \in Y - \overline{\{\eta\}}$, y is contained in an affine open subset V not containing η ; since Y is quasi-separated, V is retrocompact, so $\overline{V}$ is the union of the $\overline{\{\xi\}}$ as ξ ranges over the maximal points of Y belonging to V , and therefore $\eta \notin \overline{V}$, i.e. $Y - \overline{V}$ is a closed neighborhood of η not containing y .

Proposition 2.6.4. *Let X be a G -scheme satisfying $(\star)$, and let U be an open subset of X .*

- (i) *C^0U is an open subset of X , equal to the union of the irreducible components of X whose generic point belongs to U .*
- (i') *Consequently, if X is irreducible, it is quasi-compact.*
- (ii) *If moreover U is retrocompact in X , C^0U equals $\overline{U}$, hence is an open and closed subset of X .*

PROOF. (i) First, C^0U is open, since it is the union, for $g \in C^0$, of the projections of the open subsets $g \cdot U_{\kappa(g)} \subset X_{\kappa(g)}$, and each projection $X_{\kappa(g)} \rightarrow X$ is open.

To prove the second assertion of (i), one can replace k by k' (since $\text{Spec } k' \rightarrow \text{Spec } k$ is a universal homeomorphism), hence suppose k perfect. One can then suppose G and X reduced, hence geometrically reduced.

Let η be a maximal point of X contained in U , and Z its closure. Consider the morphism $\mu' : C^0 \times Z \rightarrow X$. Since C^0 is geometrically irreducible, $C^0 \times Z$ is irreducible; denote its generic point by γ . Since μ' sends the point $\varepsilon(\eta)$ (where ε denotes the unit section of C^0) to η , γ is sent to a generalization of η , hence to η . Thus μ' sends the underlying space of $C^0 \times Z$ into Z , and hence, since $C^0 \times Z$ is reduced, μ' factors through Z .

Now let $z \in Z$. Put $K = \kappa(z)$; then the morphism $\mu_z : G_K \rightarrow X_K$, $h \mapsto h \cdot z$, is surjective; let α be a maximal point of X_K ; the local ring $\mathcal{O}_{X_K, \alpha}$ is a field, since X_K is reduced, so μ_z is flat at every point of G_K above α , and therefore μ_z is flat by Lemma 2.5.3.

On the other hand, μ_z sends the generic point ω of C_K^0 to a point $t \in Z_K$. Let β be a maximal point of Z_K such that $t \in \overline{\{\beta\}}$; since μ_z is flat, there exists a generalization ξ of ω such that $\mu_z(\xi) = \beta$, and, since ω is a maximal point of G_K , necessarily $\xi = \omega$, so $\mu_z(\omega)$ equals β , which lies above η (since $Z_K \rightarrow Z$ is flat).

Put $L = \kappa(\omega)$ and let ω_L, z_L be the L -points obtained from ω, z ; then $\omega_L \cdot z_L = \beta'$ is a point of Z_L above β , and therefore $z_L = \omega_L^{-1} \cdot \beta' \in C_L^0 \cdot U_L$, whence $z \in C^0 \cdot U$. This proves (i).

Since C^0 is quasi-compact by 2.6.2, point (i') follows: if X is irreducible and U a nonempty affine open subset, X equals $C^0 U$, i.e. is the image of the morphism $C^0 \times U \rightarrow X$, hence is quasi-compact. Finally, if U is retrocompact in X , by 2.6.3, $\overline{U}$ is the union of the $\{\eta\}$ as η ranges over the maximal points of X contained in U , hence equals $C^0 U$. This proves (ii). $\square$

One then obtains the following result ([Per75] II Th. 2.4; see also [Per76], Prop. 4.1.1):

Theorem 2.6.5. *Let k be a field and G a k -group scheme.*

- (i) *There exists a unique subgroup scheme G^0 of G , called the neutral component of G , such that:*
 - (a) *The underlying space of G^0 is the irreducible component of the neutral element.*
 - (b) *$G^0 \rightarrow G$ is a flat closed immersion, i.e. $\mathcal{O}_{G,g} = \mathcal{O}_{G^0,g}$ for every $g \in G^0$.*
- (ii) *Moreover, G^0 is quasi-compact, geometrically irreducible, and a characteristic subgroup of G .*
- (iii) *If G is connected, then $G = G^0$.*

PROOF. (i) First recall that G is separated (0.3), hence a fortiori quasi-separated. Let U be an affine open subset of G containing the generic point ω of C^0 . By 2.6.3 and 2.6.4, $\overline{U} = C^0 U$ is both open and closed, and $C^0 = \overline{\{\omega\}}$ is set-theoretically the intersection of these open and closed subsets.

As U ranges over the affine open subsets containing ω , one obtains a projective system of G -schemes $\overline{U}$, whose transition morphisms are affine (since they are closed immersions). One can therefore form its projective limit G^0 (cf. EGA IV₃, 8.2.2), i.e. for every affine open subset V of G , $G^0 \cap V$ is the spectrum of the algebra

$$\varinjlim_{\overline{U}} \mathcal{O}_G(V \cap \overline{U}) = \mathcal{O}_G(V) / \sum_{\overline{U}} I_{\overline{U}}(V),$$

where $I_{\overline{U}}(V)$ denotes the kernel of $\mathcal{O}_G(V) \rightarrow \mathcal{O}_G(V \cap \overline{U})$. It follows that G^0 has underlying space C^0 and $G^0 \rightarrow G$ is a closed immersion. Moreover, for every $g \in G^0$, $\mathcal{O}_{G^0,g}$ is the inductive limit, as V ranges over the affine open subsets of G containing g , of the k -algebras $\mathcal{O}_{G^0}(V \cap G^0) = \varinjlim_{\overline{U}} \mathcal{O}_G(V \cap \overline{U})$, and this double inductive limit is identified with

$$\varinjlim_{\overline{U}} \varinjlim_{\substack{V \\ g \in V \subset \overline{U}}} \mathcal{O}_G(V) = \mathcal{O}_{G,g},$$

i.e. one has $\mathcal{O}_{G^0,g} = \mathcal{O}_{G,g}$ (see also EGA IV₂, 5.13.3 (ii)). Thus $i : G^0 \rightarrow G$ is a flat closed immersion. Conversely, this condition implies $i^*(\mathcal{O}_G) = \mathcal{O}_{G^0}$, and hence G^0 is uniquely determined by conditions (a) and (b). This proves (i).

The first two assertions of (ii) follow from 2.6.2. Finally, let S be a k -scheme and φ an automorphism of the S -group G_S . For every $s \in S$, φ_s sends $G_{\kappa(s)}^0$ into itself, hence $\varphi(G_S^0) \subset G_S^0$. Moreover, the closed immersion $i_S : G_S^0 \hookrightarrow G_S$ obtained from i by base change is flat, so $\mathcal{O}_{G_S, \varphi(z)} = \mathcal{O}_{G_S^0, \varphi(z)}$ for every $z \in G_S^0$, and hence $\varphi \circ i_S$ factors through G_S^0 . This proves that G^0 is a characteristic subgroup of G , whence (ii). Finally, (iii) is a special case of point (i) of the following proposition. $\square$

(33)

Proposition 2.6.6. *Let k be a field and G a k -group acting on a k -scheme X so that the morphism $G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$, is surjective. Suppose X quasi-separated. Then:*

- (i) *Every connected component C of X is irreducible.*
- (ii) *Let η be the generic point of C and L the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme, geometrically irreducible over L , and the morphism*

$$G_L^0 \times_L C \rightarrow C \times_L C$$

is surjective.

- (iii) *In particular, if C contains a rational point x , C is geometrically irreducible over k , and the morphism $\varphi : G^0 \rightarrow C$, $g \mapsto gx$, is surjective.*

PROOF. (i) Let C be a connected component of X , Y an irreducible component of X contained in C , η the generic point of Y , and U an affine open subset of X containing η . Since X is quasi-separated, U is retrocompact in X , so, by 2.6.4, $\bar{U} = G^0 U$ is an open and closed subset of X meeting C , hence contains C . Now, by 2.6.3, the intersection of the $\bar{U}$, as U ranges over a fundamental system of affine open neighborhoods of η , equals $\{\eta\}$. It follows that $C = \{\eta\}$. This proves (i).

The first assertion of (ii) (and also of (iii)) follows from 2.6.1. Let us begin by showing the second assertion of (iii). Denote by ω (resp. η) the generic point of G^0 (resp. C). Let $z \in C$ and $K = \kappa(z)$. Since G^0 (resp. C) is geometrically irreducible over k , G_K^0 (resp. C_K) is irreducible; denote its generic point by ξ (resp. β). We have seen in the proof of 2.6.4 that the morphism $\mu_z : G_K \rightarrow C_K$, $h \mapsto h \cdot z$, sends ξ to β , and similarly $\varphi(\omega) = \eta$.

Let $L = \kappa(\xi)$ and let ξ_L, z_L be the L -points obtained from ξ, z ; then $\xi_L \cdot z_L = \beta'$ is a point of C_L above $\beta \in C_K$, hence also above $\eta \in C$. Consider the cartesian square:

$$\begin{array}{ccc} G_L^0 & \xrightarrow{\varphi_L} & C_L \\ \pi_{G^0} \downarrow & & \downarrow \pi_C \\ G^0 & \xrightarrow{\varphi} & C, \end{array}$$

since $\varphi_L(\pi_{G^0}^{-1}(\omega)) = \pi_C^{-1}(\eta)$ (cf. EGA I, 3.4.8), there exists $g \in G_L^0$ such that $\varphi_L(g) = \beta'$. Thus $z_L = \xi_L^{-1} \cdot \varphi_L(g) = \varphi_L(\xi_L^{-1}g)$, whence $\varphi(\pi_{G^0}(\xi_L^{-1}g)) = \pi_C(z_L) = z$. This proves that φ is surjective.

Now let us prove the second assertion of (ii). It suffices to show that, for every $z \in C$, the morphism $\mu_z : G_L^0 \otimes_L \kappa(z) \rightarrow C \otimes_L \kappa(z)$ is surjective, but this follows from (iii), since z is a rational point of $C \otimes_L \kappa(z)$. $\square$

Remark 2.6.7. Under the hypotheses of 2.6.6, if $C(k) = \emptyset$, the morphism $G^0 \times_k C \rightarrow C \times_k C$ is not necessarily surjective. For example, for $k = \mathbf{R}$ and $G = \{\pm 1\}_{\mathbf{R}}$, the G -torsor $X = \text{Spec } \mathbf{R}[X]/(X^2 + 1)$ is connected, but the morphism $G^0 \times_{\mathbf{R}} X \rightarrow X \times_{\mathbf{R}} X$ is not surjective. (But one has $L = \mathbf{C}$, and the morphism $G^0 \times_{\mathbf{R}} X \rightarrow X \times_{\mathbf{C}} X$ is an isomorphism.)

3. Construction of quotients $F \backslash G$ (for G, F of finite type)

3.1. — Let A be an artinian local ring and $u : F \rightarrow G$ a homomorphism of A -groups. If $\mu : F \times_A F \rightarrow F$ and $\nu : G \times_A G \rightarrow G$ denote the multiplication morphisms, and λ the composite morphism

$$F \times_A G \xrightarrow{u \times G} G \times_A G \xrightarrow{\nu} G,$$

(33) **N.D.E.** This proposition (as well as the preceding results) was communicated to us by O. Gabber; it will be used to correct the proof of Theorem 5.3 of VIB.

recall that the left quotient $F \backslash G$ of G by F is the cokernel of the (Sch/A) -groupoid G_* described below:

$$F \times_A F \times_A G \begin{array}{c} \xrightarrow{F \times \lambda} \\ \xrightarrow[\text{pr}_{2,3}]{-\mu \times G} \end{array} F \times_A G \xrightarrow[\text{pr}_2]{\lambda} G$$

(pr_2 and $\text{pr}_{2,3}$ are the projections of $F \times_A G$ and $F \times_A (F \times_A G)$ onto the second factors). We shall say that G_* is the groupoid with base G defined by u (cf. Exp. V, §2.a; as in Exposé V, we do not follow in this exposé the convention of IV, 4.6.15).

Since the unique A -morphism $F \rightarrow \text{Spec } A$ is universally open (EGA IV₂, 2.4.9), pr_2 is an open morphism; the same therefore holds for λ , the composite of pr_2 and the automorphism σ of $F \times_A G$ defined by the following formulas: $\sigma(S)(x, y) = (x, u(S)(x) \cdot y)$, where S is a varying A -scheme and x, y belong to $F(S), G(S)$. One sees in the same way that pr_2 and λ are flat when F is flat over A .

To finish these preliminaries, note also that every A -morphism $s : \text{Spec } A \rightarrow G$ defines an automorphism of the groupoid G_* inducing on G , $F \times_A G$ and $F \times_A F \times_A G$ the automorphisms r_s , $\text{id}_F \times_A r_s$ and $\text{id}_F \times_A \text{id}_F \times_A r_s$ respectively. We shall again denote this automorphism of G_* by r_s , and shall say that r_s is the right translation defined by s (cf. 0.4).

Theorem 3.2. *Let F and G be flat groups locally of finite type over an artinian local ring A . Let $u : F \rightarrow G$ be a quasi-compact homomorphism of A -groups with kernel finite over A . Then.⁽³⁴⁾*

- (i) *The left quotient $F \backslash G$ of G by F exists in (Sch/A) , and the sequence*

$$F \times_A G \xrightarrow[\text{pr}_2]{\lambda} G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *The canonical morphism $p : G \rightarrow F \backslash G$ is surjective and open, and $F \backslash G$ is a direct sum of schemes of finite type over A .*
- (ii') *More precisely, $X = F \backslash G$ has a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$; consequently, the connected components of X are of finite type over A , irreducible and all of dimension $\dim G - \dim F$.*
- (iii) *The canonical morphism $F \times_A G \rightarrow G \times_{(F \backslash G)} G$ is surjective.*
- (iv) *If u is a monomorphism,⁽³⁵⁾ then:*
- (a) *$F \times_A G \xrightarrow{(\lambda, \text{pr}_2)} G \times_{(F \backslash G)} G$ is an isomorphism, and $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation.*
 - (a') *$F \backslash G$ represents the quotient fppf sheaf $\widetilde{F \backslash G}$, and $G \rightarrow F \backslash G$ is an F -torsor locally trivial for the fppf topology.*
 - (b) *$F \backslash G$ is flat over A , and is smooth over A if G is.*
 - (c) *$u : F \rightarrow G$ is a closed immersion, and $F \backslash G$ is separated.*
 - (d) *If moreover F is an invariant subgroup of G , there exists one and only one A -group structure on $F \backslash G$ such that $p : G \rightarrow F \backslash G$ is a morphism of A -groups.*

In the proof of this theorem, A' will denote a local A -algebra finite and free over A . If R is a relation involving A' , we shall say that “ $R(A')$ is true when A' is sufficiently large” if there exists an algebra A_1 , local, finite and free over A , such that the relation $R(A')$ holds for every algebra A' , local, finite and free over A_1 .

We shall first prove the theorem when F and G are of finite type over A .

3.2.1. — Suppose for a moment that every point of G has a saturated open neighborhood W such that the groupoid induced by G_* on W has a quasi-section (cf. V §6). Then, by V 6.1,

⁽³⁴⁾**N.D.E.** Point (ii') has been added and point (iv) expanded, taking account of the additions made in 2.5.2, 2.5.4 and Exp. V, 6.1.

⁽³⁵⁾**N.D.E.** In this case, the hypothesis that G is flat can be dropped, cf. subsection 3.3.

one has assertions (i), (ii), (iii) and (iv)(a), and $F \backslash G$ is of finite type over k . Moreover, under the hypothesis of (iv), since $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, assertion (b) follows from EGA IV, 2.2.14 and 17.7.7. On the other hand, assertion (iv)(a') follows from (iv)(a), by Exp. IV, 3.4.3.1, 5.2.2 and 5.1.6. Finally, we shall prove (iv)(c) in 3.2.5, and (ii') and (iv)(d) in section 5.

Let us now show the following assertion:

(†) every finite subset of $F \backslash G$ is then contained in an affine open subset.

(36) If U is a quasi-section of the groupoid induced by G_* on a saturated open subset W of G , $U \otimes_A A'$ is a quasi-section of the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $W \otimes_A A'$. Moreover, if U_* is the (Sch/A) -groupoid induced by G_* on U , $U_* \otimes_A A'$ is identified with the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $U \otimes_A A'$.⁽³⁷⁾

It then follows from the proofs in Exposé V that construction of the quotient $X = F \backslash G$ commutes with an extension $A \rightarrow A'$ of the base of the type considered here.⁽³⁸⁾

Let, then, $x_1, \dots, x_n$ be points of $X = F \backslash G$, which we can suppose closed,⁽³⁹⁾ and $g_1, \dots, g_n$ closed points of G projecting to $x_1, \dots, x_n$. Let V be an everywhere dense affine open subset of X ,⁽⁴⁰⁾ and U its inverse image in G . By 1.2, there exists a local A -algebra A' , finite and free over A , such that the points $g'_1, \dots, g'_p$ of $G' = G \otimes_A A'$ above $g_1, \dots, g_n$ are strictly rational over A' .⁽⁴¹⁾ Since the morphisms $G' \rightarrow G$ and $G \rightarrow X$ are open, $U' = U \otimes_A A'$ is dense in G' , so the open subset $\bigcap_{i=1}^p (U')^{-1} \cdot g'_i$ is nonempty, and therefore contains a closed point x . Thus, by 1.2 (and 0.4.1), one can suppose, enlarging A' if necessary, that x is strictly rational over A' . Then, since $x \in (U')^{-1} \cdot g'_i$, one has $g'_i \in U' \cdot x$.

Denote by V' the inverse image of V in $X' = X \otimes_A A'$; it is an affine open subset of X' , and also the image of U' under the projection $G' \rightarrow X'$. Since right translation r_x is an automorphism of the groupoid $G_* \otimes_A A'$, it induces an automorphism, again denoted r_x , of the quotient X' . Consequently, the image $V' \cdot x = r_x(V')$ of $U' \cdot x$ in X' is an affine open subset of X' containing the images $x'_1, \dots, x'_p$ of $g'_1, \dots, g'_p$.

Now consider the equivalence relation on $X' = X \otimes_A A'$ defined by the projection $X \otimes_A A' \rightarrow X$:

$$X \otimes_A A' \otimes_A A' \begin{matrix} \xrightarrow{d_1} \\ \xrightarrow{d_0} \end{matrix} X \otimes_A A' \longrightarrow X,$$

where d_0 and d_1 are induced by the two canonical injections of A' into $A' \otimes_A A'$. Since A' is a finite free A -algebra, say of rank n , d_0 and d_1 are finite locally free of rank n ; consequently, one can apply the argument in Exp. V, 5.b (taken from the proof of SGA 1, VIII.7.6). One thus obtains that $x'_1, \dots, x'_p$ are contained in a saturated affine open subset W' contained in the affine open subset $V' \cdot x$. The image of W' in X then contains $x_1, \dots, x_n$ and is an affine open subset of X , by V, 4.1 (ii).

(36) **N.D.E.** Let us mention here that if $A = k$ is a field, every quasi-compact open subset of $F \backslash G$ is quasi-projective (a result due to Chow for smooth algebraic groups), cf. [Ray70], VI 2.6. On the other hand, over the artinian local ring $A = \mathbf{C}[\varepsilon]/(\varepsilon^2)$ there exist abelian A -schemes G which are not projective (loc. cit., XII 4.2).

(37) **N.D.E.** What precedes holds for every base change $A \rightarrow A'$.

(38) **N.D.E.** This is expanded in 4.6 below: one must see that formation of the direct image under the morphisms p, λ, pr_2 commutes with flat base changes $A \rightarrow A'$. Since F and G are of finite type over artinian A , the morphisms f in question are all quasi-compact and quasi-separated, and the equality $f_*(\mathcal{O}_X) \otimes_A A' = f'_*(\mathcal{O}_{X'})$ (with obvious notation) follows from EGA IV₁, 1.7.21.

(39) **N.D.E.** Indeed, let $y_1, \dots, y_n$ be arbitrary points of X ; since X is of finite type over A , each y_i has in its closure a closed point x_i , and every open subset containing x_i contains y_i .

(40) **N.D.E.** Such an open subset exists, since X is of finite type over k : X has finitely many irreducible components $C_1, \dots, C_p$, and it suffices to take, for each i , a nonempty affine open subset contained in $C_i - \bigcup_{j \neq i} C_j$. (Here one further knows, by (ii'), that the C_i are disjoint...)

(41) **N.D.E.** The original has been expanded in what follows.

3.2.2. — For every algebra A' local, finite and free over A , denote now by $U(A')$ the set of points of $G \otimes_A A'$ having a saturated open neighborhood W such that the groupoid induced by $G_* \otimes_A A'$ on W has a quasi-section. It is quite clear that $U(A')$ is saturated for the actions of $G(\text{Spec } A')$ on $G \otimes_A A'$. We shall see that, when A' is sufficiently large, $U(A')$ equals $G \otimes_A A'$.

By Theorem V 8.1, $U(A)$ is nonempty, hence contains a closed point y . The proof then proceeds by induction on $\dim(G - U(A))$. Let $g_1, \dots, g_n$ be closed points belonging to the various irreducible components of $G - U(A)$. By 1.2, there exists A' local, finite and free over A such that the points $g'_1, \dots, g'_p$ (resp. $x = x_1, \dots, x_r$) of $G' = G \otimes_A A'$ projecting to $g_1, \dots, g_n$ (resp. y) are strictly rational over A' . Then $U(A')$ contains $(U(A) \otimes_A A') \cdot x^{-1}g'_i$ for every i ; hence $U(A')$ contains $g'_1, \dots, g'_p$, and one has

$$\dim(G' - U(A')) < \dim(G - U(A)).$$

The induction hypothesis then implies the existence of an algebra A'' local, finite and free over A' such that $U(A'') = G' \otimes_{A'} A'' = G \otimes_A A''$.

3.2.3. — We are now in a position to prove the existence of $F \backslash G$ when F and G are of finite type over A . Let A' be sufficiently large over A for $U(A')$ to coincide with $G \otimes_A A'$ (cf. 3.2.2). We shall put $A'' = A' \otimes_A A'$, and, for every A -scheme X , denote by X' and X'' the fibered products $X \otimes_A A'$ and $X \otimes_A A''$. By 3.2.1 and 3.2.2, the quotients $F' \backslash G'$ and $F'' \backslash G''$ exist, and one has the following commutative diagram, in which the first two rows and columns are exact:

$$(*) \quad \begin{array}{ccccc} F'' \times_{A''} G'' & \xrightarrow[\lambda'']{\text{pr}_2''} & G'' & \xrightarrow{p''} & F'' \backslash G'' \\ \downarrow w_2 \quad \downarrow w_1 & & \downarrow v_2 \quad \downarrow v_1 & & \downarrow u_2 \quad \downarrow u_1 \\ F' \times_{A'} G' & \xrightarrow[\lambda']{\text{pr}_2'} & G' & \xrightarrow{p'} & F' \backslash G' \\ \downarrow h & & \downarrow g & & \\ F \times_A G & \xrightarrow[\lambda]{\text{pr}_2} & G & & \end{array} .$$

In this diagram, pr_2' and λ' (resp. pr_2'' and λ'') are obtained from pr_2 and λ by obvious base changes; the morphisms g and h are induced by the canonical injection $A \rightarrow A'$. The canonical morphisms are denoted by p' and p'' ; the morphisms v_1, v_2 and w_1, w_2 are induced by the two canonical injections of A' into A'' . Finally, since construction of the quotient $F' \backslash G'$ commutes with the two base changes $f_1, f_2 : \text{Spec } A'' \rightrightarrows \text{Spec } A'$, one has, denoting by $\pi' : F' \backslash G' \rightarrow \text{Spec } A'$ the structure morphism, canonical isomorphisms for $i = 1, 2$:

$$\tau_i : F'' \backslash G'' \xrightarrow{\sim} (F' \backslash G') \times_{\pi', f_i} \text{Spec } A'',$$

and the morphism u_i is the composite of τ_i and the projection $(F' \backslash G') \times_{\pi', f_i} \text{Spec } A'' \rightarrow F' \backslash G'$.

Now, when one has a diagram of type $(*)$, the first two rows and columns being exact, one easily verifies that $\text{Coker}(\text{pr}_2, \lambda)$ exists if and only if $\text{Coker}(u_1, u_2)$ does, and these two cokernels are identified. The existence of $F \backslash G$ will therefore follow from that of $\text{Coker}(u_1, u_2)$.

Now it follows from compatibility of formation of $F \backslash G$ with the extensions of the base considered here (cf. N.D.E. (37) in 3.2.1, and 4.6 below) that the composite morphism

$$(F' \backslash G') \times_{\pi', f_1} \text{Spec } A'' \xrightarrow{\tau_1^{-1}} F'' \backslash G'' \xrightarrow{\tau_2} (F' \backslash G') \times_{\pi', f_2} \text{Spec } A''$$

is a descent datum on $F' \backslash G'$ relative to $f : \text{Spec } A' \rightarrow \text{Spec } A$. By 3.2.1 (†) and SGA 1, VIII 7.6, this descent datum is effective, that is, $\text{Coker}(u_1, u_2)$ exists (one could moreover use directly Theorem 4.1 of Exp. V).

3.2.4. — To finish the proof of assertions (i), (ii), (iii) and (iv)(a) of 3.2 in the case where F and G are of finite type over A , it remains to study the quotient $F \backslash G$. By V 6.1, assertions

(ii), (iii) and (iv)(a) “become true” after the base change $f : \operatorname{Spec} A' \rightarrow \operatorname{Spec} A$; by EGA IV₂, 2.6.1, 2.6.2 and 2.7.1, these assertions were therefore true before the base change. Finally, to prove the second assertion of (i), i.e. that $F \backslash G$ is the cokernel of $(\operatorname{pr}_2, \lambda)$ in the category of all ringed spaces, one need only refer to V §6.c).

3.2.5. — ⁽⁴²⁾Let us now show assertion (iv)(c) of 3.2, using the proof of VIB, 9.2.1. Denote by $X = F \backslash G$ and by d the morphism $F \times_A G \rightarrow G \times_A G$ with components λ and pr_2 .

Since u is a closed immersion by 2.5.2, and $d = \sigma \circ (u \times \operatorname{id}_G)$, where σ is the automorphism of $G \times_A G$ defined by $\sigma(x, y) = (xy, y)$, d is a closed immersion. On the other hand, by (iv)(a), one has the cartesian square below

$$\begin{array}{ccc} F \times_A G & \xrightarrow{d} & G \times_A G \\ \downarrow & & \downarrow p \times p \\ X & \xrightarrow{\Delta_X} & X \times_A X \end{array}$$

and p , hence also $p \times p$, is faithfully flat and locally of finite presentation. Thus, by fppf descent, since d is a closed immersion, so is Δ_X , i.e. X is separated.

3.3. — ⁽⁴³⁾In Theorem 3.2, the hypothesis that G is flat can be dropped when $u : F \rightarrow G$ is a monomorphism. This generalization is mentioned in Remark 9.3 b) of Exp. VIB, and also in [Ray67a], Example a) i), p. 82. The proof, found in Theorem 4 of [An73], follows from Theorem 3.2 and the following theorem of Grothendieck (mentioned in [Ray67a], Th. 1 ii) and proved in [DG70], §III.2, 7.1). If X is a scheme and R an equivalence relation on X , $\widetilde{X/R}$ will denote the quotient fppf sheaf of X by R (cf. IV 4.4.9).

Theorem 3.3.1 (Grothendieck). *Let A be a ring, X an A -scheme, and $d_0, d_1 : R \rightarrow X$ an A -equivalence relation on X such that d_i is faithfully flat and of finite presentation. Let X_0 be a saturated closed subscheme of X defined by a nilpotent ideal, and R_0 the equivalence relation induced by R on X_0 . Then, if the quotient fppf sheaf $\widetilde{X_0/R_0}$ is represented by an A -scheme, the same holds for $\widetilde{X/R}$.*

For the proof, we refer to [DG70], §III.2, 7.1. Now return to the case where A is an artinian local ring. Let $u : F \rightarrow G$ be a quasi-compact morphism between A -groups locally of finite type, and further suppose u a monomorphism. Then, by 2.5.2, u is a closed immersion.

One can now state the following variant of Theorem 3.2.

Theorem 3.3.2. *Let A be an artinian local ring, G an A -group locally of finite type, and F a closed subgroup of G , flat over A .⁽⁴⁴⁾ Then:*

- (i) *The quotient fppf sheaf $\widetilde{F \backslash G}$ is represented by a separated A -scheme $F \backslash G$ locally of finite type; moreover, the sequence*

$$F \times_A G \rightrightarrows G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *$F \times_A G \xrightarrow{(\lambda, \operatorname{pr}_2)} G \times_{(F \backslash G)} G$ is an isomorphism, and $p : G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, so p is an F -torsor locally trivial for the fppf topology.*
- (iii) *If G is flat (resp. of finite type, resp. smooth) over A , $F \backslash G$ is also so.*

⁽⁴²⁾**N.D.E.** This paragraph has been added.

⁽⁴³⁾**N.D.E.** This subsection has been added.

⁽⁴⁴⁾**N.D.E.** For an example where F is not flat and $\widetilde{F \backslash G}$ is not representable, see [DG70], §III.3, no. 3.3.

- (iv) $X = F \backslash G$ has a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$; consequently the connected components of X are of finite type over A , irreducible and all of dimension $\dim G - \dim F$.
- (v) If moreover F is an invariant subgroup of G , there exists one and only one A -group structure on $F \backslash G$ such that $p : G \rightarrow F \backslash G$ is a morphism of A -groups.

Assertions (i) and (ii) follow from 3.2 and 3.3.1, and, since $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, assertion (iii) follows from EGA IV, 2.2.14, 2.7.1 and 17.7.7. We shall prove assertions (iv) and (v) in section 5. Let us immediately note the following corollary.

Corollary 3.3.3. *Let A be an artinian local ring, G an A -group locally of finite type, and H a closed subgroup of G , flat over A . Denote by p the morphism $G \rightarrow G/H$, and by λ (resp. pr_1) the morphism $G \times H \rightarrow G$ defined by $\lambda(g, h) = gh$ (resp. the projection $G \times H \rightarrow G$). Then, for every open subset U of G/H , one has*

$$\mathcal{O}(U) = \{\varphi \in \mathcal{O}(p^{-1}(U)) \mid \varphi \circ \lambda = \varphi \circ \text{pr}_1\},$$

i.e. $\mathcal{O}(U)$ is the set of $\varphi \in \mathcal{O}(p^{-1}(U))$ such that $\varphi(gh) = \varphi(g)$ for every A -scheme S and $g \in G(S)$, $h \in H(S)$.

Indeed, since $p : G \rightarrow G/H$ is faithfully flat and locally of finite presentation, hence a covering for the fppf topology, this follows from IV, 3.3.3.2.

4. Construction of quotients $F \backslash G$ (general case)

We now suppose the hypotheses of Theorem 3.2 satisfied, F and G not necessarily being of finite type over A .

4.1. — First consider a connected component G^α of G , and let us show that the saturation $\mathcal{S}(G^\alpha)^{(45)}$ of G^α for the equivalence relation defined by the groupoid G_* is an open and closed subset of G (in other words, the union of certain connected components of G).

This saturation is the image of $F \times_A G^\alpha$ under λ , hence is open in G (cf. §3.1). If k is the residue field of A and $\bar{k}$ an algebraic closure of k , it remains to show that the image of $(F \times_A G^\alpha) \otimes_A k$ under $\lambda \otimes_A k$ is closed in $G \otimes_A k$, or, by SGA 1, VIII.4.4, that the image of $(F \times_A G^\alpha) \otimes_A \bar{k}$ under $\lambda \otimes_A \bar{k}$ is closed. Since $G^\alpha \otimes_A \bar{k}$ is the union of finitely many connected components of $G \otimes_A \bar{k}$, one reduces to the case where A is an algebraically closed field, which we shall suppose. In this case, $\mathcal{S}(G^\alpha)$ is the union of the images of G^α under left translations $\ell_{u(x)}$, where x ranges over the closed points of F ; the assertion therefore follows from the fact that these images are connected components of G .

4.2. — Take, in particular, for G^α the connected component G^0 of the origin of G . Then $\mathcal{S}(G^0)$ obviously contains the image of F under u , which is none other than the equivalence class of the origin. On the other hand, if F^β is a connected component of F , $F^\beta \times_A G^0$ is connected (2.1.2), so the image of $F^\beta \times_A G^0$ under λ is contained in the connected component of $u(F^\beta)$ in G . In other words, $\mathcal{S}(G^0)$ is the union of the connected components meeting the image of F .

One will also note that the open subscheme of G whose underlying space is $\mathcal{S}(G^0)$ is a subgroup of G (which we again denote by $\mathcal{S}(G^0)$): indeed, the inversion morphism of G preserves the image of F and permutes the connected components of G meeting this image; it therefore suffices to show that $\nu : G \times_A G \rightarrow G$ sends $\mathcal{S}(G^0) \times_A \mathcal{S}(G^0)$ into $\mathcal{S}(G^0)$, and for this one can suppose A an algebraically closed field (with the notation of 4.1, $\mathcal{S}(G^0) \otimes_A \bar{k}$ is indeed identified with the saturation of $(G \otimes_A \bar{k})^0$ for the equivalence relation defined by the homomorphism $u \otimes_A \bar{k}$); if G^γ and G^δ are then connected components of $\mathcal{S}(G^0)$, $G^\gamma \times_A G^\delta$

⁽⁴⁵⁾**N.D.E.** The saturation of G^α has been denoted by $\mathcal{S}(G^\alpha)$ instead of $\overline{G}^\alpha$.

is connected and its image under ν meets the image of F ; consequently, $u(G^\gamma \times_A G^\delta)$ is contained in a connected component of G meeting $u(F)$.

4.3. — It follows from what precedes that the groupoid G_* with base G defined by u is the direct sum of the groupoids $\mathcal{S}(G_*^\alpha)$ induced by G_* on the various open and closed subsets of G of the form $\mathcal{S}(G^\alpha)$. The cokernel of G_* is therefore the direct sum of the cokernels of these groupoids $\mathcal{S}(G_*^\alpha)$, which one is led to study separately.

First consider the groupoid $\mathcal{S}(G_*^0)$ induced by G_* on $\mathcal{S}(G^0)$. It is clear that $\mathcal{S}(G_*^0)$ is the groupoid with base $\mathcal{S}(G^0)$ defined by the homomorphism from F to $\mathcal{S}(G^0)$ induced by u (§3.1). The cokernel whose existence we wish to prove is therefore identified with $F \setminus \mathcal{S}(G^0)$. Consider, on the other hand, the groupoid

$$G_2^0 \begin{array}{c} \xrightarrow{\ell_2'} \\ \xrightarrow{\ell_1'} \\ \xrightarrow{\ell_0'} \end{array} G_1^0 \begin{array}{c} \xrightarrow{\ell_1} \\ \xrightarrow{\ell_0} \end{array} G_0^0 = G^0$$

induced by $\mathcal{S}(G_*^0)$ on G^0 . If one refers to the construction made explicit in V §3.b), the object then denoted $Y_0 \times_{X_0} X_1$ is none other than $F \times_A G^0$, so G_1^0 is the inverse image of G^0 under the morphism $F \times_A G^0 \rightarrow \mathcal{S}(G^0)$ induced by λ .

I say that this inverse image is $F_0 \times_A G^0$, where F_0 denotes the inverse image of G^0 under u . Indeed, if F^β is a connected component of F_0 , $F^\beta \times_A G^0$ is connected (2.1.2), and $\lambda(F \times_A G^0)$ is contained in G^0 ; conversely, if F^β is a connected component of F not contained in F_0 , the image of $F^\beta \times_A G^0$ is again connected and contains $u(F^\beta)$; if $u(F^\beta)$ is not contained in G^0 , $\lambda(F^\beta \times_A G^0)$ does not meet G^0 .

It follows from what precedes that the groupoid G_*^0 induced by G_* on G^0 is the groupoid with base G^0 defined by the homomorphism $F_0 \rightarrow G^0$ induced by u . Since G^0 , and hence F_0 , are of finite type over A , by paragraph 4, G_*^0 has a cokernel which is none other than $F_0 \setminus G^0$.

I now say that $F_0 \setminus G^0$ is identified with $F \setminus \mathcal{S}(G^0)$. Indeed, the proof is analogous to that of the first part of assertion (i) of Lemma V §6.1; consider the diagram:

$$\mathcal{S}(G^0) \xleftarrow{v} F \times_A G^0 \xrightarrow{\text{pr}_2} G^0,$$

where v is the morphism induced by λ . Since pr_2 has a section, pr_2 is an effective universal epimorphism, so $F_0 \setminus G^0$ coincides with $\text{Coker}(v_0, v_1)$, where

$$V_2 \begin{array}{c} \xrightarrow{v_2'} \\ \xrightarrow{v_1'} \\ \xrightarrow{v_0'} \end{array} V_1 \begin{array}{c} \xrightarrow{v_1} \\ \xrightarrow{v_0} \end{array} V = F \times_A G^0$$

is the inverse image under pr_2 of the groupoid G_*^0 (cf. V §3.a), that is, also the inverse image of $\mathcal{S}(G_*^0)$ under the composite morphism

$$F \times_A G^0 \xrightarrow{\text{inclusion}} F \times_A \mathcal{S}(G^0) \xrightarrow{\text{pr}_2} \mathcal{S}(G^0).$$

Similarly, since v is faithfully flat and quasi-compact, $F \setminus \mathcal{S}(G^0)$ coincides with the cokernel of the inverse image of $\mathcal{S}(G_*^0)$ under the base change v . Now this inverse image is isomorphic to V_* by Exp. V, §3.c; it follows that the canonical inclusion of G_*^0 into $\mathcal{S}(G_*^0)$ induces an isomorphism from $F_0 \setminus G^0$ onto $F \setminus \mathcal{S}(G^0)$.

Finally, note that construction of $F \setminus \mathcal{S}(G^0)$ commutes with finite locally free base changes, since the same holds for $F_0 \setminus G^0$ (cf. N.D.E. (37) and 4.6 below).

4.4. — It remains to construct the cokernel of the groupoid $\mathcal{S}(G_*^\alpha)$ when G^α is any connected component of G . If A' is a sufficiently large local A -algebra finite and free (cf. 3.2), $G^\alpha \otimes_A A'$ is the union of finitely many connected components $C_1, \dots, C_n$ of $G \otimes_A A'$, all having a strictly rational point. For every i , there therefore exists a right translation r_i of $G \otimes_A A'$ sending

$G^0 \otimes_A A'$ onto C_i ; this translation induces an isomorphism from the groupoid $\mathcal{S}(G_*^0) \otimes_A A'$ onto $\mathcal{S}(C_*^i)$, so the groupoid induced by $G_* \otimes_A A'$ on the saturation of C_i has a cokernel.

Since $\mathcal{S}(G_*^\alpha) \otimes_A A'$ is the direct sum of a certain number of the $\mathcal{S}(C_*^i)$, $\mathcal{S}(G_*^\alpha) \otimes_A A'$ has a cokernel; this cokernel is the direct sum of a certain number of copies of $(F_0 \otimes_A A') \setminus (G^0 \otimes_A A')$, so every finite subset of this cokernel is contained in an affine open subset; moreover, construction of this cokernel commutes with finite locally free extensions of the base (cf. N.D.E. (37) and 4.6 below). One therefore sees, as in 3.2.3, that this cokernel has the form $Y \otimes_A A'$, where Y is a cokernel of $\mathcal{S}(G_*^\alpha)$.

4.5. — We have therefore constructed $F \setminus G$ and shown that it is a direct sum of schemes of finite type over A . The other assertions of Theorem 3.2 reduce directly to assertions concerning the groupoids $\mathcal{S}(G_*^\alpha)$. As in V §6, the second assertion of (i) follows from the first and from (ii) and (iii), so it suffices to prove (ii), (iii) and (iv)(a). Since A' is a local A -algebra finite and free, the morphism $A \rightarrow A'$ is faithfully flat and of finite presentation, so, by SGA 1, VIII (3.1, 4.6, 5.4), it suffices to verify the corresponding assertions for the groupoid $\mathcal{S}(G_*^\alpha) \otimes_A A'$. Now the latter is isomorphic to the direct sum of finitely many copies of $\mathcal{S}(G_*^0) \otimes_A A'$ (cf. 4.4), so one is reduced to the groupoid $\mathcal{S}(G_*^0)$.

For the latter one continues to follow the proof established in V §6, as we began to do in 4.3.

4.6. — To finish this paragraph, let us add some remarks concerning Lemma 6.1 and §9.a of Exposé V: with the hypotheses and notation of V §9.a, we seek a condition under which construction of the cokernel of the (Sch/S) -groupoid X_* commutes with an extension $\pi : S' \rightarrow S$ of the base. Since the cokernels of X_* and X'_* are identified with the cokernels of the groupoids U_* and U'_* induced by X_* and X'_* on the quasi-sections U and U' , one is reduced to the case of a (Sch/S) -groupoid satisfying the hypotheses of Theorem V 4.1.

⁽⁴⁶⁾If Y denotes the cokernel of U_* , $Y' = Y \times_S S'$, and Y_1 the cokernel of U'_* , we have seen in V §9.a that the canonical morphism $Y_1 \rightarrow Y'$ is a homeomorphism (and even a universal homeomorphism); one can therefore identify Y_1 and Y' as topological spaces. If $p : U \rightarrow Y$ is the canonical morphism and $p' : U' \rightarrow Y'$ is obtained from it by base change, we then want the sequence of $\mathcal{O}_{Y'}$ -modules

$$(*) \quad \mathcal{O}_{Y'} \longrightarrow p'_*(\mathcal{O}_{U'}) \rightrightarrows p'_*u'_{1*}(\mathcal{O}_{U'_1}) = p'_*u'_{0*}(\mathcal{O}_{U'_1})$$

to be exact.⁽⁴⁷⁾ Since we have placed ourselves under the hypotheses of V 4.1, u_0 and u_1 are finite locally free; and, by V.4.1 (ii), p is integral. Then p and $p \circ u_i$ are affine, hence separated and quasi-compact.

Consequently, if S' is flat over S , it follows from EGA III₁, 1.4.15 (taking account of correction ErrIII 25 in EGA III₂) that sequence $(*)$ is identified with the inverse image of the sequence

$$(**) \quad \mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_U) \rightrightarrows p_*u_{1*}(\mathcal{O}_{U_1}) = p_*u_{0*}(\mathcal{O}_{U_1}),$$

which is an exact sequence.⁽⁴⁸⁾ An analogous argument applies when the groupoid X_* “locally” has quasi-sections (cf. the proof of Theorem V 7.1). One therefore obtains:

Proposition 4.6.1. *Construction of the cokernel of X_* commutes with flat extensions of the base when X_* locally has quasi-sections.*

⁽⁴⁶⁾N.D.E. The following sentence has been added.

⁽⁴⁷⁾N.D.E. In what follows, the original has been modified, the additional hypotheses made on X_* being superfluous.

⁽⁴⁸⁾N.D.E. The following sentence and the numbering 4.6.1 below have been added to bring out the stated result.

4.7. — Now consider the case of the groupoid G_* of Theorem 3.2, temporarily supposing F and G of finite type over A .

By 3.2.2, there exists an algebra A' local, finite and free over A such that the groupoid $G_* \otimes_A A'$ “locally” has quasi-sections. For every extension $T \rightarrow \text{Spec } A$ of the base, the sequence

$$(F'' \backslash G'') \times_{\text{Spec } A} T \rightrightarrows (F' \backslash G') \times_{\text{Spec } A} T \longrightarrow (F \backslash G) \times_{\text{Spec } A} T$$

obtained from diagram (*) of 3.2.3 is exact. If one further supposes T flat over $\text{Spec } A$, $(F'' \backslash G'') \times_{\text{Spec } A} T$ and $(F' \backslash G') \times_{\text{Spec } A} T$ are identified respectively, by 4.6, with the cokernels of the groupoids

$$(G_* \otimes_A A'') \times_{\text{Spec } A} T \quad \text{and} \quad (G_* \otimes_A A') \times_{\text{Spec } A} T.$$

The diagram obtained from 3.2.3 (*) by the base change $T \rightarrow \text{Spec } A$ then shows that $(F \backslash G) \times_{\text{Spec } A} T$ is identified with the cokernel of $G_* \times_{\text{Spec } A} T$. An analogous argument holds in the general case (i.e. when G and F are locally of finite type over A). One therefore obtains:⁽⁴⁹⁾

Proposition 4.7.1. *Under the hypotheses of Theorem 3.2, for every flat A -scheme T , $(F \backslash G) \times_{\text{Spec } A} T$ is identified with the left quotient of $G \times_{\text{Spec } A} T$ by $F \times_{\text{Spec } A} T$.*

5. Connections with Exposé IV and consequences

5.1. — ⁽⁵⁰⁾We return to the notation of §3 and the hypotheses of Theorem 3.2; one then has the following commutative diagram

$$\begin{array}{ccc} F \times_A G \times_A G & \xrightarrow{F \times \nu} & F \times_A G \\ \lambda \times G \downarrow & \text{pr}_2 \times G & \downarrow \lambda \\ G \times_A G & \xrightarrow{\nu} & G \\ p \times G \downarrow & & \downarrow p \\ (F \backslash G) \times_A G & \xrightarrow[\rho]{} & F \backslash G, \end{array}$$

which satisfies $\text{pr}_2 \circ (F \times \nu) = \nu \circ (\text{pr}_2 \times G)$ and $\lambda \circ (F \times \nu) = \nu \circ (\lambda \times G)$. Moreover, since G is supposed flat over A , the vertical sequence on the left is exact by 4.7, so ν induces a morphism of A -schemes:

$$\rho : (F \backslash G) \times_A G \longrightarrow F \backslash G.$$

This morphism ρ makes G act on $F \backslash G$ on the right, as one verifies immediately; moreover, the canonical morphism $G \rightarrow F \backslash G$ commutes with the right actions of G on G and $F \backslash G$.

⁽⁵¹⁾This proves the first assertion of point (ii') of 3.2. By 2.5.4, one then obtains that the connected components of $X = F \backslash G$ are of finite type, irreducible and all of the same dimension. To evaluate this dimension, one can suppose $A = k$ and k algebraically closed. By I, 2.3.3.1, the stabilizer of the k -point $p(e)$ is represented by the fiber $H = p^{-1}(p(e))$, and, since $F \backslash G$ is the quotient of G by F in the category of ringed spaces, this fiber has underlying space $u(F)$; since $\text{Ker}(u)$ is finite, one therefore has $\dim H = \dim u(F) = \dim F$. By 2.5.4 (ii), one therefore obtains $\dim X = \dim G - \dim F$. This proves point (ii') of Theorem 3.2 (and hence also point (iv) of 3.3.2).

5.2. — When the homomorphism of A -groups $u : F \rightarrow G$ is a monomorphism, one can recover 5.1 by using the results of Exposé IV. Indeed, the canonical morphism $p : G \rightarrow F \backslash G$

⁽⁴⁹⁾**N.D.E.** The numbering 4.7.1 below has been added to bring out the stated result.

⁽⁵⁰⁾**N.D.E.** The title of this section (called “Supplements” in the original) has been changed.

⁽⁵¹⁾**N.D.E.** What follows has been added.

is faithfully flat and open by 3.2; it is therefore covering for the fpqc topology (IV 6.3.1), and one can apply Corollaries IV.5.2.2 and IV.5.2.4.

In particular, if we suppose, in addition to the hypotheses of 3.2, that u is the inclusion of an invariant subgroup F into G , there exists one and only one A -group structure on $F \backslash G$ such that the canonical morphism $p : G \rightarrow F \backslash G$ is a homomorphism of A -groups.⁽⁵²⁾ This proves point (v) of 3.3.2.

5.3. — We shall now review some statements of Exposé IV.

5.3.1. — Statements IV 5.2.7 and IV 5.3.1 translate as follows. Let F and G be two groups locally of finite type and flat over A , F being an invariant closed subgroup of G . The maps $H \mapsto F \backslash H$ and $H' \mapsto H' \times_{(F \backslash G)} G$ define a bijective correspondence between flat A -subgroups of G containing F and flat A -subgroups of $F \backslash G$. In this bijection, closed (resp. invariant) subgroups of G containing F correspond to closed (resp. invariant) subgroups of $F \backslash G$.⁽⁵³⁾

5.3.2. — Proposition IV 5.2.9 implies the following result. Let F, H, G be groups locally of finite type and flat over A ; suppose $F \subset H \subset G$, with F closed in G and invariant in H . Under these conditions, $F \backslash H$ acts freely on $F \backslash G$ on the left, the quotient scheme $(F \backslash H) \backslash (F \backslash G)$ exists, and one has a canonical isomorphism of schemes with operator group G :

$$(F \backslash H) \backslash (F \backslash G) = H \backslash G.$$

5.3.3. — Finally, from IV 5.2.8 follows the assertion below. Let F, H, G be groups locally of finite type and flat over A ; suppose F contained, closed and invariant in G , H contained in G , and $F \cap H$ flat over A . Let, then, $F \times_A^\tau H$ be the A -group with underlying scheme the product $F \times_A H$, multiplication being defined by the morphism “ $((x, h), (y, h')) \mapsto (xhyh^{-1}, hh')$ ”; similarly, let $u : H \cap F \rightarrow F \times_A^\tau H$ be the monomorphism $x \mapsto (x^{-1}, x)$, and $F \cdot H$ the quotient $(F \cap H) \backslash (F \times_A^\tau H)$. Under these conditions, there exists a canonical isomorphism

$$F \backslash (F \cdot H) = (F \cap H) \backslash H.$$

5.4. —⁽⁵⁴⁾ Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, whose kernel N is flat over A . In this case, by 3.3.2 and 5.2, the quotient A -group $C = N \backslash G$ exists, and the morphism $p : G \rightarrow C$ is faithfully flat and locally of finite presentation. On the other hand, by IV 5.2.6, u induces a monomorphism $v : C \rightarrow H$, which is quasi-compact (since u is and $G \rightarrow C$ is surjective, cf. EGA IV₁, 1.1.3), hence is a closed immersion by 2.5.2. One has therefore obtained the following proposition:

Proposition 5.4.1. *Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, such that $N = \text{Ker } u$ is flat over A . Then one has the factorization:*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \downarrow & \nearrow i & \\ N \backslash G & & \end{array}$$

where p is faithfully flat and locally of finite presentation, and i a closed immersion.

Further suppose G flat over A . Then, by 3.3.2, $C = N \backslash G$ is flat over A , and hence the quotient $X = C \backslash H$ exists in (Sch/A) and represents the quotient fppf sheaf $\widetilde{C \backslash H}$, and

⁽⁵²⁾**N.D.E.** The following sentence has been added.

⁽⁵³⁾**N.D.E.** In addition to the cited statements of Exp. IV, one uses the fact that, since $G \rightarrow F \backslash G$ is faithfully flat, an A -subgroup H of G is flat over A if and only if $F \backslash H$ is.

⁽⁵⁴⁾**N.D.E.** The original has been expanded in what follows; in particular, Proposition 5.4.1 has been added.

$q : H \rightarrow X$ is a C -torsor. Consequently, denoting by $e : \operatorname{Spec} A \rightarrow G$ the unit section of G , v induces an isomorphism of fppf sheaves between $\tilde{C}$ and the fibered product of q and $q \circ e : \operatorname{Spec} A \rightarrow X$, represented by a closed subscheme of H . Consequently, v is an isomorphism of C onto a closed subgroup scheme K of G (equal to the stabilizer of the A -point $q \circ e$ of X). (This provides another proof that every quasi-compact monomorphism $v : C \rightarrow H$ between A -groups locally of finite type is a closed immersion, cf. 2.5.2 and VIB 1.4.2.)

Further suppose C an invariant subgroup of H ; in this case, the A -group $\overline{H} = C \backslash H$ is the cokernel in the category of A -groups of the morphism $u : G \rightarrow H$, and K is the kernel of the morphism $H \rightarrow \overline{H}$. When G and H are abelian A -groups, K is the image of u in the category of abelian A -groups, whereas $C = (\operatorname{Ker} u) \backslash G$ is the coimage of u . Taking account of the isomorphism $C \xrightarrow{\sim} K$ just established, one obtains:⁽⁵⁵⁾

Theorem 5.4.2. *Let k be a field. The category of commutative algebraic k -groups is abelian.*

Indeed, when k is a field, $\operatorname{Ker}(u)$ is flat over k whatever u may be.

⁽⁵⁶⁾Note that the full subcategory of affine commutative algebraic k -groups is thick. Indeed, consider an exact sequence of commutative algebraic k -groups:

$$1 \longrightarrow N \longrightarrow G \longrightarrow G/N \longrightarrow 1.$$

If G is affine, it is clear that N is so, and G/N is also so by a theorem of Chevalley, cf. VIB, 11.17. Conversely, if N and G/N are affine, G is also so by VIB, 9.2 (viii). One therefore obtains:

Corollary 5.4.3. *Let k be a field. The category of affine commutative algebraic k -groups is abelian.*

Let us further mention that the category of all affine commutative k -groups (not necessarily of finite type) is abelian; this follows from VIB, 11.17 and 11.18.2 (cf. [DG70], §III.3, 7.4); see also VIIB, 2.4.2 for a proof using formal groups.

5.5. — Let G be a group locally of finite type and flat over an artinian local ring A . One knows (2.3) that the connected component G^0 of the origin is an invariant open subgroup scheme of G , hence also flat over A . Then, by 3.2 and 5.2, $G^0 \backslash G$ is an A -group scheme flat over A . Moreover, since each connected component G^α of G is saturated for the equivalence relation defined by G^0 , $G^0 \backslash G$ is the direct sum of the $G^0 \backslash G^\alpha$ (cf. 4.3). In particular, the connected component of the origin in $G^0 \backslash G$ is none other than $G^0 \backslash G^0 \simeq \operatorname{Spec} A$, and hence $G^0 \backslash G \rightarrow \operatorname{Spec} A$ is a local isomorphism at the origin. Consequently, $G^0 \backslash G$ is étale over $\operatorname{Spec} A$, by VIB, 1.3.⁽⁵⁷⁾ One therefore obtains the following proposition (for point (ii), we refer to [DG70], §II.5, 1.7–1.10):

Proposition 5.5.1. *Let A be an artinian local ring and G an A -group locally of finite type and flat.*

- (i) $G^0 \backslash G$ is an étale A -group.
- (ii) Consequently, if $A = k$ is an algebraically closed field, $G^0 \backslash G$ is a constant k -group, acting simply transitively on the set of connected components of G ; hence, if G is algebraic, $G^0 \backslash G$ is finite.

5.6. — Let now k be a perfect field and G a k -group locally of finite type. We have seen (0.2) that G_{red} is then a subgroup scheme of G . Moreover, the equivalence class of the origin

⁽⁵⁵⁾**N.D.E.** The number 5.4.2 has been added to this theorem.

⁽⁵⁶⁾**N.D.E.** What follows has been added.

⁽⁵⁷⁾**N.D.E.** The original has been expanded in what precedes, and Proposition 5.5.1 has been added to bring out this result.

of G for the left action of G_{red} on G is the whole underlying space of G . Thus, by Theorem 3.2, one obtains:

Proposition 5.6.1. *Let k be a perfect field and G a k -group locally of finite type. Then the k -scheme $G_{\text{red}} \backslash G$ is the spectrum of a finite local k -algebra with residue field k .*

⁽⁵⁸⁾Indeed, by 3.2, $G_{\text{red}} \backslash G$ has only one point, with residue field k , and is a k -scheme of finite type; it is therefore the spectrum of a finite-dimensional local k -algebra (cf. EGA I, 6.4.4).

Proposition 5.6.2. *Let $u : F \rightarrow G$ be a morphism between groups locally of finite type over a perfect field k . The following assertions are equivalent:*

- (i) u is flat.
- (ii) $u^0 : F^0 \rightarrow G^0$ is dominant, and the morphism $v : F_{\text{red}} \backslash F \rightarrow G_{\text{red}} \backslash G$ induced by u is flat.

⁽⁵⁹⁾Indeed, consider the following commutative diagram:

$$\begin{array}{ccc} F & \xrightarrow{p} & F_{\text{red}} \backslash F \\ u \downarrow & & \downarrow v \\ G & \xrightarrow{q} & G_{\text{red}} \backslash G, \end{array}$$

where p and q denote the canonical projections. By 3.2 (iv), p and q are faithfully flat; consequently, if u is flat, $q \circ u = v \circ p$ is flat, and therefore so is v .

Conversely, suppose v flat and u^0 dominant. Since u^0 is quasi-compact (F^0 being of finite type over k by 2.4, hence noetherian), it sends the generic point ξ of F^0 to the generic point η of G^0 . Let R be the finite local k -algebra whose spectrum is $G_{\text{red}} \backslash G$, and $\mathfrak{m}$ its maximal ideal. One has local morphisms of local rings $R \rightarrow \mathcal{O}_{G,\eta} \rightarrow \mathcal{O}_{F,\xi}$. Note that one has a cartesian square:

$$\begin{array}{ccc} G_{\text{red}} & \longrightarrow & G \\ \downarrow & & \downarrow q \\ \text{Spec}(R/\mathfrak{m}) & \longrightarrow & \text{Spec}(R) \end{array}$$

and therefore $\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta} \simeq \mathcal{O}_{G_{\text{red}},\eta} = \kappa(\eta)$, so $\mathcal{O}_{F,\xi}/\mathfrak{m}\mathcal{O}_{F,\xi}$ is flat over $\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta}$.

On the other hand, since q and $v \circ p$ are flat, G and F are flat over R . Consequently, by the local flatness criterion (cf. EGA IV₃, 11.3.10.2), $\mathcal{O}_{F,\xi}$ is flat over $\mathcal{O}_{G,\eta}$, that is, u is flat at ξ . Thus, by 2.5.3, u is flat.

6. Supplements on k -groups not necessarily of finite type

⁽⁶⁰⁾Let us further mention the following results, which will be useful in the addition VIB, §12. Fix a base field k .

Lemma 6.1. *Let G be a k -group. For every $x \in G$, there exists a point $u \in G \times G$ such that $\mu(u) = x$ and both projections $p_1(u)$ and $p_2(u)$ are maximal points of G .*

PROOF. Put $K = \kappa(x)$. Since the projection $G_K \rightarrow G$ sends maximal points to maximal points, one reduces to the case where x is rational. Then left translation λ_x (resp. right translation ρ_x) gives a morphism $G \rightarrow G \times G$, $g \mapsto (\lambda_x(g^{-1}), g)$ (resp. $g \mapsto (g, \rho_x(g^{-1}))$),

⁽⁵⁸⁾**N.D.E.** The numbering 5.6.1 and the following proof have been added.

⁽⁵⁹⁾**N.D.E.** The proof of (ii) $\Rightarrow$ (i) has been expanded, and the diagram below simplified.

⁽⁶⁰⁾**N.D.E.** The following results, taken from [Per75], have been added. Note that Lemma 6.1 can be expressed, in Weil's language, by saying that "every point of G is a product of two generic points".

inducing an isomorphism from G onto $\mu^{-1}(x)$, inverse to p_2 (resp. p_1). Thus, if u is a maximal point of $\mu^{-1}(x)$, $p_1(u)$ and $p_2(u)$ are maximal points of G , so u has the required property. $\square$

Corollary 6.2. *Let $f : G \rightarrow H$ be a quasi-compact dominant morphism of k -groups.*

- (i) *f is surjective.*
- (ii) *If H is reduced, f is faithfully flat.*

PROOF. Denote by μ_H (resp. μ_G) multiplication in H (resp. G). Let $h \in H$. By 6.1, there exists $u \in H \times H$ such that $\mu_H(u) = h$ and $\alpha = p_1(u)$ and $\beta = p_2(u)$ are maximal points of H . Since f is quasi-compact and dominant, $f^{-1}(\alpha)$ and $f^{-1}(\beta)$ are nonempty (cf. EGA IV₁, 1.1.5), so there exists $v \in G \times G$ such that $(f \times f)(v) = u$ (cf. EGA I, 3.5.2). Then $g = \mu_G(v)$ satisfies $f(g) = h$. This shows that f is surjective.

Further suppose H reduced. Then $\mathcal{O}_{H,\alpha}$ is a field, and we have seen above that $f^{-1}(\alpha) \neq \emptyset$, so f is flat at every point ξ of $f^{-1}(\alpha)$, hence f is flat by Lemma 2.5.3. $\square$

Recall 6.3. Recall (cf. EGA IV₃, 11.10.1) that a morphism of schemes $f : X \rightarrow Y$ is said to be schematically dominant if it satisfies the following condition: for every open subset U of Y , if Z is a closed subscheme of U such that the morphism $f^{-1}(U) \rightarrow U$ factors through Z , then $Z = U$. When f is quasi-compact and quasi-separated, this amounts to saying that the closed image of X under f is Y (cf. loc. cit., 11.10.3 (iv) and EGA I, 9.5.8).

Proposition 6.4. *Let $f : H \rightarrow G$ be a quasi-compact morphism of k -groups. Then the closed image of f is a subgroup scheme H' of G , and f factors as:*

$$\begin{array}{ccc} H & \xrightarrow{f} & G \\ f' \downarrow & \nearrow i & \\ H' & & \end{array}$$

where f' is schematically dominant, quasi-compact and surjective.

PROOF. Since H is separated (0.3), f is quasi-compact and separated, so $f_*(\mathcal{O}_H)$ is a quasi-coherent $\mathcal{O}_G$ -module, and the closed image H' of f exists and is the closed subscheme of G defined by the quasi-coherent ideal $I = \text{Ker}(\mathcal{O}_G \rightarrow f_*(\mathcal{O}_H))$ (cf. EGA I, §9.5).

Denote by c_G, μ_G (resp. c_H, μ_H) the inversion and multiplication morphisms of G (resp. H). Then $c_G \circ f = f \circ c_H$ factors through the closed subscheme $c_G(H')$, whence $H' \subset c_G(H')$, and therefore $H' = c_G(H')$ (since $c_G^2 = \text{id}_G$). Similarly, since $f \circ \pi_H = \pi_G \circ (f \times f)$ factors through H' , $f \times f$ factors through the closed subscheme $\pi_G^{-1}(H')$ of $G \times G$. On the other hand, since formation of the closed image commutes with flat base changes (EGA III 1.4.15 and IV₁ 1.7.21), the closed image of $f \times \text{id}_H$ (resp. $\text{id}_{H'} \times f$) is $H' \times H$ (resp. $H' \times H'$). Thus, by “transitivity of closed images” (EGA I, 9.5.5), the closed image of $f \times f$ is $H' \times H'$, hence contained in $\pi_G^{-1}(H')$, i.e. the restriction of π_G to $H' \times H'$ factors through H' . This shows that H' is a closed subgroup scheme of G . Denote the inclusion $H' \hookrightarrow G$ by i .

Then $f = i \circ f'$, where $f' : H \rightarrow H'$ is schematically dominant and quasi-compact (since f is quasi-compact and i separated). Thus, by 6.2, f' is surjective. This proves 6.4. $\square$

One can now state the following theorem ([Per75], V 3.1 & 3.2; see also [Per76], 0.0 & 0.1).

Theorem 6.5 (D. Perrin). *Let G be a quasi-compact k -group. Then:*

- (i) *G is the projective limit of a filtered system (G_i) of k -groups of finite type (whose transition morphisms $u_{ij} : G_j \rightarrow G_i$ are affine for sufficiently large i), and the morphisms $G \rightarrow G_i$ are faithfully flat (and affine for sufficiently large i).*
- (ii) *Let H be a closed k -subgroup of G . Then the quotient fpqc sheaf $\widetilde{G/H}$ is a k -scheme in the following two cases:*

- (1) *The immersion $H \rightarrow G$ is of finite presentation; in this case, G/H is of finite type over k .*
- (2) *H is invariant in G .*

For the proof of this theorem (which rests on several intermediate theorems), we refer to [Per75]. For the reader's convenience, let us nevertheless prove the two corollaries below, cf. [Per75] V 3.3 to 3.4 or [Per76] 4.2.3 to 4.2.5.

Corollary 6.6. *Let $f : G \rightarrow H$ be a quasi-compact morphism of k -groups.*

- (i) *If f is schematically dominant, it is faithfully flat.*
- (ii) *This is the case, in particular, if H is affine and the morphism $f^\natural : \mathcal{O}(H) \rightarrow \mathcal{O}(G)$ is injective.*

PROOF. Suppose f schematically dominant. Then, by 6.2 (i), f is surjective, so, by 2.5.3, it suffices to show f flat at the neutral element of G . Since $\mathcal{O}_{H,e} = \mathcal{O}_{H^0,e}$ (cf. 2.6.5), one can replace H by H^0 , hence suppose H irreducible. Then H is quasi-compact (loc. cit.), and, since f is quasi-compact, so is G . By 6.5 (i), $H = \varprojlim_i H_i$, where each H_i is an algebraic k -group. Denote by N_i the kernel of $G \rightarrow H_i$; it is a closed invariant k -subgroup of G . Moreover, since the unit section $\{e\} \rightarrow H_i$ is of finite presentation, so is the immersion $N_i \rightarrow G$; hence, by 6.5 (ii), the fpqc quotient G/N_i is an algebraic k -group. One then has a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ q_i \downarrow & & \downarrow p_i \\ G/N_i & \xrightarrow{f_i} & H_i \end{array}$$

where f is schematically dominant and p_i, q_i faithfully flat (hence schematically dominant). Then $f_i \circ q_i = p_i \circ f$ is schematically dominant, and hence so is f_i . On the other hand, f_i is a monomorphism, hence a closed immersion, since G/N_i and H_i are algebraic (2.5.2). It follows that f_i is an isomorphism, and therefore $G \rightarrow H_i$ is faithfully flat. Then, by [BAC] I §2.7, Prop. 9, the morphism $G \rightarrow H$ is flat; on the other hand, it is surjective by 6.2 (i), hence faithfully flat. This proves point (i), and point (ii) follows, for if H is affine and $f^\natural$ injective, the closed image of f equals H , so f is schematically dominant (cf. 6.4). $\square$

Corollary 6.7. *Let $u : G \rightarrow H$ be a morphism of k -groups, and $N = \text{Ker}(u)$. Suppose u quasi-compact.*

- (i) *The quotient fpqc sheaf $\widetilde{G/N}$ is represented by a k -group scheme G/N , and u factors as:*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \downarrow & \nearrow i & \\ G/N & & \end{array}$$

where p is faithfully flat and i a closed immersion.

- (ii) *In particular, if u is a monomorphism, u is a closed immersion, and if u is schematically dominant, u is faithfully flat.*

PROOF. (i) By 6.4 and 6.6, the closed image G' of u is a closed subgroup scheme of G , and $p : G \rightarrow G'$ is faithfully flat and quasi-compact. One obviously has $\text{Ker}(p) = \text{Ker}(u) = N$, and hence, by Exp. IV, 3.3.2.1 and 5.1.7, G' represents the quotient fpqc sheaf $\widetilde{G/N}$.

(ii) The second assertion is contained in 6.6; let us show the first. If u is a monomorphism, so is p ; then p is both a monomorphism and an effective epimorphism, hence an isomorphism (cf. IV, 1.14). This proves 6.7. $\square$

Finally, let us mention the following corollaries (cf. [Per76], 4.2.6 to 4.2.8).

Corollary 6.8. *The category of quasi-compact commutative k -group schemes is abelian.*

Taking account of 6.7, the proof is analogous to that of 5.4.2.

Corollary 6.9. *If $\text{char}(k) = 0$, every k -group scheme G is geometrically reduced.*

Indeed, if $g \in G$ and K is an algebraic closure of $\kappa(g)$, one has $\mathcal{O}_{G,e} \otimes_k K \simeq \mathcal{O}_{G_K, g_K}$, so it suffices to show that $\mathcal{O}_{G,e} = \mathcal{O}_{G^0,e}$ is geometrically reduced. One is thus reduced to the case where G is connected, hence quasi-compact (2.6.5). Then the result follows from 6.5 and Cartier's theorem for algebraic groups (cf. VIB, 1.6.1 or [DG70] §II.6, Th. 1.1).

Corollary 6.10. *Let G be a quasi-compact k -group. Suppose k algebraically closed.*

- (i) *Let $f : G \rightarrow H$ be a faithfully flat morphism of k -groups. Then the induced map $G(k) \rightarrow H(k)$ is surjective.*
- (ii) *The set of rational points is dense in G .*

For the proof, we refer to [Per75], V 3.7 & 3.9.

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⁽⁶¹⁾N.D.E. Additional references cited in this exposé.

Exposé VIB. Generalities on group schemes

by J.-E. Bertin

⁽⁰⁾This exposé, which does not correspond to any oral lecture of the seminar, is intended to bring together a number of commonly used technical results concerning group schemes.*

1. Morphisms of groups locally of finite type over a field

1.1. — Let A be an artinian local ring, G and H two A -groups⁽²⁾ and $u : G \rightarrow H$ a morphism of A -groups. Then u induces a morphism of groups $u(A) : G(A) \rightarrow H(A)$.⁽³⁾ Since $H(A)$ acts on H by right translation, $u(A)$ defines by restriction an action of $G(A)$ on H . This action is compatible with the morphism u and the action of $G(A)$ on G defined by right translation. Since $G(A)$ acts transitively on the strictly rational points of G (see VIA 0.4 for the definition), one sees that these points “all behave in the same way with respect to u ”; from this spring the following properties:

Proposition 1.2. *Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type over A . Then the set $u(G)$ is closed in H and one has $\dim G = \dim u(G) + \dim \text{Ker } u$.⁽⁴⁾*

Since u commutes with the inversion morphisms of G and H , the image $u(G)$ is invariant under the inversion morphism of H ; the same therefore holds for $\overline{u(G)}$, its closure in H . On the other hand, let L be the set of points of $H \times_A H$ whose two projections belong to $u(G)$; it is clear that L is the image of the morphism $u \times_A u : G \times_A G \rightarrow H \times_A H$; thus the multiplication morphism of H sends L into $u(G)$, in other words $u(G) \cdot u(G) = u(G)$. On the other hand, Lemma 1.2.1 below shows that $\overline{L}$, the closure of L in $H \times_A H$, is the set of points of $H \times_A H$ whose two projections belong to $\overline{u(G)}$; thus $\overline{u(G)} \cdot \overline{u(G)} = \overline{u(G)}$, so that the reduced subscheme of H whose underlying space is $\overline{u(G)}$ is naturally endowed with a group structure in the category $(\text{Sch}/k)_{\text{red}}$, where k is the residue field of A (cf. VIA 0.2).

Let us prove the first assertion of 1.2: replacing A by the algebraic closure of its residue field k if necessary, we may suppose that A is an algebraically closed field k (cf. EGA IV₂, 2.3.12). Replacing u by $u_{\text{red}} : G_{\text{red}} \rightarrow H_{\text{red}}$ if necessary, one may suppose G and H reduced; in this case, as we have just seen, $u(G)$ is the underlying space of a reduced subgroup scheme of G ; we may therefore suppose u dominant. Then $G(k)$ acts transitively on the set of connected components of H and it suffices to show that $u(G) \cap H^0$ is closed: one is reduced to the case where H is connected, hence irreducible and of finite type (VIA 2.4). Then u is of finite type since it is quasi-compact and locally of finite type; since H is noetherian, $u(G)$

⁽⁰⁾**N.D.E.** Version of 13 October 2024.

*This exposé has been fairly substantially revised since its mimeographed edition; in particular, §§10 and 11 have been entirely rewritten.⁽¹⁾

⁽¹⁾**N.D.E.** And also §5, cf. N.D.E. (34) and (46).

⁽²⁾**N.D.E.** Recall the convention adopted at the beginning of VIA: an A -group is an A -group scheme (and such a scheme is separated, cf. VIA, 0.3).

⁽³⁾**N.D.E.** “Homomorphism” has been replaced by “morphism”.

⁽⁴⁾**N.D.E.** This statement also appears in VIA, 2.5.2.

is constructible (EGA IV₁, 1.8.5), hence contains an open subset V of H (EGA 0_{III}, 9.2.2), and then, by VIA 0.5, one has $H = V \cdot V \subset u(G) \cdot u(G) = u(G)$.

Let us prove the second assertion. First recall that the functor $\text{Ker } u$ (cf. I, 2.3.6.1) is representable by $u^{-1}(e)$, where e denotes the identity element of H . When u is locally of finite type, $\text{Ker } u$ is therefore locally of finite type over A . As before, this time by EGA IV₂, 4.1.4, one reduces to the case where A is an algebraically closed field k . One may moreover suppose G and H irreducible and of finite type and u dominant: indeed, since k is algebraically closed, it is clear that the connected components of G , on whose set $G(k)$ acts transitively, all have the same dimension, and that, if u^0 is the restriction of u to G^0 , then $(\text{Ker } u)^0 \subset \text{Ker } u^0$, and $\dim \text{Ker } u^0 = \dim \text{Ker } u$. One sees similarly that u is then of finite type over k . If η denotes the generic point of H , one has $\dim u^{-1}(\eta) = \dim G - \dim H$ (EGA IV₃, 10.6.1 (ii)). By EGA IV₃, 9.2.3 and 9.2.6, the set of $y \in H$ such that $\dim u^{-1}(y) = \dim u^{-1}(\eta)$ contains a nonempty open subset V . Since u is dominant, $U = u^{-1}(V)$ is then a nonempty open subset of G , and contains a closed point x of G , since G is a Jacobson scheme (EGA IV₃, 10.4.7). Then right translation r_x is an isomorphism of $\text{Ker } u$ onto $u^{-1}(u(x))$, so that:

$$\dim \text{Ker } u = \dim u^{-1}(u(x)) = \dim u^{-1}(\eta) = \dim G - \dim H.$$

Lemma 1.2.1. *Let $f : X' \rightarrow X$ and $g : Y' \rightarrow Y$ be two quasi-compact dominant morphisms of schemes over an artinian local ring A ; then $f \times_A g : X' \times_A Y' \rightarrow X \times_A Y$ is dominant (and quasi-compact).*

Indeed, one has $f \times_A g = (\text{id}_X \times_A g) \circ (f \times_A \text{id}_{Y'})$. It therefore suffices to show that $f \times_A \text{id}_{Y'}$ and $\text{id}_X \times_A g$ are dominant. For this one may replace A by its residue field k . In this case, X and Y' are flat over $A = k$, and since $f \times_A \text{id}_{Y'}$ (resp. $\text{id}_X \times_A g$) is obtained from f (resp. g) by the flat base change $Y' \rightarrow A$ (resp. $X \rightarrow A$), it is dominant (and quasi-compact), by EGA IV₂, 2.3.7.

Counterexample 1.2.2. Let k be a field of characteristic 0, G the constant k -group $\mathbf{Z}$ and H the additive k -group $\mathbf{G}_{a,k}$. Let $u : G \rightarrow H$ be a morphism of k -groups. If $u \neq 0$, $u(G)$ is not closed in H .

Proposition 1.3. ⁽⁵⁾ *Let A be an artinian local ring, k its residue field, G an A -group locally of finite type and flat, and $u : G \rightarrow H$ a morphism of A -groups. The following assertions are equivalent:*

- (i) *u is flat (resp. quasi-finite, resp. unramified, resp. smooth, resp. étale) at a point of G .*⁽⁶⁾
- (ii) *u is flat (resp. quasi-finite, resp. unramified, resp. smooth, étale).*

PROOF. It suffices to show that (i) implies (ii). First one has the following lemma:

Lemma 1.3.0. *Let $A \rightarrow B \rightarrow C$ be morphisms of commutative rings and $\mathfrak{n}$ a nilpotent ideal of A . Suppose that $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$ -algebra of finite type.*

⁽⁵⁾**N.D.E.** In the original, 1.3, 1.3.1 and 1.3.1.1 are stated for A a field. For completeness, the case of an artinian local ring has been treated and, to do so, Lemma 1.3.0 has been added.

⁽⁶⁾**N.D.E.** Recall that a morphism of schemes $f : X \rightarrow Y$ is said to be of finite type (resp. of finite presentation, resp. quasi-finite) at x if there are affine open subsets V containing $f(x)$ and U containing x such that $f(U) \subset V$ and $\mathcal{O}_X(U)$ is an $\mathcal{O}_Y(V)$ -algebra of finite type (resp. of finite presentation, resp. and if moreover the fiber $f^{-1}(f(x'))$ is finite for every $x' \in U$ (see also N.D.E. (40) of Exp. V)). One says that f is locally quasi-finite (resp. of finite type, of finite presentation) if it is so at every point x . On the other hand, one says that f is smooth (resp. unramified, resp. étale) at x if there is an open neighborhood U of x such that the morphism $f|_U : U \rightarrow Y$ is smooth (resp. unramified, resp. étale). In view of these definitions, it is clear that the set of points where f is of finite presentation, resp. of finite type, quasi-finite, smooth, unramified, étale, is an open subset of X . Moreover, since the flatness locus of a morphism locally of finite presentation is open (EGA IV₃, 11.3.1), the set of points of X where f is of finite presentation and flat is also open in X . All this will be used on numerous occasions below.

- (i) *Then C is a B -algebra of finite type.*
- (ii) *Moreover, if C is flat over A and $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$ -algebra of finite presentation, then C is a B -algebra of finite presentation.*

Indeed, let $x_1, \dots, x_n$ be elements of C whose images generate $C/\mathfrak{n}C$ as a $(B/\mathfrak{n})$ -algebra. By the nilpotent Nakayama lemma, the x_i generate C as a B -algebra. This proves (i). Let ϕ be the surjective morphism $B[X_1, \dots, X_n] \rightarrow C$ thus obtained, and let $I = \text{Ker}(\phi)$.

Now suppose that C is flat over A and $C/\mathfrak{n}C$ is of finite presentation over $B/\mathfrak{n}B$. Then, on the one hand, $I/\mathfrak{n}I$ is identified with the kernel of $\bar{\phi} = \phi \otimes_A (A/\mathfrak{n})$. On the other hand, by EGA IV₁, 1.4.4, the kernel of $\bar{\phi}$ is a finitely generated ideal. Then let $P_1, \dots, P_s$ be elements of I whose images generate $I/\mathfrak{n}I$ as an ideal; by the nilpotent Nakayama lemma, they generate I . This proves (ii).

Let us return to the proof of 1.3. Let x be an arbitrary point of G . Since G is flat over A , by the fiberwise flatness criterion, in the form of EGA IV₃, 11.3.10.2, u will be flat at x if $u \otimes_A k$ is. Similarly, by the preceding lemma, one sees that u will be of finite type (resp. of finite presentation) at x if $u \otimes_A k$ is. Since the other properties can then be checked on the fibers (cf. EGA IV₄ 17.4.1, 17.5.1 and 17.6.1 for unramified, smooth and étale), one is reduced to the case where $A = k$.

Now let x be a point of G where one of the conditions of 1.3 (i) holds. Since the properties under consideration are preserved by (fpqc) descent (cf. EGA IV, 2.5.1, 2.7.1 and 17.7.1), replacing k by an algebraic closure of $\kappa(x)$ reduces one to the case where k is algebraically closed and $x \in G(k)$.

Since G is a Jacobson scheme (cf. EGA IV₃, 10.4.7) and since the set W of points of G where u is flat (resp. quasi-finite, unramified, smooth, or étale) is stable under generization⁽⁷⁾ (resp. open), it suffices to show that every point y of $G(k)$ belongs to W . But, for such a point y , the translation $r_y \circ r_{x^{-1}}$ sends x to y , so u has the desired property at y , i.e. $y \in W$. $\square$

Corollary 1.3.1. *Let A be an artinian local ring, k its residue field, G a flat A -group. The following assertions are equivalent:⁽⁸⁾*

- (i) *G is locally quasi-finite (resp. unramified, resp. smooth, resp. étale) over A at a point.*
- (ii) *G is locally quasi-finite (resp. unramified, resp. smooth, resp. étale) over A .*

⁽⁹⁾Indeed, if G satisfies one of the conditions of (i) at a point x , there is an open neighborhood U of x which is of finite type over A . Consequently, it suffices to apply 1.3 to the case where H is the unit A -group, taking account of the following lemma:

Lemma 1.3.1.1. *Let A be an artinian local ring and G an A -group. If there is a nonempty open subset of G of finite type over A , then G is locally of finite type over A .*

By Lemma 1.3.0, one may suppose A equal to its residue field k . Moreover, by (fpqc) descent, one may suppose k algebraically closed (cf. EGA IV₂, 2.7.1). Let V be the open subset of G formed by the points where G is of finite type over k ; by hypothesis, $V \neq \emptyset$. Since G is a Jacobson scheme, V contains a closed point x and, to show that $V = G$, it suffices to show that every closed point y of G belongs to V . But, for such a point y , the translation $r_y \circ r_{x^{-1}}$ sends x to y , whence $y \in V$.

⁽⁷⁾**N.D.E.** The original indicated that W_{flat} is open, which does not seem evident a priori...

⁽⁸⁾**N.D.E.** Note that it is not sufficient to suppose that G is flat over A at a point: for example, supposing $A \neq k$, let H be the constant A -group $(\mathbf{Z}/2\mathbf{Z})_A$ and G the closed A -subgroup of H whose nonidentity component is reduced; then the structure morphism $G \rightarrow \text{Spec } A$ is a local isomorphism at the identity point e , but is not flat at the point $g \neq e$.

⁽⁹⁾**N.D.E.** The following sentence has been added, and the proof of Lemma 1.3.1.1 simplified by invoking EGA IV, 2.7.1 instead of loc. cit., 17.7.5.

Corollary 1.3.2. *Let A be an artinian local ring, $u : G \rightarrow H$ a morphism between A -groups locally of finite type. The following assertions are equivalent:*

- (i) u is universally open,
- (ii) u is open,
- (iii) u is open at a point of G ,
- (iv) the morphism $u^0 : G^0 \rightarrow H^0$ obtained from u is dominant,
- (iv') u^0 is surjective,
- (v) there is a connected component C of G such that, if D denotes the connected component of H containing $u(C)$, the morphism $u' : C \rightarrow D$ obtained from u is dominant.

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) and (iv') $\Rightarrow$ (iv) $\Rightarrow$ (v) are clear. Since G^0 is of finite type over A (VIA 2.4), hence noetherian, u^0 is quasi-compact, so $u^0(G^0)$ is closed in H^0 , by 1.2, hence (iv) $\Rightarrow$ (iv'). On the other hand, since G^0 (resp. C) is open in G (VIA 2.3) and H^0 (resp. D) is irreducible (VIA 2.4.1), one sees that (ii) implies (iv) (resp. that (iii) implies (v)). It therefore remains to show that (v) implies (i).

The open subset C (resp. D) of G (resp. H) will be endowed with its induced scheme structure, and u' will denote the morphism $C \rightarrow D$ obtained from u . Let k be the residue field of A .⁽¹⁰⁾ Since the base change $\text{Spec } k \rightarrow \text{Spec } A$ is a universal homeomorphism, one may suppose $A = k$. By hypothesis, u' is dominant and, since C is of finite type over k (VIA 2.4.1), hence noetherian, u' is quasi-compact. By EGA IV₂, 2.3.7, $u' \otimes_k \bar{k}$ is still quasi-compact and dominant, where $\bar{k}$ denotes an algebraic closure of k . Then, since $C \otimes_k \bar{k}$ is a union of connected components of $G \otimes_k \bar{k}$, the morphism $u \otimes_k \bar{k} : G \otimes_k \bar{k} \rightarrow H \otimes_k \bar{k}$ satisfies assertion (v). We are thus reduced to the case where $A = k$ is an algebraically closed field, taking account of EGA IV₂, 2.6.4.

In this case, we may moreover replace u by u_{red} , and we are reduced to the case where H is reduced. Then let ξ (resp. η) be the generic point of C (resp. D). Since u' is quasi-compact and dominant, $u'(\xi) = \eta$ (cf. EGA IV₁, 1.1.5). On the other hand, since H is reduced, the local ring $\mathcal{O}_{H,\eta}$ is a field, and hence u' is flat at the point ξ .⁽¹¹⁾ Thus, by 1.3, u is flat; moreover, since u is locally of finite type and H is locally noetherian, u is locally of finite presentation. Hence, by EGA IV₂, 2.4.6, u is universally open.

Proposition 1.4. *Let A be an artinian local ring, and $u : G \rightarrow H$ a quasi-compact morphism between A -groups locally of finite type. The following assertions are equivalent:*

- (i) u is proper,
- (ii) there is $h \in H$ such that the fiber $u^{-1}(h)$ is nonempty and proper over $\kappa(h)$,
- (iii) $\text{Ker } u$ is proper over A .

It is clear that (i) implies (iii), and that (iii) implies (ii). On the other hand, it follows from the hypotheses that u is of finite type and, since G is separated (VIA 0.3), so is u (EGA I, 5.5.1). It therefore remains to show that assertion (ii) implies that u is universally closed, so we may suppose A equal to its residue field k .⁽¹²⁾ Let k' be an algebraic closure of $\kappa(h)$, $u' : G' \rightarrow H'$ the morphism obtained from u by base change, h' a point of H' above h ; then the fiber $u'^{-1}(h') = u^{-1}(h) \times_{\kappa(h)} k'$ is nonempty and proper, and it suffices to show that u' is proper (EGA IV₂, 2.6.4). One may therefore suppose that k is algebraically closed and $h \in H(k)$.

We have seen (1.2) that $u(G)$ is then the underlying set of a closed reduced subgroup scheme of H ; since every closed immersion is proper (EGA II, 5.4.2), we may suppose u surjective, and H reduced. Since u is surjective, the group $G(k)$ acts transitively on the set of closed points of H ; thus, for every closed point y of H , $u^{-1}(y)$ is proper over $\kappa(y)$. By EGA

⁽¹⁰⁾**N.D.E.** The original has been expanded in what follows.

⁽¹¹⁾**N.D.E.** The original invoked the generic flatness theorem (EGA IV₂, 6.9.1), which is not needed here.

⁽¹²⁾**N.D.E.** The following sentence has been added.

IV₃, 9.6.1, the set of $y \in H$ such that $u^{-1}(y)$ is not proper over $\kappa(y)$ is locally constructible; since it contains no closed point, it is empty (cf. EGA IV₃, 10.3.1 and 10.4.7).

⁽¹³⁾Then consider the generic point η of H^0 ; by the preceding argument, the fiber $u^{-1}(\eta) = G \times_H \text{Spec } \kappa(\eta)$ is proper over $\kappa(\eta)$. On the other hand, since H is reduced, $\kappa(\eta)$ equals $\mathcal{O}_{H,\eta}$. Since $\mathcal{O}_{H,\eta}$ is the inductive limit of the rings $\mathcal{O}_H(V)$, as V ranges over the nonempty open subsets of H^0 , it follows from EGA IV₃, 8.1.2 a) and 8.10.5 (xii), that there is a nonempty open subset V of H^0 such that the restriction of u above the open subset V is proper. It is then clear that the $g \cdot V$, for $g \in G(k)$, form an open covering of H such that, for every $g \in G(k)$, the restriction of u above the open subset $g \cdot V$ is proper; one deduces that u is proper (cf. EGA II, 5.4.1).

Corollary 1.4.1. *Let A be an artinian local ring, and $u : G \rightarrow H$ a morphism between A -groups locally of finite type. The following assertions are equivalent:*

- (i) u is locally quasi-finite,
- (ii) u is quasi-finite at a point,
- (iii) $\text{Ker } u$ is discrete,
- (iv) the restriction of u to every connected component of G is finite.

Finally, if u is quasi-compact, these assertions are equivalent to the following:

- (v) u is finite.

It is clear that (iv) implies (iii), that (iii) implies (ii) (EGA I, 6.4.4), and that, when u is quasi-compact, assertions (iv) and (v) are equivalent. We have already seen in 1.3 that assertions (i) and (ii) are equivalent.

Finally let us show that (i) implies (iv). Let C be a connected component of G ; since C is of finite type over A (VIA 2.4.1) and G and H are separated (VIA 0.2), by EGA I, 5.5.1 and 6.3.4, the restriction u' of u to C is separated and of finite type.⁽¹⁴⁾ Since the fibers of u' are discrete, it follows that u' is quasi-finite (cf. EGA II, 6.2.2). Since every quasi-finite proper morphism is finite (cf. EGA III₁, 4.4.2), it therefore suffices to show that u' is universally closed.

For this we may suppose that A equals its residue field k . Then, by (fpqc) descent (cf. EGA IV₂, 2.6.4), it suffices to show that $u' \otimes_k \bar{k}$ is universally closed, where $\bar{k}$ denotes an algebraic closure of k . Moreover, since C is of finite type over k , $C \otimes_k \bar{k}$ is the sum of a finite number of connected components $C'_1, \dots, C'_n$ of $G \otimes_k \bar{k}$, and it suffices to show the assertion for each C'_i . One is thus reduced to the case where k is algebraically closed.

Then let g be a closed point of C ; if $u^0 : G^0 \rightarrow H$ is the restriction of u to G^0 , one has $u' = r_{u(g)} \circ u^0 \circ r_{g^{-1}}$, where r_g denotes right translation by g , and hence, to show that u' is proper, it suffices to show that u^0 is. By hypothesis, u is locally quasi-finite, so the fiber $\text{Ker } u$ is discrete (and nonempty); we have seen that u^0 is of finite type, so the fiber $\text{Ker } u^0$ is finite (cf. EGA II, 6.2.2), hence proper, and nonempty; thus u^0 is proper by 1.4.

Corollary 1.4.2. *Let A be an artinian local ring, and $u : G \rightarrow H$ a quasi-compact morphism between A -groups locally of finite type. The following assertions are equivalent:⁽¹⁵⁾*

- (i) u is a closed immersion,
- (ii) u is a monomorphism,
- (iii) $\text{Ker } u$ is trivial, i.e. isomorphic to the unit k -group.

⁽¹³⁾N.D.E. The original has been expanded in what follows.

⁽¹⁴⁾N.D.E. The original has been expanded in what follows.

⁽¹⁵⁾N.D.E. The hypothesis that G and H be locally of finite type can be removed, for, by [Per76, 4.2.4], every quasi-compact monomorphism $u : G \rightarrow H$ between group schemes over a field k is a closed immersion; see also [DG70, III.3.7.2 b)] for the case where G and H are affine (in which case every morphism $G \rightarrow H$ is affine (EGA II, 1.6.2 (v)), hence quasi-compact).

In particular, every subgroup scheme⁽¹⁶⁾ of H is closed.

It is clear that (i) implies (ii), and, considering the functors represented by G and H respectively, it is immediate that conditions (ii) and (iii) are equivalent. Finally, if $\text{Ker } u$ is the unit k -group, $\text{Ker } u$ is a nonempty proper fiber, so u is a proper monomorphism, by 1.4, and is of finite presentation since H is locally noetherian (EGA IV₁, 1.6.1), hence is a closed immersion (EGA IV₃, 8.11.5).

The last assertion follows from the fact that, since H is locally noetherian, every immersion $G \rightarrow H$ is quasi-compact (EGA I, 6.6.4).

Counterexample 1.4.3. Let k be a field of characteristic 0, G the constant k -group $\mathbf{Z}$ and H the k -group $\mathbf{G}_{a,k}$. Let $u : G \rightarrow H$ be a morphism of k -groups. If $u \neq 0$, $\text{Ker } u = 0$, but u is not a closed immersion (cf. 1.2.2).

We shall use later the following two results which should have appeared in Exposé VIA:

Lemma 1.5. *Let k be a field and G a k -group locally of finite type. Then:*

- (i) *All the irreducible components of G have the same dimension.*
- (ii) *For every $g \in G$, one has $\dim_g G = \dim G$.*

⁽¹⁷⁾Assertion (i) has already been proved in VIA, 2.4.1, and assertion (ii) follows from it. Indeed, let g be a point of G and C the connected component of G containing g . By definition (EGA 0_{IV}, 14.1.2), $\dim_g G$ is the infimum of the integers $\dim U$, as U ranges over the open neighborhoods of g ; one therefore has $\dim_g G = \dim U_0$ for some U_0 , which one may suppose contained in C (since $\dim V \leq \dim U$ if $V \subset U$). Then, since C is irreducible and of finite type over k (VIA, 2.4.1), one has $\dim U_0 = \dim C = \text{tr. deg}_k \kappa(\xi)$, where ξ is the generic point of C , by EGA IV₂, 5.2.1. Thus $\dim_g G = \dim C = \dim G$.

Proposition 1.6. ⁽¹⁸⁾ *Let S be a scheme of characteristic zero and G an S -group scheme, locally of finite presentation over S at every point of the unit section $\varepsilon(S)$. For G to be smooth over S at every point of the unit section, it is necessary and sufficient that the $\mathcal{O}_S$ -module $\omega_{G/S} = \varepsilon^*(\Omega_{G/S}^1)$ (called the conormal module of the unit section of G) be locally free.*

Recall that a scheme S is said to be of characteristic zero if, for every closed point x of S , the field $\kappa(x)$ is of characteristic zero.

Also recall (II 4.11) that, if π denotes the structure morphism $G \rightarrow S$, one has $\Omega_{G/S}^1 = \pi^*(\omega_{G/S})$, so that it is equivalent to say that the $\mathcal{O}_S$ -module $\omega_{G/S}$ is locally free or that the $\mathcal{O}_G$ -module $\Omega_{G/S}^1$ is locally free.

If there is an open neighborhood U of $\varepsilon(S)$ smooth over S , then, by EGA IV₄, 17.2.3, $\Omega_{U/S}^1$ is locally free of finite type, as is $\omega_{G/S} = \varepsilon^*(\Omega_{U/S}^1)$.

Conversely, if $\omega_{G/S}$ is locally free, so is $\Omega_{G/S}^1 = \pi^*(\omega_{G/S})$. Since S is of characteristic 0, the Jacobian criterion (EGA IV₄, 16.12.2) therefore implies that G is differentially smooth over S . Then it follows from EGA IV₄, 17.12.5, that G is smooth over S at every point of the unit section.

Corollary 1.6.1 (Cartier). *Given a field k of characteristic zero, every k -group locally of finite type over k is smooth over k .*

⁽¹⁶⁾**N.D.E.** In this particular case, see also VIA, 0.5.2, valid without finiteness hypotheses.

⁽¹⁷⁾**N.D.E.** The implication (ii) $\Rightarrow$ (i) is a general fact (cf. EGA 0_{IV}, 14.1.6), and (i) $\Rightarrow$ (ii) follows from the fact that, if X is an irreducible scheme of finite type over a field, one has $\dim X = \dim U$ for every nonempty open subset U of X ; the essential point here is therefore to establish assertion (i), which has already been done in an addition to VIA, 2.4.1. The statement and proof of the lemma have been modified accordingly.

⁽¹⁸⁾**N.D.E.** In the statement, “along the unit section” has been replaced by “at every point of the unit section”; on the other hand, at the end of the proof, the results of EGA IV₄ cited as references have been made explicit.

Indeed, it is then clear that the k -module $\omega_{G/k}$ is locally free, so, by 1.6, G is smooth over k at the identity point e , hence smooth over k , by 1.3.1.

2. “Open properties” of groups and morphisms of groups locally of finite presentation

2.0. — Throughout what follows, S will denote any scheme; an S -group scheme will be called an S -group. Given an S -group G , we shall denote by ε the unit section, by $c : G \rightarrow G$ the inversion morphism and by μ the multiplication morphism $G \times_S G \rightarrow G$. For every S -scheme X , we shall denote by π or π_X the structure morphism $X \rightarrow S$.

Given a property $\mathcal{P}(u)$ for a morphism of S -schemes $u : X \rightarrow Y$, we shall say that $\mathcal{P}(u)$ is stable under base change if, whenever u satisfies $\mathcal{P}(u)$, the same holds for the morphism $u_{(Y')}$, for every S -morphism $Y' \rightarrow Y$. One says that $\mathcal{P}(u)$ is local for the topology $\mathcal{T}$ (cf. Exp. IV, §§4 and 6) if $\mathcal{P}(u)$ satisfies the following two conditions:

- (a) $\mathcal{P}(u)$ is stable under base change,
- (b) whenever there is a family of S -morphisms $Y_i \rightarrow Y$ covering for the topology $\mathcal{T}$ and such that each of the morphisms $u_{(Y_i)}$ satisfies $\mathcal{P}(u)$, then u satisfies $\mathcal{P}(u)$.

Proposition 2.1. *Let $\mathcal{P}(u)$ be a property for a morphism of S -schemes, local for the (fpqc) topology. Let $u : G \rightarrow H$ be a morphism of S -groups. Suppose G flat and universally open over S .*

Let W be the largest open subset of H above which u satisfies the property $\mathcal{P}(u)$ and let $V = u^{-1}(W)$. Then $U = \pi_G(V)$ is an open subset of S and V is an open subgroup scheme of $G|_U$.

The existence of a largest open subset W of H above which u satisfies $\mathcal{P}(u)$ follows from the fact that $\mathcal{P}(u)$ is local for the Zariski topology. Since π_G is universally open, $\pi_G(V)$ is an open subset U of S . It suffices to show that V is a subgroup scheme of $G|_U$. We may therefore suppose that $U = S$.

Then put $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$, $W' = W \times_S V$ and $u' = u_{(V)}$; let W'_1 be the largest open subset of H' above which u' satisfies $\mathcal{P}(u)$; since V is flat and universally open over S , the same holds for H' over H , and Lemma 2.1.1 below shows that $W'_1 = W'$. Then consider the automorphism of V -schemes a (resp. b) of G' (resp. H'), right translation by the inverse of the diagonal section δ (resp. by the inverse of $u(\delta)$), defined by

$$a(g, v) = (g \cdot v^{-1}, v), \quad (\text{resp. } (h, v) = (h \cdot u(v^{-1}), v)),$$

for every morphism $T \rightarrow S$, $g \in G(T)$, $v \in V(T)$ and $h \in H(T)$. It is clear that $u' \circ a = b \circ u'$, which shows that W' is stable under b , hence that V' is stable under a , so that V is a subgroup scheme of G .

Lemma 2.1.1. *Let $\mathcal{P}(u)$ be a property for an S -morphism u , local for the (fpqc) topology. Consider a cartesian square of morphisms of S -schemes:*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ \downarrow & & \downarrow g \\ X & \xrightarrow{f} & Y \end{array},$$

where g is flat and open. Let W (resp. W'_1) be the largest open subset of Y (resp. Y') above which f (resp. $f' = f_{(Y')}$) satisfies $\mathcal{P}(u)$. Then $W'_1 = W \times_Y Y'$.

Put $W' = W \times_Y Y'$; since $\mathcal{P}(u)$ is stable under base change, one has $W' \subset W'_1$. Since g is open, $W_1 = g(W'_1)$ is an open subset of Y . Put $V_1 = f^{-1}(W_1)$ and $V'_1 = V_1 \times_{W_1} W'_1$; it is clear that $V'_1 = f'^{-1}(W'_1)$. Since g is flat and open, the morphism $W'_1 \rightarrow W_1 = g(W'_1)$ obtained from g is faithfully flat and open, hence covering for the (fpqc) topology (cf. IV 6.3.1

(iv)). Since the morphism $V'_1 \rightarrow W'_1$ obtained from f' satisfies $\mathcal{P}(u)$, the same holds for the morphism $V_1 \rightarrow W_1$ obtained from f , hence $W_1 \subset W$, and $W'_1 \subset g^{-1}(W_1) \subset g^{-1}(W) = W'$; thus $W' = W'_1$.

Remark 2.1.2. A great many properties for a morphism are local for the (fpqc) topology; let us mention those of being flat, (universally) open, (locally) of finite type, of finite presentation, quasi-finite (cf. EGA IV₂, 2.5.1, 2.6.1 and 2.7.1), smooth, étale, unramified (EGA IV₄, 17.7.3).

The proof of 2.1 in fact uses only base changes by flat morphisms; the proposition will therefore apply to a property satisfying condition b) of 2.0 for the (fpqc) topology, and stable under base changes by flat morphisms (example: that of being quasi-compact and dominant).

Of course, one may state an analogous proposition concerning properties local for a topology $\mathcal{T}$ finer than the Zariski topology, the condition to be checked on G then being that π_G is universally open and covering for the topology $\mathcal{T}$.

In particular, if G is flat and locally of finite presentation over S , one has an analogous statement for properties stable under base changes by flat morphisms locally of finite presentation, and satisfying condition b) of 2.0 for the (fpqc) topology (for example, those of being regular, reduced, Cohen–Macaulay, etc. (EGA IV₂, 6.8)).

Proposition 2.2. *Let G and H be two S -groups and $u : G \rightarrow H$ a morphism of S -groups. Then.⁽¹⁹⁾*

- (i) *Suppose G or H flat over S , and G or H locally of finite presentation over S , and let V be the largest open subset of G such that the restriction of u to V is flat and locally of finite presentation (resp. smooth, resp. étale). Then $U = \pi_V(V)$ is an open subset of S , and V is an open subgroup scheme of $G|_U$.*
- (ii) *Suppose G or H universally open over S , and let V be the largest open subset of G such that the restriction of u to V is universally open. Then $U = \pi_V(V)$ is an open subset of S , and V is an open subgroup scheme of $G|_U$.*

Let us first prove (i). Let us show that the restriction π_V of π_G to V is flat and locally of finite presentation:

- (a) If π_G is flat (resp. locally of finite presentation), so is π_V .
- (b) If π_H is flat (resp. locally of finite presentation), so is π_V , as the composite of the restriction of u to V and π_H .

Thus, in the four cases considered in the statement, π_V is flat and locally of finite presentation, hence universally open (EGA IV₂, 2.4.6). Put $U = \pi_V(V)$; U is therefore an open subset of S . It then suffices to show that V is an open subgroup scheme of $G|_U$; we may therefore suppose that $U = S$.

Then put $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$ and $u' = u|_{V'}$. Then, since V is flat and locally of finite presentation over S , the same holds for H' over H . By EGA IV₄, 17.7.4, V' is then the largest open subset of G' such that the restriction of u' to V' is flat and locally of finite presentation (resp. smooth, resp. étale). With the automorphisms a and b defined as in the proof of 2.1, it is then clear that V' is stable under a , hence that V is a subgroup scheme of G .

Let us prove (ii). The restriction π_V of π_G to V is a universally open morphism, either because this holds for π_G , or as the composite of the restriction of u to V and π_H in the case where π_H is universally open. Put $U = \pi_V(V)$; U is then an open subset of S . It suffices to show that V is an open subgroup scheme of $G|_U$. We may therefore suppose that $U = S$.

⁽¹⁹⁾**N.D.E.** In case (i), the open subset V consists of all the points of G at which u is smooth, resp. étale, resp. of finite presentation and flat, cf. N.D.E. (6). On the other hand, in case (ii), V is the largest open subset contained in the set E of points of G at which u is universally open, but E is not necessarily open, as the following example shows (EGA IV₃, 14.1.3 (i)): let k be a field, $H = S = \text{Spec } A$, where $A = k[x]$, and G the S -group $\text{Spec } A[y]/(xy)$; then E equals the unit section of G , which is not open.

As before put $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$ and $u' = u_{(V)}$. Then $\pi_V : V \rightarrow S$ is surjective and universally open, and so is $G' \rightarrow G$, so that V' is the largest open subset of G' such that the restriction of u' to V' is universally open, by (EGA IV₃, 14.3.4 (i) and (ii)). With the automorphisms a and b defined as before, it is then clear that V' is stable under a , hence that V is a subgroup scheme of G .

Corollary 2.3. *Let G be an S -group and V the largest open subset of G flat and locally of finite presentation (resp. smooth, étale, universally open) over S . Then $U = \pi_V(V)$ is an open subset of S , and V is an open subgroup scheme of $G|_U$.*

It suffices to apply 2.2 to the case where H is the unit S -group and u is the unique morphism of S -groups $G \rightarrow H$, for then π_H is an isomorphism and $\pi_G = \pi_H \circ u$.

Corollary 2.4. *Let G be an S -group; if there is a neighborhood X of the unit section having one (or several) of the following properties:*

X is flat and locally of finite presentation (resp. smooth, étale, universally open) over S ,

then there is an open subgroup scheme of G having the same properties.

It suffices to apply 2.3, observing that here, with the notation of 2.2, one has $\varepsilon(S) \subset V$, hence $U = S$.

Proposition 2.5. ⁽²⁰⁾ *Let $u : G \rightarrow H$ be a morphism of S -groups.*

- (i) *Suppose that G (resp. H) is of finite presentation and flat (resp. of finite type) over S at the points of its unit section. Then the sets*

$$U_{\text{flat}} \supset U_{\text{smooth}} \supset U_{\text{ét.}},$$

formed by the points $s \in S$ such that u_s is flat (resp. smooth, étale), are open in S .

If moreover G (resp. H) is locally of finite presentation and flat (resp. locally of finite type) over S , then the set V_{flat} (resp. V_{smooth} , $V_{\text{ét.}}$) of points of G where u is flat (resp. smooth, resp. étale) is the inverse image under π_G of U_{flat} (resp. U_{smooth} , $U_{\text{ét.}}$).

- (ii) *Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$ (a condition satisfied if G is of finite type over S at the points of the unit section (1.3.1.1)), and that u is locally of finite type (resp. locally of finite presentation) at the points of the unit section of G . Then the sets*

$$U_{\text{l.q.f.}} \supset U_{\text{unr.}}$$

formed by the $s \in S$ such that u_s is locally quasi-finite (resp. unramified) are open in S .

If moreover u is locally of finite type (resp. locally of finite presentation), then the set $V_{\text{l.q.f.}}$ (resp. $V_{\text{unr.}}$) of points of G where u is quasi-finite (resp. unramified) is the inverse image under π_G of $U_{\text{l.q.f.}}$ (resp. $U_{\text{unr.}}$).

Let us prove (i). First note (1.3.1.1) that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Let Y be the open subset of H formed by the points where π_H is of finite type, and let X_1 be the open subset of $u^{-1}(Y)$ formed by the points where π_G is of finite presentation. By EGA IV₃, 11.3.1, the set X of points of X_1 where π_G is of finite presentation and flat is open in X_1 , hence in G .

Let $u_X : X \rightarrow Y$ be the morphism obtained from u . Since Y (resp. X) contains the unit section of H (resp. G), the restriction π_X of π_G to X is surjective, and the morphism $S \rightarrow X$ obtained from ε_G is an S -section of X .

⁽²⁰⁾**N.D.E.** The statement and proof have been expanded, and in (i) the hypothesis on H has been weakened by replacing “of finite presentation” by “of finite type”.

Consider the sets $W_{\text{flat}} \supset W_{\text{smooth}} \supset W_{\text{ét.}}$, formed by the points of X where u_X is flat (resp. smooth, resp. étale). It is clear that W_{smooth} and $W_{\text{ét.}}$ are open in X , cf. N.D.E. (6). On the other hand, since π_Y is locally of finite type and $\pi_X = \pi_Y \circ u_X$ locally of finite presentation, by EGA IV₁, 1.4.3 (v), u_X is locally of finite presentation. Consequently, the flatness locus W_X of u_X is open in X (EGA IV₃, 11.3.1).

Let $x \in X$ and put $s = \pi_X(x)$; then, by EGA IV₂, 11.3.10 (combined with EGA IV₄, 17.5.1, resp. 17.6.1), x belongs to W_{flat} (resp. W_{smooth} , $W_{\text{ét.}}$) if and only if u_s is flat (resp. smooth, étale) at the point x , or, equivalently by 1.3, if and only if u_s is flat (resp. smooth, étale).

Consequently, for “ $\flat$ = flat, smooth or étale”, one has $U_{\flat} = \varepsilon_G^{-1}(W_{\flat})$ and $W_{\flat} = \pi_X^{-1}(U_{\flat})$. Since $W_{\flat}$ is open in X , hence in G , the first equality shows that $U_{\flat}$ is open in S .

The second assertion of (i) follows from what has just been seen, for then $Y = H$, $X = G$, and $V_{\flat} = W_{\flat} = \pi_G^{-1}(U_{\flat})$.

Let us prove assertion (ii). Let $V_{\text{l.f.t.}} \supset V_{\text{l.q.f.}}$ be the open subsets of G formed by the points at which u is of finite type (resp. quasi-finite). Put $X = V_{\text{l.f.t.}}$ and denote by u_X the restriction of u to X . By hypothesis, X contains $\varepsilon(S)$. Let $x \in X$ and put $s = \pi_X(x)$.

If u is quasi-finite at x , then, by base change, u_s is quasi-finite at x , and hence, since G_s is supposed locally of finite type, u_s is locally quasi-finite, by 1.3.1.

Conversely, if u_s is locally quasi-finite, then $u_X^{-1}(u_X(x)) \subset u_s^{-1}(u_s(x))$ is a finite set. Since $u_X : X \rightarrow H$ is locally of finite type, it follows from Chevalley’s semicontinuity theorem (EGA IV₃, 13.1.3) that there is an open neighborhood W of x in X such that the fiber $u_X^{-1}(u_X(x'))$ is discrete for every $x' \in W$. Thus, by EGA II, 6.2.2 (and EGA III₂, ErrIII 20), u_X is quasi-finite at x .

Consequently, one has $U_{\text{l.q.f.}} = \varepsilon_G^{-1}(V_{\text{l.q.f.}})$ and $V_{\text{l.q.f.}} = \pi_X^{-1}(U_{\text{l.q.f.}})$, and the first equality shows that $U_{\text{l.q.f.}}$ is open in S .

If moreover G is locally of finite type over S , then $X = G$, and hence $V_{\text{l.q.f.}}$ is the inverse image under π_G of $U_{\text{l.q.f.}}$. This proves the first half of (ii).

Now consider the open subsets $V_{\text{l.f.p.}} \supset V_{\text{unr.}}$, formed by the points where u is of finite presentation (resp. unramified), and suppose that $Z = V_{\text{l.f.p.}}$ contains the unit section of G . Let $z \in Z$; put $s = \pi_Z(z)$ and $h = u_Z(z)$.

If u is unramified at z , then, by base change, u_s is unramified at z , and hence, since G_s is supposed locally of finite type, u_s is unramified, by 1.3.1.

Conversely, if u_s is unramified at the point z , then the fiber $u_s^{-1}(h) = u^{-1}(h)$ is unramified over $\kappa(h)$ at the point z . Since Z is an open subset of G , the fiber $u_Z^{-1}(h) = u_s^{-1}(h) \cap Z$ is unramified over $\kappa(h)$ at the point z . Since u_Z is locally of finite presentation, this implies, by EGA IV₄, 17.4.1, that u_Z is unramified at the point z .

Thus one has $U_{\text{unr.}} = \varepsilon_G^{-1}(V_{\text{unr.}})$ and $V_{\text{unr.}} = \pi_Z^{-1}(U_{\text{unr.}})$, and the first equality shows that $U_{\text{unr.}}$ is open in S .

If moreover G is locally of finite presentation over S , then $Z = G$, and hence $V_{\text{unr.}}$ is the inverse image under π_G of $U_{\text{unr.}}$. This completes the proof of assertion (ii).

Corollary 2.6. *Let $u : G \rightarrow H$ be a morphism of S -groups which is a radicial morphism (as is the case if u is a monomorphism (EGA I, 3.5.4)). Suppose G (resp. H) locally of finite presentation and flat (resp. locally of finite type) over S . Then the set U of $s \in S$ such that u_s is an open immersion is open in S , and the restriction of u above U is an open immersion.*

By 2.5 (i), the set U' of points $s \in S$ such that u_s is étale is open in S . Since u is radicial, so is u_s , for every $s \in S$, hence, by EGA IV₄, 17.9.1, one has $U = U'$, which shows that U is open. Finally, by 2.5 (i), the restriction of u above U is étale; since u is radicial, this restriction is an open immersion (EGA IV₄, 17.9.1).

Proposition 2.7. *Let G be an S -group. The following conditions are equivalent:*

- (i) G is unramified over S at the points of the unit section,

- (ii) *the unit section is an open immersion,*
- (iii) *G is of finite presentation over S at the points of the unit section, and, for every $s \in S$, G_s is unramified over $\kappa(s)$.*

If moreover G is supposed locally of finite presentation over S , each of the preceding three conditions is equivalent to the following:

- (iv) *G is unramified over S .*

The equivalence of assertions (i) and (ii) follows from the more general Lemma 2.7.1 below. Observe (1.3.1.1) that either of conditions (i) or (iii) implies that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Then (EGA IV₄, 17.4.1), assertion (i) is equivalent to the fact that, for every $s \in S$, G_s is unramified over $\kappa(s)$ at the point e_s , the identity of G_s , or again (1.3.1), to the fact that G_s is unramified over $\kappa(s)$, so assertions (i) and (iii) are equivalent. Finally, if G is locally of finite presentation over S , assertions (iii) and (iv) are equivalent (EGA IV₄, 17.4.1).

Lemma 2.7.1. *Let G be an S -scheme endowed with a section ε . For G to be unramified over S at the points of this section, it is necessary and sufficient that ε be an open immersion.*

The condition is necessary, by EGA IV₄, 17.4.1 a) $\Rightarrow$ b''). Conversely, if ε is an open immersion, the restriction to $\varepsilon(S)$ of the structure morphism $G \rightarrow S$ is an isomorphism, so G is unramified over S at the points of $\varepsilon(S)$.

Corollary 2.8. *Let $u : G \rightarrow H$ be a morphism of S -groups. Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$.⁽²¹⁾*

- (a) *If u is locally of finite type, the following conditions are equivalent:*
 - (i) *u is locally quasi-finite,*
 - (ii) *for every $s \in S$, $u_s : G_s \rightarrow H_s$ is locally quasi-finite,*
 - (iii) *$\text{Ker } u$ is locally quasi-finite over S ,*
 - (iv) *the fibers of $\text{Ker } u$ are discrete.*
- (b) *If u is locally of finite presentation, the following conditions are equivalent:*
 - (v) *u is unramified,*
 - (vi) *for every $s \in S$, $u_s : G_s \rightarrow H_s$ is unramified,*
 - (vii) *$\text{Ker } u$ is unramified,*
 - (viii) *the unit section $S \rightarrow \text{Ker } u$ is an open immersion.*

The equivalences (i) $\Leftrightarrow$ (ii) and (v) $\Leftrightarrow$ (vi) follow from 2.5 (ii), and the reminder 2.8.1 below shows that (i) $\Rightarrow$ (iii) and (v) $\Rightarrow$ (vii).

For every $s \in S$, denote by e_s the identity element of H_s . Then (iii) (resp. (vii)) implies that, for every $s \in S$,

$$(\text{Ker } u)_s = \text{Ker } u_s = u_s^{-1}(e_s)$$

is locally quasi-finite (resp. unramified) over $\kappa(s) = \kappa(e_s)$, hence that u_s is quasi-finite (resp. unramified) at the identity point of G_s , hence, by 1.3, that u_s is locally quasi-finite (resp. unramified). Thus (iii) $\Rightarrow$ (ii) and (vii) $\Rightarrow$ (vi). Finally, (ii) $\Leftrightarrow$ (iv) by 1.4.1, and (vii) $\Leftrightarrow$ (viii) by 2.7.

Reminder 2.8.1. Recall (cf. I 2.3.6.1) that, given a morphism $u : G \rightarrow H$ of S -groups, one calls the kernel of u , and denotes by $\text{Ker } u$, the subgroup functor of G defined by putting, for every morphism $f : T \rightarrow S$,

$$(\text{Ker } u)(T) = \{a \in G(T) \mid u \circ a = \varepsilon_H \circ f\}.$$

⁽²¹⁾**N.D.E.** The presentation has been modified, separating the assertions concerning the “quasi-finite” case from those concerning the “unramified” case.

By loc. cit. (or EGA I, 4.4.1), this functor is representable by the S -group $G \times_H S = u^{-1}(\varepsilon_H(S))$, denoted simply by $\text{Ker } u$. In particular, the structure morphism $\text{Ker } u \rightarrow S$ is obtained from u by base change.

Lemma 2.9. *Let $\pi : X \rightarrow S$ be a morphism admitting an S -section ε .*

- (i) *If π is injective, it is integral.**
- (ii) *If π is locally of finite type, and if, for every $s \in S$, π_s is an isomorphism, then π is an isomorphism.***

First observe that, by Lemma 2.9.1 below, π , being a homeomorphism, is an affine morphism.

If π is injective, ε is surjective. Since ε is a surjective immersion, $\varepsilon(S)$ is isomorphic to the closed subscheme of X defined by a nilideal $\mathcal{I}$ of $\mathcal{O}_X$. Since ε is an S -section of the morphism π , one has a direct sum decomposition $\mathcal{O}_X = \mathcal{O}_S \oplus \mathcal{I}$ of $\mathcal{O}_S$ -modules. Since $\mathcal{I}$ is a nilideal of $\mathcal{O}_X$, $\mathcal{I}$ is evidently integral over $\mathcal{O}_S$, so $\mathcal{O}_X$ is integral over $\mathcal{O}_S$, and π is integral.

Now suppose π locally of finite type. Since $\pi \circ \varepsilon = \text{id}_S$, ε is locally of finite presentation (cf. EGA IV₁, 1.4.3 (v)), so $\mathcal{I}$ is a finitely generated ideal of $\mathcal{O}_X$ (EGA IV₁, 1.4.1). For every $s \in S$, one has $\mathcal{O}_{X_s} = \kappa(s) \oplus \mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s)$. By hypothesis, ε_s is an isomorphism, so $\mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s) = 0$ for every $s \in S$, hence a fortiori $\mathcal{I} \otimes_{\mathcal{O}_X} \kappa(x) = 0$ for every $x \in X$, which implies, by Nakayama's lemma, that $\mathcal{I} = 0$, hence that π is an isomorphism.

Lemma 2.9.1. *Let $f : X \rightarrow Y$ be a morphism of schemes which is a homeomorphism; then f is an affine morphism.****

It suffices to show that every point $y \in Y$ has an open neighborhood W such that the restriction of f above W is an affine morphism. Then let $y \in Y$, and let V be an affine open neighborhood of y in Y . Let $V' = f^{-1}(V)$. Then V' is an open neighborhood of $x = f^{-1}(y)$ in X . There is an affine open neighborhood W' of x in X contained in V' . Then put $W = f(W')$. Then W is an open neighborhood of y in Y contained in the affine scheme V , so W is separated. Since W' is an affine scheme, the restriction of f above W is then an affine morphism (EGA II, 1.2.3).

Corollary 2.10. *Let G be an S -group locally of finite type. Suppose that, for every $s \in S$, G_s is the unit $\kappa(s)$ -group; then G is the unit S -group.*

More generally:

Corollary 2.11. *Let $f : X \rightarrow Y$ be an S -morphism locally of finite type. For f to be a monomorphism, it is necessary and sufficient that, for every $s \in S$, f_s be a monomorphism.*

It is clear that the condition is necessary; let us show that it is sufficient. If, for every $s \in S$, f_s is a monomorphism, a fortiori for every $y \in Y$, f_y is a monomorphism; we may therefore suppose that $Y = S$.

By EGA I, 5.3.8, to show that f is a monomorphism, it suffices to show that $\Delta_f : X \rightarrow X \times_S X$ is an isomorphism, or, equivalently, that the first projection $p : X \times_S X \rightarrow X$ is an isomorphism. But if f_s is a monomorphism, it similarly follows from EGA I, 5.3.8 that the first projection $X_s \times_{\kappa(s)} X_s \rightarrow X_s$ (which is identified with p_s) is an isomorphism. Now p has the S -section Δ_f , so Lemma 2.9 asserts that if, for every $s \in S$, p_s is an isomorphism, then so is p .

Corollary 2.12. *Let G be an S -scheme having an S -section ε and such that the structure morphism $\pi : G \rightarrow S$ is closed. Let $s \in S$ be such that π is of finite presentation at the point $\varepsilon(s)$, and $\varepsilon_s : \text{Spec}(\kappa(s)) \rightarrow G_s$ is an isomorphism (or, equivalently, $\pi_s : G_s \rightarrow \text{Spec } \kappa(s)$ is*

*This is also a special case of EGA IV₄, 18.12.11, for π is evidently a universal homeomorphism.

**This is also an immediate consequence of EGA IV₄, 18.12.6.

***Cf. EGA IV₄, 18.12.7.1 for a slightly more general result, proved by the same argument.

an isomorphism). Then there is an open neighborhood U of s in S such that $\varepsilon|_U : U \rightarrow G \times_S U$ is an isomorphism.

Let V be the set of points of G where π is unramified; one knows that V is open (cf. N.D.E. (6)) and contains $\varepsilon(s)$. Thus $W = \varepsilon^{-1}(V)$ is an open subset of S containing s , and such that, for every $t \in W$, π is unramified at $\varepsilon(t)$. Since π is closed, so is $\pi|_W$, so we may suppose that $W = S$.

Then, by 2.7.1, $X = G - \varepsilon(S)$ is a closed subset of G not meeting G_s ; thus, since π is closed, $F = \pi(X)$ is a closed subset of S not meeting s ; put $U = S - F$; then U is an open subset of S such that $\varepsilon|_U$ is an isomorphism of U onto $G|_U$.

3. Identity component of a group locally of finite presentation

3.0. — Given a subset A (resp. B) of an S -scheme X (resp. Y), by abuse of notation, $A \times_S B$ will denote the subset of $X \times_S Y$ formed by the points whose first projection belongs to A and second to B . Given a subset A of an S -group G , we shall say that A is stable under the group law of G if one has $c(A) \subset A$ and $\mu(A \times_S A) \subset A$.

Definition 3.1. Let G be an S -group functor satisfying the following condition:

(+) for every $s \in S$, the functor $G_s = G \otimes_S \kappa(s)$ is representable.

One then denotes by G_s^0 the connected component of the identity element of the $\kappa(s)$ -group G_s .⁽²²⁾ One defines a subgroup S -functor of G , called the *identity component of G* , denoted by G^0 , by putting, for every morphism $T \rightarrow S$,

$$G^0(T) = \{u \in G(T) \mid \forall s \in S, u_s(T_s) \subset G_s^0\}.$$

One has thus defined the functor $G \mapsto G^0$ from $(\widehat{\mathbf{Sch}}/S)$ -gr. to $(\widehat{\mathbf{Sch}}/S)$ -gr.

Remark 3.2. (i) Let G be an S -group functor satisfying condition (+); then $\underline{\mathrm{Lie}}(G^0/S) = \underline{\mathrm{Lie}}(G/S)$, by Exposé II.⁽²³⁾ Indeed, for every S -scheme T , denote by I_T the T -scheme of dual numbers over T and by $\tau : T \rightarrow I_T$ the zero section (cf. II, 2.1). By definition, one has

$$\underline{\mathrm{Lie}}(G/S)(T) = \{u \in G(I_T) \mid u \circ \tau = e\},$$

where $e : T \rightarrow G$ denotes the composite of $T \rightarrow S$ and the unit section $S \rightarrow G$, and similarly

$$\begin{aligned} \underline{\mathrm{Lie}}(G^0/S)(T) &= \{u \in G^0(I_T) \mid u \circ \tau = e\} \\ &= \{u \in G(I_T) \mid u \circ \tau = e \text{ and } u_s((I_T)_s) \subset G_s^0, \forall s \in S\}. \end{aligned}$$

Now, for every $s \in S$, T_s and $(I_T)_s = I_{T_s}$ have the same underlying set, so if $u \in \underline{\mathrm{Lie}}(G/S)(T)$, one has $u_s(I_{T_s}) = e_s$, where e_s denotes the identity point of G_s , whence $u \in \underline{\mathrm{Lie}}(G^0/S)(T)$. Thus the inclusion $\underline{\mathrm{Lie}}(G^0/S)(T) \subset \underline{\mathrm{Lie}}(G/S)(T)$ is an equality for every T , whence $\underline{\mathrm{Lie}}(G^0/S) = \underline{\mathrm{Lie}}(G/S)$.

(ii) Let G and G' be two S -group functors satisfying (+); then:

- (a) if $G \subset G'$, then $G^0 \subset G'^0$,
- (b) if $G \subset G'$ and $G'^0 \subset G$, then $G^0 = G'^0$,
- (c) if, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$, then G^0 satisfies property (+), by VIA.2.3, and one has $(G^0)^0 = G^0$.

⁽²²⁾N.D.E. This is a slightly abusive notation, but is compatible with the notation of VIA, 2.3 when this connected component is the underlying topological space of an open subgroup scheme G^0 of G , cf. VIA, 2.2.bis.

⁽²³⁾N.D.E. What follows has been added.

Proposition 3.3. *Let G be an S -group functor satisfying condition (+) and let S' be an S -scheme. Then $(G \times_S S')^0 = G^0 \times_S S'$; in other words, the functor $G \mapsto G^0$ commutes with base changes, i.e. the following diagram is commutative:*

$$\begin{array}{ccc} (\widehat{\mathbf{Sch}}/S)\text{-gr.} & \xrightarrow{G \mapsto G^0} & (\widehat{\mathbf{Sch}}/S)\text{-gr.} \\ \downarrow & & \downarrow \\ (\widehat{\mathbf{Sch}}/S')\text{-gr.} & \longrightarrow & (\widehat{\mathbf{Sch}}/S')\text{-gr.} \end{array}$$

Indeed, it suffices to check that, for every $s' \in S'$, whose image in S is denoted by s , $((G \times_S S') \otimes_{S'} \kappa(s'))^0 = (G_s \otimes_{\kappa(s)} \kappa(s'))^0$ equals $G_s^0 \otimes_{\kappa(s)} \kappa(s')$; this follows from VIA, 2.1.2. Note, for later use in 4.2, that the group structure of G_s has not been used, but only the fact that G_s^0 has a rational point, namely $\varepsilon(s)$, hence is geometrically connected (see also EGA IV₂, 4.5.14).

Special case 3.4. Let G be an S -group scheme; denote by $\underline{G}^0$ the subset of G which is the union of the $\underline{G}_s^0$ as s ranges over S . Then $\underline{G}^0$ is a subset of G stable under the group law of G (cf. 3.0), and, for every morphism $S' \rightarrow S$, one has:

$$G^0(S') = \{u \in G(S') \mid u(S') \subset \underline{G}^0\}.$$

When $\underline{G}^0$ is an open subset of G , it is clear that G^0 is representable by the subscheme of G induced on the open subset $\underline{G}^0$.

Proposition 3.5. *Let S be a quasi-compact quasi-separated scheme, and G an S -group with fibers locally of finite type. There is then a quasi-compact open subset U of G containing $\underline{G}^0$.*

Since the unit section ε is an immersion, $\varepsilon(S)$ is a quasi-compact subspace of G , so there is a quasi-compact open subset V of G containing $\varepsilon(S)$. Since S is quasi-separated and V is quasi-compact, V is quasi-compact over S (EGA IV₁, 1.2.4), hence $V \times_S V$ is quasi-compact over S , hence quasi-compact. Thus $V \cdot V = \mu(V \times_S V)$ is quasi-compact. Put $V_s = V \cap G_s$ and $V_s^0 = V \cap G_s^0$. Then V_s^0 is an open subset of G_s^0 , dense in G_s^0 since G_s^0 is irreducible (VIA 2.4), so $V_s^0 \cdot V_s^0 = G_s^0$ (VIA 0.5), which shows that $V_s \cdot V_s \supset G_s^0$, hence that $V \cdot V \supset \underline{G}^0$. Finally, since $V \cdot V$ is quasi-compact, there is a quasi-compact open subset U of G containing $V \cdot V$ and a fortiori $\underline{G}^0$.

Corollary 3.6. *Let G be an S -group with fibers locally of finite type and connected. Then G is quasi-compact over S .*

Our assertion being local on S (EGA I, 6.6.1), one is reduced to the case where S is affine. By 3.5, there is then a quasi-compact open subset U of G containing $\underline{G}^0 = G$, so G is quasi-compact, hence quasi-compact over the affine scheme S (EGA I, 6.6.4 (v)).

Proposition 3.7. ⁽²⁴⁾ *Let G be an S -group locally of finite presentation; then:*

- (i) $\underline{G}^0$ is ind-constructible in G .

⁽²⁴⁾**N.D.E.** Recall the following definitions and results (cf. EGA 0_{III}, §9.1 and EGA IV₁, §§1.8 and 1.9). Let X be a topological space. (i) An open subset U of X is said to be retrocompact if the inclusion $U \hookrightarrow X$ is quasi-compact, i.e. if $U \cap V$ is quasi-compact for every quasi-compact open subset $V \subset X$. (ii) A subset C of X is said to be constructible if it is a finite union of subsets of the form $U - V$, where U and V are retrocompact open subsets of X . (iii) A subset L of X is said to be locally constructible if, for every $x \in X$, there is an open neighborhood U of x in X such that $L \cap U$ is constructible in U . (N.B. If X is quasi-compact and quasi-separated, L is then constructible.) (iv) A subset I of X is said to be ind-constructible if, for every $x \in X$, there is an open neighborhood U of x in X such that $I \cap U$ is a union of locally constructible subsets in U . Now let $f : X \rightarrow Y$ be a morphism of schemes. By Chevalley's constructibility theorem (cf. EGA IV₁, 1.8.4 and 1.9.5 (viii)), if f is of finite presentation (resp. locally of finite presentation), then the image under f of every locally constructible subset is locally constructible (resp. ind-constructible).

- (ii) *If moreover G is quasi-separated over S , and S quasi-compact and quasi-separated, then $\underline{G}^0$ is constructible.*
- (iii) *Consequently, if G is quasi-separated over S , $\underline{G}^0$ is locally constructible.*

Let us first prove the first assertion. Since $\pi : G \rightarrow S$ is locally of finite presentation, given $s \in S$, there is an open subset U of G containing $\varepsilon(s)$ and an open subset V of S containing s such that $\pi(U) \subset V$ and the morphism $\pi' : U \rightarrow V$ obtained from π is of finite presentation. Then $T = \varepsilon^{-1}(U)$ is an open subset of S and, writing $W = \pi'^{-1}(T)$ and $\pi'' = \pi'|_W$, $\pi'' : W \rightarrow T$ is of finite presentation, and admits as section the morphism $\varepsilon'' : T \rightarrow W$ obtained from ε .

For every $t \in T$, since G_t^0 is irreducible (VIA 2.4), $W \cap G_t^0$ is dense in G_t^0 , hence irreducible, hence connected: it is therefore the connected component of $\pi''^{-1}(t)$ containing $\varepsilon''(t)$. It then follows from EGA IV₃, 9.7.12 that the union W^0 of the $W \cap G_t^0$, for $t \in T$, is locally constructible in W . On the other hand, it follows from VIA 0.5 that $\underline{G}^0 \cap \pi^{-1}(T) = W^0 \cdot W^0$, i.e. $\underline{G}^0 \cap \pi^{-1}(T)$ is the image of $W^0 \times_T W^0$ under the morphism $\mu'' : W \times_T W \rightarrow \pi^{-1}(T)$ obtained from μ .⁽²⁵⁾ Since $W \times_T W$ (resp. $\pi^{-1}(T)$) is of finite presentation (resp. locally of finite presentation) over T , μ'' is locally of finite presentation and quasi-separated, by EGA IV₁, 1.4.3 and 1.2.2; if moreover $\pi^{-1}(T)$ is quasi-separated over T , μ'' is quasi-compact (loc. cit. 1.2.4), hence of finite presentation. Since $W^0 \times_T W^0$ is locally constructible in $W \times_T W$ (since W^0 is so in W), it follows from Chevalley's constructibility theorem (loc. cit., 1.8.4 and 1.9.5 (viii)) that $\underline{G}^0 \cap \pi^{-1}(T)$ is ind-constructible in $\pi^{-1}(T)$, and is locally constructible in $\pi^{-1}(T)$ if G (and hence $\pi^{-1}(T)$) is quasi-separated over T . This proves assertions (i) and (iii).

Now suppose G quasi-separated over S , and S quasi-compact and quasi-separated. Then, by 3.5, there is a quasi-compact open subset U of G containing $\underline{G}^0$. Since G is quasi-separated over S , G is quasi-separated, hence the open subset U is retrocompact (EGA IV₁, 1.2.7), and it suffices to show that $\underline{G}^0$ is constructible in U (EGA 0_{III}, 9.1.8). Moreover, since U is quasi-compact, hence quasi-compact over S (EGA IV₁, 1.2.4), and quasi-separated over S , the restriction of π to U is of finite presentation, so, by EGA IV₃, 9.7.12, $\underline{G}^0$ is locally constructible in U , hence constructible in U , since U is quasi-compact and quasi-separated (EGA IV₁, 1.8.1). This proves (ii), and it follows that, for every quasi-compact quasi-separated open subset T of S (for example, for every affine open subset of S), $\underline{G}^0 \cap \pi^{-1}(T)$ is constructible.

Corollary 3.8. *Let S_0 be a quasi-compact quasi-separated scheme, I an increasingly filtering preordered set, $(\mathcal{A}_i)_{i \in I}$ an inductive system of commutative quasi-coherent $\mathcal{O}_{S_0}$ -algebras, $\mathcal{A} = \varinjlim \mathcal{A}_i$, $S_i = \operatorname{Spec} \mathcal{A}_i$ for $i \in I$, and $S = \operatorname{Spec} \mathcal{A}$ (cf. EGA II, 1.3.1).*

Let G be an S_0 -group scheme locally of finite presentation. Then the canonical map $\varinjlim G^0(S_i) \rightarrow G^0(S)$ is bijective.

Since G is locally of finite presentation over S , the canonical map $\varinjlim G(S_i) \rightarrow G(S)$ is bijective, by EGA IV₂, 8.14.2 c). It follows immediately that the canonical map $\varinjlim G^0(S_i) \rightarrow G^0(S)$ is injective. Let us show that it is surjective. Let $g \in G^0(S) \subset G(S)$. There is $i \in I$ such that g factors by means of $g_i \in G(S_i)$ through $S \rightarrow S_i$; by hypothesis, $g^{-1}(\underline{G}^0) = S$. But, by 3.7, $\underline{G}^0$ is ind-constructible in G , so $g_i^{-1}(\underline{G}^0)$ is so in S_i . It then follows from EGA IV₂, 8.3.4 that there is an index $j \geq i$ such that $g_j^{-1}(\underline{G}^0) = S_j$, where g_j is the map obtained from g_i by the base change $S_j \rightarrow S_i$. This shows that $g_j \in G^0(S_j)$, hence that g comes from an element of $\varinjlim G^0(S_i)$.

Proposition 3.9. *Let G be an S -group locally of finite presentation. Suppose G^0 representable; then the canonical morphism $i : G^0 \rightarrow G$ is an open immersion; moreover, G^0 is quasi-compact over S .*

⁽²⁵⁾**N.D.E.** The original has been expanded in what follows.

Since G^0 is a subfunctor of the functor G , the morphism i is a monomorphism, hence is radicial. By the definition of the functor G^0 , one checks immediately from the definition (EGA IV₄, 17.1.1) that i is a formally étale morphism (observing that $\underline{G^0}$ is the image of i in G). Finally, it follows from the characterization (EGA IV₃, 8.14.2 c)) of S -schemes locally of finite presentation by means of the functor they represent, and from 3.8, that, since G is locally of finite presentation over S , so is G^0 . Thus i is locally of finite presentation (cf. EGA IV₁, 1.4.3); it is therefore a radicial étale morphism, hence an open immersion (EGA IV₄, 17.9.1).

Finally, the last assertion follows from 3.6.

Theorem 3.10. *Let G be an S -group. The following conditions are equivalent:*

- (i) G is smooth over S at the points of the unit section.
- (ii) G is flat and locally of finite presentation over S at the points of the unit section, and, for every $s \in S$, G_s is smooth over $\kappa(s)$.
- (iii) There is an open subgroup scheme G' of G , smooth over S .
- (iv) G^0 is representable by an open subscheme of G , smooth over S .

It is clear that (iv) $\Rightarrow$ (iii) $\Rightarrow$ (i) and, by 1.3.1 and 2.4, (i) implies (ii) and (iii). Moreover, (ii) $\Rightarrow$ (i) by EGA IV₄, 17.5.1.

Finally let us show that (iii) implies (iv). Lemma 3.10.1 below shows that G' contains $\underline{G^0}$, and that $\underline{G'^0} = \underline{G^0}$. It therefore suffices to show that $\underline{G^0}$ is open in G , for we have already seen (3.4) that G^0 will then be representable by the smooth subgroup scheme induced by G' on the open subset $\underline{G'^0} = \underline{G^0}$. One may therefore suppose that $G' = G$.

To show that $\underline{G^0}$ is open, it suffices to show that every $s \in S$ has a neighborhood T in S such that $\underline{G^0} \cap \pi^{-1}(T)$ is open in $\pi^{-1}(T)$. Let $s \in S$. Since $G = G'$, π is locally of finite presentation, so one may construct, as in the proof of 3.7, an open subset T of S containing s , and an open subset W of G containing $\varepsilon(s)$, such that the morphism $\pi'' : W \rightarrow T$ obtained from π is of finite presentation and admits as section the morphism $\varepsilon'' : T \rightarrow W$ obtained from ε . For every $t \in T$, $W \cap G_t^0$ is then the connected component of $\pi''^{-1}(t)$ containing $\varepsilon''(t)$. Since π is smooth, so is π'' , which is therefore smooth and of finite presentation. Then, by EGA IV₃, 15.6.5, the union W^0 of the $W \cap G_t^0$ for $t \in T$ is open in W .

On the other hand, by VIA 0.5, one has $W^0 \cdot W^0 = \underline{G^0} \cap \pi^{-1}(T)$, and one must show that this is open in $\pi^{-1}(T)$. We may therefore now suppose that $T = S$; it remains to show that $W^0 \cdot W^0$ is open in G . Since π is universally open, so is μ (VIA 0.1). Thus, since W^0 is open in G , so is $W^0 \cdot W^0 = \mu(W^0 \times_S W^0)$.

This result will be improved in 4.4.

Lemma 3.10.1. *Let G be an S -group with fibers locally of finite type. Then every open subgroup scheme H of G contains $\underline{G^0}$, and satisfies $\underline{G^0} = \underline{H^0}$.*

Let $s \in S$; put $G'_s = H_s \cap G_s^0$; then G'_s is an open subset of G_s^0 , dense in G_s^0 since G_s^0 is irreducible (VIA 2.4), so $G'_s \cdot G'_s = G_s^0$ (VIA 0.5), which shows that $G_s^0 = G'_s \cdot G'_s \subset H_s \cdot H_s = H_s$. Thus one has $G_s^0 = H_s^0$ for every s , whence $\underline{G^0} \subset H$ and $\underline{G^0} = \underline{H^0}$.

Proposition 3.11. *Let $u : G \rightarrow H$ be a morphism between S -groups locally of finite presentation. If u is flat, the map $u^0 : \underline{G^0} \rightarrow \underline{H^0}$ obtained from u is surjective; the converse holds if G is flat over S and H has reduced fibers.⁽²⁶⁾*

Suppose u flat; then, for every $s \in S$, u_s is flat and locally of finite presentation, hence open (EGA IV₂, 2.4.6), so the morphism $u_s^0 : G_s^0 \rightarrow H_s^0$ is surjective, by 1.3.2.⁽²⁷⁾ Thus the map $u^0 : \underline{G^0} \rightarrow \underline{H^0}$ is surjective.

⁽²⁶⁾N.D.E. Compare with VIA, 5.6.

⁽²⁷⁾N.D.E. The original has been shortened by referring here to 1.3.2. On the other hand, the original has been simplified below by removing a reference to the generic flatness theorem (EGA IV₂, 6.9.1), which is not necessary here.

Conversely, suppose the map $u^0 : \underline{G}^0 \rightarrow \underline{H}^0$ surjective, G flat over S and H with reduced fibers. Then, for every $s \in S$, the morphism $u_s^0 : G_s^0 \rightarrow H_s^0$ is surjective, hence flat at every point above the generic point η of H_s^0 (since $\mathcal{O}_{H_s^0, \eta}$ is a field), so u_s is flat by 1.3. Thus u is flat, by the “fiberwise flatness criterion” (EGA IV₃, 11.3.11).

4. Dimension of the fibers of groups locally of finite presentation

Proposition 4.1. *Let G be an S -scheme locally of finite type, endowed with an S -section ε , and such that, for every $s \in S$, one has $\dim G_s = \dim_{\varepsilon(s)} G_s$ (as is the case if G is an S -group (1.5)).*

- (i) *The function $s \mapsto \dim G_s$ is upper semicontinuous on S .*
- (ii) *If moreover G is locally of finite presentation over S , this function is locally constructible.*

Let $\pi : G \rightarrow S$ be the structure morphism. Chevalley’s semicontinuity theorem (EGA IV₃, 13.1.3) asserts that the function $x \mapsto \dim_x \pi^{-1}(\pi(x))$ is upper semicontinuous on G . Now, for every $s \in S$, one has

$$\dim G_s = \dim \pi^{-1}(s) = \dim_{\varepsilon(s)} \pi^{-1}(\pi(\varepsilon(s)));$$

and, since the function $s \mapsto \varepsilon(s)$ is continuous on S , the composite function $s \mapsto \dim G_s$ is upper semicontinuous on S .

Suppose G locally of finite presentation over S . To show that the function $s \mapsto \dim G_s$ is locally constructible, arguing as before one sees that it suffices to show that the function $x \mapsto \dim_x \pi^{-1}(\pi(x))$ is locally constructible on G , which follows from EGA IV₃, 9.9.1.

Proposition 4.2. *Let $\pi : G \rightarrow S$ be an S -scheme locally of finite presentation, endowed with an S -section ε and satisfying the following two conditions (which hold if G is an S -group, by 1.5 and VIA 2.4.1):*

- (a) *For every $s \in S$ and every $x \in G_s$, one has $\dim G_s = \dim_x G_s$ (or, equivalently (1.5), for every $s \in S$, all the irreducible components of G_s have the same dimension).*
- (b) *For every $s \in S$, denoting by G_s^0 the connected component of G_s containing $\varepsilon(s)$, G_s^0 is geometrically irreducible.*

Let $s \in S$. The following conditions are equivalent:

- (i) *G is universally open over S at the points of G_s^0 .*
- (i bis) *G is universally open over S at every point of a neighborhood of $\varepsilon(s)$ in G_s^0 .⁽²⁸⁾*
- (ii) *The function $t \mapsto \dim G_t$ is constant in a neighborhood of s in S .*
- (iii) *$\underline{G}^0$ is “universally open over S at the points of G_s^0 ”, that is, given $S' \rightarrow S$, $s' \in S'_s$, $g \in G_{s'}$ and an open neighborhood V of g in $G' = G_{S'}$, $\pi(V \cap \underline{G}^0)$ is an open neighborhood of s' in S' .⁽²⁹⁾*

Evidently (i) $\Rightarrow$ (i bis). By EGA IV₃, 14.3.3.1 (ii), the set of points of G_s^0 where π_G is universally open is closed in G_s^0 . Thus, since G_s^0 is irreducible, one has (i bis) $\Rightarrow$ (i).

Let us show that (i) implies (ii). Let T be the set of $t \in S$ such that $\dim G_t = \dim G_s$. By 4.1, T is locally constructible, so, by EGA IV₁, 1.10.1, to show that T is a neighborhood of s , it suffices to show that every generization s' of s belongs to T .

Let x be the generic point of G_s^0 and U an open subset of G containing x . Since π_G is universally open at ξ , by EGA IV₃, 14.3.13, for every generization s' of s , one has $\dim(U \cap G_{s'}) \geq \dim_x(U \cap G_s)$. In view of our hypothesis a), this implies $\dim G_{s'} \geq \dim G_s$. Now, since the function $s \mapsto \dim G_s$ is upper semicontinuous, by 4.1, one also has $\dim G_{s'} \leq \dim G_s$, whence $s' \in T$. This proves that (i) $\Rightarrow$ (ii).

⁽²⁸⁾**N.D.E.** The original reads: “at the points of $\varepsilon(s)$ ”; it is not sufficient to suppose that G is universally open over S at $\varepsilon(s)$, as the example given in N.D.E. (19) shows. The proof has been corrected accordingly.

⁽²⁹⁾**N.D.E.** Condition (iii), pointed out by O. Gabber, has been added; it will be useful later (5.7).

It is clear that (iii) $\Rightarrow$ (i); let us show that (ii) $\Rightarrow$ (iii). Since the dimension of the fibers is unchanged by extension of the base field and the formation of $\underline{G}^0$ commutes with base change (cf. the proof of 3.3), one may suppose that $S' = S$ and $s' = s$. Moreover, since every open subset V of G meeting G_s^0 contains the generic point η of G_s^0 , one may suppose that $g = \eta$.

One may moreover suppose S affine. Let U be an affine open neighborhood of $\varepsilon(s)$; it is then of finite presentation over S . Replacing S by $\varepsilon^{-1}(U)$ and then U by $U \cap \pi^{-1}(S)$, one is reduced to the case where $\pi : U \rightarrow S$ is of finite presentation and admits $\varepsilon : S \rightarrow U$ as section. Then, by EGA IV₃, 9.7.12, $\underline{G}^0 \cap U$ is constructible in U . Then, replacing V by an affine open subset contained in $V \cap U$, one obtains that $\underline{G}^0 \cap V$ is constructible in V . Since $\pi : V \rightarrow S$ is of finite presentation, $\pi(\underline{G}^0 \cap V)$ is locally constructible in S , by Chevalley's constructibility theorem (cf. EGA IV₁, 1.8.4).

Thus, by loc. cit., 1.10.1, to show that $\pi(\underline{G}^0 \cap V)$ is an open neighborhood of s , it suffices to show that, for every generization t of s , there is a generization ξ of η belonging to $\underline{G}^0$ (and hence to $\underline{G}^0 \cap V$). Now the generic point ξ of G_t^0 is a generization of η . Indeed, $\varepsilon(s)$ belongs to the closure X of $\{\xi\}$ in G , so, by Chevalley's semicontinuity theorem (cf. EGA IV₃, 13.1.3), one has $\dim_{\varepsilon(s)} X_s \geq \dim_{\xi} X_t$; on the other hand, hypothesis (ii) implies that $\dim_{\xi} G_t^0 = \dim_{\varepsilon(s)} G_s$. It follows that one of the irreducible components of X_s containing $\varepsilon(s)$ equals G_s^0 , whence $\eta \in \overline{\{\xi\}}$. This proves that (ii) $\Rightarrow$ (iii), which proves the proposition.

⁽³⁰⁾One may also prove the implication (ii) $\Rightarrow$ (i) as follows. Since π is locally of finite presentation, there is an open subset U of G containing $\varepsilon(s)$ and an open subset V of S containing s such that $\pi(U) \subset V$ and the morphism $\pi' : U \rightarrow V$ obtained from π is of finite presentation. Then put $T = \varepsilon^{-1}(U)$ and $W = \pi'^{-1}(T) = U \cap \pi^{-1}(T)$. Then the morphism $\pi'' : W \rightarrow T$ obtained from π' is of finite presentation and admits as section the morphism $\varepsilon'' : T \rightarrow W$ obtained from ε . Moreover, for every $t \in T$, since G_t^0 is irreducible, $W \cap G_t^0$ is dense in G_t^0 , hence irreducible, hence connected: it is therefore the connected component of $\pi''^{-1}(t)$ containing $\varepsilon''(t)$.

Since $W \cap G_t^0$ is a dense open subset of G_t^0 , by 1.5 and EGA IV₂, 5.2.1, one has $\dim(W \cap G_t^0) = \dim G_t^0 = \dim G_t$, so the function $t \mapsto \dim W \cap G_t^0$ is constant in a neighborhood of s in T . Finally let us show that, for every $t \in T$, $W \cap G_t^0$ is geometrically irreducible. Let K be an extension of $\kappa(t)$; then $(W \cap G_t^0) \otimes_{\kappa(t)} K = (W \otimes_{\kappa(t)} K) \cap (G_t^0 \otimes_{\kappa(t)} K)$ is a nonempty open subset of $G_t^0 \otimes_{\kappa(t)} K$, hence is irreducible since $G_t^0 \otimes_{\kappa(t)} K$ is.

We are then in the situation to apply EGA IV₃, 15.6.6 (ii), which asserts that π'' (hence π) is universally open at the points of $W \cap G_s^0$. But, by EGA IV₃, 14.3.3.1 (ii), the subset of G_s formed by the points where π is universally open is closed in G_s ; since it contains $W \cap G_s^0$, it therefore contains its closure G_s^0 .

Corollary 4.3. *Let G be a flat S -group locally of finite presentation. Then the function $s \mapsto \dim G_s$ is locally constant on S .*

This follows immediately from 4.2, for every flat morphism locally of finite presentation is universally open (EGA IV₂, 2.4.6).

Corollary 4.4. *Let G be an S -group locally of finite presentation over S at the points of the unit section. Consider the conditions:*

- (i) *G is smooth over S at the points of the unit section (cf. 3.10).*
- (ii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and the function $s \mapsto \dim G_s$ is locally constant on S .*
- (iii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and there is a neighborhood V of the unit section such that $\pi_V : V \rightarrow S$ is universally open.*
- (iv) *For every $s \in S$, G_s^0 is smooth over $\kappa(s)$, and G^0 is representable by an open subgroup scheme of G , universally open over S .*

⁽³⁰⁾**N.D.E.** The preceding argument was communicated to us by O. Gabber; the proof of the implication (ii) $\Rightarrow$ (i) given in the original has also been retained.

Then one has the following implications: (i) $\Rightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv).

If moreover S is supposed reduced, conditions (i) to (iv) are equivalent, and imply that G^0 is smooth over S .⁽³¹⁾

Let us show that (i) implies (ii). For every $x \in G$, one has $\dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$, by 1.5. Consequently, by EGA IV₄, 17.10.2, the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$$

is continuous in a neighborhood of the unit section; thus the function $s \longmapsto \dim G_s$ is continuous on S , hence locally constant on S . On the other hand, for every $s \in S$, G_s is smooth over $\kappa(s)$, by 1.3.1.

Let us show that (ii) implies (iv). It suffices to show that $\underline{G}^0$ is open in G , for then, by 3.4, G^0 will be representable by the subgroup scheme induced on $\underline{G}^0$, and the properties of G^0 cited in the statement will follow from 2.4 and 4.2. Given $s \in S$, construct W , T , π'' , ε'' and W^0 as in the proof of 3.10. Then, by EGA IV₃, 15.6.7, W^0 is open in W . On the other hand, under hypothesis 4.4 (ii), it follows from 4.2 that π is universally open at every point of W^0 , hence (VIA 0.1) μ is universally open at every point of $W^0 \times_S W^0$, which shows that $W^0 \cdot W^0$ is open, and one concludes as in the proof of Theorem 3.10.

It is clear that (iv) $\Rightarrow$ (iii), and (iii) $\Rightarrow$ (ii) follows from 4.2 applied to V .

Finally, suppose (ii)–(iv) satisfied and let us show that G^0 is smooth over S if S is reduced.⁽³²⁾ For this one may suppose $G = G^0$. Then G is of finite presentation over S by 5.5 below. Thus π_G is of finite presentation, with geometrically integral fibers of locally constant dimension on S . Then, by EGA IV₃, 15.6.7, the morphism $G \times_S S_{\text{red}} \rightarrow S_{\text{red}}$ obtained from π_G is flat, hence π_G is flat if S is reduced. In this case, $G = G^0$ is smooth over S , by Theorem 3.10.

5. Separation of groups and homogeneous spaces

Proposition 5.1. *For an S -group G to be separated, it is necessary and sufficient that the unit section of G be a closed immersion.*

The condition is necessary (EGA I, 5.4.6); it is sufficient by the following cartesian diagram (cf. VIA 0.3):

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\mu \circ (\text{id}_G \times c)} & G \\ \Delta_{G/S} \uparrow & & \uparrow \varepsilon \\ G & \xrightarrow{\pi} & S \end{array} .$$

Proposition 5.2. *If S is discrete, every S -group is separated.*

Indeed, S then equals $\coprod_{s \in S} \text{Spec } \mathcal{O}_{S,s}$, and, by EGA I, 5.5.5, it suffices to show that, for every $s \in S$, $G \times_S \text{Spec } \mathcal{O}_{S,s}$ is separated, which follows from VIA 0.3, since $\mathcal{O}_{S,s}$ is a zero-dimensional local ring.⁽³³⁾

Theorem 5.3. *Let S be a scheme, G an S -group scheme locally of finite presentation over S and universally open over S in a neighborhood of the unit section, X an S -scheme on which*

⁽³¹⁾**N.D.E.** One cannot dispense here with the hypothesis that S be reduced: if k is a field, $S = \text{Spec } A$, where $A = k[\delta]$ with $\delta^2 = 0$, and G is the closed subgroup $\text{Spec } A[X]/(\delta X)$ of $\mathbf{G}_{a,S}$ (which associates to every A -algebra R the subgroup of $r \in R$ such that $\delta r = 0$), then G is an S -group satisfying (ii)–(iv), but is not flat, hence not smooth, over S .

⁽³²⁾**N.D.E.** The original has been expanded in what follows.

⁽³³⁾**N.D.E.** “Artinian” has been replaced by “zero-dimensional” (and this generalization has been mentioned in VIA 0.3).

G acts so that the morphism

$$\Phi : G \times_S X \longrightarrow X \times_S X, \quad (g, x) \longmapsto (gx, x)$$

is surjective. Suppose moreover that, for every $s \in S$:

- (i) there is an open subscheme U of X , separated over S , such that U_s is dense in X_s ,
- (ii) the fiber X_s is locally of finite type over $\kappa(s)$.⁽³⁴⁾

Then X is separated over S .

Corollary 5.4. *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover X has fibers locally of finite type and connected. Then:*

- (i) X is separated over S .
- (ii) If there is an open subset V of X , quasi-compact over S and meeting every nonempty fiber X_s ,⁽³⁵⁾ then X is quasi-compact over S .

PROOF. (i) Indeed, let $s \in S$ be such that $X_s \neq \emptyset$. Since the morphism $\Phi_s : G_s \times_{\kappa(s)} X_s \longrightarrow X_s \times_{\kappa(s)} X_s$ obtained from Φ by base change is surjective, and since X_s is connected, X_s is irreducible, by VIA, 2.5.4. Thus, if U is an affine open subset of X such that U_s is nonempty, U_s is dense in X_s , and the theorem applies.

To prove (ii), one may suppose S affine. Then V is quasi-compact and, by 3.5, there is a quasi-compact open subset U of G containing G^0 . Let $s \in S$ be such that $X_s \neq \emptyset$. Then X_s is irreducible (VIA, 2.6.6) and hence, since U_s contains G_s^0 , the morphism $U_s \times_{\kappa(s)} V_s \longrightarrow X_s$, $(g, x) \longmapsto gx$ is surjective (VIA, 2.6.4). Consequently the morphism $U \times_S V \longrightarrow X$ is surjective and hence X is quasi-compact (since U and V are); since we have supposed S affine, hence separated, it follows that X is quasi-compact over S (cf. EGA I, 6.6.4 (v)). $\square$

Remark 5.4.1. ⁽³⁶⁾ One will see in the course of the proof that the conclusion of Theorem 5.3 holds if one makes only the hypothesis: (i') there is an open subscheme U of X , separated over S , such that U_s is dense in every irreducible component of X_s . (This follows from (i) if X_s locally has a finite number of irreducible components, as is the case under hypothesis (ii).) On the other hand, one will see later that 5.4 also holds under the hypothesis that each fiber X_s is quasi-separated and connected.

Corollary 5.5. *Let S be a scheme, G an S -group scheme, locally of finite presentation, with connected fibers, and universally open over S . Then G is separated and of finite presentation over S .⁽³⁷⁾*

⁽³⁴⁾N.D.E. The original has been corrected by supposing G universally open over S in a neighborhood of the unit section and adding hypothesis (ii); see below for examples due to O. Gabber, which show that statements 5.3 and 5.4 of the original are false without additional hypotheses. On the other hand, let us point out that Theorem 5.3 is a revised version of Theorem 5.3A below, which appears in the 1965 edition of SGAD, and is due to M. Raynaud, cf. the Notes (*) in Exp. X, 8.5 and 8.8.

Theorem 5.3A (Raynaud). *Let G be an S -group locally of finite presentation, universally open over S , and with connected fibers. Then G is separated over S .*

More generally, every S -scheme X locally of finite presentation over S , endowed with an action of G such that the morphism $\Phi : G \times_S X \longrightarrow X \times_S X$, $(g, x) \longmapsto (gx, x)$ is surjective, is separated over S .

⁽³⁵⁾N.D.E. Without this hypothesis, one has the following counterexample, pointed out by O. Gabber: let $G = S$ be a local scheme with closed point s , such that $S^* = S - \{s\}$ is not quasi-compact, and X the disjoint union of $\{s\}$ and S^* .

⁽³⁶⁾N.D.E. This remark has been added, cf. N.D.E. (34).

⁽³⁷⁾N.D.E. Let us point out here the following result ([Ray70a], VI 2.5): if S is normal, G smooth with connected fibers, and X is an (fppf) homogeneous G -space (i.e. the morphisms $X \longrightarrow S$ and $G \times_S X \longrightarrow X \times_S X$ are covering for the (fppf) topology), locally of finite type over S , then X is locally quasi-projective over S . In particular, G is quasi-projective over S . See also N.D.E. (35) in VIA.

Indeed, by 3.6 and 5.4, G is quasi-compact and separated over S , and, since G is locally of finite presentation over S , it is therefore of finite presentation over S .

5.6. — Proof of Theorem 5.3. Before establishing 5.3, let us prove a few lemmas.

Lemma 5.6.0. ⁽³⁸⁾ (i) *Let $A \subset B$ be integral domains, with B integral over A , and let $\mathfrak{q} \in \operatorname{Spec}(B)$ be such that A is unibranch at the point $\mathfrak{p} = \mathfrak{q} \cap A$. Then the morphism $\pi : \operatorname{Spec}(B) \rightarrow \operatorname{Spec}(A)$ is open at the point $\mathfrak{q}$.*

(ii) *Let X, Y be two irreducible preschemes, $f : X \rightarrow Y$ a dominant morphism, x a point of X such that f is quasi-finite at x and $y = f(x)$ is a unibranch point of Y . Then f is open at the point x . In particular, if $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$ is a local prescheme with closed point y , then $f(U) = Y$ for every neighborhood U of x .*

(i) Let K (resp. L) be the field of fractions of A (resp. B), A' the normalization of A , and B' the subring of L generated by A' and B . Then B' is integral over A' . Put $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $Y' = \operatorname{Spec}(A')$, $X' = \operatorname{Spec}(B')$, so that one has a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\pi'} & Y' \\ \downarrow & & \downarrow \phi \\ X & \xrightarrow{\pi} & Y \end{array}$$

in which all the morphisms are integral and surjective.

Since A is unibranch at $\mathfrak{p}$, Y' has a unique point $\mathfrak{p}'$ above $\mathfrak{p}$; consequently, if U is an open neighborhood of $\mathfrak{p}'$ in Y' , $\phi(Y' - U)$ is a closed subset not containing $\mathfrak{p}$, so that its open complement is contained in $\phi(U)$. This shows that $\phi : Y' \rightarrow Y$ is open at $\mathfrak{p}'$, and it therefore suffices to show that π' is open. One is thus reduced to the case where $A = A'$ is normal.

Let N be a quasi-Galois extension of K containing L , let $\tilde{X} = \operatorname{Spec}(\tilde{B})$, where $\tilde{B}$ is the integral closure of B in N , and let $G = \operatorname{Aut}(N/K)$. Since $\tilde{X} \rightarrow X$ is surjective, it suffices to show that $\tilde{\pi} : \tilde{X} \rightarrow Y$ is open. Let U be an open subset of $\tilde{X}$ and $U' = \bigcup_{g \in G} gU$. Since G acts transitively on the fibers of $\tilde{X} \rightarrow Y$ (cf. [BAC], Chap. V, §2.3, Prop. 6), $\tilde{\pi}(U)$ equals $\tilde{\pi}(U')$, and the latter equals the open complement of the closed subset $\tilde{\pi}(\tilde{X} - U')$. This proves (i).

(ii) One may suppose Y and X reduced. By “Zariski’s main theorem” (cf. EGA IV₃, 8.12.9), there are affine open neighborhoods U of x and $V = \operatorname{Spec}(A)$ of y , such that $f(U) \subset V$, and a factorization:

$$\begin{array}{ccc} U & \xrightarrow{j} & V' \\ & \searrow f & \downarrow u \\ & & V \end{array}$$

where j is an open immersion and u is finite. Replacing V' by the closure of $j(U)$, one may suppose V' irreducible, hence $V' = \operatorname{Spec}(B)$, where B is an integral A -algebra, finite over A . Moreover, since f is dominant, so is u , so that the morphism $A \rightarrow B$ is injective. Since, by hypothesis, A is unibranch at the point y , it follows from (i) that u is open at the point $j(x)$, and hence $f = u \circ j$ is open at the point x . This proves the first assertion of (ii). The second follows from it, for, if Y is a local prescheme with closed point y , every open subset containing y equals Y . (In the case where $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$, one may also use, instead of EGA IV₃, 8.12.9, the local form of Zariski’s main theorem, found for example in [Pes66], or [Ray70b], Ch. IV, Th. 1.)

⁽³⁸⁾**N.D.E.** This lemma, communicated by O. Gabber, has been added; it improves EGA IV₃, 14.4.1.2 and corrects the proof of loc. cit., 14.4.1.3 without changing its hypotheses (compare with the erratum (ErrIV, 38) in EGA IV₄).

Lemma 5.6.1.0. ⁽³⁹⁾ *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $s \in S$, and $x \in X_s$. Let $n = \dim_x(X_s)$ and let q be the structure morphism $\mathbb{A}_S^n \rightarrow S$.*

Suppose given a quasi-finite morphism $u : X \rightarrow \mathbb{A}_S^n$ such that $f = q \circ u$, and suppose f universally open at the generic point z of an irreducible component $\mathfrak{z}$ of X_s , containing x and of dimension n . Then u is universally open at the point $x \in X$.

First observe that (†) the hypotheses are preserved by every base change $\pi : S' \rightarrow S$ covering s (i.e. such that $\pi^{-1}(s) \neq \emptyset$). Indeed, let $S' \rightarrow S$ be such a morphism, let

$$\begin{array}{ccc} X' & \xrightarrow{u'} & \mathbb{A}_{S'}^n \\ & \searrow f' & \downarrow q' \\ & & S' \end{array}$$

be the diagram obtained by base change, and let s' be a point of S' above s and x' a point of $X'_{s'}$ above x . Since $X'_{s'} = X_s \otimes_{\kappa(s)} \kappa(s')$, by lifting of generizations and invariance of dimension under field extension (EGA IV₂, 2.3.4 (i) and 4.1.4), x' is contained in an irreducible component $\mathfrak{z}'$ of $X'_{s'}$ whose generic point z' lies above z , and one has $n \leq \dim \mathfrak{z}' \leq \dim_{x'} X'_{s'} \leq n$, whence $\dim \mathfrak{z}' = n = \dim_{x'} X'_{s'}$. Since f is universally open at x , f' is universally open at x' .

Put $Y = \mathbb{A}_S^n$. By EGA IV₃, 14.3.3.1 (i), to prove that u is universally open at x , it suffices to prove that, for every integer $r \geq 0$ and every point x' of $X' = X[T_1, \dots, T_r]$ above x , the morphism $u' : X' \rightarrow Y' = \mathbb{A}_S^n[T_1, \dots, T_r]$ is open at the point x' . Now, writing $S' = S[T_1, \dots, T_r]$ and q' for the projection $Y' \simeq \mathbb{A}_{S'}^n \rightarrow S'$, one is in the situation obtained by the base change $S' \rightarrow S$. Thus, by the preceding argument, one is reduced to showing that u is open at the point x .

Put $y = u(x)$. Since u is locally of finite presentation (since f and q are, cf. EGA IV₁, 1.4.3), by EGA IV₁, 1.10.3, it suffices to show that $u(\text{Spec } \mathcal{O}_{X,x}) = \text{Spec } \mathcal{O}_{Y,y}$. For this one may suppose S affine and integral. Then let $\pi : S' \rightarrow S$ be its normalization; denote by $u' : X' \rightarrow Y'$ the morphism obtained from u by base change. Since the morphism $\pi_Y : Y' \rightarrow Y$ is integral and surjective, one has

$$\text{Spec}(\mathcal{O}_{Y,y}) = \bigcup_{y'} \pi_Y(\text{Spec}(\mathcal{O}_{Y',y'})),$$

the union being taken over all the points of Y' above y ; it therefore suffices to show that, for each such y' , and every $x' \in X'_{y'}$ above x , one has $u'(\text{Spec } \mathcal{O}_{X',x'}) = \text{Spec } \mathcal{O}_{Y',y'}$. Since the hypotheses are preserved by the base change $\pi : S' \rightarrow S$, one is thus reduced to the case of S' , i.e. one may suppose S an integral normal scheme.

Now the hypotheses on f imply, by EGA IV₃, 14.3.13, that there is an irreducible component Z of X containing z (and hence x), dominating S and such that

$$\dim_z(Z_s) = n = \dim(Z_\eta),$$

where η is the generic point of S . Let ξ be the generic point of Z . Since u , hence also $u_\eta : Z_\eta \rightarrow \mathbb{A}_\eta^n$, is quasi-finite, the closure in $\mathbb{A}_\eta^n$ of the point $u_\eta(\xi) = u(\xi)$ has dimension n , hence $u(\xi)$ is the generic point of $\mathbb{A}_\eta^n$, which is also the generic point of $Y = \mathbb{A}_S^n$. Consequently, denoting by g the restriction of u to Z , the morphism $g : Z \rightarrow Y$ is quasi-finite and dominant. Since Y is normal, it follows from Lemma 5.6.0 that g is open, so u is open at every point of Z , in particular at the point x . Consequently, one has $u(\text{Spec } \mathcal{O}_{X,x}) = \text{Spec } \mathcal{O}_{Y,y}$, and this completes the proof of Lemma 5.6.1.0.

The following lemma advantageously replaces EGA IV₃, 14.5.10,⁽⁴⁰⁾ in that it is independent of noetherian hypotheses.

⁽³⁹⁾**N.D.E.** This lemma, used in the proof of Lemma 5.6.1, has been made explicit.

⁽⁴⁰⁾**N.D.E.** The original, which indicated 19.5.10, has been corrected.

Lemma 5.6.1. *Let S be a scheme, $f : X \rightarrow S$ an S -scheme locally of finite presentation, s a point of S , x a closed point of X_s . Suppose f universally open at the generic point z of an irreducible component $\mathfrak{z}$ of X_s , containing x and such that $\dim_x(X_s) = \dim \mathfrak{z}$. Then there is a commutative diagram:*

$$\begin{array}{ccc} X & \xrightarrow{f} & S \\ h \uparrow & \nearrow \phi & \uparrow w \\ S'' & \xrightarrow{\pi} & S' \end{array},$$

where S' is an affine scheme, w an étale morphism, π a finite surjective morphism of finite presentation, and $\phi^{-1}(s)$ consists of a single point s'' , such that $h(s'') = x$ and $\kappa(s'') = \kappa(x)$.⁽⁴¹⁾

PROOF. First one may suppose $S = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$, where B is an A -algebra of finite presentation. Let $\mathfrak{p}$ and $\mathfrak{q}$ be the prime ideals of A and B corresponding to s and x , respectively, so that $\mathfrak{p} = A \cap \mathfrak{q}$. Put $n = \dim_x X_s$, and let $t_1, \dots, t_n$ be elements of B whose images in $\mathcal{O}_{X_s, x} = B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ form a system of parameters. Then $\mathcal{O}_{X_s, x}/(t_1, \dots, t_n)$ is finite-dimensional over $\kappa(x)$ and hence also over $\kappa(s)$, since x is a closed point of the algebraic $\kappa(s)$ -scheme X_s . Consequently, the S -morphism

$$u : X \rightarrow \mathbb{A}_S^n = \operatorname{Spec}(A[T_1, \dots, T_n]),$$

defined by $T_i \mapsto t_i$, is of finite presentation, x is isolated in its fiber $u^{-1}(u(x))$, and one has $u(x) = \tau_0(s)$, where $\tau_0 : S \rightarrow \mathbb{A}_S^n$ denotes the “zero section” of $\mathbb{A}_S^n \rightarrow S$, corresponding to the morphism of A -algebras $A[T_1, \dots, T_n] \rightarrow A$ which sends every T_i to 0. Since the set of points of X isolated in their fiber above $\mathbb{A}_S^n$ is open (EGA IV₃, 13.1.4), one may suppose, shrinking X if necessary, that u is quasi-finite and $u^{-1}(u(x)) = \{x\}$. By Lemma 5.6.1.0, u is universally open at the point x .

Let $B_0 = B/(t_1, \dots, t_n)$ and let $X_0 = X \times_{\mathbb{A}_S^n} \tau_0(S) = \operatorname{Spec} B_0$ and $u_0 : X_0 \rightarrow S$ be the morphism obtained from u by the base change $\tau_0 : S \hookrightarrow \mathbb{A}_S^n$. Then u_0 is quasi-finite and of finite presentation, universally open at the point x , and x is the unique point of X_0 above s .

Let A' be the henselization of the local ring $A_{\mathfrak{p}} = \mathcal{O}_{S, s}$, and let $S' = \operatorname{Spec}(A')$, $B'_0 = B_0 \otimes_A A'$, and $X'_0 = X_0 \times_S S' = \operatorname{Spec}(B'_0)$. Then the closed point s' of S' is the unique point of S' above s , one has $\kappa(s') = \kappa(s)$, and X'_0 has a unique point x' above s' ; one has $\kappa(x') = \kappa(x)$ and x' is also the unique point of X'_0 above s and x . Since A' is henselian, by EGA IV₄, 18.5.11, X'_0 is the disjoint sum of two open and closed subsets:

$$(*) \quad X'_0 = V \sqcup W, \quad \text{where } V = \operatorname{Spec}(\mathcal{O}_{X'_0, x'});$$

and the local ring $\mathcal{O}_{X'_0, x'}$ is finite and of finite presentation over A' . The restriction π of u'_0 to V is therefore finite and of finite presentation. Moreover, since $u'_0 : X'_0 \rightarrow S'$ is open at x' , one has $u'_0(V) = \operatorname{Spec}(\mathcal{O}_{S', s}) = S'$, so π is surjective.

This proves the desired result when $S = S'$. In the general case, A' is a filtering inductive limit of subalgebras A_i étale over A , and such that $S_i = \operatorname{Spec}(A_i)$ has a unique point s_i above s (and one has $\kappa(s_i) = \kappa(s)$). Then $B'_0 = \varinjlim B_i$, where $B_i = B_0 \otimes_A A_i$. Put $X_i = \operatorname{Spec}(B_i) = X_0 \times_S S_i$ and $C = \mathcal{O}_{X'_0, x'}$. By (*) above, one has $C \simeq B'_0/fB'_0$, for some idempotent $f \in B'_0$, and there is an index i and $f_i \in B_i$ such that f is the image of f_i in B'_0 . Put $C_i = B_i/f_i B_i$ and $V_i = \operatorname{Spec}(C_i)$.

⁽⁴¹⁾**N.D.E.** Consequently, h induces a section σ of $X \times_S S'' \rightarrow S''$ such that $\sigma(s'')$ lies above x ; compare with EGA IV₃, 14.5.10.

Then $C = C_i \otimes_{A_i} A'$, whence $V = V_i \times_{S_i} S'$, and C_i is an A_i -algebra of finite presentation (since this holds for B_i). Consequently, the morphisms

$$\begin{array}{ccc} & X'_0 & \\ h' \uparrow & \searrow u'_0 & \\ V & \xrightarrow{\pi} S' & \end{array}$$

come, by the base change $S' \rightarrow S_i$, from morphisms of finite presentation:

$$\begin{array}{ccc} & X_i & \\ h_i \uparrow & \searrow u_i & \\ V_i & \xrightarrow{\pi_i} S_i & \end{array}.$$

For every $j \geq i$, let $V_j = V_i \times_{S_i} S_j$ and let $\pi_j : V_j \rightarrow S_j$ be the morphism (of finite presentation) obtained from π_i by base change. Since $\pi : V \rightarrow S'$ is finite and surjective, by EGA IV₂, 8.10.5, there is an index j such that $\pi_j : V_j \rightarrow S_j$ is finite and surjective. Then $w : S_j \rightarrow S$ is affine étale, S_j has a unique point s_j above s , and V_j has a unique point x_j above s_j (since x' is the unique point of $V = V_j \times_{S_j} S'$ above s'):

$$\begin{array}{ccccccc} X_j & \longrightarrow & X_0 & \longrightarrow & X & & \\ h_j \uparrow & \searrow u_j & & \searrow u_0 & \downarrow f & \searrow u & \\ V_j & \xrightarrow{\pi_j} & S_j & \xrightarrow{w} & S & \xrightarrow{\tau_0} & \mathbb{A}_S^n \end{array}.$$

Thus x_j is the unique point of V_j above s , its image under the morphism $V_j \rightarrow X_j \rightarrow X$ is x , and one has $\kappa(x_j) = \kappa(x)$. $\square$

Lemma 5.6.2.0. ⁽⁴²⁾ *Let k be a field and G a nonempty k -scheme of finite type.*

(i) *Let K be an extension of k and W a dense open subset of G_K . Denote by π the projection $G_K \rightarrow G$; then*

$$U = \{g \in G \mid W \text{ contains every maximal point of } \pi^{-1}(g)\}$$

is a dense open subset of G . (N.B. If g is a closed point of G belonging to U , one therefore has $\pi^{-1}(g) \subset W$.)

(ii) *Suppose moreover G geometrically irreducible. Let $\mu : G \times X \rightarrow Y$ be a morphism of k -schemes, x a point of X , and Ω an open subset of Y such that $\mu_x^{-1}(\Omega_{\kappa(x)}) \neq \emptyset$, where μ_x denotes the morphism $G_{\kappa(x)} \rightarrow Y_{\kappa(x)}$, $g \mapsto \mu(g, x)$. Then*

$$U = \{g \in G \mid \mu \text{ sends every maximal point of } g \times x \text{ into } \Omega\}$$

is a dense open subset of G , and, for every closed point g of G belonging to U , one has $\mu(g \times x) \subset \Omega$ and hence $\mu(g', x) \in \Omega_{\kappa(x)}$ (resp. $\mu(g, x') \in \Omega_{\kappa(g)}$), for every point $g' \in G_{\kappa(x)}$ above g (resp. $x' \in X_{\kappa(g)}$ above x).

(iii) *Suppose that $G = H^0$, where H is a k -group scheme locally of finite type, acting on a nonempty k -scheme X , so that the morphism $H \times_S X \rightarrow X \times_S X$, $(h, x) \mapsto (hx, x)$ is surjective. Let U be an open subset of X . Then:*

- (a) *$G \cdot U$ is an open subset of X , equal to the union of the irreducible components of X whose generic point belongs to U .*
- (b) *Suppose U dense in every irreducible component of X . Then, for every finite subset A of X , there is a closed point $g \in G$ such that $g \cdot A' \subset U_{\kappa(g)}$, where A' is the inverse image of A in $X_{\kappa(g)}$. In particular, the morphism $G \times U \rightarrow X$ is surjective.*

⁽⁴²⁾**N.D.E.** This lemma, communicated by O. Gabber, has been added. It makes it possible to simplify the proof of 5.6.2, and to prove Theorem 5.3, as well as 5.4, in a more general form; see 5.7 and 5.8 below.

PROOF. (i) Let $\bar{k}$ be a separable closure of k and let L be an extension of k containing a copy of $\bar{k}$ and of K . Denote by π_L (resp. ϕ) the projection $G_L \rightarrow G$ (resp. $G_L \rightarrow G_K$). Since, for every $g \in G$, ϕ sends the maximal points of $\pi_L^{-1}(g)$ surjectively onto those of $\pi^{-1}(g)$, one sees that $U = \{g \in G \mid W_L \text{ contains every maximal point of } \pi_L^{-1}(g)\}$. Thus, replacing K by L , one is reduced to the case where K contains $\bar{k}$.

Since the projection $p : G_K \rightarrow G_{\bar{k}}$ is surjective and open, $V = p(W)$ is a dense open subset of $G_{\bar{k}}$, and, since $p^{-1}(g) = \text{Spec}(\kappa(g) \otimes_{\bar{k}} K)$ is irreducible for every $g \in G_{\bar{k}}$ (cf. EGA IV₂, 4.3.3), for every $g \in V$ the generic point of $p^{-1}(g)$ belongs to W . On the other hand, let $\mathcal{G} = \text{Gal}(\bar{k}/k)$; since the projection $q : G_{\bar{k}} \rightarrow G$ is surjective and $\mathcal{G}$ acts transitively on the fibers of q , $U = q(V')$, where V' is the intersection of the $\mathcal{G}$ -conjugates of V .

Now let Z be the closed subset $G_{\bar{k}} - V$, endowed with the reduced closed subscheme structure. Since $G_{\bar{k}}$, and hence Z , is of finite presentation over $\bar{k}$, Z comes by base change from a reduced closed subscheme Z_1 of $G \otimes_k k_1$, for some finite Galois extension k_1 of k , so the $\mathcal{G}$ -conjugates of Z are finite in number, so their union is still a nowhere dense closed subset F of $G_{\bar{k}}$, and $V' = G_{\bar{k}} - F$ is a dense open subset of $G_{\bar{k}}$. Thus, since the projection $q : G_{\bar{k}} \rightarrow G$ is surjective and open, $U = q(V')$ is a dense open subset of G . Moreover, for every closed point g of G , the fiber $\pi^{-1}(g)$ consists of a finite number of closed points of G_K , so if $g \in U$, then $\pi^{-1}(g) \subset W$. This proves (i), and (ii) follows from it.

On the other hand, (iii)(a) has been proved in VIA, 2.6.4. Finally, if U is dense in every irreducible component of X , $X = G \cdot U$, so, for every $x \in X$, $\mu_x^{-1}(U_{\kappa(x)})$ is a nonempty open subset of $G_{\kappa(x)}$, and then (iii)(b) follows from (ii). $\square$

Corollary 5.6.2. *Let S, G, X be as in the preliminary hypotheses of 5.3, and let U be an open subset of X , $s \in S$, and A a finite subset of X_s . Suppose U_s dense in X_s and X_s locally of finite type over $\kappa(s)$.⁽⁴³⁾ Then there is a morphism $f : S'' \rightarrow S$, the composite of a finite surjective morphism $S'' \rightarrow S'$ and an étale morphism $S' \rightarrow S$, and a morphism $h : S'' \rightarrow G$, such that the inverse image A'' of A in $X \times_S S''$ (i.e. in $X_s \times_{\text{Spec } \kappa(s)} S''_s$) is contained in $\ell_h^{-1}(U_{S''})$, where ℓ_h denotes the translation of $X_{S''} = X \times_S S''$ defined by the element $h \in G(S'')$.*

PROOF. Since X_s is locally of finite type over $\kappa(s)$, the connected components of X_s are open, and irreducible (cf. VIA, 2.5.4), so U_s is dense in every irreducible component of X_s .

Thus, by Lemma 5.6.2.0, there is a closed point $g \in G_s^0$ such that $g \cdot a' \in U_s \otimes_{\kappa(s)} \kappa(g)$ for every $a' \in X_{\kappa(g)}$ above a point of A .

By Lemma 5.6.1, there is a morphism $f : S'' \rightarrow S$, the composite of a finite surjective morphism $S'' \rightarrow S'$ and an étale morphism $S' \rightarrow S$, and a morphism $h : S'' \rightarrow G$, such that $f^{-1}(s)$ consists of a single point s_0 , and $h(s_0) = g$ and $\kappa(s_0) = \kappa(g)$. Then, denoting by A'' the inverse image of A in $X_s \times_{\text{Spec } \kappa(s)} S''_s = X_s \otimes_{\kappa(s)} \kappa(s_0) = X_s \otimes_{\kappa(s)} \kappa(g)$, the translation ℓ_h of $X_{S''}$ sends A'' into $U_s \otimes_{\kappa(s)} \kappa(s_0)$. $\square$

Lemma 5.6.3. *Let X be an S -scheme. The following conditions are equivalent:*

- (i) X is separated over S .
- (ii) For every S -scheme T , every section $\sigma : T \rightarrow X_T$ is a closed immersion.
- (iii) For every reduced S -scheme T , two S -morphisms f_1 and $f_2 : T \rightarrow X$ coinciding on a dense open subset U of T are equal.
- (iv) For every $s \in S$ and every pair of points x_1, x_2 of X_s , there is a morphism $\phi : S'' \rightarrow S'$ and an open subscheme V of $X_{S''}$, separated over S'' , such that:
 - (a) $S' \rightarrow S$ is open, $S'' \rightarrow S'$ closed and surjective, and $\phi^{-1}(s) \neq \emptyset$.
 - (b) The inverse image of $\{x_1, x_2\}$ in $X_{S''}$ is contained in V .

⁽⁴³⁾**N.D.E.** The hypothesis on X_s has been added and the proof simplified (and corrected) by taking account of the addition 5.6.2.0. On the other hand, the proof shows that the conclusion holds if one supposes only that U_s is dense in every irreducible component of X_s .

(iv') For every $s \in S$, every pair of points $x_1, x_2 \in X_s$ is contained in an open subset V of X , separated over S .⁽⁴⁴⁾

The implication (ii) $\Rightarrow$ (i) is clear (take $T = X$ and σ = the diagonal section), as is (i) $\Rightarrow$ (iv') $\Rightarrow$ (iv). On the other hand, one has (i) $\Rightarrow$ (ii) by EGA I, 5.4.6.

(iii) $\Rightarrow$ (ii). Let $\sigma : T \rightarrow X_T$ be a section of $p : X_T \rightarrow T$. By EGA I, 5.3.13, σ is an immersion, i.e. an isomorphism of T onto a locally closed subscheme E of X_T . To show that E is closed, one may suppose T and E reduced. Let $\bar{E}$ be the reduced closed subscheme of X_T with the closure of E as underlying space, so that E is a dense open subscheme of $\bar{E}$. Then the immersion $i : \bar{E} \hookrightarrow X_T$ and $\sigma \circ p \circ i$ coincide on E , hence on $\bar{E}$ by hypothesis (iii). Thus every point of $\bar{E}$ belongs to $\sigma(T) = E$, whence $\bar{E} = E$.

(i) $\Rightarrow$ (iii). Suppose X separated over S , and let T be a reduced S -scheme, f_1, f_2 two S -morphisms $T \rightarrow X$ coinciding on a dense open subset U of T , and g the morphism $T \rightarrow X \times_S X$ with components f_1 and f_2 . Since $D = \Delta_{X/S}(X)$ is closed, its inverse image under g is a closed subset of T containing U , hence equal to T , and, since T is reduced, g factors through D (cf. EGA I, 5.2.2); consequently $f_1 = p_1 \circ g$ equals $p_2 \circ g = f_2$.

(iv) $\Rightarrow$ (iii). Let T be a reduced S -scheme and f_1, f_2 two S -morphisms $T \rightarrow X$ coinciding on a dense open subset U . Since T is reduced, to see that $f_1 = f_2$, it suffices to see that $f_1 = f_2$ as maps of sets. Indeed, suppose this established, and let $t \in T$, V an affine open subset of X containing $f_1(t) = f_2(t)$, and W the open neighborhood of t equal to the inverse image of V under the continuous map underlying f_1 and f_2 ; then the morphisms $f_i|_W : W \rightarrow V$ coincide on the dense open subset $U \cap W$ of W . Since V is separated and W reduced, this implies that $f_1|_W = f_2|_W$, whence $f_1 = f_2$.

Then let $t \in T$ and s be its image in S ; let us show that the points $x_1 = f_1(t)$ and $x_2 = f_2(t)$ of X_s are equal. Let $\phi : S'' \rightarrow S' \rightarrow S$ and V be an open subset of $X \times_S S''$ as in (iv); put $T' = T \times_S S'$ and $T'' = T \times_S S''$ and denote by $g : T'' \rightarrow T$ and $f'_i : T'' \rightarrow X_{S''}$ ($i = 1, 2$) the morphisms obtained by base change.

Since U is dense in T and $T' \rightarrow T$ is open, the inverse image U' of U in T' is dense in T' . Let U'' be the inverse image of U' in T'' and let F be the reduced subscheme of T'' with $\bar{U''}$ as underlying space. Since $T'' \rightarrow T'$ is surjective and closed, the image of F contains U' and is closed, hence equals T' . Consequently, $F \cap g^{-1}(t)$ contains a point u .

For $i = 1, 2$, denote by h_i the restriction of f'_i to F . Then $h_i(u)$ is a point of $X_{S''}$ above $f_i(t) = x_i$, hence belongs to V , since V contains the inverse image of $\{x_1, x_2\}$ in $X_{S''}$.

Then $W = h_1^{-1}(V) \cap h_2^{-1}(V)$ is an open subset of F containing u , and the S'' -morphisms $h_i|_W : W \rightarrow V$ coincide on the dense open subset $U'' \cap W$ of W . Since V is separated over S'' , one has $h_1|_W = h_2|_W$, whence $h_1(u) = h_2(u)$, hence $f_1(t) = f_2(t)$. This proves (iv) $\Rightarrow$ (iii).

Theorem 5.3 then follows from 5.6.2 and the implication (iv) $\Rightarrow$ (i) of 5.6.3.

Counterexample 5.6.4. Not every S -group G is separated. Let S be a scheme having a nonisolated closed point s ; let G be the scheme obtained by gluing two copies of S along the open subset $S - \{s\}$; one easily sees that G is not separated over S , and is endowed with a natural S -group structure, all of whose fibers are trivial except the fiber G_s , which is isomorphic to the two-element group $\mathbf{Z}/2\mathbf{Z}$.⁽⁴⁵⁾

⁽⁴⁴⁾**N.D.E.** The formulation of condition (iv) has been simplified and condition (iv') added. On the other hand, the proof of the implication (i) $\Rightarrow$ (iii), used in the proof of (iv) $\Rightarrow$ (iii), has been added. Observe moreover that if $T = \operatorname{Spec} k[\varepsilon, x]/(\varepsilon^2, \varepsilon x)$ (k a field), $X = T_{\text{red}} = \operatorname{Spec} k[x]$, the morphisms $\phi_\lambda : T \rightarrow X$ defined by $x \mapsto x + \lambda\varepsilon$ ($\lambda \in k$) coincide on the dense open subset $\operatorname{Spec} B_x$ but are not equal.

⁽⁴⁵⁾**N.D.E.** This example has been reproduced in VIA 0.3, N.D.E. (5).

Theorem 5.7. ⁽⁴⁶⁾ *Let S be a scheme, G an S -group locally of finite presentation over S such that the function $s \mapsto \dim G_s$ is locally constant on S , X an S -scheme on which G acts, and U an open subset of X , separated over S . Then $\underline{G}^0 \cdot U$ is an open subset of X , separated over S .*

PROOF. Denote by $\mu : G \times_S X \rightarrow X$ the action of G on X ; it is the composite of the automorphism $(g, x) \mapsto (g, gx)$ of $G \times_S X$ and the projection onto X . Since $G \times_S U$ is an open subset of $G \times_S X$, it follows from 4.2 (iii) that $V = \underline{G}^0 \cdot U$ is open in X .

Then, proceeding as in the proof of 5.6.2, one deduces from the implication (iv) $\Rightarrow$ (i) of 5.6.3 that V is separated over S . $\square$

Corollary 5.7.1. *Let S and G be as in 5.7 and let σ, τ be two S -sections of G^0 (i.e. $\sigma, \tau \in G^0(S)$). Then the coincidence subscheme $S(\sigma, \tau) \subset S$ of σ and τ (i.e. the inverse image of the diagonal of $G \times_S G$ under the morphism (σ, τ)) is closed.*

Indeed, for every $s \in S$, let U be an affine open subset of G containing $\varepsilon(s)$; then $V = \varepsilon^{-1}(U)$ is an open neighborhood of s in S , and, since $G^0 U$ is separated, $S(\sigma, \tau) \cap V$ is closed in V .

Remark 5.7.2. Gabber points out to us that one can show that if S is henselian local with closed point s , the intersection of the open subsets $G^0 U$, as U ranges over the affine open neighborhoods of $\varepsilon(s)$, is an open subgroup scheme of G , separated over S .

5.8. — Complements. ⁽⁴⁷⁾ Let us begin by recalling Proposition VIA 2.6.6:

Proposition 5.8.1. *Let k be a field, G a k -group locally of finite type acting on a k -scheme X so that the morphism $G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$ is surjective. Suppose X quasi-separated. Then the connected components of X are irreducible.*

Corollary 5.8.2. *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that every fiber X_s is quasi-separated and connected. Then X is separated and quasi-compact over S .*

Indeed, by the preceding proposition, every fiber X_s is irreducible, and the rest of the proof is identical to that of 5.4.

Example 5.8.3. Fix an algebraically closed field k . First recall that every “locally Boolean” topological space X (i.e. having a basis of compact open subsets), endowed with the sheaf of locally constant functions with values in k , is a k -scheme (cf. [DG70], I §1, 2.12). One then says that X is a locally Boolean k -scheme.

Observe on the other hand that every topological space E admitting a basis of separated open subsets (and similarly every scheme X) admits a dense separated open subset. Indeed, since every increasing union of separated open subsets is a separated open subset (for a scheme X this follows from 5.6.3), there is such an open subset U which is maximal. But then U is dense, for if there were a nonempty open subset V such that $U \cap V = \emptyset$, V would contain a separated open subset $W \neq \emptyset$, and $U \cup W$ would still be separated, contradicting the maximality of U .

Now let C be the Cantor space, which one may regard as the space underlying the group $(\mathbf{Z}/2\mathbf{Z})^{\mathbf{N}}$ endowed with the product topology. For every point p of C , let $C(p)$ be another

⁽⁴⁶⁾**N.D.E.** On the one hand, Corollary 5.6.5, which repeated 5.5, has been removed. On the other hand, the original stated Corollary 5.7.1 below as Remark 5.7, referring for the proof to 4.7, a number which does not exist in Lect. Notes 151 (but which appeared in the 1965 edition of SGAD, whose nos. 4.5 and 4.6 have become 5.6.1 and 5.6.2); Theorem 5.7, communicated by O. Gabber, has been added; it makes the aforementioned SGAD statement more precise.

⁽⁴⁷⁾**N.D.E.** This paragraph of complements and counterexamples, all communicated by O. Gabber, has been added.

copy of C , and let X be the space obtained by gluing each $C(p)$ to C along $C - \{p\}$; then X is a nonseparated locally Boolean k -scheme, and C is a dense open subset of X .

Let G be the group of automorphisms of the k -scheme X (i.e. of homeomorphisms of X). Then G acts transitively on X . Indeed, X is the union of C and, for each point $p \in C$, of a second point p' , which can be characterized as the unique point x of $X - \{p\}$ such that $(C - \{p\}) \cup \{x\}$ is separated. It follows that every automorphism ϕ of C extends to an automorphism ϕ_X of X such that $\phi_X(p') = \phi_X(p)$ for every p . On the other hand, the map $\theta_p : X \rightarrow X$ which exchanges p and p' and fixes the other points is an automorphism of X . Finally, the group of automorphisms of C acts transitively on C : for example, using the group structure of C , it suffices to consider translations.

Thus the discrete k -group G (hence locally of finite type) acts on the k -scheme X so that the morphism $G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$ is surjective, but X is not separated (although C is a dense separated open subset).

Example 5.8.4. Retain the notation of the preceding example. Using the description of C as the set of sequences $(u_n)_{n \in \mathbf{N}}$ of elements of $\mathbf{Z}/2\mathbf{Z}$, one sees that C with one point p removed is homeomorphic to a countable disjoint union of copies of C . Indeed, using for example the group structure of C , one reduces by translation to the case where p is the element 0, i.e. the zero sequence; then $C - \{0\}$ is the disjoint union of the subspaces $C_i = \{(u_n)_{n \in \mathbf{N}} \mid u_i = 1 \text{ and } u_j = 0 \text{ for } j < i\}$, for $i \in \mathbf{N}$, each homeomorphic to C . For every nonempty finite subset F of C , proceeding by induction on $|F|$ one deduces that $C - F$ is homeomorphic to a countable disjoint union of copies of C , hence to C with one point removed.

For each F of cardinality 2, let $C(F)$ be another copy of C , denote by q_F the point 0 of $C(F)$ and choose a homeomorphism $\phi_F : C(F) - \{q_F\} \xrightarrow{\sim} C - F$; then let X' be the space obtained by gluing each $C(F)$ to C by means of ϕ_F . Then X' is a nonseparated locally Boolean k -scheme. Moreover, it follows from the construction that every locally constant function $f : X' \rightarrow k$ is constant. Indeed, if $x, y \in C$ and $F = \{x, y\}$, every neighborhood of q_F meets every neighborhood of x or y , so if $f : X \rightarrow k$ is locally constant, one has $f(x) = f(q_F) = f(y)$, and if $F' = \{z, t\}$, with $z, t \in C$, one similarly has $f(q_{F'}) = f(z) = f(q_{\{z, x\}}) = f(x)$. Consequently, X' is connected.

Moreover, every point $x \in X'$ has a pointed neighborhood homeomorphic to $(C, 0)$. More precisely, fix for each F a homeomorphism of pointed topological spaces $h_F : (C, 0) \xrightarrow{\sim} (C(F), q_F)$ and denote by T the group of translations of C . Then, if $x = q_F$, one has the homeomorphism q_F , and if $x \in C$, the translation $t_x \in T$ is a homeomorphism $(C, 0) \xrightarrow{\sim} (C, x)$ (and it is the unique element of T with this property).

Denote by L the free group generated by the h_F and let $G = T * L$ be the “free product” (= coproduct) of T and L . For every $h \in \{h_F\}_{|F|=2} \cup T$, let $\sigma(h)$ be the generator of G corresponding to h and let $S(h)$ (resp. $B(h)$) be the source (resp. target) of h . It is convenient also to put $\sigma(h_F^{-1}) = \sigma(h_F)^{-1}$ and $S(h_F^{-1}) = B(h_F)$ (resp. $B(h_F^{-1}) = S(h_F)$), and to denote by E the set consisting of T , and the h_F and h_F^{-1} .

On the product space $P = G \times X'$ (where G is endowed with the discrete topology), consider the equivalence relation generated by the relations:

$$(g\sigma(h), x) \sim (g, h(x)) \quad \text{when } x \in S(h)$$

for every $h \in E$, and let Z be the quotient space. Then Z is obtained from the disjoint union $\coprod_{g \in G} \{g\} \times X'$ by gluing open subsets, and hence, for every open subset Ω of P , its saturation $\tilde{\Omega}$ is open (cf. [BTop], I §5.1, Example 2). Explicitly, since every open subset of P is the union of its intersections with the “slices” $\{g\} \times X'$, it suffices to consider an open subset of the form $\{1\} \times W$, where W is an open subset of X' . In this case, the saturation is the union

of $\{1\} \times W$ and of

$$\begin{aligned} & \{\sigma(h_1)\} \times h_1^{-1}(W \cap B(h_1)), \\ & \{\sigma(h_1)\sigma(h_2)\} \times h_2^{-1}(h_1^{-1}(W \cap B(h_1)) \cap B(h_2)), \end{aligned}$$

etc., for all finite sequences $h_1, \dots, h_n$ of elements of E , hence is open. Thus the projection $\pi : P \longrightarrow Z$ is open.

Note moreover that the word $\sigma(h_1) \cdots \sigma(h_n)$ is a reduced word of G , unless one of the h_i is the identity element of T , or two consecutive h_i belong to T , or (h_i, h_{i+1}) equals (h_F, h_F^{-1}) or (h_F^{-1}, h_F) . Thus, if $x \in X'$ and an element $\beta = (\sigma(h_1) \cdots \sigma(h_n), h_n^{-1} \cdots h_1^{-1}(x))$ belongs to $\{1\} \times X'$, one may suppose that every h_i is a translation t_i , and in this case the equality $\sigma(t_1 \cdots t_n) = 1$ implies that $t_1 \cdots t_n = 1$, hence $\beta = (1, x)$. Since the equivalence relation is compatible with the action of G (acting on $P = G \times X'$ by left translations on the first factor), one deduces that, for every $g \in G$, the restriction of π to $\{g\} \times X'$ is injective.

Then let $z \in Z$ be arbitrary and let $(g, x) \in P$ be a representative of z . By the preceding argument, $U = \pi(\{g\} \times X')$ is an open neighborhood of z , and the continuous map $\{g\} \times X' \longrightarrow U$ induced by π is open and bijective, hence a homeomorphism. This shows that Z is locally isomorphic to X' (hence also to C), so is again a locally Boolean k -scheme.

Finally, G acts transitively on Z . Indeed, since every $z \in Z$ is G -conjugate to an element of the form $\pi((1, x))$, it suffices to see that every element $(1, x) \in P$ is equivalent to an element $(\sigma(h), 0)$; if $x \in C$, one may take for h the translation t_x , and if $x = q_F$, one may take $h = h_F$. Thus Z is a locally Boolean k -scheme endowed with a transitive action of the discrete group G , but Z is not separated.

6. Subfunctors and subgroup schemes

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Definition 6.1. (i) Let X be an S -functor (that is, a functor from $(\mathbf{Sch}/S)^\circ$ to $(\mathbf{Ens})$), G an S -group functor, u and v two S -morphisms from X to G . One calls the *transporter of u into v* , and denotes by $\underline{\text{Transp}}(u, v)$, the sub- S -functor of G defined as follows:

$$\begin{aligned} \underline{\text{Transp}}(u, v)(S') &= \{g \in G(S') \mid (\text{int } g) \circ u_{S'} = v_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} u_{S''}(x) g_{S''}^{-1} = v_{S''}(x), \quad \forall x \in X(S''), S'' \longrightarrow S'\}. \end{aligned}$$

In particular, $\underline{\text{Transp}}(u, u)$ is a subgroup S -functor of G ; one calls it the *centralizer of u* and denotes it by $\underline{\text{Centr}}(u)$.

(ii) Let G be an S -group functor, X and Y two sub- S -functors of G ; one calls the *transporter of X into Y* (resp. *strict transporter of X into Y*), and denotes by $\underline{\text{Transp}}_G(X, Y)$ (resp. $\underline{\text{Transp}}_G^{\text{str}}(X, Y)$), the sub- S -functors of G defined as follows:

$$\begin{aligned} \underline{\text{Transp}}_G(X, Y)(S') &= \{g \in G(S') \mid (\text{int } g)(X_{S'}) \subset Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} \subset Y(S''), \quad \forall S'' \longrightarrow S'\}, \\ \text{resp. } \underline{\text{Transp}}_G^{\text{str}}(X, Y)(S') &= \{g \in G(S') \mid (\text{int } g)(X_{S'}) = Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} = Y(S''), \quad \forall S'' \longrightarrow S'\}. \end{aligned}$$

Note that one has

$$\underline{\text{Transp}}_G^{\text{str}}(X, Y) = \underline{\text{Transp}}_G(X, Y) \cap c(\underline{\text{Transp}}_G(Y, X)),$$

*On this same subject, see also the results of XI 6, whose natural place would be in the present Exposé VIB. In particular one will find there representability criteria for certain subgroup functors of a given group scheme.⁽⁴⁸⁾

⁽⁴⁸⁾**N.D.E.** These results have been inserted in what follows (nos. 6.5.2 to 6.5.5).

where c denotes the inversion morphism of G .⁽⁴⁹⁾

(iii) Let G be an S -group functor, H a sub- S -functor of G , i the canonical S -morphism $H \rightarrow G$; one calls the *centralizer* and *normalizer* of H in G the following subgroup S -functors of G :

$$\underline{\text{Centr}}_G H = \underline{\text{Centr}}(i) = \underline{\text{Transp}}(i, i), \quad \underline{\text{Norm}}_G H = \underline{\text{Transp}}^{\text{str}}_G(H, H).$$

Finally, one calls the *center* of G the S -group functor $\underline{\text{Centr}}(\text{id}_G) = \underline{\text{Centr}}_G G$; it will be denoted by $\underline{\text{Centr}}G$.

Remark 6.1.1. ⁽⁵⁰⁾ It follows from the definitions that the functors $\underline{\text{Transp}}(u, v)$ and $\underline{\text{Transp}}_G(X, Y)$ (and hence also $\underline{\text{Transp}}^{\text{str}}_G(X, Y)$) commute with base change: for every $S' \rightarrow S$, if G', X', Y', u', v' are obtained from G, X, Y, u, v by base change, one has:

$$\underline{\text{Transp}}(u, v)_{S'} = \underline{\text{Transp}}(u', v') \quad \text{and} \quad \underline{\text{Transp}}(X, Y)_{S'} = \underline{\text{Transp}}(X', Y').$$

Proposition 6.2. ⁽⁵¹⁾ Let G be an S -group. Consider, for a subfunctor T of the functor G , the following property:

(+_f) for every $s \in S$, T_s is representable by a closed subscheme of G_s .

Let $u, w : X \rightarrow G$ and $v : Y \rightarrow G$ be morphisms of S -schemes. Then:

- (i) $\underline{\text{Transp}}(u, w)$ and $\underline{\text{Centr}}(u) = \underline{\text{Transp}}(u, u)$ satisfy condition (+_f).
- (ii) $\underline{\text{Transp}}_G(u(X), v(Y))$ and $\underline{\text{Norm}}_G v(Y)$ satisfy condition (+_f) if, for every $s \in S$, v_s is a closed immersion.
- (iii) $\underline{\text{Transp}}^{\text{str}}_G(X, Y)$ satisfies condition (+_f) if, for every $s \in S$, u_s and v_s are closed immersions.

This follows from Remark 6.1.1 and Corollary 6.2.5 below.⁽⁵²⁾

Definition 6.2.1. Let $f : X \rightarrow S$ be a morphism of schemes. One says that f is *essentially free*, or that X is *essentially free over S* , if one can find a covering of S by affine open subsets S_i , for every i an S_i -scheme S'_i affine and faithfully flat over S_i , and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine open subsets X'_{ij} , such that, for every (i, j) , the ring of X'_{ij} is a projective module⁽⁵³⁾ over the ring of S'_i .

Proposition 6.2.2.

- (a) If X is essentially free over S , it is flat over S , the converse holding if S is artinian.
- (b) If S is the spectrum of a field, every S -scheme is essentially free over S .

⁽⁴⁹⁾**N.D.E.** If $u : X \rightarrow G$ and $v : Y \rightarrow G$ are two arbitrary morphisms of S -functors, let $u(X)$ be the image functor of u , defined by $u(X)(S') = u(S')(X(S')) \subset G(S')$; it is a subfunctor of G , as is the image functor $v(Y)$; one may then consider the transporter of the image of u into the image of v , $\underline{\text{Transp}}_G(u(X), v(Y))$. One thus sees that, in definition (ii), it is not necessary to restrict oneself to subfunctors, i.e. to the case where u and v are monomorphisms. This restriction sometimes imposed repetitions in the hypotheses in the original, such as: “Let $u, v : X \rightarrow G$ be morphisms of S -functors, $w : H \rightarrow G$ and $w' : K \rightarrow G$ monomorphisms; then $\underline{\text{Transp}}(u, v)$, $\underline{\text{Centr}}_G(w) = \underline{\text{Centr}}_G H$ and $\underline{\text{Transp}}_G(H, K)$ satisfy...”, which can be avoided by considering $\underline{\text{Transp}}_G(u(X), v(Y))$. Such modifications have been made in 6.2 and, later, in 10.11.

⁽⁵⁰⁾**N.D.E.** This remark, implicitly used in Proposition 6.2, has been added.

⁽⁵¹⁾**N.D.E.** The statement has been modified as indicated in N.D.E. (49).

⁽⁵²⁾**N.D.E.** The original referred to the results of Exp. VIII, §6. For the reader's convenience, these results (except VIII, 6.3 and 6.8) have been reproduced in nos. 6.2.1 to 6.2.5 below. Moreover, this was suggested by A. Grothendieck in a Note at the beginning of VIII.6: “The present number is independent of the theory of diagonalizable groups; its natural place would be in VIB.”

⁽⁵³⁾**N.D.E.** On the one hand, the word “free” has been replaced by “projective”, as indicated in VIII, Remark 6.8. On the other hand, the notion of an essentially free S -scheme is intimately related to the geometric notion of a flat and pure S -scheme, introduced and studied by M. Raynaud and L. Gruson ([RG71]), cf. addition 6.9 below.

- (c) If X is essentially free over S , $X' = X \times_S S'$ is essentially free over S' , for every $S' \rightarrow S$. The converse holds if $S' \rightarrow S$ is faithfully flat and quasi-compact.

The proof is immediate, using for the converse in a) the fact that a flat module over an artinian local ring is free.⁽⁵⁴⁾

The introduction of Definition 6.2.1 is justified by the following

Theorem 6.2.3. *Let S be a scheme, Z an essentially free S -scheme, Y a closed subscheme of Z . Consider the functor $F = \prod_{Z/S} Y/Z : (\mathbf{Sch})_S^\circ \rightarrow (\mathbf{Ens})$ defined by the following condition: $F(S') = \emptyset$ when $Y_{S'} \neq Z_{S'}$, $F(S')$ consists of one element in the opposite case.⁽⁵⁵⁾*

- (i) *This functor is representable by a closed subscheme T of S .*
- (ii) *If moreover $Y \rightarrow Z$ is of finite presentation, so is $T \rightarrow S$.*

First note that the functor under consideration is a sheaf for the (fpqc) topology: since $F(S') = \emptyset$ or $\{\text{pt}\}$ for every S' , this reduces to checking that if (S_i) is an open covering of S (resp. $S' \rightarrow S$ a faithfully and quasi-compact morphism), and each $Y_{S_i} \rightarrow Z_{S_i}$ (resp. $Y_{S'} \rightarrow Z_{S'}$) is an isomorphism, the same holds for $Y \rightarrow Z$; this is clear (resp. follows from SGA 1, VIII 5.4 or EGA IV₂, 2.7.1).

Moreover, by SGA 1, VIII 1.9, faithfully flat quasi-compact morphisms are of effective descent for the fibered category of closed-immersion arrows. With the notation of 6.2.1, this allows us to restrict ourselves to the case where $S = S'_i$.

Then let (Z_j) be a covering of Z by affine open subsets such that $\mathcal{O}(Z_j)$ is a projective module over $A = \mathcal{O}(S)$, and let $Y_j = Y \cap Z_j$ and $F_j : (\mathbf{Sch})_S^\circ \rightarrow (\mathbf{Ens})$ be the functor defined in terms of (Z_j, Y_j) as F is in terms of (Z, Y) . It is a subfunctor of the final functor, and one evidently has $F = \bigcap_j F_j$, which reduces us to proving that every F_j is representable by a closed subscheme T_j of S (for then F will be representable by the closed subscheme T which is the intersection of the T_j). One may therefore suppose Z also affine, $Z = \text{Spec}(B)$, where B is a projective A -module. Let L be a free A -module with basis $(e_\lambda)_{\lambda \in \Lambda}$, of which B is a direct summand as an A -module, and let $\varphi_\lambda : L \rightarrow A$ be the “coordinate” forms with respect to this basis. Let E be a set of generators of the ideal J of B defining the subscheme Y of Z , and let I be the ideal in A generated by the coordinates $\varphi_\lambda(x)$, for $x \in E$. For every A -algebra C , one then sees that the morphism $B \otimes_A C \rightarrow (B/J) \otimes_A C$ is an isomorphism if and only if the image of $x \otimes 1$ in $L \otimes_A C$ is zero for every $x \in E$, which is equivalent to saying that the kernel of $A \rightarrow C$ contains the ideal I . This shows that $T = \mathcal{V}(I) = \text{Spec}(A/I)$ satisfies the desired condition, which proves (i).

Moreover, if $Y \rightarrow Z$ is of finite presentation, one may take E finite, and then I is a finitely generated ideal of A , i.e. the closed immersion $T \rightarrow S$ is of finite presentation.

Examples 6.2.4. Let us give important examples of functors which reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.2.3 and for which it is useful subsequently to have representability criteria. We denote by S a scheme, by X, Y, Z , etc. schemes over S .

a) Suppose given an S -morphism

$$(x) \quad q : X \rightarrow \underline{\text{Hom}}_S(Y, Z),$$

(“ X acts on Y , with values in Z ”), i.e. a morphism

$$(xx) \quad r : X \times_S Y \rightarrow Z.$$

⁽⁵⁴⁾**N.D.E.** Indeed, let $(A, \mathfrak{m})$ be an artinian local ring, k its residue field, M an arbitrary A -module, $(x_i)_{i \in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k . Let F be the free A -module with basis $(e_i)_{i \in I}$, and $\phi : F \rightarrow M$ the A -morphism defined by $\phi(e_i) = x_i$. Then $Q = \text{Coker } \phi$ satisfies $Q = \mathfrak{m}Q$, whence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose moreover M flat over A ; then $K = \text{Ker } \phi$ satisfies $K \otimes_A k = 0$, i.e. $K = \mathfrak{m}K$, whence $K = 0$.

⁽⁵⁵⁾**N.D.E.** Cf. Exp. II §1, where this functor is denoted by $\prod_{Z/S} Y$; for every $S' \rightarrow S$, $F(S') = \Gamma(Y_{S'}/Z_{S'})$, which here equals $\{\text{id}_{Z_{S'}}\}$ if $Y_{S'} = Z_{S'}$, and is empty otherwise. On the other hand, assertion (ii), used in the proof of 6.5.3, has been added.

Consider a subscheme Z' of Z , whence a monomorphism

$$\underline{\mathrm{Hom}}_S(Y, Z') \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z)$$

which makes the first functor a subfunctor of the second; let X' be the inverse image of this subfunctor under the morphism q ; it is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \longrightarrow Z_T$ factors through Z'_T . This functor X' can be described as follows: put $P = X \times_S Y$, and let P' be the inverse image of Z' under $r : P \longrightarrow Z$; then one has an evident isomorphism

$$(xxx) \quad X' \simeq \prod_{P/X} P'/P.$$

One therefore obtains: if Y is essentially free over S and Z' closed in Z , the subfunctor X' of X is representable by a closed subscheme of X .⁽⁵⁶⁾ If moreover $Z' \longrightarrow Z$ is of finite presentation, so is $X' \longrightarrow X$.

b) Suppose given two ways of making X act on Y with values in Z , i.e. two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\mathrm{Hom}}_S(Y, Z),$$

and put $X' = \mathrm{Ker}(q_1, q_2)$: it is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms $q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$ are equal. Now giving q_1, q_2 is equivalent to giving a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z \times_S Z),$$

or a morphism $r : X \times_S Y \longrightarrow Z \times_S Z$; then put $U = Z \times_S Z$ and let U' be the diagonal subscheme of $Z \times_S Z$; then X' is none other than the inverse image of the subfunctor $\underline{\mathrm{Hom}}_S(Y, U')$ of $\underline{\mathrm{Hom}}_S(Y, U)$ under q , hence can be put in the form (xxx), with $P = X \times_S Y$, and $P' =$ inverse image of the diagonal under r , i.e. kernel of $X \times_S Y \xrightarrow[r_2]{r_1} Z$. One is therefore under the conditions of (a).

It follows that: if Y is essentially free over S and Z separated over S , the subfunctor X' of X is representable by a closed subscheme of X .⁽⁵⁶⁾ If moreover $Z \longrightarrow S$ is locally of finite type, $X' \longrightarrow X$ is of finite presentation.

c) Suppose given a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y),$$

i.e. “ X acts on Y ”. Let X' be the “kernel” of this morphism, i.e. the subfunctor X' of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \longrightarrow Y_T$ is the identity. This functor falls under b), as one sees by introducing a second homomorphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

“by making X act trivially on Y ”. Thus: if Y is essentially free and separated over S , the subfunctor X' which is the kernel of q is representable by a closed subscheme of X .⁽⁵⁶⁾ If moreover $Y \longrightarrow S$ is locally of finite type, $X' \longrightarrow X$ is of finite presentation.

d) Under the conditions of c), consider the subfunctor Y' of Y “of invariants under X ”; thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $q(y) : X_T \longrightarrow Y_T$ is “the constant T -morphism with value y ”. Introducing q' as in c), and the homomorphisms corresponding to q and q' :

$$q, q' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that Y' is precisely $\mathrm{Ker}(q, q')$, and therefore again falls under b) (with the roles of X, Y reversed and $Z = Y$).

Consequently, if X is essentially free over S , Y separated over S , the subfunctor Y' of Y of invariants under X is representable by a closed subscheme of Y .⁽⁵⁶⁾ If moreover $Y \longrightarrow S$ is locally of finite type, $Y' \longrightarrow Y$ is of finite presentation.

⁽⁵⁶⁾**N.D.E.** The following sentence has been added.

e) Constructions of the type made explicit in the preceding examples are particularly frequent in group theory. Thus, when G is an S -group scheme acting on the S -scheme X ,

$$q : G \longrightarrow \underline{\text{Aut}}_S(X),$$

the kernel of q (“the subgroup of G acting trivially”) is a closed subscheme of G provided X is essentially free and separated over S (example c)), and the subobject X^G of invariants is a closed subscheme of X provided G is essentially free over S and X separated over S (example d)).

Let Y, Z be subschemes of X ; consider the subfunctor $\underline{\text{Transp}}_G(Y, Z)$ of G (“transporter of Y into Z ”), whose points with values in a T over S are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subset Z_T$, i.e. induces a morphism $Y_T \longrightarrow X_T$ factoring through $Y_T \longrightarrow Z_T$. Thus: if Y is essentially free over S , and Z closed in X , $\underline{\text{Transp}}_G(Y, Z)$ is a closed subscheme of G (example a)).

One may also consider the strict transporter of Y into Z ,⁽⁵⁷⁾ whose points with values in a T over S are the $g \in G(T)$ such that $g(Y_T) = Z_T$, which is none other than $\underline{\text{Transp}}_G(Y, Z) \cap \sigma(\underline{\text{Transp}}_G(Z, Y))$, where σ is the inversion morphism of G . Consequently, if Y and Z are essentially free over S and closed in X , the strict transporter of Y into Z is a closed subscheme of G .

An important case is that where $X = G$, G acting on itself by inner automorphisms. If H is a subscheme of G , the strict transporter of H into H is also called the *normalizer of H in G* , and denoted by $\underline{\text{Norm}}_G H$. Thus: if H is a closed subgroup scheme of G , essentially free over S , $\underline{\text{Norm}}_G H$ is representable by a closed subgroup scheme of G .

Finally let Z be a subscheme of G ; then its centralizer $\underline{\text{Centr}}_G(Z)$ in G is the subgroup functor of G defined by the procedure of d), when one considers that “ Z acts on G ” by the actions induced by those of G ; thus, if Z is essentially free over S and G is separated over S , $\underline{\text{Centr}}_G(Z)$ is a closed subgroup scheme of G . In particular, if G is essentially free and separated over S , the center C of G , which is none other than $\underline{\text{Centr}}_G(G)$, is a closed subgroup scheme of G .

When S is the spectrum of a field, 6.2.2 b) shows that in examples a) to e) above the conditions “essentially free” are automatically satisfied; only separation conditions remain. Recalling that a group scheme over a field is necessarily separated (VIA, 0.3), one finds for example:

Corollary 6.2.5. *Let G be a group scheme over a field k and let Y, Y' be two subschemes of G . Then:*

- (i) *The centralizer of Y in G is a closed subgroup scheme of G ; this is in particular the case for the center $\underline{\text{Centr}}_G(G)$ of G .*
- (i') *More generally, if $u, v : X \longrightarrow G$ are morphisms of schemes, $\underline{\text{Transp}}_G(u, v)$ is representable by a closed subscheme of G .*
- (ii) *If Y is closed, the transporter $\underline{\text{Transp}}_G(Y', Y)$ is a closed subscheme of G . If Y' is also closed, one has the same conclusion for $\underline{\text{Transp}}_G^{\text{str}}(Y', Y)$.*
- (iii) *For every subgroup scheme⁽⁵⁸⁾ H of G , $\underline{\text{Norm}}_G(H)$ is a closed subgroup scheme of G .*

Corollary 6.2.6.⁽⁵⁹⁾ *Let k be a field, G a connected algebraic k -group. Then $\underline{\text{Centr}}_G(G)$ is representable by a closed subgroup scheme Z of G , and G/Z is an affine algebraic k -group.*

⁽⁵⁷⁾**N.D.E.** Denoted by $\underline{\text{Transp}}_G^{\text{str}}(Y, Z)$.

⁽⁵⁸⁾**N.D.E.** Indeed, over a field k , every subgroup scheme of G is closed, cf. VIA, 0.5.2. On the other hand, 6.2.5 completes the insertion of the results taken from VIII §6.

⁽⁵⁹⁾**N.D.E.** This corollary (cf. Exp. XII, 6.1), which will be useful later, has been inserted here. Moreover, in 6.3 one returns to the original text of VIB.

PROOF. Of course, the first assertion is contained in 6.2.5 (i), but one will see that it also follows from the proof of the second assertion. Indeed, G acts by the adjoint representation on the finite-dimensional k -vector spaces $V_n = \mathcal{O}_{G,e}/\mathfrak{m}_e^{n+1}$ (where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$); denote by K_n the kernel of the morphism $\rho_n : G \rightarrow \mathrm{GL}(V_n)$. By VIA 5.4.1, ρ_n induces a closed immersion $G/K_n \hookrightarrow \mathrm{GL}(V_n)$, so every G/K_n is affine. Since G is noetherian, the intersection K of the K_n equals one of the K_n , hence G/K is affine.

On the other hand, denoting by Z the center of G , it is clear that $Z \subset K$. Let us show that $Z = K$. Denote by $\widehat{\mathcal{O}}_{G,e}$ the completion of $\mathcal{O}_{G,e}$ for the $\mathfrak{m}_e$ -adic topology and by $\widehat{S}$ its spectrum (resp. $S = \mathrm{Spec} \mathcal{O}_{G,e}$). Since $\widehat{S} \rightarrow S$ is faithfully flat and the two morphisms

$$K \times_k S \rightarrow S, \quad (g, x) \mapsto gxg^{-1} \quad \text{resp.} \quad (g, x) \mapsto x$$

coincide after the base change $\widehat{S} \rightarrow S$, they coincide, i.e. K acts trivially on $\mathcal{O}_{G,e}$. Now, by 6.2.4 e), the subobject G^K of invariants of G under K (which is none other than $\mathrm{Centr}_G(K)$) is a closed subscheme of G , hence defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_G$. Since G^K contains $S = \mathrm{Spec} \mathcal{O}_{G,e}$ and $\mathcal{I}$ is of finite type (since G is noetherian), there is an open neighborhood U of e such that $\mathcal{I}|_U = 0$. Then the subgroup $G^K = \mathrm{Centr}_G(K)$ contains U , hence also $U \cdot U$, which equals G since G is irreducible (VIA 0.5). Thus $\mathrm{Centr}_G(K) = G$, whence $K \subset Z$, hence $Z = K$. $\square$

Remark 6.3. Let k be an algebraically closed field, G a k -group and H a subgroup scheme of G ; suppose G and H of finite type over k and reduced; then $\mathrm{Norm}_G H$ (resp. $\mathrm{Centr}_G H$) is representable by a subgroup scheme of G , whose associated reduced subscheme is none other than the normalizer (resp. centralizer) of H in G in the sense of the *Bible*.

Proposition 6.4. *Let G be an S -group and $u : X \rightarrow G$ a monomorphism of S -schemes. Put $T = \mathrm{Transp}_G(X, X)$. The following conditions are equivalent:*

- (i) T is a subgroup functor of G .
- (ii) $T = \mathrm{Transp}_G^{\mathrm{str}}(X, X) = \mathrm{Norm}_G X$.

These conditions hold in each of the following two cases:

- (a) X is of finite presentation over S .
- (b) T is representable by a scheme of finite presentation over S .

The equivalence of conditions (i) and (ii) follows from the fact that, for every morphism $S' \rightarrow S$ and every $t, t' \in T(S')$, one has $tt' \in T(S')$, and from the fact that $\mathrm{Transp}_G^{\mathrm{str}}(X, X) = T \cap c(T)$ (cf. 6.1 (ii)).

Consider case a). Let $t \in T(S)$; then $\mathrm{int}(t)$ is a monomorphism of X into X , hence an S -automorphism of X (EGA IV₄, 17.9.6), so t belongs to $\mathrm{Transp}_G^{\mathrm{str}}(X, X)$, whence a).

In case b), it is clear that $\mu(T \times_S T) \subset T$, and the assertion follows from the following lemma:

Lemma 6.4.2. ⁽⁶⁰⁾ *Let G be an S -scheme of finite presentation, endowed with an associative law (in the sense of EGA 0_{III} 8.2.5). Suppose that, for every S -scheme S' and every $g \in G(S')$, right and left translations by g in the set $G(S')$ are injective and $G(S) \neq \emptyset$. Then G is an S -group.*

It suffices to show that, for every S -scheme S' , the set $G(S')$ is a group; now it follows at once from the hypothesis that right and left translations by every element $g \in G(S')$ in $G_{S'}$ are S -monomorphisms from $G_{S'}$ to $G_{S'}$. They are therefore S -automorphisms, since G is of finite presentation over S (EGA IV₄, 17.9.6), so right and left translations by g in the set $G(S')$ are bijective, and one is reduced to the following lemma:

⁽⁶⁰⁾ **N.D.E.** The numbering of the original has been retained: there is no no. 6.4.1.

Lemma 6.4.3. *Let G be a nonempty set endowed with an associative law such that right and left translations are bijective. Then G is a group.*

The proof is left to the reader.

Definition 6.5. Let G be an S -group, H an S -functor, and $u : H \rightarrow G$ a monomorphism.

- (i) One calls the *connected centralizer of H in G* , and denotes by $\underline{\text{Centr}}_G^0 H$, the identity component of the functor $\underline{\text{Centr}}_G H$ (cf. 3.1 and 6.1 (iii)).
- (i') For every morphism $u : X \rightarrow G$, one similarly defines the functor $\underline{\text{Centr}}^0(u)$ (cf. 6.2 (iv)).
- (ii) When, for every $s \in S$, u_s is a closed immersion, one calls the *connected normalizer of H in G* , and denotes by $\underline{\text{Norm}}_G^0 H$, the identity component of the functor $\underline{\text{Norm}}_G H$.

N.B. By 1.4.2, the hypothesis of (ii) holds when H is an S -group scheme, G and H are locally of finite type over S and u is a quasi-compact morphism of S -groups.

Proposition 6.5.1. *Let G be an S -group locally of finite presentation and quasi-separated over S , H a smooth S -group with connected fibers and $u : H \rightarrow G$ a monomorphism. Let N be the normalizer of H in G (cf. 6.1). By 6.5.5 below, N is representable by a closed subgroup scheme of G and of finite presentation over G . This being so, the following conditions are equivalent:*

- (i) *The canonical morphism $H \rightarrow N$ is an open immersion.*
- (ii) *$N^0 = H$ (cf. 3.10).*
- (iii) *For every $s \in S$, one has $H_s = (N_s)^0$.*

Condition (i) implies (ii) by Lemma 3.10.1, since $H^0 = H$. On the other hand, it is clear that (ii) implies (iii), for $H_s = (N^0)_s = (N_s)^0$.

Finally let us show that (iii) implies (i). Since $H_s = (N^0)_s$ for every $s \in S$, $H_s \rightarrow N_s$ is an open immersion. Moreover, H and N are locally of finite presentation over S , and H is flat over S , so (EGA IV₄, 17.9.5), $H \rightarrow N$ is an open immersion.

⁽⁶¹⁾For the reader's convenience, the results 6.8 to 6.11 of Exp. XI have been included below.

Theorem 6.5.2. *Let X be a smooth scheme over S , with geometrically irreducible fibers.⁽⁶²⁾*

- (i) *For every closed subscheme Y of X , the functor $\prod_{X/S} Y/X$ is representable by a closed subscheme T of S .*
- (ii) *Moreover, if $Y \rightarrow X$ is of finite presentation, so is $T \rightarrow S$.*

Since $f : X \rightarrow S$ is faithfully flat and locally of finite presentation, it is covering for the (fpqc) topology. Since, on the other hand, $T = \prod_{X/S} Y/X$ is evidently a subsheaf of S for the (fpqc) topology, it follows that the question of the representability of T by a closed subscheme of S is local on S for the (fpqc) topology,⁽⁶³⁾ and the same holds for the question of deciding whether T is of finite presentation over S (cf. EGA IV₂, 2.7.1). Making the base change $S' \rightarrow S$ with $S' = X$, one is then reduced to the case where X admits a section ε over S . One may moreover suppose S affine and a fortiori quasi-compact. One then has:

Corollary 6.5.3. *Under the conditions of 6.5.2, suppose that S is quasi-compact, $X \rightarrow S$ admits a section ε , and $Y \rightarrow X$ is of finite presentation. Then there is an integer $n \geq 0$*

⁽⁶¹⁾N.D.E. Nos. 6.5.2 to 6.5.5, taken from Exp. XI, 6.8 to 6.11, have been inserted here. This was moreover suggested by A. Grothendieck in a Note at the beginning of XI 6: "The present number does not use the results of nos. 3, 4, 5 (of XI); its natural place would be in VIB."

⁽⁶²⁾N.D.E. On the one hand, the original has been corrected by replacing "connected" by "irreducible" (cf. the proof). On the other hand, by [RG71] I, 3.3.4 (iii) and 4.1.1, it suffices to suppose that X is flat over S , with geometrically irreducible fibers and without embedded components.

⁽⁶³⁾N.D.E. See for example addition 1.7 in Exp. VIII.

such that one has

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n is the n -th infinitesimal neighborhood of the immersion $\varepsilon : S \rightarrow X$, and $Y_n = Y \cap X_n$.

When Y is of finite presentation over X , this corollary does imply 6.5.2, for, since X is smooth over S , X_n is finite and locally free over S , hence a fortiori “essentially free” over S (cf. 6.2.1), so $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subscheme T of S , of finite presentation over S , by 6.2.3.

Let us first prove 6.5.3 (and hence 6.5.2) when S is noetherian. Let $T_n = \prod_{X_n/S} Y_n/X_n$; then the T_n form a decreasing sequence of closed subschemes of S , and, since S is noetherian, this sequence is stationary. Let $R = \bigcap_{n \geq 0} \prod_{X_n/S} Y_n/X_n$ be their common value for large n ; one evidently has $T \subset R$, and it suffices to establish that $R \subset T$. Making the base change $R \rightarrow S$, one is reduced to the case where $R = S$, i.e. $Y_n = X_n$ for every n , i.e. $Y \supset X_n$ for every n , and one must then prove that $T = S$, i.e. $Y = X$.

Now $Y \supset X_n$ for every n implies (thanks to the fact that X is locally noetherian) that Y is, in a neighborhood of each point of $\varepsilon(S)$, an induced open subscheme of X ,⁽⁶⁴⁾ so there is an induced open subset U of X , containing $\varepsilon(S)$, such that $U \subset Y$. By EGA IV₃, 11.10.10, the fibers of X/S being integral, U is schematically dense in X , hence (Y being a closed subscheme containing U) one has $Y = X$. This proves 6.5.3, hence 6.5.2, when S is noetherian. The general case proceeds by reduction to the preceding case. For every $s \in S$, there is an affine open neighborhood U of s and an affine open neighborhood V of $\varepsilon(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U , and if S' is an affine open neighborhood of s contained in $\varepsilon^{-1}(V) \cap f(V)$, and $X' = V \cap f^{-1}(S')$, then X' and S' are affine open subsets of X resp. S , and X'/S' admits a section. Since 6.5.2 and 6.5.3 are local, one may suppose $S = S'$. Then X' is an affine open subset of X containing $\varepsilon(S)$. Since every fiber X_s is supposed irreducible, X'_s is schematically dense in X_s , hence, by EGA IV₃, 11.10.10, X' is schematically dense in X ; similarly, for every base change $S_1 \rightarrow S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$.

It follows that $\prod_{X/S} Y/X = \prod_{X'/S} Y'/X'$, where $Y' = Y \cap X'$. This reduces us to the case where $X = X'$, so one may suppose S and X affine. Moreover, if $X = \text{Spec}(B)$ and J is the ideal of B defining Y , then J is the inductive limit of its finitely generated subideals, so Y is the intersection of closed subschemes Y_i of X of finite presentation over X , and consequently $\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X$, which reduces us, to prove 6.5.2, to the case where Y is of finite presentation over X , with S and X affine.

But then X and Y over S come, by base change $S \rightarrow S_0$, from an analogous situation X_0 and Y_0 over S_0 , with S_0 noetherian, which reduces us to the case where S is noetherian, already treated. This completes the proof of 6.5.2 and 6.5.3.

Corollary 6.5.4. *Let X be a smooth S -group scheme of finite presentation with connected fibers, Y a group scheme of finite presentation over S , $i : Y \rightarrow X$ a monomorphism of S -group schemes.*

- (i) *Then $\prod_{X/S} Y/X$ is representable by a closed subscheme of finite presentation of S .*
- (ii) *If moreover S is quasi-compact, for sufficiently large n one has:*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n denotes the n -th infinitesimal neighborhood of the unit section $\varepsilon : S \rightarrow X$, and $Y_n = X_n \cap Y$.

⁽⁶⁴⁾**N.D.E.** See, for example, the proof of 6.2.6.

The proof is essentially that of 6.5.3.⁽⁶⁵⁾ On the one hand, i is locally of finite presentation (cf. EGA IV₁, 1.4.3). On the other hand, the unit sections of Y and X induce bijective immersions $S \rightarrow Y_n$ and $S \rightarrow X_n$, hence isomorphisms of S_{red} with $(Y_n)_{\text{red}}$ and $(X_n)_{\text{red}}$. Consequently, i_n is quasi-compact, hence of finite type, and one has a commutative diagram:

$$\begin{array}{ccc} (Y_n)_{\text{red}} & \xrightarrow{\tau} & Y_n \\ & \searrow \sigma & \downarrow i_n \\ & & X_n \end{array}$$

where σ, τ are closed immersions and τ is surjective. Since i_n is separated (being a monomorphism), it follows that i_n is proper (cf. EGA II, 5.4.3). Thus i_n is a proper monomorphism of finite presentation, hence a closed immersion (cf. EGA IV₃, 8.11.5). Consequently, by 6.2.3, already used, $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subscheme T_n of S of finite presentation over S , and it therefore remains to prove the last assertion of 6.5.4 in the case where S is moreover supposed affine.

One again immediately reduces to the case where S is noetherian, and one is reduced to proving that one then has $R = T$ (with the notation of the proof of 6.5.3), or that $Y \supset X_n$ for every n implies $Y = X$. Now the hypothesis implies that $i : Y \rightarrow X$ is étale at the points of the unit section of Y over S , so Y is smooth over S at the points of the unit section, whence it follows that the open subset U of points of Y at which Y is smooth over S is an induced open subgroup of Y (cf. 2.3). Then $\tau : U \rightarrow X$ is an étale monomorphism by 2.5, hence an open immersion; now, since the fibers of X are connected and every open subgroup of an algebraic group is also closed, it follows that τ is a surjective open immersion, i.e. an isomorphism. Thus $U = X$ and a fortiori $Y = X$, which completes the proof of 6.5.4.

Proceeding as in 6.2.4 e), one concludes from 6.5.4:

Corollary 6.5.5. *Let G be an S -group scheme locally of finite type and quasi-separated over S , H a smooth group scheme of finite presentation over S with connected fibers, $i : H \rightarrow G$ a monomorphism of S -groups. Then:*

- (a) $\underline{\text{Centr}}_G(H)$ and $\underline{\text{Norm}}_G(H)$ are representable by closed subschemes of G , of finite presentation over G .
- (a') Similarly, for every monomorphism $j : K \rightarrow G$ of finite presentation of S -group schemes, $\underline{\text{Transp}}_G(H, K)$ is representable by a closed subscheme of G , of finite presentation over G .
- (b) If G is quasi-compact, there is an integer $n \geq 0$ such that (if H_n denotes the n -th infinitesimal neighborhood of the unit section of H) one has:

$$\begin{aligned} \underline{\text{Centr}}_G(H) &= \underline{\text{Centr}}_G(H_n), & \underline{\text{Norm}}_G(H) &= \underline{\text{Norm}}_G(H_n), \\ \underline{\text{Transp}}_G(H, K) &= \underline{\text{Transp}}_G(H_n, K) = \underline{\text{Transp}}_G(H_n, K_n). \end{aligned}$$

PROOF. ⁽⁶⁶⁾First note that the hypothesis on G implies that the monomorphism $H \rightarrow G$ is of finite presentation (EGA IV₁, 1.2.4 and 1.4.3), as is the diagonal immersion $\Delta_{G/S} : G \rightarrow G \times_S G$ (loc. cit. 1.4.3.1). The case of $\underline{\text{Norm}}_G(H)$ therefore reduces to that of the transporter by taking $H = K$. Taking account of 6.2.4 e), we shall apply 6.5.4 to the group scheme $X = H_G = H \times_S G$ over the base scheme G , and to the following subgroup scheme Y .

In the case of $\underline{\text{Transp}}_G(H, K)$, take for Y the inverse image of K_G under the morphism of G -groups $H \times_S G \rightarrow G \times_S G$, $(h, g) \mapsto (ghg^{-1}, g)$. In the case of $\underline{\text{Centr}}_G(H)$, take for Y the inverse image of the diagonal subgroup $\Delta_{G/S}(G)_G$ of $(G \times_S G)_G$ under the morphism of G -groups

$$H \times_S G \rightarrow G \times_S G \times_S G, \quad (h, g) \mapsto (h, ghg^{-1}, g).$$

⁽⁶⁵⁾N.D.E. The original has been expanded in what follows, adding references to EGA IV.

⁽⁶⁶⁾N.D.E. What follows has been expanded. On the other hand, this completes the insertion of XI, 6.8 to 6.11, i.e. in 6.6 below one returns to the original text of VIB.

□

Definition 6.6. Let G be an S -group functor, H a subgroup S -functor; one says that H is *invariant* (resp. *central*, resp. *characteristic*) in G if $\text{Norm}_G H = G$ (resp. if $\text{Centr}_G H = G$, resp. if, for every S -scheme T and every automorphism $a \in \text{Aut}_{T\text{-gr.}}(G_T)$, one has $a(H_T) \subset H_T$), in other words, if, for every S -scheme T , the subgroup $H(T)$ of $G(T)$ is invariant in $G(T)$ (resp. central in $G(T)$, resp. invariant under every automorphism of G_T).

N.B. If H is central (resp. characteristic), it is invariant.

Remark 6.7. Let G and H be two S -groups and $u : H \rightarrow G$ a monomorphism. For H to be invariant (resp. central) in G , it is necessary and sufficient that the morphism

$$\mu \circ (c \circ \text{pr}_2, \mu \circ (u \times \text{id}_G)) : H \times_S G \rightarrow G$$

(defined by $(h, g) \mapsto g^{-1}hg$ for every $g \in G(S')$ and $h \in H(S')$) factor through u (resp. equal $u \circ \text{pr}_1$), and, for H to be characteristic in G , it is necessary and sufficient that, for every S -scheme T and every T -group automorphism a of G_T , $a \circ u_{(T)}$ factor through $u_{(T)}$.

Example 6.8. Let G be an S -group functor. Then Centr_G is characteristic and central. If, for every $s \in S$, G_s is representable, G^0 is characteristic. This follows from the definitions and 3.3.

6.9. — ⁽⁶⁷⁾In [RG71], I 3.3.3, the authors introduce the geometric notion of a pure S -scheme, which is local on S for the étale topology; we refer to loc. cit. for the precise definition. Let us simply point out the following:

- (a) (loc. cit., Th. 3.3.5) If B is a flat A -algebra of finite presentation, B is a projective A -module if and only if $\text{Spec}(B)$ is pure over $\text{Spec}(A)$.
- (b) Consequently, if $X \rightarrow S$ is locally of finite presentation, flat and pure, X is essentially free over S .
- (c) (loc. cit., 3.3.4 (iii)) If $X \rightarrow S$ is locally of finite type, flat, with geometrically irreducible fibers and without embedded components, X is pure over S .

Since every group scheme locally of finite type over a field is Cohen–Macaulay (cf. VIA, 1.1.1), hence without embedded components, one obtains in particular:

- (d) Every S -group scheme G locally of finite presentation, flat and with connected fibers is pure over S , hence essentially free over S .

One may then take up all the statements of 6.2.4 e), taking account of results (b) and (d) above.

7. Generated subgroups; commutator group

In this number, k denotes a fixed field.

Proposition 7.1. Let G be a k -group, $(X_i)_{i \in I}$ a family of geometrically reduced k -schemes;⁽⁶⁸⁾ for every $i \in I$, let $f_i : X_i \rightarrow G$ be a k -morphism.

- (i) There is a smallest closed subgroup k -scheme of G containing each of the f_i , denoted by $\Gamma_G((f_i)_{i \in I})$. It is a geometrically reduced k -scheme, hence smooth if G is supposed locally of finite type over k (1.3.1).
- (ii) Put $X = \coprod_{i \in I} X_i$ and let $f : X \rightarrow G$ be the morphism whose restriction to X_i is f_i for every $i \in I$. Put $X^1 = X \amalg X$, and let $f^1 : X^1 \rightarrow G$ be the morphism whose restrictions to X are respectively f and $c \circ f$. For every $n > 1$, put

$$X^n = X^1 \times_k X^{n-1} \quad \text{and} \quad f^n = \mu \circ (f^1 \times_k f^{n-1}) : X^n \rightarrow G.$$

⁽⁶⁷⁾N.D.E. This subsection has been added.

⁽⁶⁸⁾N.D.E. Here and below, the little-used terminology “separable” has been replaced by the usual terminology “geometrically reduced”, cf. EGA IV₂, 4.6.2.

- Then $\Gamma_G((f_i)_{i \in I})$ is the reduced subscheme of G with underlying space the closure of the union of the $f^n(X^n)$, for $n \geq 1$.
- (iii) For every k -scheme S , $\Gamma_G((f_i)_{i \in I})_S$ is the smallest closed subgroup scheme of G_S containing each of the $f_{i,S} : X_S \rightarrow G_S$.
 - (iii') Moreover, $\Gamma_G((f_i)_{i \in I})_S$ is the smallest subgroup scheme of G_S containing each of the $f_{i,S}$.⁽⁶⁹⁾

First observe that, to prove (i), (iii) and (iii'), defining X and f as in (ii) reduces one to the case where I consists of one element.

Let H be the reduced subscheme of G with underlying set $\overline{\bigcup_{n \geq 1} f^n(X^n)}$. Then the family of morphisms $f^n : X^n \rightarrow H$ is schematically dominant (cf. EGA IV₃, 11.10.4), so every closed subscheme of G containing the f^n also contains H . Moreover, by loc. cit., 11.10.7, H is geometrically reduced. Thus, to prove (i) and (ii), it suffices to show that H is a subgroup scheme of G , hence that the restriction of c to H and the restriction of μ to $H \times_k H$ factor through the injection $H \rightarrow G$.

Since H is geometrically reduced, $H \times_k H$ is reduced, and it suffices to check that $c(H) \subset H$ and $\mu(H \times_k H) \subset H$ (as sets). But, by EGA IV₃, 11.10.6, the union of the $f^n_{(H)}(X^n \times_k H)$ is schematically dense in $H \times_k H$. Similarly, for every $n \geq 1$, the union of the $f^n_{(X^n)}(X^n \times_k X^m)$, for $m \geq 1$, is schematically dense in $X^n \times_k H$. Thus it suffices to show that $\mu(f^n_{(H)}(f^n_{(X^n)}(X^n \times_k X^m))) \subset H$ and $c(f^n(X^n)) \subset H$. Now

$$\begin{aligned} \mu(f^n_{(H)}(f^n_{(X^n)}(X^n \times_k X^m))) &= \mu((f^n \times_k f^m)(X^n \times_k X^m)) \\ &= f^{n+m}(X^{n+m}) \subset H; \end{aligned}$$

and, since $c(f^1(X^1)) \subset f^1(X^1)$, one has, for every n , $c(f^n(X^n)) \subset f^n(X^n) \subset H$. This proves (i) and (ii).

Now let us prove (iii). Let G' be a closed subgroup scheme of G_S containing f_S ; one must show that G' contains H_S , or, equivalently, that $H_S = H_S \times_{G_S} G'$. Put $H'_S = H_S \times_{G_S} G' = G' \cap H_S$. Since G' and H_S both contain the f'_S , so does H'_S . Now (EGA IV₃, 11.10.6), since the family of $f^n : X^n \rightarrow H$ is schematically dominant, so is the family of $f'_S : X'_S \rightarrow H'_S$, so H'_S , containing each of the f'_S , equals H_S by EGA IV₃, 11.10.1 c). This proves (iii).

Finally let us show that H_S is the smallest (not necessarily closed) subgroup scheme of G_S containing f_S .

Let G' be a subgroup scheme of G_S containing f_S . One must similarly show that, putting $H'_S = H_S \times_{G_S} G'$, one has $H'_S = H_S$. For this it suffices to show that H'_S is closed in H_S and apply (iii). It therefore suffices to show that H_S and H'_S have the same underlying set; a fortiori it suffices to show that, for every $s \in S$, H_s equals

$$H'_s := H'_S \times_S \kappa(s) = H_s \times_{G_s} G'_s.$$

Now, by VIA, 0.5.2, the subgroup $\kappa(s)$ -scheme G'_s is closed in G_s . Thus H'_s is closed in H_s , and then the preceding argument, applied to H'_s , H_s and the f'_s , shows that $H'_s = H_s$.

Corollary 7.1.1. ⁽⁷⁰⁾ *Let G be a k -group locally of finite type and let A, B be two k -subgroups of G , smooth and of finite type, and i_A (resp. i_B) the inclusion of A (resp. B) into G . Suppose B normalizes A ; then $A \cdot B = \Gamma_G(i_A, i_B)$.*

Indeed, let $H = A \rtimes B$ be the semidirect product of A and B (cf. I, 2.3.5); it is a smooth k -group of finite type. Then the morphism of groups $u : H \rightarrow G$, $(a, b) \mapsto ab$, is quasi-compact, so, by 1.2, $u(H) = A \cdot B$ is a reduced closed subscheme of G , which is a group in the category $(\mathbf{Sch}/k)_{\text{red}}$. Now, by EGA IV₃, 11.10.7, $A \cdot B = \overline{u(H)}$ is geometrically reduced

⁽⁶⁹⁾**N.D.E.** The original stated (iii') under the additional hypothesis that G be locally of finite type over k , but this can be omitted, by VIA, 0.5.2.

⁽⁷⁰⁾**N.D.E.** This corollary has been added to point out this special case of 7.1.

(since H is), so it is a closed subgroup of G . Since one evidently has $A \cdot B \subset \Gamma_G(i_A, i_B)$, the corollary follows.

Definitions and remarks 7.2. ⁽⁷¹⁾ (i) Given a k -group G , a family $(X_i)_{i \in I}$ of geometrically reduced k -schemes, and, for each $i \in I$, a k -morphism $f_i : X_i \rightarrow G$, one calls the *closed subgroup scheme of G generated by the family $(f_i)_{i \in I}$* , and in this number we shall denote by $\Gamma_G((f_i)_{i \in I})$, the smallest closed subgroup scheme of G containing each of the f_i . If X is a subscheme of G , geometrically reduced over k , and f is the immersion $X \hookrightarrow G$, one writes $\Gamma_G(X)$ instead of $\Gamma_G(f)$.

(i') With the notation of 7.1 (ii), we shall sometimes put $\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n)$. Note that $\Gamma'_G(f)$ is a subset of G stable under the group law (in the sense of 3.0).

(ii) It is clear that if X_1 and X_2 are two geometrically reduced k -schemes and $f_1 : X_1 \rightarrow G$ and $f_2 : X_2 \rightarrow G$ two k -morphisms such that the sets $\overline{f_1(X_1)}$ and $\overline{f_2(X_2)}$ are equal, $\Gamma_G(f_1) = \Gamma_G(f_2)$.

(iii) Let E be a subset of G such that the reduced subscheme $\overline{E}$ of G is geometrically reduced. One calls the *closed subgroup scheme of G generated by E* , and denotes by $\Gamma_G(E)$, the subgroup scheme $\Gamma_G(i)$, where i is the injection of the reduced subscheme $\overline{E}$ of G into G .

(iv) Since every subgroup scheme of G is closed, by VIA, 0.5.2, ⁽⁷²⁾ one will speak of the “generated subgroup scheme” instead of the “generated closed subgroup scheme”.

(v) Let X be a geometrically reduced k -scheme and $f : X \rightarrow G$ a k -morphism. Suppose $f(X)$ contains the identity element e of G . Put $X'^1 = X \times_k X$ and $f'^1 = \mu \circ (f \times_k (c \circ f))$, and, for $n > 1$,

$$X'^n = X'^1 \times_k X'^{n-1} \quad \text{and} \quad f'^n = \mu \circ (f'^1 \times_k f'^{n-1}).$$

Then $\Gamma_G(f)$ is the reduced subscheme of G whose underlying space is the closure of the union of the $f'^n(X'^n)$, for $n \geq 1$.

Indeed, recalling the notation of 7.1 (ii), $X^1 = X \sqcup X$ and $X^n = X^1 \times_k X^{n-1}$ for $n \geq 2$, one has the following inclusions, where the first follows from the hypothesis $e \in f(X)$:

$$f^n(X^n) \subset f'^n(X'^n) \subset f^{2n}(X^{2n}), \quad \text{for every } n \geq 1.$$

This shows moreover that, for there to be an integer n such that $f^n(X^n) = \Gamma_G(f)$, it is necessary and sufficient that there be an integer m such that $f'^m(X'^m) = \Gamma_G(f)$.

One deduces from Remark 7.2 (v) the following

Corollary 7.2.1. *Let X be a geometrically reduced and geometrically connected k -scheme, and $f : X \rightarrow G$ a k -morphism such that $f(X)$ contains the identity element of G . Then the k -group $\Gamma_G(f)$ is connected, hence irreducible.*

Indeed, every X'^n is then connected, so the union of the $f'^n(X'^n)$ (which all contain the identity element) is connected, and the same holds for its closure $\Gamma_G(f)$. Thus $\Gamma_G(f)$ is irreducible, by VIA, 2.4 (when G , hence also $\Gamma_G(f)$, is locally of finite type over k) and 2.6.5 (iii) (in the general case).

Definition 7.2.2. Let G be a k -group.

- (a) Let A and B be two geometrically reduced subgroup k -schemes of G . One calls the *commutator subgroup scheme of A and B in G* , and denotes by $(A, B)_G$ or simply (A, B) , the closed subgroup scheme of G generated by the morphism $\nu : A \times_k B \rightarrow G$ defined by $(a, b) \mapsto aba^{-1}b^{-1}$, for every k -scheme S and $a \in A(S)$, $b \in B(S)$.

⁽⁷¹⁾**N.D.E.** Point (viii) of the original has been put in (i'), and points (vi) and (vii) have been brought out in the form of Corollary 7.2.1 and Definition 7.2.2 below.

⁽⁷²⁾**N.D.E.** The hypothesis that G be locally of finite type over k has been removed here.

- (b) Suppose G geometrically reduced over k . One calls the *derived group* of G , and denotes by $\mathfrak{D}(G)$,⁽⁷³⁾ the group (G, G) .

N.B. For G to be commutative, it is necessary and sufficient that $\mathfrak{D}(G)$ be the unit k -group.

Reminders 7.3.0. ⁽⁷⁴⁾ Recall that if $\phi : Y \rightarrow Z$ is a morphism of S -schemes, the image presheaf $\phi(Y)$ is the S -functor which associates to every S' over S the subset $\phi(Y(S'))$ of $Z(S')$. Observe that if T is a subscheme of Z and the inclusion of presheaves $\phi(Y) \hookrightarrow Z$ factors through T , i.e. if $\phi \circ h \in T(S')$ for every S -morphism $h : S' \rightarrow Y$, one has $\phi(Y) \subset T$ as sets (take $h = \text{id}_Y$), and the converse holds if Y is reduced, for in this case ϕ factors through T .

Also recall that, by 6.2, if H is a closed subgroup k -scheme of G , $\underline{\text{Centr}}_G H$ and $\underline{\text{Norm}}_G H$ are representable by closed subgroup k -schemes of G .

Corollary 7.3. *Let G be a k -group, X a geometrically reduced k -scheme, $f : X \rightarrow G$ a k -morphism.*

- (i) *Let S be a k -scheme and u an endomorphism of the S -group G_S .*
 - (a) *If one has $u(f_S(X_S)) \subset \Gamma_G(f)_S$ (as sets), the morphism $u : \Gamma_G(f)_S \rightarrow G_S$ factors through $\Gamma_G(f)_S$.*
 - (b) *If u is an automorphism of the S -group G_S and one has $u(f_S(X_S)) = f_S(X_S)$ (as sets), u induces an automorphism of $\Gamma_G(f)_S$. In particular, if an element $g \in G(S)$ satisfies $\text{int}(g)(f_S(X_S)) = f_S(X_S)$ (as sets), $g \in \underline{\text{Norm}}_{G_S}(\Gamma_G(f)_S)(S)$.*
 - (c) *If $u \circ f_S = f_S$, the restriction of u to the subgroup $\Gamma_G(f)_S$ of G_S is the identity. In particular, if an element $g \in G(S)$ satisfies $\text{int}(g)(f_S) = f_S$, $g \in \underline{\text{Centr}}_{G_S}(\Gamma_G(f)_S)(S)$.*
- (ii) *Let H be a subgroup scheme of G ; then H centralizes (resp. normalizes) $\Gamma_G(f)$ if and only if, for every $S \rightarrow \text{Spec } k$ and $h \in H(S)$, one has $\text{int}(h) \circ f_S = f_S$ (resp. $\text{int}(h)(f_S(X_S)) \subset \Gamma_G(f)_S$), i.e. if, for every $S' \rightarrow S$ and $x \in X(S')$, the elements $h_{S'} f(x)$ and $f(x)$ of $G(S')$ commute (resp. $h_{S'} f(x) h_{S'}^{-1} \in \Gamma_G(f)(S')$).*
- (iii) *In particular, let Y be a second geometrically reduced k -scheme and $\phi : Y \rightarrow G$ a k -morphism. Suppose that, for every k -scheme S' and every $x \in X(S')$ and $y \in Y(S')$, the elements $\phi(y)$ and $f(x)$ of $G(S')$ commute (resp. $\text{int}(\phi(y))(f(x)) \in \Gamma_G(f)(S')$).*
Then $\Gamma_G(\phi)$ is a k -subgroup of $\underline{\text{Centr}}_G \Gamma_G(f)$, resp. $\underline{\text{Norm}}_G \Gamma_G(f)$.
- (iv) *If g is a k -point of G , the k -group $\Gamma_G(\{g\})$ is commutative.*
- (v) *Let A and B be two subgroup schemes of G geometrically reduced over k . If A and B are invariant (resp. characteristic), so is (A, B) .*

PROOF. (i) Let us prove (a). Put $\Gamma_S = \Gamma_G(f)_S$. Then $u^{-1}(\Gamma_S)$ is a closed subgroup S -scheme of G_S , so, by 7.1 (iii), it suffices to show that f_S factors through $u^{-1}(\Gamma_S)$. Now, since $u(f_S(X_S)) \subset \Gamma_S$ and X_S is reduced, $u \circ f_S$ factors through Γ_S , so f_S factors through $u^{-1}(\Gamma_S)$. This proves (a).

Then the first assertion of (b) is a consequence of (a), applied to u and u^{-1} (and in fact it suffices to suppose $u(f_S(X_S)) \subset \Gamma_S$ and $u^{-1}(f_S(X_S)) \subset \Gamma_S$), and the second assertion is its special case where $u = \text{int}(g)$.

Let us prove (c). Consider the morphism of S -groups $G_S \rightarrow G_S \times_S G_S$, $g \mapsto (u(g), g)$ and denote by H the inverse image of the diagonal; it is a subgroup scheme of G_S . Since G is separated over k (VIA 0.3), G_S is separated over S , so H is closed in G_S . Since, by hypothesis, H contains f_S , it contains $\Gamma_G(f)_S$, by 7.1 (iii). This proves the first assertion of (c), and the second is its special case where $u = \text{int}(g)$.

Let us prove (ii). Put $\Gamma = \Gamma_G(f)$ and denote by i the inclusion $\Gamma \hookrightarrow G$. Then H is contained in $N = \underline{\text{Norm}}_G(\Gamma)$ if and only if, for every k -scheme S and $h \in H(S)$, one has

⁽⁷³⁾N.D.E. The notation $\underline{\text{Der}}(G)$ of the original has been changed to avoid any risk of confusion with a space of derivations.

⁽⁷⁴⁾N.D.E. These reminders have been added, and the statement of 7.3 modified accordingly and expanded.

$\text{int}(h)(\Gamma_S) \subset \Gamma_S$ (this condition applied to h and h^{-1} implying that $\text{int}(h)(\Gamma_S) = \Gamma_S$); and, by (i)(a), this holds if and only if $\text{int}(h)(f_S(X_S)) \subset \Gamma_S$.

Similarly, H is contained in $C = \text{Centr}_G(\Gamma)$ if and only if, for every k -scheme S and $h \in H(S)$, one has $\text{int}(h) \circ i_S = i_S$, and, by (i)(c), this holds if and only if $\text{int}(h)(f_S) = f_S$. This proves (ii).

Let us prove (iii). In view of (ii), the hypothesis implies that ϕ factors through C (resp. N); since the latter is a closed subgroup of G , by 6.2, it therefore contains $\Gamma_G(f)$, by 7.1 (i).

Assertion (iv) follows from (iii), taking for f and ϕ the closed immersion $\text{Spec } k \hookrightarrow G$ defined by g : in this case, for every k -scheme S , $X(S)$ and $Y(S)$ have only one point, sent by f (resp. ϕ) to g .

Finally let us show (v). Let ν be the morphism $G \times G \rightarrow G$, $(g, h) \mapsto ghg^{-1}h^{-1}$ and let ν' be its restriction to $A \times B$; by definition (7.2.2), $(A, B) = \Gamma_G(\nu')$. Let S be a k -scheme, and u an inner automorphism (resp. an S -group automorphism) of G_S . One has $u \circ \nu_S = \nu_S \circ (u \times u)$. On the other hand, the hypothesis implies that u induces an automorphism u_1 of A_S (resp. u_2 of B_S), so

$$u(\nu'_S(A_S \times_S B_S)) = \nu'_S(u_1(A_S) \times_S u_2(B_S)) = \nu'_S(A_S \times_S B_S).$$

Thus, by (i)(b), u induces an automorphism of $(A, B)_S$. This proves (v). $\square$

Proposition 7.4. *Let G be a k -group locally of finite type, X a k -scheme of finite type, geometrically reduced and geometrically connected, and $f : X \rightarrow G$ a k -morphism such that $f(X)$ contains the identity element e of G . Then, with the notation of 7.1 (ii), there is an integer N such that one has (as sets):*

$$f^N(X^N) = \Gamma_G(f).$$

By 7.1 (iii) and EGA IV₂, 2.6.1, we may suppose k algebraically closed. By Corollary 7.2.1, we may suppose $G = G^0$; finally, it suffices to show that there is an integer N such that $f'^N(X'^N) = \Gamma_G(f)$, with the notation of 7.2 (v).

First case. Suppose X irreducible. Then the $\overline{f'^m(X'^m)}$ form an increasing sequence of irreducible closed subsets of the space G , which is noetherian, since $G = G^0$ is of finite type over k (VIA 2.4). Thus this sequence is stationary, and there is an integer m such that $\overline{f'^m(X'^m)} = \Gamma_G(f)$.

Moreover, since X and G are of finite type over k , the morphisms f'^m are of finite type over k . Consequently, $f'^m(X'^m)$ is constructible in G (EGA IV₁, 1.8.5), and thus contains an open subset U of its closure $\Gamma_G(f)$ (EGA 0_{IV}, 9.2.3). Then, by VIA 0.5, one has:

$$\Gamma_G(f) \subset U \cdot U \subset f'^{2m}(X'^{2m}) \subset \Gamma_G(f),$$

whence $f'^{2m}(X'^{2m}) = \Gamma_G(f)$.

Second case. Suppose that X has exactly two irreducible components A_1 and A_2 . Then, since X is connected and k algebraically closed, there is $a \in (A_1 \cap A_2)(k)$. Thus the four irreducible subsets $A_i \times_k A_j$ ($i, j = 1, 2$) cover $X \times_k X$, and the image of each of them under the morphism $f'^1 = \mu \circ (f \times_k (c \circ f))$ contains e . If f'_{ij}^1 denotes the restriction of f'^1 to the reduced subscheme $A_i \times_k A_j$, put

$$Y = (A_1 \times_k A_1) \times_k (A_1 \times_k A_2) \times_k (A_2 \times_k A_2) \times_k (A_2 \times_k A_1)$$

and

$$g = \mu \circ \left((\mu \circ (f'_{11}^1 \times_k f'_{12}^1)) \times_k (\mu \circ (f'_{22}^1 \times_k f'_{21}^1)) \right).$$

Then Y is irreducible, reduced and of finite type, so we have just seen that there is an integer m such that $\overline{g'^m(Y^m)} = \Gamma_G(g)$. Now, for every $n \geq 1$, one has $f'^n(X'^n) \subset g'^n(Y^n) \subset f'^{4n}(X'^{4n})$, whence $\Gamma_G(f) = \Gamma_G(g)$ and $f'^{4m}(X'^{4m}) = \Gamma_G(f)$.

General case. Let us argue by induction on the number of irreducible components of X (this number is finite since X , being of finite type over k , is noetherian). Suppose the proposition proved when X has at most $r - 1$ irreducible components, and suppose that

it has r , namely $A_1, \dots, A_r$. Then e belongs to the image of one of the A_i ; suppose, for example, that $e \in f(A_1)$. Put $Y = \Gamma_G(f_r)$, where f_r denotes the restriction of f to the reduced subscheme $X_r = A_1 \cup \dots \cup A_{r-1}$ (we suppose the numbering of the A_i chosen so that this scheme is connected, which is always possible). Then Y is a closed, reduced and irreducible subgroup of G , by Corollary 7.2.1.

Put $Z = \overline{f(A_r)}$, and let $T = Y \cup Z$, endowed with the reduced closed subscheme structure, and i the injection of T into G . It is clear (7.2 (ii)) that $\Gamma_G(i) = \Gamma_G(f)$ and that T is connected (for, since X is connected, $A_1 \cup \dots \cup A_{r-1}$ and A_r have a point a in common, and Y and Z have the point $f(a)$ in common). Moreover, $e \in T$, and T has at most two irreducible components, since Y and Z are irreducible. By the induction hypothesis, there is an integer m such that $\overline{f_r'^m(X_r'^m)} = \Gamma_G(f_r) = Y$. Since $X = X_r \cup A_r$ and since $Z = \overline{f(A_r)}$ is contained in $\overline{f'^m(X'^m)}$ (since $e \in f(A_1)$), one therefore has:

$$f(X) \subset Y \cup Z \subset \overline{f'^m(X'^m)}.$$

On the other hand, since T has at most two irreducible components, we have already seen that there is an integer p such that $i^p(T'^p) = \Gamma_G(i) = \Gamma_G(f)$. Now $T \subset \overline{f'^m(X'^m)}$, so $\overline{f'^{mp}(X'^{mp})} = \Gamma_G(f)$, and one shows, as in the first case, that this last equality implies $\overline{f'^{2mp}(X'^{2mp})} = \Gamma_G(f)$.

Lemma 7.5. *Let S be a scheme, G a group S -scheme, X an S -prescheme, $f : X \rightarrow G$ an S -morphism. Suppose that X and G are locally of finite presentation over S , and that for every $s \in S$ and every maximal point g of G_s , there is a point x of X such that $f(x) = g$ and f is flat at x . Then the morphism $\mu \circ (f \times_S f) : X \times_S X \rightarrow G$ is covering for the (fppf) topology.*

⁽⁷⁵⁾ By EGA IV₃, 11.3.1, the set V of points of X at which f is flat is open and $f|_V$ is an open morphism, so $U = f(V)$ is an open subset of G ; moreover, by hypothesis, $U \cap G_s$ is dense in G_s , for every $s \in S$. Denote by ϕ the restriction of $\mu \circ (f \times_S f)$ to $V \times_S V$.

It suffices to show that ϕ is covering for the (fppf) topology, and for this it suffices to show that ϕ is faithfully flat and of finite presentation (cf. IV, 6.3.1 (iv)). Now ϕ is the composite $V \times_S V \xrightarrow{f|_V \times f|_V} U \times_S U \xrightarrow{\mu} G$, where the first morphism is faithfully flat and locally of finite presentation, since $f|_V$ is. It will therefore suffice to prove that the same holds for the restriction of μ to $U \times_S U$.

Now, since $G \rightarrow S$ is locally of finite presentation and flat, the same holds for $\mu : G \times_S G \rightarrow G$ (which is isomorphic to the morphism obtained from $G \rightarrow S$ by the base change $G \rightarrow S$, cf. VIA 0.1), and thus also for the induced morphism $U \times_S U \rightarrow G$. To prove that the latter is surjective, it suffices to look fiber by fiber, where this follows from VIA 0.5, since $U \cap G_s$ is a dense open subset of G_s , for every $s \in S$.

Remark 7.6.0. ⁽⁷⁶⁾ Let S be a scheme, H an S -group, and $f : X \rightarrow H$ a morphism of S -schemes. The *subpresheaf of groups of H generated by the image of f* , denoted by $\langle \text{Im } f \rangle$, is the subgroup k -functor of H which associates to every S -scheme S' the subgroup of $H(S')$ generated by $f(X(S'))$. Since every element of this subgroup can be written as a finite product $f(x_1)^{\varepsilon_1} \cdots f(x_n)^{\varepsilon_n}$, with $n \in \mathbf{N}^*$, $x_i \in X(S')$ and $\varepsilon_i = \pm 1$, one therefore sees that, if one puts $X^1 = X \amalg X$ and defines the morphisms $f^n : X^n \rightarrow H$ as in 7.1, then $\langle \text{Im } f \rangle$ is precisely the *image presheaf* of the morphism

$$f^\infty : X^\infty \rightarrow H,$$

where $X^\infty = \coprod_{n \geq 1} X^n$ and $f^\infty : X^\infty \rightarrow H$ is the S -morphism whose restriction to each X^n is f^n .

⁽⁷⁵⁾**N.D.E.** The original has been expanded in what follows.

⁽⁷⁶⁾**N.D.E.** This definition has been added; in the original it was contained in the statement of Proposition 7.6.

Proposition 7.6. ⁽⁷⁷⁾ *Let S be a scheme, X an S -scheme flat and locally of finite presentation over S , H an S -group locally of finite presentation over S with reduced fibers, let $f : X \rightarrow H$ be a morphism of S -schemes, and let $f^\infty : X^\infty \rightarrow H$ be the S -morphism introduced above. The following conditions are equivalent:*

- (i) H represents the (fppf) S -sheaf associated to the presheaf $\langle \text{Im } f \rangle$.
- (ii) f^∞ is covering for the (fppf) topology.
- (iii) f^∞ is surjective, i.e. $H = \bigcup_{n \geq 1} f^n(X^n)$.

If moreover H is quasi-compact, these conditions are also equivalent to the following:

- (iv) *There is an integer n such that $f^n : X^n \rightarrow H$ is covering for the (fppf) topology (and a fortiori surjective).*

This applies in particular when X is a k -scheme locally of finite type and geometrically reduced, G a k -group locally of finite type, $\phi : X \rightarrow G$ a morphism of k -schemes, $H = \Gamma_G(\phi)$ and $f : X \rightarrow H$ is the morphism induced by ϕ .

PROOF. The sheaf considered in (i) is the image sheaf of X^∞ under f^∞ , so, by IV 4.4.3, saying that H represents this sheaf is equivalent to saying that f^∞ is covering, and this implies that f^∞ is surjective. Conversely, suppose f^∞ surjective and let us show that it is then covering for the (fppf) topology.

Let $s \in S$, η be a maximal point of the fiber H_s , and $x \in X_s^\infty$ such that $f^\infty(x) = \eta$ (such an x exists, since f^∞ is surjective). Since the fiber H_s is reduced, $\mathcal{O}_{H_s, \eta}$ is a field, so f_s^∞ is flat at the point x . Since X^∞ and H are locally of finite presentation over S , and X^∞ flat over S , the fiberwise criterion for flatness (EGA IV₃, 11.3.10) implies that f^∞ is flat at the point x . Thus, by Lemma 7.5, the morphism

$$\mu \circ (f^\infty \times_S f^\infty) : X^\infty \times_S X^\infty \rightarrow H$$

is covering for the (fppf) topology. Now, since $X^n \times_S X^m$ is canonically isomorphic to X^{n+m} , and in this isomorphism $\mu \circ (f^n \times_S f^m)$ corresponds to f^{n+m} , it is clear that $\mu \circ (f^\infty \times_S f^\infty)$ factors through f^∞ , so that f^∞ is covering for the (fppf) topology. This proves that (iii) $\Rightarrow$ (ii), whence the equivalence of conditions (i), (ii) and (iii).

Note moreover that, since the morphisms $X \rightarrow S$ and $H \rightarrow S$ are locally of finite presentation, the same holds for $f : X \rightarrow H$ (cf. EGA IV₁, 1.4.3 (v)), and since $\mu : H \times_S H$ is also of finite presentation (cf. VIA, 0.1), it follows that each $f^n : X^n \rightarrow H$ is too. Thus, by EGA IV₁, 1.9.5 (viii), the $f^n(X^n)$ are ind-constructible subsets of H .

Suppose moreover that H is quasi-compact (then S is too, since $H \rightarrow S$ is surjective). Then, by EGA IV₁, 1.9.9, one concludes that there is p such that $f^p(X^p) = H$. As above, one then deduces from the reducedness of the fibers of H and Lemma 7.5 that the morphism $\mu \circ (f^p \times_S f^p) : X^p \times_S X^p \rightarrow H$ is covering for the (fppf) topology; since this morphism equals $f^{2p} : X^{2p} \rightarrow H$, this finishes the proof of 7.6. $\square$

Remark 7.6.1. Obviously the equivalent conditions of 7.6 imply that the sheaf F under consideration is representable. The converse is false in general:⁽⁷⁸⁾ for example, if k has characteristic 0, let $G = \mathbf{G}_{a,k}$ and let $f : \text{Spec } k \rightarrow G$ be the morphism given by the point 1 of $G(k)$, then F is represented by the constant k -group $\mathbf{Z}_k$, whereas $\Gamma_G(f) = G$, so the monomorphism $\mathbf{Z}_k \hookrightarrow G$ is not surjective.

For simplicity, let us place ourselves under the hypotheses of the special case of 7.6, and suppose F representable. Then EGA IV₃, 8.14.2 implies that F is locally of finite presentation over k , so the question is then whether the dominant monomorphism $F \rightarrow \Gamma_G(f)$ is an isomorphism, or again, a closed immersion. This will be the case, by 1.4.2, if F is quasi-compact, and, by VIA, 0.5.1, this will hold if F is connected, hence, in particular (7.2.1), if X is connected and $f(X)$ contains the identity element of G .

⁽⁷⁷⁾**N.D.E.** The original stated this result under the hypotheses of the special case, but the more general form was used implicitly in the proof of 10.12; the statement has been rewritten accordingly.

⁽⁷⁸⁾**N.D.E.** The original has been expanded in what follows.

Lemma 7.7. *Let k be an algebraically closed field, G a k -group locally of finite type, X a geometrically reduced k -scheme locally of finite type, $f : X \rightarrow G$ a k -morphism and H a subgroup scheme of G such that $H \subset f(X)$. Put*

$$\Gamma' = \bigcup_{n \geq 1} f^n(X^n), \quad \Gamma'_0 = \Gamma' \cap G(k), \quad H_0 = H(k)$$

and suppose H_0 of finite index in Γ'_0 .

Then there is an integer m such that $f^m(X^m) = \Gamma_G(f)$ (cf. 7.6), and $\Gamma_G(f)$ is the union of finitely many translates of H .

For every $n \geq 1$, $f^n(X^n)$ is an ind-constructible subset of G (EGA IV₁, 1.9.5 (viii)), and thus so is Γ' , so that, since G is a Jacobson scheme, Γ'_0 is dense in Γ' . By hypothesis there is a finite sequence $a_1, \dots, a_r$ of points of Γ'_0 such that $\Gamma'_0 = a_1 H_0 \cup \dots \cup a_r H_0$, whence

$$\begin{aligned} \Gamma_G(f) &= \overline{\Gamma'} = \overline{\Gamma'_0} = \overline{a_1 H_0 \cup \dots \cup a_r H_0} \\ &= \overline{a_1 H_0} \cup \dots \cup \overline{a_r H_0} = a_1 \overline{H_0} \cup \dots \cup a_r \overline{H_0}, \end{aligned}$$

the last equality following from the fact that translation by a_i is a homeomorphism of G onto G . One therefore has $\Gamma_G(f) = a_1 H \cup \dots \cup a_r H$. It is also clear that there is an integer p such that each of the a_i ($1 \leq i \leq r$) belongs to $f^p(X^p)$. Finally, since $H \subset f(X)$, one has, for every i , $a_i H \subset f^{p+1}(X^{p+1})$, so that $f^{p+1}(X^{p+1}) = \Gamma_G(f)$.

Proposition 7.8. *Let G be a k -group locally of finite type, A and B two geometrically reduced subgroup k -schemes (thus smooth at the generic points of their irreducible components, hence smooth by 1.3) of G . Suppose one of the following conditions a) or b) holds:*

a) *A and B are invariant and of finite type over k .*

b) *A is connected and B is of finite type over k .*

Then (A, B) is of finite type over k , and represents the sheaf associated for the (fppf) (or (fpqc)) topology to the presheaf of groups of commutators of A and B in G . Moreover,⁽⁷⁹⁾ the k -groups (A, B^0) and (A^0, B) are connected, and one has

$$(A, B)^0 = (A, B^0) \cdot (A^0, B).$$

By 7.6, to show that (A, B) is the desired associated sheaf, it suffices to show that there is an integer n such that $\nu^n((A \times_k B)^n) = (A, B)$ (notation of 7.2.2). To show this, as well as to show the other two assertions, one may suppose k algebraically closed. Indeed, let $\bar{k}$ be an algebraic closure of k . By 7.1 (iii) and VIA 2.4, one has, with obvious notation:

$$(A, B)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}}), \quad ((A, B)^0)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}})^0, \quad (A, B^0)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}}^0), \quad \text{etc.}$$

Consequently, if one shows that $(A_{\bar{k}}, B_{\bar{k}})$ is of finite type over $\bar{k}$ (resp. that $(A_{\bar{k}}, B_{\bar{k}}^0)$ and $(A_{\bar{k}}^0, B_{\bar{k}})$ are connected, and that the morphism $(A_{\bar{k}}, B_{\bar{k}}^0) \times_{\bar{k}} (A_{\bar{k}}^0, B_{\bar{k}}) \rightarrow (A_{\bar{k}}, B_{\bar{k}})^0$ is surjective), then the analogous assertions will hold over k , by EGA IV₂, 2.7.1 and 2.6.1.

Let $B^1, \dots, B^p$ then be the connected components of B other than the identity component B^0 (there are finitely many of these since B is supposed of finite type over k , hence noetherian), and in case (a), similarly let $A^1, \dots, A^q$ be those of A . (In case (b), one will consider only $A^0 = A$.) Let ν_{ij} be the restriction of ν to $A^i \times_k B^j$. Then each A^i and each B^j is irreducible (VIA 2.4.1), and thus so are $A^0 \times_k B^j$ and $A^i \times_k B^0$. Since the identity element of G belongs to A^0 and to B^0 , it belongs to $\nu_{0j}(A^0 \times_k B^j)$ and to $\nu_{i0}(A^i \times_k B^0)$. Then each $\Gamma_G(\nu_{0j})$ and $\Gamma_G(\nu_{i0})$ is connected (7.2.1). Similarly, if u_{0j} (resp. u_{i0}) denotes the injection of $\Gamma_G(\nu_{0j})$ (resp. $\Gamma_G(\nu_{i0})$) into G , then

$$(A^0, B) = \Gamma_G((u_{0j})_{j=0}^p) \quad \text{and} \quad (A, B^0) = \Gamma_G((u_{i0})_{i=0}^q)$$

⁽⁷⁹⁾**N.D.E.** What follows has been made more precise and, in the proof, the reduction to the case where k is algebraically closed has been expanded.

are connected. Moreover, one readily deduces from 7.4 and the preceding constructions that there is an index r such that (A^0, B) and (A, B^0) are included in $\nu^r((A \times_k B)^r)$. In case b), one has $(A, B) = (A^0, B)$, and we are done.

Let us now consider case (a).⁽⁸⁰⁾ We already know that (A^0, B) and (A, B^0) are smooth connected subgroup k -schemes of G , hence of finite type (cf. VIA, 2.4). On the other hand, since A^0 is a characteristic subgroup of A (cf. VIA, 2.6.5), it is an invariant subgroup of G and hence, by 7.3 (v), (A^0, B) is an invariant subgroup of G , and similarly for (A, B^0) . Thus, by 7.1.1, the subgroup H of G generated by (A, B^0) and (A^0, B) is precisely $(A, B^0) \cdot (A^0, B)$. In particular, one therefore has $H \subset \nu^{2r}((A \times_k B)^{2r})$.

Given a subset X of G stable under the group law (cf. 3.0), we shall denote by X_0 the group of k -points of G belonging to X . Put $\Gamma' = \bigcup_{q \geq 1} \nu^q((A \times_k B)^q)$. Then, by Proposition 7.9 below, one has:

$$(A^0, B)_0 = (A_0^0, B_0), \quad (A, B^0)_0 = (A_0, B_0^0) \quad \text{and} \quad \Gamma'_0 = (A_0, B_0),$$

so that $H_0 = (A_0^0, B_0) \cdot (A_0, B_0^0)$ has finite index in Γ'_0 (Bible, Exp. 3, Appendix), since A_0 and B_0 are invariant, and A_0^0 (resp. B_0^0) has finite index in A_0 (resp. B_0). We are then under the conditions of Lemma 7.7: since $H \subset \nu^{2r}((A \times_k B)^{2r})$, there is an integer m such that $\nu^{2rm}((A \times_k B)^{2rm}) = (A, B)$, and there is a finite sequence $a_1, \dots, a_n$ of k -points of G such that $(A, B) = a_1 H \cup \dots \cup a_n H$. Then, since H is of finite type over k , each $a_i H$ is quasi-compact, so their union (A, B) is quasi-compact, hence of finite type over k . Since H is irreducible, so is each $a_i H$, and since $e \in H$, it is clear that $H = (A, B)^0$.

Proposition 7.9. *Let k be an algebraically closed field, G a k -group locally of finite type.*

(i) *Let A and B be two ind-constructible subsets of G . Denote by A_0 the set of rational points of G belonging to A . Then $(A \cdot B)_0 = A_0 \cdot B_0$, the second product being taken in the group $G(k)$.*

(ii) *Let X be a geometrically reduced k -scheme locally of finite type, and $f : X \rightarrow G$ a k -morphism. Put $\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n)$. Then $\Gamma'_G(f)_0$ is the subgroup of $G(k)$ generated by $f(X)_0$.*

(iii) *In particular, let A and B be two smooth subgroup schemes of G ; put $\Gamma' = \bigcup_{n \geq 1} \nu^n(A \times_k B)^n$ (notation of 7.2.2). Then Γ'_0 is the commutator group of $A(k)$ and $B(k)$ in $G(k)$.*

Let us prove (i). It is clear that $A_0 \cdot B_0 \subset (A \cdot B)_0$. Conversely, let $z \in (A \cdot B)_0$. Then $\mu^{-1}(z)$ is a closed subset of $G \times_k G$, and $A \times_k B$ (cf. 3.0) is an ind-constructible subset of $G \times_k G$, so that $\mu^{-1}(z) \cap (A \times_k B)$ is a nonempty ind-constructible subset of $G \times_k G$; by EGA IV₃, 10.4.8, it therefore contains a rational point of $G \times_k G$, whose projections x and y are rational points of G such that $x \in A$, $y \in B$ and $x \cdot y = z$, so that $(A, B)_0 = A_0 \cdot B_0$.

To show (ii), note that, since f^n is locally of finite type, $f^n(X^n)$ is an ind-constructible subset of G (EGA IV₄, 1.9.5 (viii)). Assertion (i) then allows one to show by induction that, putting $A = f(X)_0$, one has $f^n(X^n)_0 = (A \cup A^{-1})^n$, and consequently,

$$\Gamma'_G(f)_0 = \bigcup_{n \geq 1} f^n(X^n)_0 = \bigcup_{n \geq 1} (A \cup A^{-1})^n,$$

which is the subgroup of $G(k)$ generated by $A = f(X)_0$.

Finally, (iii) follows from (ii) and the definitions.

Corollary 7.10. *Under the conditions of 7.8, if k is algebraically closed, then $(A, B)(k)$ is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.*

Indeed, it suffices to apply 7.9 (iii), 7.8 and 7.6.

8. Solvable or nilpotent group schemes

⁽⁸⁰⁾**N.D.E.** The original has been expanded in what follows, taking account of the addition 7.1.1.

8.1. — Let $\mathcal{C}$ be a category endowed with a topology T (cf. IV § 4). For every presheaf P on $\mathcal{C}$, one shall denote by P^b the associated sheaf.

Let G be a presheaf of groups on $\mathcal{C}$, A and B two subpresheaves of groups of G , and let $\underline{\text{Comm}}(A, B)$ be the presheaf of groups of commutators of A and B in G ; i.e., for every $S \in \text{Ob } \mathcal{C}$, $\underline{\text{Comm}}(A, B)(S)$ is the subgroup of $G(S)$ generated by the commutators $aba^{-1}b^{-1}$, with $a \in A(S)$ and $b \in B(S)$. One puts

$$\underline{\text{Comm}}_T(A, B) = \underline{\text{Comm}}(A, B)^b.$$

We shall need, in the proof of 8.2, the following results.⁽⁸¹⁾

Lemma 8.1.1. *Let $A \subset G$ be sheaves of groups, with A invariant in G .*

(i) *$\underline{\text{Comm}}_T(G, A)$ is the smallest invariant subsheaf of groups C of G such that the sheaf $(A/C)^b$, associated to the quotient presheaf A/C , is central in $(G/C)^b$.*

(ii) *In particular, $\underline{\text{Comm}}_T(G, G)$ is the smallest invariant subsheaf of groups C of G such that the quotient sheaf $(G/C)^b$ is commutative.*

Obviously, (ii) is the special case $A = G$ of (i), so it suffices to show (i). Let C be a subsheaf of groups of A , invariant in G , and such that the quotient sheaf $(A/C)^b$ is central in $(G/C)^b$. By Lemma IV 4.4.8.1, the presheaves A/C and G/C are separated, and thus, by IV 4.3.11, all the morphisms in the diagram below are monomorphisms:

$$\begin{array}{ccc} A/C & \hookrightarrow & G/C \\ \downarrow & & \downarrow \\ (A/C)^b & \hookrightarrow & (G/C)^b. \end{array}$$

Since $(A/C)^b$ is central in $(G/C)^b$, then A/C is central in G/C , whence $\underline{\text{Comm}}(G, A) \subset C$, and therefore C contains $\underline{\text{Comm}}_T(G, A)$, by IV 4.3.12.

Conversely, $\underline{\text{Comm}}(G, A)$ is a subpresheaf of groups of A , invariant in G , and separated (cf. IV 4.3.1, N.D.E. (24)), so, by IV 4.4.8.2 (i) and IV 4.3.11, $C = \underline{\text{Comm}}_T(G, A)$ is a subsheaf of groups of A , invariant in G and containing $\underline{\text{Comm}}(G, A)$. Consequently, the presheaf A/C is central in G/C and thus, by IV 4.4.8.2 (ii), $(A/C)^b$ is central in $(G/C)^b$. This proves Lemma 8.1.1.

Lemma 8.1.2. *Let G be a sheaf of groups, A, B two subpresheaves of groups of G .*

(i) *The morphism $\tau : \underline{\text{Comm}}(A, B) \rightarrow \underline{\text{Comm}}(A^b, B^b)$ is a covering monomorphism.*

(ii) *Consequently, one has an isomorphism*

$$\underline{\text{Comm}}_T(A, B) \xrightarrow{\sim} \underline{\text{Comm}}_T(A^b, B^b).$$

PROOF. (i) Since A (resp. B) is a subpresheaf of G , then A^b (resp. B^b) is a subpresheaf of G containing A (resp. B), and it follows that τ is a monomorphism.

Let us show that τ is covering. Let $S \in \text{Ob } \mathcal{C}$ and $g \in \underline{\text{Comm}}(A^b, B^b)(S)$. Then there is an integer $n \geq 1$ and, for $i = 1, \dots, n$, elements $a'_i \in A^b(S)$, $b'_i \in B^b(S)$, and $\varepsilon_i \in \{\pm 1\}$, such that

$$g = (a'_1, b'_1)^{\varepsilon_1} \cdots (a'_n, b'_n)^{\varepsilon_n},$$

where (a, b) denotes the commutator $aba^{-1}b^{-1}$, and there is a refinement R of S such that $a'_i \in A(R)$ and $b'_i \in B(R)$ for every i . Then g is the composite morphism

$$R \xrightarrow{(a'_1, \dots, b'_n)} (A \times B)^n \xrightarrow{\Phi^{\varepsilon_1, \dots, \varepsilon_n}} \underline{\text{Comm}}(A, B),$$

where $\Phi = \Phi^{\varepsilon_1, \dots, \varepsilon_n}$ is the morphism defined set theoretically by:

$$\Phi(a_1, b_1, \dots, a_n, b_n) = (a_1, b_1)^{\varepsilon_1} \cdots (a_n, b_n)^{\varepsilon_n},$$

⁽⁸¹⁾N.D.E. These results are mentioned without proof in the original; they have been brought out in the form of Lemmas 8.1.1 and 8.1.2, and the proofs have been expanded.

for every $T \in \text{Ob } \mathcal{C}$ and $a_i \in A(T)$, $b_i \in B(T)$. This shows that $g \in \underline{\text{Comm}}(A, B)(R)$ and it follows, as in the proof of IV 4.3.11 (i), that $\underline{\text{Comm}}(A, B) \rightarrow \underline{\text{Comm}}(A^b, B^b)$ is covering.

Since $\underline{\text{Comm}}(A^b, B^b) \rightarrow \underline{\text{Comm}}(A^b, B^b)^b$ is also a covering monomorphism (IV 4.3.11 (iv)), the same holds for $\underline{\text{Comm}}(A, B) \rightarrow \underline{\text{Comm}}(A^b, B^b)^b$ and thus, by IV 4.3.12, one obtains an isomorphism:

$$\underline{\text{Comm}}(A, B)^b \xrightarrow{\sim} \underline{\text{Comm}}(A^b, B^b)^b.$$

This proves Lemma 8.1.2. □

Proposition 8.2. *Let $\mathcal{C}$ be a category, T a topology on $\mathcal{C}$, G a sheaf of groups on $\mathcal{C}$, n an integer ≥ 0 . The following conditions are equivalent:*

- (i) *If one puts $K_0 = G$, and for $p \geq 1$, $K_p = \underline{\text{Comm}}(K_{p-1}, K_{p-1})$ (resp. $K_p = \underline{\text{Comm}}(G, K_{p-1})$), then K_n is the identity presheaf of groups.*
- (ii) *If one puts $K'_0 = G$, and for $p \geq 1$, $K'_p = \underline{\text{Comm}}_T(K'_{p-1}, K'_{p-1})$ (resp. $K'_p = \underline{\text{Comm}}_T(G, K'_{p-1})$), then K'_n is the identity sheaf of groups.*
- (iii) *There is a sequence $G = H_0 \supset H_1 \supset \cdots \supset H_n$ of invariant subsheaves of G , such that, for every i , the quotient sheaf H_i/H_{i+1} is commutative (resp. central in G/H_{i+1}), and H_n is the identity sheaf.*

It is clear that $K_n \subset K'_n$; consequently (ii) implies (i). Let us show that (i) implies (ii). One has $K'_1 = \underline{\text{Comm}}_T(G, G) = K_1^b$, and one deduces by induction from Lemma 8.1.2 that $K'_p = K_p^b$ for every p . Consequently, if K_n is the identity presheaf, then $K'_n = K_n^b$ is the identity sheaf.

Finally, conditions (ii) and (iii) are equivalent by Lemma 8.1.1.

Definition 8.2.1. When these conditions hold, the sheaf G is said to be *solvable of class n* (resp. *nilpotent of class n*). When there is an integer n for which these conditions hold, one says that G is *solvable* (resp. *nilpotent*).

Note that, by condition (i), this does not depend on the topology T .

Proposition 8.3. *Let k be a field, S a nonempty k -scheme, Ω an algebraically closed extension of k , G a smooth k -group of finite type. The following conditions are equivalent:*

- (i) *G is solvable of class n (resp. nilpotent of class n).*
- (ii) *$G \times_k S$ is solvable of class n (resp. nilpotent of class n).*
- (iii) *The group $G(\Omega)$ is solvable of class n (resp. nilpotent of class n).*
- (iv) *If one puts $K_0 = G$ and considers, for $p \geq 1$, the k -groups $K_p = (K_{p-1}, K_{p-1})$ (resp. $K_p = (G, K_{p-1})$) (cf. 7.2.2), then K_n is the identity k -group.*

The equivalence of conditions (i) and (ii) follows from Proposition 8.2, since the formation of the presheaf of groups of commutators commutes with base changes (IV 4.1.3).

The equivalence of (i) and (iv) follows from the fact that, by 7.8, the k -group $K_p = (K_{p-1}, K_{p-1})$ (resp. $K_p = (G, K_{p-1})$) represents the sheaf $\underline{\text{Comm}}_T(K_{p-1}, K_{p-1})$ (resp. $\underline{\text{Comm}}_T(G, K_{p-1})$), where T is the (fppf) (or (fpqc)) topology.

To show that conditions (iii) and (iv) are equivalent, one may suppose that $k = \Omega$, and then the equivalence of conditions (iii) and (iv) follows from 7.10.

Proposition 8.4. *Let G be an S -group of finite presentation such that, for every $s \in S$, G_s is smooth over $\kappa(s)$. Let T be the set of $s \in S$ such that G_s is solvable (resp. nilpotent).*

- (i) *Then T is locally constructible in S .*
- (ii) *If one supposes moreover G flat and separated over S (i.e. when G is smooth, quasi-compact and separated over S), then T is closed in S .*

It is clear that one may suppose S affine with ring A . Then, by 10.1 and 10.10 b),^{*} there is a noetherian subring A' of A and an A' -group of finite type G' such that $G' \otimes_{A'} A$ is isomorphic to G . By EGA IV₃, 11.2.6 and 8.10.5,⁽⁸²⁾ if G is flat and separated over S , one may suppose G' flat and separated over $S' = \operatorname{Spec} A'$.⁽⁸³⁾ Since G is of finite presentation over S , then (EGA IV₃, 9.7.7) the set of $s \in S$ such that G_s is geometrically reduced (or, equivalently, smooth over $\kappa(s)$) is locally constructible. Thus, by EGA IV₃, 9.3.3, one may suppose that, for every $s' \in S'$, $G'_{s'}$ is smooth over $\kappa(s')$. On the other hand, if s' denotes the image of s in S' , one has $G'_{s'} \otimes_{\kappa(s')} \kappa(s) \simeq G_s$. Thus, by 8.3, for G_s to be solvable (resp. nilpotent), it is necessary and sufficient that $G'_{s'}$ be so. We are therefore reduced to the case where S is an affine noetherian scheme.

Let us then show that T is constructible. Applying the criterion (EGA 0_{III}, 9.2.3), one sees, arguing as above, that one is reduced to showing that, when S is noetherian and integral, T or $S - T$ contains a nonempty open subset of S .

Suppose therefore S integral and noetherian, with generic point η . Put, for every $s \in S$, $K_s^0 = G_s$, and $K_s^p = (K_s^{p-1}, K_s^{p-1})$ (resp. $K_s^p = (G, K_s^{p-1})$). Let us first show that the sequence of closed subschemes K_η^p is stationary. It follows from 7.3 (v) that each K_η^p is invariant, so the sequence of K_η^p is decreasing; this sequence is then stationary since G_η is noetherian; there is therefore an integer n such that, for every $p \geq n$, one has $K_\eta^p = K_\eta^n$.

On the other hand, by 10.12.1 and 10.13, there is a nonempty open subset S' of S and an S' -group of finite presentation D such that, for every $s \in S'$, one has $D_s = K_s^n$ and $(D_s, D_s) = D_s$ (resp. $(G_s, D_s) = D_s$). We may suppose that $S' = S$. Then, for every $s \in S$, and every $p \geq n$, one has $D_s = K_s^p$, so that G_s is solvable (resp. nilpotent) if and only if D_s is isomorphic to the identity $\kappa(s)$ -group.

But by EGA IV₃, 9.6.1 (xi), the set of $s \in S$ such that the structural morphism $D_s \rightarrow \operatorname{Spec} \kappa(s)$ is an isomorphism is constructible,⁽⁸⁴⁾ and thus is either nowhere dense or contains a nonempty open subset of S . We have therefore obtained that T is locally constructible.

Let us show that if, moreover, G is flat and separated over S , then T is closed. Since T is locally constructible, for T to be closed it is necessary and sufficient that T be stable under specialization (cf. EGA IV₁, 1.10.1).

Let therefore $s \in S$ and s' be a specialization of s in S . Since we have reduced to the case where S is noetherian, then, by EGA II, 7.1.9, there is a discrete valuation ring A and a morphism $\operatorname{Spec}(A) \rightarrow S$ such that s (resp. s') is the image of the generic point α (resp. the closed point a) of $\operatorname{Spec}(A)$. It then suffices to show that, putting $G' = G \otimes_S A$, if G'_α is solvable (resp. nilpotent), then G'_a is too. Note that, since G is flat and separated over S , G'_α is flat and separated over A , so that one is reduced to the case where S is the spectrum of a discrete valuation ring A .

Then, since G_α is supposed solvable (resp. nilpotent), there is an integer q such that K_α^q (with the notation introduced above) is isomorphic to the identity $\kappa(\alpha)$ -group. For every n , denote by $\overline{K}_\alpha^n$ the schematic closure (in the sense of EGA IV₂, 2.8.5) of K_α^n in G . Let us show, by induction on p , that $(\overline{K}_\alpha^p)_a \supset K_a^p$. Note first that, since G is flat over A , then $\overline{G}_\alpha$ is equal to G (EGA IV₂, 2.8.5), so $(\overline{K}_\alpha^0)_a = K_a^0$.

^{*}In the course of this proof we shall use results established in no. 10 which, like no. 9, do not depend on the present no. 8.

⁽⁸²⁾**N.D.E.** 10.8.5 has been corrected to 8.10.5.

⁽⁸³⁾**N.D.E.** What follows has been expanded.

⁽⁸⁴⁾**N.D.E.** Taking account of the fact that S is supposed affine, hence quasi-compact and quasi-separated (cf. EGA 0_{III}, 9.1.12).

Let $p \geq 0$. Suppose we have established that $K_a^p \subset (\overline{K_\alpha^p})_a$, and denote by $\nu_a, \nu_\alpha, \bar{\nu}, \bar{\nu}_a$ the following morphisms, defined as in 7.2.2:

solvable case	nilpotent case
$\nu_a : K_a^p \times_{\kappa(a)} K_a^p \longrightarrow G_a,$	resp. $G_a \times_{\kappa(a)} K_a^p \longrightarrow G_a,$
$\nu_\alpha : K_\alpha^p \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha,$	resp. $G_\alpha \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha,$
$\bar{\nu} : \overline{K_\alpha^p} \times_A \overline{K_\alpha^p} \longrightarrow G,$	resp. $G \times_A \overline{K_\alpha^p} \longrightarrow G,$
$\bar{\nu}_a : (\overline{K_\alpha^p})_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a \longrightarrow G_a,$	resp. $G_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a \longrightarrow G_a.$

Since ν_α factors through K_α^{p+1} , then $\bar{\nu}$ factors through $\overline{K_\alpha^{p+1}}$, which is obviously a subgroup scheme of G , so $\overline{K_\alpha^{p+1}}$ contains $\Gamma_G(\bar{\nu})$. By 7.1 (iii), one has $\Gamma_G(\bar{\nu})_a = \Gamma_{G_a}(\bar{\nu}_a)$; and, by the induction hypothesis, one has:

$$K_a^p \times_{\kappa(a)} K_a^p \subset (\overline{K_\alpha^p})_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a, \quad \text{resp.} \quad G_a \times_{\kappa(a)} K_a^p \subset G_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a,$$

so that $K_a^{p+1} = \Gamma_{G_a}(\nu_a) \subset \Gamma_{G_a}(\bar{\nu}_a) = \Gamma_G(\bar{\nu})_a \subset (\overline{K_\alpha^{p+1}})_a$.

But since K_α^q is isomorphic to the identity $\kappa(\alpha)$ -group, and the identity A -group is flat over A and is isomorphic to a closed subscheme of G (since G is separated over A , cf. 5.1), EGA IV₂, 2.8.5 implies that the schematic closure $\overline{K_\alpha^q}$ is isomorphic to the identity A -group. Since we have just seen that $K_a^q \subset (\overline{K_\alpha^q})_a$, this implies that K_a^q is isomorphic to the identity $\kappa(a)$ -group, so that G_a is solvable (resp. nilpotent).

9. Quotient sheaves

This section is essentially limited to recalling, in the special case of homogeneous spaces of groups, well-known general facts on taking quotients by flat equivalence relations (cf. Exp. IV).

Definition 9.1. Given a monomorphism $u : G' \longrightarrow G$ of S -groups, one denotes by G/G' (resp. $G' \backslash G$) and calls the *right* (resp. *left*) *quotient sheaf of G by G'* the sheaf (for the (fpqc) topology) quotient of G by the equivalence relation defined by the monomorphism:

$$G \times_S G' \xrightarrow{\delta \circ (\text{id}_G \times u)} G \times_S G \quad (\text{resp.} \quad G' \times_S G \xrightarrow{\gamma \circ (u \times \text{id}_G)} G \times_S G),$$

where δ (resp. γ) denotes the automorphism of $G \times_S G$ defined by $(g, h) \longmapsto (g, gh)$ (resp. $(h, g) \longmapsto (hg, g)$) for $g, h \in G(T)$.

Proposition 9.2. Let $u : G' \longrightarrow G$ be a monomorphism of S -groups. Suppose that G/G' is representable by an S -scheme G'' . Then:

- (i) The canonical morphism $p : G \longrightarrow G''$ is covering for the (fpqc) topology.
- (ii) If one puts $\varepsilon'' = p \circ \varepsilon$ (this morphism is called the *identity section of G''*), the following diagrams are cartesian:

$$\begin{array}{ccc} G \times_S G' & \xrightarrow{\mu \circ (\text{id}_G \times u)} & G \\ \text{pr}_1 \downarrow & & \downarrow p \\ G & \xrightarrow{p} & G'' \end{array}, \quad \begin{array}{ccc} G' & \xrightarrow{u} & G \\ \pi' \downarrow & & \downarrow p \\ S & \xrightarrow{\varepsilon''} & G'' \end{array}.$$

In particular, u is an immersion.

(iii) There is on G'' a unique structure of S -scheme with group of operators G on the left such that p is a morphism of S -schemes with group of operators G .

(iv) If one supposes moreover that G' is invariant in G , there is on G'' a unique structure of S -group such that p is a morphism of S -groups.

(v) Let S_0 be an S -scheme; put $G_0 = G \times_S S_0$, and $G'_0 = G' \times_S S_0$; then G_0/G'_0 is representable by $G''_0 = G'' \times_S S_0$.⁽⁸⁵⁾

(vi) Let $\mathcal{P}$ be a property of an S -morphism. Suppose $\mathcal{P}$ stable under base change; then, if $p : G \rightarrow G''$ satisfies $\mathcal{P}$, the structural morphism $\pi' : G' \rightarrow S$ does too.

(vii) Let $\mathcal{P}$ be a property of an S -morphism. Suppose $\mathcal{P}$ local for the (fpqc) topology (cf. 2.0 and 2.1.2). Then, for the morphism $p : G \rightarrow G''$ to satisfy $\mathcal{P}$, it is necessary and sufficient that $\pi' : G' \rightarrow S$ do so.

(viii) Let $\mathcal{P}$ be a property of an S -morphism; suppose $\mathcal{P}$ local for the (fpqc) topology and stable under composition; then, if the structural morphisms $G'' \rightarrow S$ and $G' \rightarrow S$ satisfy $\mathcal{P}$, the structural morphism $G \rightarrow S$ does too.

(ix) Suppose G reduced; then G'' is reduced.

(x) For G'' to be separated over S , it is necessary and sufficient that u (or, equivalently, ε'') be a closed immersion.

(xi) For G' to be flat over S , it is necessary and sufficient that p be a flat morphism (or, equivalently, faithfully flat).

In this case, for G'' to be flat over S , it is necessary and sufficient that G be flat over S .

(xii) For G' to be flat and locally of finite presentation over S , it is necessary and sufficient that $p : G \rightarrow G''$ be faithfully flat and locally of finite presentation.

In this case, for G'' to be locally of finite presentation (resp. locally of finite type, of finite type, smooth, étale, unramified, locally quasi-finite, quasi-finite) over S , it suffices that G be so over S (and the condition is also necessary in the first two cases, cf. (viii)).

(xiii) Suppose G' flat and of finite presentation over S .

a) Then p is of finite presentation and faithfully flat;

b) moreover, for G to be of finite presentation over S , it is necessary and sufficient that G'' be so.

Recall that the equivalence relation considered is universally effective (IV 4.4.9). Then assertions (i), (iii), (iv), (v) and the first assertion of (ii) follow from IV 4.4.3, 5.2.2, 5.2.4, 3.4.5 and 3.3.2 (iii). The second assertion of (ii) follows from the first, as shown by the following cartesian diagram, since $(G \times_S G') \times_G S$ is isomorphic to G' :

$$\begin{array}{ccccc} G' & \xrightarrow{((\varepsilon \circ \pi'), \text{id}_{G'})} & G' \times_S G & \xrightarrow{\mu_p(\text{id}_G \times u)} & G \\ \pi' \downarrow & & \text{pr}_1 \downarrow & & \downarrow p \\ S & \xrightarrow{\varepsilon} & G & \xrightarrow{p} & G'' \end{array}$$

Finally, it is clear that ε'' is an S -section of G'' , hence an immersion (EGA I, 5.3.13); by the preceding cartesian diagram, the same holds for u , which finishes the proof of (ii). Moreover, (vi) is an immediate consequence of the second cartesian diagram of (ii).

Let us show (vii). By (i), p is covering for the (fpqc) topology; thus, by (ii), to show that p satisfies $\mathcal{P}$, it suffices to show that the first projection $\text{pr}_1 : G \times_S G' \rightarrow G$ satisfies $\mathcal{P}$, which follows from the fact that $\mathcal{P}$ is stable under base change, since pr_1 comes from π' by base change. It is clear that (viii) follows from (vii), for $\pi = \pi'' \circ p$, where $\pi'' : G'' \rightarrow S$ denotes the structural morphism.

Let us show (ix). By (i), p is an epimorphism; since G is reduced, p factors through the immersion $G''_{\text{red}} \rightarrow G''$, which is therefore also an epimorphism, hence an isomorphism (IV 4.4.4).

Let us show (x). If G'' is separated over S , then ε'' is a closed immersion, by EGA I, 5.4.6. On the other hand, we saw in (ii) that ε'' is a closed immersion if and only if u is one. Finally, if u is a closed immersion, then so is $\delta \circ (\text{id}_G \times u) : G \times_S G' \rightarrow G \times_S G$; thus, by Lemma 9.2.1 below, G'' is separated over S .

Assertion (xi) follows from (vii) and EGA IV₂, 2.2.13.

⁽⁸⁵⁾**N.D.E.** I.e., the quotient is universal, cf. Exp. IV § 3.

Assertion (xii) follows from (vii), EGA IV₄, 17.7.5 and 17.7.7, and the fact that, since p is universally open, for every $s \in S$, if the underlying space of G_s is discrete, the same holds for the underlying space of G'_s .

Finally, assertion (xiii) follows from (vii), (viii), and EGA IV₄, 17.7.5.

Lemma 9.2.1. *Let X be an S -scheme and R an equivalence relation defined on X by the monomorphism $v : R \rightarrow X \times_S X$. Suppose R effective. Then:*

- (i) *v is an immersion, and it is a closed immersion if $Y = X/R$ is separated.*
- (ii) *Suppose moreover that Y represents the (fpqc) sheaf quotient of X by R ⁽⁸⁶⁾ and that v is a closed immersion. Then $Y = X/R$ is separated over S .*

Recall (IV Def. 3.3.2) that the hypothesis “ R effective” means that there is a morphism of S -schemes $p : X \rightarrow Y$ such that the natural morphism $R \rightarrow X \times_Y X$ is an isomorphism. One deduces (EGA I, 5.3.5) the following cartesian diagram:

$$\begin{array}{ccc} R & \xrightarrow{v} & X \times_S X \\ \downarrow & & \downarrow p \times p \\ Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y. \end{array}$$

Then, since $\Delta_{Y/S}$ is an immersion (EGA I, 5.3.9), the same holds for v . Similarly, if Y is separated over S , $\Delta_{Y/S}$ is a closed immersion, so the same holds for v .

Conversely, suppose v a closed immersion and Y represents the (fpqc) sheaf quotient of X by R . Then p is covering for the (fpqc) topology (IV 4.4.3), and therefore $p \times p$ is too (by base change, $p \times \text{id}_X$ and $\text{id}_Y \times p$ are covering, and thus also their composite $p \times p$). Hence, by (fpqc) descent (cf. EGA IV₂, 2.7.1), $\Delta_{Y/S}$ is a closed immersion, i.e. Y is separated over S .

Remark 9.2.2. Under the general hypotheses of 9.2, if one supposes G' flat and locally of finite presentation over S , then p is covering for the (fppf) topology,⁽⁸⁷⁾ by 9.2 (vii), so assertions (vii) and (viii) of 9.2 can be extended to properties $\mathcal{P}$ local for the (fppf) topology.

Remark 9.3. a) The question whether a quotient G/G' is representable or not is often delicate; in this seminar we prove representability of certain particular quotients.

In general, to be able to assert that the quotient G/G' is representable, it does not suffice to suppose G and G' of finite presentation over S and G' flat over S . Indeed, suppose moreover G smooth with connected fibers. In this case, if G/G' is a scheme, it is separated, by Corollary 5.4, and therefore $G' \hookrightarrow G$ is a closed immersion, by 9.2 (x); consequently, if G' is not closed in G , then G/G' is not representable.

To obtain such a counterexample, one may take for S the spectrum of a discrete valuation ring, and put $G = \mathbf{G}_{m,S}$. Consider moreover an integer $n > 1$, invertible on S ; then $\mu_n = \text{Ker}(G \xrightarrow{n} G)$ is a closed subgroup of G étale over S (cf. VIIA⁽⁸⁸⁾). Let G' be the open subgroup of μ_n obtained by removing from μ_n the closed part of the closed fiber of μ_n complementary to the origin. Then G' is not closed in G , so G/G' is not representable. (One can also construct such examples where G' is smooth with connected fibers.)

⁽⁸⁶⁾**N.D.E.** As pointed out by O. Gabber, this is used in the proof and must be inserted among the hypotheses.

⁽⁸⁷⁾**N.D.E.** (fpqc) has been corrected to (fppf).

⁽⁸⁸⁾**N.D.E.** See VIIA, 8.4 or VIII, 2.1.

b) On the other hand, it is not excluded that G/G' is representable when G and G' are of finite presentation over S , and G' is flat over S and closed in G .^{*(89)} Under these hypotheses, one knows that G/G' is representable in the following special cases:

- 1° – S is the spectrum of an artinian ring (cf. VIA 3.2 and 3.3.2).
- 2° – G' is proper over S and G quasi-projective over S (cf. V 7.1).
- 3° – S is locally noetherian of dimension 1 (cf. [An73], Th. 4.C).

10. Passage to the projective limit in group schemes and schemes with a group of operators

10.0. — Recall the essential result of EGA IV₃, § 8.8. Suppose the following situation given: S_0 a quasi-compact and quasi-separated scheme, I an increasing filtered preordered set, $(\mathcal{A}_i)_{i \in I}$ an inductive system of quasi-coherent commutative $\mathcal{O}_{S_0}$ -algebras, $\mathcal{A} = \varinjlim \mathcal{A}_i$, $S_i = \text{Spec } \mathcal{A}_i$ for $i \in I$, and $S = \text{Spec } \mathcal{A}$;⁽⁹⁰⁾ then the category of S -schemes of finite presentation is determined up to equivalence by the categories of S_i -schemes of finite presentation, the functors between these categories $\rho_{ji} : X_i \mapsto X_i \times_{S_i} S_j$ for $i \leq j$, and the transitivity isomorphisms $\rho_{kj} \circ \rho_{ji} \xrightarrow{\sim} \rho_{ki}$.

Let us be more precise. Given $j \in I$, and an S_j -scheme of finite presentation X_j , we shall put, for every $i \in I$ such that $i \geq j$, $X_i = X_j \times_{S_j} S_i$, and $X = X_j \times_{S_j} S$. Then (EGA IV₃, 8.8.2):

(i) Given $j \in I$, and two S_j -schemes of finite presentation X_j and Y_j , the canonical map from $\varinjlim_{i \geq j} \text{Hom}_{S_i}(X_i, Y_i)$ to $\text{Hom}_S(X, Y)$ is bijective.

(ii) For every S -scheme of finite presentation X , there is an index $j \in I$, an S_j -scheme of finite presentation X_j and an S -isomorphism $X \xrightarrow{\sim} X_j \times_{S_j} S$.

One concludes (EGA IV₃, 8.8.3) that, whenever one has a diagram D involving finitely many objects and arrows of the category of S -schemes of finite presentation, one can find an index $i \in I$ and a diagram D_i in the category of S_i -schemes of finite presentation, such that the diagram D comes, up to isomorphism, from the diagram D_i by base change $S \rightarrow S_i$. One can even find i and D_i such that every cartesian square of D comes from a cartesian square of D_i .

10.1. — Moreover, many common properties of a morphism, stable under base change, possess the following property:

Let $u : X \rightarrow Y$ be an S -morphism between S -schemes of finite presentation; by 10.0 it comes by base change from an S_j -morphism $u_j : X_j \rightarrow Y_j$ between S_j -schemes of finite presentation; then, for u to have property $\mathcal{P}$, it is necessary and sufficient that there be $i \geq j$ such that $u_i = u_j \times_{S_j} S_i$ has property $\mathcal{P}$.

This is the case when $\mathcal{P}$ is one of the following properties of a morphism: being separated, surjective, radicial, affine, quasi-affine, finite, quasi-finite, proper, projective, quasi-projective, an isomorphism, a monomorphism, an immersion, an open immersion, a closed immersion (EGA IV₃, 8.10.5), flat (EGA IV₃, 11.2.6), smooth, unramified or étale (EGA IV₄, 17.7.8).⁽⁹¹⁾

Note that this still holds when $\mathcal{P}$ is the property of being covering for the (fppf) topology; indeed, given two S -schemes of finite presentation X and Y , and an S -morphism $u : X \rightarrow Y$, it follows from IV, 6.3.1 (i)⁽⁹²⁾ that, for u to be covering for the (fppf) topology, it is necessary

^{*}This is too optimistic, as M. Raynaud shows in his thesis (loc. cit. X 14).

⁽⁸⁹⁾**N.D.E.** Remark (*) refers to counterexample X.14 in [Ray70a]. The base there is regular local of dimension 2.

⁽⁹⁰⁾**N.D.E.** Note that S , being affine over S_0 , is therefore quasi-compact and quasi-separated, cf. N.D.E. (92) below.

⁽⁹¹⁾**N.D.E.** The word “flat” has been added, and 17.7.6 corrected to 17.7.8.

⁽⁹²⁾**N.D.E.** And from the fact that Y , being of finite presentation over S , is quasi-compact.

and sufficient that there be an S -scheme Z and an S -morphism $v : Z \rightarrow Y$ faithfully flat and of finite presentation which factors through u .

The aim of this section 10 is to give variants of results of this kind for the category of S -groups of finite presentation, that of S -schemes with a group of operators, and for certain properties of group monomorphisms (being invariant, central with representable quotient sheaf, etc.).

The first two results of this type are the following. (In nos. 10.2 to 10.9 below, we retain the notation introduced in 10.0.)

Lemma 10.2. *Let G_j and H_j be two S_j -groups of finite presentation; put, for every $i \geq j$, $G_i = G_j \times_{S_j} S_i$, $G = G_j \times_{S_j} S$, and define H_i and H similarly. Then the canonical map below is bijective:*

$$\lim_{\substack{\longrightarrow \\ i \geq j}} \operatorname{Hom}_{S_i\text{-gr.}}(G_i, H_i) \rightarrow \operatorname{Hom}_{S\text{-gr.}}(G, H).$$

Lemma 10.3. *Let G be an S -group of finite presentation; then there is $j \in I$, an S_j -group of finite presentation G_j , and an isomorphism of S -groups $G \cong G_j \times_{S_j} S$.*

Assertions 10.2 and 10.3 are easy consequences of 10.0 and 10.1, taking account of the interpretation⁽⁹³⁾ of the S -group structure given in EGA 0_{III}, 8.2.5 and 8.2.6.

Lemma 10.4. *Let $u : G \rightarrow H$ be a morphism of S -groups between S -groups of finite presentation. By 10.3 and 10.2, u comes by base change from a morphism $u_j : G_j \rightarrow H_j$ between S_j -groups of finite presentation. Then, for u to be a central monomorphism (resp. an invariant monomorphism), it is necessary and sufficient that there be $i \geq j$ such that $u_i = u_j \times_{S_j} S_i$ has the same property.*

This is an immediate consequence of 10.0 and 10.1, taking account of the characterization given in 6.7 of central or invariant group monomorphisms.

Corollary 10.5. *Let G_j be an S_j -group of finite presentation. For $G_j \times_{S_j} S$ to be commutative, it is necessary and sufficient that there be $i \geq j$ such that $G_j \times_{S_j} S_i$ is so.*

Indeed, it amounts to the same thing to say that an S -group is commutative, or that, considered as a subgroup scheme of itself, it is central.

Proposition 10.6. *Let G_j be an S_j -group of finite presentation, G'_j a subgroup scheme of G_j flat and of finite presentation over S_j . For $(G_j \times_{S_j} S)/(G'_j \times_{S_j} S)$ to be representable for the (fpqc) topology, it is necessary and sufficient that there be $i \geq j$ such that $(G_j \times_{S_j} S_i)/(G'_j \times_{S_j} S_i)$ is so.*

This is a consequence of the following more general lemma:

Lemma 10.7. *Let X_j be an S_j -scheme of finite presentation, and R_j an equivalence relation on X_j flat and of finite presentation.⁽⁹⁴⁾ For the quotient sheaf $(X_j \times_{S_j} S)/(R_j \times_{S_j} S)$ for the topology $T = (\text{fppf})$ or (fpqc) to be representable, it is necessary and sufficient that there be $i \geq j$ such that the quotient sheaf $(X_j \times_{S_j} S_i)/(R_j \times_{S_j} S_i)$ for the topology T is so.*

Taking account of the statements of EGA IV₂, 8.8.2, 8.8.3, 8.10.5 and 11.2.6 recalled in 10.0, this lemma is a consequence of the following result:

Lemma 10.8. *Let T be the (fppf) or (fpqc) topology; let X be an S -scheme of finite presentation (resp. locally of finite presentation), R an equivalence relation on X defined by a monomorphism $v : R \rightarrow X \times_S X$ such that $\text{pr}_1 \circ v$ is flat and of finite presentation (resp. flat and locally of finite presentation). Then the following conditions are equivalent:*

⁽⁹³⁾**N.D.E.** In terms of commutative diagrams of S -morphisms.

⁽⁹⁴⁾**N.D.E.** I.e., such that the composite $R_j \hookrightarrow X_j \times_{S_j} X_j \xrightarrow{\text{pr}_1} X_j$ is flat and of finite presentation.

- (i) The quotient sheaf X/R for the topology T is representable.
 (ii) There is an S -scheme of finite presentation (resp. locally of finite presentation) Y and a faithfully flat morphism $p : X \rightarrow Y$ such that the diagram

$$(D) \quad \begin{array}{ccc} R & \xrightarrow{\text{pr}_1 \circ v} & X \\ \text{pr}_2 \circ v \downarrow & & \downarrow p \\ X & \xrightarrow{p} & Y \end{array}$$

is cartesian.

Note first that by IV, 3.3.2 and 4.4.3, for the sheaf X/R for the topology T to be representable by Y , it is necessary and sufficient that the diagram (D) be cartesian and p covering for the topology T .

Let us show that (i) implies (ii). Hypothesis (i) implies that the diagram (D) is cartesian, hence that $\text{pr}_1 \circ v$ is obtained from p by base change by p , and that p is covering for the (fpqc) topology. Thus, by (fpqc) descent (EGA IV₂, 2.7.1), since $\text{pr}_1 \circ v$ is faithfully flat and (locally) of finite presentation, the same holds for p . Then, by EGA IV₄, 17.7.5, since X is (locally) of finite presentation over S , the same holds for Y .

Let us show that (ii) implies (i). It suffices to show that p is covering for the (fppf) topology; now p is faithfully flat by hypothesis, and is locally of finite presentation because X and Y are locally of finite presentation over S (EGA IV₁, 1.4.3 (v)).

Lemma 10.9. *Let G_j be an S_j -group of finite presentation, and $G = G_j \times_{S_j} S$. For G^0 to be representable, it is necessary and sufficient that there be $i \geq j$ such that $(G_i)^0 = (G_j \times_{S_j} S_i)^0$ is so.*

The condition is sufficient, since the functor $G \mapsto G^0$ commutes with base change, by 3.3.

Conversely, suppose G^0 representable. Then, by 3.9, G^0 is open in G and quasi-compact over S , hence of finite presentation over S , since G is. Then, by 10.3 and 10.1, there is $i \geq j$ and an open subgroup scheme H_i of G_i such that $H_i \times_{S_i} S = G^0$. The structural morphism $G^0 \rightarrow S$ is connected, i.e. has geometrically connected fibers (VIA 2.1.1), so, by EGA IV₃, 9.3.3 and 9.7.7, increasing i if necessary, one may suppose that the structural morphism $H_i \rightarrow S$ is connected. Then, by 3.10.1, the underlying space of H_i is precisely $\underline{(G_i)^0}$, and thus H_i represents $(G_i)^0$.

10.10. — Recall two very useful special cases of the situation stated in 10.0 (cf. EGA IV₃, 8.1.2 a) and c)):

a) Given a point x of a scheme X , one puts $S_0 = \text{Spec } \mathbf{Z}$ and considers the decreasing filtered projective system $(S_i)_{i \in I}$ of affine open neighborhoods of x ; then $S = \text{Spec } \mathcal{O}_{X,x}$. In particular, if x is the generic point of an integral scheme X , one finds $S = \text{Spec } \kappa(x)$.

b) One puts $S_0 = \text{Spec } \mathbf{Z}$, and considers the family $(\mathcal{A}_i)_{i \in I}$, preordered by inclusion, of sub- $\mathbf{Z}$ -algebras of finite type of the ring of an affine scheme S . Since the $\mathcal{A}_i$ are noetherian rings, this allows one in many cases to pass from the noetherian case to the general case.

We shall now give two results concerning the special case considered in a).⁽⁹⁵⁾

Proposition 10.11. ⁽⁹⁶⁾ *Let S be an integral scheme with generic point η , G (resp. Y, Z) an S -group (resp. S -schemes) of finite presentation, $u, v, i : Y \rightarrow G$ and $j : Z \rightarrow G$ morphisms of S -schemes. Suppose that $i_\eta : Y_\eta \rightarrow G_\eta$ and $j_\eta : Z_\eta \rightarrow G_\eta$ are closed immersions.*

⁽⁹⁵⁾**N.D.E.** Note that the proof also uses case b).

⁽⁹⁶⁾**N.D.E.** The statement has been simplified, and the case of subgroups treated separately in Corollary 10.11.1.

Then there is a nonempty open subset S' of S such that the morphisms $i' : Y' \rightarrow G'$ and $j' : Z' \rightarrow G'$ obtained by base change are closed immersions, and the functors:

$$\underline{\text{Transp}}(u', v'), \quad \underline{\text{Transp}}_{G'}(i'(Y'), j'(Z')) \quad \text{and} \quad \underline{\text{Transp}}_{G'}^{\text{str}}(i'(Y'), j'(Z'))$$

resp.

$$\underline{\text{Centr}}(u') \quad \text{and} \quad \underline{\text{Norm}}_{G'} i'(Y')$$

are representable by closed sub- S' -schemes (resp. sub- S' -groups) of G' , of finite presentation over S' .

We shall apply the results of 10.1, first in the situation of 10.10 a), then in that of 10.10 b). Since G_η, Y_η, Z_η are flat over the field $\kappa(\eta)$, G_η is separated over $\kappa(\eta)$ (VIA 0.3), and i_η, j_η are closed immersions, then, by 10.1, there is an affine open subset $S' = \text{Spec}(A')$ of S , a noetherian subring A of A' , A -schemes G_A, Y_A, Z_A flat and of finite presentation over A , and morphisms $u_A, v_A, i_A : Y_A \rightarrow G_A$ and $j_A : Z_A \rightarrow G_A$, such that G_A is an A -group, separated over A , i_A and j_A are closed immersions, and $G \times_S S' = G_A \otimes A'$, etc. Since the functors considered for S' are obtained by base change from the analogous functors for $\text{Spec}(A)$, it suffices to establish the result for the latter.

By EGA IV₂, 6.9.2, replacing A by a localization A_f if necessary (and hence S' by the affine open subset S'_f), one may suppose that G_A, Y_A, Z_A are essentially free over A (in the sense of 6.2.1).⁽⁹⁷⁾ It then follows from 6.2.4 b) and e) that, under the hypotheses of the statement, the functors considered are representable by closed sub- A -schemes of G_A (hence of finite presentation over A , since A is noetherian and G_A of finite presentation over A), and they are sub- A -groups of G_A in the case of $\underline{\text{Centr}}(u_A)$ and $\underline{\text{Norm}}_{G_A} i_A(Y_A)$.

Corollary 10.11.1. *Let S be an integral scheme with generic point η , G, H, K be S -groups of finite presentation, $i : H \rightarrow G$ and $j : K \rightarrow G$ two quasi-compact monomorphisms of S -groups. Then there is a nonempty open subset S' of S such that the morphisms $i' : H' \rightarrow G'$ and $j' : K' \rightarrow G'$ obtained by base change are closed immersions, and the functors:*

$$\underline{\text{Transp}}_{G'}(H', K') \quad \text{and} \quad \underline{\text{Transp}}_{G'}^{\text{str}}(H', K') \quad (\text{resp. } \underline{\text{Centr}}_{G'} H' \text{ and } \underline{\text{Norm}}_{G'} H')$$

are representable by closed sub- S' -schemes (resp. sub- S' -groups) of G' , of finite presentation over S' .

This follows from the preceding proposition for, by 1.4.2, the hypotheses imply that i_η and j_η are closed immersions.

Proposition 10.12. *Let S be an integral scheme, G an S -group of finite presentation, A and B two subgroup schemes of G , of finite presentation over S and with smooth generic fiber. Suppose moreover that one of the following conditions holds:*

- a) A has connected generic fiber,
- b) A and B are invariant in G .

Then there is a nonempty open subset S' of S and a closed subgroup scheme D' of $G' = G|_{S'}$, of finite presentation over S' , with smooth fibers, which represents the (fppf) sheaf associated to the presheaf of groups of commutators of $A' = A|_{S'}$ and $B' = B|_{S'}$ in G' , and D' has connected fibers in case (a), and is invariant in G' in case (b).

In particular, for every $s \in S'$, one has $D'_s = (A_s, B_s)$ with the notation of 7.2.2.

Let η be the generic point of S ; put $H_\eta = (A_\eta, B_\eta)$. Since A_η and B_η are smooth, then, by 7.8 in case (a), and 7.3 (v) in case (b), H_η is connected (resp. invariant in G_η).

We are in the situation of 10.0 corresponding to 10.10 (a); thus, by 10.3 and 10.1, there is a nonempty open subset S' of S and a subgroup S' -scheme D' of finite presentation and

⁽⁹⁷⁾**N.D.E.** Indeed, since G_A is flat and of finite presentation over A , it is covered by affine open subsets $G_1, \dots, G_n$ such that each $\mathcal{O}(G_i)$ is a flat A -algebra of finite presentation; then, by EGA IV₂, 6.9.2, there is $f_i \in A$ such that $\mathcal{O}(G_i)_{f_i}$ is a free module over A_{f_i} ; one can then replace $\text{Spec}(A)$ by the affine open subset $D(f)$, where $f = f_1 \cdots f_n$, and one does the same for Y_A and Z_A .

closed in G' , such that $D'_\eta = D' \otimes_{S'} \kappa(\eta)$ equals H_η . Moreover, by EGA IV₃, 9.7.7 and 9.3.3, one may suppose that D' has geometrically reduced fibers. In case (a), one may suppose, by EGA IV₃, 9.7.7 and 9.3.3 again, that D' has connected fibers, hence geometrically connected fibers (cf. VIA, 2.1.1). In case (b), one may suppose, by 10.4, that D' is invariant in G' .

Moreover, in the course of the proof of 7.8 we saw that there is an integer n such that $\nu_\eta^n((A_\eta \times_{\kappa(\eta)} B_\eta)^n) = D'_\eta$, where ν_η and ν_η^n are defined as in 7.2.2 (a) and 7.1 (ii). We can define by the same formulas the morphisms

$$\nu' : A' \times_{S'} B' \longrightarrow G' \quad \text{and} \quad \nu'^n : (A' \times_{S'} B')^n \longrightarrow G',$$

and one has $\nu'^n \otimes_{S'} \kappa(\eta) = \nu_\eta^n$.

Consequently, by 10.1, one can choose S' such that the morphism ν'^n is flat and factors through D' , and that the resulting morphism $\nu'^n : (A' \times_{S'} B')^n \longrightarrow D'$ is surjective. Then, by 7.5, the morphism⁽⁹⁸⁾

$$\nu'^{2n} = \mu \circ (\nu'^n \times_{S'} \nu'^n) : (A' \times_{S'} B')^{2n} \longrightarrow D'$$

is covering for the (fppf) topology. Thus, by 7.6, D' represents the (fppf) sheaf associated to the presheaf of commutators of A' and B' in G' .

Moreover, ν'^n induces, for every $s \in S'$, a surjective morphism $\nu_s^n : (A_s \times_{\kappa(s)} B_s)^n \longrightarrow D'_s$.⁽⁹⁹⁾ Then D'_s is a closed subgroup of G_s containing $\nu_s(A_s \times_{\kappa(s)} B_s)$, hence also (A_s, B_s) . Since ν_s^n is surjective, then D'_s equals (A_s, B_s) and represents, by 7.6, the (fppf) sheaf of commutators of A_s and B_s in G_s .

Corollary 10.12.1. ⁽¹⁰⁰⁾ *Let S be an integral scheme with generic point η , and G an S -group of finite presentation with smooth fibers. Put $K_\eta^0 = G_\eta$ and $K_\eta^p = (K_\eta^{p-1}, K_\eta^{p-1})$ (resp. $K_\eta^p = (G, K_\eta^{p-1})$) for every $p \in \mathbf{N}^*$. Fix $n \in \mathbf{N}^*$. Then there is a nonempty open subset S' of S and a subgroup scheme D invariant in $G|_{S'}$, of finite presentation and with smooth fibers, such that $D_s = K_s^n$ for every $s \in S'$.*

This follows from 10.12, by induction on n .

Corollary 10.13. *Let S be an integral scheme with generic point η , let G be an S -group, H a subgroup S -scheme invariant in G ; suppose G and H of finite presentation over S and with smooth generic fiber.⁽¹⁰¹⁾*

If one has $(H_\eta, H_\eta) = H_\eta$ (resp. $(G_\eta, H_\eta) = H_\eta$), then there is a nonempty open subset S' of S such that, for every $s \in S'$, one has $(H_s, H_s) = H_s$ (resp. $(G_s, H_s) = H_s$).

Indeed, by the proof of 10.12, there is a nonempty open subset S' of S and a subgroup S' -scheme D of $G|_{S'}$, of finite presentation and with smooth fibers, such that $D_s = (H_s, H_s)$ (resp. $D_s = (G_s, H_s)$) for every $s \in S'$. On the other hand, since $D_\eta = H_\eta$ and D and H are of finite presentation over S' , then, by EGA IV₃, 8.8.2.5, there is a nonempty open subset S'' of S' such that $D|_{S''} = H|_{S''}$. For every $s \in S''$, one therefore has $H_s = D_s = (H_s, H_s)$ (resp. $H_s = D_s = (G_s, H_s)$).

10.14. — Statements 10.2 and 10.3 concerning the category of S -groups of finite presentation extend to the category of pairs formed by an S -group of finite presentation and an S -scheme of finite presentation with group of operators G . More precisely, in the situation recalled at the beginning of 10.0:

(i) let $j \in I$ and G_j, G'_j be two S_j -groups of finite presentation, H_j (resp. H'_j) an S_j -scheme of finite presentation with group of operators G_j (resp. G'_j). Put, for $i \in I, i \geq j$,

⁽⁹⁸⁾**N.D.E.** Below, $n + 1$ has been corrected to $2n$.

⁽⁹⁹⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁰⁰⁾**N.D.E.** This corollary, used in the proof of 8.4, has been added.

⁽¹⁰¹⁾**N.D.E.** The original supposed G, H of finite presentation and H with smooth fibers; the hypothesis has been modified in order to be able to apply 10.12. The proof has also been expanded.

$G_i = G_j \times_{S_j} S_i$ and $G = G_j \times_{S_j} S$, and define similarly G'_i, G', H_i, H, H'_i and H' . Denote by $\text{Dihom}_{S\text{-gr}}((G, H), (G', H'))$ the set of di-morphisms of S -groups and S -schemes with a group of operators from the pair (G, H) to the pair (G', H') . Then the canonical map

$$\lim_{\substack{\longrightarrow \\ i \geq j}} \text{Dihom}_{S_i\text{-gr}}((G_i, H_i), (G'_i, H'_i)) \longrightarrow \text{Dihom}_{S\text{-gr}}((G, H), (G', H'))$$

is bijective.

(ii) let G be an S -group of finite presentation and H an S -scheme of finite presentation with group of operators G ; there is then an index $j \in I$, an S_j -group of finite presentation G_j , an S_j -scheme of finite presentation H_j with group of operators G_j and a di-isomorphism of S -groups and S -schemes with groups of operators from $(G_j \times_{S_j} S, H_j \times_{S_j} S)$ onto (G, X) .

Definition 10.15. ⁽¹⁰²⁾ Let T be a topology on $(\text{Sch})/S$, coarser than the canonical topology. Given a group S -scheme G and an S -scheme X with group of operators G , one says that X is a *formally homogeneous space under G* (relative to the topology T) if the morphism $\Phi : G \times_S X \longrightarrow X \times_S X$, defined by $(g, x) \longmapsto (gx, x)$ for every $S' \longrightarrow S$ and $g \in G(S')$, $x \in X(S')$, is an epimorphism in the category of sheaves for the topology T , which is equivalent to saying that Φ is covering for the topology T (cf. IV 4.4.3).

One says that X is a *homogeneous space* if it is formally homogeneous and if, moreover, the morphism $X \longrightarrow S$ is also covering for the topology T .

In particular, one says that X is a *formally principal homogeneous space under G* if Φ is an isomorphism, and that X is a *principal homogeneous bundle* (or *G -torsor*) if Φ is an isomorphism and if, moreover, the morphism $X \longrightarrow S$ is covering for the topology T (cf. IV 5.1.5 and 5.1.6 (ii)).

Proposition 10.16. *We place ourselves in the situation considered at the beginning of 10.0. Let $j \in I$, G_j be an S_j -group and X_j an S_j -scheme with group of operators G_j . Suppose G_j and X_j of finite presentation over S_j .*

For $X = X_j \times_{S_j} S$ to be a homogeneous space (resp. a principal homogeneous bundle) under $G = G_j \times_{S_j} S$ for the (fppf) topology, it is necessary and sufficient that there be an index $i \geq j$ such that $X_i = X_j \times_{S_j} S_i$ is a homogeneous space (resp. a principal homogeneous bundle) under $G_i = G_j \times_{S_j} S_i$.

Taking account of 10.14 and EGA IV₃, 8.8.2, 8.8.3 and 8.10.5, the statement follows from the property concerning covering morphisms for the (fppf) topology recalled in 10.1. ⁽¹⁰³⁾

11. Affine group schemes

11.0. — Recollections. ⁽¹⁰⁴⁾ Let $q : X \longrightarrow S$ be a quasi-compact and quasi-separated morphism of schemes (cf. EGA IV₁, 1.1 & 1.2), and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. Recall that $q_*(\mathcal{F})$ is a quasi-coherent $\mathcal{O}_S$ -module (EGA I, 9.2.1). Moreover, by EGA III, 1.4.15 (supplemented by EGA IV₁, 1.7.21), one has point (c) below, and the proof in loc. cit. also gives points (a) and (b):

(a) If $\mathcal{F}$ is a filtered inductive limit of quasi-coherent submodules $\mathcal{F}_\alpha$, then $q_*(\mathcal{F}) = \lim_{\longrightarrow \alpha} q_*(\mathcal{F}_\alpha)$.

(b) If $\mathcal{E}$ is a flat $\mathcal{O}_S$ -module, the canonical morphism $\mathcal{E} \otimes_{\mathcal{O}_S} q_*(\mathcal{O}_X) \longrightarrow q_*q^*(\mathcal{E})$ is an isomorphism.

⁽¹⁰²⁾**N.D.E.** The original has been corrected by distinguishing the notions of a “formally homogeneous” or “homogeneous” object, see IV §§ 5.1 and 6.7.

⁽¹⁰³⁾**N.D.E.** For the limit properties of torsors in terms of Čech cohomology, see also SGA 4, VII 5.7 and its corollaries. This has been expanded in the article [Ma07].

⁽¹⁰⁴⁾**N.D.E.** This paragraph of recollections has been added.

(c) Let $p : S' \rightarrow S$ be a flat morphism, $q' : X' \rightarrow S'$ the morphism obtained from q by base change, and $\mathcal{F}'$ the inverse image of $\mathcal{F}$ on X' . Then the canonical morphism $p^*q_*(\mathcal{F}) \rightarrow q'_*(\mathcal{F}')$ is an isomorphism.

Indeed, let $U = \text{Spec}(A)$ be an arbitrary affine open subset of S . By hypothesis, $q^{-1}(U)$ is a union of affine open subsets $V_i = \text{Spec}(B_i)$, for $i = 1, \dots, n$, and each intersection $V_i \cap V_j$ is a union of finitely many affine open subsets $W_{ijk} = \text{Spec}(C_{ijk})$. Then $\Gamma(U, q_*(\mathcal{F})) = \Gamma(q^{-1}(U), \mathcal{F})$ is the kernel of the morphism

$$\bigoplus_{i=1}^n \Gamma(V_i, \mathcal{F}) \rightarrow \bigoplus_{i,j,k} \Gamma(W_{ijk}, \mathcal{F}).$$

Point (a) follows, for each of the terms above commutes with filtered inductive limits (since the V_i and W_{ijk} are affine, hence quasi-compact). Let us prove (b): $E = \Gamma(U, \mathcal{E})$ is a flat A -module, and $\Gamma(U, q_*q^*(\mathcal{E}))$ is the kernel $K(E)$ of the morphism

$$\bigoplus_{i=1}^n B_i \otimes_A E \rightarrow \bigoplus_{i,j,k} C_{ijk} \otimes_A E$$

and since E is flat over A , this kernel is identified with $K(A) \otimes_A E = \mathcal{O}_X(q^{-1}(U)) \otimes_A E$. Finally, if U' is an arbitrary affine open subset of S' lying over U , then $A' = \mathcal{O}(U')$ is a flat A -algebra, and one obtains as above that $\mathcal{F}(q^{-1}(U)) \otimes_A A' \xrightarrow{\sim} \mathcal{F}'(q'^{-1}(U))$.

Notation. — Let S be a scheme, X an S -scheme, $f : X \rightarrow S$ the structural morphism; one shall put $\mathcal{A}(X) = f_*(\mathcal{O}_X)$.

Lemma 11.1. *Let X and Y be two S -schemes quasi-compact and quasi-separated over S , $f : X \rightarrow S$ and $g : Y \rightarrow S$ the structural morphisms. Then the canonical homomorphism*

$$\varphi : \mathcal{A}(X) \otimes_{\mathcal{O}_S} \mathcal{A}(Y) \rightarrow \mathcal{A}(X \times_S Y)$$

is an isomorphism in each of the following cases:⁽¹⁰⁵⁾

- a) f and g are affine,
- b) g (or f) is flat and affine,
- c) g is flat and $f_*(\mathcal{O}_X)$ is a flat $\mathcal{O}_S$ -module.

In case (b), one shall suppose that it is g that is flat and affine. Put then $S' = \text{Spec } \mathcal{A}(X)$, $Y' = Y \times_S S'$, $g' = g \times_S S'$ and denote by v the morphism $S' \rightarrow S$:

$$\begin{array}{ccc} Y' & \xrightarrow{\quad} & Y \\ g' \downarrow & & \downarrow g \\ \text{Spec } \mathcal{A}(X) = S' & \xrightarrow{\quad} & S. \end{array}$$

In cases (a) and (b), g is affine and therefore, by EGA II, 1.5.2, one has:

$$(1) \quad g'_*(\mathcal{O}_{Y'}) = v^*g_*(\mathcal{O}_Y) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}.$$

One has the same equality in case (c), by 11.0 (c), since S' is flat over S and g is quasi-compact and quasi-separated.

On the other hand (EGA II 1.2.7), $f : X \rightarrow S$ factors through v by a morphism $p : X \rightarrow S'$, and one has $p_*(\mathcal{O}_X) = \mathcal{O}_{S'}$ and $X \times_S Y = X \times_{S'} Y'$. Since f is quasi-separated, p is too (EGA IV₁, 1.2.2), and since f is quasi-compact and v quasi-separated, p is also

⁽¹⁰⁵⁾**N.D.E.** Note that, if k is a field and X is an infinite sum of copies of $S = \text{Spec } k$ (so that X is not quasi-compact), then $\mathcal{A}(X) = k^X$ and the canonical morphism $k^X \otimes k^X \rightarrow k^{X \times X}$ is not surjective.

quasi-compact (EGA IV₁, 1.2.4). Consider then the cartesian square:

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{p'} & Y' \\ \downarrow & & \downarrow g' \\ X & \xrightarrow{p} & S'. \end{array}$$

In cases (b) and (c), Y is flat over S , so Y' is flat over S' ; applying 11.0 (c) again, one obtains:

$$(2) \quad p'_*(\mathcal{O}_{X \times_S Y}) = g'^* p_*(\mathcal{O}_X) = g'^*(\mathcal{O}_{S'}) = \mathcal{O}_{Y'},$$

and one has the same equality in case (a), for in this case p and p' are isomorphisms.

Finally, since v is affine, one has, by EGA II, 1.4.7, $v_*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(X)$ for every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. Combined with (2) and (1), this gives:

$$\begin{aligned} \mathcal{A}(X \times_S Y) &= v_* g'_* p'_*(\mathcal{O}_{X \times_S Y}) \stackrel{(2)}{=} v_* g'_*(\mathcal{O}_{Y'}) \\ &\stackrel{(1)}{=} v_*(\mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{A}(X). \end{aligned}$$

Corollary 11.2. *The functor $X \mapsto \text{Spec } \mathcal{A}(X)$, from the full subcategory of $(\text{Sch})/S$ formed by S -schemes X flat, quasi-compact and quasi-separated over S , and such that $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module, to that of S -schemes flat and affine over S , commutes with finite products, hence transforms S -groups into S -groups.*

Definition 11.3. Given an S -group G flat, quasi-compact and quasi-separated over S , such that $\mathcal{A}(G)$ is flat over $\mathcal{O}_S$,⁽¹⁰⁶⁾ we shall denote by G_{af} , and call the *affine envelope* of G , the S -group $G_{\text{af}} = \text{Spec } \mathcal{A}(G)$.

Proposition 11.3.1. ⁽¹⁰⁷⁾ *The canonical morphism $\tau_G : G \rightarrow G_{\text{af}}$ is a morphism of S -groups. Moreover, it satisfies the following universal property:*

(i) *For every morphism of S -schemes $\phi : G \rightarrow H$, where H is affine over S , there is a unique morphism of S -schemes $\phi' : G_{\text{af}} \rightarrow H$ such that $\phi = \phi' \circ \tau_G$.*

(ii) *If moreover H is an S -group and ϕ is a morphism of S -groups, then ϕ' is too.*

11.4. — ⁽¹⁰⁸⁾ Let $\mathcal{E}$ and $\mathcal{F}$ be two quasi-coherent $\mathcal{O}_S$ -modules. Consider the S -functor $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))$ (cf. I, 3.1.4), i.e. for every S -scheme $f : X \rightarrow S$,

$$\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))(X) = \text{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{E})_X, \mathbf{W}(\mathcal{F})_X).$$

Moreover, by I, 4.6.2, one has $\mathbf{W}(\mathcal{E})_X = \mathbf{W}(f^*(\mathcal{E}))$ (and similarly for $\mathcal{F}$) and

$$\text{Hom}_{\mathcal{O}_X}(\mathbf{W}(f^*(\mathcal{E})), \mathbf{W}(f^*(\mathcal{F}))) = \text{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})).$$

One therefore obtains (using the adjunction formula for the last equality):

$$\begin{aligned} \underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))(X) &= \text{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})) \\ (\dagger) \quad &= \text{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{F})). \end{aligned}$$

Proposition 11.5. *Let X be an S -scheme quasi-compact and quasi-separated over S , $f : X \rightarrow S$ the structural morphism, $\mathcal{E}$ and $\mathcal{E}'$ two quasi-coherent $\mathcal{O}_S$ -modules. Suppose that one of the following two conditions holds:*

- a) *f is affine,*
- b) *$\mathcal{E}$ is flat over $\mathcal{O}_S$.*

⁽¹⁰⁶⁾**N.D.E.** Let us point out that if S is a locally noetherian regular scheme of dimension ≤ 2 , and X an S -scheme flat, quasi-compact and quasi-separated, then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module, cf. [Ray70a], VII 3.2.

⁽¹⁰⁷⁾**N.D.E.** This proposition has been added; see also the additional paragraph 12 below for a study of the morphism $G \rightarrow G_{\text{af}}$ and its kernel.

⁽¹⁰⁸⁾**N.D.E.** In 11.4–11.6, $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))$ has been considered in place of $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{V}(\mathcal{F}), \mathbf{V}(\mathcal{E}))$ (cf. I, 4.6.3), and the original simplified taking account of the addition 11.0 (b).

Then the canonical morphism $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X) \rightarrow f_* f^*(\mathcal{E})$ is an isomorphism, and one therefore has

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}'), \mathbf{W}(\mathcal{E}))(X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X)).$$

Indeed, the second assertion follows from 11.4 and the first; the latter follows from EGA II, 1.4.7 in case (a), and 11.0 (b) in case (b).⁽¹⁰⁹⁾

11.6. — ⁽¹¹⁰⁾ Let G be an S -group and $\mathcal{E}$ a quasi-coherent $\mathcal{O}_S$ -module. Giving $\mathcal{E}$ a structure of G - $\mathcal{O}_S$ -module (i.e. an $\mathcal{O}_S$ -linear operation of G on $\mathbf{W}(\mathcal{E})$, cf. I 4.7.1) is equivalent to giving a morphism of S -functors in monoids $\rho : G \rightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$ (indeed, such a ρ necessarily sends G into $\underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$).

Now, by 11.4, giving a morphism of S -functors $\rho : G \rightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$ is equivalent to giving an element θ of $\mathrm{Hom}_{\mathcal{O}_G}(f^*(\mathcal{E}), f^*(\mathcal{E}))$, which corresponds by adjunction to an element δ of $\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{E}))$, where f denotes the projection $G \rightarrow S$.

Let $m : G \times_S G \rightarrow G$ be multiplication, δ_G the morphism $\mathcal{O}_G \rightarrow m_*(\mathcal{O}_{G \times_S G})$, and ϕ the projection $G \times_S G \rightarrow S$ (which equals $f \circ m$). It is convenient to denote by $\boxtimes$ the “external” tensor product; one thus obtains a morphism

$$\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \rightarrow \mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G})$$

and, by abuse of notation, one will again denote by $\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G$ the composite of the preceding morphism with the canonical morphism $\mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G}) \rightarrow m_* m^* f^*(\mathcal{E}) = m_* \phi^*(\mathcal{E})$.

On the other hand, denote by $h : G \rightarrow S$ a second copy of $f : G \rightarrow S$ and consider the following commutative diagram, where p, q denote the two projections:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{q} & G \\ p \downarrow & \searrow \phi & \downarrow h \\ G & \xrightarrow{f} & S. \end{array}$$

Again denoting by δ the morphism $\mathcal{E} \rightarrow h_* h^*(\mathcal{E})$, one obtains the morphism

$$\delta \boxtimes \mathrm{id}_{\mathcal{O}_G} : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \rightarrow h_* h^*(\mathcal{E}) \boxtimes_{\mathcal{O}_S} \mathcal{O}_G = f^* h_* h^*(\mathcal{E})$$

and, by abuse of notation, we shall again denote by $\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}$ the composite of this morphism with the canonical morphism $f^* h_* h^*(\mathcal{E}) \rightarrow p_* \phi^*(\mathcal{E})$.

Then the condition that ρ be compatible with multiplication is equivalent to saying that, for every open subset U of S , the diagram below is commutative:

$$(1) \quad \begin{array}{ccc} \Gamma(U, \mathcal{E}) & \xrightarrow{\delta} & \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \\ \delta \downarrow & & \downarrow \mathrm{id}_{\mathcal{E}} \boxtimes \delta_G \\ \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) & \xrightarrow[\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}]{} & \Gamma(U \times_S G \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \boxtimes_{\mathcal{O}_S} \mathcal{O}_G). \end{array}$$

Furthermore, the identity section $\varepsilon : S \rightarrow G$ induces a morphism u from $\mathcal{O}_G$ to $\varepsilon_* \varepsilon^*(\mathcal{O}_G) = \varepsilon_*(\mathcal{O}_S)$, and the condition that ρ preserve the identity elements is equivalent to

⁽¹⁰⁹⁾**N.D.E.** The original has been simplified; it used the isomorphism $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}'), \mathbf{W}(\mathcal{E})) \simeq \underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{V}(\mathcal{E}), \mathbf{V}(\mathcal{E}'))$, then the inclusion of the right-hand term into

$$\underline{\mathrm{Hom}}_S(\mathbf{V}(\mathcal{E}), \mathbf{V}(\mathcal{E}')) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', f_* f^*(\mathrm{Sym}(\mathcal{E})))$$

and applied EGA III, 4.1.15 to $\mathbf{V}(\mathcal{E}) = \mathrm{Spec}(\mathrm{Sym}(\mathcal{E}))$ to deduce 11.0 (b).

⁽¹¹⁰⁾**N.D.E.** 11.6 has been expanded, and the results obtained brought out in the form of Proposition 11.6.1.

commutativity of the diagram:

$$(2) \quad \begin{array}{ccc} \Gamma(U, \mathcal{E}) & \xrightarrow{\delta} & \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \\ & \searrow \simeq & \swarrow \text{id}_{\mathcal{E}} \boxtimes u \\ & \Gamma(U \times_S S, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_S) & \end{array}$$

One therefore sees that giving $\mathcal{E}$ a G - $\mathcal{O}_S$ -module structure is equivalent to giving a morphism of $\mathcal{O}_S$ -modules $\delta : \mathcal{E} \rightarrow f_* f^*(\mathcal{E})$ satisfying conditions (1) and (2) above, and in this case the morphism $\theta : f^*(\mathcal{E}) \rightarrow f^*(\mathcal{E})$, obtained from δ by adjunction, is an isomorphism (for it corresponds to the isomorphism $G \times_S \mathbf{W}(\mathcal{E}) \xrightarrow{\sim} G \times_S \mathbf{W}(\mathcal{E})$ defined set theoretically by $(g, x) \mapsto (g, gx)$, see also I, 6.5.4).

Suppose now that G is flat, quasi-compact and quasi-separated over S , and that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module; then, by 11.1 (c), the canonical morphism $\mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \rightarrow \mathcal{A}(G \times_S G)$ is an isomorphism, and the morphism $\delta_G : \mathcal{A}(G) \rightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ will be denoted by Δ .

If moreover $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G) = f_* f^*(\mathcal{E})$ (which is the case, by 11.5, if $G \rightarrow S$ is affine, or if $\mathcal{E}$ is flat over $\mathcal{O}_S$), one obtains that conditions (1) and (2) are equivalent to the conditions below, which express that $\delta : \mathcal{E} \rightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ makes $\mathcal{E}$ a right $\mathcal{A}(G)$ -comodule (cf. I 4.7.2):

(CM 1) Putting $\mathcal{A} = \mathcal{A}(G)$, the diagram below is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\delta} & \mathcal{E} \otimes \mathcal{A} \\ \delta \downarrow & & \downarrow \text{id}_{\mathcal{E}} \otimes \Delta \\ \mathcal{E} \otimes \mathcal{A} & \xrightarrow{\delta \otimes \text{id}_{\mathcal{A}}} & \mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A} \end{array}$$

(CM 2) Denoting by $\eta : \mathcal{A} \rightarrow \mathcal{O}_S$ the morphism $\mathcal{A}(\varepsilon)$, the diagram below is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\delta} & \mathcal{E} \otimes \mathcal{A} \\ & \searrow \simeq & \swarrow \text{id}_{\mathcal{E}} \otimes \eta \\ & \mathcal{E} \otimes \mathcal{O}_S & \end{array}$$

Remark 11.6.A. Recall that $V(\mathcal{E})$ denotes the vector fibration over S which represents the functor $\mathbf{V}(\mathcal{E})$, i.e. for every $S' \rightarrow S$, $\mathbf{V}(\mathcal{E})(S') = \text{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'})$. Since, by I, 4.6.2, one has an anti-isomorphism of S -functors in monoids $\underline{\text{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E})) \simeq \underline{\text{End}}_{\mathcal{O}_S}(\mathbf{V}(\mathcal{E}))$, one sees that, if $\mathcal{E}$ is a left G - $\mathcal{O}_S$ -module, one has a right action $\mu : V(\mathcal{E}) \times_S G \rightarrow V(\mathcal{E})$ of G on $V(\mathcal{E})$, defined set theoretically by $(\varphi g)(x) = \varphi(gx)$, for every $g \in G(S')$, $x \in \Gamma(S', \mathcal{E}_{S'})$ and $\varphi \in \text{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'})$. One therefore obtains commutative diagrams:

$$\begin{array}{ccc} V(\mathcal{E}) \times_S G \times_S G & \xrightarrow{\mu \times \text{id}_G} & V(\mathcal{E}) \times_S G \\ \text{id}_{V(\mathcal{E})} \times m \downarrow & & \downarrow \mu \\ V(\mathcal{E}) \times_S G & \xrightarrow{\mu} & V(\mathcal{E}) \end{array} \quad \begin{array}{ccc} V(\mathcal{E}) & \xleftarrow{\mu} & V(\mathcal{E}) \times_S G \\ & \swarrow \simeq & \uparrow \text{id}_{V(\mathcal{E})} \times \varepsilon \\ & V(\mathcal{E}) \times_S S & \end{array}$$

When G is flat, quasi-compact and quasi-separated over S , $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module, and one of the conditions of 11.5 holds, one similarly recovers conditions (CM 1) and (CM 2).

Consequently, we have obtained:

Proposition 11.6.1. *Let G be an S -group flat, quasi-compact and quasi-separated over S , such that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module, and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module.*

(i) *Giving an $\mathcal{A}(G)$ -comodule structure $\delta : \mathcal{E} \rightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ amounts to the same thing as giving $\mathcal{E}$ a G_{af} - $\mathcal{O}_S$ -module structure (i.e. an $\mathcal{O}_S$ -linear operation of G_{af} on $\mathcal{E}$). By*

composition with the morphism of S -groups $G \rightarrow G_{\text{af}}$, this defines a G - $\mathcal{O}_S$ -module structure on $\mathcal{E}$.

(ii) If moreover $\mathcal{E}$ is flat, every $\mathcal{O}_S$ -linear operation of G on $\mathcal{E}$ factors through G_{af} and corresponds to a unique $\mathcal{A}(G)$ -comodule structure on $\mathcal{E}$.

Lemma 11.7. *Let G be an S -group flat, quasi-compact and quasi-separated over S , such that $\mathcal{A} = \mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module. Let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module, $\delta : \mathcal{E} \rightarrow \mathcal{E} \otimes \mathcal{A}$ an $\mathcal{A}$ -comodule structure, and $\rho : G \rightarrow \underline{\text{Aut}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$ the associated operation of G on $\mathcal{E}$.*

Let $\mathcal{E}_0$ be a quasi-coherent sub- $\mathcal{O}_S$ -module of $\mathcal{E}$ such that the restriction δ_0 of μ to $\mathcal{E}_0$ factors through $\mathcal{E}_0 \rightarrow \mathcal{E}_0 \otimes \mathcal{A}$, i.e. such that one has a commutative diagram:

$$\begin{array}{ccc} \mathcal{E}_0 & \hookrightarrow & \mathcal{E} \\ \delta_0 \downarrow & & \downarrow \delta \\ \mathcal{E}_0 \otimes \mathcal{A} & \hookrightarrow & \mathcal{E} \otimes \mathcal{A}. \end{array}$$

(N.B. The morphism $\mathcal{E}_0 \otimes \mathcal{A} \rightarrow \mathcal{E} \otimes \mathcal{A}$ is injective, since $\mathcal{A}$ is flat over $\mathcal{O}_S$.)

Then δ_0 makes $\mathcal{E}_0$ an $\mathcal{A}(G)$ -comodule, and thus defines an operation ρ_0 of G on $\mathcal{E}_0$ (which will be called the operation induced on $\mathcal{E}_0$ by ρ , and one will say that $\mathcal{E}_0$ is stable under ρ).

This follows immediately from the definitions and 11.6. Note, however, that in general the canonical map $\mathbf{W}(\mathcal{E}_0) \rightarrow \mathbf{W}(\mathcal{E})$ is not a monomorphism.

Remark 11.7.bis. ⁽¹¹¹⁾ Let G be a flat S -group and $\mathcal{E}$ a quasi-coherent G - $\mathcal{O}_S$ -module. Denote by f the morphism $G \rightarrow S$ and by δ the morphism $\mathcal{E} \rightarrow f_*f^*(\mathcal{E})$ defined in 11.6. Let $\mathcal{E}_0$ be a quasi-coherent sub- $\mathcal{O}_S$ -module of $\mathcal{E}$; since f is flat, then $f_*f^*(\mathcal{E}_0)$ is a sub- $\mathcal{O}_S$ -module of $f_*f^*(\mathcal{E})$, and similarly for $\phi = f \times f$. Consequently, if the restriction δ_0 of δ to $\mathcal{E}_0$ factors through $f_*f^*(\mathcal{E}_0)$, then it makes $\mathcal{E}_0$ a G - $\mathcal{O}_S$ -module. In this case, one will say that $\mathcal{E}_0$ is a G -stable submodule of $\mathcal{E}$.

Definition 11.8.0. ⁽¹¹²⁾ Let S be a scheme. An $\mathcal{O}_S$ -coalgebra is an $\mathcal{O}_S$ -module $\mathcal{C}$ endowed with two morphisms of $\mathcal{O}_S$ -modules $\Delta : \mathcal{C} \rightarrow \mathcal{C} \otimes \mathcal{C}$ and $\varepsilon : \mathcal{C} \rightarrow \mathcal{O}_S$, satisfying the following two axioms (cf. I 4.2):

(CO 1) Δ is coassociative: the following diagram is commutative

$$\begin{array}{ccc} & \mathcal{C} \otimes \mathcal{C} & \\ \Delta \nearrow & & \searrow \text{id} \otimes \Delta \\ \mathcal{C} & & \mathcal{C} \otimes \mathcal{C} \otimes \mathcal{C} \\ \Delta \searrow & & \nearrow \Delta \otimes \text{id} \\ & \mathcal{C} \otimes \mathcal{C} & \end{array}$$

(CO 2): ε is a counit, i.e. the following two composites are the identity

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\text{id} \otimes \varepsilon} \mathcal{C} \otimes \mathcal{O}_S \xrightarrow{\sim} \mathcal{C},$$

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\varepsilon \otimes \text{id}} \mathcal{O}_S \otimes \mathcal{C} \xrightarrow{\sim} \mathcal{C}.$$

A (right) $\mathcal{C}$ -comodule is an $\mathcal{O}_S$ -module $\mathcal{E}$ endowed with a morphism of $\mathcal{O}_S$ -modules $\mu : \mathcal{E} \rightarrow \mathcal{E} \otimes \mathcal{C}$ satisfying axioms (CM 1) and (CM 2) of 11.6.

One will say that $\mathcal{C}$ (resp. $\mathcal{E}$) is a quasi-coherent coalgebra (resp. a quasi-coherent comodule) if it is a quasi-coherent $\mathcal{O}_S$ -module.

⁽¹¹¹⁾**N.D.E.** This remark, which generalizes 11.7 and will be useful in 11.10.bis, has been added.

⁽¹¹²⁾**N.D.E.** Since statements 11.8 and 11.9 concern solely the notion of a comodule over a coalgebra, Definition 11.8.0 has been introduced and 11.8 and 11.9 reformulated accordingly.

Let A be a commutative ring and C an A -coalgebra; then $C^\vee = \text{Hom}_A(C, A)$ is an A -algebra. One shall denote by ev the natural evaluation map $C \otimes_A C^\vee \longrightarrow A$.

Lemma 11.8. *Let C be an A -coalgebra, V a C -comodule, M a sub- A -module of V . Suppose that C is a projective A -module.⁽¹¹³⁾ Let $c(M)$ be the image of the morphism of A -modules*

$$\theta : M \otimes_A C^\vee \xrightarrow{\mu \otimes \text{id}} V \otimes_A C \otimes_A C^\vee \xrightarrow{\text{id} \otimes \text{ev}} V.$$

Then $c(M)$ is the smallest subcomodule of V containing M , and is an A -module of finite type if M is. One will say that $c(M)$ is the subcomodule generated by M .

Moreover, for every ring morphism $A \longrightarrow A'$, if M' denotes the image of $M \otimes_A A'$ in $V' = V \otimes_A A'$, then $c(M')$ is the image of $c(M) \otimes_A A'$ in V' , so: “the formation of $c(M)$ commutes with base change”.

First, $M \subset c(M)$ by (CM 2), and if N is a subcomodule of V containing M , one has $\mu(M) \subset N \otimes C$ and therefore $c(M) \subset N$.

By hypothesis, C is a direct summand of a free A -module L , with basis $(e_i)_{i \in I}$. Denote by φ_i the restriction to C of the linear form e_i^* , defined by $e_i^*(e_j) = \delta_{ij}$. Let $x \in M$. One can write:

$$(1) \quad \mu(x) = \sum_{i \in J} x_i \otimes e_i,$$

where $x_i \in V$ and J is a finite subset of I . Then $x_i = \theta(x \otimes \varphi_i)$ belongs to $c(Ax)$, and one therefore has $c(Ax) = \sum_{i \in J} Ax_i$. Since C is a direct summand of L , say $L = C \oplus R$, whence $V \otimes L = (V \otimes C) \oplus (V \otimes R)$, one obtains that

$$(c(Ax) \otimes L) \cap (V \otimes C) = c(Ax) \otimes C.$$

Consequently, $\mu(x)$ can also be written in the form

$$(2) \quad \mu(x) = \sum_{j \in J} x_j \otimes b_j,$$

with $b_j \in C$. One can write $\Delta(b_j) = \sum_{i \in I} b_{ij} \otimes e_i$, with $b_{ij} \in C$. Then, applying $\mu \otimes \text{id}$ to (1) (resp. $\text{id} \otimes \Delta$ to (2)) and using axiom (CM 1), one obtains, for every $i \in J$:

$$\mu(x_i) = \sum_{j \in J} x_j \otimes b_{ij} \in c(Ax) \otimes C.$$

This shows that $c(M)$ is a subcomodule of V , and thus it is the smallest subcomodule of V containing M .

It is clear that $c(M)$ is an A -module of finite type if M is: if $M = Ax_1 + \cdots + Ax_n$, and $\mu(x_k) = \sum_i x_{ik} \otimes e_i$, then $c(M)$ is generated by the x_{ik} , for $k = 1, \dots, n$ and i ranging over a finite subset of I .

Finally, let $A \longrightarrow A'$ be a ring morphism and let M' be the image of $M \otimes A'$ in $V' = V \otimes A'$. Then $c(M')$ (resp. the image of $c(M) \otimes A'$ in V') is the image of the morphism θ' below (resp. of the composite $\theta' \circ \tau$):

$$M \otimes A' \otimes C^\vee \xrightarrow{\tau} M \otimes \text{Hom}_A(C, A') \xrightarrow{\theta'} V'.$$

Now these two morphisms have the same image. Indeed, let $\psi \in \text{Hom}_A(C, A')$ and $x \in M$. Put $\mu(x) = \sum_{i \in J} x_i \otimes e_i$. Then

$$\theta'(x \otimes \psi) = \sum_{i \in J} \psi(e_i) x_i$$

is the image under $\theta' \circ \tau$ of the element $\sum_{i \in J} x \otimes \psi(e_i) \otimes \varphi_i$ of $M \otimes A' \otimes C^\vee$. This proves the lemma.

⁽¹¹³⁾**N.D.E.** In the original, C is supposed a free A -module, the generalization to the case where C is a projective A -module, pointed out by J.-P. Serre, being mentioned in Remark 11.10.1. This generalization has been included here and in 11.9, and the proofs expanded accordingly.

Furthermore, one has the following proposition:

Proposition 11.8.bis. ⁽¹¹⁴⁾ *Let A be a noetherian ring, C an A -coalgebra flat over A , V a C -comodule, and M a sub- A -module of finite type of V . Then there is a subcomodule W of V , of finite type over A , containing M .*

Indeed, since M is of finite type, so is $\Delta_V(M)$, so there is a sub- A -module of finite type M' of V such that $\Delta_V(M) \subset M' \otimes_A C$. Let π be the projection $V \rightarrow V/M'$ and $\bar{\Delta}_V = (\pi \otimes \text{id}_C)\Delta_V$, and let

$$W = \{x \in V \mid \Delta_V(x) \in M' \otimes_A C\} = \text{Ker } \bar{\Delta}_V;$$

this is a sub- A -module of V containing M and contained in M' (since $x = (\text{id}_V \otimes \varepsilon)\Delta_V(x)$), hence of finite type over A . Moreover, $(\bar{\Delta}_V \otimes \text{id}_C)\Delta_V = (\pi \otimes \Delta_C)\Delta_V$ vanishes on W , i.e. $\Delta_V(W)$ is contained in the kernel K of $(\bar{\Delta}_V \otimes \text{id}_C)$. But since C is flat over A , one has $K = W \otimes C$, so W is a subcomodule of V .

Lemma 11.8.1. ⁽¹¹⁵⁾ *Let C be an A -coalgebra, V a C -comodule, M a sub- A -module of V , and $f : A \rightarrow A'$ a faithfully flat ring morphism. Suppose that $C' = C \otimes A'$ is a projective A' -module.*

(i) *Then there is a smallest subcomodule $t(M)$ of V containing M , and $t(M)$ is an A -module of finite type if M is. Moreover, “the formation of $t(M)$ commutes with base change”.*

(ii) *More precisely, C is a projective A -module, and one has $t(M) = c(M)$.*

PROOF. (ii) By [RG71] (see Proposition 11.8.2 below), C is a projective A -module. One can therefore apply Lemma 11.8: $c(M)$ is the smallest subcomodule of V containing M , it is an A -module of finite type if M is, and its formation commutes with base change.

To avoid an anachronism ([RG71] being later than SGA 3), let us sketch a direct proof of point (i). Since $A \rightarrow A'$ is flat, $M' = M \otimes A'$ is a sub- A' -module of $V' = V \otimes A'$, and, since C' is a projective A' -module, $c(M')$ is the smallest subcomodule of V' containing M' . Denote by $V' \otimes A'$ and $A' \otimes V'$ the two $A' \otimes A'$ -comodule structures on $V'' = V' \otimes_{A'} (A' \otimes A')$ obtained by the two base changes $A' \rightrightarrows A' \otimes A'$, $a' \mapsto a' \otimes 1$ and $a' \mapsto 1 \otimes a'$. The A' -comodule V' is endowed with an isomorphism of $A' \otimes A'$ -comodules $\phi : V' \otimes A' \xrightarrow{\sim} A' \otimes V'$, $(x \otimes a') \otimes b' \mapsto b' \otimes (x \otimes a')$, which is a descent datum, i.e. which satisfies $\phi_{31} = \phi_{32} \circ \phi_{21}$.

Since $M' = M \otimes A'$, then ϕ sends $M' \otimes A'$ onto $A' \otimes M'$, and hence $c(M' \otimes A')$ onto $c(A' \otimes M')$. Since the formation of $c(M')$ commutes with base change, one has $c(M' \otimes A') = c(M') \otimes A'$ and $c(A' \otimes M') = A' \otimes c(M')$. One therefore has

$$\phi(c(M') \otimes A') = A' \otimes c(M')$$

and it follows that ϕ endows $c(M')$ with a descent datum. By (fpqc) descent, there is a unique subcomodule $t(M)$ of V such that $c(M') = t(M) \otimes A'$, and $t(M)$ contains M since $t(M) \otimes A'$ contains M' . Moreover, if N is a subcomodule of V containing M , then N contains $t(M)$, since $N \otimes A'$ contains $c(M') = t(M) \otimes A'$. Thus $t(M)$ is the smallest subcomodule of V containing M .

Finally, let $A \rightarrow B$ be a ring morphism. Let $B' = B \otimes A'$ and let M_B (resp. $M'_{B'}$) be the image of $M \otimes B$ in $V_B = V \otimes B$ (resp. of $M' \otimes_{A'} B'$ in $V \otimes B'$); then $M_B \otimes_B B' = M'_{B'}$. On the one hand, the preceding construction, applied to C_B and the morphism $B \rightarrow B'$, gives:

$$c(M_B \otimes_B B') = t(M_B) \otimes_B B' = t(M_B) \otimes A'.$$

On the other hand, since the formation of $c(M')$ commutes with base change, $c(M'_{B'})$ is the image in $V' \otimes_{A'} B' = V \otimes B \otimes A'$ of

$$c(M') \otimes_{A'} B' = t(M) \otimes B \otimes A'.$$

⁽¹¹⁴⁾**N.D.E.** This proposition, taken from [Se68], § 1.5, Prop. 2, has been added.

⁽¹¹⁵⁾**N.D.E.** This lemma, used in the proof of 11.9, has been added.

It follows that $t(M_B)$ is the image in V_B of $t(M) \otimes B$. $\square$

Proposition 11.8.2. (Gruson–Raynaud). *Let $f : A \rightarrow A'$ be a faithfully flat morphism; then f “descends projectivity”, i.e. if M is an A -module and $M \otimes_A A'$ is a projective A' -module, then M is a projective A -module.*

Indeed, by [RG71] II 2.5.1, f “descends the Mittag-Leffler condition”, so, by loc. cit. II 3.1.3, f descends projectivity.⁽¹¹⁶⁾

Proposition 11.9.⁽¹¹⁷⁾ *Let $\mathcal{C}$ be an $\mathcal{O}_S$ -coalgebra, $\mathcal{E}$ a $\mathcal{C}$ -comodule, $\mathcal{F}$ a sub- $\mathcal{O}_S$ -module of $\mathcal{E}$, all quasi-coherent. Suppose given a covering of S by affine open subsets $U_\alpha = \operatorname{Spec} A_\alpha$, and for each α a faithfully flat ring morphism $A_\alpha \rightarrow A'_\alpha$ such that $\Gamma(U_\alpha, \mathcal{C}) \otimes_{A_\alpha} A'_\alpha$ is a projective A'_α -module.**

There is then a smallest quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$, and $t(\mathcal{F})$ is an $\mathcal{O}_S$ -module of finite type if $\mathcal{F}$ is. Moreover, for every base change $S' \rightarrow S$, if $\mathcal{F}'$ denotes the image of $\mathcal{F} \otimes_{\mathcal{O}_S}$ in $\mathcal{E}' = \mathcal{E} \otimes_{\mathcal{O}_S}$, then $t(\mathcal{F}')$ is the image of $t(\mathcal{F}) \otimes_{\mathcal{O}_S}$ in $\mathcal{E}'$, i.e. “the formation of $t(\mathcal{F})$ commutes with base change”.

PROOF.⁽¹¹⁷⁾ For each α , the $\mathcal{O}_{U_\alpha}$ -module $\mathcal{T}_\alpha$ associated to the A_α -module

$$T_\alpha = t(\Gamma(U_\alpha, \mathcal{F}))$$

is, by 11.8.1 (i), the smallest quasi-coherent subcomodule of $\mathcal{E}|_{U_\alpha}$ containing $\mathcal{F}|_{U_\alpha}$, and is an $\mathcal{O}_{U_\alpha}$ -module of finite type if $\mathcal{F}|_{U_\alpha}$ is.

For every α, β , put $U_{\alpha\beta} = U_\alpha \cap U_\beta$. Since the construction of $t(M)$ commutes with base change, one has, for every α, β , canonical isomorphisms of $\mathcal{O}_{U_{\alpha\beta}}$ -modules

$$\phi_{\alpha\beta} : T_\beta \otimes_{A_\beta} \mathcal{O}_{U_{\alpha\beta}} \xrightarrow{\sim} T_\alpha \otimes_{A_\alpha} \mathcal{O}_{U_{\alpha\beta}}$$

which satisfy the cocycle condition $\phi'_{\alpha\gamma} = \phi'_{\alpha\beta} \circ \phi'_{\beta\gamma}$, where $\phi'_{\alpha\gamma}$ (resp. $\dots$) denotes the restriction of $\phi_{\alpha\gamma}$ (resp. $\dots$) to $U_\alpha \cap U_\beta \cap U_\gamma$.

Consequently, the $\mathcal{T}_\alpha$ glue to a quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$. We leave to the reader the verification that $t(\mathcal{F})$ is the smallest quasi-coherent subcomodule of $\mathcal{E}$ containing $\mathcal{F}$, and that its formation commutes with base change. $\square$

Definition 11.9.1.⁽¹¹⁸⁾ Let S be a scheme and $\mathcal{P}$ a quasi-coherent $\mathcal{O}_S$ -module. The following conditions are equivalent:

- (i) For every affine open subset U of S , $\Gamma(U, \mathcal{P})$ is a projective $\mathcal{O}_S(U)$ -module.
- (ii) There is a covering (U_α) of S by affine open subsets such that each $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$ -module.
- (iii) There is a covering (U_α) of S by affine open subsets, and faithfully flat ring morphisms $A_\alpha = \mathcal{O}_S(U_\alpha) \rightarrow A'_\alpha$, such that, for each α , $\Gamma(U_\alpha, \mathcal{P}) \otimes_{A_\alpha} A'_\alpha$ is a projective A'_α -module.

Indeed, it is clear that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). Conversely, if (iii) holds, 11.8.2 implies that each $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$ -module, whence (ii). Finally, suppose (ii) holds and let $V = \operatorname{Spec} A$ be an arbitrary affine open subset; it is covered by finitely many affine open subsets $V_1, \dots, V_n$, where each $V_i = \operatorname{Spec} A_i$ is contained in at least one $V \cap U_\alpha$, so that $\Gamma(V_i, \mathcal{P})$ is a projective A_i -module. Let $A' = A_1 \times \dots \times A_n$; then $A \rightarrow A'$ is faithfully

⁽¹¹⁶⁾**N.D.E.** Let us point out in passing that assertion II 2.5.2 of loc. cit., more general than II 2.5.1, is corrected in the article [Gr73] (this does not affect the case of faithfully flat morphisms).

⁽¹¹⁷⁾**N.D.E.** As mentioned in N.D.E. (112), the statement has been rewritten for an $\mathcal{O}_S$ -coalgebra $\mathcal{C}$ (rather than for an S -group G satisfying the hypotheses indicated in Corollary 11.10). Moreover, the proof has been expanded (the original stated: “($\dots$) the proposition is a consequence of Lemma 11.8.”).

*This is the case, for example, when $\mathcal{C} = \mathcal{A}(G)$, where G is a reductive S -group, as we shall see in Exp. XXII 5.7.8.

⁽¹¹⁸⁾**N.D.E.** This definition, taken from [RG71], bottom of p. 82, has been added. Thus, in Proposition 11.9, the hypothesis is that the coalgebra $\mathcal{C}$ be a locally projective $\mathcal{O}_S$ -module, and this terminology has been used in Corollary 11.10.

flat and $\Gamma(V, \mathcal{P}) \otimes_A A'$ is a projective A' -module. Thus, by 11.8.2, $\Gamma(V, \mathcal{P})$ is a projective A -module.

When these equivalent conditions hold, one says that $\mathcal{P}$ is a *locally projective* $\mathcal{O}_S$ -module.

Corollary 11.10. *Let S be a quasi-compact and quasi-separated scheme, G an S -group, and ρ a linear operation of G on a quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$. Suppose that:*

- (i) G satisfies one of the following conditions:
 - a) G is affine and flat over S ,
 - b) G is flat, quasi-compact and quasi-separated over S , and $\mathcal{E}$ is flat;
- (ii) $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$ -module.

Then $\mathcal{E}$ is the inductive limit of an increasing filtered family of quasi-coherent sub- $\mathcal{O}_S$ -modules of finite type of $\mathcal{E}$, stable under G .

By hypothesis (i) and 11.6.1, $\mathcal{E}$ is endowed with an $\mathcal{A}(G)$ -comodule structure. On the other hand, since S is quasi-compact and quasi-separated, $\mathcal{E}$ is the inductive limit of its quasi-coherent submodules of finite type (EGA I, 9.4.9 and EGA IV₁, 1.7.7). Consequently, the corollary follows from Proposition 11.9, applied to the coalgebra $\mathcal{A}(G)$.

Furthermore, one has the following proposition:

Proposition 11.10.bis. ⁽¹¹⁹⁾ *Let S be a noetherian scheme, G an S -group flat, quasi-compact and quasi-separated over S , $\mathcal{E}$ a quasi-coherent G - $\mathcal{O}_S$ -module, $\mathcal{M}$ a coherent sub- $\mathcal{O}_S$ -module of $\mathcal{E}$. Then $\mathcal{M}$ is contained in a coherent sub- $\mathcal{O}_S$ -module stable under G .*

PROOF. Denote by f the morphism $G \rightarrow S$ and by τ the adjunction morphism $\mathcal{E} \rightarrow f_*f^*(\mathcal{E})$. By 11.6, the G - $\mathcal{O}_S$ -module structure on $\mathcal{E}$ is given by an automorphism θ of the $\mathcal{O}_G$ -module $f^*(\mathcal{E})$ such that the morphism $\delta = \theta \circ \tau$ satisfies conditions (1) and (2) of 11.6. (The situation considered in [Th87] is more general, in that the author considers a G -scheme X and a G -equivariant $\mathcal{O}_X$ -module $\mathcal{E}$ (cf. Exp. I, Section 6); here $X = S$ endowed with the trivial action of G .)

Since S is noetherian, $\mathcal{E}$ is the filtered inductive limit of its coherent submodules $\mathcal{F}_\alpha$ (cf. EGA I, 9.4.9). Then $f^*(\mathcal{E})$ is the filtered inductive limit of the $f^*(\mathcal{F}_\alpha)$, which are submodules of $f^*(\mathcal{E})$ since f is flat. Since moreover f is quasi-compact and quasi-separated, then, by 11.0, the $\mathcal{O}_S$ -module $f_*f^*(\mathcal{E})$ is quasi-coherent and is the filtered inductive limit of the quasi-coherent submodules $f_*f^*(\mathcal{F}_\alpha)$. Consequently, $\mathcal{E}$ is the filtered inductive limit of the quasi-coherent sub- $\mathcal{O}_S$ -modules $\mathcal{E}_\alpha$, where $\mathcal{E}_\alpha$ denotes the inverse image under δ of $f_*f^*(\mathcal{F}_\alpha)$, i.e. the kernel of the composite morphism

$$\bar{\delta}_\alpha: \mathcal{E} \xrightarrow{\delta} f_*f^*(\mathcal{E}) \rightarrow f_*f^*(\mathcal{E}/\mathcal{F}_\alpha).$$

Since $\mathcal{M}$ is coherent and S noetherian, every increasing sequence of submodules of $\mathcal{M}$ is stationary, so $\mathcal{M}$ is contained in some $\mathcal{E}_\alpha$. Let us show that each $\mathcal{E}_\alpha$ is coherent and G -stable.

Let u be the morphism $f^*(\mathcal{E}) \rightarrow \varepsilon_*(\mathcal{E})$ corresponding by adjunction to the identity morphism from $\mathcal{E}$ to $f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$; then the composite morphism

$$\mathcal{E} \xrightarrow{\tau} f_*f^*(\mathcal{E}) \xrightarrow{f_*(u)} f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$$

is the identity (cf. 11.6 (2)). Since one has a commutative diagram:

$$\begin{array}{ccccc} \mathcal{E}_\alpha & \xrightarrow{\delta} & f_*f^*(\mathcal{F}_\alpha) & \xrightarrow{f_*(u)} & \mathcal{F}_\alpha \\ i_\alpha \downarrow & & \downarrow & & \downarrow j_\alpha \\ \mathcal{E} & \xrightarrow{\tau} & f_*f^*(\mathcal{E}) & \xrightarrow{f_*(u)} & \mathcal{E} \end{array}$$

⁽¹¹⁹⁾**N.D.E.** This proposition has been added; it is a special case of [Th87], 1.4–1.5. There the author refers to an argument of Deligne (cf. [Kn71], III Th. 1.1); one may also note the similarity with Serre's argument ([Se68], Prop. 2) recalled in 11.8.bis.

where i_α (resp. j_α) denotes the inclusion of $\mathcal{E}_\alpha$ (resp. $\mathcal{F}_\alpha$) into $\mathcal{E}$, one deduces that i_α factors through j_α , i.e. $\mathcal{E}_\alpha$ is a submodule of $\mathcal{F}_\alpha$, hence coherent.

Finally let us show that $\mathcal{E}_\alpha$ is G -stable (cf. 11.7.bis). Denote by $h : G \rightarrow S$ a second copy of $f : G \rightarrow S$ and consider the following commutative diagram:

$$\begin{array}{ccccc} G \times_S G & \xrightarrow{q} & G & & \\ p \downarrow & \searrow \phi & \downarrow h & & \\ G & \xrightarrow{f} & S & & \end{array}$$

Then the exact sequence

$$0 \rightarrow \mathcal{E}_\alpha \rightarrow \mathcal{E} \xrightarrow{\bar{\delta}_\alpha} h_* h^*(\mathcal{E}/\mathcal{F}_\alpha)$$

gives, since f is flat, the exact sequence

$$(\dagger) \quad 0 \rightarrow f_* f^*(\mathcal{E}_\alpha) \rightarrow f_* f^*(\mathcal{E}) \xrightarrow{f_* f^*(\bar{\delta}_\alpha)} f_* f^* h_* h^*(\mathcal{E}/\mathcal{F}_\alpha).$$

Moreover, since h is quasi-compact and quasi-separated, and f flat, the canonical morphism

$$f_* h_* h^*(\mathcal{E}/\mathcal{F}_\alpha) \rightarrow p_* q^* h^*(\mathcal{E}/\mathcal{F}_\alpha) = p_* \phi^*(\mathcal{E}/\mathcal{F}_\alpha)$$

is an isomorphism, so that the right-hand term in $(\dagger)$ is $\phi_* \phi^*(\mathcal{E}/\mathcal{F}_\alpha)$. Returning to the notation of 11.6 and denoting by π_α the projection $\mathcal{E} \rightarrow \mathcal{E}/\mathcal{F}_\alpha$, one therefore obtains the commutative diagram below, whose bottom row is exact:

$$\begin{array}{ccccc} \mathcal{E}_\alpha & \xrightarrow{\delta} & f_* f^*(\mathcal{E}) & & \\ \delta \downarrow & & \downarrow f_*(f^*(\pi) \boxtimes \delta_G) & & \\ f_* f^*(\mathcal{E}_\alpha) & \rightarrow & f_* f^*(\mathcal{E}) & \xrightarrow[f_* f^*(\bar{\delta}_\alpha)]{} & \phi_* \phi^*(\mathcal{E}/\mathcal{F}_\alpha) \end{array}$$

and since $f_*(f^*(\pi) \boxtimes \delta_G) \circ \delta$ vanishes on $\mathcal{E}_\alpha$, it follows that δ sends $\mathcal{E}_\alpha$ into $f_* f^*(\mathcal{E}_\alpha)$, i.e. $\mathcal{E}_\alpha$ is G -stable. The proposition is proved. $\square$

Remark 11.10.1. ⁽¹²⁰⁾ In 11.8, it is not sufficient to suppose that C is a flat A -module, even if A is a principal ideal ring. Indeed, one has the following counterexamples, which were pointed out to us (independently) by O. Gabber and J.-P. Serre.

(a) Let $(A, \mathfrak{m})$ be a discrete valuation ring, K its fraction field, and G the A -group “extension by zero” of the K -group $(\mathbf{Z}/2\mathbf{Z})_K$. Then the constant group $(\mathbf{Z}/2\mathbf{Z})_A$, and hence also its subgroup G , acts on the free A -module V with basis v_1, v_2 by interchanging v_1 and v_2 . Then Av_1 is not a sub- G -module of V , but it is the intersection of the sub- G -modules $Av_1 + \mathfrak{m}^n v_2$, for $n \in \mathbf{N}^*$. Thus there is no smallest sub- G -module of V containing v_1 .

(b) Let A be an integral ring distinct from its fraction field K , and let G be the affine flat A -group corresponding to the Hopf algebra

$$\mathcal{A}(G) = \{P \in K[T] \mid P(0) \in A\},$$

with comultiplication, resp. counit and antipode, defined by $\Delta(T) = T \otimes 1 + 1 \otimes T$, resp. $\varepsilon(T) = 0$ and $\tau(T) = -T$. (N.B. One therefore has $G \otimes_A K = \mathbf{G}_{a,K}$.)

Let V be the free A -module $Av_1 \oplus Av_2$ and u the endomorphism of V defined by $u(v_1) = v_2$, $u(v_2) = 0$, so that $u^2 = 0$. Then V is endowed with an $\mathcal{A}(G)$ -comodule structure, defined by

$$\mu(m) = 1 \otimes m + T \otimes u(m).$$

The sub- G -modules of V containing v_1 are precisely the sub- A -modules of the form $Av_1 \oplus Iv_2$, for I a nonzero ideal of A ; their intersection is Av_1 , which is not a sub- G -module. Thus there is no smallest sub- G -module of V containing v_1 . (Note moreover that $C = A \oplus K \cdot T$

⁽¹²⁰⁾**N.D.E.** The first part of the original Remark 11.10.1 has been incorporated into 11.8 and 11.9 (replacing “free” by “projective”); the counterexamples below correct the second part.

is a subcoalgebra of $\mathcal{A}(G)$, flat over A , and that the coaction $\mu : V \rightarrow V \otimes \mathcal{A}(G)$ factors through $V \otimes C$, so one also obtains a counterexample for the “very simple” coalgebra C .)

Finally, note that the two preceding examples are special cases of the following construction. Let A be an integral ring distinct from its fraction field K , let B be a Hopf A -algebra free over A . Denote by $\varepsilon : B \rightarrow A$ the augmentation of B and by $I = \text{Ker}(\varepsilon)$ the augmentation ideal. Since $B = A \cdot 1 \oplus I$, one easily sees that $B' = \{b \in B \otimes_A K \mid \varepsilon(b) \in A\}$ is a Hopf subalgebra of B_K . If V is a B -comodule, free with basis $(v_1, \dots, v_n)$ as an A -module, and if $\mu_V(v_1) \neq v_1 \otimes 1$, then Av_1 is not a subcomodule of V , but it is the intersection of the subcomodules $Av_1 + I \cdot V$, for I ranging over the nonzero ideals of A . Thus there is no smallest subcomodule of V containing v_1 .

Proposition 11.11. *Let G be an algebraic group over the field k . The following conditions are equivalent:*

- (i) G is affine.
- (ii) G is quasi-affine.
- (iii) G acts faithfully on a quasi-affine k -scheme X .
- (iv) G acts linearly and faithfully on a k -vector space (not necessarily finite-dimensional).
- (v) G is isomorphic to a closed subgroup of a group $\text{GL}(n)_k$.

PROOF. One has (i) $\Rightarrow$ (ii) trivially, and (ii) $\Rightarrow$ (iii), for G acts faithfully on itself by translations.

Suppose that G acts faithfully on the right on a quasi-affine k -scheme X .⁽¹²¹⁾ Since X is quasi-affine, it is separated and quasi-compact.⁽¹²²⁾ Similarly, G is separated (VIA 0.3) and quasi-compact (for it is of finite type over k). Thus, by 11.1 (c), one has canonical isomorphisms:

$$\mathcal{O}(X \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G) \quad \text{and} \quad \mathcal{O}(X \times G \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G).$$

One deduces that the morphism $\mu : \mathcal{O}(X) \rightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$ induced by the morphism $X \times G \rightarrow X$ endows $\mathcal{O}(X)$ with a right $\mathcal{O}(G)$ -comodule structure, i.e. G acts linearly on the left on the k -algebra $\mathcal{O}(X)$. Consequently, G also acts on the right on the affine envelope $X_{\text{af}} = \text{Spec } \mathcal{O}(X)$ of X , and the canonical morphism $X \rightarrow X_{\text{af}}$ is G -equivariant.

Moreover, since X is quasi-affine, $X \rightarrow X_{\text{af}}$ is an open immersion (EGA II, 5.1.2), so a fortiori G acts faithfully on X_{af} . One therefore obtains that the linear operation (on the left) of G on the k -algebra $\mathcal{O}(X) = \mathcal{O}(X_{\text{af}})$ is faithful. This proves the implication (iii) $\Rightarrow$ (iv).

Suppose now that G acts faithfully on a k -vector space V . Then, by 11.10, V is the inductive limit of finite-dimensional vector subspaces V_i stable under the action of G . If K_i is the kernel of the induced action of G on V_i , i.e. of the morphism $G \rightarrow \underline{\text{Aut}}(V_i)$, then K_i is a closed subscheme of G , and the hypothesis that G acts faithfully is expressed by the fact that the intersection of the K_i is the identity subgroup of G . Since G is noetherian, it follows that one of the K_i is already reduced to the identity group, hence that $G \rightarrow \underline{\text{Aut}}(V_i)$ is a monomorphism. It is therefore a closed immersion by 1.4.2, which proves that (iv) $\Rightarrow$ (v). Since (v) $\Rightarrow$ (i) trivially, this proves 11.11. $\square$

Remark 11.11.1. One can generalize 11.11 as follows. Let S be a locally noetherian regular scheme of dimension ≤ 1 , and G a group scheme flat, quasi-compact and quasi-separated over S . (In this case, $\mathcal{A}(G)$ is a torsion-free $\mathcal{O}_S$ -module, hence flat.)

- (a) One then has the equivalence of the following conditions:⁽¹²³⁾
 - (i) G is affine over S .
 - (ii) G is quasi-affine over S .

⁽¹²¹⁾**N.D.E.** The original has been expanded in what follows; moreover, G has been chosen to act on the right on X , in order to obtain a linear operation of G on the left on $\mathcal{O}(X)$.

⁽¹²²⁾**N.D.E.** Recall that a k -scheme X is said to be quasi-affine if it is isomorphic to a quasi-compact open subset of an affine k -scheme (EGA II, 5.1.1).

⁽¹²³⁾**N.D.E.** The equivalence of these conditions is proved in the additional section 12.

(iii) G acts faithfully on a quasi-affine flat S -scheme.

(iv) G acts linearly and faithfully on a quasi-coherent flat $\mathcal{O}_S$ -module.

(b) If moreover G is of finite type over S and S noetherian, these conditions imply that G is isomorphic to a closed subgroup of some $\underline{\text{Aut}}(\mathcal{E})$, where $\mathcal{E}$ is a locally free $\mathcal{O}_S$ -module of finite type.⁽¹²⁴⁾

Lemma 11.12. *Let k be a field, G an affine k -group. Put $A = \mathcal{A}(G)$. Given $x \in A$, there is a sub- k -algebra of finite type B of A such that $x \in B$, $\Delta(B) \subset B \otimes_k B$ and $u(B) \subset B$, where u denotes the involution of A corresponding to the inversion morphism of G .⁽¹²⁵⁾*

One may suppose $x \neq 0$; then $\Delta(x) \neq 0$ since $(\varepsilon \otimes \text{id})\Delta(x) = x$, where ε denotes the augmentation (counit) of A . Write

$$(1) \quad \Delta(x) = \sum_{j=1}^n e_j \otimes a_j \quad \text{with } n \text{ minimal,}$$

in which case the e_j (resp. a_j) are linearly independent. Complete $(e_1, \dots, e_n)$ to a basis $(e_i)_{i \in I}$ of A and put, for $j \in J = \{1, \dots, n\}$,

$$\Delta(e_j) = \sum_{i \in I} e_i \otimes b_{ij}.$$

Applying $\Delta \otimes \text{id}$ and $\text{id} \otimes \Delta$ to (1), one obtains from axiom (CO 1) of 11.8.0 (see also (HA 1) in I 4.2) the equalities:

$$\sum_{j \in J} e_j \otimes \Delta(a_j) = \sum_{\ell \in J} \Delta(e_\ell) \otimes a_\ell = \sum_{i \in I} e_i \otimes \left(\sum_{\ell \in J} b_{i\ell} \otimes a_\ell \right).$$

Since the e_i are linearly independent, it follows that

$$(2) \quad \forall j \in J, \quad \Delta(a_j) = \sum_{\ell \in J} b_{j\ell} \otimes a_\ell.$$

Let B then be the sub- k -algebra of finite type of A generated by the b_{ij} and the $u(b_{ij})$, for $i, j \in J$. It is clear that $u(B) = B$.

Applying $\Delta \otimes \text{id}$ and $\text{id} \otimes \Delta$ to (2), one again obtains from (CO 1) the equalities:

$$\sum_{\ell \in J} \Delta(b_{j\ell}) \otimes a_\ell = \sum_{i \in J} b_{ji} \otimes \Delta(a_i) = \sum_{i, \ell \in J} b_{ji} \otimes b_{i\ell} \otimes a_\ell,$$

and since the a_ℓ are linearly independent, one deduces that

$$(3) \quad \forall j, \ell \in J, \quad \Delta(b_{j\ell}) = \sum_{i \in J} b_{ji} \otimes b_{i\ell}.$$

Since $\Delta \circ u = (u \times u) \circ v \circ \Delta$, where $v(a \otimes b) = b \otimes a$, one therefore also has

$$(4) \quad \forall j, \ell \in J, \quad \Delta(u(b_{j\ell})) = \sum_{i \in J} u(b_{i\ell}) \otimes u(b_{ji}).$$

Since Δ is an algebra homomorphism, one deduces from (3) and (4) that $\Delta(B) \subset B \otimes_k B$. Finally, axiom (CO 2) of 11.8.0 (see also (HA 2) in I, 4.2) shows that $a_j = \sum_{i \in I} \varepsilon(a_j) b_{ij}$ and $x = \sum_{j \in J} \varepsilon(e_j) a_j$, so that $x \in B$.

Proposition 11.13. *Let k be a field and G an affine k -group with algebra A . Then G is the projective limit of an increasing filtered system of affine k -groups of finite type, whose transition morphisms are faithfully flat.*

⁽¹²⁴⁾**N.D.E.** This is proved, with various generalizations, in the additional section 13.

⁽¹²⁵⁾**N.D.E.** On the one hand, this is generalized in section 13 to the case where G is affine and flat over a regular base of dimension ≤ 2 . On the other hand, in what follows the original has been expanded and corrected.

If B and B' are two subalgebras of A of finite type stable under Δ and u , then so is the subalgebra generated by B and B' . Thus, by Lemma 11.12, A is the inductive limit of an increasing filtered family $(B_i)_{i \in I}$ of subalgebras of finite type stable under Δ and u . Then each B_i , endowed with the restriction of u and the morphism $B_i \rightarrow B_i \otimes_k B_i$ obtained from Δ , is a Hopf algebra, so by I 4.2 it is the algebra of an affine k -group G_i , of finite type over k . Finally, since $A = \varinjlim B_i$, one has $G = \varprojlim G_i$ (cf. EGA IV₃ 8.2.3). The transition morphisms are faithfully flat by the following lemma:

Lemma 11.14. *Let k be a field, $u : G \rightarrow H$ a morphism between affine k -groups of finite type, and $u^\flat : B \rightarrow A^{(126)}$ the corresponding morphism of k -algebras. For u to be faithfully flat, it is necessary and sufficient that $u^\flat$ be injective.*

The condition is obviously necessary (cf. EGA 0_I 6.6.1). Let us show that it is sufficient. Put $N = \text{Ker } u$. Then, by VIA, 3.3.2 and 5.4.1, G/N is a k -group of finite type and u factors as $G \xrightarrow{p} G/N \xrightarrow{v} H$, where p is faithfully flat and v is a closed immersion. Thus, since H is an affine scheme, G/N is an affine scheme and the morphism $v^\flat : B \rightarrow \mathcal{O}(G/N)$ is surjective (cf. EGA I 4.2.3). Now, since $u^\flat$ is supposed injective and $u^\flat = p^\flat \circ v^\flat$, then $v^\flat$ is also injective: it is therefore an isomorphism, as is v , and since p is faithfully flat, so is u .

Definition 11.15. ⁽¹²⁷⁾ Let k be a field, G a quasi-compact k -group and V a k -vector space endowed with a k -linear action of G , hence with an $\mathcal{A}(G)$ -comodule structure $\delta : V \rightarrow V \otimes \mathcal{A}(G)$, by 11.6.1. Let $v \in V$ be nonzero. The following conditions are equivalent:

- (i) There is $\lambda \in \mathcal{A}(G)$ (necessarily unique) such that $\delta(v) = v \otimes \lambda$.
- (ii) For every k -algebra R and every $g \in G(R)$, one has $g \cdot v \in Rv$ (i.e., there is $f \in R$, necessarily unique, such that one has in $V \otimes R$ the equality $g \cdot v = v \otimes f$).

Indeed, it is clear that (i) $\Rightarrow$ (ii). Conversely, if (ii) holds and one applies it to $R = \mathcal{A}(G)$ and $g = \text{id}_{\mathcal{A}(G)}$, one obtains that there is a unique $\lambda \in \mathcal{A}(G)$ such that $\delta(v) = v \otimes \lambda$.

If v satisfies these conditions, one says that v is a *semi-invariant vector under G* , and λ is the *weight of v* ; one will also say that “ v is a semi-invariant of weight λ ”.

Denote by Δ the comultiplication of $\mathcal{A}(G)$; then the equality

$$v \otimes \lambda \otimes \lambda = (\delta \otimes \text{id})(\delta(v)) = (\text{id} \otimes \Delta)(\delta(v)) = v \otimes \Delta(\lambda)$$

implies that $\Delta(\lambda) = \lambda \otimes \lambda$. Consequently, λ defines a morphism of Hopf algebras

$$\mathcal{A}(\mathbf{G}_{m,k}) = k[T, T^{-1}] \rightarrow \mathcal{A}(G), \quad T \mapsto \lambda,$$

and hence a morphism of k -groups $\lambda : G \rightarrow \mathbf{G}_{m,k}$, i.e. λ is a character of G , called the *character associated to the semi-invariant vector v* .

Lemma 11.16.0. ⁽¹²⁸⁾ *Let k be a field, H an affine k -group, V an H -module of dimension n and U a vector subspace of V of dimension d . Consider the line $D = \bigwedge^d U \subset \bigwedge^d V$. For U to be stable under H , it is necessary and sufficient that D be so.*

Since necessity is clear, let us prove sufficiency. One may suppose $d < n$. Let $(e_1, \dots, e_d)$ be a basis of U ; complete it to a basis $(e_1, \dots, e_n)$ of V . For every k -algebra R , $V_R = V \otimes R$ is a free R -module and one has

$$U_R = \{v \in V_R \mid v \wedge (e_1 \wedge \dots \wedge e_d) = 0\}$$

(for, for $i > d$, the $e_i \wedge e_1 \wedge \dots \wedge e_d$ are linearly independent in $\bigwedge_R^{d+1} V_R$). Since $H(R)$ acts on $\bigwedge_R^\bullet(V_R)$ by

$$h(x_1 \wedge \dots \wedge x_s) = h(x_1) \wedge \dots \wedge h(x_s),$$

⁽¹²⁶⁾**N.D.E.** The morphism $B \rightarrow A$ corresponding to $u : \text{Spec}(A) \rightarrow \text{Spec}(B)$ has been denoted by $u^\flat$ (instead of u°).

⁽¹²⁷⁾**N.D.E.** The hypothesis that G be quasi-compact has been added, and the equivalence of conditions (i) and (ii) expanded.

⁽¹²⁸⁾**N.D.E.** This lemma, taken from [DG70], § II.2, 3.5, has been added.

it follows that, if $H(R)$ stabilizes $D_R = Re_1 \wedge \cdots \wedge e_d$, it also stabilizes U_R .

Theorem 11.16. (Chevalley). *Let k be a field, G an affine algebraic k -group, H a closed subgroup scheme of G .⁽¹²⁹⁾ Then there is a finite-dimensional G -module V and a line D of V such that $H = \underline{\text{Norm}}_G(D)$, i.e. such that for every k -algebra R ,*

$$H(R) = \{g \in G(R) \mid g(D_R) = D_R\}.$$

In other words, there are finitely many elements $a_1, \dots, a_n \in \mathcal{A}(G)$ which are semi-invariants, all of the same weight λ , for the “right” action of H (i.e. $a_i(gh) = \lambda(h)a_i(g)$, for every k -algebra R and $g \in G(R)$, $h \in H(R)$), such that H is the largest closed subgroup scheme of G under which the a_i are semi-invariants.

Denote by Δ (resp. ε) the comultiplication (resp. augmentation) of $A = \mathcal{A}(G)$. Then $H = \text{Spec}(A/I)$, for some ideal I of A contained in $\text{Ker } \varepsilon$ and such that $\Delta(I) \subset I \otimes A + A \otimes I$. Let $B = A/I$ and π be the projection $A \rightarrow B$. Consider the left action of H on A given by $(h\varphi)(g) = \varphi(gh)$; the corresponding B -comodule structure is given by:

$$\bar{\Delta}: A \xrightarrow{\Delta} A \otimes A \xrightarrow{\text{id}_A \otimes \pi} A \otimes B.$$

Then I is a sub- H -module of A , since $\bar{\Delta}(I) \subset I \otimes B$.

On the other hand, A is a k -algebra of finite type, hence noetherian, so I has a finite set of generators $(x_1, \dots, x_r)$. By 11.8, the x_i are contained in a sub- G -module V finite-dimensional over k . Then $W = V \cap I$ is a finite-dimensional H -module, whose dimension we shall denote by d . Since V contains all the x_i , W generates the ideal I .

Put $E = \wedge^d V$, let $(w_1, \dots, w_d)$ be a basis of W , and let $\mathcal{B} = (e_0, \dots, e_n)$ be a basis of E containing the vector $e_0 = w_1 \wedge \cdots \wedge w_d$. The operation of G on V canonically determines an operation of G on E , hence an A -comodule structure $\rho: E \rightarrow E \otimes A$. For $j = 0, \dots, n$, put $\rho(e_j) = \sum_{i=1}^n e_i \otimes b_{ij}$; we saw in the proof of 11.12 that then

$$(*) \quad \Delta(b_{ij}) = \sum_{\ell=0}^n b_{i\ell} \otimes b_{\ell j}.$$

Put $a_i = b_{i0}$, i.e. if $(e_0^*, \dots, e_n^*)$ is the dual basis of $\mathcal{B}$, the a_i are the “matrix coefficients” $g \mapsto e_i^*(ge_0)$. On the other hand, the operation of H on E corresponds to:

$$\bar{\rho}: E \xrightarrow{\rho} E \otimes A \xrightarrow{\text{id}_E \otimes \pi} E \otimes B.$$

Since W is stable under H , then e_0 is semi-invariant under H , so $\bar{\rho}(e_0) = e_0 \otimes \pi(a_0)$ and $a_i = b_{i0}$ belongs to I for $i = 1, \dots, n$. Substituting this into (*), one obtains $\bar{\Delta}(a_i) = a_i \otimes \pi(a_0)$ for $i = 0, \dots, n$, i.e. the a_i are semi-invariants under H of weight $\pi(a_0)$. (Moreover, replacing E if necessary by the sub- G -module generated by e_0 (cf. 11.8), one may suppose that the a_i are linearly independent.)

Conversely, let $H' = \text{Spec } B'$, where $B' = A/I'$, be a closed subgroup scheme of G under which each a_i is semi-invariant, of weight $\lambda_i \in B'$ (this is the case, in particular, if e_0 is invariant under H' of weight λ'). Let us show that $H' = H$. Denote by π' the projection $A \rightarrow B'$; the hypothesis implies that

$$a_i \otimes \lambda_i = (\text{id}_A \otimes \pi')\Delta(b_{i0}) = \sum_{\ell=0}^n b_{i\ell} \otimes \pi'(a_\ell),$$

whence $\lambda_i = \pi'(a_0)$ and $a_\ell \in I'$ for $\ell = 1, \dots, n$, and therefore e_0 is semi-invariant under H' . By Lemma 11.16.0, this implies that W is stable under H' . Since the ideal I is generated by W , it is therefore also stable under H' , and hence

$$\Delta(I) \subset I \otimes A + A \otimes I'.$$

⁽¹²⁹⁾**N.D.E.** On the one hand, recall that every subgroup scheme of G is closed (1.4.2). On the other hand, the result has been stated in the usual form: “ H is the stabilizer of a line in a representation of G ”, while retaining the original formulation in terms of a sequence $a_1, \dots, a_n$ of semi-invariants in $\mathcal{A}(G)$.

Since $I \subset \text{Ker } \varepsilon$ and $(\varepsilon \otimes \text{id}_A) \circ \Delta = \text{id}_A$, it follows that $I \subset I'$, whence $H' \subset H$.

Lemma 11.17.0. ⁽¹³⁰⁾ *Let k be an algebraically closed field, G a reduced affine algebraic k -group, V a finite-dimensional G -module over k , and $v \in V$. Let E be the vector subspace of V generated by the vectors gv , for $g \in G(k)$. Then E is the smallest sub- G -module of V containing v , and thus the morphism $G \times E \rightarrow V$, $(g, x) \mapsto gx$ factors through E .*

PROOF. By 11.8, one knows that there is a smallest sub- G -module U of V containing v : if $\mu : V \rightarrow V \otimes \mathcal{A}(G)$ denotes the comodule structure and one writes $\mu(v) = \sum_{i=1}^n v_i \otimes a_i$ with the a_i linearly independent, one has $U = \text{Vect}(v_1, \dots, v_n)$. It is clear that U contains E , and the morphism $G \times U \rightarrow V$ factors through U .

Conversely, the inverse image of E under the morphism $\mu_v : g \mapsto gv$ is a closed subset of G which contains the rational points; now these are dense in G , since G is of finite type over k (cf. EGA IV₃, 10.4.8), so $\mu_v^{-1}(E) = G$ and thus, since G is reduced, μ_v factors through E , whence $E = U$. $\square$

Theorem 11.17. (Chevalley). *Let k be a field, G an affine k -group (not necessarily of finite type), and N a closed subgroup scheme of G invariant in G ; then the (fpqc) quotient sheaf G/N is representable by an affine k -group.* ⁽¹³¹⁾

Suppose first G of finite type. By VIA 3.2 and 5.2, the (fpqc) quotient sheaf G/N is representable by a k -group Q ; it is therefore a matter of showing that Q is affine. The proof proceeds in several steps; suppose first k algebraically closed. ⁽¹³²⁾

(a) Suppose moreover G reduced and connected and N reduced. By 11.16, there is a G -module V finite-dimensional over k , and a line $D = ke_0$ such that $\text{Norm}_G(D) = N$; in particular N acts on D through a character $\chi : N \rightarrow \mathbf{G}_{m,k}$.

Fix $h \in N(k)$. For every $g \in G(k)$, one has $hge_0 = g(g^{-1}hg)e_0 = \chi(g^{-1}hg)ge_0$, so $\chi(g^{-1}hg)$ is an eigenvalue of h . Thus the continuous map $\phi : G(k) \rightarrow k$, $g \mapsto \chi(g^{-1}hg)$, takes only finitely many values, and since $G(k)$ is irreducible (for it is dense in G), one therefore has $\phi(g) = \phi(e) = \chi(h)$ for every $g \in G(k)$ and hence

$$\chi(g^{-1}hg) = \chi(h), \quad \forall g \in G(k), h \in N(k).$$

Let E be the vector subspace of V generated by the vectors ge_0 , for $g \in G(k)$; by Lemma 11.17.0, it is the sub- G -module of V generated by e_0 .

By the above, the two morphisms $N \times E \rightarrow E$, $(h, x) \mapsto hx$ and $(h, x) \mapsto \chi(h)x$, coincide on the set of rational points $(N \times E)(k) = N(k) \times E(k)$, which is dense in $N \times E$. Since $N \times E$ is reduced (and E separated), these two morphisms are therefore equal, so N acts on E by homotheties. Consequently, N is contained in the kernel K of the morphism $\rho : G \rightarrow \text{GL}(\text{End}_k(E))$, defined by $\rho(g)(u) = gug^{-1}$, for every $g \in G(R)$ and $u \in \text{End}_R(E \otimes R)$ (R a k -algebra). On the other hand, if $g \in K(R)$ then $g(Re_0) = Re_0$, whence $g \in N(R)$. This shows that $N = K$. Then, by VIA 5.4.1, the morphism $G/N \rightarrow \text{GL}(\text{End}_k(E))$ is a closed immersion, and thus G/N is affine.

(b) Suppose now G and N reduced, with G not necessarily connected. Put $N' = N \cap G^0$; then G^0/N' is affine by (a). On the other hand, NG^0 is an invariant subgroup of G and G/NG^0 , being a quotient of the finite constant group G/G^0 (cf. VIA, 5.5.1), is likewise a finite constant group. Thus G/N is the direct sum of the fibers of the morphism $G/N \rightarrow G/NG^0$, all isomorphic to NG^0/N , hence to G^0/N' , by VIA, 5.3.3. Thus G/N is affine.

⁽¹³⁰⁾**N.D.E.** This lemma, taken from the proof of thm. 5.6 of [DG70], § III.3, has been added (by abuse of notation, the same letter E denotes a k -vector space and the k -scheme in modules $\mathbf{W}(E) = \text{Spec } S(E^*)$).

⁽¹³¹⁾**N.D.E.** For another proof of this theorem, which does not use the results of VIA, see [Ta72], Th. 5.2 (see also Remark 11.18.5).

⁽¹³²⁾**N.D.E.** In what follows the original has been expanded (and the erroneous assertion $(G/N)_{\text{red}} = G_{\text{red}}/N_{\text{red}}$ corrected), following [DG70], § III.3, 5.6.

(c) Suppose G reduced, and N arbitrary. The morphism $G \times N \rightarrow N$, $(g, h) \mapsto ghg^{-1}$, induces a morphism $(G \times N)_{\text{red}} \rightarrow N_{\text{red}}$; now, since G is reduced and k algebraically closed, one has $(G \times N)_{\text{red}} = G \times N_{\text{red}}$, so N_{red} is an invariant subgroup of G . (N.B. This fails when G is not reduced, cf. VIA, 0.2.)

Thus, by (a), $G' = G/N_{\text{red}}$ is affine. On the other hand, by VIA 5.6.1, $N' = N/N_{\text{red}}$ is a finite k -group, so, by Theorem 4.1 of Exp. V, the quotient $G/N = G'/N'$ is affine.

(d) For arbitrary G and N , the equivalence relation obtained from $G \times N \rightrightarrows G$ by the base change $G_{\text{red}} \rightarrow G$ is:

$$G_{\text{red}} \times N' \rightrightarrows G_{\text{red}}, \quad \text{where} \quad N' = N \cap G_{\text{red}}.$$

Since the underlying spaces are the same (and since the quotient is the quotient ringed space), the morphism $G_{\text{red}}/N' \rightarrow G/N$ is a homeomorphism. Since G_{red}/N' is reduced (for $p: G_{\text{red}} \rightarrow G_{\text{red}}/N'$ is faithfully flat), it follows that $(G/N)_{\text{red}}$ is identified with G_{red}/N' , which is affine, by (c). Since G/N is of finite type over k (cf. VIA, 3.3.2), this implies, by EGA I, 5.1.10, that G/N is affine.

Finally, for arbitrary k , let $\bar{k}$ be an algebraic closure of k . Then, by 9.2 (v), $(G \otimes_k \bar{k})/(N \otimes_k \bar{k})$ is isomorphic to $(G/N) \otimes_k \bar{k}$, so, since the former is affine, so is the latter, and thus G/N is also affine by (fpqc) descent (cf. EGA IV₂, 2.7.1). This proves 11.17 when G is of finite type. To extend this to the general case, we shall need the following lemma.⁽¹³³⁾

Lemma 11.17.1. *Let $(C_i \rightarrow A_i)_{i \in I}$ be a filtered inductive system of ring morphisms, all faithfully flat. Then $A = \varinjlim A_i$ is faithfully flat over $C = \varinjlim C_i$.*

PROOF. By [BAC] § I.3, Prop. 9, for a ring morphism $B \rightarrow B'$ to be faithfully flat, it is necessary and sufficient that it be injective and B'/B be a flat B -module. Since each $C_i \rightarrow A_i$ is faithfully flat, one therefore has exact sequences

$$0 \rightarrow C_i \rightarrow A_i \rightarrow A_i/C_i \rightarrow 0$$

and A_i/C_i is a flat C_i -module, so $(A_i/C_i) \otimes_{C_i} C$ is a flat C -module. Since inductive limits are exact and commute with tensor products, one obtains an exact sequence

$$0 \rightarrow C \rightarrow A \rightarrow A/C \rightarrow 0$$

as well as an isomorphism

$$\varinjlim ((A_i/C_i) \otimes_{C_i} C) = (A/C) \otimes_C C = A/C,$$

from which one deduces that A/C is a flat C -module. Thus A is faithfully flat over C . $\square$

Let us now return to the proof of 11.17 in the general case, i.e. when G is not supposed of finite type. Put $\mathcal{A}(G) = A$ and $\mathcal{A}(N) = B = A/J$. By 11.13, A is the inductive limit of an increasing filtered family $(A_i)_{i \in I}$ of Hopf subalgebras of finite type, so G is the projective limit of the affine algebraic k -groups $G_i = \text{Spec}(A_i)$. Denote by Δ (resp. τ) the comultiplication (resp. antipode) of A , and $\Delta^2 = (\Delta \otimes \text{id}_A) \circ \Delta$.

For every i , $B_i = A_i/(J \cap A_i)$ is a quotient Hopf algebra of A_i , so $N_i = \text{Spec}(B_i)$ is a closed subgroup of G_i . Moreover, since N is invariant in G , the morphism $G \times N \rightarrow G$ defined by $(g, n) \mapsto gng^{-1}$ factors through N , and this is equivalent to saying that the pair (A, J) satisfies the following property:

$$(m_{13} \circ (\Delta \otimes \tau) \circ \Delta)(J) \subset A \otimes_k J$$

where m_{13} denotes the map $a_1 \otimes a_2 \otimes a_3 \mapsto a_1 a_3 \otimes a_2$. It follows that $(A_i, A_i \cap J)$ satisfies the analogous property, hence that N_i is invariant in G_i . On the other hand, one has $\varinjlim B_i = B$ and thus $\varinjlim N_i = N$.

By what we saw above, each (fpqc) quotient sheaf G_i/N_i is representable by an affine k -group $Q_i = \text{Spec}(C_i)$. Put $C = \varinjlim C_i$. One therefore has a filtered projective system of

⁽¹³³⁾**N.D.E.** This lemma, taken from [DG70], § III.3, 7.1, has been added.

affine k -groups Q_i ; its projective limit $Q = \varprojlim Q_i$ is the k -group $\mathrm{Spec}(C)$ (cf. EGA IV₃, 8.2.3). One then has an exact sequence of k -groups:

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q.$$

Let us show that Q represents the (fpqc) quotient sheaf of G by N ; for this, it suffices to verify that the morphism $G \longrightarrow Q$ is covering for the (fpqc) topology (cf. IV, 3.3.2.1 and 5.1.7.1). Now each morphism $G_i \longrightarrow Q_i$ is faithfully flat (cf. 9.2 (xi)), in other words A_i is faithfully flat over C_i ; since $A = \varprojlim A_i$ and $C = \varprojlim C_i$, Lemma 11.17.1 implies that A is faithfully flat over C , so that $G \longrightarrow Q$ is a faithfully flat morphism. Since this morphism is affine, it is quasi-compact, hence covering for the (fpqc) topology. This completes the proof of Theorem 11.17.

11.18. — Complements.⁽¹³⁴⁾ Moreover, one deduces from 11.17 (and its proof) the following results, taken from [DG70], III § 3.7. Let k be a field. Let us begin with the following lemma (cf. [An73], 2.3.3.2), which will be useful later (cf. 12.10).

Lemma 11.18.1. *Let $u : G \longrightarrow G'$ be a morphism of k -groups, $N = \mathrm{Ker}(u)$. Suppose G' affine and G of finite type.*

- (i) *The morphism $\tau : G/N \longrightarrow G'$ is a closed immersion. In particular, G/N is affine.*
- (ii) *If moreover the morphism $u^\sharp : \mathcal{O}(G') \longrightarrow \mathcal{O}(G)$ is injective, then τ is an isomorphism. (And thus G' is of finite type and u faithfully flat.)*

Indeed, one knows (11.13) that G' is the projective limit of a filtered system of affine algebraic k -groups G'_i . Denote by u_i the composite morphism $G \longrightarrow G' \longrightarrow G'_i$ and by N_i its kernel. Then the N_i form a decreasing filtered system of closed subgroups of G , whose intersection is N . Since G is noetherian, there is an index i such that $N = N_i$. Since G and G_i are of finite type, then, by VIA, 3.2 and 5.4.1, the quotient G/N is a k -group of finite type and u_i is the composite of the projection $p : G \longrightarrow G/N$, which is faithfully flat, and a closed immersion $\tau_i : G/N \hookrightarrow G_i$.

Consider then the following commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{u} & G' \\ p \downarrow & \nearrow \tau & \downarrow q_i \\ G/N & \xrightarrow{\tau_i} & G_i. \end{array}$$

Since $q_i \circ \tau = \tau_i$ is a closed immersion and q_i separated (G' being separated, cf. VIA, 0.3), then τ is a closed immersion (cf. EGA I, 5.4.4). It follows that G/N is affine, and the morphism $\tau^\sharp : \mathcal{O}(G') \longrightarrow \mathcal{O}(G/N)$ is surjective, whence (i).

If moreover $u^\sharp : \mathcal{O}(G') \longrightarrow \mathcal{O}(G)$ is injective, then so is $\tau^\sharp$, so $\tau^\sharp$ is an isomorphism, and hence τ too (since G' and G/N are affine). This proves (ii).

Theorem 11.18.2. *Let G and G' be two affine k -groups with algebras A and A' , and let $u : G \longrightarrow G'$ be a morphism of k -groups, $N = \mathrm{Ker}(u)$, and $\varphi : A' \longrightarrow A$ the morphism induced by u .*

- (i) *If φ is injective, then u is faithfully flat and identifies G' with G/N .*
- (ii) *One has $G/N = \mathrm{Spec}(B)$, where $B = \varphi(A')$, and thus u is the composite of the faithfully flat morphism $G \longrightarrow G/N$, corresponding to the inclusion $B \hookrightarrow A$, and the closed immersion $G/N \hookrightarrow G'$, corresponding to the surjection $A' \longrightarrow B$. Moreover, N is defined in G by the ideal AB^+ , where B^+ denotes the augmentation ideal of B .*
- (iii) *In particular, if u is a monomorphism, it is a closed immersion.*

PROOF. (i) Suppose φ injective and identify A' with a Hopf subalgebra of A . By 11.13, A is a filtered union of Hopf subalgebras $A_i = \mathcal{O}(G_i)$ of finite type over k ; denote by

⁽¹³⁴⁾**N.D.E.** This subsection has been added.

$G'_i = \text{Spec}(A'_i)$, where $A'_i = A' \cap A_i$, and by N_i the kernel of the morphism $G_i \rightarrow G'_i$ induced by the inclusion $A'_i \hookrightarrow A_i$. By the preceding lemma, one has $G_i/N_i \simeq G'_i$, and one therefore obtains, for every i , an exact sequence

$$1 \longrightarrow N_i \longrightarrow G_i \xrightarrow{p_i} G'_i \longrightarrow 1$$

where p_i is faithfully flat. Since $G = \varprojlim_i G_i$ and $G' = \varprojlim_i G'_i$, one therefore obtains an exact sequence

$$1 \longrightarrow N \longrightarrow G \xrightarrow{p} G/N$$

where we have put $N = \varprojlim_i N_i$. Moreover, by Lemma 11.17.1, p is faithfully flat (and affine), so G' represents the (fpqc) quotient sheaf of G by N . This proves (i).

In the general case, $B = \varphi(A')$ is a Hopf subalgebra of A ; denote by H the k -group $\text{Spec}(B)$ and by N' the kernel of the morphism $G \rightarrow H$ induced by the inclusion $B \hookrightarrow A$. By (i), H is identified with G/N' , and u is therefore the composite of the projection $G \rightarrow G/N'$ and the closed immersion $G/N' \hookrightarrow G'$ induced by the surjection $A' \rightarrow B$. It follows that $N' = N$. Moreover, by 9.2 (ii), one has a cartesian square:

$$\begin{array}{ccc} N & \longrightarrow & G \\ \downarrow & & \downarrow p \\ \text{Spec}(k) & \xrightarrow{\varepsilon} & G' \end{array}$$

where ε is the identity section of G' , which corresponds to the augmentation morphism $B \rightarrow k$. It follows that N is defined in G by the ideal AB^+ . This proves (ii), and (iii) follows from it. $\square$

Remark 11.18.3. Let G be an affine k -group and N an invariant k -subgroup. Since the morphism $p : G \rightarrow G/N$ is faithfully flat and quasi-compact, then, by IV 3.3.3.2, $\mathcal{O}(G/N)$ is the subalgebra of $\mathcal{O}(G)$ formed by functions φ which are right N -invariant, i.e. satisfy $\varphi(gh) = \varphi(g)$, for every k -scheme S and $g \in G(S)$, $h \in N(S)$. Denoting by J the ideal of $A = \mathcal{O}(G)$ defining N , this is equivalent to saying that $\Delta(\varphi) - \varphi \otimes 1 \in \mathcal{O}(G) \otimes J$, where Δ is the comultiplication of A .

The preceding theorem can then be reformulated in terms of Hopf algebras as follows.

Corollary 11.18.4. *Let k be a field, A a commutative Hopf k -algebra, $G = \text{Spec}(A)$.*

(i) *If B is a Hopf subalgebra of A , then A is faithfully flat over B .*
 (ii) *The map $N \mapsto \mathcal{O}(G/N)$ is a bijection between the set of invariant subgroups of G and that of Hopf subalgebras of A ; the inverse map is given by $B \mapsto \text{Spec}(A/AB^+)$. Moreover, if J is the ideal of A defining N , one has*

$$\mathcal{O}(G/N) = \{x \in A \mid \Delta(x) - x \otimes 1 \in A \otimes J\}.$$

Remarks 11.18.5. (a) A consequence of the preceding theorem is that the category of commutative affine k -groups is abelian. For this, as well as other results on affine k -groups, we refer to [DG70], § III.3, 7.4 to 7.8.

(b) Finally, let us point out that M. Takeuchi gave another proof of results 11.17 to 11.18.4, cf. [Ta72], § 5; he moreover strengthened 11.18.4 (i) above by showing that A is even a projective B -module, cf. [Ta79], Th. 5 (see also [MW94], Th. 3.6).

12. Complements on G_{af} and “anti-affine” groups

⁽¹³⁵⁾ Let us begin with the following lemma, which extends 11.18.1 to the case where G is not supposed of finite type.⁽¹³⁶⁾

Lemma 12.1. *Let k be a field, $u : G \rightarrow H$ a monomorphism of k -groups, with H affine. Suppose u quasi-compact. Then u is a closed immersion.*

PROOF. By (fpqc) descent, one may suppose k algebraically closed. By VIA, 6.4, the closed image I of u is a closed subgroup scheme of H , hence still affine. Thus, replacing H by I , one may suppose u schematically dominant. Since k is algebraically closed, $H' = H_{\text{red}}$ is a subgroup scheme of H ; denote by $G' = G \times_H H'$; then the morphism $u' : G' \rightarrow H'$ obtained from u by base change is a quasi-compact monomorphism, and is dominant (the underlying continuous map being the same for u and u'). Thus, by VIA, 6.2, u' is faithfully flat; it is therefore a faithfully flat quasi-compact monomorphism, hence an isomorphism (cf. IV 1.14).

Thus $u : G \rightarrow H$ is a homeomorphism, hence affine by 2.9.1. Thus G is affine, and therefore u is a closed immersion by 11.18.2 (iii). $\square$

Theorem 12.2. *Let G be an algebraic k -group. Denote by ρ the canonical morphism $G \rightarrow G_{\text{af}}$ and by N its kernel.*

(i) *The canonical morphism $G/N \rightarrow G_{\text{af}}$ is an isomorphism, and thus G_{af} is an affine algebraic group, and ρ is faithfully flat.*

(ii) *One has a canonical isomorphism $(G/N)_{\text{af}} = G_{\text{af}}$.*

(iii) *N is a characteristic subgroup of G .*

(iv) *$\mathcal{O}(N) = k$.*

(v) *N is smooth, connected and commutative.*

PROOF. Point (i) is a special case of 11.18.1, and point (ii) follows from the universal properties of G_{af} and $(G/N)_{\text{af}}$.

Let us prove (iii). For an arbitrary k -scheme $\pi : S \rightarrow \text{Spec } k$, consider the cartesian square:

$$\begin{array}{ccc} G_S & \longrightarrow & G \\ q \downarrow & & \downarrow p \\ S & \xrightarrow{\pi} & \text{Spec } k, \end{array}$$

since p is quasi-compact and separated and π flat, then, by EGA III 1.4.15 and EGA IV₁ 1.7.21, one has $q_*(\mathcal{O}_{G_S}) = \pi^*(\mathcal{O}(G)) = \pi^*(\mathcal{O}(G_{\text{af}}))$, and thus, by EGA II, 1.5.2, one has $(G_S)_{\text{af}} = (G_{\text{af}})_S$, so N_S , being the kernel of the canonical morphism $G_S \rightarrow (G_S)_{\text{af}}$, is invariant under every automorphism of G_S , i.e. N is a characteristic subgroup of G .

To prove (iv), put $N' = \text{Ker}(N \rightarrow N_{\text{af}})$; by (ii), it is an invariant subgroup of G . Since N is algebraic (being a closed subgroup of G), then, by (i), $N/N' \cong N_{\text{af}}$; moreover, by VIA, 3.2 and 5.3.2, one has an isomorphism of k -groups

$$(G/N')/N_{\text{af}} \cong (G/N')/(N/N') \cong G/N.$$

Since N_{af} is affine, the projection $G/N' \rightarrow G/N$ is too, by 9.2 (vii), and since $G/N = G_{\text{af}}$ is affine, then G/N' is also. Thus, by the universal property of G_{af} , the projection $p' : G \rightarrow G/N'$ factors through $G_{\text{af}} = G/N$, whence $N \subset N'$ and therefore $N = N'$. Thus N_{af} is the trivial group, whence $\mathcal{O}(N) = k$.

Finally, assertion (v) is a consequence of the following lemma. $\square$

Lemma 12.3. *Let k be a field and N an algebraic k -group such that $\mathcal{O}(N) = k$. Then N is smooth, connected and commutative.*

⁽¹³⁵⁾**N.D.E.** This section has been added.

⁽¹³⁶⁾**N.D.E.** This lemma was communicated to us by M. Raynaud; it will be used in the proof of Proposition 12.9.

Indeed, one may suppose k algebraically closed. Then $H = N_{\text{red}}^0$ is a subgroup k -scheme of N , and the quotient k -scheme $X = N/H$ is finite (hence affine) over k , by VIA, 5.5.1 and 5.6.1. On the other hand, since $p : N \rightarrow X$ is faithfully flat, one has $\mathcal{O}(X) \subset \mathcal{O}(N) = k$. It follows that $N = N_{\text{red}}^0$, so N is smooth (VIA 1.3.1) and connected.

Let Z then be the center of N . By 6.2.6, N/Z is affine, and one obtains as above that $\mathcal{O}(N/Z) = k$, whence $N = Z$. This proves 12.3 and completes the proof of 12.2.

Let us also mention, without proof, the following theorem. (Recall that an abelian variety over a field k is a proper, smooth and connected group k -scheme.)

Theorem 12.4. (Chevalley). *Let k be a perfect field and G a smooth connected algebraic k -group. Then there is an affine smooth connected k -subgroup L , invariant in G , such that the quotient G/L is an abelian variety. Moreover, L is unique and its formation commutes with extension of the base field.*

Remarks 12.5. (1) This theorem was announced in 1953 by C. Chevalley, who published his proof in 1960 ([Ch60]). Meanwhile, other proofs had been obtained independently by I. Barsotti and M. Rosenlicht ([Ba55, Ro56]); see [Se99] for historical comments.

(2) A modern version (i.e. in the language of schemes) of Chevalley's proof was given by B. Conrad ([Co02]). (Note that in loc. cit., “algebraic group” means a smooth connected group k -scheme.)

(3) On the other hand, a modern version of Rosenlicht's proof was given by Ngô B.-C. in a course at Orsay in 2005–2006.

(4) If one omits the hypothesis that k be perfect, there still exists a smallest connected invariant affine subgroup L (not necessarily smooth) such that G/L is an abelian variety ([BLR], § 9.2, Thm. 1).

(5) One can also omit the hypothesis that G be smooth over k : indeed, by VIIA, 8.3, there is an integer $n \geq 1$ such that the quotient $G' = G/(\mathbb{F}_k^n G)$ is smooth; then G' contains a subgroup L' as in (4) above, and the inverse image of L' in G again has the same properties. Thus, for every connected algebraic group G over a field k , there is an exact sequence

$$1 \rightarrow H \rightarrow G \rightarrow A \rightarrow 1$$

where H is an affine k -group and A an abelian k -variety. Moreover, by [Per76], Cor. 4.2.9, one has such an exact sequence for every connected k -group G (not necessarily algebraic).

(6) Let k be an algebraically closed field and G the semidirect product of an elliptic curve E by the constant k -group $\{\pm 1\}_k$, for the action defined by $(-1) \cdot x = -x$; in this case, if L is a closed invariant subgroup of G such that G/L is connected, then $L = G$.

Remark 12.6. Following [Br09], one will say that a k -group N is *anti-affine* if $\mathcal{O}(N) = k$. By 12.3 and 12.4, if k is perfect, every anti-affine algebraic k -group is an extension of an abelian variety by an affine smooth connected commutative algebraic k -group. For the precise structure of anti-affine algebraic groups over a perfect field, and various consequences, see the recent articles by M. Brion and C. & F. Sancho de Salas ([Br09, SS09]).

To finish this section, we shall prove two results due to M. Raynaud, the first being Remark 11.11.1, the second Proposition 2.1 of Exp. XVII, Appendix III. We shall need the following lemma,⁽¹³⁷⁾ which improves (for a complete discrete valuation ring R) the flatness criteria given in [BAC], § III.5.

Lemma 12.7. *Let R be a discrete valuation ring, K its fraction field, π a uniformizer. Let A be a flat R -algebra and M an A -module flat over R . Suppose that:*

- (i) $M/\pi M$ is a flat module over $\bar{A} = A/\pi A$,
- (ii) $M \otimes_R K$ is a flat module over $A \otimes_R K$.

Then M is a flat A -module.

⁽¹³⁷⁾**N.D.E.** It is a version, improved by O. Gabber, of a statement communicated by M. Raynaud.

PROOF. By the flatness criterion in the nilpotent case (cf. [BAC], III § 5.2, Th. 1), $M/\pi^n M$ is a flat module over $A/\pi^n A$, for every $n \in \mathbf{N}^*$. It then follows from [RG71], II Lemma 1.4.2.1, that M is a flat A -module. For the reader's convenience, let us briefly indicate the proof. Put $s = \pi^n$ and $P = M/sM$. Since P is flat over A/sA and the latter has projective dimension 1 over A , the spectral sequence of composite functors implies that P has Tor-dimension ≤ 1 over A . Now, since M is R -flat and hence has no π -torsion, M_K/M is the inductive limit of the A -modules $\pi^{-n}M/M \simeq M/\pi^n M$, and thus M_K/M also has Tor-dimension ≤ 1 . Since one has the exact sequence $0 \rightarrow M \rightarrow M_K \rightarrow M_K/M \rightarrow 0$ and by hypothesis M_K is flat over A_K , hence over A , it follows that M is flat.

For completeness, let us also indicate the following simpler proof, pointed out by O. Gabber. Let I be an ideal of finite type of A ; we must show that the morphism $u : M \otimes_A I \rightarrow M$ is injective. By hypothesis (ii), $u \otimes_R K$ is injective, so $\text{Ker}(u)$ is an R -module of π -torsion. It therefore suffices to show that the π -torsion part of $M \otimes_A I$ is zero; now this is a quotient of $\text{Tor}_1^A(M, I/\pi I)$, as one sees by tensoring with M the exact sequence:

$$0 \rightarrow I \xrightarrow{\pi} I \rightarrow I/\pi I \rightarrow 0.$$

On the other hand, since M has no π -torsion (being flat over R), one obtains that

$$\text{Tor}_i^A(M, \bar{A}) = 0 \quad \text{for every } i \geq 1.$$

Consequently, if $(P_\bullet)$ is a projective resolution of the A -module M , then $(P_\bullet \otimes_A \bar{A})$ is a projective resolution of the $\bar{A}$ -module $\bar{M} = M/\pi M$, and thus for every $\bar{A}$ -module N , one has $\text{Tor}_i^A(M, N) = \text{Tor}_i^{\bar{A}}(\bar{M}, N)$, and this is zero for $i \geq 1$ since $\bar{M}$ is flat over $\bar{A}$. One therefore has $\text{Tor}_1^A(M, I/\pi I) = 0$, which proves the lemma. $\square$

Remark 12.8. Let S be a locally noetherian regular scheme of dimension 1, and X an S -scheme flat, quasi-separated and quasi-compact over S . Then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module. Indeed, one may suppose S local; denote by s its closed point, i the inclusion $X_s \hookrightarrow X$, and π a uniformizer of $R = \mathcal{O}(S)$; since X is flat over S , one has an exact sequence of sheaves

$$0 \rightarrow \mathcal{O}_X \xrightarrow{\pi} \mathcal{O}_X \rightarrow i_*(\mathcal{O}_{X_s}) \rightarrow 0$$

and thus, taking global sections, one obtains that $\mathcal{A}(X)$ is an R -module without π -torsion, hence flat.⁽¹³⁸⁾ One obtains moreover that the morphism from $\mathcal{A}((X_{\text{af}})_s) = \mathcal{A}(X)/\pi\mathcal{A}(X)$ to $\mathcal{A}(X_s)$, induced by the morphism $X \rightarrow X_{\text{af}}$, is injective.

We can now prove the following proposition (cf. Remark 11.11.1).

Proposition 12.9. *Let S be a locally noetherian regular scheme of dimension ≤ 1 , G a group S -scheme flat, quasi-separated and quasi-compact over S . Then the following conditions are equivalent:*

- (i) G is affine over S .
- (ii) G is quasi-affine over S .
- (iii) G acts faithfully on a quasi-affine flat S -scheme X .
- (iv) G acts linearly and faithfully on a quasi-coherent module $\mathcal{E}$ flat over S .
- (v) The morphism $\rho : G \rightarrow G_{\text{af}}$ is a monomorphism.

PROOF. The implication (i) $\Rightarrow$ (ii) is evident, as is (ii) $\Rightarrow$ (iii) (take $X = G$).

Suppose (iii) holds. Since $\mathcal{A}(G)$ and $\mathcal{A}(X)$ are flat $\mathcal{O}_S$ -modules, one obtains, proceeding as in 11.11, that G acts faithfully (on the right) on X_{af} , and thus acts linearly and faithfully (on the left) on the flat quasi-coherent $\mathcal{O}_S$ -module $\mathcal{A}(X)$.

On the other hand, if (iv) holds, then, by 11.6.1 (ii), the monomorphism $G \rightarrow \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{E})$ factors through G_{af} , so $G \rightarrow G_{\text{af}}$ is a monomorphism.

⁽¹³⁸⁾**N.D.E.** This holds more generally if S is locally noetherian regular of dimension ≤ 2 , cf. [Ray70a], VII 3.2.

Finally, suppose (v) holds and let us show that $\rho : G \rightarrow G_{\text{af}}$ is an isomorphism. Replacing S by one of its connected components, one may suppose S irreducible, with generic point η . Since the formation of G_{af} commutes with flat base changes, one has $(G_{\text{af}})_{\eta} = (G_{\eta})_{\text{af}}$, and hence the morphism $G_{\eta} \rightarrow (G_{\eta})_{\text{af}}$ is a monomorphism, hence a closed immersion, by 12.1, hence an isomorphism since $\mathcal{O}((G_{\eta})_{\text{af}}) = \mathcal{O}(G_{\eta})$. If $S = \text{Spec}(\kappa(\eta))$, we are done; one may therefore suppose $\dim S = 1$.

Let s then be a closed point of S ; let us show that $G_s \rightarrow (G_{\text{af}})_s$ is an isomorphism and that ρ is flat at all points of G_s . For this, one may suppose S local, with closed point s . The morphism $\rho_s : G_s \rightarrow (G_{\text{af}})_s$ obtained by base change is a monomorphism, hence a closed immersion by 12.1, so the morphism $\mathcal{O}((G_{\text{af}})_s) \rightarrow \mathcal{O}(G_s)$ induced by ρ_s is surjective. Now, by the preceding remark, it is also injective, hence an isomorphism. (In particular, ρ is therefore surjective.)

It then follows from Lemma 12.7 that $\rho : G \rightarrow G_{\text{af}}$ is faithfully flat. Since G is quasi-compact over S and G_{af} separated over S , then ρ is also quasi-compact (cf. EGA I, 6.6.4). Consequently, ρ is a faithfully flat quasi-compact monomorphism, hence an isomorphism. This proves the proposition. $\square$

Finally, let us prove Prop. 2.1 of Exp. XVII, Appendix III; taking account of [Per76], Cor. 4.2.5, “quasi-compact and quasi-separated” has been substituted for “of finite type” among the hypotheses (if one supposes G of finite type, one may use 12.1 instead of loc. cit.).

Proposition 12.10. *Let S be a locally noetherian regular scheme of dimension ≤ 1 , G a group S -scheme flat, quasi-compact and quasi-separated.*

(i) *The canonical morphism $\rho : G \rightarrow G_{\text{af}}$ is faithfully flat and quasi-compact. Consequently, G_{af} represents the (fpqc) quotient sheaf of G by $N = \text{Ker}(\rho)$.*

(ii) *If moreover G is of finite type over S , then ρ is of finite presentation and G_{af} represents the (fppf) quotient sheaf of G by N and is of finite type over S .*

(iii) *Suppose moreover G_{η} affine for every maximal point η of S . Then N is an étale S -group, and is the identity group if G is separated over S .*

PROOF. First, since G_{af} is affine and hence separated over S , then ρ is quasi-compact (cf. EGA I, 6.6.4) and the kernel $N = \text{Ker}(\rho)$ is a closed subgroup of G . Moreover, replacing S by one of its connected components, one may suppose S irreducible, with generic point η .

Note that to prove (i) and (ii), it suffices to show that ρ is faithfully flat; for then, by Exp. IV, 5.1.7.1, G_{af} represents the (fpqc) quotient sheaf of G by N , and if moreover G is of finite type over S , hence of finite presentation (S being locally noetherian), then, by 9.2 (xiii), ρ is of finite presentation (as is $G_{\text{af}} \rightarrow S$) and thus ρ is covering for the (fppf) topology.

One may therefore suppose $S = \text{Spec}(R)$, where R is a discrete valuation ring (if $\dim S = 1$) or the field $\kappa(\eta)$ (if $\dim S = 0$). Denote by s the closed point of S . Since the formation of G_{af} commutes with flat base changes, the canonical morphism $G_{\eta} \rightarrow (G_{\eta})_{\text{af}}$ is identified with the morphism $\rho_{\eta} : G_{\eta} \rightarrow (G_{\text{af}})_{\eta}$, and since $\mathcal{O}((G_{\text{af}})_{\eta}) = \mathcal{O}((G_{\eta})_{\text{af}}) = \mathcal{O}(G_{\eta})$, then ρ_{η} is faithfully flat by [Per76], 4.2.5 (see also the addition VIA, 6.6). If $\dim S = 1$, one similarly obtains that ρ_s is faithfully flat, since the morphism $\mathcal{O}((G_{\text{af}})_s) \rightarrow \mathcal{O}(G_s)$ is injective, by Remark 12.8. Thus, by Lemma 12.7, ρ is faithfully flat. This proves (i) and (ii). In particular, N is flat over S .

Suppose now G_{η} affine. Since ρ_{η} coincides with the canonical morphism $G_{\eta} \rightarrow (G_{\eta})_{\text{af}}$, its kernel N_{η} is the identity group. Let us show that N is étale over S . Since N is flat over S , it remains to see that N_s is étale over $\kappa(s)$, for every point $s \neq \eta$ of S . There is nothing to show if $S = \text{Spec}(\kappa(\eta))$, so one may suppose $S = \text{Spec}(R)$, where R is a discrete valuation ring. Let s be the closed point of S , K the fraction field of R , and π a uniformizer. Let $x \in N_s$ and U_x be an affine open neighborhood of x in N ; since N is flat over S , then $U_x \cap N_{\eta}$ is nonempty, hence equal to $\{\varepsilon(\eta)\}$, where ε denotes the identity section. Thus $A = \mathcal{O}(U_x)$ is a flat R -algebra such that $A_K = K$ and $\pi^{-1} \notin A$ (since π belongs to the maximal ideal of $\mathcal{O}_{U,x}$). It follows that $A = R$, and hence the projection $U_x \rightarrow S$ is an isomorphism. This

proves that N is étale over S ; if moreover G is separated over S , then the inverse isomorphism $S \rightarrow U_x$ equals the identity section (since they coincide on the dense open subset $\{\eta\}$ of $S = \operatorname{Spec}(R)$), so N is the identity group. The proposition is proved. $\square$

One obtains in particular the following corollary, two other proofs of which are found in [An73], Prop. 2.3.1 and [PY06], Prop. 3.1.

Corollary 12.10.1. *Let R be a discrete valuation ring, K its fraction field, G a group R -scheme separated, flat and of finite type over R . If G_K is affine, then G is affine.*

Remarks 12.10.2. (a) On the one hand, O. Gabber indicated to us examples where G is a flat group of finite type over a discrete valuation ring, whose generic fiber is an abelian variety, and where the kernel N of $G \rightarrow G_{\text{af}}$ is not smooth.

(b) On the other hand, let us point out that M. Raynaud gave an example, for S the affine space of dimension 2 over a field k , of a smooth quasi-affine group S -scheme, with affine connected fibers, which is not affine over S , cf. [Ray70a], § VII.3, p. 116.

13. Flat affine groups over a regular base of dimension ≤ 2

⁽¹³⁹⁾ Let us begin by noting that the well-known argument showing that every affine algebraic group over a field k is linear, as well as Lemma 11.12, extends to the case of a group scheme G affine, flat and of finite type over a noetherian regular base scheme S of dimension ≤ 2 . For S of dimension ≤ 1 , this proves point (b) of Remark 11.11.1. The extension to the case $\dim S = 2$ rests on the following lemma, which was communicated to us by O. Gabber.

Lemma 13.1. *Let S be a noetherian normal scheme, $\mathcal{A}$ a flat quasi-coherent $\mathcal{O}_S$ -module, $\mathcal{F}$ a sub- $\mathcal{O}_S$ -module of finite type of $\mathcal{A}$, $\mathcal{F}^{**}$ its bidual, and U the flatness locus of $\mathcal{F}$, i.e. the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$ -module.*

- (i) *U is an open subset of S and $\mathcal{F}^{**} = j_* j^*(\mathcal{F})$, where j denotes the inclusion $U \hookrightarrow S$.*
- (ii) *The canonical morphism $j_*(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathcal{A} \rightarrow j_*(\mathcal{E} \otimes_{\mathcal{O}_U} j^*(\mathcal{A}))$ is an isomorphism, for every quasi-coherent $\mathcal{O}_U$ -module $\mathcal{E}$.*
- (iii) *In particular, $\mathcal{V} = j_* j^*(\mathcal{F})$ is a submodule of $\mathcal{A} = j_* j^*(\mathcal{A})$, and the canonical morphism $\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A} \rightarrow j_* j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A})$ is an isomorphism.*

PROOF. Replacing S by one of its connected components, one may suppose S integral. By EGA IV₂, 2.1.12, the flatness locus of $\mathcal{F}$, i.e. the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$ -module, is an open subset U of S ; denote by j the inclusion $U \hookrightarrow S$. Since $\mathcal{A}_s$ is flat, hence torsion-free, so is $\mathcal{F}_s$, so U contains all points of codimension ≤ 1 . Consequently, by [BAC], VII, § 4.2, cor. of th. 1, one has $\mathcal{F}^{**} = j_* j^*(\mathcal{F})$, and one therefore obtains a monomorphism $\mathcal{F}^{**} \rightarrow j_* j^*(\mathcal{A})$.

The proof of (ii) is analogous to that of EGA III, 1.4.15, recalled in 11.0. On the other hand, since S is normal, the morphism $\mathcal{O}_S \rightarrow j_* j^*(\mathcal{O}_S)$ is an isomorphism (cf. EGA IV₂, 5.8.6 and 5.10.5). By (ii) applied to $\mathcal{E} = \mathcal{O}_U$, one therefore has $\mathcal{A} = j_* j^*(\mathcal{A})$. Finally, since $j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}) = j^*(\mathcal{F}) \otimes_{\mathcal{O}_U} j^*(\mathcal{A})$, the last assertion of (iii) follows from (ii) applied to $\mathcal{E} = j^*(\mathcal{F})$. The lemma is proved. $\square$

Furthermore, recall that an R -module of finite type M is said to be reflexive if the canonical morphism from M to its bidual M^{**} is an isomorphism. When R is a noetherian regular ring of dimension ≤ 2 , this implies that M is projective. Indeed, for every R -module of finite type N , consider a resolution $L_1 \rightarrow L_0 \rightarrow N \rightarrow 0$, where L_0 and L_1 are finite free R -modules; then one has an exact sequence

$$0 \rightarrow N^* \rightarrow L_0^* \rightarrow L_1^* \rightarrow Q \rightarrow 0,$$

⁽¹³⁹⁾**N.D.E.** This section has been added.

where Q denotes the cokernel of $L_0^* \rightarrow L_1^*$, and since R has homological dimension ≤ 2 (cf. [BAC], X § 4.2, cor. 1 of th. 1), it follows that N^* is projective.

Proposition 13.2. *Let S be a noetherian regular scheme of dimension ≤ 2 , G an affine flat S -group, $\mathcal{A}(G)$ its affine algebra.*

(i) *If G is of finite type over S , it is isomorphic to a closed subgroup of $H = \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{V})$, for some locally free $\mathcal{O}_S$ -module $\mathcal{V}$ of finite rank. If moreover S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d , whence $H = \text{GL}_{d,S}$.*

(ii) *$\mathcal{A}(G)$ is the filtered inductive limit of flat sub- $\mathcal{O}_S$ -Hopf algebras of finite type.*

PROOF. Let $\mathcal{B}$ be a sub- $\mathcal{O}_S$ -algebra of finite type of $\mathcal{A}(G)$. Since every coherent module on an open subset of S extends to a coherent module on S (cf. EGA I, 9.4.5), there is a coherent sub- $\mathcal{O}_S$ -module $\mathcal{M}$ of $\mathcal{B}$ which generates $\mathcal{B}$ as an $\mathcal{O}_S$ -algebra (loc. cit., 9.6.5). By 11.10.bis, $\mathcal{M}$ is contained in a coherent G -stable sub- $\mathcal{O}_S$ -module $\mathcal{F}$. (N.B. Since G is here affine over S , the proof in loc. cit. takes a simpler form: one can there replace $f_*f^*(\mathcal{E})$ by $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$, etc.)

Let j be the inclusion $U \hookrightarrow S$, where U denotes the flatness locus of $\mathcal{F}$. By Lemma 13.1 and the recollection following it, $\mathcal{V} = j_*j^*(\mathcal{F})$ is a locally free sub- $\mathcal{O}_S$ -module of $\mathcal{A}(G)$, and since the canonical morphism

$$\mathcal{V} \otimes \mathcal{A}(G) \longrightarrow j_*j^*(\mathcal{F} \otimes \mathcal{A}(G))$$

is an isomorphism, then $\mathcal{V}$ is a sub- G -module of $\mathcal{A}(G)$. The action of G on $\mathcal{V}$ then induces a morphism of affine S -groups $\rho_{\mathcal{V}} : G \rightarrow H = \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{V})$, hence a morphism of Hopf $\mathcal{O}_S$ -algebras $\phi_{\mathcal{V}} : \mathcal{A}(H) \rightarrow \mathcal{A}(G)$. Denote by $\mathcal{A}_{\mathcal{V}}$ the image of $\phi_{\mathcal{V}}$; it is the affine algebra of a closed subgroup $G_{\mathcal{V}}$ of H , which is the closed image of $\rho_{\mathcal{V}}$. Let us show that $\mathcal{A}_{\mathcal{V}}$ contains $\mathcal{B}$.

Since the question is local on S , one may suppose $S = \text{Spec}(R)$ and $V = \Gamma(S, \mathcal{V})$ a free R -module with basis $v_1, \dots, v_n$; in this case $H \simeq \text{GL}_{n,R}$ and $\mathcal{A}(H)$ is generated as an R -algebra by the “matrix coefficients” c_{ij} and the element d^{-1} , where d denotes the determinant. Let Δ (resp. ε) be the comultiplication (resp. augmentation) of A . For $j = 1, \dots, n$, write $\Delta(v_j) = \sum_{i=1}^n v_i \otimes a_{ij}$; then $a_{ij} = \phi(c_{ij})$ belongs to $\text{Im}(\phi)$. On the other hand, since V is a sub- R -module of A , one can use the equality $(\varepsilon \otimes \text{id}_A) \circ \Delta = \text{id}_A$, which implies that $v_j = \sum_i \varepsilon(v_i) a_{ij}$ belongs to $\text{Im}(\phi)$. Since V contains a set of generators of $B = \Gamma(S, \mathcal{B})$, it follows that $B \subset \text{Im}(\phi)$.

If G is of finite type over S , one may take $\mathcal{B} = \mathcal{A}(G)$, and ϕ is then surjective, so the morphism of S -groups $G \rightarrow H = \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{V})$ is a closed immersion.

If moreover S is affine, there is a locally free $\mathcal{O}_S$ -module $\mathcal{V}'$ of finite rank such that $\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^d$ as $\mathcal{O}_S$ -modules. Considering $\mathcal{V}'$ as a trivial G -module, one can replace $\mathcal{V}$ by $\mathcal{O}_S^d$, and one thus obtains that G is a closed subgroup of $\text{GL}_{d,S}$.

Finally, let us return to the case of an arbitrary affine flat S -group G . By EGA I, 9.4.9, $\mathcal{A}(G)$ is the union of its coherent sub- $\mathcal{O}_S$ -modules $\mathcal{M}$, hence also of the sub- $\mathcal{O}_S$ -Hopf algebras $\mathcal{A}_{\mathcal{V}}$ as above, whence (ii). $\square$

Example 13.3. Let R be a discrete valuation ring, with uniformizer π and fraction field K . Consider the filtered projective system of R -groups:

$$\cdots \longrightarrow \mathbf{G}_{a,R} \xrightarrow{\times \pi} \mathbf{G}_{a,R} \xrightarrow{\times \pi} \mathbf{G}_{a,R}$$

(which corresponds to the inductive system $R[X_0] \rightarrow R[X_1] \rightarrow R[X_2] \rightarrow \cdots$, whose transition morphisms are given by $X_n = \pi X_{n+1}$). Its projective limit G is a flat affine group R -scheme, not of finite type, whose special fiber is trivial and whose generic fiber is $\mathbf{G}_{a,K}$; the affine R -algebra $\mathcal{A}(G)$ is the subring of $K[X]$ formed by polynomials whose constant coefficient belongs to R . (N.B. We have already encountered this example in Remark 11.10.1.) Note that G represents the functor which associates to every R -algebra B the set of sequences

$(x_n)_{n \in \mathbf{N}}$ of elements of B , where $x_n = \pi x_{n+1}$ for every n . (In particular, each x_n is indefinitely π -divisible.)

Let S now be a noetherian scheme such that every coherent $\mathcal{O}_S$ -module is the quotient of a locally free $\mathcal{O}_S$ -module of finite type; this is the case, for example, if S is a separated noetherian regular scheme, cf. SGA 6, Exp. II, 2.1.1 and 2.2.7. (One can show that every noetherian regular scheme of dimension ≤ 1 also has this property; on the other hand, it does not hold when S is the affine plane over a field k with its origin doubled, cf. loc. cit., 2.2.7.2.)

Definition 13.4. Let G be an affine flat S -group. Following R. W. Thomason ([Th87], 2.1), let us say that the pair (G, S) possesses the *equivariant resolution property*, or satisfies (RE), if, for every coherent G - $\mathcal{O}_S$ -module $\mathcal{F}$, there is a locally free G - $\mathcal{O}_S$ -module $\mathcal{E}$ of finite rank, and a G -equivariant epimorphism $\mathcal{E} \rightarrow \mathcal{F}$.

In loc. cit., Th. 3.1, Thomason proves the result below under the hypothesis that G be essentially free over S (cf. the remark below). Gabber indicated to us the simpler proof below, which does not use this hypothesis.

Proposition 13.5. *Let S be a noetherian scheme and G an affine flat S -group of finite type, with affine algebra $\mathcal{A}(G)$. Suppose that (G, S) satisfies (RE). Then:*

(i) *G is isomorphic to a closed subgroup of $H = \underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathcal{V})$, for some locally free $\mathcal{O}_S$ -module $\mathcal{V}$ of finite rank.*

(ii) *If moreover S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some n , whence $H = \mathrm{GL}_{d,S}$.*

The proof is analogous to that of 13.2. As in loc. cit., there is a coherent G -stable sub- $\mathcal{O}_S$ -module $\mathcal{F}$ which generates $\mathcal{A} = \mathcal{A}(G)$ as an $\mathcal{O}_S$ -algebra. Replacing S by one of its connected components, one may suppose S connected. By hypothesis (RE), there is a locally free G - $\mathcal{O}_S$ -module $\mathcal{E}$ of rank n , and an epimorphism of $\mathcal{A}$ -comodules $\pi : \mathcal{E} \rightarrow \mathcal{F}$. Put $H = \underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathcal{E})$; it is a group S -scheme locally isomorphic to $\mathrm{GL}_{n,S}$. The action of G on $\mathcal{E}$ induces a morphism of affine S -groups $\rho : G \rightarrow H$, corresponding to a morphism of Hopf $\mathcal{O}_S$ -algebras $\phi : \mathcal{A}(H) \rightarrow \mathcal{A}(G)$. Let us show that ρ is a closed immersion.

Since the question is local on S , one may suppose $S = \mathrm{Spec}(R)$ and $V = \Gamma(S, \mathcal{E})$ a free R -module with basis $v_1, \dots, v_n$; in this case $H \simeq \mathrm{GL}_{n,R}$ and $B = \Gamma(S, \mathcal{A}(H))$ is generated as an R -algebra by the “matrix coefficients” c_{ij} and the element d^{-1} , where d denotes the determinant. Let Δ (resp. ε) be the comultiplication (resp. augmentation) of $A = \Gamma(S, \mathcal{A}(G))$, let $\mu : V \rightarrow V \otimes_R A$ be the A -comodule structure on V , and let $F = \Gamma(S, \mathcal{F})$. For $j = 1, \dots, n$, write

$$\mu(v_j) = \sum_{i=1}^n v_i \otimes a_{ij}$$

then $\phi(c_{ij}) = a_{ij}$. On the other hand, since $\pi : V \rightarrow F$ is a morphism of A -comodules, one has

$$\sum_{i=1}^n \pi(v_i) \otimes a_{ij} = (\pi \otimes \mathrm{id}_A)(\mu(v_j)) = \Delta(\pi(v_j))$$

and thus

$$\pi(v_j) = (\varepsilon \otimes \mathrm{id}_A)\Delta(\pi(v_j)) = \sum_{i=1}^n \varepsilon\pi(v_i)a_{ij} = \phi\left(\sum_{i=1}^n \varepsilon\pi(v_i)c_{ij}\right).$$

Thus $\phi(B)$ contains $\pi(V) = F$, which generates A as an R -algebra, and hence ϕ is surjective. This shows that ρ is a closed immersion, whence assertion (i).

If moreover S is affine, there is a locally free $\mathcal{O}_S$ -module $\mathcal{V}'$ of finite rank such that $\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^d$ as $\mathcal{O}_S$ -modules. Considering $\mathcal{V}'$ as a trivial G -module, one can replace $\mathcal{V}$ by $\mathcal{O}_S^d$, and one thus obtains that G is a closed subgroup of $\mathrm{GL}_{d,S}$. The proposition is proved.

Remarks 13.6. (a) For completeness, let us briefly sketch Thomason’s argument ([Th87], Th. 3.1), retaining the preceding notation. The G -equivariant epimorphism $\pi : \mathcal{E} \rightarrow \mathcal{F}$

induces a closed immersion $\tau : G \longrightarrow V(\mathcal{E})$ such that $\tau(g'g) = \tau(g') \cdot g$ (N.B.: G acts on the right on $V(\mathcal{E})$), and one has an isomorphism $H \simeq \underline{\text{Aut}}_{\mathcal{O}_S}(V(\mathcal{E}))^{\text{op}}$, which is compatible with the operations of G on the left on $\mathcal{E}$ and on the right on $V(\mathcal{E})$. Let N then be the strict transporter $\text{Transp}_H^{\text{str}}(\tau(G), \tau(G))$; when G is essentially free over S , it follows from 6.2.4 e) that N is a closed subgroup scheme of H , hence affine over S . Moreover, ρ factors as a morphism of S -groups $\rho' : G \longrightarrow N$. On the other hand, for every $S' \longrightarrow S$ and $h \in N(S')$, put $\pi(h) = \tau(1) \cdot h$ (where 1 is the identity element of $G(S')$); this defines a morphism of S -schemes $\pi : N \longrightarrow \tau(G)$ which is a retraction of ρ' (when one identifies G with $\tau(G)$). Since N is separated over S , it follows that ρ' is a closed immersion.

(b) It seems that the proof of [Th87], Th. 3.1 requires the hypothesis that G be essentially free over S , which does not appear in loc. cit. (the author invoking instead the fact that H is essentially free). However, this hypothesis holds when G is reductive (cf. Exp. XXII 5.7.8), hence holds in all cases considered in loc. cit., Cor. 3.2. In particular, Thomason proves in loc. cit., 2.5, that if S is separated noetherian regular of dimension ≤ 2 , and G affine, of finite presentation, and such that $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$ -module, then (G, S) satisfies (RE); by 13.5, this gives 13.2 under a slightly more restrictive hypothesis.

To conclude, let us point out that the proof of [Th87], 2.5 can be slightly simplified as follows. (For brevity, we place ourselves in the situation where $S = \text{Spec}(R)$ is affine.)

Proposition 13.7. *Let R be a noetherian regular ring of dimension ≤ 2 , C an R -coalgebra projective as an R -module, and F a C -comodule of finite type over R . Then F is the quotient of a C -comodule V projective of finite type over R .*

PROOF. Replacing $\text{Spec}(R)$ by one of its connected components, one may suppose R integral, with fraction field K . Denote by Δ (resp. ε) the comultiplication (resp. augmentation) of C and by $\rho : F \longrightarrow F \otimes C$ the comodule structure on F . Let $\pi : W \longrightarrow F$ be a surjective morphism, where W is a free R -module of finite rank. Endow $W \otimes C$ with the comodule structure defined by $\text{id}_W \otimes \Delta$, and similarly for $F \otimes C$. Then $\rho : F \longrightarrow F \otimes C$ is a morphism of C -comodules, which admits $\text{id}_F \otimes \varepsilon$ as a section.

Let W' be the C -comodule defined by the cartesian square below:

$$\begin{array}{ccc} W' & \longrightarrow & F \\ \downarrow & & \downarrow \rho \\ W \otimes C & \xrightarrow{\pi \otimes \text{id}_C} & F \otimes C \end{array}$$

i.e. W' is identified with the kernel of the morphism $W \otimes C \longrightarrow (F \otimes C)/\rho(F)$, and the projection $\pi' : W' \longrightarrow F$, given by $x \longmapsto (\pi \otimes \varepsilon)(x)$, is surjective. Since F is an R -module of finite type, there is a subcomodule V' of W' of finite type over R such that $\pi'(V') = F$. Since $W \otimes C$ has no R -torsion, neither does V' , so, replacing F by V' , one may suppose from the outset that F is torsion-free.

Applying the preceding construction to this new F , one obtains V' as above. Consider then the subcomodule V , kernel of the morphism

$$W \otimes C \longrightarrow E = \frac{W \otimes C \otimes K}{V' \otimes K}.$$

Then V contains V' and $Q = (W \otimes C)/V$ is a sub- R -module of the K -vector space E ; put $Q' = E/Q$. Since $W \otimes C$ and E are flat R -modules, one obtains, for every R -module N ,

$$\text{Tor}_1^R(V, N) \simeq \text{Tor}_2^R(Q, N) \simeq \text{Tor}_3^R(Q', N)$$

and since R is regular of dimension ≤ 2 , the right-hand term is zero. This shows that V is a flat R -module. Finally, let us show that V is an R -module of finite type. Put $M = W \otimes C$; it follows from the definition that V/V' is isomorphic to the R -torsion submodule $(M/V')_{\text{tors}}$ of M/V' .

Since M is a projective R -module, there is a projective R -module P such that $M \oplus P$ is a free R -module L . Then $(M/V')_{\text{tors}} \simeq (L/V')_{\text{tors}}$. On the other hand, since V' is of finite type, there is a direct summand $L' \simeq R^n$ of L such that $V' \subset L'$, and one therefore also has $(L/V')_{\text{tors}} \simeq (L'/V')_{\text{tors}}$, and the latter is of finite type since (L'/V') is. Consequently, V is a flat R -module of finite type, hence projective of finite type (R being noetherian). Proposition 13.7 is proved. $\square$

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⁽¹⁴⁰⁾N.D.E. Additional references cited in this Exposé.

Exposé VIIA. Infinitesimal study of group schemes

by P. Gabriel

In Exposé II we had confined ourselves to the study of differential invariants of the first order, and we had not approached certain phenomena peculiar to characteristic $p > 0$ or to characteristic 0. Our purpose in part A of this exposé is to fill this gap.

Moreover, the infinitesimal study of arbitrary order of a group scheme is related to that of the associated formal group; the purpose of the second part of this exposé is to present the first definitions and properties concerning formal groups.

A) Differential operators and Lie p -algebras^{*}

1. Differential operators

In this section, as well as in sections 2 and 3, S denotes a fixed scheme and the products considered are cartesian products in the category of S -schemes.⁽¹⁾ If X is an S -scheme, we denote by $p_{X/S}$, p_X , or simply p the structural morphism from X to S .

1.1. — Let $u : Y \rightarrow X$ be a morphism of S -schemes, and equip the direct image $u_*(\mathcal{O}_Y)$ of the structure sheaf of Y with the $\mathcal{O}_X$ -module structure induced by u . The sheaf $\mathcal{H} = \mathcal{H}om_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y))$ of homomorphisms of $p_X^{-1}(\mathcal{O}_S)$ -modules from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$ is thus naturally equipped with an $\mathcal{O}_X$ -bimodule structure: if U is an open subset of X , and f and d are sections of $\mathcal{O}_X$ and $\mathcal{H}$ over U , then fd and df are respectively the morphisms $g \mapsto fd(g)$ and $g \mapsto d(fg)$ from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$. From now on we shall write $(\text{ad } f)(d)$ instead of $fd - df$.

Definition 1.1.1. An S -deviation of order $\leq n$ is by definition a pair $D = (u, d)$ consisting of a morphism of S -schemes $u : Y \rightarrow X$ and a morphism of $p_X^{-1}(\mathcal{O}_S)$ -modules $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$ such that, for every open subset U of X and every sequence of $n + 1$ sections $f_0, \dots, f_n \in \mathcal{O}_X(U)$, the following equality holds in $\text{Hom}_{p_U^{-1}(\mathcal{O}_S)}(\mathcal{O}_U, u_*(\mathcal{O}_Y)|_U)$:

$$(*_n) \quad (\text{ad } f_0)(\text{ad } f_1) \cdots (\text{ad } f_n)(d) = 0.$$

^{*}Part A of the present exposé had not been treated seriously in the oral lectures.

⁽¹⁾**N.D.E.** In particular, if X and Y are two S -schemes, $X \times_S Y$ is denoted simply by $X \times Y$. On the other hand, let us point out that for the contents of sections 1 and 2 one may refer to [DG70], § II.4, nos. 5–6; see also [Ja03], § I.7.

(2) In this case we shall also say that d is an S -deviation of u of order $\leq n$. In particular, an S -deviation of u of order ≤ 0 is a morphism of $\mathcal{O}_X$ -modules from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$, i.e. an element of $\Gamma(Y, \mathcal{O}_Y)$.

Definition 1.1.2. A morphism of $p^{-1}(\mathcal{O}_S)$ -modules $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$ is an S -deviation of u if, for every point y of Y , there exist an open neighborhood U of $u(y)$ in X and an open neighborhood V of y in Y satisfying the following conditions:

- a) $u(V) \subset U$;
- b) if $v : V \rightarrow U$ is the morphism induced by u , there is an integer n such that the morphism $\mathcal{O}_U \rightarrow v_*(\mathcal{O}_V)$ induced by d is an S -deviation of v of order $\leq n$.⁽³⁾

If d is an S -deviation of u , we also say that the pair $D = (u, d)$ is an S -deviation, and we shall sometimes write $Y \xrightarrow{D} X$ or $Y \xrightarrow[u]{d} X$.

When d is the algebra homomorphism $u^\sharp : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$ corresponding to the morphism $u : Y \rightarrow X$, we shall also write u instead of D .

Remarks 1.1.3. ⁽⁴⁾ Let $\text{Dév}(u)$ (resp. $\text{Dév}^{\leq n}(u)$) be the set of S -deviations of u (resp. S -deviations of u of order $\leq n$). It is equipped with a natural $\mathcal{O}_Y(Y)$ -module structure: if $\lambda \in \mathcal{O}_Y(Y)$, then λd is the deviation sending f to $\lambda d(f)$, for every section f of $\mathcal{O}_X$ over an open subset U .

For every open subset V of Y , put $\mathcal{D}\text{év}(u)(V) = \text{Dév}(u|_V)$, i.e. $\mathcal{D}\text{év}(u)(V)$ is the set of

$$\begin{aligned} d_V \in \text{Hom}_{p^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, (u|_V)_*(\mathcal{O}_V)) &\simeq \text{Hom}_{p^{-1}(\mathcal{O}_S)}((u|_V)^{-1}\mathcal{O}_X, \mathcal{O}_V) \\ &\simeq \mathcal{H}om_{p^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)(V) \end{aligned}$$

such that, for every open subset U of X , the map $d_V(U) : \mathcal{O}_X(U) \rightarrow \mathcal{O}_Y(u^{-1}(U) \cap V)$ satisfies $(*)_n$. This defines a presheaf of $\mathcal{O}_Y$ -modules on Y , and one sees easily that it is a sheaf (more precisely, a subsheaf of $\mathcal{H}om_{p^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)$).

1.2. — Now consider two S -deviations $D = (u, d)$ and $E = (v, e)$:

$$Z \xrightarrow[e]{v} Y \xrightarrow[u]{d} X.$$

As U ranges over the open subsets of X , the composite maps

$$\Gamma(U, \mathcal{O}_X) \xrightarrow{d(U)} \Gamma(u^{-1}U, \mathcal{O}_Y) \xrightarrow{e(u^{-1}U)} \Gamma(v^{-1}u^{-1}U, \mathcal{O}_Z)$$

define an S -deviation of uv which we shall denote by de ; when d is of order $\leq m$ and e of order $\leq n$, de is of order $\leq m + n$. We shall also write

$$(\dagger) \quad D \circ E = (uv, de)$$

⁽²⁾**N.D.E.** One sees easily that this is equivalent to saying that, for every $x \in X$ and $f_0, \dots, f_n, g \in \mathcal{O}_{X,x}$, one has $(\text{ad } f_0)(\text{ad } f_1) \cdots (\text{ad } f_n)(d_x)(g) = 0$. On the other hand, recall that the adjunction isomorphism

$$\theta : \text{Hom}_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y)) \xrightarrow{\sim} \text{Hom}_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}(\mathcal{O}_X), \mathcal{O}_Y)$$

associates to every morphism of $p_X^{-1}(\mathcal{O}_S)$ -modules $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$ the morphism $d' = \varepsilon \circ u^{-1}(d)$, where ε is the canonical morphism $u^{-1}u_*(\mathcal{O}_Y) \rightarrow \mathcal{O}_Y$. Conversely, for every $p_Y^{-1}(\mathcal{O}_S)$ -morphism $d' : u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_Y$, one has $\theta^{-1}(d') = u_*(d') \circ \eta$, where η is the canonical morphism $\mathcal{O}_X \rightarrow u_*u^{-1}(\mathcal{O}_X)$. It follows that d satisfies $(*)_n$ if and only if d' satisfies:

$$(*'_n) \quad (\text{ad } f_0) \cdots (\text{ad } f_n)(d')(g) = 0$$

for every open subset V of Y and $f_0, \dots, f_n, g \in u^{-1}(\mathcal{O}_X)(V)$.

⁽³⁾**N.D.E.** If X and u are quasi-compact, every S -deviation of u is therefore of order $\leq n$, for some integer n .

⁽⁴⁾**N.D.E.** These remarks have been added; they will be useful in 1.3, 1.4 and 2.1.

⁽⁵⁾ and we shall say that $D \circ E$ or DE is the *composite S -deviation*. When $d = u^\natural$ (i.e. $D = u$ with the convention of 1.1), DE is also called the *image of E under u* .

The map $(D, E) \mapsto D \circ E$ which we have just defined will henceforth allow us to speak of the *category of S -deviations*, whose objects are the S -schemes and whose morphisms are the S -deviations.⁽⁶⁾

Definition 1.2.0. ⁽⁷⁾ Let $w : Z \rightarrow X$ be an S -morphism. An S -derivation of w , or S -derivation from $\mathcal{O}_X$ to $w_*(\mathcal{O}_Z)$, is a morphism of $p^{-1}(\mathcal{O}_S)$ -modules $d : \mathcal{O}_X \rightarrow w_*(\mathcal{O}_Z)$ such that, for every open subset U of X and $f, g \in \mathcal{O}_X(U)$,

$$d(fg) = w^\natural(f)d(g) + w^\natural(g)d(f).$$

Then d is a deviation of w of order ≤ 1 , which vanishes on the unit section of $\mathcal{O}_X$. The set of S -derivations of w will be denoted by $\text{Dér}_S(w)$; it is an $\mathcal{O}(Z)$ -module.

With the notation of 1.2, take Y equal to $I_Z = \text{Spec } \mathcal{O}_Z[t]$, where $t^2 = 0$, and v equal to the zero section $\tau : Z \rightarrow I_Z$, defined by the morphism of $\mathcal{O}_Z$ -algebras $\mathcal{O}_Z[t] \rightarrow \mathcal{O}_Z$ sending t to 0, and take e equal to the morphism of $\mathcal{O}_Z$ -modules $\sigma : \mathcal{O}_Z[t] \rightarrow \mathcal{O}_Z$ defined by $\sigma(1) = 0$ and $\sigma(t) = 1$,⁽⁸⁾ which it is convenient to denote by ∂_t .

If $u : I_Z \rightarrow X$ is a morphism satisfying $w = u \circ s$, then $\sigma \circ u^\natural$ is an S -derivation from $\mathcal{O}_X$ to $w_*(\mathcal{O}_Z)$. Conversely, to every S -derivation d one associates the morphism $u : I_Z \rightarrow X$ such that $u = w$ on the underlying spaces, and

$$u^\natural(f) = w^\natural(f) + d(f)t,$$

for every section f of $\mathcal{O}_X$ over an open subset U . One thus obtains:

Lemma 1.2.1. *Let $E = (\tau, \partial_t)$ be the deviation of $\tau : Z \rightarrow I_Z$ defined above. For every S -morphism $w : Z \rightarrow X$, the map $u \mapsto u \circ E$ is a bijection between the S -morphisms $u : I_Z \rightarrow X$ such that $u \circ s = w$, and the S -derivations of w .*

1.2.2. — Let d be an S -deviation of $u : Y \rightarrow X$. On the one hand, d is clearly an S' -deviation of u for every morphism $s : S \rightarrow S'$.

On the other hand, let $t : T \rightarrow S$ be a morphism with target S , and let $u_T : Y_T \rightarrow X_T$ be the morphism deduced from u by base change, and $t_Y : Y_T \rightarrow Y$ and $t_X : X_T \rightarrow X$ the canonical projections. There then exists one and only one T -deviation of u_T , which we shall denote by d_T or $d \times T$, satisfying the equality $t_X d_T = dt_Y$, in the sense of $(\dagger)$ above, i.e. for

⁽⁵⁾**N.D.E.** Care must be taken that, with this notation, de denotes the composite “ d followed by e ”.

⁽⁶⁾**N.D.E.** Often one considers only the S -deviations of the morphism id_X , which form the algebra of S -differential operators of X , cf. 1.4 below. However, the more general setting of S -deviations provides a convenient “functorial” language for proving statements such as: “if G is an S -group, the algebra of left-invariant S -differential operators on G is isomorphic to the algebra of S -deviations of the unit section $\varepsilon : S \rightarrow G$, cf. 2.1 and 2.4 below.

⁽⁷⁾**N.D.E.** This paragraph has been expanded, assigning to this definition (resp. to the following lemma) the number 1.2.0 (resp. 1.2.1).

⁽⁸⁾**N.D.E.** What follows has been added, i.e. the notation ∂_t has been introduced.

every open subset U of X there is a commutative diagram:⁽⁹⁾

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{t_X^\sharp} & \mathcal{O}(U \times T) \\ d(U) \downarrow & & \downarrow d_T(U \times T) \\ \mathcal{O}(u^{-1}U) & \xrightarrow{t_Y^\sharp} & \mathcal{O}(u^{-1}U \times T). \end{array}$$

If one puts $D = (u, d)$, one also writes $D_T = (u_T, d_T)$, and we shall say that d_T and D_T are deduced from d and D by base change.

1.2.3. — Let, for example, $u : Y \rightarrow X$ and $v : Z \rightarrow T$ be two S -morphisms, and d and e S -deviations of u and v . There is a commutative diagram

$$\begin{array}{ccc} X \times T & \xleftarrow{u_T} & Y \times T \\ v_X \uparrow & \swarrow u \times v & \uparrow v_Y \\ X \times Z & \xleftarrow{u_Z} & Y \times Z \end{array}$$

and we shall denote by $d \times e$ (product of d and e) the S -deviation of $u \times v$ equal to $d_T e_Y = e_X d_Z$ (with the convention $(\dagger)$ above), i.e. for every open subset U of $X \times T$, denoting by W the open subset $v_Y^{-1}u_T^{-1}U = u_Z^{-1}v_X^{-1}U$, there is a commutative diagram:

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{d_T(U)} & \mathcal{O}(u_T^{-1}U) \\ e_X(U) \downarrow & \searrow (d \times e)(U) & \downarrow e_Y(u_T^{-1}U) \\ \mathcal{O}(v_X^{-1}U) & \xrightarrow{d_Z(v_X^{-1}U)} & \mathcal{O}(W). \end{array}$$

If one puts $D = (u, d)$ and $E = (v, d)$, one also writes $D \times E = (u \times v, d \times e)$.

1.3. — ⁽¹⁰⁾ Let $u : Y \rightarrow X$ be a morphism of S -schemes. Recall that the adjunction isomorphism:

$$\mathrm{Hom}_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y)) \xrightarrow{\sim} \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}(\mathcal{O}_X), \mathcal{O}_Y)$$

associates to every morphism of $p^{-1}(\mathcal{O}_S)$ -modules $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$ the morphism $d' = \varepsilon \circ u^{-1}(d)$, where ε is the canonical morphism $u^{-1}u_*(\mathcal{O}_Y) \rightarrow \mathcal{O}_Y$.

Denote by $\mathcal{J}_u$ (resp. $\mathcal{J}_u$) the kernel of the algebra homomorphism $u^\sharp : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$ (resp. $u^\flat : u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_Y$), and let $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$ be a morphism of $p^{-1}(\mathcal{O}_S)$ -modules. If U is an open subset of X and $f_0, \dots, f_n, g \in \mathcal{O}_X(U)$, one sees easily by induction on n that condition $(*_n)$ is equivalent to the following equality (cf. EGA IV₄, 16.8.8.2):

$$(**_n) \quad 0 = \sum_{I \subset [0, n]} (-1)^{|I|} u^\sharp(f_{[0, n] - I}) d(f_I g),$$

where f_I denotes the product of the f_i , for $i \in I$. It follows that if d satisfies $(*_n)$, then d vanishes on the ideal $\mathcal{J}_u^{n+1}$.

⁽⁹⁾**N.D.E.** Explicitly, if V is an affine open subset of S and U (resp. U') an affine open subset of X (resp. T) over V , so that $\mathcal{O}_{X \times T}(U \times U') = \mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U')$, then $d_T(U \times U')$ is the composite:

$$\begin{aligned} \mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') & \xrightarrow{d(U) \otimes \mathrm{id}} \mathcal{O}_Y(u^{-1}U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') \\ & \longrightarrow \mathcal{O}_{Y \times T}(u^{-1}U \times U'). \end{aligned}$$

The author has left to the reader the verification that d_T is well defined, and the editors do likewise.

⁽¹⁰⁾**N.D.E.** The original has been expanded in this paragraph; see also N.D.E. (2) in 1.1.1.

Now suppose that Y is equal to S ; then $u : S \rightarrow X$ is a section of $p : X \rightarrow S$, hence is an immersion (cf. EGA I, 5.3.13). Then, on the one hand, $\varepsilon : u^{-1}u_*\mathcal{O}_S \rightarrow \mathcal{O}_S$ is an isomorphism, so that $u^{-1}(\mathcal{J}_u) = \mathcal{J}_u$. On the other hand, there is an isomorphism:

$$(\star) \quad u^{-1}(\mathcal{O}_X) \simeq \mathcal{O}_S \oplus \mathcal{J}_u.$$

Suppose that d vanishes on $\mathcal{J}_u^{n+1}$. Then $d' = \varepsilon \circ u^{-1}(d)$ vanishes on $\mathcal{J}_u^{n+1}$, and therefore d' satisfies the analogues $(**'_n)$ and $(*'_n)$ of $(**_n)$ and $(*_n)$, when $f_0, \dots, f_n \in \mathcal{J}_u(u^{-1}(U))$. Moreover, since $(\text{ad } a)(\phi) = 0$ for every $a \in \mathcal{O}_S(u^{-1}(U))$ and every morphism of $\mathcal{O}_{u^{-1}(U)}$ -modules $\phi : u^{-1}(\mathcal{O}_U) \rightarrow \mathcal{O}_{u^{-1}(U)}$, one deduces from $(\star)$ that d' satisfies the analogue $(*'_n)$ of $(*_n)$. It follows that d satisfies $(*_n)$. Consequently, one has obtained:

Lemma. *If $u : S \rightarrow X$ is a section of $p : X \rightarrow S$, then d is an S -deviation of u of order $\leq n$ if and only if d' vanishes on $\mathcal{J}_u^{n+1}$.*

This interpretation may be generalized as follows. Let $u : Y \rightarrow X$ be an arbitrary S -morphism, and Γu the graph of u , that is, the morphism $Y \rightarrow Y \times X$ with components id_Y and u . For every S -deviation d of u of order $\leq n$, one obtains by composition:

$$Y \xrightarrow{\text{diag.}} Y \times Y \xrightarrow[u_Y]{d_Y} Y \times X$$

a Y -deviation of Γu of order $\leq n$, which we shall denote by Γd (the *graph* of d).

Conversely, to every Y -deviation e of Γu one associates the composite S -deviation $e_X = \text{pr}_2 \circ e$:

$$Y \xrightarrow[\Gamma u]{e} Y \times X \xrightarrow{\text{pr}_2} X.$$

One sees immediately that $(\Gamma d)_X = d$, and the equality $\Gamma e_X = e$ follows from the fact that e is $\mathcal{O}_Y$ -linear.⁽¹¹⁾ One thus obtains an isomorphism of $\mathcal{O}_Y(Y)$ -modules:

$$\begin{array}{ccc} \{S\text{-deviations of } u \text{ of order } \leq n\} & \xrightarrow{\sim} & \{Y\text{-deviations of } \Gamma u \text{ of order } \leq n\} \\ d & \longmapsto & \Gamma d. \end{array}$$

Moreover, one sees easily that d is an S -derivation of u if and only if Γd is a Y -derivation of Γu .

Call $\mathcal{J}_{\Gamma u}$ the kernel of the algebra homomorphism $(\Gamma u)^{-1}(\mathcal{O}_{Y \times X}) \rightarrow \mathcal{O}_Y$ corresponding to Γu . Taking account of the preceding lemma, one has obtained:

Proposition. *Let $u : Y \rightarrow X$ be an S -morphism and $\Gamma u : Y \rightarrow Y \times X$ its graph. The S -deviations of u of order $\leq n$ identify with the Y -deviations of Γu of order $\leq n$, which are in bijection with*

$$\text{Hom}_{\mathcal{O}_Y}((\Gamma u)^{-1}(\mathcal{O}_{Y \times X})/\mathcal{J}_{\Gamma u}^{n+1}, \mathcal{O}_Y).$$

1.3.1. — ⁽¹²⁾ Return to the case where $u : S \rightarrow X$ is a section of $p : X \rightarrow S$. Then the homomorphism $\phi : u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_S$ admits a section, which we shall denote simply by $g \mapsto g \cdot 1$, so that, with the notation of 1.3, there is an isomorphism of $\mathcal{O}_S$ -modules:

$$(\star) \quad u^{-1}(\mathcal{O}_X) \simeq \mathcal{O}_S \oplus \mathcal{J}_u,$$

and for every section f of $u^{-1}(\mathcal{O}_X)$, $f - \phi(f) \cdot 1$ is a section of $\mathcal{J}_u$.

Let d be an S -deviation of u of order ≤ 1 , and d' the $\mathcal{O}_S$ -morphism $u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_S$ corresponding to d . If a, b are sections of $u^{-1}(\mathcal{O}_X)$, one has:

$$\begin{aligned} 0 &= d'((a - \phi(a) \cdot 1)(b - \phi(b) \cdot 1)) \\ &= d'(ab) - \phi(a)d'(b) - \phi(b)d'(a) + \phi(ab)d'(1). \end{aligned}$$

⁽¹¹⁾**N.D.E.** If λ, f are local sections of $\mathcal{O}_Y$ and $\mathcal{O}_X$, one has $(\Gamma e_X)(\lambda \otimes f) = \lambda \cdot e(1 \otimes g)$, and this equals $e(\lambda \otimes g)$ since e is $\mathcal{O}_Y$ -linear.

⁽¹²⁾**N.D.E.** This paragraph has been added.

Consequently, one sees that d is an S -derivation of u (cf. 1.2.1 and N.D.E. (2)) if and only if $d'(1) = 0$. One therefore obtains:

Lemma. *The S -derivations of u are exactly the S -deviations of u of order 1 which vanish on the unit section of $\mathcal{O}_X$; they correspond to the $\mathcal{O}_S(S)$ -module*

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}_u/\mathcal{I}_u^2, \mathcal{O}_S),$$

and there is an isomorphism of $\mathcal{O}_S(S)$ -modules $\mathrm{Dér}^{\leq 1}(u) \simeq \mathcal{O}_S(S) \oplus \mathrm{Dér}_S(u)$.

Returning to the general case, one deduces from this, with the notation of 1.3,

Corollary. *Let $u : Y \rightarrow X$ be an S -morphism and $\Gamma u : Y \rightarrow Y \times X$ its graph. There is a canonical isomorphism of $\mathcal{O}_Y(Y)$ -modules*

$$\mathrm{Dér}_S(u) \simeq \mathrm{Dér}_Y(\Gamma u) \simeq \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_{\Gamma u}/\mathcal{I}_{\Gamma u}^2, \mathcal{O}_Y).$$

Definition 1.4. Let X be an S -scheme. An S -differential operator (resp. an S -differential operator of order $\leq n$) on X means any S -deviation (resp. any S -deviation of order $\leq n$) of the identity morphism of X .

According to 1.1, an S -differential operator of order $\leq n$ is thus an endomorphism of the $p^{-1}(\mathcal{O}_S)$ -module $\mathcal{O}_X$ satisfying equalities $(*_n)$ of 1.1. We shall denote by $\mathrm{Dif}_{X/S}^n$ the $\Gamma(\mathcal{O}_S)$ -module⁽¹³⁾ consisting of the S -differential operators of order $\leq n$, and by $\mathrm{Dif}_{X/S}$ that consisting of all the S -differential operators.

As we saw in 1.2, one can compose the S -deviations of id_X , which equips $\mathrm{Dif}_{X/S}$ with a $\Gamma(\mathcal{O}_S)$ -algebra structure; we shall say that it is the *algebra of differential operators of X/S* .

Similarly, for every open subset V of X , put $\mathcal{A}_{X/S}(V) = \mathrm{Dif}_{V/S} = \mathrm{Dér}(\mathrm{id}_V)$; according to 1.1.3, this defines a sheaf of $\mathcal{O}_X$ -modules, called the *sheaf of S -differential operators on X* .⁽¹⁴⁾

1.4.1. — As we saw in 1.3, one can interpret the differential operators of X/S by means of the graph of the identity morphism of X , that is, of the diagonal morphism $\Delta = \Delta_{X/S}$ from X to $X \times X$. Let us translate the statements of 1.3 into the present context.

Equip $\mathcal{O}_{X \times X}$ with the $\mathrm{pr}_1^{-1}(\mathcal{O}_X)$ -algebra structure defined by pr_1 , so that $\Delta^{-1}(\mathcal{O}_{X \times X})$ is equipped with an algebra structure over $\mathcal{O}_X = \Delta^{-1}\mathrm{pr}_1^{-1}(\mathcal{O}_X)$. Let $\mathcal{I}_{X/S}$ be the kernel of the homomorphism

$$\Delta^{-1}(\mathcal{O}_{X \times X}) \rightarrow \mathcal{O}_X$$

adjoint to the homomorphism $\mathcal{O}_{X \times X} \rightarrow \Delta_*(\mathcal{O}_X)$, and let $\mathcal{P}_{X/S}^m$ be the $\mathcal{O}_X$ -algebra

$$\Delta^{-1}(\mathcal{O}_{X \times X})/\mathcal{I}_{X/S}^{m+1}.$$

If V is an affine open subset of S and U an affine open subset of X over V , and if one puts $k = \Gamma(V, \mathcal{O}_S)$ and $A = \Gamma(U, \mathcal{O}_X)$, one therefore has:

$$\Gamma(U, \mathcal{P}_{X/S}^m) = (A \otimes_k A)/I^{m+1},$$

where I is the ideal generated by the elements $a \otimes 1 - 1 \otimes a$, for $a \in A$. This being so, according to 1.3 there is an isomorphism of $\mathcal{O}_X(X)$ -modules:

$$j_X : \mathrm{Dif}_{X/S}^m \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^m, \mathcal{O}_X)$$

which may be defined as follows: if d belongs to $\mathrm{Dif}_{X/S}^m$ and c is a section of $\mathcal{P}_{X/S}^m$ over U of the form $a \otimes b + I^{m+1}$, one has $j_X(d)(c) = a \cdot d(b)$.⁽¹⁵⁾

⁽¹³⁾**N.D.E.** In this exposé, the ring $\Gamma(S, \mathcal{O}_S) = \mathcal{O}_S(S)$ is denoted by $\Gamma(\mathcal{O}_S)$.

⁽¹⁴⁾**N.D.E.** Here the original has been modified; it mentioned the sheaf $U \mapsto \mathrm{Dif}_{X_U/U}$, where U ranges over the open subsets of S ; this is the direct image of $\mathcal{A}_{X/S}$ under the morphism $p_X : X \rightarrow S$.

⁽¹⁵⁾**N.D.E.** Via this isomorphism, the X -derivations of $\Delta_{X/S}$ correspond, according to 1.3.1, to the S -derivations of id_X , i.e. to the $p^{-1}(\mathcal{O}_S)$ -derivations of $\mathcal{O}_X$.

1.4.2. — Let d be a differential operator and u a section of X over S . We call the composite S -deviation

$$S \xrightarrow{u} X \xrightarrow[\text{id}_X]{d} X.$$

the *value of d at u* .

According to 1.3 and 1.4.1, if d is a differential operator of order $\leq m$, then du (resp. d) is canonically associated to a morphism of $\mathcal{O}_S$ -modules $d' : u^{-1}(\mathcal{O}_X)/\mathcal{I}_u^{m+1} \rightarrow \mathcal{O}_S$ (resp. a morphism of $\mathcal{O}_X$ -modules $d'' : \mathcal{P}_{X/S}^m \rightarrow \mathcal{O}_X$).

It is clear that one can construct d' from d'' in the following way: the square

$$\begin{array}{ccc} X \simeq S \times X & \xrightarrow{u \times X} & X \times X \\ p \downarrow & & \downarrow \text{pr}_1 \\ S & \xrightarrow{u} & X \end{array}$$

is cartesian, which allows one to identify X with $S \times_X (X \times X)$, u with $S \times_X \Delta$, and hence $u^*(\mathcal{P}_{X/S}^m)$ with $u^{-1}(\mathcal{O}_X)/\mathcal{I}_u^{m+1}$. One thus identifies $u^*(d'')$ with a morphism

$$u^{-1}(\mathcal{O}_X)/\mathcal{I}_u^{m+1} \rightarrow \mathcal{O}_S,$$

which is none other than d' .

1.5. — Put, as usual, $I_S = \text{Spec } \mathcal{O}_S[T]/(T^2)$. Let $\tau : S \rightarrow I_S$ be the zero section and σ the canonical deviation of τ which we defined in 1.2.0, i.e. the homomorphism of $\mathcal{O}_S$ -modules which vanishes on the unit section of $\mathcal{O}_S[T]/(T^2)$ and sends the class t of T modulo T^2 to the unit section of $\mathcal{O}_S$.

Let X be an S -scheme. To every I_S -automorphism u of $I_S \times X$ inducing the identity on X , a differential operator D_u of X is associated by composition:

$$X \simeq S \times X \xrightarrow{\sigma \times X} I_S \times X \xrightarrow{u} I_S \times X \xrightarrow{\text{pr}_2} X.$$

According to II, 3.14, the map $u \mapsto D_u$ is an isomorphism of the $\Gamma(\mathcal{O}_S)$ -Lie algebra

$$\text{Lie}(\underline{\text{Aut}} X) := \underline{\text{Lie}}(\underline{\text{Aut}} X)(S)$$

onto the $\Gamma(\mathcal{O}_S)$ -Lie algebra of $p^{-1}(\mathcal{O}_S)$ -derivations of $\mathcal{O}_X$. The inverse isomorphism associates to every derivation D the automorphism of $I_S \times X$ corresponding to the automorphism $a + bt \mapsto a + (Da + b)t$ of $\mathcal{O}_X[T]/(T^2)$.

2. Invariant differential operators on group schemes

2.1. — Let G be an S -group scheme; we denote by ε or $\varepsilon_G : S \rightarrow G$ the unit section of G .

Definition. Let $U(G)$ be the $\Gamma(\mathcal{O}_S)$ -module of S -deviations of ε_G (or *S -deviations of the origin*) (cf. 1.1).

If d and e are two elements of $U(G)$, $d \times e$ is an S -deviation of $\varepsilon \times \varepsilon : S \simeq S \times S \rightarrow G \times G$. The image of $d \times e$ under the multiplication morphism $m : G \times G \rightarrow G$ (cf. 1.2) will be called the *product of d and e* and will be denoted by $d \cdot e$.

The $\Gamma(\mathcal{O}_S)$ -module $U(G)$ is thus equipped with an associative $\Gamma(\mathcal{O}_S)$ -algebra structure having ε_G as its unit element (1.1). We shall say that $U(G)$ is the *infinitesimal algebra of G* .⁽¹⁶⁾

As T ranges over the schemes over S , the infinitesimal algebra $U(G_T)$ of the T -group $G \times T$ clearly varies contravariantly in T , so that we shall be able to speak of the *infinitesimal algebra functor*.

⁽¹⁶⁾**N.D.E.** One now says “the algebra of distributions” (at the origin) of G , cf. [DG70], § II.4, 6.1 and [Ja03], I 7.7.

As T ranges over the open subsets of S , one therefore obtains a presheaf $T \mapsto U(G_T)$ of $\mathcal{O}_S$ -algebras; moreover, according to 1.1.3, this is a *sheaf*. We shall denote it by $\mathcal{U}(G)$ and call it the *sheaf of infinitesimal algebras of G* .

The algebra $U(G)$ is also a covariant functor in G . Indeed, if $u : G \rightarrow H$ is a homomorphism of S -groups and d an S -deviation of ε_G , the image of d under u is an element $U(u)(d) = ud$ of $U(H)$. The map $U(u) : U(G) \rightarrow U(H)$ thus defined is clearly a homomorphism of $\Gamma(\mathcal{O}_S)$ -algebras. One similarly defines a homomorphism $\mathcal{U}(u)$ from $\mathcal{U}(G)$ to $\mathcal{U}(H)$.

2.2. — Let d be an element of $U(G)$, i.e. an S -deviation of the origin of G . Consider the S -deviation $d \times G$ of $\varepsilon \times G : G \simeq S \times G \rightarrow G \times G$ obtained from d by base change (1.2.2); the image of $d \times G$ under the multiplication morphism $m : G \times G \rightarrow G$ is an S -deviation of $m \circ (\varepsilon \times \text{id}_G) = \text{id}_G$, i.e. an element of $\text{Dif}_{G/S}$, which will be denoted by d^G .

The map $d \mapsto d^G$ is clearly $\Gamma(\mathcal{O}_S)$ -linear, and the “commutative” diagram below shows that $(e \cdot d)^G = d^G \cdot e^G$.⁽¹⁷⁾

$$\begin{array}{ccccc}
 & & G \times G \times G & \xrightarrow{m \times G} & G \times G \\
 & \nearrow G \times d \times G & & \searrow G \times m & \\
 & \varepsilon \times G \times G & & & \\
 G \times G & \xrightarrow{e \times G} & G \times G & \xrightarrow{d \times G} & G \times G \\
 \nearrow \varepsilon \times G & & \searrow m & & \searrow m \\
 G & \xrightarrow{e^G} & G & \xrightarrow{d^G} & G \\
 & \text{id}_G & & \text{id}_G &
 \end{array}$$

The commutativity of the two lower triangles follows indeed from the definition of d^G and e^G ; on the other hand, the composite S -deviation of $e \times G$ and $G \times d \times G$ is $(e \times d) \times G$ (cf. 1.2.2), its image under $m \times G$ is $(e \cdot d) \times G$, and the image of the latter under m is therefore equal to $(e \cdot d)^G$.

One thus obtains an anti-homomorphism $U(G) \rightarrow \text{Dif}_{G/S}$ of $\Gamma(\mathcal{O}_S)$ -algebras, called *right translation*.⁽¹⁸⁾

If $\mathcal{A}_{G/S}$ denotes the sheaf of S -differential operators on G (cf. 1.4) and p the structural morphism $G \rightarrow S$, one similarly defines a “right translation”: $\mathcal{U}(G) \rightarrow p_*(\mathcal{A}_{G/S})$.

2.3. — We shall now characterize the differential operators of G over S of the form d^G . Let $g : S \rightarrow G$ be a section of the structural morphism of G , and g_G the right translation of G by g , that is, the composite morphism:

$$g_G : G \simeq G \times S \xrightarrow{G \times g} G \times G \xrightarrow{m} G.$$

For every differential operator D of G over S , the composite $g_G^{-1} D g_G$ (cf. 1.2) is again an S -deviation of id_G , i.e. an element of $\text{Dif}_{G/S}$; we shall write:

$$D^g = g_G^{-1} D g_G.$$

We shall say that D is *right-invariant* if, for every base change $t : T \rightarrow S$ and every section $g : T \rightarrow G \times T$, one has $(D_T)^g = D_T$.

⁽¹⁷⁾**N.D.E.** The original has been corrected by replacing $d \times G \times G$ in the diagram by $G \times d \times G$, so that the composite on the left side of the triangle is $(e \times d) \times G$, and the map $d \mapsto d^G$ is an *anti-isomorphism* from $U(G)$ onto the *right-invariant* differential operators (cf. 2.3, 2.4 below); on the other hand, defining ${}^G d$ as the image under m of $G \times d$, one would similarly obtain an isomorphism from $U(G)$ onto the *left-invariant* differential operators (cf. [DG70], § II.4, Th. 6.5). Sections 2.4 and 2.5 have been corrected accordingly.

⁽¹⁸⁾**N.D.E.** It would be preferable to call it *left operation*. Indeed, let d be, for example, an S -derivation of the origin; according to 1.2.1, d is the composite of the S -derivation $(\tau, \partial_t) : S \rightarrow I_S$ and a morphism $x : I_S \rightarrow G$ such that $x \circ \tau = \varepsilon$ (i.e. $x \in \text{Lie}(G/S)(S)$), and then d^G is the derivation of $\mathcal{O}_G$ sending a local section ϕ to the section $g \mapsto \partial_t \phi(xg)$. Moreover, with this terminology one could say that: “left operation commutes with right translations”.

Lemma. *For every differential operator D of G over S , the following assertions are equivalent (where m is the multiplication morphism of G):*

(i) *D is right-invariant.*

(ii) *The following two deviations of m are equal: $Dm = m(D \times G)$.*

(ii) $\Rightarrow$ (i): since condition (ii) is stable under base change, it suffices to show that (ii) implies the equality $D^g = D$ for every section $g : S \rightarrow G$. Let h be the morphism $G \times g : G \times S \rightarrow G \times G$, so that $m \circ h$ is the right translation g_G . The equality $D^g = D$ is equivalent to the equality $g_G \circ D = D \circ g_G$, and the latter follows from the commutative diagram:

$$\begin{array}{ccccc} G & \xleftarrow{m} & G \times G & \xleftarrow{h} & G \\ \text{id}_G \downarrow D & & \text{id}_{(G \times G)} \downarrow D \times G & & \text{id}_G \downarrow D \\ G & \xleftarrow{m} & G \times G & \xleftarrow{h} & G \end{array}$$

(i) $\Rightarrow$ (ii): take, indeed, for $t : T \rightarrow S$ the structural morphism $p : G \rightarrow S$, and for the section $g : T \rightarrow G \times T$ the diagonal morphism $\Delta : G \rightarrow G \times G$. The right translation

$$\Delta_{G \times G} : G \times G \rightarrow G \times G$$

is then the morphism from $G \times G$ to $G \times G$ with components m and pr_2 . The equality $(D_G)^\Delta = D_G$ is then equivalent to the commutativity of the first square of the following diagram:

$$\begin{array}{ccccc} G \times G & \xrightarrow{\Delta_{G \times G}} & G \times G & \xrightarrow{\text{pr}_1} & G \\ \text{id}_{G \times G} \downarrow D_G & & \text{id}_{G \times G} \downarrow D_G & & \text{id}_G \downarrow D \\ G \times G & \xrightarrow{\Delta_{G \times G}} & G \times G & \xrightarrow{\text{pr}_1} & G \end{array}$$

Equality (ii) therefore follows from the fact that $m = \text{pr}_1 \circ \Delta_{G \times G}$.

Consider, for example, an element d of the infinitesimal algebra $U(G)$. The squares of the diagram

$$\begin{array}{ccccccc} G \times G & \xlongequal{\quad} & S \times G \times G & \xrightarrow[\varepsilon \times G \times G]{d \times G \times G} & G \times G \times G & \xrightarrow{m \times G} & G \times G \\ m \downarrow & & S \times m \downarrow & & G \times m \downarrow & & m \downarrow \\ G & \xlongequal{\quad} & S \times G & \xrightarrow[\varepsilon \times G]{d \times G} & G \times G & \xrightarrow{m} & G \end{array}$$

are then commutative. Since one has

$$m \circ (d \times G) = d^G \quad \text{and} \quad (m \times G) \circ (d \times G \times G) = d^G \times G,$$

one also has $d^G \circ m = m \circ (d^G \times G)$. Thus: *for every S -deviation d of the origin, d^G is a right-invariant differential operator.*

Theorem 2.4.

- (i) *The map $d \mapsto d^G$ is an anti-isomorphism⁽¹⁹⁾ from the infinitesimal algebra $U(G)$ onto the subalgebra $\text{Dif}_{G/S}^G$ of $\text{Dif}_{G/S}$ consisting of the right-invariant differential operators.*
- (ii) *Similarly, the map $d \mapsto {}^G d$ is an isomorphism from $U(G)$ onto the subalgebra of $\text{Dif}_{G/S}$ consisting of the left-invariant differential operators.*

⁽¹⁹⁾**N.D.E.** “Isomorphism” has been corrected to “anti-isomorphism”, and assertion (ii) has been added, cf. N.D.E. (17).

Let, indeed, D be an arbitrary differential operator of G over S , and denote by D_0 its *value at the origin*, that is, the composite deviation $S \xrightarrow{\varepsilon} G \xrightarrow[\text{id}_G]{D} G$. The right-invariant differential operator $(D_0)^G$ is then obtained by composition:

$$G \simeq S \times G \xrightarrow{\varepsilon \times G} G \times G \xrightarrow[\text{id}_{G \times G}]{D \times G} G \times G \xrightarrow{m} G.$$

If D is right-invariant, one has $Dm = m(D \times G)$, whence

$$D = Dm(\varepsilon \times G) = m(D \times G)(\varepsilon \times G) = (D_0)^G.$$

In particular, the map $d \mapsto d^G$ is surjective.

Conversely, let d be an S -deviation of the origin. There is then a commutative square

$$\begin{array}{ccc} G \times G & \xleftarrow{d \times G} & G \\ G \times \varepsilon \uparrow & & \uparrow \varepsilon \\ G \times S \simeq G & \xleftarrow{d} & S \end{array}$$

from which it follows that $d = m(G \times \varepsilon)d = m(d \times G)\varepsilon = (d^G)_0$. A fortiori, the map $d \mapsto d^G$ is injective. This proves the theorem.

As S varies, theorem 2.4 clearly implies that the right translation $\mathcal{U}(G) \rightarrow p_*(\mathcal{A}f_{G/S})$ is an anti-isomorphism of $\mathcal{O}_S$ -algebras from $\mathcal{U}(G)$ onto the sheaf of $\mathcal{O}_S$ -algebras $p_*(\mathcal{A}f_{G/S})^G$, which associates $\text{Dif}_{G/U}^{GU}$ to every open subset U of S .

Remark 2.4.1. Consider the commutative diagram

$$\begin{array}{ccc} G & \xleftarrow{\eta} & G \times G \\ \uparrow p\varepsilon & & \uparrow \text{pr}_1 \Delta \\ S & \xleftarrow{p} & G, \end{array}$$

where η denotes the morphism “ $(x, y) \mapsto yx^{-1}$ ”.⁽²⁰⁾ The latter induces morphisms

$$\eta' : \eta^{-1}(\mathcal{O}_G) \rightarrow \mathcal{O}_{G \times G} \quad \text{and} \\ \Delta^{-1}(\eta') : p^{-1}\varepsilon^{-1}(\mathcal{O}_G) \rightarrow \Delta^{-1}(\mathcal{O}_{G \times G}).$$

For every integer $n \geq 1$, put $\mathfrak{p}_{G/S}^n = \varepsilon^{-1}(\mathcal{O}_G)/\mathcal{I}_\varepsilon^{n+1}$ (confer 1.3 and 1.4 for the notation).⁽²¹⁾ Since the square formed by the morphisms $\varepsilon, \eta, \Delta$, and p is cartesian, $\Delta^{-1}(\eta')$ induces an isomorphism of $\mathcal{O}_G$ -modules:

$$p^*(\mathfrak{p}_{G/S}^n) \xrightarrow{\sim} \mathcal{P}_{G/S}^n.$$

The differential operators of G over S of order $\leq n$ therefore correspond bijectively to the morphisms of $\mathcal{O}_G$ -modules $p^*(\mathfrak{p}_{G/S}^n) \rightarrow \mathcal{O}_G$, that is, to the morphisms of $\mathcal{O}_S$ -modules

$$\mathfrak{p}_{G/S}^n \rightarrow p_*(\mathcal{O}_G).$$

In this bijection, the right-invariant differential operators are associated to the composite arrows

$$\mathfrak{p}_{G/S}^n \rightarrow \mathcal{O}_S \xrightarrow{\text{can.}} p_*(\mathcal{O}_G).$$

One thus recovers the isomorphism of theorem 2.4.

⁽²⁰⁾**N.D.E.** i.e. G acts on the left on itself by right translations.

⁽²¹⁾**N.D.E.** In what follows, the original has been corrected; it referred to the square formed by the morphisms p, p, η , and pr_1 , instead of $\varepsilon, \eta, \Delta$, and p .

2.5. — ⁽²²⁾ Let $\text{Lie}(G)$ be the Lie algebra of G ; ⁽²³⁾ we shall define a morphism of $\Gamma(\mathcal{O}_S)$ -Lie algebras $\alpha : \text{Lie}(G) \rightarrow U(G)$.

Let $s : S \rightarrow I_S$ be the zero section of $I_S \rightarrow S$, and σ the deviation of s defined in 1.2.0. Recall (cf. II, 4.1) that $\text{Lie}(G)$ is the set of morphisms $x : I_S \rightarrow G$ such that $x \circ s = \varepsilon_G$. Then the composite

$$S \xrightarrow[\sigma]{s} I_S \xrightarrow{x} G$$

is an S -deviation of ε_G , i.e. an element of $U(G)$; with the notation of 1.2 (†), it is denoted by σx . Moreover, according to 1.2.1, the map $\alpha : x \mapsto \sigma x$ is an isomorphism of $\mathcal{O}_S(S)$ -modules from $\text{Lie}(G)$ onto the submodule $\text{Dér}(\varepsilon_G)$ of $U(G)$ consisting of the S -derivations of ε_G . We shall see that α is a morphism of Lie algebras. ⁽²⁴⁾ Let

$$\rho' : U(G) \rightarrow \text{Dif}_{G/S}$$

be the algebra morphism which associates to an S -deviation d of ε_G the left-invariant differential operator ${}^G d \in \text{Dif}_{G/S}$, cf. 2.2, N.D.E. (17).

Let $\rho : G \rightarrow \underline{\text{Aut}}_S(G)$ be the homomorphism of group functors which associates to an S -morphism $g : T \rightarrow G$ the right translation of G_T by g , i.e. the morphism:

$$G_T \simeq T \times_T G_T \xrightarrow{G_T \times g} G_T \times_T G_T \xrightarrow{m_T} G_T.$$

Recall also (cf. 1.5 and II, 3.14) that $\text{Lie}(\underline{\text{Aut}} G) = \underline{\text{Lie}}(\underline{\text{Aut}}_S(G)/S)(S)$ identifies with the infinitesimal automorphisms of G , i.e. with the automorphisms of $I_S \times G$ inducing the identity on G . Since ρ is a monomorphism, so is the morphism $\underline{\text{Lie}}(\rho) : \underline{\text{Lie}}(G/S) \rightarrow \underline{\text{Lie}}(\underline{\text{Aut}}_S(G)/S)$ (see, for example, Exp. II, N.D.E. (50)); hence $\text{Lie}(\rho) : \text{Lie}(G) \rightarrow \text{Lie}(\underline{\text{Aut}} G)$ is injective.

On the other hand, according to 1.5, the map β which associates to every infinitesimal automorphism u of G the differential operator D_u of G :

$$G \simeq S \times G \xrightarrow{\sigma \times G} I_S \times G \xrightarrow{u} I_S \times G \xrightarrow{\text{pr}_2} G$$

is an isomorphism from $\text{Lie}(\underline{\text{Aut}} G)$ onto the Lie subalgebra of $\text{Dif}_{G/S}$ consisting of the $p^{-1}(\mathcal{O}_S)$ -derivations of $\mathcal{O}_G$.

For every $x \in \text{Lie}(G)$, there is the following commutative square determining the image of x under $\text{Lie}(\rho)$:

$$\begin{array}{ccc} I_S \times G & \xrightarrow{\text{Lie}(\rho)(x)} & I_S \times G \\ x \times G \downarrow & & \downarrow \text{pr}_2 \\ G \times G & \xrightarrow{m} & G. \end{array}$$

Taking this diagram into account, the image of $\text{Lie}(\rho)(x)$ under β is the composite deviation

$$G \simeq S \times G \xrightarrow[\sigma \times G]{\sigma \times G} I_S \times G \xrightarrow{G \times x} G \times G \xrightarrow{m} G$$

⁽²²⁾**N.D.E.** In this paragraph the order has been modified, beginning by defining the map $\alpha : \text{Lie}(G) \rightarrow U(G)$, and the original has been corrected as indicated in N.D.E. (17).

⁽²³⁾**N.D.E.** In this exposé, if G (resp. X) is an S -group scheme (resp. an S -scheme), the “Lie algebra” $\text{Lie}(G)$ (resp. $\text{Lie}(\underline{\text{Aut}} X)$) denotes, with the notation of Exposé II, $\underline{\text{Lie}}(G/S)(S)$ (resp. $\underline{\text{Lie}}(\underline{\text{Aut}}_S(X)/S)(S)$); this is a $\Gamma(\mathcal{O}_S)$ -Lie algebra, according to II, 4.11 and 3.14.

⁽²⁴⁾**N.D.E.** See also II, 4.11.

which, according to 2.2 N.D.E. (17), is none other than $G(\sigma x) = \rho'(\alpha(x))$. One therefore obtains a commutative diagram:

$$\begin{array}{ccc} \mathrm{Lie}(G) & \xrightarrow{\mathrm{Lie}(\rho)} & \mathrm{Lie}(\mathrm{Aut} G) \\ \alpha \downarrow & & \downarrow \beta \\ U(G) & \xrightarrow{\rho'} & \mathrm{Dif}_{G/S} \end{array}$$

where $\mathrm{Lie}(\rho)$, β , and ρ' are morphisms of Lie algebras. Since ρ' is injective, it follows that α too is a morphism of Lie algebras. Consequently, one has obtained:

Proposition. α is an isomorphism of $\mathcal{O}_S(S)$ -Lie algebras from $\mathrm{Lie}(G)$ to the Lie algebra of S -derivations of ε_G , itself isomorphic via $\mathrm{Lie}(\rho)$ to the Lie algebra of left-invariant S -derivations of G .⁽²⁵⁾

3. Coalgebras and Cartier duality

3.1. — Let S be a scheme (or, more generally, a ringed space). An $\mathcal{O}_S$ -coalgebra⁽²⁶⁾ is a pair $(\mathcal{U}, \Delta_{\mathcal{U}})$ consisting of an $\mathcal{O}_S$ -module $\mathcal{U}$ and a morphism of $\mathcal{O}_S$ -modules $\Delta_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U}$ (called the *diagonal morphism* or *comultiplication*) such that:

- (i) $\sigma \circ \Delta_{\mathcal{U}} = \Delta_{\mathcal{U}}$, where $\sigma(a \otimes b) = b \otimes a$.
- (ii) The square

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}}} & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \\ \Delta_{\mathcal{U}} \downarrow & & \downarrow \mathrm{id}_{\mathcal{U}} \otimes \Delta_{\mathcal{U}} \\ \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \end{array}$$

is commutative.

- (iii) There exists a morphism of $\mathcal{O}_S$ -modules $\varepsilon_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{O}_S$, called the *augmentation*, such that the composite morphisms

$$\begin{aligned} \mathcal{U} &\xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes \varepsilon_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{O}_S \simeq \mathcal{U} \\ \mathcal{U} &\xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\varepsilon_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{U} \simeq \mathcal{U} \end{aligned}$$

are the identity morphism of $\mathcal{U}$.

If $\varepsilon_{\mathcal{U}}$ and $\varepsilon'_{\mathcal{U}}$ are two augmentations, one has $\varepsilon_{\mathcal{U}} = (\varepsilon_{\mathcal{U}} \otimes \varepsilon'_{\mathcal{U}}) \circ \Delta_{\mathcal{U}} = \varepsilon'_{\mathcal{U}}$; the augmentation is therefore uniquely determined by (iii).

If $(\mathcal{U}, \Delta_{\mathcal{U}})$ and $(\mathcal{V}, \Delta_{\mathcal{V}})$ are two $\mathcal{O}_S$ -coalgebras, a *morphism* from the first to the second is a morphism of $\mathcal{O}_S$ -modules $f : \mathcal{U} \rightarrow \mathcal{V}$ such that the diagrams

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{f} & \mathcal{V} \\ \Delta_{\mathcal{U}} \downarrow & & \downarrow \Delta_{\mathcal{V}} \\ \mathcal{U} \otimes \mathcal{U} & \xrightarrow{f \otimes f} & \mathcal{V} \otimes \mathcal{V} \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathcal{U} & \xrightarrow{f} & \mathcal{V} \\ \varepsilon_{\mathcal{U}} \searrow & & \swarrow \varepsilon_{\mathcal{V}} \\ & \mathcal{O}_S & \end{array}$$

⁽²⁵⁾**N.D.E.** There are examples of Lie algebras $\mathfrak{g}$ over a ring A such that the map $\mathfrak{g} \rightarrow U(\mathfrak{g})$ is not injective, cf. [BLie], § I.2, Ex. 9. The result above shows (since α factors as $\mathrm{Lie}(G) \rightarrow U(\mathrm{Lie}(G)) \rightarrow U(G)$) that this cannot occur for “algebraic” Lie algebras, i.e. those of the form $\mathrm{Lie}(G)$, where G is an A -group scheme.

⁽²⁶⁾**N.D.E.** One also says “cogebra”, cf. [BALg], III § 11.1. On the other hand, note that in this exposé (as well as in VII B) one works in the category of *cocommutative* coalgebras (i.e. satisfying condition (i)), which is crucial for defining the product and the notion of coalgebra in groups (cf. 3.1.0 and 3.2).

are commutative. Coalgebra morphisms compose like morphisms of $\mathcal{O}_S$ -modules, so that we shall be able to speak of the *category of $\mathcal{O}_S$ -coalgebras*.

3.1.0. — ⁽²⁷⁾ This category has finite products: the final object is the $\mathcal{O}_S$ -module $\mathcal{O}_S$, the comultiplication being the identity; the product of two coalgebras $(\mathcal{U}, \Delta_{\mathcal{U}})$ and $(\mathcal{V}, \Delta_{\mathcal{V}})$ is the tensor product $\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{V}$, the comultiplication being the composite morphism

$$\begin{aligned} \mathcal{U} \otimes \mathcal{V} &\xrightarrow{\Delta_{\mathcal{U}} \otimes \Delta_{\mathcal{V}}} \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{V} \otimes \mathcal{V} \\ &\xrightarrow{\text{id}_{\mathcal{U}} \otimes \sigma \otimes \text{id}_{\mathcal{V}}} \mathcal{U} \otimes \mathcal{V} \otimes \mathcal{U} \otimes \mathcal{V} \end{aligned}$$

where $\sigma(a \otimes b) = b \otimes a$; the canonical projections from $\mathcal{U} \otimes \mathcal{V}$ to the factors $\mathcal{U}$ and $\mathcal{V}$ are the morphisms $\text{id}_{\mathcal{U}} \otimes \varepsilon_{\mathcal{V}}$ and $\varepsilon_{\mathcal{U}} \otimes \text{id}_{\mathcal{V}}$,⁽²⁸⁾ and the “diagonal morphism” $\mathcal{U} \rightarrow \mathcal{U} \otimes \mathcal{U}$ (corresponding to the pair of morphisms $(\text{id}_{\mathcal{U}}, \text{id}_{\mathcal{U}})$) is none other than the comultiplication $\Delta_{\mathcal{U}}$.

3.1.1. — Let $\mathcal{A}$ be a commutative $\mathcal{O}_S$ -algebra, *locally free and of finite type* as an $\mathcal{O}_S$ -module. If we put

$$\mathcal{A}^* = \mathcal{H}om_{\mathcal{O}_S\text{-Mod.}}(\mathcal{A}, \mathcal{O}_S),$$

the canonical morphism φ from $\mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}^*$ to $(\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{A})^*$ is invertible. If $m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$ is the morphism defining the multiplication of $\mathcal{A}$, one obtains by composition a diagonal morphism

$$\Delta_{\mathcal{A}^*} : \mathcal{A}^* \xrightarrow{m^*} (\mathcal{A} \otimes \mathcal{A})^* \xrightarrow{\varphi^{-1}} \mathcal{A}^* \otimes \mathcal{A}^*.$$

This diagonal morphism clearly makes $\mathcal{A}^*$ a $\mathcal{O}_S$ -coalgebra having as augmentation the transpose of the morphism $\mathcal{O}_S \rightarrow \mathcal{A}$ defined by the unit section of $\mathcal{A}$. Moreover, it is clear that:

The functor $\mathcal{A} \mapsto \mathcal{A}^$ is an anti-equivalence from the category of $\mathcal{O}_S$ -algebras which are locally free and of finite type as $\mathcal{O}_S$ -modules onto the category of $\mathcal{O}_S$ -coalgebras locally free and of finite type as $\mathcal{O}_S$ -modules.*

3.1.2. — To every $\mathcal{O}_S$ -coalgebra $\mathcal{U}$ there is canonically associated an S -functor

$$\text{Spec}^* \mathcal{U} : (\mathbf{Sch}/S)^{\circ} \rightarrow (\mathbf{Ens}).$$

Observe, indeed, that for every S -scheme $q : T \rightarrow S$, $q^*(\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U})$ identifies with $q^*(\mathcal{U}) \otimes_{\mathcal{O}_T} q^*(\mathcal{U})$, so that $q^*(\Delta_{\mathcal{U}})$ makes $\mathcal{U}_T = q^*(\mathcal{U})$ an $\mathcal{O}_T$ -coalgebra; we may therefore put, by definition and with an obvious abuse of notation:⁽²⁹⁾

$$(\text{Spec}^* \mathcal{U})(T) = \{x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1 \text{ and } \Delta_{\mathcal{U}_T}(x) = x \otimes x\}.$$

The sections x of $\mathcal{U}_T$ clearly correspond to the morphisms of $\mathcal{O}_T$ -modules $\xi : \mathcal{O}_T \rightarrow \mathcal{U}_T$; the conditions $\varepsilon(x) = 1$ and $\Delta(x) = x \otimes x$ express simply that ξ is a coalgebra morphism. One therefore also has:

$$(\text{Spec}^* \mathcal{U})(T) = \text{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{U}_T).$$

In particular, one has the following proposition:⁽³⁰⁾

⁽²⁷⁾**N.D.E.** The numbering 3.1.0 has been added for later references.

⁽²⁸⁾**N.D.E.** What follows has been added. Recall also that, to show that $\mathcal{U} \otimes \mathcal{V}$ is indeed the product of $\mathcal{U}$ and $\mathcal{V}$ in the category of cocommutative $\mathcal{O}_S$ -cogbras, one verifies that if one has an arbitrary $\mathcal{O}_S$ -cogebra $\mathcal{E}$ and cogebra morphisms $f : \mathcal{E} \rightarrow \mathcal{U}$ and $g : \mathcal{E} \rightarrow \mathcal{V}$, then every cogebra morphism $\phi : \mathcal{E} \rightarrow \mathcal{U} \otimes \mathcal{V}$ such that $\text{pr}_{\mathcal{U}} \circ \phi = f$ and $\text{pr}_{\mathcal{V}} \circ \phi = g$ is necessarily equal to $(f \otimes g) \circ \Delta_{\mathcal{E}}$, and that the latter is a cogebra morphism if and only if it equals $(g \otimes f) \circ \Delta_{\mathcal{E}}$.

⁽²⁹⁾**N.D.E.** For every $x \otimes y \in \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \Gamma(T, \mathcal{U}_T)$, its image in $\Gamma(T, \mathcal{U}_T \otimes_{\mathcal{O}_T} \mathcal{U}_T)$ is still denoted by $x \otimes y$.

⁽³⁰⁾**N.D.E.** The numbering 3.1.2.1 has been added for later references. Observe, on the other hand, that the S -functor $\text{Spec}^* \mathcal{U}$ is a sheaf for the Zariski topology (and even for the (fpqc) topology if $\mathcal{U}$ is a quasi-coherent $\mathcal{O}_S$ -module).

Proposition 3.1.2.1. *Let $\mathcal{A}$ be a commutative $\mathcal{O}_S$ -algebra which is locally free of finite type as an $\mathcal{O}_S$ -module. Then the S -functor $\mathrm{Spec}^* \mathcal{A}^*$ is represented by $\mathrm{Spec} \mathcal{A}$.*

Indeed, for every S -scheme T , there are canonical isomorphisms:

$$\begin{aligned} (\mathrm{Spec}^* \mathcal{A}^*)(T) &= \mathrm{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{A}_T^*) \\ &\simeq \mathrm{Hom}_{\mathcal{O}_T\text{-alg.}}(\mathcal{A}_T, \mathcal{O}_T) \simeq (\mathrm{Spec} \mathcal{A})(T). \end{aligned}$$

3.2. — An $\mathcal{O}_S$ -coalgebra in groups, that is, a group in the category of $\mathcal{O}_S$ -coalgebras, consists in giving a $\mathcal{O}_S$ -coalgebra $(\mathcal{U}, \Delta_{\mathcal{U}})$ and three morphisms of $\mathcal{O}_S$ -coalgebras $m_{\mathcal{U}} : \mathcal{U} \otimes \mathcal{U} \rightarrow \mathcal{U}$, $\eta_{\mathcal{U}} : \mathcal{O}_S \rightarrow \mathcal{U}$, and $c_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{U}$ satisfying conditions (ii)*, (iii)*, and (vi) below; on the other hand, the fact that $m_{\mathcal{U}}$ is a cogeбра morphism is expressed by the commutativity of diagrams (iv) and (v) below:

(iv)

$$\begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \\ \Delta_{\mathcal{U}} \otimes \Delta_{\mathcal{U}} \downarrow & & \downarrow \Delta_{\mathcal{U}} \\ \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} & & \\ \mathrm{id}_{\mathcal{U}} \otimes \sigma \otimes \mathrm{id}_{\mathcal{U}} \downarrow & & \\ \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}} \otimes m_{\mathcal{U}}} & \mathcal{U} \otimes \mathcal{U} \end{array}$$

(v)

$$\begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \\ \searrow \varepsilon_{\mathcal{U}} \otimes \varepsilon_{\mathcal{U}} & & \swarrow \varepsilon_{\mathcal{U}} \\ & \mathcal{O}_S & \end{array}$$

(ii)* The square

$$\begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} & \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes m_{\mathcal{U}}} & \mathcal{U} \otimes \mathcal{U} \\ m_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}} \downarrow & & \downarrow m_{\mathcal{U}} \\ \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \end{array}$$

is commutative.

(iii)* The two composites below equal the identity morphism of $\mathcal{U}$:

$$\begin{aligned} \mathcal{U} &\simeq \mathcal{U} \otimes \mathcal{O}_S \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes \eta_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U} \\ \mathcal{U} &\simeq \mathcal{O}_S \otimes \mathcal{U} \xrightarrow{\eta_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}. \end{aligned}$$

(vi) The composite morphism below is equal to $\eta_{\mathcal{U}} \circ \varepsilon_{\mathcal{U}}$:

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{c_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}.$$

3.2.1. — The morphisms $\eta_{\mathcal{U}}$ and $c_{\mathcal{U}}$ are uniquely determined by $m_{\mathcal{U}}$. On the other hand, conditions (ii)* and (iii)* express simply that $m_{\mathcal{U}}$ makes $\mathcal{U}$ an $\mathcal{O}_S$ -algebra having as unit section the image under $\eta_{\mathcal{U}}$ of the unit section of $\mathcal{O}_S$. Condition (iv) also expresses that the diagonal morphism $\Delta_{\mathcal{U}}$ is compatible with multiplication; and indeed, $\Delta_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{U} \otimes \mathcal{U}$ must be a homomorphism of coalgebras in groups, which also implies the commutativity of the triangle

(v*)

$$\begin{array}{ccc} & \mathcal{O}_S & \\ \eta_{\mathcal{U}} \swarrow & & \searrow \eta_{\mathcal{U}} \otimes \eta_{\mathcal{U}} \\ \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}}} & \mathcal{U} \otimes \mathcal{U}. \end{array}$$

On the other hand, as in every category, the antipode $c_{\mathcal{U}}$ is an isomorphism from $\mathcal{U}$ onto the “opposite” group object;⁽³¹⁾ in particular, $c_{\mathcal{U}}$ induces an algebra isomorphism from $\mathcal{U}$ onto the opposite algebra $\mathcal{U}^\circ$.

3.2.2. — Since the functor $\mathcal{U} \mapsto \operatorname{Spec}^* \mathcal{U}$ commutes with finite products, it transforms a coalgebra in groups into an S -group functor; and indeed, for every S -scheme T , the elements $x \in \Gamma(T, \mathcal{U}_T)$ belonging to $(\operatorname{Spec}^* \mathcal{U})(T)$ form a group for the multiplication of the algebra $\Gamma(T, \mathcal{U}_T)$; the inverse of x is none other than $c_{\mathcal{U}}(x)$. According to 3.1.2.1, one has:

Scholium 3.2.2.1. ⁽³²⁾ *Let $\mathcal{U}$ be a $\mathcal{O}_S$ -coalgebra in groups, finite and locally free as an $\mathcal{O}_S$ -module. Then the S -group functor $\operatorname{Spec}^* \mathcal{U}$ is represented by the finite and locally free S -group $\operatorname{Spec} \mathcal{U}^*$.*

Remark 3.2.2.2. Let $\mathcal{L}$ be a $\mathcal{O}_S$ -Lie algebra and $\mathcal{U}(\mathcal{L})$ the *enveloping algebra* of $\mathcal{L}$, that is, the sheaf on S associated to the presheaf assigning to every open subset V the enveloping algebra $U(\Gamma(V, \mathcal{L}))$ of the Lie algebra $\Gamma(V, \mathcal{L})$.

Every homomorphism from $\mathcal{L}$ to the Lie algebra underlying an associative $\mathcal{O}_S$ -algebra factors in one and only one way through the canonical morphism from $\mathcal{L}$ to $\mathcal{U}(\mathcal{L})$; furthermore, this universal property implies, besides the functoriality of $\mathcal{U}(\mathcal{L})$ in $\mathcal{L}$, that the enveloping algebra of a product of Lie algebras identifies with the tensor product of the enveloping algebras.

In particular, the diagonal morphism $\delta : \mathcal{L} \rightarrow \mathcal{L} \times \mathcal{L}$ induces an algebra homomorphism $\Delta : \mathcal{U}(\mathcal{L}) \rightarrow \mathcal{U}(\mathcal{L} \times \mathcal{L}) \simeq \mathcal{U}(\mathcal{L}) \otimes \mathcal{U}(\mathcal{L})$. The zero morphism $\mathcal{L} \rightarrow 0$ induces an algebra homomorphism $\varepsilon : \mathcal{U}(\mathcal{L}) \rightarrow \mathcal{U}(0) \simeq \mathcal{O}_S$. The isomorphism $x \mapsto -x$ from $\mathcal{L}$ onto the opposite Lie algebra $\mathcal{L}^\circ$ induces an anti-isomorphism c of the algebra $\mathcal{U}(\mathcal{L})$. One then verifies easily that the multiplication m of the algebra $\mathcal{U}(\mathcal{L})$ makes $(\mathcal{U}(\mathcal{L}), \Delta)$ a $\mathcal{O}_S$ -coalgebra in groups having ε as augmentation and c as antipode.⁽³³⁾

3.2.3. — ⁽³⁴⁾ Let $\mathcal{U}$ be a $\mathcal{O}_S$ -coalgebra in groups. We shall see that the S -group functor $G = \operatorname{Spec}^* \mathcal{U}$ is *very good*, in the sense of II, 4.6 and 4.10.

Let $\mathcal{M}$ be a free $\mathcal{O}_S$ -module of rank r , and let $T \rightarrow S$ be an S -scheme. Since $I_T(\mathcal{M}) = \operatorname{Spec}(\mathcal{O}_T \oplus \mathcal{M}_T)$, so that $\pi : I_T(\mathcal{M}) \rightarrow T$ is affine, one has

$$\pi_*(\mathcal{U}_{I_T(\mathcal{M})}) = \mathcal{U}_T \otimes_{\mathcal{O}_T} \pi_*(\mathcal{O}_{I_T(\mathcal{M})}) = \mathcal{U}_T \otimes_{\mathcal{O}_T} (\mathcal{O}_T \oplus \mathcal{M}_T),$$

and therefore

$$(1) \quad \Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}) \simeq \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} (\mathcal{O}(T) \oplus \Gamma(T, \mathcal{M}_T)).$$

Let $(d_1, \dots, d_r)$ be a basis of $\mathcal{M}$. Then an element $u_0 + \sum_i u_i d_i$ of $\Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})})$ belongs to $G(I_T(\mathcal{M}))$ if and only if one has:

$$1 = \varepsilon\left(u_0 + \sum_i u_i d_i\right) = \varepsilon(u_0) + \sum_i \varepsilon(u_i) d_i,$$

and

$$\begin{aligned} \left(u_0 + \sum_i u_i d_i\right) \otimes \left(u_0 + \sum_i u_i d_i\right) &= \Delta\left(u_0 + \sum_i u_i d_i\right) \\ &= \Delta(u_0) + \sum_i \Delta(u_i) d_i, \end{aligned}$$

⁽³¹⁾**N.D.E.** i.e. equipped with the multiplication $m'_{\mathcal{U}} = m_{\mathcal{U}} \circ \sigma$.

⁽³²⁾**N.D.E.** This scholium, implicit in the original, has been added.

⁽³³⁾**N.D.E.** The S -group functor $\operatorname{Spec}^* \mathcal{U}(\mathcal{L})$ is not representable in general, but we shall see below (5.5) that if S is a scheme of characteristic p , if $\mathcal{L}$ is finite locally free over $\mathcal{O}_S$, and if $\mathcal{U}_p(\mathcal{L})$ is its *restricted enveloping algebra* (cf. 5.3), then $\operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by a finite and locally free S -group.

⁽³⁴⁾**N.D.E.** This paragraph has been expanded.

that is:

$$(2) \quad \begin{cases} \varepsilon(u_0) = 1, & \Delta u_0 = u_0 \otimes u_0, & \text{i.e. } u_0 \in G(T), \\ \varepsilon(u_i) = 0, & \Delta(u_i) = u_i \otimes u_0 + u_0 \otimes u_i, & i = 1, \dots, r. \end{cases}$$

Moreover, the morphism $G(I_T(\mathcal{M})) \rightarrow G(T)$ corresponding to the zero section of $I_T(\mathcal{M}) \rightarrow T$ is given by: $u_0 + \sum_i u_i d_i \mapsto u_0$. From this, combined with (1) and (2), one deduces that if $\mathcal{N}$ is a second free $\mathcal{O}_S$ -module of finite rank, the diagram of sets

$$\begin{array}{ccc} G(I_T(\mathcal{M} \oplus \mathcal{N})) & \longrightarrow & G(I_T(\mathcal{N})) \\ \downarrow & & \downarrow \\ G(I_T(\mathcal{M})) & \longrightarrow & G(T) \end{array}$$

is cartesian, i.e. G satisfies condition (E) of II, 3.5.

Denote by $\text{Prim } \Gamma(T, \mathcal{U}_T)$ the sub- $\mathcal{O}(T)$ -module of $\Gamma(T, \mathcal{U}_T)$ consisting of *primitive elements*, i.e. elements u satisfying (with the abuse of notation pointed out in 3.1.2):

$$\Delta u = u \otimes 1 + 1 \otimes u, \quad \varepsilon(u) = 0.$$

(35) Since $(\text{Lie}G)(T)$ is the set of elements $u_0 + ud \in G(I_T)$ over the unit element $u_0 = 1$ of $G(T)$, one obtains an isomorphism of $\mathcal{O}(T)$ -modules, functorial in T :⁽³⁶⁾

$$(\text{Lie}G)(T) \simeq \text{Prim } \Gamma(T, \mathcal{U}_T).$$

On the other hand, one deduces from (1) that

$$\text{Prim } \Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}) \simeq \text{Prim } \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})),$$

and it follows that the natural morphism of $\mathcal{O}(I_T(\mathcal{M}))$ -modules:

$$(\text{Lie}G)(T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})) \longrightarrow (\text{Lie}G)(I_T(\mathcal{M}))$$

is an isomorphism, i.e. $\text{Lie}G$ is a good $\mathbf{O}_S$ -module (cf. II, Def. 4.4).

Thus G is a good S -group functor (cf. II, Def. 4.6), and according to II, 4.7.2, $\text{Lie}G$ is equipped with a $\mathbf{O}_S$ -bilinear “Lie bracket” satisfying the Jacobi identity. It remains to show that G is very good, i.e. that the “bracket” on $(\text{Lie}G)(T)$ does indeed satisfy $[u, u] = 0$ for every $u \in (\text{Lie}G)(T)$ (cf. II, 4.10).

Let u, v be two elements of $(\text{Lie}G)(T)$, i.e. two primitive elements of $\Gamma(T, \mathcal{U}_T)$. Put $I = \text{Spec } \mathcal{O}_S[d]/(d^2)$ and $I' = \text{Spec } \mathcal{O}_S[d']/(d'^2)$. Since the composition law of $G(I \times I')$ is induced by the multiplication of the algebra $\mathcal{U}_{I \times I'}$, one has in $G(I \times I')$ the equality:

$$\begin{aligned} (1 + ud)(1 + vd')(1 + ud)^{-1}(1 + vd')^{-1} \\ = (1 + ud)(1 + vd')(1 - ud)(1 - vd) \\ = 1 + (uv - vu)dd'. \end{aligned}$$

According to the description of the bracket $[u, v]$ given before Prop. 4.8 of Exp. II, one obtains that

$$[u, v] = uv - vu,$$

where the right-hand term is the commutator of u and v in the algebra $\Gamma(T, \mathcal{U}_T)$, whence $[u, u] = 0$. One has therefore obtained the following proposition:⁽³⁷⁾

Proposition. *Let $\mathcal{U}$ be a $\mathcal{O}_S$ -coalgebra in groups. The S -group functor $G = \text{Spec}^* \mathcal{U}$ is very good, and there is an isomorphism $\text{Lie}G \simeq \text{Prim } \mathbf{W}(\mathcal{U})$ of $\mathbf{O}_S$ -Lie algebras, where $\text{Prim } \mathbf{W}(\mathcal{U})$ denotes the functor which associates to every $T \rightarrow S$ the $\mathcal{O}(T)$ -Lie algebra consisting of the primitive elements of $\mathbf{W}(\mathcal{U})(T) = \Gamma(T, \mathcal{U}_T)$.*

(35) **N.D.E.** Note that the second condition is a consequence of the first, for the latter implies that $u = (\text{id} \otimes \varepsilon)\Delta(u) = u + \varepsilon(u)$, whence $\varepsilon(u) = 0$.

(36) **N.D.E.** The $\mathbf{O}_S$ -module structure on $\text{Lie}G$ is defined in II, Prop. 3.6.

(37) **N.D.E.** This proposition, which summarizes the preceding discussion, has been added.

3.3. — Finally, suppose that $\mathcal{U}$ is a $\mathcal{O}_S$ -coalgebra in commutative groups, that is, that the triangle

$$(i^*) \quad \begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} & \xrightarrow{\sigma} & \mathcal{U} \otimes \mathcal{U} \\ & \searrow m_{\mathcal{U}} \quad \swarrow m_{\mathcal{U}} & \\ & \mathcal{U} & \end{array}$$

is commutative, or again that $m_{\mathcal{U}}$ makes $\mathcal{U}$ a commutative $\mathcal{O}_S$ -algebra. Conditions (i), (ii), (iii), (iv), (v), (vi), (i)*, (ii)*, (iii)*, and (v)* then also mean that $\mathcal{U}$ is a cogroup in the category of commutative $\mathcal{O}_S$ -algebras. Thus, if in addition $\mathcal{U}$ is a quasi-coherent $\mathcal{O}_S$ -module, the affine S -scheme $\text{Spec } \mathcal{U}$ is a commutative S -group scheme.

In this case, since the diagonal morphism Δ' of $\mathcal{O}_S[T, T^{-1}]$ sends T to $T \otimes T$, the morphisms of S -groups from $\text{Spec } \mathcal{U}$ to $\mathbf{G}_{m,S}$ (I 4.3.2) correspond bijectively to the morphisms of unital $\mathcal{O}_S$ -algebras

$$\varphi : \mathcal{O}_S[T, T^{-1}] \longrightarrow \mathcal{U}$$

such that $(\varphi \otimes \varphi) \circ \Delta' = \Delta_{\mathcal{U}} \circ \varphi$ (in this case, $\varepsilon_{\mathcal{U}} \circ \varphi$ is the neutral element of $\mathbf{G}_{m,S}(S)$, i.e. the augmentation ε'). Such a morphism φ is determined by the image $\varphi(T)$, which must be an invertible element x of $\mathcal{U}$ satisfying $\Delta_{\mathcal{U}} x = x \otimes x$ and $\varepsilon_{\mathcal{U}}(x) = \varepsilon'(T) = 1$. One therefore has:

$$\text{Hom}_{S\text{-gr.}}(\text{Spec } \mathcal{U}, \mathbf{G}_{m,S}) \simeq (\text{Spec}^* \mathcal{U})(S)$$

and since this formula remains valid after every base change, this gives:

$$\text{Spec}^* \mathcal{U} \simeq \underline{\text{Hom}}_{S\text{-gr.}}(\text{Spec } \mathcal{U}, \mathbf{G}_{m,S}).$$

One has therefore obtained the

Proposition 3.3.0. *If $\mathcal{U}$ is a $\mathcal{O}_S$ -coalgebra in commutative groups, quasi-coherent as an $\mathcal{O}_S$ -module, then the affine S -scheme $G = \text{Spec } \mathcal{U}$ is a commutative S -group scheme, and there is an isomorphism of S -group functors $\text{Spec}^* \mathcal{U} \simeq \underline{\text{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S})$.*

If one assumes, in addition, that $\mathcal{U}$ is a locally free $\mathcal{O}_S$ -module of finite type, then, according to 3.1.2.1, the S -group functor $\text{Spec}^* \mathcal{U}$ is represented by $\text{Spec } \mathcal{U}^*$. One therefore obtains the

Proposition 3.3.1. (Cartier duality). *The functor*

$$\mathcal{A}(G) \mapsto \mathcal{A}(G)^* = \mathcal{H}om_{\mathcal{O}_S\text{-Mod.}}(\mathcal{A}(G), \mathcal{O}_S)$$

induces a duality of the category of commutative, finite and locally free S -group schemes; it associates to G the S -group $\underline{\text{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S})$.*

4. “Frobeniusseries”

Let p be a fixed prime number and $(\mathbf{Sch}/\mathbf{F}_p)$ the category of schemes of characteristic p , that is, schemes over the prime field $\mathbf{F}_p$. Following the general conventions of this seminar, we identify $(\mathbf{Sch}/\mathbf{F}_p)$ with a subcategory of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ by means of the functor h of I 1.1. We likewise use the isomorphism from $\text{Hom}(h_X, \mathbf{F})$ onto $\mathbf{F}(X)$ defined in I 1.1 to identify these two sets whenever X is an $\mathbf{F}_p$ -scheme and $\mathbf{F}$ an object of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$.

A duality of a category $\mathcal{C}$ is a pair (D, φ) consisting of a contravariant functor D from $\mathcal{C}$ to $\mathcal{C}$ and a functorial isomorphism $\varphi : \text{Id}_{\mathcal{C}} \longrightarrow DD$ such that the isomorphisms $\varphi D : D \longrightarrow DDD$ and $D\varphi^{-1} : DDD \longrightarrow D$ are inverse to one another.

(38) **N.D.E.** $D\varphi$ has been corrected to $D\varphi^{-1}$.

Notation 4.0. ⁽³⁹⁾ If T is an $\mathbf{F}_p$ -scheme, a T -functor is a morphism $q : \mathbf{F} \rightarrow T$ of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ with target T ; for every T -scheme $r : X \rightarrow T$, the set of T -morphisms $X \rightarrow \mathbf{F}$, i.e. $\mathbf{F}_p$ -morphisms $s : X \rightarrow \mathbf{F}$ such that $q \circ s = r$, will then be denoted by $q(r)$, $q(X/T)$, $\mathbf{F}(r)$, or $\mathbf{F}(X/T)$ (or even $\mathbf{F}(X)$ when no confusion will be possible with $\text{Hom}(h_X, \mathbf{F})$).

4.1. — For every scheme S of characteristic p , we denote by $\text{fr}(S)$, or simply fr , the endomorphism of S inducing the identity on the topological space underlying S and associating x^p to a section x of $\mathcal{O}_S$ over an open subset U .

Then the map $\text{fr} : S \mapsto \text{fr}(S)$ is an *endomorphism of the identity functor* of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$, ⁽⁴⁰⁾ which implies the following results. Let $\mathbf{E}$ be an $\mathbf{F}_p$ -functor, that is, an object of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$; the map associating to every $\mathbf{F}_p$ -scheme S the endomorphism $\mathbf{E}(\text{fr}(S))$ of $\mathbf{E}(S)$ is a functorial endomorphism of $\mathbf{E}$ which we shall denote by $\text{fr}(\mathbf{E})$ or fr ; this notation is compatible with the identification of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ with a subcategory of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$. Moreover, the map $\mathbf{E} \mapsto \text{fr}(\mathbf{E})$ is an *endomorphism of the identity functor* of $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ (which we shall again denote by fr). ⁽⁴¹⁾

For every $\mathbf{F}_p$ -scheme S and every S -functor $q : X \rightarrow S$, we denote by $X^{(p/S)}$ or $X^{(p)}$ the inverse image of X by the base change $\text{fr}(S)$:

$$\begin{array}{ccc} X^{(p/S)} & \xrightarrow{\text{pr}_X} & X \\ \downarrow & & \downarrow q \\ S & \xrightarrow{\text{fr}(S)} & S. \end{array}$$

The commutative square

$$\begin{array}{ccc} X & \xrightarrow{\text{fr}(X)} & X \\ \downarrow q & & \downarrow q \\ S & \xrightarrow{\text{fr}(S)} & S \end{array}$$

then induces an S -morphism denoted by $\text{Fr}(X/S)$ (or simply Fr) from X to $X^{(p/S)}$ such that $\text{fr}(X) = \text{pr}_X \circ \text{Fr}(X/S)$:

$$\begin{array}{ccccc} X & & \xrightarrow{\text{fr}(X)} & & X \\ & \searrow \text{Fr}(X/S) & & & \downarrow q \\ & & X^{(p/S)} & \xrightarrow{\text{pr}_X} & X \\ & \downarrow q & \downarrow & & \downarrow q \\ & & S & \xrightarrow{\text{fr}(S)} & S. \end{array}$$

We shall say that $\text{Fr}(X/S)$ is the *Frobenius morphism of X relative to S* ; it is clear that the map $\text{Fr} : X \mapsto \text{Fr}(X/S)$ is a functorial homomorphism.

⁽³⁹⁾**N.D.E.** The numbering 4.0 has been added for later references.

⁽⁴⁰⁾**N.D.E.** i.e. for every morphism of $\mathbf{F}_p$ -schemes $f : Y \rightarrow X$, the diagram below is commutative:

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ \text{fr}(Y) \downarrow & & \downarrow \text{fr}(X) \\ Y & \xrightarrow{f} & X. \end{array}$$

⁽⁴¹⁾**N.D.E.** One says that $\text{fr}(X)$ is the “absolute” Frobenius morphism of X , to distinguish it from the “relative” Frobenius morphism $\text{Fr}(X/S)$ introduced below.

(42) Let $r : T \rightarrow S$ be an S -scheme. For every $\phi \in X(r) = \text{Hom}_S(T, X)$ (cf. 4.0), there is a commutative diagram:

$$\begin{array}{ccccc} X & \xrightarrow{\text{Fr}(X/S)} & X^{(p/S)} & \xrightarrow{\text{pr}_X} & X \\ \uparrow \phi & \searrow q & \downarrow q^{(p/S)} & & \downarrow q \\ T & \xrightarrow{r} & S & \xrightarrow{\text{fr}(S)} & S. \end{array}$$

According to the definition of $X^{(p/S)}$ as a fibered product, pr_X induces a bijection:

$$X^{(p/S)}(r) = \text{Hom}_S(T, X^{(p/S)}) \xrightarrow{\sim} \text{Hom}_S(T, X) = X(\text{fr}(S) \circ r).$$

On the other hand, $r \circ \text{fr}(T) = \text{fr}(S) \circ r$, since fr is an endomorphism of the identity functor. It follows that the map $\text{Fr}(X/S)(r) : X(r) \rightarrow X^{(p/S)}(r)$ may be characterized by the commutativity of the following square:

$$\begin{array}{ccc} X(r) & \xrightarrow{\text{Fr}(X/S)(r)} & X^{(p/S)}(r) \\ \downarrow X(\text{fr}(T)) & & \downarrow \wr \\ X(r \circ \text{fr}(T)) & \xlongequal{\quad} & X(\text{fr}(S) \circ r). \end{array}$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, then $X^{(p)}$ is the subscheme of S defined by the ideal $\mathcal{I}^{[p]}$ generated by the p -th powers of sections of $\mathcal{I}$; furthermore, $\text{Fr}(X/S)$ is then the canonical immersion from $\text{Spec}(\mathcal{O}_X/\mathcal{I})$ into $\text{Spec}(\mathcal{O}/\mathcal{I}^{[p]})$.

4.1.1.1. — (43) Let $t : T \rightarrow S$ be a base change and $X_T = X \times_{q,t} T$. Consider the inverse image of X_T under $\text{fr}(T)$:

$$\begin{array}{ccccc} (X_T)^{(p/T)} & \longrightarrow & X_T & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow q \\ T & \xrightarrow{\text{fr}(T)} & T & \xrightarrow{t} & S. \end{array}$$

Since $t \circ \text{fr}(T) = \text{fr}(S) \circ t$, $(X_T)^{(p/T)}$ identifies with the inverse image of $X^{(p/S)}$ under t ; in other words, there is a canonical isomorphism:

$$X_T^{(p/T)} \xrightarrow{\sim} (X^{(p/S)})_T.$$

It is clear that, under this identification, $\text{Fr}(X_T/T)$ identifies with the inverse image $\text{Fr}(X/S)_T$ of $\text{Fr}(X/S)$.

4.1.1.1.1. — In particular, if S is the spectrum of the prime field $\mathbf{F}_p$, $X^{(p/S)}$ is equal to X and $\text{Fr}(X/S)$ to $\text{fr}(X)$. Consequently, $X_T^{(p/T)}$ identifies with X_T and $\text{Fr}(X_T/T)$ with $\text{fr}(X)_T$.

For example, if E is a set and E_T the constant T -scheme of type E , one has $E_T^{(p/T)} \simeq E_T$ and $\text{Fr}(E_T/T) \simeq \text{id}_{E_T}$.

4.1.1.2. — The functor $X \mapsto X^{(p/S)}$ clearly commutes with products; it therefore transforms an S -group G into an S -group $G^{(p/S)}$; moreover, since Fr is a functorial homomorphism,

$$\text{Fr}(G/S) : G \rightarrow G^{(p/S)}$$

is a homomorphism of S -groups. We shall denote its kernel by $_{\text{Fr}}G$.

(42) **N.D.E.** The original has been expanded in what follows.

(43) **N.D.E.** The original has been expanded in what follows.

If $r : T \rightarrow S$ is a scheme over S , it follows from diagram (†) of 4.1 that the value of $\mathrm{Fr}G$ at r is the kernel of the homomorphism

$$G(\mathrm{fr}(T)) : G(r) \rightarrow G(r \circ \mathrm{fr}(T)).$$

Now, when T is the scheme I_R of dual numbers over an S -scheme R , $\mathrm{fr}(I_R)$ factors as follows:

$$I_R \xrightarrow{\mathrm{can.}} R \xrightarrow{\mathrm{fr}(R)} R \xrightarrow{s} I_R,$$

where s is the zero section. It follows that $(\mathrm{Fr}G)(I_R)$ contains the kernel $\underline{\mathrm{Lie}}(G/S)(R)$ of the morphism $G(s) : G(I_R) \rightarrow G(R)$, and that one therefore has: $\underline{\mathrm{Lie}}(G/S) = \underline{\mathrm{Lie}}(\mathrm{Fr}G/S)$.

4.1.3. — More generally, for every S -functor X , we define the S -functor $X^{(p^n)}$ by induction on n using the formulas:

$$X^{(p)} = X^{(p/S)} \quad \text{and} \quad X^{(p^n)} = (X^{(p^{n-1})})^{(p)}.$$

Similarly, $\mathrm{Fr}^n(X/S)$ or Fr^n denote the composite functorial homomorphism

$$\begin{aligned} X &\xrightarrow{\mathrm{Fr}(X/S)} X^{(p)} \xrightarrow{\mathrm{Fr}(X^{(p)}/S)} X^{(p^2)} \\ &\rightarrow \dots \rightarrow X^{(p^{n-1})} \xrightarrow{\mathrm{Fr}(X^{(p^{n-1})}/S)} X^{(p^n)}. \end{aligned}$$

Note that, according to 4.1.1, $\mathrm{Fr}(X^{(p)}/S)$ coincides with $\mathrm{Fr}(X/S)^{(p)}$, i.e. the following diagram is commutative:

$$\begin{array}{ccc} X^{(p)} & \xrightarrow{\quad} & X \\ \mathrm{Fr}(X^{(p)}/S) \downarrow & & \downarrow \mathrm{Fr}(X/S) \\ X^{(p^2)} & \xrightarrow{\quad} & X^{(p)}. \end{array}$$

If G is an S -group functor, $G^{(p^n)}$ is one also, and $\mathrm{Fr}^n(G/S)$ is a homomorphism of S -group functors.

Definition. We shall denote by $\mathrm{Fr}^n G$ the kernel of $\mathrm{Fr}^n(G/S)$ and shall say that G is of *height* $\leq n$ if $\mathrm{Fr}^n(G/S)$ is zero, that is, if $\mathrm{Fr}^n G = G$.

Lemma. The group subfunctor $\mathrm{Fr}^n G$ of G is characteristic, i.e. for every S -scheme T , every endomorphism ϕ of the T -group functor G_T induces an endomorphism of $(\mathrm{Fr}^n G)_T$.

Indeed, since the construction of $G^{(p^n)}$ and of $\mathrm{Fr}^n(G/S)$ commutes with base changes according to 4.1.1, one may suppose $T = S$; in this case, the assertion follows from the fact that $\mathrm{Fr}^n(G/S)$ is a functorial homomorphism.

4.1.4. — Here are a few examples.

a) First consider an “abstract” abelian group M and the diagonalizable group $G = D_S(M)$ of type M (I 4.4): for every S -scheme T , $G(T)$ is therefore the abelian group

$$\mathrm{Hom}_{(\mathrm{Ab})}(M, \Gamma(T, \mathcal{O}_T)^\times).$$

Since G is the inverse image of the diagonalizable group $D(M)$ over $\mathbf{F}_p$, $G^{(p)}$ identifies with G , and $\mathrm{Fr}(G/S)(T)$ identifies with the endomorphism $x \mapsto x^p$ of $G(T)$ (4.1.1). In particular, when M is equal to $\mathbf{Z}$, one has $D_S(M) = \mathbf{G}_{m,S}$, so that:

$\mathrm{Fr} \mathbf{G}_{m,S}$ is the S -group $\mu_{p,S}$ which associates to every S -scheme T the group of p -th roots of unity in $\Gamma(T, \mathcal{O}_T)^*$.

b) Now consider a scheme S of characteristic p and a sheaf of modules $\mathcal{E}$ on S . According to I 4.6.2, there is a canonical isomorphism

$$\mathbf{W}(\mathcal{E})^{(p)} \simeq \mathbf{W}(\mathcal{E}^{(p)}),$$

where $\mathcal{E}^{(p)}$ is the inverse image of $\mathcal{E}$ under $\mathrm{fr}(S)$. For every S -scheme $\pi : T \rightarrow S$, the map $\mathrm{Fr}(\mathbf{W}(\mathcal{E}))(\pi)$ is determined, according to 4.1 (†), by the commutative triangle

$$\begin{array}{ccc} \Gamma(T, \pi^* \mathrm{fr}(S)^* \mathcal{E}) & \xrightarrow[\mathrm{can.}]{\sim} & \Gamma(T, \mathrm{fr}(T)^* \pi^* \mathcal{E}) \\ & \nwarrow \mathrm{Fr}(\mathbf{W}(\mathcal{E})/S)(\pi) \quad \nearrow f' & \\ & \Gamma(T, \pi^* \mathcal{E}) & \end{array},$$

where f' is the map induced by $\mathrm{fr}(T)$.

In particular, if $\mathcal{E}$ is equal to $\mathcal{O}_S$, $\mathbf{W}(\mathcal{E})$ identifies with the additive group $\mathbf{G}_{a,S}$. In this case, one has $\mathcal{E}^{(p)} = \mathcal{E} = \mathcal{O}_S$, and the Frobenius morphism $\mathrm{Fr}(\mathbf{G}_{a,S}/S)$ sends $x \in \Gamma(T, \mathcal{O}_T)$ to x^p . Thus:

$\mathrm{Fr} \mathbf{G}_{a,S}$ is the S -group $\alpha_{p,S}$ which associates to every S -scheme T the group: $\{x \in \Gamma(T, \mathcal{O}_T) \mid x^p = 0\}$.

c) One would see similarly that, for every quasi-coherent $\mathcal{O}_S$ -algebra $\mathcal{A}$, $(\mathrm{Spec} \mathcal{A})^{(p)}$ identifies with the spectrum $\mathrm{Spec} \mathcal{A}^{(p)}$ of the inverse image of $\mathcal{A}$ under $\mathrm{fr}(S)$. If π denotes the endomorphism $x \mapsto x^p$ of the sheaf of rings $\mathcal{O}_S$, one therefore has

$$\mathcal{A}^{(p)} = \mathcal{A} \otimes_{\pi} \mathcal{O}_S$$

(44) and $\mathrm{Fr}((\mathrm{Spec} \mathcal{A})/S)$ is induced by the morphism of $\mathcal{O}_S$ -algebras $\mathcal{A} \otimes_{\pi} \mathcal{O}_S \rightarrow \mathcal{A}$ defined by $a \otimes_{\pi} x \mapsto a^p x$.

Finally, for every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$, there are canonical isomorphisms

$$\mathbf{V}(\mathcal{E})^{(p)} \simeq \mathbf{V}(\mathcal{E}^{(p)}) \quad \text{and} \quad S(\mathcal{E})^{(p)} \simeq S(\mathcal{E}^{(p)}),$$

where $S(\mathcal{E})$ denotes the symmetric algebra of the $\mathcal{O}_S$ -module $\mathcal{E}$.

d) Let $\mathcal{U}$ be a $\mathcal{O}_S$ -coalgebra (3.1), and T a scheme of characteristic p . If $\mathcal{U}^{(p/S)}$ or $\mathcal{U}^{(p)}$ denote the inverse image of the coalgebra $\mathcal{U}$ under $\mathrm{fr}(S)$, there is, as in b), a canonical isomorphism:

$$(\mathrm{Spec}^* \mathcal{U})^{(p)} \simeq \mathrm{Spec}^* \mathcal{U}^{(p)}.$$

If $\mathcal{U}$ is a coalgebra in groups, the value of $\mathrm{Fr}(\mathrm{Spec}^* \mathcal{U})$, i.e. of the kernel of the Frobenius morphism $\mathrm{Spec}^* \mathcal{U} \rightarrow (\mathrm{Spec}^* \mathcal{U})^{(p)}$, at an S -scheme T is therefore the set of elements γ of

$$(\mathrm{Spec}^* \mathcal{U})(T) = \{x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1, \Delta_{\mathcal{U}_T} x = x \otimes x\}$$

such that the image in $\Gamma(T, \mathcal{U}_T \otimes_{\mathrm{fr}(T)} \mathcal{O}_T)$ of the element $\gamma \otimes_{\mathrm{fr}(T)} 1$ of $\Gamma(T, \mathcal{U}_T) \otimes_{\mathrm{fr}(T)} \mathcal{O}(T)$ is equal to 1.

4.2. — (45) We shall now deal with a construction close to the preceding one: let S be a scheme of characteristic p , X an S -scheme, and X_S^p the product in the category $(\mathbf{Sch}/S)$ of p copies of X .

We then denote by $U^p(X)$ the open subscheme of X_S^p which is the union of the products U_S^p , as U ranges over the affine open subsets of X . A point x of X_S^p therefore belongs to $U^p(X)$ if and only if the projections $\mathrm{pr}_i x$ of x onto the factors of X_S^p belong to the same affine open subset of X . For example, if every finite subset of X is contained in an affine open subset, one has $U^p(X) = X_S^p$.

The symmetric group $\mathcal{S}_p$ of order p acts on X_S^p by permutation of the factors and leaves the open subset $U^p(X)$ stable. We shall call the quotient of X_S^p by $\mathcal{S}_p$ in the category of ringed spaces the p -fold symmetric product of X , and denote it by $\Sigma^p X$. Let $q(X)$, or simply q , be the canonical projection $X_S^p \rightarrow \Sigma^p X$.

Then q maps $U^p(X)$ onto an open subset $V^p(X)$ of the symmetric product, which may be described as follows (cf. V 4.1). The structure sheaf of $\Sigma^p X$ induces on $V^p(X)$ a scheme

(44) **N.D.E.** $\mathcal{A} \otimes_{\pi} \mathcal{O}_S$ denotes the $\mathcal{O}_S$ -algebra obtained by the extension of scalars $\pi : \mathcal{O}_S \rightarrow \mathcal{O}_S$, i.e. one has: $ax \otimes_{\pi} 1 = a \otimes_{\pi} x^p$, and $x \cdot (a \otimes_{\pi} 1) = a \otimes_{\pi} x$.

(45) **N.D.E.** For the contents of nos. 4.2 and 4.3, one may also refer to [DG70], § IV.3, nos. 4–6.

structure; the morphism $q'(X) : U^p(X) \rightarrow V^p(X)$ induced by $q(X)$ is affine and even integral; as U ranges over the affine open subsets of X which project into a varying affine open subset V of S , the $\Sigma^p U$ form an affine covering of $V^p(X)$; if R denotes the affine algebra of V and A that of U , $\Sigma^p U$ has as affine algebra the subalgebra $\Sigma^p A$ of $\bigotimes_R^p A$ consisting of symmetric tensors.

Now consider the diagonal morphism δ from X to $U^p(X)$. If $V = \operatorname{Spec} R$ is an affine open subset of S and $U = \operatorname{Spec} A$ an affine open subset of X over V , the restriction of δ to U is defined by the algebra morphism

$$\eta : \bigotimes_R^p A \longrightarrow A, \quad a_1 \otimes \cdots \otimes a_p \longmapsto a_1 a_2 \cdots a_p.$$

One therefore has, if N is the symmetrization operator:

$$\begin{aligned} \eta(N(a_1 \otimes \cdots \otimes a_p)) &= \eta\left(\sum_{\sigma \in \mathcal{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}\right) \\ &= p! a_1 \cdots a_p = 0. \end{aligned}$$

In other words, η vanishes on the subspace $N(\bigotimes_R^p A)$ of $\Sigma^p A$ consisting of symmetrized tensors. Moreover, if f is a symmetric tensor, one clearly has $N(fa) = fN(a)$, which shows that $N(\bigotimes_R^p A)$ is an ideal of $\Sigma^p A$. From now on we shall write

$$U^{[p/S]} = \operatorname{Spec}\left(\Sigma^p A / N\left(\bigotimes_R^p A\right)\right);$$

this is a closed subscheme of $\Sigma^p(U) = V^p(U)$. The union of the $U^{[p/S]}$, as U ranges over the affine open subsets of X which project into a varying affine open subset V of S , is a closed subscheme of $V^p(X)$, denoted by $X^{[p/S]}$.

Moreover, if $i(X)$ denotes the inclusion of $X^{[p/S]}$ in $V^p(X)$, we have just seen that $q'(X) \circ \delta$ factors through $X^{[p/S]}$, whence a morphism $F^{[p]}(X/S) : X \rightarrow X^{[p/S]}$; ⁽⁴⁶⁾

$$\begin{array}{ccc} X_S^p & \supset & U^p(X) \xleftarrow{\delta(X)} X \\ q(X) \downarrow & & \downarrow q'(X) \quad \downarrow F^{[p]}(X/S) \\ \Sigma^p(X) & \supset & V^p(X) \xleftarrow{i(X)} X^{[p/S]}. \end{array}$$

It is clear that $X^{[p/S]}$ is functorial in X and that the map $F^{[p]} : X \mapsto F^{[p]}(X/S)$ is a functorial homomorphism.

4.2.1. — The schemes $X^{[p/S]}$ and $X^{(p/S)}$ are clearly related: let V be an affine open subset of S with affine ring R , and U an affine open subset of X over V ; let A be the affine algebra of U . If π denotes the endomorphism $x \mapsto x^p$ of R , then $U^{(p/S)}$ has $A \otimes_\pi R$ as affine algebra. One verifies, moreover, that the map

$$a \otimes_\pi \lambda \longmapsto \left(\lambda a \otimes \cdots \otimes a \quad \text{mod } N\left(\bigotimes_R^p A\right) \right)$$

defines a morphism of R -algebras from $A \otimes_\pi R$ to $\Sigma^p A / N(\bigotimes_R^p A)$, and the latter induces a morphism $\varphi(U) : U^{[p/S]} \rightarrow U^{(p/S)}$ such that $\varphi(U) \circ F^{[p]}(U/S) = \operatorname{Fr}(U/S)$.

⁽⁴⁶⁾**N.D.E.** In the original, this morphism (resp. the relative Frobenius morphism) was denoted by F' (resp. F).

“Gluing the pieces together”, one then obtains a commutative triangle

$$\begin{array}{ccc} & X & \\ F^{[p]}(X/S) \swarrow & & \searrow \text{Fr}(X/S) \\ X^{[p/S]} & \xrightarrow{\varphi(X)} & X^{(p/S)}. \end{array}$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, $F^{[p]}(X/S)$ identifies with the identity morphism of X , so that $\varphi(X)$ is the canonical immersion from $\text{Spec}(\mathcal{O}_S/\mathcal{I})$ into $\text{Spec}(\mathcal{O}_S/\mathcal{I}^{\{p\}})$. One thus sees that $\varphi(X)$ is not an isomorphism in general.

However, when M is a free R -module, it is clear that the map

$$\begin{aligned} M \otimes_{\pi} R &\longrightarrow \Sigma^p M / N \left(\bigotimes_R^p M \right), \\ m \otimes_{\pi} \lambda &\longmapsto \left(\lambda m \otimes \cdots \otimes m \quad \text{mod } N \left(\bigotimes_R^p M \right) \right) \end{aligned}$$

is bijective; this map therefore remains bijective when M is R -flat, because every flat module is a filtered inductive limit of free modules (Lazard⁽⁴⁷⁾). It follows that

$\varphi(X) : X^{[p/S]} \longrightarrow X^{(p/S)}$ is an isomorphism if X is a flat S -scheme.

4.3. — Finally, consider an S -scheme in abelian groups G . Then the composite morphism $\mu(G) : U^p(G) \hookrightarrow G_S^p \rightarrow G$, which is defined by multiplication, factors through $V^p(G)$, i.e. there exists a morphism $\nu(G) : V^p(G) \rightarrow G$ such that $\nu(G) \circ q'(G) = \mu(G)$, so that one has the following commutative diagram:

$$\begin{array}{ccccc} G & \xleftarrow{\mu(G)} & U^p(G) & \xleftarrow{\delta(G)} & G \\ & \searrow \nu(G) & \downarrow q'(G) & & \downarrow F^{[p]}(G/S) \\ & & V^p(G) & \xleftarrow{i(G)} & G^{[p/S]} \\ & & & \searrow \varphi(G) & \downarrow \text{Fr}(G/S) \\ & & & & G^{(p/S)} \end{array}$$

When G is S -flat, $\varphi(G)$ is an isomorphism and one can define a morphism (called *Verschiebung*)

$$\text{Ver}(G/S) : G^{(p/S)} \longrightarrow G$$

by means of the formula $\text{Ver}(G/S) = \nu(G) \circ i(G) \circ \varphi(G)^{-1}$. As G ranges over the flat S -schemes in abelian groups, the map $\text{Ver} : G \mapsto \text{Ver}(G/S)$ is clearly a functorial homomorphism; consequently, $\text{Ver}(G/S)$ is a *group homomorphism*. Finally, for every S -scheme T , the composite map

$$G(T) \xrightarrow{\delta(G)(T)} U^p(G)(T) \xrightarrow{\mu(G)(T)} G(T)$$

sends $x \in G(T)$ to $p \cdot x$. We may write $p \cdot \text{id}_G$ instead of $\mu(G) \circ \delta(G)$, thus obtaining the classical formula:

$$(*) \quad \text{Ver}(G/S) \circ \text{Fr}(G/S) = p \cdot \text{id}_G.$$

Examples 4.3.1. (a) When G is a constant S -scheme in abelian groups, we know that $\text{Fr}(G/S)$ identifies with the identity morphism of G (cf. 4.1.1.1). One therefore has $\text{Ver}(G/S) = p \text{id}_G$.

^{*}D. Lazard, *C. R. Acad. Sc. Paris* **258**, 1964, pp. 6313–6316.

⁽⁴⁷⁾N.D.E. See also: D. Lazard, *Bull. Soc. Math. France* **97** (1969), 81–128, or: [BA]g, X § 1.6, Th. 1.

(b) When G is the diagonalizable S -group of type M , $\mathrm{Fr}(G/S)$ is equal to $p \mathrm{id}_G$ according to 4.1.4 (a); one then sees easily that $\mathrm{Ver}(G/S)$ is the identity morphism of G .

(c) When $\mathcal{E}$ is a flat $\mathcal{O}_S$ -module and G is the S -group $\mathbf{V}(\mathcal{E})$, the morphism $\mathrm{Ver}(G/S)$ is zero, as is $p \cdot \mathrm{id}_G$. We shall see in Exposé VII B that a commutative algebraic group G over a field k is “unipotent” if and only if the composite homomorphism

$$G^{(p^n)} \xrightarrow{\mathrm{Ver}(G^{(p^{n-1})}/S)} G^{(p^{n-1})} \longrightarrow \cdots \longrightarrow G^{(p)} \xrightarrow{\mathrm{Ver}(G/S)} G$$

is zero for some n (one has put $G^{(p^n)} = (G^{(p^{n-1})})^{(p)}$, cf. 4.1.3).⁽⁴⁸⁾

4.3.2. — Since the map $\mathrm{Ver} : G \mapsto \mathrm{Ver}(G/S)$ is a functorial homomorphism as G ranges over the flat S -schemes in commutative groups, the square

$$\begin{array}{ccc} G^{(p)} & \xrightarrow{\mathrm{Ver}(G/S)} & G \\ \mathrm{Fr}(G/S)^{(p)} \downarrow & & \downarrow \mathrm{Fr}(G/S) \\ G^{(p^2)} & \xrightarrow{\mathrm{Ver}(G^{(p)}/S)} & G^{(p)} \end{array}$$

is commutative, where $\mathrm{Fr}(G/S)^{(p)}$ denotes the inverse image of $\mathrm{Fr}(G/S)$ by the base change $\mathrm{fr}(S)$. According to 4.1.1, one has $\mathrm{Fr}(G/S)^{(p)} = \mathrm{Fr}(G^{(p)}/S)$, hence, according to 4.3 (*) applied to $G^{(p)}$, one obtains:

$$\begin{aligned} (**) \quad \mathrm{Fr}(G/S) \circ \mathrm{Ver}(G/S) &= \mathrm{Ver}(G^{(p)}/S) \circ \mathrm{Fr}(G^{(p)}/S) \\ &= p \cdot \mathrm{id}_{G^{(p)}}. \end{aligned}$$

4.3.3. — Finally, suppose that G is a commutative, finite and locally free S -group; let $\mathcal{A}$ be the affine $\mathcal{O}_S$ -algebra of G , and π the endomorphism of the sheaf of rings $\mathcal{O}_S$ sending a section x of $\mathcal{O}_S$ to x^p .⁽⁴⁹⁾ Denote by $\Sigma^p \mathcal{A}$ the subalgebra of $\bigotimes_{\mathcal{O}_S}^p \mathcal{A}$ consisting of sections invariant under the action of the symmetric group, and by $i(\mathcal{A})$ the inclusion of $\Sigma^p \mathcal{A}$ into the tensor product. Let $\Delta^p(\mathcal{A}) : \mathcal{A} \rightarrow \bigotimes_{\mathcal{O}_S}^p \mathcal{A}$ be the morphism obtained by iterating the diagonal morphism of the coalgebra $\mathcal{A}$ (it corresponds to the multiplication morphism from $U^p(G) = G_S^p$ to G); according to the beginning of paragraph 4.3, $\Delta^p(\mathcal{A})$ factors through $\Sigma^p \mathcal{A}$, i.e. induces a morphism

$$a(\mathcal{A}) : \mathcal{A} \rightarrow \Sigma^p \mathcal{A}$$

such that $i(\mathcal{A}) \circ a(\mathcal{A}) = \Delta^p(\mathcal{A})$.

On the other hand, let $S^p(\mathcal{A})$ be the degree p component of the symmetric algebra of $\mathcal{A}$, and $q(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^p \mathcal{A} \rightarrow S^p(\mathcal{A})$ the canonical projection. The multiplication $m^p(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^p \mathcal{A} \rightarrow \mathcal{A}$ factors through $S^p(\mathcal{A})$, i.e. induces a map

$$b(\mathcal{A}) : S^p(\mathcal{A}) \rightarrow \mathcal{A}$$

such that $b(\mathcal{A}) \circ q(\mathcal{A}) = m^p(\mathcal{A})$.

Since $\Sigma^p \mathcal{A}$ is the affine algebra of $V^p(\mathcal{A})$, according to the beginning of 4.3 again, the composite morphism $i(G) \circ \varphi(G)^{-1}$ induces an algebra homomorphism

$$r(\mathcal{A}) : \Sigma^p \mathcal{A} \rightarrow \mathcal{A} \otimes_{\pi} \mathcal{O}_S;$$

this homomorphism vanishes on sections of the form

$$\sum_{\sigma \in \mathcal{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}$$

⁽⁴⁸⁾**N.D.E.** Since this does not appear explicitly in VII B, we refer to [DG70], § IV.3, Prop. 4.11.

⁽⁴⁹⁾**N.D.E.** The order has been modified by introducing first the objects occurring in the diagram which is to follow.

and sends $a \otimes \cdots \otimes a$ to $a \otimes_\pi 1$. Similarly, $j(\mathcal{A})$ is the morphism of $\mathcal{O}_S$ -modules $a \otimes_\pi 1 \mapsto q(a \otimes \cdots \otimes a)$. One therefore obtains the commutative diagram:

$$(\mathcal{A}) \quad \begin{array}{ccccc} \mathcal{A} & \xrightarrow{\Delta^p(\mathcal{A})} & \bigotimes_{\mathcal{O}_S}^p \mathcal{A} & \xrightarrow{m^p(\mathcal{A})} & \mathcal{A} \\ & \searrow a(\mathcal{A}) & \nearrow i(\mathcal{A}) & \searrow q(\mathcal{A}) & \nearrow b(\mathcal{A}) \\ & \Sigma^p \mathcal{A} & & S^p(\mathcal{A}) & \\ & \searrow r(\mathcal{A}) & & \nearrow j(\mathcal{A}) & \\ & \mathcal{A} \otimes_\pi \mathcal{O}_S & & & \end{array} .$$

The composite $r(\mathcal{A}) \circ a(\mathcal{A})$ is associated to the Verschiebung morphism $\text{Ver}(G/S)$, while $b(\mathcal{A}) \circ j(\mathcal{A})$ is associated to the Frobenius morphism $\text{Fr}(G/S)$.

The commutative diagram $(\mathcal{A})$ above is self-dual; let, indeed, D be the functor associating to every $\mathcal{O}_S$ -module $\mathcal{M}$ the dual $\mathcal{O}_S$ -module $\mathcal{H}om_{\mathcal{O}_S}(\mathcal{M}, \mathcal{O}_S)$; it is clear that the image of diagram $(\mathcal{A})$ under the functor D is none other than diagram $(D\mathcal{A})$, the morphisms $Dr(\mathcal{A})$, $Da(\mathcal{A})$, $Dj(\mathcal{A})$, and $Db(\mathcal{A})$ identifying respectively with $j(D\mathcal{A})$, $b(D\mathcal{A})$, $r(D\mathcal{A})$, and $a(D\mathcal{A})$. According to 3.3.1, one therefore sees that:

In the category of commutative, finite and locally free S -groups, Cartier duality interchanges the Frobenius morphism and Verschiebung.⁽⁵⁰⁾

5. Lie p -algebras

Let us first recall a few results from the Séminaire Sophus Lie.⁽⁵¹⁾

5.1. — Let p be a prime number, R a commutative ring of characteristic p , and A an associative, but not necessarily commutative, R -algebra. If a and b are two elements of A , we put $[a, b] = ab - ba$ and $ab = L_a(b) = R_b(a)$. One then has:

$$\begin{aligned} (\text{ad } x^p)(y) &= [x^p, y] = (L_x^p - R_x^p)(y) \\ &= (L_x - R_x)^p(y) = (\text{ad } x)^p(y) \end{aligned}$$

whence the *first Jacobson formula*:

$$(i) \quad \text{ad}(x^p) = (\text{ad } x)^p.$$

If $a_1, \dots, a_p$ are p arbitrary elements of A , then, denoting by N the symmetrization operator (cf. 4.2), one has the equalities:

$$\begin{aligned} (*) \quad N(a_1 \otimes \cdots \otimes a_p) &= \sum_{\sigma} a_{\sigma(1)} \cdots a_{\sigma(p)} \\ &= \sum_{\tau} [a_{\tau(1)} [a_{\tau(2)} [\cdots [a_{\tau(p-1)}, a_p] \cdots]]] \end{aligned}$$

where σ ranges over the permutations of p letters and τ over those of $(p-1)$ letters. Indeed, the last term is equal to

$$\sum_{\tau} \sum_{r=0}^{p-1} \sum_{i_1 < \cdots < i_r} (-1)^s a_{\tau(i_1)} a_{\tau(i_2)} \cdots a_{\tau(i_r)} a_p a_{\tau(j_s)} \cdots a_{\tau(j_1)}$$

where τ ranges over the permutations of $p-1$ letters, $i_1, \dots, i_r$ over the strictly increasing sequences of integers in the interval $[1, p-1]$, and where $j_1, \dots, j_s$ denotes the strictly

⁽⁵⁰⁾**N.D.E.** See also [DG70], § IV.3, 4.9.

⁽⁵¹⁾**N.D.E.** Cf. P. Cartier, *Exemples d'hypercentres*, Sémin. Sophus Lie 1955/56, Exp. 3 (accessible on the Numdam site: <http://www.numdam.org>).

increasing sequence whose values are the integers of $[1, p-1]$ different from $i_1, \dots, i_r$. For a fixed value of r , the sum of the terms $(-1)^s a_{\tau(i_1)} \cdots a_{\tau(i_r)} a_p a_{\tau(j_s)} \cdots a_{\tau(j_1)}$ is clearly equal to

$$(-1)^s \binom{p-1}{s} \sum_{\rho} a_{\rho(1)} \cdots a_{\rho(r)} a_p a_{\rho(r+1)} \cdots a_{\rho(p-1)}$$

where ρ ranges over the permutations of $p-1$ letters. Now $(-1)^s \binom{p-1}{s} = 1$ in $\mathbf{F}_p$, since in $\mathbf{F}_p[x]$ (x an indeterminate) one has: $(x-1)^p = x^p - 1 = (x-1)(x^{p-1} + \cdots + 1)$, and hence $(x-1)^{p-1} = x^{p-1} + \cdots + 1$. This proves (*).

On the other hand, if x_0 and x_1 are two elements of A , one has

$$(x_0 + x_1)^p = x_0^p + x_1^p + \sum x_{z(1)} x_{z(2)} \cdots x_{z(p)},$$

where z ranges over the nonconstant maps from $[1, p]$ to $\{0, 1\}$. From this one obtains

$$(x_0 + x_1)^p = x_0^p + x_1^p + \sum_{0 < r < p} \frac{1}{r!(p-r)!} N(\underbrace{x_0, \dots, x_0}_r, \underbrace{x_1, \dots, x_1}_{p-r}).$$

(52) Now, according to (*), one has:

$$\begin{aligned} & N(\underbrace{x_0, \dots, x_0}_r, \underbrace{x_1, \dots, x_1}_{p-r}) \\ &= r!(p-1-r)! \sum_t [x_{t(1)} [x_{t(2)} [\cdots [x_{t(p-1)}, x_1] \cdots]]] \end{aligned}$$

where t ranges over the maps $[1, p-1] \rightarrow \{0, 1\}$ taking the value 0 r times. One deduces from this the *second Jacobson formula*:

$$(ii) \quad (x_0 + x_1)^p = x_0^p + x_1^p - \sum_{0 < r < p} \sum_t \frac{1}{r} [x_{t(1)} [x_{t(2)} [\cdots [x_{t(p-1)}, x_1] \cdots]]]$$

where t ranges over the maps $[1, p-1] \rightarrow \{0, 1\}$ taking the value 0 r times.

5.2. — Now let $\mathfrak{g}$ be an R -Lie algebra. One says that a map $x \mapsto x^{(p)}$ from $\mathfrak{g}$ to $\mathfrak{g}$ makes $\mathfrak{g}$ a *Lie p -algebra over R* if the following conditions hold:

- (0) $(\lambda x)^{(p)} = \lambda^p \cdot x^{(p)}$, for $\lambda \in R$, $x \in \mathfrak{g}$.
- (i) $\text{ad } x^{(p)} = (\text{ad } x)^p$, for $x \in \mathfrak{g}$.
- (ii)

$$(x_0 + x_1)^{(p)} = x_0^{(p)} + x_1^{(p)} - \sum_{0 < r < p} \sum_t \frac{1}{r} [x_{t(1)} [x_{t(2)} [\cdots [x_{t(p-1)}, x_1] \cdots]]]$$

where t ranges over the maps $[1, p-1] \rightarrow \{0, 1\}$ taking the value 0 r times ($x_0, x_1 \in \mathfrak{g}$).

The map $x \mapsto x^{(p)}$ will then be called the “symbolic p -th power”.

For example, if A is an associative R -algebra, we saw in 5.1 that one obtains a Lie p -algebra, which will be denoted by A_{Lie} , by taking the R -module underlying A and putting, for $x, y \in A$,

$$[x, y] = xy - yx \quad \text{and} \quad x^{(p)} = x^p.$$

We shall say that A_{Lie} is the *Lie p -algebra underlying A* .

In what follows we shall chiefly consider Lie p -subalgebras of p -algebras of the form A_{Lie} ; here is an example: let S be a scheme of characteristic $p > 0$, and X an S -scheme. Recall that a derivation of X over S is an endomorphism D of the sheaf of abelian groups $\mathcal{O}_X$ such that

$$D(\lambda \cdot s) = \lambda \cdot D(s) \quad \text{and} \quad D(st) = (Ds)t + s(Dt)$$

(52) **N.D.E.** The following explanation, taken from [DG70], § II.7, 3.2, has been inserted.

as λ and s, t range over sections of $\mathcal{O}_S$ and $\mathcal{O}_X$ over open subsets such that the formulas make sense. The Leibniz formula

$$D^n(st) = \sum_{i=0}^n \binom{n}{i} (D^i s)(D^{n-i} t)$$

shows that D^p is again a derivation of X over S , taking account of the equality $\binom{p}{i} \equiv 0 \pmod{p}$ for $i \neq 0, p$. It follows that:

The algebra $\text{Dér}_{X/S}$ of derivations of X over S is a Lie p -subalgebra of the $\Gamma(S, \mathcal{O}_S)$ -algebra of differential operators of X over S .

5.2.1. — If $\mathfrak{g}$ and $\mathfrak{h}$ are two Lie p -algebras, a homomorphism $h : \mathfrak{g} \rightarrow \mathfrak{h}$ is an R -linear map from $\mathfrak{g}$ to $\mathfrak{h}$ such that $h([x, y]) = [h(x), h(y)]$ and $h(x^{(p)}) = h(x)^{(p)}$ if $x, y \in \mathfrak{g}$. The composite map of two homomorphisms is again a homomorphism, so that we shall be able to speak of the category of Lie p -algebras over R .

If $(X, \mathcal{R})$ is a ringed space, we shall say that an $\mathcal{R}$ -module $\mathfrak{g}$ is equipped with a Lie p -algebra structure over $\mathcal{R}$ if, for every open subset U , $\Gamma(U, \mathfrak{g})$ is equipped with a Lie p -algebra structure over $\Gamma(U, \mathcal{R})$ and the restrictions are homomorphisms.

5.3. — We are now interested in the functor left adjoint to the functor $A \mapsto A_{\text{Lie}}$ of 5.2. Let $\mathfrak{g}$ be a Lie p -algebra over the ring R of characteristic p , $U(\mathfrak{g})$ the enveloping algebra of the Lie algebra underlying $\mathfrak{g}$ (cf. [BLie], I § 2.1), and $i_{\mathfrak{g}}$ (or simply i) the canonical map $\mathfrak{g} \rightarrow U(\mathfrak{g})$.

Let A be an associative unital R -algebra. One knows that, for every homomorphism of Lie algebras $\phi : \mathfrak{g} \rightarrow A_{\text{Lie}}$, there exists a unique homomorphism of unital R -algebras $\psi : U(\mathfrak{g}) \rightarrow A$ such that $\psi \circ i = \phi$. Furthermore, ϕ is a homomorphism of Lie p -algebras if and only if ψ vanishes on the elements $i(x)^p - i(x^{(p)})$, as x ranges over $\mathfrak{g}$.

Definition. One denotes by $U_p^R(\mathfrak{g})$, or simply $U_p(\mathfrak{g})$, the quotient of $U(\mathfrak{g})$ by the two-sided ideal generated by the elements $i(x)^p - i(x^{(p)})$, and by $j_{\mathfrak{g}}$ (or simply j) the map $\mathfrak{g} \rightarrow U_p(\mathfrak{g})$ which is the composite of $i : \mathfrak{g} \rightarrow U(\mathfrak{g})$ and the canonical map $U(\mathfrak{g}) \rightarrow U_p(\mathfrak{g})$. One says that $U_p(\mathfrak{g})$ is the *restricted enveloping algebra* of $\mathfrak{g}$.

According to what precedes, one has the

Proposition. *For every associative unital R -algebra and every morphism of Lie p -algebras $\phi : \mathfrak{g} \rightarrow A_{\text{Lie}}$, there exists a unique homomorphism of unital algebras $\psi : U_p(\mathfrak{g}) \rightarrow A$ such that $\psi \circ j = \phi$. In other words, the functor $\mathfrak{g} \mapsto U_p(\mathfrak{g})$ is left adjoint to the forgetful functor $A \mapsto A_{\text{Lie}}$.*

5.3.1. — With the notation of 5.3, now put $\beta(x) = i(x)^p - i(x^{(p)})$. For every element y of $\mathfrak{g}$, one has, according to 5.1 (i) and 5.2 (i):

$$\begin{aligned} \beta(x)i(y) &= i(y)\beta(x) + [\beta(x), i(y)] \\ &= i(y)\beta(x) + (\text{ad } i(x))^p i(y) - i((\text{ad } x)^p y) \\ &= i(y)\beta(x), \end{aligned}$$

so that $\beta(x)$ belongs to the center of $U(\mathfrak{g})$; in particular, the left ideal generated by the elements $\beta(x)$ is already two-sided.

On the other hand, it is clear that $\beta(\lambda x) = \lambda^p \beta(x)$, for $\lambda \in R$, and it follows from 5.1 (ii) and 5.2 (ii) that, for $x, y \in \mathfrak{g}$,

$$\beta(x + y) = \beta(x) + \beta(y).$$

In particular, if (x_{α}) is a family of generators of the R -module $\mathfrak{g}$, the left ideal generated by the elements $\beta(x)$ is already generated by the $\beta(x_{\alpha})$.

Proposition 5.3.2. ⁽⁵³⁾ *Let $\mathfrak{g}$ be an R -Lie algebra whose underlying R -module is free with basis (x_α) . Then the Lie p -algebra structures on $\mathfrak{g}$ correspond bijectively to the families (y_α) of $\mathfrak{g}$ such that $\text{ad } y_\alpha = (\text{ad } x_\alpha)^p$.*

Indeed, if $\mathfrak{g}$ is equipped with a Lie p -algebra structure $x \mapsto x^{(p)}$, then according to 5.2 (i) and (0), (ii), the $y_\alpha = x_\alpha^{(p)}$ satisfy $\text{ad } y_\alpha = (\text{ad } x_\alpha)^p$ and determine the Lie p -algebra structure.

Let us prove the converse. Since $\mathfrak{g}$ is a free R -module, the canonical map $i : \mathfrak{g} \rightarrow U(\mathfrak{g})$ is injective, according to the Poincaré–Birkhoff–Witt theorem (cf. [BLie], I § 2.7), so one may identify $\mathfrak{g}$ with a sub- R -module of $U(\mathfrak{g})$. Suppose that (y_α) is a family of elements of $\mathfrak{g}$ such that $\text{ad } y_\alpha = (\text{ad } x_\alpha)^p$. Let π be the map $r \mapsto r^p$ from R to R , and let $\mathfrak{g} \otimes_\pi R$ be the R -Lie algebra obtained by the extension of scalars $\pi : R \rightarrow R$.⁽⁵⁴⁾

There then exists an R -linear map γ from $\mathfrak{g} \otimes_\pi R$ to $U(\mathfrak{g})$ sending $x_\alpha \otimes_\pi 1$ to $x_\alpha^p - y_\alpha$; moreover, since one has, for every $x \in \mathfrak{g}$,

$$(\text{ad } x_\alpha^p)(x) = (\text{ad } x_\alpha)^p(x) = (\text{ad } y_\alpha)(x),$$

γ maps $\mathfrak{g} \otimes_\pi R$ into the center of $U(\mathfrak{g})$. Put, for every $x \in \mathfrak{g}$:

$$x^{(p)} = x^p - \gamma(x \otimes_\pi 1).$$

Then, for every α , one has $x_\alpha^{(p)} = y_\alpha$. If $x = \sum \lambda_\alpha x_\alpha$, one deduces from 5.1 (ii) (by induction on the number of indices α such that $\lambda_\alpha \neq 0$) that

$$x^p - \sum_\alpha \lambda_\alpha^p x_\alpha^p \in \mathfrak{g};$$

denoting this element by z , one then has $x^{(p)} = \sum \lambda_\alpha^p y_\alpha + z$ and hence $x^{(p)} \in \mathfrak{g}$.

It is clear that the map $x \mapsto x^{(p)}$ satisfies $(\lambda x)^{(p)} = \lambda^p x^{(p)}$. Moreover, since $\gamma(x \otimes_\pi 1)$ is central, $\text{ad } x^{(p)} = \text{ad } x^p$ and hence, according to the first Jacobson formula (5.1 (i)), one has

$$\text{ad } x^{(p)} = (\text{ad } x)^p.$$

Finally, according to the second Jacobson formula (5.1 (ii)), the map $x \mapsto x^{(p)}$ satisfies condition (ii) of 5.2. It therefore makes $\mathfrak{g}$ a Lie p -algebra. This proves the proposition.

Proposition 5.3.3. *Let $\mathfrak{g}$ be a Lie p -algebra over R whose underlying module is free with basis (x_α) . Then the map $j : \mathfrak{g} \rightarrow U_p(\mathfrak{g})$ is injective and, if one puts $z_\alpha = j(x_\alpha)$, $U_p(\mathfrak{g})$ has as basis the monomials*

$$\prod_\alpha z_\alpha^{n_\alpha} \quad \text{where } 0 \leq n_\alpha < p,$$

(the n_α are assumed to be zero except for finitely many of them; the basis is assumed totally ordered and the products taken in increasing order).

Indeed, let us identify $\mathfrak{g}$ with a submodule of the enveloping algebra $U(\mathfrak{g})$ by means of the canonical map i . For every family $n = (n_\alpha)$ of natural integers, zero except for finitely many of them, put

$$|n| = \sum_\alpha n_\alpha \quad \text{and} \quad x^n = \prod_\alpha x_\alpha^{n_\alpha}.$$

Writing $n_\alpha = m_\alpha + p\ell_\alpha$, with $0 \leq m_\alpha < p$, also put

$$T_n = \prod_\alpha x^{m_\alpha} \beta(x_\alpha)^{\ell_\alpha}$$

where $\beta(x) = x^p - x^{(p)}$ is the map $\mathfrak{g} \rightarrow U(\mathfrak{g})$ defined in 5.3.1. For every $r \in \mathbb{N}$, denote by U^r the sub- R -module of $U(\mathfrak{g})$ generated by the x^n such that $|n| \leq r$. Since the graded ring

⁽⁵³⁾**N.D.E.** In this paragraph the order has been modified, stating the result first and then expanding the proof.

⁽⁵⁴⁾**N.D.E.** i.e. $xr \otimes_\pi 1 = x \otimes_\pi r^p$ and $r \cdot (x \otimes_\pi 1) = x \otimes_\pi r$, for $x \in \mathfrak{g}$, $r \in R$.

$\bigoplus_r U^r/U^{r-1}$ is commutative (cf. [BLie], I § 2.6), one sees that, for every n :

$$T_n - \prod_{\alpha} x^{n_{\alpha}} \in U^{|n|-1}.$$

For every $s \in \mathbf{N}$, the x^n such that $|n| = s$ form, according to the Poincaré–Birkhoff–Witt theorem (loc. cit., § 2.7), a basis of U^s/U^{s-1} , and therefore the same is true of the T_n such that $|n| = s$.

Thus, as $s = |n|$ varies, the T_n form a basis of $U(\mathfrak{g})$. Now the kernel J of the canonical map $U(\mathfrak{g}) \rightarrow U_p(\mathfrak{g})$ is the left ideal of $U(\mathfrak{g})$ generated by the central elements $\beta(x_{\alpha})$ (5.3.1). Consequently, the T_n such that $\ell = (\ell_{\alpha}) \neq 0$ form a basis of J , and the T_n such that $n_{\alpha} < p$ for every α form a basis of $U_p(\mathfrak{g}) = U(\mathfrak{g})/J$.

5.3.3 bis. — Let $\mathfrak{g}$ be a Lie p -algebra over R and $f : R \rightarrow R'$ an extension of the base ring. I say that *there exists one and only one Lie p -algebra structure on the R' -module $R' \otimes_R \mathfrak{g}$ such that*

$$(*) \quad [\lambda \otimes x, \mu \otimes y] = \lambda \mu \otimes [x, y] \quad \text{and} \quad (\lambda \otimes x)^{(p)} = \lambda^p \otimes x^{(p)}.$$

It will follow, in particular, that the functor $\mathfrak{g} \mapsto R' \otimes_R \mathfrak{g}$ is left adjoint to the functor “restriction of scalars from R' to R ”.

The uniqueness of the Lie p -algebra structure defined by $(*)$ being clear, let us prove existence. When $\mathfrak{g}$ is free with basis (x_{α}) , there exists, according to 5.3.2, one and only one Lie p -algebra structure on the Lie algebra $R' \otimes_R \mathfrak{g}$ such that

$$(1 \otimes x_{\alpha})^{(p)} = 1 \otimes x_{\alpha}^{(p)};$$

this structure is the one we seek.

When $\mathfrak{g}$ is an arbitrary Lie p -algebra, there exists a Lie p -algebra L_0 , free (as an R -module), and a surjective homomorphism $q_0 : L_0 \rightarrow \mathfrak{g}$; it suffices, for example, to take for L_0 the Lie p -algebra $R \otimes_{\mathbf{F}_p} \mathfrak{g}$, where $\mathbf{F}_p$ denotes the prime field of characteristic p , and for q_0 the homomorphism $\lambda \otimes x \mapsto \lambda x$ ($\mathfrak{g}$ is free over $\mathbf{F}_p$!). The kernel of q_0 is then a p -ideal of L_0 , that is, an ideal of the Lie algebra L_0 stable under the endomorphism $x \mapsto x^{(p)}$; there is therefore also a Lie p -algebra L_1 , free (as an R -module), and a homomorphism $q_1 : L_1 \rightarrow L_0$ whose image is $\text{Ker } q_0$, whence the exact sequence:

$$L_1 \xrightarrow{q_1} L_0 \xrightarrow{q_0} \mathfrak{g} \rightarrow 0.$$

One deduces from this an exact sequence of R' -Lie algebras

$$R' \otimes_R L_1 \xrightarrow{R' \otimes_R q_1} R' \otimes_R L_0 \xrightarrow{R' \otimes_R q_0} R' \otimes_R \mathfrak{g} \rightarrow 0.$$

Since $R' \otimes_R q_1$ is manifestly a homomorphism of Lie p -algebras, the kernel of $R' \otimes_R q_0$ is a p -ideal, so that the symbolic p -th power operation of $R' \otimes_R L_0$ induces, by passage to the quotient, a map from $R' \otimes_R \mathfrak{g}$ to $R' \otimes_R \mathfrak{g}$ (use formula (ii) of 5.2.); the latter equips $R' \otimes_R \mathfrak{g}$ with the desired Lie p -algebra structure.

5.3.4. — The canonical map $j_{\mathfrak{g}} : \mathfrak{g} \rightarrow U_p(\mathfrak{g})$ induces, for every extension $R \rightarrow R'$ of the base ring, a homomorphism

$$R' \otimes_R j_{\mathfrak{g}} : R' \otimes_R \mathfrak{g} \rightarrow R' \otimes_R U_p(\mathfrak{g}),$$

whence a homomorphism

$$h : U_p(R' \otimes_R \mathfrak{g}) \rightarrow R' \otimes_R U_p(\mathfrak{g})$$

such that $h \circ j_{R' \otimes_R \mathfrak{g}} = R' \otimes_R j_{\mathfrak{g}}$. It clearly follows from the universal properties of $R' \otimes_R \mathfrak{g}$ and of the restricted enveloping algebra that h is an isomorphism, which will allow us to identify $U_p(R' \otimes_R \mathfrak{g})$ with $R' \otimes_R U_p(\mathfrak{g})$.

In particular, if r is an element of R and R' is the localized ring R_r , one sees that $\mathfrak{g}_r = R_r \otimes_R \mathfrak{g}$ is canonically equipped with a Lie p -algebra structure over R_r , so that the sheaf $\tilde{\mathfrak{g}}$ on $\text{Spec } R$ is a quasi-coherent Lie p -algebra on $\text{Spec } R$. Moreover, the restricted

enveloping algebra $U_p^{Rr}(\mathfrak{g}_r)$ identifies with $U_p^R(\mathfrak{g})_r$, so that the sheaf associated to the presheaf $V \mapsto U_p(\Gamma(V, \mathfrak{g}))$ is quasi-coherent.

Definition. More generally, if S is a scheme of characteristic p and $\mathcal{G}$ a quasi-coherent Lie p -algebra over $\mathcal{O}_S$, the sheaf associated to the presheaf $V \mapsto U_p(\Gamma(V, \mathcal{G}))$ is quasi-coherent; it will be denoted by $\mathcal{U}_p(\mathcal{G})$ and called the *restricted enveloping algebra of $\mathcal{G}$* . If V is affine, $U_p(\Gamma(V, \mathcal{G}))$ identifies with $\Gamma(V, \mathcal{U}_p(\mathcal{G}))$.

5.4. — The universal character of $U_p(\mathfrak{g})$ implies that $U_p(\mathfrak{g})$ is functorial in $\mathfrak{g}$: every homomorphism of Lie p -algebras $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ induces one and only one homomorphism of unital algebras $U_p(\phi)$ such that $j_{\mathfrak{h}} \circ \phi = U_p(\phi) \circ j_{\mathfrak{g}}$. Here are a few examples:

a) If $\mathfrak{h} = 0$, $U_p(\mathfrak{h})$ identifies with the base ring and $U_p(\phi)$ is an algebra homomorphism $\varepsilon_{\mathfrak{g}} : U_p(\mathfrak{g}) \rightarrow R$ called the *augmentation*.

b) Now take for $\mathfrak{h}$ the algebra $\mathfrak{g}^\circ$ opposite to $\mathfrak{g}$, i.e. $\mathfrak{g}^\circ$ has the same underlying module as $\mathfrak{g}$, the same symbolic p -th power, the bracket of two elements in $\mathfrak{g}^\circ$ being the negative of the bracket in $\mathfrak{g}$. It is clear that we may identify $U_p(\mathfrak{g}^\circ)$ with the algebra opposite to $U_p(\mathfrak{g})$. Moreover, the isomorphism $x \mapsto -x$ from $\mathfrak{g}$ to $\mathfrak{g}^\circ$ induces an isomorphism $c_{\mathfrak{g}}$ from $U_p(\mathfrak{g})$ to $U_p(\mathfrak{g}^\circ) \simeq U_p(\mathfrak{g})^\circ$. One says that $c_{\mathfrak{g}}$ is the *antipode* of $U_p(\mathfrak{g})$.

c) Finally, let $\mathfrak{f}$ and $\mathfrak{g}$ be two Lie p -algebras and $\mathfrak{h}$ the *product Lie p -algebra* $\mathfrak{f} \times \mathfrak{g}$ having as underlying R -module the direct product $\mathfrak{f} \times \mathfrak{g}$, the bracket and symbolic p -th power being defined by the formulas

$$[(x, y), (x', y')] = ([x, x'], [y, y']) \quad \text{and} \quad (x, y)^{(p)} = (x^{(p)}, y^{(p)}).$$

If $h_1 : \mathfrak{f} \rightarrow \mathfrak{k}$ and $h_2 : \mathfrak{g} \rightarrow \mathfrak{k}$ are two homomorphisms of Lie p -algebras such that $[h_1(x), h_2(y)] = 0$ for every x of $\mathfrak{f}$ and every y of $\mathfrak{g}$, the map $h_1 + h_2 : (x, y) \mapsto h_1(x) + h_2(y)$ is a homomorphism of Lie p -algebras; conversely, every homomorphism from $\mathfrak{f} \times \mathfrak{g}$ to $\mathfrak{k}$ is of this type, which allows one to characterize $\mathfrak{f} \times \mathfrak{g}$ as the solution to a universal problem. For example, the maps

$$h_1 : x \mapsto i_{\mathfrak{f}}(x) \otimes 1 \quad \text{and} \quad h_2 : y \mapsto 1 \otimes i_{\mathfrak{g}}(y)$$

induce a homomorphism $h_1 + h_2$ from $\mathfrak{f} \times \mathfrak{g}$ to the Lie p -algebra underlying $U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g})$. It follows from the universal characters of $\mathfrak{f} \times \mathfrak{g}$ and of the restricted enveloping algebras that $h_1 + h_2$ extends to an isomorphism:

$$\varphi : U_p(\mathfrak{f} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g}).$$

Definition. If $\mathfrak{f} = \mathfrak{g}$, the diagonal map $\delta : x \mapsto (x, x)$ from $\mathfrak{g}$ to $\mathfrak{g} \times \mathfrak{g}$ induces a homomorphism from $U_p(\mathfrak{g})$ to $U_p(\mathfrak{g} \times \mathfrak{g})$. We shall denote by $\Delta_{\mathfrak{g}}$ the composite of this homomorphism with the isomorphism $\varphi : U_p(\mathfrak{g} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{g}) \otimes U_p(\mathfrak{g})$.⁽⁵⁵⁾

One then sees easily that $\Delta_{\mathfrak{g}}$ and the multiplication of the algebra $U_p(\mathfrak{g})$ make $U_p(\mathfrak{g})$ an R -coalgebra in groups (cf. 3.2) having $\varepsilon_{\mathfrak{g}}$ as augmentation and $c_{\mathfrak{g}}$ as antipode.

5.5. — ⁽⁵⁶⁾ Now let S be a scheme of characteristic p . First, if $\mathcal{U}$ is a $\mathcal{O}_S$ -coalgebra in groups and G the S -group functor $\text{Spec}^* \mathcal{U}$, we saw (3.2.3) that, for every $T \rightarrow S$, $(\underline{\text{Lie}} G)(T)$ is the Lie subalgebra of $\Gamma(T, \mathcal{U}_T)$ consisting of primitive elements. Now, if x is such an element, one has $\Delta(x^p) = x^p \otimes 1 + 1 \otimes x^p$ (since $\binom{p}{i} = 0 \pmod p$ for $0 < i < p$), i.e. x^p is again a primitive element. It follows, according to 5.1 and 5.2, that the map $x \mapsto x^p$ equips $(\underline{\text{Lie}} G)(T)$ with an $\mathcal{O}(T)$ -Lie p -algebra structure.

⁽⁵⁵⁾**N.D.E.** i.e. $\Delta_{\mathfrak{g}}(x) = x \otimes 1 + 1 \otimes x$ for every $x \in \mathfrak{g}$; in particular, the comultiplication $\Delta_{\mathfrak{g}}$ is indeed cocommutative. . .

⁽⁵⁶⁾**N.D.E.** Section 5.4.1 of the original has been transformed into this § 5.5: on the one hand, proposition 5.5.1 brings together the results of Section 5 and proposition 3.2.3, and contains lemma 7.3 of the original; on the other hand, the proof of 5.5.3 (ii) takes up and expands that of the implication (i) $\Rightarrow$ (ii) in theorem 7.4 below.

Now let $\mathcal{L}$ be a $\mathcal{O}_S$ -Lie p -algebra, quasi-coherent over $\mathcal{O}_S$. As V ranges over the open subsets of S , the structures of coalgebras in groups previously defined on the sets $U_p(\Gamma(V, \mathcal{L}))$ induce on the associated sheaf, i.e. on the restricted enveloping algebra $\mathcal{U}_p(\mathcal{L})$, a structure of $\mathcal{O}_S$ -coalgebra in groups. Moreover, for every S -scheme T , there is an isomorphism $\mathcal{U}_p(\mathcal{L}_T) \xrightarrow{\sim} \mathcal{U}_p(\mathcal{L})_T$.

Denote by $\text{Prim } \mathcal{U}_p(\mathcal{L})$ the subpresheaf of $\mathcal{U}_p(\mathcal{L})$ associating to every open subset V the set of primitive elements of $\mathcal{U}_p(\mathcal{L})(V)$; one sees easily that this is a sheaf. As V ranges over the open subsets of S , the composite maps

$$\Gamma(V, \mathcal{L}) \xrightarrow{j} \text{Prim } U_p(\Gamma(V, \mathcal{L})) \longrightarrow \text{Prim } \mathcal{U}_p(\mathcal{L})(V)$$

define a morphism $\mathcal{L} \longrightarrow \text{Prim } \mathcal{U}_p(\mathcal{L})$, which we shall still denote by j or $j_{\mathcal{L}}$, and the latter again defines a morphism of $\mathbf{O}_S$ -Lie p -algebras $\mathbf{W}(\mathcal{L}) \longrightarrow \text{Prim } \mathbf{W}(\mathcal{U}_p(\mathcal{L}))$ (cf. 3.2.3).

Proposition 5.5.1. *Let $\mathcal{L}$ be a $\mathcal{O}_S$ -Lie p -algebra, locally free as an $\mathcal{O}_S$ -module. Then $j_{\mathcal{L}}$ induces an isomorphism of $\mathbf{O}_S$ -Lie p -algebras:*

$$\mathbf{W}(\mathcal{L}) \xrightarrow{\sim} \text{Prim } \mathbf{W}(\mathcal{U}_p(\mathcal{L})).$$

PROOF. Let T be an S -scheme; taking account of the identification $\mathcal{U}_p(\mathcal{L}_T) = \mathcal{U}_p(\mathcal{L})_T$, one must show that the map $\Gamma(T, \mathcal{L}_T) \longrightarrow \text{Prim } \Gamma(T, \mathcal{U}_p(\mathcal{L}_T))$ is bijective. Replacing S by T , one is reduced to the case where $T = S$, and it then suffices to show that the sheaf morphism $j_{\mathcal{L}} : \mathcal{L} \longrightarrow \text{Prim } \mathcal{U}_p(\mathcal{L})$ is an isomorphism. This question being local on S , we may suppose that S is affine with ring R and that $\mathcal{L}$ is the sheaf associated to an R -Lie p -algebra L with basis (x_{α}) . As in 5.3.3, denote by z_{α} the image of x_{α} in $U = U_p(L)$, and, for every family $n = (n_{\alpha})$ of integers between 0 and $p - 1$, zero except for finitely many of them, denote by $z^{(n)}$ the product

$$\prod_{\alpha} \frac{z_{\alpha}^{n_{\alpha}}}{n_{\alpha}!}$$

(the basis (x_{α}) is assumed totally ordered and the products taken in increasing order).

Since $\Delta(z_{\alpha}) = z_{\alpha} \otimes 1 + 1 \otimes z_{\alpha}$, one sees easily that

$$\Delta(z^{(n)}) = \sum_r z^{(n-r)} \otimes z^{(r)}$$

the sum being taken over the (finite!) set of r such that $0 \leq r_{\alpha} \leq n_{\alpha}$ for every α . Since the $z^{(n)}$ (resp. the $z^{(n)} \otimes z^{(m)}$) form a basis of U (resp. of $U \otimes U$), one deduces that an element u of U satisfies $\Delta(u) = u \otimes 1 + 1 \otimes u$ if and only if u is a linear combination of the z_{α} . This proves 5.5.1. $\square$

Remark 5.5.2. Recall (cf. 3.2.2 and 3.2.3) that the S -group functor $G = \text{Spec}^* \mathcal{U}_p(\mathcal{L})$ is very good and that $\underline{\text{Lie}}(G) = \text{Prim } \mathbf{W}(\mathcal{U}_p(\mathcal{L}))$. The preceding proposition therefore means that $j_{\mathcal{L}}$ induces an isomorphism $\mathbf{W}(\mathcal{L}) \xrightarrow{\sim} \underline{\text{Lie}}(G)$.

If one assumes, in addition, that $\mathcal{L}$ is a locally free $\mathcal{O}_S$ -module of finite rank, then $\mathcal{U}_p(\mathcal{L})$ is finite locally free over $\mathcal{O}_S$, according to 5.3.3, hence $\text{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by the S -group $G_p(\mathcal{L}) = \text{Spec } \mathcal{U}_p(\mathcal{L})^*$ (cf. 3.2.2.1), and one obtains the following more precise proposition:

Proposition 5.5.3. *Let $\mathcal{L}$ be a $\mathcal{O}_S$ -Lie p -algebra, locally free of finite rank as an $\mathcal{O}_S$ -module, let $\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*$, and let $G = G_p(\mathcal{L})$ be the affine S -group $\text{Spec } \mathcal{A}$.*

- (i) *$j_{\mathcal{L}}$ induces an isomorphism $\mathbf{W}(\mathcal{L}) \xrightarrow{\sim} \underline{\text{Lie}}(G)$ of $\mathbf{O}_S$ -Lie p -algebras.*
- (ii) *Let $\mathcal{I}$ be the augmentation ideal of $\mathcal{A}$, and $\omega_G = \mathcal{I}/\mathcal{I}^2$ (cf. II, 4.11.4). Then ω_G identifies with $\mathcal{L}^* = \mathcal{H}om_{\mathcal{O}_S}(\mathcal{L}, \mathcal{O}_S)$, hence is a locally free $\mathcal{O}_S$ -module of finite rank (and one has $\omega_{G/S}^* = \mathcal{L}$).*

PROOF. Since (i) follows from 5.5.2, let us prove (ii). Denote by $\eta_{\mathcal{U}}$ and $\varepsilon_{\mathcal{U}}$ the unit section and augmentation of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, by $\eta_{\mathcal{A}}$ and $\varepsilon_{\mathcal{A}}$ those of $\mathcal{A}$, and put $\mathcal{J} = \text{Ker } \varepsilon_{\mathcal{U}}$. One then has:

$$(1) \quad \mathcal{U} = \eta_{\mathcal{U}}(\mathcal{O}_S) \oplus \mathcal{J}.$$

Let δ be the morphism defined by the diagram below, where τ and π denote the inclusion and projection deduced from decomposition (1):

$$\begin{array}{ccc} \mathcal{J} & \xrightarrow{\delta} & \mathcal{J} \otimes \mathcal{J} \\ \tau \downarrow & & \uparrow \pi \\ \mathcal{U} & \xrightarrow{\Delta} & \mathcal{U} \otimes \mathcal{U} \end{array}$$

then there is an exact sequence:

$$(*) \quad 0 \longrightarrow \mathcal{L}^* \xrightarrow{j_{\mathcal{L}}} \mathcal{J} \xrightarrow{\delta} \mathcal{J} \otimes \mathcal{J}.$$

Moreover, according to 5.3.3, the $\mathcal{O}_S$ -module $\mathcal{J}/\mathcal{L}$ is locally free and, according to 5.5.1, sequence $(*)$ remains exact after every base change. Thus, according to [BAC], II § 3, prop. 6, δ induces an isomorphism from $\mathcal{J}/\mathcal{L}$ onto a submodule $\mathcal{Q}$ locally a direct factor of $\mathcal{J} \otimes \mathcal{J}$. It follows that $(*)$ gives by duality the exact sequence:

$$(**) \quad 0 \longleftarrow \mathcal{L} \xleftarrow{t_{j_{\mathcal{L}}}} \mathcal{J} \xleftarrow{t_{\delta}} \mathcal{J} \otimes \mathcal{J}.$$

Now decomposition (1) corresponds by duality to the decomposition:

$$(2) \quad \mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{J}$$

and the transpose of Δ is the multiplication $m_{\mathcal{A}} : \mathcal{A} \otimes \mathcal{A} \longrightarrow \mathcal{A}$. Since $\mathcal{J}$ is an ideal of $\mathcal{A}$, $m_{\mathcal{A}}$ sends $\mathcal{J} \otimes \mathcal{J}$ into $\mathcal{J}$; more precisely, taking account of decomposition (2), there is a commutative square

$$\begin{array}{ccc} \mathcal{J} & \xleftarrow{m'} & \mathcal{J} \otimes \mathcal{J} \\ \uparrow t_{\tau} & & \downarrow t_{\pi} \\ \mathcal{A} & \xleftarrow{m_{\mathcal{A}}} & \mathcal{A} \otimes \mathcal{A} \end{array}$$

which shows that the restriction m' of $m_{\mathcal{A}}$ to $\mathcal{J} \otimes \mathcal{J}$ is the transpose of δ . Exact sequence $(**)$ then gives $\mathcal{J}/\mathcal{J}^2 \simeq \mathcal{L}^*$, and the proposition follows. $\square$

6. Lie p -algebra of an S -group scheme

Let S be a scheme of characteristic $p > 0$. In paragraph 5.5 we associated to every quasi-coherent $\mathcal{O}_S$ -Lie p -algebra $\mathcal{L}$ an S -group functor $G_p(\mathcal{L}) = \text{Spec}^* \mathcal{U}_p(\mathcal{L})$. We shall now see that, for every S -group scheme G , the $\mathbf{O}_S$ -Lie algebra $\underline{\text{Lie}}(G/S)$ defined in II 4.11 is naturally equipped with a $\mathbf{O}_S$ -Lie p -algebra structure.

6.1. — First identify $\underline{\text{Lie}}(G/S)(S)$ and $\underline{\text{Lie}}(\text{Aut } G/S)(S)$ respectively with Lie subalgebras of $U(G)$ and $\text{Dif}_{G/S}$ by means of the injections α and β of 2.5; $\underline{\text{Lie}}(\text{Aut } G/S)(S)$ is therefore identified with the $\Gamma(\mathcal{O}_S)$ -Lie algebra of S -derivations of $\mathcal{O}_G$. According to 5.2, the latter is a Lie p -subalgebra of $\text{Dif}_{G/S}$.

On the other hand, the image of $L = \underline{\text{Lie}}(G/S)(S)$ under the injective algebra morphism $\ell : U(G) \longrightarrow \text{Dif}_{G/S}$, $d \mapsto {}^G d$, consists of the left-invariant derivations (cf. 2.2, N.D.E. (17), 2.4 and 2.5). If x belongs to L , $\ell(x)^p$ is none other than $\ell(x^p)$, according to loc. cit. Since $\ell(x)^p$ is again a derivation, one sees that x^p belongs to $\underline{\text{Lie}}(G/S)(S)$. Thus:⁽⁵⁷⁾

⁽⁵⁷⁾N.D.E. One can also show directly (without using $\text{Dif}_{G/S}$) that the Lie algebra of derivations of G at the origin (isomorphic to $\underline{\text{Lie}}(G/S)(S)$ according to 2.5) is stable under raising to the power p in $U(G)$: this is done in 6.2 below.

$\underline{\mathrm{Lie}}(G/S)(S)$ is a Lie p -subalgebra of the infinitesimal algebra $U(G)$.

6.1.1. — Let $\phi : G \rightarrow H$ be a homomorphism of S -group schemes. It is clear that the homomorphisms $\underline{\mathrm{Lie}}(\phi/S)(S)$ and $U(\phi)$ are compatible with the identifications of $\underline{\mathrm{Lie}}(G/S)(S)$ and $\underline{\mathrm{Lie}}(H/S)(S)$ with Lie p -subalgebras of $U(G)$ and $U(H)$. Since $U(\phi)$ is an algebra homomorphism, one therefore sees that $\underline{\mathrm{Lie}}(\phi/S)(S)$ is a homomorphism of Lie p -algebras.

Similarly, if $s : T \rightarrow S$ is a base change, the map from $\underline{\mathrm{Lie}}(G/S)(S)$ to $\underline{\mathrm{Lie}}(G/S)(T)$ induced by s is a homomorphism of Lie p -algebras. One may express this by saying that the functor $\underline{\mathrm{Lie}}(G/S)$ is equipped with a $\mathbf{O}_S$ -Lie p -algebra structure. In particular, as T ranges over the open subsets of S , one sees that the sheaf $\mathcal{L}\mathrm{ie}(G/S)$ is equipped with a $\mathcal{O}_S$ -Lie p -algebra structure.

6.2. — Following an idea of Demazure, we shall now generalize what precedes to certain S -group functors which are not necessarily representable. For this purpose, we shall first give another definition of the symbolic p -th power in the Lie algebra of an S -group scheme G .

Let D be a derivation of G at the origin;⁽⁵⁸⁾ according to 1.2.1, D is the composite of the S -derivation $\delta = (\tau, \partial_t)$ of the zero section $\tau : S \rightarrow I_S$ and a morphism $x : I_S \rightarrow G$ such that $x \circ \tau = \varepsilon$ (i.e. $x \in \underline{\mathrm{Lie}}(G/S)(S)$). According to the definition which we gave in 2.1, D^p is the following composite deviation:

$$\begin{aligned} S &\simeq \underbrace{S \times S \times \cdots \times S}_p \xrightarrow{\delta \times \cdots \times \delta} I_S \times \cdots \times I_S \\ &\xrightarrow{x \times \cdots \times x} G \times \cdots \times G \xrightarrow{m^{(p)}} G \end{aligned}$$

where $m^{(p)}$ is the morphism induced by the multiplication $m : G \times G \rightarrow G$. Since $I_S \times \cdots \times I_S$ is affine over S and has as affine algebra $\mathcal{B} = \mathcal{O}_S[d_1, \dots, d_p]/(d_1^2, \dots, d_p^2)$, the deviation $\delta \times \cdots \times \delta$ is defined by the morphism of $\mathcal{O}_S$ -modules

$$\phi : \mathcal{B} \rightarrow \mathcal{O}_S$$

sending the monomial $d_1 d_2 \cdots d_p$ to 1 and the other monomials $d_{i_1} \cdots d_{i_r}$, for $0 \leq r < p$, to 0. On the other hand, if pr_i denotes the projection from I_S^p onto the i -th factor and x_i is the image in $G(I_S^p)$ of x under $G(\mathrm{pr}_i)$, then the composite morphism $m^{(p)} \circ (x \times \cdots \times x)$ is none other than the product $x_1 x_2 \cdots x_p$. Consequently, D^p is also the following composite deviation:

$$S \xrightarrow{\delta \times \cdots \times \delta} I_S \times \cdots \times I_S \xrightarrow{x_1 x_2 \cdots x_p} G.$$

This description allows us to prove again that D^p is a derivation of G at the origin. Indeed, since G is a very good group (II 4.11), the images $G(\mathrm{pr}_1)(x)$ and $G(\mathrm{pr}_2)(x)$ of x in $G(I_S \times I_S)$ commute with one another. It follows that the elements x_i of $G(I_S^p)$ commute pairwise, and therefore, for every permutation γ of the factors of I_S^p , one has $(x_1 \cdots x_p) \circ \gamma = x_1 \cdots x_p$; it follows that $x_1 \cdots x_p$ factors through the canonical projection from I_S^p to the symmetric product $\Sigma^p I_S$ (cf. 4.2).

The symmetric product $\Sigma^p I_S$ has as affine algebra the subalgebra $\mathcal{A}$ of $\mathcal{B}$ having as basis over $\mathcal{O}_S$ the elementary symmetric functions $1 = \sigma_0, \sigma_1, \dots, \sigma_p$ of $d_1, \dots, d_p$. Denote by κ the inclusion $\mathcal{A} \hookrightarrow \mathcal{B}$ and by π the morphism of $\mathcal{O}_S$ -algebras $\mathcal{A} \rightarrow \mathcal{O}_S[t]/(t^2)$ vanishing on $\sigma_1, \dots, \sigma_{p-1}$ and sending σ_p to t ; then one has $\phi \circ \kappa = \partial_t \circ \pi$ (recall that $\partial_t : \mathcal{O}_S[t]/(t^2) \rightarrow \mathcal{O}_S$ is the morphism of $\mathcal{O}_S$ -modules vanishing on 1 and sending t to 1). Consequently, denoting

⁽⁵⁸⁾**N.D.E.** The original has been expanded in what follows.

by i the closed immersion $I_S \hookrightarrow \Sigma^p I_S$ defined by π , there is a commutative diagram:

$$\begin{array}{ccccc}
 & & D^p & & \\
 & \searrow^{\delta \times \dots \times \delta} & & \searrow^{x_1 \dots x_p} & \\
 S & \xrightarrow{\quad} & I_S^p & \xrightarrow{\quad} & G \\
 \downarrow \delta & & \downarrow \text{can.} & & \parallel \\
 I_S & \xrightarrow{\quad i \quad} & \Sigma^p I_S & \xrightarrow{\quad} & G \\
 & \searrow^y & & \searrow &
 \end{array}$$

which shows that D^p is of the form $y \circ \delta$, and thus is indeed a derivation of G at the origin.

6.3. — Let $\mathcal{S}_p$ be the symmetric group of order p , and $I_S^p \times \mathcal{S}_p$ the direct sum of a family of copies of I_S^p indexed by $\mathcal{S}_p$. We denote by $\pi : I_S^p \times \mathcal{S}_p \rightarrow I_S^p$ the canonical projection, and by

$$\mu : I_S^p \times \mathcal{S}_p \rightarrow I_S^p$$

the morphism defining the action of $\mathcal{S}_p$ on I_S^p (i.e. if τ is an element of $\mathcal{S}_p$, the restriction of μ to $I_S^p \times \tau$ has $\text{pr}_{\tau(j)}$ as j -th component). This being so, we put forward the following definition:

Definition. A functor $X : (\mathbf{Sch}/S)^\circ \rightarrow (\mathbf{Ens})$ satisfies condition (F) if:

- a) X transforms finite direct sums into direct products,
- b) for every S -scheme T , the sequence below is exact:

$$X(T \times \Sigma^p I_S) \longrightarrow X(T \times I_S^p) \xrightarrow[X(\text{id}_T \times \mu)]{X(\text{id}_T \times \pi)} X(T \times I_S^p \times \mathcal{S}_p).$$

Every S -scheme satisfies (F); if $\mathcal{F}$ is an $\mathcal{O}_S$ -module, $\mathbf{W}(\mathcal{F})$ satisfies (F); every projective limit of functors satisfying (F) also satisfies (F); if Y satisfies (F) and X is an arbitrary S -functor, $\underline{\text{Hom}}_S(X, Y)$ satisfies (F).

Let X be a *very good group* (II 4.10) satisfying condition (F). Denoting by $x : I_S \rightarrow X$ a morphism extending the unit section of X and using again the notation of 6.2, one sees as above that $x_1 \dots x_p : I_S^p \rightarrow X$ factors through $\Sigma^p I_S$:

$$\begin{array}{ccc}
 I_S^p & \xrightarrow{x_1 \dots x_p} & X \\
 \searrow \text{can.} & & \nearrow \Sigma^p(x) \\
 & \Sigma^p I_S &
 \end{array}$$

and defines by composition a morphism

$$x^{(p)} : I_S \xrightarrow{i} \Sigma^p I_S \xrightarrow{\Sigma^p(x)} X$$

which we shall call the *symbolic p -th power* of x .

The endomorphism $x \mapsto x^{(p)}$ of $\underline{\text{Lie}}(G/S)(S)$ is clearly compatible with base changes and functorial in G . It would be interesting to know for which G this endomorphism makes $\underline{\text{Lie}}(G/S)(S)$ a Lie p -algebra.

6.4. — The last definition of symbolic p -th power, which we have just given, is particularly well suited to calculation. Here are a few examples:

6.4.1. — Let M be an “abstract” abelian group and $D_S(M)$ the diagonalizable S -group of type M (I 4.4.2). For every S -scheme T , one therefore has

$$D_S(M)(T) = \text{Hom}_{(\text{Ab})}(M, \mathcal{O}(T)^\times).$$

Let x be an element of $\underline{\mathrm{Lie}}(D_S(M)/S)(S)$, that is, a homomorphism of abelian groups

$$M \xrightarrow{x} \Gamma(S, \mathcal{O}_S + d\mathcal{O}_S)^\times$$

of the form $m \mapsto 1 + d\xi(m)$, where $\xi \in \mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S))$. With the notation of 6.2 and 6.3, the product $x_1 \cdots x_p$ associates to an element m of M the expression

$$(1 + d_1\xi(m)) \cdots (1 + d_p\xi(m))$$

that is, $1 + \sigma_1\xi(m) + \sigma_2\xi(m)^2 + \cdots + \sigma_p\xi(m)^p$.

This expression does indeed belong to $\mathcal{O}(\Sigma^p I_S)$. Projecting this into $\mathcal{O}(S)[d]/(d^2)$ by annihilating $\sigma_1, \dots, \sigma_{p-1}$ and sending σ_p to d , one sees that $x^{(p)}$ is the following homomorphism from M to $\Gamma(S, \mathcal{O}_S + d\mathcal{O}_S)^\times$:

$$m \mapsto 1 + d\xi(m)^p.$$

To summarize, if one identifies $\underline{\mathrm{Lie}}(D_S(M)/S)(S)$ with $\mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S))$ as in II 5.1, the symbolic p -th power associates to ξ the homomorphism $\xi^{(p)} : m \mapsto \xi(m)^p$.

6.4.2. — Let $\mathcal{F}$ be an $\mathcal{O}_S$ -module and G the S -functor in abelian groups $\mathbf{W}(\mathcal{F})$ (cf. I, 4.6). Let y be an element of $\mathbf{W}(\mathcal{F})(S) = \Gamma(S, \mathcal{F})$ and y' its image in $\mathbf{W}(\mathcal{F})(I_S)$ under $\mathbf{W}(\mathcal{F})(I_S \rightarrow S)$.

One knows (cf. II, 4.4.2 and 4.5.1) that the map $y \mapsto dy'$ is an isomorphism of $\mathcal{O}(S)$ -modules from $\mathbf{W}(\mathcal{F})(S)$ onto $\underline{\mathrm{Lie}}(\mathbf{W}(\mathcal{F})/S)(S)$. If one puts $x = dy'$, the quantity x_i of 6.2 is none other than $d_i y''$, where y'' denotes the canonical image of $y'^{(59)}$ in $\mathbf{W}(\mathcal{F})(I_S^p)$. Consequently, the product $x_1 \cdots x_p$ is here equal to

$$x_1 + \cdots + x_p = (d_1 + \cdots + d_p)y'' = \sigma_1 y''$$

and belongs to $\mathbf{W}(\mathcal{F})(\Sigma^p I_S)$. Since the homomorphism $\mathcal{O}(\Sigma^p I_S) \rightarrow \mathcal{O}(I_S)$ defining the morphism i of 6.1 annihilates σ_1 , one sees that $x^{(p)}$ is zero. Thus:

For every $\mathcal{O}_S$ -module $\mathcal{F}$, the operation $x \mapsto x^{(p)}$ in $\underline{\mathrm{Lie}}\mathbf{W}(\mathcal{F})$ is zero.

6.4.3. — Let X be an S -scheme, G the S -group functor $\underline{\mathrm{Aut}}_S X$, and D an S -derivation of the structure sheaf $\mathcal{O}_X$. According to 1.5, D may be identified with an I_S -automorphism x of X_{I_S} inducing the identity on X , which may be described as follows. If f is a section of $\mathcal{O}_S[d]/(d^2)$ of the form $a + bd$, put $D_{I_S} f = Da + (Db)d$; in other words, D_{I_S} is deduced from D by the base change $I_S \rightarrow S$; then the automorphism in question of X_{I_S} is associated to the endomorphism $f \mapsto f + (D_{I_S} f)d = a + (D(a) + b)d$ of $\mathcal{O}_S[d]/(d^2)$.

Similarly, let $D_{I_S^p}$ be the differential operator of $X_{I_S^p}$ deduced from D by the base change $I_S^p \rightarrow S$. With the notation of 6.2, the automorphism x_i of $X_{I_S^p}$ is then associated to the endomorphism $f \mapsto f + (D_{I_S^p} f)d_i$ of $\mathcal{O}_S[d_1, \dots, d_p]/(d_1^2, \dots, d_p^2)$. The product $x_1 \cdots x_p$ is therefore associated to the endomorphism

$$(1 + d_1 D_{I_S^p})(1 + d_2 D_{I_S^p}) \cdots (1 + d_p D_{I_S^p})$$

that is, to $1 + \sigma_1 D_{I_S^p} + \sigma_2 D_{I_S^p}^2 + \cdots + \sigma_p D_{I_S^p}^p$. The coefficient of σ_p is $D_{I_S^p}^p$, which means that the Lie algebra isomorphism $\mathrm{Dér}_S(\mathcal{O}_X) \xrightarrow{\sim} \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S X)$, $D \mapsto x$ (cf. 1.5), is also an isomorphism of Lie p -algebras.

6.4.4. — Using the same method, one sees that, for every $\mathcal{O}_S$ -module $\mathcal{F}$, the isomorphism

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathbf{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F})/S)(S) \xrightarrow{\sim} (\underline{\mathrm{End}}_{\mathbf{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F}))(S)$$

of Lie algebras (cf. II 4.8) is also an isomorphism of Lie p -algebras.

6.4.5. — ⁽⁶⁰⁾ Let $\mathcal{U}$ be a $\mathcal{O}_S$ -coalgebra in groups and G the S -group functor $\mathrm{Spec}^* \mathcal{U}$; suppose that G is *representable*. In this case, for every $T \rightarrow S$, two Lie p -algebra structures have

⁽⁵⁹⁾**N.D.E.** x has been corrected to y' .

⁽⁶⁰⁾**N.D.E.** The numbering 6.4.5 has been added, and the original expanded.

been defined in 5.5 and 6.1.1 on $L(T) = \underline{\mathrm{Lie}}(G)(T)$. Since one has a commutative diagram

$$\begin{array}{ccc} L(T) & \xrightarrow{\tau} & \Gamma(T, \mathcal{U}_T) \\ \alpha \downarrow & \searrow i & \uparrow \psi \\ U(G_T) & \xleftarrow{\phi} & U(L(T)) \end{array}$$

where $U(L(T))$ is the enveloping algebra of $L(T)$ and ϕ, ψ the algebra morphisms induced by α, τ , one sees that the two Lie p -algebra structures coincide: if one identifies $x \in L(T)$ with its image in $U(G_T)$ (resp. $\Gamma(T, \mathcal{U}_T)$), then $x^{(p)}$ is the image of the element x^p of $U(L(T))$ under ϕ (resp. ψ).

7. Radicial groups of height 1

⁽⁶¹⁾ Let S be a scheme of characteristic $p > 0$. We shall say that a $\mathcal{O}_S$ -algebra $\mathcal{A}$ (resp. a $\mathcal{O}_S$ -Lie p -algebra $\mathcal{L}$) is *finite locally free* if the $\mathcal{O}_S$ -module underlying $\mathcal{A}$ (resp. $\mathcal{L}$) is locally free and of finite type. If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$ -Lie p -algebra, we know (cf. 5.5.2) that the S -group functor $\mathrm{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by a finite and locally free S -group scheme $G_p(\mathcal{L})$. We shall see that this S -group scheme is the solution to a universal problem (7.2), and we shall characterize the S -group schemes of the form $G_p(\mathcal{L})$ (7.4).

Definition 7.0. ⁽⁶²⁾ Let $H = \mathrm{Spec} \mathcal{A}$ be a finite locally free S -group scheme. One says that H is *infinitesimal* if the unit section $\varepsilon_H : S \rightarrow H$ is a homeomorphism, which is equivalent to saying that the augmentation ideal of $\mathcal{A}$ is locally nilpotent.

7.1. — ⁽⁶³⁾ Let $\mathcal{L}$ be a finite locally free $\mathcal{O}_S$ -Lie p -algebra and $G_p(\mathcal{L})$ the affine S -group $\mathrm{Spec} \mathcal{U}_p(\mathcal{L})$. According to 5.5, one knows that $\mathcal{L}$ identifies with $\mathcal{L}ie G_p(\mathcal{L})$. Now consider a very good S -group functor G satisfying condition (F) of 6.3, and let $\phi : G_p(\mathcal{L}) \rightarrow G$ be a morphism of S -group functors. According to 6.3, the morphism of $\mathcal{O}_S$ -Lie algebras $\mathcal{L}ie \phi : \mathcal{L}ie G_p(\mathcal{L}) \rightarrow \mathcal{L}ie G$ is compatible with symbolic p -th power. If we denote by $\mathrm{Hom}_p(\mathcal{L}, \mathcal{L}ie G)$ the set of morphisms of $\mathcal{O}_S$ -Lie algebras compatible with symbolic p -th power, there is therefore a map

$$\mathcal{L}ie : \mathrm{Hom}_{S\text{-Gr.}}(G_p(\mathcal{L}), G) \rightarrow \mathrm{Hom}_p(\mathcal{L}, \mathcal{L}ie G), \quad \phi \mapsto \mathcal{L}ie \phi.$$

Theorem 7.2. *If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$ -Lie p -algebra, the map*

$$\mathrm{Hom}_{S\text{-gr.}}(G_p(\mathcal{L}), G) \rightarrow \mathrm{Hom}_p(\mathcal{L}, \mathcal{L}ie G)$$

is bijective in each of the following cases:

- (i) G is an S -group scheme;
- (ii) G is of the form $\underline{\mathrm{Aut}}_S X$, where X is an S -scheme;
- (iii) G is of one of the forms $\mathbf{W}(\mathcal{F})$ or $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod}} \mathbf{W}(\mathcal{F})$, where $\mathcal{F}$ denotes a quasi-coherent $\mathcal{O}_S$ -module.

The proof of the theorem rests on the following lemma:

Lemma. *If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$ -Lie p -algebra, the S -group $G = G_p(\mathcal{L})$ is annihilated by the Frobenius morphism $\mathrm{Fr} : G \rightarrow G^{(p)}$. In particular, G is infinitesimal.*

⁽⁶¹⁾**N.D.E.** For the results of this section, one may also refer to [DG70], § II.7, nos. 3–4.

⁽⁶²⁾**N.D.E.** This definition (cf. [DG70], § II.4, 7.1), which will be used in 7.2.1, has been added.

⁽⁶³⁾**N.D.E.** The original has been simplified, taking account of the additions made in 5.5.

⁽⁶⁴⁾ Let, indeed, $\mathcal{U}$ be the restricted enveloping algebra of $\mathcal{L}$, $\mathcal{A} = \mathcal{U}^*$ the affine algebra of G , and $\mathcal{J} = \text{Ker } \varepsilon_{\mathcal{A}}$ the augmentation ideal of $\mathcal{A}$. One has

$$(1) \quad \mathcal{A} = \mathcal{J} \oplus \eta_{\mathcal{A}}(\mathcal{O}_S),$$

where $\eta_{\mathcal{A}}$ denotes the unit section of $\mathcal{A}$, and since $\varepsilon_{\mathcal{A}}$ (resp. $\eta_{\mathcal{A}}$) is the transpose of $\eta_{\mathcal{U}}$ (resp. $\varepsilon_{\mathcal{U}}$), this decomposition corresponds by duality to the decomposition

$$(2) \quad \mathcal{U} = \mathcal{J} \oplus \eta_{\mathcal{U}}(\mathcal{O}_S),$$

where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$; one therefore has $\mathcal{J} = \mathcal{J}^*$.

Denote by π the endomorphism $x \mapsto x^p$ of $\mathcal{O}_S$. We must show that the morphism $\text{Fr} : G \rightarrow G^{(p)}$ factors through the unit section of $G^{(p)}$, which is equivalent to saying (cf. 4.1.4 (c)) that the morphism $\Phi : a \otimes_{\pi} x \mapsto a^p x$ from $\mathcal{J} \otimes_{\pi} \mathcal{O}_S$ to $\mathcal{A}$ is zero. Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, it suffices to see that the transpose morphism ${}^t\Phi$ is zero.

Now Φ is none other than the following composite

$$\mathcal{J} \otimes_{\pi} \mathcal{O}_S \xrightarrow{\tau} \mathcal{A} \otimes_{\pi} \mathcal{O}_S \xrightarrow{j(\mathcal{A})} S^p \mathcal{A} \xrightarrow{b(\mathcal{A})} \mathcal{A},$$

where τ is deduced from the inclusion $\mathcal{J} \hookrightarrow \mathcal{A}$, and $b(\mathcal{A})$ and $j(\mathcal{A})$ are defined as in 4.3.3 (i.e. $b(\mathcal{A})$ is induced by the multiplication of $\mathcal{A}$ and $j(\mathcal{A})$ sends $a \otimes_{\pi} 1$ to the image of $a \otimes \cdots \otimes a$ in $S^p \mathcal{A}$). Since the dual $\mathcal{O}_S$ -module of $S^p \mathcal{A}$ is none other than the submodule $\Sigma^p \mathcal{U}$ of $\bigotimes^p \mathcal{U}$ consisting of sections invariant under the action of the symmetric group of order p , one sees that ${}^t\Phi$ is the following composite morphism:

$$\mathcal{U} \xrightarrow{a(\mathcal{U})} \Sigma^p \mathcal{U} \xrightarrow{r(\mathcal{U})} \mathcal{U} \otimes_{\pi} \mathcal{O}_S \xrightarrow{q} \mathcal{J} \otimes_{\pi} \mathcal{O}_S,$$

where q is deduced from the projection $\mathcal{U} \rightarrow \mathcal{J}$ with kernel $\eta_{\mathcal{U}}(\mathcal{O}_S)$, $a(\mathcal{U})$ is induced by the comultiplication of $\mathcal{U}$, and $r(\mathcal{U})$ vanishes on symmetrized tensors and sends a section $x \otimes \cdots \otimes x$ to $x \otimes_{\pi} 1$ (confer 4.3.3).

It is clear that ${}^t\Phi \circ \eta_{\mathcal{U}} = 0$ and hence, according to (2), it remains to see that ${}^t\Phi$ annihilates the augmentation ideal $\mathcal{J}$. Since ${}^t\Phi$ is an algebra morphism and the ideal $\mathcal{J}$ is generated by $\mathcal{L}$ (identified with its image in $\mathcal{U}$), it suffices to see that ${}^t\Phi(x) = 0$ for every section x of $\mathcal{L}$. Now $-a(\mathcal{U})(x) = (p-1)! a(\mathcal{U})(x)$ is the symmetrization of $x \otimes 1 \otimes \cdots \otimes 1$, hence its image under $r(\mathcal{U})$ is zero. This proves the first assertion of the lemma.

The second follows from it. Indeed, since every local section of $\mathcal{J}$ has zero p -th power and $\mathcal{J}$ is an $\mathcal{O}_S$ -module of finite type, $\mathcal{J}$ is locally nilpotent (explicitly, if V is an affine open subset of S such that $I = \Gamma(V, \mathcal{J})$ is generated by r elements, then $I^{r(p-1)+1} = 0$), whence $G_{\text{red}} = S_{\text{red}}$, and hence the unit section $\varepsilon_G : S \rightarrow G$ is a homeomorphism.

7.2.1. — ⁽⁶⁵⁾ We shall first prove assertion (ii) of theorem 7.2. Let $\pi : X \rightarrow S$ be an S -scheme. First consider an arbitrary infinitesimal S -group H . Morphisms ϕ from H to $\underline{\text{Aut}} X$ correspond bijectively to left actions $\mu : H \times X \rightarrow X$ of H on X . For such an action, if ε is the unit section of H , the composite morphism

$$X \simeq S \times X \xrightarrow{\varepsilon \times X} H \times X \xrightarrow{\mu} X$$

must be the identity. Since $(H \times X)_{\text{red}}$ identifies with X_{red} , one sees that μ must induce the identity on the associated reduced schemes. In particular, μ induces an action of H on every open subset of X , and one therefore obtains, for every open subset U of X affine over S , a morphism of associative unital algebras:

$$\mathcal{A}(U) \rightarrow \mathcal{A}(H) \otimes \mathcal{A}(U)$$

making $\mathcal{A}(U)$ a left $\mathcal{A}(H)$ -comodule, in such a way that the restriction maps $\mathcal{A}(U) \rightarrow \mathcal{A}(U')$, for $U' \subset U$, are comodule morphisms. Conversely, every datum of this type comes from a unique left action $\mu : H \times X \rightarrow X$. On the other hand, one has the following lemma:

⁽⁶⁴⁾**N.D.E.** The original has been expanded in what follows. For another proof, see [DG70], § II.7, 3.9.

⁽⁶⁵⁾**N.D.E.** The original has been expanded in what follows.

Lemma. *Let $X = \operatorname{Spec} \mathcal{C}$ be an affine S -scheme, $H = \operatorname{Spec} \mathcal{A}$ an infinitesimal S -group, and $\mathcal{U} = \mathcal{A}^* = \mathcal{H}om_{\mathcal{O}_S}(\mathcal{A}, \mathcal{O}_S)$. The left actions of H on X correspond bijectively to the representations of the algebra $\mathcal{U}$ in the $\mathcal{O}_S$ -module $\mathcal{C}$ such that one has:*

$$\begin{aligned} (a) \quad & u(1_{\mathcal{C}}) = \varepsilon(u) \cdot 1_{\mathcal{C}} \\ (b) \quad & u(xy) = \sum_i v_i(x)w_i(y) \quad \text{if } \Delta u = \sum_i v_i \otimes w_i. \end{aligned}$$

(In the formulas above, u denotes an arbitrary section of $\mathcal{U}$ over an affine open subset V of S , and x and y sections of $\mathcal{C}$ over V ; $1_{\mathcal{C}}$ denotes the unit section of $\mathcal{C}$, and ε and Δ the augmentation and diagonal morphism of $\mathcal{U}$.) Indeed, a left action μ of H on X is defined by a morphism of associative unital algebras:

$$\lambda : \mathcal{C} \longrightarrow \mathcal{A} \otimes \mathcal{C}$$

making $\mathcal{C}$ a left $\mathcal{A}$ -comodule. We shall denote by α the composite morphism

$$\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\mathcal{U} \otimes \lambda} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\gamma \otimes \mathcal{C}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{C} \simeq \mathcal{C}$$

where γ is the “contraction” from $\mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}$ to $\mathcal{O}_S$. Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, one knows that the map $\lambda \mapsto (\gamma \otimes \mathcal{C})(\mathcal{U} \otimes \lambda)$ is a bijection from $\operatorname{Hom}_{\mathcal{O}_S}(\mathcal{C}, \mathcal{A} \otimes \mathcal{C})$ onto $\operatorname{Hom}_{\mathcal{O}_S}(\mathcal{U} \otimes \mathcal{C}, \mathcal{C})$. Moreover, one sees easily that the condition that λ define a left $\mathcal{A}$ -comodule structure (resp. be a morphism of associative unital algebras) is equivalent, by duality, to the condition that α be a representation of $\mathcal{U}$ in $\mathcal{C}$ (resp. that α satisfy conditions (a) and (b)). This proves the lemma.

Moreover, it is clear that, for every representation of $\mathcal{U}$ in the $\mathcal{O}_S$ -module $\mathcal{C}$, the sections u of $\mathcal{U}$ satisfying conditions (a) and (b) of the lemma form a subalgebra of $\mathcal{U}$.

In the particular case $H = G_p(\mathcal{L})$ which interests us, these conditions will therefore hold for all sections u of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$ if they are true for the sections u of $\mathcal{L}$ (identifying $\mathcal{L}$ with its image in $\mathcal{U}_p(\mathcal{L})$). Now, if u is a section of $\mathcal{L}$, conditions (a) and (b) mean simply that $u(1_{\mathcal{C}}) = 0$ and $u(xy) = u(x)y + xu(y)$, i.e. that $\alpha(u)$ is a $\mathcal{O}_S$ -derivation of $\mathcal{C} = \mathcal{A}(X)$. Assertion (ii) of 7.2 follows. Indeed, every morphism ϕ from $G_p = G_p(\mathcal{L})$ to $\underline{\operatorname{Aut}} X$ defines a morphism of Lie p -algebras $\mathcal{L}ie \phi$ from $\mathcal{L} = \mathcal{L}ie G_p$ to $\pi_*(\mathcal{D}er_{\mathcal{O}_S} \mathcal{O}_X)$, and conversely every datum of this type comes, according to what precedes, from a unique action $\mu : G \times X \longrightarrow X$.

7.2.2. — Let us now show how assertion (i) of theorem 7.2 follows from (ii). Let G be an S -group scheme. If T is an S -scheme and x an element of $G(T)$, we denote by ℓ_x^T (resp. r_x^T) the left (resp. right) translation of G_T defined by x . The maps $\ell^T : x \mapsto \ell_x^T$ therefore determine a homomorphism ℓ from G to $\underline{\operatorname{Aut}} G$. Let, on the other hand, f be a T -automorphism of G_T ; one then defines ${}^x f$ to be equal to $(r_x^T)^{-1} f r_x^T$, i.e. for every $T' \longrightarrow T$ and $g \in G(T')$, $({}^x f)(g) = f(gx)x^{-1}$. In this way G acts on the left on the S -group functor $\underline{\operatorname{Aut}} G$, hence also on the functors $T \mapsto \operatorname{Hom}_{T\text{-Gr.}}(G_p(\mathcal{L}_T), \underline{\operatorname{Aut}} G_T)$ and $T \mapsto \operatorname{Hom}_p(\mathcal{L}_T, \mathcal{L}ie(\underline{\operatorname{Aut}} G_T/T))$. On the other hand, the morphism $\ell_T : G_T \longrightarrow \underline{\operatorname{Aut}} G_T$ identifies G_T with the group of automorphisms of the T -scheme G_T commuting with right translations, and the derived morphism $\mathcal{L}ie(\ell_T)$ identifies $\mathcal{L}ie G_T$ with the Lie p -algebra of $\mathcal{O}_T$ -derivations of $\mathcal{O}_{G_T}$ commuting with right translations (cf. II, 4.11.1); they therefore induce commutative squares

$$\begin{array}{ccc} \operatorname{Hom}_{T\text{-Gr.}}(G_p(\mathcal{L}_T), G_T) & \xrightarrow{\mathcal{L}ie} & \operatorname{Hom}_p(\mathcal{L}_T, \mathcal{L}ie(G_T/T)) \\ \ell_T \downarrow & & \downarrow \mathcal{L}ie \ell_T \\ \operatorname{Hom}_{T\text{-Gr.}}(G_p(\mathcal{L}_T), \underline{\operatorname{Aut}} G_T) & \xrightarrow{\mathcal{L}ie} & \operatorname{Hom}_p(\mathcal{L}_T, \mathcal{L}ie(\underline{\operatorname{Aut}} G_T/T)). \end{array}$$

The images of the two vertical arrows are the subfunctors consisting of the invariants under the action of the S -group G . Since the lower horizontal arrow is invertible according to 7.2.1 and is compatible with the action of G , the upper horizontal arrow is invertible also. This proves 7.2 (i).

7.2.3. — Now consider the case where $G = \underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F})$.⁽⁶⁶⁾ Put $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, $\mathcal{A} = \mathcal{U}^*$, and $H = G_p(\mathcal{L}) = \text{Spec } \mathcal{A}$. Since H is affine over S , according to VI B 11.6.1 a morphism of S -groups from H to $\underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F})$ is the same thing as a right $\mathcal{A}$ -comodule structure

$$\mu : \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}.$$

Moreover, since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, this is equivalent to giving a representation

$$\alpha : \mathcal{U} \longrightarrow \text{End}_{\mathcal{O}_S}(\mathcal{F})$$

of $\mathcal{U}$ in $\mathcal{F}$. Finally, according to the universal property of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, giving such a morphism α is equivalent to giving its restriction ρ to $\mathcal{L}$ (identified with its image in $\mathcal{U}$), which is a morphism of Lie p -algebras from $\mathcal{L}$ to $\text{End}_{\mathcal{O}_S}(\mathcal{F})$.

⁽⁶⁷⁾ Finally, consider the case where $G = \mathbf{W}(\mathcal{F})$, keeping the preceding notation. First, giving a morphism of S -functors $\phi : H \longrightarrow \mathbf{W}(\mathcal{F})$ is equivalent to giving an element θ of $\Gamma(H, \mathcal{F} \otimes \mathcal{O}_H)$, and since H is finite locally free over S , one has:

$$\Gamma(H, \mathcal{F} \otimes \mathcal{O}_H) = \Gamma(S, \mathcal{F} \otimes \mathcal{A}) = \text{Hom}_{\mathcal{O}_S}(\mathcal{U}, \mathcal{F}).$$

The condition that ϕ be a group morphism is then expressed by the fact that θ , considered as a morphism of $\mathcal{O}_S$ -modules $\mathcal{U} \longrightarrow \mathcal{F}$, vanishes on $1_{\mathcal{U}}$ and on $\mathcal{J} / \mathcal{J}^2$, where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$, whence

$$(1) \quad \text{Hom}_{S\text{-gr.}}(H, \mathbf{W}(\mathcal{F})) = \text{Hom}_{\mathcal{O}_S}(\mathcal{J} / \mathcal{J}^2, \mathcal{F}).$$

On the other hand, consider the quasi-coherent sheaf $[\mathcal{L}, \mathcal{L}]$, the image of the morphism $\mathcal{L} \otimes \mathcal{L} \longrightarrow \mathcal{L}$, $x \otimes y \mapsto [x, y]$; for every affine open subset V of S , one has $[\mathcal{L}, \mathcal{L}](V) = [\mathcal{L}(V), \mathcal{L}(V)]$. Then there is an exact sequence

$$(\dagger) \quad 0 \longrightarrow [\mathcal{L}, \mathcal{L}] \longrightarrow \mathcal{L} \xrightarrow{\pi} \mathcal{J} / \mathcal{J}^2 \longrightarrow 0,$$

where π is the composite of the inclusion $\mathcal{L} \hookrightarrow \mathcal{J}$ and the projection $\mathcal{J} \longrightarrow \mathcal{J} / \mathcal{J}^2$.

Indeed, the question being local on S , one may suppose that S is affine with ring R and that $L = \mathcal{L}(S)$ is free with basis $(x_1, \dots, x_r)$. Identifying L with its image in $U = U_p(L)$, let K be the sub- R -module of U which is the direct sum of $[L, L]$ and the submodule with basis the monomials $x_1^{n_1} \cdots x_r^{n_r}$ such that $n_1 + \cdots + n_r \geq 2$; one then verifies that K is a two-sided ideal of U . Since K is contained in J^2 (where J is the augmentation ideal of U) and contains all the products $x_i x_j$ (which generate J^2), one deduces that $K = J^2$, whence $J^2 \cap L = [L, L]$, and one has exact sequence $(\dagger)$.

On the other hand, one knows according to 6.4.2 that $\underline{\text{Lie}} \mathbf{W}(\mathcal{F})$ is none other than $\mathbf{W}(\mathcal{F})$, the Lie bracket and symbolic p -th power being zero. From this and what precedes one deduces that

$$(2) \quad \text{Hom}_p(\mathcal{L}, \mathcal{F}) = \text{Hom}_{\mathcal{O}_S}(\mathcal{L} / [\mathcal{L}, \mathcal{L}], \mathcal{F}) = \text{Hom}_{\mathcal{O}_S}(\mathcal{J} / \mathcal{J}^2, \mathcal{F})$$

and this, combined with (1), completes the proof of theorem 7.2.

Lemma 7.3. *If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$ -Lie p -algebra, the morphism $j_{\mathcal{L}} : \mathcal{L} \longrightarrow \text{Lie } G_p(\mathcal{L})$ of 5.5 is invertible.*

⁽⁶⁸⁾ For the proof, see 5.5.1.

⁽⁶⁶⁾**N.D.E.** In what follows, the original has been expanded (and simplified), taking account of VI B, 11.6.1.

⁽⁶⁷⁾**N.D.E.** What follows has been added. (The original indicated “the case of $\mathbf{W}(\mathcal{F})$ is analogous”.)

⁽⁶⁸⁾**N.D.E.** To avoid changing the numbering, statement 7.3 has been retained, although it was included, with its proof, in 5.5.1.

7.4. — To conclude this section, we shall give a characterization of the S -group schemes of the form $G_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$ -Lie p -algebra.

Let G be an S -group scheme, ε_G its unit section, and $\mathcal{I}'$ the kernel of the morphism $\varepsilon_G^{-1}(\mathcal{O}_G) \rightarrow \mathcal{O}_S$ corresponding to ε_G . The image of $\underline{\mathrm{Lie}}(G/S)(S)$ in $U(G)$ identifies, according to 2.5 and 1.3.1, with the morphisms of $\mathcal{O}_S$ -modules from $\varepsilon_G^{-1}(\mathcal{O}_G)$ to $\mathcal{O}_S$ which vanish on the unit section of $\varepsilon_G^{-1}(\mathcal{O}_G)$ and on $\mathcal{I}'^2$. One thus recovers the canonical isomorphism from $\underline{\mathrm{Lie}}(G/S)(S)$ onto $\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}'/\mathcal{I}'^2, \mathcal{O}_S)$ of II, 3.3 and 4.11.4.⁽⁶⁹⁾ We shall put $\omega_{G/S} = \mathcal{I}'/\mathcal{I}'^2$ as in loc. cit., so that the sheaf $\mathcal{L}\mathrm{ie}(G/S)$ identifies with $\omega_{G/S}^* = \mathcal{H}\mathrm{om}_{\mathcal{O}_S}(\omega_{G/S}, \mathcal{O}_S)$.⁽⁷⁰⁾ Moreover, if $G = G_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$ -Lie p -algebra, we saw in 5.5.3 that $\omega_{G/S} = \mathcal{L}^*$.

Theorem. *If G is a group scheme over a scheme S of characteristic $p > 0$, the following assertions are equivalent:*

- (i) *There exists a finite locally free $\mathcal{O}_S$ -Lie p -algebra $\mathcal{L}$ such that $G \simeq G_p(\mathcal{L})$.*
- (i') *The $\mathcal{O}_S$ -Lie p -algebra $\mathcal{L}\mathrm{ie}(G)$ is finite locally free and $G \simeq G_p(\mathcal{L}\mathrm{ie}(G))$.*
- (ii) *G is affine over S , $\omega_{G/S}$ is a locally free $\mathcal{O}_S$ -module of finite type, and the affine algebra of G is locally isomorphic to the quotient of the symmetric algebra $S_{\mathcal{O}_S}(\omega_{G/S})$ by the ideal generated by the p -th powers of sections of $\omega_{G/S}$.*
- (iii) *G is locally of finite presentation over S , of height ≤ 1 , and $\omega_{G/S}$ is locally free.*
- (iii') *G is locally of finite type over S , of height ≤ 1 , and $\omega_{G/S}$ is locally free.*
- (iv) *G is locally of finite presentation and flat over S , of height ≤ 1 .⁽⁷¹⁾*

7.4.1. — The equivalence (i) $\Leftrightarrow$ (i') follows from 5.5.3 (i), the implications (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iii') are clear, and one has (i) $\Rightarrow$ (iv) since $G_p(\mathcal{L})$ is finite locally free and of height ≤ 1 , according to 5.5.2 and lemma 7.2. Let us show that (i) implies (ii). Denote by $\mathcal{I}$ the augmentation ideal of $\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*$. We already saw in 5.5.3 (ii) that $\omega_{G/S} = \mathcal{I}/\mathcal{I}^2$ identifies with $\mathcal{L}^*$, hence is finite locally free.

Now suppose that S is affine. There is then a section $\sigma : \omega_{G/S} \rightarrow \mathcal{I}$ of the projection $\mathcal{I} \rightarrow \mathcal{I}/\mathcal{I}^2$; it induces an algebra morphism $\sigma' : S_{\mathcal{O}_S}(\omega_{G/S}) \rightarrow \mathcal{A}$, and, according to lemma 7.2, σ' factors into a morphism

$$\phi : S_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K} \rightarrow \mathcal{A},$$

where $\mathcal{K}$ denotes the ideal generated by the p -th powers of sections of $\omega_{G/S}$. If one filters $\mathcal{A}$ (resp. $S_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K}$) by the powers of $\mathcal{I}$ (resp. of the ideal generated by $\omega_{G/S}$), it is clear that ϕ induces an epimorphism of the associated graded objects. Thus ϕ is an epimorphism of locally free $\mathcal{O}_S$ -modules of the same rank (cf. 5.3.3); hence ϕ is an isomorphism. This proves that (i) $\Rightarrow$ (ii).

7.4.2. — Now suppose that G is of height ≤ 1 and locally of finite presentation over S .⁽⁷²⁾ Since the Frobenius morphism $\mathrm{Fr} : G \rightarrow G^{(p)}$ is integral and factors through the unit section of $G^{(p)}$, G is integral (hence affine) over S . Put $\mathcal{A} = \mathcal{A}(G)$; since G is assumed locally

⁽⁶⁹⁾**N.D.E.** If G is affine over S and $\mathcal{I}$ denotes the augmentation ideal of $\mathcal{A}(G)$, then $\mathcal{I}'/\mathcal{I}'^2$ identifies with $\varepsilon_G^*(\mathcal{I}/\mathcal{I}^2)$, cf. loc. cit.

⁽⁷⁰⁾**N.D.E.** The following sentence has been added.

⁽⁷¹⁾**N.D.E.** On the one hand, assertion (i'), implicit in the original, has been added, and on the other hand assertions (iii') and (iv), pointed out by O. Gabber; assertion (iv) takes up a footnote of the original, which indicated: "The condition on $\omega_{G/S}$ is in fact unnecessary, as one sees easily by reducing to the case where S is local with residue field k , and applying the theorem to the group G_k ." As pointed out by Gabber, this is incorrect without a flatness hypothesis: if A is a local artinian ring of characteristic $p > 0$ and J a proper nonzero ideal of A , let H be the subgroup $\mathrm{Spec} A[x]/(x^p, Jx)$ of $\alpha_{p,A}$ (i.e. for every A -algebra R , $H(R) = \{x \in R \mid x^p = 0 \text{ and } Jx = 0\}$); then H is not flat over A , hence is not of the form $G_p(\mathcal{L})$, where $\mathcal{L}$ is a free Lie p -algebra of finite rank over A .

⁽⁷²⁾**N.D.E.** The original has been expanded (and simplified) in what follows.

of finite presentation over S , it follows that G is finite and of finite presentation over S , hence that $\mathcal{A}(G)$ is a finitely presented $\mathcal{O}_S$ -module (cf. EGA IV₁, 1.4.7). Let $\mathcal{I}$ be the augmentation ideal of $\mathcal{A}$; since $\mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{I}$ (where $\eta_{\mathcal{A}}$ is the unit section of $\mathcal{A}$), $\mathcal{I}$ is a finitely presented $\mathcal{O}_S$ -module, and therefore so is $\omega_G = \mathcal{I}/\mathcal{I}^2$. When one assumes that G is of height ≤ 1 and locally of finite type over S , one obtains similarly that $\mathcal{A}(G)$, $\mathcal{I}$, and $\omega_G = \mathcal{I}/\mathcal{I}^2$ are $\mathcal{O}_S$ -modules of finite type.

Thus, under hypothesis (iii'), one obtains that $\omega_{G/S}$ is finite locally free over $\mathcal{O}_S$, as is $\mathcal{L} = \mathcal{L}e(G/S) = \omega_{G/S}^*$. Put $\mathcal{B} = \mathcal{U}_p(\mathcal{L})^*$ and $H = G_p(\mathcal{L}) = \text{Spec } \mathcal{B}$. According to theorem 7.2, the identity map of $\mathcal{L}$ corresponds to a group morphism from $H = G_p(\mathcal{L})$ to G , hence to a morphism of $\mathcal{O}_S$ -algebras $\theta : \mathcal{A} \rightarrow \mathcal{B}$. One must show that θ , which by definition induces an isomorphism from $\omega_{G/S}$ to $\omega_{H/S}$, is an isomorphism.

For this purpose one may restrict to the case where S is affine. There is then a section τ of the projection $\mathcal{I} \rightarrow \omega_{G/S}$; it induces an algebra morphism $\tau' : S_{\mathcal{O}_S}(\omega_{G/S}) \rightarrow \mathcal{A}$, and since every local section of $\mathcal{I}$ has zero p -th power (because $\text{Fr} : G \rightarrow G^{(p)}$ factors through the unit section of $G^{(p)}$), τ' induces a morphism of $\mathcal{O}_S$ -algebras ψ occurring in the commutative diagram below:

$$\begin{array}{ccc} S_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K} & \xrightarrow{\psi} & \mathcal{A} \\ & \searrow \phi & \downarrow \theta \\ & & \mathcal{B} \end{array}$$

where $\mathcal{K}$ is the ideal generated by the p -th powers of sections of $\omega_{G/S}$. On the one hand, one shows as in 7.4.1 that ψ is an epimorphism of $\mathcal{O}_S$ -modules. On the other hand, we saw in 7.4.1 that $\phi = \theta \circ \psi$ is an isomorphism. The same is therefore true of θ . This proves that (iii') $\Rightarrow$ (i).

7.4.3. — ⁽⁷³⁾ Finally, let us show that (iv) implies (iii). It suffices to show that $\omega_{G/S}$ is locally free, so one may suppose that S is affine with ring R . As observed at the beginning of 7.4.2, hypothesis (iv) then implies that $G = \text{Spec } A$, for an R -algebra A which is a finitely presented R -module, as is $\omega_{G/A} = I/I^2$ (where I is the augmentation ideal of A). Since one also assumes that G is flat over S , A is a finite locally free R -module (cf. [BAC] II, § 5.2, Th. 1 and cor. 2), and, according to loc. cit., to show that $\omega_{G/A}$ is locally free of finite rank, it suffices to show that $(\omega_{G/A})_{\mathfrak{m}}$ is flat for every maximal ideal $\mathfrak{m}$ of R . Thus one may suppose that R is local and A free of rank $n + 1$, hence I free of rank n . Let $\mathfrak{m}$ be the maximal ideal of R and $k = R/\mathfrak{m}$.

Denote by I_k the augmentation ideal of A_k and by r the dimension of $\omega_{G_k/k} = I_k/I_k^2$. Let $(e_1, \dots, e_n)$ be a basis of I_k such that $(e_{r+1}, \dots, e_n)$ is a basis of I_k^2 , and let $x_1, \dots, x_n$ be elements of I lifting the e_i . According to Nakayama's lemma, $(x_1, \dots, x_n)$ is a basis of I over R . Let N be the sub- R -module of I with basis $(x_1, \dots, x_r)$, and let B be the quotient of the symmetric algebra of N by the ideal generated by the elements x^p , for $x \in N$. Since every element of I has zero p -th power, one obtains a morphism of R -algebras

$$\psi : B \rightarrow A.$$

According to 7.4.2, $\psi \otimes k$ is an isomorphism. It follows that $\text{Coker } \psi = 0$ and that, denoting $\text{Ker } \psi$ by K , the morphism $\tau : K \otimes k \rightarrow B \otimes k$ is zero. But since ψ is surjective and A flat over R , τ is also injective, whence $K \otimes k = 0$. On the other hand, since A is a finitely presented R -module, K is an R -module of finite type (cf. [BAC] I, § 2.8, Lemma 9), whence $K = 0$ according to Nakayama. Thus ψ is an isomorphism of R -algebras, and since $\psi^{-1}(I)$ contains the augmentation ideal J of B , it follows that $\psi^{-1}(I) = J$, and hence ψ^{-1} induces an isomorphism of R -modules from I/I^2 to $J/J^2 = N$. This proves that $\omega_{G/S}$ is finite locally free, whence the implication (iv) $\Rightarrow$ (iii). This completes the proof of theorem 7.4.

⁽⁷³⁾**N.D.E.** This paragraph has been added to prove that (iv) $\Rightarrow$ (iii), cf. N.D.E. (71).

Remark 7.5. ⁽⁷⁴⁾ It clearly follows from theorems 7.2 and 7.4 that the functors $G \mapsto \mathcal{L}ie(G)$ and $\mathcal{L} \mapsto G_p(\mathcal{L})$ induce equivalences, quasi-inverse to one another, between the category of S -groups locally of finite presentation and flat, of height ≤ 1 , and the full subcategory of that of $\mathcal{O}_S$ -Lie p -algebras consisting of finite locally free $\mathcal{O}_S$ -Lie p -algebras.

8. Case of a base field

8.1. — Let us now summarize the results obtained in the case where S is the spectrum of a field k of characteristic $p > 0$. Let us say that a k -group scheme is *algebraic* if the underlying scheme is of finite type over k . In this case, according to theorem 7.2, one obtains:⁽⁷⁵⁾

Theorem 8.1.1. *The functor G_p , which associates to every Lie p -algebra $\mathcal{L}$ of finite dimension over k the k -group $G_p(\mathcal{L})$, is left adjoint to the functor which associates to every algebraic k -group G its $\mathcal{L}ie(G)$.*

Combining this with theorem 7.4, one obtains:

Theorem 8.1.2. *The functors $G_p : \mathcal{L} \mapsto G_p(\mathcal{L})$ and $G \mapsto \mathcal{L}ie(G)$ induce equivalences, quasi-inverse to one another, between the category of Lie p -algebras of finite dimension over k and that of algebraic k -groups of height ≤ 1 .*

Then, since G_p is a left adjoint functor, it commutes with inductive limits,⁽⁷⁶⁾ hence in particular with the formation of cokernels. On the other hand, if one has two morphisms $\phi : G \rightarrow H$ and $\phi' : G' \rightarrow H$ between algebraic k -groups of height ≤ 1 , the fibered product $G \times_H G'$ is again an algebraic k -group of height ≤ 1 (for the morphism $\text{Fr} : G \rightarrow G^{(p)}$ commutes with fibered products). Thus the inclusion of the category of algebraic k -groups of height ≤ 1 into that of all algebraic k -groups commutes with fibered products, hence in particular with the formation of kernels. One deduces the:

Corollary 8.1.3. *The functor G_p is exact in the following sense. If $\pi : \mathcal{L}_1 \rightarrow \mathcal{L}_2$ is a surjective morphism between Lie p -algebras of finite dimension over k and i is the inclusion of $\mathcal{L}_0 = \text{Ker } \pi$ into $\mathcal{L}_1$, there is an exact sequence of algebraic k -groups:*

$$1 \rightarrow G_p(\mathcal{L}_0) \xrightarrow{G_p(i)} G_p(\mathcal{L}_1) \xrightarrow{G_p(\pi)} G_p(\mathcal{L}_2) \rightarrow 1.$$

(77)

Indeed, according to what precedes, $G_p(i)$ induces an isomorphism from $G_p(\mathcal{L}_0)$ onto $\text{Ker}(G_p(\pi))$ (this kernel being the same in the category of all algebraic k -groups H or in that of the H of height ≤ 1), and $G_p(\pi) : G_p(\mathcal{L}_1) \rightarrow G_p(\mathcal{L}_2)$ identifies $G_p(\mathcal{L}_2)$ with the quotient of $G_p(\mathcal{L}_1)$ by $G_p(\mathcal{L}_0)$ in the category of algebraic k -groups.

Remark 8.1.4. ⁽⁷⁸⁾ Let $\phi : G \rightarrow H$ be a morphism of k -groups and $K = \text{Ker}(\phi)$. One assumes that ϕ is covering for the (fpqc) topology (this will be the case, in particular, if ϕ is covering for a coarser topology, for example the (fppf) topology). Then, on the one hand, ϕ is a K -torsor over H (cf. IV 5.1.7.1). On the other hand (cf. IV 6.3.1), there exists a covering of H by affine open subsets S_i , and for each i an affine faithfully flat morphism $T_i \rightarrow S_i$ factoring through ϕ . Then $G \times_H T_i$ is T_i -isomorphic to $K \times T_i$, hence faithfully flat over T_i , and therefore, by (fpqc) descent, $G \times_H S_i \rightarrow S_i$ is faithfully flat, so that ϕ is faithfully flat.

⁽⁷⁴⁾**N.D.E.** This remark has been added.

⁽⁷⁵⁾**N.D.E.** The numbering 8.1.1 to 8.1.3 has been added to bring out the results stated there.

⁽⁷⁶⁾**N.D.E.** What follows, as well as the proof of the corollary below, has been expanded.

⁽⁷⁷⁾**N.D.E.** Moreover, according to VI A, 3.2, $G_p(\mathcal{L}_2)$ represents the (fppf) quotient sheaf of $G_p(\mathcal{L}_1)$ by $G_p(\mathcal{L}_0)$.

⁽⁷⁸⁾**N.D.E.** This remark, pointed out by O. Gabber, has been added; it will be useful in 8.3.1.

Conversely, if ϕ is faithfully flat and quasi-compact (resp. and locally of finite presentation), it is covering for the (fpqc) (resp. (fppf)) topology, cf. IV 6.3.1. Finally, recall that a sheaf morphism is covering if and only if it is an epimorphism, cf. IV 4.4.3. One therefore obtains, in particular, that a quasi-compact morphism of k -groups is faithfully flat if and only if it is an epimorphism of (fpqc) sheaves.

Proposition 8.2. *Consider an exact sequence⁽⁷⁹⁾ of algebraic groups over a field k of characteristic $p > 0$*

$$1 \longrightarrow G' \xrightarrow{\tau} G \xrightarrow{\pi} G'' \longrightarrow 1$$

and the following assertions:

- (i) *The morphism π is smooth.*
- (ii) *G' is smooth.*
- (iii) *For every integer $n > 0$, the sequence below, induced by τ and π , is exact:*

$$1 \longrightarrow {}_{\mathrm{Fr}^n} G' \longrightarrow {}_{\mathrm{Fr}^n} G \longrightarrow {}_{\mathrm{Fr}^n} G'' \longrightarrow 1.$$

- (iv) *The morphism ${}_{\mathrm{Fr}} \pi : {}_{\mathrm{Fr}} G \longrightarrow {}_{\mathrm{Fr}} G''$ is an epimorphism of (fppf) sheaves.*
- (v) *The morphism $\mathrm{Lie}(\pi) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(G'')$ is surjective.*

Then one has the implications (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Leftrightarrow$ (v), and all the assertions are equivalent when G is smooth over k .

Indeed, (i) is equivalent to (ii) according to VI B 9.2 (vii), and it is clear that (iii) implies (iv). On the other hand, the equivalence of (iv) and (v) follows from 8.1.3.

The implication (ii) $\Rightarrow$ (iii) follows from the diagram:

$$\begin{array}{ccccccc} 1 & \longrightarrow & G' & \xrightarrow{\tau} & G & \xrightarrow{\pi} & G'' \longrightarrow 1 \\ & & \downarrow \mathrm{Fr}^n(G'/k) & & \downarrow \mathrm{Fr}^n(G/k) & & \downarrow \mathrm{Fr}^n(G''/k) \\ 1 & \longrightarrow & G'^{(p^n)} & \xrightarrow{\tau^{(p^n)}} & G^{(p^n)} & \xrightarrow{\pi^{(p^n)}} & G''^{(p^n)} \longrightarrow 1 \end{array}$$

whose two rows are exact: since $\mathrm{Fr}^n(G'/k)$ is an epimorphism of (fppf) sheaves according to corollary 8.3.1 below, π induces an epimorphism from ${}_{\mathrm{Fr}^n} G$ onto ${}_{\mathrm{Fr}^n} G''$ (generalize the snake lemma to sheaves in groups which are not necessarily commutative).

Similarly, when G is smooth over k , $\mathrm{Fr}(G/k)$ is an epimorphism, hence if in addition ${}_{\mathrm{Fr}} \pi$ is an epimorphism, the same snake lemma applied to the diagram above for $n = 1$ shows that $\mathrm{Fr}(G'/k)$ is an epimorphism, hence that G' is smooth over k , according to 8.3.1 below.

Proposition 8.3. *If G is a group locally of finite type⁽⁸⁰⁾ over a field k of characteristic $p > 0$, there exists an integer n_0 such that $G/({}_{\mathrm{Fr}^n} G)$ is smooth over k for $n \geq n_0$.*

Since the construction of $G/({}_{\mathrm{Fr}^n} G)$ commutes with extension of the base field (4.1.1 and VI A, 3.3.2), we may suppose that k is perfect. In this case G_{red} is a k -group locally of finite type (cf. VI A 0.2), and one has the following commutative and exact diagram, where H denotes the k -scheme $G_{\mathrm{red}} \backslash G$:

$$\begin{array}{ccccccc} 1 & \longrightarrow & G_{\mathrm{red}} & \longrightarrow & G & \longrightarrow & H \\ & & \downarrow \mathrm{Fr}^n(G_{\mathrm{red}}/k) & & \downarrow \mathrm{Fr}^n(G/k) & & \downarrow \mathrm{Fr}^n(H/k) \\ 1 & \longrightarrow & G_{\mathrm{red}}^{(p^n)} & \longrightarrow & G^{(p^n)} & \longrightarrow & H^{(p^n)}. \end{array}$$

⁽⁷⁹⁾**N.D.E.** i.e. π is faithfully flat and i is an isomorphism from G' onto $\mathrm{Ker} \pi$, so that G'' represents the (fppf) quotient sheaf of G by G' , cf. VI A, 3.2 and 5.2.

⁽⁸⁰⁾**N.D.E.** “Algebraic” has been replaced by “locally of finite type”.

Now H is the spectrum of a finite local k -algebra with residue field k (cf. VI A, 5.6.1). Consequently, there exists an integer n_0 such that, for every $n \geq n_0$, $\mathrm{Fr}^n(H/k)$ factors through the “unit” section of $H^{(p^n)}$. It follows that, for $n \geq n_0$, $\mathrm{Fr}^n(G/k)$ factors through $G_{\mathrm{red}}^{(p^n)}$, and hence, according to VI A, 5.4.1, there is a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{\mathrm{Fr}^n(G/k)} & G_{\mathrm{red}}^{(p^n)} \\ \pi \downarrow & \nearrow i & \\ G/(\mathrm{Fr}^n G) & & \end{array}$$

where i is a closed immersion (and π the canonical projection). Since, in addition, i induces a homeomorphism of the underlying topological spaces, it is therefore an isomorphism. Since k is perfect, $G_{\mathrm{red}}^{(p^n)}$ is smooth over k (VI A, 1.3.1), and hence $G/(\mathrm{Fr}^n G)$ is smooth over k , for every $n \geq n_0$.

Corollary 8.3.1. *Let G be a group locally of finite type⁽⁸⁰⁾ over a field k of characteristic $p > 0$, and let n be an integer ≥ 1 .⁽⁸¹⁾ The following conditions are equivalent:*

- (i) G is smooth over k .
- (ii) $\mathrm{Fr}^n(G/k) : G \rightarrow G^{(p^n)}$ is an epimorphism of (fppf) sheaves.
- (iii) $\mathrm{Fr}^n(G/k) : G \rightarrow G^{(p^n)}$ is faithfully flat.

First, since G is locally of finite type over k , $\mathrm{Fr}^n(G/k)$ is of finite presentation, so the equivalence of (ii) and (iii) follows from 8.1.4. Suppose G smooth over k , hence G reduced. Then, since $\mathrm{Fr}^n(G/k)$ is surjective, it is faithfully flat (cf. VI A, 6.2 or VI B, 1.3).

Conversely, suppose $\mathrm{Fr}^n(G/k)$ faithfully flat. Since $\mathrm{Fr}^n(G^{(p^n)}/k)$ is deduced from $\mathrm{Fr}^n(G/k)$ by base change (cf. 4.1.3), it too is faithfully flat, as is the composite:

$$\mathrm{Fr}^{2n}(G/k) : G \rightarrow G^{(p^n)} \rightarrow G^{(p^{2n})}.$$

One thus obtains that, for every $m \in \mathbf{N}$, $\mathrm{Fr}^{mn}(G/k) : G \rightarrow G^{(p^{mn})}$ is faithfully flat, hence induces an isomorphism $G/(\mathrm{Fr}^{mn} G) \simeq G^{(p^{mn})}$ (cf. VI A, 5.4.1). Now, according to proposition 8.3, $G^{(p^{mn})}$ is smooth over k for large m , hence so is G , by (fpqc) descent (cf. EGA IV₄, 17.7.1).

8.4. — In the two statements which conclude this exposé, we return to the case of a field k of arbitrary characteristic.

When k is of characteristic 0 (resp. $p > 0$), let n be an integer ≥ 1 (resp. an integer ≥ 1 prime to p); in both cases, we say simply that n is prime to the characteristic of k . Moreover, if G is a group scheme over k , we denote by $n_G : G \rightarrow G$ the morphism of k -schemes which sends an element x of $G(T)$ to $x^n \in G(T)$, as T ranges over k -schemes.

Proposition. *Let G be an algebraic group over a field k and n an integer prime to the characteristic of k . Then $n_G : G \rightarrow G$ is an étale morphism.*

⁽⁸²⁾ According to VI B 1.3, it suffices to show that n_G is étale at the origin. Let A be the local ring of G at the origin and I the maximal ideal of A . According to II 3.9.4, the map $\mathrm{Lie}(n_G) : \mathrm{Lie}(G) \rightarrow \mathrm{Lie}(G)$ induced by n_G is the homothety of ratio n . It is therefore an isomorphism, as is the endomorphism induced by n_G on $I/I^2 = \mathrm{Lie}(G)^*$. If k is of characteristic 0, G is smooth over k (VI B 1.6.1, see also VII B 3.3.1), so the canonical morphism $S(I/I^2) \rightarrow \mathrm{gr}_I(A)$ is an isomorphism, where $\mathrm{gr}_I(A)$ denotes the graded object associated to the I -adic filtration. It follows that n_G induces an automorphism of $\mathrm{gr}_I(A)$,

⁽⁸¹⁾**N.D.E.** The equivalence between (ii) and (iii) has been made explicit, and the proof expanded.

⁽⁸²⁾**N.D.E.** In the statement, “étale at the origin” has been changed to “étale”, and the following sentence added.

hence also of the completion $\hat{A}$ of A , and therefore n_G is étale at the origin (cf. EGA IV₄, 17.6.3).

If the characteristic is $p > 0$ and G is of height ≤ 1 , then A is isomorphic to the quotient of the symmetric algebra of $\omega_{G/k} = I/I^2$ by the ideal generated by the p -th powers of elements of $\omega_{G/k}$ (cf. 7.4); one can then apply the “same” argument as in characteristic 0, and one obtains that n_G induces an automorphism of A .

If G is of height $\leq r$ and we suppose our assertion proved for groups of height $\leq r - 1$, denote by B, A, A' the affine algebras of ${}_{\text{Fr}}G$, G , and $G' = {}_{\text{Fr}}G \setminus G$, and by $n_B, n_A, n_{A'}$ the endomorphisms of B, A, A' induced by $n_{{}_{\text{Fr}}G}$, n_G , and $n_{G'}$.⁽⁸³⁾ Let $I' = I \cap A$ be the maximal ideal of A' ; since there is a cartesian square

$$\begin{array}{ccc} {}_{\text{Fr}}G & \longrightarrow & G \\ \downarrow & & \downarrow \\ e & \longrightarrow & G' \end{array}$$

one has $B = A/I'A$. Observe that $n_{A'}$ (resp. n_B) is none other than the endomorphism induced by n_A on A' (resp. on B). According to VI A 3.2, A is a faithfully flat A' -module, and since A' is a local artinian ring (G' being an algebraic k -group of height $\leq r - 1$), it follows that A is a free A' -module. Since the restriction of n_A to A' is $n_{A'}$, which is an isomorphism by the induction hypothesis, it follows from Nakayama’s lemma that n_A will be an automorphism if the endomorphism it induces on $A/I'A$ is one. Now this endomorphism is none other than n_B , which is an automorphism since B is of height ≤ 1 . Thus n_A is an automorphism.

Finally, when G is an arbitrary algebraic group over a field of characteristic $p > 0$, what precedes shows that n_G induces automorphisms of the k -schemes ${}_{\text{Fr}}^r G$; these schemes are affine over k and have as algebras the quotients of the local algebra A by the ideal $I^{\{p^r\}}$ generated by the p^r -th powers of elements of I . Since n_G defines automorphisms of the algebras $A/I^{\{p^r\}}$, one sees, by passage to the projective limit, that n_G induces an automorphism of $\hat{A}$, hence n_G is étale at the origin (EGA IV₄, 17.6.3).

Proposition 8.5. *Let G be a finite algebraic group of rank n over the field k . Then $n_G : G \rightarrow G$ is the zero morphism of G .*

Let us point out at once the following corollary, obtained by combining 8.4 and 8.5.⁽⁸⁴⁾

Corollary 8.5.1. *Let G be a finite algebraic group of rank n over the field k . If n is prime to the characteristic of k , then G is étale over k .*

Let us now prove 8.5. Let H be a normal subgroup of G of rank m over k . Denote by $\lambda : H \times G \rightarrow G$ the morphism induced by the multiplication of G . Then, with the notation of VI A 3.2, there is a cartesian square:

$$\begin{array}{ccc} H \times G & \xrightarrow{\lambda} & G \\ \text{pr}_2 \downarrow & & \downarrow \pi \\ G & \xrightarrow{\pi} & H \setminus G \end{array}$$

Since $\pi : G \rightarrow H \setminus G$ is faithfully flat and quasi-compact (VI A 3.2), and pr_2 is locally free of rank m , it follows from EGA IV₂, 2.5.2 that $G \rightarrow H \setminus G$ is locally free of rank m . Denoting $\text{rg}_k(H \setminus G)$ by r , one therefore has $n = \text{rg}_k(G) = rm$.

⁽⁸³⁾**N.D.E.** The original has been expanded in what follows.

⁽⁸⁴⁾**N.D.E.** This corollary, indicated implicitly in the original by “(confer 8.4)”, has been added. For another proof of the corollary, not using 8.5, see for example [T070], Lemma 5.

On the other hand, one has an exact sequence of “abstract” groups

$$1 \longrightarrow H(T) \longrightarrow G(T) \longrightarrow (H \backslash G)(T)$$

for every k -scheme T ; it is therefore clear that n_G is zero if m_H and $r_{H \backslash G}$ are. Taking for H the neutral component G^0 of G , $G^0 \backslash G$ is étale (cf. VI A 5.5.1), so that one may suppose G étale over k or infinitesimal (cf. 7.0).

If G is étale, one reduces, by extension of the base field, to the case where k is algebraically closed. In this case, G is a constant group (cf. I 4.1), and the statement is classical.

If G is infinitesimal and nonzero, k is necessarily of characteristic $p > 0$ (cf. VI B 1.6.1 or VII B 3.3.1); the subgroups ${}_{\mathbb{F}_p^n} G$ then form a composition series of G whose quotients are of height ≤ 1 .

This reduces us to the case where G is of height ≤ 1 . Let A (resp. L) be the affine algebra (resp. Lie algebra) of G , and $U = U_p(L)$. According to 7.4, one has $G = G_p(L)$, whence $A = U^*$; thus, if $\dim_k L = r$, the rank of G over k is p^r (cf. 5.3.3). We shall therefore study the morphism $p_G : G \longrightarrow G$ defined by raising to the power p ; it induces an endomorphism p_A of A and, by duality, an endomorphism p_U of U .

Let I be the augmentation ideal of A ; we shall show that $p_A(I) \subset I^p$. Assuming this established, one will therefore have $p_A^r(I) \subset I^{p^r}$. On the other hand, one knows that $I^{r(p-1)+1} = 0$ (since I is generated by r elements with zero p -th power). Since $p^r > r(p-1)$, it follows that $p_A^r(I) = 0$, hence p_G^r is the zero morphism. It therefore remains to show the assertion:

$$(*) \quad p_A(I) \subset I^p.$$

For every integer $s \geq 1$, we shall denote by $m_A^{s-1} : A^{\otimes s} \longrightarrow A$ (resp. $\Delta_U^s : U \longrightarrow U^{\otimes s}$) the map induced by the multiplication m_A of A (resp. the comultiplication Δ_U of U). Then p_A is equal to the following composite:

$$A \xrightarrow{\Delta_A^{p-1}} A^{\otimes p} \xrightarrow{m_A^{p-1}} A,$$

and since the transpose of m_A (resp. Δ_A) is Δ_U (resp. m_U), the endomorphism $p_U = {}^t p_A$ of U is the composite below:

$$U \xrightarrow{\Delta_U^{p-1}} U^{\otimes p} \xrightarrow{m_U^{p-1}} U.$$

(85) Let J be the augmentation ideal of U ; one has $U = k1_U \oplus J$, and we shall denote by π the projection $U \longrightarrow J$ with kernel $k1_U$. For every integer $s \geq 1$, denote by $(I^s)^\perp$ the orthogonal of I^s in $A^* = U$, i.e. $(I^s)^\perp$ is the set of $u \in U$ such that the composite below is zero:

$$I^{\otimes s} \xrightarrow{m_A^{s-1}} I \xrightarrow{u} k.$$

Since the transpose of m_A is Δ_U , one sees that $(I^s)^\perp$ is the vector subspace P_{s-1} consisting of the $u \in U$ such that $\Delta_U^{s-1}(u)$ vanishes on $I^{\otimes s}$, i.e. denoting by $\overline{\Delta_U^{s-1}}$ the composite of Δ_U^{s-1} and the projection $\pi^{\otimes s} : U^{\otimes s} \longrightarrow J^{\otimes s}$, one obtains that

$$(I^s)^\perp = P_{s-1} = \text{Ker } \overline{\Delta_U^{s-1}}$$

(see also VII B, 1.3.6). Thus, to show assertion (*), one must show that the transpose map $p_U = {}^t p_A$ maps P_{p-1} into $I^\perp = k1_U$. Since $p_U(1_U) = 1_U$, it suffices to show the assertion below, where P_{p-1}^+ denotes $J \cap P_{p-1}$:

$$(**) \quad p_U(P_{p-1}^+) = 0.$$

On the other hand, one shows easily, by induction on s , that P_{s-1}^+ is the vector subspace of U generated by the products $x_1 \cdots x_t$, with $1 \leq t \leq s-1$ and $x_i \in L$ (cf. VII B 4.3). Now, if

(85) **N.D.E.** The original has been expanded in the following paragraph.

$x_1, x_2, \dots, x_t$ are elements of L , one has:

$$p_U(x_1 x_2 \cdots x_t) = m_U^{p-1} \left(\prod_{j=1}^t \sum_{i=1}^p 1 \otimes \cdots \otimes \overset{i}{\underset{j}{x}} \otimes \cdots \otimes 1 \right).$$

It is clear that the expression $\prod_j \sum_i 1 \otimes \cdots \otimes x_j \otimes \cdots \otimes 1$ is a sum of p^t terms x_h indexed by the maps h from $\{1, \dots, t\}$ to $\{1, \dots, p\}$.⁽⁸⁶⁾ Such a map h defines an *ordered partition* $\mathbf{p}_h$ of $\{1, \dots, t\}$ into at most p parts. Indeed, denote by $i_1 < \cdots < i_r$ the elements of the image of h , and, for $s = 1, \dots, r$, put $I_s = h^{-1}(i_s)$ and $x_{I_s} = \prod_{j \in I_s} x_j$, the product being taken in increasing order. Then h corresponds to the p -tensor

$$1 \otimes \cdots \otimes x_{I_1} \otimes \cdots \otimes x_{I_r} \otimes \cdots \otimes 1$$

(where each x_{I_s} is in position i_s), and its image under m_U^p is the product:

$$x_{I_1} \otimes \cdots \otimes x_{I_r}$$

which depends only on the ordered partition $\mathbf{p} = (I_1, \dots, I_r)$, and which will be denoted by $x_{\mathbf{p}}$. For fixed $\mathbf{p}$, $x_{\mathbf{p}}$ is obtained for all choices of $i_1 < \cdots < i_r$ in $\{1, \dots, p\}$, and one therefore obtains the equality

$$p_U(x_1 x_2 \cdots x_t) = \sum_{\mathbf{p}} \binom{p}{n(\mathbf{p})} x_{\mathbf{p}},$$

where $\mathbf{p}$ ranges over the set of ordered partitions of $\{1, \dots, t\}$ into at most p parts, and $n(\mathbf{p})$ denotes the number of parts of $\mathbf{p}$. (One has $1 \leq n(\mathbf{p}) \leq \min(t, p)$.) When $t < p$, all the terms $\binom{p}{n(\mathbf{p})}$ are therefore zero, so that $p_U(x_1 \cdots x_t) = 0$. Thus p_U vanishes on P_{p-1}^+ , which proves assertion (**), hence (*), and completes the proof of 8.5.

Corollary 8.5.2. ⁽⁸⁷⁾ *Let S be a reduced scheme and G a finite locally free S -group of rank n . Then $n_G : G \rightarrow G$ is the zero morphism of G .*

Indeed, let S' be the sum of the $\text{Spec } \mathcal{O}_{S, \eta}$, as η ranges over the maximal points of S . Since S is reduced, the morphism $S' \rightarrow S$ is schematically dominant, and the same is true of the morphism $f : G_{S'} \rightarrow G$, since G is finite locally free over S (cf. EGA IV₃, 11.10.5). Since $G \rightarrow S$ is affine, hence separated, the coincidence locus of n_G and the zero morphism is a closed subscheme of G ; now it contains f according to 8.5, hence equals G , i.e. n_G is the zero morphism.

Remark 8.5.3. Let us also point out that, according to a theorem of P. Deligne (see [TO70], p. 4), if G is a finite locally free *commutative* S -group of rank n over an arbitrary base S , then $n_G = 0$.

⁽⁸⁶⁾**N.D.E.** The original has been expanded in what follows, replacing the notion of preorder by the equivalent notion of ordered partition.

⁽⁸⁷⁾**N.D.E.** This corollary, mentioned in Exp. VIII, Remark 7.3.1, has been added.

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Exposé VIIB. Infinitesimal study of group schemes

by P. Gabriel

B) Formal groups

⁽⁰⁾The study of formal groups is usually extremely simple. If this does not appear clearly in the pages which follow, the responsibility lies with an arithmetician who claims to know formal groups over “something other than fields”.⁽¹⁾ We have therefore developed, for formal groups “locally free over projective limits of artinian rings”, the generalities usually stated for formal groups defined over a field. For a more detailed study of the latter, we refer to the 1964/65 Heidelberg–Strasbourg seminar in algebraic geometry.⁽²⁾

0. Recollections on pseudocompact rings and modules

This section contains some technical preliminaries; in it we recall and supplement some results of [CA] (*Des catégories abéliennes*, Bull. Soc. Math. France **90**, 1962).

0.1. — A *left pseudocompact ring* is a separated and complete topological ring with identity which has a base of neighborhoods of 0 consisting of left ideals $\mathfrak{l}$ of finite colength (i.e. $\text{length}_A(A/\mathfrak{l}) < +\infty$). Here we shall suppose that A is commutative, so that there is no need to distinguish “between left and right”. In particular, the quotients $A/\mathfrak{l}$ are artinian rings and A is identified with the topological projective limit of these rings equipped with the discrete topology. A complete noetherian local ring $(A, \mathfrak{m})$ is evidently pseudocompact.⁽³⁾

0.1.1. — Every closed ideal I of A is the intersection of the open ideals containing it.⁽⁴⁾ Every maximal closed ideal is therefore open. Moreover, if $\mathfrak{l}$ is an open ideal of A , the maximal ideals of $A/\mathfrak{l}$ correspond bijectively to the maximal ideals $\mathfrak{m}$ containing $\mathfrak{l}$; the latter are therefore open and closed. Consequently, every maximal closed ideal is an open maximal ideal (and hence closed); the converse is evident. The set of these ideals will be denoted by $\Upsilon(A)$. If $\mathfrak{l}$ is an open ideal of A and $\mathfrak{m} \in \Upsilon(A)$, the localization $(A/\mathfrak{l})_{\mathfrak{m}}$ is thus a local ring if $\mathfrak{m}$ contains $\mathfrak{l}$, and is zero otherwise. Since $A/\mathfrak{l}$ is artinian, it is the direct product of finitely many local rings, which one can write

$$A/\mathfrak{l} \simeq \prod_{\mathfrak{m} \in \Upsilon(A)} (A/\mathfrak{l})_{\mathfrak{m}}.$$

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** The interest of formal groups over a complete noetherian local ring appears, for example, in the work of Lubin and Tate (cf. [LT65]). The study of formal groups over an arbitrary base, and of questions of lifting and deformation, in particular for Barsotti–Tate groups (“ p -divisible groups”), has given rise to an abundant literature; cf. for example [LT66, Ta67, Gr74, Me72, La75, Fo77, Il85, Br00]; in particular, the results of the present exposé are largely taken up in Chapter I of [Fo77].

⁽²⁾**N.D.E.** The editors found only such a seminar dated 1965/66 and entitled “Groupes algébriques linéaires”, in which the notion of a formal group does not appear; see, however, [De72].

⁽³⁾**N.D.E.** (when it is equipped with the $\mathfrak{m}$ -adic topology)

⁽⁴⁾**N.D.E.** Indeed, if $x \notin I$, there exists an open ideal $\mathfrak{l}$ such that $(x + \mathfrak{l}) \cap I = \emptyset$; then $I + \mathfrak{l}$ is an open ideal not containing x . Moreover, in what follows we have made explicit the fact that every “maximal closed” ideal is maximal and open.

From this one obtains “canonical” isomorphisms

$$\begin{aligned} A &\simeq \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} \prod_{\mathfrak{m}} (A/\mathfrak{l})_{\mathfrak{m}} \\ &\simeq \prod_{\mathfrak{m}} \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_{\mathfrak{m}} \simeq \prod_{\mathfrak{m}} A_{\mathfrak{m}}, \end{aligned}$$

where one has put

$$A_{\mathfrak{m}} = \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_{\mathfrak{m}}.$$

This *local component* $A_{\mathfrak{m}}$ is a filtered projective limit of artinian local rings equipped with the discrete topology; it is therefore a local ring which is pseudocompact for the projective limit topology.⁽⁵⁾

0.1.2. — Let $\mathfrak{r}(A)$ be the intersection of the open maximal ideals of A , that is, the cartesian product of the ideals $\mathfrak{m}A_{\mathfrak{m}}$ when A is identified with $\prod_{\mathfrak{m}} A_{\mathfrak{m}}$. For every open ideal $\mathfrak{l}$ of A , the image of $\mathfrak{r}(A)$ in $A/\mathfrak{l}$ is contained in the radical of $A/\mathfrak{l}$. Some power of this image is therefore zero, so that $\mathfrak{r}(A)^n$ is contained in $\mathfrak{l}$ when n is sufficiently large. The sequence of the $\mathfrak{r}(A)^n$ therefore tends to 0. The same holds for the sequence of the x^n when x belongs to $\mathfrak{r}(A)$. In other words, every element of $\mathfrak{r}(A)$ is topologically nilpotent, and the converse is clear. It follows that the sequence with general term $1 + x + \cdots + x^n$ converges, and converges to $1/(1 - x)$ when $x \in \mathfrak{r}(A)$. This shows that $\mathfrak{r}(A)$ is the Jacobson radical of A , i.e. the intersection of *all* the maximal ideals of A (cf. Bourbaki, *Algèbre*, Chap. 8, § 6, Th. 1).⁽⁶⁾

Remarks. ⁽⁷⁾

- (a) If $\mathfrak{p}$ is an *open* prime ideal of A , then, since $A/\mathfrak{p}$ is artinian, $\mathfrak{p}$ is a maximal ideal. Consequently, $\Upsilon(A)$ is the set of open prime ideals of A .
- (b) Each $\mathfrak{m}A_{\mathfrak{m}}$ is an *ideal of definition* of $A_{\mathfrak{m}}$, i.e. an *open* ideal I such that the sequence of the I^n tends to 0 (cf. EGA 0_I, 7.1.2). Consequently, $\text{Spec}(A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}})$, equipped with the topological ring $A_{\mathfrak{m}}$, is an *affine formal scheme* in the sense of EGA I, 10.1.2.
- (c) The topological ring A is *admissible* in the sense of EGA 0_I, 7.1.2 if and only if $\mathfrak{r}(A)$ is open (hence an ideal of definition), and this holds if and only if $\Upsilon(A)$ is *finite*. In this case the affine formal scheme $\text{Spf}(A) = \text{Spec}(A/\mathfrak{r}(A))$ (cf. EGA I, 10.1.2) has $\Upsilon(A)$, equipped with the discrete topology, as its underlying space, and the ring of sections of the structure sheaf over a subset E of $\Upsilon(A)$ is the product $\prod_{\mathfrak{m} \in E} A_{\mathfrak{m}}$.
- (d) Let A be an arbitrary pseudocompact ring. In 1.1 below, the space $\Upsilon(A)$ is equipped with the discrete topology and with the sheaf of rings whose ring of sections over any subset E is $\prod_{\mathfrak{m} \in E} A_{\mathfrak{m}}$. By b), every point then has an open neighborhood which is an affine formal scheme, so this defines a *formal scheme* (EGA I, 10.4.2), which will be denoted by $\text{Spf}(A)$. (For this formal scheme to be affine, it must be quasi-compact, hence $\Upsilon(A)$ must be finite, and in this case $\text{Spf}(A)$ coincides with the definition of EGA I, 10.1.2.)

0.1.3. — If A and B are two pseudocompact rings, a homomorphism from A to B is, by definition, a *continuous* map compatible with addition, multiplication, and the identity

⁽⁵⁾**N.D.E.** Observe that $A_{\mathfrak{m}}$ is the localization of A at the maximal ideal $\mathfrak{m}$. Indeed, the identity element $e_{\mathfrak{m}}$ of $A_{\mathfrak{m}}$ is an idempotent of A not belonging to $\mathfrak{m}$, and since $A = A_{\mathfrak{m}} \times B$, where $B = A(1 - e_{\mathfrak{m}})$, $A_{\mathfrak{m}}$ is identified with the localization $A_{e_{\mathfrak{m}}}$, and hence also with the localization $S^{-1}A$, where $S = A - \mathfrak{m}$. Moreover, since $e_{\mathfrak{m}}(1 - e_{\mathfrak{m}}) = 0$, $\mathfrak{m}$ contains $1 - e_{\mathfrak{m}}$, hence also B , and thus $\mathfrak{m}$ is identified with $\mathfrak{m}A_{\mathfrak{m}} \times B$.

⁽⁶⁾**N.D.E.** Indeed, let $x \in \mathfrak{r}(A)$; if $\mathfrak{m}$ is a maximal ideal not containing x , there exists $y \in A$ such that $1 - xy \in \mathfrak{m}$, whereas $xy \in \mathfrak{r}(A)$ and hence $1 - xy$ is invertible, a contradiction. Note the following consequence: if $\Upsilon(A)$ is a finite set $\{\mathfrak{m}_1, \dots, \mathfrak{m}_r\}$, the $\mathfrak{m}_i$ are all the maximal ideals of A .

⁽⁷⁾**N.D.E.** We have added these remarks in order to compare the definition of the formal spectrum $\text{Spf}(A)$ given in 1.1 with those of EGA I, 10.1.2 and 10.4.2.

elements. Such a homomorphism sends a topologically nilpotent element of A to a topologically nilpotent element of B ; it therefore maps the radical $\mathfrak{r}(A)$ of A into the radical $\mathfrak{r}(B)$ of B .

0.2. — Let A be a (commutative) pseudocompact ring. A *pseudocompact A -module* M is a separated and complete topological A -module which has a base of neighborhoods of 0 consisting of submodules M' such that M/M' has finite length over A . If M and N are two pseudocompact A -modules, a morphism from M to N is by definition a continuous A -linear map. The group of these morphisms will be denoted by $\mathrm{Hom}_c(M, N)$.

Proposition 0.2.B. ⁽⁸⁾

- (i) *Pseudocompact A -modules form an abelian category, which will be denoted by $\mathbf{PC}(A)$. (In particular, for every morphism $f : M \rightarrow N$, $\mathrm{Im}(f)$ is a complete submodule, hence closed in N .)*
- (ii) *Pseudocompact A -modules of finite length (whose topology is therefore discrete) form a system of cogenerators of $\mathbf{PC}(A)$.*
- (iii) *Infinite products and filtered projective limits are exact, i.e. $\mathbf{PC}(A)$ satisfies axiom $(\mathrm{AB5}^*)$.* ⁽⁹⁾

For the reader's convenience, let us briefly indicate the steps of the proof. First, one has the following lemma ([CA] § IV.3, Lemma 1; for the proof, see [BEs], III, § 7.4, Th. 1 and Example 2):

Lemma 0.2.C. *Let B be a ring, I a filtered ordered set, and (M_i) and (N_i) two projective systems of left B -modules indexed by I . Let (s_i) be a morphism of projective systems $(M_i) \rightarrow (N_i)$ such that s_i is surjective and has artinian kernel for every i . Then the map*

$$\varprojlim s_i : \varprojlim M_i \rightarrow \varprojlim N_i$$

is surjective.

Corollary 0.2.D. ([CA] § IV.3, Prop. 10 & 11). *Let M be a pseudocompact A -module.*

- (i) *Let K be a closed submodule of M . Then M/K , equipped with the quotient topology, is a pseudocompact A -module.*
- (ii) *Let (M_i) be a decreasing filtered family of closed submodules of M .*
 - (a) *The canonical map $M \rightarrow \varprojlim M/M_i$ is surjective and has kernel $\bigcap_i M_i$.*
 - (b) *For every closed submodule N of M , one has $N + \bigcap_i M_i = \bigcap_i (N + M_i)$.*

PROOF. Let (L_j) be the decreasing filtered family of open submodules of M . Equip M/K with the quotient topology, i.e. a base of neighborhoods of 0 consists of the open submodules $(K + L_j)/K$. Since K is closed, it equals the intersection of the $K + L_j$, so the map

$$\tau : M/K \rightarrow \varprojlim_j M/(K + L_j)$$

is injective. It is also open, the right-hand term being the projective limit of the discrete modules $M/(K + L_j)$. Moreover, for each j , the map $t_j : M/L_j \rightarrow M/(K + L_j)$ is surjective, with artinian kernel, so by the preceding lemma the map t in the commutative diagram below is surjective:

$$\begin{array}{ccc} M & \xrightarrow[\sim]{p} & \varprojlim_j M/L_j \\ \downarrow & & \downarrow t \\ M/K & \xrightarrow{\tau} & \varprojlim_j M/(K + L_j). \end{array}$$

⁽⁸⁾**N.D.E.** We have brought out the results of this paragraph in the following proposition, and have then indicated the steps of the proof; cf. [CA], § IV.3 or [DG70], § V.2.

⁽⁹⁾**N.D.E.** Cf. [Gr57], I § 1.5 and Prop. 1.8; one may also consult, for example, [Po73], § 2.8 or [We95], Append. A.4.

Since p is an isomorphism because M is complete, it follows that τ is surjective, hence an isomorphism. This proves (i). Let us prove (ii)(a). By the foregoing, for every i one has an isomorphism $M/M_i \xrightarrow{\sim} \varprojlim_j M/(M_i + L_j)$, and thus both horizontal arrows in the commutative diagram below are isomorphisms:

$$\begin{array}{ccc} M & \xrightarrow[p \sim]{} & \varprojlim_j M/L_j \\ \downarrow g & & \downarrow s \\ \varprojlim_i M/M_i & \xrightarrow[\sim]{} & \varprojlim_{i,j} M/(M_i + L_j). \end{array}$$

Moreover, for each j , the family of submodules $(M_i + L_j)/L_j$ has a least element, since M/L_j is artinian; hence the morphism $s_j : M/L_j \rightarrow \varprojlim_i M/(L_j + M_i)$ is surjective; therefore, by the preceding lemma, s is surjective. It follows that g is surjective. Finally, the kernel of g is the projective limit of the M_i , i.e. their intersection. This proves (a). Let us deduce (b). Since N is a closed submodule (hence separated and complete), it is a pseudocompact module for the topology induced by that of M . Hence, by (a), the morphisms f and g in the commutative exact diagram below are surjective:

$$\begin{array}{ccccccc} 0 & \longrightarrow & N & \longrightarrow & M & \longrightarrow & M/N \longrightarrow 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h \\ 0 & \longrightarrow & \varprojlim_i N/(N \cap M_i) & \longrightarrow & \varprojlim_i M/M_i & \longrightarrow & \varprojlim_i M/(N + M_i) \end{array}.$$

Then, by the “snake lemma”, the sequence $0 \rightarrow \text{Ker } f \rightarrow \text{Ker } g \rightarrow \text{Ker } h \rightarrow 0$ is exact, and the equality $N + \bigcap_i M_i = \bigcap_i (N + M_i)$ follows. $\square$

One can now prove Proposition 0.2.B. Let $f : M \rightarrow N$ be a morphism of pseudocompact A -modules. Then $K = \text{Ker}(f)$ is a closed submodule of M , hence separated and complete, so K is a pseudocompact module for the topology induced by that of M . By 0.2.D(i), M/K equipped with the quotient topology is pseudocompact. Let us show that the continuous bijective morphism $M/K \rightarrow \text{Im}(f)$ is bicontinuous. Identifying M/K with $\text{Im}(f)$, one has to show that the quotient topology $\mathcal{Q}$ is finer than the topology $\mathcal{T}$ induced by that of N . Denote by (L_j) (resp. (N_i)) the decreasing filtered set of open submodules of M (resp. N), and put $N'_i = N_i \cap \text{Im}(f)$. Let $P = (K + L_j)/K$ be a submodule of M/K open for $\mathcal{Q}$. Since $M/(K + L_j)$ is artinian, the family $N'_i + P$ has a least element $N'_{i_0} + P$. Since the N'_i are open, hence closed, for $\mathcal{T}$ and thus also for $\mathcal{Q}$, it follows from 0.2.D(ii)(b) that

$$N'_{i_0} + P = \bigcap_i (N'_i + P) = P + \bigcap_i N'_i = P,$$

hence $N'_{i_0} \subset P$. This shows that P is open for $\mathcal{T}$, and $M/K \rightarrow \text{Im}(f)$ is therefore an isomorphism of pseudocompact modules. In particular, $\text{Im}(f)$ is complete for $\mathcal{T}$, hence closed in N . Then, by 0.2.D(i) again, $\text{Coker}(f)$ equipped with the quotient topology is pseudocompact. This proves that $\mathbf{PC}(A)$ is an abelian category. Moreover, arbitrary projective limits exist in $\mathbf{PC}(A)$: if (M_i) is a projective system of pseudocompact modules, the projective limit of the M_i has as its underlying module the projective limit of the underlying modules, and as its topology the projective limit topology. Furthermore, if one has a family of exact sequences in $\mathbf{PC}(A)$:

$$0 \longrightarrow K_i \longrightarrow M_i \longrightarrow Q_i \longrightarrow 0,$$

then the sequence $0 \rightarrow \prod_i K_i \rightarrow \prod_i M_i \rightarrow \prod_i Q_i \rightarrow 0$ is exact. Part (iii) of 0.2.B follows, since in any abelian category in which arbitrary products exist, conditions (a) and (b) of 0.2.D are equivalent and equivalent to exactness of filtered projective limits (cf. [Mi65], III 1.2–1.9 or [Po73], Chap. 2, Th. 8.6). Finally, every pseudocompact module M is a submodule of the product $\prod_L M/L$, where L runs through the open submodules of M , so the objects of

finite length form a system of cogenerators of $\mathbf{PC}(A)$. (Moreover, every object of length n is isomorphic to a quotient A^n/L , where L is an open submodule of A^n of colength n ; these quotients therefore form a *set* of cogenerators.) This completes the proof of 0.2.B. ⁽¹⁰⁾Let $(\mathbf{Ab})$ be the category of abelian groups and $\mathbf{LF}(A)$ the full subcategory of $\mathbf{PC}(A)$ consisting of objects of *finite length*. For every object M of $\mathbf{PC}(A)$, denote by h_c^M the functor

$$\mathbf{LF}(A) \longrightarrow (\mathbf{Ab}), \quad N \longmapsto \mathrm{Hom}_c(M, N).$$

By [CA], § II.4, Th. 1, Lemma 4 and Cor. 1, one has the following results.⁽¹¹⁾

Proposition 0.2.E. *The functor $M \mapsto h_c^M$ is an anti-equivalence from $\mathbf{PC}(A)$ to the category $\mathbf{Lex}(\mathbf{LF}(A), (\mathbf{Ab}))$ of left exact functors $\mathbf{LF}(A) \rightarrow (\mathbf{Ab})$.*

Corollary 0.2.F.

- (i) *An object P of $\mathbf{PC}(A)$ is projective if and only if the functor h_c^P is exact (i.e. if and only if the functor $\mathrm{Hom}_c(P, -)$ is exact on $\mathbf{LF}(A)$).*
- (ii) *Let (M_i) be a filtered⁽¹²⁾ projective system of objects of $\mathbf{PC}(A)$. For every object N of $\mathbf{LF}(A)$, one has an isomorphism functorial in N :*

$$\mathrm{Hom}_c(\varprojlim M_i, N) \simeq \varinjlim \mathrm{Hom}_c(M_i, N).$$

- (iii) *Every filtered projective limit and every product⁽¹²⁾ of projective objects of $\mathbf{PC}(A)$ is a projective object of $\mathbf{PC}(A)$.*

Finally, one deduces from 0.2.F the following

Corollary 0.2.G. *Let $(M_i)_{i \in I}$ be a family of objects of $\mathbf{PC}(A)$. Then $\prod_{i \in I} M_i$ is projective if and only if every M_i is projective.*

Indeed, for every $N \in \mathrm{Ob} \mathbf{LF}(A)$, one has $\mathrm{Hom}_c(\prod_i M_i, N) \simeq \bigoplus_i \mathrm{Hom}_c(M_i, N)$.

0.2.1. — Each local component $A_{\mathfrak{m}}$ of A is a direct factor of A , hence a projective object of $\mathbf{PC}(A)$ (A is manifestly projective). Moreover, $A_{\mathfrak{m}}$ has $S_{\mathfrak{m}} = A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ as its unique simple quotient, hence is indecomposable. On the other hand, every simple object of $\mathbf{PC}(A)$ is isomorphic to a unique $S_{\mathfrak{m}}$. By [CA], IV § 3, Cor. 1 of Th. 3,⁽¹³⁾ one therefore has:

Proposition.

- (i) *Every projective object of $\mathbf{PC}(A)$ is a direct product of indecomposable projective objects, uniquely determined (up to isomorphism).*
- (ii) *Every indecomposable projective object is isomorphic to a unique $A_{\mathfrak{m}}$ ($\mathfrak{m} \in \Upsilon(A)$).*

Definition. A pseudocompact A -module M is said to be *topologically free* if it is isomorphic to the product of a family (A_i) of copies of A . In this case, a family (m_i) of elements of M is called a *pseudobasis* of M if the A -linear maps from A_i to M which send the identity element of A_i to m_i extend to an isomorphism from $\prod_i A_i$ to M .

⁽¹⁰⁾**N.D.E.** We have inserted in this paragraph Proposition 0.2.E and Corollary 0.2.F, which will be useful in 0.2.2. (In the original, these results appeared in 0.3.)

⁽¹¹⁾**N.D.E.** Indeed, $\mathbf{PC}(A)$ has a set of artinian cogenerators, filtered projective limits are exact in it, and $\mathbf{LF}(A)$ is the subcategory of artinian objects. The dual category $\mathbf{PC}(A)^0$ therefore has a set of noetherian generators, and filtered inductive limits are exact in it. By the proof of [CA] § II.4, Th. 1, the functor $M \mapsto \mathrm{Hom}_c(M, -)$ is an anti-equivalence from $\mathbf{PC}(A)$ to $\mathbf{Lex}(\mathbf{LF}(A), (\mathbf{Ab}))$. Similarly, Lemma 4 and Cor. 1 of *loc. cit.* state results “dual” to those of Corollary 0.2.F. For a “direct” proof, see [DG70], § V.2, Th. 3.1, Lemma 3.5, Cor. 3.3 & 3.4.

⁽¹²⁾**N.D.E.** Since every infinite product is a filtered projective limit of finite products, every additive functor which commutes with filtered projective limits also commutes with infinite products.

⁽¹³⁾**N.D.E.** This refers to the “dual” statements established in *loc. cit.*, § 2, Th. 2 and § 1, Prop. 2; for a “direct” proof, see [DG70], V, § 2, Th. 4.5 and Example 4.6 b).

0.2.2. — ⁽¹⁴⁾If M is a pseudocompact A -module, the A -module (without topology)

$$\mathrm{Hom}_c(M, A)$$

will be denoted by $M^\dagger$. Conversely, if N is an A -module, denote by $N^* = \mathrm{Hom}_A(N, A)$ its dual, equipped with the topology of pointwise convergence, i.e. a base of neighborhoods of 0 in N^* consists of the following submodules, where $x \in N$ and $\mathfrak{l}$ is an open ideal of A :

$$\mathcal{V}(x, \mathfrak{l}) = \{f \in N^* \mid f(x) \in \mathfrak{l}\}.$$

This makes N^* a pseudocompact A -module. Indeed, one first sees that if $N = A$, then $N^* = A$, equipped with its pseudocompact ring topology, and if N is a free module $A^{(I)}$, then N^* is the product A^I , equipped with the product topology. On the other hand, for every morphism $\phi : N_1 \rightarrow N_2$, the transpose morphism ${}^t\phi : N_2^* \rightarrow N_1^*$ is continuous, since the inverse image under ${}^t\phi$ of a submodule $\mathcal{V}(x_1, \mathfrak{l})$ of N_1^* is precisely the submodule $\mathcal{V}(\phi(x_1), \mathfrak{l})$ of N_2^* . Then, for arbitrary N , taking a presentation

$$A^{(J)} \xrightarrow{\phi} A^{(I)} \xrightarrow{\pi} N \rightarrow 0,$$

one sees that N^* is the kernel of the continuous morphism ${}^t\phi : A^I \rightarrow A^J$, so N^* is pseudocompact. When A is artinian (in which case one can take $\mathfrak{l} = 0$ above), one deduces from 0.2.F:

Proposition. *When A is artinian, the functors*

$$P \mapsto P^\dagger \quad \text{and} \quad Q \mapsto Q^*,$$

where P (resp. Q) is a projective object of $\mathbf{PC}(A)$ (resp. a projective A -module), establish an anti-equivalence between the category of projective pseudocompact A -modules and that of projective A -modules.⁽¹⁵⁾

In particular, when A is a field k , $P \mapsto P^\dagger$ is an anti-equivalence from the category of all pseudocompact k -modules (also called *linearly compact k -vector spaces*) to that of k -vector spaces.⁽¹⁶⁾

0.3. — ⁽¹⁷⁾Let L and M be two pseudocompact A -modules. The functor

$$\mathbf{LF}(A) \rightarrow (\mathrm{Ab}), \quad N \mapsto \mathrm{Bil}_c(L \times M, N),$$

where $\mathrm{Bil}_c(L \times M, N)$ denotes the abelian group of continuous A -bilinear maps from $L \times M$ to N , is left exact. By 0.2.E, there therefore exists a pseudocompact A -module $L \widehat{\otimes}_A M$, unique up to unique isomorphism, which represents this functor, i.e. such that one has a functorial isomorphism, for every object N of $\mathbf{LF}(A)$:

$$\mathrm{Hom}_c(L \widehat{\otimes}_A M, N) \simeq \mathrm{Bil}_c(L \times M, N).$$

Moreover, $L \widehat{\otimes}_A M$ is identified with the projective limit $P = \widehat{L \otimes_A M}$ of the (discrete) A -modules $(L/L') \otimes_A (M/M')$, where L' and M' run respectively through the open submodules of L and M . Indeed, let $\varphi : L \times M \rightarrow N$ be a continuous bilinear map from $L \times M$ to a (discrete) A -module N of finite length. By Lemma 0.3.1 below, there exist open submodules L' and M' of L and M such that $\varphi(L' \times M) = \varphi(L \times M') = \{0\}$. This means that the map

⁽¹⁴⁾**N.D.E.** We have detailed the results of this paragraph, which play an important role in what follows (cf. 1.2.3, 1.3.5, 2.2.1, etc.).

⁽¹⁵⁾**N.D.E.** Recall also that, over an *artinian* ring, a module is projective if and only if it is flat; see, for example, VI B, N.D.E. (54) or 0.3.7 below.

⁽¹⁶⁾**N.D.E.** Note that the *direct sum in $\mathbf{PC}(k)$* of a family $(V_i)_{i \in I}$ of linearly compact k -vector spaces is $(\prod_{i \in I} V_i^\dagger)^*$.

⁽¹⁷⁾**N.D.E.** We have modified this paragraph, taking into account the additions made in 0.2.

$\bar{\varphi} : L \otimes_A M \longrightarrow N$ induced by φ has the form $\varphi' \circ p$, where p is the canonical projection from $L \otimes_A M$ to $(L/L') \otimes_A (M/M')$. If one denotes by $\hat{\varphi}$ the composite

$$P \longrightarrow (L/L') \otimes_A (M/M') \xrightarrow{\varphi'} N,$$

one sees that the map $\varphi \longmapsto \hat{\varphi}$ is a functorial bijection from $\text{Bil}_c(L \times M, N)$ to $\text{Hom}_c(P, N)$, whence $P \simeq L \hat{\otimes}_A M$. The pseudocompact module $L \hat{\otimes}_A M$ is therefore the separated completion of $L \otimes_A M$, for the linear topology defined by the kernels of the canonical projections from $L \otimes_A M$ to $(L/L') \otimes_A (M/M')$, and will be called the *completed tensor product* of L and M . If x and y belong to L and M , the image of $x \otimes_A y$ in $L \hat{\otimes}_A M$ will be denoted by $x \hat{\otimes}_A y$.

Lemma 0.3.1. *Let L , M , and N be pseudocompact A -modules, N being of finite length. If $\varphi : L \times M \longrightarrow N$ is a continuous A -bilinear map, there exist open submodules L' and M' of L and M such that $\varphi(L' \times M) = \varphi(L \times M') = \{0\}$.*

Indeed, $\varphi^{-1}(0)$ is an open neighborhood of $(0, 0)$, hence contains an open set of the form $L_1 \times M_1$, where L_1 and M_1 are open submodules of L and M . Since L/L_1 has finite length, there exist elements $x_1, \dots, x_r$ of L such that $L_1 + Ax_1 + \dots + Ax_r = L$. If $M' \subset M_1$ is “sufficiently small”, one also has $\varphi(x_i, M') = 0$ for every i , because the map $y \longmapsto \varphi(x_i, y)$ is continuous; from this one obtains $\varphi(L, M') = \{0\}$; similarly, $\varphi(L', M) = \{0\}$ if L' is sufficiently small.

Corollary 0.3.1.1. ⁽¹⁸⁾ *Let M be a pseudocompact A -module.*

- (i) *For every open submodule M' , there exists an open ideal $\mathfrak{l}$ of A such that $\overline{\mathfrak{l}M} \subset M'$.*
- (ii) *Consequently, $M \simeq \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$, where $\mathfrak{l}$ runs through the filtered projective system of open ideals of A and each $M/\overline{\mathfrak{l}M}$ is equipped with the quotient topology (which makes it a pseudocompact module, cf. 0.2.D).*

Indeed, consider the map $\varphi : A \times M \longrightarrow M/M'$, $(a, m) \longmapsto am + M'$; by 0.3.1 there exists an open ideal $\mathfrak{l}$ of A such that $\mathfrak{l}M \subset M'$, and since M' is also closed, it also contains $\overline{\mathfrak{l}M}$. Since the intersection of the open submodules of M is zero, one therefore has $\bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M} = (0)$. Moreover, by 0.2.D, the map $\phi : M \longrightarrow \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$ is surjective; by (the proof of) 0.2.B, ϕ therefore induces an isomorphism $M/\text{Ker}(\phi) \xrightarrow{\sim} \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$, and we have just seen that $\text{Ker}(\phi) = \bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M}$ is zero.

Remark 0.3.1.2. ⁽¹⁹⁾ The completed tensor product satisfies the usual associativity condition: if L, M, P are pseudocompact A -modules, one has a canonical isomorphism

$$(L \hat{\otimes} M) \hat{\otimes} P \simeq L \hat{\otimes} (M \hat{\otimes} P);$$

indeed, these two objects represent the functor which associates to every object N of $\mathbf{LF}(A)$ the abelian group of continuous A -trilinear maps from $L \times M \times P$ to N .

0.3.2. — Let $L' \xrightarrow{f} L \xrightarrow{g} L'' \longrightarrow 0$ be an exact sequence and M an object of $\mathbf{PC}(A)$. It is clear that, for every object N of $\mathbf{LF}(A)$, the induced sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Bil}_c(L'' \times M, N) & \longrightarrow & \text{Bil}_c(L \times M, N) & \longrightarrow & \text{Bil}_c(L' \times M, N) \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & \text{Hom}_c(L'' \hat{\otimes}_A M, N) & \longrightarrow & \text{Hom}_c(L \hat{\otimes}_A M, N) & \longrightarrow & \text{Hom}_c(L' \hat{\otimes}_A M, N) \end{array}$$

are exact. By 0.2.E, this is equivalent to saying that the sequence

$$(*) \quad L' \hat{\otimes}_A M \xrightarrow{f \hat{\otimes} M} L \hat{\otimes}_A M \xrightarrow{g \hat{\otimes} M} L'' \hat{\otimes}_A M \longrightarrow 0$$

⁽¹⁸⁾ **N.D.E.** We have added this corollary.

⁽¹⁹⁾ **N.D.E.** We have added this remark.

is exact. Thus:

Corollary. *For every pseudocompact A -module M , the functor $-\widehat{\otimes}_A M$ is right exact.*

Take, in particular, the ring A for L , the inclusion of a closed ideal $\mathfrak{a}$ in A for f , and the canonical projection from A to $A/\mathfrak{a}$ for g . One can then identify $A\widehat{\otimes}_A M$ with M by means of the map $x\widehat{\otimes}_A m \mapsto xm$. Since the image of $\mathfrak{a}\widehat{\otimes}_A M$ is closed in M (cf. 0.2.B), and the image of $\mathfrak{a} \otimes_A M$ is everywhere dense in $\mathfrak{a}\widehat{\otimes}_A M$, the image of $f\widehat{\otimes}_A M$ is precisely the closure $\overline{\mathfrak{a}M}$ of $\mathfrak{a}M$ in M . The exact sequence (*) therefore yields the isomorphism

$$(A/\mathfrak{a})\widehat{\otimes}_A M \xrightarrow{\sim} M/\overline{\mathfrak{a}M}.$$

0.3.3. Nakayama's lemma. *Let A be a pseudocompact ring, M a pseudocompact A -module, and $\mathfrak{a}$ an ideal of A contained in the radical $\mathfrak{r}(A)$. The equality $\overline{\mathfrak{a}M} = M$ then implies $M = 0$.*

Indeed, suppose that $\overline{\mathfrak{a}M} = M$.⁽²⁰⁾ Let M' be an open submodule of M and M'' the quotient M/M' . Since M'' is discrete, $\mathfrak{a}M''$ is closed in M'' , hence equal to $\overline{\mathfrak{a}M''}$. By 0.3.2, the canonical map from $M/\overline{\mathfrak{a}M}$ to $M''/\overline{\mathfrak{a}M''}$ is surjective, so one has $\mathfrak{a}M'' = \overline{\mathfrak{a}M''} = M''$. Since M'' is a finitely generated A -module and $\mathfrak{a} \subset \mathfrak{r}(A)$, this implies $M'' = 0$ by the usual Nakayama lemma. Consequently, every open submodule M' of M equals M , so M is zero.⁽²¹⁾

0.3.4. — From Nakayama's lemma one obtains the usual consequences:

Corollary. *Let $\mathfrak{a}$ be a closed ideal contained in $\mathfrak{r}(A)$ and $f : M \rightarrow N$ a morphism of pseudocompact A -modules.*

- (i) *f is surjective if the induced map $f' : M/\overline{\mathfrak{a}M} \rightarrow N/\overline{\mathfrak{a}N}$ is surjective.*⁽²²⁾
- (ii) *If N is projective, f is invertible if f' is invertible.*

Indeed, (i) follows from Lemma 0.3.3 applied to $\text{Coker } f$. For (ii), suppose that f' is invertible. Then, by (i), f is surjective, hence has a section; one then applies 0.3.3 to $\text{Ker } f$. When A is local with maximal ideal $\mathfrak{m}$, one can also deduce from 0.3.3 the following *exchange theorem*:

Theorem. *Let A be a local pseudocompact ring, $\mathfrak{m}$ its maximal ideal, M a topologically free A -module with pseudobasis $(m_i)_{i \in I}$ (0.2.1), and N a (closed) direct factor of M . There exists a pseudobasis of M consisting of elements of N and of certain m_i .*

Indeed, this is clear when A is a field (use the duality of 0.2.2 and apply the usual exchange theorem); consequently, $N/\overline{\mathfrak{m}N}$ ⁽²³⁾ has as a complement a topologically free module over $A/\mathfrak{m}$ with pseudobasis $(\overline{m}_i)_{i \in J}$, where $\overline{m}_i$ is the image of m_i in $M/\overline{\mathfrak{m}M}$ and J is a subset of I . If P denotes the direct product $\prod_{i \in J} Am_i$, the canonical map from $N \oplus P$ to M is “bijective modulo $\mathfrak{m}$ ”; it is therefore bijective by the foregoing (for another proof see [CA], § IV.2, Prop. 8).

0.3.5. — Now consider three pseudocompact A -modules L, M, N , where N has finite length. Equipping the A -module $\text{Hom}_c(M, N)$ with the discrete topology, every element ψ of $\text{Hom}_c(L, \text{Hom}_c(M, N))$ defines a continuous bilinear map $\psi' : (\ell, m) \mapsto \psi(\ell)(m)$ from $L \times M$ to N . In this way one obtains a natural isomorphism

$$(1) \quad \text{Hom}_c(L, \text{Hom}_c(M, N)) \xrightarrow{\sim} \text{Hom}_c(L\widehat{\otimes}_A M, N),$$

hence another characterization of $\text{Hom}_c(L\widehat{\otimes}_A M, N)$, which we shall use when M is the projective limit of a filtered projective system of pseudocompact A -modules M_i . Then, by (1)

⁽²⁰⁾**N.D.E.** Replacing $\mathfrak{a}$ by its closure if necessary, one can suppose that $\mathfrak{a}$ is *closed*.

⁽²¹⁾**N.D.E.** since it equals the projective limit of the $M/M' = 0$.

⁽²²⁾**N.D.E.** Consequently, every pseudocompact A -module is a quotient of a topologically free A -module (one first reduces to the case where A is local); cf. the proof of 0.3.7.

⁽²³⁾**N.D.E.** We have corrected $N/\mathfrak{m}N$ to $N/\overline{\mathfrak{m}N}$, and likewise for $M/\overline{\mathfrak{m}M}$ below.

and 0.2.F(ii), one has natural isomorphisms

$$(2) \quad \begin{aligned} \mathrm{Hom}_c(L \widehat{\otimes}_A \varprojlim M_i, N) &\simeq \mathrm{Hom}_c(L, \mathrm{Hom}_c(\varprojlim M_i, N)) \\ &\simeq \mathrm{Hom}_c(L, \varinjlim \mathrm{Hom}_c(M_i, N)). \end{aligned}$$

Moreover, since the module $\varinjlim \mathrm{Hom}_c(M_i, N)$ is discrete, every continuous map with source L factors through a quotient of L of finite length. Consequently, the natural map below is an isomorphism:

$$(3) \quad \varinjlim \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) \longrightarrow \mathrm{Hom}_c(L, \varinjlim \mathrm{Hom}_c(M_i, N)).$$

Finally, by (1) and 0.2.F(ii) again, one has natural isomorphisms

$$(4) \quad \begin{aligned} \varinjlim \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) &\simeq \varinjlim \mathrm{Hom}_c(L \widehat{\otimes}_A M_i, N) \\ &\simeq \mathrm{Hom}_c(\varprojlim (L \widehat{\otimes}_A M_i), N). \end{aligned}$$

Combining isomorphisms (2), (3), (4), one obtains:

Proposition. *Let (M_i) be a filtered projective system of objects of $\mathbf{PC}(A)$, and let L (resp. N) be an object of $\mathbf{PC}(A)$ (resp. $\mathbf{LF}(A)$). One has an isomorphism functorial in N :*

$$\mathrm{Hom}_c(L \widehat{\otimes}_A \varprojlim M_i, N) \simeq \mathrm{Hom}_c(\varprojlim (L \widehat{\otimes}_A M_i), N),$$

and hence an isomorphism

$$L \widehat{\otimes}_A \varprojlim M_i \simeq \varprojlim (L \widehat{\otimes}_A M_i).$$

Thus the completed tensor product commutes with filtered projective limits.⁽²⁴⁾

0.3.6. — In particular,⁽²⁵⁾ the completed tensor product commutes with infinite products. For example, since the ring A is the product of its local components $A_{\mathfrak{m}}$ (0.1.1), every pseudocompact A -module M ($\simeq A \widehat{\otimes}_A M$) is identified with the product of the modules $M_{\mathfrak{m}} = A_{\mathfrak{m}} \widehat{\otimes}_A M$ (the local components of M). Similarly, let M and N be two pseudocompact A -modules. Recall (cf. 0.2.2) that $M^{\dagger}$ denotes $\mathrm{Hom}_c(M, A)$. Consider the map

$$\varphi : M^{\dagger} \otimes_A N^{\dagger} \longrightarrow (M \widehat{\otimes}_A N)^{\dagger}$$

which associates to an element $f \otimes g$ of $M^{\dagger} \otimes_A N^{\dagger}$ the map $m \widehat{\otimes} n \mapsto f(m)g(n)$ from $M \widehat{\otimes}_A N$ to A . This map φ is bijective when M is isomorphic to A .

Corollary. *When A is artinian, φ is an isomorphism whenever M is topologically free (or, more generally, projective).*

Indeed, for fixed N , the functor $M \mapsto (M \widehat{\otimes}_A N)^{\dagger}$ (resp. $M \mapsto M^{\dagger} \otimes_A N^{\dagger}$) transforms every direct product into a direct sum, by the foregoing and 0.2.F.

Remark 0.3.6.A. ⁽²⁶⁾Using 0.2.F in a similar way, one also obtains the following result: Let A be an artinian ring, M, Q objects of $\mathbf{PC}(A)$, and N an object of $\mathbf{LF}(A)$. Suppose that Q is projective; then one has natural isomorphisms

$$\mathrm{Hom}_c(M, Q) \xrightarrow{\sim} \mathrm{Hom}_A(Q^{\dagger}, M^{\dagger}) \quad \text{and} \quad Q^{\dagger} \otimes_A N \xrightarrow{\sim} \mathrm{Hom}_c(Q, N).$$

0.3.7. — For every $\mathfrak{m} \in \Upsilon(A)$, the functor $M \mapsto A_{\mathfrak{m}} \widehat{\otimes}_A M$ is evidently exact. Since every projective pseudocompact A -module P is a product of modules of the form $A_{\mathfrak{m}}$, it follows that the functor $M \mapsto P \widehat{\otimes}_A M$ is exact when P is projective. The converse is true:

Proposition. *Let A be a pseudocompact ring and P a pseudocompact A -module. The following conditions are equivalent:*

⁽²⁴⁾**N.D.E.** In this way one recovers 0.3.1.1: $L = L \widehat{\otimes}_A A = L \widehat{\otimes}_A \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} L/\overline{\mathfrak{l}}L$, where $\mathfrak{l}$ runs through the open ideals of A .

⁽²⁵⁾**N.D.E.** Cf. N.D.E. (12).

⁽²⁶⁾**N.D.E.** We have added this remark, which will be useful in 2.3.1.

- (i) P is a projective object of $\mathbf{PC}(A)$.
- (ii) Each local component $P_{\mathfrak{m}}$ is a topologically free $A_{\mathfrak{m}}$ -module.
- (iii) The functor $M \mapsto P \widehat{\otimes}_A M$ is exact.

Indeed, the equivalence of (i) and (ii) follows from 0.2.F(iii) and 0.2.1, and we have just seen that (ii) $\Rightarrow$ (iii). Suppose that the functor $M \mapsto P \widehat{\otimes}_A M$ is exact. Since $P \widehat{\otimes}_A M$ is the product of its local components

$$(P \widehat{\otimes}_A M)_{\mathfrak{m}} \simeq P_{\mathfrak{m}} \widehat{\otimes}_{A_{\mathfrak{m}}} M_{\mathfrak{m}},$$

one is reduced to the case where the ring A is local. We shall then prove that P is *topologically free*. Let $\mathfrak{m}$ be the maximal ideal of A ; then $P/\overline{\mathfrak{m}P}$ is a linearly compact vector space over $A/\mathfrak{m}$, hence a product of copies of $A/\mathfrak{m}$ (cf. 0.2.2). There is therefore a family $(A_i)_{i \in I}$ of copies of A and an isomorphism $\varphi : \prod_{i \in I} (A_i/\mathfrak{m}) \xrightarrow{\sim} P/\overline{\mathfrak{m}P}$. Since $\prod_{i \in I} A_i$ is projective, there is a commutative square

$$\begin{array}{ccc} \prod_{i \in I} A_i & \xrightarrow{\psi} & P \\ \downarrow p & & \downarrow q \\ \prod_{i \in I} (A_i/\mathfrak{m}) & \xrightarrow[\sim]{\varphi} & P/\overline{\mathfrak{m}P} \end{array}$$

where p and q denote the canonical projections. Applying Nakayama's lemma to Coker ψ , and noting that $(A/\mathfrak{m}) \widehat{\otimes}_A \psi$ is precisely φ , one sees that ψ is surjective.⁽²⁷⁾ Putting $B = \prod_{i \in I} A_i$ and $N = \text{Ker } \psi$, one then has the following commutative exact diagram:

$$\begin{array}{ccccccc} & & & & 0 & \text{---} & \text{---} \\ & & & & \downarrow & & \uparrow \\ & & \mathfrak{m} \widehat{\otimes}_A N & \longrightarrow & \mathfrak{m} \widehat{\otimes}_A B & \longrightarrow & \mathfrak{m} \widehat{\otimes}_A P \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A \widehat{\otimes}_A N & \longrightarrow & A \widehat{\otimes}_A B & \xrightarrow{\psi} & A \widehat{\otimes}_A P \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & \text{---} & \longrightarrow & (A/\mathfrak{m}) \widehat{\otimes}_A N & \longrightarrow & (A/\mathfrak{m}) \widehat{\otimes}_A B & \xrightarrow{\varphi} (A/\mathfrak{m}) \widehat{\otimes}_A P \longrightarrow 0. \end{array}$$

The “snake lemma” applied to the first two rows then shows that, in the bottom row, the morphism $(A/\mathfrak{m}) \widehat{\otimes}_A N \rightarrow (A/\mathfrak{m}) \widehat{\otimes}_A B$ is a monomorphism. But then, since φ is an isomorphism, $(A/\mathfrak{m}) \widehat{\otimes}_A N$ is zero; hence $N = 0$ (0.3.3) and ψ is an isomorphism.⁽²⁸⁾

Corollary 0.3.8. *Let A be a complete noetherian local ring and P a pseudocompact A -module. Then P is topologically free if and only if P is flat over A .*

Indeed, the canonical map from $M \otimes_A P$ to $M \widehat{\otimes}_A P$ is bijective when M equals A , hence also when M is noetherian (take a finite presentation of M and use right exactness of the tensor product and the completed tensor product). Now P is flat if and only if the functor $M \mapsto M \otimes_A P$ is exact when M runs through the noetherian modules. Similarly, in the proof of Proposition 0.3.7 one has seen that P is topologically free if the sequence

$$0 \longrightarrow \mathfrak{m} \widehat{\otimes}_A P \longrightarrow A \widehat{\otimes}_A P \longrightarrow (A/\mathfrak{m}) \widehat{\otimes}_A P \longrightarrow 0$$

⁽²⁷⁾**N.D.E.** This shows the result announced in N.D.E. (22): every pseudocompact A -module M is a quotient of a topologically free A -module. (Since products are exact in $\mathbf{PC}(A)$, one reduces to the case where A is local, treated above.)

⁽²⁸⁾**N.D.E.** An entirely analogous proof, using the “nilpotent Nakayama lemma”, shows that if A is artinian and P is a flat A -module, every local component of P is a free A -module (cf. [BAC], II § 3.2, Cor. 2 of Prop. 5).

is exact. Thus P is topologically free if and only if the functor $M \mapsto M \widehat{\otimes}_A P$ is exact when M runs through the noetherian modules. The corollary therefore follows from the equality $M \otimes_A P = M \widehat{\otimes}_A P$ established above.

0.4. — Let k be a pseudocompact ring; a *topological k -algebra* is a (commutative) topological ring A equipped with a morphism of topological rings $k \rightarrow A$. One says that A is a *profinite k -algebra* if the underlying topological k -module is pseudocompact. In this case, let $\mathfrak{l}$ be an open k -submodule of A . The composite map

$$\varphi : A \times A \xrightarrow{\text{mult.}} A \xrightarrow{\text{can.}} A/\mathfrak{l}$$

is continuous, so, by Lemma 0.3.1, there exists an open k -submodule $\mathfrak{n}$ of A such that $\varphi(A \times \mathfrak{n}) = 0$. This means that $\mathfrak{l}$ contains the open ideal $A\mathfrak{n}$ and implies that A is a *pseudocompact ring*. Similarly, let M be a topological A -module whose underlying k -module is pseudocompact. If M' is an open k -submodule of M , Lemma 0.3.1 applied to the map

$$A \times M \xrightarrow{\text{mult.}} M \xrightarrow{\text{can.}} M/M'$$

shows that M' contains an open A -submodule, hence that M is also a *pseudocompact A -module*.⁽²⁹⁾ Conversely:

Lemma. *Let A be a profinite k -algebra and M a pseudocompact A -module. Then the k -module $M|_k$ obtained by restriction of scalars is pseudocompact.*

Indeed, every pseudocompact A -module of finite length is isomorphic to a quotient A^n/L (where L is an open submodule of A^n), hence is a pseudocompact k -module. Since $M|_k$ is a projective limit of such modules, it is a pseudocompact k -module.

0.4.1. — If A and B are two profinite k -algebras, a *morphism* from A to B is, by definition, a continuous homomorphism of k -algebras. The category of profinite k -algebras will be denoted by $\mathbf{Alp}/k$. It evidently has projective limits: the algebra underlying a projective limit is the projective limit of the underlying algebras; the topology is the projective limit topology. It also has *finite inductive limits*:⁽³⁰⁾ for example, if $f : A \rightarrow B$ and $g : A \rightarrow C$ are two morphisms of profinite k -algebras, the amalgamated sum of B and C under A has $B \widehat{\otimes}_A C$ as its underlying topological A -module (by 0.4, f and g equip B and C with pseudocompact A -module structures); the multiplication on $B \widehat{\otimes}_A C$ evidently satisfies $(b \widehat{\otimes} c)(b' \widehat{\otimes} c') = (bb') \widehat{\otimes} (cc')$ if $b, b' \in B$ and $c, c' \in C$.

Definition 0.4.2. A profinite k -algebra C will be said to have *finite length* if the underlying k -module has finite length (hence is discrete); we shall denote by $\mathbf{Alf}/k$ the full subcategory of $\mathbf{Alp}/k$ consisting of k -algebras of finite length.⁽³¹⁾

For every profinite k -algebra A , denote by h^A the functor

$$\mathbf{Alf}/k \rightarrow (\text{Ens}), \quad C \mapsto \text{Hom}_{\mathbf{Alp}/k}(A, C).$$

It is clear that h^A is a left exact functor.⁽³²⁾ Moreover, the canonical projections $A \rightarrow A/\mathfrak{l}$ (where $\mathfrak{l}$ runs through the open ideals of A) induce, for every object C of $\mathbf{Alf}/k$, a canonical isomorphism

$$\varinjlim_{\mathfrak{l}} \text{Hom}_{\mathbf{Alf}/k}(A/\mathfrak{l}, C) \xrightarrow{\sim} \text{Hom}_{\mathbf{Alp}/k}(A, C),$$

⁽²⁹⁾**N.D.E.** We have added the following lemma; cf. N.D.E. (36) in Proposition 0.5.

⁽³⁰⁾**N.D.E.** We shall see in 0.4.2 that it has arbitrary inductive limits.

⁽³¹⁾**N.D.E.** Beware: k is an object of $\mathbf{Alf}/k$ only if k is artinian; cf. 1.2.2 below.

⁽³²⁾**N.D.E.** i.e. one which commutes with finite projective limits.

functorial in C . This means that h^A is the inductive limit of the representable functors $h^{A/\mathfrak{l}}$, i.e.

$$(*) \quad h^A \simeq \varinjlim_{\mathfrak{l}} h^{A/\mathfrak{l}}.$$

If B is another profinite k -algebra, the general properties of the bifunctor Hom and the isomorphism $\text{Hom}(h^C, h^B) = h^B(C)$ for C of finite length give isomorphisms

$$\begin{aligned} \text{Hom}_{\mathbf{Alp}/k}(B, A) &\simeq \varprojlim \text{Hom}_{\mathbf{Alp}/k}(B, A/\mathfrak{l}) \\ &\simeq \varprojlim \text{Hom}(h^{A/\mathfrak{l}}, h^B) \\ &\simeq \text{Hom}(\varinjlim h^{A/\mathfrak{l}}, h^B); \end{aligned}$$

combined with $(*)$, this shows that the contravariant functor $A \mapsto h^A$ is *fully faithful*. In fact:

Proposition. *The functor $A \mapsto h^A$ is an anti-equivalence from $\mathbf{Alp}/k$ to the category of left exact functors from $\mathbf{Alf}/k$ to (Ens) .*

Indeed, by the foregoing it suffices to show that every left exact functor $\mathbf{F} : \mathbf{Alf}/k \rightarrow (\text{Ens})$ is isomorphic to a functor of the form h^A ; for this, one can construct A as follows (cf. TDTE II, § 3). Since $\mathbf{F}$ is left exact, for every k -algebra C of finite length and every element ξ of $\mathbf{F}(C)$, there is a least subalgebra C' of C such that ξ belongs to the image of $\mathbf{F}(C')$ in $\mathbf{F}(C)$. If $C' = C$, one says that the pair (C, ξ) is *minimal*. The minimal pairs form a category if one takes as morphisms from (C, ξ) to (D, η) the homomorphisms φ from C to D such that $(\mathbf{F}\varphi)(\xi) = \eta$; it is clear that such a φ is a surjection and that the category of minimal pairs is “left filtered”. Moreover, one can restrict to pairs (C, ξ) such that C belongs to a set containing k -algebras of finite length of each isomorphism type.⁽³³⁾ Thus the functor $(C, \xi) \mapsto C$, whose source category is that of minimal pairs and whose target category is that of profinite k -algebras, has a projective limit; one takes this projective limit for A .

Corollary. *The category $\mathbf{Alp}/k$ has infinite inductive limits.*

Indeed, the category of left exact functors from $\mathbf{Alf}/k$ to (Ens) has projective limits, defined “argument by argument”, i.e. if $(\mathbf{F}_i)$ is a projective system of such functors, one has, for every object C of $\mathbf{Alf}/k$:

$$(34) \quad (\varprojlim \mathbf{F}_i)(C) = \varprojlim \mathbf{F}_i(C).$$

0.5. — ⁽³⁵⁾Let $\varphi : k \rightarrow \ell$ be a homomorphism of pseudocompact rings (cf. 0.1.3). One can generalize the construction of 0.3 as follows.

Definition 0.5.A. For every object M of $\mathbf{PC}(k)$ (resp. N of $\mathbf{PC}(\ell)$), we shall denote by $M \widehat{\otimes}_k N$ the separated completion of $M \otimes_k N$ for the linear topology defined by the kernels of the projections

$$M \otimes_k \ell \rightarrow (M/M') \otimes_k (N/N'),$$

⁽³³⁾**N.D.E.** Indeed, every discrete k -module of length n is isomorphic to k^n/L , where L is an open submodule of k^n of colength n . One can then consider the *set* of (isomorphism classes of) profinite k -algebra structures on each such module.

⁽³⁴⁾**N.D.E.** If $(A_i)_{i \in I}$ is an inductive system in $\mathbf{Alp}/k$, its inductive limit in $\mathbf{Alp}/k$ is the separated completion of the k -algebra B , the inductive limit of the underlying k -algebras, for the topology defined by the ideals I such that $I \cap A_i$ is an open ideal of A_i for every i , and $\text{length}_k(B/I) < \infty$. Note that if, for example, K/k is an algebraic field extension of infinite degree, and (k_i) denotes the filtered inductive system of subextensions of finite degree, then the inductive limit of the system (k_i) in $\mathbf{Alp}/k$ is the zero ring!

⁽³⁵⁾**N.D.E.** We have detailed this paragraph.

where M' (resp. N') is an open submodule of M (resp. N). Then $M\widehat{\otimes}_k N$ is a *pseudocompact ℓ -module*. If $m \in M$ and $x \in N$, we shall denote by $m\widehat{\otimes}_k x$ the canonical image of $m \otimes_k x$ in $M\widehat{\otimes}_k N$. This applies in particular when $N = \ell$; in this case one says that $M\widehat{\otimes}_k \ell$ is the pseudocompact ℓ -module *deduced from M by the base change $k \rightarrow \ell$* .

Remarks 0.5.B.

- (i) When considering such a base change, ℓ will not in general be a profinite k -algebra: a typical example is the case where k is a field and ℓ is an arbitrary extension K of k .
- (ii) However, if the k -module underlying N is pseudocompact (for example if ℓ is a profinite k -algebra), then, by 0.4, every open k -submodule of N contains an open ℓ -submodule of N ; consequently, $M\widehat{\otimes}_k N$ in this case coincides with the completed tensor product (cf. 0.3) of the pseudocompact k -modules M and N , and the notation therefore entails no ambiguity.

The k -module $N|_k$ obtained by restriction of scalars is in any case a topological k -module, i.e. the map $k \times N \rightarrow N$, $(t, n) \mapsto \varphi(t)n$ is continuous. Denote by $\text{Hom}_c(M, N|_k)$ the abelian group of continuous homomorphisms of k -modules from M to $N|_k$.

Proposition 0.5.C. *For every $M \in \text{Ob } \mathbf{PC}(k)$ and $N \in \text{Ob } \mathbf{PC}(\ell)$, one has a functorial isomorphism*

$$\text{Hom}_{\mathbf{PC}(\ell)}(M\widehat{\otimes}_k \ell, N) \simeq \text{Hom}_c(M, N|_k).$$

(36)

Indeed, let ϕ be a continuous homomorphism $M \rightarrow N|_k$; then the map $\phi' : M \times \ell \rightarrow N$, $(m, \lambda) \mapsto \lambda\phi(m)$ is continuous and “bilinear” (i.e. k -linear in the first factor and ℓ -linear in the second). If N' is an open ℓ -submodule of N , one shows as in Lemma 0.3.1 that there exist an open submodule M' of M and an open ideal ℓ' of ℓ such that $\phi'(M' \times \ell)$ and $\phi'(M \times \ell')$ are contained in N' . It follows that ϕ induces a continuous homomorphism of ℓ -modules $\Phi : M\widehat{\otimes}_k \ell \rightarrow N$ such that $\Phi(m\widehat{\otimes}\lambda) = \lambda\phi(m)$ for every $m \in M$ and $\lambda \in \ell$. Conversely, to every morphism $f : M\widehat{\otimes}_k \ell \rightarrow N$ one associates the morphism $f' : m \mapsto f(m\widehat{\otimes}_k 1)$ from M to $N|_k$. One then obtains, as in 0.3.2, 0.3.5, and 0.3.1.2, the following:

Corollary 0.5.D. *The functor $\mathbf{PC}(k) \rightarrow \mathbf{PC}(\ell)$, $M \mapsto M\widehat{\otimes}_k \ell$, is right exact and commutes with filtered projective limits, i.e. if (M_i) is a filtered projective system of objects of $\mathbf{PC}(k)$, one has a canonical isomorphism*

$$(\varprojlim M_i)\widehat{\otimes}_k \ell \simeq \varprojlim (M_i\widehat{\otimes}_k \ell).$$

Moreover, if M, N are pseudocompact k -modules, one has a canonical isomorphism

$$(M\widehat{\otimes}_k N)\widehat{\otimes}_k \ell \simeq (M\widehat{\otimes}_k \ell)\widehat{\otimes}_\ell (N\widehat{\otimes}_k \ell).$$

Definition 0.5.E. Finally, if A is a profinite k -algebra, there is one and only one profinite ℓ -algebra structure on $A\widehat{\otimes}_k \ell$ such that, if $a, b \in A$ and $\lambda, \mu \in \ell$,

$$(a\widehat{\otimes}_k \lambda)(b\widehat{\otimes}_k \mu) = (ab)\widehat{\otimes}_k (\lambda\mu).$$

One says that $A\widehat{\otimes}_k \ell$ is the profinite ℓ -algebra deduced from A by the *extension of scalars* (or “base change”) $k \rightarrow \ell$.

1. Formal varieties over a pseudocompact ring

1.1. — To every pseudocompact ring A one can associate a formal scheme (EGA I, 10.4.2) as follows: the underlying topological space is the set $\Upsilon(A)$ of open prime (hence maximal)

(36) **N.D.E.** Consequently, if ℓ is a profinite k -algebra, the functor $M \mapsto M\widehat{\otimes}_k \ell$ is left adjoint to the restriction-of-scalars functor $\mathbf{PC}(\ell) \rightarrow \mathbf{PC}(k)$ (cf. Lemma 0.4).

ideals of A , equipped with the discrete topology; the structure sheaf has the cartesian product $\prod_{\mathfrak{m} \in E} A_{\mathfrak{m}}$ as its space of sections over a subset E of $\Upsilon(A)$. The formal scheme thus obtained is denoted by $\mathrm{Spf}(A)$ (the *formal spectrum* of A).⁽³⁷⁾ If A and B are two pseudocompact rings, a *morphism* from $\mathrm{Spf}(B)$ to $\mathrm{Spf}(A)$ consists of a map f from $\Upsilon(B)$ to $\Upsilon(A)$ and a family of continuous homomorphisms $f_y^\sharp : A_{f(y)} \rightarrow B_y$, for $y \in \Upsilon(B)$. Such a morphism induces a continuous homomorphism $f^\sharp$ from $A = \prod_{x \in \Upsilon(A)} A_x$ to $B = \prod_{y \in \Upsilon(B)} B_y$. The converse is true:

Proposition. *The contravariant functor $A \mapsto \mathrm{Spf}(A)$ is fully faithful.*

Indeed, if $\varphi : A \rightarrow B$ is a continuous homomorphism of algebras, the inverse image $\varphi^{-1}(\mathfrak{n})$ of an open maximal ideal of B is an open prime ideal of A , hence maximal in A . One therefore obtains a map $\mathfrak{n} \mapsto \varphi^{-1}(\mathfrak{n})$ from $\Upsilon(B)$ to $\Upsilon(A)$, and φ induces a continuous homomorphism $A_{\varphi^{-1}(\mathfrak{n})} \rightarrow B_{\mathfrak{n}}$. Thus φ induces a morphism of formal schemes $\mathrm{Spf}(\varphi) : \mathrm{Spf}(B) \rightarrow \mathrm{Spf}(A)$. One easily verifies that $\mathrm{Spf}(\varphi)^\sharp = \varphi$, and that $\mathrm{Spf}(f^\sharp) = f$ for every morphism $f : \mathrm{Spf}(B) \rightarrow \mathrm{Spf}(A)$, whence the proposition. Although here we speak only of formal schemes of the form $\mathrm{Spf}(A)$, we shall use the language of formal schemes rather than that of pseudocompact rings in order to support our assertions by geometric intuition.

1.2. — Let k be a pseudocompact ring.

Definition 1.2.A. ⁽³⁸⁾ We shall call a *formal variety over k* any formal scheme X over $\mathrm{Spf}(k)$ which is isomorphic to a formal k -scheme $\mathrm{Spf}(A)$ for some profinite k -algebra A . The algebra A is then isomorphic to the *affine algebra of X* , that is, the algebra of sections of the structure sheaf $\mathcal{O}_X$ of X . Denote by $\mathbf{Vaf}/k$ the full subcategory of the category of formal schemes over $\mathrm{Spf}(k)$ whose objects are formal k -varieties.⁽³⁹⁾

By 1.1, the functor $A \mapsto \mathrm{Spf}(A)$ is an anti-equivalence from $\mathbf{Alp}/k$ (0.4.1) to $\mathbf{Vaf}/k$. Thus, by Corollary 0.4.2:

Proposition 1.2.B. *The category $\mathbf{Vaf}/k$ has projective and inductive limits.*⁽⁴⁰⁾

For example, let $X \in \mathrm{Ob} \mathbf{Vaf}/k$ and let $f : Y \rightarrow X$ and $g : Z \rightarrow X$ be two formal k -varieties over X , and let A, B, C be the affine algebras of X, Y, Z ; by 0.4.1, the affine algebra of the fibered product $Y \times_X Z$ is identified with $B \hat{\otimes}_A C$,⁽⁴¹⁾ so that the inclusion of $\mathbf{Vaf}/k$ in the category of all formal k -schemes commutes with finite projective limits (cf. EGA I, 10.7). Inductive limits of formal k -varieties correspond to projective limits of their affine algebras.

Example 1.2.C. (Cokernels). Let, for example, $d, e : X \rightrightarrows Y$ be a double arrow in $\mathbf{Vaf}/k$; the affine algebra of $\mathrm{Coker}(d, e)$ is isomorphic to the kernel of the homomorphisms induced on the affine algebras of X and Y , but one can also give the following construction of $\mathrm{Coker}(d, e)$: the topological space underlying $\mathrm{Coker}(d, e)$ is the cokernel of the underlying spaces;⁽⁴²⁾ if p is the canonical projection from the set underlying Y to the cokernel, and z belongs to the

⁽³⁷⁾N.D.E. Cf. Remarks 0.1.2.

⁽³⁸⁾N.D.E. We have added the numbering 1.2.A to 1.2.E for later references.

⁽³⁹⁾N.D.E. Note that, by the definition of morphisms (1.1), every $X \in \mathrm{Ob} \mathbf{Vaf}/k$ is the direct sum in $\mathbf{Vaf}/k$ of the one-point formal varieties $\mathrm{Spf}(\mathcal{O}_{X,x})$, for $x \in X$.

⁽⁴⁰⁾N.D.E. In particular, cokernels exist in $\mathbf{Vaf}/k$; see below. Observe, moreover, that a filtered projective limit in $\mathbf{Vaf}/k$ whose transition morphisms are all surjective may be empty; cf. N.D.E. (34).

⁽⁴¹⁾N.D.E. Note that $Y \times_X Z$ is the direct sum, for $x \in X$, of the formal varieties $B_x \hat{\otimes}_{\mathcal{O}_{X,x}} C_x$, where B_x is the product of the $\mathcal{O}_{Y,y}$ for the $y \in Y$ such that $f(y) = x$, and C_x is defined analogously. This will be used in 2.5.1.

⁽⁴²⁾N.D.E. i.e. the quotient set of Y by the identifications $d(x) = e(x)$, for every $x \in X$, equipped with the quotient topology, which here is the discrete topology.

cokernel, the local algebra of $\text{Coker}(d, e)$ at z is the kernel of the double arrow

$$d^{\sharp}, e^{\sharp} : \prod_{p(y)=z} \mathcal{O}_{Y,y} \rightrightarrows \prod_{q(x)=z} \mathcal{O}_{X,x},$$

where one has put $q = p \circ d = p \circ e$ and where $d^{\sharp}$ and $e^{\sharp}$ are induced by the homomorphisms $d_x^{\sharp}$ and $e_x^{\sharp}$ (notation of 1.1).

Definition 1.2.D. If $\varphi : k \rightarrow \ell$ is a homomorphism of pseudocompact rings and X a formal k -variety with affine algebra A , the formal scheme $X \times_{\text{Spf}(k)} \text{Spf}(\ell)$ obtained by base change is a formal ℓ -variety, which we shall also denote by $X \widehat{\otimes}_k \ell$, and whose affine algebra is the completed tensor product $A \widehat{\otimes}_k \ell$ (cf. 0.5 and EGA I, § 10).

Remark 1.2.E. Since every formal variety over k decomposes into formal varieties over the local components of k , *in certain proofs we shall suppose that k is a local pseudocompact ring.*

Let us now give some examples while fixing our terminology.

1.2.1. — A k -functor will, by definition, be a *covariant* functor from $\mathbf{Alf}/k$ to $(\mathbf{Ens})$. By 1.1 and 0.4.2, one can identify $\mathbf{Vaf}/k \simeq (\mathbf{Alp}/k)^0$ with a full subcategory of the category of k -functors, by associating to every formal k -variety X the functor

$$\mathbf{Alf}/k \rightarrow (\mathbf{Ens}), \quad C \mapsto X(C) = \text{Hom}_{\mathbf{Vaf}/k}(\text{Spf}(C), X).$$

Later we shall encounter k -functors $\mathbf{F}$ which associate to every object C of $\mathbf{Alf}/k$ a module $\mathbf{F}(C)$ over C , and to every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$ a k -linear map $\mathbf{F}(\varphi) : \mathbf{F}(C) \rightarrow \mathbf{F}(D)$ such that, if $x \in \mathbf{F}(C)$ and $\lambda \in C$,

$$\mathbf{F}(\varphi)(\lambda x) = \varphi(\lambda) \mathbf{F}(\varphi)(x).$$

By Exp. I, 3.1, such an $\mathbf{F}$ is equipped with an $\mathbf{O}_k$ -module structure, if one denotes by $\mathbf{O}_k$ the ring-valued k -functor which associates to every object C of $\mathbf{Alf}/k$ the ring underlying C .

Definitions.

- (i) An $\mathbf{O}_k$ -module $\mathbf{F}$ will be said to be *admissible* if every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$ induces a bijection from $D \otimes_C \mathbf{F}(C)$ to $\mathbf{F}(D)$.⁽⁴³⁾
- (ii) One says that $\mathbf{F}$ is *flat* if it is admissible and if, for every object C of $\mathbf{Alf}/k$, $\mathbf{F}(C)$ is a flat C -module.⁽⁴⁴⁾

For example, if M is a k -module (*not necessarily pseudocompact*), we shall denote (as in Exp. I, 4.6.1) by $\mathbf{W}(M)$ the $\mathbf{O}_k$ -module $C \mapsto C \otimes_k M$; then $\mathbf{W}(M)$ is flat when M is flat over k . Moreover, the functor $M \mapsto \mathbf{W}(M)$ has as its right adjoint the functor which associates to every $\mathbf{O}_k$ -module $\mathbf{F}$ the k -module $\varprojlim_{\mathfrak{l}} \mathbf{F}(k/\mathfrak{l})$, where $\mathfrak{l}$ runs through the open ideals of k .

1.2.2. — In what follows, an $\mathbf{O}_k$ -module will always be denoted by a boldface letter such as $\mathbf{F}$; when k is artinian, we shall then write simply F instead of $\mathbf{F}(k)$. In this case, it goes without saying that the functor $\mathbf{F} \mapsto F$ induces an equivalence from the category of flat $\mathbf{O}_k$ -modules to that of flat k -modules! Thus the terminology we have adopted has only the purpose of allowing us to reason “as if k were always artinian”. In accordance with Exposé I § 3.1, we shall use analogous conventions for other algebraic structures: thus an $\mathbf{O}_k$ -algebra (resp. an $\mathbf{O}_k$ -coalgebra, resp. an $\mathbf{O}_k$ -Lie algebra, resp. an $\mathbf{O}_k$ -Lie p -algebra) will consist of an $\mathbf{O}_k$ -module $\mathbf{M}$ and, for every object C of $\mathbf{Alf}/k$, a C -algebra structure (resp. a C -coalgebra, resp. a C -Lie algebra, resp. a C -Lie p -algebra structure) on $\mathbf{M}(C)$; one supposes further that, for every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$, the map from $D \otimes_C \mathbf{M}(C)$ to $\mathbf{M}(D)$ induced by

⁽⁴³⁾**N.D.E.** We have introduced this terminology; cf. N.D.E. (47) in 1.2.3.

⁽⁴⁴⁾**N.D.E.** and hence projective, since C is artinian.

$\mathbf{M}(\varphi)$ is a homomorphism of D -algebras (resp. of D -coalgebras, etc.). Finally, note that if $\mathbf{F}$ and $\mathbf{G}$ are two $\mathbf{O}_k$ -modules, $\mathbf{F} \otimes_k \mathbf{G}$ will denote the $\mathbf{O}_k$ -module $C \mapsto \mathbf{F}(C) \otimes_C \mathbf{G}(C)$.

1.2.3. — ⁽⁴⁵⁾Let us begin with the following lemma.

Lemma 1.2.3.A. *Let $k \rightarrow K$ be a morphism of pseudocompact rings, B a pseudocompact K -module, and M a projective k -module (without topology). One has a canonical isomorphism of pseudocompact K -modules*

$$(2) \quad \theta : \mathrm{Hom}_k(M, k) \widehat{\otimes}_k B \xrightarrow{\sim} \mathrm{Hom}_k(M, B).$$

Here $M^* = \mathrm{Hom}_k(M, k)$ is equipped with the topology defined in 0.2.2, which makes it a pseudocompact k -module, and the topology of $\mathrm{Hom}_k(M, B)$ is defined analogously: a base of neighborhoods of 0 consists of the following K -submodules, where $x \in M$ and B' is an open K -submodule of B :

$$\mathcal{H}(x, B') = \{f \in \mathrm{Hom}_k(M, B) \mid f(x) \in B'\}.$$

Finally, $M^* \widehat{\otimes}_k B$ is the pseudocompact K -module deduced from M^* by base change; cf. 0.5.A. This being so, θ is evidently an isomorphism when $M = k$; moreover, both sides of (2), considered as functors in M , transform direct sums into products (in particular, commute with finite direct sums). One therefore obtains that (2) is an isomorphism when M is a free k -module, and then when M is projective.

Definition 1.2.3.B. Let N be a pseudocompact k -module. We denote by $\mathbf{V}_k^f(N)$ or $\mathbf{N}^\dagger$ ⁽⁴⁶⁾ the $\mathbf{O}_k$ -module which associates to every $C \in \mathrm{Ob} \mathbf{AIf}/k$ the C -module $(C \widehat{\otimes}_k N)^\dagger$ dual to $C \widehat{\otimes}_k N$ (0.2.2), i.e. the C -module

$$\mathrm{Hom}_c(C \widehat{\otimes}_k N, C) \simeq \mathrm{Hom}_c(N, C|_k),$$

where $C|_k$ denotes the k -module C obtained by restriction of scalars. This $\mathbf{O}_k$ -module $\mathbf{V}_k^f(N)$ will be called the *dual* of N .

If $k_{\mathfrak{m}}$ is a local component of k , then, for every object C of $\mathbf{AIf}/k$,

$$\mathrm{Hom}_c(k_{\mathfrak{m}}, C|_k) = C_{\mathfrak{m}} = C \otimes_k k_{\mathfrak{m}}$$

is a projective C -module, and moreover one has $D \otimes_C C_{\mathfrak{m}} = D_{\mathfrak{m}}$ for every morphism $C \rightarrow D$ of $\mathbf{AIf}/k$; thus $\mathbf{V}_k^f(k_{\mathfrak{m}})$ is a flat $\mathbf{O}_k$ -module (cf. 1.2.1). Since every projective pseudocompact k -module is a product of modules $k_{\mathfrak{m}}$ (cf. Prop. 0.2.1), from Corollary 0.2.F one deduces:

Corollary 1.2.3.C. $\mathbf{V}_k^f(N)$ is a flat $\mathbf{O}_k$ -module when N is a projective pseudocompact k -module.

Definition 1.2.3.D. Conversely, if $\mathbf{M}$ is an *admissible* $\mathbf{O}_k$ -module (cf. 1.2.1),⁽⁴⁷⁾ we call the following pseudocompact k -module the *dual* of $\mathbf{M}$ and denote it by $\Gamma^*(\mathbf{M})$. For $\mathfrak{l}$ running through the open ideals of k , equip each $k/\mathfrak{l}$ -module

$$\mathbf{M}(k/\mathfrak{l})^* = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l})$$

with the topology described in 0.2.2, which makes it a pseudocompact k -module. Since $\mathbf{M}(k/\mathfrak{l}) \simeq \mathbf{M}(k/\mathfrak{l}') \otimes (k/\mathfrak{l})$ for $\mathfrak{l} \supset \mathfrak{l}'$, one has transition morphisms

$$\begin{aligned} \mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}') &\longrightarrow \mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}) \\ &= \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l}), \end{aligned}$$

⁽⁴⁵⁾**N.D.E.** We have added the following lemma and introduced the numbering 1.2.3.A to 1.2.3.F for later references.

⁽⁴⁶⁾**N.D.E.** In fact, we shall not use this second notation; see N.D.E. (47).

⁽⁴⁷⁾**N.D.E.** The editors do not understand the following definition unless $\mathbf{M}$ is assumed admissible, hence the addition of this hypothesis. Moreover, we have written $\Gamma^*(\mathbf{M})$ instead of M^* in order to avoid confusion between M^* , the module dual to the functor $\mathbf{M}$, and $\mathbf{N}^\dagger$, the functor dual to the module N ; cf. N.D.E. (46). Finally, we have detailed the construction of $\Gamma^*(\mathbf{M})$.

and by definition $\Gamma^*(\mathbf{M})$ is the projective limit of this (filtered) projective system of pseudocompact k -modules.

From now on, suppose further that $\mathbf{M}$ is a *flat* $\mathbf{O}_k$ -module (cf. 1.2.1); then each $\mathbf{M}(k/\mathfrak{l}')$ is a projective $(k/\mathfrak{l}')$ -module, so the transition morphisms above are surjective, and hence so are the projections $\Gamma^*(\mathbf{M}) \rightarrow \mathbf{M}(k/\mathfrak{l})^*$, since filtered projective limits are exact in $\mathbf{PC}(k)$. If $\tau : k \rightarrow K$ is a morphism of pseudocompact rings, denote by $\mathbf{M} \otimes_k K$ or simply $\mathbf{M}_K$ the $\mathbf{O}_K$ -functor defined as follows. If C is a K -algebra of finite length, the kernel of $k \rightarrow C$ is an open ideal $\mathfrak{l}$, and one puts $\mathbf{M}_K(C) = \mathbf{M}(k/\mathfrak{l}) \otimes_k C$; one then has $\mathbf{M}_K(C) = \mathbf{M}(k/\mathfrak{l}') \otimes_k C$ for every open ideal $\mathfrak{l}'$ contained in $\mathfrak{l}$. One then defines $\Gamma_K^*(\mathbf{M}_K)$ as the projective limit, for I running through the open ideals of K , of the pseudocompact K -modules

$$\mathrm{Hom}_{K/I}(\mathbf{M}_K(K/I), K/I) = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), K/I),$$

where in the right-hand term $\mathfrak{l}$ is any open ideal of k such that $\tau(\mathfrak{l}) \subset I$. Moreover, by 1.2.3.A, the right-hand term is identified with $\mathbf{M}(k/\mathfrak{l})^* \hat{\otimes}_k (K/I)$. Since the projections $\Gamma^*(\mathbf{M}) \rightarrow \mathbf{M}(k/\mathfrak{l})^*$ are surjective, one sees that the projective limit of the $\mathbf{M}(k/\mathfrak{l})^* \hat{\otimes}_k (K/I)$ is precisely the pseudocompact K -module $\Gamma^*(\mathbf{M}) \hat{\otimes}_k K$ (cf. 0.5.A). Thus one has obtained that, for every flat $\mathbf{O}_k$ -module $\mathbf{M}$, formation of $\Gamma^*(\mathbf{M})$ *commutes with extension of the base*, i.e. one has

$$(\star) \quad \Gamma_K^*(\mathbf{M} \otimes_k K) \simeq \Gamma^*(\mathbf{M}) \hat{\otimes}_k K.$$

Proposition 1.2.3.E.

- (i) *The functors $N \mapsto \mathbf{V}_k^f(N)$ and $\mathbf{M} \mapsto \Gamma^*(\mathbf{M})$ define an anti-equivalence between the category of projective pseudocompact k -modules and that of flat $\mathbf{O}_k$ -modules.*⁽⁴⁸⁾
- (ii) *Moreover, if $k \rightarrow K$ is a morphism of pseudocompact rings, the preceding anti-equivalence “commutes with base change” in the following sense: if $N \simeq \Gamma^*(\mathbf{M})$, then $N \hat{\otimes}_k K \simeq \Gamma_K^*(\mathbf{M} \otimes_k K)$.*

PROOF. On the one hand, one has a natural transformation $\phi_N : N \rightarrow \Gamma^*(\mathbf{V}_k^f(N))$. Since the functor $\Gamma^* \circ \mathbf{V}_k^f$ commutes with products, by 0.3.5 and 0.2.F, it suffices to verify that ϕ_N is an isomorphism when N is a local component $k_{\mathfrak{m}}$ of k . In this case, for every open ideal $\mathfrak{l}$ of k contained in $\mathfrak{m}$, the natural morphism

$$(k/\mathfrak{l})_{\mathfrak{m}} \rightarrow \mathrm{Hom}_{k/\mathfrak{l}}(\mathrm{Hom}_c(k_{\mathfrak{m}} \hat{\otimes} k/\mathfrak{l}, k/\mathfrak{l}), k/\mathfrak{l})$$

is an isomorphism, whence the result. On the other hand, let $\mathbf{M}$ be a flat $\mathbf{O}_k$ -module. Let us show that $\Gamma^*(\mathbf{M}) = \varprojlim \mathbf{M}(k/\mathfrak{l})^*$ is a projective object of $\mathbf{PC}(k)$. Let $N \rightarrow N'$ be a surjective morphism between objects of $\mathbf{LF}(k)$. By 0.2.F(i) and (ii), it suffices to show that the natural map

$$\varinjlim \mathrm{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N) \rightarrow \varinjlim \mathrm{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N')$$

is surjective. But this is clear, since N and N' are $k/\mathfrak{l}_0$ -modules for some open ideal $\mathfrak{l}_0$; hence, if $\mathfrak{l} \subset \mathfrak{l}_0$, every morphism $\mathbf{M}(k/\mathfrak{l})^* \rightarrow N'$ lifts to a morphism $\mathbf{M}(k/\mathfrak{l})^* \rightarrow N$, since $\mathbf{M}(k/\mathfrak{l})^*$ is a projective object of $\mathbf{PC}(k/\mathfrak{l})$. One has a morphism $\psi : \mathbf{M} \rightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))$ of functors from $\mathbf{AIf}/k$ to $(\mathbf{Ens})$. Let B be an object of $\mathbf{AIf}/k$; let us show that

$$(1) \quad \begin{aligned} \psi(B) : \mathbf{M}(B) &\rightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))(B) \\ &= \mathrm{Hom}_c(\varprojlim_{\mathfrak{l}} (\mathbf{M}(k/\mathfrak{l})^* \hat{\otimes}_k B), B) \end{aligned}$$

is a bijection (in the equality above, we have used the fact that $\hat{\otimes}$ commutes with filtered projective limits). Let $\mathfrak{l}_0$ be an open ideal of k contained in the kernel of $k \rightarrow B$. By Lemma

⁽⁴⁸⁾**N.D.E.** We have added part (ii) below, as well as the following proof. The original indicated: “If $\mathbf{M}$ is a flat $\mathbf{O}_k$ -module, it is clear that $\mathbf{M}$ is precisely the dual of M^* , so the functors $N \mapsto N^*$ and $\mathbf{M} \mapsto M^*$ define an anti-equivalence. . .”

1.2.3.A, for every $\mathfrak{l} \subset \mathfrak{l}_0$, one has a canonical isomorphism of pseudocompact B -modules

$$\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k B \xrightarrow{\sim} \operatorname{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), B),$$

and, since $\mathbf{M}(B) = \mathbf{M}(k/\mathfrak{l}) \otimes_{k/\mathfrak{l}} B$, the right-hand term equals $\operatorname{Hom}_B(\mathbf{M}(B), B)$. Thus the projective system in (1) is constant for $\mathfrak{l} \subset \mathfrak{l}_0$, and (1) reduces to the canonical morphism

$$\mathbf{M}(B) \longrightarrow \operatorname{Hom}_c(\operatorname{Hom}_B(\mathbf{M}(B), B), B),$$

which is an isomorphism by 0.2.2, since B is artinian and $\mathbf{M}(B)$ is a projective B -module. This proves part (i) of the proposition, and part (ii) follows from isomorphism $(\star)$ established above. $\square$

Remark 1.2.3.F. Let us return to the statement of the proposition, and suppose further that N is a topologically free pseudocompact k -module. In this case one can choose bases for the C -modules $\mathbf{V}_k^f(N)(C)$ “in a coherent manner”. Indeed, let (n_i) be a pseudobasis of N (0.2.1) and n_i^C the canonical image of n_i in $C \widehat{\otimes}_k N$, for $C \in \mathbf{AIf}/k$. If one defines the element δ_i^C of $(C \widehat{\otimes}_k N)^\dagger$ by the equalities $\delta_i^C(n_i^C) = 1$ and $\delta_i^C(n_j^C) = 0$ for $i \neq j$, the family (δ_i^C) is a basis of $\mathbf{V}_k^f(N)(C)$; moreover, for every morphism $\varphi : C \longrightarrow D$ of $\mathbf{AIf}/k$, $\mathbf{V}_k^f(N)(\varphi)$ sends δ_i^C to δ_i^D .

1.2.4. — ⁽⁴⁹⁾For every pseudocompact k -module E , let $\widehat{S}_k(E)$ be the completed symmetric algebra of E , defined in the following manner. Let $T_k(E)$ be the direct sum of the pseudocompact k -modules

$$\widehat{\bigotimes}_k^n E = E \widehat{\otimes}_k \cdots \widehat{\otimes}_k E \quad (n \geq 0; \text{ if } n = 0, \widehat{\bigotimes}_k^0 E = k);$$

one makes $T_k(E)$ a graded k -algebra by defining multiplication in the usual manner; let $S_k(E)$ be the quotient of $T_k(E)$ by the homogeneous ideal whose n th component is the closed k -submodule of $\widehat{\bigotimes}_k^n E$ generated by the elements

$$\begin{aligned} & x_1 \widehat{\otimes} \cdots \widehat{\otimes} x_i \widehat{\otimes} x_{i+1} \widehat{\otimes} \cdots \widehat{\otimes} x_n \\ & - x_1 \widehat{\otimes} \cdots \widehat{\otimes} x_{i+1} \widehat{\otimes} x_i \widehat{\otimes} \cdots \widehat{\otimes} x_n. \end{aligned}$$

If one denotes this quotient by $S_k^n(E)$, $S_k(E)$ is evidently a graded k -algebra with n th component $S_k^n(E)$. Equip $S_k(E)$ with the linear topology defined by the set Υ of ideals $\mathfrak{l}$ such that $S_k(E)/\mathfrak{l}$ is a k -module of finite length and $S_k^n(E) \cap \mathfrak{l}$ is an open submodule of $S_k^n(E)$ for every n . Then the profinite algebra $\widehat{S}_k(E)$ is the separated completion of $S_k(E)$ for this topology.⁽⁵⁰⁾ Denote by $\mathbf{V}_k^f(E)$ the formal k -variety $\operatorname{Spf}(\widehat{S}_k(E))$. It represents the k -functor $\mathbf{V}_k^f(E)$, i.e. for every object C of $\mathbf{AIf}/k$, one has a canonical isomorphism

$$\begin{aligned} \mathbf{V}_k^f(E)(C) &= \operatorname{Hom}_c(E, C|_k) \xrightarrow{\sim} \operatorname{Hom}_{\mathbf{AIf}/k}(\widehat{S}_k(E), C) \\ &= \operatorname{Hom}_{\mathbf{Vaf}/k}(\operatorname{Spf}(C), \mathbf{V}_k^f(E)). \end{aligned}$$

Throughout what follows, we identify $\mathbf{V}_k^f(E)$ with the k -functor $\mathbf{V}_k^f(E)$.

⁽⁴⁹⁾**N.D.E.** In this paragraph we have changed the order, first defining $\widehat{S}_k(E)$ and then stating that $\mathbf{V}_k^f(E)$ represents $\mathbf{V}_k^f(E)$.

⁽⁵⁰⁾**N.D.E.** For example, let k be a field, E a pseudocompact k -vector space, and $V = \operatorname{Hom}_c(E, k)$; one has $E \simeq V^*$ (cf. 0.2.2). For every finite-dimensional subspace W of V , let $\mathcal{F}(W)$ be the set of closed points of the k -scheme $\mathbb{V}(W) = \operatorname{Spec} S_k(W^*)$. Denote by $\mathcal{F}(V)$ the inductive limit of the $\mathcal{F}(W)$. Then $\widehat{S}_k(E)$ is the product, for $x \in \mathcal{F}(V)$, of the pseudocompact k -algebras

$$\widehat{S}_k(E)_x = \varprojlim_W \widehat{O}_{W,x},$$

where W runs through the finite-dimensional subspaces of V such that $x \in \mathcal{F}(W)$, and $\widehat{O}_{W,x}$ denotes the separated completion of the local ring $\mathcal{O}_{\mathbb{V}(W),x}$ for the topology defined by the ideals of finite codimension (which here coincides with the $\mathfrak{m}$ -adic topology). If k is algebraically closed and $(v_i)_{i \in I}$ is a basis of V , so that E has a pseudobasis $(e_i)_{i \in I}$, each local component $\widehat{S}_k(E)_x$ is isomorphic to the formal power series ring $k[[e_i, i \in I]]$, equipped with the topology defined by the ideals of finite codimension.

1.2.5. — By 1.2.4, the zero morphism from E to k is associated to a morphism of profinite algebras $\pi : \widehat{S}_k(E) \rightarrow k$; this morphism π induces the zero map on $S_k^n(E)$ for $n \geq 1$ and defines a section of the structural morphism $\mathbb{V}_k^f(E) \rightarrow \mathrm{Spf}(k)$. We shall denote by $\mathbb{V}_k^{f,0}(E)$ the formal variety whose points are the images of the points of $\mathrm{Spf}(k)$ under the section $\mathrm{Spf}(\pi)$ and which has the same local algebras as $\mathbb{V}_k^f(E)$ at these points.⁽⁵¹⁾ Then the affine algebra of $\mathbb{V}_k^{f,0}(E)$ is the separated completion of $S_k(E)$ for the topology defined by the ideals $\mathfrak{l} \in \Upsilon$ (cf. 1.2.4) which contain $S_k^n(E)$ for sufficiently large n . One deduces that it is the infinite product

$$k[[E]] = k \times E \times S_k^2(E) \times S_k^3(E) \times \cdots.$$

On the other hand, let C be an object of $\mathbf{AIf}/k$, u an element of $\mathbb{V}_k^f(C) = \mathrm{Hom}_c(E, C)$, and $\phi : \widehat{S}_k(E) \rightarrow C$ the morphism of profinite k -algebras corresponding to u . Then $\mathrm{Ker}(\phi)$ is an open ideal (i.e. ϕ belongs to $\mathbb{V}_k^{f,0}(C)$) if and only if $\mathrm{Ker}(\phi)$ contains $S_k^n(E)$ for sufficiently large n , i.e. if and only if $u(E)$ is contained in the radical $\mathfrak{r}(C)$ of C . Thus: *for every object C of $\mathbf{AIf}/k$, one has canonical isomorphisms*

$$\mathbb{V}_k^{f,0}(E)(C) \simeq \mathrm{Hom}_{\mathbf{AIf}/k}(k[[E]], C) \simeq \mathrm{Hom}_c(E, \mathfrak{r}(C)).$$

1.2.6. — A formal k -variety V is said to have *finite length* if its affine algebra does. Similarly, if S is a scheme, an S -scheme X is said to have *finite length* if X is finite over S and the direct image of $\mathcal{O}_X$ on S is an $\mathcal{O}_S$ -module of finite length.⁽⁵²⁾ Thus giving an S -scheme X of finite length “is the same thing” as giving a finite set $\{s_1, \dots, s_n\}$ of closed points of S and an $\mathcal{O}_{S,s_i}$ -algebra of finite length at each of these points. One therefore sees that S -schemes of finite length are identified with formal varieties of finite length over the following formal scheme $\widehat{S}$. The topological space underlying $\widehat{S}$ is the set of closed points of S equipped with the discrete topology; if s is such a closed point, the stalk $\mathcal{O}_{\widehat{S},s}$ of the structure sheaf $\mathcal{O}_{\widehat{S}}$ at s is the separated completion $\widehat{\mathcal{O}_{S,s}}$ of $\mathcal{O}_{S,s}$ for the linear topology defined by the ideals of finite colength; one therefore has $\widehat{S} = \mathrm{Spf} \mathcal{A}(\widehat{S})$, where $\mathcal{A}(\widehat{S})$ is the product of the $\widehat{\mathcal{O}_{S,s}}$, for s running through the closed points of S , equipped with the product topology.

Definition. If X is an S -scheme, denote by $\widehat{X}/\widehat{S}$ the formal variety over $k = \mathcal{A}(\widehat{S})$ defined as follows. The underlying topological space consists of the points $x \in X$ such that $[\kappa(x) : \kappa(s)] < \infty$, where s is the image of x in S ; the local ring $\mathcal{O}_{\widehat{X}/\widehat{S},x}$ is the separated completion of $\mathcal{O}_{X,x}$ for the linear topology defined by the ideals I of $\mathcal{O}_{X,x}$ such that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{S,s}$ -module (N.B. since $[\kappa(x) : \kappa(s)] < \infty$, this is equivalent to saying that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{X,x}$ -module).

Let $\mathbf{Vaf}_{/\widehat{S}}^{\mathrm{eff}}$ be the category of formal varieties of finite length over $\widehat{S}$ (identified with that of S -schemes of finite length). By 1.1 and 1.2.1, the category $\mathbf{Vaf}/\widehat{S}$ of formal varieties over $\widehat{S}$ is equivalent to that of contravariant left exact functors from $\mathbf{Vaf}_{/\widehat{S}}^{\mathrm{eff}}$ to (Ens) . In particular, for every S -scheme X , the functor $T \mapsto \mathrm{Hom}_{((\mathrm{Sch})/S)}(T, X)$, from $\mathbf{Vaf}_{/\widehat{S}}^{\mathrm{eff}}$ to (Ens) , is such a left exact functor and therefore corresponds to a formal variety over $\widehat{S}$. This is precisely $\widehat{X}/\widehat{S}$:⁽⁵³⁾

Proposition. *For every S -scheme X , the functors*

$$\mathrm{Hom}_{\mathbf{Vaf}/\widehat{S}}(-, \widehat{X}/\widehat{S}) \quad \text{and} \quad \mathrm{Hom}_{((\mathrm{Sch})/S)}(-, X)$$

⁽⁵¹⁾**N.D.E.** We have detailed the remainder of this paragraph.

⁽⁵²⁾**N.D.E.** We have added the following sentence.

⁽⁵³⁾**N.D.E.** We have expanded the following proposition by inserting the fact that the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with finite projective limits (in the original, this appeared in the proof of 1.3.4—the proof given here is more direct than the original). Moreover, this result will be useful in Section 2 and in 3.3.1.

have the same restriction to $\mathbf{Vaf}_{\widehat{S}}^{\text{eff}}$. In this way one obtains a functor $X \mapsto \widehat{X}/\widehat{S}$ from $((\text{Sch})/S)$ to $\mathbf{Vaf}/\widehat{S}$ which commutes with finite projective limits.

Indeed, one easily sees that the formal variety $\widehat{X}/\widehat{S}$ has the required property and that the correspondence $X \mapsto \widehat{X}/\widehat{S}$ is functorial. Let us prove the second assertion. Put $\mathcal{S} = ((\text{Sch})/S)$ and $\mathcal{V} = \mathbf{Vaf}/\widehat{S}$, and write $\widehat{X}$ instead of $\widehat{X}/\widehat{S}$. One knows (1.2.B) that $\mathcal{V}$ has arbitrary projective limits. Let $(X_i)_{i \in I}$ be a projective system in $\mathcal{S}$, and suppose that $X = \varprojlim X_i$ exists in $\mathcal{S}$ (which is the case if I is finite). Since the functor which associates to every object Y of $\mathcal{S}$ (resp. V of $\mathcal{V}$) the functor $h_Y = \text{Hom}_{\mathcal{S}}(-, Y)$ (resp. $h_V = \text{Hom}_{\mathcal{V}}(-, V)$) commutes with projective limits, for every S -scheme T of finite length one has functorial isomorphisms

$$\begin{aligned} \text{Hom}_{\mathcal{S}}(T, X) &\simeq \varprojlim \text{Hom}_{\mathcal{S}}(T, X_i) \\ &\simeq \varprojlim \text{Hom}_{\mathcal{V}}(T, \widehat{X}_i) \simeq \text{Hom}_{\mathcal{V}}(T, \varprojlim \widehat{X}_i). \end{aligned}$$

Consequently, the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with projective limits when they exist in $((\text{Sch})/S)$; in particular, it commutes with finite projective limits.

1.3. Proposition. *Let $f : X \rightarrow Y$ be a morphism of formal varieties over k , A and B the affine algebras of X and Y , and $g : B \rightarrow A$ the morphism induced by f . Then f is a monomorphism in $\mathbf{Vaf}/k$ if and only if g is a surjection.*

⁽⁵⁴⁾By 1.1, $A \mapsto \text{Spf}(A)$ is an anti-equivalence from $\mathbf{Alp}/k$ to $\mathbf{Vaf}/k$, so f is a monomorphism if and only if g is an epimorphism, and this is indeed the case if g is surjective. Conversely, suppose that g is an epimorphism and let us show that it is surjective. For every $\mathfrak{n} \in \Upsilon(B)$, put $A_{\mathfrak{n}} = A \widehat{\otimes}_B B_{\mathfrak{n}}$; by 0.4, A is a pseudocompact B -module, hence is the product of the $A_{\mathfrak{n}}$ (cf. 0.3.6). Then g is the product of the morphisms $g_{\mathfrak{n}} : B_{\mathfrak{n}} \rightarrow A_{\mathfrak{n}}$ deduced from g by base change. These are still epimorphisms, which reduces us to proving the result when B is local, with maximal ideal $\mathfrak{n}$. Put $K = B/\mathfrak{n}$. By Nakayama's lemma 0.3.3, it suffices to show that the morphism $g \widehat{\otimes}_B K$ is surjective; it is deduced from g by base change, hence is an epimorphism in $\mathbf{Alp}/K$. One can therefore suppose that $B = K$ is a field. Now f is a monomorphism if and only if the diagonal morphism $X \rightarrow X \times_Y X$ is an isomorphism, that is, if the homomorphism $x \widehat{\otimes}_K x' \mapsto xx'$ is an isomorphism from $A \widehat{\otimes}_K A$ to A . Since K is a field, this implies $A = K$.

Remark. ⁽⁵⁵⁾ It follows from the proposition that every monomorphism $f : X \rightarrow Y$ of formal varieties is an isomorphism from X to a formal subvariety (necessarily closed!) of Y .

1.3.1. — The preceding proposition implies, in particular, that every monomorphism $f : X \rightarrow Y$ in $\mathbf{Vaf}/k$ is effective (cf. IV 1.3).⁽⁵⁶⁾ The same does not hold for epimorphisms, as one easily sees by slightly modifying the counterexample of Exp. V, § 2.c);⁽⁵⁷⁾ this is why we shall consider a pleasant class of effective epimorphisms.

⁽⁵⁴⁾**N.D.E.** We have detailed the beginning of the proof.

⁽⁵⁵⁾**N.D.E.** We have added this remark.

⁽⁵⁶⁾**N.D.E.** i.e. in the opposite category $(\mathbf{Vaf}/k)^0 = \mathbf{Alp}/k$, the morphism $g : B \rightarrow A$ corresponding to f is an effective epimorphism. This is indeed the case, since g is surjective and hence induces (cf. the proof of 0.2.B) an isomorphism of profinite k -algebras $B/I \xrightarrow{\sim} A$, where $I = \text{Ker } g$. Consequently, every morphism $\phi : B \rightarrow C$ of $\mathbf{Alp}/k$ which vanishes on I descends to a morphism $\bar{\phi} : A \rightarrow C$ such that $\phi = \bar{\phi} \circ g$.

⁽⁵⁷⁾**N.D.E.** i.e. let k be a field, $X = \text{Spf}(k[[T]])$, and $Y = \text{Spf}(B)$, where B is the k -subalgebra of $A = k[[T]]$ topologically generated by T^3 and T^4 (i.e. B consists of the formal power series $\sum a_n T^n$ such that $a_n = 0$ for $n = 1, 2, 5$). Then $X \rightarrow Y$ is an epimorphism which is not effective; indeed, the cokernel of $X \times_Y X \rightarrow X$ is $\text{Spf}(B')$, where B' is the subalgebra of A consisting of the a such that $a \widehat{\otimes} 1 = 1 \widehat{\otimes} a$, and B' contains T^5 .

Lemma. ⁽⁵⁸⁾ *Let $f : X \rightarrow Y$ be a morphism of formal k -varieties, let A, B be the affine algebras of X, Y , and let $f^\sharp : B \rightarrow A$ be the morphism induced by f . The following conditions are equivalent:*

- (i) *For every $x \in X$, the homomorphism $f_x^\sharp : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ makes $\mathcal{O}_{X, x}$ a topologically free pseudocompact $\mathcal{O}_{Y, f(x)}$ -module.*
- (ii) *For every $y \in Y$, the local component $A_y = \prod_{f(x)=y} \mathcal{O}_{X, x}$ is a topologically free pseudocompact B_y -module.*
- (iii) *$f^\sharp : B \rightarrow A$ makes A a projective pseudocompact B -module.*
- (iv) *The functor $\mathbf{PC}(B) \rightarrow \mathbf{PC}(A)$, $M \mapsto M \hat{\otimes}_B A$, is exact.*

If these conditions hold, one says that f is topologically flat.

The implications (i) $\Rightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv) follow from 0.2.F(iii) and 0.3.7. Conversely, suppose that (ii) holds, and let $x \in X$ and $y = f(x)$. Since $\mathcal{O}_{X, x}$ is a direct factor of A_y , it is a projective pseudocompact B_y -module, hence topologically free by 0.2.1 (since B_y is local). On the other hand, a morphism $f : X \rightarrow Y$ of formal k -varieties is said to be *surjective* if it induces a surjection of the underlying sets.

Proposition. *Let $f : X \rightarrow Y$ be a surjective and topologically flat morphism of formal k -varieties. Then f is an effective epimorphism (cf. IV 1.3).*

Indeed, let A, B be the affine algebras of X, Y , and $g : B \rightarrow A$ the morphism induced by f . One must show that Y is identified with the cokernel of $X \times_Y X \rightrightarrows X$, i.e. that, for every $\mathfrak{n} \in \Upsilon(B)$, $B_{\mathfrak{n}}$ is identified with the subring of $A_{\mathfrak{n}} = A \hat{\otimes}_B B_{\mathfrak{n}}$ consisting of the a such that $a \hat{\otimes} 1 = 1 \hat{\otimes} a$. One can therefore suppose that B is local, with maximal ideal $\mathfrak{n}$. Our hypothesis then means that g makes A a nonzero topologically free B -module. By Nakayama's lemma 0.3.3, $A/\mathfrak{n}A$ is nonzero, so the morphism $g' : B/\mathfrak{n} \rightarrow A/\mathfrak{n}A$ deduced from g is injective. By Lemma 1.3.2 below, B is a direct factor of A as a B -module, say $A = B \oplus A'$; it follows that B is identified with the part of A consisting of the a such that $a \hat{\otimes}_B 1 = 1 \hat{\otimes}_B a$.

1.3.2. Lemma. *Let B be a local pseudocompact ring, $\mathfrak{n}$ its maximal ideal, M, N two projective pseudocompact B -modules, and g a morphism $M \rightarrow N$. If $(B/\mathfrak{n}) \hat{\otimes}_B g$ is injective, g is an isomorphism from M to a direct factor of N .*

Indeed, suppose that $g' = (B/\mathfrak{n}) \hat{\otimes}_B g$ is injective. Since $B/\mathfrak{n}$ is a field, g' then has a retraction r' . Denote by p and q the canonical projections from M and N to $M/\mathfrak{n}M$ and $N/\mathfrak{n}N$; since N is projective, there exists a morphism $r : N \rightarrow M$ such that $p \circ r = r' \circ q$; consequently, r' is deduced from r by passage to the quotient. Then, since $r' \circ g'$ is an isomorphism, so is $r \circ g$, by 0.3.4 (since M is projective). Let s be the inverse isomorphism of $r \circ g$; then $s \circ r$ is a retraction of g .

1.3.3. Proposition. *Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms of formal k -varieties.*

- (i) *If f and g are topologically flat, so is $g \circ f$.*
- (ii) *If f and $g \circ f$ are topologically flat and f is surjective, g is topologically flat.*
- (iii) *If f is topologically flat, so is $f \times_Y Y'$ for every base change $Y' \rightarrow Y$.*

Assertions (i) and (iii) are clear. To prove (ii), call A, B, C the affine algebras of X, Y, Z , and $f' : B \rightarrow A$, $g' : C \rightarrow B$ the morphisms induced by f and g . Since $g \circ f$ is topologically flat, $f' \circ g'$ makes A a projective pseudocompact C -module; similarly, f' makes A a faithful projective pseudocompact B -module. When P runs through the pseudocompact C -modules and N through the pseudocompact B -modules, the functors $P \mapsto P \hat{\otimes}_C A$ and $N \mapsto N \hat{\otimes}_B A$ are therefore exact; since the second is moreover faithful, the functor $P \mapsto P \hat{\otimes}_C B$ is exact; by 0.3.7, B is therefore a projective pseudocompact C -module.

⁽⁵⁸⁾ **N.D.E.** We have added this lemma, which explains the terminology “topologically flat”.

1.3.4. Proposition. *Let S be a scheme, Y a locally noetherian S -scheme, and $f : X \rightarrow Y$ a faithfully flat S -morphism locally of finite type, so that f is an effective epimorphism, i.e. the sequence below is exact (cf. IV 6.3.1(iv) and IV 1.3):*

$$(*) \quad X \times_Y X \xrightarrow[\text{pr}_2]{\text{pr}_1} X \xrightarrow{f} Y.$$

Then the morphism of formal $\widehat{S}$ -varieties $\widehat{f} : \widehat{X}/\widehat{S} \rightarrow \widehat{Y}/\widehat{S}$ (cf. 1.2.6) is surjective and topologically flat, and the sequence below deduced from $()$ is exact:*

$$(\widehat{*}) \quad \widehat{X \times_Y X} / \widehat{S} \xrightarrow[\widehat{\text{pr}_2}]{\widehat{\text{pr}_1}} \widehat{X}/\widehat{S} \xrightarrow{\widehat{f}} \widehat{Y}/\widehat{S}.$$

Indeed, let y be a point of Y with projection s on S such that $\kappa(y)$ is a finite extension of the residue field $\kappa(s)$ of s . Since f is surjective and locally of finite type, $f^{-1}(y)$ is nonempty and locally of finite type over $\kappa(y)$; the closed points of $f^{-1}(y)$ are then the points of $\widehat{X}/\widehat{S}$ projecting to y . This shows that $\widehat{f}$ is surjective. ⁽⁵⁹⁾Let x be a closed point of $f^{-1}(y)$. Since Y is locally noetherian and f locally of finite type, the local ring $\mathcal{O}_{Y,y}$ (resp. $\mathcal{O}_{X,x}$) is noetherian, so the powers of the maximal ideal have finite colength, and thus the local ring of $\widehat{Y}/\widehat{S}$ at y (resp. of $\widehat{X}/\widehat{S}$ at x) is the completion $\widehat{\mathcal{O}}_{Y,y}$ of $\mathcal{O}_{Y,y}$ (resp. $\widehat{\mathcal{O}}_{X,x}$ of $\mathcal{O}_{X,x}$) for the $\mathfrak{m}$ -adic topology. Then, since f is flat, $\widehat{\mathcal{O}}_{X,x}$ is flat over $\widehat{\mathcal{O}}_{Y,y}$ by SGA 1, IV 5.8. Thus, by 0.3.8, $\widehat{\mathcal{O}}_{X,x}$ is a topologically free $\widehat{\mathcal{O}}_{Y,y}$ -module. This shows that $\widehat{f}$ is topologically flat. Thus, by Proposition 1.3.1, $\widehat{f}$ is an effective epimorphism, i.e. the sequence below (where we have written $\widehat{X}$ instead of $\widehat{X}/\widehat{S}$) is exact:

$$\widehat{X} \times_{\widehat{Y}} \widehat{X} \xrightarrow[\widehat{\text{pr}_2}]{\widehat{\text{pr}_1}} \widehat{X} \xrightarrow{\widehat{f}} \widehat{Y}.$$

Moreover, by 1.2.6, one has a natural isomorphism (in particular, commuting with the projections to $\widehat{X}$)

$$\widehat{X \times_Y X} \simeq \widehat{X} \times_{\widehat{Y}} \widehat{X}.$$

Consequently, sequence $(\widehat{*})$ is exact.

1.3.5. — Let k be a pseudocompact ring. A formal variety X over k is said to be *topologically flat* if its affine algebra A is a *projective* pseudocompact k -module, i.e. if the structural morphism $X \rightarrow \text{Spf}(k)$ is topologically flat. ⁽⁶⁰⁾First note that 0.2.2 and 0.3.6 imply the following result (analogous to VII A, 3.1.1).

Lemma 1.3.5.A. *Suppose that k is artinian. The functors $A \mapsto A^\dagger = \text{Hom}_c(A, k)$ and $C \mapsto C^* = \text{Hom}_k(C, k)$ define an anti-equivalence between the category of topologically flat profinite k -algebras and that of flat k -coalgebras.*

Indeed, if A is a topologically flat profinite k -algebra, then, by 0.3.6, one has an isomorphism of k -modules

$$A^\dagger \otimes_k A^\dagger \xrightarrow{\sim} (A \widehat{\otimes} A)^\dagger,$$

so multiplication $A \widehat{\otimes} A \rightarrow A$ induces by duality a k -coalgebra structure on $A^\dagger$. The rest then follows from Proposition 0.2.2. Let us return to the case of an arbitrary pseudocompact ring k .

Definition 1.3.5.B. To every formal k -variety X whose affine ring A is a *projective* pseudocompact k -module, one associates a flat $\mathbf{O}_k$ -coalgebra $\mathbf{H}(X)$, defined as follows. Denote by

⁽⁵⁹⁾**N.D.E.** We have detailed what follows; we have then taken advantage of the addition made in 1.2.6.

⁽⁶⁰⁾**N.D.E.** We have added the following lemma, used implicitly in the original; moreover, we have introduced the numbering 1.3.5.A to 1.3.5.D for later references.

$\mathbf{H}(X)$ the $\mathbf{O}_k$ -module $\mathbf{V}_k^f(A)$ “dual to A ”; it is a flat $\mathbf{O}_k$ -module, since the pseudocompact k -module underlying A is projective (cf. 1.2.3.C). Moreover, by 0.3.6, one has

$$\mathbf{V}_k^f(A \hat{\otimes} A) \simeq \mathbf{V}_k^f(A) \otimes \mathbf{V}_k^f(A),$$

and thus multiplication on A induces by transposition a diagonal morphism

$$\mathbf{H}(X) = \mathbf{V}_k^f(A) \longrightarrow \mathbf{V}_k^f(A \hat{\otimes} A) = \mathbf{H}(X) \otimes \mathbf{H}(X)$$

which makes $\mathbf{H}(X)$ a flat $\mathbf{O}_k$ -coalgebra. We shall say that $\mathbf{H}(X)$ is *the coalgebra of X over $\mathbf{O}_k$* .

Definition 1.3.5.C. Conversely, to every $\mathbf{O}_k$ -coalgebra $\mathbf{C}$ one can associate a k -functor (cf. 1.2.1) $\mathrm{Spf}^*(\mathbf{C})$, defined as follows. For every object B of $\mathbf{AIf}/k$, put (with the notation of VII A 3.1)

$$(1) \quad \begin{aligned} \mathrm{Spf}^*(\mathbf{C})(B) &= \mathrm{Hom}_{B\text{-coalg.}}(B, \mathbf{C}(B)) \\ &= \{x \in \mathbf{C}(B) \mid \varepsilon_{\mathbf{C}(B)}(x) = 1 \text{ and } \Delta_{\mathbf{C}(B)}(x) = x \otimes x\}. \end{aligned}$$

⁽⁶¹⁾ Suppose further that the $\mathbf{O}_k$ -module underlying $\mathbf{C}$ is *admissible* (cf. 1.2.1), and put

$$(2) \quad A = \Gamma^*(\mathbf{C}) = \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^*.$$

Then the algebra structures on each $\mathbf{C}(k/\mathfrak{l})^*$ equip A with a profinite k -algebra structure. For every object B of $\mathbf{AIf}/k$, one has

$$(3) \quad \begin{aligned} \mathrm{Hom}_{\mathbf{Vaf}/k}(\mathrm{Spf}(B), \mathrm{Spf}(A)) &= \mathrm{Hom}_{\mathbf{AIf}/k}(A, B) \\ &= \mathrm{Hom}_{\mathbf{AIf}/B}(A \hat{\otimes} B, B). \end{aligned}$$

Finally, suppose that $\mathbf{C}$ is a flat $\mathbf{O}_k$ -module. Then, by 1.2.3.E, $A = \Gamma^*(\mathbf{C})$ is a *projective* pseudocompact k -module. Moreover, in the proof of *loc. cit.* one has seen that if $\mathfrak{l}_0$ is an open ideal of k contained in the kernel of $k \longrightarrow B$, one has isomorphisms

$$(4) \quad \begin{aligned} A \hat{\otimes} B &= \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^* \hat{\otimes}_k B \simeq \mathrm{Hom}_{k/\mathfrak{l}_0}(\mathbf{C}(k/\mathfrak{l}_0), B) \\ &\simeq \mathrm{Hom}_B(\mathbf{C}(B), B), \end{aligned}$$

and we shall denote the right-hand term by $\mathbf{C}(B)^*$. Finally, by Lemma 1.3.5.A applied to the artinian ring B , one has a natural isomorphism

$$(5) \quad \mathrm{Hom}_{B\text{-coalg.}}(B, \mathbf{C}(B)) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{AIf}/B}(\mathbf{C}(B)^*, B).$$

Then, combining (1), (5), (4), (3), and (2), when $\mathbf{C}$ is a flat $\mathbf{O}_k$ -coalgebra one obtains an isomorphism of functors

$$(\star) \quad \mathrm{Spf}^*(\mathbf{C}) \simeq \mathrm{Spf}(A) = \mathrm{Spf}(\Gamma^*(\mathbf{C})).$$

Consequently, if one denotes by $\mathscr{A}(X)$ the affine algebra of a formal k -variety X , taking 1.2.3.E into account one obtains:

Proposition 1.3.5.D.

- (i) *The functors $X \longmapsto \mathbf{H}(X) = \mathbf{V}_k^f(\mathscr{A}(X))$ and $\mathbf{C} \longmapsto \mathrm{Spf}^*(\mathbf{C}) = \mathrm{Spf}(\Gamma^*(\mathbf{C}))$ define an equivalence between the category of topologically flat formal k -varieties and that of flat $\mathbf{O}_k$ -coalgebras.*
- (ii) *Moreover, this equivalence “commutes with base change”: if $k \longrightarrow K$ is a morphism of pseudocompact rings, $X \hat{\otimes}_k K$ corresponds to $\mathbf{H}(X) \otimes_k K$.*

⁽⁶¹⁾ **N.D.E.** We have detailed what follows.

1.3.6. — In the remainder of this exposé, we shall repeatedly define a topologically flat formal k -variety X by exhibiting the coalgebra $\mathbf{H}(X)$. We shall then have to interpret certain geometric properties of X by means of $\mathbf{H}(X)$. Here we give an example of this situation: suppose that a section σ of the structural morphism $X \rightarrow \mathrm{Spf}(k)$ is given, and ask under what condition σ induces an isomorphism of the underlying topological spaces.⁽⁶²⁾ To begin with, suppose that k is artinian. Let (H, Δ, ε) be a flat k -coalgebra, $H^+ = \mathrm{Ker}(\varepsilon)$, and $A = H^*$ the profinite k -algebra dual to H . Suppose that a morphism of k -coalgebras $k \rightarrow H$ is given, i.e. an element φ of H such that $\varepsilon(\varphi) = 1$ and $\Delta(\varphi) = \varphi \otimes \varphi$. On the one hand, φ defines a continuous morphism of algebras $\Phi : A \rightarrow k$, and hence a section $\sigma : \mathrm{Spf}(k) \rightarrow \mathrm{Spf}(A)$ of the projection $\mathrm{Spf}(A) \rightarrow \mathrm{Spf}(k)$. On the other hand, one defines k -submodules of H by putting $H_0 = k\varphi$ and, for $n \geq 1$,

$$H_n = \{x \in H \mid \Delta(x) - x \otimes \varphi \in H_{n-1} \otimes H^+\};$$

this also holds for $n = 0$ if one puts $H_{-1} = (0)$. One sees by induction on n that $H_{n-1} \subset H_n$. One calls $H_0 \subset H_1 \subset \dots$ the *filtration of H defined by φ* .

Remark. Since $\Delta(H_n) \subset H_n \otimes H_0 \oplus H_{n-1} \otimes H^+$, one has $\Delta(H_n) \subset H_n \otimes H$. Since Δ is cocommutative (i.e. $\sigma \circ \Delta = \Delta$, where $\sigma(a \otimes b) = b \otimes a$), one also has $\Delta(H_n) \subset H \otimes H_n$. When H/H_n is flat over k , it follows that H_n is a subcoalgebra of H (see also 1.3.6.A(iii) below). But in general $\Delta : H_n \rightarrow H_n \otimes H$ does not factor through $H_n \otimes H_n$.⁽⁶³⁾

Lemma 1.3.6.A. *Let k be an artinian ring, H a flat k -coalgebra, $A = H^*$ the dual profinite k -algebra, and φ an element of H such that $\varepsilon(\varphi) = 1$ and $\Delta(\varphi) = \varphi \otimes \varphi$. Let $\Phi : A \rightarrow k$ be the continuous morphism of algebras, $\sigma : \mathrm{Spf}(k) \rightarrow \mathrm{Spf}(A)$ the section of $\mathrm{Spf}(A) \rightarrow \mathrm{Spf}(k)$, and (H_n) the filtration of H defined by φ . Denote by I the kernel of Φ .*

- (i) *For every $n \geq 1$, H_{n-1} is the orthogonal in H of the closure $\overline{I^n}$ of I^n .*
- (ii) *Consequently, σ induces a bijection of the underlying sets if and only if $H = \bigcup_n H_n$.⁽⁶⁴⁾*
- (iii) *If, moreover, each H/H_n is flat over k , then, for every $n \geq 0$,*

$$(*) \quad \Delta(H_n) \subset \sum_{i=0}^n H_i \otimes H_{n-i};$$

in particular, each H_n is then a subcoalgebra of H .

PROOF. Note that, for every $x \in H$, the map $A \rightarrow k$, $f \mapsto f(x)$ is continuous, so if I^n is annihilated by x , the same holds for its closure $\overline{I^n}$. For every $n \geq 1$, one then puts

$$(\overline{I^n})^\perp = \{x \in H \mid f(x) = 0 \text{ for every } f \in \overline{I^n}\}.$$

Suppose that $\sigma : \mathfrak{m} \mapsto \Phi^{-1}(\mathfrak{m})$ is a bijection from $\mathrm{Spf}(k)$ to $\mathrm{Spf}(A)$. Since I is contained in the intersection of the $\Phi^{-1}(\mathfrak{m})$, it follows from 0.1.2 that the sequence of ideals (I^n) tends to 0. Let $x \in H$; since $J(x) = \{f \in A \mid f(x) = 0\}$ is an open and closed k -submodule of A , it contains I^n for sufficiently large n ; in other words, $x \in (\overline{I^n})^\perp$ for sufficiently large n . Conversely, suppose that $H = \bigcup_n (\overline{I^n})^\perp$. Let $\mathfrak{p}$ be an open prime ideal of A ; by the definition of the topology of A (0.2.2), $\mathfrak{p}$ contains an open k -submodule of the form

$$\mathcal{V}(x_1, \dots, x_s) = \{f \in A \mid f(x_1) = \dots = f(x_s) = 0\}.$$

⁽⁶²⁾**N.D.E.** In Lemma 1.3.6.A which follows, we have detailed the proof of parts (i) and (ii), and have added part (iii).

⁽⁶³⁾**N.D.E.** For example, let k_0 be a field, $k = k_0[T]/(T^n)$, where $n \geq 4$, and $H = k\varphi \oplus kx$, with $\varepsilon(x) = 0$ and $\Delta(x) = x \otimes \varphi + \varphi \otimes x + tx \otimes x$, where t is the image of T in k . Then $H_i = k\varphi \oplus t^{n-i}x$ for $i = 0, \dots, n$, but for $2 \leq i \leq n-2$, $\Delta(t^{n-i}x)$ does not belong to the image of $H_i \otimes H_i$ in $H \otimes H$.

⁽⁶⁴⁾**N.D.E.** In this case one says that the coalgebra H is connected; cf. addition 2.9.

By hypothesis, there exists an integer n such that $x_1, \dots, x_s \in (\overline{I^n})^\perp$, hence $I^n \subset \mathfrak{p}$. Moreover, since k is artinian, $\mathrm{Spf}(k)$ is a finite set $\{\mathfrak{m}_1, \dots, \mathfrak{m}_r\}$ and there exists an integer $t \geq 1$ such that $(\prod_i \mathfrak{m}_i)^t = 0$, whence

$$\prod_i \Phi^{-1}(\mathfrak{m}_i)^t \subset I.$$

Thus $\mathfrak{p}$ contains the product of the $\Phi^{-1}(\mathfrak{m}_i)^{tn}$; since $\mathfrak{p}$ is prime, it follows that $\mathfrak{p}$ contains, hence equals, one of the $\Phi^{-1}(\mathfrak{m}_i)$. One has thus proved that

$$\sigma \text{ is a bijection} \iff H = \bigcup_n (\overline{I^n})^\perp.$$

(65) On the other hand, one has $H = k\varphi \oplus H^+$; denote by π the projection $H \rightarrow H^+$ with kernel $k\varphi$. For every $n \geq 0$, let Δ^n be the “iterated” comultiplication $H \rightarrow H^{\otimes(n+1)}$, $\overline{\Delta}^n$ the composite of Δ^n with the projection $\pi^{\otimes(n+1)} : H^{\otimes(n+1)} \rightarrow (H^+)^{\otimes(n+1)}$, and

$$H'_n = \mathrm{Ker}(\overline{\Delta}^n) = \{x \in H \mid \Delta^n(x) \in \sum_{i=0}^n H^{\otimes(n-i)} \otimes H_0 \otimes H^{\otimes i}\}.$$

(One puts $\Delta^0 = \mathrm{id}_H$, whence $H'_0 = H_0$.) One easily sees, by induction on n , that

$$(*) \quad H_n \subset H'_n \subset (\overline{I^{(n+1)}})^\perp.$$

Until now, one has not used the hypothesis that H is flat over k . Now suppose that H is flat, hence projective over k , so that $A^\dagger = H$ by 0.2.2, and let us show that $H_n = (\overline{I^{(n+1)}})^\perp$. This is clear for $n = 0$. Suppose therefore that it holds for $n < r$. Then H_{r-1} is the kernel of the morphism $H \rightarrow (\overline{I^r})^\dagger$, and thus, since H^+ is flat, the morphism

$$(H/H_{r-1}) \otimes H^+ \rightarrow (\overline{I^r})^\dagger \otimes H^+$$

is injective. On the other hand, the hypothesis implies that I is a projective pseudocompact k -module (since it is a direct factor of $A = H^*$), whence, by 0.3.6,

$$(\overline{I^r} \widehat{\otimes} I)^\dagger \simeq (\overline{I^r})^\dagger \otimes I^\dagger = (\overline{I^r})^\dagger \otimes H^+.$$

Then the exact sequence

$$\overline{I^r} \widehat{\otimes} I \rightarrow A \rightarrow A/\overline{I^{r+1}} \rightarrow 0$$

gives by duality the exact sequence

$$(1) \quad (\overline{I^r})^\dagger \otimes H^+ \xleftarrow{\delta} H \leftarrow (A/\overline{I^{r+1}})^\dagger \leftarrow 0,$$

where δ is obtained by composing Δ_H with the projection

$$(2) \quad H \otimes H \rightarrow H \otimes H^+ \rightarrow (H/H_{r-1}) \otimes H^+ \hookrightarrow (\overline{I^r})^\dagger \otimes H^+.$$

Now, for every $u \in H$, the projection of $\Delta(u)$ to $H \otimes H^+$ is $\Delta(u) - u \otimes \varphi$. Then (1) and (2) show that, if u belongs to $(A/\overline{I^{r+1}})^\dagger = (\overline{I^{r+1}})^\perp$, $\Delta(u) - u \otimes \varphi$ belongs to the kernel of the map $H \otimes H^+ \rightarrow (H/H_{r-1}) \otimes H^+$, that is, to $H_{r-1} \otimes H^+$, and hence $u \in H_r$. This completes the proof of parts (i) and (ii), and also shows that $H_n = \mathrm{Ker}(\overline{\Delta}^n)$. Let us prove (iii). For every $i \geq 0$, put $H_i^+ = H_i \cap H^+$. Let $n \geq 1$. For every $x \in H_n$, $\bar{x} = x - \varepsilon(x)\varphi$ belongs to H_n^+ , and one has

$$\Delta(x) = \varepsilon(x)\varphi \otimes \varphi + \bar{x} \otimes \varphi + \varphi \otimes \bar{x} + \overline{\Delta}(\bar{x}).$$

Thus it suffices to show that

$$\overline{\Delta}(H_n^+) \subset \sum_{i=1}^{n-1} H_i^+ \otimes H_{n-i}^+.$$

(65) **N.D.E.** This also holds without assuming H cocommutative (k remaining a commutative artinian ring): in this case, a base of neighborhoods of 0 in $A = H^*$ is given by the two-sided ideals J such that A/J has finite length over k , and the preceding proof shows that $H = \bigcup_n (\overline{I^n})^\perp$ if and only if the $\Phi^{-1}(\mathfrak{m}_i)$ are the only open prime ideals of A .

For every $i = 0, \dots, n-1$, $\overline{\Delta}^n$ factors as

$$\begin{array}{ccc} H^+ & \xrightarrow{\overline{\Delta}} & H^+ \otimes H^+ \xrightarrow{\overline{\Delta}^i \otimes \overline{\Delta}^{n-i-1}} (H^+)^{\otimes(i+1)} \otimes (H^+)^{\otimes(n-i)} \\ & \downarrow & \uparrow g \\ \frac{H^+}{H_i^+} \otimes \frac{H^+}{H_{n-i-1}^+} & \xrightarrow{f} & \frac{H^+}{H_i^+} \otimes (H^+)^{\otimes(n-i)}. \end{array}$$

Moreover, since H^+/H_i^+ and $(H^+)^{\otimes(n-i)}$ are flat, the maps f and g above are injective. It follows that $\overline{\Delta}(H_n^+)$ is contained in $H_i^+ \otimes H^+ + H^+ \otimes H_{n-i-1}^+$ for every $i = 0, \dots, n-1$. Part (iii) then follows from the lemma below applied to $M = H^+$ and $E_i = H_{i-1}^+$. $\square$

Lemma 1.3.6.B. *Let k be a ring and $0 = E_0 \subset E_1 \subset \dots \subset E_n \subset M$ be k -modules. Suppose that M/E_i is flat for every i . Then one has the equality*

$$\bigcap_{i=0}^n (E_i \otimes M + M \otimes E_i) = \sum_{i=1}^n E_i \otimes E_{n-i+1}.$$

Denote by K (resp. S) the left-hand (resp. right-hand) term. One easily sees that $S \subset K$; let us prove the converse. For $i = 0, \dots, n$, put $K_i = K \cap (E_i \otimes M)$. For every $i = 1, \dots, n$, since M/E_{n-i+1} and E_i/E_{i-1} are flat, the map τ_i below is injective, and the composite map

$$\begin{aligned} (E_i/E_{i-1}) \otimes M &\longrightarrow (E_i/E_{i-1}) \otimes (M/E_{n-i+1}) \\ &\xrightarrow{\tau_i} (M/E_{i-1}) \otimes (M/E_{n-i+1}) \end{aligned}$$

has kernel $(E_i/E_{i-1}) \otimes E_{n-i+1}$. Since the image of K_i in $(E_i \otimes M)/(E_{i-1} \otimes M)$ is contained in, and contains, this kernel, one deduces that

$$K_i = K_{i-1} + E_i \otimes E_{n-i+1},$$

whence the lemma. ⁽⁶⁶⁾To finish this paragraph, let us return to an arbitrary pseudocompact ring k . Let $(\mathbf{H}, \Delta, \varepsilon)$ be a flat $\mathbf{O}_k$ -coalgebra, $\mathbf{H}^+ = \text{Ker}(\varepsilon)$, $A = \Gamma^*(\mathbf{H})$ the dual profinite k -algebra, and $X = \text{Spf}(A)$, so that $\mathbf{H} = \mathbf{H}(X)$ (cf. 1.3.5). Suppose that a morphism of $\mathbf{O}_k$ -coalgebras $\varphi : \mathbf{O}_k \rightarrow \mathbf{H}$ is given; it defines a continuous morphism of k -algebras $A \rightarrow k$, and hence a section σ of the structural morphism $X \rightarrow \text{Spf}(k)$. For every object B of $\mathbf{Alf}/k$, write $\mathbf{H}_0(B) = \varphi(B) = B\varphi_B$, where φ_B is the element $\varphi(1_B)$ of $\mathbf{H}(B)$, and define $\mathbf{O}_k$ -submodules $\mathbf{H}_n$ of $\mathbf{H}$ by putting, for $n \geq 1$,

$$\mathbf{H}_n(B) = \{u \in \mathbf{H}(B) \mid \Delta(u) - u \otimes \varphi_B \in \mathbf{H}_{n-1}(B) \otimes \mathbf{H}^+(B)\}.$$

In this way one obtains a filtration $\mathbf{H}_0 \subset \mathbf{H}_1 \subset \dots$ of $\mathbf{H}(X)$. By the foregoing, one has:

Proposition. *For σ to induce an isomorphism on the underlying topological spaces, it is necessary and sufficient that $\mathbf{H}(X)$ be the union of the $\mathbf{H}_n$.*

1.4. Theorem. *Let k be a pseudocompact ring and $d_0, d_1 : X_1 \rightrightarrows X$ an equivalence pair in $\mathbf{Vaf}/k$ (cf. Exp. V, § 2.b) such that d_1 is topologically flat.*

- (i) *The canonical projection from X to X/X_1 ($= \text{Coker}(d_0, d_1)$) is surjective and topologically flat, and the morphism $X_1 \rightarrow X \times_{X/X_1} X$ with components d_0 and d_1 is an isomorphism.*
- (ii) *If X is topologically flat over k , so is X/X_1 .*

First note that (ii) follows from (i), by 1.3.3(ii). The proof of (i) occupies paragraphs 1.4.1, 1.4.2, and 1.4.3.

1.4.1. — Let us first show that one can reduce to the case where X has a single point. Since we are dealing with an equivalence pair, one sees as in Exp. V, § 3.e) that one defines an

⁽⁶⁶⁾**N.D.E.** We have added what follows; the original was restricted to the case where k is artinian.

equivalence relation on the set underlying X by declaring two points x, y equivalent if there exists a point z of X_1 such that $d_0(z) = y$ and $d_1(z) = x$. One can evidently suppose without loss that X contains a single equivalence class for this relation, in other words that X/X_1 has a single point (see the construction of X/X_1 given in 1.2). In this case, let x be any point of X and U the formal variety which has x as its sole point and has the same local ring as X at x . One then sees as in Exp. V, § 6 that the equivalence relation induced by (d_0, d_1) on U still satisfies the hypotheses of the theorem, and that it suffices to prove it for this latter equivalence relation (U is a “quasi-section”). Let us briefly recall the principle of the proof given in Exp. V, § 6. Put $V = d_0^{-1}(U) = U \times_{i, d_0} X_1$, where i is the inclusion of U in X ; let u and v be the morphisms with source V induced respectively by d_0 and d_1 :

$$X \xleftarrow{v} V \xrightarrow{u} U.$$

It is clear that u and v are topologically flat and u is surjective; since X contains a single equivalence class, v is surjective. If (v_0, v_1) is the inverse image of the equivalence pair (d_0, d_1) under v (cf. V, 3.a)), it follows from V, 3.c) and 3.d) that X/X_1 and the quotient of U by the equivalence relation induced by (d_0, d_1) are both identified with $\text{Coker}(v_0, v_1)$. One then sees, as in the proof of V, 6.1, that if the conclusion of Theorem 1.4 holds for U , it also holds for X .

1.4.2. — One is thus reduced to the case where X has a single point.⁽⁶⁷⁾ Consider then the following commutative diagram (cf. V, § 1, (0,1,2)):

$$\begin{array}{ccccc} X_2 & \xrightarrow{d'_1} & X_1 & \xrightarrow{d_0} & X \\ d'_2 \downarrow & & d'_0 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_1} & X & \longrightarrow & X/X_1 \\ & & d_0 \downarrow & & \end{array}$$

where X_2 is the fibered product $X_1 \times_{d_1, d_0} X_1$, and d'_0, d'_1, d'_2 are respectively the morphisms “ $(x, y, z) \mapsto (x, y)$ ”, “ $(x, y, z) \mapsto (x, z)$ ”, and “ $(x, y, z) \mapsto (y, z)$ ”.⁽⁶⁸⁾ If B, A, A_1, A_2 denote respectively the affine algebras of $X/X_1, X, X_1, X_2$, the preceding diagram induces a commutative diagram

$$\begin{array}{ccccc} A_2 & \xleftarrow{j_1} & A_1 & \xleftarrow{i_0} & A \\ j_2 \uparrow & & j_0 \uparrow & & \uparrow i \\ A_1 & \xleftarrow{i_1} & A & \xleftarrow{i} & B \\ & & i_0 \downarrow & & \end{array}$$

in which both rows are exact and the squares determined by (i_0, j_0) and (i_1, j_1) are cocartesian. Since the morphism $X_1 \rightarrow X \times X$ with components d_0, d_1 is a monomorphism by hypothesis, the morphism $A \hat{\otimes}_k A \rightarrow A_1$ with components i_0, i_1 is surjective, by Proposition 1.3. This means that i_1 makes A_1 a pseudocompact A -module (assumed topologically free) generated by $i_0(A)$. Since A is local, Lemma 1.4.3 below implies the existence of a topologically free k -module V and a morphism of pseudocompact k -modules $f : V \rightarrow A$ such that the morphism

$$\alpha_1 : A \hat{\otimes}_k V \rightarrow A_1, \quad a \hat{\otimes} v \mapsto i_1(a) \cdot i_0(f(v))$$

is invertible. Let $\alpha : B \hat{\otimes}_k V \rightarrow A$ and $\alpha_2 : A_1 \hat{\otimes}_k V \rightarrow A_2$ be the morphisms

$$b \hat{\otimes} v \mapsto i(b) \cdot f(v) \quad \text{and} \quad a_1 \hat{\otimes} v \mapsto j_2(a_1) \cdot j_0 i_0(f(v)).$$

⁽⁶⁷⁾N.D.E. One can therefore suppose that k is local.

⁽⁶⁸⁾N.D.E. Cf. Exp. V, § 2.b).

In the following commutative diagram, both rows are therefore exact and the two left-hand squares are cocartesian. Since α_1 is invertible, so is α_2 , hence so is α .

$$\begin{array}{ccccc} A_2 & \xleftarrow{j_1} & A_1 & \xleftarrow{i_0} & A \\ \alpha_2 \uparrow & & \alpha_1 \uparrow & & \alpha \uparrow \\ A_1 \widehat{\otimes}_k V & \xleftarrow{i_1 \widehat{\otimes} V} & A \widehat{\otimes}_k V & \xleftarrow{i \widehat{\otimes} V} & B \widehat{\otimes}_k V \\ & i_0 \widehat{\otimes} V & & & \end{array}$$

This shows, on the one hand, that A is topologically free over B with pseudobasis $f(V)$ (cf. 0.2.1), and that one obtains a pseudobasis of A_1 over A (where A_1 is considered as an A -module by means of i_1) by taking the image of $f(V)$ under i_0 ; this implies that the morphism $A \widehat{\otimes}_B A \rightarrow A_1$ with components i_1, i_0 is invertible:

$$A \widehat{\otimes}_B A \simeq A \widehat{\otimes}_B B \widehat{\otimes}_k V \simeq A \widehat{\otimes}_k V \simeq A_1.$$

This proves Theorem 1.4, modulo the following Lemma 1.4.3.

1.4.3. Lemma. *Let k be a pseudocompact ring, A a local profinite k -algebra, A_1 a topologically free A -module, and $i_0 : M \rightarrow A_1$ a morphism of pseudocompact k -modules. Suppose that the map*

$$A \widehat{\otimes}_k M \rightarrow A_1, \quad a \widehat{\otimes} m \mapsto a \cdot i_0(m)$$

is surjective. There then exist a topologically free k -module V and a morphism of pseudocompact k -modules $f : V \rightarrow M$ such that the map

$$A \widehat{\otimes}_k V \rightarrow A_1, \quad a \widehat{\otimes} v \mapsto a \cdot i_0(f(v))$$

is bijective.

Since every pseudocompact k -module is a quotient of a topologically free k -module (cf. N.D.E. (27)), one can suppose without loss that M is topologically free; take therefore for M the direct product of a family $(M_i)_{i \in I}$ of copies of k . In this case $A \widehat{\otimes}_k M$ is precisely the product $\prod_{i \in I} A \widehat{\otimes}_k M_i$. Since the map $a \widehat{\otimes} m \mapsto a \cdot i_0(m)$ is surjective and A_1 is projective, the kernel of this map is a direct factor of $A \widehat{\otimes}_k M$; since A is local, it follows from the exchange theorem (0.3.4) that this kernel has as a complement a partial product $\prod_{i \in J} A \widehat{\otimes}_k M_i$, where J denotes some subset of I . One can therefore take $V = \prod_{i \in J} M_i$.

1.5. — Let k be a pseudocompact ring.

Definition. We shall say that a family of morphisms $f_i : X_i \rightarrow X$ of $\mathbf{Vaf}/k$ is a *topologically flat surjective family* if the morphism $\coprod_i X_i \rightarrow X$ induced by the f_i is surjective and topologically flat; this means that each f_i is topologically flat and every point of X belongs to the image of at least one of the X_i .

It follows from 1.3.3 that topologically flat surjective families define a pretopology on $\mathbf{Vaf}/k$ (IV 4.2.5); the corresponding topology will be called the *flat topology* on $\mathbf{Vaf}/k$. By IV, 4.3.5, a functor $\mathbf{F} : (\mathbf{Vaf}/k)^0 \rightarrow (\mathbf{Ens})$ is a sheaf for the flat topology if and only if $\mathbf{F}$ transforms every direct sum into a direct product and the sequence

$$\mathbf{F}(Y) \xrightarrow{\mathbf{F}(f)} \mathbf{F}(X) \xrightarrow[\mathbf{F}(\text{pr}_2)]{\mathbf{F}(\text{pr}_1)} \mathbf{F}(X \times_Y X)$$

is exact for every surjective topologically flat morphism $f : X \rightarrow Y$. ⁽⁶⁹⁾By IV, 4.5, Proposition 1.3.1 implies that the flat topology is less fine than the canonical topology, i.e. for every object X of $\mathbf{Vaf}/k$, the functor $h_X : T \mapsto \text{Hom}_{\mathbf{Vaf}/k}(T, X)$ is a sheaf for the flat

⁽⁶⁹⁾**N.D.E.** We have modified the original in what follows. In particular, we have replaced the statement “if $d_0, d_1 : X_1 \rightrightarrows X$ is an equivalence relation such that d_1 is topologically flat, formation of the quotient commutes with h ” by the theorem below.

topology. (In what follows, one identifies X with h_X as usual (cf. Exp. I).) By IV, 4.6.5, one can reformulate Theorem 1.4 as follows.

Theorem. *Let k be a pseudocompact ring, $d_0, d_1 : X_1 \rightrightarrows X$ an equivalence pair in $\mathbf{Vaf}/k$, and X/X_1 the quotient formal variety (i.e. $\text{Coker}(d_0, d_1)$, cf. 1.2). If d_1 is topologically flat, X/X_1 represents the quotient sheaf for the flat topology.*

1.6. — To conclude these generalities on formal varieties, it remains to define briefly formal varieties étale over k .⁽⁷⁰⁾

Recollections 1.6.A.

- (i) First recall (cf. EGA 0_{IV}, 19.10.2) that a topological k -algebra A is said to be *formally étale over k* (for the given topologies on k and A , resp. for the discrete topologies) if, for every discrete topological k -algebra C (not necessarily artinian), and every nilpotent ideal J of C , every continuous morphism of k -algebras $A \rightarrow C/J$ lifts uniquely to a continuous morphism $A \rightarrow C$ (A being equipped with the given topology, resp. the discrete topology). One sees at once that this property is preserved by base change, i.e. for every morphism $k \rightarrow k'$ of pseudocompact rings, $A \widehat{\otimes}_k k'$ is then formally étale over k' . Moreover, one easily sees that it suffices to verify the lifting condition for every square-zero ideal J ; cf. EGA IV₄, 17.1.2(ii). One says that A is *étale over k* if it is formally étale over k for the discrete topologies and if, in addition, A is a finitely presented k -algebra (cf. EGA IV₄, 17.3.2(ii)). In what follows, k being a pseudocompact ring and A a profinite k -algebra, “formally étale” will be used in the sense of the given topologies (unless otherwise specified).
- (ii) Recall also that if $F \in k[T]$ is a monic polynomial of degree $d \geq 1$ which is separable (i.e. such that the ideal generated by F and its derivative F' is $k[T]$), then the k -algebra $B = k[T]/(F)$ (which is free of rank d over k , and equipped with the product topology) is formally étale over k . Indeed, let C be a discrete k -algebra (so that the kernel of $k \rightarrow C$ is an open ideal $\mathfrak{l}$ of k), J a square-zero ideal of C , and $\phi : B \rightarrow C/J$ a continuous morphism of k -algebras. Note that, since B is a free k -module of finite rank, $\mathfrak{l}B$ is an open ideal of B , so every lifting $\Phi : B \rightarrow C$ of ϕ is automatically continuous. Let t be the image of T in B and u_0 any lifting of $\phi(t)$ in C ; then $F(u_0) \in J$ (since $\phi(t)$ is a root of F); moreover, there exist $G, H \in k[T]$ such that $GF + HF' = 1$, whence $H(u_0)F'(u_0) = 1 - G(u_0)F(u_0)$, and the right-hand term is invertible, since $F(u_0)$ has square zero, so $F'(u_0)$ is invertible. Seek $h \in J$ such that $x = x_0 + h$ is a root of F ; this is equivalent to $0 = F(x) = F(u_0) + F'(u_0)h$, and since $F'(u_0)$ is invertible, this has the unique solution $h = -F'(u_0)^{-1}F(u_0) \in J$. Of course, the same proof (without making continuity assumptions for the morphisms $k \rightarrow C$, ϕ , and Φ) shows that B is also an étale k -algebra.
- (iii) Finally, recall that if A is a finite product $A_1 \times \cdots \times A_n$, then A is formally étale over k if and only if the A_i are.⁽⁷¹⁾ Indeed, it suffices to see this for $n = 2$; in this case let $e = 1_{A_1}$ be the idempotent such that $A_1 = Ae$ and $A_2 = A(1 - e)$, and suppose that a continuous morphism $A \rightarrow C/J$ is given, where J is a square-zero ideal. Since the polynomial $F = X^2 - X$ is separable (one has $F' = 2X - 1$ and $(F')^2 - 4F = 1$), the idempotent $\phi(e)$ of C/J lifts uniquely to an idempotent f of C ,

⁽⁷⁰⁾**N.D.E.** The original continued as follows: “A formal variety X over k is said to be étale if the diagonal morphism $\Delta_X : X \rightarrow X \times X$ is a local isomorphism, that is, if Δ_X induces an isomorphism from $\mathcal{O}_{X \times X, \Delta_X(x)}$ to $\mathcal{O}_{X, x}$ for every point x of X . One easily sees with the aid of SGA I that this formulation is equivalent to the following two: the formal variety X is topologically flat, and, for every point $x \in X$ projecting to $s \in \text{Spf}(k)$, $\mathcal{O}_{X, x} \widehat{\otimes}_k \kappa(s)$ is a finite separable extension of the residue field $\kappa(s)$ of s ; or again, if A denotes the affine algebra of X , the local components (0.1) of $A \widehat{\otimes}_k (k/\mathfrak{l})$ are finite étale algebras over $k/\mathfrak{l}$, for every open ideal $\mathfrak{l}$ of k .” In what follows, we have rectified the omission of the flatness hypothesis in the first condition above, and detailed the equivalence of these conditions.

⁽⁷¹⁾**N.D.E.** Let us point out in passing that the proof given in EGA 0_{IV}, 19.3.5(v) is erroneous.

whence $C = Cf \oplus C(1 - f)$, and then giving a lifting of Φ is equivalent to giving two morphisms $\Phi_1 : A_1 \rightarrow Cf$ and $\Phi_2 : A_2 \rightarrow C(1 - f)$ lifting the restrictions of ϕ to A_1 and A_2 . The same argument shows that if e is an idempotent of k such that $A = Ae$, then A is formally étale over k if and only if it is so over the localization k_e (which is identified with ke).

Now let X be a formal k -variety and A its affine algebra. If x is a point of X (i.e. an open maximal ideal $\mathfrak{m}$ of A) with image s in $\mathrm{Spf}(k)$, the local component of k corresponding to s will be denoted by $k_{\mathfrak{m}}$ or k_s .

Definition 1.6.B. The following conditions are equivalent:

- (a) A is *formally étale over k* .
- (b) For every $\mathfrak{m} \in \Upsilon(A)$, $A_{\mathfrak{m}}$ is formally étale over k (or over $k_{\mathfrak{m}}$).
- (c) For every open ideal $\mathfrak{l}$ of k , $A \widehat{\otimes}_k (k/\mathfrak{l}) = A/\overline{A\mathfrak{l}}$ is formally étale over $k/\mathfrak{l}$.
- (d) For every $\mathfrak{m} \in \Upsilon(A)$ and every open ideal $\mathfrak{l}$ of $k_{\mathfrak{m}}$, $A_{\mathfrak{m}}/\overline{A_{\mathfrak{m}}\mathfrak{l}}$ is formally étale over $k_{\mathfrak{m}}/\mathfrak{l}$.

One says that X is *étale over k* if it satisfies these conditions, and denotes by $\mathbf{Vaf}_{/k}^{\text{ét}}$ the full subcategory of $\mathbf{Vaf}/k$ consisting of formal varieties étale over k .⁽⁷²⁾

Observe that if $\phi : A \rightarrow C/J$ is a continuous morphism of k -algebras, where C is a discrete k -algebra and J a square-zero ideal, then $I = \mathrm{Ker}(\phi)$ is an open ideal of A , hence A/I is artinian, so I is contained in only finitely many open maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_r$, and therefore contains the product of the components $A_{\mathfrak{m}}$ for $\mathfrak{m} \neq \mathfrak{m}_i$, which equals $A(1 - e)$, where e denotes the idempotent of A such that $Ae = \prod_{i=1}^r A_{\mathfrak{m}_i}$. Thus $\phi(e) = 1_{C/J}$, and giving a continuous lifting of ϕ is equivalent to giving one of the morphism from $Ae \simeq A/A(1 - e)$ to C/J induced by ϕ . Moreover, one knows (cf. N.D.E. (24)) that $A \simeq \varprojlim_{\mathfrak{l}} A/\overline{A\mathfrak{l}}$. In view of these remarks and the preceding recollections, one easily obtains the equivalence of the indicated conditions.

Definitions 1.6.C. Write $\kappa(k) = \prod_{s \in \mathrm{Spf}(k)} \kappa(s)$, equipped with the product topology, i.e. the formal variety $\mathrm{Spf}(\kappa(k))$ is the direct sum of the $\mathrm{Spec} \kappa(s)$, for $s \in \mathrm{Spf}(k)$. Moreover, denote by $S_{\kappa(k)}$ the scheme which is the direct sum of the $\mathrm{Spec} \kappa(s)$, for $s \in \mathrm{Spf}(k)$. For every formal variety X over k , denote by $X_{\kappa} = X \widehat{\otimes}_k \kappa(k)$ the formal variety over $\kappa(k)$ obtained by base change, i.e. X_{κ} has the same points as X and for every $x \in X$ projecting to s on $\mathrm{Spf}(k)$, one has $\mathcal{O}_{X_{\kappa}, x} = \mathcal{O}_{X, x} \otimes_k \kappa(s)$. This base-change functor $\mathbf{Vaf}/k \rightarrow \mathbf{Vaf}/\kappa(k)$ sends $\mathbf{Vaf}_{/k}^{\text{ét}}$ into $\mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ (cf. 1.6.A(i)).

One then has (cf. SGA 1, I 6.2):

Lemma 1.6.D. For every $Y \in \mathrm{Ob} \mathbf{Vaf}_{/k}^{\text{ét}}$ and $X \in \mathrm{Ob} \mathbf{Vaf}/k$, the canonical map

$$\mathrm{Hom}_{\mathbf{Vaf}/k}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{Vaf}/\kappa(k)}(X_{\kappa}, Y_{\kappa})$$

is bijective. In particular, the functor $\mathbf{Vaf}_{/k}^{\text{ét}} \rightarrow \mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ is fully faithful (and we shall see below that it is an equivalence).

Indeed, denote by A (resp. B) the affine algebra of X (resp. Y), and by $\mathfrak{r}$ the radical of k , and suppose that a morphism $B \widehat{\otimes}_k \kappa(k) \rightarrow A \widehat{\otimes}_k \kappa(k)$ is given, or, equivalently, a morphism $\phi : B \rightarrow A/\mathfrak{r}A$. For every open ideal $\mathfrak{l}$ of k , there exists $n \in \mathbf{N}^*$ such that $\mathfrak{r}^n \subset \mathfrak{l}$, whence $(\mathfrak{r}A)^n \subset \mathfrak{l}A \subset \overline{\mathfrak{l}A}$, and since the multiplication map $m_{n-1} : A^n \rightarrow A$ is continuous, one therefore also has $(\mathfrak{r}A)^n \subset \overline{\mathfrak{l}A}$, i.e. $\mathfrak{r}A/\overline{\mathfrak{l}A}$ is a nilpotent ideal of $A/\overline{\mathfrak{l}A}$. Consequently, ϕ lifts uniquely to a morphism $\phi_{\mathfrak{l}} : B \rightarrow A/\overline{\mathfrak{l}A}$. By uniqueness, these morphisms form a projective

⁽⁷²⁾N.D.E. Note that $\mathbf{Vaf}_{/k}^{\text{ét}}$ is stable under finite projective limits.

system, hence give a continuous morphism $\Phi : B \rightarrow \varprojlim_{\mathfrak{l}} A/\overline{\mathfrak{l}A} = A$. Moreover, Φ is unique, since if Φ' is a second lifting of ϕ , Φ' and Φ agree modulo $\overline{\mathfrak{l}A}$ for every $\mathfrak{l}$, hence are equal.

Proposition 1.6.E. (a) *Let X be a formal variety over k and A its affine algebra. The following conditions are equivalent:*

- (i) X is étale over k .
- (ii) X is topologically flat over k and the diagonal morphism $\Delta : X \rightarrow X \times X$ is a local isomorphism, i.e. Δ_X induces an isomorphism $\mathcal{O}_{X \times X, \Delta(x)} \xrightarrow{\sim} \mathcal{O}_{X, x}$ for every point x of X .
- (iii) For every $x \in X$ projecting to s on $\mathrm{Spf}(k)$, $\mathcal{O}_{X, x}$ is isomorphic to $k_s[T]/(F)$, where $F \in k_s[T]$ is a monic separable polynomial (cf. 1.6.A(ii)).
- (iv) X is topologically flat over k , and for every point $x \in X$ projecting to s on $\mathrm{Spf}(k)$, $\mathcal{O}_{X, x} \widehat{\otimes}_k \kappa(s)$ is a finite separable extension of $\kappa(s)$.
- (v) For every open ideal $\mathfrak{l}$ of k , each local component of $A \widehat{\otimes}_k (k/\mathfrak{l})$ is a finite étale algebra over the artinian ring $k/\mathfrak{l}$.

(b) $\mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ is identified with the category of étale schemes over $S_{\kappa(k)}$ (cf. 1.6.C), and the functor $X \mapsto X_{\kappa}$ induces an equivalence of categories $\mathbf{Vaf}_{/k}^{\text{ét}} \xrightarrow{\sim} \mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ (cf. SGA 1, I 6.1).

PROOF. (a) Denote by I the kernel of the multiplication morphism $m : A \widehat{\otimes}_k A \rightarrow A$. Suppose that X is étale over k , i.e. A is formally étale over k . Then, by EGA 0_{IV}, 20.7.4, the separated completion of I/I^2 for the quotient topology of that of I is zero, i.e. one has $I = \overline{I^2}$. Now, for every $x \in X$, I is contained in the maximal ideal $\mathfrak{m}_{\Delta(x)}$ of $A \widehat{\otimes}_k A$, so the localization $I_{\mathfrak{m}_{\Delta(x)}}$ is contained in the maximal ideal of $\mathcal{O}_{X \times X, \Delta(x)}$, and hence, by Nakayama's lemma 0.3, one has $I_{\mathfrak{m}_{\Delta(x)}} = 0$, and thus $\Delta : X \rightarrow X \times X$ is a local isomorphism. Now suppose that Δ is a local isomorphism and that k is a field κ , and let us show that each $\mathcal{O}_{X, x}$ is a finite-dimensional étale κ -algebra. Replacing X by $\mathrm{Spf}(\mathcal{O}_{X, x})$, one can suppose that $A = \mathcal{O}_{X, x}$ is local. One then proceeds as in the proof of EGA IV₄, 17.4.1, (b) $\Rightarrow$ (d''). Let K be a finite normal extension of κ containing the residue field $\kappa(x)$, and let $B = A \otimes_{\kappa} K = A \widehat{\otimes}_{\kappa} K$ (since $[K : \kappa] < \infty$, $-\widehat{\otimes}_{\kappa} K$ and $-\otimes_{\kappa} K$ coincide) and $X_K = \mathrm{Spf}(B) = X \widehat{\otimes}_{\kappa} K$. Then $\Delta_K : X_K \rightarrow X_K \widehat{\otimes}_K X_K$ is still a local isomorphism, so for every $y \in X_K$, multiplication $B_y \widehat{\otimes}_K B_y \rightarrow B_y$ induces an isomorphism $(B_y \widehat{\otimes}_K B_y)_{\mathfrak{m}_{\Delta_K(y)}} \xrightarrow{\sim} B_y$. Now, since the residue field of B_y is K (cf. for example VI A, 1.1.1, N.D.E. (11)), $C = B_y \widehat{\otimes}_K B_y$ is already a local ring (indeed, $\mathfrak{n} = \mathfrak{m}_y \widehat{\otimes}_K B_y + B_y \widehat{\otimes}_K \mathfrak{m}_y$ consists of topologically nilpotent elements, hence is contained in the radical of C , and $C/\mathfrak{n} = K \widehat{\otimes}_K K = K$ is a field), so one obtains that the multiplication morphism $B_y \widehat{\otimes}_K B_y \rightarrow B_y$ is an isomorphism. Taking a pseudobasis of B_y over K containing the identity element 1, one deduces that $B_y = K$. Since, moreover, B is finite over A , X_K is a finite set, so $B = A \otimes_{\kappa} K$ is the product of finitely many copies of K , and this implies that A is a finite étale κ -algebra. One thus obtains that, if κ is a field, every profinite κ -algebra A étale over κ is the product of finite separable extensions K_i of κ , equipped with the product topology, so the formal variety $\mathrm{Spf}(A)$ is the direct sum of the $\mathrm{Spf}(K_i) = \mathrm{Spec}(K_i)$, and one deduces that $\mathbf{Vaf}_{/\kappa}^{\text{ét}}$ is identified with the category of étale κ -schemes. The foregoing shows implication (ii) $\Rightarrow$ (iv) (since $\mathcal{O}_{X, x} \widehat{\otimes}_k \kappa(s)$ is formally étale over $\kappa(s)$), and implies part (b) of 1.6.E. Indeed, let k again be an arbitrary pseudocompact ring. By the foregoing, $\mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ is identified with the category of étale schemes over $S_{\kappa(k)}$. Let us show that $X \mapsto X_{\kappa}$ induces an equivalence $\mathbf{Vaf}_{/k}^{\text{ét}} \xrightarrow{\sim} \mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$. In view of 1.6.D, it suffices to show that, for every $s \in \mathrm{Spf}(k)$ and every finite separable extension K of $\kappa(s)$, there exists an étale k_s -algebra A such that $A \widehat{\otimes}_k \kappa(s) \simeq K$. Let ξ be a primitive element of the extension $K/\kappa(s)$, n its degree, and $F \in k_s[T]$ a monic polynomial of degree n whose image $\overline{F}$ in $\kappa(s)[T]$ is the minimal polynomial of ξ . Since $\overline{F}'$ is invertible in $\kappa(s)[T]/(\overline{F})$, it follows from Nakayama's lemma that F' is invertible in $k_s[T]/(F)$, so F is a separable polynomial,

and hence, by 1.6.A(ii), $A = k_s[T]/(F)$ is an étale k_s -algebra such that $A \hat{\otimes}_k \kappa(s) \simeq K$. One thus obtains that every local étale profinite k_s -algebra is free of finite rank over k_s (hence a fortiori topologically free over k), and therefore every étale profinite k -algebra is topologically free over k . Moreover, condition (v) implies condition (d) of 1.6.B, hence implies (i). One has therefore obtained that (i), (iii), and (v) are equivalent, and imply (ii), which implies (iv). Finally, let A be a profinite k -algebra satisfying (iv); let us show that A is formally étale over k . For this one can suppose that A and k are local; denote by κ the residue field of k ; by hypothesis, $K = A \hat{\otimes}_k \kappa$ is a finite separable extension of κ , say of degree n . By what has been seen above (and taking Lemma 1.6.D into account), there then exist a k -algebra B , free of rank n , formally étale over k , and a continuous morphism $\phi : B \rightarrow A$ such that $\phi \hat{\otimes}_k \kappa$ is an isomorphism. Since A is topologically flat over k , this implies $\text{Coker}(\phi) \hat{\otimes}_k \kappa = 0$ and also $\text{Ker}(\phi) \hat{\otimes}_k \kappa = 0$; by Nakayama's lemma 0.3, one therefore has $\text{Coker}(\phi) = 0 = \text{Ker}(\phi)$, so ϕ is an isomorphism (cf. the proof of 0.2.B). This completes the proof of 1.6.E. $\square$

⁽⁷³⁾Let X be a formal variety over k . For every $x \in X$ projecting to s on $\text{Spf}(k)$, the residue field $\kappa(x)$ is a finite extension of $\kappa(s)$, and one denotes by $\kappa_e(x)$ the separable closure of $\kappa(s)$ in $\kappa(x)$.

Proposition 1.6.F.

- (i) *The inclusion of $\mathbf{Vaf}_{/k}^{\text{ét}}$ in $\mathbf{Vaf}/k$ has a left adjoint $X \mapsto X_e$: the variety X_e has the same points as X ; for every $x \in X$ projecting to s on $\text{Spf}(k)$, let ξ be a primitive element of $\kappa_e(x)$, n its degree, u an arbitrary lifting of ξ in $\mathcal{O}_{X,x}$, and $F \in k[T]$ monic of degree n annihilating u ; put $\mathcal{O}_{X_e,x} = k[T]/(F)$. Then for every $Y \in \text{Ob } \mathbf{Vaf}_{/k}^{\text{ét}}$, the canonical maps below are bijective:*

$$\begin{array}{ccc} \text{Hom}_{\mathbf{Vaf}/k}(X, Y) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{Vaf}/\kappa(k)}(X_\kappa, Y_\kappa) \\ & & \downarrow \sim \\ \text{Hom}_{\mathbf{Vaf}/k}(X_e, Y) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{Vaf}/\kappa(k)}((X_e)_\kappa, Y_\kappa), \end{array}$$

the vertical arrow being induced by the inclusions $\kappa_e(x) \hookrightarrow \kappa(x)$ for every $x \in X$. This defines, in particular, a morphism $p : X \rightarrow X_e$, and every morphism of formal k -varieties $\phi : X \rightarrow Y$, with Y étale over k , factors uniquely through p .

- (ii) *The functor $X \mapsto X_e$ commutes with finite products.*

Indeed, (i) follows from 1.6.E(b) and 1.6.D. Let us prove (ii). In view of the equivalence of categories 1.6.E(b), one can suppose that k is a field. In this case, one easily sees that if X is a semilocal formal k -variety, i.e. whose affine algebra A is semilocal, the affine algebra of X_e is the largest subalgebra of A which is étale over k ; denote it by A_e . One is thus reduced to seeing that if K, L are two extensions of k of finite degree, the inclusion $K_e \otimes L_e \subset (K \otimes L)_e$ is an equality. Let p be the characteristic exponent of k (i.e. $p = 1$ if $\text{char}(k) = 0$ and $p = \text{char}(k)$ otherwise); then for every $x \in K \otimes L$, there exists $n \in \mathbf{N}^*$ such that $x^{p^n} \in K_e \otimes L_e$, so every subalgebra B of $K \otimes L$ is radicial over $K_e \otimes L_e$, and it follows that $K_e \otimes L_e = (K \otimes L)_e$. Observe, retaining the preceding notation, that $\mathcal{O}_{X_e,x}$ is not necessarily a subalgebra of $\mathcal{O}_{X,x}$, but this is the case when X is topologically flat over k , by the following proposition.

1.6.1. — Let Y be an étale formal k -variety and $f : X \rightarrow Y$ a morphism in $\mathbf{Vaf}/k$. One then has the cartesian square below, where Γ_f is the graph morphism $X \rightarrow X \times Y$ with

⁽⁷³⁾**N.D.E.** In what follows, we have used the preceding additions to detail the construction of the functor $X \mapsto X_e$ and show that it commutes with finite products (this is used in 2.5.1).

components id_X and f :

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times Y \\ f \downarrow & & \downarrow f \otimes \text{id}_Y \\ Y & \xrightarrow{\Delta_Y} & Y \times Y. \end{array}$$

It follows that Γ_f is a local isomorphism, hence that $f = \text{pr}_Y \circ \Gamma_f$ is topologically flat if pr_Y is, for example if X is topologically flat over k . Conversely, since $Y \rightarrow k$ is topologically flat, $X \rightarrow k$ will also be so if f is (cf. 1.3.3). Taking, in particular, for f the canonical morphism $p : X \rightarrow X_e$ of 1.6, one obtains:

Proposition. *Let X be a formal variety over k . The morphism $X \rightarrow X_e$ is topologically flat if and only if X is topologically flat over k .*

Remark 1.6.2. ⁽⁷⁴⁾ When k is a perfect field, the functor $X \mapsto X_e$ is also right adjoint to the inclusion $\mathbf{Vaf}_{/k}^{\text{ét}} \hookrightarrow \mathbf{Vaf}/k$. Indeed, in this case one has $\mathcal{O}_{X_e, x} = \kappa(x)$ for every $x \in X$, and the canonical projections $\mathcal{O}_{X, x} \rightarrow \kappa(x)$ define a morphism $s : X_e \rightarrow X$ which is a section of $p : X \rightarrow X_e$. For every morphism $f : X \rightarrow Y$, the diagrams below commute:

$$\begin{array}{ccc} \mathcal{O}_{Y, f(x)} & \longrightarrow & \mathcal{O}_{X, x} \\ \downarrow & & \downarrow \\ \kappa(f(x)) & \longrightarrow & \kappa(x) \end{array} \quad \begin{array}{ccc} X & \xrightarrow{f} & Y \\ s_X \uparrow & & \uparrow s_Y \\ X_e & \xrightarrow{f_e} & Y_e \end{array}$$

so s is functorial in X , and it follows that $X \mapsto X_e$ is right adjoint to the inclusion $\mathbf{Vaf}_{/k}^{\text{ét}} \hookrightarrow \mathbf{Vaf}/k$. Thus, being a right adjoint, $X \mapsto X_e$ commutes with projective limits when they exist in $\mathbf{Vaf}_{/k}^{\text{ét}}$, ⁽⁷⁵⁾ hence in particular with finite products. (This can also be verified directly: for every formal k -variety X with affine algebra A , the affine algebra of X_e is the quotient of A by its radical $\mathfrak{r}(A)$, and since k is perfect, the quotient of $\mathcal{O}_{X, x} \hat{\otimes} \mathcal{O}_{Y, y}$ by its radical is the algebra $\kappa(x) \otimes_k \kappa(y)$, since the latter is semisimple.)

2. Generalities on formal groups

2.1. — Let k be a pseudocompact ring and G a formal k -group, that is, a group in the category $\mathbf{Vaf}/k$ of formal varieties over k . Let A be the affine algebra of G . The composition law of G evidently defines a *diagonal morphism*, i.e. a homomorphism of profinite k -algebras $\Delta_A : A \rightarrow A \hat{\otimes}_k A$; this homomorphism satisfies the following conditions:

(i) The diagram

$$\begin{array}{ccc} A & \xrightarrow{\Delta_A} & A \hat{\otimes}_k A \\ \Delta_A \downarrow & & \downarrow \Delta_A \hat{\otimes} \text{id}_A \\ A \hat{\otimes}_k A & \xrightarrow{\text{id}_A \hat{\otimes} \Delta_A} & A \hat{\otimes}_k A \hat{\otimes}_k A \end{array}$$

commutes.

(ii) There exists a (necessarily unique) *augmentation*, that is, a homomorphism of profinite k -algebras $\varepsilon_A : A \rightarrow k$ such that the composite maps

$$A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{\varepsilon_A \hat{\otimes} \text{id}_A} k \hat{\otimes}_k A \simeq A$$

and $A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{\text{id}_A \hat{\otimes} \varepsilon_A} A \hat{\otimes}_k k \simeq A$

⁽⁷⁴⁾**N.D.E.** We have inserted this remark here; in the original it appeared in 2.5.2.

⁽⁷⁵⁾**N.D.E.** This is the case for finite projective limits.

are the identity maps of A .

- (iii) There exists a (necessarily unique) *antipode*, that is, a homomorphism of profinite k -algebras $c_A : A \rightarrow A$ such that the composite map

$$A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{c_A \hat{\otimes} \text{id}_A} A \hat{\otimes}_k A \xrightarrow{m_A} A$$

equals $\eta_A \circ \varepsilon_A$, if m_A denotes the linear map which sends $a \hat{\otimes} b$ to ab , and η_A the map $\lambda \mapsto \lambda 1_A$ from k to A .

Conversely, the data $(\Delta_A, \varepsilon_A, c_A)$ satisfying (i)–(iii) equip G with a formal k -group structure.⁽⁷⁶⁾ Explicitly, for every profinite k -algebra B , the set $\text{Hom}_c(A, B)$ of continuous morphisms of k -modules $\phi : A \rightarrow B$ is equipped with a group structure, functorial in B , defined by

$$\phi \cdot \phi' = m_B \circ (\phi \hat{\otimes} \phi') \circ \Delta_A,$$

the identity element being $\eta_B \circ \varepsilon_A$ (where m_B is multiplication on B and η_B the map $\lambda \mapsto \lambda 1_B$ from k to B), and $\phi \circ c_A$ being the inverse of ϕ ; and the set $\text{Hom}_{\mathbf{Alf}/k}(A, B)$ of continuous morphisms of k -algebras $A \rightarrow B$ is a subgroup of it (since the algebra B is commutative).

Definition. A morphism of formal k -groups $\theta : K \rightarrow G$ is, by definition, a morphism of formal k -varieties which respects the group structures. If B (resp. A) is the affine algebra of K (resp. G), and $f : A \rightarrow B$ is the morphism corresponding to θ , this is equivalent to saying that f is compatible with comultiplications, i.e.

$$(f \hat{\otimes} f) \circ \Delta_A = \Delta_B \circ f$$

(the conditions $\varepsilon_B \circ f = \varepsilon_A$ and $c_B \circ f = f \circ c_A$ then being automatically satisfied). Denote by $\mathbf{Grf}/k$ the category of formal k -groups.

Notation. In what follows, we shall call the ideal $I_A = \text{Ker}(\varepsilon_A)$ the *augmentation ideal* of A , and denote by $\omega_{G/k}$ the pseudocompact k -module $I_A/\overline{I_A^2}$, that is, the quotient of I_A by the closed ideal generated by the products xy , for $x, y \in I_A$.

2.2. — Let $\mathbf{H}$ be a group in the category of coalgebras over $\mathbf{O}_k$, i.e. for every object C of $\mathbf{Alf}/k$, $\mathbf{H}(C)$ is equipped with a group-coalgebra structure over C (cf. VII A 3.2; following Manin, we shall say *bialgebra*⁽⁷⁷⁾ instead of group-coalgebra); moreover, if $\varphi : C \rightarrow D$ is a morphism of $\mathbf{Alf}/k$, the map $D \otimes_C \mathbf{H}(C) \rightarrow \mathbf{H}(D)$ is a homomorphism of D -bialgebras.

Definition. We shall summarize the above properties by saying that $\mathbf{H}$ is a *bialgebra over* $\mathbf{O}_k$.

It is clear that the functor $\mathbf{H} \mapsto \text{Spf}^*(\mathbf{H})$ of 1.3.5 commutes with finite products. It therefore transforms a bialgebra over $\mathbf{O}_k$ into a *group-valued k -functor*, that is, a (covariant) functor from $\mathbf{Alf}/k$ to the category of groups. Indeed, for every k -algebra C of finite length, the elements of

$$\text{Spf}^*(\mathbf{H})(C) = \text{Spf}^*(\mathbf{H}(C)) = \{x \in \mathbf{H}(C) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$$

form a group for multiplication in the algebra $\mathbf{H}(C)$ (cf. VII A 3.2.2). Note, moreover, that the condition $\Delta(x) = x \otimes x$ implies the equality $\varepsilon(x) = \varepsilon(x)^2$, hence also $\varepsilon(x) = 1$ if C is local and $x \neq 0$.⁽⁷⁸⁾

⁽⁷⁶⁾**N.D.E.** One also says that A is a cogroup in the category of profinite k -algebras. Moreover, we have detailed what follows; in particular, we have made explicit what a morphism of formal groups $K \rightarrow G$ is; cf. Proposition 2.3.1.

⁽⁷⁷⁾**N.D.E.** One also says “bigèbre” (cf. [BAlg], III § 11.4); recall (cf. VII A, 3.1, N.D.E. (26)) that all the “bialgebras” considered here are assumed cocommutative and equipped with an antipode, i.e. they are in fact cocommutative Hopf algebras.

⁽⁷⁸⁾**N.D.E.** Since $\Delta(x) = x \otimes x$, one has $x = \varepsilon(x)x$, so $\varepsilon(x) = 0$ can occur only if $x = 0$.

2.2.1. — A bialgebra $\mathbf{H}$ over $\mathbf{O}_k$ is said to be *flat* if the underlying module is flat (cf. 1.2.1).⁽⁷⁹⁾ If $\mathbf{H}$ is flat, then, by 1.3.5, $A = \Gamma^*(\mathbf{H})$ is a topologically flat profinite k -algebra, and $\mathrm{Spf}^*(\mathbf{H})$ is isomorphic, as a functor from $(\mathbf{A}lf/k)^0 = \mathbf{V}af/k$ to $(\mathbf{Ens})$, to the functor

$$\mathrm{Spf}(A) : C \longmapsto \mathrm{Hom}_{\mathbf{V}af/k}(\mathrm{Spf}(C), \mathrm{Spf}(A)).$$

The group structure of $\mathrm{Spf}^*(\mathbf{H})$ therefore equips $\mathcal{G}(\mathbf{H}) = \mathrm{Spf}(A)$ with a formal group structure, explicitly described as follows. For every object C of $\mathbf{A}lf/k$, since the C -module underlying $\mathbf{H}(C)$ is projective, one deduces from Lemma 1.2.3.A by induction on n natural isomorphisms

$$\begin{aligned} (\mathbf{H}(C)^*)^{\widehat{\otimes}(n+1)} &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^*)^{\widehat{\otimes}n}) \\ &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^{\otimes n})^*) \simeq (\mathbf{H}(C)^{\otimes(n+1)})^*. \end{aligned}$$

From this one deduces (for $n = 1, 2$) that the C -algebra structure of $\mathbf{H}(C)$ equips $\mathbf{H}(C)^*$ with a diagonal map satisfying conditions 2.1(i)–(iii), all this functorially in C . Consequently, $A = \Gamma^*(\mathbf{H}) = \varprojlim \mathbf{H}(k/l)^*$ is equipped with a cogroup structure in $\mathbf{A}lp/k$, which defines the announced formal group structure on $\mathcal{G}(\mathbf{H})$. Conversely, let G be a topologically flat formal k -group with affine algebra A , and denote by $\mathbf{H}(G)$ the $\mathbf{O}_k$ -coalgebra $\mathbf{V}_k^f(A)$ (cf. 1.2.3). The diagonal morphism $\Delta_A : A \longrightarrow A \widehat{\otimes}_k A$ then induces, for every k -algebra C of finite length, a C -linear map

$$\mathbf{V}_k^f(A)(C) \otimes_C \mathbf{V}_k^f(A)(C) \longrightarrow \mathbf{V}_k^f(A)(C)$$

which makes the coalgebra $\mathbf{V}_k^f(A)(C)$ a C -bialgebra. One says that $\mathbf{H}(G)$ is *the covariant bialgebra of the formal group G* .⁽⁸⁰⁾ Thus, by Proposition 1.3.5.D:

Proposition.

- (i) *The functors $G \longmapsto \mathbf{H}(G)$ and $\mathbf{H} \longmapsto \mathcal{G}(\mathbf{H})$ define an equivalence between the category of topologically flat formal k -groups and that of flat $\mathbf{O}_k$ -bialgebras.*⁽⁸¹⁾
- (ii) *This equivalence “commutes with base change”: if $k \longrightarrow K$ is a morphism of pseudocompact rings, then $\mathbf{H}(G \widehat{\otimes}_k K) = \mathbf{H}(G) \otimes_k K$ and $\mathcal{G}(\mathbf{H} \otimes_k K) = \mathcal{G}(\mathbf{H}) \widehat{\otimes}_k K$.*

When k is an artinian ring and G a topologically flat formal k -group, the functor $\mathbf{H}(G)$ is evidently determined by its value $H(G) = \mathbf{H}(G)(k)$ at k . One also says that $H(G)$ is the (covariant) bialgebra of G .⁽⁸²⁾ Consequently, denoting by $\mathcal{H}_k$ the category of k -Hopf algebras and by $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{cocom}}$ the full subcategory consisting of k -Hopf algebras flat over k and cocommutative, one therefore obtains:

Corollary. *Let k be an artinian ring.*

- (i) *The functors $G \longmapsto H(G)$ and $H \longmapsto \mathcal{G}(H) = \mathrm{Spf}^* H(G)$ define an equivalence between the category of topologically flat formal k -groups and $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{cocom}}$.*
- (ii) *This equivalence “commutes with base change”: if $k \longrightarrow K$ is a morphism of artinian rings, then $H(G \widehat{\otimes}_k K) = H(G) \otimes_k K$ and $\mathcal{G}(H \otimes_k K) = \mathcal{G}(H) \widehat{\otimes}_k K$.*

Moreover, denote also by $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{com}}$ the full subcategory of $\mathcal{H}_k$ consisting of k -Hopf algebras flat over k and commutative, and recall that the functor $K \longmapsto \mathcal{O}(K)$ is an anti-equivalence from the category of affine group k -schemes to $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{com}}$.

2.2.2. — For simplicity suppose that k is artinian, and let G be a topologically flat formal k -group.⁽⁸³⁾ Then G is commutative if and only if its affine algebra $\mathcal{A}(G)$ has a cocommutative

⁽⁷⁹⁾**N.D.E.** We have detailed the original in what follows.

⁽⁸⁰⁾**N.D.E.** We have added the adjective “covariant” to make clear that the functor $G \longmapsto \mathbf{H}(G)$ is covariant; this terminology is used in [Di73], I § 2.14.

⁽⁸¹⁾**N.D.E.** Recall that in this exposé “bialgebra” means “cocommutative Hopf algebra”.

⁽⁸²⁾**N.D.E.** We have introduced here the notation $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{cocom}}$, which will be useful below.

⁽⁸³⁾**N.D.E.** We have detailed the original in what follows, in order to introduce the notation $D'(G)$ and $D(K)$; cf. [Ca62], § 14.

comultiplication, which is equivalent to saying that the bialgebra $H(G)$ has commutative multiplication. In this case $H(G)$ is a commutative and cocommutative Hopf algebra flat over k , so if one puts $D'(G) = \operatorname{Spec} H(G)$, $D'(G)$ is a *commutative group k -scheme, affine and flat* over k . Conversely, if T is such a group k -scheme, its affine algebra $\mathcal{O}(T)$ is a commutative group in the category of flat cocommutative k -coalgebras, and hence, by 1.3.5.C, one obtains a topologically flat formal k -group $D(T)$ by putting

$$D(T) = \operatorname{Spf}^* \mathcal{O}(T) = \operatorname{Spf}(\mathcal{A}), \quad \text{where } \mathcal{A} = \mathcal{O}(T)^*.$$

Since, by 1.3.5.D, one has canonical isomorphisms $G = \operatorname{Spf}^* H(G)$ and $H(D(T)) = \mathcal{O}(T)$, one obtains canonical isomorphisms

$$\begin{aligned} D(D'(G)) &= \operatorname{Spf}^* \mathcal{O}(D'(G)) = \operatorname{Spf}^* H(G) = G, \\ D'(D(T)) &= \operatorname{Spec} H(D(T)) = \operatorname{Spec} \mathcal{O}(T) = T. \end{aligned}$$

Moreover, denoting by $k\text{-Gr.}$ the category of group k -schemes, by Corollary 2.2.1 one has functorial isomorphisms

$$\operatorname{Hom}_{\mathbf{Grf}/k}(G, D(T)) \simeq \operatorname{Hom}_{\mathcal{H}_k}(H(G), \mathcal{O}(T)) \simeq \operatorname{Hom}_{k\text{-Gr.}}(T, D'(G)).$$

One therefore obtains the following:

Proposition. (*Cartier duality*). *Let k be an artinian ring.*

- (i) *The functors $G \mapsto D'(G) = \operatorname{Spec} H(G)$ and $T \mapsto D(T) = \operatorname{Spf}^* \mathcal{O}(T)$ induce an anti-equivalence between the category $\mathcal{FC}_k$ of topologically flat commutative formal k -groups and the category $\mathcal{AC}_k$ of commutative group k -schemes, affine and flat, i.e. G and $D'(G)$ (resp. T and $D(T)$) are related by the equalities⁽⁸⁴⁾*

$$H(G) = \mathcal{O}(D'(G)) \quad \text{and} \quad \mathcal{O}(T) = H(D(T)).$$

- (ii) *This anti-equivalence “commutes with base change”: if $k \rightarrow K$ is a morphism of artinian rings, then $D'(G \hat{\otimes}_k K) = D'(G) \otimes_k K$ and $D(T \otimes_k K) = D(T) \hat{\otimes}_k K$.*
- (iii) *In particular, if k is a field, one obtains an anti-equivalence between the category of commutative formal k -groups and that of affine commutative group k -schemes which commutes with extension of the base field.*

2.3. — Now consider an arbitrary⁽⁸⁵⁾ formal k -group G with affine algebra A . Continue to denote by $\mathbf{H}(G)$ the $\mathbf{O}_k$ -module $\mathbf{V}_k^f(A)$ dual to A , and denote by φ_G the functorial homomorphism

$$\varphi_G : \mathbf{H}(G) \otimes_k \mathbf{H}(G) \rightarrow \mathbf{H}(G \times G)$$

induced by the natural map (0.3.6), for every object C of $\mathbf{Alf}/k$:

$$(A \hat{\otimes}_k C)^\dagger \otimes_C (A \hat{\otimes}_k C)^\dagger \rightarrow ((A \hat{\otimes}_k C) \hat{\otimes}_C (A \hat{\otimes}_k C))^\dagger.$$

If $m : G \times G \rightarrow G$ is multiplication on G , the composite map

$$\mathbf{H}(G) \otimes_k \mathbf{H}(G) \xrightarrow{\varphi_G} \mathbf{H}(G \times G) \xrightarrow{\mathbf{H}(m)} \mathbf{H}(G)$$

makes $\mathbf{H}(G)$ an algebra over $\mathbf{O}_k$; for every $C \in \mathbf{Alf}/k$, the identity element of $\mathbf{H}(G)(C) = (A \hat{\otimes}_k C)^\dagger$ is the augmentation of $A \hat{\otimes}_k C$ (cf. 2.1).⁽⁸⁶⁾ If G is not topologically flat over k , φ_G is not necessarily an isomorphism, and thus the morphism $\delta_G : \mathbf{H}(G) \rightarrow \mathbf{H}(G \times G)$ induced by the diagonal morphism “ $x \mapsto (x, x)$ ” from G to $G \times G$ does not necessarily factor

⁽⁸⁴⁾**N.D.E.** If one denotes by $\mathcal{FC}_k^f$ (resp. $\mathcal{AC}_k^f$) the full subcategory of $\mathcal{FC}_k$ (resp. $\mathcal{AC}_k$) consisting of objects G (resp. T) such that $H(G)$ (resp. $\mathcal{O}(T)$) is a finite (and hence finite locally free) k -module, then $\mathcal{FC}_k^f$ and $\mathcal{AC}_k^f$ both have the same objects as the category $\mathcal{C}$ of commutative and cocommutative Hopf k -algebras finite and flat over k , the correspondence $G \mapsto H(G)$ (resp. $T \mapsto \mathcal{O}(T)$) being covariant (resp. contravariant), and in this way one recovers the “Cartier duality” of the category $\mathcal{AC}_k$ already seen in VII A, 3.3.1.

⁽⁸⁵⁾**N.D.E.** i.e. not necessarily topologically flat over k .

⁽⁸⁶⁾**N.D.E.** We have changed the order of the sentences in what follows.

through $\mathbf{H}(G) \otimes_k \mathbf{H}(G)$, i.e. $\mathbf{H}(G)$ is not necessarily an $\mathbf{O}_k$ -bialgebra. For this reason, in the general case we shall simply say that $\mathbf{H}(G)$ is “the covariant algebra” of the formal group G . Of course, when G is topologically flat over k , φ_G is an isomorphism and one recovers the $\mathbf{O}_k$ -bialgebra structure on $\mathbf{H}(G)$ defined in 2.2.1.

2.3.1. Proposition. *Let K, G be two formal k -groups with affine algebras B, A . Suppose that K is topologically flat over k . There then exists a canonical bijection from $\mathrm{Hom}_{\mathbf{Grf}/k}(K, G)$ to the set of homomorphisms of unital $\mathbf{O}_k$ -algebras $h : \mathbf{H}(K) \rightarrow \mathbf{H}(G)$ such that the diagram*

$$(*) \quad \begin{array}{ccc} \mathbf{H}(K) \otimes_k \mathbf{H}(K) & \xrightarrow{h \otimes h} & \mathbf{H}(G) \otimes_k \mathbf{H}(G) \\ \uparrow \Delta_{\mathbf{H}(K)} & & \searrow \varphi_G \\ \mathbf{H}(K) & \xrightarrow{h} & \mathbf{H}(G) \end{array} \quad \begin{array}{c} \nearrow \delta_G \\ \mathbf{H}(G \times G) \end{array}$$

commutes.

Since K is topologically flat, $\mathbf{H}(K)$ is equipped with a bialgebra structure (cf. 2.2), and $\Delta_{\mathbf{H}(K)}$ is its diagonal morphism; in other words, with the notation of 2.3, one has $\Delta_{\mathbf{H}(K)} = \varphi_K^{-1} \circ \delta_K$. When G is also topologically flat over k , our proposition follows from the equivalence of categories established in 2.2.1. In the general case one can suppose that k is artinian and reason with the algebras $H(K) = B^\dagger$ and $H(G) = A^\dagger$. Let $\mathrm{Hom}_c(A, B)$ be the set of continuous k -linear maps from A to B , and $\mathrm{Hom}_k(B^\dagger, A^\dagger)$ the set of k -linear maps from $B^\dagger$ to $A^\dagger$.⁽⁸⁷⁾ By 0.3.6.A one knows that, if M, P are pseudocompact k -modules and P is projective, the canonical map

$$\mathrm{Hom}_c(M, P) \rightarrow \mathrm{Hom}_k(P^\dagger, M^\dagger), \quad f \mapsto {}^t f$$

(where ${}^t f$ denotes the transpose of f) is bijective. (One will apply this to $M = A \hat{\otimes} A$ and $P = B$, or to $M = A$ and $P = B \hat{\otimes} B$.) Let $f \in \mathrm{Hom}_c(A, B)$. Consider the diagrams below, in which the squares (0) commute and the two unnamed vertical arrows are ${}^t(f \hat{\otimes} f)$.

$$\begin{array}{ccccc} A \hat{\otimes} A & \xrightarrow{m_A} & A & \xrightarrow{\Delta_A} & A \hat{\otimes} A \\ f \hat{\otimes} f \downarrow & (1) & \downarrow f & (2) & \downarrow f \hat{\otimes} f \\ B \hat{\otimes} B & \xrightarrow{m_B} & B & \xrightarrow{\Delta_B} & B \hat{\otimes} B \end{array}$$

$$\begin{array}{ccccccc} A^\dagger \otimes A^\dagger & \xrightarrow{\varphi_G} & (A \hat{\otimes} A)^\dagger & \xleftarrow{\delta_G = m_A^t} & A^\dagger & \xleftarrow{\Delta_A^t} & (A \hat{\otimes} A)^\dagger \xleftarrow{\varphi_G} A^\dagger \otimes A^\dagger \\ {}^{tf} \otimes {}^{tf} \uparrow & (0) & \uparrow & (1') & {}^{tf} \uparrow & (2') & \uparrow (0) \uparrow {}^{tf} \otimes {}^{tf} \\ B^\dagger \otimes B^\dagger & = & (B \hat{\otimes} B)^\dagger & \xleftarrow{\delta_K = m_B^t} & B^\dagger & \xleftarrow{\Delta_B^t} & (B \hat{\otimes} B)^\dagger = B^\dagger \otimes B^\dagger \end{array}$$

If $f : A \rightarrow B$ corresponds to a morphism of formal groups $K \rightarrow G$, the squares (1) and (2) commute, and $\varepsilon_B \circ f = \varepsilon_A$; consequently, the squares (1') and (2') commute and ${}^t f$ sends the identity of $B^\dagger = H(K)$ to that of $A^\dagger = H(G)$, i.e. ${}^t f$ is a morphism of unital k -algebras $H(K) \rightarrow H(G)$ such that diagram (*) of the proposition commutes. Conversely, if ${}^t f$ satisfies these conditions, $\varepsilon_B \circ f = \varepsilon_A$ and the squares (1') and (2') commute. Since, for $M = A \hat{\otimes} A$ and $P = B$ (resp. $M = A$ and $P = B \hat{\otimes} B$), the map $g \mapsto {}^t g$ is injective, one deduces that the squares (1) and (2) commute, so f is compatible with the multiplications and diagonal morphisms of A and B . It remains to see that $f(1_A) = 1_B$. Now it follows

⁽⁸⁷⁾ **N.D.E.** We have detailed the remainder of the proof.

from the foregoing that $\varepsilon_B f(1) = 1$, $\Delta_B f(1) = f(1) \hat{\otimes} f(1)$, and $f(1) \cdot f(1) = f(1)$. The first two conditions imply, by 2.1(iii), that $f(1)$ has $c_B f(1)$ as its inverse in B ; consequently, $f(1) \cdot f(1) = f(1)$ implies $f(1) = 1$. Thus $f : A \rightarrow B$ is a morphism in $\mathbf{Alf}/k$ compatible with the comultiplications of A and B .

2.3.2. — For simplicity, now suppose that the ring k is artinian. When G is topologically flat over k , the algebra $H(G) = \mathbf{H}(G)(k)$ can be characterized by a universal property (due to Cartier). Recall (cf. 1.2.1) that if U is a k -module, $\mathbf{W}(U)$ denotes the functor which associates to every k -algebra C of finite length the C -module $U \otimes_k C$.⁽⁸⁸⁾ If U is a k -algebra (associative, with identity), the same holds for $U \otimes_k C$; we shall denote by $\mathbf{W}(U)^\times$ the group-valued k -functor which associates to every $C \in \text{Ob } \mathbf{Alf}/k$ the multiplicative group of invertible elements of the algebra $U \otimes_k C$:

$$\mathbf{W}(U)^\times(C) = (U \otimes_k C)^\times.$$

Moreover, identify G with the group-valued k -functor $C \mapsto \text{Hom}_{\mathbf{Vaf}/k}(\text{Spf}(C), G)$ and denote by $\text{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times)$ the set of homomorphisms of group-valued k -functors from G to $\mathbf{W}(U)^\times$. One has the following

Proposition. *Let k be an artinian ring. For every formal group G topologically flat over k and every k -algebra U , there is a canonical isomorphism*

$$\text{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times) \xrightarrow{\sim} \text{Hom}_{k\text{-Alg.}}(H(G), U).$$

Denote by A the affine algebra of G ; by hypothesis it is a projective object of $\mathbf{PC}(k)$, and $H(G) = A^\dagger$. For every object P of $\mathbf{PC}(k)$, denote by $U \hat{\otimes}_k P$ the projective limit of the k -modules $U \otimes_k (P/N)$, where N runs through the open submodules of P . One has linear maps

$$U \otimes_k (P/N) \rightarrow \text{Hom}_k((P/N)^*, U)$$

which send $u \otimes x$ to the k -linear map $f \mapsto f(x)u$ and form a filtered projective system. One therefore obtains, by passage to the projective limit, a morphism

$$(1) \quad U \hat{\otimes}_k P \xrightarrow{\psi_P} \text{Hom}_k(P^\dagger, U).$$

When $P = k$, ψ_k is evidently an isomorphism; moreover, both sides of (1), considered as functors in P , commute with infinite products (every product being a filtered projective limit of finite products). One therefore obtains that (1) is an isomorphism when P is a product of copies of k , and then when P is a projective object of $\mathbf{PC}(k)$ (both sides of (1) commuting with finite direct sums). Now denote by $\text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$ the set of morphisms of k -functors from G to $\mathbf{W}(U)$. Since $G = \text{Spf}(A) = \varprojlim \text{Spf}(A/\mathfrak{l})$, where $\mathfrak{l}$ runs through the open ideals of A , one has canonical isomorphisms

$$\begin{aligned} \text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) &= \varprojlim \text{Hom}_{\mathcal{F}}(\text{Spf}(A/\mathfrak{l}), \mathbf{W}(U)) \\ &= \varprojlim U \otimes_k (A/\mathfrak{l}) = U \hat{\otimes}_k A. \end{aligned}$$

By the foregoing, one therefore obtains a canonical isomorphism

$$(2) \quad \text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) = U \hat{\otimes}_k A \xrightarrow{\psi_A} \text{Hom}_k(H(G), U).$$

For every k -algebra C of finite length, multiplication makes $U \otimes_k C$ a monoid with identity, and every morphism of monoids with identity $G(C) \rightarrow U \otimes_k C$ is necessarily a morphism of groups $G(C) \rightarrow (U \otimes_k C)^\times$. Consequently, one obtains that $\text{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times)$ is the subset of $\text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$ consisting of morphisms of k -functors in monoids with identity. It remains to see that these morphisms correspond to the k -linear maps from

⁽⁸⁸⁾**N.D.E.** We have added the preceding sentence, and in what follows we have written $\mathbf{W}(U)^\times$ instead of U_m . Recall, moreover (cf. 1.2.1), that one calls a covariant functor $\mathbf{Alf}/k \rightarrow (\text{Ens})$ a “ k -functor”.

$H(G)$ to U which preserve multiplication and the identity elements.⁽⁸⁹⁾ To simplify the notation, $H(G) = A^\dagger$ will be denoted by H , and we shall write $\widehat{\otimes}$ instead of $\widehat{\otimes}_k$. Denote by $\Delta_A, m_A, \varepsilon_A$ (resp. $\Delta_H, m_H, \varepsilon_H$) the comultiplication, multiplication, and augmentation of A (resp. H). Let $\theta \in \text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$; denote by γ its image in $U \widehat{\otimes} A$, and by $\phi : H \rightarrow U$ the associated k -linear map. Then θ sends the identity section $s \in G(k)$ to an element u of U , and since s corresponds to the augmentation $\varepsilon : A \rightarrow k$, which is the identity element 1_H of H , one sees that $\theta(s) = 1_U$ if and only if $\phi(1_H) = 1_U$. Moreover, the morphism $\theta \circ m_G : G \times G \rightarrow \mathbf{W}(U)$ corresponds to the element $(\text{id}_U \widehat{\otimes} \Delta_A)(\gamma)$, and this corresponds, by duality, to the map $\phi \circ m_H : H \otimes H \rightarrow U$. On the other hand, the morphism $\theta \circ \text{pr}_1 : G \times G \rightarrow \mathbf{W}(U)$ corresponds to the element $\gamma \widehat{\otimes} 1_A$ of $U \widehat{\otimes} A \widehat{\otimes} A$, which corresponds by duality to $\phi \circ (\text{id}_H \otimes \varepsilon) : H \otimes H \rightarrow U$. Similarly, $\theta \circ \text{pr}_2$ corresponds to the element $\tau(\gamma \widehat{\otimes} 1_A)$ of $U \widehat{\otimes} A \widehat{\otimes} A$ (where $\tau(u \widehat{\otimes} a \widehat{\otimes} b) = u \widehat{\otimes} b \widehat{\otimes} a$), which corresponds by duality to $\phi \circ (\varepsilon \otimes \text{id}_H) : H \otimes H \rightarrow U$. Finally, the multiplication map $\mu = m_U \widehat{\otimes} A \widehat{\otimes} A$ below,

$$\begin{aligned}
 & (U \widehat{\otimes} A \widehat{\otimes} A) \times (U \widehat{\otimes} A \widehat{\otimes} A), \\
 & (u \widehat{\otimes} a_1 \widehat{\otimes} a_2, u' \widehat{\otimes} a_3 \widehat{\otimes} a_4) \mapsto uu' \widehat{\otimes} a_1 a_3 \widehat{\otimes} a_2 a_4,
 \end{aligned}$$

can be viewed as the composite of the endomorphism σ_{23} of $(U \otimes U) \widehat{\otimes} A \widehat{\otimes} A$ which “interchanges the factors a_2 and a_3 ”, and the map

$$(U \otimes U) \widehat{\otimes} A \widehat{\otimes} A \xrightarrow{m_U \widehat{\otimes} m_A \widehat{\otimes} m_A} U \widehat{\otimes} A \widehat{\otimes} A.$$

One deduces that the map $\mu \circ (\phi \circ \text{pr}_1, \phi \circ \text{pr}_2) : G \times G \rightarrow \mathbf{W}(U)$ corresponds to the composite map β below:

$$\begin{array}{ccc}
 & H \otimes H & \xrightarrow{\beta} U \\
 \text{id}_H \circ \sigma_{23} \circ \text{id}_H \swarrow & & \uparrow m_U \\
 H \otimes H \otimes H & & U \otimes U \\
 \searrow (\text{id}_H \otimes \varepsilon_H) \otimes (\varepsilon_H \otimes \text{id}_H) & \nearrow \phi \otimes \phi & \\
 & H \otimes H &
 \end{array}$$

Finally, θ is compatible with the laws of G and $\mathbf{W}(U)^\times$ if and only if $\theta \circ m_G$ equals $\mu \circ (\phi \circ \text{pr}_1, \phi \circ \text{pr}_2)$, which is equivalent, by the foregoing, to $\phi \circ m_H = \beta$. Now it is clear that $(\phi \circ m_H)(x \otimes y) = \phi(xy)$, and one easily sees that $\beta(x \otimes y) = \phi(x)\phi(y)$.

2.4. — Let us now return to any pseudocompact ring k to apply to formal groups the results of 1.4–1.5 on passage to the quotient by a topologically flat equivalence relation.⁽⁹⁰⁾ Let $u : H \rightarrow G$ be a monomorphism of formal k -groups, $\mu : G \times G \rightarrow G$ the “multiplication” morphism of G , and λ the composite morphism

$$\lambda : G \times H \xrightarrow{\text{id}_G \times u} G \times G \xrightarrow{\mu} G.$$

Since u is a monomorphism, the pair

$$G \times H \xrightleftharpoons[\lambda]{\text{pr}_1} G$$

is an equivalence pair in $\mathbf{Vaf}/k$ (cf. V, 2.b)). Recall (cf. 1.2.C) that the cokernel G/H of this pair is defined as follows. Let $\mathcal{O}(G)$ and $\mathcal{O}(H)$ be the affine algebras of G and H , $\Delta : \mathcal{O}(G) \rightarrow \mathcal{O}(G) \widehat{\otimes}_k \mathcal{O}(G)$ the diagonal morphism of $\mathcal{O}(G)$, and I the kernel of the morphism $u^\sharp : \mathcal{O}(G) \rightarrow \mathcal{O}(H)$. (By Proposition 1.3 one knows that $u^\sharp$ induces an

⁽⁸⁹⁾**N.D.E.** The original indicated that “(this) follows from the functorial character of ψ_A ”. We have detailed this in what follows.

⁽⁹⁰⁾**N.D.E.** We have detailed the original in what follows.

isomorphism $\mathcal{O}(G)/I \xrightarrow{\sim} \mathcal{O}(H)$.) Then the affine algebra $\mathcal{O}(G/H)$ of G/H is the kernel of the pair of morphisms

$$\mathcal{O}(G) \xrightarrow[(\text{id} \hat{\otimes} u^\sharp)_\Delta]{\tau_1} \mathcal{O}(G) \hat{\otimes}_k \mathcal{O}(H),$$

where $\tau_1(x) = x \hat{\otimes} 1$, i.e.

$$\mathcal{O}(G/H) = \{x \in \mathcal{O}(G) \mid \Delta(x) - x \hat{\otimes} 1 \in \mathcal{O}(G) \hat{\otimes} I\}.$$

If, moreover, H is *topologically flat over k* , pr_1 is topologically flat and from Theorem 1.4 one deduces the following theorem.

Theorem. *Let $u : H \rightarrow G$ be a monomorphism of formal k -groups. Suppose that H is topologically flat over k . Then the projection $p : G \rightarrow G/H$ is surjective and topologically flat, one has an isomorphism*

$$(*) \quad G \times H \xrightarrow{\sim} G \times_{G/H} G,$$

and G/H represents the quotient sheaf for the flat topology. Consequently, G/H is equipped with a canonical structure of an object with group of operators G , such that $p : G \rightarrow G/H$ is a morphism of objects with operators. If, moreover, u identifies H with a normal subgroup of G , then G/H is equipped with a canonical formal k -group structure such that $p : G \rightarrow G/H$ is a morphism of formal k -groups, and H is the kernel of p .

Indeed, the first assertion follows from 1.4; the other two from IV, Corollaries 5.2.2 and 5.2.4.

Corollary. ⁽⁹¹⁾ *Let G be a formal k -group, H a formal subgroup of G , A (resp. $A/J, B$) the affine algebra of G (resp. $H, G/H$), I_A the augmentation ideal of A , and $I_B = B \cap I_A$. Suppose that H is topologically flat over k . Then J equals $\overline{AI_B}$, the closed ideal generated by I_B .*

Indeed, the projection $B \rightarrow B/I_B$ corresponds to the “identity section” $e : \text{Spf}(k) \rightarrow G/H$ of G/H . By (*), H is identified with the fibered product $\text{Spf}(k) \times_{G/H} G$, so its affine algebra A/J is identified with $(B/I_B) \hat{\otimes}_B A \simeq A/\overline{AI_B}$.

2.4.A. — ⁽⁹²⁾Let G, Q be topologically flat formal k -groups; suppose that there exist homomorphisms $\sigma : Q \rightarrow G$ and $\pi : G \rightarrow Q$ such that $\pi \circ \sigma = \text{id}_Q$. In particular, σ is a monomorphism, so Q is a formal subgroup of G (cf. Remark 1.3). Let $N = \text{Ker}(\pi)$ and σ' be the inclusion $N \hookrightarrow G$. Then G is the *semidirect product* of N by Q (cf. I, 2.3.8), i.e. for every $B \in \text{Ob } \mathbf{Alf}/k$, identifying $N(B)$ and $Q(B)$ with their images in $G(B)$ under σ' and σ , $N(B)$ is a normal subgroup of $G(B)$ and the map

$$(1) \quad \mu : N(B) \times Q(B) \rightarrow G(B), \quad (x, q) \mapsto xq$$

is bijective. Then the morphism of formal k -varieties

$$(2) \quad \theta : G(B) \rightarrow N(B), \quad g \mapsto g \cdot \sigma\pi(g^{-1})$$

is a retraction of σ' , the inverse of μ is the map

$$(3) \quad g \mapsto (\theta(g), \pi(g)),$$

and $\theta \circ \sigma' : N \rightarrow G/Q$ is an isomorphism of formal k -varieties. In particular, N is topologically flat over k , by 1.4(ii). Denote by α (resp. β) the map from $Q \times N$ (resp. $G \times N = N \times Q \times N$)

⁽⁹¹⁾**N.D.E.** We have made explicit this corollary, which is used, for example, in 5.2.1/5.2.3.

⁽⁹²⁾**N.D.E.** We have added paragraphs 2.4.A and 2.4.B, which will be useful in 5.1.3 and 2.9 respectively.

to N defined set-theoretically by $\alpha(q, y) = qyq^{-1}$ (resp. $\beta(x, q, y) = x\alpha(q, y)$). Then one has the commutative diagram below:

$$(4) \quad \begin{array}{ccccc} G \times G & \xrightarrow{\text{id} \times \theta} & G \times N & \xrightarrow{\theta \times \text{id}} & N \times N \\ m_G \downarrow & & \downarrow \beta & \swarrow m_N & \\ G & \xrightarrow{\theta} & N & & \end{array} .$$

This can be expressed as follows in terms of the affine algebras A, A_0, A' of G, Q, N (cf. 5.1.3 below). Let $\rho' : A \rightarrow A'$, $\rho : A \rightarrow A_0$, and $\tau : A_0 \rightarrow A$ be the homomorphisms of k -bialgebras corresponding to σ', σ, π , and let $I = \text{Ker}(\rho)$. Then, by the preceding corollary, A' is identified with $A/\overline{A\tau(J_0)}$, where J_0 denotes the augmentation ideal of A_0 . On the other hand, let B be the affine algebra of G/Q ; it is the kernel of the pair of morphisms

$$A \xrightarrow[\text{(id} \otimes \rho) \Delta_A]{\tau_1} A \hat{\otimes}_k A_0,$$

i.e. $B = \{x \in A \mid \Delta_A(x) - x \hat{\otimes} 1 \in A \hat{\otimes} I\}$. Denote by γ the continuous morphism of k -algebras $\theta^\sharp : A' \rightarrow A$; it is a section of ρ' and an isomorphism of profinite k -algebras from A' to B . Moreover, $\tau = \pi^\sharp$ identifies A_0 with a sub-bialgebra of A , which is precisely the affine algebra of the quotient $N \backslash G$. One then deduces from (1) and (3) that one has an isomorphism of profinite k -algebras

$$(*) \quad A' \hat{\otimes}_k A_0 \xrightarrow{\sim} A, \quad a' \hat{\otimes} a_0 \mapsto \gamma(a')\tau(a_0),$$

whose inverse is the map $a \mapsto (\rho' \otimes \rho)\Delta_A(a)$. Finally, identify A' with its image in A under γ , so that the projection $A \rightarrow A'$ is then $\gamma\rho'$. Denoting by $\Delta_{A'}$ the comultiplication of A' , one then deduces from (4) that $\Delta_A(A') \subset A \hat{\otimes} A'$ and that the diagram below commutes:

$$\begin{array}{ccc} A' & \xrightarrow{\Delta_A} & A \hat{\otimes} A' \\ \parallel & & \downarrow \gamma\rho' \otimes \text{id} \\ A' & \xrightarrow{\Delta_{A'}} & A' \hat{\otimes} A' \end{array}$$

(one therefore also has $\gamma\rho' \circ c_A = c_{A'}$, where c_A (resp. $c_{A'}$) is the antipode of A (resp. A')). Moreover, denoting by m_A multiplication on A , one deduces from (2) that, for every $a \in A$,

$$\gamma\rho'(a) = (m_A \circ (\text{id} \otimes \tau\rho c_A) \circ \Delta_A)(a).$$

2.4.B. — ⁽⁹²⁾For simplicity, suppose that k is artinian. Then the foregoing can be expressed more simply in terms of the covariant bialgebras of G, Q, N . Indeed, since $G = N \times Q$ as formal k -varieties, $H(G) = H(N) \otimes_k H(Q)$ as k -coalgebras. Moreover, since multiplication on G is given by

$$(x, q) \cdot (x', q') = (x\alpha(q, x'), qq'), \quad \text{where } \alpha(q, x') = qxq^{-1},$$

multiplication on $H(G)$ is given as follows: for every $x \in H(N)$ and $q \in H(Q)$,

$$(x \otimes q) \cdot (x' \otimes q') = x\phi(q, x') \otimes qq',$$

where $\phi : H(Q) \otimes_k H(N) \rightarrow H(N)$ is the morphism of k -coalgebras induced by α . Since α is the composite morphism below (where δ_G (resp. m_G) is the diagonal morphism (resp. multiplication) of G , c_Q the inversion morphism of Q , and $v(q \otimes q' \otimes x) = q \otimes x \otimes q'$),

$$Q \times N \xrightarrow{v \circ (\delta_G \times \text{id})} Q \times N \times Q \xrightarrow{\text{id} \times \text{id} \times c_Q} Q \times N \times Q \xrightarrow{m_G} G,$$

one obtains, still denoting by c_Q the antipode of $H(Q)$, that

$$(*) \quad \phi(q \otimes x') = \sum_i q_i x' c_Q(q'_i) \quad \text{if} \quad \Delta_{H(Q)}(q) = \sum_i q_i \otimes q'_i.$$

In particular, if M is an abstract group and $H(Q)$ is the group algebra kM (i.e. $Q = \mathrm{Spf}^* kM$ is the *constant formal k -group* M_k), for every $\gamma \in M$ and $x' \in H(N)$ one has $\phi(\gamma \otimes x') = \gamma x' \gamma^{-1}$, and this defines an action of M on $H(N)$ by automorphisms of Hopf algebras.

2.4.1. Proposition. *Let $f : G \rightarrow K$ be a morphism of formal k -groups. If $H = \mathrm{Ker}(f)$ is topologically flat over k , the homomorphism $f' : G/H \rightarrow K$ induced by f is a monomorphism.*

This is a consequence of the results of Exposé IV;⁽⁹³⁾ nevertheless, we give a direct proof. Let T be a formal variety of finite length over k and t an element of $(G/H)(T)$ such that $f' \circ t$ is the identity element of $K(T)$. We must show that t is the identity element of $(G/H)(T)$. Denote by p the projection $G \rightarrow G/H$ and by X the fibered product $T \times_{G/H} G$. By 2.4, p is surjective and topologically flat, hence so is the morphism $\mathrm{pr}_1 : X \rightarrow T$, so pr_1 is an epimorphism by Proposition 1.3.1; it therefore suffices to show that $t \circ \mathrm{pr}_1$ is the identity element of $(G/H)(X)$. Denote by pr_2 the projection $X \rightarrow G$; one has $t \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2$, whence the equality $1 = f' \circ t \circ \mathrm{pr}_1 = f' \circ p \circ \mathrm{pr}_2 = f \circ \mathrm{pr}_2$; then the exact sequence

$$1 \rightarrow H \rightarrow G \xrightarrow{f} K$$

shows that pr_2 factors through H , so $p \circ \mathrm{pr}_2$ is the zero morphism. Since $p \circ \mathrm{pr}_2 = t \circ \mathrm{pr}_1$, this proves the proposition. From the proposition one deduces the following corollary. Denote by $\mathcal{O}(G), \mathcal{O}(K), \mathcal{O}(G/H)$ the affine algebras of $G, K, G/H$; one has seen (2.4) that p induces an injection from $\mathcal{O}(G/H)$ to $\mathcal{O}(G)$. Moreover, by Proposition 1.3, since $f' : G/H \rightarrow K$ is a monomorphism, the morphism $\mathcal{O}(K) \rightarrow \mathcal{O}(G/H)$ is surjective, whence:

Corollary. *Let $f : G \rightarrow K$ be a morphism of formal k -groups and $H = \mathrm{Ker}(f)$. If H is topologically flat over k , $\mathcal{O}(G/H)$ is the image of $\mathcal{O}(K)$ in $\mathcal{O}(G)$.*

2.4.2. — Retain the preceding notation and suppose that H and G are topologically flat over k . Then G is topologically flat over k and over G/H , so, by 1.3.3, G/H is topologically flat over k . Consequently, by 2.4, the canonical projection q from K to $\mathrm{Coker}(f')$ is topologically flat, and f' is an isomorphism from G/H to $\mathrm{Ker}(q)$. Moreover, it is clear that $\mathrm{Coker}(f) = \mathrm{Coker}(f')$. Thus, under the hypothesis that G and $H = \mathrm{Ker}(f)$ are topologically flat over k , one has obtained an isomorphism between $\mathrm{Ker}(q)$, the *image* of f , and $G/\mathrm{Ker}(f)$, the *coimage* of f . This implies the theorem below.

Theorem. *Let k be a field. Commutative formal k -groups form an abelian category.*

Corollary. *Let k be a field. Affine commutative group k -schemes form an abelian category.⁽⁹⁴⁾*

This follows from the theorem and the equivalence of categories 2.2.2.

2.5. — A formal k -group is said to be *étale* if the underlying formal variety is étale (cf. 1.6); these formal groups have a very simple structure. Indeed, suppose that k is local; let κ be the residue field of k , κ_s a separable closure of κ , and Γ the topological Galois group of κ_s over κ . A Γ -set consists of a set E and a *continuous* action of Γ on E (i.e. the isotropy group of every element $x \in E$ is an open subgroup of Γ). For every formal k -variety X , put

$$X(\kappa_s) = \varinjlim_{\ell} X(\ell),$$

where ℓ runs through the finite extensions of κ contained in κ_s .⁽⁹⁵⁾ Then Γ acts continuously on each $X(\ell)$, hence also on $X(\kappa_s)$. Moreover, let $X_{\kappa} = X \widehat{\otimes}_k \kappa$ (cf. 1.6.C); for every ℓ one has $X(\ell) = X_{\kappa}(\ell)$, whence $X(\kappa_s) = X_{\kappa}(\kappa_s)$. Now suppose that X is étale over k ; then X_{κ} is the formal κ -variety which is the direct sum of the $\mathrm{Spec} \kappa(x)$, for $x \in X$, and if one

⁽⁹³⁾**N.D.E.** Cf. IV, 5.2.6.

⁽⁹⁴⁾**N.D.E.** See also the remarks following VI A, 5.4.3.

⁽⁹⁵⁾**N.D.E.** Note that if $[\kappa_s : \kappa] = \infty$, κ_s is not a profinite κ -algebra, so the preceding notation is an abuse of notation. Moreover, we have detailed the original in what follows.

denotes by X'_κ the κ -scheme which is the direct sum of the $\text{Spec } \kappa(x)$, one sees that $X_\kappa(\kappa_s)$ is precisely $X'_\kappa(\kappa_s) = \text{Hom}_{((\text{Sch})/\kappa)}(\text{Spec } \kappa_s, X'_\kappa)$. Denote by $\mathcal{C} = ((\text{Sch})/\kappa)^\text{ét}$ the full subcategory of $((\text{Sch})/\kappa)$ consisting of étale κ -schemes. One knows that the functor $X' \mapsto X'(\kappa_s)$ is an equivalence from $\mathcal{C}$ to the category $\mathcal{C}'$ of Γ -sets (cf. SGA 1, V §§ 7–8 or [DG70], § I.4 6.4); it therefore induces an equivalence between the category of $\mathcal{C}$ -groups and that of $\mathcal{C}'$ -groups, and one sees at once that a $\mathcal{C}'$ -group is the same thing as an abstract group G equipped with a continuous action of Γ by group automorphisms (one then says that G is a Γ -group). In view of the equivalences $\mathbf{Vaf}_{/k}^\text{ét} \xrightarrow{\sim} \mathbf{Vaf}_{/\kappa}^\text{ét} \xrightarrow{\sim} ((\text{Sch})/\kappa)^\text{ét}$ of 1.6.E, one therefore obtains the following:

Proposition. *Let k be a local pseudocompact ring, κ its residue field, κ_s a separable closure of κ , and $\Gamma = \text{Gal}(\kappa_s/\kappa)$.*

- (i) *The functor $X \mapsto X(\kappa_s)$ is an equivalence from the category of étale formal k -varieties to that of Γ -sets.*
- (ii) *It induces an equivalence from the category of étale formal k -groups to that of Γ -groups.*

Remark 2.5.A. ⁽⁹⁶⁾ Let k be a field, k_s a separable closure of k , G an étale formal k -group, and M the abstract group $G(k_s)$. Denote by X a set of representatives of the orbits of $\Gamma = \text{Gal}(k_s/k)$ in M , and for every $x \in X$ let Γ_x be its stabilizer, which is a subgroup of Γ of finite index, and $L_x = k_s^{\Gamma_x}$, which is an extension of k of degree $[\Gamma : \Gamma_x]$ (see, for example, [BA1g], V § 10.10). Then, by the equivalence of categories above, the affine algebra $\mathcal{A}(G)$ of G is the product of the L_x , equipped with the product topology, so the L_x are precisely the simple quotients $\mathcal{A}(G)/\mathfrak{m}$, where $\mathfrak{m}$ is an open maximal ideal of $\mathcal{A}(G)$. Since these correspond by duality to subcoalgebras of $H(G)$, one obtains that $H(G)$ is *pointed* (i.e. $\dim_k(C) = 1$ for every simple subcoalgebra C) if and only if $L_x = k$ for every x ; in this case $\mathcal{A}(G)$ is the topological algebra k^M , hence $H(G)$ is the group algebra kM , and one therefore has $M = \{x \in H(G) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\} = G(k)$.

2.5.1. — Again suppose that the pseudocompact ring k is arbitrary. Since the functor $X \mapsto X_e$ of 1.6.F commutes with finite products, it transforms every formal group G into an étale formal group G_e , and since the morphism $p : X \rightarrow X_e$ of *loc. cit.* is functorial in X , $p : G \rightarrow G_e$ in this case is a morphism of formal groups. ⁽⁹⁷⁾ Consider the kernel $\text{Ker}(p)$; it is the fibered product of the diagram

$$\begin{array}{ccc} & G & \\ & \downarrow p & \\ \text{Spf}(k) & \xrightarrow{\varepsilon} & G_e \end{array}$$

where ε is the identity section of G_e . Since p induces a bijection of the underlying sets, one deduces (cf. 1.2, N.D.E. (41)) that the set underlying $\text{Ker}(p)$ is the image of $\text{Spf}(k)$ under ε , and that for every point s of $\text{Spf}(k)$, the local algebra of $\text{Ker}(p)$ at the point $\varepsilon(s)$ is $\mathcal{O}_{G, \varepsilon(s)} \hat{\otimes} \mathcal{O}_{G_e, \varepsilon(s)} k_s$. Moreover, at each point $\varepsilon(s)$ the residue field of $\mathcal{O}_{G, \varepsilon(s)}$ is $\kappa(s)$, and hence $\mathcal{O}_{G_e, \varepsilon(s)} = k_s$, whence $\mathcal{O}_{\text{Ker}(p), \varepsilon(s)} = \mathcal{O}_{G, \varepsilon(s)}$. For these reasons we shall say that $\text{Ker}(p)$ is the *infinitesimal neighborhood of the origin of G* , and shall write $\text{Ker}(p) = G_0$, thus obtaining an exact sequence

$$1 \longrightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e.$$

In what follows, we shall say that G is *infinitesimal* if $G = G_0$. ⁽⁹⁸⁾ This is equivalent to saying that, for every $s \in \text{Spf}(k)$, or again for every continuous morphism $k \rightarrow \kappa$, where κ

⁽⁹⁶⁾ **N.D.E.** We have added this remark.

⁽⁹⁷⁾ **N.D.E.** We have detailed the original in what follows.

⁽⁹⁸⁾ **N.D.E.** We have added the following sentence.

is a field equipped with the discrete topology, the identity is the unique element of $G(\kappa(s))$, resp. of $G(\kappa)$. Suppose further that G is *topologically flat over k* .⁽⁹⁹⁾ Then, by 1.6.1, the morphism $p : G \rightarrow G_e$ is topologically flat, and since it is bijective, it is therefore an effective epimorphism by Proposition 1.3.1. Consequently, G_e is identified with the quotient G/G_0 . One has therefore obtained:

Corollary. *Let G be a formal group topologically flat over k . Then G_e is identified with the quotient G/G_0 ; i.e. one has an exact sequence of formal groups*

$$1 \rightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e \rightarrow 1.$$

2.5.2. — Suppose that k is a perfect field. In this case one has seen (cf. 1.6.2) that the morphism $p : X \rightarrow X_e$ has a section $s : X_e \rightarrow X$ depending functorially on X ; this section is therefore a morphism of formal groups when X is a formal group. One therefore obtains the following:

Proposition. *When k is a perfect field, every formal k -group G is canonically isomorphic to the semidirect product of an infinitesimal group G_0 and an étale group G_e acting on G_0 .*⁽¹⁰⁰⁾

If, moreover, G is *commutative*, G is the product of G_0 and G_e . By 2.2.2, this canonical decomposition of commutative formal k -groups corresponds to an analogous decomposition of affine commutative group k -schemes; see paragraph 2.5.3 below.

Remark 2.5.2.A.⁽¹⁰¹⁾ The exact sequence $1 \rightarrow G_0 \rightarrow G \rightarrow G_e \rightarrow 1$ is not necessarily split when k is not perfect: let us give the following example taken from [DG70], § III.6, 8.6. Let k be a nonperfect field of characteristic $p > 0$. Let $\lambda \in k - k^p$, let L_λ be the abelian Lie p -algebra with basis (x, y) , the symbolic p th power (cf. VII A, 5.2) being given by $x^{(p)} = x$ and $y^{(p)} = \lambda x$, let $U_p(L_\lambda) = k[x, y]/(x^p - x, y^p - \lambda x)$ be the restricted enveloping algebra of L_λ (cf. VII A, 5.3), and let G_λ be the commutative formal k -group with affine algebra $U_p(L_\lambda)$. Then G_λ is a nonsplit extension of the constant étale k -group $(\mathbf{Z}/p\mathbf{Z})_k$ by the infinitesimal k -group $\alpha_{p,k} = \text{Spf } k[x]/(x^p - x)$, i.e. one has a nonsplit exact sequence of commutative formal k -groups:⁽¹⁰²⁾

$$0 \rightarrow \alpha_{p,k} \rightarrow G_\lambda \rightarrow (\mathbf{Z}/p\mathbf{Z})_k \rightarrow 0.$$

By duality, it corresponds to a nonsplit exact sequence of commutative algebraic k -groups

$$0 \rightarrow \mu_{p,k} \rightarrow D'(G_\lambda) \rightarrow \alpha_{p,k} \rightarrow 0,$$

where $\mu_{p,k} = \text{Spec } k[t]/(t^p - 1)$ (and in this way one obtains all the extensions of $\alpha_{p,k}$ by $\mu_{p,k}$; cf. [DG70], § III.6, 8.6).

2.5.3. —⁽¹⁰³⁾ Let k be a field.

Definition 2.5.3.A. One says that a commutative group k -scheme is *unipotent* if it is isomorphic to $\text{Spec } H(G)$, where G is a commutative *infinitesimal* formal k -group.⁽¹⁰⁴⁾

⁽⁹⁹⁾**N.D.E.** We have detailed what follows.

⁽¹⁰⁰⁾**N.D.E.** The same argument also shows that, for an arbitrary field k , if all the residue fields of G equal k , then G_e is the constant k -group $(M)_k$, where $M = G(k) = \text{Spf}^* H(G)$, and $H(G)$ is the semidirect product of $H(G_0)$ by the group algebra kM ; cf. addition 2.9 below.

⁽¹⁰¹⁾**N.D.E.** We have added this remark.

⁽¹⁰²⁾**N.D.E.** $\alpha_{p,k}$, $(\mathbf{Z}/p\mathbf{Z})_k$, and G_λ are also algebraic k -groups finite and flat over k ; cf. N.D.E. (84) in 2.2.2.

⁽¹⁰³⁾**N.D.E.** We have introduced the numbering 2.5.3 and 2.5.3.A to 2.5.3.C.

⁽¹⁰⁴⁾**N.D.E.** This coincides with the “usual” notion of a unipotent group k -scheme; cf. addition 2.8 at the end of Section 2.

⁽¹⁰⁵⁾Moreover, following Exp. IX, Definition 1.1, one says that a group k -scheme H is of *multiplicative type* if there exists a scheme S , faithfully flat and quasi-compact over $\operatorname{Spec} k$, such that H_S is a diagonalizable S -group, i.e. is isomorphic to $D_S(M)$ for some “abstract” abelian group M . By descent (fpqc), this implies that H is *affine* and commutative. Moreover, since one can replace S by the residue field of one of its points, one sees that H is of multiplicative type if and only if there exists an extension K of k such that H_K is a diagonalizable K -group.

Proposition 2.5.3.B. *Let T be an affine commutative group k -scheme and k_s a separable closure of k .*

- (i) *For T to be of multiplicative type, it is necessary and sufficient that its dual $D(T)$ be a commutative étale formal k -group.*
- (ii) *Consequently, T is of multiplicative type if and only if $T \otimes_k k_s$ is diagonalizable.*

PROOF. Denote by $\mathcal{A}$ the affine algebra of $D(T)$ and by $\mathcal{A}_0$ its local component at the identity; then $D(T)$ is étale over k if and only if $\mathcal{A}_0 = k$. If K is an extension of k , the algebra $\mathcal{A}_0 \hat{\otimes}_k K$ is local (since it is a projective limit of artinian local rings), hence coincides with the local component $(\mathcal{A} \hat{\otimes}_k K)_0$; moreover, since formation of $D(T)$ commutes with base change (cf. 2.2.2), one also has $\mathcal{A} \hat{\otimes}_k K \simeq D(T_K)$. Suppose that T is of multiplicative type; then there exists an extension K of k such that $\mathcal{O}(T) \otimes_k K$ is isomorphic, as a Hopf K -algebra, to the group algebra KM for some abelian group M . Then $\mathcal{A} \hat{\otimes}_k K$ is isomorphic to the algebra K^M , equipped with the product topology, so one has $K = (\mathcal{A} \hat{\otimes}_k K)_0 = \mathcal{A}_0 \hat{\otimes}_k K$, and this implies $\mathcal{A}_0 = k$, hence $D(T)$ is étale. Conversely, suppose that $D(T)$ is étale. Then $D(T) \hat{\otimes}_k k_s = D(T_{k_s})$ is a constant k_s -group M , so its covariant bialgebra is the group algebra $k_s M$ (cf. Remark 2.5.A), hence $\mathcal{O}(T) \otimes_k k_s = k_s M$, which proves that T is of multiplicative type and split by the extension $k \rightarrow k_s$. The proposition follows. $\square$

By 2.2.2, 2.5.1, and 2.5, one obtains:

Corollary 2.5.3.C. *Let k be a field and G an affine commutative group k -scheme.*

- (i) *G contains a subgroup of multiplicative type G_m such that G/G_m is unipotent.*
- (ii) *When k is perfect, there also exists a canonical retraction from G to G_m , so that G is the product of a unipotent group and a group of multiplicative type.*
- (iii) ⁽¹⁰⁶⁾*Let k_s be a separable closure of k and $\Gamma = \operatorname{Gal}(k_s/k)$. The category of group k -schemes of multiplicative type is anti-equivalent to the category of continuous Γ -modules.*

2.6. — We shall now study infinitesimal formal groups, to which the following paragraphs are devoted. Lie algebras play a primordial role in this study. First suppose that the base ring k is artinian, and let G be a formal group over k . One can give three different interpretations of the Lie algebra of G , all of which we shall use:

- (a) Let D be the algebra $k[d]/(d^2)$ of dual numbers over k , and δ the homomorphism from D to k which annihilates d . For every formal group G over k , $\operatorname{Lie}(G)$ is the kernel of $G(\delta)$, so by definition one has an *exact sequence of groups*

$$1 \longrightarrow \operatorname{Lie}(G) \longrightarrow G(D) \xrightarrow{G(\delta)} G(k) \longrightarrow 1.$$

- (b) Let A be the affine algebra of G and $I_A = \operatorname{Ker} \varepsilon_A$ its augmentation ideal. The elements of the group $G(D)$ are the morphisms of profinite k -algebras $f : A \rightarrow D$. The condition $G(\delta)(f) = 1$ is equivalent to $\delta \circ f = \varepsilon_A$. Since $x - \varepsilon_A(x) \cdot 1_A \in I_A$ for

⁽¹⁰⁵⁾**N.D.E.** The original indicated: “let us say that a commutative group k -scheme is of multiplicative type if it is isomorphic to $\operatorname{Spec} H(G)$, where G is a commutative étale formal k -group”. Here we have given the “usual” definition, taken from Exp. IX, 1.1, and shown its equivalence to the preceding condition; see also [DG70], § IV.1, Th. 2.2.

⁽¹⁰⁶⁾**N.D.E.** We have added part (iii), a consequence of Proposition 2.5.

every $x \in A$, this is equivalent to $f(I_A) \subset k \cdot d$, so $\text{Ker}(f)$ contains I_A^2 , hence also $\overline{I_A^2}$, and thus f induces a continuous linear map f' from $I_A/\overline{I_A^2} = \omega_{G/k}$ to k such that for every $x \in A$ one has the equality

$$f(x) = \varepsilon_A(x) \cdot 1_D + f'(\bar{x}) \cdot d,$$

where $\bar{x}$ denotes the image of $x - \varepsilon_A(x) \cdot 1_A$ in $I_A/\overline{I_A^2}$. One then sees that the map $f \mapsto f'$ defines a bijection from $\text{Lie}(G)$ to the topological dual $\omega_{G/k}^\dagger$ of $\omega_{G/k}$ (cf. 0.2.2). This bijection respects the group structures. Indeed, let f, g be two elements of $\text{Lie}(G)$; their product $f \cdot g$ is the composite map $h \circ \Delta_A$, where $h : A \hat{\otimes} A \rightarrow D$ satisfies $h(b \hat{\otimes} b') = f(b)g(b')$. Now, if $a \in I_A$, by 2.1(ii) one has $\Delta_A(a) - a \hat{\otimes} 1 - 1 \hat{\otimes} a \in I_A \hat{\otimes} I_A$, whence $(f \cdot g)(a) = f(a) + g(a)$ (cf. also II 3.10). In what follows, we identify $\text{Lie}(G)$ with $\omega_{G/k}^\dagger$ by means of the bijection $f \mapsto f'$ described above. *The group $\text{Lie}(G)$ is thus equipped with a k -module structure.*

- (c) Let $A^\dagger, D^\dagger$ be the k -modules dual to A, D , and $\{1_D^\dagger, d^\dagger\}$ the basis dual to the basis $\{1_D, d\}$ of D over k (one has $1_D^\dagger = \delta$). Since D is free of finite rank over k , the canonical map

$$\text{Hom}_c(A, D) \longrightarrow \text{Hom}_k(D^\dagger, A^\dagger), \quad f \longmapsto {}^t f$$

is bijective. On the other hand, f is determined by the values ${}^t f(1_D^\dagger)$ and ${}^t f(d^\dagger) = x$. The condition $G(\delta)(f) = 1$ is equivalent to the equality ${}^t f(1_D^\dagger) = \varepsilon_A$. Moreover, one easily sees that f is compatible with multiplication if and only if one has (cf. 2.3)

$$(*) \quad \delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x).$$

Finally, it is clear that a continuous linear map $f : A \rightarrow D$ which is compatible with multiplication and satisfies $\delta \circ f = \varepsilon_A$ sends the identity element of A to that of D .⁽¹⁰⁷⁾ The map $f \mapsto x$ therefore allows us to identify $\text{Lie}(G)$ with the set of “primitive elements” of $\mathbf{H}(G)$ (i.e. the $x \in \mathbf{H}(G)$ satisfying relation $(*)$). If x, y are two such elements, one has

$$\begin{aligned} \delta_G(xy) &= \delta_G(x)\delta_G(y) = \varphi_G((x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y)) \\ &= \varphi_G(xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy), \end{aligned}$$

whence $\delta_G(xy - yx) = \varphi_G((xy - yx) \otimes 1 + 1 \otimes (xy - yx))$. This shows that the k -module $\text{Lie}(G)$ is identified with a Lie subalgebra of $\mathbf{H}(G)$: we shall say that $\text{Lie}(G)$ is the Lie algebra of G .⁽¹⁰⁸⁾

2.6.1. — When k is an arbitrary pseudocompact ring and G a formal group over k , we call the functor $\underline{\text{Lie}}(G)$ which associates to every object C of $\mathbf{AIf}/k$ the C -Lie algebra of the formal C -group $G' = G \hat{\otimes}_k C$ the $\mathbf{O}_k$ -Lie algebra of G .⁽¹⁰⁹⁾ put $A' = A \hat{\otimes}_k C$; since I_A is a direct factor of A , $I_{A'}$ equals $I_A \hat{\otimes}_k C = I_A \hat{\otimes}_A A'$, and since $\omega_{G/k} = I_A \hat{\otimes}_A k$ and likewise $\omega_{G'/C} = I_{A'} \hat{\otimes}_{A'} C$, one obtains $\omega_{G'/C} = \omega_{G/k} \hat{\otimes}_A C$, and hence

$$\underline{\text{Lie}}(G)(C) = \text{Hom}_c(\omega_{G/k} \hat{\otimes}_A C, C), \quad \text{i.e.} \quad \underline{\text{Lie}}(G) = \mathbf{V}_k^t(\omega_{G/k})$$

⁽¹⁰⁷⁾**N.D.E.** Indeed, $\delta \circ f = \varepsilon_A$ implies that $f(1) = 1 + \lambda d$, with $\lambda \in k$, and then $f(1) = f(1)^2 = 1 + 2\lambda d$ gives $\lambda = 0$.

⁽¹⁰⁸⁾**N.D.E.** Comparing with VII A 2.5, one sees that if K is a group k -scheme of finite type and G is the formal completion of K at the origin (i.e. $\mathcal{A}(G)$ is the completion of the local ring $\mathcal{O}_{K,e}$ for the $\mathfrak{m}$ -adic topology, where $\mathfrak{m}$ is the kernel of the augmentation $\varepsilon : \mathcal{O}_{K,e} \rightarrow k$), then $\mathbf{H}(G)$ is identified with the algebra of distributions $U(K)$, and $\text{Lie}(G)$ with $\text{Lie}(K)$. (The condition that K be of finite type over k is used to ensure that $\mathfrak{m}/\mathfrak{m}^2$ is a k -module of finite length, hence discrete, so that its topological dual coincides with its ordinary dual.) In particular, when G is a finite formal k -group (i.e. such that $\mathcal{A}(G)$ is a finite k -module), in which case G can also be considered as the group k -scheme $\text{Spec } \mathcal{A}(G)$, the two definitions of $\text{Lie}(G)$ coincide.

⁽¹⁰⁹⁾**N.D.E.** We have detailed the original in what follows.

(with the notation of 1.2.3.B). Thus, by Proposition 1.2.3.E, $\underline{\text{Lie}}(G)$ is flat over $\mathbf{O}_k$ if and only if $\omega_{G/k}$ is a projective pseudocompact k -module.

2.6.2. — Conversely, every Lie algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a group-valued k -functor. Indeed, denote by $\mathbf{U}(\mathbf{L})$ the functor which associates to every object C of $\mathbf{AIf}/k$ the enveloping algebra $U(\mathbf{L}(C))$ of the C -Lie algebra $\mathbf{L}(C)$. By VII A, 3.2.2, $\mathbf{U}(\mathbf{L})$ is a bialgebra over $\mathbf{O}_k$ and therefore determines, by 2.2, a group-valued k -functor $\text{Spf}^* \mathbf{U}(\mathbf{L})$, which we shall henceforth denote by $\mathcal{G}(\mathbf{L})$. Thus $\mathcal{G}(\mathbf{L})(C)$ is the group of elements $z \in U(\mathbf{L}(C))$ of augmentation 1 such that $\Delta_{U(\mathbf{L}(C))}(z) = z \otimes z$. Moreover, when $\mathbf{L}$ is flat over $\mathbf{O}_k$, one has the following proposition.

Proposition. ⁽¹¹⁰⁾ *Let $\mathbf{L}$ be a flat $\mathbf{O}_k$ -Lie algebra.*

- (i) $\mathcal{G}(\mathbf{L})$ is a formal group topologically flat over k , whose $\mathbf{O}_k$ -bialgebra is $\mathbf{U}(\mathbf{L})$.
- (ii) $\mathcal{G}(\mathbf{L})$ is infinitesimal.
- (iii) For every object C of $\mathbf{AIf}/k$, $\underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))(C)$ is identified with the set

$$\text{Prim } U(\mathbf{L}(C)) = \{x \in U(\mathbf{L}(C)) \mid \varepsilon(x) = 0 \text{ and } \Delta(x) = x \otimes 1 + 1 \otimes x\}$$

of primitive elements of $U(\mathbf{L}(C))$. In particular, one has a natural morphism of $\mathbf{O}_k$ -Lie algebras $\tau_{\mathbf{L}} : \mathbf{L} \rightarrow \underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))$.

Indeed, the hypothesis that $\mathbf{L}$ is flat over $\mathbf{O}_k$ means that for every morphism $C \rightarrow D$ of $\mathbf{AIf}/k$, one has $\mathbf{L}(D) = \mathbf{L}(C) \otimes_C D$, and that for each local component C' of C , $\mathbf{L}(C')$ is a free C' -module. The first condition implies $U(\mathbf{L}(D)) = U(\mathbf{L}(C)) \otimes D$ (by the universal property of the tensor product and that of the functor $L \mapsto U(L)$), and the second condition implies, by the Poincaré–Birkhoff–Witt theorem (cf. Bourbaki, *Groupes et algèbres de Lie*, I 2.7), that $U(\mathbf{L}(C'))$ is a free C' -module. Thus the bialgebra $\mathbf{U}(\mathbf{L})$ is flat over $\mathbf{O}_k$. To prove (ii) and (iii), one can suppose that k is artinian. Put $L = \mathbf{L}(k)$, $U = U(L)$, $U_0 = k \cdot 1_U$, and let U^+ be the two-sided ideal of U generated by the image of L . Put, moreover, for every $n \geq 1$,

$$U_n = \{u \in U \mid \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+\}.$$

By 1.3.6 it suffices to show that U is the union of the U_n . Now, if one identifies L with its canonical image in U , L is evidently contained in U_1 . If $x_1, \dots, x_n$ are elements of L , one therefore has $\Delta_U(x_1 \cdots x_n) = (x_1 \otimes 1 + 1 \otimes x_1) \cdots (x_n \otimes 1 + 1 \otimes x_n)$, which shows, by induction on n , that the product $x_1 \cdots x_n$ belongs to U_n , hence that $U = \bigcup_n U_n$. This proves (ii). On the other hand, let $D = k[d]/(d^2)$ be the algebra of dual numbers over k . By hypothesis, one has $\mathbf{L}(D) \simeq L \otimes D$, whence $U(\mathbf{L}(D)) \simeq U \otimes D$ by the universal properties of the tensor product and the enveloping algebra. It follows that $\underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))(k)$ is identified with the set of elements $z = 1 + xd$ of $U \oplus Ud$ (where $x \in U$) such that $\varepsilon(z) = 1$ and $\Delta(z) = z \otimes z$, which is equivalent to $\varepsilon(x) = 0$ and $\Delta(x) = x \otimes 1 + 1 \otimes x$, i.e. $x \in \text{Prim } U$. In particular, the map $\tau_{\mathbf{L}} : x \mapsto 1 + dx$ is a morphism of $\mathbf{O}_k$ -Lie algebras from $\mathbf{L}$ to $\underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))$.

2.6.3. — If $\mathbf{L}$ is a flat Lie algebra over $\mathbf{O}_k$, the formal group $\mathcal{G}(\mathbf{L})$ can be characterized by a universal property.⁽¹¹¹⁾ Indeed, every morphism ϕ from $\mathcal{G}(\mathbf{L})$ to a formal group G induces a morphism $\underline{\text{Lie}}(\phi) : \underline{\text{Lie}}(\mathcal{G}(\mathbf{L})) \rightarrow \underline{\text{Lie}}(G)$; composing it with the morphism $\tau_{\mathbf{L}} : \mathbf{L} \rightarrow \underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))$ (cf. 2.6.2), one obtains a morphism $\phi' : \mathbf{L} \rightarrow \underline{\text{Lie}}(G)$, and one has:

Proposition. *If G is a formal k -group and $\mathbf{L}$ a flat $\mathbf{O}_k$ -Lie algebra, the map $\phi \mapsto \phi'$ defined above is a bijection*

$$\text{Hom}_{\mathbf{Grf}/k}(\mathcal{G}(\mathbf{L}), G) \xrightarrow{\sim} \text{Hom}_{\text{Lie}}(\mathbf{L}, \underline{\text{Lie}}(G)),$$

where the right-hand term denotes the set of morphisms of $\mathbf{O}_k$ -Lie algebras from $\mathbf{L}$ to $\underline{\text{Lie}}(G)$.

⁽¹¹⁰⁾**N.D.E.** We have brought out parts (i) and (ii), and added part (iii), which will be useful in 2.6.3 and 3.3.2.

⁽¹¹¹⁾**N.D.E.** We have modified what follows, taking advantage of the addition made in 2.6.2.

Indeed, one immediately reduces to the case where k is artinian. Put $L = \mathbf{L}(k)$. By 2.3.1, $\mathrm{Hom}_{\mathbf{Grf}/k}(\mathcal{G}(\mathbf{L}), G)$ is in bijection with the set of morphisms of unital algebras $\phi : U(L) \rightarrow H(G)$ such that the following diagram commutes:

$$\begin{array}{ccccc} U(L) & \xrightarrow{h} & H(G) & & \\ \Delta_{U(L)} \downarrow & & \searrow \delta_G & & \\ U(L) \otimes U(L) & \xrightarrow{h \otimes h} & H(G) \otimes H(G) & \xrightarrow{\varphi_G} & H(G \times G) \end{array} .$$

Now h is defined by its restriction to L , which is a morphism of Lie algebras from L to the Lie algebra underlying $H(G)$, and commutativity of the diagram means that h maps L into the part of $H(G)$ consisting of the x such that $\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x)$, which is precisely $\mathrm{Lie}(G)$; cf. 2.6 c).

2.7. — We conclude these generalities with a statement going back to S. Lie which will serve us in paragraph 5.1. A *formal monoid over k* is by definition a pair (M, m) consisting of a formal variety M and a morphism $m : M \times M \rightarrow M$ such that $m(C)$ makes $M(C)$ a monoid for every object C of $\mathbf{Alf}/k$.⁽¹¹²⁾ In particular, the “identity section”, which associates to every object C the identity element of $M(C)$, defines a section ε_M of the canonical projection $M \rightarrow \mathrm{Spf}(k)$. We shall say that the formal monoid M is *infinitesimal* if ε_M induces a bijection of the underlying sets.

Proposition. *Every topologically flat infinitesimal formal k -monoid M is a formal k -group.*⁽¹¹³⁾

We must show that $M(C)$ is a group for every object C of $\mathbf{Alf}/k$. One therefore immediately reduces to the case where k is artinian. Let then $U = H(M)$ be the coalgebra of M (1.3.5); the multiplication $m : M \times M \rightarrow M$ induces a morphism of coalgebras $m_U : U \otimes U \rightarrow U$, which makes U an associative algebra over k ; the identity element of this algebra is the image of the identity element of k under the map from k to U induced by the identity section ε_M of M . Similarly, the projection $M \rightarrow \mathrm{Spf}(k)$ induces a homomorphism ε_U from U to k ; we shall denote the kernel of ε_U by U^+ .

We must show that there exists an antipodism, that is, a morphism of coalgebras $c_U : U \rightarrow U$ such that, for every $u \in U$, one has:

$$(*) \quad (m_U \circ (c_U \otimes \mathrm{id}_U) \circ \Delta_U)(x) = \varepsilon_U(u) \cdot 1_U.$$

Let (U_n) be the filtration of U defined in 1.3.6; put $U_n^+ = U^+ \cap U_n$. Since M is infinitesimal, U^+ is the union of the submodules U_n^+ .⁽¹¹⁴⁾ One then puts $c_0(1) = 1$ and $c_1(x) = -x$ if $x \in U_1^+$, i.e. if x is a primitive element. Suppose that $c_{n-1} : U_{n-1} \rightarrow U$ has been constructed so as to satisfy $(*)$ for every $x \in U_{n-1}$, and let $x \in U_n^+$. By the proof of Lemma 1.3.6.A,

⁽¹¹²⁾**N.D.E.** Here and in what follows we have written “monoid” instead of “monoid with identity” (recall that a monoid is by definition a set equipped with an associative composition law and having an identity element).

⁽¹¹³⁾**N.D.E.** By the equivalence of categories of 1.3.5.D, a monoid in the category of topologically flat formal k -varieties “is the same thing” as a monoid in the category of flat cocommutative $\mathbf{O}_k$ -coalgebras, i.e. a cocommutative $\mathbf{O}_k$ -bialgebra (in the usual sense, i.e. not necessarily equipped with an antipode). Moreover, by 1.3.6, the hypothesis that M be infinitesimal is equivalent to saying that the corresponding bialgebra is connected. Thus, if k is an artinian ring, the proposition is equivalent to saying that: every connected cocommutative k -bialgebra, flat over k , is a k -Hopf algebra, i.e. possesses an antipode (and the cocommutativity hypothesis is in fact superfluous, cf. the proof).

⁽¹¹⁴⁾**N.D.E.** The original continued thus: “one then shows easily, by induction on n , that there exists one and only one linear map $c_n : U_n \rightarrow U_n$ such that the composite map $m_U \circ (c_n \otimes \mathrm{id}_U) \circ \Delta_U : U_n^+ \rightarrow U$ is zero”; we have detailed the proof, which rests on that of Lemma 1.3.6.A.

one has $\Delta_U(x) - x \otimes 1 \in U_{n-1} \otimes U^+$ (this is where the hypothesis that U be flat over k enters), so one can write $\Delta_U(x) = x \otimes 1 + \sum_i y_i \otimes z_i$, with $y_i \in U_{n-1}$; one then puts $c_n(x) = -\sum_i c_{n-1}(y_i)z_i$. One thus obtains a k -linear map $c : U \rightarrow U$, which is the left inverse of id_U for the monoid law on $\text{End}_k(U)$ defined by $f \cdot g = m_U \circ (f \otimes g) \circ \Delta_U$ (the identity element being the map $\eta : u \mapsto \varepsilon(u) \cdot 1_U$). It follows that c is uniquely determined, and is also the right inverse of id_U , i.e. one also has $m_U \circ (c_n \otimes \text{id}_U) \circ \Delta_U = \eta$ (without assuming U cocommutative).

2.8. — Unipotent group schemes over a field. — ⁽¹¹⁵⁾Let k be a field. “Recall” that a group k -scheme G is said to be *unipotent* if it satisfies the following two conditions (cf. [DG70], § IV.2, Prop. 2.5):

- (a) G is affine.
- (b) Every simple $\mathcal{O}(G)$ -comodule is trivial, i.e. if $\rho : V \rightarrow V \otimes_k \mathcal{O}(G)$ is an $\mathcal{O}(G)$ -comodule structure on a k -vector space $V \neq 0$, and if there is no nonzero subspace $W \neq V$ such that $\rho(W) \subset W \otimes_k \mathcal{O}(G)$, then V has dimension 1 and $\rho(v) = v \otimes 1$ for every $v \in V$.

According to loc. cit., when G is of finite type over k , this is equivalent to the definition given in Exp. XVII, § 1, namely that G possesses a finite composition series whose successive quotients are isomorphic to k -subgroups of $\mathbf{G}_{a,k}$.

Now, for every affine group k -scheme G , the comultiplication of $\mathcal{O}(G)$ equips the pseudocompact k -module $A = \mathcal{O}(G)^*$ with a structure of profinite k -algebra, not necessarily commutative, the identity element 1_A being the augmentation $\varepsilon : \mathcal{O}(G) \rightarrow k$. On the other hand, let $I = \{f \in A \mid f(1_{\mathcal{O}(G)}) = 0\}$; this is a two-sided ideal of A , and one has $A = k \cdot 1_A \oplus I$, cf. 1.3.6.

Let V be a subspace of A of finite codimension; consider the continuous k -bilinear map $\varphi : A \times A \rightarrow A/V$, $(a, b) \mapsto ab + V$; by Lemma 0.3.1, there exist two subspaces L_1, L_2 of finite codimension in A such that V contains AL_2 and L_1A ; then $L = L_1 \cap L_2$ is a subspace of A of finite codimension, and V contains the two-sided ideal ALA , which has finite codimension. This shows that the two-sided ideals of finite codimension form a basis of neighborhoods of 0. One deduces that an $\mathcal{O}(G)$ -comodule “is the same thing” as a *continuous A -module*, i.e. an A -module V such that the map $A \times V \rightarrow V$ is continuous, V being equipped with the discrete topology. Such a module is clearly the union of submodules V_i of finite dimension over k , each being a module over a quotient k -algebra A_i of A , of finite dimension over k . It follows that if M is a simple continuous module, it has finite dimension over k , and is a faithful simple module over the finite-dimensional k -algebra $A/\text{Ann}(M)$; the latter is therefore a simple finite-dimensional k -algebra, i.e. $\text{Ann}(M)$ is an open maximal ideal of A . Conversely, let P be an open prime ideal⁽¹¹⁶⁾ of A ; then A/P is a finite-dimensional k -algebra in which the ideal (0) is prime, so it is a simple finite-dimensional k -algebra, and hence there exists, up to isomorphism, a unique simple continuous A -module whose annihilator is P . It follows that the map $M \mapsto \text{Ann}(M)$ defines a bijection between the isomorphism classes of simple continuous A -modules and the open prime ideals of A . In particular, one calls the A -module A/I , which has dimension 1 over k , the “trivial module”; it corresponds to the trivial $\mathcal{O}(G)$ -comodule V of dimension 1, i.e. such that $\rho(v) = v \otimes 1_{\mathcal{O}(G)}$ for every $v \in V$. One therefore obtains the following proposition:

Proposition 2.8.1. *Let k be a field, G an affine group k -scheme, and $A = \mathcal{O}(G)^*$.*

⁽¹¹⁵⁾**N.D.E.** We have added this subsection.

⁽¹¹⁶⁾**N.D.E.** Recall that a two-sided ideal P of a ring A is said to be prime if in A/P the product of two nonzero two-sided ideals is nonzero.

- (i) Then G is unipotent if and only if I is the unique open prime ideal of A .⁽¹¹⁷⁾
- (ii) In particular, if G is commutative, so that $\mathcal{O}(G) = H(D(G))$, where $D(G) = \mathrm{Spf}(A)$ denotes the Cartier dual of G , then G is unipotent if and only if $D(G)$ is infinitesimal.

2.9. — Pointed cocommutative Hopf algebras over a field. — ⁽¹¹⁸⁾Let k be a field, C a k -coalgebra, and A the algebra C^* equipped with the structure of profinite k -algebra, not necessarily commutative, described in 2.8; by 0.2.2, one has $C = A^\dagger = \mathrm{Hom}_c(A, k)$. One deduces that the map $D \mapsto D^\perp = \{f \in A \mid f(D) = 0\}$ is a bijection from the set of subcoalgebras of C onto that of closed two-sided ideals (in what follows, we shall simply say “ideals”) of A ; the inverse bijection is given by $I \mapsto I^\perp = \{x \in C = A^\dagger \mid x(I) = 0\}$. Since every maximal closed ideal is an open maximal ideal (cf. 0.2.1), every subcoalgebra therefore contains a simple subcoalgebra, necessarily of finite dimension.

Recall that a subcoalgebra D of C is said to be *irreducible* if it contains only one simple subcoalgebra S_0 , which is equivalent to saying that $\mathfrak{m}_0 = S_0^\perp$ is the unique open maximal ideal containing $D^\perp$, i.e. that $D^\perp + \mathfrak{m} = A$ for every open maximal ideal $\mathfrak{m} \neq \mathfrak{m}_0$. This is in particular the case if $D = S_0$. Then the sum Σ_0 of all the irreducible subcoalgebras C_i containing S_0 is clearly a subcoalgebra, and it is irreducible since, for every $\mathfrak{m} \neq \mathfrak{m}_0$, one has, by 0.2.D:

$$\mathfrak{m} + \bigcap_i C_i^\perp = \bigcap_i (\mathfrak{m} + C_i^\perp) = A.$$

One says that Σ_0 is the *irreducible component* of C corresponding to C_0 .

On the other hand, one says that C is *pointed* if every simple sub- k -coalgebra of C has dimension 1; this is equivalent to saying that for every open maximal ideal $\mathfrak{m}$ of A , one has $A/\mathfrak{m} = k$. Recall also that C is said to be *connected* if it is irreducible and pointed. (Let us note in passing that if C is a bialgebra, it is connected if and only if it is irreducible, since $k \cdot 1_C$ is a simple subcoalgebra.)

Suppose henceforth that C is *cocommutative*. Then A is commutative and is therefore the product of its local components $A_{\mathfrak{m}}$, for $\mathfrak{m} \in \Upsilon(A)$ (cf. 0.1.1); denote by $S_{\mathfrak{m}}$ the simple subcoalgebra $\mathfrak{m}^\perp \simeq (A/\mathfrak{m})^*$ and by $\Sigma_{\mathfrak{m}}$ its irreducible component. One can describe $\Sigma_{\mathfrak{m}}$ as follows. Denote by $J_{\mathfrak{m}}$ the kernel of the projection $A \rightarrow A_{\mathfrak{m}}$; it is contained in $\mathfrak{m}$ and is the smallest closed ideal I of $\mathfrak{m}$ such that $I + \mathfrak{m}' = A$ for every $\mathfrak{m}' \neq \mathfrak{m}$. Indeed, if I has this property, then I contains $A_{\mathfrak{m}'}$ for every $\mathfrak{m}' \neq \mathfrak{m}$, hence contains $J_{\mathfrak{m}}$. Since $A = J_{\mathfrak{m}} \oplus A_{\mathfrak{m}}$, it follows that $\Sigma_{\mathfrak{m}} = J_{\mathfrak{m}}^\perp$ is identified with $A_{\mathfrak{m}}^\dagger$. One can now prove the:

Proposition 2.9.1. *Let k be a field.*

- (i) *Let G be a formal k -group such that all the residue fields of G equal k . Then G_e is the constant k -group M_k , where $M = G(k) = \{x \in H(G) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$, and $H(G)$ is the semidirect product of $H(G_0)$ by kM (cf. 2.4.B).*
- (ii) *Equivalently: let H be a pointed cocommutative k -Hopf algebra. Then H is the semidirect product of the irreducible component H_0 of the identity element 1_H by kM , where $M = \{x \in H \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$.*

Let us prove (i). Since all the residue fields of G equal k , the projection $\pi : G \rightarrow G_e$ admits the section $s : G_e \rightarrow G$ defined by $\mathcal{O}_{G,g} \rightarrow \kappa(g) = k$, for every $g \in G$; moreover, for every $g, h \in G$, $\mathcal{O}_{G,g} \hat{\otimes}_k \mathcal{O}_{G,h}$ is local with residue field k , and one therefore obtains that $s \times s$ is a section of the projection $\pi \times \pi : G \times G \rightarrow (G \times G)_e = G_e \times G_e$. Since π is a morphism of groups, it follows that $\pi \circ m_G \circ (s \times s) = m_{G_e} = \pi \circ s \circ m_{G_e}$, and since π is an epimorphism this implies that s is a morphism of groups. One therefore obtains that $G = G_0 \rtimes G_e$, and

⁽¹¹⁷⁾**N.D.E.** This is also equivalent to saying that $k \cdot 1_{\mathcal{O}(G)}$ is the unique simple k -subcoalgebra of $\mathcal{O}(G)$; see for example [Ab80], 3.1.4; let us mention in passing a misprint in loc. cit., p. 130, line 4: $M \simeq C^*/\mathrm{ann} M$ is to be replaced by $C^*/\mathrm{ann} M \simeq \mathrm{End}_k(M)$.

⁽¹¹⁸⁾**N.D.E.** We have added this paragraph, to translate Proposition 2.5.2, as well as the variant mentioned in N.D.E. (100), into the language of cocommutative Hopf algebras.

hence $H(G)$ is the semidirect product of $H(G_0)$ by $H(G_e)$. Moreover, since all the residue fields of G equal k , $H(G_e)$ is the group algebra kM , where $M = G_e(k)$ (cf. 2.5.A). Finally, since G_0 is infinitesimal, the morphism $G(k) \rightarrow G_e(k)$ is injective; it is therefore bijective (since it admits a section), so $M = G(k)$. Point (i) follows.

To prove (ii), it only remains to see that H_0 equals $H(G_0)$. Now the identity element 1_H of H is none other than the augmentation $\varepsilon_A : A \rightarrow k$, which is the identity section of $G(k)$; thus the local component of $\mathcal{A}(G)$ corresponding to H_0 is none other than $\mathcal{A}(G_0)$ and hence, by what we have seen above, one has $H_0 = \mathcal{A}(G_0)^\dagger = H(G_0)$. This proves the proposition.

Remarks 2.9.2. (a) The above proposition, implicitly contained in 2.5.2, was obtained independently by B. Kostant (cf. [Sw69], Preface). Combined with Cartier's Theorem 3.3 below (cf. [Ca62], § 12, Th. 3), also obtained by Kostant (cf. [Sw69], loc. cit.), this result is often called the “Cartier–Gabriel–Kostant theorem”.

(b) In the form (ii), 2.9.1 was extended by R. G. Heyneman and M. E. Sweedler to the case where one assumes that H is pointed and a direct sum of its irreducible components (but is not necessarily cocommutative), cf. [HS69], Th. 3.5.8.

3. Phenomena particular to characteristic 0

Throughout Section 3, we assume that the pseudocompact ring k contains the field of rational numbers $\mathbf{Q}$.

Lemma 3.1. *Let C be a commutative unital $\mathbf{Q}$ -algebra and L a Lie algebra over C whose underlying C -module is free. Then the canonical map $L \rightarrow U(L)$ is an isomorphism from L onto the set of primitive elements of $U(L)$.*

Indeed, identify L with its canonical image in $U(L)$; let I be a totally ordered set and $(x_i)_{i \in I}$ a basis of L indexed by I ; denote by $\mathbf{N}^{(I)}$ the set of families $n = (n_i)_{i \in I}$ of natural numbers such that n_i is zero except perhaps for a finite number of indices $i_1 < i_2 < \dots < i_s$ (these indices depend on n); finally, put $x^n = x_{i_1}^{n_{i_1}} x_{i_2}^{n_{i_2}} \dots x_{i_s}^{n_{i_s}}$ and $n! = (n_{i_1}!)(n_{i_2}!) \dots (n_{i_s}!)$.

One then knows that the x^n form a basis of $U(L)$ (the Poincaré–Birkhoff–Witt theorem) and one sees easily that one has

$$(*) \quad \Delta_{U(L)} \left(\frac{x^n}{n!} \right) = \sum \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!},$$

the sum being taken over all the elements m of $\mathbf{N}^{(I)}$ such that $0 \leq m \leq n$ (i.e. such that $0 \leq m_i \leq n_i$ for every i). It clearly follows that one has $\Delta_{U(L)}(u) = u \otimes 1 + 1 \otimes u$ if and only if u is a linear combination of the x_i .

3.2. — Suppose now that C is *artinian*, with radical $\mathfrak{r}$. For every (associative, unital) C -algebra U , the ideal $\mathfrak{r}U$ therefore consists of nilpotent elements; if x belongs to $\mathfrak{r}U$, we shall put

$$\exp_U x = 1 + x + \frac{x^2}{2!} + \dots \quad (119)$$

One thus obtains a bijection from $\mathfrak{r}U$ onto $1 + \mathfrak{r}U$; the inverse bijection maps an element $1 + y$ of $1 + \mathfrak{r}U$ to

$$\log_U(1 + y) = y - \frac{y^2}{2} + \frac{y^3}{3} - \dots$$

Moreover, it is clear that the map $\exp_U$ is functorial in U .⁽¹²⁰⁾

⁽¹¹⁹⁾**N.D.E.** If $x, x' \in \mathfrak{r}U$ commute, one therefore has $\exp_U(x + x') = (\exp_U x)(\exp_U x')$.

⁽¹²⁰⁾**N.D.E.** i.e. for every morphism $\phi : U \rightarrow V$ of C -algebras, one has $\phi(\exp_U(x)) = \exp_V \phi(x)$.

The ring C still being artinian, suppose that U is equipped with a bialgebra structure over C (cf. 2.2). For every primitive element x of $\mathfrak{r}U$ (cf. VII A 3.2.3), one then has

$$\begin{aligned}\Delta_U(\exp_U x) &= \exp_{U \otimes U}(\Delta_U(x)) \\ &= \exp_{U \otimes U}(x \otimes 1 + 1 \otimes x) \\ &= \exp_{U \otimes U}(x \otimes 1) \cdot \exp_{U \otimes U}(1 \otimes x) \\ &= ((\exp_U x) \otimes 1) \cdot (1 \otimes (\exp_U x)) \\ &= (\exp_U x) \otimes (\exp_U x).\end{aligned}$$

One therefore sees that the bijection $\exp_U$ transforms a primitive element of $\mathfrak{r}U$ into an element z of $1 + \mathfrak{r}U$ such that $\Delta_U(z) = z \otimes z$. Conversely, if z satisfies these conditions then, putting $x = \log_U(z)$, the preceding calculation shows that $\exp_{U \otimes U}(x \otimes 1 + 1 \otimes x)$ equals $z \otimes z = \Delta(\exp_U x) = \exp_{U \otimes U}(\Delta_U(x))$, whence $x \otimes 1 + 1 \otimes x = \Delta_U(x)$.⁽¹²¹⁾ Note further that if $z \in 1 + \mathfrak{r}U$ satisfies $\Delta_U(z) = z \otimes z$, then $\varepsilon_U(z)^2 = \varepsilon_U(z)$ and since $\varepsilon_U(z)$ is invertible (since z is, $\mathfrak{r}$ being nilpotent), it follows that $\varepsilon_U(z) = 1$.

Consider in particular a Lie algebra L flat over C , take for U the enveloping algebra $U(L)$ of L over C and identify L with its canonical image in U . By Lemma 3.1, L is therefore the set of primitive elements of U (indeed L is a product of free modules over the local components of C). Then consider the formal C -group $\mathcal{G}(L) = \mathrm{Spf}^* U(L)$, which has $U = U(L)$ for its covariant bialgebra (cf. 2.6.2). Let $\mathfrak{m}$ be a maximal ideal of C and $\overline{C} = C/\mathfrak{m}$. Since $\mathcal{G}(L)$ is infinitesimal (loc. cit.), the identity element of $\overline{U} = U \otimes_C \overline{C}$ is the only element $\bar{z}$ of $\overline{U}$ such that⁽¹²²⁾ $\varepsilon_{\overline{U}}(\bar{z}) = 1$ and $\Delta_{\overline{U}}(\bar{z}) = \bar{z} \otimes \bar{z}$. It follows that the elements z of U such that $\varepsilon_U(z) = 1$ and $\Delta_U(z) = z \otimes z$ necessarily belong to $1 + \mathfrak{r}U$.

Finally, since $L \cap \mathfrak{r}U$ is identified with $\mathfrak{r}L = L \otimes_C \mathfrak{r}$, one sees at last that: $\exp_U$ defines a bijection from $L \otimes_C \mathfrak{r}$ onto the group $\mathcal{G}(L)(C)$. We summarize:

Proposition. *Let k be a pseudocompact ring containing $\mathbf{Q}$ and $\mathbf{L}$ a flat $\mathbf{O}_k$ -Lie algebra.*

- (i) *For every object C of $\mathbf{Alf}/k$, denote its radical by $\mathfrak{r}(C)$; then the map*

$$\exp_{U(\mathbf{L}(C))} : \mathbf{L}(C) \otimes_C \mathfrak{r}(C) \longrightarrow \mathcal{G}(\mathbf{L})(C)$$

is bijective and functorial in C and $\mathbf{L}$.

- (ii) *The natural morphism $\mathbf{L} \longrightarrow \underline{\mathrm{Lie}}(\mathcal{G}(\mathbf{L}))$ (cf. 2.6.2) is an isomorphism of $\mathbf{O}_k$ -Lie algebras.*⁽¹²³⁾

3.2.1. — The bijection $\exp_{U(\mathbf{L}(C))}$ allows one to define, by transport of structure, a group law on the set $\mathbf{L}(C) \otimes_C \mathfrak{r}(C)$ (which one identifies with a subset of $U(\mathbf{L}(C))$ as in 3.2). If x and y are two elements of $\mathbf{L}(C) \otimes_C \mathfrak{r}(C)$, this law is such that

$$\begin{aligned}x \cdot y &= \log((\exp x)(\exp y)) = \log \left(1 + \sum_{p+q>0} \frac{x^p y^q}{p! q!} \right) \\ &= \sum_{m \geq 1} \sum_{p_i + q_i > 0} \frac{(-1)^{m-1}}{m} \frac{x^{p_1} y^{q_1}}{p_1! q_1!} \cdots \frac{x^{p_m} y^{q_m}}{p_m! q_m!} = \sum_{\ell \geq 1} P_\ell(x, y),\end{aligned}$$

⁽¹²¹⁾**N.D.E.** We have detailed the preceding passage and added the following sentence.

⁽¹²²⁾**N.D.E.** We have added the condition $\varepsilon_{\overline{U}}(\bar{z}) = 1$, omitted in the original.

⁽¹²³⁾**N.D.E.** We have added point (ii), an immediate consequence of 3.1.

where $P_\ell(x, y)$ denotes the sum of the monomials of total degree ℓ in x and y . One has for example:

$$\begin{aligned} P_1(x, y) &= x + y, \\ P_2(x, y) &= \underbrace{\frac{x^2}{2} + xy + \frac{y^2}{2}}_{m=1} - \underbrace{\frac{1}{2}(x^2 + xy + yx + y^2)}_{m=2} \\ &= \frac{1}{2}(xy - yx) = \frac{1}{2}[x, y], \end{aligned}$$

and $P_3(x, y)$ is the sum of the following three terms:

$$\begin{aligned} &\underbrace{\frac{x^3}{6} + \frac{x^2y}{2} + \frac{xy^2}{2} + \frac{y^3}{6}}_{m=1} \\ &- \frac{1}{2} \underbrace{\left(x^3 + \frac{3}{2}x^2y + \frac{1}{2}yx^2 + xyx + yxy + \frac{1}{2}y^2x + \frac{3}{2}xy^2 + y^3 \right)}_{m=2} \\ &+ \frac{1}{3} \underbrace{(x^3 + x^2y + yx^2 + xyx + yxy + y^2x + xy^2 + y^3)}_{m=3}, \end{aligned}$$

whence

$$\begin{aligned} P_3(x, y) &= \frac{1}{12}(x^2y + yx^2 - 2xyx - 2yxy + y^2x + xy^2) \\ &= \frac{1}{12}([y, x], x) + [y, [y, x]]. \end{aligned}$$

One can show more generally that one has the *Campbell–Hausdorff formula*:⁽¹²⁴⁾

$$\begin{aligned} P_\ell(x, y) &= \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m \cdot \ell} \sum_{\substack{p_1, \dots, p_m \\ q_1, \dots, q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!} \frac{(\operatorname{ad} y)^{q_i}}{q_i!} \right) \frac{(\operatorname{ad} x)^{p_m}}{p_m!} (y) \\ &\quad + \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m \cdot \ell} \sum_{\substack{p_1, \dots, p_{m-1} \\ q_1, \dots, q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!} \frac{(\operatorname{ad} y)^{q_i}}{q_i!} \right) (x), \end{aligned}$$

where the $p_j, q_i \in \mathbf{N}$ satisfy $p_i + q_i \geq 1$ for $i = 1, \dots, m-1$ and $p_m + \sum_{i=1}^{m-1} (p_i + q_i) = \ell - 1$ (i.e. in the above sums, every nonzero “Lie monomial” has total degree ℓ); for a proof, see N. Jacobson, *Lie Algebras* (Interscience, 1962), § V.5, or [BLie], II § 6.4, Th. 2.

3.3. — If G is a formal k -group with affine algebra A , recall that one denotes by I_A the augmentation ideal of A and by $\omega_{G/k}$ the pseudocompact k -module $I_A/\overline{I_A^2} \simeq I_A \hat{\otimes}_A k$.

Theorem. (Cartier). *Let k be a pseudocompact ring⁽¹²⁵⁾ containing $\mathbf{Q}$ and G a formal k -group. The following assertions are equivalent:*

- (i) *There exists a flat $\mathbf{O}_k$ -Lie algebra $\mathbf{L}$ such that G is isomorphic to $\mathcal{G}(\mathbf{L})$ (and in this case $\mathbf{L} = \underline{\operatorname{Lie}}(G)$ by 3.2).*
- (ii) *There exists a projective pseudocompact k -module ω such that the formal variety underlying G is isomorphic to the formal variety $V_k^{f,0}(\omega)$ (cf. 1.2.5) with affine algebra $k[[\omega]]$ (and in this case $\omega \simeq \omega_{G/k}$).*
- (iii) *G is infinitesimal and $\omega_{G/k}$ is a projective pseudocompact k -module.*
- (iv) *G is infinitesimal and topologically flat over k .*

⁽¹²⁴⁾ **N.D.E.** We have corrected the formula given, which was erroneous, and added the reference [BLie].

⁽¹²⁵⁾ **N.D.E.** We have removed the hypothesis that k be local (the proof reduces to this case).

PROOF. (i) $\Rightarrow$ (ii): Let $\omega = \Gamma^*(\mathbf{L})$ be the pseudocompact k -module dual to $\mathbf{L}$ (cf. 1.2.3.D). For every object C of $\mathbf{AIf}/k$, we must exhibit an isomorphism from $V_k^{f,0}(\omega)(C)$ onto $\mathcal{G}(\mathbf{L})(C)$ which is functorial in C . By 1.2.5, $V_k^{f,0}(\omega)(C)$ is identified with the set $\text{Hom}_c(\omega, \mathfrak{r}(C))$ of continuous k -linear maps from ω to the radical of C . This set is naturally isomorphic to the set $\text{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$ of continuous C -linear maps from $\omega \widehat{\otimes}_k C$ to $\mathfrak{r}(C)$; finally, since $\omega \widehat{\otimes}_k C$ is a projective pseudocompact C -module, the canonical map

$$(\omega \widehat{\otimes}_k C)^\dagger \otimes_C \mathfrak{r}(C) \longrightarrow \text{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$$

is bijective (cf. 0.3.6.A). Since, by 1.2.3.E, $\mathbf{L}(C)$ is identified with $V_k^f(\Gamma^*(\mathbf{L}))(C) = (\omega \widehat{\otimes}_k C)^\dagger$, one obtains that $V_k^{f,0}(\omega)(C)$ is canonically isomorphic to $\mathbf{L}(C) \otimes_C \mathfrak{r}(C)$, which is canonically isomorphic to $\mathcal{G}(\mathbf{L})(C)$ by Proposition 3.2. This proves the implication (i) $\Rightarrow$ (ii).

(ii) $\Rightarrow$ (iii): Let ω be a projective object of $\mathbf{PC}(k)$ and h an isomorphism from $k[[\omega]]$ onto the affine algebra A of G . Composing h with the augmentation $\varepsilon_A : A \rightarrow k$, one obtains a homomorphism $\varepsilon_A \circ h : k[[\omega]] \rightarrow k$, which is determined by its restriction λ to ω ; the latter sends ω into the radical $\mathfrak{r}$ of k . Thus the map $x \mapsto x - \lambda(x)$, from ω to the radical of $k[[\omega]]$, extends, by the universal property of $k[[\omega]]$ (cf. 1.2.5), to an endomorphism ℓ_λ of $k[[\omega]]$. The equalities $\ell_\lambda \circ \ell_{-\lambda} = \ell_{-\lambda} \circ \ell_\lambda = \text{id}$ show that ℓ_λ is an automorphism of $k[[\omega]]$. Consequently, $h \circ \ell_\lambda$ is, like h , an isomorphism from $k[[\omega]]$ onto A and moreover $\varepsilon_A \circ h \circ \ell_\lambda$ maps ω to 0. Replacing h by $h \circ \ell_\lambda$ if necessary, one can therefore suppose that $\varepsilon_A \circ h$ vanishes on the closed ideal I of $k[[\omega]]$ generated by ω . In this case, h induces an isomorphism from $I/\overline{I^2}$ onto $I_A/\overline{I_A^2}$; since $I/\overline{I^2} \simeq \omega$, it follows that $\omega_{G/k} = I_A/\overline{I_A^2}$ is isomorphic to ω , hence projective. It is clear, on the other hand, that $V_k^{f,0}(\omega)$ is infinitesimal, as is G .

(iii) $\Rightarrow$ (i): Suppose that G is infinitesimal and that $\omega_{G/k}$ is projective. Let $\mathbf{L}$ be the $\mathbf{O}_k$ -Lie algebra of G ; its underlying $\mathbf{O}_k$ -module is $V_k^f(\omega_{G/k})$, by 2.6.1. Consequently, $\mathbf{L}$ is flat over $\mathbf{O}_k$, and $\Gamma^*(\mathbf{L}) \simeq \omega_{G/k}$, by Proposition 1.2.3.E. Thus, by the proof of (i) $\Rightarrow$ (ii), the affine algebra of the formal k -group $\mathcal{G}(\mathbf{L})$ is identified with $k[[\omega_{G/k}]]$. On the other hand, by 2.6.3, the identity morphism of $\mathbf{L}$ is canonically associated with a morphism of formal groups $\mathcal{G}(\mathbf{L}) \rightarrow G$, hence with a continuous morphism of k -algebras

$$\phi : A \rightarrow k[[\omega_{G/k}]].$$

Let I be the closed ideal of $k[[\omega_{G/k}]]$ generated by $\omega_{G/k}$; filter $k[[\omega_{G/k}]]$ (resp. A) by the closures of the ideals I^n (resp. I_A^n). We have to show that ϕ , which by definition induces the identity on $\omega_{G/k} = I_A/\overline{I_A^2} = I/\overline{I^2}$, is an isomorphism.

Since $\omega_{G/k}$ is a projective object of $\mathbf{PC}(k)$, there exists a section τ of the canonical projection $I_A \rightarrow \omega_{G/k}$. By the universal property of $k[[\omega_{G/k}]]$ (cf. 1.2.5), τ defines a continuous morphism of algebras

$$\psi : k[[\omega_{G/k}]] \rightarrow A$$

and $\phi \circ \psi$ induces the identity map on $\omega_{G/k}$, hence also on the associated graded of $k[[\omega_{G/k}]]$. It follows that $\phi \circ \psi$ is an isomorphism, by [CA], § V.7, Lemma 1.⁽¹²⁶⁾

Moreover, ψ induces an isomorphism from $I/\overline{I^2}$ onto $I_A/\overline{I_A^2}$, hence a *surjection* of the associated graded of $k[[\omega_{G/k}]]$ and A . On the other hand, since G is radicial, I_A is contained in the radical of A , so that the intersection of the $\overline{I_A^n}$ is zero. Thus, by loc. cit., ψ is surjective. Then, since $\phi \circ \psi$ is an isomorphism and ψ a surjection, ψ and ϕ are isomorphisms. This proves that (iii) $\Rightarrow$ (i).

Note finally that it is clear that (i) or (ii) imply (iv), so that it remains to prove the implication (iv) $\Rightarrow$ (ii). For this, one can suppose that k is local, with residue field k_0 . Put then $G_0 = G \widehat{\otimes}_k k_0$, $\omega = \omega_{G/k}$, $\omega_0 = \omega \widehat{\otimes}_k k_0$, etc.⁽¹²⁷⁾ Since k_0 is a field, the pseudocompact

⁽¹²⁶⁾**N.D.E.** Cf. 5.1.5 below; see also [BAC], III, § 2.8, Th. 1 and corollaries. On the other hand, we have detailed the original in what follows.

⁽¹²⁷⁾**N.D.E.** We have detailed the original in what follows.

k_0 -module ω_0 possesses a pseudobasis $(y_\lambda)_{\lambda \in \Lambda}$; denoting by ω' the topologically free k -module which is the product of copies k_λ of k , for $\lambda \in \Lambda$, and lifting each y_λ to an element x_λ of ω , one obtains a continuous k -linear map $f : \omega' \rightarrow \omega$ such that $f_0 = f \otimes_k k_0$ is invertible.⁽¹²⁸⁾ Since ω' is a projective pseudocompact k -module, f lifts to a continuous k -linear map $g : \omega' \rightarrow I_A$ such that $\pi \circ g = f$, where π is the projection $I_A \rightarrow \omega$. By the universal property of $k[[\omega']]$ (cf. 1.2.5), g induces a morphism of topological algebras $\phi : k[[\omega']] \rightarrow A$.

Now $\omega' \otimes_k k_0$ is identified with $\omega \otimes_k k_0 = \omega_{G_0/k_0}$ and hence $k[[\omega']] \otimes_k k_0$ is identified with $k_0[[\omega_{G_0/k_0}]]$. Since the hypotheses (iii) hold for k_0 and G_0 , the proof of (iii) $\Rightarrow$ (i) shows that $\phi_0 = \phi \otimes_k k_0$ is invertible. Since, on the other hand, $k[[\omega']]$ and A are projective pseudocompact k -modules, ϕ is invertible by 0.3.4. (In particular, denoting by I the augmentation ideal of $k[[\omega']]$, ϕ induces an isomorphism from $\omega' = I/\overline{I^2}$ onto $I_A/\overline{I_A^2} = \omega$.) $\square$

Corollary 3.3.1. *Let S be a locally noetherian scheme over $\mathbf{Q}$ and G a flat group S -scheme locally of finite type.⁽¹²⁹⁾ If $\omega_{G/S}$ is a locally free $\mathcal{O}_S$ -module, G is smooth over S at every point of the identity section.*

Indeed, let x be a point of the identity section, s its image in S , and $\mathcal{O}_x$ and $\mathcal{O}_s$ the local algebras of x and s .⁽¹³⁰⁾ Since, by hypothesis, $\mathcal{O}_s$ and $\mathcal{O}_x$ are noetherian local rings, the $\mathfrak{m}$ -adic topology on each of these rings coincides with the “pseudocompact” topology defined by the ideals of finite codimension. Denote by $\widehat{\mathcal{O}}_x$ and $\widehat{\mathcal{O}}_s$ the completions for this topology. By EGA IV₄, 17.5.3, G is smooth over S at x if and only if $\widehat{\mathcal{O}}_x$ is formally smooth over $\widehat{\mathcal{O}}_s$, both these algebras being equipped with the $\mathfrak{m}$ -adic topology; it therefore suffices to show this last property. Now $\widehat{\mathcal{O}}_x$ and $\widehat{\mathcal{O}}_s$ are the local rings of x and s in the formal varieties $\widehat{G}/\widehat{S}$ and $\widehat{S}$ defined in 1.2.6. Put $k = \widehat{\mathcal{O}}_s$ and $H = \mathrm{Spf}(\widehat{\mathcal{O}}_x)$; then H is an infinitesimal formal k -variety and, since the formation of $\widehat{G}/\widehat{S}$ commutes with products (loc. cit.), H is an infinitesimal formal k -group. Denote by I the augmentation ideal of $\mathcal{O}_x$. By the hypotheses, $\omega_{G/S,x} = I/I^2$ is a locally free $\mathcal{O}_s$ -module of finite rank n . Since $\mathcal{O}_s$ is noetherian, it follows that $\omega_{H/k}$, which is the completion of $\omega_{G/S,x}$, is identified with $\omega_{G/S} \otimes_{\mathcal{O}_s} \widehat{\mathcal{O}}_s$, hence is a topologically free k -module of rank n . Thus, by the implication (iv) $\Rightarrow$ (ii) of 3.3, $\widehat{\mathcal{O}}_x$ is isomorphic to $k[[\omega_{H/k}]]$, i.e. to a formal power series algebra $k[[t_1, \dots, t_n]]$. Finally, the latter is formally smooth over k , by EGA 0_{IV}, 19.3.3. This proves the corollary.

We thus recover a result obtained elsewhere for group schemes locally of finite presentation over an arbitrary scheme S , cf. VI B, 1.6.

Corollary 3.3.2. *Let k be a field of characteristic 0. The functor $L \mapsto \mathcal{G}(L)$ is an equivalence from the category of k -Lie algebras onto that of infinitesimal formal k -groups.⁽¹³¹⁾*

⁽¹²⁸⁾**N.D.E.** One does not know a priori that ω is a projective pseudocompact k -module, but this will follow from what comes next: compare with the proof of (iv) $\Rightarrow$ (iii) in VII A, 7.4.

⁽¹²⁹⁾**N.D.E.** We have added the flatness hypothesis, omitted in the original.

⁽¹³⁰⁾**N.D.E.** We have detailed the original in what follows.

⁽¹³¹⁾**N.D.E.** Since, by 2.7, 2.2.1 and 1.3.6, an infinitesimal formal k -group G is “the same thing” as a connected cocommutative k -bialgebra H (cf. 2.9), this statement is equivalent to the theorem below, obtained independently by Kostant (cf. 2.9.2). This theorem had previously been obtained by J. W. Milnor and J. C. Moore ([MM65]), under the additional hypothesis that H be generated as an algebra by its primitive elements (although published in 1965, this text had circulated before 1960, cf. the review [Ja65]), so that it is often called the “Cartier–Kostant–Milnor–Moore theorem”.

Theorem (Cartier–Kostant–Milnor–Moore). — *Let k be a field of characteristic 0. The functors $L \mapsto U(L)$ and $H \mapsto \mathrm{Prim}(H)$ define an equivalence between the category of k -Lie algebras and that of connected cocommutative k -bialgebras.*

On the other hand, if k is an artinian ring containing $\mathbf{Q}$, then 3.2 (ii) and 3.3 (combined with 2.7, 2.2.1 and 1.3.6) likewise show that the functor $L \mapsto \mathcal{G}(L)$ (resp. $L \mapsto U(L)$) is an equivalence from the category of flat k -Lie algebras onto that of topologically flat infinitesimal formal k -groups (resp. onto that of flat connected cocommutative k -bialgebras).

Indeed, when G ranges over the infinitesimal formal k -groups, the functor $G \mapsto \mathcal{G}(\text{Lie } G)$ is isomorphic, by Theorem 3.3, to the identity functor of the category of infinitesimal k -groups. Similarly, by 3.2 (ii), the functor $L \mapsto \text{Lie}(\mathcal{G}(L)) = \text{Prim } U(L)$ is isomorphic to the identity functor of the category of k -Lie algebras.

3.3.3. — Still suppose that k is a field of characteristic 0. Let $\bar{k}$ be an algebraic closure of k and Γ the topological Galois group of $\bar{k}$ over k .

For every formal k -group G , we denote by $\underline{\text{Aut}}_k(G)$ the k -group functor which associates to every finite-dimensional commutative k -algebra C the group of automorphisms of the formal C -group $G \hat{\otimes}_k C$.⁽¹³²⁾ Since G is topologically flat over k (since k is a field), i.e. its affine algebra $A = A(G)$ is topologically flat over k , or, equivalently, its covariant bialgebra $H = H(G)$ is flat over k , this k -functor is a formal k -group. Indeed, since $\underline{\text{Aut}}_k(G)$ is identified with the fibered product of the following diagram (cf. Exp. I, 1.7.3):

$$\begin{array}{ccc} & \underline{\text{End}}_k(G) \times \underline{\text{End}}_k(G) & \\ & \downarrow & \\ \text{Spf}(k) & \longrightarrow & \underline{\text{End}}_k(G) \times \underline{\text{End}}_k(G), \end{array}$$

where the vertical (resp. horizontal) arrow is given by $(\phi, \psi) \mapsto (\phi \circ \psi, \psi \circ \phi)$ (resp. is the identity section), it suffices to see that the k -functor $\underline{\text{End}}_k(G)$ is represented by an element of $\mathbf{Vaf}/k$, i.e. (cf. 1.1 and 0.4.2) that it is left exact, i.e. commutes with fibered products of k -algebras. Now giving an element of $\underline{\text{End}}_k(G)(C)$ is equivalent to giving, say, a morphism of k -algebras $\phi : H \rightarrow H \otimes_k C$ which also respects the coalgebra structure, i.e. such that the well-known diagrams commute; since H is flat over k , $H \otimes_k -$ commutes with fibered products of k -algebras, and one deduces that the k -functor $\underline{\text{End}}_k(G)$ is left exact, so that we can treat it as a formal k -group.

If L is a k -Lie algebra and G the formal group $\mathcal{G}(L)$, Theorem 3.3 shows that $\underline{\text{Aut}}_k(G)$ is isomorphic to the k -group functor $\underline{\text{Aut}}_k(L)$ which associates to a finite-dimensional k -algebra C the group of automorphisms of the C -Lie algebra $C \otimes_k L$.

If G is an arbitrary formal k -group, we have seen in 2.5.2 that G decomposes canonically into a semidirect product of an étale formal group G_e and an infinitesimal formal group G_0 . This semidirect product is determined by the data of $L = \text{Lie}(G) = \text{Lie}(G_0)$, of the Γ -group $M = G_e(\bar{k}) = G(\bar{k})$, and of a morphism of formal k -groups

$$\Phi : G_e \rightarrow \underline{\text{Aut}}_k(G_0) \simeq \underline{\text{Aut}}_k(L).$$

Such a morphism sends G_e into the “étale part” $\underline{\text{Aut}}_k(L)_e$ of $\underline{\text{Aut}}_k(L)$, cf. 2.5.1. It is therefore determined by the data of a morphism of Γ -groups:

$$\phi = \Phi(\bar{k}) : M \rightarrow (\underline{\text{Aut}}_k L)(\bar{k}) = \text{Aut}_{\bar{k}}(L \otimes_k \bar{k}).$$

If one makes Γ act on $L \otimes_k \bar{k}$ by means of the formula $\gamma(\ell \otimes t) = \ell \otimes \gamma(t)$, then ϕ makes M act on $L \otimes_k \bar{k}$ by automorphisms of $\bar{k}$ -Lie algebra in such a way that one has $\phi(\gamma(m)) = \gamma \circ \phi(m) \circ \gamma^{-1}$, i.e. :

$$\gamma(m)(\ell \otimes t) = \gamma(m(\ell \otimes \gamma^{-1}(t)))$$

for every $m \in M$, $\gamma \in \Gamma$ and $\ell \otimes t \in L \otimes_k \bar{k}$. We express this last condition by saying that M acts on $L \otimes_k \bar{k}$ compatibly with Γ .

One can summarize the situation with a “learned” statement: call a Γ -Lie algebra over k the data of a triple (L, M, ϕ) consisting of a k -Lie algebra L , a Γ -group M , and an action ϕ of M on $L \otimes_k \bar{k}$ which is “compatible with the action of Γ on M and on $L \otimes_k \bar{k}$ ”.

If (L, M, ϕ) and (L', M', ϕ') are two such Γ -Lie algebras, a morphism from the former to the latter is a pair (f, θ) consisting of a morphism $f : L \rightarrow L'$ of k -Lie algebras and a

⁽¹³²⁾**N.D.E.** The original indicated: “This k -functor is manifestly left exact (H is topologically flat over k !)”. We have detailed this in what follows.

morphism $\theta : M \rightarrow M'$ of Γ -groups such that

$$(f \otimes 1)(m \cdot x) = \theta(m) \cdot (f \otimes 1)(x)$$

for every $x \in L \otimes_k \bar{k}$ and $m \in M$. One then obtains:

Theorem. *Let k be a field of characteristic 0. Then the functor which associates to G the triple $(\text{Lie}(G), G(\bar{k}), \Phi(\bar{k}))$ is an equivalence from the category of formal k -groups onto that of Γ -Lie algebras.⁽¹³³⁾*

4. Phenomena particular to characteristic $p > 0$

Throughout Section 4, we denote by p a prime number, by k a pseudocompact ring of characteristic p , and by π the continuous endomorphism $x \mapsto x^p$ of k .

4.1. — Let X be a formal k -variety with affine algebra A ; one denotes by $X^{(p/k)}$ or $X^{(p)}$ the formal k -variety $X \hat{\otimes}_{\pi} k$ obtained by the base change $\pi : k \rightarrow k$ (cf. 1.2.D); its affine algebra is the completed tensor product $A \hat{\otimes}_{\pi} k$. Then the continuous morphism $A \hat{\otimes}_{\pi} k \rightarrow A$ which maps $a \hat{\otimes}_{\pi} \lambda$ to $a^p \lambda$ induces a morphism of formal k -varieties

$$\text{Fr}(X/k) : X \rightarrow X^{(p/k)}.$$

In what follows, we shall say that $\text{Fr}(X/k)$ is the *Frobenius morphism of X relative to k* and we shall often write Fr instead of $\text{Fr}(X/k)$.

4.1.1. — ⁽¹³⁴⁾Now consider a scheme S over the prime field $\mathbf{F}_p$ and an S -scheme $\eta : X \rightarrow S$; let $\text{fr}(S)$ be the “absolute” Frobenius morphism of S (it induces the identity on the underlying topological space and maps every section f of the structure sheaf to f^p ; cf. VII A 4.1) and let $X^{(p)}$ be the S -scheme $X \times_{\text{fr}(S)} S$ (VII A, loc. cit.), i.e. the structure morphism $X^{(p)} \rightarrow S$ is $\text{fr}(S) \circ \eta$.

Let $\hat{S}$ be the formal scheme whose underlying topological space is the set of closed points of S , equipped with the discrete topology, the local ring $\mathcal{O}_{\hat{S},s}$ at such a closed point s being the separated completion $\hat{\mathcal{O}}_{S,s}$ of $\mathcal{O}_{S,s}$ for the linear topology defined by the ideals of finite colength (cf. 1.2.6); its affine algebra $k = \mathcal{A}(\hat{S})$ is therefore the product of the $\hat{\mathcal{O}}_{S,s}$, for s ranging over the closed points of S . Recall (loc. cit.) that one denotes by $\hat{X}/\hat{S}$ the formal k -variety consisting of the points $x \in X$ such that $[\kappa(x) : \kappa(s)] < \infty$, where s is the image of x in S , the local ring $\mathcal{O}_{\hat{X}/\hat{S},x}$ being the separated completion $\hat{\mathcal{O}}_{X,x}$ of $\mathcal{O}_{X,x}$ for the linear topology defined by the ideals I such that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{X,x}$ -module (and hence also as an $\mathcal{O}_{S,s}$ -module). One can therefore form, by base change, the formal k -variety $(\hat{X}/\hat{S})^{(p)} = (\hat{X}/\hat{S}) \hat{\otimes}_{\pi} k$.

One can also consider the formal k -variety $\widehat{X^{(p)}}/\hat{S}$: the underlying set consists of the $x \in X^{(p)} = X$ such that, denoting by s the image of x in S , the morphism

$$\kappa(s) \xrightarrow{\text{fr}} \kappa(s) \xrightarrow{\eta^{\sharp}} \kappa(x)$$

makes $\kappa(x)$ a finite-degree extension of $\kappa(s)$; in this case, the same holds for $\eta^{\sharp} : \kappa(s) \rightarrow \kappa(x)$, i.e. x is a point of $\hat{X}/\hat{S}$, and one then has

$$\mathcal{O}_{\widehat{X^{(p)}}/\hat{S},x} = \hat{\mathcal{O}}_{X,x} \hat{\otimes}_{\pi} \hat{\mathcal{O}}_{S,s} = \hat{\mathcal{O}}_{X^{(p)},x}$$

(the second equality because $X \mapsto \hat{X}/\hat{S}$ commutes with products, cf. 1.2.6). One therefore sees that: $\widehat{X^{(p)}}/\hat{S}$ is identified with a formal subvariety of $(\hat{X}/\hat{S})^{(p)}$. Moreover, one has

⁽¹³³⁾**N.D.E.** In particular, when $\bar{k} = k$, one thus recovers the “Cartier–Gabriel–Kostant theorem” mentioned in 2.9.2.

⁽¹³⁴⁾**N.D.E.** We have corrected the original, which gave the inclusion $(\hat{X}/\hat{S})^{(p)} \subset \widehat{X^{(p)}}/\hat{S}$ instead of the reverse inclusion. Let us mention, on the other hand, that this paragraph is not used in what follows.

equality if and only if for every closed point s of S , $\kappa(s)$ has finite degree over $\kappa(s)^p$, which is the case, for example, if S is a scheme locally of finite type over a field κ such that $[\kappa : \kappa^p] < +\infty$.

4.1.2. — Let G be a formal k -group. Since the functor $X \mapsto X^{(p)} = X \widehat{\otimes}_{\pi} k$ commutes with finite products, it transforms a formal k -group into a formal k -group. Moreover, since $\text{Fr}(X/k)$ is functorial in X , the morphism

$$\text{Fr} : G \longrightarrow G^{(p)}$$

is a homomorphism of formal k -groups. If one puts $G^{(p^{n+1})} = (G^{(p^n)})^{(p)}$, the same holds for the composite morphism

$$\text{Fr}^n = \text{Fr}^n(G/k) : G \xrightarrow{\text{Fr}} G^{(p)} \xrightarrow{\text{Fr}} G^{(p^2)} \xrightarrow{\text{Fr}} \dots \xrightarrow{\text{Fr}} G^{(p^n)}.$$

Denote by A the affine algebra of G and by I_A its augmentation ideal.

Definitions. (a) The kernel of Fr^n will be denoted by ${}_{\text{Fr}^n}G$. It follows directly from the definitions that ${}_{\text{Fr}^n}G$ is infinitesimal and has affine algebra the quotient $A/I_A^{\{p^n\}}$, where $I_A^{\{p^n\}}$ denotes the closed ideal of A generated by the p^n th powers of the elements of I_A .

(b) One says that G has *height* $\leq n$ if $I_A^{\{p^n\}} = 0$, that is, if one has ${}_{\text{Fr}^n}G = G$.

4.2. — It is clear that the Lie algebra $\underline{\text{Lie}}(G)$ of a formal k -group G is a Lie p -subalgebra of the algebra $H(G)$ (cf. 2.3). Indeed, one immediately reduces to the case where k is artinian; in this case, $\underline{\text{Lie}}(G)$ is the subset of $H(G)$ consisting of the elements x such that $\varphi_G(x \otimes 1 + 1 \otimes x) = \Delta_G(x)$ with the notation of 2.3 and 2.6 (c). One then has

$$\begin{aligned} \varphi_G(x^p \otimes 1 + 1 \otimes x^p) &= \varphi_G((x \otimes 1 + 1 \otimes x)^p) \\ &= \varphi_G(x \otimes 1 + 1 \otimes x)^p = \Delta_G(x)^p = \Delta_G(x^p). \end{aligned}$$

4.2.1. — Conversely, every Lie p -algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a k -group functor. Indeed, denote by $U_p(\mathbf{L})$ the functor which associates to every object C of $\mathbf{A}l/k$ the restricted enveloping algebra $U_p(\mathbf{L}(C))$ of the Lie p -algebra $\mathbf{L}(C)$ over C (VII A 5.3). By VII A 5.4, $U_p(\mathbf{L})$ is an $\mathbf{O}_k$ -bialgebra and therefore determines, by 2.2, a k -group functor $\text{Spf}^* U_p(\mathbf{L})$ which we shall henceforth denote by $\mathcal{G}_p(\mathbf{L})$.

Thus, $\mathcal{G}_p(\mathbf{L})(C)$ is the group of elements $z \in U_p(\mathbf{L}(C))$ of augmentation 1 and such that $\Delta_{U_p(\mathbf{L}(C))}(z) = z \otimes z$.

4.2.2. — Suppose that $\mathbf{L}$ is a Lie p -algebra flat over $\mathbf{O}_k$. Then, taking account of VII A 5.3.3, one shows as in point (i) of Proposition 2.6.2 that $U_p(\mathbf{L})$ is flat over $\mathbf{O}_k$; thus $\mathcal{G}_p(\mathbf{L})$ is a topologically flat formal k -group which has $U_p(\mathbf{L})$ for its covariant $\mathbf{O}_k$ -bialgebra.

⁽¹³⁵⁾By the proof of 2.6.2 (iii), for every k -algebra C of finite length, $\underline{\text{Lie}}(\mathcal{G}_p(\mathbf{L}))(C)$ is the set of primitive elements of $U_p(\mathbf{L})(C) = U_p(\mathbf{L}(C))$ (see also VII A 3.2.3); moreover, by VII A 5.5.3, the canonical morphism $\mathbf{L} \longrightarrow U_p(\mathbf{L})$ induces an isomorphism of Lie p -algebras

$$\tau_{p,\mathbf{L}} : \mathbf{L} \xrightarrow{\sim} \underline{\text{Lie}}(\mathcal{G}_p(\mathbf{L}))$$

(compare with 3.1 in characteristic 0).

The formal group $\mathcal{G}_p(\mathbf{L})$ can be characterized by a universal property. Indeed, every morphism h from $\mathcal{G}_p(\mathbf{L})$ to a formal group G induces a morphism $\underline{\text{Lie}}(h) : \underline{\text{Lie}}(\mathcal{G}_p(\mathbf{L})) \longrightarrow \underline{\text{Lie}}(G)$. Composing the latter with the isomorphism $\tau_{p,\mathbf{L}}$, one obtains a morphism $h' : \mathbf{L} \longrightarrow \underline{\text{Lie}}(G)$.

Proposition. If k is a pseudocompact ring of characteristic $p > 0$, G a formal k -group and $\mathbf{L}$ a Lie p -algebra flat over $\mathbf{O}_k$, the map $h \mapsto h'$ defined above is a bijection

$$\text{Hom}_{\mathbf{Grf}/k}(\mathcal{G}_p(\mathbf{L}), G) \xrightarrow{\sim} \text{Hom}_{p\text{-Lie}}(\mathbf{L}, \underline{\text{Lie}}(G)),$$

where the right-hand term denotes the set of morphisms of Lie p -algebras from $\mathbf{L}$ to $\underline{\text{Lie}}(G)$.

⁽¹³⁵⁾**N.D.E.** We have detailed what follows.

(136) Indeed, one immediately reduces to the case where k is artinian. Put $L = \mathbf{L}(k)$. By 2.3.1, $\text{Hom}_{\mathbf{Grf}/k}(\mathcal{G}_p(\mathbf{L}), G)$ is in bijection with the set of morphisms of unital algebras $h : U_p(L) \rightarrow H(G)$ such that the following diagram commutes:

$$\begin{array}{ccc} U_p(L) & \xrightarrow{h} & H(G) \\ \Delta_{U_p(L)} \downarrow & & \searrow \delta_G \\ U_p(L) \otimes U_p(L) & \xrightarrow{h \otimes h} & H(G) \otimes H(G) \end{array} \quad \begin{array}{c} \\ \\ \nearrow \varphi_G \end{array} \quad \begin{array}{c} \\ \\ H(G \times G) \end{array}.$$

Now h is defined by its restriction to L , which is a morphism of Lie p -algebras from L to the Lie p -algebra underlying $H(G)$, and commutativity of the diagram means that h maps L into the subset of $H(G)$ consisting of the x such that $\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x)$, which is none other than $\underline{\text{Lie}}(G)$, cf. 2.6 (c).

4.3. — We now propose to study the formal k -group $\mathcal{G}_p(\mathbf{L})$ in greater detail when $\mathbf{L}$ is a Lie p -algebra flat over $\mathbf{O}_k$.

For this, we first consider a ring C of characteristic p and a Lie p -algebra L over C whose underlying module is free with basis $(x_i)_{i \in I}$. Denote, moreover, by P the set of families $n = (n_i)_{i \in I}$ consisting of natural numbers such that $0 \leq n_i < p$ and such that the n_i are zero except perhaps for a finite number of them. If we equip I with a total order and call $i_1, i_2, \dots, i_r$ ($i_1 < i_2 < \dots < i_r$) the indices i such that $n_i \neq 0$, we can therefore put $|n| = n_{i_1} + \dots + n_{i_r}$ and

$$x^n = x_{i_1}^{n_{i_1}} \cdot x_{i_2}^{n_{i_2}} \cdot \dots \cdot x_{i_r}^{n_{i_r}}, \quad n! = (n_{i_1}!) \cdot \dots \cdot (n_{i_r}!).$$

One knows that the monomials $x^n/n!$ form a basis of $U_p(L)$ (VII A 5.3.3) and it is clear that one has

$$(*) \quad \Delta \left(\frac{x^n}{n!} \right) = \sum_{m \leq n} \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!},$$

the sum being taken over all $m \in P$ such that one has $m \leq n$ (i.e. $m_i \leq n_i$ for every $i \in I$). Formula (*) allows an easy determination of the natural filtration of $U_p(L)$ (cf. 1.3.6). Put $U = U_p(L)$, let U^+ be the two-sided ideal generated by L , and let $U_0 = C \cdot 1_U$. As in 1.3.6, one defines a filtration of U (by sub- C -coalgebras) by putting, for $n \geq 1$:

$$U_n = \{u \in U \mid \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+\}.$$

Formula (*) then implies that U_n is the free C -module with basis the monomials x^m such that $|m| \leq n$.

4.3.1. — With the notation of 4.3, let us determine the elements ξ of $U = U_p(L)$ such that $\varepsilon_U(\xi) = 1$ and $\Delta_U(\xi) = \xi \otimes \xi$. Every element ξ of U is written $\xi = \sum_{n \in P} \xi_n (x^n/n!)$, with $\xi_n \in C$. The condition $\varepsilon_U(\xi) = 1$ implies the equality $\xi_0 = 1$, where 0 denotes the element of P all of whose components are zero. The condition $\Delta_{U_p(L)}(\xi) = \xi \otimes \xi$ implies:

$$\sum_{m \leq n} \xi_n \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!} = \sum_{q,r} \xi_q \xi_r \frac{x^q}{q!} \otimes \frac{x^r}{r!},$$

that is,

$$\xi_{q+r} = \xi_q \xi_r \quad \text{if } q_i + r_i < p, \quad \xi_q \xi_r = 0 \quad \text{otherwise.}$$

If one denotes by ξ_i the value of ξ_n when $n_i = 1$ and $n_j = 0$ for $j \neq i$, these conditions mean that one has

$$\xi_n = \prod_i \xi_i^{n_i} \quad \text{if } n \in P, \quad \xi_i^p = 0 \quad \forall i \in I.$$

(136) **N.D.E.** The proof is identical to that of 2.6.3.

One derives from this:

Proposition. *Let k be a local pseudocompact ring⁽¹³⁷⁾ of characteristic $p > 0$, $\mathbf{L}$ a Lie p -algebra flat over $\mathbf{O}_k$, C an object of $\mathbf{Alf}/k$ and $\sqrt[p]{0_C}$ the ideal of C consisting of the elements x such that $x^p = 0$. There exists a bijection “functorial in C ”:*

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathcal{G}_p(\mathbf{L})(C).$$

By Remark 1.2.3.F, one can indeed choose a basis $({}^C x_i)_{i \in I}$ of $\mathbf{L}(C)$ over C in such a way that, for every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$, $\mathbf{L}(\varphi)$ maps ${}^C x_i$ to ${}^D x_i$. By the preceding, the map

$$\sum_i {}^C x_i \otimes \xi_i \mapsto \sum_{n \in P} \left(\prod_i \xi_i^{n_i} \right) \frac{{}^C x^n}{n!}$$

is a functorial bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathcal{G}_p(\mathbf{L})(C)$.

4.3.2. — “There is no Campbell–Hausdorff formula in characteristic $p > 0$.” Let us explain: the functorial isomorphism of 4.3.1 depends on the choice of the bases $({}^C x_i)_{i \in I}$. Unlike what happens in 3.2 (the characteristic 0 case), there is, in general, no bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathcal{G}_p(\mathbf{L})(C)$ which is “functorial in both C and $\mathbf{L}$ ”. The argument which follows shows, for example, that such an isomorphism does not exist when k is a field containing the $(p^2 - 1)$ th roots of unity.

Indeed, choose $\mathbf{L}$ in such a way that, for every $C \in \mathbf{Alf}/k$, $\mathbf{L}(C)$ is the abelian Lie p -algebra over C generated by an element x subject to the relation $x^{(p^2)} = 0$. One therefore has

$$\mathbf{L}(C) = Cx \oplus Cx^{(p)}, \quad U_p(\mathbf{L}(C)) \simeq k[x]/(x^{p^2}).$$

We shall show that the only functorial morphism

$$\chi_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \rightarrow U_p(\mathbf{L}(C))$$

which is compatible with the endomorphisms of $\mathbf{L}$ and maps $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ into $\mathcal{G}_p(\mathbf{L})(C)$ is the constant morphism with value 1.

Indeed, let ψ_C be the bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathcal{G}_p(\mathbf{L})(C) = \text{Prim } U_p(\mathbf{L}(C))$ given by 4.3.1, i.e. :

$$x \otimes a + x^{(p)} \otimes b \mapsto \sum_{0 \leq i, j < p} a^i b^j x^{i+pj}.$$

Composing χ_C with ψ_C^{-1} , one obtains a morphism functorial (in C):

$$\begin{aligned} \theta_C : \quad \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} &\longrightarrow \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \\ x \otimes a + x^{(p)} \otimes b &\longmapsto x \otimes P(a, b) + x^{(p)} \otimes Q(a, b). \end{aligned}$$

Functoriality in C implies that $P(a, b)$ and $Q(a, b)$ are the values at (a, b) of two polynomials $P, Q \in k[x, y]$ which one can suppose to have degree $< p$ in X as well as in Y .⁽¹³⁸⁾ Let α be an element of k and ℓ_α the endomorphism of Lie p -algebra of $\mathbf{L}$ which maps x to αx (and hence $x^{(p)}$ to $\alpha^p x^{(p)}$). Then $U_p(\ell_\alpha)$ is the algebra endomorphism which sends x to αx (and hence each x^i to $\alpha^i x^i$, for $i < p^2$), and one sees easily that the square below

$$\begin{array}{ccc} \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} & \xrightarrow{\psi_C} & U_p(\mathbf{L}(C)) \\ \ell_\alpha(C) \otimes \text{id} \downarrow & & \downarrow U_p(\ell_\alpha)(C) \\ \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} & \xrightarrow{\psi_C} & U_p(\mathbf{L}(C)) \end{array}$$

⁽¹³⁷⁾**N.D.E.** We have added the hypothesis that k be local, so that every topologically flat pseudocompact k -module is topologically free, cf. the proof.

⁽¹³⁸⁾**N.D.E.** We have detailed the original in what follows.

commutes. The hypothesis $\chi_C \circ (\ell_\alpha(C) \otimes \text{id}) = U_p(\ell_\alpha)(C) \circ \chi_C$ then implies the equalities:

$$P(\alpha a, \alpha^p b) = \alpha P(a, b) \quad \text{and} \quad Q(\alpha a, \alpha^p b) = \alpha^p Q(a, b).$$

Writing $P = \sum_{i,j < p} \lambda_{ij} X^i Y^j$ and $Q = \sum_{i,j < p} \mu_{ij} X^i Y^j$, and taking for C the algebra $k[X, Y]/(X^p, Y^p)$, one deduces that if $\lambda_{ij} \neq 0$ (resp. $\mu_{ij} \neq 0$) then $\alpha^{i-1+pj} = 1$ (resp. $\alpha^{i+p(j-1)} = 1$). Thus, taking for α a primitive root of unity of order $p^2 - 1$, one deduces that $P = \lambda X$ and $Q = \mu Y$, with $\lambda, \mu \in k$. Consequently, one has:

$$\chi_C(x \otimes a + x^{(p)} \otimes b) = \sum_{0 \leq i, j < p} (\lambda a)^i (\mu b)^j x^{i+pj}.$$

(138) Finally, consider the endomorphism f of $\mathbf{L}$ which sends x to $x + x^{(p)}$; taking $C = k[a]/(a^3)$ and comparing the coefficients of x^p and x^{p+1} in $(\chi_C \circ f(C))(x \otimes a)$ and in $(U_p(f)(C) \circ \chi_C)(x \otimes a)$, one obtains that $\lambda = \mu$ and $\lambda^2 a^2 = 0$, whence $\lambda = \mu = 0$.

Theorem 4.4. *Let k be a pseudocompact ring of characteristic $p > 0$ and G a formal k -group. The following assertions are equivalent:*

- (i) *There exists a Lie p -algebra $\mathbf{L}$ flat over $\mathbf{O}_k$ such that G is isomorphic to $\mathcal{G}_p(\mathbf{L})$ (and in this case $\mathbf{L} = \underline{\text{Lie}}(G)$ by 4.2.2).*
- (ii) *There exists a projective pseudocompact k -module ω such that the affine algebra of G is isomorphic to the quotient of $k[[\omega]]$ by the closed ideal generated by the x^p , for $x \in \omega$ (and in this case $\omega \simeq \omega_{G/k}$).*
- (iii) *G has height ≤ 1 and $\omega_{G/k}$ is a projective pseudocompact k -module.*
- (iv) *G has height ≤ 1 and is topologically flat over k .*

PROOF. (i) $\Rightarrow$ (ii): Let $\omega = \Gamma^*(\mathbf{L})$ (cf. 1.2.3.D) and let A be the quotient of $k[[\omega]]$ by the closed ideal generated by the x^p , for $x \in \omega$. Every continuous morphism $h : A \rightarrow C$, where C is an object of $\mathbf{AIf}/k$, is determined by its restriction h' to ω ; this restriction h' sends ω into $\sqrt[p]{0_C}$. One thus obtains a canonical bijection from $\text{Hom}_{\mathbf{AIf}/k}(A, C)$ onto the set $\text{Hom}_c(\omega, \sqrt[p]{0_C})$ of continuous k -linear maps from ω to $\sqrt[p]{0_C}$. This last set is canonically isomorphic to $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ (see the proof of 3.3). The implication (i) $\Rightarrow$ (ii) therefore follows from the functorial bijection $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathcal{G}_p(\mathbf{L})(C)$ established in 4.3.1.

For the other implications, consult the proofs of Theorems 3.3 and VII A 7.4.1, which are analogous. $\square$

Remark 4.4.A. (139) Let G be an infinitesimal formal k -group, with affine algebra A , such that $\omega_{G/k} = I_A/\overline{I_A^2}$ is a projective pseudocompact k -module. Then there exists a section $\tau : \omega_{G/k} \rightarrow I_A$ of the projection $I_A \rightarrow \omega_{G/k}$, and τ induces a continuous morphism of algebras $\psi : k[[\omega_{G/k}]] \rightarrow A$ which is *surjective*, cf. the proof of the implication (iii) $\Rightarrow$ (i) in 3.3.

Corollary 4.4.1. *If k is a field of characteristic $p > 0$, the functor $L \mapsto \mathcal{G}_p(L)$ is an equivalence from the category of Lie p -algebras over k onto that of formal k -groups of height ≤ 1 .*

Indeed, when G ranges over the formal groups of height ≤ 1 , the functor $G \mapsto \mathcal{G}_p(\text{Lie } G)$ is isomorphic to the identity functor by Theorem 4.4; similarly, we have seen in 4.2.2 that the functor $L \mapsto \text{Lie } \mathcal{G}_p(L)$ is isomorphic to the identity functor (see also VII A, 5.5.3).⁽¹⁴⁰⁾

4.4.2. — Still take for k a field of characteristic p . Let G be an infinitesimal formal k -group, with affine algebra A . Since G is infinitesimal, every open ideal of A contains $I_A^{\{p^n\}}$ for n large enough, so G is the projective limit of the affine algebras $A/I_A^{\{p^n\}}$ of the groups $\mathbb{F}_n G$ (cf.

(139) **N.D.E.** We have added this remark, used in 4.4.2.

(140) **N.D.E.** If k is an artinian ring of characteristic $p > 0$, the same proof gives an equivalence between the category of flat Lie p -algebras over k and that of topologically flat formal k -groups of height ≤ 1 .

4.1.2). *Every infinitesimal formal k -group is therefore an inductive limit of formal k -groups of finite height.*

Suppose that G has height $\leq n$ and put $H = G/\mathrm{Fr}G$.⁽¹⁴¹⁾ By 2.4 and 2.4.1, $\mathrm{Fr} : G \rightarrow G^{(p)}$ factors into an epimorphism $\pi : G \rightarrow H$ followed by a monomorphism $i : H \rightarrow G^{(p)}$. One then has the following commutative diagram:

$$\begin{array}{ccccc} G & \xrightarrow{\pi} & H & \xrightarrow{\mathrm{Fr}^{n-1}} & H^{(p^{n-1})} \\ & \searrow \mathrm{Fr} & \downarrow i & & \downarrow i^{(p^{n-1})} \\ & & G^{(p)} & \xrightarrow{\mathrm{Fr}^{n-1}} & G^{(p^n)} \end{array}$$

and since the functor $X \mapsto X^{(p)}$ commutes with fibered products, $i^{(p^{n-1})}$ is again a monomorphism. Since $\mathrm{Fr}^n(G/k)$ is zero and π is an epimorphism, $\mathrm{Fr}^{n-1}(G^{(p)}/k) \circ i = i^{(p^{n-1})} \circ \mathrm{Fr}^{n-1}(H/k)$ is zero and hence, since $i^{(p^{n-1})}$ is a monomorphism, $\mathrm{Fr}^{n-1}(H/k)$ is zero, so H has height $\leq n-1$. One therefore sees that: *every formal k -group of finite height possesses a composition series whose quotients have height ≤ 1 , hence can be described by Lie p -algebras over k .*

Finally, the affine algebra A of G is a quotient of $k[[\omega_{G/k}]]$, cf. 4.4.A, so if $\omega_{G/k}$ has finite dimension over k , all the algebras $A/I_A^{(p^n)}$ are finite-dimensional k -vector spaces. One therefore sees that: *every infinitesimal formal group G over a field of characteristic $p > 0$, such that $\omega_{G/k}$ has finite dimension over k , is an inductive limit of finite formal groups (i.e. of finite length, cf. 1.2.6).*

5. Homogeneous spaces of infinitesimal formal groups over a field

5.0. — ⁽¹⁴²⁾Suppose, for simplicity, that k is a field. Let G be a formal k -group with affine algebra $A = \mathcal{A}(G)$. Let A^+ be the augmentation ideal of A ; for every $a \in A$, denote by $\bar{a} = a - \varepsilon_A(a)1_A$ its projection onto A^+ . Let F be a formal subgroup of G , defined by the closed ideal J of A , and let π (resp. $\bar{\pi}$) be the projection $A \rightarrow A/J$ (resp. the composite of the projections $A \rightarrow A^+ \rightarrow A^+/J$). Note that, for every $a \in A$, the projection of $\Delta_A(a)$ onto $A \hat{\otimes} A^+$ is $\Delta_A(a) - a \hat{\otimes} 1$.

By Theorem 2.4, one can form the quotient formal k -variety $X = G/F$; its affine algebra B is

$$\begin{aligned} B &= \mathrm{Ker} \left(A \begin{array}{c} \xrightarrow{\tau_1} \\ \xrightarrow{(\mathrm{id}_A \hat{\otimes} \pi) \Delta_A} \end{array} A \hat{\otimes}_k (A/J) \right) \\ &= \{a \in A \mid (\mathrm{id}_A \hat{\otimes} \pi) \Delta_A(a) = a \hat{\otimes} \pi(1)\} \\ &= \mathrm{Ker}((\mathrm{id}_A \hat{\otimes} \bar{\pi}) \Delta_A). \end{aligned}$$

It is also the subalgebra of A consisting of the elements ϕ such that, for every $C \in \mathrm{Ob} \mathbf{AIf}/k$ and $g \in G(C)$, $h \in F(C)$, one has $\phi(gh) = \phi(g)$. For every $g, g' \in G(C)$ and $h \in F(C)$, one has $\phi(gg'h) = \phi(gg')$, so $\Delta(\phi)$ belongs to the kernel of $\mathrm{id}_A \hat{\otimes}_k (\mathrm{id}_A \hat{\otimes} \bar{\pi}) \Delta_A$, which equals $A \hat{\otimes}_k B$ since A is topologically flat over k . One therefore has $\Delta_A(B) \subset A \hat{\otimes}_k B$, i.e. the closed subalgebra B is also a *left coideal*.

On the other hand, B determines F since, by Corollary 2.4.1, one has $J = \overline{AB^+}$, i.e. J is the closed ideal generated by $B^+ = B \cap A^+$. One thus obtains an *injective* map $\mathcal{Q}$ from the set $\mathcal{F}$ of formal subgroups of G into the set $\mathcal{B}$ of closed subalgebras B of A which are left coideals. There then arises the question of determining the image of this map, and

⁽¹⁴¹⁾**N.D.E.** We have detailed the original in what follows.

⁽¹⁴²⁾**N.D.E.** We have added subsection 5.0, to express the following Proposition 5.1 in the language of cocommutative Hopf algebras, and to cite the results obtained since in this direction.

Proposition 5.1 below shows that $\mathcal{Q}$ is bijective when G is *infinitesimal*. In fact, this is true for *every* formal k -group G .

Indeed, recall (cf. 2.2.1) that the functor $G \rightarrow H(G)$ is an *equivalence* between the category of formal k -groups and that of cocommutative k -Hopf algebras; if F is a formal subgroup of G , defined by the closed ideal J of A , then the Hopf subalgebra $H(F)$ of $H = H(G)$ is the orthogonal of J for the duality between $A = H^*$ and $H = A^\dagger$ (cf. 0.2.2). On the other hand, if B is a closed subalgebra of A which is also a left coideal, then its orthogonal $I = B^\perp$ is a coideal of H (i.e. $\Delta_H(I) \subset I \otimes H + H \otimes I$ and $\varepsilon_H(I) = 0$) and a left ideal. Denote by $\mathcal{H}$ (resp. $\mathcal{I}$) the set of Hopf subalgebras (resp. left ideals which are coideals) of H . For every $I \in \mathcal{I}$, denote by π_I (resp. $\bar{\pi}_I$) the projection $H \rightarrow H/I$ (resp. the composite of the projections $H \rightarrow H^+ \rightarrow H^+/I$), where H^+ is the augmentation ideal of H .

Let K be a Hopf subalgebra of H and $K^+ = K \cap H^+$. If F is the formal subgroup corresponding to K , then $J = K^\perp$ and A^+/J is identified with the dual of K^+ , and since $B = \mathcal{Q}(F)$ is the kernel of the map

$$A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{\text{id} \hat{\otimes} \bar{\pi}} A \hat{\otimes}_k (A^+/J),$$

one obtains that $\mathcal{Q}$ corresponds by duality to the map Φ which associates to K the image of

$$H \xleftarrow{m_H} H \otimes_k H \xleftarrow{\text{id} \hat{\otimes} \text{can.}} H \otimes_k K^+,$$

i.e. the left ideal HK^+ , which is also a coideal. One sees similarly that the map which associates to $B \in \mathcal{B}$ the ideal $\overline{AB^+}$ corresponds by duality to the map Ψ which associates to every $I \in \mathcal{I}$ the kernel of $(\text{id} \otimes_k \bar{\pi}_I) \circ \Delta_H$, i.e. one has

$$\Psi(I) = \{x \in H \mid (\text{id} \otimes_k \pi_I)\Delta_H(x) = x \otimes \pi_I(1)\}.$$

One then has the following theorem:

Theorem 5.0.1. *Let k be a field and H a cocommutative Hopf algebra. Then the maps Φ and Ψ above are inverse bijections between the set of Hopf subalgebras of H and that of left ideals which are coideals.*

This theorem was first proved by K. Newman, cf. [Ne75], Th. 4.1 (where the word “cocommutative” was omitted). His proof uses “the Cartier–Gabriel–Kostant theorem” (cf. 2.9) to reduce to the case where H is connected, then the existence in this case of a “Sweedler basis” (cf. [Sw67], Th. 3), a result dual to the Dieudonné–Cartier Theorem 5.2.2 below. Another, shorter, proof was given by H. J. Schneider [Sch90], Th. 4.15. A generalization was subsequently obtained by A. Masuoka when one assumes only that the coradical H_0 of H (i.e. the sum of the simple subcoalgebras) is commutative, [Ma91], Th. 1.3 (3).

Let us mention finally that for a commutative k -Hopf algebra, thus corresponding to an affine group k -scheme G , one cannot expect an analogue of 5.0.1 without additional hypotheses, since for a k -subgroup F of G the quotient G/F is not necessarily affine. But M. Takeuchi established in [Tak72], Th. 4.3 (resp. [Tak79], Th. 3), an analogous bijection between the set of k -subgroups F of G which are invariant (resp. such that G/F is affine), and that of subalgebras B of $\mathcal{O}(G)$ such that $\Delta(B) \subset A \otimes B$ and which are stable under the antipode (resp. and such that $B \rightarrow A$ is faithfully flat).

5.1. — Let k be a pseudocompact ring.⁽¹⁴³⁾ Let G be a topologically flat infinitesimal formal k -group, A its affine algebra, B a closed subalgebra of A , $X = \text{Spf}(B)$ and $r : G \rightarrow X$ the

⁽¹⁴³⁾**N.D.E.** In the original, k is assumed in 5.1 to be a field. In fact, this hypothesis can be replaced by flatness hypotheses; we have modified nos. 5.1 to 5.1.5 accordingly.

epimorphism induced by the inclusion of B into A . One proposes to see under what condition r makes X the right quotient of G by a subgroup H (cf. 2.4).⁽¹⁴⁴⁾

Proposition. *Let G be a topologically flat infinitesimal formal k -group, A its affine algebra, I_A the augmentation ideal of A , B a closed subalgebra of A , and $J_B = \overline{AI_B}$, where $I_B = B \cap I_A$. Suppose that A is topologically flat over k , as is $\overline{J_B^n}/\overline{J_B^{n+1}}$ for every $n \geq 0$. Then the following two assertions are equivalent:*

- (i) *For every $x \in B$, $\Delta_A(x) - x \hat{\otimes} 1$ belongs to $A \hat{\otimes}_k J_B$.*
- (ii) *The sequence below (where $\tau_1(a) = a \hat{\otimes} 1$ and π is the projection $A \rightarrow A/J_B$) is exact:*

$$(*) \quad B \longrightarrow A \xrightarrow[\substack{\tau_1 \\ (\pi \hat{\otimes} \text{id}_A) \Delta_A}]{} A \hat{\otimes}_k (A/J_B),$$

i.e. B is the set of all $x \in A$ such that $\Delta_A(x) - x \hat{\otimes} 1$ belongs to $A \hat{\otimes}_k J_B$.

In this case, $H = \text{Spf}(A/J_B)$ is a formal subgroup of G , and the sequence below (where λ is the restriction to $G \times H$ of the multiplication of G) is exact:

$$(**) \quad G \times H \xrightarrow[\lambda]{\text{pr}_1} G \longrightarrow \text{Spf}(B),$$

i.e. $\text{Spf}(B)$ is isomorphic to G/H .

Put $\overline{A} = A/J_B$ and $H = \text{Spf}(\overline{A})$; then H is a formal subvariety of G . Since $J_B \subset I_A$, the augmentation ε_A induces a continuous morphism of k -algebras $\overline{\varepsilon} : \overline{A} \rightarrow k$.

If (i) holds, then $\Delta_A(I_B) \subset I_B \hat{\otimes} 1 + A \hat{\otimes}_k J_B$, and hence Δ_A induces, on passing to the quotient, a diagonal morphism $\overline{\Delta}$. Then $\overline{\Delta}$ and $\overline{\varepsilon}$ equip H with a structure of formal submonoid of G . Since G is infinitesimal, the same holds for H ; thus, by Proposition 2.7, H is a formal subgroup of G . It then follows from the definition of G/H (cf. 2.4) that (ii) implies the last assertion of the proposition.

On the other hand, it is clear that (ii) implies (i). The proof of the converse occupies paragraphs 5.1.1 to 5.1.5.

5.1.1. — First consider the following category $\mathcal{C}$: an object of $\mathcal{C}$ is a pair (A, J) consisting of a profinite k -algebra A and a closed ideal J of A ; a morphism $\psi : (A, J) \rightarrow (A', J')$ of $\mathcal{C}$ is a continuous homomorphism of k -algebras $A \rightarrow A'$ which maps J into J' . If one associates to (A, J) the pair $(\text{Spf}(A/J), \text{Spf}(A))$, one clearly obtains an antiequivalence from $\mathcal{C}$ onto the category of pairs (Z, Y) consisting of a formal k -variety Y and a formal subvariety Z , a morphism $\phi : (Z, Y) \rightarrow (Z', Y')$ being a morphism of formal k -varieties $Y \rightarrow Y'$ which maps Z into Z' .

A *cogroup* structure on an object (A, J) of $\mathcal{C}$ consists of the data of a formal group structure on $\text{Spf}(A)$ such that the following conditions hold (notation of 2.1):

- (1) $\Delta_A(J) \subset J \hat{\otimes}_k A + A \hat{\otimes}_k J$;
- (2) $\varepsilon_A(J) = 0$;
- (3) $c_A(J) \subset J$.

These conditions also mean that $H = \text{Spf}(A/J)$ is a formal subgroup of $G = \text{Spf}(A)$.⁽¹⁴⁵⁾

Suppose, moreover, that A is *local*, i.e. that $\text{Spf}(A)$ is an infinitesimal formal group. Then, if $J \neq A$, conditions (2) and (3) are consequences of (1). Indeed, if J is a closed ideal distinct from A , it is contained in the augmentation ideal I_A , so (2) holds, and $M = \text{Spf}(A/J)$ is

⁽¹⁴⁴⁾**N.D.E.** We have replaced “left” by “right” and modified the statement of Proposition 5.1, to bring out more clearly, on the one hand, the equivalent conditions (i), (ii), and, on the other hand, the conclusion $\text{Spf}(B) \simeq G/H$.

⁽¹⁴⁵⁾**N.D.E.** Note that if (A', J') is a second cogroup of $\mathcal{C}$, corresponding to a pair $H' \subset G'$ of formal k -groups, then giving a morphism of cogroups $(A', J') \rightarrow (A, J)$ is equivalent to giving a morphism of formal k -groups $G \rightarrow G'$ which maps H into H' .

a formal submonoid of G . Since G is infinitesimal, it follows from 2.7 that M is a formal subgroup of G , i.e. condition (3) holds.

5.1.2. — Denote by $\mathbf{Alpg}/k$ the category of “graded profinite” k -algebras: an object of this category consists of the data of a sequence $A_0, A_1, \dots, A_n, \dots$ of pseudocompact k -modules and a profinite algebra structure on the product $\prod_{n \geq 0} A_n$ such that one has $A_n \cdot A_m \subset A_{m+n}$ (A_n is identified with a subset of $\prod_{i \geq 0} A_i$ by means of the canonical injection); a morphism $\psi : (A_n) \rightarrow (B_n)$ is a sequence of continuous linear maps $\psi_n : A_n \rightarrow B_n$ such that one has $\psi_{m+n}(a \cdot a') = \psi_m(a) \cdot \psi_n(a')$ if $a \in A_m$ and $a' \in A_n$.

Definitions. It is clear that two graded profinite k -algebras (A_n) and (B_n) have a *coproduct*⁽¹⁴⁶⁾ in $\mathbf{Alpg}/k$, whose n th component is the product $\prod_{i=0}^n A_i \hat{\otimes}_k B_{n-i}$ of the pseudocompact k -modules $A_i \hat{\otimes}_k B_{n-i}$. This coproduct will be denoted by $(A_n) \hat{\otimes}_k (B_n)$.

Then a cogroup structure on an object (A_n) of $\mathbf{Alpg}/k$ is the data of continuous k -linear maps $\Delta_n : A_n \rightarrow \prod_{i=0}^n A_i \hat{\otimes}_k A_{n-i}$ and $\varepsilon : A_0 \rightarrow k$ which induce on $\prod_{n \geq 0} A_n$ (putting $\varepsilon(A_i) = 0$ for $i \geq 1$) a cogroup structure in $\mathbf{Alp}/k$.

Finally, for every object (A, J) of $\mathcal{C}$, one denotes by $\mathrm{Gr}_J(A)$ the graded profinite algebra associated with the filtration of A by the closures $\overline{J^n}$ of the powers of J ; one therefore has $\mathrm{Gr}_J(A)_n = \overline{J^n} / \overline{J^{n+1}}$ and the multiplication of $\mathrm{Gr}_J(A)$ is induced by that of A .

Lemma.⁽¹⁴⁷⁾ *Let U, V be two pseudocompact k -modules, with U topologically flat, and let $U = U_0 \supset U_1 \supset \dots$ and $V = V_0 \supset V_1 \supset \dots$ be two decreasing sequences of closed sub- k -modules. Filter the completed tensor product $W = U \hat{\otimes}_k V$ by means of the closed submodules*

$$W_n = U_n \hat{\otimes}_k V_0 + U_{n-1} \hat{\otimes}_k V_1 + \dots + U_0 \hat{\otimes}_k V_n.$$

Suppose that each U_i/U_{i+1} is topologically flat over k (so that U/U_n , hence also U_n , is as well, for every n). Then, for every n , one has an isomorphism

$$W_n/W_{n+1} \simeq \bigoplus_{i+j=n} (U_i/U_{i+1}) \hat{\otimes}_k (V_j/V_{j+1}).$$

PROOF. Put $W_{i,j} = U_i \hat{\otimes}_k V_j$ and $\overline{W}_{i,j} = (U_i/U_{i+1}) \hat{\otimes}_k (V_j/V_{j+1})$, for every $i, j \geq 0$. Let us show by induction on n that the natural map

$$\pi_n : W_n \rightarrow \bigoplus_{i+j=n} \overline{W}_{i,j}$$

is surjective and that the inclusion $W_{n+1} \subset \mathrm{Ker}(\pi_n)$ is an equality. For $n = 0$, the projection

$$\pi_0 : U_0 \hat{\otimes}_k V_0 \rightarrow (U_0/U_1) \hat{\otimes}_k (V_0/V_1)$$

is surjective and, since $U_0, U_0/U_1$ and hence U_1 are topologically flat over k , one sees that $\mathrm{Ker}(\pi_0) = U_0 \hat{\otimes}_k V_1 + U_1 \hat{\otimes}_k V_0$ and that, moreover, $U_0 \hat{\otimes}_k V_1 \cap U_1 \hat{\otimes}_k V_0 = U_1 \hat{\otimes}_k V_1$.

Suppose therefore that $n > 0$ and the result is established for $n - 1$. Put $M_0 = U_0 \hat{\otimes}_k V_n$ and $S_0 = \sum_{i=1}^n U_i \hat{\otimes}_k V_{n-i}$. One has $S_0 \subset U_1 \hat{\otimes}_k V_0$ and hence, by the preceding passage applied to $V_0 \supset V_n$ instead of $V_0 \supset V_1$, one has

$$M_0 \cap S_0 \subset U_0 \hat{\otimes}_k V_n \cap U_1 \hat{\otimes}_k V = U_1 \hat{\otimes}_k V_n,$$

whence one deduces that $M_0 \cap S_0 = U_1 \hat{\otimes}_k V_n$. Since $W_n = M_0 + S_0$, one obtains a commutative diagram with exact rows, where one has put $U'_i = U_{i+1}$ and $\overline{W}'_{i,n-1-i} = \overline{W}_{i+1,n-1-i}$ for

⁽¹⁴⁶⁾**N.D.E.** We have replaced “direct sum” by “coproduct”.

⁽¹⁴⁷⁾**N.D.E.** In the original, the lemma is stated when k is a field, the proof being left to the reader in this case.

$i = 0, \dots, n-1$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & S_0 & \longrightarrow & W_n & \longrightarrow & (U_0/U_1) \hat{\otimes}_k V_n \longrightarrow 0 \\ & & \downarrow \pi'_{n-1} & & \downarrow \pi_n & & \downarrow p \\ 0 & \longrightarrow & \bigoplus_{i=0}^{n-1} \overline{W}'_{i,n-1-i} & \longrightarrow & \bigoplus_{i=0}^n \overline{W}_{i,n-1} & \longrightarrow & \overline{W}_{0,n} \longrightarrow 0. \end{array}$$

Then p is surjective, with kernel $(U_0/U_1) \hat{\otimes}_k V_{n+1}$. Moreover, by the induction hypothesis applied to the sequence (U'_i) , π'_{n-1} is surjective, with kernel equal to $W'_n = \sum_{i=1}^n W_{i,n+1-i}$. It follows that π_n is surjective, and that the inclusion $W_{n+1} \subset \text{Ker}(\pi_n)$ is an equality. This proves the lemma. $\square$

Return to an object (A, J) of $\mathcal{C}$ and note that, by 0.2.G, the hypothesis that each $\overline{J^n}/\overline{J^{n+1}}$ be topologically flat over k is equivalent to saying that $\text{Gr}_J(A)$ is topologically flat over k .

Corollary. *Let $\mathcal{P}$ be the full subcategory of $\mathcal{C}$ consisting of the objects (A, J) such that A and $\text{Gr}_J(A)$ are topologically flat over k . Then the functor $\mathcal{P} \rightarrow \mathbf{Alpg}/k$, $(A, J) \mapsto \text{Gr}_J(A)$ commutes with finite coproducts, hence transforms a cogroup of $\mathcal{P}$ into a cogroup of $\mathbf{Alpg}/k$.*

In particular, if k is a field then, for every cogroup (A, J) of $\mathcal{C}$, $\text{Gr}_J(A)$ is a cogroup of $\mathbf{Alpg}/k$.⁽¹⁴⁸⁾

5.1.3. — Identify every profinite k -algebra Γ with the graded profinite k -algebra $(\Gamma_n)_{n \geq 0}$ such that $\Gamma_0 = \Gamma$ and $\Gamma_n = 0$ if $n > 0$. In particular, if $(A_n)_{n \geq 0}$ is a graded profinite k -algebra, we shall consider A_0 indifferently as a profinite k -algebra or as a graded profinite k -algebra. We shall then denote by $\rho : (A_n) \rightarrow A_0$ the morphism of $\mathbf{Alpg}/k$ such that $\rho_0 = \text{id}_{A_0}$ and $\rho_n = 0$ if $n > 0$. Similarly, $\tau : A_0 \rightarrow (A_n)$ will denote the section of ρ such that $\tau_0 = \text{id}_{A_0}$ and $\tau_n = 0$ if $n > 0$.

Every cogroup structure on $(A_n)_{n \in \mathbf{N}}$ induces a cogroup structure on A_0 such that ρ and τ are homomorphisms of cogroups. In this case, denote by I_0 the augmentation ideal of A_0 and put $A'_n = A_n / \overline{I_0 A_n}$ for every $n \geq 0$ (so that $A'_0 = k$).

Then $(A'_n)_{n \in \mathbf{N}}$ is a cogroup in $\mathbf{Alpg}/k$ (note that, since $A_0 = I_0 \oplus k \cdot 1$, $I_0 \hat{\otimes}_k A_n \simeq \overline{I_0 A_n}$ is a direct summand of A_n , for every n). Since τ is a section of ρ , by 2.4.A the cogroup $(A_n)_{n \in \mathbf{N}}$ is the “semidirect coproduct” of A_0 and the cogroup $(A'_n)_{n \in \mathbf{N}}$. More precisely, $(A'_n)_{n \in \mathbf{N}}$ is isomorphic, as an object of $\mathbf{Alpg}/k$, to the kernel of the pair:

$$(A_n) \xrightarrow[\text{(id} \hat{\otimes} \rho) \Delta]{\tau_1} (A_n) \hat{\otimes}_k A_0$$

(where $\Delta : (A_n) \rightarrow (A_n) \hat{\otimes} (A_n)$ is the comultiplication of (A_n) and $\tau_1(x) = x \hat{\otimes} 1$), and, identifying A'_n with its image in A_n , the map

$$(A'_n \hat{\otimes}_k A_0) \rightarrow (A_n), \quad a'_n \hat{\otimes} a_0 \mapsto a'_n a_0$$

is an isomorphism of $\mathbf{Alpg}/k$. (N.B. This is not an isomorphism of cogroups, but $\Delta(A') \subset A \hat{\otimes} A'$ and $(\gamma \rho' \hat{\otimes} \text{id}) \circ \Delta|_{A'} = \Delta'$, where Δ' is the comultiplication of A' and $\gamma \rho'$ the projection $A \rightarrow A'$, cf. 2.4.A.)

5.1.4. — Let (A, J) be an object of $\mathcal{C}$ and $(A_n) = \text{Gr}_J(A)$ the associated object of $\mathbf{Alpg}/k$, i.e. $A_n = \overline{J^n}/\overline{J^{n+1}}$ for every $n \geq 0$. It is clear that the algebra $\mathcal{A} = \prod_{n \geq 0} A_n$ is generated by A_0 and A_1 , that is, for $n \geq 1$, the map

$$(1) \quad A_1 \hat{\otimes}_{A_0} A_1 \hat{\otimes}_{A_0} \cdots \hat{\otimes}_{A_0} A_1 \rightarrow A_n$$

⁽¹⁴⁸⁾**N.D.E.** To a pair $H \subset G$ of formal k -groups, one therefore associates the “formal completion $\widehat{G}_H$ of G along H ”, which is a formal k -group; moreover, we shall see in 5.1.3–5.1.4 that the inclusion $\sigma : H \hookrightarrow \widehat{G}_H$ possesses a retraction $\pi : \widehat{G}_H \rightarrow H$ and that the formal k -group $N = \text{Ker}(\pi)$ is identified, as a formal variety, with the completion of the homogeneous space G/H along the identity section. This will be useful in 5.2.2.

defined by multiplication is surjective.

Suppose, moreover, that (A, J) is a cogroup of $\mathcal{C}$ and that A and the quotients $\overline{J^n}/\overline{J^{n+1}}$ are flat over k . Then, by Corollary 5.1.2, $\text{Gr}_J(A)$ is a cogroup of $\mathbf{Alpg}/k$. Thus, by 5.1.3, if one puts

$$(2) \quad A'_n = \left\{ x \in \overline{J^n}/\overline{J^{n+1}} \mid \Delta(x) - x \hat{\otimes} 1 \in \bigoplus_{i=1}^n (\overline{J^{n-i}}/\overline{J^{n-i+1}}) \hat{\otimes} (\overline{J^i}/\overline{J^{i+1}}) \right\},$$

then the map $(A'_n \hat{\otimes} A_0) \longrightarrow (A_n)$, $a'_n \hat{\otimes} a'_n \mapsto a'_n a_0$ is an isomorphism of $\mathbf{Alpg}/k$.⁽¹⁴⁹⁾

Denoting by I_0 the augmentation ideal of A_0 , one deduces from (1) and from the commutative diagram below, where $A_1^{\hat{\otimes} n}$ denotes $A'_1 \hat{\otimes}_k \cdots \hat{\otimes}_k A'_1$ (n factors):

$$\begin{array}{ccccc} A_1 \hat{\otimes}_{A_0} \cdots \hat{\otimes}_{A_0} A_1 & \xleftarrow{\sim} & A_1^{\hat{\otimes} n} \hat{\otimes}_k A_0 & \xleftarrow{\sim} & (A_1^{\hat{\otimes} n} \hat{\otimes}_k I_0) \oplus A_1^{\hat{\otimes} n} \\ \downarrow m & & \downarrow m' \hat{\otimes} \text{id} & & \downarrow m' \hat{\otimes} \text{id} \oplus m' \\ A_n & \xleftarrow{\sim} & A'_n \hat{\otimes}_k A_0 & \xleftarrow{\sim} & (A'_n \hat{\otimes}_k I_0) \oplus A'_n \end{array}$$

that the map

$$(3) \quad m' : A'_1 \hat{\otimes}_k \cdots \hat{\otimes}_k A'_1 \longrightarrow A'_n$$

induced by multiplication is surjective; in other words, the profinite k -algebra $\mathcal{A}' = \prod_{n \geq 0} A'_n$ is generated by its degree 1 terms.

⁽¹⁵⁰⁾Return now to hypothesis (i) of Proposition 5.1: let G be a topologically flat infinitesimal formal k -group, A its affine algebra, I_A the augmentation ideal of A , B a closed subalgebra of A , and $J = \overline{AI_B}$, where $I_B = B \cap I_A$. Denote by H the formal subgroup $\text{Spf}(A/J)$ and by π the projection $A \longrightarrow A/J$. Suppose that $A/\overline{J^n}$ is topologically flat over k , for every $n \geq 1$, and that B is contained in the kernel $\tilde{B}$ of the pair:

$$A \xrightarrow[\text{(id}_A \hat{\otimes}_k \pi) \Delta_A]{\tau_1} A \hat{\otimes}_k (A/J).$$

Let $(A_n) = \text{Gr}_J(A)$, let $I_0 = I_A/J$ be the augmentation ideal of $A_0 = A/J$, and define $(A'_n)_{n \in \mathbf{N}}$ as in (2) above. Denote by $(B_n)_{n \in \mathbf{N}}$ (resp. $(\tilde{B}_n)_{n \in \mathbf{N}}$) the object of $\mathbf{Alpg}/k$ associated with the filtration of B (resp. $\tilde{B}$) induced by that of A , i.e. defined by the ideals $B \cap \overline{J^n}$ (resp. $\tilde{B} \cap \overline{J^n}$). Then it is clear that $B_n \subset \tilde{B}_n \subset A'_n$ for every n , and that

$$\mathcal{B} = \prod_{n \geq 0} B_n \subset \tilde{\mathcal{B}} = \prod_{n \geq 0} \tilde{B}_n$$

are subalgebras of $\mathcal{A}' = \prod_{n \geq 0} A'_n$.

On the other hand, J (resp. $\overline{J^2}$) is the image in A of $I_B \hat{\otimes}_k A$ (resp. $\overline{I_B^2} \hat{\otimes}_k A$). Consequently, the map

$$(I_B/\overline{I_B^2}) \hat{\otimes}_k A \longrightarrow J/\overline{J^2} = A_1 \simeq A'_1 \hat{\otimes}_k A_0$$

⁽¹⁴⁹⁾**N.D.E.** Let G, H and $\hat{G}_H$ be the formal k -groups corresponding to $A, A_0 = A/J$ and $\text{Gr}_J(A)$; then $\tau : A_0 \hookrightarrow \text{Gr}_J(A)$ corresponds to a retraction $\pi : \hat{G}_H \longrightarrow H$ of the inclusion $H \hookrightarrow \hat{G}_H$, and the preceding passage means that $\hat{G}_H$ is the semidirect product of $N = \text{Ker}(\pi)$ by H .

⁽¹⁵⁰⁾**N.D.E.** We have detailed the original in what follows, and placed at the end the “supplement” $\overline{I_B^n} = B \cap \overline{J_B^n}$ (which is not necessary to establish Proposition 5.1).

is surjective, and factors through $(I_B/\overline{I_B^2}) \hat{\otimes}_k A_0$. Since $A_0 = k \cdot 1 \oplus I_0$, one deduces from the commutative exact diagram:

$$\begin{array}{ccc} (I_B/\overline{I_B^2}) \hat{\otimes}_k A_0 & \xleftarrow{\sim} & (I_B/\overline{I_B^2}) \hat{\otimes}_k I_0 \oplus (I_B/\overline{I_B^2}) \\ \downarrow & & \downarrow \\ J/\overline{J^2} & \xleftarrow{\sim} & A'_1 \hat{\otimes}_k I_0 \oplus A'_1 \\ \downarrow & & \downarrow \\ 0 & & 0 \end{array}$$

that A'_1 is the image of $I_B/\overline{I_B^2}$, so that one has $B_1 = A'_1$. Since $\mathcal{A}'$ is generated by A'_1 , it follows that, for every $n \geq 1$, one has

$$A'_n \subset B_n \subset \tilde{B}_n \subset A'_n,$$

whence $B_n = \tilde{B}_n = A'_n$.⁽¹⁵¹⁾

Finally, since the formal group G is *infinitesimal*, one has $\mathfrak{r}(A) = I_A$ and hence the ideals $\overline{I_A^n}$ tend to 0 (cf. 0.1.2); a fortiori, the ideals $\overline{J^n}$ tend to 0, and hence the induced filtrations on $\tilde{B}$ and B are separated. Moreover, since B is a *closed* subalgebra of A , it is complete for the topology defined by the ideals $B \cap \overline{J^n}$. Consequently, it follows from [CA], § V.7, Lemma 1 (see also 5.1.5 below) that $B = \tilde{B}$. This completes the proof of Proposition 5.1.

One has, moreover, the following supplement. For every n , $\overline{J^n} = \overline{AI_B^n}$ is the image in A of $A \hat{\otimes}_B \overline{I_B^n}$ and also of $A \hat{\otimes}_B (B \cap \overline{J^n})$. Now, by hypothesis, the affine algebra A/J of the formal subgroup H is topologically flat over k . Thus, by Theorem 2.4, the morphism $G = \mathrm{Spf}(A) \rightarrow G/H = \mathrm{Spf}(B)$ is *surjective and topologically flat*; one therefore has

$$A \hat{\otimes}_B \overline{I_B^n} = \overline{J^n} = A \hat{\otimes}_B (B \cap \overline{J^n}),$$

and this implies that $\overline{I_B^n} = B \cap \overline{J^n}$ for every n . This also follows from the fact that the maps

$$\overline{I_B^n}/\overline{I_B^{n+1}} \rightarrow A'_i = (B \cap \overline{J^n})/(B \cap \overline{J^{n+1}})$$

are surjective, and from 5.1.5 (ii) below, applied to $B'_n = \overline{I_B^n}$ and $B_n = B \cap \overline{J^n}$.

Lemma 5.1.5. ⁽¹⁵²⁾ (i) Let M and N be two abelian groups filtered by decreasing sequences of subgroups $(M_n)_{n \in \mathbf{Z}}$ and $(N_n)_{n \in \mathbf{Z}}$. Suppose that the union of the M_n (resp. N_n) equals M (resp. N), that the intersection of the M_n (resp. N_n) is zero, and that M is complete for the topology defined by the M_n . Let $f : M \rightarrow N$ be a morphism of filtered groups.

a) If f induces a surjection of the associated graded, then f is a surjection and N is complete for the topology defined by the N_n .

b) If f induces an injection of the associated graded, then f is an injection.

(ii) Let B be an abelian group, and let $B = B'_0 \supset B'_1 \supset \cdots$ and $B = B_0 \supset B_1 \supset \cdots$ be two separated filtrations of B by subgroups such that $B'_n \subset B_n$ for every n . Suppose that B is complete for the topology defined by the filtration (B'_n) .

If the map $B'_i/B'_{i+1} \rightarrow B_i/B_{i+1}$ is surjective for every i , then $B'_n = B_n$ for every n .

Indeed, (i) is Lemma 1 of [CA], § V.7 (see also [BAC], III, § 2.8), and (ii) follows by taking $M = B'_n \supset B'_{n+1} \supset \cdots$ and $N = B_n \supset B_{n+1} \supset \cdots$.

5.2. — Throughout the remainder of Section 5, k denotes a perfect field of characteristic $p > 0$.

⁽¹⁵¹⁾ **N.D.E.** With the notation of N.D.E. (149), this implies that the formal completion of G/H along the identity section (whose affine algebra is $\prod_n B_n$) is isomorphic, as a formal variety, to the formal k -group N .

⁽¹⁵²⁾ **N.D.E.** The original stated only point (ii); for the reader's convenience, we have stated Lemma 1 of [CA], § V.7, in (i).

We put $\overline{\mathbf{N}} = \mathbf{N} \cup \{\infty\}$. If B is a profinite k -algebra and $r \in \overline{\mathbf{N}}$, we denote by $((x^{p^r}))_{x \in \mathfrak{r}(B)}$ the closed ideal of B generated by the elements x^{p^r} , where x ranges over the radical $\mathfrak{r}(B)$ of B . If $r = \infty$, we use the same notation, agreeing that $((x^{p^\infty}))_{x \in \mathfrak{r}(B)}$ is the zero ideal. In both cases, B_r denotes the quotient $B/((x^{p^r}))_{x \in \mathfrak{r}(B)}$.

We say that B has *height* $\leq r$ if $((x^{p^r}))_{x \in \mathfrak{r}(B)}$ is the zero ideal; if this holds and r is finite, we say that B has *finite height*.

Consider in particular the case where B is of the form $k[[\omega]]$, ω being a pseudocompact k -vector space (cf. 1.2.5).⁽¹⁵³⁾ We then say that B is a *formal power series algebra* and that B_r is a *truncated formal power series algebra* ($r \in \overline{\mathbf{N}}$; we therefore agree to say that $B = B_\infty$ is also “truncated”). If $B = k[[\omega]]$, we also write $((x^{p^r}))_{x \in \omega}$ instead of $((x^{p^r}))_{x \in \mathfrak{r}(B)}$.

Notation. Let ω be a pseudocompact k -vector space filtered by an increasing sequence of closed vector subspaces

$$0 = \omega_0 \subset \omega_1 \subset \omega_2 \subset \omega_3 \subset \cdots.$$

(a) The closed ideal of $k[[\omega]]$ generated by the elements x^{p^r} , where r ranges over $\mathbf{N}$ and x ranges over ω_r , will be denoted by $((x^{p^r}))_{r \in \omega_r}$.

(b) On the other hand, we shall denote by ${}_r\omega$ the filtered pseudocompact vector space such that

$${}_r\omega_i = \omega_i \quad \text{if } i < r, \quad {}_r\omega_i = \omega \quad \text{if } i \geq r.$$

Theorem. (Dieudonné–Cartier). *Let $H \rightarrow G$ be a monomorphism of infinitesimal formal groups over a perfect field k of characteristic $p > 0$. Let B be the affine algebra of the homogeneous space G/H and suppose that one of the following three conditions holds:*⁽¹⁵⁴⁾

- (i) *B has finite height (this holds in particular if G has finite height).*
- (ii) *B is a complete noetherian local ring.*
- (iii) *B is a reduced ring.*

Then B is isomorphic to the completed tensor product of a finite family of truncated formal power series algebras.

The proof of this theorem occupies paragraphs 5.2.1 to 5.2.5.

5.2.1. — Let A be the affine algebra of G , I_A its augmentation ideal, and $I = B \cap I_A$. By 2.4, one has $H = \text{Spf}(A/\overline{AI})$ and $B = \{x \in A \mid \Delta(x) - 1 \hat{\otimes} x \in A \hat{\otimes} \overline{AI}\}$. Put $\omega = I/\overline{I^2}$. One denotes by I_r the closed subideal of I consisting of the x such that $x^{p^r} = 0$, and by ω_r the canonical image of I_r in ω . We shall prove:

Proposition. *If there exists a continuous section $\sigma : \omega \rightarrow I$ of the projection $I \rightarrow I/\overline{I^2}$ such that $\sigma(\omega_r) \subset I_r$ for every r , then B is isomorphic to $k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$.*

Such a section indeed extends to a continuous morphism $k[[\omega]] \rightarrow B$, which factors through $B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$. We prove in 5.2.2 to 5.2.5 that the morphism

$$\phi : B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r} \rightarrow B$$

thus obtained is an isomorphism.

⁽¹⁵³⁾**N.D.E.** In the original, the author uses “linearly compact vector space”, which is equivalent to “pseudocompact vector space” (cf. [BAC], § III.2, Exercises 15 a), 19 a) and 20 d)). We have preferred to retain the terminology “pseudocompact”, used up to this point.

⁽¹⁵⁴⁾**N.D.E.** On the one hand, we have replaced $H \setminus G$ by G/H , and likewise in the proof; on the other hand, we have added condition (iii).

5.2.1.A. — ⁽¹⁵⁵⁾For each $r \in \mathbf{N}$, put $\mathcal{I}_r = \overline{I^2} + I_r$, so that $\mathcal{I}_r/\overline{I^2} \simeq \omega_r$; one then has a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{I}_{r-1} & \longrightarrow & \mathcal{I}_r & \longrightarrow & \mathcal{I}_r/\mathcal{I}_{r-1} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \wr \\ 0 & \longrightarrow & \omega_{r-1} & \longrightarrow & \omega_r & \longrightarrow & \omega_r/\omega_{r-1} \longrightarrow 0 \end{array}$$

and since k is a field, the rows are split: one can extend a pseudobasis $\mathcal{B}_{r-1}$ of $\mathcal{I}_{r-1}$ to a pseudobasis $\mathcal{B}_{r-1} \cup \mathcal{B}'_r$ of $\mathcal{I}_r$, and then the closed subspace S_r with pseudobasis $\mathcal{B}'_r$ is a complement to $\mathcal{I}_{r-1}$ in $\mathcal{I}_r$, and the projection $\pi : I \rightarrow \omega$ induces an isomorphism from S_r onto a complement ω'_r to ω_{r-1} in ω_r . Denote by $\mathcal{I}_\infty$ the closed ideal $\bigcup_r \mathcal{I}_r$; it likewise admits a complement S_∞ in I , and π induces an isomorphism from $\mathcal{I}_\infty/\overline{I^2}$ (resp. from S_∞) onto the closure ω_∞ of the union of the ω_r (resp. onto a complement ω'_∞ to ω_∞ in ω). Denote by η the isomorphism $S_\infty \xrightarrow{\sim} \omega'_\infty$. One then obtains continuous linear maps:

$$\begin{array}{c} \overline{I^2} \times S_\infty \times \widehat{\bigoplus_r S_r}^\phi \longrightarrow I \\ \eta \times \theta \downarrow \\ \omega = \omega'_\infty \times \omega_\infty \end{array}$$

where $\widehat{\bigoplus_r S_r}$ is the direct sum of the S_r in $\mathbf{PC}(k)$, i.e. $(\prod_r S_r^\dagger)^*$ (cf. N.D.E. (16) of 0.2.2), and where $\theta : \widehat{\bigoplus_r S_r} \rightarrow \omega_\infty$ is induced by the maps $S_r \xrightarrow{\sim} \omega'_r \hookrightarrow \omega_\infty$. One therefore sees that a sufficient condition (but not a necessary one, see below) for obtaining a section $\sigma : \omega \rightarrow I$ as desired is that θ be an isomorphism. By duality (cf. 0.2.2), this is equivalent to saying that the linear map $\omega_\infty^\dagger \rightarrow \prod_r S_r^\dagger$ is bijective.

5.2.1.B. — Put, as before, $\omega_\infty = \overline{\bigcup_{n \in \mathbf{N}} \omega_n}$. A second case where a section $\sigma : \omega \rightarrow I$ as desired exists is the case where ω_∞ possesses a pseudobasis $\mathcal{B}_\infty$ which is a union of pseudobases of the ω_n/ω_{n-1} , for $n \in \mathbf{N}^*$ (one can then extend it by a pseudobasis $\mathcal{B}'_\infty$ of ω/ω_∞ to obtain a pseudobasis of ω compatible with the filtration). Putting $V = \omega_\infty^\dagger$ and denoting by V_n the orthogonal in V of ω_n , this is equivalent to saying that, in the category of “ordinary” k -vector spaces, the separated decreasing filtration

$$V = V_0 \supset V_1 \supset V_2 \supset \cdots$$

is split, i.e. that V is the direct sum, for $n \in \mathbf{N}$, of subspaces F_n such that $F_n \simeq V_n/V_{n+1}$. This is not necessarily the case: for example, if V is the space $\mathcal{S} = k^{\mathbf{N}}$ of sequences of elements of k and $\mathcal{S}_n$ the subspace of sequences (u_i) such that $u_i = 0$ for $i < n$, so that $\dim \mathcal{S}_n/\mathcal{S}_{n+1} = 1$, then $\mathcal{S}$ is not isomorphic to the direct sum of the $\mathcal{S}_n/\mathcal{S}_{n+1}$ since $\mathcal{S}$ does not have countable dimension (on the other hand, $\mathcal{S}$ here is the *product* of the $\mathcal{S}_n/\mathcal{S}_{n+1}$, cf. 5.2.1.A). This is, however, the case if V has countable dimension.⁽¹⁵⁶⁾ Indeed, let $(e_n)_{n \in \mathbf{N}}$ be a basis of V ; we shall construct by induction on n an increasing function $g : \mathbf{N} \rightarrow \mathbf{N}$ and subspaces F_i , for $i = 0, \dots, g(n)$, such that $F_i \simeq V_i/V_{i+1}$ and such that $F_{\leq g(n)} = \bigoplus_{i=0}^{g(n)} F_i$ is a complement to $V_{g(n)+1}$ containing $e_0, \dots, e_n$; one will then have $V = \bigoplus_{i \geq 0} F_i$. Let $n+1 \in \mathbf{N}$; one can suppose that the assertion is established for n (the assertion being empty for $n = -1$). If $e_{n+1} \in F_{\leq g(n)}$, one puts $g(n+1) = g(n)$; otherwise one writes $e_{n+1} = f + x$ with $f \in F_{\leq g(n)}$ and $x \in V_{g(n)+1}$ nonzero. Let then j be the smallest integer such that $x \in V_j - V_{j+1}$; for $i = g(n) + 1, \dots, j$, choose a complement F_i to V_{i+1} in V_i , so that $e_{n+1} \in F_j$, and put $g(n+1) = j$.

⁽¹⁵⁵⁾**N.D.E.** We have added paragraphs 5.2.1.A and 5.2.1.B.

⁽¹⁵⁶⁾**N.D.E.** This paragraph is the fruit of discussions with J.-M. Fontaine and E. Bouscaren; in particular, Bouscaren indicated the following proof to us.

5.2.1.C. — ⁽¹⁵⁷⁾In particular, the two preceding conditions (5.2.1.A and B) hold when the filtration of ω is *stationary*, i.e. when there exists an integer n_0 such that $\omega_n = \omega_{n_0}$ for $n_0 \leq n < +\infty$. In this case, one obtains an isomorphism from $k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$ onto the completed tensor product:

$$\frac{k[[\omega_1]]}{((x^p))_{x \in \omega_1}} \widehat{\otimes} \frac{k[[\omega'_2]]}{((x^{p^2}))_{x \in \omega'_2}} \widehat{\otimes} \cdots \widehat{\otimes} \frac{k[[\omega'_{n_0}]]}{((x^{p^{n_0}}))_{x \in \omega'_{n_0}}} \widehat{\otimes} k[[\omega'_\infty]],$$

where ω'_n (resp. ω'_∞) is a complement to ω_{n-1} in ω_n (resp. to $\omega_\infty = \omega_{n_0}$ in ω). The filtration of ω is clearly stationary in case (i), i.e. if $\omega_r = \omega$ for r large enough, and in case (ii), i.e. if ω has finite dimension, and also in case (iii), i.e. if $I_r = 0$ for every r (and in this case B will be isomorphic to the formal power series algebra $k[[\omega]]$). The above remarks therefore imply our theorem, modulo points 5.2.2–5.2.5 below.

5.2.2. — *First suppose that B has height ≤ 1* , i.e. that $x^p = 0$ if $x \in I$. By 5.1.4, the graded $\text{Gr}_I(B)$ associated with B for the filtration $I \supset \overline{I}^2 \supset \overline{I}^3 \supset \cdots$ is equipped with a cogroup structure in the category **Alg**/ k , i.e. the profinite algebra $C = \prod_{n \geq 0} \text{Gr}_I(B)_n$ is the affine algebra of a formal k -group N . It is clear that one has $\omega_{N/k} = I/\overline{I}^2$ and that N is infinitesimal of height ≤ 1 . By 4.4, the identity map of $\omega_{N/k}$ therefore induces an isomorphism from $B' = k[[\omega_{N/k}]]/((x^p))_{x \in \omega_{N/k}}$ onto C . This implies in particular that the map ϕ of 5.2.1 induces an isomorphism of the associated gradeds of B' and B when one filters B' and B by the powers of the augmentation ideal. Thus ϕ is an isomorphism, by [CA], § V.7, Lemma 1 (see also 5.1.5).

5.2.3. — *Suppose now that B has finite height $\leq r$* . Let π be the isomorphism $x \mapsto x^p$ from k onto k . The linear map from $B \widehat{\otimes}_\pi k$ to B which sends $b \widehat{\otimes}_\pi x$ to $b^p x = (bx^{1/p})^p$ has a closed image which is none other than the closed subalgebra $B^p = \{b^p \mid b \in B\}$ of B . Put $J = B^p \cap I = B^p \cap I_A$.

⁽¹⁵⁸⁾Denote by G_1 the kernel of the Frobenius morphism $\text{Fr} : G \rightarrow G^{(p)}$ and by HG_1 the formal subgroup of G which is the inverse image of the formal subgroup $H^{(p)}$ of $G^{(p)}$. Then HG_1 is defined by the closed ideal generated by the p th powers of elements of $\overline{AI}$, which equals $\overline{AJ}$. On the other hand, since the formation of G/H commutes with base change (since G/H represents the quotient sheaf for the flat topology, cf. 2.4), $(G/H)^{(p)} = G^{(p)}/H^{(p)}$ and one therefore has commutative diagrams:

$$\begin{array}{ccc} G & \xrightarrow{\text{Fr}} & G^{(p)} \\ \downarrow & & \downarrow \\ G/H & \xrightarrow{\text{Fr}} & G^{(p)}/H^{(p)} \end{array} \quad \begin{array}{ccc} A & \xleftarrow{a \widehat{\otimes} 1 \mapsto a^p} & A \widehat{\otimes}_\pi k \\ \uparrow & & \uparrow \\ B & \xleftarrow{b \widehat{\otimes} 1 \mapsto b^p} & B \widehat{\otimes}_\pi k \end{array}$$

from which one deduces that B^p is the affine algebra of the quotient G/HG_1 .⁽¹⁵⁹⁾ Temporarily denote by C the affine algebra of the quotient HG_1/H . Since the formation of G/H commutes with base change, one has a cartesian square:

$$\begin{array}{ccc} HG_1 & \longrightarrow & G \\ \downarrow & & \downarrow \\ HG_1/H & \longrightarrow & G/H \end{array}$$

whence an isomorphism $A \widehat{\otimes}_B C \simeq A/\overline{AJ} = A \widehat{\otimes}_B (B/\overline{BJ})$, and since A is topologically free over B (by 2.4, since A and B are local), it follows that the natural morphism $B/\overline{BJ} \rightarrow C$

⁽¹⁵⁷⁾**N.D.E.** We return here to the original, which we have shortened in view of the preceding additions.

⁽¹⁵⁸⁾**N.D.E.** We have added what follows.

⁽¹⁵⁹⁾**N.D.E.** As indicated in the original, this also follows from Proposition 5.1, but we have preferred to give the argument above, which does not use the implication (i) $\Rightarrow$ (ii) of loc. cit.

is an isomorphism, so $B/\overline{B\overline{J}}$ is the affine algebra of $HG_1/H^{(159)}$ and of course $B/\overline{B\overline{J}} = B_1$ has height ≤ 1 since $J = ((x^p))_{x \in \tau(B)}$.

Let $B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$, $\phi : B' \rightarrow B$ the morphism introduced in 5.2.1, B'^p the subalgebra $\{x^p \mid x \in B'\}$, and J' the augmentation ideal of B'^p . Then one has a commutative diagram:

$$\begin{array}{ccc} B' & \xrightarrow{\phi} & B \\ \downarrow & & \downarrow \\ B'_1 = B'/\overline{B'J'} & \xrightarrow[\phi_1]{} & B_1 = B/\overline{B\overline{J}} \end{array}$$

and, by 5.2.2, ϕ_1 is an isomorphism.

On the other hand, by 2.4, A is topologically flat over $B = \mathcal{A}(G/H)$ and over $B^p = \mathcal{A}(G/HG_1)$, hence, by 1.3.3, B is topologically flat over B^p . Moreover, by 5.2.4 below, the morphism $B'^p \rightarrow B^p$ induced by ϕ is an isomorphism. One can then apply 0.3.4 to the pseudocompact ring $B'^p = B^p$ and the pseudocompact B^p -modules $M = B'$, $N = B$: by the preceding, $\phi_1 = \phi \hat{\otimes}_{B^p} k$ is an isomorphism, and it follows that ϕ is an isomorphism. This proves 5.2 when B has finite height, modulo point 5.2.4 below.

5.2.4. — For every pseudocompact vector space V over k , we denote by πV the space $V \hat{\otimes}_{\pi k}$ obtained from V by the extension $x \mapsto x^p$ of the scalar field.⁽¹⁶⁰⁾ One then has a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \pi \overline{I^2} & \xrightarrow{\alpha} & \pi I & \xrightarrow{\beta} & \pi \omega \longrightarrow 0 \\ & & \downarrow u & & \downarrow v & & \downarrow w \\ 0 & \longrightarrow & \overline{J^2} & \xrightarrow{\gamma} & J & \xrightarrow{\delta} & \overline{\omega} \longrightarrow 0, \end{array}$$

where one has put $\overline{\omega} = J/\overline{J^2}$ and where the maps u, v, w are induced by the linear map $x \hat{\otimes} a \mapsto x^p a$ from πB to B^p . Since u and v are surjections, w is a surjection and has kernel the image $\pi \omega_1$ of $\pi I_1 = \text{Ker}(v)$.

Then, putting $J_n = \{x \in J \mid x^{p^n} = 0\}$ and $\overline{\omega}_n = \delta(I_n)$, one has $J_n = v(\pi I_{n+1})$ and $\overline{\omega}_n = w(\pi \omega_{n+1})$, for every $n \geq 0$. The section $\pi \sigma : \pi \omega \rightarrow \pi I$, induced by the section σ of 5.2.1, therefore defines, on passing to the quotient, a section $\tau : \overline{\omega} \rightarrow J$ compatible with the filtrations of J and $\overline{\omega}$. Since B^p has height $\leq r - 1$, this section induces, by the induction hypothesis, an isomorphism

$$\psi : B'' = k[[\overline{\omega}]]/((x^{p^n}))_{x \in \overline{\omega}_n} \xrightarrow{\sim} B^p.$$

Now B'' is identified with B'^p and ψ with the morphism $B'^p \rightarrow B^p$ induced by ϕ , and hence *our theorem is proved when B has finite height*.

Remark 5.2.4.A. ⁽¹⁶¹⁾ Suppose that B has height $\leq r + 1$ (with $r \in \mathbf{N}^*$); then $I_{r+1} = I$ and one has an isomorphism

$$(1) \quad B \simeq \frac{k[[S_1]]}{((x_1^p))_{x_1 \in S_1}} \hat{\otimes}_k \cdots \hat{\otimes}_k \frac{k[[S_r]]}{((x_r^{p^r}))_{x_r \in S_r}} \hat{\otimes}_k \frac{k[[S_{r+1}]]}{((x_{r+1}^{p^{r+1}}))_{x_{r+1} \in S_{r+1}}}$$

where each S_n is a complement to $\overline{I^2} + I_{n-1}$ in $\overline{I^2} + I_n$. Then $\omega = I/\overline{I^2}$ is identified with $\prod_{i=1}^{r+1} S_i$, and one sees easily that, for $n = 1, \dots, r + 1$, the image ω_n of I_n in ω is identified with $\prod_{i=1}^n S_i$.

⁽¹⁶⁰⁾**N.D.E.** Since k is perfect, one can identify πV with the abelian group V on which k acts by $\lambda \cdot v = \lambda^{1/p} v$.

⁽¹⁶¹⁾**N.D.E.** We have added this remark, used in 5.2.5.

This has the following consequence. Let $B_r = B/J_r$, where $J_r = ((x^{p^r}))_{x \in \mathfrak{t}(B)}$, and let $\mathfrak{m} = I/J_r$ be the augmentation ideal of B_r ; since $J_r \subset \overline{I^2}$, $\omega(r) = \mathfrak{m}/\overline{\mathfrak{m}^2}$ is identified with ω . For $n = 1, \dots, r$, denote by $\omega(r)_n$ the image in $\omega(r)$ of $\mathfrak{m}_n$; it is also the image in ω of $\{x \in I \mid x^{p^n} \in J_r\}$, so $\omega(r)_n$ contains ω_n . On the other hand, it follows from isomorphism (1) that one has $J_r = ((x_{r+1}^{p^r}))_{x_{r+1} \in S_{r+1}}$, whence

$$(2) \quad B_r \simeq \frac{k[[S_1]]}{((x_1^p))_{x_1 \in S_1}} \widehat{\otimes}_k \cdots \widehat{\otimes}_k \frac{k[[S_r]]}{((x_r^{p^r}))_{x_r \in S_r}} \widehat{\otimes}_k \frac{k[[S_{r+1}]]}{((x_{r+1}^{p^r}))_{x_{r+1} \in S_{r+1}}}$$

and hence, by the preceding, $\omega(r)_n$ is identified with $\prod_{i=1}^n S_i$ for $n = 1, \dots, r-1$. One therefore obtains that the inclusion $\omega_n \subset \omega(r)_n$ is an equality, for $n = 1, \dots, r-1$.

5.2.5. — *It remains to consider the case where B has infinite height, and where the projection $I \rightarrow \omega$ possesses a section σ compatible with the filtrations of ω and I . Consider the morphism*

$$\phi : B' = k[[\omega]]/((x^{p^n}))_{x \in \omega_n} \rightarrow B$$

induced by σ ; it suffices to show that for every $r \in \mathbf{N}$, the map $\phi_r : B'_r \rightarrow B_r$ induced by ϕ is invertible.⁽¹⁶²⁾

For every $r \in \mathbf{N}^*$, denote by G_r the kernel of the iterated Frobenius morphism $G \rightarrow G^{(p^r)}$ and by HG_r the formal subgroup of G which is the inverse image of the formal subgroup $H^{(p^r)}$ of $G^{(p^r)}$, so that HG_r is defined by the closed ideal generated by the p^r th powers of elements of $\overline{AI}$, which equals $\overline{AJ_r}$, where $J_r = \{x^{p^r} \mid x \in I\}$. One then obtains, exactly as in 5.2.3, that $B_r = B/\overline{BJ_r}$ is the affine algebra of HG_r/H (and has, of course, height $\leq r$).

Denote by $\mathfrak{m}(r) = I/\overline{BJ_r}$ the augmentation ideal of B_r ; the canonical projection from B onto B_r clearly induces an isomorphism from $\omega = I/\overline{I^2}$ onto $\omega(r) = \mathfrak{m}(r)/\overline{\mathfrak{m}(r)^2}$, which allows us to identify these two spaces. Let $\omega(r)_n$ be the image in $\omega(r)$ of the closed ideal $\mathfrak{m}(r)_n = \{y \in \mathfrak{m}(r) \mid y^{p^n} = 0\}$; it is also the image in ω of the closed ideal $I(r)_n = \{x \in I \mid x^{p^n} \in \overline{BJ_r}\}$.⁽¹⁶³⁾ It is clear that $\omega_n(r) = \omega$ if $n \geq r$; let us show that $\omega_n(r) = \omega_n$ if $n < r$. For every r, n , the sequence below is exact:

$$0 \rightarrow I(r)_n \cap \overline{I^2} \rightarrow I(r)_n \rightarrow \omega(r)_n \rightarrow 0.$$

Moreover, for fixed n , one has $\bigcap_r I(r)_n = I_n$, since $\bigcap_r \overline{BJ_r} = 0$. Since in $\mathbf{PC}(k)$ filtered projective limits are exact (cf. 0.2), it follows that, for every n , one has

$$(*) \quad \omega_n = \bigcap_r \omega(r)_n.$$

On the other hand, by Remark 5.2.4.A, one has $\omega(r)_n = \omega(r+1)_n$ if $n \leq r$. Combined with (*), this implies that $\omega(r)_n = \omega_n$ if $n < r$.

Consequently, the vector space ω filtered by the subspaces $(\omega(r)_n)_{n \geq 0}$ is none other than ${}_r\omega$ (Notation 5.2). A fortiori, the map $\sigma(r)$ which is the composite of $\sigma : \omega \rightarrow I$ and the projection $I \rightarrow \mathfrak{m}(r)$ is compatible with the filtrations $(\omega(r)_n)$ and $(\mathfrak{m}(r)_n)$ of ω and $\mathfrak{m}(r)$. Since $k[[{}_r\omega]]/((x^{p^n}))_{x \in {}_r\omega_n}$ is none other than B'_r and $\phi_r : B'_r \rightarrow B_r$ is the morphism induced by $\sigma(r)$, the result already established for algebras of finite height shows that ϕ_r is an isomorphism.

Definition 5.2.6. ⁽¹⁶⁴⁾ Let $(A_\lambda)_{\lambda \in \Lambda}$ be a family of profinite k -algebras, each equipped with an augmentation $\varepsilon_\lambda : A_\lambda \rightarrow k$ (this is the case, in particular, if each A_λ is local with residue field k). One then defines the infinite completed tensor product $\mathcal{A} = \widehat{\bigotimes}_{\lambda \in \Lambda} A_\lambda$ as the projective limit in $\mathbf{Alp}/k$ of the $A_F = \widehat{\bigotimes}_{\lambda \in F} A_\lambda$, for F ranging over the finite subsets of Λ , the transition morphisms $A_{F'} \rightarrow A_F$, for $F' = F \cup \{\lambda\}$, being $\text{id} \otimes \varepsilon_\lambda$. In particular,

⁽¹⁶²⁾**N.D.E.** Indeed, B' (resp. B) is the projective limit of the B'_r (resp. B_r). On the other hand, we have modified the original in what follows, taking account of the addition made in 5.2.3.

⁽¹⁶³⁾**N.D.E.** We have detailed the original in what follows.

⁽¹⁶⁴⁾**N.D.E.** We have added this number, to define the infinite tensor products used in 5.3 (a).

if $\Lambda = \mathbf{N}^*$ and one denotes by X_n the formal k -variety $\mathrm{Spf}(A_n)$, then $\mathrm{Spf}(\mathcal{A})$ represents the functor which associates to every $C \in \mathbf{Alf}/k$ the set of “finite” sequences of elements of $\prod_{n \geq 1} X_n(C)$, i.e. sequences

$$(x_1, x_2, \dots) \in \prod_{n \geq 1} X_n(C)$$

such that $x_n = \varepsilon_n$ for n large enough, where ε_n denotes, by abuse of notation, the composite of $\varepsilon_n : A_n \rightarrow k$ and the structure morphism $k \rightarrow C$. (If, moreover, each A_n is a quotient of an algebra $k[[\omega'_n]]$, one can denote by 0 the unique morphism $A_n \rightarrow C$ which vanishes on ω'_n , and one therefore obtains the set of sequences such that “ $x_n = 0$ ” for n large enough.)

Remarks 5.3. (a) Call *stationary* every profinite k -algebra which is the completed tensor product of a family of truncated formal power series algebras.⁽¹⁶⁵⁾

If G is an infinitesimal formal k -group and B the affine algebra of a homogeneous space of G , it follows from Theorem 5.2 that the algebra $B/((x^{p^r}))_{x \in \mathfrak{r}(B)}$ is stationary for every integer $r \in \mathbf{N}$. This implies in particular that B is a projective limit of stationary algebras.⁽¹⁶⁶⁾

(b) I do not know whether, with the notation of 5.2.1, one can choose A and B in such a way that there is no section $\sigma : \omega \rightarrow I$ compatible with the filtrations.⁽¹⁶⁷⁾ One will note, however, that one can have for ω any pseudocompact vector space filtered by an increasing sequence of closed subspaces. Indeed, if $\omega_1 \subset \omega_2 \subset \dots \subset \omega$ is such a filtered space, one can define in the algebra $B = A = k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$ a diagonal morphism $\Delta_A : A \rightarrow A \hat{\otimes}_k A$ satisfying conditions (i), (ii), (iii) of 2.1; it suffices to put $\Delta_A(y) = y \hat{\otimes} 1 + 1 \hat{\otimes} y$ when y is the image in A of an element of ω .

Corollary 5.4. Let G be an algebraic group over a perfect field k of characteristic $p > 0$, H an algebraic subgroup of G , e the image of the identity element of G in G/H and A the local algebra of G/H at e . Then $\hat{A}$ is isomorphic to an algebra of the form

$$k[[X_1, \dots, X_r, \dots, X_s]]/(X_1^{p^{n_1}}, \dots, X_r^{p^{n_r}}).$$

Indeed, consider the infinitesimal formal groups $\hat{G} = \mathrm{Spf}(\hat{\mathcal{O}}_{G,e})$ and $\hat{H} = \mathrm{Spf}(\hat{\mathcal{O}}_{H,e})$; by 1.3.4, the completion $\hat{A}$ of $A = \mathcal{O}_{G/H,e}$ is isomorphic to $\mathcal{A}(\hat{G}/\hat{H})$, and the corollary therefore follows from Theorem 5.2 (ii).⁽¹⁶⁸⁾

5.5. — Complements. — ⁽¹⁶⁹⁾Recall the following definitions. On the one hand, one says that a noetherian local ring A is a *complete intersection* if the completion $\hat{A}$ is a quotient of

⁽¹⁶⁵⁾**N.D.E.** The author was doubtless thinking of a tensor product $\mathcal{A} = \widehat{\bigotimes}_{n \in \mathbf{N}^*} k[[\omega'_n]]/((x^{p^n}))_{x \in \omega'_n}$, where the ω'_n are arbitrary pseudocompact k -vector spaces. In this case, one sees without difficulty that $\omega = \omega_{\mathcal{A}/k}$ is identified with the product of the ω'_n , and the filtration $(\omega_n)_{n \in \mathbf{N}^*}$ is given by $\omega_n = \prod_{i=1}^n \omega'_i$, i.e. one is in case 5.2.1.B. For this reason, it would be preferable to call these algebras *stable* (rather than “stationary”), cf. [Di73], II § 2.9, p. 75. If, for example, $\dim \omega'_n = 1$ for every $n \in \mathbf{N}^*$, then $\mathcal{A}$ represents the functor which associates to every $C \in \mathbf{Alf}/k$ the set of sequences $(x_n)_{n \in \mathbf{N}^*}$ of elements of C such that $x_n^{p^n} = 0$, and $x_n = 0$ for n large enough. Note finally that this case (i.e. the case where $\omega = \prod_{n \in \mathbf{N}^*} \omega'_n$) corresponds to the case studied, in the dual situation of connected cocommutative Hopf algebras, by M. E. Sweedler, cf. [Sw67], Th. 3.

⁽¹⁶⁶⁾**N.D.E.** But such a projective limit is not necessarily a stable profinite k -algebra (in the sense of the preceding N.D.E.). For example, let $\mathcal{S}$ be the “ordinary” k -vector space of sequences $(u_1, u_2, \dots)$ of elements of k and let $\omega = \mathcal{S}^*$; then ω is the direct sum in $\mathbf{PC}(k)$ of copies k_n of k , for $n \in \mathbf{N}^*$, i.e. one is in case 5.2.1.A. If one denotes by x_n the element of ω defined by $x_n(u) = u_n$, for every sequence $u = (u_i)_{i \in \mathbf{N}^*}$, then the k -algebra $A = k[[\omega]]/((x_n^{p^n}))_{n \in \mathbf{N}^*}$ is such that $\omega_{A/k} = \omega$ and $\omega_n = \prod_{i=1}^n kx_i$, but is not stable: $\mathrm{Spf}(A)$ represents the functor which associates to every $C \in \mathbf{Alf}/k$ the set of “infinite” sequences $(x_n)_{n \in \mathbf{N}^*}$ of elements of C such that $x_n^{p^n} = 0$ for every $n \in \mathbf{N}^*$.

⁽¹⁶⁷⁾**N.D.E.** The editors do not know either, outside the cases considered in 5.2.1.A and B.

⁽¹⁶⁸⁾**N.D.E.** See also [DG70], § III.3, Th. 6.1.

⁽¹⁶⁹⁾**N.D.E.** We have added this subsection, to give some consequences of 5.4, mentioned in Exposés III and VI A.

a complete regular noetherian local ring B by an ideal I generated by a regular sequence of elements of B (cf. EGA IV₄, 19.3.1). On the other hand, let $\tau : Y \hookrightarrow X$ be a closed immersion of schemes. If $y \in Y$, one says that τ is a *regular immersion at y* if the kernel of $\mathcal{O}_{X,y} \rightarrow \mathcal{O}_{Y,y}$ is generated by a regular sequence; if, moreover, X is locally noetherian and τ is a regular immersion at every point, one says that τ is a *regular immersion* (cf. loc. cit., Prop. 16.9.10 and Def. 16.9.2).

Corollary 5.5.1. *If G is an algebraic group over a field k , the local ring $\mathcal{O}_{G,e}$ is a complete intersection.*

Indeed, by EGA IV₄, 19.3.4, one can suppose that k is algebraically closed. If $\text{char}(k) = 0$, one already knows that G is smooth (cf. 3.3.1 or VI B, 1.6.1) and hence $\mathcal{O}_{G,e}$ is a formal power series k -algebra, by EGA IV₄, 17.5.3 (d''). If $\text{char}(k) = p > 0$, it follows from 5.4, applied to $H = \{e\}$, that $\mathcal{O}_{G,e}$ is a complete intersection.

Remarks 5.5.2. Let k be a field, G a smooth algebraic k -group, and H a closed subgroup of G .

a) We have seen in Exp. III, 4.15, that the immersion $H \hookrightarrow G$ is regular; this can also be deduced from 5.4, as follows. As in loc. cit., one can suppose that k is algebraically closed, and it suffices to show that the kernel I of $\mathcal{O}_{G,e} \rightarrow \mathcal{O}_{H,e}$ is generated by a regular sequence. Put $A = \mathcal{O}_{G,e}$, $\hat{G} = \text{Spf}(\hat{A})$ and $\hat{H} = \text{Spf}(\hat{\mathcal{O}}_{H,e})$. Since A is noetherian, one has an exact sequence

$$0 \rightarrow I \otimes_A \hat{A} \rightarrow \hat{A} \xrightarrow{\pi} \mathcal{A}(H) \rightarrow 0$$

and $\hat{A}$ is faithfully flat over A . Thus, by EGA IV₄, 16.9.10 (ii) and 19.1.5 (ii), it suffices to show that the kernel $\hat{I} = I \otimes_A \hat{A}$ of π is generated by a regular sequence of elements of $\hat{A}$.

Now, since G is smooth, $\hat{A}$ is reduced; by 5.4, the subalgebra $B = \mathcal{A}(\hat{G}/\hat{H})$ is therefore isomorphic to a formal power series algebra $k[[x_1, \dots, x_n]]$, and hence the identity section of $\hat{G}/\hat{H}$ is defined in B by the regular sequence $(x_1, \dots, x_n)$. Since $\hat{A}$ is noetherian, the ideal J of $\hat{A}$ generated by $x_1, \dots, x_n$ is closed and hence equal to $\hat{I}$, by Corollary 1.4. Moreover, since $\hat{A}$ is topologically flat, hence flat over B (cf. 0.3.8), $(x_1, \dots, x_n)$ is a regular sequence in $\hat{A}$, by EGA IV₄, 19.1.5 (ii).

b) One can also deduce from 5.2 (ii) the following more precise result. Suppose that k is perfect. By 5.2 (ii) applied to the algebra $C = \mathcal{A}(H)$, there exist a basis $(y_1, \dots, y_{r+s})$ of ω_H and integers $1 \leq n_1 \leq \dots \leq n_r$ such that $\mathcal{A}(H)$ is isomorphic to the quotient of $k[[y_1, \dots, y_{r+s}]]$ by the ideal generated by the $y_i^{p^{n_i}}$ for $i = 1, \dots, r$. Lift the y_i to elements x_i of ω_G and extend $(x_1, \dots, x_{r+s})$ to a basis $(x_1, \dots, x_n)$ of ω_G . Since $\mathcal{A}(G)$ is reduced, the morphism $k[[x_1, \dots, x_n]] \rightarrow \mathcal{A}(G)$ is an isomorphism, by 5.2 (iii). One therefore obtains: *there exists a "coordinate system" $(x_1, \dots, x_n)$ of G (i.e. an isomorphism $\mathcal{A}(G) \simeq k[[x_1, \dots, x_n]]$) such that H is defined by the equations $x_i^{p^{n_i}} = 0$ for $i = 1, \dots, r$ and $x_i = 0$ for $i > r + s$ ("Dieudonné's theorem", compare with [Di55], § 19, Th. 6, and [Di73], II § 3.2, Prop. 3, and what precedes it).*

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⁽¹⁷⁰⁾N.D.E. To this reference, which appeared in the original, we have added the references which follow.

Exposé VIII. Diagonalizable groups

by A. Grothendieck

1. Biduality

Let $\mathcal{C}$ be a category, which we identify as usual with a full subcategory of $\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}, (\text{Ens}))$ (cf. Exp. I). Let I be a commutative group functor, i.e. an object of $\widehat{\mathcal{C}}$ endowed with a commutative group structure (cf. I, 2.1).⁽¹⁾ For every $X \in \text{Ob}(\widehat{\mathcal{C}})$, the object $\underline{\text{Hom}}(X, I)$ is endowed with a commutative group structure, induced by that of I . For every group G in $\widehat{\mathcal{C}}$, let

$$D(G) = \underline{\text{Hom}}_{\text{gr.}}(G, I)$$

be the subobject of $\underline{\text{Hom}}(G, I)$ defined, for every $S \in \text{Ob}(\mathcal{C})$, by:

$$(x) \quad D(G)(S) = \text{Hom}_{S\text{-gr.}}(G_S, I_S),$$

where $G_S = G \times S$ and $I_S = I \times S$ are regarded as S -groups, i.e. groups in $\widehat{\mathcal{C}}/S$. Then $D(G)$ is a sub- $\widehat{\mathcal{C}}$ -group of $\underline{\text{Hom}}(G, I)$. In this way one obtains a contravariant functor D from the category of $\widehat{\mathcal{C}}$ -groups to the category of commutative $\widehat{\mathcal{C}}$ -groups.

The right-hand side of (x) may also be interpreted as the subset of $\text{Hom}(G \times S, I)$ consisting of the morphisms $G \times S \rightarrow I$ that are “multiplicative with respect to the first argument G ”. Moreover, the preceding formulas hold more generally when S is any object of $\widehat{\mathcal{C}}$, not necessarily coming from $\mathcal{C}$.

If one now takes for S a group in $\widehat{\mathcal{C}}$, which we shall denote by G' , then in the left-hand side $\text{Hom}(G', D(G))$ of (x) one may distinguish the subset $\text{Hom}_{\text{gr.}}(G', D(G))$ consisting of morphisms that respect the group structures of G' and $D(G)$. It then corresponds to the subset of $\text{Hom}(G \times G', I)$ consisting of morphisms that are multiplicative with respect to the first and with respect to the second argument, — which one may call bilinear morphisms from $G \times G'$ to I , or pairings of G and G' with values in I . One thus finds

$$(xx) \quad \text{Hom}_{\text{gr.}}(G', D(G)) \xrightarrow{\sim} \text{Hom}_{\text{bil.}}(G \times G', I),$$

which is an isomorphism functorial in the pair (G, G') . As the right-hand side is symmetric in G and G' , one deduces a functorial bijection:

$$(xxx) \quad \text{Hom}_{\text{gr.}}(G', D(G)) \xrightarrow{\sim} \text{Hom}_{\text{gr.}}(G, D(G')).$$

In other words, “it amounts to the same thing” to give a group homomorphism $G' \rightarrow D(G)$, or a group homomorphism $G \rightarrow D(G')$, both indeed amounting to giving a pairing $G \times G' \rightarrow I$.

Applying this to the case where $G' = D(G)$ and to the identity homomorphism $G' \rightarrow D(G)$, one then finds a *canonical homomorphism*

$$(xxxx) \quad G \rightarrow D(D(G)).$$

⁽⁰⁾Version 1.1 of November 8, 2009: additions in 1.2, 1.4, 1.7, 3.1, 3.4, 4.5.1, 6.4, 6.8 — 1.5.1 and section 7 to be reviewed.

⁽¹⁾**N.D.E.** The original has been expanded in what follows.

Definition 1.0. ⁽²⁾ One shall say that G is *reflexive* (with respect to I) if the preceding homomorphism is an isomorphism. One will note that this implies that G is commutative.

One thus sees that:

Proposition 1.0.1. *The functor D induces an anti-equivalence of the category of reflexive $\widehat{\mathcal{C}}$ -groups with itself.*

In particular, if G, H are two reflexive groups, D induces an isomorphism:

$$\mathrm{Hom}_{\mathrm{gr.}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr.}}(D(H), D(G))$$

(it even suffices for H to be reflexive, as one sees from formula (xxx)).

Definition 1.0.2. As usual, we shall then say that a $\mathcal{C}$ -group G is reflexive if it is reflexive as a $\widehat{\mathcal{C}}$ -group (without concerning ourselves with whether $D(G)$ is representable or not).

One thus obtains, by D , an anti-equivalence of the category of reflexive $\mathcal{C}$ -groups G such that $D(G)$ is representable, with itself.

Remark 1.0.3. Let us point out, to conclude these generalities, that the formation of the duals $D(G)$ commutes with extension of the base, which therefore transforms reflexive groups into reflexive groups.

We shall henceforth be interested in the case where $\mathcal{C} = (\mathrm{Sch})/S$, the category of preschemes over S , and $I = \mathbf{G}_{\mathrm{m},S}$, the “multiplicative group over S ”, cf. exposé I. For every ordinary group M , we consider the S -group M_S . One sees immediately that for every group prescheme J over S , one has a canonical isomorphism (functorial in M and J , and compatible with extension of the base):

$$\mathrm{Hom}_{S\text{-gr.}}(M_S, J) = \mathrm{Hom}_{\mathrm{gr.}}(M, J(S)).$$

Applying this to $J = I = \mathbf{G}_{\mathrm{m},S}$ and to a variable S' over S , one finds a functorial isomorphism:

$$(1.0.4) \quad D(M_S)(S') \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr.}}(M, \mathbf{G}_{\mathrm{m}}(S')).$$

One thus recovers the functor already considered in I, 4.4, also denoted by $D_S(M)$, which is representable for commutative M , since

$$D_S(M) = D(M_S) = \mathrm{Spec} \mathcal{O}_S(M),$$

where $\mathcal{O}_S(M)$ denotes the group algebra of M with coefficients in $\mathcal{O}_S$. (Let us also note, in the general case, that $D(M)$ does not change if one replaces M by its abelianization, so that nothing is lost by supposing M commutative.)

Definition 1.1. A group prescheme G over S is said to be *diagonalizable* if it is isomorphic to a scheme of the form $D_S(M) = D(M_S) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, \mathbf{G}_{\mathrm{m}})$ for a suitable commutative group M .

One says that G is *locally diagonalizable* if every point of S admits an open neighborhood U such that $G|U$ is diagonalizable.

Theorem 1.2. *Let Γ be a constant commutative group scheme over S , i.e. isomorphic to a group scheme of the form M_S , where M is an ordinary commutative group. Then Γ is reflexive, i.e. the canonical homomorphism*

$$\Gamma \longrightarrow D(D(\Gamma))$$

is an isomorphism. ⁽³⁾ *The diagonalizable group $D(M_S)$ is therefore also reflexive.*

⁽²⁾**N.D.E.** The numbering 1.0, 1.0.1, ... has been added in order to bring out the definitions and statements found there.

⁽³⁾**N.D.E.** The following sentence has been added.

In view of the definitions, this follows from the following statement (which one shall apply to a prescheme S' over S):

Corollary 1.3. *Let $G = D(M_S)$. Then every homomorphism of S -groups*

$$u : G \longrightarrow \mathbf{G}_{m,S}$$

is defined by a uniquely determined section of M_S over S , i.e. by a uniquely determined locally constant map from S to M .

PROOF. Since by definition one has

$$\mathbf{G}_{m,S} = \mathrm{GL}(1)_S = \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\mathcal{O}_S),$$

one sees that giving a group homomorphism $G \longrightarrow \mathbf{G}_{m,S}$ is equivalent to giving on $\mathcal{O}_S$ a G - $\mathcal{O}_S$ -module structure compatible with the natural $\mathcal{O}_S$ -module structure on $\mathcal{O}_S$ (cf. I, 4.7). By I, 4.7.3, this also amounts to giving a grading of type M on $\mathcal{O}_S$, i.e. a decomposition of $\mathcal{O}_S$ as a direct sum of modules $\mathcal{L}_m$ ($m \in M$). Now it is well known that a direct summand of a locally free module of finite type is locally free of finite type, so each $\mathcal{L}_m$ is, in a neighborhood of each point of S , either zero or free of rank 1, and in the latter case identical to $\mathcal{O}_S$ in this neighborhood. Let S_m be the open subset of S consisting of the points where this second alternative occurs. Expressing the fact that $\mathcal{O}_S$ is the direct sum of the $\mathcal{L}_m$, one sees that the union of the S_m is S and that the S_m are pairwise disjoint. Thus giving a group homomorphism $G \longrightarrow \mathbf{G}_{m,S}$ is equivalent to giving a decomposition of S as a union of pairwise disjoint open subsets S_m ($m \in M$), i.e. to giving a locally constant map from S to M . This establishes 1.3 and hence 1.2. $\square$

Corollary 1.4. *A diagonalizable group is reflexive; the same is therefore true of a locally diagonalizable group. If M, N are two ordinary commutative groups, the natural homomorphism*

$$\mathrm{Hom}_{S\text{-gr.}}(M_S, N_S) \longrightarrow \mathrm{Hom}_{S\text{-gr.}}(D(N_S), D(M_S))$$

is bijective.

(4) As the preceding isomorphism is compatible with extension of the base, one deduces an isomorphism of S -group functors:

$$(1) \quad \underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, N_S) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(D(N_S), D(M_S)).$$

For every S -prescheme T , one has

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, N_S)(T) = \mathrm{Hom}_{\mathrm{gr.}}(M, \Gamma(N_T/T)),$$

and, by I 1.8, $\Gamma(N_T/T)$ is the abelian group of locally constant maps $T \longrightarrow N$. On the other hand, let $\mathrm{Hom}_{\mathrm{gr.}}(M, N)_S$ be the constant S -group associated with the ordinary abelian group $\mathrm{Hom}_{\mathrm{gr.}}(M, N)$. One has an evident homomorphism of commutative S -group functors:

$$(2) \quad \mathrm{Hom}_{\mathrm{gr.}}(M, N)_S \xrightarrow{\theta} \underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, N_S),$$

which is always a monomorphism. Moreover, it is an *isomorphism* if M is of finite type.⁽⁵⁾

⁽⁴⁾**N.D.E.** What follows has been expanded.

⁽⁵⁾**N.D.E.** Indeed, denoting by $F(M)$ (resp. $G(M)$) the left-hand (resp. right-hand) side of (2), if $M = M_1 \oplus M_2$, one has a canonical isomorphism $F(M) = F(M_1) \oplus F(M_2)$, and similarly for G . This reduces us to checking that θ is an isomorphism when $M = \mathbf{Z}/r\mathbf{Z}$, for an integer $r \geq 0$. In this case, $F(M) = ({}_rN)_S$, where ${}_rN$ is the kernel of $r \cdot \mathrm{id}_N$, and, for every $T \longrightarrow S$, the homomorphism

$$F(M)(T) = \Gamma({}_rN_T/T) \longrightarrow G(M)(T) = {}_r\Gamma(N_T/T)$$

is bijective, whence the desired result.

Part (a) of the following corollary follows from the above; part (b) follows from it by the descent results “recalled” in 1.7.⁽⁶⁾

Corollary 1.5. *a) Let M, N be two ordinary commutative groups, M of finite type; then one has an isomorphism*

$$\mathrm{Hom}_{\mathrm{gr.}}(M, N)_S \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(D(N_S), D(M_S));$$

consequently $\underline{\mathrm{Hom}}_{S\text{-gr.}}(D(N_S), D(M_S))$ is representable.

b) More generally, if G, H are locally diagonalizable, H being of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable.

⁽⁷⁾ One concludes from 1.5:

Corollary 1.6. *Under the conditions of 1.5, if S is connected, one has*

$$\mathrm{Hom}_{S\text{-gr.}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr.}}(M, N)$$

and

$$\mathrm{Isom}_{S\text{-gr.}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Isom}_{\mathrm{gr.}}(M, N).$$

1.7. — Descent of representability. — ⁽⁸⁾ In this paragraph one “recalls” some descent results which will frequently be used in what follows.

Scholium 1.7.1. *Let S be a prescheme and X, Y, T be S -preschemes. If (T_i) is an open covering of T , and if one puts $T_{ij} = T_i \cap T_j = T_i \times_T T_j$, then, since giving a morphism of T -preschemes $X_T \rightarrow Y_T$ is local on T , one has an exact sequence of sets:*

$$(1) \quad \mathrm{Hom}_T(X_T, Y_T) \rightarrow \prod_i \mathrm{Hom}_{T_i}(X_{T_i}, Y_{T_i}) \rightrightarrows \prod_{i,j} \mathrm{Hom}_{T_{ij}}(X_{T_{ij}}, Y_{T_{ij}})$$

i.e. $\underline{\mathrm{Hom}}_S(X, Y)$ is a local S -functor, that is, a sheaf on $(\mathrm{Sch})_S$ endowed with the Zariski topology.

More generally, by IV 4.5.13, $\underline{\mathrm{Hom}}_S(X, Y)$ is a sheaf on $(\mathrm{Sch})_S$ for every topology coarser than the canonical topology, for example for the (fpqc) topology.

If G, H are S -group preschemes, one deduces that the subfunctor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X, Y)$ is a sheaf for the (fpqc) topology (and hence a fortiori a local functor).

Lemma 1.7.2. ⁽⁹⁾ *Let F be a local S -functor.*

(i) Suppose that there exists an open covering (S_i) of S such that the restriction $F_i = F \times_S S_i$ of F to each S_i is representable by an S_i -prescheme X_i . Then F is representable by an S -prescheme X .

(ii) Suppose that F is an (fpqc) sheaf and that there exists a faithfully flat and quasi-compact morphism $S' \rightarrow S$ such that the restriction $F' = F \times_S S'$ of F is representable by an S' -prescheme X' . Then X' is endowed with a descent datum (cf. IV 2.1) with respect to $S' \rightarrow S$.

If, in addition, this descent datum is effective (which is the case if X' is affine over S'), then F is representable by an S -prescheme X .

⁽⁶⁾**N.D.E.** After 1.5, the original stated: “One concludes more generally that if G, H are locally diagonalizable, H being of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable”. This assertion has been included in the statement of 1.5, and its proof has been made explicit in 1.7.

⁽⁷⁾**N.D.E.** A statement 1.5.1 dealing with the functor $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H)$, considered in X 5.10 and 5.11, ought to be added. . .

⁽⁸⁾**N.D.E.** This paragraph has been added to make explicit the “local on S ” character of the representability of certain sheaves over S , used many times in what follows (and implicitly in the original).

⁽⁹⁾**N.D.E.** See also remark XI 3.4.

PROOF. (i) It follows from the hypothesis that $X_i \times_S S_j$ and $X_j \times_S S_i$ both represent the restriction of F to $S_{ij} = S_i \times_S S_j$, so, by Yoneda's lemma, there exists a unique isomorphism of S_{ij} -preschemes

$$c_{ji} : X_i \times_S S_j \xrightarrow{\sim} X_j \times_S S_i;$$

one then has isomorphisms of preschemes over $S_{ijk} = S_i \times_S S_j \times_S S_k$:

$$\begin{array}{ccc} X_i \times_S S_j \times_S S_k & \xrightarrow{c_{ji} \times \text{id}_{S_k}} & X_j \times_S S_i \times_S S_k = X_j \times_S S_k \times_S S_i \\ \parallel & & \downarrow c_{kj} \times \text{id}_{S_i} \\ X_i \times_S S_k \times_S S_j & \xrightarrow{c_{ki} \times \text{id}_{S_j}} & X_k \times_S S_i \times_S S_j = X_k \times_S S_j \times_S S_i \end{array}$$

and as all these objects represent the restriction of F to S_{ijk} , this diagram is commutative, i.e. the c_{ji} satisfy the usual cocycle relation $c_{kj} \circ c_{ji} = c_{ki}$.

It follows that the X_i glue into an S -prescheme X such that $X \times_S S_i = X_i$ for every i . Thus, for every Y over S_i , one has

$$(*) \quad F(Y) = F_i(Y) = \text{Hom}_{S_i}(Y, X \times_S S_i) = \text{Hom}_S(Y, X) = h_X(Y).$$

Then, for arbitrary $Y \rightarrow S$, the $Y_i = Y \times_S S_i$ form an open covering of Y ; put $Y_{ij} = Y_i \times_Y Y_j = Y \times_S S_{ij}$. Since F (resp. h_X) is a local functor by hypothesis (resp. since the Zariski topology is coarser than the canonical topology), $F(Y)$ and $h_X(Y)$ are both identified, in view of (*), with the kernel of the double arrow:

$$\begin{array}{ccc} \prod_i F(Y_i) & \rightrightarrows & \prod_{i,j} F(Y_{ij}) \\ \parallel & & \parallel \\ \prod_i h_X(Y_i) & \rightrightarrows & \prod_{i,j} h_X(Y_{ij}). \end{array}$$

This proves (i).

(ii) It follows from the hypothesis that $F'_1 = F' \times_{S'} S''_1$ (where $S''_1 = S'' = S' \times_S S'$ regarded as an S' -prescheme via the first projection) is represented by $X''_1 = X' \times_{S'} S''_1$; similarly, $F''_2 = F' \times_{S'} S''_2$ is represented by $X''_2 = X' \times_{S'} S''_2$. Now $F'_1 = F' \times_S S'' = F''_2$, so there exists a (unique) S'' -isomorphism $c : X''_1 \xrightarrow{\sim} X''_2$; then, denoting by q_i (resp. p_{ji}) the projection of $S''' = S' \times_S S' \times_S S'$ onto the i -th factor (resp. onto the factors i and j), by $X'''_i = X' \times_{S'} S'''_i$ (where $S'''_i = S'''$ regarded as an S' -prescheme via q_i), and by $p^*_{ji}(c) : X'''_i \xrightarrow{\sim} X'''_j$ the isomorphism of S''' -preschemes deduced from c by base change, one obtains a diagram of isomorphisms of S''' -preschemes:

$$\begin{array}{ccc} X'''_1 & \xrightarrow{p^*_{21}(c)} & X'''_2 \\ & \searrow p^*_{31}(c) & \downarrow p^*_{32}(c) \\ & & X'''_3 \end{array}$$

and as all these objects represent the restriction of F to S''' , this diagram is commutative, i.e. one has the usual cocycle relation $p^*_{32}(c) \circ p^*_{21}(c) = p^*_{31}(c)$, i.e. c is a descent datum on X' with respect to $S' \rightarrow S$ (cf. IV 2.1). Suppose, in addition, that this descent datum is effective, i.e. that there exists an S -prescheme X such that $X' \simeq X \times_S S'$ (by SGA 1, VIII 2.1, this is the case if X' is affine over $S'^{(10)}$). Then, for every $Y \rightarrow S'$, one has

$$(**) \quad F(Y) = F'(Y) = \text{Hom}_{S'}(Y, X \times_S S') = \text{Hom}_S(Y, X) = h_X(Y).$$

⁽¹⁰⁾ **N.D.E.** For another criterion of effectiveness, see below X 5.4–5.6.

Then, for arbitrary $Y \rightarrow S$, put $Y' = Y \times_S S'$ and $Y'' = Y' \times_Y Y' \simeq Y \times_S S''$. Then $Y' \rightarrow Y$ is, like $S' \rightarrow S$, faithfully flat and quasi-compact, hence an $\mathcal{M}$ -effective epimorphism (where $\mathcal{M}$ = the family of faithfully flat quasi-compact morphisms), i.e. the equivalence relation

$$Y' \times_Y Y' \rightrightarrows Y'$$

is $\mathcal{M}$ -effective and has quotient Y . Since F (resp. h_X) is an (fpqc) sheaf by hypothesis (resp. since the (fpqc) topology is coarser than the canonical topology), $F(Y)$ and $h_X(Y)$ are both identified, in view of (**), with the kernel of the double arrow:

$$\begin{array}{ccc} F(Y') & \rightrightarrows & F(Y' \times_Y Y') \\ \parallel & & \parallel \\ h_X(Y') & \rightrightarrows & h_X(Y' \times_Y Y'). \end{array}$$

This proves (ii). $\square$

Corollary 1.7.3. *Let F be an (fpqc) sheaf on $(\text{Sch})_S$. Suppose that there exists an open covering (S_i) of S and, for each i , a faithfully flat and quasi-compact morphism $S'_i \rightarrow S_i$ such that the restriction $F'_i = F \times_S S'_i$ is representable by an S'_i -prescheme X'_i affine over S'_i . Then F is representable by an S -prescheme X affine over S (such that $X \times_S S'_i = X'_i$ for every i).*

If, in addition, each $X'_i \rightarrow S'_i$ is a closed immersion (resp. a finite étale morphism), the same is true of $X \rightarrow S$.

The first assertion follows from 1.7.2. For the second, it suffices to check that each morphism $X \times_S S_i \rightarrow S_i$ is a closed immersion (resp. finite and étale), which follows from EGA IV₂, 2.7.1 (resp. and IV₄, 17.7.3).

Remark 1.7.4. Assertion 1.5 (b) follows, as announced, from 1.7.1 and 1.7.2 (i).

2. Schematic properties of diagonalizable groups

They are summarized in the

Proposition 2.1. *Let S be a nonempty prescheme, M an ordinary commutative group, $G = D(M_S)$ the diagonalizable S -group defined by M . One has the following:*

- a) G is faithfully flat over S , and affine over S (a fortiori quasi-compact over S).
- b) M of finite type $\iff G$ of finite type over $S \iff G$ of finite presentation over S .
- c) M finite $\iff G$ finite over $S \iff G$ of finite type over S and annihilated by an integer $n > 0$. Then $\deg(G/S) = \text{Card}(M)$.
- c') M a torsion group $\iff G$ integral over S .
- d) $M = 0 \iff G =$ the unit S -group.
- e) M of finite type, and the order of its torsion subgroup prime to the residue characteristics of $S \iff G$ smooth over S .

The verification of a) to d) is trivial and left to the reader. Let us prove e). If G is smooth over S , it is locally of finite presentation over S , hence of finite presentation over S since it is affine over S , so M is of finite type. One may therefore already suppose M of finite type, hence G of finite presentation over S . Then⁽¹¹⁾ G is smooth over S if and only if its geometric fibers are so, which reduces us to the case where S is the spectrum of an algebraically closed field k . Writing $M = T \times L$, with T the torsion subgroup and L free, $L \simeq \mathbf{Z}^r$, one has $D(M) = D(T) \times D(L)$, where $D(L) \simeq \mathbf{G}_m^r$ is smooth over k . Thus $G = D(M)$ is smooth over k if and only if $D(T)$ is so, which means, since $D(T)$ is finite over k of degree equal to the order n of T , that $D(T)(k)$ has n elements. Now T is isomorphic to a sum of groups

⁽¹¹⁾**N.D.E.** (since G is flat over S , by a))

$\mathbf{Z}/n_i\mathbf{Z}$, n being the product of the n_i , so $D(T)$ is a product of the $D(\mathbf{Z}/n_i\mathbf{Z}) = \mu_{n_i}$ (group schemes of n_i -th roots of unity), hence

$$\text{card}(D(T)(k)) = \prod_i \text{card } \mu_i(k)$$

where $\text{card } \mu_i(k) = (\text{number of } n_i\text{-th roots of unity in } k) \leq n_i$, equality being attained if and only if n_i is prime to the characteristic p of k . Thus one has $\text{card}(D(T)(k)) = n$ (where $n = \prod_i n_i$) if and only if all the n_i are prime to p , i.e. if and only if n is prime to p . Q.E.D.

3. Exactness properties of the functor D_S

Theorem 3.1. *Let S be a prescheme, and*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

an exact sequence of ordinary commutative groups. Consider the sequence of transposed homomorphisms:

$$0 \longrightarrow D_S(M'') \xrightarrow{v^t} D_S(M) \xrightarrow{u^t} D_S(M') \longrightarrow 0.$$

(i) v^t induces an isomorphism of $D_S(M'')$ with the kernel of u^t , and u^t is faithfully flat and quasi-compact.

(ii)⁽¹²⁾ $D_S(M')$ represents the (fpqc) quotient sheaf $D_S(M)/D_S(M'')$.

Denote by $\mathcal{M}$ the family of faithfully flat quasi-compact morphisms. First, (ii) follows from (i) (cf. IV, 4.6.5.1). Indeed, the equivalence relation in $D_S(M)$ defined by u^t is the same as that defined by the subgroup $\text{Ker}(u^t) = D_S(M'')$; as $u^t \in \mathcal{M}$, this equivalence relation is $\mathcal{M}$ -effective (cf. IV, 3.3.2.1), and therefore $D_S(M')$ represents the quotient sheaf for the (fpqc) topology (cf. IV, 4.6.5).

The first assertion of (i) is a trivial consequence of the definition of the functors $D_S(-)$; more generally, for every exact sequence

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

(without a zero on the left), one shall have an exact transposed sequence:

$$0 \longrightarrow D_S(M'') \longrightarrow D_S(M) \longrightarrow D_S(M').$$

(This holds more generally in the context of the beginning of no. 1.) On the other hand, as $D_S(M)$ and $D_S(M')$ are affine over S , u^t is necessarily an affine morphism, a fortiori quasi-compact (whatever the homomorphism $u : M' \longrightarrow M$). Thus the second assertion of (i) will follow from part a) in the

Corollary 3.2. *Let S be a nonempty prescheme, $u : M' \longrightarrow M$ a homomorphism of ordinary commutative groups, $u^t : G \longrightarrow G'$ the transposed homomorphism. Then:*

- a) *For u to be a monomorphism, it is necessary and sufficient that u^t be faithfully flat.*
- b) *For u to be an epimorphism, it is necessary and sufficient that u^t be a monomorphism (and then u^t is even a closed immersion).*

To prove a), one notes that if u is a monomorphism, then $\mathcal{O}_S(M)$ is a module over $\mathcal{O}_S(M')$ admitting a nonempty basis (namely the system of sections defined by any system of representatives of M modulo M'), a fortiori it is faithfully flat. Conversely, if this is so, then $u^t : \mathcal{O}_S(M') \longrightarrow \mathcal{O}_S(M)$ is injective, which (for $S \neq \emptyset$) implies that $u : M' \longrightarrow M$ is injective.

To prove b), one notes that if u is an epimorphism, then $\mathcal{O}_S(M') \longrightarrow \mathcal{O}_S(M)$ is surjective, so u^t is a closed immersion and a fortiori a monomorphism. Conversely, if this is so, then $\text{Ker } u^t = \text{the unit group}$; now putting $M'' = \text{Coker } u$, one has seen that $\text{Ker } u^t \simeq D_S(M'')$, so by 2.1 d) one has $M'' = 0$, hence u is an epimorphism.

⁽¹²⁾**N.D.E.** The following has been added, and the proof has been expanded accordingly.

One concludes from 3.1 in the usual way:

Corollary 3.3. *Let $M' \xrightarrow{u} M \xrightarrow{v} M''$ be an exact sequence of ordinary commutative groups; consider the transposed sequence*

$$G'' \xrightarrow{v^t} G \xrightarrow{u^t} G'.$$

Then v^t induces a faithfully flat and quasi-compact morphism from G'' to $\text{Ker } u^t$, and the latter is a diagonalizable group isomorphic to $D_S(v(M)) = D_S(\text{Coker } u)$.

Corollary 3.4. *Let S be a prescheme, $u : G \rightarrow H$ a homomorphism of locally diagonalizable S -group preschemes, with H of finite type over S . Put $G' = \text{Ker } u$. Then:*

- a) G' is locally diagonalizable; it is of finite type over S if G is so.*
- b) The quotient G/G' “exists”; more precisely, the equivalence relation defined by G' in G is $\mathcal{M}$ -effective (where $\mathcal{M}$ = the set of faithfully flat quasi-compact morphisms, cf. IV, 3.4). Moreover, G/G' is locally diagonalizable, of finite type over S .*
- c) The homomorphism $u : G \rightarrow H$ factors uniquely as*

$$G \xrightarrow{v} G/G' \xrightarrow{w} H,$$

where v is the canonical homomorphism (so v is faithfully flat and quasi-compact). Moreover, w is a closed immersion, and a fortiori a monomorphism.

Finally, the quotient $H' = H/\text{Im } w = \text{Coker } w = \text{Coker } u$ exists; more precisely, the equivalence relation defined by G/G' in H is $\mathcal{M}$ -effective, and H' is of finite type over S .

The first assertion of c) is a consequence of b), by definition of the quotient G/G' (cf. IV, 3.2.3).⁽¹³⁾ Let us show that the (fpqc) quotient sheaf $\widetilde{G}/\widetilde{G}'$ is representable. This is checked locally on S , as are all the other assertions; one may therefore suppose G and H diagonalizable, of the form $D_S(M)$ and $D_S(N)$.

As H is of finite type over S , N is of finite type by 2.1 b), so by 1.5, u is defined by a homomorphism $u' : N \rightarrow M$. Then, by 3.1 and 3.2, G' is isomorphic to $D_S(\text{Coker } u')$ and $\widetilde{G}/\widetilde{G}'$ is representable by $D_S(\text{Im } u')$; moreover, considering the exact sequence

$$0 \rightarrow \text{Ker } u' \rightarrow N \xrightarrow{w'} \text{Im } u' \rightarrow 0,$$

one obtains that w is a closed immersion and that the quotient $H' = H/\text{Im } w$ is $D_S(\text{Ker } u')$; this is of finite type over S since N , and hence $\text{Ker } u'$, is of finite type.

Remarks. The existence result for quotients 3.4 will be substantially generalized in no. 5.

On the other hand, one will note that in the present no. and the preceding one, the hypothesis $S \neq \emptyset$ entered only to ensure the validity of certain converses, allowing properties of the corresponding ordinary groups to be deduced from certain hypotheses on diagonalizable S -groups. The results in the “direct” sense hold without restriction on S , and the proofs given here apply in the general case.

Corollary 3.5. *Let G be a diagonalizable group prescheme over S , and let n be an integer $\neq 0$. Then the subgroup ${}_nG$ of G , kernel of the homomorphism $n \cdot \text{id}_G : G \rightarrow G$, is integral over S , and finite over S if G is of finite type over S .*

Indeed, if $G = D_S(M)$, then ${}_nG = D_S(M/nM)$ by 3.1, and one concludes by 2.1 b), c), c').

4. Torsors under a diagonalizable group

Let S be a prescheme, and $G = D_S(M)$ a diagonalizable group over S . We propose to determine the G -torsors (or principal homogeneous G -bundles) over S , in the sense of the “faithfully flat quasi-compact topology”, cf. Exp. IV, 5.1. Recall that a prescheme P

⁽¹³⁾**N.D.E.** The original has been expanded in what follows.

over S , with group of operators G , is said to be a *torsor* or *principal homogeneous* if every point of S admits an open neighborhood U and a faithfully flat and quasi-compact morphism $S' \rightarrow U$ such that $P' = P \times_S S'$ is a bundle with operators isomorphic to $G' = G \times_S S'$ (acting on itself by right translations). Since G is affine over S , it follows by SGA 1, VIII 5.6 that P is necessarily affine over S . Let us also note that since G is itself faithfully flat and quasi-compact over S , P is principal homogeneous under G if and only if it is “formally principal homogeneous” and is, in addition, faithfully flat and quasi-compact over S (cf. IV, 5.1.6).

Recall, on the other hand (Exp. I, 4.7.3), that giving an S -prescheme P affine over S with group of operators $G = D_S(M)$ amounts to giving a quasi-coherent commutative algebra graded of type M over S , i.e. a quasi-coherent algebra $\mathcal{A}$ over S endowed with a decomposition as a direct sum (as a module):

$$\mathcal{A} = \coprod_{m \in M} \mathcal{A}_m,$$

with

$$\mathcal{A}_m \cdot \mathcal{A}_{m'} \subseteq \mathcal{A}_{m+m'} \quad \text{for } m, m' \in M.$$

This being so, the answer to the problem posed above is given by the

Proposition 4.1. *For the prescheme P with group of operators $G = D_S(M)$, defined by the algebra $\mathcal{A}$ graded of type M , to be a principal homogeneous bundle under G , it is necessary and sufficient that $\mathcal{A}$ satisfy the following conditions:*

- a) *For every $m \in M$, $\mathcal{A}_m$ is an invertible module over S .*
- b) *For $m, m' \in M$, the homomorphism*

$$\mathcal{A}_m \otimes_{\mathcal{O}_S} \mathcal{A}_{m'} \rightarrow \mathcal{A}_{m+m'}$$

induced by multiplication in $\mathcal{A}$ is an isomorphism.

The necessity of the conditions is immediate by descent, since they are satisfied in the case where P is the trivial principal homogeneous bundle, i.e. $\mathcal{A} = \mathcal{O}_S(M)$. For sufficiency, one notes that a) already implies that P is faithfully flat over S ; it is in any case quasi-compact over S (being affine over S), so it remains to check that it is formally principal homogeneous under G , i.e. that the well-known homomorphism

$$P \times_S G \rightarrow P \times_S P$$

is an isomorphism. Now on the affine algebras this homomorphism is made explicit as the homomorphism

$$\mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}(M) = \mathcal{A} \otimes \mathcal{O}_S(M)$$

which in bidegree (m, n) (where $m, n \in M$) is given by

$$x_m \otimes y_n \mapsto x_m y_n \otimes e_n.$$

From the point of view of degrees, this homomorphism is compatible with the homomorphism $M \times M \rightarrow M \times M$ given by

$$(m, n) \mapsto (m + n, n),$$

which is an isomorphism. This shows that b) expresses precisely (independently of a)) that P is formally principal homogeneous, and establishes 4.1.

Let us also note that one obtains, by faithfully flat descent:

Corollary 4.2. *The conditions of 4.1 imply that the homomorphism*

$$\mathcal{O}_S \rightarrow \mathcal{A}_0$$

is an isomorphism.

If, for example, $M = \mathbf{Z}$, then under the conditions of 4.1 one sees that $\mathcal{A}$ is essentially known when one knows $\mathcal{A}_1 = \mathcal{L}$, namely

$$\mathcal{A} \simeq \coprod_{n \in \mathbf{Z}} \mathcal{L}^{\otimes n}$$

(isomorphism of graded algebras). One thus recovers the well-known result:

Corollary 4.3. *There is an equivalence between the category of principal homogeneous bundles P over S with group $\mathbf{G}_{m,S}$ and the category of invertible modules $\mathcal{L}$ over S (taking as morphisms to define both categories the isomorphisms for the structures involved). One obtains two functors quasi-inverse to one another by associating with every P the degree 1 component of its $\mathbf{Z}$ -graded affine algebra, and by associating with every $\mathcal{L}$ the spectrum of the $\mathbf{Z}$ -graded algebra $\coprod_{n \in \mathbf{Z}} \mathcal{L}^{\otimes n}$.*

In particular:

Corollary 4.4. *The group of classes of principal homogeneous bundles over S with group $\mathbf{G}_{m,S}$ is isomorphic to the group $\text{Pic}(S)$ of classes of invertible modules over S , i.e. to $H^1(S, \mathcal{O}_S^\times)$.*

In view of the fact that $\mathbf{G}_{m,S}$ is the scheme of automorphisms of the module $\mathcal{O}_S$, one sees that 4.4 is equivalent to the following statement, which is one of the variants of Hilbert's "Theorem 90".

Corollary 4.5. *Every principal homogeneous bundle over S with group $\mathbf{G}_m$ is locally trivial (in the sense of the Zariski topology).*

Remark 4.5.1. One will note that the preceding statement is no longer true in general for a group such as μ_n , or for a "twisted form" of $\mathbf{G}_m$; for example, the unique twisted form of $\mathbf{G}_m$ over the field $\mathbf{R}$ of real numbers gives a 1-cohomology group equal to $\mathbf{Z}/2\mathbf{Z}$.

⁽¹⁴⁾ Indeed, let S^1 be the kernel of the norm morphism $N : \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}} \rightarrow \mathbf{G}_{m,\mathbf{R}}$; it is a $\mathbf{C}/\mathbf{R}$ -twisted form of $\mathbf{G}_{m,\mathbf{R}}$. The equation $N(z) = -1$ in $\prod_{\mathbf{C}/\mathbf{R}} (\mathbf{G}_{m,\mathbf{C}})$ defines an S^1 -torsor X over $\text{Spec}(\mathbf{R})$, locally trivial for the étale topology, but nontrivial since $X(\mathbf{R}) = \emptyset$. Let us show that $H_{\text{ét}}^1(\mathbf{R}, S^1) \simeq \mathbf{Z}/2\mathbf{Z}$. One has an exact sequence of smooth commutative algebraic $\mathbf{R}$ -groups:

$$1 \rightarrow S^1 \rightarrow \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}} \rightarrow \mathbf{G}_{m,\mathbf{R}} \rightarrow 1$$

which gives rise to a long exact sequence of étale cohomology (or Galois cohomology):

$$0 \rightarrow S^1(\mathbf{R}) \rightarrow \mathbf{C}^\times \xrightarrow{N} \mathbf{R}^\times \rightarrow H_{\text{ét}}^1(\mathbf{R}, S^1) \rightarrow H_{\text{ét}}^1(\mathbf{R}, \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}}) \rightarrow \dots$$

Now (see, for example, XXIV, 8.4), $H_{\text{ét}}^1(\mathbf{R}, \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}}) \simeq H_{\text{ét}}^1(\mathbf{C}, \mathbf{G}_{m,\mathbf{C}})$, and the latter is zero by 4.5 (or, here, since $\mathbf{C}$ is algebraically closed). One therefore obtains an isomorphism $H_{\text{ét}}^1(\mathbf{R}, S^1) \simeq \mathbf{R}^\times / N(\mathbf{C}^\times) \simeq \{\pm 1\}$.

We shall need the following result in the next no. :

Proposition 4.6. *Under the conditions of 4.1, conditions a) and b) are equivalent to the following conditions:*

a') $\mathcal{O}_S \rightarrow \mathcal{A}_0$ is an isomorphism.

b') For every m in M (it suffices: in a system of generators of M), one has:

$$\mathcal{A}_m \cdot \mathcal{A}_{-m} = \mathcal{A}_0.$$

⁽¹⁴⁾**N.D.E.** The following has been added.

Necessity being evident, in view of 4.2⁽¹⁵⁾, we shall reduce to proving the

Corollary 4.7. *Let $A = \coprod_{n \in \mathbf{Z}} A_n$ be a $\mathbf{Z}$ -graded ring such that*

$$A_1 \cdot A_{-1} = A_0.$$

Then the A_n are invertible A_0 -modules, and, for $n, n' \in \mathbf{Z}$, the homomorphism

$$A_n \otimes_{A_0} A_{n'} \longrightarrow A_{n+n'}$$

induced by multiplication in A is an isomorphism.

By hypothesis, there exist $f_i \in A_1, g_i \in A_{-1}$ such that

$$(*) \quad \sum_i f_i g_i = 1.$$

As the conclusion to be established is local on $\text{Spec}(A_0)$,⁽¹⁶⁾ and as, by (*), $\text{Spec}(A_0)$ is covered by the affine open subsets $D(f_i g_i)$, one is reduced to the case where there exists an element $f \in A_1$ invertible in A . Then, for every $n \in \mathbf{Z}$, f^n is an element of A_n invertible in A , and therefore defines an isomorphism $h \mapsto f^n h$ from A_0 onto A_n . Moreover, this shows that one obtains an isomorphism $A_0[t, t^{-1}] \rightarrow A$ of graded A_0 -algebras by sending t to f , which completes the proof of 4.7.

Then, under the conditions of 4.6, 4.7 already implies that the $\mathcal{A}_m$ ($m \in M$) are invertible. To prove condition 4.1 b), one may therefore suppose that $\mathcal{A}_m$ and $\mathcal{A}_{m'}$ have bases f_m and $f_{m'}$, with inverses $f_m^{-1} \in \Gamma(\mathcal{A}_{-m})$ and $f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m'})$. Then multiplication by $f_m^{-1} f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m-m'})$ defines a homomorphism $\mathcal{A}_{m+m'} \rightarrow \mathcal{A}_0 \simeq \mathcal{O}_S$, transforming the image of $f_m \otimes f_{m'}$ into the section 1 of $\mathcal{O}_S$. In the diagram

$$\begin{array}{ccccc} & & w & & \\ & \searrow & & \nearrow & \\ \mathcal{A}_m \otimes \mathcal{A}_{m'} & \xrightarrow{u} & \mathcal{A}_{m+m'} & \xrightarrow{v} & \mathcal{A}_0 \simeq \mathcal{O}_S \end{array}$$

w and v are therefore epimorphisms of invertible sheaves, hence isomorphisms, so u is an isomorphism. Q.E.D.

5. Quotient of an affine scheme by a diagonalizable group acting freely

⁽¹⁷⁾ Denote by $\mathcal{M}$ the set of faithfully flat quasi-compact morphisms, and recall that we consider torsors in the sense of the (fpqc) topology.

Theorem 5.1. *Let S be a prescheme, M an ordinary commutative group, $G = D_S(M)$ the diagonalizable group over S that it defines, P an S -prescheme affine over S on which G acts freely on the right.*

Then the equivalence relation defined by G in P is $\mathcal{M}$ -effective (cf. IV, 3.4), i.e. the quotient $X = P/G$ exists and P is a torsor over X with group $G_X = D_X(M)$. Moreover, P/G is affine over S ; more precisely, if P is defined by the algebra $\mathcal{A}$ graded of type M , then P/G is isomorphic to $\text{Spec}(\mathcal{A}_0)$, where $\mathcal{A}_0 = \mathcal{A}^G$ is the degree 0 component of $\mathcal{A}$.

PROOF. Put $X = \text{Spec}(\mathcal{A}_0)$; then one has a natural morphism $P \rightarrow X$, deduced from $\mathcal{A}_0 \rightarrow \mathcal{A}$, which is invariant under the operations of G . In this way P becomes an X -prescheme, affine over X , with group of operators $G_X = D_X(M)$, and the hypothesis that G acts freely on P/S implies that G_X acts freely on P/X . Everything amounts to showing that P is a principal homogeneous bundle under G_X , using the fact that $\mathcal{B}_0 = \mathcal{O}_X$, where $\mathcal{B}$ is the algebra graded of type M over X that defines P/X . One may then suppose $X = S$ and S affine, so P is affine, given by a ring A graded of type M , whose homogeneous part of degree

⁽¹⁵⁾ **N.D.E.** 4.1 has been corrected to 4.2.

⁽¹⁶⁾ **N.D.E.** The following passage has been expanded.

⁽¹⁷⁾ **N.D.E.** The following sentence has been added.

m will be denoted by A_m , so that $S = \text{Spec}(A_0)$. In view of 4.6, it remains to check that one has:

$$(x) \quad A_m \cdot A_{-m} = A_0 \quad \text{for every } m \in M.$$

One also verifies by an immediate calculation that (x) is equivalent to saying that $P \times_S G \rightarrow P \times_S P$ is a *closed immersion*, and not merely a monomorphism (according to the hypothesis that G acts freely), i.e. that the homomorphism on affine rings

$$\theta : A \otimes_{A_0} A \rightarrow A(M)$$

is surjective.⁽¹⁸⁾ This gives 5.1 when one supposes that the equivalence relation defined by G in P is closed. We shall, however, show that this hypothesis is already a consequence of the fact that G acts freely (which is, moreover, implicitly contained in theorem 5.1, since $G \times_S P$ must be isomorphic to $P \times_X P$, which is closed in $P \times_S P$ since X is affine over S and hence separated over S).

Let $R = P \times_S G$. The hypothesis that G acts freely, i.e. that $R \rightarrow P \times_S P$ is a monomorphism, is expressed by saying that the diagonal morphism

$$R \rightarrow R \times_{(P \times_S P)} R = R'$$

is an isomorphism. One has $R = \text{Spec}(A(M))$ and

$$R' = \text{Spec}(A(M \times M)/K)$$

⁽¹⁹⁾ where K is the ideal generated by the elements of the form

$$x_m(e_{m,0} - e_{0,m}), \quad \text{with } m \in M, x_m \in A_m;$$

let $\phi : A(M \times M) \rightarrow A(M)$ be the surjective ring homomorphism defined by

$$xe_{m,n} \mapsto xe_{m+n} \quad (m, n \in M, x \in A)$$

(where the e_m , resp. $e_{m,n} = e_m \otimes e_n$, are the elements of the canonical basis of $A(M)$, resp. $A(M \times M)$). Then the diagonal morphism $R \rightarrow R'$ corresponds to the homomorphism

$$\bar{\phi} : A(M \times M)/K \rightarrow A(M)$$

obtained by passing to the quotient by K . Now the kernel of ϕ is the ideal K' generated by the

$$d_m = e_{m,0} - e_{0,m}.$$

One has $K \subseteq K'$, and the hypothesis that G acts freely on P , i.e. that $\bar{\phi}$ is an isomorphism, is equivalent to the equality $K' = K$, which is expressed by the relations

$$(xx) \quad d_m \in K = \sum_p A(M \times M) A_p d_p, \quad \text{for every } m \in M.$$

Using the natural trigrading of $A(M \times M)$, and the fact that the first degree of d_m is zero, this means that one can write d_m as a sum of elements of the form

$$fe_{r,s}(e_{p,0} - e_{0,p}) \quad \text{with } f \in A_{-p} \cdot A_p,$$

and, using the fact that the total degree of d_m is m , one may restrict to terms such that

$$r + s + p = m.$$

⁽¹⁸⁾**N.D.E.** Indeed, for every $b \in A$, $a_m \in A_m$, one has $\theta(b \otimes a_m) = ba_m \otimes e_m$, so surjectivity of θ is equivalent to (x).

⁽¹⁹⁾**N.D.E.** The ideal denoted by J in the original has been denoted by K , to distinguish it from the ideals J_p of A_0 which appear in (xxx). On the other hand, it has been made explicit below that relations (xx) are equivalent to the equality $K = K'$ (see what follows).

Thus one must have, for every $m \in M$, an expression:

$$(xxx) \quad \begin{cases} d_m = e_{m,0} - e_{0,m} = \sum_{r,s} \lambda_{r,s} (e_{m-s,s} - e_{r,m-r}) \\ \text{with } \lambda_{r,s} \in J_p = A_p \cdot A_{-p} \subseteq A_0, \quad p = m - (r + s). \end{cases}$$

One must deduce relation (x), i.e. the relations

$$(xxxx) \quad J_n = A_0 \quad \text{for every } n \in M.$$

Now for this it suffices to establish the same relation modulo every maximal ideal of A_0 . As the hypotheses made are invariant under such a reduction, one may already suppose that A_0 is a *field*.

Lemma 5.2. *Under the preceding conditions (with A_0 a field), if $M \neq 0$, there exists a $p \in M - \{0\}$ such that $J_p = A_0$.*

Indeed, if this were not so, the sum in the last member of (xxx) would be zero for every $m \in M$, which is absurd.

Corollary 5.3. *Under the preceding conditions, but without any longer supposing that A_0 is a field, there exists a finite number of elements $p_i \in M - \{0\}$ such that $\sum_i J_{p_i} = A_0$.*

Indeed, one applies result 5.2 to the situations deduced from A/A_0 by reduction modulo the maximal ideals of A_0 .⁽²⁰⁾

Corollary 5.4. *Suppose again that A_0 is a field. Then for every subgroup N of M such that $N \neq M$, there exists a $p \in M - N$ such that $J_p = A_0$.*

Indeed, let $M' = M/N$, and consider the ring A' graded of type M' whose underlying ring is A and whose grading is given by

$$A'_{m'} = \coprod_{m \in h^{-1}(m')} A_m,$$

where $h : M \rightarrow M' = M/N$ is the canonical homomorphism. Geometrically, this construction amounts to considering the subgroup $G' = D_S(M')$ of G , and the structure on P of a scheme with group of operators G' induced by the operations of G . It is then evident that G' acts freely on P' , i.e. the pair (M', A') satisfies the hypotheses of 5.3. One therefore obtains

$$1 = \sum_i f_i g_i \quad \text{with } f_i \in A'_{m'_i}, \quad g_i \in A'_{-m'_i} \quad \text{and } m'_i \in M' - \{0\},$$

whence immediately conclusion 5.4 by taking the components of the right-hand side in A_0 , and using the fact that A_0 is a field.

Let us now note that

$$J_p \cdot J_q \subseteq J_{p+q} \quad \text{and } J_p = J_{-p},$$

so, if N denotes the set of $m \in M$ such that $J_p = A_0$, one sees that N is a subgroup of M . Using 5.4 one sees that it equals M . This completes the proof of theorem 5.1. $\square$

As we pointed out in the course of the proof, theorem 5.1 implies:

Corollary 5.5. *Under the conditions of 5.1, the graph morphism*

$$P \times_S G \rightarrow P \times_S P$$

is a closed immersion.

One concludes immediately:

Corollary 5.6. *Let σ be a section of P over S . Then the morphism $g \mapsto \sigma \cdot g$ from G to P defined by σ is a closed immersion.*

⁽²⁰⁾**N.D.E.** It follows from 5.2 that $\sum_{p \neq 0} J_p = A_0$, so 1 is written as a finite sum $\sum_i x_i$, with $x_i \in J_{p_i}$.

Corollary 5.7. *Let G, H be two S -groups, with G diagonalizable, H affine over S , and let $u : G \rightarrow H$ be a homomorphism of S -groups which is a monomorphism. Then u is a closed immersion, $H/G = X$ exists and H is a principal homogeneous bundle over X with group G_X ; finally, X is affine over S .*

Corollary 5.8. *Under the conditions of 5.1, if P is of finite type (resp. of finite presentation) over S , the same is true of $X = P/G$.*

Indeed, it follows from the hypothesis that the fibers of G_X are of finite type, so G_X is of finite presentation over X by 2.1 b); hence P , being a torsor under G_X , is of finite presentation over X .⁽²¹⁾ Since it is also faithfully flat over X , our conclusion then follows from Exp. V, Prop. 9.1.

6. Essentially free morphisms, and representability of certain functors of the form $\prod_{Y/S} Z/Y$

(22)

Definition 6.1. Let $f : X \rightarrow S$ be a morphism of preschemes. One says that f is *essentially free*, or again that X is *essentially free over S* , if one can find a covering of S by affine open subsets S_i , for every i an S_i -prescheme S'_i affine and faithfully flat over S_i , and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine open subsets X'_{ij} , such that for every (i, j) the ring of X'_{ij} is a free module over the ring of S'_i .⁽²³⁾

Proposition 6.2. *a) If X is essentially free over S , it is flat over S , the converse being true if S is artinian.*

b) If S is the spectrum of a field, every S -prescheme is essentially free over S .

c) If X is essentially free over S , and if $S' \rightarrow S$ is a base change morphism, $X' = X \times_S S'$ is essentially free over S' . The converse is true if $S' \rightarrow S$ is faithfully flat and quasi-compact.

The proof is immediate, using for the converse in a) the fact that a flat module over an artinian local ring is free.⁽²⁴⁾

Proposition 6.3. *Let H be a diagonalizable S -group prescheme (more generally, one which becomes diagonalizable by a suitable faithfully flat quasi-compact extension of every affine open subset of S , i.e. H is “of multiplicative type”, cf. IX 1.1). Then H is essentially free over S .*

Indeed, if H is diagonalizable, it is affine over S and defined by an algebra which is a free $\mathcal{O}_S$ -module.

The introduction of definition 6.1 is justified by the

⁽²¹⁾**N.D.E.** Cf. EGA II, 2.7.1 (vi).

⁽²²⁾The present no. is independent of the theory of diagonalizable groups; its natural place would be in VI_B.

⁽²³⁾**N.D.E.** One may replace “free” by “projective”, cf. 6.8 below. On the other hand, this notion should be compared with that of a flat and pure S -prescheme, introduced and developed in [RG71]; see in particular loc. cit., Part 1, 3.3.12 and Part 2, 3.1.4.1.

⁽²⁴⁾**N.D.E.** Indeed, let $(A, \mathfrak{m})$ be an artinian local ring, k its residue field, M an arbitrary A -module, $(x_i)_{i \in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k . Let F be the free A -module with basis $(e_i)_{i \in I}$, and $\phi : F \rightarrow M$ the A -morphism defined by $\phi(e_i) = x_i$. Then $Q = \text{Coker } \phi$ satisfies $Q = \mathfrak{m}Q$, whence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose in addition that M is flat over A ; then $K = \text{Ker } \phi$ satisfies $K \otimes_A k = 0$, i.e. $K = \mathfrak{m}K$, whence $K = 0$.

Theorem 6.4. *Let S be a prescheme, Z an essentially free S -prescheme, Y a closed subprescheme of Z . Consider the following functor⁽²⁵⁾*

$$F = \prod_{Z/S} Y/Z : (\text{Sch})_{/S}^{\circ} \longrightarrow (\text{Ens}), \quad F(S') = \Gamma(Y_{S'}/Z_{S'}) = \begin{cases} \emptyset & \text{if } Z_{S'} \neq Y_{S'}; \\ \{\text{id}_{Z_{S'}}\} & \text{if } Z_{S'} = Y_{S'}. \end{cases}$$

This functor is representable by a closed subprescheme of S .⁽²⁶⁾

⁽²⁷⁾ Let us first note that F is a sheaf for the (fpqc) topology: since $F(S') = \emptyset$ or $\{\text{pt}\}$ for every S' , this amounts to checking that if (S_i) is an open covering of S (resp. $S' \rightarrow S$ a faithfully flat and quasi-compact morphism), and if each $Y_{S_i} \rightarrow Z_{S_i}$ (resp. if $Y_{S'} \rightarrow Z_{S'}$) is an isomorphism, the same is true of $Y \rightarrow Z$; now this is clear (resp. follows from SGA 1, VIII 5.4 or EGA IV₂, 2.7.1). Moreover, by SGA 1, VIII 2.1 and 5.5, faithfully flat and quasi-compact morphisms are of effective descent for the fibered category of closed immersion arrows. This allows us to restrict, with the notation of 6.1, to the case where $S = S'_i$.

Let then (Z_j) be a covering of Z by affine open subsets such that $\mathcal{O}(Z_j)$ is a free module over $A = \mathcal{O}(S)$, and let $Y_j = Y \cap Z_j$ and $F_j : (\text{Sch})_{/S}^{\circ} \rightarrow (\text{Ens})$ be the functor defined in terms of (Z_j, Y_j) as F is in terms of (Z, Y) . It is a subfunctor of the final functor, and one evidently has $F = \bigcap_j F_j$, which reduces us to proving that each F_j is representable by a closed subscheme T_j of S (for then F will be representable by the closed subscheme T which is the intersection of the T_j). One may therefore suppose Z also affine, $Z = \text{Spec}(B)$, where B is a free A -module. Let J be a subset of B defining the subscheme Y of Z , and let K be the ideal in A generated by the $u_i(J) \subseteq A$, where the $u_i : B \rightarrow A$ are the coordinate forms with respect to the chosen basis. One verifies immediately that $T = \mathcal{V}(K) = \text{Spec}(A/K)$ satisfies the desired condition, which completes the proof.

Examples 6.5. Let us give important examples of functors that reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.4 and for which it will be useful in what follows to have criteria of representability. Denote by S a prescheme, by X, Y, Z , etc., preschemes over S .

a) Let us give ourselves an S -morphism

$$(x) \quad q : X \longrightarrow \underline{\text{Hom}}_S(Y, Z),$$

(“ X acts on Y , with values in Z ”), i.e. a morphism

$$(xx) \quad r : X \times_S Y \longrightarrow Z.$$

Consider a subprescheme Z' of Z , whence a monomorphism

$$\underline{\text{Hom}}_S(Y, Z') \longrightarrow \underline{\text{Hom}}_S(Y, Z)$$

which makes the first functor a subfunctor of the second; let X' be the inverse image of this subfunctor by (x). It is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \rightarrow Z_T$ factors through Z'_T . This functor X' may be described as follows: put $P = X \times_S Y$, let P' be the inverse image of Z' by $r : P \rightarrow Z$; then one has an evident isomorphism

$$(xxx) \quad X' \simeq \prod_{P/X} P'/P.$$

One therefore obtains: *if Y is essentially free over S and Z' closed in Z , the subfunctor X' of X is representable by a closed subprescheme of X .*

b) Let us give ourselves two ways of making X act on Y with values in Z , i.e. two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\text{Hom}}_S(Y, Z),$$

⁽²⁵⁾**N.D.E.** Cf. II.1, where this functor is denoted by $\prod_{Z/S} Y$.

⁽²⁶⁾**N.D.E.** Z has been corrected to S .

⁽²⁷⁾**N.D.E.** The original has been expanded in what follows; see also 1.7.3.

and put $X' = \text{Ker}(q_1, q_2)$: it is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms $q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$ are equal. Now giving q_1, q_2 is equivalent to giving a morphism

$$q : X \longrightarrow \underline{\text{Hom}}_S(Y, Z \times_S Z),$$

or again a morphism $r : X \times_S Y \longrightarrow Z \times_S Z$; put then $U = Z \times_S Z$, let U' be the diagonal subprescheme of $Z \times_S Z$; then X' is precisely the inverse image of the subfunctor $\underline{\text{Hom}}_S(Y, U') \longrightarrow \underline{\text{Hom}}_S(Y, U)$ by q , and so can be put in the form (xxx), with $P = X \times_S Y$, and $P' =$ inverse image of the diagonal by r , i.e. kernel of $X \times_S Y \xrightarrow[r_2]{r_1} Z$. One is therefore under the conditions of (a).

One thus sees that: *if Y is essentially free over S and Z separated over S , then the subfunctor X' of X is representable by a closed subprescheme of X .*

c) Let us give ourselves a morphism

$$q : X \longrightarrow \underline{\text{Hom}}_S(Y, Y),$$

i.e. “ X acts on Y ”. Let X' be the “kernel” of this morphism, i.e. the subfunctor X' of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \longrightarrow Y_T$ is the identity. This functor is covered by b), as one sees by introducing a second homomorphism

$$q' : X \longrightarrow \underline{\text{Hom}}_S(Y, Y)$$

“by making X act trivially on Y ”. Thus: *if Y is essentially free over S and separated over S , the kernel subfunctor of q is representable by a closed subprescheme of X .*

d) Under the conditions of c), consider the subfunctor Y' of Y “of invariants under X ”, so $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $\bar{q}(y) : X_T \longrightarrow Y_T$ is “the constant T -morphism with value y ”. Introducing q' as in c), and the homomorphisms corresponding to q and q' :

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\text{Hom}}_S(X, Y),$$

one sees that Y' is precisely $\text{Ker}(\bar{q}, \bar{q}')$, and is therefore again covered by b) (with the roles of X, Y reversed and $Z = Y$).

Consequently, *if X is essentially free over S , Y separated over S , then the subfunctor Y' of Y of invariants under X is representable by a closed subprescheme of Y .*

e) Constructions of the type made explicit in the preceding examples are especially frequent in group theory. Thus, when G is an S -group prescheme acting on the S -prescheme X :

$$q : G \longrightarrow \underline{\text{Aut}}_S(X),$$

the kernel of q (“the subgroup of G acting trivially”) is a closed subscheme of G provided that X is essentially free and separated over S (example c)), and the subobject X^G of invariants is a closed subprescheme of X , provided that G is essentially free over S and X separated over $S^{(28)}$ (example d)).

Let Y, Z be subpreschemes of X ; consider the subfunctor $\underline{\text{Transp}}_G(Y, Z)$ of G (“transporter of Y into Z ”) whose points with values in a T over S are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subseteq Z_T$, i.e. induces a morphism $Y_T \longrightarrow Z_T$ factoring as $Y_T \longrightarrow Z_T$. Thus: *if Y is essentially free over S and Z closed in X , then $\underline{\text{Transp}}_G(Y, Z)$ is a closed subprescheme of G (example a)).*

One may also consider the *strict transporter of Y into Z* ,⁽²⁹⁾ whose points with values in a T over S are the $g \in G(T)$ such that $g(Y_T) = Z_T$, which is precisely $\underline{\text{Transp}}_G(Y, Z) \cap \sigma(\underline{\text{Transp}}_G(Z, Y))$, where σ is the symmetry of G . Consequently, *if Y and Z are essentially free over S and closed in X , the strict transporter of Y into Z is a closed subprescheme of G .*

An important case is that where $X = G$, G acting on itself by inner automorphisms. If H is a subprescheme of G , the strict transporter of H into H is also called the *normalizer of H*

⁽²⁸⁾N.D.E. S_X has been corrected to S .

⁽²⁹⁾N.D.E. Denoted by $\underline{\text{Transp}}_G^{\text{str}}(Y, Z)$.

in G , and denoted by $\underline{\text{Norm}}_G H$. Thus: if H is a closed subgroup prescheme of G , essentially free over S , then $\underline{\text{Norm}}_G H$ is representable by a closed subgroup prescheme of G .

Finally, let Z be a subprescheme of G ; then its *centralizer* $\underline{\text{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure of d), when one considers that “ Z acts on G ” by the operations induced by those of G ; thus if Z is essentially free over S and G separated over S , $\underline{\text{Centr}}_G(Z)$ is a closed subgroup prescheme of G . In particular, if G is essentially free and separated over S , then the center C of G , which is precisely $\underline{\text{Centr}}_G(G)$, is a closed subgroup prescheme of G .

When S is the spectrum of a field, 6.3 b) shows that in examples a) to e) above the “essentially free” conditions are automatically satisfied; only separation conditions remain. Recalling that a group prescheme over a field is necessarily separated, one finds, for example:

Corollary 6.7. ⁽³⁰⁾ *Let G be a group prescheme over a field k . Then:*

- *For every subprescheme Z of G , the centralizer of Z in G is a closed subgroup prescheme of G ; this is in particular the case for the center $\underline{\text{Centr}}_G(G)$ of G .*
- *More generally, if $u, v : X \rightarrow G$ are morphisms of preschemes, $\underline{\text{Transp}}_G(u, v)$ is representable by a closed subprescheme of G .*
- *For two subpreschemes Y, Z of G , with Z closed, $\underline{\text{Transp}}_G(Y, Z)$ is a closed subprescheme of G . If Y is also closed, one has the same conclusion for $\underline{\text{Transp}}_G^{\text{str}}(Y, Z)$.*
- *For every subgroup prescheme⁽³¹⁾ H of G , the normalizer $\underline{\text{Norm}}_G(H)$ is a closed subgroup scheme of G .*

Remark 6.8. ⁽³²⁾ Let A be a commutative ring, M an A -module, $M^\vee = \text{Hom}_A(M, A)$; endow the A -module $\text{End}_A(M)$ with the topology of discrete pointwise convergence, i.e. a neighborhood basis of 0 consists of the following A -submodules, where $n \in \mathbf{N}$ and $m_1, \dots, m_n \in M$:

$$K(m_1, \dots, m_n) = \{u \in \text{End}_A(M) \mid u(m_i) = 0 \text{ for } i = 1, \dots, n\}.$$

One says that M is a *quasi-free A -module* if the image of the canonical morphism $\Theta : M \otimes_A M^\vee \rightarrow \text{End}_A(M)$ contains id_M in its closure, i.e. if the following condition is satisfied:

$$(*) \quad \text{for every } m_1, \dots, m_n \in M, \text{ there exist } x_1, \dots, x_r \in M \text{ and } f_1, \dots, f_r \in M^\vee \\ \text{such that } m_i = \Theta \left(\sum_{s=1}^r x_s \otimes f_s \right) (m_i) = \sum_{s=1}^r f_s(m_i) x_s \text{ for } i = 1, \dots, n.$$

(In this case, $\text{Im } \Theta$ is dense in $\text{End}_A(M)$ since for every $u \in \text{End}_A(M)$ one has $u(m_i) = \sum_{s=1}^r f_s(m_i) u(x_s) = \Theta(\sum_{s=1}^r u(x_s) \otimes f_s)(m_i)$.)

Let us first note that this property is stable under base change. Indeed, let $\phi : A \rightarrow A'$ be a ring morphism, $M' = M \otimes_A A'$, and $m'_1, \dots, m'_n \in M'$; then $m'_i = \sum_j m_{ij} \otimes b_{ij}$ ($m_{ij} \in M$, $b_{ij} \in A'$); by hypothesis, there exist $x_1, \dots, x_r \in M$ and $f_1, \dots, f_r \in \text{Hom}_A(M, A)$ such that $m_{ij} = \sum_s x_s f_s(m_{ij})$ for every i, j . Denote by $\phi \circ f_s$ the image of f_s in $\text{Hom}_A(M, A') = \text{Hom}_{A'}(M, A')$; then for every $i = 1, \dots, n$ one has:

$$\left(\sum_s x_s \otimes \phi \circ f_s \right) (m'_i) = \sum_{s,j} x_s \otimes \phi(f_s(m_{ij})) b_{ij} = \sum_j \left(\sum_s x_s f_s(m_{ij}) \right) \otimes b_{ij} = m'_i,$$

which proves that M' is quasi-free over A' . Let us also note that every projective A -module P is quasi-free (there exist A -morphisms $A^{(I)} \xrightarrow{\pi} P \xrightarrow{\tau} A^{(I)}$ such that $\pi \circ \tau = \text{id}_P$; denote by (e_i) the canonical basis of $A^{(I)}$ and by f_i the linear form $e_i^* \circ \tau$ on P ; if $m_1, \dots, m_n \in P$, there exists a finite subset J of I such that $m_k = \sum_{i \in J} f_i(m_k) \pi(e_i)$ for $k = 1, \dots, n$).

⁽³⁰⁾ **N.D.E.** The original numbering has been preserved: there is no 6.6.

⁽³¹⁾ **N.D.E.** Indeed, over a field k , every subgroup prescheme of G is closed, cf. VI_A, 0.6.1.

⁽³²⁾ **N.D.E.** The original has been expanded in what follows; in particular lemma 6.8.1 has been added.

Then theorem 6.4 remains valid on replacing, in the statement of definition 6.1, the word “free” by “projective” or, more generally, by “quasi-free”. Indeed, proceeding as in the proof of 6.4, one is reduced to proving the

Lemma 6.8.1. *Let M be a quasi-free A -module, N a submodule, F the covariant functor (A -algebras) $\longrightarrow$ (Ens) such that $F(B) = \{\text{pt}\}$ if $M_B = (M/N)_B$, and $F(B) = \emptyset$ otherwise. Then there exists an ideal K of A such that $F(B) = \{\text{pt}\}$ if and only if the morphism $A \longrightarrow B$ factors through A/K , i.e. one has an isomorphism functorial in B :*

$$F(B) \simeq \text{Hom}_{A\text{-alg.}}(A/K, B).$$

PROOF. Let (n_j) be a system of generators of N , and let F_j be the subfunctor of the final functor $\underline{e}$ ($\underline{e}(B) = \{\text{pt}\}$ for every B) corresponding to the submodule An_j . One has $F(B) = \{\text{pt}\}$ if and only if the image of each n_j in M_B is zero, so F is the intersection of the functors F_j . This reduces us to the case where N is generated by an element n .

Let $\phi : A \longrightarrow B$ be an A -algebra; if the image $n \otimes 1$ of n in M_B is zero, then for every $f \in M^\vee = \text{Hom}_A(M, A)$ one has $0 = f(n) \otimes 1 = \phi(f(n))$. On the other hand, since M is quasi-free, there exist $x_1, \dots, x_r \in M$ and $f_1, \dots, f_r \in M^\vee$ such that $n = \sum_s f_s(n)x_s$, whence $n \otimes 1 = \sum_s x_s \otimes \phi(f_s(n))$. It follows that $n \otimes 1 = 0$ if and only if ϕ factors through A/K , where K is the ideal generated by the $f_s(n)$ for $s = 1, \dots, r$. This proves the lemma. $\square$

7. Appendix: On monomorphisms of group preschemes

The result proved in the present no. is unnecessary for the rest of the seminar, except X 8.8 and XV and XVI. It relies essentially on the existence theorem for quotient groups of Exposé VI_A.⁽³³⁾

Corollary 5.7 leads one to ask under what conditions one can assert that a monomorphism $u : G \longrightarrow H$ of S -groups is an immersion, or even a closed immersion.

We have seen in VI_B 1.4.2 that this is so if S is the spectrum of a field, provided G is of finite type over k and H locally of finite type over k . One easily concludes that the same result remains valid if one merely supposes S artinian.⁽³⁴⁾

On the other hand, it is easy to give examples of bijective monomorphisms which are not immersions, S being, for example, the affine line over a field or the spectrum of a discrete valuation ring. One will take, for example, $H = (\mathbf{Z}/2\mathbf{Z})_S$, $G = G_1 \times_S G_2$, where G_1 is the open subgroup of H complementary to the closed point x distinct from 0 of the fiber $H_s \simeq \mathbf{Z}/2\mathbf{Z}$ (where s denotes a fixed closed point of S), and G_2 the closed subscheme of H which is the sum of the subscheme reduced to the unit section and of the reduced closed subscheme defined by the closed point x . One easily verifies that G_2 is indeed stable under multiplication in H , and hence is a group scheme. The immersions $G_i \longrightarrow H$ ($i = 1, 2$) then define a homomorphism of S -groups $G = G_1 \times G_2 \longrightarrow H$, which is evidently a bijective monomorphism (and in addition a local immersion), but is not an immersion. (One verifies that G and H are reduced, G having three disjoint irreducible components, whereas H has only two irreducible components.) One will note that $G_1 \longrightarrow H$ also gives an example of an open subgroup G of H which is not closed (contrary to what occurs for algebraic groups over a field). The theory of degeneration of elliptic curves supplies other examples of this latter phenomenon, with H in addition smooth over S of relative dimension 1, and G with connected fibers. It is possible,⁽³⁵⁾ on the other hand, that as soon as one supposes G flat over S , and (say) G and H of finite presentation over S , a monomorphism $u : G \longrightarrow H$ of

⁽³³⁾N.D.E. Cf. Exp. VI_A, Theorems 3.2 and 3.3.2.

⁽³⁴⁾N.D.E. Take account of the additions made in VI_B...

⁽³⁵⁾In fact, M. Raynaud has constructed a counterexample, with G smooth with connected fibers, cf. XVI 1.1 c). If one does not suppose G to have connected fibers, one may take $S = \text{Spec}(\mathbf{Z}_2)$, $G = (\mathbf{Z}/2\mathbf{Z})_S$ with the non-neutral closed point removed, $H = (\mu_2)_S$.

S -groups is automatically an immersion. We shall prove a result of this kind, under additional hypotheses.

Let us first note that one may suppose S affine, and (thanks to the finite presentation hypothesis on G and H , which allows one to reduce to the case of the spectrum of a ring of finite type over $\mathbf{Z}$ ⁽³⁶⁾) S noetherian. Then G and H are noetherian. Saying that $u : G \rightarrow H$ is a closed immersion (resp. an immersion) then amounts to saying that u is a monomorphism (which is true by hypothesis) and that u is proper (resp. and that u is proper at every point of $u(G)$).⁽³⁷⁾ The valuative criterion of properness assures us that it suffices to check that for every base change $S' \rightarrow S$, with S' the spectrum of a discrete valuation ring, complete if one wishes, the morphism $u' : G' \rightarrow H'$ has the same properness property. (The case of local properness was forgotten in EGA II 7.3, and will appear as an afterthought in EGA IV.⁽³⁸⁾) This therefore reduces us to the case where S is itself the spectrum of a complete discrete valuation ring, — provided that the additional hypotheses on G, H, u which we shall be led to formulate are stable under base change.

Let then s (resp. s') be the closed point (resp. the generic point) of S . Then the homomorphisms induced on the fibers

$$u_s : G_s \rightarrow H_s \quad \text{and} \quad u_{s'} : G_{s'} \rightarrow H_{s'}$$

are closed immersions, since they are monomorphisms of group schemes of finite type over fields (VI_B 1.4.2). We may therefore identify $G_{s'}$ with a closed subscheme of $H_{s'}$. Now one has the following result:

Lemma 7.1. *Let S be the spectrum of a discrete valuation ring, s' its generic point, H an S -prescheme, L' a closed subprescheme of the generic fiber $H_{s'}$, so that L' is also a subprescheme of H .*

Then the schematic closure $\overline{L'}$ in H (i.e. the smallest closed subprescheme of H containing L' , cf. EGA I 9.5) exists and is also the unique closed subprescheme of H , flat over S , whose generic fiber is L' . Moreover, formation of L is functorial with respect to a variable pair (H, L') and commutes with formation of cartesian products over S .

In particular, if H is an S -group and L' a subgroup of $H_{s'}$, then L is a subgroup of H .

The proof is immediate and left to the reader.⁽³⁹⁾ Applying this to the situation $H, G_{s'}$, one sees that the monomorphism $u : G \rightarrow H$ factors as $G \rightarrow L \rightarrow H$, where L is a subgroup of H which is a closed subprescheme, flat over S , and where $G \rightarrow L$ induces an isomorphism on the generic fibers. Then $u : G \rightarrow H$ is an immersion (resp. a closed immersion) if and only if $u' : G \rightarrow L$ is so. This therefore reduces us to the case where H is flat over S and $u_{s'} : G_{s'} \rightarrow H_{s'}$ an isomorphism (provided that the additional hypotheses which we shall have to formulate on G, H, u are respected when one replaces H by a closed subgroup prescheme). As H is then the schematic closure of $H_{s'}$, if u is an immersion, $u(G)$ will be a subprescheme⁽⁴⁰⁾ of H containing $H_{s'}$, so its schematic closure will also be H , and consequently $u(G)$ will be an open subprescheme of H . Consequently, we shall in fact have to prove that u is an open immersion (resp. an isomorphism if we wished to establish that u is a closed immersion). As G and H are flat over S , it amounts to the same thing to say that the morphisms induced on the fibers are open immersions, resp. isomorphisms (cf. SGA 1, I 5.7), and since this is already so for $u_{s'}$, one is reduced to proving that u_s is an open immersion, resp. an isomorphism. Let us first note that, since G and H are flat over S , the dimension of their fibers remains constant (VI_B 4.3). As G and H have the same generic fiber, it follows that this dimension is the same for G and H , so $u_s : G_s \rightarrow H_s$ is a monomorphism

⁽³⁶⁾N.D.E. Cf. EGA IV₃, §8, and Exp. VI_B, §10.

⁽³⁷⁾N.D.E. Cf. EGA IV₃, 8.11.5 and 15.7.1.

⁽³⁸⁾Cf. EGA IV₃, 15.7.

⁽³⁹⁾Cf. EGA IV₂, 2.8.

⁽⁴⁰⁾N.D.E. The word “closed” has been deleted.

of algebraic groups of the same dimension. One easily concludes that G_s is open in H_s , and in fact is set-theoretically a union of connected components of H_s (one is reduced to the case where the base field is algebraically closed and G and H reduced, hence smooth over k , where this is immediate. . .). Thus $H_s - G_s$ is closed in H_s , hence in H , so its complement H' in H is open, and is evidently an open subset stable under the group law of H . Thus, replacing H by H' if necessary, to prove that u is an immersion one may reduce (with the usual proviso) to the case where, in addition to the preceding hypotheses, one supposes u bijective, i.e. $u_s : G_s \rightarrow H_s$ bijective. One is therefore in any case reduced to proving that u , or again u_s , is an isomorphism, with the bijectivity hypothesis when appropriate.

Let us therefore first suppose that u is bijective. If H_s is reduced, one may evidently conclude that u_s is an isomorphism, for G_s is identified with a closed subscheme of H_s with the same underlying set. In particular, if $k = \kappa(s)$ is of characteristic zero, every algebraic group over k is reduced by Cartier (cf. VI_B, 1.6.1, or VII_B, 3.3.1, or EGA IV₄, 16.12.2 and 17.12.5), and one has thus obtained:

Proposition 7.2. *Let $u : G \rightarrow H$ be a homomorphism of group preschemes of finite presentation over S . Suppose that u is a monomorphism, G flat over S , and the residue fields of S of characteristic zero. Then u is an immersion.*

When $\kappa(s)$ is of characteristic $p > 0$, we shall restrict to the case where G is commutative. Then (with the reductions made) H is also commutative, since it is the schematic closure of $H_{s'}$, which is isomorphic to $G_{s'}$ and hence commutative. For every integer $n \geq 0$, we put $S_n = \text{Spec}(V/\mathfrak{m}^{n+1})$ (where V is the valuation ring defining S and $\mathfrak{m}$ its maximal ideal), $G_n = G \times_S S_n$, $H_n = H \times_S S_n$. For every integer $m > 0$, we also introduce the subgroups ${}_m G$ and ${}_m H$ of G and H , kernels of the m -th power. One defines similarly ${}_m(G_n) = ({}_m G)_n$, which one will denote simply by ${}_m G_n$, and similarly for H .

By VI_A 3.2, one may form the quotients $Q_n = H_n/G_n$; then Q_n is a commutative group scheme over S_n , flat over S_n , and $H_n \rightarrow Q_n$ is a faithfully flat morphism with kernel G_n . As formation of quotients commutes with extension of the base,⁽⁴¹⁾ one has

$$Q_n \simeq Q_m \times_{S_m} S_n \quad \text{for } m \geq n,$$

in particular the fiber $Q_n \times_{S_n} S_0$ is precisely $Q_0 = H_0/G_0$. As G_0 has the same underlying set as H_0 , Q_0 is set-theoretically reduced to a single point: it is a purely infinitesimal group. Consequently, each Q_n is finite and flat over S_n . Thus Q_n is defined by an algebra C_n over $V_n = V/\mathfrak{m}^{n+1}$ which is a free module of finite type over this ring, and, for $m \geq n$, one has $C_n = C_m \otimes_{V_m} V_n$, an isomorphism also respecting the diagonal map. One therefore obtains a free module of finite type $C = \varprojlim C_n$ over $V = \varprojlim V_n$, and the diagonal maps of the C_n define a diagonal map of C , so that $Q = \text{Spec}(C)$ becomes a finite and flat group scheme over S such that

$$Q \times_S S_n = Q_n$$

for every n .

Lemma 7.3. ⁽⁴²⁾ *Let K be a field, Q a finite group scheme over K , of degree n . Then the n -th power morphism in Q is zero.*

Cf. VII_A 8.5.

Remark 7.3.1. Statement 7.3 still makes sense for a group scheme Q/S finite and locally free over S , S being any base prescheme. It would be interesting to find a proof in this general case.

⁽⁴¹⁾**N.D.E.** Bring this out in Exp. V and VI_A...

⁽⁴²⁾**N.D.E.** The hypothesis that Q is commutative has been deleted here, and 7.3.1 has been modified accordingly. Note that if Q is not commutative, the n -th power morphism is not in general a group homomorphism.

One will note that 7.3 (i.e. VII_A, 8.5) proves in any case that the contemplated statement is true if S is reduced, as one sees by applying 7.3 to the fibers of Q at the maximal points (i.e. generic points of the irreducible components) of S .⁽⁴³⁾

In particular, under the conditions of the preceding proof, where S is the spectrum of a discrete valuation ring and Q is commutative, one finds that $n \cdot \text{id}_Q = 0$. Moreover, here n is a power p^{ν_0} of the residue characteristic, and one finds the

Corollary 7.4. *Q and the Q_n being as above, and their common degree being p^{ν_0} , one shall have $p^{\nu_0} \cdot \text{id}_Q = 0$ and consequently $p^{\nu_0} \cdot \text{id}_{Q_n} = 0$ for every n .*

Corollary 7.5. *Suppose in addition G_0 smooth over k , and the homomorphism $p \cdot \text{id}_{G_0}$ flat (which amounts to saying, by the structure of algebraic groups over an algebraically closed field $\bar{k}$, that $G_{\bar{k}}$ contains no subgroup isomorphic to the additive group). Then for every $\nu \geq \nu_0$ and $n > 0$, the group $p^\nu H_n$ is flat over S_n .*

Indeed, it follows from $p^{\nu_0} \cdot \text{id}_{Q_n} = 0$ that $p^\nu \cdot \text{id}_{H_n}$ factors through G_n , so one shall have a commutative diagram:

$$\begin{array}{ccc} H_n & \xrightarrow{p^\nu \cdot \text{id}_{H_n}} & H_n \\ u_n \uparrow & \searrow v_n & \uparrow u_n \\ G_n & \xrightarrow{p^\nu \cdot \text{id}_{G_n}} & G_n \end{array}$$

I say that v_n is flat. Indeed, since H_n and G_n are flat over S_n , one is reduced to checking that v_0 is flat (SGA 1, IV 5.9), so one may suppose $n = 0$, whence $S_n = \text{Spec}(k)$. As $p \cdot \text{id}_{G_0}$, and hence $p^\nu \cdot \text{id}_{G_0}$, is flat, its image is an induced open subgroup G'_0 of G_0 , and as $u_0 : G_0 \rightarrow H_0$ is surjective, it follows that v_0 takes its values in G'_0 , and hence may be regarded as a homomorphism into G'_0 . As its composite with u_0 is an epimorphism, it is an epimorphism, hence a flat homomorphism into G'_0 , and hence a flat homomorphism into G_0 . Thus v_n is flat, so $\text{Ker } v_n = p^\nu H$ is flat over S . Q.E.D.

Remark. We have not explicitly used the fact that G_0 is smooth over k ; but it is easy to see that this is a consequence of the fact that $p \cdot \text{id}_G$ is flat, which is why we made this condition explicit in the hypothesis of corollary 7.5.

Lemma 7.6. *Let $u : G \rightarrow H$ be a surjective monomorphism of commutative algebraic groups over a field k of characteristic $p > 0$; consider the (purely infinitesimal) group $Q = H/G$. Then there exists an integer ν_1 such that for $\nu \geq \nu_1$ the sequence*

$$0 \rightarrow p^\nu G \rightarrow p^\nu H \rightarrow Q \rightarrow 0$$

is exact.

It suffices to ensure exactness at Q , and for this to ensure that the homomorphism

$$(x) \quad \mathcal{O}_{Q,e} \rightarrow \mathcal{O}_{p^\nu H,e}$$

of local rings at the neutral elements is injective (N.B. recall that Q is set-theoretically reduced to the element e). Now one has a natural homomorphism

$$(xx) \quad \mathcal{O}_{p^\nu H,e} \rightarrow \mathcal{O}_{H,e}/\mathfrak{m}^{p^\nu},$$

where $\mathfrak{m}$ is the augmentation ideal (i.e. the maximal ideal) of $\mathcal{O}_{H,e}$, as one sees by noting that $p^\nu \cdot \text{id}_H$ vanishes on the kernel of the “iterated Frobenius homomorphism” F^ν ; the composite of homomorphisms (x) and (xx) also equals the natural composite

$$(xxx) \quad \mathcal{O}_{Q,e} \rightarrow \mathcal{O}_{H,e} \rightarrow \mathcal{O}_{H,e}/\mathfrak{m}^{p^\nu}.$$

⁽⁴³⁾**N.D.E.** Add this in VII_A — On the other hand, the statement is true for every S if Q is commutative, cf. Deligne’s theorem in [TO70], p. 4.

Now $\mathcal{O}_{Q,e} \rightarrow \mathcal{O}_{H,e}$ is injective, since $H \rightarrow Q$ is an epimorphism and hence flat; on the other hand $\mathcal{O}_{Q,e}$ is artinian; finally, the intersection in $\mathcal{O}_{H,e}$ of the $\mathfrak{m}^{p^\nu}$ is reduced to 0, and hence so is the intersection of their traces on $\mathcal{O}_{Q,e}$. Consequently, one of these traces is reduced to 0, which proves that (xxx) is injective, and a fortiori (x) is injective.

Lemma 7.7. *Under the conditions of 7.5 there exists a ν_1 such that, for $\nu \geq \nu_1$ and every n , the sequence of S_n -groups*

$$0 \rightarrow {}_{p^\nu}G_n \rightarrow {}_{p^\nu}H_n \xrightarrow{w_n} Q_n \rightarrow 0$$

is exact (more precisely, w_n is faithfully flat and its kernel is ${}_{p^\nu}G_n$).

One takes ν_1 as in 7.6 applied to $u_0 : G_0 \rightarrow H_0$, and $\nu_1 \geq \nu_0$ (where $p^{\nu_0} = \text{rank } Q_0$). One need only check that w_n is faithfully flat. Now by 7.5, ${}_{p^\nu}H$ is flat over S_n , and since Q_n is also so, one is reduced to checking that $w_0 : {}_{p^\nu}H_0 \rightarrow Q_0$ is faithfully flat, i.e. is an epimorphism, which is true by the choice of ν_1 .

Corollary 7.8. *Suppose in addition H separated over S , more generally that ${}_{p^\nu}H$ is separated over S for every ν , and that ${}_{p^\nu}G$ is finite over S for every ν . Then the group schemes ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are finite and flat over S for $\nu \geq \nu_1$, and one has an exact sequence*

$$0 \rightarrow {}_{p^\nu}G \rightarrow {}_{p^\nu}H \xrightarrow{w} Q \rightarrow 0.$$

As $u : G \rightarrow H$ is surjective, so is the induced morphism ${}_{p^\nu}G \rightarrow {}_{p^\nu}H$, and as the first member is finite over S and the second separated over S , it follows that the second member is finite over S . To check then that ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are also flat over S , it suffices to check that this is so for ${}_{p^\nu}G_n$ and ${}_{p^\nu}H_n$ for every n , which is contained in 7.5. Finally, the exact sequence of 7.8 comes from the exact sequences 7.7 for variable n .

Taking the generic fibers in exact sequence 7.8 and recalling that $u_{s'} : G_{s'} \rightarrow H_{s'}$ is an isomorphism, one finds $Q_{s'} =$ the unit group, whence (since Q is flat over S) $Q =$ the unit group, whence $Q_s =$ the unit group, so $u_s : G_s \rightarrow H_s$ is an isomorphism. Thus:

Proposition 7.9. *Let $u : G \rightarrow H$ be a homomorphism of group preschemes of finite presentation over the prescheme S . Suppose:*

- a) *u is a monomorphism.*
- b) *G is flat over S .*
- c) *For every $s \in S$ such that $\kappa(s)$ is of residue characteristic $p > 0$, one requires that the following conditions be satisfied for the homomorphism $u' : G' \rightarrow H'$ of group preschemes over $S' = \text{Spec}(\mathcal{O}_{S,s})$ deduced from $u : G \rightarrow H$ by the base change $S' \rightarrow S$:*

- *G' is commutative,*
- *the special fiber $G'_0 = G'_s$ is smooth over $\kappa(s)$,*
- *for every integer $\nu > 0$, ${}_{p^\nu}G'$ is finite over S' and ${}_{p^\nu}H'$ is separated over S' .*

Under these conditions, u is an immersion.

It suffices to note that in c), the condition that ${}_{p^\nu}G'$ be finite over S' implies that ${}_{p^\nu}G'_0$ is finite over $\kappa(s)$, which already implies that $G'_0 \otimes_{\kappa(s)} \overline{\kappa(s)}$ has no subgroup isomorphic to the additive group, so one is under the hypotheses of 7.5.

Remark 7.10. Examples of M. Raynaud (XVI 1.1 a) and b)) show that in c) one can drop neither the hypothesis that the ${}_{p^\nu}G'$ are finite over S' nor that the ${}_{p^\nu}H'$ are separated over S' .

We now want conditions ensuring that u is a closed immersion. We therefore retain the hypotheses preceding 7.9, but no longer suppose u surjective (only that u is an isomorphism on the generic fibers). We already know that $u : G \rightarrow H$ is an open immersion; the same is therefore true of $u_0 : G_0 \rightarrow H_0$, whose image thus contains the connected component

of the neutral element $(H_0)^0$. Note that since G_0 is smooth over k and “without additive component” over the algebraic closure of k , the same is true of H_0 . Now one has the

Lemma 7.11. *Let H be a commutative algebraic group over a field k , such that $H \otimes_k \bar{k}$ contains no subgroup isomorphic to $\mathbf{G}_a$. Let $n = \text{degree of } H/H^0$; then the homomorphism*

$${}_nH \longrightarrow H/H^0$$

is surjective.

Indeed, one may suppose k algebraically closed, and then this follows from the well-known fact that $H^0(k)$ is a divisible group.

Suppose now (returning to our situation $u : G \longrightarrow H$) that the ${}_nG$ ($n > 0$) are proper over S , and the ${}_nH$ are separated over S . Let us note, on the other hand, that the ${}_nH$ are flat over S . Indeed, it suffices to see this at the points over s ; one is then reduced to proving that $n \cdot \text{id}_H$ is flat at the points over s , and for this one is reduced to checking that $n \cdot \text{id}_{H_0}$ is flat, which is equivalent (as we have already noted) to the fact that $(H_0)^0$ is smooth⁽⁴⁴⁾ over k and has no $\mathbf{G}_a$ component over the algebraic closure of k . As $(H_0)^0 = (G_0)^0$, this follows from the analogous hypothesis made on G_0 . On the other hand the ${}_nH$ are separated over S since H is so, hence the morphism ${}_nG \longrightarrow {}_nH$ is proper, hence its image is closed. As this image contains the generic fiber of ${}_nH$ (since $u_{s'} : G_{s'} \longrightarrow H_{s'}$ is an isomorphism) and ${}_nH$ is flat over S , hence identical to the closure of its generic fiber, it follows that ${}_nG \longrightarrow {}_nH$ is surjective, hence ${}_nG_0 \longrightarrow {}_nH_0$ is surjective. Recalling that $G_0 \supseteq (H_0)^0$ and applying 7.11, one finds that $G_0 \longrightarrow H_0$ is surjective, hence u is surjective. One thus obtains:

Proposition 7.12. *With the notation of 7.9, suppose that conditions a) and b) are satisfied, as well as condition c') (stronger than c)), obtained by requiring that for every integer $n > 0$ (not only of the form p^ν), ${}_nG'$ be finite over S' and ${}_nH'$ separated over S' . Under these conditions, u is a closed immersion.*

Remark 7.13. a) One easily checks that the separation hypothesis made on the ${}_nH$ in fact implies that H is separated over S . This is moreover formally contained in 7.12 by taking for G the unit S -group. Let us also note that when S is locally noetherian, one may in 7.9 restrict to supposing H locally of finite type over S (instead of: of finite type over S), the proof given applying as it stands; for 7.12 one shall suppose in addition that the fibers of H are of finite type.

b) Using 7.12, it is not difficult to prove that if $u : G \longrightarrow H$ is a monomorphism of S -group preschemes of finite presentation, with G diagonalizable (or more generally “of multiplicative type”) and H separated over S , then u is a closed immersion, — which constitutes a satisfactory generalization of the first conclusion stated in 5.7. When G is smooth over S , it suffices to apply 7.12, and the general case reduces easily to that one. When one no longer supposes H separated over S , one can still show that u is an immersion; in the case where G is a torus, this fact follows, moreover, also from what follows.

c) When in 7.9 one supposes G to have connected fibers, one may in condition 7.9 c) drop the hypothesis that the ${}_p H'$ are separated over S' . Indeed, with the reductions made in the proof, one may suppose H flat over S and u bijective, hence H with connected fibers, hence H separated over S by Raynaud’s theorem (VI_B 5.5).⁽⁴⁵⁾

⁽⁴⁴⁾**N.D.E.** The terminology has been updated, replacing “simple” by “smooth” (see, for example, note (1) of A. Grothendieck in SGA 1, Exp. II).

⁽⁴⁵⁾**N.D.E.** Specify this attribution in VI_B §5, which appeared in SGAD 1965, indicating the modifications between SGAD and Lect. Notes 151.

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Exposé IX. Groups of multiplicative type: Homomorphisms into a group scheme

by A. Grothendieck

1. Definitions

Definition 1.1. Let S be a prescheme, G an S -group prescheme. One says that G is a *group of multiplicative type* if G is locally diagonalizable in the sense of the faithfully flat quasi-compact topology (cf. IV 6.3), i.e. if for every $s \in S$ there exists an open neighborhood U of s and a faithfully flat and quasi-compact morphism $S' \rightarrow U$ such that $G' = G \times_S S'$ is a diagonalizable S' -group (I 4.4 and VIII 1.1).

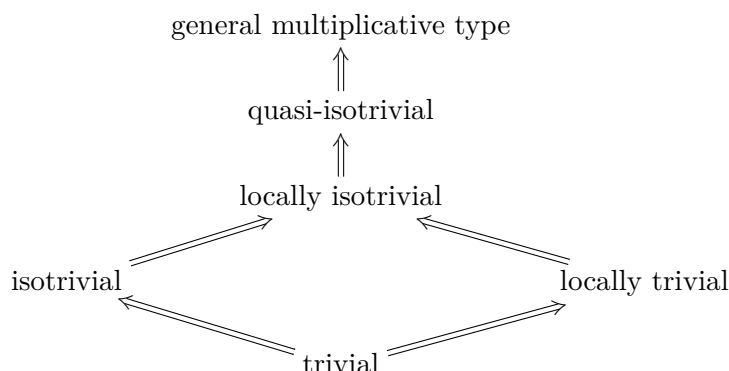
- One says that G is of *quasi-isotrivial* multiplicative type if it is even locally diagonalizable in the sense of the étale topology (IV 6.3), i.e. if, in the preceding definition, one may even take $S' \rightarrow U$ étale surjective, or again (which amounts to the same thing, as one sees by taking the sum scheme of the S' attached to the various $s \in S$) if there exists $S' \rightarrow S$ étale and surjective such that $G' = G \times_S S'$ is a locally diagonalizable S' -group.
- If one may even choose $S' \rightarrow S$ étale surjective and finite, one says that G is of *isotrivial* multiplicative type.
- Finally, one says that G is a group of *locally trivial* (resp. *locally isotrivial*) multiplicative type if every $s \in S$ admits an open neighborhood U such that $G \times_S U$ is a diagonalizable U -group (resp. a group of isotrivial multiplicative type, i.e. there exists $S' \rightarrow U$ a finite surjective étale morphism such that $G \times_S S'$ is a diagonalizable S' -group).

Comments 1.2. One will note that the preceding five notions are all deduced from the notion of a diagonalizable group by the process of localization, in the sense of five different “topologies” associated with (Sch).

One agrees in general that when the word “locally” is not made explicit by specifying the topology considered, it refers to the Zariski topology, in conformity with the received terminology. In the terminology introduced here, “of locally trivial multiplicative type” is equivalent to “locally diagonalizable” of VIII 1.1; similarly, “trivial” translates as “diagonalizable”. Between the five notions introduced, one has the following diagram of implications,

⁽⁰⁾Version 1.0 of November 8, 2009: additions in the proof of 3.6 bis, 4.4–7, 5.0–6, 6.1, 7.1, 8.2. Review 8.1 and 4.5.

which follows from the corresponding relations between the topologies involved:



From the practical point of view, let us point out immediately that all the groups of multiplicative type which we shall encounter will be quasi-isotrivial; thus we shall see in the next exposé (X 4.5) that when G is of finite type over S , G is automatically quasi-isotrivial, but we shall give examples where it is not locally isotrivial. We shall likewise see there that G may be locally trivial without being isotrivial or a fortiori trivial (which easily implies that the inclusions in the preceding diagram are strict).

On the other hand, we shall also see there (X 5.16) that when S is locally noetherian and normal (more generally geometrically unibranch), every group of multiplicative type of finite type over S is necessarily isotrivial, and moreover trivial as soon as it is locally trivial (or merely trivial over a dense open subset). This explains why most of the groups of multiplicative type which one will encounter in practice will doubtless be isotrivial, all the more so since we shall see later that the maximal tori of semisimple group schemes are automatically isotrivial.

Definition 1.3. Let S, G be as in 1.1. One says that G is a *torus* if it is locally isomorphic, in the sense of the faithfully flat quasi-compact topology, to a group of the form $\mathbf{G}_m^r$ (where r is an integer ≥ 0).

With the notation of 1.1, this therefore means that one may choose S' such that G' is isomorphic to a group of the form $(\mathbf{G}_{m,S'})^r$. One will note that the integer r depends on $s \in S$; it is the dimension of the fiber $G_s = G \otimes_S \kappa(s)$. It is a locally constant function of s , as one verifies immediately. There is occasion to generalize this remark:

Definition 1.4. Let G be a diagonalizable group scheme over a field k ; thus G is isomorphic to a group $D_k(M)$, where M is an ordinary commutative group, defined up to isomorphism by this condition, more precisely $M \simeq \text{Hom}_{k\text{-gr.}}(G, \mathbf{G}_{m,k})$ (VIII 1.3). The isomorphism class of M is called the *type* of the diagonalizable group G ; it is evidently invariant under extension of the base field.

If now G is a group scheme of multiplicative type over an arbitrary prescheme S , then for every $s \in S$ there exists an extension k of $\kappa(s)$ such that $G \times_S \text{Spec}(k)$ is a diagonalizable k -group;⁽¹⁾ its type is then independent of the chosen extension by the preceding remark, and will be called the *type of G at s* , or *type of G_s* . In particular, if G itself is diagonalizable, hence isomorphic to a group of the form $D_S(M)$, then for every $s \in S$ the type of G at s equals the class of the group M .

In general, if G is a group of multiplicative type over S , M an ordinary commutative group, one shall say that G is *of type M* if it is of type M at every point $s \in S$, in other words if G is locally isomorphic to $D_S(M)$ in the sense of the faithfully flat and quasi-compact topology.

⁽¹⁾**N.D.E.** Indeed, by hypothesis there exists an open neighborhood U of s and a faithfully flat and quasi-compact morphism $S' \rightarrow U$ such that $G_{S'}$ is diagonalizable; then for every $s' \in S'$ over s , $\kappa(s')$ is an extension of $\kappa(s)$ and $G_{s'} = G \times_S \text{Spec}(\kappa(s'))$ is diagonalizable.

Remark 1.4.1. ⁽²⁾ a) One sees immediately that for a given group of multiplicative type G over S , the function $s \mapsto \text{type of } G_s$ is locally constant on S : indeed, with the notation of 1.1, if G' is of type M , then G is of type M on the neighborhood U of s . It follows that one has a canonical partition of S as a sum of preschemes S_i , such that for every i , G_{S_i} is of type M_i , where the M_i are pairwise nonisomorphic commutative groups.

b) In particular, if S is connected, the type of the fibers of G is constant, i.e. there exists a commutative group M such that G is of type M . Finally, if G is a torus, the type of G at s is characterized by the integer $\dim G_s$ (indeed, G_s is of type $\mathbf{Z}^n$, where $n = \dim G_s$).

Remark 1.5. a) It is trivial from definitions 1.1 and 1.3 that they are “compatible with extension of the base”. Thus, if G is a group prescheme over S and $S' \rightarrow S$ a base change morphism, then if G is of multiplicative type (resp. of isotrivial multiplicative type, etc.), the same is true of G' .

b) When in addition $S' \rightarrow S$ is faithfully flat and quasi-compact, then if G' is of multiplicative type, resp. a torus, the same is true of G . If in addition $S' \rightarrow S$ is étale (resp. finite étale), and G' quasi-isotrivial (resp. isotrivial), the same is true of G .

c) Finally, returning to an arbitrary base change morphism $f : S' \rightarrow S$, if $s' \in S'$ and $s = f(s')$, then the type of G' at s' equals that of G at s , since $G'_{s'} = G_s \otimes_{\kappa(s)} \kappa(s')$.

2. Extension of certain properties of diagonalizable groups to groups of multiplicative type

The extensions in question are essentially trivial consequences of the results of the preceding exposé, in view of definitions 1.1 and the “local” nature (in the sense of the faithfully flat and quasi-compact topology) of the results in question.

Definition 2.0. ⁽³⁾ Denote by $\mathcal{M}$ the set of faithfully flat and quasi-compact morphisms.

Proposition 2.1. *Let G be a group of multiplicative type over a prescheme S . One has the following:*

- a) G is faithfully flat over S , and affine over S (a fortiori quasi-compact over S).
- b) G of finite type over $S \iff G$ of finite presentation over $S \iff$ for every $s \in S$, the type of G at s is given by a commutative group of finite type.
- c) G finite over $S \iff$ for every $s \in S$, the type of G at s is given by a finite commutative group $\iff$ (if S quasi-compact) G is of finite type over S and is annihilated by an integer $n > 0$.
- c') G integral over $S \iff$ for every $s \in S$, the type of G at s is given by a commutative torsion group.
- d) G is the unit group $\iff$ for every $s \in S$, the type of G at s is given by the zero commutative group.
- e) G is smooth over $S \iff$ for every $s \in S$, the type of G at s is given by a commutative group of finite type whose torsion subgroup has order prime to the characteristic of $\kappa(s)$.

These statements follow from VIII 2.1, in view of the fact that the properties in question descend by faithfully flat and quasi-compact morphisms (cf. SGA 1, VIII or EGA IV₂, §2).

Using VIII 3.5, one likewise obtains:

Proposition 2.2. *Let G be a group of multiplicative type and of finite type over S ; then for every integer $n \neq 0$, the kernel ${}_nG$ of $n \cdot \text{id}_G$ is a group of multiplicative type finite over S .*

Proposition 2.3. *Let G be a group of multiplicative type over the prescheme S , acting freely on the right on the S -prescheme X affine over S . Then:*

⁽²⁾**N.D.E.** The following remarks have been given the number 1.4.1, for later references.

⁽³⁾**N.D.E.** This definition has been added; it appears in propositions 2.3 and 2.7.

a) The equivalence relation defined by G in X is $\mathcal{M}$ -effective (IV 3.4); moreover $Y = X/G$ is affine over S .

b) If in addition X is of finite presentation (resp. of finite type) over S , the same is true of Y .

The first assertion follows from VIII 5.1, treating the case where G is diagonalizable, and from IV 3.5.2, which allows one to reduce to it, taking account of the fact that faithfully flat and quasi-compact morphisms $S' \rightarrow S$ are morphisms of effective descent for the fibered category of schemes affine over others, i.e. for every Y' affine over S' , endowed with a descent datum relative to $S' \rightarrow S$, this descent datum is effective, i.e. Y' comes from a Y affine over S , cf. SGA 1, VIII 2.1.

For the second assertion one is also reduced to the diagonalizable case VIII 5.8, since the finiteness conditions considered descend by faithfully flat and quasi-compact morphisms (SGA 1, VIII 3.3 and 3.6⁽⁴⁾). Proceeding as in VIII, corollaries 5.5 to 5.7, one draws from 2.3:

Corollary 2.4. *Under the conditions of 2.3, the graph morphism*

$$X \times_S G \rightarrow X \times_S X$$

is a closed immersion. For every section σ of X over S , the corresponding morphism $G \rightarrow X$, $g \mapsto \sigma \cdot g$, is a closed immersion.

In particular:

Corollary 2.5. *Let $u : G \rightarrow H$ be a monomorphism of S -group preschemes, with G of multiplicative type and H affine over S . Then u is a closed immersion, $H/G = Y$ exists and is affine over S ; finally, H is a principal homogeneous bundle over Y with group G_Y .*

Remark 2.6. Let $u : G \rightarrow H$ be a monomorphism of S -group preschemes, G being of multiplicative type and of finite type over S , H being of finite presentation over S and separated over S . Then one can show that u is a closed immersion, using VIII 7.12 (see remark VIII 7.13 b)).

Proposition 2.7. *Let $u : G \rightarrow H$ be a homomorphism of S -groups of multiplicative type, with H of finite type over S . Then:*

(i) $G' = \text{Ker } u$ is an S -group of multiplicative type; the equivalence relation defined by G' in G is $\mathcal{M}$ -effective, so (IV 3.4) u factors as

$$G \rightarrow G/G' = I \rightarrow H,$$

where $I \rightarrow H$ is a monomorphism⁽⁵⁾ of S -groups; I is of multiplicative type, the equivalence relation in H defined by I is $\mathcal{M}$ -effective, so $H' = H/I$ exists, and moreover H' is of multiplicative type.

(ii) H' and I are of finite type, and the same is true of G' if G is so.

PROOF. Proceeding as for 2.3, one is reduced to the case where G and H are diagonalizable, and then 2.7 reduces to VIII 3.4. $\square$

Let us note the following corollaries:

Corollary 2.8. a) *Let S be a prescheme. Then the category of S -groups of multiplicative type and of finite type is an abelian category.*

b) *Let $u : G \rightarrow H$ be a homomorphism in this category; for it to be a monomorphism (resp. an epimorphism, resp. an isomorphism) in this category, it is necessary and sufficient that u be a monomorphism of preschemes (resp. that u be faithfully flat, resp. that u be an isomorphism of preschemes).*

⁽⁴⁾**N.D.E.** Cf. also V, 9.1 or EGA IV₂, 2.7.1.

⁽⁵⁾**N.D.E.** And even a closed immersion.

It suffices to note that the product of two S -groups of multiplicative type is again of multiplicative type (which is immediate), evidently of finite type over S if both factors are so. The rest of 2.8 follows immediately from 2.7; the details are left to the reader.

Corollary 2.9. *Let $u : G \rightarrow H$ be a homomorphism of S -groups of multiplicative type and of finite type. Let U be the set of points $s \in S$ such that $u_s : G_s \rightarrow H_s$ is a monomorphism (resp. faithfully flat, resp. an isomorphism). Then U is both open and closed, and the induced homomorphism $u_U : G_U \rightarrow H_U$ is a monomorphism (resp. faithfully flat, resp. an isomorphism).*

Let P (resp. Q) be the kernel (resp. cokernel) of u . Thanks to 2.7, Q exists and P and Q are of multiplicative type; moreover formation of P and Q commutes with every base change $S' \rightarrow S$, in particular with formation of the fibers. On the other hand u is a monomorphism (resp. faithfully flat, resp. an isomorphism) if and only if $P = 0$ (resp. $Q = 0$, resp. $P = Q = 0$). Thanks to these remarks one is therefore reduced to checking this: if R is an S -group of multiplicative type, then the set U of $s \in S$ such that $R_s = 0$ is open and closed, and $R_U = 0$. Now this is contained in remark 1.4.1 a).

Corollary 2.10. *Let G be an S -group of multiplicative type and of finite type, H and H' two subgroups of multiplicative type.*

- a) $H'' = H \cap H'$ is a subgroup of multiplicative type of G .
- b) Let U be the set of $s \in S$ such that $H_s \subseteq H'_s$ (resp. $H_s = H'_s$). Then U is open and closed, and $H_U \subseteq H'_U$ (resp. $H_U = H'_U$).

Of course, the intersection sign in $H \cap H'$ denotes intersection in the functorial sense, i.e. $H \times_G H'$, which is evidently a sub- S -group of G ; similarly the inclusion (resp. equality) sign denotes the order relation (resp. equality) between subfunctors of H (and not inclusion (resp. equality) of the underlying sets).

First applying 2.7 to the inclusion morphism $H \rightarrow G$, one finds that H is of finite type; similarly H' is of finite type. It then follows from 2.8 that $H \cap H'$ is of multiplicative type and of finite type (one will note that the canonical functor from the category considered in 2.8 to the category $(\text{Sch})_S$ commutes with finite projective limits).

Formation of $H \cap H'$ commutes with extension of the base, in particular with formation of the fibers. On the other hand $H \subseteq H'$ (resp. $H = H'$) is equivalent to $H'' = H$ (resp. $H'' = H$ and $H'' = H'$). Consider then the canonical homomorphisms $H'' \rightarrow H$ and $H'' \rightarrow H'$; then U is the set of $s \in S$ such that the induced homomorphism $H''_s \rightarrow H_s$ is an isomorphism (resp. such that $H''_s \rightarrow H_s$ and $H''_s \rightarrow H'_s$ are isomorphisms). By 2.9, it follows that U is open and closed, and that $H''_U \rightarrow H_U$ (resp. $H''_U \rightarrow H_U$ and $H''_U \rightarrow H'_U$) are isomorphisms. Whence the desired conclusion.

Proposition 2.11. *Let S be a prescheme, G an S -group of multiplicative type and of finite type over S , H a subgroup of multiplicative type of G , and $K = G/H$ (which is a quotient group of multiplicative type of G).*

(i) *Suppose G trivial, i.e. diagonalizable. Then there exists a partition of S into open and closed subsets S_i such that for every i , H_{S_i} and K_{S_i} are diagonalizable. In particular, if S is connected, H and K are diagonalizable.*

(ii) *Same statement as in (i), replacing “diagonalizable” by “isotrivial”, provided S is connected, or locally connected, or quasi-compact.*

(iii) *Suppose G locally trivial (resp. locally isotrivial, resp. quasi-isotrivial); then the same is true of K and H .*

PROOF. (i) By hypothesis one has $G = D_S(M)$, where M is a commutative group of finite type. For every quotient group M_i of M , let $H_i = D_S(M_i)$ be the corresponding diagonalizable subgroup of G (VIII 3.1). Let S_i be the set of $s \in S$ such that $H_s = (H_i)_s$; by 2.10, S_i is open and closed, and one has $H_{S_i} = (H_i)_{S_i}$, so H_{S_i} is diagonalizable, and hence so

is K_{S_i} (VIII 3.1). Evidently the S_i are pairwise disjoint; I say that they cover S . This follows from the fact that for every s , H_s is diagonalizable, as a subgroup of the diagonalizable group G_s (cf. 8.1 below⁽⁶⁾), and hence H_s is of the form $D_{\kappa(s)}(M_i)$ by VIII, 1.5 and 3.2 b). Restricting to the family of nonempty S_i , conclusion (i) appears.

(ii) By hypothesis, there exists $S' \rightarrow S$ étale finite surjective such that $G_{S'}$ is diagonalizable. Thus every point of S' has a neighborhood U' , both open and closed, such that $H_{U'}$ and $K_{U'}$ are diagonalizable. Then the image U of U' in S is both open and closed, and $S' \rightarrow S$ still induces a finite surjective étale morphism $U' \rightarrow U$; one therefore sees that every point s of S has an open and closed neighborhood U such that H_U and K_U are isotrivial. Conclusion (ii) follows immediately.

(iii) Follows immediately from (i) and (ii) and the definitions. Note also that the “quasi-isotrivial” case will also follow from the more general fact “finite type $\Rightarrow$ quasi-isotrivial”, announced in 1.2 (cf. X 4.5). $\square$

3. Infinitesimal properties: Lifting and conjugacy theorems

The fundamental infinitesimal properties of groups of multiplicative type follow from the following theorem:

Theorem 3.1. *Let S be an affine scheme, H an S -group of multiplicative type, $\mathcal{F}$ a quasi-coherent sheaf over S on which H acts (I 4.7). Then one has*

$$H^i(H, \mathcal{F}) = 0 \quad \text{for } i > 0,$$

where H^i is the Hochschild cohomology studied in Exp. I, §5.

Indeed, by loc. cit., 5.3, Hochschild cohomology is made explicit, in terms of the affine rings A of S and B of G , and of the module M over A defining $\mathcal{F}$, as the cohomology of a complex of A -modules $C^\bullet(H, M)$, whose formation commutes with every base change $A \rightarrow A'$. Consequently, for a base change $A \rightarrow A'$ with A' flat over A , one shall have

$$H^i(H', \mathcal{F}') = H^i(H, \mathcal{F}) \otimes_A A',$$

and therefore, if one even supposes A' faithfully flat over A , to prove that $H^i(H, \mathcal{F})$ is zero it suffices to prove that $H^i(H', \mathcal{F}')$ is so. This remark reduces us to checking 3.1 in the case where G is diagonalizable; in this case it was proved in I 5.3.3. Using the results of Exposé III, we shall draw from 3.1 various consequences of a geometric nature:

Theorem 3.2. *Let S be an affine scheme, S_0 an affine subscheme defined by an ideal $\mathcal{J}$, H a group of multiplicative type over S , G any group prescheme over S , $u, v : H \rightarrow G$ two group homomorphisms, $u_0, v_0 : H_0 \rightarrow G_0$ the homomorphisms deduced from u, v by the base change $S_0 \rightarrow S$. If $\mathcal{J}^2 = 0$ and $u_0 = v_0$, then there exists a $g \in G(S)$ such that $v = \text{int}(g) \circ u$ and $g_0 = e$.*

This follows from III 2.1 (ii) and from 3.1 (vanishing of H^1).

Corollary 3.3. *Under the preliminary conditions of 3.2, suppose in addition G smooth over S , but, on the other hand, suppose only $\mathcal{J}$ nilpotent (not necessarily of square zero). If there exists $g_0 \in G_0(S_0)$ such that $v_0 = \text{int}(g_0) \circ u_0$, then g_0 lifts to a $g \in G(S)$ such that $v = \text{int}(g) \circ u$.*

By induction on the integer n such that $\mathcal{J}^n = 0$, one is reduced to the case where $\mathcal{J}$ has square zero. Moreover, G being smooth over S , one can lift g_0 to an element g' of $G(S)$. Replacing v by $v' = \text{int}(g') \circ u$ if necessary, one is then reduced to situation 3.2.

⁽⁶⁾**N.D.E.** The erroneous reference to IV 4.7.5 has been corrected. Note that no. 8 of the present exposé is independent of nos. 3 to 7.

Corollary 3.4. *Under the preliminary conditions of 3.2, with $\mathcal{J}$ nilpotent, suppose in addition u central (for example G commutative). Then $u_0 = v_0$ implies $u = v$.*

Indeed, reduction to the case of $\mathcal{J}$ of square zero is again immediate, and then it suffices to apply 3.2. In particular:

Corollary 3.5. *Let S, S_0, H, G be as in 3.2, with $\mathcal{J}$ nilpotent. Let $u : H \rightarrow G$ be a homomorphism of S -groups such that $u_0 : H_0 \rightarrow G_0$ is the unit homomorphism; then u is the unit homomorphism.*

Theorem 3.6. *Let S be an affine scheme, S_0 a subscheme defined by a nilpotent ideal $\mathcal{J}$, H a group of multiplicative type over S , G a group prescheme smooth over S , $u_0 : H_0 \rightarrow G_0$ a homomorphism of the S_0 -groups deduced by the base change $S_0 \rightarrow S$.*

Then there exists a homomorphism $u : H \rightarrow G$ of S -groups lifting u_0 (and by 3.3 two such liftings u, u' are conjugate by an element of $G(S)$ reducing to the unit element of $G(S_0)$).

This follows from III 2.1 and 2.3, and from 3.1 (vanishing of H^2 for the existence of the lifting of u_0 , vanishing of H^1 for uniqueness up to conjugacy).

One can similarly prove the following variants of 3.2 to 3.6 (which in fact imply the preceding statements, by applying them to the graph subgroups of the homomorphisms considered in 3.2 to 3.6):

Theorem 3.2 bis. *Let S be an affine scheme, S_0 a subscheme defined by an ideal $\mathcal{J}$, G an S -group prescheme, H, H' subgroup preschemes of multiplicative type, G_0, H_0, H'_0 the groups over S_0 deduced by the base change $S_0 \rightarrow S$. If $\mathcal{J}^2 = 0$ and $H_0 = H'_0$, then there exists $g \in G(S)$ such that $H' = \text{int}(g)(H)$ and $g_0 = e$.*

Corollary 3.3 bis. *Under the preliminary conditions of 3.2 bis, suppose in addition G smooth over S , and $\mathcal{J}$ nilpotent (not necessarily of square zero). Let $g_0 \in G_0(S_0)$ be such that $H'_0 = \text{int}(g_0)(H_0)$; then g_0 lifts to $g \in G(S)$ such that $H' = \text{int}(g)(H)$.*

Corollary 3.4 bis. *Under the preliminary conditions of 3.2 bis, suppose $\mathcal{J}$ nilpotent, H central (for example G commutative). Then $H_0 = H'_0$ implies $H = H'$.*

Theorem 3.6 bis. *Let S be an affine scheme, S_0 a subscheme defined by a nilpotent ideal $\mathcal{J}$, G a smooth S -group prescheme, H_0 a subgroup prescheme of $G_0 = G \times_S S_0$, of multiplicative type. Then:*

- (a) *There exists a subgroup prescheme H of G , flat over S , such that $H \times_S S_0 = H_0$.*
- (b) *Such an H is necessarily of multiplicative type.*
- (c) *Finally, by 3.2 bis, two such liftings H, H' of H_0 are conjugate by an element $g \in G(S)$ reducing to the unit element of $G(S_0)$.*

Let us show how one can prove 3.2 bis (of which 3.3 bis and 3.4 bis are an immediate consequence) and assertion (a) of 3.6 bis. The latter follows from 3.1 (vanishing of H^2) and III 4.37 (ii). Similarly, 3.2 bis follows from 3.1 (vanishing of H^1) and III 4.37 (i), at least when G and H are smooth over S .

But using the results of the next exposé, one can very simply deduce 3.2 bis and 3.6 bis, in the general form stated here, from 3.2 resp. 3.6. For 3.2 bis, it indeed suffices to note that by X 2.1, H and H' are isomorphic, which reduces us to 3.2.

For 3.6 bis, one notes that H_0 is necessarily of finite type over S_0 :⁽⁷⁾ indeed, as a subprescheme of G_0 , which is smooth over S_0 , H_0 is locally of finite type over S_0 ; now by 2.1 a), H_0 is affine over S_0 , and hence of finite type over S_0 . Then, by X 4.5, H_0 is quasi-isotrivial and therefore comes, by X 2.1, from a group of multiplicative type H over S .⁽⁸⁾ Then, by 3.6,

⁽⁷⁾**N.D.E.** The original indicated: “by 2.1 b), since its fibers are so”. The editors did not understand this argument and have substituted the following one for it.

⁽⁸⁾**N.D.E.** The following sentence has been expanded.

there exists a homomorphism of S -groups $u : H \rightarrow G$ lifting the immersion $H_0 \hookrightarrow G_0$; since H and H_0 (resp. G and G_0) have the same underlying topological space, and since for every $h \in H$ the morphism $\mathcal{O}_{G,u(h)} \rightarrow \mathcal{O}_{H,h}$ is surjective (for it is so after reduction modulo $\mathcal{J}$, which is nilpotent), u is also an immersion (cf. EGA I, 4.2.2).

Finally, for every lifting H of H_0 to a flat subgroup of G , H is necessarily of multiplicative type by X 2.3,⁽⁹⁾ which also proves assertion (b) of 3.6 bis. (The reader will check that results 3.2 bis to 3.6 bis are not used in Exposé X, so there is no vicious circle!)

Proposition 3.8. ⁽¹⁰⁾ *Let S be a prescheme, S_0 a closed subprescheme defined by a locally nilpotent ideal $\mathcal{J}$, G an S -group of multiplicative type, X an S -prescheme locally of finite presentation. Suppose that G acts on X in such a way that $G_0 = G \times_S S_0$ acts trivially on $X_0 = X \times_S S_0$. Under these conditions, G acts trivially on X .*

The proof is that of III 2.4 b); one can also reduce to loc. cit. by noting that 3.8 reduces immediately to the case where G is diagonalizable, which is the case considered in loc. cit.

Corollary 3.9. *Let S, S_0 be as above, and $u : G \rightarrow H$ a homomorphism of S -groups, with G of multiplicative type and H locally of finite presentation over S . Suppose that the homomorphism $u_0 : G_0 \rightarrow H_0$ deduced by base change $S_0 \rightarrow S$ is central. Then u is central.*

Indeed, it suffices to apply 3.8 by putting $X = H$ and making G act on H by $(g, h) \mapsto \text{int}(u(g)) \cdot h = u(g)hu(g)^{-1}$.

4. The density theorem

This theorem (cf. 4.7 below),⁽¹¹⁾ together with the theorem⁽¹²⁾ of no. 7, will be the essential tool in the present exposé and the following two for passing from the infinitesimal properties of groups of multiplicative type which we have just developed to the “finite” properties.

Definition 4.1. Let X be a prescheme. One says that a family $(Z^i)_{i \in I}$ of subpreschemes of X is *schematically dense* if for every open subset U of X and every closed subprescheme U' of U containing the $Z^i \cap U$, one has $U' = U$.

One says that a subprescheme Z of X is schematically dense in X if the family consisting of Z is so.

One sees immediately (cf. EGA IV₃, 11.10.1) that the definition is equivalent to saying that for every open subset U of X , every section f of $\mathcal{O}_U$ which is zero on the $Z^i \cap U$ is zero, which also means that the intersection of the kernels of the canonical homomorphisms

$$u_i : \mathcal{O}_X \rightarrow (v_i)_*(\mathcal{O}_{Z^i})$$

is zero, where $v_i : Z^i \rightarrow X$ is the canonical immersion.⁽¹³⁾ When X is over a prescheme S , this is also equivalent to saying that for every open subset U of X and every pair (u, v) of morphisms from U to an S -prescheme Y separated over S , coinciding on the $Z^i \cap U$, one has $u = v$. (Indeed, the relation $u = v$ is equivalent to the relation $U' = U$, where U' is the inverse image of the diagonal of $Y \times_S Y$ by the S -morphism $X \rightarrow Y \times_S Y$ defined by (u, v) ; this diagonal is a closed subprescheme of $Y \times_S Y$, so U' is a closed subprescheme of U containing the $Z^i \cap U$ by the hypothesis on (u, v) , so if the family (Z^i) is schematically dense,

⁽⁹⁾**N.D.E.** One will note that X 2.3 depends essentially on theorem 3.6 of the present exposé.

⁽¹⁰⁾**N.D.E.** The original numbering has been preserved: there is no no. 3.7.

⁽¹¹⁾All results from 4.1 to 4.6 are contained in EGA IV₃, 11.10, to which we refer the reader for a formal exposition of the notion of schematic density.

⁽¹²⁾**N.D.E.** The “algebraicity” theorem 7.1.

⁽¹³⁾**N.D.E.** All the schematic density results stated in EGA IV₃, §11.10 follow from the results on “separating families of homomorphisms” proved in loc. cit., §11.9, to which we shall refer in some of the following N.D.E.

one shall have $U' = U$, whence $u = v$; one sees the converse implication by simply taking $Y = \operatorname{Spec} \mathcal{O}_S[T]$.)

With the terminology introduced in EGA I, end of no. 9.5, saying that the subscheme Z of X is schematically dense also means that X is identical to the closure subscheme of Z in X .

Lemma 4.2. *Let X be a flat S -prescheme, $(Z^i)_{i \in I}$ a family of subschemes of X , flat over S . Let S_0 be a subscheme of S , defined by a nilpotent ideal $\mathcal{J}$; suppose the modules $\mathcal{J}^n/\mathcal{J}^{n+1}$ locally free over S_0 .*

Let X_0, Z_0^i be deduced from X, Z^i by the base change $S_0 \rightarrow S$. Then if the family $(Z_0^i)_{i \in I}$ is schematically dense in X_0 , the family $(Z^i)_{i \in I}$ is so in X .

Suppose $\mathcal{J}^{n+1} = 0$ (where $n \geq 0$), and argue by induction on n , the assertion being trivial for $n = 0$. Defining S_m, X_m, Z_m^i by reduction modulo $\mathcal{J}^{m+1}$ as usual, the induction hypothesis already implies that $(Z_{n-1}^i)_{i \in I}$ is schematically dense in X_{n-1} . Replacing X by an induced open subset if necessary, we are reduced to proving that every section f of $\mathcal{O}_X$ vanishing on the Z^i is zero.

Now the section f_{n-1} of $\mathcal{O}_{X_{n-1}} = \mathcal{O}_X/\mathcal{J}^n \mathcal{O}_X$ defined by f vanishes on the Z_{n-1}^i , and is therefore zero, so f is a section of $\mathcal{J}^n \mathcal{O}_X$. Since X is flat over S , one has

$$\mathcal{J}^n \mathcal{O}_X \xleftarrow{\sim} \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{X_0}, \quad \text{where } \mathcal{E} = \mathcal{J}^n = \mathcal{J}^n/\mathcal{J}^{n+1}.$$

Similarly, since each Z^i is flat over S , the restriction f_i of f to Z^i may be regarded as a section of:

$$\mathcal{J}^n \mathcal{O}_{Z^i} \xleftarrow{\sim} \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Z_0^i}.$$

By hypothesis the f_i are zero. Now $\mathcal{E}$ is locally free by hypothesis, and hence so is $\mathcal{F} = \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{X_0}$. Thus f is a section of the locally free module $\mathcal{F}$ over X_0 , such that for every i its restriction to Z_0^i is zero. Since $(Z_0^i)_{i \in I}$ is schematically dense in X_0 , it follows immediately that f is zero. Q.E.D.

Corollary 4.3. *Let X be a locally noetherian S -prescheme flat over S , $(Z^i)_{i \in I}$ a family of subschemes of X flat over S . Suppose that for every $s \in S$, the family $(Z_s^i)_{i \in I}$ of fibers at s is schematically dense in X_s . Then the family $(Z^i)_{i \in I}$ is schematically dense in X .*

Replacing X by an induced open subset if necessary, one is reduced to proving that every section f of $\mathcal{O}_X$ whose restriction to the Z^i is zero is zero.⁽¹⁴⁾ For this it suffices to prove that for every $x \in X$ the image of f in $\mathcal{O}_{X,x}$ is zero. Denote by $\mathfrak{m}_x$ the maximal ideal of $\mathcal{O}_{X,x}$, and by $\mathfrak{m}_s$ that of $\mathcal{O}_{S,s}$, where s is the image of x in S . As $\mathcal{O}_{X,x}$ is noetherian, one has $\bigcap_{n \in \mathbf{N}} \mathfrak{m}_x^n = 0$, whence, a fortiori, $\bigcap_{n \in \mathbf{N}} \mathcal{O}_{X,x} \mathfrak{m}_s^n = 0$.

Thus it suffices to show that for every integer n the section induced by f on $X \times_S \operatorname{Spec}(\mathcal{O}_{S,s}/\mathfrak{m}_s^{n+1})$ is zero. By hypothesis the family of fibers $Z_s^i = Z^i \otimes_{\mathcal{O}_{S,s}} \kappa(s)$ is schematically dense in the fiber $X_s = X \otimes_{\mathcal{O}_{S,s}} \kappa(s)$. This therefore reduces us to the case where S is the spectrum of a local ring whose maximal ideal J is nilpotent, $S_0 = \operatorname{Spec} \kappa(s)$, and the hypotheses of 4.2 are satisfied, whence the conclusion.

Lemma 4.4. *Let S be a locally noetherian prescheme, X a locally noetherian S -prescheme flat over S , $(Z^i)_{i \in I}$ a family of subschemes of X flat over S . Suppose that for every $s \in S$, the family $(Z_s^i)_{i \in I}$ of fibers at s is schematically dense in X_s .*

Then for every base change $S' \rightarrow S$ (S' not necessarily locally noetherian), the family $(Z_n^i)_{i \in I}$ is schematically dense in X' (i.e. the family $(Z^i)_{i \in I}$ is universally schematically dense in X relative to S , cf. EGA IV₃, Def. 11.10.8).

⁽¹⁴⁾**N.D.E.** The original has been expanded in what follows.

PROOF. ⁽¹⁵⁾ Let f' be a section of $\mathcal{O}_{X'}$ over an open subset U' of X' which vanishes on the $Z_n^{i'}$; one must show that f' is zero. Let $s' \in S'$; it suffices to prove that f' is zero at all points of U' over s' . Let s be the image of s' in S ; replacing S, S' by the spectra of the local rings at s, s' if necessary, one may suppose S, S' local and s, s' their closed points. One may further suppose X affine, hence

$$S = \operatorname{Spec}(A), \quad S' = \operatorname{Spec}(A'), \quad X = \operatorname{Spec}(B), \quad X' = \operatorname{Spec}(B'), \quad B' = B \otimes_A A'$$

where A, A' are local, $A \rightarrow A'$ is a local homomorphism, and A, B are noetherian. One may also suppose U' affine, of the form $X'_{g'}$, with $g' \in B'$.

For every A -subalgebra A'' of A' , one considers $S'' = \operatorname{Spec}(A'')$ and $X'', Z_n^{i''}$ deduced from X, Z^i by the base change $S'' \rightarrow S$, whence the diagram:

$$\begin{array}{ccccc} Z^i & \longleftarrow & Z_n^{i''} & \longleftarrow & Z_n^{i'} \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & X'' & \longleftarrow & X' \\ \downarrow & & \downarrow & & \downarrow \\ S & \longleftarrow & S'' & \longleftarrow & S' \end{array}$$

We shall henceforth restrict to those A'' which are localizations of A -subalgebras of finite type A''_1 of A' at $\mathfrak{m}' \cap A''_1$ (where $\mathfrak{m}'$ is the maximal ideal of A'), so that the homomorphisms $A \rightarrow A'' \rightarrow A'$ are local. Note that these A'' form an increasing filtering family of subalgebras whose union is A' , so their inductive limit is also A' . One therefore has similarly $B' = \varinjlim_{A''} B \otimes_A A''$. Thus there exists an A'' and an element $g'' \in B'' = B \otimes_A A''$ whose image in B' is g' .

⁽¹⁶⁾ By lemma 4.5 below, the property that the family of fibers $(Z_s^i)_{i \in I}$ is schematically dense in X_s is preserved by every base change $S'' \rightarrow S$; consequently, since each S'' is locally noetherian, if one replaces S by a suitable S'' and X, Z^i by $X'', Z^{i''}$, the hypotheses of 4.4 will be preserved.

Thus, replacing S by S'' (and X, Z^i by $X'', Z^{i''}$) if necessary, one may suppose that g' comes from $g \in B$. Replacing X by the open subset $X_g = U$ if necessary, one may therefore suppose that $U' = X'$. One sees similarly that one may suppose that f' comes from a section f of $\mathcal{O}_X$ over X .

Let $Y = \mathcal{V}(f)$ be the subscheme of X defined by f , and $Y^i = Z^i \times_X Y$ its intersection with Z^i , which is a closed subscheme of Z^i , equal to $\mathcal{V}(f^i)$, where f^i is the section of $\mathcal{O}_{Z^i}$ induced by f . Denoting by $Y', Y^{i'}$ the S' -schemes deduced from the preceding ones by base change, one shall still have

$$Y' = \mathcal{V}(f'), \quad Y^{i'} = Z^{i'} \times_{X'} Y' = \mathcal{V}(f^{i'}),$$

and one has the analogous relations for $Y'', Y^{i''}$. The hypothesis that f' vanishes on the $Z^{i'}$ is expressed by the relations $Y^{i'} = Z^{i'}$ for every i .⁽¹⁷⁾

Let $\mathfrak{m}$ be the maximal ideal of A . For every integer $n \geq 0$, introduce the subscheme $S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1})$ of S and the schemes X_n, Y_n, Z_n^i, Y_n^i deduced from X, Y, Z^i, Y^i by the base change $S_n \rightarrow S$. In general, for every S -prescheme P , one shall put $P_n = P \times_S S_n$.

⁽¹⁵⁾**N.D.E.** For another proof, using a reduction to the case where S' is locally noetherian (and 4.3 and 4.5 as here), see EGA IV₃, 11.9.16 and 11.9.12 (N.B. in the last line of the proof of 11.9.16, replace 11.9.5 by 11.9.12).

⁽¹⁶⁾**N.D.E.** The following sentence has been added, and the original expanded in what follows.

⁽¹⁷⁾**N.D.E.** The original added: “at least at every point x' of X' over s' ”, but this restriction seems unnecessary.

For every n and $i \in I$, consider the functor

$$F_n^i = \prod_{Z_n^i/S_n} Y_n^i/Z_n^i : (\text{Sch})_{/S_n}^\circ \longrightarrow (\text{Ens})$$

defined by (cf. VIII, 6.4): for every P over S_n ,

$$F_n^i(P) = \Gamma((Y_n^i)_P/(Z_n^i)_P) = \begin{cases} \emptyset & \text{if } (Y_n^i)_P \neq (Z_n^i)_P; \\ \{\text{id}\} & \text{if } (Y_n^i)_P = (Z_n^i)_P. \end{cases}$$

Since S_n is local artinian and Z_n^i flat, hence essentially free over S_n , by VIII, 6.4, each F_n^i is representable by a closed subscheme T_n^i of S_n .

(18) The completion $\hat{A}$ of the local ring A is noetherian since A is so, and it is the projective limit of the A_n . Denote by K_n^i the ideal of $A_n = A/\mathfrak{m}^{n+1}$ defining T_n^i , and by K^i the projective limit of the K_n^i ; it is an ideal of $\hat{A}$.

For fixed i , $m \geq n$, and every S_n -prescheme P , one has

$$(Y_n^i)_P = Y^i \times_S S_n \times_{S_n} P = Y^i \times_S P = (Y_m^i)_P,$$

and similarly $(Z_n^i)_P = (Z_m^i)_P$; it follows that F_n^i is the restriction $F_m^i \times_{S_m} S_n$ of F_m^i to $(\text{Sch})_{/S_n}$, whence $T_n^i = T_m^i \times_{S_m} S_n$. One therefore has a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & K_m^i & \longrightarrow & A_m & \longrightarrow & \mathcal{O}(T_m^i) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & K_n^i & \longrightarrow & A_n & \longrightarrow & \mathcal{O}(T_n^i) \longrightarrow 0 \end{array}$$

where all the vertical arrows are surjective. This has the following consequences: on the one hand, the projective system $(K_n^i)_{n \in \mathbf{N}}$ satisfies the Mittag-Leffler condition, and hence the projective limit of the $\mathcal{O}(T_n^i)$ is identified with the topological quotient ring $\hat{A}/K^i$ (cf. EGA 0III, §13.2). On the other hand, the map $K^i \rightarrow K_n^i$ is surjective (cf. [BEns], III, §7.4, Prop. 5), whence it follows that $K_n^i \simeq (K^i + \mathfrak{m}^{n+1}\hat{A})/\mathfrak{m}^{n+1}\hat{A}$ and $\mathcal{O}(T_n^i) \simeq (\hat{A}/K) \otimes_A (A/\mathfrak{m}^{n+1})$. In other words, $(T_n^i)_n$ is an inductive system of artinian affine schemes, and the inductive limit $T^i = \varinjlim_n T_n^i$ is a closed formal subscheme of the formal scheme $\hat{S} = \text{Spf}(\hat{A})$ (cf. EGA I, §10), whose reduction modulo $\mathfrak{m}^{n+1}$ is T_n^i .

Let T be the closed formal subscheme of $\hat{S}$ which is the intersection of the T^i , i.e. defined by $K = \sum_i K^i$. Since $\hat{A}$ is noetherian, there exists a finite subset J of I such that one has $K = \sum_{i \in J} K^i$ (i.e. $T = \bigcap_{i \in J} T^i$). Let us then note that for every n one has $K_n = \sum_{i \in J} K_n^i$, where K_n denotes the image of K in A_n .

Recall that f^i denotes the image of $f \in B = \mathcal{O}(X)$ in $\mathcal{O}(Z^i)$, and $f^{i'}$ its image in $\mathcal{O}(Z_n^{i'}) = \mathcal{O}(Z^i) \otimes_A A'$; the hypothesis $Y^{i'} = Z^{i'}$ is equivalent to the vanishing of $f^{i'}$. As $\mathcal{O}(Z^i) \otimes_A A'$ is the inductive limit of the subalgebras $\mathcal{O}(Z^i) \otimes_A A''$ (where A'' satisfies the conditions made explicit above), there therefore exists an A'' such that $Y^{i''} = Z^{i''}$. A priori, A'' depends on i , but one can find an A'' which works for all $i \in J$, since J is finite. Put $S'' = \text{Spec}(A'')$ and $S''_{(n)} = S'' \times_S S_n$.⁽¹⁹⁾

Since $Y^{i''} \times_{S''} S''_{(n)} = (Y_n^i)_{S''_{(n)}}$ equals $Z^{i''} \times_{S''} S''_{(n)} = (Z_n^i)_{S''_{(n)}}$ for every $i \in J$, it follows from the definition of the T_n^i that $S''_{(n)} \rightarrow S_n$ factors through T_n^i for every $i \in J$, and hence also through $T_n = \bigcap_{i \in J} T_n^i$, and hence also through T_n^i for every $i \in I$. Denoting by f'' the image of f in $B'' = B \otimes_A A''$, and by $f''_{(n)}$ its image in $B''_{(n)} = B''/\mathfrak{m}^{n+1}B''$, this means that for every n , $f''_{(n)}$ vanishes on the $Z^i \times_S S''_{(n)}$.

(18) **N.D.E.** The original has been expanded and corrected to show that the projective limit of the rings $\mathcal{O}(T_n^i)$ defines a closed formal subscheme of the formal scheme $\text{Spf}(\hat{A})$.

(19) **N.D.E.** The original has been expanded and corrected, distinguishing $S''_{(n)}$ from the subscheme $S''_n = S''_n(s'')$ introduced below.

Since the morphisms $A \rightarrow A'' \rightarrow A'$ are local, the image s'' of s' in S'' is the closed point of S'' , corresponding to the maximal ideal $\mathfrak{m}''$ of B'' , and the image of s'' in S is s . Fix n and now denote by S''_n the n -th infinitesimal neighborhood of s'' in S'' , i.e. $\text{Spec}(A''/\mathfrak{m}''^{n+1})$. Put $B''_n = B''/\mathfrak{m}''^{n+1}B''$ and $Z''_n = Z^i \times_S S''_n = \text{Spec}(B''_n)$. Then the image f''_n of $f''_{(n)}$ in B''_n vanishes on Z''_n for every i . Now, by lemma 4.5 below, the family of fibers

$$Z''_0 = Z''_n \times_{S''_n} \kappa(s'') = Z^i_s \times_{\kappa(s)} \kappa(s'')$$

is schematically dense in the fiber $X''_0 = X''_n \times_{S''_n} \kappa(s'') = X_s \times_{\kappa(s)} \kappa(s'')$. Thus $S''_n, X''_n, (Z''_n)_{i \in I}$, and $S''_0 = \text{Spec } \kappa(s'')$ satisfy the hypotheses of lemma 4.2; it therefore follows that $f''_n = 0$, i.e. $f'' \in \mathfrak{m}''^{n+1}B''$, for every n . Since B is noetherian and $B'' = B \otimes_A A''$ is a localization of a B -algebra of finite type, B'' is noetherian, and hence its local rings are separated for their usual topology. It follows that f'' is zero at the points x'' of X'' such that $\mathfrak{m}''B''$ is contained in the maximal ideal of $\mathcal{O}_{X'', x''}$, i.e. at the points of X'' over s'' . A fortiori, f' is zero at the points x' of X' over s' . $\square$

Lemma 4.5.0. ⁽²⁰⁾ *Let X be an S -prescheme, $(Z^i)_{i \in I}$ a family of subpreschemes of X , and $S' \rightarrow S$ a faithfully flat morphism. If the family $(Z'^i)_{i \in I}$ is schematically dense in X' , then $(Z^i)_{i \in I}$ is schematically dense in X .*

This follows from EGA IV₃, 11.10.5 (i) and 11.9.10 (i). It remains to set out the proof of the

Lemma 4.5. *Let X be a locally noetherian⁽²¹⁾ prescheme over a field k , $(Z^i)_{i \in I}$ a family of subpreschemes of X , k' an extension of k , X' and Z'^i the preschemes deduced from X and Z^i by the base change k'/k . For $(Z^i)_{i \in I}$ to be schematically dense in X , it is necessary and sufficient that $(Z'^i)_{i \in I}$ be so in X' .*

The “sufficient” part follows from 4.5.0; let us prove the “necessary” part.⁽²²⁾ First one may suppose $X = \text{Spec}(B)$ affine. For every i , let $(Z^{ij})_{j \in J_i}$ be a covering of Z^i by affine open subsets; replacing $(Z^i)_{i \in I}$ by the family $(Z^{ij})_{(i,j) \in J}$, where $J = \coprod_{i \in I} J_i$, one may also suppose the Z^i affine.

Let $g \in B' = B \otimes_k k'$ and $t' \in B'_g$ be a section of $\mathcal{O}_{X'}$ on the affine open subset $U' = X'_g$, zero on the $Z'^i \cap U'$, i.e. whose image in each $(\mathcal{O}(Z^i) \otimes_k k')_g$ is zero. There exists a k -subalgebra of finite type A of k' such that $g \in B_A = B \otimes_k A$ and $t' \in (B_A)_g$. Since the map $\mathcal{O}(Z^i) \otimes_k A \rightarrow \mathcal{O}(Z^i) \otimes_k k'$ is injective, the same is true of the map

$$(\mathcal{O}(Z^i) \otimes_k A)_g \rightarrow (\mathcal{O}(Z^i) \otimes_k k')_g;$$

in view of the hypothesis, this implies that the image of t' in each $(\mathcal{O}(Z^i) \otimes_k A)_g$ is zero. This reduces us to showing that the family $(Z^i_A)_{i \in I}$ is schematically dense in X_A ,⁽²³⁾ which follows from EGA IV₃, 11.9.10 b).

Let us note in passing the following result, which will serve in a later exposé:⁽²⁴⁾

Corollary 4.6. *Let X be a flat S -prescheme, U an open subset of X . Suppose that for every $s \in S$, U_s is schematically dense in X_s , and that X is locally noetherian, or locally of finite presentation over S . Then U is schematically dense in X .*

⁽²⁰⁾**N.D.E.** This lemma has been inserted here; it is used in the proof of 4.5 and 4.7.

⁽²¹⁾**N.D.E.** This hypothesis is in fact superfluous, cf. EGA IV₃, 11.10.6 and 11.9.13.

⁽²²⁾**N.D.E.** The original has been expanded in what follows.

⁽²³⁾**N.D.E.** The original continued: “Using 4.3 for the $X_{A'}$, where A' is a quotient of A by a power of a maximal ideal, and Hilbert’s Nullstellensatz, one is reduced to the case where k' is a finite extension of k ”. One could expand this reduction and proceed along these lines, since the case of a finite extension k'/k is somewhat simpler than the more general case treated in EGA IV₃, 11.9.10 b).

⁽²⁴⁾**N.D.E.** Specify this. . .

Suppose for simplicity U retrocompact in X (cf. EGA IV₃, 11.10.10 and 11.9.17 for the general case). The case of X locally noetherian is mentioned only for the record; it is contained in 4.3.

In the second case considered, one may evidently suppose S and X affine, so X and U are of finite presentation over $S = \operatorname{Spec}(A)$. The ring A is the inductive limit of its $\mathbf{Z}$ -subalgebras A_i of finite type. The “patented procedure” already used (cf. EGA IV₃, 8.8.2 and 8.10.5 (iii)) shows that there exists an i , an affine scheme X_i over $S_i = \operatorname{Spec}(A_i)$, and an open subset U_i in X_i , from which X, U are deduced by base change $S \rightarrow S_i$. Let E_i be the subset of S_i consisting of the $s \in S_i$ such that $(U_i)_s$ is schematically dense in $(X_i)_s$.

(25) By EGA IV₂, 5.9.9 and 5.10.2, E_i is the set of $s \in S_i$ such that $(U_i)_s$ contains the set $\operatorname{Ass} \mathcal{O}_{(X_i)_s}$ of points “associated” with the structure sheaf of $(X_i)_s$, and by EGA IV₃, 11.9.17.1 this condition is of constructible nature, i.e. E_i is a constructible subset of S_i .

(25) By 4.5, the inverse image of E_i by $S_j \rightarrow S_i$ (resp. by $S \rightarrow S_i$) is E_j (resp. the set E of $s \in S$ such that U_s is schematically dense in X_s). Moreover, by hypothesis $E = S$, which is also the inverse image by each $S \rightarrow S_i$ of S_i . By EGA IV₃, 8.3.11, this implies that there exists a $j \geq i$ such that $E_j = S_j$, i.e. such that for every $s \in S_j$, $(U_j)_s$ is schematically dense in $(X_j)_s$.

Then, by 4.4 applied to (S_j, X_j, U_j) and to the base change $S \rightarrow S_j$, it follows that U is schematically dense in X . One will note moreover that here one uses 4.4 only in the case of a family whose index set is finite (in fact reduced to one element), in which case the proof of 4.4 simplifies considerably, as the reader will verify for himself.

Theorem 4.7. *Let S be a prescheme, G a group of multiplicative type and of finite type over S . For every integer $n > 0$, let ${}_nG$ be the kernel of $n \cdot \operatorname{id}_G$. Then the family of subpreschemes $({}_nG)_{n>0}$ of G is schematically dense (def. 4.1).*

We distinguish two cases.

a) Case of S locally noetherian. Then by 4.3 one is reduced to the case where S is the spectrum of a field k . By 4.5.0, one may suppose k algebraically closed and G diagonalizable, i.e. of the form $D_k(M)$, where M is a commutative group of finite type. Then M is of the form $\Gamma \times \mathbf{Z}^r$, with Γ finite, so G is of the form $G_1 \times G_2$, with $G_1 = D(\Gamma)$ and $G_2 = \mathbf{G}_m^r$, and for n multiplicatively large (namely n a multiple of the order of Γ) one shall have ${}_nG = G_1 \times {}_nG_2$ since one shall have ${}_nG_1 = G_1$. Applying 4.3 again to the projection $G \rightarrow G_1$, one is reduced to the case of G_2 , i.e. to the case where $G = \mathbf{G}_m^r$. As G is then reduced, it amounts to the same thing to say that $({}_nG)_{n>0}$ is schematically dense in G or that the union of the ${}_nG$ is dense in G for the usual topology. Since ${}_nG = ({}_n\mathbf{G}_m)^r$, one is reduced to the case of $G = \mathbf{G}_m$, hence G irreducible of dimension 1. Then this follows from the fact that the union of the ${}_nG$ (equal to the set of roots of unity in k) is infinite.

b) General case. For every point s of S , there exists an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \rightarrow U$ such that $G' = G_{S'}$ is diagonalizable, i.e. of the form $D_{S'}(M)$. By 4.5.0 again, one may reduce to the case of G' , so one may suppose G diagonalizable, hence it comes from the absolute group $H = D_{\mathbf{Z}}(M)$ over $\operatorname{Spec}(\mathbf{Z})$ by base change $S \rightarrow \operatorname{Spec}(\mathbf{Z})$. By the proof of a), for every $s \in \operatorname{Spec}(\mathbf{Z})$ the family $({}_nH_s)_{n>0}$ is schematically dense in H ; it now suffices to apply 4.4.⁽²⁶⁾

Corollary 4.8. *a) Let $u, v : H \rightarrow G$ be two homomorphisms of S -group preschemes, with H of multiplicative type⁽²⁷⁾ and of finite type, and suppose that for every integer $n > 0$ the restrictions of u and v to ${}_nH$ are identical. Then $u = v$.*

(25) **N.D.E.** The original has been expanded in what follows.

(26) **N.D.E.** This result will be needed in X 4.3 for S not locally noetherian (namely $S = \operatorname{Spec}(\widehat{A} \otimes_A \widehat{A})$, where $\widehat{A}$ is the completion of the noetherian local ring A).

(27) **N.D.E.** “Diagonalizable” has been replaced by “of multiplicative type”.

b) Let H_1, H_2 be two subgroups of multiplicative type and of finite type of a group prescheme G , and suppose that for every integer $n > 0$ one has ${}_nH_1 = {}_nH_2$; then $H_1 = H_2$.

The first assertion follows from the second by considering the subgroups H_1 and H_2 of $H \times G$, graphs of u and v . To prove b), let $H = H_1 \cap H_2$; it is a subgroup prescheme of H_i ($i = 1, 2$), and one must show that it is identical to H_i ; now the hypothesis means that it contains the ${}_nH_i$. One is therefore reduced to proving the

Corollary 4.9. *Let G be an S -group of multiplicative type and of finite type, and H a subgroup prescheme containing the ${}_nG$, $n > 0$. Then $H = G$.*

By 4.7 one is reduced to proving that the subprescheme H is closed, or again that it equals G set-theoretically. This reduces us to the case where S is the spectrum of a field, but then every subgroup prescheme of G is closed (VI_B 1.4.2), whence the conclusion.

Remark 4.10. Under the conditions of 4.7, let m be an integer > 0 with the following properties: for every $s \in S$, m is not a power of the characteristic of $\kappa(s)$, and if G_s is of type M , the prime divisors of the torsion of M divide m . (N.B. this second condition is always satisfied if G is a torus.) Then the proof given shows that in the statement of 4.7 and corollaries 4.8 and 4.9 one may restrict to considering the subgroups of the form $(m^r)G$, with $r > 0$.

Moreover, one sees immediately that when G is smooth over S , for every point $s \in S$ there exists an open neighborhood U of s and an integer $m > 0$ prime to all the residue characteristics of U , and satisfying the preceding conditions for G_U . (Take, for example, for m the order of the torsion of the type of G at s , cf. 1.4.1 a) and 2.1 e).) Under these conditions one finds $(m^r)G$ which are finite and étale over S , and whose family is schematically dense, provided one restricts over U . This remark allows one in certain cases (in particular those involving theorems 3.2 and 3.2 bis, cf. XI 6, but not those involving theorems 3.6 and 3.6 bis) to dispense with theorem 3.1, which involves Hochschild cohomology.

5. Central homomorphisms of groups of multiplicative type

Lemma 5.0. ⁽²⁸⁾ *Let $(A, \mathfrak{m})$ be a noetherian local ring, $S = \operatorname{Spec}(A)$, and H a finite scheme over S , hence $H = \operatorname{Spec}(B)$, where B is finite over A . Let K be a subscheme of H such that by reduction modulo $\mathfrak{m}^{n+1}$ one has $K_n = H_n$ for every n . Then $K = H$.*

Let s be the closed point of S . Let us first note that K is a closed subscheme of H . Indeed, it is a priori a closed subscheme of an open subset U of H . But K , and hence U , contains the fiber $K_s = H_s$; since the morphism $H \rightarrow S$ is finite, hence closed, it follows that the complement of U is empty, i.e. $U = H$. Thus K is defined by an ideal I of B . The hypothesis implies that I is contained in $\mathfrak{m}^n B$ for every n ; since B is a finite A -module, it is separated for the $\mathfrak{m}$ -adic topology, whence $I = 0$.

Theorem 5.1. *Let $u, v : H \rightarrow G$ be two homomorphisms of S -group preschemes, with H of multiplicative type and of finite type, and S locally noetherian or G of finite presentation over S . Let $s \in S$ be such that $u_s = v_s$, and suppose u_s central. Then there exists an open neighborhood U of s such that $u_U = v_U$.*

Let us distinguish the two cases:

a) S locally noetherian.⁽²⁹⁾ Let $K = \operatorname{Ker}(u, v)$ be the inverse image of the diagonal of $G \times_S G$ by the morphism (u, v) ; it is a subgroup prescheme of H . We want to find U such that $K_U = H_U$. Note that, as S is locally noetherian and H of finite type over S , H is

⁽²⁸⁾**N.D.E.** This lemma has been made explicit; it is used several times in what follows (implicitly in the original).

⁽²⁹⁾**N.D.E.** The original has been expanded in what follows.

locally noetherian, so the immersion $K \hookrightarrow H$ is of finite type (cf. EGA I, 6.3.5). Thus K is of finite type over S , hence of finite presentation over S , since S is locally noetherian. Consequently, by EGA IV₃, 8.8.2.4, to show that there exists an open neighborhood U of s such that $K_U = H_U$, it suffices to show that $K_{S'} = H_{S'}$, where S' is the spectrum of $A = \mathcal{O}_{S,s}$. One may therefore suppose S local with closed point s . Replacing the noetherian local ring A by its completion $\hat{A}$ if necessary, which introduces a faithfully flat and quasi-compact base change $\hat{S} \rightarrow S$, one may even suppose⁽³⁰⁾ A complete if one wishes.

By 3.4 one has $u_n = v_n$ for every n , where, as usual, the subscript n indicates reduction modulo $\mathfrak{m}^{n+1}$ ($\mathfrak{m}$ being the maximal ideal of A). For every integer $m > 0$, denoting by ${}_m u, {}_m v$ the homomorphisms ${}_m H \rightarrow G$ induced by u, v , one therefore also has $({}_m u)_n = ({}_m v)_n$. This being true for every n , and ${}_m H$ being finite over S by 2.2, it follows that ${}_m u = {}_m v$ by 5.0. This being true for every m , one therefore has $u = v$ by 4.7.

b) G of finite presentation.⁽³¹⁾ Since H is also of finite presentation over S , by EGA IV₃, 8.8.2, we may again suppose S local with closed point s and prove that then $u = v$. If $f : S' \rightarrow S$ is a faithfully flat quasi-compact morphism, and if one denotes by f_H the morphism $H' \rightarrow H$ and by $u', v' : H' \rightarrow G'$ the morphisms deduced from u, v , then the equality $u' = v'$ implies $u' \circ f_H = v' \circ f_H$, whence $u = v$, since f_H is an epimorphism. Thus, making a faithfully flat and quasi-compact extension of the base if necessary, one may suppose in addition H diagonalizable, hence of the form $D_S(M)$, with M a commutative group of finite type. Introduce, as in the proof of 4.6, the increasing filtering family of $\mathbf{Z}$ -subalgebras of finite type A_i of $A = \mathcal{O}_{S,s}$, and $S_i = \text{Spec}(A_i)$.⁽³¹⁾ Note that $H = D_S(M)$ comes, for every i , from the diagonalizable group $H_i = D_{S_i}(M)$. Since G is of finite presentation over S , by EGA IV₃, 8.8.2 (see also VI_B, 10.2 and 10.3), there exists an index i , a group prescheme G_i of finite presentation over S_i , and morphisms of S_i -groups $u_i, v_i : H_i \rightarrow G_i$ from which u, v come by base change. Let s_i be the image of s in S_i , and let $\rho_i : H_i \times_{S_i} G_i \rightarrow G_i$ be the morphism of S_i -preschemes defined by $\rho_i(h, g) = u_i(h)g u_i(h)^{-1}$. Then, since u_s is central, $\rho_s = \rho_i \times_{\kappa(s_i)} \kappa(s)$ equals the second projection, and hence the same is true of ρ_i (since $\kappa(s_i) \rightarrow \kappa(s)$ is faithfully flat and quasi-compact), i.e. $(u_i)_{s_i}$ is central. Similarly, since $u_s = v_s$, one has $(u_i)_{s_i} = (v_i)_{s_i}$. One may then apply a) to the situation over S_i , whence immediately the announced conclusion.

Corollary 5.2. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, and S locally noetherian or G of finite presentation over S . Let $s \in S$ and suppose that u_s is the unit homomorphism. Then there exists an open neighborhood U of s such that u_U is the unit homomorphism.*

Corollary 5.3. *Let u, H, G be as in 5.2, and suppose in addition G separated over S . Let U be the set of $s \in S$ such that u_s is the unit homomorphism. Then U is an open and closed subset of S , and $u_U : H_U \rightarrow G_U$ is the unit homomorphism.*

It remains only to prove that U is closed. Now let $K = \text{Ker}(u)$; since G is separated over S , K is a closed subscheme of H , and U is the set of $s \in S$ such that $K_s = H_s$. One then easily sees that U is closed, for example as an application of VIII 6.4 (H being essentially free over S by VIII 6.3), or by noting that one may suppose S reduced and U dense in S , hence schematically dense in S , which implies H_U schematically dense in H since H is flat over S ,⁽³²⁾ and as u and the unit homomorphism coincide on H_U , they coincide on H .

⁽³⁰⁾**N.D.E.** For $K = H$ if $K_{\hat{S}} = H_{\hat{S}}$, cf. SGA 1, VIII 5.4; but what follows does not use the hypothesis “ A complete”.

⁽³¹⁾**N.D.E.** The original has been expanded in what follows.

⁽³²⁾**N.D.E.** And of finite presentation over S , cf. EGA IV₃, 11.10.5 (ii) and 11.9.10 (ii).

Corollary 5.4. *Let S be a prescheme, H and G two S -groups, with H of multiplicative type and of finite type and G separated and of finite presentation over S , $\pi : S' \rightarrow S$ a⁽³³⁾ universal effective epimorphism (for example a faithfully flat and quasi-compact morphism, or a morphism admitting a section, cf. IV 1.12) with geometrically connected fibers.⁽³⁴⁾*

Let $u' : H' \rightarrow G'$ be a central homomorphism of the S' -groups deduced from H, G by the base change $S' \rightarrow S$. Then there exists a unique homomorphism $u : H \rightarrow G$ of S -groups such that $u \times_S S' = u'$. When S'/S admits a section g , u is the morphism deduced from u' by the base change $g : S \rightarrow S'$.

Since π is a universal effective epimorphism, the same is true of $H' \rightarrow H$, whence uniqueness of u , cf. the beginning of the proof of 5.1 b). If π admits a section g , then $u' = \pi^*(u)$ implies $u = g^*\pi^*(u) = g^*(u')$.

For the existence of u , one is reduced, by IV 2.3, to showing that the two homomorphisms $u'_1, u'_2 : H'' \rightarrow G''$ of S'' -groups deduced from u' by the two base changes $\text{pr}_1, \text{pr}_2 : S'' = S' \times_S S' \rightarrow S'$ are identical. Now they coincide on the diagonal of S'' ; more precisely, the inverse images of u'_1 and u'_2 by the diagonal morphism $S' \rightarrow S''$ are identical (being equal to u').⁽³⁵⁾ Since u'_1 and u'_2 are central, one may apply 5.3 to the morphism $u'_1(u'_2)^{-1}$. There therefore exists an open and closed subset U of S'' , containing the diagonal of S'' , such that u'_1 and u'_2 coincide over U .

Now the fibers of S'/S being geometrically connected, the same is true of those of S''/S , which are therefore a fortiori connected, whence it follows that U (containing the diagonals of the said fibers) contains the said fibers, and hence equals S'' , which completes the proof.

Corollary 5.5. *Let S be a prescheme, K an S -group prescheme locally of finite type⁽³⁶⁾ and with connected fibers, H an invariant subgroup of multiplicative type and of finite type of K . Then H is a central subgroup of K .*

⁽³⁷⁾ Let us first note that $\pi : K \rightarrow S$ has geometrically connected fibers, since for a group scheme locally of finite type over a field, connected implies geometrically connected (cf. VI_A 2.4). One may then apply 5.4 by putting $G = H$ and $S' = K$ to the homomorphism of K -groups $u' : H_K \rightarrow H_K$ defined set-theoretically by $(h, k) \mapsto (khk^{-1}, k)$, which is central since H is commutative. The inverse image of u' by the unit section $\varepsilon : S \rightarrow K$ is the identity homomorphism of K , so by 5.4 the same is true of $H_K \rightarrow H_K$, hence H is central in K . Let us state the variants of the preceding results for central subgroups of multiplicative type. One obtains, proceeding as for the preceding results (and using 3.2 bis):

Theorem 5.1 bis. *Let G be an S -group prescheme, H_1 and H_2 two subgroup preschemes of multiplicative type and of finite type, with $(H_1)_s$ central. Suppose S locally noetherian or G of finite presentation over S . Then for every $s \in S$ such that $(H_1)_s = (H_2)_s$ there exists an open neighborhood U of s such that $H_1|U = H_2|U$.*

Corollary 5.3 bis. *Under the preceding conditions, suppose in addition G separated over S . Let U be the set of $s \in S$ such that $(H_1)_s = (H_2)_s$. Then U is an open and closed subset of S , and $H_1|U = H_2|U$.*

⁽³³⁾**N.D.E.** “Covering morphism for the faithfully flat and quasi-compact topology” has been replaced by “universal effective epimorphism”, in order to be able to apply this to the morphism $K \rightarrow S$ of 5.5, and the beginning of the proof has been modified accordingly.

⁽³⁴⁾**N.D.E.** This is the case, for example, if $S = \text{Spec } k$ and $S' = \text{Spec } k'$, where k is a field and k' a radical extension of k , cf. X, Prop. 1.4.

⁽³⁵⁾**N.D.E.** The following sentence has been added.

⁽³⁶⁾**N.D.E.** The hypothesis “locally of finite type” has been added to agree with the reference VI_A, 2.4. In fact, one can show that over a field every connected group scheme is geometrically connected.

⁽³⁷⁾**N.D.E.** The original has been expanded in what follows.

Corollary 5.4 bis. *Let S be a prescheme, G a group prescheme separated and of finite presentation over S , $S' \rightarrow S$ a covering morphism for the faithfully flat quasi-compact topology, with geometrically connected fibers, H' a subgroup of multiplicative type and of finite type of $G' = G \times_S S'$.*

Then there exists a unique subgroup H of G such that $H' = H \times_S S'$, and H is of multiplicative type and of finite type over S , by 1.1 and 2.1 b). If $S' \rightarrow S$ admits a section g , H is the inverse image of the subgroup H' of G' by $g : S \rightarrow S'$.

Theorem 5.6. *Let S be a prescheme, $u : H \rightarrow G$ a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, G of finite presentation over S .*

a) Suppose G to have connected fibers, and let $s \in S$ be such that $u_s : H_s \rightarrow G_s$ is a central homomorphism. Then there exists an open neighborhood U of s such that the homomorphism $H|_U \rightarrow G|_U$ induced by u is central.

b) Suppose that for every $s \in S$, $u_s : H_s \rightarrow G_s$ is central. Then u is central.

⁽³⁸⁾ Let us prove (a). Proceeding exactly as in the proof of 5.1 (b), one may suppose S local with closed point s , then $H = D_S(M)$ diagonalizable, then that there exists a $\mathbf{Z}$ -subalgebra of finite type A_i of $A = \mathcal{O}_{S,s}$ and a morphism $u_i : D_{S_i}(M) \rightarrow G_i$ such that u_{s_i} is central (where s_i is the image of s in S_i) and from which u comes by base change. This reduces us to the case where S is local noetherian with closed point s , and one must then prove that u is central.

Let K be the subprescheme $\text{Ker}(v, w)$, where $v, w : H \times_S G \rightrightarrows G$ are defined by

$$v(h, g) = g, \quad w(h, g) = \text{int}(u(h)) \cdot g = u(h)gu(h)^{-1}.$$

Then K is a sub- G -group of the G -group $H_G = H \times_S G$, and one wants to show that it equals H_G itself. By 4.9 one is reduced to proving that it contains the $_m(H_G) = (_m H) \times_S G$ for every integer $m > 0$, which reduces us to the case where $H = _m H$, hence H finite over S .

Let e be the unit element of the fiber G_s ; then $S' = \text{Spec}(\mathcal{O}_{G,e})$ is a noetherian local scheme (G being of finite presentation over noetherian S); put $S'_n = S' \times_S S_n$, where $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$. Then $K_{S'} = K \times_G S'$ is a subprescheme of $H_{S'} = H \times_S S'$, and by 3.9 one has $K_{S'_n} = H_{S'_n}$ for every n .⁽³⁹⁾ Since $H_{S'}$ is finite over S' , one concludes from 5.0 (applied to the noetherian local ring B) that $K_{S'} = H_{S'}$.

On the other hand, since $H \times_S G$ is noetherian (being of finite presentation over noetherian S), the immersion $K \hookrightarrow H \times_S G$ is of finite type (cf. EGA I, 6.3.5), so K is of finite type, hence of finite presentation over G . Then the equality $K_{S'} = H_G \times_G S'$ implies, by EGA IV₃, 8.8.2.4, that there exists an open neighborhood W of e in G such that $K \times_G W = H_G \times_G W = H \times_S W$. Thus K contains the open neighborhood $V = H \times_S W$ of the unit section of G_H over H .

For every $t \in H$, the fiber G_t (being an algebraic $\kappa(t)$ -group) is Cohen-Macaulay (VI_A, 1.1.1), and hence without embedded components; as it is in addition connected, hence irreducible (VI_A, 2.4), it has its generic point as its unique associated point. Thus by EGA IV₂, 3.1.8, the open subset V_t is schematically dense in G_t . By 4.3, V is therefore schematically dense in G_H . Moreover, since K is a sub- H -group of G_H , it induces on each fiber G_h a subgroup K_h , and as the latter contains an open neighborhood of the neutral element and G_h is connected, it follows that $K_h = G_h$, whence $K = G_H$ set-theoretically. Thus K is a closed subprescheme of G_H containing the schematically dense subprescheme V , hence $K = G_H$, which proves a).

Finally, b) is a direct consequence of 3.9, in view of the fact that to check that a subprescheme K of $P = G \times_S H$ is identical to P , it suffices to check that for every base change $S' \rightarrow S$, where S' is an artinian quotient of a local ring of S , one has $K' = P'$.

⁽³⁸⁾**N.D.E.** The original has been expanded in the proof of a).

⁽³⁹⁾**N.D.E.** The original has been expanded and corrected, replacing “unit section of $(G_H)_n$ over H_n ” by $H_{S'_n}$.

6. Monomorphisms of groups of multiplicative type, and canonical factorization of a homomorphism of such a group

Lemma 6.1. *Let S be a quasi-compact prescheme, G a commutative S -group prescheme, of finite presentation and quasi-finite over S . Then there exists an integer $n > 0$ such that $n \cdot \text{id}_G = 0$, i.e. $G = {}_n G$.*

If S is the spectrum of a field k , G is finite over k , and by VII_A 8.5 it suffices to take $n = \deg(G/k)$. Suppose now $S = \text{Spec}(A)$ local artinian; let k be the residue field of A , $G_0 = G \otimes_A k$, and $n_0 = \deg(G_0/k)$. Distinguish two cases.

a) k is of characteristic zero. Then G_0 is separable over k , hence G unramified over S . Then the unit section of G is an open immersion, hence ${}_n G = \text{Ker}(n \cdot \text{id}_G)$ is an open subscheme of G , so for it to equal G it suffices that it do so set-theoretically; one may therefore then take $n = n_0$.

b) k is of characteristic $p > 0$. Denote by $\mathfrak{m}$ the maximal ideal of A and put $S_r = \text{Spec}(A/\mathfrak{m}^{r+1})$ for every r . Let m be an integer such that $\mathfrak{m}^{m+1} = 0$; I say that one may take $n = p^m n_0$. Indeed, proceeding by induction on m and putting $n' = p^{m-1} n_0$, one may suppose that $n' \cdot \text{id}_G$ induces the zero endomorphism in $G_{m-1} = G \times_S S_{m-1}$.

⁽⁴⁰⁾ Denote by E the set of endomorphisms u of G having this property and by K the kernel of the morphism of group functors $G \rightarrow \prod_{S_{m-1}/S} G$ (cf. III 0.1.2). Then E is identified with $\text{Hom}_{S\text{-gr.}}(G, K)$; in particular the abelian group law on E is induced by that of K . Now by III 0.9, K is the S -group functor associating with every $g : T \rightarrow S$ the k -vector space

$$\text{Hom}_{\mathcal{O}_{T_0}}(g_0^*(\Omega_{G_0/S_0}^1), \mathfrak{m}^m \mathcal{O}_T);$$

one therefore has $pu = 0$ for every $u \in E$. Thus one has $pn' \cdot \text{id}_G = 0$, i.e. $p^m n_0 \cdot \text{id}_G = 0$.

Suppose now S noetherian (one reduces to this in 6.1 by the usual reduction to the noetherian case⁽⁴¹⁾). It then suffices to combine the preceding with the

Lemma 6.2. *Let X be a prescheme quasi-finite over noetherian S , and consider an increasing filtering family of subpreschemes $(X^i)_{i \in I}$ having the following property:*

for every $s \in S$ and $n \geq 0$, putting $S_{s,n} = \text{Spec}(\mathcal{O}_{S,s}/\mathfrak{m}^{n+1})$, there exists $i \in I$ such that $X^i \times_S S_{s,n} = X \times_S S_{s,n}$.

Under these conditions there exists an $i \in I$ such that $X^i = X$.

Since S is noetherian, there exists a maximal open subset U such that one has $X|U = X^i|U$ for large i ; one shall show that $U = S$. In other words, one shall show that if $U \neq S$, one can find a U' strictly larger than U , and an i such that $X|U' = X^i|U'$. Localizing at a maximal point s of $S - U$, one is reduced to the case where S is local with closed point s and $U = S - \{s\}$. (Indeed, if one denotes by $S' = \text{Spec}(\mathcal{O}_{S,s})$ and if there exists i such that $X \times_S S' = X^i \times_S S'$, then⁽⁴²⁾ there exists an open neighborhood V of s such that $X|V = X^i|V$; thus, taking i sufficiently large that $X^i|U = X|U$, one shall have $X|W = X^i|W$, where $W = U \cup V$.)

Then for large i , since $X|U = X^i|U$, one sees that X^i is a closed subprescheme of X defined by an ideal $\mathcal{J}^{(i)}$ with support contained in $X_s = X_0 = X \times_S S_0$ (where $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, $S = \text{Spec}(A)$). As X_0 is quasi-finite over S_0 , X_0 is a finite closed subset of noetherian X ,

⁽⁴⁰⁾**N.D.E.** The original has been expanded in what follows, specifying the results of Exp. III that are used.

⁽⁴¹⁾**N.D.E.** Since S is quasi-compact, it is covered by finitely many affine open subsets, so one is reduced to the case where $S = \text{Spec}(A)$. Then, since G is of finite presentation over S , one may apply EGA IV₃, 8.8.2.

⁽⁴²⁾**N.D.E.** $X - \{s\}$ has been corrected above to $S - \{s\}$. On the other hand, recall that a quasi-finite morphism is supposed of finite type, cf. EGA II, 6.2.3 (and EGA III₂, Err_{III} 20 for the definition of “locally quasi-finite”). Thus here (S being noetherian), X is of finite presentation over S , each immersion $X^i \hookrightarrow X$ is of finite type (by EGA I, 6.3.5), so X^i is of finite presentation over S , and one may apply EGA IV₃, 8.8.2.

so $\mathcal{J}^{(i)}$ is a module of finite length. It follows that there exists an integer $n \geq 0$ such that $\mathcal{J}^{(i)} \cap \mathfrak{m}^{n+1}\mathcal{O}_X = 0$. On the other hand, by the hypothesis in 6.2 one may suppose (on increasing i) that the image of $\mathcal{J}^{(i)}$ in $\mathcal{O}_{X_n} = \mathcal{O}_X/\mathfrak{m}^{n+1}\mathcal{O}_X$ is zero. This implies $\mathcal{J}^{(i)} = 0$, hence $X^i = X$. Q.E.D.

Lemma 6.3. *Let S be a prescheme, K a group prescheme over S , locally of finite presentation, $s \in S$ such that K_s is quasi-finite (resp. unramified) over $\kappa(s)$ at the unit element. Then there exists an open neighborhood U of s such that $K|U$ is locally quasi-finite (resp. unramified) over U .⁽⁴³⁾*

Indeed, let V be the set of points x of K such that, denoting by t the image of x in S , the fiber K_t is quasi-finite (resp. unramified) over $\kappa(t)$ at x , i.e. such that x is isolated in K_t (resp. and its local ring in K_t a separable extension of $\kappa(t)$). One knows that V is open since K is locally of finite presentation over S ,⁽⁴⁴⁾ so if ε denotes the unit section of K , $\varepsilon^{-1}(V)$ is open. By hypothesis it contains s , so it is an open neighborhood U of s . The latter does the job; in other words $t \in U$ implies that K_t is locally quasi-finite (resp. unramified) over $\kappa(t)$: indeed, since K_t is a group locally of finite type over $\kappa(t)$, this follows from the fact that it is quasi-finite (resp. unramified) over $\kappa(t)$ at the point $\varepsilon(t)$, cf. VI_B 1.3.

Combining 6.1 and 6.3 one finds the

Theorem 6.4. *Let S be a prescheme, H an S -group of multiplicative type and of finite type, K a closed subgroup prescheme, of finite presentation over S . Let $s \in S$ be such that K_s is finite.*

Then there exists an open neighborhood U of s such that $K|U$ is contained in ${}_nH|U$ for some n , and a fortiori (${}_nH$ being finite over S) such that $K|U$ is finite over U .

Using Nakayama's lemma, one deduces:

Corollary 6.5. *With the preceding notation, suppose that K_s is the unit group. Then there exists an open neighborhood U of s such that $K|U$ is the unit group.*

Corollary 6.6. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, and G separated over S . Suppose in addition S locally noetherian, or G locally of finite presentation⁽⁴⁵⁾ over S . Let $s \in S$ be such that $u_s : H_s \rightarrow G_s$ is a monomorphism; then there exists an open neighborhood U of s such that u induces a monomorphism $H|U \rightarrow G|U$.*

Indeed, let $K = \text{Ker}(u)$; the hypothesis on u_s means that K_s is the unit group, the conclusion that $K|U$ is the unit group for suitable U . Now G being separated over S , K is a closed subgroup of H , and in the case where one does not suppose S locally noetherian, but on the other hand G locally of finite presentation over S , K is locally of finite presentation over S ,⁽⁴⁵⁾ and hence of finite presentation over S since it is separated over S (H being so) and quasi-compact over S (being closed in H , which is quasi-compact over S). One may therefore apply 6.5, whence 6.6.

Remark 6.6.1.⁽⁴⁶⁾ Under the hypotheses of 6.6, one will note that when G is in addition affine over S (resp. of finite presentation over S), $H|U \rightarrow G|U$ is even a closed immersion, as pointed out in 2.5 (resp. 2.6).

⁽⁴³⁾Cf. VI_B 2.5 (ii) for a more general statement.

⁽⁴⁴⁾**N.D.E.** Cf. EGA IV₃, 13.1.4 and EGA IV₄, 17.4.1. Moreover, the image of x has been denoted by t (instead of s).

⁽⁴⁵⁾**N.D.E.** In fact it suffices to suppose G locally of finite type over S ; by EGA IV₄, 1.4.3 (v), this implies ($H \rightarrow S$ being of finite presentation) that $H \rightarrow G$ is locally of finite presentation, and hence also $K \rightarrow S$, which is deduced from it by base change.

⁽⁴⁶⁾**N.D.E.** The numbering 6.6.1 has been added, for later references.

Corollary 6.7. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type; suppose S locally noetherian or G locally of finite presentation over S .*

If all the homomorphisms induced on the fibers $u_s : H_s \rightarrow G_s$ are monomorphisms, then u is a monomorphism.

The reasoning is the same as in 6.6, the hypothesis that G is separated over S being unnecessary here to ensure that K is closed in H , since the hypothesis that the u_s are monomorphisms implies that K is set-theoretically reduced to the unit section of G .

Theorem 6.8. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, with H of multiplicative type and of finite type, and G separated over S . Suppose in addition S locally noetherian or G of finite presentation over S .*

Then $K = \text{Ker}(u)$ is a subgroup of multiplicative type and of finite type of H , and u factors as

$$H \xrightarrow{u'} H/K \xrightarrow{u''} G,$$

where H/K is of multiplicative type and of finite type, u' is the canonical homomorphism and is faithfully flat (and affine), and u'' is a monomorphism.

(N.B. As remarked in 6.6.1, u'' is a closed immersion if G is affine over S or of finite presentation over S .) It suffices to prove that K is of multiplicative type, the rest of the proposition then following from 2.7 and IV 5.2.6.

Suppose first G of finite presentation over S . This hypothesis being stable under base change, to prove that K is of multiplicative type one is reduced to the case where H is diagonalizable, i.e. $H = D_S(M)$, where M is a commutative group of finite type. Let $s \in S$; then K_s is a closed subgroup of $H_s = D_{\kappa(s)}(M)$, hence by 8.1⁽⁴⁷⁾ is of the form $D_{\kappa(s)}(N)$, where N is a quotient group of M . Put $K' = D_S(N)$; then K' is a subgroup of multiplicative type of H ; let $v' : K' \rightarrow G$ be induced by u .⁽⁴⁸⁾ Then v'_s is the unit homomorphism by construction, so by 5.2 there exists an open neighborhood U of s such that $v'|_U : K'|_U \rightarrow G|_U$ is the unit homomorphism. Thus, replacing S by U if necessary, one may suppose that v' is the unit homomorphism, and hence that u factors as

$$H \xrightarrow{u'} H/K' \xrightarrow{u''} G.$$

Now, since u''_s is deduced from u_s by factoring through $H_s \rightarrow H_s/\text{Ker}(u_s)$, u''_s is a monomorphism (IV 5.2.6), so by 6.6 there exists an open neighborhood U of s such that $u''|_U : (H/K')|_U \rightarrow G|_U$ is a monomorphism. Thus, restricting S if necessary, one sees that u'' is a monomorphism, hence $\text{Ker}(u) = \text{Ker}(u') = K'$, which proves that $\text{Ker } u$ is of multiplicative type.

The same proof is valid if, instead of supposing G of finite presentation over S , one supposes S locally noetherian, at least in the case where H is diagonalizable. In the case where one does not make this hypothesis on H , one must show that one can find a covering base change $S' \rightarrow S$, with S' locally noetherian, which trivializes H . This is indeed what one will see in the next exposé (X 4.6).

7. Algebraicity of formal homomorphisms into an affine group

Theorem 7.1. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/I^{n+1})$, H, G S -group preschemes, with H of isotrivial multiplicative type and G affine.*

⁽⁴⁷⁾N.D.E. The erroneous reference to IV 4.7.5 has been corrected. Note that no. 8 of the present exposé is independent of nos. 2 to 7.

⁽⁴⁸⁾N.D.E. “Induced by G ” has been corrected to “induced by u ”.

Then the canonical map

$$(x) \quad \theta : \operatorname{Hom}_{S\text{-gr.}}(H, G) \longrightarrow \varprojlim_n \operatorname{Hom}_{S_n\text{-gr.}}(H_n, G_n)$$

is bijective (where H_n, G_n are the S_n -groups deduced from H, G by the base change $S_n \longrightarrow S$).

Suppose first H diagonalizable, hence of the form

$$H = \operatorname{Spec} A(M),$$

where $B = A(M)$ is the group algebra of the commutative group M with coefficients in A . One also has $G = \operatorname{Spec}(C)$, where C is an A -algebra endowed with a diagonal map (satisfying the known axioms). Then the homomorphisms of S -groups $H \longrightarrow G$ correspond bijectively to the homomorphisms of A -algebras $\varphi : C \longrightarrow B$ compatible with the diagonal maps, i.e. such that for every $f \in C$,

$$\Delta_H(\varphi(f)) = (\varphi \otimes \varphi)(\Delta_G(f))$$

where Δ_H and Δ_G are the diagonal maps. One has an analogous description for the homomorphisms of S_n -groups $H_n \longrightarrow G_n$, defined by certain homomorphisms of A_n -algebras $\varphi_n : C_n \longrightarrow B_n$ (where one puts $A_n = A/I^{n+1}$, $B_n = B \otimes_A A_n$, $C = C \otimes_A A_n$). Put

$$\widehat{B} = \varprojlim_n B_n \quad \text{and} \quad \widehat{C} = \varprojlim_n C_n;$$

then $\widehat{B} = \varprojlim_n A_n(M)$ is identified with a submodule of the product A^M (namely that consisting of the families $(a_m)_{m \in M}$ of elements of A tending to 0 (for the I -adic topology) along the filter of complements of finite subsets of M). This already implies that the canonical homomorphism $B \longrightarrow \widehat{B}$ is injective,⁽⁴⁹⁾ since the projective system $\theta(\varphi) = (\varphi_n)_{n \geq 0}$ defines a homomorphism $\widehat{\varphi} = \widehat{C} \longrightarrow \widehat{B}$ such that the diagram below is commutative:

$$\begin{array}{ccc} C & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ \widehat{C} & \xrightarrow{\widehat{\varphi}} & \widehat{B}. \end{array}$$

Let us prove that θ is surjective: take a projective system $(\varphi_n)_{n \geq 0}$ and prove that it comes by reduction from a φ in the left-hand side. A priori, (φ_n) defines a homomorphism on the completed algebras

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B},$$

and everything amounts to seeing that its composite $\Phi : C \longrightarrow \widehat{C} \xrightarrow{\widehat{\varphi}} \widehat{B}$ with the canonical homomorphism $C \longrightarrow \widehat{C}$ maps C into B . Indeed, if this is so, one finds a homomorphism of A -algebras $\varphi : C \longrightarrow B$ reducing to the φ_n , whence one concludes immediately that it is compatible with the diagonal maps (since the φ_n are so and $B \otimes_A B \longrightarrow \widehat{B \otimes_A B}$ is injective, as one sees as above on replacing M by $M \times M$).

Let us note that the diagonal homomorphism Δ_H of H defines, on passing to completions, a homomorphism

$$\widehat{\Delta}_H : \widehat{B} \longrightarrow \widehat{B \otimes_A B},$$

and one has a commutative diagram (deduced by passing to the projective limit from the analogous diagrams defined by the φ_n):

$$\begin{array}{ccc} C & \xrightarrow{\Phi} & \widehat{B} \\ \Delta_G \downarrow & & \downarrow \widehat{\Delta}_H \\ C \otimes_A C & \xrightarrow{\Psi} & \widehat{B \otimes_A B}, \end{array}$$

⁽⁴⁹⁾**N.D.E.** What follows has been expanded.

where Ψ is the composite

$$C \otimes_A C \xrightarrow{\Phi \otimes \Phi} \widehat{B} \otimes_A \widehat{B} \longrightarrow \widehat{B \otimes_A B}$$

(the last arrow is the evident canonical homomorphism). It follows that for every $f \in C$, $\Phi(f)$ is an element of $\widehat{B}$ whose image under $\widehat{\Delta}_H$ is a “decomposable” element of $\widehat{B \otimes_A B}$, i.e. is in the image of $\widehat{B} \otimes_A \widehat{B}$.

Denote by $(e_m)_{m \in M}$ the canonical basis of $A(M)$ and by $(e_{m,m'})$ that of $A(M \times M) = A(M) \otimes_A A(M)$. Since $\Delta_H(e_m) = e_m \otimes e_m = e_{m,m}$ for every m , it now suffices to apply the

Lemma 7.2. *Let A be a noetherian ring, M a set, $(a_{m,m'})$ a family of elements of A indexed by $M \times M$, such that*

(i) $a_{m,m'} = 0$ if $m \neq m'$ (i.e. the support of the family is contained in the diagonal of $M \times M$),

(ii) One has

$$a_{m,m'} = \sum_i b_m^i c_{m'}^i$$

where the b^i, c^i are finitely many elements of A^M (i.e. $(a_{m,m'})$ belongs to the image of the canonical homomorphism $A^M \otimes_A A^M \longrightarrow A^{M \times M}$).

Under these conditions, the support of the family $(a_{m,m'})$ is finite.

By (i), the family $(a_{m,m'})$ is determined by knowing the $a_m = a_{m,m}$. Put, for every $x = (x_n)_{n \in M} \in A^{(M)}$:

$$(u \cdot x)_m = \sum_{m'} a_{m,m'} x_{m'},$$

i.e. interpret $(a_{m,m'})$ as the matrix of a homomorphism $u : A^{(M)} \longrightarrow A^M$. Then by (i) one simply has

$$(u \cdot x)_m = a_m x_m.$$

On the other hand, by (ii) one has

$$u \cdot x \in \sum_i A \cdot b^i,$$

so $u(A^{(M)})$ stays in an A -module of finite type. Consequently, denoting by $(e^m)_{m \in M}$ the canonical basis of $A^{(M)} \subseteq A^M$, the $a_m e^m$ stay in an A -module of finite type. As A is noetherian, the module they generate is itself of finite type, which implies (since the e^m are linearly independent) that all but finitely many of the a_m are zero. This proves 7.2 and hence 7.1 in the case where H is diagonalizable.

Let us now prove the general case of 7.1, where one merely supposes H isotrivial, i.e. that there exists a finite surjective étale morphism $S' \longrightarrow S$ such that $H' = H \times_S S'$ is diagonalizable. We shall use only the fact that $S' \longrightarrow S$ is finite and covering (for the faithfully flat and quasi-compact topology, or simply for the canonical topology of (Sch)) — thus the hypothesis “étale” could be replaced by “flat”.

Let $S'' = S' \times_S S'$; introduce similarly S'_n and $S''_n = S'' \times_S S_n = S'_n \times_{S_n} S'_n$, and $H', G', H'', G'', H'_n, G'_n, H''_n, G''_n$ deduced from H and G by the base changes which one can guess. Note that H' and H'' are now diagonalizable. Note also that S' , and hence S'' , is affine, and that if $S' = \text{Spec}(A')$, $S'' = \text{Spec}(A'')$, then A' and A'' are separated and complete for the topology defined by IA' resp. by IA'' (since A' and A'' are finite over A). Since $S' \longrightarrow S$ and $S'_n \longrightarrow S_n$ are covering, one obtains a commutative diagram of maps of sets whose two rows are exact:

$$\begin{array}{ccccc} \text{Hom}_{S\text{-gr.}}(H, G) & \longrightarrow & \text{Hom}_{S'\text{-gr.}}(H', G') & \rightrightarrows & \text{Hom}_{S''\text{-gr.}}(H'', G'') \\ \downarrow u & & \downarrow u' & & \downarrow u'' \\ \varprojlim_n \text{Hom}_{S_n\text{-gr.}}(H_n, G_n) & \longrightarrow & \varprojlim_n \text{Hom}_{S'_n\text{-gr.}}(H'_n, G'_n) & \rightrightarrows & \varprojlim_n \text{Hom}_{S''_n\text{-gr.}}(H''_n, G''_n). \end{array}$$

(N.B. The second row is exact as a projective limit of exact diagrams relative to the various $S'_n \rightarrow S_n$.) By what has already been proved (diagonalizable case), u' and u'' are bijective. The same is therefore true of u , which completes the proof of 7.1.

Corollary 7.3. *Under the conditions of 7.1, suppose in addition G smooth over S , and let $u_0 : H_0 \rightarrow G_0$ be a homomorphism of S_0 -groups. Then there exists a homomorphism of S -groups $u : H \rightarrow G$ lifting u_0 . Two such liftings u, u' are conjugate by an element g of $G(S)$ reducing to the unit element of $G_0(S_0)$.*

Follows from the conjunction of 3.6 and 7.1. To construct a u , one constructs the u_n successively, which is possible by 3.6, and by 7.1 the system (u_n) comes from a u . Given two liftings u and u' , to construct g such that $u' = \text{int}(g)u$, $g_0 = 1$, one constructs successively g_n such that $g_0 = 1$, g_n is deduced from g_{n+1} by reduction, $u'_n = \text{int}(g_n)u_n$; this is possible thanks to 3.6. Since A is separated and complete, the g_n come from a $g \in G(S)$, and to prove that $u' = \text{int}(g)u$ it suffices to use injectivity in assertion 7.1.

Remark 7.4. One will compare 7.1 with EGA III 5.4.1, which implies that statement 7.1 is valid if, instead of supposing H of multiplicative type and G affine, one supposes H proper over S , and G separated and locally of finite type over S . Having a statement like 7.1 without a properness hypothesis is rather exceptional, and must here be interpreted as one of the aspects of the great “rigidity” of the structure of a group of multiplicative type. The analogous statement with $G = H = \mathbf{G}_a$ (additive group) is false in general, as one sees by taking A of characteristic $p > 0$ and defining the u_n by reduction modulo I^{n+1} from an additive formal series

$$u(T) = \sum_{n \geq 0} a_n T^{p^n},$$

where the a_n are elements of A tending to 0 for the I -adic topology, but not for the discrete topology. Statement 7.1 also becomes false if one suppresses the hypothesis G affine, even for $H = \mathbf{G}_m$ (multiplicative group); one sees an example (with A a complete discrete valuation ring) by starting with an elliptic curve over the field of fractions K of A which reduces (in the Néron-Kodaira reduction theory, say) to the group $\mathbf{G}_m$ over the residue field k .⁽⁵⁰⁾ one shall therefore have a smooth commutative group scheme G over S whose special fiber is $\mathbf{G}_{m,k}$ (which allows one, thanks to 3.6, to define a projective system of isomorphisms $u_n : H_n \xrightarrow{\sim} G_n$, where $H = \mathbf{G}_{m,A}$), but whose generic fiber is an abelian variety, so that there is no homomorphism of S -groups $H \rightarrow G$ other than 0.

8. Subgroups, quotient groups and extensions of groups of multiplicative type over a field

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Scholium 8.0. ⁽⁵²⁾ *Let k be a field, H an affine k -group scheme. Denote by $k[H]$ its affine algebra and by $X(H) = \text{Hom}_{k\text{-gr.}}(H, \mathbf{G}_{m,k})$ the group of characters of H . By the independence lemma for characters, $X(H)$ is a linearly independent subset of $k[H]$. It follows that H is diagonalizable if and only if $X(H)$ generates $k[H]$ as a k -vector space.*

Proposition 8.1. *Let G be a diagonalizable (resp. multiplicative type) group scheme over a field k .⁽⁵³⁾ Then for every subgroup scheme H of G , H and G/H are diagonalizable (resp. of multiplicative type).*

⁽⁵⁰⁾N.D.E. One could make such an example explicit. . .

⁽⁵¹⁾Added in July 1969. This afterthought no. is independent of nos. 3 to 7.

⁽⁵²⁾N.D.E. This scholium has been added.

⁽⁵³⁾N.D.E. “Over a field k ”, implicit in the original, has been added; on the other hand, “algebraic group” has been replaced by “group scheme”, since G is not supposed of finite type.

Definition 1.1 reduces us immediately to the case without “resp.” An easy passage to the limit, using VI_B 11.13 and VIII 3.1, reduces us to the case where G is of finite type over k .⁽⁵⁴⁾ By VI_B 11.16,⁽⁵⁵⁾ one can find a finite family of nonzero elements

$$(8.1.1) \quad f_i = \sum a_{im} m, \quad a_{im} \in k$$

of the affine ring $k(M)$ of $G = D_k(M)$ such that the points of H (with values in an arbitrary k -algebra k') are the points $g \in G(k')$ such that one has

$$(8.1.2) \quad \tau_g f_i = \lambda_i(g) f_i, \quad \text{with } \lambda_i(g) \in k',$$

where τ_g denotes translation by g . Now one has

$$(8.1.3) \quad \tau_g f_i = \sum a_{im} \chi_m(g) m,$$

$\chi_m : G \rightarrow \mathbf{G}_{m,k}$ being the character associated with m , so that (8.1.2) is equivalent to the relation

$$(8.1.4) \quad \chi_m(g) = \chi_{m'}(g) \quad \text{if } m, m' \in Z_i,$$

Z_i denoting the set of $m \in M$ such that $a_{im} \neq 0$. This relation may also be written

$$\chi_{m'-m}(g) = 1 \quad \text{if } m, m' \in Z_i.$$

Denoting by N the subgroup of M generated by all the $m' - m$ (i variable, and $m, m' \in Z_i$), one deduces from the definition of $D_k(M)$ ($\simeq G$) that H is identified with $D_k(M/N)$. It then follows from VIII 3.1 that G/H is identified with $D_k(N)$. Q.E.D.

Proposition 8.2. ⁽⁵⁶⁾ *Let k be a field, H, K k -group schemes of multiplicative type and of finite type, and G a k -group scheme such that one has an exact sequence*

$$1 \rightarrow H \rightarrow G \rightarrow K \rightarrow 1$$

(which implies that G is of finite type over k).

- (i) *If one supposes G commutative or K connected, then G is of multiplicative type.*
- (ii) *If K and H are diagonalizable, K being a torus, then G is diagonalizable.*

For the proof of (i), the reader will refer to XVII 7.1.1, of which 8.2 (i) is a particular case; the case of a field is treated in part (i) of the proof of XVII 7.1.1 (not using the results of the following exposés).

(ii) Suppose now K and H diagonalizable:

$$K \simeq D_k(M) \quad \text{and} \quad H \simeq D_k(N),$$

⁽⁵⁶⁾ and suppose G of multiplicative type (which is the case by (i) if K is a torus). Then, by X 1.4, G is isotrivial over k , i.e. there exists a finite separable extension k'/k such that $G' = G \times_k k'$ is diagonalizable, hence $G' = D_{k'}(E)$ for a commutative group E , and one has an exact sequence

$$(1) \quad 0 \rightarrow D_{k'}(N) \rightarrow D_{k'}(E) \rightarrow D_{k'}(M) \rightarrow 0.$$

Thus, by VIII, 3.1 and 3.2, M is a subgroup of E , and one has an exact sequence

$$(2) \quad 0 \rightarrow M \rightarrow E \rightarrow N \rightarrow 0.$$

For a given extension E_0 of N by M , consider the diagonalizable k -group $G_0 = D_k(E_0)$; then the k -group functor A of automorphisms of the extension

$$(3) \quad 1 \rightarrow H \rightarrow G_0 \rightarrow K \rightarrow 1$$

⁽⁵⁴⁾**N.D.E.** Expand this “passage to the limit”: this uses 8.0 and also the equality $G/H = \varprojlim_i G_i/H_i$, cf. VI_B, proof of 11.17. To see that $H = \varprojlim_i (H \cap G_i)$, does one use the fact that H is closed in G ? Using this, one can give a direct proof. . .

⁽⁵⁵⁾**N.D.E.** VI_B 11.16 ought doubtless to be rewritten in the usual, more pleasant form. In particular, a single f_i suffices below. . .

⁽⁵⁶⁾**N.D.E.** The statement, as well as its proof, has been expanded.

i.e. the group subfunctor of $\underline{\mathrm{Aut}}_{k\text{-gr.}}(G_0)$ whose points over a k -prescheme T are the $\phi \in \mathrm{Aut}_{T\text{-gr.}}(G_T)$ inducing the identity on H_T and on K_T , is identified with $\underline{\mathrm{Hom}}_{k\text{-gr.}}(K, H)$, which by VIII 1.5 is the constant k -group with value $L = \mathrm{Hom}_{\mathrm{gr.}}(N, M)$.

One therefore sees that the classification of extensions G of K by H which over a separable closure k_s of k become isomorphic to extension (3) is the same as that of k -torsors for the étale topology under the constant group L_k . Denoting by $\Gamma = \mathrm{Gal}(k_s/k)$, these are classified by the Galois cohomology group (L being a trivial Γ -module):

$$H_{\mathrm{\acute{e}t}}^1(k, L_k) = H^1(\Gamma, L) = \mathrm{Hom}_{\mathrm{gr. top.}}(\Gamma, L).$$

If, in addition, K is a torus, M is torsion-free, and hence so is L , whence $\mathrm{Hom}_{\mathrm{gr. top.}}(\Gamma, L) = 0$; it follows that every extension of K by H is already diagonalizable.

Remark 8.3. As already pointed out in the proof of 8.2, the first assertion is stated and proved over an arbitrary base in XVII 7.1.1. On the other hand, the second assertion generalizes, with essentially the same proof, to the case of a normal integral base scheme (or more generally a geometrically unibranch one), using X 5.13 below.

Finally, one may also consider 8.1 as a corollary of the (considerably less trivial) result 6.8, also valid over an arbitrary base scheme.⁽⁵⁷⁾

⁽⁵⁷⁾**N.D.E.** Note, however, that the proof of 6.8 uses 8.1!

Bibliography

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⁽⁵⁸⁾**N.D.E.** Additional references cited in this Exposé.

Exposé X. Characterization and classification of groups of multiplicative type

by A. Grothendieck

1. Classification of isotrivial groups. Case of a base field

Recall (IX 1.1) that the group of multiplicative type H over the prescheme S is said to be *isotrivial* if there exists a finite surjective étale morphism $S' \rightarrow S$ such that $H' = H \times_S S'$ is diagonalizable. When S is connected, S' decomposes as a finite sum of connected components S'_i , and one may therefore (replacing S' by one of the S'_i if necessary) suppose S' connected. Finally, one knows that one may dominate S' by an S'_1 finite étale connected, which is *Galois*, i.e. a principal homogeneous bundle over S with group Γ_S , where Γ is an ordinary finite group (cf. SGA 1, V nos. 7 & 3 when S is locally noetherian, and EGA IV₄, 18.2.9 in the general case).⁽¹⁾ We suppose S' chosen in this way, and propose to determine the groups of multiplicative type H over S which are “trivialized” by S' , i.e. such that $H' = H \times_S S'$ is diagonalizable.⁽²⁾ By descent theory (cf. SGA 1, VIII 5.4), the category of these H is equivalent to the category of diagonalizable groups H' over S' , endowed with operations of Γ on H' compatible with the operations of Γ on S' . (N.B. Since the groups considered are affine over the base, the question of effectiveness of a descent datum is resolved in the affirmative, cf. SGA 1, VIII 2.1.) Now S' being connected, the contravariant functor

$$M \mapsto D_{S'}(M)$$

is an anti-equivalence of the category of ordinary commutative groups with the category of diagonalizable groups over S' (cf. VIII 1.6), a quasi-inverse functor being $H \mapsto \text{Hom}_{S'\text{-gr.}}(H, \mathbf{G}_{m,S'})$.⁽³⁾

Proposition 1.1. *Let S be a connected prescheme, S' a connected principal covering of S with (finite) group Γ . Then the category of groups of multiplicative type over S split⁽⁴⁾ by S' is anti-equivalent to the category of Γ -modules, i.e. of ordinary commutative groups M endowed with a homomorphism $\Gamma \rightarrow \text{Aut}_{\text{gr.}}(M)$.*

One concludes in the standard manner:

Corollary 1.2. *Let S be a connected prescheme, $\xi : \text{Spec}(\Omega) \rightarrow S$ a “geometric point” of S , i.e. a homomorphism into S from the spectrum of an algebraically closed field Ω ; consider the*

⁽⁰⁾Version xy of November 6, 2009: Addenda placed in Sect. 9, revised through 8.8.

⁽¹⁾**N.D.E.** More precisely, this follows from the “axiomatic conditions of a Galois theory”, cf. SGA 1, V no. 4 g); when S is locally noetherian, the verification of the axioms is made in loc. cit., nos. 7 & 3, in particular 3.7, which rests on SGA 1, I 10.9. This latter result is proved, without noetherian hypotheses, in EGA IV₄, 18.2.9.

⁽²⁾**N.D.E.** In what follows one shall say that an S -group of multiplicative type is “split” if it is “trivial” in the sense of IX 1.2, i.e. if it is a diagonalizable S -group.

⁽³⁾**N.D.E.** S has been corrected to S' .

⁽⁴⁾**N.D.E.** Here and in what follows, “splittés” has been replaced by “déployés”, “splitte” by “déplote”, etc.

fundamental group of S at ξ (cf. SGA 1 V, no. 7):

$$\pi = \pi_1(S, \xi).$$

Then the category of isotrivial groups of multiplicative type H over S is anti-equivalent to the category of “Galois modules” under π , i.e. of π -modules M such that the stabilizer in π of every point of M is an open subgroup.

In this correspondence, with the isotrivial group of multiplicative type H is associated the group $M = \text{Hom}_{\Omega\text{-gr.}}(H_\xi, \mathbf{G}_{m, \Omega})$, where H_ξ is the fiber of H at ξ ; this group is naturally endowed with operations of $\pi_1(S, \xi)$.

Remark 1.3. We shall see below (cf. 5.16) that if S is normal, or more generally geometrically unibranch, every group of multiplicative type and of finite type over S is necessarily isotrivial, so classification principle 1.2 applies to groups of multiplicative type and of finite type over S , which correspond to the Galois π -modules of finite type over $\mathbf{Z}$. For the moment, let us restrict to the following result:

Proposition 1.4. *Let k be a field, H a group of multiplicative type and of finite type over k ; then H is isotrivial, i.e. there exists a finite separable extension k' of k which splits H .*

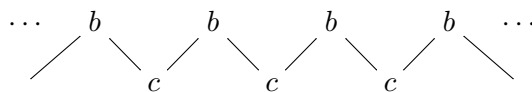
Consequently, by 1.2, if π is the topological Galois group of an algebraic closure Ω of k , the category of groups of multiplicative type and of finite type H over k is anti-equivalent to the category of Galois modules M under π of finite type as $\mathbf{Z}$ -modules.

It follows first from the fact that H is of finite type over k and from the “finite extension principle” (cf. EGA IV₃, 9.1.4) that there exists a finite extension k' of k which splits H . Recall the principle of the proof: by hypothesis there exists a diagonalizable group of finite type G over k , an S' faithfully flat over $S = \text{Spec}(k)$, and an isomorphism of S' -groups $H_{S'} \simeq G_{S'}$. Replacing S' by the residue field of a point of S' if necessary, one may suppose that S' is the spectrum of an extension K of k . The latter is the inductive limit of its subalgebras A_i of finite type, whence it follows easily (cf. EGA IV₃, 8.8.2.4) that u comes from an A_i -isomorphism $u_i : H_{A_i} \simeq G_{A_i}$ for sufficiently large i . By the Nullstellensatz there exists a quotient ring k' of A_i which is a finite extension of k . This therefore splits H .

Then k' is a radicial extension of a separable extension k'_s of k . By IX 5.4, the isomorphism $u' : H_{k'} \simeq G_{k'}$ comes from an isomorphism $H_{k'_s} \simeq G_{k'_s}$, which proves that k'_s splits H and establishes 1.4.

Remark 1.5. Statement 1.2 supplies in particular a characterization of isotrivial tori over S of relative dimension n : putting $\pi = \pi_1(S, \xi)$, they correspond to the classes (up to “equivalence”) of representations $\pi \rightarrow \text{GL}(n, \mathbf{Z})$ with kernel an open subgroup of π .

1.6. — Even when S is an algebraic curve, there may exist tori (of relative dimension 2) over S which are not locally isotrivial (and a fortiori not isotrivial); there may also exist locally trivial tori which are not isotrivial. (Note, however, that such phenomena can occur only if S is not normal, as already pointed out in 1.3.) Let, for example, S be an irreducible algebraic curve (over an algebraically closed field to fix ideas) having an ordinary double point a ; let S' be its normalization, and b, c the two points of S' over a . One then constructs a connected principal homogeneous bundle P over S with structural group $\mathbf{Z}$, by joining together infinitely many copies of S' according to the diagram



(N.B. This is a principal bundle in the sense of the étale topology.) Now one has a homomorphism

$$\varphi : \mathbf{Z} \rightarrow \text{GL}(2, \mathbf{Z}), \quad \varphi(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix},$$

which allows one to construct a torus T over S of relative dimension 2 from the trivial torus $\mathbf{G}_m^2$ over P and the descent datum on the latter deduced from φ . (N.B. One will note that the projection $P \rightarrow S$ is covering for the étale topology and a fortiori for the canonical topology of (Sch), and that the descent datum considered is necessarily effective, since $\mathbf{G}_{m,P}^2$ is affine over P .) It is not difficult to prove that T is not isotrivial in a neighborhood of $a^{(5)}$ (it is nevertheless trivial on $S - \{a\}$).

One finds a variant of this construction by taking for S a curve having two irreducible components S_1, S_2 meeting at two points a', a'' , which allows one to construct a connected and locally trivial principal homogeneous bundle P over S with group $\mathbf{Z}_S$, whence an associated torus T which is locally trivial but not isotrivial.

2. Infinitesimal variations of structure

⁽⁶⁾ Let us begin by recalling the following result (cf. SGA 1, I 8.3 in the case of S locally noetherian, EGA IV₄, 18.1.2 in general).

Recollection 2.0. *Let S be a prescheme, S_0 a subprescheme with the same underlying set. Then the functor*

$$X \longmapsto X_0 = X \times_S S_0$$

is an equivalence between the category of preschemes étale over S and the analogous category over S_0 .

Proposition 2.1. *Let S be a prescheme, S_0 a subprescheme with the same underlying set (i.e. defined by a nil ideal $\mathcal{I}$). Then the functor*

$$H \longmapsto H_0 = H \times_S S_0$$

from the category of group preschemes of multiplicative type over S to the analogous category over S_0 is fully faithful.

Moreover, it induces an equivalence between the category of quasi-isotrivial groups of multiplicative type over S and the analogous category over S_0 .

Let us first prove full faithfulness, i.e. that if H, G over S are of multiplicative type, then

$$\mathrm{Hom}_{S\text{-gr.}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0)$$

is bijective. The question being local on S , one may suppose S affine; there then exists a faithfully flat and quasi-compact morphism $S' \rightarrow S$ splitting H and G . Let $S'' = S' \times_S S'$; denote by H', G' , resp. H'', G'' , the groups deduced from H, G by the base change $S' \rightarrow S$, resp. $S'' \rightarrow S$; define similarly S'_0, S''_0 , the latter being also isomorphic to $S'_0 \times_{S_0} S'_0$. One then finds a commutative diagram with exact rows:

$$\begin{array}{ccccc} \mathrm{Hom}_{S\text{-gr.}}(H, G) & \longrightarrow & \mathrm{Hom}_{S'\text{-gr.}}(H', G') & \rightrightarrows & \mathrm{Hom}_{S''\text{-gr.}}(H'', G'') \\ \downarrow u & & \downarrow u' & & \downarrow u'' \\ \mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0) & \longrightarrow & \mathrm{Hom}_{S'_0\text{-gr.}}(H'_0, G'_0) & \rightrightarrows & \mathrm{Hom}_{S''_0\text{-gr.}}(H''_0, G''_0), \end{array}$$

so to prove that u is bijective, it suffices to prove that u' and u'' are so, which reduces us to the case where H and G are diagonalizable, hence of the form $D_S(M)$ and $D_S(N)$, where M, N are ordinary commutative groups. One shall therefore similarly have $H_0 = D_{S_0}(M)$,

⁽⁵⁾**N.D.E.** See 7.3 below.

⁽⁶⁾**N.D.E.** This “recollection” has been added, for later references.

$G_0 = D_{S_0}(N)$. One then has a commutative diagram⁽⁷⁾

$$\begin{array}{ccc} \mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) & \xrightarrow{\sim} & \mathrm{Hom}_{S\text{-gr.}}(D_S(M), D_S(N)) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0}) & \xrightarrow{\sim} & \mathrm{Hom}_{S_0\text{-gr.}}(D_{S_0}(M), D_{S_0}(N)), \end{array}$$

where the horizontal arrows are isomorphisms by VIII 1.4, so one is reduced to proving that the homomorphism

$$(x) \quad \mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0})$$

is bijective, i.e. to proving that the functor $M_S \mapsto M_{S_0}$ from constant commutative group preschemes over S to constant commutative group preschemes over S_0 is fully faithful. Now (x) is also identified with the natural map

$$\mathrm{Hom}_{\mathrm{gr.}}(N, \Gamma(M_S)) \longrightarrow \mathrm{Hom}_{\mathrm{gr.}}(N, \Gamma(M_{S_0}))$$

deduced from $\Gamma(M_S) \longrightarrow \Gamma(M_{S_0})$; now the latter map is evidently bijective (since $\Gamma(M_S) =$ the set of locally constant maps from S to M depends only on the underlying topological space of S), whence the desired conclusion.

To prove the second assertion of 2.1, it remains to see that every quasi-isotrivial group of multiplicative type H_0 over S_0 comes from a quasi-isotrivial group of multiplicative type H over S . To see this, let $S'_0 \longrightarrow S_0$ be a surjective étale morphism splitting H_0 .

One knows (cf. 2.0) that there exists an étale morphism $S' \longrightarrow S$ and an S_0 -isomorphism $S' \times_S S_0 \simeq S'_0$, so one may suppose that S'_0 comes from S' by reduction. Since H'_0 is diagonalizable, one sees immediately that it is isomorphic to the group deduced by base change $S'_0 \longrightarrow S'$ from a diagonalizable group H' over S' (N.B. if $H'_0 = D_{S'_0}(M)$, one takes $H' = D_{S'}(M)$). Put, as usual, $S'' = S' \times_S S'$, $S''' = S' \times_S S' \times_S S'$, and define similarly S''_0, S'''_0 , deduced from the preceding ones by base change $S_0 \longrightarrow S$ and also isomorphic to the fibered square resp. cube of S'_0 over S_0 . Using the full faithfulness already proved in the cases (S'', S''_0) and (S''', S'''_0) , one sees that the natural descent datum on H'_0 relative to $S'_0 \longrightarrow S_0$ (cf. IV 2.1) comes from a well-determined descent datum on H' relative to $S' \longrightarrow S$. This descent datum is effective since H' is affine over S' (SGA 1, VIII 2.1); there therefore exists an S -group H such that $H \times_S S' = H' = D_{S'}(M)$, and H is thus of quasi-isotrivial multiplicative type.

One then easily checks, now using the full faithfulness result for (S', S'_0) , that the given isomorphism between H'_0 and $H' \times_{S'} S'_0$ comes from an isomorphism between H_0 and $H \times_S S_0$. (For a more formal exposition of results of this type, see Giraud's article in preparation⁽⁸⁾ on descent theory.)

Corollary 2.2. *Let H be an S -group of multiplicative type, and $H_0 = H \times_S S_0$. For H to be quasi-isotrivial (resp. locally isotrivial, resp. isotrivial, resp. locally trivial, resp. trivial), it is necessary and sufficient that H_0 be so.*

The “necessary” part is trivial; the “sufficient” part has already been seen in the quasi-isotrivial case, since thanks to full faithfulness it suffices to know that every quasi-isotrivial group over S_0 lifts to a quasi-isotrivial group over S . The same argument works for “trivial”. For the “isotrivial” case one repeats the argument establishing the second assertion of 2.1, but taking $S'_0 \longrightarrow S_0$ étale surjective and finite. The “locally isotrivial” and “locally trivial” cases follow immediately from the “isotrivial” and “trivial” cases.

One may generalize 2.2 somewhat when $\mathcal{I}$ is nilpotent, by not supposing a priori H of multiplicative type:⁽⁹⁾

⁽⁷⁾**N.D.E.** The original has been corrected by interchanging M and N in the right-hand terms.

⁽⁸⁾Cf. J. Giraud, *Méthode de la descente*, Bull. Soc. Math. France, Mémoire 2 (1964).

⁽⁹⁾**N.D.E.** Recall that 2.3 is used to prove theorem IX 3.6 bis.

Corollary 2.3. *Suppose the ideal $\mathcal{I}$ defining S_0 locally nilpotent. Let H be a group prescheme over S , flat over S , and $H_0 = H \times_S S_0$. For H to be of quasi-isotrivial multiplicative type, it is necessary and sufficient that H_0 be so.*

Indeed, suppose that H_0 is of quasi-isotrivial multiplicative type, and let us prove that H is so. The question being local for the étale topology, and the category of preschemes étale over S being equivalent to the category of preschemes étale over S_0 by the functor $S' \mapsto S' \times_S S_0$ (cf. 2.0), one is reduced immediately to the case where H_0 is diagonalizable, hence isomorphic to a group $D_{S_0}(M)$. Let $G = D_S(M)$; one therefore has an isomorphism $u_0 : H_0 \xrightarrow{\sim} G_0$; I say that it comes from a unique homomorphism $u : H \rightarrow G$, which will therefore be an isomorphism since u_0 is so (H and G being flat over S and $\mathcal{I}$ locally nilpotent⁽¹⁰⁾), which will establish 2.3. Now one has (cf. VIII 1 (xxx))

$$\mathrm{Hom}_{S\text{-gr.}}(H, G) \simeq \mathrm{Hom}_{S\text{-gr.}}(M_S, \underline{\mathrm{Hom}}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})),$$

and the right-hand side is also identified with

$$\mathrm{Hom}_{\mathrm{gr.}}(M, \mathrm{Hom}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})),$$

so the homomorphism

$$(x) \quad \mathrm{Hom}_{S\text{-gr.}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0)$$

is isomorphic to the homomorphism

$$\mathrm{Hom}_{\mathrm{gr.}}(M, \mathrm{Hom}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})) \longrightarrow \mathrm{Hom}_{\mathrm{gr.}}(M, \mathrm{Hom}_{S_0\text{-gr.}}(G_0, \mathbf{G}_{m,S_0}))$$

deduced from the restriction homomorphism

$$(xx) \quad \mathrm{Hom}_{S\text{-gr.}}(H, \mathbf{G}_{m,S}) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(H_0, \mathbf{G}_{m,S_0}),$$

so to prove that (x) is bijective it suffices to prove that (xx) is so. The question is local on S , so one may suppose S affine. Now $\mathbf{G}_{m,S}$ being commutative and smooth over S , the situation is covered by IX 3.6, which completes the proof.

Corollary 2.4. *Let A be an artinian local ring with residue field k , $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(k)$.*

(i)⁽¹¹⁾ *Let H be a flat S -group prescheme locally of finite type, such that $H_0 = H \times_S S_0$ is of multiplicative type. Then H is of multiplicative type, of finite type, and isotrivial. In particular, every S -group prescheme H of multiplicative type and of finite type is isotrivial.*

(ii) *The functor $H \mapsto H_0$ is an equivalence between the category of groups of multiplicative type of finite type over A and the analogous category over k .*

Indeed, let H be as in (i). Then H_0 is of multiplicative type and locally of finite type, hence of finite type, hence isotrivial by 1.4. Thus, by 2.3 and 2.2, H is of multiplicative type (hence of finite type) and isotrivial. Assertion (ii) then follows from 2.1.

Corollary 2.5. *Let $S' \rightarrow S$ be a faithfully flat and radicial morphism.*

(i) *The functor $H \mapsto H' = H \times_S S'$ from the category of groups of multiplicative type over S to the analogous category over S' is fully faithful.*

Moreover, it induces an equivalence between the subcategories consisting of quasi-isotrivial groups of multiplicative type.

(ii) *If H is of multiplicative type, for it to be quasi-isotrivial (resp. locally isotrivial, resp. isotrivial, resp. locally trivial, resp. trivial), it is necessary and sufficient that H' be so.*

⁽¹⁰⁾**N.D.E.** Since u_0 is an isomorphism, it suffices to see that for every $h \in H$ the morphism $\phi : \mathcal{O}_{G, u(h)} \rightarrow \mathcal{O}_{H, h}$ is bijective. It is so by reduction modulo $I = \mathcal{I}_s$ (where s is the image of h in S), so its cokernel C satisfies $C = IC$, whence $C = 0$ since I is nilpotent. Then, since $\mathcal{O}_{H, h}$ is flat over $\mathcal{O}_{S, s}$, the kernel K of ϕ also satisfies $K = IK$, whence $K = 0$.

⁽¹¹⁾**N.D.E.** The statement has been rewritten retaining only the nontrivial implication, in order to bring it out.

Indeed, let $S'' = S' \times_S S'$ and $S''' = S' \times_S S' \times_S S'$; then the hypothesis that $S' \rightarrow S$ is radicial implies that the diagonal immersions $S' \rightarrow S''$ and $S' \rightarrow S'''$ are surjective, so base change by either of these immersions is covered by 2.1 and 2.2. In view of the fact that $S' \rightarrow S$ is a morphism of effective descent for the fibered category of groups of multiplicative type over preschemes (since it is faithfully flat and quasi-compact⁽¹²⁾), our assertion follows formally from 2.1 and 2.2 (cf. for a formal argument the already cited article of Giraud).

3. Finite variations of structure: Complete base ring

Lemma 3.1. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/I)$, G, H two S -group schemes, with G of multiplicative type and isotrivial, H affine over S , flat over S at the points of $H_0 = H \times_S S_0$, H_0 of quasi-isotrivial multiplicative type. Then the natural map*

$$\operatorname{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \operatorname{Hom}_{S_0\text{-gr.}}(G_0, H_0)$$

is bijective.

For every integer $n \geq 0$, let $S_n = \operatorname{Spec}(A/I^{n+1})$, and let G_n, H_n be the groups deduced from G, H by the base change $S_n \rightarrow S$. Since G/S is of isotrivial multiplicative type and H affine over S , by IX 7.1 the natural homomorphism

$$\operatorname{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \varprojlim_n \operatorname{Hom}_{S_n\text{-gr.}}(G_n, H_n)$$

is bijective. On the other hand, by 2.3 the H_n are of quasi-isotrivial multiplicative type, and by 2.1 the transition homomorphisms in the projective system $(\operatorname{Hom}_{S_n\text{-gr.}}(G_n, H_n))_n$ are isomorphisms, whence immediately 3.1.

Theorem 3.2. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/I)$. Then:*

(i) *The functor*

$$H \longmapsto H_0 = H \times_S S_0$$

is an equivalence between the category of isotrivial groups of multiplicative type over S and the analogous category over S_0 .

(ii) *Let H be an S -group of multiplicative type and of finite type; for H to be isotrivial, it is necessary and sufficient that H_0 be so.*

PROOF. For (i), one may either repeat the proof of 2.1 or use 1.2, using in either case the fact that the functor

$$S' \longmapsto S'_0 = S' \times_S S_0$$

from the category of finite étale schemes over S to the category of finite étale schemes over S_0 is an equivalence (SGA 1, I 8.4), which one may also state (reducing to the case of S connected, i.e. S_0 connected), choosing a geometric point ξ of S_0 , by saying that the canonical homomorphism

$$\pi_1(S_0, \xi) \longrightarrow \pi_1(S, \xi)$$

is an isomorphism.

Let us prove (ii), i.e. that if H_0 is isotrivial, H is so. By (i), there exists an isotrivial group of multiplicative type G over S and an S_0 -isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

⁽¹³⁾ Since H is of finite type, so are H_0 and G_0 ; thus by IX 2.1 b), the type of G at each point of S is an abelian group of finite type, and hence G is of finite type over S . On the other

⁽¹²⁾N.D.E. Specify this point...

⁽¹³⁾N.D.E. The original has been expanded in what follows.

hand, by 3.1, u_0 comes from a homomorphism of S -groups

$$u : G \longrightarrow H.$$

Finally, since G, H are of multiplicative type and of finite type over S , and since u_0 is an isomorphism, by IX 2.9, u is an isomorphism (taking account of the fact that every neighborhood of S_0 in S equals S). $\square$

Corollary 3.3. *Let A be a complete noetherian local ring with residue field k .*

- (i) *Every group of multiplicative type and of finite type over A is isotrivial.*
- (ii) *The functor $H \longmapsto H \otimes_A k = H_0$ is an equivalence between the categories of groups of multiplicative type and of finite type over A and over k .*

⁽¹³⁾ First, (i) follows from 3.2 (ii) and 1.4; then (ii) follows from 3.2 (i), in view of the fact that H is of finite type if and only if H_0 is so (cf. the proof of 3.2 (ii)).

Remark 3.3.1. One will note that 3.3 gives, by 1.4, a complete classification of groups of multiplicative type and of finite type over A in terms of the topological Galois group of an algebraic closure Ω of k .

Remarks 3.4. Under the hypotheses of 3.2 (i.e. A noetherian, separated and complete for the I -adic topology, but without any other hypothesis on A/I), it will follow from no. 5 that the functor $H \longmapsto H_0$ from the category of groups of multiplicative type and of finite type over S to the analogous category over S_0 is still fully faithful (without isotriviality hypotheses).⁽¹⁴⁾

However, it is not in general essentially surjective; in fact there may exist an S_0 -group H_0 of multiplicative type and of finite type, locally trivial if one wishes (but not isotrivial), which does not come by reduction from a group of multiplicative type H over S .

To see this, let us return to either of examples 1.6 of a nonisotrivial group of multiplicative type over a nonnormal curve. One may evidently take this curve affine, say S_0 , and suppose it embedded in affine space of dimension 2, hence defined by an ideal J in $k[X, Y]$. We shall take for A the completion of this ring for the J -preadic topology, so that A is a regular ring of dimension 2, a fortiori normal. We shall see in 5.16 that it follows that every group of multiplicative type and of finite type over $S = \text{Spec}(A)$ is isotrivial; hence H_0 , which is of finite type and nonisotrivial, does not come from a group of multiplicative type H over S (for H would necessarily be of finite type, and hence isotrivial).

Lemma 3.5. ⁽¹⁵⁾ *Let S be a prescheme, $u : G \longrightarrow H$ a homomorphism of S -group preschemes locally of finite presentation and flat over S , U the set of $s \in S$ such that the homomorphism induced on the fibers $u_s : G_s \longrightarrow H_s$ is flat (resp. smooth, resp. unramified, resp. étale, resp. quasi-finite).*

Then U is open, and the restriction $u|_U : G|_U \longrightarrow H|_U$ is a flat morphism (resp. smooth, resp. unramified, resp. étale, resp. quasi-finite).

Indeed, let V be the set of points at which u is flat (resp...). One knows that V is open (cf. SGA 1, I to IV in the locally noetherian case, EGA IV in general⁽¹⁶⁾), and that for $x \in G$ over $s \in S$ one has $x \in V$ if and only if $u_s : G_s \longrightarrow H_s$ is flat (resp...) at x (same reference; in the flat, smooth, étale cases one uses here the flatness of G and H over S). Since u_s is a homomorphism of locally algebraic groups, this also means that u_s is flat (resp...) everywhere (Exp. VI_B, 1.3), i.e. $s \in U$. Thus U is the inverse image of the open subset V by the unit section of G , hence open, and $V = G|_U$, so $G|_U \longrightarrow H|_U$ is flat (resp...), which

⁽¹⁴⁾**N.D.E.** Specify this reference, possibly adding a corollary in no. 5...

⁽¹⁵⁾Cf. also VI_B 2.5 for more systematic developments of this nature.

⁽¹⁶⁾**N.D.E.** Specify these references: Since G, H are locally of finite presentation, u is locally of finite presentation (IV₁, 1.4.3 (v)); then one applies IV₃, 11.3.10 and 13.1.4 for flat and quasi-finite, IV₄, 17.4.1, 17.5.1 and 17.6.1 for unramified, smooth and étale; compare with VI_B 2.5.

completes the proof. (N.B. In the “unramified” or “quasi-finite” case, the flatness hypothesis on G, H is unnecessary.)

Lemma 3.6. *Let H be a commutative algebraic group over a field k admitting an open subgroup G of multiplicative type. Then the family of subschemes ${}_nH$ ($n > 0$) of H is schematically dense; in particular, if one has ${}_nH = {}_nG$ for every $n > 0$, then $H = G$.*

Here ${}_nH$ denotes the kernel of $n \cdot \text{id}_H$.⁽¹⁷⁾ Let $\bar{k}$ be an algebraic closure of k ; it suffices to show that the family $({}_nH_{\bar{k}})_{n>0}$ is schematically dense in $H_{\bar{k}}$, for then the family $({}_nH)_{n>0}$ will be so in H (cf. IX 4.5). Thus one may suppose k algebraically closed, and hence G of the form $D_k(M)$, M an ordinary commutative group of finite type. Let $M_0 = M / \text{Torsion}(M)$; then $T = D_k(M_0)$ is the largest torus contained in G , and H/T is finite, so $H(k)/T(k)$ is annihilated by an integer $\nu > 0$. One may find finitely many elements $g_i \in H(k)$ such that

$$H = \coprod_i g_i \cdot G,$$

and one shall have $g_i^\nu \in T(k)$. Since k is algebraically closed, $\nu \cdot \text{id}_T$ is surjective in $T(k) \simeq k^{*d}$; thus, replacing the g_i by $g_i t_i^{-1}$, where $t_i \in T(k)$ is such that $t_i^\nu = g_i^\nu$, one may suppose that $g_i^\nu = 1$. If then n is a multiple of ν , one shall have

$${}_nH \supseteq g_i \cdot {}_nG,$$

and as (for variable $n > 0$) the family of the ${}_nG$ is schematically dense in G by IX 4.7, conclusion 3.6 appears.

Theorem 3.7. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \text{Spec}(A)$, $S_0 = \text{Spec}(A/I)$, and let H be an affine S -group scheme of finite type, flat over S at the points of H_0 , such that $H_0 = H \times_S S_0$ is of multiplicative type and isotrivial.*

Then there exists an open and closed subgroup G of H , of isotrivial multiplicative type (and of finite type), such that $G_0 = H_0$.

By 3.2 (i), there exists an isotrivial group of multiplicative type G over S and an isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

By 3.1, u_0 comes from a unique homomorphism of S -groups

$$u : G \longrightarrow H.$$

Using IX 6.6 one sees that u is a monomorphism (for if U is the set of $s \in S$ such that $u_s : G_s \longrightarrow H_s$ is a monomorphism, U is an open neighborhood of S_0 , hence identical to S , and $G|U \longrightarrow H|U$ is a monomorphism). By IX 2.5, u is even a closed immersion.

⁽¹⁸⁾ Thus G is of finite type, hence of finite presentation over S . Then, by lemma 3.5 in the “étale” case, one sees that there exists an open subset U which is a neighborhood of S_0 , hence identical to S , such that $G|U \longrightarrow H|U$ is étale; hence u is étale, hence an open immersion (since it is an étale monomorphism⁽¹⁹⁾), which completes the proof.

Corollary 3.8. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \text{Spec}(A)$, $S_0 = \text{Spec}(A/I)$, and let H be an S -group prescheme of finite type, affine and flat over S .*

For H to be of multiplicative type and isotrivial, it is necessary and sufficient that H_0 be so, and that H satisfy one of the following conditions a), b) (which are therefore equivalent under the preceding conditions):

⁽¹⁷⁾N.D.E. The following sentence has been added.

⁽¹⁸⁾N.D.E. The following sentence has been added.

⁽¹⁹⁾N.D.E. Cf. EGA IV₄, 17.9.1.

a) The fibers of H are of multiplicative type and of constant type on each connected component of S .

b) H is commutative and the ${}_nH$ ($n > 0$) are finite over S .

These latter conditions are also implied by the following:

c) The fibers of H are connected.

Of course, if H is of multiplicative type (and isotrivial), conditions a), b) are satisfied by IX 1.4.1 a) and 2.1 c), so only the “sufficient” part requires a proof. We shall use the subgroup G indicated in 3.7. When condition c) is satisfied, one evidently has $G = H$ and is finished.

In case b), one notes that the immersion $u : G \rightarrow H$ induces an open immersion

$${}_nu : {}_nG \rightarrow {}_nH$$

which induces an isomorphism $({}_nG)_0 \xrightarrow{\sim} ({}_nH)_0$; as ${}_nH$ is finite over S , this implies immediately that ${}_nu$ is an isomorphism (the complement of its image is finite over S and reduces to $\emptyset$). By 3.6 it follows that the morphisms induced on the fibers $u_s : G_s \rightarrow H_s$ are isomorphisms, hence u is surjective, hence an isomorphism.

Finally, in case a) one may suppose S connected,⁽²⁰⁾ and it follows that for every $s \in S$, $u_s : G_s \rightarrow H_s$ is a monomorphism of algebraic groups of multiplicative type and of the same type over $\kappa(s)$.⁽²¹⁾ I say that such a homomorphism is necessarily an isomorphism (which will again complete the proof⁽²²⁾). Indeed, one may suppose, extending the base field if necessary, that the two groups over $\kappa(s)$ are diagonalizable, and then this follows from VIII 3.2 b) and from the fact that a surjective homomorphism of isomorphic $\mathbf{Z}$ -modules of finite type $M \rightarrow N$ is necessarily bijective.

Corollary 3.9. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, and let H be an S -group prescheme of finite type, affine and flat over S , with connected fibers.*

For H to be an isotrivial torus, it is necessary and sufficient that H_0 be so.

4. Case of an arbitrary base. Quasi-isotriviality theorem

Let A be a local ring. Recall that one says with Nagata that A is *henselian* if every algebra B finite over A is a product of local algebras B_i finite over A .

Recollection 4.0. ⁽²³⁾ *Let A be a henselian local ring, k its residue field, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(k)$, and ξ a geometric point of S_0 . Then the functor*

$$X \mapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale coverings over S and the analogous category over S_0 (cf. EGA IV₄, §18.5). Consequently (cf. SGA 1, V), one has $\pi_1(S_0, \xi) = \pi_1(S, \xi)$.

Suppose in addition A noetherian and denote by A' its completion, $S' = \operatorname{Spec}(A')$. Then A' is a complete noetherian local ring, hence henselian (loc. cit., 18.5.14), and the functor

$$X \mapsto X' = X \times_S S'$$

⁽²⁰⁾**N.D.E.** The original indicated: “i.e. S_0 connected”. Note that, as A is separated for the I -adic topology, S_0 meets every connected component of S (and as A is complete, the connected components of S_0 and of S are in bijection).

⁽²¹⁾**N.D.E.** Since G_s and H_s are of the same type for every $s \in S_0$, and hence for every s .

⁽²²⁾**N.D.E.** For u will again be a surjective open immersion.

⁽²³⁾**N.D.E.** This recollection has been expanded, and remark 4.0.1 added.

from the category of étale coverings over S to the analogous category over S' fits into a commutative diagram

$$\begin{array}{ccc} \text{Cov. ét.}(S) & \xrightarrow{\quad} & \text{Cov. ét.}(S') \\ & \searrow \simeq \quad \swarrow \simeq & \\ & \text{Cov. ét.}(S_0) & \end{array}$$

and is therefore also an equivalence of categories, whence $\pi_1(S_0, \xi) = \pi_1(S', \xi)$.

Remark 4.0.1. Since S is connected (A being local), it follows from 1.2 that the category of isotrivial groups of multiplicative type over S is equivalent to the analogous category over S_0 (and also over S' if A is in addition noetherian).

Lemma 4.1. Let A be a henselian local ring with residue field k , $S = \text{Spec}(A)$, $S_0 = \text{Spec}(k)$.

(i) The functor

$$H \mapsto H_0 = H \times_S S_0$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S_0 .

(ii) If, in addition, A is noetherian, denoting by A' its completion and by $S' = \text{Spec}(A')$, the functor

$$H \mapsto H' = H \times_S S'$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S' .

As in 4.0, the second assertion is a consequence of the first; let us prove the latter. One already knows that the functor considered is essentially surjective, since every group of multiplicative type H_0 over $S_0 = \text{Spec}(k)$, finite and hence of finite type over k , is isotrivial by 1.4, and hence comes from an isotrivial group of multiplicative type over S by 4.0.1.

It remains to prove full faithfulness, i.e. that for two finite groups of multiplicative type G, H over S the map below is bijective:

$$\text{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \text{Hom}_{S_0\text{-gr.}}(G_0, H_0),$$

(24) i.e. denoting by F the S -functor $\underline{\text{Hom}}_{S\text{-gr.}}(G, H)$, that the natural map

$$\text{Hom}_S(S, F) \longrightarrow \text{Hom}_{S_0}(S_0, F_0)$$

induced by the base change $S_0 \longrightarrow S$ is bijective. For this, in view of recollection 4.0, it suffices to prove the

Lemma 4.2. Let G, H be two finite groups of multiplicative type over a prescheme S . Then $F = \underline{\text{Hom}}_{S\text{-gr.}}(G, H)$ is representable by a finite étale scheme over S .

(25) Let $f : S' \longrightarrow S$ be a faithfully flat and quasi-compact morphism such that G', H' are diagonalizable. It suffices to show that $F_{S'}$ is representable by a prescheme X' étale and finite (hence affine) over S' , for the descent datum on X' relative to f (cf. VIII 1.7.2) will then be effective (by SGA 1 VIII, 2.1), whence the existence of an S -prescheme X such that $X \times_S S' = X'$, representing F and étale and finite over S (cf. SGA 1, V 5.7 and EGA IV₄, 17.7.3 (ii)).

One may therefore suppose $G = D_S(M)$ and $H = D_S(N)$, where M, N are finite commutative groups (cf. VIII 2.1 c)). Then $K = \text{Hom}_{\text{gr.}}(N, M)$ is a finite abelian group, and by VIII 1.5 one has an isomorphism

$$\underline{\text{Hom}}_{S\text{-gr.}}(G, H) \simeq K_S,$$

(24) **N.D.E.** The following has been added.

(25) **N.D.E.** The original has been expanded in what follows.

which completes the proof of 4.2 and hence of 4.1.

Proposition 4.3.0. ⁽²⁶⁾ *Let S be a henselian local scheme, s its closed point, $g : X \rightarrow S$ a morphism locally of finite type, x an isolated point of the fiber X_s .*

(i) Then $\mathcal{O}_{X,x}$ is finite over $\mathcal{O}_{S,s}$. (In particular, if the residue extension $\kappa(x)/\kappa(s)$ is trivial, $\mathcal{O}_{S,s} \rightarrow \mathcal{O}_{X,x}$ is surjective, by Nakayama's lemma.)

(ii) If in addition g is separated, $X' = \text{Spec}(\mathcal{O}_{X,x})$ is an open and closed subscheme of X , i.e. one has a decomposition as a disjoint sum $X = X' \amalg X''$.

PROOF. By the local form of Zariski's main theorem found in [Pes66] or [Ray70, Ch. IV, Th. 1], x has an affine open neighborhood $U = \text{Spec}(B)$ of finite type and quasi-finite over $A = \mathcal{O}_{S,s}$, and there exists an open immersion $U \hookrightarrow Y = \text{Spec}(C)$, where C is an algebra finite over A . (N.B. To reach this conclusion one may also use Chevalley's semicontinuity theorem (EGA IV₃, 13.1.4), then the form of Zariski's main theorem given in loc. cit., 8.12.8.)

Since A is henselian, Y is the disjoint sum of local schemes $Y_1, \dots, Y_n$, each finite over S , and the points of Y over s are the closed points $y_1, \dots, y_n$. Thus $x = y_i$ for some i , and $\mathcal{O}_{X,x} = \mathcal{O}_{U,x} = C_i$ is finite over A . Moreover, $X' = C_i$ is an open subscheme of U , hence of X . Suppose in addition g separated. Then, since the morphism $X' \rightarrow S$ is finite (C_i being finite over A), the same is true of the immersion $X' \rightarrow X$ (cf. EGA II, 6.1.5 (v)), so X' is also closed in X . $\square$

Lemma 4.3. *Let A be a noetherian henselian local ring, A' its completion, $S = \text{Spec}(A)$, $S' = \text{Spec}(A')$, s the closed point of S , G, H two S -group preschemes, with G of multiplicative type and of finite type over S , H locally of finite type and separated over S , H_s of multiplicative type, and H flat over S at the points of H_s .*

Let G', H' be deduced from G, H by the base change $S' \rightarrow S$. Then the natural map below is bijective:

$$\text{Hom}_{S\text{-gr.}}(G, H) \xrightarrow{\sim} \text{Hom}_{S'\text{-gr.}}(G', H').$$

Since S' is faithfully flat and quasi-compact over S , one knows by SGA 1, VIII 5.2 that this map is a bijection of the left-hand side onto the subset of the right-hand side consisting of the $u' : G' \rightarrow H'$ such that the two inverse images $u'_1, u'_2 : G'' \rightarrow H''$ of u' over $S'' = S' \times_S S'$ (by the projections pr_1, pr_2 of S'' onto S') are equal.

Thus everything amounts to proving that for every homomorphism of S' -groups $u' : G' \rightarrow H'$ one has $u'_1 = u'_2$. By density theorem IX 4.8 it suffices to prove that u'_1, u'_2 coincide on ${}_n G''$ for every integer $n > 0$. (N.B. The density theorem is needed here essentially in a case where the base prescheme, here S'' , is not locally noetherian.)

This reduces us, replacing G by ${}_n G$, to the case where there exists an integer $n > 0$ such that $n \cdot \text{id}_G = 0$, hence where G is finite over S . Similarly, let ${}_n H$ be the kernel of the n -th power morphism ϕ_n in H . (N.B. We have not supposed H commutative, so ϕ_n (resp. ${}_n H$) is not necessarily a group homomorphism (resp. a subgroup of H).) Since ${}_n H$ is defined as the fibered product of $\phi_n : H \rightarrow H$ and the unit section $\varepsilon : S \rightarrow H$, its formation commutes with every base change $T \rightarrow S$, i.e. one has $({}_n H)_T = {}_n(H_T)$.

⁽²⁷⁾ Denote by a subscript m on the right the reductions modulo $\mathfrak{m}^{m+1}$, where $\mathfrak{m}$ is the maximal ideal of A . Then H_m is flat over S_m (since H is so over S at the points of H_0); thus by 2.4, since H_0 is of multiplicative type and of finite type, the same is true of H_m . Hence each $({}_n H)_m = {}_n(H_m)$ is a subgroup of H_m , of multiplicative type, finite and flat over S_m .

In particular, $({}_n H)_0 = {}_n(H_0)$ is finite over S_0 ; since S is local henselian and ${}_n H$ is (like H) locally of finite type and separated over S , by 4.3.0 the local rings of ${}_n H$ at the points of

⁽²⁶⁾**N.D.E.** This recollection has been added; it is used in the proof of 4.3 (the reference to EGA IV₄, 18.5.11 not being entirely satisfactory).

⁽²⁷⁾**N.D.E.** The original has been expanded in what follows, on the basis of indications of M. Raynaud.

$({}_nH)_0$ are finite over S , and one has a decomposition as a disjoint sum of open subsets

$$(*) \quad {}_nH = {}_nH^+ \coprod {}_nH^-,$$

where $Z = {}_nH^+$ is finite over S , and ${}_nH^-$ lies over $S - \{s\}$.

Note that for every finite S -scheme Y (such as G), every S -morphism $Y \rightarrow {}_nH$ factors through Z , and that formation of decomposition $(*)$ commutes with the base change $S' \rightarrow S$ (where $S' = \text{Spec}(A')$, A' the completion of A).

Then $Z' = Z \times_S S'$ is a finite scheme over S' , as is $P' = Z' \times_{S'} Z'$. Denote by ν the restriction to P' of multiplication in H' , and by σ the automorphism of P' interchanging the two factors. Since P' is finite over S' and H' separated and locally of finite type over S' , by EGA II 5.4.3 and IV₁ 1.1.3, $Y = \nu(P')$ is a closed subscheme of H' , universally closed and quasi-compact, hence of finite type, hence proper over S' . Moreover $Y \rightarrow S'$ has finite fibers (since $P' \rightarrow S'$ does). Thus, since S' is noetherian, the morphism $Y \rightarrow S'$ is finite (cf. EGA III, 4.4.2). Since $Z' \subseteq Y$ and $Z'_m = Y'_m$ for every m , one has $Z' = Y$ by lemma IX 5.0, hence Z' is a subgroup of H' . Similarly, the kernel $K = \text{Ker}(\nu, \nu \circ \sigma)$ is a closed subscheme of P' such that $K_m = P'_m$ for every m (since $Z'_m = {}_nH_m$ is commutative), so $K = P'$, i.e. Z' is a commutative subgroup of H' . On the other hand, since each reduction $Z'_m = {}_nH_m$ is flat over S_m , by the “local criterion of flatness” (cf. EGA 0III, 10.2.2 or [BAC], III §5, Example 1 and Th. 1), Z' is flat over S' . Since, in addition, $Z'_0 = {}_nH_0$ is a finite group of multiplicative type over S_0 , hence isotrivial by 1.4, Z' is of multiplicative type (and isotrivial) over S' by 3.8 b). Since $S' \rightarrow S$ is faithfully flat and quasi-compact, one deduces that multiplication $Z \times_S Z \rightarrow H$ factors through Z and makes Z a subgroup of H , finite over S and of multiplicative type (since Z' is so).

Finally, by the remarks made above concerning decomposition $(*)$, Z' is the “finite part” of ${}_nH'$, and therefore the morphism $u' : G' \rightarrow {}_nH'$ takes its values in Z' . Since Z is of multiplicative type and finite over S , as is $G = {}_nG$, by 4.1 u' comes from a $u : G \rightarrow Z$, and hence $u'_1 = u''_2$. This completes the proof of 4.3.

Lemma 4.4.0. ⁽²⁸⁾ *Let A be a ring, $S = \text{Spec}(A)$, H an S -group scheme of multiplicative type and of finite type, quasi-isotrivial (resp. isotrivial), (A_i) a filtering family of subrings of A whose inductive limit is A , $S_i = \text{Spec}(A_i)$.*

Then there exists an index i and an S_i -group scheme H_i of multiplicative type and of finite type, quasi-isotrivial (resp. isotrivial), such that $H = H_i \times_{S_i} S$.

Theorem 4.4. *Let S be a prescheme, H an S -group prescheme affine and of finite presentation over S , and $s \in S$. Suppose:*

$\alpha)$ H is flat over S at the points of H_s .

$\beta)$ H_s is of multiplicative type.

Then there exists an étale morphism $S' \rightarrow S$, a point s' of S' over s such that the extension $\kappa(s')/\kappa(s)$ is trivial, and an open and closed subgroup G' of $H' = H \times_S S'$, of multiplicative type and of finite type, isotrivial, such that $G'_{s'} = H'_{s'}$.

⁽²⁹⁾ a) Denote temporarily by $T = \text{Spec}(\mathcal{O}_{S,s})$ and let us show that by “descent of conclusions” one may reduce to the case where $S = T$. Indeed, suppose one has found an étale morphism $g : T' \rightarrow T$, a point $s' \in T'$ and a subgroup G' of $H' = H_{T'}$ satisfying the conditions of the statement; replacing T' by an affine open neighborhood of s' , one may suppose T' of finite presentation over T .

Since G' is isotrivial, there exists a finite surjective étale morphism $f : \tilde{T} \rightarrow T'$ such that $\tilde{G} = G' \times_{T'} \tilde{T}$ is isomorphic to $D_{\tilde{T}}(M)$, where M is an abelian group of finite type. Since

⁽²⁸⁾**N.D.E.** This lemma has been added; it is used several times in what follows.

⁽²⁹⁾**N.D.E.** The following reductions have been expanded (the original indicated: “One reduces immediately to the case where S is the spectrum of a noetherian local ring A , and s is the closed point of S .”).

T' , H , and $\tilde{T}$, G' are of finite presentation over $T = \text{Spec}(\mathcal{O}_{S,s})$, and $\mathcal{O}_{S,s}$ is the inductive limit of its subalgebras $A_i = \mathcal{O}_S(S_i)$, where S_i ranges over the affine open neighborhoods of s in S , by EGA IV₃, 8.8.2 (and Exp. VI_B, 10.2 and 10.3) there exists an index i , S_i -preschemes (resp. an S_i -group prescheme) of finite presentation $S'_i, \tilde{S}_i$ (resp. G'_i), and morphisms $g_i : S'_i \rightarrow S_i$, $f_i : \tilde{S}_i \rightarrow S'_i$ (resp. a morphism of S'_i -group preschemes $u_i : G'_i \rightarrow H \times_S S'_i$), from which f, g (resp. $u : G' \rightarrow H'$) come by base change $T' \rightarrow S_i$. Moreover, taking i sufficiently large, g_i will be étale, f_i finite surjective étale, and u_i an open and closed immersion (cf. EGA IV, 8.10.5 and 17.7.8).

Then $\tilde{G}$ comes from the groups $\tilde{G}_i = G_i \times_{S_i} \tilde{S}_i$ and $D_{\tilde{S}_i}(M)$; thus by EGA IV₃, 8.8.2 (i) (and VI_B, 10.2), there exists an index $j \geq i$ such that $\tilde{G}_j \simeq D_{\tilde{S}_j}(M)$, so G_j is of isotrivial multiplicative type. Denote by s'_j the image of s' in S'_j . Then the étale morphism $S'_j \rightarrow S_j \hookrightarrow S$, the point s'_j and the open and closed subgroup G'_j of $H \times_S S'_j$ satisfy the conditions of the statement. This shows that one may suppose $S = \text{Spec}(A)$ local with closed point s . b) Then A is the inductive limit of local rings A_i which are localizations of $\mathbf{Z}$ -algebras of finite type; put $S_i = \text{Spec}(A_i)$. Let us show that hypotheses α), β) “descend” to some A_i . Since $H \rightarrow S$ is of finite presentation, the set of $x \in H$ such that H is flat over S at x is an open subset W , containing H_s by hypothesis, and hence containing a quasi-compact open subset V containing H_s (for H_s being affine, one may cover it by finitely many affine open subsets contained in W). Then the open immersion $\tau : V \hookrightarrow H$ is of finite presentation, so $V \rightarrow S$ is also so.

Thus by EGA IV₃, 8.8.2 (and Exp. VI_B, 10.2 and 10.3) there exists an index i , an S_i -prescheme V_i (resp. an S_i -group prescheme H_i) of finite presentation over S_i , and an S_i -morphism $\tau_i : V_i \rightarrow H_i$ from which V, H, τ come by base change $S \rightarrow S_i$; moreover, taking i sufficiently large, τ_i will be an open immersion and V_i flat over S_i , by EGA IV₃, 8.10.5 and 11.2.6. Denote by s_i the image of s in s_i ; since the open immersion $(V_i)_{s_i} \hookrightarrow (H_i)_{s_i}$ gives, by the base change $\kappa(s_i) \rightarrow \kappa(s)$, the equality $V_s = H_s$, one already has $(V_i)_{s_i} = (H_i)_{s_i}$, i.e. $(H_i)_{s_i} \subseteq V_i$, so H_i is flat over S_i at the points of $(H_i)_{s_i}$. Finally, $(H_i)_{s_i}$ is of multiplicative type, since $H_s = (H_i)_{s_i} \otimes_{\kappa(s_i)} \kappa(s)$ is so. Thus the triple (S_i, H_i, s_i) satisfies the hypotheses of 4.4, and if the desired assertion is satisfied for this triple, it will also be so, by base change, for (S, H, s) . This reduces us to the case where A is local and noetherian. Let us now distinguish two cases.

1°) A is local noetherian and henselian. Let $\hat{A}$ be its completion, $\hat{S} = \text{Spec}(\hat{A})$ and $\hat{H} = H \times_S \hat{S}$. Apply theorem 3.7: one finds a $\hat{S}$ -group $\hat{G}$ of multiplicative type, isotrivial and of finite type, and a homomorphism of $\hat{S}$ -groups

$$\hat{u} : \hat{G} \rightarrow \hat{H}$$

which is an open immersion and a closed immersion, such that $\hat{u}$ induces an isomorphism $\hat{u}_0 : \hat{G}_0 \xrightarrow{\sim} \hat{H}_0$.

By remark 4.0.1, the base change functor by $\hat{S} \rightarrow S$ induces an equivalence between the category of isotrivial groups of multiplicative type over S and over $\hat{S}$; in particular $\hat{G}$ “comes from” an S -group of multiplicative type G , isotrivial and of finite type. By 4.3, $\hat{u}$ comes from a homomorphism

$$u : G \rightarrow H;$$

moreover, u is an open and closed immersion and induces an isomorphism $u_0 : G_0 \rightarrow H_0$, since this is so after the faithfully flat quasi-compact base change $\hat{S} \rightarrow S$. This therefore proves 4.4 in this case (taking of course, in the conclusion of 4.4, $S' = S$ and $s' = s$).

2°) A is local noetherian. Reduction to case 1°) is immediate by applying 1°) to the “henselization” A^h of A . More precisely, one sees easily (using SGA 1, I §5⁽³⁰⁾) that the local rings $\mathcal{O}_{S',s'}$ of étale S -preschemes S' endowed with a point s' over s such that $\kappa(s')/\kappa(s)$ is trivial form a filtering inductive system whose inductive limit is a noetherian henselian local

⁽³⁰⁾Or EGA IV₄, §18.6.

ring A^h (called the “henselization” of A); for details of this construction (due to Nagata in the normal case), cf. SGA 4, VIII §4.⁽³⁰⁾ The sorites of EGA IV₃, §8 then allow one, as in part a) of the proof, to deduce from a result known over the inductive limit A^h of the $\mathcal{O}_{S',s'}$ an analogous result over one of the (S', s') , which proves 4.4.

Corollary 4.5. *Let S be a prescheme, H an S -group prescheme of multiplicative type and of finite type. Then H is quasi-isotrivial, i.e. is split by a surjective étale morphism $S' \rightarrow S$.*

Indeed, let $s \in S$. By 4.4, there exists an étale morphism $S' \rightarrow S$, an $s' \in S'$ over s such that $\kappa(s) = \kappa(s')$, and a subgroup G' of H' , of isotrivial multiplicative type and of finite type, such that $G'_{s'} = H'_{s'}$. Since G', H' are of multiplicative type and of finite type, by IX 2.9 there exists an open neighborhood U' of s' such that $G'|U' = H'|U'$.

⁽³¹⁾ Suppose in addition S local henselian with closed point s ; then by EGA IV₄, 18.5.11 b) there exists a section σ of $S' \rightarrow S$ such that $\sigma(s) = s'$. (N.B. One may see directly that $B = \mathcal{O}_{S',s'}$ equals $A = \mathcal{O}_{S,s}$ as follows: by 4.3.0 (i), one has $B \simeq A/I$, and since B is an A -algebra of finite presentation and flat, I is an ideal of finite type (cf. EGA IV₁, 1.4.7), and $I = \text{Im}$ (where $\mathfrak{m}$ is the maximal ideal of A), whence $I = 0$.) Thus H is already isotrivial. One therefore obtains:

Corollary 4.6. *Let A be a henselian local ring, k its residue field, and π the topological Galois group of an algebraic closure of k .*

- (i) *Every group of multiplicative type and of finite type over $S = \text{Spec}(A)$ is isotrivial.*
- (ii) *The category of these groups over S is equivalent (via the functor $H \mapsto H \otimes_A k$) to the analogous category over $S_0 = \text{Spec}(k)$; it is therefore anti-equivalent, by 1.4, to the category of Galois modules under π of finite type as $\mathbf{Z}$ -modules.*

Corollary 4.7. *Under the conditions of 4.4, suppose in addition one of the following conditions satisfied:*

- a) *For every generization t of s , H_t is of multiplicative type and of the same type as H_s .*
 - b) *H is commutative and the ${}_n H$ ($n > 0$) are finite over S in a neighborhood of s .*
 - c) *For every generization t of s , the fiber H_t is connected.*
- Then there exists an open neighborhood U of s such that $H|U$ is of multiplicative type.*

Taking account of the fact that an étale morphism is open, one is reduced to proving that there exists (with the notation of the conclusion of 4.4) an open neighborhood U' of s' such that $G'|U' = H'|U'$. Put $S'' = \text{Spec}(\mathcal{O}_{S',s'})$; as G', H' are of finite presentation over S' , it suffices to show, by EGA IV₃, 8.8.2, that $G'' = H''$. One may therefore suppose $S = S''$, and then hypotheses (a), (b), (c) become those of the lemma below, whose proof is the same as that of 3.8:

Lemma 4.7.1. *Let S be a local scheme, s its closed point, H an S -group scheme of finite type, G an open and closed subgroup of H , of multiplicative type, such that $G_s = H_s$. Suppose in addition one of the following conditions satisfied:*

- a) *The fibers of H are of multiplicative type and of the same type as H_s .*
 - b) *H is commutative and the ${}_n H$ ($n > 0$) are finite over S .*
 - c) *The fibers of H are connected.*
- Then $G = H$.*

Corollary 4.8. *Let S be a prescheme, H an S -group prescheme affine, flat and of finite presentation over S , with fibers of multiplicative type.*

For H to be of multiplicative type, it is necessary and sufficient that it satisfy one of the following two conditions (equivalent under the preceding conditions):

- a) *The type of H_s (cf. IX 1.4) is a locally constant function of $s \in S$.*
- b) *H is commutative, and the ${}_n H$ ($n > 0$) are finite over S .*

⁽³¹⁾N.D.E. The original has been expanded in the preceding and following passages.

Moreover, these conditions a), b) are implied by the following:

c) The fibers of H are connected.

In particular, one finds:

Corollary 4.9. *Let S be a prescheme, H an S -group prescheme flat and of finite presentation over S . Suppose in addition H affine over S with connected fibers.⁽³²⁾ If $s \in S$ is such that H_s is a torus, there exists an open neighborhood U of s such that $H|_U$ is a torus.*

In particular, if all the fibers of H are tori, H is a torus.

5. Scheme of homomorphisms from one group of multiplicative type into another. Twisted constant groups and groups of multiplicative type

Definition 5.1. a) Let S be a prescheme, R a group prescheme over S . One says that R is a *twisted constant group over S* if it is locally isomorphic, in the sense of the faithfully flat quasi-compact topology, to a constant group scheme, i.e. of the form M_S with M an ordinary group.

b) One says that the twisted constant group R over S is *quasi-isotrivial*, resp. *isotrivial*, resp. *locally isotrivial*, resp. *locally trivial*, resp. *trivial*, if in the above definition one can replace the faithfully flat quasi-compact topology by the étale topology, resp. the global finite étale topology, resp. the finite étale topology, resp. the Zariski topology, resp. the coarsest (or “chaotic”) topology, cf. IX 1.1 and 1.2, and IV 6.6.

Saying that R is quasi-isotrivial (resp. isotrivial) therefore means that there exists a surjective étale (resp. and finite) morphism $S' \rightarrow S$ such that $R' = R \times_S S'$ is a constant group over S ; saying that it is trivial means that R is a constant group. c) One defines as in VIII 1.4 the *type* of a twisted constant group R over S at an $s \in S$; it is an isomorphism class of ordinary groups which, for variable s , is a locally constant function of s , and hence constant if S is connected. One will also say that R is “of type M ” if all the fibers of R are of type M .

One should take care that R is quasi-compact over S only if it is finite over S , i.e. if its type at every $s \in S$ is a finite group.⁽³³⁾

d) The case of greatest interest to us is that where R is commutative. We shall then say that R is “finitely generated” if its type at every point $s \in S$ is given by a $\mathbf{Z}$ -module of finite type, a notion which must not be confused with the schematic notion “ R of finite type over S ” (cf. above).

Remark 5.2. We shall also have to consider S -preschemes X which are locally isomorphic (for the faithfully flat quasi-compact topology) to constant preschemes, independently of any group structure. We shall then say that X is a *twisted constant bundle over S* , and extend to these preschemes the terminology introduced in 5.1. Of course, one should take care that when X is endowed with an S -group structure, the meaning of the expressions “twisted constant”, “isotrivial”, etc. changes according to whether one takes account of the group structure over S . The same is true if one considers on X any other kind of algebraic structure (for example that of a Galois principal bundle, which will be considered in the next no.).

Proposition 5.3. *Let R be a commutative twisted constant group over S .*

(i) *The functor $H = D_S(R)$ (cf. VIII 1) is representable and is a group of multiplicative type over S .*

(ii) *For every $s \in S$, the type of R at s equals that of H at s .*

⁽³²⁾**N.D.E.** This result is generalized in 8.1: in fact it suffices to suppose the fibers of H affine and connected.

⁽³³⁾**N.D.E.** See lemma 5.12 below.

(iii) For R to be quasi-isotrivial (resp. isotrivial, resp. trivial, resp. locally isotrivial, resp. locally trivial), it is necessary and sufficient that H be so.

Remark 5.3.1. ⁽³⁴⁾ One may be concerned to see the term “type” used in 5.3 in two different senses according to whether R or H is involved; fortunately, when an S -group prescheme G is both a twisted constant group and of multiplicative type, its type in either sense is the same, thanks to the fact that an ordinary finite commutative group is isomorphic to its dual!

PROOF. As covering families for the faithfully flat quasi-compact topology are families of effective descent for the fibered category of affine group preschemes over variable base preschemes (SGA 1, VIII 2.1), one sees that H is representable (and affine over S), since it is so “locally”⁽³⁵⁾ (for it is so when R is constant, and then H is a diagonalizable group).

The fact that H is of multiplicative type is then evident by definition, as is the fact that the type of R and H at $s \in S$ is the same. Finally, since $H_{S'} = D_{S'}(R_{S'})$, the last assertion is reduced to the “trivial” case, i.e. to checking that R is trivial if and only if H is so, which follows immediately from biduality theorem VIII 1.2. $\square$

To finish specifying the correspondence between twisted constant groups and groups of multiplicative type, one must start with a group of multiplicative type H and study $R = D_S(H)$. If the latter is representable, it is evidently a twisted constant group, and one shall have $H \simeq D_S(R)$. In other words:

Scholium 5.4.0. The functor $R \mapsto D_S(R)$ is an anti-equivalence⁽³⁶⁾ between the category of twisted constant groups over S and that of groups of multiplicative type H over S such that $D_S(H)$ is representable.

I do not know whether this condition on H is always satisfied; we shall nevertheless see that it is when H is quasi-isotrivial, in particular when H is of finite type.

Lemma 5.4. Let $S' \rightarrow S$ be a faithfully flat morphism locally of finite presentation, X' an S' -prescheme separated, locally of finite presentation and locally quasi-finite over S' .

Then every descent datum on X' relative to $S' \rightarrow S$ is effective, i.e. there exists an S -prescheme X and an S' -isomorphism $X \times_S S' \simeq X'$ compatible with the descent datum.

One has already noted that the hypothesis on $S' \rightarrow S$ implies that it is a covering morphism for the faithfully flat quasi-compact topology (IV 6.3.1 (iv)), a fortiori a morphism of effective descent.

When $X' \rightarrow S'$ is quasi-compact, hence of finite presentation and quasi-finite, it is a quasi-affine morphism (cf. SGA 1, VIII 6.2⁽³⁷⁾), and effectiveness follows in this case from SGA 1, VIII 7.9. In the general case one reduces immediately to the case where S, S' are affine. One covers X' by affine open subsets U'_i ; let V'_i be the saturation of U'_i for the equivalence relation in X' defined by the descent datum, i.e. $V'_i = q_2(q_1^{-1}(U'_i))$, where q_1, q_2 are the two projections of $X'_1 = X' \times_{S'} (S'', p_1)$ onto X' ($q_1 = \text{pr}_1$, and q_2 is deduced from the first projection of $X'_2 = X' \times_{S'} (S'', p_2)$ by the given descent isomorphism $X'_1 \simeq X'_2$). Since $S' \rightarrow S$ is faithfully flat quasi-compact locally of finite presentation, so is $p_1 : S'' = S' \times_S S' \rightarrow S'$, hence so are q_1, q_2 , which are consequently open morphisms (SGA 1 IV 6.6⁽³⁸⁾). Consequently V'_i is an open and quasi-compact subset of X' . By what has already been seen, the descent data induced on the V'_i are effective, whence it follows that the datum on X' is also so (SGA 1, VIII 7.2).

⁽³⁴⁾N.D.E. This remark, which appeared after the proof in the original, has been moved here.

⁽³⁵⁾N.D.E. Cf. VIII §1.7.

⁽³⁶⁾N.D.E. “Equivalence” has been corrected to “anti-equivalence”.

⁽³⁷⁾N.D.E. (When S' is locally noetherian, and EGA IV₃, 8.11.2 in general.)

⁽³⁸⁾N.D.E. (When S' is locally noetherian, and EGA IV₂, 2.4.6 in general.)

Corollary 5.5. *A faithfully flat morphism of finite presentation $S' \rightarrow S$ is a morphism of effective descent for the fibered category of twisted constant groups (over variable base preschemes).*

Indeed, this amounts to asserting the effectiveness of a descent datum under the conditions of 5.4 when X' is a constant S' -prescheme.

Theorem 5.6. *Let S be a prescheme, G, H two S -group preschemes of quasi-isotrivial multiplicative type, with $G^{(39)}$ of finite type.*

Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$ is representable by, and is, a quasi-isotrivial twisted constant group over S ;⁽⁴⁰⁾ for every $s \in S$, if the type of G (resp. H) at s is M (resp. N), that of $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$ is $\mathrm{Hom}_{\mathrm{gr.}}(M, N)$.

One proceeds as in 4.2, using the fact that the assertion is established (VIII 1.5) when G, H are trivial. The necessary effectiveness criterion is supplied by 5.5 (in the case of a surjective étale morphism $S' \rightarrow S$).

In particular, putting $G = \mathbf{G}_{m,S}$, one finds:

Corollary 5.7. (i) *Let H be an S -group of quasi-isotrivial multiplicative type; then the S -group $D_S(H)$ is representable and is a quasi-isotrivial twisted constant group over S .*

(ii) *The functors $R \mapsto D_S(R)$ and $H \mapsto D_S(H)$ are anti-equivalences, quasi-inverse to one another, between the category of quasi-isotrivial twisted constant S -groups and that of S -groups of quasi-isotrivial multiplicative type.*

(iii) *These functors induce anti-equivalences between the subcategories consisting of isotrivial, resp. locally isotrivial, resp. locally trivial, resp. trivial groups.*

The last assertion is mentioned only for the record, being already contained in 5.3.

Moreover, since every S -group of multiplicative type and of finite type is quasi-isotrivial by 4.5, one deduces from 5.6:

Corollary 5.8. *Let S be a prescheme, G, H two S -group preschemes of multiplicative type and of finite type; then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$ is representable and is a finitely generated quasi-isotrivial twisted constant S -group over S .*

Note also that in 5.3, R is finitely generated if and only if $H = D_S(R)$ is of finite type (IX 2.1 b)). By 4.5, H is then quasi-isotrivial, so R is quasi-isotrivial. One thus finds:

Corollary 5.9. *The functors of 5.7 induce anti-equivalences quasi-inverse to one another between the category of S -groups H of multiplicative type of finite type and that of finitely generated twisted constant S -groups R ; moreover, every such group R is quasi-isotrivial.*

Corollary 5.10. *Let H, G be two S -group preschemes of multiplicative type and of finite type.*

(i) *Then $\underline{\mathrm{Isom}}_{S\text{-gr.}}(H, G)$ is representable by an open and closed subprescheme of $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$, and is a twisted constant S -prescheme.*

(ii) *In particular, $\underline{\mathrm{Aut}}_{S\text{-gr.}}(H)$ is representable and is a twisted constant S -group (in general noncommutative).*

This follows from 5.8 and VIII 1.6.⁽⁴¹⁾

Recollection 5.11.0.⁽⁴²⁾ Recall that if X is a locally noetherian prescheme, its connected components (which are always closed) are open. Indeed, let C be a connected component

⁽³⁹⁾**N.D.E.** H has been corrected to G .

⁽⁴⁰⁾**N.D.E.** The following has been added.

⁽⁴¹⁾**N.D.E.** Correct this reference by treating the case of the functor $\underline{\mathrm{Isom}}_{S\text{-gr.}}(H, G)$ in an addition 1.5.1 in VIII...

⁽⁴²⁾**N.D.E.** This recollection has been added; it is used several times in what follows. (See also EGA I, 6.1.9; note, however, that the proof in loc. cit. seems unnecessarily complicated.)

of X , $x \in C$ and U a noetherian open neighborhood of x ; then U has only finitely many connected components, so the connected component U_x of x in U is open in U , hence in X ; as $U_x \subseteq C$, this shows that C is open in X .

Proposition 5.11. *Let S be a prescheme, R a commutative twisted constant group over S , $H = D_S(R)$ the group of multiplicative type it defines. Consider the following conditions:*

- (i) H is isotrivial (i.e. R is isotrivial).
 - (ii) R is a union of subpreschemes R_i which are both open and closed and quasi-compact over S (and then necessarily finite over S).
 - (iii) The connected components of R are finite over S .
- a) Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii).⁽⁴³⁾
 b) One has (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) if S is locally noetherian.
 c) Finally, (i) $\Leftrightarrow$ (ii) if R is finitely generated (i.e. if H is of finite type over S), at least if S is quasi-compact or its connected components are open.

First decomposing S as a sum prescheme of preschemes S_i on each of which H has constant type (cf. IX 1.4.1), one is reduced to the case where H , and hence R , has constant type M . We shall need the

Lemma 5.12. *Let S be a prescheme, R a twisted constant prescheme over S . Then every closed subprescheme Z of R which is quasi-compact over S is finite over S .*

Indeed, one is reduced to the case where R is constant, hence of the form I_S , where I is a set, hence the increasing filtering union of the J_S , where J ranges over the finite subsets of I . One may suppose in addition S affine; then Z is quasi-compact, hence contained in one of the J_S . Since J_S is finite over S , so is Z .

Lemma 5.12 already establishes the assertion in parentheses in 5.11 (ii).⁽⁴⁴⁾ The implication (ii) $\Rightarrow$ (iii) is then clear, since the connected components of R are closed; by 5.11.0 they are open and closed if S is locally noetherian (R being étale, hence locally of finite type over S), whence (iii) $\Rightarrow$ (ii) in this case.

Let us prove (i) $\Rightarrow$ (ii). For this, let $S' \rightarrow S$ be a finite surjective étale morphism splitting H , and hence R (cf. 5.3), so that R' is isomorphic to $M_{S'} = \coprod_{m \in M} R'_m$, where the R'_m are disjoint open subsets of R' , S' -isomorphic to S' . Let $g: R' \rightarrow R$ be the projection, which is a finite surjective étale morphism, hence an open and closed morphism; then the $R_m = g(R'_m)$ are subsets both open and closed in R , and evidently quasi-compact over S since the R'_m are so.

Finally, suppose H of finite type over S , and let us prove (ii) $\Rightarrow$ (i). The case where the connected components of S are open reduces immediately to the case where S is connected, so one may suppose S quasi-compact or connected. Since M is finitely generated, one may write $M = \mathbf{Z}^r \times N$, where r is an integer ≥ 0 and N a finite abelian group. Let $G = D_S(M)$; consider the preschemes

$$P = \underline{\text{Isom}}_{S\text{-gr.}}(H, G) \subset \underline{\text{Hom}}_{S\text{-gr.}}(H, G) = Q,$$

cf. 5.8 and 5.10. One has isomorphisms

$$\begin{aligned} Q &\simeq \underline{\text{Hom}}_{S\text{-gr.}}(M_S, R) \\ &\simeq \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{Z}_S^r, R) \times \underline{\text{Hom}}_{S\text{-gr.}}(N_S, R) \simeq R^r \times E, \end{aligned}$$

where $E = \underline{\text{Hom}}_{S\text{-gr.}}(N_S, R)$ is finite over S .⁽⁴⁵⁾ It follows that Q is a union of subpreschemes Q_i both open and closed and finite over S . Thus P is the union of the subpreschemes $P_i = P \cap Q_i$, both open and closed and finite over S . Since they are étale over S , their images in S are subsets S_i both open and closed, and they cover S . If S is connected or

⁽⁴³⁾**N.D.E.** Compare with examples 1.6. . .

⁽⁴⁴⁾**N.D.E.** The original has been expanded in what follows.

⁽⁴⁵⁾**N.D.E.** For it is a twisted constant group of type $\text{End}_{\text{gr.}}(N)$ (cf. 5.6 and 5.8).

quasi-compact, there therefore exists a finite set of indices i such that the S_i cover S ; let S' be the union of the corresponding P_i . Then $S' \rightarrow S$ is finite surjective étale, and putting $P' = P \times_S S' = \underline{\text{Isom}}_{S' \text{-gr.}}(H', G')$, one sees that P' has a section over S' , so there exists an isomorphism of S' -groups

$$H' = H \times_S S' \xrightarrow{\sim} G' = G \times_S S' = D_{S'}(M),$$

which proves that S' splits H . This completes the proof of 5.11.⁽⁴⁶⁾

Remark 5.11 bis. Let us note moreover that when H is of finite type over S and of constant type one may give the following isotriviality criterion (where it is no longer necessary to impose any restriction on S): $H = D_S(R)$ is isotrivial if and only if R is a union of a sequence of subsets both open and closed and finite over S .⁽⁴⁷⁾

Lemma 5.13. *Let S be a locally noetherian connected prescheme, P a quasi-isotrivial twisted constant S -prescheme, Z an open and closed subset of P such that there exists an $s \in S$ with Z_s finite. Then Z is finite over S .*

Let us not initially suppose S connected: let U be the set of $s \in S$ such that Z_s is finite; we shall prove that U is both open and closed, and that $Z|_U$ is finite over U . This is evidently a statement essentially equivalent to 5.13 but with the advantage of being “of local nature” on S for the étale topology (say), which reduces us to the case where P is constant, i.e. of the form I_S , where I is a set. (N.B. The locally noetherian hypothesis is not lost by an étale base change; this is where one uses the quasi-isotriviality of P over S .)

One may suppose in addition S connected, since its connected components are open (S being locally noetherian). But then one necessarily has $Z = J_S$, where J is a subset of I , and hence $U = \emptyset$ or $U = S$ according as J is infinite or finite, which gives the desired conclusion.

Recollections 5.14.0.⁽⁴⁸⁾ Let S be a locally noetherian prescheme, and let $\tilde{S}$ be the normalization of S_{red} (cf. EGA II, 6.3.8). Recall that S is said to be *geometrically unibranch* (cf. EGA 0_{IV}, §23.2 and IV₂, §6.15) if the canonical morphism $\tilde{S} \rightarrow S$ is radicial (and hence a universal homeomorphism); in particular, the connected components of S are irreducible.

Suppose then S connected, hence irreducible; let η be its generic point and let $f : P \rightarrow S$ be a flat and locally quasi-finite morphism. Let P_i be the irreducible components of P , and ξ_i the generic point of P_i . Since P is flat over S , each ξ_i lies over η (cf. EGA IV₂, 2.3.4), and therefore $(P_i)_\eta = P_i \cap P_\eta$ is the closure of ξ_i in P_η . Since the fiber P_η is discrete, one therefore has $(P_i)_\eta = \{\xi_i\}$. This applies in particular when f is étale; in this case P is also locally noetherian and geometrically unibranch (cf. EGA IV₄, 17.5.7), so its irreducible components are its connected components and are open (and closed).

Corollary 5.14. *Let S be a locally noetherian geometrically unibranch prescheme, P a quasi-isotrivial twisted constant S -prescheme. Then the connected components of P are finite over S .*

⁽⁴⁶⁾**N.D.E.** Modulo verification that (ii) $\Rightarrow$ (i) when S is locally noetherian and R is not finitely generated...

⁽⁴⁷⁾**N.D.E.** Expand this point: M being a $\mathbf{Z}$ -module of finite type, it is countable, and the proof of 5.11 (i) $\Rightarrow$ (ii) shows that R is a union of a countable family of open and closed subsets finite over S . One should see the converse...

⁽⁴⁸⁾**N.D.E.** The original treated in 5.14 the case where S is locally noetherian and normal, and pointed out in Remark 5.15 that the argument applies more generally if one merely supposes S geometrically unibranch instead of normal. The statement of 5.14 (and also 5.16) has been modified accordingly, and these “recollections” taken from EGA IV₄, 18.10.6 and 18.10.7 have been added, showing that the proof of 5.14 applies verbatim to the geometrically unibranch case.

One may evidently suppose S connected, hence irreducible; let η be its generic point. By the preceding, each connected component P_i of P is open and closed and meets the fiber P_η in a single point. Thus 5.13 applies and shows that each P_i is finite over S .

Theorem 5.16. ⁽⁴⁹⁾ *Let S be a locally noetherian geometrically unibranch prescheme. Then every S -group of multiplicative type and of finite type is isotrivial.*

Indeed, one may suppose S connected, hence H of constant type M . It suffices to apply 5.14 to $P = \underline{\text{Isom}}_{S\text{-gr.}}(H, G)$, where $G = D_S(M)$, then to argue as in the proof of 5.11 (ii) $\Rightarrow$ (i). One may also apply 5.14 to $P = R = D_S(H)$ (cf. 5.9), then use 5.11.

6. Infinite Galois principal coverings and the enlarged fundamental group

(The results of the present no. and the next will no longer be used in the rest of this Seminar.)

Let S be a prescheme; we propose to determine the principal homogeneous bundles P over S with structural group of the form G_S , the constant S -group defined by an ordinary group G (not necessarily finite), which one shall also call *Galois principal bundles over S with group G* . We take “principal bundle” in the sense of the faithfully flat and quasi-compact topology (cf. Exp. IV, Def. 5.1.5), but one will note that for such a P , the structural morphism $P \rightarrow S$ is necessarily étale and surjective, hence covering for the étale topology, so P is also locally trivial for the étale topology (cf. IV, Prop. 5.1.6).⁽⁵⁰⁾

We shall suppose that S is a sum of connected preschemes, i.e. that its connected components are open, which reduces us immediately to the case where S is connected. We shall then choose a “geometric point” ξ of S , i.e. an S -scheme ξ which is the spectrum of an algebraically closed field $\Omega = \kappa(\xi)$. Then for every Galois principal bundle P over S with group G , P_ξ is a Galois principal bundle over the algebraically closed field $\kappa(\xi)$, and hence trivial. We shall therefore specify the initial problem by proposing to determine the category of Galois principal bundles over S pointed over ξ , i.e. endowed with an S -homomorphism $\xi \rightarrow P$, i.e. with a trivialization of P_ξ . For fixed G , the set of classes of such bundles up to an isomorphism respecting the base point will be denoted by $\pi^1(S, \xi; G)$. Then the set $\pi^1(S; G)$ of isomorphism classes of Galois principal bundles over S with group G (without specified base point) is isomorphic to the set of orbits of G in $\pi^1(S, \xi; G)$ (taking account of the natural operations of G on this set, corresponding to translation by G of the marked point in a pointed Galois principal bundle P):

$$\pi^1(S; G) = \pi^1(S, \xi; G)/G.$$

For every morphism $S' \rightarrow S$ which is a morphism of universal effective descent for the fibered category of twisted constant preschemes over a variable base (for example $S' \rightarrow S$ faithfully flat and locally of finite presentation, cf. 5.4; for other examples cf. SGA 1 IX), we propose to determine the subsets of the preceding sets, denoted by $\pi^1(S'/S, \xi; G)$ and $\pi^1(S'/S; G)$, consisting of Galois principal bundles over S which become trivial over S' (or, as one says, are “split” by S'). One will in fact determine the category of Galois principal bundles P over S split by S' . Of course, one shall then have

$$\pi^1(S, \xi; G) = \varinjlim_{S'} \pi^1(S'/S, \xi; G),$$

where on the right-hand side S' ranges over a cofinal system in the set of S'/S covering for the étale topology (for example, when S is quasi-compact, the set of S' over S which are quasi-compact with étale and surjective structural morphism). Similarly, the category of

⁽⁴⁹⁾**N.D.E.** Remark 5.15, rendered obsolete by addition 5.14.0 (cf. the preceding N.D.E.), has been deleted, and in 5.16 “normal” has been replaced by “geometrically unibranch”.

⁽⁵⁰⁾**N.D.E.** Note that P is supposed to be a prescheme — in the a priori more general case of an (fpqc) sheaf P which is a G_S -torsor, is P necessarily representable?

Galois principal bundles over S will be the inductive limit of the subcategories defined by the S' (consisting of bundles split over S').

Thanks to the hypothesis on S'/S , the category of Galois principal bundles over S split by S' is equivalent to the category of trivial Galois principal bundles over S' (hence of the form $G_{S'}$, where G acts by right translations), endowed with a descent datum relative to $S' \rightarrow S$. Giving a base point on a Galois principal bundle P over S split by S' amounts, in terms of the corresponding trivial bundle P' over S' and its descent datum, to giving a trivialization of $P' \times_{S'} S'_\xi$ compatible with the induced descent datum relative to $S'_\xi \rightarrow \xi$ (N.B. One has put $S'_\xi = S' \times_S \xi$), i.e. a section σ of P'_ξ over S'_ξ compatible with the descent datum. It is then useful, for any S -prescheme S' (for which one no longer supposes that $S' \rightarrow S$ is a morphism of universal effective descent for the fibered category of twisted constant bundles...), to define $\pi^1(S'/S; G)$ and $\pi^1(S'/S, \xi; G)$ as the set of isomorphism classes of the structures with descent data just specified. One then obtains for these functors in G a very simple simplicial description, in terms of the fibered square and cube S'', S''' of S' over S , which we shall sketch below (cf. 6.3).

The important conclusion to retain will be the following:

Proposition 6.1. *Suppose that the connected components of S' and S'' are open, and, for example, that the quotient set of $\pi_0(S')$ by the equivalence relation induced by the two projections $\pi_0(S'') \rightarrow \pi_0(S')$ consists of one point.⁽⁵¹⁾*

(i) *The functor $G \mapsto \pi^1(S'/S, \xi; G)$ from the category of groups to the category of sets is representable by a group, denoted by $\pi_1(S'/S, \xi)$ and called the fundamental group of S at ξ relative to $S' \rightarrow S$. One therefore has a functorial bijection:*

$$\pi^1(S'/S, \xi; G) \simeq \text{Hom}_{\text{gr.}}(\pi_1(S'/S, \xi), G).$$

(ii) *This group has a set of generators in bijection with $\pi_0(S'')$, and is described in terms of these generators by relations in bijection with the elements of $\pi_0(S''')$.⁽⁵²⁾ In particular, $\pi^1(S'/S, \xi)$ is finitely generated (resp. of finite presentation) if $\pi_0(S'')$ (resp. as well as $\pi_0(S''')$) is finite. (iii) *The category of Galois principal bundles over S split by S' , with base point over ξ , is equivalent to the category of ordinary groups G endowed with a homomorphism from $\pi_1(S'/S, \xi)$ to G .**

The proof is given below, cf. ...

6.2. — When S is a connected locally noetherian prescheme, which implies that every étale prescheme S' over S is locally noetherian and hence has open connected components, one concludes from the preceding⁽⁵³⁾ that the functor $G \mapsto \pi^1(S, \xi; G)$ from the category of ordinary groups to the category of sets is strictly pro-representable (cf. Séminaire Bourbaki, February 1960, no. 195, §§ A.2 and A.3), i.e. there exists a projective system

$$\Pi = \Pi_1(S; \xi) = (\pi_i)_{i \in I}$$

⁽⁵¹⁾**N.D.E.** The hypothesis that the connected components of S' are open has been added, as has the second hypothesis. The latter means that the simplicial set $\mathbf{K}_\bullet = \pi_0(S_\bullet)$ defined in 6.3 is connected; in fact a weaker hypothesis suffices, namely that the cone $\bar{\mathbf{K}}_\bullet$ of a certain morphism of simplicial sets $\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet$ be connected (cf. loc. cit.).

⁽⁵²⁾**N.D.E.** The superfluous hypothesis that the connected components of S''' are open has been deleted. On the other hand, the description given below (cf. ...) gives as natural set of generators the set $\pi_0(S'') \amalg \pi_0(S'_\xi)$; one then reduces to $\pi_0(S'')$ by means of the relations between these generators coming from the 2-cells. See, for example, [Kan58], §19 for a finer description.

⁽⁵³⁾**N.D.E.** One could expand this deduction; I is in bijection with a cofinal set of morphisms $S' \rightarrow S$ covering for the étale topology...

of ordinary groups over an increasing filtering index set I which is “strict” (i.e. with surjective transition morphisms $\pi_j \rightarrow \pi_i$), and an isomorphism of functors in G

$$\pi^1(S, \xi; G) \simeq \varinjlim_i \operatorname{Hom}_{\operatorname{gr.}}(\pi_i, G).$$

The right-hand side is also simply denoted by $\operatorname{Hom}_{\operatorname{pro-gr.}}(\Pi, G)$ (cf. loc. cit.).

In the case where the projective limit $\pi = \varprojlim \pi_i$ is “sufficiently large”, more precisely when the canonical homomorphisms $\Pi \rightarrow \Pi_i$ are surjective, there is occasion to endow Π with the projective limit topology of the discrete topologies of the Π_i , and the preceding isomorphism is also written:

$$\pi^1(S, \xi; G) \simeq \operatorname{Hom}_{\operatorname{gr. top.}}(\pi, G),$$

where the right-hand side denotes the set of homomorphisms of topological groups, it being understood that G is endowed with the discrete topology. The hypothesis just formulated on the projective system Π is satisfied, as is well known, when the π_i are finite groups (cf. [BEns], III §7.4, Th. 1). This latter condition evidently also means that every Galois principal bundle over S is isotrivial, i.e. split by a finite surjective étale morphism. This is the case when S is geometrically unibranch (for example normal), as follows immediately from 5.14.⁽⁵⁴⁾ In the case where the π_i are finite, the group π also coincides with the fundamental group $\pi_1(S, \xi)$ introduced in SGA 1, V.

Thus, in the favorable case ($\pi \rightarrow \pi_i$ surjective), one could call π the *enlarged fundamental group* of S at ξ . Outside this favorable case, π itself is of little interest, and the role of the usual fundamental group is played by the projective system Π itself, which one shall call the *enlarged fundamental pro-group* of S at ξ . (Any terminological suggestion better than “enlarged” is welcome!⁽⁵⁵⁾) One will note that knowledge of this pro-group is more precise than that of the usual fundamental group $\pi_1(S, \xi)$ of SGA 1 V; more precisely, the latter is the projective limit of the projective system consisting of the finite quotients of the π_i .

6.3. — Let us indicate briefly the “calculation” of $\pi_1(S'/S, \xi)$. Let S_i be the $(i+1)$ -st fibered power of S' over S (i.e. $S_0 = S'$, $S_1 = S''$, etc.). There are evident simplicial operations between the S_i , making $(S_i)_{i \in \mathbb{N}}$ a simplicial object of $(\operatorname{Sch})/S$.

Transforming this simplicial object by the functor “set of connected components”

$$\pi_0 : (\operatorname{Sch})/S \rightarrow (\operatorname{Ens}),$$

one finds a simplicial set $\mathbf{K}_\bullet = (K_i)_{i \in \mathbb{N}}$, with $K_i = \pi_0(S_i)$.

Similarly, the $S_{i\xi}$ (= the $(i+1)$ -st fibered power of S'_ξ over ξ) form a simplicial object of $(\operatorname{Sch})/\xi$, hence of $(\operatorname{Sch})/S$, moreover endowed with a natural homomorphism of simplicial objects into $(S_i)_{i \in \mathbb{N}}$, whence a simplicial set $\mathbf{k}_\bullet$ (with $k_i = \pi_0(S_{i\xi})$) and a canonical homomorphism

$$\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet.$$

We may form a new simplicial set by taking the cone of this morphism (cf. 9.5.1):

$$\tilde{\mathbf{K}}_\bullet = \operatorname{Cone}(\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet).$$

⁽⁵⁶⁾ In this way one obtains a “pointed simplicial set” $\tilde{\mathbf{K}}_\bullet$ (i.e. a simplicial set endowed with a homomorphism $\tilde{\xi} : \mathbf{e}_\bullet \rightarrow \tilde{\mathbf{K}}_\bullet$, where $\mathbf{e}_\bullet$ is the final simplicial set). We may construct its well-known combinatorial invariants $\pi_0(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$ and $\pi_1(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$, whose construction moreover involves only the components of degree ≤ 1 resp. of degree ≤ 2 . These invariants are defined without restriction on S or S' . One verifies without difficulty, when the connected

⁽⁵⁴⁾N.D.E. 5.13 has been corrected to 5.14.

⁽⁵⁵⁾N.D.E. Some authors speak of the “true” fundamental group.

⁽⁵⁶⁾N.D.E. The original has been corrected: it considers $\tilde{\mathbf{K}}_\bullet = \mathbf{K}_\bullet/\mathbf{k}_\bullet$, whereas on the one hand $\mathbf{k}_\bullet$ is already contractible (cf. 9.7.1), and on the other hand the morphism $\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet$ is not in general injective. It is an epimorphism if S'/S is finite étale (cf. 9.7.3).

components of S_0, S_1 are open and $\tilde{\mathbf{K}}_\bullet$ is connected,⁽⁵⁷⁾ that $\pi_1(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$ represents the functor $G \mapsto \pi^1(S'/S, \xi; G)$, i.e. that one has:

$$\pi_1(S'/S; \xi) \simeq \pi_1(\tilde{\mathbf{K}}_\bullet, \tilde{\xi}).$$

Let us also point out that when the morphism $S' \rightarrow S$ is “universally submersive” (cf. SGA 1, IV 2.1), and the connected components of S' are open,⁽⁵⁸⁾ the simplicial set $\mathbf{K}_\bullet$, and hence also $\tilde{\mathbf{K}}_\bullet$, is connected.

Examples 6.4. It remains to give examples of enlarged fundamental groups. Return to examples 1.6, i.e. let k be an algebraically closed field, C_1 a complete rational curve over k having exactly one singular point, this point being an ordinary double point, and C_2 a curve which is a union of two irreducible components isomorphic to $\mathbb{P}_k^1$ and meeting at exactly two points, which are ordinary double points of C_2 . In either case the enlarged fundamental group of the curve is isomorphic to $\mathbf{Z}$.⁽⁵⁹⁾

In general, there would be occasion to reconsider (simplifying and correcting them) the results of SGA 1 IX 5 in the setting of the enlarged fundamental group. The examples of loc. cit., 5.5 would give as many examples of enlarged fundamental pro-groups which are not profinite. Thus, if instead of an ordinary double point one took in the first example a double point with n distinct branches,⁽⁶⁰⁾ one would find as enlarged fundamental group the free (discrete!) group on $n - 1$ generators.

7. Classification of twisted constant preschemes and groups of multiplicative type of finite type in terms of the enlarged fundamental group

7.0. — Let S be a prescheme, which we again suppose locally noetherian, to ensure that S and certain preschemes over S which we shall consider (in particular those étale over S , more generally those locally of finite type over S) are locally connected.

Proposition 7.0.1. *Every twisted constant prescheme X over S which is locally trivial for the (fppf) topology (i.e. split by a faithfully flat morphism locally of finite presentation $S' \rightarrow S$) is quasi-isotrivial (i.e. one may even choose $S' \rightarrow S$ étale surjective).*

Indeed, one may suppose S connected, hence X of type I , where I is a fixed set. Thus $X' = X \times_S S'$ is isomorphic to $I_{S'}$, so $I_{S'}$ is endowed with a descent datum relative to $S' \rightarrow S$, i.e. one has an isomorphism $I_{S''} \xrightarrow{\sim} I_{S''}$ satisfying the usual transitivity condition. Now $S'' = S' \times_S S'$ is locally noetherian, hence locally connected, whence it follows that the automorphisms of $I_{S''}$ correspond to the sections of $G_{S''}$, where $G = \text{Aut}(I)$ is the permutation group of I .

In this way one obtains a descent datum on $G_{S'}$ (regarded as a trivial Galois principal bundle) relative to $S' \rightarrow S$. By 5.4 this descent datum is effective, whence a Galois principal bundle P over S with group G . By construction it represents the functor $\underline{\text{Isom}}_S(I_S, X)$ in

⁽⁵⁷⁾**N.D.E.** The hypothesis that $\tilde{\mathbf{K}}_\bullet$ is connected has been added; for the proof, see the addenda below (section 9).

⁽⁵⁸⁾**N.D.E.** What follows has been modified, taking account of the correction made above, cf. N.D.E. (53). (The original was: “ $\pi_0(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$ is also canonically isomorphic to the set $\pi_0(S, \xi)$ of connected components of S , pointed by the connected component of ξ in S ”.)

⁽⁵⁹⁾**N.D.E.** One could expand this: first, in view of 5.14, the enlarged fundamental group of the projective line $\mathbb{P}_k^1$ is zero, i.e. $\mathbb{P}_k^1$ is “truly” simply connected. Then one should extend to the case of the enlarged fundamental group and of the category of principal bundles (not necessarily finite) Corollary 5.4 of SGA 1 IX and the discussion following it. (Let Γ_1, Γ_2 be two copies of $\mathbb{P}_k$, a_i, b_i two distinct points of Γ_i , C_2'' the disjoint union of Γ_1, Γ_2 , C_2' the curve obtained by identifying a_1 with a_2 ; then C_2 is obtained from C_2' by further identifying b_1 with b_2 . The discussion following loc. cit., extended to the enlarged case, then shows that the enlarged fundamental group of C_2' (resp. C_2) is zero (resp. $\mathbf{Z}$).)

⁽⁶⁰⁾**N.D.E.** This is an n -fold point with n distinct tangents, for example the curve $X^n - Y^n = X^{n+1}$.

the category of locally noetherian preschemes over S . Consequently, the surjective étale base change $P \rightarrow S$ splits X , so X is indeed quasi-isotrivial.

Remark 7.0.2. One should take care that even if S is the spectrum of a field, it is not true in general that every twisted constant bundle over S is quasi-isotrivial. It suffices, for example, to take for X the sum scheme of a sequence of schemes of the form $\mathrm{Spec}(k_i)$, where the k_i are separable extensions of k of strictly increasing degrees.

The proof given above shows at the same time that classification of quasi-isotrivial twisted constant bundles X over S of type I is equivalent to that of Galois principal bundles over S with group $G = \mathrm{Aut}(I)$. This is even an equivalence of categories.

It may be put in a more convenient form as in SGA 1 V. For this, suppose S connected and endowed with a geometric point ξ . Consequently the enlarged fundamental pro-group $\Pi = \Pi_1(S, \xi)$ is defined. On the other hand, for every quasi-isotrivial twisted constant bundle X over S , let $I = X(\xi)$ be its set-theoretic fiber at ξ . Thus X is of type I , and consequently associated, as just stated, with a Galois principal bundle $P = \underline{\mathrm{Isom}}_S(I_S, X)$ over S with group $G = \mathrm{Aut}(I)$.

By the definition of Π , one therefore obtains a canonical homomorphism from Π to G , i.e. from one of the Π_i to G . Since G is the permutation group of $I = X(\xi)$, this means that Π “acts continuously on $I = X(\xi)$ ”, it being understood by this that the π_i (large i) act on I compatibly with the transition morphisms.

We leave it to the reader to check that every S -morphism $X \rightarrow Y$ between quasi-isotrivial twisted constant bundles over S induces a map $X(\xi) \rightarrow Y(\xi)$ compatible with the operations of Π , and that the functor so obtained is an equivalence of categories:

Proposition 7.0.3. *Let S be a connected locally noetherian prescheme, ξ a geometric point of S , $\Pi = \Pi_1(S, \xi)$ the enlarged fundamental pro-group of S at ξ . Then the functor*

$$X \mapsto X(\xi)$$

is an equivalence between the category of quasi-isotrivial twisted constant bundles over S and the category of sets on which Π acts continuously.

This functor is compatible with the operations of finite sums and finite projective limits. It follows, for example, that quasi-isotrivial twisted constant groups (or rings, etc.) over S correspond to ordinary groups (resp. rings, etc.) on which the pro-group Π acts continuously. In particular:

Corollary 7.0.4. *The category of quasi-isotrivial twisted constant commutative groups over S is equivalent to the category of “ Π -modules”, i.e. of commutative groups M on which Π acts continuously.*

Using now 5.7, one concludes the:

Theorem 7.1. *Let S be a connected locally noetherian prescheme, ξ a geometric point of S , $\Pi = \Pi_1(S, \xi)$ the enlarged fundamental pro-group of S at ξ . Then the functor*

$$G \mapsto \mathrm{Hom}_{\kappa(\xi)\text{-gr.}}(G_\xi, \mathbf{G}_{m, \xi})$$

induces an anti-equivalence of the category of quasi-isotrivial groups of multiplicative type over S with the category of Π -modules.

Using 4.5 one concludes:

Corollary 7.2. *The preceding functor induces an anti-equivalence of the category of groups of multiplicative type and of finite type over S and the category of Π -modules of finite type over $\mathbf{Z}$.*

Example 7.3. Take again, for example, for S a complete rational curve over an algebraically closed field, having exactly one multiple point with $n+1$ distinct branches. By 6.4 the enlarged fundamental group $\Pi(S, \xi)$ is a free group on n generators. Thus by 7.2, classification of tori of relative dimension m over S is equivalent to classification of systems of n endomorphisms $A_1, \dots, A_n$ of the $\mathbf{Z}$ -module $\mathbf{Z}^m$, up to automorphism of $\mathbf{Z}^m$. Except for $n \leq 1$ or $m \leq 1$, explicit classification of such systems seems hopeless. One may at least define a host of nontrivial invariants for such a system, such as the characteristic polynomials of the A_i .⁽⁶¹⁾

Remark 7.4. If no hypothesis is made on S , it remains true that for a given ordinary commutative group M of finite type over $\mathbf{Z}$, the category of groups of multiplicative type of type M over S is anti-equivalent to the category of Galois principal bundles over S with group $G = \text{Aut}_{\text{gr.}}(M)$. This follows easily from 5.9 and 5.10.

Remark 7.5. The theory of the fundamental pro-group which we have sketched in the present two nos. will advantageously be written in the setting of general sites. In this form it also applies, for example, to ordinary topological spaces, and gives a satisfactory theory at least for a locally connected topological space (not necessarily locally simply connected). In this case too it seems that one cannot content oneself with defining a fundamental group and that a pro-group is needed. Finally, let us note that once one has the language of topologies and descent (which is really at the heart of these questions), the exposition sketched here is also technically simpler than that of SGA 1 V, and should therefore in principle replace it.

8. Appendix: Elimination of certain affineness hypotheses

Our object is to prove the following generalization of 4.9.

Theorem 8.1. *Let S be a prescheme, H an S -group prescheme flat and of finite presentation, with connected and affine fibers.⁽⁶²⁾ Let $s \in S$; suppose H_s is a torus. Then there exists an open neighborhood U of s such that $H|U$ is a torus.*

One concludes immediately:

Corollary 8.2. *Let H be an S -group prescheme flat and of finite presentation over S . For H to be a torus, it is necessary and sufficient that its fibers be tori.*

Remark 8.3. Even when S is the spectrum of a discrete valuation ring, one cannot in 8.1 drop the hypothesis that the fibers of H (here the generic fiber) are affine, for there are examples of smooth groups over S whose generic fiber is an elliptic curve and whose special fiber is $\mathbf{G}_m$.

PROOF OF 8.1. One may evidently suppose $S = \text{Spec}(A)$ affine, which reduces us by the standard procedure (cf. EGA IV₂, 8.8.2) to the case where S is in addition noetherian. We begin by proving 8.2 in this case.

By 4.9 one is reduced to proving that H is affine over S . One may therefore suppose⁽⁶³⁾ A noetherian local, and since the completion $\hat{A}$ is faithfully flat over A , one is reduced by descent to the case where A is a complete noetherian local ring. Let S' be the normalization of S_{red} ; one knows by Nagata that it is finite over S (EGA 0_{IV}, 23 1.5); moreover $S' \rightarrow S$ is surjective, so $H' = H \times_S S' \rightarrow H$ is finite and surjective; hence, to prove that H is affine, it suffices to show that H' is so (EGA II, 6.7.1).

⁽⁶⁴⁾ Replacing S by a connected component of S' reduces us to the case where A is a noetherian (semilocal) normal integral ring. Moreover, replacing A by its normalization in

⁽⁶¹⁾**N.D.E.** In particular, this shows that the tori of relative dimension 2 considered in 1.6 are nonisotrivial.

⁽⁶²⁾**N.D.E.** Note that the hypotheses imply that H is separated over S , by VI_B, Th. 5.3 or Cor. 5.5.

⁽⁶³⁾**N.D.E.** By EGA IV₂, 8.8.2 again.

⁽⁶⁴⁾**N.D.E.** The original has been expanded in what follows.

a finite separable extension of its field of fractions if necessary, one may suppose that the generic fiber H_η of H is diagonalizable, i.e. that one has an isomorphism

$$u_\eta : H_\eta \xrightarrow{\sim} T_\eta,$$

where $T = \mathbf{G}_{m,S}^r$. Now one has the

Lemma 8.4. *Let S be a locally noetherian normal irreducible prescheme with generic point η , H a group prescheme over S smooth with connected fibers, T a group prescheme of multiplicative type and of finite type over S , $u_\eta : H_\eta \rightarrow T_\eta$ a homomorphism of algebraic groups over $\kappa(\eta)$.*

Then u_η extends to a group homomorphism $u : H \rightarrow T$.

Replacing S by its normalization in a finite separable extension of its function field if necessary, one may suppose T diagonalizable (which is moreover the case in the application we have in mind). Then T is a closed subgroup of a group of the form $\mathbf{G}_{m,S}^r$, which reduces us to the case where $T = \mathbf{G}_{m,S}$.

Everything amounts to proving that u_η , regarded as a rational map from H to $T = \mathbf{G}_{m,S}$, is everywhere defined (for the morphism $u : H \rightarrow T$ extending it is then necessarily a group homomorphism). One may regard u_η as an invertible section f of the structure sheaf of H_η , and must show that it extends to an invertible section of the structure sheaf of H . Now H , being smooth over normal S , is normal (SGA 1, II 3.1), so it suffices to find a closed subset Z of H of codimension ≥ 2 such that f extends to an invertible section of the structure sheaf of $H - Z$. This reduces us immediately to the case where S is the spectrum of a discrete valuation ring A (by localizing at the codimension 1 points of S).

Let t be a uniformizer of A , t' the section of $\mathcal{O}_H$ it defines, so that the special fiber H_0 equals $\mathcal{V}(t')$. By hypothesis H_0 is smooth over the residue field k and connected. Then f is a rational function on H with neither zeros nor poles in $H - H_0$; since $H_0 = \mathcal{V}(t')$ is an irreducible divisor, there exists an integer $n \in \mathbf{Z}$ such that $t'^n f = t^n f$ has neither zeros nor poles, i.e. is an invertible section of $\mathcal{O}_H$. It therefore defines a morphism $v : H \rightarrow \mathbf{G}_{m,S}$, and since $v_\eta = t'^n u_\eta$ and u_η is a group homomorphism, v transforms the unit section of H into a section of $\mathbf{G}_{m,S}$ whose value at the generic point of S is t^n ; since it is a section of $\mathbf{G}_{m,S}$, t^n must be a unit, i.e. $n = 0$, so v extends u_η , which finishes proving 8.4.

Applying this lemma to the present case, one finds a group homomorphism

$$u : H \rightarrow T = \mathbf{G}_{m,S}^r$$

inducing an isomorphism on the generic fibers. Let us prove that u is an isomorphism.

Lemma 8.5. *For every integer $n > 0$ prime to the residue characteristic of A , ${}_n H = \text{Ker}(n \cdot \text{id}_H)$ is finite over S .*

If n is prime to the residue characteristic of A , it is prime to all the residue characteristics of the points of S . Thus $n \cdot \text{id}_H$ induces on every fiber of H an étale morphism, so $n \cdot \text{id}_H$ is étale (SGA 1, I 5.9), hence its kernel ${}_n H$ is étale over S . On the other hand, ${}_n H$ is separated over S since H is so.⁽⁶⁵⁾⁽⁶⁶⁾ Moreover, all its fibers have the same rank n^r , since the fibers of H are tori, all of the same dimension r (H being smooth over S). One concludes that ${}_n H$ is finite over S (SGA 1, I 10.9 or EGA IV₄, 18.2.9).

Thus by the preceding lemma, $u({}_n H)$ is a closed subset of T . Since on the generic fiber T_η this subset is identical to ${}_n(T_\eta)$, it follows that it contains the closure of this subset, namely ${}_n T$. Now for every $s \in S$, the ${}_n(T_s)$ are dense in T_s ; as $u_s(H_s)$ is a closed subset (VI_B 1.4.2) containing them, one sees that $u_s(H_s) = T_s$, hence u is surjective. Since a surjective

⁽⁶⁵⁾By Raynaud's theorem VI_B 5.3.

⁽⁶⁶⁾**N.D.E.** Expand this point, in connection with the modifications in VI_B §5.

homomorphism of tori of the same dimension over a field is flat,⁽⁶⁷⁾ it follows that u induces a flat morphism on every fiber, so u is flat (SGA 1 I 5.9). Consequently $K = \text{Ker}(u)$ is flat over S ,⁽⁶⁸⁾ hence equal to the closure of its generic fiber K_η . Now K_η is the unit group by construction, and since K is separated over S (since H is so), its unit section is closed, whence it follows that K is the unit group. Thus u is a monomorphism; as one has seen it to be faithfully flat, it is therefore an isomorphism (cf. SGA 1, I 5.1 or EGA IV₄, 17.9.1). This proves that H is a torus, and hence completes the proof of 8.2.⁽⁶⁹⁾

Remark 8.5.1. Instead of invoking 8.5, one may also invoke Zariski's "Main Theorem", which directly implies that u is an open immersion, hence an isomorphism.⁽⁷⁰⁾

To prove 8.1, one is reduced, thanks to the quasi-isotriviality theorem, to the case where S is local and s its closed point,⁽⁷¹⁾ and to proving that under the other hypotheses made H is then a torus. By 8.2 already proved, one is reduced to proving that the fibers of H are tori. One may suppose S the spectrum of a complete noetherian local ring A . By 3.3 there exists for every $n > 0$ a finite group of multiplicative type T_n over S and an isomorphism $(T_n)_s \simeq {}_n(H_s)$, where s is the closed point of S . Proceeding as in 3.1 and using the fact that T_n is finite over S ,⁽⁷²⁾ one sees that the preceding isomorphism comes from a homomorphism $u_n : T_n \rightarrow H$, moreover uniquely determined. (Pass to the limit over the artinian quotients of A .)

Moreover, by the uniqueness properties, $u_n : T_n \rightarrow H$ is deduced from $u_m : T_m \rightarrow H$ by restriction to ${}_n(T_m) \simeq T_n$ when m is a multiple of n . It follows from IX, 6.6 that the u_n are monomorphisms, so the T_n are subgroups of H , and for m a multiple of n one has $T_n = {}_n(T_m)$.

Thus for every $t \in S$, the $(T_n)_t$ are subgroups of H_t of type $(\mathbf{Z}/n\mathbf{Z})^r$ (where $r = \dim H_s = \dim H_t$), such that for m a multiple of n one has $(T_n)_t = {}_n(T_m)_t$. The fact that H_t is a torus now follows from the

Lemma 8.6. *Let H be a smooth affine algebraic group over an algebraically closed field k , $(T_n)_{n>0}$ a family of subgroups of multiplicative type of H such that for every integer $n > 0$, T_n is of type $(\mathbf{Z}/n\mathbf{Z})^r$, and for every multiple m of n one has $T_n = {}_n(T_m)$.*

Under these conditions, H contains a torus of dimension $\geq r$ containing the T_n , so if H is connected of dimension $\leq r$, H is a torus of dimension r .

This is an exercise on affine algebraic groups which we shall treat by references to *Bible*. We restrict to considering the T_n for n prime to the characteristic. Let K be the closure of the union of the T_n in H , endowed with the induced reduced structure; then standard arguments show that K is a commutative algebraic subgroup of H . By *Bible*, §4.5, th. 4, K is therefore isomorphic to a product $K_u \times K_s$, with K_u "unipotent" and K_s diagonalizable. Every diagonalizable subgroup of K is contained in K_s , so the T_n are subgroups of K_s , hence $K = K_s$. Write $K = D(M)$ with M an ordinary commutative group of finite type over $\mathbf{Z}$; then $T_n \subseteq K$ means that M admits a quotient group isomorphic to $(\mathbf{Z}/n\mathbf{Z})^r$. This being true for every integer n prime to the characteristic of k (it would suffice for powers of a fixed prime), it follows that M has rank $\geq r$, so K contains a torus of dimension r , say T . When H is connected of dimension r , it follows that $T = H$, which completes the proof of 8.6. Thus 8.1 is proved. $\square$

⁽⁶⁷⁾**N.D.E.** This follows, for example, from VIII 3.2 a); more generally, if $f : G \rightarrow H$ is a surjective morphism of algebraic groups over a field k and H is reduced, f is flat (cf. VI_B 1.3).

⁽⁶⁸⁾**N.D.E.** The original has been expanded in what follows.

⁽⁶⁹⁾**N.D.E.** At least in the case considered so far, namely S locally noetherian.

⁽⁷⁰⁾**N.D.E.** The editors did not understand this remark, not understanding why u would a priori have finite fibers and be surjective. . .

⁽⁷¹⁾**N.D.E.** Expand this reduction. . .

⁽⁷²⁾**N.D.E.** Expand this point. . .

Remarks 8.7. Using 8.1, it should not be difficult to give analogous generalizations of 4.7 and 4.8. A more interesting study would be that of situation 8.1 where one drops the hypothesis that the fibers of H are affine. One can show that there then exists an open neighborhood U of s such that $H|_U$ is commutative and for every $t \in U$ the geometric fiber H_t is an extension of an abelian variety by a torus.⁽⁷³⁾ Of course, in questions of this kind one may restrict to the case where S is the spectrum of a discrete valuation ring, s its closed point, t its generic point.

One may generalize this result as follows. For every smooth connected algebraic group G over an algebraically closed field k , a well-known theorem of Chevalley tells us that G is (uniquely) an extension of an abelian variety A by a smooth connected affine group V . Denote by the *abelian rank* (resp. *reductive rank*, resp. *nilpotent rank*, resp. *semisimple dimension*) of G , and write $\rho_{\text{ab}}(G)$ (resp. $\rho_r(G)$, resp. $\rho_n(G)$, resp. $d_s(G)$), the dimension of A , resp. the dimension of the maximal tori of G , resp. the dimension of the Cartan subgroups⁽⁷⁴⁾ of G , resp. the dimension of the quotient of G (or again of V) by its radical (cf. *Bible* for all these notions). Introduce also the unipotent rank $\rho_u(G) = \rho_u(V) = \rho_n(G) - \rho_r(G) - \rho_{\text{ab}}(G)$. When G is not algebraically closed, we still use the same names and the same notation for the corresponding invariants of $G_{\bar{k}}$, where $\bar{k}$ is the algebraic closure of k . This being so, let G be a smooth group scheme over the spectrum S of a discrete valuation ring; let ρ_{ab} , etc. (resp. ρ'_{ab} , etc.) be the invariants associated with the special fiber (resp. generic fiber). Then one has the inequalities:

$$\begin{cases} \rho_{\text{ab}} \leq \rho'_{\text{ab}} \\ \rho_r + \rho_{\text{ab}} \leq \rho'_r + \rho'_{\text{ab}} \\ d_s \leq d'_s \end{cases} \quad \begin{cases} \rho_n \geq \rho'_n \\ \rho_u \geq \rho'_u. \end{cases}$$

It amounts to the same thing to say that if G is smooth of finite type over an arbitrary base S , the functions $s \mapsto \rho_{\text{ab}}(s)$, $\rho_{\text{ab}}(s) + \rho_r(s)$, $d_s(s)$ are lower semicontinuous, and the functions $s \mapsto \rho_n(s)$, $\rho_u(s)$ are upper semicontinuous.⁽⁷⁵⁾ The same results probably still hold without supposing G smooth over S , but merely flat of finite presentation over S , agreeing to denote, for an algebraic group G over an algebraically closed field k , by $\rho_{\text{ab}}(G)$, etc., the corresponding invariants of G_{red}^0 .

In this Seminar we present some results of this type for G affine over S , or more generally with affine fibers: in this case we shall verify the semicontinuity properties for ρ_r, ρ_n , hence for $\rho_u = \rho_n - \rho_r$, and continuity of ρ_s in a neighborhood of a point s whose fiber is a reductive group.⁽⁷⁵⁾

One may generalize 8.2 when one already supposes G commutative, as follows:

Theorem 8.8. ⁽⁷⁶⁾ *Let G be a commutative S -group prescheme flat and of finite presentation over S , with affine connected fibers. Let $s \in S$, and suppose:*

- a) If $\bar{k}$ denotes an algebraic closure of $\kappa(s)$, $(G_{\bar{k}})_{\text{red}}$ is a torus.*
- b) There exists a generization t of s such that G_t is smooth over $\kappa(t)$.*

Under these conditions there exists an open neighborhood U of s such that $G|_U$ is a torus.

(Note moreover that if one merely supposes that for every generization t of s , G_t is affine resp. connected, one easily deduces that the same conclusion holds for t in an open neighborhood of s .)

⁽⁷³⁾**N.D.E.** Give a reference here?

⁽⁷⁴⁾**N.D.E.** Recall the definition of “Cartan subgroup”...

⁽⁷⁵⁾**N.D.E.** Give a reference for these results?

⁽⁷⁶⁾**N.D.E.** G. Prasad and J.-K. Yu have generalized this result (under some additional hypotheses), without supposing G commutative and replacing “torus” by “reductive group” in hypothesis a) and the conclusion, cf. [PY06], Th. 6.2.

PROOF OF 8.8. It suffices to prove that G_s is smooth over $\kappa(s)$. Indeed, as G is flat of finite presentation over S , it then follows that G is smooth over S above a neighborhood of s (cf. 3.5), but then one is under the conditions of 8.1.

To prove that G_s is smooth over $\kappa(s)$, the usual procedure reduces us to the case where S is affine noetherian. Choosing a homomorphism $S' \rightarrow S$ from the spectrum S' of a discrete valuation ring into S , sending its closed point to s and its generic point η to t , one is reduced to the case where S is itself the spectrum of a discrete valuation ring, which one may in addition suppose complete with algebraically closed residue field, and where s and $t = \eta$ are respectively the closed point and generic point of S .

Thus G is flat, separated, of finite type over S , the generic fiber G_η is smooth and connected, and the special fiber G_0 is such that $T_0 = (G_0)_{\text{red}}$ is a torus. Let m be an integer > 0 prime to the residue characteristic at s , hence also to that at η ; one then knows that $m \cdot \text{id}_G$ is an étale morphism fiber by fiber (VII_A 8.4), hence an étale morphism since G is flat over S (SGA I I 5.9); hence its kernel ${}_mG$ is étale over S , and since G is separated⁽⁷⁷⁾⁽⁷⁸⁾ of finite type over S , so is ${}_mG$. Since $({}_mG)_0 = {}_m(T_0)$, its degree is m^r , where $r = \dim T_0 = \dim G_0 = \dim G$. It follows that the rank of $({}_mG)_\eta = {}_m(G_\eta)$ is $\geq m^r$, which already proves (using 8.6) that G_η is a torus of dimension r , since the two fibers of ${}_mG$ have the same rank, and hence as in 8.5 that ${}_mG$ is finite over S .⁽⁷⁹⁾

Note that since S is complete with algebraically closed residue field, the finite étale covering ${}_mG$ splits completely, so through every point of ${}_mG_0$ passes a section of G over S ; in particular the set of points of G_0 through which passes a section of G over S is everywhere dense. Now one has the following:

Lemma 8.9. *Let S be a locally noetherian regular prescheme of dimension 1, G an S -group prescheme flat and locally of finite type such that G_η is smooth over $\kappa(\eta)$ for every maximal point η of S (which implies that G is reduced). Suppose that the normalization scheme X of G is finite over G (this is the case by Nagata if S is the spectrum of a complete discrete valuation ring, cf. EGA IV₂, 7.7.4), and let G' be the open subset of X consisting of the points at which X is smooth over S . With this notation:*

a) *If the projection $G' \rightarrow S$ is surjective, there exists on G' a unique S -group prescheme structure such that the canonical morphism $G' \rightarrow G$ is a group homomorphism.*

b) *Suppose that for every closed point s of S , the set of points of G'_s through which passes an étale quasi-section is dense in G'_s for the Zariski topology. Then X is regular, one is under the conditions of a), and the map $\Gamma(G'/S) \rightarrow \Gamma(G/S)$ is bijective.*

In the case of interest to us this lemma applies and gives a group homomorphism $u : G' \rightarrow G$, where G' is smooth over S and u_η is an isomorphism $G'_\eta \xrightarrow{\sim} G_\eta$. Replacing G' by an open subgroup if necessary, one may suppose G'_0 connected, and since $u_0 : G'_0 \rightarrow G_0$ induces a surjective morphism $G'_0 \rightarrow T_0$, where T_0 is a torus of the same dimension as G'_0 , one easily concludes that G'_0 is a torus (for example using 8.6, or referring to *Bible*, §7.3, th. 3 a)). Consequently by 8.2 G' is a torus, but then by IX 6.8, $\text{Ker } u$ is a subgroup of multiplicative type of G' , and since its generic fiber is reduced to the unit group, it is the unit group, hence u is a monomorphism. Using now VIII 7.9 it follows that u is an immersion. Being surjective, and G being reduced, it follows that u is an isomorphism, which finishes proving 8.8. $\square$

It remains to prove 8.9. To prove a), note that uniqueness of the group law on G' making $u : G' \rightarrow G$ a group homomorphism is clear, since the group law of G' is known on the generic fiber (supposing S irreducible, which is permitted). For existence, one easily reduces to the case where S is local, the spectrum of a discrete valuation ring A , and thanks to

⁽⁷⁷⁾By Raynaud's theorem VI_B 5.3.

⁽⁷⁸⁾N.D.E. Cf. N.D.E. (62) in 8.5.

⁽⁷⁹⁾N.D.E. Review the preceding sentence. . .

uniqueness and to the fact that the operation of integral closure commutes with an étale extension of the base, one may make étale extensions of the base on S , which reduces us to the case where A is “strictly local”, i.e. henselian with separably closed residue field. The same reduction applies for b), but in the hypothesis made in b) one may now replace “étale quasi-section” by “section”.

Note that G' being smooth over S and X normal, $G' \times_S X$ is normal (being smooth over normal X), so the composite morphism $G' \times_S X \rightarrow G \times_S G \rightarrow G$ factors as

$$p : G' \times_S X \rightarrow X.$$

Let us prove that this latter morphism induces on the open subset $G' \times_S G'$ of $G' \times_S X$ a morphism

$$G' \times_S G' \rightarrow G'.$$

One must therefore show that $G'_0 \times_k G'_0$ is mapped into the open subset G'_0 of X_0 ; it suffices to see that for every point g'_0 of G'_0 with values in k , the morphism

$$(x) \quad h'_0 \mapsto p_0(g'_0, h'_0)$$

from G'_0 to X_0 takes its values in G'_0 . Now since G' is smooth over S and S henselian, every g'_0 as above is induced by a section g' of G' over S , and one verifies immediately that morphism (x) above is then induced by the morphism $h' \mapsto p(g', h')$ from G' to X , itself deduced by transport of structure from the automorphism $h \mapsto g \cdot h$ of G , left translation by the section g of G which is the image of g' . Thus $h' \mapsto p(g', h')$ is itself an automorphism of X , hence maps G' to G' , which proves our assertion.

It remains to prove that the composition law so obtained on G' is a group law. Associativity follows immediately from associativity of the generic fiber (isomorphic to that of G). On the other hand, the symmetry automorphism of the S -prescheme G induces an automorphism of X , leaving G' stable and inducing an automorphism σ of G' . One then verifies that the latter has the properties of an inverse for the composition law on G' , for this again amounts to checking commutativity of certain diagrams involving fibered powers of G' over S , and these being smooth over S , it suffices to check their commutativity on the generic fiber, which is clear. This proves part a) of 8.9.

Let us prove b). Let Z' be the set of $x \in X$ such that $\mathcal{O}_{X,x}$ is not regular; it is a closed subset by a theorem of Nagata (EGA IV₂, 6.12.6), evidently contained in X_0 ; let Z be its image in G , hence a closed subset of G_0 . Then Z is a nowhere dense subset of G_0 , i.e. contains no maximal point y of G_0 . Indeed, since G_0 is defined in G by an equation $t = 0$ (where t is a uniformizer of the discrete valuation ring A), $\mathcal{O}_{G,y}$ is of dimension 1 by the Hauptidealsatz, so for every x of X over y , $\mathcal{O}_{X,x}$ is of dimension 1, hence a discrete valuation ring since X is normal and thus regular in codimension 1. On the other hand, it is evident that for every section g of G over S , Z' is stable under the automorphism of X defined by transport of structure from left translation by g in G , so Z is stable under left translation in G_0 defined by g_0 . Now by hypothesis the set of these g_0 is dense in G_0^0 . Since Z is stable under translation by these g_0 and is closed and nowhere dense, it follows immediately that $Z = \emptyset$, whence $Z' = \emptyset$, so X is regular. Consequently, X is smooth over S at every point through which passes a section. Now every section of G over S lifts uniquely to a section of X over S , and hence of G' over S . This is so in particular for the unit section, which proves that the image of G' in S is S , i.e. one is under the conditions of (a). This completes the proof of 8.9, and hence of 8.8.

9. Addenda

(80)

(80)**N.D.E.** This additional section was written by Fabrice Orgogozo (following indications of Ofer Gabber).

9.1. — Simplicial sets, topoi, groupoids and topological spaces. —

Notations 9.1.1. (1) Let E be a simplicial set. One may associate with it the following objects:

(2) A topos $T = E^\sim$, obtained from the topoi naturally associated with the sets E_i , according to the procedure described in [Del74], 6.3.1 (see also [Ill72], VI.5.2 and SGA 4 VI.7);

(3) A groupoid $G = \Pi(E)$ whose objects are the elements of E_0 (the “vertices”) and whose arrows are defined in [GZ67], II.7;

(4) A topological space $X = |E|$ (a cell complex), called the “geometric” (or “topological”) “realization” (cf. loc. cit., III.1).

Let us note that a sheaf on T is nothing other than a simplicial set over E .

9.2. — Locally constant sheaves; descent data. —

Definitions 9.2.1. One calls a *locally constant sheaf* on:

(1) A simplicial set E , every morphism of simplicial sets $E' \rightarrow E$ such that for every $i \in \mathbf{N}$ and every $e \in E_i$ the face operators d induce isomorphisms $E'_e \xrightarrow{\sim} E'_{d(e)}$ between the fibers (cf. [AM69], §10);

(2) A topos T , every object F of T such that there exists an epimorphism $U \rightarrow 1$ and an isomorphism $F \times U \simeq f^*I \times U$, where I is a set and $f : T \rightarrow (\mathbf{Ens})$ is the final morphism (cf. SGA 4, IX.2);

(3) A groupoid G , every presheaf on G , i.e. every contravariant functor from G to the category of sets (cf. [GZ67], append. I.1.2);

(4) A topological space X , every sheaf of sets on X locally constant in the usual sense.

Finally:

(5) One calls a *descent datum on a simplicial set E* the data of a sheaf F on $E_0^\sim$ (i.e. a set-theoretic function on E_0) and an isomorphism $\alpha : p_1^*F \xrightarrow{\sim} p_2^*F$, where $p_1, p_2 : E_1^\sim \rightarrow E_0^\sim$ are the morphisms (deduced from the) faces, satisfying the usual cocycle relation (cf. Exp. IV, 2.1⁽⁸¹⁾ and below).

The morphisms between these five types of objects, as well as the associated inverse image functors, are defined in the evident way.

9.3. — Some equivalences of categories. — Let E be a simplicial set, and respectively T, G, X the associated topos, groupoid and topological space.

Proposition 9.3.1. *The categories of locally constant sheaves on E, T, G, X , as well as the category of descent data on E , are equivalent.*

SKETCH OF PROOF. Denote by (1) to (5) the categories of objects defined in the preceding paragraph.

- (1) $\Leftrightarrow$ (5). This is a particular case of [AM69], 10.6 (see also [GZ67], append. I.2.3, [Fri82], ?5.6 or [Ill72], VI.8.1.6). An equivalence of categories is given by the functor associating with the object $(E' \rightarrow E)$ the pair (E'_0, α) , where E'_0 is regarded as a sheaf on E_0 , and where α is the unique isomorphism $p_1^*E'_0 \xrightarrow{\sim} p_2^*E'_0$ whose fiber at each $y \in E_1$ — with images denoted by x_1, x_2 under the two projections — is given by the isomorphisms $(E'_0)_{x_1} \xleftarrow{\sim} (E'_1)_y \xrightarrow{\sim} (E'_0)_{x_2}$.
- (5) $\Leftrightarrow$ (3). Evident: one of the two relations defining the morphisms in the groupoid G associated with E is a cocycle relation.
- (1) $\Leftrightarrow$ (4). Cf. [GZ67], append. I.3.2.1.
- (1) $\Leftrightarrow$ (2). One must show that an object over E is a locally constant sheaf in the simplicial sense if and only if it is so as a sheaf on $T = E^\sim$. This follows from the

⁽⁸¹⁾PP: I have replaced “FGA, Technique de descente I, 1.6” by “Exp. IV, 2.1”.

fact that locally constant sheaves $F_\bullet$ on T are precisely cartesian sheaves, i.e. those for which the arrows $([n] \rightarrow [m])^* F_m \rightarrow F_n$ are isomorphisms. This last point is a particular case of a general fact about simplicial topoi, together with the fact that every sheaf on $E_0^\sim$ is locally constant. (See also [Ill72], VI.8.1.6.) $\square$

For every group H , the equivalences of categories above induce equivalences between the categories of H -torsors, the latter being locally constant sheaves endowed with an action of H , with fibers isomorphic to H acting on itself by translation.

9.4. — Fundamental groups and groupoids. — It follows from the preceding equivalences of categories that for every group H and every simplicial set E , the sets of isomorphism classes of H -torsors $H^1(E, H)$, $H^1(E^\sim, H)$, $\pi_0(\text{Hom}(\Pi(E), BH))$ and $H^1(|E|, H)$ are naturally in bijection. Recall that BH denotes the one-object groupoid associated with H , and π_0 the functor associating with a category the set of isomorphism classes of its objects.

Similarly, if e is a point of E , the preceding equivalences induce bijections between the sets of isomorphism classes of H -torsors trivialized over e , denoted respectively by $H^1(E \text{ rel } e, H)$, $H^1(E^\sim \text{ rel } e^\sim, H)$, $\text{Hom}(\pi_1(\Pi(E), e), H)$ and $H^1(|E| \text{ rel } |e|, H)$. Recall that $\pi_1(\Pi(E), e)$ denotes the group $\text{Isom}_{\Pi(E)}(e, e)$. For variable H , these functors are represented, in the connected case, by a group denoted by $\pi_1(E, e)$. The group $\pi_1(E, e)$ is isomorphic to $\pi_1(\Pi(E), e)$ and $\pi_1(|E|, |e|)$, hence also to the fundamental group of a simplicial set as defined by Kan (cf. e.g. [May67], 16.1 or [Ill72], I.2.1.1). (Recall that the set $H^1(E, H)$ is for its part isomorphic to the set $H^1(\pi_1(E, e), H)$ of morphisms to H modulo conjugacy, also denoted by $\text{Hom ext}(\pi_1(E, e), H)$.)

9.5. — Cones. —

Definitions 9.5.1. (1) Let $f : E' \rightarrow E$ be a morphism of simplicial sets. Recall (cf. for example [Del74], 6.3.1) that $C(f)$ denotes the pointed simplicial set whose underlying set in degree $n \geq 0$ is

$$E_n \amalg \left(\coprod_{i < n} E'_i \right) \amalg \star,$$

where $\star$ is a singleton. We leave it to the reader to define the simplicial maps, the definition of the faces in degrees at most two being recalled below. (See also [GZ67], VI.2 for a pointed variant.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on E endowed with a trivialization of the inverse image over E' . The simplicial set $C(f)$ is naturally pointed by $\star \rightarrow C(f)$.

(2) Let $f : T' \rightarrow T$ be a morphism of topoi. Denote by $C(f)$ the topos whose objects are the quintuples $(F, F', A, \alpha : f^*F \rightarrow F', \beta : A \rightarrow \Gamma(T', F'))$, where F (resp. F') is an object of T (resp. T'), A is a set, and α (resp. β) is a morphism in T' (resp. (Ens)). (See also [Del80], 4.3.4 and [Ill72], III.4 for a variant of this construction.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on T endowed with a trivialization of the inverse image over T' . The topos $C(f)$ is naturally pointed by the fiber functor sending the quintuple $(F, F', A, \alpha : f^*F \rightarrow F', \beta : A \rightarrow \Gamma(T', F'))$ to the set A .

(3) Let $f : G' \rightarrow G$ be a morphism of groupoids. Denote by $C(f)$ the colimit of the diagram

$$G \leftarrow G' \rightarrow B1$$

where $B1$ is the one-object category (one object, one arrow). The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on G endowed with a trivialization of the inverse image over G' .

(4) Let $f : X' \rightarrow X$ be a morphism of topological spaces. Denote by $C(f)$ the colimit of the diagram

$$\star \leftarrow X' \times \{0\} \rightarrow X' \times [0, 1] \leftarrow X' \times \{1\} \rightarrow X.$$

The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on X endowed with a trivialization of the inverse image over X' . The topological space $C(f)$ is naturally pointed by $\star \rightarrow C(f)$.

9.5.2. — Descent data on the cone of a simplicial map. Let $f : E' \rightarrow E$ be a morphism of simplicial sets and $C(f)$ its cone (cf. above, (1)). Use the letter p (resp. q, r) to denote the face maps of E' (resp. $E, C(f)$). Before stating the proposition below, let us make explicit the faces r in degrees at most two in terms of p, q . By convention, p_{ij} (resp. p_i) is the face map $E'_2 = E'_{\{1,2,3\}} \rightarrow E'_1 = E'_{\{1,2\}}$ (resp. $E'_1 = E'_{\{1,2\}} \rightarrow E'_0 = E'_{\{1\}}$) corresponding to the increasing map $\{1, 2\} \rightarrow \{1, 2, 3\}$ (resp. $\{1\} \rightarrow \{1, 2\}$) with image $\{i, j\}$ (resp. $\{i\}$).⁽⁸²⁾ One adopts the same convention for the faces of E and $C(f)$. The morphism

$$r_1 : C(f)_1 = E_1 \coprod E'_0 \coprod \star \rightarrow C(f)_0 = E_0 \coprod \star$$

(resp. r_2) is:

- $q_1 : E_1 \rightarrow E_0$ (resp. $q_2 : E_1 \rightarrow E_0$) on E_1 ;
- $E'_0 \rightarrow \star$ (resp. $f_0 : E'_0 \rightarrow E_0$) on E'_0 ;
- $\star \rightarrow \star$ on $\star$.

Similarly, the morphism

$$r_{21} : C(f)_2 = E_2 \coprod E'_1 \coprod E'_0 \coprod \star \rightarrow C(f)_1 = E_1 \coprod E'_0 \coprod \star$$

(resp. r_{32}, r_{31}) is:

- $q_{21} : E_2 \rightarrow E_1$ (resp. q_{32}, q_{31}) on E_2 ;
- $p_1 : E'_1 \rightarrow E'_0$ (resp. $f_1 : E'_1 \rightarrow E_1, p_2 : E'_1 \rightarrow E'_0$) on E'_1 ;
- $E'_0 \rightarrow \star$ (resp. $\text{Id} : E'_0 \rightarrow E'_0, \text{Id} : E'_0 \rightarrow E'_0$) on E'_0 .

Let H be a descent datum on $C(f)$, i.e. a sheaf H_0 on $C(f)_0 = E_0 \coprod \star$ endowed with an isomorphism $\gamma : r_1^* H_0 \xrightarrow{\sim} r_2^* H_0$ between its inverse images on $C(f)_1 = E_1 \coprod E'_0 \coprod \star$ satisfying the cocycle relation $r_{31}^* \gamma = r_{32}^* \gamma \circ r_{21}^* \gamma$.

The restriction of the isomorphism γ to E_1 (resp. E'_0) is a descent datum $\beta : q_1^* G_0 \xrightarrow{\sim} q_2^* G_0$, where G_0 is the restriction of H_0 to E_0 (resp. a trivialization $t : \underline{H_0(\star)} \xrightarrow{\sim} f^* G_0 =: F_0$, where $\underline{H_0(\star)}$ is the constant sheaf with stalk $H_0(\star)$). The restriction of the cocycle relation satisfied by γ to E_2 (resp. E'_1) is the cocycle relation for β (resp. $p_2^*(t) = f_1^*(\beta) \circ p_1^*(t)$). This latter relation means that the trivialization t is compatible with the descent datum $f_1^*(\beta)$ induced by β on F_0 . It follows that:

Proposition 9.5.3. *Let $f : E' \rightarrow E$ be a morphism of simplicial sets and $C(f)$ its cone, pointed by $\star$. The category of descent data on $C(f)$ trivialized over $\star$ is equivalent to the category of descent data on E endowed with a trivialization of the descent datum induced on E' .*

9.6. — Representability of the functor $\pi^1(S'/S, \xi; -)$. — The notation is that of page 90, and one henceforth supposes the connected components of $S_0 = S'$ and $S_1 = S' \times_S S'$ open. Under this hypothesis one has the following explicit combinatorial description, which follows immediately from the recollections above.

Proposition 9.6.1. *For every group G , the set $\pi^1(S'/S, \xi; G)$ of isomorphism classes of descent data endowed with a trivialization over ξ is in natural bijection with the relative nonabelian cohomology set $H^1(K \text{ rel } k, G) := H^1(\tilde{K} \text{ rel } \star, G)$.*

It follows that this functor is representable by a group, denoted by $\pi_1(K, k)$ (the “relative fundamental group”), as soon as the simplicial set $\tilde{K} = C(k \rightarrow K)$ is connected. One verifies immediately that this is so if and only if the map $\pi_0(k) \rightarrow \pi_0(K)$ is surjective. In particular

⁽⁸²⁾This convention departs from the current simplicial standard but is closer to the notation used by Grothendieck in his descent theory.

it suffices that the simplicial set K be connected. As indicated in the text, this is the case if the scheme S is connected and the morphism $S' \rightarrow S$ universally submersive.

9.7. — Contractibility of k , counterexample to injectivity of $\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet$. — The following two propositions make N.D.E. (53) more precise.

Proposition 9.7.1. *Let κ be an algebraically closed field, and X a connected κ -scheme. For every integer $i \geq 0$, denote by X_i the $(i+1)$ -st fibered power of X over κ . For every integer $i \geq 0$, the canonical map*

$$\pi_0(X_i) \longrightarrow \pi_0(X)^{i+1}$$

is a bijection. In particular, the simplicial set $\pi_0(X_\bullet)$ is contractible.

It suffices to prove the following lemma, which follows by passage to the limit from the Künneth formula. [Expand?]

Lemma 9.7.2. *Let κ be an algebraically closed field and X, Y two connected κ -schemes. Then the fibered product $X \times_\kappa Y$ is connected.*

Proposition 9.7.3. *Let X be a connected scheme, $X' \rightarrow X$ a finite étale morphism and $\bar{x}$ a geometric point of X located at a point x . The canonical morphism $\pi_0(X'_\bar{x}) \rightarrow \pi_0(X')$ is surjective.*

Indeed, every connected component of X' has as image an open and closed subset of X , and hence meets the fiber X'_x . The morphism $\pi_0(X'_x) \rightarrow \pi_0(X')$ is therefore surjective, as is the morphism $\pi_0(X'_\bar{x}) \rightarrow \pi_0(X'_x)$.

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⁽⁸³⁾**N.D.E.** Additional references cited in this Exposé; references [AM69] and following concern the Addenda.

Exposé XI. Representability criteria. Applications to subgroups of multiplicative type of affine group schemes

by A. Grothendieck

0. Introduction

As we have already seen examples in Exp. X, Nos. 4, 5, the representability of certain functors, such as certain functors of the type $\underline{\mathrm{Hom}}_S(X, Y)$ and various variants, plays an important role in many questions concerning group preschemes.

Among the results particularly useful in this theory, let us mention (in addition to the questions of representability of quotients, studied in exposés V and VI and in Exp. VIII 5) the question of representability of functors of the form $\prod_{X/S} Y/X$ (Y a subobject of X), studied in Exp. VIII 6 in a very elementary case, of which variants will be given in no. 6 of the present exposé; these results provide us with the representability of various centralizers, normalizers, transporters.

Less elementary representability criteria, using results that will appear in EGA VI, are indicated in 6.12 and in Exp. XV, XVI, where a criterion for representability of quotients G/H in cases not covered by the preceding exposés will be given (a criterion that was not developed in the oral exposés).

Our main object in the present exposé is the proof of theorems 4.1 and 4.2, which provide a typical example of a nonprojective construction technique (close to that which will be developed in EGA VI). It has moreover become apparent, since the oral exposé and the writing of the present text, that the affine hypotheses made in 4.1 and 4.2 can be eliminated to a large extent (cf. XV), and that, on the other hand, for the essential part of the theory developed in the following exposé, one can dispense with 4.1 and 4.2. Finally, it might be interesting to prove the analogue of these results for a general reductive (for example semisimple) group prescheme instead of a group of multiplicative type, in which case 4.1 and 4.2 will doubtless be the key result for the proof.

1. Review of smooth, étale, unramified morphisms

The reader is referred to EGA IV §§ 17 & 18, and, pending its publication, to SGA 1 I, II, III (where, however, certain noetherian hypotheses, inconvenient in applications, should be replaced by hypotheses of finite presentation).

Definition 1.1. Let S be a prescheme, F a functor $(\mathrm{Sch})_S^\circ \rightarrow (\mathrm{Ens})$. One says that F is *formally smooth* (resp. *formally unramified*^{*}, resp. *formally étale*) if for every S -prescheme S' , affine (in the absolute sense), and every subscheme S'_0 of S' defined by a nilpotent ideal $\mathcal{J}$, the map

$$F(S') \longrightarrow F(S'_0)$$

is surjective (resp. injective, resp. bijective). A prescheme X over S is said to be *formally smooth over S* (resp. *formally unramified over S* , resp. *formally étale over S*) if the

⁰_{xy} version of 5/12/08.

^{*}One will now rather say “net” instead of “unramified”.

corresponding functor is formally smooth (resp. formally unramified, resp. formally étale); one says that X is *smooth* over S (resp. *unramified* over S , resp. *étale* over S) if it satisfies the preceding condition and if, in addition, X is locally of finite presentation over S .

The interest of these definitions for X lies in the fact that, on the one hand, they are expressed in a remarkably simple way in terms of the functor represented by X (and in practice, X is often given as the object over S representing an explicitly described functor), and that, on the other hand, they are also expressed by remarkable properties concerning the local structure of X , which we shall recall in the following statements (for the proof we refer to *loc. cit.*).

Proposition 1.2. *Let X be a prescheme locally of finite presentation over S . Then:*

- (i) *For X to be smooth over S , it is necessary and sufficient that X be flat over S , and that its geometric fibers $X \otimes_S \text{Spec}(\overline{\kappa(s)})$ be regular schemes. More generally, for X to be smooth over S in a neighborhood of the point $x \in X$ (one then says that X is smooth over S at x), it is necessary and sufficient that X be flat over S at x and X_s be smooth over $\kappa(s)$ at x , i.e. $X_s \otimes_{\kappa(s)} \overline{\kappa(s)}$ be regular at the points (or simply, one point) above x .*
- (ii) *Suppose S , hence X , locally noetherian, let $x \in X$ and $s \in S$ be its image in S ; then the smoothness of X over S at x can be recognized on the local homomorphism $A = \mathcal{O}_{S,s} \rightarrow B = \mathcal{O}_{X,x}$ of noetherian local rings (or even on the local homomorphism $\hat{A} \rightarrow \hat{B}$ of their completions), by the following characteristic property: B is flat over A , and $B \otimes_A k$ is geometrically regular over k (the residue field of A), i.e. for every finite extension k' of k , $(B \otimes_A k) \otimes_k k' = B \otimes_A k'$ is a regular semilocal ring. When the residue extension $k(B)/k(A)$ is trivial, these conditions are also equivalent to the following: $\hat{B}$ is isomorphic as an $\hat{A}$ -algebra to a formal power series algebra $\hat{A}[[t_1, \dots, t_n]]$.*

Thus, from the “formal” point of view, the structure of X over S is that of the *standard affine space* $S[t_1, \dots, t_n]$ over S .

Proposition 1.3. *Let X be a prescheme locally of finite presentation over S . Then:*

- (i) *The following conditions are equivalent:*
 - a) *X is unramified over S .*
 - b) *The diagonal morphism $X \rightarrow X \times_S X$ is an open immersion.*
 - c) *$\Omega^1_{X/S} = 0$.*
 - d) *The geometric fibers of X/S are discrete and reduced, i.e. isomorphic to sums of copies of the base field.*
 - e) *For every $s \in S$, the fiber $X \otimes_S \text{Spec} \kappa(s) = X_s$ is unramified over $\kappa(s)$, or again is isomorphic to a sum of spectra of finite separable extensions of $\kappa(s)$.*
- (ii) *One has analogous conditions of a pointwise nature for X to be unramified over S at a given point x (i.e. in a neighborhood of the said point); for example, it is necessary and sufficient that $\mathcal{O}_{X_s,x}$ be a finite separable extension of $\kappa(s)$, which can also be expressed in terms of the local homomorphism $A = \mathcal{O}_{S,s} \rightarrow B = \mathcal{O}_{X,x}$ by the condition that $B \otimes_A k$ be a finite separable extension of k (the residue field of A), i.e. $\mathfrak{r}(A)B = \mathfrak{r}(B)$ and $k(B)$ is a finite separable extension of $k(A)$. When A , hence also B , is noetherian, and $k(A) \xrightarrow{\sim} k(B)$, this also means that $\hat{A} \rightarrow \hat{B}$ is surjective.*

Thus, from the “formal” point of view, saying that X is unramified over S means that X is essentially a *sub-prescheme* of S . Observe also that conditions d) and e) are expressed solely in terms of the fibers of X/S .

Proposition 1.4. *Let X be a prescheme locally of finite presentation over S . For X to be étale over S , it is necessary and sufficient that it be smooth over S and unramified over*

S (trivial by definition), which allows the criteria 1.2 and 1.3 to be applied. One finds in particular:

- (i) For X to be étale over S , it is necessary and sufficient that it be flat over S and unramified over S . An analogous local criterion for X to be étale over S at a given point x .
- (ii) Suppose in addition S , hence X , locally noetherian. Then the fact that X is étale over S at a point x can be recognized on the local homomorphism $A = \mathcal{O}_{S,s} \rightarrow B = \mathcal{O}_{X,s}$ (and even on the local homomorphism $\hat{A} \rightarrow \hat{B}$) by the following characteristic property: B is flat over A , and $B \otimes_A k$ is a finite separable extension of k (the residue field⁽¹⁾ of A). When $k(A) \xrightarrow{\sim} k(B)$, this condition simply means that $\hat{A} \rightarrow \hat{B}$ is an isomorphism.

Thus, from the “formal” point of view, saying that X is étale over S simply means that X is locally isomorphic to S .

Remark 1.5. When X is locally of finite presentation over S , and a point $x \in X$ is given, then the fact that X is smooth (resp. unramified, resp. étale) over S at x , i.e. near x , can still be recognized on the functor $X : (\text{Sch})^\circ_S \rightarrow (\text{Ens})$ by the following property: for every S' over S , S' the spectrum of a local ring, every subscheme S'_0 of S' defined by a nilpotent ideal, and every S -morphism $u_0 : S'_0 \rightarrow X$ taking the closed point s' of S' to x , there exists at least one (resp. at most one, resp. exactly one) S -morphism $u : S' \rightarrow X$ extending it. This statement shows in particular that in definition 1.1 one can restrict oneself to S' that are local schemes (provided the functor considered is representable by a prescheme locally of finite presentation over S). On the other hand, when S is locally noetherian, in the preceding pointwise criterion one can even restrict oneself to S' that are artinian local schemes, and one can therefore make the same restriction on S' in definition 1.1 (provided S is locally noetherian and the functor considered is represented by a prescheme locally of finite type over S).

Remark 1.6. Of course, given a morphism $f : X \rightarrow Y$ of preschemes, one will say that f is smooth (resp. ...) if f makes X a smooth Y -prescheme (resp. ...). When X and Y are S -preschemes and f an S -morphism of finite presentation, these properties of f are expressed immediately in terms of the morphism of functors $F, G : (\text{Sch})^\circ_S \rightarrow (\text{Ens})$ defined by f : f is smooth (resp. unramified, resp. étale) if for every S -prescheme S' , S' affine (in the absolute sense), and every subscheme S'_0 of S' defined by a nilpotent ideal $\mathcal{J}$, the map

$$F(S') \rightarrow F(S'_0) \times_{G(S'_0)} G(S')$$

deduced from the commutative square

$$\begin{array}{ccc} F(S') & \longrightarrow & G(S') \\ \downarrow & & \downarrow \\ F(S'_0) & \longrightarrow & G(S'_0) \end{array}$$

is surjective (resp. injective, resp. bijective), i.e. for every commutative square of morphisms of functors over S (where S', S'_0 are as above and $i : S'_0 \rightarrow S'$ is the canonical immersion):

$$(Q) \quad \begin{array}{ccc} S'_0 & \xrightarrow{i} & S' \\ u_0 \downarrow & & \downarrow v \\ F & \xrightarrow{f} & G \end{array}$$

⁽¹⁾**N.D.E.** “Residue extension” has been replaced by “residue field”.

there exists at least one (resp. at most one, resp. exactly one) morphism

$$\begin{array}{ccc} & & S' \\ & \swarrow u & \\ F & & \end{array}$$

making the two corresponding triangles commutative:

$$u_0 = ui \quad \text{and} \quad v = fu.$$

This property for a homomorphism of functors $f : F \rightarrow G$ over S still makes sense even if the functors considered are not representable; one will say (if it holds) that the homomorphism of functors $f : F \rightarrow G$ is formally smooth (resp. formally unramified, resp. formally étale). One will note that it depends only on the homomorphism of functors $(\text{Sch})^\circ \rightarrow (\text{Ens})$ defined by F, G (cf. I 1.4.1), and not on the structural morphisms $F \rightarrow S$ and $G \rightarrow S$. An equivalent way of expressing the preceding definition, rather easier to handle in applications, is the following: *the morphism of functors $F \rightarrow G$ over S is said to be formally smooth (resp. formally unramified, resp. formally étale) when, for every S' over S and every morphism $S' \rightarrow G$, the functor $F' = F \times_G S'$ over S' is formally smooth (resp. formally unramified, resp. formally étale).*

Remark 1.7. Under the conditions of 1.6, when a “point of F ” with values in a field k is given, i.e. an element x of $F(\text{Spec}(k))$ or, what amounts to the same thing, a morphism $\text{Spec}(k) \rightarrow F$, one likewise says that f is formally smooth (resp. formally unramified, resp. formally étale) *at x* if the condition of the preceding definition holds whenever S' is a local scheme and the morphism $S'_0 \rightarrow F$ in diagram (Q) above is “compatible with the marked points” in the following sense: if k' denotes the residue field of S' and S'_0 at their closed point, the diagram

$$\begin{array}{ccc} \text{Spec}(k') & & \text{Spec}(k) \\ j \downarrow & & \downarrow x \\ S'_0 & \xrightarrow{u_0} & F \end{array}$$

(where $j : \text{Spec}(k') \rightarrow S'_0$ denotes the canonical immersion) can be completed to a *commutative diagram*

$$\begin{array}{ccc} & \text{Spec}(k'') & \\ & \swarrow \quad \searrow & \\ \text{Spec}(k') & & \text{Spec}(k) \\ \downarrow & & \downarrow \\ S'_0 & \xrightarrow{\quad} & F, \end{array}$$

where k'' is the spectrum of a field. When F, G are representable by S -preschemes X, Y and the S -morphism $f : X \rightarrow Y$ is locally of finite presentation, this condition means precisely (by 1.5) that $f : X \rightarrow Y$ is smooth at the point of X that is the image of $\text{Spec}(k)$ under $x : \text{Spec}(k) \rightarrow X$.

Remark 1.8. When the condition of the preceding definition holds on restricting oneself to artinian local S' , one will say that $F \rightarrow G$ is *infinitesimally smooth* (resp. *infinitesimally unramified*, resp. *infinitesimally étale*) at x , and one says that $F \rightarrow G$ is infinitesimally smooth (resp. ...) if it is so at every point x , in other words if the condition considered in 1.6 holds whenever S' is artinian local. This variant of the preceding notions is technically useful, for it is often more convenient to check, being a weaker notion, while frequently

being sufficient (for example if $F \rightarrow G$ is a morphism locally of finite presentation, with G representable by a locally noetherian prescheme...) to imply the strong condition.

Smooth morphisms of preschemes behave remarkably simply with respect to differential calculus. We shall restrict ourselves here to recalling the following property:

Proposition 1.9. *Let $f : X \rightarrow S$ be a smooth morphism of preschemes. Then $\Omega_{X/S}^1$ is a locally free module of finite type on X , its rank at a point $x \in X$ is equal to the dimension of the fiber X_s (where $s = f(x)$) near the point x .*

One calls this dimension the *relative dimension* of X over S at x . One will note that it is zero (when f is smooth at x) if and only if f is étale at x . In practice, this dimension can also be calculated in terms of the functor F represented by X , in the following way. Let ξ be a point of F with values in a field k , “localized at x ”, i.e. a morphism $\text{Spec}(k) \rightarrow F = X$ whose image is x . Consider the algebra $D(k) = k[t]/(t^2)$ of dual numbers over k , considered as an S -prescheme, and consider the map

$$F(D(k)) \rightarrow F(k)$$

deduced from the augmentation $D(k) \rightarrow k$; finally, let $F(D(k), \xi)$ be the inverse image of $\xi \in F(k)$ under this map. Then this set is naturally endowed with a vector space structure over k (in fact, it is the vector space dual to $\Omega_{X/S}^1 \otimes_{\mathcal{O}_{X,x}} k$), whose dimension is the relative dimension of X over S at x .

To describe the vector space law on $F(D(k), \xi)$ explicitly, it is useful to introduce more generally, as in exposé II, for every vector space V over k , the algebra $D_k(V) = k + V$ (V an ideal of square zero), and to consider $F(D_k(V), \xi)$, the inverse image of ξ under $F(D_k(V)) \rightarrow F(k)$, as a covariant functor in V with values in (Ens). It then suffices that this functor commute with products of two factors (which means that F transforms certain amalgamated sums of a very particular type into fibered products, compare exposé II, a condition always satisfied if F is representable) to conclude that the $F(D_k(V), \xi)$ and in particular $F(D(k), \xi)$ are endowed with vector space structures over k . One can thus define the relative dimension of F over X at the “point” ξ under conditions substantially broader than the representability of F .

In the present exposé, the fact that certain functors which we shall describe explicitly are representable by smooth preschemes over S will serve us chiefly through the following result, which will be for us the technical intermediary for passing from constructions on the completion of the local ring A of a point s of a noetherian scheme S to a local ring A' over A étale over A , which will give in particular a means of passing from s to neighboring points:

Proposition 1.10. *Let $f : X \rightarrow S$ be a smooth morphism of preschemes, s a point of S , and x a point of X above s such that $\kappa(x)$ is a finite separable extension of $\kappa(s)$. Then there exists a subscheme S' of S , étale over S , passing through x . Thus one can find an étale morphism $S' \rightarrow S$, a point s' of S' above s with residue extension equal to $\kappa(x)/\kappa(s)$, and an S -morphism $S' \rightarrow X$ taking s' to x .*

To construct S' , one simply takes a system $g_1, \dots, g_n$ of sections of $\mathcal{O}_X$ on a neighborhood of x that induce at x a regular system of parameters of the local ring of the fiber X_s at x ; the subscheme S' defined by the g_i is then étale over S at x , and, provided one shrinks S' , it will therefore be étale over S .

We shall use 1.10 when $\kappa(x) = \kappa(s)$, i.e. x is rational over $\kappa(s)$, i.e. x can be considered as a section of X_s over $\text{Spec}(\kappa(s))$. Then 1.10 has the nature of a theorem on extending sections (after an étale extension of the base). It takes a particularly simple form in the following special case:

Corollary 1.11. *Under the conditions of 1.10, suppose that S is the spectrum of a henselian local ring, and that $\kappa(x) = \kappa(s)$. Then there exists a section of X over S passing through x (uniquely determined if X is even étale over S at x).*

Indeed, since S is henselian, it follows under the conditions of 1.10 that S' contains an open subscheme that is finite over S and whose fiber at s is reduced to x . Since it is étale over S , it follows that it is isomorphic to S , whence the conclusion.—One will note that when S is the spectrum of a complete local ring, 1.10 or 1.11 is more or less equivalent to the classical “Hensel’s lemma”, and it sometimes happens that it is referred to by this name.

2. Examples of formally smooth functors drawn from the theory of groups of multiplicative type

We shall interpret, in the language introduced in the preceding no. , the results stated in IX 3 concerning infinitesimal extensions of a homomorphism from a group of multiplicative type (consequences of the vanishing of the Hochschild cohomology of such a group, established in Exposé I).

Proposition 2.1. *Let S be a prescheme, H a group of multiplicative type over S , G an S -group prescheme smooth over S ; consider the functor over S*

$$M_H = \underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$$

(whose value at S' over S is the set of homomorphisms of S' -groups from $H_{S'}$ to $G_{S'}$). This functor is formally smooth over S .

Cf. IX 3.6. More generally:

Corollary 2.2. *Let S, G be as above, consider a homomorphism $u : H_1 \rightarrow H_2$ of group schemes of multiplicative type over S , whence, with the preceding notation, a morphism of functors over S :*

$$M_{H_2} \rightarrow M_{H_1} \quad (M_{H_i} = \underline{\mathrm{Hom}}_{S\text{-gr}}(H_i, G)),$$

given by $w \mapsto w \circ u$. This homomorphism is formally smooth.

Indeed, by definitions 1.6, this is equivalent to the following statement: when S is affine, S_0 a subscheme defined by a nilpotent ideal, let

$$v : H_1 \rightarrow G$$

be a homomorphism of S -groups, and

$$w_0 : (H_2)_{S_0} \rightarrow G_{S_0}$$

a homomorphism of S_0 -groups such that $w_0 u_{S_0} = v_{S_0}$; there then exists a homomorphism of S -groups

$$w : H_2 \rightarrow G$$

extending w_0 , and such that $wu = v$. To see this, one first extends w_0 to a homomorphism of S -groups $w' : H_2 \rightarrow G$, which is possible by 2.1; consider then $v' = w'u : H_1 \rightarrow G$, which satisfies $v'_{S_0} = v_{S_0}$ by the hypothesis on w_0 ; thus, by VIII 3.6, there exists an element g of $G(S)$, whose image in $G(S_0)$ is the identity element, and such that $v = \mathrm{int}(g)v'$, whence $v = (\mathrm{int}(g)w')u$, and it will therefore suffice to take $w = \mathrm{int}(g)w'$.

Corollary 2.3. *With the notation of 2.1, consider $M_H = M$ as a functor with group of operators G (G acting by $(v, g) \mapsto \mathrm{int}(g) \circ v$). Then the corresponding morphism*

$$R : G \times_S M \rightarrow M \times_S M$$

defined by $(g, v) \mapsto (\mathrm{int}(g) \circ v, v)$ is a formally smooth morphism.

By means of a base change $S' \rightarrow S$, this is equivalent to the following statement:

Corollary 2.4. *With the notation of 2.1, let $v_1, v_2 : H \rightarrow G$ be two morphisms of S -groups, let $\text{Transp}(v_1, v_2)$ be the subfunctor of G formed by the g such that $\text{int}(g) \circ v_1 = v_2$. Then this functor is formally smooth over S . In particular (if $v_1 = v_2 = v$) the functor $\text{Centr}(v)$, a subgroup of G formed by the h such that $\text{int}(h)v = v$, is formally smooth over S .*

(N.B. The pair (v_1, v_2) can be considered as a section over S of the second member in the morphism R of 2.3, and $\text{Transp}(v_1, v_2)$ as the inverse image functor of the said section under R). Statement 2.4 itself is equivalent to the following: when S is affine and S_0 is a subscheme of S defined by a nilpotent ideal, for every $g_0 \in G(S_0)$ such that $\text{int}(g_0)(v_1)_{S_0} = (v_2)_{S_0}$, g_0 extends to a $g \in G(S)$ such that $\text{int}(g)v_1 = v_2$. To prove it, one first extends g_0 to a section g' of G over S , which is possible since G is smooth over S ; one puts $v'_2 = \text{int}(g')v_1$, notes that v_2 and v'_2 have the same restriction over S_0 , so by the result IX 3.6 already invoked there exists a $g'' \in G(S)$ inducing the identity section on S_0 and such that $v_2 = \text{int}(g'')v'_2$, whence $v_2 = \text{int}(g'')\text{int}(g')v_1 = \text{int}(g''g')v_1$, so it suffices to take $g = g''g'$.

Proposition 2.1 bis. *Let S be a prescheme, G an S -group prescheme smooth over S ; consider the functor $M : (\text{Sch})^\circ_S \rightarrow (\text{Ens})$:*

$$M(S') = \text{set of subgroups of multiplicative type of } G_{S'}.$$

Then M is formally smooth over S .

Cf. IX 3.6 bis.

Corollary 2.2 bis. *Let n be an integer, and consider the morphism of functors*

$$\varphi_n : M \rightarrow M$$

defined by

$$\varphi_n(H) = {}_nH = \text{Ker}(n \cdot \text{id}_H).$$

Then φ_n is a formally smooth morphism. If for every integer $p > 0$, M_p denotes the subfunctor of M such that $M_p(S')$ is the set of subgroups H of multiplicative type of $G_{S'}$ such that ${}_pH = H$, then the morphism induced by φ_n :

$$M_{np} \rightarrow M_n$$

is formally smooth.

The second assertion is trivially contained in the first and is included only for the convenience of a later reference. The proof of the first is analogous to that of 2.2, invoking this time IX 3.6 bis.

Corollary 2.3 bis. *With the notation of 2.1 bis, consider M as a functor with group of operators G (G acting by $(g, H) \mapsto \text{int}(g)(H)$). Then the corresponding morphism*

$$R : G \times_S M \rightarrow M \times_S M$$

defined by $(g, v) \mapsto (\text{int}(g)v, v)$ is formally smooth.

This is equivalent to the following statement:

Corollary 2.4 bis. *With the notation of 2.1 bis, let H_1, H_2 be two subgroups of multiplicative type of G , and let $\text{Transp}_G(H_1, H_2)$ be the subfunctor of G formed by the g such that $\text{int}(g)H_1 = H_2$. Then this functor is formally smooth over S . In particular, if $H_1 = H_2 = H$, the subfunctor $\text{Norm}_G(H)$ of G , the normalizer of H , is formally smooth over S .*

The proof is analogous to that of 2.4, again invoking IX 3.6 bis.

Proposition 2.5. *Let S be a prescheme, G an S -group prescheme smooth over S , K a sub- S -group prescheme, smooth over S or of multiplicative type, H an S -group prescheme of multiplicative type, $u : H \rightarrow G$ a homomorphism of S -groups, and denote by $\text{Transp}_G(u, K)$*

the subfunctor of G whose value, at an S' over S , is formed by the $g \in G(S')$ such that $\text{int}(g)u_{S'} : H_{S'} \rightarrow G_{S'}$ factors through $K_{S'}$. Then this functor is formally smooth over S .

The proof is analogous to that of 2.2, invoking IX 3.6 and X 2.1 (the latter in the case where K is of multiplicative type).

3. Auxiliary representability results

Proposition 3.1. *Let $F \rightarrow S$ be a functor over the prescheme S . The following conditions are equivalent:*

- (i) F is representable, and $F \rightarrow S$ is an open immersion (one also says simply that $F \rightarrow S$ is an open immersion).
- (ii) F is a sheaf for the faithfully flat quasi-compact topology, “commutes with inductive limits of rings”, $F \rightarrow S$ is a monomorphism, and finally the following condition holds: for every local prescheme S' over S , with residue field k , and every S -morphism $\text{Spec}(k) = S'_0 \rightarrow F$, there exists an S -morphism $S' \rightarrow S$ extending it.
- (iii) (When S is locally noetherian). F is a sheaf for the faithfully flat quasi-compact topology, “commutes with inductive limits of rings”, “commutes with adic projective limits of artinian local rings”, $F \rightarrow S$ is a monomorphism, and is infinitesimally étale (cf. 1.8).

Let us first clarify two points of terminology.

Remark 3.2. One says that a functor⁽²⁾ F over S “commutes with inductive limits (understood: filtered) of rings” if for every filtered projective system $(S'_i)_{i \in I}$ of affine S -schemes over an affine open subset of S , with rings A'_i , the natural homomorphism

$$(*) \quad \varinjlim_i F(S'_i) \rightarrow F(S') \quad (\text{where } S' = \text{Spec } A', \ A' = \varinjlim_i A'_i)$$

is bijective. One will note that S' is nothing other than the projective limit of (S'_i) in the category of preschemes (and even of all ringed spaces), so the condition considered has the nature of a right exactness condition (commutation with certain inductive limits in $(\text{Sch})^\circ_S$), just like the condition of being a sheaf for some topology. One should take care that the condition considered is essentially *relative*, i.e. involves the morphism $F \rightarrow S$ and not only the functor $F : (\text{Sch})^\circ \rightarrow (\text{Ens})$; precisely, in $(*)$ $F(S'_i)$ and $F(S')$ denote $\text{Hom}_S(S'_i, F)$, $\text{Hom}_S(S', F)$. Thus, when F is representable, the condition considered means that F is locally of finite presentation over S . (And we have used several times, in the last two exposés, the fact that a functor represented by an S -prescheme locally of finite presentation commutes with inductive limits of rings).

Remark 3.3. One says that a functor⁽²⁾ F over S *commutes with adic projective limits of artinian local rings* if for every S' over S that is the spectrum of a complete noetherian local ring A' , putting $S'_n = \text{Spec}(A'/\text{rad}(A')^{n+1})$, the natural map

$$(xx) \quad F(S') \rightarrow \varprojlim_n F(S'_n)$$

is bijective. One will note that this condition, which has the nature of a left exactness condition, is *satisfied whenever F is representable*. One sees easily that, unlike the condition of commutation with inductive limits of rings, it is intrinsic to F as an element of $\text{Ob}(\widehat{(\text{Sch})})$, i.e. does not involve the morphism $F \rightarrow S$.

Remark 3.4. Let F be a functor over S that is a sheaf for the Zariski topology, or, as one also says, that is “of a local nature”. (For this it suffices that F be a sheaf for a finer topology, such as the faithfully flat quasi-compact topology). Let (S_i) be a covering of S by open

⁽²⁾**N.D.E.** Contravariant.

subsets; then one verifies easily (by a method of gluing pieces) that F is representable if and only if the $F_i = F \times_S S_i$ are, which allows one, for example, to reduce to the case where S is affine. Suppose that the functor F of a local nature commutes with inductive limits of rings. Then, for F to be representable, it is necessary and sufficient that its restriction to the category of preschemes locally of finite presentation over S be representable. The “necessary” was pointed out in 3.2, the “sufficient” amounts to this: if X is a prescheme locally of finite presentation over S and $X \rightarrow F$ a morphism such that, for every S' locally of finite presentation over S , the induced morphism

$$\mathrm{Hom}_S(S', X) \rightarrow \mathrm{Hom}_S(S', F)$$

is bijective, then $X \rightarrow F$ is an isomorphism. Now this follows easily from the fact that X and F are two functors of a local nature that commute with inductive limits of rings.

Let us now prove 3.1. The implications (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iii) are obvious; let us prove the converse implications.

One has (ii) $\Rightarrow$ (i). Indeed, let U be the set of $s \in S$ such that the canonical monomorphism $\mathrm{Spec}(\kappa(s)) \rightarrow S$ factors through F . By the last condition (ii), for every $s \in U$, the canonical monomorphism $\mathrm{Spec}(\mathcal{O}_{S,s}) \rightarrow U$ factors through F . Noting that $\mathcal{O}_{S,s}$ is the inductive limit of the rings of the affine neighborhoods of s , it follows from the fact that F commutes with inductive limits of rings that for every $s \in S$, there exists an open neighborhood U^s such that the canonical immersion $U^s \rightarrow S$ factors through F . This implies $U^s \subset U$, so U is open. Since $F \rightarrow S$ is a monomorphism, and F is of a local nature, the S -morphisms $U^s \rightarrow F$ glue in the $U^s \cap U^{s'}$ ($s, s' \in U$), hence come from an S -morphism $U \rightarrow F$. It remains to prove that this is an isomorphism, hence that every S -morphism $S' \rightarrow F$ factors uniquely through U (where S' is an S -prescheme). Since $F \rightarrow S$ and $U \rightarrow S$ is a monomorphism, it amounts to the same thing to say that the structural morphism $S' \rightarrow S$ factors through U , which reduces us to the case where S' is the spectrum of a field, hence reduced to a single point s' . Let s be the point of S below s' ; I say that the S -morphism $S' \rightarrow F$ factors through $\mathrm{Spec}(\kappa(s)) = S_0 \rightarrow F$ (which implies $s \in U$ and will prove what we want).

It amounts to the same thing, since $S' \rightarrow S_0$ is covering for fpqc and F is a sheaf for this topology, that the two composites

$$S'' = S' \times_{S_0} S' \rightrightarrows S' \rightarrow F$$

are the same, which follows from the fact that $F \rightarrow S$ is a monomorphism.

One has (iii) $\Rightarrow$ (ii) (when S is locally noetherian). It suffices to prove the last condition of (ii), and moreover (by the preceding proof) it suffices to do so when S' is of the form $\mathrm{Spec}(\mathcal{O}_{S,s})$, with $s \in S$. Let $A = \mathcal{O}_{S,s}$, $A_n = A/\mathfrak{m}^{n+1}$, $S_n = \mathrm{Spec}(A_n)$; it then follows from the hypothesis that $F \rightarrow S$ is infinitesimally smooth that the given morphism $S_0 \rightarrow F$ extends to morphisms $S_n \rightarrow F$. Since $F \rightarrow S$ is a monomorphism, one thus obtains an element of $\varprojlim_n F(S_n)$, and since F commutes with adic projective limits of artinian local rings, the $S_n \rightarrow F$ come from a morphism $\mathrm{Spec}(\hat{A}) = \hat{S}' \rightarrow F$. Since $F \rightarrow S$ is a monomorphism, F a sheaf for fpqc, and $\hat{S}' \rightarrow S'$ covering for the said topology, this morphism $\hat{S}' \rightarrow F$ factors through $S' \rightarrow F$, which completes the proof.

Proposition 3.5. *Let S be a locally noetherian prescheme, $F \rightarrow S$ a functor over S , $(X_i, u_i)_{i \in I}$ a family of S -morphisms $u_i : X_i \rightarrow F$, where the X_i are preschemes locally of finite type over S . Suppose the following conditions hold:*

- a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, commutes with adic projective limits of artinian local rings.*
- b) *The $u_i : X_i \rightarrow F$ are monomorphisms, and are infinitesimally étale (cf. 1.8).*
- c) *The family of u_i is “set-theoretically surjective”.*

Under these conditions, F is representable by a prescheme locally of finite type over S (and the u_i are open immersions, which make the family of X_i an open covering of F).

Remark 3.6. Proceeding as at the end of remark 1.7, one defines, for every functor $F : (\text{Sch})^\circ \rightarrow (\text{Ens})$, an “underlying set” $\text{ens}(F)$ as a quotient set of the set of points of F with values in fields (for the equivalence relation specified in 1.7). When F is representable by X , one indeed recovers the underlying set of X . Obviously $\text{ens}(F)$ depends functorially on F , so if $G \rightarrow F$ is a morphism of functors, one can say that this morphism is set-theoretically surjective if the induced map $\text{ens}(G) \rightarrow \text{ens}(F)$ is surjective. This therefore also means that every point of F with values in a field k “comes from” a point of G with values in a suitable extension of k . This definition extends at once to the case of a family of morphisms $G_i \rightarrow F$, which specifies the meaning of c).

Let us prove 3.5. For this, introduce, for $(i, j) \in I \times I$,

$$X_{i,j} = X_i \times_F X_j,$$

and consider the projections

$$v_{i,j} : X_{i,j} \rightarrow X_i \quad \text{and} \quad w_{i,j} : X_{i,j} \rightarrow X_j.$$

I say that these latter are representable by open immersions. To see this, one applies criterion 3.1 (iii): $X_{i,j}$ satisfies the three exactness conditions (being a sheaf, commuting with $\varinjlim$ of rings and with adic $\varprojlim$ of artinian local rings), for F, X_i, X_j satisfy them, and these conditions are stable under finite projective limits, in particular under fibered products; since $X_i \rightarrow F$ is a monomorphism, so is $v_{i,j} : X_{i,j} \rightarrow X_i$, which is deduced from it by the base change $X_j \rightarrow F$, and symmetrically $w_{j,i}$ is a monomorphism; finally the condition “infinitesimally étale” is also preserved by base change. This proves that one is under conditions 3.1 (iii).

We can now use the $X_i, X_{i,j}, v_{i,j}$ and $w_{i,j}$ to construct in the usual way an S -prescheme X , such that the X_i are identified with open subsets of X , the $X_{i,j}$ with the intersections $X_i \cap X_j$ and the $v_{i,j}, w_{i,j}$ with the canonical immersions. Note that X is also the quotient of

$$\mathbf{X} = \coprod_i X_i$$

by the equivalence relation $R = \coprod_{i,j} X_{i,j}$ (the two projections $v, w : R \rightrightarrows Y$ being defined by the $v_{i,j}$, resp. the $w_{i,j}$). More precisely, since F is a sheaf for fpqc, the $u_i : X_i \rightarrow F$ come from a $u : \mathbf{X} \rightarrow F$, and R is nothing other than the equivalence relation defined by u , $R = \mathbf{X} \times_F \mathbf{X}$; finally the quotient $X = Y/R$ is also a quotient in the category of sheaves for fpqc (and even in the category of sheaves for the Zariski topology): it suffices to use the definitions of “quotient” and “sheaf” to convince oneself of this. Consequently u factors uniquely through a morphism

$$u : X \rightarrow F,$$

and this morphism is a monomorphism. It remains to show that it is an isomorphism. Since F is of a local nature, one can suppose S affine, and since moreover F commutes with inductive limits of rings, it suffices to verify that for every T affine of finite type over S , every morphism $T \rightarrow F$ factors through X (cf. 3.4). For this, consider $G = X \times_F T \rightarrow T$; it suffices to prove that this is an isomorphism. Now, since T is noetherian, one sees as above that it is an open immersion (N.B. $X \rightarrow F$ is infinitesimally étale, as follows at once from the fact that the induced morphisms $u_i : X_i \rightarrow F$ are). But by hypothesis $X \rightarrow F$ is set-theoretically surjective, and one sees immediately that this is a condition stable under base change, so $G \rightarrow T$ is set-theoretically surjective, hence an isomorphism since it is an open immersion. Q.E.D.

Proposition 3.7. *Let S be a locally noetherian prescheme, I an increasingly filtered index set, $(T_i)_{i \in I}$ a projective system of S -preschemes locally of finite type, $T = \varprojlim T_i$ the projective limit functor, F a functor over S , $u : F \rightarrow T$ an S -morphism. Suppose the following conditions hold:*

- a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, and with adic projective limits of artinian local rings.*

- b) The morphism $u : F \longrightarrow T$ is a monomorphism.
- b') The morphism $u : F \longrightarrow T$ is infinitesimally étale.
- c) For every point ξ of F with values in the spectrum of a field k , denoting by $\xi_i \in T_i(\text{Spec}(k))$ its image and by t_i the corresponding element of T_i , there exists an $i \in I$ such that for $j \geq i$ the transition morphism $p_{i,j} : T_j \longrightarrow T_i$ is étale at t_j .
- d) For every prescheme X locally of finite type over S , and every S -morphism $X \longrightarrow F$, the set of $x \in X$ at which this morphism is infinitesimally étale is open.

Under these conditions, F is representable by a prescheme locally of finite type over S .

Let us note at once that in the case which will concern us in the following no. , conditions c) and d) will be verified by means of the following corollary:

Corollary 3.8. Assuming a), b), b'), conditions c) and d) are implied by the following:

- c') The T_i are smooth over S , and the transition morphisms $p_{i,j} : T_j \longrightarrow T_i$ are smooth.
- d') For every point ξ of F with values in the spectrum of a field k , let $t_i(\xi)$ be the element of T_i defined by ξ , $d_i(\xi)$ the relative dimension of T_i over S at $t_i(\xi)$, and $d(\xi) = \sup d_i(\xi)$. Then:

- 1°) for every ξ as above, one has $d(\xi) < +\infty$, and
- 2°) for every prescheme X locally of finite type over S , and every S -monomorphism $v : X \longrightarrow F$, the function $x \longmapsto d(\xi_x)$ on X is locally constant (where, for $x \in X$, ξ_x denotes the point of F with values in $\kappa(x)$ induced by v).

Let us prove 3.7. Place ourselves under the conditions of c), let t be the element of $\text{ens}(F)$ defined by ξ (cf. 3.6) and put

$$\mathcal{O}_t = \varinjlim_i \mathcal{O}_{T_i, t_i}.$$

Then, using the condition stated in c), one sees easily that $\mathcal{O}_t$ is a *noetherian* local ring (EGA 0_{IV} 10.3.1.3). Its residue field $\kappa(t)$ is the inductive limit of the residue fields $\kappa(t_i)$, and k is an extension of it. If η denotes the spectrum of $\kappa(t)$, ξ that of k , one has a commutative diagram

$$\begin{array}{ccc} \xi & \longrightarrow & F \\ \downarrow & & \downarrow \\ \eta & \longrightarrow & T, \end{array}$$

whose definition is obvious and left to the reader. Since $\xi \longrightarrow \eta$ is covering for the fpqc topology, F is a sheaf for that topology by a), and $F \longrightarrow T$ a monomorphism by b), one sees at once that the morphism $\eta \longrightarrow T$ factors (uniquely) through a morphism $\eta \longrightarrow F$. Let us now note the

Lemma 3.9. Under the conditions of 3.7, let $T' = \text{Spec}(A)$ be a *noetherian* local scheme, $T'_0 = \text{Spec}(k(A))$, $i : T'_0 \longrightarrow T'$ the canonical immersion, and suppose given a commutative square of morphisms

$$\begin{array}{ccc} T'_0 & \longrightarrow & F \\ i \downarrow & & \downarrow \\ T' & \longrightarrow & T. \end{array}$$

Then there exists a unique T -morphism $T' \longrightarrow F$.

The proof is that of 3.1 (iii) $\Rightarrow$ (ii) (where the fact that the S of the cited statement is representable was not used), using that F is a sheaf for the fpqc topology, commutes with adic projective limits of artinian local rings (in this case the $A/\text{rad}(A)^{n+1}$), and that $F \longrightarrow T$ is a monomorphism and is infinitesimally étale.

We shall apply lemma 3.9 to the case where $T' = \text{Spec}(\mathcal{O}_t)$, hence $T'_0 = \text{Spec}(\kappa(t))$, noting that we have just constructed $T'_0 \longrightarrow F$, and that the canonical homomorphisms

$\mathcal{O}_{T_i, t_i} \longrightarrow \mathcal{O}_t$ define a projective system of morphisms $\mathrm{Spec}(\mathcal{O}_t) \longrightarrow T_i$, whence a canonical morphism $\mathrm{Spec}(\mathcal{O}_t) \longrightarrow T$. The commutativity of the corresponding square (x) is trivial (for, by definition of T , it suffices to verify it with T replaced by T_i), whence a unique T -morphism

$$v' : T' = \mathrm{Spec}(\mathcal{O}_t) \longrightarrow F.$$

Since F commutes with inductive limits of rings, this morphism factors through

$$v'_i : T'_i = \mathrm{Spec}(\mathcal{O}_{T_i, t_i}) \longrightarrow F$$

for large i . For such an i , one has

$$T' \xrightarrow{\sim} T'_i \quad \text{i.e.} \quad \mathcal{O}_{T_i, t_i} \xrightarrow{\sim} \mathcal{O}_t.$$

Indeed, since $T' \longrightarrow T'_i$ is faithfully flat quasi-compact and hence an effective epimorphism, it suffices to see that it is a monomorphism. Now if one has two morphisms $R \rightrightarrows T'$ (with R representable) having the same composite with $T' \longrightarrow T'_i$, they have the same composite with $T' \longrightarrow T$ (which factors through $T' \longrightarrow T'_i$), hence the same composite with $T' \longrightarrow T_j$ for every j , hence with $T' \longrightarrow T'_j$ for every j , and are therefore equal since T' is the projective limit of the T'_j in the category (Sch).

Consider the diagram

$$\begin{array}{ccc} T' & \xrightarrow{v'} & F \\ \downarrow & \nearrow v'_i & \downarrow \\ T'_i & \longrightarrow & T_i, \end{array}$$

where the unspecified arrows are the obvious arrows. The square is commutative and the upper triangle also, so since $T' \longrightarrow T'_i$ is an epimorphism, it follows that the lower triangle is commutative. Now $\mathcal{O}_{T_i, t_i}$ is the inductive limit of the affine rings of the affine open neighborhoods of t_i in T_i , so since F commutes with inductive limits of rings, v'_i comes from a morphism

$$v_t : U_t \longrightarrow F$$

from an open neighborhood U_t of t_i in T_i . One sees immediately that, on restricting this neighborhood if necessary, v_t is necessarily a T_i -morphism (T_i being locally of finite type over S , hence also commuting with inductive limits of rings). Then the composite of v_t with $F \longrightarrow T_i$ is a monomorphism, so v_t is a monomorphism. I say that it is *infinitesimally étale at t_i* . Since $F \longrightarrow T$ is infinitesimally étale, it amounts to the same thing to say that the composite $T_i \longrightarrow F \longrightarrow T$ is infinitesimally étale at t_i , or again that the induced morphism $T'_i \longrightarrow T$ is infinitesimally étale, or finally (since $T' \xrightarrow{\sim} T'_i$) that $T' \longrightarrow T$ is infinitesimally étale, which is immediate, this morphism being the projective limit of the infinitesimally étale morphisms $T'_i \longrightarrow T_i$.

Let us now apply condition d) (which had not yet been used); it follows that v_t is infinitesimally étale near t , so on replacing U_t by a smaller open subset one can suppose that v_t is an infinitesimally étale monomorphism.

For variable $t \in \mathrm{ens}(F)$, the family of morphisms v_t falls under 3.5, which implies the conclusion of 3.7.

Let us now prove 3.8, supposing that conditions c') and d') of 3.8 hold. Then, with the notation of d'), one will have $d_i(\xi) = \text{constant}$ for large i , so the relative dimension of T_j over T_i at t_j , equal to $d_j(\xi) - d_i(\xi)$, is zero, hence $T_j \longrightarrow T_i$ is étale at t_j , which proves condition c). Consequently the preceding proof applies to give, for every $t \in \mathrm{ens}(F)$, an index i , an open neighborhood U_t of t_i and a morphism $U_t \longrightarrow F$ that is a monomorphism, infinitesimally étale at t_i , and everything amounts to proving that this morphism is infinitesimally étale near t_i . Now with the notation of d') 2°, where one takes $X = U_t$, one can suppose, on replacing U_t by the connected component of t_i in U_t , that for every $x \in U_t$ one has

$$d(\xi_x) = d(\xi_{t_i}) \quad \text{for every } x \in U_t.$$

But $d(\xi_{t_i}) = \text{relative dimension of } T_i \text{ over } S \text{ at } t_i$, and since the relative dimension of T_i over S remains constant on U_t , one will also have, for every $x \in U_t$:

$$(*) \quad d(\xi_x) = \text{relative dimension of } U_t \text{ over } S \text{ at } x.$$

On the other hand, with the notation of the proof of 3.7, one sees at once that for every noetherian local prescheme $R' = \text{Spec}(A)$, putting $R'_0 = \text{Spec } k(A)$, every morphism $R' \rightarrow F$ such that the induced morphism $R'_0 \rightarrow F$ factors through F (i.e. taking the closed point of R' to $t \in \text{ens}(F)$) factors (obviously uniquely) through T' . (The proof is that of 3.1 or 3.9: one uses that $T' \rightarrow F$ is a monomorphism infinitesimally étale at t , and that T' is a sheaf for fpqc commuting with $\varprojlim$...). Applying this result to the morphism $\text{Spec}(\mathcal{O}_{U_t, x}) = R' \rightarrow F$ induced by v_t and to the point $t_x \in \text{ens}(F)$ that is the image of the closed point of R' , i.e. the image of x under v_t , one finds a factorization

$$\text{Spec}(\mathcal{O}_{U_t, x}) \rightarrow \text{Spec}(\mathcal{O}_{t_x}) \rightarrow F$$

and since the second arrow is infinitesimally étale at t_x , to prove that the composite is so at x , it suffices to prove that the first is so at t_x .

Now, thanks to formula (x) above, it is a local S -homomorphism of the localizations, at two points x, t_x , of smooth S -preschemes U_t, U_{t_x} of the same relative dimension d over S at t, t_x . This morphism is induced by an S -morphism

$$w : U \rightarrow V$$

where U is an open neighborhood of x in U_t , and where $V = U_{t_x}$ (EGA I 6.5.1). Everything amounts to proving that this morphism is étale at x . Moreover U and V are endowed with monomorphisms $U \rightarrow F, V \rightarrow F$, and one sees at once that, on restricting U further, w is an F -morphism, so w is a monomorphism. It now suffices to prove the

Lemma 3.10. *Let U, V be two smooth S -preschemes, of the same relative dimension d over S , and $w : U \rightarrow V$ an S -morphism that is a monomorphism; then w is an open immersion (and a fortiori is étale).*

By SGA I 5.7 one is reduced to the case where S is the spectrum of a field, which one can suppose algebraically closed. By SGA I 5.1 it suffices to prove that w is étale, and it suffices to prove it at the closed points of U . Let x be such a point, $y = w(x)$; then, taking a regular system of parameters $f_1, \dots, f_d$ of $\mathcal{O}_{V, y}$, one sees that $A = \mathcal{O}_{U, x} / \sum f_i \mathcal{O}_{U, x}$ is the trivial extension of $k = \mathcal{O}_{V, y} / \sum f_i \mathcal{O}_{V, y}$ (for, since w is a monomorphism, so is the structural morphism $\text{Spec}(A) \rightarrow \text{Spec}(k)$ deduced from it by base change). Since $\mathcal{O}_{U, x}$ is a regular local ring of dimension d , it follows that the f_i form a regular system of parameters of this ring, which implies at once that w is étale at x and completes the proof of 3.10, hence of 3.8.

Corollary 3.11. *Under the conditions of 3.8, for every quasi-compact open subset U of F separated over S , there exists an $i \in I$ such that for every $j \geq i$ the morphism $u_j|_U : U \rightarrow T_j$ is an open immersion. In particular, if the T_i are quasi-affine over S , then every open subset U of F quasi-compact over S , i.e. of finite type over S , is quasi-affine over S .*

The proof of 3.7 shows that for every $t \in F$, there exists an $i \in I$ such that $u_i : F \rightarrow T_i$ is a local isomorphism at t , and then u_j is a local isomorphism at x for every $j \geq i$. By quasi-compactness, one can choose i independent of $x \in U$. It remains to prove that for large i , $u_i|_U : U \rightarrow T_i$ is a monomorphism. Now, since $U \rightarrow T$ is a monomorphism, one sees that the intersection of the equivalence relations $U \times_{T_i} U \subset U \times_S U$ is reduced to the diagonal, and since $U \times_S U$ is a noetherian prescheme and the $U \times_{T_i} U$ closed sub-preschemes, it follows that one of these $U \times_{T_i} U$ is already reduced to the diagonal, i.e. $u_i|_U$ is a monomorphism. This proves the first assertion in 3.11, and the second is an immediate consequence of it.

Proposition 3.12. *Let S be a prescheme, G an affine S -group prescheme.*

a) Let F be the functor $(\text{Sch})/S \longrightarrow (\text{Ens})$ such that, for every T over S ,

$$F(T) = \text{set of subgroups of multiplicative type of } G_{T_S} \\ \text{that are finite over } T.$$

Suppose S locally noetherian or G of finite presentation over S . Then the functor F is representable and is affine over S . If G is of finite presentation over S , then F is locally of finite presentation over S .

b) Let H be an S -group prescheme of multiplicative type, and finite over S . Then $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable. It is affine over S , and if G is of finite type (resp. of finite presentation) over S , then so is $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$.

Remark 3.13. Except for the precision that $\underline{\text{Hom}}_{S\text{-gr}}$ is affine, and in the case where G is of finite presentation over S (which will suffice for us), 3.12 is an immediate consequence of the theory of Hilbert Schemes (A. Grothendieck, *Techniques de construction et théorèmes d'existence en Géométrie Algébrique: IV Les Schémas de Hilbert*, Séminaire Bourbaki, May 1961, no. 221). It even suffices that G be quasi-projective over S ; in case a), one can also represent the larger functor

$$F'(T) = \left\{ \begin{array}{l} \text{set of group sub-preschemes of } G_T, \\ \text{flat, proper and of finite presentation over } T, \end{array} \right\}$$

(the canonical monomorphism $F \longrightarrow F'$ is an “open immersion”, as follows from criterion 3.1 and X 4.7 b), so the representability of F' implies that F is representable by an open subset of F'); in case b), one can restrict oneself to supposing that H is projective and of finite presentation over S . In both cases, one obtains a functor locally of finite presentation over S . In the present exposé, 3.12 is only a technical lemma for proving a key result in the following no. , so we shall sketch an easy direct proof of 3.12, not using Hilbert schemes.

Let us first prove b). It will suffice that we prove that $\underline{\text{Hom}}_S(H, G)$ is representable (independently of any group structure on H, G), and has the additional properties stated for $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$, for, using also the same result for $\underline{\text{Hom}}_S(H \times_S H, G)$, one describes the subfunctor $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ of $\underline{\text{Hom}}_S(H, G)$ by a finite projective limit (in fact, by means of fibered products) involving G , $\underline{\text{Hom}}_S(H, G)$ and $\underline{\text{Hom}}_S(H \times_S H, G)$, which we leave the reader to describe explicitly. On the other hand, one will have

$$H = \text{Spec}(\mathcal{B}),$$

where $\mathcal{B}$ is a sheaf of algebras on S that is locally free as a sheaf of modules (this is the only hypothesis on H that we shall have to retain). Since the representability question considered is local on S , we can suppose S affine with ring A . On the other hand, one will have $G = \text{Spec}(C)$, where C is an A -algebra. When $G = S[t] = G_0$, t an indeterminate, the functor $\underline{\text{Hom}}$ is nothing other than

$$T \longmapsto \Gamma(T, \mathcal{B}_T),$$

which is representable ($\mathcal{B}$ being locally free) by the vector bundle $\mathbf{V}(\mathcal{B}^\vee)$, where $\mathcal{B}^\vee$ is the sheaf of modules dual to $\mathcal{B}$. When $G = S[(t_i)]$, with (t_i) a family (not necessarily finite) of indeterminates, one will therefore have $G = G_0^I$ (product over S of a family of copies of G_0), which is representable by the affine scheme

$$\underline{\text{Hom}}_S(H, G)^I = \mathbf{V}(\mathcal{B}^\vee)^I = \mathbf{V}(\mathcal{B}^{\vee(I)}).$$

In the general case, G will be isomorphic to a closed subscheme of a scheme of the form $S[(t_i)]$, i.e. C will be a quotient of an A -algebra of the form $A[(t_i)]$. Let (F_j) be a system of generators of the ideal by which one takes the quotient. Suppose $\mathcal{B}$ free of rank n , which is permissible on covering S by smaller affine open subsets. Choose a basis $(e_k)_{1 \leq k \leq n}$ of $\mathcal{B}$; then, writing the n components with respect to this basis of $F_j((x_i))$, for $x_i = \sum x_{ik} e_k$ (x_{ik} indeterminate coefficients, taken in an unspecified algebra A' over A), one finds, for every F_j ,

n polynomials $F_{j,k}$ in the $(x_{i,k})_{i \in I, 1 \leq k \leq n}$, with coefficients in A . One observes at once that $\underline{\text{Hom}}_S(H, G)$ is represented by the spectrum of the quotient of the polynomial ring $A[(x_{i,k})]$ by the ideal generated by the $F_{j,k}$. This proves b) at once.

Let us prove a). For every ordinary finite commutative group M , let F_M be the subfunctor of F obtained by restricting oneself to subgroups of G_T that are of multiplicative type and of type M . One sees easily, by a gluing argument like the one used in 3.5 (which we should have stated by breaking things down a little more!), that it suffices to verify that the F_M are representable; then F will be representable by the sum prescheme of the F_M , where M runs through the set of classes of finite commutative groups up to isomorphism. (F is indeed the sum of the F_M in the category of sheaves...).

Henceforth we suppose M fixed, and shall write F instead of F_M . Let $H = D_S(M)$, consider the subfunctor $F' = \underline{\text{Imm}}_{S\text{-gr}}(H, G)$ of $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ whose value at T is the set of homomorphisms of T -groups $H_T \rightarrow G_T$ that are closed immersions. One already knows by b) that $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable by an affine S -prescheme, and using IX 6.8 one sees at once that F' is representable by an open and closed sub-prescheme of the latter, hence is also affine over S . Finally, consider the canonical morphism $F' \rightarrow F$, which associates to each monomorphism $H_T \rightarrow G_T$ the image group. One observes at once, by the definitions, that in this way F' becomes a principal homogeneous bundle (in the category of sheaves for fpqc) over F , with group $\Gamma_F = \Gamma_S \times_S F$, where $\Gamma = \text{Aut}_{\text{gr}}(M)$, whence it follows “by descent” that this morphism is representable (i.e. for every morphism $T \rightarrow F$, with T representable, $T \times_F F'$ over T is representable—in fact, representable by a principal Galois bundle under Γ). Thus, by IV 4.6.6, F is representable if and only if the quotient F'/F'' , where $F'' = F' \times_F F'$, exists in (Sch) and is universally effective for faithfully flat quasi-compact morphisms, or, what amounts to the same thing, if and only if the quotient F'/Γ exists and is universally effective for the said morphisms. Now, since F' is affine over S , we saw in V 4.1 that the condition in question is indeed satisfied. This proves the representability of F in a).

As for the addition, relating to the case where one supposes G of finite presentation over S , it follows at once from the preceding proof, taking into account that by b), F' is then locally of finite presentation over S , so that one can apply Exposé V⁽³⁾.

4. The scheme of subgroups of multiplicative type of a smooth affine group

The main result of the present exposé consists of the

Theorem 4.1. *Let S be a prescheme, G an S -group prescheme smooth and affine over S , F the functor $(\text{Sch})_S^\circ \rightarrow (\text{Ens})$ defined by*

$$F(T) = \text{set of subgroups of multiplicative type of } G_T.$$

Then the functor F is representable, and is smooth and separated over S .

Let us point out at once the following variant:

Corollary 4.2. *Let G, H be two S -group preschemes, with G smooth and affine over S , H of multiplicative type and of finite type. Then $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable, and is smooth and separated over S .*

Indeed, when H is smooth over S , so is $H \times_S G$, and one can apply 4.1 to the latter. By considering graph groups, $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ becomes a subfunctor F' of the functor F : “subgroups of multiplicative type of $H \times_S G$ ”, namely $F'(T) = \text{set of subgroups } K \text{ of multiplicative type of } (H \times_S G)_T = H_T \times_T G_T \text{ such that the homomorphism induced by the first projection}$

$$K \rightarrow H_T$$

⁽³⁾**N.D.E.** Reference to be checked/made precise.

is an isomorphism. By IX 2.9 the canonical morphism $F' \rightarrow F$ is an open immersion, so, since F is representable, so is F' , which will be representable by an open subset of F ; and since F is separated and smooth over S , so will F' be. In the case where H is finite over S , it suffices to apply 3.12 b) for the representability of $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$. When H is a product $H_1 \times_S H_2$, with H_1 smooth over S and H_2 finite over S , then $\mathrm{Hom}_{T\text{-gr}}(H_T, G_T)$ is identified with the subset of $\mathrm{Hom}_{T\text{-gr}}((H_1)_T, G_T) \times \mathrm{Hom}_{T\text{-gr}}((H_2)_T, G_T)$ formed by pairs (u_1, u_2) such that $u_1 \cdot u_2 = u_2 \cdot u_1$, whence it follows that $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$ is representable by a closed sub-prescheme of $\underline{\mathrm{Hom}}_{S\text{-gr}}(H_1, G) \times_S \underline{\mathrm{Hom}}_{S\text{-gr}}(H_2, G) = X$, as one sees by applying VIII 6.5 b), where one takes $Y = H$, $Z = G$, q_1 and q_2 being defined respectively by $(u_1, u_2) \mapsto u_1 \cdot u_2$ and $(u_1, u_2) \mapsto u_2 \cdot u_1$.

In the general case, since the question is local on S for the Zariski topology, one can suppose S affine, and H of constant type over S . Then, since H is quasi-isotrivial (X 4.5), one can find an étale surjective morphism $S' \rightarrow S$, S' affine, which splits H , i.e. such that $H' = H_{S'}$ is diagonalizable. Then the preceding result applies, for a diagonalizable group is the product of a diagonalizable torus with a finite diagonalizable group. It remains to see that the descent datum obtained on the S' -prescheme $\underline{\mathrm{Hom}}_{S'\text{-gr}}(H', G') = X'$ is effective. This is seen by the argument of X 5.4, noting that in *loc. cit.*, the hypothesis that $X' \rightarrow S'$ was separated and locally quasi-finite served only to ensure that every open subset of X' quasi-compact over S' was quasi-affine over S' . Now this property is still satisfied in the present case, as follows easily from the fact that it holds for the functor F of 4.1 (which will be seen in the course of the proof of 4.1). This (or any other variant of this little reduction) establishes the representability of $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$, and at the same time the fact that it is separated over S . It is locally of finite presentation over S , as one sees, for example (as was pointed out in 3.2), thanks to the fact that this functor “commutes with inductive limits of rings”. Finally, since this functor is formally smooth over S (by 2.1), it is smooth over S .

Remark 4.3. We have here deduced 4.2 from 4.1, which is really immediate only when H is also smooth over S . For the deduction to be made without contortions in the general case, the representability result 4.1 would have to be established without supposing G smooth over S , but only affine of finite presentation over S . (Of course, then F will in general no longer be smooth over S !). There is little doubt that 4.1 remains true under these more general hypotheses, but the proof seems to have to be more delicate (for lack of being able to invoke 3.8)^{*(4)}. Let us point out, however, that when G is a closed subgroup of an affine smooth group G' over S , then the functor F representing subgroups of multiplicative type of G is representable by a closed sub-prescheme of the prescheme representing the analogous functor F' for G' (falling under 4.1), as one sees easily by applying VIII 6.4. This also raises the question: is a group scheme G over an affine S , which is affine and of finite presentation over S , isomorphic to a group subscheme of a $\mathrm{GL}(n)_S$, for suitable n ? This is true when S is the spectrum of a field, cf. VI_B 11.11, but unfortunately false in general, even for tori, cf. 4.6. Finally, note that one could also prove 4.2 directly by exactly the same method as 4.1.

Let us now prove 4.1. Since the functor F is obviously of a local nature, one can suppose S affine, hence $S = \mathrm{Spec}(A)$, A a ring. Considering A as the inductive limit of its subrings of finite type over $\mathbf{Z}$, and noting that G comes from a smooth affine group over such a subring (EGA IV 8), one is reduced to the case where S is noetherian. For every integer $n > 0$, let T_n be the functor defined like F , but restricting oneself to subgroups H of multiplicative type of G_T such that $n \cdot \mathrm{id}_H = 0$, i.e. such that ${}_n H = H$. Order the set I of integers > 0 by the

* It is indeed proved for G flat and quasi-affine over S with connected fibers, provided one restricts oneself to central subtori of G (XV 8.8).

⁽⁴⁾**N.D.E.** In the case where H is smooth (not necessarily affine) over a locally noetherian normal base S , M. Raynaud showed that the largest representable open subset of the functor of subtori of H is a disjoint sum of smooth affine open subsets over S . This is theorem IX.9.26 in *Faisceaux amples sur les schémas en groupes et sur les espaces homogènes*, Lecture Notes Maths. 119 (1970).

divisibility relation. When m is a multiple of n , define

$$p_{n,m} : T_m \longrightarrow T_n$$

by $p_{n,m}(H) = {}_nH = \text{Ker}(n \cdot \text{id}_H)$.

In this way the T_n form a projective system of functors over S . By 3.12 a), the functors T_n are representable, affine and of finite type over S . Define likewise morphisms

$$u_n : F \longrightarrow T_n,$$

by the relation $u_n(H) = {}_nH$. In this way one obtains a morphism

$$u : F \longrightarrow T = \varprojlim_n T_n,$$

where the $\varprojlim$ is taken in the category of functors over S . But let us point out at once that, since the T_n are representable and affine over S , so is T (it will be the spectrum of the inductive limit of the quasi-coherent algebras over S defining the T_n). Of course T is not in general of finite type over S .

We shall apply 3.7 and are reduced to verifying conditions a) to d) of 3.7, which will imply that F is representable by a prescheme locally of finite type over S . It then follows from 2.1 bis that F is even smooth over S , and since F is a subfunctor of T , which is affine over S , it follows that F is separated over S (being separated over T , which is separated over S). Let us prove at once the addition invoked above, namely that every open subset U of F quasi-compact over S is quasi-affine over S : this follows from 3.11 and the fact that the T_u are affine over S .

Let us therefore verify the conditions of 3.7.

a) F is a sheaf for the faithfully flat quasi-compact topology, by descent theory SGA 1 VIII, which applies here since groups of multiplicative type over T are affine over S . It commutes with inductive limits of rings by the general groundwork EGA IV 8. Let us show that it commutes with adic projective limits of artinian local rings. When one is dealing, instead of the functor F , with the analogous functor considered in 4.2, this property is nothing other than that of IX 7.1 in the special case where A is a complete noetherian local ring, endowed with an ideal of definition for its usual topology (N.B. It is precisely here that the hypothesis G affine enters essentially). In the present case, we are reduced to proving the

Lemma 4.4. *Let A be a complete noetherian local ring, with maximal ideal $\mathfrak{m}$, G an affine group scheme over $S = \text{Spec}(A)$. For every integer $n \geq 0$, let $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, $G_n = G \times_S S_n$. For every n , let H_n be a subgroup of multiplicative type and of finite type of G_n , such that for $m \geq n$, H_n is deduced from H_m by reduction. Under these conditions, there exists a unique subgroup H of multiplicative type in G that reduces to the H_n .*

By X 3.2 there exists a group H of multiplicative type over S , necessarily of finite type and isotrivial, determined up to unique isomorphism, endowed with an isomorphism $H \times_S S_0 \simeq H_0$ (using the fact that H_0 is isotrivial, being of finite type over a field, cf. X 1.4). By X 2.1, for every n , the isomorphism $H \times_S S_0 \simeq H_0$ lifts to a unique isomorphism $H \times_S S_n \simeq H_n$. This being said, by IX 7.1 already cited, the homomorphisms $H_n \longrightarrow G_n$ come from a unique homomorphism of S -groups $u : H \longrightarrow G$.

By IX 6.6 the latter is a monomorphism since $u_0 : H_0 \longrightarrow G_0$ is. This completes the proof of 4.4.

b) The morphism $u : F \longrightarrow T$ is a monomorphism. This follows from the density theorem IX 4.7, in the form of corollary 4.8 b). One should take care that it is essential, for the application we make of it here, to have this result over a base that is not necessarily noetherian. (N.B. Since the functor T does not commute with inductive limits of rings, it is not possible to reduce to that case a priori).

b') The morphism $u : F \longrightarrow T$ is infinitesimally étale, in other words one has the

Lemma 4.5. *Let A be an artinian local ring with residue field k , $S = \operatorname{Spec}(A)$, I an ideal $\subset \operatorname{rad}(A)$, $S' = \operatorname{Spec}(A/I)$, G an S -group prescheme smooth over S , $G' = G \times_S S'$; give ourselves a subgroup H' of multiplicative type of G' and, for every integer $n > 0$, a subgroup $H(n)$ of multiplicative type of G , in such a way that*

- 1°) *for every multiple m of n , $H(n) = {}_n H(m)$, and*
- 2°) *$H(n)' = {}_n H'$.*

Under these conditions, there exists one and only one subgroup H of multiplicative type of G such that ${}_n H = H(n)$ for every n .

Uniqueness is already contained in b). For existence, an immediate induction reduces us to the case where $J\mathfrak{m} = 0$, $\mathfrak{m}$ being the maximal ideal of A . Let $k = A/\mathfrak{m}$, $S_0 = \operatorname{Spec}(k)$, $G_0 = G \times_S S_0$, $\mathfrak{g}_0$ the Lie algebra of G_0 , $\mathfrak{h}_0$ that of H_0 . One has a canonical group isomorphism:

$$\mathfrak{g}_0 \otimes_k J \simeq \operatorname{Ker}(G(S) \longrightarrow G(S_0)),$$

cf. Exp. III. By IX 3.6 bis and 3.7, there exists a subgroup H of multiplicative type of G reducing to H' , and such an H is determined modulo inner automorphism by an element of $N = \operatorname{Ker}(G(S) \longrightarrow G(S_0))$. Thus the set P of liftings H of H_0 is a principal homogeneous set under the group N/M , where M is the group of $g \in N$ such that $\operatorname{int}(g) \cdot H = H$.

Now one sees easily (Exp. III) that this subgroup is nothing other than the vector subspace of $\mathfrak{g}_0 \otimes_k J$ formed by the invariants under H_0 , when H_0 acts on $\mathfrak{g}_0 \otimes_k J$ by the representation induced by the adjoint representation of G_0 . Likewise, the set $P(n)$ of liftings of $H(n)_0 = {}_n H_0$ is a principal homogeneous set under $N/M(n)$, where $M(n)$ is the vector subspace of $\mathfrak{g}_0 \otimes_k J = N$ formed by the invariants under ${}_n H_0$. Using the density theorem Exp. IX 4.7, one sees easily that for large n (in the sense of the order relation put on the set of integers $n > 0$, namely the divisibility relation) one has $M = M(n)$. Consequently the natural map $H \mapsto {}_n H$ from P to $P(n)$, which is compatible with the actions of N and hence with the homomorphism $N/M \longrightarrow N/M(n)$ on the groups of operators, is bijective for large n . The conclusion of 4.5 follows at once.

To verify c) and d) of 3.7, we use 3.8, which reduces us to the verification of c') and d') below.

c') The T_n are smooth over S , and the transition morphisms $p_{n,m} : T_m \longrightarrow T_n$ are smooth. This is nothing other than 2.2 bis.

d') With the notation of 3.8, a point ξ of F with values in a field k over S is nothing other than a subgroup H_0 of multiplicative type of $G_0 = G_k$. Taking up again the argument of b') above, one sees that the integer $d(\xi)$ considered in 3.8 is nothing other than the dimension of $\mathfrak{g}_0/\mathfrak{g}_0^{H_0}$, where $\mathfrak{g}_0$ is the Lie algebra of G_0 and $\mathfrak{g}_0^{H_0}$ is the vector subspace of invariants under H_0 . It is therefore a finite integer, i.e. condition d') 1°) of 3.8 holds. With the notation of d') 2°, giving a morphism $v : X \longrightarrow F$ amounts to giving a subgroup H of multiplicative type of G_X . For $x \in X$, the integer $d(\xi_x)$ is then nothing other than the dimension of $(\mathfrak{g} \otimes \kappa(x))^{H_x}$, where $\mathfrak{g}$ is the sheaf of Lie algebras of G_X (which is locally free of finite type on X , for G_X is smooth over X , and $\mathfrak{g} \otimes \kappa(x)$ is nothing other than the Lie algebra of the fiber G_x of G_X at x), and H_x is the fiber of H at x . Now, since $\mathfrak{g}$ is a locally free module over S on which the group H of multiplicative type acts, one sees at once that the subfunctor $\mathfrak{g}^H$ of invariants under H is given by a subsheaf locally a direct summand, hence locally free, of $\mathfrak{g}$. (By descent, one is reduced to the case where H is diagonalizable, and where one applies Exp. I 4.7.3, noting that the subsheaf of invariants corresponds to the component of degree zero). Consequently $d(\xi_x) = \operatorname{rank}$ at x of $\mathfrak{g}^H$, hence it is a locally constant function of x . This completes the proof of condition d').

We have thus verified the conditions of 3.7, which completes the proof of 4.1.

Remark 4.6. When $G = \operatorname{GL}(n)_S$, one can give a direct proof of 4.1 that is distinctly simpler and more explicit, using I 4.7.3. The proof shows in addition that, in this case, the moduli scheme is a sum scheme of a family of affine schemes over S . Proceeding as was said in 4.3,

one deduces the same result whenever G is a closed subgroup of a group of the form $\mathrm{GL}(n)_S$. One should beware, however, of believing that the preschemes representing the functors in 4.1 or 4.2 are always sums of a family of affine schemes over S . Let, for example, H be a group of multiplicative type and of finite type over a locally noetherian prescheme S ; then by X 5.11, H is isotrivial if and only if $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, \mathbf{G}_m)$ is a sum of a family of affine preschemes over S , but it was pointed out (X 1.6) that non-isotrivial tori H (of relative dimension 2) can exist; for such a torus, $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, \mathbf{G}_m)$ is therefore not a sum of affine S -preschemes over S , and one sees likewise that the “dual” twisted constant group $\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbf{G}_m, H)$ is not such a sum either (for if two commutative twisted constant groups of finite presentation R, R' are dual to each other, $R' = \underline{\mathrm{Hom}}_{S\text{-gr}}(R, \mathbf{Z}_S)$, one sees easily that one is isotrivial if and only if the other is). This last point also shows that such an H is not isomorphic to a subgroup of a group of the form $\mathrm{GL}(n)_S$; more precisely, one can show that a group of multiplicative type and of finite type over a connected locally noetherian S is isomorphic to a subgroup of a group of the form $\mathrm{Aut}_{(\mathrm{Mod})}(\mathcal{E})$ (with $\mathcal{E}$ a locally free module of finite type over S) if and only if it is isotrivial. Finally, taking $G = H \times \mathbf{G}_m$ in the two preceding examples, one finds an example where the prescheme representing the functor F of 4.1 is not a sum of a family of affine S -preschemes over S (with G a torus of relative dimension 3, if desired).

5. First corollaries of the representability theorem

Let H, G be as in 4.2. Put

$$M = \underline{\mathrm{Hom}}_{S\text{-gr}}(H, G),$$

which is a smooth S -prescheme, separated over S , by 4.2. Note that G acts on the functor $M = \underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$ by

$$(g, u) \longmapsto \mathrm{int}(g) \circ u,$$

whence a canonical morphism

$$(x) \quad G \times_S M \longrightarrow M \times_S M,$$

whose components are the preceding morphism $G \times_S M \longrightarrow M$, and the second projection $G \times_S M \longrightarrow M$.

Corollary 5.1. *The preceding morphism (x) is smooth.*

This follows from 4.2 and 2.3. This statement is equivalent to the following:

Corollary 5.2. *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S -groups. Then the subfunctor $\underline{\mathrm{Transp}}(u_1, u_2)$ of G (cf. 2.4) is representable by a closed sub-prescheme of G , smooth over S .*

It remains only to prove that $\underline{\mathrm{Transp}}(u_1, u_2) \longrightarrow G$ is indeed a closed immersion, which follows from the fact that it is the kernel of a pair of morphisms $G \rightrightarrows M$ (namely $g \longmapsto \mathrm{int}(g)u_1$ and the “constant” morphism $g \longmapsto u_2$), and from the fact that M is separated over S . In particular:

Corollary 5.3. *Let $u : H \longrightarrow G$ be a morphism of S -groups. Then $\underline{\mathrm{Centr}}_G(u) = \underline{\mathrm{Transp}}(u, u)$ is representable by a closed group sub-prescheme of G , smooth over S . In addition,*

$$G/\underline{\mathrm{Centr}}_G(u)$$

is representable by an open sub-prescheme of M .

It remains to prove this last point. Now the morphism

$$g \longmapsto \mathrm{int}(g) \circ u$$

from G to M is smooth of finite type by 5.2, hence is an open morphism (EGA IV 6.6), and if U denotes its image, with the structure induced by M , the induced morphism $G \longrightarrow U$ is smooth, surjective, of finite type, hence covering for the faithfully flat quasi-compact topology. Moreover, it is obvious that the preceding morphism $G \longrightarrow M$ makes G a formally principal

homogeneous sheaf under $\underline{\text{Centr}}_G(u)_M$, which implies that the sheaf $G/\underline{\text{Centr}}_G(u)$ is indeed representable by U .

Corollary 5.4. *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S -groups. The following conditions are equivalent:*

- (i) u_1 and u_2 are locally conjugate for the étale topology.
- (i bis) u_1, u_2 are locally conjugate in the sense of the faithfully flat quasi-compact topology.
- (ii) For every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, the morphisms $u_{1,\bar{s}}, u_{2,\bar{s}} : H_{\bar{s}} \rightrightarrows G_{\bar{s}}$ are conjugate by an element of $G(\overline{\kappa(s)})$.
- (ii bis) The structural morphism $\underline{\text{Transp}}(u_1, u_2) \rightarrow S$ is surjective.
- (iii) $\underline{\text{Transp}}(u_1, u_2)$ is a torsor under the action of the smooth S -group prescheme of finite type $\underline{\text{Centr}}_G(u_1)$.

(i) $\Rightarrow$ (i bis) and (ii) $\Rightarrow$ (ii bis) are trivial (the second thanks to the Nullstellensatz); moreover (i bis) $\Rightarrow$ (ii) by the “finite extension principle” (EGA IV 9). On the other hand, (ii bis) $\Rightarrow$ (iii) thanks to the fact that $\underline{\text{Transp}}(u_1, u_2)$ is smooth over S , hence flat over S , and of finite type, hence quasi-compact over S , so it is faithfully flat quasi-compact over S if and only if its structural morphism is surjective. Since, on the other hand, it is formally principal homogeneous under $\underline{\text{Centr}}_G(u_1)$, which is faithfully flat quasi-compact over S , one sees that this last condition is also equivalent to saying that $\underline{\text{Transp}}(u_1, u_2)$ is a torsor under $\underline{\text{Centr}}_G(u_1)$ (understood: in the sense of the faithfully flat quasi-compact topology). Finally (iii) $\Rightarrow$ (i) thanks to the fact that $\underline{\text{Transp}}(u_1, u_2)$ is smooth over S and to “Hensel’s lemma” in the form 1.10.

Remark 5.5. For fixed $u_1 = u : H \rightarrow G$, the functor $(\text{Sch})/S \rightarrow (\text{Ens})$ that associates to every T over S the set of homomorphisms of T -groups $u_2 : H_T \rightarrow G_T$ that are conjugate to $u_T : H_T \rightarrow G_T$ locally for the étale topology is precisely representable by the open subset of M , isomorphic to $G/\underline{\text{Centr}}_G(u)$, considered in 5.3.

Let us sketch the variants of the preceding results obtained by applying 4.1 instead of 4.2. Thus let G be a group prescheme smooth and affine over S , and now denote by M the smooth S -prescheme, separated over S , representing the functor considered in 4.1. One again has actions of G on M :

$$(g, H) \mapsto \text{int}(g)(H),$$

whence, as above, a morphism

$$(x \text{ bis}) \quad G \times_S M \rightarrow M \times_S M.$$

Corollary 5.1 bis. *The preceding morphism is smooth.*

Follows from 4.1 and 2.3 bis. One again concludes:

Corollary 5.2 bis. *Let H_1, H_2 be two subgroups of multiplicative type of G . Then the subfunctor $\underline{\text{Transp}}_G(H_1, H_2)$ of G is representable by a closed sub-prescheme of G , smooth over S .*

In particular:

Corollary 5.3 bis. *Let H be a subgroup of multiplicative type of G . Then the group subfunctor $\underline{\text{Norm}}_G(H)$ of G is representable by a closed sub-prescheme of G , smooth over S . In addition, the quotient $G/\underline{\text{Norm}}_G(H)$ is representable by an open sub-prescheme of M .*

Corollary 5.4 bis. *Let H_1, H_2 be two subgroups of multiplicative type of G . The following conditions are equivalent:*

- (i) H_1 and H_2 are locally conjugate for the étale topology.
- (i bis) H_1 and H_2 are locally conjugate for the faithfully flat quasi-compact topology.

- (ii) For every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, the subgroups $H_{1,\bar{s}}, H_{2,\bar{s}}$ of $G_{\bar{s}}$ are conjugate by an element of $G(\overline{\kappa(s)})$.
- (ii bis) The structural morphism $\text{Transp}(H_1, H_2) \rightarrow S$ is surjective.
- (iii) $\text{Transp}(H_1, H_2)$ is a principal homogeneous bundle under the action of the smooth S -group prescheme of finite type $\text{Norm}_G(H_1)$.

Remark 5.5 bis. Remark 5.5 also carries over to the present case.

Remark 5.6. Note that to establish result 5.2, and consequently also the first assertion in 5.3, as well as 5.4, the reference to 4.2 can be replaced by a reference to VIII 6.4, whose proof is much easier. This also shows that the hypothesis G affine over S is unnecessary there. In addition, in no. 6 we shall show how a variant of this method allows these results to be extended to the case of certain groups H more general than groups of multiplicative type. These same observations extend to variants 5.2 bis etc. On the other hand, result 5.8 below uses essentially all the hypotheses made (in particular G affine and smooth over S , H of multiplicative type), and the lecturer knows no other proof of it than via the representability theorems 4.1 or 4.2*.

5.7. — Since the morphism (x), resp. (x bis), is smooth, hence open, its image is open. Let R be this image, endowed with the structure induced by $M \times_S M$; one observes easily that R is an *equivalence relation* in M , which is moreover nothing other than that described explicitly in 5.4, resp. 5.4 bis. It would be interesting to know whether the quotient sheaf M/R (which is formally étale over S) is representable (it is then representable by an étale prescheme over S); it is so, for example, if S is the spectrum of a field. One will note, moreover, that it can happen that R is not closed in $M \times_S M$ (even if G is quasi-finite over $S \dots$), which means (when M/R is representable) that M/R is then not separated over S .

Theorem 5.8. Let S be a prescheme, G an S -prescheme smooth and affine over S , $s \in S$. Then:

- a) For every subgroup H_0 of multiplicative type of G_s , there exists an étale morphism $S' \rightarrow S$, a point s' of S' above s such that the residue extension $\kappa(s')/\kappa(s)$ is trivial, and a subgroup H' of multiplicative type of $G' = G \times_S S'$ such that $H'_{s'} = H_0 \otimes_{\kappa(s)} \kappa(s')$.
- b) For every group homomorphism $u_0 : H_s \rightarrow G_s$, where H is an S -group prescheme of multiplicative type and of finite type, there exists an étale morphism $S' \rightarrow S$, a point s' of S' above s such that the residue extension $\kappa(s')/\kappa(s)$ is trivial, and a group homomorphism $u' : H' \rightarrow G'$ such that $u'_{s'} : H'_{s'} \rightarrow G'_{s'}$ is equal to $u_0 \otimes_{\kappa(s)} \kappa(s')$.

This follows from 4.1, resp. 4.2, and Hensel's lemma in the form 1.10.

Proposition 5.9. Let S be a prescheme, G an S -prescheme smooth and affine over S , H a subgroup of multiplicative type of G . Consider $\text{Centr}_G(H) \hookrightarrow \text{Norm}_G(H)$, which by 5.3 and 5.3 bis are closed group sub-preschemes of G , smooth over S . Then the first group is an open and closed sub-prescheme of the second, and the quotient sheaf

$$W_G(H) = \text{Norm}_G(H) / \text{Centr}_G(H)$$

is representable by an open group sub-prescheme of $\text{Aut}_{S\text{-gr}}(H)$; hence it is an S -group prescheme quasi-finite, étale and separated over S .

Indeed, consider the obvious homomorphism

$$\theta : \text{Norm}_G(H) \rightarrow \text{Aut}_{S\text{-gr}}(H),$$

whose kernel is by definition $\text{Centr}_G(H)$. Since $\text{Aut}_{S\text{-gr}}(H)$ is representable by an S -group prescheme étale and separated over S (X 5.10), its identity section is an open and closed

*The situation has changed since the writing of this text, cf. XV and XIX no. 6.

immersion, so its inverse image under θ is an open and closed subgroup of $\underline{\text{Norm}}_G(H)$. I say, in addition, that θ is a *smooth morphism*: this follows indeed formally from the definitions, and from the fact that $\underline{\text{Norm}}_G(H)$ is smooth over S and $\underline{\text{Aut}}_{S\text{-gr}}(H)$ is étale over S . One concludes as in 5.3 that the image of θ is an open subset U of $\underline{\text{Aut}}_{S\text{-gr}}(H)$ and that, endowed with the induced structure, U represents the quotient sheaf $\underline{\text{Norm}}_G(H)/\underline{\text{Centr}}_G(H)$. The latter is therefore étale and separated over S , since $\underline{\text{Aut}}_{S\text{-gr}}(H)$ is, and it is quasi-finite over S , being quasi-compact over S as the image of $\underline{\text{Norm}}_G(H)$, which is so. This completes the proof of 5.9.

Corollary 5.10. *For every $s \in S$, let*

$$w(s) = \text{rank } \underline{\text{Norm}}_{G_s}(H_s)/\underline{\text{Centr}}_{G_s}(H_s)$$

(which is also the index of $\underline{\text{Centr}}_{G(k)} H(k)$ in $\underline{\text{Norm}}_{G(k)} H(k)$, where k is an algebraic closure of $\kappa(s)$). Then the function $s \mapsto w(s)$ is lower semicontinuous. For it to be constant near the point s , it is necessary and sufficient that $W_G(H)$ be finite over S near s .

Indeed, for every S -prescheme W that is étale, of finite type and separated over S , it is true that the function $s \mapsto w(s) = \text{rank } [W_s : \kappa(s)]$ is lower semicontinuous, and that it is constant near the point s if and only if W is finite over S near s (a fact pointed out in SGA 1, I 10.9, whose proof, which offers no difficulty, will be found in EGA IV^{*}).

Remark 5.11. ⁽⁵⁾Let G be a group prescheme affine and smooth over S , H a subgroup of multiplicative type; then 5.3 and 5.3 bis imply that the quotients

$$G/\underline{\text{Centr}}_G(H) \quad \text{and} \quad G/\underline{\text{Norm}}_G(H)$$

are smooth preschemes over S and *quasi-affine* over S , since in either case the moduli prescheme M into which the quotient is embedded is such that every open subset U of M quasi-compact over S is quasi-affine over S (3.11). We shall moreover see in the following exposé⁽⁶⁾ that for every U as above, the schematic closure $\overline{U}$ of U in M is even affine over S , which shows that the quotients considered are affine over S provided the open subset U in the corresponding moduli scheme (of group homomorphisms from H to G , resp. of subgroups of multiplicative type of G) is closed in M . This is, for example, the case if H is a “maximal torus” in the second case considered, or when S is the spectrum of a field, cf. XII.5.4⁽⁷⁾. I do not know whether the quotients considered are affine over S in general.

6. On a rigidity property for homomorphisms of certain group schemes, and the representability of certain transporters^{*}

Proposition 6.1. *Let S be a locally noetherian prescheme, H a commutative S -group prescheme of finite type over S , E a set of integers > 0 stable under multiplication, and suppose the following conditions hold:*

- a) *For every $n \in E$, the subgroup ${}_n H = \text{Ker}(n \cdot \text{id}_H)$ is finite and flat over S .*
- b) *For every $s \in S$, the family of ${}_n H_s$ ($n \in E$) is schematically dense in H .*

^{*}EGA IV 15.5.1 and 18.10.7.

⁽⁵⁾**N.D.E.** For a generalization to the non-affine case, see M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes Math. 119 (1970), IX.2.8.

⁽⁶⁾**N.D.E.** Cf. remark XII.5.7. Let us mention here the following generalization. Without any affineness hypothesis on G , if H is a torus and S is locally noetherian, U is affine and smooth by a theorem of Raynaud (*loc. cit.*, IX.2.6 and IX.2.8). Indeed, $G/\underline{\text{Norm}}_G(H)$ is an open subset of M and the largest representable open subset of M is a disjoint sum of smooth affine open subsets over S .

⁽⁷⁾**N.D.E.** Under the hypothesis that G has locally constant reductive rank.

^{*}The present no. does not use the results of Nos. 3, 4, 5; its natural place would be in VI_B.

Let Y be a closed sub-prescheme of H . Then the subfunctor $\prod_{H/S} Y/H$ of S (compare VIII 6) is representable by a closed sub-prescheme T of S , and if S is noetherian, there exists an $n \in E$ such that

$$(x) \quad \prod_{H/S} Y/H = \prod_{nH/S} Y \cap_n H/S.$$

Indeed, by VIII 6.4, since by condition a) $_nH$ is finite and flat and a fortiori “essentially free” over S , it follows that the second member of (x) is representable by a closed sub-prescheme T_n of T . Of course, for $n \in E$, E ordered by divisibility, the T_n form a decreasing family of closed sub-preschemes of S , so if S is noetherian (which we can suppose), it is stationary for large n .

Let T be the value of T_n for large n ; I say that T does indeed represent the first member of (x), which will complete the proof. One is reduced to proving that if S' is a prescheme over S such that $(Y \cap_n H)_{S'} = (_nH)_{S'}$ for every $n \in E$, i.e. such that $Y_{S'} \supset _nH_{S'}$ for every $n \in E$, then $Y_{S'} = H_{S'}$. Now this is indeed the case, for by IX 4.4 the family of $_nH_{S'}$ is schematically dense in $H_{S'}$, taking account of conditions a) and b).

Theorem 6.2. *Let S be a prescheme, H, E as in 6.1 satisfying conditions a), b), $u : H \rightarrow G$ a homomorphism of S -groups from H to an S -group prescheme locally of finite type over S , and finally K a closed group sub-prescheme of H . Consider the subfunctor $\text{Transp}_G(u, K)$ of G (cf. 2.5). Then the latter is representable by a closed sub-prescheme of G , and if G is noetherian (for example S noetherian and G of finite type over S), there exists an integer $n \in E$ such that $\text{Transp}_G(u, K) = \text{Transp}_G(u|_{_nH}, K)$. If finally G is smooth over S , and K smooth over S or of multiplicative type, and if H, E satisfy the following condition, stronger than a):*

a') For every $n \in E$, the subgroup $_nH$ of H is of multiplicative type, then $\text{Transp}_G(u, K)$ is smooth over S .

Indeed, consider the G -group $H_G = H \times_S G$, which obviously satisfies the conditions of 6.1 (S replaced by G , and H by H_G), and the closed sub-prescheme Y of H_G , inverse image of $K \subset G$ under

$$H_G = H \times_S G \rightarrow G$$

defined by $(h, g) \mapsto \text{int}(g) \cdot u(h)$. Then $\text{Transp}_G(u, K)$ is nothing other than $\prod_{H_G/G} Y/H_G$ (compare VIII, examples 6.5 e)). Thus the first assertions follow from 6.1, and moreover one sees that for every quasi-compact open subset U of G , there exists $n \in E$ such that $\text{Transp}_G(u, K)$ and $\text{Transp}_G(u|_{_nH}, K)$ have the same trace on U . To verify the last assertion of 6.2, one can therefore replace H by an $_nH$, and then it suffices to apply 2.5, which applies since $_nH$ is supposed of multiplicative type over S .

Remark 6.3. The preceding proof uses only the very elementary result VIII 6.4, and in addition (for the last part) 2.5, that is, when K is smooth over S , the infinitesimal result IX 3.6, hence the vanishing of the cohomology of groups of multiplicative type. One will note that in the most important cases (cf. 6.7) one can even suppose the $n \in E$ prime to the residue characteristics of S , i.e. invertible in $\mathcal{O}_S$, hence the $_nH$ finite étale over S , and then the cohomological result invoked is practically trivial, so 6.2 is then independent of the theory of groups of multiplicative type.

6.4. — One sees as usual that 6.1 extends to the case where one has an S -prescheme S' over S (not necessarily locally noetherian), and a closed sub-prescheme Y' of $H' = H_{S'}$, provided $Y' \rightarrow H'$ is of finite presentation, i.e. the ideal defining Y' of finite type: then $\prod_{H'/S'} Y'/H'$ is representable by a closed sub-prescheme T' of S' , such that $T' \rightarrow S'$ is of finite presentation, and if S' is quasi-compact, there exists an $n \in E$ such that the relation analogous to (x) holds. It also follows that the first statement in 6.2 holds without supposing G locally of finite type over S , provided the immersion $K \rightarrow G$ is of finite presentation.

6.5. — As announced in 5.6, theorem 6.2 allows one to extend to group preschemes satisfying a') and b) above certain results established by another method and under more restrictive conditions for groups of multiplicative type. This is so for results 5.2, the beginning of 5.3, 5.4 and the bis variants of the preceding results 5.9 and 5.10. It is likewise so for the results of IX, Nos. 5 and 6, excluding IX 6.8 (already false for a homomorphism of abelian schemes $u : H \rightarrow G$ over the spectrum S of a discrete valuation ring with residue characteristic $p > 0$: it can indeed happen that $\text{Ker } u$ has as generic fiber the identity group, and as special fiber a radicial group not reduced to the identity group).

6.6. — We have just given an example of a rigidity property for groups of multiplicative type that is not shared by abelian schemes. Another example, extremely important, is the fact that the existence theorem for infinitesimal extensions of homomorphisms IX 3.6 no longer holds when H is an abelian scheme. Thus, an abelian scheme $\neq 0$ over a field admits nontrivial infinitesimal variations, unlike what happens for a group of multiplicative type—which is the infinitesimal aspect of the fact that there exists a “moduli theory” (moreover far from complete) for abelian varieties, whereas moduli theory for groups of multiplicative type is empty. Another “global” aspect of this infinitesimal difference is that if H is an abelian scheme over a locally noetherian S , and G a commutative S -group prescheme locally of finite type over S , then one can show that $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable by a prescheme locally of finite type over S , but, unlike what happens for H of multiplicative type, this prescheme is *not étale* over S , but only unramified over S . Thus, if S is, for example, the spectrum of a complete discrete valuation ring, H and G abelian schemes over S , there can exist homomorphisms $H_0 \rightarrow G_0$ on the special fibers that do not come “by specialization” from a homomorphism on the generic fibers.

6.7. — Theorem 6.2 applies whenever H is an abelian scheme over S , or more generally an extension of such a scheme by a torus. Indeed, since the question is local on S , one can suppose that there exists a prime number ℓ prime to the residue characteristics of S , and one sees that it then suffices to take for E the set of powers of ℓ to satisfy conditions a') and b).

An argument analogous to that of 6.1 gives us the

Theorem 6.8. *Let X be a prescheme smooth over S , with nonempty connected geometric fibers. Then for every closed sub-prescheme Y of X , the functor $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme T of S . If Y is of finite presentation over X , then T is of finite presentation over S .*

Since $f : X \rightarrow S$ is faithfully flat locally of finite presentation, it is covering for the fpqc topology. Since, on the other hand, $T = \prod_{X/S} Y/X$ is obviously a subsheaf of S (for the fpqc topology), it follows that the question of representability of T by a closed sub-prescheme of S is of a local nature on S for the fpqc topology, and the same is true of the question of deciding whether T is of finite presentation over S . On making then the base change $S' \rightarrow S$, with $S' = X$, one is reduced to the case where X admits a section e over S . One can in addition suppose S affine and a fortiori quasi-compact. One then has:

Corollary 6.9. *Under the conditions of 6.8, suppose that S is quasi-compact and that X admits a section e over S . For every integer $n \geq 0$, let X_n be the sub-prescheme of X that is the infinitesimal neighborhood of order n of the section e . Suppose Y of finite presentation over X . Then there exists an integer $n \geq 0$ such that one has $\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n$ (where $Y_n = Y \cap X_n$).*

This corollary does indeed imply 6.8 when Y is of finite presentation over X , thanks to VIII 6.4, for, since X is smooth over S , X_n is finite and locally free over S and a fortiori is “essentially free” over S in the sense of VIII 6.1, hence $\prod_{X_n/S} Y_n/X_n$ is representable by a closed sub-prescheme of S . In addition, the proof in *loc. cit.*, or the reduction to the

noetherian case, shows us immediately that the said closed sub-prescheme of S is of finite presentation over S .

Let us first prove 6.9, hence 6.8, when S is noetherian. Let $T_n = \prod_{X_n/S} Y_n/X_n$; then the T_n form a decreasing sequence of closed sub-preschemes of S , and since S is noetherian, this sequence is stationary. Let $R = \bigcap_{n \geq 0} \prod_{X_n/S} Y_n/X_n$ be their common value for large n ; one obviously has $T \subset R$, and it suffices to establish that one has $R \subset T$. On making the base change $R \rightarrow S$, one is reduced to the case where $R = S$, i.e. $Y_n = X_n$ for every n , i.e. $Y \supset X_n$ for every n , and to proving then $T = S$, i.e. $Y = X$. Now $Y \supset X_n$ for every n implies (thanks to the fact that X is locally noetherian) that Y is, near every point of $e(S)$, an induced open sub-prescheme of X , so there exists an induced open subset U of X , containing $e(S)$, such that $U \subset Y$. By IX 4.3, since the fibers of X/S are integral, U is schematically dense in X , hence (Y being a closed sub-prescheme containing U) one has $Y = X$. This proves 6.9, hence 6.8, in this case.

The general case proceeds by reduction to the preceding case. For every $s \in S$, there exists an affine open neighborhood U of s and an affine open neighborhood V of $e(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U , and if S' is an affine open neighborhood of s contained in $e^{-1}(V) \cap f(V)$, and $X' = V \cap f^{-1}(S')$, then X' and S' are affine open subsets of X , resp. S , and X'/S' admits a section. Because of the local nature of 6.8 and 6.9, one can suppose $S = S'$. I say that one then has $\prod_{X/S} Y/X = \prod_{X'/S} Y'/X'$, where $Y' = Y \cap X'$; indeed, by IX 4.6, X' is schematically dense in X (at least when X is quasi-separated over S , hence X' is retrocompact in X ; but in fact one can show without difficulty that IX 4.6 remains valid without the retrocompactness hypothesis), and likewise for every base change $S_1 \rightarrow S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$, whence at once the asserted equality. This reduces us to the case where $X = X'$, so one can suppose S and X affine.

In addition, if $X = \text{Spec}(B)$ and if J is the ideal of B defining Y , J is the inductive limit of its subideals of finite type, so Y is an intersection of closed sub-preschemes Y_i of X that are of finite presentation over S , and consequently $\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X$, which reduces us, to prove 6.8, to the case where Y is of finite presentation over X . It then suffices to prove 6.9 with S and X affine. But then X and Y over S come by the base change $S \rightarrow S_0$ from an analogous situation X_0 and Y_0 over S_0 , with S_0 noetherian, which reduces us to the case where S is noetherian, which has already been treated. This completes the proof of 6.8 and 6.9.

Corollary 6.10. *Let X be a smooth S -group prescheme of finite presentation, with connected fibers, Y a group prescheme of finite presentation over S , $i : Y \rightarrow X$ a monomorphism of S -group preschemes, thus making Y a subgroup of X . Then $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme of finite presentation of S . If S is quasi-compact, denoting for every integer $n \geq 0$ by X_n the infinitesimal neighborhood of order n of the identity section of X , and putting $Y_n = X_n \cap Y$, one has for large enough n : $\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n$.*

The proof is essentially that of 6.9. Note that the identity sections of X and Y induce bijective immersions $S \rightarrow X_n$ and $S \rightarrow Y_n$, hence induce isomorphisms of S_{red} with $(X_n)_{\text{red}}$ and $(Y_n)_{\text{red}}$, which implies that the injection $Y_n \rightarrow X_n$ is proper, hence, being a monomorphism of finite presentation, is a closed immersion. Consequently, by VIII 6.4 already used, $\prod_{X_n/S} Y_n/X_n$ is representable by a closed sub-prescheme of S of finite presentation over S , and it therefore remains to prove the last assertion of 6.10 in the case where one supposes in addition S affine. One is again immediately reduced to the case where S is noetherian, and one is reduced to proving that one then has $R = T$ (with the notation of the proof of 6.9), or again that $Y \supset X_n$ for every n implies $Y = X$. Now the hypothesis implies that $i : Y \rightarrow X$ is étale at the points of the identity section of Y over S , so Y is smooth over S at the points of the identity section, whence it follows that the open subset Y' of points of Y at which Y is smooth over S is an induced open subgroup of Y . Then $Y' \rightarrow X$ is

an étale monomorphism by X 3.5, hence an open immersion; but since the fibers of X are connected and every open subgroup of an algebraic group is also closed, it follows that this is a surjective open immersion, i.e. an isomorphism. Thus $Y' = X$ and a fortiori $Y = X$, which completes the proof of 6.10.

Proceeding as in VIII 6.5, one concludes from 6.10:

Corollary 6.11. *Let G be an S -group prescheme locally of finite type and quasi-separated over S , H a group prescheme smooth of finite presentation over S with connected fibers, $i : H \rightarrow G$ a monomorphism of S -groups, thus making H a subgroup of G . Then:*

- a) $\underline{\text{Centr}}_G(H)$ and $\underline{\text{Norm}}_G(H)$ are representable by closed sub-preschemes of G of finite presentation over G , and likewise, for every monomorphism $j : K \rightarrow G$ of finite presentation of S -group preschemes, $\underline{\text{Transp}}_G(H, K)$ is representable by a closed sub-prescheme of G of finite presentation over G .
- b) If G is quasi-compact, in the various cases considered in a), there exists an integer $n \geq 0$ such that (if H_n denotes the infinitesimal neighborhood of order n of the identity section of H) one has

$$\begin{aligned}\underline{\text{Centr}}_G(H) &= \underline{\text{Centr}}_G(H_n), \\ \underline{\text{Norm}}_G(H) &= \underline{\text{Norm}}_G(H_n), \\ \underline{\text{Transp}}_G(H, K) &= \underline{\text{Transp}}_G(H_n, K) = \underline{\text{Transp}}_G(H_n, K_n).\end{aligned}$$

One applies 6.10 to the group prescheme $X = H_G = H \times_S G$ over the base prescheme G , and to the group sub-prescheme Y that is the inverse image of the diagonal subgroup of $(G \times_S G)_G$ over G under a suitable homomorphism of G -groups from X to $(G \times_S G)_G$ (in the case of $\underline{\text{Centr}}$), resp. the inverse image of K_G under a suitable homomorphism of G -groups from X to G_G (in the case of $\underline{\text{Transp}}$). The case of $\underline{\text{Norm}}$ reduces to the transporter by taking $K = H$, the hypothesis on G ensuring that $H \rightarrow G$ is of finite presentation (hence $Y \rightarrow X$ is of finite presentation); in the case of $\underline{\text{Centr}}$, the hypothesis made on G ensures that the diagonal group of $G \times_S G$ is of finite presentation over $G \times_S G$, whence again the fact that Y is of finite presentation over X .

Remark 6.12. One can prove (using rather delicate results of EGA VI^{*}) that if X is a flat prescheme of finite presentation over S , which is proper over S or has nonempty connected geometric fibers, then for every closed sub-prescheme Y of X of finite presentation over S , $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme of S of finite presentation over S . Likewise, if X is a flat S -group prescheme of finite presentation with connected fibers, and $i : Y \rightarrow X$ a monomorphism of S -groups, with Y an S -group prescheme of finite presentation, then $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme of S of finite presentation over S . In particular, 6.11 a) remains valid on replacing the hypothesis “ H smooth over S ” by “ H flat over S ”.

^{*}Cf. also J. P. Murre, *Representation of unramified functors. Applications*, Sémin. Bourbaki no. 294 (May 1965), th. 3 (p. 13).

Exposé XII. Maximal tori, the Weyl group, Cartan subgroups, the reductive center of smooth affine group schemes

by A. Grothendieck

Starting with the present exposé, unlike the preceding exposés, we shall use the known results on the structure of smooth algebraic groups over an algebraically closed field k , and above all Borel's theory of affine algebraic groups, presented in the Chevalley Seminar 56/57: "Classification des groupes de Lie algébriques"⁽¹⁾. Following the usage in the theory of algebraic groups, we refer to this Seminar by the abbreviation BIBLE. For the coming exposés, we shall need in particular the results of BIBLE 4, 5, 6, 7 (the numeral after BIBLE refers to the number of the exposé). It seems moreover that scheme theory brings no notable simplification to Borel's theory as presented in BIBLE. This is why it did not seem useful to reproduce it in the present Seminar, our aim here being to deduce, from known results over algebraically closed fields, analogous results valid over any base prescheme. (This will no longer be so for Chevalley's structure theory of semisimple groups, which, it seems, is treated advantageously *ab ovo* over an arbitrary base prescheme).

In the oral exposés (which we have followed in Nos. 1 to 4), we had restricted ourselves to group preschemes affine over S , relying essentially on the representability theorems of Exp. XI no. 4. In the present notes (cf. Nos. 6 to 8), we show briefly how the affine hypotheses can be eliminated by a simpler method not using the most delicate results of Exposé XI. For other generalizations of the results contained in the present exposé, see also exposés XV and XVI. (It is obvious that the contents of exposés XI, XII, XV, XVI ought to be completely recast).

1. Maximal tori

1.0. — First let G be an algebraic group over an *algebraically closed* field k . One calls a *maximal torus* of G an algebraic subgroup T of G that is a torus (which means here, since k is algebraically closed, that it is isomorphic to a group of the form $\mathbf{G}_m^r$), and that is maximal for this property. Note that, since k is perfect, G_{red} is a subgroup of G , smooth over k , hence is essentially an algebraic group in the sense of BIBLE. Moreover, every reduced subgroup of G (such as a torus) is automatically a subgroup of G_{red} , so the maximal tori of G coincide with the maximal tori of G_{red} . When G is affine, hence G_{red} affine, a fundamental theorem of Borel tells us that two maximal tori of G are conjugate by an element of $G(k) = G_{\text{red}}(k)$ (BIBLE 6, th. 4 c)), in particular they have the same dimension. We shall call the common dimension of the maximal tori of G the *reductive rank* of G . Note moreover that the restriction G affine is unnecessary, as follows from a known theorem of Chevalley asserting that every smooth connected algebraic group over k is an extension of an abelian variety by an affine group; cf. no. 6. In Nos. 1 to 4, we shall most often restrict ourselves to group preschemes affine over the base.

Let G be a smooth algebraic group over k , and T a maximal torus of G ; the centralizer $C(T)$ of T in G will be called the *Cartan subgroup* of G associated to T . For us this is a group sub-prescheme defined thanks to VIII 6.7, but one will note that since G is smooth over k , so

⁰_{xy} version of 2/12/08.

⁽¹⁾**N.D.E.** This seminar has now been published, in a form revised by P. Cartier, as volume 3 of the *Oeuvres de Chevalley* (Springer, 2005).

is the Cartan subgroup C , by XI 5.3, so in this case C is the unique group sub-prescheme of G smooth over k (i.e. an algebraic subgroup of G in the sense of BIBLE) such that $C(k)$ is the subgroup of $G(k)$ centralizing $T(k)$, i.e. it is essentially the centralizer in the sense of BIBLE. By the conjugacy theorem already cited, the Cartan subgroups associated to the various maximal tori are conjugate to each other, hence have the same dimension. We shall call their common dimension the *nilpotent rank* of G ; it is equal to that of G_{red} . Let $\rho_r(G)$ and $\rho_n(G)$ be the reductive and nilpotent ranks of G ; then one has

$$\rho_r(G) \leq \rho_n(G),$$

and the difference

$$\rho_u(G) = \rho_n(G) - \rho_r(G) = \dim C/T$$

could be called, when G is affine, the *unipotent rank* of G . When G is smooth, affine and connected, then C is a *nilpotent and connected* algebraic group (BIBLE 6, th. 6 a) and c)), hence (BIBLE 6, th. 2) isomorphic to the product $C_s \times C_u$, where $C_s = T$ is the original maximal torus (which is a fortiori a maximal torus in C), and where C_u is a smooth unipotent subgroup, i.e. a successive extension of groups isomorphic to $\mathbf{G}_a$ (BIBLE 6, th. 1 cor. 1 and 7, th. 4). One therefore also has, in this case:

$$\rho_u(G) = \dim C_u.$$

Remark 1.1. In addition to the three notions of rank that we have just specified for an affine algebraic group, there are two others that are useful, namely the *semisimple rank* $\rho_s(G)$, which is by definition the reductive rank of the quotient G/R , where R is the radical of G , and finally the *infinitesimal rank* $\rho_i(G)$, which is by definition the nilpotent rank of the Lie algebra of G (which will be defined and studied in the following exposé). We shall not use them in the present exposé. Let us only point out the inequalities:

$$\rho_s \leq \rho_r \leq \rho_n \leq \rho_i.$$

Lemma 1.2. *Let G be an algebraic group over the algebraically closed field k , T an algebraic subgroup of G , k' an algebraically closed extension of k , G' and T' the groups deduced from G, T by extension of the base. For T to be a maximal torus of G , it is necessary and sufficient that T' be a maximal torus of G' .*

This is an immediate consequence of the “finite extension principle” EGA IV 9.1.1.

Definition 1.3. Let S be a prescheme, G an S -group prescheme of finite type, T a group sub-prescheme of G . One says that T is a *maximal torus* of G if

- a) T is a torus (IX 1.3) and
- b) for every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, $T_{\bar{s}}$ is a maximal torus in $G_{\bar{s}}$.

Remarks 1.4. It follows from 1.2 that when S is the spectrum of an algebraically closed field one recovers the usual definition, and that property 1.3 is stable under every base change. Note that by X 8.8, in definition 1.3 one can replace condition a) by the condition:

- a') T is of finite presentation and flat over S .

One should take care that a maximal torus in the sense of definition 1.3 is indeed maximal in the set of subtori of G (as follows at once from IX 2.9), but that the converse can be exact, the base S being, say, connected, only if G actually admits a maximal torus in the sense of 1.3, which is not the case in general (even if $S = \text{Spec}(\mathbf{Z})$ and if G is “semisimple”; see also 1.6). We shall see, however, in XIV that the converse is true when S is artinian, or when S is a local scheme and G is “reductive”: in this case, every torus of G is contained in a maximal torus.

Definition 1.5. Let G be an algebraic group over a field k . One calls the *reductive rank* (resp. *nilpotent rank*, resp. *unipotent rank*, etc.) of G the reductive rank (resp. ...) of $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k .

One will note that, by 1.2 and the commutation of the formation of $\text{Centr}_G(T)$ with extension of the base, the notions of rank introduced in 1.5 are *invariant under extension of the base field*; on the other hand, for algebraically closed k , they coincide with those introduced at the beginning of the present number.

Remark 1.6. It is not difficult to construct an affine smooth group scheme G over the spectrum S of a discrete valuation ring, whose generic fiber is isomorphic to $\mathbf{G}_m$, and whose special fiber is isomorphic to $\mathbf{G}_a$. Such a G contains no torus except the trivial torus T (reduced to the identity subgroup), which is obviously not a maximal torus. More precisely, in the special fiber $G_0 = \mathbf{G}_{a,k}$, T_0 is indeed a maximal torus, but in the generic fiber $G_1 = \mathbf{G}_{m,K}$, T_1 is no longer a maximal torus (k is the residue field, K the fraction field of the valuation ring considered). One also sees from this example that the reductive rank of G_s ($s \in S$) is not a continuous function of s . One nevertheless has the following results:

Theorem 1.7. Let S be a prescheme, G an S -group prescheme affine and smooth over S . For every $s \in S$, consider $\rho_r(s) = \rho_r(G_s)$ and $\rho_n(s) = \rho_n(G_s)$, the reductive and nilpotent ranks of G_s (1.5). With this notation, one has the following:

- a) The function ρ_r on S is lower semicontinuous, the function ρ_n on S is upper semicontinuous, hence the function $\rho_u = \rho_n - \rho_r$ is upper semicontinuous.
- b) The following conditions (stable under arbitrary base change) are equivalent:
 - (i) The function ρ_r on S is locally constant.
 - (ii) There exists locally, in the sense of the étale topology, a maximal torus in G .
 - (ii bis) There exists locally, in the sense of the faithfully flat quasi-compact topology, a maximal torus in G .
- c) Let T_1, T_2 be two maximal tori in G (which implies that one is under the conditions considered in b)). Then T_1, T_2 are locally conjugate in the sense of the étale topology, i.e. there exists an étale surjective morphism $S' \rightarrow S$ such that the subgroups $(T_1)_{S'}$ and $(T_2)_{S'}$ of $G_{S'}$ are conjugate by a section of $G_{S'}$ over S' .
- d) Under the conditions of b), i.e. when ρ_r is locally constant, so is ρ_n (hence also $\rho_u = \rho_n - \rho_r$).

PROOF. a) Note that for every morphism $S' \rightarrow S$, if $G' = G \times_S S'$, the functions ρ'_r , etc., on S' , defined in terms of G' as ρ_r , etc., in terms of G , are obtained simply by composing the latter with $S' \rightarrow S$. When $S' \rightarrow S$ is faithfully flat quasi-compact, it follows that ρ' is continuous, resp. upper semicontinuous, resp. lower semicontinuous, if and only if ρ is, the Zariski topology of S being indeed the quotient of that of S' (SGA 1 VIII 4.3). Consequently, the assertions of a) are local for the faithfully flat quasi-compact topology. Let then $s \in S$; one wants to show that the set U of $t \in S$ such that $\rho_r(t) \geq \rho_r(s)$ (resp. $\rho_n(t) \leq \rho_n(s)$) is a neighborhood of s . By the finite extension principle, there exists a finite extension k of $\kappa(s)$ such that G_k admits a maximal torus. There then exists an open neighborhood U of s , and a finite surjective locally free morphism $S' \rightarrow U$, such that the fiber S'_s is $\kappa(s)$ -isomorphic to $\text{Spec}(k)$ (cf. EGA III 10.3.2, where the noetherian hypothesis is manifestly unnecessary). Since $S' \rightarrow S$ is an open morphism (SGA 1 IV 6.6), one is reduced to the case where $S' = S$, i.e. to the case where there exists a maximal torus T_s in G_s . In addition, thanks to XI 5.8 a), on replacing S further by an S' étale over S and endowed with a point s' above s , one can suppose that T_s is the fiber at s of a subtorus T of G . Then for every $t \in S$, $\rho_r(t) = \rho_r(G_t) \geq \dim T_t = \dim T_s = \rho_r(G_s) = \rho_r(s)$, which proves that ρ_r is lower semicontinuous. On the other hand, by XI 5.3, the functor

$$C = \text{Centr}_G(T)$$

is representable by $C = \text{Centr}_G(T)^{(2)}$, a closed group sub-prescheme of G smooth over S . Thus there exists a neighborhood U of s such that $t \in U$ implies $\dim C_t = \dim C_s = \rho_n(s)$. The upper semicontinuity of ρ_n is then a consequence of the relation

$$\rho_n(t) \leq \dim C_t \quad \text{for every } t \in S,$$

which is itself contained in the following purely geometric lemma:

Lemma 1.8. *Let k be a field, G a smooth affine algebraic group over k , T a torus in G , C its centralizer; then one has*

$$\rho_n(G) \leq \dim C.$$

Indeed, one can suppose k algebraically closed, so that T is contained in a maximal torus T' . Let C' be the centralizer of the latter; then one has $C' \subset C$, hence $\dim C' \leq \dim C$. Q.E.D.

b) If ρ_r is locally constant, then for every torus T in G , and every $s \in S$, if T_s is a maximal torus in G_s , then there exists an open neighborhood U of s such that $T|_U$ is a maximal torus in $G|_U$. Using now the argument of a), one sees that (i) $\Rightarrow$ (ii bis). On the other hand (ii bis) $\Rightarrow$ (i), for when G admits a maximal torus T , then it is obvious that $\rho_r(s) = \dim T_s$ is a locally constant function of s , but we pointed out at the beginning of the proof of a) that the question of the continuity of ρ_r was local for the faithfully flat topology. Thus (i) $\Leftrightarrow$ (ii bis), and obviously (ii) $\Rightarrow$ (ii bis); it remains to prove the converse implication (i) $\Rightarrow$ (ii). For this, introduce the functor F of XI 4.1⁽³⁾, which is thus a smooth separated prescheme over S , and consider the subfunctor $\mathcal{T}$ of F , whose value at an S' over S is the set of maximal tori in $G_{S'}$. Using the preceding remark that, assuming (i), a torus in $G_{S'}$ that is maximal in the fiber of a point $s \in S'$ is maximal over an open neighborhood of s , one sees that $\mathcal{T}$ is representable by an open sub-prescheme of F , and consequently is smooth and separated over S . Since the structural morphism $\mathcal{T} \rightarrow S$ is obviously surjective, it therefore admits a section locally in the sense of the étale topology by XI 1.10, which proves (i) $\Rightarrow$ (ii).

c) This is an immediate consequence of XI 5.4 bis, taking account of Borel's conjugacy theorem recalled at the beginning of the number.

d) Assuming the remark at the beginning of the proof in a), and taking account of b), one can suppose that there exists a maximal torus T in G . If C is its centralizer, then C is representable and is smooth over S by XI 5.3, so the function $s \mapsto \rho_n(s) = \dim C_s$ is indeed locally constant.

The proof of 1.7 is complete. We shall refer to the conditions considered in 1.7 b) by saying that, in this case, G has *locally constant reductive rank*. Note: $\square$

Corollary 1.9. *Let G be as in 1.7, and let $s \in S$ be such that $\rho_u(s) = 0$, i.e. $\rho_r(s) = \rho_n(s)$ (i.e. the Cartan subgroups of $G_{\bar{\kappa}}$ are tori, where $\bar{\kappa}$ is an algebraic closure of $\kappa(s)$). Then there exists an open neighborhood U of s on which ρ_r and ρ_n are constant; in particular, for every $t \in U$, the unipotent rank $\rho_u(t)$ of G_t is zero.*

Indeed, this follows immediately from 1.7 a) and the inequality $\rho_r(t) \leq \rho_n(t)$ for every $t \in S$.

Note also that we proved, at the same time as b), the

Corollary 1.10. *Let G be as in 1.7, and suppose G of locally constant reductive rank. Consider the functor*

$$\mathcal{T} : (\text{Sch})^\circ \longrightarrow (\text{Ens}),$$

such that for every S' over S , one has

$$\mathcal{T}(S') = \text{set of maximal tori of } G_{S'}.$$

⁽²⁾**N.D.E.** Modification made to introduce the prescheme $C = \text{Centr}_G(T)$ representing the functor $\text{Centr}_G(T)$.

⁽³⁾**N.D.E.** This is the functor of subgroups of multiplicative type of G .

Then $\mathcal{T}$ is representable by a prescheme smooth, separated and of finite type over S .

It remains to verify that $\mathcal{T}$ is of finite type over S . Since the question is local for the faithfully flat quasi-compact topology, one can suppose that G admits a maximal torus T . By XI 5.3 bis and the bis version of 5.5, $\text{Norm}_G(T)$ and $G/\text{Norm}_G(T)$ are representable by preschemes $\text{Norm}_G(T)$ and $G/\text{Norm}_G(T)$, and $\mathcal{T}$ is isomorphic to $G/\text{Norm}_G(T)^{(4)}$. Since the morphism $G \rightarrow \mathcal{T}$ defined by $g \mapsto \text{int}(g)(T)$ is surjective, and G quasi-compact over S , so is $\mathcal{T}$, which completes the proof.

Remarks 1.11. a) The prescheme $\mathcal{T}$ of 1.10 will be called, as is natural, the *prescheme of maximal tori* of G . One will see in no. 5 that it is in fact *affine* over S . One can verify this directly when G is isomorphic to a closed group sub-prescheme of a prescheme of the form $\text{GL}(n)$, using XI 4.6 and observing that, by the argument of the proof of 1.7 b), $\mathcal{T}$ is identified with a sub-prescheme, both *open and closed*, of the prescheme F representing the subgroups of multiplicative type of G .

b) It is possible to give a proof of 1.7, hence of 1.9, not using the delicate results of XI, but only the easy results XI 3.12 and 6.2, working solely with groups of multiplicative type finite over the base (morally, the groups ${}_nT$ where T is a maximal torus), compare no. 7. The same remark holds for the proof of 1.10.

We finish this number by giving examples where there exists a unique maximal torus.

Proposition 1.12. *Let G be an S -group prescheme of multiplicative type and of finite type. Then G admits a unique maximal torus, and every torus in G is contained in this maximal torus.*

Uniqueness follows obviously from this last assertion, which characterizes the maximal torus as the largest subtorus of G . From uniqueness it follows that the existence question is local for the faithfully flat quasi-compact topology, which allows us to suppose G diagonalizable, i.e. of the form $D_S(M)$, M a commutative group of finite type. Let M_0 be the quotient of M by its torsion subgroup; I say that the torus $T = D_S(M_0)$ in G is a maximal torus and a largest subtorus. Indeed, a subtorus T' of G is locally diagonalizable for the fpqc topology, so to prove $T' \subset T$, one can suppose T' diagonalizable, hence of the form $D_S(N)$, where N is a free quotient of M (VIII 1.4 and 3.2 b)), hence N is a quotient of M_0 , so $T' \subset T$. Since the construction of T as $D_S(M_0)$ is compatible with every extension of the base, this shows at the same time that T is a maximal torus of G , and completes the proof. In the case where G is smooth over S , one can generalize 1.12:

Proposition 1.13. *Let G be an S -group prescheme of finite presentation over S . Suppose that G admits locally for the faithfully flat quasi-compact topology a central maximal torus. Then G admits (globally) a unique maximal torus⁽⁵⁾, and it is the largest subtorus of G .*

Here again, uniqueness is trivial from the last assertion, and makes existence a local question for fpqc, which allows us to suppose that G admits a central maximal torus T . Let us show that every torus R of G is contained in T .

This follows from the

Lemma 1.14. *Let G be an S -group prescheme of finite presentation over S , T a maximal torus of G , R a subtorus of G , and suppose that T and R commute. Then $R \subset T$.*

Indeed, since R and T commute, the morphism $R \times_S T \rightarrow G$ defined by $(r, t) \mapsto r \cdot t$ is a group homomorphism, so by IX 6.8 it admits an image subgroup T' in G , which is a group of multiplicative type quotient of $R \times_S T$, hence a torus, obviously containing T . Since T is a

⁽⁴⁾**N.D.E.** Modification made to introduce the preschemes $\text{Norm}_G(T)$ and $G/\text{Norm}_G(T)$.

⁽⁵⁾**N.D.E.** Which is central!

maximal torus, one has $T' = T$ (1.4), so $R \subset T$, which proves 1.14, hence 1.13. In particular, using 1.7 b):

Corollary 1.15. *Let G be a commutative S -group prescheme, smooth and affine over S , with locally constant reductive rank. Then G admits a unique maximal torus, and the latter contains every subtorus of G .*

Corollary 1.16. *Let G be an S -group prescheme smooth and affine over S . Suppose that for every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure k of $\kappa(s)$, the geometric fiber $G_{\bar{s}}$ is a connected and nilpotent algebraic group (in the sense of BIBLE, i.e. the group $G(k)$ of its points with values in k is nilpotent). Suppose in addition the reductive rank of G locally constant. Then G admits a unique maximal torus T ; in addition T is central and is the largest subtorus of G .*

By 1.7 b), G admits locally for fpqc a maximal torus, and by 1.13 one is reduced to proving that a maximal torus of G is central. By IX 5.6 b), one is reduced to the case where S is the spectrum of a field, which one can suppose algebraically closed. Then $T(k)$ is in the center of $G(k)$ by BIBLE 6 th. 2, which implies that T is in the center of G , for $\text{Centr}_G(T)$ is a closed subscheme of G (VIII 6.6), containing the points of $G(k)$, hence is identical to G by the Nullstellensatz (G being reduced).

Finally, for later reference, let us point out the following trivial proposition (which we have already used implicitly):

Proposition 1.17. *Let $G \supset H \supset T$ be group preschemes of finite type over S . Equivalent conditions:*

- (i) T is a maximal torus of G .
- (ii) T is a maximal torus of H , and for every $s \in S$, one has equality of the reductive ranks $\rho_r(G_s) = \rho_r(H_s)$.

2. The Weyl group

First let k be an algebraically closed field, and let G be an algebraic group over k , smooth and affine over k . If T is a maximal torus, C its centralizer and N its normalizer, then by XI 5.9 these are smooth closed subgroups of G , and C is an open subgroup of N , so $W = N/C$ is a finite étale group over k , hence determined by the group $W(k)$ of its points with values in k , as $W = W(k)_k$ (the constant group defined by the ordinary finite group $W(k)$). The finite group $W(k)$ will be called the *geometric Weyl group* (or simply Weyl group if no confusion is to be feared) of G relative to T . By Borel's conjugacy theorem, the Weyl groups relative to the various maximal tori are isomorphic to each other, which is why one sometimes speaks of "the" Weyl group of G , without specifying a maximal torus. Since the formation of C, N , and N/C commutes with every extension of the base, one sees that if k' is an algebraically closed extension of k , the geometric Weyl group of $G_{k'}$ relative to $T_{k'}$ is canonically isomorphic to that of G relative to T ; consequently "the" geometric Weyl group of G (which is properly speaking an isomorphism class of ordinary finite groups) coincides with that of G' .

This allows one, when G is a smooth affine algebraic group over an arbitrary field k , to speak of the *geometric Weyl group* of G as the isomorphism class of the geometric Weyl group of $G_{k'}$, where k' is any algebraically closed extension of k . When G admits the maximal torus T , one can obviously form as above $C = \text{Centr}_G(T)$, $N = \text{Norm}_G(T)$, $W = N/C$, which is a finite étale group over k , called the *Weyl group of G relative to T* ; the geometric Weyl group is then nothing other than the class of the group of points of W with values in any algebraically closed extension k' of k . Here, knowledge of the geometric Weyl group $W(k')$ obviously no longer suffices in general to reconstruct the algebraic group W : one must know in addition the action on $W(k')$ of the Galois group of the separable algebraic closure of k in k' .

When finally G is a group prescheme over an arbitrary base S , G smooth and affine over S , and if T is a maximal torus of G , then XI 5.9 again allows us to form the group

$$W(T) = N(T)/C(T),$$

which is an S -group prescheme étale, separated and quasi-finite over S . Its geometric fibers (relative to the algebraic closures of the residue fields $\kappa(s)$, $s \in S$) are the geometric Weyl groups of the fibers G_s . Consequently XI 5.10 gives us information on the variation of these groups with $s \in S$. We can make this information more precise and generalize it in the following way:

Theorem 2.1. *Let S be a prescheme, G an S -group prescheme smooth and affine over S . For every $s \in S$, let $w(s)$ be the geometric Weyl group of G_s , which is thus an isomorphism class of finite groups. In the set E of such classes, introduce the following preorder relation: $w \leq w'$ if and only if w and w' are represented by finite groups W and W' such that W is isomorphic to a quotient of a subgroup of W' . With this convention:*

- a) *The function $s \mapsto w(s)$ from S to E is lower semicontinuous.*
- b) *Suppose that the reductive rank of G is locally constant. Then the following conditions are equivalent:*
 - (i) *The function $s \mapsto w(s)$ is locally constant.*
 - (i bis) *The function $s \mapsto \text{card } w(s)$ is locally constant.*
 - (ii) *There exists, locally for the étale topology (or merely for the fpqc topology), a maximal torus T such that $W(T)$ is finite over its base.*
 - (ii bis) *For every S' over S , and every maximal torus T of $G_{S'}$, the associated Weyl group $W(T)$ is finite over S' .*

PROOF. a) Proceeding as in 1.7 a), to prove that for every $s \in S$ there exists an open neighborhood U of s such that $t \in U$ implies $w(t) \geq w(s)$, one is reduced to the case where there exists a torus R in G such that R_s is a maximal torus in G_s . Let

$$W(R) = \underline{\text{Norm}}_G(R)/\underline{\text{Centr}}_G(R)$$

as in XI 5.9; it is a group prescheme étale, separated, quasi-finite over S . For every $t \in S$, let $w'(t) \in E$ be its geometric fiber at t . Since R_s is a maximal torus in G_s , and the formation of Norm, Centr, Norm / Centr is compatible with every extension of the base, in particular passage to fibers, one sees that one has

$$w(s) = w'(s);$$

I say in addition that, for t near s , one will have the inequalities

$$w(t) \geq w'(t) \quad \text{and} \quad w'(t) \geq w'(s),$$

which will suffice to establish a). These two inequalities are contained in the following two lemmas:

Lemma 2.2. *Let S be a prescheme, W an S -group prescheme that is étale, separated and quasi-finite over S . For every $s \in S$, let $f(s)$ be the class of the geometric fiber of W at s , which is an element of the ordered set E of isomorphism classes of finite groups introduced in 2.1. Then the function $f : S \rightarrow E$ is lower semicontinuous. For it to be constant near $s \in S$, it is necessary and sufficient that the same hold for the function $s \mapsto \text{card } f(s)$, and for this it is necessary and sufficient that W be finite over S above an open neighborhood U of s .*

This result is a refinement, in terms of groups, of that invoked in the proof of XI 5.10; we restrict ourselves to a sketch of the proof (which is of the most standard type). One is reduced as usual to the case of affine noetherian S . One sees immediately that the function f is constructible (EGA 0_{III} 9.3.1 and 9.3.2, and the sorites of EGA IV 9), and by EGA 0_{III} 9.3.4, for semicontinuity one is reduced to proving that if t is a generalization of s , one has $f(t) \geq f(s)$. This reduces us, thanks to EGA II 7.1.7, to the case where S is the spectrum of a

discrete valuation ring, which one can suppose complete with algebraically closed residue field. But then, s denoting the closed point of S , since G is étale and separated over S , it contains a subscheme G' both open and closed, finite over S , such that $G'_s = G_s$ (EGA II 6.2.6), and one sees immediately that G' is here a subgroup of G . Moreover, since G' is finite étale over $S = \text{Spec}(V)$, V complete with algebraically closed residue field, it is a constant group, hence of the form A_S , where $A = G'(\kappa(s)) = G(\kappa(s))$ has class $f(s)$. If B is the geometric fiber of G at the generic point t of S , one therefore has a canonical monomorphism $A \rightarrow B$, which proves $f(s) \leq f(t)$. (N.B. This proof in fact proves semicontinuity for an order relation on E finer than that indicated in 2.1). The fact that f is continuous at s if and only if $s \mapsto \text{card } f(s)$ is follows from the fact that for $w, w' \in E$, the relations $w \leq w'$ and $\text{card } w = \text{card } w'$ imply $w = w'$. The fact that this condition is equivalent to the finiteness of W over a neighborhood of s is then independent of the group structure on W , and was pointed out after XI 5.10; moreover its proof is easily given by the preceding arguments, using the valuative criterion of properness EGA II 7.3.8.

Lemma 2.3. *Let G be a smooth affine algebraic group over an algebraically closed field k , $R \subset T$ two subtori, $W(R)$ and $W(T)$ the two finite groups associated as in XI 5.9, quotients of the normalizer by the centralizer. Then $W(R)$ is isomorphic to a quotient of a subgroup of $W(T)$.*

Indeed, consider the diagram

$$\begin{array}{ccccc} T & \hookrightarrow & C(T) & \hookrightarrow & C(R) \\ & & \downarrow & & \downarrow \\ & & N(T) \cap N(R) & \hookrightarrow & N(R) \\ & & \downarrow & & \\ & & N(T) & & \end{array}$$

then $(N(T) \cap N(R))/C(T)$ is a subgroup of $W(T) = N(T)/C(T)$, and one has an obvious homomorphism:

$$(N(T) \cap N(R))/C(T) \rightarrow W(R) = N(R)/C(R),$$

and everything amounts to proving that the latter is surjective, hence that for every point g of $N(R)$ with values in k , there exists a point c of $C(R)$ with values in k such that cg normalizes T , i.e. such that

$$\text{int}(c)(\text{int}(g)T) = T.$$

Now for this, it suffices to note that $\text{int}(g)T$ is a torus of $N(R)$, hence of $C(R)$ (which is an open subgroup of it). Then T and $\text{int}(g)T$ are maximal tori of $C(R)$, since they are maximal in G , and one concludes by Borel's conjugacy theorem.

This proves 2.3 and thereby 2.1 a).

b) We have already pointed out that (i) and (i bis) are trivially equivalent; they imply (ii bis) by the converse of 2.2 or XI 5.10, as one chooses; (ii bis) $\Rightarrow$ (ii) thanks to 1.7 b), and finally (ii) $\Rightarrow$ (i bis), for one sees as in 1.7 a) that condition (i) is local for fpqc, which allows us to suppose that G admits a maximal torus T such that $W(T)$ is finite over S , and one again concludes by XI 5.10. This completes the proof of 2.1. $\square$

3. Cartan subgroups

Definition 3.1. Let G be a smooth group prescheme of finite type over the prescheme S . One calls a *Cartan subgroup* of G a group sub-prescheme C of G , smooth over S , such that for every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, $C_{\bar{s}}$ is a Cartan subgroup (cf. no. 1) of $G_{\bar{s}}$.

It is immediate that if C is a Cartan subgroup of G , then for every S' over S , $C_{S'}$ is a Cartan subgroup of $G_{S'}$. If S is the spectrum of an algebraically closed field, one recovers the notion recalled in no. 1. Finally, one verifies at once that the fact that a group sub-prescheme C of G is a Cartan subgroup is *of a local nature* for the faithfully flat quasi-compact topology.

Theorem 3.2. *Let G over S be as in 3.1, and suppose that G is affine over S and of locally constant reductive rank. Then the map*

$$T \longmapsto \text{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G to the set of Cartan subgroups^{} of G . If C corresponds to T , then T is the unique maximal torus of C .*

If T is a maximal torus of G , the functor $\text{Centr}_G(T)$ is representable, by XI 5.3 or 6.2, by a closed smooth sub-prescheme of G , $C = \text{Centr}_G(T)$, and it follows from the definitions that C is a Cartan subgroup of G . In addition, T is obviously a maximal torus of C , and being central, is the unique maximal torus of C (1.13). Thus the map $T \longmapsto \text{Centr}_G(T)$ from the set of maximal tori to the set of Cartan subgroups is injective; it remains to prove that it is surjective. Thus let C be a Cartan subgroup of G ; let us prove that it is of the form $\text{Centr}_G(T)$, for T a maximal torus of G . For this it suffices to find a maximal torus T of C , for then T will be a maximal torus of G (for, for every $s \in S$, G_s and C_s have the same reductive rank). By IX 5.6 b), T is in the center of C , so $C \subset C' = \text{Centr}_G(T)$, and then C is a smooth subgroup of the smooth group C' over S' , coinciding with C' fiber by fiber, whence $C = C'$. Now, since G is of locally constant reductive rank, so is C , hence C admits a maximal torus *locally* for the étale topology by 1.7 b), and since this torus is central by the preceding argument, it follows by 1.13 that C does indeed admit a maximal torus, which completes the proof.

Corollary 3.3. *Let G be a group prescheme smooth and affine over S of locally constant reductive rank, let $\mathcal{C} : (\text{Sch})^\circ_S \longrightarrow (\text{Ens})$ be the functor defined by*

$$\mathcal{C}(S') = \text{set of Cartan subgroups of } G_{S'}.$$

Then the functor $\mathcal{C}$ is isomorphic to the functor $\mathcal{T}$ of 1.10, hence is representable by the same prescheme smooth, separated and of finite type over S .

Corollary 3.4. *Under the conditions of 3.2, if $C = \text{Centr}_G(T)$, one has*

$$\text{Norm}_G(C) = \text{Norm}_G(T).$$

Remark 3.5. When G is not of locally constant reductive rank, it is nevertheless possible that G admits Cartan subgroups (for example if G has connected nilpotent fibers, G is a Cartan subgroup of itself, but is not necessarily of locally constant reductive rank, cf. 1.6). In XV, we develop the theory of Cartan subgroups without supposing G affine over S or of locally constant reductive rank, using the theory of regular elements of the next exposé. See also Nos. 6 and 7 for the elimination of the affine hypothesis.

4. The reductive center

Definition 4.1. Let S be a prescheme, G an S -group of finite presentation over S with affine fibers, Z a group sub-prescheme of G . One says that Z is a *reductive center* of G if

- (i) Z is central, and of multiplicative type, and
- (ii) for every base change $S' \longrightarrow S$, and every central homomorphism $u : H \longrightarrow G_{S'}$, where H is a group of multiplicative type and of finite type over S , u factors through $Z_{S'}$.

^{*}The proof given here in fact proves 3.2 only for closed Cartan subgroups of G . However, 7.1 a) establishes 3.2 in the form stated, and implies that the Cartan subgroups of G are closed.

(For a variant of this notion when the fibers of G are not supposed affine, cf. 8.6).

Note that such a Z is necessarily of finite type over S (since its fibers are), hence is uniquely determined as the largest central subgroup of multiplicative type of G . It is easy to give examples where G (smooth and affine over S) admits a largest central subgroup Z of multiplicative type, but where Z is not a reductive center (i.e. G admits no reductive center), cf. for example 1.6; it nevertheless follows easily from IX 6.8 that a subgroup Z of G is a reductive center of G if and only if it is a largest central subgroup of multiplicative type, and retains this property under every base change; see also 4.3 below.

It is obvious from 4.1 that if Z is the reductive center of G , then for every base change $S' \rightarrow S$, $Z_{S'}$ is the reductive center of $G_{S'}$. From this, and from the uniqueness of the reductive center, there follows, thanks to faithfully flat quasi-compact descent theory (SGA 1 VIII):

Proposition 4.2. *Let G be an S -group prescheme of finite presentation over S with affine fibers. If G admits a reductive center locally for the faithfully flat quasi-compact topology, it admits a reductive center. For Z to be a reductive center of G , it is necessary and sufficient that it be so locally for the fpqc topology.*

Note also:

Proposition 4.3. *Let G be an S -group prescheme of finite presentation with affine fibers, Z a group sub-prescheme. For Z to be a reductive center of G , it is necessary and sufficient that it be of multiplicative type, and that for every $s \in S$, Z_s be a reductive center of G_s .*

Indeed, it follows first from IX 5.6 b) that Z is then central. Since the conditions considered are stable under base change, it remains to prove that every central homomorphism $u : H \rightarrow G$, with H of multiplicative type and of finite type, factors through Z . Now, since Z is central, u and the canonical immersion $v : Z \rightarrow G$ define a group homomorphism

$$w : H \times_S Z \rightarrow G.$$

By IX 6.8 the latter admits an image group K , which is a subgroup of multiplicative type of G , and everything amounts to proving that one has $K = Z$. Now this is true fiber by fiber by the hypothesis on Z , and it now suffices to apply IX 5.1 bis, which completes the proof of 4.3.

One will note that in criterion 4.3, the hypothesis that for every $s \in S$, Z_s is the reductive center of G_s , is in fact purely geometric, i.e. it suffices to verify it over the algebraic closure of $\kappa(s)$, as follows from the second assertion of 4.2.

Theorem 4.4. *Let G be an affine algebraic group over a field k . Then G admits a reductive center.*

Since the center of G is representable by a closed subgroup of G (VIII 6.7), one is at once reduced to the case where G is commutative. In addition, by 4.2 one can suppose k algebraically closed. We shall see in XVII that in this case G is written as a product $Z \times U$, where Z is of multiplicative type and U “unipotent”, whence it will follow at once that Z is indeed a reductive center of G .

We shall give here a proof independent of the structure theorem for commutative algebraic groups in the general form just indicated.

Note that it follows from IX 6.8 that the set of subgroups H of multiplicative type of G is increasingly filtered. Let us show that it admits a largest element.

When G is smooth over k , one applies the structure theorem BIBLE 4 th. 4,

$$G = G_s \times G_u,$$

with G_s of multiplicative type and G_u “unipotent”, which means here that G_u admits a composition series whose factors are subgroups (smooth, if one wishes) of $\mathbf{G}_a$. (Indeed, G_u/G_u^0 is a unipotent group by BIBLE 4, cor. to th. 3, hence a p -group, where $p = \text{characteristic exponent}$,

thanks to BIBLE 4 prop. 4; on the other hand G_u^0 admits a composition series with smooth connected quotients of dimension 1 by BIBLE 6 th. 1, cor. 1, and the latter are isomorphic to $\mathbf{G}_a$ by BIBLE 7 th. 4). Now one sees easily (cf. the lemma below) that every homomorphism from a group of multiplicative type to $\mathbf{G}_a$, hence also to G_u , is trivial, which does indeed prove that every subgroup of multiplicative type of G is contained in G_s .

In the general case, consider the subgroup

$$G_0 = G_{\text{red}}$$

of G , which is smooth over k , hence by the preceding admits a largest subgroup Z_0 of multiplicative type. The subgroups of multiplicative type of G containing Z_0 correspond to subgroups of multiplicative type of $G' = G/Z_0$. Now one has an exact sequence

$$0 \longrightarrow G_0/Z_0 \longrightarrow G/Z_0 \longrightarrow G/G_0 \longrightarrow 0,$$

where, by the preceding, G_0/Z_0 has no subgroup of multiplicative type except the identity group. Since a subgroup of a group of multiplicative type is of multiplicative type (IX.8), it follows that for every subgroup H of multiplicative type of G/Z_0 , one has $H \cap (G_0/Z_0) = 0$, hence $H \longrightarrow G/G_0$ is injective. Since G/G_0 is a radicial algebraic group, hence finite over k , this implies that H is itself radicial and of rank bounded above by that of G/G_0 . This implies that the family of subgroups of multiplicative type of G admits a largest element, say Z .

I say that G/Z has no subgroup of multiplicative type other than the identity subgroup. This follows from the fact (proved in 7.1.1) that a commutative extension of two algebraic groups of multiplicative type is of multiplicative type: thus, if Z' is the inverse image in G of a subgroup of multiplicative type of G/Z , then Z' is of multiplicative type, hence $Z' = Z$ by the maximal character of Z , so $Z'/Z = 0$.

It now follows easily from the “finite extension principle” that for every extension K of k , $(G/Z)_K = G_K/Z_K$ has no subgroup of multiplicative type except the identity group either.

We can now prove that Z is a reductive center of G . Indeed, let $u : H \longrightarrow G_S$ be a homomorphism of S -groups, where S is a prescheme over k and H a group of multiplicative type and of finite type over S ; let us prove that u factors through Z_S . It amounts to the same thing to say that the composite homomorphism $H \longrightarrow G_S \longrightarrow (G/Z)_S = G_S/Z_S$ is zero.

Now I say that, putting $U = G/Z$, every homomorphism $u : H \longrightarrow U_S$ is zero. Indeed, by IX 5.2 one is reduced to the case where S is the spectrum of a field, and by IX 6.8 this then follows from the fact that U_K has no subgroup of multiplicative type other than 0. This completes the proof of 4.4. It remains only to set down the proof of the

Lemma 4.4.1. *Let H be an S -group prescheme of multiplicative type; then every homomorphism of S -groups $u : H \longrightarrow \mathbf{G}_a$ is trivial.*

Indeed, consider the module $\mathcal{O}_S^2 = E$ as an extension of $\mathcal{O}_S = E'$ by $\mathcal{O}_S = E''$; then $\mathbf{G}_a$ is identified with the prescheme of automorphisms of this extension, so a homomorphism $u : H \longrightarrow \mathbf{G}_a$ is identified with an H -module structure on E respecting the extension structure, i.e. such that E' is stable under H and that the actions induced by H on E' and E'' are trivial. By I 4.7.3 it follows that H acts trivially on E , hence u is trivial. Q.E.D.

Remark 4.4.2. If G is a nonzero abelian variety over k , then G admits no reductive center in the sense of 4.1 with the restriction “ G affine” omitted, for, for n prime to the characteristic, ${}_nG$ is étale over k of order prime to the characteristic, hence of multiplicative type; but the family of ${}_nG$ is schematically dense in G , so if there were a reductive center, it would be identical to G , which is absurd, G not being of multiplicative type. This is the reason for imposing in 4.1 the restriction that G have affine fibers.

Lemma 4.5. *Let S be a prescheme, G an S -group prescheme smooth over S , affine over S , with connected fibers, T a maximal torus of G , $u : H \longrightarrow G$ a central homomorphism, with H an S -group prescheme of multiplicative type and of finite type. Then u factors through T .*

Indeed, let C be the centralizer of T , which is a closed group sub-prescheme of G smooth over S (XI 5.3), hence affine over S . Since T is in the center of G , it is invariant, and one can consider the quotient group $G/T = U$, which is representable (VIII 5.1). Since u is central, it factors through C , and everything amounts to proving that the composite homomorphism $H \rightarrow C \rightarrow U = C/T$ is trivial. By IX 5.2 one is reduced to the case where S is the spectrum of a field, which one can suppose algebraically closed. But then by BIBLE 6, th. 2, U is a connected algebraic group (smooth over k) “unipotent”, which means, as we have already observed, that it admits a composition series with quotients isomorphic to $\mathbf{G}_a$. Thus every homomorphism from a group H of multiplicative type and of finite type to U is trivial. This proves 4.5.

Corollary 4.6. *Let G be as in 4.5. If G admits a reductive center, the latter is contained in every maximal torus of G .*

Theorem 4.7. *Let S be a prescheme, G an S -group prescheme, smooth over S , affine over S , and with connected fibers.*

- a) *For every $s \in S$, let $z(s)$ be the type of the reductive center of G_s (which is defined by 4.4). Order the set E of isomorphism classes of $\mathbf{Z}$ -modules of finite type by declaring that the class of M is greater than that of M' if M' is isomorphic to a quotient of M . Then the map $S \rightarrow E$, $s \mapsto z(s)$ is lower semicontinuous.*
- b) *For G to admit a reductive center Z , it is necessary and sufficient that the preceding function $z : S \rightarrow E$ be locally constant. If this is so, G/Z is representable (cf. VIII 5.1), and G/Z admits the identity subgroup as reductive center.*
- c) *Suppose that the reductive rank of G is locally constant (cf. 1.7 b)). Then G admits a reductive center Z . If G/Z is representable (for example G affine over S), then in addition the maximal tori T of G (resp. the Cartan subgroups C of G) and T' (resp. C') of $G' = G/Z$ are in one-to-one correspondence, T (resp. C) corresponding to $T' = T/Z$ (resp. $C' = C/Z$), and T' (resp. C') corresponding to $T = \varphi^{-1}(T')$ (resp. $C = \varphi^{-1}(C')$), where $\varphi : G \rightarrow G'$ is the canonical homomorphism.*
- d) *Let T be a maximal torus of G , denote by $\mathfrak{g}$ the Lie algebra of G , and consider the homomorphism*

$$\theta : T \rightarrow \mathrm{GL}(\mathfrak{g}),$$

induced by the adjoint representation of G (II 4). Then the kernel of θ is a reductive center of G .

PROOF. a) and b). The proof of a) and of the first assertion in b) is essentially identical to that of 1.7 a) and b), and we dispense with reproducing the argument here. Let us only point out that in a) one must appeal to IX 5.6 (using the fact that G has connected fibers). Let us prove the second assertion of b), i.e. that if Z is a reductive center of G , then $G' = G/Z$ admits the identity subgroup as reductive center. Note at once that by VIII 5.1 the quotient group G/Z is indeed representable; it is affine over S , of finite presentation over S (VIII 5.8) and even smooth over S (for it suffices to see this for the fibers, G' being flat of finite presentation over S , but for the fibers it was pointed out in VI_B.9.2.(xii) that a quotient of a smooth algebraic group over a field k is smooth); finally, since G has connected fibers, so does G' . Thus G' satisfies the same initial hypotheses as G . To see that the identity subgroup of G' is a reductive center, one can restrict oneself if one wishes to the case where S is the spectrum of a field (4.3). Let Z' be a central subgroup of multiplicative type of G' ; everything amounts to proving that Z' is reduced to the identity group. Let Z_1 be the inverse image of Z' in G , and consider the action of Z_1 on G induced by inner automorphisms of G . Since Z is central, Z acts trivially, so Z_1 acts through an action of Z' on G . Moreover, since Z is in the center of G , Z_1 , hence Z' , acts trivially on Z ; in addition, since Z' is in the center of G' , the action of Z' on the quotient $G/Z = G'$ is also trivial. Since Z is central, it follows

at once that the action of Z' on G comes from a group homomorphism

$$u : Z' \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr}}(G', Z),$$

by

$$r(z') \cdot g = g \cdot u(z')(\varphi(g)),$$

where $r : Z' \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G)$ is the representation considered of Z' by automorphisms of G , $\varphi : G \longrightarrow G'$ the canonical homomorphism. Now giving a group homomorphism u as above amounts to giving a group homomorphism

$$v : G' \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr}}(Z', Z),$$

and, on the other hand, by X 5.8 the second member is representable and is an étale group over S , hence its identity section is an open immersion, so $\mathrm{Ker} v$ is an open subgroup of G' . Since G' has connected fibers, it is equal to G' , hence v is zero, so u is zero, so Z' acts trivially on G , hence so does Z_1 , which is therefore *central* in G . Thus Z_1 is a *commutative* extension of a group Z' of multiplicative type by a group Z of multiplicative type, hence (since we are over a field) is a group of multiplicative type (7.1.1). Being central in G , it is contained in the reductive center Z , i.e. $Z_1 = Z$, whence $Z' = \text{identity group}$, which completes the proof of b).

d) Let $Z = \mathrm{Ker} \theta$, which is a subgroup of multiplicative type of T (for example by IX 6.8). By 4.5, every central homomorphism $u : H \longrightarrow G$, with H of multiplicative type and of finite type, factors through T , hence through Z . Since the formation of Z is compatible with every base change, it remains to prove that Z is central, i.e. that the centralizer C of Z is equal to G . Now, by XI 5.3, C is a smooth closed subgroup of G ; on the other hand, since Z acts trivially on $\mathfrak{g}$, one sees that $\mathrm{Lie}(C) = \mathfrak{g}$. One concludes easily that $C = G$: indeed, the immersion $C \longrightarrow G$ is étale, for it is so fiber by fiber (as an unramified homomorphism of smooth algebraic groups of the same dimension, VI_B.1.3), and one can apply X 3.5. Thus $C \longrightarrow G$ is an étale closed immersion, hence an open immersion (SGA 1 I 5.1), and since it is also a closed immersion and G has connected fibers, it is an isomorphism.

c) By 1.7 b), G admits locally for the fpqc topology a maximal torus, so by 4.2 and d) just proved, G admits a reductive center Z . We saw in d) that every maximal torus T of G contains Z . One observes at once that $T' = T/Z$ is a torus; to prove that it is a maximal torus in G' , one is reduced by definition 1.3 to the case where S is the spectrum of an algebraically closed field, and then the assertion is contained in BIBLE 7 th. 3a). Conversely, let T' be a maximal torus of G' , and let $T = \varphi^{-1}(T')$; let us prove that T is a maximal torus of G . Since the question is local for the fpqc topology, one can suppose that G already admits a maximal torus, say T_0 , so that by the preceding, $T'_0 = T_0/Z_0$ is a maximal torus of G' . By conjugacy theorem 1.7 c), T' and T'_0 are locally conjugate for the fpqc topology, hence (since $\varphi : G \longrightarrow G'$ is covering for this topology, so every section of G' lifts locally to a section of G) T and T_0 are likewise locally conjugate. Since T is a maximal torus, so is T_0 . One proceeds analogously for Cartan subgroups. This completes the proof. $\square$

Let us give a useful translation of d) in the case where T is diagonalizable, hence of the form $D_S(M)$, where M is a free $\mathbf{Z}$ -module of finite type. Then (I 4.7.3) the T -module $\mathfrak{g}$ decomposes as a direct sum of sub- T -modules $\mathfrak{g}_m$ ($m \in M$):

$$\mathfrak{g} = \coprod_{m \in M} \mathfrak{g}_m,$$

which are necessarily locally free.

Suppose that for every $m \in M$, the rank of $\mathfrak{g}_m$ is constant (which is the case in particular if S is connected). Then the set R of $m \in M$ such that $\mathfrak{g}_m \neq 0$ ("roots") is finite. This being said:

Corollary 4.8. *Under the preceding conditions and with the preceding notation, the reductive center of G is the intersection of the kernels of the root characters $m \in R$ on T . One therefore*

has an isomorphism

$$Z \simeq D_S(N),$$

where N is the quotient of M by the subgroup generated by R .

Corollary 4.9. *Let G be an algebraic group over an algebraically closed field k . Suppose G smooth over k , connected affine, with reductive center reduced to the identity group, and that the Lie algebra $\mathfrak{g}$ of G is nilpotent. Then G is “unipotent”, i.e. admits a composition series with quotients isomorphic to $\mathbf{G}_a$.*

By BIBLE 6, th. 4, cor. 3, it suffices to prove that a maximal torus T of G is reduced to the identity group, or, what amounts to the same thing, that the Lie algebra $\mathfrak{t}$ of T is reduced to 0. Now it follows from the fact that the T -module $\mathfrak{g}$ decomposes according to the characters α of T (I 4.7.3), that for every $t \in \mathfrak{t}$, the operation $\text{ad}_{\mathfrak{g}}(t)$ on $\mathfrak{g}$ is semisimple. Thus if $\mathfrak{g}$ is nilpotent, $\text{ad}_{\mathfrak{g}}(t)$ is zero. But by 4.7 d), since the reductive center of G is reduced to the neutral element, the homomorphism $T \rightarrow \text{GL}(\mathfrak{g})$ is a monomorphism, hence induces an injective map on the Lie algebras, which means (II 4.5.) that for $t \in \mathfrak{t}$, the relation $\text{ad}_{\mathfrak{g}}(t) = 0$ implies $t = 0$. This proves that $\mathfrak{t} = 0$ and completes the proof.

Proposition 4.10. *Let G be a smooth connected affine algebraic group over an algebraically closed field k . Then the reductive center of G is the intersection of the maximal tori of G .*

Of course, this is the intersection in the schematic sense (or, what amounts to the same thing, in the sense of subfunctors of G), i.e. the largest closed sub-prescheme of G contained in the maximal tori of G . It follows from the noetherian character of G that it is also the intersection of a suitable *finite* set of maximal tori of G .

Let Z be the intersection in question; Z is a closed subgroup of a torus, hence is of multiplicative type, and, on the other hand, by 4.5 it contains the reductive center of G . To prove that it is equal to it, it remains to prove that it is central. Since G is connected, it suffices by IX 5.5 to prove that Z is *invariant*. Now by construction Z is invariant under the $\text{int}(g)$, with $g \in G(k)$, so the normalizer N of Z (cf. VIII 6.7) is a closed subgroup of G containing the rational points of G . Since G is reduced, one has $N = G$, which completes the proof.

Proposition 4.11. *Let S be a prescheme, G an S -group prescheme smooth over S , affine over S , with connected fibers, and of zero unipotent rank, which implies (1.9) that G has locally constant reductive rank, hence (1.10) that the “prescheme of maximal tori of G ” is defined, say $\mathcal{T}$, and is smooth, separated, of finite type over S . Make G act on $\mathcal{T}$ via inner automorphisms, whence a homomorphism of group functors*

$$u : G \rightarrow \underline{\text{Aut}}_S(\mathcal{T}).$$

Under these conditions, the following three subfunctors of G are identical:

- (i) *The reductive center Z of G .*
- (ii) *The center Z' of G .*
- (iii) *The kernel $Z'' = \text{Ker } u$ of the preceding homomorphism.*

In particular, the center of G is representable by a subgroup of multiplicative type of G .

PROOF. One trivially has $Z \subset Z' \subset Z''$; it remains to prove that $Z'' \subset Z$, which amounts to proving (the hypotheses being stable under base change) that every section g of G over S that acts trivially on $\mathcal{T}$ is a section of Z . Introducing $G' = G/Z$ and using 4.7 b) and c), which imply in particular that the prescheme $\mathcal{T}'$ of maximal tori of G' is canonically isomorphic to $\mathcal{T}$, one is reduced to the case where $G = G'$, i.e. to the case where the reductive center Z of G is zero. (N.B. Note that by 4.7 c), the unipotent rank of G' is equal to that of G , hence is zero since that of G is). Thus in this case one must prove that g is the *identity section* of G . The usual reduction procedure reduces us to the case where S is noetherian, and even to the case where S is *artinian local* (since to verify that g is the identity section, it

suffices to verify, when S is locally noetherian, that this is so after every base change $S' \rightarrow S$, with S' artinian local). Now in this case the kernel Z'' of u is representable (VIII 6.2 a) and 6.5 c)) (Note that Z'' is representable by XI 6.8). To prove that Z'' is reduced to the identity subgroup, it suffices by Nakayama to prove that this is so for its fiber Z''_0 . This therefore reduces us to the case where S is the spectrum of a field k , which one can obviously suppose algebraically closed. Now Z'' is contained in the stabilizer of every point of $\mathcal{T}(k)$, i.e. in the normalizer $N(T)$ of every maximal torus T of G . Since the reductive rank of G is zero, it follows by XI 5.9 that T is an open subgroup of $N(T)$, hence that the Lie algebra of $N(T)$ is identical to that of T . Thus the Lie algebra of Z'' is contained in that of T . On the other hand, it follows from 4.10 that the intersection of the Lie algebras of the maximal tori T of G is nothing other than the Lie algebra of the reductive center Z , hence here zero, since we supposed $Z = 0$. Consequently the Lie algebra of Z'' is zero, i.e. Z'' is étale over k . Moreover, Z'' is obviously invariant in G , and since G is connected, it follows easily that Z'' is in the center of G . It is therefore in $T = \text{Centr}_G(T)$ for every maximal torus T , hence in the intersection of the maximal tori, i.e. in $Z = 0$ by 4.10, which completes the proof. $\square$

5. Application to the scheme of subgroups of multiplicative type*

Theorem 5.1. *Let S be a prescheme, G an S -group prescheme smooth and affine over S , M the “scheme of subgroups of multiplicative type of G ”, representing the functor described explicitly in XI 4.1. For every integer $n > 0$, let T_n be the subfunctor of M such that $T_n(S') =$ set of group sub-preschemes H of multiplicative type of $G_{S'}$ such that $\text{nid}_H = 0$, so T_n is representable and is affine over S (XI 3.12 a)). Let $u_n : M \rightarrow T_n$ be the morphism defined by $u_n(H) = {}_nH$ (where ${}_nH = \text{Ker}(\text{nid}_H)$). With this notation, one has the following:*

- a) *Every sub-prescheme U of M of finite type over S is contained in a closed sub-prescheme of finite type over S , and every closed sub-prescheme X of M of finite type over S is affine over S .*
- b) *Suppose S quasi-compact, and let X be a closed sub-prescheme of M of finite type over S . Then there exists an integer $n > 0$ such that for every multiple m of n , the induced morphism*

$$u_m|_X : X \rightarrow T_m$$

is a closed immersion.

PROOF. a) To prove the first assertion of a), one will take for X the schematic closure of U (EGA I 9.5.1 and 9.5.3), which is defined since the immersion $i : U \rightarrow M$ is quasi-compact (for U is of finite type over S and M separated over S (4.1)). Thus one must prove that such an X is affine over S , which will prove at the same time the second assertion of a). In the preceding form, one sees that the question is local on S , which one can therefore suppose affine. Then, since U is of finite type over S , it is contained in a quasi-compact open subset of M , and this reduces us to the case where U is itself an open subset of finite type over S , i.e. quasi-compact. (N.B. A closed subscheme of a scheme affine over S is affine over S).

It suffices to prove that such a U is contained in a closed sub-prescheme of M that is affine. In this form, the usual reduction procedure reduces us at once to the case where S is noetherian. One proceeds likewise for b), which reduces to the case where S is affine noetherian.

Take up again the projective limit T of the T_n used in the proof of XI 4.1, which is an affine prescheme (but not of finite type) over S , and whose local rings are noetherian rings, as was seen at the beginning of the proof of IX 3.7. (N.B. For $t \in T$, $t = (t_n)$, since the transition morphisms $T_m \rightarrow T_n$ are smooth and the dimension of the T_m is bounded above,

*For a generalization of the results of the present no. to the case where G is not supposed affine over S , cf. M. Raynaud: *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Chap. IX.

there exists an n_0 such that the $T_n \rightarrow T_{n_0}$ are étale at t_n for every n a multiple of n_0). The proof in *loc. cit.*, or XI 3.11, shows that the canonical morphism

$$u : M \rightarrow T$$

is an *immersion*, and induces isomorphisms on the local rings (but one should take care that u is not in general an open immersion or a quasi-compact morphism). Let U be a quasi-compact open subset of M ; we shall prove that its closure in T is contained in M , which proves (if X denotes the schematic closure of U in M) that $u|_X : X \rightarrow T$ is a closed immersion, hence that X is affine, and this will prove a).

Since U is noetherian, it has only finitely many irreducible components, and every element $t \in T$ in the closure of U is a specialization of an element x of U . Since the local ring of t in T is noetherian, so is the local ring A of t in $\bar{x}$ endowed with the induced reduced structure.

The canonical morphism from the integral noetherian local scheme $S' = \operatorname{Spec}(A)$ to T then sends the closed point of S' to t , the generic point to $x \in M$, and under these conditions one must show that $t \in M$. On replacing A by the quotient of a suitable complete local A -algebra flat over A (EGA 0_{III} 10.3.1) by a minimal prime ideal, one can suppose that A is complete with algebraically closed residue field (and one could even reduce to the case where it is in addition a discrete valuation ring, thanks to EGA II 7.1.7). One is thus reduced (making the change of notation: S' denoted by S) to the

Lemma 5.2. *Let S be the spectrum of a complete integral noetherian local ring with algebraically closed residue field, G an S -group prescheme smooth and affine over S , $(H(n))_{n>0}$ a system of subgroups of multiplicative type of G , indexed by the integers $n > 0$, η the generic point of S . Suppose:*

- a) *If m is a multiple of n , one has $H(n) = {}_n H(m)$.*
- b) *There exists a subgroup H_η of multiplicative type of G_η such that one has $H(n)_\eta = {}_n(H_\eta)$ for every $n > 0$.*

Under these conditions, there exists a subgroup H of multiplicative type of G such that, for every $n > 0$, one has $H(n) = {}_n H$.

For every n , let $C_n = \operatorname{Centr}_G(H(n))$, which is a closed group sub-prescheme of G , smooth over S (XI 5.3). The set of integers $n > 0$ being ordered by divisibility, the C_n form a decreasing family of closed sub-preschemes of G , and since G is noetherian, this family is stationary. Let C be the common value of the C_n for large n . Then C satisfies the same conditions as G ; in addition the $H(n)$ are in fact subgroups of C . Thus, on replacing G by C , one can suppose that the $H(n)$ are *central* subgroups of G .

Let then s be the closed point of S , and Z_s the reductive center of G_s (defined thanks to 4.4). By the definition (4.1), Z_s contains the $H(n)_s$. Since A is complete, Z_s comes from a group sub-prescheme Z of multiplicative type of G (XI 5.8). I say that Z contains the $H(n)$. Indeed, since $H(n)$ is central, hence commutes with Z in G , one deduces a group homomorphism $Z \times H(n) \rightarrow G$, which, by IX 6.8, admits an image group that is a subgroup K of multiplicative type of G containing Z , and everything amounts to proving that $K = Z$, which follows from $K_s = Z_s$ and IX 5.1 bis.

One is thus reduced to proving the analogue of 5.2, but with G replaced by a group Z of multiplicative type and of finite type over S (not necessarily smooth over S). Since A is complete with algebraically closed residue field, Z is diagonalizable (X 3.3 and 1.4), hence of the form $D_S(M)$, M a commutative group of finite type. Consequently every subgroup H of multiplicative type of Z is diagonalizable (IX 2.11 (i)), hence of the form $D_S(N)$, where N is a quotient group of M (VIII 3.2). Thus the $H(n)$ correspond to quotient groups $M(n)$ of M , and the existence of a subgroup H of multiplicative type of Z such that $H(n) = {}_n H$ for every n amounts to that of a quotient group N of M such that $M(n) = N/nN$ for every n .

Now this follows at once from the fact that there exists a subgroup H_η of G_η such that $H(n)_\eta = {}_n(H_\eta)$ for every n . This completes the proof of 5.2, hence of a).

b) We already know that $u|_X : X \rightarrow T$ is a closed immersion. It follows easily that for large m , the composite $u_m|_X : X \rightarrow T_m$ is also a closed immersion. Since we shall not need this fact later, we dispense with detailing its proof here. $\square$

Corollary 5.3. *With the notation of 5.1, let U be a subset both open and closed of M , of finite type over S . Then U is affine over S for the structure induced by M , and if S is quasi-compact, there exists an integer $n > 0$ such that for every multiple m of n , the induced morphism $u_m|_U : U \rightarrow T_m$ is an open and closed immersion (i.e. an isomorphism onto an open and closed subset of T_m , endowed with the induced structure).*

The first assertion follows from 5.1 a), the second from 5.1 b), taking into account that $u_m : M \rightarrow T_m$ is smooth (XI 2.2 bis) and that a smooth, i.e. étale, immersion is an open immersion.

Corollary 5.4. *Let S be a prescheme, G an S -group prescheme smooth and affine over S , of locally constant reductive rank (1.7 b)), $\mathcal{T}$ the “prescheme of maximal tori of G ” (1.10). Then $\mathcal{T}$ is smooth and affine over S . If T is a maximal torus of G , $N(T)$ its normalizer, then $G/N(T)$ (XI 5.3 bis) is affine over S . The same holds for $G/C(T)$ (where $C(T)$ is the centralizer of T), provided $W(T) = N(T)/C(T)$ is finite over S (cf. 2.1 b)).*

The second assertion is contained in the first, since by the conjugacy theorem, $G/N(T)$ is isomorphic to $\mathcal{T}$ (XI 5.5 bis). For the first assertion, one notes that by construction $\mathcal{T}$ is isomorphic to an open and closed sub-prescheme of M (for one can suppose the reductive rank of G constant and equal to r , and then $\mathcal{T}$ is the sub-prescheme of M corresponding to subtori of relative dimension r , i.e. the largest sub-prescheme of M over which the “universal subgroup of multiplicative type” $H \subset G_M$ is a torus of relative dimension r). One can therefore apply 5.3. Finally, for the last assertion, one notes that $G/C(T)$ is finite over $G/N(T) \simeq (G/C(T))/W(T)$, hence is affine since $G/N(T)$ is.

Corollary 5.5. *Let G be a smooth affine algebraic group over a field k . Then the scheme M of subgroups of multiplicative type of G is a direct sum of affine schemes over S . For every subgroup H of multiplicative type of G , if C and N denote respectively its centralizer and its normalizer, the quotients G/C and G/N are affine.*

Using XI 5.1 bis, one sees that the saturation under G acting on M of every finite closed subset of M is open: indeed, one is reduced to the case where k is algebraically closed, hence to the case of the orbit of a point x rational over k , but then by *loc. cit.* the morphism $g \mapsto g \cdot x$ from G to M is smooth, hence open, so its image is open.

Let U be the union of the classes of closed points of M for the equivalence relation defined by the action of G . Then U is open and contains every closed point of M , hence by the Nullstellensatz is identical to M . Thus M is a disjoint union of open subsets, which are therefore necessarily closed, so M is the sum prescheme of preschemes M_i , each of which is an orbit under G of a closed point, hence is quasi-compact, hence of finite type. By 5.3 the M_i are therefore affine. If H is a subgroup of multiplicative type of G , it corresponds to a point x of M rational over k , and G/N is identified with the orbit of x under G (XI 5.5 bis). It is therefore affine by the preceding. Since C is an open subgroup of N (XI 5.9), G/C is therefore finite over G/N (for the latter is isomorphic to $(G/C)/(N/C)$), hence affine since G/C is. This proof shows at the same time:

Corollary 5.6. *Under the conditions of 5.5 for G and H , the subscheme U of M “of subgroups of multiplicative type of G that are locally conjugate to H ” is an open and closed sub-prescheme of G .*

In pictorial terms, every subgroup H' of multiplicative type of G that is a limit of subgroups of G conjugate to H is itself conjugate to H .

Remarks 5.7. Let S be a prescheme, G an S -prescheme smooth and affine over S , H an S -prescheme of multiplicative type and of finite type; put $M = \underline{\text{Hom}}_{S\text{-gr}}(H, G)$, which by XI 4.2 is representable and is smooth over S and separated over S . One can then prove for M a result entirely analogous to 5.1, either by reducing to 5.1 by a proof analogous to that of XI 4.2, or by proceeding directly by a proof modeled on that of 5.1. One deduces corresponding variants for 5.3, 5.5, 5.6, which the reader will formulate.

6. Maximal tori and Cartan subgroups of algebraic groups not necessarily affine (algebraically closed base field)

Lemma 6.1. *Let G be a connected algebraic group over a field k , Z an algebraic subgroup of finite index of its center; then G/Z is affine.*

One can suppose that Z is the center of G , for a scheme finite over an affine scheme is affine. Consider the vector spaces

$$P^n = P^n(G) = \mathcal{O}_{G,e}/\mathfrak{m}_{G,e}^{n+1} \quad (n \text{ an integer } \geq 0),$$

where $\mathfrak{m}_{G,e}$ is the maximal ideal; then G acts on the $P^n(G)$ by the adjoint representation, and if Z_n is the kernel of the corresponding homomorphism

$$\text{ad}_n : G \longrightarrow \text{GL}(P^n),$$

one verifies easily (using the fact that G is connected) that Z is the intersection of the Z_n , hence (since G is noetherian) equal to one of the Z_n . But ad_n defines, by passage to the quotient, a monomorphism $G/Z_n \longrightarrow \text{GL}(P^n)$, which is therefore a closed immersion, hence G/Z_n is affine, and consequently so is G/Z .

Lemma 6.2. *Let G be a smooth algebraic group over the field k , Z a central algebraic subgroup, $G' = G/Z$, $u : G \longrightarrow G'$ the canonical homomorphism, T a subgroup of multiplicative type in G , $T' = u(T)$ the image group, C the centralizer of T in G , C' that of T' in G' . Then one has*

$$C'^0 \subset u(C).$$

One can suppose k algebraically closed. Let

$$C_1 = (u^{-1}(C')^0)_{\text{red}};$$

it suffices to prove that one has $C_1 \subset C$, for $u(C_1)$ is connected of finite index in C' , hence equal set-theoretically to C'^0 , hence equal to C'^0 since C' and consequently C'^0 are smooth. Consider the morphism

$$(g, t) \longmapsto gtg^{-1}t^{-1}$$

from $G \times G$ to G ; it induces a morphism

$$\varphi : C_1 \times T \longrightarrow Z_1, \quad \text{where } Z_1 = Z_{\text{red}},$$

for, since the first member is reduced, it suffices to see that for $g \in C_1(k)$, $t \in T(k)$, one has $gtg^{-1}t^{-1} \in Z(k)$, which comes from the fact that g centralizes T modulo Z . One sees easily (by calculating on points rational over k) that $\varphi(g, t)$ is additive in g , and additive in t , hence is “bilinear”; I say that (since Z_1 is smooth and C_1 connected) this homomorphism is identically zero, which will indeed prove that $C_1 \subset C$. Using the density theorem for T , one is reduced to the case where T is finite, i.e. where there exists an integer $n > 0$ such that $n \text{id}_T = 0$. Note that φ is defined by a group homomorphism

$$C_1 \longrightarrow \underline{\text{Hom}}_{S\text{-gr}}(T, Z_1),$$

but, since Z_1 is commutative and smooth, the second member is representable by an étale algebraic group over k , and since C_1 is connected, every group homomorphism from C_1 to the latter is zero. Q.E.D.

Corollary 6.3. *Under the preceding conditions, suppose C' connected; then one has*

$$C' = u(C), \quad C = u^{-1}(C').$$

Indeed, then $C' = C'^0$; on the other hand C obviously contains Z , hence is equal to $u^{-1}(u(C))$.

Lemma 6.4. *Let G be an algebraic group over the field k , Z a central algebraic subgroup such that $G/Z = G'$ is a torus. Then G is commutative, and if k is algebraically closed, there exists a torus T in G such that $u(T) = G'$, where u is the canonical homomorphism $G \rightarrow G/Z = G'$.*

One can suppose k algebraically closed. Consider again the morphism $G \times G \rightarrow G$ defined by $(g, h) \mapsto ghg^{-1}h^{-1}$; then (since Z is central) this morphism factors through a morphism $G' \times G' \rightarrow G$, and, since G' is commutative, the latter takes its values in Z , and even in $Z_1 = Z_{\text{red}}$, since $G' \times G'$ is reduced. One sees as above that the morphism $\varphi : G' \times G' \rightarrow Z_{\text{red}}$ thus obtained is bilinear, hence zero, which proves that G is commutative. To find a torus T of G such that $u(T) = G'$, one can (on replacing G by G_{red}^0) suppose G smooth and connected, and in addition (on replacing Z by Z_{red}^0) that Z is smooth and connected. By a well-known theorem of Chevalley⁽⁶⁾, Z is an extension of an abelian variety by an affine group, which reduces us at once to proving our assertion in each of the following two cases: 1°) Z is affine, 2°) Z is an abelian variety. In case 1°), G is affine and the result is well known (BIBLE 7 th. 3 a)). Thus suppose that Z is an abelian variety. Since every homomorphism from the additive group $\mathbf{G}_a$ to the torus $G' = R$ or to the abelian variety Z is trivial, it follows that the same holds for every homomorphism from $\mathbf{G}_a$ to G , hence G contains no subgroup isomorphic to $\mathbf{G}_a$. By the structure theorem of Chevalley already invoked, one has an exact sequence

$$(*) \quad 0 \rightarrow T \rightarrow G \rightarrow A \rightarrow 0,$$

where A is an abelian variety, and T a smooth connected affine group. Since the latter is commutative and contains no additive subgroup, it follows that T is a torus (and it is obviously the unique maximal torus of G). Everything amounts to proving that every epimorphism

$$u : G \rightarrow R,$$

where R is a torus, satisfies $u(T) = R$. Put

$$\text{Hom}_{\text{gr}}(T, \mathbf{G}_m) = M, \quad \text{Hom}_{\text{gr}}(R, \mathbf{G}_m) = P,$$

(these are free $\mathbf{Z}$ -modules of finite type recovering T, R up to isomorphism by $T = D_k(M)$, $R = D_k(P)$), and let

$$B = \text{Ext}_{k\text{-gr}}^1(A, \mathbf{G}_m),$$

(B is also the set of points rational over k of the abelian variety dual to A). One obviously has

$$\begin{aligned} \text{Ext}^1(A, T) &= \text{Hom}_{\text{gr}}(M, B), \\ \text{Ext}^1(A, R) &= \text{Hom}_{\text{gr}}(P, B), \end{aligned} \quad (\text{xx})$$

in particular the extension G of A by T is given by a homomorphism

$$\theta : M \rightarrow B.$$

On the other hand, exact sequence $(*)$ gives the exact sequence

$$0 \rightarrow \text{Hom}(A, R) \rightarrow \text{Hom}(G, R) \rightarrow \text{Hom}(T, R) \rightarrow \text{Ext}^1(A, R),$$

and moreover $\text{Hom}(A, R) = 0$, and $\text{Hom}(T, R) \rightarrow \text{Ext}^1(A, R)$ is identified with the homomorphism

$$\text{Hom}(P, M) \rightarrow \text{Hom}(P, B)$$

⁽⁶⁾**N.D.E.** Give references here.

deduced from $\theta : M \longrightarrow B$. Putting

$$N = \text{Ker}(\theta),$$

one therefore finds a canonical bijection

$$\text{Hom}(G, R) \simeq \text{Hom}(P, N) \simeq \text{Hom}(S, R), \quad \text{where } S = D_k(N),$$

which one can immediately describe geometrically as follows:

Lemma 6.5. *Let G be an extension of an abelian variety A by a torus $T = D_k(M)$, defined by a homomorphism $\theta : M \longrightarrow B = \text{Ext}^1(A, \mathbf{G}_m)$ (algebraically closed base field k). Let $N = \text{Ker } \theta$, $S = D_k(N)$ the corresponding torus, isomorphic to T/U where $U = D_k(M/N)$; consider the extension $H = G/U$ of A by S . This extension splits⁽⁷⁾, so one has a unique projection from H to S , whence a unique homomorphism*

$$v : G \longrightarrow S$$

extending the canonical homomorphism $T \longrightarrow S$. This being said, for every torus R and every homomorphism $u : G \longrightarrow R$, there exists a unique homomorphism $u' : S \longrightarrow R$ such that $u = u'v$. In particular, one has $\text{Im}(u) = \text{Im}(u') = u(T)$.

This shows in particular that if u is an epimorphism, so is its restriction to T , which completes the proof of 6.4.

Theorem 6.6. *Let G be a smooth connected algebraic group over an algebraically closed field k .*

- a) *The maximal tori of G are conjugate.*
- b) *Let T be a torus of G ; then its centralizer is smooth and connected.*
- c) *The map $T \longmapsto \text{Centr}_G(T)$ establishes a one-to-one correspondence between the set of maximal tori T of G and the set of Cartan subgroups (no. 1) of G . For an algebraic subgroup C of G to be a Cartan subgroup, it is necessary and sufficient that it be smooth, nilpotent and set-theoretically equal to its connected normalizer (and then it is even equal to its connected normalizer); one then has $C = \text{Centr}_G(T)$, where T is the unique maximal torus of C , and $\text{Norm}_G(C) = \text{Norm}_G(T)$.*
- d) *Let $v : G \longrightarrow H$ be an epimorphism of smooth connected algebraic groups; then the maximal tori (resp. the Cartan subgroups) of H are the images of the maximal tori (resp. the Cartan subgroups) of G . If T is a maximal torus of G , C its centralizer, then $v(C)$ is the centralizer of $v(T)$.*
- e) *Under the conditions of d), suppose that $\text{Ker } v$ is a central subgroup of G ; then $T \longmapsto v(T)$ (resp. $C \longmapsto v(C)$) establishes a one-to-one correspondence between the set of maximal tori (resp. of Cartan subgroups) of G and the set of maximal tori (resp. of Cartan subgroups) of H . The Cartan subgroups of G contain the center of G and a fortiori $\text{Ker } v$, and are the groups of the form $v^{-1}(C')$, where C' is a Cartan subgroup of H .*

Let us also describe explicitly at once the following immediate consequence of c):

Corollary 6.7. *For G to be nilpotent (i.e. the group $G(k)$ nilpotent), it is necessary and sufficient that the tori in G be central, or again that G have only one maximal torus (and then the latter is the largest subtorus of G).*

PROOF OF 6.6. Let Z be a central algebraic subgroup of G , let $G' = G/Z$, and $u : G \longrightarrow G'$ the canonical homomorphism. Then G' is a smooth connected group. If T' is a subtorus of G' , it follows from 6.4 that $u^{-1}(T')$ is commutative, and that T' is the image of a subtorus of $u^{-1}(T')$, hence of a subtorus of G .

⁽⁷⁾**N.D.E.** Is split!

Since $u^{-1}(T')$ is commutative, it obviously admits a largest subtorus T (for the sum of two subtori gives a third containing both), and one therefore has $u(T) = T'$. From this it follows immediately that for every maximal torus T of G , its image $T' = u(T)$ is a maximal torus of G' , and that $T \mapsto u(T)$ is a one-to-one correspondence between maximal tori of G and maximal tori of G' .

We now take

$$Z = \text{Centr}(G)_{\text{red}}^0,$$

then G' is affine by 6.1. Since the maximal tori of G' are then conjugate, so are those of G , which proves a). Moreover, for G to be nilpotent, resp. to have only one maximal torus, it is necessary and sufficient that G' satisfy the same condition; but since G' is affine, the two conditions in question on G' are equivalent (BIBLE 6 th. 4 cor. 2), so the same holds for the conditions in question for G . Moreover, if G has only one maximal torus T , the latter is invariant, hence central, and since every torus in G is contained in a maximal torus, it is central. Conversely, if every torus is central, so are the maximal tori, and by conjugacy theorem a) there is only one maximal torus. This proves 6.7.

Let T be any torus of G , $T' = u(T)$; then $C' = \text{Centr}_{G'}(T')$ is connected (BIBLE 6 th. 6 a)), so by 6.3 the centralizer C of T is equal to $u^{-1}(C')$, hence connected (since Z is connected), which proves b). If T is maximal, hence T' maximal, then one knows that C' is nilpotent, hence C (which is a central extension of C') is nilpotent. In addition, T is a maximal torus of C , hence by 6.7 it is the unique maximal torus of C , so the map $T \mapsto \text{Centr}_G(T)$ from the set of maximal tori of G to the set of Cartan subgroups is bijective. Moreover one has

$$\text{Centr}_G(T) = C \subset \text{Norm}_G(C) \subset \text{Norm}_G(T)$$

and since one knows that the centralizer of T is of finite index in its normalizer (cf. XI 5.9, whose argument is valid without an affine hypothesis, using only the representability of the two functors in question, as was pointed out in XI 6.5), and that C is smooth and connected, one concludes

$$C = \text{Norm}_G(C)^0.$$

In addition, by the one-to-one correspondence between maximal tori and Cartan subgroups, one sees that $\text{Norm}_G(C)$ and $\text{Norm}_G(T)$ have the same points with values in k , and since the second is smooth, one has

$$\text{Norm}_G(T) = \text{Norm}_G(C).$$

To complete the establishment of c), it remains to prove that if C is a smooth connected nilpotent subgroup of G of finite index in its normalizer, then C is a Cartan subgroup. Now since Z is central, the normalizer of C contains Z , and since Z is smooth and connected, one concludes $Z \subset C$, whence $C = u^{-1}(C')$, where $C' = u(C)$. One then has

$$\text{Norm}_G(C) = u^{-1}(\text{Norm}_{G'}(C')),$$

which proves that C' is connected nilpotent of finite index in its normalizer, hence by BIBLE 7 th. 1 is a Cartan subgroup of G' , whence at once that C is a Cartan subgroup of G .

Let us prove e): one is under the conditions of the beginning of the proof (putting $\text{Ker } u = Z$, $G' = H$); we have already seen that $T \mapsto u(T)$ is a one-to-one correspondence between maximal tori of G and maximal tori of G' . Taking account of the one-to-one correspondence between maximal tori and Cartan subgroups just proved, one deduces a one-to-one correspondence between Cartan subgroups C of G and Cartan subgroups C' of G' , by associating to $C = \text{Centr}_G(T)$ the group $C' = \text{Centr}_{G'}(T')$, where $T' = u(T)$. Since C' is connected by b), it follows by 6.3 that $C' = u(C)$, and $C = u^{-1}(C')$, which proves e).

It remains to prove d). Let T be a maximal torus of G ; let us prove that $v(T)$ is a maximal torus of H (which, taking account of conjugacy theorem a), will imply that the maximal tori of H are all of the form $v(T)$ as above). Thus let R be a torus in H containing $v(T)$, and let us prove $R = v(T)$. On replacing H by R , G by $v^{-1}(R)_{\text{red}}^0$, one can suppose

$R = H$, i.e. that H is a torus, and one is reduced to proving that then $v(T) = H$. Let again $Z = \text{Centr}(G)_{\text{red}}^0$, $G' = G/Z$, and $H' = H/v(Z)$:

$$(D) \quad \begin{array}{ccccccc} e & \longrightarrow & Z & \longrightarrow & G & \xrightarrow{u} & G' \longrightarrow e \\ & & \downarrow v'' & & \downarrow v & & \downarrow v' \\ e & \longrightarrow & v(Z) & \longrightarrow & H & \xrightarrow{u'} & H' \longrightarrow e. \end{array}$$

We already know that $u(T)$ is a maximal torus of G' , and since G' is affine, hence $v' : G' \rightarrow H'$ an epimorphism of smooth connected affine groups, $v'(u(T))$ is a maximal torus of H' (BIBLE 7 th. 3 a)), hence equal to H' , i.e. $u'(v(T)) = H'$, so to prove $v(T) = H$, it suffices to show that $v(T) \supset v(Z)$. Now $v(Z)$ is a smooth connected subgroup of H , hence a torus, and $v'' : Z \rightarrow v(Z)$ is an epimorphism, so by 6.4 one has $v(Z) = v''(S)$, where S is a torus of Z . But S , being central in G , is obviously contained in the maximal torus T , whence $v(Z) \subset v(T)$. This proves assertion d) in the case of maximal tori.

Taking account of the one-to-one correspondence between maximal tori and Cartan subgroups, it remains to prove that if T is a maximal torus of G , C its centralizer, then $v(C)$ is the centralizer of $v(T)$. For this, take up again diagram (D) above (where of course H is no longer supposed to be a torus), let $T' = u(T)$, $C' = u(C)$; we already saw in e) that C' is the centralizer of the maximal torus T' , hence (since G' is affine, hence H' affine) $v'(C')$ is the centralizer of the maximal torus $v'(T')$ of H' (BIBLE 7 th. 3 a)), i.e. $u'(v(C))$ is the centralizer of $u'(v(T))$; since C contains Z , hence $v(C)$ contains $v(Z)$, $v(C)$ is therefore the inverse image under u' of $u'(v(C))$, i.e. of the centralizer of $u'(v(T))$, hence is the centralizer of $v(T)$, as follows from e) applied to $u' : H \rightarrow H'$ and to the maximal torus $v(T)$ of H . This completes the proof of 6.6. $\square$

7. Application to smooth group preschemes not necessarily affine

Theorem 7.1. *Let S be a prescheme, G an S -group prescheme smooth, separated and of finite type over S . Suppose that G admits locally for the faithfully flat quasi-compact topology a maximal torus. Then:*

a) *The map*

$$T \longmapsto \text{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G to the set of Cartan subgroups of G . If C corresponds to T , then T is the unique maximal torus of C .

b) *Let T, T' be two maximal tori of G , C, C' the corresponding Cartan subgroups; then one has*

$$\underline{\text{Transp}}_G(T, T') = \underline{\text{Transp}}_G(C, C') = \underline{\text{Transp}}_G(T, C'),$$

the first two terms are also identical to the strict transporters, and finally the functor in question is representable by a closed sub-prescheme of G smooth over S . The tori T, T' and the Cartan subgroups C, C' are locally conjugate for the étale topology.

c) *There exists locally for the étale topology a maximal torus of G and a Cartan subgroup of G .*

d) *Suppose that every finite subset of a fiber G_s of G is contained in an affine open subset of G (for example G quasi-projective over S , or S artinian); then the functor $\mathcal{T} : (\text{Sch})_{/S}^\circ \rightarrow (\text{Ens})$ defined in 1.10 (the functor of maximal tori of G), isomorphic to the functor $\mathcal{T} : (\text{Sch})_{/S}^\circ \rightarrow (\text{Ens})$ of Cartan subgroups of G , is representable by a prescheme smooth, separated, of finite type over S , which is quasi-projective over S when G is, and is affine over S when G is affine over S , or when S is artinian.*

e) *Let $u : G \rightarrow G'$ be a morphism of S -group preschemes, where G' is smooth, separated, of finite type over S , and suppose that for every $s \in S$, one has $u_s(G_s) = G'_s$, i.e. u_s*

is faithfully flat (Exp. VI⁽⁸⁾). Then for every maximal torus T of G , $u(T)$ is a maximal torus of G' ; if G' has connected fibers, then for every Cartan subgroup C of G , $u(C)$ is a Cartan subgroup of G' , and if C is the centralizer of T , $u(C)$ is the centralizer of $u(T)$. In either case, the induced morphism $T \rightarrow u(T)$, resp. $C \rightarrow u(C)$, is faithfully flat.

f) Under the conditions of e), suppose in addition that $\text{Ker } u$ is a central subgroup of G . Then $T \mapsto u(T)$ is a bijective map from the set of maximal tori of G to the set of maximal tori of G' , and if G' has connected fibers, $C \mapsto u(C)$ is likewise a bijective map from the set of Cartan subgroups of G to the set of Cartan subgroups of G' ; if $C' = u(C)$, one has $C = u^{-1}(C')$.

Remarks 7.2. We shall see in XV that the conclusion of d) remains valid without the restrictive condition on finite subsets of the G_s . We shall also prove there the conclusions concerning Cartan subgroups alone contained in b) c) d) e), when one supposes only that G admits locally for the faithfully flat quasi-compact topology a Cartan subgroup (but not necessarily a maximal torus). For this, we shall moreover have to use 7.1 in the case where S is artinian. Moreover, the proof of c) and d) (in the case of non-artinian S) is considerably simplified by using the method of XV.

PROOF OF 7.1. a) Proceeding as in 3.2, one sees that for every maximal torus T of G , $C = \text{Centr}_G(T)$ is indeed a Cartan subgroup of G , and T is determined in terms of C as the unique maximal torus of C , so it remains only to show that every Cartan subgroup C of G is defined by a maximal torus of G , or again that C admits a central maximal torus. Since the question is again local for the fpqc topology, one can suppose that G admits a maximal torus T . Then by XI 6.2, $\text{Transp}_G(T, C)$ is representable by a closed sub-prescheme S' of G smooth over S ; moreover $S' \rightarrow S$ is surjective, as follows from 6.6 c). Thus on making the base change $S' \rightarrow S$, one can suppose that there exists a section g of G such that $\text{ad}(g) \cdot T \subset C$, but then $\text{ad}(g) \cdot T$ is a maximal torus of C , which is central fiber by fiber by 6.6 c), hence central by IX 5.6 b). Q.E.D.

b) Let T, T' be two maximal tori of G , and C, C' the corresponding Cartan subgroups of G . Then the following conditions are equivalent:

$$\begin{aligned} (1) \quad T \subset T' \quad (2) \quad T = T' \quad (3) \quad T \subset C' \\ (4) \quad C \subset C' \quad (5) \quad C = C'. \end{aligned}$$

This follows trivially from a). Using the same result after arbitrary base change, one concludes the identity stated in b) between various transporters and strict transporters. Moreover, it was already observed in a) that $\text{Transp}_G(T, C')$ is representable by a closed sub-prescheme of G , smooth over S , and that its structural morphism is surjective. Consequently, by Hensel there exists locally for the étale topology a section of this prescheme over S , so T and T' on the one hand, C and C' on the other, are locally conjugate for the étale topology, which proves b).

c) Suppose first that S is artinian local. When G admits a maximal torus T , then it follows from the conjugacy theorem proved in b) that the functor $\mathcal{T}$ of maximal tori of G is representable by the homogeneous space G/N , where N is the normalizer of T in G , which is smooth by b). Moreover, as was observed in VI_A.3.2.1, since $\mathcal{T}$ is a homogeneous space under G , every finite subset of $\mathcal{T}$ is contained in an affine open subset. In the general case, there exists a finite extension k' of the residue field k of S such that $G_{k'}$ has a maximal torus; then k' comes from k by a finite flat base change $S' \rightarrow S$, and the maximal torus of $G_{k'}$ lifts to a maximal torus $G_{S'}$ (XI 2.1 bis), so the functor $\mathcal{T}_{S'}$ is representable by a prescheme over S' , smooth, separated and of finite type over S' , every finite subset of which is contained in an affine open subset. Thus the natural descent datum on $\mathcal{T}_{S'}$ is effective, hence $\mathcal{T}$ is

⁽⁸⁾**N.D.E.** Reference not located in VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), proposition II.5.1.(c).

representable, and by descent one sees that $\mathcal{T}$ is smooth over S , separated and of finite type over S . From smoothness it follows, thanks to Hensel, that $\mathcal{T}$ admits a section locally for the étale topology. This proves c) and d) in the case of artinian S (N.B. We shall prove below that in this case $\mathcal{T}$ is in fact affine over S).

Suppose now S arbitrary. To prove c) and d), which are local assertions on S for the Zariski topology, one can suppose that there exists a prime number ℓ prime to the residue characteristics of S , that S is affine, and that the reductive rank of the fibers of G (which is obviously locally constant, thanks to the hypothesis of local existence for fpqc of a maximal torus) is constant, say r . Let $S' \rightarrow S$ be a faithfully flat quasi-compact morphism, S' affine, such that $G_{S'}$ admits a maximal torus T' . Let C' be its centralizer; I say that there exists a suitable power n of ℓ such that one also has $C = \text{Centr}_G({}_nT)$. Indeed, to see this, one is reduced at once to the case of noetherian S' , where this was seen in XI 6.2. Fix n thus; put

$$M = (\mathbf{Z}/n\mathbf{Z})^r, \quad P = \underline{\text{Hom}}_{S\text{-gr}}(M_S, G),$$

then P is obviously representable as a closed sub-prescheme of finite presentation of $({}_nG)^r$, where

$${}_nG = \underline{\text{Hom}}_{S\text{-gr}}((\mathbf{Z}/n\mathbf{Z})_S, G),$$

which is also the kernel of the n th-power morphism in G , hence representable by a closed sub-prescheme of finite presentation of G . Moreover P is smooth over S by XI 2.1. Let $s \in S$; by what was seen above, there exists a finite separable extension k'' of $\kappa(s)$ such that there exists a maximal torus T'' in $G_{k''}$; in addition, since ${}_nT''$ is étale over k'' (n being prime to the characteristic of k''), one can suppose (on replacing k'' by a finite separable extension) that ${}_nT''$ is isomorphic to $M_{k''}$. Let $S'' \rightarrow S$ be an étale morphism such that there exists an $s'' \in S''$ above $s \in S$ giving rise to the residue extension $\kappa(s'') \simeq k''$. One therefore has a section of $P_{S''} \otimes \kappa(s'')$ over $\text{Spec}(\kappa(s''))$, so by Hensel, on replacing (S'', s'') by (S''', s''') étale over it, one can suppose that there exists a section of $P_{S''}$ over S'' , i.e. an element of $P(S'')$, extending the given section. In other words, one has a homomorphism $M_{S''} \rightarrow G_{S''}$ inducing an isomorphism $M_{\kappa(s'')} \simeq {}_nT''$. By IX 6.4 (which here reduces to a simple application of Nakayama's lemma), this homomorphism is a closed immersion over an open neighborhood of s' , which one can suppose equal to S'' . Let H'' be its image, and consider its centralizer C'' in $G_{S''}$, which is a smooth closed sub-prescheme of G , by XI 6.2. Note moreover:

Lemma 7.3. *Under the preceding conditions for ℓ, n, r , for every prescheme S'' over S and every maximal torus T'' of $G_{S''}$, one has*

$$\text{Centr}_{G_{S''}}(T'') = \text{Centr}_{G_{S''}}({}_nT'').$$

Indeed, by faithfully flat descent from $S' \times_S S''$, one is reduced to the case where $G_{S''}$ admits a maximal torus T''_1 for which the preceding relation is true. But since T'' is locally conjugate to T''_1 for the étale topology by b), it follows that the same relation is true for T'' .

Apply the preceding result with $\text{Spec}(\kappa(s''))$ instead of S'' ; one sees that the fiber $C''_{s''}$ is a Cartan subgroup of G . Note now that since G is smooth over S , the union of the neutral components of the fibers G_s is an open subset of G (by a general result of EGA IV on smooth morphisms⁽⁹⁾), obviously stable under the group law of G , hence is a subgroup of G for the prescheme structure induced by G .

In addition, G^0 satisfies the preliminary hypotheses on G , and there is an obvious one-to-one correspondence between the maximal tori of G and those of G^0 . Thus to prove c) and d), one can suppose G with connected fibers, which we shall do. Then the Cartan subgroups of G have connected fibers (6.6 a)). This being said, I say that C'''^0 is a Cartan subgroup of $G_{S''}$ over an open neighborhood of s'' in S'' (which will complete the proof of c)). This follows indeed from the

⁽⁹⁾N.D.E. Make this reference precise.

Lemma 7.4. *Under the conditions of 7.1, suppose G with connected fibers; let C be a closed group sub-prescheme of G , smooth over S , and s an element of S such that C_s is a Cartan subgroup of G_s ; then C^0 is a Cartan subgroup of G over an open neighborhood of s .*

One reduces easily to the case where S is local and s its closed point, and to proving that C is then a Cartan subgroup of G , then by flat descent to the case where G admits a maximal torus, say T . Then by XI 6.2, $\text{Transp}_G(T, C)$ is representable by a closed sub-prescheme of G smooth over S . In addition, by the hypothesis on C_s , the fiber of the transporter at s is nonempty (taking account of conjugacy theorem 6.6 a)). This reduces us by faithfully flat descent to the case where this transporter admits a section over S , hence to the case where C contains a maximal torus T of G . But then T is central in C^0 by IX 5.6 a), hence $C^0 \subset \text{Centr}_G(T)$, and since this is an inclusion of group schemes smooth over S , having the same relative dimension (namely the dimension of their common fiber at s) and connected fibers, it is an equality, which completes the proof.

d) We retain the preceding notation and hypotheses for ℓ, n, r , and the connectedness of the fibers of G . Let $Q : (\text{Sch})_{/S}^\circ \rightarrow (\text{Ens})$ be the functor defined by

$$Q(S') = \text{set of subgroups of multiplicative type of } G_{S'} \\ \text{of type equal to } (\mathbf{Z}/n\mathbf{Z})^r = M \quad (\text{IX 1.4}).$$

Then

$$T \mapsto {}_nT$$

is a morphism

$$\varphi : \mathcal{T} \rightarrow Q,$$

which is a monomorphism by 7.3. I say that Q is representable by a separated prescheme of finite presentation over S . Indeed, as was pointed out in the proof of XI 3.12 a), one has an isomorphism

$$Q \simeq P'/\Gamma$$

where P' is the open and closed sub-prescheme of the prescheme $P = \underline{\text{Hom}}_{S\text{-gr}}(M_S, G)$ introduced in c) corresponding to monomorphisms $M_{S'} \rightarrow G_{S'}$ (cf. IX 6.8), and where $\Gamma = (\text{Aut}_{\text{gr}}(M))_S$. The hypothesis that every finite subset of a fiber G_s is contained in an affine open subset of G , being stable under passage to a closed sub-prescheme and under cartesian products, is obviously inherited by $({}_nG)^r$, hence by P , hence by P' , so Q is representable by a prescheme of finite presentation over S (cf. V)⁽¹⁰⁾.

One sees in the same way that if G is quasi-projective (resp. affine) over S , so is Q . In any case, Q is separated over S . Now one has the

Lemma 7.5. *The preceding homomorphism $\varphi = \mathcal{T} \rightarrow Q$ is representable by an open immersion.*

In other words, one must prove that if H is a subgroup of multiplicative type of G , of type M , then there exists an open subset U of S such that for every S' over S , $H_{S'}$ is of the form ${}_nT_{S'}$, for a suitable maximal torus $T_{S'}$ of $G_{S'}$, if and only if $S' \rightarrow S$ factors through U . One can obviously suppose that the nilpotent rank of the fibers of G is constant (for, thanks to b), it is locally constant), say r' . Let $C = \text{Centr}_G(H)$, which is a closed group sub-prescheme of G smooth over S (XI 6.2). Then, on replacing S by the open and closed subset of points at which C is of relative dimension r' , one can suppose C of relative dimension r' everywhere. One then sees at once that H is of the form ${}_nT$, for a maximal torus T of G , if and only if C^0 admits a central maximal torus T of relative dimension r everywhere and $H = {}_nT$, which gives another expression for the subfunctor U of S that we want to represent (replacing S in the preceding criterion by an S' over S). Moreover, by flat descent one can suppose that G admits a maximal torus T_1 . Let R be the subfunctor $\text{Transp}_G(T_1, \text{Centr}_{C^0}(C^0))$ of $\text{Transp}_G(T_1, C)$; the latter is representable by a smooth prescheme over S by XI 6.2, and the

⁽¹⁰⁾**N.D.E.** Make this reference precise.

former is representable by an induced open sub-prescheme, as follows at once from IX 5.6 a); in particular it is smooth over S . Consequently its structural morphism to S is open, hence its image is open, and on replacing S by the said image (endowed with the induced structure), one can suppose the structural morphism surjective. Then, by 1.13 applied to C^0 , C^0 admits a central maximal torus T , since it admits one locally for fpqc, which will obviously be of relative dimension equal to that of T_1 , i.e. r . Thus the condition to express for H is the equality $H = {}_nT$, which by IX 2.10 amounts again to taking a suitable open (and closed) subset of S .

Lemma 7.5 therefore implies that $\mathcal{T}$ is representable by a separated prescheme locally of finite presentation over S , and even of finite presentation over S , as one sees by taking up again the proof of 7.5 to ensure that φ is in fact a quasi-compact open immersion, or by reducing by faithfully flat quasi-compact descent to the case where G admits a maximal torus, and where $\mathcal{T}$ is therefore isomorphic to $G/\mathrm{Norm}_G(T)$. This last expression, or alternatively XI 2.1, shows in addition that $\mathcal{T}$ is smooth over S . Finally, if G is quasi-projective over S , so is Q , hence also $\mathcal{T}$. If G is affine over S , the assertion that $\mathcal{T}$ is then affine over S is included as a reminder, having been established in 5.4 (N.B. I do not know whether, without an affine hypothesis for G/S , it is possible to choose n so that in 7.5 the open immersion is also a closed immersion). For the assertion that $\mathcal{T}$ is affine over S if S is artinian, one is reduced to the case where S is the spectrum of a field (EGA I 6.1.7), which one can suppose algebraically closed. Then thanks to f), which will be proved below, it suffices to prove the same assertion for $G/\mathrm{Centr}(G)$; but the latter is affine by 6.1, so one is under the preceding conditions.

This completes the proof of d).

e) By IX 6.8, one knows that there exists a subtorus T' of G' such that u induces a faithfully flat morphism $T \rightarrow T'$ (which characterizes T' as the subsheaf $u(T)$ of G'). Let C be the centralizer of T , C' that of T' ; let us prove that the morphism $C \rightarrow C'$ is flat, and faithfully flat if G' has connected fibers. Since C, C' are flat of finite presentation over S , one is reduced to the case of a base field (SGA 1 I 5.9), which one can obviously suppose algebraically closed. In addition one can suppose G, G' connected (on replacing them by G^0 and G'^0 , which does not change C^0 and C'^0), and it then suffices to apply 6.6 d), taking account of a).

f) Taking account of a) and e), one can restrict oneself to proving the assertion concerning Cartan subgroups. Now since a Cartan subgroup of G is the centralizer of a maximal torus, it contains the center of G and a fortiori $\mathrm{Ker} u$, so it is of the form $u^{-1}(C')$, where $C' = u(C)$ is the Cartan subgroup of G' considered in e). Thus the map $C \mapsto u(C)$ is injective; to show that it is bijective, it suffices to see that for every Cartan subgroup C' of G' , $u^{-1}(C')$ is a Cartan subgroup of G . Since the question is local for fpqc, one can suppose that G admits a Cartan subgroup C_1 , hence $u(C_1) = C'_1$ is a Cartan subgroup of G' , hence locally conjugate to C' for fpqc by b), and since $u^{-1}(C'_1) = C_1$ is a Cartan subgroup of G , it follows that $u^{-1}(C')$ is a Cartan subgroup of G . Q.E.D.

One can also, to prove that $u^{-1}(C')$ is a Cartan subgroup, note that it is flat over S , for u is (SGA 1 I 5.9), which reduces us by definition to the case of a base field, and one can apply 6.6 e). □

Corollary 7.6. *Under the conditions of 7.1 e), G' admits locally for the étale topology a maximal torus, hence satisfies the preliminary conditions for G . If in addition $\mathrm{Ker} u$ is central, then the functors $\mathcal{T}_G, \mathcal{C}_G$ of maximal tori of G and Cartan subgroups of G are isomorphic to the analogous functors $\mathcal{T}_{G'}, \mathcal{C}_{G'}$ for G' (hence, in the case where they are representable, they are represented by isomorphic S -preschemes).*

Remark 7.7. a) Unlike what happens in the case where G is affine over S (it suffices, in fact, that G have affine fibers, as one will see in XVI), it is not true that the fact that G has locally constant reductive rank implies that G admits locally for fpqc a maximal torus. Let,

for example, S be the spectrum of a discrete valuation ring, G_1 and G_2 smooth separated group schemes of finite type over S , such that the generic fiber of G_1 is an elliptic curve, the special fiber a group $\mathbf{G}_m$, and the generic fiber of G_2 a group $\mathbf{G}_m$, the special fiber a group $\mathbf{G}_a$, and take $G = G_1 \times_S G_2$. Then the two fibers of G have reductive rank 1, but one sees at once that G admits no maximal torus locally for fpqc. On the other hand, it is very plausible that the following condition (for a smooth separated group of finite type over a prescheme S) is sufficient for the existence of a maximal torus locally for the étale topology: the reductive rank and the abelian rank of the fibers of G are locally constant functions.

b) In the proof of 7.1 (in particular a)), we invoked XI 6.2 in cases where S is not supposed locally noetherian. However, in the cases considered of application of XI 6.2, reduction to the case of affine noetherian S is immediate.

Here is a variant of 7.1 b):

Proposition 7.8. *Let G be a smooth S -group prescheme of finite presentation with connected fibers, H a group prescheme smooth over S , $i : H \rightarrow G$ a monomorphism of S -groups (making H a sub- S -group of G). Then for every Cartan subgroup^{*} C of G , $\underline{\text{Transp}}_G(C, H)$ is representable by a closed sub-prescheme of G smooth over S .*

The fact that the transporter is representable by a closed sub-prescheme of G of finite presentation is contained in XI 6.11, taking into account that C has connected fibers since G does (6.6 b)). To show that the transporter is smooth over S , one is reduced by the standard procedure to the case where S is affine noetherian, then to the case where S is artinian local, and by descent to the case where the residue field of S is algebraically closed. But then C admits a maximal torus T , which is a maximal torus of G . One can suppose that the reductive rank and the nilpotent rank of the fiber H_0 are equal to those of G_0 (otherwise the transporter would be empty), but then one sees at once (using the connectedness of C and the fact that the centralizer in H of a maximal torus of H is smooth) that one has

$$\underline{\text{Transp}}_G(C, H) = \underline{\text{Transp}}_G(T, H)$$

and since one knows that the second member is smooth (XI 2.5), so is the first. The preceding argument proves more generally part b) of the

Proposition 7.9. *Let G and $i : H \rightarrow G$ be as in 7.8; suppose in addition that for every $s \in S$, H_s is connected, and has the same reductive rank and the same nilpotent rank as G_s (i.e. H_s contains a Cartan subgroup of G_s). Then one has the following:*

- a) $\underline{\text{Norm}}_G(H)$ is representable by a closed sub-prescheme $\text{Norm}_G(H)$ of G of finite presentation over S , and the canonical monomorphism $H \rightarrow \text{Norm}_G(H)$ is an open immersion; consequently i is an immersion, and one has

$$H = \text{Norm}_G(H)^0.$$

- b) For every Cartan subgroup C of G , $\underline{\text{Transp}}_G(C, H)$ is a closed sub-prescheme of G , smooth over S , with surjective structural morphism. If C is the centralizer of a maximal torus T of G , one has in addition

$$\underline{\text{Transp}}_G(T, H) = \underline{\text{Transp}}_G(C, H).$$

- c) Let C be a group sub-prescheme of H . For it to be a Cartan subgroup of H , it is necessary and sufficient that it be a Cartan subgroup of G .
 d) Suppose that G admits locally for the étale topology, or for the fpqc topology, a Cartan subgroup (resp. a maximal torus); then the same holds for H .

PROOF. a) The representability of $\underline{\text{Norm}}_G(H)$ by a closed sub-prescheme $\text{Norm}_G(H)$ of G of finite presentation over S is contained in XI 6.11. Since H is smooth, hence flat locally

^{*}The proof shows that it suffices to suppose that C is a smooth subgroup of G each of whose geometric fibers is the connected centralizer of a subgroup of a maximal torus of $G_{\bar{s}}$.

of finite presentation over S , to verify that $H \rightarrow \text{Norm}_G(H)$ is an open immersion, one is reduced to verifying it on the fibers (VI_B.2.6), which reduces us to the case where S is the spectrum of an algebraically closed field k . One is then reduced (Exp. VI⁽¹¹⁾) to verifying that the corresponding homomorphism on the Lie algebras is an isomorphism, or, what amounts to the same thing, that $(\mathfrak{g}/\mathfrak{h})^H = 0$, where $\mathfrak{g}$ and $\mathfrak{h}$ are the Lie algebras of G and H , and the exponent H denotes the invariants under H (cf. II 5.2.3 (i)). Now by hypothesis H contains a Cartan subgroup C of G , the centralizer of the maximal torus T of G , and it therefore suffices to prove that one has

$$(\mathfrak{g}/\mathfrak{h})^T = 0,$$

which follows, taking account of the complete reducibility of representations of T (I 4.7.3), from the analogous relation $(\mathfrak{g}/\mathfrak{c})^T$, where $\mathfrak{c} = \text{Lie}(C)$. As for the latter, equivalent to

$$\mathfrak{g}^T = \mathfrak{c},$$

it means that the centralizer C and the normalizer N of T have the same Lie algebra, which follows from the fact that C is an open subgroup of N (XI 5.9). This completes the proof of a).

b) As was pointed out, the proof was given in 7.8.

c) Taking account of the fact that H is a sub-prescheme of G , the assertion reduces trivially to the case where S is the spectrum of an algebraically closed field, in which case it follows at once from the hypothesis made on H .

d) One uses b), c) and “Hensel’s lemma” XI 1.10. $\square$

Corollary 7.10. *Let G and H be as in 7.9, and suppose that for every algebraically closed field k over S , every element of $G_k(k)$ normalizing H_k is in $H_k(k)$. Then H is a closed sub-prescheme of G , and is its own normalizer.*

This follows trivially from 7.9 a). We shall apply 7.10 in particular to the Borel subgroups (more generally, to the parabolic subgroups) of G .

Corollary 7.11. *Let G, H be as in 7.9. Then H contains every group sub-prescheme Z of G , central in G and flat and of finite presentation over S .*

One can as usual reduce to the case of affine noetherian S , then to the case of artinian S , which implies that one is under the conditions of 7.1. By 7.9 c), one can suppose that H contains a Cartan subgroup C of G , and one is reduced to proving that $C \supset Z$. Now since S is artinian, $G' = G/Z$ is representable by a group prescheme of finite type over S , the canonical morphism $u : G \rightarrow G'$ being faithfully flat and its kernel being Z (VI_A.3.2). Obviously G' is smooth over S , and one can apply 7.1 f), which implies that C is of the form $u^{-1}(C')$, hence contains Z .

Corollary 7.12. *Let G, G' be two smooth S -group preschemes of finite presentation, $u : G \rightarrow G'$ a faithfully flat group homomorphism (i.e. for every $s \in S$, u_s is faithfully flat); suppose $\text{Ker } u$ central and G with connected fibers. Then the map*

$$H' \mapsto H = u^{-1}(H')$$

establishes a one-to-one correspondence between group sub-preschemes H' of G' , smooth of finite presentation over S , having the same reductive rank and the same nilpotent rank as G' at every $s \in S$, and the set of group sub-preschemes H of G , smooth of finite presentation over S , having the same reductive rank and the same nilpotent rank as G at every $s \in S$. For H to have connected fibers, it is necessary and sufficient that H' do so.

Let H be a group sub-prescheme of G having the properties just specified. Then by 7.10, H contains $Z = \text{Ker } u$, hence by faithfully flat descent theory is of the form $u^{-1}(H')$, where

⁽¹¹⁾**N.D.E.** Reference not located in VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), corollary II.5.6.

H' is a well-determined group sub-prescheme of G' , and one observes at once, taking account of 6.6 e), that the latter has the properties stated above. Moreover, if H has connected fibers, so obviously does $H' = H/Z$. Thus it remains to prove that if one starts from a subgroup H' of G' having the stated properties, then $u^{-1}(H') = H$ has the same properties in G ; and that if H' has connected fibers, so does H . Taking account of 6.6 e), one is reduced to proving that H is smooth over S (resp. and has connected fibers). Now H is already flat over S as the inverse image of H' , which is so, under the flat morphism u , hence one is reduced to verifying that the geometric fibers of H are smooth (resp. and connected), which reduces us to the case where S is the spectrum of an algebraically closed field. Then H' contains a Cartan subgroup C' of G' , hence H contains the inverse image C of C' , which is a Cartan subgroup of G by 6.6 e), hence C is smooth and connected. Consequently 7.11 follows from the following lemma:

Lemma 7.13. *Let G, G' be two flat group preschemes of finite presentation over S , $u : G \rightarrow G'$ a group homomorphism that is faithfully flat, C' a group sub-prescheme of G' of finite presentation over S , such that $C = u^{-1}(C')$ is smooth over S (resp. has connected fibers). Then for every group sub-prescheme H' of G' of finite presentation over S , containing C' , and such that H' is smooth over S (resp. has connected fibers), its inverse image $H = u^{-1}(H')$ is smooth over S (resp. has connected fibers).*

As we observed above, this statement reduces at once to the case where S is the spectrum of a field. Note then that H is a principal bundle with base H/C and group C (Exp. VI_B.9); on the other hand $H/C \simeq H'/C'$ (cf. Exp. IV), and since H' is smooth (resp. connected), so is H'/C' , hence H/C . Since by hypothesis the same holds for C , it follows at once that it holds again for the bundle H . Q.E.D.

8. Semisimple elements, union and intersection of maximal tori in group schemes not necessarily affine

Throughout this no. , G denotes a smooth group prescheme of finite presentation over S , with connected fibers.

Suppose first that S is the spectrum of an algebraically closed field. When G is affine, the notion of a *semisimple element* of $G(k)$ was defined in BIBLE 4 no. 4; one observes at once that this notion is invariant under an algebraically closed extension k'/k of the base field. In addition, one saw in BIBLE 6 th. 5 (c) that $g \in G(k)$ is semisimple if and only if it is contained in a maximal torus of G . When G is no longer supposed affine, G is written canonically (thanks to Chevalley) as an extension of an abelian variety by a smooth connected affine algebraic group:

$$(*) \quad e \longrightarrow V \longrightarrow G \longrightarrow A \longrightarrow e.$$

We shall say that an element g of $G(k)$ is *semisimple* if it is a semisimple element of $V(k)$. Since the maximal tori of V are obviously identical to the maximal tori of G , it amounts to the same thing to say that g belongs to a maximal torus of G . This is obviously again a notion invariant under an algebraically closed extension k'/k of the base field k .

Suppose now that S is the spectrum of an arbitrary field k , and let $g \in G$. Then (choosing an algebraically closed extension K of $\kappa(g)$) one sees that g is the image of a geometric point g' of G with values in an algebraically closed extension K of k , and we shall say that g is *semisimple* if g' is semisimple, which is independent of the particular choice of g' , thanks to what was said above. If k' is an extension of k , then the set of semisimple elements of $G_{k'}$ is the inverse image of the set G^{ss} of semisimple elements of G .

Suppose finally S arbitrary; then a point $g \in G$ is said to be semisimple if it is semisimple in its fiber G_s . If $S' \rightarrow S$ is any morphism, then $G_{S'}^{\text{ss}}$ is therefore the inverse image of G^{ss} . Suppose that the functor $\mathcal{T}$ defined in 1.10 (the functor of maximal tori) is representable by a prescheme of finite presentation over S (which is the case, for example, if G admits locally

for fpqc a maximal torus, by 7.1 d), at least if G is quasi-projective over S). Consider the canonical maximal torus $\underline{T}$ of $G_{\mathcal{T}}$ (“universal maximal torus of G ”), and the morphism

$$u : \underline{T} \longrightarrow G$$

induced by the projection $G_{\mathcal{T}} \longrightarrow G$. Then it follows at once from the definition that G^{ss} is nothing other than the image of the preceding morphism. One will conclude:

Proposition 8.1. *The set G^{ss} of semisimple elements of G is locally constructible (hence constructible if S , hence G , is quasi-compact and quasi-separated).*

One reduces as usual to the case where S is affine noetherian; in addition one can suppose (by the usual noetherian criterion of constructibility (EGA 0_{III} 9.2.3)) that S is integral, and restrict oneself to proving that there exists a nonempty open subset U of S such that $G^{\text{ss}}|_U$ is constructible. Taking U small enough, and replacing it if necessary by a finite covering, one can suppose that G is separated over S and contains a maximal torus T . But then by 7.1 d), the functor $\mathcal{T}$ is representable by a prescheme of finite presentation over S , and so therefore is $\underline{T}$, whose image in G is consequently constructible. Q.E.D.

Suppose again that k is a field, and consider the subgroup H of G generated by the preceding morphism (cf. VI_B.1.2). It is a smooth group subscheme of G , connected since $\underline{T}$ is, whose formation is obviously compatible with every extension of the base field (cf. VI), that of $\underline{T}$ being so. When k is algebraically closed, one sees immediately that H is also the algebraic subgroup of G generated by the maximal tori of G , or, what amounts to the same thing, by the tori of G , and it is also the smallest algebraic subgroup of G containing the semisimple elements of $G(k)$. (In fact, these characterizations of H remain valid as soon as k is an infinite field, thanks to the fact, proved in XIV, that the set of points of $\underline{T}$ rational over k is dense in $\underline{T}$). Moreover, H is invariant in G , for to see this one can restrict oneself to the case where k is algebraically closed, and then (H and G being smooth over k) it suffices to verify that H is stable under the inner automorphisms $\text{int}(g)$, $g \in G(k)$, which is obvious. (One could even show that H is a characteristic subgroup of G , i.e. stable under $\text{Aut}_{k\text{-gr}}(G)$). It then follows at once from 6.6 d) that the reductive rank of G/H is zero; more precisely, if K is an invariant algebraic subgroup of G , it follows at once from the fact that for algebraically closed k the maximal tori of G/K are the direct images of the maximal tori of G , that the reductive rank of G/K is zero if and only if K contains all the maximal tori of G (k being supposed algebraically closed), or again if and only if K contains H (k arbitrary): hence H is the smallest invariant algebraic subgroup of G such that G/H has zero reductive rank. Another obvious characterization of H is the following: it is the smallest algebraic subgroup of G having the same reductive rank as G . Note finally that H is affine: indeed, to see this one can again suppose k algebraically closed, and taking up again exact sequence (*) from the beginning of the no. , one notes that the maximal tori of G are contained in V , hence so is H , so since V is affine, H is. Let us summarize the results obtained:

Proposition 8.2. *Let G be a smooth connected algebraic group over the field k , and let $\bar{k}$ be an algebraic closure of k . There exists an algebraic subgroup H of G such that $H_{\bar{k}}$ is the algebraic subgroup of $G_{\bar{k}}$ generated by the maximal tori (or again by the tori) of $G_{\bar{k}}$. The group H is also characterized as the smallest algebraic subgroup of G having the same reductive rank as G , or the smallest invariant algebraic subgroup of G such that the reductive rank of G/H is zero. It is a smooth connected invariant affine subgroup of G , whose formation commutes with every extension of the base field.*

To use the characterization of H in terms of G/H , it is appropriate to state explicitly the

Corollary 8.3. *Let G be a smooth connected algebraic group over an algebraically closed field k . For G to have zero reductive rank (i.e. with the notation of 8.2, for one to have $H = (e)$), it is necessary and sufficient that G be an extension of an abelian variety by a*

smooth connected unipotent algebraic group (i.e. a successive extension of groups isomorphic to the additive group $\mathbf{G}_a$).

Indeed, thanks to Chevalley's exact sequence (*), one is reduced to proving that the reductive rank of the smooth connected affine group V is zero if and only if V is unipotent, which is contained in BIBLE 6.4. th. 4 cor. 3.

Thus, with the notation of 8.2, H is the smallest invariant algebraic subgroup of G such that $(G/H)_{\bar{k}}$ is an extension of an abelian variety by a smooth connected affine unipotent algebraic group. One also concludes:

Corollary 8.4. *With the notation of 8.2, for one to have $G = H$ (i.e. $G_{\bar{k}}$ generated by its maximal tori), it is necessary and sufficient that G be affine and that every homomorphism from $G_{\bar{k}}$ to the additive group be trivial.*

Remarks 8.5. a) Let $\bar{V}$ be the largest smooth connected affine algebraic subgroup of $G_{\bar{k}}$ (so $G_{\bar{k}}$ is an extension of an abelian variety by $\bar{V}$). It is well known (Rosenlicht⁽¹²⁾), if k is not perfect, that $\bar{V}$ is not in general “defined over k ”, i.e. there does not in general exist an algebraic subgroup V of G such that $V_{\bar{k}} = \bar{V}$. However, when $\bar{V}$ is generated by its maximal tori, i.e. when $\bar{V}$ has no quotient group isomorphic to the additive group $\mathbf{G}_{a,\bar{k}}$, there exists such a V , namely the group H of 8.2. One therefore sees that in this question of rationality, as in many others (cf. for example XIV no. 6), all the troubles come from unipotent groups, i.e. from the additive group, whereas it is the presence of (sufficiently many) groups of multiplicative type that ensures, on the contrary, that things work well.

b) One can also introduce, with the notation of 8.2, the inverse image H' in G of the commutator subgroup of G/H ; then H' is the smallest invariant algebraic subgroup of G such that $(G/H')_{\bar{k}}$ is a commutative extension of an abelian variety by a smooth connected affine unipotent algebraic group. H' is again a smooth connected affine subgroup of G . Let H'' be the inverse image in G of $p^n(G/H')$ for large n , p being the characteristic (supposed > 0); then one sees easily that H''/H' is an abelian variety, and H'' is the smallest invariant algebraic subgroup of G such that G/H'' is a smooth connected affine unipotent commutative algebraic group. This being said, one sees easily that for $\bar{V} = H_{\bar{k}}$ (notation of a)), i.e. for every homomorphism from $\bar{V}$ to the additive group to be trivial, it is necessary and sufficient that $H'' = G$, or again that every homomorphism from $G_{\bar{k}}$ to the additive group be trivial.

To finish, we shall generalize to smooth groups with connected fibers the notion of reductive center developed in no. 4, taking inspiration from 4.10. Let Z be the subfunctor of G defined by

$$\begin{aligned} Z(S') &= \text{Set of sections } g' \text{ of } G_{S'} \text{ over } S' \text{ such that} \\ &\quad \text{for every } S'' \text{ over } S' \text{ and every maximal torus } T'' \\ &\quad \text{of } G_{S''}, \text{ the inverse image } g'' \text{ of } g' \text{ under } S'' \rightarrow S' \\ &\quad \text{is a section of } T'' \text{ over } S''. \end{aligned}$$

Introducing the functors $\mathcal{T}(S') = \text{set of maximal tori of } G_{S'}$, and $\underline{T}(S') = \text{set of pairs } (T', g')$, where T' is a maximal torus of $G_{S'}$ and g' a section of T' over S' , one sees that $\underline{T}$ is a subfunctor of $\mathcal{T} \times_S G = \mathcal{T}_G$, and with this notation one can also write

$$Z = \prod_{\mathcal{T}_G/G} \underline{T}/\mathcal{T}_G.$$

Using XI 6.8, and 7.1 d), which ensures the representability of $\mathcal{T}$ by a smooth S -prescheme under certain conditions, one could conclude the representability of Z under certain conditions, which we shall, however, obtain by a more direct route below.

⁽¹²⁾N.D.E. Indicate references here.

Definition 8.6. Let G be a smooth S -group prescheme of finite presentation with connected fibers. One says that G *admits a reductive center* if the preceding functor Z (which is obviously a subgroup of G) is representable by a group of multiplicative type. One then says that Z is the *reductive center* of G .

One will note that if Z is a reductive center of G , then for every base change $S' \rightarrow S$, $Z_{S'}$ is a reductive center of $G_{S'}$; on the other hand, the existence of a reductive center is obviously a local question for the fpqc topology. As for the terminology “reductive center”, note that Z is in any case central, for obviously Z is invariant under

$$\underline{\mathrm{Aut}}_S(G)$$

and a fortiori is an invariant subgroup of G , and one applies IX 5.5.

Lemma 4.5 must here be replaced by:

Lemma 8.7. *Let $u : H \rightarrow G$ be a central homomorphism, where H is of multiplicative type and of finite type over S ; suppose that for every algebraically closed field k over S , $u_k(H_k)$ is contained in the largest smooth connected affine subgroup of G_k . Then u factors through every maximal torus T of G (hence u in fact factors through the subfunctor Z of G defined above).*

Using an easy variant of IX 5.1 bis (where the sign $=$ would be replaced by an inclusion sign), one is reduced to the case where S is the spectrum of a field k , which one can suppose algebraically closed. (Reduce to the case of affine noetherian S , then artinian, then use IX 3.6). Since T is contained in the largest smooth connected affine subgroup V of G , one is then reduced to the case where $G = V$, i.e. where G is affine, where the result was proved in 4.5.

Proposition 8.8. *Let G be a smooth S -group prescheme of finite presentation with connected fibers.*

- a) *If S is the spectrum of a field k , then G admits a reductive center Z . When G is an extension of an abelian variety by a smooth connected affine algebraic group V (for example if k is algebraically closed), then Z is also the reductive center of V , and it is the largest central subgroup of multiplicative type of V .*
- b) *Let Z be a group sub-prescheme of multiplicative type of G . Then Z is a reductive center of G if and only if for every $s \in S$, Z_s is a reductive center of G_s . Then Z is the largest subgroup K of multiplicative type of G such that for every $s \in S$, K_s is contained in the reductive center of G_s ; more generally, for every homomorphism $u : H \rightarrow G$, with H of multiplicative type and of finite type over S , such that for every algebraically closed field k over S , $u_k : H_k \rightarrow G_k$ factors through the largest smooth connected affine subgroup of G_k , u factors through Z (and in particular u is central).*
- c) *If G admits locally for the fpqc topology a maximal torus, then G admits a reductive center.*
- d) *Let T be a maximal torus of G . Then $T \cap \mathrm{Centr}(G) = \mathrm{Ker}(T \rightarrow \mathrm{GL}(\mathfrak{g}))$ and this is a reductive center of G .*
- e) *Let Z be a reductive center of G , and suppose $G' = G/Z$ representable (for example S artinian); then G' admits the identity subgroup as reductive center, and $T' \mapsto T = u^{-1}(T')$ establishes a one-to-one correspondence between maximal tori of G' and maximal tori of G .*

PROOF. a) Suppose that S is the spectrum of a field k . To prove then the existence of a reductive center, one can suppose k algebraically closed, and consequently one is reduced to the case where G is an extension of an abelian variety by a smooth connected affine algebraic group V . Since for every S' over k , the maximal tori of $G_{S'}$ are those of $V_{S'}$ (by IX 5.2 and the fact that over an algebraically closed field a homomorphism from a torus to an abelian variety is trivial), it follows that the functor Z defined above in terms of G is the same as that

defined in terms of V . One is thus reduced to the case of affine G . Since $\mathcal{T}$ is representable, and necessarily “essentially free” over k (VIII 6.1), it follows that $\mathcal{T}_G$ is essentially free over G , and since $\underline{T}$ is a closed subscheme of $\mathcal{T}_G = G_{\mathcal{T}}$ (by, for example, VIII 5.7), it follows by VIII 6.4 that $Z = \prod_{\mathcal{T}_G/G} \underline{T}/\mathcal{T}_G$ is representable by a closed subscheme of G . Thus it is a group subscheme of G ; I say that it is of multiplicative type: indeed, one can suppose k algebraically closed; then G admits a maximal torus T , and by definition one will have $Z \subset T$, so Z is of multiplicative type as an algebraic subgroup of a group of multiplicative type (IX.8). This proves that Z is a reductive center of G . The fact that it is the largest central subgroup of multiplicative type of V is contained in 8.7.

b) The “only if” being trivial, let us prove that if Z is a subgroup of multiplicative type of G such that for every $s \in S$, Z_s is the reductive center of G_s , then Z is a reductive center of G . One must first prove that Z is contained in every maximal torus of G (which will therefore remain true after every base change): this is an immediate consequence of 8.7. Next, one must prove that if g is a section of G over S such that for every S' over S , and every maximal torus T' of $G_{S'}$, $g_{S'}$ is a section of T' , then g is a section of Z . Note that one could reduce as usual (taking account of the fact that $\prod_{\mathcal{T}_G/G} \underline{T}/\mathcal{T}_G = Z'$ is a sheaf for fpqc that commutes with inductive limits of rings) to the case of affine S , then noetherian S , and finally artinian S . Then $\mathcal{T}$ is representable by a smooth prescheme over S by 7.1 d), so proceeding as in a), one sees that $\prod_{\mathcal{T}_G/G} \underline{T}/\mathcal{T}_G$ is representable by a closed subscheme Z' of G . We have already seen that $Z \subset Z'$; on the other hand, by the hypothesis on Z , one has $Z_k = Z'_k$ (where k is the residue field), but since Z is flat over S , it follows that $Z = Z'$, which proves that Z is a reductive center of G . The other assertions of b) are contained in 8.7.

c) Reduces immediately to d). d) Of course, $\mathfrak{g}$ denotes the Lie algebra of G , and $T \rightarrow \mathrm{GL}(\mathfrak{g})$ the homomorphism induced by the adjoint representation of G . One trivially has

$$T \cap \mathrm{Centr}_G \subset \mathrm{Ker}(T \rightarrow \mathrm{GL}(\mathfrak{g}));$$

the proof of the reverse inclusion is the same as in 4.7 d), and we do not repeat it here. Let Z be the group in question; as $\mathrm{Ker}(T \rightarrow \mathrm{GL}(\mathfrak{g}))$ it is of multiplicative type (cf. for example IX 6.8). To prove that it is a reductive center of G , one is reduced by b) to the case where S is the spectrum of a field. Let then Z' be the reductive center (which exists by a)); one obviously has $Z' \subset T \cap \mathrm{Centr}(G) = Z$; on the other hand, since Z is a central subgroup of multiplicative type contained in the smooth connected affine subgroup T , it follows from a) that $Z' \supset Z$, hence $Z' = Z$. Q.E.D.

e) Under the conditions of 7.1 f), put $Z = \mathrm{Ker} u$ and suppose that for every algebraically closed field k over S , Z_k is contained in a maximal torus of G_k . Then one observes easily that the map $T' \mapsto u^{-1}(T')$ induces a one-to-one correspondence between the set of maximal tori of G' and the set of maximal tori of G . Applying this to situation 8.8 e), the desired conclusion follows at once. □

Remarks 8.9. a) The proof given of 8.8 is independent of the results of no. 4, and in particular that of 8.8 a) does not use 4.4 (whose proof is somewhat laborious).

b) One sees easily that the subgroup Z of G considered in 8.6 is always central (whether representable or not), and it may be tempting to call it the reductive center of G in all cases.

c) One can also generalize 4.9; one finds the following statement: let G be a smooth connected algebraic group over an algebraically closed field; for G to be an extension of an abelian variety by a smooth connected unipotent algebraic group (i.e. for the reductive rank of G to be zero, cf. 8.3), it is necessary and sufficient that the reductive center of G be reduced to the identity group, and that the Lie algebra of G be nilpotent.

9. Supplement: action of a group scheme and fixed points

The aim of this section, added in January 2008, is to study the fixed points under the action of a group scheme.

9.1. — Representability of the fixed-point functor.—Let S be a scheme, G an S -group scheme acting on an S -scheme X . One defines the subfunctor X^G of X of fixed points of X under G : for every S -scheme T , $X^G(T)$ is the subset of $X(T)$ formed by the fixed points under G (VIII.6.e).

Let us recall the notion of an “ S -pure scheme” introduced in ([G-R], § 3.3, [R]). Let G be a flat S -group scheme of finite presentation. Since the fibers of G have no embedded components, the notion of purity for G takes the following form. The scheme G is S -pure if for every strictly henselian local S -scheme T , with closed point t , every generic point $\tilde{x}$ of a fiber of $G \times_S T$ specializes to a point of G_t , i.e. the closure of $\tilde{x}$ in $G \times_S T$ meets G_t .

Examples.—

- (1) If G is quasi-finite and separated over S , G is S -pure if and only if G is finite over S .
- (2) If $S = \operatorname{Spec}(R)$ and $G = \operatorname{Spec}(A)$, G is S -pure if and only if A is a projective R -module ([G-R], 3.3.5). In particular G is S -pure if G is diagonalizable.
- (3) A group scheme of multiplicative type is S -pure, for the notion of purity is local for the étale topology on S .
- (4) The scheme G is S -pure if G is S -proper or if the fibers of G are irreducible or if S is semilocal artinian.
- (5) In particular, a reductive S -group scheme G is S -pure.

Proposition 9.2. *Let G be an S -group scheme of finite presentation, S -flat and S -pure, acting on an S -scheme X of finite presentation. Then the functor X^G of fixed points of X under G is representable by a closed subscheme of X , of finite presentation over S .*

Indeed, let a be in $X(S)$ and let H be the group subscheme of G that is the stabilizer of a . Then a is fixed under G if and only if $H = G$. Now the subfunctor C of S of coincidences (more precisely, if T is an S -scheme, $C(T) = \{\emptyset\}$ if $H \times_S T \rightarrow G \times_S T$ is bijective, and $C(T) = \emptyset$ otherwise) of H with G is representable by a closed subscheme of S , defined by a sheaf of ideals of finite type ([G-R], 4.1.1). One applies this result with $S = X$, taking for a the universal point of X (i.e. the identity in $X(X) = \operatorname{Hom}_S(X, X)$).

9.3. — Infinitesimal obstruction.—Under the hypothesis that G is a flat S -group scheme of finite presentation acting (on the left) on a smooth S -prescheme X , we shall now study the formal smoothness of the functor X^G .

One supposes here that S is endowed with a closed subscheme S_0 defined by a sheaf of quasi-coherent ideals $\mathcal{I}$ of square zero. One writes $j : S_0 \rightarrow S$, $G_0 = G \times_S S_0$, $X_0 = X \times_S S_0$.

Let ϵ_0 be a point of $X^G(S_0)$ lifting to a point of $X(S)$. We shall study the obstruction to lifting ϵ_0 to a point of $X^G(S)$. Since X is smooth, one knows that the liftings of ϵ_0 to $X(S)$ form a trivial principal homogeneous space under the abelian group

$$\operatorname{Hom}_{\mathcal{O}_{S_0}}(\epsilon_0^*(\Omega_{X_0/S_0}^1), \mathcal{I})$$

(III.0.2, 0.3). One then puts

$$L_0 = \underline{\operatorname{Hom}}_{\mathcal{O}_{S_0}}(\epsilon_0^*(\Omega_{X_0/S_0}^1), \mathcal{I}),$$

which is an $\mathcal{O}_{S_0}$ -module. Moreover, since ϵ_0 is fixed under G , L_0 is naturally a G_0 - $\mathcal{O}_{S_0}$ -module. One denotes by $\rho_0 : G_0 \rightarrow \underline{\operatorname{Aut}}_{S_0}(L_0)$ (resp. $\rho : G \rightarrow \underline{\operatorname{Aut}}_S(j_*L_0)$) the associated representation.

Lemma 9.4. *There exists a certain class $c(\epsilon_0) \in H^1(G, j_*L_0) \simeq H^1(G_0, L_0)$, defined canonically by ϵ_0 , such that ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.*

The adjunction $H^1(G, j_*L_0) \simeq H^1(G_0, L_0)$ is that of lemma III.1.1.2. We shall work with the small fppf site over S . For every scheme T flat and of finite presentation over S , write $G_T = G \times_S T$, $X_T = X \times_S T$ for the corresponding objects over T . Consider the sheaf A on

the small fppf site of S such that for every T flat and of finite presentation over S , one has:

$$A(T) = \{\text{set of liftings of } \epsilon_0 \times_S T \text{ to } X(T)\}.$$

Since the formation of L_0 commutes with flat base changes, the preceding considerations indicate that $A(T)$ is a trivial principal homogeneous space under the abelian group $H^0(T, j_* L_0)$. Again because ϵ_0 is fixed, for every T flat and of finite presentation over S , and every g in $G(T)$, g acts by affine automorphisms on $A(T)$, compatibly with the action of G on $j_* L_0$. That is:

$$g(a_T + v) = g(a_T) + \rho(g)(v), \quad v \in H^1(T, j_* L_0).$$

Since G is flat of finite type, one can apply these considerations by taking $T = G$ and for g the universal point id_G of G . One thus obtains an action of the fppf sheaf G on A .

Let a be a lifting of ϵ_0 to $X(S)$. Denote by $g^\sharp$ the universal point of G and define $v^\sharp \in H^0(G, j_* L_0)$ by

$$\rho(g^\sharp)(v^\sharp) = g^\sharp \cdot a_G - a_G.$$

For every S -prescheme Y , put $z(Y) : G(Y) \rightarrow H^0(Y, j_* L_0)$, $g \mapsto v^\sharp(g)$. This defines the 1-cocycle z in $Z^1(G, L)$ (exposé I, § 5).

Its class $c(\epsilon_0)$ in $H^1(G, j_* L_0)$ does not depend on the choice of a . In particular, if ϵ_0 lifts to $X^G(S)$, one has $c(\epsilon_0) = 0$. Conversely, if $c(\epsilon_0) = 0$, then there exists $w \in H^0(S, j_* L_0)$ such that $z(Y)(g) = g \cdot w_Y - w_Y$ for every S -prescheme Y and every $g \in G(Y)$. Applying this to $Y = G$ and to $g^\sharp$, one concludes that $a - w \in X^G(S)$. We have therefore established that ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

Remark 9.5. In the case where G is affine and flat of finite type over affine S , one can refine this obstruction to a class $\tilde{c}(\epsilon_0) \in H_{G_0}^1(X_0, L_0)$, where $H_{G_0}^1(X_0, L_0)$ is the G_0 -equivariant cohomology group defined by Wevers ([W], app. C). In this case, it is not necessary to suppose that ϵ_0 lifts to $X(S)$.

9.6. — Smoothness of fixed points.—

Theorem 9.7. *Let G be a flat S -group scheme of finite type over a noetherian base S , acting on a smooth S -scheme X . Suppose that G is S -pure, and for every geometric point s over S (with algebraically closed residue field), one has*

$$H^1(G_s, (\Omega_{X_s}^1)^*) = 0.$$

Then the functor X^G of fixed points of X under G is representable by a closed subscheme of X smooth over S .

Let us prove theorem 9.7. One knows that X^G is representable by a closed subscheme of X of finite presentation over S . One can suppose $S = \text{Spec}(A)$ local. Following the smoothness criterion (SGA 1 III.3.1.iii.bis), it suffices to verify the formal smoothness of X^G for $S' = \text{Spec}(A')$ and the closed subscheme $S'_0 = \text{Spec}(A'/\mathcal{I}') = \text{Spec}(A'_0)$, where A' is an artinian local ring and $\mathcal{I}'$ a square-zero ideal of A' . By reduction, one is reduced to the case where $\mathcal{I}'$ is annihilated by the maximal ideal of A' . In particular, if k denotes the residue field of A' , $\mathcal{I}'$ is a k -vector space.

One therefore gives oneself an element $\epsilon'_0 \in X^G(S'_0)$ which one wishes to lift to $X^G(S')$. One then writes $X' = X \times_S S'$, $G' = G \times_S S'$, $X'_0 = X \times_S S'_0$, $G'_0 = G \times_S S'_0$. Since S' is affine and X' is smooth over S' , $\epsilon'_0 \in X^G(S'_0)$ lifts to $X(S')$. According to lemma 9.4, the class

$$c(\epsilon_0) \in H^1(G'_0, L'_0)$$

is the obstruction to lifting ϵ'_0 to $X^G(S')$, where $L'_0 = \underline{\text{Hom}}_{\mathcal{O}_{S'_0}}(\epsilon'_0{}^*(\Omega_{X'_0/S'_0}^1), \mathcal{I}')$. Since L'_0 is a k -vector space, one has a canonical isomorphism $H^1(G'_0, L'_0) \xrightarrow{\sim} H^1(G'_0 \times_{A'_0} k, L'_0)$. It

suffices to show that $H^1(G'_0 \times_{A'_0} k, L'_0) = 0$. Writing $\bar{k}$ for an algebraic closure of k , one has an isomorphism (IX.3.1 in the affine case, lemma 9.11 in general)

$$H^1(G'_0 \times_{A'_0} k, L'_0) \otimes_k \bar{k} \xrightarrow{\sim} H^1(G'_0 \times_{A'_0} \bar{k}, L'_0 \otimes_k \bar{k}).$$

One then observes that the representation $L'_0 \otimes_k \bar{k}$ is a direct sum of $(\Omega^1_{X'_0 \times_k \bar{k}})^*$. By hypothesis one has $H^1(G'_0 \times_{\bar{k}} A'_0, (\Omega^1_{X'_0 \times_k \bar{k}})^*) = 0$, which implies $H^1(G'_0 \times_{A'_0} k, L'_0) = 0$.

Corollary 9.8. *Let G be a flat S -group scheme of finite presentation acting on a smooth S -scheme X of finite presentation. Suppose that G admits a composition series whose factors are of one of the following types:*

- (1) *S -abelian scheme (i.e. G is smooth over S and its fibers are abelian varieties),*
- (2) *S -group scheme of multiplicative type,*
- (3) *finite étale S -group scheme of degree invertible on S ,*
- (4) *reductive S -group if S is a $\mathbf{Q}$ -scheme.*

Then the functor X^G of fixed points of X under G is representable by a closed subscheme of X , of finite presentation, smooth over S .

Indeed, if $1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1$ is an exact sequence of S -groups satisfying the hypotheses of the theorem, the S -group G'' acts on the functor $X^{G'}$ and the functor X^G is nothing other than that of fixed points of $X^{G'}$ under G'' . Thus one is reduced to the case of an S -group of one of the four types above. By passage to the limit, one can suppose S noetherian. One will verify the cohomological criterion stated above for a fiber G_s over a geometric point s of S and a finite-dimensional linear representation V of G_s .

In the case of an abelian scheme, every map from G_s^n to V is constant and consequently the $H^i(G_s, V)$ vanish for $i > 0$.

In the case of a group scheme of multiplicative type, G_s is a diagonalizable group. Theorem I.5.3.3 shows that the $H^i(G_s, V)$ vanish for $i > 0$.

In the next two cases, the argument is the same, since it rests on the semisimplicity of linear representations of G_s . Indeed, in the case of a finite étale S -group scheme of degree invertible on S , G_s is a finite constant group of degree invertible in the residue field $\kappa(s)$; it is well known that every linear representation of G_s is semisimple.

For the reductive case, the residue field $\kappa(s)$ is supposed (algebraically closed) of characteristic zero. One then knows that every linear representation of G_s is semisimple (see [T-Y], theorem 27.3.3).

Remark 9.9. This applies in particular to the case of an action of G/S on a smooth S -group scheme H . If G is a group of multiplicative type acting by inner transformations on H smooth and affine over S , one then recovers corollary XI.5.3 asserting the smoothness of the centralizer of G in H .

Remark 9.10. In the case where $S = \text{Spec}(k)$ and G is a linearly reductive algebraic group defined over the algebraically closed field k , this result is due independently to Fogarty ([F], theorem 5.4) and Iversen ([I], proposition 1.3).

The proof of theorem 9.7 uses the following lemma, well known in the case of an affine group scheme (IX.3.1).

Lemma 9.11. *Let G be a group scheme over an affine scheme $S = \text{Spec}(A)$, and $f : S' = \text{Spec}(A') \rightarrow S$ a flat morphism. Put $G' = G \times_S S'$. Then for every quasi-coherent G - $\mathcal{O}_S$ -module M , one has canonical isomorphisms*

$$H^i(G, M) \otimes_A A' \xrightarrow{\sim} H^i(G', M \otimes_A A') \quad (i \geq 0).$$

Since chains of degree $n \geq 1$ for Hochschild cohomology are determined by their value at the point $(\text{id}, \dots, \text{id}) \in G(G) \times \dots \times G(G)$, one has an isomorphism of A -modules $C^n(G, M) \simeq$

$\Gamma(G^n, M \otimes_A \mathcal{O}_{G^n})$. The cohomology group $H^n(G, M)$ is the n th cohomology group of the complex

$$\begin{aligned} L \longrightarrow \Gamma(G, M \otimes_A \mathcal{O}_G) &\longrightarrow \Gamma(G^2, M \otimes_A \mathcal{O}_{G^2}) \\ &\longrightarrow \Gamma(G^3, M \otimes_A \mathcal{O}_{G^3}) \longrightarrow \cdots \end{aligned}$$

Since A' is flat over A , by EGA IV₁, 1.7.21, one has

$$\Gamma(G^n, M \otimes_A \mathcal{O}_{G^n}) \otimes_A A' \xrightarrow{\sim} \Gamma(G'^n, (M \otimes_A A') \otimes_{A'} \mathcal{O}_{G'^n}).$$

This implies the desired assertion on taking cohomology.

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Exposé XIII. Regular elements of algebraic groups and of Lie algebras

by A. Grothendieck

1. An auxiliary lemma on varieties with operators

Let S be a prescheme, G an S -group prescheme acting on the left on an S -prescheme V , W a closed sub- S -prescheme of V , and N its stabilizer in G , the subgroup of G whose points with values in an S' over S are the $g \in G(S')$ such that $g \cdot W_{S'} = W_{S'}$. One equips $((\text{Sch})/S)$ with the faithfully flat quasi-compact topology, and identifies G, V, W with the corresponding sheaves (cf. IV). One shall therefore reason in the category of sheaves on $((\text{Sch})/S)$, and in this section the expression “locally” refers to the topology just specified on $((\text{Sch})/S)$. Note that N is a sheaf; consider the quotient sheaf G/N . One sees immediately that it is isomorphic to the following functor: to each S' over S , one associates the set of subsheaves W' of $V_{S'}$ which are locally conjugate to $W_{S'}$ under the group G . Let X be the subsheaf of $G/N \times_S V$ whose value, for every S' over S , is the set of (W', v) , where W' is as above and v is a section of W' over S' (thus a section of $V_{S'}$ over S'). Let Z be the inverse image of X in $G \times_S V$, so that one has the cartesian diagram

$$(x) \quad \begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow i \\ G \times_S V & \longrightarrow & G/N \times_S V, \end{array}$$

where i is the canonical immersion, and the second horizontal arrow comes from the canonical morphism $G \rightarrow G/N$. Since this associates to the point $g \in G(S')$ the subsheaf $g \cdot W_{S'}$ of $V_{S'}$, one sees that $Z(S')$ is the set of pairs $(g, v) \in G(S') \times V(S')$ such that $v \in g \cdot W_{S'}(S')$. Consequently, Z is isomorphic to the sheaf $G \times_S W$, by means of the isomorphism

$$G \times_S W \xrightarrow{\sim} Z$$

defined by $(g, w) \mapsto (g, g \cdot w)$. Thus the preceding cartesian diagram gives the cartesian diagram

$$(xx) \quad \begin{array}{ccc} G \times_S W & \xrightarrow{q} & X \\ \lambda \downarrow & & \downarrow i \\ G \times_S V & \longrightarrow & G/N \times_S V \end{array}$$

where $\lambda(g, w) = (g, g \cdot w)$, hence $q(g, w) = (\bar{g}, g \cdot w)$, where $\bar{g}$ denotes the image of g under the canonical map $G(S') \rightarrow (G/N)(S')$. Finally, one sees from diagram (x) that $Z \rightarrow X$ makes Z a principal bundle with base X and group N acting on the right by $(g, v) \cdot n = (gn, v)$, so that in (xx), $q : G \times_S W \rightarrow X$ makes $G \times_S W$ a principal bundle with base X and group N acting on the right by

$$(g, w) \cdot n = (gn, n^{-1} \cdot w).$$

One summarizes the main preceding morphisms in the following diagram:

$$(D) \quad \begin{array}{ccccc} & & G \times_S W & & \\ & & \downarrow q & \searrow \varphi & \\ G/N & \xleftarrow{p} & X & \xrightarrow{\psi} & V \\ & \swarrow \text{pr}_1 & \downarrow i & \searrow \text{pr}_2 & \\ & & G/N \times_S V & & \end{array}$$

where $\psi = \text{pr}_2 \circ i$ and $\varphi = \psi \circ q$, i.e. $\varphi(g, w) = g \cdot w$. If v is a section of V over S , the subsheaf X_v of X which is the inverse image of this section under ψ is given by $X_v(S') =$ the set of subsheaves W' of $V_{S'}$ which are locally conjugate to $W_{S'}$ under G , and which contain the section $v_{S'}$ of $V_{S'}$, while the subsheaf of $G \times_S W$ which is the inverse image of v under φ is isomorphic to the subsheaf M_v of G given by $M_v(S') =$ the set of $g \in G(S')$ such that $v_{S'} \in g \cdot W(S')$, i.e. such that $g^{-1}v_{S'} \in W(S')$. If v is a section of W and not merely of V , then M_v clearly contains N .

In these descriptions, one has not used the fact that G, V, W were representable (nor that the site on which one works is defined in terms of preschemes!). But suppose now that N is representable and faithfully flat and quasi-compact over S , and that G/N is representable. When S is the spectrum of a field, and G is of finite type over k , one knows that this hypothesis is necessarily satisfied (VIII 6 and VI_B.11.18). One then sees from the cartesian diagram (xx), using faithfully flat quasi-compact descent theory and the fact that $Z \rightarrow G \times_S V$ is a closed immersion, that X is representable (it is obtained by descent of the closed subscheme Z of $G \times_S V$ along the faithfully flat and quasi-compact morphism $G \times_S V \rightarrow G/N \times_S V$). Thus diagram (D) is a diagram of morphisms of preschemes over S .

From now on one assumes that S is the spectrum of a field k , and that G, V, W are of finite type over k . Let $\mathfrak{n}$ be the Lie algebra of N , so that $\dim N \leq \text{rank } \mathfrak{n}$, equality holding if and only if N is smooth over k (Exp VI⁽¹⁾). Let $a \in W(k)$, and consider the subscheme M_a of G defined above, containing N , and isomorphic to $\varphi^{-1}(a)$; one shall denote by $\mathfrak{m}_a$ its Zariski tangent space at the identity element e of N , so that one has

$$(1) \quad \mathfrak{n} \subset \mathfrak{m}_a, \quad \dim N \leq \text{rank } \mathfrak{n} \leq \mathfrak{m}_a.$$

Lemma 1.1. *With the preceding notation:*

a) *Consider the following conditions:*

- (i) $\mathfrak{n} = \mathfrak{m}_a$ and N is smooth over k .
- (i bis) $\dim N = \text{rank}_k \mathfrak{m}_a$.
- (ii) $\psi : X \rightarrow V$ is unramified at $(\bar{e}, a)$.
- (iii) M_a and N coincide in a neighborhood of e .

Then one has the implications (i) $\Leftrightarrow$ (i bis) $\Rightarrow$ (ii) $\Leftrightarrow$ (iii).

b) Suppose that $\varphi : G \times_S W \rightarrow V$ is smooth at (e, a) . Then M_a is smooth over k at e , and ψ is smooth at $(\bar{e}, a)$.

PROOF. a) The equivalence of (i) and (i bis) follows at once from relations (1) and from the fact noted above that N is smooth over k if and only if $\dim N = \text{rank}_k \mathfrak{n}$. On the other hand, consider the inclusion morphism $N \rightarrow M_a$; it is well known⁽²⁾ that if N is smooth over k at e and the tangent map at e is surjective, then $N \rightarrow M_a$ is smooth at e , and hence (being an immersion) is an isomorphism at e , which shows that (i) implies (iii). To prove the equivalence of (ii) and (iii), consider, as above, $X_a = \psi^{-1}(a)$ and use the isomorphism

⁽¹⁾**N.D.E.** Having been unable to identify this reference, we refer to Theorem II.5.2.1 of the book: M. Demazure & P. Gabriel, *Groupes algébriques I*, Masson (1970).

⁽²⁾**N.D.E.** See, for example, EGA IV₄, Th. 17.11.1 d).

$M_a \simeq \varphi^{-1}(a) \simeq q^{-1}(X_a)$ to obtain a morphism $p_a : M_a \rightarrow X_a$ which makes M_a a principal homogeneous bundle with group N_{X_a} . Consider the following diagram

$$\begin{array}{ccc} M_a & \xleftarrow{j_a} & N \\ p_a \downarrow & & \downarrow \\ X_a & \xrightarrow{j'_a} & S = \operatorname{Spec}(k), \end{array}$$

where $\operatorname{Spec}(k) \rightarrow X_n$ is defined by the point $(\bar{e}, a)$ of X , and $j_a : N \rightarrow M_a$ is the canonical immersion. Saying that ψ is unramified at $(\bar{e}, a)$ means that j'_a is an open immersion, or again that it induces an isomorphism $S \xrightarrow{\sim} \operatorname{Spec}(\mathcal{O}_{X_a, a})$. Since p_a is flat, it is equivalent to say that the morphism obtained from the preceding one by the base change $\operatorname{Spec}(\mathcal{O}_{M_a, e}) \rightarrow \operatorname{Spec}(\mathcal{O}_{X_a, b})$ is an isomorphism; now this resulting morphism is precisely the morphism $\operatorname{Spec}(\mathcal{O}_{N, e}) \rightarrow \operatorname{Spec}(\mathcal{O}_{M_a, e})$, which proves the equivalence of (ii) and (iii).

One will moreover note that the proof shows that conditions (ii), (iii) imply the following condition, apparently stronger than (iii):

(iii bis) N is an open and closed subscheme of M .

b) The first assertion follows from the fact that M_a is isomorphic to $\varphi^{-1}(a)$, the second from the fact that q is flat and $q(e, a) = (\bar{e}, a)$. $\square$

2. Density theorem and theory of regular points of G

One shall apply the constructions and notation of the preceding section in the case where G is a smooth connected algebraic group over k , where $V = G$ on which G acts by inner automorphisms, and where W is a smooth connected algebraic subgroup H of G . One shall denote by $\mathfrak{g}$ the Lie algebra of G , by $\mathfrak{h}$ that of H , by N the normalizer of H in G , and by $\mathfrak{n}$ the Lie algebra of N . If $a \in G(k)$, one shall again denote, as in no. 1, by M_a the inverse of its transporter into H , so that if $a \in H(k)$, one has $N \subset M_a$; in this case, one denotes by $\mathfrak{m}_a$ the Zariski tangent space of M_a at the identity element e of G . Note that

$$\mathfrak{h} \subset \mathfrak{n} \subset \mathfrak{m}_a \subset \mathfrak{g} \quad \text{for } a \in H(k).$$

One shall need the

Lemma 2.0. *For $H = N^0$ to hold, it is necessary and sufficient that $(\mathfrak{g}/\mathfrak{h})^H = 0$ (where the left-hand side denotes the subspace of invariants under the action of H deduced from the adjoint representation). When this condition is satisfied, N is smooth and one has $\dim X = \dim G$. In all cases, $\dim X \leq \dim G$, and this inequality is an equality if and only if H has finite index in N .*

Indeed, one has seen (II 5.2.3 (i)) that $\mathfrak{n}$ is equal to the inverse image of $(\mathfrak{g}/\mathfrak{h})^H$ under the morphism $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$, hence $(\mathfrak{g}/\mathfrak{h})^H = 0$ is equivalent to $\mathfrak{h} = \mathfrak{n}$, which is also equivalent (H being a smooth connected algebraic subgroup of the algebraic group N) to $H = N^0$ (cf. VI.2). This clearly implies that N is smooth. On the other hand, one has

$$\dim X = (\dim G - \dim N) + \dim H = \dim G - (\dim N - \dim H),$$

so that one has

$$\dim X \leq \dim G,$$

equality being attained if and only if $\dim H = \dim N$, i.e. if and only if H has finite index in N . This is the case in particular if $H = N^0$, which completes the proof of 2.0.

Theorem 2.1. *Let G be a smooth connected algebraic group over the algebraically closed field k , H a smooth connected algebraic subgroup, N its normalizer, $\mathfrak{g}, \mathfrak{h}, \mathfrak{n}$ the Lie algebras, $X = G \times^N H$ the scheme (fibered over G/N with typical fiber H) introduced in no. 1, and $\psi : X \rightarrow G$ the canonical morphism (whose image is also the image of $\varphi : G \times H \rightarrow G$ defined by $\varphi(g, h) = \operatorname{int}(g)h = ghg^{-1}$). The following conditions are all equivalent:*

- (i) H contains a Cartan subgroup (XII 1) C of G .
- (i bis) H has the same reductive rank and the same nilpotent rank (XII 1) as G .
- (ii) H contains a maximal torus T of G , and $(\mathfrak{g}/\mathfrak{h})^T = 0$.
- (iii) The set of conjugates of H containing a given maximal torus is finite and nonempty, and H has finite index in its normalizer.
- (iv) There exists $a \in H(k)$ which is contained in only finitely many conjugates of H (or merely such that $\psi^{-1}(a)$ has an isolated point), and H has finite index in its normalizer.
- (iv bis) The morphism $\psi : X \rightarrow G$ is generically quasi-finite (i.e. there exists a dense open subset of X on which ψ is quasi-finite), and H has finite index in its normalizer.
- (v) There exists a dense open subset U of G such that for every $x \in U(k)$, the set of conjugates of H containing x is finite and nonempty, i.e. $\psi : X \rightarrow G$ is dominant and generically quasi-finite.
- (vi) There exists a dense open subset U of G such that every $x \in U(k)$ is contained in a conjugate of H , i.e. $\psi : X \rightarrow G$ is dominant.
- (vii) There exists $a \in H(k)$ such that the subspace of $\mathfrak{g}/\mathfrak{h}$ of fixed points of $\text{ad}_{\mathfrak{g}/\mathfrak{h}}(a)$ is zero.

Moreover, these conditions imply that H is its own connected normalizer, i.e. N is smooth and $\dim H = \dim N$, and that $\psi : X \rightarrow G$ is generically étale.

PROOF. By 2.0, one has $\dim X \leq \dim G$, with equality if and only if $\dim H = \dim N$, i.e. H has finite index in N . From the inequality $\dim X \leq \dim G$ it follows that ψ is dominant if and only if it is dominant and generically quasi-finite, or again if and only if ψ is generically quasi-finite and $\dim X = \dim G$. Since this last equality also means, by 2.0, that H has finite index in N , one has proved the equivalence of (vi), (v), (iv bis). The equivalence of (iv) and (iv bis) is immediate.

The equivalence of (i) and (i bis) is immediate from the definitions, and is left to the reader. On the other hand, if H contains a Cartan subgroup C of G , it contains the maximal torus T of C , which is a maximal torus of G . Since C is the centralizer of T , its Lie algebra $\mathfrak{c}$ is given by

$$\mathfrak{c} = \mathfrak{g}^T$$

(II, 5.2.3 (ii)). Thus, since $H \supset C$, whence $\mathfrak{h} \supset \mathfrak{n}$, it follows that $\mathfrak{h} \subset \mathfrak{g}^T$, which is equivalent, by I, 4.7.3, to the relation

$$(x) \quad (\mathfrak{g}/\mathfrak{h})^T = 0.$$

Conversely, suppose that H contains the maximal torus T and that the preceding relation holds, i.e. that one has $\mathfrak{h} \supset \mathfrak{n}$; I say that H contains the centralizer C of T (which will establish (i) $\Leftrightarrow$ (ii)). This follows from the

Lemma 2.1.1. *Let G be a smooth algebraic group over the field k , T a subgroup of multiplicative type of G , C its connected centralizer (equal to the centralizer of T if G is connected and T a torus, (XII 6.6 b)), and H a smooth subgroup of G containing T . For H to contain C , it is necessary and sufficient that its Lie algebra $\mathfrak{h}$ contain that $\mathfrak{c}$ of C .*

Indeed, one knows (XI 2.4) that $\text{Centr}_G(T)$ is smooth over k , hence C is smooth over k ; likewise $\text{Centr}_H(T)$ is smooth over k ; now $\text{Centr}_H(T) = \text{Centr}_G(T) \cap H$ has $\mathfrak{h} \cap \mathfrak{c}$ as Lie algebra, so the hypothesis implies that the smooth subgroup $\text{Centr}_H(T)$ of the smooth group $\text{Centr}_G(T)$ has the same Lie algebra, hence it contains the connected component C of the latter, so H contains C . Q.E.D.

Let us prove the equivalence of (i) and (iii), which amounts to proving that if H contains the maximal torus T of G , then the condition $H \supset C$ (which is also equivalent to (x) above, as one has just seen) is equivalent to the fact that H has finite index in its normalizer and that the set of conjugates of H containing T is finite. If H contains C , hence if one has (x),

then a fortiori

$$(xx) \quad (\mathfrak{g}/\mathfrak{h})^H = 0,$$

now one knows that $\mathfrak{n}$ is the inverse image of the left-hand side of the preceding relation under the canonical homomorphism $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$ (II 5.2.3 (i)), so the preceding relation also means, by 2.0, that $H = N^0$; a fortiori H has finite index in its normalizer. Consider now the diagram of subgroups

$$\begin{array}{ccccc} T & \longrightarrow & N(T) \cap H & \longrightarrow & H \\ & & \downarrow & & \downarrow \\ & & N(T) \cap N(H) & \longrightarrow & N(H) \\ & & \downarrow & & \\ & & N(T) & & \end{array}$$

Using the conjugacy theorem for maximal tori in H (XII 6.6 a)), one sees that every conjugate of H containing T is conjugate to H by an element of $N(T)(k)$, hence that the set of conjugates of H containing T is in one-to-one correspondence with the set of points of $N(T)/N(T) \cap N(H)$ with coefficients in k ; now, since $H \supset C$, one has $N(T) \cap H \supset C$, so the preceding set is a quotient of $(N(T)/C)(k)$, which is a finite set, and is therefore finite. This proves that (i) $\Rightarrow$ (iii). Conversely, suppose (iii), i.e. $N(T)/N(T) \cap N(H)$ is finite and $N(H)/H$ is finite. Using the conjugacy theorem in H again, one again sees that the homomorphism

$$N(T) \cap N(H)/N(T) \cap H \rightarrow N(H)/H$$

induced by the preceding diagram is bijective on the points with values in k (in fact, it is an isomorphism), so, since the second is finite, so is the first; hence $N(T) \cap H$ has finite index in $N(T)$, and thus contains $C = N(T)^0$, so $H \supset C$. Thus (i), (i bis), (ii), (iii) are equivalent conditions.

Let us prove that (ii) $\Rightarrow$ (vii). One sees at once that conditions (ii) and (vii) are each invariant under an extension $k \rightarrow k'$ of the base field, with k' algebraically closed, which allows one to suppose that k has infinite transcendence degree over its prime subfield. Then it is well known (and one verifies immediately) that there exists an element a of $T(K)$ such that the subgroup of $T(K)$ which it generates is dense in T for the Zariski topology. One easily concludes that $(\mathfrak{g}/\mathfrak{h})^T = (\mathfrak{g}/\mathfrak{h})^{\text{ad}(a)}$, and since by hypothesis the left-hand side is zero, one concludes (vii).

Let us prove (vii) $\Rightarrow$ (vi). This implication is contained in the following result, which makes 2.1 more precise:

Corollary 2.2. *Let G be a smooth algebraic group over a field k , H a smooth algebraic subgroup, N its normalizer in G , $\varphi : G \times H \rightarrow G$ the morphism defined by $\varphi(g, h) = \text{ad}(g)h = ghg^{-1}$, $\psi : X = G \times^N H \rightarrow G$ the morphism deduced from φ by passage to the quotient (cf. no. 1), and $a \in H(k)$. The following conditions are equivalent:*

- (i) φ is smooth at (e, a) .
- (ii) ψ is étale at $(\bar{e}, a)$, and N is smooth over k .
- (iii) $(\mathfrak{g}/\mathfrak{h})^{\text{ad}(a)} = 0$ (where $\mathfrak{g}, \mathfrak{h}$ are the Lie algebras of G, H).

These conditions imply $H^0 = N^0$.

One knows that the smoothness of φ (which is a morphism of smooth k -preschemes) at a point rational over k is equivalent to the surjectivity of the tangent map at that point. Now an immediate calculation shows that this tangent map is written (by means of the usual identifications of the tangent spaces at points of G and H with the Lie algebra of G and H)

$$d\varphi(\xi, \eta) = (\text{id} - \text{ad}(a)) \cdot \xi + \eta,$$

considered as a map from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Surjectivity is therefore equivalent to the surjectivity of $(\text{id} - \text{ad}(a))$ on $\mathfrak{g}/\mathfrak{h}$, i.e. to (iii). Now (iii) implies a fortiori $(\mathfrak{g}/\mathfrak{h})^H = 0$, i.e. (cf. 2.0, where the connectedness hypothesis at the beginning of the section is unnecessary) $H^0 = N^0$. One deduces that N is smooth, and $\dim H = \dim N$, whence $\dim X = \dim G$. Now, since $q : G \times H \rightarrow X$ is flat and $\psi = \varphi \circ q$ smooth at (e, a) , it follows that ψ is smooth at $q(e, a) = (\bar{e}, a)$, hence étale at this point for reasons of dimension. Thus one has proved (i) $\Leftrightarrow$ (iii) $\Rightarrow$ (ii); on the other hand (ii) $\Rightarrow$ (i), since the smoothness of N implies that of q . Q.E.D.

Finally, let us prove (vi) $\Rightarrow$ (i), which, together with the implications already established, will prove the theorem. Suppose first that G is affine. Let U be a nonempty open subset of G such that $x \in U(k)$ implies that x is contained in a conjugate of H . Let C be a Cartan subgroup of G . Using the implication (i) $\Rightarrow$ (iv) for C in place of H (this is Borel's "density theorem"), it follows that one can find a conjugate of C which meets U ; hence one may suppose that $U \cap C \neq \emptyset$, i.e. that there exists a nonempty open subset V of C such that for every $x \in V(k)$, x is contained in a conjugate of H . Write C as a product

$$C = T \cdot C_u$$

where T is the maximal torus of C (which is a maximal torus of G) and C_u the unipotent part of C , T being in the center of C (BIBLE 6 th. 2). One may again suppose that k has infinite transcendence degree over its prime subfield, which allows one to find an element t of $T(k)$ belonging to the projection of V onto T (which is a nonempty open subset of T), i.e. $t \cdot C_u \cap V \neq \emptyset$, and such that t "generates" T . Since every algebraic subgroup of G which contains a product $t \cdot u$ ($t \in T(k)$, $u \in C_u(k)$) contains both factors (BIBLE 4 th. 3), it follows, with the preceding choice of t , and taking $t \cdot u \in V(k)$, that there exists a conjugate of H which contains t , hence T . Thus one may already suppose that one has

$$T \subset H.$$

If W is the open subset of C_u which is the inverse image of V under $u \mapsto t \cdot u$, one therefore sees that for every element x of $(T \cdot W)(k)$, there exists a conjugate of H which contains T and x . As one has already noted, such a conjugate is of the form $\text{int}(g) \cdot H$, where $g \in N(T)(k)$. Consider then the morphism

$$f : N(T) \times H \rightarrow G$$

defined by $f(g, h) = \text{int}(g) \cdot h = ghg^{-1}$; then the image of f contains $T \cdot W$, so, since $N(T)$ is a finite union of translates $C \cdot g_i$ (where $g_i \in N(T)(k)$), as C has finite index in $N(T)$, it follows that there exists a dense open subset V' of $C = T \cdot C_u$ which is contained in the image of $(C \cdot g_i) \times H$ under f . Replacing H by $\text{int}(g_i) \cdot H$, one may suppose $g_i = e$, i.e. $f(C \times H) \supset V'$. Thus for every $u \in V'(k)$, there exist $v \in C(k)$ and $h \in H(k)$ such that

$$v^{-1}hv = u \quad \text{whence } vuv^{-1} \in H(k),$$

whence, setting

$$C' = C \cap H = \text{Centr}_H(T),$$

$vuv^{-1} \in C'$, whence $\text{int}(v) \cdot C' \rightarrow u$. This proves that the union of the conjugates of C' in C (by elements of $C(k)$) is dense, which implies (as one has seen for the pair (G, H) in place of (C, C')) that C' has finite index in its normalizer in C . By BIBLE 7, the lemma of no. 1, it follows that $C' = C$, hence $C = H$, which proves (vi) $\Rightarrow$ (i) when G is affine.

In the general case, we proceed by induction on $n = \dim G$, the assertion being trivial if $n = 0$. Let Z be the center of G , and distinguish two cases:

1°) $\dim Z \cap H > 0$; then, setting $G' = G/Z \cap H$, one has $\dim G' < n$; on the other hand, hypothesis (vi) on H implies the same condition for the image H' of H in G' , so H' contains a Cartan subgroup C' of G' , hence H contains the inverse image C of C' , which is a Cartan subgroup by XII 6.6 e).

2°) $\dim Z \cap H = 0$, so the canonical morphism $H \rightarrow G/Z$ is a finite morphism, and since G/Z is affine by XII 6.1, it follows that H is affine, hence every homomorphism from H to an abelian variety is zero (and even every morphism of preschemes from H to an abelian

variety is zero): this follows from the fact that a smooth connected affine algebraic group over an algebraically closed field is a rational variety, or merely that it is the union of its Borel subgroups (BIBLE 6 th. 5 b)); now it follows very easily from the structure theorems, BIBLE 6.2 and 6.3, that a smooth connected solvable affine group is a rational variety. Let us now use Chevalley's structure theorem for G , according to which G is an extension of an abelian variety A by a smooth affine group. Then the image of H in A is zero, H being affine; on the other hand it is identical to A , since the union of its conjugates in A must be dense. Thus $A = 0$, so G is affine, and one is reduced to the case already treated. This completes the proof of 2.1. $\square$

Corollary 2.3. *Suppose that the equivalent conditions of 2.1 hold.*

- a) *Let $k(X)$ (resp. $k(G)$) be the field of rational functions of X (resp. G); then $k(X)$ is a finite separable extension of $k(G)$. Denote its degree by d .*
- b) *Let T be a maximal torus of G contained in H (which exists by form 2.1 (ii)), and let C be the corresponding Cartan subgroup of G . Then $C \subset H$. On the other hand, $\text{Norm}_G(T)$ is a smooth subgroup of G and $\text{Norm}_G(T) \cap \text{Norm}_G(H) = \text{Norm}_{\text{Norm}_G(H)}(T)$ is a smooth subgroup of it of finite index equal to d (defined in a)). The number of conjugates of H containing a given maximal torus or a given Cartan subgroup is equal to d .*
- c) *Let U be the largest open subset of G such that $\psi : X \rightarrow G$ induces a morphism $\psi^{-1}(U) \rightarrow U$ which is finite and étale. Then U is a dense open subset, and for $g \in G(k)$, one has $g \in U(k)$ if and only if there exist exactly d conjugates of H containing g , or again if and only if there exist at least d distinct conjugates H_i of H containing g such that for every i , $(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0$ (where $\mathfrak{h}_i = \text{Lie}(H_i)$).*

Assertion a) follows from the fact that ψ is generically étale (which was stated at the end of 2.1); this also implies that the open subset U introduced in c) is nonempty, i.e. dense, and the two stated characterizations of the elements of $U(k)$ (taking into account that ψ is separated, X integral and G integral and normal, SGA1 I 10.11, and the fact that $(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0$ means that ψ is étale at the point x_i of $\psi^{-1}(g)$ corresponding to H_i). If H contains the maximal torus T of G , then the centralizers of T in H and G have the same dimension and are smooth and connected (XII 6.6 b)), hence are equal, which proves that $C \subset G$. Moreover, one knows that the normalizer of T in a smooth group which contains it is smooth (XI, 2.4 bis), so $\text{Norm}_G(T)$ and $\text{Norm}_{\text{Norm}_G(H)}(T)$ are smooth (N.B. it was noted that $N = \text{Norm}_G(H)$ is smooth, at the end of statement 2.1); moreover, $\text{Norm}_N(T)$ contains C , which has finite index in $N(T)$, hence it has finite index in $N(T)$. Using the conjugacy theorem for maximal tori of H , one sees that the index in question is equal to the number of conjugates of H which contain T , or, what amounts to the same thing, which contain C . Now, since the union of the conjugates of C in G is dense (by 2.1 (i) $\Rightarrow$ (vi) applied to C in place of H), and the open subset U defined in c) is clearly stable under inner automorphisms, one sees that $C \cap U \neq \emptyset$. Proceeding as in the proof of implication (vi) $\Rightarrow$ (ii) of 2.1, one concludes that (making, if necessary, a harmless change of base field) there exists a $g \in (C \cap U)(k)$ such that every conjugate of H which contains g contains T , and consequently also C . Thus the conjugates of H containing C are those containing g , and since $g \in U(k)$, their number is equal to d , which finishes establishing b). One will note that we have in fact established that the set of conjugates of H containing T is a homogeneous set under the group of rational points of

$$W_G(T) = \text{Norm}_G(T) / \text{Centr}_G(T),$$

which proves in particular that

$$d \leq \text{order of the Weyl group of } G.$$

Corollary 2.4. *With the notation of 2.1, the following conditions are equivalent:*

- (i) $\psi : X \rightarrow G$ is a birational morphism.
- (ii) *There is one and only one conjugate of H containing a given Cartan subgroup of G .*

- (iii) H contains a Cartan subgroup C of G , and $\text{Norm}_G(H) \supset \text{Norm}_G(C)$.
- (iv) There exists a nonempty open subset V of G such that $g \in V(k)$ implies that g is contained in exactly one conjugate of H .

This is clear by 2.1 and 2.3.

Corollary 2.5. *Suppose that the conditions of 2.4 hold, and let $g \in G(k)$. The following conditions are equivalent:*

- (i) $g \in U(k)$, where U is defined in 2.3 c), i.e. g is contained in one and only one conjugate of H .
- (ii) The set of conjugates of H containing g is finite and nonempty.
- (iii) The scheme $\psi^{-1}(g)$ “of conjugates of H containing g ” contains an isolated point.
- (iv) There exists a conjugate H' of H containing g , and one has $(\mathfrak{g}/\mathfrak{h}')^{\text{ad}(g)} = 0$, where $\mathfrak{h}' = \text{Lie}(H')$.

Finally, U is also the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$.

The equivalence of (i), (ii), (iii), as well as the last assertion, follows from the “Main Theorem”⁽³⁾ applied to the birational morphism $\psi : X \rightarrow G$, taking into account that G is normal. The equivalence of these conditions with (iv) follows at once from the last assertion of 2.1 characterizing the set of elements of X at which ψ is étale.

Theorem 2.6. *Let G be a smooth connected algebraic group over an algebraically closed field k , C a Cartan subgroup, associated to a maximal torus T , $N = \text{Norm}_G(C) = \text{Norm}_G(T)$ (cf. XII 8.4), and let $X = G \times^N C$, where N acts on the left factor G by right translations, and on the right factor C by inner automorphisms; let $\psi : X \rightarrow G$ be the canonical morphism.*

- a) *The morphism ψ is birational.*
- b) *Let U be the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$ (cf. 2.5). Let*

$$\rho = \rho_\nu(G) = \dim C$$

be the nilpotent rank of G . Then for every $g \in G(k)$, the multiplicity of the eigenvalue 1 in $\text{ad}(g)$ acting on $\mathfrak{g}$ is at least ρ , and for it to equal ρ , it is necessary and sufficient that one have $g \in U(k)$.

PROOF. Since condition 2.1 (i) is verified, one may apply 2.4 (iii) $\Rightarrow$ (i), which establishes (a). In BIBLE 7 (in the case where G is affine) the points of $U(k)$ are called the *regular points* of $G(k)$, and we shall follow this terminology by calling U the open subset of regular points of G . (N.B. The proof given in BIBLE of the fact that the set in question is itself open is incorrect, but we have obtained it in the present exposé under more general conditions.)

Let us prove b), and for this purpose introduce for every $g \in G(k)$ the characteristic polynomial

$$P(\text{ad}(g), t) = t^n + c_1(g)t^{n-1} + \cdots + c_n(g);$$

one sees at once (replacing k by an arbitrary algebra over k) that the $c_i(g)$ come from well-determined sections

$$c_i \in \Gamma(G, \mathcal{O}_G).$$

When $g \in G(k)$ is an element contained in a Cartan subgroup (for example, a regular element), which we may suppose to be C , then by 2.5 (iv) one sees that one has $(\mathfrak{g}/\mathfrak{c})^{\text{ad}(g)} = 0$ if and only if g is regular (where $\mathfrak{c}$ denotes the Lie algebra of C); on the other hand, since C is nilpotent one sees immediately that $\text{ad}_{\mathfrak{c}}(g)$ has only the eigenvalue 1, which proves that the multiplicity of the eigenvalue 1 in $\text{ad}_{\mathfrak{g}}(g)$ is $\geq \rho$, and equal to exactly $\dim C = \rho$ if and only if g is regular. In particular, the above polynomial is divisible by $(t - 1)^\rho$. Since the divisibility relation by $(t - 1)^\rho$ is expressed by linear relations (with integer coefficients) between the

⁽³⁾N.D.E. Of Zariski!

coefficients of the polynomial, and these relations hold for $g \in U(k)$, U being a dense open subset, it follows (G being reduced) that they hold for every g ; in fact one has a relation

$$(\dagger) \quad t^n + c_1 t^{n-1} + c_0 = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho})$$

in the polynomial ring over $\Gamma(G, \mathcal{O}_G)$; in particular for every $g \in G(K)$, $\text{ad}(g)$ has the eigenvalue 1 with multiplicity at least ρ . Moreover, one has seen that one has equality if g is regular; let us prove the converse. For this, suppose first that G is affine, and write g as the product

$$g = g_s g_u$$

of its semisimple part and its unipotent part (BIBLE 4 no. 4); then

$$\text{ad}(g) = \text{ad}(g_s) \text{ad}(g_u)$$

is the analogous decomposition of $\text{ad}(g)$ (*loc. cit.* cor. to th. 3), and consequently $\text{ad}(g)$ and $\text{ad}(g_s)$ have the same eigenvalues (counted with their multiplicities); in particular the eigenvalue 1 occurs with the same multiplicity in $\text{ad}(g)$ and in $\text{ad}(g_s)$.

On the other hand, by BIBLE 7 th. 2 cor. 1, g is regular if and only if g_s is. Thus, to prove b), one may suppose $g = g_s$, i.e. g semisimple, hence contained in a maximal torus by BIBLE 6 th. 5 c), and a fortiori in a Cartan subgroup, a case which has already been treated. This proves b) in the case where G is affine. In the general case, let $Z = \text{Centr}(G)_{\text{red}}$; then by XII 6.6 e) the Cartan subgroups of G are the inverse images of those of $G' = G/Z$, so g is regular in G if and only if its image g' in G' is regular in G' . On the other hand, since Z is smooth, the Lie algebra $\mathfrak{g}'$ of G' is precisely $\mathfrak{g}/\mathfrak{z}$, where $\mathfrak{z} = \text{Lie}(Z)$, and $\text{ad}(g')$ is precisely $\text{ad}(g)_{\mathfrak{g}/\mathfrak{z}}$, so the multiplicity of the eigenvalue 1 in $\text{ad}(g)$ is equal to $d = \dim Z$ plus the multiplicity of the eigenvalue 1 in $\text{ad}(g')$, whence it follows at once that the first is equal to the nilpotent rank of G if and only if the second is equal to the nilpotent rank of G' . Thus one is reduced to the case of G' ; now, G' being affine by XII 6.1, this case has already been treated. This completes the proof of 2.6. $\square$

Corollary 2.7. *With the notation of the preceding proof⁽⁴⁾, let*

$$b = 1 + b_1 + \cdots + b_{n-\rho} \in \Gamma(G, \mathcal{O}_G).$$

Then the open subset of regular points of G is given by

$$U = G_b$$

(the set of points of G at which b is invertible); in particular U is an affine open subset if G is affine.

Corollary 2.8. *Let H be a smooth connected algebraic subgroup of G containing a Cartan subgroup of G .*

a) Let C be an algebraic subgroup of H . For C to be a Cartan subgroup of H , it is necessary and sufficient that it be a Cartan subgroup of G .

b) Let $g \in G(k)$, and let d be the integer introduced in 2.3. For g to be a regular point of G , it is necessary and sufficient that there exist exactly d conjugates H_i of H containing g , and that for each i , g be a regular element of H_i , or again that there be at most d conjugates of H containing g , and that g be regular in one of them. If this is so, and if C is the unique Cartan subgroup of G containing g , then the conjugates of H containing g are the conjugates of H containing C .

c) Let $g \in H(k)$; for g to be regular in G , it is necessary and sufficient that it be regular in H , and that one have

$$(\mathfrak{g}/\mathfrak{h})^{\text{ad}(g)} = 0.$$

⁽⁴⁾**N.D.E.** Cf. ($\dagger$) above.

PROOF. Let us prove a). Under either hypothesis on C , the unique maximal torus T of C is a maximal torus of H and of G (H having the same reductive rank as G), so, since $\text{Centr}_H(T) \subset \text{Centr}_G(T)$ are smooth connected groups of the same dimension, they are equal; hence it amounts to the same thing to say that C is equal to one or the other of these two groups, which proves a).

Let us prove b). Suppose first that g is regular for G , and let C be the unique Cartan subgroup of G containing g ; then by 2.3 b) there exist exactly d conjugates H_i of H containing C . Since $(\mathfrak{g}/\mathfrak{c})^{\text{ad}(g)} = 0$, i.e. $\text{ad}(g)$ does not have the eigenvalue $+1$ on $\mathfrak{g}/\mathfrak{c}$, one has a fortiori $(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0$, so by 2.3 c) there are exactly d conjugates of H containing g , namely the H_i . For such an H_i , a Cartan subgroup of H_i containing g is a Cartan subgroup of G containing g , by a), and is therefore equal to C , which proves that g is regular in H_i . Conversely, suppose that there exist at most d conjugates H_i of H containing g , and that g is regular in one of them, which may be supposed equal to H . Let us prove that g is regular in G . Since g is regular in H , it is contained in a unique Cartan subgroup C of H ; by a), this is a Cartan subgroup of G . Let C' be a Cartan subgroup of G containing g ; let us prove $C' = C$ (which will prove that g is regular in C). Indeed, by 2.3 b), there exist exactly d conjugates of H containing C' , and since these contain g , they are necessarily the H_i , so the H_i , and in particular H , contain C' . Thus C, C' are two Cartan subgroups of H (by a)) which contain the same regular element g of H , and are therefore equal. Q.E.D.

Let us prove c): denoting by $\nu(u)$ the nullity⁽⁵⁾ of $\text{id} - u$, for an endomorphism of a finite-dimensional vector space, one will have

$$\nu(\text{ad}(g)_{\mathfrak{g}}) = \nu(\text{ad}(g)_{\mathfrak{h}}) + \nu(\text{ad}(g)_{\mathfrak{g}/\mathfrak{h}}),$$

now the two terms on the right-hand side are respectively $\geq$ the nilpotent rank of H (equal to the nilpotent rank ρ of G , by a)) and ≥ 0 , so one has $\nu(\text{ad}(g)_{\mathfrak{g}}) = \rho$ if and only if $\nu(\text{ad}(g)_{\mathfrak{h}}) = \rho$ and $\nu(\text{ad}(g)_{\mathfrak{g}/\mathfrak{h}}) = 0$, i.e. g is regular in G if and only if g is regular in H and $\text{ad}(g)_{\mathfrak{g}/\mathfrak{h}}$ has no nonzero invariants. $\square$

Remarks 2.9. In statement 2.1, one cannot weaken condition (iii) by supposing merely that H contains a maximal torus and has finite index in its normalizer, even if one requires that this normalizer moreover be smooth, i.e. that one have $H = N^0$, and even when G is affine and solvable. An example is provided by the group G of matrices of the form

$$g = \begin{pmatrix} t & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

and the subgroup H of matrices of the preceding form, with $b = c = 0$ (N.B. the Cartan subgroup of G here consists of the matrices g with $a = c = 0$).

Remarks 2.10. Let G be a smooth algebraic group over k , H a smooth algebraic subgroup, but no longer suppose that H and G are connected. Suppose that H^0 contains a Cartan subgroup of G^0 . Then $(\mathfrak{g}/\mathfrak{h})^H \subset (\mathfrak{g}/\mathfrak{h})^{H^0} = 0$, so $H^0 = N^0$ (N is the normalizer of H); in particular N is smooth. However, one easily constructs examples, with G connected, where H has a connected component H_i such that for no $h \in H_i(k)$ does one have $(\mathfrak{g}/\mathfrak{h})^{\text{ad}(h)} = 0$, i.e. the morphism $(g, h) \mapsto \text{ad}(g) \cdot h$ from $G \times H_i$ to G is étale (or even quasi-finite) at no point (take, for example, for H the normalizer of the maximal torus in $\text{SL}(2)_k$). Likewise, even if the image H' of H in the finite group $G' = G/G^0$ is equal to G' , it is not necessarily true that the union of the conjugates of H in G is dense (take for G the semidirect product of $\mathbf{Z}/2\mathbf{Z}$ with $G^0 = \text{SL}(2)_k$, on which it acts by “symmetry”, and for H the semidirect product $\mathbf{Z}/2\mathbf{Z} \cdot T$, where T is a maximal torus of G^0). On the other hand, if one does not assume a priori that H^0 contains a Cartan subgroup of G^0 , but that the union of the conjugates of H in G is dense, then H^0 necessarily contains a Cartan subgroup of G^0 : to verify this, one may

⁽⁵⁾N.D.E. That is, the dimension of its nilspace.

clearly suppose G connected, and it suffices to repeat the proof of 2.1 (vi) $\Rightarrow$ (i), which is valid without supposing H connected.

3. Case of an arbitrary base prescheme

Suppose first that one is over a base field k , not necessarily algebraically closed. Since conditions 2.1 (i bis), (iv bis), (v), (vi) are invariant under extension of the base field, one sees by passing to the algebraic closure $\bar{k}$ of k that they are equivalent to one another, and equivalent to the fact that $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$. When this condition is satisfied, then (with the notation of 2.3) it will still be true that $k(X)$ is a finite separable extension of $k(G)$, of degree d independent of any extension of the base. If U is the largest open subset of G such that ψ induces a morphism $\psi^{-1}(U) \rightarrow U$ which is finite and étale, then the formation of U commutes with extension of the base field. If ψ is birational, then U is also the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$, and if then $g \in U(k)$, there exists one and only one subgroup H' of G , conjugate to H over the algebraic closure $\bar{k}$ of k , such that $H' \ni g$.

A point $g \in G(k)$ is said to be *regular* if it is regular as an element of $G(\bar{k}) = G_{\bar{k}}(\bar{k})$. More generally, the construction of 2.7 gives one an open subset of G , whose formation commutes with every extension of the base field, called the *open subset of regular points* of G , which is also characterized by the fact that for every algebraically closed extension K of k and every point $g \in G(K)$, g is a regular point of G_K if and only if $g \in U(K)$. If $g \in U(k)^{(6)}$, one sees that g is contained in one and only one Cartan subgroup of G , as we shall show under more general conditions below.

Let G be a smooth separated group prescheme of finite type with connected fibers over the prescheme S ; consider the functor $\mathcal{C} : ((\text{Sch}))_S^\circ \rightarrow (\text{Ens})$ defined by

$$\mathcal{C}(S') = \text{set of Cartan subgroups (XII 3.1) of } G_{S'}.$$

Suppose that this functor is representable by a prescheme smooth over S ; we give in XV a necessary and sufficient condition for this to be so, but we already know that this hypothesis is satisfied if G is affine over S of locally constant reductive rank (XII 3.3), or more generally if G admits a maximal torus locally for the fpqc topology (XII 7.1 a)), for example if S is the spectrum of a field. Let X be the Cartan subgroup of the $\mathcal{C}$ -group prescheme $G_{\mathcal{C}}$, the “universal Cartan subgroup” of G . As a prescheme over S , X therefore represents the functor

$$X(S') = \{(C, g) : C \text{ a Cartan subgroup of } G_{S'} \text{ and } g \text{ a section of } C \text{ over } S'\}.$$

Consider the canonical projection morphism $(C, g) \mapsto g$

$$\psi : X \rightarrow G.$$

One then has the

Theorem 3.1. *Under the preceding conditions on G , and with the preceding notation, let U be the set of $g \in G$ such that g is a regular element of its fiber G_s . Then U is open, and it is also the largest open subset U of G such that ψ induces an isomorphism*

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

PROOF. Let us first prove that U is open. From the hypothesis of representability of $\mathcal{C}$ as a prescheme smooth over S , since its structural morphism is clearly surjective, one concludes at once that G admits a Cartan subgroup locally for the étale topology, and that the nilpotent rank of the fibers of G is locally constant. The same holds for the dimension of the fibers of

⁽⁶⁾**N.D.E.** Correction of $G(k)$ to $U(k)$.

G , and, localizing on S , one may suppose that both are constant, say ρ and n . Consider then the Killing polynomial

$$P_G(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t], \quad \text{where } A = \Gamma(G, \mathcal{O}_G).$$

The restriction of this polynomial to the fibers G_s of G , and in particular to the fibers at the maximal points of S , is divisible by $(t-1)^\rho$, which is expressed by the fact that certain linear combinations with integer coefficients of the c_i are zero on the fibers G_s . When S is reduced (which one may suppose in order to establish that U is open), it follows that they are themselves zero, hence that the Killing polynomial itself is divisible by $(t-1)^\rho$, say

$$P_G(t) = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho}).$$

Let b be the sum of the coefficients $b_0 = 1, b_1, \dots, b_{n-\rho}$ of the second factor; then by 2.7 applied to the fibers of G , one sees that

$$U = G_b,$$

which indeed proves that U is open.

To prove that $\psi^{-1}(U) \rightarrow U$ is an isomorphism, one is reduced by SGA1 I 5.7 to verifying it fiber by fiber, which reduces one to the case of a base field, which may be supposed algebraically closed. Then there exists a Cartan subgroup C of G , and if N is its normalizer, C is identified by the conjugacy theorem XII 7.1 a) and b) with G/N , and the morphism $\psi : X \rightarrow G$ considered here is precisely that defined in no. 2. One then concludes by 2.6 b). The same argument also shows that U is the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \rightarrow U$. $\square$

Corollary 3.2. *Under the conditions of 3.1, let g be a regular section of G , i.e. such that for every $s \in S$, $g(s)$ is a regular point of G_s . Then there exists one and only one Cartan subgroup C of G such that g is a section of C .*

Indeed, the hypothesis on g means that g is a section of U , and the conclusion that there exists a unique section of X lifting it, which is merely another way of expressing that $\psi^{-1}(U) \rightarrow U$ is an isomorphism.

Note now that the open subset $\psi^{-1}(U)$ of the Cartan subgroup X of G_C is precisely the open subset of X consisting of the points of X which are regular in G_C (by which one means: regular in their fiber). One thus obtains a natural “fibration” of the dense open subset U of regular points of G over the prescheme $\mathcal{C}$, the fibers being dense open subsets of Cartan subgroups of the fibers of G_C (namely the open subsets of regular points in G). One finds, for example, the following result (which will be made considerably more precise in the next exposé):

Corollary 3.3. *Let G be a smooth connected algebraic group over the field k , $\mathcal{T}$ the scheme of maximal tori of G ($\simeq$ the scheme of Cartan subgroups of G). Then the function field $k(G)$ of G is isomorphic to the function field of a smooth connected affine nilpotent algebraic group C over the function field $k(\mathcal{T})$ of $\mathcal{T}$, namely $C =$ “the generic Cartan subgroup of G ”. If G is affine of unipotent rank zero, i.e. if the Cartan subgroups of G_K are tori, then $k(G)$ is a unirational extension of $k(\mathcal{T})$.*

Of course, by the generic Cartan subgroup of G , one means (by abuse of language) the Cartan subgroup of $G_{k(\mathcal{T})}$ which is the generic fiber of X over $\mathcal{T}$. It only remains to prove the last assertion of 3.3, which is contained in the following well-known result (due to Chevalley):

Lemma 3.4. *Let k be a field, T a torus over k , and $k(T)$ the field of rational functions on T ; then $k(T)$ is a unirational extension of k , i.e. is contained in a purely transcendental extension of k .*

Indeed, let k' be a finite separable extension of k which splits⁽⁷⁾ T (X 1.4); then $T \otimes_k k'$ is a rational variety, i.e. admits a dense open subset isomorphic to a dense open subset of affine space $\mathbb{A}_{k'}^n$, hence $T' = \prod_{\text{Spec}(k')/\text{Spec}(k)} T_{k'}/\text{Spec}(k')$ is a rational variety (for it admits a dense open subset isomorphic to a dense open subset of $\prod_{\text{Spec}(k')/\text{Spec}(k)} \mathbb{A}_{k'}^n$, which is isomorphic to affine space of dimension mn over k , where $m = [k' : k]$). Consider the norm homomorphism from T' to T (defined as soon as T is a commutative group scheme over k); the composite $T \rightarrow T' \rightarrow T$ is the m th power on T , hence dominant, so $T' \rightarrow T$ is dominant, which proves that T is unirational.

Let us return to the conditions of 3.1, but supposing even that G admits a maximal torus locally for the fpqc topology (XII 7.1). Let Y be the maximal torus of the Cartan subgroup X of G , so that the morphism $\psi : X \rightarrow G$ induces a morphism $Y \rightarrow G$ whose image consists, as a set, of the semisimple elements of the fibers of G (XII 8). Finally, it follows from 3.1 that the restriction of ψ to the open subset Y^{reg} of regular points of Y induces a closed immersion

$$Y^{\text{reg}} \rightarrow U = G^{\text{reg}}.$$

Making explicit the meaning of $Z = Y^{\text{reg}}$ as a functor over S , one finds:

Corollary 3.5. *Let G be a smooth separated S -group prescheme of finite type with connected fibers over the prescheme S , admitting a maximal torus locally for the fpqc topology. Let $Z : ((\text{Sch}))_S^\circ \rightarrow (\text{Ens})$ be the functor defined by*

$$Z(S') = \{\text{regular sections of } G_{S'} \text{ over } S' \text{ which are contained in a maximal torus of } G_{S'}\}.$$

Then Z is representable by a closed subprescheme, smooth over S , of the open subset $U = G^{\text{reg}}$ of G introduced in 3.1.

Finally, note the following result, which makes the density theorem 2.1 (i) $\Rightarrow$ (vi) more precise:

Corollary 3.6. *Under the conditions of 3.5, let C be a Cartan subgroup of G , and consider the morphism*

$$\varphi : Z \times C \rightarrow G$$

defined by $\varphi(g, h) = \text{ad}(g)h = ghg^{-1}$. Then φ is dominant.

It clearly suffices to prove this fiber by fiber, which reduces one to the case where S is the spectrum of an algebraically closed field. Let T be the maximal torus of C , t_0 an element of $T(k)$ regular in G , and c_0 an element of $C(k)$ regular in G ; consider $\varphi^{-1}(\varphi(t_0, c_0))$, whose points rational over k are the pairs (t, c) , with $t \in Z(k)$, $c \in C(k)$, such that

$$\text{ad}(t)c = \text{ad}(t_0)c_0 \quad \text{i.e.} \quad c = \text{ad}(t^{-1}t_0)c_0,$$

which are therefore in one-to-one correspondence with the $t \in Z(k)$ such that $\text{ad}(t^{-1}t_0)c_0 \in C$, or, what amounts to the same thing, c_0 being regular, such that $t^{-1}t_0 \in N$ (the normalizer of C), i.e. $t \in N$. One obtains an open and closed part of this fiber by restricting to the $t \in Z(k)$ such that $t \in C(k)$. Thus one has found a connected component of $\varphi^{-1}(\varphi(t_0, c_0))$ isomorphic to T (N.B. that the preceding set-theoretic argument does indeed give an isomorphism of schemes is seen by replacing points with values in k by points with values in an arbitrary k -prescheme), so the generic fiber of φ has dimension $\leq \dim T$, hence $\text{Im}(\varphi)$ has dimension $\geq \dim Z \times C - \dim T = \dim Z + \dim C - \dim T$; now one has $\dim Z = \dim Y = \dim \mathcal{C} + \dim T = \dim G - \dim C + \dim T$, whence finally $\dim \text{Im}(\varphi) \geq \dim G$, so φ is dominant. Q.E.D.

Remarks 3.7. One will note that the argument shows in addition that the connected component at (t_0, c_0) of the fiber of $\varphi(t_0, c_0)$ is isomorphic to T , in particular is smooth over k , and has the same dimension as the generic fiber, which implies that φ is in fact smooth at

⁽⁷⁾**N.D.E.** Splits (?), or: “over which T is split”.

(t_0, c_0) (which one should also be able to verify by calculating the tangent map). It follows that under the conditions of 3.6 the induced morphism $Z \times_S C^{\text{reg}} \rightarrow G^{\text{reg}}$ (where one has set $C^{\text{reg}} = C \cap G^{\text{reg}}$) is a smooth morphism. One sees likewise that the analogous morphism $Z \times T^{\text{reg}} \rightarrow Z$ (where T is a maximal torus of G) is smooth; more generally, for every smooth connected invariant algebraic subgroup H of C containing a regular element c_0 of $G(k)$, the image of $Z \times H \rightarrow G$ is dense in that of $G \times H \rightarrow G$.

4. Lie algebras over a field: rank, regular elements, Cartan subalgebras

In the remainder of this exposé, we take up again the theory developed by Chevalley in his book “Théorie des Groupes de Lie III” (Act. Sc. Ind. 1226, Paris 1955), the technique of schemes allowing us to eliminate the characteristic-zero hypothesis. We begin by recalling in this section certain well-known notions and results.

Let $\mathfrak{g}$ be a Lie algebra over a ring k . For every $a \in \mathfrak{g}$, one denotes by $\text{ad}(a)$ the endomorphism

$$\text{ad}(a) \cdot x = [a, x]$$

of $\mathfrak{g}$, which is a derivation of the Lie algebra $\mathfrak{g}$. Now, for every derivation D of the Lie algebra $\mathfrak{g}$, the nilspace of D , i.e. the union of the kernels of the iterates of D , is a Lie subalgebra of $\mathfrak{g}$, as one sees from Leibniz’s formula

$$D^m([x, y]) = \sum_{0 \leq p \leq m} \binom{m}{p} [D^p x, D^{m-p} y].$$

We shall set

$$\text{Nil}(a, \mathfrak{g}) = \text{nilspace of } \text{ad}(a) = \bigcup_{m \geq 0} \text{Ker } \text{ad}(a)^m;$$

when no confusion is to be feared, we shall denote it simply by $\text{Nil}(a)$, and call it the nilspace of a (in $\mathfrak{g}$).

Proposition 4.1. *For every $a \in \mathfrak{g}$, its nilspace $\text{Nil}(a, \mathfrak{g})$ is a Lie subalgebra of $\mathfrak{g}$, equal to its own normalizer.*

It remains to prove that it is its own normalizer, i.e. that every element of $\mathfrak{g}/\text{Nil}$ annihilated by the adjoint representation of Nil in $\mathfrak{g}/\text{Nil}$ is zero, which is trivial (for every element in this quotient annihilated by $\text{Ad}(a)$ is zero).

In the rest of this section, we suppose that k is a field, and $\mathfrak{g}$ finite-dimensional over k . We shall denote by $W(\mathfrak{g})$ the scheme over k defined by $\mathfrak{g}$, whose points in the k -algebra A are the elements of $\mathfrak{g} \otimes_k A$. If $a \in \mathfrak{g}$, the characteristic polynomial of $\text{ad}(a)$ is also called the characteristic polynomial or *Killing polynomial* of a in $\mathfrak{g}$, say

$$P_{\mathfrak{g}}(a, t) = t^n + c_1(a)t^{n-1} + \cdots + c_n(a),$$

where $n = \text{rank}_k \mathfrak{g}$, the $c_i(a) \in k$. Taking this polynomial also for $a \in \mathfrak{g} \otimes_k A^{(8)}$, where A is an arbitrary k -algebra, one sees that the $c_i(a)$ come from well-determined sections c_i of the structure sheaf of $W(\mathfrak{g})$, i.e. elements of the symmetric algebra $A = \text{Sym}_k(\mathfrak{g}^\vee)$, where $\mathfrak{g}^\vee$ is the dual of the k -module $\mathfrak{g}$. (When k is an infinite field, the c_i are determined by knowledge of the corresponding polynomial functions $\mathfrak{g} \rightarrow k$, but this is no longer so if k is a finite field.) Let r be the largest integer such that the Killing polynomial

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t]$$

is divisible by t^r , i.e. one has:

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_{n-r} t^r, \quad c_{n-r} \neq 0.$$

The integer r is called the *nilpotent rank* of the Lie algebra $\mathfrak{g}$. It is invariant under extension of the base field.

⁽⁸⁾**N.D.E.** $W \otimes_k A$ has been corrected to $\mathfrak{g} \otimes_k A$.

Proposition 4.2. *Let r be the nilpotent rank of $\mathfrak{g}$, and let $a \in \mathfrak{g}$. Then one has*

$$\text{rank}_k \text{Nil}(a, \mathfrak{g}) \geq r,$$

and one has equality if and only if one has

$$c_{n-r}(a) \neq 0.$$

In this case, $\text{Nil}(a, \mathfrak{g})$ is a nilpotent Lie algebra (and we shall see in 5.7 b) the converse, when $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G over k).

The first assertion is trivial, for by definition one has $\text{rank}_k \text{Nil}(a, \mathfrak{g}) =$ the multiplicity of the zero root in $P_{\mathfrak{g}}(a, t)$. Let us prove that if $c_{n-r}(a) \neq 0$, then $\text{Nil}(a)$ is nilpotent, which also means that for every $x \in \text{Nil}(a)$, $\text{ad}_{\text{Nil}(a)}(x)$ is a nilpotent endomorphism. One may suppose k algebraically closed; then, since $\text{ad}(a)_{\mathfrak{g}/\text{Nil}(a)}$ is injective, there exists a nonempty open subset U of $W(\text{Nil}(a))$ such that for every $x \in U(k)$, $\text{ad}(x)_{\mathfrak{g}/\text{Nil}(a)}$ is injective, hence $\text{Nil}(x) \subset \text{Nil}(a)$; one may moreover suppose U contained in the open subset of points where c_{n-r} does not vanish (since this open subset is nonempty by $c_{n-r}(a) \neq 0$), and then $\text{Nil}(x)$ having the same dimension as $\text{Nil}(a)$, one will have $\text{Nil}(x) = \text{Nil}(a)$. Consequently, for every $x \in U(k)$, $\text{ad}(x)_{\text{Nil}(a)}$ is nilpotent, and by the principle of extension of algebraic identities, this will remain true for every $x \in \text{Nil}(a)$, so $\text{Nil}(a)$ is nilpotent.

One says that the element a of $\mathfrak{g}$ is *regular* if $c_{n-r}(a) \neq 0$, i.e. if $\text{rank}_k \text{Nil}(a, \mathfrak{g}) = r$. When k is infinite, this therefore also means that $\text{rank}_k \text{Nil}(a, \mathfrak{g})$ is as small as possible (for a varying in $\mathfrak{g}$). In all cases, the notion of regular element of $\mathfrak{g}$ is invariant under extension of the base field, and the set of points of $W(\mathfrak{g})$ which are regular (i.e. which come from regular points of $W(\mathfrak{g})$ with values in a suitable extension of k) is open, for it is identical to $W(\mathfrak{g})_{c_{n-r}}$ (the set of points where c_{n-r} is invertible).

Corollary 4.3. *Let a be a regular element of $\mathfrak{g}$, and $\mathfrak{h}$ a Lie subalgebra of $\mathfrak{g}$ containing a . Then $\mathfrak{h}$ is nilpotent if and only if $\mathfrak{h} \subset \text{Nil}(a, \mathfrak{g})$; in particular, $\text{Nil}(a, \mathfrak{g})$ is a maximal nilpotent subalgebra of $\mathfrak{g}$.*

Since $\text{Nil}(a)$ is nilpotent, the relation $\mathfrak{h} \subset \text{Nil}(a)$ indeed implies that $\mathfrak{h}$ is nilpotent, and conversely, if $\mathfrak{h}$ is nilpotent, it is contained in the nilspace of its element a , i.e. $\mathfrak{h} \subset \text{Nil}(a)$.

Proposition 4.4. *Suppose k infinite. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. Consider the following conditions:*

- (i) $\mathfrak{d}$ is maximal nilpotent and contains a regular element of $\mathfrak{g}$.
- (i bis) $\mathfrak{d}$ is of the form $\text{Nil}(a, \mathfrak{g})$, where a is a regular element of $\mathfrak{g}$.
- (ii) $\mathfrak{d}$ is nilpotent and of the form $\text{Nil}(a, \mathfrak{g})$, where $a \in \mathfrak{g}$.
- (ii bis) $\mathfrak{d}$ is nilpotent, and there exists $a \in \mathfrak{d}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.
- (iii) $\mathfrak{d}$ is nilpotent and identical to its own normalizer.

One has the implications:

$$(i) \Leftrightarrow (i \text{ bis}) \Rightarrow (ii) \Leftrightarrow (ii \text{ bis}) \Leftrightarrow (iii)$$

(and we shall see in 5.7 a) that if $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group, then all the preceding conditions are equivalent).

The equivalence of (i) and (i bis) is trivial by 4.3, and these conditions trivially imply (ii). The equivalence of (ii) and (ii bis) is also trivial, as is $(ii \text{ bis}) \Rightarrow (iii)$ (cf. 4.1). It remains to prove implication $(iii) \Rightarrow (ii \text{ bis})$, the only one, moreover, which uses the fact that k is infinite, and which follows at once from the

Lemma 4.5. *Let $\mathfrak{d}$ be a nilpotent Lie algebra over an infinite field k , acting on a finite-dimensional vector space V . Suppose that for every $x \in \mathfrak{d}$, the endomorphism $u(x)$ is not injective. Then there exists a nonzero element v of V annihilated by $\mathfrak{d}$.*

One may suppose k algebraically closed and $\mathfrak{d}$ finite-dimensional. One then knows that V is a direct sum of finitely many nonzero stable vector subspaces V_i ($1 \leq i \leq n$), such that for every i , and every $x \in \mathfrak{d}$, $u(x)|_{V_i}$ has a single eigenvalue $\lambda_i(x)$ (cf. Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Exercise 22). Let $c_i(x)$ be the constant term of the characteristic polynomial of $u(x)|_{V_i}$, so that $\lambda_i(x) = 0$ if and only if $c_i(x) = 0$. Then c_i is a polynomial function on $\mathfrak{d}$, and the hypothesis means that $\mathfrak{d}$ is the union of the zero sets of the c_i . Thus one of the c_i is zero, which reduces one (replacing V by V_i) to the case where V is such that the $u(x)$ ($x \in \mathfrak{d}$) are nilpotent. But then Engel's theorem (Bourbaki *loc. cit.* th. 1) implies that there exists a nonzero v in V annihilated by $\mathfrak{d}$. Q.E.D.

One easily sees that (k still being an infinite field) conditions (i), (i bis) of 4.4 are invariant under every extension of the base field. If they hold, one will say that $\mathfrak{d}$ is a *Cartan subalgebra* of $\mathfrak{g}$; in the general case (k not necessarily infinite) one will say that $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{g}$ if it becomes a Cartan subalgebra for one (and consequently every) extension of the base field $k \rightarrow k'$, with k' infinite. This therefore implies that $\mathfrak{d}$ is nilpotent and equal to its own normalizer.

Proposition 4.6. *a) Let a be an element of $\mathfrak{g}$. If a is regular, it is contained in one and only one Cartan subalgebra of $\mathfrak{g}$ (and we shall see in 6.1 d) the converse when k is algebraically closed and $\mathfrak{g}$ the Lie algebra of a smooth algebraic group).*

b) Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, and a an element of $\mathfrak{d}$; then a is regular in $\mathfrak{g}$ if and only if $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.

Indeed, for a) one notes that if a is regular, then $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ (for this is true over an infinite extension k' of k), and it then follows at once from 4.3 that every Cartan subalgebra of $\mathfrak{g}$ containing a is identical to the preceding one. For b) one notes that the nullity⁽⁹⁾ of $\text{ad}(a)_{\mathfrak{g}}$ is equal to the sum of the nullities of $\text{ad}(a)_{\mathfrak{d}}$ and of $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$, and since the first is r , the sum is equal to r if and only if $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective. Q.E.D.

Corollary 4.7. *Let a be a regular element of $\mathfrak{g}$, $\mathfrak{d}$ a Cartan subalgebra of $\mathfrak{g}$ containing a , A an algebra over k , $\mathfrak{g}_A$ and $\mathfrak{d}_A \subset \mathfrak{g}_A$ the A -Lie algebras obtained from $\mathfrak{g}, \mathfrak{d}$ by base change, and a_A the image of a in $\mathfrak{g}_A$. Let u be an automorphism of $\mathfrak{g}_A$. For $u(\mathfrak{d}_A) = \mathfrak{d}_A$ to hold, it is necessary and sufficient that one have $u(a_A) \in \mathfrak{d}_A$.*

The condition is trivially necessary; let us prove that it is also sufficient. If it holds, then $\mathfrak{d}' = u(\mathfrak{d}_A)$ is a Lie subalgebra containing a_A , and every element b of it is such that $\text{ad}(b)_{\mathfrak{d}}$ is nilpotent (for $\mathfrak{d}'$ is isomorphic to $\mathfrak{d}_A$, which has this property, as follows at once from the definition of “nilpotent” in Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, def. 1). Taking $b = a_A$, one sees that the nilspace $\text{Nil}(b, \mathfrak{g}_A)$ contains $\mathfrak{d}'$; on the other hand, it is equal to $\mathfrak{d}_A$, and since $\mathfrak{d}'$ is locally a direct summand in the module $\mathfrak{g}_A$ ($\mathfrak{d}$ being so), hence in $\mathfrak{d}_A$, and it is a projective module of the same rank r as the latter, one concludes that it is equal to it. Q.E.D.

Proposition 4.8. *Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$.*

a) The following conditions are equivalent if k is infinite:

- (i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (ii) $\mathfrak{h}$ contains a regular element a of $\mathfrak{g}$, and an element b such that $\text{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ is injective.
- (iii) $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$, and contains a regular element of $\mathfrak{g}$.

These conditions are invariant under extension of the base field k .

b) Suppose that these conditions hold over a suitable infinite extension k' of k . Let $a \in \mathfrak{h}$; then a is regular in $\mathfrak{g}$ if and only if it is regular in $\mathfrak{h}$ and $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, i.e. if and only if $\text{Nil}(a, \mathfrak{h}) = \text{Nil}(a, \mathfrak{g})$ and this is a Cartan subalgebra of $\mathfrak{h}$.

c) Under the conditions of b), let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$; for it to be a Cartan subalgebra of $\mathfrak{h}$, it suffices that it be a Cartan subalgebra of $\mathfrak{g}$ (and we shall see in 5.8 that the condition

⁽⁹⁾**N.D.E.** That is, the dimension of the nilspace.

is also necessary if $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G , and $\mathfrak{h}$ the Lie algebra of a smooth algebraic subgroup H of G).

One sees immediately that conditions (ii) and (iii) of a) are invariant under extension of the base field k (supposed infinite), and that in statements b) and c), one may suppose k infinite, which we shall do. If $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$, then a is a regular element of $\mathfrak{g}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, hence (i) $\Rightarrow$ (ii). Conversely, if (ii) holds, then for a “sufficiently general” element a of $\mathfrak{h}$, a simultaneously satisfies the two conditions considered in (ii), hence $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ and is contained in $\mathfrak{h}$, so one has (i). Thus (i) and (ii) are equivalent. Suppose they hold, and let a be a variable element of $\mathfrak{h}$; then

$$(x) \quad \begin{aligned} \text{rank}_k \text{Nil}(a, \mathfrak{g}) &= \text{rank}_k \text{Nil}(a, \mathfrak{h}) \\ &\quad + \text{rank}_k \text{Nil}(\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}), \end{aligned}$$

on the other hand, the two terms on the right-hand side are respectively $\geq r' =$ the nilpotent rank of $\mathfrak{h}$, and ≥ 0 , the equalities moreover being attained⁽¹⁰⁾ for a “sufficiently general” element of $\mathfrak{h}$. Moreover, one also has $\text{rank}_k \text{Nil}(a, \mathfrak{g}) \geq r =$ the nilpotent rank of $\mathfrak{g}$, equality being attained for a “sufficiently general” element of $\mathfrak{h}$, and being attained if and only if a is regular in $\mathfrak{g}$. One concludes from this that one has $r = r'$, and that a is regular if and only if the two terms on the right-hand side of (x) are equal respectively to r' and to 0, i.e. if and only if a is regular in $\mathfrak{h}$ and $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, which proves b); and c) follows trivially by taking an element a in $\mathfrak{d}$ regular in $\mathfrak{g}$, so that $\text{Nil}(a, \mathfrak{g}) = \mathfrak{d}$. Moreover, the preceding result shows that (i) $\Rightarrow$ (iii); finally (iii) $\Rightarrow$ (i), for under (iii), a sufficiently general element a of $\mathfrak{h}$ is regular in $\mathfrak{h}$ and in $\mathfrak{g}$, so $\text{Nil}(a, \mathfrak{h}) \subset \text{Nil}(a, \mathfrak{g})$ are respectively Cartan subalgebras of $\mathfrak{h}$ and of $\mathfrak{g}$, and since they have the same rank over k , they are identical, which proves (i). This completes the proof of 4.8.

5. Case of the Lie algebra of a smooth algebraic group: density theorem

Let G be a smooth algebraic group over the field k , and $\mathfrak{g}$ its Lie algebra. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Let G act on $W(\mathfrak{g})$ by the adjoint representation, and consider the subscheme $W(\mathfrak{h})$. The construction of no. 1 leads us to introduce

$$N = \text{Norm}_G(\mathfrak{h}) = \text{Norm}_G(W(\mathfrak{h})),$$

which is an algebraic subgroup of G (not necessarily smooth),

$$\mathfrak{n} = \text{Lie}(N),$$

and the scheme

$$X = G \times^N W(\mathfrak{h})$$

the quotient of $X = G \times W(\mathfrak{h})$ by N acting on the right by $(g, x) \cdot n = (gn, \text{Ad}(n^{-1})x)$. We consider the canonical morphisms

$$\begin{array}{ccc} G \times W(\mathfrak{h}) & & \\ q \downarrow & \searrow \varphi & \\ X & \xrightarrow{\psi} & W(\mathfrak{g}). \end{array}$$

Theorem 5.1. *With the preceding notation, suppose k infinite. Consider the following conditions:*

- (i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (ii) There exists $a \in \mathfrak{h}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective.
- (iii) $\varphi : G \times W(\mathfrak{h}) \longrightarrow W(\mathfrak{g})$ is generically smooth.

⁽¹⁰⁾**N.D.E.** “The equality” has been replaced by “the equalities”.

- (iv) The preceding morphism φ (or again $\psi : X \rightarrow W(\mathfrak{g})$) is dominant, and $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$.
- (v) $\psi : X \rightarrow W(\mathfrak{g})$ is generically smooth and $\mathfrak{h} = \mathfrak{n}$.
- (vi) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and $\mathfrak{h} = \mathfrak{n}$.
- (vii) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant.
- (viii) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and N is smooth.
- (ix) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and $\mathfrak{h} = \mathfrak{n}$, and $\mathfrak{h}$ is the Lie algebra of a smooth algebraic subgroup H of G .
- (x) $\psi : X \rightarrow W(\mathfrak{g})$ is generically quasi-finite, and $\mathfrak{n} = \mathfrak{h}$.
- (xi) $\psi : X \rightarrow W(\mathfrak{g})$ is generically étale, and $\mathfrak{n} = \mathfrak{h}$.
- (xii) There exists a smooth algebraic subgroup H of G with Lie algebra $\mathfrak{h}$, and $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (xiii) There exists $a \in \mathfrak{h}$ such that N and the transporter M_a of a into $\mathfrak{h}$ (cf. no. 1) coincide in a neighborhood of e , and $\mathfrak{n} = \mathfrak{h}$.

One then has the following diagram of implications:

$$\begin{array}{ccccccc}
 \text{(xi)} & \Longleftrightarrow & \text{(xii)} & \Leftrightarrow & \text{(xiii)} & \Rightarrow & \text{(viii)} \Leftrightarrow \text{(ix)} \Leftrightarrow \text{(x)} \\
 & & \Downarrow & & & & \Downarrow \\
 \text{(i)} & \Leftrightarrow & \text{(ii)} & \Leftrightarrow & \text{(iii)} & \Leftrightarrow & \text{(iv)} \Rightarrow \text{(v)} \Rightarrow \text{(vi)} \Rightarrow \text{(vii)}.
 \end{array}$$

When k has characteristic zero, all the conditions considered are equivalent. Finally, one has

$$\text{(xi)} \Leftrightarrow [\text{(i) and (viii)}] \Leftrightarrow [\text{(v) and (viii)}].$$

PROOF. Note first the trivial implications:

$$\text{(v)} \Rightarrow \text{(vi)} \Rightarrow \text{(vii)}, \quad \text{(ix)} \Rightarrow \text{(vi)}, \quad \text{(xi)} \Leftrightarrow [\text{(x) and (v)}].$$

Let us prove the equivalence of conditions (i) to (iv) and the fact that they imply (v). Implication (i) $\Rightarrow$ (ii) is trivial. On the other hand, (iii) means, when k is algebraically closed, that there exists a point of $G \times W(\mathfrak{h})$ rational over k at which the tangent map to φ is surjective, and one sees at once that one can take this point of the form (e, a) , where $a \in \mathfrak{h}$ (transforming it, if necessary, by an operation of $G(k)$). One concludes that if k is infinite (not necessarily algebraically closed) this condition (clearly sufficient) of generic smoothness is still necessary. Now the tangent map is easily calculated: identifying the tangent space to $W(\mathfrak{h})$ at a with $\mathfrak{h}$, it is the map

$$(\xi, x) \mapsto [\xi, a] + x$$

from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Saying that it is surjective also means that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is surjective, or, what amounts to the same thing, injective. This proves the equivalence of conditions (ii) and (iii). Moreover, (ii) clearly implies $\mathfrak{h} = \mathfrak{n}$, and (iii) implies that ψ is generically smooth, for if φ is smooth at a point u , it follows (q being flat) that ψ is smooth at $q(u)$. Thus (ii), (iii) imply (v). Let us prove that they imply (i). For this, note that since ψ is dominant, and the set of regular points of $\mathfrak{g}$ is dense open, it follows that $\mathfrak{h}$ contains regular elements of $\mathfrak{g}$, so that a “sufficiently general” element b of $\mathfrak{h}$ is regular in $\mathfrak{g}$ and satisfies $\text{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ injective, hence $\text{Nil}(b, \mathfrak{g}) \subset \mathfrak{h}$, so $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = \text{Nil}(b, \mathfrak{g})$. Thus (i), (ii), (iii) are equivalent; finally (i) $\Leftrightarrow$ (iv), for one has already noted that if $\mathfrak{h}$ contains a Cartan subalgebra, it has the same rank as $\mathfrak{g}$ (4.6), hence (i) $\Rightarrow$ (iv); conversely, if (iv) holds, then $\mathfrak{h}$ contains a regular element of $\mathfrak{g}$, and since it has the same rank as $\mathfrak{g}$, it contains a Cartan subalgebra by 4.6.

Let us prove the equivalence of conditions (viii) to (x). Note first the following facts:

Lemma 5.2. *a) If N is smooth, then*

$$\dim X \leq \dim G,$$

equality holding if and only if $\mathfrak{h} = \mathfrak{n}$.

b) If $\mathfrak{h} = \mathfrak{n}$, then

$$\dim X \geq \dim G,$$

equality holding if and only if N is smooth.

These assertions follow at once from the formula

$$\dim X = \dim G - \dim N + \dim_k \mathfrak{h},$$

and from the fact that $\dim N = \dim_k \mathfrak{n}$ is equivalent to N smooth.

This being established, (viii) $\Rightarrow$ (x), for (viii) implies $\dim X \geq \dim G$, hence, by 5.2 a), equality of these dimensions and $\mathfrak{h} = \mathfrak{n}$, hence (x), and one similarly sees (x) $\Rightarrow$ (viii) by applying 5.2 b). On the other hand, (viii) implies (ix), for it implies $\mathfrak{h} = \mathfrak{n}$, so $\mathfrak{h}$ is the Lie algebra of the smooth algebraic subgroup N of G , and conversely (ix) $\Rightarrow$ (viii), for since H normalizes its Lie algebra $\mathfrak{h}$, and it is contained in N , and since H is smooth and has the same Lie algebra as N , it follows that N is smooth.

Finally, let us prove the equivalence of conditions (xi), (xii), (xiii) and the fact that they imply (iii) (which will finish establishing our diagram of implications). One has (xi) $\Leftrightarrow$ (xiii), for if $\mathfrak{n} = \mathfrak{h}$, then by 5.2 b) one has $\dim X \geq \dim G$, hence (xi) is then equivalent (taking into account that $W(\mathfrak{g})$ is normal) to the fact that ψ is generically unramified, which is also equivalent to (xiii) by (1.1 (ii) $\Leftrightarrow$ (iii)), proceeding as above for the proof of (ii) $\Leftrightarrow$ (iii). Since (xi) $\Rightarrow$ (x) $\Rightarrow$ (viii) by what we have already seen, one sees that (xi) implies that N is smooth, i.e. $q : G \times W(\mathfrak{h}) \rightarrow X$ is smooth, so the composite $\varphi = \psi \circ q$ is generically smooth, i.e. one has (iii). Since (iii) $\Rightarrow$ (i), it also follows that (xi) $\Rightarrow$ (xii)⁽¹¹⁾. Finally (xii) $\Rightarrow$ (xi), for one clearly has (xii) $\Rightarrow$ (i), so, since one has seen (i) $\Rightarrow$ (iii) $\Rightarrow$ (v), one has (xii) $\Rightarrow$ (v); it follows that one also has (xii) $\Rightarrow$ (ix), and since one has seen (ix) $\Rightarrow$ (x), it follows that (xii) $\Rightarrow$ ((v) and (x)), hence (xii) $\Rightarrow$ (xi), since generically étale = generically smooth + generically quasi-finite.

Finally, when k has characteristic 0, then (vii) $\Rightarrow$ (viii), for by a theorem of Cartier, N is automatically smooth (VI_B 1.6.1.), and [(viii) and (x)] $\Rightarrow$ (xi), since in characteristic zero, for a morphism of integral preschemes, generically étale = dominant and generically quasi-finite. This shows that in this case, all conditions (i) to (xiii) are equivalent. $\square$

Corollary 5.3. *Under the equivalent conditions (viii) to (x), there exists a unique smooth connected algebraic subgroup H of G whose Lie algebra is $\mathfrak{h}$, and one has*

$$\text{Norm}_G(H) = \text{Norm}_G(\mathfrak{h}) = N, \quad H = N^0.$$

Indeed, $H = N^0$ will satisfy the desired conditions; on the other hand, if H satisfies them, then (since H normalizes its Lie algebra $\mathfrak{h}$) one has $H \subset N$, so, since this is an inclusion of smooth groups with the same Lie algebra, with H connected, one will have $H = N^0$. For the identity $\text{Norm}_G(H) = \text{Norm}_G(\mathfrak{h})$, one may suppose k algebraically closed; then from what one has just seen it follows immediately that the points of the two groups with values in k are the same; on the other hand the inclusions $H \subset \text{Norm}_G(H) \subset N$ show that $\text{Norm}_G(H)$ and N have the same Lie algebra, hence they are identical.

Corollary 5.4. *Under the equivalent conditions (i) to (iv), let $a \in \mathfrak{h}$. Then the following conditions are equivalent, and hold if a is regular in $\mathfrak{g}$:*

- (i) φ is smooth at (e, a) .
- (ii) $M_a = \text{Transp}_G(a, \mathfrak{h})$ is smooth at e , and $\dim_e(M_a) = \text{rank}_k \mathfrak{h}$.
- (iii) $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (or again, bijective).

When one is under the equivalent conditions (xi) to (xiii), let H be the algebraic subgroup of G considered in 5.3. Then the preceding conditions are also equivalent to the following conditions:

⁽¹¹⁾ **N.D.E.** For $H = N$ will do.

- (iv) ψ is étale at $(\bar{e}, a)$.
- (v) Denoting by M_a^0 the connected component of e in the transporter M_a of a into $\mathfrak{h}$, equipped with the structure induced by M_a , one has

$$H = M_a^0.$$

Clearly (i) $\Rightarrow$ (ii), since M_a is isomorphic to the fiber $\varphi^{-1}(a)$, the point e corresponding to (e, a) , and one has (ii) $\Rightarrow$ (i), for (ii) implies that φ is “equidimensional” at (e, a) (i.e. the dimension of the fiber through this point is that of the generic fiber), which implies ($G \times W(\mathfrak{h})$ and $W(\mathfrak{g})$ being regular) that it is flat at (e, a) , hence smooth since its fiber is so at this point. The equivalence of (i) and (iii) was seen in the proof of 5.1 as resulting from the simple calculation of the tangent map. Moreover, one has seen in 4.8 b) that “ a regular in $\mathfrak{g}$ ” $\Rightarrow$ (iii). Under conditions (xi) to (xiii), since $q : G \times W(\mathfrak{h}) \rightarrow X$ is smooth (N being smooth), it follows that (i) is equivalent to ψ smooth at $(\bar{e}, a)$, and since ψ is generically étale, this is equivalent to (iv). Finally, as was noted at the end of the proof of 1.1, (iv) implies that N is the prescheme induced on M_a by an open and closed part of M_a , whence (v); finally (v) $\Rightarrow$ (ii) trivially (or again (v) $\Rightarrow$ (iv) by 1.1, for ψ being dominant and $W(\mathfrak{g})$ normal, “unramified” is here equivalent to “étale”); this completes the proof of 5.4.

Corollary 5.5. *Let G be a smooth algebraic group over a field k , H a smooth algebraic subgroup such that its Lie algebra $\mathfrak{h}$ contains (at least over a suitable extension of k) a Cartan subalgebra of the Lie algebra $\mathfrak{g}$ of G . Let K be a connected algebraic subgroup of G (not necessarily smooth), with Lie algebra $\mathfrak{k}$; suppose that $\mathfrak{k}$ contains a regular element a of $\mathfrak{g}$ (at least over a suitable extension of k). Then H contains K if and only if $\mathfrak{h}$ contains $\mathfrak{k}$.*

Indeed, by 5.4 (iii) $\Rightarrow$ (v) one has $H = M_a^0$ (N.B. of course, this relation being invariant under extension of the base field, it is valid without the hypothesis that the latter be infinite!); on the other hand $\mathfrak{k} \subset \mathfrak{h}$ clearly implies $K \subset M_a$, so, since K is connected, $K \subset M_a^0$, whence $K \subset H$. Q.E.D.

Corollary 5.6. *Let G, H be as in 5.5 and let K be an algebraic subgroup of G ; one supposes H connected and K smooth. Then K contains H if and only if $\mathfrak{k}$ contains $\mathfrak{h}$.*

Indeed, if $\mathfrak{k} \supset \mathfrak{h}$, then K satisfies the hypothesis considered in 5.5 for H ; on the other hand H clearly satisfies the condition considered for K in 5.5. The conclusion then follows from 5.5.

Corollary 5.7. *Let $\mathfrak{g}$ be the Lie algebra of a smooth algebraic group G over a field k . Then:*
a) Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. For $\mathfrak{d}$ to be a Cartan subalgebra, it is necessary and sufficient that $\mathfrak{d}$ be nilpotent and equal to its own normalizer.
b) Let a be an element of $\mathfrak{g}$; for a to be regular, it is necessary and sufficient that $\text{Nil}(a, \mathfrak{g})$ be nilpotent.

Extending the base field, one may suppose k infinite. Taking 4.4 into account, one is reduced for a) to proving that if $\mathfrak{d}$ is nilpotent and contains an element a such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective, then $\mathfrak{d}$ is a Cartan subalgebra. Now by 5.1 (ii) $\Rightarrow$ (i), $\mathfrak{d}$ contains a Cartan subalgebra $\mathfrak{d}' = \text{Nil}(a, \mathfrak{g})$, and by 4.3 one concludes from the fact that $\mathfrak{d}$ is nilpotent that $\mathfrak{d} = \mathfrak{d}'$. To prove b), one notes that $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ by a), hence a is regular.

Corollary 5.8. *Let G be a smooth algebraic group over a field k , H a smooth algebraic subgroup, and $\mathfrak{g}, \mathfrak{h}$ the Lie algebras; suppose that after a suitable extension of the base field, $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$. Let $\mathfrak{d}$ be a subalgebra of $\mathfrak{h}$; then it is a Cartan subalgebra of $\mathfrak{h}$ if and only if it is a Cartan subalgebra of $\mathfrak{g}$.*

Taking 4.8 c) into account, it remains to show that if $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$, it is a Cartan subalgebra of $\mathfrak{g}$, and for this one is reduced to showing that $\mathfrak{d}$ contains an element a regular in $\mathfrak{g}$, supposing (as one may) k algebraically closed. But since there is a dense open

subset of $\mathfrak{h}$ consisting of regular points of $\mathfrak{g}$, our assertion follows from 5.1 (i) $\Rightarrow$ (vii) applied to $(\mathfrak{h}, \mathfrak{d})$.

6. Cartan subalgebras and subgroups of type (C), relative to a smooth algebraic group

For simplicity, we confine ourselves in the following theorem to the case of an algebraically closed base field (the case of an arbitrary base prescheme being treated in the next exposé):

Theorem 6.1. *Let G be a smooth algebraic group over an algebraically closed field k , and $\mathfrak{g}$ its Lie algebra. Then:*

- a) *The Cartan subalgebras of $\mathfrak{g}$ are conjugate.*
- b) *Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then its normalizer N in G is smooth, and $D = N^0$ is the only smooth connected subgroup of G whose Lie algebra is $\mathfrak{d}$. One has*

$$\text{Norm}_G(\mathfrak{d}) = \text{Norm}_G(D) = N, \quad \text{hence } D = \text{Norm}_G(D)^0.$$

- c) *With $\mathfrak{d}$ as in b), set, as in no. 5, $X = G \times^N W(\mathfrak{d})$, and consider the canonical morphism*

$$\psi : X \longrightarrow W(\mathfrak{g}),$$

(whose fiber at $a \in \mathfrak{g}$ has as its points with values in k the set of Cartan subalgebras of $\mathfrak{g}$ which contain a). Then ψ is a birational morphism.

- d) *With the notation of c), let U be the largest open subset of $W(\mathfrak{g})$ such that ψ induces an isomorphism*

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Then for $a \in \mathfrak{g}$, the following conditions are equivalent:

- (i) $a \in U(k)$.
- (ii) a is contained in one and only one Cartan subalgebra of $\mathfrak{g}$.
- (iii) The set of Cartan subalgebras of $\mathfrak{g}$ containing a is finite and nonempty.
- (iv) The fiber $\psi^{-1}(a)$ has an isolated point.
- (v) (If $a \in \mathfrak{d}$) The morphism ψ is étale (or merely: quasi-finite) at the point $(\bar{e}, a)$.
- (vi) a is a regular element of $\mathfrak{g}$.
- (vii) With the notation of 5.4 (iii), one has $N = M_a$.

PROOF. Let us apply 5.1 when $\mathfrak{h}$ is a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$; let us show that the strongest conditions (xi) to (xiii) then hold. This is clear, either in form (xiii), taking 4.7 into account (which implies that for a regular element a of $\mathfrak{g}$ contained in $\mathfrak{d}$, one has $N = M_a$), or in form (xi) = (x) + (i), for condition (i) is trivial and condition (x) follows from the fact that a regular point of $\mathfrak{g}$ is contained in only one Cartan subalgebra of $\mathfrak{g}$ (4.6 a)), and a fortiori in only one conjugate of $\mathfrak{d}$. Then b) follows from 5.3, and c) follows from the fact that ψ is generically étale and that a sufficiently general point (more precisely, a regular point) of $\mathfrak{g}$ is contained in only one Cartan subalgebra of $\mathfrak{g}$. Under these conditions, the equivalence of conditions (i) to (v) on a is an immediate consequence of Zariski's Main Theorem for the separated birational morphism ψ , taking into account that $W(\mathfrak{g})$ is normal and X integral. The equivalence of (v) and (vi) is a particular case of 5.4 (iii) $\Leftrightarrow$ (iv) (reducing to the case where $a \in \mathfrak{d}$ by transforming a by a suitable element $g \in G(k)$), taking 4.6 b) into account. Moreover, by 5.4, (v) and (vi) are also equivalent to $N \supset M_a^0$, and by 4.7, already invoked, this even implies $N = M$. This proves d).

Of course, b), c), and the equivalence of (i), (iv), (v), (vi), (vii) remain valid over an arbitrary base field. Let us now prove a), using the fact that k is algebraically closed. By 5.1 (i) $\Rightarrow$ (vii), $\psi : X \longrightarrow W(\mathfrak{g})$ is dominant, hence there exists a dense open subset V of $W(\mathfrak{g})$ such that every $a \in V(k)$ is the image of an element of $X(k)$, i.e. is contained in a conjugate of $\mathfrak{d}$. Applying this result to a second Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}$, one sees that one may take V such that every element of $V(k)$ is conjugate to an element of $\mathfrak{d}$ and to an element of $\mathfrak{d}'$. Taking a regular element in $V(k)$, it follows that there is a conjugate $\mathfrak{d}''$ of $\mathfrak{d}'$ which contains

an element of $\mathfrak{d}$ regular in $\mathfrak{g}$, and hence is equal to $\mathfrak{d}$ by 4.6 a). This completes the proof of 6.1. $\square$

Definition 6.2. Let G be a smooth algebraic group over a field k . One calls a *subgroup of type (C)* of G any smooth connected algebraic subgroup whose Lie algebra is a Cartan subalgebra of $\mathfrak{g}$. One calls the *infinitesimal rank* of G the nilpotent rank of its Lie algebra $\mathfrak{g}$, equal to the dimension of every subgroup of type (C) of G .

From 6.1 b) one concludes at once:

Corollary 6.3. *Under the conditions of 6.2, the map $D \mapsto \mathfrak{d} = \text{Lie } D$ establishes a one-to-one correspondence between subgroups of type (C) of G and Cartan subalgebras $\mathfrak{d}$ of $\mathfrak{g}$. If D is a subgroup of type (C) of G , D is its own connected normalizer:*

$$D = \text{Norm}_G(D)^0.$$

Combining 6.1 (a) and 6.3, one finds:

Corollary 6.4. *If k is algebraically closed, the subgroups of type (C) of G are conjugate to one another.*

Corollary 6.5. *Let G be a smooth algebraic group over algebraically closed k , H a smooth algebraic subgroup of G , and $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras. For $\mathfrak{h}$ to contain a Cartan subalgebra of $\mathfrak{g}$, it is necessary and sufficient that H contain a subgroup D of type (C) of G .*

This is clearly sufficient, and it is also necessary, for $H \supset D$ holds if and only if $\mathfrak{h} \supset \mathfrak{d}$, by 5.5.

Remarks 6.6. a) One will note that the connected subgroups H of G described in 6.5 correspond one-to-one to the Lie subalgebras $\mathfrak{h}$ of $\mathfrak{g}$ which satisfy the strongest condition (xii) of 5.1 (and by 5.5, the inclusion relations between such subgroups can already be recognized on their Lie algebras).

b) Still suppose k algebraically closed, and let D be a subgroup of type (C) of G ; then it is easy to show that D contains a Cartan subgroup C of G : indeed, let $V = \mathfrak{g}/\mathfrak{d}$; then for a “general” element a of $\mathfrak{d}$, $\text{ad}(a)$ acting on V is injective (it suffices that a be regular in $\mathfrak{g}$), whence one easily concludes that for “general” $g \in D(k)$, $\text{Ad}(g)$ acting on V has no nonzero invariant, which allows one to apply 2.1 (vii) $\Rightarrow$ (i). In fact, we shall see in the next exposé a more precise result: every Cartan subgroup C of G is contained in one and only one subgroup of type (C) of G .

c) One should take care not to confuse Cartan subgroups and subgroups of type (C) in general: the subgroups of type (C) of G are Cartan subgroups if and only if they are nilpotent (Cartan subgroups being indeed maximal nilpotent subgroups); now it may happen that $\mathfrak{g}$ is nilpotent without G being so (example: $\text{SL}(2)_k$ for k of characteristic 2); then the Cartan subalgebras of $\mathfrak{g}$ are identical to $\mathfrak{g}$, i.e. G is a subgroup of type (C) of itself if G is connected, but it is not a Cartan subgroup of G ! On the other hand, we shall see in XIV that if G is an adjoint semisimple group, then its subgroups of type (C) are its Cartan subgroups. Likewise, in characteristic 0, without restriction on the smooth algebraic group G over k , the same conclusion holds: this follows at once from the fact that in characteristic 0, a connected algebraic group is nilpotent if (and only if) its Lie algebra is. This follows at once from the fact that in characteristic zero, the center Z of the connected algebraic group G has as Lie algebra the center of $\mathfrak{g}$ (for one obtains a priori the space of invariants of $\mathfrak{g}$ under the adjoint representation of G ; now G being connected and k of characteristic zero, these are also the “invariants” in the infinitesimal sense of the adjoint representation of $\mathfrak{g}$, i.e. the center of $\mathfrak{g}$), and that by Cartier’s theorem (cf. VI_B 1.6.1) Z is smooth.

Exposé XIV. Regular elements, continued. Application to algebraic groups

by A. Grothendieck
with an Appendix by J.-P. Serre

1. Construction of Cartan subgroups and maximal tori for a smooth algebraic group

Theorem 1.1. *Let G be a smooth algebraic group over a field k . Then G admits a maximal torus T , hence a Cartan subgroup $C = \text{Centr}_G(T)$.*

PROOF. By XII 3.2, it amounts to the same thing to find a maximal torus T of G , or a Cartan subgroup C of G . Moreover, since the maximal tori of G are those of G^0 , one may suppose G connected. We distinguish two cases:

1°) The field k is finite. Let $\mathcal{T}$ be the scheme of maximal tori of G (Exp XII 1.10.), which is a smooth scheme over k . Note that G acts on $\mathcal{T}$ by inner automorphisms, and by the conjugacy theorem (XII 6.6 a)), two points of $\mathcal{T}_{\bar{k}}$ rational over $\bar{k}$ are congruent under $G_{\bar{k}}(\bar{k})$. Taking into account that $\mathcal{T}$ is smooth over k , hence $\mathcal{T}_{\bar{k}}$ smooth over $\bar{k}$, it follows that $\mathcal{T}_{\bar{k}}$ is isomorphic to $G_{\bar{k}}/\bar{N}$, where $\bar{N}$ is the stabilizer of an element of $\mathcal{T}_{\bar{k}}(\bar{k})$, i.e. the normalizer of a maximal torus $\bar{T}$ of $G_{\bar{k}}$. Consequently, $\mathcal{T}$ is a “homogeneous space” under the action of the group G . A well-known theorem of Lang (Amer. J. Math. 78, 1956, pp. 555–563) tells us that every homogeneous space under a smooth connected algebraic group over a finite field k admits a rational point. In particular, $\mathcal{T}$ admits a rational point, i.e. G admits a maximal torus T . Q.E.D.

2°) The field k is infinite. We shall use the

Lemma 1.2. *Let G be a smooth algebraic group over a field k . Then G admits a subgroup of type (C) (XIII 6.2).*

By XIII 6.3 it amounts to the same thing to say that $\mathfrak{g}$ contains a Cartan subalgebra $\mathfrak{d}$. This is trivial if k is infinite, for then $\mathfrak{g}$ contains a regular element a , and one will take $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$. The case of finite k is treated exactly as in proof 1°) above, but requires the preliminary construction of the scheme $\mathfrak{d}$ of Cartan subalgebras of $\mathfrak{g}$ and the fact that the latter is smooth over k , which will be seen below (2.16). To establish 1.1 in case 2°) in which we have placed ourselves, it will in any event suffice to know 1.2 for infinite k . Let us also record:

Lemma 1.3. *Let G be a smooth connected affine algebraic group whose reductive center (XII 4.1 and 4.4) is reduced to the unit group. Then G is nilpotent if (and only if) its Lie algebra $\mathfrak{g}$ is nilpotent.*

This is contained in XII 4.9.

We can now give a method of constructing Cartan subgroups of G (also valid when k is finite, admitting 1.2 in this case). Suppose first G affine. We proceed by induction on

⁰_{xy} version of 1/12/08

$n = \dim G$, the assertion being trivial if $n = 0$. Thus suppose $n > 0$ and the assertion proved for dimensions $n' < n$. Let Z be the reductive center of G , and let

$$u : G \longrightarrow G' = G/Z$$

be the canonical homomorphism. By XII 4.7 c), one has a one-to-one correspondence $C' \mapsto C = u^{-1}(C')$ between Cartan subgroups of G' and Cartan subgroups of G . Thus, replacing G by G' , one may suppose that the reductive center Z of G is reduced to the identity element (for this holds for that of G' by XII 4.7 b)). By 1.3, G admits a subgroup D of type (C). We know that over the algebraic closure $\bar{k}$ of k , $D_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$ (XIII 6.6 b)), so every Cartan subgroup of D is a Cartan subgroup of G (XIII, 2.8 a)). Thus one is reduced to finding a Cartan subgroup of D . If $\dim D = \dim G$, i.e. $D = G$, then the Lie algebra of G is a Cartan subalgebra of itself, hence is nilpotent, so by 1.3 G is nilpotent, hence is a Cartan subgroup of itself. If $\dim D < \dim G$, then by the induction hypothesis there exists a Cartan subgroup of D , which is therefore a Cartan subgroup of G , which completes the proof in the case where G is affine. In the general case, let Z be the center of G ; then $G/Z = G'$ is affine (XII 6.1), and for every Cartan subgroup C' of G' , its inverse image C in G is a Cartan subgroup of G (XII 6.6 e)). One is reduced to finding a Cartan subgroup of the smooth affine algebraic group G' , a case which has already been treated. $\square$

Corollary 1.4. *Let G be a smooth group scheme of finite type over an artinian scheme S . Then G admits a maximal torus T , hence a Cartan subgroup $C = \text{Centr}_G(T)$. Every torus S in G is contained in a maximal torus.*

One may indeed suppose S local with residue field k ; then by 1.1, $G_0 = G \times_S \text{Spec}(k)$ admits a maximal torus T_0 ; by IX 3.6 bis and X 2.3, T_0 comes from a torus T of G , which is clearly a maximal torus. The last statement follows by applying the preceding result to the centralizer of S , which is indeed smooth over k (XI 2.4).

Remarks 1.5. a) We shall give below (3.20, 3.21 and XV) variants of 1.4 in the case where S is not assumed artinian.

b) I do not know whether every algebraic group G (not necessarily smooth) over a field k admits a maximal torus. The question only arises in characteristic $p > 0$, and using 1.1 for a smooth quotient group of the form $G' = G/I$, where I is a suitable radicial subgroup of G (for example, the kernel of a sufficiently high power of the Frobenius homomorphism), by taking the inverse image in G of a maximal torus of G' , one is reduced to the case where $(G_{\bar{k}})_{\text{red}}$ is a torus ($\bar{k}$ still denoting the algebraic closure of k). It is easy to see that the answer is affirmative when G is commutative (or more generally nilpotent): then G admits a unique maximal torus, which one may construct, for example, by descent from the maximal torus of $G_{\bar{k}}^*$.

c) In the case where G is affine, and k perfect or G solvable, 1.1 is known and due to Rosenlicht; his proof is very different from that given here.

d) When k is infinite, 1.1 is a consequence of the much more precise result that the scheme $\mathcal{T}$ of maximal tori of G is a rational variety, proved below (6.1). The method is essentially a combination of the proof of 1.1 and the explicit description of the structure of the scheme $\mathfrak{d}$ of Cartan subalgebras of $\mathfrak{g}$. To arrive at the desired result, we must first generalize to the case of an arbitrary base prescheme certain results of XIII 4 to 6 (this is the purpose of the next two sections), and make the preceding construction proving 1.1 more precise, using the fact that every Cartan subgroup of G is contained in one and only one subgroup of type (C) of G (Nos. 4, 5).

* M. Raynaud has given a negative answer to the question raised here; cf. XVII Example 5.9.c).

2. Lie algebras over an arbitrary prescheme: regular sections and Cartan subalgebras

Lemma 2.1. *Let A be a ring, $\mathfrak{d}$ a Lie algebra over A , and for every $s \in \operatorname{Spec}(A)$, let $\mathfrak{d}(s) = \mathfrak{d} \otimes_A k(s)$ be the corresponding Lie algebra over the residue field $k(s)$. One assumes the A -module $\mathfrak{d}$ finitely presented. The following conditions are equivalent:*

- (i) *For every $s \in \operatorname{Spec}(A)$, $\mathfrak{d}(s)$ is nilpotent.*
- (ii) *For every $x \in \mathfrak{d}$, $\operatorname{ad}(x)$ is nilpotent.*
- (iii) *There exists an integer $N \geq 0$ such that for every sequence $x_1, \dots, x_N$ of elements of $\mathfrak{d}$, one has*

$$\operatorname{ad}(x_1) \operatorname{ad}(x_2) \dots \operatorname{ad}(x_N) = 0.$$

When A is a field, the equivalence of (i) and (iii) is the definition of “nilpotent”, and that of (ii) and (iii) is a well-known consequence of Engel’s theorem (Bourbaki, *Groupes et algèbres de Lie*, Chap. I, §4, no. 2). In the general case, one trivially has (iii) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (i) by the preceding result and the fact that (ii) is stable under passage to a quotient and localization. It remains to prove (i) $\Rightarrow$ (iii). When A is artinian local with maximal ideal $\mathfrak{m}$, let $n > 0$ be an integer such that $\mathfrak{m}^n = 0$, and N an integer such that condition (iii) holds for $\mathfrak{d}(s) = \mathfrak{d} \otimes_A A/\mathfrak{m}$; take $N' = nN$; one sees at once that this integer satisfies (iii). When A is noetherian, there then exist finitely many elements $s_i \in \operatorname{Spec}(A)$ and ideals of definition $\mathfrak{q}_i \subset A_{s_i}$ such that the natural map

$$\mathfrak{d} \longrightarrow \prod_i \mathfrak{d} \otimes_A A_i, \quad \text{where } A_i = A_{s_i}/\mathfrak{q}_i,$$

is injective; then by what precedes there exists for every i an integer N_i satisfying (iii) for the Lie algebra $\mathfrak{d} \otimes_A A_i$, and taking for N the largest of the N_i , one satisfies (iii) for $\mathfrak{d}$. Finally, the general case reduces to the noetherian case by the usual procedure explained in EGA IV 8.

Definition 2.2. Let S be a prescheme, $\mathfrak{d}$ a quasi-coherent Lie algebra over S which is a finitely presented module. One says that $\mathfrak{d}$ is locally nilpotent if for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ over $k(s)$ is nilpotent. One says that $\mathfrak{d}$ is strictly locally nilpotent if it is locally free, and if on every open subset U of S on which it has constant rank r , its Killing polynomial reduces to $P_{\mathfrak{d}}(t) = t^r$.

Of course, if $\mathfrak{d}$ is a locally free Lie algebra (as a module) over S , one defines its Killing polynomial as a polynomial

$$P_{\mathfrak{d}} \in A[t], \quad A = \Gamma(\underline{\operatorname{Sym}}_{\mathcal{O}_S}(\mathfrak{d})) \simeq \Gamma(W(\mathfrak{d}))$$

where A is the ring of sections of the structure sheaf of the vector bundle $W(\mathfrak{d})$ defined by $\mathfrak{d}$. It is clear that the two notions just introduced in 2.2 are stable under base change and local for the fpqc topology. Note:

Proposition 2.3. *If $\mathfrak{d}$ is strictly locally nilpotent, it is locally nilpotent. The converse is true when $\mathfrak{d}$ is locally free and S reduced.*

The proof is immediate.

Definition 2.4. Let S be a prescheme, $\mathfrak{g}$ a Lie algebra over S which is a locally free module of finite type. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$; one says that it is a Cartan subalgebra if it satisfies the following conditions:

- (i) $\mathfrak{d}$ is a submodule locally a direct summand (hence also locally free of finite type).
- (ii) For every $s \in S$, $\mathfrak{d}(s)$ is a Cartan subalgebra of $\mathfrak{g}(s)$.

Definition 2.5. Let $S, \mathfrak{g}$ be as in 2.4. A section a of $\mathfrak{g}$ is said to be quasi-regular if for every $s \in S$, $a(s) \in \mathfrak{g}(s)$ is a regular element of the Lie algebra $\mathfrak{g}(s)$ over $k(s)$. One says that a is a regular section if it is quasi-regular and contained in a Cartan subalgebra of $\mathfrak{g}$.

Notions 2.4 and 2.5 are again stable under base change and local for the fpqc topology. Only the last assertion, and for the case of the notion “regular section”, requires a proof, and follows from the fact that the Cartan subalgebra containing a given regular section is uniquely determined. More precisely:

Proposition 2.6. *Let $S, \mathfrak{g}$ be as in 2.4, and let a be a quasi-regular section of $\mathfrak{g}$. Then there exists at most one Cartan subalgebra of $\mathfrak{g}$ containing a . For one to exist, i.e. for a to be a regular section, it is necessary and sufficient that a satisfy the following condition: $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ is a submodule locally a direct summand of $\mathfrak{g}$, and $\text{ad}(a)$ induces an automorphism of $\mathfrak{g}/\mathfrak{d}$. In this case, $\mathfrak{d}$ is the unique Cartan subalgebra of $\mathfrak{g}$ containing a .*

Indeed, suppose that a is contained in the Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$. Then $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is bijective on every fiber, hence (since $\mathfrak{g}/\mathfrak{d}$ is locally free of finite type) it is an automorphism of $\mathfrak{g}/\mathfrak{d}$; on the other hand, by 2.1, $\text{ad}(a)_{\mathfrak{d}}$ is locally nilpotent, hence $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$, which proves the uniqueness of $\mathfrak{d}$ and the necessity of the announced regularity criterion. For sufficiency, note that the hypothesis made on a implies that the formation of $\text{Nil}(a, \mathfrak{g})$ commutes with extension of the base and in particular with passage to the fibers, which shows in particular that the fibers $\mathfrak{d}(s)$ of $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ are Cartan subalgebras of the $\mathfrak{g}(s)$; moreover $\mathfrak{d}$ is a Lie subalgebra of $\mathfrak{g}$ by XIII 4.1, hence it is a Cartan subalgebra.

Corollary 2.7. *Under the conditions of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, a a section of $\mathfrak{d}$ which is regular in $\mathfrak{g}$, and u an automorphism of $\mathfrak{g}$. For u to leave $\mathfrak{d}$ invariant, it is necessary and sufficient that $u(a)$ be a section of $\mathfrak{d}$.*

Indeed, then by transport of structure $u(a)$ is a regular section of $\mathfrak{g}$, contained in both Cartan subalgebras $\mathfrak{d}$ and $u(\mathfrak{d})$, which are therefore identical.

Corollary 2.8. *Under the conditions of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then for every $s \in S$ such that $k(s)$ is infinite, there exists an open neighborhood V of s and a regular section a of $\mathfrak{g}$ over V such that $\mathfrak{d} = \text{Nil}(a, \mathfrak{g}|_V)$ (i.e. such that a is a section of $\mathfrak{d}|_V$).*

Indeed, the fact that $k(s)$ is infinite ensures the existence of a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}(s)$; one may extend it to a section a of $\mathfrak{d}$ over an open neighborhood U of s , and since $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ induces an automorphism of $\mathfrak{g}(s)/\mathfrak{d}(s)$, it follows that, provided one restricts V , $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is an automorphism, which implies that a is a quasi-regular section of $\mathfrak{g}|_V$ satisfying the desired condition.

Let $\mathfrak{g}$ be as in 2.4; then examination of its Killing polynomial implies at once that the function

$$s \longmapsto \text{nilpotent rank of } \mathfrak{g}(s)$$

on S is upper semicontinuous. We are mainly interested in the case where this function is in fact continuous, i.e. locally constant. Here are some variants of this property:

Proposition 2.9. *Let $S, \mathfrak{g}$ be as in 2.4. Consider the following conditions:*

(C₀): *The nilpotent rank of the $\mathfrak{g}(s)$ ($s \in S$) is a locally constant function of s .*

(C₁): *There exists a Cartan subalgebra of $\mathfrak{g}$ locally for the fpqc topology.*

(C'₁): *As (C₁), replacing “fpqc topology” by “étale topology”.*

(C₂): *Condition (C₀) holds, and for every S' over S , every quasi-regular section of $\mathfrak{g}_{S'} = \mathfrak{g} \otimes_S S'$ is regular.*

(C₃): *Every $s \in S$ has an open neighborhood V on which the Killing polynomial of $\mathfrak{g}$ is of the form*

$$t^r(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r})$$

where for every $s \in V$, $c_{n-r}(s) \in \text{Sym}(\mathfrak{g}(s)^\vee)$ is nonzero.

With this notation, one has the implications

$$(C_3) \Rightarrow (C_2) \Rightarrow (C'_1) \Rightarrow (C_1) \Rightarrow (C_0),$$

and when S is reduced, the five conditions considered are equivalent.

PROOF. Note also that the conditions considered are manifestly stable under base change and local for the fpqc topology.

The implications $(C'_1) \Rightarrow (C_1) \Rightarrow (C_0)$ are trivial, and implication $(C_0) \Rightarrow (C_3)$ when S is reduced is immediate. Note moreover:

Corollary 2.10. *Suppose condition (C_0) holds. Let U be the set of elements of $W(\mathfrak{g})$ which are regular in their fiber. Then U is open; in particular, for every section a of $\mathfrak{g}$ over S , the set V of $s \in S$ such that $a(s) \in \mathfrak{g}(s)$ is regular is open.*

Indeed, the first assertion follows from the second (applied to $\mathfrak{g}_{S'}$ for every base change S' over S). For the latter, since one may here suppose S reduced, hence (C_3) verified, it suffices to examine the Killing polynomial of a in $\mathfrak{g}$.

Implication $(C_2) \Rightarrow (C'_1)$ now follows easily from b) in the following more precise statement:

Corollary 2.11. *Suppose condition (C_2) holds. Then:*

- a) *For every $s \in S$ and every Cartan subalgebra $\mathfrak{d}_0$ of $\mathfrak{g}(s)$ such that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$ (a condition automatically verified if $k(s)$ is infinite), there exists an open neighborhood V of s and a subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ whose fiber at s is $\mathfrak{d}_0$. If S_1 is a subscheme of S containing s , and if one has already extended $\mathfrak{d}_0$ to a Cartan subalgebra $\mathfrak{d}_1$ of $\mathfrak{g} \otimes_S S_1$, then one may find an open neighborhood V of s in S and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ such that $\mathfrak{d} \otimes_V (S_1 \cap V)$ is equal to $\mathfrak{d}_1|_{(S_1 \cap V)}$.*
- b) *For every $s \in S$ such that $\mathfrak{g}(s)$ contains a regular element (a condition automatically satisfied if $k(s)$ is infinite), there exists an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$.*

Statement b) follows from a) by taking $\mathfrak{d}_0 = \text{Nil}(a_0, \mathfrak{g}(s))$, a_0 being a regular element of $\mathfrak{g}(s)$. To prove a), say the second formulation, one considers a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$, extends it in a neighborhood of s in S_1 to a section of $\mathfrak{d}_1$, and extends the latter to a section of $\mathfrak{g}$ in a neighborhood of s . By 2.10, this section is quasi-regular in an open neighborhood V of s , hence regular by (C_2) , so by 2.6, $\mathfrak{d} = \text{Nil}(a, \mathfrak{g}|_V)$ satisfies the desired conditions.

Remark 2.12. I do not know whether statement 2.11 a) and b) is valid without a hypothesis of existence of regular points (when $k(s)$ is finite).

It remains to prove implication $(C_3) \Rightarrow (C_2)$. Note also the following equivalent form of (C_3) :

(C'_3) One has (C_0) , i.e. the nilpotent rank of the $\mathfrak{g}(s)$ ($s \in S$) is locally constant, and on every open subset V of S where this rank has value r , the Killing polynomial of $\mathfrak{g}|_V$ is divisible by t^r .

One must show that this condition implies that every quasi-regular section a of $\mathfrak{g}$ is regular. Taking 2.6 into account, this is contained in the following lemma (applied to the endomorphism $\text{ad}(a)$ of $\mathfrak{g}$), (iv) $\Rightarrow$ (iii):

Lemma 2.13. *Let A be a ring, M a projective A -module of finite type, and u an endomorphism of M . The following conditions are equivalent:*

- (i) *M is the direct sum of two stable modules M', M'' such that $u|_{M'}$ is nilpotent, and $u|_{M''}$ is an automorphism of M'' .*
- (ii) *There exists an integer $n > 0$ such that $\text{Im } u^n + \text{Ker } u^n = M$.*

- (iii) The nilspace $\text{Nil}(u) = \bigcup_{n>0} \text{Ker } u^n$ is a direct summand of M , and $M = \text{Nil}(u) + u(M)$.

These conditions are implied by the following one (and are equivalent to it when A is reduced):
 (iv) Locally on $\text{Spec}(A)$ (for the Zariski topology) the characteristic polynomial $P_u(t)$ of u can be put in the form

$$t^r(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}),$$

where c_{n-r} is invertible.

The equivalence of (i), (ii), (iii) is immediate and is recorded for reference. The fact that (i) implies (iv) when A is reduced follows from the fact that in this case a nilpotent endomorphism of a projective module of rank r has characteristic polynomial t^r , whereas in all cases the characteristic polynomial of an automorphism of a projective module of finite type has as constant term the determinant of u up to sign (locally on $\text{Spec}(A)$), hence an invertible element of A . Finally, to prove (iv) $\Rightarrow$ (i), one notes that M is a module over the polynomial ring $A[t]$, letting t act by u , and the well-known identity

$$P(u) = 0$$

shows that M is annihilated by $PA[t]$, hence can be considered as a module over $A[t]/PA[t]$. Now, writing $P = t^r Q$, where the constant term of Q is invertible, one sees at once that

$$PA[t] = t^r A[t] \cap QA[t],$$

so $A[t]/PA[t]$ decomposes as the product of the rings $A[t]/t^r A[t]$ and $A[t]/QA[t]$, whence a corresponding decomposition of M as a sum of two $A[t]$ -modules, i.e. as a sum of two sub- A -modules M' and M'' stable under u ; this is the decomposition considered in (i).

This completes the proof of 2.13, hence of 2.9. $\square$

Corollary 2.14. *When condition (C_3) of 2.9 holds, the Cartan subalgebras of $\mathfrak{g}$ are strictly nilpotent (2.2).*

The proof is immediate.

Remark 2.15. a) Let us point out that one can prove a converse of 2.14: (C_3) is equivalent to the fact that for every S' over S , every quasi-regular section of $\mathfrak{g}_{S'}$ is regular and every Cartan subalgebra of $\mathfrak{g}_{S'}$ strictly nilpotent, or again every quasi-regular section of $\mathfrak{g}_{S'}$ contained in a strictly nilpotent Cartan subalgebra of $\mathfrak{g}_{S'}$.

b) Unlike the other conditions (C_0) to (C_2) , condition (C_3) is of “infinitesimal nature”; more precisely, when S is locally noetherian, $\mathfrak{g}$ satisfies condition (C_3) if and only if for every base change $S' \rightarrow S$, with S' artinian local (if one wishes, S' the spectrum of an artinian quotient of a local ring of S), $\mathfrak{g}_{S'}$ satisfies the same condition. Likewise, when (C_0) holds, the condition for a section of $\mathfrak{g}$ to be regular is of infinitesimal nature.

c) When $\mathfrak{g}$ is the Lie algebra of a smooth group prescheme of finite presentation over S , then we shall see that conditions (C_0) , (C_1) , (C'_1) , (C_2) on $\mathfrak{g}$ are equivalent (5.2 a)); I do not know what happens in general (except that, even for S artinian local, (C_0) does not imply (C_1)). However, even in the case where $\mathfrak{g}$ comes from a G , and S is artinian local, it is not true in general that (C_2) implies (C_3) , for $\mathfrak{g}$ can be nilpotent without being strictly nilpotent. One obtains an example of this fact by starting from a smooth affine group scheme G over the spectrum S of a discrete valuation ring, such that the Lie algebra of the general fiber is not nilpotent (for example, the general fiber is an adjoint semisimple group), and that of the special fiber is nilpotent (for example, the special fiber being a vector group): then for n sufficiently large, the Lie algebra of $G_n = G \times_S S_n$ is not strictly nilpotent, but it is nilpotent.

Let us still place ourselves under the conditions of 2.4, and let

$$\mathcal{D} : ((\text{Sch}))^\circ_S \rightarrow (\text{Ens})$$

be the functor defined by

$$\mathcal{D}(S') = \text{set of Cartan subalgebras of } \mathfrak{g}_{S'}.$$

Introduce also the functor

$$X(S') = \{(\mathfrak{d}, a) : \mathfrak{d} \text{ is a Cartan subalgebra of } \mathfrak{g}_{S'} \\ \text{and } a \text{ a section of } \mathfrak{d}\}.$$

One thus has two projections $(\mathfrak{d}, a) \mapsto \mathfrak{d}$ and $(\mathfrak{d}, a) \mapsto a$:

$$p : X \longrightarrow \mathfrak{d} \quad \text{and} \quad \psi : X \longrightarrow W(\mathfrak{g}).$$

When (C_0) holds, we also consider the open subset U of regular points of $W(\mathfrak{g})$ (cf. 2.10), and condition (C_2) is then expressed by the fact that the morphism

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism (a priori, it is a monomorphism by 2.6). Note that it is trivial that the morphism $p : X \longrightarrow \mathfrak{d}$ is representable by a projection of vector bundles (i.e. for every S -morphism $S' \longrightarrow \mathfrak{d}$, corresponding to a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}_{S'}$, $X \times_{\mathfrak{d}} S'$ is representable by a vector bundle over S' , namely $W(\mathfrak{d})$); hence if $\mathfrak{d}$ is representable, so is X . Now one has:

Theorem 2.16. *Let S be a prescheme, $\mathfrak{g}$ a Lie algebra over S which is a locally free $\mathcal{O}_S$ -module of finite type; suppose condition (C_0) of 2.9 holds.*

- a) *The functor $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ defined above is representable by a quasi-projective prescheme of finite presentation over S . The same holds for the functor X defined above.*
- b) *When condition (C_2) of 2.9 holds, $\mathcal{D}$ and X are smooth over S , and the morphism $\psi^{-1}(U) \longrightarrow U$ induced by ψ is an isomorphism.*
- c) *Still supposing condition (C_2) holds, let $s \in S$, and let $\mathfrak{d}_0$ be a Cartan subalgebra of $\mathfrak{g}(s)$ corresponding to a point d of $\mathcal{D}$ rational over $k(s)$. Suppose that $\mathfrak{d}_0$ contains a regular point of $\mathfrak{g}(s)$ (a condition automatically verified if $k(s)$ is infinite). Let r be the infinitesimal rank of $\mathfrak{g}(s)$, and n its rank over $k(s)$; then there exists an open neighborhood V of d in $\mathcal{D}$ which is S -isomorphic to an open subset V' of $S[t_1, \dots, t_{n-r}]$.*

PROOF. One may suppose that $\mathfrak{g}$ has constant rank n and constant nilpotent rank r . The assertions made about X in a) and b) follow at once from the assertions made about $\mathcal{D}$ and the fact that X is a vector bundle over $\mathcal{D}$ defined by a locally free module, and are included merely for reference.

a) The functor $\mathcal{D}$ is a subfunctor of the functor $\text{Grass}_{n-r}(\mathfrak{g})$ whose value at S' is the set of locally free quotient modules of rank $n-r$ of $\mathfrak{g}_{S'}$, and it is well known that the latter functor is representable by a projective prescheme smooth over S (cf., for example, Séminaire Cartan 1960/61, Exp. 12, Nos. 2 and 3, whose constructions carry over unchanged to the case of preschemes)^{*(1)}. One is therefore reduced to a relative problem, namely the following: given a locally free quotient module of rank $n-r$ of $\mathfrak{g}$, or, what amounts to the same thing, a locally free submodule $\mathfrak{d}$ of rank r which is locally a direct summand, represent the following functor: $F(S') = \emptyset$ if $\mathfrak{d}$ is not a Cartan subalgebra of $\mathfrak{g}_{S'}$, $F(S') = \{\emptyset\}$ otherwise. In fact, we shall see that F is representable by a subprescheme of finite presentation of S (which will show that $\mathcal{D} \longrightarrow \text{Grass}$ is representable by an immersion of finite presentation, and will finish proving a)). One begins by expressing the condition that $\mathfrak{d}_{S'}$ be a Lie subalgebra of $\mathfrak{g}_{S'}$; one sees immediately that this is expressed by the fact that $S' \longrightarrow S$ factors through a certain closed subprescheme S_1 of S , of finite presentation over S (whose local equations over S are written down immediately using a basis of $\mathfrak{g}$ adapted to the submodule $\mathfrak{d}$). One may therefore suppose that one already has $S = S_1$. One must next express that $\mathfrak{d}_{S'}$ contains, locally for fpqc, a quasi-regular section of $\mathfrak{g}_{S'}$, and for this one considers $V = W(\mathfrak{d}) \cap U$, where U is

^{*}Cf. also EGA I, 2nd edition (to appear in North Holland Publishing Cie).

⁽¹⁾**N.D.E.** See §I.1.3 of M. Demazure and P. Gabriel, *Groupes algébriques*, Masson (1970).

the open subset of regular points of $W(\mathfrak{g})$ (2.10); then the structural morphism $V \rightarrow S$ being smooth and quasi-compact, its image S_1 is an open part of S , and the immersion morphism $S_1 \rightarrow S$ is quasi-compact, i.e. of finite presentation. The condition considered on S' is then expressed by saying that $S' \rightarrow S$ factors through S_1 . Thus one is reduced to the case $S = S_1$, and, using descent theory, to the case where $\mathfrak{d}$ admits a section a which is a quasi-regular section of $\mathfrak{g}$. One must finally express that the section $a_{S'}$ of $\mathfrak{g}_{S'}$ deduced from a satisfies $\text{ad}(a_{S'})_{\mathfrak{g}_{S'}/\mathfrak{d}_{S'}}$ bijective, which again amounts to saying that $S' \rightarrow S$ factors through a certain open subscheme of finite presentation of S , namely S_D , where D is the determinant of $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$. But then one sees immediately that $\mathfrak{d}|_{S_D}$ is a Cartan subalgebra of $\mathfrak{g}|_{S_D}$, so S_D represents the functor F , which proves a).

b) Is immediate by 2.11 a) and XI 1.5. Of course, b) is also a consequence of the more precise statement c).

c) Let a_0 be a regular point of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$; extend it to a section a of $\mathfrak{g}$ over an open neighborhood V of s ; one may clearly suppose $V = S$. On the other hand, let M_0 be a complement of the vector space $\mathfrak{d}_0$ in $\mathfrak{g}(s)$; then in an open neighborhood V of S there exists a submodule M of $\mathfrak{g}$, a direct summand of $\mathfrak{g}|_V$, such that $M(s) = M_0$, and one may again suppose $V = S$. Let now $\mathbf{V}$ be the subfunctor of $\mathcal{D}$ such that $\mathbf{V}(S')$ is the set of Cartan subalgebras $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ satisfying the following two conditions:

1°) $\mathfrak{d}'$ is complementary to $M_{S'}$, and

2°) the unique section of $(a_{S'} + M_{S'}) \cap \mathfrak{d}$ is a regular section of $\mathfrak{g}_{S'}$.

Condition 1°) corresponds to an open subset V_1 of $\mathcal{D}$ (induced by the open subset of $\text{Grass}_{n-r}(\mathfrak{g})$ defined by the same condition 1°); the conjunction of 1° and 2° corresponds to an open subset of V_1 by 2.10 and (C₂). Thus $\mathbf{V}$ is represented by an open subscheme V of $\mathcal{D}$, clearly containing a .

On the other hand, let $\mathbf{V}'$ be the subfunctor of $W(M)$ defined by

$$\mathbf{V}'(S') = \{\text{sections } u' \text{ of } M_{S'} \text{ such that:}$$

(i) $a_{S'} + u'$ is a regular section of $\mathfrak{g}_{S'}$, and

(ii) the unique Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$

containing $a_{S'} + u'$ is complementary to $M_{S'}\}$.

Then condition 1°) corresponds to an open subscheme V'_1 of $W(M)$, namely the inverse image of the open subset U of regular points of $W(\mathfrak{g})$ (cf. 2.10) under the translation morphism $m \mapsto a + m$. The conjunction of (i) and (ii) corresponds to an open subset V' of V'_1 , namely the inverse image of V under the obvious morphism $V'_1 \rightarrow \mathcal{D}$ (associating to u' the unique Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ containing $a_{S'} + u'$). The restriction of this last morphism to V' is a morphism

$$V' \rightarrow V,$$

which is clearly an isomorphism. This proves c). $\square$

Corollary 2.17. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k . Then the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ (2.16 a)) is quasi-projective, smooth and irreducible. When $\mathfrak{g}$ contains a regular element (for example when k is infinite), $\mathfrak{d}$ is a rational variety, i.e. its function field is a pure extension of k .*

The fact that $\mathcal{D}$ is irreducible follows from the fact that one has a surjective morphism $\psi^{-1}(U) \rightarrow \mathcal{D}$, and $\psi^{-1}(U)$ is irreducible, being isomorphic to the open subset U of $W(\mathfrak{g})$. The assertion about the function field is an immediate consequence of c).

Remarks 2.18. I do not know whether this conclusion remains true if k is finite, without supposing that $\mathfrak{g}$ contains a regular point; compare 2.12. One can prove that it is so when $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G over k , at least when $G/\text{radical}$ is an “adjoint” semisimple group, using a result of Chevalley mentioned below (cf. Appendix). It is plausible

that this result remains valid without restriction on G ; for this it would suffice that the cited result of Chevalley be proved for every semisimple algebraic group (not necessarily adjoint).

3. Subgroups of type (C) of group preschemes over an arbitrary prescheme

Theorem 3.1. *Let S be a prescheme, G a smooth S -group prescheme, $\mathfrak{g}$ its Lie algebra (which is a locally free module of finite type over S), and $\mathfrak{h}$ a Lie subalgebra of $\mathfrak{g}$ which is (as a module) locally a direct summand in $\mathfrak{g}$, and such that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let a be a quasi-regular section of $\mathfrak{g}$ (2.5). Then*

$$M_a = \underline{\text{Transp}}_G(a, \mathfrak{h})$$

(the subfunctor of G whose points with values in S' are the $g \in G(S')$ such that $\text{ad}(g) \cdot a_{S'} \in \Gamma(S', \mathfrak{h}_S)$) is representable by a closed subprescheme⁽²⁾ of G smooth over S , whose structural morphism to S is surjective.

Consider the canonical morphism

$$\varphi : G \times_S W(\mathfrak{h}) \longrightarrow W(\mathfrak{g})$$

given by $(g, x) \mapsto \text{ad}(g) \cdot x$; then M_a is S -isomorphic to $\varphi^{-1}(a)$, the inverse image of a (considered as a section of $W(\mathfrak{g})$ over S) under φ . It will therefore suffice for the smoothness of M_a that we show that φ is smooth at the points of $G \times_S W(\mathfrak{h})$ over $\text{Im}(a)$; more generally, φ is smooth at every point over a point of $W(\mathfrak{g})$ which is regular in its fiber $W(\mathfrak{g}(s))$ over S . To see this, since the source and target of φ are smooth, hence flat and locally of finite presentation over S , one is reduced to verifying it fiber by fiber, which reduces one to the case where S is the spectrum of an algebraically closed field, G thus being a locally algebraic group over k , $\mathfrak{h}$ a subalgebra of its Lie algebra $\mathfrak{g}$ containing a Cartan subalgebra of $\mathfrak{g}$, and a a regular point of $\mathfrak{g}$. One may clearly suppose (taking into account that φ is a G -morphism) that the point considered in $G \times W(\mathfrak{h})$ is of the form (e, a) . One may clearly suppose G connected, hence of finite type over k , but then our assertion is precisely XIII 5.4. Moreover, the fact that M_a is a closed subprescheme of G (of finite presentation over S) is trivial, since M_a is the inverse image of $W(\mathfrak{h})$ under the morphism $g \mapsto \text{ad}(g) \cdot a$ from G to $W(\mathfrak{g})$. The surjectivity of the structural morphism $M_a \rightarrow S$ also reduces to the case of an algebraically closed base field, but then $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ by hypothesis, which is therefore conjugate to the Cartan subalgebra $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ by the conjugacy theorem XIII 6.1 a), hence $\mathfrak{h}$ contains a conjugate of a . This completes the proof of 3.1.

Corollary 3.2. *Let $G, \mathfrak{g}$ be as in 3.1, with G of finite type over S , and let $\mathfrak{k}, \mathfrak{h}$ be two Lie subalgebras of $\mathfrak{g}$, locally direct summands (as modules); suppose that one is under one of the following two hypotheses:*

- a) *For every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent and contains a regular element of $\mathfrak{g}_{\bar{s}}$; the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*
- b) *$\mathfrak{k}$ is a Cartan subalgebra of $\mathfrak{g}$.*

Then $\underline{\text{Transp}}_G(\mathfrak{k}, \mathfrak{h})$ is a closed subprescheme of G smooth over S ; moreover, in case a), its structural morphism to S is surjective.

The fact that the transporter is representable by a closed subprescheme of finite presentation of G is immediate, and is left to the reader. Let us first prove smoothness in case a). Suppose first that there exists a section a of $\mathfrak{k}$ which is quasi-regular in $\mathfrak{g}$. Then it suffices to apply 3.1 and the

⁽²⁾**N.D.E.** One will denote it by $\text{Transp}_G(a, \mathfrak{h})$.

Lemma 3.3. *Under the conditions of 3.2 a), if a is a section of $\mathfrak{k}$ quasi-regular in $\mathfrak{g}$, then one has*

$$\underline{\text{Transp}}_G(\mathfrak{k}, \mathfrak{h}) = \underline{\text{Transp}}_G(a, \mathfrak{h}).$$

Indeed, taking the definitions into account, this amounts to showing that if a is moreover a section of $\mathfrak{h}$, then one has $\mathfrak{k} \subset \mathfrak{h}$. Now, since by hypothesis $\mathfrak{k}$ is locally nilpotent, it follows from 2.1 that $\mathfrak{k} \subset \text{Nil}(a, \mathfrak{g})$; on the other hand, $\text{Nil}(a, \mathfrak{g}) \subset \mathfrak{h}$ for $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (being so fiber by fiber by XIII 4.8 b)). Whence the conclusion. In the general case, one reduces to the case where S is affine noetherian by the usual standard procedure, then to the case where S is artinian local (smoothness being a property of infinitesimal nature), and by flat descent to the case where its residue field is infinite, hence the fiber $\mathcal{X}_0$ admits an element regular in $\mathfrak{g}_0$. One lifts this element to an element of $\mathcal{X} = \Gamma(\mathfrak{k})$, which reduces one to the preceding case. Thus one has proved, in case a), smoothness of the transporter; as for the fact that its structural morphism is surjective, it reduces to the case where S is the spectrum of an algebraically closed field, hence where $\mathcal{X}$ contains a regular point of $\mathfrak{g}$, and one applies 3.3 and 3.1.

To prove b), one is reduced by the definition of smoothness (XI 1.1) to proving that if S is affine, S_0 a subscheme defined by a nilpotent quasi-coherent Ideal J , and g_0 an element of $G(S_0)$ transporting $\mathfrak{k}_0$ into $\mathfrak{g}_0$, then g_0 lifts to an element g of $G(S)$ transporting $\mathfrak{k}$ into $\mathfrak{g}$. Now the hypothesis made on g_0 implies that one is under the conditions of a), already treated. This completes the proof.

Of course, when in 3.2 b) $\mathfrak{h}$ satisfies the stronger hypothesis of 3.1, then (and only then) the structural morphism $\underline{\text{Transp}}_G(\mathfrak{d}, \mathfrak{h}) \rightarrow S$ is surjective. Using Hensel's lemma XI 1.10, one concludes from 3.1 and 3.2:

Corollary 3.4. *Under the conditions of 3.1 for G and $\mathfrak{h}$, and supposing G of finite type over S : a) For every quasi-regular section a of $\mathfrak{g}$, there exists, locally for the étale topology, a conjugate of a which is a section of $\mathfrak{h}$. b) For every Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$, $\mathfrak{d}$ is, locally for the étale topology, conjugate to a subalgebra of $\mathfrak{h}$.*

In particular, when $\mathfrak{h}$ is itself a Cartan subalgebra of $\mathfrak{g}$, one finds:

Corollary 3.5. *Let G be a smooth S -group prescheme of finite type, $\mathfrak{g}$ its Lie algebra, and $\mathfrak{d}, \mathfrak{d}'$ two Cartan subalgebras of G . Then $\underline{\text{Transp}}_G(\mathfrak{d}, \mathfrak{d}')$ is identical to the strict transporter of $\mathfrak{d}$ into $\mathfrak{d}'$, and is a closed subprescheme of G smooth over S , with surjective structural morphism. Locally for the étale topology, $\mathfrak{d}$ and $\mathfrak{d}'$ are conjugate.*

The fact that the transporter is here identical to the strict transporter follows trivially from the fact that $\mathfrak{d}$ and $\mathfrak{d}'$ are locally direct summands in $\mathfrak{g}$ and have the same rank at each point. Thus 3.5 is a particular case of 3.4. In particular, if $\mathfrak{d} = \mathfrak{d}'$:

Corollary 3.6. *Let G be as in 3.5, and let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then $\text{Norm}_G(\mathfrak{d})$ is a closed group subprescheme of G smooth over S , whose Lie algebra is identical to $\mathfrak{d}$.*

Indeed, this last assertion amounts to saying that $\mathfrak{d}$ is its own normalizer in $\mathfrak{d}$, which follows at once from the fact that this is true fiber by fiber.

Corollary 3.7. *Let $G, \mathfrak{g}$ be as in 3.5. Then conditions $(C_2), (C'_1), (C_1)$ of 2.9 are equivalent; in other words, if $\mathfrak{g}$ admits a Cartan subalgebra locally for the fpqc topology, then every quasi-regular section of $\mathfrak{g}$ is regular.*

Indeed, let a be a quasi-regular section; let us prove that it is regular. The question being local for the fpqc topology, one may suppose that $\mathfrak{g}$ admits a Cartan subalgebra $\mathfrak{d}$. By 3.4 a), a is then, locally for the étale topology, conjugate to a section of $\mathfrak{d}$, which reduces one to the case where a is a section of $\mathfrak{d}$, where the conclusion is trivial from the definition.

Definition 3.8. Let G be a smooth group prescheme over a prescheme S . One calls a subgroup of type (C) of G a group subscheme D of G , smooth over S , with connected fibers, such that $\mathfrak{d} = \text{Lie}(D)$ is a Cartan subalgebra (2.4) of $\mathfrak{g} = \text{Lie}(G)$, i.e. such that for every $s \in S$, D_s is a subgroup of type (C) of the algebraic group G_s (XIII 6.2).

Theorem 3.9. *Let G be a smooth S -group prescheme of finite presentation, and $\mathfrak{g}$ its Lie algebra. Then:*

a) *The map*

$$D \longmapsto \mathfrak{d} = \text{Lie}(D)$$

establishes a one-to-one correspondence between subgroups of type (C) of G and Cartan subalgebras of $\mathfrak{g}$.

b) *If D and $\mathfrak{d}$ correspond, one has*

$$\text{Norm}_G(D) = \text{Norm}_G(\mathfrak{d}),$$

which is a closed subscheme of G smooth over S , and one has

$$D = \text{Norm}_G(D)^0 = \text{Norm}_G(\mathfrak{d})^0.$$

c) *Two subgroups of type (C), D and D' , of G are conjugate locally for the étale topology.*

PROOF. Let D be a subgroup of type (C) of G , and $\mathfrak{d}$ its Lie algebra; then $D \subset \text{Norm}_G(\mathfrak{d})$, and by definition 3.8 and 3.6 this is an inclusion of group preschemes smooth over S , inducing an isomorphism on the Lie algebras. Since D has connected fibers, one therefore has $D = \text{Norm}_G(\mathfrak{d})^0$. Thus the map considered in a) is injective; let us prove it is surjective. Let then $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$; then by 3.6, $\text{Norm}_G(\mathfrak{d}) = N$ is a closed group subscheme of G smooth over S , admitting $\mathfrak{d}$ as Lie algebra. Since G is of finite presentation over S , the same holds for N , so (as was noted in XII after 7.3) the union of the connected components of the fibers of N is the underlying set of an open group subscheme N° of N , which is clearly a subgroup of type (C) of G with Lie algebra $\mathfrak{d}$. This proves a); the first assertion b) follows at once, and the formula $D = \text{Norm}_G(\mathfrak{d})^0$ has already been proved. Finally, c) follows from a) and 3.5. $\square$

Corollary 3.10. *Suppose that $\mathfrak{g}$ admits a Cartan subalgebra locally for the fpqc topology (or, what amounts to the same thing by 3.9 a), that G admits a subgroup of type (C) locally for the fpqc topology). Consider the functor $\mathcal{D} : ((\text{Sch}))_S^\circ \rightarrow (\text{Ens})$ defined by $\mathcal{D}(S') =$ the set of subgroups of type (C) of $G_{S'}$. Then this functor is representable by a quasi-projective prescheme smooth over S , with connected geometric fibers. When S is the spectrum of a field k , hence G a smooth algebraic group over k , and $\mathfrak{g}$ admits a regular point (a condition automatically verified if k is infinite), then $\mathcal{D}$ is a rational variety over k .*

Indeed, by 3.9 a), the functor $\mathcal{D}$ is canonically isomorphic to the functor considered in 2.16; on the other hand, by 3.7, condition (C₂) holds. Thus 3.10 follows from 2.16 and 2.17.

Corollary 3.11. *Let D be a subgroup of type (C) of G , and N its normalizer in G . Then the quotient sheaf G/N is canonically isomorphic to the functor $\mathcal{D}$ of 3.10, hence representable by a quasi-projective prescheme smooth over S , with connected geometric fibers.*

Proposition 3.12. *Let G be an S -group prescheme smooth and of finite presentation over S , H, K two group subschemes smooth and of finite presentation over S , K having connected fibers, and $\mathfrak{g}, \mathfrak{k}, \mathfrak{h}$ the corresponding Lie algebras. One assumes that one of the following two conditions holds for the geometric fibers of the latter:*

a) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$, and $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

b) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Under these conditions, for $H \supset K$ to hold, it is necessary and sufficient that one have $\mathfrak{h} \supset \mathfrak{k}$.*

Of course, one need only prove that if $\mathfrak{h} \supset \mathfrak{k}$, then $H \supset K$. In case b), the inclusion $\mathfrak{h} \supset \mathfrak{k}$ shows that one is in fact under the conditions of a), so it suffices to prove a). Proceeding as in 3.2 a) by reduction to the case of S artinian local, one is reduced to the case where there exists a section a of $\mathfrak{k}$ which is quasi-regular in $\mathfrak{g}$. In this case, proceeding as in XIII 5.5, one is reduced to the following statement:

Corollary 3.13. *Let G be an S -group prescheme smooth and of finite presentation over S , H a group subprescheme of G smooth and of finite presentation over S , $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras, and a a section of $\mathfrak{h}$ which is quasi-regular in $\mathfrak{g}$. One assumes that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let*

$$M_a = \text{Transp}_G(a, \mathfrak{h}),$$

which is a closed subprescheme of G smooth over S (cf. 3.1), so that M_a^0 (the union of the connected components of the identity element in the fibers of M_a) is an open part of M_a , which we shall equip with the structure induced by M_a . One then has $H^0 = M_a^0$.

Clearly, one has $H \subset M_a$, whence $H^0 \subset M_a^0$. Since this is an inclusion of preschemes smooth over S , to prove that it is an equality, one is reduced to verifying it on the fibers, which reduces one to the case where S is the spectrum of an algebraically closed field, a case seen in XIII 5.4.

Corollary 3.14. *Let G be a smooth S -group prescheme of finite presentation, D a subgroup of type (C) of G , and H a group subprescheme of G smooth over S and of finite presentation over S , and $\mathfrak{d}$ and $\mathfrak{h}$ their Lie algebras. Then one has*

$$\underline{\text{Transp}}_G(D, H) = \underline{\text{Transp}}_G(\mathfrak{d}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G smooth over S .

Indeed, the identity between the two transporters follows from 3.12 b), which allows one to apply 3.2.

Corollary 3.15. *Let G, H be as in 3.14, and suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Suppose moreover that $\mathfrak{g}$ admits a Cartan subalgebra locally for the fpqc topology. Then locally for the étale topology, H contains a subgroup of type (C) of G .*

By 3.7 and 3.9 a), G admits a subgroup of type (C) locally for the étale topology, so one may suppose that G admits such a subgroup, say D . Then the hypothesis on $\mathfrak{h}$ also means that the structural morphism of the transporter considered in 3.14 is surjective (taking into account the conjugacy theorem XIII 6.1 a)). One then concludes by Hensel's lemma XI 1.10.

Corollary 3.16. *Let G, H, K be as in 3.12 a); suppose moreover that for every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent (i.e. $\mathfrak{k}$ is locally nilpotent). Then one has*

$$\underline{\text{Transp}}_G(K, H) = \underline{\text{Transp}}_G(\mathfrak{k}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G smooth over S , with surjective structural morphism to S . Locally for the étale topology, H contains a subgroup conjugate to K .

The identity of the two transporters is again contained in 3.12 a); the assertion about its structure is then precisely 3.2 a), and the last assertion of 3.16 is then a consequence of Hensel's lemma.

Remarks 3.17. a) In 3.12 and 3.16, one may replace the hypothesis that K is smooth over S by the following weaker hypothesis: the sheaf $e_K^*(\Omega_{K/S}^1) = \omega_K^1$ of relative 1-differentials along the unit section is locally free. In this way, 3.12 contains XIII 5.5.

b) Let $G, \mathfrak{g}, \mathfrak{h}$ be as in 3.1, with G of finite presentation over S . Then $N = \text{Norm}_G(\mathfrak{h})$ is not necessarily smooth over S along the unit section, or, what amounts to the same thing, there does not necessarily exist a group subscheme H of S smooth over S whose Lie algebra is $\mathfrak{h}$, even if S is the spectrum of a field. When such an H exists, so that one then has (taking H with connected fibers) $N = \text{Norm}_G(H)$, I do not know whether N is smooth over S . In this question, one may clearly reduce to the case where S is artinian local.

Finally, to end this section, let us examine the case where G is “semisimple” over S :

Theorem 3.18. *Let S be a prescheme, G an S -group prescheme smooth over S , whose geometric fibers are “adjoint” semisimple groups, i.e. semisimple with reductive center (XII 4.1 and 4.4) reduced to the unit group. Then the subgroups of type (C) of G are identical to its maximal tori, hence also to its Cartan subgroups (XII 3.1).*

Taking the definitions into account, one is reduced to the case where S is the spectrum of an algebraically closed field, and to proving then that for a maximal torus T of G , the Lie algebra $\mathfrak{t}$ of T is a Cartan subalgebra of $\mathfrak{g}$, that is (taking into account the inequality: nilpotent rank of $\mathfrak{g} \geq \text{nilpotent rank of } G = \dim T = \text{rank } \mathfrak{t} = r$) that there exists $x \in \mathfrak{t}$ with $\dim \text{Nil}(x, \mathfrak{g}) = r$. Since $\mathfrak{t}$ is abelian and a fortiori nilpotent, it amounts to the same thing to say that there exists an $a \in \mathfrak{t}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{t}}$ is injective (XIII 5.7 a)). Now consider the characters α of T which occur in the representation induced on T by the adjoint representation of G . The structure theory of the semisimple group G (BIBLE, 13 th. 1 a) and th. 3, cor. 2), more precisely of the “big cell” of G , a semidirect product of T and subgroups P_α isomorphic to the additive group $\mathbf{G}_a$, invariant under T and corresponding to the “root” characters of G for the torus T , shows that the eigenspace of $\mathfrak{g}$ for the unit character is precisely $\mathfrak{t}$, and the other eigenspaces have dimension 1, the associated characters α being precisely the roots of G for T . By the calculation of the reductive center of G as the intersection of the kernels of the characters of T which occur in the adjoint representation of G (XII 4.8), one sees that G being adjoint is interpreted by the fact that the roots generate the lattice $M = \text{Hom}(T, \mathbf{G})$. Now a well-known lemma of root theory tells us that every root belongs to a system of simple roots, hence to a basis of the group generated by the roots, and consequently to a basis of the dual M of $T^{(3)}$. One concludes:

Corollary 3.19. *If G is an adjoint semisimple algebraic group over an algebraically closed field k , and T a maximal torus of G , then for every root α of G with respect to T , $\alpha : T \rightarrow \mathbf{G}_m$, the corresponding homomorphism $\alpha' : \mathfrak{t} \rightarrow k$ is nonzero.*

This result is essentially equivalent to theorem 3.18, for, for a $t \in \mathfrak{t}$, $\text{ad}(t)$ is semisimple, and its eigenvalues on $\mathfrak{g}/\mathfrak{t}$ are precisely the $\alpha'(t)$; hence $\text{ad}(t)_{\mathfrak{g}/\mathfrak{t}}$ is injective if and only if the $\alpha'(t)$ are $\neq 0$, and there exists a $t \in \mathfrak{t}$ with this property if and only if all the α' are $\neq 0$.

Corollary 3.20. *Let S be a prescheme, G an S -group prescheme, smooth and of finite presentation over S , with geometric fibers connected reductive algebraic groups (i.e. extensions of a semisimple group by a torus). Then for every $s \in S$ there exists an open neighborhood U of s such that $G|_U$ admits a maximal torus.**

Indeed, we shall see in XVI that the hypothesis just made on G implies that G is affine over S and has locally constant reductive rank; hence (XII 4.7 c)) G admits a reductive center Z , $G' = G/Z$ is a smooth affine group over S whose reductive center is reduced to the unit group, and finally the maximal tori of G and of G' are in one-to-one correspondence. Moreover, one sees at once that the geometric fibers of G' are connected semisimple groups, and moreover adjoint by definition (their reductive center being trivial). Thus it suffices to

⁽³⁾**N.D.E.** $\hat{T}$ has been replaced by T .

*This result, as well as 2.11 on which it rests, generalizes immediately to the case where s is replaced by a finite part of S contained in an affine open subset.

confine oneself to the case where G is semisimple and adjoint, and by 3.18 one is reduced to finding an open neighborhood U of s and a subgroup of type (C) of $G|_U$, or, what amounts to the same thing (3.9 a)), a Cartan subalgebra of $\mathfrak{g}|_U$. Now this is possible if $k(s)$ is infinite, for by 3.7, $\mathfrak{g}$ satisfies condition (C₂) of 2.9, hence one may apply 2.11 b). In fact, statement 3.20 remains valid without supposing $k(s)$ infinite. Indeed, by the preceding proof, it suffices to know that for every adjoint semisimple group G over a finite field k , the Lie algebra $\mathfrak{g}$ of G contains a regular element. Now this statement has been proved by Chevalley (using the properties of the Coxeter element of the Weyl group...); cf. the Appendix below by J.-P. Serre.

Remarks 3.21. a) Statement 3.20 remains valid, with essentially the same proof, replacing G by a closed group subscheme H , smooth over S , having everywhere the same rank as G (for example, a “parabolic subgroup” of G), provided $k(s)$ is infinite. I do not know whether here again the hypothesis that $k(s)$ be infinite is superfluous. One can show that it is so at least if H is parabolic, thanks to the construction of the radical U of H and of the quotient H/U , which reduces one to the semisimple case. Unfortunately, the method of regular elements seems powerless here, for one easily constructs examples (for example, with the projective group and its “standard” Borel subgroup) where the Lie algebra $\mathfrak{h}$ of H contains no regular element.

b) The proof of 3.20 in fact shows a more precise result (invoking 3.9 b)) in the case where $k(s)$ is infinite, namely that every maximal torus T_0 of G_s comes from a maximal torus T over an open neighborhood of s . I do not know whether this statement remains valid when $k(s)$ is no longer supposed infinite, the difficulty clearly coming from the fact that the Lie algebra $\mathfrak{t}_0$ of T_0 generally contains no regular element of the Lie algebra $\mathfrak{g}_0$ of the fiber G_s . An affirmative answer to this problem would imply the following statement (which is proved only in the case of an infinite residue field or when A is separated and complete): Let A be a local ring with residue field k , M an “Azumaya algebra” over A , i.e. an algebra such that M is a free module of finite type over A and $M_0 = M \otimes_A k$ a central simple algebra over k , and D_0 a commutative subalgebra of M_0 separable over k , such that $[M_0 : k] = ([D_0 : k])^2$; then there exists a commutative subalgebra D of M which is a module direct summand in M and such that $D \otimes_A k = D_0$ (?). (Note that giving M is equivalent to giving a principal homogeneous bundle under the projective group $\mathrm{PGL}(n)_A$, whence an “inner” twisted form G of $\mathrm{PGL}(n)$, and the maximal tori of G correspond one-to-one to the commutative subalgebras D of M , étale over A , of rank n over A .)

c) Applying 3.20 to the centralizer of a subtorus Q of G (G a reductive S -group prescheme), one deduces that every such Q is contained, locally for the Zariski topology, in a maximal torus of G .

4. A digression on Borel subgroups

Definition 4.1. Let G be a smooth algebraic group over an algebraically closed field. One calls a Borel subgroup of G a smooth solvable connected algebraic subgroup which is maximal for these properties.

When G is affine, one thus recovers the terminology of BIBLE 6 def. 1. Note immediately that if Z is a smooth connected subgroup of G contained in the center (or, more generally, solvable and invariant), then for every Borel subgroup B of G , the image BZ of $B \times Z$ under the morphism $(b, z) \mapsto bz$ from $B \times Z$ to G is a smooth solvable connected subgroup of G containing B , hence identical to B , so B contains Z , hence B is the inverse image of an algebraic subgroup B' of $G' = G/Z$, and it is immediate that B' is a Borel subgroup of G' . Taking $Z = \mathrm{Centr}(G^0)_{\mathrm{red}}^0$, $G' = G/Z$ is affine (XII 6.1), so the Borel subgroups of G' are conjugate, and for such a B' , G'/B' is a projective variety (BIBLE 6 th. 4). Consequently:

Proposition 4.2. *Let G be as in 4.1. Then the Borel subgroups of G are conjugate. If B is a Borel subgroup, then G/B is a projective variety. The maximal tori of B (resp. the Cartan subgroups of B , G being connected) are maximal tori of G (resp. Cartan subgroups of G).*

It remains to prove the last assertion, and one may clearly suppose $G = G^0$. For Cartan subgroups, it follows from the analogous assertion in G' (BIBLE 6 th. 4 cor. 4) and from XII 6.6 e). For maximal tori, it follows from the preceding assertion, since by XII 6.6 c) the maximal tori of a smooth algebraic group are the maximal tori of its Cartan subgroups.

Corollary 4.3. *Suppose G connected. Then every element of G is contained in a Borel subgroup of G .*

One is reduced to the same statement in G' , which is well known (BIBLE 6 th. 5 d)).

Corollary 4.4. *Let B be a Borel subgroup of G , C a Cartan subgroup of B , and N its normalizer in G ; then $N \cap B = C$.*

Indeed, $N \cap B = \text{Norm}_B(C)$, so one is reduced to showing that when G is connected and solvable, a Cartan subgroup C is its own connected normalizer. Now, with the preceding notation, C is the inverse image of a Cartan subgroup C' of G' , so one is reduced to the case where G is affine. Since one knows that the normalizer of a Cartan subgroup is smooth (C being its own connected normalizer; cf., for example, XII 6.6 c)), it suffices to see that C and N have the same points with values in k , which is precisely BIBLE th. 6 d).

Definition 4.5. Let G be a smooth group prescheme of finite presentation over a prescheme S . One calls a Borel subgroup of G any smooth group subprescheme B of finite presentation of G such that for every $s \in S$, the geometric fiber $B_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$.

Thus this is, as one verifies at once, a notion stable under base change and local for the fpqc topology (for if k' is an algebraically closed extension of an algebraically closed field k , then an algebraic subgroup B of the smooth algebraic group G over k is a Borel subgroup of G if and only if $B_{k'}$ is one of $G_{k'}$). It follows from this definition that if G is a smooth algebraic group over an arbitrary field k , and B a Borel subgroup of G , then G/B is a projective variety, every maximal torus T of B is a maximal torus of G , its normalizer in B is identical to its centralizer C , and is a Cartan subgroup of G when G is connected.

Remarks 4.6. Unfortunately, it is no longer true in general (even if G is affine over S and S is the spectrum of the algebra of dual numbers over an algebraically closed field k) that two Borel subgroups of G are conjugate locally for the fpqc topology. As a consequence of this regrettable fact, let us point out that if G is a smooth affine connected algebraic group over an imperfect field k , it is not in general possible to define naturally a homogeneous space D under G , playing the role of a flag variety, i.e. of the variety of Borel subgroups of G (which over the algebraic closure $\bar{k}$ of k would therefore be isomorphic to $G_{\bar{k}}/\bar{B}$, where $\bar{B}$ is a Borel subgroup of $G_{\bar{k}}$). Indeed, when the quotient of $G_{\bar{k}}$ by its radical $\bar{R}$ is an adjoint semisimple group, then the kernel of $G_{\bar{k}} \rightarrow \underline{\text{Aut}}_{\bar{k}}(\bar{D})$ is the radical of $G_{\bar{k}}$; hence if $\bar{D}$ comes from a homogeneous space D under G , the radical $\bar{R}$ comes from a subgroup R of G . Now one easily constructs examples where $G_{\bar{k}}/\bar{R}$ is adjoint but $\bar{R}$ is not “defined over k ”. It is easy to see that under these conditions, the functor $\mathcal{B} : ((\text{Sch}))_S^\circ \rightarrow (\text{Ens})$ such that $\mathcal{B}(S') =$ the set of Borel subgroups of $G_{S'}$ is not representable by a smooth S -prescheme. From the infinitesimal point of view (III §3), the failure of the conjugacy theorem is expressed by the fact that if B is a Borel subgroup of the smooth algebraic group G , the cohomology group $H^1(B, \mathfrak{g}/\mathfrak{b})^{(4)}$ may be nonzero.

On the other hand, we shall see in a later exposé that when G is semisimple, or more generally reductive, such unpleasant phenomena do not occur. It is doubtless these phenomena,

⁽⁴⁾**N.D.E.** Where $\mathfrak{b}$ is the Lie algebra of B .

as well as the absence of good existence theorems, which explain why Borel subgroups play only a relatively unobtrusive role in the study of general group schemes from the scheme-theoretic point of view, whereas they will dominate the theory of semisimple group schemes in later exposés.

Proposition 4.8. ⁽⁵⁾ *Let G be a smooth S -group prescheme of finite presentation with connected fibers, and B a Borel subgroup of G ; then B is identical to its own normalizer, and it is a closed subprescheme of G .*

Indeed, by XII 7.10, one is reduced to proving that over an algebraically closed field k , every element of $G(k)$ which normalizes B is in $B(k)$, which for affine G is a fundamental result of Chevalley (BIBLE 9 th. 1); the general case reduces to it at once by the reduction already used in 4.2.

Remarks 4.8.1. One may generalize definition 4.5 by introducing also the notion of parabolic subgroup of G : one thus calls a group subprescheme P of G , smooth and of finite presentation over S , such that for every $s \in S$, the geometric fiber $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$, i.e. contains a Borel subgroup of $G_{\bar{s}}$. Proposition 4.8 extends (with the same proof of reduction to the “set-theoretic” statement, which is known) to the case of a parabolic subgroup P of G . Note the following consequence of this result (cf. XVI). If P is a parabolic subgroup of G , then G/P is representable by a quasi-projective prescheme of finite presentation over S (N.B. one assumes G has connected fibers). Moreover, G/P is clearly smooth over S , and moreover has connected proper geometric fibers, whence one may easily conclude, using EGA III 5.5.1, that $D = G/P$ is in fact proper, hence projective, over S . Moreover, if its relative dimension is n , it is known that the invertible sheaf $\Omega_{D/S}^n$ is such that its inverse induces ample sheaves on the geometric fibers of D/S , hence (EGA III 4.7.1) $(\Omega_{D/S}^n)^{-1}$ is ample on D relative to S .

One easily sees, by reduction to the affine case and to the case of an algebraically closed base field, that if $u : G \rightarrow G'$ is an epimorphism of smooth algebraic groups, then for every Borel subgroup B of G , $u(B) = B'$ is a Borel subgroup of G' . We are interested in the case where one thus obtains a bijective correspondence between Borel subgroups of G and of G' :

Proposition 4.9. *Let G, G' be two smooth S -group preschemes of finite presentation with connected fibers, and $u : G \rightarrow G'$ a faithfully flat homomorphism of groups, i.e. surjective⁽⁶⁾. One assumes that one is in one of the following two cases (where one has set $N = \text{Ker } u$):*

a) N is central in G .

b) S is the spectrum of a field k , and if $\bar{k}$ denotes an algebraic closure of it, $N_{\bar{k}} = \text{Ker } u_{\bar{k}}$ is contained in the radical of $G_{\bar{k}}$, i.e. in the largest smooth connected solvable invariant subgroup of $G_{\bar{k}}$.

Then the map $B' \mapsto u^{-1}(B')$ induces a bijection from the set of Borel subgroups of G' onto the analogous set for G .

Case b) follows at once from the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N , and the fact that when k is algebraically closed, the Borel subgroups of G contain the radical of G (which is immediate by the argument preceding 4.2).

Let us prove case a). By XII 7.12, the map $H' \mapsto H = u^{-1}(H')$ establishes a one-to-one correspondence between group subpreschemes H' of G' which are smooth and of finite presentation over S , with connected fibers, and which have at every $s \in S$ the same reductive rank and the same nilpotent rank as G' , and group subpreschemes H of G having the analogous properties. Now the Borel subgroups (of G' , or of G) have the properties in question. It remains to prove that if H', H correspond, then H' is a Borel subgroup of G'

⁽⁵⁾N.D.E. There is no number 4.7.

⁽⁶⁾N.D.E. Add a reference here?

if and only if H is one in G . By definition, this question reduces to the case where S is the spectrum of an algebraically closed field. Now, since N is central in G , hence in H , it follows that H is solvable if and only if H' is. Finally, taking into account the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N , one sees at once that H' possesses the maximality property of definition 4.1 if and only if H possesses it, which completes the proof.

Corollary 4.10. *With the notation of 4.7, if B' and B are Borel subgroups of G' and G which correspond, one has*

$$\mathfrak{b} = \text{Lie}(u)^{-1}(\mathfrak{b}')$$

where $\mathfrak{g}, \mathfrak{g}', \mathfrak{b}, \mathfrak{b}'$ are the Lie algebras of G, G', B, B' , and where $\text{Lie}(u) : \mathfrak{g} \rightarrow \mathfrak{g}'$ is the homomorphism deduced from u .

This statement follows trivially from the definitions and the relation $u^{-1}(B') = B$.

We can now prove the main result of the present section:

Theorem 4.11. *Let G be a smooth algebraic group over an algebraically closed field k , and $\mathfrak{g}$ its Lie algebra. Then $\mathfrak{g}$ is equal to the union of the Lie algebras $\mathfrak{b}$ of the Borel subgroups B of G .*

One may clearly suppose G connected. Let R be the radical of G , and let $G' = G/R$. Then 4.9 b) and 4.10 reduce one to proving theorem 4.11 for G' in place of G , i.e. one may suppose G semisimple. Let then Z be the center of G , identical to the reductive center, and let $G' = G/Z$. The same argument (now using 4.9 a)) reduces one to proving the theorem for G' , i.e. one may suppose G adjoint semisimple. Let B be a Borel of G , T a maximal torus of B , hence of G , and $\mathfrak{b}, \mathfrak{t}$ the Lie algebras. By 3.18, T is a subgroup of type (C) of G , i.e. $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$, so the union of the conjugates of $\mathfrak{t}$ is dense in $\mathfrak{g}$ (XIII 5.1 (i) $\Rightarrow$ (vii)). A fortiori the union of the conjugates of $\mathfrak{b}$ is dense in $\mathfrak{g}$. Now let X be the closed subscheme of $G/B \times W(\mathfrak{g})$ whose points with values in k are the (g', x) such that $x \in \text{Ad}(g) \cdot \mathfrak{b}$ (cf. XIII 1). Then the morphism $\psi : X \rightarrow W(\mathfrak{g})$ induced by the second projection is proper since G/B is proper over k ; on the other hand one has just seen that it is dominant, so it is surjective, which proves 4.11.

The only result of the present section that we shall use in the rest of this exposé is the following corollary:

Corollary 4.12. *Let k be an infinite field, G a smooth algebraic group over k , T a maximal torus of G , $\mathfrak{g}$ and $\mathfrak{t}$ the Lie algebras, and $u : G \rightarrow \text{GL}(V)$ a linear representation (V a finite-dimensional vector space over k), whence a representation of Lie algebras $u' : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ making V a $\mathfrak{g}$ -module. Then the minimum of the nullity of $u'(x)$ ($x \in \mathfrak{g}$) is attained for an element $x \in \mathfrak{t}$.*

One is reduced at once to the case where k is algebraically closed. One may clearly suppose G connected, and, dividing G by $(\text{Ker } u)_{\text{red}}$, one may suppose G affine. Using 4.11 and the conjugacy theorem for maximal tori of G , one is reduced to the case where G is moreover solvable. Then G is a semidirect product $T \cdot V$, where V is the “unipotent part” of G , which is a smooth connected unipotent group (BIBLE 6 th. 3). Thus $\mathfrak{g}$ is a direct sum (as a vector space) of the Lie subalgebras $\mathfrak{t}$ and $\mathfrak{v} = \text{Lie}(V)$. By the Lie–Kolchin theorem (BIBLE 6 th. 1), $\mathfrak{v}$ admits a composition series by stable subspaces $\mathfrak{v}_i$ such that $\mathfrak{v}_i/\mathfrak{v}_{i+1} = \mathfrak{w}_i$ has dimension 1. Then for each i , one has an induced representation $u_i : G \rightarrow \text{GL}(\mathfrak{w}_i) = \mathbf{G}_m$ and the corresponding homomorphism of Lie algebras $u'_i : \mathfrak{g} \rightarrow k$, so that for every $x \in \mathfrak{g}$, the nullity of $u'(x)$ is equal to the number of i such that $u'_i(x) = 0$. Since V is unipotent, the u_i are trivial on V , hence the u'_i are trivial on $\mathfrak{v}$, which proves that if $x = t + v$ ($t \in \mathfrak{t}$, $v \in \mathfrak{v}$), then $u'_i(x) = u'_i(t)$ for every i , hence the nullity of $u'(x)$ is equal to that of $u'(t)$. Assertion 4.12 follows at once.

5. Relations between Cartan subgroups and Cartan subalgebras

Applying 4.12 to the adjoint representation of G , one finds:

Theorem 5.1. *Let G be a smooth algebraic group over an infinite field, T a maximal torus of G , and $\mathfrak{g} \supset \mathfrak{t}$ the Lie algebras; then $\mathfrak{t}$ contains a regular element of $\mathfrak{g}$.*

Corollary 5.2. *Let G be a smooth S -group prescheme of finite presentation, and $\mathfrak{g}$ its Lie algebra.*

a) *Conditions (C_0) to (C_2) of 2.9 on $\mathfrak{g}$ are equivalent; in particular, if the infinitesimal rank of the fibers of $\mathfrak{g}$ at the points of S is locally constant, then $\mathfrak{g}$ admits a Cartan subalgebra locally for the étale topology, hence (by 3.9 a)) G admits a subgroup of type (C) locally for the étale topology.*

b) *Let H be a group subprescheme of G smooth and of finite presentation over S , with connected fibers having the same reductive rank as G at each $s \in S$ (for example, H is a maximal torus or a Cartan subgroup of G), and D a subgroup of type (C) of G ; then one has $H \subset D$ if and only if one has $\mathfrak{h} \subset \mathfrak{d}$.*

c) *Suppose condition (C_0) holds, i.e. the infinitesimal rank of G is locally constant. Let H be a group subprescheme of G smooth and of finite presentation over S , with nilpotent connected fibers, having at every point the same reductive rank as G (for example, H is a maximal torus or a Cartan subgroup of G). Then H is contained in one and only one subgroup D of type (C) of G .*

d) *Suppose that G admits a Cartan subgroup locally for fpqc; then so does every subgroup D of type (C) of G .*

PROOF. a) Suppose condition (C_0) holds, and let us prove that the same holds for (C_2) , i.e. that every quasi-regular section a of $\mathfrak{g}$ is regular. One reduces as usual to the case of S affine noetherian, then, the question being “infinitesimal” (2.15 b)), to the case where S is artinian local (then (C_0) is moreover trivially satisfied). One may suppose in addition that the residue field k of S is infinite. Note that by 3.7 it suffices to establish that $\mathfrak{g}$ admits a Cartan subalgebra. Let T be a maximal torus of G ; by 5.1 there exists a quasi-regular element t contained in the Lie algebra $\mathfrak{t}$ of T ; let us prove that it is regular, which will finish the proof. Consider the linear representation of T in $\mathfrak{g}$ induced by the adjoint representation of G ; there thus exists a finite set $(u_i)_{i \in I}$ of characters of T such that $\mathfrak{g}$ decomposes as a direct sum of submodules $\mathfrak{g}_i$ stable under T , T acting on $\mathfrak{g}_i$ by u_i (cf. I §4.7.3). Let $u'_i : \mathfrak{t} \rightarrow A$ be the homomorphism of Lie algebras deduced from $u_i : T \rightarrow \mathbf{G}_m$ (N.B. A denotes the ring of S). Consider the homomorphisms u_{i0} and u'_{i0} deduced from the preceding ones by passage to the fibers, i.e. by the base change $A \rightarrow k$. Let I' be the set of $i \in I$ such that $u'_{i0} \neq 0$, and let $I'' = I - I'$. The fact that t is regular is expressed by the condition $u'_{i0}(t_0) \neq 0$ for every $i \in I'$, hence $u'_i(t)$ invertible for $i \in I'$. The Cartan subalgebra of $\mathfrak{g}_0$ defined by t , i.e. the kernel subspace of the semisimple endomorphism $\text{ad}(t_0)$ of $\mathfrak{g}_0$, is $\sum_{i \in I''} (\mathfrak{g}_i)_0$. Consider

$$\mathfrak{d} = \sum_{i \in I''} \mathfrak{g}_i,$$

then $\text{ad}(t)$ is nilpotent on $\mathfrak{d}$; on the other hand, it is an automorphism of $\mathfrak{g}/\mathfrak{d} \simeq \sum_{i \in I'} \mathfrak{g}_i$. By 2.6, t is therefore regular.

b) Follows from 3.12 a), H satisfying the hypothesis that every geometric fiber of $\mathfrak{h}$ contains a regular element of that of $\mathfrak{g}$, by 5.1.

c) By b), one is reduced to proving that $\mathfrak{h}$ is contained in one and only one Cartan subalgebra of $\mathfrak{d}$. By a), one knows moreover that $\mathfrak{g}$ satisfies (C_2) . One is therefore reduced to the

Lemma 5.3. *Let $\mathfrak{g}$ be a Lie algebra over S which is a locally free module of finite type and satisfies condition (C_2) of 2.9. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ satisfying the following conditions: it is a module locally a direct summand, it is locally nilpotent, and for every $s \in S$*

the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$. Then $\mathfrak{h}$ is contained in one and only one Cartan subalgebra of $\mathfrak{g}$.

(N.B. in the case of interest to us, $\mathfrak{h}$ satisfies the stated conditions: it is locally nilpotent for H has nilpotent fibers, and the condition on regular elements follows from 5.1.) Since 5.3 is local for the fpqc topology, it suffices to prove that at a point $s \in S$ such that $\kappa(s)$ is infinite, there exists an open neighborhood U of s such that existence and uniqueness hold for every base change $S' \rightarrow S$ factoring through U . Take a regular element of $\mathfrak{g} \otimes \kappa(s)$ contained in $\mathfrak{h} \otimes \kappa(s)$; extend it to a section a of $\mathfrak{h}$ over an open neighborhood U of s ; by (C₂) one may suppose that this section is regular (2.10). One may suppose $U = S$. A Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$ contains a , hence is identical to $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ (2.6), whence uniqueness. Moreover, since $\mathfrak{h}$ is locally nilpotent, one indeed has $\mathfrak{h} \subset \mathfrak{d}$, which proves existence.

d) This is a particular case of XII 7.9 d). $\square$

Corollary 5.4. *Let G be a smooth connected algebraic group over an algebraically closed field k , H a connected algebraic subgroup such that $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$, i.e. such that H contains a subgroup of type (C) of G . Then the number of conjugates of H containing a regular element of $G(k)$ is equal to the number of conjugates of $\mathfrak{h}$ containing a regular element of $\mathfrak{g}$.*

Indeed, let g be a regular element of $G(k)$, and C the unique Cartan subgroup of G containing g (XIII 2); then the conjugates of H containing g are those containing C (XIII 2.8 b)). Likewise, let x be a regular element of $\mathfrak{g}$; then if a conjugate $\mathfrak{h}'$ of $\mathfrak{h}$ contains x , then $\text{ad}(x)_{\mathfrak{g}/\mathfrak{h}'}$ is injective (XIII 5.4), so $\mathfrak{h}'$ contains $\text{Nil}(x, \mathfrak{g}) = \mathfrak{d}$; hence the number of conjugates of $\mathfrak{h}$ containing x is equal to the number of conjugates containing the Cartan subalgebra $\mathfrak{d}$. Moreover, $\mathfrak{d}$ is the Lie algebra of a subgroup D of type (C) of G . One may clearly suppose $C \subset D$, and assertion 5.4 will follow from this: for H to contain C , it is necessary and sufficient that $\mathfrak{h}$ contain $\mathfrak{d}$. Indeed, by XIII 5.5 the relation $\mathfrak{h} \supset \mathfrak{d}$ implies $H \supset D$, and a fortiori $H \supset C$. Conversely, by 5.1, $\mathfrak{t}$ contains a regular element x of $\mathfrak{g}$; then $H \supset C$ implies $\mathfrak{h} \ni x$, hence, as already noted, this implies $\mathfrak{h} \supset \mathfrak{d}$. This completes the proof.

Let G be as in 5.2, and suppose that the infinitesimal rank of the fibers of G remains locally constant (condition (C₀)). Then by 5.2 c), one finds a homomorphism of functors on $((\text{Sch}))_S^\circ$:

$$\mathcal{C} \longrightarrow \mathcal{D}$$

where

$$\mathcal{C}(S') = \text{set of Cartan subgroups of } G_{S'},$$

$$\mathcal{D}(S') = \text{set of subgroups of type (C) of } G_{S'}.$$

By 3.10 and 5.2 a), $\mathcal{D}$ is representable by a smooth quasi-projective prescheme over S . Consider the subgroup D of type (C) of $G_{\mathcal{D}}$ playing the universal role relative to G/S ; one may then consider the functor $\mathcal{C}_D : ((\text{Sch}))_{\mathcal{D}}^\circ \rightarrow (\text{Ens})$ defined in terms of the $\mathcal{D}$ -group D as $\mathcal{C}$ in terms of the S -group G . One then has the following result:

Proposition 5.5. *Under the preceding conditions, consider $\mathcal{C}$ as a functor over the prescheme $\mathcal{D}$; then $\mathcal{C}$ is $\mathcal{D}$ -isomorphic to the functor $\mathcal{C}_D$ “of Cartan subgroups of D ”.*

This follows at once from the

Corollary 5.6. *Let G be a smooth S -group prescheme of finite presentation, and D a subgroup of type (C) of G . Then there is a one-to-one correspondence between the Cartan subgroups of G contained in D and the Cartan subgroups of D (more precisely, for a subgroup H of G , H is a Cartan subgroup of G if and only if it is a Cartan subgroup of D).*

Indeed, this is a particular case of XII 7.9 c), taking into account that over an algebraically closed field a subgroup of type (C) of G contains a Cartan subgroup of G .

For the next section, the main result obtained here is 5.2 c) for H a Cartan subgroup of G , which allows one to state 5.5 and thus provides a useful “dévissage” of $\mathcal{C}$.

6. Applications to the structure of algebraic groups

Theorem 6.1. *Let G be a smooth algebraic group over a field k . Consider the scheme $\mathcal{T}$ of maximal tori of G , isomorphic to the scheme $\mathcal{C}$ of Cartan subgroups of G , which is a homogeneous space under G^0 , and a smooth connected affine algebraic scheme (XII 7.1 d)). Then $\mathcal{C}$ is a rational variety, i.e. the field of rational functions on $\mathcal{C}$ is a pure extension of k .*

PROOF. We shall first give the proof in the case where k is infinite. One may clearly suppose G connected, for $\mathcal{T}$, hence $\mathcal{C}$, does not change when G is replaced by G^0 . Moreover, by XII 7.6, $\mathcal{C}$ does not change when G is divided by a central subgroup. This allows us, first dividing by the center of G , to suppose G affine (XII 6.1), then, dividing by its reductive center (XII 4.1 and 4.4), to suppose that the reductive center of G is trivial (XII 4.7 b)). Furthermore, we proceed by induction on $n = \dim G$, supposing the theorem proved for dimensions $n' < n$. If G is nilpotent, then $\mathcal{C}$ is reduced to a point rational over k , and 6.1 is trivial. Otherwise, the Lie algebra of G is not nilpotent (1.3), hence the Cartan subalgebras of $\mathfrak{g}$ have dimension $n' < n$, hence the subgroups of type (C) of G have dimension $n' < n$. Consider then the morphism

$$\mathcal{C} \longrightarrow \mathcal{D}$$

considered in 5.5. One knows by 3.10 (k being an infinite field, hence $\mathfrak{g}$ containing a regular element) that $\mathcal{D}$ is a rational variety, i.e. the field K of rational functions on $\mathcal{D}$ is a pure extension of k . Consider the fiber of $\mathcal{C}$ at the generic point x of $\mathcal{D}$; by 5.5 it is the scheme of Cartan subgroups of a certain smooth connected algebraic group D_x over $K = \kappa(x)$ (namely $D_x =$ “the generic subgroup of type (C) of G ”). The field L of rational functions on $\mathcal{C}$ is therefore isomorphic to the field of rational functions on $\mathcal{C}_{D_x}$, which by the induction hypothesis (since $\dim D_x = n' < n$) is a pure extension of K . Thus by transitivity L is a pure extension of k .

When k is finite, a different proof is needed. One may again suppose that G is affine and connected. Note that k is perfect; it follows at once that the radical $\bar{R}$ of $G_{\bar{k}}$ is “defined over k ”, i.e. comes from a subgroup R of G . Suppose first that $R \neq G$, i.e. G is not solvable, and let

$$u : G \longrightarrow G' = G/R$$

be the canonical morphism. Consider the corresponding morphism $C \longmapsto u(C)$

$$v : \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}$$

(whose definition is immediate by XII 7.1 e)). Let x be the generic point of $\mathcal{C}_{G'}$; then the fiber $v^{-1}(x)$ is identified with the scheme of Cartan subgroups of G_K (where $K = \kappa(x)$) whose image in G'_K is a certain Cartan subgroup C'_x (namely, “the generic Cartan subgroup of G' ”). It is therefore also the scheme of Cartan subgroups of $H = u_K^{-1}(C')$ (XII 7.9 c)), and since K is here an infinite extension of k , it follows from the part of 7.1 already proved that the field L of rational functions on $\mathcal{C}_G$, equal to that of $\mathcal{C}_H$, is a purely transcendental extension of K . To prove that it is a purely transcendental extension of k , it therefore suffices to prove that this is so for K , i.e. one is reduced to the case where G is semisimple. One may moreover suppose that G is adjoint (dividing G by its reductive center). But then by 3.18 one has $\mathcal{C}_G \simeq \mathcal{D}_G$, and by 3.10 it suffices to prove that $\mathfrak{g}$ admits a regular point, which (as was noted in 3.20) is an unpublished result of Chevalley⁽⁷⁾.

It therefore only remains to treat the case where k is finite and G connected affine solvable. One in fact has a more general result:

⁽⁷⁾**N.D.E.** Cf. the Appendix, by J.-P. Serre.

Corollary 6.2. *Let G be a smooth solvable algebraic group over a field k ; then the variety $\mathcal{C}$ of Cartan subgroups of G is isomorphic to an affine space $\operatorname{Spec} k[t_1, \dots, t_N]$.*

One may again suppose G connected and affine. Let G_∞ be the smallest of the groups occurring in the descending central series of G (by G_i such that $G_{i+1} = [G, G_i]$); it is thus the smallest invariant algebraic subgroup of G such that G/G_∞ is nilpotent. Let C be a Cartan subgroup of G (one exists by 1.1); then the image of C in G/G_∞ is a Cartan subgroup of it, and is therefore equal to G/G_∞ ; consequently the morphism $g \mapsto \operatorname{Ad}(g) \cdot C$ from G_∞ to $\mathcal{C}$ is an epimorphism, and identifies $\mathcal{C}$ with the homogeneous space $G_\infty/N \cap G_\infty$, where N is the normalizer of C in G (moreover equal to C , as recalled in 4.4, but that does not matter here). Note that $U = G_\infty$ is clearly a smooth connected “unipotent” algebraic group (for over the algebraic closure of k , it is contained in the unipotent part of G , by the known structure of smooth solvable affine groups, BIBLE 6 th. 3). When k is perfect (the only case needed to establish 6.1), it follows very easily that U is even k -unipotent, i.e. admits a composition series by algebraic subgroups U_i such that U_i/U_{i+1} is isomorphic to $\mathbf{G}_a$. In fact, Rosenlicht has proved that this result remains valid for a group of the form G_∞ as above, without restriction on k (M. Rosenlicht, *Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Matematica 1963, pp. 97–120, theorem 4 cor. 2), a considerably more delicate result which we shall admit here. It now suffices to apply the following lemma, doubtless well known to specialists:

Lemma 6.3. *Let U be a smooth connected algebraic group over a field k , $X = U/V$ a homogeneous space under U having a point rational over k . Suppose U k -unipotent. Then as a k -scheme, X is isomorphic to an affine space $\operatorname{Spec} k[t_1, \dots, t_N]$.*

Indeed, let $(U_i)_{0 \leq i \leq n}$ be a composition series of U by smooth connected subgroups, with $U_n = (e)$, $U_0 = U$, $U_i/U_{i+1} \simeq \mathbf{G}_a$, the U_i invariant in U . Then the $K_i = U_i V$ are algebraic subgroups of U (not necessarily smooth or connected if V is not, but that matters little to us), and K_{i+1} is invariant in K_i . One has a canonical morphism $U_i/U_{i+1} \rightarrow K_i/K_{i+1}$ which is an epimorphism, which proves that K_i/K_{i+1} is either reduced to the unit group or isomorphic to $\mathbf{G}_a/H_i$, where H_i is a finite subgroup of $\mathbf{G}_a$, which by Rosenlicht (*loc. cit.* th. 2) implies that $K_i/K_{i+1} \simeq \mathbf{G}_a$ (a result moreover immediate if k is perfect). Set now $X_i = U/K_i$; let us prove by induction on i that X_i is isomorphic to an affine space. Indeed, if this holds for X_i , let us prove that the same holds for X_{i+1} . If $K_i/K_{i+1} = e$, one has $X_i = X_{i+1}$, and this is trivial. Otherwise, X_{i+1} is a principal bundle with base X_i and structural group $\mathbf{G}_a \simeq K_i/K_{i+1}$. Thus X_i being affine, X_{i+1} is a trivial bundle, hence isomorphic to $X_i \times \mathbf{G}_a$, which again proves that X_{i+1} is isomorphic to a standard affine space. This proves 6.2 and consequently finishes proving 6.1. $\square$

Corollary 6.4. *Let G be a smooth algebraic group over an infinite field k . Then the set of points of $\mathcal{C}$ (notation of 6.1) rational over k is dense for the Zariski topology. The union of the Cartan subgroups of G is dense in G .*

The first assertion holds for every unirational variety over an infinite field, and is moreover here for us the most important “arithmetic” consequence of the unirationality results. The second assertion follows from the first and from the density theorem XIII 2.1.

Corollary 6.5. *Let G be a smooth connected algebraic group over k . Then the variety Z of regular semisimple points of G (XIII 3.5) is a unirational variety. In particular, if k is infinite, the set of points of Z rational over k is dense in Z .*

Indeed, Z is an open subset of a torus over $\mathcal{C}$, hence its function field L is the function field of a torus defined over the function field K of $\mathcal{C}$ (namely the “generic maximal torus” of G); it is therefore a unirational extension of K (XIII 3.4), and since K is a pure extension of k by 6.1, L is a unirational extension of k .

Corollary 6.6. *Let G be a smooth connected algebraic group over k , and let H be the subgroup of G generated by the subvariety Z of regular semisimple points, i.e. (XII 8.2) the smallest invariant algebraic subgroup of G such that G/H has reductive rank zero (i.e. over the algebraic closure of k is an extension of an abelian variety by a smooth connected unipotent group). Then H is a unirational variety. In particular, if $G = H$, i.e. (XII 8.4) if G is affine and if over the algebraic closure $\bar{k}$ there exists no nontrivial homomorphism from $G_{\bar{k}}$ to $\mathbf{G}_a$, then G is a unirational variety, hence if k is infinite, the set of its points rational over k is dense.*

This follows at once from 6.5, for it is immediate that if one has smooth connected k -preschemes Z_i which are unirational varieties and morphisms $u_i : Z_i \rightarrow G$, then the algebraic subgroup of G generated by the u_i (VI_B 7.1) is a unirational variety.

As an interesting particular case of 6.5 or 6.6 (as one prefers), note:

Corollary 6.7. *Let G be a smooth connected affine algebraic group of unipotent rank zero; then G is a unirational variety.*

One may make 6.4 more precise as follows:

Corollary 6.8. *Let G be a smooth connected algebraic group over the infinite field k , and C a Cartan subgroup of G . Then the union of the conjugates of C by regular semisimple elements of $G(k)$ is dense in G .*

This follows at once from 6.4 and XIII 3.6, which says that the morphism $\varphi : Z \times C \rightarrow G$ defined by $\varphi(t, c) = \text{ad}(t)c$ is dominant. This result also implies (without supposing k infinite):

Corollary 6.9. *Let G be a smooth connected algebraic group over k , H a smooth connected algebraic subgroup of G such that H has the same reductive rank and the same nilpotent rank as G (i.e. over the algebraic closure $\bar{k}$ of k , $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$). If H is a unirational variety, so is G . If $H(k)$ is dense in H , $G(k)$ is dense in G .*

Indeed, the morphism $\varphi : Z \times H \rightarrow G$ defined by $\varphi(t, h) = \text{ad}(t)h$ is dominant; now by 6.5, Z is a unirational variety, and by hypothesis so is H , hence $Z \times H$, whence the first result. The second is proved similarly.

We now recover the following well-known result (due to Chevalley in characteristic 0, to Rosenlicht in characteristic $p > 0$):

Corollary 6.10. *Let G be a smooth connected affine algebraic group over a perfect field k . Then G is a unirational variety, hence if k is moreover infinite, $G(k)$ is dense in G .*

Indeed, by 1.1, G admits a Cartan subgroup C . By 6.9 it suffices to prove that the latter is a unirational variety. Now k being perfect, one sees at once by Galois descent from the case of algebraically closed k (BIBLE 6 th. 2) that one has $C = T \times C_u$, where T is the maximal torus of C and C_u a smooth connected unipotent group. One already knows that T is a unirational variety (XIII 3.4); it remains to see that the same holds for C_u . Now k being perfect, C_u is even k -unipotent, and one may apply 6.3.

Remarks 6.11. a) There are known examples (Rosenlicht) of twisted forms of $\mathbf{G}_a$ over an imperfect field which have only finitely many rational points, hence a fortiori are not unirational varieties. On the other hand, Chevalley has given an example of a torus over a field of characteristic zero which is not a rational variety. On the other hand, it follows from Chevalley's theory of semisimple groups that over an algebraically closed field, every smooth connected affine algebraic group is a rational variety. Note moreover that the question of unirationality arises in any event only for affine algebraic groups, a unirational algebraic group being necessarily affine by Chevalley's structure theorem.

b) With the notation of 6.6, it is tempting to try to give a condition of unirationality of G in terms of the group G/H (which is unipotent if G is assumed affine). The latter must clearly

be unirational; is this condition also sufficient? Note that an example of Rosenlicht (*loc. cit.*) shows that a smooth connected unipotent algebraic group U can be a rational variety without being k -unipotent.

c) It would be interesting to study, over a finite field k , questions of the “density” type, such as the following (raised by Rosenlicht): Let G be a smooth connected algebraic group over k ; is G then generated by its Cartan subgroups^{*(8)?}

In this question, one can reduce to the case of affine G , by dividing by the center. The answer would be affirmative in the semisimple case if one could make the result of Chevalley on existence of regular points mentioned in 3.20 more precise, so as to obtain a regular element of $\mathfrak{g}$ which does not belong to $\mathfrak{h} = \text{Lie}(H)$, where H is a smooth algebraic subgroup of G with $H \neq G$ given in advance (G an adjoint semisimple group over the finite field k).

7. APPENDIX: Existence of regular elements over finite fields

by J.-P. Serre

Throughout what follows, k denotes a finite field and $\bar{k}$ its algebraic closure; the Galois group of $\bar{k}/k$ is denoted by $\mathcal{G}$; recall that if $q = \text{Card}(k)$, $\mathcal{G}$ is topologically generated by the “Frobenius” element $F : x \mapsto x^q$.

One proposes to prove the following theorem⁽¹⁾:

Theorem. *Let G be an adjoint semisimple group defined over k , and $\mathfrak{g}$ its Lie algebra. The k -Lie algebra $\mathfrak{g}(k)$ contains a regular element.*

Remarks. (1) It is well to recall that “adjoint” means that the center of G is trivial (as a group subscheme of G). In view of the Chevalley seminar, this also means that if T is a maximal torus of G , the group $X(T)$ of characters of T (defined over $\bar{k}$) is generated by the set R of roots. It follows in particular that the rank of the Lie algebra $\mathfrak{g}$ is equal to the dimension of T (i.e. to the rank of G).

(2) The writer does not know whether the hypothesis that G is adjoint is indispensable.

Lemma 1. *It suffices to prove the theorem when G is geometrically simple.*

^{*}This question has since been answered in the affirmative by Steinberg.

(8)**N.D.E.** Having been unable to identify this result in the *Collected Papers* of R. Steinberg, let us give a proof based on the Bruhat decomposition. Let G/k be a semisimple group defined over the finite field k . One must show that G is generated by its k -Cartan subgroups. This question is stable under central isogeny and restriction of scalars; one is therefore reduced to the case where G is geometrically simple (in the same way as in lemma 1 of the appendix below). Denote by $G^\sharp$ the subgroup of G generated by its k -subtori. The group G is quasi-split (Lang) and therefore admits a Killing pair (T, B) . By definition, one has $T \subset G^\sharp$; in particular T normalizes $G^\sharp$. The big cell $R_u(B^-) T R_u(B)$ of G indicates that it suffices to verify that $R_u(B) \subset G^\sharp$. Let S be the maximal split k -torus of T . One considers the relative root system $\Phi(G, S)$ and a base Δ_k . Given $\alpha \in \Phi(G, S)$, one denotes by $U_{(\alpha)}$ the unipotent subgroup associated to α (cf. A. Borel, *Linear Algebraic Groups*, second edition (1991), Springer, Prop. 21.9). Since the k -group $R_u(B)$ is generated by the unipotent k -groups $U_{(\alpha)}$ ($\alpha \in \Delta_k$), one is reduced to verifying that $U_{(\alpha)} \subset G^\sharp$. A glance at the classification shows that there exists a semisimple group G_α of quasi-split type A_1 , 2A_1 , 3A_1 or 2A_2 such that $U_{(\alpha)} \subset G_\alpha \subset G$. It is therefore permissible to suppose that $G = \text{PGL}_2$ or $G = \text{SU}_3(K)$, where K denotes the unique quadratic field extension of k . The group $\text{PGL}_2^\sharp$ containing the standard split torus T , the possibilities up to conjugacy under $G(k)$ are the following: $\text{PGL}_2^\sharp = T$, $\text{PGL}_2^\sharp = B$, or $\text{PGL}_2^\sharp = \text{PGL}_2$. The case $\text{PGL}_2^\sharp = T$ is excluded since $\text{PGL}_2^\sharp$ contains the k -torus $R_{K/k}(\mathbf{G}_m)/\mathbf{G}_m$. The preceding discussion indicates that if $\text{PGL}_2^\sharp = B$, then $\text{PGL}_2^\sharp = \text{PGL}_2$. One therefore has $\text{PGL}_2^\sharp = \text{PGL}_2$. If $G = \text{SU}_3(K)$, $\Delta_k = \{\alpha\}$ and the possibilities for $G^\sharp$ (up to conjugacy) are the following: $G^\sharp = T$, $G^\sharp = B$, $G^\sharp = U_{(2\alpha)} \rtimes T$, $G^\sharp = \langle U_{(2\alpha)}, U_{(-2\alpha)}, T \rangle = \text{SL}_2 \cdot T$, or $G^\sharp = G$. The case $G^\sharp = B$ is excluded as for PGL_2 . The case $\text{SL}_2 \cdot T$ is excluded because $G^\sharp$ contains the k -torus $R_{K/k}^\sharp(R_{K_3/K}^1(\mathbf{G}_m))$, K_3/K denoting the extension of degree 3 of K , and the isogeny class of this torus is irreducible. Moreover, this torus also excludes the cases $G^\sharp = T$ and $G^\sharp = U_{(2\alpha)} \rtimes T$. One concludes that $G^\sharp = G$.

(1) This theorem is due to Chevalley. The writer wishes to express his gratitude to American Express which, by mislaying a trunk of Chevalley’s manuscripts, obliged him to reconstruct the proof.

(One says that G is geometrically simple if $G \otimes \bar{k}$ contains no normal group subscheme smooth over $\bar{k}$ apart from G and $\{e\}$; an equivalent condition: the associated root system R is irreducible.)

One may first suppose G indecomposable over k . The group $G \otimes \bar{k}$ is then a product of geometrically simple components S which are permuted transitively by the Galois group $\mathcal{G}$. If $\mathcal{H}$ is the subgroup of $\mathcal{G}$ fixing one of these components S , this component is defined over the field K corresponding to $\mathcal{H}$ (i.e. comes by extension of scalars from a subscheme of $G \otimes K$), and a standard argument shows that $G = \prod_{K/k}(S)$ (i.e. $R_{K/k}(S)$, for readers accustomed to Weil's notation). Likewise, the Lie algebra $\mathfrak{g}$ of G is identified with $\prod_{K/k} \mathfrak{s}$, where $\mathfrak{s}$ is the Lie algebra of S . If the theorem is true for S , there exists $x \in \mathfrak{s}(K) = \mathfrak{g}(k)$ which is regular in $\mathfrak{s}$; one then easily verifies that it is regular in $\mathfrak{g}$. Q.E.D.

From now on, one supposes G geometrically simple and chooses a maximal torus T of G (one knows that this is possible). Denote by X the character group of $T \otimes \bar{k}$, by R its root system, by W its Weyl group, and by E the group of automorphisms of X preserving R . The group W is a normal subgroup of E .

If T' is another torus of G , denote by X', R', W', E' the corresponding character group, root system, If one chooses $y \in G(\bar{k})$ such that $y \cdot (T \otimes \bar{k}) \cdot y^{-1} = T' \otimes \bar{k}$, one may identify X', R', W', E' with X, R, W, E by means of $\text{int}(y)$. Changing y changes this identification by an automorphism of X corresponding to an element of W .

The canonical generator $x \mapsto x^q$ of $\mathcal{G}$ acts on X' leaving R' stable; it therefore defines an element $f_{T'}$ of E' . One will in particular set $f = f_T$. When one identifies E' with E as just described, the element $f_{T'}$ becomes an element f' of E , defined up to replacement by $wf'w^{-1}$, with $w \in W$. One will compare this element to f :

Lemma 2. *One has $f' \equiv f \pmod{W}$; conversely, every $f' \in E$ satisfying this condition can be obtained from a maximal torus T' of G .*

Set $\bar{T} = T \otimes \bar{k}$, $\bar{T}' = T' \otimes \bar{k}$, and let $y \in G(\bar{k})$ be such that $y\bar{T}y^{-1} = \bar{T}'$; since $y^q\bar{T}y^{-q} = \bar{T}'$, one concludes that $n = y^{-1}y^q$ belongs to the normalizer $N(\bar{T})$ of $\bar{T}$. The effect of f' on the points of $\bar{T}(\bar{k})$ is then as follows:

$$f'(t) = y^{-1}(yty^{-1})^q y = y^{-1}y^q t^q y^{-q} y = nt^q n^{-1}.$$

If $w \in W$ is the element defined by n , this shows that f' and $f \circ w$ have the same effect on the points of $\bar{T}$, hence also on its characters, and one has $f' = f \circ w$, whence $f' \equiv f \pmod{W}$. Conversely, if $w \in W$ is given, one represents it by an element $n \in N(\bar{T})(\bar{k})$; by a classical theorem of Lang, one may write n in the form $n = y^{-1}y^q$, with $y \in G(\bar{k})$; the torus $\bar{T}' = y\bar{T}y^{-1}$ is then defined over k , and the preceding calculation shows that the corresponding f' is equal to $f \circ w$.

Lemma 3. *Let X, R, W, E be as above (R being irreducible), and let $\varphi \in E/W$. There then exists an element $f \in E$ representing φ , and a family $\theta_1, \dots, \theta_n$ of roots having the following two properties:*

- (1) $(\theta_1, \dots, \theta_n)$ is a basis of X .
- (2) R is the union of the orbits of the θ_i under the powers of f (i.e. every $\alpha \in R$ is written $\alpha = f^m \theta_i$, with suitable m and i).

The proof will be given a little later.

Here is now a linear algebra lemma:

Lemma 4. *Let V be a vector space over k , and $(\theta_1, \dots, \theta_n)$ a basis of the dual of $V \otimes \bar{k}$. There exists $x \in V$ such that $\theta_i(x) \neq 0$ for every i .*

Let V^* be the dual of V , and $\bar{W}_i$ the subspace of $V^* \otimes \bar{k}$ generated by the conjugates of θ_i ; this subspace is defined over k , i.e. of the form $W_i \otimes \bar{k}$. The obvious map $W_1 \otimes \dots \otimes W_n \longrightarrow \bigwedge^n V^*$ is nonzero (otherwise its extension to $\bar{k}$ would also be zero, which is absurd since

$\theta_i \in \bar{W}_i$, and $\theta_1 \wedge \cdots \wedge \theta_n \neq 0$). There therefore exists a basis of V^* consisting of elements $u_i \in W_i$. Let $x \in V$ be such that $u_i(x) \neq 0$ for every i (for example $u_i(x) = 1$). The element x answers the question, for if one had $\theta_i(x) = 0$, the conjugates of θ_i would also vanish at x , and the same would hold for u_i , which is not the case.

End of the proof of the theorem. Combining lemmas 2 and 3, one may choose a torus T whose element f satisfies the properties of lemma 3. If Y is the dual of X , the Lie algebra $\mathfrak{t}(\bar{k})$ of $\bar{T}^{(9)}$ is canonically identified with $Y \otimes \bar{k}$, and this operation is compatible with the action of the Galois group (the latter acting on $Y \otimes \bar{k}$ by its action on Y and on $\bar{k}$). Let $V = \mathfrak{t}(k)$ be the Lie algebra of T over k . An element $x \in V$ is regular if and only if it is annihilated by no root $\alpha \in R$, or rather by none of the linear forms $\bar{\alpha} \in V^* \otimes \bar{k}$ defined canonically by the $\alpha \in R$. By lemma 4, one may find such an x which is annihilated by none of the roots θ_i ; but every root is conjugate to a θ_i (this is what condition (2) of lemma 3 expresses); it follows that x is annihilated by no root, and it is indeed a regular element.

Proof of lemma 3. It relies on the properties of *Coxeter transformations*. Let us briefly recall what this consists of:

Let $a_1, \dots, a_n$ be a simple system of roots of R , and for every i let r_i be the reflection corresponding to a_i . Set:

$$c = r_1 \cdots r_n.$$

One has $c \in W$; of course, the element c depends on the choice of the simple system $(a_1, \dots, a_n)$ as well as on the order of the a_i ; however, one proves that its conjugacy class depends on none of these choices. One calls c the *Coxeter element* of the system under consideration. One proves (we shall admit it) that c does not have 1 as eigenvalue.

Lemma 5. Set $\theta_i = r_n \cdots r_{i+1}(a_i)$. Then:

- (a) The θ_i form a basis of the group X generated by the roots.
- (b) One has $\theta_i > 0$ and $c(\theta_i) < 0$ for every i .⁽²⁾
- (c) Every root $a \in R$ such that $a > 0$ and $c(a) < 0$ is equal to one of the θ_i .
- (d) R is the union of the orbits of the θ_i under the powers of c .

It is clear that θ_i is of the form $a_i + \sum_{j>i} m_{ij}a_j$, with $m_{ij} \in \mathbf{Z}$, which proves (a). Assertions (b) and (c) are consequences of the following observation: the reflection r_i preserves the sign of every root distinct from $\pm a_i$, and changes the sign of $\pm a_i$.

Finally, for (d), one observes that an orbit of c cannot consist entirely of positive (resp. negative) roots, for by taking the sum of these roots one would find a nonzero element of X invariant under c , and one has been willing to admit that c does not have 1 as eigenvalue. There is therefore necessarily in every orbit an element $a > 0$ such that $c(a) < 0$, and one applies (c).

Remark. The preceding proof has been sketched only to facilitate the reader's task; one could have confined oneself to referring to the canonical texts on Coxeter (cf., for example, Koszul, Séminaire Bourbaki, 1959/1960, exposé 191). These texts contain other results: the orbits of c all have the same number h of elements, and each contains only one θ_i . In particular, $h = \text{Card}(R)/n$.

Let us now return to the proof of lemma 3. Distinguish three cases:

- (1) The given element $\varphi \in E/W$ is equal to the identity element.

One must then take $f \in W$; one chooses $f = c$; this works by lemma 5.

- (2) The system R has type A_n , with even $n \geq 2$, and φ is the unique nontrivial element of E/W .

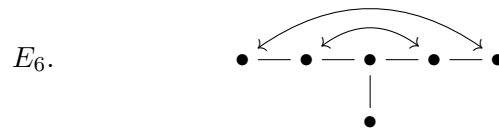
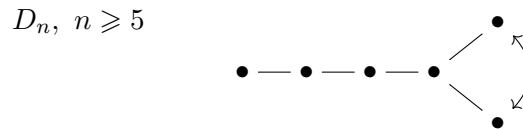
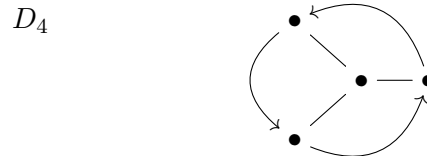
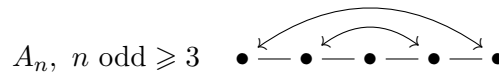
One knows that $-1 \notin W$; one then takes $f = -c$. A simple calculation shows that c has order $h = n + 1$; its order is therefore odd. If $a \in R$ is any root, one has $a = c^m \theta_i$ for a pair

⁽⁹⁾**N.D.E.** T has been changed to $\bar{T}$.

⁽²⁾For the order relation defined by $(a_1, \dots, a_n)$.

(m, i) ; cf. lemma 5; adding h to m if necessary, one may suppose m even, and one sees that one then has $a = (-c)^m \theta_i = f^m \theta_i$. The orbits of the θ_i therefore indeed fill R .

(3) The element $\varphi \in E/W$ is nontrivial, and R has one of the following types:



(A glance at the classification shows that these are indeed all the cases (with A_n , even n) where E/W is nontrivial.)

Let S be a simple system of roots, which we do not number for the moment. One knows that E is the semidirect product of W and the group Ψ of permutations of S which leave the Cartan matrix invariant (or the Dynkin diagram; it is the same thing). The group E/W is thus identified with Ψ , and in particular φ corresponds to an element $\psi \in \Psi$. One observes by inspecting the Dynkin diagrams (cf. the figures above) that every orbit of ψ in S consists of pairwise orthogonal roots (i.e. not joined in the diagram). [Note that this would not be the case for A_n (even n), which obliged us to treat this case separately.] If σ is such an orbit, the reflections r_a , $a \in \sigma$, commute with one another; their product will be denoted by ρ_σ . It is clear that ρ_σ commutes with ψ .

This being so, let us choose a total order on S such that every orbit is a segment for this order relation; this amounts to numbering the elements of S : $S = \{a_1, \dots, a_n\}$. Let $\theta_1, \dots, \theta_n$ be the roots defined above, and $c = r_1 \cdots r_n$ the corresponding Coxeter element. The element c is a product of the ρ_σ , the σ being arranged in a certain order; it follows that it commutes with ψ . One then sets $f = c\psi$. One observes in addition that ψ permutes the θ_i among themselves. Indeed, one has $\psi(\theta_i) > 0$ since $\theta_i > 0$, and $c(\psi(\theta_i)) = \psi(c(\theta_i)) < 0$, hence (lemma 5, (c)), $\psi(\theta_i)$ is equal to a θ_j . It is now immediate that $f = c\psi$ answers the question. Indeed, if $a \in R$, one has $a = c^m \theta_i$ for a pair (m, i) , whence $a = f^m(\psi^{-m} \theta_i) = f^m \theta_j$ for a certain j . Q.E.D.

Remark. One can prove that, except in case (2), every orbit of f has exactly $h = \text{Card}(R)/n$ elements and contains one and only one θ_i . In case (2), some of the θ_i are unnecessary.

Exposé XV. Complements on the subtori of a group prescheme. Application to smooth groups

by M. Raynaud*

0. Introduction

This exposé complements and partly takes up again exposés XI and XII; knowledge of exposés XIII and XIV is not indispensable. Continuing the effort begun in XII, we shall work with group S -preschemes which are not necessarily affine and not necessarily separated over S .

Sections 1, 2, 3, 4 are devoted to the study of subtori of a group prescheme. One obtains infinitesimal (§ 2) and global (§ 4) lifting theorems, in which points of finite order play an essential role (§ 1).

Sections 5, 6 and 7 are independent of the preceding sections. The consideration of infinitesimal neighborhoods leads to the representability of the functor of smooth subgroups equal to their connected normalizer (§ 5). In §§ 6 and 7, one is interested more particularly in Cartan subgroups.

Finally, in § 8 one gives a necessary and sufficient condition for the functor of subtori of a smooth group, or that of maximal tori, to be representable.

1. Lifting finite subgroups

1. Finite, smooth and central subgroups of multiplicative type

Proposition 1.1. *Let S be an affine scheme, S_0 a closed subscheme of S defined by an ideal of square zero, G a group S -prescheme, H_0 a group subscheme of $G_0 = G \times_S S_0$, smooth over S_0 , of finite type, and of multiplicative type. Then, for there to exist a group subscheme of G , of multiplicative type, lifting H_0 , it is necessary and sufficient that there exist a subscheme H' of G , flat over S , lifting H_0 .*

PROOF. The necessity of the condition is quite clear; let us prove its sufficiency. The group H_0 of multiplicative type is quasi-isotrivial (X 4.5); by Exp. X 2.1, there exist an S -group H of multiplicative type and an S_0 -isomorphism of groups:

$$u_0 : H \times_S S_0 \xrightarrow{\sim} H_0.$$

Since H' is flat over S and $H' \times_S S_0$ is of finite presentation over S_0 (Exp. IX 2.1 b)), H' is of finite presentation over S ; moreover, its fibers are smooth, so H' is smooth over S (EGA IV 17.5.1). Since the scheme S is affine, u_0 therefore lifts to an S -morphism of preschemes:

$$u : H \longrightarrow H'.$$

*This exposé and the next two (Exp. XVI and XVII) do not correspond to oral lectures in the seminar. With a few complements, they develop the substance of (succinct) handwritten notes by A. Grothendieck, written in 1964 on the occasion of the present seminar.

⁰_{xy} version of 1/12/08

It then follows from Exp. III 2.1 and Exp. IX 3.1 that the composite morphism $v_0 : H \times_{S_0} S \xrightarrow{\sim} H_0 \longrightarrow G_0$ also lifts to an S -morphism of groups:

$$v : H \longrightarrow G.$$

Since v_0 is an immersion, the same is true of v . The image of H under v is therefore a group subscheme of G , of multiplicative type, lifting H_0 . $\square$

Proposition 1.2. *Let S be a prescheme, S_0 a subprescheme of S defined by a locally nilpotent sheaf of ideals, G a group S -prescheme, flat and of finite presentation over S , and H_0 a group subscheme of $G_0 = G \times_S S_0$ which is smooth, finite over S_0 , of multiplicative type and central. Then there exists a unique group subscheme H of G , of multiplicative type, lifting H_0 . Moreover, H is central. (See XVII App. III, 1.)*

Proposition 1.2 bis. *Let S, G, S_0, G_0 be as above, H a group S -scheme, of multiplicative type, smooth and finite over S , and $u_0 : H_0 = H \times_S S_0 \longrightarrow G_0$ a central homomorphism. Then u lifts uniquely to a homomorphism $u : H \longrightarrow G$. Moreover, u is central.*

The existence of the lifting u in 1.2 bis follows easily from 1.2 by considering the graph of u_0 . The lifting u is unique and central by Exp. IX 3.4 and Exp. IX 5.1.

PROOF OF 1.2. The uniqueness of H and the fact that H is central follow from Exp. IX 5.6 b) and Exp. IX 3.4 bis. Taking uniqueness into account, to prove the existence of H one may suppose S affine and S_0 defined by an ideal of square zero, and it suffices (1.1) to find a subscheme of G , flat over S , lifting H_0 .

Since H_0 is smooth and finite over S_0 , one may suppose, after restricting S , that there exists an integer $n > 0$, invertible on S , such that $H_0 = {}_nH_0$. Consider the morphism of raising to the n th power in G :

$$u : G \longrightarrow G, \quad x \longmapsto x^n.$$

Again denote by ${}_nG$ the “kernel of u ”, i.e. the inverse image under u of the unit section of G (N.B. u is not in general a group morphism). Let us grant for a moment the following lemma:

Lemma 1.3. *Let k be a field, G a group scheme locally of finite type over k , n an integer prime to the characteristic of k , u the morphism of raising to the n th power in G . Then u is étale at every point x of G which belongs to the center of G .*

Since G is flat and of finite presentation over S , it follows from the preceding lemma and EGA IV 17.8.2 that if x is a point of G projecting to s in S and belonging to the center of G_s , the morphism u is étale at x . If in addition x is a point of ${}_nG$, then ${}_nG$ is étale over S at x . By hypothesis, the group H_0 is central and contained in ${}_nG_0$, so it is in fact contained in the largest open subset V of ${}_nG$ which is étale over S . Since H_0 and $V \times_S S_0 = V_0$ are étale over S_0 , H_0 is an open subset of V_0 (EGA IV 17.8.7 and 17.9.1). But then, the open subprescheme of V having the same underlying space as H_0 is a subscheme of G , flat over S , lifting H_0 .

It remains to prove Lemma 1.3. For this, note that the largest open subset of G on which u is étale is invariant under extension of the base field (EGA IV 17.8.2); this allows one to reduce to the case where x is rational over k . Denote by t_x translation by x , which is a k -automorphism of the scheme G . Since x is in the center of G , one has the commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{u} & G \\ t_x \downarrow & & \downarrow t_x^n \\ G & \xrightarrow{u} & G. \end{array}$$

It therefore suffices to show that u is étale at the origin, but this was seen in VIIA 8.4. $\square$

2. Global lifting of finite groups

Lemma 1.4. *Let A be a local ring, separated and complete for the topology defined by its maximal ideal $\mathfrak{m}$, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/\mathfrak{m}^n)$. Then for every prescheme X (resp. for every S -prescheme X), the canonical map:*

$$(x) \quad \text{Hom}(S, X) \longrightarrow \varprojlim_n \text{Hom}(S_n, X)$$

(resp.:

$$(xx) \quad \Gamma(X/S) \longrightarrow \varprojlim_n \Gamma(X_n/S_n), \quad \text{where } X_n = X \times_S S_n$$

is bijective.

PROOF. (xx) is an easy consequence of (x). Let us prove (x).

Let $u_n : S_n \rightarrow X$ ($n \in \mathbf{N}$) be a coherent system of morphisms. The image y of the closed point of S_n is thus independent of n , and u_n factors through $\text{Spec } \mathcal{O}_y$. By passage to the projective limit, the morphisms u_n define a ring morphism:

$$\tilde{u} : \mathcal{O}_y \longrightarrow \varprojlim_n (A/\mathfrak{m}^n) = A.$$

This shows that (x) is surjective; it is injective as soon as A is separated for the $\mathfrak{m}$ -adic topology. $\square$

Corollary 1.5. *Let A be a complete noetherian local ring, $\mathfrak{m}$ its maximal ideal, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/\mathfrak{m}^n)$, X a scheme finite over S and Y an S -prescheme. Then the canonical map:*

$$\text{Hom}_S(X, Y) \longrightarrow \varprojlim_n \text{Hom}_{S_n}(X_n, Y_n)$$

(where $X_n = X \times_S S_n$ and likewise $Y_n = \dots$) is bijective.

PROOF. Indeed, it follows from EGA II 6.2.5 that X is a finite sum of local S -schemes finite over S . This reduces one to the case where X itself is the spectrum of a complete noetherian local ring. But $\text{Hom}_S(X, Y) = \Gamma(Z/X)$ where Z is the X -prescheme $Y \times_S X$, and one applies the preceding proposition. $\square$

Proposition 1.6. *Let A, S, S_n be as above and let G and M be two group S -preschemes, with M finite over S . Then:*

a) *The canonical map:*

$$\text{Hom}_{S\text{-gr}}(M, G) \longrightarrow \varprojlim_n \text{Hom}_{S_n\text{-gr}}(M_n, G_n)$$

is bijective.

b) *If M is of multiplicative type and G is smooth over S , the canonical map:*

$$\varphi : \text{Hom}_{S\text{-gr}}(M, G) \longrightarrow \text{Hom}_{S_0\text{-gr}}(M_0, G_0)$$

is surjective. Moreover, if $\varphi(u) = \varphi(u') = u_0$, then u and u' are conjugate by an element of $G(S)$ reducing to the unit element of $G(S_0)$.

c) *If M is of multiplicative type and smooth over S , if G is flat of finite type over S , and if $u_0 : M_0 \rightarrow G_0$ is a central homomorphism, u_0 lifts uniquely to a homomorphism*

$$u : M \longrightarrow G.$$

Moreover, u is central if G has connected fibers.

PROOF. a) Follows from 1.5, from the fact that $M \times_S M$ is finite over S , and from the characterization of group morphisms as those making the well-known diagram commute:

$$\begin{array}{ccc} M \times_S M & \xrightarrow{u \times u} & G \times_S G \\ \downarrow & & \downarrow \\ M & \xrightarrow{u} & G. \end{array}$$

b) By Exp. IX 3.6, one can construct a coherent system of homomorphisms $u_n : M_n \rightarrow G_n$ lifting a homomorphism $u_0 : M_0 \rightarrow G_0$. Hence the first assertion of b), taking a) into account.

If now u and u' are two liftings of u_0 , then u_n and u'_n are conjugate by an element g_n of $G(S_n)$ lifting the unit element of $G(S_0)$ (Exp. IX 3.6); *loc. cit.* also implies that one can choose the g_n coherently, hence (1.4) arising from a section g of $G(S)$. The morphisms u and $\text{int}(g)u'$ then agree modulo $\mathfrak{m}^n$ for every n , and therefore agree (1.5).

c) The existence and uniqueness of u follow from a) and 1.2 bis. If G has connected fibers, u is central by Exp. IX 5.6 a). $\square$

2. Infinitesimal lifting of subtori

1. Statement of the theorem

We shall give an infinitesimal lifting theorem for subtori of a group prescheme which does not call upon smoothness hypotheses (unlike Exp. IX 3.6 bis) and which moreover answers a very natural question⁽¹⁾: is it enough to be able to lift “sufficiently many” points of finite order of a subtorus T_0 , to be assured of being able to lift T_0 (infinitesimally, of course)?

Theorem 2.1. *Let S be a noetherian affine scheme, S_0 a closed subscheme of S defined by an ideal J of square zero, G a group S -prescheme of finite type, $G_0 = G \times_S S_0$, T_0 a subtorus of G_0 , q an integer > 0 invertible on S . Suppose that for every integer n equal to a power of q , there exists a subscheme M_n of G , flat over S , such that $M_n \times_S S_0 = {}_nT_0$. Then there exists a subtorus T of G such that $T \times_S S_0 = T_0$.*

Theorem 2.1 will be useful to us through the following two corollaries:

Corollary 2.2. *Let S be a locally noetherian prescheme, S_0 a closed subprescheme of S defined by a locally nilpotent sheaf of ideals, G a group S -prescheme of finite type, T_0 a subtorus of $G_0 = G \times_S S_0$, q an integer > 0 invertible on S ; finally, with the integer n ranging over the powers of q , let (M_n) be a coherent system of group S -subschemes of G , of multiplicative type, lifting the ${}_nT_0$ (N.B. The system of subgroups of multiplicative type (M_n) is said to be coherent if $M_m = {}_m(M_n)$ when the integer m divides n .) Then there exists one and only one subtorus T of G such that $T \times_S S_0 = T_0$ and ${}_nT = M_n$ for every n .*

Corollary 2.3. *Let G be a group S -prescheme, flat and of finite presentation over S , S_0 a closed subprescheme of S defined by a sheaf of ideals of finite type and locally nilpotent, T_0 a central torus of $G_0 = G \times_S S_0$. Then there exists one and only one subtorus T of G lifting T_0 . Moreover, T is central.*

Remark 2.4. We leave to the reader the task of formulating the analogues of statements 2.1, 2.2, 2.3, where instead of lifting a subtorus of G_0 , one is given a torus T over S and proposes to lift a morphism

$$u_0 : T_0 \rightarrow G_0$$

(one reduces to the preceding cases by considering the graph of u_0).

Let us show how one deduces Corollaries 2.2 and 2.3 from 2.1.

Proof of Corollary 2.2. The uniqueness of T follows from Exp. IX 4.8 b) and Exp. IX 4.10. To prove the existence of T , one may therefore reduce to the case where S is affine, hence noetherian, and to the case where S_0 is defined by an ideal of square zero.

Lemma 2.5. *Let G be a group S -prescheme, of finite presentation, and H a group subscheme of G , of multiplicative type, smooth over S . Then $\text{Centr}_G(H)$ is representable by a group subprescheme of G , of finite presentation.*

⁽¹⁾**N.D.E.** The idea of approximating a torus by its finite subgroups appears in Grothendieck's proof of the connectedness of centralizers of tori, see § 4.6 of BIBLE.

The lemma follows from Exp. VIII 6.5 e), without a smoothness hypothesis on H , when G is separated over S . In the present case, one notes that the assertion to be proved is local for the fpqc topology, which allows one to suppose H diagonalizable, hence of the form $D_S(M)$. One may also suppose S affine, then S noetherian thanks to EGA IV 8. Since H is smooth over S and of finite type, the order q of the torsion subgroup of M is invertible on S (Exp. VIII 2.1 e)). It is then immediate (cf. Exp. IX 4.10) that the subgroups ${}_nH$, with n ranging over the powers of q , are schematically dense in H (Exp. IX 4.1). But ${}_nH$ is a completely split covering of S (that is, is isomorphic to a finite direct sum of copies of S), so $\text{Centr}_G({}_nH) = Z_n$ is representable by the intersection of the centralizers in G of the sections of ${}_nH$ over S . It then suffices to apply the lemma:

Lemma 2.5 bis. *Let S be a noetherian prescheme, G a group S -prescheme of finite type, H a subgroup of multiplicative type of G , H_i an increasing filtered family of subgroups of multiplicative type of G , and suppose that $Z_i = \text{Centr}_G(H_i)$ is representable by a group subprescheme of G . Then the family of the Z_i is stationary. If in addition the H_i are schematically dense in H , one has $Z_i = \text{Centr}_G(H)$ for i sufficiently large.*

To see that the family of the Z_i is stationary, it suffices to show that the family of the underlying sets $\text{ens}(Z_i)$ is stationary. Indeed, the stationary value will be a closed subset of an open subset U of G , and after replacing G by U , one is reduced to studying a decreasing filtered family of closed subpreschemes of a noetherian prescheme. An easy constructibility argument reduces one to the case where S is integral. One must then show that the family of the $\text{ens}(Z_i)$ is stationary over a nonempty open subset of S . Now the generic fiber of G is separated (Exp. VIA 0.3), so, after restricting S , one may suppose G separated over S (EGA IV 8). But then Z_i is closed in G (Exp. VIII 6.5 e)). To establish the last assertion of the lemma, denote by Z the stationary value of the family Z_i . It is clear that $\text{Centr}_G(H)$ is a subfunctor of Z ; let us show that Z centralizes H . Now let E be the subprescheme of $H \times_S Z$ which is the kernel of the pair of morphisms:

$$\begin{aligned} H \times_S Z &\rightrightarrows G \\ (h, c) &\longmapsto c \\ (h, c) &\longmapsto hch^{-1}. \end{aligned}$$

The prescheme E contains $H_i \times_S Z$ for every i . On the other hand, the H_i are flat over S , so (EGA IV 11.10.9) for every point s of S , the $(H_i)_s$ are schematically dense in H_s , and the $H_i \times_S Z$ are schematically dense in $H \times_S Z$. Since G_s is separated, E_s is closed in $(H \times_S Z)_s$ and consequently is equal to it. But then E is closed in $H \times_S Z$, and is therefore equal to $H \times_S Z$. This says that Z centralizes H .

Let us return to the proof of 2.2. By 2.5, $Z_n = \text{Centr}_G(M_n)$ is representable, and by 2.5 bis the decreasing family of subpreschemes Z_n is stationary; let Z be the stationary value. The group Z contains T_0 and the M_n . After replacing G by Z , one may therefore suppose the M_n central.

One is then in the conditions for applying Theorem 2.1, and there exists a subtorus T of G lifting T_0 . The groups ${}_nT$ and M_n are then two liftings of ${}_nT_0$, so they are conjugate (Exp. IX 3.2 bis), and consequently agree, since M_n is central. The torus T answers the question.

PROOF OF COROLLARY 2.3. The uniqueness of T follows from Exp. IX 5.1 bis, and the fact that T is central follows from IX 5.6 b). This remark allows one to reduce by the usual procedure to the case where S is affine (hence S_0 defined by a nilpotent ideal of finite type), then to the case where S is noetherian. Restricting S if necessary, one may suppose that there exists an integer q invertible on S . Corollary 2.3 is then a consequence of 2.2 and 1.2. $\square$

Remark 2.6. One shows easily that Corollary 2.3 remains true if one replaces the torus T_0 by a smooth central subgroup of multiplicative type of G_0 .

2. Proof of 2.1

a) Reduction to the case where $T_0 = \mathbf{G}_{m, S_0}$.

Thanks to 1.1, one may suppose that M_n is a subgroup of multiplicative type. Using Exp. IX 3.2 bis, one may suppose that the family of the M_n is coherent (2.2). The centralizer Z_n of M_n in G is representable (2.5), and the filtered family of the Z_n is stationary (2.5 bis). After replacing G by Z_n for n sufficiently large, one may therefore suppose T_0 and the M_n central. The uniqueness of the lifting of T_0 is then assured (Exp. IX 5.1 bis).

Proceeding as in the proof of 1.1, one may suppose that there exist an S -torus T and an S_0 -isomorphism:

$$u_0 : T \times_S S_0 \xrightarrow{\sim} T_0$$

and it is equivalent to lift u_0 or to lift T_0 . In view of uniqueness, it suffices to prove the existence of a lifting of u_0 after making a faithfully flat affine extension $S' \rightarrow S$ of finite type (fpqc descent), which allows one to suppose $T = \mathbf{G}_{m, S}^r$ (Exp. X 4.5). If the restriction of u_0 to each factor $\mathbf{G}_m$ lifts to an S -morphism, necessarily central, one immediately deduces a lifting of u_0 . In brief, one may suppose $T_0 = \mathbf{G}_{m, S_0}$.

b) Definition of the obstruction to the existence of a lifting of T_0 .

To prove 2.1, it suffices by 1.1 to find a subscheme of G , flat over S , lifting T_0 . We shall see that one can define the obstruction to the existence of such a lifting as an element of a certain $\text{Ext}^1(,)$ of sheaves of modules.

Let U be an open subset of G such that T_0 is closed in U , and again denote by U (resp. U_0) the open subscheme of G (resp. G_0) having U as underlying space. The sheaf $\mathcal{O}_{T_0}$, regarded as a sheaf on U , is therefore a quotient of $\mathcal{O}_{U_0}$. Let h be the canonical epimorphism:

$$h : \mathcal{O}_{U_0} \rightarrow \mathcal{O}_{T_0}.$$

Lemma 2.7. *The canonical map:*

$$h' = \text{id}_J \otimes h : J \otimes_{S_0} \mathcal{O}_{U_0} \rightarrow J \otimes_{S_0} \mathcal{O}_{T_0}$$

factors (obviously uniquely) as $i \circ j_U$, where j_U is the canonical epimorphism:

$$J \otimes_{S_0} \mathcal{O}_{U_0} \rightarrow J\mathcal{O}_U \simeq J\mathcal{O}_{U_0}.$$

PROOF. One must show that $h'(K) = 0$, where K is the kernel of j_U . Now for every integer n equal to a power of q , one has an epimorphism:

$$h_n : \mathcal{O}_{T_0} \rightarrow \mathcal{O}_{nT_0},$$

and since nT_0 lifts to a scheme M_n flat over S , the canonical morphism:

$$j_n : J \otimes_{S_0} \mathcal{O}_{nT_0} \rightarrow J\mathcal{O}_{M_n}$$

is an isomorphism.

The commutative diagram below:

$$\begin{array}{ccccc} K & \longrightarrow & J \otimes_{S_0} \mathcal{O}_{U_0} & \xrightarrow{j_U} & J\mathcal{O}_U \subset \mathcal{O}_U \\ & & \downarrow h' & \swarrow i & \\ & & J \otimes_{S_0} \mathcal{O}_{T_0} & & \\ & & \downarrow h'_n & & \\ & & J \otimes_{S_0} \mathcal{O}_{nT_0} & \xrightarrow[\cong]{j_n} & J\mathcal{O}_{M_n} \subset \mathcal{O}_{M_n} \end{array}$$

proves that $h'(K)$ is contained in $\text{Ker } h'_n$ for every n , hence is contained in $\bigcap_n \text{Ker } h'_n$, and it suffices to show that the latter is zero. Now the sheaf $\mathcal{O}_{T_0}$ is equal to the sheaf $\mathcal{O}_{S_0}[\mathbf{Z}]$, the group algebra of $\mathbf{Z}$ with coefficients in $\mathcal{O}_{S_0}$, whereas $\mathcal{O}_{nT_0}$ is the quotient algebra $\mathcal{O}_{S_0}[\mathbf{Z}/n\mathbf{Z}]$.

Let $a = \sum_{m \in \mathbf{Z}} a_m \otimes m$ be an element of $J \otimes_{S_0} \mathcal{O}_{T_0}$. Thus the a_m are sections of J , almost all zero. Take n sufficiently large that the indices m for which a_m is nonzero have

distinct images in $\mathbf{Z}/n\mathbf{Z}$. Then if $a \in \text{Ker } h'_n$, one necessarily has $a = 0$. This proves that $\bigcap_n \text{Ker } h'_n = 0$, and proves 2.7. $\square$

Let then K_0 be the kernel of h ; consider the following diagram:

$$\begin{array}{ccccccc}
 & & & & 0 & & \\
 & & & & \downarrow & & \\
 & & J\mathcal{O}_U & & K_0 & & \\
 & & \downarrow \wr & & \downarrow & & \\
 0 & \longrightarrow & J\mathcal{O}_{U_0} & \longrightarrow & \mathcal{O}_U & \longrightarrow & \mathcal{O}_{U_0} \longrightarrow 0 \\
 & & \downarrow i & & & & \downarrow h \\
 & & J \otimes_{S_0} \mathcal{O}_{T_0} & & & & \mathcal{O}_{T_0} \\
 & & & & & & \downarrow \\
 & & & & & & 0
 \end{array}$$

The sheaf $\mathcal{O}_U$ defines an element Φ of the group $\text{Ext}_{\mathcal{O}_U}^1(\mathcal{O}_{U_0}, J\mathcal{O}_{U_0})$. Let Ψ be the element of $\text{Ext}_{\mathcal{O}_U}^1(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$ deduced from Φ by the bifactoriality of $\text{Ext}^1(,)$ thanks to the morphisms $K_0 \longrightarrow \mathcal{O}_{U_0}$ and $i : J\mathcal{O}_{U_0} \longrightarrow J \otimes_{S_0} \mathcal{O}_{T_0}$. It follows from Exp. III 4.1 and the infinitesimal flatness criterion (cf. Exp. III 4.3) that there exists a subscheme of U , flat over S , lifting T_0 if and only if Ψ is zero.

But note that T_0 is affine, so it suffices (Exp. III 4.5 and 4.6) to show that there exists locally on U a subscheme flat over S lifting T_0 . In brief, it suffices to show that the image of Ψ in the sheaf:

$$\mathcal{E} = \mathcal{E}xt_{\mathcal{O}_U}^1(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$$

is zero. We shall again denote by Ψ this element of $\Gamma(U, \mathcal{E})$.

c) Reduction to the case of S local artinian, with algebraically closed residue field.

Since K_0 and $J \otimes_{S_0} \mathcal{O}_{T_0}$ are coherent sheaves, the same holds for the sheaf $\mathcal{E}$; moreover, $\mathcal{E}$ has its support in T_0 , since this is so for $J \otimes_{S_0} \mathcal{O}_{T_0}$. To show that the section f of $\mathcal{E}$ is zero, it suffices to see that for every point u of T_0 , the image of f in the fiber of $\mathcal{E}$ at u is zero. But the formation of $\mathcal{E}xt^i(,)$ of coherent sheaves commutes with flat extensions of the base⁽²⁾, so one is reduced to proving the existence of a lifting of $T_0 \cap \text{Spec } \mathcal{O}_u$ which is flat over S . Let s be the projection of u to S ; one may then replace S by $\text{Spec } \mathcal{O}_s$ and G by $G \times_S \text{Spec } \mathcal{O}_s$.

After again making a faithfully flat extension, one may suppose that $\mathcal{O}_s$ has an algebraically closed residue field (EGA 0_{III} 10.3.1).

Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_s$, and $S_n = \text{Spec } \mathcal{O}_s/\mathfrak{m}^n$. Suppose one has shown that for every $n > 0$, the obstruction to lifting $T_0 \times_S S_n = (T_0)_n$ to a subscheme of $G_n = G \times_S S_n$, flat over S_n , is zero, and let T_n be the unique lifting of $(T_0)_n$, flat over S_n , which is a subtorus of G_n . It is clear that if $n > n'$, one has:

$$(T_n) \times_{S_n} S_{n'} = T_{n'}.$$

If then u is a point of T_0 projecting to s , it follows from the lemma below, applied to the coherent system of liftings $(T_n \cap \text{Spec } \mathcal{O}_u)$ of $(T_0)_n \cap \text{Spec } \mathcal{O}_u$, that there does indeed exist a lifting of $T_0 \cap \text{Spec } \mathcal{O}_u$ flat over $\mathcal{O}_s$. One is therefore reduced to proving that Ψ is zero when $S = S_n$ is the spectrum of a local artinian ring with algebraically closed residue field, and to proving:

Lemma 2.8. *Let $A \longrightarrow B$ be a local homomorphism of noetherian local rings, $\mathfrak{m}$ the maximal ideal of A , J an ideal of square zero of A , M a B -module of finite type, $A_0 = A/J$, $B_0 = B/JB$, $M_0 = M/JM$, N_0 a quotient B_0 -module of M_0 which is flat over A_0 . For*

⁽²⁾N.D.E. Cf. EGA 0_{III}, 12.3.5.

PROOF OF 2.8. Denote by P_0 the kernel of the epimorphism $M_0 \rightarrow N_0$. For every $n > 0$, one therefore has the following commutative diagram, where the horizontal rows are exact:

$$(*)_n$$

$$\begin{array}{ccccccc} & & & & & & 0 \\ & & & & & & \downarrow \\ & & & & & & P_{0,n} \\ & & & & & & \downarrow \\ & & & & & & M_{0,n} \\ & & & & & & \downarrow \\ 0 & \longrightarrow & J_n M_n & \longrightarrow & M_n & \longrightarrow & M_{0,n} \longrightarrow 0 \\ & \nearrow & \downarrow & & \downarrow & & \downarrow \\ J_n \otimes_{A_{0,n}} M_{0,n} & & & & & & \\ & \searrow & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & J_n \otimes_{A_{0,n}} N_{0,n} & \longrightarrow & N_n & \longrightarrow & N_{0,n} \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array}$$

The Artin–Rees lemma (Bourbaki, *Commutative Algebra*, Chap. 3 § 3 Cor. 1) shows that the filtration defined on JM (resp. $J \otimes_{A_0} M_0$ and $J \otimes_{A_0} N_0$) by the kernels of the morphisms:

$$JM \longrightarrow J_n M_n,$$

(resp. $J \otimes_{A_0} M_0 \longrightarrow J_n \otimes_{A_{0,n}} M_{0,n}$ and
 $J \otimes_{A_0} N_0 \longrightarrow J_n \otimes_{A_{0,n}} N_{0,n}$)

is $\mathfrak{m}B$ -good, so that by taking the projective limit of the diagrams $(*)_n$ and denoting by $\widehat{Q}$ the separated completion of a B -module Q for the $\mathfrak{m}B$ -adic topology, one obtains the following

commutative diagram where the two horizontal rows are again exact:

$$\begin{array}{ccccccc}
 & & & & & & 0 \\
 & & & & & & \downarrow \\
 & & & & & & \widehat{P}_0 \\
 & & & & & & \downarrow \\
 & & & & & & \widehat{M}_0 \\
 & & & & & & \downarrow \\
 0 & \longrightarrow & \widehat{JM} & \longrightarrow & \widehat{M} & \longrightarrow & \widehat{M}_0 \longrightarrow 0 \\
 & \nearrow & \downarrow & & \downarrow & & \downarrow \\
 J \widehat{\otimes}_{A_0} M_0 & & & & & & \\
 & \searrow & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & J \widehat{\otimes}_{A_0} N_0 & \longrightarrow & \varprojlim_n N_n & \longrightarrow & \widehat{N}_0 \longrightarrow 0 \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

Now $(B, \mathfrak{m}B)$ is a Zariski ring, and $J \otimes_{A_0} N_0$ is of finite type over B , hence is separated for the $\mathfrak{m}B$ -adic topology. Diagram $(*)$ then shows that the morphism:

$$J \otimes_{A_0} M_0 \longrightarrow J \otimes_{A_0} N_0$$

deduced from the epimorphism $M_0 \longrightarrow N_0$ factors through JM :

$$\begin{array}{ccc}
 J \otimes_{A_0} M_0 & \xrightarrow{\text{can.}} & JM \\
 & \searrow & \swarrow \\
 & J \otimes_{A_0} N_0 &
 \end{array}$$

Under these conditions, it follows from Exp. III 4.1 and Exp. III 4.3 that there exists a quotient B -module N of M , flat over A , lifting N_0 , if and only if a certain element g of $E = \text{Ext}_B^1(P_0, J \otimes_{A_0} N_0)$ is zero. More precisely, the exact sequence:

$$0 \longrightarrow JM \longrightarrow M \longrightarrow M_0 \longrightarrow 0$$

defines an element f of F , where F is the B -module $\text{Ext}_B^1(M_0, JM)$, and g is the image of f under the natural morphism $F \longrightarrow E$ arising by bifactoriality from the morphisms:

$$P_0 \longrightarrow M_0 \quad \text{and} \quad JM \longrightarrow J \otimes_{A_0} N_0.$$

It follows from diagram $(*)$ and Exp. III 4.1 that the image of g in $\widehat{E}$, canonically identified with $\text{Ext}_{\widehat{B}}^1(\widehat{P}_0, J \widehat{\otimes}_{A_0} N_0)$, is zero. But E is of finite type over B , hence is separated for the $\mathfrak{m}B$ -adic topology, and consequently g is indeed equal to 0, which completes the proof of 2.8. $\square$

d) Study of $\mathcal{E}$. We therefore suppose that S is the spectrum of a local artinian ring A , with algebraically closed residue field k . Let A_0 be the ring of S_0 , $\mathfrak{m}_0$ the maximal ideal of A_0 . Since S_0 is artinian, T_0 is closed in G (Exp. VIB 1.4.2); one may therefore take the open subset U equal to G , so that:

$$\mathcal{E} = \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, J \otimes_{S_0} \mathcal{O}_{T_0}).$$

Let B_0 be the ring of the affine S_0 -scheme $T_0 = \mathbf{G}_{\mathfrak{m}, S_0}$, and $\overline{B}_0$ the ring of the special fiber $T_0 \times_{S_0} \text{Spec}(k) = \mathbf{G}_{\mathfrak{m}, k}$ of T_0 . The sheaf $\mathcal{E}$ is a coherent $\mathcal{O}_{T_0}$ -module, hence is defined by a B_0 -module of finite type which we shall denote by E .

Consider the graded module associated with E and the $\mathfrak{m}_0 B_0$ -adic filtration:

$$E_r = \mathfrak{m}_0^r E / \mathfrak{m}_0^{r+1} E.$$

Each E_r is therefore a $\overline{B}_0$ -module of finite type, and $E_r = 0$ for r sufficiently large, since S_0 is artinian.

Let then g be a section of G over S inducing on S_0 a section g_0 of T_0 . Translation (on the left, to fix ideas) by the element g defines an “automorphism of the situation” from the point of view of the obstruction problem under consideration. In particular, to g corresponds an S_0 -automorphism of the sheaf $\mathcal{E}$ leaving the obstruction Ψ fixed. More precisely, g defines a semilinear automorphism of the B_0 -module E (relative to the A_0 -automorphism of B_0 defined by translation by g_0 in the group T_0). By reduction modulo $\mathfrak{m}_0^{r+1}$, g thus defines a semilinear automorphism of E_r (relative to the k -automorphism of $\overline{B}_0$ defined by translation by $g_0 \times_{S_0} \text{Spec}(k)$ in $T_0 \times_{S_0} \text{Spec}(k)$).

Lemma 2.9. *For every integer $r \geq 0$, E_r is a locally free $\overline{B}_0$ -module.*

PROOF. Let x be a point of T_0 , $\kappa(x)$ its residue field, $(E_r)_x$ “the fiber” of E_r at x , equal to $E_r \otimes_{\overline{B}_0} \kappa(x)$, $\ell(x)$ the rank of $(E_r)_x$ over $\kappa(x)$, ℓ the maximum value of $\ell(x)$ as x ranges over the points of T_0 . Denote by L the largest closed subscheme of $\text{Spec } \overline{B}_0 = \mathbf{G}_{m,k}$ over which E_r is locally free of rank ℓ (TDTE IV Lemma 3.6). Let β be a point of $L(k)$ (there are such points, since L is of finite type over the algebraically closed field k), and let α be a point of $\mathbf{G}_{m,k}(k)$ of order equal to a power n of q . The point α is therefore rational over k , and since by hypothesis ${}_n T_0$ lifts to a subscheme M_n étale over S , there exists a section a of G over S lifting α and which, over S_0 , is a section of T_0 . The remarks made above then show that the fibers of E_r are k -isomorphic at the points β and $\alpha\beta$ of $\mathbf{G}_{m,k}(k)$. But the points of order a power of q are dense in $\mathbf{G}_{m,k}$, and the same holds for their translates by β . Since L is closed in $\mathbf{G}_{m,k}$ and $\mathbf{G}_{m,k}$ is reduced, L equals $\mathbf{G}_{m,k}$ and E_r is locally free over $\overline{B}_0$, of rank ℓ . $\square$

e) End of the proof of Theorem 2.1. Let $K_0(n)$ be the sheaf of ideals of $\mathcal{O}_{G_0}$ defining the closed subscheme ${}_n T_0$. The sheaf $K_0^{(3)}$ is therefore a subsheaf of $K_0(n)$. To simplify, put:

$$R = J \otimes_{S_0} \mathcal{O}_{T_0} \quad \text{and} \quad R(n) = J \otimes_{S_0} \mathcal{O}_{{}_n T_0}.$$

The sheaf $R(n)$ is therefore a quotient of R , and one has the diagram of maps:

$$\begin{array}{ccccccc} & & & & K_0 & & \\ & & & & \downarrow & & \\ & & & & K_0(n) & & \\ & & & & \downarrow & & \\ 0 & \longrightarrow & J\mathcal{O}_G & \longrightarrow & \mathcal{O}_G & \longrightarrow & \mathcal{O}_{G_0} \longrightarrow 0 \\ & & \downarrow & & & & \\ & & R & & & & \\ & & \downarrow & & & & \\ & & R(n) & & & & \end{array}$$

⁽³⁾**N.D.E.** Introduced after Lemma 2.7.

Using the bifactoriality of $\mathcal{E}xt_{\mathcal{O}_G}^1(,)$, one then obtains the following commutative diagram:

$$\begin{array}{ccc} \mathcal{E}xt_{\mathcal{O}_G}^1(\mathcal{O}_{G_0}, J\mathcal{O}_G) & & \\ \searrow & & \\ \mathcal{E}xt_{\mathcal{O}_G}^1(K_0(n), R) & \longrightarrow & \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) = \mathcal{E} \\ \downarrow & & \downarrow \\ \mathcal{E}xt_{\mathcal{O}_G}^1(K_0(n), R(n)) & \longrightarrow & \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n)). \end{array}$$

Again denote by Φ the element of $\mathcal{E}xt_{\mathcal{O}_G}^1(\mathcal{O}_{G_0}, J\mathcal{O}_G)$ defined by the exact sequence:

$$0 \longrightarrow J\mathcal{O}_G \longrightarrow \mathcal{O}_G \longrightarrow \mathcal{O}_{G_0} \longrightarrow 0,$$

so that Ψ is the image of Φ in $\mathcal{E}$. Since ${}_nT_0$ lifts to a subscheme M_n of G , flat over S , the image of Φ in the sheaf $\mathcal{E}xt_{\mathcal{O}_G}^1(K_0(n), R(n))$ is zero (Exp. III 4.1); a fortiori the image of Φ in $\mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n))$, which is also the image of Ψ , is zero.

Lemma 2.10. *The canonical morphism:*

$$\mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) \otimes_{B_0} \mathcal{O}_{{}_nT_0} \longrightarrow \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n))$$

is injective for every integer n .

PROOF. Indeed, the affine scheme $T_0 = \mathbf{G}_{m, S_0}$ has ring:

$$B_0 = A_0[T, T^{-1}].$$

The subscheme ${}_nT_0$ is defined by the vanishing of the following section of B_0 :

$$h(n) = T^n - 1,$$

which is regular (EGA 0_{IV} 15.2.2) and remains regular after every base change $S'_0 \longrightarrow S_0$. One therefore has an exact sequence of sheaves:

$$0 \longrightarrow \mathcal{O}_{T_0} \xrightarrow{h(n)} \mathcal{O}_{T_0} \longrightarrow \mathcal{O}_{{}_nT_0} \longrightarrow 0.$$

Since ${}_nT_0$ is flat over S , by tensoring with J over $\mathcal{O}_{S_0}$ one obtains the exact sequence:

$$0 \longrightarrow R \xrightarrow{h(n)} R \longrightarrow R(n) \longrightarrow 0,$$

then the exact sequence of $\mathcal{E}xt$:

$$\cdots \longrightarrow \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) \xrightarrow{h(n)} \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) \longrightarrow \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n)),$$

which completes the proof of the lemma. $\square$

What precedes shows that for every integer n equal to a power of q , the image of Ψ in $E/h(n)E$ is zero. To show that Ψ is zero, it suffices to see that if $\Psi \in \mathfrak{m}_0^r E$, then $\Psi \in \mathfrak{m}_0^{r+1} E$. Let $\bar{\Psi}$ be the image of Ψ in E_r . There exists an element $\Psi(n)$ of E such that:

$$\Psi = \Psi(n)h(n) \quad (n \text{ equal to a power of } q).$$

We noted that the image $\overline{h(n)}$ of $h(n)$ in $\bar{B}_0$ is again $\bar{B}_0$ -regular. Since E_s is locally free over $\bar{B}_0$ for every s (2.9), multiplication by $\overline{h(n)}$ in E_s is injective. One immediately deduces that $\Psi(n)$ is in $\mathfrak{m}_0^r E$. Let $\overline{\Psi(n)}$ be its image in E_r , so that one has the relation:

$$\bar{\Psi} = \overline{h(n)} \overline{\Psi(n)} \quad (n \text{ equal to a power of } q).$$

This shows that the set F of points of $\mathbf{G}_{m, k}$ at which $\bar{\Psi}$ takes the value 0 contains the dense set of points of order a power of q . Moreover, F is closed (since E_r is locally free over $\bar{B}_0$), and $\mathbf{G}_{m, k}$ is reduced, so $\bar{\Psi}$ is zero.

This completes the proof of Theorem 2.1.

3. Characterization of a subtorus by its underlying set

1. Statement of the theorem

Notation. If X is a prescheme, $\text{ens}(X)$ denotes the underlying set of X . If X and S' are two S -preschemes, $X_{S'} = X \times_S S'$, $u : X_{S'} \rightarrow X$ the canonical morphism, E a subset of $\text{ens}(X)$, one denotes by $E_{S'}$ (or $E \times_S S'$) the subset of $\text{ens}(X_{S'})$ equal to $u^{-1}(E)$.

Theorem 3.1. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type, q an integer > 0 invertible on S , E a subset of $\text{ens}(G)$. Consider the following assertions concerning E :*

- (i) *The set E is the underlying set of a subtorus T of G .*
- (ii) *a) For every point s of S , there exists a subtorus T_s of G_s such that $E \cap \text{ens}(G_s) = E_s$ is the underlying set of T_s .*
b) With the integer n ranging over the powers of q , there exists a coherent family (cf. 2.2) M_n of group subschemes of G , of multiplicative type, such that for every point s of S , one has:

$$(M_n)_s = {}_n T_s.$$

- (iii) *a) As in (ii) a).*
b) The set E is locally closed in $\text{ens}(G)$, and the dimension of the fibers of E , relative to S , is locally constant.
- (iv) *a) As in (ii) a).*
b) For every S -scheme S' which is the spectrum of a complete discrete valuation ring with algebraically closed residue field, $E_{S'}$ is the underlying set of a subtorus of $G_{S'}$.

Then one has the following implications:

A) (i) $\iff$ (ii) $\implies$ (iii) $\implies$ (iv).

B) If G is separated over S , one has (iii) $\iff$ (iv), and moreover E is closed.

C) Conditions (i), (ii), (iii) (and also (iv), if G is separated over S) are equivalent in the following two cases:

First case: a) The prescheme S is reduced, or else G is flat over S , and

b) For every point s of S , T_s is a central torus of G_s .

Second case: S is normal.

Moreover, in the two cases above, the torus T with underlying space E is unique.

Remarks 3.2. a) When S is reduced, it is unnecessary in (ii) to suppose that the family M_n is coherent, this condition being implied by the other hypotheses. Indeed, if the integer m divides n , the subgroups ${}_m M_n$ and M_m are étale over S , hence reduced, and have the same underlying space, so they agree.

b) If S is no longer supposed normal, it is no longer true in general that (iii) $\implies$ (i), even when S is reduced, geometrically unibranch, and G is a group scheme smooth over S . Indeed, consider the Borel subgroup of $\text{SL}_{2,S}$ formed by matrices of the form:

$$\begin{pmatrix} t & u \\ 0 & t^{-1} \end{pmatrix},$$

where S is the affine curve over a field k with ring:

$$k[x, y]/(y^2 - x^3).$$

Consider then the set E obtained as follows: over the “cusp of S ” ($x = y = 0$), we take the diagonal torus ($u = 0$). Over the complementary open subset ($x \neq 0$), we take the torus deduced from the diagonal torus by conjugation by the element:

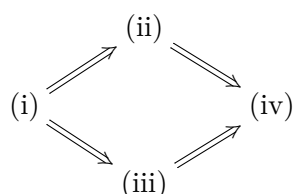
$$\begin{pmatrix} 1 & y/x \\ 0 & 1 \end{pmatrix}.$$

The set E obtained satisfies (iii) a); on the other hand, it is closed in G , and the reduced subscheme with underlying set E has equations:

$$\begin{cases} xu + y(t - t^{-1}) = 0, \\ u^2 - x(t - t^{-1})^2 = 0. \end{cases}$$

It is therefore not a subtorus of G , since the fiber over the cusp is not reduced.

Plan of the proof of 3.1. In a first part, we shall establish the following “easy” implications:



and (iv) $\implies$ [(iii) and E closed] if G is separated over S .

The proof of the more delicate implications will proceed in three stages:

- I) Reduction of the implication (iii) $\implies$ (i) (under the hypotheses of C), first case) to the case where S is normal.
- II) (iii) $\implies$ (ii) if S is normal.
- III) (ii) $\implies$ (i).

2. Proof of the “easy” assertions contained in 3.1

(i) $\implies$ (ii) and (iii) is trivial.

(iii) $\implies$ (iv). After replacing S by S' , we may suppose that S is the spectrum of a discrete valuation ring. Let t be the generic point of S , and s the closed point.

Since E is locally closed, there exists a subprescheme of G which is reduced and whose underlying space is E ; we shall denote it by $\overline{E}$. The generic fiber $\overline{E}_t$ of $\overline{E}$ is therefore equal to the subtorus T_t of rank r of G_t (by (iii) a)). Let E' be the schematic closure of $\overline{E}_t$ in $\overline{E}$. The prescheme E' is irreducible, and its closed fiber E'_s is nonempty (it contains the unit section of G_s), so E'_s has dimension r (EGA IV 14.3.10). But E'_s is then a closed subscheme of $\overline{E}_s$, and the latter has dimension r (by (iii) b)) and is irreducible (since it has the same underlying space as T_s), so E'_s has the same underlying space as $\overline{E}_s$, and consequently $E' = \overline{E}$, which proves that $\overline{E}$ is flat over S .

Let now E'' be the schematic closure of $\overline{E}_t$ in G . Then E'' is a group subprescheme of G , flat over S (Exp. VIII 7.1), containing E' , hence $\overline{E}$. The closed fiber E''_s is an algebraic subgroup of G_s , of dimension r (*loc. cit.*). Since T_s is an irreducible closed subset of G_s , of dimension r , E''_s has the same underlying set as the neutral component $(E''_s)^0$ of E''_s . Let $(E'')^0$ be the “neutral component” of E'' , i.e. the open subgroup of E'' complementary to the union of the irreducible components of E''_s not containing the origin. One therefore has:

$$E = \text{ens}[(E'')^0].$$

Since $\overline{E}$ and E'' are reduced, one even has $\overline{E} = (E'')^0$. Finally, $\overline{E}$ is a group subprescheme of G , flat and of finite type over S , with connected fibers, hence separated (Exp. VIB 5.2), whose generic fiber is a torus T_t , and whose reduced closed fiber is a torus T_s ; but then $\overline{E}$ is a torus (Exp. X 8.8).

(ii) $\implies$ (iv). One may again suppose that S is the spectrum of a discrete valuation ring, and we retain the notation introduced above. The schematic closure in G of T_t is a group subprescheme $\overline{T}$ of G , flat over S , containing M_n for every integer n equal to a power of q . Consequently, the closed fiber of $\overline{T}$ is a closed subscheme of G_s containing $(M_n)_s$ for every n , hence containing T_s . For reasons of dimension, the “neutral component” $\overline{T}^0$ of $\overline{T}$ has E as underlying set, and one concludes as above that $\overline{T}^0$ is a subtorus of G .

(iv) $\implies$ [(iii) and E closed], if G is separated over S . Let us show that E is closed.⁽⁴⁾
Let us first prove the lemma:

Lemma 3.3. *If the conditions stated in 3.1 iv) are satisfied, E is a locally constructible subset of $\text{ens}(G)$.*

PROOF. By the usual method, one is reduced to studying the case where S is noetherian, integral, with generic point η . After restricting S , one may suppose (Exp. VIB 10.10) that there exists a group subscheme of G , say H , flat over S , with connected fibers, whose generic fiber H_η is equal to T_η . To prove 3.3, it then suffices to show that $\text{ens}(H) = E$. Now if s is a point of S , by EGA II 7.1.9 there exists an S -scheme S' , the spectrum of a discrete valuation ring, which one may suppose complete with algebraically closed residue field, whose generic point t' projects to η and whose closed point s' projects to s . By (iv) b), there exists a subtorus $T_{S'}$ of $G_{S'}$ with underlying space $E_{S'}$. The two group subschemes $T_{S'}$ and $H_{S'} = H \times_S S'$ of $G_{S'}$ are flat over S' , have connected fibers, and have the same generic fiber $T_{t'}$, so they agree with the neutral component of the schematic closure of $T_{t'}$ in $G_{S'}$. Consequently $\text{ens}(H_s) = E_s$, hence $\text{ens}(H) = E$, which proves the lemma. $\square$

This being so, knowing that E is locally constructible, to see that E is closed it suffices to show that every specialization in G of a point of E is a point of E . By the usual technique, one is reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. But then, the subtorus of G with underlying space E ((iv) b)) is closed in G , since G is separated (Exp. VIII 7.12).

3. Continuation of the proof of 3.1

I) Reduction of (iii) $\implies$ (i) (C, first case) to the case where S is normal.

a) Reduction to the case of S affine reduced.

We therefore suppose that for every point s of S , E_s is the underlying space of a central subtorus of G_s . The uniqueness of T then follows from Exp. IX 5.1 bis, and moreover T will necessarily be central (*loc. cit.*). In view of uniqueness, to prove the existence of T , we may suppose S noetherian, affine with ring A . If S is not reduced, by hypothesis G is flat over S . By 2.3, it then suffices to solve the problem for S_{red} and $G \times_S S_{\text{red}}$. We may therefore suppose in addition that S is reduced.

b) Reduction to the case where the ring A is of finite type over $\mathbf{Z}$.

Let us first prove two lemmas:

Lemma 3.4. *Let k be a field, G an algebraic k -group, E a subset of G , k' an extension of k , $T_{k'}$ a subtorus of $G_{k'}$ with underlying space $E_{k'}$. Then if $T_{k'}$ is central or k is perfect, E is the underlying set of a subtorus of G_k .*

PROOF. Indeed, by fpqc descent, it suffices to show that the two inverse images of $T_{k'}$ in $G_{k''}$, where $k'' = k' \otimes_k k'$, agree. Now they have the same underlying space, namely the inverse image of E . If $T_{k'}$ is central, the lemma is then a consequence of Exp. IX 5.1 bis. If k is perfect, k'' is reduced, and the two inverse images of $T_{k'}$, being smooth over k'' , are reduced, hence agree. $\square$

Remark 3.5. It follows from the preceding lemma that in the statement of 3.1 (iv), property (iv) a) is a consequence of (iv) b) in the following two cases:

- (1) One supposes that the residue fields of the points of S are perfect.
- (2) For every S' as in (iv) b), one supposes that the torus with underlying space equal to $E_{S'}$ is central in $G_{S'}$.

Lemma 3.6. *Let S be a prescheme which is a projective limit of affine schemes S_i (cf. EGA IV 8), H a group S -scheme of multiplicative type and of finite type. Then there exist an index*

⁽⁴⁾**N.D.E.** The fact that (iv) $\implies$ (iii) will appear in the course of the proof.

i , a group S_i -scheme H_i of multiplicative type and of finite type, and an S -isomorphism:

$$H_i \times_{S_i} S \xrightarrow{\sim} H.$$

If in addition H is isotrivial, one may suppose H_i isotrivial.

PROOF. Since H is of finite type over S , H is even of finite presentation over S (Exp. IX 2.1 b)), so there exist an index ℓ and a group S_ℓ -scheme H_ℓ such that $H_\ell \times_{S_\ell} S$ is isomorphic to H (Exp. VIB 10). Putting $H_i = \prod H_{\ell S_i} S_\ell$, one therefore has $H \simeq \prod H_i S_i$ for every $i \geq \ell$.

Since H is of finite type over S , H is quasi-isotrivial (Exp. X 4.5), hence trivialized by an étale surjective morphism $S' \rightarrow S$. Using the quasi-compactness of S , one sees easily that there exists a covering of S by finitely many affine open subsets S_α and, for every α , an étale, surjective morphism of finite presentation $S'_\alpha \rightarrow S_\alpha$ trivializing $H|_{S_\alpha}$. This covering (S_α) of S then comes from a covering ($S_{i,\alpha}$) of S_i for i sufficiently large (EGA IV 8). After replacing S_i by $S_{i,\alpha}$ and S by S_α , one may therefore suppose that H is trivialized by an étale surjective morphism $S' \rightarrow S$ of finite presentation. For i sufficiently large, there then exist a prescheme S'_i , an étale surjective morphism $S'_i \rightarrow S_i$ of finite presentation, and an S -isomorphism $S'_i \times_{S_i} S \rightarrow S'$ (EGA IV 17.16). Put then, for j sufficiently large:

$$S'_j = S'_i \times_{S_i} S_j, \quad H'_j = H_j \times_{S_j} S'_j, \quad H' = H \times_S S'.$$

In view of the choice of S' , there exist an abelian group M of finite type and an S -isomorphism of group schemes $D_{S'}(M) \xrightarrow{\sim} H'$. Since the S'_i are quasi-compact and $S' = \varprojlim S'_i$, it follows from Exp. VIB 10 that there exist an index j and an S'_j -isomorphism of group schemes:

$$D_{S'_j}(M) \xrightarrow{\sim} H'_j.$$

But this says that H_j is a quasi-isotrivial group of multiplicative type.

When H is isotrivial, one proceeds in an analogous manner, using a trivialization of H by a finite étale morphism $S' \rightarrow S$. $\square$

This being so, we can carry out the announced reduction b). The ring A of S is the filtered inductive limit of its subrings A_i of finite type over $\mathbf{Z}$. Let $S_i = \text{Spec } A_i$, $u_j : S \rightarrow S_j$ and $u_{jk} : S_k \rightarrow S_j$ ($k \geq j$) be the transition morphisms. Since S is noetherian and G is of finite presentation over S , there exist an index i and a group S_i -prescheme G_i , of finite type over S_i , such that G is S -isomorphic to $G_i \times_{S_i} S$. Likewise, since E is locally closed in G , one may suppose that there exists a locally closed subset E_i of G_i such that $E = E_i \times_{S_i} S$ (EGA IV 8.3.11). For every $j > i$, denote by $G_j = G_i \times_{S_i} S_j$ and $E_j = E_i \times_{S_i} S_j$, and let Q_j be the set of points s of S_j such that $(E_j)_s$ is the underlying set of a central subtorus of $(G_j)_s$.

It follows from 3.4 that $Q_k = u_{jk}^{-1}(Q_j)$ for $k \geq j$, and by hypothesis $\text{ens}(S) = u_j^{-1}(Q_j)$ for $j \geq i$. Moreover, I say that Q_j is ind-constructible (EGA IV 1.9.4). Indeed, since S_j is noetherian, it suffices (EGA IV 1.9.10) to see that if S is an integral noetherian scheme with generic point η , and if E_η is the underlying set of a central subtorus T_η of G_η , there exists a neighborhood U of η such that for every point s of U , E_s has the same property. Now after restricting S , one may suppose that the center Z of G is representable and that there exists a group subscheme T of G whose generic fiber is T_η (VIB § 10). But then $\text{ens}(T)$ and E are two locally closed subsets of $\text{ens}(G)$ which agree on the generic fiber, so one can find a neighborhood U of η such that, over U , $\text{ens}(T) = E$ (EGA IV 8.3.11). For the same reasons, one may suppose that over U , T is central, since T is a torus (3.6).

Knowing now that Q_j is ind-constructible, it follows from EGA IV 8.3.4 that $Q_j = \text{ens}(S_j)$ for j sufficiently large. We may then replace S, G, E by S_j, G_j, E_j , which reduces one to the case where S is a reduced affine scheme of finite type over $\mathbf{Z}$.

c) Reduction to the case where S is the spectrum of a reduced complete noetherian local ring.

Because of the uniqueness of the torus with underlying space E , the usual technique (EGA IV 8) and Lemma 3.6 allow one to replace S by the spectrum of the local ring A of a point of S . Let $\hat{S}$ be the spectrum of the completion $\hat{A}$ of A for the topology defined by

the maximal ideal. Since A is the localization of an algebra of finite type over $\mathbf{Z}$, $\widehat{S}$ is again reduced (EGA IV 7.6.5). I say that it suffices to solve the problem after extension of the base $\widehat{S} \rightarrow S$. Indeed, if $\widehat{T}$ denotes the subtorus of $G_{\widehat{S}}$ with underlying space $E_{\widehat{S}}$, its two inverse images in $G_{\widehat{S} \times_S \widehat{S}}$ are two central subtori having the same underlying space, hence agreeing (Exp. IX 5.1 bis); and by fpqc descent, $\widehat{T}$ comes from a torus T of G answering the question (cf. 3.4).

d) A descent lemma.

Recall the following properties of finite morphisms, which were pointed out in TDTE I: let S and S' be two preschemes and $u : S' \rightarrow S$ a finite morphism. Then:

- (1) The morphism u is an epimorphism if and only if the canonical sheaf morphism:

$$\mathcal{O}_S \rightarrow u_*(\mathcal{O}_{S'})$$

is injective.

- (2) The morphism u is an effective epimorphism (Exp. IV 1.3) if and only if the canonical diagram:

$$0 \rightarrow \mathcal{O}_S \rightarrow u_*(\mathcal{O}_{S'}) \rightrightarrows u_*(\mathcal{O}_{S'}) \otimes_{\mathcal{O}_S} u_*(\mathcal{O}_{S'})$$

is exact.

- (3) If in addition S is noetherian and u is an epimorphism, u is the composite of a finite sequence of finite effective epimorphisms.

We are then in a position to prove the following lemma, whose proof uses a technique of nonflat descent:

Lemma 3.7. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type, S' a prescheme, and $u : S' \rightarrow S$ a finite morphism. For every S -prescheme T , denote by $M(T)$ the set of subgroups of multiplicative type of G_T (resp. the set of central subgroups of multiplicative type of G_T), so that M is naturally a contravariant functor defined on $(\text{Sch})/S$. Then if u is an effective epimorphism (resp. an epimorphism), the diagram of sets:*

$$(*) \quad M(S) \rightarrow M(S') \rightrightarrows M(S' \times_S S')$$

is exact.

Proof. i) *Reduction to the case where u is an effective epimorphism.* We are therefore interested in the functor of central subgroups of multiplicative type of G . The injectivity of $M(S) \rightarrow M(S')$ is a question local in nature on S , and once this injectivity is granted, the exactness of $(*)$ becomes a problem local in nature on S . One may therefore suppose S affine noetherian.

Let us study the case where $u : S \rightarrow S'$ is the composite of two finite epimorphisms:

$$S' \xrightarrow{v} S'' \xrightarrow{w} S.$$

I say that if the two diagrams:

$$(*)' \quad M(S'') \rightarrow M(S') \rightrightarrows M(S' \times_{S''} S'),$$

$$(*)'' \quad M(S) \rightarrow M(S'') \rightrightarrows M(S'' \times_S S'')$$

are exact, the same holds for $(*)$.

Indeed, the injectivity of $M(S) \rightarrow M(S')$ is clear. If now T' is an element of $\text{Ker}[M(S') \rightrightarrows M(S' \times_S S')]$, a fortiori T' belongs to $\text{Ker}[M(S') \rightrightarrows M(S' \times_{S''} S')]$, so by the exactness of $(*)'$, it comes from a unique element T'' of $M(S'')$. It suffices to show that T'' belongs to $\text{Ker}[M(S'') \rightrightarrows M(S'' \times_S S'')]$, for in view of the exactness of $(*)''$, T'' will descend to S . Let therefore T''_1 and T''_2 be the two inverse images of T'' in $M(S'' \times_S S'')$. Since they are two central subgroups of multiplicative type of $G_{S'' \times_S S''}$, to show that $T''_1 = T''_2$, it suffices to see

that they have the same fibers (Exp. IX 5.1 bis). Consider the commutative diagram:

$$\begin{array}{ccc} S' & \xleftarrow{\quad} & S' \times_S S' \\ v \downarrow & & \downarrow v \times v \\ S'' & \xleftarrow{\quad} & S'' \times_S S'' \end{array}$$

The morphism v is a finite epimorphism, hence is dominant (property (a) above) and closed, so is surjective, and the same holds for $v \times v$. Let then x'' be a point of $S'' \times_S S''$, and x' a point of $S' \times_S S'$ over x'' . It follows from the commutativity of the diagram above that the two inverse images of $(T_1'')_{x''}$ and $(T_2'')_{x''}$ in $G_{x'}$ agree, hence $(T_1'')_{x''} = (T_2'')_{x''}$ (EGA IV 2.2.15).

What precedes and an immediate induction on the number of factors of a factorization of $u : S' \rightarrow S$ into effective epimorphisms (property c) recalled above) show that to prove the exactness of $(*)$ in the case of central subgroups of multiplicative type, one may restrict to the case where u is an effective epimorphism. Finally, using Exp. IX 5.1 bis once more, one sees that it suffices to prove 3.7 when M is the functor of subgroups of multiplicative type of G and u an effective epimorphism.

ii) *Injectivity of $M(S) \rightarrow M(S')$.* Since $u : S' \rightarrow S$ is an epimorphism, the canonical morphism:

$$\mathcal{O}_S \rightarrow u_*(\mathcal{O}_{S'})$$

is injective; moreover, an S -group of multiplicative type is flat over S ; the injectivity of $M(S) \rightarrow M(S')$ is therefore a consequence of the following more general lemma:

Lemma 3.8. *Let $f : X \rightarrow S$ and $g : S' \rightarrow S$ be two morphisms of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, $X' = X \times_S S'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, $Q(\mathcal{F})$ the set of quotient $\mathcal{O}_X$ -modules $\mathcal{G}$ of $\mathcal{F}$ which are quasi-coherent and flat over S , $Q(\mathcal{F}')$ the analogue relative to $\mathcal{F}'$, X' and S' . Suppose g quasi-compact and $\mathcal{O}_S \rightarrow g_*(\mathcal{O}_{S'})$ injective; then the canonical map:*

$$\begin{aligned} Q(\mathcal{F}) &\rightarrow Q(\mathcal{F}') \\ \mathcal{G} &\mapsto \mathcal{G} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \end{aligned}$$

is injective.

Indeed, one may suppose S affine, then X affine. Since the morphism g is quasi-compact, S' is then the union of finitely many affine open subsets S'_i . After replacing S' by $\coprod S'_i$ (an operation preserving the injectivity of $\mathcal{O}_S \rightarrow g_*(\mathcal{O}_{S'})$), one may suppose S' affine. One is then reduced to proving the following lemma, whose proof is immediate:

Lemma 3.9. *Let $A \rightarrow A'$ be an injective ring homomorphism, M an A -module, $N = M/P$ a quotient A -module flat over A , $M' = M \otimes_A A'$, $N' = N \otimes_A A' = M'/P'$. Then P is the inverse image of P' under the canonical homomorphism $M \rightarrow M'$ (so N and P are known when one knows N' and P').*

iii) *Exactness of 3.7 (*) at $M(S')$.* Let T' be an element of $\text{Ker}[M(S') \rightrightarrows M(S' \times_S S')]$. Suppose one has proved iii) when S is the spectrum of a noetherian local ring. For every point s of S , there therefore exists a subgroup of multiplicative type T_s of $G \times_S \text{Spec}(\mathcal{O}_{S,s})$ arising by descent from $T' \times_{S'} u^{-1} \text{Spec}(\mathcal{O}_{S,s})$. By Exp. VIB § 10 and 3.6, there exist a neighborhood U of s in S and a group subscheme T of G over U , of multiplicative type, extending T_s . Let U' be the inverse image of U in S' . One then knows two subschemes of $G'|_{U'}$: $T'|_{U'}$ and $T \times_U U'$, agreeing on $u^{-1}(\text{Spec}(\mathcal{O}_{S,s}))$. If one regards $u^{-1}(\text{Spec}(\mathcal{O}_{S,s}))$ as the projective limit of the schemes $u^{-1}(V)$, where V ranges over the open neighborhoods of s in U , it follows from EGA IV 8 that there exists an open subset V of U containing s such that $T \times_U U'$ and T' agree over $u^{-1}(V)$. Thus, under the hypotheses made, T' descends locally on S , but because of the uniqueness proved in ii), T' then descends globally on S . In brief, it suffices to prove iii) when S is the spectrum of a noetherian local ring.

Denote then by $\widehat{S}$ the spectrum of the completion of the ring of S , and let $S'' = S' \times_S S'$, $\widehat{S}' = \widehat{S} \times_S S'$, $\widehat{S}'' = \widehat{S} \times_S S'' = \widehat{S}' \times_S \widehat{S}'$. I say that it suffices to show that the diagram:

$$(*) \quad M(\widehat{S}) \longrightarrow M(\widehat{S}') \rightrightarrows M(\widehat{S}'')$$

is exact at $M(\widehat{S}')$. This follows from the commutative diagram below, where the second row is exact at $M(\widehat{S}')$ by hypothesis, where the first two columns are exact (fpqc descent), and where the map f is injective, as follows from ii) applied to the finite epimorphism:

$$\widehat{S}' \times_{S'} \widehat{S}' \longrightarrow \widehat{S} \times_S \widehat{S}$$

deduced from $S' \longrightarrow S$ by the flat base change $\widehat{S} \times_S \widehat{S} \longrightarrow S$:

$$\begin{array}{ccccc} M(S) & \longrightarrow & M(S') & \rightrightarrows & M(S'') \\ \downarrow & & \downarrow & & \downarrow \\ M(\widehat{S}) & \longrightarrow & M(\widehat{S}') & \rightrightarrows & M(\widehat{S}'') \\ \Downarrow & & \Downarrow & & \\ M(\widehat{S} \times_S \widehat{S}) & \xrightarrow{f} & M(\widehat{S}' \times_{S'} \widehat{S}') & & \end{array}$$

(the diagram chasing is left to the reader). It follows from characterization b) of finite effective epimorphisms that the morphism $\widehat{S}' \longrightarrow \widehat{S}$, deduced from $S' \longrightarrow S$ by the flat base change $\widehat{S} \longrightarrow S$, is again a finite effective epimorphism. One is therefore reduced to the case where in addition S is the spectrum of a complete noetherian local ring. Denote by S_0 the reduced subscheme of S whose underlying space is the closed point of S . Let T' be an element of $\text{Ker}[M(S') \rightrightarrows M(S'')]$, T'' its image in $M(S'')$. Since $S'_0 = S' \times_S S_0$ is faithfully flat over S_0 , there exists an S -subgroup of multiplicative type T_0 of G_{S_0} whose inverse image in $G_{S'_0}$ is $T'_{S'_0}$ (fpqc descent). But S is local noetherian complete, so there exist an S -group T of multiplicative type and an S_0 -morphism $u_0 : T \times_S S_0 \longrightarrow T_0$ (Exp. X 3.3). The inverse image u'_0 of u_0 over S'_0 extends uniquely to an S' -isomorphism $u' : T_{S'} \longrightarrow T'$, again by Exp. X 3.3 (note that S' , being finite over S local complete, is the sum of finitely many complete local schemes). The two inverse images of u' over S'' are two morphisms from $T_{S''}$ to T'' which agree on $S_0 \times_S S''$, hence agree (*loc. cit.*). Since T is flat over S and $S' \longrightarrow S$ is a finite effective epimorphism, it follows from TDTE I page 8 that the diagram:

$$\text{Hom}_S(T, G) \longrightarrow \text{Hom}_{S'}(T_{S'}, G_{S'}) \rightrightarrows \text{Hom}_{S''}(T_{S''}, G_{S''})$$

is exact. Thus u' comes from a morphism of preschemes $u : T \longrightarrow G$. Likewise, $T \times_S T$ is flat over S , and consequently the map:

$$\text{Hom}_S(T \times_S T, G) \longrightarrow \text{Hom}_{S'}(T_{S'} \times_{S'} T_{S'}, G_{S'})$$

is injective. One immediately deduces that u is a group morphism, since this is so for u' . Moreover, u is a monomorphism. Indeed, note first that $\text{Ker } u$ is flat over S , for to establish this fact one may suppose S local artinian (EGA 0_{III} 10.2.2), hence G separated (Exp. VIA 0.3), but then $\text{Ker } u$ is of multiplicative type (Exp. IX 6.8), hence is flat over S . Since $\text{Ker}(u) \times_S S' = \text{Ker}(u') = 0$, one has $\text{Ker}(u) = 0$ (3.8). But u , being a monomorphism, is an immersion (Exp. VIII, Remarks 7.13), and the image group $u(T)$ is indeed an element of $M(S)$ whose image in $M(S')$ is T' , which completes the proof of 3.7.

e) End of the proof of I).

By reduction c), one is reduced to the case where S is the spectrum of a reduced complete noetherian local ring A . Let S' be the spectrum of the normalization A' of A , which is finite over A by Nagata (EGA 0_{IV} 23.1.5); $S' \longrightarrow S$ is an epimorphism, since A maps injectively into A' . Suppose then that there exists a torus T' of $G_{S'}$ with underlying space $E_{S'}$. The two inverse images of T' in $G_{S' \times_S S'}$ are two central subtori having the same underlying space, hence agreeing (Exp. IX 5 bis); so by 3.7, T' comes from a central subtorus T of G_S which

obviously has E as underlying space. It therefore suffices to prove the existence of T' , which reduces one to the case where S is normal and completes the proof of I).

II) Proof of (iii) $\implies$ (ii) when S is normal.

We may restrict to the case where S is integral; let t be its generic point. For every integer n equal to a power of q , let ${}_nG$ be the subprescheme of G which is the “kernel” of raising to the n th power in G . Since E is locally closed in G (by (iii) b)), the intersection of $\text{ens}({}_nG)$ with E is locally closed in $\text{ens}(G)$; denote then by $E(n)$ the reduced subprescheme of G with underlying space $\text{ens}({}_nG) \cap E$.

Let us show that the structural morphism $E(n) \longrightarrow S$ is separated and universally open. For these two properties, one has a valuative criterion (EGA II 7.2.3 and EGA IV 14.5.8). Let therefore S' be an S -scheme which is the spectrum of a complete discrete valuation ring with algebraically closed residue field. We have proved that 3.1 (iii) $\implies$ 3.1 (iv), so there exists a subtorus T' of $G_{S'}$ with underlying space $E_{S'}$. Now ${}_nT'$ is finite and étale over S' , hence is separated and universally open over S' , and the same holds for $E(n) \times_S S'$, which has the same underlying space as ${}_nT'$.

Moreover, the fibers of $E(n)$ have the same number of geometric points, namely r^n if r is the rank of the torus T_t . Finally, the generic fiber $E(n)_t$, being reduced, is equal to ${}_nT_t$, hence is étale over $\text{Spec}(t)$. Since S is normal, it then follows from SGA I 10.11 that $E(n)$ is an étale covering of S .

If s is a point of S , $E(n)_s$ is étale over $\text{Spec } \kappa(s)$, hence is reduced and consequently agrees with the group of multiplicative type ${}_nT_s$. Let us show that $E(n)$ is a group subprescheme of G . Indeed, let m be the morphism

$$E(n) \times_S E(n) \longrightarrow G$$

induced by multiplication in G . The map $\text{ens}(m)$ underlying m factors through $E(n)$, so m factors through the prescheme $E(n)$, since the first member $E(n) \times_S E(n)$ is étale over S , hence reduced. It then follows from Exp. X 4.8 a) that $E(n)$ is a subgroup of multiplicative type. As already noted (3.2 a)), the family of subgroups $E(n)$ is necessarily coherent. We have therefore proved that (iii) $\implies$ (ii) when S is normal.

III) Proof of (ii) $\implies$ (i).

In fact, we shall show that there exists a unique subtorus T of G with underlying space equal to E and such that ${}_nT = M_n$ for every n equal to a power of q . The uniqueness of T follows simply from Exp. IX 4.8 b). To establish the existence of T , taking uniqueness into account, we may suppose successively that:

a) S is noetherian.

b) The M_n are central subgroups. Indeed, it suffices to replace G by $Z_n = \text{Centr}_G(M_n)$ for n sufficiently large (2.5 and 2.5 bis).

c) S is the spectrum of a local ring. Suppose indeed that the problem is solved after every base change $\text{Spec}(\mathcal{O}_{S,s}) \longrightarrow S$, where s ranges over the points of S . Let T_s be the subtorus of $G \times_S \text{Spec}(\mathcal{O}_{S,s})$ thus obtained. For every s , there then exist an open neighborhood U of s and a subtorus T of $G|_U$ extending T_s (Exp. VIB § 10 and 3.6). We have proved that 3.1 (ii) $\implies$ 3.1 (iv); since S is noetherian, E is therefore constructible (3.3). Consequently, after restricting U , one may suppose that $\text{ens}(T) = E \times_S U$ (EGA IV 9.5.2). But then, for every integer n equal to a power of q , ${}_nT$ and $M_n \times_S U$ are two subgroups of multiplicative type of $G|_U$ having the same fibers, hence agreeing, since M_n is central (Exp. IX 5.3 bis). In brief, under the hypotheses made, there exists a solution locally on S , hence, because of uniqueness, there exists a global solution.

d) S is the spectrum of a complete noetherian local ring and, if s is the closed point, T_s is trivial. This follows from EGA 0_{III} 10.3.1 and fpqc descent.

e) S is reduced. One applies 2.2.

f) S is normal. One applies Nagata's finiteness theorem (EGA 0_{IV} 23.1.5) and Lemma 3.7 in the case of central subtori.

These reductions having been made, since T_s is trivial, $(M_n)_s$ is trivial, hence M_n is trivial (Exp. X 3.3). If t is a point of S , the subgroups ${}_nT_t$ of T_t are therefore trivial for every n equal to a power of q , and it follows easily from Exp. X 1.4 that T_t itself is trivial. It then suffices to prove the lemma:

Lemma 3.10. *Under the hypotheses of 3.1 (ii), suppose in addition that S is noetherian and normal, and that for every point s of S , T_s is a trivial torus. Then there exists a unique subtorus T of G with underlying space equal to E , such that ${}_nT = M_n$ for every n equal to a power of q . Moreover, T is trivial.*

PROOF. The uniqueness of T follows from the fact that T , being smooth over S , is reduced. To prove existence, one may suppose S irreducible with generic point η . Let r be the rank of T_η , $T' = \mathbf{G}_{m,S}^r$, u_η an isomorphism of T'_η onto T_η , ${}_nu_\eta$ the restriction of u_η to ${}_nT'_\eta$. Since ${}_nT'$ and M_n are trivial, there exists a unique extension ${}_nu$ of ${}_nu_\eta$ to an S -isomorphism of ${}_nT'$ onto M_n . I say that for every point s of S , there exists a group isomorphism, necessarily unique:

$$u_s : T'_s \xrightarrow{\sim} T_s$$

extending ${}_nu_s$ for every n equal to a power of q . Indeed, let S_1 be an S -scheme, the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point t_1 projects to η and whose closed point s_1 projects to s (EGA II 7.1.9). It follows from the proof of 3.1 (ii) $\implies$ 3.1 (iv) that there exists a subtorus T^1 of G_{S_1} such that $(M_n)_{S_1} = {}_nT^1$ for every n equal to a power of q . Since S_1 is normal, T^1 is isotrivial (Exp. X 5.16), and it follows from the classification of isotrivial tori (Exp. X 1.2) and SGA1 V 8.2 that T^1 , having trivial generic fiber, is trivial. One can therefore extend the isomorphism $u_\eta \times_\eta t_1$ to an S_1 -isomorphism of group schemes:

$$u^1 : T' \times_S S_1 \xrightarrow{\sim} T^1.$$

The restriction of u^1 to ${}_nT' \times_S S_1$, on the one hand, and ${}_nu \times_S S^1$, on the other hand, agree on the generic fiber by construction, hence agree. The restriction $u^1_{s_1}$ of u^1 to the closed fiber therefore gives the required extension after extension of the residue field $\kappa(s) \longrightarrow \kappa(s_1)$. It then follows from Exp. IX 4.8 a) and fpqc descent that $u^1_{s_1}$ descends to $\kappa(s)$ as a group isomorphism $u_s : T'_s \xrightarrow{\sim} T_s$ answering the question.

Moreover, T' is smooth over S , which is normal, hence is normal. It then follows from an easy criterion for extending rational maps (EGA IV₄ 20.4.6) that, for there to exist an S -morphism $u : T' \longrightarrow G$ whose restriction to T'_s , for every point s of S , is the composite morphism:

$$T'_s \xrightarrow{u_s} T_s \longrightarrow G_s,$$

it is necessary and sufficient that this be so after every base change $S_1 \longrightarrow S$, where S_1 is the spectrum of a complete discrete valuation ring with algebraically closed residue field.

Now in the present case, an argument analogous to the one just made shows that the extension condition is indeed satisfied. Denote by u the S -morphism $T' \longrightarrow G$ thus obtained.

Let us show that u is indeed a group morphism. Let $m_{T'}$ (resp. m_G) be the morphism defining multiplication in T' (resp. G). One must verify that $m_G \circ (u \times_S u) = u \circ m_{T'}$. Now the coincidence subscheme of these two morphisms is a subscheme of $T' \times_S T'$ containing the fibers (since u_S is a group morphism), hence having the same underlying space as $T' \times_S T'$, hence equal to the latter, since $T' \times_S T'$ is smooth over S , thus reduced.

Finally, note that u is a monomorphism (since this is so on the fibers), hence is an immersion (Exp. VIII 7.9). The image of T' under u is then a subtorus of G having E as underlying set. $\square$

This completes the proof of Theorem 3.1.

4. Characterization of a subtorus T by the subgroups ${}_nT$

1. Statement of the main theorem

Theorem 4.1. *Let S be a connected locally noetherian prescheme, G a group S -prescheme of finite type over S , q an integer > 1 invertible on S , r a positive integer. For every integer n equal to a power of q , let $M(n)$ be a group subscheme of G , of multiplicative type and of type $(\mathbf{Z}/n\mathbf{Z})^r$. One supposes that:*

a) *The family of subgroups $M(n)$ is coherent, that is, if the integer m divides n , one has:*

$${}_mM(n) = M(m).$$

b) *There exist a point s of S and a subtorus T_s of G_s such that:*

$$M(n)_s = {}_nT_s \quad \text{for every } n.$$

c) *For every point t of S , there exists a closed affine subscheme $F_{\bar{t}}$ of $G_{\bar{t}}$ containing $M(n)_{\bar{t}}$ for every n .*

Then there exists one and only one subtorus T of G such that ${}_nT = M(n)$ for every n equal to a power of q .

One has an analogous theorem concerning the lifting of morphisms:

Theorem 4.1 bis. *Let S, G, q be as above, T an S -torus, and for every integer n equal to a power of q , let $u(n)$ be a group S -morphism ${}_nT \rightarrow G$. One supposes that:*

a) *The family of morphisms $u(n)$ is coherent, i.e. if m divides n , one has:*

$$u(m) = u(n)|_{{}_mT}.$$

b) *There exist a point s of S and a group morphism:*

$$u_s : T_s \rightarrow G_s$$

such that $u_s|_{{}_nT_s} = u(n)_s$ for every n equal to a power of q .

c) *For every point t of S , there exists a closed affine subscheme $F_{\bar{t}}$ of $G_{\bar{t}}$ containing $u(n)_{\bar{t}}({}_nT_{\bar{t}})$ for every n .*

Then there exists a unique group morphism $u : T \rightarrow G$ such that for every n equal to a power of q , the restriction of u to ${}_nT$ is equal to $u(n)$.

Remark 4.2. Using the lower semicontinuity of the abelian rank of a group prescheme flat of finite type over the spectrum of a discrete valuation ring (cf. Exp. X 8.7), one may, in the statements of 4.1 and 4.1 bis, weaken condition c) by merely asking that the required closed affine subscheme $F_{\bar{t}}$ exist for every maximal point t of S .

Let us show how 4.1 bis follows from 4.1. Let $G' = G \times_S T$. For every integer n equal to a power of q , consider the group morphism:

$$v(n) : {}_nT \rightarrow G'$$

whose projections onto G and T are respectively $u(n)$ and the canonical immersion ${}_nT \rightarrow T$. The morphism $v(n)$ is therefore an immersion; let $M(n)$ be the image subgroup. It is clear that the family of subgroups $M(n)$ is coherent in the sense of 4.1, that the group $M(n)_s$ is equal to ${}_nT'_s$, where T'_s is the subtorus of G'_s which is the graph of u_s , and that for every point t of S , the closed affine subscheme $F_{\bar{t}} \times_{\bar{t}} T_{\bar{t}}$ of $G'_{\bar{t}}$ contains the subgroups $M(n)_{\bar{t}}$ for every n . By 4.1, there therefore exists a subtorus T' of G' such that ${}_nT' = M(n)$ for every n equal to a power of q . Let f be the restriction to T' of the projection of G' onto T , and $f(n)$ the restriction of f to $M(n)$. One has:

$$f(n) \circ v(n) = \text{id}_{{}_nT}.$$

The fiber at s of T' is the torus already denoted by T'_s , equal to the graph of u_s (this follows from Exp. IX 4.8 b)), so f_s is an isomorphism. But $\text{Ker } f$ and $\text{Coker } f$ are groups of

multiplicative type (Exp. IX 2.7), of constant type since S is connected, hence reduced to the unit group, and f is an isomorphism. Let v be the inverse isomorphism of f . One has:

$$v|_{nT} = v(n).$$

Consequently, the composite of v and the projection of G' onto G is a morphism $u : T \rightarrow G$ answering the question. What precedes proves the existence of the morphism u ; as for uniqueness, it follows in any case from Exp. IX 4.8 a).

2. Application

We propose to generalize Theorem 7.1 of Exp. IX.

Let A be a complete noetherian local ring, I its maximal ideal, $S = \operatorname{Spec} A$, $S_m = \operatorname{Spec}(A/I^m)$. For every prescheme X , put $X_m = X \times_S S_m$.

Let then G be a group S -prescheme of finite type, T an S -torus, q an integer invertible on S , and $u_m : T_m \rightarrow G_m$ ($m \in \mathbf{N}$) a coherent family of group morphisms. With the integer n ranging over the powers of q , denote by $u_m(n)$ the restriction of u_m to ${}_nT_m$, and by $u(n)$ the unique group morphism:

$$u(n) : {}_nT \rightarrow G$$

extending the morphisms $u_m(n)$ for every m (1.6 a)). We shall say that the family (u_m) , $m \in \mathbf{N}$, is admissible if for every point t of S , there exists a closed affine subprescheme $F_{\bar{t}}$ of $G_{\bar{t}}$ containing $u(n)_{\bar{t}}({}_nT_{\bar{t}})$ for every n equal to a power of q (this property is independent of q , as we shall see).

Proposition 4.3. *With the notation above, the canonical map:*

$$\operatorname{Hom}_{S\text{-gr}}(T, G) \rightarrow \varprojlim_m \operatorname{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

induces an isomorphism from the first member onto the subset of the second member formed by the “admissible” coherent families.

Indeed, it suffices to apply 4.1 bis, taking as s the closed point of S .

Corollary 4.4. *With the notation above, suppose in addition that G has affine fibers; then the canonical map:*

$$\operatorname{Hom}_{S\text{-gr}}(T, G) \rightarrow \varprojlim_m \operatorname{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

is an isomorphism.

Indeed, if G has affine fibers, every coherent family is “admissible”.

Remark 4.5. When G is separated, in 4.3 one may replace the ring A by a noetherian I -adic ring; indeed, one may use EGA III 5.4.1 instead of 1.6 a).

3. Proof of 4.1

Lemma 4.6. *Let k be a field, G an algebraic k -group, q an integer prime to the residual characteristic of k , r an integer > 0 , $M(n)$ (with n ranging over the powers of q) a coherent family of subgroups of G , of multiplicative type and of type $(\mathbf{Z}/n\mathbf{Z})^r$. Then there exists a smallest closed subscheme H of G containing the $M(n)$ for every n . Moreover, H is an algebraic subgroup of G , smooth, connected and commutative, “whose formation is compatible with extension of the base field”.*

Let M be the subset of $\operatorname{ens}(G)$ which is the union of the underlying sets of the subgroups $M(n)$ for every n , and H the reduced closed subscheme of G whose underlying space is the closure of M . Since $M(n)$ is étale over k , hence reduced, $M(n)$ is contained in H , and consequently H is the smallest closed subscheme of G containing $M(n)$ for every n . Let now $\bar{k}$ be an algebraically closed extension of k . By construction, the subschemes $M(n)$ are schematically dense in H (Exp. IX 4.1), so the $M(n)_{\bar{k}}$ are schematically dense in $H_{\bar{k}}$ (Exp. IX 4.5); moreover, $M(n)_{\bar{k}}$ is reduced, and it follows that $H_{\bar{k}}$ is necessarily equal to the closed

reduced subscheme of $G_{\bar{k}}$ whose underlying space is the closure of $M_{\bar{k}}$. This proves that H is geometrically reduced (EGA IV 4.6.1). Since the family $M(n)$ is coherent, M is stable under the group law; moreover, $M \times_k M$ is dense in $\text{ens}(H \times_k H)$ and $H \times_k H$ is reduced (EGA IV 4.6.5); one immediately deduces that H is an algebraic subgroup of G . Moreover, H is smooth over S , since it is geometrically reduced (Exp. VIA 1.3.1), and H is commutative since the $M(n)$ are commutative. It remains to see that H is connected. Let H^0 be the neutral component of H , m the number of geometric points of H/H^0 , q^s the exponent of q in the prime factorization of m . For every integer n equal to a power of q , $q^s M(n)$ is then contained in H^0 . But the family $M(n)$ is coherent and $M(n)$ is of type $(\mathbf{Z}/n\mathbf{Z})^r$, so $q^s M(nq^s) = M(n)$. Thus H^0 contains $M(n)$ for every n , and consequently is equal to H .

This being so, we shall make condition c) of 4.1 more precise. Indeed, by what precedes, we may consider the smallest closed subscheme H_t of G_t containing $M(n)_t$ for every n . Since the formation of H_t commutes with extension of the base field (4.6), $H_{\bar{t}} = H_t \times_t \bar{t}$ is contained in the affine closed subscheme $F_{\bar{t}}$, hence is affine. In brief, H_t is affine. On the other hand, we know (4.6) that H_t is a smooth, connected, commutative algebraic group. It then follows from the structure of smooth commutative connected affine algebraic groups over an algebraically closed field (BIBLE 4 Th. 4) that $H_{\bar{t}}$ is the direct product of a torus $T_{\bar{t}}$ and a unipotent group $U_{\bar{t}}$. But then $M(n)_{\bar{t}}$ is necessarily contained in $T_{\bar{t}}$, so $H_{\bar{t}} = T_{\bar{t}}$, and consequently H_t is a torus.

We may now begin the proof of 4.1.

- a) Uniqueness of the solution: it suffices to apply Exp. IX 4.8 b).
- b) Case where S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field k . Denote by K the fraction field of A , s the closed point of S , t the generic point, J the maximal ideal of A , $S_m = \text{Spec}(A/J^m)$, $X_m = X \times_S S_m$.

Let us distinguish two cases:

First case: The point s of 4.1 b) is the generic point t of S . Let then T' be the schematic closure in G of the torus T_t . The closed fiber T'_s is therefore an algebraic subgroup of G_s , of dimension r , containing $M(n)_s$ for every n , so T'_s contains H_s . But H_s is a torus containing $M(n)_s$, so H_s has rank at least r . Consequently H_s has the same underlying space as the neutral component of T'_s . The “neutral component” $(T')^0$ of T' is then a group subscheme of G , flat, separated (VIB 5.2), whose generic fiber T_t is a torus and whose reduced closed fiber H_s is a torus. But then $(T')^0$ is a torus (Exp. X 8.8), which we denote by T . The groups ${}_n T$ and M_n are smooth over S , hence reduced, and since they have the same underlying space, they agree. Thus the torus T is the solution of the problem.

Second case: The point s of 4.1 b) is the closed point s of S . After replacing G by the schematic closure in G of the smallest algebraic subgroup H_t containing the family $M(n)_t$ (4.6), we may suppose G_t affine.

For every integer $m \geq 0$, it follows from 2.2 that there exists a unique subtorus T_m of G_m lifting T_s and such that for every integer n equal to a power of q , one has ${}_n T_m = M(n)_m$. Moreover, let $T' = \mathbf{G}_{m,S}^r$. Since k is algebraically closed, T_s is trivial, and there exists a k -isomorphism $u_s : T'_s \xrightarrow{\sim} T_s$. The morphism u_s lifts uniquely to an S_m -isomorphism $u_m : T'_m \rightarrow T_m$ (Exp. IX 3.3). The family of morphisms u_m , $m \in \mathbf{N}$, defines in the limit a morphism $\hat{u}$ of the formal completions $\widehat{T'}$ and $\widehat{G}$ of T' and G along their closed fiber:

$$\begin{array}{ccc} \widehat{T'} & \xrightarrow{\hat{u}} & \widehat{G} \\ i \downarrow & & \downarrow j \\ T' & & G, \end{array}$$

where i and j denote the canonical morphisms.

We shall show that the morphism $\hat{u}$ is algebraizable. For this, we shall reduce to the case where the group G is affine.

Lemma 4.7. *Let S be the spectrum of a discrete valuation ring A , X and Y two S -preschemes, quasi-compact, quasi-separated and flat over S . Then the canonical map:*

$$\Gamma(X, \mathcal{O}_X) \otimes_A \Gamma(Y, \mathcal{O}_Y) \longrightarrow \Gamma(X \times_S Y, \mathcal{O}_{X \times_S Y})$$

is an isomorphism.

PROOF. Let $f : X \rightarrow S$ (resp. $g : Y \rightarrow S$) be the structural morphism. Since X (resp. Y) is quasi-compact and quasi-separated, it follows from EGA I 9.2.2 and EGA IV 1.7.4 that $f_*(\mathcal{O}_X)$ (resp. $g_*(\mathcal{O}_Y)$) is a quasi-coherent $\mathcal{O}_S$ -algebra, hence corresponds to an affine S -scheme $\tilde{X}$ (resp. $\tilde{Y}$). By hypothesis Y is flat over S ; it then follows from EGA III 1.4.15, taking EGA IV 1.7.21 into account, that the canonical map (deduced from the canonical morphism $X \rightarrow \tilde{X}$):

$$\Gamma(\tilde{X} \times_S Y, \mathcal{O}_{\tilde{X} \times_S Y}) \longrightarrow \Gamma(X \times_S Y, \mathcal{O}_{X \times_S Y})$$

is an isomorphism. But since X is flat over S , $\tilde{X}$ is flat over S (for flat over A is equivalent to torsion-free). Applying EGA III 1.4.15 once more, interchanging the roles of X and Y , one obtains an isomorphism:

$$\begin{aligned} \Gamma(\tilde{X}, \mathcal{O}_{\tilde{X}}) \otimes_A \Gamma(\tilde{Y}, \mathcal{O}_{\tilde{Y}}) &\simeq \Gamma(\tilde{X} \times_S \tilde{Y}, \mathcal{O}_{\tilde{X} \times_S \tilde{Y}}) \\ &\xrightarrow{\sim} \Gamma(\tilde{X} \times_S Y, \mathcal{O}_{\tilde{X} \times_S Y}), \end{aligned}$$

whence the lemma. $\square$

In the present case, the S -group G is flat over S and of finite type, hence quasi-compact and quasi-separated. One may therefore apply the lemma to $G \times_S G$:

$$\Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G) \xrightarrow{\sim} \Gamma(G \times_S G, \mathcal{O}_{G \times_S G}).$$

To the morphism $m_g : G \times_S G \rightarrow G$ defining multiplication in G , there therefore corresponds a morphism:

$$\begin{aligned} \Gamma(G, \mathcal{O}_G) &\longrightarrow \Gamma(G \times_S G, \mathcal{O}_{G \times_S G}) \\ &\xrightarrow{\sim} \Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G), \end{aligned}$$

hence an S -morphism:

$$m_{\tilde{G}} : \tilde{G} \times_S \tilde{G} \longrightarrow \tilde{G},$$

where $\tilde{G}$ denotes the affine S -scheme with A -algebra $\Gamma(G, \mathcal{O}_G)$. From there, it is formal to verify that $m_{\tilde{G}}$ gives $\tilde{G}$ a structure of group S -scheme such that the canonical morphism $v : G \rightarrow \tilde{G}$ is a morphism of S -groups.

Remark 4.8. $\tilde{G}$ plays the role of a largest “affine quotient” of G ; moreover, one can show that $\tilde{G}$ is indeed a quotient of G for fpqc, hence is of finite type over S (XVII App. III, 2).

In the case under consideration, the generic fiber G_t of G is affine, so it follows from EGA I 9.3.3 that v_t is an isomorphism. Since $\tilde{G}$ is affine, the coherent family of morphisms:

$$w_m = v_m u_m : T'_m \longrightarrow \tilde{G}_m \quad (m \in \mathbf{N})$$

comes from a unique morphism of S -groups (Exp. IX 7.1):

$$w : T' \longrightarrow \tilde{G}.$$

Let then T_t be the subtorus of G_t equal to $v_t^{-1} w_t(T'_t)$. The torus T_t therefore has rank at most r (as the image of a torus of rank r). Let us show that T_t contains $M(n)_t$ for every n . Indeed, let $u(n)_m$ be the S_m -isomorphism $({}_n T')_m \xrightarrow{\sim} M(n)_m$ obtained by restricting u_m to $({}_n T')_m$. The coherent family of morphisms $u(n)_m$ comes from a unique S -isomorphism $u(n) : {}_n T' \xrightarrow{\sim} M(n)$ (for $M(n)$ is finite over S). For every integer $m \geq 0$, one then has the equalities:

$$w_m|_{({}_n T')_m} = (v_m u_m)|_{({}_n T')_m} = (v \circ u(n))_m.$$

Consequently $w|_{nT'} = v \circ u(n)$ (1.6 a)). In particular, one has:

$$w_t|_{(nT')_t} = v_t \circ u(n)_t,$$

hence $v_t^{-1}w_t(nT')_t = u(n)_t(nT')_t = M(n)_t$.

This indeed proves that T_t contains $M(n)_t$, and implies that T_t has rank r . One concludes as in the first case already studied, by considering the schematic closure in G of T_t , say T'' . Since T_t contains $M(n)_t$, T''_s contains $M(n)_s$, hence contains T_s (density theorem). On the other hand, since T'' is flat over S and T_t has dimension r , T''_s has dimension r (Exp. VIB 4). In brief, T_s has the same underlying space as the neutral component of T''_s , and one concludes as in the first case that the neutral component of T'' is a subtorus of G answering the question.

c) End of the proof of 4.1.

To prove the existence of the subtorus T , one may suppose S reduced (2.2). Taking 3.1 (ii) $\implies$ (i) into account, it then suffices to prove that the set U of points x of S such that there exists a subtorus T_x of G_x with $nT_x = M(n)_x$ for every n equal to a power of q is equal to $\text{ens}(S)$. The torus T_x in question is necessarily unique, and by fpqc descent it suffices to prove its existence after extension of the residue field of x (Exp. IX 4.8 b)).

This being so, since S is locally noetherian and connected, and U is nonempty (it contains the point s of 4.1 b)), one is reduced, by an immediate argument, to proving that if x and x' are two points of S , with x a specialization of x' , and if one of the two points belongs to U , then both points are in U . By the usual technique (EGA II 7.1.9), one reduces to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, a case treated in b).

This completes the proof of Theorem 4.1.

5. Representability of the functor: smooth subgroups identical to their connected normalizer

Proposition 5.1. *Let S be a prescheme, G a group S -prescheme of finite presentation, H a group subprescheme of G , smooth with connected fibers. Then:*

a) *The normalizer N of H in G is representable by a closed subprescheme of G , of finite presentation over S .*

b) *The following conditions are equivalent:*

- i) *The canonical immersion $H \longrightarrow N$ is an open immersion.*
- ii) *The group N is smooth along the unit section, and its neutral component, which is then representable (Exp. VIB 3.10), is equal to H .*
- iii) *For every point s of S , one has $H_s = (N_s)^0$.*

PROOF. The group prescheme H is locally of finite presentation over S (H is smooth over S) and has connected fibers, hence is of finite presentation over S (Exp. VIB 5.3.3). Assertion a) is then a consequence of Exp. XI 6.11. The equivalence of the conditions appearing in b) is recorded here for reference, and was proved in VIB 6.5.1. $\square$

This being so, we may state the main theorem of this section:

Theorem 5.2. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , $\mathcal{L}_G$ (or simply $\mathcal{L}$ if there is no ambiguity) the S -functor such that for every S -prescheme S' one has:*

$$\begin{aligned} \mathcal{L}_G(S') = \text{set of group subpreschemes of } G_{S'}, \text{ smooth over } S', \\ \text{with connected fibers, identical to their connected normalizer.} \end{aligned}$$

Then the functor $\mathcal{L}$ is representable by an S -prescheme, the union of an increasing family $(U_i)_{i \in \mathbb{N}}$ of open subpreschemes, quasi-projective and of finite presentation over S , hence a fortiori separated over S .

First reductions. For every integer $r \geq 0$, let $\mathcal{L}_r$ be the subfunctor of $\mathcal{L}$ such that for every S -prescheme S' one has:

$$\begin{aligned} \mathcal{L}_r(S') = & \text{set of group subpreschemes of } G_{S'}, \text{ smooth,} \\ & \text{with connected fibers, identical to their connected normalizer} \\ & \text{and of relative dimension } r. \end{aligned}$$

Since the dimension of the fibers of a smooth group is a locally constant function, the canonical monomorphism $\mathcal{L}_r \rightarrow \mathcal{L}$ is representable by an immersion both open and closed. It therefore suffices to show that for every r , $\mathcal{L}_r$ is representable by an S -prescheme having the properties above, for $\mathcal{L}$ will then be representable by the sum S -prescheme $\coprod_{r \in \mathbb{N}} \mathcal{L}_r$.

For every integer $n \geq 0$, every S -prescheme S' and every group subprescheme H of $G_{S'}$, we shall denote by $H^{(n)}$ the n th normal invariant of the unit section $S' \rightarrow H$ of H (EGA IV 16.1.2), so that $H^{(n)}$ is a sheaf of $\mathcal{O}_{S'}$ -modules, of finite type, corresponding to the n th infinitesimal neighborhood of the unit section of H . If H is smooth over S' of relative dimension r , $H^{(n)}$ is a locally free $\mathcal{O}_S$ -module whose rank $\varphi(n, r)$ depends only on n and r . Moreover, since H is a subprescheme of $G_{S'}$, one has a canonical epimorphism, compatible with extension of the base:

$$G^{(n)} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \simeq G_{S'}^{(n)} \rightarrow H^{(n)}.$$

Introduce then the projective S -scheme:

$$P_{\varphi(n, r)} = \text{Grass}_{\varphi(n, r)}((G)^{(n)})$$

(EGA I 2nd ed. 9.7; cf. also Séminaire Cartan, 1960/61, Exp. No. 14 by A. Grothendieck). It then follows from the preceding remarks that the map:

$$H \mapsto H^{(n)}$$

defines a canonical morphism:

$$u_{n, r} : \mathcal{L}_r \rightarrow P_{\varphi(n, r)}.$$

The group G acts naturally on $G^{(n)}$, hence on $P_{\varphi(n, r)}$, by means of the representation:

$$\text{int} : G \rightarrow \underline{\text{Aut}}_{S\text{-gr}}(G), \quad g \mapsto \text{int}(g).$$

Moreover, if S' is a quasi-compact S -prescheme and H an element of $\mathcal{L}_r(S')$, one knows (Exp. XI 6.11) that for n sufficiently large, one has:

$$N = \underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}}(H^{(n)}).$$

For each integer $n \geq 0$, introduce the subfunctor $\mathcal{L}_r^n$ of $\mathcal{L}_r$ such that for every S -prescheme S' one has:

$$\begin{aligned} \mathcal{L}_r^n(S') = & \text{set of subgroups } H \text{ of } G_{S'} \text{ belonging to } \mathcal{L}_r(S') \\ & \text{and such that } \underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}}(H^{(n)}). \end{aligned}$$

Representability of $\mathcal{L}_r^n$. Since the integer r is fixed until the end of the proof of 5.2, we shall omit it from the notation. Thus we shall write $\mathcal{L}, \mathcal{L}^n, P_n, u_n$ instead of $\mathcal{L}_r, \mathcal{L}_r^n, P_{\varphi(n, r)}, u_{n, r}$.

Let v_n be the restriction of u_n to the subfunctor $\mathcal{L}^n$ of $\mathcal{L}$. It follows from the definition of $\mathcal{L}^n$ and 5.1 b) ii) that v_n is a monomorphism. In fact one has the following lemma:

Lemma 5.3. *The morphism v_n is an immersion of finite presentation. A fortiori, $\mathcal{L}^n$ is representable by an S -prescheme, quasi-projective and of finite presentation over S .*

After replacing S by P_n , one is reduced by the usual technique to proving the following assertion: let $Q \in P_n(S)$, and consider the subfunctor $\mathcal{F}$ of the functor h_S represented by

the final object S of $(\text{Sch})/S$, such that for every S -prescheme S' one has:

$$\begin{aligned} \mathcal{F}(S') &= h_S(S') \text{ (a set with one element) if there exists } H \in \mathcal{L}^n(S') \\ &\quad \text{such that } H^{(n)} = Q_{S'}, \\ \mathcal{F}(S') &= \emptyset \text{ otherwise.} \end{aligned}$$

Then the canonical monomorphism $\mathcal{F} \rightarrow S$ is an immersion of finite presentation.

Let us begin by transforming the definition of the functor $\mathcal{F}$. For this, note that the normalizer of Q in G is representable by a group subprescheme N of finite presentation over S (namely the inverse image of the point Q of $P_n(S)$ under the morphism:

$$G \rightarrow P_n, \quad g \mapsto g \cdot Q.$$

) I say that the functor $\mathcal{F}$ agrees with the following subfunctor of h_S :

$$\begin{cases} \mathcal{F}(S') = h_S(S') \text{ if one has:} \\ \quad \text{(a) } N_{S'} \text{ is smooth along the unit section and of relative dimension } r. \\ \quad \text{(b) } (N_{S'})^{(n)} \text{ (then canonically an element of } P_n(S')) \text{ equals } Q_{S'}. \\ \mathcal{F}(S') = \emptyset \text{ otherwise.} \end{cases}$$

Indeed, let us temporarily denote by $\mathcal{F}_1$ the functor $\mathcal{F}$ in its first form, and by $\mathcal{F}_2$ the functor $\mathcal{F}$ in its second form. Then:

$$\text{i) } \mathcal{F}_1(S') = h_S(S') \implies \mathcal{F}_2(S') = h_S(S').$$

Indeed, let $H \in \mathcal{F}^n(S')$ be the subgroup of $G_{S'}$ such that $H^{(n)} = Q_{S'}$. One therefore has:

$$\underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}}(H^{(n)}) = \underline{\text{Norm}}_{G_{S'}}(Q_{S'}) = N_{S'}.$$

Thus by 5.1 b) ii), $N_{S'}$ is smooth along the unit section and its neutral component is H . Consequently $N_{S'}$ has relative dimension r , and since H is open in $N_{S'}$ (by 5.1 b) i)), one has $(N_{S'})^{(n)} = H^{(n)} = Q_{S'}$. In brief, $\mathcal{F}_2(S') = h_S(S')$.

$$\text{ii) } \mathcal{F}_2(S') = h_S(S') \implies \mathcal{F}_1(S') = h_S(S').$$

By hypothesis, $N_{S'}$ is smooth along the unit section, of relative dimension r ; its neutral component is therefore representable (Exp. VIB 3.10) by a group subprescheme H , smooth over S' , with connected fibers of dimension r . Since H is invariant in $N_{S'}$ and open in $N_{S'}$, one has the following inclusions:

$$\begin{aligned} N_{S'} &\subset \underline{\text{Norm}}_{G_{S'}}(H) \subset \underline{\text{Norm}}_{G_{S'}}(H^{(n)}) \\ &= \underline{\text{Norm}}_{G_{S'}}(N_{S'}^{(n)}) = \underline{\text{Norm}}_{G_{S'}}(Q_{S'}) = N_{S'}. \end{aligned}$$

The inclusions above are therefore equalities. The first inclusion then shows that H is equal to its connected normalizer, and the second shows that H is an element of $\mathcal{L}^n(S')$. This says that $\mathcal{F}_1(S') = h_S(S')$.

Implications i) and ii) give $\mathcal{F}_1 = \mathcal{F}_2$. We retain the second definition of the functor $\mathcal{F}$, and shall first “represent condition a)” by an immersion of finite presentation. For this, it suffices to apply:

Lemma 5.4. *Let S be a prescheme, X an S -prescheme locally of finite presentation over S , $\sigma : S \rightarrow X$ a section of X , r an integer ≥ 0 , and $L : ((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ the subfunctor of S defined as follows:*

$$\begin{cases} L(S') = h_S(S') \text{ if } X_{S'} \text{ is smooth along the section } \sigma_{S'} \\ \quad \text{and of relative dimension } r \text{ at the points of } \sigma_{S'}(S'). \\ L(S') = \emptyset \text{ otherwise.} \end{cases}$$

Then:

a) *The monomorphism $L \rightarrow S$ is an immersion of finite presentation.*

b) Let J be the conormal sheaf relative to the immersion $S \rightarrow X$ (EGA IV 16.1.2); suppose that for every point s of S , $J \otimes_{\mathcal{O}_{S,s}} \kappa(s)$ has rank at most r ; then the immersion $L \rightarrow S$ is a closed immersion.

PROOF. The functor L is local in nature on S , which reduces one to the case where S is affine. After replacing X by a neighborhood of $\sigma(S)$, one may suppose X of finite presentation over S , then (EGA IV 8.9) S noetherian (note, in case b), that formation of the conormal sheaf commutes with extension of the base (EGA IV 16.6.4), and that the rank of the fibers of J is a constructible function on S). This being so, for every S -prescheme S' , denote by $J' = J_{S'}$ the conormal sheaf relative to the section $\sigma_{S'}$; let $S^\bullet(J')$, canonically isomorphic to $S^\bullet(J)_{S'}$, be the symmetric algebra of J' over $\mathcal{O}_{S'}$, $\text{Gr}_\bullet(\sigma_{S'})$ the sheaf of graded $\mathcal{O}_{S'}$ -algebras associated with $\sigma_{S'}$ (EGA IV 16), and for every integer $n \geq 0$ let $\sigma_{n,S'} : S^n(J') \rightarrow \text{Gr}_n(\sigma_{S'})$ be the canonical epimorphism.

It follows from EGA IV 17.12.3 and EGA 0_{IV} 19.5.4 that, for $X_{S'}$ to be smooth along the section $\sigma_{S'}$ and of relative dimension r at the points of $\sigma_{S'}(S')$, it is necessary and sufficient that:

- i) J' be a locally free $\mathcal{O}_{S'}$ -sheaf of rank r .
- ii) For every integer $n \geq 1$, $\sigma_{n,S'}$ be an isomorphism.

Now it follows from TDTE IV, Lemma 3.6, that “the functor making J locally free of rank r ” is representable by a subprescheme S_1 , closed in S in case b). Replacing S by S_1 , one is reduced to the case where J is locally free of rank r . Let us then proceed by induction on the integer n . Suppose one has represented by a closed subprescheme S_{n-1} of S the “functor making the morphisms $\sigma_{q,S'}$ injective for every integer $q \leq n-1$ ”, and let us show that the “subfunctor⁽⁵⁾ making $\sigma_{n,S'}$ injective” is representable by a closed subprescheme S_n of S_{n-1} . Replacing S by S_{n-1} , we may suppose that $\sigma_{q,S}$ is bijective for $q \leq n-1$. But then, the $(n-1)$ st normal invariant $X^{(n-1)}$ relative to the section σ_s admits a composition series $X^{(0)}, \dots, X^{(n-1)}$ whose successive quotients $\text{Gr}_0(\sigma_s), \dots, \text{Gr}_{n-1}(\sigma_s)$ are locally free over $\mathcal{O}_{S'}$, hence flat, so $X^{(n-1)}$ is flat over $\mathcal{O}_S$. Since formation of $X^{(i)}$ commutes with extension of the base (EGA IV 16), one has for every S -prescheme S' :

$$\begin{aligned} \text{Gr}_n(\sigma_{S'}) &= \text{Ker}[X_{S'}^{(n)} \rightarrow X_{S'}^{(n-1)}] = (\text{Gr}_n(\sigma))_{S'}, \\ \text{and} \quad \sigma_{n,S'} &= (\sigma_{n,S})_{S'}. \end{aligned}$$

Thus the functor of interest is “the one making the morphism $\sigma_{n,S}$ injective”. This functor is local in nature on S , which allows one to suppose S affine and $S^n(J)$ free over $\mathcal{O}_S$. But it is then clear that the functor in question is representable by the closed subscheme of S defined by the ideal generated by the coordinates of $\text{Ker } \sigma_{n,S}$ relative to a basis of $S^n(J)$.

Since S is noetherian, the decreasing sequence $\{S_n\}$ of closed subpreschemes of S_1 is stationary, and the stationary value represents the functor L , which completes the proof of 5.4. $\square$

Let us return to the question of the representability of $\mathcal{L}^n$. Replacing S by a suitable subprescheme S_1 , one may therefore suppose N smooth along the unit section and of relative dimension r . The functor $\mathcal{L}^n$ is then the “coincidence functor” of two sections of P_n over S , h and g , corresponding to the sheaves Q and $N^{(n)}$ (condition b) occurring in the definition of the functor $\mathcal{F}_2$ above). It is therefore representable by the closed subprescheme of S , the inverse image of the diagonal of $P_n \times_S P_n$ under the morphism $h \times_S g$ of finite presentation. This completes the proof of 5.3.

Study of the transition morphisms $\mathcal{L}^n \rightarrow \mathcal{L}^m$. If a subgroup H of $G_{S'}$ belongs to $\mathcal{L}^n(S')$, it belongs a fortiori to $\mathcal{L}^m(S')$ for every $m \geq n$, whence natural monomorphisms:

$$u_n^m : \mathcal{L}^n \rightarrow \mathcal{L}^m \quad \text{for } m \geq n.$$

⁽⁵⁾**N.D.E.** “Functor” has been replaced by “subfunctor”. Should one write “making $\sigma_{q,S'}$ injective for $q \leq n$ ”?

Lemma 5.5. *The morphism u_n^m is an open immersion.*

PROOF. After changing S , one is reduced to the following problem: let $H \in \mathcal{L}^m(S)$,

$$N = \underline{\text{Norm}}_G(H) = \underline{\text{Norm}}_G(H^{(m)}), \quad N' = \underline{\text{Norm}}_G(H^{(n)}),$$

and let $\mathcal{D} : ((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ be the coincidence functor of N and N' , defined by $\mathcal{D}(S') = h_S(S')$ if $N_{S'} = N'_{S'}$, and $\mathcal{D}(S') = \emptyset$ otherwise. One must show that $\mathcal{D} \rightarrow S$ is an open immersion. Now I say that $\mathcal{D}$ is also the subfunctor of S which “makes the immersion $H \rightarrow N'$ open”. Indeed, if $N_{S'} = N'_{S'}$, then $H_{S'} \rightarrow N'_{S'}$ is indeed an open immersion, since this is so for $H_{S'} \rightarrow N_{S'}$ (Prop. 5.1). Conversely, if $H_{S'} \rightarrow N'_{S'}$ is an open immersion, since H has connected fibers, $H_{S'}$ is the neutral component of $N'_{S'}$ (Exp. VIB 3.10), and consequently is invariant in $N'_{S'}$, hence $N'_{S'} \subset N_{S'}$. Since in any case N' contains N , one indeed has $N_{S'} = N'_{S'}$. The group preschemes H and N' are of finite presentation over S , and H is flat over S ; the fact that $\mathcal{D} \rightarrow S$ is an open immersion then follows from Exp. VIB 2.6. $\square$

End of the proof of 5.2. Since the functors $\mathcal{L}^n$ are representable, and the transition morphisms u_n^m are compatible with each other and representable by open immersions, there exists an S -prescheme X , the union of an increasing sequence of open subsets $(X_i)_{i \in \mathbb{N}}$, such that X_i represents the functor $\mathcal{L}^i$, and such that if one identifies $\mathcal{L}^i$ with X_i , the inclusion $X_i \rightarrow X_j$ ($j \geq i$) is identified with u_i^j . To conclude that X represents the functor $\mathcal{L}$, it then suffices to note that, in the category of sheaves on $(\text{Sch})/S$ endowed with the Zariski topology (Exp. IV 6.1), one has:

$$X = \varinjlim X_i \quad \text{and} \quad \mathcal{L} = \varinjlim \mathcal{L}^i.$$

Remark 5.6. With the preceding notation, suppose in addition that S has all its residual characteristics zero. Then for every integer $r \geq 0$, the functor $\mathcal{L}_r$ is equal to $\mathcal{L}_r^1$, hence is representable by an S -prescheme of finite presentation and quasi-projective over S . Indeed, it suffices to show that if $H \in \mathcal{L}_r(S')$, the canonical immersion:

$$H \rightarrow N = \underline{\text{Norm}}_{G_{S'}}(H^{(1)})$$

is an open immersion (for this implies $N = \underline{\text{Norm}}_{G_{S'}}(H)$, hence $H \in \mathcal{L}_r^1(S')$). Since H is flat over S , and H and N are of finite presentation over S , it suffices (Exp. VIB 2.6) to show that for every point s of S , $H_s \rightarrow N_s$ is an open immersion. Now it follows easily⁽⁶⁾ from Cartier’s theorem (Exp. VIB 1.6) that if G is an algebraic group over a field of characteristic 0, and H is a connected algebraic subgroup, one has:

$$\underline{\text{Norm}}_G(H) = \underline{\text{Norm}}_G(\text{Lie } H) = \underline{\text{Norm}}_G(H^{(1)}).$$

On the other hand, if S has nonzero residual characteristics, the subfunctors $\mathcal{L}_r^n$ of $\mathcal{L}_r$ may form a strictly increasing sequence (even when S is quasi-compact), and in this case $\mathcal{L}_r$ is not representable by an S -prescheme quasi-compact over S . Take, for example, the algebraic group G , defined over a field k of characteristic $p > 0$, equal to the semidirect product of the torus $T = \mathbf{G}_m \times \mathbf{G}_m$ by the unipotent group $U = \mathbf{G}_a \times \mathbf{G}_a$, with the action of T on U defined by $(t, t', u, u') \rightarrow (tu, t'u')$. For every integer $n > 0$, consider then the smooth connected subgroup U_n of U with equation $u' = u^{p^n}$, and the subtorus T_n of T with equation $t' = t^{p^n}$. It is immediate to verify that T_n acts on U_n , and that the subgroup G_n of G equal to $T_n \cdot U_n$ is smooth, connected and identical to its normalizer in G . Now all the groups G_n , for $n \geq m$, are distinct and have the same infinitesimal neighborhood of order p^m .

Remark 5.7. There exists on $\mathcal{L}$ a canonical invertible sheaf L whose restriction to every open subset U of $\mathcal{L}$, quasi-compact over S , is S -ample. Indeed, consider the group subprescheme H of $G_{\mathcal{L}}$, smooth over $\mathcal{L}$, with connected fibers, equal to its connected normalizer, and

⁽⁶⁾**N.D.E.** One can apply, for example, Proposition II.6.1 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North Holland (1970).

universal for these properties. I say that one may take for L the sheaf $(\det(\mathrm{Lie} H))^{-1}$ (recall that if F is a sheaf of $\mathcal{O}_S$ -modules on a prescheme S , locally free of finite rank, $\det(F)$ denotes the invertible $\mathcal{O}_S$ -module whose restriction to the open and closed subprescheme S_r of S ($r \geq 0$) where F has rank r is equal to $\bigwedge^r(F)$). We retain the notation of the proof of 5.2. To prove the assertion made about L , we may restrict to the functor $\mathcal{L}_r^n$ and prove that $L|_{\mathcal{L}_r^n}$ is S -ample. Consider the canonical immersion $v_n : \mathcal{L}_r^n \rightarrow P_n$, and let Q be the locally free sheaf on P_n , universal for the Grassmannian P_n . By construction, one has $v_n^*(Q) = H^{(n)}$ (where H now denotes the group subprescheme of $G_{\mathcal{L}_r^n}$ universal for the functor $\mathcal{L}_r^n$). Now $\det(Q)$ is the canonical ample sheaf on P_n (EGA I 2nd ed., 9.7), so $\det H^{(n)}$ is ample relative to $\mathcal{L}_r^n$ (EGA II 4.6.13 i) bis). Again denote by J the conormal sheaf, equal to $H^{(1)}$, and by $S^q(J)$ the homogeneous part of degree q of the symmetric algebra of J over $\mathcal{O}_{\mathcal{L}_r^n}$. Since H is smooth over $\mathcal{L}_r^n$, it is immediate that one has a canonical isomorphism:

$$\det H^{(n)} \simeq \prod_{1 \leq q \leq n} \det S^q(J).$$

On the other hand, one proves that for every locally free sheaf J of rank r and every integer $q > 0$, there exists a canonical isomorphism:

$$\det S^q(J) \simeq (\det J)^{\otimes s},$$

where $s > 0$ is an integer depending only on r and q . Finally, one obtains $\det H^{(n)} \simeq (\det J)^{\otimes s}$ for a suitable integer $s > 0$, so (EGA II 4.5.6) $\det J = (\det \mathrm{Lie} H)^{-1}$ is indeed S -ample.

Remark 5.8. Let S be a prescheme, G and H two group S -preschemes of finite presentation over S , $i : H \rightarrow G$ a group S -homomorphism which is a monomorphism. If H is smooth over S , with connected fibers, one knows (Exp. XI 6.11) that $N = \underline{\mathrm{Norm}}_G(H)$ is representable by a closed group subprescheme of G , of finite presentation over S . Suppose in addition that N is smooth along the unit section and has the same relative dimension over S as H . The neutral component N^0 of N is then representable by an open group subprescheme of N , smooth over S (Exp. VIB 3.10). The monomorphism i obviously factors through N^0 . In fact one has $H = N^0$. Indeed, for every point s of S , one has $H_s = (N^0)_s$, these two algebraic groups being connected, smooth and of the same dimension. Since H is flat over S , one deduces that $H \rightarrow N^0$ is an isomorphism (EGA IV 17.9.5). Finally, H is a group subprescheme of G . One has therefore shown that the functor $\mathcal{L}$ introduced in this section is identical to the functor of subgroups H of G , smooth over S , with connected fibers, and equal to their connected normalizer.

6. Functor of Cartan subgroups and functor of parabolic subgroups

When G is a smooth connected algebraic group defined over an algebraically closed field k , one has defined the subtori of G , maximal subtori, Cartan subgroups (Exp. XII 1), Borel subgroups (Exp. XIV 4.1), and parabolic subgroups (Exp. XIV 4.8 bis). We extend these notions to the case of a group prescheme over an arbitrary base as follows:

Definition 6.1. Let S be a prescheme, G a group S -prescheme of finite presentation over S , H a group S -prescheme, $i : H \rightarrow G$ an S -monomorphism making H a subgroup of G . We shall say that H is a subtorus of G (resp. a maximal subtorus of G , a Cartan subgroup, a Borel subgroup, a parabolic subgroup) if:

- i) H is smooth over S .
- ii) For every geometric point $\bar{s}$ over S , $H_{\bar{s}}$ is a subtorus of $(G_{\bar{s}})_{\mathrm{red}}^0$ (resp. a maximal subtorus, a Cartan subgroup, a Borel subgroup, a parabolic subgroup).

Remarks 6.1 bis. a) If the S -group H is a subtorus of G (resp. ...), its fibers are connected, and consequently H is of finite presentation over S (Exp. VIB 5.3.3).

b) If H is a subtorus of G , then H is a torus in the sense of Exp. IX, as follows immediately from Exp. X 8.1. Moreover, the monomorphism $i : H \rightarrow G$ is an immersion (cf. 8.3 below).

c) If G is smooth over S , with connected fibers, and H is a Cartan subgroup of G (resp. a Borel subgroup, a parabolic subgroup), the monomorphism $i : H \rightarrow G$ is an immersion, so that our definitions agree with those introduced in Exp. XII and Exp. XIV. Indeed, H is then identical to its connected normalizer (by XII 6.6 c), XIV 4.8 and 4.8 bis), and it suffices to apply 5.8.

Definition 6.1 ter. Let S be a prescheme, G a group S -prescheme locally of finite type, s a point of S . The dimension of the Cartan subgroups of $(G_{\bar{s}})_{\text{red}}^0$ is called the nilpotent rank of G at the point s , and denoted by $\rho_n(s)$. One defines analogously the reductive rank $\rho_r(s)$, the unipotent rank $\rho_u(s)$, the abelian rank $\rho_{\text{ab}}(s)$ (cf. Exp. X 8.7).

If now G is a smooth connected algebraic group defined over an algebraically closed field k , recall that the radical of G , denoted by $\text{rad}(G)$, is the largest algebraic subgroup of G which is invariant, smooth, connected and solvable; $G/\text{rad}(G)$ is then semisimple (use Exp. XII 6.1 to reduce to the case of G affine). If G is in addition affine, one defines the unipotent radical $\text{rad}^u(G)$ of G as the largest invariant, smooth, connected and unipotent algebraic subgroup of G : $G/\text{rad}^u(G)$ is then reductive.

Proposition 6.2. *Let S be a prescheme, G and H two group S -preschemes of finite presentation over S , $i : H \rightarrow G$ a group S -monomorphism making H a subgroup of G , and let $P(s)$ be one of the following properties concerning the point s of S :*

- i) $(G_{\bar{s}})_{\text{red}}^0$ is an abelian variety (resp. is affine, is a torus, is unipotent).
- ii) $(H_{\bar{s}})_{\text{red}}^0$ is a maximal torus of $G_{\bar{s}}$.
- iii) $(H_{\bar{s}})_{\text{red}}^0$ is the centralizer in $(G_{\bar{s}})_{\text{red}}^0$ of a torus of $G_{\bar{s}}$ (resp. is a Cartan subgroup of $G_{\bar{s}}$).
- iv) $(H_{\bar{s}})_{\text{red}}^0$ is a Borel subgroup (resp. a parabolic subgroup) of $G_{\bar{s}}$.
- v) $(H_{\bar{s}})_{\text{red}}^0$ is the radical of $G_{\bar{s}}$ (resp. $(G_{\bar{s}})_{\text{red}}^0$ is semisimple).
- vi) $G_{\bar{s}}$ is affine, and $(H_{\bar{s}})_{\text{red}}^0$ is the unipotent radical of $G_{\bar{s}}$ (resp. $(G_{\bar{s}})_{\text{red}}^0$ is reductive).

Then the set E of points s of S such that $P(s)$ is true is locally constructible (EGA 0III 9.1.11).

Remark 6.2.1. This proposition complements Exp. VIB § 10. Moreover, one can make the structure of E more precise using semicontinuity theorems (cf. Exp. X 8.7); we shall see an example a little further on.

Proof of 6.2. Note that if S is the spectrum of a field, E is invariant under extension of this field. A standard reduction (EGA IV 9) then allows one to reduce to the case where S is noetherian integral with generic point η . One must show that E or $\text{ens}(S) \setminus E$ contains a neighborhood of η (EGA IV 9.2.1). One may suppose S affine with ring A and fraction field K . If L is a finite extension of K , it is immediate that there exists a sub- A -algebra B of L , finite over A , with fraction field L . The canonical morphism $S' \rightarrow S$, where $S' = \text{Spec } B$, is dominant, of finite presentation, so the image of a nonempty open subset of S' contains a nonempty open subset of S (EGA IV 1.8.4). From the point of view of interest to us, we may therefore replace S by S' , hence replace K by a finite extension L . Thus we may choose L so that $(G_L)_{\text{red}}$ and $(H_L)_{\text{red}}$ are smooth over L (EGA IV 4.6.6). After restricting S' , we may suppose that G_{red} and H_{red} are group preschemes smooth over S (Exp. VIB § 10 and EGA IV 17). In view of the properties to be proved, we may replace G and H by their reduced neutral components (Exp. VIB 10.9), hence suppose G and H smooth over S , with connected fibers. Finally, we may suppose that H is a closed group subprescheme of G (Exp. VIB 10.4). *Proof of i).* After making a finite extension of K , we may suppose that G_{η} possesses a ‘‘Chevalley decomposition’’, i.e. is an extension of an abelian variety D_{η} by a smooth connected linear algebraic group F_{η} (Séminaire Bourbaki 1956/57 No. 145). By Exp. VIB

10.16, there exists an open neighborhood U of η such that this generic extension comes from an extension:

$$1 \longrightarrow F \longrightarrow G|_U \longrightarrow D \longrightarrow 1.$$

One may in addition suppose F and D smooth over U , with connected fibers, F affine over U and D proper over U (EGA IV 8, 9 and 17). For every point s of U , (D_s, F_s) is then the “Chevalley decomposition” of G_s . Consequently G_s is an abelian variety (resp. is affine) if and only if F_s (resp. D_s) is the unit group, which is a constructible property (EGA IV 9.2.6.1).

To establish the last two assertions of i), in view of what precedes, we may suppose G affine over S . Let q be a prime number invertible on S , and ${}_qG$ the “kernel” of raising to the q th power in G . It follows easily from the structure of affine algebraic groups that G_s is a torus (resp. is unipotent) if and only if a) ${}_qG_s$ is quasi-finite, which is a constructible property (EGA IV 9.3.2), and b) ${}_qG_s$ has r^q geometric points, where r denotes the relative dimension of G over S (resp. ${}_qG_s$ has a single point). Now the function $s \mapsto$ (number of geometric points of ${}_qG_s$) is constructible (EGA IV 9.7.9). This completes the proof of (i).

Proof of iii). a) Case of a centralizer of a torus. Suppose that $H_\eta = \text{Centr}_{G_\eta}(T_\eta)$, where T_η is a torus of G_η , and let us show that H_s is the centralizer of a subtorus of G_s for s in a neighborhood of η . By i), after restricting S , one may suppose that T_η comes from a subtorus T of G . But then $Z = \text{Centr}_G(T)$ is representable (Exp. XI 6.11) by a group subscheme of G . Since H and Z agree generically, they agree over a neighborhood of η . This proves that the set E of points s of S such that H_s is the centralizer in G_s of a subtorus of G_s is ind-constructible (EGA IV 1), and this result will suffice to establish, in Lemma 6.6 below, that E is an open subset of S ; a fortiori, E will indeed be a locally constructible subset of S .
iii) b) Case of a Cartan subgroup. Suppose that H_η is a Cartan subgroup of G_η , and let us show that H_s is a Cartan subgroup of G_s at every point s of a neighborhood of η . The group H_η is the centralizer in G_η of a torus of G_η and is nilpotent (Exp. XII 6.6). By a) and Exp. VIB 8.4, H_s has the same properties at every point s of a neighborhood U of η . For every point s of U , the group H_s therefore has the same reductive rank as G_s , and its unique maximal torus is central (Exp. XII 6.7), so it is the centralizer of a maximal torus of G_s , i.e. a Cartan group of G_s .

Suppose now that H is not a Cartan subgroup of G , and let us show that H_s is not a Cartan subgroup of G_s for s in a neighborhood U of η . Taking into account the assertion provisionally granted in (a) above, we may restrict to the case where H is the centralizer in G of a subtorus T . But then H_η contains a Cartan subgroup C_η of H_η . We have just seen that, after restricting S , C_η extends to a Cartan subgroup C of G , which one may suppose contained in H . By hypothesis H_η strictly contains C_η , so H_s strictly contains C_s for s in a neighborhood U of η (EGA IV 9.5.2); a fortiori, H_s is not a Cartan subgroup of G_s for s in U .

Proof of ii). Suppose that H_η is a maximal torus of G_η , and let C_η be its centralizer in G_η . By i) and iii), H is a torus over a neighborhood U of η , and $C = \text{Centr}_{G|_U}(H|_U)$ is a Cartan subgroup of $G|_U$. To prove that H is a maximal torus of G over a neighborhood of η , one may then replace G by C , then by the linear component F of a Chevalley decomposition of C (cf. i)). Let q be an integer invertible on S , ${}_qF$ the kernel of raising to the q th power in F . Since F_η is affine, nilpotent, smooth and connected, F_η is the direct product of its maximal torus T_η by a unipotent group (BIBLE 6-04), so ${}_qF_\eta = {}_qT_\eta$. Since H is a maximal torus, ${}_qH_\eta = {}_qF_\eta$, and consequently ${}_qH = {}_qF$ over a neighborhood V of η . For every point s of V , ${}_qH_s = {}_qT_s$, so $H_s = T_s$ is a maximal torus.

Suppose now that H_η is not a maximal torus of G_η . By i), we may restrict to the case where H_η is a torus, then suppose that it is contained in a strictly larger torus T_η . The latter extends to a torus T strictly containing H over a neighborhood U of η . A fortiori, H_s is not a maximal torus for $s \in U$.

Proof of iv). After restricting S , one may suppose that the center Z of G is representable (Exp. VIB 10.11) and flat over S , as is the quotient G/Z (*loc. cit.*). The property “ H_s

contains Z_s is constructible (EGA IV 9.5.2), and every parabolic subgroup of G_s contains Z_s (Exp. XIV 4.9 a)); this allows one to replace G by G/Z , hence suppose G affine over S (Exp. XII 6.1 and i)). One may again suppose that G/H is representable, but then H_s is a parabolic subgroup of G_s if and only if $(G/H)_s$ is proper (BIBLE 6. Th. 4 b)), which is an ind-constructible property (EGA IV 9.3.5). Thus E is ind-constructible, and this will suffice to prove that E is open (Lemma 6.6), hence locally constructible.

Let us now examine the case of Borel subgroups. If H_η is a Borel subgroup of G_η , i.e. a solvable parabolic subgroup of G_η , what precedes and Exp. VIB 8.4 imply that these properties are still true at every point s of a neighborhood of η . If now H_η is not a Borel subgroup of G_η , to prove that the same is true at the points s of a neighborhood of η , we may restrict (in view of what precedes) to the case where H_η is a parabolic subgroup, then suppose that H_η contains a Borel subgroup B_η . We have just shown that the latter extends to a Borel subgroup B of G over a neighborhood U of η . Since H_η strictly contains B_η , H_s strictly contains B_s at every point of an open subset V , and H_s is not a Borel subgroup of G_s for $s \in V$.

Proof of v). Suppose that H_η is the radical of G_η . The group H_η is therefore invariant in G_η , solvable (smooth and connected); the same therefore holds for H_s for s belonging to a neighborhood U of η (Exp. VIB 8 and 10), so for $s \in U$, H_s is contained in the radical of G_s . Replacing G by G/H (Exp. VIB 10), one must prove that if G_η is semisimple, G_s is semisimple at every point of a neighborhood V of η . Thanks to i) and ii), one may suppose that G is affine over S and possesses a maximal torus T . Let W be the Weyl group of T (Exp. XII 2), which is quasi-finite and étale over S , hence finite and étale over an open subset V . It then follows from the elementary properties of roots (Exp. XIX 1.12) that G is semisimple over V .

Suppose now that H_η is not the radical of G_η . After replacing K by a finite extension L , one may suppose that G_η possesses a radical R_η . By what precedes, R_η extends to a group subscheme R of $G|_U$ such that for every $s \in U$, R_s is the radical of G_s . By hypothesis $R_\eta \neq H_\eta$. Thus $R_s \neq H_s$ for $s \in V$. It remains to prove that if G_η is not semisimple, the same holds for G_s at neighboring points, but this is a particular case of what precedes (take $H = \text{unit group}$).

Proof of vi). The proof is entirely analogous to that of v), taking i) into account, and is left to the reader.

Corollary 6.3. *Let S_0 be a quasi-compact prescheme, S_i ($i \in L$) a projective system of S_0 -preschemes affine over S_0 , $S = \varprojlim S_i$ (EGA IV 8.2), G_0 a group prescheme of finite presentation over S_0 , $G_i = G_0 \times_{S_0} S_i$, $G = G_0 \times_{S_0} S$, H a subgroup of G . Then if H is a subtorus of G (resp. a maximal subtorus, a Cartan subgroup, a Borel subgroup, a parabolic subgroup), there exist an index $i \in L$ and a subgroup H_i of G_i such that $H = H_i \times_{S_i} S$ and H_i is a subtorus of G_i (resp. ...).*

Indeed, H is smooth, with connected fibers, hence of finite presentation over S (Exp. VIB 5.3.3). By Exp. VIB § 10, there exist $i \in L$ and a subgroup H_i of G_i , smooth over S , such that $H = H_i \times_{S_i} S$. Corollary Exp. 6.3 then follows from Definition 6.1, 6.2 and EGA IV 9.3.3.

Corollary 6.3 bis. *Let S be a prescheme, G a group S -prescheme of finite presentation over S . Then the functions $\rho_n, \rho_r, \rho_u, \rho_{ab}$ (cf. 6.1 ter) are locally constructible functions on S .*

It suffices to show (EGA IV 9) that if S is an integral noetherian scheme with generic point η , the functions in question are constant on a neighborhood of η . After replacing S by a scheme S' , finite over S , dominating S , we may suppose that G_η possesses a Cartan subgroup C_η with a Chevalley decomposition $1 \rightarrow L_\eta \rightarrow C_\eta \rightarrow A_\eta \rightarrow 1$. The argument made in 6.2 i) proves that this decomposition extends to a Chevalley decomposition over a

neighborhood of η :

$$1 \longrightarrow L \longrightarrow C \longrightarrow A \longrightarrow 1.$$

Moreover, one may suppose that C is a Cartan subgroup of G (6.3), and that the maximal torus T_η of L_η extends to a maximal torus T of L (6.3). The corollary follows immediately from this and from the definitions.

6.4.0. — Let then S be a prescheme, G a group S -prescheme of finite presentation over S , $\mathcal{L}$ the functor of subgroups of G , smooth, with connected fibers, and equal to their connected normalizer (cf. § 5); $\mathcal{L}$ is representable (5.2 and 5.8). The rest of this section is devoted to the study of certain subfunctors of $\mathcal{L}$. More precisely, introduce the subfunctors of $\mathcal{L}$ denoted by $\mathcal{LC}$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$), defined as follows: for every S -prescheme S' , $\mathcal{LC}(S')$ (resp. $\mathcal{CT}(S')$, resp. $\mathcal{P}(S')$) is the set of subgroups H of $G_{S'}$, smooth over S' , with connected fibers, equal to their connected normalizer, and such that for every point s of S' , H_s contains a Cartan subgroup (6.1) of G_s (resp. is the centralizer in $(G_{\bar{s}})_{\text{red}}^0$ of a subtorus of $G_{\bar{s}}$, resp. is a parabolic subgroup (6.1) of G_s).

Theorem 6.4. *Let S be a prescheme, G a group S -prescheme of finite presentation over S . Then the S -functors $\mathcal{LC}, \mathcal{CT}, \mathcal{P}$ above (6.4.0) are representable by S -preschemes of finite presentation over S , quasi-projective over S .*

Remark 6.5. If G has smooth fibers, for example if G is smooth over S or the residual characteristics of S are zero (Exp. VIB 1.6.1), every subgroup H of G , smooth over S , with connected fibers, such that for every point s of S , H_s contains a Cartan subgroup of $(G_s)^0$, is necessarily equal to its connected normalizer (and consequently is an element of $\mathcal{LC}(S)$). Indeed, thanks to 5.1 iii), one may suppose that S is the spectrum of a field, in which case the property was pointed out at the end of the statement of Exp. XIII 2.1.

Note that one has natural monomorphisms:

$$\begin{array}{ccc} \mathcal{CT} & & \\ & \searrow & \\ & \mathcal{LC} & \longrightarrow \mathcal{L} \\ & \nearrow & \\ \mathcal{P} & & \end{array}$$

Let us show that these monomorphisms are open immersions of finite presentation (which will already prove, taking 5.2 into account, that $\mathcal{LC}, \mathcal{CT}, \mathcal{P}$ are representable by S -preschemes, unions of increasing sequences of open subpreschemes, quasi-projective and of finite presentation over S). Now this will follow, by the usual technique, from the following lemma:

Lemma 6.6. *Let S be a prescheme, G a group S -prescheme of finite presentation, H a subgroup of G , smooth with connected fibers. Then the set of points s of S such that $H_{\bar{s}}$ contains a Cartan subgroup of $G_{\bar{s}}$ (resp. is equal to the centralizer in $(G_{\bar{s}})_{\text{red}}^0$ of a torus of $G_{\bar{s}}$) is an open subset of S . If in addition H is equal to its connected normalizer^{*}, the set of points s of S such that H_s is a parabolic subgroup of G_s is also an open subset of S .*

The assertion to be proved is local on S , which allows one to suppose S affine, then, by EGA IV 8.1 and 6.3, S noetherian. Denote by E the set of points of S having the property in question. It then follows from the assertions actually proved in 6.2 that E is ind-constructible. But S is noetherian, so to prove that E is open, it then suffices to show that E is stable under generalization (EGA IV 1.10.1). Using EGA II 7.1.9, one is finally led to prove that if S is the spectrum of a discrete valuation ring and the closed point s belongs to E , then the same holds for the generic point t . We shall need the following lemma:

^{*}A hypothesis which is in fact superfluous, cf. XVII App. III 3.

Lemma 6.7. *Let A be a complete noetherian local ring, $S = \operatorname{Spec} A$, s the closed point of S , H a group S -prescheme, smooth with connected fibers, T_s a subtorus of H_s . Then:*

i) There exists a closed group subscheme C of H , smooth with connected fibers, such that $C_s = \operatorname{Centr}_{H_s}(T_s)$.

ii) For every point t of S , C_t is the centralizer in H_t of a subtorus T_t of H_t .

Proof of 6.7. Let T' be an S -torus such that there exists an isomorphism $u_0 : T'_s \xrightarrow{\sim} T_s$ (Exp. X 4.6). Let $\mathfrak{m}$ be the maximal ideal of A , $A_n = A/\mathfrak{m}^n$, $S_n = \operatorname{Spec} A_n$, $H_n = H \times_S S_n$, etc. Since H is smooth over S , for every integer $n > 0$, there exists a group S_n -morphism:

$$u_n : T'_n \longrightarrow H_n$$

lifting u_0 (Exp. IX 3.6), and by induction on n one may suppose that u_n lifts u_{n-1} . On the other hand, let q be a prime number invertible on S ; for every integer ℓ equal to a power of q , denote by ${}_\ell u_n$ the restriction of u_n to the subgroup ${}_\ell T'_n$. For fixed ℓ and variable n , the morphisms ${}_\ell u_n$ form a projective system, hence come from a unique group S -morphism ${}_\ell u : {}_\ell T' \longrightarrow H$ (1.6 a)). Since H is separated (Exp. VIB 5.2) and u_0 is a monomorphism, ${}_\ell u$ is a monomorphism (Exp. IX 6.8), and even a closed immersion, since it is finite, ${}_\ell T'$ being finite over S . Denote by $M(\ell)$ the image group. It is clear that the family of subgroups of multiplicative type $M(\ell)$ is coherent in the sense of 4.1. Let $C_\ell = \operatorname{Centr}_H(M(\ell))$, which is representable by a group subscheme (2.5), closed (H is separated), smooth over S (Exp. XI 2.4). The C_ℓ form a decreasing filtered family of closed subschemes, hence stationary, since H is noetherian. The stationary value is a subgroup C , smooth and closed, such that $C_s = \operatorname{Centr}_{H_s}(T_s)$ by the density theorem (Exp. IX 4.7). It remains to show that for every point t of S , C_t is the centralizer in H_t of a subtorus T_t (which will imply that C_t is connected). But this will follow from the following more precise lemma, applied to the family $M(\ell)_t$ of subgroups of H_t :

Lemma 6.8. *Let G be a connected algebraic group defined over a field k , r an integer > 0 , q an integer prime to the characteristic⁽⁷⁾ of k , $M(\ell)$ (ℓ ranging over the powers of q) a coherent family of subgroups of multiplicative type of G , of type $(\mathbf{Z}/\ell\mathbf{Z})^r$ (cf. 4.6), M the algebraic subgroup of G generated by the $M(\ell)$ (loc. cit.), T the unique maximal torus of M (cf. 3.4). Then one has*

$$\operatorname{Centr}_G(T) = \operatorname{Centr}_G(M) = \operatorname{Centr}_G(M(\ell)) \quad \text{for } \ell \text{ sufficiently large.}$$

The last equality is quite clear. To prove the first, introduce the center Z of G , $G' = G/Z$, M' (resp. T') the image of M (resp. T) in G' , K the inverse image of T' in G (i.e. the algebraic subgroup of G generated by T and Z). It plainly suffices to prove that $K \supset M$, hence that $T' = M'$. Now M is smooth and connected (4.6) and G' is affine (Exp. XII 6.1), so M' is the direct product of its maximal torus T' (Exp. XII 6.6 d)) and a unipotent group (BIBLÉ 4 Th. 4) (one may suppose k algebraically closed). The image of $M(\ell)$ in M' is therefore necessarily contained in T' . Thus the inverse image of T' in M contains $M(\ell)$ for every ℓ , hence is equal to M . Consequently $M' = T'$. This proves 6.8 and therefore 6.7.

This being so, let us prove 6.6. We have reduced to the case where S is the spectrum of a discrete valuation ring A , which one may moreover suppose complete with algebraically closed residue field. Replacing A by its normalization in a finite extension of its field of fractions, one may suppose that $(G_t)_{\text{red}}$ is smooth⁽⁸⁾. It is clear that, to prove 6.6, one may replace G by the identity component of the schematic closure in G of $(G_t)_{\text{red}}^0$, hence suppose that G is flat over S , with connected fibers, and that G_t is smooth.

a) Suppose that H_s is the centralizer in $(G_s)_{\text{red}}$ of a torus T_s and let us show that H_t is then the centralizer in G_t of a subtorus of G_t . By Lemma 6.7, there exists a group subscheme C of H , smooth over S , whose closed fiber is H_s and such that $C_t = \operatorname{Centr}_{H_t}(T_t)$, where T_t

⁽⁷⁾**N.D.E.** The word “residual” has been deleted.

⁽⁸⁾**N.D.E.** Details or references to be given here. . .

is a subtorus of H_t . Since H is smooth over S , with connected fibers, one concludes, for reasons of dimension, that $C = H$. Keeping the notation of 6.7, one has $H = C = C_\ell$ for large ℓ . Similarly consider $C'_\ell = \underline{\text{Centr}}_G(M(\ell))$ (2.5), and let C' be the stationary value of C'_ℓ for large ℓ (2.5 bis). The group scheme C' contains H and is such that $C'_t = \text{Centr}_{G_t}(T_t)$ (6.8) and $C'_s = \text{Centr}_{G_s}(T_s)$. The hypothesis on H_s implies $\dim H_s = \dim C'_s$. Moreover, $\dim H_s = \dim H_t$ (since H is smooth over S) and $\dim C'_t \leq \dim C'_s$ (Exp. VIB 4.1), so one has $\dim H_t = \dim C'_t$. But since G_t is smooth and connected, $C'_t = \text{Centr}_{G_t}(T_t)$ is smooth and connected, so finally one has $H_t = C'_t = \text{Centr}_{G_t}(T_t)$.

b) Suppose that H_s contains a Cartan subgroup C_s of G_s , i.e. the centralizer in $(G_s)_{\text{red}}$ of a maximal torus T_s of G_s . By 6.7, there exists a group subscheme C of H , smooth over S , with connected fibers, which lifts C_s . It plainly suffices to prove that C_t contains a Cartan subgroup of G_t . Now by a), applied with $H = C$, C_t is the centralizer in G_t of a subtorus of G_t , hence contains a Cartan subgroup of G_t .

c) Suppose that H_s is a parabolic subgroup of G_s . Let $N = \underline{\text{Norm}}_G(H)$. By hypothesis, H is equal to its connected normalizer, so N_s is smooth, and consequently is equal to $\text{Norm}_{(G_s)_{\text{red}}}(H_s) = H_s$ (Exp. XII 8 bis). But then N is flat over S . We shall see in Exp. XVI that under these conditions G/N is representable. Since $H_s = N_s$ is a parabolic subgroup of G , $(G/N)_s$ is proper. Since (G/N) has connected fibers and is flat over S , it follows from EGA III 5.5.1 that (G/N) is proper over S . Thus $(G/N)_t = G_t/N_t$ is proper over t , and so is G_t/H_t , since N_t/H_t is finite. It then suffices to apply the following lemma:

Lemma 6.9. *Let k be a field, G an algebraic k -group, H a smooth connected algebraic subgroup of G , $N = \text{Norm}_G H$; then if $\dim H = \dim N$ and G/H is proper, H is a parabolic subgroup of G .*

Indeed, one may suppose k algebraically closed and G smooth and connected. The center Z of G is contained in N , and the hypothesis $\dim H = \dim N$ implies that $Z' = (Z)_{\text{red}}^0$ is contained in H , so $G' = G/Z'$ is affine (Exp. XII 6.1). Replacing G by G' and H by its image in G' , one reduces to the case where G is affine (Exp. XIV 4.9), and Lemma 6.9 then follows from BIBLE 6 Th. 4. We have therefore proved Lemma 6.6.

To finish proving 6.4, we must prove that the S -preschemes representing $\mathcal{LC}$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$) are of finite presentation over S . This assertion is local on S , which allows one to suppose S affine, then, since G is of finite presentation, S noetherian (EGA IV 8.9). We have just seen that the natural inclusions $\mathcal{CT} \rightarrow \mathcal{L}$ (resp. $\mathcal{P} \rightarrow \mathcal{L}$) are immersions, so the same holds for the inclusions $\mathcal{CT} \rightarrow \mathcal{LC}$ (resp. $\mathcal{P} \rightarrow \mathcal{LC}$), and consequently it suffices to prove that $\mathcal{LC}$ is representable by an S -prescheme of finite presentation.

Let us return to the notation introduced in 5.2. For every integer $n \geq 0$, let $\mathcal{L}^n$ therefore be the subfunctor of $\mathcal{L}$ such that

$$\mathcal{L}^n(S') = \{H \in \mathcal{L}(S') \text{ such that } \underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}} H^{(n)}\}.$$

The S -functor $\mathcal{L}^n$ is thus representable by an open subprescheme of $\mathcal{L}$, the sum of the $\mathcal{L}_r^n$. Each $\mathcal{L}_r^n$ is of finite presentation over S (5.3) and is empty for $r > \sup_{s \in S} \dim G_s$ (which is a finite number, S being quasi-compact), so $\mathcal{L}^n$ is of finite presentation over S . It suffices to prove that $\mathcal{LC}$ is contained in $\mathcal{L}^n$ for sufficiently large n .

For every point s of S , let $d(s)$ be the smallest integer n (finite or infinite) such that $\mathcal{LC}_{G_s} \subset \mathcal{L}_{G_s}^n$. It suffices to show that the function d is bounded on S , for if M is an upper bound, $\mathcal{LC}$ will be set-theoretically contained in $\mathcal{L}^M$, hence $\mathcal{LC}$ will be contained in $\mathcal{L}^M$, since the latter is an open subset of $\mathcal{L}$. An immediate constructibility argument reduces us to proving that if S is noetherian and integral with generic point t , then the function d is bounded on a neighborhood of t .

a) *Reduction to the case where G is smooth over S .* Proceeding as in 6.2, one sees that, after changing S , one may suppose that $(G)_{\text{red}}$ is a group prescheme smooth over S , which we shall denote by G' . Put $X = \mathcal{LC}_G$, $X' = \mathcal{LC}_{G'}$, and let H (resp. H') be the group subprescheme of G_X (resp. $G'_{X'}$) universal for the functor $\mathcal{LC}_G$ (resp. $\mathcal{LC}_{G'}$). Since H is smooth over

X , $H \times_X (X_{\text{red}})$ is reduced, hence contained in $G'_{X_{\text{red}}}$, and it is an element of $\mathcal{LC}_{G'}(X_{\text{red}})$, whence a canonical morphism $p : X_{\text{red}} \rightarrow X'$. It is clear that p is a monomorphism; let us show that p is even an immersion. Let N' be the normalizer of H' in $G_{X'}$. The set of points s of X' such that the immersion $H'_s \rightarrow N'_s$ is an open immersion is an open subset U , and $H'|_U \rightarrow N|_U$ is an open immersion (Exp. VIB 2.5 and EGA IV 17.9.5). It follows from 5.1 iii) that U is the largest open subset of X' over which H' is equal to its connected normalizer in $G_{X'}$, so $H'|_U \in \mathcal{LC}_G(U)$. One immediately deduces that p is an isomorphism of X_{red} onto U_{red} . If one can show that $\mathcal{LC}_G$ is of finite type when G is smooth, X' will be of finite type over S , hence X_{red} will be of finite type (S is noetherian), and consequently will be contained in $\mathcal{L}_G^M$ for sufficiently large M ; the same will hold for X .

b) *Case where S is the spectrum of an algebraically closed field k of characteristic $p > 0$ and G is a smooth algebraic group over k .* Instead of using the infinitesimal neighborhoods $H^{(n)}$ of a group subscheme H of G , we shall use the radicial subgroups $F^n(H)$, kernels of the iterates of the Frobenius morphism in H (Exp. VIIA 4), which is legitimate here, since $F^n(H)$ is contained in the infinitesimal neighborhood of order p^n of the unit section of H . If T is a subtorus of G , $F^n(T) = p^n(T)$, and it is immediate by duality that $\text{Centr}_G(p^n(T))$ is equal to $\text{Centr}_G(T)$ for sufficiently large n . One then has the following more precise proposition:

Proposition 6.10. *Let k be a field of characteristic $p > 0$, G a smooth algebraic k -group, T a maximal torus of G_k , m the smallest integer such that*

$$\text{Centr}_G(p^m(T))^0 = \text{Centr}_G(T)^0.$$

Then, for every k -prescheme S and every $H \in \mathcal{LC}(S)$, one has

$$\text{Norm}_{G_S}(H) = \text{Norm}_{G_S}(F^m(H)).$$

A fortiori, $\mathcal{LC}$ is contained in $\mathcal{L}^{p^m}$.

Since H is smooth over S , of finite presentation over S , and has constant nilpotent rank (namely that of G), we shall see in the next paragraph (7.3) that the functor $\mathcal{C}_H$ of Cartan subgroups of H is representable by an S -prescheme smooth over S (the reader will check that the proof given of this property does not use the fact that $\mathcal{LC}_H$ is of finite type over S). It then follows from Exp. XIII 3.1 that one may consider the open subset U of regular points of H .

Let S' be an S -prescheme, g an element of $G(S')$ normalizing $F^m(H)_{S'}$. To prove that g normalizes $H_{S'}$, it suffices to prove that $\text{int}(g)U_{S'}$ is contained in $H_{S'}$; indeed, $H_{S'} \cap \text{int}(g)H_{S'}$ will then contain an open subgroup of $\text{int}(g)H$, hence will be equal to $\text{int}(g)H$, since the latter has connected fibers. Replacing S' by a suitable S'' , then S'' by S , one reduces to proving that if $g \in G(S)$ normalizes $F^m(H)$ and $u \in U(S)$, then $\text{int}(g)u \in H(S)$.

Now let C be the unique Cartan subgroup of H_S which “contains” u (Exp. XIII 3.2). It suffices for us to show that $C' = \text{int}(g)C \subset H$. Since $H \in \mathcal{LC}_G(S)$, C is also a Cartan subgroup of G , but the latter has maximal tori, so (Exp. XII 7.1 (a)) C is the centralizer in G_S^0 of its unique maximal torus T . It follows from the definition of m (and the fact that two maximal tori of G are locally conjugate for fpqc (Exp. XII 7.1)) that one has

$$C = (\text{Centr}_G(T))^0 = \text{Centr}_G(p^m(T))^0,$$

whence, by conjugation by g ,

$$C' = \text{Centr}_G(\text{int}(g)(p^m(T)))^0.$$

But $\text{int}(g)(p^m(T))$ is a subgroup of multiplicative type of $\text{int}(g)(F^m(H))$, which is equal to $F^m(H)$ (g normalizes $F^m(H)$), hence is contained in H . It then follows from Exp. XIII 2.1 (which is proved when the base is a field, but extends immediately to the case of an arbitrary base) that, for C' to be contained in H , it suffices that one have

$$\text{Lie } C' \subset \text{Lie } H.$$

Now $\text{Lie } C' = \text{int}(g)(\text{Lie } C) \subset \text{int}(g)(\text{Lie } H)$.

On the other hand, if $m \geq 1$, which one is free to suppose, one has (using Exp. VIIA 4.1.2)

$$\mathrm{Lie} H = \mathrm{Lie}({}_{F^m}(H)) = \mathrm{Lie}(\mathrm{int}(g)({}_{F^m}(H))) = \mathrm{int}(g)(\mathrm{Lie} H).$$

Thus $\mathrm{Lie} C' \subset \mathrm{Lie} H$, which completes the proof of 6.10.

We shall need another definition of the integer m introduced in 6.10.

Lemma 6.11. *Let G be a smooth algebraic group defined over an algebraically closed field k of characteristic $p > 0$, T a maximal torus of G , $\mathfrak{g} = \mathrm{Lie} G$, and R the family of nonzero characters of T occurring in the representation of T in $\mathfrak{g}$ induced by the adjoint representation of G . For every element $r \in R$, denote by e_r the largest integer n such that p^n divides r in the group of characters of T . Then $m = \sup_{r \in R} (e_r + 1)$ if $R \neq \emptyset$, and $m = 0$ otherwise.*

Indeed, $\mathrm{Centr}_G(T)$ is smooth and contained in $\mathrm{Centr}_G({}_{p^m}(T))$, so

$$\begin{aligned} (\mathrm{Centr}_G T)^0 &= (\mathrm{Centr}_G({}_{p^m}(T)))^0 \iff \mathrm{Lie}(\mathrm{Centr}_G T) = \mathrm{Lie}(\mathrm{Centr}_G({}_{p^m}(T))) \\ &\iff \mathfrak{g}^T = \mathfrak{g}^{p^m(T)} \quad (\text{Exp. II 5.2.3}). \end{aligned}$$

Now with the usual notation, one has

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \coprod_{r \in R} \mathfrak{g}_r.$$

Thus $\mathfrak{g}^T = \mathfrak{g}_0$ and $\mathfrak{g}^{p^m(T)} = \mathfrak{g}_0 + \coprod_{r \in R'} \mathfrak{g}_r$, where R' is the part of R formed by the characters of T whose restriction to ${}_{p^m}(T)$ is zero. But a nonzero character r of T has zero restriction to ${}_{p^m}(T)$ if and only if $m \leq e_r$, whence the lemma.

c) Let us return to the proof of 6.4. We have reduced (according to point a) and the paragraph preceding it) to the case where S is a noetherian integral scheme and G is smooth over S . We must show that the function d is bounded on a neighborhood of the generic point t of S . After changing S , we may suppose that G has a split maximal torus (6.2) $T = D_S(M)$. Let $\mathfrak{g} = \coprod_{\lambda \in M} \mathfrak{g}^\lambda$ be the decomposition of the Lie algebra of G according to the characters of T , and let R be the finite set of nonzero characters of M such that $\mathfrak{g}^\lambda \neq 0$. Let us then distinguish two cases:

1st case: the point t , and consequently all points of S , have residue characteristic $p > 0$. It is clear, in view of the foregoing, that the function d is then bounded above by p^m , where m is defined as in 6.11.

2nd case: the point t has residue characteristic zero. For every $\lambda \in R$, let n_λ be the largest integer dividing λ in the group M , and put $n = \prod n_\lambda$, $\lambda \in R$. For every prime number q dividing n , denote by S_q the closed subset of S formed by the points of S whose residue characteristic is equal to q , and let U be the nonempty open subset (it contains t) complementary in S to the union of the S_q . If now s is a point of U , either s has residue characteristic zero, and then $d(s) = 1$ (5.6), or s has residue characteristic $p > 0$ which does not divide n , so the integer m for the group G_s , defined in 6.11, is less than or equal to one. Moreover, it follows from Exp. VII⁽⁹⁾ that one has

$$\underline{\mathrm{Norm}}_{G_{S'}}({}_{F^1}(H)) = \underline{\mathrm{Norm}}_{G_{S'}}(\mathrm{Lie} H) = \underline{\mathrm{Norm}}_{G_{S'}}(H^{(1)}).$$

Finally, it follows from 6.10 that if $H \in \mathcal{LC}_{G_s}(S')$, one has $\underline{\mathrm{Norm}}_{G_{S'}}(H) = \underline{\mathrm{Norm}}_{G_{S'}}(H^{(1)})$, and consequently $d(s) \leq 1$, so d is bounded by 1 on U .

This completes the proof of 6.4.

Corollary 6.12. *Let S be a prescheme and G a group S -prescheme of finite presentation. Suppose that the nilpotent rank (resp. the dimension of the Borel subgroups) of the fibers of G is a locally constant function on S ; then the functor $\mathcal{C}$ of Cartan subgroups of G (resp. the functor $\mathcal{B}$ of Borel subgroups of G) is representable by an S -prescheme of finite presentation over S .*

⁽⁹⁾**N.D.E.** Argument to be made explicit...

Indeed, after restricting S , one may suppose that the nilpotent rank ν of the fibers is constant. But then it is clear that $\mathcal{C}$ is represented by the subprescheme, both open and closed, of the prescheme X representing $\mathcal{LC}$ (6.4), over which the universal subgroup of G_X for the functor $\mathcal{LC}$ has relative dimension ν . The proof is analogous for the functor $\mathcal{B}$, taking account of the representability of the functor $\mathcal{P}$.

7. Cartan subgroups of a smooth group

Proposition 7.1. *Let S be a prescheme, G a group S -prescheme of finite presentation, smooth over S , with connected fibers.*

i) Let $\mathcal{CT}$ be the S -functor $((\text{Sch})/S)^\circ \longrightarrow (\text{Ens})$ such that for every S -prescheme S' , one has

$\mathcal{CT}(S')$ = set of group subpreschemes H of $G_{S'}$, smooth over S' , such that for every point s' of S' , $H_{s'}$ is the centralizer in $G_{s'}$ of a subtorus of $G_{s'}$.

Then $\mathcal{CT}$ is representable by an S -prescheme of finite presentation over S , smooth and quasi-projective over S .

ii) If S is artinian and $H \in \mathcal{CT}(S)$, H is the centralizer in G of a subtorus of G . If S is the spectrum of a henselian local ring and $H \in \mathcal{CT}(S)$, H is the centralizer in G of a subgroup of multiplicative type of G , étale over S .

iii) If L is a group S -prescheme, smooth and of finite presentation over S , $i : L \longrightarrow G$ a group S -monomorphism, and H an element of $\mathcal{CT}(S)$, then $\text{Transp}_G(H, L)$ and $\text{Transp}_G^{\text{str}}(H, L)$ (cf. Exp. VIII 6.5 e)) are representable by closed subpreschemes of G , smooth over S .

iv) If $H \in \mathcal{CT}(S)$, H is closed in G , $N = \text{Norm}_G(H)$ is representable by a group subprescheme of G , closed and smooth over S ; N/H is representable by a group S -prescheme, separated over S , étale and of finite type over S ; G/N is representable by an S -prescheme smooth and quasi-projective over S .

v) Let G' be a group S -prescheme of finite presentation over S , and $u : G \longrightarrow G'$ a faithfully flat group S -morphism, so that G' satisfies the same hypotheses as G (Exp. VIB 9). Then if $H \in \mathcal{CT}_G(S)$, the image of H under u is representable by a group subprescheme H' of G' which is an element of $\mathcal{CT}_{G'}(S)$. Moreover, $H \longrightarrow H'$ is faithfully flat, and if H is the centralizer in G of a torus T , H' is the centralizer in G' of the torus $T' = u(T)$.

vi) Under the conditions of v), consider the S -morphism

$$\tilde{u} : \mathcal{CT}_G \longrightarrow \mathcal{CT}_{G'}, \quad H \longmapsto H' = u(H).$$

Then $\tilde{u}$ is a faithfully flat quasi-compact morphism; if moreover $\text{Ker } u$ is central, $\tilde{u}$ is an isomorphism, the inverse isomorphism being $H' \mapsto u^{-1}(H')$.

Proof of ii). For the first assertion, one may suppose S local artinian with closed point s . Let $H \in \mathcal{CT}(S)$ and T_s the maximal central torus of H_s , which is already defined over $\kappa(s)$ (cf. 3.4). Since $H_s \in \mathcal{CT}(s)$, one has $H_s = \text{Centr}_{G_s} T_s$. The group H is smooth, so T_s lifts (uniquely) to a subtorus T of H , central in H (Exp. IX 3.6 bis and Exp. IX 5.6). But then $H' = \text{Centr}_G T$ contains H and has the same fiber as H ; since H is flat over S , one has $H = H'$ (Exp. VIB 2.5).

Suppose now that S is the spectrum of a henselian local ring, which one may suppose noetherian by the usual reductions. Denote by s the closed point of S , T_s the maximal central torus of H_s , q an integer invertible on S , ℓ a power of q , T an S -torus whose closed fiber is isomorphic to T_s (Exp. X 4.6). On the other hand, let ${}_\ell H$ be the “kernel” of raising to the ℓ th power in H , and let U_ℓ be the largest open subset of ${}_\ell H$ which is étale over S . It then follows from 1.3 and the fact that H is flat over S that U_ℓ contains ${}_\ell T_s$. Since S is henselian, there exists a unique S -morphism

$$u_\ell : {}_\ell T \longrightarrow U_\ell$$

which, on the closed fiber, induces the canonical immersion ${}_\ell T_s \longrightarrow (U_\ell)_s$. By uniqueness, one easily sees that u_ℓ is a central morphism of S -groups (Exp. IX 5.6 a)). Proceeding as in

6.6 and 6.7, one shows that u_ℓ is an immersion and that, if M_ℓ is the image group $u_\ell(\ell T)$, then $\underline{\text{Centr}}_G(M_\ell)$ is equal to H for sufficiently large ℓ (this is where the hypothesis that S is noetherian is used).

Proof of i). The group G is smooth over S , with connected fibers, so if $H \in \mathcal{CT}(S)$, H has connected fibers (Exp. XII 6.6 b)) and is equal to its connected normalizer (6.5), so that the functor $\mathcal{CT}$ defined in 7.1 i) coincides with the functor also denoted by $\mathcal{CT}$ which was defined in 6.4.0. Thus, by Theorem 6.4, $\mathcal{CT}$ is representable by an S -prescheme of finite presentation and quasi-projective over S . It remains to show that this prescheme is smooth over S . One first reduces, by EGA IV 8, to the case where S is affine noetherian. By Exp. XI 1.5, it then suffices to prove that if S is the spectrum of an artinian local ring, S_0 a subscheme defined by a nilpotent ideal, H_0 an element of $\mathcal{CT}(S_0)$, then H_0 lifts to a group subscheme H of G , smooth over S . Now by ii), $H_0 = \underline{\text{Centr}}_{G_0} T_0$, where T_0 is a subtorus of G_0 . Since G is smooth, T_0 lifts to a subtorus T of G (Exp. IX 3.6 bis), and it suffices to take $H = \underline{\text{Centr}}_G(T)$, which is indeed smooth over S (Exp. XI 2.4 and Lemma 2.5).

Proof of iii). Since H is smooth, with connected fibers, by Exp. XI 6.11, $\underline{\text{Transp}}_G(H, L)$ is representable by a closed subscheme of G , of finite presentation over S . To show that this transporter is smooth, one reduces, as above, to proving that if S is local artinian, S_0 a closed subscheme of S , $g_0 \in G(S_0)$ such that $\text{int}(g_0)H_0 \subset L_0$, then g_0 lifts to $g \in G(S)$ such that $\text{int}(g)H \subset L$. Since the group G is smooth over S , there exists a section g_1 of G lifting g_0 ; let $H' = \text{int}(g_1)H$. It is therefore an element of $\mathcal{CT}(S)$ such that $H'_0 \subset L_0$. By ii), H' is the centralizer in G of a torus T' of G . Since L is smooth, the torus T'_0 of L_0 lifts to a torus T'' of L (Exp. XI 3.6 bis). The group $\underline{\text{Centr}}_L T''$ is contained in $\underline{\text{Centr}}_G T''$, has the same fiber as the latter (namely H'_0), and is smooth, hence is equal to $\underline{\text{Centr}}_G T'' = H''$. The subtori T' and T'' of G are two liftings of T'_0 , hence are conjugate by an element h of $G(S)$ reducing to the unit section of G_0 (Exp. IX 3.3 bis); the same therefore holds for their centralizers H' and H'' in G . The section $g = hg_1$ lifts g_0 , and one indeed has $\text{int}(g)H \subset L$.

If now $g \in \underline{\text{Transp}}_G(H, L)(S)$, for $\text{int}(g)H = L$, it is necessary and sufficient that for every $s \in S$, $\dim H_s = \dim L_s$. It follows that if U denotes the subscheme of S , both open and closed, over which the fibers of H have the same dimension as those of L , the strict transporter of H into L , $\underline{\text{Transp}}_G^{\text{str}}(H, L)$, is representable by the S -prescheme

$$U \times_S \underline{\text{Transp}}_G(H, L).$$

Proof of iv). To see that if $H \in \mathcal{CT}(S)$, H is closed in G , one may suppose S affine noetherian, then S the spectrum of a complete local ring (EGA IV 8), but then H is the centralizer in G of a subgroup of multiplicative type (by ii)), hence is closed since G is separated over S (Exp. VIB 5.2).

By iii), $N = \underline{\text{Norm}}_G(H) = \underline{\text{Transp}}_G^{\text{str}}(H, H)$ is representable by a group subscheme smooth over S and closed in G . Consider the S -morphism

$$G \longrightarrow \mathcal{CT}, \quad g \longmapsto \text{int}(g)H.$$

It follows from iii) that this morphism is smooth, so its image is an open subset U of $\mathcal{CT}$. One then proves, as in Exp. XI 5.3, that G/N is representable by U , hence in particular is quasi-projective.

Let us now study the quotient N/H . By EGA IV 8, to prove that N/H is representable, one may suppose S affine noetherian, then S the spectrum of a local ring A , hence of finite dimension⁽¹⁰⁾. We shall proceed by increasing induction on the dimension of S . If $\dim S = 0$, the property follows from Exp. VIA § 4. Note now that if N/H is representable, it is separated over S (since H is closed in N by iv)), of finite type and étale over S (since N is smooth over S , of finite type, and H is open in N), so N/H is necessarily quasi-affine over S (SGA1 VIII.6.2). By effective descent of quasi-affine schemes (loc. cit. 7.9), one may replace A by its completion, hence suppose S the spectrum of a complete noetherian local ring. Let s

⁽¹⁰⁾ **N.D.E.** (Krull dimension)

be its closed point and $U = S \setminus s$. By the induction hypothesis, $(N|_U)/(H|_U)$ is representable by a U -group K . Let then T_s be the maximal central torus of H_s , q an integer invertible on S , ℓ a power of q , M_ℓ the unique central subgroup of multiplicative type of H lifting ${}_\ell T_s$ (cf. ii)). Choose ℓ sufficiently large that $\text{Centr}_G(M_\ell) = H$, and let $N' = \text{Norm}_G(M_\ell)$. Since $\text{Centr}_G(M_\ell) = H$, one has $N' \subset \text{Norm}_G(H)$. Moreover, one immediately checks that T_s (hence also $(M_\ell)_s$) is a characteristic subgroup of H_s (i.e. stable under $\text{Aut}_{\text{gr}}(H_s)$), so N_s normalizes $(M_\ell)_s$, and consequently $N_s = N'_s$. The proof of Exp. XI 5.9 then proves that the quotient $N'/H = \text{Norm}_G(M_\ell)/\text{Centr}_G(M_\ell)$ is representable by an S -group K' . Since N' is smooth over S and $N'_s = N_s$, N' is an open subgroup of N (Exp. VIB 2.5) containing H , so the image of $N'|_U$ in K is an open subgroup, isomorphic to $K'|_U$. Let L be the S -prescheme obtained by gluing K and K' by means of the preceding isomorphism, and let p be the S -morphism $N \rightarrow L$ obtained by gluing the canonical projections $N|_U \rightarrow K$ and $N' \rightarrow K'$. It is clear that (L, p) represents the quotient N/H .

Proof of v). Suppose first that S is the spectrum of a field k . The image H' of H is then a smooth subgroup of G' . We must show that $H' \in \mathcal{CT}_{G'}(S)$, which will follow from the following more precise lemma:

Lemma 7.2. *Let $u : G \rightarrow G'$ be an epimorphism of smooth connected algebraic k -groups, T a torus of G , T' its image in G' ; then*

$$u(\text{Centr}_G T) = \text{Centr}_{G'} T'.$$

Write $H = \text{Centr}_G T$, $H' = \text{Centr}_{G'} T'$, $H'' = u(H)$. One has $H'' \subset H'$. To prove that $H' = H''$, one may suppose the base field algebraically closed, and it suffices to prove that every Cartan subgroup of H' is contained in H'' . Indeed, H'' will then contain the open subset of regular points of H' , hence H'' will be an open subgroup of H' and consequently will be equal to H' , since the latter has connected fibers. Let therefore C' be a Cartan subgroup of H' ; C' is also a Cartan subgroup of G' , since H' is the centralizer of a torus T' , hence has the same reductive rank and the same nilpotent rank as G' . Put $K = (u^{-1}(C'))_{\text{red}}^0$. Since T' is in the center of H' , C' contains T' , so K contains T . Let then C be a Cartan subgroup of K containing T . The torus T is contained in the unique maximal torus of C , which is central in C (Exp. XII 6.6 c)), so C is contained in $H = \text{Centr}_G T$. Using now the fact that two Cartan subgroups of G are conjugate and that the image of a Cartan subgroup of G is a Cartan subgroup of G' (Exp. XII 6.6), one deduces that C is also a Cartan subgroup of G , so its image is a Cartan subgroup of G' ; since it is contained in C' , one has $u(C) = C'$, so C' is indeed contained in H'' .

We have therefore established v) when S is the spectrum of a field k . Let us now study the general case. Since G , H and G' are of finite presentation over S , to prove that $u(H)$ is representable and is an element of $\mathcal{CT}_{G'}(S)$, one reduces by the usual technique to the case where S is affine noetherian, then to the case where S is the spectrum of a local ring. By fpqc descent of subpreschemes of G' , one may even suppose that S is the spectrum of a complete noetherian local ring A .

Let us return to the notation of ii), i.e.: let T_s be the maximal central torus of H_s (s is the closed point of S), M_ℓ a subgroup of multiplicative type of H lifting ${}_\ell T_s$ and such that $H = \text{Centr}_G M_\ell$. Let T'_s be the image of T_s in G'_s . Since G' is separated over S (Exp. VIB 5.2), the image of M_ℓ under u is a subgroup of multiplicative type M'_ℓ of G' (Exp. IX 6.8). Put then $H' = \text{Centr}_{G'} M'_\ell$, which is a smooth group subprescheme of G' . For every integer ℓ' equal to a power of q , there exists ℓ such that $(M'_\ell)_s$ contains ${}_{\ell'} T'_s$, so one may suppose ℓ chosen sufficiently large that $H'_s = \text{Centr}_{G'_s} T'_s = u_s(H_s)$, where the last equality follows from 7.2. The restriction of u to H , say v , plainly factors through H' . Let us prove that $v : H \rightarrow H'$ is a flat morphism. Since H and H' are flat over S and $H_s \rightarrow H'_s$ is flat, v is flat on a neighborhood of H_s (EGA IV 11.3.10 and 11.3.1). The morphism v is therefore flat on an open subgroup of H (Exp. VIB 2.2), hence v is flat, H having connected fibers. The set-theoretic image of H is thus an open subset of H' (necessarily equal to $(H')^0$) which,

equipped with its induced structure, represents the image sheaf $u(H)$ (for the fpqc topology). The fact that $u(H)$ is an element of $\mathcal{CT}_{G'}(S)$ then follows from 7.2.

Suppose now that H is the centralizer in G of a torus T , and let T' be the image of T under u , which is a subtorus of G' (Exp. IX 6.8). The image of H under u is contained in $\text{Centr}_{G'}(T')$, coincides fiber by fiber with the latter (7.2), and is smooth over S , so $u(H) = \text{Centr}_{G'}(T')$.

Proof of vi). To show that $\tilde{u}$ is a faithfully flat S -morphism, knowing that $\mathcal{CT}_G$ and $\mathcal{CT}_{G'}$ are smooth over S (by i)), it suffices to check this on the geometric fibers. We are therefore reduced to the case where S is the spectrum of an algebraically closed field k . Let $H' \in \mathcal{CT}_{G'}(k)$, T' its maximal central torus, T a subtorus of G whose image is T' , $H = \text{Centr}_G(T)$, so that $H' = u(H)$ (7.2), $N = \underline{\text{Norm}}_G(H)$, $N' = \underline{\text{Norm}}_{G'}(H')$. We have shown in iv) that G/N (resp. G'/N') is canonically identified with an open neighborhood of H in $\mathcal{CT}_G$ (resp. of H' in $\mathcal{CT}_{G'}$). Under these identifications, the restriction of $\tilde{u}$ to G/N coincides with the natural morphism

$$w : G/N \longrightarrow G'/N'$$

deduced from u by passage to the quotient. Now w is an epimorphism of homogeneous spaces under G , hence is faithfully flat. This proves that $\tilde{u}$ is a flat morphism such that $u(\mathcal{CT}_G)$, which is therefore an open subset of $\mathcal{CT}_{G'}$, contains every point of $\mathcal{CT}_{G'}(k)$. Since $\mathcal{CT}_{G'}$ is of finite type over k , one deduces that $\tilde{u}$ is surjective, hence is faithfully flat.

The additional assertions in vi) in the case where $\text{Ker } u$ is central follow from Exp. XII 7.12.

The following theorem generalizes Theorem 7.1 of Exp. XII:

Theorem 7.3. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , smooth, with connected fibers, and consider the S -functor $\mathcal{C} : ((\text{Sch})/S)^\circ \longrightarrow (\text{Ens})$ such that*

$$\mathcal{C}(S') = \text{set of Cartan subgroups of } G_{S'}.$$

i) *The following conditions are equivalent:*

- a) *The functor $\mathcal{C}$ is representable.*
- b) *The functor $\mathcal{C}$ is representable by a smooth quasi-projective S -prescheme of finite presentation over S , with affine fibers.*
- c) *The group G has, locally for the étale topology, a Cartan subgroup.*
- d) *The group G has, locally for the faithfully flat topology, a Cartan subgroup.*
- e) *The nilpotent rank of the fibers of G is a locally constant function on S .*

ii) *If the preceding conditions hold, two Cartan subgroups of G are locally conjugate for the étale topology. The set of regular points of the fibers of G (Exp. XIII 2.7) is an open subset G_{reg} , of finite presentation over S , and every section of G_{reg} over S is contained in one and only one Cartan subgroup of G .*

iii) *Let G' be a group S -prescheme of finite presentation over S , and $u : G \longrightarrow G'$ a faithfully flat group S -morphism, so that G' is smooth over S , with connected fibers. Then if C is a Cartan subgroup of G , $u(C) = C'$ is representable by a Cartan subgroup of G' , and $C \longrightarrow C'$ is faithfully flat.*

iv) *Under the conditions of i) and iii), the morphism*

$$\tilde{u} : \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}, \quad C \longmapsto u(C) = C'$$

is faithfully flat. If moreover $\text{Ker}(u)$ is central, $\tilde{u}$ is an isomorphism.

v) *For further information concerning transporters and relations with maximal tori, one may consult 7.1 and Exp. XII 7.1.*

PROOF. i) We shall show that b) $\Rightarrow$ c) $\Rightarrow$ d) $\Rightarrow$ e) $\Rightarrow$ b) $\Rightarrow$ a) $\Rightarrow$ d).

b) $\Rightarrow$ c). Let s be a point of S . Since $\mathcal{C}_s$ is smooth over $\kappa(s)$, there exist points of $\mathcal{C}_s$ whose residue field is a finite separable extension of $\kappa(s)$. Applying Exp. XI 1.10, one sees

that there exists an open neighborhood U of s and a surjective étale morphism $U' \rightarrow U$ such that $G_{U'}$ has a Cartan subgroup.

c) $\Rightarrow$ d) is quite clear.

d) $\Rightarrow$ e). Let $s \in S$. By hypothesis, there exists an S -prescheme S' , flat over S , whose image contains s , such that $G_{S'}$ has a Cartan subgroup. Let s' be a point of S' above s . The nilpotent rank of the fibers of $G_{S'}$ is therefore constant on $\text{Spec } \mathcal{O}_{S',s'}$, and consequently the nilpotent rank of the fibers of G is constant on $\text{Spec } \mathcal{O}_{S,s}$, which is the image of $\text{Spec } \mathcal{O}_{S',s'}$ (EGA IV 2.3.4 ii). Let r be its value. It follows from 6.3 bis that the set E_r formed by the points x of S such that the nilpotent rank of G_x is equal to r is an ind-constructible subset of S , hence contains a neighborhood of s (EGA IV 1.10.1).

e) $\Rightarrow$ b). The assertion is local on S , so one may suppose that the nilpotent rank of the fibers of G is constant and equal to r . For every S -prescheme S' , the Cartan subgroups of $G_{S'}$ are then precisely the group subschemes of $G_{S'}$, smooth over S' , of relative dimension r , whose geometric fibers are the centralizers of a torus. Since $\mathcal{CT}$ is representable by 7.1, $\mathcal{C}$ is representable by the subscheme, both open and closed, of $\mathcal{CT}$ representing the subfunctor of $\mathcal{CT}$ formed by the groups of relative dimension r . The other assertions in b) are contained in 7.1 i), except the fact that the fibers of $\mathcal{C}$ are affine, which follows from Exp. XII 7.1 d).

It is clear that b) $\Rightarrow$ a). Let us show that a) $\Rightarrow$ d). It follows from 6.2 iii) that the functor $\mathcal{C}$ commutes with filtered inductive limits of rings, so if $\mathcal{C}$ is representable, it is necessarily representable by an S -prescheme locally of finite presentation (EGA IV 8.14.2). To prove that $\mathcal{C}$ is smooth over S , one reduces to showing that if S is affine and S_0 is the closed subscheme defined by a nilpotent ideal J , then every Cartan subgroup C_0 of $G_0 = G \times_S S_0$ lifts to a Cartan subgroup C of G . But the existence of C_0 implies that condition e) is satisfied for S_0 , hence for S , which has the same underlying space, and one concludes from the fact that d) $\Rightarrow$ b). Since $\mathcal{C}$ is smooth over S and $\mathcal{C} \rightarrow S$ is surjective, one sees that a) $\Rightarrow$ d).

ii) Let C and C' be two Cartan subgroups of G . Then $\text{Transp}_G(C, C')$ is representable by a prescheme smooth over S (7.1 iii)), with nonempty fibers (cf. Exp. XII 6.6 a) and c)). The fact that C and C' are locally conjugate for the étale topology is then a consequence of Hensel's lemma (Exp. XI 1.10).

The other assertions of ii) are consequences of XIII 3.1 and XIII 3.2, taking account of i).

iii) Let C be a Cartan subgroup of G . One knows (7.1 v)) that $u(C)$ is representable by a smooth subgroup C' of G' . Since the fibers of C' are Cartan subgroups of the fibers of G' (Exp. XII 6.6 d)), C' is a Cartan subgroup of G' .

iv) To prove that the morphism $\tilde{u}$ is faithfully flat, one proceeds as in 7.1 vi).

If now $\text{Ker } u$ is central and C' is a Cartan subgroup of G' , $u^{-1}(C') = C$ is smooth, with connected fibers (7.1 vi)), and its fibers are Cartan subgroups (Exp. XII 6.6 f)), so C is a Cartan subgroup of G , which completes the proof of iv), taking account of 7.1 vi). $\square$

8. Representability criterion for the subtori functor of a smooth group

8.0. — In this paragraph, if S is a prescheme and G a group S -prescheme, $\mathcal{T}_G$ (or simply $\mathcal{T}$ if there is no ambiguity) denotes the S -functor $((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ such that, for every S -prescheme S' , one has

$$\mathcal{T}(S') = \text{set of subtori of } G_{S'}.$$

One similarly defines $\mathcal{TC}$ as the functor of central subtori of G .

Proposition 8.1. *Let S be a locally noetherian prescheme and G a group S -prescheme of finite type. Then the following conditions are equivalent:*

i) *The functor $\mathcal{T}_G$ “commutes with adic limits of artinian local rings”, i.e. for every S -prescheme of the form $\text{Spec}(A) = S'$, where A is a noetherian local ring complete for the topology defined by its maximal ideal $\mathfrak{m}$, the canonical map*

$$\mathcal{T}(S') \rightarrow \varprojlim_n \mathcal{T}(S'_n) \quad (\text{where } S'_n = \text{Spec}(A/\mathfrak{m}^n))$$

is bijective.

ii) As in i), but one restricts to the case where A is a complete discrete valuation ring with algebraically closed residue field.

iii) As in ii), but one restricts to the subfunctor $\mathcal{T}^{(1)}$ of $\mathcal{T}$ corresponding to the subtori of G of relative dimension 1.

i bis) For every S -prescheme S' as in i) and every S' -torus T , the canonical map

$$\mathrm{Hom}_{S'-\mathrm{gr}}(T, G_{S'}) \longrightarrow \varprojlim_n \mathrm{Hom}_{S'_n-\mathrm{gr}}(T_{S'_n}, G_{S'_n})$$

is bijective.

ii bis) As in i bis), but one restricts to the case where A is a complete discrete valuation ring with algebraically closed residue field.

iii bis) As in ii bis), but one restricts to the case where T is the multiplicative group $\mathbf{G}_m$.

Remark 8.2. One has an analogous proposition on restricting to the central subtori of G and to central homomorphisms from a torus to G .

Proof. We shall use the following lemma:

Lemma 8.3. Let S be a prescheme, G a group S -prescheme, T an S -torus, $u : T \rightarrow G$ a group S -morphism. Suppose moreover that G is of finite presentation over S , or that S is locally noetherian and G locally of finite type. Then:

a) $\mathrm{Ker} u$ is a subgroup of multiplicative type of T .

b) The quotient $T' = T / \mathrm{Ker} u$ is a torus.

c) The canonical monomorphism $T' \rightarrow G$, deduced from u by passage to the quotient, is an immersion.

This lemma is a consequence of Exp. IX 6.8 when G is separated over S . In the general case, one reduces as usual to the case where S is noetherian. Let us first show that $K = \mathrm{Ker} u$ is flat. We may suppose that S is the spectrum of an artinian local ring (EGA 0_{III} 10.2.6), in which case G is separated (Exp. VIB 5.2), so K is of multiplicative type (Exp. IX 6.8), and a fortiori flat over S . Let us now prove that $\mathrm{Ker} u$ is closed in T , which reduces us to the case where S is the spectrum of a discrete valuation ring (EGA II 7.2.1). Since T is flat, with connected fibers, u factors through the identity component of the schematic closure in G of the generic fiber of G . We may therefore suppose G flat with connected fibers, but then G is separated (Exp. VIB 5.2), so K is closed. This being so, it follows from Exp. X 4.8 b) that K is a subgroup of multiplicative type of T . The quotient $T' = T/K$ is then representable, and T' is a group of multiplicative type (Exp. IX 2.7) whose fibers are tori, hence is a torus. The fact that the monomorphism $T' \rightarrow G$ is an immersion then follows from Exp. VIII 7.9.

Proof of 8.1. i) $\Rightarrow$ i bis). Put $T_n = T_{S'_n}$, $G_n = G_{S'_n}$, and let $(u_n)_{n \in \mathbf{N}'}$ be an element of $\varprojlim_n \mathrm{Hom}_{S'_n-\mathrm{gr}}(T_n, G_n)$. For every integer n , $u_n(T_n)$ is therefore a subtorus T'_n of G_n (Lemma 8.3). By hypothesis, there exists a unique subtorus T' of G lifting T'_n for every n . Since T' has affine fibers, one concludes by 4.4.

i bis) $\Rightarrow$ i). Let $(T_n)_{n \in \mathbf{N}}$ be an element of $\varprojlim_n \mathcal{T}(S_n)$. By Exp. X 4.6, there exists an S' -torus T' and an S'_0 -isomorphism

$$u_0 : T'_0 \xrightarrow{\sim} T_0.$$

Since T_n is smooth over S'_n , u_0 lifts to an S'_n -morphism

$$u_n : T'_n \longrightarrow T_n \quad (\text{Exp. IX 3.6}),$$

and one may suppose the family $(u_n)_{n \in \mathbf{N}}$ coherent, hence coming from a morphism $u : T' \rightarrow G$. The image of T' under u is then a subtorus of G (Lemma 8.3) lifting T_n for every n .

The implications i) $\Rightarrow$ ii) $\Rightarrow$ iii), on the one hand, and i bis) $\Rightarrow$ ii bis) $\Rightarrow$ iii bis), on the other, are evident. The implication iii) $\Rightarrow$ iii bis) is proved as i) $\Rightarrow$ i bis). It therefore suffices for us to prove iii bis) $\Rightarrow$ ii bis) $\Rightarrow$ i bis).

ii bis) $\Rightarrow$ i bis). With the terminology introduced in 4.3, assertion i bis) is true if and only if every element $(u_n)_{n \in \mathbf{N}}$ of $\varprojlim_n \text{Hom}_{S_n\text{-gr}}(T_n, G_n)$ is “admissible”. For every point t of S' distinct from the closed point s of S' , there exists an S -scheme S'' , the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point maps to t and whose closed point maps to s (EGA II 7.1.9). One immediately deduces a valuative criterion for a family of morphisms to be admissible. This says that ii bis) $\Rightarrow$ i bis).

iii bis) $\Rightarrow$ ii bis). Let T be an S -torus, where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. The torus T is then split (Exp. X 4.6), i.e. isomorphic to $(\mathbf{G}_{mS})^r$ for a suitable integer r . Let $(u_n)_{n \in \mathbf{N}}$ be an element of $\varprojlim_n \text{Hom}_{S_n\text{-gr}}((\mathbf{G}_m)_{S_n}^r, G_n)$. By hypothesis, the restrictions of the u_n to each factor of $(\mathbf{G}_m)_S^r$ come from a group morphism

$$\mathbf{G}_{mS} \longrightarrow G.$$

Whence a product morphism $(\mathbf{G}_m)_S^r \longrightarrow G^r$ which, composed with the morphism $G^r \longrightarrow G$ defined by the composition law in G , provides a morphism

$$v : (\mathbf{G}_m)_S^r \longrightarrow G.$$

In view of the existence of the group morphism u_n , it is clear that $u_n = v_n$. It remains to see that v is a group morphism, which means that two evident morphisms $f, g : X = (\mathbf{G}_m)_S^r \times_S (\mathbf{G}_m)_S^r \longrightarrow G$ coincide. Let Z be the coincidence subscheme of f and g . Since $(\mathbf{G}_m)_S^r$ has connected fibers and $v(\mathbf{G}_m)_S^r$ contains the unit section, one sees as in 8.3 that v factors through the identity component of G , which allows us to suppose G separated (Exp. VIB 5.2), hence Z closed. Moreover, since v_n is a group morphism, one has $f_n = g_n$ for every n , so Z contains a neighborhood of the closed fiber of X (EGA I 10.9.4), hence is schematically dense in X , X having smooth irreducible fibers (Exp. IX 4.6), and consequently $Z = X$, so $f = g$.

Definition 8.4. Let S be a locally noetherian prescheme, G a group S -prescheme of finite type, and $\mathcal{S}$ the set of S -schemes S' , spectra of complete discrete valuation rings with algebraically closed residue field, with closed point s , generic point t , such that $(G_t)_{\text{red}}^0$ is smooth and has a Chevalley decomposition, i.e. is an extension of an abelian variety by a smooth connected linear algebraic subgroup L_t (this decomposition is then unique). If $S' \in \mathcal{S}$, denote by G' (resp. L') the schematic closure in $G_{S'}$ of G_t (resp. L_t).

Under these conditions, we shall say that

“the abelian part of G does not degenerate into a toric part”, or more briefly that G satisfies property AT,

if for every $S' \in \mathcal{S}$, L_s has the same reductive rank as G'_s . (Intuitively, suppose that the quotient $A = G'/L$ is representable, in which case A is a flat group prescheme such that $(A_t)_{\text{red}}^0$ is an abelian variety. Condition “AT” then means that A_s has reductive rank zero, hence that $(A_s)_{\text{red}}^0$ is an extension of an abelian variety by a unipotent group.)

Similarly, supposing moreover that G has connected fibers, we shall say that

G satisfies property ATC

if for every $S' \in \mathcal{S}$, the schematic closure Z of the center Z_t of G_t satisfies AT.

These technical definitions are justified by the following proposition:

Proposition 8.5. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type. Then:*

- a) *For the functor $\mathcal{T}$ of subtori of G to “commute with adic limits of artinian local rings” (cf. 8.1), it is necessary and sufficient that G satisfy property AT (8.4).*
- b) *If G has connected fibers, for the functor $\mathcal{TC}$ of central subtori of G to commute with adic limits of artinian local rings, it is necessary and sufficient that G satisfy property ATC (8.4).*

We shall need the following technical lemma:

Lemma 8.6. *Let S be the spectrum of a discrete valuation ring, s the closed point of S , G a group S -prescheme flat and of finite type, T_s a subtorus of G_s . Then there exists an S -scheme S' , the spectrum of a discrete valuation ring, faithfully flat over S , with closed point s' , and a group subscheme C of $G_{S'}$, flat over S' , commutative, with connected fibers, such that $C_{s'}$ contains $T_s \times_s s'$.*

Replacing S by S' , the spectrum of a discrete valuation ring faithfully flat over S , one may suppose that T_s is equal to $(\mathbf{G}_m)_s^r$ and that the transcendence degree of $\kappa(s)$ over the prime field is $\geq r$ (EGA 0_{III} 10.3.1 and EGA II 7.1.9). There then exists an element x of $T_s(s)$ such that every algebraic subgroup of G_s which “contains” x contains T_s (cf. Exp. XIII proof of 2.1 (ii) $\Rightarrow$ (vii)). Since G is flat over S , a quasi-section passes through x (EGA IV 14.5.8), and consequently, replacing S by the spectrum of a discrete valuation ring faithfully flat over S , one may suppose that there exists a section x of G over S lifting x . Let C_t be the commutative algebraic subgroup of G_t generated by x_t (Exp. VIB 7), and let C be the schematic closure of C_t in G . It is clear that C_s contains x , hence contains T_s , and consequently the “identity component” of C will be a flat commutative group scheme answering the question.

Proof of 8.5 a). Suppose that the functor $\mathcal{T}$ commutes with adic limits of artinian rings, and let us show that G satisfies property AT. Let therefore $S' \in \mathcal{S}$, and let T_s be a maximal torus of G'_s . We must prove that T_s is contained in L_s . The formation of L and of G' plainly commutes with faithfully flat extensions $S'' \rightarrow S'$ of discrete valuation rings. After changing S' , we may therefore, by Lemma 8.6, suppose that there exists a flat commutative group subscheme C of $G'_{S'}$ such that C_s contains T_s . But then T_s is a central subtorus of C_s , and consequently lifts infinitesimally to a central subtorus (Cor. 2.3). In view of the hypothesis on G , T_s lifts to a subtorus T of G . Plainly T_t is contained in the linear component L_t of G_t , so T is contained in L .

Suppose now that G satisfies property AT, and let us show that condition 8.1 iii) bis is verified. Let therefore S be the spectrum of a discrete valuation ring A , complete, with algebraically closed residue field, $\mathfrak{m}$ the maximal ideal of A , $S_n = \text{Spec}(A/\mathfrak{m}^n)$, u_n , $n \in \mathbf{N}$, a coherent system of group morphisms $u_n : (\mathbf{G}_{mS})_n \rightarrow G_n = G \times_S S_n$. Let q be a prime number invertible on S . With the integer ℓ ranging over the powers of q , there exists a unique group S -morphism

$$\ell u : \ell \mathbf{G}_{mS} \rightarrow G$$

lifting $u_n|_{\ell(\mathbf{G}_{mS})_n}$ for every n (Prop. 1.6 a)). Consequently, if there exists a group S -morphism $u : \mathbf{G}_{mS} \rightarrow G$ lifting u_n for every n , its restriction to $\ell \mathbf{G}_{mS}$ is uniquely determined. Taking account of the density theorem (Exp. IX 4.8 (a)), this proves uniqueness of u and the fact that, to prove existence of u , we may allow ourselves a faithfully flat extension of the base.

Now $(G_t)_{\text{red}}^0$ has a Chevalley decomposition, and this is already defined over a finite extension L of the field of fractions K of A . Replacing S by the normalization of S in L , we may therefore suppose that $S \in \mathcal{S}$. The morphism ℓu_t factors through $(G_t)_{\text{red}}$, so ℓu factors through the schematic closure G'' of $(G_t)_{\text{red}}$ in G' . Again by the density theorem, one deduces that u_n factors through G''_n for every n . Since G has property AT, every subtorus of G'_s , and a fortiori of G''_s , is contained in L_s , so u_s factors through L_s . I say that for every n , u_n factors through L_n . Indeed, since L_t is invariant in $(G'')_t$, L is invariant in G'' ; moreover, L is flat over S , so for every integer n , the quotient $H_n = G''_n/L_n$ is representable (Exp. VIA § 4). The image of $(\mathbf{G}_{mS})_n$ in H_n is a subtorus of H_n (Exp. IX 6.8), whose closed fiber is zero, so this image is zero, and consequently u_n factors through L_n . But since L_t is affine, one deduces that the family u_n is admissible (Prop. 4.3), which completes the proof.

The proof of 8.5 b) is entirely analogous to that of 8.5 a), taking account of 8.2 and Exp. IX 5.6 a).

Proposition 8.7. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type. Then:*

- i) If G has connected fibers and satisfies AT, it satisfies ATC.
- ii) If G satisfies AT, every group subprescheme of G satisfies AT.
- iii) If G is flat over S and the abelian rank of the fibers of G is a locally constant function on S , then G satisfies AT.

i) follows immediately from 8.5 and Exp. IX 5.6 a).

ii) Let H be a group subprescheme of G . To prove that H satisfies property AT, we may suppose that S is the spectrum of a complete discrete valuation ring, and we must show that every coherent family of group S_n -morphisms

$$u_n : (\mathbf{G}_m S)_n \longrightarrow H_n$$

is admissible (8.5 and 8.1 iii bis)). Now by hypothesis, G satisfies property AT, so there exists a group S -morphism

$$u : (\mathbf{G}_m)_S \longrightarrow G$$

lifting u_n for every n . Proceeding as in the proof of 8.5, one sees that $u|_{\ell \mathbf{G}_m S}$ (where ℓ ranges over the powers of a prime number q invertible on S) factors through H . By density, one deduces that on the generic fiber, u_t factors through H_t . Since $(\mathbf{G}_m)_S$ is reduced, it indeed follows that u factors through H .

iii) Let $S' \in \mathcal{S}$ (notation of 8.4). The group L is flat over S , and its generic fiber L_t is affine; under these conditions, one can show that L_s is necessarily affine (XVII App. III, 2). Since G and L are flat, one has $G = G'$, and the dimension of the fibers of G and L is constant on S (Exp. VIB 4), so that one has the inequalities

$$\text{abelian rank } G_s \leq \dim G_s - \dim L_s = \dim G_t - \dim L_t = \text{abelian rank } G_t.$$

The hypothesis implies that these inequalities are equalities. It follows that $(G_s/L_s)_{\text{red}}^0$ is an abelian variety, hence has reductive rank zero, and consequently L_s has the same reductive rank as G_s .

We are now in a position to prove the main theorems of this paragraph:

Theorem 8.8. *Let G be a group prescheme of finite type over a locally noetherian prescheme S . Suppose G flat over S and with connected fibers. Then:*

- a) *For the functor $\mathcal{T}\mathcal{C}$ of central subtori of G to be representable, it is necessary and sufficient that G have property ATC (8.4). Moreover, in this case, $\mathcal{T}\mathcal{C}$ is representable by an S -prescheme étale and separated over S .*
- b) *Under the conditions of a), for every S -torus, the functor $\underline{\text{Hom}}_{S\text{-gr}}^{\text{cent}}(T, G)$ of central homomorphisms from T to G is representable by an S -prescheme étale and separated over S .*

Theorem 8.9. *Let S be a locally noetherian prescheme and G a group S -prescheme of finite type, smooth over S .*

- a) *For the functor $\mathcal{T}$ of subtori of G to be representable, it is necessary and sufficient that G have property AT (8.4). Moreover, in this case, $\mathcal{T}$ is representable by an S -prescheme smooth and separated over S ; more precisely, the structural morphism $\mathcal{T} \longrightarrow S$ admits a canonical factorization*

$$\mathcal{T} \xrightarrow{u} Y \xrightarrow{v} S$$

where v is a smooth quasi-projective morphism (hence of finite type) and u is a separated étale morphism.

- b) *Under the conditions of a), for every S -torus T , the functor $\underline{\text{Hom}}_{S\text{-gr}}(T, G)$ of homomorphisms from T to G is representable by an S -prescheme smooth and separated over S .*

Proof of 8.8 a). If the functor $\mathcal{T}\mathcal{C}$ is representable, it commutes with adic limits of artinian rings, and consequently (8.2 and 8.5 b)) G has property ATC. To establish the converse, we shall use the following result, which will be proved in EGA VI, and is also found in Murre's exposé: Sémin. Bourbaki, May 1965, no. 294, Theorem 1, Corollary 2.

Lemma 8.10. *Let S be a locally noetherian prescheme, and $\mathcal{F}$ a contravariant functor defined on $(\text{Sch})/S$, with values in the category of sets. For $\mathcal{F}$ to be representable by an étale separated S -prescheme, (it is necessary and) it suffices that $\mathcal{F}$ satisfy the following five properties:*

- i) $\mathcal{F}$ is a sheaf for the fpqc topology (Exp. IV).
- ii) $\mathcal{F}$ commutes with inductive limits of rings (Exp. XI 3.2).
- iii) $\mathcal{F}$ commutes with adic limits of artinian local rings (8.1 i)).
- iv) $\mathcal{F}$ satisfies the “valuative criterion of separation”, i.e. for every S -scheme S' which is the spectrum of a discrete valuation ring, with generic point t , the canonical map

$$\mathcal{F}(S') \longrightarrow \mathcal{F}(t)$$

is injective.

- v) $\mathcal{F}$ is infinitesimally étale (Exp. XI 1.8).

Let us show that the functor $\mathcal{TC}$ of 8.8 satisfies the five conditions of 8.10.

i) The functor $\mathcal{T}$ (resp. $\mathcal{TC}$) is a sheaf for the fpqc topology as soon as G is of finite presentation over S . Indeed, every monomorphism from a torus to G is then an immersion (Exp. VIII 7.9), and the property follows from fpqc descent of subpreschemes.

ii) Corollary 6.3 proves that the functor $\mathcal{T}$ commutes with inductive limits of rings if G is of finite presentation over S ; one immediately deduces that the same holds for $\mathcal{TC}$.

iii) By 8.5, condition iii) is precisely equivalent to property ATC.

iv) follows simply from the fact that if S is the spectrum of a discrete valuation ring, two subtori of G having the same generic fiber coincide, and more precisely coincide with the identity component of the schematic closure in G of their generic fiber.

v) follows from 2.3, since G is flat over S .

Proof of 8.8 b). Proceeding as in Exp. XI 4.2, one sees that it suffices to prove that the product group $T \times_S G$ still satisfies property ATC, which is immediate from the definition.

Proof of 8.9 a). Replacing G by its identity component (Exp. VIB 3.10), one may suppose G with connected fibers. If T is a subtorus of G , its centralizer $\text{Centr}_G(T)$ is then representable (Exp. XI 6.11) by a group subprescheme of G , smooth over S (Exp. XI 2.4), with connected fibers (Exp. XII 6.6 b)), hence is an element of $\mathcal{CT}(S)$, where $\mathcal{CT}$ is the functor defined in 7.1 i). It is clear that the map

$$T \longmapsto \text{Centr}_G(T)$$

defines an S -morphism

$$u : \mathcal{T} \longrightarrow \mathcal{CT}.$$

Since $\mathcal{CT}$ is representable by an S -prescheme smooth and quasi-projective over S (7.1 i)), it suffices for us to prove that the morphism u is representable by a separated étale morphism.

After a suitable base change $S' \longrightarrow S$ (with $S' = \mathcal{CT}$, hence S' locally noetherian), we are reduced to the following problem: let S be a locally noetherian prescheme, G a group S -prescheme, smooth and of finite type over S , with connected fibers, H a subgroup of G , smooth over S and with connected fibers. Consider the subfunctor X of $\mathcal{TC}_H$ such that, for every S -prescheme S' , $X(S')$ is the set of central subtori T of $H_{S'}$ such that $\text{Centr}_{G_{S'}}(T) = H_{S'}$. We shall show that X is representable by an S -prescheme étale and separated over S .

Indeed, by hypothesis, G satisfies property AT, so the same holds for H (8.7 ii)), and since H has connected fibers, H also satisfies property ATC (8.7 i)). On the other hand, H is smooth over S , hence flat. By 8.8 a), $\mathcal{TC}_H$ is representable by an S -prescheme étale and separated over S . It then suffices for us to show that the canonical monomorphism $X \longrightarrow \mathcal{TC}_H$ is representable by an open immersion.

Put $S' = \mathcal{TC}_H$, and let K be the centralizer in $G_{S'}$ of the “universal” central torus of $H_{S'}$. The group K is a group scheme smooth over S' , with connected fibers, containing $H_{S'}$. By definition, one has $X = \prod_{K/S'} H_{S'}/K$, which is indeed representable by the subprescheme, both open and closed, of S' over which $H_{S'}$ and K have the same relative dimension.

One proves 8.9 b) analogously to 8.8 b).

Corollary 8.11. *Let S be a prescheme, G a group S -prescheme smooth over S and of finite presentation. Then, if the abelian rank of the fibers of G is a locally constant function on S (in particular if the fibers of G are affine), the functor of subtori of G is representable by an S -prescheme smooth and separated over S , and the same holds for the functor $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, G)$, for every S -torus T .*

Indeed, the assertion is local on S , so we may suppose S affine and the abelian rank of the fibers of G constant. One can then find (Exp. VIB § 10) an affine noetherian scheme S_0 and a group S_0 -prescheme G_0 , smooth and of finite type over S_0 , such that G is S -isomorphic to $G_0 \times_{S_0} S$. Moreover, the abelian rank of the fibers of a group S -prescheme of finite presentation over S is an ind-constructible function (6.3 bis). By a standard argument (EGA IV 8), one concludes that in the present case, one may suppose the abelian rank of the fibers of G_0 constant on S_0 . But then G_0 has property AT (8.7 iii)), and consequently (8.10) the functor of subtori of G_0 is representable by an S_0 -prescheme smooth and separated over S_0 , whence the asserted property for G . As for the functor $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, G)$, one proceeds analogously.

Generalization of 8.9. Since the functor $\mathcal{T}$ of subtori of a smooth group G is not necessarily representable, we shall state sufficient conditions for a subfunctor of $\mathcal{T}$ to be representable.

Proposition 8.12. *Let S be a locally noetherian prescheme, G a group S -prescheme, smooth over S , with connected fibers, and $\mathcal{F}$ a sub- S -functor of the functor $\mathcal{T}$ of subtori of G , satisfying the following properties:*

- i) $\mathcal{F}$ is a sheaf for the fpqc topology (Exp. IV).
- ii) $\mathcal{F}$ commutes with inductive limits of rings (Exp. XI 3.2).
- iii) $\mathcal{F}$ commutes with adic limits of artinian local rings (Exp. XI 3.3).
- iv) $\mathcal{F}$ is infinitesimally smooth over S (Exp. XI 1.8).
- v) $\mathcal{F}$ is stable under inner automorphisms of G , i.e. for every S -prescheme S' , one has

$$T \in \mathcal{F}(S') \text{ and } g \in G(S') \Rightarrow \mathrm{int}(g)T \in \mathcal{F}(S').$$

Then $\mathcal{F}$ is representable by an S -prescheme smooth and separated over S .

Proposition 8.12 bis. *Let S and G be as above, T an S -torus and $\mathcal{F}$ a subfunctor of $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, G)$, satisfying the following properties:*

- i), ii), iii), iv) as above.
- v) $\mathcal{F}$ is stable under inner automorphisms of G , i.e. for every S -prescheme S' , one has

$$u \in \mathcal{F}(S') \text{ and } g \in G(S') \Rightarrow \mathrm{int}(g)u \in \mathcal{F}(S').$$

Then $\mathcal{F}$ is representable by an S -prescheme smooth and separated over S .

Proof of 8.12. (The proof of 8.12 bis is entirely analogous and is left to the reader.)

Let $u : \mathcal{F} \rightarrow \mathcal{CT}$ (7.1 i)) be the S -morphism associating to every element T of $\mathcal{F}(S')$ the element $\underline{\mathrm{Centr}}_{G_{S'}}(T)$ of $\mathcal{CT}(S')$. Since $\mathcal{CT}$ is representable by a smooth quasi-projective S -prescheme (7.1 i)), we are reduced to proving representability of u . After the base change $\mathcal{CT} \rightarrow S$, we are reduced to the following problem: given S and G as above, H a smooth subgroup of G with connected fibers, we must represent the functor $\mathcal{F}_H$ such that $\mathcal{F}_H(S') =$ set of elements $T \in \mathcal{F}(S')$ such that

$$H_{S'} = \underline{\mathrm{Centr}}_{G_{S'}}(T).$$

We shall show that $\mathcal{F}_H$ is étale and separated over S . To do so, it suffices for us to verify the five conditions of 8.10.

Conditions i), ii) and iii) of 8.10 follow easily from 8.12 i), ii) and iii). It has already been remarked that $\mathcal{TC}_H$ is a separated infinitesimally étale functor, so $\mathcal{F}_H$, which is a subfunctor of $\mathcal{TC}_H$, is separated and infinitesimally unramified (Exp. XI 1.8). It therefore

suffices for us to show that $\mathcal{F}_H$ is infinitesimally smooth (loc. cit.). Let S be the spectrum of an artinian local ring, S_0 a subscheme of S defined by a nilpotent ideal, T_0 an element of $\mathcal{F}_H(S_0)$; let us prove that T_0 lifts to an element T of $\mathcal{F}_H(S)$. By hypothesis (8.12 iv)), T_0 lifts to an element T' of $\mathcal{F}(S)$. On the other hand, H being smooth, T_0 lifts to a subtorus T'' of H_S (Exp. IX 3.6 bis) conjugate to T' by an element $g \in G(S)$ (loc. cit.), so $T'' \in \mathcal{F}(S)$ (8.12 v)). Since $\text{Centr}_{G_S}(T'')$ is smooth over S and coincides with H_{S_0} over S_0 , T'' is in the center of H_S (Exp. IX 5.6 a)), and its centralizer in G is equal to H_S ; in brief, $T'' \in \mathcal{F}_H(S)$, and one takes $T = T''$.

Corollary 8.13. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, and let $\mathcal{T}\mathcal{D}$ be the subfunctor of $\mathcal{T}$ whose set of points with values in an S -prescheme S' is the set of subtori T of $G_{S'}$ such that, for every point s' of S' , $T_{s'}$ is contained in the derived subgroup (Exp. VIB 7) of $G_{s'}$. Then $\mathcal{T}\mathcal{D}$ is representable by an S -prescheme smooth and separated over S .*

Corollary 8.13 bis. *Let S and G be as above, T an S -torus, and let*

$$\underline{\text{Hom}}_{S\text{-gr}}^{\text{der}}(T, G)$$

be the subfunctor of $\underline{\text{Hom}}_{S\text{-gr}}(T, G)$ whose set of points with values in S' is the set of S' -morphisms $u : T_{S'} \rightarrow G_{S'}$ such that, for every point s' of S' , $u_{s'}$ factors through the derived group of $G_{s'}$. Then this functor is representable by an S -prescheme smooth and separated over S .

Corollary 8.13 and Corollary 8.13 bis are proved analogously; let us prove, for example, 8.13 bis. By the usual procedure, we reduce to the case where S is noetherian. Let us verify that the five conditions of 8.12 bis are satisfied:

Conditions i) and iv) follow immediately from the corresponding properties of the functor $\underline{\text{Hom}}_{S\text{-gr}}(T, G)$. Condition v) is satisfied since the derived group of an algebraic group is invariant (Exp. VIB § 7). To establish ii), we are reduced by a standard reduction (EGA IV 8) to proving that if S is a noetherian integral scheme with generic point η and $u : T \rightarrow G$ is a group S -morphism which on the generic fiber factors through the derived group of G_η , then there exists a neighborhood U of η such that, for every point s of U , u_s factors through the derived group of G_s . But this follows immediately from Exp. VIB 10.12. To establish iii), let us return to the notation of 4.3. To show that an element $(u_m)_{m \in \mathbf{N}}$ of $\varprojlim_m \underline{\text{Hom}}_{S_m\text{-gr}}^{\text{der}}(T_m, G_m)$ is “admissible” in the sense of 4.3 and comes from an element of $\underline{\text{Hom}}_{S\text{-gr}}^{\text{der}}(T, G)$, one immediately reduces to the case where S is the spectrum of a complete discrete valuation ring (cf. 8.1). Let t be the generic point and s the closed point of S , D the schematic closure in G of the derived group D_t of G_t .

a) D_s contains the derived group of G_s . Indeed, since D_t is invariant in G_t and G is flat over S , D is invariant in G . Moreover, the morphism

$$G \times_S G \rightarrow G, \quad (x, y) \mapsto xyx^{-1}y^{-1}$$

factors through D_t on the generic fiber, so it factors through D . Consequently the algebraic group G_s/D_s is commutative, whence assertion a).

b) If $u_m \in \underline{\text{Hom}}_{S_m\text{-gr}}^{\text{der}}(T_m, G_m)$, u_m factors through D_m . Indeed, by hypothesis, u_s factors through the derived group of G_s , hence a fortiori factors through D_s , by a). Since D_m is flat over S_m and invariant in G_m , the quotient group $H_m = G_m/D_m$ is representable (Exp. VIA § 4). Since the image of T_m in H_m is a torus (Exp. IX 6.8) whose closed fiber is zero, the image of T_m is zero; this says that u_m factors through D_m .

c) The family $(u_m)_{m \in \mathbf{N}}$ is “admissible” and lifts to a morphism $T \rightarrow G$ belonging to $\underline{\text{Hom}}_{S\text{-gr}}^{\text{der}}(T, G)$. In view of the foregoing and 4.1 bis, it suffices to prove that D_t is an affine algebraic group, which follows from the following lemma:

Lemma 8.14. *Let G be a smooth connected algebraic group defined over a field k . Then the derived group D of G is affine.*

Since formation of D commutes with extension of the base field (Exp. VIB 7), one may suppose k algebraically closed. But then G is an extension of an abelian variety A by a linear group L . Since A is commutative, D is necessarily contained in L , hence is affine.

Maximal tori.

Theorem 8.15. *Let S be a locally noetherian prescheme and G a group S -prescheme smooth and of finite type over S . Then the following conditions are equivalent:*

- i) *The S -functor $\mathcal{T}\mathcal{M}$ whose set of points with values in an S -prescheme S' is the set of maximal tori of $G_{S'}$ (Exp. XII 1.3) is representable.*
- ii) *The preceding functor $\mathcal{T}\mathcal{M}$ is representable by an S -prescheme smooth and quasi-projective over S , with affine fibers.*
- iii) *The group G has, locally for the étale topology, a maximal torus.*
- iv) *The group G has, locally for the faithfully flat topology, a maximal torus.*
- v) *The group G has property AT (8.4), and the reductive rank of its fibers is a locally constant function on S .*

Proof: ii) $\Rightarrow$ i) is clear.

i) $\Rightarrow$ iii). Indeed, since G is of finite presentation over S , it follows from 6.3 that $\mathcal{T}\mathcal{M}$ commutes with inductive limits of rings, hence is locally of finite presentation if it is representable (EGA IV 8.14). Moreover, $\mathcal{T}\mathcal{M}$ is formally smooth (Exp. XI 2.1 bis). Thus if it is representable, it is representable by a prescheme smooth over S , and iii) then follows from Hensel's lemma (Exp. XI 1.10).

iii) $\Rightarrow$ iv) is clear.

iv) $\Rightarrow$ v). Let s be a point of S . By hypothesis, there exists an S -prescheme S' , flat over S , whose image contains s , such that $G_{S'}$ has a maximal torus T' . Let s' be a point of S' above s . The reductive rank of the fibers of $G_{S'}$ is therefore constant on $\text{Spec } \mathcal{O}_{S',s'}$, and consequently the reductive rank of the fibers of G is constant on the image of $\text{Spec } \mathcal{O}_{S',s'}$, which is $\text{Spec } \mathcal{O}_{S,s}$ (EGA IV 2.3.4 ii)). Now the reductive rank of the fibers of G is a locally constructible function on S (6.3 bis), hence this rank is constant on a neighborhood of s (EGA IV 1.10.1).

To see that G has property AT, consider an S -scheme S_1 , the spectrum of a discrete valuation ring, the closed point s_1 of S_1 mapping to the preceding point s . The prescheme $S'_1 = S_1 \times_S S'$ is faithfully flat over S_1 , and $G_{S'_1}$ has a maximal torus. Let A (resp. A') be the ring of S_1 (resp. S'_1). Considering A' as the inductive limit of its A -subalgebras of finite type, it follows from 6.3 that there exists an S_1 -scheme S''_1 such that $G_{S''_1}$ has a maximal torus and such that the structural morphism $S''_1 \rightarrow S_1$ is of finite type and surjective. Using now EGA II 7.1.9, we may suppose that S'_1 is the spectrum of a discrete valuation ring. But then it is clear that $G_{S'_1}$, hence also G_{S_1} , has property AT. Since this is true for every S -prescheme S_1 , the spectrum of a discrete valuation ring, G has property AT.

v) $\Rightarrow$ i). Indeed, by 8.9 the functor $\mathcal{T}$ of subtori of G is representable, and it is clear that $\mathcal{T}\mathcal{M}$ is representable by the subprescheme, both open and closed, of $\mathcal{T}$ representing the subfunctor of tori of rank r , where r denotes the reductive rank of G (which one may suppose constant). iii) $\Rightarrow$ ii). Indeed, if condition iii) holds, we may use the results of Exp. XII 7.1. The functor $\mathcal{T}\mathcal{M}$ is therefore canonically isomorphic to the functor of Cartan subgroups of G , and it suffices to apply 7.3 i).

Remark 8.16. One can show that the prescheme $\mathcal{T}\mathcal{M}$ of maximal tori of G is affine over S^* , which generalizes Exp. XII 5.4.

* Cf. M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes* (thesis, to appear)(N.D.E.: see Lecture Notes Math. 119 (1970), Springer), especially IX 2.9.

Corollary 8.17. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S . Suppose that the abelian rank and the reductive rank of the fibers of G are locally constant functions on S ; then G satisfies the (equivalent) properties i) to iv) of 8.15.*

We may suppose that the abelian rank and the reductive rank of the fibers of G are constant. Proceeding as in 8.11 and taking account of 6.3 bis, one reduces to the case where S is noetherian. But then G has property AT (8.7), and one concludes by 8.15 v).

Corollary 8.18. *Let S be a prescheme, G a group S -prescheme smooth over S . Then the following conditions are equivalent:*

- a) *The unipotent rank ρ_u and the abelian rank ρ_{ab} (6.1 ter) of the fibers of G are locally constant functions on S .*
- b) *The unipotent rank ρ_u is locally constant, and G has, locally for the fpqc topology, maximal tori.*
- c) *The reductive rank ρ_r (6.1 ter) and the abelian rank ρ_{ab} of the fibers of G are locally constant functions on S .*

Remarks 8.19. Under the hypotheses of 8.18, a more precise argument, using the lower semicontinuity of the abelian rank (announced in Exp. X 8.7), shows that if two of the three ranks ρ_u , ρ_r , ρ_{ab} are locally constant, then the same holds for the third.

Proof of 8.18. Replacing G by its identity component, we may suppose G of finite presentation over S (Exp. VIB 5.3.3).

a) $\Rightarrow$ c). Let s be a point of S . Since ρ_{ab} is locally constant, it follows from 8.11 that, after making an étale extension covering S , one may suppose that there exists a subtorus T of G whose fiber T_s is a maximal torus of G_s . Let $C = \underline{\text{Centr}}_G(T)$, which is a group subprescheme of G , smooth over S , with connected fibers. For every point t of S , C_t plainly contains a Cartan subgroup C'_t of G_t . After restricting S , one may suppose ρ_u , ρ_{ab} and the relative dimension of C over S constant.

One then has the inequalities

$$\begin{aligned} \dim C_s = \dim C_t &\geq \dim C'_t \geq \rho_u(t) + \rho_{ab}(t) + \dim T_t \\ &= \rho_u(s) + \rho_{ab}(s) + \dim T_s = \rho_\nu(s) = \dim C_s. \end{aligned}$$

One deduces that $C_t = C'_t$, hence that T is a maximal torus of G , and a fortiori $\rho_r(t) = \rho_r(s)$.

c) $\Rightarrow$ b) by 8.17.

b) $\Rightarrow$ a). Indeed, since G has, locally for the fpqc topology, maximal tori, it has, locally for the fpqc topology, Cartan subgroups, and consequently (Exp. XII 7.3) the nilpotent rank $\rho_\nu = \rho_u + \rho_r + \rho_{ab}$ is locally constant. Since ρ_r and ρ_{ab} are locally constant, ρ_u is locally constant.

Exposé XVI. Groups of unipotent rank zero

by M. Raynaud*

1. An immersion criterion

1.1. — Examples of monomorphisms of group preschemes which are not immersions. We shall construct a prescheme S , two group S -preschemes G and H , and a group S -monomorphism $u : G \rightarrow H$ which is not an immersion. The groups G and H will be of finite presentation over S , flat over S , and G will even be smooth over S (cf. Exp. VIII, the paragraph and note (*) preceding 7.1, see also XVII App. III, 4).

First take for S the spectrum of a discrete valuation ring A of mixed characteristic and with residue characteristic equal to 2. Let t be the generic point of S and s the closed point. a) Take for G the open subgroup of the constant group $G' = (\mathbf{Z}/2\mathbf{Z})_S$ obtained by removing the point of the closed fiber $(\mathbf{Z}/2\mathbf{Z})_s$ distinct from the identity element, and take for H the group of multiplicative type $(\mu_2)_S$ of second roots of unity (Exp. I 4.4.4). In view of the choice of S , H_t is isomorphic to $(\mathbf{Z}/2\mathbf{Z})_t$, whereas H_s is a radicial group. One has an evident morphism $G' \rightarrow H$, defined by the section (-1) of H , whence a morphism $u : G \rightarrow H$, which is a monomorphism but is not an immersion (otherwise it would necessarily be an isomorphism (cf. Exp. VIII 7)). Here Exp. VIII 7.9 does not apply, since ${}_2G = G$ is not finite over S .

b) Denote by K the étale nonseparated group S -prescheme obtained from the unit group by “doubling” the unique point of the closed fiber. Take for H the product group $(\mu_2)_S \times_S K$. Let a (resp. b) be the unique section of μ_2 (resp. K) over S distinct from the unit section, and let $h = (a, b)$ be the corresponding section of H . The datum h defines a group S -morphism

$$u : G = (\mathbf{Z}/2\mathbf{Z})_S \rightarrow H.$$

It is clear that u is a monomorphism, and it is not an immersion. Here Exp. VIII 7.9 does not apply, since ${}_2H = H$ is not separated over S .

c) More interesting is the following example, where G is a smooth group with connected fibers (and even $G = (\mathbf{G}_a)_S$) (cf. Exp. VIII 7.10).

Take for S the spectrum of a discrete valuation ring A of equal characteristic 2, and let π be a uniformizer of A , $\mathfrak{m}$ the maximal ideal of A .

Consider the additive group $(\mathbf{G}_a)_S$, the product group $(\mathbf{G}_a \times \mathbf{G}_a)_S$ with ring $A[X, Y]$, and let G_1 be the closed subgroup with equation

$$(1) \quad X^2 + X - \pi Y = 0.$$

The group G_1 is smooth over S , its generic fiber is isomorphic to $(\mathbf{G}_a)_t$, and its closed fiber is isomorphic to the product of $(\mathbf{G}_a)_s$ with $(\mathbf{Z}/2\mathbf{Z})_s$. One may take as factor $(\mathbf{Z}/2\mathbf{Z})_s$ the subgroup N_s of $(G_1)_s$ with equation

$$(2) \quad Y = X^2 + X = 0.$$

Let N' and N'' be two group subschemes of G_1 , isomorphic to $(\mathbf{Z}/2\mathbf{Z})_S$, whose closed fibers coincide with N_s and whose generic fibers are distinct. The groups N' and N'' are therefore

* Cf. the note on page 1 of Exp. XV.

⁰xy version of 1/12/08

defined by the datum of two sections of G_1 over S , with coordinates (a', b') (resp. (a'', b'')), such that

$$(3) \quad \begin{cases} a'^2 + a' - \pi b' = a''^2 + a'' - \pi b'' = 0, \\ a' \text{ and } a'' \equiv 1 \pmod{\mathfrak{m}}, \quad a' \neq a'', \\ b' \text{ and } b'' \equiv 0 \pmod{\mathfrak{m}}. \end{cases}$$

(One may take, for example, $(1, 0)$ and $(1 + \pi^2, \pi + \pi^3)$.)

The quotient groups $G' = G_1/N'$ and $G'' = G_1/N''$ are representable (Exp. V 4.1). Let u be the canonical morphism

$$u : G_1 \longrightarrow G' \times_S G'',$$

and let H be the group subscheme of $G' \times_S G''$ equal to the schematic closure in $G' \times_S G''$ of $u_t(G_1)_t$, so that u factors through H . The kernel of u is $K = N' \cap N''$, whose generic fiber is the unit group and whose closed fiber is equal to N_s ; in particular, K is not flat over S . On the generic fiber, u_t is therefore an isomorphism of $(G_1)_t$ onto H_t . On the other hand, I say that H_s is not smooth. Indeed, if H_s were smooth, since u_s is a morphism with finite kernel and $(G_1)_s$ and H_s have the same dimension (namely 1), u_s would be a flat morphism; since G_1 and H are flat, u would be a flat morphism, and consequently $\text{Ker } u$ would be flat over S , which it is not. It is clear that the restriction of u to the identity component G of G_1 is a monomorphism (the fibers of $K \cap G$ are unit groups, so $K \cap G$ is the unit group (Exp. VIB 2.9)), but it is not an immersion (otherwise it would be an open immersion, and H_s would be smooth). Here Exp. VIII 7.9 does not apply, since ${}_2G = G$ is not finite over S .

For enthusiasts of equations, let us say that in the above example, one may take for H the closed subgroup of $(\mathbf{G}_a \times_S \mathbf{G}_a)_S$ with ring $A[V, W]/(F)$, where

$$F = (a''b' - b''a')(a''^2V - a'^2W) - (a''V - a'W)^2$$

(where a', a'', b', b'' satisfy (3)). For G , one takes the group $(\mathbf{G}_a)_S$ with ring $A[T]$, and as morphism $u : G \longrightarrow H$, the S -morphism defined by the maps

$$V \longmapsto a'(a'T + \pi T^2), \quad W \longmapsto a''(a''T + \pi T^2).$$

Remark 1.2. The preceding construction is inspired by the method of Koizumi–Shimura⁽¹⁾. It does not apply when S is the spectrum of a ring of mixed characteristic, the points of finite order of G then being “too rigid”.

1.3. — Statement of the immersion criterion.

Theorem 1.3. *Let S be a locally noetherian prescheme, G a group S -prescheme smooth over S , of finite type, having locally for the fpqc topology Cartan subgroups (Exp. XV 6.1 and 7.3 (i)), H a group S -prescheme locally of finite type, $u : G \longrightarrow H$ a group S -monomorphism. Then:*

a) *If G has connected fibers, for u to be an immersion, (it is necessary and) it suffices that for every S -scheme S' which is the spectrum of a complete discrete valuation ring with algebraically closed residue field, and for every Cartan subgroup C of $G_{S'}$, the restriction of $u_{S'}$ to C be an immersion.*

b) *For u to be an immersion (resp. a closed immersion), (it is necessary and) it suffices that for every S' as above and every Cartan subgroup C of the identity component $(G_{S'})^0$ of $G_{S'}$, the restriction of $u_{S'}$ to the normalizer (Exp. XI 6.11) N of C in $G_{S'}$ be an immersion (resp. a closed immersion).*

Before proving 1.3, let us state a few applications:

Corollary 1.4. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation over S , of unipotent rank (Exp. XV 6.1 ter) zero, and having, locally for the*

⁽¹⁾**N.D.E.** S. Koizumi & G. Shimura, *Specialization of abelian varieties*, Sci. Papers Coll. Gen. Ed. Univ. Tokyo 9 (1959), 187–211.

fpqc topology, maximal tori, H a group S -prescheme, $u : G \rightarrow H$ a group S -monomorphism. Suppose moreover either that H is of finite presentation over S , or that S is locally noetherian and H locally of finite type. Then:

- a) If G has connected fibers, u is an immersion.*
- b) If H is separated over S and if for every S' over S and every maximal torus T of $G_{S'}$, the Weyl group*

$$W = \underline{\mathrm{Norm}}_{G_{S'}}(T) / \underline{\mathrm{Centr}}_{G_{S'}^0}(T)$$

is representable (a condition always satisfied if G has connected fibers (Exp. XV 7.1 (iv))) and is finite over S , then u is a closed immersion.

Corollary 1.5. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation over S , with connected fibers, H a group S -prescheme, $u : G \rightarrow H$ a group S -monomorphism. Suppose either that H is of finite presentation over S , or that S is locally noetherian and H locally of finite type. Then:*

- a) If G is reductive, i.e. affine over S with reductive fibers (cf. Exp. XIX), u is an immersion, and a closed immersion if H is separated.*
- b) If G has affine solvable connected fibers of unipotent rank zero, u is an immersion, and a closed immersion if H is separated.*

Proof of 1.4 from 1.3.

Reduction to the case where S is noetherian. If S is locally noetherian, the reduction is immediate, the properties to be proved being local on S . In the second case, G and H are of finite presentation over S . After restricting S , we may suppose S affine. By Exp. VIB § 10, there exist a noetherian scheme S_0 , a group S_0 -prescheme G_0 , smooth over S_0 (with connected fibers in case a)), of finite type over S_0 , a group S_0 -prescheme H_0 of finite type (separated in case b)), a group S_0 -morphism

$$u_0 : G_0 \rightarrow H_0,$$

such that G, H, u are obtained from G_0, H_0, u_0 by an extension of the base $S \rightarrow S_0$. The fact that u is a monomorphism means that $\mathrm{Ker} u = \text{unit group}$; one may therefore suppose that u_0 is a monomorphism. Since G has unipotent rank ρ_u zero and has, locally for the fpqc topology, maximal tori, the abelian rank ρ_{ab} of the fibers of G is locally constant (Exp. XV 8.18). But ρ_u and ρ_{ab} are locally constructible functions (Exp. XV 6.3 bis). A standard argument (cf. EGA IV 8.3.4) shows that one can choose S_0 and G_0 so that the unipotent rank (resp. the abelian rank) of the fibers of G_0 is zero (resp. locally constant). But then G_0 has locally maximal tori for the fpqc topology (Exp. XV 8.18), and the functor $\mathcal{TM}_{G_0}$ of maximal tori of G_0 is representable by an S_0 -scheme X_0 of finite type over S_0 (Exp. XV 8.15). Let T_0 be the “universal” maximal torus of $(G_0)_{X_0}$, X the S -prescheme $X_0 \times_{S_0} S$, $T = T_0 \times_{S_0} S$ the universal maximal torus for G . By hypothesis, in case b), the Weyl group corresponding to T is representable and finite over X . These two properties are compatible with projective limits of preschemes (Exp. VIB 10.1 iii) and EGA IV 8.10.5). One can therefore choose S_0 so that the Weyl group of T_0 is finite over X_0 . It is clear under these conditions that, to prove 1.4, one may replace S, G, u, H by S_0, G_0, u_0, H_0 , hence suppose S noetherian.

Let us use the valuative criterion provided by 1.3. We are reduced to the case where S is the spectrum of a discrete valuation ring and to the case where G^0 has a Cartan subgroup C . *Proof of 1.4 a).* We must show that the restriction of u to C is an immersion (1.3 a)), which reduces us to the case where $G = C$ has connected nilpotent fibers. Standard reductions (cf. Exp. VIII proof of 7.1) reduce us to the case where H is flat over S , u is an isomorphism on the generic fiber, and u is an isomorphism of the underlying spaces on the closed fiber. The group H then has connected fibers, hence is separated over S (Exp. VIB 5.2). Let us admit for a moment the lemma:

Lemma 1.6. *Let S be a prescheme, G a group S -prescheme smooth over S , with connected nilpotent fibers of unipotent rank zero. Then:*

- i) G is commutative.*
- ii) For every integer $n > 0$, ${}_nG$ is a group prescheme flat and quasi-finite over S , finite over S if and only if the abelian rank, or the reductive rank, of G is locally constant on S .*
- iii) For every integer $q > 0$ invertible on S , the family of subgroups ${}_qG$ is universally schematically dense in G relative to S (EGA IV 11.10.8).*

Lemma 1.6 applies to the group G , and since G has locally maximal tori, the reductive rank of the fibers of G is locally constant, so (1.6 ii)), for every integer $n > 0$, ${}_nG$ is finite over S . The fact that u is an immersion then follows from Exp. VIII 7.9.

Proof of 1.6. Let us first examine the case where S is the spectrum of a field:

Lemma 1.7. *Let k be a field of characteristic p , G an algebraic k -group, smooth, connected, nilpotent, of unipotent rank zero. Then G is a commutative group, an extension of an abelian variety A by a torus T . For every integer $n > 0$, ${}_nG$ is a finite group, étale if $(n, p) = 1$, defined by a k -algebra of rank $n^{\rho_r + 2\rho_{ab}}$. If $(q, p) = 1$, the family of subgroups ${}_qG$ ($q \in \mathbf{N}$) is schematically dense in G .*

Proof of 1.7. Let Z be the center of G . The group G/Z is affine (Exp. XII 6.1), of unipotent rank zero, smooth and connected, hence is a torus (BIBLE 4 Th. 4), but then G is commutative (Exp. XII 6.4). If T is the unique maximal torus of G (Exp. XV 3.4), it follows immediately from Chevalley's theorem (Sém. Bourbaki 1956/57, no. 145) that G is an extension of an abelian variety A by T . For every integer $n > 0$, raising to the n th power is an epimorphism in T , and one deduces an exact sequence

$$0 \longrightarrow {}_nT \longrightarrow {}_nG \longrightarrow {}_nA \longrightarrow 0.$$

A classical theorem of Weil (A. Weil: *Variétés abéliennes et courbes algébriques*, § IX Th. 33 Cor. 1) tells us that ${}_nA$ is a finite group defined by a k -algebra of rank $n^{2\rho_{ab}}$. Since ${}_nT$ is a finite group of rank n^{ρ_r} , one deduces the asserted structure of ${}_nG$.

Let then H be the smallest closed subscheme of G containing ${}_qG$ for every q . It follows from the foregoing and Exp. XV 4.6 that H is a smooth connected subgroup, hence of the same type as G . Raising to the n th power in H is an epimorphism (since ${}_qH$ is finite), so that one has the exact sequence

$$0 \longrightarrow {}_qH \longrightarrow {}_qG \longrightarrow {}_q(G/H) \longrightarrow 0.$$

It follows that ${}_q(G/H) = 0$, hence $G/H = 0$. This says that the subgroups ${}_qG$ are schematically dense in G .

Continuation of the proof of 1.6. i). To show that G is commutative, one reduces by the usual procedure to the case where S is noetherian, then to the case where S is the spectrum of a local ring with closed point s . The center of G is representable by a closed group subscheme Z of G (Exp. XI 6.11). To show that $Z = G$, it suffices to show that $Z = G$ after reduction by every power of the maximal ideal of the ring of S (for Z will then be an open subgroup of G , hence will be equal to G , since G has connected fibers). This reduces us to the case where S is local artinian.

Let q be an integer invertible on S . For every integer n , ${}_qG_s$ is an étale central subgroup of multiplicative type of G_s (1.7). Since G is smooth, ${}_qG_s$ lifts to an étale central group S -subscheme M_n of G (Exp. XV 1.2). The family of flat subgroups M_n , $n \in \mathbf{N}$, is then schematically dense in G (1.7 and EGA IV 11.10.9). Since Z is closed in G and contains all the M_n , one indeed has $Z = G$.

1.6 ii). To see that ${}_nG$ is flat and quasi-finite over S , it suffices to show that raising to the n th power in G is a flat quasi-finite morphism. Since G is flat over S , one reduces to the case where S is the spectrum of a field (EGA IV 11.3.10), and it was remarked in the proof of 1.7 that raising to the n th power was an epimorphism. Moreover, ${}_nG$ is separated over S , since G is separated over S (Exp. VIB 5.2). For ${}_nG$ to be finite over S , it is then necessary and

sufficient that the fibers of ${}_nG$ be spectra of finite algebras of locally constant rank (one sees this immediately by reducing to the case where S is the spectrum of a henselian local ring). But by 1.7, this condition is equivalent to saying that the abelian rank, or the reductive rank, of the fibers of G is locally constant.

1.6 iii). To see that the family of subgroups ${}_nG$ ($n \in \mathbf{N}$) is universally schematically dense in G , one again reduces to the case where S is noetherian. Taking account of 1.7 and 1.6 b), it then suffices to apply EGA IV 11.10.9.

Proof of 1.4 b). We have reduced to the case where S was the spectrum of a discrete valuation ring and to the case where G^0 had a Cartan subgroup C . Let $N = \underline{\text{Norm}}_G C = \underline{\text{Norm}}_G T$, where T is the unique maximal torus of C (Exp. XII 7.1 a) and b)). Since H is separated over S , it follows from 1.6 ii) and Exp. VIII 7.12 that the restriction of u to C is a closed immersion. On the other hand, to prove that u is a closed immersion, it suffices to show that this holds for $u|_N$ (1.3 b)). Since by hypothesis $W = N/C$ is representable by a finite group scheme, this will follow from the following lemma, applied to the exact sequence

$$0 \longrightarrow C \longrightarrow N \longrightarrow W \longrightarrow 1$$

(note that a proper immersion is a closed immersion).

Lemma 1.8. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , an extension of a group prescheme G'' , proper and of finite presentation over S , by a group prescheme G' , of finite presentation and flat over S :*

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1.$$

On the other hand, let H be a group S -prescheme of finite presentation over S (or locally of finite presentation if S is locally noetherian) and $u : G \longrightarrow H$ a group S -morphism. Then if the restriction of u to G' is proper, u is proper.

Proof of 1.8. We reduce as usual to the case where S is noetherian (Exp. VIB § 10 and Exp. XV 6.2), and we can then apply the valuative criterion of properness (EGA II 7.3.8). We therefore suppose that S is the spectrum of a discrete valuation ring A , with closed point s and generic point t . Let $\bar{x} \in G(t)$ and $h \in H(S)$ be such that $u_t(\bar{x}) = h(t)$. We must show that $\bar{x}$ comes from a unique element x of $G(S)$. It even suffices to prove existence and uniqueness of x after a faithfully flat extension of the discrete valuation ring A . Now let $\bar{y}$ be the projection of $\bar{x}$ in $G''(t)$. Since G'' is proper over S , $\bar{y}$ comes from a unique element y of $G''(S)$. Let X be the inverse image of y in G . The prescheme X is faithfully flat over S (since G' is faithfully flat over S , as is the morphism $G \longrightarrow G''$ (Exp. VIB § 9)). After making a faithfully flat change of discrete valuation ring, one may suppose that X has a section ℓ over S (EGA IV 14.5.8). Replacing $\bar{x}$ by $\bar{x}\ell_t^{-1}$, one may suppose that $\bar{x} \in G'(t)$. But then existence and uniqueness of x follow from the fact that $u|_{G'}$ is proper.

Proof of 1.5. a) We shall see in Exp. XIX that if G is reductive, G has locally for the faithfully flat topology maximal tori, has a finite Weyl group and unipotent rank zero. Assertion a) follows, taking account of 1.4.

b) If G has affine fibers of unipotent rank zero, G has maximal tori locally for the fpqc topology (Exp. XV 8.18). It then suffices to apply 1.4, taking account of the fact that, since G has connected smooth affine solvable fibers, its Weyl group is the unit group (BIBLE 6 Th. 1 Cor. 3).

Proof of 1.3 a). Since the morphism u is already a monomorphism, to see that u is an immersion it suffices to show that u is proper at the points of $u(G)$ (EGA IV 15.7.1), and for this we can use the valuative criterion of local properness (EGA IV 15.7.5). We are therefore reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with closed point s and generic point t . Since G is flat over S , standard reductions (cf. Exp. VIII proof of 7.1) allow us to reduce to the case where H is flat over S , u_t is an isomorphism and u_s is an isomorphism on the underlying spaces. To say that u is an immersion is then equivalent to saying that u is an isomorphism.

By hypothesis, the group G has locally Cartan subgroups for the fpqc topology, so we can speak of the open subset U of regular points of G (Exp. XV 7.3 i) and ii)).

Lemma 1.9. *With the preceding hypotheses, for u to be an immersion, it suffices that $u|_U$ be an immersion.*

Indeed, to say that $u|_U$ is an immersion means that there exist an open subset V of H and a closed subset F of V such that $u|_U$ factors through F and induces an isomorphism of U onto F . Since H_t , hence also V_t , is irreducible, one necessarily has $V_t = F_t$. But then F_t contains the schematic closure of V_t in V , which is equal to V , since H is flat over S . In brief, one has $V = F$. It follows that $u|_U$ is an open immersion, hence u is an open immersion (VIB 2.6).

It therefore remains for us to show that $u|_U$ is an immersion, and for this let us apply the valuative criterion of local properness. Replacing S by the spectrum of a discrete valuation ring faithfully flat over S , we must show that if h is a section of H over S whose image $h(S)$ is contained in $u(U)$, then h is the image of a section g of U over S . It suffices to show that h is contained in the image of a Cartan subgroup C of G . Indeed, by hypothesis, $u|_C$ is an immersion, so h is the image of a section g of C , which is necessarily a section of U .

Let $a \in U(s)$ (resp. $b \in U(t)$) be such that $u_s(a) = h(s)$ (resp. $u_t(b) = h(t)$). Since u_t is an isomorphism, $h(t)$ is a regular point of H_t , hence is contained in a unique Cartan subgroup D_t of H_t (Exp. XIII 3.2). Let D be the schematic closure of D_t in H .

Lemma 1.10. *i) D_s is a nilpotent algebraic group (Exp. VIB § 8).*

ii) $\dim D_s = \nu = \text{nilpotent rank of } G = \text{nilpotent rank of } (H_s)_{\text{red}}$.

iii) $h(s) \in D_s(\kappa(s))$.

i) Since H is separated (Exp. VIB 5.2), so is D . Moreover, D is flat over S , and its generic fiber is nilpotent. The fact that D_s is nilpotent then follows from Exp. VIB 8.4.

ii) Since $(H_s)_{\text{red}} \simeq G_s$, these two groups have the same nilpotent rank. The group G having locally Cartan subgroups for the fpqc topology, G_s and G_t have the same nilpotent rank ν . Finally, the group D being flat over S , one has $\dim D_s = \dim D_t = \nu$ (Exp. VIB § 4).

c) Since h_t factors through D_t , h plainly factors through D , whence iii).

This being so, the following lemma proves that $(D_s)_{\text{red}}$ is a Cartan subgroup of $(H_s)_{\text{red}} \simeq G_s$:

Lemma 1.11. *Let G be a smooth connected algebraic group defined over an algebraically closed field k , D a smooth nilpotent algebraic subgroup containing a regular element a of $G(k)$. Then D^0 is contained in a Cartan subgroup of G . If moreover $\dim D$ is equal to the nilpotent rank of G , then D is a Cartan subgroup of G (hence is connected).*

Let Z be the center of G , $G' = G/Z$, which is affine (Exp. XII 6.1), a' , D' the images of a , D in G' . It follows immediately from the correspondence between Cartan subgroups of G and Cartan subgroups of G' (Exp. XII 6.6 e)) that a' is a regular element of G' and that it suffices to prove the lemma for the pair D', G' ; this allows us to suppose G affine.

Let then s be the semisimple component of a , which belongs to $D(k)$ (BIBLE 4 Th. 3) and is regular (BIBLE 7 Th. 2 Cor. 1). By BIBLE 6 Th. 2, s centralizes D^0 , so D^0 is contained in the connected centralizer of s , which is a Cartan subgroup of G (BIBLE 7 Th. 2). If now $\dim D = \text{nilpotent rank of } G$, D^0 is therefore a Cartan subgroup of G , equal to $\text{Centr}_G(T)$, where T is the unique maximal torus of D^0 (Exp. XII 6.6). But D is nilpotent, hence centralizes T (BIBLE 6 Th. 2), so $D = D^0$.

Let us now work with the group G . The group $C_t = u_t^{-1}(D_t)$ is the unique Cartan subgroup of G_t containing b . Let C be the schematic closure of C_t in G .

By functoriality of schematic closure, $u|_C$ factors through D , so $u_s(C_s) \subset D_s$. Since u_s is an isomorphism of G_s onto $(H_s)_{\text{red}}$, $\dim C_s = \dim C_t = \nu = \dim D_s$ (1.10), and D_s is

connected (1.11), u_s provides an isomorphism of $(C_s)_{\text{red}}$ onto $(D_s)_{\text{red}}$, so $(C_s)_{\text{red}}$ is a Cartan subgroup of G_s . The following lemma shows that in fact C is a Cartan subgroup of G .

Lemma 1.12. *Let S be the spectrum of a discrete valuation ring, G a smooth group S -prescheme of finite type, C a group subscheme of G , flat over S , such that the generic fiber C_t is a Cartan subgroup of G_t and the reduced geometric closed fiber $(C_{\bar{s}})_{\text{red}}$ is a Cartan subgroup of $G_{\bar{s}}$. Then C is a Cartan subgroup of G .*

We must show that C is smooth over S , and it suffices to establish this after a faithfully flat extension of the base. The hypotheses on C imply that the nilpotent rank of the fibers of G is constant, and we can therefore speak of the open subset U of regular points of G (Exp. XV 7.3). Since $(C_{\bar{s}})_{\text{red}}$ is a Cartan subgroup of $G_{\bar{s}}$, $C_{\bar{s}}$ contains a regular point $a_{\bar{s}}$ of $G_{\bar{s}}$. But C is flat over S , so (EGA IV 14.5.8), after making a faithfully flat extension of the base, we may suppose that $a_{\bar{s}}$ is an element of $G(s)$ and lifts to a section a of $C(S)$. Since U is open and contains $a(s)$, U contains a . Let C' be the unique Cartan subgroup of G containing a (Exp. XV 7.3). One necessarily has $C_t = C'_t$, and consequently C' and C coincide with the identity component of the schematic closure of C_t in G (note that C' and C are flat over S). *End of the proof of 1.3 a).* By hypothesis, the restriction of u to the Cartan subgroup C of G is an immersion. It is clear that $h(S)$ is contained in $u(C) = D$, so h is the image of a section g of $C(S)$. Q.E.D.

Proof of 1.3 b). We shall again use the valuative criterion of local properness (EGA IV 15.5) in the case of an immersion (resp. the valuative criterion of properness (EGA II 7.3.6) in the case of a closed immersion). This reduces us to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. Let s be the closed point and t the generic point of S . Replacing H by the schematic closure in H of $u_t(G_t)$, we may suppose that H is flat over S and u_t is an isomorphism.

Let G^0 be the connected component of G . By 1.2 a), $u|_{G^0}$ is an immersion; denote by H' the group subscheme of H which is the image of G^0 . One therefore has $H'_s = (H_s)_{\text{red}}^0$.

To verify the valuative criterion, we must show that every section h of H over S such that $h(S)$ is contained in $u(G)$ in the case of an immersion (resp. every section h of H over S in the case of a closed immersion) is the image of a section g of G over S .

Let C be a Cartan subgroup of G , $D = u(C)$ the Cartan subgroup of H' which is the image of C . The S -group $D' = \text{int}(h)D$ has connected fibers, hence has underlying space contained in that of H' ; moreover, being smooth over S , it is reduced, and consequently D' is a smooth group subscheme of H' . The fibers of D' are Cartan subgroups of the fibers of H' , so D' is a Cartan subgroup of H' , and $C' = u^{-1}(D')$ is a Cartan subgroup of G . But $\text{Transp}_G^{\text{str}}(C, C')$ is smooth and surjective over S (Exp. XII 7.1 b)). Since S is henselian with algebraically closed residue field, there exists $g \in G(S)$ such that $\text{int}(g)C = C'$. Replacing h by $u(g)^{-1}h$, we may suppose that $h(t)$ normalizes D_t (and that $h(s)$ is the image of an element of the normalizer of C_s in G_s in the case of an immersion). Let $N = \text{Norm}_G(C)$. By hypothesis, $u|_N$ is an immersion (resp. a closed immersion), so h is indeed the image of a section of N over S , which completes the proof.

2. A theorem on representability of quotients

Let us “recall” the following result:

Theorem 2.1. *Let S be a prescheme, X and Y two S -preschemes, $f : X \rightarrow Y$ an S -morphism. Suppose that one is in one of the following two cases:*

- a) *The morphism f is locally of finite presentation.*
- b) *The prescheme S is locally noetherian, and X is locally of finite type over S .*

Then the following conditions are equivalent:

i) There exists an S -prescheme X' and a factorization of f

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S -morphism locally of finite presentation, and f'' is a monomorphism.

ii) The (first) projection

$$p_1 : X \times_Y X \longrightarrow X$$

is a flat morphism.

Moreover, if the preceding conditions hold, (X', f') is a quotient of X by the equivalence relation defined by f (for the fpqc topology), so that the factorization $f = f'' \circ f'$ of i) is unique up to isomorphism.

The proof of this delicate theorem will be found in EGA V; one may also consult the exposé of J.-P. Murre, Séminaire Bourbaki, May 1965, no. 294 Th. 2 Cor. 2, where the case of Y locally noetherian and X of finite type over Y is treated. We shall see that one can reduce to this case.

First let us make a few remarks:

a) i) $\Rightarrow$ ii) is trivial. Indeed, the first projection

$$p'_1 : X \times_{X'} X \longrightarrow X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism, since f'' is a monomorphism, and p'_1 is flat, since f' is flat, so p_1 is flat.

b) The assertions of 2.1 are local on Y (hence are local on S); they are also local on X , as follows easily from the fact that a flat morphism locally of finite presentation is open (EGA IV 11.3.1).

c) Under the hypotheses of 2.1 a), in view of the foregoing, we are reduced to the case where X and Y are affine and f is of finite presentation. Replacing S by Y , one may suppose X and Y of finite presentation over S . One then reduces to the case where S is noetherian by EGA IV 11.2.6.

d) Under the hypotheses of 2.1 b), one may suppose S, X, Y affine, S noetherian and X of finite type over S . Consider Y as a filtered projective limit of affine schemes Y_i of finite type over S . The schemes $X \times_{Y_i} X$ form a decreasing filtered family of closed subschemes of $X \times_S X$, whose projective limit is $X \times_Y X$. Since $X \times_S X$ is noetherian, one has $X \times_{Y_i} X = X \times_Y X$ for sufficiently large i , so $f_i : X \xrightarrow{f} Y \rightarrow Y_i$ satisfies the hypotheses of 2.1 ii) if f does. Since the equivalence relation defined by f on X coincides with that defined by f_i , it is clear that it suffices to prove ii) $\Rightarrow$ i) for f_i , which reduces us to the case where Y is of finite type over S .

Application to group preschemes

Theorem 2.2. *Let S be a prescheme, G a group S -prescheme locally of finite presentation over S , acting on an S -prescheme X . Let $\xi \in X(S)$ be a section of X over S such that the stabilizer H of ξ in G is a group subprescheme of G , flat over S . Then, if X is locally of finite type over S , or if S is locally noetherian, the quotient homogeneous space G/H is representable by an S -prescheme locally of finite presentation over S , and the S -morphism*

$$f : G \longrightarrow X, \quad g \longmapsto g \cdot \xi$$

factors as

$$\begin{array}{ccc} G & & \\ p \downarrow & \searrow f & \\ G/H & \xrightarrow{i} & X, \end{array}$$

where p is the canonical projection, which is a faithfully flat morphism locally of finite presentation, and i is a monomorphism. (To fix ideas, we have supposed that G acts on X on the left.)

Proof. The morphism f makes G an X -prescheme. By definition of the stabilizer of ξ , the morphism

$$G \times_S H \longrightarrow G \times_X G, \quad (g, h) \longmapsto (g, gh)$$

is an isomorphism. Since H is flat over S , $G \times_S H$ is flat over G , so the first projection $p_1 : G \times_S G \longrightarrow G$ is a flat morphism. Moreover, if X is locally of finite type over S , f is locally of finite presentation (EGA IV 1.4.3 v)), and otherwise S is supposed locally noetherian. It then suffices to apply 2.1 to the morphism f .

It remains to see that G/H is locally of finite presentation over S , but this follows immediately from Exp. V 9.1.

Corollary 2.3. *Let S be a prescheme, G and H two group S -preschemes, $u : G \longrightarrow H$ an S -homomorphism. Suppose G locally of finite presentation over S , and either H locally of finite type over S , or S locally noetherian. Then, if $K = \text{Ker}(u)$ is flat over S , the quotient group G/K is representable by a group S -prescheme locally of finite presentation over S , and u factors as*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ & \searrow p & \nearrow i \\ & G/K & \end{array}$$

where p is the canonical projection and i a monomorphism.

Proof: one applies 2.2, taking $X = H$ and for ξ the unit section of H .

Corollary 2.4. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , H a group S -prescheme smooth over S , with connected fibers (hence of finite presentation over S , by VIB 5.5), $i : H \longrightarrow G$ a monomorphism of S -groups, so that H is a subgroup of G .*

Suppose that $N = \underline{\text{Norm}}_G(H)$ (which is representable by a closed group subprescheme of G , of finite presentation over S (Exp. XI 6.11)) is flat over S . Then G/N is representable by an S -prescheme of finite presentation over S and quasi-projective over S .

All the assertions to be proved, except the last, are local on S . To establish them, we may therefore suppose S quasi-compact and the relative dimension of H over S constant and equal to r . Proceed as in XV § 5. For every integer $n > 0$, let $G^{(n)}$ (resp. $H^{(n)}$) be the n th normal invariant of the unit section of G (resp. of H) (EGA IV 16). The sheaf of $\mathcal{O}_S$ -modules $H^{(n)}$ is a quotient of $G^{(n)}$, and, H being smooth over S of dimension r , $H^{(n)}$ canonically defines an element ξ_n of $X_n = \text{Grass}_{\varphi(n,r)} G^{(n)}$ (EGA I 2nd ed. § 9) for a suitable integer $\varphi(n, r)$. On the other hand, G acts naturally on $G^{(n)}$ (hence also on X_n) through the representation

$$G \longrightarrow \text{Aut}_{S\text{-gr}}(G), \quad g \longmapsto \text{int}(g).$$

Since S is quasi-compact, there exists an integer m such that for $n \geq m$, one has

$$N = \underline{\text{Norm}}_G(H) = \underline{\text{Norm}}_G H^{(n)} \quad (\text{Exp. XI 6.11 b)).$$

This says that N is the stabilizer of ξ_n for large n . Representability of G/N therefore follows from 2.2. The fact that G/N is of finite presentation over S is a consequence of Exp. V 9.1. It remains to see that G/N is quasi-projective over S , and for this we shall exhibit a canonical S -ample invertible sheaf on G/N . Consider the functor $\mathcal{F} : ((\text{Sch})/S)^\circ \longrightarrow (\text{Ens})$

such that for every S -prescheme T , one has

$$\mathcal{F}(T) = \text{set of } T\text{-subgroups } H \text{ of } G_T, \text{ representable, smooth over } T, \\ \text{with connected fibers.}$$

Such a group H is of finite presentation over S (Exp. VIB 5.5), and $H \rightarrow G_T$ is a quasi-affine morphism (EGA IV 8.11.2). By effective descent of quasi-affine morphisms (SGA1 VIII 7.9), one deduces that $\mathcal{F}$ is a sheaf for the fpqc topology. Since G/N is the sheaf associated to a subsheaf of $\mathcal{F}$, one sees that there exists a canonical monomorphism $G/N \rightarrow \mathcal{F}$. There therefore exists a subgroup H' of $G \times_S (G/N)$, representable, smooth over G/N , with connected fibers, “universal” for the functor G/N . I say that the invertible sheaf $\mathcal{L} = (\det(\text{Lie } H'))^{-1}$ is S -ample. In this form, the assertion becomes local on S , and the proof is analogous to that given in (Exp. XV 5.8).

The following corollary was announced in Exp. XIV 4.8 bis.

Corollary 2.5. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, P a parabolic subgroup of G (Exp. XIV 4.8 bis). Then G/P is representable by an S -prescheme smooth and projective over S .*

It remains only to show (loc. cit.) that G/P is representable by a quasi-projective prescheme of finite presentation, which follows from 2.4, taking account of the fact that $P = \underline{\text{Norm}}_G(P)$ (loc. cit.), hence is flat over S .

3. Groups with flat center

Proposition 3.1. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers. Suppose that the center Z of G (which is representable by Exp. XI 6.11) is flat over S . For every integer $n > 0$, denote by $G^{(n)}$ the locally free $\mathcal{O}_S$ -module equal to the n th normal invariant of G along the unit section, and let ρ_n be the natural “adjoint representation” of G in $\text{GL}_S(G^{(n)})$. Then:*

- a) *The quotient $G' = G/Z$ is representable by a group S -prescheme smooth, of finite presentation over S , quasi-affine, with affine connected fibers.*
- b) *The representation ρ_n factors through G' :*

$$\begin{array}{ccc} G & \xrightarrow{\rho_n} & \text{GL}_S(G^{(n)}) \\ & \searrow & \nearrow i_n \\ & G/Z & \end{array}$$

If S is quasi-compact, i_n is a monomorphism for sufficiently large n .

- c) *The functor $\mathcal{T}_{G'}$ of subtori of G' is representable by an S -prescheme smooth over S , which is a sum of a family of preschemes affine over S if S is quasi-compact.*

Proof. The assertions in a) are local on S , which allows us to suppose S quasi-compact. By Exp. XI 6.11 b), Z is equal to $\text{Ker } \rho_n$ for sufficiently large n , so the latter is flat over S . The fact that G/Z is representable and that i_n is a monomorphism therefore follows from 2.3. Since G is smooth and of finite presentation over S , the same holds for G' (Exp. VIB § 9). The group $\text{GL}_S(G^{(n)})$ is affine over S , and a monomorphism of finite presentation is quasi-affine (EGA IV 8.11.2), so G' is quasi-affine over S and has affine fibers. Since G' is smooth over S , the functor $\mathcal{T}_{G'}$ is formally smooth (Exp. XI 2.1 bis). Taking account of Exp. XI 4.6 and 4.3, the assertions in 3.1 c) will therefore follow from the following lemma:

Lemma 3.2. *Let S be a prescheme, G and H two group S -preschemes of finite presentation over S , $u : G \rightarrow H$ a group S -monomorphism, $\mathcal{T}_G$ (resp. $\mathcal{T}_H$) the functors of subtori of G (resp. of H) (cf. Exp. XV § 8). Then the map $T \mapsto u(T)$ defines a monomorphism*

(XV 8.3 c))

$$\tilde{u} : \mathcal{T}_G \longrightarrow \mathcal{T}_H$$

which is representable by a closed immersion of finite presentation.

By the usual procedure, we are reduced to the following problem: let T be a subtorus of H , $T' = G \times_T H$ its inverse image in G , $u_T : T' \longrightarrow T$ the group S -monomorphism deduced from u . One must show that the S -functor $\prod_{T/S}(T'/T)$ is representable by a closed subscheme of S , of finite presentation. But T , being a torus, is smooth over S with connected fibers, and it suffices to apply Exp. XI 6.10.

Theorem 3.3. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, Z the center of G . Suppose that Z is flat over S and that the unipotent rank (Exp. XV 6.1 ter) of G is equal to that of Z . Then:*

- a) *The group $G' = G/Z$ is representable, and if S is quasi-compact, the canonical morphism $i_n : G' \longrightarrow \mathrm{GL}_S(G^{(n)})$ (cf. 3.1 b)) is an immersion for large n .*
- b) *The group G' is quasi-affine over S , with affine fibers, the center of G' is the unit group, and G' has unipotent rank zero.*
- c) *The group G' has locally for the étale topology maximal tori, and these are also Cartan subgroups of G' . The functor $\mathcal{T}_{\mathcal{M}_{G'}}$ of maximal tori of G' (Exp. XV § 8) is representable by an S -prescheme smooth and affine over S .*
- d) *The group G has locally for the étale topology Cartan subgroups, and the functor $\mathcal{C}_G$ of Cartan subgroups of G is representable by an S -prescheme smooth and affine over S .*

Proof. By 3.1, the group G' is representable by an S -prescheme smooth and quasi-affine over S , with affine fibers. In view of the correspondence between Cartan subgroups of the fibers of G and of the fibers of G' (Exp. XII 6.6 e)), the hypothesis on the unipotent rank of Z implies that G' has unipotent rank zero. Using now Exp. XV 8.18, one sees that G' has locally for the étale topology maximal tori. The fact that i_n is an immersion for large n then follows from 3.1 b) and 1.4 a). This finishes proving a).

Let us now show c). Since G' has unipotent rank zero, it is clear that every maximal torus of G' is also a Cartan subgroup of G' . The functor of maximal tori of G' is representable by a subscheme, both open and closed, of the functor of subtori of G' ; moreover, it is of finite type over S (Exp. XV 8.15); it then follows from 3.1 c) that this functor is representable by an S -prescheme smooth and affine over S .

Since G' has locally for the étale topology Cartan subgroups, the same holds for G , and the functor of Cartan subgroups of G is canonically isomorphic to that of G' (Exp. XV 7.3 iv)), so c) $\Rightarrow$ d).

It remains for us to show b), and more precisely it remains to prove that the center Z' of G' is the unit group. Since Z' is representable (Exp. XI 6.11) and of finite presentation over S , it suffices to show that the fibers of Z' are reduced to the unit group (Exp. VIB 2.9), which reduces us to the case where S is the spectrum of an algebraically closed field. The center Z' is plainly contained in every Cartan subgroup of G' , hence in every maximal torus of G' by c), so is of multiplicative type. Moreover, we shall see in Exp. XVII that if G is a connected algebraic group with center Z , the center Z' of G/Z is unipotent. In the present case, Z' , being both of multiplicative type and unipotent, is reduced to the unit group (cf. Exp. XVII).

Examples of groups whose center is flat

Proposition 3.4. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , smooth, with connected fibers, Z the center of G . Then Z is flat over S in the following two cases:*

- a) *The unipotent rank ρ_u (Exp. XV 6.1 ter) of the fibers of G is zero.*
- b) i) *S is reduced.*
 ii) *The dimension of the fibers of Z is a locally constant function on S .*

iii) The unipotent rank of G is equal to the unipotent rank of Z .

Proof of 3.4 a). We shall prove at the same time the

Proposition 3.5. *Under the hypotheses of 3.4 a), suppose moreover that G has a maximal torus T , and let $C = \text{Centr}_G(T)$ be the Cartan subgroup of G associated to T (Exp. XII 7.1). Then:*

(i) $Z \cap T$ is a subgroup of multiplicative type of G .

(ii) C is commutative and equal to $T \cdot Z$.

(iii) If the quotients $Z/Z \cap T$ and C/T are representable, they are representable by abelian preschemes (i.e. group S -preschemes smooth over S whose fibers are abelian varieties), and the canonical monomorphism $Z/Z \cap T \rightarrow C/T$ is an isomorphism.

Using the general properties of passage to the limit proved in Exp. VIB § 10 and Exp. XV 6.2, 6.3 and 6.3 bis, one reduces as usual to the case where S is noetherian (note that the assertions in 3.4 and 3.5 are local on S). We remarked in 3.1 that the hypotheses on G imply that Z is representable. To show that Z is flat over S , one then reduces by EGA 0_{III} 10.2.6 to the case where S is local artinian. But then, G being smooth, G has locally for the étale topology maximal tori (Exp. XV 8.17). After making a finite flat extension of S (which is permissible for proving that Z is flat), we may therefore suppose that G has a maximal torus T .

Let us show, in the same way, that to establish 3.5, we can reduce to the case where S is artinian.

i) Since Z is closed in G (Exp. XI 6.11), $Z \cap T$ is a closed group subscheme of T . It then follows from Exp. X 4.8 b) that $Z \cap T$ is of multiplicative type if and only if it is flat over S . As above, it suffices to establish that $Z \cap T$ is flat when S is artinian.

ii) Since G has unipotent rank zero, every Cartan subgroup C of G satisfies the hypotheses of Lemma 1.6, hence is commutative. The fact that $C = T \cdot Z$ will follow from iii).

iii) If C/T is representable, C/T is smooth over S (Exp. VIB 9), and its fibers are abelian varieties (1.7), so C/T is an abelian prescheme. To show that the canonical monomorphism

$$i : Z/Z \cap T \rightarrow C/T$$

is an isomorphism, it suffices to verify this when S is local artinian. Indeed, one will successively deduce that $Z/Z \cap T$ is flat over S (EGA 0_{III} 10.2.6), then that i is flat (EGA IV 11.3.10), then that i is an open immersion (EGA IV 17.9.1), then that i is an isomorphism (since C/T has connected fibers).

We henceforth suppose that S is local artinian with closed point s and that G has a maximal torus T . Let $C = \text{Centr}_G(T)$. The group C is a Cartan subgroup of G (Exp. XII 7.1) and contains Z .

The algebraic group $M_s = Z_s \cap T_s$ is a subgroup of multiplicative type of G_s which is central. Since T is smooth over S , M_s lifts to a group subscheme M of T , of multiplicative type (Exp. IX 3.6 bis) and contained in the center of G (Exp. IX 3.9), hence contained in $Z \cap T$. Since S is artinian and M and T flat over S , the quotient groups $Z \cap T/M$, $Z' = Z/M$ and $A = C/T$ are representable (Exp. VIA §§ 4 and 5). By construction, $(Z \cap T/M)_s$ is the unit group, so $Z \cap T = M$ (Exp. VIB 2.9); a fortiori, it is flat over S , which proves 3.5 i).

By passage to the quotient, one has a canonical monomorphism $i : Z' \rightarrow A$, which is therefore a closed immersion (Exp. VIB 1.4.2). We have already remarked that A is an abelian scheme. It remains to show that i is an isomorphism. Indeed, this will prove 3.5 iii) and imply that Z' is flat over S , hence that Z is flat over S (as an extension of Z' by the flat group M (Exp. VIB § 9)). Since Z_s has the same abelian rank as G_s (Exp. XII 6.1), i_s is an epimorphism, hence is an isomorphism. Let q be an integer > 0 invertible on S . By the density theorem of 1.6 iii), to see that $i(Z') = A$, it suffices to show that for every integer n equal to a power of q , $i(S')$ contains ${}_n A$. Now let M_s^0 be the identity component of M_s . It is immediate by duality that raising to the q th power in M_s^0 is an epimorphism. One immediately deduces

that if m_0 is the exponent of q in the prime factorization of $\text{card}(M_s/M_s^0)$, the image of ${}_{nm_0}Z_s$ in $A_s \simeq Z_s/M_s$ contains ${}_nA_s$. There therefore exists a subgroup of multiplicative type $M_s(n)$ of Z_s whose image in A_s is ${}_nA_s$. As above, one sees that $M_s(n)$ lifts to a central subgroup of multiplicative type $M(n)$ of G . The image of $M(n)$ in A is a subgroup of multiplicative type (Exp. IX 6.8), hence necessarily equal to ${}_nA$, since this is so on the reduced fiber (Exp. IX 5.1 bis). This completes the proof of 3.4 a) and 3.5.

Proof of 3.4 b). The assertion to be proved is local on S , so we may suppose S affine with ring A . Considering A as the inductive limit of its $\mathbf{Z}$ -subalgebras of finite type, one reduces as above to the case where S is noetherian and reduced.

To show that Z is flat, one then has a valuative criterion of flatness (EGA IV 11.8.1), which allows us to reduce to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with generic point t and closed point s . Let Z' be the schematic closure in Z of Z_t . We must show that $Z' = Z$, and it even suffices to show that $Z_s = Z'_s$ (Exp. VIB 2.6). By (Exp. VIB § 4), one has the inequalities

$$\dim Z_t = \dim Z'_t = \dim Z'_s \leq \dim Z_s.$$

But by hypothesis $\dim Z_t = \dim Z_s$, so $\dim Z'_s = \dim Z_s$, and consequently Z'_s contains $(Z_s)_{\text{red}}^0$. It follows that $G'_s = G_s/Z'_s$ is affine (Exp. XII 6.1). In view of the correspondence between Cartan subgroups of G and G' (Exp. XII 6.6 e)), hypothesis 3.4 iii) implies that the unipotent rank of G'_s is zero, and consequently its Cartan subgroups are also its maximal tori. The image Z''_s of Z_s in G'_s is a central algebraic subgroup of G'_s , hence contained in every Cartan subgroup of G'_s , that is to say, in every maximal torus; Z''_s is therefore a subgroup of multiplicative type of G'_s . Under these conditions, we shall show in (Exp. XVII § 7) that there exists a finite subgroup of multiplicative type M_s of Z_s whose image in Z_s/Z'_s is Z''_s (we used this fact in the proof of 3.4 a) when Z''_s is étale). Since G is smooth over S and S is the spectrum of a complete local ring, M_s lifts to an S -subgroup of multiplicative type M of G (Exp. IX 3.6 bis and Exp. XV 1.6 b)) which is central (Exp. IX 5.6 a)). Thus M_t is contained in $Z'_t = Z_t$, and since M is flat, M is contained in Z' . The group M_s is therefore contained in Z'_s , but this implies that Z''_s is the unit group, i.e. that $Z_s = Z'_s$. This completes the proof of 3.4 b).

Example of a smooth group prescheme with connected fibers whose center is not flat Let S be the spectrum of a discrete valuation ring A , π a uniformizer of A , t the generic point of S , s the closed point. Let G be the S -group smooth and affine over S , with ring $B = A[T, T^{-1}, U]/F$, where $F = 1 - T + \pi U$, the composition law being defined by

$$(t, u), (t', u') \longmapsto (tt', \pi uu' + u + u').$$

The generic fiber G_t is therefore isomorphic to the multiplicative group $(\mathbf{G}_m)_t$, whereas the closed fiber G_s is isomorphic to the additive group $(\mathbf{G}_a)_s$. The function T is invertible in $\Gamma(G, \mathcal{O}_G)$ and defines a group S -morphism $\varphi : G \rightarrow (\mathbf{G}_m)_S$ which is an isomorphism on the generic fiber and the zero morphism on the closed fiber. The datum φ allows one to construct the semidirect product group $H = G \cdot (\mathbf{G}_a)_S$ (note that $(\mathbf{G}_m)_S$ acts on $(\mathbf{G}_a)_S$). The center Z of H is the unit group on the generic fiber and is equal to H_s on the closed fiber, hence is not flat over S .

4. Groups with affine fibers of unipotent rank zero

Theorem 4.1. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation, with affine connected fibers of unipotent rank (Exp. XV 6.1 ter) zero. Then:*

- a) *The center Z of G is a group of multiplicative type (and consequently is a reductive center of G (Exp. XII 4.1)).*
- b) *G satisfies conditions a) to d) of 3.3.*

c) G has locally for the étale topology maximal tori, and these are also Cartan subgroups of G . The functor $\mathcal{T}\mathcal{M}_G$ of maximal tori of G is representable by an S -prescheme smooth and affine over S .

d) G is quasi-affine over S . Moreover, if T is a maximal torus of G , $N = \underline{\text{Norm}}_G T$ its normalizer in G , $W = N/T$ the Weyl group corresponding to T , the following conditions are equivalent:

- (i) G is affine over S .
- (ii) $G' = G/Z$ is affine over S .
- (iii) N is affine over S .
- (iv) W is affine over S .

These conditions always hold if S is locally noetherian of dimension ≤ 1 .

Proof. Since G has affine fibers (hence abelian rank zero) and unipotent rank zero, G has locally for the étale topology maximal tori (Exp. XV 8.18), and every maximal torus of G is plainly a Cartan subgroup of G , which proves the first part of c). To see that Z is of multiplicative type, one may suppose, by the foregoing, that G has a maximal torus T . Since T is also a Cartan subgroup, T contains Z (for $T = \underline{\text{Centr}}_G(T)$ by Exp. XII 7.1), and $Z = Z \cap T$ is of multiplicative type by 3.5 i). This proves a). Assertion b) is clear, taking account of a). On the other hand, the functor $\mathcal{T}\mathcal{M}_G$ is isomorphic to the functor of Cartan subgroups of G (Exp. XII 7.1), hence is smooth and affine over S (3.3 c)), which finishes proving c). It remains to prove d).

Proof of d). Since G' is quasi-affine (3.5 b)) and Z is of multiplicative type, hence affine over S , G is quasi-affine (Exp. VIB § 9).

i) $\Leftrightarrow$ ii). If G is affine, G' is affine by Exp. IX 2.3. If G' is affine, G is affine as an extension of an affine group by an affine group (Exp. VIB § 9).

i) $\Leftrightarrow$ iii). If G is affine, N is affine since it is closed in G (Exp. XI 6.11 a)). Moreover, G/N is isomorphic to the functor $\mathcal{T}\mathcal{M}_G$ (conjugacy of maximal tori, cf. Exp. XII 7.1 b)), hence is affine over S by c). Thus if N is affine over S , G is affine as an “extension” of an affine homogeneous space by an affine group (Exp. VIB § 9).

iii) $\Leftrightarrow$ iv). If N is affine, $W = N/T$ is affine (Exp. IX 2.3), and conversely by Exp. VIB § 9.

Moreover, N/T is representable by a group S -prescheme étale and separated over S , of finite type (Exp. XV 7.1 iv)), hence quasi-finite over S . The last assertion of d) will therefore follow from the following lemma, applied with $X = W$ and $Y = S$:

Lemma 4.2. *Let S be a locally noetherian prescheme of dimension 1, X and Y two S -preschemes locally of finite type over S , $u : X \rightarrow Y$ a quasi-finite separated S -morphism. Suppose that for every point s of S , the morphism $u_s : X_s \rightarrow Y_s$ deduced from u by the base change $\text{Spec } \kappa(s) \rightarrow S$ is finite. Then u is an affine morphism.*

The assertion to be proved is local on Y , which allows us to suppose Y (hence also X) of finite type over S . By EGA IV 8, one sees that it suffices to prove 4.2 when S is the spectrum of a local ring A . By fpqc descent, one may suppose A complete, then A reduced (EGA II 1.6.4), then A normal (Nagata’s theorem (EGA 0_{IV} 22) and Chevalley’s theorem (EGA II 6.7.1)). If A is a field, u is finite by hypothesis, hence affine. Otherwise A is a discrete valuation ring; let s be the closed point of S , t the generic point, π a uniformizer of A . Let y be a point of Y_s . Applying EGA IV 8 once again, one may replace Y by the spectrum Y' of $\mathcal{O}_{Y,y}$ and X by $X' = X \times_Y Y'$; finally, one may suppose $\mathcal{O}_{Y,y}$ complete. Since u is quasi-finite and separated, by EGA II 6.2.6, X' is the sum of two schemes X_1 and X_2 , with X_1 finite over Y' (hence affine) and X_2 such that $u(X_2)$ does not contain the closed point y of Y' . I say that under these conditions, X_2 does not meet the closed fiber X'_s . Indeed, by hypothesis, $X'_s \rightarrow Y'_s$ is finite, so the restriction of this morphism to the closed subset $(X_2)_s$ is finite, and its image in $(Y')_s$ is closed. Since this image does not contain the closed point of the local scheme $(Y')_s$, $(X_2)_s$ is empty. Since u_t is finite, the restriction to $(X_2)_t = X_2$ of the

morphism $X'_t \rightarrow Y'_t$ is finite. Moreover, one has $Y'_t = Y'_\pi$, so the open immersion $Y'_t \rightarrow Y'$ is affine. In brief, the composite morphism $X_2 \rightarrow Y'_t \rightarrow Y'$ is affine, and it indeed follows that the morphism $X' = X_1 \amalg X_2 \rightarrow Y'$ is affine.

Corollary 4.3. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with affine solvable connected fibers of unipotent rank zero. Then G is affine over S . If moreover the center of G is the unit group and S is quasi-compact, the canonical morphism $i_n : G \rightarrow \mathrm{GL}_S(G^{(n)})$ (cf. 3.1 b)) is a closed immersion for sufficiently large n .*

To prove that G is affine over S , one may suppose that G has a maximal torus T (4.1 c)). The group G having solvable fibers, the Weyl group corresponding to T is the unit group (BIBLE 6 Th. 1 Cor. 3), and condition 4.1 d) iv) is satisfied. If the center of G is the unit group, i_n is a monomorphism for sufficiently large n (3.1), hence is a closed immersion (1.5 b)).

5. Application to reductive and semisimple groups

Definition 5.1. A group S -prescheme G is said to be reductive (resp. semisimple) if G is smooth and affine over S , with reductive (resp. semisimple) fibers.

5.1.1. — Reductive groups will be systematically studied beginning with Exp. XIX. In this paragraph, we shall need the following properties, which will be proved in Exp. XIX (without using the developments of the present exposé).

- a) Let S be a prescheme, G a group S -prescheme smooth and affine over S , with connected fibers, s a point of S such that G_s is reductive (resp. semisimple). Then there exists a neighborhood U of s such that $G|_U$ is reductive (resp. semisimple).
- b) If G is reductive, G has locally for the étale topology maximal tori, and if T is a maximal torus of G , the Weyl group $W = \mathrm{Norm}_G(T)/T$ is finite over S .
- c) A reductive group has unipotent rank zero.

This being admitted, we propose to improve assertion a) above.

Theorem 5.2. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, s a point of S . Then:*

- (i) *If the fibers of G are affine and G_s is reductive, there exists an open neighborhood U of s such that $G|_U$ is reductive.*
- (ii) *If G_s is semisimple, there exists an open neighborhood U of s such that $G|_U$ is semisimple.*

Using Exp. XV 6.2 i) and Exp. VIB § 10, one reduces to the case where S is noetherian.

In case ii), consider the center Z of G , which is representable (Exp. XI 6.11 a)). Since G_s is semisimple, it is well known that Z_s is finite. Consequently (Exp. VIB § 4), there exists a neighborhood U of s such that Z is quasi-finite over U . For every point t of U , G_t is then affine (Exp. XII 6.1). After restricting S , we may therefore suppose that the fibers of G are affine, in case ii) as in case i).

To prove 5.2, it suffices, taking account of a), to prove that G is affine over S . Since G has affine fibers, we know that the functor of subtori of G is representable by a smooth S -prescheme (Exp. XV 8.11 and 8.9). After making an étale extension covering s , which is permissible, we may therefore suppose that there exists a subtorus T of G such that T_s is a maximal torus of G_s . But then $C = \mathrm{Centr}_G(T)$ is a smooth subgroup of G with connected fibers, containing T , such that $C_s = T_s$ (since C_s is a Cartan subgroup of G_s and G_s has unipotent rank zero by c)). It follows that $C = T$, so T is a Cartan subgroup and a maximal torus of G ; a fortiori, the unipotent rank of G is zero, and one can apply 4.1.

By 4.1 d), it suffices to show that the Weyl group W corresponding to T is affine over a neighborhood U of s . In fact, we shall see that W is even finite over a neighborhood of s . Since W is quasi-finite over S (Exp. XV 7.1 iv)), to say that W is finite over a neighborhood

U of s is equivalent to saying that the morphism $W \rightarrow S$ is proper at s (EGA IV 15.7 and EGA III 4.4.2), and to establish this one has the valuative criterion of local properness (EGA IV 15.7), which reduces us to the case where S is the spectrum of a discrete valuation ring with closed point s . But then, by the last assertion of 4.1 d), G is affine over S , and one can apply property a) recalled above to conclude that G is reductive (resp. semisimple). Using now property b), one concludes that W is indeed finite over S , which completes the proof of 5.2.

6. Applications: Extension of certain rigidity properties of tori to groups of unipotent rank zero

Proposition 6.1. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation, with affine connected fibers, and whose center is of multiplicative type (for example, G of unipotent rank zero, cf. 4.1 a)). Then every invariant closed subgroup K of G , of finite presentation over S , quasi-finite over S , is finite over S .*

Indeed, for every geometric point $\bar{s}$ over S , $(K_{\bar{s}})_{\text{red}}$ is a finite étale subgroup of $G_{\bar{s}}$, invariant, hence central (since $G_{\bar{s}}$ is connected). Consequently $K' = Z \cap K$ (where Z denotes the center of G) has the same underlying space as K , and it suffices to show that K' is finite over S . Now K' is a closed quasi-finite subgroup of the group of multiplicative type Z , hence is finite, as one sees immediately (locally on S , K' will be contained in ${}_nZ$ for a suitable integer n , and ${}_nZ$ is finite over S).

Corollary 6.2. *Let S and G be as above, K a group subscheme of G , of finite presentation over S , invariant and closed in G , s a point of S . If K_s is finite (resp. is the unit group), there exists an open neighborhood U of s such that $K|_U$ is finite over S (resp. is the unit group).*

In view of the upper semicontinuity of the dimension of the fibers of K (Exp. VIB § 4), one may already suppose, after restricting S , that K is quasi-finite over S , hence finite (6.1). If moreover K_s is the unit group, one easily deduces by Nakayama's lemma that K is the unit group over $\text{Spec } \mathcal{O}_s$, hence over a neighborhood of s .

Proposition 6.3. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with affine connected fibers of unipotent rank zero, H a group S -prescheme, s a point of S , $u : G \rightarrow H$ a group S -morphism such that $\text{Ker } u_s$ is central. Suppose moreover that H is of finite presentation over S , or that S is locally noetherian and G locally of finite type over S . Then there exists an open neighborhood U of s such that, if $K = \text{Ker } u$, $K|_U$ is central, of multiplicative type. Moreover, $G' = (G|_U)/(K|_U)$ is representable, and the morphism $u' : G' \rightarrow H|_U$ deduced from u by passage to the quotient is an immersion.*

Proof. Let Z be the center of G , and $K' = K \cap Z$.

a) K' is a subgroup of multiplicative type over a neighborhood U of s . Indeed, after making an étale extension over a neighborhood U of s , which is legitimate, we may suppose that G has a maximal torus T (4.1 c)). But then $T \cap K$ is a group of multiplicative type (Exp. XV 8.3) whose fiber at s is central by hypothesis, so $T \cap K$ is central over a neighborhood of s (Exp. IX 5.6 a)), and consequently coincides with K' .

b) Let us show that $K' = K$ over a neighborhood U of s . By a), one may already suppose that K' is of multiplicative type, hence flat over S . Since $K'_s = K_s$, the natural immersion $K' \rightarrow K$ is then open over a neighborhood U of s (Exp. VIB 2.6). If $t \in U$, the image of K_t in the group $G'_t = G_t/Z_t$ is therefore a finite étale group invariant in G'_t (since K_t is invariant in G_t), hence central, hence reduced to the unit group (3.3 b)). This says that K_t is contained in Z_t , hence is equal to K'_t , whence $K = K'$ over U .

c) Representability of G' then follows from 2.3; the fact that u' is an immersion is contained in 1.3 a), taking account of 4.1 c) applied to G' .

Proposition 6.4. *Let S be a prescheme, G a group S -prescheme smooth, of finite presentation over S , with connected fibers, s a point of S such that G_s is generated by its subtori (Exp. XII 8.2) (for example, G_s affine of unipotent rank zero), H a group S -prescheme, u and $v : G \rightarrow H$ two S -homomorphisms such that $u_s = v_s$ and u_s is central. Suppose moreover that H is of finite presentation over S , or that S is locally noetherian. Then there exists a neighborhood U of s such that $u|_U = v|_U$.*

One reduces as usual to the case where S is noetherian (to study the condition “ G_s is generated by its subtori”, one uses Exp. VIB 7.4). Since G has connected fibers, to show that $u = v$ over a neighborhood U of s , it suffices to show that $u = v$ after reduction by every power of the maximal ideal of the local ring $\mathcal{O}_{S,s}$, which reduces us to the case where S is local artinian.

But then the functor of maximal subtori of G is representable by an S -scheme X , smooth and of finite type over S (Exp. XV 8.17). Let T be the maximal subtorus of G_X , universal for the functor X . The hypothesis on G_s means that the algebraic subgroup of G_s generated (Exp. VIB § 7) by the $\kappa(s)$ -morphism $f : T_s \rightarrow G_s$ (the composite of the immersion $T_s \rightarrow G_{X_s}$ and the canonical projection $G_{X_s} \rightarrow G_s$) is equal to the whole of G_s . But X_s is geometrically connected (for if T is a maximal torus of G_s and N its normalizer in G_s , X_s is isomorphic to G_s/N), and the image of T_s under f contains the unit section of G_s ; it then follows from Exp. VIB 7.4 that there exist an integer $N > 0$ and a certain S -morphism $f^N : T^N \rightarrow G$ (where $T^N = T \times_S \cdots \times_S T$ (N factors)), and f^N depends only on f and on the multiplication law of G) such that $(f^N)_s$ is surjective.

Consider the base change $X \rightarrow S$ and the restrictions of u_X and v_X to the subtorus T of G_X . By hypothesis, one has $u_{X_s} = v_{X_s}$, and $u_{X_s} : T_{X_s} \rightarrow H_{X_s}$ is a central homomorphism. It then follows from Exp. IX 5.1 that one has $u_X|_T = v_X|_T$. Making the definition of f^N explicit and taking account of the fact that u and v are homomorphisms, one immediately deduces that $uf^N = vf^N$. But the $\kappa(s)$ -morphism f_s^N is surjective, and G_s is smooth, hence reduced, so f_s^N is generically flat. Since T is smooth over S , hence flat, there exists a nonempty open subset V of T such that $f^N|_V$ is flat (EGA IV 11.3.10). The image of V is an open subset W of G (EGA IV 11.3.1), and $f^N : V \rightarrow W$ is faithfully flat, of finite presentation, hence a covering for the fpqc topology. The equality $uf^N = vf^N$ then implies $u|_W = v|_W$. Since W is schematically dense in G (EGA IV 11.10.10) and H separated over S (Exp. VIA 0.3), one immediately deduces that $u = v$.

Exposé XVII. Unipotent algebraic groups. Extensions between unipotent groups and groups of multiplicative type

by M. Raynaud*

0. Some notation

In the present chapter, we shall mainly have to consider algebraic groups defined over a field k . The number $p \geq 0$ will always denote the characteristic of k , $\mathbf{F}_p$ the prime field with p elements if $p > 0$, $\bar{k}$ an algebraically closed extension of k , q a prime number distinct from p .

For every S -prescheme, $(\mathbf{G}_a)_S$ (resp. $(\mathbf{G}_m)_S$) denotes the additive group (resp. the multiplicative group) over S (cf. Exp. I 4.3). For every integer $n > 0$, $(\mu_n)_S$ (resp. $(\mathbf{Z}/n\mathbf{Z})_S$) denotes the group of n th roots of unity (Exp. I 4.4.4) (resp. the constant group over S associated with the abstract group $\mathbf{Z}/n\mathbf{Z}$ (Exp. I 4.1)). The group $(\mathbf{Z}/n\mathbf{Z})_S$ is finite and étale over S ; the group $(\mu_n)_S$ is flat and finite over S , and is étale over S if and only if n is invertible on S (Exp. VIII 2.1).

If S is a prescheme of characteristic $p > 0$, for every integer $n > 0$ and every group S -prescheme G , we denote by $_{F^n}(G)$ the radicial group S -subprescheme of G equal to the kernel of the n th iterate of the Frobenius morphism relative to G (Exp. VIIA). In particular, if $G = (\mathbf{G}_a)_S$, we put $_{F^n}(G) = (\alpha_{p^n})_S$, which is a radicial S -group, flat and finite over S , representing the following functor: for every S -prescheme S' , $(\alpha_{p^n})_S(S')$ is the set of $x' \in \Gamma(S', \mathcal{O}_{S'})$ such that $x'^{p^n} = 0$.

The group $(\mu_{p^n})_S$, already defined, is canonically isomorphic to $_{F^n}(\mathbf{G}_m)_S$.

When there is no ambiguity about the base scheme S , we shall simply write $\mathbf{G}_a, \mathbf{G}_m, \alpha_{p^n}$, etc., instead of $(\mathbf{G}_a)_S, (\mathbf{G}_m)_S, (\alpha_{p^n})_S$, etc.

If G is a commutative group S -prescheme, for every integer $n > 0$, $_nG$ is the group subprescheme of G equal to the kernel of raising to the n th power in G .

For the reader's convenience, we have collected in an appendix some properties of algebraic groups proved in Exp. VI and VII, as well as elementary properties of Hochschild cohomology, which will be useful to us in this chapter.

1. Definition of unipotent algebraic groups

Definition 1.1. An algebraic group G defined over an algebraically closed field k is said to be *unipotent* if G admits a composition series whose successive quotients are isomorphic to algebraic subgroups of $(\mathbf{G}_a)_k$.

Proposition 1.2. *Let k be an algebraically closed field, K an algebraically closed extension of k , G an algebraic group defined over k . Then, for G to be unipotent, it is necessary and sufficient that G_K be unipotent.*

The necessity of the condition is clear, since a composition series gives a composition series by extension of the base.

* Cf. the note on page 1 of Exposé XV.

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The proof of sufficiency is standard: K is the inductive limit of its finitely generated k -subalgebras. By Exp. VIB § 10, one can find a finitely generated k -subalgebra A of K , a composition series G_i of G_S ($S = \operatorname{Spec} A$), and immersions $u_i : H_i = G_i/G_{i+1} \rightarrow (\mathbf{G}_a)_S$. To prove that G is unipotent, it then suffices to make a k -extension of the base, $A \rightarrow k$, which is possible since $\operatorname{Hom}_{k\text{-alg}}(A, k)$ is nonempty, A being a nonzero finitely generated k -algebra over the algebraically closed field k .

Definition 1.3. Let G be an algebraic group defined over a field k . We shall say that G is *unipotent* if there is an algebraically closed extension $\bar{k}$ of k such that $G_{\bar{k}}$ is unipotent (Definition 1.1).

By 1.2, the property is independent of the chosen algebraically closed extension $\bar{k}$.

Definition 1.4. Let G be an algebraic group defined over a field k , and H an algebraic group defined over an extension k' of k . We shall say that H is a *form* of G over k' if the algebraic groups $G_{k'}$ and $H_{k'}$ become isomorphic over $\bar{k}'$ (as above, one sees that the property does not depend on the choice of the algebraically closed extension $\bar{k}'$ of k'). We shall also say that H is a “twisted” group G .

We are now in a position to describe the algebraic subgroups of $\mathbf{G}_a$.

Proposition 1.5. *Let k be a field of characteristic $p \geq 0$. Then an algebraic subgroup H of $(\mathbf{G}_a)_k$ is of one of the following types:*

- (i) $H = 0$.
- (ii) $H = \mathbf{G}_a$.
- (iii) (If $p > 0$) H is an extension of a twisted constant group $(\mathbf{Z}/p\mathbf{Z})^r$ by a radicial group α_{p^n} (r and n integers ≥ 0 , $r + n > 0$). If, moreover, k is perfect, this extension is necessarily trivial.

Proof. If H is of dimension 1, it is clear that $H = \mathbf{G}_a$. Otherwise H is of dimension 0 and consequently is an extension⁽¹⁾ of an étale group H'' by its neutral component H' , which is a radicial group. To describe H''_k , it suffices to know the abstract group $H''(\bar{k})$, isomorphic to $H(\bar{k})$. Now the latter is a finite subgroup of $\mathbf{G}_a(\bar{k})$, hence is zero if $p = 0$ and is of the form $(\mathbf{Z}/p\mathbf{Z})^r$ otherwise, since it is annihilated by p . The group H' is closed in $\mathbf{G}_a$ and is defined by a single equation (the ring $k[T]$ of $\mathbf{G}_a$ is principal), which over $\bar{k}$ has 0 as its only root; hence this equation is of the form $T^n = 0$. Compatibility with the group law implies that $(T + T')^n$ belongs to the ideal generated by T^n and T'^n in the ring $k[T, T']$; hence: if $p = 0$, one has $n = 1$ and H' is the unit group; if $p > 0$, one has $n = p^m$ and $H' = \alpha_{p^m}$.

The last assertion of 1.5 follows more generally from the lemma:

Lemma 1.6. *If k is a perfect field, every extension H of an étale algebraic group H'' by a radicial group H' is trivial. Moreover, there is a unique lifting of H'' into H , namely H_{red} .*

Indeed, since k is perfect, the reduced k -scheme H_{red} is an algebraic subgroup of H (Exp. VIA 0.2), geometrically reduced, hence smooth over k (Exp. VIA, 1.3.1), hence étale, since H is of dimension 0. To see that the canonical projection $H_{\text{red}} \rightarrow H''$ is an isomorphism, it suffices to see it after extension of the base field $k \rightarrow \bar{k}$, in which case it suffices to show that one has an isomorphism on the points with values in $\bar{k}$, which is quite clear. The last assertion follows from the fact that every lifting of H'' into H , being étale over k , is reduced, hence is necessarily contained in H_{red} .

Note that α_{p^n} is a multiple extension of groups isomorphic to α_p . One then deduces from 1.5 the corollary:

⁽¹⁾**N.D.E.** Reference to VIIA?

Corollary 1.7. *For an algebraic group G defined over an algebraically closed field k to be unipotent, it is necessary and sufficient that it have a composition series whose successive quotients are isomorphic to $\mathbf{G}_a$ if $p = 0$, and to one of the groups $\mathbf{G}_a, \mathbf{Z}/p\mathbf{Z}, \alpha_p$ if $p > 0$. (We shall call these groups the elementary unipotent groups.)*

2. First properties of unipotent groups

Proposition 2.1. *A unipotent algebraic group defined over a field k is affine over k .*

By (fpqc) descent of affine morphisms, it suffices to prove 2.1 when k is algebraically closed. In this case, G is, by definition, a multiple extension of affine algebraic groups, hence is affine, by Exp. VIB 9.2(viii) applied to affine morphisms.⁽²⁾

Proposition 2.2.

- i) *The property for an algebraic group of being unipotent is invariant under extension of the base field.*
- ii) *Every algebraic subgroup of a unipotent group is unipotent.*
- iii) *Every algebraic quotient group of a unipotent group is unipotent.*
- iv) *Every extension of a unipotent algebraic group by a unipotent algebraic group is likewise unipotent.*

Proof. i) follows immediately from 1.3 and 1.2. To establish the other properties, we can assume that the field k is algebraically closed. Then iv) is obvious from Definition 1.3.

Let therefore G be a unipotent algebraic group, G' an algebraic subgroup of G , G'' an algebraic quotient group of G , G_i ($i = 1, \dots, n$) a composition series of G such that $H_i = G_i/G_{i+1}$ is an elementary unipotent group (1.7).

To prove ii), consider the composition series of G' induced by that of G : $G'_i = G_i \cap G'$. The group G'_i/G'_{i+1} is identified with an algebraic subgroup of H_i , hence is isomorphic to a subgroup of $\mathbf{G}_a$, and consequently G' is unipotent.

To prove iii), consider the composition series of G'' that is the image of that of G : $G''_i = \text{image of } G_i \text{ in } G''$. The group G''_i/G''_{i+1} is then a quotient of H_i , and it suffices to prove the lemma:

Lemma 2.3. *If H is an elementary unipotent group (1.7) defined over a field k , every algebraic quotient group H'' of H is zero or is isomorphic to H over k .*

Proof. If $H = \mathbf{G}_a$ in characteristic 0, or if $H = \alpha_p$ or $\mathbf{Z}/p\mathbf{Z}$ ($p > 0$), it suffices to note that it follows from 1.5 that H has no algebraic subgroups other than 0 and H (observe that a nonzero algebraic subgroup of α_p is defined by a k -algebra of rank over k at least equal to p (1.5 iii), hence is equal to α_p).

Let now $H = \mathbf{G}_a$ ($p > 0$), so that one has an exact sequence:

$$0 \longrightarrow N \longrightarrow \mathbf{G}_a \longrightarrow H'' \longrightarrow 0.$$

If $N = \mathbf{G}_a$, then $H'' = 0$. Otherwise, proceeding by induction on the length of a composition series of N , one can assume that $N = \alpha_p$, or that N is a form of $(\mathbf{Z}/p\mathbf{Z})^r$ (1.5).

a) If $N \simeq \alpha_p$, the proof of 1.5 iii) shows that N is necessarily the kernel of the Frobenius morphism F in $\mathbf{G}_a$, and one concludes with the aid of the exact sequence:

$$0 \longrightarrow \alpha_p \longrightarrow \mathbf{G}_a \xrightarrow{F} \mathbf{G}_a \longrightarrow 0.$$

b) If $N \simeq \mathbf{Z}/p\mathbf{Z}$, it is immediate that there is $a \in k^*$ such that N is the closed subscheme of $\mathbf{G}_a = \text{Spec } k[X]$ defined by the equation $X^p - aX = 0$. One then concludes with the aid of the exact sequence:

$$0 \longrightarrow N \longrightarrow \mathbf{G}_a \xrightarrow{\mathcal{P}} \mathbf{G}_a \longrightarrow 0,$$

⁽²⁾**N.D.E.** Check this reference.

where $\mathcal{P}$ is the Artin–Schreier morphism: $x \mapsto x^p - ax$.

c) If N is a form of $(\mathbf{Z}/p\mathbf{Z})^r$, there is a finite Galois extension k' of k trivializing N . By b) and an obvious induction on r , $\mathbf{G}_a/N$ is a form of $\mathbf{G}_a$ trivialized by k' . It then suffices to apply the following lemma:

Lemma 2.3 bis. *Let k be a field, G an algebraic k -group that is a form of $\mathbf{G}_a$, trivialized by a finite separable extension k' of k . Then G is isomorphic to $\mathbf{G}_a$.*

Indeed, the group of k' -automorphisms of the algebraic group $(\mathbf{G}_a)_{k'}$ is the group of nonzero homotheties $(k')^\times$ (*Bible*, Exp. 9, Lemma 1), and the Galois cohomology group $H^1(k'/k, \mathbf{G}_m)$ is zero (one can assume that k' is a Galois extension of k), by Hilbert's Theorem 90. Lemma 2.3 bis then follows from the classification of k -forms of an algebraic group (cf. J.-P. Serre, *Cohomologie galoisienne*, Chap. III, 1.3).

This completes the proof of 2.2.

Proposition 2.4. *Let k be a field, M a k -group of multiplicative type (Exp. VIII) and of finite type, U a unipotent algebraic k -group. Then:*

- i) $\underline{\mathrm{Hom}}_{k\text{-gr}}(M, U) = \underline{e}$ (a fortiori $\mathrm{Hom}_{k\text{-gr}}(M, U) = e$).
- ii) $\mathrm{Hom}_{k\text{-gr}}(U, M) = e$.

To prove i), we must establish that for every prescheme S over k , $\mathrm{Hom}_{S\text{-gr}}(M_S, U_S) = e$. But this follows from the following lemma:

Lemma 2.5. *Let S be a prescheme, M an S -group of multiplicative type and of finite type over S , U a group S -prescheme, of finite presentation over S , with unipotent fibers; then $\mathrm{Hom}_{S\text{-gr}}(M, U) = e$.*

Indeed, under the hypotheses made, for a group S -morphism $u : M \rightarrow U$ to be the unit homomorphism, it suffices that the restriction of u to the fibers of M over the points of S be the zero morphism (Exp. IX 5.2). We are therefore reduced to the case where S is the spectrum of a field, which one can moreover assume to be algebraically closed. In view of Definition 1.1, one can restrict oneself to $U = \mathbf{G}_a$, in which case the property has already been proved (Exp. XII 4.4.1).

Let us now prove 2.4 ii). Let therefore $u : U \rightarrow M$ be a group k -morphism. The image $u(U)$ is representable by an algebraic subgroup U'' of M (Exp. VIB 5.4).⁽³⁾ The group U'' is unipotent as a quotient of a unipotent group (2.2 iii)) and is of multiplicative type as a subgroup of a group of multiplicative type (cf. *Bible*, Exp. 4, Th. 2 cor. 1, or Exp. IX 6.8), hence U'' is the unit group by 2.4 i).

Remark 2.6. Keeping the notation of 2.4, it is no longer true in general that the functor $\underline{\mathrm{Hom}}_{k\text{-gr}}(U, M)$ is equal to $\underline{e}$. Thus, take a prescheme S such that $\Gamma(S, \mathcal{O}_S)$ contains a nonzero element ε such that $\varepsilon^2 = 0$ (for example, the spectrum of the algebra of dual numbers of a ring A). For every S' over S , the map $u \mapsto 1 + \varepsilon_{S'} u$ defines a homomorphism, functorial in S' , from the additive group $\Gamma(S', \mathcal{O}_{S'})$ to the multiplicative group $\Gamma(S', \mathcal{O}_{S'}^\times)$, hence defines a group S -morphism $(\mathbf{G}_a)_S \rightarrow (\mathbf{G}_m)_S$, and since $\varepsilon \neq 0$, this morphism is nonzero.

Recall (Exp. VIIA § 3) that when G is a commutative group S -prescheme, finite and flat over S , one has Cartier duality, and G is reflexive in the sense of Exp. VIII § 1. More precisely, the functor $G \mapsto \underline{\mathrm{Hom}}_{S\text{-gr}}(G, \mathbf{G}_m)$ is representable by a commutative group S -prescheme $D(G)$, finite and flat over S , and the canonical morphism $G \rightarrow D(D(G))$ is an isomorphism. In particular, one obtains:

- (i) $D(\mu_n) \simeq \mathbf{Z}/n\mathbf{Z}$, and consequently $D(\mathbf{Z}/n\mathbf{Z}) \simeq \mu_n$.
- (ii) If S is of characteristic $p > 0$, $D(\alpha_p) \simeq \alpha_p$.

⁽³⁾N.D.E. Check this reference.

3. Unipotent groups acting on a vector space

Recall (Exp. I, 4.6.1) that if S is a prescheme and M a sheaf of $\mathcal{O}_S$ -modules, one denotes by $W(M)$ the S -functor $((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ defined by the condition $W(M)(S') = \Gamma(S', M \otimes_{\mathcal{O}_{S'}} \mathcal{O}_{S'})$ for every S -prescheme S' .

Moreover, recall that if an S -group acts on an S -functor $\mathbf{V}$, one defines the S -functor $\mathbf{V}^G$ of invariants of $\mathbf{V}$ under G to be the subfunctor of $\mathbf{V}$ whose set of points with values in an S' over S is the set of $x \in \mathbf{V}(S')$ such that $x_{S''}$ is fixed under $G(S'')$ for every S'' over S' .

This being so, one has the following lemma:

Lemma 3.1. *Let S be a prescheme, G a group S -prescheme, affine over S , defined by the quasi-coherent $\mathcal{O}_S$ -algebra A . Suppose that G acts on a quasi-coherent sheaf of $\mathcal{O}_S$ -modules M , and let $\mu : M \rightarrow A \otimes_{\mathcal{O}_S} M$ be the comorphism defining the action of G on M (Exp. I 4.7.2), and $\nu : M \rightarrow A \otimes_{\mathcal{O}_S} M$ the morphism $x \mapsto \mu(x) - 1 \otimes x$. Then:*

- i) $\underline{M}^G(S) = \Gamma \text{Ker}(\nu)$.
- ii) If S is the spectrum of a field k , $\underline{M}^G$ is of the form $W(N)$, where N is a vector subspace of M .
- iii) If S is the spectrum of a field k , every element x of M is contained in a vector subspace of M , finite-dimensional over k , stable under the action of G .

Proof. iii) is included as a reminder and has already been proved in Exp. VIB 11.2.

i) It is clear that $\underline{M}^G(S)$ contains $\Gamma \text{Ker}(\nu)$. To establish the converse, one can assume that S is affine with ring B . Let $m \in \underline{M}^G(S)$. Then, for every B -algebra B' and every element u of $\text{Hom}_{B\text{-alg}}(A, B')$ (u corresponds to an element of $G(\text{Spec } B')$), one has:

$$(u \otimes 1_M)\nu(m) = 0 \quad \text{in } B' \otimes_B M.$$

Take, in particular, $B' = A$ and u the identity of A ; one does indeed find $\nu(m) = 0$.

ii) Let N be the kernel of ν , equal to $\underline{M}^G(k)$ by i). Every k -prescheme S is flat over k , hence one has:

$$\underline{M}^G(S) = \Gamma \text{Ker}(\nu \otimes_k S) = \Gamma(N \otimes_k S).$$

Thus $\underline{M}^G$ is isomorphic to the functor $W(N)$.

Proposition 3.2. *Let G be a unipotent algebraic group defined over a field k , acting on a k -vector space V . Then, if $V \neq 0$, one has $V^G \neq 0$.*

In view of 3.1 ii) and iii), one can assume that k is algebraically closed and V is finite-dimensional over k .

Let $0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0$ be an exact sequence of algebraic groups, and suppose that G acts on a sheaf $\mathbf{V}$ (for the fpqc topology). One has, of course, $\mathbf{V}^G \subset \mathbf{V}^{G'}$, and the quotient presheaf G/G' acts naturally on $\mathbf{V}^{G'}$. But $\mathbf{V}^{G'}$ is a sheaf (as the kernel of the well-known pair of morphisms $\mathbf{V} \rightrightarrows \underline{\text{Hom}}(G', \mathbf{V})$); consequently, the sheaf associated with G/G' , that is, G'' , acts on $\mathbf{V}^{G'}$, and it is immediate to check that $(\mathbf{V}^{G'})^{G''} = \mathbf{V}^G = (\mathbf{V}^{G'})^{G/G'}$.

This remark allows us, in order to prove 3.2, to reduce to the case where G is an elementary unipotent group (1.7).

a) $G = \mathbf{G}_a$, $p = 0$. It follows from (BIBLE 4 prop. 4) that a morphism from $\mathbf{G}_a$ to the linear group $\text{GL}(V)$ is given by an exponential map:

$$T \mapsto \sum_{q=0}^{\infty} T^q \frac{n^q}{q!},$$

where n is a nilpotent endomorphism of V . But then $V \neq 0 \Rightarrow \text{Ker } n \neq 0$, and it is clear that every vector of V annihilated by n is left fixed by G .

Suppose now that $p > 0$.

b) $G = \alpha_p$. Since the group α_p is a radicial group of height 1 (Exp. VIIA § 7), giving a representation of α_p in V amounts to giving a representation of the Lie p -algebra α_p in $\mathfrak{gl}(V)$

(App. II 2.2), that is, here, to giving an element X of $\text{End}(V)$ such that $X^p = 0$ (App. II 2.1). But then $V \neq 0 \Rightarrow W = \text{Ker}(X) \neq 0$, and, again by (App. II 2.2), one has $W = V^{\alpha_p}$.

c) $G = \mathbf{Z}/p\mathbf{Z}$. A representation of G in V is equivalent to giving an element x of $\text{Aut}(V)$ such that $x^p = 1$, i.e. $(1 - x)^p = 0$, hence x is of the form $1 + n$, with n nilpotent, and $W = \text{Ker } n$ is left fixed by x .

d) $G = \mathbf{G}_a$. Let G_i , $i \in I$, be the increasing filtered family of étale algebraic subgroups of $\mathbf{G}_a$, hence isomorphic to $(\mathbf{Z}/p\mathbf{Z})^{r_i}$ (Prop. 1.5), and let $V_i = V^{G_i}$. Since V is finite-dimensional and nonzero, and V_i is nonzero by c), the decreasing filtered family of the V_i is stationary, and $W = \bigcap_{i \in I} V_i \neq 0$. Now one has the lemma:

Lemma 3.3. *The family of étale subgroups of $(\mathbf{G}_a)_S$ (S a prescheme of characteristic $p > 0$) is schematically dense in G (Exp. IX 4.1).*

By Exp. IX 4.4, it suffices to prove the lemma when S is the spectrum of the prime field $\mathbf{F}_p$. In this case, it suffices to consider the family of étale subgroups G_n ($n \geq 1$) with equation $X^{p^n} - X = 0$, which is schematically dense in $(\mathbf{G}_a)_{\mathbf{F}_p}$, since it contains every closed point.

This being so, let us return to the proof of 3.2 d). If $w \in W$, its stabilizer in G is an algebraic subgroup of $\mathbf{G}_a$ containing G_i for every $i \in I$, hence is equal to $\mathbf{G}_a$ (3.3), and consequently $W = V^G$.

One immediately deduces from 3.2 the

Corollary 3.4. *Let k be a field, G a unipotent algebraic k -group acting on a finite-dimensional k -vector space V . Then V has a sequence of vector subspaces V_i , defined over k , stable under G ,*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V,$$

such that G acts trivially on V_{i+1}/V_i . One can moreover assume that V_{i+1}/V_i is of dimension 1.

We shall now summarize and complete the already proved properties of unipotent groups in the following theorem:

Theorem 3.5. *Let G be an algebraic group defined over a field k . The following properties are equivalent:*

- i) G is unipotent.
- ii) G has a composition series, defined over k , whose successive quotients are isomorphic to $\mathbf{G}_a$ if $p = 0$ (resp. to α_p , $\mathbf{G}_a$, or a twisted $(\mathbf{Z}/p\mathbf{Z})^r$ (1.4) if $p > 0$).
- iii) As in ii), but one moreover assumes that the composition series is central.
- iv) G has a characteristic composition series (Exp. VIB) defined over k , whose successive quotients are isomorphic to $(\mathbf{G}_a)^r$ if $p = 0$ (resp. to $(\alpha_p)^r$, a twisted $(\mathbf{G}_a)^s$ or a twisted $(\mathbf{Z}/p\mathbf{Z})^t$, taken in this order, if $p > 0$).
- v) G is isomorphic to an algebraic subgroup of the group $\text{Trigstr}(n)_k$ of strict upper triangular matrices of the linear group $\text{GL}(n)_k$, for a suitable integer $n \geq 0$.
- vi) G is affine, and for every linear representation of G in a nonzero finite-dimensional k -vector space V , one has $V^G \neq 0$.

Proof.

i) $\Rightarrow$ vi) by 2.1 and 3.2.

vi) $\Rightarrow$ v). Since the algebraic group G is affine, G is an algebraic subgroup of a suitable linear group $\text{GL}(V)$ (Exp. VIB 11.3). Apply 3.4 to the representation of G in V defined by this embedding; one finds v).

v) $\Rightarrow$ iii). One knows that the algebraic group $\text{Trigstr}(n)$ has a central composition series with successive quotients isomorphic to $\mathbf{G}_a$. The composition series induced on G gives property iii), in view of 1.5.

iii) $\Rightarrow$ ii) $\Rightarrow$ i), and iv) $\Rightarrow$ i), are clear.

We shall prove $i) \Rightarrow iv)$ in a moment, but let us already note a few consequences of what has been proved.

Definition 3.6. We shall say that a Lie p -algebra $\mathfrak{g}$ ($p > 0$) (cf. Exp. VIIA § 5) is *unipotent* if the map $x \mapsto x^{(p)}$ is nilpotent, that is, if for every $x \in \mathfrak{g}$ there is an integer $n > 0$ such that $x^{(p^n)} = 0$.

Corollary 3.7. *A unipotent algebraic group G is nilpotent (Exp. VIB § 8); its Lie algebra $\mathfrak{g}$ is nilpotent (Bourbaki, Groupes et algèbres de Lie, Chap. 1 § 4) and is isomorphic to a Lie algebra of nilpotent endomorphisms of a finite-dimensional vector space. In characteristic $p > 0$, $\mathfrak{g}$ is a unipotent Lie p -algebra (3.6).*

Since $i) \Rightarrow v)$, it suffices to prove 3.7 when $G = \text{Trigstr}(n)$. We have already used the fact that $\text{Trigstr}(n)$ is a nilpotent algebraic group. Moreover, the Lie algebra $\mathfrak{h}$ of $\text{Trigstr}(n)$ consists of the upper triangular endomorphisms of V that have zeros on the main diagonal. They are therefore nilpotent, and consequently $\mathfrak{h}$ is nilpotent (Bourbaki, *loc. cit.*, Chap. 1 § 4, cor. 3). If $p > 0$, since the p th power in the Lie p -algebra $\mathfrak{gl}(V) = \text{End}(V)$ coincides with the p th power of endomorphisms of V (Exp. VIIA 6.4.4), one sees that $\mathfrak{h}$ is unipotent.

Corollary 3.8. *Let k be an algebraically closed field, and let G be an algebraic group smooth and affine over k . Then the following properties are equivalent:*

- i) G is unipotent.
- ii) $G(k)$ consists of unipotent elements (BIBLE 4 prop. 4 cor. 1), that is, G is unipotent in the sense of BIBLE.

$i) \Rightarrow ii)$, since G is isomorphic to an algebraic subgroup of a group $\text{Trigstr}(n)$ by 3.2 $i) \Rightarrow v)$. $ii) \Rightarrow i)$. Let therefore G be a unipotent algebraic group in the sense of BIBLE, and denote its neutral component by G^0 . Since the maximal tori of G^0 consist of unipotent elements, they are trivial, hence G^0 is equal to its Cartan subgroups. Consequently, G^0 is solvable (BIBLE 6 Th. 6), hence triangularizable (BIBLE 6, Th. 1). Thus G^0 is an algebraic subgroup of a group $\text{Trigstr}(n)$, and is therefore unipotent in the sense of this exposé.

The group $(G/G^0)(k)$ is a finite group consisting of unipotent elements; it is therefore zero if $p = 0$ and equal to a finite p -group if $p > 0$ (BIBLE 4 prop. 4). But then G/G^0 is unipotent in the sense of this exposé, as a multiple extension of groups isomorphic to $\mathbf{Z}/p\mathbf{Z}$. This proves that G is unipotent.

End of the proof of 3.5. Let us prove $i) \Rightarrow iv)$.

a) $p > 0$. Consider the increasing sequence of algebraic subgroups of G :

$$\{e\} \subset {}_F(G) \subset {}_{F^2}(G) \subset \cdots \subset {}_{F^n}(G) \subset G^0 \subset G.$$

One thus obtains a characteristic composition series of G (App. II 1), and, for n sufficiently large, $G/{}_{F^n}(G)$ is smooth (App. II 3.1), so that the successive quotients are, in order:

- (1) radicial groups of height 1,
- (2) a smooth connected group,
- (3) an étale group.

To prove $i) \Rightarrow iv)$, it therefore suffices for us to prove:

Lemma 3.9. *Let G be a unipotent algebraic group defined over a field k of characteristic $p > 0$. Then G has a characteristic composition series, defined over k , whose successive quotients are isomorphic to:*

- i) $(\alpha_p)^r$ if G is radicial.
- ii) a twisted $(\mathbf{G}_a)^r$ if G is smooth and connected.
- iii) a twisted $(\mathbf{Z}/p\mathbf{Z})^r$ if G is étale.

Proof. i) The group G is radicial. Filtering G by the ${}_{F^n}(G)$, one reduces to the case where G is radicial of height 1. Since G is nilpotent (3.5 $i) \Rightarrow iii)$), and since the center of an

algebraic group is representable (Exp. VIII 6.5 e)), one can consider the ascending central series of G , which is obviously characteristic in G , reducing us to the case where G is moreover commutative.

Let $\mathfrak{g} = \text{Lie}(G)$. The p th-power morphism π is therefore additive in $\mathfrak{g}$ (Exp. VIIA); we shall reduce to the case where it is zero. For every prescheme S over k , put $\mathfrak{g}_S = \mathfrak{g} \otimes_k \mathcal{O}_S$, and let $\mathfrak{h}_S$ be the subsheaf of abelian groups of $\mathfrak{g}_S$ that is the image of $\mathfrak{g}_S$ under π_S . Finally, let $\bar{\mathfrak{h}}_S$ be the subsheaf of $\mathcal{O}_S$ -modules of $\mathfrak{g}_S$ generated by $\mathfrak{h}_S$. It is clear that $\bar{\mathfrak{h}}_S = \bar{\mathfrak{h}}_k \otimes_k \mathcal{O}_S$, and that $\bar{\mathfrak{h}}_S^{(4)}$ is a characteristic Lie p -subalgebra of $\mathfrak{g}_S$ (that is, it is stable under the S -functor $\text{Aut}_{p\text{-Lie}}(\mathfrak{g})$). It then follows from App. II 2.2 that $\bar{\mathfrak{h}}_k$ is the Lie algebra of an algebraic subgroup H of G characteristic in G .

Moreover, in view of 3.7 and Lemma 3.9 bis below, if $G \neq \{e\}$, H is distinct from G , since $\bar{\mathfrak{h}}_k$ is distinct from $\mathfrak{g}$. Furthermore, if $G/H = G''$, one has $\text{Lie } G'' = \mathfrak{g}/\bar{\mathfrak{h}}_k$ (App. II 2.2), and consequently the p th power is zero in $\text{Lie } G''$. Proceeding by induction on $\dim \text{Lie } G$, one is therefore reduced to the case where $\text{Lie } G$ is a Lie p -algebra in which the p th power is zero. But then $\text{Lie } G''$ is isomorphic to $\text{Lie}(\alpha_p)^r$ for a suitable integer $r \geq 0$ (App. II 2.1), and consequently (App. II 2.2) G'' is isomorphic to $(\alpha_p)^r$. It remains to prove

Lemma 3.9 bis. *Let k be a field of characteristic $p > 0$, $\mathfrak{g}$ a commutative unipotent Lie p -algebra (3.6), finite-dimensional over k , and $\mathfrak{h}$ the Lie p -subalgebra of $\mathfrak{g}$ generated by the image of the p th power in $\mathfrak{g}$. Then, if $\mathfrak{g} \neq 0$, one has $\mathfrak{h} \neq \mathfrak{g}$.*

Indeed, since $\mathfrak{g}$ is commutative, $\mathfrak{h}$ is simply the k -vector subspace of $\mathfrak{g}$ generated by the $X^{(p)}$ ($X \in \mathfrak{g}$). If $\mathfrak{g} \neq 0$ and is unipotent, there is $X \in \mathfrak{g}$, $X \neq 0$, such that $X^{(p)} = 0$. Let $X_1, \dots, X_n$ be a basis of a complement in $\mathfrak{g}$ to the line kX . The Lie algebra $\mathfrak{h}$ is then the k -vector subspace of $\mathfrak{g}$ generated by $X_1^{(p)}, \dots, X_n^{(p)}$, hence is of dimension at most $n = \dim \mathfrak{g} - 1$.

Proof of 3.9 ii). G is smooth and connected. In this case, the descending central series of G is representable by smooth connected characteristic algebraic subgroups G_i (Exp. VIB 8.3 and 7.4), and $G_i = 0$ for i sufficiently large, since G is nilpotent (3.5 i) $\Rightarrow$ iii)). It suffices to prove 3.9 for the groups G_i/G_{i+1} , reducing us to the case where G is moreover commutative. For every integer $n > 0$, let G_n be the algebraic subgroup of G that is the image of G under the morphism of raising to the p^n th power. The group G_n is therefore smooth, connected and characteristic, and it follows from Definition 1.1 of unipotent groups that $G_n = 0$ for n sufficiently large. Replacing G by G_n/G_{n+1} , one can moreover assume that G is annihilated by raising to the p th power. But then, by (J.-P. Serre, *Groupes algébriques et corps de classes*, chap. VII, prop. 11), G is a form of $(\mathbf{G}_a)^r$, for a suitable integer r . *Proof of 3.9 iii).* G is étale. Proceeding as in ii), one reduces to the case where G is commutative, then to the case where G is annihilated by p ; but then $G_{\bar{k}}$ is isomorphic to $(\mathbf{Z}/p\mathbf{Z})^r$.

Proof of 3.5 i) $\Rightarrow$ iv) in case b), $p = 0$. The group G is then smooth and connected, and, proceeding as in 3.9 ii), one reduces to the case where G is moreover commutative. One then has the following more precise result:

Lemma 3.9 ter. *Let k be a field of characteristic 0, G a commutative unipotent algebraic k -group, $\mathfrak{g} = \text{Lie } G$. Then there is a canonical isomorphism:*

$$\exp : W(\mathfrak{g}) \xrightarrow{\sim} G.$$

The morphism $\exp$ is the unique homomorphism $W(\mathfrak{g}) \rightarrow G$ inducing the identity on the Lie algebras.

Since G is unipotent, G can be realized as an algebraic subgroup of $\text{Trigstr}(n)$ for a suitable integer n (3.5 i) $\Rightarrow$ v)). The choice of such an embedding allows one to identify $\mathfrak{g}$

(4) **N.D.E.** One has changed $\mathfrak{h}$ to $\bar{\mathfrak{h}}$.

with a Lie subalgebra of $\mathfrak{gl}(n)$ consisting of nilpotent endomorphisms. Hence a k -morphism:

$$\exp : W(\mathfrak{g}) \longrightarrow \mathrm{GL}(n), \quad T \longmapsto \sum_{i \geq 0} \frac{T^i}{i!}.$$

Since G is commutative, the morphism $\exp$ is a homomorphism. Let G' be the algebraic group that is the image of $W(\mathfrak{g})$ under the morphism $\exp$. If one canonically identifies $\mathrm{Lie} W(\mathfrak{g})$ with $\mathfrak{g}$, the linear tangent map to $\exp$ is simply the injection $\mathfrak{g} \longrightarrow \mathfrak{gl}(n)$. Consequently, $\mathrm{Lie}(G \cap G') = \mathfrak{g} \cap \mathfrak{g} = \mathfrak{g}$. Since $G \cap G'$ is smooth (Exp. VIB 1.6.1), and G is connected (since it is a multiple extension of groups $\mathbf{G}_a$ (3.5 i) $\Rightarrow$ ii)), one necessarily has $G = G'$. The kernel $\mathrm{Ker}(\exp)$ is an étale unipotent group, hence is the unit group, and consequently $\exp$ is an isomorphism from $W(\mathfrak{g})$ onto G .

If $h : W(\mathfrak{g}) \longrightarrow G$ is another homomorphism such that $\mathrm{Lie}(h)$ is the identity map of $\mathfrak{g}$, the morphism $h - \exp$ is a homomorphism (G is commutative) whose linear tangent map is zero. Since k is of characteristic 0 and $W(\mathfrak{g})$ is connected, it again follows from Cartier's theorem that one necessarily has $h = \exp$.

Proposition 3.10. *Let G be an algebraic group defined over an algebraically closed field k . Then the following properties are equivalent:*

- i) Every morphism from a group of multiplicative type M to G is the zero morphism.
- i bis) G has no nonzero subgroups of multiplicative type.
- ii) a) if $p = 0$: $G(k)$ contains no points of finite order other than e ;
 b) if $p \neq 0$: for every prime number $q \neq p$, $x \in G(k)$ and $x^q = e$ imply $x = e$,
 for every $X \in \mathfrak{g} = \mathrm{Lie} G$ such that $X^{(p)} = X$, one has $X = 0$.
- ii bis) a) if $p = 0$: as in ii) a);
 b) if $p \neq 0$: every finite subgroup of $G(k)$ is a p -group,
 for every $X \in \mathfrak{g} = \mathrm{Lie} G$ such that $X^{(p)} = X$, one has $X = 0$.

Proof.

i) $\Leftrightarrow$ i bis), since the image of a group of multiplicative type is of multiplicative type (Exp. IX 2.7).

ii) $\Leftrightarrow$ ii bis). For if an ordinary finite group H has an order that is not a power of p , there are a prime number $q \neq p$ and an element x of H distinct from e such that $x^q = e$ (Sylow's theorem, cf. J.-P. Serre, *Corps locaux*, chap. IX § 2).

i) $\Rightarrow$ ii) follows from the following lemma:

Lemma 3.11. *Let G be an algebraic group defined over a field k .*

a) *If k contains the n th roots of unity, n an integer prime to p , one has:*

$$\mathrm{Hom}_{k\text{-gr}}(\mu_n, G) \simeq \mathrm{Hom}_{k\text{-gr}}(\mathbf{Z}/n\mathbf{Z}, G) \simeq {}_nG(k)$$

(points of order n of $G(k)$).

b) *If $p > 0$, $\mathrm{Hom}_{k\text{-gr}}(\mu_p, G) \simeq \{X \in \mathfrak{g} = \mathrm{Lie} G \mid X^{(p)} = X\}$.*

Indeed, to prove a), one notes that μ_n is then isomorphic over k to $(\mathbf{Z}/n\mathbf{Z})$, and b) follows from App. II 2.1.

ii) $\Rightarrow$ i bis). By Lemma 3.11, ii) is equivalent to the fact that for every prime number r , G contains no subgroups μ_r , which implies i bis) by:

Lemma 3.12. *Let G be a diagonalizable S -group, of finite type over S and distinct from the unit group. Then there are a prime number r and a subgroup of G isomorphic to $(\mu_r)_S$.*

Let $G = D_S(M)$, where M is a finitely generated abelian group, hence an extension of a free group M'' by a finite group M' . If $M'' \neq 0$, it is clear that M admits quotients isomorphic to $\mathbf{Z}/r\mathbf{Z}$ for every integer r . If $M'' = 0$, M' admits a quotient isomorphic to $\mathbf{Z}/r\mathbf{Z}$ for every prime number r dividing the order of M . One deduces the lemma by duality.

We have seen (Prop. 2.4) that a unipotent group satisfies the equivalent conditions of 3.10. The purpose of the following paragraph is to prove the converse.

4. A characterization of unipotent groups

As announced, we shall show that an algebraic group G defined over an algebraically closed field k , containing no nonzero subgroup of multiplicative type, is unipotent. In fact, it suffices that it contain no “elementary” subgroups of multiplicative type of certain very particular kinds, depending on the hypotheses made on G . Before stating the general theorem, let us study a few particular cases in detail.

4.1. Smooth connected affine algebraic groups. —

Proposition 4.1.1. *Let k be a field, G a smooth connected affine algebraic k -group, $\mathfrak{g} = \text{Lie } G$. Then the following properties are equivalent:*

- i) G is unipotent.
- ii) G has a central composition series whose successive quotients are forms of $\mathbf{G}_a$.
- iii) G has a central characteristic composition series whose successive quotients are forms of $(\mathbf{G}_a)^r$.
- iv) There is an integer $n > 1$ such that $G_{\bar{k}}$ contains no subgroup isomorphic to μ_n .
- v) Every maximal torus of G is the unit group.

Suppose moreover that G is an algebraic subgroup of a linear group $\text{GL}(n)$. Then the preceding conditions are also equivalent to:

- vi) $\mathfrak{g} \subset \mathfrak{gl}(n)$ consists of nilpotent endomorphisms.
- vii) $\mathfrak{g}$ is nilpotent, and its center contains no nonzero semisimple endomorphism.

Proof. ii) $\Rightarrow$ i) is clear, and i) $\Rightarrow$ iii) was seen in 3.9. The implication iii) $\Rightarrow$ ii) will follow from the following lemma:

Lemma 4.1.2. *Let k be a field, G an algebraic k -group that is a form of $(\mathbf{G}_a)^r$. Then:*

- a) G can be realized as an algebraic subgroup of the group $(\mathbf{G}_a)^n$ for a suitable integer n .
- b) G has a composition series whose successive quotients are forms of $\mathbf{G}_a$.

Indeed, by hypothesis, there is an extension k' of k such that $G_{k'}$ is isomorphic to $(\mathbf{G}_{a,k'})^r$. By the principle of finite extension (EGA IV 9.1.1), one can assume that k' is a finite extension of k . But then, for a), it suffices to consider the canonical closed immersion (EGA V⁽⁵⁾):

$$G \longrightarrow \prod_{k'/k} (G_{k'})/k' \xrightarrow{\sim} (\mathbf{G}_{a,k})^n \quad (\text{with } n = r \deg(k'/k)).$$

To prove b), in view of a), one can assume that G is a closed subgroup of $G' = (\mathbf{G}_a)^n$. If $G \neq 0$, there is a hyperplane $\mathfrak{h}$ of $\mathfrak{g}' = \text{Lie } G'$ that does not contain $\mathfrak{g}$. Let H be the vector subgroup $W(\mathfrak{h})$ of $W(\mathfrak{g}') = G'$. Since H is defined by one equation in G' , $H \cap G$ is defined by one equation in G , and one has the inequalities:

$$\begin{aligned} \dim G - 1 &\leq \dim(G \cap H) \\ &\leq \dim \text{Lie}(G \cap H) = \dim_k(\mathfrak{g} \cap \mathfrak{h}) \\ &= \dim_k(\mathfrak{g}) - 1 = \dim G - 1. \end{aligned}$$

Hence $\dim(G \cap H) = \dim \text{Lie}(G \cap H)$, and consequently $G \cap H$ is smooth. The group $G_1 = (G \cap H)^0$ is a smooth connected algebraic subgroup of G such that G/G_1 is smooth, connected, of dimension 1, hence is a form of $\mathbf{G}_a$ (4.1 i) $\Rightarrow$ iii)). One finishes by induction on the dimension of G .

Before continuing the proof of 4.1, note that the equivalence i) $\Leftrightarrow$ ii) and 2.3 bis imply the following corollary:

⁽⁵⁾**N.D.E.** Give another reference here. . .

Corollary 4.1.3. *If k is a perfect field, a smooth connected algebraic k -group is unipotent if and only if it has a composition series with successive quotients isomorphic to $\mathbf{G}_a$.*

Continuation of the proof of 4.1.

i) $\Rightarrow$ iv) by 2.4 i).

iv) $\Rightarrow$ v). By Exp. XIV 4.1, G has a maximal torus T defined over k . Now, if $r = \dim T$, $({}_nT)_{\bar{k}}$ is isomorphic to $(\mu_n)^r$. Hence $r = 0$.

v) $\Rightarrow$ i), as one observed in the proof of 3.8.

i) $\Rightarrow$ vi). By 3.4, G is in fact contained in an algebraic subgroup of $\mathrm{GL}(n)$ isomorphic to $\mathrm{Trigstr}(n)$, hence $\mathfrak{g}$ consists of nilpotent endomorphisms.

vi) $\Rightarrow$ vii). Indeed, $\mathfrak{g}$ is nilpotent by Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, chap. I § 4 cor. 3).

vii) $\Rightarrow$ v). Let T be a maximal subtorus of G (Exp. XIV 1.1), $\mathfrak{t}$ its Lie algebra. The embedding of G in $\mathrm{GL}(n)$ defines a representation of T that is necessarily semisimple (this is seen over an algebraic closure $\bar{k}$ of k , and one applies Exp. I 4.7.3). Thus, if $X \in \mathfrak{t}$, X is a semisimple endomorphism in $\mathfrak{gl}(n)$. One immediately sees that this implies that the map:

$$\mathrm{ad} X : Y \mapsto [X, Y]$$

is a semisimple endomorphism of $\mathfrak{gl}(n)$, hence of $\mathfrak{g}$. Since, moreover, this endomorphism is nilpotent, $\mathfrak{g}$ being nilpotent, $\mathrm{ad} X$ is zero, hence X is central. But then $\mathfrak{t}$ is central and consists of semisimple endomorphisms, hence is zero by hypothesis; a fortiori, T is the unit group.

Remark 4.1.4. a) We shall give later (4.3.1) an infinitesimal characterization of unipotent groups in characteristic $p > 0$, independent of an embedding in $\mathrm{GL}(n)$.

b) When k is perfect, conditions ii) and iii) of 4.1.1 simplify by virtue of the following lemma:

Lemma 4.1.5. *If k is a perfect field, every algebraic k -group G that is a form of $(\mathbf{G}_a)^r$ is isomorphic to $(\mathbf{G}_a)^r$.*

The lemma follows from 3.9 ter if the characteristic p of k is zero, and from 2.3 bis if $r = 1$. In the general case ($p > 0$), realize G as an algebraic subgroup of $(\mathbf{G}_a)^n$ for a suitable integer n (4.1.2), and argue by induction on the integer $n - r$. If $r = n$, one does indeed have $G = (\mathbf{G}_a)^r$. Otherwise, the quotient group $\mathbf{G}_a^n/G$ is a nonzero smooth connected unipotent group which, in view of 4.1.1 i) $\Rightarrow$ ii) and 2.3 bis, has a composition series with quotients isomorphic to $\mathbf{G}_a$. One deduces that there is a smooth connected algebraic subgroup G_1 of $(\mathbf{G}_a)^n$, containing G , such that $\mathbf{G}_a^n/G_1 = \mathbf{G}_a$. By induction, it suffices to show that G_1 is isomorphic to $(\mathbf{G}_a)^{n-1}$. Now it is immediate to check that a homomorphism from $\mathrm{Spec} k[X_1, \dots, X_n] = \mathbf{G}_a^n$ to $\mathbf{G}_a = \mathrm{Spec} k[T]$ is defined by an additive polynomial of the form:

$$\sum_{i,j} a_{i,j} X_i^{p^j}.$$

Since G_1 is smooth, the linear part of this polynomial is nonzero. After making a linear change in the coordinates X_i , we can assume that G_1 is an algebraic subgroup of $(\mathbf{G}_a)^n$ defined by the equation:

$$(*) \quad P(X) = -X_1 + \sum_{j=1}^q a_j X_1^{p^j} + Q(X_2, \dots, X_n) = 0,$$

$$\text{with } Q(X_2, \dots, X_n) = \sum_{\substack{i \geq 1 \\ j \geq 0}} b_{i,j} X_i^{p^j}.$$

Let us then proceed by induction on the degree of P . If $\deg P = 1$, it is clear that $G_1 \xrightarrow{\sim} \mathbf{G}_a^{n-1}$. Otherwise, since k is perfect, one has $Q(X) = Q_1(X)^p$, and we can define an endomorphism

v of $\mathbf{G}_a^n$ by the formulas:

$$X_i \mapsto X_i \quad \text{for } i > 1, \quad X_1 \mapsto \sum_{j=1}^q a_j^{1/p} X_1^{p^{j-1}} + Q_1(X).$$

It is clear that v induces an isomorphism on G_1 and that $v(G_1)$ has equation in $\mathbf{G}_a^n$:

$$(*)_1 \quad P_1(X) = -X_1 + \sum_{j=1}^q a_j^{1/p} X_1^{p^j} + Q_1(X_2, \dots, X_n) = 0.$$

Let us then distinguish two cases:

First case. $\deg(Q) = \deg P > p^q$. Then one has $\deg P_1 < \deg P$, and the induction hypothesis gives the result.

Second case. $\deg P = p^q$ ($a_q \neq 0$). One then has $\deg P_1 = \deg P = p^q$ and $\deg Q_1 < p^q$. (One cannot have $Q = 0$, for then G_1 would not be connected.) If the polynomial Q_1 has no linear part, one can repeat the preceding transformation. Continuing the process, one finally obtains an equation of the form:

$$(*)_s \quad P_s(X) = -X_1 + \sum_{j=1}^q a_j^{1/p^s} X_1^{p^j} + Q_s(X),$$

where $Q_s(X) = Q(X_2, \dots, X_n)^{1/p^s}$ is an additive polynomial with a nonzero linear part and, moreover, $\deg Q_s < p^q$. Suppose, for example, that the coefficient of X_2 in Q_s is nonzero, and let $-L$ be the linear part of $P_s(X)$. After making a linear change of coordinates, the equation of G_1 becomes:

$$P'(X) = -L + \sum_{j=1}^q a_j^{1/p^s} X_1^{p^j} + Q'(L, X_3, \dots, X_n),$$

where Q' is an additive polynomial, without linear part, and $\deg(Q') < p^q$. But then we are reduced to the first case⁽⁶⁾.

4.2. Radicial groups. —

Proposition 4.2.1. *Let G be a radicial algebraic group defined over a field k of characteristic $p > 0$. Then the following conditions are equivalent:*

- i) G is unipotent.
- ii) G has a central composition series with successive quotients isomorphic to α_p .
- iii) G has a central characteristic composition series with successive quotients isomorphic to $(\alpha_p)^r$.
- iv) $G_{\bar{k}}$ contains no subgroup isomorphic to μ_p .
- v) $\mathfrak{g} = \text{Lie } G$ is a unipotent Lie p -algebra (3.6).

Proof. iii) $\Rightarrow$ ii) $\Rightarrow$ i) is clear, i) $\Rightarrow$ iii) is 3.9 i), and i) $\Rightarrow$ iv) by 2.4 i).

We shall need the following lemma on abelian Lie p -algebras:

Lemma 4.2.2. *Let $\mathfrak{g}$ be an abelian Lie p -algebra, finite-dimensional over a perfect field k . Then $\mathfrak{g}$ can be written uniquely as the direct sum of a Lie p -subalgebra $\mathfrak{r}$ on which the p th power is bijective and a unipotent Lie p -subalgebra $\mathfrak{u}$ (3.6). (The algebra $\mathfrak{r}$ will be called the reductive part of $\mathfrak{g}$, and $\mathfrak{u}$ the unipotent part.) The formation of $\mathfrak{r}$ and $\mathfrak{u}$ is compatible with extension of the field k . If, moreover, k is algebraically closed, $\mathfrak{r}$ admits a basis e_i such that $e_i^{(p)} = e_i$.*

The proof of this lemma is easy and left to the reader (cf. Bourbaki, *Groupes et algèbres de Lie*, chap. I § 1 exercise 23). Let us simply say that $\mathfrak{u}$ is the kernel of a suitable iterate of the map $X \mapsto X^{(p)}$, and that $\mathfrak{r}$ is the image of the same iterate.

⁽⁶⁾**N.D.E.** By replacing X_1 by L .

This being so, let us prove $\text{iv}) \Rightarrow \text{v})$. After extending the base field, one can assume that k is algebraically closed.

Let then X be an element of $\mathfrak{g}$, and $\mathfrak{h}$ the Lie p -subalgebra generated by X in $\mathfrak{g}$. The algebra $\mathfrak{h}$ is obviously commutative, and its reductive part (4.2.2) is zero, for otherwise, by *loc. cit.*, $\mathfrak{h}$ would contain an element $Y \neq 0$ such that $Y^{(p)} = Y$, and consequently (App. II 2.1 and 2.2) G would contain a subgroup isomorphic to μ_p , contrary to the hypothesis. Thus $\mathfrak{h}$, and consequently $\mathfrak{g}$, is a unipotent Lie p -algebra.

$\text{v}) \Rightarrow \text{i})$. This is the least trivial implication of 4.2.1.

a) *Case where G is of height 1* (Exp. VIIA 4.1.3). Since G is radicial, it is affine, hence isomorphic to an algebraic subgroup of a linear group $\text{GL}(V)$ (Exp. VIB § 11). This embedding identifies $\mathfrak{g}$ with a Lie p -subalgebra of $\text{End}(V)$, the p th power of X in $\mathfrak{g}$ coinciding with the p th power of the endomorphism X (Exp. VIIA 6.4.4). Since $\mathfrak{g}$ is unipotent by hypothesis, $\mathfrak{g}$ is therefore an algebra of nilpotent endomorphisms of V , and by *Engel's theorem* (Bourbaki, *Groupes et algèbres de Lie*, chap. I § 4 th. 1) is a Lie subalgebra of the Lie algebra $\mathfrak{h}$ of the group of strict upper triangular matrices $\text{Trigstr}(n)$ with respect to a suitable basis of V . Since G is of height 1, one then deduces from App. II 2.2 that G itself is an algebraic subgroup of $\text{Trigstr}(n)$, hence is unipotent (3.5 v) $\Rightarrow$ i)).

b) *General case.* Let us proceed by induction on the height h of G^* . The case $h = 1$ has just been treated. Suppose that $h > 1$, and put $G' = {}_F G$ and $G'' = G/G'$. The group G' is of height 1, and $\text{Lie } G' = \text{Lie } G$ is unipotent, hence G' is unipotent by a). To show that G is unipotent, it therefore suffices to prove that G'' is unipotent (2.2). But G'' is of height $h - 1$, hence, by the induction hypothesis, it suffices to show that $\text{Lie } G''$ is unipotent. Since $\text{iv}) \Rightarrow \text{v})$, it suffices to show that $G''_{\bar{k}}$ contains no groups isomorphic to μ_p . Let therefore a subgroup of $G''_{\bar{k}}$ isomorphic to μ_p be given, and let H be its inverse image in $G'_{\bar{k}}$. Since the group G' is unipotent, we shall prove in § 5 that the extension:

$$e \longrightarrow G'_{\bar{k}} \longrightarrow H \longrightarrow \mu_p \longrightarrow e$$

is necessarily trivial (the proof given of this fact is independent of the results of the present paragraph). Thus the group μ_p lifts into H , but, being of height 1, it is necessarily contained in $G'_{\bar{k}} = {}_F G_{\bar{k}}$; hence a contradiction, since G' is unipotent.

4.3. Connected affine groups in characteristic $p > 0$. —

Proposition 4.3.1. *Let G be a connected affine algebraic group over a field k of characteristic $p > 0$. Then the following conditions are equivalent:*

- i) G is unipotent.
- ii) G has a composition series with successive quotients isomorphic to α_p and $\mathbf{G}_a$ (taken in this order).
- iii) G admits a characteristic composition series with successive quotients isomorphic to $(\alpha_p)^r$ and $(\mathbf{G}_a)^s$ (taken in this order).
- iv) $G_{\bar{k}}$ contains no subgroup isomorphic to μ_p .
- v) $\mathfrak{g} = \text{Lie } G$ is unipotent (3.6).
- vi) $\mathfrak{g}$ is nilpotent, and the reductive part of the center of $\mathfrak{g}$ (4.2.2) is trivial.
- vi bis) $\mathfrak{g}$ is nilpotent, and every subgroup of multiplicative type of the neutral component of the center of G is zero.
- vi ter) G is nilpotent, and every subgroup of multiplicative type of the neutral component of the center of G is zero.

Proof. It is clear that $\text{iii}) \Rightarrow \text{ii}) \Rightarrow \text{i})$. To establish $\text{i}) \Rightarrow \text{iii})$, we shall need the following lemma:

Lemma 4.3.2. *Let k be a field of characteristic $p > 0$, $n \in \mathbf{N}$, k' a radicial extension of k such that $(k')^{p^n}$ is contained in k ; for every k -prescheme X (resp. every k' -prescheme X'),*

*I.e. the least integer such that $F^h = \text{id}_G$, cf. App. II 1.

denote by $X^{(p^n)}$ (resp. X'_φ) the k -prescheme obtained from X (resp. X') by the base change:

$$F^n : k \longrightarrow k, \quad x \longmapsto x^{p^n}, \quad (\text{resp. } \varphi : k' \longrightarrow k, \quad x' \longmapsto x'^{p^n}).$$

Then, for every k -prescheme X , there is a functorial isomorphism:

$$(X_{k'})_\varphi \xrightarrow{\sim} X^{(p^n)}.$$

Consequently, if X and Y are two k -preschemes such that there is a k' -isomorphism $u' : X_{k'} \xrightarrow{\sim} Y_{k'}$, then there is a k -isomorphism $v : X^{(p^n)} \xrightarrow{\sim} Y^{(p^n)}$. If, moreover, X and Y are endowed with structures of group k -preschemes, and if u' is a k' -homomorphism, then v is a k -homomorphism.

The lemma simply follows from the transitivity of base changes and the fact that the composite morphism $k \longrightarrow k' \xrightarrow{\varphi} k$ is equal to F^n .

Continuation of the proof of 4.3.1.

i) $\Rightarrow$ iii). Let us proceed by induction on $\dim G$. If $\dim G = 0$, since G is connected, it is radicial, and one applies 3.9 i). If $\dim G > 0$, there is an integer $m \geq 0$ such that the quotient $G/F^m G$ is a smooth group (App. II 3.1), obviously connected and nonzero. Applying 4.1.1 i) $\Rightarrow$ iii) to the latter, one sees that there is an algebraic subgroup G' of G that is characteristic and connected and such that the quotient $G'' = G/G'$ is a form of $\mathbf{G}_a^r$ ($r > 0$). By 4.1.5, if K is a perfect closure of k , one has $G''_K \xrightarrow{\sim} (\mathbf{G}_{a,K})^r$. Since G'' is of finite type over k , there is a finite radicial extension k' of k such that $G''_{k'} \xrightarrow{\sim} (\mathbf{G}_{a,k'})^r$ (Exp. VIB § 10). Let $n > 0$ be such that $(k')^{p^n} \subset k$. Keeping the notation of 4.3.2, one deduces that there is a k -isomorphism of algebraic groups:

$$(\mathbf{G}_{a,k})^r = (\mathbf{G}_{a,k'})^r_\varphi \xrightarrow{\sim} (G'')^{(p^n)}.$$

Consider then the Frobenius homomorphism relative to G'' (Exp. VIIA § 4)

$$F^n : G'' \longrightarrow (G'')^{(p^n)}.$$

Since G'' (hence also $(G'')^{(p^n)}$) is smooth over k , and F^n is radicial, F^n is an epimorphism for the fpqc topology, so that $(G'')^{(p^n)}$ is identified with $G''/F^n(G'')$. Finally, we have shown that $G''/F^n(G'')$ is isomorphic, as an algebraic group, to $(\mathbf{G}_a)^r$. The inverse image G'_n of $F^n(G'')$ in G is a connected characteristic subgroup of G of dimension strictly less than that of G , to which we can apply the induction hypothesis.

i) $\Rightarrow$ iv) by 2.4 i).

iv) $\Rightarrow$ i). Consider G as an extension of a smooth connected group G'' by a radicial group G' (App. II 3.1). The group G' is unipotent (4.2.1 iv) $\Leftrightarrow$ i)). It suffices to see that G'' is unipotent, and for this it suffices to show that $G''_{\bar{k}}$ contains no subgroup isomorphic to μ_p (4.1.1 i) $\Leftrightarrow$ iv)). Now, if $G''_{\bar{k}}$ contained a subgroup μ_p , the latter would lift into $G_{\bar{k}}$ by the result (5.1) already used and proved in § 5; hence a contradiction with iv).

i) $\Rightarrow$ v) by 3.7.

v) $\Rightarrow$ vi). Indeed, since $(\text{ad } X)^{p^r} = \text{ad}(X^{(p^r)})$ (VIIA 5.2), $\text{ad } X$ is nilpotent if $\mathfrak{g}$ is unipotent, hence $\mathfrak{g}$ is nilpotent by *Engel's theorem* (Bourbaki, *Groupes et algèbres de Lie*, chap. I § 4). Moreover, if $\mathfrak{g}$ is unipotent, the same is obviously true of its center, whose reductive part is then trivial (4.2.2).

vi) $\Rightarrow$ iv). Indeed, if $G_{\bar{k}}$ contains a subgroup isomorphic to μ_p , there is an element $X \neq 0$ of its Lie algebra such that $X^{(p)} = X$ (App. II 2.1), hence $X^{(p^r)} = X$ for every $r > 0$. Since $\text{ad } X$ is nilpotent because $\mathfrak{g}$ is nilpotent, and $(\text{ad } X)^{p^r} = \text{ad } X^{(p^r)}$, one necessarily has $\text{ad } X = 0$, hence X belongs to the reductive part of the center of $\mathfrak{g}_{\bar{k}}$; hence a contradiction with vi).

i) $\Rightarrow$ vi ter) follows from 2.4 i) and 3.5 i) $\Rightarrow$ iii).

vi ter) $\Rightarrow$ vi bis). Indeed, if G is nilpotent, so is its subgroup ${}_F G$. Moreover, it follows from App. II 2.2 that ${}_F G$ is nilpotent if and only if $\text{Lie } {}_F G = \text{Lie } G$ (Exp. VIIA) is nilpotent.

vi bis) $\Rightarrow$ vi). Let Z be the neutral component of the center of G , and let $\mathfrak{r}$ be the reductive component of the center of $\mathfrak{g}$. We must show that $\mathfrak{r} = 0$. Now it is immediate that $\mathfrak{r}$ is a characteristic Lie p -subalgebra of $\mathfrak{g} = \text{Lie } {}_F G$ (that is, stable under the functor $\underline{\text{Aut}}_{p\text{-Lie}}(\mathfrak{g})$); hence $\mathfrak{r}$ is the Lie algebra of a characteristic radicial subgroup R of ${}_F G$ (App. II 2.2). Moreover, it follows from the last assertion in 4.2.2 and from App. II 2.1 that R is a form of $(\mu_p)^r$. Since the group R is characteristic in ${}_F G$, which is itself a characteristic subgroup of G (App. II 1), R is a fortiori invariant in G , hence is central, since G is connected (Exp. IX 5.5). Thus, by hypothesis vi bis), R is zero, and therefore so is $\mathfrak{r}$.

4.4. Étale groups. — The following proposition is an easy consequence of Sylow's theorems and of the structure of finite q -groups (cf. J.-P. Serre, *Corps locaux*, chap. IX § 1).

Proposition 4.4.1. *Let G be a finite étale algebraic group defined over an algebraically closed field k . Then, for G to be unipotent, it is necessary and sufficient that for every prime number q distinct from the characteristic p of k , G contain no subgroup isomorphic to μ_q .*

4.5. Abelian varieties. — Let G be an abelian variety defined over an algebraically closed field k . Then the following conditions are equivalent:

- i) G is unipotent.
- ii) $G = 0$.
- iii) There is an integer n , prime to the characteristic p of k , such that G contains no subgroup isomorphic to μ_n .

Indeed, if G is an abelian variety of dimension d , one knows (cf. S. Lang, *Abelian varieties*, chap. IV § 3, th. 6) that the group ${}_n G(k)$ (n an integer prime to p) is isomorphic to $(\mathbf{Z}/n\mathbf{Z})^{2d}$, hence is isomorphic to $(\mu_n)^{2d}$. Hence iii) $\Rightarrow$ ii), and ii) $\Rightarrow$ i) $\Rightarrow$ iii) are obvious.

4.6. General case. — If G and H are two algebraic groups defined over an algebraically closed field k , we shall denote by $P(G, H)$ the property: “there is no algebraic subgroup of G isomorphic to H ”. One then obtains the following characterizations of unipotent groups:

Theorem 4.6.1. *Let G be an algebraic group defined over an algebraically closed field k of characteristic p . Then:*

- i) *If G is smooth, affine and connected:*

$$\begin{aligned} G \text{ is unipotent} &\iff \exists n > 1 \text{ such that } P(G, \mu_n) \text{ is true} \\ &\iff P(G, \mathbf{G}_m) \text{ is true.} \end{aligned}$$

- ii) *If G is smooth and connected:*

$$G \text{ is unipotent} \iff \exists n \text{ prime to } p \text{ such that } P(G, \mu_n) \text{ is true.}$$

- iii) *If G is smooth:*

$$G \text{ is unipotent} \iff \text{for every prime number } n \neq p, P(G, \mu_n) \text{ is true.}$$

- iv) *G is affine and connected, and $p > 0$:*

$$G \text{ unipotent} \iff P(G, \mu_p) \text{ is true.}$$

- v) *G is connected and $p > 0$:*

$$\begin{aligned} G \text{ unipotent} &\iff \exists n \text{ prime to } p \text{ such that } P(G, \mu_n) \text{ is true} \\ &\quad \text{and } P(G, \mu_p) \text{ is true.} \end{aligned}$$

- vi) *G any algebraic group:*

$$G \text{ is unipotent} \iff \text{for every prime number } n, P(G, \mu_n) \text{ is true.}$$

Proof. i) follows from 4.1.1, and iv) from 4.3.1. We shall prove vi); ii), iii), v) are proved similarly and are left to the reader.

Let therefore G be an algebraic group. If G is unipotent, $P(G, \mu_n)$ is true for every $n > 1$ (2.4 i)). Conversely, suppose that $P(G, \mu_n)$ is true for every prime number n , and let us show that G is unipotent. Let G^0 be the neutral component of G . If G^0 is smooth, it follows from a classical theorem of Chevalley* that G^0 is an extension of an abelian variety A by a smooth connected affine group L . If G is not smooth, which supposes $p > 0$ (Exp. VIB 1.6.1), there is an integer $n > 0$ such that $G'' = G^0 /_{F^n}(G)$ is smooth (App. II 3.1). Thus G'' is an extension of an abelian variety A by a smooth connected linear group L'' . Denote by L the inverse image of L'' in G^0 , which is again affine and connected, since $_{F^n}(G)$ is radicial. In all cases, G therefore has a composition series:

$$0 \subset L \subset G^0 \subset G$$

such that L is affine and connected, $G^0/L = A$ is an abelian variety, and G/G^0 is an étale group.

If $P(G, \mu_n)$ is true, a fortiori $P(L, \mu_n)$ is true, hence L is unipotent (4.1.1 and 4.3.1). If A is nonzero, there are a prime number n and a subgroup of A isomorphic to μ_n (4.5); by 5.1 below, this subgroup lifts into G , contradicting hypothesis $P(G, \mu_n)$; hence $A = 0$. Finally, if G/G^0 is not unipotent, there are an integer q and a subgroup of G/G^0 isomorphic to μ_q (4.4.1). One deduces, as above, that $P(G, \mu_q)$ is not true; hence G/G^0 is unipotent, and consequently the same is true of G .

5. Extension of a group of multiplicative type by a unipotent group

5.1. Statement of the theorem. —

Definition 5.1.0. Let k be a field, G an algebraic k -group. Following the terminology introduced by Rosenlicht (*Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Math. 61 (1963)), we shall say that G is “ k -solvable” if it satisfies the following equivalent conditions:

- i) G has a composition series with successive quotients isomorphic to $\mathbf{G}_a$.
- ii) G has a characteristic composition series (G_i) such that the successive quotients G_i/G_{i+1} are commutative and have a composition series with successive quotients isomorphic to $\mathbf{G}_a$.

The fact that i) $\Rightarrow$ ii) is proved in *loc. cit.*. In fact, for the composition series (G_i) , one can take the composition series introduced in the proof of 3.9 ii).

Theorem 5.1.1. Let k be a field, U a unipotent algebraic k -group, H a k -group of multiplicative type, E an algebraic k -group that is an extension of H by U , so that one has the exact sequence:

$$1 \longrightarrow U \longrightarrow E \longrightarrow H \longrightarrow 1.$$

Then:

- i) The extension E is trivial in each of the following cases:
 - a) k is algebraically closed.
 - b) k is perfect, and one of the groups U or H is connected.
 - c) U is k -solvable.
 - d) U is smooth and H is connected.
- ii) If H' and H'' are two liftings of H into E , then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:
 - a) k is algebraically closed and H is smooth.
 - b) k is algebraically closed and U is smooth.

*Sém. Bourbaki no. 145, 1956/57.

(We point out, during the proof, other cases where the conclusion of ii) is true without assuming that k is algebraically closed.)

If U is an algebraic group (resp. a commutative algebraic group) defined over a field k , we denote by $H^1(k, U)$ (resp. $H^i(k, U)$, $i \geq 0$) the pointed set (resp. the i th group) of Galois cohomology of k with values in U (cf. J.-P. Serre, *Cohomologie Galoisienne*, Lecture Notes Maths. no. 5, Springer).

If S is a prescheme, H a commutative group S -prescheme, G a group S -prescheme acting on H , we denote by $H^i(G, H)^{(7)}$ the i th Hochschild cohomology group of G with values in H (App. I 1).

To prove 5.1.1, we shall proceed in several steps.

5.2. Proof of 5.1.1 i) and ii) in the case where U is smooth and H is étale. —

Lemma 5.2.1. *With the notation of 5.1.1, if H is étale, the canonical morphism $E \rightarrow H$ has a section $s : H \rightarrow E$, defined over k , in the following cases:*

- a) k is algebraically closed.
- b) k is perfect, and U is smooth and connected.
- c) U is “ k -solvable”.

a) simply follows from the fact that $E(k) \rightarrow H(k)$ is surjective. One has $b) \Rightarrow c)$ by 4.1.2 b) and 2.3 (bis). It therefore suffices to treat case c). Now let x be a point of H ; x is therefore a subset of H that is both open and closed, and the induced subscheme is isomorphic to $\text{Spec } K$, where K is a finite separable extension of k . Let X be the inverse image in E of the scheme x . The K -scheme X is a principal homogeneous K -space under the group U_K . But U_K has a composition series with successive quotients isomorphic to $(\mathbf{G}_a)_K$, hence $H^1(K, U_K) = 0$ (J.-P. Serre, *Cohomologie Galoisienne*, chap. III, prop. 6), and consequently X is trivial, hence has a rational point over K . One thus obtains a k -section of $E \rightarrow H$ over x , for every point x of H , whence the existence of a section $H \rightarrow E$.

Lemma 5.2.3. ⁽⁸⁾ *With the notation of 5.1.1, suppose that H is étale. Then:*

- i) *The extension E is trivial in each of the following cases:*
 - a) U is commutative and $E \rightarrow H$ has a section.
 - b) k is algebraically closed.
 - c) k is perfect and U is connected.
 - d) U is “ k -solvable”.

Moreover, in each of the above cases, two liftings H' and H'' of H into E are conjugate by an element of $U(k)$.

- ii) *Let R be the k -functor $((\text{Sch})/k)^\circ \rightarrow (\text{Ens})$ such that, for every k -prescheme S , $R(S)$ is the set of liftings of H_S into E_S . Then, if U is commutative, R is representable by a nonempty affine k -scheme. The group U acts by inner automorphisms on R , and this action makes R a homogeneous space under U (for the fpqc topology (cf. Exp. IV)).*

Proof of i). We shall reduce cases b), c) and d) to case a).

Case b). Since U has a characteristic composition series with commutative successive quotients (3.5 i) $\Leftrightarrow$ iv)), one immediately reduces to the case where U is commutative. Moreover, $E \rightarrow H$ has a section by 5.2.1 a), and we are reduced to case a). *Case d).* One proceeds in the same way, using 5.1.0 ii) and 5.2.1 c).

Case c). Let E_{red} be the associated reduced subgroup of E . Since H is smooth, E_{red} is an extension of H by $U' = U \cap E_{\text{red}}$, and one can replace E by E_{red} . But U , hence also U' , is connected, and it is clear that U' is the neutral component of E_{red} , hence is smooth. But

⁽⁷⁾**N.D.E.** One has replaced H_0^i by H^i , to agree with the notation of App. I and Exp. I.

⁽⁸⁾**N.D.E.** There is no no. 5.2.2.

then U' is smooth and connected, k is perfect, hence U' is k -solvable (4.1.4 b)), and we are reduced to case d).

The preceding reductions show that in cases b), c) and d), one can assume that U has a characteristic composition series (U_i) such that U_i/U_{i+1} is commutative and such that the maps $U_i(k) \rightarrow (U_i/U_{i+1})(k)$ are surjective (in cases c) and d), this last point follows from $H^1(k, \mathbf{G}_a) = 0$). An immediate dévissage then shows that it suffices to prove the conjugacy of two liftings H' and H'' into E when U is commutative. Thus it suffices to prove i) a). In this case, the triviality of the extension E is assured if $H^2(H, U) = 0$ (App. I 3.1), and the conjugacy of H' and H'' if $H^1(H, U) = 0$. Now we have the following lemma:

Lemma 5.2.4. *Let S be a prescheme, U a commutative group S -prescheme, H an étale finite group S -prescheme of rank n , acting on U . Then the groups $H^i(H, U)$ ($i > 0$) are annihilated by n in the following two cases:*

- a) H is a constant S -group (Exp. I 4.1).
- b) S is the spectrum of a field.

Proof of a). By hypothesis, the group H is the constant group associated with an ordinary group $\{H\}$ of order n . It is then clear⁽⁹⁾ that $H^i(H, U)$ is isomorphic to the i th cohomology group $H^i(\{H\}, U(S))$ of the group $\{H\}$, with values in the ordinary group $U(S)$, and it is classical that these groups are annihilated by n (J.-P. Serre, *Corps locaux*, Chap. VIII, prop. 4 cor. 1).

Proof of b). Let $x \in H^i(H, U)$ ($i > 0$) be represented by an i -cocycle $f : H^{(i)} \rightarrow U$ (where $H^{(i)}$ denotes the product, over k , of i copies of H). If K is a finite Galois extension of k splitting H , it follows from a) that nf_K is a coboundary. More precisely, an easy calculation shows that if one defines the K -morphism $F_K : H_K^{(i-1)} \rightarrow U_K$ by the formula:

$$F_K(h_1, \dots, h_{i-1}) = \sum_{h \in H(K)} f_K(h_1, \dots, h_{i-1}, h),$$

one has, up to sign:

$$d(F_K) = nf_K \quad (d \text{ the coboundary operator}).$$

Now an immediate Galois argument shows that F_K comes from a k -morphism $F : H^{(i-1)} \rightarrow U$, and consequently one has $d(F) = nf$.

Corollary 5.2.4 bis. *With the notation of 5.2.4, suppose moreover that U is flat and of finite presentation over S , with unipotent fibers, and that n is prime to the residue characteristics of S . Then, in cases a) and b) above, one has $H^i(H, U) = 0$ for $i > 0$.*

It suffices to show that raising to the n th power in U is an isomorphism, for this will imply that multiplication by n in $H^i(H, U)$ is both an isomorphism and the zero morphism, hence $H^i(H, U) = 0$. Now, under the hypotheses made on U , it suffices to check that raising to the n th power is an isomorphism on the fibers of U (EGA IV 17.9.5), reducing us to the case where S is the spectrum of a field k of characteristic p . Since $(n, p) = 1$, raising to the n th power in U is an étale morphism (Exp. VII), and it is a monomorphism (2.4 i)), hence an open immersion (EGA IV 17.9.1). This already proves that the restriction to the neutral component U^0 is an isomorphism; since U/U^0 is a finite p -group, one has finished.

This completes the proof of 5.2.3 i) a), since H , being an étale group of multiplicative type, is of order prime to p .

Proof of 5.2.3 ii). It is clear that R is a sheaf for the fpqc topology. Moreover, in view of descent for affine schemes, the assertions of 5.2.3 ii) are local for the fpqc topology. We can therefore assume that k is algebraically closed. The group H is then split and the extension E is trivial (i b)); let H' be a lifting of H into E . For every k -prescheme S and every lifting

⁽⁹⁾**N.D.E.** This is Proposition III.6.4.2 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

H'' of H_S into E_S , H'_S and H'' are conjugate by an element of $U(S)$, since $H^1(H_S, U_S) = 0$ (5.2.4 bis). Let then $U^{H'}$ be the sheaf of invariants of U under H' , which is representable by an algebraic subgroup of U (Exp. VIII 6.5 d)). It follows from the preceding remarks that the k -morphism:

$$U \longrightarrow R, \quad u \longmapsto \text{int}(u)H' \quad (u \in U(S))$$

defines a k -isomorphism $U/U^{H'} \xrightarrow{\sim} R$. This proves the representability of R and the fact that R is affine (2.1).

Remark 5.2.5. One can show that the assertions of 5.2.3 ii) are still true when U is not commutative, but we shall not need this to prove 5.1.1.

5.3. Study of the case where H is smooth. —

Proposition 5.3.1. *The assertions in 5.2.3 i) remain true when one replaces the hypothesis “ H étale” by “ H smooth”.*

Proceeding as in the proof of 5.2.3 i), one reduces to the case where U is moreover commutative.

Let $\mathbf{N}'$ be the set of integers > 0 prime to p , ordered by divisibility. For every $n \in \mathbf{N}'$, ${}_nH^0$ is an étale group, and the family of the ${}_nH^0$ ($n \in \mathbf{N}'$) is schematically dense in H , since H is smooth (Exp. IX 4.10). Denote by E_n the inverse image of ${}_nH$ in E , so that E_n is an extension of ${}_nH$ by U ; finally, let R_n be the k -functor of liftings of ${}_nH$ into E_n (cf. 5.2.3 ii)). If n divides m , it is clear that one has a natural k -morphism $R_m \longrightarrow R_n$, so that the R_n form a projective system of k -functors. Since R_n is representable by a nonempty affine k -scheme (5.2.3 ii)), and since a filtered inductive limit of nonzero rings is nonzero, the functor $R = \varprojlim R_n$ is representable by a nonempty affine k -scheme (EGA IV 8 and 1.9.1). There is therefore an extension K of k and a point $u \in R(K)$. The image u_n of u in $R_n(K)$ corresponds to a lifting H'_n of $({}_nH)_K$ into $(E_n)_K$. By construction, $H'_n = {}_n(H'_m)$ if n divides m . Put $U_n = (U_K)^{H'_n}$. The choice of H'_n allows one to identify $(R_n)_K$ with U_K/U_n . But the family of the H'_n is increasing and filtered, hence the family of the U_n is decreasing and filtered and consequently is stationary for sufficiently large n (U_K is noetherian). It follows that the family of the $(R_n)_K$ is stationary, and consequently the same is true of the family of the R_n . In brief, one has $R_n = R$ for sufficiently large n .

Under the hypotheses made, it follows from 5.2.3 i) that $R_n(k)$ is nonempty. One can therefore find a coherent system of liftings H'_n of ${}_nH$ for $n \in \mathbf{N}'$. Denote by H' the smallest closed subscheme of E containing H'_n for every n (Exp. VIB §7). The argument made in Exp. XV 4.6 shows that H' is a smooth commutative algebraic group whose formation commutes with extension of the base field. To show that H' is a lifting of H into E , we can therefore assume that k is algebraically closed. By BIBLE 4 th. 4, H' is then the direct product of a (smooth) group of multiplicative type M and a unipotent group V . The groups H'_n are then necessarily contained in M (2.4), and in view of the definition of H' this implies $H' = M$. Thus H' is of multiplicative type and consequently $H' \cap U = 0$. The morphism $H' \longrightarrow H$ is therefore a monomorphism; moreover, it follows from the density theorem (Exp. IX 4.10) that it is an epimorphism, so it is indeed an isomorphism.

Let now H' and H'' be two liftings of H into E . For every $n \in \mathbf{N}'$, denote by $T_n = \text{Transp}({}_nH', {}_nH'')$ the transporter of ${}_nH'$ into ${}_nH''$, which is representable by a closed subscheme of U (Exp. VIII 6.5 e)). The T_n form a decreasing filtered family of closed subschemes of U , nonempty by 5.2.3 i a). Let T be the stationary value. Under the hypotheses of 5.2.3 i), $T_n(k)$ is nonempty. There is therefore an element u of $U(k)$ such that ${}_nH'' = \text{int}(u){}_nH'$ for every $n \in \mathbf{N}'$. But then $H'' = \text{int}(u)H'$ (Exp. IX 4.8 b).

5.4. Study of the case where U is radicial. —

Proposition 5.4.1. *If k is a perfect field of characteristic $p > 0$, and if U is radicial, the extension E of 5.1.1 is trivial.*

Using a characteristic composition series of U , we can restrict ourselves to the case where U is equal to $(\alpha_p)^r$ (3.9).

It follows from App. II 2.2 and 2.1 that one has isomorphisms of k -functors:

$$\underline{\text{Aut}}_{k\text{-gr}}(\alpha_p)^r \xrightarrow{\sim} \underline{\text{Aut}}_{p\text{-Lie}}(\text{Lie}(\alpha_p)^r) \xrightarrow{\sim} \text{GL}(\text{Lie}(\alpha_p)^r).$$

Consider then a k -vector space V of rank r , the vector group k -scheme $W(V)$ (Exp. I 4.6), and identify $(\alpha_p)^r$ with ${}_F V$. The preceding remarks then show that the action of H on ${}_F V$, defined by the extension E , comes from a linear representation v of H in V . Consider then the exact sequence:

$$(*) \quad 0 \longrightarrow {}_F V \longrightarrow V \xrightarrow{F} V^{(p)} \longrightarrow 0.$$

where F is the Frobenius morphism. Then $(*)$ is an exact sequence of H -groups, provided that H acts on the factor $V^{(p)}$ through the linear representation $H \xrightarrow{v} \text{GL}(V) \xrightarrow{F} \text{GL}(V^{(p)})$.

Since the field k is perfect, the morphism $F : V \longrightarrow V^{(p)}$ induces a surjective map $V(k) \longrightarrow V^{(p)}(k)$. It then follows from App. I 2.1 that the exact sequence $(*)$ defines an exact sequence:

$$(**) \quad H^1(H, V^{(p)}) \longrightarrow \text{Ext}_{\text{alg}}(H, {}_F V) \longrightarrow \text{Ext}_{\text{alg}}(H, V).$$

Let us show that $\text{Ext}_{\text{alg}}(H, V) = 0$. Let therefore E' be an algebraic group that is an extension of H by V :

$$1 \longrightarrow V \longrightarrow E' \xrightarrow{h} H \longrightarrow 1.$$

The scheme E' is a torsor with base H and group $\mathbf{G}_a^r$, hence defines an element of $H^1(H, \mathcal{O}_H^r)$ (in the sense of coherent sheaf cohomology). Since H is affine, one has $H^1(H, \mathcal{O}_H^r) = 0$ (EGA III §1). This says that $E' \longrightarrow H$ has a section. Consequently, the group $\text{Ext}_{\text{alg}}(H, V)$ is isomorphic to $H^2(H, V)$ (App. I 3.1). Now $H^i(H, V) = H^i(H, W(V)) = 0$ for $i > 0$ (Exp. IX 3.1). One then concludes, by the exact sequence $(**)$, that $\text{Ext}_{\text{alg}}(H, {}_F V) = 0$, hence that E is a trivial extension.

5.5. Proof of 5.1.1 i). —

If k has characteristic 0, U is k -solvable (4.1.3) and H is smooth; the fact that E is a trivial extension therefore follows from 5.3.1 and 5.2.3 d). One proves in the same way that two liftings of H into E are conjugate by an element of $U(k)$.

Henceforth, we therefore assume that k is a field of characteristic $p > 0$.

Proof of i) b): Case where k is perfect, U connected.

We shall reduce to the case where U is radicial. For this, note that since k is perfect, H_{red} is smooth; let E' be its inverse image in E . It then follows from 5.3.1 and 5.2.3 i) c) that the extension:

$$1 \longrightarrow U \longrightarrow E' \longrightarrow H_{\text{red}} \longrightarrow 1$$

is trivial. Let H_1 be a lifting of H_{red} into E . By App. II 3.1, there is an integer $n > 0$ such that $E^{(n)} = E/_{F^n}(E)$ is smooth; let E'' be the algebraic subgroup of E generated by H_1 and $_{F^n}(E)$ (i.e. the inverse image in E of the image of H_1 in $E^{(n)}$). Denote by H'' the image of E'' in H . Then I say that $H'' = H$. Indeed, denote by R the image of $_{F^n}(E)$ in H , so that H'' is generated by R and H_{red} . The group H/R is a quotient of $E^{(n)}$, hence is smooth; consequently the canonical morphism

$$H_{\text{red}} \longrightarrow H/R$$

is an epimorphism, hence H is generated by R and H_{red} , hence is equal to H'' . One thus obtains an exact sequence:

$$(\dagger) \quad 1 \longrightarrow U'' = U \cap E'' \longrightarrow E'' \longrightarrow H \longrightarrow 1.$$

But E'' has the same underlying space as H_1 , hence U'' is radicial. Moreover, it is clear that it suffices to prove that the extension $(\dagger)$ is trivial, which follows from 5.4.1.

Proof of i) b): Case where k is perfect, H connected.

The group U is an extension of an étale group by its neutral component U^0 . The case where U is connected having just been treated, it suffices to examine the case where U is étale. One then has the more precise lemma:

Lemma 5.5.1. *With the notation of 5.1.1, suppose moreover that U is étale. Then:*

- i) *If H is connected, there is a unique lifting of H into E , namely the neutral component E^0 of E .*
- ii) *If k is algebraically closed, E is trivial, and two liftings of H into E are conjugate by an element of $U(k)$.*

i) Since formation of the neutral component commutes with extension of the base field, we can restrict ourselves to the case where k is algebraically closed. If H is a torus, E is trivial (5.3.1 and 5.2.3 i b)), and it is clear that E^0 is the unique lifting of H . This already proves that in the general case, $E^0 \cap U$ is radical; since it is moreover étale (U being étale), it is zero. The morphism $E^0 \rightarrow H$ is therefore a monomorphism, flat (since E^0 is open in E) and surjective (H is connected), hence is an isomorphism. If now H' is another lifting of H , H' is connected, hence contained in E^0 and consequently is equal to E^0 .

ii) Let H^0 be the neutral component of H . By i), E^0 is the unique lifting of H^0 into E . Put $E' = E/E^0$, $H' = H/H^0$, so that one has the extension:

$$1 \rightarrow U \rightarrow E' \rightarrow H' \rightarrow 1.$$

Since H' is étale, this extension is trivial (5.2.3 i b)). If H'_1 is a lifting of H' into E' , and H_1 its inverse image in E , it is clear that H_1 lifts H into E . If H_2 is a second lifting of H into E , it contains E^0 ; by 5.2.3 i b), the image of H_2 in E' is conjugate to H'_1 by an element u of $U(k)$, whence immediately $H_2 = \text{int}(u)H_1$.

Remark 5.5.2. Under the hypotheses of 5.5.1 i), it is easy to see that E^0 centralizes U .

Proof of 5.1.1 i) a). Using the composition series

$$1 \rightarrow U^0 \rightarrow U \rightarrow U/U^0 \rightarrow 1,$$

i) a) follows from the conjunction of i) b) and 5.5.1 ii).

Before proving 5.1.1 c) and d), we shall first establish 5.1.1 ii).

Proof of 5.5.1 ii) a). For lack of a satisfactory general statement, we shall describe a certain number of cases where, when H is smooth, two liftings of H into E are conjugate:

Proposition 5.6.1. *With the notation of 5.1.1, suppose moreover that H is smooth, and let H' and H'' be two liftings of H into E . Then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:*

- a) *k is algebraically closed.*
- b) *U is commutative.*
- c) *k is perfect and U is connected.*
- d) *U is k -solvable.*
- e) *The centralizer of H' in U is k -solvable.*
- f) *The group of multiplicative type H is split by a finite Galois extension K of k of degree prime to p .*

Proof. a), b), c), d) follow from 5.3.1.

e) Let Z be the centralizer of H' in U , and T the transporter of H' into H'' . By a), T is nonempty, hence T is a principal homogeneous space under Z , and the hypothesis made on Z implies that it is trivial.

f) Proceeding as in 5.3.1, one sees that it suffices to consider the case where H is étale. Suppose first that H is diagonalizable, defined by the ordinary group M , of order m prime to p . The datum of the two liftings H' and H'' defines a 1-cocycle of M with values in $U(k)$, that is, a map h from M into $U(k)$ such that $h(mn) = h(m)^m h(n)$ for every pair m, n of

elements of M . The groups H' and H'' are conjugate by an element of $U(k)$ if and only if there exists $a \in U(k)$ such that

$$h(m) = a^{-1}({}^ma).$$

Now the abstract group $U(k)$ has a composition series with successive quotients commutative and annihilated by a power of p (one may assume $p > 0$ in view of 5.6.1 c)). One immediately deduces in this case that h is a coboundary.

Let us now examine the general case. Denote again by Z the centralizer of H' in U , and by T the transporter of H' into H'' , which is a torsor under Z . By hypothesis, there is a finite Galois extension K of k , with Galois group G , of order prime to p , which splits H . By the study made above, H'_K and H''_K are conjugate by an element of $U(K)$, hence T_K is trivial. Consequently T is defined by an element of $H^1(G, Z)$. For the same reasons as above, the hypothesis made on G implies that $H^1(G, Z) = e$, hence T is trivial.

5.7. Proof of 5.1.1 ii) b). —

Lemma 5.7.1. *Let S be a prescheme, G a group S -prescheme, separated and smooth over S , and H an S -group of multiplicative type acting on G . Then the S -functor $Z = G^H$ of invariants of G under H is representable by a subgroup S -prescheme of G , closed and smooth over S .*

The fact that G^H is representable by a closed subgroup prescheme of G follows from Exp. VIII 6.5 d). Consider then the semidirect product $K = G \times_S H$. The centralizer of H in K is then equal to $Z \times_S H$. To prove that Z is smooth, we must check that if S is affine, if S_0 is a closed subscheme of S defined by an ideal of square zero, and if u_0 is an element of $Z(S_0)$, then there is an element u of $Z(S)$ lifting u_0 . Now since G is smooth over S , there is an element u of $G(S)$ lifting u_0 . Let i be the canonical immersion of H into K , and consider the two group S -morphisms

$$H \rightrightarrows K, \quad i \text{ and } \text{int}(u)i = j.$$

Since u_0 belongs to $Z(S_0)$, one has $i_{S_0} = j_{S_0}$. By Exp. IX 3.2, there exists $v \in K(S)$ lifting the identity section of $K(S_0)$ and such that

$$i = \text{int}(v)j = \text{int}(vu)i.$$

Thus $vu = (u', v')$ belongs to $(Z \times_S H)(S)$. Thus u' belongs to $Z(S)$ and lifts u_0 .

Lemma 5.7.2. *Let k be a field, H an algebraic k -group, and let $P(H)$ be the following property:*

“for every smooth unipotent k -group U , and for every extension E of H by U , two liftings of H into E are conjugate by an element of $U(k)$ ”.

Then if H is an algebraic group that is an extension of a group of multiplicative type H'' by a group of multiplicative type H' , and if $P(H')$ and $P(H'')$ are true, $P(H)$ is true.

Indeed, let U be a smooth unipotent k -group, E an extension of H by U , H_1 and H_2 two liftings of H into E , and H'_1 and H'_2 the corresponding liftings of H' . Since $P(H')$ is true, there exists $u \in U(k)$ such that $H'_2 = \text{int}(u)H'_1$. Replacing H_1 by $\text{int}(u)H_1$, we can assume $H'_1 = H'_2$, which we allow ourselves to denote simply by H' . Let $E' = \text{Centr}_E H'$, which is equal to

$$U^{H'} \cdot H_1 = U^{H'} \cdot H_2.$$

By 5.6.1, $U^{H'} = U'$ is smooth. Consider then the extension

$$1 \longrightarrow U' \longrightarrow E' \longrightarrow H \longrightarrow 1.$$

By construction H' is central in E' , hence invariant. Passing to the quotient, one obtains the exact sequence:

$$1 \longrightarrow U' \longrightarrow E'' \longrightarrow H'' \longrightarrow 1.$$

Since U' is smooth, and since $P(H'')$ is true, the two images of H_1 and H_2 in E'' are conjugate by an element u of $U'(k)$, but then $H_1 = \text{int}(u)H_2$.

To prove 5.1.1 ii) b), note then that, since k is algebraically closed, H has a composition series whose successive quotients are smooth or isomorphic to μ_p when $p > 0$. By repeated use of 5.7.2, we are reduced to the case where H is smooth or equal to μ_p . In the first case, it suffices to apply 5.1.1 ii) a). There remains the case $H = \mu_p$. Since U is smooth, U has a characteristic composition series with successive quotients étale or isomorphic to $(\mathbf{G}_a)^r$ (3.9). If U is étale, one applies 5.5.1. There remains finally the case $H = \mu_p$, $U = \mathbf{G}_a^r$.

We must show that $H^1(\mu_p, U) = 0$. The method used in 5.4.1 no longer applies here, for μ_p does not in general act linearly on $(\mathbf{G}_a)^r$. Let us fix the notation: H' denotes a lifting of H into E , $\mathfrak{e} = \text{Lie } E$, $\mathfrak{u} = \text{Lie } U$, $\mathfrak{h} = \text{Lie } H$, $\mathfrak{h}' = \text{Lie } H'$. Let X be a nonzero element of $\mathfrak{h}$ such that $X^{(p)} = X$ (App. II 3.1), and let X' be its lifting in $\mathfrak{h}'$.

Since μ_p is a radicial group of height 1, there is a one-to-one correspondence between the set of liftings of H into E and the set A of $Y \in \mathfrak{e}$ such that $Y^{(p)} = Y$ and which project onto X (App. II 2.2). Likewise, if $Y \in A$ corresponds to the lifting H'' of H into E , and if $u \in U(k)$, then $\text{int}(u)H' = H''$ if and only if $\text{Ad}(u)X = Y$. Let therefore B be the subset of $Y \in \mathfrak{e}$ of the form $\text{Ad}(u)X$, where $u \in U(k)$. One obviously has $B \subset A$, and everything reduces to showing that $A = B$ if k is algebraically closed.

a) Study of A . Since $\mathfrak{u}$ is commutative and normalized by $\mathfrak{h}$, the Jacobson formula (Exp. VIIA 5.2) gives simply here, for $u \in \mathfrak{u}$:

$$(X' + u)^{(p)} = X'^{(p)} + u^{(p)} + (\text{ad } X')^{p-1}(u) = X' + (\text{ad } X')^{p-1}(u),$$

so that $X' + u \in A \Leftrightarrow u = (\text{ad } X')^{p-1}(u)$.

Now let $\mathfrak{u} = \coprod_{n \in \mathbf{Z}/p\mathbf{Z}} \mathfrak{u}_n$ be the canonical decomposition of $\mathfrak{u}$ under the action of H' , which one can also write:

$$\mathfrak{u} = \mathfrak{u}_0 \oplus \coprod_{n \in (\mathbf{Z}/p\mathbf{Z})^\times} \mathfrak{u}_n.$$

If $u \in \mathfrak{u}_0$, one has $\text{ad } X'(u) = 0$. If $u \in \mathfrak{u}_n$ ($n \in (\mathbf{Z}/p\mathbf{Z})^\times$), one has $(\text{ad } X')^{p-1}(u) = u$. Finally, $Y = X' + u$ is an element of A if and only if $u \in \coprod_{n \in (\mathbf{Z}/p\mathbf{Z})^\times} \mathfrak{u}_n$. Let us retain that A is the set of rational points over k of an irreducible subscheme of $W(\mathfrak{e})$ of dimension equal to $\text{rk } \mathfrak{u} - \text{rk } \mathfrak{u}_0 = \text{rk } \mathfrak{u} - \text{rk } \text{Centr}_{\mathfrak{u}}(X')$.

b) Study of B . We shall need the following lemma:

Lemma 5.7.3. (Rosenlicht). *Let U be a unipotent algebraic group over a field k acting on a quasi-affine k -scheme X . Then the orbit of every point $x \in X(k)$ is closed in X (by orbit of x we mean the subset of $\text{ens}(X)$ that is the image of G under the morphism $g \mapsto g \cdot x$).*

By functoriality, G acts on the affine envelope of X (that is, $\text{Spec } \Gamma(X, \mathcal{O}_X)$), which allows us to assume that X is affine. One can then assume that k is algebraically closed, X reduced and U smooth (note that U_{red} acts on X_{red} if k is perfect). Let Y be the schematic image of G (EGA I 9.5.1) under the morphism $g \mapsto g \cdot x$, which is a closed reduced subscheme of X on which G acts. It follows easily from EGA IV 1.8.6 that the orbit of x is an open subset Z of Y , dense in Y . We must show that $Z = \text{ens}(Y)$. Let F be the reduced closed subscheme of Y having $Y \setminus Z$ as underlying space. One therefore has $F = V(J)$, where J is a nonzero ideal of $\Gamma(Y, \mathcal{O}_Y)$. Since G is smooth, G acts on F , hence on J and consequently (3.2) $J^G \neq 0$. If a is a nonzero element of J^G , a is necessarily constant on the orbit Z , hence is constant on Y , since Z is dense in Y . But then the ideal J contains k and $F = \emptyset$.

This being so, let us apply the preceding lemma to the group U acting on the affine space $W(\mathfrak{e})$ through the adjoint representation. One obtains that the orbit of X' is the underlying set of a closed subscheme of $W(\mathfrak{e})$. Moreover, the stabilizer Z of X' is the centralizer of X' in U , and one has a closed immersion:

$$U/Z \longrightarrow W(\mathfrak{e}).$$

Let us retain that the orbit of X' is the underlying space of a closed subscheme of $W(\mathfrak{e})$ of dimension equal to $\dim U - \dim Z$.

c) End of the proof of 5.1.1 ii) b). When k is algebraically closed, the canonical map $U(k) \rightarrow (U/Z)(k)$ is surjective, so that, by point b) above, B is the set of rational points of a closed subscheme of $W(\mathfrak{e})$ of dimension $\dim U - \dim Z$. In view of point a), to prove that $A = B$, it then suffices to show that one has:

$$\mathrm{rk} \, u - \mathrm{rk} \, \mathrm{Centr}_u(X') \leq \dim U - \dim \mathrm{Centr}_U(X').$$

Now since U is smooth, one has $\dim U = \mathrm{rk} \, u$. On the other hand, one has (Exp. II 5.3.3):

$$\dim \mathrm{Centr}_U(X') \leq \mathrm{rk} \, \mathrm{Centr}_u(X'),$$

whence the result (note that one has in fact $\dim \mathrm{Centr}_U(X') = \mathrm{rk} \, \mathrm{Centr}_u(X')$, which proves anew that Z is smooth (5.7.1)).

5.8. End of the proof of 5.1.1 i). —

It remains for us to prove i) c) and i) d).

Proof of 5.1.1 i) d) (U smooth, H connected). We shall need the following lemma:

Lemma 5.8.1. *With the notation of 5.1.1, suppose that U is smooth and H radicial, and let H_1 be an algebraic subgroup of H having a lifting H'_1 into E . Then H has a lifting H' into E containing H'_1 .*

By induction on the height of H/H_1 , one can assume that H/H_1 has height 1. Let $C' = \mathrm{Centr}_E(H'_1)$. I say that the canonical morphism $C' \rightarrow H$ is an epimorphism. To establish this point, we can assume that k is algebraically closed; but then the extension E is trivial (5.1.1 i a)); let H'' be a lifting of H into E , and H''_1 the inverse image of H_1 in H'' . The groups H'_1 and H''_1 are two liftings of H_1 into E , hence are conjugate by an element of $U(k)$, since k is algebraically closed and U smooth (5.1.1 ii b)). It is then clear that to prove the assertion about C' , it suffices to prove it for $C'' = \mathrm{Centr}_E(H''_1)$. But in this case, C'' contains H'' , and the property is clear. Moreover, it follows from 5.7.1 that $C \cap U$ is smooth. It is then clear that we can replace E by C , hence assume that H'_1 is central. But then, passing to the quotient by H'_1 , we can assume $H_1 = 0$ and H of height 1.

Since U is smooth, one has the exact sequence of Lie p -algebras (App. II 3.2):

$$(*) \quad 0 \rightarrow \mathrm{Lie} \, U \rightarrow \mathrm{Lie} \, E \rightarrow \mathrm{Lie} \, H \rightarrow 0.$$

In view of App. II 2.2, saying that E is trivial is equivalent to saying that $(*)$ is a trivial extension of Lie p -algebras. Suppose $H \neq 0$ (hence $\mathrm{Lie} \, H \neq 0$), and suppose that we have found a Lie p -subalgebra $\mathfrak{h}_1$ of $\mathfrak{h} = \mathrm{Lie} \, H$ that is nonzero and lifts to a Lie p -subalgebra $\mathfrak{h}'_1$ of $\mathrm{Lie} \, E$. By *loc. cit.*, there is a subgroup H_1 of H such that $\mathrm{Lie} \, H_1 = \mathfrak{h}_1$, and a lifting H'_1 of H_1 into E such that $\mathrm{Lie} \, H'_1 = \mathfrak{h}'_1$. Applying again the reduction described above, one reduces to the same problem, where H has been replaced by H/H_1 . Since H has height 1, $\mathrm{Lie}(H/H_1) = \mathrm{Lie} \, H / \mathrm{Lie} \, H_1$ (*loc. cit.*), hence $\mathrm{rk} \, \mathrm{Lie}(H/H_1) \leq \mathrm{rk} \, \mathrm{Lie}(H) - 1$. In brief, proceeding by induction on the rank of $\mathrm{Lie} \, H$, one sees that it suffices, when $\mathfrak{h} \neq 0$, to find a nonzero Lie subalgebra of $\mathrm{Lie} \, H$ that lifts into $\mathrm{Lie} \, E$. Now one has the following lemma:

Lemma 5.8.2. *Let k be a field of characteristic $p > 0$, and φ a surjective morphism of Lie p - k -algebras of finite rank $\mathfrak{g} \rightarrow \mathfrak{h}$. Then:*

- i) *If $\mathfrak{h}_{\bar{k}}$ is reductive (4.2.2) and $\neq 0$, there is a reductive Lie subalgebra $\mathfrak{h}_1$ of $\mathfrak{g}$ whose image in $\mathfrak{h}$ is nonzero.*
- ii) *If k is perfect and if u is a unipotent element of $\mathfrak{h}$ (i.e. there is n such that $u^{(p^n)} = 0$), then u lifts to a unipotent element of $\mathfrak{g}$.*

Take for X an element $\neq 0$ of $\mathfrak{h}$ in case i) and u in case ii), and let X' be a lifting of X in $\mathfrak{g}$. The Lie p -subalgebra of $\mathfrak{g}$ generated by X' is an abelian Lie p -algebra (Exp. VII) $\mathfrak{h}'$.

Case i). It is clear from the description given in 4.2.2 that the reductive part (*loc. cit.*) of $\mathfrak{h}'_{\bar{k}}$ is already defined over k ; denote it by $\mathfrak{r}'$. I say that the image of $\mathfrak{r}'$ in $\mathfrak{h}$ is nonzero. To establish

this point, we can assume that k is algebraically closed, so that $\mathfrak{h}' = \mathfrak{r}' \oplus \mathfrak{u}'$ ($\mathfrak{u}'$ the unipotent part of $\mathfrak{h}'$). If the image of $\mathfrak{r}'$ in $\mathfrak{h}$ were zero, the image of $\mathfrak{h}'$ in $\mathfrak{h}$ would be unipotent, hence would be zero since $\mathfrak{h}$ is reductive (cf. 2.4 ii)), whereas by construction it contains X .

Case ii). One proceeds in the same way, interchanging the roles of $\mathfrak{r}'$ and $\mathfrak{u}'$.

End of the proof of 5.8.1. Assuming $H \neq 0$, there is, by 5.8.2, a nonzero reductive Lie subalgebra $\mathfrak{h}'_1$ of $\text{Lie } E$. Since $\text{Lie } U$ is unipotent (4.3 i)), one necessarily has $(\text{Lie } U) \cap \mathfrak{h}'_1 = 0$, so that $\mathfrak{h}'_1$ is a lifting of a Lie p -subalgebra of $\text{Lie } H$.

End of the proof of 5.1.1 i d). By 5.8.1, there is a family of algebraic subgroups H'_n ($n \in \mathbf{N}$) of E , such that H'_{n+1} contains H'_n and such that H'_n lifts ${}_{F^n}H$. The decreasing sequence of subgroups $\text{Centr}_E(H'_n)$ is stationary; let C be its stationary value. The center Z of C contains H'_n for every n , hence the image of Z in H contains ${}_{F^n}H$ for every n , and consequently is an open subgroup of H (Exp. VIIA §4), hence is equal to H since H is connected. To prove that E is trivial, one can therefore replace E by Z , hence assume that E is commutative. We shall then see in 7.2.1 that E contains a maximal subgroup of multiplicative type M whose formation commutes with extension of the base field. Since $E_{\bar{k}}$ is a trivial extension (5.1.1 i) a)) and U is unipotent, it is clear that M is the unique lifting of H into E .

Proof of 5.1.1 i) c) (U k -solvable). Since $(H/H^0)(k)$ is of order prime to p , it is immediate by duality that there is an integer n such that ${}_nH$ is an étale subgroup and such that the canonical morphism ${}_nH \rightarrow H/H^0$ is an epimorphism, so that $H/{}_nH$ is connected. By 5.2.3 d), there is a lifting H' of ${}_nH$ into E . One shows, as at the beginning of the proof of 5.8.1, that $C = \text{Centr}_E(H')$ is a subgroup of E such that $C \rightarrow H$ is an epimorphism and such that $C \cap U$ is smooth. Replacing E by C and passing to the quotient by H' , one is reduced to the case where U is smooth and H connected, that is, to case i) d).

This completes the proof of 5.1.1.

5.9. Counterexamples. —

Let us first indicate a procedure for obtaining nontrivial extensions of a k -group of multiplicative type H by a unipotent k -group U . Suppose that an action of H on U is given, and let E be the semidirect product $E = U \cdot H$. Let on the other hand an element of $H^1(k, U)$ be represented by a 1-cocycle a . The group U acts by inner automorphisms on E . The datum of a therefore defines a k -form of E , denoted by E_a . Suppose moreover that U is commutative; then U acts trivially on U and on the quotient $E/U = H$, so that E_a is again an extension of H by U . Suppose, for simplicity, that H is an étale diagonalizable group, so that the Galois group $\mathcal{G}$ of $\bar{k}/k$ acts trivially on $H(k)$; the action of $\mathcal{G}$ on the points of $E_a(\bar{k})$ is then given by the formula:

$${}^g(u, h) = ({}^gu + a(g) - {}^ha(g), h) \quad (g \in \mathcal{G}, u \in U(\bar{k}), h \in H(\bar{k})).$$

If h is a point of $H(k)$, and X its inverse image in E_a , X is therefore a torsor under U defined by the class of the 1-cocycle of $\mathcal{G}$ with values in U , $g \mapsto a(g) - {}^ha(g)$. It follows that if there is a point h of $H(k)$ such that the preceding 1-cocycle is nontrivial, the extension E_a is nontrivial. We shall apply this construction in two particular cases:

a) *Nontrivial extension of an étale diagonalizable group H by $\mathbf{Z}/p\mathbf{Z}$.*

Take for H the group $(\mathbf{Z}/p\mathbf{Z})^\times$, which one makes act by multiplication on $U = \mathbf{Z}/p\mathbf{Z}$. Let on the other hand k be a field of characteristic p , K an extension of k with Galois group $\mathcal{G}$ isomorphic to $\mathbf{Z}/p\mathbf{Z}$, and a a nonzero element of $H^1(\mathcal{G}, U) = \text{Hom}(\mathcal{G}, U)$. The group E_a then answers the question.

b) *Example of a nontrivial extension of an étale diagonalizable group H by a smooth connected unipotent group U .*

Take for the field k a nonperfect field such that there exists a k -form U of $\mathbf{G}_a$ such that $H^1(k, U) \neq 0$. For example (cf. J.-P. Serre, *Cohomologie Galoisienne*), one can take for k the fraction field of a discrete valuation ring of equal characteristic $p > 0$, and for U the algebraic subgroup of $\mathbf{G}_a \times_k \mathbf{G}_a$ with equation $X^p + X + tY^p = 0$, where t denotes a uniformizer of k . Indeed, assuming for simplicity that k contains the $(p-1)$ th roots of unity, one has an exact

sequence of algebraic groups:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbf{Z}/p\mathbf{Z} & \longrightarrow & U & \longrightarrow & \mathbf{G}_a \longrightarrow 0 \\ & & & & (x, y) & \longmapsto & y, \end{array}$$

hence an exact cohomology sequence:

$$\mathbf{G}_a(k) \xrightarrow{d} H^1(k, \mathbf{Z}/p\mathbf{Z}) \longrightarrow H^1(k, U) \longrightarrow 0,$$

where d associates to x the homogeneous space under $\mathbf{Z}/p\mathbf{Z}$ with equation

$$X^p + X + tx^p = 0.$$

Moreover, one knows that $H^1(k, \mathbf{Z}/p\mathbf{Z})$ is isomorphic to $k/\mathfrak{P}(k)$ (where $\mathfrak{P}(x) = x^p + x$), hence $H^1(k, U)$ is isomorphic to $k/(\mathfrak{P}(k) + tk^p)$. Suppose moreover $p \neq 2$; it is then clear that t^2 is an element of k that does not belong to $\mathfrak{P}(k) + tk^p$, hence $H^1(k, U) \neq 0$. On the other hand μ_{p-1} acts on U by the formula:

$$(h, x, y) \longmapsto (hx, hy).$$

Denote by $\mathcal{G}$ the Galois group of the extension K defined by the equation

$$X^p + X + t^2 = 0,$$

and let $a \in H^1(\mathcal{G}, U)$ be the nonzero element described above. One immediately checks that E_a is then a nontrivial extension of μ_{p-1} by U .

c) *Nontrivial extension of $\mathbf{G}_m$ by α_p .*

By 5.1.1 i) b), such an extension can exist only over a nonperfect field k . Let therefore k be a nonperfect field, G the semidirect product of $U = \mathbf{G}_a$ by $H = \mathbf{G}_m$, with $\mathbf{G}_m$ acting on $\mathbf{G}_a$ by homotheties. Since U is invariant in G , ${}_F U$ is also. Let $G' = G/{}_F U$. The group G' is isomorphic to $\mathbf{G}_a \cdot \mathbf{G}_m$, where $\mathbf{G}_m$ acts on $\mathbf{G}_a$ by the formula:

$$(h, u) \longmapsto h^p u.$$

The functor $\mathcal{T}_G$ (resp. $\mathcal{T}_{G''}$) of subtori of G (resp. G'') (cf. Exp. XV) is isomorphic to $\mathbf{G}_a$, and the morphism $\mathcal{T}_G \longrightarrow \mathcal{T}_{G''}$ deduced from the morphism $G \longrightarrow G''$ is identified with the morphism $u \longmapsto u^p$. It follows that if T'' is a subtorus of G'' corresponding to a point x of $k \simeq \mathbf{G}_a(k)$ such that $x^{1/p}$ is not in k , the inverse image E of T'' in G will be an extension of a torus T'' by ${}_F U = \alpha_p$, will have no maximal tori defined over k , hence will be nontrivial. One finds for E the subgroup of $G = \text{Spec}[U, T, T^{-1}]$ with equation $U^p = x - xT^p$.

Remark 5.9.1. This last example shows that a nonsmooth algebraic group defined over a nonperfect field does not necessarily have maximal tori, and thus answers the question asked in Exp. XIV 1.5 b).

d) *Let us now give an example of a trivial extension E of a group of multiplicative type H by a unipotent group U , and of two liftings H' and H'' of H that are not conjugate by an element of $U(k)$.*

Take for E the semidirect product of $U = \mathbf{G}_a$ by $\mu_p = \text{Spec } k[T]/(T^p - 1)$, the action of μ_p on $\mathbf{G}_a = \text{Spec } k[U]$ being defined by the comorphism:

$$U \longmapsto (U + U^p)T - U^p.$$

The centralizer of μ_p in U is then the étale group Z with equation $U + U^p = 0$. It follows that if k is not algebraically closed, the canonical map $U(k) \longrightarrow (U/Z)(k)$ is not in general surjective, hence, in view of 5.1.1 ii) b), two liftings of μ_p into E are not necessarily conjugate by an element of $U(k)$.

Here is another example, with k algebraically closed of characteristic $p > 0$. Let G be the radicial group that is the semidirect product of μ_p by α_p , where μ_p acts on α_p by “homotheties”. One then has an exact sequence of algebraic groups with operator group μ_p :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \alpha_p & \longrightarrow & \mathbf{G}_a & \longrightarrow & \mathbf{G}_a \longrightarrow 0 \\ & & & & x & \longmapsto & x^p, \end{array}$$

where μ_p acts by homotheties on the first term $\mathbf{G}_a$, and trivially on the second. The exact cohomology sequence (App. I, prop. 11) gives here the exact sequence:

$$0 \longrightarrow \mathbf{G}_a(k) \longrightarrow H^1(\mu_p, \alpha_p) \longrightarrow H^1(\mu_p, \mathbf{G}_a).$$

Since the last term is zero (I 5.3.3), one sees that $H^1(\mu_p, \alpha_p)$ is nonzero, hence two liftings of μ_p into G are not necessarily conjugate.

6. Extension of a unipotent group by a group of multiplicative type

6.1. Statement of the theorem. —

Theorem 6.1.1. *Let k be a field, U a unipotent algebraic k -group, H a k -group of multiplicative type, and E an algebraic k -group that is an extension of U by H , so that one has the exact sequence*

$$1 \longrightarrow H \longrightarrow E \longrightarrow U \longrightarrow 1.$$

Then the extension E is trivial and there is a unique lifting of U into E in each of the following cases:

- A) *The group U is smooth and one of the following conditions is satisfied:*
 - i) *U is connected and the canonical morphism $E \longrightarrow U$ has a section.*
 - ii) *U has a composition series with successive quotients isomorphic to $\mathbf{G}_a$.*
 - iii) *H is étale.*
 - iv) *k is perfect.*
- B) *$U = \alpha_p$ and k is perfect.*
- C) *E is commutative and k is perfect.*

6.2. Proof of 6.1.1 A). —

Let us first establish three lemmas.

Lemma 6.2.1. *Let S be a prescheme, E a group S -prescheme that is an extension of a group S -prescheme U with connected fibers by an S -group of multiplicative type and of finite type H (i.e. U is the quotient of E by H for the fpqc topology). Then E is a central extension.*

Indeed, since H is commutative, the group U acts by inner automorphisms on H , through a group S -morphism

$$u : U \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(H).$$

The functor $\underline{\mathrm{Aut}}_{S\text{-gr}}(H)$ is representable by an étale S -scheme (Exp. X 5.10), and consequently the identity section is an immersion that is both open and closed. Since U has connected fibers, one deduces that u is the unit morphism.

Lemma 6.2.2. *With the notation of 6.1.1, if E is trivial, there is a unique lifting of U into E .*

Let therefore U' and U'' be two liftings of U into E . To show that $U' = U''$, we can assume that k is algebraically closed, and it suffices to show that $H^1(U, H) = 0$. If U is connected, U centralizes H (6.2.1), hence

$$H^1(U, H) = \mathrm{Hom}_{k\text{-gr}}(U, H) = 0$$

by 2.4 ii). In the general case, denote by U'_1 the unique lifting into E of the connected component U^0 of U , and let $N = \underline{\mathrm{Norm}}_E(U'_1)$. If $g \in E(k)$, $\mathrm{int}(g)U'_1$ is a lifting of U^0 , hence is equal to U'_1 , and consequently $N(k) \supset E(k)$. Moreover, N contains H (6.2.1) and U'_1 , whence immediately the fact that $N = E$. Passing to the quotient by U'_1 , one is reduced to the case where U is étale. In this case, since k is algebraically closed, $H^1(U, H)$ is identified

with the ordinary cohomology group $H^1(U(k), H(k))^{(10)}$ and consequently is zero, since $U(k)$ is a finite p -group and $H(k)$ is uniquely p -divisible.

Corollary 6.2.3. *To prove that E is trivial, it suffices to show that E becomes trivial after finite separable extension of the base field; in particular, one can assume that H is diagonalizable.*

Lemma 6.2.4. *For every central extension E of U by a diagonalizable k -group H to be trivial, it suffices that every central extension of U by $\mathbf{G}_m$ be trivial.*

Indeed, by induction on r , one first notes that the hypothesis made implies that E is trivial if $H = \mathbf{G}_m^r$. In the general case, H embeds into $\mathbf{G}_m^r$ for a suitable integer r (this is immediate by duality); let $H'' = \mathbf{G}_m^r/H$. One obtains the exact sequence (App. I 2.1):

$$Z^1(U, H'') \longrightarrow \text{Ext}_{\text{alg}}(U, H) \longrightarrow \text{Ext}_{\text{alg}}(U, \mathbf{G}_m^r) = 0$$

(where U acts trivially on $H, H'', \mathbf{G}_m^r$). But $Z^1(U, H'') = \text{Hom}_{k\text{-gr}}(U, H'') = 0$ (2.4 ii)), hence $\text{Ext}_{\text{alg}}(U, H) = 0$.

Proof of 6.1.1 A) i). Since $E \rightarrow U$ has a section, E defines an element $\bar{f}$ of $H^2(U, H)$ (App. I 3.1). We must show that $\bar{f}$ is zero, and for this it suffices to show that a 2-cocycle $f : U \times_k U \rightarrow H$ is a constant morphism, which will follow from the following lemma:

Lemma 6.2.5. *Let U be a smooth connected unipotent algebraic k -group, and H a k -group of multiplicative type; then every k -morphism (of preschemes):*

$$f : U \longrightarrow H$$

is constant.

To prove this lemma, we can assume that k is algebraically closed. We proceed by increasing induction on $\dim U$. If $\dim U > 0$, U has a composition series (cf. 3.9):

$$1 \longrightarrow U' \longrightarrow U \xrightarrow{\pi} U'' \longrightarrow 1$$

with $U' \simeq \mathbf{G}_a$ and $\dim U'' < \dim U$. It suffices to show that f factors through U'' . Since the graph of the equivalence relation defined by π is smooth over k , hence reduced, it suffices to show that if $x, y \in U(k)$ have the same image z in $U''(k)$, then $f(x) = f(y)$. Now $\pi^{-1}(z)$ is isomorphic to $\mathbf{G}_a$, hence the restriction of f to $\pi^{-1}(z)$ factors through a reduced irreducible component of H , hence through a k -scheme isomorphic to $(\mathbf{G}_m)^r$. It then suffices to note that every morphism from $\mathbf{G}_a$ to $(\mathbf{G}_m)^r$ is constant, since every invertible regular function on $\mathbf{G}_a$ is constant.

Proof of 6.1.1 A) ii). By 6.2.1, 6.2.3 and 6.2.4, we can assume that $H = \mathbf{G}_m$. By hypothesis, U has a composition series $U \supset U_1 \supset \cdots$ such that U_i/U_{i+1} is isomorphic to $\mathbf{G}_a$. Let E_1 be the inverse image of U_1 in E . By induction on $\dim U$, one can assume that the extension E_1 is trivial; let U'_1 be the unique lifting of U_1 . Proceeding as in the proof of 6.2.4, one shows that U'_1 is invariant in E . After passing to the quotient by U'_1 , one reduces to the case where $U = \mathbf{G}_a$. The S -scheme E (where $S = U$) is then a torsor under the S -group $\mathbf{G}_m \times_k S$, hence has a section, since $\text{Pic}(S) = 0$ (Exp. VIII 4.3). The extension E is then trivial by A) i).

Proof of 6.1.1 A) iii) (U smooth, H étale). Suppose first that U is connected. The group $U_{\bar{k}}$ then has a composition series with successive quotients isomorphic to $\mathbf{G}_a$, so $E_{\bar{k}}$ is trivial by A) ii). Since H is étale, it is clear that the unique lifting of $U_{\bar{k}}$ in $E_{\bar{k}}$ is the connected component of $E_{\bar{k}}$, so this lifting is already defined over k . In the general case, after passing to the quotient by the connected component of E , one reduces to the case where E is étale, then to the case where E is completely split (6.2.3). Since $U(k)$ is a p -group and $H(k)$ has order prime to p , one may take as lifting of $U(k)$ the Sylow p -subgroup of $E(k)$.

⁽¹⁰⁾**N.D.E.** See, for example, Proposition III.6.4.2 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

Proof of 6.1.1 A) iv) (U smooth, k perfect). If U is connected, U has a composition series with successive quotients isomorphic to $\mathbf{G}_a$ (4.1.2 b)), and one applies A) ii). In the general case, the foregoing allows us to reduce to the case where U is étale, then to the case where U is completely split and H diagonalizable (6.2.3). Using now a characteristic composition series of H ($H \supset H^0 \supset_{F^n} (H) \supset 0$), one reduces to the case where H is of one of the following three types:

- a) H is étale.
- b) $H = (\mathbf{G}_m)^r$.
- c) H is radicial.

In case a), one applies A) iii); in case c), one applies 1.6. Finally, in case b), one remarks that by Hilbert's Theorem 90, $E \rightarrow U$ has a section, so that it suffices to show that $H^2(U, (\mathbf{G}_m)^r) = H^2(U(k), (\mathbf{G}_m)^r(k)) = 0$. Now $U(k)$ is a finite p -group, whereas $(\mathbf{G}_m)^r(k)$ is uniquely p -divisible (since k is perfect).

6.3. — Proof of 6.1.1 B) and C). By 6.2.1, 6.2.3 and 6.2.4, one sees that it suffices to prove B) when $H = \mathbf{G}_m$. One therefore has an exact sequence

$$1 \longrightarrow \mathbf{G}_m \longrightarrow E \longrightarrow \alpha_p \longrightarrow 1.$$

Since $\mathbf{G}_m$ is smooth, one deduces an exact sequence of Lie p -algebras (App. II 3.2):

$$(*) \quad 0 \longrightarrow \text{Lie } \mathbf{G}_m \longrightarrow \text{Lie } E \longrightarrow \text{Lie } \alpha_p \longrightarrow 0.$$

The group α_p being of height 1, the extension E is trivial if and only if (App. II 2.2) the exact sequence (*) of Lie p -algebras is split. One knows that $\text{Lie } \alpha_p$ is generated by an element $X \neq 0$ such that $X^{(p)} = 0$ (App. II 2.1); it therefore suffices to show that X lifts to an element Z of $\text{Lie } E$ such that $Z^{(p)} = 0$. Now by 5.8.2 ii), there exists a unipotent element Z of $\text{Lie } E$ lifting X . Since the unipotent part of $\text{Lie } E$ clearly has dimension at most 1, one necessarily has $Z^{(p)} = 0$.

Proof of 6.1.1 C). If U'_1 is a lifting in E of an algebraic subgroup U_1 of U , U'_1 is invariant in E , since E is supposed commutative, and we may pass to the quotient by U'_1 . Using a composition series of U (3.5 ii)), the preceding remark allows us to reduce to the case where U is smooth or equal to α_p . But then E is trivial by A) iv) and B).

6.4. — Examples of extensions of a unipotent group U by a group of multiplicative type H which are not trivial. In view of 6.1.1 A) iv), the problem arises only in characteristic $p > 0$.

- a) $H = \mathbf{G}_m$, U is a nontrivial form of $\mathbf{G}_a$ (cf. App. III, § 5).

Let k be an imperfect field of characteristic 2, u an element of k such that $u^{1/2}$ does not belong to k . Consider the affine group E with ring $k[X, Y, (X^2 + uY^2)^{-1}]$, where multiplication is given by the comorphism

$$(X, Y) \longmapsto (XX' + uYY', XY' + YX').$$

The group E is smooth, connected, commutative, of dimension 2; the subscheme $Y = 0$ defines a subgroup $H \simeq \mathbf{G}_m$. The kernel K of raising to the second power in E has equation

$$X^2 + uY^2 = 1,$$

hence has dimension 1. The group K contains the unipotent radical of E (defined over $\bar{k}$), but also the contribution of H , which is isomorphic to μ_2 . Since K is reduced over k , the unipotent radical of E is not defined over k , and $U = E/H$ does not lift in E . (One immediately checks that U is the form of $\mathbf{G}_a$ with ring $k[V, W]/(V + uV^2 + W^2)$, the morphism $E \rightarrow U$ corresponding to the comorphism $V \mapsto Y^2/(X^2 + Y^2u)$, $W \mapsto XY/(X^2 + Y^2u)$.)

- b) $H = \mathbf{G}_m$, $U = \mathbf{Z}/2\mathbf{Z}$, k imperfect of characteristic 2.

Choosing k and u as in a), consider the subgroup E of GL_2 generated by the element X such that

$$X = \begin{pmatrix} 0 & 1 \\ u & 0 \end{pmatrix} = \begin{pmatrix} u^{1/2} & 0 \\ 0 & u^{1/2} \end{pmatrix} \begin{pmatrix} 0 & u^{-1/2} \\ u^{1/2} & 0 \end{pmatrix}.$$

The group E is an extension of $\mathbf{Z}/2\mathbf{Z}$ by $\mathbf{G}_m$, but this extension is not trivial, since the unipotent part of X is not defined over k .

c) $H = \mu_p$, $U = \alpha_p$, k imperfect.

Let $\mathfrak{e}$ be the commutative Lie p -algebra generated by two elements X and Y such that $X^{(p)} = X$ and $Y^{(p)} = aX$. By App. II 2.2, $\mathfrak{e}$ is the Lie p -algebra of an algebraic group E , an extension of α_p by μ_p , but this extension is trivial if and only if there exists $b \in k$ such that $b^p = a$ (for one then has $(bX + Y)^{(p)} = 0$).

d) $H = \mu_2$, $U = \alpha_2 \times \alpha_2$, E noncommutative, k a field of characteristic 2.

Consider the special linear group $\mathrm{SL}_{2,k}$ and let $E = {}_F(\mathrm{SL}_{2,k})$. The group E is a radicial group of height 1, whose Lie algebra is generated by three elements X, Y, Z satisfying the following relations:

$$[X, Y] = Z, \quad [X, Z] = [Y, Z] = 0,$$

$$X^{(p)} = Y^{(p)} = 0, \quad Z^{(p)} = Z.$$

Consequently, E is a central extension of $U \simeq \alpha_2 \times \alpha_2$ by μ_2 . Each factor α_2 of U lifts uniquely in E , but U itself does not lift in E , since $[X, Y] \neq 0$.

7. Nilpotent affine algebraic groups

7.1. — Extensions of groups of multiplicative type.

Proposition 7.1.1. *Let S be a prescheme, H and K two group S -preschemes of multiplicative type and of finite type, E a group prescheme which is an extension of K by H (i.e. K is the quotient of E by H for the fpqc topology). Then E is of multiplicative type in the following two cases:*

a) E is commutative.

b) The fibers of K are connected.

Proof. i) Case where S is the spectrum of a field k . The assertion to be proved is local for the fpqc topology, which allows us to suppose k algebraically closed, hence H and K diagonalizable. Note that K acts trivially on H by inner automorphisms: this is clear in case a), and follows from 6.2.1 in case b). By induction on the length of a suitable composition series of K , one reduces to the case where K is of one of the following three types: a) $K = \mathbf{G}_m$, b) $K = \mu_p$, c) $K = \mu_q$ with $(q, p) = 1$ and, in this case, E is commutative. Using now an embedding of H in $(\mathbf{G}_m)^r$, one deduces that E embeds in an extension of K by $(\mathbf{G}_m)^r$. One may therefore suppose $H = (\mathbf{G}_m)^r$.

a) If $K = \mathbf{G}_m$, E is a smooth connected affine algebraic group (Exp. VIB 9.2) of unipotent rank zero, hence is a torus.

b) $K = \mu_p$. In this case, E is a trivial extension. Indeed, since H is smooth, the canonical morphism $\mathrm{Lie} E \rightarrow \mathrm{Lie} \mu_p$ is surjective (App. II 3.2), and it suffices to apply 5.8.2 i), taking account of App. II 2.2.

c) $K = \mu_q$, with $(q, p) = 1$ and E commutative. Here again, the extension E is trivial. Indeed, let $x \in E(k)$ be a lifting of a generator $\bar{x}$ of $\mu_q(k)$. The element x^q is an element of $H(k)$, hence is of the form y^q , $y \in H(k)$ (note that $(\mathbf{G}_m)^r(k)$ is q -divisible). Since E is commutative, $y^{-1}x$ is a lifting of $\bar{x}$ of order q .

ii) General case. The groups H and K are flat, affine and of finite presentation over S (Exp. IX 2.1), and consequently the same holds for E (Exp. VIB 9.2). Using then the general technique of VIB § 10, we reduce to the case where S is noetherian. To show that E is of

multiplicative type, it then suffices to prove that E is commutative and that ${}_nE$ is finite over S for every n (Exp. X 4.8 b)).

a) E is commutative. One must verify that the morphism

$$E \times_S E \longrightarrow E, \quad (x, y) \longmapsto [x, y] = xyx^{-1}y^{-1}$$

factors through the unit section of E , and it suffices to verify this when S is the spectrum of an artinian local ring. But then E is of multiplicative type by i) and Exp. X 2.3.

b) ${}_nE$ is finite over S . Indeed, one has the exact sequence

$$0 \longrightarrow {}_nH \longrightarrow {}_nE \longrightarrow {}_nK \xrightarrow{u} H_n$$

(where H_n is the cokernel of raising to the n th power in H). One knows that H_n is of multiplicative type (Exp. IX 2.7), hence separated, and that ${}_nH$ and ${}_nK$ are finite over S ; $\text{Ker } u$ is a closed subgroup of ${}_nK$, hence is finite over S , and ${}_nE$ is finite over S as an extension of a finite group by a finite group (Exp. VIB 9.2).

7.2. — Structure of commutative affine algebraic groups.

Theorem 7.2.1. *Let k be a field, G a commutative affine algebraic k -group. Then:*

- a) *G contains a largest subgroup of multiplicative type M . The group M is characteristic in G and G/M is unipotent, and its formation commutes with extension of the field k .*
- b) *If k is perfect, G is the direct product of M and a unipotent algebraic subgroup U , and this uniquely.*

Proof. i) k algebraically closed. When G is smooth, 7.2.1 b) is well known (BIBLE § 4 Th. 4). If G is radicial of height 1, the decomposition of $\text{Lie } G$ described in 4.2.2 corresponds, taking account of 4.3.1 v) and App. II 2.2, to a decomposition of G of the type 7.2.1 b). In the general case, G admits a composition series whose successive quotients are smooth or radicial of height 1 (App. II 3.1). To prove 7.2.1 b), it then suffices to note that if one has an exact sequence of commutative algebraic groups

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0,$$

where G' (resp. G'') is the product of a group of multiplicative type and a unipotent group $G' = M' \cdot U'$ (resp. $G'' = M'' \cdot U''$), then the same holds for G . Indeed, consider the exact sequence

$$0 \longrightarrow G'/U' \longrightarrow G/U' \longrightarrow G'' \longrightarrow 0.$$

By 7.1.1 a), the inverse image in G/U' of M'' is a subgroup of multiplicative type M_1 . The group M_1 lifts to a subgroup M of G (5.1.1 i) a)). Similarly, using this time 6.1.1 C), one proves that there exists a unipotent subgroup U of G which is an extension of U'' by U' . It is clear that $G = M \cdot U$.

ii) k arbitrary. By i), $G_{\bar{k}}$ is the direct product of a group of multiplicative type $M_{\bar{k}}$ with a unipotent group $U_{\bar{k}}$. Put $S = \bar{k} \times_k \bar{k}$, and let M_1 and M_2 be the two inverse images of $M_{\bar{k}}$ by the two projections $S \rightrightarrows \bar{k}$. The group $G_S/M_2 = (G/M_{\bar{k}})_S$ has unipotent fibers, so the image of M_1 in G_S/M_2 is zero (2.4 i)), and M_2 contains M_1 . Similarly, M_1 contains M_2 , and finally $M_1 = M_2$. By fpqc descent, it follows that $M_{\bar{k}}$ comes from an algebraic subgroup M of G . It is clear that M is of multiplicative type, that G/M is unipotent, and that formation of M is compatible with every extension of the field k . For every k -prescheme S , every subgroup of multiplicative type H of G_S is contained in M_S . Indeed, by 2.5 its image in the group with unipotent fibers $(G/M)_S = G_S/M_S$ is zero. Taking in particular for H the transform of M_S by an automorphism of G_S , one deduces that M is characteristic in G .

Finally, if k is perfect, G/M lifts in G to a unipotent group, and this uniquely by 6.1.1 C).

Remark 7.2.2. i) If k is imperfect, the unipotent component of $G_{\bar{k}}$ is not necessarily defined over k , as shown by example 6.4 a).

- ii) Contrary to what happens for the component of multiplicative type M , the unipotent component U is in general not characteristic in G (and this regardless of the characteristic of k). Of course, uniqueness of the decomposition 7.2.1 b) implies that U is invariant under every k -automorphism of G . But if U and M are such that there exist a k -prescheme S and a nonzero S -homomorphism $h : U_S \rightarrow M_S$ (cf. 2.6), one deduces an S -automorphism of G_S , $(u, m) \mapsto (u, h(u) + m)$, which does not leave U_S invariant.
- iii) If G is finite over k , G/M corresponds by Cartier duality (2.6) to the connected component of the dual $D(G)$ of G .

7.3. — Structure of nilpotent affine algebraic groups.

Theorem 7.3.1. *Let k be a field, G a nilpotent algebraic k -group (Exp. VIB § 8), affine, connected. Then G has a largest subgroup of multiplicative type M . The group M is central and characteristic, and G/M is a unipotent algebraic group.*

Let Z be the center of G , M the largest subgroup of multiplicative type of Z (7.2.1). Since Z is characteristic in G and M characteristic in Z , M is characteristic in G . It suffices to show that G/M is unipotent. By induction on the length of the ascending central series of G , this will follow more generally from the following lemma:

Lemma 7.3.2. (Rosenlicht). *Let G be a connected algebraic k -group, Z its center. Then the center Z' of $G' = G/Z$ is unipotent.*

Proof. We may suppose k algebraically closed. It then suffices to show that Z' contains no subgroup isomorphic to μ_ℓ for any prime number ℓ (4.6.1 vi)). Let therefore μ_ℓ be a subgroup of Z' , N its inverse image in G . Since μ_ℓ is central in G' , N is invariant in G .

i) *Case where $(\ell, p) = 1$.* One can find an element x of $N(k)$ and an integer n having the following properties:

a) x lifts a generator $\bar{x}$ of μ_ℓ .

b) $x^{\ell^n} \in Z^0(k)$;

(it suffices to choose a lifting of $\bar{x}$ whose image in $N/Z^0(k)$ belongs to the Sylow ℓ -subgroup).

Raising to the ℓ th power in the commutative group Z^0 is an étale morphism, so $Z^0(k)$ is ℓ -divisible. Consequently, after multiplying x by an element of $Z^0(k)$, one may suppose that $x^{\ell^n} = 0$. The group N is then generated by two commutative groups which commute (Z and the group generated by x), hence is commutative. The group ${}_{\ell^n}N$ is a group of multiplicative type, characteristic in N , hence invariant in G , and consequently central, G being connected (Exp. IX 5.5). Thus ${}_{\ell^n}N$ is contained in Z , which contradicts the fact that its image in G' contains μ_ℓ .

ii) $\ell = p$. There then exists an integer n such that the image of ${}_{F^n}N$ in G' contains μ_p , so that one has the exact sequence

$$(*) \quad 1 \longrightarrow K \longrightarrow {}_{F^n}N \longrightarrow \mu_p \longrightarrow 1.$$

The group K is contained in Z , hence is commutative; it then follows from 7.2.1 and 7.1.1 b) and 5.5.1 that there exists a subgroup of multiplicative type contained in ${}_{F^n}N$ whose image in the quotient by K is μ_p . One deduces as in i) that ${}_{F^n}N$ is commutative. The component of multiplicative type of ${}_{F^n}N$ (7.2.1) is characteristic in ${}_{F^n}N$, hence invariant in G , hence central (Exp. IX 5.5); since its image in G' contains μ_p , one obtains a contradiction.

A. Appendix I. Hochschild cohomology and extensions of algebraic groups

A.1. — Definition of the cohomology groups. Let k be a field, G an algebraic k -group, $(\text{Ab})_G$ the abelian category of commutative algebraic k -groups on which G acts. If $A \in \text{Ob}(\text{Ab})_G$, the functor $h_A : ((\text{Sch})/k)^\circ \rightarrow (\text{Ens})$ canonically defined by A is a G - $\mathbf{Z}$ -module in the sense of I 3.2. We may therefore consider the standard complex $C^\bullet(G, A)$

of algebraic cochains of G with values in A (Exp. I 5.1), as well as the group of i -cocycles $Z^i(G, A)$, of i -coboundaries $B^i(G, A)$ and the i th cohomology group $H^i(G, A)$. As usual, $H^0(G, A)$ is identified with the group $A^G(k)$, where A^G is the k -functor of invariants of A under G . The group $H^1(G, A)$ classifies trivial principal homogeneous spaces under A on which G acts.

The functor $A \mapsto H^\bullet(G, A)$ is in general not a cohomological functor from the category $(\text{Ab})_G$ with values in (Ab) ; however, one has the following proposition:

Proposition A.1.1. *Let $0 \rightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \rightarrow 0$ be an exact sequence in $(\text{Ab})_G$. Then:*
a) If v has a section (that is to say, if there exists a k -morphism of preschemes $s : A'' \rightarrow A$ such that $vs = 1_{A''}$), one has the usual exact cohomology sequence

$$\cdots \rightarrow H^i(G, A) \rightarrow H^i(G, A'') \xrightarrow{d} H^{i+1}(G, A') \rightarrow \cdots$$

b) If $A(k) \rightarrow A''(k)$ is surjective, one has the exact sequence

$$(1) \quad 0 \rightarrow A'^G(k) \rightarrow A^G(k) \rightarrow A''^G(k) \xrightarrow{d} H^1(G, A') \rightarrow H^1(G, A) \rightarrow H^1(G, A'').$$

Proof. a) One notes that existence of a section implies exactness of the sequence of complexes

$$0 \rightarrow C^\bullet(G, A') \rightarrow C^\bullet(G, A) \rightarrow C^\bullet(G, A'') \rightarrow 0.$$

b) If $x'' \in A''^G(k)$, its inverse image in A is a trivial principal homogeneous space under A' (since $A(k) \rightarrow A''(k)$ is supposed surjective) on which G acts, hence defines an element $d(x'') \in H^1(G, A')$. Exactness of sequence (1) is then immediate.

A.2. — The group $\text{Ext}_{\text{alg}}(G, A)$. Let A and G be two algebraic k -groups, E and E' two algebraic k -groups which are extensions of G by A . These two extensions are said to be isomorphic if there exists a group k -morphism $u : E \rightarrow E'$ making the diagram commutative:

$$\begin{array}{ccccc} & & E & & \\ & \nearrow & \downarrow u & \searrow & \\ A & & & & G \\ & \searrow & \downarrow & \nearrow & \\ & & E' & & \end{array}$$

The group E acts on A by inner automorphisms, and if A is commutative, this action factors through G , so G acts on A . Conversely, if $A \in \text{Ob}(\text{Ab})_G$, we denote by $\text{Ext}_{\text{alg}}(G, A)$ the set of classes of algebraic extensions E of G by A for which the action of G on A defined by E and that coming from the structure of an object of $(\text{Ab})_G$ coincide.

$\text{Ext}_{\text{alg}}(G, A)$ is naturally a bifunctor covariant in A and contravariant in G . More precisely:
 a) If $f : A \rightarrow B$ is a morphism in $(\text{Ab})_G$ and E represents an element of $\text{Ext}_{\text{alg}}(G, A)$, one defines $f_*(E) \in \text{Ext}_{\text{alg}}(G, B)$ as the class of the extension of G by B equal to the quotient of the semidirect product $B \cdot E$ (E acting on B through G) by the algebraic subgroup which is the image of A under the morphism (f, i) (this quotient is representable by Exp. VIA § 5), so that one has a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & A & \xrightarrow{i} & E & \longrightarrow & G \longrightarrow 1 \\ & & \downarrow f & & \downarrow & & \downarrow 1_G \\ 1 & \longrightarrow & B & \longrightarrow & f_*(E) & \longrightarrow & G \longrightarrow 1. \end{array}$$

b) If $g : H \rightarrow G$ is a k -morphism of algebraic k -groups and E is an extension of G by A , the fibered product $E \times_G H$ is naturally an extension of H by A , denoted by $g^*(E)$. One

therefore has a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & A & \longrightarrow & g^*(E) & \longrightarrow & H \longrightarrow 1 \\ & & \downarrow 1_A & & \downarrow & & \downarrow g \\ 1 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & G \longrightarrow 1. \end{array}$$

Adapting the proofs given in J.-P. Serre, *Groupes algébriques et corps de classe*, chap. VII, one equips $\text{Ext}_{\text{alg}}(G, A)$ with a natural abelian group structure, functorial in A and G .

Proposition A.2.1. *Let $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ be an exact sequence in $(\text{Ab})_G$; then:*
 a) *One has a canonical exact sequence of abelian groups*

$$Z^1(G, A) \rightarrow Z^1(G, A'') \xrightarrow{d} \text{Ext}_{\text{alg}}(G, A') \rightarrow \text{Ext}_{\text{alg}}(G, A) \rightarrow \text{Ext}_{\text{alg}}(G, A'').$$

b) *If $A(k) \rightarrow A''(k)$ is surjective, one deduces from a) the exact sequence*

$$(2) \quad H^1(G, A) \rightarrow H^1(G, A'') \xrightarrow{d} \text{Ext}_{\text{alg}}(G, A') \rightarrow \text{Ext}_{\text{alg}}(G, A) \rightarrow \text{Ext}_{\text{alg}}(G, A'').$$

The exact sequence of a) generalizes the usual exact sequence of the $\text{Hom}(\cdot, \cdot)$ valid in the context of commutative extensions (loc. cit.) and is proved in the same way. Let us simply recall the definition of the coboundary $d : Z^1(G, A'') \rightarrow \text{Ext}_{\text{alg}}(G, A')$. For this, consider the extension

$$1 \rightarrow A' \rightarrow A \cdot G \rightarrow A'' \cdot G \rightarrow 1,$$

deduced in an evident way from the exact sequence $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$. If $u \in Z^1(G, A'')$, u defines in the usual way a section homomorphism $u : G \rightarrow A'' \cdot G$. One then has $d(u) = u^*(A \cdot G)$.

A.3. — Comparison of $H^2(G, A)$ and $\text{Ext}_{\text{alg}}(G, A)$. It is well known, in the case of abstract groups, that there exists a functorial isomorphism between the abelian groups $H^2(G, A)$ and $\text{Ext}(G, A)$. Similarly, in the present case, if A is an element of $(\text{Ab})_G$, to every 2-cocycle $u \in Z^2(G, A)$ one can associate an algebraic group structure on the prescheme $A \times_k G$ making it an element of $\text{Ext}_{\text{alg}}(G, A)$. Moreover, this extension is trivial if and only if $u \in B^2(G, A)$ (cf. Exp. III 1.2.2). Recall that the composition law on $A \times G$ is defined by the formula

$$(a, g)(a', g') = a + {}^g a' + u(g, g').$$

It is clear that the extensions of G by A thus obtained are not arbitrary, since they have a section. But conversely, if $E \in \text{Ext}_{\text{alg}}(G, A)$ has a section s , E is isomorphic to the extension of G by A associated to the 2-cocycle u such that

$$u(g, g') = s(g)s(g')s(gg')^{-1}.$$

One finally obtains the following proposition:

Proposition A.3.1. *There exists a functorial isomorphism between the bifunctors with values in abelian groups*

$$(G, A) \longmapsto H^2(G, A) \quad \text{and} \quad (G, A) \longmapsto \text{Ext}_s(G, A),$$

where $\text{Ext}_s(G, A)$ denotes the subgroup of $\text{Ext}_{\text{alg}}(G, A)$ formed by the classes of extensions of G by A which have a section.

B. Appendix II. Recollections and complements on radicial groups

Let p be a prime number > 1 , and let S be an $\mathbf{F}_p$ -prescheme.

B.1. — The Frobenius morphism. For every S -prescheme X and every integer $n > 0$, denote by $P^n : X \rightarrow X$ the $\mathbf{F}_p$ -endomorphism corresponding to raising to the p^n th power in $\mathcal{O}_X$, and denote by $X^{(n)}$ the S -prescheme which is the inverse image of X under the

morphism $P^n : S \rightarrow S$. There then exists a unique S -morphism $F^n : X \rightarrow X^{(n)}$ making the diagram commutative:

$$\begin{array}{ccc}
 & & X \\
 & \swarrow P^n & \searrow F^n \\
 X & \xleftarrow{\quad} & X^{(n)} \\
 \downarrow & & \downarrow \\
 S & \xleftarrow{P^n} & S
 \end{array}$$

It is clear that F^n is identified with the “ n th iterate” of $F^1 = F$, called the Frobenius morphism of X/S .

If G is a group S -prescheme, $G^{(n)}$ is a group S -prescheme, and $F^n : G \rightarrow G^{(n)}$ is a group S -morphism. Its kernel ${}_{F^n}(G)$ is a characteristic group subscheme of G (i.e. stable under the functor $\text{Aut}_{S\text{-gr}}(G)$), radical over S . If G is a radical group S -prescheme, one says that G has height $\leq h$ if ${}_{F^h}(G) = G$.

B.2. — Groups and Lie p -algebras. If G is a group S -prescheme, $\text{Lie}(G)$ (Exp. II) is naturally a restricted Lie p -Algebra (Exp. VIIA §§ 5 and 6). In particular, one has the following result (cf. Exp. VIIA):

Proposition B.2.1. *i) $\text{Lie}(\alpha_p)_S = \text{Lie}(\mathbf{G}_a)_S^{(11)}$ is a sheaf of $\mathcal{O}_S$ -modules, free over $\mathcal{O}_S$ of rank 1, generated by an element X such that $X^{(p)} = 0$.
ii) $\text{Lie}(\mu_p)_S = \text{Lie}(\mathbf{G}_m)_S$ is a sheaf of $\mathcal{O}_S$ -modules, free over $\mathcal{O}_S$ of rank 1, with a canonical basis X such that $X^{(p)} = X$.*

Let us now recall the fundamental result proved in Exp. VIIA § 7:

Theorem B.2.2. *Suppose S affine with ring A . Then the functor*

$$G \mapsto \text{Lie } G$$

establishes an equivalence of categories between, on the one hand, the category of group S -preschemes G of finite presentation and flat over S , of height 1, whose Lie algebra is locally free over $\mathcal{O}_S$, and, on the other hand, the category of restricted Lie p - A -algebras locally free of finite rank.

Moreover, if G is as above and H is a group S -prescheme of finite presentation, the canonical morphism

$$\text{Hom}_{S\text{-gr}}(G, H) \rightarrow \text{Hom}_{p\text{-}A\text{-Lie}}(\text{Lie } G(S), \text{Lie } H(S))$$

is an isomorphism.

B.3. — Radical groups and smooth groups. We now suppose that S is the spectrum of a field k of characteristic p .

Recall that in Exp. VIA § 5, it was shown that if G is an algebraic k -group and H an algebraic subgroup of G , then the sheaf G/H (a sheaf for the fpqc topology) is representable. Let us then recall (VIIA 8.3):

Proposition B.3.1. *Let G be an algebraic k -group. Then there exists an integer m such that for every $n \geq m$, the algebraic group $G/{}_{F^n}(G)$ is smooth over k .*

Proposition B.3.2. *Consider an exact sequence of algebraic k -groups*

$$1 \rightarrow G' \rightarrow G \xrightarrow{u} G'' \rightarrow 1$$

⁽¹¹⁾**N.D.E.** The definition of α_p , already given at the beginning of this exposé, has been deleted here.

and the following assertions:

- i) The morphism u is smooth.
- ii) G' is smooth over k .
- iii) For every integer $n > 0$, one has the exact sequence

$$1 \longrightarrow F^n(G') \longrightarrow F^n(G) \longrightarrow F^n(G'') \longrightarrow 1.$$

- iv) The morphism ${}_FG \longrightarrow {}_FG''$ is an epimorphism.
- v) The morphism $(\operatorname{Lie} G)(k) \longrightarrow (\operatorname{Lie} G'')(k)$ is surjective.

Then one has the following implications:

$$i) \Longleftrightarrow ii) \implies iii) \implies iv) \Longleftrightarrow (v).$$

Moreover, if G is smooth, the five assertions are equivalent.

- i) $\Leftrightarrow$ ii) by Exp. VIB 9.2 vii).
- ii) $\Rightarrow$ iii). If G' is smooth, $F^n : G' \longrightarrow G'^{(n)}$ is an epimorphism, and iii) follows from the “snake” diagram.
- iii) $\Rightarrow$ iv) is clear.
- iv) $\Leftrightarrow$ v) by Theorem B.2.2.
- v) $\Rightarrow$ ii) when G is smooth. Indeed, G'' is then smooth, and v) implies that one has $\dim G' = \dim_k(\operatorname{Lie} G')(k)$, so G' is smooth.

C. Appendix III. Remarks and complements concerning exposés XV, XVI, XVII

C.1. — It may be that Propositions 1.2 and 1.2 bis of XV remain true if one removes the hypothesis that H_0 is smooth over S_0 . This is in particular the case if G is finite, flat and commutative.

C.2. — Complement to XV 4.8. The following proposition, as well as Theorems C.3.1 and C.4.1 below, will appear in an article in preparation by M. Raynaud on group schemes over a discrete valuation ring.⁽¹²⁾

Proposition C.2.1. *Let S be the spectrum of a discrete valuation ring, t its generic point, G a group S -prescheme of finite type and flat, $\tilde{G} = \operatorname{Spec} \Gamma(G, \mathcal{O}_G)$, $u : G \longrightarrow \tilde{G}$ the canonical morphism. Then:*

- (1) $\tilde{G}$ is naturally a group S -scheme, and u is a homomorphism.
- (2) $\operatorname{Ker}(u)$ is flat over S , and $(\tilde{G}, u)$ is an fpqc quotient of G by $\operatorname{Ker}(u)$, so that $\tilde{G}$ is the largest affine quotient of G .
- (3) If G_t is affine, $\operatorname{Ker}(u)$ is an étale group over S , equal to the unit group if and only if G is separated over S . In particular, a group S -scheme G , flat, of finite type, separated, with affine generic fiber, is affine.

C.3. — In the statement of XV 6.6, the hypothesis that H is equal to its connected normalizer is unnecessary for the set of points s of S such that H_s is a parabolic subgroup of G_s to be an open set. Indeed, let us return to the proof given in paragraph c) preceding Lemma 6.9, and denote by $\bar{N}$ the schematic closure of N_t in G . Thus $\bar{N}$ is a closed group subscheme of G , flat over S , containing H and contained in N . It then follows from the theorem below that $G/\bar{N}$ is representable. Since $G_s/\bar{N}_s$ is proper and connected, one finishes as in loc. cit.

⁽¹²⁾**N.D.E.** Comments to be added here, including references to VIB...

Theorem C.3.1. ⁽¹³⁾ *Let S be the spectrum of a henselian discrete valuation ring, G a group S -prescheme locally of finite type, H a closed group subscheme of G , flat over S ; then G/H is representable.*

C.4. — Complement to (XVI 1.1). The following theorem makes (VIII 7.9) more precise.

Theorem C.4.1. *Let S be the spectrum of a discrete valuation ring of residue characteristic $p > 0$, and let G be a group S -scheme, commutative, smooth, of finite type and separated over S . Then the following conditions are equivalent:*

- (i) ${}_pG$ is finite over S .
- (ii) *For every S -prescheme S' and every group S' -prescheme H' of finite presentation over S' , separated over S' , every S' -monomorphism $u : G_{S'} \rightarrow H'$ is an immersion.*

C.5. — Example (XVII 6.4 a)) provides an example of a smooth group over a field k whose unipotent radical is not defined over k . The following proposition gives a general method for obtaining such groups:

Proposition C.5.1. *Let k be a field, K a finite radical extension of k of degree > 1 , H a connected smooth algebraic K -group of dimension r , $G = \prod_{K/k} H/K$, which is a smooth connected algebraic k -group of dimension $r[K : k]$. Let $u : G_K \rightarrow H$ be the canonical homomorphism, and $R = \text{Ker}(u)$. Then:*

- i) *The morphism u is an epimorphism, and R is a smooth unipotent connected algebraic group.*
- ii) *If U is a smooth algebraic subgroup of G such that U_K contains R , then $U = G$.*

Corollary. *Keep the notation of the preceding proposition.*

- a) *If H is not unipotent, the unipotent radical of $G_{\bar{k}}$ is not defined over k .*
- b) *If H is a nonzero abelian variety, G is not an extension of an abelian variety by a smooth linear group.*
- c) *If H is not solvable, the solvable radical of $G_{\bar{k}}$ is not defined over k , and G has no Borel subgroup defined over k .*

Let us first show how the corollary follows from the proposition.

- a) Let U be a unipotent radical of G . Then U_K is the unipotent radical of G_K , hence contains R , since R is smooth unipotent connected by i), so $U = G$ by ii). Now G_K admits H as a quotient, and H is not unipotent by hypothesis, so G is not unipotent, whence a contradiction.
- b) If G is an extension of an abelian variety by a linear group L , smooth over k , L_K necessarily contains R , so $L = G$. But G cannot be a linear group, since G_K has a quotient which is a nonzero abelian variety.
- c) Let S (resp. B) be a solvable radical of G (resp. a Borel subgroup of G); then S_K (resp. B_K) contains R , so $S = G$ (resp. $B = G$). But then G is solvable, which contradicts the fact that G_K has a quotient which is not solvable.

Proof of Proposition 5.1: i) Let us begin by describing the morphism u . The datum of a K -scheme

$$f : S \rightarrow \text{Spec}(K)$$

allows one to construct the diagram

$$\begin{array}{ccc} \text{Spec}(K) & \xleftarrow{h} & T \\ \downarrow j & \swarrow f & \downarrow j_T \\ \text{Spec}(k) & \xleftarrow{g} & S \end{array} \quad \begin{array}{c} \nearrow s \\ \searrow \end{array}$$

⁽¹³⁾**N.D.E.** See Theorem 4C in: S. Anantharaman, *Schémas en groupes, espaces homogènes et espaces algébriques sur une base de dimension 1*, Bull. Soc. Math. France, Mém. 33 (1976), 5–79.

where $g = j \circ f$, $T = \operatorname{Spec}(K) \times_{\operatorname{Spec}(k)} S$, h and j_T are the two projections, and s is the section of T over S such that $h \circ s = f$. The map

$$u(S) : G_K(S) \longrightarrow H(S)$$

is simply the composite map

$$G_K(S) \xrightarrow{\sim} H(T) \longrightarrow H(S),$$

where the last arrow is defined by the section s .

Take in particular for S the spectrum of an algebraic closure $\bar{k}$ of k , and for f the unique k -morphism $K \longrightarrow \bar{k}$, so that T is a local artinian scheme. To prove i), it suffices to do so after the base field extension $K \longrightarrow \bar{k}$. Now it is clear that $G_S = \prod_{T/S} H_T/T$ represents the Greenberg functor of H_T relative to S (M. J. Greenberg, *Schemata over local rings*, Ann. of Maths. 73, 1961, pp. 624–648). The description above then shows that, under this last identification, u_S is the canonical transition morphism

$$\operatorname{Green}(H_T) \longrightarrow H_S = H_T \times_T S.$$

Assertion i) then follows from the fact that H is smooth over K and from (M. J. Greenberg, *Schemata over local rings II*, Ann. of Maths. 78, 1963, pp. 256–266).

ii) To establish ii), we may suppose k separably closed. Let therefore U be a smooth algebraic subgroup of G such that $U_K \supset R$, and let us show that $U = G$. Since G is connected, one may suppose U connected. Let V be the smooth connected algebraic subgroup $u(U_K)$ of H , and $V' = \prod_{K/k} V/K$, which is a smooth connected algebraic subgroup of G . The group V'_K is an algebraic subgroup of G_K , and the canonical morphism $V'_K \longrightarrow V$ is simply the restriction of u to V'_K . By hypothesis, U_K contains R ; a fortiori, U_K contains $\operatorname{Ker}(V'_K \longrightarrow V)$; on the other hand, by construction, the image of U_K in H is equal to V . One deduces that U_K contains V'_K , hence that U contains V' . On the other hand, the canonical isomorphism

$$G(k) \hookrightarrow G(K) \xrightarrow{u(K)} H(K)$$

plainly sends $U(k)$ into $V(K) = V'(k)$; this says that $U(k)$ is contained in $V'(k)$. Since U is smooth and k separably closed, this implies $U \subset V'$, whence $U = V'$. One then has the equalities

$$([K : k] - 1) \dim V = \dim \operatorname{Ker}(V'_K \longrightarrow V) = \dim R = ([K : k] - 1) \dim H.$$

Since $K \neq k$, one concludes that $\dim V = \dim H$, whence $V = H$ and finally $U = V' = G$.

Exposé XVIII. Weil's theorem on the construction of a group from a rational law

by Michael Artin

0. Introduction

This exposé is devoted to the well-known theorem of Weil [1] which gives the construction of a group variety from a birational law. It seems that the generalization of this result to the case where the base is the spectrum of a discrete valuation ring was already known to several people; one may, for example, see [2]. Here we shall prove the theorem for a flat group of finite presentation over an arbitrary base prescheme S . Since one must make the hypotheses with a little more care, the statement is not in the form given by Weil; but when S is the spectrum of a field, one sees that the form given here is essentially equivalent to Weil's. We use the following suggestion of Grothendieck: given a “group germ” X over a prescheme S , one must construct the group as the quotient of $X \times_S X$ by a suitable equivalence relation.

1. “Recollections” on rational maps

Let X/S be a relative prescheme and $U \subseteq X$ an open subset. One says that U is *schematically dense in X relative to S* , or that $U \hookrightarrow X$ is *relatively schematically dense*, if for every base change $S' \rightarrow S$ the open subset $U \times_S S'$ is schematically dense in $X \times_S S'$. For the definition and properties of this notion one refers to Exp. IX, § 4. ^(*)

Proposition 1.1. (i) *A finite intersection, as well as the union of a nonempty family of open subsets schematically dense relative to S , is schematically dense relative to S .*

(ii) *If $U \subseteq X$ is schematically dense relative to S and $S \rightarrow T$ is a morphism, then $U_T \subseteq X_T$ is schematically dense relative to T .*

(iii) *Let $U \subseteq V \subseteq X$ be open immersions. For U to be schematically dense in X relative to S , it is necessary and sufficient that it be so in V and that V be so in X .*

(iv) *If $U \subseteq X$ and $V \subseteq Y$ are relatively schematically dense, then $U \times_S V$ is schematically dense in $X \times_S Y$ relative to S .*

In Exp. IX one finds the following criterion:

Proposition 1.2. *Let X/S be locally of finite presentation and flat, and let U be an open subset of X . If, for each $s \in S$, the fiber U_s is schematically dense in X_s , then U is schematically dense in X relative to S .*

In particular:

Corollary 1.3. *If X/S is locally of finite presentation and flat, and each fiber X_s is “without embedded components”, then U is schematically dense in X relative to S if and only if, for each $s \in S$, U_s is dense in X_s in the topological sense.*

⁽⁰⁾Version of 8 May 2009.

^(*)Cf. also EGA IV₃, 11.9 and 11.10 (in particular 11.10.8), where one says “universally schematically dense relative to S ”.

One easily deduces from the definition the

Proposition 1.4. *Let $U \subseteq X$ be schematically dense relative to S , and let $f, g : X \rightarrow Y$ be two morphisms of functors over S , where Y satisfies one of the following conditions:*

- (i) *Y is a prescheme over S , and each fiber Y_s is separated.*
- (ii) *Y is a presheaf⁽¹⁾ over S such that the diagonal morphism $Y \rightarrow Y \times_S Y$ is represented by a closed immersion.*⁽²⁾

Then, if $f = g$ on U , one has $f = g$.

PROOF. In case (i), the standard argument after IX 4.1 shows that f, g are equal on each fiber; hence the subprescheme $\text{Ker}(f, g)$ of X is set-theoretically equal to X , hence is closed, and since it contains U , it is equal to X , i.e. $f = g$. Case (ii) is proved as in *loc. cit.*⁽³⁾ $\square$

Definition 1.5.^(*) Let X be a prescheme over S , and Y a presheaf over S . One calls a rational map $f : X \rightarrow Y$ over S an equivalence class of morphisms $f : U \rightarrow Y$ over S , where U is an open subset of X schematically dense relative to S , and where one sets $(f : U \rightarrow Y) \sim (f' : U' \rightarrow Y)$ if and only if there exists an open subset $U'' \subseteq U' \cap U$, schematically dense relative to S , such that $f = f'$ on U'' .

This definition is made in such a way that one can define in an evident manner the rational map $f \times_S S'$ for every base extension $S' \rightarrow S$.

If U is an open subset of X , one says that the rational map f is *defined on U* if there exists a morphism representing f whose set of definition contains U .

Definition 1.5.1.⁽⁴⁾ If Y satisfies one of conditions (i), (ii) of 1.4, it is clear that there exists a largest open subset U of X such that f is defined on U , and this U is schematically dense relative to S . One calls it the *domain of definition of f over S* , and denotes it by $\text{Dom}(f)$.

This notion does not commute with base changes, but one has

Proposition 1.6. *Let X be an S -prescheme and Y an S -functor satisfying one of hypotheses (i), (ii) of 1.4. Let $f : X \rightarrow Y$ be a rational map over S , $S' \rightarrow S$ a flat morphism locally of finite presentation, and $f' = f \times_S S'$. Then*

$$\text{Dom}(f') = \text{Dom}(f) \times_S S'.$$

PROOF. Set $U = \text{Dom}(f)$. It is clear that $V' = \text{Dom}(f')$ contains $U \times_S S'$. Let V be the image of V' , which is an open subset of X , since $X' \rightarrow X$ is open. One must prove that $V = U$, that is, one must find a morphism $V \rightarrow Y$ representing f . Set $S'' = S' \times_S S'$,

$$X'' = X \times_S S'', \quad U'' = U \times_S S'', \quad V'' = V' \times_{V'} V'.$$

Then U'' is schematically dense in X'' relative to S'' , hence U'' is schematically dense in V'' relative to S , since $U'' \subseteq V'' \subseteq X''$. The restriction of $f' : V' \rightarrow Y'$ to U' is deduced from $f : U \rightarrow Y$ by base change. The two morphisms $V'' \rightarrow Y$ deduced from g by base change are equal on U'' , hence equal. Now, since $V' \rightarrow V$ is flat and locally of finite presentation, it is fppf-covering (Exp. IV 6.3), and one finds the morphism $V \rightarrow Y$ by descent. $\square$

⁽¹⁾**N.D.E.** Specify for which topology: a priori (fpqc).

⁽²⁾**N.D.E.** Let us recall here the definition of a closed immersion. A morphism of S -functors $F \rightarrow G$ is relatively representable if, for every S -morphism $T \rightarrow G$, where T is an S -prescheme, the T -functor $F_T = F \times_S T$ is represented by a T -prescheme. A morphism of S -functors $F \rightarrow G$ is an open (resp. closed) immersion if it is relatively representable and, for every S -morphism $T \rightarrow G$, where T is an S -prescheme, the T -morphism $F_T \rightarrow T$ is an open (resp. closed) immersion.

⁽³⁾**N.D.E.** Give details of this point...

^(*)Cf. EGA IV₄, 20.5, where one says "pseudo-morphism from X to Y relative to S ", so as not to conflict with EGA I, 7.12.

⁽⁴⁾**N.D.E.** The number 1.5.1 has been added.

We shall frequently use the following triviality:

Proposition 1.7. *Let X/S be faithfully flat, locally of finite presentation, and let $U \subseteq X$ be a relatively schematically dense open subset. Then there exists a base extension $S' \rightarrow S$ which is fppf-covering, and a section $x \in X(S')$ which is contained in $U(S')$.*

PROOF. Indeed, U/S is fppf-covering. Set $S' = U$, and take as section the S -graph of the inclusion of U in X . $\square$

2. Local determination of a morphism of groups

Let G, H be groups, and let $U \subseteq G$ be a subset such that $U \cdot U = G$. Then, if f, g are homomorphisms from G to H such that $f = g$ on U , one has $f = g$. Similarly:

Proposition 2.1. *Let S be a site (cf. Exp. IV⁽⁵⁾), G a sheaf of groups on S , and $U \subseteq G$ a subsheaf of sets such that the morphism $U \times U \rightarrow G$ induced by multiplication is an epimorphism. Then, if f, g are homomorphisms from G to a sheaf of groups H , equal on U , one has $f = g$.*

Corollary 2.2. *Let G be an S -group prescheme locally of finite presentation and flat, let $U \subseteq G$ be a relatively schematically dense open subset, and H a sheaf of groups on S for the fppf topology. Then every homomorphism $f : G \rightarrow H$ is determined by its restriction to U .*

PROOF. Indeed, since G is flat over S , the composition law in G is a flat morphism (VIB.9.2.xi), and it follows that $U \times_S U \rightarrow G$ is faithfully flat and locally of finite presentation, hence fppf-covering, hence an epimorphism. $\square$

Proposition 2.3. *Let G be a group prescheme locally of finite presentation and flat over S , $U \subseteq G$ a relatively schematically dense open subset, and H a sheaf of groups for the fppf topology. Denote by m_G the multiplication of G , and by m_H that of H .*

Let $\bar{f} : U \rightarrow H$ be an S -morphism, and suppose that there exists a relatively schematically dense open subset V of $m_G^{-1}(U) \cap (U \times_S U)$ such that the diagram

$$\begin{array}{ccc} V & \xrightarrow{(\bar{f} \times \bar{f})|_V} & H \times_S H \\ m_G \downarrow & & \downarrow m_H \\ U & \xrightarrow{\bar{f}} & H \end{array}$$

is commutative. Then there exists a morphism of S -groups $f : G \rightarrow H$ (necessarily unique) extending $\bar{f}$, in each of the following situations:

- (i) H is representable.
- (ii) The diagonal morphism $H \rightarrow H \times_S H$ is represented by a closed immersion. ⁽⁶⁾
- (iii) For every section $a \in U(S)$, the open subset $\text{pr}_2((a \times_S U) \cap V)$ is relatively schematically dense in U , ^(*) and this statement remains true after every base change $S' \rightarrow S$.

PROOF. First note that V is relatively schematically dense in $G \times_S G$. Indeed, V is relatively schematically dense in

$$m_G^{-1}(U) \cap (U \times_S G) \cap (G \times_S U),$$

and the three factors are deduced from $U \subseteq G$ by an evident base change $G \times_S G \rightarrow G$, which implies the assertion by 1.1.

⁽⁵⁾**N.D.E.** That is, a category endowed with a topology, cf. IV, § 4.2.

⁽⁶⁾**N.D.E.** See the note in 1.4.

^(*)Or in G , which amounts to the same by virtue of 1.1 (iv).

To construct a morphism $f : G \rightarrow H$, it suffices, since $U \times_S U \rightarrow G$ is fppf-covering, to find a morphism from $U \times_S U$ to H such that the two induced morphisms on $(U \times_S U) \times_G (U \times_S U)$ are the same. Take $m_H \circ (\bar{f} \times \bar{f}) : U \times_S U \rightarrow H$. One must verify that whenever one has sections $a, b, c, d \in U(S')$, with $S' \rightarrow S$ arbitrary, such that $ab = cd$, one also has $\bar{f}(a)\bar{f}(b) = \bar{f}(c)\bar{f}(d)$. By hypothesis, this is true if (a, d) and (c, d) are contained in $V(S')$, since in this case one has $\bar{f}(a)\bar{f}(b) = \bar{f}(ab)$ and $\bar{f}(c)\bar{f}(d) = \bar{f}(cd)$. Thus it is true whenever $(a, b, c, d) \in (V \times_G V)(S')$.

I say that $V \times_G V$ is an open subset of $(U \times_S U) \times_G (U \times_S U)$ schematically dense relative to S . This will complete the proof in cases (i) and (ii), by 1.4, the facts that the morphism f thus constructed extends $\bar{f}$, and that f is a homomorphism, being evident, also by 1.4. Now write

$$V \times_G V = (V \times_G (G \times_S G)) \cap ((G \times_S G) \times_G V).$$

By symmetry and 1.1, it suffices to verify that $V \times_G (G \times_S G)$ is relatively schematically dense in $(G \times_S G) \times_G (G \times_S G)$. But this last prescheme is S -isomorphic to $G \times_S G \times_S G$, the morphism being given by $(a, b, c, d) \mapsto (a, b, c)$. Thus what one must prove is that $V \times_S G$ is schematically dense in $G \times_S G \times_S G$ relative to S , which is a consequence of the fact that V is relatively schematically dense in $G \times_S G$.

It remains to treat case (iii). To prove $\bar{f}(a)\bar{f}(b) = \bar{f}(c)\bar{f}(d)$, one may make an fppf-covering base extension. Suppose that one has a section $x \in G(S')$, with $S' \rightarrow S$ fppf-covering, such that $(b, x), (a, bx), (d, x), (c, dx)$ are all in V . Then one will have

$$\begin{aligned} \bar{f}(a)\bar{f}(b)\bar{f}(x) &= \bar{f}(a)\bar{f}(bx) = \bar{f}(abx) = \bar{f}(cdx) \\ &= \bar{f}(c)\bar{f}(dx) = \bar{f}(c)\bar{f}(d)\bar{f}(x), \end{aligned}$$

whence the desired equality.

(7) To find such an x , set, for every $z \in U(S')$,

$$V_z = \text{pr}_2((z \times_{S'} U_{S'}) \cap V_{S'}).$$

Then the hypotheses on x mean that $x \in W$, where

$$W = V_b(S') \cap V_d(S') \cap b^{-1}V_a(S') \cap d^{-1}V_c(S').$$

By (iii), W is relatively schematically dense in G . Thus the existence of a section x after an fppf-covering extension follows from Proposition 1.7.

Let us verify that the morphism thus constructed is multiplicative: let $a, b \in G(S')$, with $S' \rightarrow S$ arbitrary, and write $S' = S$. Choose an fppf-covering base extension $S' \rightarrow S$ and a section $x \in G(S')$ such that $x \in U(S')$ and $ax^{-1} \in U(S')$, that is, $x \in (a^{-1}U)^{-1}(S')$. Choose moreover another fppf-covering extension $S'' \rightarrow S'$ and a section $y \in G(S'')$ such that

$$y, by^{-1}, aby^{-1} \text{ belong to } U(S''),$$

and

$$by^{-1} \in V_x, \quad xby^{-1} \in V_{ax^{-1}}^{-1}.$$

Then, by definition of f , one has over S''

$$\begin{aligned} f(a) &= \bar{f}(ax^{-1})\bar{f}(x), & f(b) &= \bar{f}(by^{-1})\bar{f}(y), \\ f(ab) &= \bar{f}(aby^{-1})\bar{f}(y). \end{aligned}$$

Moreover,

$$\begin{aligned} f(a)f(b) &= \bar{f}(ax^{-1})\bar{f}(x)\bar{f}(by^{-1})\bar{f}(y) \\ &= \bar{f}(ax^{-1})\bar{f}(xby^{-1})\bar{f}(y) \\ &= \bar{f}(aby^{-1})\bar{f}(y) = f(ab), \end{aligned}$$

whence multiplicativity.

(7) **N.D.E.** The beginning of the following sentence has been added.

The fact that f extends $\bar{f}$ is now easy: let $a \in U(S')$, write $S' = S$, and choose $S' \rightarrow S$ fppf-covering and sections $x, y \in G(S')$ such that $(x, y), (ax, y), (a, xy)$ are in $V(S')$. Then

$$\begin{aligned} f(xy) &= \bar{f}(x)\bar{f}(y) = \bar{f}(xy), \\ f(axy) &= \bar{f}(ax)\bar{f}(y) = \bar{f}(axy). \end{aligned}$$

Thus

$$\begin{aligned} f(a) &= f(axy)f((xy)^{-1}) = f(axy)f(xy)^{-1} \\ &= \bar{f}(axy)\bar{f}(xy)^{-1} = \bar{f}(a). \end{aligned}$$

□

Remark 2.3.1. In many cases hypothesis (iii) is true, for it will even be true if one replaces U by a smaller open subset U' , still relatively schematically dense in G . For example, one has:

Proposition 2.4. *With the situation as in 2.3, suppose that each geometric fiber of G/S is irreducible. Let $U' = \text{pr}_1 V$, and*

$$V' = V \cap m_G^{-1}(U') \cap (U' \times_S U').$$

Then $U' \subseteq U$ is a relatively schematically dense open subset of G , and the objects U' , $\bar{f}|_{U'}$, and V' satisfy hypothesis (iii).

PROOF. U' is open, since $G \times_S G \rightarrow G$ is flat and locally of finite presentation. All the other verifications are trivial, except for hypothesis (iii). Let $a \in U(S')$, and write $S' = S$. To verify that $\text{pr}_2((a \times U') \cap V')$ is relatively schematically dense in U' , it suffices to do so fiber by fiber by Corollary 1.3, that is, it suffices to treat the case where S is the spectrum of a field, and in this case it suffices to prove that U' is nonempty, since G is irreducible and “without embedded components” (cf. VIA, 1.1.1). Now

$$\begin{aligned} \text{pr}_2((a \times U') \cap V') &= \text{pr}_2((a \times U') \cap V) \\ &\quad \cap \text{pr}_2((a \times U') \cap m_G^{-1}(U')), \end{aligned}$$

and the second term of the right-hand member is dense in G . Thus it suffices to prove that $\text{pr}_2((a \times U') \cap V)$ is dense in G , that is, nonempty, which is clear since $a \in U' = \text{pr}_1 V$. □

3. Construction of a group from a rational law

3.0. — Suppose that a prescheme X/S and a rational map $X \times_S X \rightarrow X$ over S are given, and seek a group G/S and a map birational relative to S ⁽⁸⁾

$$X \dashrightarrow G$$

which commutes with the composition laws. We treat only the case where X/S satisfies the following hypothesis:

- ($\diamond$) X/S is faithfully flat, of finite presentation,
and has separated fibers “without embedded components”.

(Note that the last two hypotheses are properties true for a group prescheme.⁽⁹⁾)

We shall often suppress the symbol S in fibered products. Let X/S be a prescheme having properties ($\diamond$) above, and let W be a subprescheme of finite presentation of $X \times X \times X$ having the following property:

- (*) The three morphisms $W \rightarrow X \times X$ given by the projections of X^3 onto X^2
are open immersions schematically dense relative to S .

⁽⁸⁾**N.D.E.** Make explicit the notion of birational map (relative to S), perhaps in an addition 1.5.2?

⁽⁹⁾**N.D.E.** Cf. VIA, to be specified. . .

Notations. We shall use the following terminology: given sections $a, b, c \in X(S')$, with $S' \rightarrow S$ arbitrary, such that $(a, b, c) \in W(S')$, we write:

$$c = ab, \quad b = a^{-1}c, \quad a = cb^{-1}.$$

Definition 3.0.1. ⁽¹⁰⁾ We say, given a section $(a, b) \in X^2(S')$, that ab is defined if and only if there exists a section $c \in X(S')$ such that $(a, b, c) \in W(S')$, i.e. if and only if (a, b) is in $\text{pr}_{12}W(S')$. Similarly, saying that $a^{-1}b$ or ab^{-1} is defined has the analogous meaning, and one also extends this terminology to products of several factors.

Remark 3.0.2. Let us immediately note the following fact: by (i), W defines a rational map $X^2 \rightarrow X$ over S (that given by $(a, b) \mapsto ab$). It may well happen that this rational map has a domain of definition larger than $\text{pr}_{12}W$. Nevertheless, we say that ab is defined only if $(a, b) \in \text{pr}_{12}W(S')$.

Definition 3.1. A *group germ* is a prescheme X/S having properties $(\diamond)$ above, and a subprescheme W of $X \times X \times X$ of finite presentation having the following properties:

- (i) W satisfies property $(*)$ above.
- (ii) For every section $a \in X(S)$, the sets

$$\begin{aligned} \text{pr}_i((a \times X \times X) \cap W), & \quad i = 2, 3, \\ \text{pr}_i((X \times a \times X) \cap W), & \quad i = 3, 1, \\ \text{pr}_i((X \times X \times a) \cap W), & \quad i = 1, 2, \end{aligned}$$

are schematically dense in X relative to S , and this statement remains true after every base change $S' \rightarrow S$. (Hypothesis (i) implies that these sets are open subsets of X .) Intuitively, this means that “ ax is defined for sufficiently general x ”, etc.

(iii) The law is *associative*, that is, if $(a, b, c) \in X^3(S')$, with $S' \rightarrow S$ arbitrary, is such that $(ab)c$ and $a(bc)$ are defined, one has $(ab)c = a(bc)$.

Remarks. (a) In (i), one may replace the condition that W be schematically dense in X^2 relative to S by the condition that, for every $s \in S$, the fiber W_s be dense in X_s^2 in the topological sense, thanks to the hypotheses on X and to 1.2.

(b) Condition (iii) is equivalent to the following: let V_1 (resp. V_2) be the open subset of X^3 where the rational map $(a, b, c) \mapsto a(bc)$ (resp. $(a, b, c) \mapsto (ab)c$) is defined. ^(*) Then there exists an open subset $V \subseteq V_1 \cap V_2$ schematically dense in X^3 relative to S such that the two preceding maps agree on V . This is a consequence of 1.4, since, of course, V_1, V_2 are schematically dense in X^3 relative to S .

(c) Hypothesis (ii) will serve below to ensure that X will be a subobject of the group prescheme which it defines. In many cases, one can deduce (ii) from (i), provided that one replaces X by a relatively schematically dense open subset X' , and W by a relatively schematically dense open subset of $W \cap X'^3$. In fact, one has:

Proposition 3.2. *Suppose that each geometric fiber of X/S is irreducible, and let W be a subprescheme of X^3 satisfying condition (i) of 3.1. Then there exists a relatively schematically dense open subset X' of X , and a relatively schematically dense open subset W' of $W \cap (X' \times X')$, such that the pair (X', W') satisfies conditions (i) and (ii). If (iii) holds for (X, W) , it holds for (X', W') .*

PROOF. Set $X' = \bigcap_{i=1}^3 \text{pr}_i W$. Each $\text{pr}_i W$ is open in X , since $W \rightarrow X^2$ is an open immersion, and the projections $X^2 \rightarrow X$ are flat and of finite presentation. Moreover, $\text{pr}_i W$

⁽¹⁰⁾**N.D.E.** The numbering 3.0.1 and 3.0.2 has been introduced.

^(*)In the sense explained in 3.0.1; one could also replace these open subsets by the domains of definition (cf. 1.5) of the rational maps under consideration.

is relatively schematically dense in X , since W is so in X^2 , and the projections $X^2 \rightarrow X$ are surjective (it suffices to verify density in the topological sense). Take $W' = W \cap X'^3$.

To verify that (i) is true, note that

$$\begin{aligned} W' &= (W \cap (X' \times X' \times X)) \cap (W \cap (X \times X' \times X')) \\ &\quad \cap (W \cap (X' \times X \times X')). \end{aligned}$$

Now $W \cap (X' \times X' \times X) \simeq \text{pr}_{12}W \cap (X' \times X')$, hence is relatively schematically dense in W . Similarly, the other terms of the right-hand member are relatively schematically dense in W , and consequently W' is relatively schematically dense in W . Thus $\text{pr}_{ij}W'$ is relatively schematically dense in $\text{pr}_{ij}W$, hence in $X \times X$, hence in $X' \times X'$.

To verify condition (ii), let $a \in X'(S')$, and write $S' = S$. We must prove that, for example, $\text{pr}_2((a \times X' \times X') \cap W')$ is schematically dense in X' relative to S . By 1.3, it suffices to verify this fiber by fiber, that is, it suffices to treat the case where S is the spectrum of a field, and in this case it suffices to verify that the open subset is nonempty, since the fibers of X/S are irreducible and “without embedded components”. Since $\text{pr}_2((a \times X \times X) \cap W)$ is nonempty, a being a section of X' , this open subset is dense in X . One has

$$\text{pr}_2((a \times X \times X) \cap W) = \text{pr}_2((a \times X) \cap \text{pr}_{12}W).$$

Thus

$$\begin{aligned} \text{pr}_2((a \times X) \cap \text{pr}_{12}W) \cap X' &= \text{pr}_2((a \times X') \cap \text{pr}_{12}W) \\ &= \text{pr}_2((a \times X' \times X) \cap W) \end{aligned}$$

is dense in X , hence in X' . Similarly, $\text{pr}_3(a \times X \times X' \cap W)$ is dense in X , hence in $\text{pr}_3(a \times X \times X \cap W)$; that is, $(a \times X \times X' \cap W)$ is dense in $(a \times X \times X \cap W)$; that is, $\text{pr}_2(a \times X \times X' \cap W)$ is dense in $\text{pr}_2(a \times X \times X \cap W)$, hence dense in X , hence dense in X' . Now, since

$$\begin{aligned} \text{pr}_2(a \times X' \times X' \cap W') &= \text{pr}_2(a \times X' \times X' \cap W) \\ &= \text{pr}_2(a \times X' \times X \cap W) \cap \text{pr}_2(a \times X \times X' \cap W), \end{aligned}$$

it is indeed dense in X' , which is what had to be proved. The other assertions of (ii) follow by symmetry, and the fact that condition (iii) is preserved is trivial. ⁽¹¹⁾ Proposition 3.2 is proved. $\square$

(12)

3.2.1. — Now fix a group germ (X, W) over S . We must make some preliminary remarks on the situation, which we have gathered below. We shall often use these rules without explicit mention in what follows.

Let $a \in X(S')$, and write $S' = S$. Then one obtains a (bi)rational map φ over S from X to itself by associating to a section x the section ax , if it is defined. By 3.1 (ii), the domain of definition of φ contains the relatively schematically dense open subset $\text{pr}_2((a \times X \times X) \cap W)$, and φ defines an isomorphism from this open subset onto the open subset (where $a^{-1}x$ is defined) $\text{pr}_3((a \times X \times X) \cap W)$. This remark is generalized in an evident manner in Rule 1.

Rule 1. Let $P = P(x, t_1, \dots, t_n)$ be a “product” of the symbols $x, t_1, \dots, t_n$, obtained recursively as follows: $P_0 = x$; P_{i+1} is one of the following expressions:

$$P_i t, \quad t P_i, \quad P_i^{-1} t, \quad t P_i^{-1}, \quad P_i t^{-1}, \quad t^{-1} P_i,$$

where t is one of the t_j ; $P_r = P$. Let $a_1, \dots, a_n \in X(S')$. Then there exists a relatively schematically dense open subset U of $X_{S'}$ such that the product $P(x, a_1, \dots, a_n)$ is defined (in the sense of Remark 3.0.2) for a section $x \in X(S'')$ if and only if $x \in U(S'')$, and the map

⁽¹¹⁾**N.D.E.** The following sentence has been added.

⁽¹²⁾**N.D.E.** The numbering 3.2.1, as well as 3.2.2, has been added.

$x \mapsto P(x, (a))$ gives an isomorphism from U onto another relatively schematically dense open subset of $X_{S'}$, denoted $P(U, (a))$.

Rule 2. Let $a, b \in X(S')$. Then:

- If ab is defined, so is $a^{-1}(ab)$, and $a^{-1}(ab) = b$.
- If $a^{-1}b$ is defined, so is $a(a^{-1}b)$, and $a(a^{-1}b) = b$.
- If ba^{-1} is defined, so is $(ba^{-1})a$, and $(ba^{-1})a = b$.

Rule 3. Let $a, b, b' \in X(S')$. If $ab = ab'$, if $ba = b'a$, if $a^{-1}b = a^{-1}b'$, or if $ba^{-1} = b'a^{-1}$, then $b = b'$. Here it is understood that the equality relation implies that both sides are defined.

Rule 4. Let $a, b, c \in X(S')$. Then, whenever both sides are defined, one has

$$a((ba)^{-1}c) = b^{-1}c.$$

Similarly:

$$\begin{aligned} (c(ab)^{-1})a &= cb^{-1}, & a^{-1}((ab^{-1})c) &= b^{-1}c, \\ (c(b^{-1}a))a^{-1} &= cb^{-1}. \end{aligned}$$

Rule 5. All the following associativity laws are true whenever both sides are defined:

$$\begin{aligned} (a^{-1}b)c &= a^{-1}(bc), & (ab^{-1})c &= a(b^{-1}c), & (ab)c^{-1} &= a(bc^{-1}), \\ (ab)^{-1}c &= b^{-1}(a^{-1}c), & (a^{-1}b)c^{-1} &= a^{-1}(bc^{-1}), \\ (ab^{-1})c^{-1} &= a(cb)^{-1}. \end{aligned}$$

3.2.2. — Verification of the rules. (1) is proved by evident induction on the length r of P , the case $r = 1$ being a direct consequence of (3.1) (ii).

(2) This is trivial by definition.

(3) Indeed, by Rule 2, for example in the first case, one has

$$b = a^{-1}(ab) = a^{-1}(ab') = b'.$$

(4) Let us verify, for example, the first relation: on the right-hand side, left multiplication by b is defined and gives c , by Rule 2. Suppose that it is also defined on the left-hand side. Then one will have

$$b(a((ba)^{-1}c)) = (ba)((ba)^{-1}c) = c.$$

Indeed, (ba) is defined by hypothesis, since it occurs in the expression. Thus the middle member is defined and equal to c by Rule 2, and equal to the left-hand member by associativity (cf. 3.1 (iii)). Thus Rule 3 implies that the desired equality is true if this multiplication by b is defined. Now fix a, b . Then Rule 1 implies that $b(a((ba)^{-1}c))$ is indeed defined for c “in” an open subset U of X relatively schematically dense; hence, in this relatively schematically dense open subset, the two rational maps $c \mapsto b^{-1}c$ and $c \mapsto a((ba)^{-1}c)$ are equal. By 1.4, they are equal on every common domain of definition, whence the desired result.

(5) This is the same kind of argument as the preceding one. For example, one verifies $(ab^{-1})c^{-1} = a(cb)^{-1}$ as follows: if right multiplication by c is defined on the right-hand side, one has equality by Rule 4. Since it suffices to verify such a formula on a relatively schematically dense open subset, one is reduced to the case where this multiplication is indeed defined.

3.2.3. — Now consider the relation R on $X \times X$ obtained by setting, for $a, b, a', b' \in X(S')$, $(a, b) \sim (a', b')$ modulo $R(S')$ if and only if there exists an $S' \rightarrow S$ covering for fppf, and a section $x \in X(S')$ such that $(xa)b, (xa')b'$ are defined and equal. Then R is an equivalence relation.

Indeed, this relation is evidently symmetric. By Rule 1, the product $(xa)b$ is defined if x is “in” a suitable relatively schematically dense open subset. Thus 1.7 asserts that there exists

an $S' \rightarrow S$ covering for fppf, and an $x \in X(S')$, such that $(xa)b$ is defined. The relation is therefore reflexive, and transitivity is a consequence of the following lemma:

Lemma 3.3. *Let x, y, a, b, a', b' be sections of $X(S')$ such that $(xa)b, (xa')b', (ya)b, (ya')b'$ are defined. If $(xa)b = (xa')b'$, then $(ya)b = (ya')b'$.*

Indeed, the lemma says that one can test $(a, b) \sim (a', b')$ with an arbitrary $x \in X(S')$, $S' \rightarrow S$ fppf-covering, such that both products are defined. Given $a, b, a', b', a'', b'' \in X(S)$, one may, by Rule 1, 1.1, and 1.7, find an fppf-covering extension $S' \rightarrow S$ and a section $x \in X(S')$ such that the three products in question are defined, whence transitivity.

PROOF OF THE LEMMA. To begin with, write formally:

$$\begin{aligned} (za)b &= (((zx^{-1})x)a)b \\ &= ((zx^{-1})(xa))b = (zx^{-1})((xa)b), \\ (za')b' &= (((zx^{-1})x)a')b' \\ &= ((zx^{-1})(xa'))b' = (zx^{-1})((xa')b'). \end{aligned}$$

One verifies that these equalities are indeed true if the members are defined, by the appropriate rules. ⁽¹³⁾ It follows that $(za)b = (za')b'$ if all these expressions are defined. Moreover, by Rule 1 and the hypotheses already made, these expressions are indeed defined if $z \in X(S'')$ is in $V(S'')$, where V is a certain relatively schematically dense open subset of X (we have taken $S' = S$). Thus the two rational maps from X to itself given by $z \mapsto (za)b$ and $z \mapsto (za')b'$ are equal, whence $(ya)b = (ya')b'$. $\square$

Lemma 3.4. *Consider the rational map $\varphi : X^3 \rightarrow X$ over S , defined by $(a, b, c) \mapsto c^{-1}(ab)$. Let U be the domain of definition of φ , and consider the graph Γ of the morphism $f : U \rightarrow X$ induced by φ , which is a subscheme of X^4 . Then a section $(a, b, c, d) \in X^4(S)$ is in $\Gamma(S)$ if and only if $(a, b) \sim (c, d)$.*

PROOF. First note that the rational map φ is the same as that given by the formula $(a, b, c) \mapsto (xc)^{-1}((xa)b)$ for an arbitrary section $x \in X(S)$. It amounts to the same to say that $c^{-1}(ab) = (xc)^{-1}((xa)b)$ whenever both sides are defined. We leave the verification to the reader.

Let us then show that the map φ is defined for a section $(a, b, c) \in X^3(S)$ if and only if there exists d with $(a, b) \simeq (c, d)$, and that then $d = \varphi(a, b, c)$. Indeed, suppose that one has $(a, b) \sim (c, d)$. To verify that the map φ is defined, one may make an fppf-covering base extension (Proposition 1.6), and may therefore suppose that there exists a section $x \in X(S')$ such that $(xa)b, (xc)d$ are defined and equal. It follows (Rule 2) that $(xc)^{-1}((xa)b)$ is defined and equal to d .

Conversely, suppose that the map φ is defined at the section (a, b, c) , and let $d = \varphi(a, b, c)$. Choose $S' \rightarrow S$ fppf-covering and a section $x \in X(S')$ such that $(xa)b, (xc)d$ are defined. We wish to show that they are equal. For this, it suffices to prove that the two rational maps over S from X to itself, given by $b \mapsto (xa)b$ and $b \mapsto (xc)\varphi(a, b, c)$, are the same, which follows from the remark in the first paragraph. $\square$

Proposition 3.5. *The equivalence relation $R \rightrightarrows X^2$ is representable, and is a flat relation of finite presentation, that is, the projections of R onto X^2 are flat morphisms of finite presentation.*

PROOF. One may suppose S affine. Since (X, W) is of finite presentation over S , one can descend the entire situation to an S of finite type over $\text{Spec } \mathbf{Z}$, hence noetherian. We may therefore suppose that S is noetherian. Then it is trivial that the graph Γ of (3.4) is of

⁽¹³⁾**N.D.E.** To be specified...

finite presentation over X^2 . The projection $\mathrm{pr}_{12}|_\Gamma$ is flat, since $\Gamma \xrightarrow{\sim} \mathrm{pr}_{123}\Gamma$, which is an open subset of X^3 , and the projection $\mathrm{pr}_{12} : X^3 \rightarrow X^2$ is flat because X is flat over S .

I say that Γ represents R . Note that there is something to prove, since the domain of definition of a rational map does not in general commute with base extensions. Let $S'' \rightarrow S$. What is clear, by (3.4) applied to S'' , is that $R(S'') \supset \Gamma(S'')$, since $\varphi \times_S S''$ is certainly defined on $U \times_S S''$. Thus let $(a, b, c, d) \in R(S'')$. One must prove that $(a, b, c, d) \in \Gamma(S'')$. The verification is made locally for, say, the étale topology. One may therefore suppose that S'' is strictly local, i.e. the spectrum of a henselian ring with separably closed residue field. Moreover, applying (1.6) and the customary sorites of passage to the limit, one is reduced to the case where S is strictly local and $S'' \rightarrow S$ is local. Suppose that one has a section $x \in X(S)$ such that over S'' the products $(xa)b, (xc)d$ are defined. This will imply that $(xc)^{-1}((xa)b)$ is defined and equal to d . Now there exists a relatively schematically dense open subset V of X^3 such that $(xc)^{-1}((xa)b)$ is defined if and only if $(a, b, c) \in V(S'')$, and one has $V \subseteq U$. Thus $(a, b, c, d) \in \Gamma(S'')$ if such an x exists. By (1.6), one may make an fppf-covering base extension $S' \rightarrow S$ to find such an x . Since S is strictly local, one can (*) find $S' \rightarrow S$ faithfully flat, local, and finite, and a section $x \in X(S')$ which “passes through” an arbitrary closed point of the closed fiber of X/S . Now saying that $(xa)b$ and $(xc)d$ are defined means that, over S'' , x is in a certain relatively schematically dense open subset, which is checked on the closed fiber of $X_{S''}$. Thus it works. $\square$

Now let G be the quotient of X^2 by R as a sheaf for the fppf topology. We shall define a composition law on G as follows: let $(g, g') \in G(S')$ be represented by a section $((a, b), (c, d))$ of $X^2 \times X^2(S'')$, with $S'' \rightarrow S'$ fppf-covering. Suppose moreover that X admits a section x over S'' such that $a(b(cx))$ and $x^{-1}d$ are defined, which is permitted by Rule 1 and (1.7), and call gg' the class in $G(S')$ represented by the section $(a(b(cx)), x^{-1}d)$ of $X^2(S'')$.

Let us verify that gg' does not depend on the choice of the section x and the representative $((a, b), (c, d))$: indeed, let $(a', b') \sim (a, b)$, $(c', d') \sim (c, d)$, and x be such that $a'(b'(c'x'))$ and $x'^{-1}d'$ are defined. We may suppose that all are sections over S'' . One must prove that

$$(a(b(cx)), x^{-1}d) \sim (a'(b'(c'x')), x'^{-1}d'),$$

that is, for a suitable section $z \in X(S''')$, with $S''' \rightarrow S''$ suitably fppf-covering, one has

$$(z(a(b(cx))))(x^{-1}d) = (z(a'(b'(c'x'))))(x'^{-1}d').$$

Whenever all the products are defined, one has

$$\begin{aligned} (z(a(b(cx))))(x^{-1}d) &= ((za)(b(cx)))(x^{-1}d) \\ &= (((za)b)(cx))(x^{-1}d) \\ &= (((za)b)c)x(x^{-1}d) \\ &= (((za)b)c)(x(x^{-1}d)) = (((za)b)c)d, \end{aligned}$$

and the same identities hold with the primes. Now, by Rule 1 and (1.7), such a z exists. One must therefore prove that

$$(((za)b)c)d = (((za')b')c')d'.$$

But $(za)b = (za')b'$, since $(a, b) \sim (a', b')$ (3.3), and one therefore has the desired equality, since $(c, d) \sim (c', d')$.

Consider the natural morphism $i : X \rightarrow G$ defined as follows: for $a \in X(S')$, choose $S'' \rightarrow S'$ fppf-covering and a section $b \in X(S'')$ such that ab^{-1} is defined, and set

$$i(a) = \text{class in } G \text{ of } (ab^{-1}, b).$$

One easily verifies that this class, which is a priori in $G(S'')$, does not depend on the choice of b , and thus gives a well-defined element of $G(S')$. The reader will take pleasure in verifying the

(*)Combining Exp. VIA, 1.1.1 and EGA IV₄, 17.16.2.

Proposition 3.6. *The morphism i commutes with the composition laws of X and G , that is, if $a, b \in X(S')$ are sections such that ab is defined, one has $i(a)i(b) = i(ab)$.*

The aim of this number is the following theorem:

Theorem 3.7. ⁽¹⁴⁾ *Let (X, W) be a group germ over S , with X/S faithfully flat, of finite presentation, and with separated fibers without embedded components. Then, with the notation above, one has*

- (i) G is a sheaf of groups.
- (ii) $i : X \rightarrow G$ is represented by an open immersion.
- (iii) G is representable locally on S for the fppf topology.
- (iv) If G/S is representable, then it is a flat group of finite presentation, and $i : X \rightarrow G$ is schematically dense relative to S .

Note that G is evidently characterized by properties (i), $\dots$, (iv); one may therefore forget the explicit construction of G . The proof proceeds in several steps:

Lemma 3.8. (i) *Let c be a section of $X(S)$; then the morphism from the prescheme $c \times X$ (which is S -isomorphic to X) to G , given by $c \times X \hookrightarrow X \times X \rightarrow G$, is a monomorphism.*

(ii) *Let $\{c_i\}$, $i \in I$, be sections of $X(S)$, let $Z = \coprod_i c_i \times X$, and call $R' \rightrightarrows Z$ the equivalence relation induced on Z by the evident morphism $Z \rightarrow X^2$. Then R' is a “gluing” of the $c_i \times X \simeq X$.*

PROOF. (i) For two sections $(c, a), (c, a')$ of $(c \times X)(S')$ to have the same image in G , one must have $(c, a) \sim (c, a')$, that is, $(xc)a \sim (xc)a'$ for a suitable $x \in X(S'')$, with $S'' \rightarrow S'$ fppf-covering, whence $a = a'$ by Rule 3.

(ii) Let c_i, c_j be two sections, and consider the birational map ψ_{ji} from X to itself over S , given by the formula $x \mapsto c_j^{-1}(c_i x)$. This is the same map as that given by $x \mapsto (yc_j)^{-1}((yc_i)x)$, if $y \in X(S)$, as one easily sees. Moreover, one verifies that φ_{ji} is defined for a $b \in X(S')$ if and only if there exists b' such that $(c_i, b) \sim (c_j, b')$, and then $b' = \varphi_{ji}(b)$. Let U_{ji} be the domain of definition over S of φ_{ji} . It remains to prove that this domain of definition is universal, that is, if $b \in X(S'')$, with $S'' \rightarrow S$ arbitrary, and φ_{ji} is defined at b , then $b \in U_{ji}(S'')$. It amounts to the same to prove that if $b, b' \in X(S'')$ are such that $(c_i, b) \simeq (c_j, b')$, then $b \in U_{ji}(S'')$. We leave the verification of this fact, analogous to that of (3.5), to the reader. $\square$

Lemma 3.9. *Suppose that $\{c_i\}$, $i \in I$, are sections of $X(S)$ such that $\coprod_i c_i \times X \rightarrow G$ is surjective as a morphism of sheaves. Then G is representable and flat, of finite presentation over S , and the structural morphism $X^2 \rightarrow G$ is flat and of finite presentation.*

PROOF. The fact that G is representable is an immediate consequence of (3.8), and it follows that $\coprod_i c_i \times X \rightarrow G$ is an open covering. To prove that $X^2 \rightarrow G$ is flat, it suffices to do so locally, hence to prove that the rational map $X^2 \rightarrow c_i \times X$ induces a flat morphism on its domain of definition. Now this rational map is given by $(a, b) \mapsto (c_i, ((xc_i)^{-1}(xa))b)$, with x an arbitrary section, and, if it is defined at $(a, b) \in X^2(S')$, one can find $S'' \rightarrow S'$ fppf-covering and a section $x \in X(S'')$ such that $((xc_i)^{-1}(xa))b$ is defined. One easily sees that it is therefore a flat morphism. It follows similarly that it is locally of finite presentation, hence fppf-covering. Now, by construction, the relation R is effective. Thus, by (3.5), the morphism $X^2 \rightarrow G$ becomes of finite presentation after the base change $G \leftarrow X^2$, which is fppf-covering; hence $X^2 \rightarrow G$ is of finite presentation. Let us show that $G \rightarrow S$ is flat and of finite presentation. It is flat and locally of finite presentation, since G is covered by the $c_i \times X \simeq X$. Now X^2/S is quasi-compact, and $X^2 \rightarrow G$ is surjective. This proves that G/S is quasi-compact. To prove that $G \rightarrow S$ is quasi-separated, note that one has the following

⁽¹⁴⁾**N.D.E.** If X is smooth, separated over S , faithfully flat of finite presentation over S , then G/S is represented by a smooth S -group scheme of finite type over S . This is Theorem 6.6.1 of the book *Néron models* by Bosch–Lütkebohmert–Raynaud, Springer (1990).

cartesian diagram:

$$\begin{array}{ccc} R & \xrightarrow{\alpha} & X^2 \times_S X^2 \\ \gamma \downarrow & & \downarrow \beta \\ G & \xrightarrow[\Delta]{} & G \times_S G \end{array}$$

One has γ surjective and α, β quasi-compact, hence $\beta\alpha$ is quasi-compact, hence Δ is quasi-compact. ⁽¹⁵⁾ $\square$

Lemma 3.10. *Let $\{c_i\}$, $i \in I$, be sections of $X(S)$. For $\coprod_i c_i \times X \rightarrow G$ to be surjective as a morphism of fppf-sheaves, it suffices that the following condition hold:*

For every $S' \rightarrow S$ and $(a, b) \in X^2(S')$, there exists an open covering $\{S'_\nu\}$, $\nu \in N$, of S' , and a function $N \rightarrow I$, $\nu \mapsto i(\nu)$, such that $(c_{i(\nu)}^{-1}a)b$ is defined over S'_ν .

PROOF. Let $S'' \rightarrow S$ be arbitrary, and $g \in G(S'')$. Choose $S' \rightarrow S''$ fppf-covering, and a section $(a, b) \in X^2(S')$ representing g . Take the open covering $\{S'_\nu\}$ of S' which exists by the hypothesis of the lemma. Then, over each S'_ν , one has $(a, b) \sim (c_{i(\nu)}, (c_{i(\nu)}^{-1}a)b)$; hence g is represented by a section of $[c_{i(\nu)} \times X](S')$ over S'_ν , which proves surjectivity, since the family of morphisms $\{S'_\nu \rightarrow S''\}$ is fppf-covering. $\square$

Lemma 3.11. *Let Y/S be a prescheme of finite presentation, and let $\{a_i\}$, $i \in I$, be sections of $Y(S)$. Let s_0, s_1 be points of S such that s_0 is a specialization of s_1 , and Y_j the fiber of Y/S at s_j . Let C_j be the closure in Y_j of the set of points $\{a_i(s) \cap Y_j\}$. Then one has $\dim C_1 \geq \dim C_0$.*

PROOF. It suffices to make the verification after a base change $S' \rightarrow S$ with chosen points s'_0, s'_1 such that $s'_j \mapsto s_j$ and s'_0 is a specialization of s'_1 . One is therefore (EGA II 7.1.4) reduced to the case where S is the spectrum of a valuation ring A , s_0 the closed point of S , and s_1 the generic point of S . Now let V be the closure of C_1 in Y . It is clear that $C_0 \subseteq V$, and thus the lemma is a consequence of the “well-known” fact that an irreducible closed subscheme V of a prescheme Y/S of finite presentation satisfies $\dim(V \times s_1) \geq \dim(V \times s_0)$ if S is the spectrum of a valuation ring and $V \times s_1 \neq \emptyset$ (EGA IV 13.1.6). $\square$

Lemma 3.12. *Suppose that S is the spectrum of a local ring, with closed point s_0 , and let $\{c_i\}$, $i \in I$, be sections such that the closure C_0 of the set $\{c_i(s) \cap X_0\}$ in the closed fiber X_0 has dimension equal to $\dim X_0 = n$. Then the condition of Lemma 3.9 is satisfied.*

PROOF. First note that the fibers of X/S all have the same dimension n , which follows from EGA IV 12.1.1 (i), and the fact that X has a rational composition law. Lemma (3.11) therefore implies that, for every morphism $S_1 \rightarrow S$ with S_1 the spectrum of a field, the dimension of the closure of the set $\{c_i \times_S S_1\}$ in X_{S_1} is equal to n . Let us verify the condition of (3.10): let $(a, b) \in X^2(S')$. For $(c_i^{-1}a)b$ to be defined, it is necessary and sufficient that c_i be contained in a certain open subset $U \subseteq X_{S'}$ schematically dense relative to S' (Rule 1). One must prove that this is true for a suitable i , locally on S' . It therefore suffices to treat the case where S' is the spectrum of a local ring; then the fact that $c_i \in U(S')$ is checked on the closed fiber. One is therefore reduced to the case $S' = \text{Spec } k$, with k a field. Now, with the above notation, take $S' = S_1$. One has $\dim C_1 = \dim X_{S_1}$, and U is relatively schematically dense in X_{S_1} . Thus $U \cap C_1 \neq \emptyset$, whence $U \cap \{c_p \times_S S_1\} \neq \emptyset$, and one has won. $\square$

The proof of the theorem is now easy. First note the following consequence of the finiteness assertion of Lemma (3.9): if $\{A_i\}$ is an inductive system of rings over S , if $\underline{A} = \varinjlim A_i$, and if the hypotheses of (3.9) hold for $S = \text{Spec } \underline{A}$, then one can descend the object representing the

⁽¹⁵⁾**N.D.E.** Another way to conclude here is by faithfully flat descent (EGA IV₂, 2.7.1), since β is covering for fppf.

quotient G of $R \rightrightarrows X^2$ to one of the $S_i = \operatorname{Spec} A_i$, with the finiteness and flatness properties stated in (3.9). This is the customary passage to the limit (EGA IV 8 and 11). It follows that, for the proof of (iii) and (iv) of (3.7), one may restrict oneself to the case $S = \operatorname{Spec} A$, with A a strictly local ring. Then let x_0 be a closed point of the closed fiber X_0 of X/S . There exists ^(*) an extension A' of A , local, free and finite, and a section of $X' = X \times_S S'$ passing through the unique point x'_0 of X'_0 over x_0 . Note that $X'_0 \rightarrow X_0$ is radicial, since S is strictly local, hence the residue field of A is separably closed. It follows that there exists an inductive system $\{A_i\}$ of local rings, flat and finite over S , such that, setting

$$\underline{A} = \varinjlim A_i, \quad \underline{X} = X \times_S \operatorname{Spec} \underline{A},$$

$\underline{X}$ has a set of sections inducing a dense set on the closed fiber $\underline{X}_0$. For every closed point x_0 of X_0 , take an extension $A(x_0)$ such that the corresponding base extension of X admits a section “passing through x_0 ”, and take as inductive system the system of finite tensor products of the $A(x_0)$. Note that $\underline{A}$ is local, being a limit of local rings. Thus, by (3.12) and (3.10), one has the quotient G over $\operatorname{Spec} \underline{A}$, hence over one of the $\operatorname{Spec} A_i$ by the remarks above, hence locally for the fppf topology.

In fact, it follows from the constructions that locally for the fppf topology one can find a finite set of sections $\{c_i\}$, $i = 1, \dots, r$, such that G is covered by the $c_i \times X$. (ii) and (iv) follow easily from this fact, and we leave the verification of (i) to the reader.

Corollary 3.13. *The sheaf of groups G determined by a group germ (X, W) over S is representable in each of the following situations:*

- (i) S is artinian.
- (ii) For every local scheme $S' \rightarrow S$ of a closed point s of S , $X_{S'}$ has a set of sections inducing on the closed fiber a set whose closure has dimension $\dim X_s$.
- (iii) S is strictly local, and X/S is smooth.
- (iv) There exists $S' \rightarrow S$ fppf-covering such that $G_{S'}$ is representable and affine over S .

PROOF. Indeed, (ii) is a consequence of (3.10) and (3.12); (iii) follows directly from (ii) and “Hensel’s lemma”; (iv) from descent of affine schemes; and (i) from descent of group schemes, which is possible here since one knows that every finite subset of a group over a field is contained in an affine open subset (Exp. VI). $\square$

^(*)Cf. the note at the bottom of page 15, Exp. VII.

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Exposé XIX. Reductive groups—generalities

by *M. Demazure*

⁽⁰⁾The remainder of this Seminar is devoted to the study of reductive groups. Its principal aim is the generalization of the classical results of Chevalley (*Bible* and *Tôhoku*)⁽¹⁾ to arbitrary base schemes, the two central results being the *uniqueness theorem* (Exp. XXIII, th. 4.1 and cor. 5.1 to 5.10) and the *existence theorem* (Exp. XXV, th. 1.1) for “pinned” reductive group schemes corresponding to prescribed “root data”. The proofs used are inspired by those of Chevalley, the technique of schemes allowing them to be made more effective.

The results of the first volume of the *Bible* (Exp. 1 to 13) will be used systematically. On the other hand, we shall prove directly over an arbitrary scheme the results of the second volume (in particular, the isomorphism theorem); knowledge of the proofs over an algebraically closed field is therefore not absolutely indispensable.

In the proof of these two fundamental results, we shall use only the most elementary results of the theory of groups of multiplicative type, contained for the most part in Exp. VIII and IX; on the other hand, we shall make essential use of the results of Exposé XVIII.⁽²⁾ The reader especially interested in the existence and uniqueness theorems may, on a first reading, skip Exposés X to XVII.

1. Reminders on groups over an algebraically closed field

1.1. — *In this section, k will always denote an algebraically closed field.* As announced above, the only results of *Bible* used in what follows are in volume 1 (Exposés 1 to 13). All the results of *Bible* (loc. cit.) concerning *semisimple groups* are valid more generally for *reductive groups* (definition below), and their proof is identical, with the following harmless modifications:

- Exp. 9, § 4, definition 3: see 1.6.1 below.
- Exp. 12, § 4, theorem 2, e): delete “finite”.
- Exp. 13, § 3, theorem 2: replace “rank” by “semisimple rank”.
- Exp. 13, § 4, corollary 2 to theorem 3: replace “rank” by “reductive rank”.

1.2. — Let G be a smooth affine connected k -group. The *radical* of G (*Bible*, § 9.4, prop. 2)⁽³⁾ is the reduced subgroup associated with the identity component of the intersection of the Borel subgroups of G ; it is also the largest smooth connected solvable invariant subgroup of G ; we shall denote it by $\text{rad}(G)$.

The *unipotent radical* of G is the unipotent part of the radical of G ; it is also the largest smooth connected unipotent invariant subgroup of G ; we shall denote it by $\text{rad}^u(G)$.

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** See the bibliographical references at the end of this Exposé. In particular, a new edition of the Chevalley Seminar 1956–58, cited [Bible], revised by P. Cartier, was published in 2005, cf. [Ch05].

⁽²⁾**N.D.E.** More precisely, proposition XVIII 2.3 (extension of a “generic homomorphism” between groups) is used in XXII 4.1.11 and then in Exp. XXIII (proof of the uniqueness theorem), and also in Exp. XXIV; theorem XVIII 3.7 (construction of a group from a group germ) is used only in Exp. XXV.

⁽³⁾**N.D.E.** The reference 9-06 (= Exp. 9, p. 6) has been replaced by § 9.4 (= Exp. 9, § 4), which applies equally to [Bible] and [Ch05]. When the numbering of [Ch05] differs from [Bible], as is the case in Exp. 6, both references will be indicated explicitly.

1.3. — Let G be a smooth affine connected k -group, and Q a torus of G . Then the centralizer $\text{Centr}_G(Q)$ of Q in G is a closed subgroup of G (Exp. VIII, 6.7), smooth (Exp. XI, 2.4) and connected (cf. *Bible*, § 6.6, th. 6 a) or [Ch05], § 6.7, th. 7 a)).

One has the fundamental relation:

$$(1.3.1) \quad \text{rad}^u(\text{Centr}_G(Q)) = \text{rad}^u(G) \cap \text{Centr}_G(Q).$$

⁽⁴⁾First, $U = \text{rad}^u(G) \cap \text{Centr}_G(Q)$ is a unipotent invariant subgroup of $\text{Centr}_G(Q)$; let us show that it is smooth and connected. If one makes Q act on $\text{rad}^u(G)$ by inner automorphisms, U is precisely the scheme of invariants of this action. Now this is smooth and connected, by lemma 1.4 below.

Consequently, U is a closed subgroup of $\text{rad}^u(\text{Centr}_G(Q))$. On the other hand, by *Bible*, Exp. 12, § 3, cor. to th. 1, one has $\text{rad}^u(\text{Centr}_G(Q))(k) \subset \text{rad}^u(G)(k)$. Equality (1.3.1) follows.

Let us point out a particular case of the preceding result: if G is a smooth affine connected k -group and T is a maximal torus of G ,

$$(1.3.2) \quad \text{Centr}_G(T) = T \cdot (\text{Centr}_G(T) \cap \text{rad}^u(G)).$$

Indeed (cf. *Bible*, § 6.6, th. 6 c) or [Ch05], § 6.7, th. 7 c)), $\text{Centr}_G(T)$ is a solvable group, hence the semidirect product of its maximal torus T and its unipotent radical.

By the density theorem (cf. *Bible*, § 6.5, th. 5 a) or [Ch05], § 6.6, th. 6 a)), the union of the $\text{Centr}_G(T)$, as T runs through the set of maximal tori of G , contains a dense open subset of G ; it therefore follows from (1.3.2):

Corollary 1.3.3. *If G is a smooth affine connected k -group, the union of the $T \cdot \text{rad}^u(G)$, where T runs through the set of maximal tori of G , contains a dense open subset of G .*

Notation 1.4.0. ⁽⁵⁾We shall systematically use the following notation: if S is a scheme and s a point of S , $\bar{s}$ will denote the spectrum of an algebraic closure $\kappa(\bar{s})$ of $\kappa(s)$.

Lemma 1.4. ⁽⁵⁾Let S be a scheme, Q an S -torus acting on a group S -scheme H , separated and smooth over S .

- (i) *The functor of invariants H^Q is representable by a closed subscheme of H , smooth over S .*
- (ii) *If in addition H is affine over S , and has connected fibers, then so does H^Q .*

First, by VIII 6.5 (d), ⁽⁶⁾ since H is separated over S and Q essentially free over S , H^Q is representable by a closed subscheme of H . In particular, if H is affine over S , then so is H^Q .

Consider now the semidirect product $G = H \cdot Q$; it is smooth and separated over S . Then the centralizer $\text{Centr}_G(Q)$ is representable by a closed subscheme of G , of finite presentation over G , by Exp. XI 6.11 (a), ⁽⁶⁾ and it is smooth over S by Exp. XI 2.4.

Since $\text{Centr}_G(Q) = H^Q \cdot Q$, it follows that H^Q is smooth over S . (Indeed, using the section $\sigma : H^Q \rightarrow \text{Centr}_G(Q)$ deduced from the unit section ε_Q of Q , one sees at once that H^Q is formally smooth over S ; since moreover ε_Q is of finite presentation, H^Q is locally of finite presentation over S .) This proves (i).

Finally, suppose that H is affine over S and has connected fibers. Then each geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa(\bar{s})$ -group, hence, by the first assertion of 1.3, the same is true of

$$\text{Centr}_{G_{\bar{s}}}(Q_{\bar{s}}) = (\text{Centr}_G(Q))_{\bar{s}} = (H^Q)_{\bar{s}} \cdot Q_{\bar{s}},$$

and this implies that $(H^Q)_{\bar{s}}$ is connected.

⁽⁴⁾**N.D.E.** The following has been expanded.

⁽⁵⁾**N.D.E.** These notations, which appeared in 2.3 of the original, have been inserted here; on the other hand, the statement and proof of 1.4 have been expanded.

⁽⁶⁾**N.D.E.** See also the additions made in VI_B, 6.2.4 (d) and 6.5.5 (a).

1.5. — Recall that the *reductive rank* of the smooth affine k -group G is the common dimension of the maximal tori of G . We shall denote it by $\mathrm{rgred}(G/k)$ or $\mathrm{rgred}(G)$. For $\mathrm{rgred}(G/k) = 0$, it is necessary and sufficient that G be unipotent (i.e. that $G = \mathrm{rad}^u(G)$), by *Bible*, § 6.4, cor. 1 to th. 4 or [Ch05], § 6.5, cor. 1 to th. 5.

If H is an invariant subgroup of the smooth affine k -group G , the quotient G/H is affine and smooth (Exp. VI_B, 11.17 and VI_A, 3.3.2 (iii)). Moreover (*Bible*, § 7.3, th. 3, a) and c)), one has

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/H) + \mathrm{rgred}(H).$$

Definition 1.6.1. ⁽⁷⁾One says that the k -group G is *reductive* if it is smooth, affine and connected, and if $\mathrm{rad}(G)$ is a torus, that is, if $\mathrm{rad}^u(G) = \{e\}$.

Lemma 1.6.2. *Let G be a reductive k -group.*

- (i) *If Q is a torus of G , then $\mathrm{Centr}_G(Q)$ is reductive.*
- (ii) *In particular, if T is a maximal torus of G , then $\mathrm{Centr}_G(T) = T$.*
- (iii) *The center of G is contained in every maximal torus, hence is diagonalizable.*
- (iv) *The radical of G is the (unique) maximal torus of $\mathrm{Centr}(G)$.*

Indeed, (i) follows from (1.3.1), (ii) from (1.3.2), and (iii) follows from (ii). Finally, the maximal torus of $\mathrm{Centr}(G)$ (i.e. the identity component $\mathrm{Centr}(G)^0$) is a smooth solvable connected subgroup, invariant in G , hence contained in $\mathrm{rad}(G)$; conversely, $\mathrm{rad}(G)$ is a torus invariant in G , hence central (*Bible*, § 4.3, cor. to prop. 2), whence (iv).

Remark 1.6.3. If G is reductive and $\dim(G) \neq \mathrm{rgred}(G)$, then $\dim(G) - \mathrm{rgred}(G) \geq 2$. Indeed, this difference is always even (cf. 1.10 below).

1.7. — Let G be a smooth affine connected k -group and H a smooth connected invariant subgroup. Then

$$(1.7.1) \quad \mathrm{rad}(H) = (\mathrm{rad}(G) \cap H)_{\mathrm{red}}^0 \quad \text{and} \quad \mathrm{rad}^u(H) = (\mathrm{rad}^u(G) \cap H)_{\mathrm{red}}^0,$$

as one sees at once. In particular, if G is reductive, so is H .

Let $f : G \rightarrow G'$ be a surjective morphism of smooth affine connected k -groups. Then

$$(1.7.2) \quad f(\mathrm{rad}^u(G)) = \mathrm{rad}^u(G').$$

In particular, if G is reductive, so is G' .

Let us prove (1.7.2). First, f sends $\mathrm{rad}^u(G)$ into $\mathrm{rad}^u(G')$. Introducing

$$H = (f^{-1}(\mathrm{rad}^u(G'))_{\mathrm{red}})^0,$$

which contains $\mathrm{rad}^u(G)$, one has $\mathrm{rad}^u(H) = \mathrm{rad}^u(G)$ and one is reduced to the case where $G = H$, i.e. where G' is unipotent. Since the union of the $T \cdot \mathrm{rad}^u(G)$ (T running through the set of maximal tori of G) is dense in G , the union of the $f(T)f(\mathrm{rad}^u(G))$ is dense in G' ; but $f(T)$ consists of semisimple elements, hence $f(T) = \{e\}$, since G' is unipotent; this shows that $f(\mathrm{rad}^u(G))$ is dense in G' . Therefore, by *Bible*, § 5.4, lemma 4 or [Ch05], § 6.1, lemma 1,⁽⁸⁾ $f(\mathrm{rad}^u(G))$ is an open subgroup of G' ; since the latter is connected, it follows that $f(\mathrm{rad}^u(G)) = G'$.

(N. B.: under the same hypotheses one can prove that $f(\mathrm{rad}(G)) = \mathrm{rad}(G')$.)

1.8. — One says that the k -group G is *semisimple* if it is smooth, affine and connected and if $\mathrm{rad}(G) = \{e\}$, that is, if G is reductive and $\mathrm{Centr}(G)$ finite. If G is any smooth affine connected k -group, then $G/\mathrm{rad}(G)$ is semisimple (*Bible*, § 9.4, prop. 2), and $G/\mathrm{rad}^u(G)$ is reductive. One calls *semisimple rank* of G , and denotes by $\mathrm{rgss}(G/k)$ or $\mathrm{rgss}(G)$, the reductive rank of $G/\mathrm{rad}(G)$.

⁽⁷⁾**N.D.E.** Paragraph 1.6 has been changed into paragraphs 1.6.1 to 1.6.3. Note that in this Seminar every reductive (resp. semisimple, cf. 1.8) k -group is, by definition, connected.

⁽⁸⁾**N.D.E.** The following has been expanded.

If G is reductive, one therefore has

$$\operatorname{rgred}(G) = \operatorname{rgss}(G) + \dim(\operatorname{rad}(G)).$$

If G is a smooth affine connected k -group and Q is a central subtorus of G , then G/Q is semisimple if and only if G is reductive and $Q = \operatorname{rad}(G)$. Indeed, if G/Q is semisimple, $Q \supset \operatorname{rad}(G)$, hence $\operatorname{rad}(G)$ is a torus, hence G is reductive, hence $\operatorname{rad}(G)$ is the maximal torus of $\operatorname{Centr}(G)$, hence $\operatorname{rad}(G) = Q$. If G is reductive and Q is a central subgroup, then (G/Q is reductive and) $\operatorname{rgss}(G) = \operatorname{rgss}(G/Q)$.

1.9. — If K is an algebraically closed extension of k and G is a smooth affine connected k -group, G is reductive (resp. semisimple) if and only if G_K is, and one has

$$\operatorname{rgred}(G/k) = \operatorname{rgred}(G_K/K), \quad \operatorname{rgss}(G/k) = \operatorname{rgss}(G_K/K).$$

1.10. — Let G be a smooth connected k -group and T a torus of G .⁽⁹⁾ One denotes by $\mathfrak{g}$ the Lie k -algebra of G , i.e. $\mathfrak{g} = \operatorname{Lie}(G) = \operatorname{Lie}(G/k)(k)$; similarly, one puts $\mathfrak{t} = \operatorname{Lie}(T)$. Then $\mathfrak{g}$ decomposes under the action of T (by $T \rightarrow G$ and the adjoint action of G) into

$$(1.10.1) \quad \mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha,$$

where the $\alpha \in R$ are nontrivial characters of T and the $\mathfrak{g}^\alpha$ are $\neq 0$. The characters $\alpha \in R$ are called the *roots* of G with respect to T . By Exp. II, 5.2.3, (i), one has

$$(1.10.2) \quad \mathfrak{g}^0 = \operatorname{Lie}(\operatorname{Centr}_G(T)).$$

In particular,⁽¹⁰⁾ since $\operatorname{Centr}_G(T)$ is connected (1.3 when G is affine, and Exp. XII 6.6 b) in the general case), T is its own centralizer if and only if $\mathfrak{g}^0 = \operatorname{Lie}(T)$.

This condition implies that T is maximal and that $\operatorname{Centr}(G) \subset T$, hence by Exp. XII 8.8 d) one has:

$$(1.10.3) \quad \operatorname{Centr}(G) = \operatorname{Ker}(T \rightarrow \operatorname{GL}(\mathfrak{g})) = \bigcap_{\alpha \in R} \operatorname{Ker}(\alpha);$$

moreover, $\operatorname{Centr}(G)$ is then affine, hence $G \rightarrow G/\operatorname{Centr}(G)$ affine; since the latter group is affine (Exp. XII 6.1),⁽¹¹⁾ so is G .

When G is reductive and T maximal, the roots in the preceding sense coincide with the roots in the sense of *Bible*, § 12.2, def. 1; the latter are indeed roots in the sense of this paragraph (*Bible*, § 13.2, th. 1, c)) and there are $\dim(G) - \dim(T)$ of them (*Bible*, § 13.4, cor. 2 to th. 3). Moreover, if α is a root, so is $-\alpha$ (*Bible*, § 12.2, cor. to prop. 1). (As usual, one writes the group structure of $\operatorname{Hom}_{k\text{-gr.}}(T, \mathbf{G}_{m,k})$ indifferently additively or multiplicatively.) It follows that, for G reductive,

$$(1.10.4) \quad \dim(G) - \operatorname{rgred}(G) = \operatorname{Card}(R)$$

is always even.

The semisimple rank of the reductive group G is the rank of the subset R of the $\mathbf{Q}$ -vector space $\operatorname{Hom}_{k\text{-gr.}}(T, \mathbf{G}_{m,k}) \otimes \mathbf{Q}$ (*Bible*, § 13.3, th. 2).

Lemma 1.11. *Let k be an algebraically closed field, G a smooth affine connected k -group, T a torus of G , $W(T) = \operatorname{Norm}_G(T)/\operatorname{Centr}_G(T)$ the Weyl group of G with respect to T . The following conditions are equivalent:*

- (i) G is reductive, T is maximal, $\operatorname{rgss}(G) = 1$.
- (ii) G is reductive, T is maximal, $G \neq T$; there exists a subtorus Q of T , of codimension 1 in T , central in G .
- (iii) G is not solvable and $\dim(G) - \dim(T) \leq 2$.

⁽⁹⁾**N.D.E.** The following sentence has been added.

⁽¹⁰⁾**N.D.E.** The following has been expanded.

⁽¹¹⁾**N.D.E.** See also the addition VI_B, 6.2.6.

- (iv) $W(T) \neq \{e\}$ and $\dim(G) - \dim(T) \leq 2$.
- (v) $W(T) = (\mathbf{Z}/2\mathbf{Z})_k$ and $\dim(G) - \dim(T) = 2$.

Moreover, under these conditions, there are exactly two roots of G with respect to T ; they are opposite. Under the conditions of (ii), $Q = \text{rad}(G)$.

One clearly has (v) $\Rightarrow$ (iv). One has (iv) $\Rightarrow$ (iii) by *Bible*, § 6.1, cor. 3 to th. 1 or [Ch05], § 6.2, cor. 3 to th. 2. Let us prove (iii) $\Rightarrow$ (ii): let U be the unipotent radical of G ; one knows that G/U is reductive and is not a torus (otherwise G would be solvable); hence, by (1.10.4), one has

$$\dim(G/U) - \text{rgred}(G/U) = \text{Card}(R) \geq 2;$$

but

$$\text{rgred}(G) = \text{rgred}(G/U) + \text{rgred}(U) = \text{rgred}(G/U),$$

whence

$$\begin{aligned} \dim(G) - \dim(U) - \text{rgred}(G) &= \dim(G/U) - \text{rgred}(G/U) \\ &\geq 2 \geq \dim(G) - \dim(T) \geq \dim(G) - \text{rgred}(G). \end{aligned}$$

This gives $\dim(U) = 0$, hence G is reductive, $\text{rgred}(G) = \dim(T)$, hence T is maximal, $\dim(G) - \dim(T) = 2$. By *Bible*, § 10.4, prop. 8, there exists a singular torus Q of codimension 1 in T ; then $\text{Centr}_G(Q)$ is reductive (1.6.2 (i)), not solvable (definition of a singular torus), hence $\dim(\text{Centr}_G(Q)) - \text{rgred}(G) \geq 2$, which proves that $G = \text{Centr}_G(Q)$ and Q is central in G .

Let us prove (ii) $\Rightarrow$ (i). One knows that G/Q is reductive (1.7) and that $\text{rgred}(G/Q) = 1$, hence $\text{rgss}(G/Q) = 0$ or 1. The first case is impossible, for it would imply $\text{rgss}(G) = 0$, hence $G = T$; one therefore has $\text{rgred}(G/Q) = \text{rgss}(G/Q) = 1$, which proves that G/Q is semisimple; hence Q is the radical of G and $\text{rgss}(G) = 1$. Let us finally prove (i) $\Rightarrow$ (v). If Q is the radical of G , one has $\dim(T) - \dim(Q) = 1$ and Q is central in G , hence $G = \text{Centr}_G(Q)$, which proves that Q is a singular torus; by *Bible*, § 11.3, th. 2, one has $W_G(T) = (\mathbf{Z}/2\mathbf{Z})_k$; by *Bible*, § 12.1, lemma 1, one has $\dim(G) - \dim(T) = 2$. There are therefore at most two roots of G with respect to T , whereas there are at least two, opposite (1.10).

Proposition 1.12. *Let k be an algebraically closed field, G a smooth connected k -group, T a torus of G , R the set of roots of G with respect to T , and*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha, \quad \text{with } \mathfrak{g}^\alpha \neq 0,$$

the decomposition of $\mathfrak{g}$ under $\text{Ad}(T)$. For each $\alpha \in R$, let T_α be the maximal torus of $\text{Ker}(\alpha)$,⁽¹²⁾ and $Z_\alpha = \text{Centr}_G(T_\alpha)$. The following conditions are equivalent:

- (i) G is affine, reductive; T is maximal.
- (ii) $\mathfrak{g}^0 = \mathfrak{t}$, each Z_α ($\alpha \in R$) is reductive.
- (iii) $\mathfrak{g}^0 = \mathfrak{t}$, each $\mathfrak{g}^\alpha$ ($\alpha \in R$) has dimension 1; and if $\alpha, \beta \in R$ and $q \in \mathbf{Q}$ are such that $\beta = q\alpha$, then $q = \pm 1$; for each $\alpha \in R$, there exists a $w_\alpha \in G(k)$ which normalizes T , centralizes T_α , but does not centralize T .

Moreover, under these conditions, each Z_α has semisimple rank 1 and one has $\text{Lie}(Z_\alpha) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}$.

If G is affine and reductive and T is maximal, each Z_α is affine and reductive (1.6.2 (i)), with maximal torus T ; moreover, T is its own centralizer (1.6.2 (ii)), hence $\mathfrak{g}^0 = \mathfrak{t}$, which proves (i) $\Rightarrow$ (ii).

On the other hand, $\mathfrak{g}^0 = \mathfrak{t}$ implies that T is maximal and G affine (cf. 1.10), hence each Z_α is affine, smooth and connected, by 1.3. In any case, by Exp. II, 5.2.2, one has

$$\text{Lie}(Z_\alpha) = \mathfrak{g}^{T_\alpha} = \mathfrak{g}^0 \oplus \coprod_{\beta \in R \cap \mathbf{Q}\alpha} \mathfrak{g}^\beta.$$

⁽¹²⁾**N.D.E.** I.e. the connected component of $\text{Ker}(\alpha)$.

One therefore has

$$\mathrm{Lie}(Z_\alpha) \supset \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

which implies $Z_\alpha \neq T$. Since T_α is a subtorus of codimension 1 in T , central in Z_α , one obtains by 1.11, applied to Z_α , the equivalence (ii) $\Leftrightarrow$ (iii) and the supplementary assertions.

It remains to prove (ii) $\Rightarrow$ (i); one already knows that (ii) implies that T is maximal and G affine. Let U be its unipotent radical; it is invariant in G , its Lie algebra is invariant under $\mathrm{Ad}(T)$. One therefore has

$$\mathrm{Lie}(U) = (\mathrm{Lie}(U) \cap \mathfrak{g}^0) + \coprod_{\alpha \in R} (\mathrm{Lie}(U) \cap \mathfrak{g}^\alpha).$$

By (1.3.1), one has

$$\begin{aligned} U \cap T &= U \cap \mathrm{Centr}_G(T) = \mathrm{rad}^u(\mathrm{Centr}_G(T)) = \mathrm{rad}^u(T) = \{e\}, \\ U \cap Z_\alpha &= U \cap \mathrm{Centr}_G(T_\alpha) = \mathrm{rad}^u(\mathrm{Centr}_G(T_\alpha)) = \mathrm{rad}^u(Z_\alpha) = \{e\}. \end{aligned}$$

One therefore has⁽¹³⁾

$$\begin{aligned} \mathrm{Lie}(U) \cap \mathfrak{g}^0 &= \mathrm{Lie}(U) \cap \mathfrak{t} = \mathrm{Lie}(U \cap T) = 0, \\ \mathrm{Lie}(U) \cap \mathfrak{g}^\alpha &\subset \mathrm{Lie}(U) \cap \mathrm{Lie}(Z_\alpha) = \mathrm{Lie}(U \cap Z_\alpha) = 0; \end{aligned}$$

whence $\mathrm{Lie}(U) = 0$ and $U = \{e\}$, i.e. G is reductive.

Corollary 1.13. *Let k be an algebraically closed field, G a smooth connected k -group, T a torus of G , R the set of roots of G with respect to T , and*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha$$

as above. For each $\alpha \in R$, let T_α be the maximal torus of $\mathrm{Ker}(\alpha)$ and $Z_\alpha = \mathrm{Centr}_G(T_\alpha)$. The following conditions are equivalent:

- (i) *G is affine, semisimple; T is maximal.*
- (ii) *$\mathfrak{g}^0 = \mathfrak{t}$, each Z_α is reductive and $\bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$ is finite.*

⁽¹⁴⁾This follows from the equalities $\mathrm{rad}(G) = \mathrm{Centr}(G)^0$ and $\mathrm{Centr}(G) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$, established in 1.6.2 (iv) and (1.10.3).

2. Reductive group schemes. Definitions and first properties

Scholium 2.1. *If G is a group scheme over S , the following properties are equivalent:*

- (i) *G is affine and smooth over S , and has connected fibers.*
- (ii) *G is affine and flat over S , of finite presentation over S , and has geometrically integral fibers.*

These properties are stable under base change and local for (fpqc).

⁽¹⁵⁾Indeed, suppose (i) holds. Since G is affine and smooth over S , it is of finite presentation over S ; and since its fibers are smooth and connected, they are geometrically integral, by VI_A, 2.4.

Conversely, if (ii) holds, the fibers of G are connected and geometrically reduced, hence smooth (VI_A, 1.3.1); then G is smooth over S , by EGA IV₄, 17.5.2.

Of course, these properties are stable under base change: cf. EGA II, 1.5.1 for “affine”, IV₁, 1.6.2 (iii) for “of finite presentation”, IV₂, 2.1.4 for “flat”, and IV₂, 4.6.5 (i) for “with geometrically integral fibers”.

⁽¹³⁾**N.D.E.** Note that if U, L are two subgroup schemes of G , one has $\mathrm{Lie}(U) \cap \mathrm{Lie}(L) = \mathrm{Lie}(U \cap L)$.

⁽¹⁴⁾**N.D.E.** The following sentence has been added.

⁽¹⁵⁾**N.D.E.** The following has been added.

On the other hand, these properties are clearly local for the Zariski topology, so it suffices to check that if $S' \rightarrow S$ is a faithfully flat quasi-compact morphism and $G_{S'} \rightarrow S'$ has the indicated properties, the same is true of $G \rightarrow S$. This follows from EGA IV₂, 2.5.1 for “flat”, 2.7.1 (vi) and (xiii) for “of finite presentation” and “affine”, and 4.6.5 (i) for “with geometrically integral fibers” (and also EGA IV₄, 17.7.3 (ii) for “smooth”).

2.2. — Let G be a group S -scheme satisfying the preceding conditions, and Q a torus (cf. IX, Def. 1.3) of G .⁽¹⁶⁾ Then, by XI, 6.11 a) and XI, 2.4, $\text{Centr}_G(Q)$ is representable by a closed subgroup scheme of G (hence affine over S), of finite presentation and smooth over S ; moreover, since each geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa(\bar{s})$ -group, by the first assertion of 1.3 the same is true of

$$\text{Centr}_{G_{\bar{s}}}(Q_{\bar{s}}) = (\text{Centr}_G(Q))_{\bar{s}}.$$

Lemma 2.3. *Let S be a scheme, G a group S -scheme smooth and affine over S , with connected fibers, and T a torus of G . The set of $s \in S$ such that $G_{\bar{s}}$ is a reductive $\bar{s}$ -group, of semisimple rank 1 and with maximal torus $T_{\bar{s}}$, is an open subset U of S .*

Since G and T are flat over S , the function

$$s \mapsto \dim(G_{\bar{s}}/\bar{s}) - \dim(T_{\bar{s}}/\bar{s})$$

is locally constant; let U_1 be the open subset of points of S where its value is 2.

⁽¹⁷⁾By 6.3, the Weyl group

$$W_G(T) = \text{Norm}_G(T)/\text{Centr}_G(T)$$

is representable by a group S -scheme étale and separated over S , and the function

$$s \mapsto \text{number of points of } W_G(T)_{\bar{s}}$$

is lower semicontinuous. Let U_2 be the set of points of S where this function is > 1 ; it is open. By 1.11, the set of s such that $G_{\bar{s}}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}}$, is $U = U_1 \cap U_2$; moreover, for every $s \in U$, $W_G(T)_{\bar{s}}$ has exactly two points.

Consequently (cf. SGA 1, I 10.9 and EGA IV₃, 15.5.1 and IV₄, 18.12.4), $W_G(T)_U$ is étale and finite over U .

Remark 2.4. The group $W_G(T)_U$ is isomorphic to $(\mathbf{Z}/2\mathbf{Z})_U$. Indeed, by what precedes, it is étale and finite over U ; since the functor of automorphisms of $(\mathbf{Z}/2\mathbf{Z})_U$ is the unit group, it suffices to check the assertion locally, and it is immediate if the algebra $\mathcal{A}$ defining $W_G(T)_U$ is a free $\mathcal{O}_U$ -module.⁽¹⁸⁾

Notations and reminders 2.5.0. ⁽¹⁹⁾Let S be a scheme, G a group S -scheme, $\varepsilon : S \rightarrow G$ the unit section of G . We saw in II, § 4.11 that the functor $\text{Lie}(G/S)$ is representable by the vector bundle $V(\omega_{G/S}^1)$ (where $\omega_{G/S}^1 = \varepsilon^*(\Omega_{G/S}^1)$), and one denotes by

$$\mathfrak{g} = \text{Lie}(G/S) = (\omega_{G/S}^1)^\vee$$

the sheaf of sections of this vector bundle. Suppose in addition that G is smooth over S ; then $\omega_{G/S}^1$ and hence $\text{Lie}(G/S)$ are locally free $\mathcal{O}_S$ -modules of finite type, and one has (cf. I

⁽¹⁶⁾**N.D.E.** The following has been expanded.

⁽¹⁷⁾**N.D.E.** The following has been expanded. Note, on the other hand, that section 6 is independent of the rest of this Exposé.

⁽¹⁸⁾**N.D.E.** The hypothesis implies that $\mathcal{A}$ is locally free of rank 2, and since the augmentation ideal $\mathcal{I}$ is a direct summand of $\mathcal{A}$, replacing U by a sufficiently small affine open subset $\text{Spec}(R)$, one may suppose that $I = \Gamma(U, \mathcal{I})$ is a free R -module of rank 1. If x is a generator of I , one then has $x^2 = ax$ for some $a \in R$, and the hypothesis that $A = \Gamma(U, \mathcal{A})$ is étale implies that a is invertible in R , and one then easily sees that A is the affine algebra of the constant R -group $\mathbf{Z}/2\mathbf{Z}$ (compare with the first lines of [TO70]).

⁽¹⁹⁾**N.D.E.** This paragraph of notations and reminders has been added.

4.6.5.1):

$$\underline{\mathrm{Lie}}(G/S) = \mathrm{W}(\mathcal{L}ie(G/S)),$$

i.e. for every S -scheme S' ,

$$\underline{\mathrm{Lie}}(G/S)(S') = \Gamma(S', \mathcal{L}ie(G/S) \otimes \mathcal{O}_{S'}).$$

By II 4.1.1.1, the adjoint action of G equips $\underline{\mathrm{Lie}}(G/S) = \mathrm{W}(\mathcal{L}ie(G/S))$ with a structure of G - $\mathcal{O}_S$ -module, hence $\mathcal{L}ie(G/S)$ is a G - $\mathcal{O}_S$ -module (cf. I 4.7.1). If in addition G is *affine* over S , this amounts to saying, by I 4.7.2, that $\mathcal{L}ie(G/S)$ is an $\mathcal{A}(G)$ -comodule.

If T is a torus over S (IX Def. 1.3), one will say that T is *split* (“trivial” in the original) if it is isomorphic to $\mathbf{G}_{m,S}^r$ for some integer $r \geq 0$, and one will say that T is *trivialized* if one has fixed such an isomorphism or, more generally, an isomorphism $T \simeq \mathrm{D}_S(M)$, where M is a free abelian group of rank r .

Finally, recall (XII Def. 1.3) that a torus T of G is said to be *maximal* if, for every $s \in S$, $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Theorem 2.5. *Let S be a scheme, G a group S -scheme affine and of finite presentation over S , with connected fibers, and s_0 a point of S . Suppose that G is flat over S at $\varepsilon(s_0)$ and that the geometric fiber $G_{\bar{s}_0}$ is (reduced and) reductive (resp. semisimple). Then there exist an open subset U of S containing s_0 , and a surjective étale morphism $S' \rightarrow U$, such that:*

- (i) G_U is smooth over S , with reductive (resp. semisimple) geometric fibers, of constant reductive rank and semisimple rank.
- (ii) $G_{S'}$ has a split maximal torus⁽²⁰⁾ T , and the Weyl group (cf. 6.3)

$$W_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T) / \underline{\mathrm{Centr}}_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T) / T$$

is finite over S' .

⁽²¹⁾Let $\pi : G \rightarrow S$ be the structural morphism and $\varepsilon : S \rightarrow G$ the unit section. Since G is flat over S at $\varepsilon(s_0)$ and $G_{\bar{s}_0}$ reduced (hence smooth over $\bar{s}_0$, cf. VI_A 1.3.1), G is smooth over S at the point $\varepsilon(s_0)$ (EGA IV₄, 17.5.1), i.e. there exists an open neighborhood V of $\varepsilon(s_0)$ such that $\pi|_V$ is smooth. Then $S' = \varepsilon^{-1}(V)$ is an open subset of S , and $G_{S'}$ is smooth over S' at the points of $\varepsilon(S')$. Since G has connected fibers, $G_{S'}$ is smooth over S' , by VI_B, 3.10. Thus, replacing S by S' , one may suppose that G is smooth over S .

By Exp. XI, th. 4.1, the functor of subgroups of multiplicative type of G is representable by an S -scheme $\mathcal{M}$, smooth and separated over S . Let r_0 be the rank of $G_{\bar{s}_0}$ and consider the open subscheme $\mathcal{M}_{r_0}$ of $\mathcal{M}$ which represents the functor of subtori of rank r_0 of G (IX 1.4). Smoothness implies that $\mathcal{M}_{r_0}$ has a rational point over a finite separable extension of $\kappa(s_0)$ (cf. EGA IV₄, 17.15.10 (iii)). Thus there exists an étale $S' \rightarrow S$ equipped with a point s'_0 mapping to s_0 such that $G_{s'_0}$ has a torus of rank r_0 . Therefore, replacing S by S' , one may suppose that G_{s_0} has a torus of rank r_0 .

By “Hensel’s lemma” (cf. XI, 1.10), the smoothness of $\mathcal{M}_{r_0}$ allows one to lift this torus to an S' -torus T of G , where $S' \rightarrow S$ is étale and equipped with a point s'_0 mapping to s_0 . By Exp. X, 4.5 (see also 6.1,⁽²²⁾), there exists an étale morphism $f : S'' \rightarrow S'$ splitting T and such that $f^{-1}(s'_0) \neq \emptyset$. Since an étale morphism is open and the assertions of (i) are local for the étale topology, one may therefore suppose that G has a split torus T ,⁽²³⁾ maximal at s_0 . Let us therefore write $T = \mathrm{D}_S(M)$ and let

$$\mathfrak{g} = \coprod_{m \in M} \mathfrak{g}^m$$

be the decomposition of $\mathfrak{g} = \mathcal{L}ie(G/S)$ under $\mathrm{Ad}(T)$ (Exp. I, 4.7.3).

⁽²⁰⁾**N.D.E.** “Split” has been added.

⁽²¹⁾**N.D.E.** The proof has been expanded.

⁽²²⁾**N.D.E.** Which is independent of the rest of this exposé.

⁽²³⁾**N.D.E.** Throughout this volume, “trivial torus” has been replaced by “split torus”.

One puts $\mathfrak{t} = \mathcal{L}ie(T)$ and, for every $m \in M$, denotes by $\mathfrak{g}^m(s_0) = \mathfrak{g}^m \otimes_{\mathcal{O}_S} \kappa(s_0)$.⁽²⁴⁾ Let R be the set of $m \in M$, $m \neq 0$, such that $\mathfrak{g}^m(s_0) \neq 0$.⁽²⁵⁾ Since $G_{\bar{s}_0}$ is reductive, one has $\mathfrak{g}^0(s_0) = \mathfrak{t}(s_0)$, hence

$$\mathfrak{g}(s_0) = \mathfrak{t}(s_0) \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha(s_0).$$

Since the modules in question are locally free, one may, restricting S if necessary, suppose that the $\mathfrak{g}^\alpha$ are free and

$$\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha.$$

Recall, cf. Exp. XII, 1.12, that a group of multiplicative type has a unique maximal torus (this is, moreover, almost trivial by descent, the diagonalizable case being obvious). Let then T_α be the maximal torus of the kernel of α and $Z_\alpha = \underline{\text{Centr}}_G(T_\alpha)$.⁽²⁶⁾ By 2.2, Z_α is affine and smooth over S , with connected fibers, and by 1.12 its fiber $(Z_\alpha)_{\bar{s}_0}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}_0}$. By 2.3, there therefore exists an open subset U_α of S containing s_0 , such that $Z_\alpha|_{U_\alpha}$ has reductive fibers.

Put $U = \bigcap_{\alpha \in R} U_\alpha$. By 1.12, for every $s \in U$, $G_{\bar{s}}$ is reductive, with maximal torus $T_{\bar{s}}$, and the set of roots of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ is identified with R , so that

$$\text{rgred}(G_{\bar{s}}) = \dim(T) = \text{rg}(M), \quad \text{rgss}(G_{\bar{s}}) = \text{rg}(R) \quad (\text{cf. 1.10}).$$

One has therefore proved (i) and the first assertion of (ii); it remains to prove that the Weyl group $W_{G_U}(T_U)$ is *finite* over U , i.e. “that it has the same number of points in each geometric fiber” (cf. SGA 1, I 10.9 and EGA IV₃, 15.5.1 and IV₄, 8.12.4).

For this, it suffices to note that the geometric fiber of this group at $s \in U$ is determined by the situation $R \subset M$, as the constant group associated with the “abstract Weyl group of this root system”, and in particular is independent of the point s , cf. *Bible*, § 11.3, th. 2 (see also Exp. XXII, no. 3).

Corollary 2.6. *Let G be an S -group affine and smooth over S , with connected fibers. The set of points $s \in G$ such that $G_{\bar{s}}$ is reductive (resp. semisimple) is an open subset U of S , and the functions*

$$s \longmapsto \text{rgred}(G_{\bar{s}}/\bar{s}), \quad s \longmapsto \text{rgss}(G_{\bar{s}}/\bar{s})$$

are locally constant on U .

Definition 2.7. An S -group (= group S -scheme) G is said to be *reductive* (resp. *semisimple*) if it is affine and smooth over S , with connected, reductive (resp. semisimple) geometric fibers.

The property of being reductive (resp. semisimple) for an S -group G is stable under base change and local for the (fpqc) topology.

2.8. — Let G be a reductive S -group. For every torus (resp. maximal torus) Q of G , $Z(Q) = \underline{\text{Centr}}_G(Q)$ is reductive (resp. is Q). This follows at once from 2.2 and 1.6. Applying 2.5 to $\underline{\text{Centr}}_G(Q)$, one deduces that Q is contained (locally for the étale topology) in a maximal torus.

Remark 2.9. Using, as usual, the technique of EGA IV₃, § 8, the hypotheses of finite presentation, and theorem 2.5, one sees that if G is a reductive group over S , there exists an open covering $\{U_i\}$ of S such that each G_{U_i} comes by base change from a reductive group over a noetherian ring (in fact, an algebra of finite type over $\mathbf{Z}$). Similarly, if G has a split maximal torus T , one may in addition suppose that T_{U_i} comes from a split maximal torus of the preceding group,

⁽²⁴⁾**N.D.E.** The preceding sentence has been added.

⁽²⁵⁾**N.D.E.** Note that, since $\mathfrak{g}$ is a finite locally free $\mathcal{O}_S$ -module, R is a finite set.

⁽²⁶⁾**N.D.E.** The following sentence has been expanded.

3. Roots and root systems of reductive group schemes

3.1. — Let S be a scheme, T an S -torus acting linearly on a locally free $\mathcal{O}_S$ -module $\mathcal{F}$ of finite type (cf. I, § 4.7). For every character α of T (i.e. $\alpha \in \text{Hom}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$), one defines a subfunctor of $W(\mathcal{F})$ by

$$W(\mathcal{F})^\alpha(S') = \{x \in W(\mathcal{F})(S') \mid tx = \alpha(t)x \text{ for every } t \in T(S''), S'' \longrightarrow S'\}.$$

Lemma. ⁽²⁷⁾ Then $W(\mathcal{F})^\alpha = W(\mathcal{F}^\alpha)$, where $\mathcal{F}^\alpha$ is a submodule of $\mathcal{F}$, locally a direct summand of $\mathcal{F}$, hence also locally free.

Indeed, the assertion is local for the (fpqc) topology, and one may suppose $T = D_S(M)$, where M is a finitely generated (free) abelian group. Then α is identified with a locally constant function from S to M (Exp. VIII 1.3), and, restricting S if necessary, one may suppose that this function is constant. One is then reduced to Exp. I, 4.7.3.

Definition 3.2. Let S be a scheme, G a group S -scheme smooth with connected fibers,⁽²⁸⁾ and T a torus of G . One puts $\mathfrak{g} = \text{Lie}(G/S)$ and makes T act on $\mathfrak{g}$ through the adjoint representation of G .

One says that the character α of T is a *root* of G with respect to T if the following equivalent conditions are satisfied:

- (i) For each $s \in S$, $\alpha_{\bar{s}}$ is a root of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ (1.10).
- (ii) α is nontrivial on each fiber and $\mathfrak{g}^\alpha(s) \neq 0$ for each $s \in S$.

Lemma 3.3. Let S be a scheme, T an S -torus, α a character of T . The following conditions are equivalent:

- (i) α is nontrivial on each fiber, that is: for every $s \in S$, $\alpha_{\bar{s}}$ is distinct from the unit character of $T_{\bar{s}}$.
- (ii) For every $S' \longrightarrow S$, $S' \neq \emptyset$, $\alpha_{S'}$ is distinct from the unit character of $T_{S'}$.
- (iii) The morphism α is faithfully flat.

⁽²⁹⁾It is clear that (ii) $\Rightarrow$ (i), and one easily sees that (iii) $\Rightarrow$ (i). One has (i) $\Rightarrow$ (ii), for if $s' \in S'$ lies above s and $\alpha_{s'}$ is the unit character, then so is α_s . Finally, to prove (i) $\Rightarrow$ (iii), one reduces to the case where T is diagonalizable and concludes by Exp. VIII 3.2 a).

3.4. — ⁽³⁰⁾Let G be a *reductive* group S -scheme, T a maximal torus of G . Let α be a root of G with respect to T . Then, by 2.5.0 and 1.12, $\mathfrak{g}^\alpha$ is a locally free $\mathcal{O}_S$ -module of rank one. Moreover, by 1.10, $-\alpha$ is also a root of G with respect to T . In particular, if G has semisimple rank 1, by 1.11 and 1.12 one has:

Lemma 3.5. Let S be a scheme, G a reductive S -group of semisimple rank 1, T a maximal torus of G . If α is a root of G with respect to T , then $-\alpha$ is also one and one has

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank 1.

⁽²⁷⁾**N.D.E.** This statement has been made into a lemma, to emphasize it.

⁽²⁸⁾**N.D.E.** The original added the hypothesis that G be of finite presentation over S , which does not seem to be used in what follows. In any case, since G is smooth over S and has connected fibers, it is quasi-compact and separated over S (VI_B, 5.5), hence of finite presentation over S .

⁽²⁹⁾**N.D.E.** The original, which invoked Exp. IX, 5.2, has been simplified. Note, however, that loc. cit., 5.3 shows that if S is connected, the conditions of the lemma hold as soon as (i) holds at one point s of S .

⁽³⁰⁾**N.D.E.** The following has been modified, to recall the hypotheses of 3.2 and add that T is a maximal torus.

Definition 3.6. Let S be a scheme, G a reductive S -group, T a maximal torus of G , R a set of roots of G with respect to T . One says that R is a *root system* of G with respect to T if the following equivalent conditions are satisfied:

- (i) For each $s \in S$, $\alpha \mapsto \alpha_s$ is a bijection from R onto the set of roots of G_s with respect to T_s .
- (ii) The elements of R are distinct on each fiber (i.e. if $\alpha, \alpha' \in R$, $\alpha \neq \alpha'$, then $\alpha - \alpha'$ ($= \alpha\alpha'^{-1}$) is nontrivial on each fiber) and, for each $s \in S$, one has

$$\dim(G_s/\kappa(s)) - \dim(T_s/\kappa(s)) = \text{Card}(R).$$

- (iii) One has $\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha$.

The equivalence of these various conditions is trivial.

Lemma 3.7. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , R a root system of G with respect to T . Every root of G with respect to T is locally on S equal to an element of R .*

This is apparent from condition (iii).

Put $\mathcal{M} = \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$; this is a twisted constant group S -scheme (Exp. X 5.6). If G has a root system R with respect to T , the inclusion $R \hookrightarrow \mathcal{M}(S)$ defines a morphism $R_S \rightarrow \mathcal{M}$, where R_S is the constant S -scheme defined by R ; thanks to condition (ii), one easily sees that this morphism is an open and closed immersion whose image is precisely $\bigcup_{\alpha \in R} \alpha(S)$ (each $\alpha \in R$ being considered as a section $S \rightarrow \mathcal{M}$).

Let $\mathcal{R}$ be the *functor of roots* of G with respect to T : by definition, $\mathcal{R}(S')$ is the set of roots of $G_{S'}$ with respect to $T_{S'}$ for every $S' \rightarrow S$; if $S' = \emptyset$, one puts $\mathcal{R}(\emptyset) = \{e\}$, and if $S' \neq \emptyset$, the inclusion $R \hookrightarrow \mathcal{M}(S')$ identifies R with a root system of $G_{S'}$ with respect to $T_{S'}$, and hence, by 3.7, one has

$$\mathcal{R}(S') = \text{Hom}_{\text{loc. const.}}(S', R),$$

which shows that $\mathcal{R}$ is representable by R_S .

If now one no longer necessarily supposes that G has a root system relative to T , $\mathcal{R}$ is in any case a subsheaf of $\mathcal{M}$ for (fpqc). Locally for this topology, G has a root system with respect to T (take, for example, T split and use the proof of Th. 2.5). By Exp. IV 4.6.8 and the theory of descent of open (resp. closed) subschemes,⁽³¹⁾ one obtains:

Proposition 3.8. *Let S be a scheme, G an S -group, T a maximal torus of G . The functor $\mathcal{R}$ of roots of G with respect to T is representable by a finite twisted constant S -scheme (Exp. X 5.1) which is an open and closed subscheme of $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$.*

For $R \subset \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$ to be a root system of G with respect to T , it is necessary and sufficient that the morphism $R_S \rightarrow \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$ defined by the preceding inclusion induce an isomorphism $R_S \xrightarrow{\sim} \mathcal{R}$.

3.9. — Let S be a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T (i.e. a section of $\mathcal{R}$). Consider the kernel $\text{Ker}(\alpha)$ of R , its unique maximal torus T_α , and the centralizer of the latter, $Z_\alpha = \underline{\text{Centr}}_G(T_\alpha)$. This is an S -group closed in G , reductive (2.8), of semisimple rank 1 (1.12). Moreover,

$$\mathcal{L}ie(Z_\alpha/S) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

hence $\{\alpha, -\alpha\}$ is a root system of Z_α with respect to T .

3.10. — Let S be a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T . If $q \in \mathbf{Q}$ and $q\alpha$ is a root of G with respect to T , then $q = 1$ or $q = -1$. This follows at once from 1.12.

4. Roots and vector group schemes

⁽³¹⁾N.D.E. See SGA 1, VIII 4.4 or EGA IV₂, 2.7.1.

4.1. — Let S be a scheme, $\mathcal{F}$ a *locally free* $\mathcal{O}_S$ -module of finite type. The S -scheme $W(\mathcal{F})$ is *smooth* over S . Its Lie algebra is canonically isomorphic to $\mathcal{F}$. Indeed, one has a canonical isomorphism

$$W(\mathcal{F}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(W(\mathcal{F})/S) = W(\mathrm{Lie}(W(\mathcal{F})/S)).$$

(Exp. II, 4.4.1 and 4.4.2). We shall always identify $\mathcal{F}$ and $\mathrm{Lie}(W(\mathcal{F})/S)$.

Lemma 4.2. *Let S be a scheme, V a vector bundle over S , smooth over S . There then exists a unique isomorphism of $\mathcal{O}_S$ -modules*

$$\exp : W(\mathrm{Lie}(V/S)) \xrightarrow{\sim} V$$

inducing the identity on Lie algebras.⁽³²⁾

Indeed, one has $V = V(\mathcal{F})$ for some quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. Since V is smooth over S , $\mathcal{F} \simeq \omega_{V/S}^1$ is locally free of finite type, and hence

$$\underline{\mathrm{Lie}}(V/S) = V(\omega_{V/S}^1) \simeq W(\mathrm{Lie}(V/S)).$$

Moreover, by Exp. II loc. cit., one has a canonical isomorphism

$$V \xrightarrow{\sim} \underline{\mathrm{Lie}}(V/S) \simeq W(\mathrm{Lie}(V/S)),$$

and the uniqueness of $\exp$ follows at once, since W is fully faithful.

4.3. — Notations. If V is a vector bundle over S , $V^\times$ will denote the open subset of V obtained by removing the zero section. Let us write the group law of V in multiplicative notation. The action of $\mathbf{O}_S$ on V defining the module structure will then be written exponentially:

$$\mathbf{O}_S \times_S V \longrightarrow V, \quad (x, v) \longmapsto v^x.$$

One has $(vv')^x = v^x v'^x$, $v^{x+x'} = v^x v^{x'}$, $v^{xx'} = (v^x)^{x'}$. In particular, if one restricts the operator law to $\mathbf{G}_{m,S}$, $V^\times$ is stable and is therefore equipped with the structure of an object with operator group $\mathbf{G}_{m,S}$. One will again denote this law by $(z, v) \longmapsto v^z$.

Definition 4.4.1. ⁽³³⁾Let $\mathcal{L}$ be an invertible module over S and $W(\mathcal{L})$ the associated vector bundle. Then $W(\mathcal{L})^\times$ is a (locally trivial) principal homogeneous bundle under $\mathbf{G}_{m,S}$. One puts $\Gamma(S, \mathcal{L})^\times = W(\mathcal{L})^\times(S)$.

⁽³²⁾**N.D.E.** The proof of the original has been retained; one may also expand it as follows. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$ -module, $V = V(\mathcal{F})$. Let π denote the projection $V \longrightarrow S$ and ε the zero section $S \longrightarrow V$. Then $\Omega_{V/S}^1 = \pi^* \mathcal{F}$, whence

$$(1) \quad \omega_{V/S}^1 = \varepsilon^* \Omega_{V/S}^1 \simeq \varepsilon^* \pi^* \mathcal{F} \simeq \mathcal{F},$$

and hence $\mathrm{Lie}(V/S) = (\omega_{V/S}^1)^\vee \simeq \mathcal{F}^\vee$. If one supposes V smooth over S , then, by (1), $\mathcal{F}$ is locally free of finite type, and hence

$$(2) \quad V = V(\mathcal{F}) \simeq W(\mathcal{F}^\vee) \simeq W(\mathrm{Lie}(V/S)).$$

Now, if U is an S -scheme equipped with a section $\tau : S \longrightarrow U$, and one has given an isomorphism $\phi : V \xrightarrow{\sim} U$ of “pointed” S -schemes, i.e. such that $\phi \circ \varepsilon = \tau$ (for example if U is an S -group), then ϕ induces an isomorphism $d\phi : \mathcal{F}^\vee \xrightarrow{\sim} \mathrm{Lie}(U/S)$, whence an isomorphism ψ :

$$\begin{array}{ccc} W(\mathcal{F}^\vee) & \xlongequal{\quad} & V(\mathcal{F}) \xrightarrow{\phi} U \\ & \nwarrow \simeq & \nearrow \simeq \\ & W(d\phi^{-1}) & W(\mathrm{Lie}(U/S)) \end{array}$$

which allows one to identify U with $W(\mathrm{Lie}(U/S))$. Since the functor W is fully faithful, there exists a unique isomorphism $\exp : W(\mathrm{Lie}(U/S)) \xrightarrow{\sim} U$ such that $\mathrm{Lie}(\psi^{-1} \circ \exp) = \mathrm{id}$. In fact, one can see directly that $\mathrm{Lie}(\psi) = \mathrm{id}$, so that $\exp = \psi$.

⁽³³⁾**N.D.E.** For subsequent references, 4.4 has been changed into 4.4.1 and 4.4.2.

Scholium 4.4.2. *There is a bijective correspondence between isomorphisms of $\mathcal{O}_S$ -modules $\mathcal{O}_S \simeq \mathcal{L}$, isomorphisms of $\mathbf{O}_S$ -modules $\mathbf{O}_S \simeq \mathbf{W}(\mathcal{L})$,⁽³⁴⁾ and sections $S \rightarrow \mathbf{W}(\mathcal{L})^\times$.*

This correspondence is given by $f \mapsto \mathbf{W}(f) \mapsto \mathbf{W}(f)(1)$. It is compatible with extension of the base. One may therefore regard $\mathbf{W}(\mathcal{L})^\times$ as the “scheme of trivializations of $\mathbf{W}(\mathcal{L})$ ”.

4.5. — Let S be a scheme, T a torus over S , P an S -group with operator group $\mathbf{G}_{\mathbf{m},S}$ (for example, a vector bundle), α a character of T . One denotes by $T \cdot_\alpha P$ the semidirect product of P by T , where T acts on P by the composite morphism

$$T \xrightarrow{\alpha} \mathbf{G}_{\mathbf{m},S} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(P).$$

Definition 4.6. Let S be a scheme, G a group S -scheme, T a subgroup of G , α a character of T , $\mathcal{L}$ an $\mathcal{O}_S$ -module. Let

$$p : \mathbf{W}(\mathcal{L}) \longrightarrow G$$

⁽³⁵⁾be a morphism of group S -functors. One says that p is *normalized by T with multiplier α* if it satisfies the following equivalent conditions:

- (i) p is a morphism of objects with operator group T , if one makes T act on $\mathbf{W}(\mathcal{L})$ by α and on G by inner automorphisms. In other words, for every $S' \rightarrow S$ and all $t \in T(S')$ and $x \in \mathbf{W}(\mathcal{L})(S') = \Gamma(S', \mathcal{L} \otimes \mathcal{O}_{S'})$, one has

$$\mathrm{int}(t)p(x) = p(\alpha(t)x).$$

- (ii) The morphism $T \cdot_\alpha \mathbf{W}(\mathcal{L}) \rightarrow G$ defined by the product in G (i.e. by $(t, x) \mapsto t \cdot p(x)$) is a morphism of groups.

Lemma 4.7. *Under the conditions of 4.6, suppose in addition that G is smooth and has connected fibers,⁽³⁶⁾ T is a maximal torus of G , and $\mathcal{L}$ invertible. If p is a monomorphism and α nonzero on each fiber, then α is a root of G with respect to T .*

Indeed, $\mathrm{Lie}(p) : \mathcal{L} \rightarrow \mathfrak{g}$ is a monomorphism which factors through $\mathfrak{g}^\alpha$.

Proposition 4.8. *Under the conditions of 4.7, suppose that G is reductive and that p is a monomorphism. Then α is a root of G with respect to T , and $\mathrm{Lie}(p)$ induces an isomorphism*

$$\mathrm{Lie}(p) : \mathcal{L} \xrightarrow{\sim} \mathfrak{g}^\alpha.$$

Indeed, by 4.7 and the fact that $\mathfrak{g}^\alpha$ is invertible, it suffices to prove that α is nonzero on each fiber. Let therefore $s \in S$ be such that $\alpha_{\bar{s}} = 0$ ($= 1$ in multiplicative notation). If X is a nonzero section of $\mathcal{L} \otimes_S \kappa(\bar{s})$, $p(X)$ is a unipotent element $\neq e$ of $G(\bar{s})$ which centralizes $T_{\bar{s}}$, which is impossible, since the latter is its own centralizer.

Corollary 4.9. *Under the conditions of 4.8, there exists a monomorphism of groups with operators T (i.e. normalized by T with multiplier α)*

$$\mathbf{W}(\mathfrak{g}^\alpha) \longrightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^\alpha \rightarrow \mathfrak{g}$.

We shall soon see that 4.9 in fact holds whenever G is a reductive group and α a root of G with respect to T , and that such a morphism is unique.

⁽³⁴⁾**N.D.E.** Here the vector bundle $\mathbf{W}(\mathcal{L})$ is identified with the functor in $\mathbf{O}_S$ -modules which it represents.

⁽³⁵⁾**N.D.E.** Here $\mathbf{W}(\mathcal{L})$ has been written (with W in bold), since, for an arbitrary $\mathcal{O}_S$ -module $\mathcal{L}$, $\mathbf{W}(\mathcal{L})$ is not necessarily representable. But in what follows, $\mathcal{L}$ will be supposed locally free of finite rank (and even invertible), in which case the functor is representable by the vector bundle $\mathbf{V}(\mathcal{L}^\vee)$, and it will simply be denoted $\mathbf{W}(\mathcal{L})$.

⁽³⁶⁾**N.D.E.** Cf. N.D.E. (28).

Reminder 4.10. Let k be an algebraically closed field, G a reductive k -group, T a maximal torus of G , α a root of G with respect to T . There exists a monomorphism

$$p : \mathbf{G}_{a,k} \longrightarrow G$$

normalized by T with multiplier α .

See *Bible*, § 13.1, th. 1.

4.11. — Let us finish this section with a technical result which will be useful to us. Let S be a scheme and $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module. Let q be an integer > 0 such that $x \mapsto x^q$ is an endomorphism of the S -group $\mathbf{G}_{a,S}$. (If $S \neq \emptyset$, one has $q = 1$, or $q = p^n$, p being a prime number zero on S ; this follows at once from the following elementary fact: the gcd of the binomial coefficients $\binom{q}{i}$, for $i \neq 0, q$, is p if $q = p^n$, p prime, and 1 otherwise.) The morphism defined by the q th power

$$\mathcal{L} \longrightarrow \mathcal{L}^{\otimes q}$$

is a morphism of sheaves of abelian groups. By base change, it defines a morphism of group S -schemes:

$$W(\mathcal{L}) \longrightarrow W(\mathcal{L}^{\otimes q}).$$

In particular, if $\mathcal{L}'$ is another invertible module and one has a morphism of $\mathcal{O}_S$ -modules

$$h : \mathcal{L}^{\otimes q} \longrightarrow \mathcal{L}',$$

one deduces a morphism of group S -schemes:

$$W(\mathcal{L}) \longrightarrow W(\mathcal{L}'), \quad x \mapsto h(x^q).$$

With these notations, one has

Proposition 4.12. *Let S be a scheme, T (resp. T') an S -torus, $\mathcal{L}$ (resp. $\mathcal{L}'$) an invertible $\mathcal{O}_S$ -module, α (resp. α') a character of T (resp. T').⁽³⁷⁾ Let $f : T \longrightarrow T'$ be a morphism of groups and $g : W(\mathcal{L}) \longrightarrow W(\mathcal{L}')$ a morphism of S -schemes (not necessarily a morphism of S -groups) satisfying the following condition:*

$$(\star) \quad g(\alpha(t)x) = \alpha'(f(t))g(x)$$

for all $x \in W(\mathcal{L})(S')$, $t \in T(S')$, $S' \longrightarrow S$. Let $s_0 \in S$ be such that $\alpha_{\bar{s}_0} \neq 0$.

- Suppose that g sends the zero section to the zero section and that for every integer $n > 0$ one has $(\alpha' \circ f)_{\bar{s}_0} \neq n\alpha_{\bar{s}_0}$. Then $g = 0$ in a neighborhood of s_0 .
- Suppose that g is a morphism of groups such that $g_{\bar{s}_0} \neq 0$. There then exist an open subset U of S containing s_0 and an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,U}$ and $(\alpha' \circ f)_U = q\alpha_U$.
- Suppose that $(\alpha' \circ f)_{\bar{s}_0} = q\alpha_{\bar{s}_0}$, where q is an integer > 0 such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$. There then exist an open subset U of S containing s_0 and a unique morphism of $\mathcal{O}_S$ -modules

$$h : \mathcal{L}^{\otimes q}|_U \longrightarrow \mathcal{L}'|_U$$

such that g_U is the composite morphism

$$W(\mathcal{L})_U \xrightarrow{x \mapsto x^q} W(\mathcal{L}^{\otimes q})_U \xrightarrow{W(h)} (\mathcal{L}')_U.$$

Let us prove (a). Since the conclusion is local on S , one may suppose that $W(\mathcal{L}) = W(\mathcal{L}') = \mathbf{G}_{a,S}$ and hence that g is expressed as a polynomial

$$g(X) = \sum_{n \geq 0} a_n X^n, \quad a_n \in \Gamma(S, \mathcal{O}_S).$$

⁽³⁷⁾**N.D.E.** “Consider the semidirect product $T \cdot_{\alpha} W(\mathcal{L})$, resp. $T' \cdot_{\alpha'} W(\mathcal{L}')$ ” has been deleted. On the other hand, the hypothesis in (b) that g be a morphism of groups, combined with $(\star)$, amounts to saying that the morphism $(t, x) \mapsto (f(t), g(x))$ is a morphism of groups from $T \cdot_{\alpha} W(\mathcal{L})$ to $T' \cdot_{\alpha'} W(\mathcal{L}')$.

Condition $(\star)$, which relates f and g , is written as an identity in $\Gamma(S', \mathcal{O}_{S'})[X]$:

$$\sum_{n \geq 0} a_n \alpha'(f(t)) X^n = \sum_{n \geq 0} a_n \alpha(t)^n X^n,$$

i.e. for every $n \geq 0$, every $S' \rightarrow S$ and every $t \in T(S')$,

$$a_n(\alpha'(f(t)) - \alpha(t)^n) = 0.$$

For each $n \geq 0$, let S_n be the set of $s \in S$ such that $(\alpha' \circ f)_{\bar{s}} = n\alpha_{\bar{s}}$. One knows (Exp. IX 5.3) that the S_n are open and closed, and that $(\alpha' \circ f)_{S_n} = n\alpha_{S_n}$. Moreover, since $\alpha_{\bar{s}_0} \neq 0$, one may, restricting S if necessary, suppose that α is nonzero on each fiber (same reference), which implies that the S_n are disjoint. Restricting S if necessary, one may therefore suppose that one is in one of the following two cases: there exists an n such that $S = S_n$, or all the S_n are empty.

Let $m \geq 0$ be such that $S_m = \emptyset$; I say that then $a_m = 0$; indeed, $\alpha' \circ f$ and $m\alpha$ are distinct on each fiber of S , and one has:

Lemma 4.13. *Let S be a scheme, T an S -torus, α and α' two characters of T distinct on each fiber; there exists a family $\{S_i \rightarrow S\}$ covering for (fpqc), and, for each i , a $t_i \in T(S_i)$ such that $\alpha(t_i) - \alpha'(t_i) = 1$.*

The lemma is trivial, by reduction to the diagonalizable case, then to the case $T = \mathbf{G}_{m,S}$.

Let us resume the proof of proposition 4.12; we have just proved (a). In cases (b) and (c), there exists an n such that $S = S_n$ ($n = q$ in (c)). By the preceding result, one therefore has $a_m = 0$ for $m \neq n$, which proves that g is written

$$g(X) = a_n X^n, \quad a_n \in \Gamma(S, \mathcal{O}_S).$$

This proves (c) at once. In case (b), one knows that $a_n(s_0) \neq 0$, and one may therefore suppose a_n invertible on S , which implies that $x \mapsto x^n$ is an endomorphism of $\mathbf{G}_{a,S}$ (by the hypothesis of (b)) and completes the proof.

5. An instructive example

5.1. — Let k be an algebraically closed field of characteristic 0. Put $A = k[t]$, the polynomial ring in one variable over k , and $S = \text{Spec}(A)$. Consider the following Lie algebra $\mathfrak{g}$ over $\mathcal{O}_S$: as a module, it is free of dimension 3, with basis $\{X, Y, H\}$; its multiplication table is

$$[X, Y] = 2tH, \quad [H, X] = X \quad \text{and} \quad [H, Y] = -Y.$$

For $s \in S$, $s \neq s_0$ (the point defined by $t = 0$), the fiber $\mathfrak{g}(s) = \mathfrak{g} \otimes_A \kappa(s)$ is isomorphic to the Lie algebra of the group $\text{PGL}_{2, \kappa(s)}$. For $s = s_0$, it is a solvable Lie algebra.

5.2. — Let G_1 be the group scheme of automorphisms of $\mathfrak{g}$: for every $S' \rightarrow S$, $G_1(S')$ is the group of automorphisms of the Lie $\mathcal{O}_{S'}$ -algebra $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$. This is a closed subscheme of the group $\text{GL}(\mathfrak{g})$ of automorphisms of the $\mathcal{O}_S$ -module $\mathfrak{g}$. Let $S' \rightarrow S$ and $u \in \text{M}_3(\Gamma(S', \mathcal{O}_{S'}))$, considered as an endomorphism of the $\mathcal{O}_S$ -module $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$:

$$\begin{aligned} u(X) &= aX + bY + eH, \\ u(Y) &= b'X + a'Y + e'H, \\ u(H) &= cX + c'Y + dH. \end{aligned}$$

One sees at once that u is a section of G_1 if and only if $\det(u)$ is invertible and one has the relations:⁽³⁸⁾

$$\begin{aligned} (1) \quad a(d-1) &= ec, & (1') \quad a'(d-1) &= e'c', \\ (2) \quad b(d+1) &= ec', & (2') \quad b'(d+1) &= e'c, \\ (3) \quad e &= 2t(bc - ac'), & (3') \quad e' &= 2t(b'c' - a'c), \\ (4) \quad 2tc &= eb' - ae', & (4') \quad 2tc' &= be' - ea', \\ (5) \quad 2t(aa' - bb') &= 2td. \end{aligned}$$

Lemma 5.3. *Relations (1), (1'), (2), (2') imply*

$$\det(u) = aa'(2-d) + bb'(d+2), \quad aa' - bb' = d \cdot \det(u).$$

Indeed, the first assertion follows at once by substituting relations (1), (1'), (2), (2') into the expansion of $\det(u)$:

$$\begin{aligned} \det(u) &= aa'd + be'c + b'c'e - a'ec - ae'c' - bb'd \\ &= aa'd + bb'(d+1) + bb'(d+1) - aa'(d-1) - bb'd - aa'(d-1) \\ &= aa'(d-d+1-d+1) + bb'(d+1+d+1-d) \\ &= aa'(2-d) + bb'(d+2). \end{aligned}$$

Multiplying this relation by d , one obtains

$$d \cdot \det(u) = aa'(2d-d^2) + bb'(d^2+2d).$$

But the relation $(1) \times (1') = (2) \times (2')$ gives at once

$$aa'(d-1)^2 = bb'(d+1)^2.$$

Combining the preceding two relations, one finds at once the second desired formula.

5.4. — Consider then $G = G_1 \cap \mathrm{SL}(\mathfrak{g})$. This is the closed subgroup of G_1 defined by the equation $\det(u) = 1$. It is therefore an affine group over S .

Proposition 5.5. *The group G is smooth over S .*

To prove the proposition, one will need the following lemmas.

Lemma 5.6. *Let U be the open subset of $\underline{\mathrm{End}}_A(\mathfrak{g}) \simeq \mathrm{W}(\mathrm{M}_3(\mathcal{O}_S))$ defined by the condition “the product ad is invertible”, i.e. the open subset $\underline{\mathrm{End}}_A(\mathfrak{g})_f$, where f is the function defined by $f(u) = ad$. Then $U \cap G$ is the closed subscheme of U defined by the six equations:*

$$(1), \quad (2), \quad (2'), \quad (3), \quad (3') \quad \text{and} \quad (D) : aa' - bb' = d.$$

First, it is clear that these six relations are satisfied by every “point” of G (lemma 5.3), in particular by every “point” of $U \cap G$. Conversely, one must show that if $u \in U(S')$ (for every $S' \rightarrow S$), and u satisfies the six conditions of the statement, then $\det(u) = 1$ and u also satisfies (1'), (4), (4') and (5).

First, one has $(D) \Rightarrow (5)$. By (2) and (2'), one has

$$bb'(d+1) = bce' = b'c'e.$$

But by (3) and (3'), writing $2t(bc - ac')(b'c' - a'c)$ in two ways, one has

$$(bc - ac')e' = (b'c' - a'c)e.$$

Combining this with the preceding relation gives $ac'e' = a'ce$. But by (1), $a'ce = a'a(d-1)$, which proves $a(a'(d-1) - e'c') = 0$ and implies (1'), since a is assumed invertible.

⁽³⁸⁾**N.D.E.** The equality $[u(X), u(Y)] = 2tu(H)$ (resp. $[u(H), u(X)] = u(X)$, resp. $[u(H), u(Y)] = -u(Y)$) gives relations (4), (4') and (5) (resp. (1–3), resp. (1'–3')).

Thus (1), (2), (2') and (1') hold, hence by lemma 5.3 and (D) one has $d(\det(u) - 1) = 0$. Since d is assumed invertible, this implies $\det(u) = 1$.

Let us prove (4) and (4'). Let us do so, for example, for (4'), the other calculation being deduced from it by symmetry. By (3), (3') and (D), one has at once

$$a'e + be' = -2t(aa' - bb')c' = -2tdc'.$$

Combining (1') and (2), one also has⁽³⁹⁾

$$a'e + be' = -d(be' - ea'),$$

which finishes the proof of (4'), since d is assumed invertible.

Lemma 5.7. *G is smooth over S along the unit section.*

By 5.6 and SGA 1, II 4.10, it suffices to prove that the differentials of the functions

$$\begin{aligned} a(d-1) - ec, & \quad b(d+1) - ec', & \quad b'(d+1) - e'c, \\ e - 2t(bc - ac'), & \quad e' - 2t(b'c' - a'c), & \quad aa' - bb' - d, \end{aligned}$$

at the points of the unit section of G are linearly independent. Now, denoting by a capital letter the differential of the corresponding lower-case letter, these are⁽⁴⁰⁾

$$D, \quad 2B, \quad 2B', \quad E + 2tC', \quad E' + 2tC, \quad A + A' - D,$$

which are indeed linearly independent modulo every $(t - \lambda)$, $\lambda \in k$.⁽⁴¹⁾

Lemma 5.8. *For $s \in S$, $s \neq s_0$, the fiber G_s is connected and semisimple.*

⁽⁴²⁾Indeed, since $s \neq s_0$, $\mathfrak{g}(s)$ is isomorphic to the Lie algebra of $\mathrm{PGL}_{2,\kappa(s)}$, and, on the other hand, one has $G_s = (G_1)_s$; now it is known that the group of automorphisms of the Lie algebra of PGL_2 over a field of characteristic 0 is PGL_2 itself, which is connected and semisimple.

Lemma 5.9. *The fiber G_{s_0} is solvable and has two connected components of the following form:*

$$G_{s_0}^0 = \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & a^{-1} & 0 \\ c & c' & 1 \end{pmatrix} \right\} \quad \text{and} \quad G_{s_0}^- = \left\{ \begin{pmatrix} 0 & b & 0 \\ b^{-1} & 0 & 0 \\ c & c' & -1 \end{pmatrix} \right\}.$$

Indeed, one has $e = e' = 0$, since $t = 0$ at s_0 . One then immediately solves equations (1), (1'), ..., (5) and (D).

Lemma 5.10. $w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ is a section of G over S such that $w(s_0) \in G_{s_0}^-$.

Let us now prove 5.5.⁽⁴³⁾ Denote by G^0 the union of the identity components of the fibers of G (that is, the complement of $G_{s_0}^-$); since G is smooth over S along the unit section (5.7), G^0 is an open subgroup of G , smooth over S , by VI_B, 3.10. Since, by translation, G is clearly smooth at the points of $w(S)$, G is smooth over S .

⁽³⁹⁾**N.D.E.** The sign has been corrected.

⁽⁴⁰⁾**N.D.E.** $A + A' + B$ has been corrected to $A + A' - D$.

⁽⁴¹⁾**N.D.E.** “ $(t - a)$, $a \in A$ ” has been corrected to “ $(t - \lambda)$, $\lambda \in k$ ”.

⁽⁴²⁾**N.D.E.** The following sentence has been slightly modified.

⁽⁴³⁾**N.D.E.** The following sentence has been modified, adding the reference to VI_B, 3.10.

5.11. — Consider the morphism $\mathbf{G}_{m,S} \rightarrow G^0$ defined by $z \mapsto \begin{pmatrix} z & 0 & 0 \\ 0 & 1/z & 0 \\ 0 & 0 & 1 \end{pmatrix}$. It is a monomorphism defining a torus T of G^0 . I say that one has

$$T = \underline{\text{Centr}}_G(T) = \underline{\text{Centr}}_{G^0}(T).$$

Indeed, it suffices to check the first equality. Since these are subgroups of G smooth over S , it suffices to check that they have the same geometric points. For the fibers at points $s \neq s_0$, this follows from the fact that $\text{PGL}_{2,\kappa(s)}$ is reductive and that T_s is a maximal torus of it for dimension reasons (cf. 1.11). On the fiber at s_0 , the calculation is immediate. In particular, it follows that T is a maximal torus of G and of G^0 .

5.12. — The section w of G defined in 5.9 normalizes T . It follows at once (cf. 2.4) that the Weyl group of G is isomorphic to $(\mathbf{Z}/2\mathbf{Z})_S$, and in particular finite over S .

$$W_G(T) = \underline{\text{Norm}}_G(T)/T = (\mathbf{Z}/2\mathbf{Z})_S.$$

On the other hand, $W_{G^0}(T)$ is *not finite* over S : it “lacks a point” above s_0 .

5.13. — The open immersion $G^0 \rightarrow G$ is not a closed immersion (since G^0 is dense in G); it is nevertheless an affine morphism (and hence G^0 is affine over S). Indeed, since $G_{s_0}^0$ is closed in G_{s_0} , which is closed in G , the complement U of $G_{s_0}^0$ in G is open; G^0 and U form an open covering of G , and it suffices to check that the immersions $G^0 \rightarrow G^0$ and $G^0 \cap U \rightarrow U$ are affine; for the first this is trivial, for the second one notes that $U \cap G^0$ is defined in U by the equation $b \neq 0$.

One has therefore constructed a smooth affine S -group G^0 , with connected fibers, having a maximal torus T which is its own centralizer and whose Weyl group $W_{G^0}(T)$ is *not finite* (compare with theorem 2.5).

6. Local existence of maximal tori. The Weyl group

In the course of the proof of 2.5, we used a result of Exp. XI on the local existence, for the étale topology, of maximal tori; the proof of Exp. XI uses a delicate representability result (XI 4.1). In the particular case which concerns us, one can give another proof, based on the ideas of Exp. XII no. 7, and of a much more elementary nature.

Proposition 6.1. *Let S be a scheme, G a group S -scheme smooth, affine and with connected fibers over S , and s_0 a point of S such that the maximal tori of the geometric fiber $G_{\bar{s}_0}$ are their own centralizer. There exists an étale morphism $S' \rightarrow S$ covering s_0 , and a split maximal torus T of $G_{S'}$.*

First, one may suppose S affine.⁽⁴⁴⁾ Since G is of finite presentation over S , one may suppose S noetherian, then local, then henselian with separably closed residue field (cf. EGA IV, 8.12, § 8.8, and § 18.8). Put therefore $S = \text{Spec}(A)$, A henselian with separably closed residue field $k = \kappa(s_0)$. Choose a maximal torus T_0 of $G_0 (= G_k)$ (one exists, for example, because the scheme of maximal tori of G_0 is smooth over k , Exp. XII, 7.1 c)); since k is separably closed, T_0 is split (cf. X 1.4), and is therefore given by a monomorphism of groups

$$f_0 : \mathbf{G}_{m,k}^r \rightarrow G_0.$$

Let m be an integer > 1 prime to the characteristic of k . By Exp. VIII 6.7, for every $h > 0$, $\underline{\text{Centr}}_{G_0}(m^h T_0)$ is representable by a closed subscheme of G_0 . Since the $m^h T_0$ are schematically dense in T_0 (cf. Exp. IX 4.10) and G_0 is noetherian, there exists an h such that

$$\underline{\text{Centr}}_{G_0}(m^h T_0) = \underline{\text{Centr}}_{G_0}(T_0) = T_0.$$

⁽⁴⁴⁾**N.D.E.** The preceding sentence has been added, as well as the reference to EGA IV in what follows.

Put $n = m^h$; since n is invertible on S , ${}_n\mathbf{G}_{m,S}$ is isomorphic to $(\mathbf{Z}/n\mathbf{Z})_S$; f_0 therefore defines a monomorphism of groups

$$u_0 : (\mathbf{Z}/n\mathbf{Z})_k^r \longrightarrow G_0$$

such that $\underline{\text{Centr}}_G(u_0) = T_0$. Now the S -functor

$$P = \underline{\text{Hom}}_{S\text{-gr.}}((\mathbf{Z}/n\mathbf{Z})_S^r, G)$$

is representable by an S -scheme of finite type (as a closed subscheme of r copies of ${}_nG = \underline{\text{Hom}}_{S\text{-gr.}}((\mathbf{Z}/n\mathbf{Z})_S, G) = \text{Ker}(G \xrightarrow{n} G)$). But P is smooth over S (Exp. IX 3.6), hence $u_0 \in P(k)$ lifts to a section $u \in P(S)$ (Hensel's lemma, Exp. XI 1.11):

$$u : (\mathbf{Z}/n\mathbf{Z})_S^r \longrightarrow G.$$

Consider $H = \underline{\text{Centr}}_G(u)$; it is a closed subgroup scheme of G , by Exp. VIII 6.5 e), and one has $H_0 = T_0$ by hypothesis.⁽⁴⁵⁾ Moreover, H is smooth over S : indeed, let $S' = \text{Spec}(A)$ be an affine scheme over S , $u' : ({}_n\mathbf{G}_{m,S'})^r \longrightarrow G_{S'}$ the morphism deduced from u by base change, $S'_J = \text{Spec}(A/J)$, where J is a square-zero ideal, and let $x \in H(S'_J)$; since G is smooth, x lifts to an element g of $G(S')$; then $v = \text{int}(g)(u')$ satisfies $v_J = u'_J$, and hence, by IX 3.2, there exists an element g' of $G(S')$ such that $g'_J = e$ and $\text{int}(g')(v) = u'$; then $h = g'g$ belongs to $H(S')$ and satisfies $h_J = x$.

Let then H^0 be the identity component of H ; it is a subgroup scheme of G , smooth with connected fibers, whose special fiber is a torus. By Exp. X 8.1, it is a torus, necessarily split (Exp. X 4.6). Put $H^0 = T$, and let $C = \underline{\text{Centr}}_G(T)$, which is a closed subgroup of G (Exp. VIII 6.5 e)), smooth (Exp. XI 2.4). Consider C^0 (in fact one has $C^0 = C$, but we do not need to know this); then $C^0 \supset T$, and these are two smooth groups with connected fibers. They coincide at s_0 , hence in a neighborhood. Restricting S if necessary, one may therefore suppose $C^0 = T$, hence a fortiori T maximal.

Remark 6.2. The proof shows in particular that the reductive rank of $G_{\bar{s}}$ is constant in a neighborhood of $s = s_0$.

Proposition 6.3. *Let S be a scheme, G a group S -scheme smooth and of finite presentation over S , Q a subtorus of G .*

- (i) $\underline{\text{Centr}}_G(Q)$ and $\underline{\text{Norm}}_G(Q)$ are representable by closed subgroup schemes, smooth (and hence of finite presentation) over S .
- (ii) $\underline{\text{Centr}}_G(Q)$ is an open and closed subscheme of $\underline{\text{Norm}}_G(Q)$. The quotient $W_G(Q) = \underline{\text{Norm}}_G(Q)/\underline{\text{Centr}}_G(Q)$ is representable by an open subgroup scheme of $\underline{\text{Aut}}_{S\text{-gr.}}(Q)$, and is therefore a group S -scheme quasi-finite, étale and separated over S .
- (iii) For every $s \in S$, put

$$w(s) = \left| \underline{\text{Norm}}_{G(\bar{s})}(Q(\bar{s}))/\underline{\text{Centr}}_{G(\bar{s})}(Q(\bar{s})) \right|.$$

Then $s \mapsto w(s)$ is lower semicontinuous, and is constant in a neighborhood of s if and only if $W_G(Q)$ is finite over S in a neighborhood of s .

By Exp. XI 6.11, $\underline{\text{Centr}}_G(Q)$ and $\underline{\text{Norm}}_G(Q)$ are representable by closed subschemes of finite presentation of G . These are smooth by Exp. XI 2.4 and 2.4 bis, which proves (i). Assertions (ii) and (iii) are then proved as in Exp. XI 5.9 and 5.10, whose proof in fact uses only (i), and not the delicate theorems Exp. XI 4.1 and 4.2.

⁽⁴⁵⁾**N.D.E.** The reference VIII 6.5 e) has been added above, and the following sentence has been expanded.

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Exposé XX. Reductive groups of semisimple rank 1

by M. Demazure

1. Elementary systems. The groups U_α and $U_{-\alpha}$

(0)

Reminder 1.1. Let $S = \operatorname{Spec}(k)$, where k is an algebraically closed field, and let G be a reductive S -group of semisimple rank 1, T a maximal torus (necessarily split) of G . One then has

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where α and $-\alpha$ are the roots of G with respect to T . Moreover, there exist two monomorphisms of groups

$$p_\alpha : \mathbf{G}_{a,S} \longrightarrow G \quad \text{and} \quad p_{-\alpha} : \mathbf{G}_{a,S} \longrightarrow G$$

such that

$$tp_\alpha(x)t^{-1} = p(\alpha(t)x) \quad \text{and} \quad tp_{-\alpha}(x)t^{-1} = p_{-\alpha}(\alpha(t)^{-1}x),$$

for every $S' \longrightarrow S$ and all $t \in T(S')$, $x \in \mathbf{G}_a(S')$, and such that the morphism

$$\mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \longrightarrow G,$$

defined by $(y, t, x) \longmapsto p_{-\alpha}(y)tp_\alpha(x)$, is radicial and dominant (*Bible*, § 13.4, cor. 2 to th. 3).

Since the tangent map at the identity element is bijective, this morphism is also separable, hence birational; by Zariski's "Main Theorem" (EGA III₁, 4.4.9), it is therefore an open immersion.

Lemma 1.2. Let S be a scheme, G a group S -scheme, T a torus of G , Q a subtorus of T , α a character of T inducing on Q a character nontrivial on each fiber. Let $p_\alpha : \mathbf{G}_{a,S} \longrightarrow G$ (resp. $p_{-\alpha} : \mathbf{G}_{a,S} \longrightarrow G$) be a morphism of groups normalized by T with multiplier α (resp. $-\alpha$). Suppose that the morphism

$$u : \mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \longrightarrow G$$

defined set-theoretically by $u(y, t, x) = p_{-\alpha}(y)tp_\alpha(x)$ is an open immersion. Finally, let q be an integer ≥ 0 and

$$p : \mathbf{G}_{a,S} \longrightarrow G$$

a morphism of groups such that for every $S' \longrightarrow S$ and all $t \in Q(S')$, $x \in \mathbf{G}_a(S')$, one has

$$\operatorname{int}(t)^q(p(x)) = p(\alpha(t)x).$$

There then exists a unique $\nu \in \mathbf{G}_a(S)$ such that $p(x) = p_\alpha(\nu x^q)$.

Indeed, let Ω be the image of u and $U = p^{-1}(\Omega)$. This is an open subset of $\mathbf{G}_{a,S}$ containing the zero section. For every section t of Q , the automorphism of $\mathbf{G}_{a,S}$ defined by multiplication by $\alpha(t)$ preserves U globally. One has $U = \mathbf{G}_{a,S}$; indeed, it suffices to check this when S is the spectrum of an algebraically closed field k ; then $\alpha : Q(k) \longrightarrow k^*$ is surjective, which proves at once that $U(k) \supset k^*$, hence $U = \mathbf{G}_{a,k}$. There therefore exist three morphisms

$$a : \mathbf{G}_{a,S} \longrightarrow \mathbf{G}_{a,S}, \quad b : \mathbf{G}_{a,S} \longrightarrow T, \quad c : \mathbf{G}_{a,S} \longrightarrow \mathbf{G}_{a,S},$$

(0)**N.D.E.** Version of 13/10/2024

such that

$$p(x) = p_{-\alpha}(a(x))b(x)p_{\alpha}(c(x)).$$

The condition on p becomes

$$\begin{aligned} a(\alpha(t)x) &= \alpha(t)^{-q}a(x), \\ b(\alpha(t)x) &= b(x), \\ c(\alpha(t)x) &= \alpha(t)^q c(x). \end{aligned}$$

For the same reason as above, for every $S' \rightarrow S$ and every $z \in \mathbf{G}_m(S')$, one therefore has

$$a(zx) = z^{-q}a(x), \quad b(zx) = b(x), \quad c(zx) = z^q c(x),$$

hence

$$z^q a(z) = a(1), \quad b(z) = b(1), \quad c(z) = z^q c(1).$$

Since $\mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}$, one has at once, for every $x \in \mathbf{G}_a(S')$, $S' \rightarrow S$:

$$\begin{aligned} x^q a(x) &= a(1) = a(0) = 0, & \text{whence } a &= 0, \\ c(x) &= x^q c(1) = \nu x^q, & \text{for some } \nu &\in \mathbf{G}_a(S), \\ b(x) &= b(1) = b(0) = e, & \text{whence } b &= e, \end{aligned}$$

which completes the proof.

Definition 1.3. Let S be a scheme. An *elementary S -system* is a triple (G, T, α) where

- (i) G is a reductive S -group of semisimple rank 1 (Exp. XIX 2.7),
- (ii) T is a maximal torus of G ,
- (iii) α is a root of G with respect to T (Exp. XIX 3.2).

One therefore has a direct sum decomposition (Exp. XIX 3.5)⁽¹⁾

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha},$$

$\mathfrak{g}^{\alpha}$ and $\mathfrak{g}^{-\alpha}$ being locally free of rank one.

1.4. — If (G, T, α) is an elementary S -system, then $(G_{S'}, T_{S'}, \alpha_{S'})$ is an elementary S' -system for every $S' \rightarrow S$. If (G, T, α) is an elementary S -system, then $(G, T, -\alpha)$ is also one.

If S is a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T , then (Exp. XIX 3.9) (Z_{α}, T, α) is an elementary S -system.

Let (G, T, α) be an elementary S -system. The invertible module $\mathfrak{g}^{\alpha}$ is canonically equipped with a T -module structure. One therefore also has a T -module structure on the vector bundle $W(\mathfrak{g}^{\alpha})$. On the other hand, the inner automorphisms of T define on G a structure of group with operator group T .

Theorem 1.5. Let (G, T, α) be an elementary S -system.

- (i) There exists a unique morphism of groups with operator group T

$$\exp : W(\mathfrak{g}^{\alpha}) \rightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^{\alpha} \rightarrow \mathfrak{g}$.⁽²⁾

In other words, $\exp$ is the unique morphism satisfying the following conditions: for every $S' \rightarrow S$ and all $t \in T(S')$, $X, X' \in W(\mathfrak{g}^{\alpha})(S')$, one has

$$\begin{aligned} \exp(X + X') &= \exp(X) \exp(X'), \\ \text{int}(t)(\exp(X)) &= \exp(\alpha(t)X), \\ \underline{\text{Lie}}(\exp)(X) &= X. \end{aligned}$$

⁽¹⁾**N.D.E.** Of $\mathcal{O}_S$ -modules.

⁽²⁾**N.D.E.** We shall see later (Cor. 5.9) that $\exp$ is an isomorphism from $W(\mathfrak{g}^{\alpha})$ onto a closed subgroup of G .

(ii) If one defines similarly (in the elementary S -system $(G, T, -\alpha)$)

$$\exp : W(\mathfrak{g}^{-\alpha}) \longrightarrow G,$$

then the morphism

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^{\alpha}) \longrightarrow G$$

defined set-theoretically by $(Y, t, X) \mapsto \exp(Y) \cdot t \cdot \exp(X)$ is an open immersion.

Suppose we have proved the existence of the required morphisms $\exp$, and let us prove the other assertions of the theorem. Let us first prove (ii). Since both sides are of finite presentation and flat over S , it suffices to do so when S is the spectrum of an algebraically closed field (SGA 1, I 5.7 and VIII 5.5). Let then $S = \operatorname{Spec} k$. Let $X \in \Gamma(S, \mathfrak{g}^{\alpha})^{\times}$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^{\times}$. It suffices to prove that the morphism

$$\mathbf{G}_{a,k} \times_k T \times_k \mathbf{G}_{a,k} \longrightarrow G, \quad (y, t, x) \mapsto \exp(yY)t \exp(xX)$$

is an open immersion. Now, by 1.1 and 1.2, there exist $a, b \in k$ such that

$$\exp(yY) = p_{-\alpha}(ay) \quad \text{and} \quad \exp(xX) = p_{\alpha}(bx).$$

Since $\exp : W(\mathfrak{g}^{-\alpha}) \longrightarrow G$ induces a monomorphism on Lie algebras, one has $a \neq 0$; similarly $b \neq 0$, and one is reduced to 1.1.

The uniqueness of the morphism $\exp$ can be proved locally on S ; one then reduces to the case where $\mathfrak{g}^{\alpha}$ and $\mathfrak{g}^{-\alpha}$ are free, and one need only apply 1.2 (with $Q = T$ and $q = 1$).

It therefore remains to prove the existence of the required morphism $\exp$. Note first that, by the theory of faithfully flat descent and the preceding uniqueness assertion, it suffices to prove this existence locally on S for the (fpqc) topology. By the usual arguments using finite presentation, one reduces to the case where S is noetherian, then to the case where it is local noetherian. By the preceding remark, one may therefore content oneself with proving the existence of the desired morphism $\exp$ when $S = \operatorname{Spec}(A)$, A local noetherian complete with algebraically closed residue field k . Let then $p_0 : \mathbf{G}_{a,k} \longrightarrow G_k$ be a monomorphism of k -groups normalized by T_k with multiplier $\alpha_0 = \alpha \otimes_A k$ (one exists by 1.1). One knows (1.1 and 1.2) that the corresponding morphism $T_k \cdot_{\alpha_0} \mathbf{G}_{a,k} \longrightarrow G_k$ is an immersion, hence in particular a monomorphism. Let us provisionally admit the following two lemmas:

Lemma 1.6. *Let S be a scheme, G an S -group of finite presentation, T an S -torus, α a character nontrivial on each fiber of T , s_0 a point of S . Let*

$$f : T \cdot_{\alpha} \mathbf{G}_{a,S} \longrightarrow G$$

be a morphism of S -groups such that f_{s_0} is a monomorphism and the restriction of f to T is a monomorphism. There exists an open neighborhood U of s_0 such that $f|_U$ is a monomorphism.

Lemma 1.7. *Let A be a complete local noetherian ring with algebraically closed residue field k , (G, T, α) an elementary A -system, $p_0 : \mathbf{G}_{a,k} \longrightarrow G_k$ a morphism of k -groups normalized by T_k with multiplier $\alpha \otimes_A k$. There exists a morphism of groups $p : \mathbf{G}_{a,A} \longrightarrow G$ normalized by T with multiplier α .*

Let p be the morphism whose existence is asserted by 1.7. Let $f : T \cdot_{\alpha} \mathbf{G}_{a,S} \longrightarrow G$ be the corresponding morphism. It satisfies the hypotheses of 1.6, hence is a monomorphism; in particular p is a monomorphism. One then concludes by Exp. XIX 4.9.

Proof of 1.6. Let $\varepsilon : S \longrightarrow T \cdot_{\alpha} \mathbf{G}_{a,S}$ denote the unit section. Since f is unramified at $\varepsilon(s_0)$, it is so at $\varepsilon(s)$ for all s in an open neighborhood U of s_0 ; $f|_U$ is therefore unramified (Exp. X 3.5),⁽³⁾ hence its kernel $\operatorname{Ker}(f)|_U$ is unramified over U . To prove that $f|_U$ is a monomorphism, it therefore suffices⁽⁴⁾ to prove that $\operatorname{Ker}(f)|_U$ is radicial over U , which is a set-theoretic question. One is therefore reduced to proving:

⁽³⁾**N.D.E.** See also VI_B, 2.5.

⁽⁴⁾**N.D.E.** By EGA IV₄, 17.9.1.

Lemma 1.8. *Let k be an algebraically closed field; let N be an invariant subgroup of $T \cdot_\alpha \mathbf{G}_{a,k}$ (α a nontrivial character of the torus T), étale over k and such that $N \cap T = \{e\}$. Then $N = \{e\}$.*

One has $\text{int}(t')(t, x) = (t, \alpha(t')x)$. If (t, x) is a point of N , with $x \neq 0$, then (t, zx) is also a point of N for $z \in k^*$, and (t, x) is not isolated, hence N is not quasi-finite. One therefore has set-theoretically $N \subset T$, and the proof is finished.

Proof of 1.7. Let $\mathfrak{m}$ be the radical of A and $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, $n \geq 0$. Let us first show, by induction on n , that p_0 can be extended for each n to a morphism of S_n -groups

$$p_n : \mathbf{G}_{a,S_n} \longrightarrow G_{S_n}$$

normalized by T_{S_n} with multiplier α_{S_n} , the p_n satisfying in addition $p_{n+1} \times_{S_{n+1}} S_n = p_n$.

Let $H = T \cdot_\alpha \mathbf{G}_{a,S}$. The morphism $H_{S_n} \longrightarrow G_{S_n}$ defined by p_n is denoted f_n . Let us admit the following lemma:

Lemma 1.9. *If (G, T, α) is an elementary k -system, k algebraically closed, and $p : \mathbf{G}_{a,k} \longrightarrow G$ is a monomorphism normalized by T with multiplier α , one has*

$$H^2(T \cdot_\alpha \mathbf{G}_{a,k}, \mathfrak{g}) = 0.$$

(One makes $T \cdot_\alpha \mathbf{G}_{a,k}$ act on $\mathfrak{g}$ through the morphism $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow G$ defined by p , and the adjoint representation of G .)

Then, by Exp. III 2.8, f_n will extend to a morphism of S_{n+1} -groups

$$f'_{n+1} : H_{S_{n+1}} \longrightarrow G_{S_{n+1}}.$$

Now f'_{n+1} and the canonical immersion of $T_{S_{n+1}}$ into $G_{S_{n+1}}$ have the same restriction to T_{S_n} . By Exp. III 2.5, there exists an element $g \in G(S_{n+1})$ such that $g \times_{S_{n+1}} S_n = e$ and such that $f_{n+1} = \text{int}(g) \circ f'_{n+1}$ restricts to T_{n+1} by the canonical immersion of T_{n+1} . Let p_{n+1} be the restriction of f_{n+1} to $\mathbf{G}_{a,S_{n+1}}$. It is a morphism normalized by $T_{S_{n+1}}$ with multiplier $\alpha_{S_{n+1}}$, extending p_n .

One has therefore constructed a coherent system (f_n) , and one must now algebraize it. Now one has:

Lemma 1.10. *Let A be a complete local noetherian ring, $\mathfrak{m}$ its maximal ideal, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, T an S -torus, α a nonzero character of T , X an affine S -scheme on which T acts. Let T act on $\mathbf{G}_{a,S}$ through α . Let q be an integer ≥ 0 , and let $(f_n)_{n \geq 0}$ be a coherent system of morphisms*

$$f_n : \mathbf{G}_{a,S_n}^q \longrightarrow X_{S_n}$$

of objects with operators T_{S_n} . There exists a unique morphism of objects with operators T

$$f : \mathbf{G}_{a,S}^q \longrightarrow X$$

inducing the f_n (compare with Exp. IX 7.1).

Corollary 1.11. *If X is a group with operator group T and the f_n are morphisms of groups, f is also one.*

It suffices to apply the uniqueness assertion of the lemma to the two morphisms $\mathbf{G}_{a,S}^{2q} \longrightarrow X$ deduced from f in the usual way.

Proof of 1.10. Suppose T split, which is moreover the case in the application of 1.10 to the proof of 1.5. One knows (Exp. I 4.7.3.1) that $X \longmapsto \mathcal{A}(X)$ gives an equivalence between the category of affine S -schemes equipped with an action of T and the opposite of the category of S -algebras graded of type $M = \text{Hom}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$.

One therefore has gradings

$$B = \mathcal{A}(X) = \coprod_{m \in M} B_m \quad \text{and} \quad C = \mathcal{A}(\mathbf{G}_{a,S}^q) = \coprod_{m \in M} C_m.$$

One sees at once that each C_m is *free of finite type* over A . (Indeed, $C_m = 0$ if m is not a multiple of α , and if $m = d\alpha$, C_m is isomorphic to the A -module of homogeneous polynomials of degree d in q variables.) Put

$$\begin{aligned}\widehat{B}_m &= \varprojlim_n B_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{C}_m &= \varprojlim_n C_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{B} &= \prod_{m \in M} \widehat{B}_m, \quad \widehat{C} = \prod_{m \in M} \widehat{C}_m.\end{aligned}$$

One then has canonical morphisms of algebras graded of type M

$$g_B : B \longrightarrow \widehat{B} \quad \text{and} \quad g_C : C \longrightarrow \widehat{C}.$$

It follows from the remark above that g_C is an isomorphism. Giving a coherent system (f_n) as in the statement is equivalent to giving a morphism of graded A -algebras

$$\widehat{F} : \widehat{B} \longrightarrow \widehat{C}.$$

Finding a morphism f as in the statement is equivalent to finding a morphism of graded A -algebras $F : B \longrightarrow C$ making the diagram commute:

$$\begin{array}{ccc} B & \xrightarrow{F} & C \\ g_B \downarrow & & \downarrow g_C \\ \widehat{B} & \xrightarrow{\widehat{F}} & \widehat{C}. \end{array}$$

Since g_C is an isomorphism, the existence and uniqueness of F are immediate. This proves 1.10.

To finish the proof of 1.5, it therefore remains only to prove 1.9.

1.12. — *Proof of 1.9.* One has $\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}$. As explained in 1.9, consider $\mathfrak{g}$ as a $(T \cdot_\alpha \mathbf{G}_{a,k})$ -module. Clearly $\mathfrak{t} \oplus \mathfrak{g}^\alpha$ is a submodule of $\mathfrak{g}$, the quotient being isomorphic to $\mathfrak{g}^{-\alpha}$ as a k -vector space and even as a T -module. Clearly $\mathbf{G}_{a,k}$ acts trivially on this one-dimensional quotient (since every group morphism from $\mathbf{G}_{a,k}$ to $\mathbf{G}_{m,k}$ is trivial). Similarly, $\mathfrak{g}^\alpha$ is a submodule of $\mathfrak{t} \oplus \mathfrak{g}^\alpha$, the quotient being isomorphic to $\mathfrak{t}$ as a T -module, $\mathbf{G}_{a,k}$ acting trivially on it. To summarize:

Lemma 1.13. *Under the conditions of 1.9, $\mathfrak{g}$ has a composition series as a $(T \cdot_\alpha \mathbf{G}_{a,k})$ -module whose successive quotients are*

$$\mathfrak{g}^{-\alpha}, \quad \mathfrak{t}, \quad \mathfrak{g}^\alpha,$$

considered as $(T \cdot_\alpha \mathbf{G}_{a,k})$ -modules by the projection $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow T$.

One is therefore reduced to calculating the cohomology of $T \cdot_\alpha \mathbf{G}_{a,k}$ acting through the projection $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow T$ and the character β of T (here $\beta = 0$, α or $-\alpha$) on $W(k)$.⁽⁵⁾ Denote by $k[x_1, \dots, x_n]$ the polynomial algebra over k in n variables, and by $k_q[x_1, \dots, x_n]$ the subspace of homogeneous polynomials of degree q .

Lemma 1.14. *With the preceding notations, one has $H^n(T \cdot_\alpha \mathbf{G}_{a,k}, k) = H^n(C_{\alpha, \beta}^*)$, where the complex $C_{\alpha, \beta}^*$ is defined by*

$$C_{\alpha, \beta}^n = \begin{cases} k_q[x_1, \dots, x_n] & \text{if } \beta = q\alpha, \text{ with } q \in \mathbf{N}^*; \\ 0 & \text{otherwise,} \end{cases}$$

⁽⁵⁾**N.D.E.** The following sentence has been added.

and

$$\begin{aligned} \delta f(x_1, x_2, \dots, x_{n+1}) &= f(x_2, \dots, x_{n+1}) \\ &+ \sum_{i=1}^n (-1)^i f(x_1, \dots, x_i x_{i+1}, \dots, x_{n+1}) \\ &+ (-1)^{n+1} f(x_1, \dots, x_n). \end{aligned}$$

Indeed, the functor $M \mapsto H^0(T, M)$ is exact on the category of T -modules (and the $H^q(T, -)$ vanish), by Exp. I 5.3.2. It follows, as in the usual case of group cohomology, that $H^n(T \cdot_\alpha \mathbf{G}_{a,k}, k)$ can be calculated as the n th cohomology group of the complex of cochains from $\mathbf{G}_{a,k}$ to k , invariant under T , i.e. satisfying

$$f(\alpha(t)x_1, \dots, \alpha(t)x_n) = \beta(t)f(x_1, \dots, x_n).$$

This indeed gives the announced complex.

To prove 1.9, it therefore suffices to prove that $H^2(C_{\alpha,\beta}^*) = 0$ for $\beta = 0, \alpha, -\alpha$, which is immediate.

Remark 1.15. One can explicitly calculate the groups $H^n(C_{\alpha,\beta}^*)$ for $\beta = q\alpha$ (see M. Lazard, *Lois de groupes et analyseurs*, Annales E.N.S., 1955). In particular, one finds $H^n(C_{\alpha,q\alpha}^*) = 0$ for $n > q$.

Notation 1.16. The image of the canonical immersion

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^\alpha) \longrightarrow G$$

will be denoted Ω . It is an open subset of G containing the unit section. The image of

$$W(\mathfrak{g}^{-\alpha}), \quad \text{resp. } W(\mathfrak{g}^\alpha), \quad \text{resp. } W(\mathfrak{g}^{-\alpha}) \times_S T, \quad \text{resp. } T \times_S W(\mathfrak{g}^\alpha)$$

will be denoted⁽⁶⁾

$$U_{-\alpha}, \quad \text{resp. } U_\alpha, \quad \text{resp. } U_{-\alpha} \cdot T, \quad \text{resp. } T \cdot U_\alpha.$$

Then U_α (resp. $U_{-\alpha}$) is a subgroup of G canonically equipped with a vector bundle structure, and one has

$$\text{int}(t)(x) = x^{\alpha(t)} \quad (\text{resp. } x^{-\alpha(t)}),$$

for all $S' \longrightarrow S$, $t \in T(S')$, $x \in U_\alpha(S')$ (resp. $x \in U_{-\alpha}(S')$).

One has canonical isomorphisms

$$T \cdot U_\alpha \simeq T \cdot_\alpha U_\alpha \quad \text{and} \quad T \cdot U_{-\alpha} \simeq T \cdot_{-\alpha} U_{-\alpha}.$$

The open subset Ω is stable under $\text{int}(T)$: one has

$$\text{int}(t')(y \cdot t \cdot x) = y^{-\alpha(t')} \cdot t \cdot x^{\alpha(t')}.$$

Corollary 1.17. *One has $\mathcal{L}ie(U_\alpha/S) = \mathfrak{g}^\alpha$ and $\mathcal{L}ie(U_{-\alpha}/S) = \mathfrak{g}^{-\alpha}$. The isomorphisms*

$$W(\mathfrak{g}^\alpha) \xrightarrow[\sim]{\exp} U_\alpha \quad \text{and} \quad W(\mathfrak{g}^{-\alpha}) \xrightarrow[\sim]{\exp} U_{-\alpha}$$

are those of Exp. XIX 4.2.

Corollary 1.18. *The open subset Ω is relatively schematically dense in G (cf. XVIII, § 1).*

Clear by Exp. XVIII, 1.3.⁽⁷⁾

Corollary 1.19. *The center of G is $\text{Centr}(G) = \text{Ker}(\alpha)$. It is therefore a closed subgroup of G , of multiplicative type and of finite type.*

The second assertion follows from the first by Exp. IX 2.7. Let us therefore prove the latter. The inner automorphism defined by a section of $\text{Ker}(\alpha)$ acts trivially on Ω (last

⁽⁶⁾**N.D.E.** $P_{-\alpha}$ and P_α have been replaced by $U_{-\alpha}$ and U_α .

⁽⁷⁾**N.D.E.** See also EGA IV₃, 11.10.10.

formula of 1.16), hence on G by 1.18. Conversely, if $g \in G(S)$ centralizes G , it centralizes T and U_α , hence is a section of T (Exp. XIX 2.8) annihilating α ; since this also holds after every base change, one indeed has $\text{Centr}(G) = \text{Ker}(\alpha)$.

Corollary 1.20. *For there to exist a monomorphism $p_\alpha : \mathbf{G}_{a,S} \rightarrow G$ normalized by T with multiplier α , it is necessary and sufficient that the $\mathcal{O}_S$ -module $\mathfrak{g}^\alpha$ be free. More precisely, one has a bijection given by*

$$X_\alpha \mapsto (x \mapsto \exp(xX_\alpha)) \quad \text{and} \quad p_\alpha \mapsto \text{Lie}(p_\alpha)$$

between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of monomorphisms p_α as above (which is also the set of isomorphisms of vector group schemes $\mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$).⁽⁸⁾

Corollary 1.21. *The subgroups U_α and $U_{-\alpha}$ of G do not commute on any fiber.*

Indeed, if $(U_\alpha)_s$ and $(U_{-\alpha})_s$ commute, Ω_s is a subgroup of G_s , hence $\Omega_s = G_s$ ⁽⁹⁾ and G_s is solvable, which contradicts the hypothesis that G_s is reductive of semisimple rank 1.

2. Structure of elementary systems

Theorem 2.1. *Let S be a scheme, (G, T, α) an elementary S -system. There exist a morphism of $\mathcal{O}_S$ -modules*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \rightarrow \mathcal{O}_S, \quad (X, Y) \mapsto \langle X, Y \rangle,$$

and a morphism of S -groups

$$\alpha^* : \mathbf{G}_{m,S} \rightarrow T$$

such that for every $S' \rightarrow S$ and all $X \in \Gamma(S', \mathfrak{g}^\alpha \otimes \mathcal{O}_{S'})$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha} \otimes \mathcal{O}_{S'})$ one has the equivalence

$$\exp(X) \cdot \exp(Y) \in \Omega(S') \iff 1 + \langle X, Y \rangle \in \mathbf{G}_m(S'),$$

and under these conditions one has the formula:

$$(F) \quad \boxed{\exp(X) \cdot \exp(Y) = \exp\left(\frac{Y}{1 + \langle X, Y \rangle}\right) \alpha^*(1 + \langle X, Y \rangle) \exp\left(\frac{X}{1 + \langle X, Y \rangle}\right).}$$

Moreover, the morphisms $(X, Y) \mapsto \langle X, Y \rangle$ and α^ are uniquely determined, the first is an isomorphism, hence puts the modules $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ in duality, and one has $\alpha \circ \alpha^* = 2$ (squaring in $\mathbf{G}_{m,S}$).*

In view of the uniqueness assertions of the theorem, it suffices to give the proof locally on S . One may therefore suppose that $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free over S . Take then $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$, and put $p_\alpha(x) = \exp(xX)$, $p_{-\alpha}(y) = \exp(yY)$ for $x, y \in \mathbf{G}_a(S')$, $S' \rightarrow S$. By 1.5 and 1.21, it suffices to prove:

Proposition 2.2. *Let S be a scheme, G an S -group, T a torus of G , α a character of T nontrivial on each fiber, $p_\alpha : \mathbf{G}_{a,S} \rightarrow G$ (resp. $p_{-\alpha} : \mathbf{G}_{a,S} \rightarrow G$) a monomorphism of groups normalized by T with multiplier α (resp. $-\alpha$). Suppose that:*

- (i) *The morphism $\mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \rightarrow G$ defined by $(y, t, x) \mapsto p_{-\alpha}(y)tp_\alpha(x)$ is an open immersion. Denote its image by Ω .*
- (ii) *For every $s \in S$, $(p_\alpha)_s(\mathbf{G}_{a,\kappa(s)})$ and $(p_{-\alpha})_s(\mathbf{G}_{a,\kappa(s)})$ do not commute.*

Then there exist uniquely determined $a \in \mathbf{G}_a(S)$ and $\alpha^ \in \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$ with the following properties: for every $S' \rightarrow S$ and all $x, y \in \mathbf{G}_a(S')$, one has*

$$p_\alpha(x)p_{-\alpha}(y) \in \Omega(S') \iff 1 + axy \in \mathbf{G}_m(S'),$$

⁽⁸⁾**N.D.E.** Indeed, on the one hand, $\text{Lie}(\mathbf{G}_{a,S}) = \mathcal{O}_S$, and $\text{Lie}(p_\alpha)$ is an element of $\text{Hom}_{\mathcal{O}_S}(\mathcal{O}_S, \mathfrak{g}^\alpha) = \Gamma(S, \mathfrak{g}^\alpha)$.

⁽⁹⁾**N.D.E.** By 1.18.

and, under this condition, one has the formula

$$p_\alpha(x)p_{-\alpha}(y) = p_{-\alpha}\left(\frac{y}{1+axy}\right)\alpha^*(1+axy)p_\alpha\left(\frac{x}{1+axy}\right).$$

Moreover, a is invertible (i.e. $a \in \mathbf{G}_m(S)$) and $\alpha \circ \alpha^* = 2$.

Proof:

A) Consider the morphism

$$\mathbf{G}_{a,S}^2 \longrightarrow G$$

defined by $(x, y) \mapsto p_\alpha(x)p_{-\alpha}(y)$. Let U be the inverse image of Ω by this morphism. It is an open subset of $\mathbf{G}_{a,S}^2$ containing $0 \times_S \mathbf{G}_{a,S}$ and $\mathbf{G}_{a,S} \times_S 0$. There therefore exist uniquely determined morphisms of S -schemes

$$A : U \longrightarrow \mathbf{G}_{a,S}, \quad C : U \longrightarrow \mathbf{G}_{a,S}, \quad B : U \longrightarrow T$$

satisfying the set-theoretic relation:

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(A(u, v))B(u, v)p_\alpha(C(u, v)).$$

One immediately has the relations

$$\begin{aligned} A(0, v) &= v, & A(u, 0) &= 0, & C(u, 0) &= u, & C(0, v) &= 0, \\ B(u, 0) &= B(0, v) = e. \end{aligned}$$

Let S' be a separated S -scheme and $t \in T(S')$ a point of T . Since $\Omega_{S'}$ is stable under $\text{int}(t)$, by the last formula of 1.16, $U_{S'}$ is stable under the automorphism $(x, y) \mapsto (\alpha(t)x, \alpha(t)^{-1}y)$ of $\mathbf{G}_{a,S'}^2$, and one has the relations:

$$\begin{aligned} A(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)^{-1}A(u, v), \\ C(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)C(u, v), \\ B(\alpha(t)u, \alpha(t)^{-1}v) &= B(u, v). \end{aligned}$$

Since α is faithfully flat, one deduces that for every $S' \rightarrow S$ and every $z \in \mathbf{G}_m(S')$, $U_{S'}$ is stable under the transformation $(x, y) \mapsto (zx, z^{-1}y)$ and one has

$$\begin{aligned} A(zu, z^{-1}v) &= z^{-1}A(u, v), & C(zu, z^{-1}v) &= zC(u, v), \\ B(zu, z^{-1}v) &= B(u, v). \end{aligned}$$

Suppose first that v is invertible; putting $z = v$, one deduces that if (u, v) is a section of U , then $(uv, 1)$ is also one, and one has

$$A(uv, 1) = v^{-1}A(u, v), \quad B(uv, 1) = B(u, v).$$

Let then V be the open subset of $\mathbf{G}_{a,S}^2$ defined by⁽¹⁰⁾

$$(u, v) \in V(S') \iff (u, v), (uv, 1) \text{ and } (1, uv) \text{ belong to } U(S').$$

Since U is an open subset of $\mathbf{G}_{a,S}^2$ containing $0 \times_S \mathbf{G}_{a,S}$ and $\mathbf{G}_{a,S} \times_S 0$, V is a neighborhood of the zero section of $\mathbf{G}_{a,S}^2$, and we have just seen that the morphisms

$$\begin{aligned} (u, v) &\mapsto A(u, v) \quad \text{and} \quad (u, v) \mapsto vA(uv, 1), \\ \text{resp. } (u, v) &\mapsto B(u, v) \quad \text{and} \quad (u, v) \mapsto B(uv, 1) \end{aligned}$$

coincide in $V \cap (\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S})$. Since $\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}^2$, these morphisms therefore coincide in V .

One knows that $A(0, 1) = 1$; it follows that there exists an open subset W_1 of $\mathbf{G}_{a,S}$ containing the zero section, such that for every section x of W_1 , $A(x, 1)$ is invertible; putting $A(x, 1)^{-1} = F(x)$, one obtains that if $(u, v) \in V(S')$ ⁽¹¹⁾ and $uv \in W_1(S')$, $S' \rightarrow S$, then

⁽¹⁰⁾**N.D.E.** The condition “ $(1, uv) \in U(S')$ ” has been added.

⁽¹¹⁾**N.D.E.** Here and in what follows, $U(S')$ has been replaced by $V(S')$.

$A(u, v) = vA(uv, 1) = vF(uv)^{-1}$. Reasoning similarly for C , one obtains that there exist an open subset W_2 of $\mathbf{G}_{a,S}$ containing the zero section and an element $E^{(12)}$ of $\mathcal{O}(W_2)^\times$ such that $C(u, v) = uC(1, uv) = uE(uv)^{-1}$ if $(u, v) \in V(S')$ and $uv \in W_2(S')$. Consequently, putting $W = W_1 \cap W_2$, one obtains:

There exist an open subset W of $\mathbf{G}_{a,S}$ containing the zero section, and S -morphisms

$$\begin{aligned} F : W &\longrightarrow \mathbf{G}_{m,S}, & F(0) &= 1, \\ H : W &\longrightarrow T, & H(0) &= e, \\ E : W &\longrightarrow \mathbf{G}_{m,S}, & E(0) &= 1, \end{aligned}$$

such that if $(u, v) \in V(S')$ and $uv \in W(S')$, $S' \longrightarrow S$, one has

$$(+) \quad p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(vF(uv)^{-1})H(uv)p_\alpha(uE(uv)^{-1}).$$

B) Let us now use associativity in G to write

$$p_\alpha(u)p_{-\alpha}(v)p_{-\alpha}(w) = p_\alpha(u)p_{-\alpha}(v+w).$$

There exists an open subset L of $\mathbf{G}_{a,S}^3$, containing the unit section, such that $(u, v, w) \in L(S')$ is equivalent to

$$\begin{aligned} (u, v) \in V(S'), \quad (uE(uv)^{-1}, w) \in V(S'), \quad (u, v+w) \in V(S'), \\ uv \in W(S'), \quad uwE(uv)^{-1} \in W(S'), \quad u(v+w) \in W(S'). \end{aligned}$$

Using formula (+), one writes at once, for $(u, v, w) \in L(S')$, the relations:

$$\begin{aligned} (1) \quad E(uv + uw) &= E(uwE(uv)^{-1})E(uv), \\ (2) \quad H(uv + uw) &= H(uwE(uv)^{-1})H(uv), \\ (3) \quad (v+w)F(uv + uw)^{-1} &= \alpha(H(uv)^{-1})wF(uwE(uv)^{-1})^{-1} \\ &\quad + vF(uv)^{-1}. \end{aligned}$$

It is immediate from the definition of L that $(1, 0, 0) \in L(S)$. Consider therefore

$$L \cap (1 \times_S \mathbf{G}_{a,S} \times_S \mathbf{G}_{a,S}) = 1 \times_S M;$$

M is an open subset of $\mathbf{G}_{a,S}^2$ containing the section $(0, 0)$, and, for $(v, w) \in M(S')$, one has $v, wE(v)^{-1}, v+w \in W(S')$ and

$$\begin{aligned} (1') \quad E(v+w) &= E(wE(v)^{-1})E(v), \\ (2') \quad H(v+w) &= H(wE(v)^{-1})H(v), \\ (3') \quad (v+w)F(v+w)^{-1} &= \alpha(H(v))^{-1}wF(wE(v)^{-1})^{-1} + vF(v)^{-1}. \end{aligned}$$

Finally, consider the morphism from M to $\mathbf{G}_{a,S}^2$ defined set-theoretically by $(v, w) \mapsto (v, wE(v)^{-1})$.⁽¹³⁾ It preserves the section $(0, 0)$ and induces an isomorphism from M onto an open subset N of $\mathbf{G}_{a,S}^2$ containing the zero section (the inverse isomorphism being given by $(x, y) \mapsto (x, yE(x))$).⁽¹⁴⁾ One has therefore proved the following assertion:

There exists an open subset N of $\mathbf{G}_{a,S}^2$ containing the zero section such that if $(x, y) \in N(S')$, then x, y and $x + yE(x)$ ⁽¹⁵⁾ belong to $W(S')$ and:

$$\begin{aligned} (1'') \quad E(x + yE(x)) &= E(x)E(y), \\ (2'') \quad H(x + yE(x)) &= H(x)H(y), \\ (3'') \quad (x + yE(x))F(x + yE(x))^{-1} &= xF(x)^{-1} + \alpha(H(x))^{-1}yE(x)F(y)^{-1}. \end{aligned}$$

⁽¹²⁾**N.D.E.** The element denoted G in the original has been denoted E , since G already denotes the S -group under consideration.

⁽¹³⁾**N.D.E.** The following has been corrected.

⁽¹⁴⁾**N.D.E.** I.e. one has made the “change of variables” $x = v, y = wE(v)^{-1}$, that is, $v = x, w = yE(x)$.

⁽¹⁵⁾**N.D.E.** $yE(x)$ has been corrected to $x + yE(x)$.

C) Reasoning similarly with associativity on the left, one proves the following assertion:⁽¹⁶⁾

There exists an open subset N' of $\mathbf{G}_{a,S}^2$ containing the zero section such that if $(x, y) \in N'(S')$, then x, y and $x + yF(x)$ ⁽¹⁷⁾ belong to $W(S')$, and

$$(4'') \quad F(x + yF(x)) = F(x)F(y),$$

$$(5'') \quad H(x + yF(x)) = H(x)H(y),$$

$$(6'') \quad (x + yF(x))E(x + yF(x))^{-1} = xE(x)^{-1} + \alpha(H(x))^{-1}yF(x)E(y)^{-1}.$$

We are therefore led to solve the “functional equation” (1'').

Lemma 2.3. *Let S be a scheme, W an open subset of $\mathbf{G}_{a,S}$ containing the unit section, $F : W \rightarrow \mathbf{G}_{m,S}$ an S -morphism. Suppose that $F(0) = 1$ and that there exists an open subset N of $\mathbf{G}_{a,S}^2$ containing the zero section such that for $(x, y) \in N(S')$, x, y and $x + yF(x)$ ⁽¹⁷⁾ belong to $W(S')$ and one has:*

$$(\dagger) \quad F(x + yF(x)) = F(x)F(y).$$

(i) *If S is the spectrum of a field k , there exists $a \in k$ such that $F(x) = 1 + ax$.*

(ii) *If $a = F'(0) \in \Gamma(S, \mathcal{O}_S)$ is invertible, then $F(x) = 1 + ax$.*

By the hypotheses, we can differentiate the given equation at $x = 0$ (resp. at $y = 0$), and find that

$$(*) \quad F'(y)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (0, y) \in N(S'),$$

resp.

$$F'(x)F(x) = F(x)F'(0) \quad \text{for } (x, 0) \in N(S').$$

Since F takes values in $\mathbf{G}_m$, the second relation gives

$$(*') \quad F'(x) = F'(0) \quad \text{for } (x, 0) \in N(S');$$

whence, by the first,

$$F'(0)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (y, 0), (0, y) \in N(S').$$

If $a = F'(0)$ is invertible, this gives

$$F(y) = 1 + ay,$$

for y a section of an open subset of W containing the unit section, hence schematically dense in W , which proves (ii). This also proves (i) when $F'(0) \neq 0$.

If $F'(0) = 0$, then, by (*'), $F'(x) = 0$ when x is “near 0”, hence $F' = 0$ by schematic density. If k has characteristic 0, F is a rational fraction with zero derivative, hence constant and equal to $F(0) = 1$.

If k has characteristic p and F is not constant,⁽¹⁸⁾ there exist an integer $n > 0$ and a rational fraction $F_1 \in k(X)$ such that $F'_1(X) \neq 0$ and

$$F(X) = F_1(X^{p^n}) = F_1(X)^{p^n}.$$

Substituting in the functional equation, one finds

$$(\dagger_1) \quad F_1(x + yF_1(x)^{p^n}) = F_1(x)F_1(y).$$

Differentiating at $x = 0$, one finds

$$(*_1) \quad F'_1(y) = F'_1(0)F_1(y),$$

and differentiating $(\dagger_1)$ at $y = 0$, one obtains

$$(*'_1) \quad F'_1(x)F_1(x)^{p^n} = F_1(x)F'_1(0).$$

⁽¹⁶⁾**N.D.E.** I.e. one writes the equalities resulting from $p_\alpha(t)p_\alpha(u)p_{-\alpha}(v) = p_\alpha(t+u)p_{-\alpha}(v)$, and puts $v = 1$ and $x = u$, $t = yF(u)$ (i.e. $y = tF(u)^{-1}$).

⁽¹⁷⁾**N.D.E.** $yF(x)$ has been corrected to $x + yF(x)$.

⁽¹⁸⁾**N.D.E.** The following has been corrected.

Since, by hypothesis, $F'_1(X)$ is an invertible element of $k(X)$, one deduces from these two equalities that

$$F_1(X)^{p^n} = 1,$$

hence F_1 is a constant, contradicting the initial hypothesis. This shows that F is constant and equal to $1 = F(0)$.

D) Suppose that S is the spectrum of a field. If $F'(0) = 0$, then $F = 1$. Formula (5'') then gives $H(x+y) = H(x)H(y)$, which shows that H extends to a morphism of groups $\mathbf{G}_{a,S} \rightarrow T$ (Exp. XVIII 2.3), which is necessarily constant with value e . On the other hand, by lemma 2.3, one will also have $E(x) = 1 + bx$ for some $b \in k$. But then (6'') gives, for $(x, y) \in N(S')$,

$$(x+y)E(x+y)^{-1} = xE(x)^{-1} + yE(y)^{-1},$$

hence, ⁽¹⁹⁾ by Exp. XVIII 2.3 again, $x \mapsto xE(x)^{-1}$ extends to a morphism of k -groups $\mathbf{G}_{a,k} \rightarrow \mathbf{G}_{a,k}$, hence $x/(1+bx) = cx$ for some $c \in k$, whence $b = 0$ (and $c = 1$).

This shows that F, H, E are constant with values $(1, e, 1)$ in a neighborhood of the unit section, hence everywhere, which by (+) shows that U_α and $U_{-\alpha}$ commute, contrary to hypothesis (ii).

If S is now arbitrary, one has therefore proved that $F'(0)$ vanishes on no fiber, hence is invertible. The same is clearly true of $E'(0)$, which, by lemma 2.3, shows that there exist $a, b \in \mathbf{G}_m(S)$ such that

$$(\diamond_1) \quad F(x) = 1 + ax, \quad E(x) = 1 + bx, \quad \text{for } x \in W(S').$$

E) The rest is now easy. Substituting the preceding results in (3''), one finds

$$y\alpha(H(x))(1+ay) = y(1+ax+ay(1+bx))(1+ax).$$

This formula is valid for every section (x, y) of N . But since $\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}^2$, one deduces

$$(1+ay)\alpha(H(x)) = (1+ax+ay(1+bx))(1+ax).$$

Putting $y = 0$ gives $\alpha(H(x)) = (1+ax)^2$. Substituting this in the preceding equality, one finds ⁽²⁰⁾

$$a^2xy = abxy.$$

Since $\mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}$, one deduces $a^2 = ab$, whence, since a is invertible,

$$(\diamond_2) \quad a = b \quad \text{and} \quad \alpha(H(x)) = (1+ax)^2.$$

Since a is invertible, $x \mapsto 1+ax$ is an automorphism of $\mathbf{G}_{a,S}$; one can therefore find an open subset W' of $\mathbf{G}_{a,S}$ containing the section 1 and a morphism

$$P : W' \rightarrow T$$

such that $P(1+ax) = H(x)$. ⁽²¹⁾

Substituting in relation (2'), one finds at once, for $(x, y) \in N(S')$,

$$P(1+ax+ay) = P\left(\frac{1+ax+ay}{1+ax}\right)P(1+ax),$$

which proves that there exists an open neighborhood of 1 in $\mathbf{G}_{m,S}$ such that for x and y in this neighborhood one has $P(x)P(y) = P(xy)$. By Exp. XVIII 2.3, there exists a morphism of groups

$$(\diamond_3) \quad \alpha^* : \mathbf{G}_{m,S} \rightarrow T$$

⁽¹⁹⁾**N.D.E.** The following sentence has been added. One can also see by a direct calculation that the preceding equality implies $0 = xyb(2 + (x+y)b)$, whence $0 = b(2 + (x+y)b)$, and finally $b = 0$.

⁽²⁰⁾**N.D.E.** In the preceding three equalities, the original has been corrected by replacing the factor $(1+bx)$ on the right by $1+ax$. Substituting the third equality in the second and taking account of the invertibility of $1+ax$, one indeed obtains $a^2xy = abxy$.

⁽²¹⁾**N.D.E.** I.e. one has made the change of variables $x' = 1+ax$, that is, $x = (x' - 1)/a$.

extending P . Since $\alpha(H(x)) = (1 + ax)^2$ in a neighborhood of the zero section, one has $\alpha(\alpha^*(z)) = z^2$ in a neighborhood of the section 1, hence

$$(\diamond_4) \quad \alpha \circ \alpha^* = 2.$$

F) ⁽²²⁾Collecting results (+) and $(\diamond_1 - \diamond_4)$, one sees that there exist $a \in \mathbf{G}_m(S)$ and $\alpha^* \in \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$ such that $\alpha \circ \alpha^* = 2$, and such that, if $(u, v) \in V(S')$ and $uv \in W(S')$, then $1 + auv$ is invertible and

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\left(\frac{v}{1 + auv}\right)\alpha^*(1 + auv)p_\alpha\left(\frac{u}{1 + auv}\right).$$

Consider the open subset V' of $\mathbf{G}_{a,S}^2$ defined by “ $1 + auv$ invertible”, i.e. $V' = (\mathbf{G}_{a,S}^2)_f$, where $f(u, v) = 1 + auv$. Both sides of the preceding formula define morphisms from V' to G which coincide in a neighborhood of the zero section, hence coincide in V' . The preceding formula is therefore valid for every section (u, v) of V' . In particular, it follows that $V' \subset U$, where U is the open subset introduced at the beginning of A).

Let us prove that $U = V'$. Returning to the notations of A), one has a morphism

$$A : U \longrightarrow \mathbf{G}_{a,S}$$

which, on V' , is defined by $A(u, v) = v(1 + auv)^{-1}$. To show that $U = V'$, which is a set-theoretic question, one reduces to the case where S is the spectrum of a field k , hence to the following obvious assertion: the domain of definition of the rational map $\mathbf{G}_{a,k}^2 \longrightarrow \mathbf{G}_{a,k}$ defined by the rational fraction $Y/(1 + aXY)$ is the open subset defined by the function $1 + aXY$.

G) One has therefore proved the existence of a and α^* , as well as the two supplementary properties announced. It remains to prove uniqueness. Let therefore a' and $\alpha^{*'} also satisfy the required conditions. If $u, v \in \mathbf{G}_a(S')^2$, one has at once:$

$$1 + auv \text{ invertible} \Rightarrow 1 + a'uv \text{ invertible and } \frac{v}{1 + auv} = \frac{v}{1 + a'uv};$$

one therefore has for every section u of $\mathbf{G}_a(S')$

$$1 + au \text{ invertible} \Rightarrow 1 + au = 1 + a'u,$$

which proves at once that $a = a'$.

With the same notations, one then has

$$1 + au \text{ invertible} \Rightarrow \alpha^*(1 + au) = \alpha^{*'}(1 + au),$$

hence also $\alpha^* = \alpha^{*'}$.

Corollary 2.4. *Let $\exp(Y)t\exp(X)$ and $\exp(Y')t'\exp(X')$ be two elements of $\Omega(S')$. Then their product is in $\Omega(S')$ if and only if $u = 1 + \langle X, Y' \rangle$ is invertible, and one then has*

$$(F') \quad \begin{aligned} \exp(Y)t\exp(X) \cdot \exp(Y')t'\exp(X') &= \exp(Y + u^{-1}\alpha(t)^{-1}Y') \\ &\quad \cdot tt'\alpha^*(u) \cdot \exp(u^{-1}\alpha(t')^{-1}X + X'). \end{aligned}$$

Remark 2.5. One can also write formula (F) of theorem 2.1 without using the morphisms $\exp$. Indeed, transporting by these morphisms the duality $\mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} \longrightarrow \mathcal{O}_S$, one obtains a canonical pairing of vector bundles:

$$U_\alpha \times_S U_{-\alpha} \longrightarrow \mathbf{G}_{a,S},$$

which we shall again denote by $(x, y) \longmapsto \langle x, y \rangle$. One therefore has

$$\langle \exp X, \exp Y \rangle = \langle X, Y \rangle.$$

⁽²²⁾**N.D.E.** The following has been slightly modified, since an open subset V has already been introduced in A).

If $x \in U_\alpha(S')$, $y \in U_{-\alpha}(S')$ and $1 + \langle x, y \rangle \in \mathbf{G}_m(S')$, one has

$$(F) \quad x \cdot y = y^{(1+\langle x, y \rangle)^{-1}} \cdot \alpha^*(1 + \langle x, y \rangle) \cdot x^{(1+\langle x, y \rangle)^{-1}}.$$

Corollary 2.6. *The pairing*

$$W(\mathfrak{g}^\alpha) \times_S W(\mathfrak{g}^{-\alpha}) \longrightarrow \mathbf{G}_{a,S}$$

defines a pairing of principal bundles under $\mathbf{G}_{m,S}$

$$W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times \longrightarrow \mathbf{G}_{m,S}.$$

This pairing will be denoted $(X, Y) \mapsto \langle X, Y \rangle$, or more simply $(X, Y) \mapsto XY$.

For every section $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, there therefore exists a unique section X^{-1} of $\Gamma(S, \mathfrak{g}^{-\alpha})^\times$ such that $XX^{-1} = 1$. One has $(zX)^{-1} = z^{-1}X^{-1}$. The morphism

$$s : W(\mathfrak{g}^\alpha)^\times \longrightarrow W(\mathfrak{g}^{-\alpha})^\times$$

thus defined is therefore an isomorphism of schemes, compatible with the isomorphism $s : z \mapsto z^{-1}$ on the operator groups.

Definition 2.6.1. One will say that X and $s(X) = X^{-1}$ are *paired*.

Apply corollary 2.4 to $Y = 0 = X'$ and $Y' = aX^{-1}$, $a \in \mathcal{O}_S(S)$. Then $u = 1 + a$ and $u^{-1}Y' = u^{-1}(u - 1)X^{-1} = (1 - u^{-1})X^{-1}$, whence:

Corollary 2.7. *Let $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and $u \in \Gamma(S, \mathcal{O}_S)^\times$. One has*

$$\alpha^*(u) = \exp((u^{-1} - 1)X^{-1}) \exp(X) \exp((u - 1)X^{-1}) \exp(-u^{-1}X).$$

Definition 2.8. The morphism α^* is called the *coroot* associated with the root α .

Remark 2.9. If (G, T, α) is an elementary S -system, $(G, T, -\alpha)$ is also one. One therefore has by theorem 2.1 a duality between $\mathfrak{g}^{-\alpha}$ and $\mathfrak{g}^\alpha$, and a coroot $(-\alpha)^*$. Taking the inverse of formula (F), one proves at once

$$\langle X, Y \rangle = \langle Y, X \rangle, \quad (-\alpha)^* = -\alpha^*.$$

Let us now pass to the Lie algebra of G . The root α and the coroot α^* define the linear forms

$$\mathcal{O}_S \xrightarrow{\bar{\alpha}^*} \mathfrak{t} \xrightarrow{\bar{\alpha}} \mathcal{O}_S.$$

One puts $H_\alpha = \bar{\alpha}^*(1)$. One calls $\bar{\alpha}$ the *infinitesimal root* associated with α , and H_α the corresponding *infinitesimal coroot*.

Lemma 2.10. *Let $S' \longrightarrow S$, and $X, X' \in W(\mathfrak{g}^\alpha)(S')$, $H \in W(\mathfrak{t})(S')$, $Y, Y' \in W(\mathfrak{g}^{-\alpha})(S')$, $t \in T(S')$. One has*

- (1) $\text{Ad}(t)H = H, \quad \text{Ad}(t)X = \alpha(t)X, \quad \text{Ad}(t)Y = \alpha(t)^{-1}Y.$
- (2) $\begin{cases} \text{Ad}(\exp(X))H = H - \bar{\alpha}(H)X, & \text{Ad}(\exp(X))X' = X', \\ \text{Ad}(\exp(X))Y = Y + \langle X, Y \rangle H_\alpha - \langle X, Y \rangle X. \end{cases}$
- (2') $\begin{cases} \text{Ad}(\exp(Y))H = H + \bar{\alpha}(H)Y, & \text{Ad}(\exp(Y))Y' = Y', \\ \text{Ad}(\exp(Y))X = X + \langle X, Y \rangle H_{-\alpha} - \langle X, Y \rangle Y. \end{cases}$
- (3) $[H, X] = \bar{\alpha}(H)X, \quad [H, Y] = -\bar{\alpha}(H)Y, \quad [X, Y] = \langle X, Y \rangle H_\alpha.$
- (4) $H_{-\alpha} = -H_\alpha.$
- (5) $\bar{\alpha}(H_\alpha) = 2.$

The proof of these various formulas is either trivial or an immediate consequence of formula (F) of 2.1.

Corollary 2.11. *Suppose that H_α is nonzero on every fiber (which is in particular the case if 2 is invertible on S , by (5)). Then $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ are paired if and only if $[X_\alpha, X_{-\alpha}] = H_\alpha$.*

2.12. — Let (G, T, α) be an elementary S -system. We know (1.19) that the center of G is $\text{Centr}(G) = \text{Ker}(\alpha)$, a group of multiplicative type and of finite type. If Q is a subgroup of multiplicative type of $\text{Centr}(G)$, the quotient G/Q is affine over S (Exp. IX 2.5), smooth over S (Exp. VI_B 9.2), with connected reductive fibers of semisimple rank 1 (Exp. XIX 1.8).

Put $G' = G/Q$; it is a reductive S -group of semisimple rank 1; $T' = T/Q$ is a maximal torus of it. The open subset $U_{-\alpha} \cdot T \cdot U_\alpha$ of G is stable under Q , and one sees at once that the quotient is isomorphic to $U_{-\alpha} \times_S (T/Q) \times_S U_\alpha$. If α' denotes the character of T' induced by α , it follows that the derived morphism of the canonical morphism $G \rightarrow G'$ induces isomorphisms

$$\mathfrak{g}^\alpha \xrightarrow{\sim} \mathfrak{g}'^{\alpha'} \quad \text{and} \quad \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}.$$

In particular, α' is a root of G' with respect to T' . Thus, denoting by α/Q the character $T/Q \rightarrow \mathbf{G}_{m,S}$ induced by α , one has:

Lemma 2.13. *If Q is a subgroup of multiplicative type of $\text{Ker}(\alpha)$, then*

$$(G/Q, T/Q, \alpha/Q)$$

is an elementary system.

Lemma 2.14. *Under the preceding conditions, the following diagrams commute:*

$$\begin{array}{ccccc} W(\mathfrak{g}^\alpha) & \xrightarrow{\text{exp}} & G & \xleftarrow{\text{exp}} & W(\mathfrak{g}^{-\alpha}) \\ \text{can} \downarrow \simeq & & \downarrow \text{can} & & \simeq \downarrow \text{can} \\ W(\mathfrak{g}'^{\alpha'}) & \xrightarrow{\text{exp}} & G' & \xleftarrow{\text{exp}} & W(\mathfrak{g}'^{-\alpha'}) \\ \\ \mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} & \xrightarrow{\sim} & \mathcal{O}_S & & \\ \text{can} \downarrow \simeq & & \downarrow \text{id} & & \\ \mathfrak{g}'^{\alpha'} \otimes \mathfrak{g}'^{-\alpha'} & \xrightarrow{\sim} & \mathcal{O}_S & & \\ \\ \begin{array}{ccc} & T & \\ \alpha^* \nearrow & \downarrow \text{can} & \searrow \alpha \\ \mathbf{G}_{m,S} & & \mathbf{G}_{m,S} \\ \alpha'^* \searrow & \downarrow & \nearrow \alpha' \\ & T' & \end{array} & & \end{array}.$$

3. The Weyl group

Notation 3.0. ⁽²³⁾If (G, T, α) is an elementary S -system, one puts

$$N = \text{Norm}_G(T), \quad W = \text{Norm}_G(T)/T$$

(cf. Exp. XIX 6.3); N is a closed subgroup of G , smooth over S . One denotes by $N^\times = N - T$ the open subscheme of N induced on the complement of T .⁽²⁴⁾ Let R be the (unique) maximal torus of $\text{Ker}(\alpha)$, and T' the image of $\alpha^* : \mathbf{G}_{m,S} \rightarrow T$, which is a one-dimensional subtorus of T .

⁽²³⁾**N.D.E.** The numbering 3.0 has been added for subsequent references.

⁽²⁴⁾**N.D.E.** Q has been replaced by the more suggestive notation $N^\times$.

The morphism

$$T' \times_S R \longrightarrow T$$

induced by the product in T is surjective (hence faithfully flat); indeed, one reduces to checking this on geometric fibers, and it follows at once from the formula $\alpha \circ \alpha^* = 2$.

Theorem 3.1. *With the preceding notations:*

- (i) W is isomorphic to the constant group $(\mathbf{Z}/2\mathbf{Z})_S$.
- (ii) $N^\times$ is a locally trivial principal homogeneous bundle under T , on the left by the law $(t, q) \mapsto tq$ (resp. on the right by the law $(q, t) \mapsto qt$).
- (iii) One has the formula

$$\text{int}(w)t = t \cdot \alpha^*(\alpha(t)^{-1})$$

for $w \in N^\times(S')$, $t \in T(S')$, $S' \rightarrow S$. In the decomposition $T_{S'} = T'_{S'} \cdot R_{S'}$, $\text{int}(w)$ induces the identity on $R_{S'}$ and reflection on $T'_{S'}$. One has the relations

$$\alpha \circ \text{int}(w) = \alpha^{-1}, \quad \text{int}(w) \circ \alpha^* = (\alpha^*)^{-1}.$$

- (iv) For $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$, put

$$w_\alpha(X) = \exp(X) \exp(-X^{-1}) \exp(X).$$

Then $w_\alpha(X) \in N^\times(S')$, and the morphism $w_\alpha : W(\mathfrak{g}^\alpha)^\times \rightarrow N^\times$ thus defined satisfies

$$w_\alpha(zX) = \alpha^*(z)w_\alpha(X) = w_\alpha(X)\alpha^*(z)^{-1},$$

for $z \in \mathbf{G}_m(S')$, $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$.

- (v) ⁽²⁵⁾For $X, Y \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$, with the notations of 2.6, one has

$$w_\alpha(X)w_\alpha(Y) = \alpha^\vee(-XY^{-1}).$$

In particular,

$$w_\alpha(X)^2 = \alpha^*(-1) \in {}_2T(S) \cap \underline{\text{Centr}}(G)(S),$$

$$w_\alpha(X)^{-1} = w_\alpha(-X) = \alpha^*(-1)w_\alpha(X).$$

- (vi) If one similarly defines, for $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$,

$$w_{-\alpha}(Y) = \exp(Y) \exp(-Y^{-1}) \exp(Y),$$

one has (in addition to formulas analogous to the preceding ones)

$$w_{-\alpha}(X^{-1}) = w_\alpha(X)^{-1} = w_\alpha(-X),$$

$$w_\alpha(X)w_{-\alpha}(Y) = \alpha^*(XY).$$

Proof. (i) was already seen in Exp. XIX 2.4; it follows at once that $N^\times$ is indeed a principal homogeneous bundle under T for the laws defined in (ii); its local triviality (for the Zariski topology) follows, in particular, from (iv).

Let us prove (iii); if $w \in N^\times(S)$, clearly $\alpha \circ \text{int}(w)$ is a root of G with respect to T , hence is locally equal to α or $-\alpha$; since on each fiber it is $-\alpha$ (*Bible*, 12-05, proof of the cor. to prop. 1), one has $\alpha \circ \text{int}(w) = -\alpha$. By transport of structure, one deduces

$$-\alpha^* = \text{int}(w)^{-1} \circ \alpha^* = \text{int}(w) \circ \alpha^*,$$

since $\text{int}(w)^2 = \text{int}(w^2)$ and w^2 is a section of T . Thus $\text{int}(w)$ induces reflection on T' ; since R is central, $\text{int}(w)$ induces the identity on R . The formula of (iii) defines a morphism $T \rightarrow T$ with the same properties, hence coincides with $\text{int}(w)$.

Let us prove (iv). One has successively

$$\begin{aligned} w_\alpha(X)tw_\alpha(X)^{-1} &= \exp(X) \exp(-X^{-1}) \exp(X)t \exp(-X) \exp(X^{-1}) \exp(-X) \\ &= \exp(X) \exp(-X^{-1}) \exp(X - \alpha(t)X) \exp(\alpha(t)^{-1}X^{-1}) \exp(-\alpha(t)X)t. \end{aligned}$$

⁽²⁵⁾**N.D.E.** The first formula of the original has been corrected.

Applying formula (F), one has

$$\begin{aligned} \exp(-X^{-1}) \exp((1 - \alpha(t))X) &= \exp((\alpha(t)^{-1} - 1)X) \alpha^*(\alpha(t)^{-1}) \\ &\quad \cdot \exp(-\alpha(t)^{-1}X^{-1}). \end{aligned}$$

Substituting in the preceding relation, one finds

$$\begin{aligned} \text{int}(w_\alpha(X))t &= \exp(\alpha(t)^{-1}X) \alpha^*(\alpha(t)^{-1}) \exp(-\alpha(t)X)t \\ &= \exp(aX) \alpha^*(\alpha(t)^{-1})t, \end{aligned}$$

where

$$a = \alpha(t)^{-1} - (\alpha \circ \alpha^*)(\alpha(t)^{-1})\alpha(t),$$

but $\alpha \circ \alpha^* = 2$, which gives at once $a = 0$ and $w_\alpha(X) \in N^\times(S')$.

Let us now prove the second assertion of (iv). One has⁽²⁶⁾

$$\begin{aligned} \alpha^*(z)w_\alpha(X) &= \exp(z^2X) \exp(-z^{-2}X^{-1}) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp((z^2 - z)X) \exp(-z^{-2}X^{-1}) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp(-z^{-1}X^{-1}) \alpha^*(z)^{-1} \exp((z^3 - z^2)X) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp(-z^{-1}X^{-1}) \exp(zX) = w_\alpha(zX). \end{aligned}$$

Let us prove (v). By the preceding result, the first formula of (v) follows at once from the second; let us prove the latter:

$$\begin{aligned} w_\alpha(X)^2 &= \exp(X) \exp(-X^{-1}) \exp(2X) \exp(-X^{-1}) \exp(X) \\ &= \exp(X) \exp(-X^{-1}) \exp(X^{-1}) \alpha^*(-1) \exp(-2X) \exp(X) \\ &= \exp(X) \alpha^*(-1) \exp(-X) = \alpha^*(-1), \end{aligned}$$

since $\alpha(\alpha^*(-1)) = (-1)^2 = 1$, which proves that $\alpha^*(-1) \in \text{Centr}(G)(S)$.

Finally, let us prove (vi). The first assertion is a particular case of the second; let us prove the latter. Both sides of this formula define morphisms from $W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times$ to G . To prove that they coincide, it suffices to do so on an open subset nonempty on each fiber (Exp. XVIII 1.4); it therefore suffices to check the relation when $1 + XY$ is invertible. One then has successively:

$$\begin{aligned} w_\alpha(X)w_{-\alpha}(Y) &= \exp(X) \exp(-X^{-1}) \exp(X) \exp(Y) \exp(-Y^{-1}) \exp(Y) \\ &= \exp(X) \exp(-X^{-1}) \exp\left(\frac{Y}{1+XY}\right) \alpha^*(1+XY) \\ &\quad \exp\left(\frac{X}{1+XY}\right) \exp(-Y^{-1}) \exp(Y) \\ &= \exp(X) \exp\left(\frac{-X^{-1}}{1+XY}\right) \alpha^*(1+XY) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \exp(-X^{-2}Y^{-1}) \alpha^*\left(\frac{XY}{1+XY}\right) \exp(X+Y^{-1}) \\ &\quad \alpha^*(1+XY) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \exp(-X^{-2}Y^{-1}) \alpha^*(XY) \exp\left(\frac{Y^{-1}+X}{(1+XY)^2}\right) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \alpha^*(XY) \exp(-Y) \exp(Y) = \alpha^*(XY). \end{aligned}$$

⁽²⁶⁾**N.D.E.** The first equality follows from 1.5 (i), which, combined with $\alpha \circ \alpha^* = 2$, gives the formulas

$$\begin{aligned} \alpha^*(z) \exp(X) \alpha^*(z)^{-1} &= \exp(z^2X), \\ \alpha^*(z) \exp(X^{-1}) \alpha^*(z)^{-1} &= \exp(z^{-2}X), \end{aligned}$$

the third equality follows from formula (F), and the fourth from (†), again. Finally, a similar calculation shows that $w_\alpha(X) \alpha^*(z^{-1}) = w_\alpha(zX)$.

Corollary 3.2. *Let $n \in \mathbf{Z}$, $n \neq 0$. For every $w \in G(S)$, the following conditions are equivalent:*

- (i) $w \in N^\times(S)$,
- (ii) *one has $\text{int}(w) \circ n\alpha^* = -n\alpha^*$ (recall that $(n\alpha^*)(z) = \alpha^*(z)^n$).*

One has (i) $\Rightarrow$ (ii) (assertion (iii) of theorem 3.1); conversely, one may suppose that $N^\times$ has a section, and one is reduced to proving:

Lemma 3.3. *One has $\underline{\text{Centr}}_G(n\alpha^*) = T$ for $n \neq 0$.*

Indeed, the image T' of $n\alpha^*$ is a subtorus of G . It follows (Exp. XIX 2.8) that $\underline{\text{Centr}}_G(n\alpha^*)$ is a reductive subgroup of G , containing T . Since on each fiber one has $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) \neq G_{\bar{s}}$, one has $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) = T_{\bar{s}}$ (Exp. XIX 1.6.3⁽²⁷⁾), hence $\underline{\text{Centr}}_G(n\alpha^*) = T$, since these are smooth subgroups of G .

Remark 3.4. The construction of w_α and the fact that $w_\alpha(X)$ normalizes T rely only on formula (F). In particular, if G is an S -group satisfying the conditions of 2.2, $\underline{\text{Norm}}_G(T)$ is different from T on each fiber. It follows that if G is an affine S -group with connected fibers satisfying the conditions of 2.2, it is reductive of semisimple rank 1. Indeed, it is smooth in a neighborhood of the unit section, hence smooth, and one can apply the criterion of Exp. XIX 1.11.

3.5. — Before stating the following theorem, let us make a few remarks. As usual, we identify $\mathfrak{g}^{-\alpha}$ with $(\mathfrak{g}^\alpha)^{\otimes -1}$. Similarly, we shall identify $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathfrak{g}^{-\alpha}, \mathfrak{g}^\alpha)$ with $(\mathfrak{g}^\alpha)^{\otimes 2}$, and hence

$$\underline{\text{Isom}}_{\mathcal{O}_S\text{-mod.}}(\mathbf{W}(\mathfrak{g}^{-\alpha}), \mathbf{W}(\mathfrak{g}^\alpha)) \simeq \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times.$$

If $w \in N^\times(S)$, $\text{Ad}(w)$ interchanges $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ (3.1, (iii)), hence defines an isomorphism

$$a_\alpha(w) : \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}^\alpha,$$

which we shall therefore identify with a section $a_\alpha(w) \in \Gamma(S, (\mathfrak{g}^\alpha)^{\otimes 2})^\times$. This construction is compatible with base change and therefore defines a morphism

$$a_\alpha : N^\times \longrightarrow \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times,$$

such that $a_\alpha(w)Y = \text{Ad}(w)Y$ for all $w \in N^\times(S')$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})^\times$, $S' \longrightarrow S$.

Theorem 3.6.

- (i) *One has*

$$\text{int}(w) \exp(Y) = \exp(a_\alpha(w)Y)$$

for every $S' \longrightarrow S$ and all $w \in N^\times(S')$, $Y \in \mathbf{W}(\mathfrak{g}^{-\alpha})(S')$.

- (ii) *One has*

$$a_\alpha(tw) = \alpha(t)a_\alpha(w), \quad a_\alpha(wt) = \alpha(t)^{-1}a_\alpha(w).$$

- (iii) *If one similarly defines $a_{-\alpha} : N^\times \longrightarrow \mathbf{W}((\mathfrak{g}^{-\alpha})^{\otimes 2})^\times$, one has*

$$a_{-\alpha}(w) = a_\alpha(w)^{-1}.$$

(28)

- (iv) *For every $X \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$, $S' \longrightarrow S$, one has*

$$a_\alpha(w_\alpha(X)) = -X^2.$$

⁽²⁷⁾**N.D.E.** The hypothesis $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) \neq G_{\bar{s}}$ implies that $\dim \underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) - \dim T_{\bar{s}} < 2$, whereas this difference is even, by loc. cit.

⁽²⁸⁾**N.D.E.** I.e. $a_{-\alpha}(w)$ and $a_{-\alpha}(w)$ are paired, cf. 2.6.1.

Assertion (i) is trivial by the characterization of the morphisms $\exp$ given in 1.5. Assertion (ii) is immediate, as is (iii). Let us prove (iv): let $X \in \Gamma(S', \mathfrak{g}^\alpha)^\times$, $Z \in \Gamma(S', \mathfrak{g}^\alpha)$; by definition, one has⁽²⁹⁾

$$\begin{aligned} a_\alpha(w_\alpha(X))^{-1}(Z) &= \text{Ad}(w_\alpha(X))(Z) \\ &= \text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1})) \text{Ad}(\exp(X))(Z). \end{aligned}$$

Applying formulas (2') and (2) of lemma 2.10, as well as the equalities $H_{-\alpha} = -H_\alpha$, $\bar{\alpha}(H_\alpha) = 2$ (loc. cit. (4) and (5)) and $\langle X, X^{-1} \rangle = 1$ (2.6), one obtains that the right-hand term equals, successively:

$$\begin{aligned} \text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1}))(Z) &= \text{Ad}(\exp(X))(Z + \langle X^{-1}, Z \rangle (H_\alpha - X^{-1})) \\ &= Z + \langle X^{-1}, Z \rangle (H_\alpha - 2X - X^{-1} - H_\alpha + X) \\ &= Z - \langle X^{-1}, Z \rangle X - \langle X^{-1}, Z \rangle X^{-1}. \end{aligned}$$

But $Z = \langle X^{-1}, Z \rangle X$ and $\langle X^{-1}, Z \rangle X^{-1} = X^{-2}Z$, hence this shows that $a_\alpha(w_\alpha(X))^{-1} = -X^{-2}$, whence $a_\alpha(w_\alpha(X)) = -X^2$.

Corollary 3.7. *In particular, one has*

$$\text{int}(w_\alpha(X)) \exp(X) = \exp(-X^{-1}),$$

whence (by the definition of $w_\alpha(X)$):

$$w_\alpha(X) \exp(X) w_\alpha(X)^{-1} = \exp(-X) w_\alpha(X) \exp(-X),$$

i.e. by an immediate calculation,

$$(w_\alpha(X) \exp(X))^3 = e.$$

Corollary 3.8. *Let $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and $n \in \mathbf{Z}$, $n \neq 0$. Then $w_\alpha(X)$ is the unique section $w \in G(S)$ satisfying*

- (i) $\text{int}(w) \circ n\alpha^* = -n\alpha^*$.
- (ii) $(w \exp(X))^3 = e$.

One knows that $w_\alpha(X)$ indeed satisfies these conditions. Conversely, let $w \in G(S)$ satisfy (i) and (ii). By 3.2 and 3.1 (ii), one knows that there exists $t \in T(S)$ such that $w = w_\alpha(X)t$. Put $u = \exp(X)$. One then has

$$wuw^{-1} = w_\alpha(X)t \exp(X)t^{-1}w_\alpha(X)^{-1} = \exp(-\alpha(t)X^{-1}),$$

and, on the other hand,

$$\begin{aligned} u^{-1}wu^{-1} &= \exp(-X)w_\alpha(X)t \exp(-X) \\ &= \exp(-X)w_\alpha(X) \exp(-X) \exp(X - \alpha(t)X)t \\ &= \exp(-X^{-1}) \exp(X - \alpha(t)X)t = \exp(-X^{-1})t \exp(H). \end{aligned}$$

Now $(wu)^3 = e \iff wuw^{-1} = u^{-1}wu^{-1}$; comparing the two decompositions of this element on $U_{-\alpha} \cdot T \cdot U_\alpha$, one obtains $t = e$.

Remark 3.9. One can summarize a number of the results of this section by the following diagram of principal homogeneous bundles (on the left):

$$\begin{array}{ccc} \mathbf{W}(\mathfrak{g}^\alpha)^\times & \xrightarrow{w_\alpha} & N^\times \xrightarrow{a_\alpha} \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times \\ & & \\ \mathbf{G}_{\mathbf{m},S} & \xrightarrow{\alpha^*} & T \xrightarrow{\alpha} \mathbf{G}_{\mathbf{m},S}. \end{array}$$

⁽²⁹⁾**N.D.E.** The following has been expanded.

Note that a_α is faithfully flat (since α is) and that w_α is a monomorphism if and only if α^* is a monomorphism. We leave it to the reader to write the corresponding diagrams for the structures of principal bundles on the right, as well as diagrams of the same kind for the root $-\alpha$, and to study the relations between these various diagrams.

Lemma 3.10. *Let S be a scheme, q an integer > 0 such that $x \mapsto x^q$ defines an endomorphism of $\mathbf{G}_{a,S}$, (G, T, α) and (G', T', α') two elementary S -systems, $f : G \rightarrow G'$ a morphism of S -groups. Let*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

be an isomorphism of $\mathcal{O}_S$ -modules, and

$$h^\vee : (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}$$

the contragredient isomorphism. For every $S' \rightarrow S$ and every $X \in W(\mathfrak{g}^\alpha)(S')$, suppose that:

$$f(\exp(X)) = \exp(h(X^q)).$$

Then the following conditions are equivalent:

- (i) $f(\alpha^*(z)) = \alpha'^*(z)^q$.
- (ii) $f(w_\alpha(Z)) = w_{\alpha'}(h(Z^q))$.
- (iii) $f(\exp(Y)) = \exp(h^\vee(Y^q))$.

(Each condition is to be read: for every $S' \rightarrow S$ and every $z \in \mathbf{G}_m(S')$, $Z \in W(\mathfrak{g}^\alpha)^\times(S')$, $Y \in W(\mathfrak{g}^{-\alpha})(S')$, one has)

Indeed, (i) $\Rightarrow$ (ii) by 3.8, (ii) $\Rightarrow$ (iii) by 3.7, (iii) $\Rightarrow$ (i) by 2.7.

Proposition 3.11. *Let S be a scheme, $q \in \mathbf{Z}$, $q > 0$, such that $x \mapsto x^q$ defines an endomorphism of $\mathbf{G}_{a,S}$, (G, T, α) and (G', T', α') two elementary S -systems, $f : G \rightarrow G'$ a morphism of S -groups. The following conditions on f are equivalent:*

- (i) *The restriction of f to T factors as a morphism $f_T : T \rightarrow T'$ making the diagram commute:*

$$\begin{array}{ccccc} \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\ q \downarrow & & \downarrow f_T & & \downarrow q \\ \mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S} \end{array}$$

- (ii) *There exists a (unique) isomorphism of $\mathcal{O}_S$ -modules*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \rightarrow \mathfrak{g}'^{\alpha'}$$

such that

$$f(\exp(X)) = \exp(h(X^q)), \quad f(\exp(Y)) = \exp(h^\vee(Y^q))$$

for all $X \in W(\mathfrak{g}^\alpha)(S')$, $Y \in W(\mathfrak{g}^{-\alpha})(S')$, $S' \rightarrow S$ (it follows that f also satisfies the equivalent conditions of 3.10).

One has (ii) $\Rightarrow$ (i). Indeed, by 3.10, condition (ii) implies $f \circ \alpha^* = q\alpha'^*$, hence, by 3.3, $f|_T$ factors through T' . It remains to prove that $\alpha'(f(t)) = \alpha(t)^q$, which follows at once from the fact that f induces a morphism of groups $T \cdot U_\alpha \rightarrow T' \cdot U_{\alpha'}$.

Let us prove (i) $\Rightarrow$ (ii). Let $X \in \Gamma(S, \mathfrak{g}^\alpha)$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$. Put $p_+(x) = f(\exp(xX))$ and $p_-(x) = f(\exp(yY))$; these are morphisms of groups

$$p_+, p_- : \mathbf{G}_{a,S} \rightarrow G.$$

Now one has

$$\begin{aligned} \text{int}(\alpha'^*(z))^q(p_+(x)) &= \text{int}(f_T(\alpha^*(z)))(f(\exp(xX))) \\ &= f(\text{int}(\alpha^*(z))(\exp(xX))) \\ &= f(\exp(z^2xX)) = p_+(z^2x). \end{aligned}$$

Applying lemma 1.2 (with $Q = \alpha'^*(\mathbf{G}_{m,S})$), one deduces that there exists a section $X' \in \Gamma(S, \mathfrak{g}^{\alpha'})$ such that

$$f(\exp(xX)) = p_+(x) = \exp(x^q X').$$

Similarly, there exists a section $Y' \in \Gamma(S, \mathfrak{g}^{-\alpha'})$ such that

$$f(\exp(yY)) = \exp(y^q Y').$$

Writing now that f is a morphism of groups, hence respects formula (F), one obtains at once

$$X^q Y^q = (XY)^q = X' Y'.$$

One easily concludes that $X^q \mapsto X'$ and $Y^q \mapsto Y'$ define isomorphisms h and $h^\vee$ as announced.

Proposition 3.12. *Let (G, T, α) be an elementary S -system, $w \in N^\times(S)$; put*

$$\Omega_0 = \Omega \cap \text{int}(w^{-1})(\Omega).$$

Let d be the function on Ω defined by

$$d(\exp(Y) \cdot t \cdot \exp(X)) = \alpha(t)^{-1} + XY.$$

Then $\Omega_0 = \Omega_d$, and for $\exp(Y) \cdot t \cdot \exp(X) \in \Omega_0(S')$ one has the following formula (put $z = d(\exp(Y) \cdot t \cdot \exp(X))$):

$$(\star) \quad \begin{aligned} \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(z^{-1} a_\alpha(w)^{-1} X) \\ &\quad \cdot t \alpha^*(z) \cdot \exp(z^{-1} a_\alpha(w) Y). \end{aligned}$$

Moreover, $d \circ \text{int}(w) = d^{-1}$.

Indeed, one has at once⁽³⁰⁾

$$\begin{aligned} \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(a_\alpha(w) Y) \cdot t \alpha^*(\alpha(t)^{-1}) \cdot \exp(a_\alpha(w)^{-1} X) \\ &= \exp(a_\alpha(w) Y) \cdot \exp(\alpha(t) a_\alpha(w)^{-1} X) \cdot t \alpha^*(\alpha(t)^{-1}). \end{aligned}$$

By 2.1, this is a section of Ω if and only if $1 + \alpha(t)XY$ is invertible, which indeed proves $\Omega_0 = \Omega_d$; then applying formula (F) of loc. cit., one deduces by an immediate calculation the announced formula $(\star)$. Finally, it follows from $(\star)$ that one has

$$\begin{aligned} (d \circ \text{int}(w))(\exp(Y) \cdot t \cdot \exp(X)) &= \alpha(t \alpha^*(z))^{-1} + z^{-2} XY \\ &= z^{-2} (\alpha(t)^{-1} + XY) = z^{-1}, \end{aligned}$$

whence the last assertion.

N. B. One will note that the function d is independent of the choice of w .

4. The isomorphism theorem

Theorem 4.1. *Let S be a scheme, $q \in \mathbf{Z}$, $q > 0$, such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$, (G, T, α) and (G', T', α') two elementary S -systems. Let*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \longrightarrow \mathfrak{g}'^{\alpha'} \quad \text{and} \quad h^\vee : (\mathfrak{g}^{-\alpha})^{\otimes q} \longrightarrow \mathfrak{g}'^{-\alpha'}$$

⁽³⁰⁾**N.D.E.** The original has been corrected by interchanging $a_\alpha(w)$ and $a_\alpha(w)^{-1}$, and the proof of $d \circ \text{int}(w) = d^{-1}$ has been expanded.

be two mutually contragredient isomorphisms. Let $f_T : T \rightarrow T'$ be a morphism of S -groups making the diagram commute:

$$\begin{array}{ccc} \mathbf{G}_{m,S} & \xrightarrow{q} & \mathbf{G}_{m,S} \\ \alpha^* \downarrow & & \downarrow \alpha'^* \\ T & \xrightarrow{f_T} & T' \\ \alpha \downarrow & & \downarrow \alpha' \\ \mathbf{G}_{m,S} & \xrightarrow{q} & \mathbf{G}_{m,S}. \end{array}$$

There exists a unique morphism of S -groups $f : G \rightarrow G'$ extending f_T and satisfying

$$f(\exp(X)) = \exp(h(X^q))$$

for every $X \in W(\mathfrak{g}^\alpha)(S')$, $S' \rightarrow S$. Moreover, this morphism also satisfies

$$f(\exp(Y)) = \exp(h^\vee(Y^q)) \quad \text{and} \quad f(w_\alpha(Z)) = w_\alpha(h(Z^q)),$$

for every $S' \rightarrow S$ and all $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})$, $Z \in \Gamma(S', \mathfrak{g}^\alpha)^\times$.

If $f : G \rightarrow G'$ extends f_T , then $f \circ \alpha^* = (\alpha'^*)^q$. If in addition f satisfies the second condition, it also satisfies the other two by 3.10. It follows that f is determined on Ω by the relation

$$f(\exp(Y)t\exp(X)) = \exp(h^\vee(Y^q))f_T(t)\exp(h(X^q)).$$

Since Ω is schematically dense in G , this already proves the uniqueness of f . To prove its existence, it suffices, by Exp. XVIII 2.3, to prove that the preceding formula defines a “generically multiplicative” morphism from Ω to G' . Now, by 2.4, this amounts to checking that $\alpha' \circ f = \alpha^q$, which follows from the fact that f extends f_T .

Scholium 4.2. One can also interpret 4.1 as follows: consider the category $\mathcal{E}$ of elementary S -systems and the category $\mathcal{D}$ of pairs

$$\left(\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathcal{L} \right),$$

where T is a torus, α and α^* morphisms of groups such that $\alpha \circ \alpha^* = 2$, and $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module (the reader will specify the morphisms of the two categories considered). One defines a functor $\mathcal{E} \rightarrow \mathcal{D}$ by

$$(G, T, \alpha) \mapsto \left(\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathfrak{g}^\alpha \right).$$

The preceding theorem says that this functor is fully faithful. It is in fact an equivalence of categories, as will be seen in the next section. One already has:

Corollary 4.3. If $q = 1$ and f_T is an isomorphism, then f is an isomorphism.

Corollary 4.4. If $q = 1$ and f_T is faithfully flat with kernel Q (cf. Exp. IX 2.7), then f is faithfully flat (quasi-compact) with kernel Q , hence identifies G' with G/Q .

Indeed, if f_T is faithfully flat with kernel Q , then

$$Q = \text{Ker}(f_T) \subset \text{Ker}(f_T \circ \alpha') = \text{Ker}(\alpha).$$

Introducing the elementary S -system $(G/Q, T/Q, r/Q)$ of 2.13, one is reduced by 2.14 to proving that f/Q induces an isomorphism from G/Q onto G' , which follows at once from 4.3.

5. Examples of elementary systems, applications

5.1. — Let S be a scheme, $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module. Consider the group $G_{\mathcal{L}}$ over S defined by

$$G_{\mathcal{L}}(S') = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left| \begin{array}{l} a, d \in \mathbf{G}_a(S'), \quad b \in W(\mathcal{L})(S'), \\ c \in W(\mathcal{L}^{-1})(S'), \quad ad - bc \in \mathbf{G}_m(S') \end{array} \right. \right\},$$

equipped with the usual matrix multiplication law. It is locally isomorphic to $\mathrm{GL}_{2,S}$. It is therefore a group S -scheme, affine and smooth over S , with connected fibers.

Remark. Let $\mathcal{L}'$ and $\mathcal{L}''$ be two invertible sheaves on S such that $\mathcal{L} = \mathcal{L}' \otimes \mathcal{L}''^{-1}$.⁽³¹⁾ Then one has an isomorphism of S -groups:

$$G_{\mathcal{L}} \xrightarrow{\sim} \mathrm{GL}(\mathcal{L}' \oplus \mathcal{L}'')$$

defined as follows: if x (resp. y) is a section of $\mathcal{L}'$ (resp. $\mathcal{L}''$) over an open subset V of S , one has

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

5.2. — One denotes by $S_{\mathcal{L}}$ the closed subgroup of $G_{\mathcal{L}}$ defined by $ad - bc = 1$. It is also a group S -scheme, affine and smooth over S , with connected fibers (isomorphic to $\mathrm{SL}(\mathcal{L}' \oplus \mathcal{L}'')$ by the preceding isomorphism).

Similarly, consider the morphism $\mathbf{G}_{m,S} \rightarrow G_{\mathcal{L}}$ defined by $z \mapsto \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}$. It is a central monomorphism; passing to the quotient, one deduces a group $P_{\mathcal{L}}$, smooth and affine over S , with connected fibers (cf. Exp. VIII 5.7).

One can see that, passing to the quotient from the isomorphism of the preceding remark, $P_{\mathcal{L}}$ is identified with the group of automorphisms of the projective bundle $\mathbf{P}(\mathcal{L}' \oplus \mathcal{L}'')$ (cf. EGA, II 4.2.7). One denotes by i and p the canonical morphisms

$$S_{\mathcal{L}} \xrightarrow{i} G_{\mathcal{L}} \xrightarrow{p} P_{\mathcal{L}};$$

i is a closed immersion, p is faithfully flat and affine.

5.3. — Consider the morphisms of groups

$$\begin{aligned} t_G : \mathbf{G}_{m,S}^2 &\rightarrow G_{\mathcal{L}}, & t_G(z, z') &= \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix}; \\ t_S : \mathbf{G}_{m,S} &\rightarrow S_{\mathcal{L}}, & t_S(z) &= \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}; \\ t_P : \mathbf{G}_{m,S} &\rightarrow P_{\mathcal{L}}, & t_P(z) &= p(t_G(z, 1)). \end{aligned}$$

These are monomorphisms of groups defining in each group a (split) torus of relative codimension 2. For every $s \in S$, let

$$X \in \Gamma(\bar{s}, \mathcal{L} \otimes \bar{s})^{\times};$$

then the section $\begin{pmatrix} 0 & X \\ X^{-1} & 0 \end{pmatrix}$ of $G_{\mathcal{L},\bar{s}}$ normalizes $t_G(\mathbf{G}_{m,\bar{s}}^2)$ and does not centralize it; one then concludes from Exp. XIX 1.6 that $G_{\mathcal{L}}$ is *reductive*, of *semisimple rank 1*, with *maximal torus* $t_G(\mathbf{G}_{m,S}^2)$.

One reasons similarly for $S_{\mathcal{L}}$ and $P_{\mathcal{L}}$, and sees that $S_{\mathcal{L}}$ (resp. $P_{\mathcal{L}}$) is *reductive*, of *semisimple rank 1*, with maximal torus $t_S(\mathbf{G}_{m,S})$ (resp. $t_P(\mathbf{G}_{m,S})$).

5.4. — Reasoning as usual, one immediately determines the Lie algebra of these various groups and the adjoint action of the chosen maximal torus. Let us do so for $G_{\mathcal{L}}$; this is immediate by Exp. II 4.8: $\mathrm{Lie}(G_{\mathcal{L}}/S)$ is the Lie algebra of the matrices below:

$$\mathrm{Lie}(G_{\mathcal{L}}/S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left| \begin{array}{l} a \text{ and } d \text{ sections of } \mathcal{O}_S, \quad b \text{ a section of } \mathcal{L}, \\ c \text{ a section of } \mathcal{L}^{-1} \end{array} \right. \right\}$$

⁽³¹⁾**N.D.E.** $\mathcal{L}'' \otimes \mathcal{L}'^{-1}$ has been corrected to $\mathcal{L}' \otimes \mathcal{L}''^{-1}$, and the following sentence has been expanded.

with the usual bracket; one has

$$\mathrm{Ad}(t_G(z, z')) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & zz'^{-1}b \\ z'z^{-1}c & d \end{pmatrix}.$$

Put $\mathcal{L}ie(G_{\mathcal{L}}/S) = \mathfrak{g}$. Let $\alpha_G : t_G(\mathbf{G}_{m,S}^2) \rightarrow \mathbf{G}_{m,S}$ be the character defined by

$$\alpha_G(t_G(z, z')) = zz'^{-1}.$$

One sees at once from the preceding relation that α_G is a root of $G_{\mathcal{L}}$ with respect to $t_G(\mathbf{G}_{m,S}^2)$, and that the morphism

$$u : \mathcal{L} \rightarrow \mathfrak{g} \quad (\text{resp. } u_- : \mathcal{L}^{-1} \rightarrow \mathfrak{g})$$

defined by $u(X) = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}$ (resp. $u_-(X) = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}$) is an isomorphism from $\mathcal{L}$ onto $\mathfrak{g}^{\alpha_G}$ (resp. from $\mathcal{L}^{-1}$ onto $\mathfrak{g}^{-\alpha_G}$).

One has therefore proved that $(G, t_G(\mathbf{G}_{m,S}^2), \alpha_G)$ is an elementary S -system.

Putting similarly

$$\alpha_S(t_S(z)) = z^2, \quad \alpha_P(t_P(z)) = z,$$

one proves that $(S_{\mathcal{L}}, t_S(\mathbf{G}_{m,S}), \alpha_S)$ and $(P_{\mathcal{L}}, t_P(\mathbf{G}_{m,S}), \alpha_P)$ are elementary systems, and defines isomorphisms of $\mathcal{L}$ (resp. $\mathcal{L}^{-1}$) with the corresponding direct summands of the Lie algebras of $S_{\mathcal{L}}$ and $P_{\mathcal{L}}$.

5.5. — Put $\exp \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}$. One has thus defined a morphism

$$W(\mathfrak{g}^{\alpha_G}) \rightarrow G_{\mathcal{L}}$$

which induces on Lie algebras the canonical morphism, hence is the unique morphism of this type (1.5). Similarly, put $\exp \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}$. Calculating formula (F) explicitly, one finds

$$\left\langle \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} \right\rangle = XY, \quad \alpha_G^*(z) = \begin{pmatrix} z & 0 \\ 0 & 1/z \end{pmatrix} = t_G(z, z^{-1}).$$

⁽³²⁾The open subset $N^\times = N_G^\times$ (defined before 3.1) is:

$$N_G^\times(S') = \left\{ \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \mid P \in W(\mathfrak{g}^{\alpha})^\times(S'), Q \in W(\mathfrak{g}^{-\alpha})^\times(S') \right\},$$

the morphism w_{α_G} (cf. 3.1 (iv)) is given, for every $X \in W(\mathfrak{g}^{\alpha})^\times(S')$, by

$$w_{\alpha_G}(X) = \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix};$$

the morphism a_{α_G} (cf. 3.5) is given by:

$$\text{if } w = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \in N_G^\times(S'), \quad \text{then } a_{\alpha_G}(w) = PQ^{-1} \in W((\mathfrak{g}^{\alpha})^{\otimes 2})^\times(S'),$$

i.e. for every $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$, one has $a_{\alpha_G}(w)(Y) = PQ^{-1}Y \in W(\mathfrak{g}^{\alpha})^\times(S')$.

5.6. — We leave it to the reader to perform the same calculations in $S_{\mathcal{L}}$ and $P_{\mathcal{L}}$. One finds the same duality formula and the coroots

$$\alpha_S^*(z) = t_S(z), \quad \alpha_P^*(z) = t_P(z^2).$$

Let p_T be the morphism induced by $p : G_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$ on $t_S(\mathbf{G}_{m,S})$, i.e.

$$p_T(t_S(z)) = t_P(z^2).$$

⁽³²⁾**N.D.E.** The following has been expanded.

One therefore has the commutative diagram:⁽³³⁾

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \text{id} & \searrow \alpha_S^* & \swarrow \alpha_P^* & \searrow 2 \\
 \mathbf{G}_{m,S} & \xrightarrow{t_S} & t_S(\mathbf{G}_{m,S}) & \xrightarrow{p_T} & t_P(\mathbf{G}_{m,S}) \xrightarrow{t_P} \mathbf{G}_{m,S} \\
 & \searrow 2 & \swarrow \alpha_S & \swarrow \alpha_P & \searrow \text{id} \\
 & & \mathbf{G}_{m,S} & &
 \end{array}$$

One recognizes in the central part the commutative diagram of 4.1⁽³⁴⁾ relative to the canonical morphism $p \circ i : S_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$, which induces a morphism of the preceding elementary S -systems.

5.7. — Let now (G, T, α) be any elementary S -system. Consider the commutative diagram:

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \text{id} & \searrow \alpha^* & \swarrow 2 & \\
 \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\
 & \searrow 2 & \swarrow \alpha & \swarrow \text{id} & \\
 & & \mathbf{G}_{m,S} & &
 \end{array}$$

Combining the preceding two diagrams, one obtains a commutative diagram:

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \alpha_S^* & \searrow \alpha^* & \swarrow \alpha_P^* & \\
 t_S(\mathbf{G}_{m,S}) & \xrightarrow{\alpha^* \circ t_S^{-1}} & T & \xrightarrow{t_P \circ \alpha} & t_P(\mathbf{G}_{m,S}) \\
 & \searrow \alpha_S & \swarrow \alpha & \swarrow \alpha_P & \\
 & & \mathbf{G}_{m,S} & &
 \end{array}$$

Using 4.1, one therefore has:

Proposition 5.8. *Let S be a scheme, (G, T, α) an elementary S -system. Put $\mathcal{L} = \mathfrak{g}^{\alpha}$ (and hence $\mathcal{L}^{-1} = \mathfrak{g}^{-\alpha}$).*

(i) *There exists a unique morphism of groups $f : S_{\mathcal{L}} \rightarrow G$ satisfying the following equivalent conditions:*

$$\begin{aligned}
 (a) \quad & f \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = \alpha^*(z), & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X); \\
 (b) \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \exp(Y); \\
 (c) \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix} = w_{\alpha}(X).
 \end{aligned}$$

(ii) *There exists a unique morphism of groups $g : G \rightarrow P_{\mathcal{L}}$ satisfying*

$$g(t) = \begin{pmatrix} \alpha(t) & 0 \\ 0 & 1 \end{pmatrix}, \quad g(\exp(X)) = p \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

⁽³³⁾**N.D.E.** Where t_S and t_P are isomorphisms.

⁽³⁴⁾**N.D.E.** With $q = 1$.

Moreover, one has

$$g(\exp(Y)) = p \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \quad g(w_\alpha(X)) = p \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

The morphism g is faithfully flat quasi-compact with kernel $\text{Ker}(\alpha) = \underline{\text{Centr}}(G)$, and $g \circ f$ is the canonical morphism $S_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$.

Note that conditions (b) of (i) give an explicit description of the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$.

Corollary 5.9. *Let (G, T, α) be an elementary S -system. The subgroups $T \cdot U_\alpha$, $T \cdot U_{-\alpha}$, U_α and $U_{-\alpha}$ are closed.*

Since U_α is a closed subgroup scheme of $T \cdot U_\alpha$, it suffices to check this for the latter. By Noether's theorem (Exp. IV 5.3.1 and 6.4.1), it suffices to prove that $(T \cdot U_\alpha)/\text{Ker}(\alpha)$ is a closed subgroup of $G/\text{Ker}(\alpha)$. By 5.8, one is therefore reduced to proving that the subgroup of $P_{\mathcal{L}}$ (or of $G_{\mathcal{L}}$, which amounts to the same thing by another application of Noether's theorem) defined by $c = 0$ is closed, which is trivial.

Consequently, the morphisms $\exp$ of theorem 1.5 (i) are closed immersions.

N. B. The corollary also follows from the fact that $T \cdot U_\alpha$ and $T \cdot U_{-\alpha}$ are "Borel subgroups" of G (cf. Exp. XII 7.10).

5.10. — Let $\mathcal{L}$ be an invertible $\mathcal{O}_S$ -module and

$$\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}$$

a diagram of groups⁽³⁵⁾ such that $\alpha \circ \alpha^* = 2$. Let R be the maximal torus of $\text{Ker}(\alpha)$ and $K = \alpha^{*-1}(R)$. Then K is a subgroup of multiplicative type of $\mathbf{G}_{m,S}$; by $\alpha \circ \alpha^* = 2$, it is even a subgroup of $\mu_{2,S}$. In particular, the morphism

$$K \rightarrow S_{\mathcal{L}}, \quad z \mapsto \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}$$

is central. One therefore has a central monomorphism of groups:

$$K \rightarrow R \times S_{\mathcal{L}}, \quad z \mapsto \left(\alpha^*(z), \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} \right).$$

Consider the group $G = (R \times S_{\mathcal{L}})/K$ obtained by passing to the quotient. It is a group affine and smooth over S , with connected fibers. It is immediate that the sequence

$$1 \rightarrow K \rightarrow R \times t_S(\mathbf{G}_{m,S}) \rightarrow T \rightarrow 1$$

where $u(x, t_S(z)) = x\alpha^*(z)$ is exact. The image of $R \times t_S(\mathbf{G}_{m,S})$ in G is therefore a torus T' isomorphic to T . One now shows without difficulty that if α' is the character of T' deduced from α by the preceding isomorphism, (G, T', α') is an elementary S -system, that $\mathfrak{g}^{\alpha'}$ is isomorphic to $\mathcal{L}$, and that α'^* is obtained from α^* by the isomorphism $T \xrightarrow{\sim} T'$. One has therefore constructed an elementary S -system (G, T', α') such that the corresponding object $(\mathbf{G}_{m,S} \xrightarrow{\alpha'^*} T' \xrightarrow{\alpha'} \mathbf{G}_{m,S}, \mathfrak{g}^{\alpha'})$ of the category $\mathcal{D}$ defined in 4.2 is isomorphic to $(\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathcal{L})$. One has therefore proved:

Theorem 5.11. *With the notations of 4.2, the functor*

$$(G, T, \alpha) \mapsto (\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathfrak{g}^\alpha)$$

is an equivalence of categories between $\mathcal{E}$ and $\mathcal{D}$.

6. Generators and relations for an elementary system

⁽³⁵⁾N.D.E. T being a torus.

6.1. — Let S be a scheme, (G, T, α) an elementary S -system. Let $X \in W(\mathfrak{g}^\alpha)^\times(S)$ and $u = \exp(X)$; we saw in 3.8 that the element $w = w_\alpha(X)$ satisfies, in particular, the relation

$$(wu)^3 = e.$$

⁽³⁶⁾Let s_α denote the automorphism of T induced by $\text{int}(w)$; by theorem 3.1 (iii), for every $S' \rightarrow S$ and $t \in T(S')$, one has

$$s_\alpha(t) = \text{int}(w)(t) = t \cdot \alpha^*(\alpha(t)^{-1}).$$

Theorem 6.2. *Let H be a sheaf of S -groups for (fppf). Let*

$$f_T : T \rightarrow H, \quad f_\alpha : U_\alpha \rightarrow H$$

be morphisms of groups and $h \in H(S)$ a section of H . For there to exist a morphism of groups (necessarily unique)

$$f : G \rightarrow H$$

extending f_T and f_α and satisfying $f(w) = h$, it is necessary and sufficient that the following conditions hold:

(i) *For every $S' \rightarrow S$, every $t \in T(S')$ and every $x \in U_\alpha(S')$, one has*

$$(1) \quad f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(txt^{-1}) = f_\alpha(x^{\alpha(t)}).$$

(in other words, f_T and f_α extend to a morphism of groups from the semidirect product $T \cdot U_\alpha$ to H).

(ii) *For every $S' \rightarrow S$ and every $t \in T(S')$, one has*

$$(2) \quad hf_T(t)h^{-1} = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^*(\alpha(t)^{-1})).$$

(iii) *One has the two relations in $H(S)$:*

$$(3) \quad h^2 = f_T(\alpha^*(-1)),$$

$$(4) \quad (hf_\alpha(u))^3 = e.$$

Proof. Let us write U_α and $U_{-\alpha}$ additively, and their vector structure multiplicatively. If f satisfies the conditions of the statement, one necessarily has, for every $y \in U_{-\alpha}(S')$,

$$f(y) = f(w^{-1}yw^{-1}w) = hf_\alpha(w^{-1}yw)h^{-1}.$$

Let therefore $f_{-\alpha} : U_{-\alpha} \rightarrow H$ be the morphism defined by

$$(*_1) \quad f_{-\alpha}(y) = hf_\alpha(w^{-1}yw)h^{-1}.$$

It is a morphism of groups. On the other hand, f is determined on the big cell Ω by

$$f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x).$$

This shows the uniqueness of f ; since the conditions of the statement are manifestly necessary, let us show that they are sufficient.

By (4), one has

$$hf_\alpha(u)h^{-1}h^2 = f_\alpha(-u)h^{-1}f_\alpha(-u).$$

Now, by (3) and (1), $h^2 = h^{-2}$ commutes with $f_\alpha(-u)$, which gives

$$hf_\alpha(u)h^{-1} = f_\alpha(-u)hf_\alpha(-u).$$

But, by definition, $hf_\alpha(u)h^{-1} = f_{-\alpha}(wuw^{-1})$; by 3.7, since $u = \exp(X)$ and $w = w_\alpha(X)$, one has

$$(*_2) \quad wuw^{-1} = -\tilde{u},$$

where $\tilde{u}$ denotes the element paired with u . One therefore obtains:

$$(*_3) \quad f_{-\alpha}(-\tilde{u}) = f_\alpha(-u)hf_\alpha(-u).$$

⁽³⁶⁾**N.D.E.** The following sentence has been added.

Let now t be a section of T over a variable $S' \rightarrow S$. Let $\text{int}(f_T(t))$ act on the preceding formula. On the left-hand side, one obtains⁽³⁷⁾

$$\begin{aligned} f_T(t)f_{-\alpha}(-\tilde{u})f_T(t)^{-1} &= f_T(t)hf_{\alpha}(u)h^{-1}f_T(t)^{-1} \\ &= h(h^{-1}f_T(t)h)f_{\alpha}(u)(h^{-1}f_T(t)^{-1}h)h^{-1} \\ &= hf_T(s_{\alpha}(t))f_{\alpha}(u)f_T(s_{\alpha}(t))^{-1}h^{-1} \\ &= hf_{\alpha}(\alpha(s_{\alpha}(t))u)h^{-1} \end{aligned}$$

by (2) and (1); then, since $s_{\alpha}(t) = t \cdot \alpha^*(\alpha(t)^{-1})$ and $\alpha \circ \alpha^* = 2$, this equals

$$hf_{\alpha}(\alpha(t)^{-1}u)h^{-1}.$$

Finally, by $(*_1)$ and $(*_2)$, one has

$$hf_{\alpha}(\alpha(t)^{-1}u)h^{-1} = f_{-\alpha}(\alpha(t)^{-1}uwu^{-1}) = f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}).$$

The right-hand side of $(*_3)$ gives

$$f_{\alpha}(-\alpha(t)u) \cdot f_T(t)hf_T(t)^{-1}h^{-1} \cdot h \cdot f_{\alpha}(-\alpha(t)u),$$

and since $hf_T(t)^{-1}h^{-1} = f_T(s_{\alpha}(t^{-1})) = f_T(t \cdot \alpha^*(\alpha(t)))$, this equals

$$f_{\alpha}(-\alpha(t)u) \cdot f_T(\alpha^*(\alpha(t))) \cdot h \cdot f_{\alpha}(-\alpha(t)u).$$

Comparing the two expressions obtained, one obtains

$$f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}) = f_{\alpha}(-\alpha(t)u) \cdot f_T(\alpha^*(\alpha(t))) \cdot h \cdot f_{\alpha}(-\alpha(t)u).$$

Since $\alpha : T \rightarrow \mathbf{G}_{m,S}$ is faithfully flat and H is a separated presheaf, one may replace $-\alpha(t)^{-1}$ by an arbitrary section of $\mathbf{G}_{m,S}$ and obtains:

Lemma 6.2.1. *For every $z \in \mathbf{G}_m(S')$, $S' \rightarrow S$, one has*

$$f_{-\alpha}(z\tilde{u}) = f_{\alpha}(z^{-1}u) \cdot f_T(\alpha^*(-z^{-1})) \cdot h \cdot f_{\alpha}(z^{-1}u).$$

Let now $x, y \in \mathbf{G}_a(S')$, $S' \rightarrow S$; suppose y and $(1 + xy)$ invertible. First applying the lemma with $z = y$, one obtains⁽³⁸⁾

$$f_{\alpha}(xu)f_{-\alpha}(y\tilde{u}) = f_{\alpha}((x + y^{-1})u) \cdot f_T(\alpha(-y^{-1})) \cdot h \cdot f_{\alpha}(y^{-1}u).$$

Now $x + y^{-1} = y^{-1}(1 + xy)$. Applying the lemma with $z = y/(1 + xy)$, one finds

$$\begin{aligned} f_{\alpha}((x + y^{-1})u) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right)f_{\alpha}(-(x + y^{-1})u) \\ &\quad \cdot h^{-1} \cdot f_T\left(\alpha^*\left(\frac{-y}{1 + xy}\right)\right). \end{aligned}$$

Substituting this in the preceding equality, one obtains

$$\begin{aligned} f_{\alpha}(xu)f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right)f_{\alpha}(-(x + y^{-1})u) \\ &\quad \cdot h^{-1} \cdot f_T(\alpha^*(1 + xy)^{-1}) \cdot h \cdot f_{\alpha}(y^{-1}u). \end{aligned}$$

Since $h^{-1}f_T(t)h = f_T(s_{\alpha}(t))$ by (2) (note that $s_{\alpha}^2 = \text{id}$), and since $s_{\alpha} \circ \alpha^* = -\alpha^*$ (cf. 6.2.1), this equals

$$f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right)f_{\alpha}(-(x + y^{-1})u) \cdot f_T(\alpha^*(1 + xy)) \cdot f_{\alpha}(y^{-1}u).$$

Finally, since for all $x' \in U_{\alpha}(S')$ and $z \in \mathbf{G}_m(S')$, one has

$$f_{\alpha}(x')f_T(\alpha^*(z)) = f_T(\alpha^*(z))f_{\alpha}(z^{-2}x'),$$

⁽³⁷⁾**N.D.E.** The following has been corrected.

⁽³⁸⁾**N.D.E.** The following calculations have been expanded.

one obtains

$$\begin{aligned} f_\alpha(xu)f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1+xy}\tilde{u}\right) \cdot f_T(\alpha^*(1+xy)) \cdot f_\alpha\left(\left(\frac{-y^{-1}(1+xy)}{(1+xy)^2} + y^{-1}\right)u\right) \\ &= f_{-\alpha}\left(\frac{y}{1+xy}\tilde{u}\right) \cdot f_T(\alpha^*(1+xy)) \cdot f_\alpha\left(\frac{x}{1+xy}u\right). \end{aligned}$$

One has therefore proved:

Lemma 6.2.2. *Let $S' \rightarrow S$. If $a \in U_\alpha(S')$, $b \in U_{-\alpha}^\times(S')$, and $1+ab \in \mathbf{G}_m(S')$, one has*

$$f_\alpha(a)f_{-\alpha}(b) = f_{-\alpha}\left(\frac{b}{1+ab}\right) f_T(\alpha^*(1+ab)) f_\alpha\left(\frac{a}{1+ab}\right).$$

By schematic density, this formula remains valid when $b \in U_{-\alpha}(S')$, $1+ab$ still being invertible. Consider then the morphism

$$f : \Omega \rightarrow H$$

defined by $f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x)$.

It follows at once from 6.2.2, condition 6.2 (i), and formula (F') of 2.4, that if $g, g' \in \Omega(S')$ and $gg' \in \Omega(S')$, one has $f(gg') = f(g)f(g')$. By Exp. XVIII 2.3 (iii) and 2.4,⁽³⁹⁾ there therefore exists a morphism of groups $G \rightarrow H$ extending f . Denote it also by f ; it answers the question; indeed, it suffices to prove that $f(w_\alpha) = h$. Now $w_\alpha = u \cdot (-\tilde{u}) \cdot u$, whence, by $(*)_3$:⁽⁴⁰⁾

$$f(w_\alpha) = f_\alpha(u)f_{-\alpha}(-\tilde{u})f_\alpha(u) = h.$$

Remark 6.3. We shall supplement these results in Exp. XXIII 3.5.

⁽³⁹⁾**N.D.E.** Note that each geometric fiber of G is connected, for example by 1.1.

⁽⁴⁰⁾**N.D.E.** The original has been simplified by invoking $(*)_3$.

Exposé XXI. Root data

by *M. Demazure*

⁽⁰⁾This exposé collects, in the absence of a suitable reference,⁽¹⁾ known results on root data (= “abstract” root systems), most of which will be used in what follows.

Notations. One denotes by $\mathbf{Q}_+$ the set of positive (or zero) rational numbers; one has $\mathbf{Z} \cap \mathbf{Q}_+ = \mathbf{N}$. Let V be a $\mathbf{Q}$ -vector space; if A (resp. B) is a subset of $\mathbf{Q}$ (resp. V), one denotes by $A \cdot B$ the image of $A \otimes B$ by the morphism $\mathbf{Q} \otimes V \rightarrow V$, in other words the set of linear combinations of elements of B with coefficients in A . One puts $-B = \{-1\}B$. One denotes by $E - F$ the set of elements of E which do not belong to F .

1. Generalities

1.1. Definitions, first properties

Definition 1.1.1. Let M and M^* be two free $\mathbf{Z}$ -modules of finite type in duality. Put $V = M \otimes \mathbf{Q}$, $V^* = M^* \otimes \mathbf{Q}$; these are two $\mathbf{Q}$ -vector spaces in duality. Identify M (resp. M^*) with a subset of V (resp. V^*). The canonical bilinear form on $M^* \times M$ (resp. $V^* \times V$) is denoted $(\ , \)$.

Let R be a finite subset of M . Give a map $\alpha \mapsto \alpha^*$ from R to M^* ; the set of α^* for $\alpha \in R$ is denoted R^* . With each $\alpha \in R$, associate the endomorphism s_α (resp. s_α^*) of M and V (resp. M^* and V^*) given by the formulas:

$$(1) \quad s_\alpha(x) = x - (\alpha^*, x)\alpha, \quad \text{i.e. } s_\alpha = \text{id} - \alpha^* \otimes \alpha;$$

$$(1^*) \quad s_\alpha^*(u) = u - (u, \alpha)\alpha^*, \quad \text{i.e. } s_\alpha^* = \text{id} - \alpha \otimes \alpha^*.$$

One says that the pair (R, R^*) (more precisely the pair $(R, R \rightarrow M^*)$) is a *root datum* in (M, M^*) , or that (M, M^*, R, R^*) is a root datum, if the following axioms hold:

(DR I) For each $\alpha \in R$, one has $(\alpha^*, \alpha) = 2$.

(DR II) For each $\alpha \in R$, one has $s_\alpha(R) \subset R$, $s_\alpha^*(R^*) \subset R^*$.

One says that R is the *root system* of the root datum $\mathcal{R} = (M, M^*, R, R^*)$. The elements of R (resp. R^*) are called the *roots* (resp. *coroots*) of the root datum.

Remark 1.1.2. Axiom (DR I) is equivalent to any one of the following properties:

$$\begin{aligned} (2) \quad s_\alpha s_\alpha &= \text{id}, & (2^*) \quad s_\alpha^* s_\alpha^* &= \text{id}, \\ (3) \quad s_\alpha(\alpha) &= -\alpha, & (3^*) \quad s_\alpha^*(\alpha^*) &= -\alpha^*. \end{aligned}$$

Remark 1.1.3. Axioms (DR I) and (DR II) imply

$$R = -R, \quad R^* = -R^*, \quad 0 \notin R, \quad 0 \notin R^*.$$

Lemma 1.1.4. *The map $R \rightarrow R^*$ is a bijection. More generally, if $\alpha, \beta \in R$ and $(\alpha^*, x) = (\beta^*, x)$ for every $x \in R$, then $\alpha = \beta$.*

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** For the results on root systems (§§ 1–5), one may consult [BLie], Chap. VI.

Indeed, one then has $s_\beta(\alpha) = \alpha - 2\beta$, $s_\alpha(\beta) = \beta - 2\alpha$. One deduces at once

$$\begin{aligned}s_\beta s_\alpha(\alpha) &= 2\beta - \alpha = \alpha + 2(\beta - \alpha), \\ s_\beta s_\alpha(\beta - \alpha) &= s_\beta(\beta - \alpha) = \beta - \alpha,\end{aligned}$$

whence $(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha) \in R$ by (DR II). Since R is finite, one has $\beta - \alpha = 0$.

Corollary 1.1.5. *The inverse map $R^* \longrightarrow R$ defines a root datum*

$$\mathcal{R}^* = (M^*, M, R^*, R)$$

called the dual of $\mathcal{R}$.⁽²⁾

Definition 1.1.6. Denote by $\Gamma_0(R)$ the subgroup of M generated by R . Denote by $\mathcal{V}(R)$ the vector subspace of V generated by R , that is, $\Gamma_0(R) \otimes \mathbf{Q}$. Applying these definitions to $\mathcal{R}^*$, one similarly constructs $\Gamma_0(R^*)$ and $\mathcal{V}(R^*)$.

One calls the *reductive rank* of $\mathcal{R}$ the number

$$\text{rgred}(\mathcal{R}) = \text{rank}(M) = \dim(V) = \dim(V^*) = \text{rank}(M^*) = \text{rgred}(\mathcal{R}^*).$$

One calls the *semisimple rank* of $\mathcal{R}$ the number

$$\text{rgss}(\mathcal{R}) = \text{rank}(R) = \text{rank}(\Gamma_0(R)) = \dim(\mathcal{V}(R)).$$

One therefore has $\text{rgss}(\mathcal{R}) \leq \text{rgred}(\mathcal{R})$.

We shall see below that $\text{rgss}(\mathcal{R}) = \text{rgss}(\mathcal{R}^*)$, i.e. that $\mathcal{V}(R)$ and $\mathcal{V}(R^*)$ have the same dimension.

Definition 1.1.7. One says that $\mathcal{R}$ is *semisimple* (resp. *trivial*) if $\text{rgss}(\mathcal{R}) = \text{rgred}(\mathcal{R})$ (resp. $\text{rgss}(\mathcal{R}) = 0$). For $\mathcal{R}$ to be trivial, it is therefore necessary and sufficient that R be empty. The trivial root datum of reductive rank zero is denoted $0 = (\{0\}, \{0\}, \emptyset, \emptyset)$.

Definition 1.1.8. Denote by $W(\mathcal{R})$ the group of transformations of M generated by the s_α , $\alpha \in R$. It is called the *Weyl group* of $\mathcal{R}$. Put

$$W^*(\mathcal{R}) = W(\mathcal{R}^*).$$

Then $W(\mathcal{R})$ acts on R , $\Gamma_0(R)$, $\mathcal{V}(R)$, M and V . If $w \in W(\mathcal{R})$ and $x \in M$ (resp. $x \in V$), one has $wx - x \in \Gamma_0(R)$ (resp. $wx - x \in \mathcal{V}(R)$); this is immediate from formula (1). Similarly for $W^*(\mathcal{R})$.

Lemma 1.1.9. *For all $\alpha \in R$, $x \in V$, $u \in V^*$, one has*

$$(4) \quad (s_\alpha^*(u), s_\alpha(x)) = (u, x).$$

Indeed, taking the scalar product of (1) and (1*), one finds that the left-hand side is equal to $(u, x) + (u, \alpha)(\alpha^*, x)((\alpha^*, \alpha) - 2) = (u, x)$.

Remark 1.1.10. If one supposes $0 \notin R$ and $0 \notin R^*$, then 1.1.9 is equivalent to (DR I).

Corollary 1.1.11. *Let $h \longmapsto h^\vee$ denote the isomorphism from $\text{GL}(M)$ onto $\text{GL}(M^*)$ associating with h its contragredient. Then formula (4) is also written*

$$(5) \quad (s_\alpha)^\vee = s_\alpha^*.$$

Corollary 1.1.12. *The preceding isomorphism induces an isomorphism from $W(\mathcal{R})$ onto $W^*(\mathcal{R})$.*

Scholium 1.1.13. *By the preceding result, we shall identify W and W^* , and consider W as a group of transformations of R , R^* , M , M^* , $\Gamma_0(R)$, $\Gamma_0(R^*)$, V , V^* , $\mathcal{V}(R)$, $\mathcal{V}(R^*)$. We shall write s_α for s_α^* .*

⁽²⁾**N.D.E.** One will note that $(\alpha^*)^* = \alpha$.

1.2. The map p

Lemma 1.2.1. *Let $p : M \longrightarrow M^*$ (resp. $V \longrightarrow V^*$) be the linear map defined by*

$$(6) \quad p(x) = \sum_{u \in R^*} (u, x)u.$$

Put $\ell(x) = (p(x), x)$. One has the following properties:

$$(7) \quad \ell(x) \geq 0, \quad \ell(\alpha) > 0 \quad \text{for } \alpha \in R.$$

$$(8) \quad \ell(wx) = \ell(x), \quad \text{for } w \in W.$$

$$(9) \quad (p(x), y) = (p(y), x), \quad \text{for } x, y \in V.$$

$$(10) \quad \ell(\alpha)\alpha^* = 2p(\alpha), \quad \text{for } \alpha \in R.$$

The first three relations are immediate.⁽³⁾ Let us prove the last. By (1), for $u \in V^*$, one has:

$$\begin{aligned} (u, \alpha)^2 \alpha^* &= (u, \alpha)u - (u, \alpha)s_\alpha(u) \\ &= (u, \alpha)u + (u, s_\alpha(\alpha))s_\alpha(u) \\ &= (u, \alpha)u + (s_\alpha(u), \alpha)s_\alpha(u) \quad (\text{since } s_\alpha^2 = \text{id}). \end{aligned}$$

Since s_α is a permutation of R^* (by (DR II)), one need only sum over $u \in R^*$ to conclude.

Scholium 1.2.2. *Relation (10) says that $\alpha \longmapsto \ell(\alpha)\alpha^*$ is the restriction to R of a linear map from M to M^* . In particular, one has*

$$(-\alpha)^* = -\alpha^*.$$

Corollary 1.2.3. *The map p induces an isomorphism from $\mathcal{V}(R)$ onto $\mathcal{V}(R)^*$.*

Indeed, p sends $\mathcal{V}(R)$ onto $\mathcal{V}(R^*)$. One therefore has

$$\dim(\mathcal{V}(R)) \geq \dim(\mathcal{V}(R^*)).$$

Applying this formula to the dual root datum, one deduces

$$\dim(\mathcal{V}(R)) = \dim(\mathcal{V}(R^*)),$$

hence p , being surjective, is also bijective.

Corollary 1.2.4. *One has $\dim(\mathcal{V}(R)) = \dim(\mathcal{V}(R^*))$, hence*

$$\text{rgss}(\mathcal{R}) = \text{rgss}(\mathcal{R}^*).$$

Corollary 1.2.5. *The bilinear form (u, x) is nondegenerate on $\mathcal{V}(R^*) \times \mathcal{V}(R)$, hence puts these $\mathbf{Q}$ -vector spaces in duality.*

Indeed, if $(u, x) = 0$ for every $u \in R^*$, then $p(x) = 0$.

Corollary 1.2.6. *The symmetric bilinear form $(p(x), y)$ is positive nondegenerate on $\mathcal{V}(R)$.*

Corollary 1.2.7. *W acts faithfully on R (and hence on the other sets of 1.1.13).*

Indeed, let $u \in V^*$. Let $w \in W$; suppose $w(\alpha) = \alpha$ for every $\alpha \in R$, and let us prove $w(u) = u$. One has

$$(w(u) - u, \alpha) = (w(u), \alpha) - (u, \alpha) = (u, w^{-1}(\alpha)) - (u, \alpha) = 0.$$

But $w(u) - u \in \mathcal{V}(R^*)$. If it is orthogonal to all the roots, it is zero by 1.2.5.

Corollary 1.2.8. *The group W is finite.*

⁽³⁾**N.D.E.** (8) follows from (4), (2) and (DR II).

Proposition 1.2.9. *The actions of W respect the correspondence between roots and coroots. In other words, for $\alpha \in R$ and $w \in W$, one has*

$$w(\alpha^*) = w(\alpha)^*.$$

It suffices to check this for $w = s_\beta$, $\beta \in R$, i.e. to check the formula

$$s_\beta(\alpha^*) = s_\beta(\alpha)^*.$$

Now $\ell(s_\beta(\alpha))s_\beta(\alpha)^*/2$ equals:

$$\begin{aligned} p(s_\beta(\alpha)) &= \sum_{u \in R^*} (u, s_\beta(\alpha))u = \sum_{u \in R^*} (s_\beta(u), \alpha)u \\ &= \sum_{u \in R^*} (u, \alpha)s_\beta(u) = s_\beta(p(\alpha)); \end{aligned}$$

since $\ell(s_\beta(\alpha)) = \ell(\alpha)$, one obtains $s_\beta(\alpha)^* = s_\beta(2p(\alpha)/\ell(\alpha)) = s_\beta(\alpha^*)$.

Corollary 1.2.10. *If $\alpha \in R$ and $w \in W$, one has*

$$ws_\alpha w^{-1} = s_{w(\alpha)}.$$

Indeed, $ws_\alpha w^{-1}(x) = x - (\alpha^*, w^{-1}(x))w(\alpha) = x - (w(\alpha^*), x)w(\alpha)$, and this last term equals, by 1.2.9:

$$x - (w(\alpha)^*, x)w(\alpha) = s_{w(\alpha)}(x).$$

Corollary 1.2.11. *Let $R' \subset R$. For (M, M^*, R', R'^*) to be a root datum, it is necessary and sufficient that $\alpha, \beta \in R'$ imply $s_\alpha(\beta) \in R'$.*

2. Relations between two roots

2.1. Proportional roots

Proposition 2.1.1. *Let α and β be two roots. The following conditions are equivalent:*

- (i) *There exists $k \in \mathbf{Q}$ such that $\alpha = k\beta$.*
- (ii) *$s_\alpha = s_\beta$.*

Moreover, under these conditions, one has $\alpha^* = k^{-1}\beta^*$, and k is equal to one of the numbers 1, -1, 2, -2, 1/2, -1/2.

Suppose first (i). First, one has

$$\alpha^* = \ell(\alpha)^{-1}2p(\alpha) = k^{-2}\ell(\beta)^{-1}2kp(\beta) = k^{-1}\beta^*.$$

This implies at once $s_\alpha = s_\beta$. Conversely, if one has $s_\alpha = s_\beta$, then

$$\alpha = s_\alpha(-\alpha) = s_\beta(-\alpha) = (\beta^*, \alpha)\beta - \alpha,$$

whence $2\alpha = (\beta^*, \alpha)\beta$, with $(\beta^*, \alpha) \in \mathbf{Z}$, hence (ii) implies (i). Finally, if $\alpha = k\beta$, then $\alpha^* = k^{-1}\beta^*$, whence

$$(\alpha^*, \beta) = 2k^{-1}, \quad (\beta^*, \alpha) = 2k,$$

hence $2k$ and $2k^{-1}$ are integers, and the proof is finished.

Application 2.1.2. Root data $\mathcal{R}$ such that $\text{rgss}(\mathcal{R}) = 1$ are of one of the following two types:

- (i) Type A_1 : there exists $\alpha \in M$ such that the roots are α and $-\alpha$. The coroots are then α^* and $-\alpha^*$.
- (ii) Type A'_1 : there exists $\alpha \in M$ such that the roots are $\alpha, -\alpha, 2\alpha, -2\alpha$. The coroots are then $\alpha^*, -\alpha^*, \alpha^*/2, -\alpha^*/2$.

Definition 2.1.3. One says that $\alpha \in R$ is *indivisible* if $\alpha/2 \notin R$. One says that $\mathcal{R}$ is *reduced* if every root is indivisible.

For $\mathcal{R}$ to be reduced, it is necessary and sufficient that $\mathcal{R}^*$ be so. If α is indivisible and $w \in W$, then $w(\alpha)$ is indivisible.

Definition 2.1.4. Let $\alpha \in R$. If α is indivisible, put $\text{ind}(\alpha) = \alpha$. Otherwise, put $\text{ind}(\alpha) = \alpha/2$.

Corollary 2.1.5. If $\alpha \in R$ is indivisible and $k\alpha \in R$, where $k \in \mathbf{Q}$, then $k \in \mathbf{Z}$.

Proposition 2.1.6. Let $\mathcal{R}$ be a root datum. Then

$$\text{ind}(\mathcal{R}) = (M, M^*, \text{ind}(R), \text{ind}(R)^*)$$

is a reduced root datum, and one has

$$W(\text{ind}(\mathcal{R})) = W(\mathcal{R}).$$

Indeed, $\text{ind}(\mathcal{R})$ is a root datum by 1.2.11, since the Weyl group permutes the indivisible roots. The second assertion follows from 2.1.1.

Remark 2.1.7. If $\mathcal{R}$ is not reduced, one has $\text{ind}(R)^* \neq \text{ind}(R^*)$, and hence $\text{ind}(\mathcal{R}^*) \neq \text{ind}(\mathcal{R})^*$.

2.2. Orthogonal roots

Lemma 2.2.1. Let α and β be two roots. One has

$$(11) \quad \ell(\alpha)(\alpha^*, \beta) = \ell(\beta)(\beta^*, \alpha).$$

This follows at once from 1.2.1, formulas (9) and (10).

Corollary 2.2.2. Let $\alpha, \beta \in R$. The following conditions are equivalent:

- (i) $(\alpha^*, \beta) = 0$, (i bis) $(\beta^*, \alpha) = 0$,
- (ii) $(p(\alpha), \beta) = 0$,
- (iii) $s_\alpha(\beta) = \beta$, (iii bis) $s_\beta(\alpha) = \alpha$,
- (iv) $s_\alpha(\beta^*) = \beta^*$, (iv bis) $s_\beta(\alpha^*) = \alpha^*$,
- (v) $s_\alpha \neq s_\beta$, and s_α and s_β commute.

All the equivalences are immediate except those involving (v). Let us show that (i) (and (i bis)) imply (v). One has

$$s_\alpha s_\beta(x) = x - (\beta^*, x)\beta - (\alpha^*, x)\alpha + (\beta^*, x)(\alpha^*, \beta)\alpha.$$

If $(\alpha^*, \beta) = 0$, then $(\beta^*, \alpha) = 0$, and $s_\alpha s_\beta(x) = s_\beta s_\alpha(x)$.

Conversely, suppose (v). One then has

$$s_\alpha = s_\beta s_\alpha s_\beta = s_{s_\beta(\alpha)} \quad (\text{by 1.2.10}).$$

By 2.1, there therefore exists $k \in \mathbf{Q}^*$ such that $\alpha = ks_\beta(\alpha) = k\alpha - k(\beta^*, \alpha)\beta$. Since $s_\alpha \neq s_\beta$, β and α are not proportional by 2.1.1, hence $(\beta^*, \alpha) = 0$.

Definition 2.2.3. Two roots satisfying the equivalent conditions of 2.2.2 are called *orthogonal*.

Remark 2.2.4. The roots α and β are orthogonal if and only if the coroots α^* and β^* are orthogonal.

Lemma 2.2.5. If α and β are two orthogonal roots, then $\alpha + \beta \in R$ if and only if $\alpha - \beta \in R$.

Indeed, $s_\beta(\alpha + \beta) = s_\beta(\alpha) + s_\beta(\beta) = \alpha - \beta$.

Lemma 2.2.6. *Let α and β be two nonorthogonal roots. If one defines $\ell(\gamma^*)$ for a coroot γ^* as ℓ of the root γ^* of $\mathcal{R}^*$, one has the relation*

$$(12) \quad \ell(\alpha)\ell(\alpha^*) = \ell(\beta)\ell(\beta^*).$$

Indeed, multiplying formula (11) by the corresponding formula for $\mathcal{R}^*$, and taking account of $(\gamma^*)^* = \gamma$ for every $\gamma \in R$, one finds:

$$(\beta^*, \alpha)(\alpha^*, \beta)\ell(\alpha)\ell(\alpha^*) = (\beta^*, \alpha)(\alpha^*, \beta)\ell(\beta)\ell(\beta^*).$$

2.3. General case

Proposition 2.3.1. *If α and β are any two roots, one has*

$$(13) \quad 0 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 4.$$

If α and β are neither proportional nor orthogonal, one has

$$1 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 3.$$

Indeed, one has $4(p(\alpha), \beta)^2 = \ell(\alpha)\ell(\beta)(\alpha^*, \beta)(\beta^*, \alpha)$. On the other hand, by 1.2.6, the symmetric bilinear form $(p(x), y)$ is positive nondegenerate on $\mathcal{V}(R)$, whence $(p(x), y)^2 \leq \ell(x)\ell(y)$.⁽⁴⁾

Corollary 2.3.2. *Let α and β be two nonorthogonal roots. If $\ell(\alpha) = \ell(\beta)$, there exists $w \in W$ such that $w(\alpha) = \beta$.*

Indeed, if α and β are proportional, since $\ell(\alpha) = \ell(\beta)$, one has $\alpha = \beta$ or $\alpha = -\beta$; in this case take $w = 1$ or $w = s_\alpha$. If α and β are neither proportional nor orthogonal, by formula (11) and 2.3.1 one has:

$$(\beta^*, \alpha) = (\alpha^*, \beta) = \pm 1.$$

If $(\beta^*, \alpha) = (\alpha^*, \beta) = 1$, take $w = s_\beta s_\alpha s_\beta$. If $(\beta^*, \alpha) = (\alpha^*, \beta) = -1$, take $w = s_\alpha s_\beta$.

Corollary 2.3.3. *If α and β are two roots, if $\alpha \neq \beta$ and $(\beta^*, \alpha) > 0$ (resp. if $\alpha \neq -\beta$ and $(\beta^*, \alpha) < 0$), then $\alpha - \beta$ (resp. $\alpha + \beta$) is a root.*

The second case is deduced from the first by changing β to $-\beta$. If α and β are proportional and $(\beta^*, \alpha) > 0$, one has $\beta = \alpha$, $2\beta = \alpha$, or $\beta = 2\alpha$. The first case is excluded. In the others, one respectively has $\alpha - \beta = \beta \in R$ or $\alpha - \beta = -\alpha \in R$. If α and β are not proportional, (α^*, β) and (β^*, α) are two strictly positive integers whose product is at most 3. One of them is therefore 1. If $(\beta^*, \alpha) = 1$, one has $\alpha - \beta = s_\beta(\alpha) \in R$; if $(\alpha^*, \beta) = 1$, one has $\alpha - \beta = -s_\alpha(\beta) \in R$.

Lemma 2.3.4. *Let α and β be two nonproportional roots. If $\beta - \alpha$ is not a root, then $\beta + k\alpha \in R$ for $k = -(\alpha^*, \beta)$, but not for $k = -(\alpha^*, \beta) + 1$.*

Indeed, one has $\beta - (\alpha^*, \beta)\alpha = s_\alpha(\beta) \in R$, but $\beta + (-(\alpha^*, \beta) + 1)\alpha = s_\alpha(\beta - \alpha) \notin R$.

Proposition 2.3.5. *Let α and β be two nonproportional roots. The set of integers k such that $\beta + k\alpha \in R$ is an interval $[p, q]$ ($p \leq 0$, $q \geq 0$), and one has $p + q = -(\alpha^*, \beta)$.*

For the first assertion, it suffices, for example, to prove that if $\beta + k\alpha \in R$, k an integer > 0 , then $\beta + (k - 1)\alpha \in R$. If $k = 1$, this is trivial. If $k \geq 2$, one has

$$(\alpha^*, \beta + k\alpha) = (\alpha^*, \beta) + 2k \geq -3 + 4 > 0,$$

and one concludes by 2.3.3. Let therefore $[p, q]$ be the interval in question. Applying 2.3.4 to $\beta + p\alpha$, one finds

$$q - p = -(\alpha^*, \beta + p\alpha) = -(\alpha^*, \beta) - 2p.$$

⁽⁴⁾**N.D.E.** With equality if and only if x and y are proportional.

(5)

Remark 2.3.6. The preceding formula contains the qualitative statements 2.2.5 and 2.3.3.

Supplements 2.3.7. ⁽⁶⁾By 1.2.1 (9) and 1.2.6, the bilinear form on $\mathcal{V}(R)$ defined by

$$\langle x, y \rangle = (p(x), y)$$

is symmetric and positive definite. By 1.2.1 (10), for every $\alpha \in R$ and $y \in \mathcal{V}(R)$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y),$$

where $\ell(\alpha) = \langle \alpha, \alpha \rangle$. Consequently, one deduces from 2.3.3 the following corollary.

Corollary. *Let $\alpha \neq \beta$ in R . If $\alpha - \beta \notin R$, then $\langle \alpha, \beta \rangle \leq 0$.*

3. Simple roots, positive roots

3.1. Systems of simple roots

Lemma 3.1.1. *Let α and α_i , $i = 1, \dots, n$, be roots. Suppose α distinct from the α_i . If one has a relation*

$$\alpha = \sum_{i=1}^n q_i \alpha_i, \quad q_i \in \mathbf{Q}_+,$$

there exists an index i such that $q_i \neq 0$, $(\alpha^, \alpha_i) > 0$, $\alpha - \alpha_i \in R$.*

Indeed, one writes $2 = (\alpha^*, \alpha) = \sum_{i=1}^n q_i (\alpha^*, \alpha_i)$, which proves the first two assertions. The third then follows from 2.3.3.

Proposition 3.1.2. *Let α and α_i , $i = 1, \dots, n$, be roots. If*

$$\alpha = \sum_{i=1}^n m_i \alpha_i, \quad m_i \in \mathbf{N}_+,$$

there exists a sequence $\beta_1, \dots, \beta_m$ of roots taken from among the α_i (not necessarily pairwise distinct) such that if one puts

$$\gamma_p = \sum_{i=1}^p \beta_i, \quad p = 1, \dots, m,$$

one has $\gamma_p \in R$ and $\gamma_m = \alpha$.

Let us argue by induction on the integer $\sum m_i = m'$. If α is equal to one of the α_i , say α_{i_0} (which is automatic if $m' = 1$), take $m = 1$, $\beta_1 = \alpha_{i_0}$. Otherwise, apply the preceding lemma, and there exists an index i such that $m_i \neq 0$ and $\alpha - \alpha_i$ is a root. One then has $(m_i - 1) \in \mathbf{N}$ and

$$\alpha - \alpha_i = (m_i - 1)\alpha_i + \sum_{j \neq i} m_j \alpha_j.$$

One need only apply the induction hypothesis to $\alpha - \alpha_i$.

Corollary 3.1.3. *Let $R' \subset R$. The following conditions are equivalent:*

- (i) *If $\alpha, \beta \in R'$ and $\alpha + \beta \in R$, then $\alpha + \beta \in R'$.*
- (ii) *$(\mathbf{N} \cdot R') \cap R \subset R'$.*

⁽⁵⁾**N.D.E.** $+2p$ has been corrected to $-2p$.

⁽⁶⁾**N.D.E.** These supplements, useful for the proof of 3.1.5, have been added.

Indeed, clearly (ii) $\Rightarrow$ (i). The converse follows at once from the proposition.

Definition 3.1.4. A set of roots satisfying the conditions of 3.1.3 is said to be *closed*.

Proposition 3.1.5. Let $\Delta \subset R$ be a set of roots. The following conditions are equivalent:

(i) The elements of Δ are indivisible, linearly independent, and

$$R \subset (\mathbf{Q}_+ \cdot \Delta) \cup (-\mathbf{Q}_+ \cdot \Delta).$$

(ii) The elements of Δ are linearly independent and

$$R \subset (\mathbf{N} \cdot \Delta) \cup (-\mathbf{N} \cdot \Delta).$$

(iii) Every root has a unique expression as a linear combination of the elements of Δ , with integer coefficients all of the same sign.

(7) Clearly (ii) $\Rightarrow$ (iii). Denote by $\alpha_1, \dots, \alpha_n$ the (distinct) elements of Δ . Let us prove (i) $\Rightarrow$ (ii). Let $\alpha \in R$. One therefore has a unique expression

$$\varepsilon\alpha = \sum q_i \alpha_i, \quad q_i \in \mathbf{Q}_+, \quad \varepsilon = \pm 1.$$

Let us show that the q_i are integers. This is certainly true if they are all zero except one (cf. 2.1.5). Otherwise, α is distinct from the α_i and, applying 3.1.1, one finds an i_0 such that $\alpha' = \varepsilon\alpha - \alpha_{i_0} \in R$ and $q_{i_0} \neq 0$. This gives

$$\alpha' = (q_{i_0} - 1)\alpha_{i_0} + \sum_{i \neq i_0} q_i \alpha_i.$$

Since at least one of the q_i , $i \neq i_0$, is nonzero, (i) implies $q_{i_0} - 1 \geq 0$. One repeats the operation for α' , and after finitely many steps one has proved that the q_i are integers.

Let us show (iii) $\Rightarrow$ (i). Let us show that $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly independent over $\mathbf{Q}$. Otherwise, one would have an equality

$$x = \sum_{i \in I} a_i \alpha_i = \sum_{j \in J} b_j \alpha_j,$$

where I, J are two disjoint subsets of $\{1, \dots, n\}$, at least one of them, say I , being nonempty, and $a_i, b_j \in \mathbf{N}^*$. By corollary 2.3.7, one has $\langle \alpha_i, \alpha_j \rangle \leq 0$ if $i \in I$ and $j \in J$, whence $\langle x, x \rangle \leq 0$, and hence $x = 0$. Let $i_0 \in I$; then the equalities

$$\alpha_{i_0} = \alpha_{i_0} + \sum_{i \in I} a_i \alpha_i = \alpha_{i_0} + \sum_{j \in J} b_j \alpha_j$$

imply (by (iii)) $a_i = 0 = b_j$ for all i, j , a contradiction. This shows that the elements of Δ are linearly independent over $\mathbf{Q}$; let us show that they are also indivisible.

Let therefore $\beta \in \Delta$ be divisible. One has $\beta/2 \in R$, whence

$$\frac{\beta}{2} = \varepsilon \sum_{\alpha \in \Delta} m_\alpha \alpha, \quad m_\alpha \in \mathbf{N}, \quad \varepsilon = \pm 1,$$

and hence also $\beta = 2\varepsilon \sum_{\alpha} m_\alpha \alpha$, whence, by uniqueness, $m_\alpha = 0$ if $\alpha \neq \beta$ and $2\varepsilon m_\beta = 1$, a contradiction.

Definition 3.1.6. A set Δ of roots satisfying the conditions of 3.1.5 is called a *system of simple roots*, or a *basis* of R .

If $w \in W$ and Δ is a system of simple roots, then $w(\Delta)$ is a system of simple roots.

Remark 3.1.7. This definition involves only R , and not $\mathcal{R}$; in fact, it even involves only $\text{ind}(R)$.

(7) **N.D.E.** In what follows, the order has been modified and the proof of the implication (iii) $\Rightarrow$ (i) expanded.

Remark 3.1.8. If Δ is a system of simple roots, Δ is a basis of the free abelian group $\Gamma_0(R)$. One therefore has $\text{Card}(\Delta) = \text{rgss}(\mathcal{R})$.

Remark 3.1.9. The conditions of 3.1.5 are then clearly equivalent to the following:

- (i') The elements of Δ are indivisible, number $\leq \text{rgss}(R)$, and $R \subset (\mathbf{Q}_+ \cdot \Delta) \cup (-\mathbf{Q}_+ \cdot \Delta)$.
- (ii') One has $\text{Card}(\Delta) \leq \text{rgss}(R)$, and $R \subset (\mathbf{N} \cdot \Delta) \cup (-\mathbf{N} \cdot \Delta)$.

Corollary 3.1.10. *If Δ is a system of simple roots, $\text{ind}(\Delta^*)$ is a system of simple coroots (i.e. a system of simple roots of R^*).*

Indeed, if $\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha$ ($a_\alpha \in \mathbf{N}$), then $p(\alpha) = \sum_{\alpha \in \Delta} a_\alpha p(\alpha)$, whence, by 1.2.1 (10):

$$\beta^* = \sum_{\alpha \in \Delta} a_\alpha \frac{\ell(\alpha)}{\ell(\beta)} \alpha^*,$$

which proves that $\text{ind}(\Delta^*)$ satisfies (i').

⁽⁸⁾By 2.1.1, if α^* is not indivisible, then $\alpha^* = 2u^*$, where $u^* \in \text{ind}(\Delta^*)$ (and $u = 2\alpha$). One deduces:

Corollary 3.1.11. *If Δ is a system of simple roots and $\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha$ ($a_\alpha \in \mathbf{Z}$) is the expression of β in Δ , then $2a_\alpha \ell(\alpha)$ is divisible by $\ell(\beta)$ (and even by $2\ell(\beta)$ if α^* is indivisible).*

Before continuing to state the properties of systems of simple roots, let us show that they exist.

3.2. Systems of positive roots

Definition 3.2.1. A set $R_+ \subset R$ is called a *system of positive roots* of $\mathcal{R}$ (or of R , cf. remark 3.2.2) if it satisfies the following conditions:

- (P 1) $R_+ \cap -R_+ = \emptyset$.
- (P 2) $R_+ \cup -R_+ = R$.
- (P 3) $(\mathbf{Q}_+ \cdot R_+) \cap R \subset R_+$.

In particular, such a set is closed. We shall see later (3.3.8) that in fact a closed set satisfying (P 1) and (P 2) also satisfies (P 3), hence is a system of positive roots. If $w \in W$ and R_+ is a system of positive roots, $w(R_+)$ is a system of positive roots.

Remark 3.2.2. This definition involves only R . One will also say that R_+ is a system of positive roots of R .

Remark 3.2.3. From (P 1) and (P 2), one deduces at once

$$\text{Card}(R_+) = \text{Card}(R)/2.$$

It follows that if R_+ and R'_+ are two systems of positive roots and $R_+ \subset R'_+$, then $R_+ = R'_+$.

Remark 3.2.4. If R_+ is a system of positive roots, R_+^* is a system of positive coroots (i.e. a system of positive roots of R^*).

This follows at once from 1.1.4 and 1.2.2.

Definition 3.2.5. Let Δ be a system of simple roots. Put

$$\mathcal{P}(\Delta) = (\mathbf{Q}_+ \cdot \Delta) \cap R = (\mathbf{N}_+ \cdot \Delta) \cap R.$$

⁽⁸⁾**N.D.E.** The following sentence has been added, and in 3.1.11 $4a_\alpha \ell(\alpha)$ has been replaced by $2a_\alpha \ell(\alpha)$.

Proposition 3.2.6. *If Δ is a system of simple roots, $\mathcal{P}(\Delta)$ is a system of positive roots. If Δ is a system of simple roots and R_+ a system of positive roots, one has the equivalence:*

$$\Delta \subset R_+ \iff R_+ = \mathcal{P}(\Delta).$$

The first assertion is immediate. If $\Delta \subset R_+$, then $\mathcal{P}(\Delta) \subset R_+$ by (P 3), hence $\mathcal{P}(\Delta) = R_+$ by 3.2.3. The rest is trivial.

Remark 3.2.7. There exist systems of positive roots: let $\geq$ be a totally ordered vector space structure on $\mathcal{V}(R)$. The set of roots ≥ 0 for this order relation is a system of positive roots.

Theorem 3.2.8. *Let R_+ be a system of positive roots. There exists a unique system of simple roots $\mathcal{S}(R_+)$ such that $\mathcal{S}(R_+) \subset R_+$, i.e. such that $R_+ = \mathcal{P}(\mathcal{S}(R_+))$.*

Uniqueness follows at once from:

Lemma 3.2.9. *Let Δ be a system of simple roots. Then for $\alpha \in \mathcal{P}(\Delta)$ to belong to Δ , it is necessary and sufficient that α not be the sum of two elements of $\mathcal{P}(\Delta)$.*

This lemma follows at once from the definitions and 3.1.2.

Let us now prove the existence of $\mathcal{S}(R_+)$. Consider the set of subsets T of R_+ such that $T = \text{ind}(T)$ and $(\mathbf{Q}_+ \cdot T) \supset R_+$. This set is nonempty, for it contains $\text{ind}(R_+)$. Let Δ be a minimal element of this set for inclusion. Let us show that Δ is a system of simple roots, i.e., by 3.1.5 (i), that Δ is a linearly independent subset of $M \otimes \mathbf{Q}$.

Lemma 3.2.10. *If $\alpha, \beta \in \Delta$ and $q\alpha + q'\beta \in R$, where $q, q' \in \mathbf{Q}$, then $qq' \geq 0$.*

Indeed, if $qq' < 0$, one can write (interchanging α and β if necessary)

$$a\alpha - b\beta \in R_+, \quad a, b \in \mathbf{Q}_+^*,$$

hence by hypothesis there exists a relation

$$a\alpha - b\beta = \sum_{\gamma \in \Delta} c(\gamma)\gamma, \quad c(\gamma) \in \mathbf{Q}_+.$$

If $a \leq c(\alpha)$, it is written

$$-\beta = \frac{c(\alpha) - a}{b}\alpha + \sum_{\gamma \neq \alpha} \frac{c(\gamma)}{b}\gamma.$$

Then this element belongs to $(\mathbf{Q}_+ \cdot \Delta) \cap R$, which is contained in R_+ by (P 3). But then $\beta \in R_+ \cap -R_+$, which contradicts (P 1).

If, on the contrary, $a > c(\alpha)$, one writes

$$(a - c(\alpha))\alpha = b\beta + \sum_{\gamma \neq \alpha} c(\gamma)\gamma,$$

which proves that $\Delta \subset \mathbf{Q}_+ \cdot (\Delta - \{\alpha\})$, contrary to the minimality of Δ . Recall (cf. 2.3.7) that the bilinear form on $\mathcal{V}(R)$ defined by $\langle x, y \rangle = (p(x), y)$ is a Euclidean scalar product; moreover, for $\alpha \in R$, one has $\langle \alpha, y \rangle = \ell(\alpha)(\alpha^*, y)/2$, so that $\langle \alpha, y \rangle$ and $\langle \alpha^*, y \rangle$ have the same sign.

Lemma 3.2.11. *If $\alpha, \beta \in \Delta$, then $\langle \beta^*, \alpha \rangle \leq 0$, and hence $\langle \beta, \alpha \rangle \leq 0$.*

Indeed, $s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta \in R$, whence $(\beta^*, \alpha) \leq 0$ by 3.2.10.

Let us now prove that Δ is linearly independent. Otherwise, 3.2.11 implies, as in the proof of 3.1.5, that there exists a nontrivial relation

$$\sum_{\alpha \in \Delta} m(\alpha)\alpha = 0, \quad m(\alpha) \in \mathbf{N},$$

whence $-\alpha_0 = (m(\alpha_0) - 1)\alpha_0 + \sum_{\alpha \neq \alpha_0} m(\alpha)\alpha$ if $m(\alpha_0) \geq 1$. Then, by (P 3), α_0 belongs to $R_+ \cap -R_+$, contradicting (P 1).

This shows that Δ is a basis of R and completes the proof of theorem 3.2.8.

Corollary 3.2.12. *Let R_+ be a system of positive roots, Δ a basis of R , and $w \in W$. One has:*

$$\begin{aligned}\Delta \subset R_+ &\iff R_+ = \mathcal{P}(\Delta) \iff \Delta = \mathcal{S}(R_+), \\ \mathcal{P}(\text{ind}(\Delta^*)) &= \mathcal{P}(\Delta)^*, \quad \mathcal{S}(R_+^*) = \text{ind}(\mathcal{S}(R_+)^*); \\ \mathcal{S}(w(R_+)) &= w(\mathcal{S}(R_+)), \quad \mathcal{P}(w(\Delta)) = w(\mathcal{P}(\Delta)).\end{aligned}$$

Definition 3.2.13. If one has chosen a system of simple roots Δ , the elements of $\mathcal{P}(\Delta)$ will be called *positive*. If one has chosen a system of positive roots R_+ , the elements of $\mathcal{S}(R_+)$ will be called *simple*.

Corollary 3.2.14. *Let R_+ be a system of positive roots. Let $\alpha \in R_+$. The following conditions are equivalent:*

- (i) α is simple (i.e. $\alpha \in \mathcal{S}(R_+)$).
- (ii) α is not the sum of two elements of R_+ .
- (iii) $R_+ - \{\alpha\}$ is closed (cf. 3.1.4).

The equivalence of (i) and (ii) follows at once from 3.2.9. The equivalence of (ii) and (iii) is immediate.

Definition 3.2.15. Let Δ be a system of simple roots. The sum of the coefficients of the decomposition of a root α in Δ is called the *order* of α relative to Δ , and is denoted $\text{ord}_\Delta(\alpha)$.

One has the equivalences:

$$\alpha \in \mathcal{P}(\Delta) \iff \text{ord}_\Delta(\alpha) > 0, \quad \alpha \in \Delta \iff \text{ord}_\Delta(\alpha) = 1.$$

Lemma 3.2.16. *Let Δ be a system of simple roots and $\alpha \in \mathcal{P}(\Delta)$. There exists a sequence $\alpha_i \in \Delta$, $i = 1, \dots, m$, such that*

$$\alpha_1 + \alpha_2 + \dots + \alpha_p \in \mathcal{P}(\Delta) \quad \text{for } p = 1, \dots, m,$$

$$\alpha_1 + \alpha_2 + \dots + \alpha_m = \alpha.$$

Moreover, for every sequence α_i satisfying these conditions, one has $m = \text{ord}_\Delta(\alpha)$.

Trivial by 3.1.2.

3.3. Characterization and conjugacy of systems of positive roots

Lemma 3.3.1. *If $\alpha \in \mathcal{P}(\Delta)$, $\beta \in \Delta$, and α and β are not proportional, then $s_\beta(\alpha) \in \mathcal{P}(\Delta)$.*

Indeed, $s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta$. Since at least one simple root other than β occurs in the decomposition of α with a nonzero (hence strictly positive) coefficient, and therefore also in the decomposition of $s_\beta(\alpha)$ with the same coefficient, $s_\beta(\alpha)$ is also positive.

Corollary 3.3.2. *If $\beta \in \Delta$, the reflection s_β permutes the elements of $\mathcal{P}(\Delta)$ not proportional to β .*

Lemma 3.3.3. *If $\alpha \in \mathcal{P}(\Delta) - \Delta$, α indivisible, there exists $\beta \in \Delta$ such that $s_\beta(\alpha) \in \mathcal{P}(\Delta)$ and $\text{ord}_\Delta(s_\beta(\alpha)) < \text{ord}_\Delta(\alpha)$.*

Indeed, by 3.1.1 there exists $\beta \in \Delta$ such that $(\beta^*, \alpha) > 0$. Since $\alpha \notin \Delta$ and is indivisible, α cannot be proportional to β . Thus $s_\beta(\alpha) \in \mathcal{P}(\Delta)$ by 3.3.1, and one has $\text{ord}_\Delta(s_\beta(\alpha)) = \text{ord}_\Delta(\alpha) - (\beta^*, \alpha) < \text{ord}_\Delta(\alpha)$.

Corollary 3.3.4. *If $\alpha \in \mathcal{P}(\Delta)$ is indivisible, there exists a sequence $\beta_p \in \Delta$, $p = 1, \dots, q$, such that*

$$\alpha_p = s_{\beta_p} \cdots s_{\beta_1}(\alpha) \in \mathcal{P}(\Delta) \quad \text{for } p = 1, \dots, q,$$

and $\alpha_q \in \Delta$.

This follows from 3.3.3 by induction on $\text{ord}_\Delta(\alpha)$.

Proposition 3.3.5. *The Weyl group is generated by the s_α for $\alpha \in \Delta$. Every indivisible root is conjugate to a simple root by an element of the Weyl group.*

The second assertion follows at once from 3.3.4. The first follows from it by 1.2.10 and 2.1.1.

Proposition 3.3.6. *Let R_+ be a system of positive roots. Let $P' \subset R$ satisfy (P 2) and be closed. Then there exists $w \in W$ such that $w(R_+) \subset P'$.*

Let us state the corollaries immediately.

Corollary 3.3.7. *The Weyl group acts transitively on the set of systems of positive roots (resp. of systems of simple roots).*

Corollary 3.3.8. *For a subset of R to be a system of positive roots, it is necessary and sufficient that it satisfy (P 1) and (P 2) and be closed.*

Corollary 3.3.9. *If one equips $\Gamma_0(R)$ with an ordered group structure such that every root is > 0 or < 0 , the set of roots positive for this order structure is a system of positive roots.*

Let us now prove 3.3.6. One can find $w \in W$ such that $\text{Card}(w(R_+) \cap P')$ is maximal, hence, replacing R_+ by $w(R_+)$, one may suppose that

$$(*) \quad \text{Card}(R_+ \cap P') \geq \text{Card}(R_+ \cap s_\alpha(P'))$$

for every $\alpha \in \mathcal{S}(R_+) = \Delta$. Let us prove that $R_+ \subset P'$. Otherwise, since P' is closed, there exists $\alpha \in \Delta$, $\alpha \notin P'$. But since P' satisfies (P 2), one then has $-\alpha \in P'$ (hence $-2\alpha \in P'$ if 2α is a root). For every $\beta \in R_+ \cap P'$, one has $\beta \neq \alpha$; if 2α is not a root, one then has $s_\alpha(\beta) \in R_+$ (by 3.3.1), hence

$$s_\alpha(R_+ \cap P') \subset R_+ \cap s_\alpha(P');$$

but $R_+ \cap s_\alpha(P')$ also contains $\alpha = s_\alpha(-\alpha)$, which contradicts inequality (*). If 2α is a root, one reasons similarly.

To study the sets of roots satisfying (P 2) and closed, one is therefore reduced to the case where they contain a set of positive roots.

Proposition 3.3.10. *Let R_+ be a system of positive roots and P' a closed subset of R containing R_+ . If one puts $\Delta = \mathcal{S}(R_+)$ and $\Delta' = \Delta \cap -P'$, then P' is the union of R_+ and the set of roots which are linear combinations with negative coefficients of the elements of Δ' .*

Let us prove the assertion by induction on the order of a root $\gamma \in P' - R_+$. If $\text{ord}_\Delta(\gamma) = -1$, then $-\gamma \in \Delta$ and $\gamma \in -\Delta'$. If $\text{ord}_\Delta(\gamma) < -1$, there exists, by 3.1.2, $\beta \in \Delta$ such that $-\gamma - \beta \in R$; then

$$0 < \text{ord}_\Delta(-\gamma - \beta) = \text{ord}_\Delta(-\gamma) - 1 < \text{ord}_\Delta(-\gamma),$$

and the first inequality shows that $-\gamma - \beta \in R_+$. Thus, since $R_+ \cap -R_+ = \emptyset$, and since $\gamma + \beta$ is the sum of two roots of P' , it is an element of $P' - R_+$ such that $\text{ord}_\Delta(\gamma + \beta) > \text{ord}_\Delta(\gamma)$. Hence, by the induction hypothesis, $\gamma + \beta$ is a linear combination with negative coefficients of the elements of Δ' . Since $\gamma = (\gamma + \beta) - \beta$, it suffices to check that $\beta \in -P'$. Now $\gamma \in P'$ and $-(\gamma + \beta) \in R_+ \subset P'$, so that $-\beta = \gamma - (\gamma + \beta)$ belongs to P' .

Definition 3.3.10.1. ⁽⁹⁾One says that a subset R' of R is *symmetric* if $-R' = R'$.

⁽⁹⁾**N.D.E.** This definition has been inserted here.

Scholium 3.3.11. Let P' be a set of roots satisfying (P 2) and closed. There exist a system of simple roots Δ and a subset Δ' of Δ such that

$$P' = R \cap (\mathbf{N} \cdot \Delta \cup -\mathbf{N} \cdot \Delta').$$

If one puts $R_{\Delta'} = (\mathbf{Z} \cdot \Delta') \cap R$, which is a closed symmetric subset of R , then P' is therefore the disjoint union of $R_{\Delta'}$ and the closed subset of $\mathcal{P}(\Delta)$ formed by the $\alpha \in \mathcal{P}(\Delta) - \mathbf{N} \cdot \Delta'$, i.e. the positive roots which in their decomposition in Δ “contain at least one root of $\Delta - \Delta'$ ”.

3.4. Closed symmetric sets of roots

Proposition 3.4.1. Let $\mathcal{R} = (M, M^*, R, R'^*)$ be a root datum and R' a closed symmetric subset of R . Then:

- (i) $\mathcal{R}' = (M, M^*, R', R'^*)$ is a root datum;
- (ii) for every system of positive roots R_+ of R , $R'_+ = R_+ \cap R'$ is a system of positive roots of R' ;
- (iii) the Weyl group $W(\mathcal{R}')$ of $\mathcal{R}'$ is the subgroup of $W(\mathcal{R})$ generated by the s_α for $\alpha \in R'$.

The first assertion is trivial by 1.2.11, the second follows from 3.3.8, and the third is obvious.

Corollary 3.4.2. Let $w \in W(\mathcal{R}')$. The order of w is the least integer $n > 0$ such that $w^n(\alpha') = \alpha'$ for every $\alpha' \in R'$.

It suffices to apply 1.2.7 to the root datum $\mathcal{R}'$.

Corollary 3.4.3. Let α and β be two nonproportional roots. Let n be the least integer > 0 such that $(s_\alpha s_\beta)^n(\alpha) = \alpha$ and $(s_\alpha s_\beta)^n(\beta) = \beta$. Then the subgroup $W_{\alpha, \beta}$ of W generated by s_α and s_β is defined by the relations:

$$s_\alpha^2 = 1, \quad s_\beta^2 = 1, \quad (s_\alpha s_\beta)^n = 1.$$

Taking account of 3.4.2, it suffices to check:

Lemma 3.4.4. Let G be the group generated by two elements x and y subject to $x^2 = y^2 = 1$. Every invariant subgroup of G containing neither x nor y is generated (as an invariant subgroup) by an element of the form $(xy)^n$, where $n > 0$.

⁽¹⁰⁾Indeed, every element of G is written $(xy)^n$, or $(yx)^n = (xy)^{-n}$, or:

$$(a) \quad x(yx)^{2n} \quad \text{or} \quad y(xy)^{2n+1},$$

$$(b) \quad y(xy)^{2n} \quad \text{or} \quad x(yx)^{2n+1},$$

where $n \in \mathbf{N}$. Now the elements of type (a), resp. (b), are conjugate to x , resp. to y .

Remark 3.4.5. One immediately calculates the integer n : if one puts

$$(\alpha^*, \beta) = p, \quad (\beta^*, \alpha) = q,$$

one has⁽¹¹⁾

$$(s_\alpha s_\beta)(\alpha) = (pq - 1)\alpha - q\beta, \quad (s_\alpha s_\beta)(\beta) = p\alpha - \beta.$$

The integer n is therefore the order of the matrix

$$\begin{pmatrix} pq - 1 & p \\ -q & -1 \end{pmatrix}.$$

If $pq = 0$, then $p = q = 0$ by 2.2.2, whence $n = 2$. Otherwise, by 2.3.1, pq is 1, 2 or 3, and one finds, respectively, $n = 3, 4$ or 6.

⁽¹⁰⁾**N.D.E.** The original has been corrected in what follows.

⁽¹¹⁾**N.D.E.** The original has been corrected in what follows.

N. B. Writing that the order of the preceding matrix is finite, one recovers inequality (13) of 2.3.1.

Definition 3.4.6. Let Δ be a system of simple roots and $\Delta' \subset \Delta$. Put

$$R_{\Delta'} = R \cap (\mathbf{Q} \cdot S') = R \cap (\mathbf{Z} \cdot \Delta').$$

Lemma 3.4.7. $R_{\Delta'}$ is closed and symmetric, Δ' is a system of simple roots of the root datum $(M, M^*, R_{\Delta'}, R_{\Delta'}^*)$, whose Weyl group is the subgroup $W_{\Delta'}$ of W generated by the s_α for $\alpha \in \Delta'$. One has $\Delta \cap R_{\Delta'} = \Delta'$.

Trivial.

Proposition 3.4.8. Let $R' \subset R$. The following conditions are equivalent:

- (i) There exists a vector subspace V' of V (or of $\mathcal{V}(R)$) such that $R' = R \cap V'$.
- (ii) There exist a system of simple roots Δ of R and a subset Δ' of Δ such that $R' = R_{\Delta'}$.

More precisely, under these conditions, every system of simple roots Δ' of (M, M^*, R', R'^*) is contained in a system of simple roots Δ of R , and one has $R' = R_{\Delta'}$.

Clearly (ii) $\Rightarrow$ (i). Suppose (i) holds: then R' is closed and symmetric, hence

$$(M, M^*, R', R'^*)$$

is a root datum. Let Δ' be a system of simple roots of this datum and $R'_+ = \mathcal{P}(\Delta')$. If $V' = V$, then Δ' is a system of simple roots of R , and the proof is finished. Otherwise, there exists $x \in V^*$ such that

$$(x, V') = \{0\}, \quad (x, \alpha) \neq 0 \quad \text{for every } \alpha \in R - R'.$$

Put $R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}$ and $R_+ = R_+(x) \cup R'_+$. For every $\alpha \in R$, one has the equivalences

$$\begin{aligned} (x, \alpha) > 0 &\iff \alpha \in R_+(x), \\ (x, \alpha) < 0 &\iff \alpha \in -R_+(x), \\ (x, \alpha) = 0 &\iff \alpha \in R'. \end{aligned}$$

It follows at once from 3.3.8 that R_+ is a system of positive roots of R . Put $\Delta = \mathcal{S}(R_+)$. Clearly, it suffices to prove $\Delta' \subset \Delta$. Otherwise, let $\alpha \in \Delta' - \Delta$. Then, by 3.2.14, there exist $\beta, \gamma \in R_+$ such that $\alpha = \beta + \gamma$. If $\beta, \gamma \in R_+(x)$, one has $\alpha \in R_+(x)$, which is absurd since $(x, S') = 0$. If β or γ , for example β , belongs to R'_+ , then, since R' is symmetric and closed, $\gamma = \alpha - \beta$ belongs to $R_+ \cap R' = R'_+$; but then α is not simple in R'_+ .

Lemma 3.4.9. Under the preceding conditions, let $\alpha \in \mathcal{P}(\Delta) - R'$. For every $w \in W_{\Delta'}$, one has $w(\alpha) \in \mathcal{P}(\Delta) - R'$.

Indeed, it suffices to check this for $w = s_\beta$, $\beta \in \Delta'$, in which case it follows from 3.3.1 and $s_\beta(R') = R'$.

Lemma 3.4.10. Let $w \in W$. Under the conditions of 3.4.8, the following conditions are equivalent:

- (i) $w \in W_{\Delta'}$.
- (ii) For every $m \in M$, $w(m) - m \in V'$.
- (iii) For every $\alpha \in R$, $w(\alpha) - \alpha \in V'$.

Clearly (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). Let us prove (iii) $\Rightarrow$ (i). Let therefore $w \in W$ be such that $w(\alpha) - \alpha \in V'$ for every $\alpha \in R$. Write $w = s_{\alpha_n} \cdots s_{\alpha_1}$, with $\alpha_i \in \Delta$, and let us show by induction on n that each $\alpha_i \in \Delta'$. One may suppose that $w' = s_{\alpha_{n-1}} \cdots s_{\alpha_1} \in W_{\Delta'}$. One then has, for every $\alpha \in R$,

$$w(\alpha) - \alpha = w'(\alpha) - \alpha - (\alpha_n^*, w'(\alpha))\alpha_n.$$

Taking $\alpha = w'^{-1}(\alpha_n)$, one finds $2\alpha_n \in V'$, whence $\alpha_n \in \Delta'$, hence $w = s_{\alpha_n} w'$ belongs to $W_{\Delta'}$.

3.5. Various remarks

Proposition 3.5.1. *Let R_+ be a system of positive roots. Put*

$$\rho_{R_+} = \frac{1}{2} \sum_{\alpha \in \text{ind}(R_+)} \alpha.$$

Then $(\beta^, \rho_{R_+}) = 1$ for every $\beta \in \mathcal{S}(R_+)$.*

Indeed, one can write

$$2\rho_{R_+} = \beta + \sum_{\substack{\alpha \in \text{ind}(R_+) \\ \alpha \neq \beta}} \alpha,$$

hence $s_\beta(2\rho_{R_+}) = 2\rho_{R_+} - 2\beta$, by 3.3.2.

Corollary 3.5.2. *Put $\rho_{R_+}^* = \frac{1}{2} \sum_{\alpha \in \text{ind}(R_+^*)} \alpha^*$. Then:*

- (i) $(\rho_{R_+}^*, \alpha) > 0$ for every $\alpha \in R_+$ (i.e. $\rho_{R_+}^* \in \mathcal{C}(R_+)$, cf. 3.6.8).
- (ii) For every $\alpha \in R$, one has $\text{ord}_{\mathcal{S}(R_+)}(\alpha) = (\rho_{R_+}^*, \alpha)^{(12)}$

Remark 3.5.3. If $w \in W$, one has $\rho_{w(R_+)} = w(\rho_{R_+})$ and $\rho_{w(R_+)}^* = w(\rho_{R_+}^*)$.

Proposition 3.5.4. *Let α and γ be two nonproportional roots, α being supposed indivisible. There exists a system of simple roots containing α and a root β such that $\gamma = a\alpha + b\beta$, with $a, b \in \mathbf{N}$.*

Indeed, construct a basis of the vector space $\mathcal{V}(R)$ containing $\alpha_1 = \alpha$, $\alpha_2 = \gamma$. Consider the lexicographic order relative to this basis. Denoting the set of roots > 0 by R_+ , it is clear that R_+ is a system of positive roots, and that⁽¹³⁾ the least element of R_+ not proportional to α is simple. This element has the form

$$\beta = p\alpha + q\gamma, \quad 0 < q \leq 1.$$

One therefore has $\gamma = q^{-1}\beta - q^{-1}p\alpha$, and, since $q^{-1} > 0$, one has $q^{-1} \in \mathbf{N}^*$ and $-q^{-1}p \in \mathbf{N}$.

Finally, let us make two remarks on the group $\Gamma_0(R)$.

Proposition 3.5.5. *Let G be an abelian group and $f : R \rightarrow G$ a map satisfying the following two conditions:*

- (i) If $\alpha \in R$, $f(-\alpha) = -f(\alpha)$.
- (ii) If $\alpha, \beta, \alpha + \beta \in R$, $f(\alpha + \beta) = f(\alpha) + f(\beta)$.

Then there exists a unique group homomorphism $\bar{f} : \Gamma_0(R) \rightarrow G$ such that $\bar{f}(\alpha) = f(\alpha)$ for $\alpha \in R$.

Indeed, if Δ is a system of simple roots of R , and $\beta \in R$ is written $\sum_{\alpha \in \Delta} a(\alpha)\alpha$, it follows at once from 3.2.16 that $f(\beta) = \sum_{\alpha \in \Delta} a(\alpha)f(\alpha)$. Now Δ is a basis of $\Gamma_0(R)$.

Proposition 3.5.6. *Let Δ be a system of simple roots. There exists on $\Gamma_0(R)$ a totally ordered group structure such that the roots > 0 are the elements of $\mathcal{P}(\Delta)$ and $\alpha \mapsto \text{ord}_\Delta(\alpha)$ is an increasing function.*

Indeed, let $\alpha_1, \alpha_2, \dots, \alpha_n$ ($n = \text{rgss}(\mathcal{R})$) be the elements of Δ . For $x \in \Gamma_0(R)$, one has a decomposition

$$x = \sum_{i=1}^n m_i(x)\alpha_i.$$

It suffices to take the lexicographic order relative to the functions $\sum m_i, m_n, m_{n-1}, \dots, m_2$.

⁽¹²⁾**N.D.E.** Indeed, by 3.1.5, it suffices to check the formula for $\alpha \in \Delta$. Now, by 3.5.1 applied to R^* , one has in this case $(\rho_{R_+}^*, \alpha) = 1 = \text{ord}_{\mathcal{S}(R_+)}(\alpha)$.

⁽¹³⁾**N.D.E.** The original has been modified to take account of the case where 2α is a root.

Remark 3.5.7. The first roots are, in order:

$$\alpha_1, \alpha_2, \dots, \alpha_n;$$

then one has (if they are roots) $2\alpha_1, \alpha_1 + \alpha_2, \alpha_1 + \alpha_3, \dots$

3.6. Weyl chambers

Lemma 3.6.1. *Let V be a finite-dimensional $\mathbf{R}$ -vector space.⁽¹⁴⁾ Let f_i be independent linear forms. Put*

$$C = \{x \in V \mid f_i(x) > 0\}.$$

Then C is a maximal convex subset of $X = V - \bigcup_i f_i^{-1}(0)$.

Trivial.

Definition 3.6.2. A subset C of V which can be described by the procedure of 3.6.1 will be called (here) a *chamber* of V .

Definition 3.6.3. One says that the hyperplane H of V is a *wall* of C if $H \cap (\overline{C} - C)$ contains a nonempty open subset of H .

Remark 3.6.4. For a convex subset, the closure can be described without appealing to the topology of V : it is the set of endpoints of all open segments contained in the given subset.

Lemma 3.6.5. *Under the conditions of 3.6.1, one has*

$$\overline{C} = \{x \in V \mid f_i(x) \geq 0\}.$$

The walls of C are the hyperplanes $f_i^{-1}(0)$.

This is clear for the first assertion. The second then follows from $\overline{C} - C \subset \bigcup_i f_i^{-1}(0)$ and the fact that the $f_i^{-1}(0)$ are clearly walls of C .

Proposition 3.6.6. *Let C be a chamber of V . If $H_i, i = 1, 2, \dots, n$, are the distinct walls of C , then for every system of linear forms $\{u_i\}$ such that $H_i = u_i^{-1}(0)$, there exist $\varepsilon_i \in \{-1, +1\}$ such that C is defined by*

$$C = \{x \in V \mid \varepsilon_i u_i(x) > 0\}.$$

For every wall H of C , one has $H \cap C = \emptyset$ and $H \cap \overline{C} = \overline{C_H}$, where C_H is a chamber in H . The walls of C_{H_i} are the $H_j \cap H_i$ for $j \neq i$.

This follows trivially from the lemma.

Definition 3.6.7. The C_{H_i} are the *faces* of C .

Let now $\mathcal{R} = (M, M^*, R, R^*)$ be a root datum. Put $V_{\mathbf{R}}^* = M^* \otimes \mathbf{R}$.

Definition 3.6.8. ⁽¹⁵⁾For every $\alpha \in R$, put

$$\mathcal{H}_\alpha = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) = 0\}.$$

Put $X = V_{\mathbf{R}}^* - \bigcup_{\alpha \in R} \mathcal{H}_\alpha$. For every $x \in X$, put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}.$$

For every system of positive roots R_+ , put

$$\mathcal{C}(R_+) = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in R_+\}.$$

Proposition 3.6.9.

⁽¹⁴⁾**N.D.E.** $\mathbf{Q}$ has been replaced by $\mathbf{R}$.

⁽¹⁵⁾**N.D.E.** The definition of the hyperplanes $\mathcal{H}_\alpha$ has been added.

(i) For every $x \in X$, $R_+(x)$ is a system of positive roots. For every system of positive roots R_+ , $\mathcal{C}(R_+)$ is a chamber in $V_{\mathbf{R}}^*$. The $\mathcal{C}(R_+)$ are the maximal convex subsets of X .

(ii) Let Δ be a system of simple roots. One has

$$\mathcal{C}(\mathcal{P}(\Delta)) = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in \Delta\}.$$

The walls of $\mathcal{C}(\mathcal{P}(\Delta))$ are the hyperplanes $\mathcal{H}_\alpha = \alpha^{-1}(0)$, for $\alpha \in \Delta$; its faces are the

$$C_\alpha = \{x \in V_{\mathbf{R}}^* \mid (x, \alpha) = 0, (x, \beta) > 0 \text{ for } \beta \in \Delta, \beta \neq \alpha\}.$$

(iii) One has the equivalence

$$R_+(x) = R_+ \iff x \in \mathcal{C}(R_+).$$

First, it is clear that $R_+(x)$ is a system of positive roots. Since $x \in \mathcal{C}(R_+(x))$, the union of the $\mathcal{C}(R_+(x))$ is X . Property (iii) is immediate; it follows that the $\mathcal{C}(R_+)$ form a partition of X . The first assertion of (ii) is obvious. It follows at once that $\mathcal{C}(R_+)$ is a chamber in V , which proves the rest of (i). Putting $C = \mathcal{C}(\mathcal{P}(\Delta))$, one need only note that

$$\overline{C} \bigcup_{\alpha \in R} \alpha^{-1}(0) = \overline{C} \bigcup_{\alpha \in \Delta} \alpha^{-1}(0)$$

to complete the proof of (ii) by 3.6.1.

Definition 3.6.10. The $\mathcal{C}(R_+)$ are called the *Weyl chambers* of the root datum. On the other hand, for every Weyl chamber C , put $R_+(C) = R_+(x)$ for any $x \in C$.

Corollary 3.6.11. The maps $R_+ \mapsto \mathcal{C}(R_+)$ and $C \mapsto R_+(C)$ give a bijective correspondence between systems of positive roots and Weyl chambers.

This correspondence is invariant under the Weyl group:

Lemma 3.6.12. If $w \in W(\mathcal{R})$, one has $\mathcal{C}(w(R_+)) = w(\mathcal{C}(R_+))$.

Corollary 3.6.13. The correspondences $\Delta \leftrightarrow R_+ \leftrightarrow C$ are isomorphisms of homogeneous spaces under $W(\mathcal{R})$.

We shall see later that these homogeneous spaces are principal (5.5).

Remark 3.6.14. If C is a Weyl chamber, $-C$ is also one, called the *opposite* of C . There therefore exists $w_0 \in W$ (and in fact only one, cf. 5.5) such that $w_0(C) = -C$; it is called the *reflection* of the root datum relative to the Weyl chamber C (or relative to $R_+(C)$ or $\mathcal{S}(R_+(C)), \dots$).

4. Reduced root data of semisimple rank 2

4.0. — ⁽¹⁶⁾Let $\mathcal{R}$ be a root datum of semisimple rank 2. Let $\{\alpha, \beta\}$ be a system of simple roots. Suppose $\ell(\alpha) \leq \ell(\beta)$. By 2.3.1 and 3.2.1, one then has four possibilities:

Type	$\ell(\beta)/\ell(\alpha)$	$\ell(\beta^*)/\ell(\alpha^*)$	(β^*, α)	(α^*, β)
$A_1 \times A_1$	—	—	0	0
A_2	1	1	−1	−1
B_2	2	1/2	−1	−2
G_2	3	1/3	−1	−3

It then follows from 3.4.5 that the order of $s_\alpha s_\beta$ is respectively 2, 3, 4, 6.

Let us study each of these systems separately and give the list of indivisible roots.

Type $A_1 \times A_1$. The indivisible roots are $\alpha, \beta, -\alpha, -\beta$. The corresponding coroots are $\alpha^*, \beta^*, -\alpha^*, -\beta^*$.

⁽¹⁶⁾**N.D.E.** The numbering 4.0 has been added for subsequent references.

Type A_2 . The positive indivisible roots are as follows:

root γ	α	β	$\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	1	1
coroot γ^*	α^*	β^*	$\alpha^* + \beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	1	1	1

The half-sum of the positive indivisible roots is $\rho = \alpha + \beta$.

Type B_2 . The positive indivisible roots are the following:

root γ	α	β	$\alpha + \beta$	$2\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	2	1	2
coroot γ^*	α^*	β^*	$2\alpha^* + \beta^*$	$\alpha^* + \beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	2	1	2	1

The half-sum of the positive indivisible roots is $\rho = (4\alpha + 3\beta)/2$.

Type G_2 . The positive indivisible roots are the following:

root γ	α	β	$\alpha + \beta$	$2\alpha + \beta$	$3\alpha + \beta$	$3\alpha + 2\beta$
$\ell(\gamma)/\ell(\alpha)$	1	3	1	1	3	3
coroot γ^*	α^*	β^*	$\alpha^* + 3\beta^*$	$2\alpha^* + 3\beta^*$	$\alpha^* + \beta^*$	$\alpha^* + 2\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	3	1	3	3	1	1

The half-sum of the positive indivisible roots is $\rho = 5\alpha + 3\beta$.

Proposition 4.1. ^(*)Let n be the order of $s_\alpha s_\beta$. Put $u_0 = 0$ and, for $p \in \mathbf{N}$,

$$\begin{cases} u_{2p+1} = u_{2p} + (s_\alpha s_\beta)^p(\alpha); \\ u_{2p+2} = u_{2p+1} + (s_\alpha s_\beta)^p s_\alpha(\beta). \end{cases}$$

Then:⁽¹⁷⁾

- (i) $u_{k+2n} = u_k$ for every $k \in \mathbf{N}$.
- (ii) $u_0 = 0$, $u_1 = \alpha$, $u_{2n-1} = \beta$, $u_{2n} = 0$.
- (iii) If $1 < k < 2n - 1$, one has $u_k = a_k \alpha + b_k \beta$ with $a_k, b_k \in \mathbf{N}^*$.

Assertion (i) follows from $(s_\alpha s_\beta)^n = 1$ and $u_{2n} = 0$.

Let us prove (ii) and (iii). Calculation gives at once, in the four cases, the sequences of values:⁽¹⁸⁾

$$\begin{aligned} (A_1 \times A_1) & \quad 0, \alpha, \beta + \alpha, \beta, 0. \\ (A_2) & \quad 0, \alpha, \beta + 2\alpha, 2\beta + 2\alpha, 2\beta + \alpha, \beta, 0. \\ (B_2) & \quad 0, \alpha, \beta + 3\alpha, 2\beta + 4\alpha, 3\beta + 4\alpha, 3\beta + 3\alpha, 2\beta + \alpha, \beta, 0. \\ (G_2) & \quad 0, \alpha, \beta + 4\alpha, 2\beta + 6\alpha, 4\beta + 9\alpha, 5\beta + 10\alpha, 6\beta + 10\alpha, \\ & \quad 6\beta + 9\alpha, 5\beta + 6\alpha, 4\beta + 4\alpha, 2\beta + \alpha, \beta, 0. \end{aligned}$$

Lemma 4.2. Put $w_{2p} = (s_\beta s_\alpha)^p$, $w_{2p+1} = s_\alpha (s_\beta s_\alpha)^p$, so that $w_0, \dots, w_{2n-1}$ are the distinct elements of W . Let $u_0, \dots, u_{2n-1}$ be the elements of V defined in 4.1. For every $x \in V^*$, one has

$$x - w_k(x) = n_k \alpha^* + m_k \beta^*,$$

with $n_k, m_k \in \mathbf{Q}$ and $n_k + m_k = (x, u_k)$.⁽¹⁹⁾

^(*)The following paragraphs 4.1 to 4.4 are used in the proof of 5.1. There are now simpler proofs of 5.1 (see [BLie], § V.3, Th. 1). **N.D.E.** The reference has been specified, and this Note, which in the original appeared in 4.2, has been placed here.

⁽¹⁷⁾**N.D.E.** Let ρ be the half-sum of the positive roots (cf. 3.5.1); if one puts, as in 4.2 below, $w_0 = 1$, $w_1 = s_\alpha$, $w_2 = s_\beta s_\alpha$, $w_3 = s_\alpha s_\beta s_\alpha$, etc., then u_k is precisely $\rho - w_k^{-1}(\rho)$, which proves (i), since $w_{2n} = \text{id}$.

⁽¹⁸⁾**N.D.E.** The original has been corrected to agree with the convention $\ell(\alpha) \leq \ell(\beta)$ of 4.0.

⁽¹⁹⁾**N.D.E.** Taking account of N.D.E. (17), this follows immediately from 3.5.1 and 1.1.9: one has $n_k + m_k = (x - w_k(x), \rho) = (x, \rho - w_k^{-1}(\rho)) = (x, u_k)$.

The proof is by induction on k . If $k = 0$, the formula is trivially satisfied. Let us carry out, for example, the step from w_{2p} to w_{2p+1} . One has $w_{2p+1} = s_\alpha w_{2p}$, whence

$$\begin{aligned} x - w_{2p+1}(x) &= x - w_{2p}(x) + w_{2p}(x) - s_\alpha w_{2p}(x) \\ &= n_{2p}\alpha^* + m_{2p}\beta^* + (w_{2p}(x), \alpha)\alpha^*. \end{aligned}$$

Thus

$$\begin{aligned} n_{2p+1} + m_{2p+1} &= n_{2p} + m_{2p} + ((s_\beta s_\alpha)^p(x), \alpha) \\ &= (x, u_{2p}) + (x, (s_\alpha s_\beta)^p(\alpha)) = (x, u_{2p+1}). \end{aligned}$$

Corollary 4.3. ⁽²⁰⁾ *Let $x \in V^*$. For every $w \in W$, put*

$$x - w(x) = a_w\alpha^* + b_w\beta^*.$$

If $(x, \alpha) \geq 0$ and $(x, \beta) \geq 0$, then $a_w + b_w \geq 0$. If in addition $(x, \alpha) > 0$ (resp. $(x, \beta) > 0$), then $a_w + b_w > 0$ for $w \neq 1, s_\beta$ (resp. for $w \neq 1, s_\alpha$).

This follows at once from 4.1 and 4.2.

Corollary 4.4. *Let $\mathcal{R}$ be any root datum and Δ a system of simple roots. Let γ be a positive root, α, β two simple roots, $W_{\alpha, \beta}$ the subgroup of W generated by s_α and s_β . If*

$$\text{ord}_\Delta(s_\alpha(\gamma)) < \text{ord}_\Delta(\gamma), \quad \text{ord}_\Delta(s_\beta(\gamma)) \leq \text{ord}_\Delta(\gamma),$$

then, for every $w \in W_{\alpha, \beta}$, one has $\text{ord}_\Delta(w(\gamma)) \leq \text{ord}_\Delta(\gamma)$; moreover, $\text{ord}_\Delta(w(\gamma)) < \text{ord}_\Delta(\gamma)$ if $w \neq 1, w \neq s_\beta$.

Indeed, consider the dual root datum $\mathcal{R}^*$, and then the datum (M^*, M, R'^*, R') , where R' is the set of roots which are rational linear combinations of α and β . Applying 4.3 to this datum, one obtains the announced corollary. ⁽²¹⁾

5. The Weyl group: generators and relations

Let $\mathcal{R}$ be a root datum. Since the Weyl group is the same for this datum and for the corresponding reduced datum, one may suppose $\mathcal{R}$ reduced to study the Weyl group.

Let $\Delta = \{\alpha_1, \dots, \alpha_n\}$ be a system of simple roots ($n = \text{rgss}(\mathcal{R})$). Let n_{ij} be the order of the element $s_{\alpha_i} s_{\alpha_j}$ of W . In particular, one has $n_{ii} = 1$, and we saw in 3.4.2 and 3.4.3 that the subgroup W_{ij} of W generated by s_{α_i} and s_{α_j} was defined by the relations:

$$s_{\alpha_i}^2 = s_{\alpha_j}^2 = (s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1.$$

Theorem 5.1. *The group W is the group generated by the elements s_{α_i} , $i = 1, 2, \dots, n$, subject to the relations $(s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1$.*

We have already seen that the theorem is true when $n = 2$; we shall use this remark in the course of the proof. Introduce the group $\overline{W}$ generated by elements T_i , $i = 1, 2, \dots, n$, subject to the relations $(T_i T_j)^{n_{ij}} = 1$. In particular, one has $n_{ii} = 1$, whence $T_i^2 = 1$. Let $p : \overline{W} \rightarrow W$ be the morphism of groups sending T_i to s_{α_i} . One knows that p is surjective; we shall show that it is injective.

Lemma 5.2. *One can define uniquely, for each $\alpha \in \mathcal{P}(\Delta)$, an element $T_\alpha \in \overline{W}$ in such a way that the following properties hold:*

- (i) $p(T_\alpha) = s_\alpha$,
- (ii) $T_{\alpha_i} = T_i$,
- (iii) if β and α are two positive roots such that $s_{\alpha_i}(\alpha) = \beta$, then $T_i T_\alpha T_i = T_\beta$.

⁽²⁰⁾**N.D.E.** The statement has been corrected.

⁽²¹⁾**N.D.E.** Indeed, the hypotheses amount to saying that $(\gamma, \alpha^*) > 0$ and $(\gamma, \beta^*) \geq 0$; then $\gamma - w(\gamma)$ belongs to $\mathbf{N}\alpha + \mathbf{N}\beta$ for every $w \in W_{\alpha, \beta}$, and is $\neq 0$ if $w \notin \{1, s_\beta\}$.

Note first that it follows from 1.2.10 and 3.3.6 that (i) is a consequence of (ii) and (iii), and that (ii) and (iii) determine the T_α completely. We shall make the construction by induction on $\text{ord}_\Delta(\alpha)$. If $\text{ord}_\Delta(\alpha) = 1$, then $\alpha \in \Delta$, and put $T_\alpha = T_i$ if $\alpha = \alpha_i$. Consider the hypothesis:

- (H_p) there exist T_α , for $\alpha \in \mathcal{P}(\Delta)$, $\text{ord}_\Delta(\alpha) \leq p$, satisfying (ii) and condition (iii) whenever $\text{ord}_\Delta(\alpha) \leq p$, $\text{ord}_\Delta(\beta) \leq p$.

This holds for $p = 1$: indeed, if α and $s_{\alpha_i}(\alpha) = \beta$ are simple, α_i and α are orthogonal; thus, putting $\alpha_j = \alpha = \beta$, one has $n_{ij} = 2$, whence

$$T_i T_j T_i = T_j.$$

Suppose $p > 1$ and (H_{p-1}) holds.

A) *Construction of the T_α for $\text{ord}_\Delta(\alpha) \leq p$.* Clearly, it suffices to do this for $\text{ord}_\Delta(\alpha) = p$. There then exists $\alpha_i \in \Delta$ such that $s_{\alpha_i}(\alpha) \in \mathcal{P}(\Delta)$ and $\text{ord}_\Delta(s_{\alpha_i}(\alpha)) < p$ (3.3.3). One will then put

$$(\star) \quad T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

Let us check that T_α depends only on α . Let therefore $\alpha_j \in \Delta$ be such that $s_{\alpha_j}(\alpha) \in \mathcal{P}(\Delta)$ and $\text{ord}_\Delta(s_{\alpha_j}(\alpha)) < p$. Let us prove that

$$(+)\quad T_i T_{s_{\alpha_i}(\alpha)} T_i = T_j T_{s_{\alpha_j}(\alpha)} T_j.$$

Distinguish two cases.

(1) Suppose α a linear combination of α_i and α_j . The same is then true of $s_{\alpha_i}(\alpha)$ and $s_{\alpha_j}(\alpha)$, and, by (H_{p-1}), $T_{s_{\alpha_i}(\alpha)}$ and $T_{s_{\alpha_j}(\alpha)}$ are written as words in T_i and T_j . Since the projection of (+) into W holds, and the theorem is true for $n = 2$, hence p is injective on the subgroup of $\overline{W}$ generated by T_i and T_j , (+) indeed holds.

(2) Suppose α is not a linear combination of α_i and α_j . Then if $w \in W_{\alpha_i, \alpha_j}$, the $w(\alpha)$ are all positive (cf. 3.4.9). The relation to be checked is also written

$$(++)\quad (T_i T_j)^{n_{ij}-1} T_{s_{\alpha_i}(\alpha)} (T_j T_i)^{n_{ij}-1} = T_{s_{\alpha_j}(\alpha)}.$$

Now it follows from 4.4 that the $w(\alpha)$ all have order $< p$ for $w \in W_{\alpha_i, \alpha_j}$, $w \neq 1$. One may therefore apply hypothesis (H_{p-1}) $2(n_{ij} - 1)$ times, and the proof is finished.

B) *Verification of (H_p).*⁽²²⁾ One must check that if $\alpha_j \in \Delta$ and $\beta = s_{\alpha_j}(\alpha)$ satisfies $\text{ord}_\Delta(\beta) \leq p$, then $T_j T_\alpha T_j = T_\beta$. If $\text{ord}_\Delta(\beta) < p$, this follows from what precedes (since $T_j^2 = 1$); thus one may suppose $\text{ord}_\Delta(\beta) = p = \text{ord}_\Delta(\alpha)$. In this case, α and α_j are orthogonal, hence $\beta = \alpha$, and one must see that

$$(\dagger) \quad T_j T_\alpha T_j = T_\alpha.$$

By ($\star$) above, one has $T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i$, and one therefore need only check the following equality:

$$(+++) \quad T_j T_i T_{s_{\alpha_i}(\alpha)} T_i T_j = T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

Put $m = n_{ij}$ and $s_i = s_{\alpha_i}$, $s_j = s_{\alpha_j}$. One has $T_j T_i = (T_i T_j)^{m-1}$ and, by 4.4, $\text{ord}_\Delta(w(\alpha)) < p$ for every $w \in W_{ij}$ distinct from $1 = (s_i s_j)^m$ and from $s_j = s_i (s_i s_j)^{m-1}$. One deduces, by the induction hypothesis, that

$$\begin{aligned} T_j (T_i T_j)^{m-2} T_{s_i(\alpha)} (T_j T_i)^{m-2} T_j &= T_{s_j(s_i s_j)^{m-2} s_i(\alpha)} \\ &= T_{s_i s_j(\alpha)} = T_{s_i(\alpha)} \end{aligned}$$

(the last equality since $s_j(\alpha) = \alpha$), whence finally

$$(T_i T_j)^{m-1} T_{s_i(\alpha)} (T_j T_i)^{m-1} = T_i T_{s_i(\alpha)} T_i,$$

which proves (+++).

⁽²²⁾**N.D.E.** The original has been expanded in what follows.

Lemma 5.3. *Let $h \in \overline{W}$. Write it*

$$h = T_{\alpha_1} \cdots T_{\alpha_m}$$

with $\alpha_i \in \Delta$, not necessarily distinct, in such a way that m is minimal. Then

$$p(h)(\alpha_m) \in -\mathcal{P}(\Delta).$$

Indeed, since $p(T_{\alpha_m})(\alpha_m) = s_{\alpha_m}(\alpha_m) = -\alpha_m$, if $p(h)(\alpha_m)$ were positive, there would exist an index k , $1 \leq k \leq m-1$, such that

$$u = s_{\alpha_{k+1}} \cdots s_{\alpha_m}(\alpha_m) = -s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) \in -\mathcal{P}(\Delta),$$

and $s_{\alpha_k}(u) \in \mathcal{P}(\Delta)$. But then necessarily $u = -\alpha_k$ (3.3.1), whence

$$s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) = \alpha_k,$$

which implies by (iii)

$$T_{\alpha_k} T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}} T_{\alpha_m} = T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}},$$

and this contradicts the minimality of m .

Let now $h \in \overline{W}$ be such that $p(h)(\mathcal{P}(\Delta)) \subset \mathcal{P}(\Delta)$. By lemma 5.3, one has $p(h) = 1$, which proves theorem 5.1 and, in addition:

Corollary 5.4. *If R_+ is a system of positive roots and $w \in W$ is such that $w(R_+) = R_+$, then $w = 1$.*

Corollary 5.5. *The Weyl group acts freely and transitively on the set of systems of positive roots (resp. systems of simple roots, resp. Weyl chambers).*

Now choose a system of simple roots Δ . Put $R^+ = \mathcal{P}(\Delta)$.

⁽²³⁾For every pair of simple roots $(\alpha, \beta) \in \Delta \times \Delta$, denote by $R_{\alpha, \beta}$ the set of roots which are linear combinations of α and β . Put $R_{\alpha, \beta}^+ = R^+ \cap R_{\alpha, \beta}$, and let $W_{\alpha, \beta}$ be the Weyl group of $R_{\alpha, \beta}$, that is, the subgroup of W generated by s_α and s_β .

Theorem 5.6 (Tits). *Let α and β be two simple roots, and $w \in W$ such that $w(\alpha) = \beta$. There exist a sequence of simple roots $\alpha_0, \dots, \alpha_m$ and a sequence $w_0, \dots, w_{m-1}$ of elements of W satisfying the following conditions:*

- (i) $\alpha_0 = \alpha$, $\alpha_m = \beta$.
- (ii) $w = w_{m-1} w_{m-2} \cdots w_0$.
- (iii) $w_i(\alpha_i) = \alpha_{i+1}$ for $0 \leq i \leq m-1$.
- (iv) For every i , $0 \leq i \leq m-1$, such that $\alpha_i \neq \alpha_{i+1}$, one has $w_i \in W_{\alpha_i, \alpha_{i+1}}$.
- (v) For every i , $0 \leq i \leq m-1$, such that $\alpha_i = \alpha_{i+1}$, there exists a simple root β_i such that $w_i \in W_{\alpha_i, \beta_i}$.

Put⁽²⁴⁾

$$M(w) = \text{Card}(R^+ \cap w^{-1}(-R^+)) = \text{Card}\{\alpha \in R^+ \mid w(\alpha) \in -R^+\}.$$

If $M(w) = 0$, then $w(R^+) = R^+$, hence $w = 1$ by 5.4, and the theorem is trivial ($m = 0$, assertions (iii) to (v) are empty). Argue by induction on $M(w)$. If $M(w) > 0$, there exists $\beta_0 \in \Delta$ such that $w(\beta_0) \in -R^+$. Put $\alpha_0 = \alpha$. Consider the set

$$A = w^{-1}(R^+) \cap R_{\alpha_0, \beta_0}.$$

It is a system of positive roots of R_{α_0, β_0} . There therefore exists $w_0 \in W_{\alpha_0, \beta_0}$ such that

$$w_0^{-1}(R_{\alpha_0, \beta_0}^+) = A.$$

⁽²³⁾**N.D.E.** R^+ has been written here instead of R_+ , so as to be able to write $R_{\alpha, \beta}^+ = R_{\alpha, \beta} \cap R^+$.

⁽²⁴⁾**N.D.E.** In what follows, the original has been expanded, and equality (1) corrected.

Put $w' = ww_0^{-1}$. By 3.4.9, one has at once

$$R^+ - R_{\alpha_0, \beta_0}^+ = w_0(R^+ - R_{\alpha_0, \beta_0}^+),$$

whence

$$(1) \quad (R^+ - R_{\alpha_0, \beta_0}^+) \cap w^{-1}(-R^+) = w_0((R^+ - R_{\alpha_0, \beta_0}^+) \cap w'^{-1}(-R^+)).$$

On the other hand,

$$(2) \quad \beta_0 \in R_{\alpha_0, \beta_0}^+ \cap w^{-1}(-R^+),$$

and, since $w_0(R_{\alpha_0, \beta_0}) = R_{\alpha_0, \beta_0}$, one has $w_0(-A) = R_{\alpha_0, \beta_0} \cap w'^{-1}(-R^+)$, whence

$$(2') \quad R_{\alpha_0, \beta_0}^+ \cap w'^{-1}(-R^+) = R_{\alpha_0, \beta_0}^+ \cap w_0(-A) = R_{\alpha_0, \beta_0}^+ \cap -R_{\alpha_0, \beta_0}^+ = \emptyset.$$

It follows from (1), (2) and (2') that $M(w') < M(w)$.

Put $\alpha_1 = w_0(\alpha_0)$; let us show that $\alpha_1 \in \Delta$, i.e. $\alpha_0 \in w_0^{-1}(\Delta)$. One knows that $w(\alpha_0) \in \Delta$, hence $\alpha_0 \in w^{-1}(\Delta)$, hence also $\alpha_0 \in w^{-1}(\Delta) \cap R_{\alpha_0, \beta_0}$, hence α_0 is a simple root of $A = w^{-1}(R^+) \cap R_{\alpha_0, \beta_0} = w_0^{-1}(R_{\alpha_0, \beta_0}^+)$, hence belongs to

$$w_0^{-1}(\Delta \cap R_{\alpha_0, \beta_0}^+) = w_0^{-1}(\{\alpha_0, \beta_0\})$$

(see 3.4.8). Thus $\alpha_1 = w_0(\alpha_0)$ equals α_0 or β_0 . If $\alpha_1 \neq \alpha_0$, one has $\alpha_1 = \beta_0$ and $w_0 \in W_{\alpha_0, \alpha_1}$.

Finally, one has $\beta = w'(\alpha_1)$, with $M(w') < M(w)$, and one concludes by induction.

6. Morphisms of root data

6.1. — Definition. Let $\mathcal{R} = (M, M^*, R, R)$ and $\mathcal{R}' = (M', M'^*, R', R'^*)$ be two root data. Let $f : M' \rightarrow M$ be a linear map and ${}^t f : M^* \rightarrow M'^*$ its transpose.

Definition 6.1.1. One says that f is a *morphism* from $\mathcal{R}'$ to $\mathcal{R}$, and writes $f : \mathcal{R}' \rightarrow \mathcal{R}$, if f induces a bijection from R' onto R and ${}^t f$ a bijection from R^* onto R'^* .

Then ${}^t f$ is a morphism of the dual root data:

$${}^t f : \mathcal{R}^* \rightarrow \mathcal{R}'^*.$$

One easily sees that if f is a morphism from $\mathcal{R}'$ to $\mathcal{R}$, and one puts $\alpha = f(\alpha')$ for $\alpha' \in R'$, then $\alpha'^* = {}^t f(\alpha^*)$. Indeed, one sees immediately that if p and p' denote the maps of 1.2.1 relative to $\mathcal{R}$ and $\mathcal{R}'$, respectively, one has $p' = {}^t f \circ p \circ f$, and the desired assertion follows at once. We leave to the reader the proofs of the following statements, almost all of which are trivial.

Proposition 6.1.2. Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism of root data. If $\alpha' \in R'$ and $\alpha = f(\alpha')$, then $\alpha'^* = {}^t f(\alpha^*)$. Moreover, f induces isomorphisms:

$$R' \xrightarrow{\sim} R, \quad \Gamma_0(R') \xrightarrow{\sim} \Gamma_0(R), \quad \mathcal{V}(R') \xrightarrow{\sim} \mathcal{V}(R),$$

and ${}^t f$ induces isomorphisms:

$$R^* \xrightarrow{\sim} R'^*, \quad \Gamma_0(R^*) \xrightarrow{\sim} \Gamma_0(R'^*), \quad \mathcal{V}(R^*) \xrightarrow{\sim} \mathcal{V}(R'^*),$$

the last being the transpose of the corresponding morphism induced by f . The map $s_{\alpha'} \mapsto s_{f(\alpha')}$ extends to an isomorphism $W(\mathcal{R}') \xrightarrow{\sim} W(\mathcal{R})$ compatible with the actions of these two groups on the sets of 1.1.13.

Proposition 6.1.3. The maps

$$\Delta' \mapsto f(\Delta'), \quad R'_+ \mapsto f(R'_+), \quad C' \mapsto ({}^t f \otimes \mathbf{R})^{-1}(C')$$

define bijective correspondences between systems of simple roots, systems of positive roots, and Weyl chambers for $\mathcal{R}'$ and $\mathcal{R}$. These correspondences are compatible with the action of the Weyl groups and with the correspondences

$$\mathcal{S}(R_+) \leftrightarrow R_+ \leftrightarrow \mathcal{C}(R_+).$$

Lemma 6.1.4. *Morphisms compose. For the morphism $f : \mathcal{R}' \rightarrow \mathcal{R}$ to be an isomorphism, it is necessary and sufficient that $f : M' \rightarrow M$ be bijective.*

6.2. Isogenies

Definition 6.2.1. A morphism $f : \mathcal{R}' \rightarrow \mathcal{R}$ of root data is called an *isogeny* if $f : M' \rightarrow M$ is injective with finite cokernel.

If f is an isogeny, ${}^t f$ is an isogeny.

Definition 6.2.2. Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be an isogeny. Put $K(f) = \text{Coker}(M' \xrightarrow{f} M)$.

Lemma 6.2.3. *One has a natural pairing*

$$K(f) \times K({}^t f) \rightarrow \mathbf{Q}/\mathbf{Z},$$

putting these two finite groups in duality.

This is classical.

Lemma 6.2.4. *If $f : \mathcal{R}' \rightarrow \mathcal{R}$ is a morphism, then $\text{rgss}(\mathcal{R}') = \text{rgss}(\mathcal{R})$. If in addition f is an isogeny, one also has $\text{rgred}(\mathcal{R}') = \text{rgred}(\mathcal{R})$.*

Trivial.

Lemma 6.2.5. *Every morphism of semisimple root data is an isogeny.*

This follows at once from the fact that f must induce an isomorphism from $V' = \mathcal{V}(R')$ onto $V = \mathcal{V}(R)$.

If $\mathcal{R}'$ and $\mathcal{R}$ are semisimple, every isogeny $f : \mathcal{R}' \rightarrow \mathcal{R}$ defines a commutative diagram:

$$\begin{array}{ccc} \Gamma_0(R') & \xrightarrow{\sim} & \Gamma_0(R) \\ \downarrow & & \downarrow \\ M' & \xrightarrow{f} & M. \end{array}$$

If $M = \Gamma_0(R)$, f is necessarily an isomorphism.

Definition 6.2.6. A root datum is called *adjoint* (resp. *simply connected*) if $M = \Gamma_0(R)$, resp. $M^* = \Gamma_0(R^*)$.

An adjoint or simply connected root datum is therefore semisimple. On the other hand, $\mathcal{R}$ is adjoint (resp. simply connected) if and only if $\mathcal{R}^*$ is simply connected (resp. adjoint). By the preceding result, one has:

Proposition 6.2.7. *Let $\mathcal{R}$ be a semisimple root datum. The following conditions are equivalent:*

- (i) $\mathcal{R}$ is adjoint (resp. simply connected).
- (ii) Every isogeny $\mathcal{R}' \rightarrow \mathcal{R}$ (resp. $\mathcal{R} \rightarrow \mathcal{R}'$) is an isomorphism.

Proposition 6.2.8. *Let $\mathcal{R}$ be an adjoint (resp. simply connected) root datum. Every indivisible root (resp. coroot) is an indivisible element of M (resp. M^*).*

Indeed, every indivisible root belongs to a basis of $\Gamma_0(R)$ by 3.3.5.

6.3. Radical and coradical

Let $\mathcal{R}$ be a root datum. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha^* \in R^*\}; \quad N^* = M^*/\mathcal{V}(R^*) \cap M^*.$$

Lemma 6.3.1. *Consider the canonical morphisms:*

$$N \longrightarrow M, \quad M^* \longrightarrow N^*.$$

They are transposes of one another, and N^ is identified with the dual of N .*

This is immediate, taking account of 1.2.5.

Definition 6.3.2. One calls the *coradical* of $\mathcal{R}$, and denotes by $\text{corad}(\mathcal{R})$, the trivial root datum

$$\text{corad}(\mathcal{R}) = (N, N^*, \emptyset, \emptyset).$$

If one puts $\mathcal{R}^0 = (M, M^*, \emptyset, \emptyset)$ (this is a trivial root datum), one therefore has a morphism

$$\text{corad}(\mathcal{R}) \longrightarrow \mathcal{R}^0.$$

Definition 6.3.3. One calls the *radical* of $\mathcal{R}$, and denotes by $\text{rad}(\mathcal{R})$, the trivial root datum:

$$\text{rad}(\mathcal{R}) = \text{corad}(\mathcal{R}^*)^*.$$

(25)

One therefore has a diagram

$$\begin{array}{ccc} \text{corad}(\mathcal{R}) & \xrightarrow{\quad} & \text{rad}(\mathcal{R}), \\ & \searrow \quad \nearrow & \\ & \mathcal{R}^0 & \end{array}$$

whose transpose is the corresponding diagram for $\mathcal{R}^*$.

Lemma 6.3.4. *The canonical morphism $u : \text{corad}(\mathcal{R}) \longrightarrow \text{rad}(\mathcal{R})$ is an isogeny.*

Definition 6.3.5. Put $N(\mathcal{R}) = K(u) = M/((\mathcal{V}(R) \cap M) + \bigcap_{\alpha \in R} \text{Ker}(\alpha^*))$. One then has a canonical pairing

$$N(\mathcal{R}) \times N(\mathcal{R}^*) \longrightarrow \mathbf{Q}/\mathbf{Z}.$$

Lemma 6.3.6. *One has $\text{rgred}(\text{rad}(\mathcal{R})) = \text{rgred}(\text{corad}(\mathcal{R})) = \text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R})$, and the following conditions are equivalent:*

- (i) $\mathcal{R}$ is semisimple,
- (ii) $\text{rad}(\mathcal{R}) = 0$,
- (iii) $\text{corad}(\mathcal{R}) = 0$.

6.4. Products of root data

Definition 6.4.1. Let $\mathcal{R} = (M, M^*, R, R^*)$ and $\mathcal{R}' = (M', M'^*, R', R'^*)$ be two root data. One calls the *product root datum* of $\mathcal{R}$ and $\mathcal{R}'$, and denotes by $\mathcal{R}'' = \mathcal{R} \times \mathcal{R}'$, the root datum (M'', M''^*, R'', R''^*) where

$$\begin{aligned} M'' &= M \times M', & M''^* &= M^* \times M'^*, \\ R'' &= (R \times 0) \cup (0 \times R'), & R''^* &= (R^* \times 0) \cup (0 \times R'^*), \end{aligned}$$

the map $\alpha \mapsto \alpha^*$ being the obvious map.

(25) **N.D.E.** I.e. if one puts $P = \{y \in M^* \mid (y, \alpha) = 0 \text{ for every } \alpha \in R\}$ and $P^* = M/\mathcal{V}(R) \cap M$, one has $\text{rad}(\mathcal{R}) = (P^*, P, \emptyset, \emptyset)$.

Proposition 6.4.2. *Under the preceding conditions, one has canonical isomorphisms*

$$\Gamma_0(R'') \simeq \Gamma_0(R) \times \Gamma_0(R'), \quad \mathcal{V}(R'') \simeq \mathcal{V}(R) \times \mathcal{V}(R'),$$

$$W(\mathcal{R}'') \simeq W(\mathcal{R}) \times W(\mathcal{R}'),$$

etc., and the equalities

$$\text{rgred}(\mathcal{R}'') = \text{rgred}(\mathcal{R}) + \text{rgred}(\mathcal{R}'),$$

$$\text{rgss}(\mathcal{R}'') = \text{rgss}(\mathcal{R}) + \text{rgss}(\mathcal{R}').$$

One also has a canonical isomorphism of root data

$$(\mathcal{R} \times \mathcal{R}')^* \simeq \mathcal{R}^* \times \mathcal{R}'^*.$$

The preceding definitions generalize at once to a product of several factors. One has at once:

Proposition 6.4.3. *Let $\mathcal{R} = \mathcal{R}_1 \times \mathcal{R}_2 \times \cdots \times \mathcal{R}_n$ be a product of root data. The following conditions are equivalent:*

- (i) $\mathcal{R}$ is semisimple (resp. simply connected, resp. adjoint, resp. reduced).
- (ii) Each $\mathcal{R}_i$ is semisimple (resp. simply connected, resp. adjoint, resp. reduced).

Consider the following particular case: let $\mathcal{R}_0$ be a trivial root datum and $\mathcal{R}_1$ a semisimple root datum. One then has a commutative diagram

$$\begin{array}{ccc} & \mathcal{R}_0 \times \mathcal{R}_1 & \\ \nearrow & & \searrow \\ \mathcal{R}_1 & \xrightarrow{\text{id}} & \mathcal{R}_1. \end{array}$$

Lemma 6.4.4. *One has canonical isomorphisms*

$$\begin{array}{ccc} \text{corad}(\mathcal{R}_0 \times \mathcal{R}_1) & \xrightarrow{\sim} & \text{rad}(\mathcal{R}_0 \times \mathcal{R}_1) \\ & \searrow \simeq & \nearrow \simeq \\ & \mathcal{R}_0 & \end{array}$$

In particular, $N(\mathcal{R}_0 \times \mathcal{R}_1) = 0$.

We shall see later that, conversely, if $N(\mathcal{R}) = 0$, the root datum $\mathcal{R}$ is the product of a semisimple datum and a trivial datum.

6.5. Induced and coinduced root data

Let $\mathcal{R} = (M, M^*, R, R^*)$ be a root datum. Let $N \subset M$ be a subgroup containing the roots, i.e. such that

$$\Gamma_0(R) \subset N \subset M.$$

The canonical linear map $i_N : N \rightarrow M$ gives by transposition a linear map

$${}^t i_N : M^* \rightarrow N^*.$$

Put $R_N = R$ and $R_N^* = {}^t i_N(R^*)$.

Lemma 6.5.1. $\mathcal{R}_N = (N, N^*, R_N, R_N^*)$ is a root datum, and i_N a morphism.

Let us first show that ${}^t i_N$ induces an isomorphism from R^* onto R_N^* . If $\alpha, \beta \in R$ and ${}^t i_N(\alpha^*) = {}^t i_N(\beta^*)$, one has $(\alpha^*, x) = (\beta^*, x)$ for every $x \in N$, in particular for $x \in R$, which gives $\alpha = \beta$ by 1.1.4. The rest follows without difficulty.

Definition 6.5.2. $\mathcal{R}_N$ is called the root datum *induced* by $\mathcal{R}$ on N .

Lemma 6.5.3. *Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism. Put $N = f(M') \subset M$. Then f factors uniquely through i_N .*

In particular, isogenies $\mathcal{R}' \rightarrow \mathcal{R}$, up to isomorphism, correspond bijectively to subgroups of finite index of $M/\Gamma_0(R)$, which makes 6.2.7 more precise.

Let now N^* be a subgroup of M^* containing R^* . One defines the datum *coinduced* by $\mathcal{R}$ on N^* by

$$\mathcal{R}^{N^*} = (\mathcal{R}_{N^*}^*)^*,$$

and one has a canonical morphism:

$$p^{N^*} : \mathcal{R} \rightarrow \mathcal{R}^{N^*}.$$

Lemma 6.5.4. *Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism. There exist subgroups $N \subset M$ and $N'^* \subset M'^*$ such that f factors as*

$$\begin{array}{ccc} \mathcal{R}' & \xrightarrow{f} & \mathcal{R} \\ p^{N'} \downarrow & & \uparrow i_N \\ \mathcal{R}'^{N'^*} & \xrightarrow[\sim]{f_0} & \mathcal{R}_N, \end{array}$$

where f_0 is an isomorphism.

Indeed, take $N = f(M')$ as in 6.5.3. The resulting morphism $M' \rightarrow N$ is surjective, hence its transpose injective. Take the image of the latter as N'^* .

Let us now treat certain particular cases. If one takes $N = \Gamma_0(R)$, one will write $\mathcal{R}_N = \text{ad}(\mathcal{R})$. If one takes $N = \mathcal{V}(R) \cap M$, one will write $\mathcal{R}_N = \text{ss}(\mathcal{R})$. One therefore has a diagram:

$$\text{ad}(\mathcal{R}) \rightarrow \text{ss}(\mathcal{R}) \rightarrow \mathcal{R}.$$

Put $\text{dér}(\mathcal{R}) = \text{ss}(\mathcal{R}^*)^*$ and $\text{sc}(\mathcal{R}) = \text{ad}(\mathcal{R}^*)^*$; by duality, one obtains a diagram:

$$\mathcal{R} \rightarrow \text{dér}(\mathcal{R}) \rightarrow \text{sc}(\mathcal{R}).$$

Proposition 6.5.5.

(i) *In the first row of the diagram*

$$\begin{array}{ccccccc} \text{ad}(\mathcal{R}) & \rightarrow & \text{ss}(\mathcal{R}) & \xrightarrow{\quad} & \text{dér}(\mathcal{R}) & \rightarrow & \text{sc}(\mathcal{R}) \\ & & & \searrow & \nearrow & & \\ & & & \mathcal{R} & & & \end{array}$$

the four data are semisimple and the three morphisms are isogenies.

- (ii) $\text{ad}(\mathcal{R})$ is an adjoint datum, and $\mathcal{R}$ is adjoint if and only if $\text{ad}(\mathcal{R}) \rightarrow \mathcal{R}$ is an isomorphism.
- (iii) $\text{sc}(\mathcal{R})$ is a simply connected datum, and $\mathcal{R}$ is simply connected if and only if $\mathcal{R} \rightarrow \text{sc}(\mathcal{R})$ is an isomorphism.
- (iv) The following conditions are equivalent:
 - (a) $\mathcal{R}$ is semisimple,
 - (b) $\text{ss}(\mathcal{R}) \rightarrow \mathcal{R}$ is an isomorphism,
 - (c) $\mathcal{R} \rightarrow \text{dér}(\mathcal{R})$ is an isomorphism.

Let us pause a moment on the morphism $\text{ss}(\mathcal{R}) \rightarrow \text{dér}(\mathcal{R})$. Returning to the construction of $\text{ss}(\mathcal{R})$ and $\text{dér}(\mathcal{R})$, it is easy to prove:

Lemma 6.5.6. *Let $h : \text{ss}(\mathcal{R}) \rightarrow \text{dér}(\mathcal{R})$ be the canonical isogeny. One has $K(h) \simeq N(\mathcal{R})$.*

6.5.7. — Consider other particular cases of induced data. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for } \alpha \in R\} \times \Gamma_0(R);$$

one knows that the sum is direct by 1.2.5. It follows that the root datum $\mathcal{R}_N$ is identified with the product $\text{ad}(\mathcal{R}) \times \text{corad}(\mathcal{R})$.

One can do the same replacing $\Gamma_0(R)$ by $\mathcal{V}(R) \cap M$, then dualize these two constructions. One thus obtains a diagram of root data:

$$\begin{array}{ccccccc}
 \text{ad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) & \longrightarrow & \text{dér}(\mathcal{R}) & \longrightarrow & \text{sc}(\mathcal{R}) & ; \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & \\
 \text{ad}(\mathcal{R}) \times \text{corad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) & \longrightarrow & \mathcal{R} & \longrightarrow & \text{dér}(\mathcal{R}) \times \text{rad}(\mathcal{R}) & \longrightarrow \text{sc}(\mathcal{R}) \times \text{rad}(\mathcal{R}) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & \\
 \text{ad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) & \longrightarrow & \text{dér}(\mathcal{R}) & \longrightarrow & \text{sc}(\mathcal{R})
 \end{array}$$

which is commutative, as one checks at once. This diagram is self-dual in an obvious sense. The horizontal morphisms are isogenies. The composites of the vertical arrows are the identity.

Lemma 6.5.8. *Let h_1 and h_2 be the canonical isogenies:*

$$\text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) \xrightarrow{h_1} \mathcal{R} \xrightarrow{h_2} \text{dér}(\mathcal{R}) \times \text{rad}(\mathcal{R}).$$

One has $K(h_1) \simeq K(h_2) \simeq N(\mathcal{R})$.

This is trivial from the definitions.

Corollary 6.5.9. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

- (i) $N(\mathcal{R}) = 0$, i.e. $\text{corad}(\mathcal{R}) \longrightarrow \text{rad}(\mathcal{R})$ is an isomorphism.
- (ii) $h : \text{ss}(\mathcal{R}) \longrightarrow \text{dér}(\mathcal{R})$ is an isomorphism.
- (iii) $h_1 : \text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) \longrightarrow \mathcal{R}$ is an isomorphism.
- (iv) $h_2 : \mathcal{R} \longrightarrow \text{dér}(\mathcal{R}) \times \text{rad}(\mathcal{R})$ is an isomorphism.
- (v) $\mathcal{R}$ is the product of a semisimple datum and a trivial datum.

Let us also state a trivial consequence of the preceding remarks:

Corollary 6.5.10. *For every root datum $\mathcal{R}$, there exist isogenies*

$$\text{ad}(\mathcal{R}) \times \mathcal{R}_0 \longrightarrow \mathcal{R} \longrightarrow \text{sc}(\mathcal{R}) \times \mathcal{R}_0,$$

where $\mathcal{R}_0$ is “the” trivial root datum of rank $\text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R})$.

Finally, let us point out a result which may be useful:

Lemma 6.5.11. *Let $\mathcal{R}$ be a root datum, Δ a system of simple roots, Δ' a subset of Δ ; consider the root datum (cf. 3.4.7)*

$$\mathcal{R}_{\Delta'} = (M, M^*, R_{\Delta'}, R_{\Delta'}^*).$$

- (i) *If $\mathcal{R}$ is simply connected, $\text{dér}(\mathcal{R}_{\Delta'})$ is simply connected.*
- (ii) *If $\mathcal{R}$ is adjoint, $\text{ss}(\mathcal{R}_{\Delta'})$ is adjoint.*

The two assertions are clearly equivalent by duality. The second reduces to checking the formula:

$$M \cap \mathcal{V}(R_{\Delta'}) = \Gamma_0(R_{\Delta'});$$

now, if $M = \Gamma_0(R)$, both sides are equal to the subgroup of M generated by Δ' .

6.6. Weights

Definition 6.6.1. Let $\mathcal{R}$ be a root datum. Put⁽²⁶⁾

$$\Lambda(\mathcal{R}) = \{x \in V(R) \mid (\alpha^*, x) \in \mathbf{Z} \text{ for every } \alpha^* \in R^*\}.$$

The elements of $\Lambda(\mathcal{R})$ are called the *weights* of $\mathcal{R}$. The weights of $\mathcal{R}^*$ are called the *coweights* of $\mathcal{R}$.

One has $\Gamma_0(R) \subset \Lambda(\mathcal{R})$, and $\Lambda(\mathcal{R})$ is stable under $W(\mathcal{R})$.

Lemma 6.6.2. The bilinear map $V^* \times V \longrightarrow \mathbf{Q}$ induces a duality

$$\Gamma_0(R^*) \times \Lambda(\mathcal{R}) \longrightarrow \mathbf{Z}.$$

Trivial.

Corollary 6.6.3. Let $\Delta^* = (\alpha_1^*, \dots, \alpha_n^*)$ be a system of simple coroots. Let p_i , $i = 1, 2, \dots, n$, be the elements of $\mathcal{V}(R)$ defined by

$$(\alpha_i^*, p_j) = \delta_{ij},$$

(whence $s_{\alpha_i}(p_i) = p_i - \alpha_i$ and $s_{\alpha_i}(p_j) = p_j$ for $i \neq j$).^(*) Then $\Lambda(\mathcal{R})$ is the free abelian group generated by the p_i .

The p_i are called the *fundamental weights* corresponding to the system of simple coroots Δ^* .

Corollary 6.6.4. For every $\alpha^* \in \Delta^*$, one therefore has $(\alpha^*, \sum_i p_i) = 1$, hence $\sum_i p_i = \rho_{R_+}$ (cf. 3.5.1), where $R_+ = \mathcal{P}(\text{ind}(\Delta))$.

Corollary 6.6.5. For every $x \in \mathcal{V}(R)$, one has $x = \sum_i (\alpha_i^*, x) p_i$.

Note that $R^* \subset \Gamma_0(R^*)$ and $R \subset \Lambda(\mathcal{R})$, hence $(\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*)$ is a root datum.

Corollary 6.6.6. The canonical morphism $\Gamma_0(R^*) \longrightarrow M^*$ is the transpose of the morphism $x \longmapsto \sum_i (\alpha_i^*, x) p_i$, which defines a morphism of root data, and one has a commutative diagram:

$$\begin{array}{ccc} & \text{sc}(\mathcal{R}) & \\ & \uparrow & \downarrow \simeq \\ \mathcal{R} & \longrightarrow & (\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*). \end{array}$$

One therefore has an explicit description of $\text{sc}(\mathcal{R})$ in terms of the weights of $\mathcal{R}$. Similarly, one finds a commutative diagram:

$$\begin{array}{ccc} \text{ad}(\mathcal{R}) & & \\ \downarrow \simeq & \searrow & \\ (\Gamma_0(R), \Lambda(\mathcal{R}^*), R, R^*) & \longrightarrow & \mathcal{R} \end{array}$$

Corollary 6.6.7. For $\mathcal{R}$ to be simply connected, it is necessary and sufficient that $M = \Lambda(\mathcal{R})$.

Remark 6.6.8. One has $\Lambda(\mathcal{R}) \cap M = \mathcal{V}(R) \cap M$. For $\mathcal{R}$ to be semisimple, it is therefore necessary and sufficient that $M \subset \Lambda(\mathcal{R})$.

It also follows from the results of 6.5 that:

Corollary 6.6.9. For $\mathcal{R}$ to be the product of a simply connected datum and a trivial datum, it is necessary and sufficient that $M \supset \Lambda(\mathcal{R})$.

⁽²⁶⁾**N.D.E.** $\Gamma(\mathcal{R})$ has been replaced by $\Lambda(\mathcal{R})$ to avoid any risk of confusion with $\Gamma_0(R)$.

^(*)Attention: if the datum is not reduced, the α_i do not form a system of simple roots.

Consider now the canonical isogeny

$$f : \text{ad}(\mathcal{R}) \longrightarrow \text{sc}(\mathcal{R}),$$

and put $Z(\mathcal{R}) = K(f)$. One has $Z(\mathcal{R}) \simeq Z(\text{sc } \mathcal{R}) \simeq Z(\text{ad } \mathcal{R})$.

Corollary 6.6.10. *One has a canonical isomorphism $Z(\mathcal{R}) = \Lambda(\mathcal{R})/\Gamma_0(R)$. More precisely, one has an exact sequence of $W(\mathcal{R})$ -modules:*

$$0 \longrightarrow \Gamma_0(R) \longrightarrow \Lambda(\mathcal{R}) \longrightarrow Z(\mathcal{R}) \longrightarrow 0.$$

Corollary 6.6.11. *One has a canonical pairing*

$$Z(\mathcal{R}^*) \times Z(\mathcal{R}) \longrightarrow \mathbf{Q}/\mathbf{Z}$$

putting these groups in duality.

Remark 6.6.12. One has $Z(\mathcal{R} \times \mathcal{R}') \simeq Z(\mathcal{R}) \times Z(\mathcal{R}')$. Consider, in particular, simply connected data $\mathcal{R}_i$, $i = 1, 2, \dots, n$, and a trivial datum $\mathcal{R}_0$. Put $\mathcal{R} = \mathcal{R}_0 \times \mathcal{R}_1 \times \dots \times \mathcal{R}_n$. Let $\mathcal{R} = (M, M^*, R, R^*)$, $\mathcal{R}_0 = (M_0^*, M_0^*, \emptyset, \emptyset)$. One has

$$M/\Gamma_0(R) \simeq M_0 \times Z(\mathcal{R}_1) \times \dots \times Z(\mathcal{R}_n).$$

6.7. Automorphisms

An automorphism of $\mathcal{R}$ is, by 6.1.4, an automorphism u of M such that $u(R) = R$, ${}^t u(R^*) = R^*$. In particular, every element w of $W(\mathcal{R})$ defines an automorphism of $\mathcal{R}$.

Lemma 6.7.1. *$W(\mathcal{R})$ is an invariant subgroup of $\text{Aut}(\mathcal{R})$. More precisely, if $u \in \text{Aut}(\mathcal{R})$ and $\alpha \in R$, one has*

$$us_\alpha u^{-1} = s_{u(\alpha)}.$$

The proof is the same as that of 1.2.10.

Proposition 6.7.2. *Let Δ be a system of simple roots. Put*

$$E_\Delta(\mathcal{R}) = \{u \in \text{Aut}(\mathcal{R}) \mid u(\Delta) = \Delta\}.$$

Then $\text{Aut}(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta(\mathcal{R})$.

This follows at once from the fact that $W(\mathcal{R})$ acts simply transitively on the systems of simple roots, and that if Δ is a system of simple roots of R , $u(\Delta)$ is a system of simple roots for every automorphism u of $\mathcal{R}$.

We shall see later a simpler description of $E_\Delta(\mathcal{R})$ in the case of reduced irreducible root data.⁽²⁷⁾

Definition 6.7.3. Denote by $\text{Aut}^s(\mathcal{R})$ ⁽²⁸⁾ the set of $u \in \text{Aut}(\mathcal{R})$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{u} & \mathcal{R} \\ & \searrow & \swarrow \\ & \text{rad}(\mathcal{R}) & \end{array}$$

Put $E_\Delta^s(\mathcal{R}) = E_\Delta(\mathcal{R}) \cap \text{Aut}^s(\mathcal{R})$.

Remark 6.7.4. If $u \in \text{Aut}(\mathcal{R})$, one therefore has $u \in \text{Aut}^s(\mathcal{R})$ if and only if $(u - \text{id})(M) \subset \mathcal{V}(R)$. In particular, $W(\mathcal{R}) \subset \text{Aut}^s(\mathcal{R})$. It follows at once that:

Proposition 6.7.5. *The group $\text{Aut}^s(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta^s(\mathcal{R})$, for every system of simple roots Δ .*

⁽²⁷⁾**N.D.E.** When $\mathcal{R}$ is simply connected or adjoint, see 7.4.5.

⁽²⁸⁾**N.D.E.** The exponent s is intended to suggest “semisimple”.

With every automorphism of $\mathcal{R}$ there is associated by functoriality an automorphism of $\text{ad}(\mathcal{R})$. One therefore has a canonical morphism

$$\text{Aut}(\mathcal{R}) \longrightarrow \text{Aut}(\text{ad}(\mathcal{R})).$$

Lemma 6.7.6. *The morphism $\text{Aut}^s(\mathcal{R}) \longrightarrow \text{Aut}(\text{ad}(\mathcal{R}))$ is injective.*

Indeed, let u be an automorphism of M such that $(u - \text{id})(M) \subset \mathcal{V}(R)$ and ${}^t u(\alpha^*) = \alpha^*$ for $\alpha^* \in R^*$. For every $x \in M$, one has

$$(\alpha^*, u(x) - x) = ({}^t u(\alpha^*) - \alpha^*, x) = 0,$$

hence $u(x) - x = 0$ by 1.2.5.

Lemma 6.7.7. *The group $\text{Aut}^s(\mathcal{R})$ is finite.*

Indeed, it suffices to prove that $\text{Aut}(\mathcal{R})$ is finite if $\mathcal{R}$ is adjoint. Since M is then generated by R , every automorphism of $\mathcal{R}$ is determined by the permutation of R which it defines.

Remark 6.7.8. One sees at once that $\text{Aut}(\mathcal{R})$ (resp. $E_\Delta(\mathcal{R})$) is finite if and only if $\text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R}) \leq 1$.

6.8. p -morphisms of reduced root data

In this section, p is an integer > 0 , fixed once and for all.

Definition 6.8.1. Let $\mathcal{R} = (M, M^*, R, R^*)$ and $\mathcal{R}' = (M', M'^*, R', R'^*)$ be two reduced root data. One says that a morphism of groups

$$f : M' \longrightarrow M$$

is a p -morphism from $\mathcal{R}'$ to $\mathcal{R}$ if the following conditions hold: there exist a bijection

$$u : R \xrightarrow{\sim} R'$$

and a map $q : R \longrightarrow \{p^n, n \in \mathbf{N}\}$ such that:

- (i) $f(u(\alpha)) = q(\alpha)\alpha$ for every $\alpha \in R$.
- (ii) ${}^t f(\alpha^*) = q(\alpha)u(\alpha)^*$ for every $\alpha \in R$.⁽²⁹⁾

Corollary 6.8.2. *A 1-morphism is precisely a morphism.*

Corollary 6.8.3. *The transpose of a p -morphism is a p -morphism.*

Lemma 6.8.4. *If $w \in W(\mathcal{R})$, $\alpha \in R$, one has $q(w(\alpha)) = q(\alpha)$. The map $s_\alpha \longmapsto s_{u(\alpha)}$ extends to an isomorphism $\bar{u} : W(\mathcal{R}) \longrightarrow W(\mathcal{R}')$ such that*

$$u(w(\alpha)) = \bar{u}(w)(u(\alpha)).$$

It suffices to prove that for $\alpha, \beta \in R$, one has $u(s_\alpha(\beta)) = s_{u(\alpha)}u(\beta)$ and $q(s_\alpha(\beta)) = q(\beta)$. Now one has successively:

$$\begin{aligned} f(s_{u(\alpha)}u(\beta)) &= f(u(\beta)) - (u(\alpha)^*, u(\beta))f(u(\alpha)) \\ &= q(\beta)\beta - q(\beta)q(\alpha)^{-1}(\alpha^*, \beta)q(\alpha)\alpha \\ &= q(\beta)(\beta - (\alpha^*, \beta)\alpha) = q(\beta)s_\alpha(\beta). \end{aligned}$$

If $\gamma = u^{-1}(s_{u(\alpha)}u(\beta))$, one therefore has $q(\gamma)\gamma = f(u(\gamma)) = q(\beta)s_\alpha(\beta)$. The two roots γ and $s_\alpha(\beta)$ are therefore proportional (over $\mathbf{Q}$), hence equal or opposite, but $q(\gamma)$ and $q(\beta)$ are positive. One therefore has $q(\gamma) = q(\beta)$ and $\gamma = s_\alpha(\beta)$.

Definition 6.8.5. The $q(\alpha)$ are called the *root exponents* of f .

⁽²⁹⁾**N.D.E.** Note that these two conditions imply $q(\alpha)(u(\alpha^*), u(\beta)) = q(\beta)(\alpha^*, \beta)$ for all $\alpha, \beta \in R$.

Example 6.8.6. Let $\mathcal{R}$ be a reduced root datum and $q = p^n$ ($n \in \mathbf{N}$). Then multiplication by $q : M \rightarrow M$, $x \mapsto qx$, is a p -morphism all of whose root exponents are q (and $u = \text{id}$); it is denoted

$$q : \mathcal{R} \rightarrow \mathcal{R}.$$

Proposition 6.8.7. *With the notations of 6.8.1, u gives an isomorphism from the set of systems of simple roots (resp. positive roots) of R onto the corresponding set for R' .*

This follows at once from 3.1.5 (resp. 3.2.1).

7. Structure

7.1. Decomposition of a root datum

Proposition 7.1.1. *Let $\mathcal{R}$ be a root datum, Δ a system of simple roots.*

- (i) *Let R' and R'' be two closed symmetric sets of roots forming a partition of R . If one puts $\Delta' = \Delta \cap R'$, $\Delta'' = \Delta \cap R''$, then $R' = R_{\Delta'}$, $R'' = R_{\Delta''}$, and every root of Δ' is orthogonal to every root of Δ'' .*
- (ii) *Let Δ' and Δ'' be two orthogonal subsets of Δ forming a partition of Δ . Then $R' = R_{\Delta'}$ and $R'' = R_{\Delta''}$ form a partition of R .*

Let us first prove (i).

Lemma 7.1.2. *Under the conditions of (i), if α , β and $\alpha + \beta$ are roots, all three belong to R' or all three to R'' .*

Suppose, for example, $\alpha + \beta \in R'$. Then one cannot have $\alpha, \beta \in R''$, since R'' is closed; suppose therefore $\alpha \in R'$. Then $-\alpha \in R'$, and $\beta = (\beta + \alpha) - \alpha \in R'$.

Let us now show that $R' = R_{\Delta'}$ by induction on the order of a positive root $\alpha \in R' \cap \mathcal{P}(\Delta)$. If $\text{ord}_{\Delta}(\alpha) = 1$, then $\alpha \in R' \cap \Delta = \Delta'$. If $\text{ord}_{\Delta}(\alpha) > 1$, there exists $\beta \in \Delta$ such that $\alpha - \beta \in R$. By the lemma, one has $\beta \in \Delta'$, $\alpha - \beta \in R'$, hence $\alpha - \beta \in R_{\Delta'}$ by induction, and finally $\alpha = (\alpha - \beta) + \beta \in R_{\Delta'}$.

Finally, let us show that Δ' and Δ'' are orthogonal. If $\alpha \in \Delta'$ and $\beta \in \Delta''$, then $(\beta^*, \alpha) \leq 0$. If $(\beta^*, \alpha) \neq 0$, $\beta + \alpha$ is a root, contrary to the lemma.

Let us prove (ii). If Δ' or Δ'' is empty, this is immediate. Otherwise, if $R_{\Delta'}$ and $R_{\Delta''}$ do not form a partition of R , there exists a root α of the form

$$\alpha = \sum m'_i \alpha'_i + \sum m''_j \alpha''_j, \quad m'_i \in \mathbf{Z}_+, \quad m''_j \in \mathbf{Z}_+,$$

where α'_i (resp. α''_j) denote elements of Δ' (resp. Δ''). Applying 3.1.2, one deduces a relation of the form (interchanging Δ' and Δ'' if necessary):

$$\delta = \gamma + \beta, \quad \gamma \in R_{\Delta'}, \quad \beta \in \Delta'', \quad \delta \in R.$$

But since $(\beta^*, \gamma) = 0$, $\gamma - \beta$ is also a root by 2.2.5, which is impossible.

Proposition 7.1.3. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

- (i) *There exists no nontrivial partition of R into two closed symmetric subsets.*
- (ii) *For one (resp. every) system of simple roots Δ of R , there exists no partition of Δ into two nonempty orthogonal subsets.*
- (iii) *The natural representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible.*
- (iv) *For every pair (α, β) of roots, there exists a sequence of roots $\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_n$, with $\alpha = \alpha_0$, $\alpha_n = \beta$, the roots α_i and α_{i+1} ($i = 0, \dots, n-1$) being nonorthogonal.*

One has (i) $\Leftrightarrow$ (ii) by 7.1.1. Clearly (iv) $\Rightarrow$ (ii). Conversely, if (ii) holds for Δ , condition (iv) holds if $\alpha, \beta \in \Delta$. Now, for every root there exists a simple root not orthogonal to it

(3.1.1, for example). On the other hand, (iii) $\Rightarrow$ (i). Indeed, under the conditions of 7.1.1, $\mathcal{V}(R')$ is stable under $W(\mathcal{R})$. It remains to prove (i) $\Rightarrow$ (iii).

Let therefore H be a vector subspace of $\mathcal{V}(R)$ stable under $W(\mathcal{R})$. For every $\alpha \in R$, the equation $s_\alpha(H) = H$ gives at once $\alpha \in H$ or $\alpha^* \in H^\perp$ (the orthogonal of H in $\mathcal{V}(R^*)$, which is in duality with $\mathcal{V}(R)$). Putting $R' = \{\alpha \in R \mid \alpha \in H\}$ and $R'' = \{\alpha \in R \mid \alpha^* \in H^\perp\}$, one has obtained a partition of R into two closed symmetric subsets.

Definition 7.1.4. A root datum (resp. root system) satisfying the equivalent conditions of 7.1.3 and of semisimple rank $\neq 0$ is called *irreducible*.

Corollary 7.1.5. For every root datum $\mathcal{R}$, there exists a unique partition (up to order) of R into closed symmetric irreducible subsets.

Corollary 7.1.6. Every adjoint (resp. simply connected) root datum is a product of irreducible adjoint (resp. simply connected) root data.

It suffices to see this in the adjoint case. The assertion then follows from the fact that under the conditions of 7.1.1 one has

$$\Gamma_0(R) = \Gamma_0(R') \times \Gamma_0(R'').$$

Corollary 7.1.7. For every root datum (resp. reduced root datum) $\mathcal{R}$, there exists an isogeny $\mathcal{R} \rightarrow \mathcal{R}'$, where $\mathcal{R}'$ is a product of a trivial root datum and irreducible (resp. reduced) simply connected root data.

7.2. Properties of irreducible root data

Definition 7.2.1. Let $\mathcal{R}$ be an irreducible root datum. For every $\alpha \in R$, put

$$\text{long}(\alpha) = \ell(\alpha)/\ell(\alpha_0);$$

where $\alpha_0 \in R$ is such that $\ell(\alpha_0)$ is minimal; one says that $\text{long}(\alpha)$ is the *length* of α .

Lemma 7.2.2. Let $\mathcal{R}$ be an irreducible root datum. The Weyl group acts transitively on the set of roots of the same length.

Indeed, let $\alpha, \beta \in R$. Since the representation of W on $\mathcal{V}(R)$ is irreducible, α cannot be orthogonal to all the $w(\beta)$, $w \in W$. There therefore exists $w \in W$ with $w(\beta)$ nonorthogonal to α . Now $\ell(w(\beta)) = \ell(\beta)$, and one concludes by 2.3.2.

Lemma 7.2.3. If $\mathcal{R}$ is irreducible and reduced, $\text{long}(R)$ is $\{1\}$, $\{1, 2\}$ or $\{1, 3\}$.

By the remark used above, for every $\alpha, \beta \in R$ there always exists $w \in W$ such that $w(\beta)$ is not orthogonal to α . One therefore has $\ell(\alpha)/\ell(\beta) = 1, 2, 3, 1/2$ or $1/3$ (by 2.3.1). One therefore has $\text{long}(\alpha) = 1, 2$ or 3 , but if $\text{long}(\alpha) = 2$, $\text{long}(\beta) = 3$, then $\ell(\alpha)/\ell(\beta) = 2/3$, which is impossible.

Remark 7.2.4. Reasoning similarly, one proves the following result: if R is irreducible and nonreduced with $\text{rgss}(\mathcal{R}) > 1$, one has $\text{long}(R) = \{1, 2, 4\}$. Putting $\text{long}^{-1}(i) = R_i$, one has $\text{ind}(R) = R_1 \cup R_2$, $R_4 = 2R_1$, and two nonproportional roots of R_1 are orthogonal. Conversely, if R is an irreducible reduced system such that $\text{long}(R) = \{1, 2\}$, put $\text{long}^{-1}(i) = R_i$ and suppose that two nonproportional roots of R_1 are orthogonal; then $R \cup 2R_1$ is irreducible, nonreduced, and $\text{ind}(R \cup 2R_1) = R$.

Lemma 7.2.5. If $\mathcal{R}$ is an irreducible root datum, $\mathcal{R}^*$ is also one, and the product

$$\text{long}(\alpha) \text{long}(\alpha^*)$$

is constant as α runs through R .

This follows at once from 7.1.3 (iv) and 2.2.6.

Definition 7.2.6. Let $\mathcal{R}$ be any root datum. One calls the *length* of $\alpha \in R$, and denotes by $\text{long}(\alpha)$, the length of α in its irreducible component.

Lemma 7.2.7. *There exists a unique group homomorphism $u : \Gamma_0(R) \longrightarrow \Gamma_0(R^*)$ such that $u(\alpha) = \text{long}(\alpha)\alpha^*$ for $\alpha \in R$.*

By 3.5.5, it suffices to check that if $\alpha, \beta, \alpha + \beta \in R$, one has

$$\text{long}(\alpha)\alpha^* + \text{long}(\beta)\beta^* = \text{long}(\alpha + \beta)(\alpha + \beta)^*.$$

But α, β and $\alpha + \beta$ are in the same irreducible component of R by 7.1.2, and one is reduced to 1.2.2.

Remark 7.2.8. Let u be as in 7.2.7. For $\alpha, \beta \in R$, one has $(u(\alpha), \beta) = (u(\beta), \alpha)$. Indeed, this amounts to seeing that

$$\text{long}(\alpha)(\alpha^*, \beta) = \text{long}(\beta)(\beta^*, \alpha),$$

which clearly holds if α and β are orthogonal. If α and β are not orthogonal, they are in the same irreducible component of R , and one is reduced to 1.2.1, formula (9).

Remark 7.2.9. The symmetric bilinear form $(u(x), y)$ is positive nondegenerate on $\Gamma_0(R)$.

Indeed, let R_i be the irreducible components of R . One has

$$\Gamma_0(R) = \prod_i \Gamma_0(R_i),$$

and the bilinear form $(u(x), y)$ is the product of the forms

$$2^{-1}\ell(\alpha_i)^{-1}(p(x), y)$$

on the $\Gamma_0(R_i)$, where $\ell(\alpha_i)$ is the minimum of $\ell(\alpha)$ for $\alpha \in R_i$. Now these various symmetric bilinear forms are positive nondegenerate (1.2.6).

7.3. Cartan matrix

Let $\mathcal{R}$ be a root datum. If Δ is a system of simple roots, one calls the *Cartan matrix* of $\mathcal{R}$ relative to Δ the square matrix indexed by Δ defined by

$$a_{\alpha, \beta} = (\alpha^*, \beta) \quad \text{for } \alpha, \beta \in \Delta.$$

Note first that if Δ' is another system of simple roots and w an element of $W(\mathcal{R})$ such that $w(\Delta) = \Delta'$, one has

$$(w(\alpha)^*, w(\beta)) = (\alpha^*, \beta),$$

hence the Cartan matrix of $\mathcal{R}$ relative to Δ' is obtained from that relative to Δ by the isomorphism $\Delta \longrightarrow \Delta'$ of index sets defined by w . It follows that, up to an isomorphism of the index set, the Cartan matrix depends only on $\mathcal{R}$.

Proposition 7.3.1. *The Cartan matrix has the following properties:*

- (i) $a_{\alpha, \alpha} = 2$, $a_{\alpha, \beta} \leq 0$ for $\alpha \neq \beta$.
- (ii) $a_{\alpha, \beta} = 0$ implies $a_{\beta, \alpha} = 0$.
- (iii) *There exist strictly positive integers m_α ($= \text{long}(\alpha)$) such that the matrix $(m_\alpha a_{\alpha, \beta})$ is symmetric, positive and nondegenerate.*
- (iv) *The diagonal minors of the matrix $(a_{\alpha, \beta})_{\alpha, \beta \in \Delta}$, i.e. the determinants*

$$\det(a_{\alpha, \beta})_{\alpha, \beta \in \Delta'} \quad \text{for } \Delta' \subset \Delta,$$

are strictly positive.

- (v) *One has $s_\alpha(\beta) = \beta - a_{\alpha, \beta}\alpha$ and $s_\alpha(\beta^*) = \beta^* - a_{\beta, \alpha}\alpha^*$.*

Indeed, (v) is a definition, (i) follows from 3.2.11, (ii) from 2.2.2, (iii) from 7.2.9, and (iv) follows at once from (iii) by the relation

$$\det(m_\alpha a_{\alpha,\beta})_{\alpha,\beta \in \Delta'} = \left(\prod_{\alpha \in \Delta'} m_\alpha \right) \det(a_{\alpha,\beta})_{\alpha,\beta \in \Delta'}.$$

Proposition 7.3.2. *Let $\mathcal{R}$ and $\mathcal{R}'$ be two simply connected (resp. adjoint) reduced root data, Δ (resp. Δ') a system of simple roots of $\mathcal{R}$ (resp. $\mathcal{R}'$), and $u : \Delta \rightarrow \Delta'$ an isomorphism such that if $(a_{\alpha,\beta})$ and $(a'_{\alpha',\beta'})$ denote the Cartan matrices of $\mathcal{R}$ and $\mathcal{R}'$ relative to Δ and Δ' , one has:*

$$a'_{u(\alpha),u(\beta)} = a_{\alpha,\beta}.$$

Then there exists a unique isomorphism from $\mathcal{R}$ onto $\mathcal{R}'$ inducing u on Δ .

Clearly, it suffices to give the proof in the adjoint case. Then $M = \Gamma_0(R)$ and $M' = \Gamma_0(R')$ are the free abelian groups generated by Δ and Δ' . There therefore exists a unique group isomorphism from M onto M' inducing u on Δ . Denote it also by u . Let us show that $u(R) \subset R'$. Every root α of $\mathcal{R}$ is written $s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1})$, with $\alpha_i \in \Delta$. Clearly,

$$u(\alpha) = s_{u(\alpha_1)} \cdots s_{u(\alpha_n)}(u(\alpha_{n+1})),$$

by the hypothesis on u and relations (v) of 7.3.1. It remains to prove that ${}^t u(R'^*) \subset R^*$, which follows from the fact that the elements of M^* (resp. M'^*) are determined by duality with R or Δ (resp. R' or Δ'), by 1.2.5.

Corollary 7.3.3. *A reduced simply connected or adjoint root datum is determined up to isomorphism by its Cartan matrix.*

Corollary 7.3.4. *Let $\mathcal{R}$ be a reduced simply connected (resp. adjoint) root datum, and Δ a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the set Δ which leave the Cartan matrix invariant.*

Remark 7.3.5. The question of the existence of a root datum corresponding to a given Cartan matrix satisfying (i), (ii) and (iv) (for example) is not easily solved directly, without using the classification.

7.4. Dynkin diagrams

Definition 7.4.1. A *Dynkin diagram structure* (the word “scheme” has been banished for obvious reasons) on a finite set Δ is the datum of a set of pairs of distinct elements of Δ , called *linked pairs*, and of a map from Δ to the set $\{1, 2, 3\}$. The notion of isomorphism of such structures is obvious.

Definition 7.4.2. Let $\mathcal{R}$ be a root datum and Δ a system of simple roots. One calls the *Dynkin diagram* of $\mathcal{R}$ relative to Δ the set Δ , two simple roots being linked if and only if they are not orthogonal, each root being assigned its length.

Proposition 7.4.3. *Dynkin diagram and Cartan matrix determine one another bijectively.*

Indeed, the equivalence

$$\alpha \text{ and } \beta \text{ are not linked} \iff a_{\alpha,\beta} = 0,$$

and the relation

$$\text{long}(\alpha)a_{\alpha,\beta} = \text{long}(\beta)a_{\beta,\alpha}$$

(with $\inf \text{long}(\alpha) = 1$ in each connected component of the diagram) determine the $a_{\alpha,\beta}$ in terms of the links and lengths, and conversely (the details of the verification are left to the reader).

Corollary 7.4.5. *Let $\mathcal{R}$ be a reduced simply connected or adjoint root datum and Δ a system of simple roots. The group $E_{\Delta}(\mathcal{R})$ is identified with the automorphism group of the Dynkin diagram of $\mathcal{R}$ relative to Δ , that is, with the group of permutations of Δ preserving lengths and links.*

$$\begin{array}{c} 1 \quad 3 \\ \bullet - \bullet \end{array} G_2.$$
$$\begin{aligned} E_{\Delta}(\mathcal{R}) &= \{e\} && \text{for } A_1, B_n, C_n, E_7, E_8, F_4, G_2. \\ E_{\Delta}(\mathcal{R}) &= \mathbf{Z}/2\mathbf{Z} && \text{for } A_n \ (n \geq 2), D_n \ (n \geq 5), E_6. \\ E_{\Delta}(\mathcal{R}) &= \mathfrak{S}_3 && \text{for } D_4. \end{aligned}$$

Let $f : \mathcal{R} \rightarrow \mathcal{R}'$ be a p -morphism (cf. 6.8). It is clear from the definitions that the associated bijection $u : R \xrightarrow{\sim} R'$ makes the systems of simple roots, systems of positive roots, irreducible components (etc.) of R and R' correspond. Suppose therefore, for simplicity, that R and R' are *irreducible*.

$$\text{long}(u(\alpha))(u(\alpha)^*, u(\beta)) = \text{long}(u(\beta))(u(\beta)^*, u(\alpha)).$$

⁽³⁰⁾**N.D.E.** For a slightly different proof, see also [De80].

One deduces at once that for nonorthogonal α and β , one has

$$\frac{q(\alpha)^2 \text{long}(\alpha)}{\text{long}(u(\alpha))} = \frac{q(\beta)^2 \text{long}(\beta)}{\text{long}(u(\beta))},$$

and one then concludes by 7.1.3 (iv).

Remark 7.5.2. It follows from 7.2.2 and 6.8.4 that $q(\alpha)$ depends only on $\text{long}(\alpha)$. One then easily sees that if $q(\alpha)$ is not constant, $q(\alpha) \text{long}(\alpha)$ is constant, which shows that then $p = 2$ or 3 . A glance at the diagrams of the preceding section shows that there are four possible cases (the same letter denotes a Dynkin diagram and the corresponding reduced simply connected root datum):

$$\begin{aligned} p = 2, \quad B_n &\xrightarrow{f_1} C_n, \quad C_n \xrightarrow{f_2} B_n \quad (\text{with } C_2 = B_2). \\ p = 2, \quad F_4 &\xrightarrow{g} F_4. \\ p = 3, \quad G_2 &\xrightarrow{h} G_2. \end{aligned}$$

The reader will note that $f_1 \circ f_2$, $f_2 \circ f_1$, $g \circ g$ and $h \circ h$ are p -morphisms of the form described in 6.8.6.

7.5.3. — One sees at once from the preceding description that if $\mathcal{R}$ and $\mathcal{R}'$ are two reduced root data of semisimple rank ≤ 2 , and one has a p -morphism from $\mathcal{R}'$ to $\mathcal{R}$, then $\mathcal{R}$ and $\mathcal{R}'$ are of the same type. More precisely, one has the following table.

Notations: Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a p -morphism. Denote by q (resp. q_1) any positive power of p . For systems of rank 2, use the notations of section 4 (denote the simple roots by α, β , with $\ell(\alpha) \leq \ell(\beta)$).

Type	p	values of f	values of ${}^t f$
Trivial	arbitrary	—	—
A_1	arbitrary	$f(\alpha') = q\alpha$	${}^t f(\alpha^*) = q\alpha'^*$
$A_1 \times A_1$	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q_1\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q_1\beta'^*$
A_2, B_2, G_2	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q\beta'^*$
B_2	$p = 2$	$f(\alpha') = q\beta$ $f(\beta') = 2q\alpha$	${}^t f(\alpha^*) = 2q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$
G_2	$p = 3$	$f(\alpha') = q\beta$ $f(\beta') = 3q\alpha$	${}^t f(\alpha^*) = 3q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$

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Exposé XXII. Reductive groups: splittings, subgroups, quotient groups

by *M. Demazure*

⁽⁰⁾This exposé comprises two parts. The first (1 to 5.5) collects the technical results necessary for the proof of the uniqueness and existence theorems. The second (5.6 to the end) will not be used in this proof; the end of no. 5 will be used in particular in exposé XXVI devoted to parabolic subgroups; no. 6 establishes in the framework of schemes the classical results on the derived group of a reductive group.

1. Roots and coroots. Split groups and root data

Theorem 1.1. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T .*

(i) *There exists a unique morphism of groups with operator group T*

$$\exp_\alpha : W(\mathfrak{g}^\alpha) \longrightarrow G$$

which induces on the Lie algebras the canonical morphism $\mathfrak{g}^\alpha \longrightarrow \mathfrak{g}$. This morphism is a closed immersion. The corresponding morphism

$$T \cdot_\alpha W(\mathfrak{g}^\alpha) \longrightarrow G$$

is also a closed immersion.

If $p_\alpha : \mathbf{G}_{a,S} \longrightarrow G$ is a monomorphism normalized by T with multiplier α , there exists a unique $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^{\times(1)}$ such that $p_\alpha(x) = \exp_\alpha(xX_\alpha)$; one has $\text{Lie}(p_\alpha)(1) = X_\alpha$ and the two preceding formulas establish a bijective correspondence between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of monomorphisms $\mathbf{G}_{a,S} \longrightarrow G$ normalized by T with multiplier α .

(ii) *There exists a unique duality (denoted $(X, Y) \longmapsto XY$)*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathcal{O}_S,$$

and a unique morphism of groups

$$\alpha^* : \mathbf{G}_{m,S} \longrightarrow T,$$

such that one has formula (F) of Exp. XX 2.1. One has

$$\alpha \circ \alpha^* = 2, \quad (-\alpha)^* = -\alpha^*,$$

and α^ is given by the formula of Exp. XX 2.7.*

Indeed, a morphism normalized by T with multiplier α necessarily factors through the closed subgroup $Z_\alpha = \text{Centr}_G(T_\alpha)$ of G (cf. Exp. XIX 3.9). Now (Z_α, T, α) is an elementary S -system (Exp. XX 1.4), and one is reduced to the results of exposé XX (1.5, 2.1 and 5.9).

Remark 1.2. Part (i) of theorem 1.1 remains valid if one assumes only that α is a character of T , nontrivial on each fiber. Indeed, one then has a decomposition $S = S' \amalg S''$, such that

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** The set $\Gamma(S, \mathfrak{g}^\alpha)^\times$ is defined in XIX 4.4.1.

$\alpha|_{S'}$ is a root of $G_{S'}$ with respect to $T_{S'}$ and $\mathfrak{g}^\alpha|_{S''} = 0$. If $S = S'$, one is reduced to 1.1; if $S = S''$ the result is trivial; the general case follows at once.

Notations 1.3. As in exposé XX, one denotes by U_α the image of $W(\mathfrak{g}^\alpha)$; it is a closed subgroup of G , canonically endowed with a vector structure. One will say that it is the *vector group associated with the root α* . One says that α^* is the *coroot associated with α* . Sections $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)$ and $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})$ are said to be *paired* if $X_\alpha X_{-\alpha} = 1$. Then $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and likewise for $X_{-\alpha}$. The corresponding morphisms p_α and $p_{-\alpha}$ are contragredient to one another in the sense of XX, 1.5⁽²⁾ and one has

$$p_\alpha(x) p_{-\alpha}(y) = p_{-\alpha} \left(\frac{y}{1+xy} \right) \alpha^*(1+xy) p_\alpha \left(\frac{x}{1+xy} \right).$$

Proposition 1.4. *Under the conditions of 1.1, let $w \in \text{Norm}_G(T)(S)$. Then $\beta = \alpha \circ \text{int}(w)^{-1} : T \rightarrow \mathbf{G}_{m,S}$ is a root of G with respect to T , $\beta^* = \text{int}(w) \circ \alpha^*$ is the corresponding coroot, and the following diagram is commutative:*

$$\begin{array}{ccc} W(\mathfrak{g}^\alpha) & \xrightarrow{\exp_\alpha} & G \\ \text{Ad}(w) \downarrow & & \downarrow \text{int}(w) \\ W(\mathfrak{g}^\beta) & \xrightarrow{\exp_\beta} & G. \end{array}$$

Trivial: transport of structure.

Definitions 1.5. (a) Under the conditions of 1.1, one denotes by s_α the automorphism of T defined by

$$s_\alpha(t) = t \cdot \alpha^*(\alpha(t))^{-1}.$$

One denotes by $(\ , \)$ the canonical pairing:

$$\begin{aligned} & \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T) \times \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S}) \\ & \longrightarrow \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, \mathbf{G}_{m,S}) = \mathbf{Z}_S. \end{aligned}$$

Then s_α acts on $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$, resp. $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$, by the following formulas, where χ (resp. u) denotes an arbitrary section of $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$ (resp. of $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$):

$$s_\alpha(\chi) = \chi - (\alpha^*, \chi)\alpha, \quad s_\alpha(u) = u - (u, \alpha)\alpha^*.$$

One has $s_\alpha \circ s_\alpha = \text{id}$ and $s_{-\alpha} = s_\alpha$.

(b) If $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then the inner automorphism $w_\alpha(X)$ of T defined by

$$w_\alpha(X) = \exp_\alpha(X) \exp_{-\alpha}(-X^{-1}) \exp_\alpha(X)$$

(cf. Exp. XX 3.1) coincides with s_α (*loc. cit.*). One then concludes from 1.4:

Corollary 1.6. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , α and β two roots of G with respect to T . Then*

$$s_\alpha(\beta) = \beta - (\alpha^*, \beta)\alpha$$

is a root of G with respect to T , the corresponding coroot being

$$s_\alpha(\beta)^* = s_\alpha(\beta^*) = \beta^* - (\beta^*, \alpha)\alpha^*.$$

Corollary 1.7. *Under the preceding conditions, $\alpha^* = \beta^*$ implies $\alpha = \beta$.*

Indeed, if $\alpha^* = \beta^*$, one has cf. XXI.1.4

$$s_\beta(\alpha) = \alpha - 2\beta, \quad s_\alpha(\beta) = \beta - 2\alpha,$$

⁽²⁾**N.D.E.** That is, if one denotes by $\langle \ , \ \rangle$ the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ and identifies $\mathfrak{g}^\alpha$ with U_α via $p_\alpha(x) = \exp(xX_\alpha)$, and likewise for $-\alpha$, then $\langle p_\alpha(x), p_{-\alpha}(y) \rangle = xy$.

and one deduces at once

$$(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha).$$

If $\beta \neq \alpha$, there exists an $s \in S$ such that $\alpha_s \neq \beta_s$. But then the preceding formula shows that there exist infinitely many distinct roots of G_s with respect to T_s , which is impossible.

Definitions 1.8.0. ⁽³⁾If $u : \mathbf{G}_{m,S} \rightarrow T$ is a morphism of groups, one will say that u is a *coroot* of G with respect to T if there exists a root α of G with respect to T such that $\alpha^* = u$. Consider the functor $\mathcal{R}^*$ of coroots of G with respect to T defined as follows:

$$\mathcal{R}^*(S') = \text{set of coroots of } G_{S'} \text{ with respect to } T_{S'}.$$

If $\mathcal{R}$ is the functor of roots of G with respect to T (Exp. XIX 3.8.), one has a canonical morphism $\mathcal{R} \rightarrow \mathcal{R}^*$. By 1.7 and Exp. XIX 3.8, one has:

Corollary 1.8. *The canonical morphism $\mathcal{R} \rightarrow \mathcal{R}^*$ is an isomorphism. In particular, $\mathcal{R}^*$ is representable by a twisted constant finite S -scheme which is an open and closed subscheme of $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$.*

This leads one to give the following definition:

Definition 1.9. Let S be a scheme, T an S -torus. One calls a *twisted root datum* in T the datum:

- (i) of a finite subscheme $\mathcal{R}$ of $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$,
- (ii) of a finite subscheme $\mathcal{R}^*$ of $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$,
- (iii) of an isomorphism $\mathcal{R} \xrightarrow{\sim} \mathcal{R}^*$ denoted $\alpha \mapsto \alpha^*$,

satisfying the following conditions:

- (DR 1) For every $S' \rightarrow S$ and every $\alpha \in \mathcal{R}(S')$, one has $\alpha \circ \alpha^* = 2$.
- (DR 2) For every $S' \rightarrow S$ and all $\alpha, \beta \in \mathcal{R}(S')$, one has

$$\alpha - (\beta^*, \alpha)\beta \in \mathcal{R}(S'), \quad \alpha^* - (\alpha^*, \beta)\beta^* \in \mathcal{R}^*(S').$$

Moreover, if $\alpha \in \mathcal{R}(S')$ ($S' \neq \emptyset$) implies $2\alpha \notin \mathcal{R}(S')$, one says that the root datum is *reduced*.

Proposition 1.10. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , $\mathcal{R}$ (resp. $\mathcal{R}^*$) the scheme of roots (resp. of coroots) of G with respect to T . Then $(\mathcal{R}, \mathcal{R}^*)$ is a reduced twisted root datum in T .*

The only fact which remains to be verified is that this twisted root datum is reduced. This is what was done in Exp. XIX 3.10.

1.11. — Let $T = D_S(M)$ be a trivialized torus. If one denotes by M^* the dual abelian group of M , one has canonical isomorphisms (cf. Exp. VIII 1.5):

$$\begin{aligned} \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S}) &\xrightarrow{\sim} M_S, \\ \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T) &\xrightarrow{\sim} M_S^*, \end{aligned}$$

hence isomorphisms of groups:

$$\begin{aligned} \text{Hom}_{S\text{-gr.}}(T, \mathbf{G}_{m,S}) &\longrightarrow \text{Hom}_{\text{loc.const.}}(S, M), \\ \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T) &\longrightarrow \text{Hom}_{\text{loc.const.}}(S, M^*). \end{aligned}$$

A character of T (resp. a morphism of groups $\mathbf{G}_{m,S} \rightarrow T$) will be called *constant* (relative to the given trivialization) if the preceding isomorphism transforms it into a constant map from S to M (resp. M^*).

1.12. — With the same notations, let (M, M^*, R, R^*) be a root datum (Exp. XXI). Then (R_S, R_S^*) is a twisted root datum in T . Conversely, if $(\mathcal{R}, \mathcal{R}^*)$ is a twisted root datum in a

⁽³⁾**N.D.E.** The numbering 1.8.0 has been added to bring out these definitions.

torus T , one will call a *splitting* of this root datum the datum of an ordinary root datum (M, M^*, R, R^*) and of an isomorphism $T \simeq D_S(M)$ which transforms $(\mathcal{R}, \mathcal{R}^*)$ into (R_S, R_S^*) .

Definition 1.13. Let S be a scheme, G a reductive S -group, T a maximal torus of G . One calls a *splitting of G relative to T* the datum

- (i) of an abelian group M and an isomorphism $T \simeq D_S(M)$,
- (ii) of a system of roots R of G with respect to T (Exp. XIX 3.6),

satisfying the following two conditions:

(D₁) S is nonempty and the roots $\alpha \in R$ (resp. the corresponding coroots) are identified with constant functions from S to M (resp. M^*).

(D₂) The $\mathfrak{g}^\alpha$ ($\alpha \in R$) are free $\mathcal{O}_S$ -modules.

One says that G is *splittable relative to T* if there exists a splitting of G relative to T . One calls a *splitting of G* the datum of a maximal torus T of G and of a splitting of G with respect to T . One says that G is *splittable* if there exists a splitting of G . One calls a *split S -group* a reductive S -group endowed with a splitting; one will denote it by a symbol of the form (G, T, M, R) , or simply G if no confusion is possible.

Condition (D 1) implies that R (resp. R^*) is canonically identified with a subset of M (resp. M^*).

Proposition 1.14. Let S be a (nonempty) scheme, (G, T, M, R) a split S -group; then

$$\mathcal{R}(G, T, M, R) = (M, M^*, R, R^*)$$

is a reduced root datum (Exp. XXI 1.1 and 2.1.3); it is a splitting of the twisted root datum of 1.10.

This is a trivial consequence of 1.10 and of Exp. XIX 3.7.

One will sometimes write for simplicity $\mathcal{R}(G, T, M, R) = \mathcal{R}(G)$. One will systematically use the notations V , $\mathcal{V}(R)$, W , ... of Exp. XXI.

Remark 1.15. a) If S is connected and nonempty (resp. if $\text{Pic}(S) = 0$), condition (D 1) (resp. (D 2)) is automatically satisfied.

b) If (G, T, M, R) is a split S -group, then for every $S' \rightarrow S$, $S' \neq \emptyset$, $(G_{S'}, T_{S'}, M, R)$ is a split S' -group and $\mathcal{R}(G, T, M, R) = \mathcal{R}(G_{S'}, T_{S'}, M, R)$.

1.16. — Let $T = D_S(M)$ be a trivialized torus. The Lie algebra $\mathfrak{t}$ of T is canonically identified (Exp. II 5.1.1) with

$$\mathfrak{t} \simeq M^* \otimes \mathcal{O}_S.$$

For every morphism of groups $u : T \rightarrow \mathbf{G}_{m,S}$, $\underline{\text{Lie}}(u)$ is a linear form

$$\underline{\text{Lie}}(u) : \mathfrak{t} \rightarrow \mathcal{O}_S = \underline{\text{Lie}}(\mathbf{G}_{m,S}/S).$$

In particular, if u is defined by an element $\alpha \in M$, then $\underline{\text{Lie}}(u)$ is the linear form $\bar{\alpha}$ on $M^* \otimes \mathcal{O}_S$ defined by α :

$$\bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Symmetrically, for every morphism of groups $h : \mathbf{G}_{m,S} \rightarrow T$, $\underline{\text{Lie}}(h)$ is an $\mathcal{O}_S$ -morphism $\mathcal{O}_S = \underline{\text{Lie}}(\mathbf{G}_{m,S}/S) \rightarrow \mathfrak{t}$, canonically defined by the section

$$H = \underline{\text{Lie}}(h)(1) \in \Gamma(S, \mathfrak{t}).$$

In particular, if h is defined by an element $m \in M^*$, one has

$$H = \underline{\text{Lie}}(h)(1) = m \otimes 1.$$

Comparing the two definitions, one finds in particular

$$\bar{\alpha}(H) = (h, \alpha) \cdot 1 \in \Gamma(S, \mathcal{O}_S).$$

1.17. — These definitions apply in particular to the case where T is the maximal torus of a split group. Every root $\alpha \in R$ defines an *infinitesimal root* $\bar{\alpha} \in \text{Hom}_{\mathcal{O}_S}(\mathfrak{t}, \mathcal{O}_S)$ with

$$\bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Each coroot $\alpha \in R$ defines an *infinitesimal coroot*

$$H_\alpha \in \Gamma(S, \mathfrak{t}), \quad H_\alpha = \alpha^* \otimes 1.$$

For $\alpha, \beta \in R$, one has the relation

$$\bar{\alpha}(H_\beta) = (\beta^*, \alpha) \cdot 1,$$

and in particular

$$\bar{\alpha}(H_\alpha) = 2.$$

In particular, if 2 is invertible on S , then $\bar{\alpha}$ and H_α are nonzero on each fiber.

2. Existence of a splitting. Type of a reductive group

Proposition 2.1. *Let S be a scheme, G a reductive S -group, T a maximal torus of G . Assume T split. Then G is locally splittable with respect to T : for every $s_0 \in S$, there exists an open neighborhood U of s_0 such that the U -group G_U is splittable relative to T_U .*

Indeed, write $T \simeq D_S(M)$ and

$$\mathfrak{g} = \coprod_{m \in M} \mathfrak{g}^m.$$

Let $R = \{m \in M \mid m \neq 0, \mathfrak{g}^m(s_0) \neq 0\}$. Restricting S and replacing it by an open neighborhood of s_0 , one may assume the $\mathfrak{g}^\alpha$, $\alpha \in R$, free, and the $\mathfrak{g}^m$, $m \neq 0$, $m \notin R$, zero. One then has

$$\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha,$$

the $\mathfrak{g}^\alpha$ being free of rank 1. It follows that R is a system of roots of G with respect to T (Exp. XIX 3.6). The coroots α^* corresponding to the $\alpha \in R$ are then identified with locally constant functions on S with values in M^* . Restricting S further, one may assume them constant and one is done.

Note that the proof gives at once:

Proposition 2.2. *Let S be a nonempty connected scheme such that $\text{Pic}(S) = 0$, for example $\text{Spec}(\mathbf{Z})$ or a local scheme (in particular the spectrum of a field). If G is a reductive S -group possessing a split maximal torus T , then G is splittable relative to T .*

One deduces at once from 2.1 and the fact that a reductive group possesses maximal tori locally for the étale topology (Exp. XIX 2.5):

Corollary 2.3. *Let S be a scheme, G a reductive S -group (resp. and T a maximal torus of G). Then G is locally splittable (resp. locally splittable relative to T) for the étale topology on S .*

Corollary 2.4. *Let k be a field, G a reductive k -group. There exists a finite separable extension K/k such that G_K is splittable.*

Remark 2.5. Using 2.1 and the remark Exp. XIX 2.9, one proves at once the following result: let $G = (G, T, M, R)$ be a split S -group; there exists a covering of S by open subsets U_i such that each split group G_{U_i} comes by base change from a split group over a noetherian ring (and in fact a $\mathbf{Z}$ -algebra of finite type). We will moreover prove that every split group over S already comes from a split $\mathbf{Z}$ -group (Exp. XXV).

2.6. — Let k be an algebraically closed field and G a reductive k -group. One knows (2.4 for example) that splittings of G exist. Let (G, T, M, R) and (G, T', M', R') be two splittings of G ; the root data $\mathcal{R}(G, T, M, R)$ and $\mathcal{R}(G, T', M', R')$ are then isomorphic.

Indeed, one sees first that one may reduce to the case where $T = T'$ (for there exists $g \in G(k)$ such that $T' = \text{int}(g)T$, and one easily verifies that if one transports a splitting by an automorphism of G , one obtains a root datum isomorphic to the initial datum); but since $S = \text{Spec}(k)$ is connected, the isomorphism $D_k(M) \xrightarrow{\sim} T \xrightarrow{\sim} D_k(M')$ comes from a unique isomorphism $M \simeq M'$; for the same reason, there exists at most one system of roots of G with respect to T .

Definition 2.6.1. ⁽⁴⁾If G is a reductive k -group (k an algebraically closed field), one will call the *type of G* the isomorphism class of the root datum defined by any splitting of G ; if G is a torus, of type M in the sense of Exp. IX 1.4, then the type of G as a reductive group is given by the trivial root datum $(M, M^*, \emptyset, \emptyset)$.

By 1.15 b) ⁽⁵⁾, the type is invariant under (algebraically closed) extension of the base field.

Definition 2.7. If G is a reductive S -group and if $s \in S$, one calls the *type of G at s* the type of the reductive $\bar{s}$ -group $G_{\bar{s}}$.

For every $S' \rightarrow S$ and every $s' \in S'$ projecting to $s \in S$, the type of $G_{S'}$ at s' is equal to the type of G at s .

If G is splittable, and if (G, T, M, R) is a splitting of G , then the type of G at s is the isomorphism class of $\mathcal{R}(G, T, M, R)$ by 1.15 b) ⁽⁵⁾. One then obtains at once from 2.3 the

Proposition 2.8. *Let G be a reductive S -group ($S \neq \emptyset$). The function*

$$s \longmapsto \text{type of } G \text{ at } s$$

is locally constant on S . In particular, there exists a partition of S into nonempty open subschemes such that on each of them G has constant type. More precisely, let E be the set of types of the fibers of G ; for every $t \in E$, let S_t be the set of points $s \in S$ where G has type t ; then $(S_t)_{t \in E}$ is a partition of S and each S_t is open and closed (and nonempty).

3. The Weyl group

3.1. — Let S be a scheme, G a reductive S -group, T a maximal torus of G . Then

$$W_G(T) = \underline{\text{Norm}}_G(T)/T$$

is a finite étale S -group (Exp. XIX 2.5). The morphism $n \mapsto \text{int}(n)$ induces on passing to the quotient a canonical monomorphism (which is moreover an open immersion):

$$W_G(T) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(T).$$

3.2. — Assume now that G is splittable relative to T . Choose a splitting, say (G, T, M, R) . One then has a canonical isomorphism (Exp. VIII 1.5)

$$\underline{\text{Aut}}_{S\text{-gr.}}(T) \simeq (\text{Aut}_{\text{gr.}}(M))_S.$$

In particular, if W is the Weyl group of the root datum $\mathcal{R}(G)$ (Exp. XXI 1.1.8), one has a monomorphism

$$W_S \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(T).$$

3.3. — For each root $\alpha \in R$, the reflection $s_\alpha \in W$ acts on M by

$$s_\alpha(x) = x - (\alpha^*, x)\alpha,$$

⁽⁴⁾**N.D.E.** The number 2.6.1 has been added, for later references.

⁽⁵⁾**N.D.E.** The original, which referred to 1.17, has been corrected.

hence on T (by the preceding morphism), by

$$s_\alpha(t) = t \cdot \alpha^*(\alpha(t))^{-1}.$$

On the other hand, since $\mathfrak{g}^\alpha$ is assumed free, there exists an $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$. Consider then $w_\alpha(X) \in \underline{\text{Norm}}_G(T)(S)$ (Exp. XX 3.1). One has (*loc. cit.*)

$$\text{int}(w_\alpha(X))(t) = s_\alpha(t).$$

Since W is generated by the s_α , $\alpha \in R$, it follows from the preceding remarks that if one considers W and $\underline{\text{Norm}}_G(T)(S)/T(S)$ as groups of automorphisms of T , one has

$$W \subset \underline{\text{Norm}}_G(T)(S)/T(S) \subset W_G(T)(S).$$

By definition of the constant group W_S associated with W (cf. I 1.8), one therefore has a commutative diagram

$$\begin{array}{ccc} W_S & \xrightarrow{\quad} & W_G(T) \\ & \searrow \quad \swarrow & \\ & \underline{\text{Aut}}_{S\text{-gr.}}(T) & \end{array}$$

Proposition 3.4. *Let S be a scheme, (G, T, M, R) a split S -group, W the Weyl group of the root datum $\mathcal{R}(G)$. Then the canonical monomorphism*

$$W_S \longrightarrow W_G(T) = \underline{\text{Norm}}_G(T)/T$$

is an isomorphism.

These are indeed étale groups over S ; it therefore suffices to verify that for every $s \in S$, $W_S(\bar{s}) \longrightarrow W_G(T)(\bar{s})$ is an isomorphism.⁽⁶⁾ Now this last assertion follows, for example, from *Bible*, § 11.3, th. 2.

Remark 3.5. Using 2.3, the preceding proposition gives a new proof of the fact that the Weyl group of a maximal torus of a reductive S -group G is finite over S (Exp. XIX 2.5 (ii)).⁽⁷⁾

3.6. — Under the conditions of 3.1, for every $w \in W_G(T)(S)$, one denotes by N_w ⁽⁸⁾ the fibered product of the following diagram:

$$\begin{array}{ccc} N_w & \longrightarrow & \underline{\text{Norm}}_G(T) \\ \downarrow & & \downarrow \\ S & \xrightarrow{\quad w \quad} & W_G(T). \end{array}$$

It is an open and closed subscheme of $\underline{\text{Norm}}_G(T)$, which is a principal homogeneous bundle under T on the left (resp. on the right) by the law $(t, q) \mapsto tq$ (resp. $(q, t) \mapsto qt$). If $n \in N_w(S)$, one has

$$N_{ww'} = n \cdot N_{w'}, \quad N_{w'w} = N_{w'} \cdot n.$$

3.7. — In particular, if α is a root of G with respect to T , N_{s_α} is none other than what had been denoted by $N^\times$ in Exp. XX 3.0. If $\mathfrak{g}^\alpha$ is free over S , one therefore has $N_{s_\alpha}(S) \neq \emptyset$.

By 3.4 and condition (D 2) of the splitting, one deduces the

⁽⁶⁾**N.D.E.** Indeed, since W_S and $W_G(T) = \underline{\text{Norm}}_G(T)/T$ are étale over S , the morphism $f : W_S \longrightarrow W_G(T) = \underline{\text{Norm}}_G(T)/T$ is étale (EGA IV₄, 17.3.4); if moreover each $f_{\bar{s}}$ is an isomorphism then, according to *loc. cit.*, 17.9.1, f will be a surjective open immersion, hence an isomorphism.

⁽⁷⁾**N.D.E.** Indeed, let T be a maximal torus of G . The fact that $W_G(T)$ is finite over S is local for the (fpqc) topology (EGA IV₂, 2.7.1), hence a fortiori for the étale topology. By 2.3, one may therefore assume that G is split relative to T , in which case the assertion follows from 3.4.

⁽⁸⁾**N.D.E.** Q_w has been replaced by N_w , just as in XX 3.0 Q had been replaced by $N^\times$.

Corollary 3.8. *Under the conditions of 3.4, the morphism*

$$\underline{\mathrm{Norm}}_G(T)(S) \longrightarrow W_G(T)(S) = \mathrm{Hom}_{\mathrm{loc. cons.}}(S, W)$$

is surjective. In particular, for every $w \in W$, there exists an $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ such that $\mathrm{int}(n_w)|_T = w$.

4. Homomorphisms of split groups

4.1. The “big cell”

4.1.1. — Let (G, T, M, R) be a split reductive S -group. Choose a *system of positive roots* (Exp. XXI 3.2.1) R_+ of the root datum $\mathcal{R}(G)$. Put $R_- = -R_+$.

Choose a total order on R_+ (resp. R_-) and consider the morphism induced by the product in G

$$u : \prod_{\alpha \in R_-} U_\alpha \times_S T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow G.$$

It is an *open immersion*. Indeed, since both sides are flat and of finite presentation over S , it suffices to verify this on each geometric fiber (SGA 1, I 5.7 and VIII 5.4); one is therefore reduced to the case where S is the spectrum of an algebraically closed field; but, by *Bible*, § 13.4, cor. 2 to th. 3, u is radicial and dominant; since the tangent map to u at the origin is an isomorphism (definition of a system of roots), u is birational; but since G is normal, one may apply Zariski’s “Main Theorem” (EGA III₁, 4.4.9) and u is an open immersion.

Let us show that the image Ω of this open immersion is independent of the order chosen on R_+ (resp. R_-). Since this amounts to comparing open subsets of G , one is reduced to proving that they have the same geometric points, so one may again assume that S is the spectrum of an algebraically closed field. But then the assertion is none other than *Bible*, § 13, prop. 1 (c) and th. 1 (a).

One has therefore proved:

Proposition 4.1.2. *Let (G, T, M, R) be a split S -group. Let R_+ be a system of positive roots of R . There exists an open subset Ω_{R_+} of G such that for every total order on R_+ (resp. R_-), the morphism induced by the product in G*

$$\prod_{\alpha \in R_-} U_\alpha \times_S T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow G$$

is an open immersion with image Ω_{R_+} .

Remark 4.1.3. One may express 4.1.2 as follows: choose for every $\alpha \in R$ an isomorphism of vector groups $p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$ (cf. 1.19); then the morphism (one puts $N = \mathrm{Card}(R_+) = \mathrm{Card}(R_-)$)

$$\mathbf{G}_{a,S}^N \times_S T \times_S \mathbf{G}_{a,S}^N \longrightarrow G$$

defined set-theoretically by

$$((x_\alpha)_{\alpha \in R_-}, t, (x_\alpha)_{\alpha \in R_+}) \longmapsto \prod_{\alpha \in R_-} p_\alpha(x_\alpha) \cdot t \cdot \prod_{\alpha \in R_+} p_\alpha(x_\alpha)$$

is an open immersion whose image depends only on R_+ (and not on the choice of the p_α and of the orders on R_+ and R_-).

Notation 4.1.4. One denotes $\Omega_{R_+} = \prod_{\alpha \in R_-} U_\alpha \cdot T \cdot \prod_{\alpha \in R_+} U_\alpha$.⁽⁹⁾

Proposition 4.1.5. *The scheme Ω_{R_+} is of finite presentation over S (hence retrocompact in G) and is universally schematically dense in G relative to S (cf. Exp. XVIII 1).*

⁽⁹⁾**N.D.E.** And one calls it the “big cell” corresponding to R_+ .

The first assertion is trivial. Then,⁽¹⁰⁾ Ω_{R_+} is flat and of finite presentation over S , and contains the identity section, hence meets each fiber of G in a nonempty, hence dense, open subset; the second assertion therefore follows from Exp. XVIII 1.3⁽¹¹⁾.

Corollary 4.1.6. *Let (G, T, M, R) be a split reductive S -group. Then*

$$\underline{\text{Centr}}(G) = \bigcap_{\alpha \in R} \text{Ker}(\alpha).$$

⁽¹²⁾ *Consequently, $\underline{\text{Centr}}(G)$ is representable by a diagonalizable closed subgroup of G .*

The second assertion follows at once from the first. To prove the latter, one may invoke Exp. XII 4.8 and 4.11; one may also proceed directly as follows.⁽¹³⁾ Let $S' \rightarrow S$. If $t \in T(S')$ and if $\alpha(t) = 1$ for every $\alpha \in R$, then $\text{int}(t)$ induces the identity on $T_{S'}$ and on each $(U_\alpha)_{S'}$, $\alpha \in R$, hence also on $(\Omega_{R_+})_{S'}$, hence on $G_{S'}$ by schematic density, whence $t \in \underline{\text{Centr}}(G)(S')$.

Conversely, since $\underline{\text{Centr}}_{G_{S'}}(T_{S'}) = T_{S'}$ (cf. Exp. XIX 2.8), if $g \in G(S')$ centralizes $T_{S'}$ and the $(U_\alpha)_{S'}$, it is a section of $T_{S'}$ which annihilates the $\alpha \in R$.

Corollary 4.1.7. *Let S be a scheme, G a reductive S -group. Then the center of G is representable by a closed subgroup of G , of multiplicative type; it is also “the intersection of the maximal tori of G ” in the following sense: for every $S' \rightarrow S$, $\underline{\text{Centr}}(G)(S')$ is the set of $g \in G(S')$ whose inverse image in $G(S'')$, for every $S'' \rightarrow S'$, is contained in all the $T(S'')$, where T ranges over the set of maximal tori of $G_{S''}$.*

Taking 2.3 into account, the first assertion follows from 4.1.6 by descent.⁽¹⁴⁾ Let us prove the second assertion. Let H be “the intersection of the maximal tori of G ” in the preceding sense. One clearly has $\underline{\text{Centr}}(G) \subset H$.⁽¹⁵⁾ Then, by descent, it suffices to prove $\underline{\text{Centr}}(G) = H$ in the case where G is split. Since H is contained in the intersection of the maximal tori of G in the usual sense, this then follows from the following remark: if (G, T, M, R) is a splitting, $\alpha \in R$ and $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then $\text{int}(\exp_\alpha(X))(T) \cap T = \text{Ker}(\alpha)$, as a trivial calculation shows. (Cf. also Exp. XII 8.6 and 8.8 for a more general statement).

Remark 4.1.8. In what follows, in the split case we will systematically identify T with $D_S(M)$. Then $\underline{\text{Centr}}(G)$ is none other than $D_S(M/\Gamma_0(R))$, where $\Gamma_0(R)$ is the subgroup of M generated by R (cf. Exp. XXI 1.1.6). If $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a system of simple roots of R , one has at once (cf. Exp. XX 1.19):

$$\underline{\text{Centr}}(G) = \bigcap \text{Ker}(\alpha_i) = \bigcap \underline{\text{Centr}}(Z_{\alpha_i}).$$

Proposition 4.1.9. *Let S be a scheme, (G, T, M, R) a split S -group, Q an S -torus, α_0 a character of Q , $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module,*

$$f : Q \rightarrow T, \quad p : W(\mathcal{L}) \rightarrow G$$

morphisms of groups satisfying the set-theoretic relation

$$p(\alpha_0(q)x) = \text{int}(f(q)) \cdot p(x),$$

⁽¹⁰⁾**N.D.E.** The original has been expanded in what follows.

⁽¹¹⁾**N.D.E.** See also EGA IV₃, 11.10.10.

⁽¹²⁾**N.D.E.** The following sentence has been added.

⁽¹³⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁴⁾**N.D.E.** Indeed, representability of the center by a closed subscheme of G is local for the (fpqc) topology (SGA 1, VIII 5.2 and 5.4), hence a fortiori for the étale topology, and the same holds for the property “of multiplicative type”.

⁽¹⁵⁾**N.D.E.** The original has been expanded in what follows.

for all $q \in Q(S')$, $x \in W(\mathcal{L})(S')$, $S' \rightarrow S$. Assume that f separates the elements of R in the following sense: if $\alpha, \alpha' \in R$ and if $m, m' \in \mathbf{Z}$, then $m\alpha \circ f = m'\alpha' \circ f$ implies $m\alpha = m'\alpha'$.⁽¹⁶⁾ Finally let $s \in S$ be such that $(\alpha_0)_{\bar{s}} \neq e$ and $p_{\bar{s}} \neq e$.

There then exist an open subset U of S containing s , an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,U}$, a root $\alpha \in R$ and an isomorphism of $\mathcal{O}_U$ -modules

$$h : (\mathcal{L}|_U)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}^\alpha|_U$$

such that

- (i) $(\alpha \circ f)_U = (q\alpha_0)_U$,
- (ii) $p(X) = \exp_\alpha(h(X^q))$ for every $X \in W(\mathcal{L})(S')$, $S' \rightarrow U$.

Moreover, once U has been chosen, q , α and h are uniquely determined.

Restricting S , we may assume that α_0 is nonzero on each fiber of S . Choose a system of positive roots R_+ of R and let $V = p^{-1}(\Omega_{R_+})$. It is an open subset of $W(\mathcal{L})$ containing the zero section and stable under multiplication by every $\alpha_0(q)$, $q \in Q(S')$, $S' \rightarrow S$. Since α_0 is nontrivial on each fiber, it follows immediately that $V = W(\mathcal{L})$, hence that p factors through Ω_{R_+} . Choose any order on R_+ and R_- ; all products will be assumed taken in this order. One therefore has unique morphisms

$$a_\alpha : W(\mathcal{L}) \rightarrow U_\alpha, \quad \alpha \in R, \quad b : W(\mathcal{L}) \rightarrow T$$

such that

$$p(x) = \prod_{\alpha \in R_-} a_\alpha(x) \cdot b(x) \cdot \prod_{\alpha \in R_+} a_\alpha(x).$$

Writing the covariance condition under Q , one obtains at once

$$\begin{aligned} a_\alpha(\alpha_0(q)x) &= \alpha(f(q))a_\alpha(x), & \alpha \in R, \\ b(\alpha_0(q)x) &= b(x) \end{aligned}$$

for all $x \in W(\mathcal{L})(S')$, $q \in Q(S')$, $S' \rightarrow S$. The second condition gives at once $b = e$.

Let now $\alpha \in R$ be such that $(a_\alpha)_{\bar{s}} \neq e$ (we know that such an α exists, since $p_{\bar{s}}$ is assumed $\neq e$). Applying Exp. XIX 4.12 (a), one deduces that there exists an integer $n > 0$ such that $(\alpha \circ f)_{\bar{s}} = (n\alpha_0)_{\bar{s}}$. Restricting S , one may assume $\alpha \circ f = n\alpha_0$ (Exp. IX 5.3). But then, for every $\alpha' \in R$, $\alpha' \neq \alpha$, one has $(\alpha' \circ f)_{\bar{s}} \neq m\alpha_0$ for every integer $m > 0$ by the hypothesis on f (and the fact that the only roots proportional to α are α and $-\alpha$). Applying Exp. XIX 4.12 (a) again, this time to $a_{\alpha'}$, one deduces that $a_{\alpha'}$ is zero in a neighborhood of S ; since R is finite, one may, restricting S further, assume the $a_{\alpha'}$ zero for $\alpha' \in R$, $\alpha' \neq \alpha$. One then has $p = a_\alpha$, and one may apply to it Exp. XIX 4.12 (b), then (c), which gives the stated result (the uniqueness assertions are obvious).

Remark 4.1.10. The condition imposed on f in 4.1.9 holds in particular if f is surjective (= faithfully flat).

Proposition 4.1.11. Let (G, T, M, R) be a split S -group, R_+ a system of positive roots of R , Ω_{R_+} the corresponding “big cell”.

- (i) Let H be an S -group functor, separated⁽¹⁷⁾ for (fppf). If $f, g : G \rightrightarrows H$ are two morphisms of groups which coincide on Ω_{R_+} , then $f = g$.

⁽¹⁶⁾**N.D.E.** Note that this is equivalent to the hypothesis: if $\alpha, \alpha' \in R$ and $m, m' \in \mathbf{Z}$ and if $(m\alpha \circ f)_{\bar{s}} = (m'\alpha' \circ f)_{\bar{s}}$ for every geometric point $\bar{s}$ of S , then $m\alpha = m'\alpha'$. In particular, this separation hypothesis is stable under base change.

⁽¹⁷⁾**N.D.E.** Recall (cf. Exp. IV, 4.3.5) that an S -presheaf H is separated for a topology $\mathcal{T}$ if for every $S' \rightarrow S$ and every family of S -morphisms $(S'_i \rightarrow S')_{i \in I}$ covering for $\mathcal{T}$, the map $H(S') \rightarrow \prod_i H(S'_i)$ is injective.

- (ii) Let H be an S -sheaf of groups for (fppf) and $f : \Omega_{R_+} \rightarrow H$ an S -morphism satisfying the following condition: for every $S' \rightarrow S$ and all $x, y \in \Omega_{R_+}(S')$ such that $xy \in \Omega_{R_+}(S')$, one has $f(xy) = f(x)f(y)$. There then exists a (unique, by (i)) morphism of groups $\bar{f} : G \rightarrow H$ extending f .

Indeed, by 4.1.5, (i) (resp. (ii)) follows at once from Exp. XVIII 2.2 (resp. 2.3 and 2.4).

Remark 4.1.12. If $\alpha \in R_+$, one has

$$(\dagger) \quad \Omega_{R_+} \cap Z_\alpha = U_{-\alpha} \cdot T \cdot U_\alpha.$$

⁽¹⁸⁾Indeed, for every $S' \rightarrow S$, if $g = \prod_{\beta \in R_-} p_\beta(x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(x_\beta)$ is an element of $\Omega_{R_+}(S')$ and if $t' \in T_\alpha(S'')$, then

$$t'gt'^{-1} = \prod_{\beta \in R_-} p_\beta(\beta(t')x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(\beta(t')x_\beta)$$

and since α and $-\alpha$ are the only two elements of R which take the value 1 on T_α , one obtains that g belongs to $Z_\alpha = \text{Centr}(T_\alpha)$ if and only if $x_\beta = 0$ for $\beta \neq \pm\alpha$.

By $(\dagger)$, one deduces from XX 2.1 that if $X \in \Gamma(X, \mathfrak{g}^\alpha)$ and $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$, one has:

$$\exp_\alpha(X) \exp_{-\alpha}(Y) \in \Omega_{R_+}(S) \iff 1 + XY \text{ invertible.}$$

4.2. Morphisms of split groups

Definition 4.2.1. Let S be a (nonempty) scheme, (G, T, M, R) and (G', T', M', R') two split S -groups. One says that the morphism of S -groups $f : G \rightarrow G'$ is *compatible with the splittings*, or defines a *morphism of split groups*, if the restriction of f to T factors as a morphism $f_T : T \rightarrow T'$ of the form $f_T = D_S(h)$, where $h : M' \rightarrow M$ is a morphism of groups satisfying the following condition:

there exists a bijection $d : R \xrightarrow{\sim} R'^{(19)}$ and for each $\alpha \in R$ an integer $q(\alpha) > 0$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbf{G}_{a,S}$ and such that

$$h(d(\alpha)) = q(\alpha)\alpha, \quad {}^th(\alpha^*) = q(\alpha)d(\alpha)^*.$$

Remark 4.2.2. It is immediate that $h, d, q(\alpha)$ for $\alpha \in R$ are uniquely determined by f . One writes $h = \mathcal{R}(f)$. The $q(\alpha)$ are the *radicial exponents* of f (or of h).

Let p be the prime number (if it exists) which is zero on S ; put $p = 1$ if no prime number is zero on S . Then $\mathcal{R}(f)$ is a p -morphism of reduced root data in the sense of Exp. XXI 6.8. One has therefore defined a functor $\mathcal{R}$ from the category of split S -groups to that of reduced root data (endowed with the p -morphisms).

Proposition 4.2.3. *Under the conditions of 4.2.1, one has the following properties:*

- (i) For every $\alpha \in R$, there exists a unique isomorphism of $\mathcal{O}_S$ -modules

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)} \xrightarrow{\sim} \mathfrak{g}'^{d(\alpha)}$$

such that

$$f(\exp_\alpha(X)) = \exp_{d(\alpha)}(f_\alpha(X^{q(\alpha)}))$$

for every $X \in W(\mathfrak{g}^\alpha)(S')$, $S' \rightarrow S$.

- (ii) For every $\alpha \in R$, one has $q(-\alpha) = q(\alpha)$ and f_α and $f_{-\alpha}$ are contragredient to one another.

- (iii) For every $\alpha \in R$, every $Z \in W(\mathfrak{g}^\alpha)^*(S')$, $S' \rightarrow S$, one has

$$f(w_\alpha(Z)) = w_{d(\alpha)}(Z^{q(\alpha)}).$$

⁽¹⁸⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁹⁾**N.D.E.** The notation $u : R \rightarrow R'$ has been replaced by $d : R \rightarrow R'$.

By hypothesis the diagram

$$\begin{array}{ccccc}
 \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\
 q(\alpha) \downarrow & & \downarrow f_T & & \downarrow q(\alpha) \\
 \mathbf{G}_{m,S} & \xrightarrow{d(\alpha)^*} & T' & \xrightarrow{d(\alpha)} & \mathbf{G}_{m,S}
 \end{array}$$

is commutative. It follows that f maps $\text{Ker}(\alpha)$ into $\text{Ker}(d(\alpha))$, hence T_α into $T'_{d(\alpha)}$, hence Z_α into $Z'_{d(\alpha)}$. One then has only to apply Exp. XXI 3.10 and 3.11 to the groups Z_α and $Z'_{d(\alpha)}$.

Proposition 4.2.4. *The morphism f induces a morphism f_N from $\underline{\text{Norm}}(T)$ to $\underline{\text{Norm}}_{G'}(T')$, hence a morphism f_W from $W_G(T)$ to $W_{G'}(T')$; the latter is an isomorphism. More precisely, if one denotes by $\bar{d} : W(\mathcal{R}(G)) = W \rightarrow W' = W(\mathcal{R}(G'))$ the isomorphism extending $s_\alpha \mapsto s_{d(\alpha)}$ (Exp. XXI 6.8.4), one has a commutative diagram of isomorphisms:*

$$\begin{array}{ccc}
 W_G(T) & \xrightarrow[\sim]{f_W} & W_{G'}(T') \\
 \sim \uparrow & & \uparrow \sim \\
 W_S & \xrightarrow[\bar{d}_S]{\sim} & W'_S.
 \end{array}$$

This follows at once from 3.4, Exp. XXI 6.8.4, and (iii) above.

Remark 4.2.5. With the notations of 4.2.3, the restriction of f to Ω_{R_+} (for a system of positive roots R_+) is written explicitly: it maps Ω_{R_+} into $\Omega'_{d(R_+)}$ ($d(R_+)$ is a system of positive roots of R' by Exp. XXI 6.8.7) and is given by the set-theoretic formula:

$$\begin{aligned}
 & f \left(\prod_{\alpha \in R_-} \exp_\alpha(X_\alpha) \cdot t \cdot \prod_{\alpha \in R_+} \exp_\alpha(X_\alpha) \right) \\
 &= \prod_{\alpha \in R_-} \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)})) \cdot f_T(t) \cdot \prod_{\alpha \in R_+} \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)})).
 \end{aligned}$$

Proposition 4.2.6.

- (i) f is surjective (= faithfully flat in the present case, cf. VIB 3.11) if and only if f_T is.
- (ii) One has $\text{Ker}(f) \subset \Omega_{R_+}$.

Let us prove (i): if f is surjective, then $f_T(T) = f(T)$ is a maximal torus of G' (indeed $f(T)$ is a subtorus of a maximal torus T' (Exp. IX 6.8); to verify that $f(T) = T'$, one is reduced to the case of an algebraically closed field, where this is *Bible*, § 7.3, th. 3 (a)).

If f_T is surjective, then the preceding formula shows that f induces a surjection from $\Omega = \Omega_{R_+}$ onto $\Omega' = \Omega'_{d(R_+)}$.⁽²⁰⁾ Since the fibers of G' are connected, it follows (cf. Exp. VIA, 0.5) that f is surjective.

Let us prove (ii), and for this admit a result which will be proved below (5.7.4): choose for each $w \in W$ an $n_w \in \underline{\text{Norm}}_G(T)(S)$ representing it; then the open subsets $n_w \Omega$ ($w \in W$) form a covering of G . It then suffices to prove that $\text{Ker}(f) \cap n_w \Omega \neq \emptyset$ implies $w = 1$. If $x \in \Omega(S')$, $S' \rightarrow S$ and $f(n_w x) = 1$, one has $f(x) = f(n_w)^{-1}$; now $f(x) \in \Omega'(S')$ and $f(n_w)^{-1} \in \underline{\text{Norm}}_{G'}(T')(S')$. By 4.2.4, one is reduced to proving:

Lemma 4.2.7. *Under the conditions of 4.1.2, one has $\Omega \cap \underline{\text{Norm}}_G(T) = T$.*

⁽²⁰⁾**N.D.E.** The reference to Exp. VI has been expanded in what follows.

Let

$$x = \prod_{\alpha \in R_-} p_\alpha(x_\alpha) \cdot t \cdot \prod_{\alpha \in R_+} p_\alpha(x_\alpha) = vt u \in \Omega(S').$$

If x normalizes $T_{S'}$, one has for every $t' \in T(S')$,

$$xt'x^{-1} = t'' \in T(S'),$$

that is, $xt' = t''x$, which is written

$$v(tt')(t'^{-1}ut') = (t''vt''^{-1})(t''t)u,$$

which gives $t'^{-1}ut' = u$, hence $u \in \underline{\text{Centr}}_G(T)(S') = T(S')$, that is $u = 1$. Likewise $v = 1$.

Corollary 4.2.8. *One has*

$$\text{Ker}(f) = \prod_{\alpha \in R_-} K_\alpha \cdot \text{Ker}(f_T) \cdot \prod_{\alpha \in R_+} K_\alpha,$$

where for each $\alpha \in R$, K_α denotes the finite S -group

$$K_\alpha = \text{Ker}(U_\alpha \longrightarrow U_\alpha^{\otimes q(\alpha)}) \simeq \alpha_{q(\alpha), S}.$$

To apply this corollary, put:

Definition 4.2.9. Let S be a scheme, G and G' two reductive S -groups. A faithfully flat finite morphism of S -groups $f : G \longrightarrow G'$ (i.e. surjective and with kernel finite over S) is called an *isogeny*. If moreover $\text{Ker}(f)$ is a central subgroup of G , one says that f is a *central isogeny*.

Proposition 4.2.10. *Let $f : G \longrightarrow G'$ be a morphism of split groups. For f to be an isogeny (resp. a central isogeny), it is necessary and sufficient that f_T be an isogeny, i.e. that $\mathcal{R}(f)$ be injective with finite cokernel (resp. and that for every $\alpha \in R$, one have $q(\alpha) = 1$).*

Indeed, by 4.2.8, $\text{Ker}(f)$ is finite over S if and only if $\text{Ker}(f_T)$ is finite over S , and $\text{Ker}(f) \subset T$ if and only if each $q(\alpha)$ is 1 ($\text{Ker}(f)$ is then central since it is of multiplicative type and invariant, cf. Exp. IX 5.5).

Remark 4.2.11. a) One therefore sees that $f : G \longrightarrow G'$ is a central isogeny if and only if $\mathcal{R}(f) : \mathcal{R}(G') \longrightarrow \mathcal{R}(G)$ is an isogeny in the sense of Exp. XXI 6.2; moreover one has in this case (with the notations of *loc. cit.*):

$$\text{Ker}(f) = \text{D}_S(K(\mathcal{R}(f))), \quad K(\mathcal{R}(f)) = \text{Coker}(\mathcal{R}(f)).$$

b) If G and G' are semisimple, every morphism of split groups $f : G \longrightarrow G'$ is an isogeny⁽²¹⁾.

c) If $f : G \longrightarrow G'$ is faithfully flat and finite and if G is reductive (resp. semisimple), then G' is also. It is indeed of finite presentation over S (Exp. V 9.1), affine over S (EGA II 6.7.1), smooth over S (Exp. VI 9.2), with connected reductive (resp. semisimple) geometric fibers by Exp. XIX 1.7.

Definition 4.2.1 may seem arbitrary. It is justified by the following proposition (which we will state, for simplicity, for semisimple groups).

Let us say that a morphism $f : G \longrightarrow G'$ of reductive S -groups is *splittable* if there exist splittings of G and G' with which f is compatible. One then has the

Proposition 4.2.12. *Let S be a scheme, G and G' two semisimple S -groups, $f : G \longrightarrow G'$ a morphism of groups. Let $s \in S$. The following conditions are equivalent:*

- (i) $\text{Ker } f_{\bar{s}}$ is finite ($\Leftrightarrow e$ is isolated in $\text{Ker } f(\bar{s})$) and $f_{\bar{s}}$ is surjective, i.e. $f_{\bar{s}}$ is an isogeny.
- (ii) $f_{\bar{s}}$ is splittable.

⁽²¹⁾**N.D.E.** Indeed, since G and G' are semisimple, f_T is surjective with finite kernel, hence f is faithfully flat with finite kernel by 4.2.6 (i) and 4.2.8.

- (iii) *There exists an étale morphism $S' \rightarrow S$ covering s such that $f_{S'} : G_{S'} \rightarrow G'_{S'}$ is splittable.*

One clearly has (iii) $\Leftrightarrow$ (ii); (ii) $\Rightarrow$ (i) follows from 4.2.11 (b) (it is here that the hypothesis that G and G' are semisimple enters—the other implications hold for reductive groups).

Let us now prove (i) $\Rightarrow$ (iii). One may assume G and G' split in such a way that f induces a morphism $f_T : T \rightarrow T'$ (2.3 and Exp. XIX 2.8); restricting S , one may assume that $f_T = D_S(h)$, where h is a morphism of groups $M' \rightarrow M$. Let $\alpha \in R$, consider the composite morphism

$$p : W(\mathfrak{g}^\alpha) \xrightarrow{\exp_\alpha} G \xrightarrow{f} G'.$$

Since $\text{Ker}(p_{\bar{s}})$ is finite, $p_{\bar{s}} \neq e$. On the other hand $f_{T_{\bar{s}}}$ is surjective; one may therefore apply 4.1.9 and there exist an open subset V_α of S containing s , a root $\alpha' \in R'$, an integer $q(\alpha)$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbf{G}_{a, V_\alpha}$, and an isomorphism of $\mathcal{O}_{V_\alpha}$ -modules

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)}|_{V_\alpha} \rightarrow \mathfrak{g}'^{\alpha'}|_{V_\alpha}$$

such that $f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)}))$ and $\alpha' \circ f_T = h(\alpha') = q(\alpha)\alpha$. One may replace S by the intersection of the V_α , for $\alpha \in R$. Put $\alpha' = d(\alpha)$. It is clear that $d : R \rightarrow R'$ is a bijection, for the kernel of h is finite ($f_{T_{\bar{s}}}$ being surjective). It only remains to prove that $f_T \circ \alpha^* = q(\alpha)\alpha'^*$, which is done by a trivial modification of the argument used in Exp. XX 3.11.

In any case, as was seen in the course of the proof, one has (i) $\Rightarrow$ (iii). One therefore has:

Corollary 4.2.13. *Let S be a scheme, $f : G \rightarrow G'$ an isogeny of reductive groups. Then f is locally splittable for the étale topology.*

4.3. Central quotients of reductive groups

Consider first a particular case.

Proposition 4.3.1. *Let S be a scheme, (G, T, M, R) a split S -group, N a subgroup of M containing R , $Q = D_S(M/N) \subset \text{Centr}(G)$. Then:*

- (i) $G' = G/Q$ is a reductive S -group, $T' = T/Q$ is a maximal torus of it;
- (ii) if one identifies T' with $D_S(N)$, then $R \subset N$ is a system of roots of G' with respect to T' , (G', T', N, R) is a splitting of G , and $\mathcal{R}(G')$ is canonically identified with the induced root datum (Exp. XXI 6.5) $\mathcal{R}(G)_N$;
- (iii) the canonical morphism $G \rightarrow G'$ is compatible with the splittings, with radicial exponents 1, and gives by functoriality the canonical morphism (loc. cit.) $\mathcal{R}(G)_N \rightarrow \mathcal{R}(G)$.

One knows that $G' = G/Q$ is representable by a group scheme affine over S (Exp. VIII 5.7), smooth over S (Exp. VIB 9.2), with connected reductive geometric fibers (as quotients of reductive groups, cf. Exp. XIX 1.7); G' is therefore a reductive S -group.

It is clear that $T' = T/Q \simeq D_S(N)$ is a maximal torus of it. Note next that on choosing a system of positive roots R_+ of R , the open subset Ω_{R_+} of 4.2 is stable under Q and that one has a canonical isomorphism

$$\Omega_{R_+}/Q \simeq \prod_{\alpha \in R_-} U_\alpha \times_S (T/Q) \times_S \prod_{\alpha \in R_+} U_\alpha,$$

and that Ω_{R_+}/Q is an open subset of G' , containing the identity section (cf. Exp. IV, 4.7.2 and 6.4.1).

It follows that if one denotes by $\mathfrak{g}'$ the Lie algebra of G' and by α the character of T/Q induced by α (or, equivalently, defined by $\alpha \in N$ in the identification $T/Q = D_S(N)$), the canonical morphism $\mathfrak{g} \rightarrow \mathfrak{g}'$ induces for each $\alpha \in R$ an isomorphism

$$\mathfrak{g}^\alpha \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}.$$

One has therefore proved that R is a system of roots of G' with respect to T' , and one finishes the proof without difficulty.

Corollary 4.3.2. *Let S be a scheme, G a reductive S -group, Q an invariant subgroup of multiplicative type of G . Then Q is central in G , the quotient G/Q is representable by a reductive S -group, and the canonical morphism $G \rightarrow G/Q$ is locally splittable for the étale topology (with radicial exponents equal to 1).*

The first assertion follows from Exp. IX 5.5; the others are local for the étale topology and one is reduced to 4.3.1.

Definition 4.3.3. Let G be a reductive S -group. One says that G is *adjoint* (resp. *simply connected*) if for every $s \in S$, the type of G at s is given by an adjoint (resp. simply connected) root datum, i.e. (Exp. XXI 6.2.6) such that M is generated by R (resp. M^* generated by R^*).

Proposition 4.3.4.

- (i) *An adjoint (resp. simply connected) reductive group is semisimple.*
- (ii) *If T is a maximal torus of the adjoint (resp. simply connected) reductive group G and if α is a root of G with respect to T , then the infinitesimal root $\bar{\alpha}$ is nonzero on each fiber (resp. α^* is a monomorphism and the infinitesimal coroot H_α is nonzero on each fiber).*

Indeed, (i) is trivial; (ii) is verified on the geometric fibers and follows at once from Exp. XXI 6.2.8.

Proposition 4.3.5.

- (i) *For the reductive group G to be adjoint, it is necessary and sufficient that $\underline{\text{Centr}}(G) = \{e\}_S$.*
- (ii) *For every reductive group G , the quotient group $G/\underline{\text{Centr}}(G)$ is an adjoint reductive group.*

Indeed, one may assume G split; then (i) is trivial (for $\underline{\text{Centr}}(G) = D_S(M/\Gamma_0(R))$), and (ii) follows from 4.3.1.

Definition 4.3.6. Let G be a reductive S -group. One calls the *adjoint group* of G , and denotes by $\text{ad}(G)$, the group $G/\underline{\text{Centr}}(G)$. One calls the *radical* of G , and denotes by $\text{rad}(G)$, the maximal torus (unique by Exp. XII 1.12) of $\underline{\text{Centr}}(G)$. One calls the *semisimple group associated with G* the quotient $G/\text{rad}(G)$.

The preceding definitions are compatible with base change. If $s \in S$, $\text{rad}(G)_{\bar{s}}$ is indeed the radical of $G_{\bar{s}}$ in the usual sense (Exp. XIX 1.6).

4.3.7. — If (G, T, M, R) is a split group, then $\text{rad}(G) = D_S(M/N)$, where $N = M \cap \mathcal{V}(R)$, hence the semisimple group associated with G (as indeed the adjoint group of G) is endowed with a canonical splitting (4.3.1) and one has a diagram of split groups

$$G \rightarrow \text{ss}(G) \rightarrow \text{ad}(G)$$

corresponding to the canonical diagram of root data (Exp. XXI 6.5.5)

$$\text{ad}(\mathcal{R}(G)) \rightarrow \text{ss}(\mathcal{R}(G)) \rightarrow \mathcal{R}(G).$$

Remark 4.3.8. Let (G, T, M, R) be an adjoint (resp. simply connected) split S -group, Δ a system of simple roots of R . Then the family $\{\alpha\}_{\alpha \in \Delta}$, resp. $\{\alpha^*\}_{\alpha \in \Delta}$, induces an isomorphism

$$T \xrightarrow{\sim} (\mathbf{G}_{m,S})^\Delta, \quad \text{resp.} \quad (\mathbf{G}_{m,S})^\Delta \xrightarrow{\sim} T.$$

Indeed, $M = \Gamma_0(R)$ (resp. ...) and Δ is a basis of the free abelian group $\Gamma_0(R)$ (Exp. XXI 3.1.8).

Remark 4.3.9. The radical of a reductive group is a *characteristic* subgroup (i.e. stable under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$), in view of its definition.

5. Subgroups of type (R)

We are particularly interested in reductive groups, but some of the results we are going to establish hold more generally for a larger class of groups: groups of type (RR).

5.1. Groups of type (RR)

Definition 5.1.1. Let S be a scheme, G an S -group scheme. One says that G is of *type (RR)* if it satisfies the following conditions:

- (i) G is smooth and of finite presentation over S and has connected fibers.
- (ii) G possesses maximal tori locally for the (fpqc) topology.
- (iii) For every $s \in S$, every maximal torus T of $G_{\bar{s}}$ and every root of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ (Exp. XIX 1.10), $\underline{\mathrm{Lie}}(G_{\bar{s}})^{\alpha}$ has rank 1 (as a vector space over $\kappa(\bar{s})$).
- (iv) For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, denote by R the set of roots of $G_{\bar{s}}$ with respect to T and by $\Gamma_0(R)$ the subgroup of $\mathrm{Hom}_{\bar{s}\text{-gr.}}(T, \mathbf{G}_{m, \bar{s}})$ generated by R ; then the content⁽²²⁾ of every root $\alpha \in R$ in the free abelian group $\Gamma_0(R)$ (which is thus an integer > 0) is invertible on S .

Recall 5.1.1.1. ⁽²³⁾Recall that if G is a smooth connected algebraic group over an algebraically closed field k , a *Cartan subgroup* of G is the centralizer of a maximal torus of G (XII 1.0), and such a subgroup is *smooth and connected*: for this, as well as for other characterizations of Cartan subgroups, see *Bible*, § 7.1, Th. 1 in the case G affine and Exp. XII Th. 6.6 in the general case. If S is an arbitrary scheme and G a smooth S -group of finite type, one calls a *Cartan subgroup* of G a smooth S -subgroup C of G such that, for every $s \in S$, $C_{\bar{s}}$ is a Cartan subgroup of $G_{\bar{s}}$ (XII Def. 3.1).

Remark 5.1.2. a) By Exp. XII 7.1 (where the hypothesis G separated holds here since G has connected fibers, cf. Exp. VIB 5.5), (i) and (ii) imply that G possesses maximal tori (resp. Cartan subgroups) locally for the étale topology, conjugate locally for the étale topology.

b) The Cartan subgroups of G have connected fibers (Exp. XII 6.6).

c) If G is affine over S , (i) and (ii) are equivalent respectively to

(i') G is smooth over S and has connected fibers.

(ii') The reductive rank of the fibers of G is locally constant (Exp. XII

1.7).

d) By Exp. XII 8.8 (c) and (d), G possesses a reductive center Z and for every $s \in S$ one has, with the notations of (iv),⁽²⁴⁾ $Z_{\bar{s}} = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$, whence

$$\mathrm{Hom}_{\bar{s}\text{-gr.}}((T/Z)_{\bar{s}}, \mathbf{G}_{m, \bar{s}}) \simeq \Gamma_0(R).$$

e) Condition (iv) holds in particular in the following two cases:

(1) S has characteristic 0;

(2) every root is an indivisible element of the group generated by the roots.

f) Being of type (RR) is stable under base change and is local for the (fpqc) topology.

From remarks (c) and (e), one obtains at once the

⁽²²⁾**N.D.E.** The content of the root α is the positive generator of the ideal $\{f(\alpha), f \in \Gamma_0(R)^*\}$ of $\mathbf{Z}$; it is the largest integer $c > 0$ such that $\alpha/c \in \Gamma_0(R)$.

⁽²³⁾**N.D.E.** This reminder has been added.

⁽²⁴⁾**N.D.E.** The original indicated “under the conditions of (iv)”, but the last condition of (iv) does not seem to be used here.

Proposition 5.1.3. *A reductive group is of type (RR).*

Proposition 5.1.4. *Let S be a scheme, G an S -group of type (RR), Q a central subgroup of G of finite presentation over S such that the quotient G/Q is representable (for example G affine over S and Q of multiplicative type (IX 2.3), or S artinian (VIA 3.3.2)); then G/Q is an S -group of type (RR).*

Indeed G/Q is smooth over S (Exp. VIB 9.2), of finite presentation over S (Exp. V 9.1) and has connected fibers, so condition (i) holds. On the other hand, condition (ii) follows from Exp. XII 7.6; it remains to verify conditions (iii) and (iv).

Write $G' = G/Q$, let $u : G \rightarrow G'$ be the canonical morphism, $T' = u(T)$ the maximal torus of G' which is the image of T (cf. Exp. XII 7.1 (e)); for each $\alpha \in R$, denote again by α the character of T' defined by α (one has $Q \cap T \subset \bigcap_{\alpha \in R} \text{Ker}(\alpha)$ by 5.1.2 (d)). Let us first prove:

Lemma 5.1.5. *Under the conditions of 5.1.4, let $T = D_S(M)$ be a trivialized maximal torus of G , assume that the decomposition of $\mathfrak{g} = \underline{\text{Lie}}(G)$ under $\text{Ad}(T)$ is of the form*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha, \quad R \subset M - \{0\},$$

where for every $s \in S$, $\mathfrak{g}^\alpha(s) \neq 0$ for $\alpha \in R$ (so $\mathfrak{g}^\alpha$ is an invertible $\mathcal{O}_S$ -module for every $\alpha \in R$ and R is the set of roots of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ for every $s \in S$).

Then the Lie algebra $\mathfrak{g}'$ of G' decomposes under $\text{Ad}(T')$ as follows:

$$\mathfrak{g}' = \mathfrak{g}'^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}'^\alpha$$

and $\underline{\text{Lie}}(u)$ induces an isomorphism of $\mathfrak{g}^\alpha$ onto $\mathfrak{g}'^\alpha$.

Indeed, let $p = \underline{\text{Lie}}(u) : \mathfrak{g} \rightarrow \mathfrak{g}'$. One has at once $p(\mathfrak{g}^\alpha) \subset \mathfrak{g}'^\alpha$ for every $\alpha \in R$, and $p(\mathfrak{g}^0) \subset \mathfrak{g}'^0$. Since

$$\text{Ker}(p) = \underline{\text{Lie}}(Q) \subset \underline{\text{Lie}}(\text{Centr}_G(T)) = \mathfrak{g}^0,$$

p induces a monomorphism of $\mathfrak{g}^\alpha$ into $\mathfrak{g}'^\alpha$, for every $\alpha \in R$.

To prove the lemma, it suffices to do so when S is the spectrum of an algebraically closed field, and by the preceding remarks it then suffices to prove that $\text{rg}(\mathfrak{g}') = \text{rg}(\mathfrak{g}'^0) + \text{Card}(R)$. Now put $C = \underline{\text{Centr}}_G(T)$, $C' = \underline{\text{Centr}}_{G'}(T')$; by Exp. XII 7.1 (e), u induces a faithfully flat morphism $C \rightarrow C'$ with kernel Q . One therefore has

$$\dim C' + \dim Q = \dim C.$$

But G , G' , C and C' are smooth, hence

$$\begin{aligned} \dim G &= \text{rg}(\mathfrak{g}) = \text{rg}(\mathfrak{g}^0) + \text{Card}(R) = \dim C + \text{Card}(R) \\ &= \dim Q + \dim C' + \text{Card}(R) \\ &= \dim Q + \text{rg}(\mathfrak{g}'^0) + \text{Card}(R), \\ \text{rg}(\mathfrak{g}') &= \dim G' = \dim G - \dim Q, \end{aligned}$$

which implies

$$\text{rg}(\mathfrak{g}') = \text{rg}(\mathfrak{g}'^0) + \text{Card}(R),$$

that is, the desired relation.

Let us return to the proof of 5.1.4; one may assume that S is the spectrum of an algebraically closed field. Let T be a maximal torus of G ; applying 5.1.5, one has at once (iii) and (iv) for G/Q .

To use the preceding proposition, let us introduce a definition:

Definition 5.1.6. One says that the S -group scheme G is of type (RA) if it is of type (RR) and satisfies in addition condition (iv') (stronger than (iv)):

(iv') For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, every root of $G_{\bar{s}}$ with respect to T has a content in $\text{Hom}_{\bar{s}\text{-gr.}}(T, \mathbf{G}_{m, \bar{s}})$ which is invertible on S .

Remarks 5.1.7. a) An adjoint reductive S -group is of type (RA).

b) If S has characteristic 0, every group of type (RR) is of type (RA).

c) Being of type (RA) is stable under base change and is local for the (fpqc) topology.

The preceding remark (a) generalizes as:

Proposition 5.1.8. *Let S be a scheme, G an S -group of type (RR), Z its reductive center; assume the quotient G/Z representable (for example G affine over S , or S artinian); then G/Z is of type (RA).*

Indeed, this follows at once from 5.1.4, 5.1.5 and 5.1.2 (d).

Remark 5.1.9. If G is of type (RR) and if T is a maximal torus of G , one may apply Exp. XIX 6.3. In particular $W_G(T)$ is étale, quasi-finite and separated over S .

5.2. Subgroups of type (R)

Definition 5.2.1. Let S be a scheme, G a smooth S -group scheme of finite presentation with connected fibers,⁽²⁵⁾ H a group subscheme of G . One says that H is of type (R) if:

- (i) H is smooth, of finite presentation over S and has connected fibers.⁽²⁵⁾
- (ii) For every $s \in S$, $H_{\bar{s}}$ contains a Cartan subgroup of $G_{\bar{s}}$.

This notion is stable under base change and local for the (fpqc) topology.

Recall 5.2.2. (cf. Exp. XII 7.9) Under the preceding conditions:

- a) $H = \underline{\text{Norm}}_G(H)^0$.
- b) If G contains Cartan subgroups (resp. maximal tori) locally for the étale topology, the same holds for H , and the Cartan subgroups (resp. maximal tori) of H are Cartan subgroups (resp. maximal tori) of G .

Examples 5.2.3. a) *Borel subgroups:* a Borel subgroup of G is a subgroup H of type (R) whose geometric fibers are Borel subgroups of those of G .⁽²⁶⁾

b) *Parabolic subgroups:* a parabolic subgroup of G is a subgroup of type (R) whose geometric fibers contain Borel subgroups.

Other examples are given by the following propositions.

Proposition 5.2.4. *Let G be as in 5.2.1, $K \subset H$ two group subschemes of G , H being assumed of type (R). Then K is a subgroup of type (R) of H if and only if it is a subgroup of type (R) of G .*

⁽²⁷⁾Indeed, let $s \in S$. Since H is of type (R), every maximal torus of $H_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$, and therefore the same holds for Cartan subgroups.

Proposition 5.2.5. *Let G be as in 5.2.1, T a maximal torus of G , Q a subtorus of T , $Z = \underline{\text{Centr}}_G(Q)$. If H is a subgroup of type (R) of G containing T , then $H \cap Z$ is a subgroup of type (R) of Z .*

⁽²⁵⁾**N.D.E.** The hypothesis that G (resp. H) is of finite presentation over S holds automatically since G (resp. H), being smooth over S and with connected fibers, is quasi-compact and separated over S (VIB 5.5), hence of finite presentation over S .

⁽²⁶⁾**N.D.E.** It amounts to the same to say that H is a smooth subgroup of G , each of whose geometric fibers $H_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$ (since every Borel subgroup of $G_{\bar{s}}$ is connected and contains a Cartan subgroup of $G_{\bar{s}}$).

⁽²⁷⁾**N.D.E.** What follows has been expanded.

Recall first that Z is a closed group subscheme of G , of finite presentation (Exp. XI 6.11), with connected fibers (Exp. XII 6.6), smooth over S (Exp. XI 2.4), hence satisfies the conditions imposed in definition 5.2.1. Likewise, $H \cap Z$ is of finite presentation, smooth and with connected fibers (for $H \cap Z = \underline{\text{Centr}}_H(Q)$); moreover $H \cap Z \supset \underline{\text{Centr}}_G(T)$.

Proposition 5.2.6. *Let S be a scheme, G an S -group of type (RR) (resp. (RA)), H a subgroup of type (R) of G . Then H is an S -group of type (RR) (resp. (RA)).*

Indeed (i) is clear, (ii) follows from 5.2.2 (b), (iii) and (iv) (resp. (iv')) are to be verified when S is the spectrum of an algebraically closed field. Then H contains a maximal torus T of G (and hence also $C = \underline{\text{Centr}}_G(T)^{(28)}$) and the assertions to be proved follow at once from:

Lemma 5.2.7. *Let S be a scheme, G an S -group of type (RR), T a maximal torus of G endowed with a trivialization $T \simeq \text{D}_S(M)$; assume that*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha$$

(the $\mathfrak{g}^\alpha$ being then invertible $\mathcal{O}_S$ -modules).

Let H be a subgroup of type (R) containing $C = \underline{\text{Centr}}_G(T)$ (i.e. containing T). Then $\mathfrak{h} = \underline{\text{Lie}}(H/S)$ is locally over S of the form

$$\mathfrak{g}^0 + \coprod_{\alpha \in R'} \mathfrak{g}^\alpha = \mathfrak{g}_{R'};$$

more precisely, let for each $s \in S$, $R'(s) = \{\alpha \in R \mid \mathfrak{g}^\alpha(s) \subset \mathfrak{h}(s)\}$. Then $R'(s)$ is a locally constant function of s ; if U is an open subset of S on which $R'(s) = R'$, one has

$$\mathfrak{h}_U = \mathfrak{g}_U^0 \oplus \coprod_{\alpha \in R'} \mathfrak{g}_U^\alpha.$$

Indeed, $\mathfrak{h}$ is a submodule of $\mathfrak{g}$, locally a direct summand, containing $\mathfrak{g}^0$ and stable under T .

5.3. Strict transporter of two subgroups of type (R). Applications

Recall 5.3.0. ⁽²⁹⁾Let S be a scheme, G a smooth S -group, $\mathfrak{g} = \underline{\text{Lie}}(G/S)$ and $\mathfrak{h}$ an $\mathcal{O}_S$ -submodule of $\mathfrak{g}$ locally a direct summand. The $\mathcal{O}_S$ -algebra $\mathcal{A} = \text{Sym}(\omega_{G/S}^1)$ is locally free, hence the S -scheme $\underline{\text{Lie}}(G/S) = \text{W}(\mathfrak{g}) = \text{Spec}(\mathcal{A})$ is essentially free in the sense of Exp. VIII, 6.1. Since $\text{W}(\mathfrak{h})$ is a closed subscheme of $\underline{\text{Lie}}(G/S)$, of finite presentation over $\underline{\text{Lie}}(G/S)$, $N = \underline{\text{Norm}}_G(\mathfrak{h})$ is representable by a closed group subscheme of G , of finite presentation over G , by Exp. VIII, 6.5 (a). (See also additions 6.2.3 and 6.2.4 (a) in Exp. VIB.) On the other hand, by Exp. II 5.3.1, one has $\underline{\text{Lie}}(N/S) = \underline{\text{Norm}}_{\underline{\text{Lie}}(G/S)}(\mathfrak{h})$.

Finally, by Exp. VIB 3.10, if N is smooth over S at the points of the identity section, then the subgroup functor N^0 (defined in VIB 3.1) is represented by an open group subscheme of N , smooth over S .

Proposition 5.3.1. *Let S be a scheme, G an S -group of type (RA) (5.1.6), H a subgroup of type (R) of G , $\mathfrak{g} \supset \mathfrak{h}$ their Lie algebras.*

Then $\underline{\text{Norm}}_G(\mathfrak{h})$ (which is representable by a closed subscheme of G of finite presentation over S by 5.3.0) is smooth over S at every point of the identity section and one has

$$\underline{\text{Norm}}_G(\mathfrak{h})^0 = H.$$

⁽²⁸⁾**N.D.E.** By hypothesis, H contains $\underline{\text{Centr}}_G(T')$ for some maximal torus T' of G ; then T and T' are conjugate in H , hence H also contains $C = \underline{\text{Centr}}_G(T)$.

⁽²⁹⁾**N.D.E.** This reminder, which is used in the proof of 5.3.1 and 5.3.4, has been added.

PROOF. ⁽³⁰⁾Put $N = \underline{\text{Norm}}_G(\mathfrak{h})$ and $\mathfrak{n} = \underline{\text{Lie}}(N/S)$. One has $H \subset N$ and, by Exp. II 5.3.1, one has for every $s \in S$

$$\mathfrak{h}(s) \subset \mathfrak{n}(s) = \underline{\text{Norm}}_{\mathfrak{g}(s)}(\mathfrak{h}(s)).$$

Now, by 5.3.2 below, one has $\mathfrak{h}(s) = \underline{\text{Norm}}_{\mathfrak{g}(s)}(\mathfrak{h}(s))$, and since H is smooth over S , one has $\dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s$ (cf. [DG70], § II.5, Th. 2.1). One therefore obtains

$$\dim_{\kappa(s)} \mathfrak{n}(s) = \dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s \leq \dim N_s$$

whence $N_s^0 = H_s^0 = H_s$ (H having connected fibers). It follows that the subgroup functor N^0 (defined in VIB, 3.1) is represented by the smooth S -group H . This proves 5.3.1, modulo the following lemma: $\square$

Lemma 5.3.2. *Under the conditions of 5.2.7, if G is of type (RA), one has, for every $s \in S$,*

$$\underline{\text{Norm}}_{\mathfrak{g}(s)}(\mathfrak{h}(s)) = \mathfrak{h}(s).$$

Indeed, one is reduced to the case where S is the spectrum of a field, hence where $\mathfrak{h} = \mathfrak{g}_{R'}$ for some $R' \subset R$. But one already has

$$\underline{\text{Transp}}_{\mathfrak{g}}(\mathfrak{t}, \mathfrak{h}) = \mathfrak{h}.$$

Indeed, if $H \in \mathfrak{t}$ and $X \in \mathfrak{g}^\alpha$, one has $[H, X] = \bar{\alpha}(H)X$, where $\bar{\alpha} : \mathfrak{t} \rightarrow \mathcal{O}_S$ is the derivative morphism of α . Now condition (iv') says precisely that $\bar{\alpha} \neq 0$ for every $\alpha \in R$.

Corollary 5.3.3. *Let S be a scheme, G an S -group of type (RA), H and H' two subgroups of type (R) of G , $\mathfrak{h}$ and $\mathfrak{h}'$ their Lie algebras. Then*

$$H = H' \iff \mathfrak{h} = \mathfrak{h}'.$$

Corollary 5.3.4. *Under the conditions of 5.2.7, with G of type (RA), the maps*

$$H \mapsto \underline{\text{Lie}}(H/S), \quad \mathfrak{h} \mapsto \underline{\text{Norm}}_G(\mathfrak{h})^0$$

establish a bijective correspondence between the set of subgroups of type (R) of G containing T , and the set of Lie subalgebras of $\mathfrak{g}$ containing $\mathfrak{g}^0$, stable under T , and whose normalizer in G is smooth over S at every point of the identity section.

⁽³¹⁾Indeed, let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ satisfying the above properties. By 5.3.0, $H = \underline{\text{Norm}}_G(\mathfrak{h})^0$ is a smooth S -group scheme. Moreover, since $C = \underline{\text{Centr}}_G(T)$ stabilizes each $\mathfrak{g}^\alpha$ and has connected fibers (XII 6.6), one has $C \subset H$. Hence H is a subgroup of G of type (R). By Exp. II 5.3.1, one has $\underline{\text{Lie}}(H) = \underline{\text{Norm}}_{\mathfrak{g}}(\mathfrak{h})$. Finally, by the proof of 5.3.2, one has $\underline{\text{Norm}}_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$.

Corollary 5.3.5. *Let S be a scheme, G an S -group of type (RR) (5.1.1), T a maximal torus of G , H and H' two subgroups of type (R) of G , containing T . Then*

$$H = H' \iff \mathfrak{h} = \mathfrak{h}'.$$

By the finite presentation hypotheses, one reduces as usual (cf. EGA IV₃, § 8 and Exp. VIB § 10) to the case where S is noetherian; it then suffices to verify that $\mathfrak{h} = \mathfrak{h}'$ implies $H_{S'} = H'_{S'}$ for every S' which is the spectrum of an artinian quotient of a local ring of S ; ⁽³²⁾one is therefore reduced to the case where S is artinian and where one may apply 5.1.8. Let then $u : G \rightarrow G' = G/Z$ be the canonical morphism and $T' = T/Z$ the maximal torus of G' corresponding to T . By Exp. XII 7.12, there exist subgroups of type (R) H_1 and H'_1 of G' ,

⁽³⁰⁾N.D.E. The original has been expanded in what follows.

⁽³¹⁾N.D.E. The following proof has been added.

⁽³²⁾N.D.E. Indeed, let $g \in G$, s its image in S , $\mathfrak{m}$ the maximal ideal of $\mathcal{O}_{G,g}$, $\mathfrak{n}$ that of $\mathcal{O}_{S,s}$, and I (resp. I') the kernel of the morphism from $\mathcal{O}_{G,g}$ to $\mathcal{O}_{H,g}$ (resp. $\mathcal{O}_{H',g}$) (the latter being the zero ring if $g \notin H$, resp. $g \notin H'$). Since $\mathcal{O}_{G,g}$ is noetherian, I and I' are closed for the $\mathfrak{m}$ -adic topology, hence a fortiori for the $\mathfrak{n}$ -adic topology, so it suffices to show that $I + \mathfrak{n}^n \mathcal{O}_{G,g} = I' + \mathfrak{n}^n \mathcal{O}_{G,g}$ for every $n \in \mathbb{N}$.

containing T' , such that $H = u^{-1}(H_1)$ and $H' = u^{-1}(H'_1)$. It suffices to prove that $H_1 = H'_1$. But by 5.2.7 and 5.1.5, one has

$$\underline{\mathrm{Lie}}(H_1) = \underline{\mathrm{Lie}}(H'_1),$$

and one is reduced to 5.3.3.

Remark 5.3.6. The fact that H and H' contain the same maximal torus is essential for the validity of 5.3.5 when G is not of type (RA). Example: maximal tori of $\mathrm{SL}_{2,k}$, for k of characteristic 2.⁽³³⁾

Corollary 5.3.7. *Let S be a scheme, G an S -group of type (RR), T a maximal torus of G , H and H' two subgroups of type (R) of G , containing T . The set U of $s \in S$ such that $H_s = H'_s$ is open and closed in S and $H_U = H'_U$.*

Indeed, this follows at once from 5.3.5 and 5.2.7.

Corollary 5.3.8. *The “functor of subgroups of type (R) containing T ”, where T is a given maximal torus in a group G of type (RR), is formally unramified (Exp. XI 1.1).*

Theorem 5.3.9. *Let G be an S -group of type (RR) (5.1.1), H and H' two subgroups of type (R) (5.2.1). Let $\underline{\mathrm{Transt}}_G(H, H')$ be the strict transporter of H into H' defined by*

$$\underline{\mathrm{Transt}}_G(H, H')(S') = \{g \in G(S') \mid \mathrm{int}(g)H_{S'} = H'_{S'}\}.$$

Then $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G , which is smooth and of finite presentation over S .

The fact that $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G , of finite presentation over S , follows from Exp. XI 6.11 (a). To prove that it is smooth over S , one must prove that if S is affine and if S_0 is the closed subscheme defined by a nilpotent ideal J , and if $g_0 \in G(S_0)$ and $\mathrm{int}(g_0)H_0 = H'_0$, there exists a $g \in G(S)$ projecting to g_0 and such that $\mathrm{int}(g)H = H'$. Since the question of smoothness is local for the étale topology, one may assume that H contains a maximal torus T of G .

Then T_0 is a maximal torus of H_0 , hence $\mathrm{int}(g_0)T_0$ a maximal torus of H'_0 . By Exp. IX 3.6 bis, there exists a torus T' of H' such that $T'_0 = \mathrm{int}(g_0)T_0$; by Exp. IX 3.3 bis, there therefore exists a $g \in G(S)$ projecting to g_0 and such that $\mathrm{int}(g)T = T'$. Replacing H by $\mathrm{int}(g)H$, one may therefore assume that H and H' contain the same maximal torus T and that $H_0 = H'_0$. But then $H = H'$ by 5.3.7. Q.E.D.

Corollary 5.3.10. *Let G be an S -group of type (RR), H a subgroup of type (R) of G . Then $\underline{\mathrm{Norm}}_G(H)$ is representable by a closed group subscheme of G , of finite presentation and smooth over S .*

Using now the argument which served, in Exp. XI, to deduce 5.4 bis from 5.2 bis, one obtains:

Corollary 5.3.11. *Under the hypotheses of 5.3.9, the following conditions are equivalent:*

- (i) H and H' are conjugate locally in G for the étale topology.
- (i bis) *idem* for the (fpqc) topology.
- (ii) For every $s \in S$, $H_{\bar{s}}$ and $H'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$.
- (ii bis) The structural morphism $\underline{\mathrm{Transt}}_G(H, H') \rightarrow S$ is surjective.
- (iii) $\underline{\mathrm{Transt}}_G(H, H')$ is a principal homogeneous bundle under the action of the smooth S -group scheme of finite presentation $\underline{\mathrm{Norm}}_G(H)$.

Let us simply note that the nontrivial assertion (iii) $\Rightarrow$ (i) is Hensel's lemma.

⁽³³⁾**N.D.E.** Indeed, if k is algebraically closed, all maximal tori of $G = \mathrm{SL}_{2,k}$ are conjugate under $G(k)$, and all have as Lie algebra the line $k \cdot \mathrm{id} \subset M_2(k)$ (which is invariant under the adjoint action).

Using now *Bible*, § 6.4, th. 4 (= [Ch05], § 6.5 th. 5) and § 9.3, th. 1, one obtains by 5.3.10 and 5.3.11:

Corollary 5.3.12. *Let G be an S -group of type (RR) . The Borel subgroups of G are closed in G , equal to their normalizer, and conjugate locally for the étale topology.*

Definition 5.3.13. Let S be a scheme, G a smooth S -group scheme of finite presentation and with connected fibers.⁽³⁴⁾ One calls a *Killing pair* of G a pair $T \subset B$, where T is a maximal torus of G and B a Borel subgroup of G containing T .

Using now the conjugacy of maximal tori in B (cf. 5.1.2 (a) and 5.2.6, for example), one has:

Corollary 5.3.14. *Let G be an S -group of type (RR) . The Killing pairs of G are conjugate locally for the étale topology.*

Corollary 5.3.15. *Let G be an S -group of type (RR) . Let T be a maximal torus of G , $W_G(T) = \underline{\text{Norm}}_G(T)/\underline{\text{Centr}}_G(T)$ the corresponding Weyl group (Exp. XIX 6.3). The “functor of Borel subgroups of G containing T ” is formally principal homogeneous under $W_G(T)$.*

This follows at once from 5.3.14 and the fact that if B is a Borel subgroup of G containing T , one has

$$\underline{\text{Norm}}_G(T) \cap B = \underline{\text{Centr}}_G(T),$$

cf. Exp. XIV 4.4.

Proposition 5.3.16. *Let G be an S -group of type (RR) , H a subgroup of type (R) , $N = \underline{\text{Norm}}_G(H)$ its normalizer (5.3.10). Let T be a maximal torus of H , $W_H(T)$ and $W_N(T)$ the corresponding Weyl groups (étale quasi-finite separated by Exp. XIX 6.3). One has the following exact sequence of sheaves (for the étale topology) (the morphisms are induced by the morphisms $\underline{\text{Norm}}_H(T) \rightarrow \underline{\text{Norm}}_N(T) \rightarrow N/H$):*

$$1 \rightarrow W_H(T) \rightarrow W_N(T) \rightarrow N/H \rightarrow 1.$$

The only nontrivial point is the fact that the last arrow is an epimorphism. Let therefore $n \in N(S')$, $S' \rightarrow S$. The two maximal tori T and $\text{int}(n)T$ of H are conjugate in H locally for the étale topology. There therefore exist a covering family $\{S'_i \rightarrow S'\}$ and for each i an $h_i \in H(S'_i)$ such that $\text{int}(h_i)T = \text{int}(n)T$. One therefore has $nh_i^{-1} \in \underline{\text{Norm}}_N(T)$, which implies the desired result.

Remark 5.3.17. One may describe $W_N(T)$ as follows: assume we have reduced to the situation of 5.2.7, with $\mathfrak{h} = \mathfrak{g}_{R'}$. Then $W_N(T)$ equals $\underline{\text{Norm}}_W(R')$, the sheaf of sections of $W = W_G(T)$ which, acting on R , normalize R' . Indeed, by 5.3.5, one has

$$\underline{\text{Norm}}_N(T) = \underline{\text{Norm}}_G(H) \cap \underline{\text{Norm}}_G(T) = \underline{\text{Norm}}_G(\mathfrak{h}) \cap \underline{\text{Norm}}_G(T).$$

Corollary 5.3.18. *Let G be an S -group of type (RR) , H a subgroup of type (R) . Assume “the Weyl groups of G finite”, i.e. that for every $S' \rightarrow S$ and every maximal torus T of $G_{S'}$, the étale S' -scheme $\underline{\text{Norm}}_{G_{S'}}(T)/\underline{\text{Centr}}_{G_{S'}}(T)$ is finite (cf. Exp. XIX 6.3 (iii)). The following conditions are equivalent:*

- (i) H is closed in G .
- (ii) $\underline{\text{Norm}}_G(H)/H$ is representable by a finite étale S -scheme.
- (iii) “the Weyl groups of H are finite”.

Indeed, one may assume that H possesses a maximal torus T . By 5.3.10, $N = \underline{\text{Norm}}_G(H)$ is closed in G , hence $W_N(T)$ is closed in $W_G(T)$ and therefore finite over S . One clearly has

⁽³⁴⁾**N.D.E.** cf. N.D.E. (25).

(i) $\Rightarrow$ (iii), and one has (iii) $\Rightarrow$ (ii) by the exact sequence of 5.3.16. Finally, one has (ii) $\Rightarrow$ (i) for if N/H is finite, it is separated, hence H is closed in N , hence in G .

Remark 5.3.19. When G is reductive, the preceding conditions on H seem always to hold. We prove them below in most cases.

5.4. Subgroups of type (R) of a split reductive group (generalities)

5.4.1. — If H is a subgroup of type (R) of the reductive group G , then H contains a maximal torus of G locally for the étale topology (5.2.2). By 2.3, one may, locally for the étale topology, assume G split with respect to this torus. Let therefore (G, T, M, R) be a split S -group, H a subgroup of type (R) of G containing T . By 5.3.5 such a subgroup is characterized by its Lie algebra, which (5.2.7) is locally over S of the form $\mathfrak{g}_{R'}$:

$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha.$$

Definition 5.4.2. Let (G, T, M, R) be a split S -group. One will say that the subset R' of R is of type (R) if $\mathfrak{g}_{R'}$ is the Lie algebra of a subgroup of type (R) of G containing T . This subgroup, uniquely determined by R' , is denoted $H_{R'}$.

Lemma 5.4.3. Under the preceding conditions, one has the following equivalences:

$$\begin{aligned} H \cap Z_\alpha = T &\iff \alpha \notin R', \quad -\alpha \notin R', \\ H \supset U_\alpha &\iff \alpha \in R', \\ H \cap U_\alpha = e &\iff \alpha \notin R', \\ H \supset Z_\alpha &\iff \alpha \in R', \quad -\alpha \in R'. \end{aligned}$$

Indeed, $H \cap Z_\alpha$ is a subgroup of type (R) of Z_α , by 5.2.5; but a subgroup of type (R) of Z_α , containing T , is locally equal to one of the following subgroups: T , $T \cdot U_\alpha$, $T \cdot U_{-\alpha}$, Z_α , by 5.3.5.

Lemma 5.4.4. Under the conditions of 5.4.2, let R_+ be a system of positive roots; choose orders on $R' \cap R_+$ and $R' \cap -R_+$. The morphism

$$\Omega_{R_+, R'} = \prod_{\alpha \in R' \cap -R_+} U_\alpha \times_S T \times_S \prod_{\alpha \in R' \cap R_+} U_\alpha \longrightarrow G$$

induced by the product in G induces an open immersion

$$\Omega_{R_+, R'} \longrightarrow H_{R'}.$$

Indeed, by 5.4.3, this morphism factors through $H_{R'}$ and therefore induces an immersion $\Omega_{R_+, R'} \longrightarrow H_{R'}$. One then argues as in 4.1.1.

Proposition 5.4.5. Let (G, T, M, R) be a split S -group. Let R' and R'' be two subsets of R , of type (R).

(i) $H_{R'} \cap H_{R''}$ is smooth at every point of the identity section, $R' \cap R''$ is of type (R), and one has

$$(H_{R'} \cap H_{R''})^0 = H_{R' \cap R''}.$$

(ii) One has the equivalence

$$H_{R'} \subset H_{R''} \iff R' \subset R''.$$

Indeed, (ii) follows at once from (i). To prove (i), it suffices to prove that $H_{R'} \cap H_{R''}$ is smooth at every point of the identity section: its identity component (cf. Exp. VIB 3.10) will then be a subgroup of type (R) containing T , hence equal to $H_{R' \cap R''}$; but $\Omega_{R_+, R'} \cap \Omega_{R_+, R''} = \Omega_{R_+, R' \cap R''}$ is an open subset of $H_{R'} \cap H_{R''}$ containing the identity section and smooth over S .

Corollary 5.4.6. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , s a point of S . If H and H' are two subgroups of type (R) of G containing T such that $H_s \subset H'_s$, there exists an open subset U of S containing s such that $H_U \subset H'_U$.*

One may indeed assume G split with respect to T . The assertion then follows at once from 5.4.5 (ii).

One is led to ask which are the subsets R' of type (R) of R . One may assume the group adjoint; one must then verify that $\mathfrak{g}_{R'}$ is a Lie algebra and that its normalizer is smooth at every point of the identity section. The most important case is given by the

Theorem 5.4.7. *Every closed subset R' of R is of type (R). (Recall, cf. Exp. XXI 3.1.4, that $R' \subset R$ is said to be closed if $\alpha, \beta \in R'$, $\alpha + \beta \in R$ implies $\alpha + \beta \in R'$).*

Remark 5.4.8. We will see later (Exp. XXIII 6.6) that if $6 \cdot 1_S \neq 0^{(35)}$ (for example, if S possesses a residue characteristic distinct from 2 and 3), the fact that $\mathfrak{g}_{R'}$ is a Lie algebra already implies that R' is closed, hence R' is of type (R) if and only if it is closed. Theorem 5.4.7 therefore gives all subsets of type (R) “independent of the characteristic”.

Let us first prove:

Lemma 5.4.9. *Choose for each $\alpha \in R$ an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$. Let $\alpha, \beta \in R$, with $\alpha + \beta \neq 0$, and let q be the largest integer i such that $\alpha + i\beta \in R$. There exist uniquely determined sections $M_{\alpha, \beta, i} \in \Gamma(S, \mathcal{O}_S)$ such that*

$$\mathrm{Ad}(\exp_\alpha(xX_\alpha))(X_\beta) = X_\beta + \sum_{i=1}^q M_{\alpha, \beta, i} x^i X_{\beta+i\alpha},$$

for every $x \in \mathbf{G}_a(S')$, $S' \rightarrow S$.

Indeed, $x \mapsto \mathrm{Ad}(\exp_\alpha(xX_\alpha))(X_\beta)$ defines a morphism $\mathbf{G}_{a,S} \rightarrow \mathrm{W}(\mathfrak{g}) \simeq \mathbf{G}_{a,S}^m$. There therefore exist uniquely determined sections $Y_n \in \Gamma(S, \mathfrak{g})$ such that

$$\mathrm{Ad}(\exp_\alpha(xX_\alpha))(X_\beta) = \sum_{n \geq 0} x^n Y_n.$$

Applying the inner automorphism defined by a section t of T , one finds at once

$$\mathrm{Ad}(t)(Y_n) = \beta(t)\alpha(t)^n Y_n,$$

which implies $Y_n \in \Gamma(S, \mathfrak{g}^{\beta+n\alpha})$. Since α and β are not proportional, none of the $\beta + n\alpha$ is zero; one therefore has $Y_n = 0$ for $n > q$, $Y_n = M_{\alpha, \beta, n} X_{\beta+n\alpha}$ for $0 \leq n \leq q$, where $M_{\alpha, \beta, n} \in \mathbf{G}_a(S)$ is uniquely determined. Putting $x = 0$ in the formula obtained, one finds $Y_0 = X_\beta$, which completes the proof.

Remark 5.4.10. Differentiating the preceding formula at $x = 0$, one finds

$$[X_\alpha, X_\beta] = \begin{cases} N_{\alpha, \beta} X_{\alpha+\beta}, & \text{where } N_{\alpha, \beta} = M_{\alpha, \beta, 1}, \quad \text{if } \alpha + \beta \in R, \\ 0 & \text{if } \alpha + \beta \notin R, \quad \alpha + \beta \neq 0. \end{cases}$$

Let us now prove 5.4.7. If R' is a closed subset of R , then $\mathfrak{g}_{R'}$ is a Lie subalgebra of $\mathfrak{g}$, by 5.4.10 and Exp. XX 2.10, formula (3). By 5.4.9 and Exp. XX 2.10, formula (2), U_α normalizes $\mathfrak{g}_{R'}$ for each $\alpha \in R'$. Choose a system of positive roots R_+ and consider the open subset Ω_{R_+} of 4.1.2; let $\Omega_{R_+, R'}$ be the closed subscheme of Ω_{R_+} defined as follows:

$$\Omega_{R_+, R'} = \left(\prod_{\alpha \in R' \cap -R_+} U_\alpha \right) \cdot T \cdot \left(\prod_{\alpha \in R' \cap R_+} U_\alpha \right).$$

The canonical immersion $\Omega_{R_+, R'} \rightarrow G$ factors through $i : \Omega_{R_+, R'} \rightarrow \underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'})$. Assume G adjoint; the tangent map to i at the points of the identity section is bijective by 5.3.2; in

⁽³⁵⁾**N.D.E.** In fact, it suffices (cf. *loc. cit.*) that 2 and 3 be nonzero on S .

particular, the morphism i is étale at every point of the identity section, hence is a local immersion⁽³⁶⁾ at every point of the identity section, hence $\underline{\text{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the identity section, as was to be proved.

5.5. Borel subgroups of a split reductive group

Proposition 5.5.1. *Let (G, T, M, R) be a split S -group. For every system of positive roots R_+ of R , H_{R_+} (which exists by 5.4.7) is a Borel subgroup of G and, for every order on R_+ , the morphism induced by the product in G*

$$T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow G$$

is a closed immersion with image H_{R_+} . One writes $B_{R_+} = H_{R_+}$.

By definition of Borel subgroups, the first assertion may be verified on replacing S by the spectrum of an algebraically closed field. Let then B be the Borel subgroup of G containing T and corresponding to the system of positive roots R_+ (Bible, § 10.4, prop. 9); the Lie algebra of B is $\mathfrak{g}_{R_+}$; one therefore has $B = H_{R_+}$ by 5.3.5.

Let us prove the second assertion: the morphism of the statement induces an open immersion $i : T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow H_{R_+}$ (5.4.4). Now i is surjective (Bible, § 15.1, cor. 1 to prop. 1).

Corollary 5.5.2. *Choose any order on R_+ and for each $\alpha \in R_+$ an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$. Let $\alpha, \beta \in R_+$. For each pair $(i, j) \in \mathbf{N}^* \times \mathbf{N}^*$ such that $i\alpha + j\beta \in R$, there exists a unique section*

$$C_{i,j,\alpha,\beta} \in \Gamma(S, \mathcal{O}_S)$$

such that, for all $x, y \in \mathbf{G}_a(S')$, $S' \longrightarrow S$, one has

$$\begin{aligned} & \exp_\alpha(xX_\alpha) \exp_\beta(yX_\beta) \exp_\alpha(xX_\alpha)^{-1} = \\ & \exp_\beta(yX_\beta) \prod_{\substack{i,j \in \mathbf{N}^* \\ i\alpha + j\beta \in R}} \exp_{i\alpha + j\beta}(C_{i,j,\alpha,\beta} x^i y^j X_{i\alpha + j\beta}). \end{aligned}$$

If $\alpha = \beta$, the assertion is trivial. Assume therefore $\alpha \neq \beta$; then, by the proposition, there exist unique morphisms

$$F_0 : \mathbf{G}_{a,S}^2 \longrightarrow T, \quad F_\gamma : \mathbf{G}_{a,S}^2 \longrightarrow \mathbf{G}_{a,S} \quad (\gamma \in R_+)$$

such that one has

$$\exp(xX_\alpha) \exp(yX_\beta) \exp(xX_\alpha)^{-1} = F_0(x, y) \prod_{\gamma \in R_+} \exp(F_\gamma(x, y)X_\gamma).$$

Let $t \in T(S')$, $S' \longrightarrow S$. Apply $\text{int}(t)$ to this formula; one has at once the relations

$$(1) \quad F_0(\alpha(t)x, \beta(t)y) = F_0(x, y),$$

$$(2) \quad F_\gamma(\alpha(t)x, \beta(t)y) = \gamma(t)F_\gamma(x, y).$$

Since α and β are two linearly independent characters (over $\mathbf{Q}$) of T , one concludes as usual from the first relation that F_0 is constant, hence $F_0(x, y) = e$. Write next

$$F_\gamma(x, y) = \sum a_{ij} x^i y^j, \quad \text{with } a_{ij} \in \Gamma(S, \mathcal{O}_S).$$

Substituting in relation (2) and identifying the polynomials on both sides, one finds

$$a_{ij}(\alpha(t)^i \beta(t)^j - \gamma(t)) = 0.$$

⁽³⁶⁾**N.D.E.** By EGA IV₄, 17.11.2, i is étale at every point of the identity section (and $\underline{\text{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the identity section). Moreover, let V be the largest open subset of $\Omega_{R_+, R'}$ on which i is étale; since i is a monomorphism, i_V is an open immersion (ibid., 17.9.1).

If $\gamma \neq i\alpha + j\beta$, one knows (Exp. XIX 4.13) that there exist a faithfully flat quasi-compact $S' \rightarrow S$ and a $t \in T(S')$ such that $\alpha(t)^i \beta(t)^j - \gamma(t) = 1$. One therefore has $a_{ij} = 0$ on S' , hence on S . If $\gamma = i\alpha + j\beta$, one puts $a_{ij} = C_{i,j,\alpha,\beta}$. Putting $x = 0$ (resp. $y = 0$), one finds $C_{0,1,\alpha,\beta} = 1$ (resp. $C_{1,0,\alpha,\beta} = 0$).

Remark 5.5.3. Differentiating at $y = 0$ and comparing with 5.4.9, one finds

$$C_{i,1,\alpha,\beta} = M_{\alpha,\beta,i}.$$

Corollary 5.5.4. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , $\alpha \neq \beta$ two roots of G with respect to T such that $\alpha + \beta$ is nontrivial on each fiber. Order the set of $i\alpha + j\beta$ ($i, j \in \mathbf{N}^*$) in any way. For all $i, j \in \mathbf{N}^*$ such that $i\alpha + j\beta \in R$, there exists a unique morphism of $\mathcal{O}_S$ -modules*

$$f_{\alpha,\beta,i,j} : (\mathfrak{g}^\alpha)^{\otimes i} \otimes (\mathfrak{g}^\beta)^{\otimes j} \longrightarrow \mathfrak{g}^{i\alpha+j\beta}$$

such that for every $S' \rightarrow S$ and all $X \in W(\mathfrak{g}^\alpha)(S')$, $Y \in W(\mathfrak{g}^\beta)(S')$ one has (the exp on the right-hand side being taken in the sense of 1.2⁽³⁷⁾):

$$\begin{aligned} \exp_\alpha(X) \exp_\beta(Y) \exp_\alpha(-X) = \\ \exp_\beta(Y) \prod_{(i,j)} \exp_{i\alpha+j\beta}(f_{\alpha,\beta,i,j}(X^i \otimes Y^j)). \end{aligned}$$

The assertion is local for (fpqc). By § 2, one may therefore assume G split relative to T , α and β constant in the splitting. Since $\alpha + \beta \neq 0$, there exists a system of positive roots R_+ containing α and β (Exp. XXI 3.5.4) and one is reduced to 5.5.2.

Corollary 5.5.5. *Let S be a scheme, G a reductive S -group.*

- (i) *G possesses Borel subgroups locally for the étale topology. If T is a maximal torus of G , then G also possesses Borel subgroups containing T locally for the étale topology.*
- (ii) *If T is a maximal torus of G , the “functor of Borel subgroups of G containing T ” is representable by a principal homogeneous bundle under $W_G(T)$.*
- (iii) *If (G, T, M, R) is split, every Borel subgroup B of G containing T is locally over S of the form B_{R_+} , where R_+ is a system of positive roots of R .*
- (iv) *If $T \subset B$ is a Killing pair of G , there exists a covering family $\{S_i \rightarrow S\}$ for the étale topology, and for each i a splitting $(G_{S_i}, T_{S_i}, M_i, R_i)$ and a system of positive roots R_{i+} of R_i such that $B_{S_i} = B_{R_{i+}}$.*

Indeed, (i) follows from 2.3 and 5.5.1, (ii) from (i) and 5.3.15, (iii) from (ii) and 5.5.1, (iv) from (iii) and 2.3.

Lemma 5.5.6. *Choose on the group $\Gamma_0(R)$ generated by the roots a totally ordered group structure such that the roots > 0 are the elements of R_+ (cf. Exp. XXI 3.5.6)⁽³⁸⁾. Let $\alpha_1 < \dots < \alpha_N$ be the elements of R_+ . Consider the isomorphism*

$$f : T \times_S U_{\alpha_1} \times_S \dots \times_S U_{\alpha_N} \longrightarrow B_{R_+}$$

induced by the product in G . Put for $i = 1, \dots, N$,

$$U_{\geq i} = f(U_{\alpha_i} \times_S \dots \times_S U_{\alpha_N}).$$

- (i) *Each $U_{\geq i}$ is an invariant subgroup of B_{R_+} .*
- (ii) *For $1 \leq i \leq N-1$, $U_{\geq i}$ is identified with the semidirect product*

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1}.$$

⁽³⁷⁾**N.D.E.** That is, if $\mathfrak{g}^{i\alpha+j\beta} = 0$ on a connected component of S , the corresponding exponential is 1 on this component.

⁽³⁸⁾**N.D.E.** Note that such an order is necessarily compatible with ord_Δ , where $\Delta = \mathcal{S}(R_+)$ (cf. XXI 3.2.15).

(iii) B_{R_+} is identified with the semidirect product

$$B_{R_+} = T \cdot U_{\geq 1}.$$

(iv) For $1 \leq i \leq N-1$, the inner automorphisms of $U_{\geq 1}$ act trivially on $U_{\geq i}/U_{\geq i+1}$ (which is identified with U_{α_i} by (ii)).

Let us first prove by induction on i the following assertion:

$U_{\geq i}$ is an invariant subgroup of B_{R_+} , a semidirect product of U_{α_i} and $U_{\geq i+1}$.

The assertion is true for $i = N$; assume it true for $i+1$ and let us prove it for i . One has (as schemes)

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1};$$

it is first clear that $U_{\geq i}$ is stable under the inner automorphisms of B_{R_+} . This is clear for $\text{int}(t)$, $t \in T(S)$; it suffices to verify it for $\text{int}(x)$, $x \in U_{\alpha}(S)$, $\alpha \in R_+$. Now $U_{\geq i+1}$ is assumed invariant, so it suffices to see that $\text{int}(x)U_{\alpha_i} \subset U_{\geq i}$. By 5.5.2, if $y \in U_{\alpha_i}(S')$, one has $y^{-1}xyx^{-1} \in U_{\geq i+1}(S')$, which implies $\text{int}(x)(y) \in U_{\geq i}(S')$.

Let us now prove that $U_{\geq i}$ is a subgroup of B_{R_+} . If $x, y \in U_{\geq i}(S)$, one may write $x = px'$, $y = qy'$, with $p, q \in U_{\alpha_i}(S)$, and $x', y' \in U_{\geq i+1}(S)$. One has

$$xy = px'qy' = pq(q^{-1}x'q)y' \in U_{\alpha_i}(S')U_{\geq i+1}(S');$$

likewise $x^{-1} = p^{-1}(px'^{-1}p^{-1}) \in U_{\alpha_i}(S')U_{\geq i+1}(S')$. We have therefore proved (i) and (ii), as well as (iv) along the way. As for (iii), it is a trivial consequence of 5.5.1.

Lemma 5.5.7. *With the preceding notations, choose for each $1 \leq i \leq N$ an $X_i \in \Gamma(S, \mathfrak{g}^{\alpha_i})^\times$ and consider the isomorphism*

$$a : \mathbf{G}_{a,S}^N \longrightarrow U_{\geq 1}$$

defined set-theoretically by

$$a(x_1, \dots, x_N) = \exp_{\alpha_1}(x_1 X_1) \cdots \exp_{\alpha_N}(x_N X_N).$$

There exists a unique family (Q_i) , $i = 1, \dots, N$, of polynomials

$$Q_i = Q_i(x_1, \dots, x_N, y_1, \dots, y_N)$$

with coefficients in $\Gamma(S, \mathcal{O}_S)$ such that one has set-theoretically

$$a(x_1, \dots, x_N) a(y_1, \dots, y_N) = a(Q_1(x_1, \dots, y_N), \dots, Q_N(x_1, \dots, y_N)).$$

Moreover, the Q_i have coefficients in the subring of $\Gamma(S, \mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$ of 5.5.2 ($\alpha, \beta \in R_+$, $i, j \in \mathbf{N}^*$) and each Q_i has the form

$$Q_i(x_1, \dots, y_N) = x_i + y_i + Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}).$$

The existence and uniqueness of the Q_i follow at once from the fact that a is an isomorphism of schemes. Denoting by z, z', z'' sections of $U_{\geq i+1}$, one has

$$\begin{aligned} a(x_1, \dots, x_N) a(y_1, \dots, y_N) &= a(x_1, \dots, x_{i-1}, 0, \dots, 0) \exp(x_i X_i) z \\ &\quad \cdot a(y_1, \dots, y_{i-1}, 0, \dots, 0) \exp(y_i X_i) z'; \end{aligned}$$

using 5.5.6 (i) and (iv), the right-hand term is written

$$\begin{aligned} &a(x_1, \dots, x_{i-1}, 0, \dots, 0) a(y_1, \dots, y_{i-1}, 0, \dots, 0) \\ &\quad \cdot \exp((x_i + y_i) X_i) z''; \end{aligned}$$

which gives, using 5.5.6 again,

$$Q_i(x_1, \dots, x_N, y_1, \dots, y_N) = x_i + y_i + Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}),$$

with

$$\begin{aligned} &Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}) \\ &= Q_i(x_i, \dots, x_{i-1}, 0, \dots, 0, y_1, \dots, y_{i-1}, \dots, 0); \end{aligned}$$

that is, the precise form requested.

Let us finally prove the assertion on the coefficients of the polynomials Q_i . Let A be the subring of $\Gamma(S, \mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$ ($\alpha, \beta \in R_+$, $i, j \in \mathbf{N}^*$). Let us prove by descending induction on i that if $x_1 = \cdots = x_{i-1} = 0$ and $y_1 = \cdots = y_{i-1} = 0$, that is, if $a(x_1, \dots, x_N)$ and $a(y_1, \dots, y_N)$ are sections of $U_{\geq i}$, then the polynomials

$$R_j(x_i, \dots, x_N, y_i, \dots, y_N) = Q_j(x_1, \dots, x_N, y_1, \dots, y_N),$$

have coefficients in A . This is trivial for $i = N$ and also for $j < i$ (for $R_j = 0$ for $j < i$). Let $i < N$, assume the assertion holds for $i + 1$ and let us prove it for i (and $j \geq i$). One has

$$\begin{aligned} a(0, \dots, 0, x_i, \dots, x_N) &= \exp(x_i X_i) a(0, \dots, 0, x_{i+1}, \dots, x_N) \\ &= \exp(x_i X_i) Z_i. \end{aligned}$$

Write likewise

$$a(0, \dots, y_i, \dots, y_N) = \exp(y_i X_i) T_i.$$

One has

$$\begin{aligned} a(0, \dots, x_i, \dots, x_N) a(0, \dots, y_i, \dots, y_N) \\ = \exp((x_i + y_i) X_i) \text{int}(\exp(-y_i X_i))(Z_i) \cdot T_i. \end{aligned}$$

Now

$$\begin{aligned} &\text{int}(\exp(-y_i X_i))(Z_i) \\ &= \text{int}(\exp(-y_i X_i))(\exp(x_{i+1} X_{i+1}) \cdots \exp(x_N X_N)) \end{aligned}$$

is a product of $N - i - 1$ sections of $U_{\geq i+1}$ whose coefficients in the decomposition $U_{\geq i+1} = U_{\alpha_{i+1}} \cdots U_{\alpha_N}$ are polynomials in y_i and $x_{i+1}, \dots, x_N$ with coefficients in A (by 5.5.2). Applying the induction hypothesis, one deduces that the coefficients of

$$\text{int}(\exp(-y_i X_i))(Z_i) \cdot T_i$$

are also polynomials with coefficients in A , which completes the proof.

Note that the preceding induction gives at once a proof of the

Lemma 5.5.8. *With the notations of 5.5.6, let for each $i = 1, \dots, N$ a morphism of groups*

$$f_i : U_{\alpha_i} \longrightarrow H,$$

where H is an S -group functor. For the morphism

$$f : U_{\geq 1} \longrightarrow H$$

defined by

$$f(\exp(x_1 X_1) \cdots \exp(x_N X_N)) = f_1(\exp(x_1 X_1)) \cdots f_N(\exp(x_N X_N))$$

to be a morphism of groups, it is necessary and sufficient that for every pair $i < j$, one have

$$\begin{aligned} f_j(\exp(x_j X_j)) f_i(\exp(x_i X_i)) f_j(\exp(-x_j X_j)) = \\ f(\exp(x_j X_j) \exp(x_i X_i) \exp(-x_j X_j)). \end{aligned}$$

5.6. Subgroups of type (R) with solvable fibers

Proposition 5.6.1. *Let (G, T, M, R) be a split S -group, R' a subset of R of type (R) (5.4.2), $H_{R'}$ the corresponding subgroup of G . The following conditions are equivalent:*

- (i) $H_{R'}$ has solvable geometric fibers.
- (ii) There exists a system of positive roots R_+ such that $R' \subset R_+$, hence $H_{R'} \subset B_{R_+}$ (cf. 5.4.5).
- (iii) $R' \cap -R' = \emptyset$.
- (iv) For every order on R' , the morphism induced by the product in G

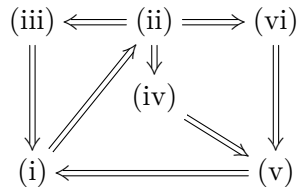
$$T \times_S \prod_{\alpha \in R'} U_{\alpha} \longrightarrow H_{R'}$$

is an isomorphism.

- (v) $H_{R'} \cap \underline{\text{Norm}}_G(T) = T$.
 (vi) For every subset R'' of R , of type (R), one has (cf. 5.4.5)

$$H_{R'} \cap \underline{\text{Norm}}_G(H_{R''}) = H_{R' \cap R''}.$$

We are going to prove these equivalences according to the logical diagram



One clearly has $(ii) \Rightarrow (iii)$ and $(vi) \Rightarrow (v)$ (take $R'' = \emptyset$). By 5.4.6, it suffices to verify $(i) \Rightarrow (ii)$ on the geometric fibers; now if S is the spectrum of an algebraically closed field, $H_{R'}$ is contained in a Borel group containing T , hence of the form H_{R_+} (5.5.5 (iii)).

Likewise $(iii) \Rightarrow (i)$ is verified on the geometric fibers; assume (iii) satisfied; if $H_{R'}$ were not solvable, there would exist a subtorus Q of T , of codimension 1 in T , such that $\underline{\text{Centr}}_{H_{R'}}(Q)$ is not solvable (*Bible*, § 10.4, prop. 8); now $\underline{\text{Centr}}_{H_{R'}}(Q)$ has as Lie algebra $\mathfrak{g}_{R''}$, where R'' is the set of roots of R' vanishing on Q , hence $R'' = \emptyset$ or $\{\alpha\}$ (by (iii)); hence $\underline{\text{Centr}}_{H_{R'}}(Q)$, which is a subgroup of type (R) of G , is T or $T \cdot U_\alpha$, hence solvable contrary to the hypothesis.

Likewise $(ii) \Rightarrow (iv)$ is verified on the geometric fibers (for these are flat S -schemes of finite presentation); by *Bible*, § 13.2, th. 1 d), the morphism in question is bijective; it induces an isomorphism on the tangent spaces at the origin, and one concludes as usual (cf. 4.1.1).

One has $(iv) \Rightarrow (v)$ by 4.2.7. To prove $(v) \Rightarrow (i)$, one is again reduced to the case where S is the spectrum of an algebraically closed field and one concludes by *Bible*, § 10.3, cor. to prop. 6 and § 9.3, cor. 3 to th. 1. It therefore only remains to prove assertion $(ii) \Rightarrow (vi)$. One may reduce to the case where G is adjoint. One then has, by 5.3.3

$$\underline{\text{Norm}}_G(H_{R''}) = \underline{\text{Norm}}_G(\mathfrak{g}_{R''}) \subset \underline{\text{Transp}}_G(\mathfrak{t}, \mathfrak{g}_{R''}).$$

⁽³⁹⁾By 5.4.5, it suffices to prove

$$(x) \quad H_{R'}(S) \cap \underline{\text{Transp}}_{G(S)}(\mathfrak{t}, \mathfrak{g}_{R''}) \subset H_{R' \cap R''}(S).$$

Let us first prove a lemma.

Lemma 5.6.2. *In the notations of 5.5.7, let*

$$u = \exp(x_1 X_1) \cdots \exp(x_N X_N)$$

where $x_i \in \mathbf{G}_a(S)$. Let m be an integer, $1 \leq m \leq N$, such that $x_i = 0$ for $i < m$.

a) If $H \in \Gamma(S, \mathfrak{t})$, the component of $\text{Ad}(u)H$ on $\mathfrak{g}^{\alpha_m}$ is

$$-\bar{\alpha}_m(H)x_m X_m.$$

b) If $Y \in \Gamma(S, \mathfrak{g}^{-\alpha_m})$, the component of $\text{Ad}(u)Y$ on $\mathfrak{t}$ is (with the notations of Exp. XX 2.6)

$$x_m \langle X_m, Y \rangle H_{\alpha_m}.$$

Denote indeed by $\mathfrak{u} = \mathfrak{g}^{\alpha_{m+1}} + \cdots + \mathfrak{g}^{\alpha_N}$. By 5.4.9, one has

$$\text{Ad}(\exp(x_i X_i))\mathfrak{u} \subset \mathfrak{u}, \quad \text{for } i > m.$$

By Exp. XX 2.10, one has

$$\text{Ad}(\exp(x_i X_i))H = H - \bar{\alpha}_i(H)x_i X_i.$$

This gives at once, modulo $\mathfrak{u}$,

$$\text{Ad}(u)H = \text{Ad}(\exp(x_m X_m))H = H - \bar{\alpha}_m(H)x_m X_m,$$

which implies the first result.

⁽³⁹⁾**N.D.E.** The inclusion follows from $\mathfrak{t} \subset \mathfrak{g}_{R''}$.

Likewise, denote⁽⁴⁰⁾ by $\mathfrak{n} = \mathfrak{g}^{\alpha_1} + \cdots + \mathfrak{g}^{\alpha_N}$ and $u_1 = \exp(x_{m+1}X_{m+1}) \cdots \exp(x_N X_N)$. For $i > m$, one has $\alpha_i > \alpha_m$, hence, by 5.4.9, one has, modulo $\mathfrak{n}$,

$$\mathrm{Ad}(u_1)Y \equiv Y \quad \text{whence} \quad \mathrm{Ad}(u)Y \equiv \mathrm{Ad}(\exp(x_m X_m))Y.$$

Applying Exp. XX 2.10, one therefore obtains, modulo $\mathfrak{n}$,

$$\mathrm{Ad}(u)Y - Y \equiv x_m \langle X_m, Y \rangle H_{\alpha_m}$$

whence the second result.

Let us return to the proof of inclusion (x). Assume there exists $h \in H_{R'}(S)$, $h \notin H_{R' \cap R''}(S)$, such that

$$\mathrm{Ad}(h)\mathfrak{t} \subset \mathfrak{g}_{R''}.$$

One may write

$$h = t \exp(x_1 X_1) \cdots \exp(x_N X_N).$$

Since $h \notin H_{R' \cap R''}(S)$, there exists a smallest m such that

$$\begin{aligned} t \exp(x_1 X_1) \cdots \exp(x_{m-1} X_{m-1}) &\in H_{R' \cap R''}(S) \\ \text{and } \alpha_m &\notin R'', \quad x_m \neq 0. \end{aligned}$$

Then $h' = \exp(x_m X_m) \cdots \exp(x_N X_N)$ also satisfies the conditions imposed on h above. But by 5.6.2, for every $H \in \Gamma(S, \mathfrak{t})$ the component of $\mathrm{Ad}(h')H$ on $\mathfrak{g}^{\alpha_m}$ is $-\bar{\alpha}_m(X)x_m X_m$. By the hypothesis on h and on m , one therefore has $\bar{\alpha}_m(H) = 0$ for every $H \in \Gamma(S, \mathfrak{t})$, which is impossible (for G is assumed adjoint and $\bar{\alpha}_m$ is therefore nonzero on each fiber).

Remark 5.6.2 bis. Let us return to the notations of 5.6.2. If $\mathrm{Ad}(u)$ is the identity on $\mathfrak{t}$ and on $\mathfrak{g}^{-\alpha_m}$, one has $x_m = 0$. Indeed, one has $x_m \bar{\alpha}_m = 0$ and $x_m H_{\alpha_m} = 0$; if $\alpha_m \notin 2M$, then $\bar{\alpha}_m$ is nonzero on each fiber; if $\alpha_m \in 2M$, then $\alpha_m^* \notin 2M^*$ and H_{α_m} is nonzero on each fiber; in each case, this implies $x_m = 0$. It follows that $u = e$ if $\mathrm{Ad}(u)$ acts trivially on $\mathfrak{g}$.

Remark 5.6.3. If H is a subgroup of type (R) of the reductive group G , with solvable geometric fibers, then H is closed in G and $\mathrm{Norm}_G(H)/H$ is representable by a separated finite étale S -scheme.

This follows from 5.3.18 and, as one chooses, 3.5 or Exp. XIX 2.5 (ii).

Corollary 5.6.4. *Let (G, T, M, R) be a split reductive group. If $R' \subset R$ is closed and if $R' \cap -R' = \emptyset$, then R' is contained in a system of positive roots.⁽⁴¹⁾*

Indeed, R' is of type (R) by 5.4.7, hence the result follows from 5.6.1.

Corollary 5.6.5. *Under the conditions of 5.6.1, the product in G induces an isomorphism*

$$\prod_{\alpha \in R'} U_{\alpha} \xrightarrow{\sim} U_{R'},$$

where $U_{R'}$ is a closed group subscheme of G , smooth over S , with connected unipotent geometric fibers, independent of the choice of the order on R' . Moreover, $H_{R'}$ is the semidirect product $T \cdot U_{R'}$ ($U_{R'}$ invariant).

Indeed, if $R' \subset R_+$, then $H_{R'} \cap U_{\geq 1}$ (notations of 5.5.6) is a closed subgroup of G of finite presentation, invariant in $H_{R'}$. By 5.6.1 (iv), one has $H_{R'} = T \cdot U_{R'}$, which implies the other assertions.

Remark 5.6.6. In particular, U_{R_+} is the group $U_{\geq 1}$ of 5.5.6.

Let us extract some corollaries from the preceding results concerning groups of the type $U_{R'}$.

⁽⁴⁰⁾**N.D.E.** The original has been corrected in what follows.

⁽⁴¹⁾**N.D.E.** This is in fact a statement on root systems, supplementing exposé XXI (cf. [BLie], VI § 1.7, Prop. 22), and proved here by an indirect route.

Corollary 5.6.7. *Let (G, T, M, R) be a split reductive group, R' and R'' two subsets of R of type (R) , with $R' \cap -R' = \emptyset$.*

(i) *One has*

$$U_{R'} \cap \underline{\text{Norm}}_G(H_{R''}) = U_{R' \cap R''}.$$

(ii) *Assume R' closed. If the conditions $\alpha \in R'$, $\beta \in R''$ and $\alpha + \beta \in R$ imply $\alpha + \beta \in R'$, then $H_{R''}$ normalizes $U_{R'}$.*

Indeed, (i) follows at once from 5.6.5 and 5.6.1 (vi). To prove (ii), it suffices, in view of 5.4.4, to show that T and each U_β , $\beta \in R''$, normalize $U_{R'}$.⁽⁴²⁾ For T , this is trivial; for U_β , this follows from 5.5.2 and Exp. XXI 3.1.2.⁽⁴³⁾

Corollary 5.6.8. *Let (G, T, M, R) be a split S -group, R_+ a system of positive roots, α a simple root of R_+ (i.e. an element of R_+ such that $R_+ - \{\alpha\}$ is closed). Write*

$$U_{\hat{\alpha}} = U_{R_+ - \{\alpha\}}.$$

Then

- (i) $U_{\hat{\alpha}}$ *is an invariant subgroup of B_{R_+} .*
- (ii) U_{R_+} *is the semidirect product of $U_{\hat{\alpha}}$ by U_α .*
- (iii) $U_{-\alpha}$ *normalizes $U_{\hat{\alpha}}$.*
- (iv) Z_α *normalizes $U_{\hat{\alpha}}$.*

If one similarly defines $U_{-\hat{\alpha}} = U_{R_- - \{-\alpha\}}$ (where $R_- = -R_+$), one has

$$\Omega_{R_+} = U_{-\hat{\alpha}} \cdot U_{-\alpha} \cdot T \cdot U_\alpha \cdot U_{\hat{\alpha}}.$$

Indeed, (ii) follows from 5.6.5, and (i) from 5.6.7 (ii). Likewise, (iii) follows from 5.5.2 (indeed, if $\beta \in R_+$, $\beta \neq \alpha$, no combination $i(-\alpha) + j\beta$, with $i, j > 0$, can be negative, for β contains at least one simple root $\neq \alpha$). Then (iv) follows from (i) and (iii), for $U_{-\alpha} \cdot T \cdot U_\alpha$ is schematically dense in Z_α . Finally, the last assertion follows from (ii) and its analogue for U_{R_-} .

Let us return to the general situation.

Proposition 5.6.9. *Let S be a scheme, G a reductive S -group, H a subgroup of type (R) , with solvable geometric fibers.*

- (i) $D_S(H) = \underline{\text{Hom}}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})$ *is representable by a twisted constant S -group, whose type at $s \in S$ is $\mathbf{Z}^{\text{rgred}(G_s)}$. The biduality morphism (Exp. VIII § 1)*

$$f : H \longrightarrow D_S(D_S(H))$$

is smooth and surjective.

- (ii) *The kernel H^u of f is the largest invariant closed group subscheme of H , smooth over S , with connected unipotent geometric fibers. One says that it is the unipotent part of H and one also writes $H^u = \text{rad}^u(H)$.*

Then H^u is also the commutator sheaf of H : every morphism of groups from H into an S -presheaf of commutative groups, separated for (fppf), vanishes on H^u and therefore factors through $H/H^u = D_S(D_S(H))$.

- (iii) *If T is a maximal torus of H , the morphism $T \longrightarrow H$ induces isomorphisms $D_S(H) \xrightarrow{\sim} D_S(T)$ and $T \xrightarrow{\sim} D_S(D_S(H))$. Moreover, H is identified with the semidirect product of H^u by T .*

⁽⁴²⁾**N.D.E.** Indeed, G has connected fibers, hence is separated over S by VIB 5.5, so by XI 6.11 (see also addition 6.5.5 in VIB), $\underline{\text{Norm}}_G(U_{R'})$ is represented by a closed group subscheme N of G . If N contains T and the U_β , for $\beta \in R''$, it contains the big cell $\Omega_{R_+, R''}$; this is schematically dense in $H_{R''}$ by 5.4.4 and EGA IV₃, 11.10.10 (the fibers of $H_{R''}$ are integral and $\Omega_{R_+, R''}$ contains a nonempty open subset of each fiber). It follows that $H_{R''} \subset N$, hence $H_{R''}$ normalizes $U_{R'}$.

⁽⁴³⁾**N.D.E.** One must see that, under the hypotheses of (ii), if $\alpha \in R'$ and $\beta \in R''$, then all roots of the form $i\alpha + j\beta$ with $i, j \in \mathbf{N}^*$ belong to R' , and for this the reference XXI 2.3.5 has been replaced by XXI 3.1.2. This may also be seen directly by inspection of the root systems of rank 2.

(iv) In the situation of 5.6.1, if $H = H_{R'}$, then $H^u = U_{R'}$.

The assertions of the proposition are local for the étale topology (Exp. X 5.5). One may therefore reduce to the case of 5.6.1. One then knows (5.6.5) that $H_{R'}$ is the semidirect product of $U_{R'}$ by T . Let us show that $U_{R'}$ is the commutator sheaf of $H_{R'}$: since $H_{R'}/U_{R'} = T$ is commutative, it suffices to prove that every morphism of groups $\varphi : H_{R'} \rightarrow V$ as in (ii) vanishes on $U_{R'}$. It suffices to prove that φ vanishes on each U_α , $\alpha \in R'$. Now if $t \in T(S')$, $X \in \mathbf{W}(\mathfrak{g}^\alpha)(S')$, one has

$$1 = \varphi(t \exp_\alpha(X) t^{-1} \exp_\alpha(-X)) = \varphi(\exp_\alpha((\alpha(t) - 1)X)).$$

Since $\alpha : T \rightarrow \mathbf{G}_{m,S}$ is faithfully flat, one deduces at once that φ vanishes on $U_\alpha^\times$; but every section of U_α is locally a sum of two sections of $U_\alpha^\times$. One therefore has

$$\mathrm{Hom}_{S\text{-gr.}}(H, V) = \mathrm{Hom}_{S\text{-gr.}}(H/U_{R'}, V)$$

for every V as above. Applying this result to $V = \mathbf{G}_{m,S}$, one deduces at once (i) and (iii), then (iv) and the second assertion of (ii). It now suffices to prove the first assertion of (ii); the only nontrivial fact is that every invariant closed subgroup U of H , smooth over S with connected unipotent geometric fibers, is a subgroup of H^u . Now one first has:

Lemma 5.6.10. *Let G be a reductive S -group, T a maximal torus, U a group subscheme of G , smooth over S , with unipotent geometric fibers, normalized by T . Then $U \cap T = e$.*

Indeed, since $T = \mathrm{Centr}_G(T)$, one has $U \cap T = U^T$ (invariants under $\mathrm{int}(T)$). Applying Exp. XIX 1.4, one deduces that $U \cap T$ is smooth over S , but it is also radicial over S : for every $s \in S$, $U(\bar{s}) \cap T(\bar{s})$ consists of elements which are both unipotent and semisimple. This proves the lemma.

Let us return to the proof of 5.6.9 (ii). If U is an invariant subgroup of H as above, then the semidirect product $T \cdot U$ is a subgroup of type (R) of G , with solvable geometric fibers. One may therefore assume it of the form $H_{R''}$, with $R'' \subset R'$. It suffices to prove $U = U_{R''}$ and one is therefore reduced to the case where $H = T \cdot U$; but since the quotient H/U is commutative, U is a subsheaf of the commutator sheaf of H , which is H^u . Q.E.D.

Note that we have in fact just proved:

Proposition 5.6.11. *Let S be a scheme, G a reductive S -group, T a maximal torus of G . The maps*

$$H \mapsto H^u, \quad U \mapsto T \cdot U$$

are mutually inverse bijections between the set of subgroups H of type (R) of G , containing T and with solvable geometric fibers, and the set of subgroups U of G , smooth over S , normalized by T , with connected unipotent geometric fibers.⁽⁴⁴⁾

In particular, when (G, T, M, R) is split, the groups $H_{R'}$ and $U_{R'}$ correspond to one another.

Corollary 5.6.12. *Let S be a scheme, (G, T, M, R) a split S -group (resp. and R_+ a system of positive roots of R defining the Borel subgroup B).*

Every smooth group subscheme of G , with connected unipotent geometric fibers (resp. every smooth group subscheme of B^u), normalized by T , is locally over S of the form $U_{R'}$, where R' is a subset of R contained in a system of positive roots (resp. a subset of R_+) of type (R).

For the “resp.” case, it suffices to note that the geometric fibers of the given group are unipotent and connected by *Bible*, § 13.2, th. 1 (d).

⁽⁴⁴⁾**N.D.E.** The hypothesis that U is closed has been deleted, since it holds automatically. Indeed, for such a U , one has $U \cap T = e$ by 5.6.10, hence the semidirect product $H = T \cdot U$ is a subgroup of type (R) of G (cf. 5.2.1), with solvable geometric fibers. Hence, by 5.6.3, H is closed in G , and since U is closed in H , it is closed in G .

Proposition 5.6.9 has moreover the following corollary:

Corollary 5.6.13. *Let S be a scheme, G a reductive S -group, H a subgroup of type (R) with solvable geometric fibers, $\underline{\mathrm{Tor}}(H)$ the functor of maximal tori of H :*

$$\underline{\mathrm{Tor}}(H)(S') = \{\text{maximal tori of } H_{S'}\}.$$

Then $\underline{\mathrm{Tor}}(H)$ is representable by an affine smooth S -scheme, which is a principal homogeneous bundle under H^u for the law $(h, T) \mapsto \mathrm{int}(h)T$.⁽⁴⁵⁾

Indeed, if T and T' are two maximal tori of $H_{S'}$, there exists a unique section $h \in H^u(S')$ such that $\mathrm{int}(h)T = T'$. The uniqueness of h follows at once from the equality

$$\underline{\mathrm{Norm}}_G(T) \cap H^u = e$$

(cf. for example 5.6.1); it therefore suffices to prove the existence of h locally for the étale topology. By 5.2.6 and 5.1.2 (a), one may assume T and T' conjugate by a section of H , whence the desired conclusion since $H = H^u \cdot T$ by 5.6.9 (iii). It follows that $\underline{\mathrm{Tor}}(H)$ is a principal homogeneous sheaf under H^u , which is affine and smooth over S , which implies the statement at once⁽⁴⁶⁾.

5.7. Bruhat's theorem

Recall 5.7.1. Let k be an algebraically closed field, G a reductive k -group, B a Borel subgroup of G , T a maximal torus of B , $N = \underline{\mathrm{Norm}}_G(T)$. Then

$$G(k) = B(k)N(k)B(k);$$

this is Bruhat's theorem (*Bible*, § 13.4, cor. 1 to th. 3); more precisely, with the notations of 3.6, the sets

$$B(k)N_w(k)B(k) = B^u(k)N_w(k)B^u(k)$$

form, as w ranges over $(N/T)(k)$, a partition of $G(k)$. If B' is another Borel subgroup of G containing T , the sets $B'(k)N_w(k)B(k)$ also form a partition of $G(k)$. Indeed, if $y \in N(k)$ is such that $\mathrm{int}(y)B = B'$, one has

$$yB(k)N_w(k)B(k) = B'(k)N_{yw}(k)B(k).$$

Definition 5.7.2. Let (G, T, M, R) be a split S -group, R_- a system of positive roots of R , $B' = B_{R_-}$ the Borel group it defines. For $w \in W$, one writes (cf. 5.6.5):

$$R_-^w = R_- \cap w(R_-), \quad B'_w{}^u = U_{R_-^w} = \prod_{\alpha \in R_-^w} U_\alpha.$$

If $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ is a representative of w (3.8), one may also write

$$B'_w{}^u = B'^u \cap \mathrm{int}(n_w)B'^u.$$

Lemma 5.7.3. *Let (G, T, M, R) be a split S -group, R_+ a system of positive roots of R , $R_- = -R_+$, B (resp. B') the Borel subgroup of G defined by R_+ (resp. R_-). Let $w \in W$, N_w and $B'_w{}^u$ be the corresponding subschemes of G (3.8 and 5.7.2).*

(i) *The sheaf $B' \cdot N_w \cdot B$, image of the morphism*

$$B' \times_S N_w \times_S B \longrightarrow G$$

induced by the product in G , is representable by a subscheme of G (and in fact a closed subscheme of the open subset $n_w\Omega_{R_+}$).

⁽⁴⁵⁾**N.D.E.** This proves again and sharpens XII 1.10 (for G reductive).

⁽⁴⁶⁾**N.D.E.** By SGA 1, VIII 2.1 and EGA IV₄, 17.7.1.

(ii) *The morphism*

$$B'_w \times_S N_w \times_S B^u \longrightarrow G,$$

induced by the product in G , is an immersion with image the preceding subscheme.

Let us first show that the morphism of (ii) is an immersion. By definition, $\text{int}(n_w)^{-1}$ induces a closed immersion of B'^u_w into B'^u , hence the morphism

$$(u, b) \longmapsto n_w^{-1} u n_w b$$

induces a closed immersion

$$B'^u_w \times_S B \longrightarrow \Omega_{R_+}.$$

This implies immediately that the morphism of (ii) induces a closed immersion of the first side into the open subset $n_w \Omega_{R_+}$. To prove (i), it suffices to see that

$$B'(S)N_w(S)B(S) = B'^u_w(S)N_w(S)B^u(S).$$

Now, if $\alpha \in R$, one has $\text{int}(n_w)U_\alpha(S) = U_{w(\alpha)}(S)$, hence if $w^{-1}(\alpha) \in R_+$,

$$\begin{aligned} U_\alpha(S)N_w(S)B(S) &= U_\alpha(S)n_w T(S)B^u(S) \\ &= n_w U_{w^{-1}(\alpha)}(S)T(S)B^u(S) \\ &= n_w B(S) = N_w(S)B^u(S). \end{aligned}$$

This implies at once, in view of the definition of R_-^w , the desired assertion.

Theorem 5.7.4. *Let S be a scheme, (G, T, M, R) a split S -group, B the Borel subgroup defined by the system of positive roots R_+ , B' the Borel subgroup defined by $R_- = -R_+$.*

- (i) *(Bruhat's theorem) The schemes $B'^u_w \cdot N_w \cdot B$ form, as w ranges over W , a partition of the underlying set of G .*
- (ii) *For each $w \in W$, let n_w be a representative of w in $\text{Norm}_G(T)(S)$ (3.8); then the open subsets $n_w \Omega = n_w B'^u \cdot T \cdot B^u$ form, as w ranges over W , a covering of G .*

The two assertions are indeed verified on the geometric fibers, where one concludes by 5.7.1 and 5.7.3.

Remark 5.7.5. (i) implies that if S is the spectrum of a field, $G(S)$ is the disjoint union of the $B'^u_w(S) \cdot T(S) \cdot B^u(S)$. The corresponding assertion for arbitrary S (even local or artinian) is clearly false. Note however that (ii) implies that if S is local, $G(S)$ is indeed the union of the $n_w \Omega(S)$. In fact:

Corollary 5.7.6. *Let Δ be a system of simple roots of the split group G over the local scheme S .*

- (i) *Then $G(S)$ is generated by $T(S)$ and the $U_\alpha(S)$, $\alpha \in \Delta \cup -\Delta$.*
- (ii) *If G is simply connected (4.3.3), $G(S)$ is already generated by the $U_\alpha(S)$, $\alpha \in \Delta \cup -\Delta$.*

Indeed, let H be the subgroup of $G(S)$ generated by the $U_\alpha(S)$, $\alpha \in \Delta \cup -\Delta$. Note first that H contains a representative of each s_α ($\alpha \in \Delta$) in $\text{Norm}_G(T)(S)$ (Exp. XX 3.1), hence a representative n_w of each $w \in W$.

Since every $\alpha \in R$ is written $w(\alpha_0)$ with $w \in W$, $\alpha_0 \in \Delta$, one has

$$U_\alpha(S) = \text{int}(n_w)U_{\alpha_0}(S) \subset H.$$

The subgroup generated by H and $T(S)$ therefore contains $\Omega(S)$ and is therefore all of $G(S)$, by the remark made earlier.

If now G is simply connected, let us prove that $H \supset T(S)$. By Exp. XX 2.7, H contains $\alpha^*(\mathbf{G}_m(S))$ for every $\alpha \in \Delta$, and it suffices to apply 4.3.8.

Remark 5.7.6.1. Instead of taking, for each $\alpha \in \Delta$, $U_\alpha(S)$ and $U_{-\alpha}(S)$, one may content oneself with taking $U_\alpha(S)$ and a representative w_α of the reflection s_α , or $U_\alpha(S)$ and a section of $U_{-\alpha}^\times, \dots$

Corollary 5.7.7. *If G has semisimple rank 1, choose a $u_\alpha \in U_\alpha^\times(S)$. Then Ω and $u_\alpha\Omega$ form a covering of G .*

Indeed, if $u_{-\alpha}$ is the section of $U_{-\alpha}$ paired with u_α (cf. 1.3), one has, by 5.7.4 (ii),

$$G = \Omega \cup u_{-\alpha}^{-1} u_\alpha u_{-\alpha}^{-1} \Omega,$$

whence

$$G = u_{-\alpha} G = u_{-\alpha} \Omega \cup u_\alpha u_{-\alpha}^{-1} \Omega = \Omega \cup u_\alpha \Omega.$$

Corollary 5.7.8. *Let S be a scheme, G a reductive S -group. Then G is essentially free over S (Exp. VIII 6.1).⁽⁴⁷⁾*

Indeed, the assertion is local for the (fpqc) topology; one may assume G split. Then G admits a covering by open subsets isomorphic to $\mathbf{G}_{a,S}^N \times_S \mathbf{G}_{m,S}^n$, hence essentially free.

Lemma 5.7.9. *Under the conditions of 5.7.4, let α be a simple root of R_+ and $u_\alpha \in U_\alpha^\times(S)$. For every $v \in U_{-\alpha}(S)$, one has*

$$\Omega \cdot v \subset \Omega \cup u_\alpha \cdot \Omega.$$

One must compare two open subsets of G ; it suffices to do so when S is the spectrum of a field k . One must therefore prove

$$\Omega(k)v \subset \Omega(k) \cup u_\alpha \Omega(k).$$

Now

$$\begin{aligned} \Omega(k)v &= B'^u(k)T(k)B^u(k)v \\ &= U_{-\hat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\hat{\alpha}}(k)v \\ &\subset U_{-\hat{\alpha}}(k)Z_\alpha(k)U_{\hat{\alpha}}(k)v. \end{aligned}$$

(One uses the decomposition of 5.6.8). Applying now 5.6.8 (iii) and using 5.7.7 for the group Z_α , one obtains

$$\begin{aligned} \Omega(k)v &\subset U_{-\hat{\alpha}}(k)Z_\alpha(k)vU_{\hat{\alpha}}(k) \subset U_{-\hat{\alpha}}(k)Z_\alpha(k)U_{\hat{\alpha}}(k) \\ &\subset U_{-\hat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\hat{\alpha}}(k) \\ &\quad \cup U_{-\hat{\alpha}}(k)u_\alpha U_{-\alpha}(k)T(k)U_\alpha(k)U_{\hat{\alpha}}(k). \end{aligned}$$

Using 5.6.8 (iii) again (for R_- instead of R_+), one obtains the result.

Proposition 5.7.10. *Under the conditions of 5.7.4, choose for each simple root α a $u_\alpha \in U_\alpha^\times(S)$. Let U_1 be the submonoid of $B^u(S)$ generated by the u_α . The open subsets $u\Omega$, for $u \in U_1$, form a covering of G .*

Once again, one may assume that S is the spectrum of a field k ; by 5.7.6, it suffices to prove that $\bigcup_{u \in U_1} u\Omega(k)$ is stable under right multiplication by $T(k)$, $U_\alpha(k)$, $U_{-\alpha}(k)$ (for α simple). In the first two cases, this is trivial. In the last, this follows from the lemma.

Remark 5.7.11. Let us point out a particular case of 5.7.2. If $w = s_\alpha$ is the reflection with respect to the simple root α , then

$$R_- \cap s_\alpha(R_-) = R_- - \{-\alpha\}$$

(Exp. XXI 3.3.1), and, in the notations of 5.6.8, one therefore has

$$B'^u_{s_\alpha} = U_{-\hat{\alpha}}.$$

Remark 5.7.12. In fact, the proof of 5.7.10 gives at once the following statement: under the conditions of 5.7.10, let Γ be a submonoid of $G(S)$; for the open subsets $g\Omega$ ($g \in \Gamma$) to form a covering of G , it is necessary and sufficient that for every $s \in S$ and every simple root α , one have

$$(u_\alpha)_{\bar{s}} B'^u(\bar{s}) \subset \Gamma \cdot B'^u(\bar{s}) \cdot T(\bar{s}) \cdot B^u(\bar{s}).$$

⁽⁴⁷⁾**N.D.E.** See also the additions made in VIB, 6.2.1 to 6.2.6 and 6.5.2 to 6.5.5.

Remark 5.7.13. By 5.5.5 (iii), arguing as in 5.7.1, one obtains at once the following variant of 5.7.4: let (G, T, M, R) be a split S -group, B and B' two Borel subgroups of G containing T ; for every $w \in W$, the sheaf $B' \cdot N_w \cdot B$ is representable by a subscheme of G ; these subschemes form, for $w \in W$, a partition of the underlying set of G . One may also give the analogue of 5.7.3 (ii): one must put

$$B'^u_w = B'^u \cap \text{int}(n_w) \tilde{B}^u,$$

where $\tilde{B}$ is the Borel subgroup “opposite” to B relative to T (cf. 5.9.2).

Proposition 5.7.14. *Let S be a scheme, G a reductive S -group, and*

$$\text{Ad} : G \longrightarrow \text{GL}_{\mathcal{O}_S}(\mathfrak{g})$$

its adjoint representation. Then $\text{Ker}(\text{Ad}) = \underline{\text{Centr}}(G)$ (in other words, the canonical homomorphism deduced from Ad on passing to the quotient:

$$\overline{\text{Ad}} : G/\underline{\text{Centr}}(G) = \text{ad}(G) \longrightarrow \text{GL}_{\mathcal{O}_S}(\mathfrak{g})$$

is a monomorphism).⁽⁴⁸⁾

One may assume G split. Choose on $\Gamma_0(R)$ a total order structure compatible with the group structure and let R_+ be the set of positive roots. By 5.7.4 (ii) and 4.1.6, it suffices to prove that if n_w is a representative of the element w of W , if $u \in U(S)$, $t \in T(S)$, $v \in U_-(S)$, and if $\text{Ad}(n_w v t u) = \text{id}$, then $w = e$, $v = e$, $u = e$. For each $m \in R \cup \{0\}$, put

$$\mathfrak{g}^{>m} = \coprod_{n>m} \mathfrak{g}^n, \quad \mathfrak{g}^{<m} = \coprod_{n<m} \mathfrak{g}^n.$$

Let $X \in \Gamma(S, \mathfrak{g}^m)$; write $\text{Ad}(tu)X = \text{Ad}(v^{-1}n_w^{-1})X$. Now

$$\text{Ad}(t) \text{Ad}(u)X - m(t)X \in \Gamma(S, \mathfrak{g}^{>m}),$$

$$\text{Ad}(v^{-1}n_w^{-1})X - \text{Ad}(n_w^{-1})X \in \Gamma(S, \mathfrak{g}^{<w^{-1}(m)}).$$

If $w \neq e$, there exists an $\alpha \in R$ such that $w^{-1}(\alpha) < \alpha$, and putting $m = \alpha$, one obtains a contradiction, for

$$\text{Ad}(tu)X \in \Gamma(S, \mathfrak{g}^\alpha + \mathfrak{g}^{>\alpha}) \cap \Gamma(S, \mathfrak{g}^{w^{-1}(\alpha)} + \mathfrak{g}^{<w^{-1}(\alpha)}) = 0.$$

One therefore has $w = e$, and one may choose $n_w = e$; one then has

$$\text{Ad}(v^{-1})X - X \in \Gamma(S, \mathfrak{g}^{<m} \cap (\mathfrak{g}^m + \mathfrak{g}^{>m})) = 0,$$

whence $\text{Ad}(v)X = X$ for every $X \in \Gamma(S, \mathfrak{g}^m)$, hence $\text{Ad}(v) = \text{id}$. Likewise $\text{Ad}(u) = \text{id}$. One then concludes by 5.6.2 bis.

5.8. Schemes associated with a reductive group

Theorem 5.8.1. *Let S be a scheme, G a reductive S -group. Let $\mathcal{H}$ be the functor of subgroups of type (R) of G : for every $S' \rightarrow S$, $\mathcal{H}(S')$ is the set of subgroups of type (R) of $G_{S'}$ (cf. 5.2.1). Then $\mathcal{H}$ is representable by a quasi-projective S -scheme, of finite presentation over S .*

⁽⁴⁹⁾Let $G' = G/\underline{\text{Centr}}(G)$ be the adjoint group of G (4.3.6) and u the morphism $G \rightarrow G'$. By Exp. XII 7.12, the map $H' \mapsto u^{-1}(H')$ establishes a bijection between the set of subgroups of type (R) of G' and of G (and this remains valid after every base change). Hence, replacing G by G' , one may assume that G is adjoint. Consider then the morphism

$$u : \mathcal{H} \longrightarrow \underline{\text{Grass}}(\mathfrak{g})$$

⁽⁴⁸⁾**N.D.E.** It is even a closed immersion, by Exp. XVI 1.5 (a).

⁽⁴⁹⁾**N.D.E.** The original has been expanded in what follows.

which associates with each subgroup of type (R) its Lie algebra (which is a submodule locally a direct summand of $\mathfrak{g}^{(50)}$). Then u is a monomorphism by 5.3.3. It suffices to prove that it is representable by an immersion of finite presentation, in other words to prove the following assertion: for every $S_1 \rightarrow S$, given a submodule $\mathfrak{h}$ locally a direct summand of $\mathfrak{g}_{S_1}$, the $S' \rightarrow S_1$ such that $\mathfrak{h}_{S'}$ is the Lie algebra of a subgroup of type (R) of $G_{S'}$ are exactly those which factor through a certain subscheme Σ of finite presentation of S_1 . Replacing S_1 by S , one reduces to $S_1 = S$, and one may moreover assume S affine; then there exists an affine noetherian scheme S_0 such that G (resp. $\mathfrak{h}$) comes by base change from an adjoint reductive S_0 -group G_0 (resp. a submodule $\mathfrak{h}_0$ locally a direct summand of $\mathfrak{g}_0 = \underline{\mathrm{Lie}}(G_0/S_0)$). It suffices to show that there exists a subscheme Σ_0 of S_0 having the required properties (for then one will have $\Sigma = \Sigma_0 \times_{S_0} S$). Replacing S by S_0 , one may therefore assume S affine and noetherian (note that then every subscheme of S is of finite presentation over S). Finally, replacing S by a sufficiently small open subset, one may assume that $\mathfrak{g}$ is free of rank n and that $\mathfrak{h}$ is a direct summand, free of rank r .

One must first express that $\mathfrak{h}_{S'}$ is a Lie subalgebra of $\mathfrak{g}_{S'}$, i.e. that the morphism induced by the Lie bracket:

$$\varphi : \mathfrak{h} \otimes \mathfrak{h} \xrightarrow{[\cdot, \cdot]} \mathfrak{g}$$

factors through $\mathfrak{h}$. If $(e_1, \dots, e_n)$ is a basis of $\mathfrak{g}$ such that $(e_1, \dots, e_r)$ is a basis of $\mathfrak{h}$, then φ is given by sections a_k^{ij} of $\mathcal{O}_S$ (where $i, j = 1, \dots, r$ and $k = 1, \dots, n$), and the preceding condition is equivalent to saying that $S' \rightarrow S$ factors through the closed subscheme of S defined by the equations $a_k^{ij} = 0$ for $k = r+1, \dots, n$ and $i, j = 1, \dots, r$. Replace S by this closed subscheme.

Then, by 5.3.0, $N = \underline{\mathrm{Norm}}_G(\mathfrak{h})$ is a closed group subscheme of G , of finite presentation over G . One must now express (cf. 5.3.1) that $N_{S'}$ is smooth at every point of the identity section, of relative dimension $r = \mathrm{rank}(\mathfrak{h})$, and that the inclusion of $\mathfrak{h}_{S'}$ into $\underline{\mathrm{Lie}}(N_{S'}/S')$ is an equality.

⁽⁵¹⁾Since N is affine over S (being closed in G), the identity section $\varepsilon : S \rightarrow N$ is a closed immersion, so $\varepsilon(S)$ is defined by a quasi-coherent ideal $\mathcal{J}$ of $\mathcal{A}(N)$. Note that $\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2$ is identified with $\varepsilon^*(\Omega_{N/S}^1)$, hence its formation commutes with every base change $S' \rightarrow S$. By equivalence (c') $\Leftrightarrow$ (a) in EGA IV₄, 17.12.1 (applied to $f : N \rightarrow S$ and $j = \varepsilon$), $N_{S'} \rightarrow S'$ is smooth, of relative dimension r , at every point of $\varepsilon(S')$ if and only if $\mathfrak{n}_{S'/N'} = \mathfrak{n}_{S/N} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ is locally free of rank r and the morphism $\varphi_n(S') : \mathrm{Sym}^n(\mathfrak{n}_{S'/N'}) \rightarrow \mathcal{J}_{S'}^n/\mathcal{J}_{S'}^{n+1}$ is an isomorphism for every $n \geq 1$. Put $K_n(S') = \mathrm{Ker} \varphi_n(S')$. By TDTE I, Lemma 3.6, $\mathfrak{n}_{S/N} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ is locally free of rank r if and only if $S' \rightarrow S$ factors through a certain subscheme Z of S . Replacing S by Z , one may therefore assume that $\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2$ is locally free of rank r . Then, for every $S' \rightarrow S$, one has $(\mathcal{J}^2/\mathcal{J}^3) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} = \mathcal{J}_{S'}^2/\mathcal{J}_{S'}^3$, and hence $K_2(S') = K_2(S) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$. It follows that $\varphi_2(S')$ is an isomorphism if and only if $S' \rightarrow S$ factors through the closed subscheme S_2 of S defined by the ideal generated by the image of $\mathrm{Sym}^2(\mathfrak{n}_{S/N})^* \otimes K_2(S)$ in $\mathcal{O}_S$. Then, over S_2 , $\mathcal{J}^2/\mathcal{J}^3$ is isomorphic to $\mathrm{Sym}^2(\mathfrak{n}_{S/N})$, hence locally free, and the same argument shows that $\varphi_3(S')$ is an isomorphism if and only if $S' \rightarrow S$ factors through a certain closed subscheme S_3 of S , etc. One thus obtains that $N_{S'} \rightarrow S'$ is smooth, of relative dimension r , at every point of $\varepsilon(S')$ if and only if $S' \rightarrow S$ factors through the closed subscheme Z which is the intersection of the S_n . But then, for every $S' \rightarrow Z$, $\underline{\mathrm{Lie}}(N_{S'}/S')$ is locally a direct summand of rank r of $\mathfrak{g}_{S'}$, and hence the inclusion $\mathfrak{h}_{S'} \subset \underline{\mathrm{Lie}}(N_{S'}/S')$ is an equality. One then puts $H = N^0$.

⁽⁵⁰⁾**N.D.E.** cf. Exp. II 4.11.8.

⁽⁵¹⁾**N.D.E.** The original text indicated that, denoting by $\mathfrak{n}_{N/G}$ the inverse image of the conormal sheaf $\mathcal{N}_{N/G}$ by the identity section $S \rightarrow N$, the condition that $\mathfrak{n}_{N_{S'}/G_{S'}} \rightarrow \omega_{G_{S'}/S'}^1$ be universally injective is equivalent to the fact that $S' \rightarrow S$ factors through a certain open subscheme of S . We have not succeeded in justifying this point, owing to the fact that the formation of $\mathcal{N}_{N/G}$ does not commute with base change, and we have replaced this argument by the one which follows, indicated by O. Gabber.

Replacing S by Z , it only remains to express that $H_{s'}$ has the same reductive rank as $G_{s'}$ at every point $s' \in S'$, or, equivalently, that H_s has the same reductive rank as G_s at every point s of the (set-theoretic) image of S' in S . Now this condition defines an open subset of S (Exp. XIX 6.2).

Remark. In general, the scheme $\mathcal{H}$ is not smooth over S . It is however if 6 is invertible on S , or if there exists a prime number p such that $p \cdot 1_S = 0$ (i.e. if S has characteristic $p > 0$).

Corollary 5.8.2. *Let S be a scheme, G a reductive S -group, H a subgroup of type (R) of G . (Recall (5.3.10) that $\underline{\text{Norm}}_G(H)$ is representable by a closed group subscheme of G , smooth over S).*

Then the quotient sheaf $G/\underline{\text{Norm}}_G(H)$ is representable by a quasi-projective S -scheme, smooth and of finite presentation over S (which is in fact an open subset of $\mathcal{H}$).

Indeed, consider the morphism

$$f : G \longrightarrow \mathcal{H},$$

defined set-theoretically by $f(g) = \text{int}(g)H$. By 5.3.9, this morphism is smooth and of finite presentation, hence open. Let $V = f(G)$ be endowed with its structure of open subscheme of $\mathcal{H}$. The morphism $f : G \longrightarrow V$ is covering and has kernel $\underline{\text{Norm}}_G(H)$, which proves that $G/\underline{\text{Norm}}_G(H)$ is representable by V (cf. Exp. IV 4.6.5).

Corollary 5.8.3. *Let S be a scheme, G a reductive S -group. Consider the functors $\underline{\text{Tor}}(G)$, $\underline{\text{Bor}}(G)$, $\underline{\text{Kil}}(G)$ defined by*

$$\begin{aligned} \underline{\text{Tor}}(G)(S') &= \{\text{maximal tori of } G_{S'}\}, \\ \underline{\text{Bor}}(G)(S') &= \{\text{Borel subgroups of } G_{S'}\}, \\ \underline{\text{Kil}}(G)(S') &= \{\text{Killing pairs of } G_{S'} \text{ (cf. 5.3.13)}\}. \end{aligned}$$

- (i) $\underline{\text{Tor}}(G)$, $\underline{\text{Bor}}(G)$, $\underline{\text{Kil}}(G)$ are representable by smooth S -schemes of finite presentation, with integral geometric fibers, and respectively affine, projective, affine over S .
- (ii) The canonical morphism $\underline{\text{Kil}}(G) \longrightarrow \underline{\text{Tor}}(G)$ (resp. $\underline{\text{Kil}}(G) \longrightarrow \underline{\text{Bor}}(G)$) is finite étale surjective (resp. affine smooth surjective).
- (iii) Let T be a maximal torus of G (resp. B a Borel subgroup of G , resp. $B \supset T$ a Killing pair of G). The morphism

$$G \longrightarrow \underline{\text{Tor}}(G), \quad \text{resp. } G \longrightarrow \underline{\text{Bor}}(G), \quad \text{resp. } G \longrightarrow \underline{\text{Kil}}(G)$$

defined by

$$g \longmapsto \text{int}(g)T, \quad \text{resp. } g \longmapsto \text{int}(g)B, \quad \text{resp. } g \longmapsto (\text{int}(g)B, \text{int}(g)T)$$

induces an isomorphism

$$\begin{aligned} G/\underline{\text{Norm}}_G(T) &\xrightarrow{\sim} \underline{\text{Tor}}(G), & \text{resp. } G/B &\xrightarrow{\sim} \underline{\text{Bor}}(G), \\ & & \text{resp. } G/T &\xrightarrow{\sim} \underline{\text{Kil}}(G). \end{aligned}$$

One sees first that (iii) follows from the conjugacy theorem for maximal tori (resp. Borel subgroups, resp. Killing pairs) and the fact that

$$\underline{\text{Norm}}_G(B) = B, \quad \underline{\text{Norm}}_G(B) \cap \underline{\text{Norm}}_G(T) = T,$$

all results established previously (5.1.2, 5.3.12, 5.3.14, 5.6.1).

It follows first that the canonical morphisms

$$\underline{\text{Tor}}(G) \longrightarrow \mathcal{H}, \quad \underline{\text{Bor}}(G) \longrightarrow \mathcal{H}$$

are representable, locally for the étale topology, by open immersions (5.8.2 and 5.1.2 resp. 5.5.5), hence by descent that $\underline{\text{Tor}}(G)$ and $\underline{\text{Bor}}(G)$ are representable by open subsets of $\mathcal{H}$.

Likewise $\underline{\mathrm{Kil}}(G)$ is locally (for the étale topology) representable by a scheme affine over the base (Exp. IX 2.3), hence representable by an affine S -scheme, by descent of affine schemes.⁽⁵²⁾

The assertions of (ii) follow at once from 5.5.5 (ii) and 5.6.13. It already follows that $\underline{\mathrm{Tor}}(G)$ is affine over S (EGA II 6.7.1). It therefore only remains to prove the fact that $\underline{\mathrm{Bor}}(G)$ is projective over S . One already knows that it is quasi-projective; it remains to prove that it is proper;⁽⁵³⁾ now it has connected fibers, hence, by EGA IV₃, 15.7.10, one is reduced to proving it on the geometric fibers; if S is the spectrum of an algebraically closed field, one has $\underline{\mathrm{Bor}}(G) = G/B$ by (iii) and one concludes by *Bible*, § 6.4, th. 4 (or [Ch05], § 6.5, th. 5).

Remark 5.8.4. Under the conditions of 5.8.3, let Q be a central subgroup of multiplicative type of G . The obvious morphisms define isomorphisms

$$\begin{aligned}\underline{\mathrm{Tor}}(G) &\simeq \underline{\mathrm{Tor}}(G/Q), & \underline{\mathrm{Bor}}(G) &\simeq \underline{\mathrm{Bor}}(G/Q), \\ \underline{\mathrm{Kil}}(G) &\simeq \underline{\mathrm{Kil}}(G/Q).\end{aligned}$$

Corollary 5.8.5. *Let S be a scheme, G a reductive S -group, P a group subscheme of G , smooth and of finite presentation over S . The following conditions are equivalent:*

- (i) *For each $s \in S$, $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$ (i.e. the quotient scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper over $\bar{s}$, or $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$, cf. *Bible*, § 6.4, th. 4 or [Ch05], § 6.5, th. 5).*
- (ii) *The quotient sheaf G/P is representable by a smooth S -scheme projective over S .*

Moreover, under these conditions, P is closed in G , has connected fibers and one has $P = \underline{\mathrm{Norm}}_G(P)$.

One clearly has (ii) $\Rightarrow$ (i). If (i) holds, then $P(\bar{s}) = \underline{\mathrm{Norm}}_{G(\bar{s})}(P_{\bar{s}})$ and $P_{\bar{s}}$ is connected (for the first point, cf. *Bible*, § 12.3, lemma 4;⁽⁵⁴⁾ the second point follows from it, for $P' = P_{\bar{s}}^0$ is a parabolic subgroup of $G_{\bar{s}}$ normalized by $P_{\bar{s}}$, whence $P'(\bar{s}) = P(\bar{s})$ and hence $P' = P_{\bar{s}}$); it follows that P is of type (R), and that P equals $\underline{\mathrm{Norm}}_G(P)$, hence is closed in G . By 5.8.2, $G/P = G/\underline{\mathrm{Norm}}_G(P)$ is representable by a quasi-projective S -scheme. Its fibers are connected and proper, hence it is projective by the argument of 5.8.3.

Remark 5.8.6. Statements 5.8.1, 5.8.2, 5.8.5 hold for an S -group of type (RA), or for an S -group of type (RR) satisfying 5.1.8.⁽⁵⁵⁾

Remark 5.8.7. Through the inner automorphisms of G , one has canonical actions:

$$\begin{aligned}G &\longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Tor}}(G)), & G &\longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Bor}}(G)), \\ & & G &\longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G)),\end{aligned}$$

which, in the situation of 5.8.3 (iii), are identified with the canonical actions

$$\begin{aligned}G &\longrightarrow \underline{\mathrm{Aut}}_S(G/\underline{\mathrm{Norm}}_G(T)), & G &\longrightarrow \underline{\mathrm{Aut}}_S(G/B), \\ & & G &\longrightarrow \underline{\mathrm{Aut}}_S(G/T).\end{aligned}$$

One concludes in particular that

$$\begin{aligned}\mathrm{Ker}(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Tor}}(G))) &= \mathrm{Ker}(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Bor}}(G))) \\ &= \mathrm{Ker}(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G))) \\ &= \underline{\mathrm{Centr}}(G).\end{aligned}$$

⁽⁵²⁾**N.D.E.** cf. SGA 1, VIII 2.1.

⁽⁵³⁾**N.D.E.** One may assume S affine and, since $\underline{\mathrm{Bor}}(G)$ is of finite presentation over S by (i), reduce to the case where S is noetherian; one is then under the hypotheses of EGA IV₃, 15.7.10.

⁽⁵⁴⁾**N.D.E.** What follows has been expanded.

⁽⁵⁵⁾**N.D.E.** Indeed, the proof of 5.8.1 uses only 5.3.3 (valid for a group of type (RA)) and XIX 6.2 which, by XII 1.7 (b), also holds for groups of type (RR).

It is indeed clear that $\text{Centr}(G)$ acts trivially on each of the three schemes. Conversely, the kernel of $G \longrightarrow \underline{\text{Aut}}_S(\underline{\text{Kil}}(G))$ is “the intersection of the maximal tori of G ” in the sense of 4.1.7, hence equals $\text{Centr}(G)$ (*loc. cit.*). For $\text{Bor}(G)$, one notes that “the intersection of the Borel subgroups of G ” is also “the intersection of its maximal tori” (see the next number). For $\text{Tor}(G)$, one uses Exp. XII 4.11.

5.9. Properties peculiar to Borel subgroups

Most of these properties will be generalized in Exp. XXVI to parabolic subgroups.

Definition 5.9.1. Let S be a scheme, G a reductive S -group, B and B' two Borel subgroups of G . One says that B and B' are *in general position* (or that B' is in general position relative to B) if $B \cap B'$ is a torus (necessarily maximal) of G .

If T is a maximal torus of G contained in B and B' , one says that B and B' are *opposite* (relative to T) if $B \cap B' = T$.

Proposition 5.9.2. Let S be a scheme, G a reductive S -group, B a Borel subgroup of G , T a maximal torus of B . There exists a unique Borel subgroup B' of G , opposite to B relative to T .

If (G, T, M, R) is a splitting of G with respect to T and if $B = B_{R_+}$ (5.5.1), then $B' = B_{-R_+}$.

By faithfully flat descent, it suffices to prove the proposition in the split case, when $B = B_{R_+}$ (5.5.5 (iv)). Then B_{-R_+} is indeed opposite to B (4.1.2); let us show that it is the only Borel subgroup of G containing T which is opposite to B . If B' is a Borel subgroup of G containing T , then B' is locally over S of the form $B_{R'_+}$, where R'_+ is a second system of positive roots of R (5.5.5 (iii)). If $R'_+ \neq -R_+$, there exists $\alpha \in R'_+ \cap R_+$, hence such that $U_\alpha \subset B_{R_+} \cap B_{R'_+}$.

Proposition 5.9.3. Let S be a scheme, G a reductive S -group, B a Borel subgroup of G .

- (i) If B' is a Borel subgroup of G , the following conditions are equivalent:
 - (a) B' is in general position with respect to B (5.9.1).
 - (b) $B'^u \cap B^u = e$.
 - (b') $B'^u \cap B = e$.
 - (c) The product in G induces an open immersion $B'^u \times_S B \longrightarrow G$.
 - (c') The canonical morphism $B'^u \longrightarrow G/B$ is an open immersion.
- (ii) The functor $\underline{\text{Opp}}(B)$:

$$S' \longmapsto \left\{ \begin{array}{l} \text{Borel subgroups of } G_{S'} \text{ in} \\ \text{general position with respect to } B_{S'} \end{array} \right\}$$

is representable by an open subscheme of $\text{Bor}(G)$ (5.8.3). The morphism

$$\underline{\text{Opp}}(B) \longrightarrow \text{Tor}(B)$$

defined by $B' \longmapsto B \cap B'$ is an isomorphism. In particular (5.6.13), the inner automorphisms of B^u endow $\underline{\text{Opp}}(B)$ with a principal homogeneous bundle structure under B^u .

Let us first examine (i). One has (a) $\Rightarrow$ (c); indeed, (c) is local for the étale topology; by 5.5.5 (iv), one reduces to the case where G is split with respect to $B \cap B'$ and B has the form B_{R_+} ; by 5.9.2, one then has $B'^u = U_{-R_+}$ and one is reduced to 4.1.2.

One has trivially (c') $\Leftrightarrow$ (c) $\Rightarrow$ (b') $\Rightarrow$ (b). It therefore remains to prove (b) $\Rightarrow$ (a). Let us first prove (ii); the second assertion is a formal consequence of 5.9.2, the third follows at once by 5.6.13; let us then prove the first; it is local for the étale topology and one may therefore assume that B possesses a maximal torus T ; let B'_0 be the opposite of B relative to T (5.9.2).

By what precedes the morphism $B^u \rightarrow \underline{\text{Bor}}(G)$ induced by the canonical morphism $G \rightarrow G/B'_0 \rightarrow \underline{\text{Bor}}(G)$ (5.8.3) induces an isomorphism $B^u \xrightarrow{\sim} \underline{\text{Opp}}(B)$. One therefore has a commutative diagram

$$\begin{array}{ccc} B^u & \longrightarrow & G/B'_0 \\ \sim \downarrow & & \downarrow \sim \\ \underline{\text{Opp}}(B) & \longrightarrow & \underline{\text{Bor}}(G). \end{array}$$

Now the morphism $B^u \rightarrow G/B'_0$ is an open immersion (by (i) (a) $\Rightarrow$ (c')), which completes the proof of (ii). Note at once the corollary

Corollary 5.9.4. *Let G be a reductive S -group and B and B' two Borel subgroups of G . If $s \in S$ is such that B_s and B'_s are in general position, there exists an open subset V of S containing s such that B_V and B'_V are in general position.*

It therefore only remains to prove (b) $\Rightarrow$ (a). By the preceding corollary, it suffices to do so when S is the spectrum of an algebraically closed field k . One may assume G split with respect to a maximal torus T of B . Let B'_0 be the Borel subgroup opposite to B . Since the Borel subgroups of G are conjugate under $G(k)$, there exists $g \in G(k)$ such that $\text{int}(g)B'_0 = B'$. By Bruhat's theorem (5.7.4), one may write $g = bnb'$, with $b \in B(k)$, $b' \in B'_0(k)$, $n \in \underline{\text{Norm}}_G(T)(k)$. One therefore has

$$B' = \text{int}(b) \text{int}(n)B'_0$$

and $B' \cap B = \text{int}(b)(\text{int}(n)B'_0 \cap B)$. If $n \notin T(k)$, $\text{int}(n)B'_0 \cap B^u \neq e$ (cf. proof of 5.9.2); it follows that (b) implies $B' \cap B = \text{int}(b)(B'_0 \cap B) = \text{int}(b)T$. Q.E.D.

Proposition 5.9.5. *Let S be a scheme, G a reductive S -group, B a Borel subgroup of G , B^u its unipotent part. There exists a sequence of subgroups of B :*

$$U_0 = B^u \supset U_1 \supset \cdots \supset U_n \supset \cdots$$

having the following properties:

- (i) *Each U_i is smooth, with connected fibers, characteristic in B ; the inner automorphisms of B^u act trivially on the (sheaf) quotients U_i/U_{i+1} .*
- (ii) *For each $i \geq 0$, there exists a locally free $\mathcal{O}_S$ -module $\mathcal{E}_i$ and an isomorphism of S -sheaves of groups*

$$U_i/U_{i+1} \xrightarrow{\sim} \text{W}(\mathcal{E}_i).$$

- (iii) *For every $s \in S$, $(U_n)_s = e$ for $n \geq \dim(B_s^u)$.*

Assume first that there exist a splitting (G, T, M, R) of G and a system of positive roots R_+ of R such that $B = B_{R_+}$. One denotes by Δ the set of simple roots of R_+ ; for each $\alpha \in R_+$, one denotes by $\text{ord}(\alpha)$ the sum of the coefficients of α on the basis Δ of $\Gamma_0(R)$; this is the *order of α relative to R_+* . One has $\text{ord}(\alpha) \leq \text{Card}(R_+)$. For every $i > 0$, let $R^{(i)}$ be the set of roots of order $> i$; it is a closed set of positive roots, hence one may construct (5.6.5)

$$U_i = U_{R^{(i)}}.$$

If $\alpha \in R_+$ and $\beta \in R^{(i)}$, then $\alpha + \beta \in R^{(i+1)}$. It follows, by 5.5.2, that each U_i is an invariant subgroup of B and that the inner automorphisms of B^u act trivially on U_i/U_{i+1} . This group is moreover identified with

$$\prod_{\text{ord}(\alpha)=i+1} U_\alpha$$

and is therefore endowed with a vector structure.

If B has the form $B_{R'}$ for another splitting (G, T', M', R') of G , let us show that the groups U'_i constructed as above with the aid of the new splitting coincide with the U_i and that the vector structures on the successive quotients also coincide. By 5.6.13, there exists $b \in B^u(S)$ such that $T' = \text{int}(b)T$; the assertion to be proved is local over S and one may

therefore assume that the isomorphism $T \xrightarrow{\sim} T'$ induced by $\text{int}(b)$ comes by duality from an isomorphism of root data

$$h : (M', M'^*, R', R'^*) \xrightarrow{\sim} (M, M^*, R, R^*).$$

It is clear that the roots of R'_+ are the $\alpha \circ \text{int}(b) = h(\alpha)$, $\alpha \in R_+$, and that the simple roots of R'_+ are the $\alpha \circ \text{int}(b) = h(\alpha)$, $\alpha \in \Delta$, hence that $\text{ord}(h(\alpha)) = \text{ord}(\alpha)$ for $\alpha \in R_+$. On the other hand, it is clear by transport of structure that the vector groups $U'_{h(\alpha)}$ are none other than the $\text{int}(b)U_\alpha$. One therefore has $\text{int}(b)U_i = U'_i$, but since U_i is invariant, this gives $U_i = U'_i$.

Likewise the isomorphism of vector groups

$$\text{int}(b) : U_i/U_{i+1} \xrightarrow{\sim} U'_i/U'_{i+1}$$

is the identity, by what has already been proved.

Let us now treat the general case. There exists a covering family for the étale topology $\{S_i \rightarrow S\}$ and for each i a splitting (G_i, T_i, M_i, R_i) and a system of positive roots R_{i+} of R_i such that $B \times_S S_i = B_{R_{i+}}$ (5.5.5, (iii)). For each i , one therefore has a family

$$B_{S_i} = U_{i,0} \supset U_{i,1} \supset \cdots \supset U_{i,j} \supset \cdots$$

and vector structures on the $U_{i,j}/U_{i,j+1}$. By descent, it suffices to prove that for every pair (i, i') and every j , one has

$$U_{i,j} \times_{S_i} S_{ii'} = U_{i',j} \times_{S_{i'}} S_{ii'}$$

(one writes $S_{ii'} = S_i \times_S S_{i'}$) and that the vector structures on the quotients

$$(U_{i,j}/U_{i,j+1}) \times_{S_i} S_{ii'} \quad \text{and} \quad (U_{i',j}/U_{i',j+1}) \times_{S_{i'}} S_{ii'}$$

coincide. Now if $S_{ii'} = \emptyset$, this is trivial; if $S_{ii'} \neq \emptyset$, then one is in the situation studied previously: $B \times_S S_{ii'}$ is defined by the system of positive roots R_{i+} (resp. $R_{i'+}$) in the splitting $(G_{S_{ii'}}, T_i \times_{S_i} S_{ii'}, M_i, R_i)$ (resp. in the splitting $(G_{S_{ii'}}, T_{i'} \times_{S_{i'}} S_{ii'}, M_{i'}, R_{i'})$).

Corollary 5.9.6. *If S is affine, $H^1(S, B^u) = e$, i.e. every principal bundle under B^u possesses a section.*

Indeed, S decomposes as a direct sum of subschemes on each of which B^u has constant relative dimension. One may therefore, by 5.9.5 (iii), assume that there exists an n such that $U_n = e$. Since, by TDTE I, B.1.1,⁽⁵⁶⁾

$$H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0,$$

one has $H^1(S, B^u) = 0$.

Corollary 5.9.7. *If S is affine, B possesses maximal tori. If T is a maximal torus of B , one has $H^1(S, T) = H^1(S, B)$.*

The first assertion follows at once from 5.9.6 and 5.6.13; the second is deduced from it in the standard way.⁽⁵⁷⁾

Corollary 5.9.8. *If G is a reductive S -group, the canonical morphism (cf. 5.8.3)*

$$\underline{\text{Kil}}(G) \rightarrow \underline{\text{Bor}}(G)$$

possesses sections over every affine open subset.

Corollary 5.9.9. *Under the conditions of 5.9.5, assume S affine; then there exists a locally free $\mathcal{O}_S$ -module $\mathcal{E}$ such that B^u is, as a scheme, S -isomorphic to $W(\mathcal{E})$.*

⁽⁵⁶⁾**N.D.E.** This is the corollary on page 18 of TDTE I.

⁽⁵⁷⁾**N.D.E.** Indeed, one has an exact sequence $H^1(S, B^u) \rightarrow H^1(S, B) \xrightarrow{\pi} H^1(S, B/B^u) = H^1(S, T)$, see [Se64], I § 5.5, Prop. 38 or [Gi71], III Prop. 3.3.1. Now $H^1(S, T) \rightarrow H^1(S, B)$ is a section of π , hence π is surjective; on the other hand $H^1(S, B^u) = 0$ by 5.9.6.

Let us show by induction on i that B^u/U_i is S -isomorphic to $W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1})$. This is clear for $i = 0$; assume $i \geq 1$. Then B^u/U_i is a principal homogeneous bundle with base $X = B^u/U_{i-1}$ under the group $(U_{i-1}/U_i)_X$. Since B^u/U_{i-1} is affine, by the induction hypothesis, and since $U_{i-1}/U_i = W(\mathcal{E}_{i-1})$, this bundle is trivial. One therefore has (at least) an isomorphism of S -schemes

$$B^u/U_i \xrightarrow{\sim} (B^u/U_{i-1}) \times_S W(\mathcal{E}_{i-1}) = W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

One concludes at once by condition (iii) of 5.9.5.

Corollary 5.9.10. *Let S be a semilocal scheme, $\{s_i\}$ its closed points, B a Borel subgroup of the reductive S -group G . The canonical map*

$$B^u(S) \longrightarrow \prod_i B^u(\mathrm{Spec} \kappa(s_i))$$

is surjective.

Indeed, if $S = \mathrm{Spec}(A)$, $\kappa(s_i) = A/\mathfrak{p}_i$ and if $\mathcal{E}$ is given by the A -module E , one has

$$B^u(S) = E \otimes A, \quad B^u(\mathrm{Spec} \kappa(s_i)) = E \otimes_A (A/\mathfrak{p}_i).$$

The assertion then follows at once from the fact that $A \longrightarrow \prod_i A/\mathfrak{p}_i$ is surjective.

5.10. Subgroups of type (R) with reductive fibers

Proposition 5.10.1. *Let (G, T, M, R) be a split S -group, R' a subset of R of type (R) (5.4.2), $H_{R'}$ the corresponding subgroup of G . The following conditions are equivalent:*

- (i) $H_{R'}$ is reductive (i.e. has reductive geometric fibers).
- (ii) One has $R' = -R'$, i.e. R' is symmetric.

Moreover, under these conditions, $(H_{R'}, T, M, R')$ is a splitting of $H_{R'}$; for every system of positive roots R_+ of R , $R'_+ = R' \cap R_+$ is a system of positive roots of R' and

$$B_{R_+} \cap H_{R'} = H_{R'_+}$$

is a Borel subgroup of $H_{R'}$, whose unipotent part is

$$U_{R_+} \cap H_{R'} = U_{R'_+}.$$

One clearly has (i) $\Rightarrow$ (ii) (it suffices to verify it fiber by fiber and R' is a system of roots of $H_{R'}$ with respect to T). To prove (ii) $\Rightarrow$ (i), one notes by 5.4.3 that

$$H_{R'} \cap Z_\alpha = \underline{\mathrm{Centr}}_{H_{R'}}(T_\alpha) = Z_\alpha$$

for every $\alpha \in R'$ and one applies the criterion of Exp. XIX 1.12.

If R_+ is a system of positive roots of R , then $R'_+ = R_+ \cap R'$ is clearly a closed subset of R' such that $R'_+ \cup -R'_+ = R'$ and $R'_+ \cap -R'_+ = \emptyset$, hence a system of positive roots of R' . The two other assertions follow respectively from 5.6.1 (vi) and 5.6.7 (i).

Corollary 5.10.2. *Let S be a scheme, G a reductive S -group, H a reductive S -subgroup such that for every $s \in S$, $G_{\bar{s}}$ and $H_{\bar{s}}$ have the same reductive rank. Then H is closed in G , $\underline{\mathrm{Norm}}_G(H)$ is smooth over S , $\underline{\mathrm{Norm}}_G(H)/H$ is representable by a finite étale S -scheme.*

If T is a maximal torus of H and B a Borel subgroup of G containing T , then $B \cap H$ is a Borel subgroup of H , whose unipotent part is $(B \cap H)^u = B^u \cap H$.

The first assertions follow at once from 5.3.10 and 5.3.18, through the fact that the Weyl groups of G and of H are finite (Exp. XIX 2.5). The other assertions are local for the étale topology and reduce to the case studied in 5.10.1.

Proposition 5.10.3. *Let S be a scheme, G a reductive S -group.*

a) If Q is a torus of G , $\text{Centr}_G(Q)$ is a subgroup of type (R) of G with reductive fibers. If $Q \subset Q'$ are two tori of G , then $\text{Centr}_G(Q) \supset \text{Centr}_G(Q')$.

b) If H is a subgroup of type (R) of G with reductive fibers, then $\text{rad}(H)$ (4.3.6) is a torus of G . If $H \subset H'$ are two subgroups of type (R) of G with reductive fibers, then $\text{rad}(H) \supset \text{rad}(H')$.

c) If Q is a torus of G , one has

$$\begin{aligned} \text{rad}(\text{Centr}_G(Q)) \supset Q \quad \text{and} \\ \text{Centr}_G(\text{rad}(\text{Centr}_G(Q))) = \text{Centr}_G(Q). \end{aligned}$$

d) If H is a subgroup of type (R) of G with reductive fibers, one has

$$\text{Centr}_G(\text{rad}(H)) \supset H \quad \text{and} \quad \text{rad}(\text{Centr}_G(\text{rad}(H))) = \text{rad}(H).$$

Indeed, a) follows at once from Exp. XIX 2.8. To prove b), it suffices to note that $\text{rad}(H') \subset H$, for H contains (locally for (fpqc)) a maximal torus of G , hence of H' . The first assertion of c) (resp. d)) is trivial; the second follows by the usual argument.

This proposition leads to the following definition:

Definition 5.10.4. Let S be a scheme, G a reductive S -group, H a reductive subgroup of type (R) of G , and Q a subtorus of G .⁽⁵⁸⁾

1) One says that H is a *critical subgroup* if it is the centralizer of its radical.

2) One says that Q is a *C-critical torus* if it is the radical of its centralizer.

It then follows from proposition 5.10.3:

Corollary 5.10.5.

- (i) For every subtorus Q of G , $\text{Centr}_G(Q)$ is critical.
- (ii) For every subgroup of type (R) with reductive fibers H of G , $\text{rad}(H)$ is a C-critical torus of G .
- (iii) The maps

$$Q \longmapsto \text{Centr}_G(Q), \quad H \longmapsto \text{rad}(H)$$

are mutually inverse bijections between the set of C-critical tori of G and that of its critical reductive subgroups of type (R) .

- (iv) If Q is a torus of G , $\text{rad}(\text{Centr}_G(Q))$ is the smallest C-critical torus of G containing Q .
- (v) If H is a reductive subgroup of type (R) of G , $\text{Centr}_G(\text{rad}(H))$ is the smallest critical reductive subgroup of type (R) of G containing H .

Remark 5.10.5.1. ⁽⁵⁸⁾1) A torus T of G is a critical subgroup of G if and only if it is a maximal torus.

2) In what follows, “critical torus” means “C-critical torus”.

Proposition 5.10.6. Let (G, T, M, R) be a split S -group, R' a subset of R . The following conditions are equivalent:

- (i) R' is of type (R) , $H_{R'}$ is reductive and critical.
- (ii) There exists a system of simple roots Δ of R and a subset Δ' of Δ such that R' is the set of elements of R which are linear combinations of the elements of Δ' .
- (iii) R' is closed, symmetric, and every system of simple roots of R' is the intersection with R' of a system of simple roots of R .

Indeed, by Exp. XXI 3.4.8, (ii) and (iii) are equivalent and are also equivalent to the fact that R' is the intersection of R with a $\mathbf{Q}$ -vector subspace of $M \otimes \mathbf{Q}$. Now this last condition

⁽⁵⁸⁾**N.D.E.** The original has been modified by introducing the terminology “C-critical torus” instead of “critical torus”, in order to avoid confusion in later references (cf. Exp. XXVI, 3.9). Statement 5.10.5 has also been expanded and remark 5.10.5.1 added.

is implied by (i): if $H_{R'} = \underline{\text{Centr}}_G(Q)$, then R' is the set of elements of R vanishing on Q (Exp. II 5.2.3 (ii)). Finally, this condition implies (i), for $\text{rad}(H_{R'})$ is the maximal torus of $\bigcap_{\alpha \in R'} \text{Ker}(\alpha)$, hence $\underline{\text{Centr}}_G(\text{rad}(H_{R'}))$ is none other than $H_{R''}$ where R'' is the intersection of R with the vector subspace generated by R' .

5.10.7. — Let us collect some of the preceding results: let (G, T, M, R) be a split S -group, and let Δ be a system of simple roots of R and R_+ the corresponding system of positive roots; choose a subset Δ' of Δ , denote by R' the set of elements of R which are linear combinations of the elements of Δ' , and put $R'_+ = R' \cap R_+$. Let $T_{\Delta'}$ be the maximal torus of $\bigcap_{\alpha \in \Delta'} \text{Ker}(\alpha)$ and $Z_{\Delta'} = \underline{\text{Centr}}_G(T_{\Delta'})$.

Then $Z_{\Delta'}$ is a reductive subgroup of G , with radical $T_{\Delta'}$; $(Z_{\Delta'}, T, M, R')$ is a split S -group; $B_{R_+} \cap Z_{\Delta'}$ is the Borel subgroup of $Z_{\Delta'}$ defined by the system of positive roots R'_+ (or by the system of simple roots Δ'), and its unipotent part is $U_{R_+} \cap Z_{\Delta'} = U_{R'_+}$.

Remark 5.10.8. Under the conditions of 5.10.4, let Q be a critical torus of G , $L = \underline{\text{Centr}}_G(Q)$ its centralizer. Since $Q = \text{rad}(L)$, Q is a characteristic subgroup of L ; it follows at once that

$$\underline{\text{Norm}}_G(L) = \underline{\text{Norm}}_G(Q),$$

hence also

$$\underline{\text{Norm}}_G(L)/L = \underline{\text{Norm}}_G(Q)/\underline{\text{Centr}}_G(Q) = W_G(Q).$$

By 5.10.2, one deduces

Proposition 5.10.9. *Let S be a scheme, G a reductive S -group, Q a critical torus of G . The Weyl group $W_G(Q)$ is (étale) finite over S .*

Remark 5.10.10. Under the conditions of 5.10.7, one may make explicit

$$W_G(T_{\Delta'}) = \underline{\text{Norm}}_G(Z_{\Delta'})/Z_{\Delta'}.$$

It is the constant group associated with the quotient W_1/W_2 , where W_1 is the subgroup of W consisting of the elements which normalize the subgroup of M generated by Δ' and W_2 the subgroup of W generated by the s_α , $\alpha \in \Delta'$.

5.11. Subgroups of type (RC)

Definition 5.11.1. Let S be a scheme, G a reductive S -group. A group subscheme H of G is said to be of type (RC) if it is of type (R), i.e. (5.2.1) satisfies the following two conditions:

- (i) H is smooth over S , with connected fibers;
- (ii) for every $s \in S$, $H_{\bar{s}}$ contains a maximal torus of $G_{\bar{s}}$;

and if it satisfies in addition the following condition:

- (iii) for every $s \in S$ and every maximal torus T of $H_{\bar{s}}$, the set of roots of $H_{\bar{s}}$ with respect to T is a closed subset of the set of all roots of $G_{\bar{s}}$ with respect to T .

Remark 5.11.2. As we have already pointed out in 5.4.8, condition (iii) is a consequence of the others when 6 is invertible on S .⁽⁵⁹⁾

Lemma 5.11.3. *Let (G, T, M, R) be a split S -group and R' a closed subset of R . Let*

$$R_1 = \{\alpha \in R', -\alpha \in R'\} \quad \text{and} \quad R_2 = \{\alpha \in R', -\alpha \notin R'\}.$$

Then R_1 and R_2 are closed. Consider the groups $H_{R'}$, H_{R_1} and U_{R_2} (5.4.7 and 5.6.5), which are smooth and have connected fibers.

- (i) *The group U_{R_2} is invariant in $H_{R'}$ and $H_{R'}$ is the semidirect product of U_{R_2} by H_{R_1} .*

⁽⁵⁹⁾**N.D.E.** i.e. when each residue characteristic of S is > 3 .

- (ii) H_{R_1} is reductive, U_{R_2} has connected unipotent geometric fibers; every invariant subgroup of $H_{R'}$, smooth over S and with connected unipotent geometric fibers, is contained in U_{R_2} , and every reductive subgroup of $H_{R'}$ containing T is contained in H_{R_1} .
- (iii) One has $U_{R_2} \cap \underline{\text{Norm}}_G(H_{R_1}) = e$.

One first has (iii) by 5.6.7 (i). The first assertion of (i) follows from 5.6.7 (ii). Since $U_{R_2} \cap H_{R_1} = e$ by (iii), the semidirect product $H_{R_1} \cdot U_{R_2}$ is a subgroup of $H_{R'}$; but these are two subgroups of type (R) of G , containing T , and they have the same Lie algebra $\mathfrak{g}_{R'}$; they therefore coincide by 5.3.5, which completes the proof of (i).

Let us now prove (ii); the first two assertions are none other than 5.10.1 and 5.6.5. Let U be a group subscheme of $H_{R'}$, smooth and of finite presentation, invariant (hence normalized by T), with connected unipotent geometric fibers; by 5.6.12, one has, locally over S , $U = U_{R''}$, where R'' is a subset of R' such that $R'' \cap -R'' = \emptyset$. If $U \not\subset U_{R_2}$, then $R'' \not\subset R_2$, hence there exists $\alpha \in R''$ such that $-\alpha \in R'$. Then $Z_\alpha \subset H_{R'}$ (5.4.3), hence Z_α normalizes U . But U contains U_α and Z_α possesses a section w such that $\text{int}(w)U_\alpha = U_{-\alpha}$; this implies $-\alpha \in R''$, contradicting the hypothesis $R'' \cap -R'' = \emptyset$.

Finally, if L is a reductive subgroup of $H_{R'}$ containing T , one has locally over S , $L = H_{R'''}$, with R''' symmetric and contained in R' , hence contained in R_1 .

Proposition 5.11.4. *Let S be a scheme, G a reductive S -group, H a group subscheme of G of type (RC).*

- (i) H is closed in G , $\underline{\text{Norm}}_G(H)/H$ is representable by a finite étale S -group scheme.
- (ii) H possesses a largest smooth invariant group subscheme of finite presentation over S , with connected unipotent geometric fibers; one says that it is the unipotent radical of H and denotes it by $\text{rad}^u(H)$. The quotient sheaf $H/\text{rad}^u(H)$ is representable by a reductive S -group.
- (iii) If T is a maximal torus of H , H possesses a reductive subgroup L containing T , of type (RC), having the following two properties:
 - (a) Every reductive subgroup of H containing T is contained in L .
 - (b) H is the semidirect product $H = L \cdot \text{rad}^u(H)$, i.e. the canonical morphism $L \rightarrow H/\text{rad}^u(H)$ is an isomorphism.

Moreover, L is the unique reductive subgroup of H containing T and satisfying either of the two preceding conditions. Finally, one has the following equalities:

$$\underline{\text{Norm}}_H(L) = L, \quad \underline{\text{Norm}}_H(T) = \underline{\text{Norm}}_L(T), \quad W_H(T) = W_L(T),$$

in particular $W_H(T)$ is finite over S .

PROOF. Note first that (i) is local for the étale topology. Hence, by corollary 5.3.18, (i) is a consequence of the last assertion of (iii).

The assertions of (ii) are local for the étale topology. One may therefore assume one is in the situation of 5.11.3, where one concludes at once by (i) and (ii).

By the uniqueness assertions it contains, (iii) is also local for the étale topology and one may again reduce to the situation of 5.11.3, where properties (a) and (b) have been verified. The uniqueness of an L satisfying (a) is trivial; the uniqueness of an L satisfying (b) is obvious, in view of (a). The equality $\underline{\text{Norm}}_H(L) = L$ is none other than 5.11.3 (iii); if a section of H normalizes T , then it normalizes L , by uniqueness of L , hence is a section of L by what has just been proved, which proves the second equality; the third is then trivial. $\square$

Proposition 5.11.5. *Let S be a scheme, G a reductive S -group, $\mathcal{H}_c$ the functor of subgroups of type (RC) of G , which is a subfunctor of the functor $\mathcal{H}$ of 5.8.1.*

- (i) $\mathcal{H}_c$ is representable by an open subscheme of $\mathcal{H}$, smooth, quasi-projective and of finite presentation over S .

(ii) *There exists a finite étale S -scheme Cl_c and a morphism*

$$cl : \mathcal{H}_c \longrightarrow Cl_c,$$

smooth, quasi-projective, of finite presentation, surjective and with connected geometric fibers, having the following property:

For every $S' \rightarrow S$ and all $H, H' \in \mathcal{H}_c(S')$, $cl(H) = cl(H')$ if and only if H and H' are conjugate in G locally for the étale topology (or, equivalently by 5.3.11, if for every $s \in S$, $H_{\bar{s}}$ and $H'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$).

- (iii) *Cl_c and cl are determined (up to a unique isomorphism) by the preceding conditions.*
- (iv) *If (G, T, M, R) is a splitting of G , let E be the set of conjugacy classes modulo W of closed subsets of R ; then there exists an isomorphism $Cl_c \xrightarrow{\sim} E_S$ such that, for every closed subset R' of R , $cl(H_{R'})$ corresponds to the canonical image of R' in $E_S(S) = \text{Hom}_{\text{loc.const.}}(S, E)$.*

It is first clear that $\mathcal{H}_c$ is a sheaf for the étale topology and that (ii) implies that Cl_c is none other than the quotient sheaf of $\mathcal{H}_c$ by the equivalence relation defined by conjugacy.

This implies first (iii), as well as the fact that it suffices to verify (i) and (ii) locally for the étale topology. One therefore reduces to the situation of (iv); let us first construct a morphism

$$f : \mathcal{H}_c \longrightarrow E_S.$$

It suffices to construct a map $\mathcal{H}_c(S) \rightarrow E_S(S)$ functorial in S ; let therefore H be a subgroup of type (RC) of G ; since H possesses maximal tori locally for the étale topology, and since the maximal tori of G are conjugate locally for the étale topology, there exist a covering family $\{S_i \rightarrow S\}$ and for each i a $g_i \in G(S_i)$ and a closed subset R_i of R such that $\text{int}(g_i)(H \times_S S_i) = H_{R_i} \times_S S_i$; each R_i defines a section η_i of E_{S_i} , i.e. an element of $E_S(S_i)$; it now suffices to prove that the family (η_i) comes from a section $\eta = f(H)$ of E_S over S , and that the latter depends only on H .

To do this, one is reduced to proving that $H_{R'}$ and $H_{R''}$ are conjugate locally for the étale topology if and only if R' and R'' are conjugate by an element of the Weyl group W , which is trivial.

For every $\eta \in E$, there exists an $H_0 \in \mathcal{H}_c(S)$ such that $f(H_0) = \eta$: it suffices to take $H_0 = H_{R'}$ where R' is a closed subset of R whose image in E is η . If $H \in \mathcal{H}_0(S')$, $S' \rightarrow S$, H is conjugate to H_0 locally for the étale topology if and only if $f(H) = \eta$ (as one sees at once by the preceding argument), which shows that $f^{-1}(\eta)$ is identified with the quotient $G/\underline{\text{Norm}}_G(H_0)$, which by 5.8.2 is an open subset of $\mathcal{H}$, smooth, quasi-projective of finite presentation over S , with connected nonempty fibers. Since E_S is the sum of the open subschemes which are the images of the sections corresponding to the $\eta \in E$, $\mathcal{H}_c$ is identified with the sum of the $f^{-1}(\eta)$, $\eta \in E$, which proves (i) and (ii). Finally (iv) holds by construction.

Corollary 5.11.6. *If $u \in Cl_c(S')$, $S' \rightarrow S$, $cl^{-1}(u)$ is a smooth quasi-projective S' -scheme of finite presentation with connected nonempty fibers; it is an open subset of $\mathcal{H}_c$ and a “homogeneous” scheme under $G_{S'}$ (by inner automorphisms). In particular, if $H \in cl^{-1}(u)(S')$, the morphism $G_{S'} \rightarrow (\mathcal{H}_c)_{S'}$ defined by $g \mapsto \text{int}(g)H$ identifies $G_{S'}/\underline{\text{Norm}}_{G_{S'}}(H)$ with $cl^{-1}(u)$.*

Examples 5.11.7. In particular, one has two canonical sections u_t, u_b of Cl_c corresponding respectively to maximal tori ($R' = \emptyset$) and to Borel subgroups ($R' =$ system of positive roots). The S -schemes $cl^{-1}(u_t)$ and $cl^{-1}(u_b)$ are none other than the S -schemes $\underline{\text{Tor}}(G)$ and $\underline{\text{Bor}}(G)$ introduced in 5.8.3. We will see other examples in Exp. XXVI.

Remark 5.11.8. One may construct an S -scheme Cl , of finite presentation and unramified, and a smooth surjective morphism $\mathcal{H} \rightarrow Cl$, with connected geometric fibers, enjoying properties analogous to 5.11.5 (ii) and (iii).

6. The derived group

6.1. Preliminaries

In this number, one fixes a scheme S , a split S -group (G, T, M, R) , a system of positive roots R_+ of R , and one writes

$$B = B_{R_+}, \quad B^- = B_{R_-}, \quad U = B^u, \quad U^- = (B^-)^u, \\ \Omega = \Omega_{R_+} = U^- \cdot T \cdot U.$$

6.1.1. — One denotes by T' the subtorus of T “image of the family α^* , $\alpha \in R$ ”; in other words T' is the image of the morphism of groups

$$\mathbf{G}_{m,S}^R \longrightarrow T$$

defined by $(z_\alpha)_{\alpha \in R} \mapsto \prod_{\alpha \in R} \alpha^*(z_\alpha)$. One sees at once that if Δ denotes the set of simple roots of R_+ , the morphism

$$\mathbf{G}_{m,S}^\Delta \longrightarrow T'$$

defined in the same way is surjective with finite kernel. If one identifies T with $D_S(M)$, then T' is identified with $D_S(M/N)$, where

$$N = M \cap \mathcal{V}(R^*)^\perp$$

(one denotes by $\mathcal{V}(R^*)^\perp$ the orthogonal of $\mathcal{V}(R^*)$ in the duality between V and V^*).

Lemma 6.1.2. *The morphism defined by the product in T*

$$\mathrm{rad}(G) \times_S T' \longrightarrow T$$

is an isogeny (cf. 4.2.9).

Indeed, the canonical morphism $\mathrm{rad}(T) \longrightarrow T/T'$ comes by duality from the morphism of commutative groups

$$M \cap \mathcal{V}(R^*)^\perp \longrightarrow M/M \cap \mathcal{V}(R),$$

which one sees at once to be injective with finite cokernel (cf. Exp. XXI 6.3).

Definition 6.1.3. One puts $\Omega' = U^- \cdot T' \cdot U$; it is a closed subscheme of $\Omega = U^- \cdot T \cdot U$.

Lemma 6.1.4. *Let α be a simple root and $w_\alpha \in \underline{\mathrm{Norm}}_G(T)(S)$ lifting s_α . One has*

$$\mathrm{int}(w_\alpha)\Omega' \cap \Omega \subset \Omega'.$$

It suffices to prove that if $g \in \Omega'(S)$ and if $\mathrm{int}(w_\alpha)g \in \Omega(S)$, then $\mathrm{int}(w_\alpha)g \in \Omega'(S)$. By 5.6.8, write

$$g = a \exp_{-\alpha}(Y) t \exp_{\alpha}(X) b,$$

with $a \in U_{-\hat{\alpha}}(S)$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$, $t \in T'(S)$, $X \in \Gamma(S, \mathfrak{g}^{\alpha})$, $b \in U_{\hat{\alpha}}(S)$. One then has

$$\mathrm{int}(w_\alpha)g = \mathrm{int}(w_\alpha)a \cdot \mathrm{int}(w_\alpha)(\exp_{-\alpha}(Y)t \exp_{\alpha}(X)) \cdot \mathrm{int}(w_\alpha)b.$$

By 5.6.8 (iv), one has

$$\mathrm{int}(w_\alpha)a \in U_{-\hat{\alpha}}(S), \quad \mathrm{int}(w_\alpha)b \in U_{\hat{\alpha}}(S).$$

The following equivalences result (putting $h = \exp_{-\alpha}(Y)t \exp_{\alpha}(X)$):

$$\begin{aligned} \mathrm{int}(w_\alpha)g \in \Omega(S) &\iff \mathrm{int}(w_\alpha)h \in \Omega(S), \\ \mathrm{int}(w_\alpha)g \in \Omega'(S) &\iff \mathrm{int}(w_\alpha)h \in \Omega'(S). \end{aligned}$$

One is therefore reduced to the case where $g = h$. Since one has (4.1.12)

$$Z_\alpha \cap \Omega = U_{-\alpha} \cdot T \cdot U_\alpha, \quad Z_\alpha \cap \Omega' = U_{-\alpha} \cdot T' \cdot U_\alpha,$$

one is reduced to proving the following assertion:

$$\text{int}(w_\alpha)h \in (U_{-\alpha} \cdot T \cdot U_\alpha)(S) \implies \text{int}(w_\alpha)h \in (U_{-\alpha} \cdot T' \cdot U_\alpha)(S).$$

Now the latter follows at once from Exp. XX 3.12, which shows that the component on T of $\text{int}(w_\alpha)h$ has the form $t \cdot \alpha^*(z) \in T'(S)$.

Lemma 6.1.5. *For every $w \in \underline{\text{Norm}}_G(T)(S)$, there exists an open subset V_w of G , containing the identity section, such that*

$$\text{int}(w)\Omega' \cap V_w \subset \Omega'.$$

Choose for each simple root α an $n_\alpha \in \underline{\text{Norm}}_G(T)(S)$ lifting s_α . For every point $s \in S$, there exist an open subset V of S containing s , a $t \in T(V)$, and over V a relation

$$w = n_{\alpha_1} \cdots n_{\alpha_p} t, \quad \text{with the } \alpha_i \text{ simple.}$$

One may clearly content oneself with giving the proof for $V = S$; it is done by induction on p . If $p = 0$, then $w \in T(S)$ and one takes $V_w = G$; assume therefore $w = n_\alpha \cdot w'$, with w' satisfying the conclusion of the lemma; there therefore exists an open subset $V_{w'}$ of G , containing the identity section, such that $\text{int}(w')\Omega' \cap V_{w'} \subset \Omega'$. One may then write

$$\begin{aligned} \text{int}(w)\Omega' \cap \text{int}(n_\alpha)V_{w'} \cap \Omega &= \text{int}(n_\alpha)(\text{int}(w')\Omega' \cap V_{w'}) \cap \Omega \\ &\subset \text{int}(n_\alpha)\Omega' \cap \Omega \subset \Omega', \end{aligned}$$

by 6.1.4. One then takes $V_w = \text{int}(n_\alpha)V_{w'} \cap \Omega$ and one is done.

Lemma 6.1.6. *There exists an open subset V_0 of G , containing the identity section, such that for every $S' \rightarrow S$, one has*

$$U(S')U^-(S') \cap V_0(S') \subset \Omega'(S').$$

Let indeed n_0 be an element of $\underline{\text{Norm}}_G(T)(S)$ lifting the reflection w_0 of the Weyl group,⁽⁶⁰⁾ that is, such that $\text{int}(n_0)U = U^-$ (cf. Exp. XXI 3.6.14); then $n_0^2 \in T(S)$. Let us show that the open subset $V_0 = V_{n_0}$ of 6.1.5 answers the question. Indeed

$$\begin{aligned} U(S')U^-(S') &= \text{int}(n_0)(\text{int}(n_0)^{-1}U(S') \cdot \text{int}(n_0)^{-1}U^-(S')) \\ &= \text{int}(n_0)(U^-(S') \cdot U(S')) \subset \text{int}(n_0)\Omega'(S'). \end{aligned}$$

Whence

$$U(S')U^-(S') \cap V_0(S') \subset \text{int}(n_0)\Omega'(S') \cap V_0(S') \subset \Omega'(S').$$

Lemma 6.1.7. *Consider the morphism*

$$f : \Omega = U^- \cdot T \cdot U \longrightarrow T/T'$$

composite of the second projection and the canonical morphism from T to T/T' . Then f is “generically multiplicative”: there exists an open subset V of $\Omega \times_S \Omega$, containing the identity section (and hence relatively schematically dense, Exp. XVIII 1.3), such that for every $S' \rightarrow S$ and every $(x, y) \in V(S')$, one has $xy \in \Omega(S')$ and $f(xy) = f(x)f(y)$.

Let indeed x and y be two sections of Ω over S' . Write

$$x = utv, \quad y = u't'v', \quad \text{with } u, u' \in U^-(S'), \quad t, t' \in T(S'), \quad v, v' \in U(S').$$

Let V_0 be the open subset of 6.1.6 and V the open subset of $\Omega \times_S \Omega$ defined by “ $vu' \in V_0(S')$ ” (it is the inverse image of V_0 by the morphism $\Omega \times_S \Omega$ written set-theoretically $(x, y) \mapsto vu'$). Then V answers the question. Indeed, for $(x, y) \in V(S')$, one has

$$xy = (utv)(u't'v') = (ut)(vu')(t'v').$$

But $vu' \in \Omega'(S')$, whence

$$xy \in U^-(S')t\Omega'(S')t'U(S') \subset U^-(S')tt'T'(S')U(S'),$$

⁽⁶⁰⁾**N.D.E.** The reflection w_0 is defined in XXI 3.6.14.

which shows that $xy \in \Omega(S')$ and that

$$f(xy) = f(tt') = f(t)f(t') = f(x)f(y).$$

Proposition 6.1.8. *There exists a morphism of groups*

$$f : G \longrightarrow T/T'$$

inducing on T the canonical projection. The kernel $\mathrm{Ker}(f)$ of f is a closed group subscheme of G , smooth over S and with connected fibers. Every morphism of groups from G into a presheaf of commutative groups over S , separated for (fppf), vanishes on $\mathrm{Ker}(f)$.

The first assertion follows at once from 4.1.11. One has immediately $\mathrm{Ker}(f) \cap \Omega = \Omega'$, which proves that $\mathrm{Ker}(f)$ is smooth over S at every point of the identity section.⁽⁶¹⁾ By 5.6.9 (ii), every morphism φ from G into a presheaf of commutative groups separated for (fppf) vanishes on U and U^- . By Exp. XX 2.7, φ therefore also vanishes on T' , hence on Ω' . Taking the notations of 5.7.10, one sees that the monoid U_1 is contained in $\mathrm{Ker}(f)(S)$, which shows that

$$\mathrm{Ker}(f) = \bigcup_{u \in U_1} u\Omega'.$$

It follows on the one hand that every φ as above vanishes on $\mathrm{Ker}(f)$, and on the other hand that $\mathrm{Ker}(f)$ has connected fibers, hence is smooth over S by Exp. VIB 3.10.

6.2. Derived group of a reductive group

Theorem 6.2.1. *Let S be a scheme, G a reductive S -group.*

- (i) $D_S(G) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S})$ *is representable by a twisted constant S -group, whose type at $s \in S$ is $\mathbf{Z}^{\mathrm{rgred}(G_s) - \mathrm{rgss}(G_s)}$.*
- (ii) *Write $\mathrm{corad}(G) = D_S(D_S(G))$, which is thus an S -torus. The biduality morphism (cf. Exp. VIII § 1)*

$$f_0 : G \longrightarrow \mathrm{corad}(G)$$

is smooth and surjective.

- (iii) *The composite morphism*

$$\mathrm{rad}(G) \longrightarrow G \longrightarrow \mathrm{corad}(G)$$

is an isogeny (cf. 4.2.9).

- (iv) *The kernel of f_0 , denoted*

$$\mathrm{dér}(G) = \mathrm{Ker}(f_0)$$

is a closed group subscheme of G , semisimple over S , which one calls the derived group of G . If G is semisimple, one has $\mathrm{dér}(G) = G$.

- (v) *Every morphism of groups from G into an S -presheaf of commutative groups, separated for (fppf), vanishes on $\mathrm{dér}(G)$ and therefore factors through f_0 .*

PROOF. All assertions of the theorem are local for the étale topology; one may therefore reduce to the case where G is split over S . Consider then the morphism f of 6.1.8. By the last assertion of 6.1.8, one has at once an isomorphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(T/T', \mathbf{G}_{m,S}),$$

⁽⁶¹⁾**N.D.E.** “At every point of the identity section” has been added, as well as the reference to VIB 3.10 at the end of the proof. On the other hand, in the following sentence “prescheme” has been replaced by “presheaf”.

which proves (i), then (ii), and gives a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{f_0} & \text{corad}(G) \\ & \searrow f & \downarrow \sim \\ & & T/T'. \end{array}$$

One then has (v) by 6.1.8, and (iii) by 6.1.2. One also has $\text{Ker}(f) = \text{Ker}(f_0)$, which by 6.1.8 implies that $\text{dér}(G)$ is smooth over S and has connected fibers; it remains to verify that its fibers are semisimple; now they are reductive by Exp. XIX 1.7, as invariant subgroups of reductive groups. By (iii), $\text{rad}(G) \cap \text{dér}(G)$ is finite, which indeed implies that the fibers of $\text{dér}(G)$ are semisimple. $\square$

Remark 6.2.2. a) By construction, in the case where G is split, $\text{dér}(G)$ is the (fppf) subsheaf of G generated by the U_α , $\alpha \in R$. (It even suffices to take the U_α , $\alpha \in \Delta \cup -\Delta$, where Δ is a basis of R).

b) ⁽⁶²⁾Let C be the presheaf of commutators of G , i.e. the S -group functor which associates with every $S' \rightarrow S$ the commutator group of $G(S')$ (i.e. the subgroup of $G(S')$ generated by the elements $xyx^{-1}y^{-1}$, for $x, y \in G(S')$), and let $\tilde{C}$ be the associated (fppf) sheaf. Since the quotient $G/\text{dér}(G) = T/T'$ is commutative, $\text{dér}(G)$ contains C and hence $\tilde{C}$ (cf. Exp. IV 4.3.12).

On the other hand, the quotient presheaf $G/\tilde{C}$ is separated (Exp. IV 4.4.8.1), and hence by (v) one has $\text{dér}(G) \subset \tilde{C}$, whence $\text{dér}(G) = \tilde{C}$, i.e. $\text{dér}(G)$ is the (fppf) sheaf of commutators of G .

Note finally that C , being a subpresheaf of G , is separated, but is not equal to $\text{dér}(G)$ in general: for example, $\text{dér}(\text{SL}_2) = \text{SL}_2$ but $\text{SL}_2(\mathbf{F}_2) \simeq \mathfrak{S}_3$ is not equal to its derived group.

c) When S is the spectrum of an algebraically closed field k , $\text{dér}(G)(k)$ is the commutator group of $G(k)$ (Exp. VIB 7.10).

6.2.3. — Consider now the two exact sequences

$$\begin{aligned} 1 &\longrightarrow \text{rad}(G) \longrightarrow G \longrightarrow \text{ss}(G) \longrightarrow 1, \\ 1 &\longrightarrow \text{dér}(G) \longrightarrow G \longrightarrow \text{corad}(G) \longrightarrow 1. \end{aligned}$$

Since $\text{rad}(G)$ is central in G , the product in G defines a morphism of groups

$$u : \text{rad}(G) \times_S \text{dér}(G) \longrightarrow G$$

which is covering by 6.2.1 (iii), hence surjective and flat (Exp. VIB 9.2 (xi)).⁽⁶³⁾ Its kernel is isomorphic to $\text{rad}(G) \cap \text{dér}(G)$, which is also the kernel of $\text{rad}(G) \rightarrow \text{corad}(G)$, hence is a finite subgroup of multiplicative type of $\text{rad}(G)$.

One argues in the same way for the morphism

$$G \longrightarrow \text{corad}(G) \times_S \text{ss}(G),$$

whose kernel is $\text{dér}(G) \cap \text{rad}(G)$. One therefore has the

Proposition 6.2.4. *Let G be a reductive S -group. The morphisms*

$$\begin{aligned} \text{rad}(G) \times_S \text{dér}(G) &\longrightarrow G, & G &\longrightarrow \text{corad}(G) \times_S \text{ss}(G), \\ \text{rad}(G) &\longrightarrow \text{corad}(G) \end{aligned}$$

are central isogenies, and their kernels are isomorphic.

⁽⁶²⁾**N.D.E.** In what follows, the original has been expanded, and the assertion that “ $\text{dér}(G)$ is the separated (fppf) presheaf of commutators of G ” has been deleted.

⁽⁶³⁾**N.D.E.** Indeed, G is the (fppf) quotient of $\text{rad}(G) \times_S \text{dér}(G)$ by $\text{Ker}(u)$, which is a group of multiplicative type, hence flat over S . Hence, by VIB 9.2 (xi), the morphism u is flat.

Corollary 6.2.5. *The following conditions are equivalent:*

- (i) G is the product of a semisimple group and a torus.
- (ii) $\mathrm{rad}(G) \times_S \mathrm{dér}(G) \xrightarrow{\sim} G$.
- (iii) $G \xrightarrow{\sim} \mathrm{corad}(G) \times_S \mathrm{ss}(G)$.
- (iv) $\mathrm{rad}(G) \cap \mathrm{dér}(G) = e$.

6.2.6. — Let us return temporarily to the case of a split group. Keep the notations of 6.1. Put $N = M \cap \mathcal{V}(R^*)^\perp$. One therefore has $T' = D_S(M/N)$. One has seen that $U^- \cdot T' \cdot U$ was an open neighborhood of the identity section of $\mathrm{dér}(G)$. One therefore has

$$\underline{\mathrm{Lie}}(\mathrm{dér}(G)/S) = \mathfrak{t}' \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha.$$

Since the characters induced on T' by the $\alpha \in R$ are nonzero and distinct (cf. Exp. XXI 1.2.5—one has moreover already used this fact in 6.1.2), R is a system of roots of $\mathrm{dér}(G)$ with respect to T' . It is then immediate (for $U_\alpha \subset \mathrm{dér}(G)$) that the exp morphisms of $\mathrm{dér}(G)$ “are” those of G , and likewise for the coroots.

It follows:

Proposition 6.2.7. *In the preceding notations, $(\mathrm{dér}(G), T', M/N, R)$ is a split group with root datum $\mathrm{dér}(\mathcal{R}(G))$. The canonical morphism $\mathrm{dér}(G) \rightarrow G$ gives by functoriality the canonical morphism of root data $\mathcal{R}(G) \rightarrow \mathrm{dér}(\mathcal{R}(G))$ of Exp. XXI 6.5.*

N.B. The reader may as an exercise construct the diagram of split groups corresponding to the three left-hand columns of the diagram of root data of Exp. XXI 6.5.7.

Proposition 6.2.8. *Let S be a scheme, G a reductive S -group, $\mathrm{dér}(G)$ its derived group.*

- (i) *For every maximal torus T of G , $T \cap \mathrm{dér}(G)$ is a maximal torus of $\mathrm{dér}(G)$. For every maximal torus T' of $\mathrm{dér}(G)$, $\underline{\mathrm{Centr}}_G(T') = \mathrm{rad}(G) \cdot T'$ is a maximal torus of G . The two preceding constructions are mutually inverse and establish a bijective correspondence between maximal tori of G and of $\mathrm{dér}(G)$.*
- (ii) *For every Borel subgroup B of G , $B \cap \mathrm{dér}(G)$ is a Borel subgroup B' of $\mathrm{dér}(G)$. One has $B'^u = B^u$. For every Borel subgroup B' of $\mathrm{dér}(G)$, $\underline{\mathrm{Norm}}_G(B') = \mathrm{rad}(G) \cdot B'$ is a Borel subgroup of G . The preceding maps are mutually inverse and establish a bijective correspondence between Borel subgroups of G and of $\mathrm{dér}(G)$.*

By the local conjugacy theorem for maximal tori and the construction of the derived group, the only assertion which remains to be proved in (i) is the following: if T is a maximal torus of G , then

$$T = (T \cap \mathrm{dér}(G)) \cdot \mathrm{rad}(G) = \underline{\mathrm{Centr}}_G(T \cap \mathrm{dér}(G)).$$

The first equality is trivial (for one reduces to the split case); the second follows at once, for $\mathrm{rad}(G)$ is central in G , hence $T = \underline{\mathrm{Centr}}_G(T) = \underline{\mathrm{Centr}}_G(T \cap \mathrm{dér}(G))$. One argues in the same way for (ii).

6.3. Subgroups with commutative quotients

6.3.1. — Let G be a reductive S -group. If H is a subsheaf of groups of G , the following conditions are equivalent:

- H contains $\mathrm{dér}(G)$.
- H is invariant and G/H is commutative.

In this case, the canonical morphism $f_0 : G \rightarrow \mathrm{corad}(G)$ maps H onto a subsheaf $f_0(H)$ of $\mathrm{corad}(G)$; one has

$$\begin{aligned} G/H &\simeq \mathrm{corad}(G)/f_0(H), & H/\mathrm{dér}(H) &\simeq f_0(H), \\ \mathrm{dér}(G) &= \mathrm{dér}(H), & H &= f_0^{-1}(f_0(H)). \end{aligned}$$

Since $\mathrm{dér}(G)$ is smooth over S and has connected fibers, then,⁽⁶⁴⁾ by Exp. IV, 5.3.1 and 6.3.1, and Exp. IVB 9.2, the map $H \mapsto f_0(H)$ establishes a bijective correspondence between group subschemes (resp. closed group subschemes) of G , containing $\mathrm{dér}(G)$, smooth over S and with connected fibers, and group subschemes (resp. closed group subschemes) of $\mathrm{corad}(G)$, smooth over S and with connected fibers.

Now, if H' is a group subscheme of $\mathrm{corad}(G)$, smooth over S with connected fibers, then H' is of finite presentation over S (Exp. VIB 5.5) and its fibers are tori (since those of $\mathrm{corad}(G)$ are), hence by Exp. X 8.2, H' is a subtorus of $\mathrm{corad}(G)$, hence is closed in $\mathrm{corad}(G)$ (Exp. IX 2.6).

Consequently, every subgroup of G , smooth with connected fibers and containing $\mathrm{dér}(G)$, is closed in G .⁽⁶⁴⁾

6.3.2. — If H is a closed group subscheme of G , smooth over S , with connected fibers and invariant in G , then H is reductive. If moreover $H \supset \mathrm{dér}(G)$, then $\mathrm{dér}(H) = \mathrm{dér}(G)$ and $f_0(H)$ is identified with $\mathrm{corad}(H)$. One has therefore proved the

Proposition 6.3.3. *Let G be a reductive S -group. Every group subscheme H of G , invariant in G , with commutative quotient (i.e. containing $\mathrm{dér}(G)$), smooth over S , with connected fibers,⁽⁶⁵⁾ is closed and reductive. One has $\mathrm{dér}(H) = \mathrm{dér}(G)$ and $f_0(H)$ is identified with $\mathrm{corad}(H)$; one has*

$$G/H \simeq \mathrm{corad}(G)/\mathrm{corad}(H), \quad H = (H \cap \mathrm{rad}(G)) \cdot \mathrm{dér}(G).$$

Moreover, $H \mapsto f_0(H)$ defines a bijection between the set of subgroups H of G having the preceding properties and the set of subtori of $\mathrm{corad}(H)$.

By a new application of Noether's theorem (Exp. IV, 5.3.1 and 6.3.1), one deduces the

Proposition 6.3.4. *Let S be a scheme, G a reductive S -group, T a maximal torus of G . For every subgroup H of G as above, $T \cap H$ is a maximal torus of G and one has*

$$G/H \simeq T/T \cap H, \quad H = (T \cap H) \cdot \mathrm{dér}(G).$$

Moreover, $H \mapsto T \cap H$ is a bijection between the set of subgroups H of G as above and the set of subtori of T containing $T \cap \mathrm{dér}(G)$.

⁽⁶⁴⁾**N.D.E.** The original has been expanded in what follows. In particular, the conclusion (implicit in the original) that every subgroup of G , smooth with connected fibers and containing $\mathrm{dér}(G)$, is closed in G , has been added.

⁽⁶⁵⁾**N.D.E.** The hypothesis that H is retrocompact in G has been deleted, since it holds automatically: by VIB 5.5, G and H are separated and quasi-compact over S , hence $H \hookrightarrow G$ is quasi-compact by EGA IV₁, 1.1.2 (v).

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Exposé XXIII. Reductive groups: uniqueness of pinned groups

⁽⁰⁾The aim of this exposé is to prove the uniqueness theorem (Theorem 4.1). This was proved by Chevalley in the case of an algebraically closed field; the method of reduction to rank two used here is also due to Chevalley (see *Bible*, exp. 23 and 24). Along the way, we obtain an explicit description of reductive groups by generators and relations (3.5).

1. Pinnings

Definition 1.1. Let S be a scheme, and (G, T, M, R) a split S -group (XXII, 1.13). A *pinning*⁽¹⁾ of this split group is the datum of a system Δ of *simple roots* of R and, for each $\alpha \in \Delta$, a section $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

In other words, a pinning of the reductive group G over the nonempty scheme S is the datum:

- (i) of a maximal torus T ,
- (ii) of an abelian group M and an isomorphism $T \simeq D_S(M)$,
- (iii) of a root system R of G with respect to T ,
- (iv) of a system of simple roots Δ of R ,
- (v) of an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ for every $\alpha \in \Delta$, that is, of a

$$u_\alpha = \exp_\alpha(X_\alpha) \in U_\alpha^\times(S) \quad \text{for every } \alpha \in \Delta,$$

satisfying condition (D 1) of Exp. XXII, 1.13 (indeed, condition (D 2) of *loc. cit.* is automatically satisfied⁽²⁾).

Every split group has a pinning; in particular, every reductive group is locally pinnable for the étale topology.

1.2. — If G is a *pinned* S -group, that is, a split S -group equipped with a pinning, it is canonically equipped with the system of positive roots R_+ defined by Δ , the corresponding Borel subgroup $B = B_{R_+}$, the opposite Borel subgroup $B^- = B_{R_-}$, the unipotent groups $U = B^u$, $U^- = (B^-)^u$, the open subset $U^- \cdot T \cdot U$, etc. Likewise, for each $\alpha \in \Delta$, one has a canonical isomorphism of vector groups

$$p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha, \quad x \longmapsto \exp_\alpha(xX_\alpha) = u_\alpha^x,$$

normalized by T with multiplier α , and whose datum is equivalent to that of X_α (Exp. XXII, 1.1).

By duality, one deduces from it an $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ and an isomorphism

$$p_{-\alpha} : \mathbf{G}_{a,S} \xrightarrow{\sim} U_{-\alpha}$$

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** Demazure tells us that behind this terminology lies the image of a butterfly (which Grothendieck supplied to him): the body is a maximal torus T , the wings are two opposite Borel subgroups with respect to T ; one splits the butterfly by spreading its wings, then fixes elements in the additive groups (pins) to rigidify the situation (i.e., to eliminate automorphisms).

⁽²⁾**N.D.E.** It is implied by condition (v), i.e., the existence of a section $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

which is the contragredient of the preceding one (Exp. XXII, 1.3). We shall put (Exp. XX, 3.1)

$$\begin{aligned} w_\alpha &= w_\alpha(X_\alpha) = p_\alpha(1)p_{-\alpha}(-1)p_\alpha(1) \\ &= p_{-\alpha}(-1)p_\alpha(1)p_{-\alpha}(-1). \end{aligned}$$

One then has (*loc. cit.* 3.1, 3.7)

$$\begin{aligned} w_\alpha^2 &= \alpha^*(-1), \quad \text{int}(w_\alpha)t = s_\alpha(t) = t \cdot \alpha^*(\alpha(t)^{-1}), \\ \begin{cases} \text{int}(w_\alpha)p_\alpha(x) = p_{-\alpha}(-x) = p_{-\alpha}(x)^{-1}, \\ \text{Ad}(w_\alpha)X_\alpha = -X_{-\alpha}, \end{cases} \\ \begin{cases} \text{int}(w_\alpha)p_{-\alpha}(x) = p_\alpha(-x) = p_\alpha(x)^{-1}, \\ \text{Ad}(w_\alpha)X_{-\alpha} = -X_\alpha. \end{cases} \end{aligned}$$

We shall systematically use the preceding notation in what follows.

Definition 1.3. Let S be a scheme, and $(G, T, M, R, \Delta, (X_\alpha))$ and $(G', T', M', \dots)$ two pinned S -groups. One says that the morphism of split groups (Exp. XXII, 4.2.1)

$$f : G \longrightarrow G'$$

is compatible with the pinnings, or defines a *morphism of pinned groups*, if the bijection $d : R \longrightarrow R'$ associated with it (cf. *loc. cit.*) satisfies $d(\Delta) = \Delta'$ and if, for every $\alpha \in \Delta$, one has

$$f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(X'_{d(\alpha)}), \quad \text{i.e.} \quad f(u_\alpha) = u'_{d(\alpha)}.$$

(3)

1.4. — If one denotes by $q(\alpha)$ the integer of *loc. cit.*, one therefore has

$$f(p_\alpha(x)) = p'_{d(\alpha)}(x^{q(\alpha)}) \quad \text{for } \alpha \in \Delta,$$

and hence also

$$f(p_{-\alpha}(x)) = p'_{-d(\alpha)}(x^{q(\alpha)}), \quad f(w_\alpha) = w'_{d(\alpha)}.$$

Recall (Exp. XXII, 4.2) that for every $\alpha \in R$, and all $z \in \mathbf{G}_m(S')$, $t \in T(S')$, one has:

$$\begin{aligned} f(\alpha^*(z)) &= (d(\alpha)^*(z))^{q(\alpha)} = d(\alpha)^*(z^{q(\alpha)}), \\ d(\alpha)(f(t)) &= \alpha(t)^{q(\alpha)}. \end{aligned}$$

1.5. — Let us call a *pinned root datum* a root datum equipped with a system of simple roots, and a *p-morphism of pinned root data* a p -morphism of root data (Exp. XXI, 6.8) taking simple roots to simple roots.

If G is a pinned S -group, one denotes by $\mathcal{R}(G)$ the corresponding pinned root datum (it is the root datum of Exp. XXII, 1.14 equipped with Δ). Let p be the integer defined in Exp. XXII, 4.2.2. One then has

Scholium 1.6. *The correspondence $G \mapsto \mathcal{R}(G)$ defines a functor from the category of pinned reductive S -groups to that of pinned root data (with p -morphisms as morphisms).*

1.7. — **The pinned groups Z_{Δ_1}**

Let Δ_1 be a subset of the system of simple roots Δ of the pinned group G . Let T_{Δ_1} be the maximal torus of $\bigcap_{\alpha \in \Delta_1} \text{Ker}(\alpha)$; put

$$Z_{\Delta_1} = \text{Centr}_G(T_{\Delta_1}).$$

Write $R' = \mathbf{Z} \cdot \Delta_1 \cap R$; one knows (Exp. XXII, 5.10.7) that Z_{Δ_1} is a reductive S -group with radical T_{Δ_1} , that (T, M, R') is a splitting of it and Δ_1 a system of simple roots. It follows

(3) **N.D.E.** We have denoted the bijection $R \xrightarrow{\sim} R'$ by d (instead of u), to avoid the notation $u'_{u(\alpha)}$.

that $(Z_{\Delta_1}, T, M, R', \Delta_1, (X_\alpha)_{\alpha \in \Delta_1})$ is a pinned S-group. We shall always equip Z_{Δ_1} with this pinning. In particular, one will consider the groups

$$Z_\alpha = Z_{\{\alpha\}}, \quad Z_{\alpha\beta} = Z_{\{\alpha, \beta\}}.$$

One writes $B_{\Delta_1} = B \cap Z_{\Delta_1}$; one knows (*loc. cit.*) that this is the canonical Borel subgroup⁽⁴⁾ of Z_{Δ_1} , and that its unipotent part is $U_{\Delta_1} = U \cap Z_{\Delta_1}$. In particular, one has

$$B_\alpha = T \cdot U_\alpha.$$

We shall write

$$U_{\alpha\beta} = U_{\{\alpha, \beta\}} = U \cap Z_{\alpha\beta} = \prod_{\gamma \in R_{\alpha\beta}^+} U_\gamma,$$

where $R_{\alpha\beta}^+$ denotes the set of positive roots that are linear combinations of α and β .

Now let $f : G \rightarrow G'$ be a morphism of pinned S-groups. If $d : R \rightarrow R'$ is the corresponding bijection and Δ_1 is a subset of Δ , then $d(\Delta_1) = \Delta'_1$ is a subset of Δ' , and it is clear that f sends T_{Δ_1} into $T'_{\Delta'_1}$, hence Z_{Δ_1} into $Z'_{\Delta'_1}$. The corresponding S-morphism:

$$f_{\Delta_1} : Z_{\Delta_1} \rightarrow Z'_{\Delta'_1}$$

is compatible with the canonical pinning; it defines a morphism of pinned root data

$$\mathcal{R}(f_{\Delta_1}) : \mathcal{R}(Z_{\Delta_1}, T, M, \dots) \rightarrow \mathcal{R}(Z'_{\Delta'_1}, T', M', \dots)$$

and the morphisms $M' \rightarrow M$ underlying $\mathcal{R}(f)$ and $\mathcal{R}(f_{\Delta_1})$ coincide.

1.8. — Study of the group $\text{Norm}_G(T)$

For each pair (α, β) of simple roots, denote by $n_{\alpha\beta}$ the order of the element $s_\alpha s_\beta$ of the Weyl group W . In particular, one has $n_{\alpha\alpha} = 1$. Thus one has $(w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S)$.

Definition 1.8.1. For every $(\alpha, \beta) \in \Delta \times \Delta$, put

$$t_{\alpha\beta} = (w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S).$$

Moreover, put (Exp. XX, 3.1)

$$t_\alpha = t_{\alpha\alpha} = w_\alpha^2 = \alpha^*(-1) \in T(S).$$

Proposition 1.8.2. Let H be an S-group functor taking direct sums of schemes to products (for example a sheaf for the Zariski topology). Let

$$f_T : T \rightarrow H$$

be a group morphism and h_α ($\alpha \in \Delta$) elements of $H(S)$. For there to exist a group morphism (necessarily unique)

$$f_N : \text{Norm}_G(T) \rightarrow H$$

which induces f_T on T and such that $f(w_\alpha) = h_\alpha$ for $\alpha \in \Delta$, it is necessary and sufficient that the following two conditions hold:

(i) For every $\alpha \in \Delta$, one has

$$f_T(s_\alpha(t)) = h_\alpha f_T(t) h_\alpha^{-1}$$

for every $t \in T(S')$, $S' \rightarrow S$ (i.e., $f_T \circ s_\alpha = \text{int}(h_\alpha) \circ f_T$).

(ii) For every $(\alpha, \beta) \in \Delta \times \Delta$, one has

$$f_T(t_{\alpha\beta}) = (h_\alpha h_\beta)^{n_{\alpha\beta}}.$$

⁽⁴⁾**N.D.E.** I.e., the Borel subgroup of Z_{Δ_1} corresponding to $R'_+ = R' \cap R_+$.

PROOF. Indeed, equip (Sch) with the topology $\mathcal{T}$ generated by the pretopology whose covering families are the direct sums; the hypothesis of the statement says that H is a $\mathcal{T}$ -sheaf. Let L be the free group with generators $(m_\alpha)_{\alpha \in R}$ and L_1 the invariant subgroup generated by the elements $(m_\alpha m_\beta)^{n_{\alpha\beta}}$, $(\alpha, \beta) \in \Delta \times \Delta$. Let $g : L \rightarrow W$ be the morphism defined by $g(m_\alpha) = s_\alpha$; one knows (Exp. XXI, 5.1) that g induces an isomorphism $\bar{g}$ from L/L_1 to W . Let L act on T through g (or, equivalently, by $m_\alpha \cdot t = s_\alpha(t)$). Let L_S be the constant group defined by L ; consider the semidirect product $T \cdot L_S = N$ for the preceding action. One has a morphism of S -groups

$$h : T \cdot L_S = N \rightarrow \underline{\text{Norm}}_G(T)$$

unique such that $h(m_\alpha) = w_\alpha$, $h(t) = t$ for every $t \in T(S')$, $S' \rightarrow S$. Let N_1 be the invariant group subsheaf of N generated by the

$$t_{\alpha\beta}^{-1} \cdot (m_\alpha m_\beta)^{n_{\alpha\beta}}, \quad (\alpha, \beta) \in \Delta \times \Delta.$$

One obviously has $N_1 \subset \text{Ker } h$; consider the induced morphism

$$h_1 : N/N_1 \rightarrow \underline{\text{Norm}}_G(T).$$

Let us prove that h_1 is an *isomorphism*. Since h induces on T the canonical immersion, which is a monomorphism, the canonical morphism

$$T \rightarrow N/N_1$$

is also a monomorphism, hence induces an isomorphism from T to TN_1/N_1 . For the same reason h_1 induces an isomorphism from TN_1/N_1 to T ; to prove that h_1 is an isomorphism, it therefore suffices to check that the corresponding morphism

$$h_2 : N/TN_1 \rightarrow \underline{\text{Norm}}_G(T)/T$$

is an isomorphism. Now TN_1 is the invariant group $\mathcal{T}$ -subsheaf of N generated by T and the $(m_\alpha m_\beta)^{n_{\alpha\beta}}$, that is, the $\mathcal{T}$ -subsheaf generated by T and L_1 , that is, $T \cdot (L_1)_S$. The morphism h_2 thus identifies with the morphism

$$\bar{g} : L_S/(L_1)_S \rightarrow W_S$$

which is an isomorphism by construction.

The proof of 1.8.2 is now easy; the conditions are obviously necessary; let us prove that they are sufficient. Condition (i) shows that there exists a morphism

$$u : N \rightarrow H$$

such that $u(m_\alpha) = h_\alpha$ for $\alpha \in \Delta$, and $u|_T = f_T$. Condition (ii) says that u vanishes on N_1 , which immediately gives the result. $\square$

1.9. — Faithfulness of the functor $\mathcal{R}$

Proposition 1.9.1. *The functor $\mathcal{R}$ of 1.6 is faithful: if*

$$f, g : G \rightrightarrows G'$$

are two morphisms of pinned groups such that $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.

Indeed, f and g coincide on T , U_α ($\alpha \in \Delta$) and $U_{-\alpha}$ ($\alpha \in \Delta$); it therefore suffices to apply:

Lemma 1.9.2. *Let S be a scheme, G a pinned reductive S -group, and H an S -group presheaf, separated for fppf. Let*

$$f, g : G \rightrightarrows H$$

be two morphisms of S -groups. The following conditions are equivalent:

- (i) $f = g$.
- (ii) f and g coincide on T , on each U_α ($\alpha \in \Delta$), on each $U_{-\alpha}$ ($\alpha \in \Delta$).
- (iii) f and g coincide on T , on each U_α ($\alpha \in \Delta$) and $f(w_\alpha) = g(w_\alpha)$ for each $\alpha \in \Delta$.

PROOF. Indeed, (i) $\Rightarrow$ (ii) is trivial, and (ii) $\Rightarrow$ (iii) follows immediately from the definition of w_α (1.2). It remains to prove (iii) $\Rightarrow$ (i). If $\alpha \in R$, there exists a sequence $\{\alpha_i\} \subset \Delta$ with $\alpha = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_n}(\alpha_{n+1})$, hence

$$U_\alpha = \text{int}(w_{\alpha_1} \cdots w_{\alpha_n}) U_{\alpha_{n+1}},$$

which proves that f and g coincide on each U_α . It follows that f and g coincide on Ω , hence coincide (Exp. XXII, 4.1.11). $\square$

Remark 1.9.3. If G is semisimple, one may, in (ii) and (iii), omit the hypothesis that f and g coincide on T . Indeed, G is generated as an fppf sheaf by the U_α , $\alpha \in R$ (Exp. XXII, 6.2.2 (a)).

2. Generators and relations for a pinned group

In this section, we fix a pinned S -group G . If $\alpha, \beta \in \Delta$, we shall use the notation Z_α , $Z_{\alpha\beta}$, $U_{\alpha\beta}$, $R_{\alpha\beta}^+$ of 1.7.

Theorem 2.1. *Let S be a scheme, G a pinned S -group, and H an S -group sheaf for fppf. Let*

$$f_N : \underline{\text{Norm}}_G(T) \longrightarrow H, \quad f_\alpha : U_\alpha \longrightarrow H, \quad \alpha \in R,$$

be group morphisms. For there to exist a group morphism (necessarily unique)

$$\bar{f} : G \longrightarrow H$$

extending f_N and the f_α ($\alpha \in R$), it is necessary and sufficient that the following conditions hold:

(i) *For every $\alpha \in \Delta$ and every $\beta \in R$, one has*

$$\text{int}(f_N(w_\alpha)) \circ f_\beta = f_{s_\alpha(\beta)} \circ \text{int}(w_\alpha).$$

(ii) *For every $\alpha \in \Delta$, there exists a group morphism*

$$F_\alpha : Z_\alpha \longrightarrow H$$

extending f_α , $f_{-\alpha}$ and $f_N|_{\underline{\text{Norm}}_{Z_\alpha}(T)}$.

(iii) *For every pair $(\alpha, \beta) \in \Delta \times \Delta$, there exists a group morphism $U_{\alpha\beta} \longrightarrow H$ inducing f_γ on U_γ for every $\gamma \in R_{\alpha\beta}^+$ (i.e., $U_\gamma \subset U_{\alpha\beta}$).*

2.1.1. — *Proof.* The conditions of the statement are obviously necessary. On the other hand, choose a totally ordered group structure on $\Gamma_0(R)$ so that the roots > 0 are the elements of R_+ (Exp. XXI, 3.5.6); every product indexed by a subset of R will be taken relative to this order. Denote by f_T the restriction of f_N to T and consider the morphism

$$f : \Omega \longrightarrow H$$

defined set-theoretically by

$$f \left(\prod_{\alpha \in R_-} y_\alpha \cdot t \cdot \prod_{\alpha \in R_+} x_\alpha \right) = \prod f_\alpha(y_\alpha) \cdot f_T(t) \cdot \prod f_\alpha(x_\alpha).$$

Every morphism satisfying the conditions of the statement must extend f ; on the other hand, every group morphism $f : G \longrightarrow H$ extending f also extends f_N : indeed, extending f_α and $f_{-\alpha}$, it satisfies $\bar{f}(w_\alpha) = F_\alpha(w_\alpha) = f_N(w_\alpha)$ and it extends f_T by hypothesis. By Exp. XXII, 4.1.11 (ii), one is therefore reduced to proving:

Proposition 2.1.2. *The morphism $f : \Omega \longrightarrow H$ defined above is “generically multiplicative”; more precisely, for every $S' \longrightarrow S$ and all $x, y \in \Omega(S')$ such that $xy \in \Omega(S')$, one has $f(xy) = f(x)f(y)$.*

Lemma 2.1.3. *Let $n \in \underline{\text{Norm}}_G(T)(S')$, $S' \longrightarrow S$, and let $\alpha, \beta \in R$ be such that $\text{int}(n)U_\alpha = U_\beta$ (i.e., $\bar{n}(\alpha) = \beta$); then one has*

$$\text{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \text{int}(n).$$

PROOF. Indeed, it suffices to check the formula for a system of generators of the sheaf $\underline{\text{Norm}}_G(T)$; it holds for each w_α , $\alpha \in \Delta$ (by 2.1 (i)), so it suffices to do so for $n \in T(S')$. This is trivial by 2.1 (ii) if α is simple; otherwise, one takes a $w \in W$ such that $w^{-1}(\alpha) \in \Delta$; writing w as a product of simple reflections,⁽⁵⁾ one is reduced to proving that if the formula holds for α and for every n , it also holds for $w_{\alpha_0}(\alpha)$ and $t \in T(S')$, where $\alpha_0 \in \Delta$. Now, by 2.1 (i), one has:

$$\begin{aligned} \text{int}(f_N(t)) \circ f_{w_{\alpha_0}(\alpha)} &= \text{int}(f_N(tw_{\alpha_0})) \circ f_\alpha \circ \text{int}(w_{\alpha_0}^{-1}) \\ &= f_{w_{\alpha_0}(\alpha)} \circ \text{int}(tw_{\alpha_0}) \circ \text{int}(w_{\alpha_0}^{-1}). \end{aligned}$$

□

Lemma 2.1.4. *The restriction of f to U (resp. U^-) is a group morphism. In particular, this restriction is independent of the order chosen on the roots.*

PROOF. It suffices to give the proof for U . By Exp. XXII, 5.5.8, it suffices to check that for every pair $\alpha < \beta$ of positive roots, one has, for all $x_\alpha \in U_\alpha(S')$, $x_\beta \in U_\beta(S')$, $S' \longrightarrow S$,

$$f_\beta(x_\beta)f_\alpha(x_\alpha)f_\beta(x_\beta^{-1}) = f(x_\beta x_\alpha x_\beta^{-1}).$$

According to Exp. XXII, 5.5.2, there exist $x_\gamma \in U_\gamma(S')$ ($\gamma = i\alpha + j\beta \in R$, $i > 0$, $j \geq 0$)⁽⁶⁾ such that

$$x_\beta x_\alpha x_\beta^{-1} = \prod_{\gamma} x_\gamma,$$

and one must therefore check the relation

$$f_\beta(x_\beta)f_\alpha(x_\alpha)f_\beta(x_\beta^{-1}) = \prod_{\substack{\gamma=i\alpha+j\beta \\ i>0, j\geq 0}} f_\gamma(x_\gamma).$$

By Exp. XXI, 3.5.4, there exists a $w \in W$ such that $w(\alpha) = \alpha_0 \in \Delta$, $w(\beta) \in A_{\alpha_0\beta_0}$ (notation of 1.7), where $\beta_0 \in \Delta$. Lifting w to an $n \in \underline{\text{Norm}}_G(T)(S)$ (by Exp. XXII, 3.8), it suffices to check the preceding relation after conjugation by $f_N(n)$. By 2.1.3, one is therefore reduced to the case where $\alpha, \beta \in R_{\alpha_0\beta_0}^+$, a case where one concludes by condition 2.1 (iii). □

Lemma 2.1.5. *Let $n \in \underline{\text{Norm}}_G(T)(S)$ be such that $\text{int}(n)U = U^-$ (i.e., $\bar{n}$ is the symmetry of the Weyl group⁽⁷⁾ (Exp. XXI, 3.6.14)). For every $u \in U(S')$, $S' \longrightarrow S$ (resp. $u \in U^-(S')$, $S' \longrightarrow S$), one has*

$$f(nun^{-1}) = f_N(n)f(u)f_N(n^{-1}).$$

PROOF. Immediate by 2.1.3 and 2.1.4. □

Lemma 2.1.6. *Let $u \in B(S')$, $v \in B^-(S')$, $g \in \Omega(S')$, $S' \longrightarrow S$. Then*

$$f(vgu) = f(v)f(g)f(u).$$

PROOF. Indeed, put $v = v_1t_1$, $g = v_2t_2u_2$, $u = t_3u_3$, with $v_i \in U^-(S')$, $t_i \in T(S')$, $u_i \in U(S')$. One has

$$f(v)f(g)f(u) = f(v_1)f_T(t_1)f(v_2)f_T(t_2)f(u_2)f_T(t_3)f(u_3),$$

⁽⁵⁾**N.D.E.** We have replaced “fundamental symmetries” by “simple reflections”.

⁽⁶⁾**N.D.E.** We have corrected $i, j > 0$ to $i > 0$, $j \geq 0$ (since x_α occurs in the product on the right).

⁽⁷⁾**N.D.E.** With respect to Δ .

on the one hand, and

$$\begin{aligned} f(vgu) &= f(v_1 t_1 v_2 t_1^{-1} t_1 t_2 t_3 t_3^{-1} u_2 t_3 u_3) \\ &= f(v_1 \cdot t_1 v_2 t_1^{-1}) f_T(t_1 t_2 t_3) f(t_3^{-1} u_2 t_3 \cdot u_3). \end{aligned}$$

Using 2.1.4 to decompose the two outer terms of this last expression, one obtains

$$f(vgu) = f(v_1) f(t_1 v_2 t_1^{-1}) f_T(t_1 t_2 t_3) f(t_3^{-1} u_2 t_3) f(u_3).$$

One is then reduced to the obvious formulas

$$\begin{aligned} f(t_1 v_2 t_1^{-1}) &= f_T(t_1) f(v_2) f_T(t_1)^{-1}, \\ f(t_3^{-1} u_2 t_3) &= f_T(t_3)^{-1} f(u_2) f_T(t_3), \end{aligned}$$

which follow from the definition of f and from 2.1.3. $\square$

Lemma 2.1.7. *Let $\alpha \in \Delta$ and $g \in \Omega(S') \cap \text{int}(w_\alpha)^{-1} \Omega(S')$, $S' \rightarrow S$. Then*

$$f(w_\alpha g w_\alpha^{-1}) = f_N(w_\alpha) f(g) f_N(w_\alpha)^{-1}.$$

PROOF. Indeed, let $g \in \Omega(S')$, $S' \rightarrow S$. Write, by Exp. XXII, 5.6.8,

$$g = a x_{-\alpha} t x_\alpha b,$$

with $a \in U_{-\hat{\alpha}}(S')$, $x_{-\alpha} \in U_{-\alpha}(S')$, $t \in T(S')$, $x_\alpha \in U_\alpha(S')$, $b \in U_{\hat{\alpha}}(S')$. By 2.1.3, 2.1.4 and the fact that s_α permutes the positive roots $\neq \alpha$ (cf. Exp. XXI, 3.3.2), one has

$$\text{int}(w_\alpha) a \in U_{-\hat{\alpha}}(S'), \quad \text{int}(w_\alpha) b \in U_{\hat{\alpha}}(S')$$

and the formula to be proved holds for $g = a$ or $g = b$. By 2.1.6, one is therefore reduced to proving the desired assertion when $g = x_{-\alpha} t x_\alpha \in Z_\alpha(S')$. But then “everything takes place in Z_α ” and one concludes by condition (ii) of 2.1. $\square$

Lemma 2.1.8. *Let $n \in \underline{\text{Norm}}_G(T)(S)$. For every $S' \rightarrow S$ and every $g \in \Omega(S')$ such that $\text{int}(n)g \in \Omega(S')$, one has*

$$f(n g n^{-1}) = f_N(n) f(g) f_N(n)^{-1}.$$

PROOF. This is trivial if $n \in T(S)$ (by 2.1.3). The two sides of the preceding formula define morphisms from $\Omega \cap \text{int}(n)^{-1} \Omega$ to H ; to check that they coincide, it suffices to check that there exists an open subset V_n of Ω , containing the identity section, such that $\text{int}(n)V_n \subset \Omega$, and that $f \circ \text{int}(n)$ and $\text{int}(f_N(n)) \circ f$ coincide on V_n . By the structure of $\underline{\text{Norm}}_G(T)$, it suffices to prove that if the preceding assertion holds for an $n' \in \underline{\text{Norm}}_G(T)(S)$ and if $\alpha \in \Delta$, it holds for $n = n' w_\alpha$. Put

$$V_n = \Omega \cap \text{int}(w_\alpha)^{-1} V_{n'}.$$

One has $\text{int}(n)V_n \subset \text{int}(n')V_{n'} \subset \Omega$. If $g \in V_n(S')$, one has

$$\text{int}(n)g = \text{int}(n') \text{int}(w_\alpha)g.$$

Now $\text{int}(w_\alpha)g \in V_{n'}(S')$, so by hypothesis

$$f(\text{int}(n') \text{int}(w_\alpha)g) = \text{int}(f_N(n')) f(\text{int}(w_\alpha)g);$$

since $\text{int}(w_\alpha)g \in \Omega(S')$, one may apply 2.1.7, which gives

$$f(\text{int}(w_\alpha)g) = \text{int}(f_N(w_\alpha)) f(g),$$

and one concludes immediately. $\square$

Let us now prove 2.1.2. Let $x, x' \in \Omega(S')$; write as usual

$$x = vtu, \quad x' = v't'u',$$

whence

$$xx' = vt(uv')t'u'.$$

By 2.1.6 and the relation $U^-(S')\Omega(S')U(S') = \Omega(S')$, one is reduced to proving that if $uv' \in \Omega(S')$, one has $f(uv') = f(u)f(v')$. Let $n \in \underline{\text{Norm}}_G(T)(S')$ be as in 2.1.5 (ii). One has

$$\begin{aligned} f(u) &= f_N(n)^{-1}f(nun^{-1})f_N(n), \\ f(v') &= f_N(n)^{-1}f(nv'n^{-1})f_N(n), \end{aligned}$$

by *loc. cit.*, whence

$$f(u)f(v') = f_N(n)^{-1}f(nun^{-1})f(nv'n^{-1})f_N(n).$$

But $nun^{-1} \in U^-(S')$, $nv'n^{-1} \in U(S')$, so that

$$f(nun^{-1})f(nv'n^{-1}) = f((nun^{-1})(nv'n^{-1})) = f(nuv'n^{-1}),$$

which gives

$$f(u)f(v') = f_N(n)^{-1}f(nuv'n^{-1})f_N(n).$$

If $uv' \in \Omega(S')$, one may apply 2.1.8 and the proof is complete.

Remark 2.2. Instead of giving f_N , one may give a group morphism $f_T : T \rightarrow H$, and sections $h_\alpha \in H(S)$ ($\alpha \in \Delta$) satisfying the conditions of 1.8.2. One must then replace condition (ii) of the theorem by

(ii') There exists a group morphism $F_\alpha : Z_\alpha \rightarrow H$ which extends

$$f_\alpha, f_{-\alpha}, f_T \quad \text{and satisfies } F_\alpha(w_\alpha) = h_\alpha.$$

We shall now give the preceding theorem a more explicit form. Retain the preceding notation.

Theorem 2.3. *Let S be a scheme, and G a pinned S -group. Let H be an S -group sheaf for fppf,*

$$f_T : T \rightarrow H, \quad f_\alpha : U_\alpha \rightarrow H, \quad \alpha \in \Delta,$$

group morphisms, and

$$h_\alpha \in H(S), \quad (\alpha \in \Delta),$$

sections of H . For there to exist a group morphism

$$f : G \rightarrow H$$

(necessarily unique) inducing f_T and the f_α ($\alpha \in \Delta$) and satisfying $f(w_\alpha) = h_\alpha$ for every $\alpha \in \Delta$, it is necessary and sufficient that the following conditions hold:

- (i) *For every $S' \rightarrow S$, every $t \in T(S')$, every $\alpha \in \Delta$ and every $x \in U_\alpha(S')$, one has*

$$(1) \quad f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(\text{int}(t)x).$$
- (ii) *For every $\alpha \in \Delta$, every $S' \rightarrow S$, every $t \in T(S')$, one has*

$$(2) \quad h_\alpha f_T(t)h_\alpha^{-1} = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^* \alpha(t)^{-1}).$$
- (iii) *For every $(\alpha, \beta) \in \Delta \times \Delta$, one has*

$$(3) \quad (h_\alpha h_\beta)^{n_{\alpha\beta}} = f_T(t_{\alpha\beta}).$$
- (iv) *For every $\alpha \in \Delta$, one has (recall that one writes $u_\alpha = \exp_\alpha(X_\alpha)$)*

$$(4) \quad (h_\alpha f_\alpha(u_\alpha))^3 = e.$$
- (v) *For every $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$, there exists a group morphism*

$$f_{\alpha\beta} : U_{\alpha\beta} \rightarrow H$$

satisfying the following two conditions:

a) One has

$$(5) \quad f_{\alpha\beta}|_{U_\alpha} = f_\alpha, \quad f_{\alpha\beta}|_{U_\beta} = f_\beta,$$

b) For every $\gamma \in R_{\alpha\beta}^+$, $\gamma \neq \alpha$ (resp. $\gamma \neq \beta$), and every $x \in U_\gamma(S')$, $S' \rightarrow S$, one has

$$(6) \quad \begin{aligned} \text{int}(h_\alpha)f_{\alpha\beta}(x) &= f_{\alpha\beta}(\text{int}(w_\alpha)x), \\ (\text{resp. } \text{int}(h_\beta)f_{\alpha\beta}(x) &= f_{\alpha\beta}(\text{int}(w_\alpha)x)). \end{aligned}$$

Proof. Uniqueness is clear by 1.9.2. Let us prove existence.

Lemma 2.3.1. *There exists a group morphism*

$$f_N : \underline{\text{Norm}}_G(T) \rightarrow H$$

extending f_T and satisfying $f_N(w_\alpha) = h_\alpha$.

Indeed, this is what 1.8.2 asserts, taking conditions (2) and (3) into account.

Lemma 2.3.2. *There exists a group morphism*

$$F_\alpha : Z_\alpha \rightarrow H, \quad \alpha \in \Delta,$$

extending f_T , f_α and satisfying $F_\alpha(w_\alpha) = h_\alpha$, hence extending $f_N|_{\underline{\text{Norm}}_{Z_\alpha}(T)}$.

This is clear by Exp. XX, 6.2 and conditions (1), (3) and (4).

Lemma 2.3.3. *For every $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$, every $n \in \underline{\text{Norm}}_{Z_{\alpha\beta}}(T)(S)$ such that $\bar{n}(\alpha) = \alpha$ (resp. $\bar{n}(\alpha) = \beta$), i.e., $\text{int}(n)U_\alpha = U_\alpha$ (resp. $\text{int}(n)U_\alpha = U_\beta$), every $S' \rightarrow S$ and every $x \in U_\alpha(S')$, one has*

$$\begin{aligned} \text{int}(f_N(n))f_\alpha(x) &= f_\alpha(\text{int}(n)x) \\ \text{resp. } \text{int}(f_N(n))f_\alpha(x) &= f_\beta(\text{int}(n)x). \end{aligned}$$

PROOF. Indeed, there exist a $t \in T(S)$ and a sequence $\{\alpha_i\} \subset \{\alpha, \beta\}$ such that $n = tw_{\alpha_k} \cdots w_{\alpha_1}$. The condition holds for $n = t$ by condition (1). One may therefore suppose $n = w_{\alpha_k} \cdots w_{\alpha_1}$. Let us give the proof by induction on k , i.e., suppose the assertion proved for every n which can be written as a product of fewer than $k - 1$ simple reflections (and satisfies the hypotheses of the lemma). Consider the roots

$$\gamma_i = s_{\alpha_i} \cdots s_{\alpha_1}(\alpha).$$

If all the γ_i are positive, i.e., belong to $R_{\alpha\beta}^+$, one may apply condition (v) of 2.3; adopting the notation of 2.3 (v), one concludes immediately by induction that

$$\text{int}(f_N(w_{\alpha_i} \cdots w_{\alpha_1}))f_\alpha(x) = f_{\alpha\beta}(\text{int}(w_{\alpha_i} \cdots w_{\alpha_1})x),$$

which, for $i = k$, is the desired property. If not all the γ_i are positive, there exists an index $j < k$ such that

$$\gamma_j \in R_+, \quad \gamma_{j+1} \notin R_+.$$

Since $\gamma_{j+1} = s_{\alpha_j}(\gamma_j)$, it follows immediately that $\gamma_j = \alpha_j$, by Exp. XXI, 3.3.2, hence that γ_j is α or β , and one may decompose n as

$$\begin{aligned} n &= n' \cdot n'', \\ n' &= w_{\alpha_k} \cdots w_{\alpha_{j+1}}, \\ n'' &= w_{\alpha_j} \cdots w_{\alpha_1}, \end{aligned}$$

n' and n'' satisfying the hypotheses of the lemma and being products of fewer than $k - 1$ reflections, hence satisfying the conclusion of the lemma by the induction hypothesis. $\square$

Lemma 2.3.4. *Let $\alpha \in R$. If $n, n' \in \underline{\text{Norm}}_G(T)(S)$ and $\beta, \beta' \in \Delta$ satisfy $\bar{n}(\alpha) = \beta$ and $\bar{n}'(\alpha) = \beta'$, one has*

$$\text{int}(f_T(n)^{-1})f_\beta(nxn^{-1}) = \text{int}(f_T(n')^{-1})f_{\beta'}(n'xn'^{-1}),$$

for every $x \in U_\alpha(S')$, $S' \rightarrow S$.

PROOF. It suffices to check that if $\bar{n}(\alpha) = \beta$, $\alpha, \beta \in \Delta$, then $\text{int}(f_T(n)) \circ f_\alpha = f_\beta \circ \text{int}(n)$. Now, by Tits's lemma (Exp. XXI, 5.6), there exist a sequence of simple roots $\alpha_0 = \alpha$, $\alpha_1, \dots, \alpha_m = \beta$, and a sequence of elements $w_i \in W$, $i = 0, 1, \dots, m-1$, with

$$\begin{aligned} \bar{n} &= w_{m-1} \cdots w_0, \\ w_i(\alpha_i) &= \alpha_{i+1}, \quad i = 0, 1, \dots, m-1, \end{aligned}$$

the following condition being moreover satisfied: for every $i = 0, 1, \dots, m-1$, there exists a simple root β_i such that

$$w_i \in W_{\alpha_i \beta_i}, \quad \alpha_{i+1} = \alpha_i \quad \text{or} \quad \beta_i.$$

One is then reduced by induction to the case treated in the preceding lemma. $\square$

Lemma 2.3.5. *There exists a family of group morphisms $f'_\gamma : U_\gamma \rightarrow H$, $\gamma \in R$, satisfying*

- (i) *For $\alpha \in \Delta$, one has $f'_\alpha = f_\alpha$ and $f'_{-\alpha} = F_\alpha|_{U_{-\alpha}}$, where F_α is defined in 2.3.2.*
- (ii) *For $\alpha, \beta \in \Delta$ and $\gamma \in R_{\alpha\beta}^+$, one has $f'_\gamma = f_{\alpha\beta}|_{U_\gamma}$.*
- (iii) *For every $n \in \text{Norm}_G(T)(S)$ and $\alpha, \beta \in R$ such that $\bar{n}(\alpha) = \beta$, one has*

$$\text{int}(f_N(n))f'_\alpha(x) = f'_\beta(\text{int}(n)x)$$

for every $x \in U_\alpha(S')$, $S' \rightarrow S$.

PROOF. For every root $\alpha \in R$, there exists an $n \in \text{Norm}_G(T)(S)$ such that $\bar{n}(\alpha) \in \Delta$. Define $f'_\alpha(x)$ to be the expression of 2.3.4. This is independent of the choice of n and f'_α is indeed a group morphism. Property (iii) holds by construction. The first assertion of (i) is clear (take $n = e$), and so is the second (take $n = w_\alpha$ and apply 2.3.2); if $\gamma \in R_{\alpha\beta}^+$, $(\alpha, \beta \in \Delta)$, there exists $n \in \text{Norm}_{Z_{\alpha\beta}}(T)(S)$ such that $\bar{n}(t) = \alpha$ or β ; one then applies (iii) and conditions (5) and (6) and has proved (ii). $\square$

2.3.6. — Let us now prove the theorem by proving that the conditions of 2.1 hold for the morphisms f_N and f'_α , $\alpha \in R$.

2.1 (i) follows from 2.3.5 (iii),

2.1 (ii) follows from 2.3.5 (i) and 2.3.2,

2.1 (iii) follows from 2.3.5 (ii) and the fact that $f_{\alpha\beta}$ is a group morphism.

An obvious corollary is:

Corollary 2.4. *Let S be a scheme, G a pinned S -group of semisimple rank ≥ 1 , and H an fppf S -group sheaf. For each $(\alpha, \beta) \in \Delta \times \Delta$, let*

$$F_{\alpha\beta} : Z_{\alpha\beta} \rightarrow H$$

be a group morphism. For there to exist a group morphism

$$f : G \rightarrow H$$

inducing the $F_{\alpha\beta}$, it is necessary and sufficient that for every $(\alpha, \beta) \in \Delta \times \Delta$, one have

$$F_{\alpha\beta}|_{Z_\alpha} = F_{\alpha\alpha}.$$

PROOF. The condition is obviously necessary. Suppose it satisfied. Put $f_T = F_{\alpha\alpha}|_T$ (which does not depend on α , since $F_{\alpha\alpha}|_T = F_{\alpha\beta}|_T = F_{\beta\beta}|_T$). Put

$$p_\alpha = F_{\alpha\alpha}|_{U_\alpha}, \quad h_\alpha = F_{\alpha\alpha}(w_\alpha), \quad f_{\alpha\beta} = F_{\alpha\beta}|_{U_{\alpha\beta}}.$$

The conditions of 2.3 are obviously satisfied: (1), (2), (4) “are checked” in Z_α , (3), (5) and (6) in $Z_{\alpha\beta}$. There thus exists a morphism $f : G \rightarrow H$ extending f_T , the f_α , $\alpha \in \Delta$, and satisfying $f(w_\alpha) = h_\alpha$; it coincides with $F_{\alpha\beta}$ on generators of $Z_{\alpha\beta}$, hence satisfies $f|_{Z_{\alpha\beta}} = F_{\alpha\beta}$. $\square$

One also has the following technical corollary:

Corollary 2.5. *Let S be a scheme, G and G' two pinned S -groups of semisimple rank 2, q an integer > 0 such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$, and $h : \mathcal{R}(G') \rightarrow \mathcal{R}(G)$*

a q -morphism of pinned root data. For each $\alpha \in R_+$, choose an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and an $X'_{d(\alpha)} \in \Gamma(S, \mathfrak{g}^{t_{d(\alpha)}})^\times$ (extending the canonical choices for $\alpha \in \Delta$). Suppose the following conditions hold:

- (i) If $\Delta = \{\alpha, \beta\}$, one has $D_S(h)(t_{\alpha\beta}) = t'_{d(\alpha)d(\beta)}$.
- (ii) For every $\alpha \in \Delta$ and every $\beta \in R_+$, $\beta \neq \alpha$ (whence $s_\alpha(\beta) \in R_+$), if $z \in \mathbf{G}_m(S)$ is defined by

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{s_\alpha(\beta)},$$

one also has

$$\mathrm{Ad}(w'_{d(\alpha)})X'_{d(\beta)} = z^{q(\beta)}X'_{d(s_\alpha(\beta))}.$$

- (iii) There exists a group morphism $f : U \rightarrow U'$ such that for every $\alpha \in R_+$ one has, for every $x \in \mathbf{G}_a(S')$, $S' \rightarrow S$,

$$f(\exp(xX_\alpha)) = \exp(x^{q(\alpha)}X'_{d(\alpha)}).$$

Then there exists a morphism of pinned groups $G \xrightarrow{g} G'$ such that $\mathcal{R}(g) = h$.

PROOF. Indeed, one defines $f_\alpha : U_\alpha \rightarrow G'$ by

$$f_\alpha(\exp(xX_\alpha)) = \exp(x^{q(\alpha)}X'_{d(\alpha)});$$

one puts $f_T = D_S(h)$, $h_\alpha = w'_{d(\alpha)}$. The conditions of 2.3 are satisfied (observe that $q(s_\alpha(\beta)) = q(\beta)$ (Exp. XXI, 6.8.4) and that one always has $D_S(h)(t_\alpha) = t'_{d(\alpha)}$) and one concludes immediately. $\square$

Remark 2.6. One may make condition (v) of 2.3 more precise as follows. One must first check:

(a) For every word in w_α and w_β such that the corresponding word takes α or β to α or β , the corresponding relation of type 2.3.3 holds. In fact the proof of 2.3.3 shows that it suffices to check it for the words in w_α and w_β which are minimal (in the sense that every nontrivial initial subword fails to satisfy the imposed conditions).

If condition (a) holds, one may now define, for each $\gamma \in R_{\alpha\beta}^+$, an $f_\gamma : U_\gamma \rightarrow H$ as in 2.3.5; one must then write:

(b) The morphism defined by the f_γ

$$U_{\alpha\beta} = \prod_{\gamma \in R_{\alpha\beta}^+} U_\gamma \rightarrow H$$

is a group morphism. According to Exp. XXII, 5.5.8, (b) is implied by:

(b') The preceding morphism respects the commutation relations between U_γ and U_δ for $\gamma, \delta \in R_{\alpha\beta}^+$, $\gamma > \delta$ (i.e., the relations in $C_{i,j,\gamma,\delta}$ of Exp. XII, 5.5.2).

It is clear that, conversely, the collection of conditions (a) and (b') is equivalent to (v).

One may even reduce the preceding conditions to conditions involving only the elements h_α , h_β , $f_\alpha(u_\alpha)$, $f_\beta(u_\beta)$ of H . A condition of type (a) will be written, for example, if $s_\alpha s_\beta s_\alpha(\beta) = \alpha$:

$$(1) \quad \mathrm{int}(h_\alpha h_\beta h_\alpha) f_\beta(x) = f_\alpha(\mathrm{int}(w_\alpha w_\beta w_\alpha)x);$$

for every $x \in U_\alpha(S')$, $S' \rightarrow S$. In particular, for $x = u_\beta$, one has $\mathrm{int}(w_\alpha w_\beta w_\alpha)u_\beta = u_\alpha^z$ for a certain section z of $\mathbf{G}_m(S)$, and the preceding relation gives

$$(1') \quad \mathrm{int}(h_\alpha h_\beta h_\alpha) f_\beta(u_\beta) = f_\alpha(u_\alpha)^z.$$

Let us show that, conversely, taking conditions (i) to (iv) of 2.3 into account, (1') implies (1). If $t \in T(S')$, $S' \rightarrow S$, let $\mathrm{int}(t)$ act on (1'); taking conditions (i) and (ii) into account, one obtains (1) for $x = \mathrm{int}(t)u_\beta = u_\beta^{\beta(t)}$. It now suffices to observe that $\beta : T \rightarrow \mathbf{G}_{m,S}$ is faithfully flat, hence that condition (1) certainly holds for $x \in U_\alpha(S')^\times$, $S' \rightarrow S$. Since it is additive in x and every section of U_α can locally be written as a sum of two sections of $U_\alpha^\times$, one does indeed conclude that (1') + (i) + (ii) $\implies$ (1).

One argues in the same way with conditions of type (b). One must then use the fact that if γ and γ' are two distinct positive roots (and hence linearly independent over $\mathbf{Q}$), the morphism $T \rightarrow \mathbf{G}_{m,S}^2$ with components γ and γ' is faithfully flat. We leave the details of this transposition to the reader.

3. Groups of semisimple rank 2

3.1. — Generalities

Lemma 3.1.1. *Let S be a scheme, G a pinned S -group, and α and β two roots of G , with $\alpha + \beta \neq 0$.*

(i) *If $\alpha + \beta \notin R$, one has*

$$\exp(X_\alpha) \exp(X_\beta) = \exp(X_\beta) \exp(X_\alpha)$$

for all $X_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)(S')$, $X_\beta \in \mathbf{W}(\mathfrak{g}^\beta)(S')$, $S' \rightarrow S$.

(ii) *If $\alpha + \beta$ and $\alpha - \beta$ are not roots, one has*

$$w_\alpha(z_\alpha) \exp(X_\beta) w_\alpha(z_\alpha)^{-1} = \exp(X_\beta)$$

for all $X_\beta \in \mathbf{W}(\mathfrak{g}^\beta)(S')$, $z_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$, and

$$w_\alpha(z_\alpha) w_\beta(z_\beta) = w_\beta(z_\beta) w_\alpha(z_\alpha)$$

for all $z_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$ and $z_\beta \in \mathbf{W}(\mathfrak{g}^\beta)^\times(S')$, $S' \rightarrow S$.

(iii) *Let $X_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$, $X_\beta \in \mathbf{W}(\mathfrak{g}^\beta)^\times(S')$, and $w \in \underline{\text{Norm}}_G(T)(S')$ be such that $\bar{w}(\alpha) = \beta$; define $z \in \mathbf{G}_m(S')$ by*

$$\text{Ad}(w)X_\alpha = zX_\beta.$$

Then

$$\text{int}(w) \exp(xX_\alpha) = \exp(xzX_\beta),$$

$$\text{int}(w) \exp(yX_\alpha^{-1}) = \exp(yz^{-1}X_\beta^{-1}),$$

$$\text{int}(w)w_\alpha(X_\alpha) = \beta^*(z)w_\beta(X_\beta).$$

In particular, if $z = \pm 1$, one has

$$\text{int}(w)w_\alpha(X_\alpha) = w_\beta(X_\beta)^z.$$

(iv) *If one puts $t_\alpha = \alpha^*(-1)$, $t_\beta = \beta^*(-1)$, one has*

$$s_\alpha(t_\beta) = t_\beta t_\alpha^{(\beta^*, \alpha)}, \quad \beta(t_\alpha) = (-1)^{(\alpha^*, \beta)}.$$

PROOF. (i) follows immediately from Exp. XXII, 5.5.2, (ii) from Exp. XX, 3.1 and from (i) applied to (β, α) , $(\beta, -\alpha)$, $(-\beta, \alpha)$, $(-\beta, -\alpha)$; (iii) is evident from the definitions; for the last assertion of (iii), observe that $\beta^*(-1) = w_\beta(X_\beta)^{-2}$. Finally, (iv) is trivial. $\square$

Proposition 3.1.2. (Groups of type $A_1 \times A_1$). *Let S be a scheme, G a pinned S -group of type $A_1 \times A_1$; write $\Delta = R_+ = \{\alpha, \beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^2 = t_\alpha t_\beta = (w_\beta w_\alpha)^2 = t_{\beta\alpha}.$$

(ii) *One has*

$$\text{Ad}(w_\alpha)X_\beta = X_\beta, \quad \text{Ad}(w_\beta)X_\alpha = X_\alpha.$$

(iii) *U_α and U_β commute (i.e., U is commutative).*

PROOF. Indeed, by assertion (ii) of the lemma, one has

$$w_\alpha w_\beta = w_\beta w_\alpha,$$

whence $(w_\alpha w_\beta)^2 = w_\alpha^2 w_\beta^2 = t_\alpha t_\beta$, which is (i). By assertion (ii) of the lemma, one also has (ii); finally, (iii) is assertion (i) of the lemma. $\square$

3.1.3. — Let us make condition (v) of 2.3 explicit here. Using the method explained in 2.6, one obtains the following two groups of conditions, putting $v_\alpha = f_\alpha(u_\alpha)$, for $\alpha \in \Delta$:

$$(A) \quad \begin{cases} h_\alpha v_\beta h_\alpha^{-1} = v_\beta, \\ h_\beta v_\alpha h_\beta^{-1} = v_\alpha, \end{cases} \quad (B) \quad v_\alpha v_\beta = v_\beta v_\alpha.$$

3.2. — Groups of type A_2

Proposition 3.2.1. *Let S be a scheme, G a pinned S -group of type A_2 ; write $\Delta = \{\alpha, \beta\}$, $R_+ = \{\alpha, \beta, \alpha + \beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^3 = e = (w_\beta w_\alpha)^3 = t_{\beta\alpha}.$$

(ii) *Put $\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}$. Then*

$$\begin{aligned} \text{Ad}(w_\alpha)X_\beta &= -X_{\alpha+\beta}, & \text{Ad}(w_\alpha)X_{\alpha+\beta} &= X_\beta, \\ \text{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha. \end{aligned}$$

(iii) *Put $p_{\alpha+\beta}(x) = \exp(xX_{\alpha+\beta}) = \text{int}(w_\beta)p_\alpha(x)$. One has:*

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy).$$

3.2.2. — The proof occupies nos. 3.2.2 to 3.2.7. First, one has $(\beta^*, \alpha) = (\alpha^*, \beta) = -1$, whence $\alpha(t_\beta) = \beta(t_\alpha) = -1$.

Put $\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}$; one immediately has

$$\text{Ad}(w_\beta)X_{\alpha+\beta} = \alpha(t_\beta)X_\alpha = -X_\alpha.$$

Put

$$\text{Ad}(w_\alpha)X_\beta = zX_{\alpha+\beta}, \quad z \in \mathbf{G}_m(S),$$

whence

$$\text{Ad}(w_\alpha)X_{\alpha+\beta} = \beta(t_\alpha)z^{-1}X_\beta = -z^{-1}X_\beta.$$

We know (Exp. XXII, 5.5.2) that there exists a unique section $A \in \mathbf{G}_a(S)$ such that

$$(+) \quad p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy).$$

Thus, to prove (ii) and (iii), the task is to show that $A = -z = 1$.

3.2.3. — Let $\text{int}(w_\beta)$ act on the preceding formula (+); one obtains:

$$(++) \quad p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy).$$

3.2.4. — By definition of $p_{\alpha+\beta}$, one has

$$w_\beta p_\alpha(x) w_\beta^{-1} = p_{\alpha+\beta}(x),$$

which can be written

$$\begin{aligned} p_\beta(1)p_{-\beta}(-1)p_\beta(1)p_\alpha(x)p_\beta(-1)p_{-\beta}(1)p_\beta(-1) \\ = p_{\alpha+\beta}(x). \end{aligned}$$

Since p_β and $p_{\alpha+\beta}$ commute, $\alpha + 2\beta$ not being a root, this can also be written

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Now using (+) on the left and (++) on the right, one obtains:

$$\begin{aligned} p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_\beta(-1) \\ = p_{\alpha+\beta}(x)p_{-\beta}(1)p_\alpha(Ax)p_{-\beta}(-1). \end{aligned}$$

Since $\alpha + 2\beta$ (resp. $\alpha - \beta$) is not a root, the left (resp. right) side can be written

$$p_\alpha(x)p_{\alpha+\beta}(Ax) \quad \text{resp.} \quad p_{\alpha+\beta}(x)p_\alpha(Ax)$$

and the term on the right equals $p_\alpha(Ax)p_{\alpha+\beta}(x)$, since $2\alpha + \beta$ is not a root. Thus

$$p_\alpha(x)p_{\alpha+\beta}(Ax) = p_\alpha(Ax)p_{\alpha+\beta}(x)$$

which gives (by XXII 4.1.3) $A = 1$.

3.2.5. — Now let $\text{int}(w_\alpha)$ act on formula (+); one finds, using the fact that $A = 1$,

$$(+++) \quad p_{\alpha+\beta}(zy)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(zy)p_{\beta}(-z^{-1}xy).$$

3.2.6. — Let us now write, as before,

$$w_\alpha p_\beta(y)w_\alpha^{-1} = p_{\alpha+\beta}(zy),$$

whence, since p_α and $p_{\alpha+\beta}$ commute,

$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{\alpha+\beta}(zy)p_{-\alpha}(-1).$$

Now using (+) and (+++), this can also be written

$$p_\beta(y)p_{\alpha+\beta}(-y) = p_{\alpha+\beta}(zy)p_{\beta}(-z^{-1}y),$$

and since p_β and $p_{\alpha+\beta}$ commute, this gives $z = -1$.

3.2.7. — We have thus proved (ii) and (iii); let us prove (i). One has

$$w_\alpha w_\beta w_\alpha = w_\alpha w_\beta w_\alpha^{-1} w_\alpha^2 = w_{\alpha+\beta}^{-1} t_\alpha,$$

whence

$$\begin{aligned} w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{\alpha+\beta}^{-1} t_\alpha w_\beta \\ &= w_\beta w_{\alpha+\beta}^{-1} w_\beta^{-1} \cdot s_\beta(t_\alpha) \cdot t_\beta \\ &= w_\alpha \cdot t_\alpha t_\beta \cdot t_\beta = w_\alpha t_\alpha = w_\alpha^{-1}, \end{aligned}$$

which gives

$$(w_\alpha w_\beta)^3 = (w_\beta w_\alpha)^3 = e,$$

which completes the proof.

Remark 3.2.8. Condition (v) of 2.3 is expressed here as follows (writing $v_\alpha = f_\alpha(u_\alpha)$):

$$\begin{aligned} \text{(A)} \quad & h_\alpha v_\beta^{-1} h_\alpha^{-1} = h_\beta v_\alpha h_\beta^{-1}, \\ \text{(B)} \quad & \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta \cdot h_\beta v_\alpha h_\beta^{-1}, \\ v_\beta \cdot h_\beta v_\alpha h_\beta^{-1} = h_\beta v_\alpha h_\beta^{-1} \cdot v_\beta, \\ v_\alpha \cdot h_\beta v_\alpha h_\beta^{-1} = h_\beta v_\alpha h_\beta^{-1} \cdot v_\alpha. \end{cases} \end{aligned}$$

Putting $v_{\alpha+\beta} = \text{int}(h_\beta)v_\alpha$, the last three conditions can also be written

$$\text{(B)} \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta}, \\ v_\alpha v_{\alpha+\beta} = v_{\alpha+\beta} v_\alpha, \\ v_\beta v_{\alpha+\beta} = v_{\alpha+\beta} v_\beta. \end{cases}$$

3.3. — Groups of type B_2

Proposition 3.3.1. *Let S be a scheme, G a pinned S -group of type B_2 ; write $\Delta = \{\alpha, \beta\}$, $R_+ = \{\alpha, \beta, \alpha + \beta, 2\alpha + \beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^4 = t_\alpha = (w_\beta w_\alpha)^4 = t_{\beta\alpha}.$$

(ii) *If one puts*

$$\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}, \quad \text{Ad}(w_\alpha)X_\beta = X_{2\alpha+\beta},$$

one has:

$$\begin{aligned} \text{Ad}(w_\alpha)X_{\alpha+\beta} &= -X_{\alpha+\beta}, \\ \text{Ad}(w_\alpha)X_{2\alpha+\beta} &= X_\beta, \\ \text{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha, \\ \text{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(xX_{\alpha+\beta}) = \text{int}(w_\beta)p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(xX_{2\alpha+\beta}) = \text{int}(w_\alpha)p_\beta(x). \end{aligned}$$

Then:

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(2xy). \end{aligned}$$

3.3.2. — The proof occupies nos. 3.3.2 to 3.3.6. One has $(\beta^*, \alpha) = -1$, $(\alpha^*, \beta) = -2$, whence $\alpha(t_\beta) = -1$, $\beta(t_\alpha) = 1$. Observe in passing that $\beta(t_\alpha) = \alpha(t_\alpha) = 1$, which shows that t_α is a section of $\text{Centr}(G)$. Put

$$\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}, \quad \text{Ad}(w_\alpha)X_\beta = X_{2\alpha+\beta}.$$

One immediately has

$$\begin{aligned} \text{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \text{Ad}(w_\alpha)X_{2\alpha+\beta} &= \beta(t_\alpha)X_\beta = X_\beta. \end{aligned}$$

Since $(2\alpha + \beta) + \beta$ and $(2\alpha + \beta) - \beta$ are not roots, one has

$$\text{Ad}(w_\beta)X_{2\alpha+\beta} = X_{2\alpha+\beta}.$$

There exists a scalar $k \in \mathbf{G}_m(S)$ such that

$$\text{Ad}(w_\alpha)X_{\alpha+\beta} = kX_{\alpha+\beta}.$$

On the other hand, by Exp. XXII, 5.5.2, there exist sections $A, B, C \in \mathbf{G}_a(S)$ such that

$$(1) \quad p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y),$$

$$(2) \quad p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Cxy).$$

Thus, in (ii) and (iii), the task is to prove $A = B = 1$, $C = 2$, $k = -1$.

3.3.3. — Let $\text{int}(w_\alpha)$ act on formula (2). One finds

$$(3) \quad p_{2\alpha+\beta}(y)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{2\alpha+\beta}(y)p_{\alpha+\beta}(Akxy)p_\beta(Bx^2y).$$

Transforming (2) in the same way, one obtains

$$(4) \quad p_{\alpha+\beta}(ky)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(ky)p_\beta(Cxy).$$

Transforming (1) by $\text{int}(w_\beta)$, one has

$$(5) \quad p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y).$$

3.3.4. — Let us write

$$w_\beta p_\alpha(X)w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Since $\alpha + 2\beta$ is not a root, this gives

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (1) on the left and (5) on the right, one obtains

$$\begin{aligned} p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_\beta(-1) \\ = p_{\alpha+\beta}(x)p_{-\beta}(1)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{-\beta}(-1). \end{aligned}$$

Since p_β commutes with $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ on the one hand, and $p_{-\beta}$ commutes with p_α and $p_{2\alpha+\beta}$ on the other hand, this can be written

$$p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) = p_{\alpha+\beta}(x)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2).$$

Transforming the right side by (2), one obtains

$$\begin{aligned} p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) \\ = p_\alpha(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AC - B)x^2), \end{aligned}$$

which gives

$$A = 1, \quad C = 2B.$$

3.3.5. — Let us now write

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{2\alpha+\beta}(y).$$

Since $3\alpha + \beta$ is not a root, this gives

$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{2\alpha+\beta}(y)p_{-\alpha}(-1).$$

Using (1) on the left and (3) on the right, one obtains

$$p_\beta(y)p_{\alpha+\beta}(-Ay)p_{2\alpha+\beta}(By) = p_{2\alpha+\beta}(y)p_{\alpha+\beta}(Aky)p_\beta(By).$$

Since $p_{\alpha+\beta}$, $p_{2\alpha+\beta}$ and p_β commute, this immediately gives

$$B = 1, \quad -A = Ak,$$

whence finally

$$A = 1, \quad B = 1, \quad C = 2, \quad k = -1.$$

3.3.6. — We have thus proved (ii) and (iii). Let us prove (i). Taking into account the equality $s_\beta(t_\alpha) = t_\alpha t_\beta^{(\alpha^*, \beta)} = t_\alpha$ (since $(\alpha^*, \beta) = 2$), one successively has:

$$\begin{aligned} w_\alpha w_\beta w_\alpha &= w_\alpha w_\beta w_\alpha^{-1} t_\alpha = w_{2\alpha+\beta} t_\alpha, \\ w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{2\alpha+\beta} w_\beta^{-1} \cdot s_\beta(t_\alpha) \cdot t_\beta = w_{2\alpha+\beta} t_\alpha t_\beta, \\ w_\alpha w_\beta w_\alpha w_\beta w_\alpha w_\beta w_\alpha &= w_\alpha w_{2\alpha+\beta} w_\alpha^{-1} \cdot s_\alpha(t_\alpha t_\beta) \cdot t_\alpha \\ &= w_\beta \cdot t_\alpha \cdot t_\beta t_\alpha \cdot t_\alpha = w_\beta^{-1} t_\alpha, \end{aligned}$$

whence

$$(w_\beta w_\alpha)^4 = t_\alpha,$$

and

$$(w_\alpha w_\beta)^4 = s_\beta(t_\alpha) = t_\alpha,$$

which completes the proof.

Remark 3.3.7. Condition (v) of 2.3 is expressed here in the following way, putting $v_{\alpha+\beta} = \text{int}(h_\beta)v_\alpha$ and $v_{2\alpha+\beta} = \text{int}(h_\alpha)v_\beta$:

$$\begin{aligned} \text{(A)} \quad & \begin{cases} \text{int}(h_\alpha h_\beta h_\alpha)v_\beta = v_\beta, \\ \text{int}(h_\beta h_\alpha h_\beta)v_\alpha = v_\alpha, \end{cases} \\ \text{(B)} \quad & \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta}, \\ v_{\alpha+\beta} v_\alpha = v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2, \\ v_{\alpha+\beta} v_\beta = v_\beta v_{\alpha+\beta}, \\ v_{2\alpha+\beta} v_\alpha = v_\alpha v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_\beta = v_\beta v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_{\alpha+\beta} = v_{\alpha+\beta} v_{2\alpha+\beta}. \end{cases} \end{aligned}$$

3.4. — Groups of type G_2

Proposition 3.4.1. *Let S be a scheme, G a pinned S -group of type G_2 ; write $\Delta = \{\alpha, \beta\}$, $R_+ = \{\alpha, \beta, \alpha + \beta, 2\alpha + \beta, 3\alpha + \beta, 3\alpha + 2\beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^6 = e = (w_\beta w_\alpha)^6 = t_{\beta\alpha}.$$

(ii) If one puts

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \operatorname{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, \\ \operatorname{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \operatorname{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, \end{aligned}$$

one has:

$$\begin{aligned} \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, & \operatorname{Ad}(w_\alpha)X_{3\alpha+\beta} &= X_\beta, \\ \operatorname{Ad}(w_\alpha)X_{3\alpha+2\beta} &= X_{3\alpha+2\beta}, & \operatorname{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha, \\ \operatorname{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}. \end{aligned}$$

(iii) If one puts

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(xX_{\alpha+\beta}) = \operatorname{int}(w_\beta)p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(xX_{2\alpha+\beta}) = \operatorname{int}(w_\alpha w_\beta)p_\alpha(x), \\ p_{3\alpha+\beta}(x) &= \exp(xX_{3\alpha+\beta}) = \operatorname{int}(w_\alpha)p_\beta(-x), \\ p_{3\alpha+2\beta}(x) &= \exp(xX_{3\alpha+2\beta}) = \operatorname{int}(w_\beta w_\alpha)p_\beta(-x), \end{aligned}$$

one has:

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y) \\ &\quad \cdot p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(2xy) \\ &\quad \cdot p_{3\alpha+\beta}(3x^2y)p_{3\alpha+2\beta}(3xy^2), \\ p_{2\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(3xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy), \\ p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) &= p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(3xy). \end{aligned}$$

3.4.2. — The proof occupies nos. 3.4.2 to 3.4.9. One has $(\beta^*, \alpha) = -1$, $(\alpha^*, \beta) = -3$, whence $\beta(t_\alpha) = \alpha(t_\beta) = -1$. Define $X_{\alpha+\beta}$, $X_{2\alpha+\beta}$, $X_{3\alpha+\beta}$ and $X_{3\alpha+2\beta}$ as in (ii). One immediately has

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= \alpha(t_\alpha)\beta(t_\alpha)X_{\alpha+\beta} = -X_{\alpha+\beta}, \\ \operatorname{Ad}(w_\alpha)X_{3\alpha+\beta} &= -\beta(t_\alpha)X_\beta = X_\beta, \\ \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= \alpha(t_\beta)^3\beta(t_\beta)X_{3\alpha+\beta} = -X_{3\alpha+\beta}. \end{aligned}$$

Finally, since $(3\alpha + 2\beta) \pm \alpha$ and $(2\alpha + \beta) \pm \beta$ are not roots, one has:

$$\operatorname{Ad}(w_\alpha)X_{3\alpha+2\beta} = X_{3\alpha+2\beta}, \quad \operatorname{Ad}(w_\beta)X_{2\alpha+\beta} = X_{2\alpha+\beta},$$

which completes the proof of (ii).

3.4.3. — By Exp. XXII, 5.5.2, there exist scalars⁽⁸⁾

$$A, B, C, D, E, F, G, H, J \in \mathbf{G}_a(S),$$

such that

$$(1) \quad \begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y) \\ &\quad p_{3\alpha+\beta}(Cx^3y)p_{3\alpha+2\beta}(Dx^3y^2). \end{aligned}$$

$$(2) \quad \begin{aligned} p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Exy) \\ &\quad p_{3\alpha+\beta}(Fx^2y)p_{3\alpha+2\beta}(Gxy^2). \end{aligned}$$

$$(3) \quad p_{2\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hxy).$$

⁽⁸⁾**N.D.E.** Absolute constants A, B, ..., J are introduced here. These constants will be determined in the following pages; their values are $A = B = C = D = 1$, $E = 2$, $J = -1$, $F = G = H = 3$, cf. 3.4.8.

$$(4) \quad p_{3\alpha+\beta}(y)p_{\beta}(x) = p_{\beta}(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(Jxy).$$

3.4.4. — Let $\text{int}(w_{\beta})$ act on (1), (3) and (4):

$$(5) \quad \begin{aligned} p_{-\beta}(-y)p_{\alpha+\beta}(x) &= p_{\alpha+\beta}(x)p_{-\beta}(-y)p_{\alpha}(-Axy) \\ p_{2\alpha+\beta}(Bx^2y)p_{3\alpha+2\beta}(Cx^3y)p_{3\alpha+\beta}(-Dx^3y^2). \end{aligned}$$

$$(6) \quad p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(Hxy).$$

$$(7) \quad p_{3\alpha+2\beta}(y)p_{-\beta}(-x) = p_{-\beta}(-x)p_{3\alpha+2\beta}(y)p_{3\alpha+\beta}(-Jxy).$$

Similarly letting $\text{int}(w_{\alpha})$ act on (1), one finds

$$(8) \quad \begin{aligned} p_{3\alpha+\beta}(-y)p_{-\alpha}(-x) &= p_{-\alpha}(-x)p_{3\alpha+\beta}(-y)p_{2\alpha+\beta}(Axy) \\ p_{\alpha+\beta}(-Bx^2y)p_{\beta}(Cx^3y)p_{3\alpha+\beta}(Dx^3y^2). \end{aligned}$$

3.4.5. — Let us write

$$w_{\beta}p_{\alpha}(x)w_{\beta}^{-1} = p_{\alpha+\beta}(x),$$

that is, $\alpha + 2\beta$ not being a root:

$$(9) \quad p_{\beta}(1)p_{\alpha}(x)p_{\alpha}(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Transforming the left side of (9) by (1), then (4), one obtains:

$$(10) \quad \begin{aligned} &p_{\beta}(1)p_{\alpha}(x)p_{\beta}(-1) \\ &= p_{\alpha}(x)p_{\beta}(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) \\ &\quad p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}(Dx^3)p_{\beta}(-1) \\ &= p_{\alpha}(x)p_{\beta}(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{\beta}(-1) \\ &\quad p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}((D - CJ)x^3) \\ &= p_{\alpha}(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{3\alpha+\beta}(Cx^3) \\ &\quad p_{3\alpha+2\beta}((D - CJ)x^3). \end{aligned}$$

Let us transform the right side of (9) by (5), then (7):

$$(11) \quad \begin{aligned} &p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1) \\ &= p_{\alpha+\beta}(x)p_{-\beta}(1)p_{\alpha}(Ax)p_{2\alpha+\beta}(-Bx^2) \\ &\quad p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}(-Dx^3)p_{-\beta}(-1) \\ &= p_{\alpha+\beta}(x)p_{\alpha}(Ax)p_{2\alpha+\beta}(-Bx^2) \\ &\quad p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}((CJ - D)x^3). \end{aligned}$$

Now using (2), this right side becomes

$$(12) \quad \begin{aligned} &p_{\alpha}(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}(AEx^2)p_{3\alpha+\beta}(A^2Fx^3)p_{3\alpha+2\beta}(AGx^3) \times \\ &\quad p_{2\alpha+\beta}(-Bx^2)p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}((CJ - D)x^3) \\ &= p_{\alpha}(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AE - B)x^2) \\ &\quad p_{3\alpha+\beta}((A^2F + CJ - D)x^3)p_{3\alpha+2\beta}((AG - C)x^3). \end{aligned}$$

Thus (9) can be rewritten:

$$\begin{aligned} &p_{\alpha}(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}((D - CJ)x^3) \\ &= p_{\alpha}(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AE - B)x^2) \\ &\quad p_{3\alpha+\beta}((A^2F + CJ - D)x^3)p_{3\alpha+2\beta}((AG - C)x^3) \end{aligned}$$

which gives

$$A = 1, \quad E = 2B, \quad C + D = F + CJ, \quad F = G.$$

3.4.6. — Let us now write

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{3\alpha+\beta}(-y),$$

that is, $4\alpha + \beta$ not being a root:

$$(13) \quad p_\alpha(1) p_\beta(y) p_\alpha(-1) = p_{-\alpha}(1) p_{3\alpha+\beta}(-y) p_{-\alpha}(-1).$$

Let us transform the left side by (1):

$$\begin{aligned} p_\alpha(1) p_\beta(y) p_\alpha(-1) &= p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) \\ &\quad p_{3\alpha+\beta}(-Cy) p_{3\alpha+2\beta}(-Dy^2). \end{aligned}$$

Let us transform the right side of (13) successively by (8), (6) and (4):

$$\begin{aligned} &p_{-\alpha}(1) p_{3\alpha+\beta}(-y) p_{-\alpha}(-1) \\ &= p_{3\alpha+\beta}(-y) p_{2\alpha+\beta}(Ay) p_{\alpha+\beta}(-By) p_\beta(Cy) p_{3\alpha+2\beta}(Dy^2) \\ &= p_{3\alpha+\beta}(-y) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) \\ &\quad p_{3\alpha+2\beta}(-ABHy^2) p_\beta(Cy) p_{3\alpha+2\beta}(Dy^2) \\ &= p_\beta(Cy) p_{3\alpha+\beta}(-y) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) \\ &\quad p_{3\alpha+2\beta}((D - CJ - ABH)y^2) \\ &= p_\beta(Cy) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) p_{3\alpha+\beta}(-y) \\ &\quad p_{3\alpha+2\beta}((D - CJ - ABH)y^2). \end{aligned}$$

Thus (13) can be rewritten:

$$\begin{aligned} &p_\beta(Cy) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) p_{3\alpha+\beta}(-y) p_{3\alpha+2\beta}((D - CJ - ABH)y^2) \\ &= p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) p_{3\alpha+\beta}(-Cy) p_{3\alpha+2\beta}(-Dy^2) \end{aligned}$$

whence

$$C = 1, \quad A = B, \quad D - CJ - ABH = -D.$$

Let us take into account the results already obtained:

$$\begin{aligned} A = B = C = 1, \quad E = 2, \quad F = G, \\ D + 1 = F + J, \quad 2D = H + J. \end{aligned}$$

3.4.7. — Let us write

$$w_\beta p_{3\alpha+\beta}(x) w_\beta^{-1} = p_{3\alpha+2\beta}(x),$$

that is

$$p_\beta(1) p_{3\alpha+\beta}(x) p_\beta(-1) = p_{-\beta}(1) p_{3\alpha+2\beta}(x) p_{-\beta}(-1).$$

Transforming the left side by (4), and the right side by (7), one obtains:

$$p_{3\alpha+\beta}(x) p_{3\alpha+2\beta}(-Jx) = p_{3\alpha+2\beta}(x) p_{3\alpha+\beta}(-Jx),$$

that is, $J = -1$.

3.4.8. — Finally, let us write

$$w_\alpha p_{\alpha+\beta}(y) w_\alpha^{-1} = p_{2\alpha+\beta}(y),$$

that is

$$\begin{aligned} &p_\alpha(1) p_{\alpha+\beta}(y) p_\alpha(-1) \\ &= p_{-\alpha}(1) p_\alpha(-1) p_{2\alpha+\beta}(y) p_\alpha(1) p_{-\alpha}(-1). \end{aligned}$$

Transforming the left side by (2), and the right side by (3), one obtains:

$$\begin{aligned} &p_{\alpha+\beta}(y) p_{2\alpha+\beta}(-Ey) p_{3\alpha+\beta}(Fy) p_{3\alpha+2\beta}(-Gy^2) \\ &= p_{-\alpha}(1) p_{2\alpha+\beta}(y) p_{3\alpha+\beta}(Hy) p_{-\alpha}(-1). \end{aligned}$$

It is immediate to see that if one commutes $p_{-\alpha}(-1)$ with $p_{3\alpha+\beta}(Hy)$, then $p_{2\alpha+\beta}(y)$, one introduces no new terms in $p_{3\alpha+\beta}$ on the right side. The latter can thus be written, denoting by empty parentheses the quantities whose exact value does not matter to us:

$$p_{\alpha+\beta}(\)p_{2\alpha+\beta}(\)p_{\beta}(\)p_{3\alpha+\beta}(Hy)p_{3\alpha+2\beta}(\).$$

Comparing with the left side, one immediately has $F = H$, whence, by the earlier results, $2D = D + 1$, that is, $D = 1$, and finally $F = G = H = 2D - J = 3$, which completes the determination of the coefficients $A, \dots, J$ and the proof of (iii).

3.4.9. — Finally, let us prove (i) in the usual manner. One successively has:

$$\begin{aligned} w_{\alpha}w_{\beta}w_{\alpha} &= w_{\alpha}w_{\beta}w_{\alpha}^{-1}t_{\alpha} = w_{3\alpha+\beta}^{-1} \cdot t_{\alpha}, \\ w_{\beta}(w_{\alpha}w_{\beta})^2 &= w_{\beta}w_{3\alpha+\beta}^{-1}w_{\beta}^{-1} \cdot s_{\beta}(t_{\alpha}) \cdot t_{\beta} \\ &= w_{3\alpha+2\beta}^{-1} \cdot t_{\alpha}t_{\beta} \cdot t_{\beta} = w_{3\alpha+2\beta}^{-1} \cdot t_{\alpha}, \\ w_{\alpha}(w_{\beta}w_{\alpha})^3 &= w_{\alpha}w_{3\alpha+2\beta}^{-1}w_{\alpha}^{-1} = w_{3\alpha+2\beta}^{-1}, \\ w_{\beta}(w_{\alpha}w_{\beta})^4 &= w_{\beta}w_{3\alpha+2\beta}^{-1}w_{\beta}^{-1} \cdot t_{\beta} = w_{3\alpha+\beta} \cdot t_{\beta}, \\ w_{\alpha}(w_{\beta}w_{\alpha})^5 &= w_{\alpha}w_{3\alpha+\beta}w_{\alpha}^{-1} \cdot s_{\alpha}(t_{\beta}) \cdot t_{\beta} \\ &= w_{\beta} \cdot t_{\beta}t_{\alpha} \cdot t_{\alpha} = w_{\beta}^{-1}. \end{aligned}$$

Whence

$$(w_{\alpha}w_{\beta})^6 = (w_{\beta}w_{\alpha})^6 = e.$$

Remark 3.4.10. Condition (v) of 2.4 consists of

$$(A) \quad \begin{cases} \text{int}(h_{\alpha}h_{\beta}h_{\alpha}h_{\beta}h_{\alpha})v_{\beta} = v_{\beta}, \\ \text{int}(h_{\beta}h_{\alpha}h_{\beta}h_{\alpha}h_{\beta})v_{\alpha} = v_{\alpha}, \end{cases}$$

and, putting

$$\begin{aligned} v_{\alpha+\beta} &= \text{int}(h_{\beta})v_{\alpha}, & v_{2\alpha+\beta} &= \text{int}(h_{\alpha}h_{\beta})v_{\alpha}, \\ v_{3\alpha+\beta} &= \text{int}(h_{\alpha})v_{\beta}^{-1}; & v_{3\alpha+2\beta} &= \text{int}(h_{\beta}h_{\alpha})v_{\beta}^{-1}, \end{aligned}$$

of the commutation relations:

$$(B) \quad \begin{cases} v_{\beta}v_{\alpha} = v_{\alpha}v_{\beta}v_{\alpha+\beta}v_{2\alpha+\beta}v_{3\alpha+\beta}v_{3\alpha+2\beta}, \\ v_{\alpha+\beta}v_{\alpha} = v_{\alpha}v_{\alpha+\beta}v_{2\alpha+\beta}^2v_{3\alpha+\beta}^3v_{3\alpha+2\beta}^3, \\ v_{2\alpha+\beta}v_{\alpha} = v_{\alpha}v_{2\alpha+\beta}v_{3\alpha+\beta}^3, \\ v_{3\alpha+\beta}v_{\alpha} = v_{\alpha}v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta}v_{\alpha} = v_{\alpha}v_{3\alpha+2\beta}, \end{cases}$$

$$(B) \quad \begin{cases} v_{\alpha+\beta}v_{\beta} = v_{\beta}v_{\alpha+\beta}, \\ v_{2\alpha+\beta}v_{\beta} = v_{\beta}v_{2\alpha+\beta}, \\ v_{3\alpha+\beta}v_{\beta} = v_{\beta}v_{3\alpha+\beta}v_{3\alpha+2\beta}^{-1}, \\ v_{3\alpha+2\beta}v_{\beta} = v_{\beta}v_{3\alpha+2\beta}, \\ v_{2\alpha+\beta}v_{\alpha+\beta} = v_{\alpha+\beta}v_{2\alpha+\beta}v_{3\alpha+2\beta}^3, \end{cases}$$

$$(B) \quad \begin{cases} v_{3\alpha+\beta}v_{\alpha+\beta} = v_{\alpha+\beta}v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta}v_{\alpha+\beta} = v_{\alpha+\beta}v_{3\alpha+2\beta}, \\ v_{3\alpha+\beta}v_{2\alpha+\beta} = v_{2\alpha+\beta}v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta}v_{2\alpha+\beta} = v_{2\alpha+\beta}v_{3\alpha+2\beta}, \\ v_{3\alpha+2\beta}v_{3\alpha+\beta} = v_{3\alpha+\beta}v_{3\alpha+2\beta}. \end{cases}$$

3.5. — Explicit form of the theorem on generators and relations

Theorem 3.5.1. *Let S be a scheme, G a pinned S -group, T its maximal torus, Δ its system of simple roots, and $u_\alpha \in U_\alpha(S)^\times$ and $w_\alpha \in \underline{\text{Norm}}_G(T)(S)$ the elements defined by the pinning ($\alpha \in \Delta$). Let*

$$f_T : T \longrightarrow H, \quad f_\alpha : U_\alpha \longrightarrow H, \quad \alpha \in \Delta$$

be group morphisms, H being an S -group sheaf for fppf; let $h_\alpha \in H(S)$, ($\alpha \in \Delta$) be sections of H ; put $v_\alpha = f_\alpha(u_\alpha)$, $\alpha \in \Delta$. For there to exist a group morphism

$$f : G \longrightarrow H$$

extending f_T , f_α ($\alpha \in \Delta$) and satisfying $f(w_\alpha) = h_\alpha$ ($\alpha \in \Delta$) (and then necessarily unique), it is necessary and sufficient that the following conditions hold:

(i) *For every $S' \longrightarrow S$, every $\alpha \in \Delta$, every $t \in T(S')$ and every $x \in U_\alpha(S')$, one has*

$$(1) \quad \text{int}(f_T(t))f_\alpha(x) = f_\alpha(\text{int}(t)x) = f_\alpha(x^{\alpha(t)}).$$

(ii) *For every $\alpha \in \Delta$, every $S' \longrightarrow S$ and every $t \in T(S')$, one has*

$$(2) \quad \text{int}(h_\alpha)f_T(t) = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^* \alpha(t)^{-1}).$$

(iii) *For every $\alpha \in \Delta$, one has⁽⁹⁾*

$$(3_1) \quad h_\alpha^2 = f_T(\alpha^*(-1)),$$

$$(4) \quad (h_\alpha v_\alpha)^3 = e.$$

(iv) *For all $\alpha, \beta \in \Delta$, $\alpha \neq \beta$, such that $(\alpha^*, \beta) = 0$ (resp. $(\alpha^*, \beta) = -1$, resp. $(\alpha^*, \beta) = -2$, resp. $(\alpha^*, \beta) = -3$), one has:*

(a) *the relation*

$$(3_2) \quad \begin{cases} (h_\alpha h_\beta)^2 = f_T(\alpha^*(-1)\beta^*(-1)) & \text{if } (\alpha^*, \beta) = 0; \\ (h_\alpha h_\beta)^3 = e, & \text{if } (\alpha^*, \beta) = -1; \\ (h_\alpha h_\beta)^4 = f_T(\alpha^*(-1)), & \text{if } (\alpha^*, \beta) = -2; \\ (h_\alpha h_\beta)^6 = e, & \text{if } (\alpha^*, \beta) = -3. \end{cases}$$

(b) *The relations (A) and (B) of 3.1.3 (resp. 3.2.8, resp. 3.3.7, resp. 3.4.10).*

This follows immediately from 2.3, 2.6 and the calculations made in each particular case.

Remark 3.5.2. One may present the preceding results in a slightly different manner: one gives morphisms, for $\alpha \in \Delta$,

$$a_\alpha : T \cdot U_\alpha \longrightarrow H, \quad b_\alpha : \underline{\text{Norm}}_{Z_\alpha}(T) \longrightarrow H,$$

and one puts

$$h_\alpha = b_\alpha(w_\alpha), \quad v_\alpha = a_\alpha(u_\alpha);$$

then the conditions to be checked are the following:

- (1) all the a_α ($\alpha \in \Delta$) and all the b_α ($\alpha \in \Delta$) have the same restriction to T ;
- (2) conditions (4) and (iv) of 3.5.1 above hold.

3.5.3. — Various applications of this theorem will be given in the following exposé. Let us mention one here: Theorem 3.5.1 gives a description by generators and relations of G in the category of S -sheaves for fppf; in other words, consider for each $S' \longrightarrow S$ the group $H(S')$ generated by $T(S')$, $U_\alpha(S')$, $\alpha \in R$, and w_α , $\alpha \in R$, subject to relations analogous to (1), (2), (3₁), (4), (3₂), (A), (B); then G is precisely the sheaf associated with the presheaf $S' \longmapsto H(S')$.

In particular, if S' is the spectrum of an algebraically closed field k , one has $G(S') = H(S')$ (an immediate consequence of the Nullstellensatz in the form: “an fppf covering sieve of an

⁽⁹⁾**N.D.E.** Relations (3₁) and (3₂) form the description of the normalizer of the torus, (3₁) being, like (4), in a rank 1 group, whereas (3₂) is in a rank 2 group.

algebraically closed field is trivial”), so that 3.5.1 immediately gives an explicit description by generators and relations of the “abstract” group $G(k)$.⁽¹⁰⁾

4. Uniqueness of pinned groups: fundamental theorem

Theorem 4.1. *Let S be a nonempty scheme. The functor $\mathcal{R}$ of 1.6 is fully faithful: let G and G' be two pinned S -groups, p ⁽¹¹⁾ an integer > 0 such that $x \mapsto x^p$ is an endomorphism of $\mathbf{G}_{a,S}$, and $h : \mathcal{R}(G') \rightarrow \mathcal{R}(G)$ a p -morphism of pinned root data. There exists a unique morphism of pinned groups*

$$f : G \rightarrow G'$$

such that $\mathcal{R}(f) = h$.

Uniqueness is proved in 1.9. It suffices to prove existence. By hypothesis, one has a bijection $d : R \xrightarrow{\sim} R'$ and a map $\mathbf{q} : R \rightarrow \{p^n \mid n \in \mathbf{N}\}$ such that

$$h(d(\alpha)) = \mathbf{q}(\alpha)\alpha \quad \text{and} \quad {}^th(\alpha^*) = \mathbf{q}(\alpha)d(\alpha)^*$$

for every $\alpha \in R$. In particular, the semisimple ranks of G and G' coincide.

4.1.1. — Suppose $\text{rgss}(G) = \text{rgss}(G') = 0$. Then G and G' are tori: one has $G = T = D_S(M)$, $G' = T' = D_S(M')$ and h is simply a morphism of ordinary groups $h : M' \rightarrow M$. One then takes $f = D_S(h)$.

4.1.2. — Suppose $\text{rgss}(G) = \text{rgss}(G') = 1$. Then consider

$$f_T = D_S(h) : T \rightarrow T'.$$

By hypothesis one has a commutative diagram, where $\alpha' = d(\alpha)$:

$$\begin{array}{ccccc} \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\ \mathbf{q}(\alpha) \downarrow & & \downarrow f_T & & \downarrow \mathbf{q}(\alpha) \\ \mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S}. \end{array}$$

One then applies Exp. XX, 4.1.

4.1.3. — Suppose $\text{rgss}(G) = \text{rgss}(G') = 2$. Then, by Exp. XXI, 7.5.3, one knows all the possibilities for $h : \mathcal{R}(G') \rightarrow \mathcal{R}(G)$. Let us study them successively, checking each time the conditions of 2.5.

Write $\Delta = \{\alpha, \beta\}$, $\Delta' = \{\alpha', \beta'\}$ so that $d(\alpha) = \alpha'$, $d(\beta) = \beta'$.

4.1.4. — G and G' of type $A_1 \times A_1$. One then has

$$h(\alpha') = q\alpha, \quad h(\beta') = q_1\beta.$$

Let us show that the conditions of 2.5 hold: (ii) and (iii) follow from 3.1.2 (ii) and (iii); let us prove (i). One has

$$\begin{aligned} D_S(h)t_{\alpha\beta} &= D_S(h)(t_\alpha t_\beta) = {}^th(\alpha^*)(-1) \cdot {}^th(\beta^*)(-1) \\ &= \alpha'^*((-1)^q) \cdot \beta'^*((-1)^{q_1}). \end{aligned}$$

⁽¹²⁾Now the hypothesis that $x \mapsto x^p$ is an endomorphism of $\mathbf{G}_{a,S}$ (and $S \neq \emptyset$) implies that $p = 1$ or that S has characteristic p ; in every case one has $(-1)^q = -1$ (if q is even, $p = 2$

⁽¹⁰⁾**N.D.E.** For an arbitrary field k and simply connected G , R. Steinberg gave a presentation of the group $G(k)$ in [St62], Th. 3.2; see also [St67], §6, Th. 8.

⁽¹¹⁾**N.D.E.** To avoid a problem of notation later, we have replaced q here by p , so that in what follows, q (resp. q_1) is an arbitrary power of p .

⁽¹²⁾**N.D.E.** We have expanded what follows.

and $1 = -1$). Consequently,

$$D_S(h)t_{\alpha\beta} = \alpha'^*(-1) \cdot \beta'^*(-1) = t'_{\alpha'\beta'}.$$

This shows that condition 2.5 (i) holds.

4.1.5. — G and G' of type A_2 . One then has

$$h(\alpha') = q\alpha, \quad h(\beta') = q\beta.$$

Put $X_{\alpha+\beta} = \text{Ad}(w_\beta)X_\alpha$ and $X'_{\alpha'+\beta'} = \text{Ad}(w_{\beta'})X_{\alpha'}$. Let us check the conditions of 2.5. For (i), one argues as above, using 3.2.1 (i); for (ii), this is immediate by 3.2.1 (ii); it remains to check (iii). One must check that

$$p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z) \mapsto p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(z^q)$$

is a group morphism. The only nontrivial commutation relation is that of 3.2.1 (iii), which can be written

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy), \\ p'_{\beta'}(y^q)p'_{\alpha'}(x^q) &= p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(x^qy^q). \end{aligned}$$

4.1.6. — One argues in the same way for G and G' of type B_2 (resp. G_2), when the root exponents are equal, using 3.3.1 (resp. 3.4.1); to finish the case of rank 2 groups, it therefore remains to treat the two exceptional cases of Exp. XXI, 7.5.3.

4.1.7. — G and G' are of type B_2 , S has characteristic 2, and one has $\mathbf{q}(\alpha) = 2q$, $\mathbf{q}(\beta) = q$. The positive roots are $\{\alpha, \beta, \alpha + \beta, 2\alpha + \beta\}$ and $\{\alpha', \beta', \alpha' + \beta', \alpha' + 2\beta'\}$ (observe that the “short” simple roots are α and β'). One has

$$\begin{aligned} h(\alpha') &= 2q\alpha, & h(\beta') &= q\beta, \\ h(\alpha' + \beta') &= q(2\alpha + \beta), & h(\alpha' + 2\beta') &= 2q(\alpha + \beta), \end{aligned}$$

which gives

$$\begin{aligned} d(\alpha + \beta) &= \alpha' + 2\beta', & \mathbf{q}(\alpha + \beta) &= 2q, \\ d(2\alpha + \beta) &= \alpha' + \beta', & \mathbf{q}(2\alpha + \beta) &= q. \end{aligned}$$

Put

$$\begin{aligned} X_{\alpha+\beta} &= \text{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \text{Ad}(w_\alpha)X_\beta, \\ X'_{\alpha'+\beta'} &= \text{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \text{Ad}(w'_{\beta'})X'_{\alpha'}. \end{aligned}$$

Let us now check the conditions of 2.5.

(i) Since S has characteristic 2, one has $-1 = 1$ on S , hence $t_{\alpha\beta} = t_\alpha = \alpha^*(-1) = e = \beta'^*(-1) = t'_{\beta'} = t'_{\alpha'\beta'}$ (cf. 3.3.1 (i)).

(ii) By construction one has

$$\begin{aligned} \text{Ad}(w_\alpha)X_\beta &= X_{2\alpha+\beta}, & \text{Ad}(w'_{\alpha'})X'_{\beta'} &= X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)}; \\ \text{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \text{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}. \end{aligned}$$

By 3.3.1 (ii) and the fact that $-1 = 1$ on S , one has on either side

$$\begin{aligned} \text{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{\alpha+\beta}, \\ \text{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+2\beta'} = X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}; \\ \text{Ad}(w_\alpha)X_{2\alpha+\beta} &= X_\beta, \\ \text{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+\beta'} = X'_{\beta'} = X'_{d(\beta)}; \\ \text{Ad}(w_\beta)X_{\alpha+\beta} &= X_\alpha, \\ \text{Ad}(w'_{\beta'})X'_{d(\alpha+\beta)} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+2\beta'} = X'_{\alpha'} = X'_{d(\alpha)}; \\ \text{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, \\ \text{Ad}(w'_{\beta'})X'_{d(2\alpha+\beta)} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+\beta'} = X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)}. \end{aligned}$$

(iii) By 3.3.1 (iii), one sees that the only nontrivial commutation relation in U (resp. U') is

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y),$$

resp.

$$p'_{\alpha'}(y')p'_{\beta'}(x') = p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(x'y')p'_{\alpha'+2\beta'}(x'^2y').$$

We must check that the morphism

$$\begin{aligned} p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z)p_{2\alpha+\beta}(t) &\longmapsto p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q) \\ &\quad p'_{\alpha'+2\beta'}(z^{2q})p'_{\alpha'+\beta'}(t^q) \end{aligned}$$

is a group morphism; one immediately sees that this amounts to seeing that

$$\begin{aligned} p'_{\beta'}(y^q)p'_{\alpha'}(x^{2q}) &= p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q) \\ p'_{\alpha'+2\beta'}((xy)^{2q})p'_{\alpha'+\beta'}((x^2y)^q), \end{aligned}$$

which is precisely the second relation above (putting $y' = x^{2q}$, $x' = y^q$).

4.1.8. — G and G' are of type G_2 , S has characteristic 3, and one has $\mathbf{q}(\alpha) = 3q$ and $\mathbf{q}(\beta) = q$. The positive roots are $\{\alpha, \beta, \alpha + \beta, 2\alpha + \beta, 3\alpha + \beta, 3\alpha + 2\beta\}$ on the one hand, $\{\alpha', \beta', \alpha' + \beta', \alpha' + 2\beta', \alpha' + 3\beta', 2\alpha' + 3\beta'\}$ on the other hand (as in the preceding case, the short simple roots are α and β'). One has

$$\begin{aligned} h(\alpha') &= 3q\alpha, & h(\beta') &= q\beta, & h(\alpha' + \beta') &= q(3\alpha + \beta), \\ h(\alpha' + 2\beta') &= q(3\alpha + 2\beta), & h(\alpha' + 3\beta') &= 3q(\alpha + \beta), \\ h(2\alpha' + 3\beta') &= 3q(2\alpha + \beta), \end{aligned}$$

which gives

$$\begin{aligned} d(\alpha + \beta) &= \alpha' + 3\beta', & \mathbf{q}(\alpha + \beta) &= 3q, \\ d(2\alpha + \beta) &= 2\alpha' + 3\beta', & \mathbf{q}(2\alpha + \beta) &= 3q, \\ d(3\alpha + \beta) &= \alpha' + \beta', & \mathbf{q}(3\alpha + \beta) &= q, \\ d(3\alpha + 2\beta) &= \alpha' + 2\beta', & \mathbf{q}(3\alpha + 2\beta) &= q. \end{aligned}$$

Put

$$\begin{aligned} X_{\alpha+\beta} &= \text{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \text{Ad}(w_\alpha)X_{\alpha+\beta}, \\ X_{3\alpha+\beta} &= -\text{Ad}(w_\alpha)X_\beta, & X_{3\alpha+2\beta} &= \text{Ad}(w_\beta)X_{3\alpha+\beta}; \\ X'_{\alpha'+\beta'} &= -\text{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+\beta'}, \\ X'_{\alpha'+3\beta'} &= \text{Ad}(w'_{\beta'})X'_{\alpha'}, & X'_{2\alpha'+3\beta'} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+3\beta'}. \end{aligned}$$

Let us now check the conditions of 2.5.

(i) One has $t_{\alpha\beta} = e$ and $t'_{\alpha'\beta'} = e$ by 3.4.1 (i).

(ii) By construction one has

$$\begin{aligned} \text{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \text{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{\alpha'+3\beta'} = X'_{d(\alpha+\beta)}; \\ \text{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \text{Ad}(w'_{\alpha'})X'_{\beta'} &= -X'_{\alpha'+\beta'} = -X'_{d(3\alpha+\beta)}; \\ \text{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, \\ \text{Ad}(w'_{\beta'})X'_{d(3\alpha+\beta)} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+\beta'} = X'_{\alpha'+2\beta'} = X'_{d(3\alpha+2\beta)}; \\ \text{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, \\ \text{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+3\beta'} = X'_{2\alpha'+3\beta'} = X'_{d(2\alpha+\beta)}. \end{aligned}$$

By 3.4.1 (ii), one has on either side:

$$\begin{aligned} \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, \\ \operatorname{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= \operatorname{Ad}(w'_{\alpha'})X'_{2\alpha'+3\beta'} = -X'_{\alpha'+3\beta'} = -X'_{d(\alpha+\beta)}; \\ &\dots \dots \\ \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}, \\ \operatorname{Ad}(w'_{\beta'})X'_{d(3\alpha+2\beta)} &= \operatorname{Ad}(w'_{\beta'})X'_{\alpha'+2\beta'} = -X'_{\alpha'+\beta'} = -X'_{d(3\alpha+\beta)}. \end{aligned}$$

(The dots replace four checks of the same kind.)

(iii) The only nontrivial commutation relations in U and U' are, by 3.4.1 (iii) (and taking account of $3 = 0$ on S):

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y) \\ &\quad p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy); \\ p'_{\alpha'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(-x'y') \\ &\quad p'_{\alpha'+2\beta'}(-x'^2y')p'_{\alpha'+3\beta'}(-x'^3y')p'_{2\alpha'+3\beta'}(-x'^3y'^2), \\ p'_{\alpha'+\beta'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'+\beta'}(y')p'_{\alpha'+2\beta'}(x'y'), \\ p'_{\alpha'+3\beta'}(y')p'_{\alpha'}(x') &= p'_{\alpha'}(x')p'_{\alpha'+3\beta'}(y')p'_{2\alpha'+3\beta'}(x'y'). \end{aligned}$$

We must check that the morphism φ from U to U' defined by

$$\begin{aligned} \varphi(p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(t)p_{2\alpha+\beta}(u)p_{3\alpha+\beta}(v)p_{3\alpha+2\beta}(w)) \\ = p'_{\alpha'}(x^{3q})p'_{\beta'}(y^q)p'_{\alpha'+3\beta'}(t^{3q})p'_{2\alpha'+3\beta'}(u^{3q}) \\ p'_{\alpha'+\beta'}(v^q)p'_{\alpha'+2\beta'}(w^q) \end{aligned}$$

is a group morphism. Now one immediately checks that the last three commutation relations can also be written

$$\begin{aligned} p'_{\beta'}(y^q)p'_{\alpha'}(x^{3q}) &= p'_{\alpha'}(x^{3q})p'_{\beta'}(y^q)p'_{\alpha'+3\beta'}((xy)^{3q}) \\ &\quad p'_{2\alpha'+3\beta'}((x^2y)^{3q})p'_{\alpha'+\beta'}((x^3y)^q)p'_{\alpha'+2\beta'}((x^3y^2)^q), \\ p'_{\alpha'+3\beta'}(y^{3q})p'_{\alpha'}(x^{3q}) &= p'_{\alpha'}(x^{3q})p'_{\alpha'+3\beta'}(y^{3q})p'_{2\alpha'+3\beta'}(-(xy)^{3q}), \\ p'_{\alpha'+\beta'}(y^q)p'_{\beta'}(x^q) &= p'_{\beta'}(x^q)p'_{\alpha'+\beta'}(y^q)p'_{\alpha'+2\beta'}(-(xy)^q); \end{aligned}$$

which shows that φ is indeed a group morphism and completes the proof of 4.1.7.

4.1.9. — Case where G and G' have semisimple rank > 2 . For each root $\alpha \in \Delta$, write $\alpha' = d(\alpha) \in \Delta' = d(\Delta)$. For each $(\alpha, \beta) \in \Delta \times \Delta$, consider the pinned groups of semisimple rank ≤ 2 , $Z_{\alpha\beta}$ and $Z'_{\alpha'\beta'}$. The group morphism $M' \rightarrow M$ underlying h defines a p -morphism of root data

$$h_{\alpha\beta} : \mathcal{R}(Z_{\alpha\beta}) \rightarrow \mathcal{R}(Z'_{\alpha'\beta'}).$$

By the preceding results, there thus exists a morphism of pinned groups

$$f_{\alpha\beta} : Z_{\alpha\beta} \rightarrow Z'_{\alpha'\beta'}$$

such that $\mathcal{R}(f_{\alpha\beta}) = h_{\alpha\beta}$. Let us prove that the $f_{\alpha\beta}$ satisfy the gluing condition of 2.5; indeed, $f_{\alpha\beta}|_{Z_\alpha}$ and $f_{\alpha\alpha}$ are two morphisms of pinned groups

$$Z_\alpha \rightarrow Z'_{\alpha'}$$

corresponding to the same morphism of pinned root data, and hence coincide by the uniqueness result already proved. By 2.5, there thus exists a group morphism

$$f : G \rightarrow G'$$

extending the $f_{\alpha\beta}$. This is obviously a morphism of pinned groups such that $\mathcal{R}(f) = h$, which completes the proof of Theorem 4.1.

5. Corollaries of the fundamental theorem

The most important is:

Corollary 5.1. *Let S be a nonempty scheme, G and G' two pinned S -groups, and h an isomorphism of pinned root data*

$$h : \mathcal{R}(G') \xrightarrow{\sim} \mathcal{R}(G).$$

There exists a unique isomorphism of pinned S -groups

$$f : G \xrightarrow{\sim} G'$$

such that $\mathcal{R}(f) = h$.

Observe that 5.1 also follows from 3.5.1 (the relations of 3.5.1 can be written using only the datum of $\mathcal{R}(G)$); observe also that 5.1 follows from the most elementary part of the proof of 4.1 (one need not consider the “exceptional isogenies” of 4.1.7 and 4.1.8).

Corollary 5.2. (“Uniqueness theorem”). *Let S be a scheme, G and G' two splittable S -groups (Exp. XXII, 1.13). If G and G' have the same type (Exp. XXII, 2.6), they are isomorphic.*

Corollary 5.3. *Let S be a scheme, G and G' two splittable S -groups. The following conditions are equivalent:*

- (i) G and G' are isomorphic.
- (ii) G and G' are locally isomorphic for the fpqc topology.
- (iii) There exists an $s \in S$ such that the s -groups G_s and G'_s have the same type.

PROOF. Indeed, one obviously has (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). On the other hand, (iii) implies that $\mathcal{R}(G) = \mathcal{R}(G_s) = \mathcal{R}(G'_s) = \mathcal{R}(G')$, hence that G and G' satisfy the condition of 5.2. $\square$

Corollary 5.4. (“Uniqueness of Chevalley schemes”). *Let G and G' be two reductive groups over $\mathbf{Z}$ possessing split maximal tori.* The following conditions are equivalent:*

- (i) G and G' are isomorphic.
- (ii) There exists $s \in \text{Spec}(\mathbf{Z})$ such that G_s and G'_s have the same type.
- (iii) $G_{\mathbf{C}}$ and $G'_{\mathbf{C}}$ have the same type.

Indeed, G and G' are splittable by Exp. XXII, 2.2.

Corollary 5.5. (“Existence of outer automorphisms”). *Let S be a scheme, G a pinned S -group, and h an automorphism of the pinned root datum $\mathcal{R}(G)$. There exists a unique automorphism u of G respecting its pinning and such that $\mathcal{R}(u) = h$.*

Let us make the preceding corollary explicit:

Corollary 5.5.bis. *Let S be a scheme, G a split reductive S -group, and R_+ a system of positive roots of G . For each simple root α , choose an isomorphism of vector groups $p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$. Let h be an automorphism of M permuting the positive roots and the corresponding coroots: if $\alpha \in R_+$, $h(\alpha) \in R_+$ and $h^\vee(\alpha^*) = h(\alpha)^*$. There exists a unique automorphism u of G inducing $D_S(h)$ on T and such that $u \circ p_\alpha = p_{h(\alpha)}$ for every simple root α .*

Corollary 5.6. *Let G and G' be two reductive S -groups. The following conditions are equivalent:*

*In fact, one can prove that every $\mathbf{Z}$ -torus is split.¹³

⁽¹³⁾**N.D.E.** This follows from the fact that every $\mathbf{Z}$ -torus is isotrivial (Exp. X, 5.16) and that every étale covering of $\text{Spec}(\mathbf{Z})$ is trivial.

- (i) G and G' are locally isomorphic for the fpqc topology.
- (i bis) G and G' are locally isomorphic for the étale topology.
- (ii) For every $s \in S$, $G_{\bar{s}}$ and $G'_{\bar{s}}$ are isomorphic.
- (iii) The functions $s \mapsto \text{type of } G_{\bar{s}}$ and $s \mapsto \text{type of } G'_{\bar{s}}$ are equal.

PROOF. Indeed, (i bis) $\Rightarrow$ (i) trivially, (i) $\Rightarrow$ (ii) by the principle of finite extension (EGA IV₃, 9.1.4), (ii) $\Rightarrow$ (iii) trivially; it remains to prove (iii) $\Rightarrow$ (i bis). Now one may suppose G and G' splittable (Exp. XXII, 2.3), in which case the assertion follows from 5.3. $\square$

Corollary 5.7. *Let S be a scheme, G a reductive S -group, and G' an affine, smooth S -group with connected fibers. Let $s \in S$ be such that $G_{\bar{s}}$ and $G'_{\bar{s}}$ are isomorphic; there then exists an $S' \rightarrow S$ which is étale and covers s such that $G_{S'}$ and $G'_{S'}$ are isomorphic.*

Indeed, by Exp. XIX 2.5 and Exp. XXII 2.1, one may suppose G and G' splittable reductive groups, and one is reduced to 5.3.

When S is the spectrum of a field, one deduces from 5.6 and 5.7:

Corollary 5.8. *Let k be a field and G and G' two reductive k -groups. The following conditions are equivalent:⁽¹⁴⁾*

- (i) G and G' have the same type.
- (ii) $G \otimes_k \bar{k}$ and $G' \otimes_k \bar{k}$ are isomorphic.
- (iii) There exists a finite separable extension K of k such that $G \otimes_k K$ and $G' \otimes_k K$ are isomorphic.

Corollary 5.9. *Let S be a nonempty scheme and $\mathcal{R}$ a root datum. The following conditions are equivalent:⁽¹⁵⁾*

- (i) There exists a pinned S -group of type $\mathcal{R}$.
- (ii) There exists an S -group of type $\mathcal{R}$.
- (iii) There exists locally for fpqc a reductive S -group of type $\mathcal{R}$.

PROOF. The task is obviously to prove (iii) $\Rightarrow$ (i). To simplify the proof, suppose there exist a faithfully flat quasi-compact morphism $S' \rightarrow S$ and a reductive S' -group G' of type $\mathcal{R}$. One may suppose G' splittable; fix a pinning E' of G' ; write $\mathcal{R} = \mathcal{R}(G', E')$. The two inverse images of (G', E') on $S'' = S' \times_S S'$ are pinned groups (G''_1, E''_1) , (G''_2, E''_2) ; one has canonical isomorphisms $p_i : \mathcal{R}(G''_i, E''_i) \xrightarrow{\sim} \mathcal{R}$, whence an isomorphism

$$p = p_2^{-1} \circ p_1 : \mathcal{R}(G''_1, E''_1) \rightarrow \mathcal{R}(G''_2, E''_2).$$

By the uniqueness theorem, there exists a unique isomorphism

$$f : (G''_1, E''_1) \xrightarrow{\sim} (G''_2, E''_2)$$

such that $\mathcal{R}(f) = p$. One therefore has a gluing datum on G' ; this is a descent datum.

Indeed, one must check a compatibility condition between the inverse images of f on S'' , but it suffices to make this check on the transforms of these arrows under the functor $\mathcal{R}$, since the latter is fully faithful. Now p is indeed a descent datum, by construction, which shows that f is also one. Since G' is affine, this descent datum is effective; since the pinning of G' is stable under the descent datum, one easily checks that there exists a pinned S -group (G, E) which gives (G', E') by base extension and which is therefore of type $\mathcal{R}$.

N.B. Naturally, in the language of fibered categories, the preceding proof becomes simpler (and comprehensible). $\square$

⁽¹⁴⁾**N.D.E.** Another proof of this corollary, not using reduction to the case of rank 2 groups, was given by M. Takeuchi ([Ta83], Th. 4.6); see also [Ja87], II 1.14.

⁽¹⁵⁾**N.D.E.** This corollary is made unnecessary by Exp. XXV, which shows the existence of a split group of type $\mathcal{R}$ over $\text{Spec}(\mathbf{Z})$, hence over any base S . (In fact, after XXV 1.3 one finds a reference to the present corollary to ensure that the reductive $\mathbf{Z}$ -group obtained is split, but this already follows from XXII 2.2; cf. N.D.E. (3) of XXV.)

Corollary 5.10. *Let S be a nonempty scheme. Let $\mathcal{R}$ be a pinned root datum such that there exists a reductive S -group of type $\mathcal{R}$. Then there exists a pinned S -group of type $\mathcal{R}$, unique up to a unique isomorphism.*

Definition 5.11. Under the preceding conditions, one will denote by $\acute{\text{E}}\text{p}_S(\mathcal{R})$ the unique pinned S -group of type $\mathcal{R}$, by $T_S(\mathcal{R})$ its canonical maximal torus, by $B_S(\mathcal{R})$ its canonical Borel subgroup, ...

If one has a morphism $S' \rightarrow S$ (S' nonempty), one may identify $\acute{\text{E}}\text{p}_{S'}(\mathcal{R})$ with $\acute{\text{E}}\text{p}_S(\mathcal{R}) \times_S S'$, etc. In particular, if $\acute{\text{E}}\text{p}_{\text{Spec}(\mathbf{Z})}(\mathcal{R})$ exists (we shall see that this is always the case), one denotes it by $\acute{\text{E}}\text{p}(\mathcal{R})$ and one has

$$\acute{\text{E}}\text{p}_S(\mathcal{R}) = \acute{\text{E}}\text{p}(\mathcal{R}) \times_{\text{Spec}(\mathbf{Z})} S.$$

One says that $\acute{\text{E}}\text{p}(\mathcal{R})$ is the *Chevalley group scheme of type $\mathcal{R}$* .

5.12. — It therefore amounts to the same thing to say that the S -group sheaf G is a reductive S -group of type $\mathcal{R}$ or to say that it is locally isomorphic (for the étale or fpqc topology) to $\acute{\text{E}}\text{p}_S(\mathcal{R})$. Likewise, by the conjugacy theorems, it amounts to the same thing to say that (G, T) is a reductive S -group of type $\mathcal{R}$ equipped with a maximal torus or that it is locally isomorphic to $(\acute{\text{E}}\text{p}_S(\mathcal{R}), T_S(\mathcal{R}))$; likewise with Borel subgroups or Killing pairs.

6. Chevalley systems

The explicit calculations of no. 3 have important numerical consequences. Let us first give the following definition:

Definition 6.1. Let S be a scheme, and (G, T, M, R) a split S -group. A *Chevalley system* of G is a family $(X_\alpha)_{\alpha \in R}$ of elements

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$$

satisfying the following condition:

(SC) for every pair $\alpha, \beta \in R$, one has

$$\text{Ad}(w_\alpha(X_\alpha))X_\beta = \pm X_{s_\alpha(\beta)}.$$

Recall (Exp. XX, 3.1) that

$$w_\alpha(X_\alpha) = \exp(X_\alpha) \exp(-X_\alpha^{-1}) \exp(X_\alpha).$$

Observe that (SC) implies in particular $X_{-\alpha} = \pm X_\alpha$ for $\alpha \in R$, by the relation (Exp. XX, 3.7) $\text{Ad}(w_\alpha(X_\alpha))X_\alpha = -X_\alpha^{-1}$.

Proposition 6.2. *Every split group has a Chevalley system. More precisely, let $(\Delta, (X'_\alpha)_{\alpha \in \Delta})$ be a pinning (1.1) of the split group (G, T, M, R) ; there then exists a Chevalley system $(X_\alpha)_{\alpha \in R}$ of G such that $X_\alpha = X'_\alpha$ for $\alpha \in \Delta$.*

Let us first show that it suffices to check condition (SC) for $\alpha \in \Delta$; for every $\alpha \in R$, there exists a sequence $\{\alpha_i\} \subset \Delta$ such that $\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1})$, whence, applying condition (SC) for each α_i ,

$$X_\alpha = \pm \text{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))X_{\alpha_{n+1}}.$$

By 3.1.1 (iii), one has

$$\begin{aligned} & w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}) w_{\alpha_{n+1}}(X_{\alpha_{n+1}}) \\ & \cdot w_{\alpha_n}(X_{\alpha_n})^{-1} \cdots w_{\alpha_1}(X_{\alpha_1})^{-1} = \alpha^*(\pm 1) w_\alpha(X_\alpha). \end{aligned}$$

Now it suffices to observe that $w_{\alpha_i}(X_{\alpha_i})^{-1} = \alpha_i^*(-1) w_{\alpha_i}(X_{\alpha_i})$ and that for every pair of roots (β, γ) , one has $\beta(\gamma^*(-1)) = (-1)^{(\gamma^*, \beta)} = \pm 1$, which implies that if (SC) holds for the pairs (α_i, γ) ($\gamma \in R$), it holds for every pair (α, β) , ($\beta \in R$).

Let us now construct a system $(X_\alpha)_{\alpha \in R}$ in the following manner. For every $\alpha \in R$, choose a sequence $\{\alpha_i\} \subset \Delta$ as above and define X_α by

$$X_\alpha = \text{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))X'_{\alpha_{n+1}}.$$

To check (SC), it suffices to prove:

Lemma 6.3. *Let $(G, T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ be a pinned S -group; let α_i ($0 \leq i \leq n+1$) be a sequence of simple roots such that*

$$\text{int}(s_{\alpha_1} \cdots s_{\alpha_n})(\alpha_{n+1}) = \alpha_0.$$

Then

$$\text{Ad}(w_{\alpha_1} \cdots w_{\alpha_n})X_{\alpha_{n+1}} = \pm X_{\alpha_0}.$$

PROOF. Arguing as in 2.3.4 with Tits's lemma (Exp. XXI 5.6), one is reduced to checking Lemma 6.3 in the following two cases:

a) G has semisimple rank at most 2 or b) $w_{\alpha_1} \cdots w_{\alpha_n}$ is a section of T . In case a), observe that 6.3 is a consequence of 6.2 and that 6.2 was checked in part (i) of 3.1.2 (resp. 3.2.1, resp. 3.3.1, resp. 3.4.1).

It therefore remains to prove 6.3 in case b), or, equivalently, that if $\{\alpha_i\}$ is a sequence of simple roots such that $s_{\alpha_1} \cdots s_{\alpha_n} = \text{id}$, then $t = w_{\alpha_1} \cdots w_{\alpha_n}$ satisfies $\alpha(t) = \pm 1$ for every root $\alpha \in R$. By the structure of the Weyl group (Exp. XXI, 5.1), the word $s_{\alpha_1} \cdots s_{\alpha_n}$ in the free group generated by the s_α , $\alpha \in \Delta$, lies in the invariant subgroup generated by the $(s_\alpha s_\beta)^{n_{\alpha\beta}}$, $(\alpha, \beta) \in \Delta \times \Delta$. One is therefore reduced to the case where $s_{\alpha_1} \cdots s_{\alpha_n}$ has the form

$$s_{\alpha_1} \cdots s_{\alpha_i} (s_{\alpha_{i+1}} s_{\alpha_{i+2}})^{n_{\alpha_{i+1}\alpha_{i+2}}} s_{\alpha_i} \cdots s_{\alpha_1}.$$

Then one has

$$t = s_{\alpha_1} \cdots s_{\alpha_i} (t_{\alpha_{i+1}\alpha_{i+2}}),$$

and one is reduced to checking that for every pair of simple roots (α_1, α_2) and every root $\beta \in R$, one has $\beta(t_{\alpha_1\alpha_2}) = \pm 1$, which is trivial, in view of the values of $t_{\alpha_1\alpha_2}$ calculated in no. 3 (part (i) of 3.1.2, 3.2.1, 3.3.1, 3.4.1). $\square$

Proposition 6.4. *Let (G, T, M, R) be a split S -group, and $(X_\alpha)_{\alpha \in R}$ a Chevalley system of G . Let α and β be two nonproportional roots; suppose*

$$\text{long}(\alpha) \leq \text{long}(\beta), \quad \text{i.e.} \quad |(\beta^*, \alpha)| \leq |(\alpha^*, \beta)|.$$

Let q and $p-1$ be the integers ≥ 0 such that the set of roots of the form $\beta + k\alpha$, $k \in \mathbf{Z}$, is

$$\{\beta - (p-1)\alpha, \dots, \beta, \dots, \beta + q\alpha\}.$$

(Cf. Exp. XXI, 2.3.5; by loc. cit., one therefore has $-(\alpha^, \beta) = q - p + 1$.) Then the commutation relation between U_α and U_β is given by the following table (which exhausts the possible cases, since the length of the preceding root chain is $p + q - 1 \leq 3$), where for each $\gamma \in R$ one writes $p_\gamma(x) = \exp(xX_\gamma)$:*

(p, q)	$p_\beta(y)p_\alpha(x)p_\beta(-y)p_\alpha(-x)$
$(-, 0)$	$= e$
$(1, 1)$	$= p_{\alpha+\beta}(\pm xy)$
$(1, 2)$	$= p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)$
$(1, 3)$	$= p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)$ $p_{3\alpha+\beta}(\pm x^3y)p_{3\alpha+2\beta}(\pm x^3y^2)$
$(2, 1)$	$= p_{\alpha+\beta}(\pm 2xy)$
$(2, 2)$	$= p_{\alpha+\beta}(\pm 2xy)p_{2\alpha+\beta}(\pm 3x^2y)p_{\alpha+2\beta}(\pm 3xy^2)$
$(3, 1)$	$= p_{\alpha+\beta}(\pm 3xy).$

PROOF. By Exp. XXI, 3.5.4, there exists a system of simple roots Δ of R satisfying: $\alpha \in \Delta$ and there exist $\alpha' \in \Delta$ and $a, b \in \mathbf{Q}_+$ such that $\beta = a\alpha + b\alpha'$. Consider the pinning $(\Delta, (X_\alpha)_{\alpha \in \Delta})$ of G . The commutation relation to be checked is a relation between elements of $U_{\alpha\alpha'}$; one is therefore reduced to the explicit calculations of no. 3, and one concludes immediately by condition (SC). $\square$

Corollary 6.5. (Chevalley's rule). *Let S be a scheme, and $(X_\alpha)_{\alpha \in R}$ a Chevalley system of the split S -group G . If $\alpha, \beta, \alpha + \beta \in R$, then*

$$[X_\alpha, X_\beta] = \pm p X_{\alpha+\beta},$$

where p is the smallest integer > 0 such that $\beta - p\alpha$ is not a root.

Indeed, since the assertion is symmetric in α and β , one may suppose $\text{long}(\alpha) \leq \text{long}(\beta)$, and one is reduced to 6.4.

Corollary 6.6. *Let S be a scheme such that $6 \cdot 1_S \neq 0$ and (G, T, M, R) a split S -group. If R' is a subset of R such that*

$$(*) \quad \mathfrak{g}_{R'} = \mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha$$

is a Lie subalgebra of $\mathfrak{g}$, then R' is a closed subset of R (Exp. XXI, 3.1.4).

⁽¹⁶⁾Indeed, let $(X_\gamma)_{\gamma \in R_+}$ be a Chevalley system of $\mathfrak{g}$ and let $\alpha, \beta \in R'$ be such that $\alpha + \beta \in R$. By 6.5 and 6.4, one has

$$[X_\alpha, X_\beta] = \pm p X_{\alpha+\beta}, \quad \text{with } p \in \{1, 2, 3\}$$

and since neither 2 nor 3 is zero on S , this implies, by (*), that $\mathfrak{g}_{R'}$ contains $\mathfrak{g}^{\alpha+\beta}$, and hence $\alpha + \beta \in R'$.⁽¹⁷⁾

6.7. — It is possible to specify the exact value of the various $\pm$ of this no., through the study of the group $\text{Norm}_G(T)$ and, more precisely, of the “extended Weyl group”:

$$W^* = N(\mathbf{Z}), \quad \text{where } N = \text{Norm}_{\dot{\text{E}}_{\mathbf{P}^{\mathbf{Z}}}(\mathcal{R})}(T_{\mathbf{Z}}(\mathcal{R}))$$

(cf. 5.11), which is an extension of $W(\mathcal{R})$ by an abelian group of type $(2, 2, \dots, 2)$, which is “responsible for the signs”⁽¹⁸⁾.

Remark 6.7.1. ⁽¹⁹⁾Observe that, by point (i) of 3.1.2 and 3.n.1 ($n = 2, 3, 4$), the elements w_α and w_β of $N(\mathbf{Z})$ satisfy the “braid relations”:

$$w_\alpha w_\beta \cdots = w_\beta w_\alpha \cdots \quad (n_{\alpha\beta} \text{ factors in each term}).$$

(See also [Ti66], [BLie], §IX.4, Ex. 12, and [Sp98], 9.3.2.)

⁽¹⁶⁾**N.D.E.** We have added the following proof. Observe that it suffices that 2 and 3 be nonzero on S ; for example the result holds for $S = \text{Spec}(\mathbf{Z}/6\mathbf{Z})$.

⁽¹⁷⁾**N.D.E.** On the other hand, let us point out that if $2 = 0$ on S and R is of type C_n , then the set R' of short roots (which is a root system of type D_n) is not closed in R , but is a symmetric subset of type (R) of R (cf. XXII 5.4.2 and 5.4.10), i.e., there corresponds to it a subgroup $H_{R'}$ of type (R) of G with reductive fibers: this is in particular the case for the natural embedding (in characteristic 2) of $\text{SO}(2n)$ in $\text{Sp}(2n)$. Likewise, if $2 = 0$ on S and R is of type F_4 (resp. if $3 = 0$ on S and R is of type G_2), the set R' of short roots (which is a root system of type D_4 (resp. A_2)) is not closed in R , but corresponds to a subgroup $H_{R'}$ of type (R) of G with reductive fibers.

* Cf. J. Tits: *Sur les constantes de structure et le théorème d'existence des algèbres de Lie semi-simples*, Publ. Math. I.H.É.S. **31** (1966), 21–58.

⁽¹⁸⁾**N.D.E.** See also [Ti66].

⁽¹⁹⁾**N.D.E.** We have added this remark.

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⁽²⁰⁾**N.D.E.** Additional references cited in this exposé.

Exposé XXIV. Automorphisms of reductive groups

⁽⁰⁾The first part of this exposé (nos. 1 to 5) is a direct consequence of the existence, for a reductive group, of “sufficiently many outer automorphisms”, a result which is a consequence of the weakest form of the isomorphism theorem for pinned groups. The second part (nos. 6 and 7) presents two applications of the more precise results of the preceding exposé; in particular, no. 7 uses the theorem on generators and relations in its explicit form. Finally, in an appendix (no. 8) we have given results on “Galois” cohomology used in the text.

Let us specify our cohomological notation: if S is a scheme and G an S -group scheme, we shall denote by $H^1(S, G)$ the first cohomology set of S with coefficients in G , computed for the (fpqc) topology; it is also the set of isomorphism classes of (fpqc) principal homogeneous sheaves under G . We shall denote by $H_{\text{ét}}^1(S, G)$ the corresponding set for the étale topology; it is therefore the part of $H^1(S, G)$ consisting of the classes of homogeneous sheaves under G which are quasi-isotrivial (= locally trivial for the étale topology). We shall denote by $\text{Fib}(S, G)$ the part of $H^1(S, G)$ consisting of the classes of representable sheaves (principal homogeneous bundles). We therefore have the inclusions

$$H_{\text{ét}}^1(S, G) \subset H^1(S, G), \quad \text{Fib}(S, G) \subset H^1(S, G).$$

If every principal homogeneous sheaf under G is representable (for example if G is quasi-affine over S , cf. SGA 1, VIII 7.9), we therefore have $\text{Fib}(S, G) = H^1(S, G)$.

If $S' \rightarrow S$ is a covering morphism for the (fpqc) topology, we denote by $H^1(S'/S, G)$ the kernel of the canonical map $H^1(S, G) \rightarrow H^1(S', G_{S'})$. One knows that $H^1(S'/S, G)$ can be computed simplicially (TDTE I, § A.4), which implies that when $S' \rightarrow S$ is covering for the étale topology, $H^1(S'/S, G)$ is also the kernel of $H_{\text{ét}}^1(S, G) \rightarrow H_{\text{ét}}^1(S', G_{S'})$.

Finally, following Exp. VIII, 4.5, one calls “theorem 90” the following assertion: “every principal homogeneous sheaf under $\mathbf{G}_{m,S}$ is representable and locally trivial”, an assertion equivalent to “ $H^1(S, \mathbf{G}_{m,S}) = \text{Pic}(S)$ ”, or again to “ $H^1(S, \mathbf{G}_{m,S}) = 0$ for S local (or, more generally, semi-local)”.

1. Scheme of automorphisms of a reductive group

1.0. — It is appropriate first to specify certain definitions from the preceding exposé. Let S be a nonempty scheme, G a reductive S -group, and $\mathcal{R} = (M, M^*, R, R^*, \Delta)$ a reduced pinned root datum (Exp. XXIII 1.5). One calls a *pinning* of G of type $\mathcal{R}$, or an $\mathcal{R}$ -*pinning* of G , the datum:

- (i) of an isomorphism of $D_S(M)$ onto a maximal torus T of G (or, equivalently, of a monomorphism $D_S(M) \rightarrow G$ whose image is a maximal torus T of G), identifying R with a root system of G relative to T (Exp. XIX, 3.6) and R^* with the set of corresponding coroots,
- (ii) for each $\alpha \in \Delta$, of an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

For G to possess an $\mathcal{R}$ -pinning, it is necessary and sufficient that it be splittable and of type $\mathcal{R}$ (Exp. XXII, 2.7).

If $u : G \rightarrow G'$ is an isomorphism of reductive S -groups, to every $\mathcal{R}$ -pinning $\mathcal{E}$ of G there corresponds, by “transport of structure”, an $\mathcal{R}$ -pinning $u(\mathcal{E})$ of G' . If $v : \mathcal{R}' \rightarrow \mathcal{R}$ is an

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

isomorphism of pinned root data, to every $\mathcal{R}'$ -pinning $\mathcal{E}$ of G' there corresponds by transport of structure an $\mathcal{R}$ -pinning $v(\mathcal{E})$ of G .

Let us call a *pinned group* a triple $(G, \mathcal{R}, \mathcal{E})$ where G is a reductive S -group, $\mathcal{R}$ a reduced pinned root datum, and $\mathcal{E}$ a pinning of G of type $\mathcal{R}$. One calls an *isomorphism* of the pinned group $(G, \mathcal{R}, \mathcal{E})$ onto the pinned group $(G', \mathcal{R}', \mathcal{E}')$ a pair (u, v) where u is an isomorphism $u : G \rightarrow G'$ and v an isomorphism of pinned root data $v : \mathcal{R}' \rightarrow \mathcal{R}$, such that $u(\mathcal{E}) = v(\mathcal{E}')$.⁽¹⁾

N. B. If S is nonempty, v is uniquely determined by u , and by abuse of language one will also say that u is an isomorphism of the pinned groups. In particular, if $(G, \mathcal{R}, \mathcal{E})$ is a pinned group, an automorphism of $(G, \mathcal{R}, \mathcal{E})$ is therefore an automorphism u of G such that there exists an automorphism v of $\mathcal{R}$ such that $u(\mathcal{E}) = v(\mathcal{E})$; it is therefore an automorphism of G , normalizing T , inducing on T an automorphism of the form $D_S(h)$, where h is an automorphism of M , and permuting the elements X_α , $\alpha \in \Delta$, among themselves. (As one readily sees, the preceding conditions in fact characterize the automorphisms of $(G, \mathcal{R}, \mathcal{E})$.)

One has an evident contravariant functor

$$\Phi : (G, \mathcal{R}, \mathcal{E}) \mapsto \mathcal{R}, \quad (u, v) \mapsto v$$

and the principal result of the preceding exposé (Exp. XXIII, 4.1) shows us that it is a fully faithful functor (we shall moreover see in the following exposé that it is an equivalence of categories). It follows in particular that the group of automorphisms of $(G, \mathcal{R}, \mathcal{E})$ is canonically isomorphic to the group of automorphisms of the pinned root datum $\mathcal{R}$ (cf. Exp. XXIII, 5.5).

1.1. — Let S be a scheme; equip $(\text{Sch})/S$ with the (fpqc) topology and consider the S -sheaf of groups $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, where G is an S -group scheme. One has an exact sequence of S -sheaves of groups

$$1 \longrightarrow \underline{\text{Centr}}(G) \longrightarrow G \xrightarrow{\text{int}} \underline{\text{Aut}}_{S\text{-gr.}}(G)$$

which defines a monomorphism

$$j : G/\underline{\text{Centr}}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G).$$

The image sheaf of j is the sheaf of inner automorphisms of G ; for an automorphism u of G to be inner, it is necessary and sufficient that there exist a covering family $\{S_i \rightarrow S\}$ and, for each i , a $g_i \in G(S_i)$ such that $\text{int}(g_i) = u_{S_i}$. In this case, if v is another automorphism of G , one sees at once that $\text{int}(v)u = vu v^{-1}$ is the inner automorphism defined by the family $g'_i = v(g_i)$. It follows that the image of j is normal in $\underline{\text{Aut}}_{S\text{-gr.}}(G)$. The quotient sheaf of groups, denoted by $\underline{\text{Autext}}(G)$, is the sheaf of outer automorphisms of G . One therefore has an exact sequence

$$1 \longrightarrow G/\underline{\text{Centr}}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\text{Autext}}(G) \longrightarrow 1.$$

The preceding definitions are all compatible with base change. They are of course valid in any site.

1.2. — Let S be a scheme and $(G, \mathcal{R}, \mathcal{E})$ a pinned reductive S -group. Let E be the (abstract) group of automorphisms of the pinned root datum $\mathcal{R}$, i.e. the group of automorphisms of $\mathcal{R}$ normalizing Δ . By Exp. XXIII, 5.5, one has a canonical monomorphism

$$E \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G)$$

which associates to $h \in E$ the unique automorphism u of the pinned group G such that $\Phi(u) = h$. This monomorphism canonically defines a monomorphism of sheaves

$$(*) \quad a : E_S \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G).$$

For an automorphism u of G to be a section of the image sheaf of a , it is necessary and sufficient that the following conditions hold:

⁽¹⁾**N.D.E.** Thus $\text{Lie}(u)(X_{v(\alpha)}) = X'_\alpha$, for every $\alpha \in R'$.

- (i) u normalizes T . One then knows that u permutes the roots of G relative to T . If $\alpha \in R$, then $u(\alpha) : t \mapsto \alpha(u^{-1}(t))$ is therefore locally on S of the form $t \mapsto \beta(t)$, with $\beta \in R$. The second condition can then be written as follows:
- (ii) If $\alpha \in \Delta$ and if U is an open subset of S such that $u(\alpha)_U \in R$, then $u(\alpha)_U \in \Delta$ and

$$\mathrm{Lie}(u_U)(X_\alpha)_U = (X_{u(\alpha)})_U.$$

It follows at once from the definitions that the sections of $a(E_S)$ normalize the subgroups of G defined by the pinning: T, B, B^-, U, U^- . These definitions being given, one has:

Theorem 1.3. *Let S be a scheme and G a reductive S -group. Consider the canonical exact sequence of S -sheaves of groups⁽²⁾*

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \xrightarrow{p} \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

- (i) $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable by a smooth separated S -scheme.
- (ii) $\underline{\mathrm{Autext}}(G)$ is representable by a twisted constant S -scheme of finite generation (Exp. X, 5.1).
- (iii) If G is splittable, the preceding exact sequence is split. More precisely, for every pinning of G , the morphism (cf. 1.2 (*))

$$p \circ a : E_S \longrightarrow \underline{\mathrm{Autext}}(G)$$

is an isomorphism.

Let us first show how the theorem follows from the following lemma:

Lemma 1.4. *Under the hypotheses of (iii), $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is the semidirect product $a(E_S) \cdot \mathrm{ad}(G)$.⁽³⁾*

The lemma immediately implies the theorem when G is splittable. Since G is locally splittable for the étale topology (Exp. XXII, 2.3), hence also for the (fppf) topology, and since the latter is “of effective descent” for the fibered category of twisted constant morphisms (Exp. X, 5.5), one deduces (ii) in the general case (cf. Exp. IV, 4.6.8). To deduce (i), one observes that $\mathrm{ad}(G)$ is affine over S , hence the morphism p is affine when $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable, and one concludes by descent of affine schemes.⁽⁴⁾

It therefore only remains for us to prove 1.4. To do so, it suffices to prove:

Lemma 1.5. *If $(\mathcal{R}, \mathcal{E})$ and $(\mathcal{R}', \mathcal{E}')$ are two pinning of the reductive S -group G , there exists a unique inner automorphism u of G over S taking one pinning to the other (i.e. such that there exists $v : \mathcal{R}' \xrightarrow{\sim} \mathcal{R}$ such that $u(\mathcal{E}) = v(\mathcal{E}')$, cf. 1.0).*

1.5.1. — Uniqueness. It suffices to prove that if G is a pinned S -group and if $\mathrm{int}(g)$ is an automorphism of a pinned group ($g \in G(S)$), then $\mathrm{int}(g) = \mathrm{id}$. Now one first has $\mathrm{int}(g)T = T$, $\mathrm{int}(g)B = B$, hence $g \in \underline{\mathrm{Norm}}_G(T)(S) \cap \underline{\mathrm{Norm}}_G(B)(S) = T(S)$ (cf. for example Exp. XXII, 5.6.1). It follows that $\mathrm{int}(g)$ normalizes each U_α and that

$$\mathrm{Lie}(\mathrm{int}(g))X_\alpha = \mathrm{Ad}(g)X_\alpha = \alpha(g)X_\alpha$$

for every $\alpha \in \Delta$. One therefore has $\alpha(g) = 1$ for $\alpha \in \Delta$, hence $g \in \bigcap_{\alpha \in \Delta} (\mathrm{Ker} \alpha)(S) = \underline{\mathrm{Centr}}(G)(S)$ (Exp. XXII, 4.1.8). Q.E.D.

1.5.2. — Existence. It suffices to prove it locally for the (fpqc) topology. Let

$$(T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta}) \quad \text{and} \quad (T', M', R', \Delta', (X'_{\alpha'})_{\alpha' \in \Delta'})$$

⁽²⁾**N.D.E.** Recall that $\mathrm{ad}(G) = G/\underline{\mathrm{Centr}}(G)$ denotes the adjoint group of G , cf. XXII 4.3.6.

⁽³⁾**N.D.E.** Here and in what follows we have replaced the notation $\mathrm{int}(G)$ by $\mathrm{ad}(G)$.

⁽⁴⁾**N.D.E.** Cf. SGA 1, VIII 2.1.

be the two pinnings. By conjugacy of maximal tori, one may suppose $T = T'$. After restricting S , one may suppose that the isomorphism $D_S(M) \simeq D_S(M')$ comes from an isomorphism $M \simeq M'$ taking R onto R' , and one is reduced to the situation $T = T'$, $M = M'$, $R = R'$. Since the systems of simple roots are conjugate under the Weyl group (Exp. XXI, 3.3.7), one may also suppose $\Delta = \Delta'$. There then exists for each $\alpha \in \Delta$ a scalar $z_\alpha \in \mathbf{G}_m(S)$ such that $X'_\alpha = z_\alpha X_\alpha$, and it suffices to construct locally for (fpqc) a section t of T such that $\alpha(t) = z_\alpha$ for each $\alpha \in \Delta$. But the morphism $T \rightarrow (\mathbf{G}_{m,S})^\Delta$ with components $\{\alpha, \alpha \in \Delta\}$ is the dual of an injection $\mathbf{Z}^\Delta \rightarrow M$, hence is faithfully flat, which completes the proof of 1.5.2 and therefore of 1.3.

Corollary 1.6. *Let S be a scheme and G a reductive S -group. The following conditions are equivalent:*

- (i) $\underline{\text{Aut}}_{S\text{-gr.}}(G)$ is affine (resp. of finite type, resp. of finite presentation, resp. quasi-compact) over S .
- (ii) $\underline{\text{Autext}}(G)$ is finite over S .
- (iii) For every $s \in S$, one has $\text{rkred}(G_s) - \text{rkss}(G_s) \leq 1$.

PROOF. Indeed, since $\text{ad}(G)$ is affine, flat and of finite presentation over S , the morphism $p : \underline{\text{Aut}}_{S\text{-gr.}}(G) \rightarrow \underline{\text{Autext}}(G)$ is affine, faithfully flat and of finite presentation.

If $\underline{\text{Autext}}(G)$ is finite over S , it is affine over S , hence so is $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, which proves (ii) $\Rightarrow$ (i). If $\underline{\text{Aut}}_{S\text{-gr.}}(G)$ is quasi-compact over S , it is of finite presentation over S (being in any case locally of finite presentation and separated over S); by Exp. V, 9.1, $\underline{\text{Autext}}(G)$ is then of finite presentation over S , hence finite, which proves (i) $\Rightarrow$ (ii). Finally, to prove the equivalence of (ii) and (iii), one may suppose G split, and one is reduced to Exp. XXI, 6.7.8. $\square$

Corollary 1.7. *Let S be a scheme and G a reductive S -group. Then one has $\underline{\text{Aut}}_{S\text{-gr.}}(G)^0 \simeq \text{ad}(G)$.*

Corollary 1.8. *Let S be a scheme, G a reductive S -group, and H a smooth affine S -group scheme with connected fibers over S . Then the S -functor*

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, H)$$

is representable by a smooth separated S -scheme (which is affine over S if G is semisimple).

Indeed, let U be the set of points s of S such that H_s is reductive; it is open (Exp. XIX, 2.6); if S' is an S -scheme, $H_{S'}$ is reductive if and only if $S' \rightarrow S$ factors through U . It follows that the canonical morphism $\underline{\text{Isom}}_{S\text{-gr.}}(G, H) \rightarrow S$ factors through U . One may therefore suppose $S = U$ and one is reduced to:

Corollary 1.9. *Let S be a scheme and G, G' two reductive S -groups. Then*

$$F = \underline{\text{Isom}}_{S\text{-gr.}}(G, G')$$

is representable by a smooth separated S -scheme (affine if G or G' is semisimple). Moreover, S decomposes as the sum of two open subschemes S_1 and S_2 such that $F_{S_1} = \emptyset$ and F_{S_2} is a left (resp. right) principal homogeneous bundle under $\underline{\text{Aut}}_{S_2\text{-gr.}}(G'_{S_2})$ (resp. $\underline{\text{Aut}}_{S_2\text{-gr.}}(G_{S_2})$).

PROOF. Indeed, let S_2 be the set of points of S where G and G' have the same type, and let S_1 be its complement.

Since the type of a reductive group is a locally constant function, S_1 and S_2 are open. It is clear that $F_{S_1} = \emptyset$, and one may suppose $S = S_2$. By Exp. XXIII, 5.6, F is a principal homogeneous sheaf under $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, locally trivial for the étale topology. It follows that $F_0 = F/\text{ad}(G)$ is a principal homogeneous sheaf under $\underline{\text{Autext}}(G)$, locally trivial for the étale topology, hence representable (Exp. X, 5.5). Since $\text{ad}(G)$ is affine, F is therefore also representable. $\square$

Let us observe that in the course of the proof one has obtained:

Corollary 1.10. *Let S be a scheme and G, G' two reductive S -groups of the same type at every $s \in S$. Then $\mathrm{ad}(G)$ acts freely (on the right) on $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$, and the quotient sheaf*

$$\underline{\mathrm{Isomext}}(G, G') = \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') / \mathrm{ad}(G)$$

is representable by a twisted constant S -scheme, which is a principal homogeneous bundle under $\underline{\mathrm{Autext}}(G)$ (and which is therefore finite over S if G is semisimple). Moreover, the isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \simeq \underline{\mathrm{Isom}}_{S\text{-gr.}}(G', G)$$

defined by $u \mapsto u^{-1}$ induces an isomorphism

$$\underline{\mathrm{Isomext}}(G, G') \simeq \underline{\mathrm{Isomext}}(G', G).$$

Remark 1.11. If $\underline{\mathrm{Isomext}}(G, G')(S) \neq \emptyset$, one says that G is an inner twisted form of G' ; then G' is an inner twisted form of G ; one can then reduce the structure group of $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ to $\mathrm{ad}(G)$. More precisely, let $u \in \underline{\mathrm{Isomext}}(G, G')(S)$, considered as a section $u : S \rightarrow \underline{\mathrm{Isomext}}(G, G')$. Denote by

$$\underline{\mathrm{Isomint}}_u(G, G')$$

the inverse image, under the canonical morphism $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \rightarrow \underline{\mathrm{Isomext}}(G, G')$, of the closed subscheme of $\underline{\mathrm{Isomext}}(G, G')$ which is the image of u . The natural action of $\mathrm{ad}(G)$ on $\underline{\mathrm{Isomint}}_u(G, G')$ equips this scheme with a structure of principal homogeneous bundle; by extension of the structure group⁽⁵⁾ $\mathrm{ad}(G) \rightarrow \underline{\mathrm{Aut}}(G)$, $\underline{\mathrm{Isomint}}_u(G, G')$ gives back $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$.

By Hensel's lemma (Exp. XI, 1.11), 1.8 immediately gives:

Corollary 1.12. *Let S be a henselian local scheme, G a reductive S -group, G' a smooth affine S -group with connected fibers, and s the closed point of S . If G_s and G'_s are isomorphic algebraic $\kappa(s)$ -groups, G and G' are isomorphic. More precisely, every $\kappa(s)$ -isomorphism $G_s \simeq G'_s$ comes from an S -isomorphism $G \simeq G'$.*

Applying 1.7 (resp. 1.12) to the scheme of dual numbers over a field, one deduces from Exp. III, 2.10 (resp. 3.10) part (i) (resp. (ii)) of the following corollary.

Corollary 1.13. *Let k be a field and G a reductive k -group.*

- (i) *If G is adjoint, one has $H^1(G, \mathrm{Lie}(G/k)) = 0$.⁽⁶⁾*
- (ii) *One has $H^2(G, \mathrm{Lie}(G/k)) = 0$.*

Remark 1.14. (i) The assertion concerning H^1 was known (Chevalley); that concerning H^2 was proved by Chevalley in most cases of the classification.

(ii) In fact, the conjunction of 1.13 and the uniqueness theorem over an algebraically closed field is essentially equivalent to the uniqueness theorem. A direct proof of 1.13 would therefore give a way to deduce the general uniqueness theorem from Chevalley's uniqueness theorem over a field.

The existence of reductive groups of all types over all schemes (Exp. XXV) shows that the obstructions to lifting a reductive k -group G over Artinian rings with residue field k (which, by Exp. III, 3.8, are elements of

$$H^3(G, \mathrm{Lie}(G/k) \otimes V) \simeq H^3(G, \mathrm{Lie}(G/k)) \otimes V,$$

⁽⁵⁾**N.D.E.** We have corrected $\mathrm{ad}(G) \rightarrow \underline{\mathrm{Autext}}(G)$ to $\mathrm{ad}(G) \rightarrow \underline{\mathrm{Aut}}(G)$.

⁽⁶⁾**N.D.E.** We have corrected the original, which stated (i) without assuming G adjoint. According to the characterization of the $H^i(G, -)$ as derived functors of $\mathrm{Hom}_G(k, -)$ (Exp. I, 5.3.1, see also [Ja87], I 4.16), one has $H^i(G, V) = \mathrm{Ext}_G^i(k, V)$ for every G -module V ; now if $\mathrm{char}(k) = p > 0$ and if $G = \mathrm{SL}_{p,k}$, then $\mathrm{Lie}(\mathrm{GL}_p/k)$ is a nontrivial extension of k by $\mathfrak{g} = \mathrm{Lie}(\mathrm{SL}_p/k)$, hence $H^1(G, \mathfrak{g}) \neq 0$. See also the addition 1.15.1 below.

where V is a certain k -vector space (equipped with the trivial action of G)) vanish. This seems to suggest that $H^3(G, \text{Lie}(G/k)) = 0$. Here again, a direct proof of this fact (if it is true) would doubtless give a way to deduce the general existence theorem from the existence theorem over a field (Chevalley's Tôhoku).

Corollary 1.15. *Let k be a field,⁽⁷⁾ and G a reductive k -group. Consider k as a trivial G -module. Then*

$$H^1(G, k) = H^2(G, k) = 0.$$

PROOF. ⁽⁸⁾Since $H^i(G_{\bar{k}}, \bar{k}) = H^i(G, k) \otimes \bar{k}$, one may suppose k algebraically closed. An element of $H^1(G, k)$ is nothing other than a morphism of k -groups $\phi : G \rightarrow \mathbf{G}_{a,k}$. Then $\phi(G) = G/\text{Ker}(\phi)$ is a smooth, connected and reductive subgroup (cf. XIX 1.7) of $\mathbf{G}_{a,k}$, hence trivial. Thus $H^1(G, k) = 0$.

Consider now the reductive k -group $H = G \times_k \mathbf{G}_{m,k}$. One has $\text{Lie}(H/k) = \text{Lie}(G/k) \oplus k$, a decomposition stable under H . For an arbitrary H -module V , one has

$$H^i(H, V) = H^i(G, H^0(\mathbf{G}_{m,k}, V))$$

(this follows from the characterization of the $H^i(H, -)$ as derived functors of $H^0(H, -) = H^0(G, H^0(\mathbf{G}_{m,k}, -))$, and from the fact that the functor $H^0(\mathbf{G}_{m,k}, -)$ is exact, cf. Exp. I, 5.3.1 and 5.3.3). In particular, for every i one has

$$H^i(H, \text{Lie}(H/k)) = H^i(G, \text{Lie}(G/k)) \oplus H^i(G, k)$$

hence $H^2(G, k) = 0$ by applying 1.13 (ii) to the reductive group H . $\square$

Corollary 1.15.1. ⁽⁹⁾*Let k be a field, G a reductive k -group, Z its center and $\mathcal{R} = (M, M^*, R, R^*)$ the type of G . Then one has*

$$H^1(G, \text{Lie}(G/k)) \simeq \text{Ext}_{\mathbf{Z}}^1(M/\Gamma_0(R), k).$$

In particular, $H^1(G, \text{Lie}(G/k)) = 0$ if and only if Z is smooth over k .

PROOF. Indeed, let $\mathfrak{g}$ (resp. $\mathfrak{z}$, resp. $\mathfrak{g}_{\text{ad}}$) be the Lie algebra of G (resp. Z , resp. $G_{\text{ad}} = G/Z$). It follows from 1.15 (and its proof) that

$$H^1(G, \mathfrak{g}) = H^1(G_{\text{ad}}, \mathfrak{g}) \simeq H^1(G_{\text{ad}}, \mathfrak{g}/\mathfrak{z}).$$

Put $C = \text{Coker}(\mathfrak{g} \rightarrow \mathfrak{g}_{\text{ad}})$; one has an exact sequence

$$0 \longrightarrow \mathfrak{g}/\mathfrak{z} \longrightarrow \mathfrak{g}_{\text{ad}} \longrightarrow C \longrightarrow 0.$$

Since $H^1(G_{\text{ad}}, \mathfrak{g}_{\text{ad}}) = 0$ (1.13) and $H^0(G_{\text{ad}}, \mathfrak{g}_{\text{ad}}) = 0$ (cf. Exp. II, 5.2.3), one obtains

$$H^1(G_{\text{ad}}, \mathfrak{g}/\mathfrak{z}) \simeq H^0(G_{\text{ad}}, C).$$

To compute the right-hand term, one may suppose k algebraically closed. Let (G, T, M, R) be a splitting of G ; then $Z = D_k(M/\Gamma_0(R))$ and C identifies with

$$\text{Coker}(\text{Hom}_{\mathbf{Z}}(M, k) \rightarrow \text{Hom}_{\mathbf{Z}}(\Gamma_0(R), k)) \simeq \text{Ext}_{\mathbf{Z}}^1(M/\Gamma_0(R), k),$$

equipped with the trivial action of G_{ad} . The corollary follows, since Z is not smooth over k if and only if $\text{char}(k) = p > 0$ and $M/\Gamma_0(R)$ has p -torsion (Exp. IX, 2.1). $\square$

⁽⁷⁾**N.D.E.** We have replaced “scheme” by “field”.

⁽⁸⁾**N.D.E.** Because of the correction made in 1.13, we have given another proof in the case of H^1 . Moreover, it follows from a theorem of G. Kempf that $H^i(G, k) = 0$ for every $i > 0$, cf. [Ja87], II 4.5 and 4.11.

⁽⁹⁾**N.D.E.** We have added this corollary, drawn from remarks of O. Gabber, which makes 1.13 (i) more precise.

Definition 1.16. Let S be a scheme and G a reductive S -group. One calls a *form of G over S* an S -group scheme G' locally isomorphic to G for the (fpqc) topology (it amounts to the same thing to say (cf. Exp. XXIII, 5.6) that G' is locally isomorphic to G for the étale topology, or again that G' is a reductive S -group of the same type as G at every point of S).

Corollary 1.17. *Let S be a scheme and G a reductive S -group.*

(i) *The functor*

$$G' \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$$

is an equivalence between the category of forms of G over S and the category of principal homogeneous bundles under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$.

(ii) *If $S' \rightarrow S$ is a covering morphism, forms of G trivialized by S' and bundles trivialized by S' correspond to one another.*

(iii) *Every principal homogeneous sheaf under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable and quasi-isotrivial (i.e. locally trivial for the étale topology).*

PROOF. The first assertion is formal in the category of sheaves (for (fpqc), for example). Moreover, every sheaf locally isomorphic to G (for (fpqc)) is representable (since G is affine over S) and locally isomorphic to G for the étale topology. Finally, for every form G' of G , the S -sheaf $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ is representable, by 1.8. The corollary follows at once. $\square$

Corollary 1.18. *The set of isomorphism classes of forms of the reductive group G over S is isomorphic to*

$$H^1(S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)) = H^1_{\text{ét}}(S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)) = \mathrm{Fib}(S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)).$$

If $S' \rightarrow S$ is a covering morphism, the subset consisting of forms trivialized by S' is isomorphic to $H^1(S'/S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G))$.

Corollary 1.19. *Let S be a scheme and $\mathcal{R}$ a reduced pinned root datum such that $\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})^{(10)}$ exists (a condition automatically satisfied, cf. Exp. XXV). Put*

$$\begin{aligned} A_S(\mathcal{R}) &= \underline{\mathrm{Aut}}_{S\text{-gr.}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})) \\ &= \mathrm{ad}(\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})) \cdot E(\mathcal{R})_S. \end{aligned}$$

(i) *The set of isomorphism classes of reductive S -groups of type $\mathcal{R}$ (Exp. XXII, 2.7) is isomorphic (by Exp. XXIII, 5.12) to*

$$H^1(S, A_S(\mathcal{R})) = H^1_{\text{ét}}(S, A_S(\mathcal{R})) = \mathrm{Fib}(S, A_S(\mathcal{R})).$$

(ii) *If $S' \rightarrow S$ is a covering morphism, the subset consisting of classes of groups splittable over S' is isomorphic to $H^1(S'/S, A_S(\mathcal{R}))$.*

Remark 1.20. With the preceding notation, to every reductive S -group of type $\mathcal{R}$ there is canonically associated a right principal homogeneous bundle under $A_S(\mathcal{R})$:

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R}), G) = P.$$

Observe that P can be interpreted as the “scheme of pinnings of G of type $\mathcal{R}$ ” (cf. 1.0). Moreover P is also a principal homogeneous bundle (on the left) under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$, a structure which appears immediately in the above description.

Proposition 1.21. *Let S be a henselian local scheme and s its closed point. The functor*

$$G \longmapsto G_s$$

induces a bijection from the set of isomorphism classes of reductive S -groups onto the set of isomorphism classes of reductive $\kappa(s)$ -groups.

⁽¹⁰⁾**N.D.E.** Cf. XXIII, Definition 5.11.

In particular, for every reductive S -group G , there exists a finite surjective étale morphism $S' \rightarrow S$ such that $G_{S'}$ is splittable.

PROOF. Using the existence of the $\acute{\text{E}}\text{p}_S(\mathcal{R})$ (Exp. XXV), one is reduced to proving that, if one puts $H = A_S(\mathcal{R})$, the canonical map

$$\text{Fib}(S, H) \rightarrow \text{Fib}(\kappa(s), H_s)$$

is bijective (and that every element of $\text{Fib}(S, H)$ has the property indicated above). Now every finite subset of H is contained in an affine open subset (this is indeed trivial for a constant group, and H is affine over a constant group); one can therefore use the result proved in the appendix (8.1). $\square$

2. Automorphisms and subgroups

Let us introduce some notation: if $H = \underline{\text{Aut}}_{S\text{-gr.}}(G)$, and if X is a subfunctor of G , one puts

$$\begin{aligned} \underline{\text{Aut}}_{S\text{-gr.}}(G, X) &= \underline{\text{Norm}}_H(X), \\ \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_X) &= \underline{\text{Centr}}_H(X). \end{aligned}$$

If Y is a second subfunctor of G , one likewise defines $\underline{\text{Aut}}_{S\text{-gr.}}(G, X, Y) = \underline{\text{Aut}}_{S\text{-gr.}}(G, X) \cap \underline{\text{Aut}}_{S\text{-gr.}}(G, Y)$, and if G' is a second S -group and X' a subfunctor of G , one denotes by $\underline{\text{Isom}}_{S\text{-gr.}}(G, X; G', X')$ the subfunctor of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$ defined by: for every $S' \rightarrow S$,

$$\begin{aligned} \underline{\text{Isom}}_{S\text{-gr.}}(G, X; G', X')(S') \\ = \{u \in \underline{\text{Isom}}_{S\text{-gr.}}(G, G')(S') \mid u(X_{S'}) = X'_{S'}\} \end{aligned}$$

and one similarly defines $\underline{\text{Isom}}_{S\text{-gr.}}(G, X, Y; G', X', Y')$, etc.⁽¹¹⁾

Proposition 2.1. *Let S be a scheme, G a reductive S -group, and T a maximal torus of G (resp. B a Borel subgroup of G , resp. $B \supset T$ a Killing pair of G). Denote by T^{ad} (resp. B^{ad}) the maximal torus (resp. Borel subgroup) of $\text{ad}(G)$ corresponding to T (resp. B):*

$$\begin{aligned} B^{\text{ad}} &\simeq B/\underline{\text{Centr}}(G) = B/\underline{\text{Centr}}(B), \\ T^{\text{ad}} &\simeq T/\underline{\text{Centr}}(G). \end{aligned}$$

Then $\underline{\text{Aut}}_{S\text{-gr.}}(G, T)$ (resp. $\underline{\text{Aut}}_{S\text{-gr.}}(G, B)$, resp. $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)$) is representable by a closed subscheme of $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, smooth over S , and the exact sequence of Theorem 1.3 induces exact sequences:

$$\begin{aligned} 1 &\rightarrow \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}}) \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, T) \rightarrow \underline{\text{Autext}}(G) \rightarrow 1; \\ 1 &\rightarrow B^{\text{ad}} \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, B) \rightarrow \underline{\text{Autext}}(G) \rightarrow 1; \\ 1 &\rightarrow T^{\text{ad}} \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, B, T) \rightarrow \underline{\text{Autext}}(G) \rightarrow 1. \end{aligned}$$

PROOF. By descent of closed subschemes,⁽¹²⁾ one is immediately reduced to the case where G is pinned and $B \supset T$ is its canonical Killing pair (cf. Exp. XXII, 5.5.5 (iv)). Since the group E of 1.2 normalizes B and T , the result follows at once from the normalizer theorems in $\text{ad}(G)$ (Exp. XXII, 5.3.12 and 5.6.1). $\square$

Using now the conjugacy theorems (cf. Exp. XXIII, 5.12), and reasoning as in no. 1, one deduces:

Corollary 2.2. *Let S be a scheme and G, G' two reductive S -groups of the same type at every point. Let $B \supset T$ (resp. $B' \supset T'$) be a Killing pair of G (resp. G').*

⁽¹¹⁾N.D.E. We have made the preceding definitions explicit (the original indicated “One similarly defines $\underline{\text{Aut}}_{S\text{-gr.}}(G, X, Y), \dots, \underline{\text{Isom}}_{S\text{-gr.}}(G, X; G', X'), \dots$ ”).

⁽¹²⁾N.D.E. Cf. SGA 1, VIII 1.9.

- (i) The S -functor $\underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')$ is representable by a smooth closed subscheme of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$ which is principal homogeneous under $\underline{\text{Aut}}_{S\text{-gr.}}(G, T)$.

Moreover, $\underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}})$ acts freely on this scheme, and one has a canonical isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T') / \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}}) \simeq \underline{\text{Isomext}}(G, G').$$

- (ii) The S -functor $\underline{\text{Isom}}_{S\text{-gr.}}(G, B; G', B')$ is representable by a smooth closed subscheme of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$ which is principal homogeneous under $\underline{\text{Aut}}_{S\text{-gr.}}(G, B)$.

Moreover, B^{ad} acts freely on this scheme and one has a canonical isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, B; G', B') / B^{\text{ad}} \simeq \underline{\text{Isomext}}(G, G').$$

- (iii) The S -functor $\underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T')$ is representable by a smooth closed subscheme of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$, principal homogeneous under $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)$.

Moreover, T^{ad} acts freely on this scheme and one has a canonical isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T') / T^{\text{ad}} \simeq \underline{\text{Isomext}}(G, G').$$

Reasoning again as in no. 1, one deduces:

Corollary 2.3. Let S be a scheme, G a reductive S -group, and $B \supset T$ a Killing pair of G . The functor

$$(G', T') \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T'),$$

resp.

$$(G', B') \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, B; G', B'),$$

resp.

$$(G', B', T') \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T'),$$

is an equivalence between the category of pairs (G', T') (resp. of pairs (G', B') , resp. of triples (G', B', T')), where G' is a form of G and T' a maximal torus of G' (resp. B' a Borel subgroup of G' , resp. $B' \supset T'$ a Killing pair of G'), and the category of principal homogeneous bundles under the S -group H , where $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, T)$ (resp. $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, B)$, resp. $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)$).

Moreover, every principal homogeneous sheaf under H is representable and quasi-isotrivial, so that one has

$$H^1(S, H) = H^1_{\text{ét}}(S, H) = \text{Fib}(S, H).$$

Remark 2.4. Under the conditions of 2.2, the morphism denoted set-theoretically by $u \longmapsto u(T)$ (resp. $u \longmapsto u(B)$, resp. $u \longmapsto (u(B), u(T))$) induces an isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, G') / \underline{\text{Aut}}_{S\text{-gr.}}(G, T) \simeq \underline{\text{Tor}}(G'),$$

$$\text{resp. } \underline{\text{Isom}}_{S\text{-gr.}}(G, G') / \underline{\text{Aut}}_{S\text{-gr.}}(G, B) \simeq \underline{\text{Bor}}(G'),$$

$$\text{resp. } \underline{\text{Isom}}_{S\text{-gr.}}(G, G') / \underline{\text{Aut}}_{S\text{-gr.}}(G, B, T) \simeq \underline{\text{Kil}}(G').$$

The proof is immediate: it suffices to give it locally for (fpqc), so one may suppose $G \simeq G'$, and one is reduced to Exp. XXII, 5.8.3 (iii).

Remark 2.5. The preceding results can immediately be interpreted in terms of restriction of the structure group: if G' is a form of G , corresponding to the principal bundle $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$, giving a restriction of the structure group of this bundle to $\underline{\text{Aut}}_{S\text{-gr.}}(G, T)$ amounts to giving a maximal torus T' of G' , the bijections suggested above being that of 2.4 on the one hand, and the map $T' \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')$ on the other hand. Likewise for Borel subgroups and Killing pairs.

Proposition 2.6. Let S be a scheme and G, G' two reductive S -groups of the same type at every point, and T (resp. T') a maximal torus of G (resp. G'). Then T^{ad} acts freely on

$\underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')$, and the quotient

$$P = \underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')/T^{\text{ad}}$$

is representable; it is a principal homogeneous bundle under

$$A = \underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}},$$

where A is representable by a twisted constant S -scheme, an extension of $\underline{\text{Autext}}(G)$ by $\text{W}_{\text{ad}(G)}(T^{\text{ad}}) = \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}})/T^{\text{ad}}$. Moreover, if one lets A act on T in the evident way, the bundle associated to P is nothing other than T' .⁽¹³⁾

PROOF. The first part of the proposition follows at once from the preceding results. To prove the second, one observes that there is an evident morphism $P \times_S T \rightarrow T'$ (defined by $(u, t) \mapsto u(t)$); to prove that after passage to the quotient by A it induces an isomorphism, one may once more suppose $(G, T) \simeq (G', T')$, in which case it is immediate. $\square$

In an entirely similar way, one has:

Proposition 2.7. *Let S be a scheme and G, G' two reductive S -groups of the same type at every point, and $B \supset T$ (resp. $B' \supset T'$) a Killing pair of G (resp. G'). If one lets $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)/T^{\text{ad}} \simeq \underline{\text{Autext}}(G)$ act on T in the evident way, the bundle associated to $\underline{\text{Isomext}}(G, G')$ is nothing other than T' .*

Corollary 2.8. *Let G and G' be two reductive S -groups which are inner twisted forms of one another; let $B \supset T$ (resp. $B' \supset T'$) be a Killing pair of G (resp. G'). Then T and T' are isomorphic.⁽¹⁴⁾*

Remark 2.9. It is not true in general that B and B' are isomorphic; they are, however, inner twisted forms of one another (cf. no. 5).

One can develop “ $\underline{\text{Isomint}}$ ”⁽¹⁵⁾ variants of the preceding results. Let us mention one:

Proposition 2.10. *Let S be a scheme and G, G' two reductive S -groups of the same type at every point, and $B \supset T$ (resp. $B' \supset T'$) a Killing pair of G (resp. G'). Let $u \in \underline{\text{Isomext}}(G, G')(S)$; consider*

$$\begin{aligned} & \underline{\text{Isomint}}_u(G, B, T; G', B', T') \\ &= \underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T') \cap \underline{\text{Isomint}}_u(G, G'). \end{aligned}$$

It is a smooth affine S -scheme which is a principal homogeneous bundle under T^{ad} . In particular, $\underline{\text{Isomint}}_u(G, B, T; G', B', T')(S) \neq \emptyset$ if and only if the corresponding element of $H^1(S, T^{\text{ad}})$ is zero.

To finish this section, let us prove two results which will be useful to us in what follows:

Proposition 2.11. *Let S be a scheme, G a reductive S -group, and T a maximal torus of G . The evident morphism*

$$T^{\text{ad}} \xrightarrow{\sim} \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_T)$$

is an isomorphism.

This follows from the preceding statements; moreover, a direct proof has been given in the course of the proof of 1.5.2.

⁽¹³⁾**N.D.E.** Thus, if P has a section, then T and T' are isomorphic; in particular T'_P and T_P are isomorphic. See the addition 4.5.1 where this observation is developed in the case where $G' = \acute{\text{E}}\text{p}_S(\mathcal{A})$.

⁽¹⁴⁾**N.D.E.** By 2.7, since $\underline{\text{Isomext}}(G, G')$ has a section (cf. the preceding N.D.E.).

⁽¹⁵⁾**N.D.E.** Recall that $\underline{\text{Isomint}}_u(G, G')$ is defined in 1.11.

Corollary 2.12. *Under the preceding conditions, there exists an equivalence between the category of pairs (G', f) , where G' is a form of G and f an isomorphism of T onto a maximal torus of G' , and the category of principal homogeneous bundles under T^{ad} .*

Corollary 2.13. *If $H^1(S, T^{\text{ad}}) = 0$, and if G' is a form of G possessing a maximal torus isomorphic to T , then G' is isomorphic to G .*

Corollary 2.14. *Let S be a scheme such that $\text{Pic}(S) = 0$, and G a reductive S -group of constant type. For G to be splittable, it is necessary and sufficient that G possess a split maximal torus.*

Let G be a reductive group and $\text{rad}(G)$ its radical (Exp. XXII, 4.3.9); since $\text{rad}(G)$ is central and characteristic in G , one has a canonical morphism

$$q : \underline{\text{Autext}}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(\text{rad}(G)).$$

Proposition 2.15. *Let G be a reductive S -group. The following sequence is exact:*

$$1 \longrightarrow \text{ad}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_{\text{rad}(G)})^p \longrightarrow \underline{\text{Autext}}(G) \xrightarrow{q} \underline{\text{Aut}}_{S\text{-gr.}}(\text{rad}(G))$$

and $\text{Ker}(q) = \text{Im}(p)$ is an open and closed subscheme of $\underline{\text{Autext}}(G)$, finite over S .

PROOF. One may suppose G split. The first assertion is immediate; the second follows from Exp. XXI, 6.7.5 and 6.7.7. $\square$

Putting $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_{\text{rad}(G)})$, one deduces:

Corollary 2.16. *There exists an equivalence between the category of pairs (G', f) , where G' is a form of G and f an isomorphism of $\text{rad}(G)$ onto the radical of G' , and the category of principal bundles under a certain S -group scheme H , where H is such that there exists an exact sequence*

$$1 \longrightarrow \text{ad}(G) \longrightarrow H \longrightarrow F \longrightarrow 1,$$

where the S -group F is étale and finite over S .

3. Dynkin scheme of a reductive group. Quasi-split groups

3.1. — Recall (Exp. XXI, 7.4.1) that a *Dynkin diagram* is a finite set equipped with the structure defined by a set of pairs of distinct elements (bonds) and a map into $\{1, 2, 3\}$ (lengths). To each reduced pinned root datum $\mathcal{R}$ is associated a Dynkin diagram $\Delta(\mathcal{R})$, whose underlying set is the set of simple roots.

3.2. — Let S be a scheme. A *Dynkin S -scheme* is a finite twisted constant S -scheme Δ , equipped with the structure defined by a subscheme L of $\Delta \times_S \Delta$ having empty intersection with the diagonal, and a morphism $\Delta \longrightarrow \{1, 2, 3\}_S$. If S' is a connected⁽¹⁶⁾ S -scheme, $\Delta(S')$ is naturally equipped with the structure of a Dynkin diagram.

One immediately defines the following notions: isomorphism of two Dynkin schemes, extension of the base of a Dynkin scheme, constant Dynkin scheme associated to a Dynkin diagram.

Every descent datum on a Dynkin scheme for the étale topology is effective.

3.3. — One proposes to associate to every reductive S -group G a Dynkin S -scheme. Suppose first that G is *splittable* over S ; for every pinning $\mathcal{E}$ of G , denote by $\Delta(\mathcal{E})$ the constant Dynkin scheme associated to the pinned root datum defined by $\mathcal{E}$; if $\mathcal{E}$ and $\mathcal{E}'$ are two pinnings of G , there exists by 1.5 a *unique inner automorphism* of G over S taking $\mathcal{E}$ to $\mathcal{E}'$; this automorphism of G defines an isomorphism $a_{\mathcal{E}\mathcal{E}'} : \Delta(\mathcal{E}) \xrightarrow{\sim} \Delta(\mathcal{E}')$; the $a_{\mathcal{E}\mathcal{E}'}$ clearly form a transitive system, so that one can identify the $\Delta(\mathcal{E})$ (i.e. take the inductive limit); the result

⁽¹⁶⁾**N.D.E.** We have added the hypothesis that S' be connected.

is a constant Dynkin scheme denoted by $\underline{\text{Dyn}}(G)$. If now G is an *arbitrary* reductive S -group, there exists a covering family for the étale topology $\{S_i \rightarrow S\}$ such that G_{S_i} is splittable. Reasoning as before, one therefore has a canonical descent datum on the $\underline{\text{Dyn}}(G_{S_i})$, allowing one to construct by descent a Dynkin S -scheme $\underline{\text{Dyn}}(G)$.

3.4. — This construction satisfies the following properties (which, moreover, essentially characterize it):

- (i) To every reductive S -group is associated a Dynkin scheme $\underline{\text{Dyn}}(G)$; to every isomorphism $u : G \xrightarrow{\sim} G'$ is functorially associated an isomorphism $\underline{\text{Dyn}}(u) : \underline{\text{Dyn}}(G) \xrightarrow{\sim} \underline{\text{Dyn}}(G')$.
- (ii) If S' is an S -scheme and G a reductive S -group, one has

$$\underline{\text{Dyn}}(G \times_S S') \simeq \underline{\text{Dyn}}(G) \times_S S'.$$

- (iii) If $\mathcal{E}$ is a pinning of G , defining the pinned root datum with Dynkin diagram Δ , one has

$$\underline{\text{Dyn}}(G) \simeq \Delta_S.$$

- (iv) If u is an inner automorphism of G , $\underline{\text{Dyn}}(u)$ is the identity automorphism of $\underline{\text{Dyn}}(G)$.

3.5. — Let S be a scheme and G a reductive S -group. It is clear that the functor $\underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G))$ of automorphisms of $\underline{\text{Dyn}}(G)$ for the Dynkin scheme structure is representable by a finite twisted constant S -scheme. By 3.4 (i) and (ii), one has a canonical morphism

$$\underline{\text{Aut}}_{S\text{-gr.}}(G) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G)),$$

which, by (iv), factors through a morphism

$$\underline{\text{Autext}}(G) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G)).$$

More generally, if G and G' are two reductive S -groups, one has a canonical morphism

$$\underline{\text{Isomext}}(G, G') \rightarrow \underline{\text{Isom}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G), \underline{\text{Dyn}}(G'));$$

in particular, if G' is an inner twisted form of G (1.11), the Dynkin schemes $\underline{\text{Dyn}}(G)$ and $\underline{\text{Dyn}}(G')$ are isomorphic.

3.6. — If G is semisimple (resp. adjoint or simply connected), the morphism

$$\underline{\text{Autext}}(G) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G))$$

is a monomorphism (resp. an isomorphism). Indeed, one may suppose G pinned and one is reduced to the corresponding result for reduced pinned root data (cf. Exp. XXI, 7.4.5).

One has an analogous result for the $\underline{\text{Isom}}$; it follows in particular that two adjoint (resp. simply connected) semisimple S -groups are inner twisted forms of one another if and only if their Dynkin schemes are isomorphic.

3.7. — One can give a different construction of the Dynkin scheme associated to a reductive group. Let $\mathcal{R}$ be a reduced pinned root datum and G a reductive S -group of type $\mathcal{R}$; denote by $\Delta(\mathcal{R})$ the Dynkin diagram defined by the root datum $\mathcal{R}$. One has (3.5) a canonical morphism

$$A_S(\mathcal{R}) = \underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{E}}p_S(\mathcal{R})) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\Delta(\mathcal{R})_S).$$

The reductive S -group G corresponds (1.17) to a bundle $\underline{\text{Isom}}_{S\text{-gr.}}(\dot{\text{E}}p_S(\mathcal{R}), G)$, principal homogeneous under $A_S(\mathcal{R})$. The associated bundle under $\underline{\text{Aut}}_{\underline{\text{Dyn}}}(\Delta(\mathcal{R})_S)$ corresponds to a form over S of $\Delta(\mathcal{R})_S$: it is $\underline{\text{Dyn}}(G)$; in other words, this associated bundle is nothing other than $\underline{\text{Isom}}_{\underline{\text{Dyn}}}(\Delta(\mathcal{R})_S, \underline{\text{Dyn}}(G))$. In this latter form the proof is immediate.

3.8. — **Dynkin scheme and Killing pairs.** Let S be a scheme, G a reductive S -group, and $B \supset T$ a Killing pair of G . There exists a canonical morphism

$$i : \underline{\text{Dyn}}(G) \rightarrow \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$$

which identifies $\underline{\text{Dyn}}(G)$ with the “scheme of simple roots of B relative to T ”; this morphism is immediately defined by descent from the pinned case. Observe moreover that the datum of T and i allows one to reconstruct B (“one-to-one correspondence between systems of simple roots and systems of positive roots”).

It follows from the preceding description of $\mathcal{D} = \underline{\text{Dyn}}(G)$ that there exists a *canonical* root of $B_{\mathcal{D}}$ relative to $T_{\mathcal{D}}$: this root $\alpha_{\mathcal{D}}$ is the image under $i(\mathcal{D})$ of the identity morphism of $\mathcal{D}$. One deduces a *canonical invertible* $\mathcal{O}_{\mathcal{D}}$ -module $\mathfrak{g}^{\mathcal{D}}$:

$$\mathfrak{g}^{\mathcal{D}} = (\mathfrak{g} \otimes_S \mathcal{O}_{\mathcal{D}})^{\alpha_{\mathcal{D}}}.$$

In the pinned case one has

$$\mathcal{D} = \coprod_{\alpha \in \Delta} S_{\alpha},$$

where each S_{α} is a copy of S , and $\mathfrak{g}^{\mathcal{D}}$ is then the $\mathcal{O}_{\mathcal{D}}$ -module which induces $\mathfrak{g}^{\alpha}$ on S_{α} , for every $\alpha \in \Delta$.

3.9. — Quasi-pinnings. Quasi-pinned groups. If G is a reductive S -group, one calls a *quasi-pinning* of G the datum:

- (i) of a Killing pair $B \supset T$ of G ,
- (ii) of a section $X \in \Gamma(\underline{\text{Dyn}}(G), \mathfrak{g}^{\mathcal{D}})^{\times}$.

One says that a reductive S -group is *quasi-splittable* if it possesses a quasi-pinning. One calls a *quasi-pinned group* a reductive group equipped with a quasi-pinning.

Let $B \supset T$ be a Killing pair of the reductive S -group G ; then G admits a quasi-pinning relative to this Killing pair if and only if $\mathfrak{g}^{\mathcal{D}}$ has a section nonzero at every point, i.e. if the element of $\text{Pic}(\underline{\text{Dyn}}(G))$ defined by $\mathfrak{g}^{\mathcal{D}}$ is zero. Suppose in particular that S is semi-local; then $\underline{\text{Dyn}}(G)$ is also semi-local, hence $\text{Pic}(\underline{\text{Dyn}}(G)) = 0$. One deduces:

Proposition 3.9.1. *Let S be a semi-local scheme and G a reductive S -group. For G to be quasi-splittable, it is necessary and sufficient that it possess a Borel subgroup.*

PROOF. ⁽¹⁷⁾Indeed, S is affine, hence, by the first assertion of Exp. XXII, 5.9.7, if G possesses a Borel subgroup B , it also possesses a Killing pair $B \supset T$. Then, since $\text{Pic}(\mathcal{D}) = 0$, $\mathfrak{g}^{\mathcal{D}}$ has an everywhere nonzero section X . $\square$

Let again A be a semi-local ring and $S = \text{Spec}(A)$; observe now that for every reductive S -group G the morphism $\underline{\text{Bor}}(G) \rightarrow S$ is surjective (since $G_{\bar{s}}$ possesses Borel subgroups, for every $s \in S$) and smooth and projective (Exp. XXII, 5.8.3), hence has sections after a finite surjective étale extension of the base.⁽¹⁸⁾ One deduces the

⁽¹⁷⁾N.D.E. We have given the reference to Exp. XXII, 5.9.7 in detail.

⁽¹⁸⁾N.D.E. The original referred to EGA IV, § 24, which did not appear. One must use “Bertini’s theorem”. Let us give the argument in detail, as indicated to us by O. Gabber. Let $X \rightarrow S$ be a surjective, smooth and projective morphism; replacing S by a connected component, one may suppose that X/S is everywhere of relative dimension d (cf. EGA IV₄, 17.10.2). One may also suppose that X is a closed subscheme of a projective space $\mathbf{P}_S^n = \mathbf{P}(E)$, where E is a free A -module of rank $n+1$ (cf. EGA II, 5.3.3). Let $s_1, \dots, s_r$ be the closed points of S ; by Bertini’s theorem (see for example [Jou83], I 6.10), there exists an open subset U of the product $P = \mathbf{P}(E^*)^d$, with nonempty fibers, such that for every point $u = (f_1, \dots, f_d)$ of U_{s_i} , the intersection of $X_{\kappa(u)}$ with the d hyperplanes of $\mathbf{P}_{\kappa(u)}^n$ defined by u is étale over $\kappa(u)$. After shrinking U , one may suppose that U is the complement in affine space $\mathbf{A}_S^{nd}$ of the zero locus $\mathcal{V}(Q)$ of a certain polynomial Q of degree $m > 0$. One then sees easily (by induction on the number of variables) that $\mathcal{V}(Q)$ cannot contain all the rational points of $\mathbf{A}_{s_i}^{nd}$ if $|\kappa(s_i)| > m$. Since the morphism $A \rightarrow \prod_i \kappa(s_i)$ is surjective, one can find a monic polynomial $R \in A[X]$ of degree m whose image in $\kappa(s_i)[X]$ is irreducible if $\kappa(s_i)$ is finite, and has m distinct roots if $\kappa(s_i)$ is infinite. Put $A' = A[X]/(R)$ and $S' = \text{Spec}(A')$; then $S' \rightarrow S$ is étale, finite and surjective, and the residue field at each of its closed points $s'_1, \dots, s'_t$ has cardinality $\geq 2^m$. Then U has a rational point u_i over each closed point of S' , and since $A' \rightarrow \prod_i \kappa(s_i)$ is surjective, these lift to a section u of $P_{S'}$. Denote by Z the intersection of $X_{S'}$ with the d hyperplanes of $\mathbf{P}_{S'}^n$ defined by u , and by V the open subset of Z consisting of points where Z is étale over S' . By EGA IV₃, 11.3.8, V contains the fibers $Z_{s'_i}$ for every i ; since $\pi : Z \rightarrow S'$ is proper, it follows that the closed subset $\pi(Z - V)$ is empty, whence $V = Z$.

Corollary 3.9.2. *Let S be a semi-local scheme and G a reductive S -group. There exists an étale, finite and surjective morphism $S' \rightarrow S$ such that $G_{S'}$ is quasi-splittable.*

Remark 3.9.3. Under the preceding conditions, let T be a maximal torus of G (cf. Exp. XIV, 3.20); then one may suppose in addition that $G_{S'}$ is quasi-splittable relative to $T_{S'}$: it suffices to apply the preceding reasoning to the “scheme of Borel subgroups containing T ”, which is finite and étale over S (Exp. XXII, 5.5.5 (ii)).

3.10. — Let $\mathcal{E}$ and $\mathcal{E}'$ be two quasi-pinnings of the reductive S -group G . There exists a *unique* inner automorphism of G taking $\mathcal{E}$ to $\mathcal{E}'$. Indeed, one is immediately reduced to the split case, where the assertion has already been proved (1.5; indeed, it suffices to observe that for an inner automorphism of G it amounts to the same thing to respect a pinning or the underlying quasi-pinning). One concludes, as in no. 1, that a quasi-pinning of the reductive S -group G defines a splitting h of the exact sequence:

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \xleftarrow[\substack{h \\ p}]{\substack{h \\ p}} \underline{\mathrm{Autext}}(G) \longrightarrow 1,$$

the image of h being the subgroup of $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ which leaves the quasi-pinning invariant.

Likewise, if G and G' are two quasi-pinned S -groups, one naturally defines the subfunctor

$$\underline{\mathrm{Isom}}_{S\text{-gr. q-pin.}}(G, G')$$

of $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$; the projection from $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ onto $\underline{\mathrm{Isomext}}(G, G')$ induces an isomorphism

$$(*) \quad \underline{\mathrm{Isom}}_{S\text{-gr. q-pin.}}(G, G') \xrightarrow{\sim} \underline{\mathrm{Isomext}}(G, G').$$

Theorem 3.11. *Let S be a scheme, $\mathcal{R}$ a reduced pinned root datum such that $\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})$ exists (cf. Exp. XXV), and E the group of its automorphisms. Consider the following three categories:*

- (i) *The category $\underline{\mathrm{Rev}}$ of principal Galois coverings of S with group E (the morphisms are the isomorphisms of E -bundles).*
- (ii) *The category $\underline{\mathrm{Redext}}$ whose objects are the reductive S -groups of type $\mathcal{R}$ (Exp. XXII, 2.7), the morphisms from G to G' being the elements of $\underline{\mathrm{Isomext}}(G, G')(S)$.*
- (iii) *The category $\underline{\mathrm{Qép}}$ of quasi-pinned reductive S -groups of type $\mathcal{R}$ (the morphisms are the isomorphisms respecting the quasi-pinnings).*

These three categories are equivalent: more precisely, one has a diagram of functors, commutative up to functorial isomorphisms

$$\begin{array}{ccc} \underline{\mathrm{Rev}} & \xrightarrow{\quad \mathrm{qép} \quad} & \underline{\mathrm{Qép}} \\ & \swarrow \mathrm{rev} \quad \searrow i & \\ & \underline{\mathrm{Redext}} & \end{array} .$$

We shall describe these three functors below, leaving to the reader the task of verifying the commutativity of the diagram.⁽¹⁹⁾

3.11.1. — The functor i . It is the evident functor: $i(G) = G$, $i(f) = \text{image of } f \text{ under the isomorphism of 3.10 } (*)$.

3.11.2. — The functor $\mathrm{qép}$. Let S' be a principal Galois covering of S with group E . Let $(G, T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ be a pinning of $G = \acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})$ and let $(X'_\alpha)_{\alpha \in \Delta}$ be the corresponding

Then $Z \rightarrow S'$ is surjective, étale and proper, hence finite, as is the composite $Z \rightarrow S$, and this supplies the desired section of $X \rightarrow S$.

⁽¹⁹⁾**N.D.E.** In 3.11.4 we have sketched the verification that the functors i , rev and $\mathrm{qép}$ are fully faithful, and that the composite $\mathrm{rev} \circ i \circ \mathrm{qép}$ is isomorphic to the identity functor of $\underline{\mathrm{Rev}}$.

pinning of $G' = G_{S'} = \dot{\text{Ép}}_{S'}(\mathcal{R})$. By Exp. XXIII 5.5 bis, there exists a unique group morphism

$$\theta : E \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{Ép}}_S(\mathcal{R})) = A(\mathcal{R})(S)$$

such that, for every $h \in E$, $\theta(h)$ induces $D_S(h)$ on T and $\text{Lie}(\theta(h))(X_\alpha) = X_{h(\alpha)}$ for every $\alpha \in \Delta$. Taking account of the action of E on S' , one thus obtains an action of E on G' , compatible with the action of E on S' and respecting the canonical quasi-pinning of G' .⁽²⁰⁾ Since $S' \longrightarrow S$ is covering for the (fpqc) topology, it is a morphism of effective descent for the fibered category of affine morphisms, and one denotes by

$$\text{qép}(S') = \text{Q-Ép}_{S'/S}(\mathcal{R})$$

the *quasi-pinned* S -group obtained by Galois descent.⁽²¹⁾

3.11.3. — The functor rev . Let G be a reductive S -group of type $\mathcal{R}$. One puts

$$\text{rev}(G) = \underline{\text{Isomext}}(\dot{\text{Ép}}_S(\mathcal{R}), G),$$

it is a principal homogeneous bundle under E_S (cf. 1.10 and 1.19), i.e. an object of $\underline{\text{Rev}}$.

3.11.4. — ⁽²²⁾It is clear, by the isomorphism 3.10 (*), that the functor i is fully faithful. Moreover, if G_0, G, G' are reductive S -groups, the natural morphism

$$\underline{\text{Isom}}(G_0, G) \times_S \underline{\text{Isom}}(G, G') \longrightarrow \underline{\text{Isom}}(G_0, G')$$

induces a morphism

$$(*) \quad \underline{\text{Isomext}}(G_0, G) \times_S \underline{\text{Isomext}}(G, G') \longrightarrow \underline{\text{Isomext}}(G_0, G')$$

since, for every $S' \longrightarrow S$, if $g_0 \in G_0(S')$, $g \in G(S')$, $u \in \underline{\text{Isom}}(G_0, G)(S')$, and $v \in \underline{\text{Isom}}(G, G')(S')$, one has the equality $v \circ \text{int}(g) \circ u \circ \text{int}(g_0) = v \circ u \circ \text{int}(u^{-1}(g)g_0)$. Taking $G_0 = \dot{\text{Ép}}_S(\mathcal{R})$, one obtains a map

$$\begin{aligned} \underline{\text{Isomext}}(G, G')(S) &\longrightarrow \underline{\text{Hom}}_{\text{Rev}}(\underline{\text{Isomext}}(\dot{\text{Ép}}_S(\mathcal{R}), G), \\ &\quad \underline{\text{Isomext}}(\dot{\text{Ép}}_S(\mathcal{R}), G')) \end{aligned}$$

and to show that it is a bijection, one may suppose, by (fpqc) descent, that G and G' are pinned, in which case it is evident. Likewise, if S_1, S_2 are two objects of $\underline{\text{Rev}}$ and if one denotes by $S_i \times^E \dot{\text{Ép}}_S(\mathcal{R})$ the S -group $\text{qép}(S_i)$ obtained by twisting $\dot{\text{Ép}}_S(\mathcal{R})$ by the E -torsor S_i (cf. N.D.E. (21)), one has a natural map

$$\begin{aligned} \underline{\text{Hom}}_{\text{Rev}}(S_1, S_2) &\longrightarrow \underline{\text{Hom}}_{\text{Qép}}(S_1 \times^E \dot{\text{Ép}}_S(\mathcal{R}), \\ &\quad S_2 \times^E \dot{\text{Ép}}_S(\mathcal{R})), \end{aligned}$$

and to show that it is a bijection, one may suppose, by (fpqc) descent, that S_1 and S_2 are trivial E -bundles, in which case it is evident.

⁽²⁰⁾**N.D.E.** We have expanded the original in the above, to show that the action of E on G' is obtained by combining the natural action of E on $\dot{\text{Ép}}_{S'}(\mathcal{R})$ and the given action on S' . The canonical Killing pair of G' is preserved by this action, as is the quasi-pinning given by the element $\tilde{X}'$ of $\Gamma(\Delta_{S'}, \text{Lie}(G'/S')^{\Delta_{S'}})$, equal to X'_α on the copy of S' indexed by α (indeed, since E acts on $\Delta_{S'}$ by permuting the copies of S' , one has $h(\tilde{X}') = \tilde{X}'$ for every $h \in E$).

⁽²¹⁾**N.D.E.** See, for example, TDTE I, p. 22, Example 1. One can also describe $\text{qép}(S')$ as the twist of $G = \dot{\text{Ép}}_S(\mathcal{R})$ by the E -torsor S'/S , i.e. the (fpqc) quotient sheaf of $S' \times_S G$ by the right action of E defined by $(s', g) \cdot h = (s'h, h^{-1}(g))$. Since G is affine over S and E acts on G by group automorphisms, this sheaf is representable by an S -group $G^\sharp$, which is a “twisted” form of G , and $\mathcal{D}^\sharp = \underline{\text{Dyn}}(G^\sharp)$ is the twist of the constant Dynkin scheme Δ_S by the torsor S'/S . Since E normalizes B and T , one similarly obtains a pair $(B^\sharp, T^\sharp)$ which is a Killing pair of $G^\sharp$. Moreover, let $\mathfrak{g} = \text{Lie}(G/S)$ and $\mathfrak{g}^\sharp = \text{Lie}(G^\sharp/S)$; for every $U \longrightarrow S$, the sections of $\mathfrak{g}^\sharp$ over U are the E -equivariant S -morphisms $U \times_S S' \longrightarrow \mathbf{W}(\mathfrak{g})$. Since $\mathcal{D}^\sharp \times_S S'$ is E -isomorphic to $\Delta \times S'$, equipped with the action $(\alpha, s') \cdot h = (h^{-1}(\alpha), s'h)$, one obtains that the morphism given by $(\alpha, s') \longmapsto X_\alpha$ is a section of $\mathfrak{g}^\sharp$ over $\mathcal{D}^\sharp$ which is a quasi-pinning, i.e. an everywhere nonzero section of $(\mathfrak{g}^\sharp \otimes_{\mathcal{O}_S} \mathcal{O}_{\mathcal{D}^\sharp})^{\mathcal{D}^\sharp}$ (cf. 3.8).

⁽²²⁾**N.D.E.** We have added this paragraph.

Thus the three functors of the diagram are fully faithful. Moreover, for every object S' of $\underline{\text{Rev}}$ one has a natural morphism

$$S' \longrightarrow \underline{\text{Isomext}}(\dot{\text{Ep}}_S(\mathcal{R}), S' \times^E \dot{\text{Ep}}_S(\mathcal{R}))$$

and to see that it is an isomorphism, one may suppose, by (fpqc) descent, that S' is a trivial E -bundle, in which case it is evident.

One therefore has an isomorphism of functors $\text{Id}_{\underline{\text{Rev}}} \longrightarrow \text{rev} \circ i \circ \text{qép}$. Since the functor rev is fully faithful one therefore obtains, for every reductive S -group H (not necessarily quasi-split), a bijection

$$\underline{\text{Isomext}}(\text{Q-}\dot{\text{Ep}}_{\text{rev}(H)/S}(\mathcal{R}), H)(S) \simeq \text{Aut}_{\underline{\text{Rev}}}(\text{rev}(H))$$

functorial in H . In particular, the identity morphism of $\text{rev}(H)$ corresponds to a “canonical” element u_0 of

$$\underline{\text{Isomext}}(\text{Q-}\dot{\text{Ep}}_{\text{rev}(H)/S}(\mathcal{R}), H)(S);$$

moreover, every

$$u \in \underline{\text{Isomext}}(\text{Q-}\dot{\text{Ep}}_{\text{rev}(H)/S}(\mathcal{R}), H)(S)$$

corresponds to an automorphism ϕ_u of $\text{rev}(H)$, and $\text{qép}(\phi_u)$ is the unique automorphism of the pinned S -group $\text{Q-}\dot{\text{Ep}}_{\text{rev}(H)/S}(\mathcal{R})$ such that $u_0 \circ \text{qép}(\phi_u) = u$. We have thus proved Theorem 3.11. Let us develop one of the corollaries of 3.11:

Corollary 3.12. *For every reductive S -group G , there exists a quasi-pinned S -group $G_{q\text{-pin}}$, and an “outer isomorphism” $u \in \underline{\text{Isomext}}(G_{q\text{-pin}}, G)(S)$. The pair $(G_{q\text{-pin}}, u)$ is unique up to a unique isomorphism.*

Indeed, one may suppose G of constant type $\mathcal{R}$, and one takes $G_{q\text{-pin}} = \text{Q-}\dot{\text{Ep}}_{\text{rev}(G)/S}(\mathcal{R})$.

3.12.1. — Observe that the datum of the pair $(G_{q\text{-pin}}, u)$ allows one to define canonically the S -scheme (cf. 1.11)

$$Q = \underline{\text{Isomint}}(G_{q\text{-pin}}, G),$$

⁽²³⁾ which is a principal homogeneous bundle under $\text{ad}(G_{q\text{-pin}})$, and whose datum “is equivalent” to that of the inner twisted form G of $G_{q\text{-pin}}$. Moreover Q is nothing other than the “scheme of quasi-pinnings of G ”, a definition which accounts for its structure of principal homogeneous bundle (on the left) under $\text{ad}(G)$ (3.10)—compare with 1.20.

Proposition 3.13. *Let S be a scheme, G an adjoint (resp. simply connected) semisimple S -group, $B \supset T$ a Killing pair of G , and $\underline{\text{Dyn}}(G)$ the Dynkin S -scheme of G . There exists a canonical isomorphism of S -group schemes*

$$T \xrightarrow{\sim} \prod_{\underline{\text{Dyn}}(G)/S} \mathbf{G}_{m, \underline{\text{Dyn}}(G)}.$$

(Recall (cf. Exp. II, § 1) that the right-hand side is by definition the S -functor which associates to $S' \longrightarrow S$ the group $\mathbf{G}_m(\underline{\text{Dyn}}(G) \times_S S')$, or, equivalently, $\mathbf{G}_m(\underline{\text{Dyn}}(G_{S'}))$.)

FIRST PROOF. For simplicity let us give it in the adjoint case. Consider the composite morphism

$$T \longrightarrow \prod_{\mathcal{D}/S} T_{\mathcal{D}} \longrightarrow \prod_{\mathcal{D}/S} \mathbf{G}_{m, \mathcal{D}},$$

where the first morphism is the canonical morphism and the second is $\prod_{\mathcal{D}/S} \alpha_{\mathcal{D}}$ (we have put $\mathcal{D} = \underline{\text{Dyn}}(G)$). To verify that this morphism is an isomorphism, one may suppose G split; now, in this case, it is nothing other than the morphism $T \longrightarrow (\mathbf{G}_{m, S})^{\Delta}$ with components α , for $\alpha \in \Delta$, and the latter is an isomorphism (Exp. XXII, 4.3.8). $\square$

⁽²³⁾**N.D.E.** We write $\underline{\text{Isomint}}(G_{q\text{-pin}}, G)$ instead of $\underline{\text{Isomint}}_u(G_{q\text{-pin}}, G)$ (cf. 1.11), since the pair $(G_{q\text{-pin}}, u)$ is unique up to a unique isomorphism.

SECOND PROOF. By 2.8, 3.5 and 3.11, one may suppose that $G = \mathbf{Q}\text{-}\dot{\text{E}}\text{p}_{S'/S}(\mathcal{R})$, T being the canonical maximal torus. Over S' , one has by Exp. XXII 4.3.8 an isomorphism $T_{S'} \xrightarrow{\sim} (\mathbf{G}_{m,S'})^\Delta$, defined by the simple roots (resp. the simple coroots). The group E acts on the right-hand side by permuting Δ . Now the bundle associated to S'/S by $E \rightarrow \text{Aut}(\Delta)$ is $\underline{\text{Dyn}}(G)$ (3.7), and one concludes at once. $\square$

Using Prop. 8.4 of the appendix, one derives:

Corollary 3.14. *Under the preceding conditions, one has*

$$H^1(S, T) \xrightarrow{\sim} H^1(\underline{\text{Dyn}}(G), \mathbf{G}_m) \xrightarrow{\sim} \text{Pic}(\underline{\text{Dyn}}(G)).$$

In particular, $H^1(S, T) = 0$ when S is semi-local.

Remark 3.15. Let S be a scheme, G (resp. G') a reductive S -group, $B \supset T$ (resp. $B' \supset T'$) a Killing pair of G (resp. G'), and $u \in \underline{\text{Isomext}}(G, G')(S)$. Put (cf. 2.10)

$$P = \underline{\text{Isomint}}_u(G, B, T; G', B', T');$$

it is a principal homogeneous bundle under T^{ad} (by $(f, t) \mapsto f \circ \text{int}(t)$).

Let, on the other hand, $\mathcal{D} = \underline{\text{Dyn}}(G) = \underline{\text{Dyn}}(G')$ (identified by means of u (3.5)), and let $L = \underline{\text{Isom}}_{\mathcal{D}}(\mathfrak{g}^{\mathcal{D}}, \mathfrak{g}'^{\mathcal{D}})$ be the principal homogeneous bundle under $\mathbf{G}_{m,\mathcal{D}}$ defined by the invertible $\mathcal{O}_{\mathcal{D}}$ -module

$$\underline{\text{Hom}}_{\mathcal{O}_{\mathcal{D}}}(\mathfrak{g}^{\mathcal{D}}, \mathfrak{g}'^{\mathcal{D}}) = \mathfrak{g}'^{\mathcal{D}} \otimes (\mathfrak{g}^{\mathcal{D}})^{\vee}.$$

Each $f \in P(S')$ defines an isomorphism of $\mathfrak{g}^{\mathcal{D}} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ onto $\mathfrak{g}'^{\mathcal{D}} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, whence a canonical morphism

$$P \rightarrow \prod_{\mathcal{D}/S} L.$$

This morphism is an *isomorphism*, compatible with the isomorphism of acting groups

$$T^{\text{ad}} \xrightarrow{\sim} \prod_{\mathcal{D}/S} \mathbf{G}_{m,\mathcal{D}}$$

defined above. Indeed, it suffices to verify it in the case where the two groups are pinned, where it is easy.

It follows in particular that in the isomorphism

$$H^1(S, T^{\text{ad}}) \xrightarrow{\sim} \text{Pic}(\underline{\text{Dyn}}(G))$$

of 3.14, the class of the bundle P is taken to $\text{cl}(\mathfrak{g}^{\mathcal{D}}) - \text{cl}(\mathfrak{g}'^{\mathcal{D}})$. The bundle P is therefore trivial if and only if $\mathfrak{g}'^{\mathcal{D}}$ and $\mathfrak{g}^{\mathcal{D}}$ are isomorphic.

If (G, B, T) is quasi-splittable, for example if for G one takes the group $G'_{\text{q-pin.}}$, with its canonical Killing pair, it follows that the image of the class of P is nothing other than the obstruction to the quasi-splitting of G' defined in 3.9.

3.16. — Symmetries

3.16.1. — Let S be a scheme, G a reductive S -group, and $B \supset T$ a Killing pair of G . Put $\mathcal{D} = \underline{\text{Dyn}}(G)$. Recall (Exp. XXII, 5.9.1) that there exists a unique Borel subgroup B^- of G such that $B \cap B^- = T$. If $X \in \Gamma(\mathcal{D}, \mathfrak{g}^{\mathcal{D}})^{\times}$ defines a quasi-pinning of G relative to (B, T) (3.9), then $Y = -X^{-1} \in \Gamma(\mathcal{D}, (\mathfrak{g}^{\mathcal{D}})^{\vee})^{\times}$ defines a quasi-pinning of G relative to (B^-, T) ; one says that it is the *opposite quasi-pinning*.

If $\mathcal{R}$ is a reduced pinned root datum and $\mathcal{E}$ an $\mathcal{R}$ -pinning of the reductive S -group G , one defines an $\mathcal{R}$ -pinning $\mathcal{E}^-$ said to be *opposite to $\mathcal{E}$* as follows: one keeps the same maximal torus T , takes the opposite of the isomorphism $D_S(M) \xrightarrow{\sim} T$, and “pins” by $Y_{\alpha} = -X_{\alpha}^{-1} \in \Gamma(S, \mathfrak{g}^{-\alpha})^{\times}$, for $\alpha \in \Delta$. The quasi-pinning underlying $\mathcal{E}^-$ is the quasi-pinning opposite to the quasi-pinning underlying $\mathcal{E}$.

Remark. In the notation of Exp. XIX, 3.1, if one puts

$$w_\alpha(X_\alpha) = \exp(X_\alpha) \exp(-X_\alpha^{-1}) \exp(X_\alpha),$$

one has $w_\alpha(X_\alpha) = w_{-\alpha}(Y_\alpha)$ (loc. cit., 3.1 (vi)).

Proposition 3.16.2. *Let S be a scheme and G a reductive S -group.*

(i) *Let T be a maximal torus of G ; there exists a unique*

$$i_T \in (\underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}})(S) \subset \text{Aut}_{S\text{-gr.}}(T)$$

such that $i_T(t) = t^{-1}$ for every section t of T .

(ii) *Let (B, T) be a Killing pair of G ; there exists a unique section*

$$w_{B,T} \in (\underline{\text{Norm}}_G(T)/T)(S) = W_G(T)(S)$$

such that $\text{int}(w_{B,T})(B) = B^-$ (with the evident abuse of language).

(iii) *Let $\mathcal{Q} = (B, T, X)$ be a quasi-pinning of G , and $\mathcal{Q}^- = (B^-, T, Y)$ the opposite quasi-pinning; there exists a unique inner automorphism of G*

$$n_{\mathcal{Q}} \in \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}})(S) \subset \text{Aut}_{S\text{-gr.}}(G)$$

such that $n_{\mathcal{Q}}(\mathcal{Q}) = \mathcal{Q}^-$, i.e. $n_{\mathcal{Q}}(T) = T$, $n_{\mathcal{Q}}(B) = B^-$, $n_{\mathcal{Q}}(X) = Y$.

(iv) *Let $(\mathcal{R}, \mathcal{E})$ be a pinning of G , and $(\mathcal{R}, \mathcal{E}^-)$ the opposite pinning; there exists a unique automorphism of G*

$$u_{\mathcal{E}} \in \text{Aut}_{S\text{-gr.}}(G, T) \subset \text{Aut}_{S\text{-gr.}}(G)$$

such that $u_{\mathcal{E}}(\mathcal{E}) = \mathcal{E}^-$, i.e. $u_{\mathcal{E}}(t) = t^{-1}$ for every section t of T , and $\text{Ad}(u_{\mathcal{E}})X_\alpha = Y_\alpha$ for every $\alpha \in \Delta$.

PROOF. (ii) follows from Exp. XXII, 5.5.5 (ii), then (iii) follows from 3.10, and (iv) follows from Exp. XXIII, 4.1. Finally, to prove (i), one may suppose G pinned. Existence follows from (iv), for example, and uniqueness from the fact that an automorphism of G inducing the identity on T is given by a section of T^{ad} (2.11). $\square$

Corollary 3.16.3. *One has*

$$i_T^2 = w_{B,T}^2 = n_{\mathcal{Q}}^2 = u_{\mathcal{E}}^2 = e.$$

Moreover, i_T (resp. $u_{\mathcal{E}}$) is $\neq e$ if $G \neq e$, and $w_{B,T}$ (resp. $n_{\mathcal{Q}}$) is $\neq e$ if G is not a torus.

Corollary 3.16.4. *In the situation of (iii) (resp. (iv)), $n_{\mathcal{Q}}$ projects to $w_{B,T}$ (resp. $u_{\mathcal{E}}$ projects to i_T) under the canonical morphism*

$$\underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}}) \longrightarrow W_{\text{ad}(G)}(T^{\text{ad}}) \simeq W_G(T),$$

resp.

$$\underline{\text{Aut}}_{S\text{-gr.}}(G, T) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}}.$$

Corollary 3.16.5. *The preceding definitions are compatible with extension of the base, and are functorial under isomorphism (in an evident sense).*

Proposition 3.16.6.

(i) *One can define uniquely for every reductive group G over a scheme S an element*

$$s_G \in \underline{\text{Autext}}(G)(S)$$

in such a way that this construction is functorial in G under isomorphism, is compatible with base change, and whenever T is a maximal torus of the reductive S -group G , s_G is the image of i_T under the canonical morphism

$$\underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}} \longrightarrow \underline{\text{Autext}}(G).$$

(ii) *One has $s_G^2 = e$, and s_G is a central element of $\underline{\text{Autext}}(G)$.*

- (iii) Under the conditions of 3.16.2 (ii), if one identifies $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)/T^{\text{ad}}$ with $\underline{\text{Autext}}(G)$ (2.2), one has

$$w_{B,T}i_T = i_Tw_{B,T} = s_G.$$

- (iv) Under the conditions of 3.16.2 (iv), if one identifies $\text{Aut}_{S\text{-gr.-pin}}(G, \mathcal{R}, \mathcal{E})$ with $\underline{\text{Autext}}(G)$ (1.3 (iii)), one has

$$n_{\mathcal{E}}u_{\mathcal{E}} = u_{\mathcal{E}}n_{\mathcal{E}} = s_G.$$

PROOF. (i) is proved without difficulty by descent. Moreover, since i_T is evidently a central section of square e in $\text{Aut}_{S\text{-gr.}}(T)$, (ii) follows immediately; (iii) is a consequence of (iv) by descent. Finally, under the conditions of (iv), it is clear that $n_{\mathcal{E}}u_{\mathcal{E}} = u_{\mathcal{E}}n_{\mathcal{E}}$ and that this automorphism of G respects the pinning; under the identification made, it is therefore equal to its image in $\underline{\text{Autext}}(G)$; but $n_{\mathcal{E}}$ is inner and $u_{\mathcal{E}}$ projects to s_G . $\square$

Remark 3.16.7. (i) One determines s_G explicitly in each case of the classification by means of (iii): for each irreducible pinned root datum, it suffices to compose the symmetry about the origin with the symmetry in the Weyl group (i.e. the element w_0 of the Weyl group such that $w_0(\Delta) = -\Delta$). One finds the following results: $s_G = e$ except for A_n ($n \geq 2$), D_n (n odd) and E_6 , in which case s_G is the unique nontrivial “outer automorphism”.

(ii) The automorphism $u_{\mathcal{E}}$ is the one used to construct “the compact real forms” in the theory of semisimple Lie algebras.

Remark 3.16.8. In 3.16.1 one has defined an involution on the S -scheme

$$Q = \underline{\text{Isomint}}(G_{\text{q-pin.}}, G)$$

of quasi-pinnings of G (cf. 3.12.1); by transport of structure from $G_{\text{q-pin.}}$ to G , one sees at once that this involution is given by the action of an element of $\text{ad}(G_{\text{q-pin.}})(S)$: the element n_0 defined (3.16.2 (iii)) by the canonical quasi-pinning of $G_{\text{q-pin.}}$.

In the same way, one has defined an involution on the S -scheme

$$P = \underline{\text{Isom}}_{S\text{-gr.}}(\acute{\text{E}}\text{p}_S(\mathcal{R}), G)$$

of $\mathcal{R}$ -pinnings of G (cf. 1.20). Reasoning as before, one sees that this involution is given by the action of the automorphism u_0 of $\acute{\text{E}}\text{p}_S(\mathcal{R})$ defined (3.16.2 (iv)) by the canonical pinning of $\acute{\text{E}}\text{p}_S(\mathcal{R})$.

4. Isotriviality of reductive groups and principal bundles under reductive groups

4.1. — Definitions. Isotriviality theorem

Definition 4.1.1. Let S be a scheme, G an S -group scheme, and P a principal homogeneous bundle under G . One says that P is *locally isotrivial* (resp. *semi-locally isotrivial*) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open subset) there exist an open subset U of S containing s (resp. F) and a finite surjective étale morphism $S' \rightarrow U$ such that $P_{S'}$ is trivial.

Definition 4.1.2. Let S be a scheme and G a reductive S -group. One says that G is *locally isotrivial* (resp. *semi-locally isotrivial*) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open subset) there exist an open subset U of S containing s (resp. F) and a finite surjective étale morphism $S' \rightarrow U$ such that $G_{S'}$ is splittable.

Remark 4.1.3. (i) The equivalence of categories of 1.17 respects, by definition, local (resp. semi-local) isotriviality.

(ii) Add to the conditions of 4.1.1: G locally of finite presentation over S . Then the principal bundle P (or the reductive group G) is locally isotrivial (resp. semi-locally isotrivial)

if and only if for every $S' \rightarrow S$, with S' local (resp. semi-local), $P_{S'}$ is isotrivial (or $G_{S'}$ is isotrivial), i.e. if there exists a finite surjective étale morphism $S'' \rightarrow S'$ such that $P_{S''}$ is trivial (or $G_{S''}$ split).

(iii) In the case of tori, Definition 4.1.2 coincides with that of Exp. IX, 1.1.

4.1.4. — Recall (Exp. XXII, 4.3 and 6.2) that for every reductive group G we have introduced the groups $\text{rad}(G)$, $\text{corad}(G)$ and $\text{der}(G)$. The groups $\text{rad}(G)$ and $\text{corad}(G)$ are tori, which are split when G is split; moreover, there exists an isogeny $\text{rad}(G) \rightarrow \text{corad}(G)$. The S -group $\text{der}(G)$ is semisimple, and one has $G/\text{der}(G) = \text{corad}(G)$; it follows that for every principal homogeneous bundle P under G , $P/\text{der}(G)$ is a principal homogeneous bundle under $\text{corad}(G)$. This being said, one has:

Theorem 4.1.5. *Let S be a scheme and G a reductive S -group of constant type.*

- (i) *The following conditions are equivalent:*
 - (a) *G is locally (resp. semi-locally) isotrivial.*
 - (b) *The torus $\text{rad}(G)$ is so.*
 - (b') *The torus $\text{corad}(G)$ is so.*
 - (c) *The Galois covering of S associated to G (3.11) is so.*
- If T is a maximal torus of G , these conditions are also equivalent to*
 - (d) *The torus T is locally (resp. semi-locally) isotrivial.*
- (ii) *Let P be a principal homogeneous bundle under G ; for P to be locally (resp. semi-locally) isotrivial, it is necessary and sufficient that the principal $\text{corad}(G)$ -bundle $P/\text{der}(G)$ be so.*

Corollary 4.1.6. *The conditions of (i) hold when G is semisimple or when S is locally noetherian and normal (or, more generally, geometrically unibranch). The conditions of (ii) hold when G is locally (resp. semi-locally) isotrivial.*

PROOF. For (i), the first assertion is trivial from (b), and the second follows from (c) and Exp. X, 5.14 and 5.15. For (ii), it suffices to observe that, by theorem 90, a principal bundle under a split torus is semi-locally isotrivial. $\square$

4.2. — Proof: the semisimple case Let us first prove, for later reference:

Proposition 4.2.1. *Let S be a scheme, G a reductive S -group, T a maximal torus of G (resp. B a Borel subgroup, resp. $B \supset T$ a Killing pair of G), P a principal homogeneous bundle under G , and G' the twisted form of G associated to P (via the morphism $\text{int} : G \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G)$). One has canonical isomorphisms*

$$P/\underline{\text{Norm}}_G(T) \xrightarrow{\sim} \underline{\text{Tor}}(G'), \quad P/B \xrightarrow{\sim} \underline{\text{Bor}}(G'), \quad P/T \xrightarrow{\sim} \underline{\text{Kil}}(G').$$

PROOF. By construction, G' is the quotient of $P \times_S G$ by a certain action of G ($(p, g')g = (pg, g^{-1}g'g)$); one therefore has a morphism $P \times_S G \rightarrow G'$, i.e. a morphism

$$P \rightarrow \underline{\text{Hom}}_S(G, G'),$$

which, as one sees at once, factors through a morphism

$$f : P \rightarrow \underline{\text{Isom}}_{S\text{-gr.}}(G, G'),$$

(to verify this assertion, one may suppose $G = P$, in which case one has $G' = G$, $f = \text{int}$). The set-theoretic map $p \mapsto f(p)(T)$ defines a morphism

$$P \rightarrow \underline{\text{Tor}}(G').$$

To verify that this morphism induces an isomorphism $P/\underline{\text{Norm}}_G(T) \xrightarrow{\sim} \underline{\text{Tor}}(G')$ as asserted, one may again suppose $P = G$, in which case one is reduced to Exp. XXII, 5.8.3 (iii). (In fact, loc. cit. ought to be replaced by the above statement.) One reasons likewise for $\underline{\text{Bor}}$ and $\underline{\text{Kil}}$. $\square$

Proposition 4.2.2. *Let S be a semi-local scheme and G a semisimple S -group of constant type.*

- (i) G is isotrivial.
- (ii) Every principal bundle under G is isotrivial.

Let us prove (i). After a finite surjective étale extension of the base, one may, by 3.9.2, suppose G quasi-split. But then G is isotrivial by construction (3.11, the group E is finite). To prove (ii), one may, by (i), suppose G split; one is then reduced to:

Lemma 4.2.3. *Let S be a semi-local scheme. Every principal bundle under a split reductive group is isotrivial.*

PROOF. Indeed, with the notation of 4.2.1, where $B \supset T$ denotes the canonical Killing pair of the split G , the S -scheme $\underline{\text{Kil}}(G')$ has a section, after a finite surjective étale extension of the base, by 3.9.2. One can therefore reduce the structure group of G to T ; now T is split, hence $H^1(S, T) = 0$ (theorem 90). $\square$

Corollary 4.2.4. *Let S be a scheme and*

$$1 \longrightarrow G \longrightarrow G' \longrightarrow G'' \longrightarrow 1$$

an exact sequence of S -group schemes (for (fpqc)), G being semisimple of constant type. Let P be a principal homogeneous bundle under G' , and suppose the associated sheaf⁽²⁴⁾ P/G representable (for example G'' affine over S). For P to be locally isotrivial (resp. semi-locally isotrivial), it is necessary and sufficient that P/G be so (as a bundle under $G''^{(24)}$).

PROOF. If P is trivial, P/G is also trivial, which shows that the condition is necessary. Conversely, suppose S local (resp. semi-local) and P/G isotrivial, hence trivialized by an étale finite surjective extension S' of S . Extending the base to S' , one can reduce the structure group of $P_{S'}$ to $G_{S'}$. But S' is still semi-local and $G_{S'}$ semisimple of constant type, so the bundle obtained is isotrivial (4.2.2). $\square$

4.3. — Proof: general case. Observe first that 4.1.5 (ii) follows immediately by applying 4.2.4 to the exact sequence

$$1 \longrightarrow \text{der}(G) \longrightarrow G \longrightarrow \text{corad}(G) \longrightarrow 1.$$

Let us therefore prove (i). If G is split, $\text{rad}(G)$ and $\text{corad}(G)$ are split, as is $\text{rev}(G)$; thus (a) implies (b), (b'), and (c).

4.3.1. — One has (c) $\Rightarrow$ (a). Let $\mathcal{R}$ be the type of G ; one has an exact sequence

$$1 \longrightarrow \text{ad}(\dot{\text{Ep}}_S(\mathcal{R})) \longrightarrow A_S(\mathcal{R}) \longrightarrow E_S \longrightarrow 1.$$

Applying 4.2.4 to the bundle $P = \underline{\text{Isom}}_{S\text{-gr}}(\dot{\text{Ep}}_S(\mathcal{R}), G)$ and the associated bundle $\text{rev}(G) = P/\text{ad}(\dot{\text{Ep}}_S(\mathcal{R}))$, one has (c) $\Rightarrow$ (a).

4.3.2. — One has (b) $\Rightarrow$ (a). It suffices for us to prove that if $\text{rad}(G)$ is split, G is semi-locally isotrivial. Now let $G_0 = \dot{\text{Ep}}_S(\mathcal{R})$; consider the category of pairs (G', f) , where G' is a form of G_0 and f an isomorphism of $\text{rad}(G_0)$ onto $\text{rad}(G)$.

One knows (2.16) that this category is equivalent to the category of principal homogeneous bundles under a certain S -group H , an extension of a finite twisted constant group by a semisimple group. It suffices for us to prove that every principal bundle under H is semi-locally isotrivial; now this follows at once from 4.2.4.

4.3.3. — One has (b') $\Rightarrow$ (a). One can reason as before (in fact the same group H will occur). One can also see that (b) and (b') are equivalent: a torus isogenous to a locally split torus is also locally split (cf. Exp. IX, 2.11 (iii)).

⁽²⁴⁾**N.D.E.** We have replaced “bundle” by “sheaf”, and then G' by G'' .

4.3.4. — One has (d) $\Rightarrow$ (a). It suffices to prove that a group of constant type possessing a split maximal torus is semi-locally isotrivial; now this follows at once from 2.14.

4.3.5. — One has (a) $\Rightarrow$ (d). It suffices to prove that a maximal torus of a split group is semi-locally isotrivial. More precisely, one has:

Lemma 4.3.6. *Let S be a semi-local scheme, G a reductive S -group, T_0 a split maximal torus of G , $W_0 = \underline{\text{Norm}}_G(T_0)/T_0$ (it is a locally constant S -group, and constant if G is of constant type, by 2.14), and T a maximal torus of G .*

There exists a morphism $S' \rightarrow S$ which is principal homogeneous under W_0 (hence étale, finite and surjective, and even principal Galois if G is of constant type), such that $T_{S'}$ is conjugate to $(T_0)_{S'}$ by an element of $G(S')$ (and therefore in particular split).

PROOF. Indeed, one knows that $\underline{\text{Transp}}_G(T_0, T)$ is a principal homogeneous bundle under $\underline{\text{Norm}}_G(T_0)$ (cf. for example Exp. XI, 5.4 bis). Put $S' = \underline{\text{Transp}}_G(T_0, T)/T_0$; it is a principal homogeneous bundle under W_0 . Extending the base from S to S' , one can reduce the structure group of $\underline{\text{Transp}}_G(T_0, T)$ to T_0 . Now S' is semi-local and T_0 split, so $\underline{\text{Transp}}_G(T_0, T)$ has a section over S' .⁽²⁵⁾ $\square$

4.4. — Use of the existence of maximal tori By using Grothendieck's theorem on the existence of maximal tori (Exp. XIV, 3.20), one can considerably sharpen the preceding results. Let us state at once:

Theorem 4.4.1. *Let S be a semi-local scheme, $\mathcal{R}$ a reduced pinned root datum, W its Weyl group, and E the group of its automorphisms. (Recall that E acts naturally on W and that the semidirect product $A = W \cdot E$ identifies with the group of automorphisms of the unpinned $\mathcal{R}$, cf. Exp. XXI, 6.7.2.)*

- (i) *Every principal homogeneous bundle under $\dot{\text{Ep}}_S(\mathcal{R})$ is trivialized by a principal Galois covering $S' \rightarrow S$ with group W .*
- (ii) *Let G be a reductive S -group of type $\mathcal{R}$; let $\text{rev}(G) = \underline{\text{Isomext}}(\dot{\text{Ep}}_S(\mathcal{R}), G)$ be the associated Galois covering of S with group E . Let W_0 be the form of W_S associated to $\text{rev}(G)$. There exists a morphism $S' \rightarrow S$, which is a principal homogeneous bundle under W_0 , such that $G_{S'}$ is quasi-split (i.e. possesses a Borel subgroup).*
- (iii) *Every reductive S -group of type $\mathcal{R}$ is split by a principal Galois covering $\tilde{S} \rightarrow S$ with group A .*

Let us first state:

Proposition 4.4.2. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , P a principal homogeneous bundle under G , and G' the twisted form of G associated to P (one then has, cf. 4.2.1, a canonical isomorphism $P/\underline{\text{Norm}}_G(T) \xrightarrow{\sim} \underline{\text{Tor}}(G')$). Let T' be a maximal torus of G' , defining a composite morphism*

$$S \rightarrow \underline{\text{Tor}}(G') \xrightarrow{\sim} P/\underline{\text{Norm}}_G(T).$$

Consider the canonical morphisms

$$P \rightarrow P/T \rightarrow P/\underline{\text{Norm}}_G(T)$$

and take their inverse images under the preceding morphism:

$$\begin{array}{ccccc} P & \longrightarrow & P/T & \longrightarrow & P/\underline{\text{Norm}}_G(T) \\ \uparrow & & \uparrow & & \uparrow \\ H & \longrightarrow & S' & \longrightarrow & S. \end{array}$$

⁽²⁵⁾N.D.E. By “theorem 90”.

Then S' (resp. H) is a principal homogeneous bundle over S (resp. S') under $W_G(T)$ (resp. $T_{S'}$). Moreover, if one lets $W_G(T)$ act on T in the evident way, the bundle associated to S' is isomorphic to T' .

The first two assertions are trivial, and the last is proved as is the corresponding assertion of 2.6.

Corollary 4.4.3. *Let S be a semi-local scheme, G a reductive S -group, and T a maximal torus of G . Suppose that one of the following two conditions holds:*

- (i) T is split.
- (ii) T is contained in a Borel subgroup of G , and G is either adjoint or simply connected.

Let moreover P be a principal homogeneous bundle under G . There exists an S -scheme S' , which is a principal homogeneous bundle under $W_G(T)$, such that $P_{S'}$ is trivial.

PROOF. Indeed, if G' is the form of G associated to P , then G' possesses a maximal torus T' (Exp. XIV, 3.20). Taking up the notation of the preceding proposition, one sees that $H^1(S', T_{S'}) = 0$ (by theorem 90 for (i), and by 3.14 for (ii)). The morphism $H \rightarrow S'$ has a section, so $P_{S'}$ also has a section over S' . $\square$

Let us now prove the theorem. Assertion (i) is a special case of the preceding corollary (take $G = \text{Ép}_S(\mathcal{R})$, equipped with its canonical split torus). Let us prove (ii). One knows (3.12) that G is an inner twisted form of

$$G_0 = G_{\text{rev}(G)/S}^{\text{qÉp}}(\mathcal{R}).$$

If T_0 is the canonical maximal torus of G_0 , $W_{G_0}(T_0)$ is indeed the group W_0 described in the statement. The form G of G_0 corresponds to a principal homogeneous bundle P under $\text{ad}(G_0)$ ($P = \text{Isomint}(G_0, G)$). The group $W_{\text{ad}(G_0)}(T_0^{\text{ad}})$ is canonically isomorphic to W_0 , and one obtains the desired result by applying 4.4.3 to the situation $(\text{ad}(G_0), T_0^{\text{ad}}, P)$, hypothesis (ii) of 4.4.3 being satisfied. Let us finally prove (iii). Take up the notation of (ii); one has a diagram

$$\begin{array}{ccc} & \text{rev}(G) & \\ E_S \downarrow & & \\ S & \xleftarrow{W_0} & S'. \end{array}$$

One knows that $G_{S'}$ is isomorphic to $(G_0)_{S'}$ and that $(G_0)_{\text{rev}(G)}$ is splittable. If one puts $\bar{S} = S' \times_S \text{rev}(G)$, $G_{\bar{S}}$ is indeed splittable, and it only remains to verify that $\bar{S}$ is indeed a principal Galois covering of S with group A , which follows from the more general lemma below (naturally valid in any site):

Lemma 4.4.4. *Let S be a scheme, G and H two S -group schemes, $G \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(H)$ an action of G on H , E a principal homogeneous G -bundle, and F a principal homogeneous H_0 -bundle, where H_0 is the form of H associated to E . Then $E \times_S F$ is naturally equipped with a structure of principal homogeneous bundle under the semidirect product $H \cdot G$.*

PROOF. Denote by $(e, g) \mapsto eg$ (resp. $(f, u) \mapsto fu$) the (right) action of G on E (resp. of H_0 on F). Denote by

$$p : E \times_S H \rightarrow H_0$$

the canonical projection (H_0 is by definition the quotient of $E \times_S H$ by G acting on it according to the formula $(e, h)g = (eg, g^{-1}hg)$). Consider the morphism

$$r : E \times_S F \times_S H \times_S G \rightarrow E \times_S F$$

defined set-theoretically by

$$r(e, f, h, g) = (eg, fp(e, h)).$$

The morphism r does indeed define an action of the semidirect product $H \cdot G$ on $E \times_S F$. Indeed, one has set-theoretically

$$r(r(e, f, h, g), h', g') = (egg', fp(e, h)p(eg, h'));$$

but

$$p(e, h)p(eg, h') = p(e, h)p(e, gh'g^{-1}) = p(e, hgh'g^{-1}),$$

whence

$$r(r(e, f, h, g), h', g') = r(e, f, hgh'g^{-1}, gg'),$$

which is what had to be proved.

To prove now that this law is indeed a principal homogeneous bundle law, one may suppose that E and F are trivial, in which case one sees at once that $E \times_S F$ is also a trivial bundle. $\square$

4.5. — Isotriviality of maximal tori of semisimple groups⁽²⁶⁾ Let us bring out here the following result, implicitly contained in 2.6 (cf. N.D.E. (13)) and mentioned at the end of Exp. IX 1.2. Recall (XXIII 5.11) that, for every scheme S and every reduced root datum $\mathcal{R}$, one denotes by $\dot{\text{Ep}}_S(\mathcal{R})$ the unique pinned S -group of type $\mathcal{R}$ (which exists by Exp. XXV) and by $T_S(\mathcal{R})$ its canonical maximal torus.

Proposition 4.5.1. *Let G be a semisimple S -group of constant type $\mathcal{R}$; suppose that G possesses a maximal torus T (which is the case if S is semi-local, by XIV 3.20).*

- (i) *Then $P = \underline{\text{Isom}}_{S\text{-gr.}}(\dot{\text{Ep}}_S(\mathcal{R}), T_S(\mathcal{R}); G, T)/T_S(\mathcal{R})^{\text{ad}}$ is a principal S -bundle under the finite group $\text{Aut}(\mathcal{R})$, and T_P is a split maximal torus of G_P . Consequently, T is isotrivial.*
- (ii) *Moreover, for every finite set F of points of P contained in an affine open subset, there exists an open subset U of P containing F such that G_U is split.*

PROOF. The exact sequence

$$1 \longrightarrow T_S(\mathcal{R})^{\text{ad}} \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{Ep}}_S(\mathcal{R}), T_S(\mathcal{R})) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(T_S(\mathcal{R}))$$

induces for every $S' \rightarrow S$ a morphism of bundles $P_{S'} \rightarrow \underline{\text{Isom}}_{S'\text{-gr.}}(T_{S'}(\mathcal{R}), T_{S'})$, so if $P_{S'}$ has a section, then $T_{S'}(\mathcal{R})$ and $T_{S'}$ are isomorphic. This is in particular the case for $S' = P$.

Observe moreover that $\underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{Ep}}_S(\mathcal{R}), T_S(\mathcal{R}))/T_S(\mathcal{R})^{\text{ad}}$ is nothing other than the constant S -group $\text{Aut}(\mathcal{R})_S$, and that P identifies with the $\text{Aut}(\mathcal{R})_S$ -torsor $\underline{\text{Isom}}(\mathcal{R}, \tilde{\mathcal{R}}(G, T))$, where $\tilde{\mathcal{R}}(G, T)$ denotes the twisted root datum associated to (G, T) (cf. XXII 1.10).

Then (ii) follows from (i) and from 2.14, taking account of the fact that the $\mathfrak{g}^\alpha$ (α ranging over the roots of $\mathcal{R}$) are free $\mathcal{O}_U$ -modules of rank 1 on a sufficiently small affine open subset U containing F . $\square$

5. Canonical decomposition of an adjoint or simply connected group

In this section we shall use the results of no. 1 to generalize to schemes a classical decomposition of adjoint (resp. simply connected) groups. To avoid overburdening the exposition indefinitely, the proofs are sketched and the details left to the reader; in fact these are always entirely standard proofs from the theory of principal bundles: reduction of the structure group, twisting, ...

5.1. — Recall (Exp. XXI, 7.4) that a Dynkin diagram is the disjoint union of its connected components, which are Dynkin diagrams. Moreover, every nonempty connected Dynkin diagram corresponding to a root datum is isomorphic to one of the standard diagrams $(A_n, B_n, \dots, G_2)$ exhibited in Exp. XXI, 7.4.6. In what follows we shall only be concerned with Dynkin diagrams whose connected components are of one of the preceding types. Let $\mathbf{T}$ be

⁽²⁶⁾N.D.E. We have added this no. 4.5.

the set of these standard diagrams. For every Dynkin diagram D , let $n(t) = n_D(t)$ be the number of connected components of D isomorphic to t , where $t \in \mathbf{T}$. The *type* of D is by definition $\sum_{t \in \mathbf{T}} n_D(t)t$.

A Dynkin diagram of type t is said to be *simple of type t* , and a Dynkin diagram of type nt is said to be *isotypic of type t* . Let D_0 be the set of connected components of D and let

$$a : D_0 \longrightarrow \mathbf{T}$$

be the evident map. The type of D is nothing other than $\sum_{x \in D_0} a(x)$.

5.2. — Let S be a scheme and D a *Dynkin S -scheme* (satisfying the restrictive condition stated above). The cokernel of the pair of morphisms $L \rightrightarrows D$ ($L = {}^{(27)}\text{scheme of bonds of } D$) is denoted by D_0 . It is the “scheme of connected components” of D (it exists trivially when D is constant; the general case follows by descent); it is a finite twisted constant S -scheme. One has a canonical morphism

$$a : D_0 \longrightarrow \mathbf{T}_S.$$

For $t \in \mathbf{T}$, put $a^{-1}(t) = D_{0,t}$; it is a subscheme of D_0 , whose inverse image in D , denoted by D_t , is the *isotypic component of type t* of the Dynkin scheme D . Each D_t is a subscheme of D , and one has

$$D = \coprod_{t \in \mathbf{T}} D_t.$$

Observe that the degree of $D_{0,t}$ at $s \in S$ is $n(s, t)$, if the type of D at s is $\sum_t n(s, t)t$.

5.3. — In what follows we shall consider only adjoint (resp. simply connected) semisimple groups. To simplify the language, we shall state the results for adjoint groups; *all the statements will remain valid* if one everywhere substitutes *simply connected* for *adjoint*.

Recall that a reduced adjoint root datum is determined up to isomorphism by the type of its Dynkin diagrams. One will therefore say that an adjoint root datum $\mathcal{R}$ (resp. an adjoint semisimple group G) is of *type $\sum n(t)t$* if its Dynkin diagrams are so (resp. if its type is given by an adjoint root datum of type $\sum n(t)t$). One will say that $\mathcal{R}$ or G is *simple of type t* (resp. *isotypic of type t*) if its type is t (resp. nt , $n > 0$).

If G is an adjoint semisimple group, we shall use the symbols $\underline{\text{Dyn}}_0(G)$ and $\underline{\text{Dyn}}_{0,t}(G)$ in the sense defined in 5.2.

5.4. — Let t_i , $i = 1, 2, \dots, n$, be distinct elements of $\mathbf{T}$, and let $\mathcal{R}_i$ be a pinned adjoint root datum isotypic of type t_i . Consider the product $\mathcal{R} = \mathcal{R}_1 \times \dots \times \mathcal{R}_n$ (Exp. XXI, 6.4.1). Let S be a scheme such that the various $\acute{\text{E}}\text{p}_S(\mathcal{R}_i)$ exist (cf. Exp. XXV). One verifies at once that there exists a canonical isomorphism

$$(*) \quad \acute{\text{E}}\text{p}_S(\mathcal{R}) = \acute{\text{E}}\text{p}_S(\mathcal{R}_1) \times_S \dots \times_S \acute{\text{E}}\text{p}_S(\mathcal{R}_n).$$

Moreover, if E_i denotes the group of automorphisms of $\mathcal{R}_i$, E the group of automorphisms of $\mathcal{R}$, and D_i (resp. D) the Dynkin diagram of $\mathcal{R}_i$ (resp. $\mathcal{R}$), one has isomorphisms:

$$E_i \simeq \text{Aut}(D_i), \quad D = \coprod D_i, \quad E \simeq \prod E_i \simeq \text{Aut}(D).$$

Combining with $(*)$ and 1.4, one sees that $(*)$ induces an isomorphism

$$A_S(\mathcal{R}) \simeq A_S(\mathcal{R}_1) \times_S \dots \times_S A_S(\mathcal{R}_n).$$

Proposition 5.5. *Let S be a scheme and G an adjoint semisimple S -group. There exists a unique decomposition*

$$G \simeq \prod_{t \in \mathbf{T}} G_t,$$

⁽²⁷⁾**N.D.E.** We have replaced R by L , as in 3.2.

where G_t is an adjoint semisimple S -group isotypic of type t . Moreover, the preceding decomposition induces isomorphisms

$$\underline{\mathrm{Dyn}}_t(G) \simeq \underline{\mathrm{Dyn}}(G_t), \quad \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \simeq \prod_{t \in \mathbf{T}} \underline{\mathrm{Aut}}_{S\text{-gr.}}(G_t).$$

PROOF. This has indeed been proved above when G is split. In the general case, one may suppose G of constant type $\mathcal{R}$. Using the preceding decomposition of $A_S(\mathcal{R})$ and 1.17, one deduces the desired decomposition of G . The other results are then proved by descent. $\square$

Remark 5.6. More generally, if G and H are two adjoint semisimple groups, one has canonical isomorphisms as follows (the diagram is commutative):

$$\begin{array}{ccc} \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H) & \xrightarrow{\sim} & \prod_{t \in \mathbf{T}} \underline{\mathrm{Isom}}_{S\text{-gr.}}(G_t, H_t) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Isomext}}(G, H) & \xrightarrow{\sim} & \prod_{t \in \mathbf{T}} \underline{\mathrm{Isomext}}(G_t, H_t) \\ \downarrow \wr & & \downarrow \wr \\ \underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\underline{\mathrm{Dyn}}(G), \underline{\mathrm{Dyn}}(H)) & \xrightarrow{\sim} & \prod_{t \in \mathbf{T}} \underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\underline{\mathrm{Dyn}}(G_t), \underline{\mathrm{Dyn}}(H_t)). \end{array}$$

Remark 5.7. One can give an intrinsic characterization of G_t , which we state below without proof: it is the largest normal (and indeed characteristic) reductive subgroup of G isotypic of type t .

The preceding proposition allows one to reduce the study of adjoint semisimple groups to that of isotypic adjoint semisimple groups. It is this study that we shall carry out below.

5.8. — Let $\mathcal{R}$ be a *simple* reduced pinned adjoint root datum of type t , I a nonempty finite set, and $\mathcal{R}^I$ the product root datum of copies $\mathcal{R}_i$ of $\mathcal{R}$, for $i \in I$. Denote by E the group of automorphisms of $\mathcal{R}$, which identifies with the group of automorphisms of the Dynkin diagram D of $\mathcal{R}$. The Dynkin diagram of $\mathcal{R}^I$ identifies with D^I , which shows that one has an exact sequence:

$$1 \longrightarrow \mathrm{Aut}(D)^I \longrightarrow \mathrm{Aut}(D^I) \longrightarrow \mathrm{Aut}(I) \longrightarrow 1,$$

where $\mathrm{Aut}(I)$ denotes the group of permutations of I . It follows that the canonical isomorphism

$$\dot{\mathrm{Ep}}_S(\mathcal{R})^I \simeq \dot{\mathrm{Ep}}_S(\mathcal{R}^I)$$

induces an exact sequence

$$1 \longrightarrow A_S(\mathcal{R})^I \longrightarrow A_S(\mathcal{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S) \longrightarrow 1,$$

the last group being the constant S -group associated to $\mathrm{Aut}(I)$. Observe moreover that I identifies canonically with the set of connected components of D^I . If G is a semisimple S -group of type $\mathcal{R}^I$, defining (cf. 1.17) a principal homogeneous bundle P under $A_S(\mathcal{R}^I)$, the definition of $\underline{\mathrm{Dyn}}(G)$ by descent (3.7), and that of $\underline{\mathrm{Dyn}}_0(G)$ (5.2), show that the bundle associated to P by the morphism $A_S(\mathcal{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S)$ is nothing other than $\underline{\mathrm{Isom}}_S(I_S, \underline{\mathrm{Dyn}}_0(G))$, corresponding to the form $\underline{\mathrm{Dyn}}_0(G)$ of I_S . Using again the equivalence of categories 1.17 and the preceding exact sequence, one deduces by a formal argument that there exist a reductive $\underline{\mathrm{Dyn}}_0(G)$ -group of type $\mathcal{R}$, say G_0 , and an S -isomorphism $G \simeq \prod_{\underline{\mathrm{Dyn}}_0(G)/S} G_0$.

Proposition 5.9. *Let S be a scheme and G an isotypic adjoint semisimple S -group of type t . There exist a simple adjoint semisimple $\underline{\mathrm{Dyn}}_0(G)$ -group G_0 of type t , and an S -isomorphism (unique)*

$$G \simeq \prod_{\underline{\mathrm{Dyn}}_0(G)/S} G_0.$$

Moreover, this isomorphism induces an exact sequence

$$1 \longrightarrow \prod_{\underline{\text{Dyn}}_0(G)/S} \underline{\text{Aut}}_{S\text{-gr.}}(G_0) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\text{Aut}}_S(\underline{\text{Dyn}}_0(G)) \longrightarrow 1.$$

PROOF. One may evidently suppose G of constant type nt . The first assertion has already been proved (the uniqueness assertion is evident). The second then follows from the split case by descent. $\square$

One can combine 5.5 and 5.9:

Proposition 5.10. *Let S be a scheme, G an adjoint semisimple S -group, and $\mathcal{D} = \underline{\text{Dyn}}(G)$ its Dynkin scheme.*

(i) *There exists a canonical decomposition*

$$G \simeq \prod_{t \in \mathbf{T}} \prod_{\mathcal{D}_{0,t}/S} G_{0,t} \simeq \prod_{\underline{\text{Dyn}}_0(G)/S} G_0,$$

where each $G_{0,t}$ is a simple adjoint $\underline{\text{Dyn}}_{0,t}(G)$ -group of type t (resp. where G_0 is an adjoint semisimple $\underline{\text{Dyn}}_0(G)$ -group whose type at $x \in \underline{\text{Dyn}}_0(G)$ is $a(x) \in \mathbf{T}$ (the morphism $a : \underline{\text{Dyn}}_0(G) \rightarrow \mathbf{T}_S$ was defined in 5.2)).

(ii) *The preceding decompositions induce isomorphic exact sequences (written vertically), where $\underline{\text{Aut}}_{S,a}(\mathcal{D}_0)$ denotes the scheme of automorphisms of $\underline{\text{Dyn}}_0(G)$ which commute with the morphism $a : \underline{\text{Dyn}}_0(G) \rightarrow \mathbf{T}_S$:*

$$\begin{array}{ccc} 1 & & 1 \\ \downarrow & & \downarrow \\ \prod_{t \in \mathbf{T}} \prod_{\mathcal{D}_{0,t}/S} \underline{\text{Aut}}_{\text{gr.}}(G_{0,t}) & \xrightarrow{\sim} & \prod_{\underline{\text{Dyn}}_0(G)/S} \underline{\text{Aut}}_{\text{gr.}}(G_0) \\ \downarrow & & \downarrow \\ \underline{\text{Aut}}_{S\text{-gr.}}(G) & \longrightarrow & \underline{\text{Aut}}_{S\text{-gr.}}(G) \\ \downarrow & & \downarrow \\ \prod_{t \in \mathbf{T}} \underline{\text{Aut}}_S(\underline{\text{Dyn}}_{0,t}(G)) & \xrightarrow{\sim} & \underline{\text{Aut}}_{S,a}(\underline{\text{Dyn}}_0(G)) \\ \downarrow & & \downarrow \\ 1 & & 1 \end{array}$$

Corollary 5.11. *Under the preceding conditions, the following three categories are equivalent:*

- (i) *the category of principal homogeneous bundles under G .*
- (ii) *the category of principal homogeneous bundles under G_0 .*
- (iii) *the product category, for $t \in \mathbf{T}$, of the categories of principal homogeneous bundles under the $G_{0,t}$.*

This follows formally from the preceding decompositions and from 8.4.

Corollary 5.12. *One has canonical isomorphisms*

$$\underline{\text{Tor}}(G) \simeq \prod_{t \in \mathbf{T}} \prod_{\mathcal{D}_{0,t}/S} \underline{\text{Tor}}(G_{0,t}) \simeq \prod_{\underline{\text{Dyn}}_0(G)/S} \underline{\text{Tor}}(G_0),$$

and likewise on replacing $\underline{\text{Tor}}$ by $\underline{\text{Bor}}$ (resp. $\underline{\text{Kil}}$).

Trivial from the split case.

Remark 5.13. The canonical morphism

$$\underline{\text{Dyn}}(G) \longrightarrow \underline{\text{Dyn}}_0(G)$$

allows one to consider $\underline{\text{Dyn}}(G)$ as a Dynkin $\underline{\text{Dyn}}_0(G)$ -scheme; in fact it is the Dynkin scheme $\underline{\text{Dyn}}(G_0)$ of G_0 .

Likewise, if $T \subset B$ is a Killing pair of G , canonically corresponding to the Killing pair $T_0 \subset B_0$ of G_0 , one verifies that the obstructions to quasi-splitting G and G_0 , which lie (3.9) in $\text{Pic}(\underline{\text{Dyn}}(G)) = \text{Pic}(\underline{\text{Dyn}}(G_0))$, coincide. One deduces:

Corollary 5.14. *The following conditions are equivalent:*

- (i) G is quasi-splittable,
- (ii) G_0 is quasi-splittable,
- (iii) each $G_{0,t}$, $t \in \mathbf{T}$, is quasi-splittable.

6. Automorphisms of Borel subgroups of reductive groups

Lemma 6.1. *Let S be a scheme, (G, T, M, R) a split S -group, Δ a system of simple roots of R , and B the corresponding Borel subgroup. Then B^u is generated as an (fppf) sheaf of groups by the U_α , $\alpha \in \Delta$.*

Let H be the subsheaf of groups of B^u generated by the U_α , $\alpha \in \Delta$. Let us prove $H \supset U_\beta$ ($\beta \in R_+$) by induction on the integer $\text{ord}(\beta) = \text{ord}_\Delta(\beta) > 0$ (cf. Exp. XXI, 3.2.15). The assertion holds by hypothesis for $\text{ord}(\beta) = 1$. Suppose therefore $\text{ord}(\beta) > 1$ and $U_\gamma \subset H$ whenever $\text{ord}(\gamma) < \text{ord}(\beta)$. There exists $\alpha \in \Delta$ such that $\beta - \alpha \in R_+$ (Exp. XXI, 3.1.1). Let p be the largest integer such that $\beta - p\alpha = \beta' \in R_+$. One has $U_\alpha, U_{\beta'} \subset H$, $\beta' - \alpha \notin R$. One is therefore reduced to proving:

Lemma 6.2. *Take up the notation of Exp. XXIII, 6.4. Suppose $p = 1$, i.e. $\beta - \alpha$ is not a root. If H is a subsheaf of groups of B^u such that $U_\alpha, U_\beta \subset H$, then $U_{i\alpha+j\beta} \subset H$ whenever $i\alpha + j\beta \in R$, $i > 0$, $j > 0$.*

PROOF. Let us distinguish four cases according to the value of $q = 0, 1, 2, 3$. In what follows x and y denote two arbitrary sections of $\mathbf{G}_{a,S'}, S' \rightarrow S$.

If $q = 0$, there is nothing to prove.

If $q = 1$, one has $p_{\alpha+\beta}(\pm x) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S')$, hence $U_{\alpha+\beta} \subset H$.

If $q = 2$, one similarly has

$$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S'),$$

whence, after changing certain signs,

$$F(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y) \in H(S').$$

Since $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ commute, one then has $F(x, 1)F(1, -x) = p_{2\alpha+\beta}(x^2 - x)$.⁽²⁸⁾ If $a \in \mathbf{G}_a(S)$, the equation $X^2 - X = a$ defines a free surjective extension of S (it is $\text{Spec } \mathcal{O}_S[X]/(X^2 - X - a)$); one therefore has $p_{2\alpha+\beta}(a) \in H(S')$, hence $U_{2\alpha+\beta} \subset H$, hence also $U_{\alpha+\beta} \subset H$ (since $F(1, a) \in H(S')$).

If $q = 3$, one similarly has

$$F(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2) \in H(S');$$

and since

$$p_{3\alpha+2\beta}(\pm x) = F(1, 1)^{-1}p_\beta(-x)F(1, 1)p_\beta(x) \in H(S'),$$

one obtains that $U_{3\alpha+2\beta} \subset H$ and therefore

$$K(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y) \in H(S').$$

⁽²⁸⁾**N.D.E.** We have corrected $F(-1, x)$ to $F(1, -x)$.

Then computing

$$K(x+y, 1)K(-x, 1)K(1, y)^{-1}$$

modulo $U_{3\alpha+2\beta}$, one finds

$$p_{2\alpha+\beta}(2x^2 + y^2 + 2xy - y)p_{3\alpha+\beta}(y^3 + 3xy^2 + 3x^2y - y) \in H(S').$$

If $a \in \mathbf{G}_a(S)$, the “equations”

$$x^2 = -xy - y + 1 - a$$

$$y^2 = 3y - 2 + 3a$$

define a free surjective extension of S ; one then has $y^3 + 3xy^2 + 3x^2y - y = 0$ and the preceding expression gives $p_{2\alpha+\beta}(a) \in H(S')$.⁽²⁹⁾ One has therefore proved that H contains $U_{2\alpha+\beta}$ and $U_{3\alpha+2\beta}$ and that

$$E(x, y) = p_{\alpha+\beta}(xy)p_{3\alpha+\beta}(x^3y) \in H(S').$$

Since $p_{\alpha+\beta}(xy)$ and $p_{3\alpha+\beta}$ commute, one is reduced to the preceding computation, i.e. $E(1, x)E(1, -x) = p_{3\alpha+\beta}(x^3 - x)$, whence $U_{3\alpha+\beta} \subset H$, then $U_{\alpha+\beta} \subset H$. $\square$

Remark 6.2.1. The preceding proof shows that one could have replaced the hypothesis “ H contains U_α and U_β ” by “ H contains U_α or U_β , and H is invariant under $\text{int}(U_\alpha)$ and $\text{int}(U_\beta)$ ”.

Theorem 6.3. *Let S be a scheme, G and G' two semisimple S -groups, and B (resp. B') a Borel subgroup of G (resp. G'). Every isomorphism $B \xrightarrow{\sim} B'$ is induced by a unique isomorphism $G \xrightarrow{\sim} G'$.*

PROOF. The assertion is local for the étale topology and one may suppose G split: $G = (G, T, M, R)$, B being defined by the system R_+ of positive roots of R . The given isomorphism $u : B \xrightarrow{\sim} B'$ induces an isomorphism of T onto a maximal torus T' of B' , hence of G' . The given isomorphism $T \simeq D_S(M)$ gives an isomorphism $T' \simeq D_S(M)$ in which the elements of R_+ become the constant roots of B' relative to T' , hence the elements of R the constant roots of G' relative to T' . Since G and G' are semisimple, the coroots are determined by duality (Exp. XXI, 1.2.5), which proves that (T', M, R) is a splitting of G' such that $\mathcal{R}(G) = \mathcal{R}(G')$.

Applying Exp. XXIII, 5.1 (Uniqueness theorem), one deduces that there exists a unique isomorphism $G \xrightarrow{\sim} G'$ agreeing with u on T and the U_α , $\alpha \in \Delta$. By 5.1, the restriction of this isomorphism to B is u . $\square$

Remark 6.3.1. (i) Using Exp. XXII, 4.1.9 and reasoning as in loc. cit. 4.2.12, one can replace “isomorphism” in the statement of the theorem by “isogeny” (resp. “central isogeny”).

(ii) The theorem is false for reductive groups. Take for example $G = G' = \text{SL}_2 \times \mathbf{G}_m$, identified with the group of the following matrices:

$$\left\{ \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & h \end{pmatrix} \mid ad - bc = 1, h \text{ invertible} \right\};$$

take for $B = B'$ the Borel subgroup defined by $c = 0$ and for u the following automorphism of B : $u(a, b, d, h) = (a, b, d, ah)$.

Corollary 6.4. *The functor $(G, B) \mapsto B$ from the category consisting of pairs (G, B) , where G is a semisimple S -group and B a Borel subgroup of G , to the category of S -group schemes (the morphisms are the isomorphisms) is fully faithful.*

Corollary 6.5. *Let S be a scheme, G a semisimple S -group, B a Borel subgroup of G , and B' an S -group locally isomorphic to B for (fpqc). Then B' is locally isomorphic to B for the*

⁽²⁹⁾**N.D.E.** We have corrected $p_{3\alpha+\beta}(a)$ to $p_{2\alpha+\beta}(a)$, and given the end of the argument in detail.

local finite étale topology⁽³⁰⁾ and there exists a semisimple S -group G' of which B' is a Borel subgroup; G' is unique up to a unique isomorphism inducing the identity on B' .

Corollary 6.6. *Let S be a scheme, G a semisimple S -group, and B a Borel subgroup of G . Then $\underline{\text{Aut}}_{S\text{-gr.}}(B)$ is representable by a smooth affine S -scheme, $\underline{\text{Autext}}(B)$ is representable by a finite étale S -scheme, and the evident morphisms induce an isomorphism of exact sequences*

$$\begin{array}{ccccccc} 1 & \longrightarrow & B^{\text{ad}} & \longrightarrow & \underline{\text{Aut}}_{S\text{-gr.}}(G, B) & \longrightarrow & \underline{\text{Autext}}(G) \longrightarrow 1 \\ & & \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ 1 & \longrightarrow & B^{\text{ad}} & \longrightarrow & \underline{\text{Aut}}_{S\text{-gr.}}(G) & \longrightarrow & \underline{\text{Autext}}(B) \longrightarrow 1. \end{array}$$

This follows at once from 2.1 and the preceding results. We leave to the reader the task of developing the analogues of the results of nos. 1, 2, 3, 4 in the above setting.

Remark 6.7. If S is a scheme and B an S -group, B can therefore be a Borel subgroup of only one semisimple group, completely determined by B . It therefore remains to characterize the S -groups B which are indeed Borel subgroups of semisimple groups.⁽³¹⁾

7. Representability of the functors $\underline{\text{Hom}}_{S\text{-gr.}}(G, H)$, for reductive G

7.1. — The split case

7.1.1. — Let S be a scheme, G a pinned S -group, T its maximal torus, Δ the set of simple roots, and, for $\alpha \in \Delta$,

$$u_\alpha \in U_\alpha^\times(S), \quad w_\alpha \in \underline{\text{Norm}}_G(T)(S),$$

the elements defined by the pinning. Let, on the other hand, H be an S -group scheme; we are interested in the functor $\underline{\text{Hom}}_{S\text{-gr.}}(G, H)$, and more precisely in the morphism

$$q : \underline{\text{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\text{Hom}}_{S\text{-gr.}}(T, H).$$

Let therefore

$$f_T : T \longrightarrow H$$

be a morphism of S -groups, and consider $q^{-1}(f_T) = \mathcal{F}$. It is the functor defined by

$$\mathcal{F}(S') = \{f \in \underline{\text{Hom}}_{S'\text{-gr.}}(G_{S'}, H_{S'}) \mid f = (f_T)_{S'} \text{ on } T_{S'}\}.$$

One has a morphism of S -functors

$$i : \mathcal{F} \longrightarrow H^{2n},$$

where $n = \text{Card}(\Delta)$, defined set-theoretically by $i(f) = (f(u_\alpha), f(w_\alpha))_{\alpha \in \Delta}$. By Exp. XXIII, 1.9, i is moreover a monomorphism.

Proposition 7.1.2. *If H is separated over S , $\mathcal{F}$ is representable and i is a closed immersion.*

The usual technique of relative representability⁽³²⁾ shows us that it suffices to prove that, given sections

$$v_\alpha, h_\alpha \in H(S), \quad \text{for } \alpha \in \Delta,$$

the S -schemes S' such that there exists an S' -homomorphism

$$f : G_{S'} \longrightarrow H_{S'}$$

⁽³⁰⁾**N.D.E.** Denoted by (étf) in Exp. IV 6.3. In other words, for every $s \in S$ there exist an open neighborhood U of s and a finite surjective étale morphism $V \longrightarrow U$ such that $B' \times_S V \simeq B \times_S V$.

⁽³¹⁾**N.D.E.** One may ask whether every smooth affine S -group, each of whose geometric fibers is a Borel subgroup of a semisimple group, is a Borel subgroup of a semisimple S -group. (We have removed the “counterexample” given in the original, for S = the scheme of dual numbers over a field k , which was erroneous, as M. Demazure pointed out to us.)

⁽³²⁾**N.D.E.** See for example the proof of XXII 5.8.1.

extending $(f_T)_{S'}$ and satisfying $f(u_\alpha) = v_\alpha$, $f(w_\alpha) = h_\alpha$, are precisely those which factor through a certain closed subscheme of S . One may evidently suppose S affine.

To simplify what follows, let us say that a morphism $Y \rightarrow X^{(33)}$ of affine schemes satisfies condition (L) if Y is the spectrum of an $\mathcal{O}(X)$ -algebra which is a free $\mathcal{O}(X)$ -module. It is clear that if one restricts to the category of affine schemes, a product or a composite of morphisms satisfying (L) satisfies (L), and that (L) is stable under extension of the base.

Lemma 7.1.3. *Suppose S affine, and let $\alpha \in \Delta$. Consider the morphism*

$$a : T \times_S T \rightarrow \mathbf{G}_{a,S}$$

defined set-theoretically by $a(t, t') = \alpha(t) + \alpha(t')$.

- (i) *a is faithfully flat and of finite presentation.*
- (ii) *Let R be the fibered square of a . The structure morphism $R \rightarrow S$ satisfies (L).*

PROOF. It is first clear that the morphism a is surjective, since $\alpha : T \rightarrow \mathbf{G}_{m,S}$ is so. It is trivial that a is of finite presentation. Since α satisfies (L), it suffices, to prove (i) and (ii), to show that the morphism

$$u : (\mathbf{G}_{m,S})^2 \rightarrow \mathbf{G}_{a,S}$$

defined set-theoretically by $u(x, y) = x + y$ satisfies (L). In other words, it suffices to verify that for every ring A , the ring $A[X, Y, X^{-1}, Y^{-1}]$ is a free module over its subring $A[X + Y]$.⁽³⁴⁾ Now $A[X, Y]$ is a free module with basis $\{1, X\}$ over the subring $B = A[X + Y, XY]$ (one has $X^2 - (X + Y)X + XY = 0$), hence $A[X^{\pm 1}, Y^{\pm 1}] = A[X, Y]_{XY}$ is free over $B_{XY} = A[X + Y, (XY)^{\pm 1}]$ with basis $\{1, X\}$, and free over $A[X + Y]$ with basis the elements $(XY)^p$ and $X(XY)^p$, for $p \in \mathbf{Z}$. $\square$

Lemma 7.1.4. *Let $\alpha \in \Delta$, and let $b : T \times_S T \rightarrow H$ be the morphism defined set-theoretically by*

$$b(t, t') = (\text{int}(f_T(t))v_\alpha)(\text{int}(f_T(t'))v_\alpha).$$

Let $f_\alpha : U_\alpha \rightarrow H$ be an S -morphism. The following conditions are equivalent.⁽³⁵⁾

- (i) *f_α is a group morphism; one has $f_\alpha(u_\alpha) = v_\alpha$ and*

$$f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(\alpha(t)x)$$

for all $t \in T(S')$, $x \in U_\alpha(S')$, $S' \rightarrow S$.

- (ii) *One has $f_\alpha(u_\alpha) = v_\alpha$ and the relation*

$$f_\alpha(a(t, t')u_\alpha) = b(t, t')$$

for all $t, t' \in T(S')$, $S' \rightarrow S$.

PROOF. If f_α satisfies (i), one has $f_\alpha(\alpha(t)u_\alpha) = \text{int}(f_T(t))v_\alpha$, which immediately implies (ii). Conversely, suppose (ii) holds and let us prove the various conditions of (i); let us first prove the last one. Since a is covering for (fpqc), it suffices to prove that if $t, t', t'' \in T(S')$, one has

$$f_T(t)f_\alpha(a(t', t'')u_\alpha)f_T(t)^{-1} = f_\alpha(\alpha(t)a(t', t'')u_\alpha);$$

now this can also be written

$$f_T(t)b(t', t'')f_T(t)^{-1} = b(tt', tt''),$$

⁽³³⁾**N.D.E.** We have corrected $X \rightarrow Y$ to $Y \rightarrow X$.

⁽³⁴⁾**N.D.E.** We have simplified the proof in the original.

⁽³⁵⁾**N.D.E.** In what follows, the group law of U_α is written additively, i.e. if one denotes by $p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$ the isomorphism such that $p_\alpha(1) = u_\alpha$ and if $x = p_\alpha(z)$, then $\alpha(t)x$ (resp. $a(t, t')u_\alpha$) denotes $p_\alpha(\alpha(t)z)$ (resp. $p_\alpha(a(t, t')) = \alpha(t)u_\alpha + \alpha(t')u_\alpha$).

a property evident from the definition of b . It remains to prove that f_α is a group morphism. Now the preceding property immediately gives

$$\begin{aligned} f_\alpha(\alpha(t)u_\alpha)f_\alpha(\alpha(t')u_\alpha) &= (f_T(t)f_\alpha(u_\alpha)f_T(t)^{-1}) \cdot (f_T(t')f_\alpha(u_\alpha)f_T(t')^{-1}) \\ &= b(t, t') = f_\alpha(\alpha(t)u_\alpha + \alpha(t)\alpha(t')u_\alpha). \end{aligned}$$

One therefore has $f_\alpha(x + x') = f_\alpha(x)f_\alpha(x')$ whenever x and x' are sections of the open subset $U_\alpha^\times$ of U_α , which is schematically dense; one then concludes by Exp. XVIII, 1.4. $\square$

7.1.5. — Let us provisionally fix an $\alpha \in \Delta$. The morphism a is faithfully flat and quasi-compact, hence $\mathbf{G}_{a,S}$ identifies with the quotient of $T \times_S T$ by the equivalence relation R defined by a . Let

$$R \rightrightarrows_{i_2}^{i_1} T \times_S T$$

be this equivalence relation.

For the morphism b to factor through a ,⁽³⁶⁾ it is necessary and sufficient that $b \circ i_1 = b \circ i_2$; in other words, if one denotes by K the kernel of the pair of morphisms

$$R \rightrightarrows_{b \circ i_2}^{b \circ i_1} H,$$

it is necessary and sufficient that $K = R$. Now H is assumed separated over S , hence K is a closed subscheme of R . Recall, on the other hand, that R is essentially free over S (7.1.3).

7.1.6. — According to the above, if S' is an S -scheme, for there to exist over S' an f_α satisfying the conditions of 7.1.4 (i) (and then necessarily unique), it is necessary and sufficient that $K_{S'} = R_{S'}$, and that the morphism f_α obtained satisfy $f_\alpha(u_\alpha) = v_\alpha$.

The first condition is equivalent to $S' \rightarrow S$ factoring through a certain closed subscheme of S (Exp. VIII, 6.4⁽³⁷⁾); if one replaces S by this closed subscheme, the second condition again defines a closed subscheme of S (equality of two sections of H , and H is assumed separated over S).

After replacing S by this closed subscheme, one may therefore suppose that there exists a morphism $f_\alpha : U_\alpha \rightarrow H$ satisfying the conditions of 7.1.4 (i). Taking the intersection of the subschemes of S obtained for each $\alpha \in \Delta$, one may suppose this condition satisfied for every $\alpha \in \Delta$.

7.1.7. — Likewise, consider for each $\alpha \in \Delta$ the two morphisms of S -groups

$$f_T \circ \text{int}(w_\alpha), \text{int}(h_\alpha) \circ f_T : T \rightarrow H.$$

Since H is separated over S and T essentially free over S , the same reasoning as before shows that, after replacing S by a closed subscheme, one may suppose that for every $\alpha \in \Delta$ one has

$$f_T \circ \text{int}(w_\alpha) = \text{int}(h_\alpha) \circ f_T.$$

7.1.8. — Using now the theorem on generators and relations (Exp. XXIII, 3.5), one sees that there exists a group homomorphism $f : G \rightarrow H$ satisfying the required conditions if and only if a certain finite set of algebraic relations between the sections $h_\alpha, v_\alpha, f_T(\alpha^*(-1))$ ($\alpha \in \Delta$) of H holds:

$$R_i((h_\alpha)_\alpha, (v_\alpha)_\alpha, (f_T(\alpha^*(-1)))_\alpha) = e, \quad i = 1, \dots, n.$$

Since H is separated over S , this again defines a closed subscheme of S , and the proof is complete.

⁽³⁶⁾**N.D.E.** I.e. for condition (ii) of 7.1.4 to hold.

⁽³⁷⁾**N.D.E.** See also the addition in Exp. VIB, 6.2.3.

Corollary 7.1.9. *Let S be a scheme, G a split reductive S -group, T its maximal torus, and H a separated S -group scheme. Let, on the other hand, $\mathcal{P}$ be a property of morphisms such that:*

- (i) *A closed immersion satisfies $\mathcal{P}$.*
- (ii) *The composite of two morphisms satisfying $\mathcal{P}$ also satisfies $\mathcal{P}$.*
- (iii) *$\mathcal{P}$ is stable under base change.*
- (iv) *The structure morphism $H \rightarrow S$ satisfies $\mathcal{P}$.*

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$$

is relatively representable by a separated morphism satisfying $\mathcal{P}$.⁽³⁸⁾

PROOF. Indeed, one may suppose G pinned; the structure morphism $H^{2n} \rightarrow S$ satisfies $\mathcal{P}$ and one concludes by 7.1.2. $\square$

Corollary 7.1.10. *Let S be a scheme, G a split reductive S -group, and H a smooth S -group scheme with affine fibers.⁽³⁹⁾ Then the functor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme locally of finite presentation and separated over S .*

PROOF. Indeed, since H is smooth over S , one can consider its identity component H^0 , which has affine fibers and is smooth, separated and of finite presentation over S (Exp. VIB, 3.10 and 5.5). Since G has connected fibers, one can replace H by H^0 . Since G and H are then of finite presentation, one may suppose S noetherian, and one applies 7.1.9 taking for $\mathcal{P}$ the property “of finite type”. But, by Exp. XV, 8.9, one knows that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$ is representable by an S -scheme separated and locally of finite type. $\square$

Remark 7.1.11. If H is affine over S , one can replace the reference to Exp. XV by Exp. XI, 4.2.

7.2. — General case

Proposition 7.2.1. *Let S be a scheme, G a reductive S -group, and T a maximal torus of G . Let, on the other hand, H be an S -group scheme such that the structure morphism $H \rightarrow S$ satisfies the following condition:*

(+) *Every point $s \in S$ has an open neighborhood U such that the morphism $H_U \rightarrow U$ is quasi-projective.*

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$$

is relatively representable by a morphism satisfying (+).

When G is splittable relative to T , one applies 7.1.9 taking for $\mathcal{P}$ property (+) above. When G is locally isotrivial, for example semisimple (4.2.2), one observes that the assertion of the theorem is local for the local finite étale topology (indeed, the property of being quasi-projective is local for the global finite étale topology and ensures effectiveness of descent for this topology, cf. SGA 1, VIII, 7.7). Finally, in the general case, one uses the following lemma:

Lemma 7.2.2. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , G' the derived group of G (Exp. XXII, 6.2), and $T' = T \cap G'$ the maximal torus of G' corresponding*

⁽³⁸⁾**N.D.E.** I.e. for every $S' \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$, with S' representable, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \times_S S'$ is representable and the morphism of S -schemes $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \times_S S' \rightarrow S'$ is separated and satisfies $\mathcal{P}$.

⁽³⁹⁾**N.D.E.** We have removed the hypothesis that H be quasi-separated, which is superfluous.

to T (Exp. XXII, 6.2.8). Then the diagram

$$\begin{array}{ccc} G & \longleftarrow & G' \\ \uparrow & & \uparrow \\ T & \longleftarrow & T' \end{array}$$

is an amalgamated sum in the category of S -sheaves of groups: i.e. for every S -sheaf of groups H , the following square is cartesian:

$$\begin{array}{ccc} \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) & \longrightarrow & \underline{\mathrm{Hom}}_{S\text{-gr.}}(G', H) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H) & \longrightarrow & \underline{\mathrm{Hom}}_{S\text{-gr.}}(T', H). \end{array}$$

PROOF. Indeed, if $\mathrm{rad}(G)$ is the radical of G , then $\mathrm{rad}(G) \subset T$, hence

$$\mathrm{rad}(G) \cap G' = \mathrm{rad}(G) \cap T' = K,$$

and multiplication in G induces isogenies (Exp. XXII, 6.2)

$$i : G' \times_S \mathrm{rad}(G) \longrightarrow G, \quad j : T' \times_S \mathrm{rad}(G) \longrightarrow T.$$

Let $f_{G'} : G' \longrightarrow H$, $f_T : T \longrightarrow H$, and $f_{T'} : T' \longrightarrow H$ be morphisms of S -groups such that $f_{G'}|_{T'} = f_T|_{T'} = f_{T'}$. Let us show that there exists a unique morphism of S -groups $f : G \longrightarrow H$ inducing $f_{G'}$ and f_T . Let $f_{\mathrm{rad}} = f_T|_{\mathrm{rad}(G)}$ ⁽⁴⁰⁾ and let

$$f_1 = f_{G'} \cdot f_{\mathrm{rad}} : G' \times_S \mathrm{rad}(G) \longrightarrow H.$$

For f to exist (and it will evidently be unique), it is necessary and sufficient that f_1 induce the identity on the kernel of i , i.e. that $f_{G'}$ and f_{rad} agree on K ; but the existence of f_T shows by the same reasoning that $f_{T'}$ and f_{rad} agree on K . It only remains to observe that $f_{G'}$ and $f_{T'}$ evidently agree on $K \subset T'$. $\square$

Reasoning now as in 7.1.10, one deduces from 7.2.1:

Corollary 7.2.3. *Let S be a scheme, G a reductive S -group, and H an S -group scheme, smooth and quasi-projective over S with affine fibers. Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme locally of finite presentation and separated over S .*

7.3. — Phenomena peculiar to characteristic 0 If G and H are two smooth S -groups, and $\mathfrak{g}$ and $\mathfrak{h}$ their Lie algebras, one has a canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h}),$$

where the right-hand side has an evident definition.

Proposition 7.3.1. *Let S be a scheme of characteristic 0 (i.e. a $\mathbf{Q}$ -scheme), G a reductive S -group, and H an S -group scheme smooth and quasi-projective over S with affine fibers.*

- (i) $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme smooth and separated over S .
- (ii) If G is semisimple, this scheme is affine and of finite presentation over S .
- (iii) If G is simply connected (Exp. XXII, 4.3.3), the canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$$

is an isomorphism.

Lemma 7.3.2. *Let k be a field of characteristic 0, G a reductive k -group, V a finite-dimensional k -vector space, and $G \longrightarrow \mathrm{GL}(V)$ a linear representation of G on V . One has*

$$H^0(G, V) = H^0(\mathfrak{g}, V), \quad H^i(G, V) = 0, \quad \text{for } i > 0.$$

⁽⁴⁰⁾N.D.E. We have replaced f_r by f_{rad} .

PROOF. The first equality is true in general for a smooth connected group;⁽⁴¹⁾ in the case of a reductive group, one can prove it as follows: one may suppose k algebraically closed, hence G splittable, hence G generated by subgroups isomorphic to $\mathbf{G}_{m,k}$,⁽⁴²⁾ and it suffices to verify the assertion for this group, which is easy.

From this first equality it follows that $H^0(G, V)$ is an exact functor in V when G is semisimple; $\mathfrak{g}$ is then indeed a semisimple Lie algebra and one applies [BLie], § I.6, Exercise 1 (b). The assertion remains true when G is reductive; indeed, introducing the radical R of G ⁽⁴³⁾ and the quotient G/R , which is semisimple, one has $H^0(G, V) = H^0(G/R, H^0(R, V))$, and $H^0(G, -)$ is the composite of two exact functors. Applying Exp. I, 5.3.1, one then deduces $H^i(G, V) = 0$ for $i > 0$. $\square$

Remark 7.3.3. Under the preceding conditions, if G is semisimple, one has $H^1(\mathfrak{g}, V) = H^2(\mathfrak{g}, V) = 0$, cf. [BLie], loc. cit. (b) and (d).

7.3.4. — Let us now prove the proposition. Already, by 7.2.3, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme locally of finite presentation and separated over S ; to show that it is smooth, it suffices to prove that it is infinitesimally smooth (Exp. XI, 1.8),⁽⁴⁴⁾ which follows from Exp. III, 2.8 (i) by 7.3.2. One has therefore proved (i).

Let us show that (ii) follows from (iii). It suffices first to prove that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is affine over S ; it will then be of finite presentation over S , since it is smooth over S by (i); in any case $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$ is representable by a closed subscheme of

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(\mathfrak{g}, \mathfrak{h}) \simeq \mathbf{W}(\mathfrak{g}^\vee \otimes_{\mathcal{O}_S} \mathfrak{h})$$

which is an affine S -scheme, and the desired conclusion appears when G is simply connected.

In the general case, one may suppose G split, hence $G \simeq \dot{\mathrm{E}}p_S(\mathcal{R})$; introducing the simply connected root datum $\mathrm{sc}(\mathcal{R})$ (Exp. XXI, 6.5.5), and using the existence theorem (Exp. XXV, 1.1), one constructs a simply connected S -group $\bar{G}$ and a central isogeny $\bar{G} \rightarrow G$. The kernel K of this isogeny is a finite diagonalizable S -group (hence a twisted constant S -group, S being of characteristic 0) and $\underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H)$ is (trivially) representable by a separated S -scheme (if $K \simeq (\mathbf{Z}/n\mathbf{Z})_S$, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H)$ is isomorphic to $\mathrm{Ker}(H \xrightarrow{n} H)$). One has an exact sequence of “pointed S -schemes”:

$$1 \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \xrightarrow{u} \underline{\mathrm{Hom}}_{S\text{-gr.}}(\bar{G}, H) \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H),$$

hence u is a closed immersion, hence $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is affine over S .

7.3.5. — Let us finally prove (iii). Reasoning as in the proof of (i) and using 7.3.3, one can show that the S -scheme $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$ is smooth over S . To prove (iii), one may therefore suppose that $S = \mathrm{Spec}(k)$, where k is an algebraically closed field of characteristic 0; it even suffices to prove that the morphism Lie is bijective on points with values in k and induces a bijection on the tangent spaces at two corresponding points. Let us first show that

$$\mathrm{Hom}_{k\text{-gr.}}(G, H) \rightarrow \mathrm{Hom}_{k\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$$

is bijective.

If $u : G \rightarrow H$ is a morphism of k -groups, the graph of u is a connected subgroup of $G \times_k H$ which is determined by its Lie algebra, which is the graph of $\mathrm{Lie}(u)$; the map is

⁽⁴¹⁾**N.D.E.** Indeed, it is clear that $V^G \subset V^\mathfrak{g}$. Moreover, $H = \mathrm{Centr}_G(V^\mathfrak{g})$ is a closed subgroup of G , smooth since $\mathrm{char}(k) = 0$; by Exp. II 5.3.1, one has $\mathrm{Lie}(H) = \mathrm{Centr}_\mathfrak{g}(V^\mathfrak{g}) = \mathfrak{g}$, and since H is smooth and G connected this implies $H = G$, whence $V^\mathfrak{g} \subset V^G$ (see also [DG70], § II.6, Prop. 2.1 (c)).

⁽⁴²⁾**N.D.E.** Indeed, the union of the maximal tori of G is dense in G , cf. Bible, § 6.5, Th. 5 (= [Ch05], § 6.6, Th. 6).

⁽⁴³⁾**N.D.E.** Note that R is a torus.

⁽⁴⁴⁾**N.D.E.** I.e. for every local Artinian S -scheme S' , with closed point s' , every morphism of $\kappa(s')$ -groups $G_{s'} \rightarrow H_{s'}$ lifts to a morphism of S' -groups $G_{S'} \rightarrow H_{S'}$. By Exp. III 2.8, this follows from the vanishing of $H^2(G_{s'}, V)$, where $V = \mathrm{Lie}(H_{s'}/\kappa(s'))$.

therefore injective. Conversely, if $v : \mathfrak{g} \rightarrow \mathfrak{h}$ is a Lie algebra morphism, the graph $\mathfrak{k}$ of v is a Lie subalgebra of $\mathfrak{g} \times \mathfrak{h}$, isomorphic to $\mathfrak{g}$. In particular, since $\mathfrak{g}$ is its own derived algebra, the same is true of $\mathfrak{k}$ and therefore, by a theorem of Chevalley ([Ch51], § 14, Th. 15), $\mathfrak{k}$ is the Lie algebra of a connected subgroup K of $G \times_k H$.⁽⁴⁵⁾ Moreover, since $\mathfrak{k} \simeq \mathfrak{g}$ is semisimple, G is a semisimple k -group. Since

$$\dim(K) = \dim(\mathfrak{k}) = \dim(\mathfrak{g}) = \dim(G)$$

and since $K \cap H$ is finite (because its Lie algebra is zero), the canonical morphism $\mathrm{pr}_1 : K \rightarrow G$ is finite and dominant; since G is connected, pr_1 is surjective; it is therefore an isogeny. Since G is simply connected and K semisimple, by Exp. XXI 6.2.7, pr_1 is an isomorphism, hence K is the graph of a morphism of k -groups $u : G \rightarrow H$ such that $\mathrm{Lie}(u) = v$.

Finally, let $u : G \rightarrow H$ be a morphism of k -groups. The tangent space to $\underline{\mathrm{Hom}}_{k\text{-gr.}}(G, H)$ at u identifies with $Z^1(G, \mathfrak{h})$ (cf. Exp. II, 4.2; G acts on $\mathfrak{h}$ by $\mathrm{Ad}_H \circ u$); in the same way, one can prove that the tangent space to $\underline{\mathrm{Hom}}_{k\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$ at $\mathrm{Lie}(u)$ identifies with $Z^1(\mathfrak{g}, \mathfrak{h})$. We must therefore prove that the canonical map $Z^1(G, \mathfrak{h}) \rightarrow Z^1(\mathfrak{g}, \mathfrak{h})$ is bijective. Now one has a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{h}^G & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(G, \mathfrak{h}) \longrightarrow H^1(G, \mathfrak{h}) \\ & & \downarrow & & \parallel & & \downarrow \\ 0 & \longrightarrow & \mathfrak{h}^{\mathfrak{g}} & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(\mathfrak{g}, \mathfrak{h}) \longrightarrow H^1(\mathfrak{g}, \mathfrak{h}) \end{array}$$

and by 7.3.2 and 7.3.3 one has $\mathfrak{h}^G = \mathfrak{h}^{\mathfrak{g}}$ and $H^1(G, \mathfrak{h}) = 0 = H^1(\mathfrak{g}, \mathfrak{h})$, whence the desired conclusion.

Remark 7.3.6. It is likely that if S is a locally noetherian scheme, G a simply connected semisimple S -group, H a smooth S -group scheme, and $\widehat{G}$ and $\widehat{H}$ their formal completions along the identity section, every homomorphism $\widehat{G} \rightarrow \widehat{H}$ comes from a unique homomorphism from G to H , which would generalize 7.3.1 (iii). When S is the spectrum of a field k and H is affine and of finite type, this follows from a theorem of Dieudonné ([Di57], § 15, Th. 4).⁽⁴⁶⁾

7.4. — An example As an example, we shall determine

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S}).$$

7.4.1. — Recall (Exp. XX, no. 5) that $\mathrm{SL}_{2,S}$ is the S -group scheme consisting of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ over S satisfying $ad - bc = 1$. A maximal torus T is defined by the monomorphism $\alpha^* : \mathbf{G}_{m,S} \rightarrow \mathrm{SL}_{2,S}$

$$\alpha^*(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}.$$

⁽⁴⁵⁾**N.D.E.** See also [Bo91], II 7.9. Moreover, we have added the following sentence.

⁽⁴⁶⁾**N.D.E.** See Exp. VIIB for the definition of the formal groups $\widehat{G}$ and $\widehat{H}$. Suppose that $S = \mathrm{Spec}(k)$, where k is a field. By loc. cit., 2.2.1, giving a morphism of formal k -groups $v : \widehat{G} \rightarrow \widehat{H}$ is equivalent to giving a morphism of Hopf k -algebras $\phi : \mathrm{Dist}(G) \rightarrow \mathrm{Dist}(H)$, where $\mathrm{Dist}(G)$ denotes the algebra of distributions (at the origin) of G , cf. Exp. VIIA, 2.1 or [DG70], § II.4, 6.1 or [Ja87], I 7.7 (it is called the “hyperalgebra” of G in [Di57] and [Ta75]). Theorem 4 of [Di57] (see also [Ta75], 0.3.4 (f) and (g)) generalizes in this context the theorem of Chevalley used in 7.3.5; one thus obtains that there exists a connected closed k -subgroup K of $G \times H$ such that $\widehat{K}$ equals the graph of v ; since $\mathrm{Dist}(K) \rightarrow \mathrm{Dist}(G)$ is an isomorphism, $K \rightarrow G$ is a finite étale morphism. One then deduces that K is semisimple and then, by Exp. XXI 6.2.7, that $K \rightarrow G$ is an isomorphism (since G is simply connected); see also [Ta75], 1.8 and 2.2. More generally, loc. cit. studies the k -groups G with property (SC): every finite étale morphism of k -groups $G' \rightarrow G$, with G' connected, is an isomorphism. Finally, observe that the above shows that every finite-dimensional $\mathrm{Dist}(G)$ -module V is, uniquely, a G -module; for an extension to the case of a split reductive k -group G (or even of a Borel subgroup of G), see [Ja87], II 1.20 (and the references there).

A root of G relative to T is defined by $\alpha(\alpha^*(z)) = z^2$, a corresponding monomorphism

$$p : \mathbf{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}$$

being defined by

$$p(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Finally, the representative of the Weyl group corresponding to $u = p(1)$ is

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Recall moreover (Exp. XX, 6.2) that $\mathrm{SL}_{2,S}$ is generated by T , $p(\mathbf{G}_{a,S})$ and w , subject to the relations

$$\begin{aligned} \alpha^*(z)p(x)\alpha^*(z^{-1}) &= p(z^2x), \\ w\alpha^*(z)w^{-1} &= \alpha^*(z^{-1}), \\ w^2 &= \alpha^*(-1), \\ (wu)^3 &= 1. \end{aligned}$$

7.4.2. — Let f be an endomorphism of $\mathrm{SL}_{2,S}$. It first defines a homomorphism $f \circ \alpha^* : \mathbf{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$. The kernel $\mathrm{Ker}(f \circ \alpha^*)$ is a closed subgroup of $\mathbf{G}_{m,S}$, hence is locally on S equal to a $\mu_{n,S}$ ($n \geq 1$) or to $\mathbf{G}_{m,S}$. After restricting S , one may therefore suppose that there exist an $n \in \mathbf{N}$ and a monomorphism

$$f' : \mathbf{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$$

such that $f \circ \alpha^*(z) = f'(z^n)$.

By conjugacy of the monomorphisms $\mathbf{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$, one can, after an extension of the base covering for the étale topology, find a section g of G such that $f \circ \alpha^*(z) = \mathrm{int}(g) \circ \alpha^*(z^n)$. Transforming f by $\mathrm{int}(g)$, one is therefore reduced to the case where there exists an $n \in \mathbf{N}$ such that $f \circ \alpha^*(z) = \alpha^*(z^n)$.

7.4.3. — Consider now the morphism

$$f \circ p : \mathbf{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}.$$

It satisfies the condition

$$\alpha^*(z^n)(f \circ p)(x)\alpha^*(z^n)^{-1} = (f \circ p)(z^2x).$$

If $n = 0$, it follows at once that $f \circ p$ is invariant under homotheties of $\mathbf{G}_{a,S}$, hence constant. If $n \neq 0$, one can apply Exp. XXII, 4.1.9; $x \mapsto x^n$ is an endomorphism of $\mathbf{G}_{a,S}$, and there exists a $\lambda \in \mathbf{G}_{m,S}$ such that

$$f \circ p(x) = p(\lambda x^n);$$

this relation is moreover valid for $n = 0$, taking $\lambda = 1$. After again extending S , one can find a $z \in \mathbf{G}_{m,S}$ such that $z^2 = \lambda$. Replacing f by $\mathrm{int}(\alpha^*(z)) \circ f$, one is therefore reduced to the case where one has

$$f \circ \alpha^*(z) = \alpha^*(z^n), \quad f \circ p(x) = p(x^n).$$

7.4.4. — Finally, one must have $f(w)\alpha^*(z^n)f(w)^{-1} = \alpha^*(z^n)^{-1}$ and $(f(w)u)^3 = 1$. By Exp. XX, 3.8, this implies $f(w) = w^n$. Since G is generated by T , $p(\mathbf{G}_{a,S})$ and w , this implies that for every $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(S')$, $S' \longrightarrow S$, one has

$$f \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$

7.4.5. — Summarizing the preceding discussion, one sees that locally on S for the étale topology, one can find for every endomorphism f of $\mathrm{SL}_{2,S}$ an (inner) automorphism ϕ of

$\mathrm{SL}_{2,S}$, and an integer $n \geq 0$ such that $f = \phi \circ F_n$, where

$$F_n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$

Observe that if $f = \phi \circ F_n$, the integer n is determined by a fiber f_s of f , for example by $\mathrm{Ker}(f_s)$. It follows that n is a locally constant function on S .

7.4.6. — One deduces immediately that if f is an endomorphism of $\mathrm{SL}_{2,S}$, then S decomposes canonically as a sum of open and closed subschemes S_0, S_1, S_{p^n} (where p^n ranges over the powers > 0 of prime numbers) such that:

- (i) f_{S_0} is the zero morphism,
- (ii) f_{S_1} is an isomorphism (= an inner automorphism),
- (iii) S_{p^n} is of characteristic p and $f_{S_{p^n}}$ decomposes uniquely in the form $\phi \circ F_p^n$, where ϕ is an inner automorphism and F_p the Frobenius endomorphism of $\mathrm{SL}_{2,\mathbf{F}_p}$.

7.4.7. — In other words, $\mathrm{Hom}_{\mathbf{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbf{Z}}, \mathrm{SL}_{2,\mathbf{Z}})$ is the sum scheme:

- (i) of a scheme isomorphic to $\mathrm{Spec}(\mathbf{Z})$,
- (ii) of a scheme isomorphic to $\mathrm{Aut}_{\mathbf{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbf{Z}}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbf{Z}})$,
- (iii) for each prime number p and each integer $n > 0$, of a scheme isomorphic to $\mathrm{Aut}_{\mathbf{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbf{F}_p}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbf{F}_p})$.

7.4.8. — It follows in particular that

- (i) $\mathrm{Hom}_{\mathbf{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbf{F}_p}, \mathrm{SL}_{2,\mathbf{F}_p})$ has infinitely many connected components, and therefore *is not quasi-compact*.
- (ii) If S is a scheme of mixed characteristics, $\mathrm{Hom}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S})$ *is not flat* over S .

8. Appendix: Cohomology of a smooth group over a henselian ring. Cohomology and the functor Π

Proposition 8.1. *Let S be a henselian local scheme, s its closed point, and G a smooth S -group scheme such that every finite subset of G is contained in an affine open subset.* Then*

- (i) *If P is a principal homogeneous bundle under G , there exists a finite surjective étale morphism $S' \rightarrow S$ trivializing P . One has $\mathrm{Fib}(S, G) \simeq H_{\mathrm{\acute{e}t}}^1(S, G)$.*
- (ii) *For every finite surjective étale morphism $S' \rightarrow S$, the canonical map*

$$H^1(S'/S, G) \rightarrow H^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is bijective.

- (iii) *The canonical map*

$$\mathrm{Fib}(S, G) \rightarrow \mathrm{Fib}(\kappa(s), G_s)$$

is bijective.

8.1.2. — If K is a finite separable extension of $\kappa(s)$, there exists a finite surjective étale morphism $S' \rightarrow S$ such that $K \simeq S' \otimes_S \kappa(s)$.⁽⁴⁷⁾ If P is a principal homogeneous bundle under G , then P is smooth over S , hence P_s smooth over $\kappa(s)$; there therefore exists a finite separable extension K of $\kappa(s)$ such that P_K has a section (cf. EGA IV₄, 17.15.10). Representing K as indicated above, one sees that $P_{S'}$ has a section by “Hensel’s lemma” (cf. EGA IV₄, 18.5.17), which proves the first part of (i).

Conversely, if P is a principal homogeneous *sheaf* under G for the étale topology, there exists a finite surjective étale morphism $S' \rightarrow S$ trivializing P (indeed every covering family

*This last condition is in fact unnecessary (cf. A. Grothendieck, *Groupe de Brauer III*, in *Dix Exposés sur la cohomologie des schémas*, North-Holland, 1968, Theorem 11.7 and Remarks 11.8.3).

⁽⁴⁷⁾**N.D.E.** Note that S' is still local and henselian. Moreover, we have preserved the numbering of the original: there is no no. 8.1.1.

of a henselian local scheme for the étale topology is refined by a family consisting of a single morphism $S' \rightarrow S$ of the required form).

By the descent hypothesis made on G , P is representable (SGA 1, VIII, 7.6), which completes the proof of (i), and shows that (ii) implies (iii). It therefore only remains for us to prove (ii).

8.1.3. — The map of (ii) is injective: let P and Q be two principal homogeneous bundles under G trivialized by S' . Consider the S -sheaf of groups $H = \underline{\text{Isom}}_{G\text{-bundles}}(P, Q)$; since $H_{S'}$ is isomorphic to $G_{S'}$, H is representable, by the second hypothesis on G , cf. above. If $H(\kappa(s)) \neq \emptyset$, then $H(S) \neq \emptyset$ by Hensel's lemma, hence P and Q are isomorphic.

8.1.4. — Let us finally prove that the map of (ii) is surjective, or again that the canonical map

$$Z^1(S'/S, G) \rightarrow Z^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is surjective. Let $\mathbf{Z}^1$ be the S -functor defined by

$$\mathbf{Z}^1(T) = Z^1(S' \times_S T/T, G_T);$$

the preceding map identifies with the map

$$\mathbf{Z}^1(S) \rightarrow \mathbf{Z}^1(\kappa(s));$$

by another application of Hensel's lemma, it suffices to prove that $\mathbf{Z}^1$ is representable by a smooth S -scheme.

8.1.5. — Let us prove that $\mathbf{Z}^1$ is representable by an S -scheme locally of finite presentation. Let $\mathbf{C}^i$, $i = 0, 1, \dots$, be the S -functor defined by

$$\mathbf{C}^i(T) = C^i(S' \times_S T/T, G_T),$$

i.e.

$$\mathbf{C}^i(T) = G((S' \times_S T/T)^{i+1}) = G((S'/S)^{i+1} \times_S T),$$

or again

$$\mathbf{C}^i = \underline{\text{Hom}}_S((S'/S)^{i+1}, G).$$

Since $\mathbf{Z}^1$ is obtained from $\mathbf{C}^1$ and $\mathbf{C}^2$ by fibered products, it suffices to prove that $\mathbf{C}^i$, $i = 1, 2$, is representable by an S -scheme locally of finite presentation.

8.1.6. — If $T \rightarrow S$ is a finite surjective étale morphism splitting S' , then

$$\mathbf{C}^i \times_S T = \underline{\text{Hom}}_T((S' \times_S T/T)^{i+1}, G_T)$$

is representable by a product of n copies of G_T , where n is the degree of $(S'/S)^{i+1}$. Applying the hypothesis on G once more, one deduces that $\mathbf{C}^i$ is indeed representable by an S -scheme locally of finite presentation (SGA 1, VIII, loc. cit.).

8.1.7. — To prove that $\mathbf{Z}^1$ is smooth, we must now, by definition, prove that if T is an *affine* S -scheme, and T_0 the closed subscheme defined by a square-zero ideal $\mathcal{J}$, the canonical map

$$\mathbf{Z}^1(T) \rightarrow \mathbf{Z}^1(T_0)$$

is surjective. Since G is smooth, the canonical map $G(T) \rightarrow G(T_0)$ is surjective, and it suffices to prove that the canonical map

$$H^1(S' \times_S T/T, G_T) \rightarrow H^1(S' \times_S T_0/T_0, G_{T_0})$$

is bijective.

Changing the notation slightly and generalizing the hypotheses, it now suffices for us to prove:

Lemma 8.1.8. *Let S and S' be two affine schemes, $S' \rightarrow S$ a faithfully flat morphism, $\mathcal{J}$ a square-zero ideal on S , S_0 the closed subscheme it defines, and G a smooth S -group. Put $S'_0 = S' \times_S S_0$, $G_0 = G \times_S S_0$. The canonical map*

$$H^1(S'/S, G) \rightarrow H^1(S'_0/S_0, G_0)$$

is bijective.

Remark 8.1.9. If one assumes G commutative, the same assertion holds for all the H^i , $i > 0$, with an analogous proof.

PROOF. Let $\mathcal{M}_0$ be the quasi-coherent $\mathcal{O}_{S_0}$ -module $\mathcal{M}_0 = \text{Lie}(G_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$. For each S_0 -prescheme T_0 , put $M_0(T_0) = H^0(T_0, \mathcal{M}_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0})$, and let

$$M = \prod_{S_0/S} M_0 \quad \text{and} \quad \bar{G} = \prod_{S_0/S} G_0$$

be the S -functors of groups defined by $M(T) = M_0(T_0)$ and $\bar{G}(T) = G_0(T_0)$, where $T_0 = T \times_S S_0$. By Exp. III, 0.9 and (0.6.2), there exists for every affine S -scheme T an exact sequence, functorial in T :

$$1 \rightarrow M(T) \rightarrow G(T) \rightarrow \bar{G}(T) \rightarrow 1.$$

We have to study the map

$$H^1(S'/S, G) \rightarrow H^1(S'_0/S_0, G_0) = H^1(S'/S, \bar{G}).$$

Suppose first that G is commutative. One then has an exact cohomology sequence

$$H^1(S'/S, M) \rightarrow H^1(S'/S, G) \rightarrow H^1(S'/S, \bar{G}) \rightarrow H^2(S'/S, M);$$

but

$$H^i(S'/S, M) = H^i(S'_0/S_0, M_0) = H^i(S'_0/S_0, \mathcal{M}_0),$$

and one knows (TDTE I, B, Lemma 1.1) that $H^i(S'_0/S_0, \mathcal{M}_0) = 0$ for $i \neq 0$.

If now G is not commutative, we must use the nonabelian cohomology exact sequence. If $u \in Z^1(S'/S, G)$, one knows that the elements of $H^1(S'/S, G)$ having the same image in $H^1(S'/S, \bar{G})$ as the class of u are in the image of the corresponding coboundary map:

$$H^1(S'/S, M_u) \rightarrow H^1(S'/S, G),$$

where M_u is the S -functor M “twisted by u ”. Likewise, if v is an element of $Z^1(S'/S, \bar{G})$, there exists a “coboundary”

$$\Delta(v) \in H^2(S'/S, M_v),$$

where M_v is the S -functor M “twisted by v ”, such that $\Delta(v) = 0$ if and only if the class of v is in the image of $H^1(S'/S, G)$. It suffices for us to prove that $H^1(S'/S, M_u) = H^2(S'/S, M_v) = 0$.

Now recall (Exp. III 0.8) that the action of $\bar{G}$ on M defined by the exact sequence is nothing other than the one deduced functorially from the adjoint representation of G_0 :

$$\text{ad} : G_0 \rightarrow \underline{\text{Aut}}_{\mathcal{O}_{S_0}}(\text{Lie}(G_0/S_0)).$$

The element u (resp. v) therefore acts on $M_{S' \times_S S'}$ through an $S'_0 \times_{S_0} S'_0$ -automorphism of $\text{Lie}(G_0/S_0)$. Since u (resp. v) is a cocycle, this automorphism is a descent datum; denote by $\mathcal{L}_u$ (resp. $\mathcal{L}_v$) the quasi-coherent $\mathcal{O}_{S_0}$ -module obtained. One verifies at once that for $T \rightarrow S$ one has

$$M_u(T) = H^0(T_0, \mathcal{L}_u \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0})$$

and the same relation on replacing u by v . One therefore has

$$H^1(S'/S, M_u) = H^1(S'_0/S_0, \mathcal{L}_u \otimes \mathcal{J}),$$

$$H^2(S'/S, M_v) = H^2(S'_0/S_0, \mathcal{L}_v \otimes \mathcal{J}),$$

and both do indeed vanish by the result already used. $\square$

Proposition 8.2. *Let $\mathcal{C}$ be a category possessing fibered products, equipped with a topology coarser than the canonical topology, $S' \rightarrow S$ a morphism of $\mathcal{C}$, G' an S' -sheaf of groups, and G the S -sheaf of groups $\prod_{S'/S} G'$. Let $H_S^1(S', G') \subset H^1(S', G')$ be the set of classes of principal homogeneous sheaves under G' which are trivialized by a sieve of S' obtained by base change from a suitable covering sieve of S . The canonical map $H^1(S, G) \rightarrow H^1(S', G')$ defined by the functor*

$$P \mapsto P \times_S S'$$

induces a bijection

$$H^1(S, G) \xrightarrow{\sim} H_S^1(S', G');$$

the inverse bijection is defined by the functor $P' \mapsto \prod_{S'/S} P'$.

PROOF. For every object X of $\mathcal{C}/S$, one has by definition an isomorphism functorial in X

$$\mathrm{Hom}_S(X, G) \xrightarrow{\sim} \mathrm{Hom}_{S'}(X \times_S S', G').$$

One therefore has for each S -object T a bijection functorial in T

$$H^1(T/S, G) \xrightarrow{\sim} H^1(T'/S', G').$$

Replacing now the single morphism $T \rightarrow S$ by an arbitrary covering family of S and passing to the inductive limit, one deduces the first part of the statement. The second part follows without difficulty. $\square$

Lemma 8.3. *Under the conditions of 8.2, the assertion $H_S^1(S', G') = H^1(S', G')$ is local on S : suppose that there exists a covering family $\{S_i \rightarrow S\}$ such that for every i one has $H_{S_i}^1(S' \times_S S_i, G') = H^1(S' \times_S S_i, G')$. Then $H_S^1(S', G') = H^1(S', G')$.*

PROOF. Indeed, let P' be a principal homogeneous sheaf under G' . Put

$$P'_i = P' \times_{S'} (S' \times_S S_i);$$

by the hypothesis, there exists a covering family $\{S_{ij} \rightarrow S_i\}$ such that for each j , $P' \times_{S'} (S' \times_S S_{ij})$ has a section. But $\{S_{ij} \rightarrow S\}$ is a covering family of S , and P' is indeed trivialized by the covering family of S' obtained from it by base change. $\square$

Proposition 8.4. *Let $S' \rightarrow S$ be a finite étale morphism of schemes. Let G' be an S' -sheaf of groups, and G the S -sheaf of groups $\prod_{S'/S} G'$. For the étale (resp. local finite étale, resp. (fpqc)) topology, the functors*

$$P \mapsto P \times_S S' \\ \prod_{S'/S} P' \leftarrow P'$$

induce mutually inverse bijections:

$$H^1(S, G) \simeq H^1(S', G').$$

PROOF. By 8.2, it suffices to show that $H_S^1(S', G') = H^1(S', G')$. By 8.3, it suffices to do so locally for the local finite étale topology; one may therefore suppose that S' is a finite direct sum of copies of S , say I_S , where I is a suitable finite set. Then G' is given by a family $(G_i)_{i \in I}$ of sheaves on S , and

$$H^1(S', G') \simeq \prod_{i \in I} H^1(S, G_i).$$

On the other hand

$$H^1(S, G) \simeq \prod_{i \in I} H^1(S, G_i),$$

whence, by 8.2, $H_S^1(S', G') = H^1(S', G')$. $\square$

Remarks 8.5. One can interpret 8.2 and 8.3 by means of the following exact sequence (f is the given morphism $S' \rightarrow S$)

$$1 \rightarrow H^1(S, f_*(G')) \rightarrow H^1(S', G') \rightarrow H^0(S, R^1 f_*(G')).$$

In the commutative case, this exact sequence follows from the Leray spectral sequence; it is still valid in the noncommutative case (cf. Giraud's thesis⁽⁴⁸⁾).

In this form, one sees that the result is still valid if f is merely assumed *finite*, or simply *integral*, the topology being the étale topology, since for every G' one has

$$R^1 f_*(G') = \text{final sheaf},$$

by SGA 4, VIII, 5.3.⁽⁴⁹⁾

On the other hand, this result becomes false if one takes a topology such as (fpqc) or (fppf), even if $S = \operatorname{Spec}(k)$, with k an algebraically closed field of characteristic $p \neq 0$, $S' = \operatorname{Spec}(k[t]/t^2)$, and $G' = \mu_p$ or α_p .

Likewise, 8.2 becomes false, even for the étale topology, if one removes the hypothesis that f is finite, as one sees by taking for f an open immersion; for example, if $S = \operatorname{Spec}(V)$, with V a complete discrete valuation ring with algebraically closed residue field, S' being the open subset induced at the generic point, and G' the constant group $(\mathbf{Z}/n\mathbf{Z})_{S'}$, with n prime to the residue characteristic of V , one has $H^1(S, G) = 0$, $H^1(S', G') \neq 0$. Replacing moreover S' by $S \amalg S'$, one deduces an analogous example, with $S' \rightarrow S$ étale and surjective, hence covering for the topology in question.

⁽⁴⁸⁾**N.D.E.** See paragraph V 3.1.4 of [Gi71].

⁽⁴⁹⁾**N.D.E.** This reference also points to paragraph V 3.1.4 of [Gi71].

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⁽⁵⁰⁾**N.D.E.** To these three references, appearing in the original, we have added the following references.

Exposé XXV. The existence theorem

⁽⁰⁾ For completeness, we give in this exposé a proof of the existence theorem for split groups. Like Chevalley’s original proof (“Sur certains schémas de groupes semi-simples”, Séminaire Bourbaki, May 1961, no. 219), it rests on the existence of complex semisimple algebraic groups of all possible types. The principle of a more satisfactory proof, proving directly the existence of a simply connected split semisimple $\mathbf{Z}$ -group corresponding to a given Cartan matrix, was given by Cartier (unpublished).⁽¹⁾ We point out, however, that the difficulty is not to give an explicit construction of a group scheme, but to verify that the group thus constructed does satisfy the required conditions, that is, essentially, that its fibers are indeed smooth and reductive.

1. Statement of the theorem

Theorem 1.1. *Let S be a nonempty scheme. The functor*

$$G \longmapsto \mathcal{R}(G)$$

is an equivalence from the category of pinned reductive S -groups to the category of pinned reduced root data.

By the uniqueness theorem (Exp. XXIII, 4.1), the preceding statement is equivalent to:

Corollary 1.2 (Existence of split groups). *For every reduced root datum $\mathcal{R}$, there exists a split reductive $\mathbf{Z}$ -group G such that $\mathcal{R}(G) \simeq \mathcal{R}$.⁽²⁾*

In particular:

Corollary 1.3. *Let k be a field. For every splittable reductive k -group G_k , there exists a reductive $\mathbf{Z}$ -group G such that $G \otimes_{\mathbf{Z}} k \simeq G_k$.*

Conversely, let us first note that, to prove 1.2, it suffices by Exp. XXII 4.3.1 and Exp. XXI 6.5.10 to consider the case where the root datum $\mathcal{R}$ is simply connected (and even irreducible, if one wishes, by Exp. XXI, 7.1.6).

Moreover, under the conditions of 1.3, the group scheme G has constant type (since $\mathrm{Spec}(\mathbf{Z})$ is connected), hence type $\mathcal{R}(G_k)$; by Exp. XXIII, 5.9, it follows that the validity of 1.3 for a given group G_k implies the existence of a split $\mathbf{Z}$ -group of type $\mathcal{R}(G_k)$.⁽³⁾

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** Such a principle was also outlined by B. Kostant [Ko66]; a complete proof, using quantum groups, was given recently by G. Lusztig [Lu09].

⁽²⁾**N.D.E.** We also point out that the work [BT84] of F. Bruhat and J. Tits contains a variant of Chevalley’s construction (*loc. cit.*, 2.2.3–2.2.5 and § 3.2), which provides in particular a smooth affine $\mathbf{Z}$ -group G with connected fibers, possessing a split maximal torus, and whose generic fiber is a reductive $\mathbf{Q}$ -group of type $\mathcal{R}$; the fact that the geometric fibers of G are reductive follows from the study of the unipotent radical of a special fiber made in *loc. cit.*, 4.6.12 and 4.6.15 (valid for more general G , associated with a valued root datum), but it is simpler to deduce it from the description of $\mathfrak{g} = \mathrm{Lie}(G)$, from Exp. XIX 1.12 (iii), and from the existence of the elements $w_\alpha(X)$ of Exp. XX 3.1 (iv) (cf. [BT84], 3.2.1).

⁽³⁾**N.D.E.** In fact, Exp. XXIII, 5.9 is not necessary, since the present exposé constructs, for every semisimple $\mathbf{Q}$ -group $G_{\mathbf{Q}}$, a semisimple $\mathbf{Z}$ -group G of the same type as $G_{\mathbf{Q}}$, equipped with a split maximal torus T ; thus, by Exp. XXII 2.2, G is split.

To prove 1.2 and hence 1.1, it therefore suffices to prove 1.3 when k has characteristic zero (for example $k = \mathbf{C}$) and G_k is simply connected (and in particular semisimple), as well as:

Proposition 1.4. *For every simply connected reduced root datum $\mathcal{R}$, there exists a semisimple algebraic $\mathbf{C}$ -group of type $\mathcal{R}$.*

One can prove 1.4 as follows. First, one knows that there exists a complex semisimple Lie algebra $\mathfrak{g}$ of type $\mathcal{R}$, cf. for example N. Jacobson, *Lie Algebras*, ch. VII, Th. 5.⁽⁴⁾ Then $G = \text{Aut}(\mathfrak{g})^0$ is a semisimple algebraic $\mathbf{C}$ -group of type $\text{ad}(\mathcal{R})$.⁽⁵⁾ By Bible, § 23.1, prop. 1, one deduces the existence of a semisimple $\mathbf{C}$ -group of type $\mathcal{R}$.

The remainder of this exposé is devoted to proving 1.3 for k of characteristic zero and G_k semisimple. This will be done in two stages: construction of a “chunk of a group $\mathbf{Z}$ -scheme” (no. 2), study of the group obtained by applying the “Weil theorem” (no. 3).

To avoid confusion, *we shall not use in no. 2* the abbreviated notation X_k to denote the k -scheme $X \otimes_{\mathbf{Z}} k$, where X is a $\mathbf{Z}$ -scheme.

2. Existence theorem: construction of a group chunk

2.1. — Let us choose once and for all a splitting of G_k , denoted

$$(G_k, T_k, M, R)$$

(cf. Exp. XXII, 1.13), a system of simple roots Δ of R (defining the system of positive roots R_+), and a Chevalley system $(X_{\alpha,k})_{\alpha \in R}$ of G_k (Exp. XXIII, 6.1 and 6.2) satisfying the following additional condition (cf. XX 2.6): for every $\alpha \in R$, one has

$$X_{\alpha,k} X_{-\alpha,k} = 1.$$

Finally, let us choose on the subgroup of M generated by R a total ordering compatible with the group structure, such that the roots > 0 are the elements of R_+ . The roots are then denoted

$$-\alpha_n < -\alpha_{n-1} < \cdots < -\alpha_1 < \alpha_1 < \alpha_2 < \cdots < \alpha_n.$$

For $\alpha \in R$, one denotes by $U_{\alpha,k}$ the vector group corresponding to the root α , and by

$$p_{\alpha,k} : \mathbf{G}_{a,k} \xrightarrow{\sim} U_{\alpha,k}$$

the isomorphism of vector groups defined by $X_{\alpha,k}$.

2.2. — The splitting of G_k includes in particular an isomorphism of k -groups

$$T_k \simeq D_k(M).$$

Put $T = D(M)$; this is a $\mathbf{Z}$ -torus, and one can regard the preceding isomorphism as an isomorphism

$$T_k \simeq T \otimes_{\mathbf{Z}} k.$$

⁽⁴⁾**N.D.E.** Let Δ be a base of R , and let $\tilde{\mathfrak{g}}$ be the $\mathbf{C}$ -Lie algebra generated by generators $(e_\alpha, f_\alpha, h_\alpha)_{\alpha \in \Delta}$ subject to the relations $[h_\alpha, h_\beta] = 0$, $[e_\alpha, f_\beta] = h_\alpha$ if $\beta = \alpha$ and $= 0$ otherwise, $[h_\alpha, e_\beta] = (\alpha^*, \beta)e_\beta$ and $[h_\alpha, f_\beta] = -(\alpha^*, \beta)f_\beta$. In *loc. cit.*, $\mathfrak{g}$ is defined as the quotient of $\tilde{\mathfrak{g}}$ by the intersection of the kernels of the finite-dimensional irreducible representations of $\tilde{\mathfrak{g}}$; in [Se66], § VI.5, Th. 9 (see also [BLie], VIII, § 4.3, Th. 1), it is shown that $\mathfrak{g}$ is the quotient of $\tilde{\mathfrak{g}}$ by the relations $\text{ad}(e_\alpha)^{1-(\alpha^*, \beta)}(e_\beta) = 0$ and $\text{ad}(f_\alpha)^{1-(\alpha^*, \beta)}(f_\beta) = 0$. For an explicit description of the structure constants (in particular the choice of signs), see Exp. XXIII 6.5 and 6.7, as well as [Ti66], § 4, Th. 1, and (for types A,D,E) [Sp98], 10.2.5.

⁽⁵⁾**N.D.E.** One can suppose $\mathcal{R}$ irreducible, hence $\mathfrak{g}$ simple. By a general argument, one knows that $\text{Lie}(G)$ is the $\mathbf{C}$ -Lie algebra of derivations of $\mathfrak{g}$ (cf. [DG70], § II.4, Prop. 2.3); these are all inner (cf. [BLie], I § 6.1, cor. 3 of prop. 1), so $\text{Lie}(G) = \mathfrak{g}$; consequently G has no invariant subgroup of dimension > 0 , hence G is semisimple. Its root system is then the same as that of $\mathfrak{g}$; moreover the center of G acts both trivially and faithfully on $\mathfrak{g}$, and so is trivial, hence G is adjoint.

One has

$$\mathrm{Hom}_{\mathbf{Z}\text{-gr.}}(T, \mathbf{G}_m) = M,$$

and one will regard the elements of $R \subset M$ as characters of T . Similarly, one will regard the elements of R^* as morphisms of $\mathbf{Z}$ -groups $\mathbf{G}_m \rightarrow T$.

2.3. — For each $\alpha \in R_+$, let $\mathbf{G}_a(\alpha)$ be a copy of the group $\mathbf{G}_a$; consider the $\mathbf{Z}$ -scheme

$$U = \mathbf{G}_a(\alpha_1) \times \cdots \times \mathbf{G}_a(\alpha_n).$$

If U_k denotes the unipotent part of the Borel group B_k of G_k defined by R_+ , denote by

$$a : U \otimes_{\mathbf{Z}} k \xrightarrow{\sim} U_k$$

the isomorphism of k -schemes defined by

$$a(x_1, \dots, x_n) = p_{\alpha_1, k}(x_1) \cdots p_{\alpha_n, k}(x_n).$$

2.4. — The group law of U_k is expressed by relations of the form

$$a(x_1, \dots, x_n) \cdot a(y_1, \dots, y_n) = a(z_1, \dots, z_n),$$

where each z_h ($h = 1, \dots, n$) is expressed as a polynomial

$$z_h = x_h + y_h + Q_h(x_1, \dots, x_{h-1}, y_1, \dots, y_{h-1}),$$

the coefficients of Q_h being integers (Exp. XXII, 5.5.8 and Exp. XXIII, 6.4). Moreover $Q_h(x_1, \dots, x_{h-1}, 0, \dots, 0) = 0$.

Let us equip U with the composition law defined by the preceding formulas (which are indeed “defined over $\mathbf{Z}$ ”). Since this law induces on U_k its group law, it is associative, and (0) is an identity element for it (indeed, the preceding two assertions are expressed by relations among the polynomials Q_h , and $\mathbf{Z} \rightarrow k$ is injective). Let us show that it is a group law: if (x_i) is a section of U (over any S), one computes the inverse (y_i) of (x_i) by the recurrence formulas:

$$y_i = -x_i - Q_i(x_i, \dots, x_{i-1}, y_1, \dots, y_{i-1})$$

which are again “defined over $\mathbf{Z}$ ”.

To sum up, we have constructed on U a group law such that the preceding isomorphism a is an isomorphism of groups.

For each $\alpha \in R_+$, consider the morphism

$$p_\alpha : \mathbf{G}_a \rightarrow U$$

defined by $p_\alpha(x) = (x_i)$, where

$$\begin{cases} x_i = x & \text{if } \alpha_i = \alpha, \\ x_i = 0 & \text{if } \alpha_i \neq \alpha. \end{cases}$$

This is a closed immersion and a group homomorphism; its image is denoted U_α . One has $(x_i) = p_{\alpha_1}(x_1) \cdots p_{\alpha_n}(x_n)$, which proves that U is identified with the product

$$U = U_{\alpha_1} \cdot U_{\alpha_2} \cdots U_{\alpha_n}.$$

2.5. — Let us make $T = D(M)$ act on each U_α through the character α ; one immediately verifies that this defines an action of T on the group U , and one can construct the semidirect product $B = T \cdot U$. One has a canonical isomorphism of k -groups

$$B \otimes_{\mathbf{Z}} k \xrightarrow{\sim} B_k.$$

If we now take any ordering on R_+ , the morphism

$$\prod_{\alpha \in R_+} U_\alpha \rightarrow U$$

defined by the product in U is still an isomorphism. Indeed, since both sides are flat $\mathbf{Z}$ -schemes of finite presentation, one can content oneself with verifying the assertion on the geometric fibers; one is then reduced to Lazard’s theory (Bible, § 13.1, prop. 1): one regards U as a

group with operators T , and uses the fact that the U_α are pairwise nonisomorphic as groups with operators (since the characters $\alpha \in R_+$ of T are pairwise distinct on each fiber).

2.6. — Replacing R_+ by $R_- = -R_+$, one similarly constructs groups U^- , B^- , U_α ($\alpha \in R_-$), and isomorphisms

$$p_\alpha : \mathbf{G}_a \xrightarrow{\sim} U_\alpha, \quad \alpha \in R_-.$$

Finally, introduce the product scheme

$$\Omega = U^- \times T \times U;$$

one has a canonical isomorphism of k -schemes

$$\Omega \otimes_{\mathbf{Z}} k \xrightarrow{\sim} U_k^- \times_k T_k \times_k U_k \simeq \Omega_k,$$

where Ω_k is the “big cell” of G_k (Exp. XXII, 4.1.2).

From now on, we identify $\Omega \otimes_{\mathbf{Z}} k$ with Ω_k by the preceding isomorphism; we regard U^- , T , U as subschemes of Ω , by means of the identity sections. One denotes by $\underline{e} = ((0), e, (0))$ the “identity section” of Ω .

Our aim now is to put a group-chunk law on Ω .

Lemma 2.7. *Let $\alpha \in \Delta$, and let $w_{\alpha,k}$ be the element of $\text{Norm}_{G_k}(T_k)(k)$ defined by $X_{\alpha,k}$ (recall that by definition one has*

$$w_{\alpha,k} = p_{-\alpha,k}(-1)p_{\alpha,k}(1)p_{-\alpha,k}(-1)).$$

There exists an open subset V_α of Ω , containing the section $\underline{e}$, and a morphism

$$h_\alpha : V_\alpha \longrightarrow \Omega,$$

satisfying the following conditions:

- (i) $h_\alpha(\underline{e}) = \underline{e}$,
- (ii) $(h_\alpha) \otimes_{\mathbf{Z}} k$ coincides with the restriction of $\text{int}(w_{\alpha,k})$ to $V_\alpha \otimes_{\mathbf{Z}} k \subset G_k$.
- (iii) One has $T \subset V_\alpha$, and h_α sends T into T . For every $\beta \in R$, one has $U_\beta \subset V_\alpha$, and h_α sends U_β into $U_{s_\alpha(\beta)}$.

By the definition of a Chevalley system (Exp. XXIII, 6.1), there exists for each $\beta \in R$ an integer $e_\beta = \pm 1$ such that

$$\text{int}(w_{\alpha,k})p_{\beta,k}(x) = p_{s_\alpha(\beta),k}(e_\beta x)$$

for every $x \in \mathbf{G}_a(S)$, $S \longrightarrow \text{Spec}(k)$.

Let S be any scheme; let us write any element of $\Omega(S)$ in the form

$$u = \left(\left(\prod_{\substack{\beta \in R_- \\ \beta \neq -\alpha}} p_\beta(x_\beta) \right) \cdot p_{-\alpha}(x_{-\alpha}), \quad t, \right. \\ \left. p_\alpha(x_\alpha) \cdot \left(\prod_{\substack{\beta \in R_+ \\ \beta \neq \alpha}} p_\beta(x_\beta) \right) \right),$$

where one has chosen an ordering (any ordering) on $R_- - \{-\alpha\}$ and $R_+ - \{\alpha\}$ (cf. 2.5).

One defines a morphism $d : \Omega \longrightarrow \text{Spec}(\mathbf{Z})$ by

$$d(u) = \alpha(t) + x_\alpha x_{-\alpha}.$$

Let V_α be the open subset Ω_d (that is, the open subset of Ω defined by “ $d(u)$ invertible”); it contains $\underline{e}$, T , and each U_β , $\beta \in R$. Let

$$h_\alpha : V_\alpha \longrightarrow \Omega$$

be the morphism defined by $h_\alpha(u) = (a(u), b(u), c(u))$, where

$$\begin{aligned} a(u) &= \left(\prod_{\substack{\beta \in R_- \\ \beta \neq -\alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right) \cdot p_{-\alpha}(-x_\alpha d(u)^{-1}), \\ b(u) &= t \cdot \alpha^*(d(u)), \\ c(u) &= p_\alpha(-x_{-\alpha} d(u)^{-1}) \cdot \left(\prod_{\substack{\beta \in R_+ \\ \beta \neq \alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right). \end{aligned}$$

Since s_α permutes the positive (resp. negative) roots distinct from α (resp. $-\alpha$), $c(u)$ (resp. $a(u)$) is then a section of U (resp. U^-), and the preceding morphism is well defined. It trivially satisfies (i) and (iii). As for (ii), this follows immediately from the definition of the e_β , $\beta \in R$, and from Exp. XX, 3.12.

Lemma 2.8. *There exist open subsets V and V' of Ω , and morphisms*

$$h : V \longrightarrow \Omega, \quad h' : V' \longrightarrow \Omega,$$

satisfying the following conditions:

- (i) V and V' contain $\underline{e}$, and $h(\underline{e}) = h'(\underline{e}) = \underline{e}$.
- (ii) The morphism induced by $h' \circ h : h^{-1}(V') \longrightarrow \Omega$ is the restriction of the identity morphism.
- (iii) The k -morphisms $h \otimes_{\mathbf{Z}} k$ and $h' \otimes_{\mathbf{Z}} k$ are the restrictions to $V \otimes_{\mathbf{Z}} k$ and $V' \otimes_{\mathbf{Z}} k$ of automorphisms of the group G_k .
- (iv) V and V' contain U , T and U^- ; h and h' send U into U^- , U^- into U , and T into T .

Let $\bar{w}_0$ be the element of the Weyl group of G_k which transforms R_+ into R_- . Write

$$\bar{w}_0 = s_{\alpha_n} \cdots s_{\alpha_1}, \quad \alpha_i \in \Delta$$

(no connection with the numbering of 2.1). Put

$$w_0 = w_{\alpha_n, k} \cdots w_{\alpha_1, k} \in \underline{\text{Norm}}_{G_k}(T_k)(k).$$

Define by induction on $i \leq n$ an open subset V_i of Ω and a morphism $h_i : V_i \longrightarrow \Omega$ by $V_0 = \Omega$, $h_0 = \text{id}$, and, for $i = 0, \dots, n-1$,

$$V_{i+1} = h_i^{-1}(V_{\alpha_{i+1}}), \quad h_{i+1} = h_{\alpha_{i+1}} \circ h_i,$$

where the notations V_{α_j} and h_{α_j} are those of 2.7.

Take $V = V_n$ and $h = h_n$. The conditions of (i), (iii) and (iv) concerning V and h are indeed satisfied; for (i) and (iii) this follows immediately from 2.8, and for (iv), from the fact that $h \otimes_{\mathbf{Z}} k$ is the restriction of $\text{int}(w_0)$ to $V \otimes_{\mathbf{Z}} k$.

Since $(\bar{w}_0)^2 = 1$, one also has

$$\bar{w}_0 = s_{\alpha_1} \cdots s_{\alpha_n} = (s_{\alpha_1}^3) \cdots (s_{\alpha_n}^3).$$

Putting

$$w'_0 = (w_{\alpha_1, k})^3 \cdots (w_{\alpha_n, k})^3,$$

and carrying out the same construction as above, one deduces an open subset V' and a morphism h' likewise satisfying (i), (iii), (iv). Moreover, $h' \otimes_{\mathbf{Z}} k$ is the restriction of $\text{int}(w'_0)$ to $V' \otimes_{\mathbf{Z}} k$. But for each simple root $\alpha \in \Delta$, one has $(w_{\alpha, k})^4 = e$ (cf. Exp. XX, 3.1), so $w'_0 \cdot w_0 = e$, which shows that $h' \circ h$ induces the identity morphism in a nonempty open subset of $\Omega \otimes_{\mathbf{Z}} k$. But since Ω is smooth and of finite presentation over $\mathbf{Z}$, $\Omega \otimes_{\mathbf{Z}} k$ is schematically dense in Ω , which proves (ii).

Proposition 2.9. *There exist an open subset V_1 of $\Omega \times \Omega$, an open subset V_2 of Ω , and morphisms*

$$\pi : V_1 \longrightarrow \Omega, \quad \sigma : V_2 \longrightarrow \Omega,$$

possessing the following properties:

- (i) If $x \in \Omega(S)$, then $(\underline{e}, x)$ and $(x, \underline{e})$ are sections of V_1 , and

$$\pi(\underline{e}, x) = \pi(x, \underline{e}) = x.$$

- (ii) V_2 contains $\underline{e}$, and $\sigma(\underline{e}) = \underline{e}$.

- (iii) π_k and σ_k are the restrictions of the morphisms $G_k \times_k G_k \rightarrow G_k$ and $G_k \rightarrow G_k$ defined by the product (resp. the inverse).

PROOF. Let (v, t, u) and (v', t', u') be two sections of Ω . Then $h(u)$ is a section of U^- , $h(v')$ is a section of U by 2.8 (iv), and one can therefore regard $(h(u), e, h(v'))$ as a section of Ω . Let V_1 be the open subset of $\Omega \times \Omega$ defined by the condition:

$$((v, t, u), (v', t', u')) \in V_1(S) \iff (h(u), e, h(v')) \in V'(S)$$

(notations of 2.8). If $((v, t, u), (v', t', u'))$ is a section of V_1 , then $h'(h(u), e, h(v'))$ is defined; this is a section of Ω which one can decompose:

$$h'(h(u), e, h(v')) = (v'', t'', u'').$$

One then puts

$$\pi((v, t, u), (v', t', u')) = (v \cdot tv''t^{-1}, tt''t', t'^{-1}u''t' \cdot u').$$

The verification of (i) is immediate (by 2.8 (ii)). To verify the condition of (iii) concerning π , one sees that

$$h'(h(u), e, h(v')) = uv' = v''t''u''$$

when $u \in U(S)$, $v \in U^-(S)$, $S \rightarrow k$, by 2.8 (iii) and (ii). One constructs σ in a similar manner: if (v, t, u) is a section of Ω , $h(u^{-1})$ is a section of U^- , $h(v^{-1})$ a section of U , $h(t^{-1})$ a section of T , so $(h(u^{-1}), h(t^{-1}), h(v^{-1}))$ is a section of Ω , and one can define an open subset V_2 of Ω by

$$(v, t, u) \in V_2(S) \iff (h(u^{-1}), h(t^{-1}), h(v^{-1})) \in V'(S)$$

and a morphism $\sigma : V_2 \rightarrow \Omega$ by

$$\sigma(v, t, u) = h'(h(u^{-1}), h(t^{-1}), h(v^{-1})).$$

One verifies the conditions on σ as above. □

Corollary 2.10. π is “generically associative”, and σ is a “generic inverse”: if $x, y, z \in \Omega(S)$ and if the expressions below are defined (which always occurs above an open subset of Ω containing the identity section), one has:

$$\pi(x, \pi(y, z)) = \pi(\pi(x, y), z), \quad \pi(x, \sigma(x)) = \underline{e} = \pi(\sigma(x), x).$$

Indeed, the two sides of each of these formulas define morphisms between smooth $\mathbf{Z}$ -schemes of finite presentation, which coincide on the generic fibers, by 2.10 (iii).

Corollary 2.11. Let $\alpha \in R$. For every S and all $x, y \in \mathbf{G}_a(S)$ such that $(p_\alpha(x), p_{-\alpha}(y)) \in V_1(S)$ and $1 + xy \in \mathbf{G}_m(S)$ (which defines an open subset of $\mathbf{G}_{a,S}^2$ containing the section $(0, 0)$), one has:

$$\pi(p_\alpha(x), p_{-\alpha}(y)) = \left(p_{-\alpha}\left(\frac{y}{1+xy}\right), \alpha^*(1+xy), p_\alpha\left(\frac{x}{1+xy}\right) \right).$$

The proof is the same as before, by Exp. XX, 2.1.

3. Existence theorem: end of the proof

To simplify the language, let us make the following definition.

Definition 3.1. Let S be a scheme and G a group S -scheme. One says that G is *admissible* if there exists an open immersion of S -schemes $i : \Omega_S = \Omega \times S \rightarrow G$ satisfying the following conditions:

(i) The diagram

$$\begin{array}{ccc} (V_1)_S & \longrightarrow & \Omega_S \times_S \Omega_S \xrightarrow{i \times_S i} G \times_S G \\ \pi \downarrow & & \downarrow \pi_G \\ \Omega_S & \xrightarrow{i_S} & G, \end{array}$$

where π_G denotes the multiplication morphism in G , is commutative.

(ii) There exists a finite set of sections $a_j \in \Omega(S)$ such that the $i(a_j) \cdot i(\Omega_S)$ cover G .

By the “Weil theorem” (Exp. XVIII, 3.13 (iii) and (iv)), one has:

Lemma 3.2. *If, for every scheme S étale and of finite type over $\mathbf{Z}$, every admissible S -group is affine, then there exists an admissible affine $\mathbf{Z}$ -group.*

Now one has:

Lemma 3.3. *Let S be a scheme and G an admissible S -group. Then G is smooth and of finite presentation over S , with affine connected semisimple fibers.*

Since Ω_S is smooth and of finite presentation over S , with connected fibers, the same holds for G , by condition (ii). To verify 3.3, one can therefore suppose that S is the spectrum of a field K . Identify Ω_K with its image in G . It is clear that Ω_K is the product

$$\prod_{\alpha \in R_-} U_{\alpha,K} \cdot T_K \cdot \prod_{\alpha \in R_+} U_{\alpha,K}$$

of the subgroups T_K and $U_{\alpha,K}$ ($\alpha \in R$) of G . The Lie algebra of G is therefore identified with the direct sum

$$\mathrm{Lie}(T_K) \oplus \coprod_{\alpha \in R} \mathrm{Lie}(U_{\alpha,K}).$$

Since the inner automorphism defined by a section of T_K acts on $U_{\alpha,K}$, and hence on $\mathrm{Lie}(U_{\alpha,K})$, through the character

$$\alpha \in R \subset M \simeq \mathrm{Hom}_{K\text{-gr.}}(T_K, \mathbf{G}_{m,K}),$$

the preceding decomposition of $\mathrm{Lie}(G)$ is exactly the decomposition under the adjoint action of T . The roots of G_K relative to T_K are therefore the $\alpha \in R$. Apply Exp. XIX, 1.13. Let T_α be the maximal torus of $\mathrm{Ker}(\alpha) \subset T_K$, and let $Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha)$; it suffices for us to prove that each Z_α is reductive. Now $Z_\alpha \cap \Omega_K$ is precisely

$$\prod_{\substack{\beta \in R_- \\ \beta|T_\alpha=e}} U_{\beta,K} \cdot T_K \cdot \prod_{\substack{\beta \in R_+ \\ \beta|T_\alpha=e}} U_{\beta,K};$$

but the roots vanishing on T_α are the rational multiples of α , hence α and $-\alpha$; this proves

$$Z_\alpha \cap \Omega_K = U_{-\alpha,K} \cdot T_K \cdot U_{\alpha,K}.$$

To prove that Z_α is reductive, it suffices, by Exp. XX 3.4, to prove that $U_{\alpha,K}$ and $U_{-\alpha,K}$ do not commute, which follows immediately from 2.11.

It follows from 3.2 and 3.3 that the proof will be complete if we prove:

Lemma 3.4. *If S is a locally noetherian scheme of dimension ≤ 1 , and if G is a smooth S -group of finite type, with affine connected semisimple fibers, then G is affine (and hence semisimple).*

Nota. — In Exp. XVI, one saw that 3.4 holds without any hypothesis on S , but the proof is relatively delicate; since here we need only the special case 3.4, we give a direct proof of it.

Consider the Lie algebra $\mathfrak{g}$ of G , which is a locally free $\mathcal{O}_S$ -module, and the adjoint representation of G

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}_{\mathcal{O}_S}(\mathfrak{g}).$$

To prove that G is affine over S , it suffices to prove that the morphism Ad is affine. Since G is smooth with connected fibers, it is separated over S (VIB 5.5), hence the morphism Ad is separated. Using a result proved in the appendix (see 4.1), it suffices to prove that the morphism Ad is quasi-finite. One is therefore reduced to the case where S is the spectrum of a field; in this case G is affine, hence semisimple, and one is reduced to Exp. XXII 5.7.14.

4. Appendix

We used the following proposition in the course of the proof:

Proposition 4.1. *Let S be a locally noetherian scheme of dimension ≤ 1 , G and H two group S -schemes of finite type, and $f : G \rightarrow H$ a quasi-finite separated group morphism. If G is flat over S ,⁽⁶⁾ then f is affine.*

We shall give the proof only in the case where G is smooth over S , a hypothesis which is indeed satisfied in the application we have made of the proposition.

4.2. — By EGA II, 1.6.4, one can suppose S reduced. By the usual techniques of passage to the limit,⁽⁷⁾ one can suppose S local. If $\dim(S) = 0$, the assertion is trivial;⁽⁸⁾ suppose $\dim(S) = 1$. By faithfully flat descent, one can suppose that S is complete with algebraically closed residue field. Replacing S by its normalization $\tilde{S}$, one can (EGA II, 6.7.1 and EGA 0_{IV} 23.1.5) suppose S normal.⁽⁹⁾ One is therefore reduced to the case where S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field.

4.3. — Let η (resp. s) be the generic (resp. closed) point of S . Consider the image $f_\eta(G_\eta)$ of G_η in H_η . This is a closed group subscheme of H_η . Let H' be the schematic closure in H of $f_\eta(G_\eta)$. Since $H' \rightarrow H$ is affine (it is a closed immersion), one can replace H by H' , and hence suppose H flat over S and f_η surjective. Since f_η is finite, and G and H flat over S , one has

$$\dim(G_s) = \dim(G_\eta) = \dim(H_\eta) = \dim(H_s).$$

4.4. — Let $H_s^0, \dots, H_s^n$ be the irreducible components of H_s , where H_s^0 denotes the identity component, and let $z_0, \dots, z_n$ be their generic points. Since each local ring $\mathcal{O}_{H, z_i}$ has dimension ≤ 1 , the morphism $G \times_H \mathcal{O}_{H, z_i} \rightarrow \mathcal{O}_{H, z_i}$ is affine, since it is quasi-finite and separated (cf. Exp. XVI, lemma 4.2), hence G is affine over H in a neighborhood of z_i . Denoting by V the largest open subset of H such that $G|_V$ is affine over V , it follows that V contains all the z_i ,⁽¹⁰⁾ hence contains at least one closed point $y_i \in H_s^i$ (and one has $\kappa(y_i) = \kappa(s)$, since $\kappa(s)$ is algebraically closed).

On the other hand, V is evidently stable under the translation defined by any element $g \in G(S)$. But one has $\dim(G_s) = \dim(H_s)$ and f_s is finite, so

$$f(s) : G_s^0(s) \rightarrow H_s^0(s)$$

⁽⁶⁾**N.D.E.** We have added the flatness hypothesis, which had been omitted.

⁽⁷⁾**N.D.E.** Cf. EGA IV₃, 8.10.5 (viii).

⁽⁸⁾**N.D.E.** Indeed, if $S = \text{Spec}(k)$ (k a field), then f is the composite of the projection $p : G \rightarrow G/\text{Ker}(f)$ and a closed immersion i , and since $\text{Ker}(f)$ is finite over k , p is finite (VIB 9.2).

⁽⁹⁾**N.D.E.** Indeed, the irreducible components $S_1, \dots, S_r$ of S are complete integral noetherian local schemes, hence, by a theorem of Nagata (cf. EGA 0_{IV}, 23.1.5), the normalization $\tilde{S}_i$ is finite over S_i , and so $\tilde{S}$ is finite over S . Then, by a theorem of Chevalley (cf. EGA II, 6.7.1), if $f \times_S \tilde{S}$ is an affine morphism, the same holds for f .

⁽¹⁰⁾**N.D.E.** The following phrase has been added.

is surjective. Since A is complete and G smooth over S , the canonical map $G^0(S) \rightarrow G_s^0(s)$ is surjective; since $H_s^0(s)$ acts transitively on each $H_s^i(s)$, it follows that $V \supset H_s(s)$, hence (since $\kappa(s)$ is algebraically closed) $V \supset H_s$.⁽¹¹⁾

Since one evidently has $V \supset H_\eta$, since f_η is finite, one therefore has $V = H$. Q.E.D.

⁽¹¹⁾**N.D.E.** When G is not supposed smooth over S , one can proceed as follows. Let h_0 be a rational point of H_s^0 , and g_0 a rational point of G_s^0 such that $f(g_0) = h_0$. By VIB 5.6.1, there exists a commutative diagram

$$\begin{array}{ccc} S'' & \xrightarrow{g} & G \\ \pi \downarrow & \searrow \phi & \downarrow \\ S' & \xrightarrow{w} & S \end{array}$$

where w is étale and surjective, π finite and surjective, and $\phi^{-1}(s)$ consists of a single point s'' such that $g(s'') = g_0$. Then the morphism $G_{S''} \rightarrow H_{S''}$ is affine above a neighborhood of $h_0 y_i$, and the same holds for $G_{S'} \rightarrow H_{S'}$ (EGA II, 6.7.1), then for $G \rightarrow H$ by faithfully flat descent (EGA IV₂, 2.7.1 (xiii)). Thus $h_0 y_i \in V$, and it follows that V contains $H_s(s)$ and hence H_s .

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⁽¹²⁾**N.D.E.** Additional references cited in this exposé.

Exposé XXVI. Parabolic subgroups of reductive groups

⁽⁰⁾ This exposé studies the parabolic subgroups of a reductive S -group G . Its essential result is the conjugacy theorem (5.4). The essential tool is the notion of transversal position of two parabolic subgroups, a notion studied systematically in no. 4. Another fact plays an important role: the decomposition of the unipotent radical $\mathrm{rad}^u(P)$ of a parabolic subgroup P into successive extensions of vector groups (2.1).⁽¹⁾

Various schemes associated with G are studied in no. 3; no. 6 deals with the split subtori⁽²⁾ of G and their relations with the parabolic subgroups.

Finally, in no. 7, we briefly explain how, over a semilocal base, one formulates the “relative theory” of reductive groups as it is set out in the case of fields in the article by A. Borel and J. Tits, *Groupes réductifs*, Publications Mathématiques de l’IHÉS, no. 27. In this article, cited as [BT65] below, the reader will moreover find, in the case of a base field, other results which have not been touched upon here.

1. Recollections. Levi subgroups

Definition 1.1. Let S be a scheme, G a reductive S -group, and P a group S -subscheme of G . One says that P is a *parabolic subgroup* of G if

- (i) P is smooth over S ,
- (ii) for each $s \in S$, the quotient $\bar{s}$ -scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper (i.e. Bible, § 6.4, th. 4 (= [Ch05], § 6.5, th. 5), $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$).

Proposition 1.2 (Exp. XXII, 5.8.5). *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G . Then P is closed in G , with connected fibers, and*

$$P = \underline{\mathrm{Norm}}_G(P).$$

Moreover, the quotient sheaf G/P is representable by an S -scheme smooth and projective over S .

Proposition 1.3 (Exp. XXII, 5.3.9 and 5.3.11). *Let S be a scheme, G a reductive S -group, and P and P' two parabolic subgroups of G . The following conditions are equivalent:*

- (i) P and P' are conjugate in G , locally for the étale (resp. (fpqc)) topology.
- (ii) For each $s \in S$, $P_{\bar{s}}$ and $P'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$.
- (iii) The strict transporter $\underline{\mathrm{Transt}}_G(P, P')$ of P into P' (defined by

$$\underline{\mathrm{Transt}}_G(P, P')(S') = \{g \in G(S') \mid \mathrm{int}(g)P_{S'} = P'_{S'}\}$$

for every $S' \rightarrow S$) is a closed subscheme of G , smooth and of finite presentation over S , which is a principal homogeneous bundle on the right under P , and on the left under P' .

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** Since the Levi subgroups of P form a torsor under $\mathrm{rad}^u(P)$ (1.9), this implies, when S is semilocal, that P possesses a Levi subgroup and hence a maximal torus (2.4).

⁽²⁾**N.D.E.** The terminology “trivial torus” has been replaced by “split torus”.

Proposition 1.4. *Let S be a nonempty scheme, (G, T, M, R) a split reductive S -group, and R' a subset of R . The following conditions on R' are equivalent:*

- (i) $\mathfrak{g}_{R'} = \mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha$ is the Lie algebra of a parabolic subgroup of G containing T (necessarily unique, Exp. XXII, 5.3.5).
- (ii) R' is of type (R) (Exp. XXII, 5.4.2) and contains a system of positive roots.
- (iii) R' is a closed subset of R and satisfies: if $\alpha \in R - R'$, then $-\alpha \in R'$ (that is, $R = R' \cup (-R')$).
- (iv) There exist a system of simple roots Δ and a subset A of Δ such that R' is the union of the set of positive roots and the set of negative roots which are linear combinations of the elements of A .
- (v) R' contains a system of simple roots of R ; moreover, if $\Delta \subset R'$ is a system of simple roots of R , and if one puts

$$A = (-R') \cap \Delta,$$

then R' is the union of the set of positive roots and the set of negative roots which are linear combinations of the elements of A .

One has (i) $\Leftrightarrow$ (ii) by Exp. XXII, 5.4.5 (ii) and 5.5.1. One has (iii) $\Rightarrow$ (ii) by Exp. XXI, 3.3.6 and Exp. XXII, 5.4.7. One evidently has (v) $\Rightarrow$ (iv) $\Rightarrow$ (iii). One has (iii) $\Rightarrow$ (v) by Exp. XXI, 3.3.6 and 3.3.10.

It therefore remains to prove that (i) implies that R' is a closed subset of R . Now this last assertion can be checked on any geometric fiber; one can therefore suppose that S is the spectrum of an algebraically closed field.

Let P be the parabolic subgroup of G containing T whose Lie algebra is $\mathfrak{g}_{R'}$. Since the Borel subgroups of P are the Borel subgroups of G contained in P , it follows from Bible, § 12.3, th. 1, and from Exp. XXII, 5.4.5 (i) that, if one denotes by U the unipotent radical of P , then $T \cdot U$ is the subgroup of G containing T with Lie algebra

$$\mathfrak{g}_{R''} = \mathfrak{t} \oplus \coprod_{\alpha \in R''} \mathfrak{g}^\alpha,$$

where R'' is the intersection of the systems of positive roots of R contained in R' .

It follows in particular that R'' is closed and that $R'' \cap (-R'') = \emptyset$. On the other hand, the group $H = P/U$ is reductive, the canonical image $\overline{T}$ of T is a maximal torus of it ($T \rightarrow \overline{T}$ is an isomorphism), and one has an isomorphism of T -modules, i.e. of vector spaces graded of type M ,

$$\text{Lie}(H) \simeq \mathfrak{t} \oplus \coprod_{\alpha \in R_s} \mathfrak{g}^\alpha,$$

where R_s is the complement of R'' in R' . It follows that R_s is naturally identified with the set of roots of H relative to $\overline{T}$, and in particular satisfies $R_s = -R_s$. It follows immediately that one has

$$R'' = \{\alpha \in R' \mid -\alpha \notin R'\}, \quad R_s = \{\alpha \in R' \mid -\alpha \in R'\}.$$

Let us now show that R' is closed. Let $\alpha, \beta \in R'$ be such that $\alpha + \beta \in R$; let us prove that $\alpha + \beta \in R'$. If $\alpha, \beta \in R''$, then $\alpha + \beta \in R''$, since R'' is closed. If $\alpha \in R_s$, $\beta \in R''$, and if $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$, and one has $-\alpha = -(\alpha + \beta) + \beta \in R''$, since R'' is closed, which implies $-\alpha \in R'' \cap R_s$ and contradicts the fact that $R_s \cap R'' = \emptyset$. It therefore remains to study the case where $\alpha, \beta \in R_s$. If $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$. But since $\alpha + \beta \neq 0$, there exists a system of positive roots of the root system R_s containing α and β , hence a Borel subgroup of $H = P/U$ containing the canonical images of U_α and U_β . Its inverse image in P is a Borel subgroup containing U_α , U_β and U , hence U_α , U_β and $U_{-(\alpha+\beta)}$, which is impossible.

Corollary 1.5. *A parabolic subgroup of a reductive group is of type (RC) (Exp. XXII, 5.11.1).*

Proposition 1.6 (Exp. XXII, 5.11.4). *Let S be a scheme, G a reductive S -group, and P a subgroup of type $(RC)^{(3)}$ of G .*

- (i) *P possesses a largest invariant group subscheme, smooth and of finite presentation over S , with connected unipotent geometric fibers. This is a characteristic subgroup of P , called the unipotent radical of P , denoted $\text{rad}^u(P)$. The quotient sheaf $P/\text{rad}^u(P)$ is representable by a reductive S -group.*
- (ii) *If T is a maximal torus of P , P possesses a reductive subgroup L containing T such that:*
 - (a) *Every reductive subgroup of P containing T is contained in L .*
 - (b) *P is the semidirect product $L \cdot \text{rad}^u(P)$, i.e. the canonical morphism $L \rightarrow P/\text{rad}^u(P)$ is an isomorphism.*

Moreover, L is the unique subgroup (resp. reductive subgroup) of P containing T and satisfying (b) (resp. (a)). Finally, one has

$$\underline{\text{Norm}}_P(L) = L, \quad \underline{\text{Norm}}_P(T) = \underline{\text{Norm}}_L(T).$$

1.7. — A subgroup L of P satisfying condition (b) above is called a *Levi subgroup* of P . It is a maximal reductive subgroup of P ; indeed, it is reductive, since it is isomorphic to $P/\text{rad}^u(P)$; let us show that it is maximal for this property. Let L' be a reductive subgroup of P containing L ; to prove that $L' = L$, one can reason locally for the (fpqc) topology, and hence suppose that L possesses a maximal torus T , and one is reduced to 1.6 (ii).

If L and L' are two Levi subgroups of P , L and L' are conjugate in P , locally for the (fpqc) topology. Indeed, locally for this topology, one can suppose that L (resp. L') possesses a maximal torus T (resp. T'); since T and T' are conjugate in P locally for the (fpqc) topology, one can suppose $T = T'$, and one then has $L = L'$ by 1.6 (ii). But since $P = L \cdot \text{rad}^u(P)$ and $\underline{\text{Norm}}_P(L) = L$, one immediately deduces:

Corollary 1.8. *Let P be a subgroup of type $(RC)^{(4)}$ of the reductive S -group G . If L and L' are two Levi subgroups of P , there exists a unique $u \in \text{rad}^u(P)(S)$ such that $\text{int}(u)L = L'$.*

Denote by $\underline{\text{Lev}}(P)$ the functor of Levi subgroups of P : for $S' \rightarrow S$, $\underline{\text{Lev}}(P)(S')$ is the set of Levi subgroups of $P_{S'}$. One deduces from 1.8:

Corollary 1.9. *Let P be a subgroup of type $(RC)^{(4)}$ of the reductive S -group G . Then $\underline{\text{Lev}}(P)$ is a principal homogeneous bundle under the S -group $\text{rad}^u(P)$, and in particular is representable by an S -scheme smooth and affine over S , with integral geometric fibers.*

It follows immediately from 1.6:

Corollary 1.10. *Let P be a subgroup of type $(RC)^{(4)}$ of the reductive S -group G . The functor $\underline{\text{Tor}}(P)$ of maximal tori of P is representable by a smooth affine S -scheme; the “relation $L \supset T$ ” defines a morphism*

$$\underline{\text{Tor}}(P) \rightarrow \underline{\text{Lev}}(P),$$

the fiber of this morphism above $L \in \underline{\text{Lev}}(P)(S)$ is identified with $\underline{\text{Tor}}(L)$ (Exp. XXII, 5.8.3).

The first assertion of 1.10 is a consequence of the other two and of Exp. XXII, 5.8.3.

Definition 1.11. Let S be a nonempty scheme, G a reductive S -group, P a parabolic subgroup of G , and $\mathcal{E} = (T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ a pinning of G . One says that $\mathcal{E}$ is *adapted to P* , or that $\mathcal{E}$ is a *pinning of the pair (G, P)* , if $P \supset T$ and if the Lie algebra of P is of the form $\mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha$, where R' is a subset of R containing R_+ .

⁽³⁾**N.D.E.** “Parabolic subgroup” has been replaced by “subgroup of type (RC)”, as in Exp. XXII, 5.11.4, since it will be useful later (4.5.1, 6.17) to be able to apply this statement to $P \cap P'$, when P, P' are two parabolic subgroups such that $P \cap P'$ is of type (RC).

⁽⁴⁾**N.D.E.** Cf. N.D.E. (3).

In particular, if $T \subset B$ is the Killing pair defined by the pinning, one has $T \subset B \subset P$.

Under the preceding conditions, one puts $\Delta(P) = \Delta \cap (-R')$; then, by Exp. XXII, 5.4.3, one has:

$$\begin{aligned} \alpha \in \Delta(P) &\iff \alpha \in \Delta \text{ and } U_{-\alpha} \subset P \\ &\iff \alpha \in \Delta \text{ and } U_{-\alpha} \cap P \neq e. \end{aligned}$$

It follows immediately from 1.4 (v) and Exp. XXII, 5.11.3 and 5.10.6:

Proposition 1.12. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , $(T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ a pinning of G adapted to P , and $\Delta(P)$ the subset of Δ defined above.*

(i) *The unipotent radical $\text{rad}^u(P)$ of P is precisely*

$$U_{R''} = \prod_{\alpha \in R''} U_\alpha,$$

where R'' is the set of positive roots which, in their decomposition on Δ , contain at least one element of $\Delta - \Delta(P)$ ⁽⁵⁾ with nonzero coefficient.

(ii) *The unique Levi subgroup L of P containing T is precisely*

$$Z_{\Delta(P)} = \underline{\text{Centr}}_G(T_{\Delta(P)}),$$

where $T_{\Delta(P)}$ is the maximal torus of $\bigcap_{\alpha \in \Delta(P)} \text{Ker}(\alpha)$; moreover one has $T_{\Delta(P)} = \text{rad}(L)$.

Corollary 1.13. *Every Levi subgroup L of the parabolic subgroup P of the reductive group G is a critical subgroup of G , i.e. satisfies (Exp. XXII, 5.10.4):*

$$L = \underline{\text{Centr}}_G(\text{rad}(L)).$$

This follows immediately from 1.12 and the following lemma, contained in 1.4 and Exp. XXII, 5.4.1:

Lemma 1.14. *Locally for the étale topology, every pair (G, P) , where P is a parabolic subgroup of the reductive group G , can be pinned (1.11).*

Let us note:

Proposition 1.15. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and $\mathcal{E}$ and $\mathcal{E}'$ two pinnings of G adapted to P . The unique inner automorphism of G over S which transforms $\mathcal{E}$ into $\mathcal{E}'$ (Exp. XXIV, 1.5) comes from P , by the morphism*

$$P \longrightarrow P/\underline{\text{Centr}}(P) = P/\underline{\text{Centr}}(G) \longrightarrow G/\underline{\text{Centr}}(G).$$

Indeed, it suffices to reason as in Exp. XXIV, 1.5, using:

Lemma 1.16 (Exp. XXII, 5.3.14 and 5.2.6). *The maximal tori (resp. Borel subgroups, resp. Killing pairs) of a parabolic subgroup P of the reductive S -group G are conjugate in P , locally for the étale topology.*

Proposition 1.17. *Let S be a scheme, G a reductive S -group, P and P' two parabolic subgroups of G , and B a Borel subgroup contained in P and P' . If P and P' are conjugate in G , locally for the étale topology, then $P = P'$.*

Indeed, one can suppose that there exists $g \in G(S)$ such that $\text{int}(g)P = P'$. Then B and $\text{int}(g)^{-1}B$ are two Borel subgroups of P . Extending S , one can, by 1.16, suppose that there exists $p \in P(S)$ such that $\text{int}(p)\text{int}(g^{-1})B = B$. Then

$$pg^{-1} \in \underline{\text{Norm}}_G(B)(S) = B(S),$$

and $g \in B(S) \cdot p \subset P(S)$, hence $P' = \text{int}(g)P = P$.

⁽⁵⁾**N.D.E.** $\Delta(P)$ has been corrected to $\Delta - \Delta(P)$.

Remark 1.18. If P and P' are two parabolic subgroups of G containing the same Borel subgroup, then $P \cap P'$ is again a parabolic subgroup of G . Indeed, it is smooth along the identity section (Exp. XXII, 5.4.5), and it contains a Borel subgroup.

Proposition 1.19. *Let S be a scheme, G a reductive S -group, and G' its derived group (Exp. XXII, 6.2.1).*

(i) *The maps*

$$P \mapsto P' = P \cap G' \quad \text{and} \quad P \mapsto P' \cdot \text{rad}(G) = \underline{\text{Norm}}_G(P')$$

are mutually inverse bijections between the set of parabolic subgroups of G and the set of parabolic subgroups of G' . One has $\text{rad}^u(P) = \text{rad}^u(P')$.

(ii) *Let P be a parabolic subgroup of G , and $P' = P \cap G'$. The maps*

$$L \mapsto L' = L \cap G' = L \cap P' \\ L' \mapsto L' \cdot \text{rad}(G) = \underline{\text{Centr}}_G(\text{rad}(L'))$$

are mutually inverse bijections between the set of Levi subgroups of P and the set of Levi subgroups of P' . Moreover, one has $\text{rad}(L') = (\text{rad}(L) \cap G')^0$.

The proof (by reduction to the split case, for example) presents no difficulty and is left to the reader, as is the immediate proof of:

Proposition 1.20. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and L a Levi subgroup of P . The maps*

$$Q \mapsto Q \cap L = Q', \quad Q' \mapsto Q' \cdot \text{rad}^u(P)$$

are mutually inverse bijections between the set of parabolic subgroups of G contained in P and the set of parabolic subgroups of L . Moreover, the Levi subgroups of Q' are the Levi subgroups of Q contained in L .

One can supplement 1.6 as follows:

Proposition 1.21. *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G .*

- (i) *P possesses a largest invariant subgroup, smooth and of finite presentation over S , with connected solvable geometric fibers. This is a characteristic subgroup of P , called the radical of P , and denoted $\text{rad}(P)$. The quotient sheaf $P/\text{rad}(P)$ is representable by a semisimple S -group.*
- (ii) *If L is a Levi subgroup of P , $\text{rad}(P)$ is the semidirect product of $\text{rad}^u(P)$ and $\text{rad}(L)$; one has $\text{rad}(L) = L \cap \text{rad}(P)$, hence $L = \underline{\text{Centr}}_G(L \cap \text{rad}(P))$, and $P/\text{rad}(P) \simeq L/\text{rad}(L)$.*

Indeed, since assertion (i) is local, one can suppose that G possesses a Levi subgroup L , and one is reduced to proving that $R = \text{rad}^u(P) \cdot \text{rad}(L)$ possesses the properties stated in (i), which is immediate. For (ii), it remains only to prove that $\text{rad}(L) = L \cap \text{rad}(P)$, which follows immediately from the fact that $L \cap \text{rad}(P)$ is smooth with connected fibers, L being the centralizer of a torus.⁽⁶⁾

2. Structure of the unipotent radical of a parabolic subgroup

Proposition 2.1. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and $\text{rad}^u(P)$ its unipotent radical. There exists a sequence of group subschemes of $\text{rad}^u(P)$*

$$\text{rad}^u(P) = U_0 \supset U_1 \supset U_2 \supset \cdots \supset U_n \supset \cdots$$

possessing the following properties:

⁽⁶⁾**N.D.E.** Cf. XIX 1.4.

- (i) Each U_i is smooth, with connected fibers, characteristic and closed in P . The commutator of a section of U_i and a section of U_j is a section of U_{i+j+1} (over a variable $S' \rightarrow S$).
- (ii) For each $i \geq 0$, there exist a locally free $\mathcal{O}_S$ -module $\mathcal{E}_i$ and an isomorphism of S -sheaves of groups

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i).$$

Moreover, the automorphisms of P (over a variable $S' \rightarrow S$) act linearly on $W(\mathcal{E}_i)$.

- (iii) For every $s \in S$, one has $U_{n,s} = e$ for $n > \dim(\text{rad}^u(P)_s)$.

2.1.1. — First suppose that the pair (G, P) is pinnable. Let $(T, M, R, \Delta, \dots)$ be a pinning of G adapted to P ; let $\Delta(P)$ be the subset of Δ defined by P . Let $\alpha_1, \dots, \alpha_p$ be the elements of $\Delta(P)$, and $\beta_1, \dots, \beta_q$ the elements of $\Delta - \Delta(P)$. Every root $\gamma \in R$ is written uniquely as

$$\gamma = a_1\alpha_1 + \dots + a_p\alpha_p + b_1\beta_1 + \dots + b_q\beta_q.$$

Put

$$a(\gamma) = b_1 + \dots + b_q. \quad (7)$$

The following properties follow immediately from the definitions (cf. 1.12):

- (i) $U_\gamma \subset P \Leftrightarrow a(\gamma) \geq 0$.
- (ii) $U_\gamma \subset \text{rad}^u(P) \Leftrightarrow a(\gamma) > 0$.
- (iii) $a(n\gamma + m\gamma') = na(\gamma) + ma(\gamma')$ for all $n, m \in \mathbf{Z}$.

For $i > 0$, let R_i be the set of roots $\gamma \in R$ such that $a(\gamma) > i$. Each R_i is a closed set of roots satisfying $R_i \cap (-R_i) = \emptyset$. Consider (Exp. XXII, 5.6.5) the S -group

$$U_i = U_{R_i} = \prod_{\gamma \in R_i} U_\gamma.$$

This is a closed group subscheme of G , smooth over S , with connected fibers.

Let $\alpha, \beta \in R$; consider the commutation relation of Exp. XXII, 5.5.2

$$p_\alpha(x)p_\beta(y)p_\alpha(-x) = p_\beta(y) \prod_{n,m \in \mathbf{N}^*} p_{n\alpha+m\beta}(C_{n,m,\alpha,\beta}x^n y^m),$$

where each p_γ is an isomorphism of vector groups $\mathbf{G}_{a,S} \xrightarrow{\sim} U_\gamma$. First note that, if $a(\alpha) > i$ and $a(\beta) > j$, one has

$$a(n\alpha + m\beta) = na(\alpha) + ma(\beta) \geq n(i+1) + m(j+1) > i+j+1$$

when n and m are > 0 . It follows that the commutator of a section of U_α and a section of U_β is a section of U_{i+j+1} (over a variable $S' \rightarrow S$), which indeed implies $(U_i, U_j) \subset U_{i+j+1}$. For each $i \geq 0$, the quotient U_i/U_{i+1} is therefore commutative; it is naturally identified with

$$U_i/U_{i+1} \simeq \prod_{a(\gamma)=i+1} U_\gamma \simeq W(\mathcal{E}_i),$$

where $\mathcal{E}_i$ is the direct sum of the $\mathfrak{g}^\gamma$ for $a(\gamma) = i+1$.

Let us return to the commutation formula above, and suppose $a(\alpha) \geq 0$, $a(\beta) > i$. If $n, m \in \mathbf{N}^*$,

- either $a(n\alpha + m\beta) > i+1$,
- or $a(n\alpha + m\beta) = i+1$, in which case one necessarily has $m = 1$.

This first proves that $\text{int}(p_\alpha(x))$ preserves U_i (hence also U_{i+1}), then that the expression $\text{int}(p_\alpha(x))p_\beta(y)$ involves modulo U_{i+1} only terms of the form $p_{n\alpha+\beta}(C_{n,1,\alpha,\beta}x^n y)$, which are therefore *linear* in y . It follows that the inner automorphisms defined by sections of U_α act linearly on the quotient U_i/U_{i+1} identified with $W(\mathcal{E}_i)$. Since this is also trivially true for the inner automorphisms defined by sections of T , and since P is generated by T and the U_α , $a(\alpha) \geq 0$, one deduces that:

- (i) each U_i is invariant in P ,

(7) **N.D.E.** $a_1 + \dots + a_p$ has been corrected to $b_1 + \dots + b_q$.

(ii) the inner automorphisms defined by sections of P act linearly on $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$.

2.1.2. — Now let $(T', M', R', \Delta', \dots)$ be a new pinning of G adapted to P . By 1.15, there exists an inner automorphism of G coming from P which transforms the old pinning into the new one. Extending S , one can suppose that this inner automorphism is of the form $\text{int}(p)$, $p \in P(S)$.

If one repeats the preceding constructions using the new pinning, it is clear that the groups U'_i and the isomorphisms $U'_i/U'_{i+1} \simeq W(\mathcal{E}'_i)$ obtained are deduced from U_i and $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$ by transport of structure by means of $\text{int}(p)$. It follows from remarks (i) and (ii) above that one will therefore have $U'_i = U_i$, and that the two vector structures constructed on $U_i/U_{i+1} = U'_i/U'_{i+1}$ coincide.

This shows us that the groups U_i and the vector structures on the quotients U_i/U_{i+1} are independent of the pinning considered (and in particular invariant under every automorphism of P , as one easily sees).

We have therefore proved the proposition when the pair (G, P) is pinnable (part (iii) is trivial, since by Exp. XXI 3.1.2, the set $\{a(\gamma) \mid \gamma \in R\}$ is an interval of $\mathbf{Z}$, so one can have $\dim(U_{i,s}) = \dim(U_{i+1,s})$ only if $U_{i,s} = e$).

2.1.3. — In the general case, there exists a covering family for the (fpqc) topology, $\{S_j \rightarrow S\}$, such that each pair (G_{S_j}, P_{S_j}) is pinnable (1.14). By the foregoing, one has descent data on the $\text{rad}^u(P_{S_j})_i$, compatible with the vector structures of the quotients, and one concludes by descent of closed subschemes (resp. locally free modules).⁽⁸⁾

Corollary 2.2. *Let S be an affine scheme, G a reductive S -group, and P a parabolic subgroup of G . One has*

$$H^1(S, \text{rad}^u(P)) = 0,$$

i.e. every principal homogeneous bundle under $\text{rad}^u(P)$ is trivial.

Indeed, S decomposes into a sum of subschemes on each of which $\text{rad}^u(P)$ has constant relative dimension. One can therefore, by (iii), suppose that there exists an n such that $U_n = e$. Since $H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0$ by TDTE I, B, 1.1 (or SGA 1, XI 5.1), one concludes immediately.

Corollary 2.3. *Under the preceding conditions, P possesses a Levi subgroup L . If L is a Levi subgroup of P , the canonical map*

$$H^1(S, L) \rightarrow H^1(S, P)$$

is bijective (cf. the introduction to Exp. XXIV for the definition of $H^1(S, \cdot)$).

The first assertion follows from 2.2 and 1.9. The canonical map $H^1(S, L) \rightarrow H^1(S, P)$ is surjective, since P is the semidirect product $L \cdot \text{rad}^u(P)$. To prove that it is injective, it suffices to see that, for every principal homogeneous bundle Q under L , one has $H^1(S, \text{rad}^u(P)_Q) = 0$, where the subscript Q denotes the twisting operation by the L -bundle Q . This can be proved in two ways: one can repeat the proof of 2.2, using the fact that the vector structures on the U_i/U_{i+1} are invariant under L ; one can also note that $\text{rad}^u(P)_Q$ is identified with the unipotent radical of the parabolic subgroup P_Q of G_Q , and apply 2.2 to P_Q .

Corollary 2.4. *Let S be a semilocal scheme, G a reductive S -group, and P a parabolic subgroup of G . There exists a maximal torus T of G contained in P .*

Indeed, by 2.3, P possesses a Levi subgroup L , and it suffices to prove that L possesses a maximal torus, which follows from Exp. XIV, 3.20.

⁽⁸⁾**N.D.E.** Cf. SGA 1, VIII 1.3, 1.9 and 1.10.

Corollary 2.5. *Let S be an affine scheme, G a reductive S -group, and P a parabolic subgroup of G . There exists a locally free $\mathcal{O}_S$ -module $\mathcal{E}$ such that $\mathrm{rad}^u(P)$ is isomorphic as an S -scheme to $W(\mathcal{E})$.*

Indeed, let us prove by induction on i that one has an isomorphism of S -schemes

$$\mathrm{rad}^u(P)/U_i \simeq W(\mathcal{E}_0 \oplus \mathcal{E}_1 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

This is clear for $i = 0$. Suppose $i > 0$; then $\mathrm{rad}^u(P)/U_i$ is a principal homogeneous bundle with base $\mathrm{rad}^u(P)/U_{i-1}$, under the group

$$(U_{i-1}/U_i)_{\mathrm{rad}^u(P)/U_{i-1}} \simeq W(\mathcal{E}_{i-1} \otimes \mathcal{O}_{\mathrm{rad}^u(P)/U_{i-1}}).$$

Now the base is affine (for example by the induction hypothesis), so this bundle is trivial (TDTE I or SGA 1 XI, *loc. cit.*), and there exists an isomorphism of S -schemes $\mathrm{rad}^u(P)/U_i \simeq (\mathrm{rad}^u(P)/U_{i-1}) \times_S (U_{i-1}/U_i)$, which completes the proof.

Corollary 2.6. *Let S be a semilocal scheme, $\{s_i\}$ the set of its closed points, G a reductive S -group, and P a parabolic subgroup of G . The canonical map*

$$\mathrm{rad}^u(P)(S) \longrightarrow \prod_i \mathrm{rad}^u(P)(\mathrm{Spec} \kappa(s_i))$$

is surjective.

Indeed, if $S = \mathrm{Spec}(A)$, $\kappa(s_i) = A/\mathfrak{p}_i$, and if $\mathcal{E}$ is given by the projective (hence flat) module E , we must prove that the map

$$E \longrightarrow \prod_i E \otimes_A A/\mathfrak{p}_i$$

is surjective. It suffices to do so when $E = A$, in which case this is well known (cf. Bourbaki, *Alg. Comm.* Chap. II, § 1, no. 2, proposition 5).

Corollary 2.7. *Let k be an infinite field, G a reductive k -group, and P a parabolic subgroup of G ; then $\mathrm{rad}^u(P)(k)$ is dense in $\mathrm{rad}^u(P)$.*

Corollary 2.8. *Let S be a semilocal scheme, $\{s_i\}$ the set of its closed points, G a reductive S -group, P a parabolic subgroup of G , and L_i a Levi subgroup of P_{s_i} for each i . There exists a Levi subgroup L of P inducing L_i for each i .*

Indeed, let L_0 be a Levi subgroup of P (2.3). For each i , let $u_i \in \mathrm{rad}^u(P)(\mathrm{Spec}(\kappa(s_i)))$ be such that $\mathrm{int}(u_i)L_{0,s_i} = L_i$ (1.8); if $u \in \mathrm{rad}^u(P)(S)$ induces u_i for each i (2.6), then $L = \mathrm{int}(u)L_0$ has the required property.

Corollary 2.9. *In the situation of 2.1, let moreover H be a group subscheme of G , smooth and of finite presentation over S , with connected fibers, such that $P \cap H$ locally contains a maximal torus of G for the (fpqc) topology. Then, for each $i \geq 0$, there exists a submodule $\mathcal{F}_i$ of $\mathcal{E}_i$, locally a direct summand, such that the isomorphism $U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i)$ induces an isomorphism of groups*

$$(U_i \cap H)/(U_{i+1} \cap H) \xrightarrow{\sim} W(\mathcal{F}_i).$$

Indeed, H is a subgroup of type (R) of G (Exp. XXII, 5.2.1). Moreover, the assertion to be proved is local for the (fpqc) topology, and one can suppose G split relative to a maximal torus of $P \cap H$; one can even reduce to the situation of 2.1.1, H being defined by a subset R' of R . Taking up the notations of *loc. cit.*, one sees by Exp. XXII, 5.6.7 (ii) that $U_i \cap H = \prod_{\alpha \in R_i \cap R'} U_\alpha$, hence that $(U_i \cap H)/(U_{i+1} \cap H)$ is identified with $\prod_{\alpha \in R', a(\alpha)=i+1} U_\alpha$, which implies the result.

Corollary 2.10. *In the situation of 2.9, the conclusions of 2.2, 2.5, 2.6, 2.7 are also valid on replacing $\mathrm{rad}^u(P)$ by $\mathrm{rad}^u(P) \cap H$.*

Corollary 2.11. ⁽⁹⁾ *Let G be a reductive S -group, P a parabolic subgroup, and H a subgroup of type (RC) of P such that $\text{rad}^u(H) = \text{rad}^u(P) \cap H$. Then statements 2.2 to 2.8 are also valid on replacing P by H .*

3. Scheme of parabolic subgroups of a reductive group

3.1. — Let E be a finite twisted constant S -scheme (Exp. X, 5.1). Consider the S -functor $\underline{\text{Of}}(E)$, where $\underline{\text{Of}}(E)(S')$ is the set of open and closed subschemes of $E_{S'}$ (or, equivalently, the set of open and closed subsets of $E_{S'}$); then $\underline{\text{Of}}(E)$ is representable by a finite twisted constant S -scheme. Indeed, if $E = A_S$, where A is a finite set, one immediately has $\underline{\text{Of}}(E) \simeq \mathcal{P}(A)_S$ (where $\mathcal{P}(A)$ denotes the set of subsets of A), and one concludes by descent of open and closed subschemes. One evidently has:

$$\underline{\text{Of}}(E_{S'}) = \underline{\text{Of}}(E)_{S'}, \quad \underline{\text{Of}}(E \times_S E') = \underline{\text{Of}}(E) \times_S \underline{\text{Of}}(E').$$

3.2. — Let S be a scheme and G a reductive S -group. The functor $\underline{\text{Par}}(G)$ of parabolic subgroups of G is defined by

$$\underline{\text{Par}}(G)(S') = \text{set of parabolic subgroups of } G_{S'}.$$

In particular $G \in \underline{\text{Par}}(G)(S)$, $\underline{\text{Bor}}(G) \subset \underline{\text{Par}}(G)$. We propose to define a morphism

$$t : \underline{\text{Par}}(G) \longrightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G))$$

possessing the following properties:

- (i) t is functorial in G (with respect to isomorphisms) and commutes with extension of the base.
- (ii) If $(T, M, R, \Delta, \dots)$ is a pinning of G adapted to the parabolic subgroup P (1.11), the canonical isomorphism $\underline{\text{Dyn}}(G) \simeq \Delta_S$ (Exp. XXIV, 3.4 (iii)) transforms $t(P)$ into $\Delta(P)_S$ (notations of 1.11, 1.12).

First let P be a parabolic subgroup of G , and $(T, M, R, \Delta, \dots)$ a pinning of G adapted to P . One defines $t(P)$ by (ii); the subscheme $t(P)$ of $\underline{\text{Dyn}}(G)$ thus constructed is independent of the pinning chosen. Indeed, if $(T', M', R', \Delta', \dots)$ is another pinning of G adapted to P , the unique inner automorphism of G transforming the first pinning into the second comes from P (1.15); the canonical isomorphism $\Delta \xrightarrow{\sim} \Delta'$ therefore transforms $\Delta(P)$ into $\Delta'(P)$, which implies the stated result.

If one now no longer necessarily supposes (G, P) pinnable, it follows immediately from 1.14 and from the definition of $\underline{\text{Dyn}}(G)$ (Exp. XXIV, 3.3) that one can define by descent a unique open and closed subscheme $t(P)$ of $\underline{\text{Dyn}}(G)$ such that, for every $S' \rightarrow S$ such that $(G, P)_{S'}$ is pinnable, one has $t(P)_{S'} = t(P_{S'})$.

Theorem 3.3. *Let S be a scheme, G a reductive S -group, and*

$$t : \underline{\text{Par}}(G) \longrightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G))$$

the morphism defined above.

- (i) *In order that two parabolic subgroups P and P' of G be locally conjugate for the (fpqc) topology (cf. 1.3), it is necessary and sufficient that $t(P) = t(P')$.*
- (ii) *$\underline{\text{Par}}(G)$ is representable, and the morphism t is smooth, projective, with integral geometric fibers.*

By 3.2 (i), and the fact that the inner automorphisms of G act trivially on $\underline{\text{Dyn}}(G)$ (Exp. XXIV, 3.4 (iv)), one indeed has $t(P) = t(P')$ when P and P' are conjugate. Conversely, let P and P' be two parabolic subgroups of G such that $t(P) = t(P')$; let us prove that P and P' are conjugate in G , locally for the (fpqc) topology. One can first suppose the pairs (G, P) and (G, P') pinnable (1.14); by conjugacy of pinnings in G (Exp. XXIV, 1.5),

⁽⁹⁾**N.D.E.** Corollary 2.11, which will be used in 4.5.1 and 6.17, has been added.

one can suppose that there exists a pinning (T, M, R, Δ) of G adapted to P and P' . Then $t(P) = t(P')$ implies $\Delta(P) = \Delta(P')$, hence $P = P'$ (cf. 1.4 (v)). We have therefore proved (i). To prove (ii), let us take up the notations of Exp. XXII, 5.11.5.⁽¹⁰⁾

One has a canonical morphism $\underline{\text{Par}}(G) \rightarrow \mathcal{H}_c$, and it is clear (for example by reduction to the pinned case) that it fits into a cartesian square (where the vertical arrows are monomorphisms)

$$\begin{array}{ccc} \underline{\text{Par}}(G) & \xrightarrow{t} & \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \\ \downarrow & & \downarrow \\ \mathcal{H}_c & \xrightarrow{cl} & Cl_c. \end{array}$$

Now (*loc. cit.*) $\mathcal{H}_c$ is representable, and the morphism cl is smooth, quasi-projective, of finite presentation, with integral geometric fibers, so the same holds for t .

It remains to prove that t is proper; but this is now an assertion local for the (fpqc) topology, and one can reduce to the pinned case $G = (G, T, M, R, \Delta, \dots)$. One then has $\underline{\text{Dyn}}(G) \simeq \Delta_S$, and it suffices to prove that, for every subset Δ_1 of Δ , the S -scheme $t^{-1}((\Delta_1)_S)$ is proper over S . Now, if P_1 is the parabolic subgroup of G containing T such that $\Delta(P_1) = \Delta_1$, it follows from (i) that the morphism $G \rightarrow \underline{\text{Par}}(G)$ defined set-theoretically by $g \mapsto \text{int}(g)P_1$ induces an isomorphism from $G/\underline{\text{Norm}}_G(P_1)$ onto $t^{-1}((\Delta_1)_S)$. Now, by 1.2, $G/\underline{\text{Norm}}_G(P_1) = G/P_1$ is projective over S .

Definition 3.4. $\underline{\text{Of}}(\underline{\text{Dyn}}(G))$ is called the *scheme of parabolic types* of G ; $t(P)$ is called the *type* of P .

Corollary 3.5. The S -functor $\underline{\text{Par}}(G)$ is representable by an S -scheme smooth and projective over S . The decomposition

$$\underline{\text{Par}}(G) \rightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \rightarrow S$$

is the Stein factorization (EGA III, 4.3.3) of the structural morphism $\underline{\text{Par}}(G) \rightarrow S$.

Corollary 3.6. For each $t \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$, the S -scheme

$$\underline{\text{Par}}_t(G) = t^{-1}(t)$$

of parabolic subgroups of G of type t is smooth and projective over S , homogeneous under G . If P is a parabolic subgroup of G , one has a canonical isomorphism $G/P \xrightarrow{\sim} \underline{\text{Par}}_{t(P)}(G)$. One has $\underline{\text{Par}}_{\emptyset}(G) = \underline{\text{Bor}}(G)$, $\underline{\text{Par}}_{\underline{\text{Dyn}}(G)}(G) \xrightarrow{\sim} S$.

Remark 3.7. The S -scheme $\underline{\text{Of}}(\underline{\text{Dyn}}(G))$ is equipped with a natural order structure (domination relation, here set-theoretic inclusion, between subschemes). This order structure is a lattice; in particular the infimum of two open and closed subschemes of $\underline{\text{Dyn}}(G)_S$ is evidently their intersection. Note moreover that, if B is a Borel subgroup of G , one can define the functor X of parabolic subgroups of G containing B . The morphism $X \rightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G))$ induced by t is an isomorphism (for the “ordered scheme” structure), by the assertion $P \subset Q \Rightarrow t(P) \subset t(Q)$ and by:

Lemma 3.8. Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and t' a section of $\underline{\text{Of}}(\underline{\text{Dyn}}(G))$ over S such that $t(P) \subset t'$. There exists a unique parabolic subgroup P' of G containing P and such that $t(P') = t'$.

⁽¹⁰⁾**N.D.E.** Recall (cf. *loc. cit.*) that $\mathcal{H}_c = \mathcal{H}_c(G)$ denotes the functor of subgroups of G of type (RC), $Cl_c = Cl_c(G)$ the functor of “conjugacy classes” of such subgroups, and that $cl : \mathcal{H}_c \rightarrow Cl_c$ is the canonical projection.

Extending the base, one can suppose that P contains a Borel subgroup B of G . The uniqueness of P' then follows from 1.17. To prove existence, one can place oneself in the split case, in which case the assertion is evident, cf. no. 1.

Remarks 3.8.1.

- (i) The assertion analogous to 3.8 obtained by reversing the inclusions is evidently false. It would imply, for example, that every group of type A_1 possesses a Borel subgroup, which is not so, cf. Exp. XX, no. 5.
- (ii) It follows immediately from the foregoing that $t(P) \subset t(Q)$ means that, locally for (fpqc) or (ét), P is conjugate to a subgroup of Q (it moreover suffices to verify the assertion on the geometric fibers). Furthermore, we shall see in no. 5 that one can replace the étale topology by the Zariski topology.

3.9. — The preceding discussions can be repeated in the case of critical subgroups. Recall (Exp. XXII, 5.10.4 and 5.10.5) that a reductive subgroup H of the reductive group G is *critical* if $H = \underline{\text{Centr}}_G(\text{rad}(H))$, that a subtorus Q of G is a *C-critical torus* if $Q = \text{rad}(\underline{\text{Centr}}_G(Q))$, and that critical subgroups and C-critical tori⁽¹¹⁾ are in one-to-one correspondence (by $H \mapsto \text{rad}(H)$ and $Q \mapsto \underline{\text{Centr}}_G(Q)$).

If (G, T, M, R) is a split S -group, the subgroup of G containing T corresponding to the subset R' of R (Exp. XXII, 5.4.2) is critical if and only if R' is “vectorial” (that is, the intersection of R with a vector subspace of $M \otimes \mathbf{Q}$), cf. Exp. XXII, 5.10.6.

If G is any reductive S -group, one will define, as in Exp. XXII, 5.11.5, a finite étale S -scheme $C\ell_{\text{crit}}$, which in the split case will be the constant scheme associated with the set of vectorial subsets of R modulo the action of the Weyl group. If $\underline{\text{Crit}}(G)$ denotes the “functor of critical subgroups” of G , one will have a canonical morphism $\underline{\text{Crit}}(G) \xrightarrow{cl} C\ell_{\text{crit}}$, which will fit into a cartesian diagram⁽¹²⁾

$$\begin{array}{ccc} \underline{\text{Crit}}(G) & \xrightarrow{cl} & C\ell_{\text{crit}} \\ \downarrow & & \downarrow \\ \mathcal{H}_c & \xrightarrow{cl} & C\ell_c. \end{array}$$

Proposition 3.10. *Let S be a scheme, G a reductive S -group, $\underline{\text{Crit}}(G)$ the functor of its critical subgroups, and $C\ell_{\text{crit}}$ and $cl : \underline{\text{Crit}}(G) \rightarrow C\ell_{\text{crit}}$ the finite étale S -scheme and the morphism defined above.*

- (i) *In order that the critical subgroups H and H' of G be conjugate (locally for the (fpqc) topology), it is necessary and sufficient that $cl(H) = cl(H')$.*
- (ii) *$\underline{\text{Crit}}(G)$ is representable, and the morphism cl is smooth, affine, with integral geometric fibers.*

This is proved as in 3.3, except for the assertion “ cl is affine”. It suffices to prove that $\underline{\text{Crit}}(G)$ is affine over S . Now $\underline{\text{Crit}}(G)$ is naturally identified with the S -functor of critical tori of G , and one therefore has a canonical monomorphism

$$\underline{\text{Crit}}(G) \rightarrow M,$$

where M is the scheme of subgroups of multiplicative type of G (Exp. XI, 4.1). To prove that $\underline{\text{Crit}}(G)$ is affine over S , it suffices, by Exp. XII 5.3, to show that this morphism is an open and closed immersion, or again, making the base change $M \rightarrow S$, to prove the following assertion: if Q is a subgroup of multiplicative type of the reductive group G , the $S' \rightarrow S$

⁽¹¹⁾**N.D.E.** Here “critical torus” has been replaced by “C-critical torus”, cf. *loc. cit.*. In the sequel, one will simply write “critical torus” instead of “C-critical torus”.

⁽¹²⁾**N.D.E.** Cf. 3.3, N.D.E. (10), for the notations $\mathcal{H}_c$ and $C\ell_c$.

such that $Q_{S'}$ is a critical torus of $G_{S'}$ are those which factor through a certain open and closed subscheme of S .

Now saying that Q is a critical torus means:

- (1) that Q is a torus,
- (2) Q being a torus, that $\text{rad}(\text{Centr}_G(Q))$, which is also a torus, has the same relative dimension as Q .

Now these two conditions are indeed of the type considered.

Corollary 3.11. *The S -functor $\underline{\text{Crit}}(G)$ is representable by an S -scheme smooth and affine over S .*

Corollary 3.12. *Let H be a critical subgroup of the reductive S -group G . Then $G/\underline{\text{Norm}}_G(H)$ and G/H are representable by S -schemes affine and smooth over S .*

The first assertion follows from 3.11, the second from the first and from Exp. XXII, 5.10.2.

Corollary 3.13. *Let Q be a subtorus of the reductive S -group G . Then $G/\underline{\text{Centr}}_G(Q)$ is representable by an S -scheme smooth and affine over S . The same holds for $G/\underline{\text{Norm}}_G(Q)$ if Q is a critical subtorus of G .*

Indeed, $H = \underline{\text{Centr}}_G(Q)$ is critical (Exp. XXII, 5.10.5), and one has $\underline{\text{Norm}}_G(H) = \underline{\text{Norm}}_G(Q)$ if Q is critical (*loc. cit.*, 5.10.8).

3.14. — By conjugacy of the Levi subgroups of the parabolic subgroups of G , there exists a unique morphism

$$u : \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \longrightarrow C\ell_{\text{crit}}$$

such that, for every parabolic subgroup P of G and every Levi subgroup L of P , one has $c\ell(L) = u(t(P))$, and such that this holds after every base change.

3.15. — Let S be a scheme and G a reductive S -group. Consider the S -functors:⁽¹³⁾

$$\begin{aligned} \underline{\text{PL}}(S') &= \{\text{pairs } P \supset L, P \text{ parabolic in } G_{S'}, \\ &\quad L \text{ a Levi subgroup of } P\}; \\ \underline{\text{PT}}(S') &= \{\text{pairs } P \supset T, P \text{ parabolic in } G_{S'}, \\ &\quad T \text{ a maximal torus of } P\}; \\ \underline{\text{CT}}(S') &= \{\text{pairs } C \supset T, C \text{ a critical subgroup of } G_{S'}, \\ &\quad T \text{ a maximal torus of } C\}; \\ \underline{\text{PLT}}(S') &= \{\text{triples } P \supset L \supset T, (P, T) \in \underline{\text{PT}}(S'), \\ &\quad L \text{ a Levi subgroup of } P\}. \end{aligned}$$

One has evident morphisms between these functors and the functors $\underline{\text{Par}}(G)$, $\underline{\text{Crit}}(G)$, $\underline{\text{Tor}}(G)$ already introduced, and one has a commutative diagram in the shape of a truncated cube (see the following figure):

⁽¹³⁾**N.D.E.** $\underline{\text{LT}}$ has been changed to $\underline{\text{CT}}$.

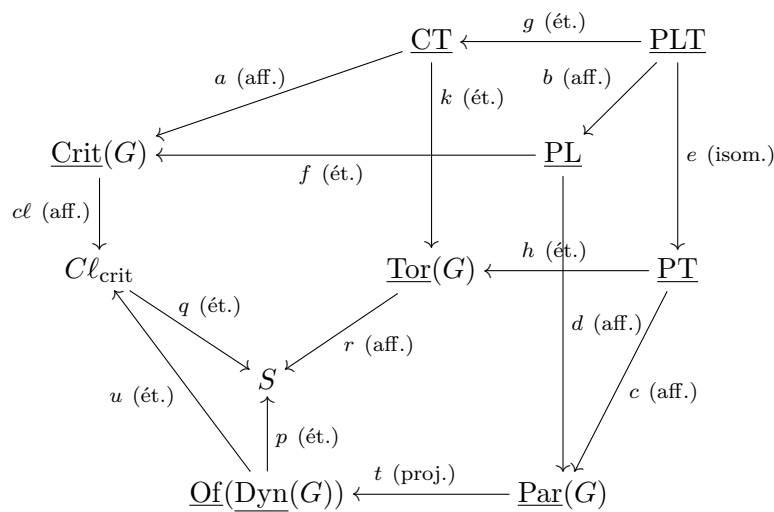


Figure 3.15.1

Theorem 3.16. (*Cf. figure 3.15.1.*)

- (i) All the morphisms of the diagram are smooth, surjective, and of finite presentation.
- (ii) All the morphisms of the diagram, except t , are affine; the morphism t is projective.
- (iii) All the morphisms of the diagram are either finite étale or have integral geometric fibers: the morphisms f , g , h , k , p , q and u are finite étale, the morphisms a , b , c , d , r and cl have integral geometric fibers, and the morphism e is an isomorphism.
- (iv) The square (a, b, f, g) is cartesian.

PROOF. First it is clear that e is an isomorphism, by 1.6 (ii). Moreover, (iv) is evident.

The morphism a is smooth, affine, with integral geometric fibers: indeed, by the base change $\underline{\text{Crit}}(G) \rightarrow S$, it suffices to verify that the morphism $\underline{\text{Tor}}(L_0) \rightarrow S$, where L_0 is the universal critical subgroup, possesses these properties; now L_0 is reductive (by definition), and one is reduced to Exp. XXII, 5.8.3. The morphism b therefore possesses the same properties, by (iv).

The morphism d is also smooth, affine, with integral geometric fibers, by 1.9; the same therefore holds for $c = db e^{-1}$. The morphism r possesses these same properties (Exp. XXII, 5.8.3), as does the morphism cl (3.10).

Moreover, we have already proved that the morphisms p and q are finite étale surjective (3.1 and 3.9). If we prove that f and k are finite étale surjective, the same properties will hold for g (by (iv)) and for h (since $h = k g e^{-1}$); since the stated properties of t were proved in 3.3 (ii), it therefore remains only to prove that f (resp. k) is finite étale surjective; let us give the proof for k , the one for f being analogous.

It suffices for us to prove that, if T is a maximal torus of G , the functor $\underline{\text{C}}$ of critical subgroups of G containing T is representable by a finite étale S -scheme with nonempty fibers; one can suppose G split relative to T ; let then E be the set of vectorial subsets of R (the root system of the splitting); $\underline{\text{C}}$ is representable by E_S (3.9), which completes the proof. $\square$

Corollary 3.17. *All the functors of the preceding diagram are representable by S -schemes smooth over S , and they are all affine over S , except for $\underline{\text{Par}}(G)$.*

Remark 3.18.

- (i) The fact that the morphism $f : \underline{\text{PL}} \rightarrow \underline{\text{Crit}}(G)$ is étale surjective implies that a subgroup of G is critical if and only if it is, locally for the étale topology, a Levi subgroup of a parabolic subgroup of G .

On the other hand, one should not believe that in general the map $f(S) : \underline{\text{PL}}(S) \rightarrow \underline{\text{Crit}}(G)(S)$ is surjective: it may very well happen that a critical subgroup

C of G does not come over S from a parabolic subgroup of G ; for example, a maximal torus is not always contained in a Borel subgroup (example: a nonsplit form of SL_2 , cf. Exp. XX, no. 5).

- (ii) Similarly, it may happen that the morphism $u : \underline{\mathrm{Of}}(\underline{\mathrm{Dyn}}(G)) \rightarrow C\ell_{\mathrm{crit}}$ is not an isomorphism: two parabolic subgroups of distinct types may have Levi subgroups of the same type; example: in a group of type A_2 , there are two types of parabolic subgroups whose Levi subgroups have semisimple rank 1 (corresponding to the two vertices of the diagram), whereas there is only one type of critical subgroups of rank 1.

The analogous example with a group of type A_3 shows that, even over an algebraically closed field, nonisomorphic parabolic subgroups may have Levi subgroups of the same type.⁽¹⁴⁾

Let us end this section with an application to the theory of principal bundles.

Lemma 3.20. ⁽¹⁵⁾ *Let S be a scheme, G a reductive S -group, F a principal homogeneous bundle under G , and G^F the corresponding twisted form of G . Identify $\underline{\mathrm{Dyn}}(G)$ and $\underline{\mathrm{Dyn}}(G^F)$ (Exp. XXIV, 3.5). Let P be a parabolic subgroup of G . One has a canonical isomorphism*

$$F/P \xrightarrow{\sim} \underline{\mathrm{Par}}_t(P)(G^F).$$

In particular, in order that the structural group of F can be reduced to P , it is necessary and sufficient that G^F possess a parabolic subgroup of type $t(P)$.

The proof proceeds exactly as in Exp. XXIV, 4.2.1.

3.21. — If S is a scheme, G a reductive S -group, and if $t \in \underline{\mathrm{Of}}(\underline{\mathrm{Dyn}}(G))(S)$, one denotes by $H_t^1(S, G)$ the subset of $H^1(S, G)$ consisting of the classes of principal bundles F under G such that the associated group G^F possesses a parabolic subgroup of type t . If G itself possesses a parabolic subgroup P of type t , $H_t^1(S, G)$ is precisely the image of $H^1(S, P)$ in $H^1(S, G)$, an image which therefore does not depend on the chosen P .

4. Relative position of two parabolic subgroups

4.1. A preliminary result

Lemma 4.1.1. *Let k be a field, G a reductive k -group, and P and P' two parabolic subgroups of G .*

- (i) *Then $P \cap P'$ is smooth, has the same reductive rank as G , and contains a maximal torus T of G .*

⁽¹⁴⁾**N.D.E.** I.e. the parabolic subgroups

$$P = \left\{ \begin{pmatrix} * & * & * & * \\ * & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \quad P' = \left\{ \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

of GL_4 are not isomorphic (since $\mathrm{rad}^u(P) \not\cong \mathrm{rad}^u(P')$), but their Levi subgroups

$$L = \left\{ \begin{pmatrix} * & * & 0 & 0 \\ * & * & 0 & 0 \\ 0 & 0 & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \quad L' = \left\{ \begin{pmatrix} * & 0 & 0 & 0 \\ 0 & * & * & 0 \\ 0 & * & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

are conjugate by the element $\begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

⁽¹⁵⁾**N.D.E.** The numbering of the original has been retained: there is no no. 3.19.

- (ii) *The set R' of roots of $P \cap P'$ relative to T is a closed subset of the set R of roots of G relative to T .⁽¹⁶⁾*

First suppose k algebraically closed. Let B (resp. B') be a Borel subgroup of P (resp. P'). One knows that there exists $g \in G(k)$ such that $\text{int}(g)B = B'$. Moreover, if T_0 is a maximal torus of B , and if one puts $N = \underline{\text{Norm}}_G(T_0)$, one knows (Bruhat's theorem, Bible, § 13.4, cor. 1 to th. 3) that $G(k) = B(k)N(k)B(k)$. One therefore sees that there exist $b, b_1 \in B(k)$ and $n \in N(k)$ such that $g = bnb_1$, hence $\text{int}(b)\text{int}(n)B = B'$. One then has

$$P \cap P' \supset B \cap B' = \text{int}(b)(B \cap \text{int}(n)B') \supset \text{int}(b)T_0.$$

Now suppose k arbitrary. Applying the preceding result, one sees that $P_{\bar{k}} \cap P'_{\bar{k}}$ contains a maximal torus of $G_{\bar{k}}$; by Exp. XXII, 5.4.5, one deduces that $(P \cap P')_{\bar{k}}$ is smooth “along the identity section”, hence smooth since one is over a field (Exp. VIA 1.3.1), hence that $P \cap P'$ is smooth. By Exp. VIA 2.3.1, the identity component $(P \cap P')^0$ of $P \cap P'$ is therefore an open subgroup of $P \cap P'$, smooth over S . One can then apply Exp. XIV, 1.1 to it, whence (i).

Finally, the set R_P (resp. $R_{P'}$) of roots of P (resp. P') relative to T is closed, and one has $R' = R_P \cap R_{P'}$, whence (ii).

Remark 4.1.2. One can prove ([BT65], 4.5) that $P \cap P'$ is connected;⁽¹⁷⁾ this will be used in 4.5.1.

Remark 4.1.3. The preceding lemma does not hold over an arbitrary scheme. Indeed, let for example G be a reductive group over an algebraically closed field k , and let B be a Borel group of G . Take $G = X$ as base, and consider the Borel subgroups B_1 and B_2 of G_X , where $B_1 = B_X$ and $B_2 = \text{int}(g_0)B_1$, g_0 being the canonical (diagonal) section of G_X . For each $g \in X(k)$, the fiber of $B_1 \cap B_2$ at g is precisely $B \cap \text{int}(g)B$. If one supposes $B \neq G$, the dimension of this fiber varies with g , so $B_1 \cap B_2$ cannot be smooth over X .

4.2. Transversal position

Theorem 4.2.1. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G . The following conditions on the pair (P, Q) are equivalent:*

- (i) $\text{Lie}(P/S) + \text{Lie}(Q/S) = \text{Lie}(G/S)$.
- (ii) *The canonical morphism $P \times_S Q \rightarrow G$ is smooth.*
- (ii') *The canonical morphism $P \rightarrow G/Q$ is smooth.*
- (iii) *The canonical morphism $P \times_S Q \rightarrow G$ is open.*
- (iii') *The canonical morphism $P \rightarrow G/Q$ is open.*
- (iv) *The canonical morphism $P \times_S Q \rightarrow G$ is dominant fiber by fiber.*
- (iv') *The canonical morphism $P \rightarrow G/Q$ is dominant fiber by fiber.*
- (v) *For every $s \in S$, “ $P_{\bar{s}} \cap Q_{\bar{s}}$ has minimum dimension”, i.e. one has*

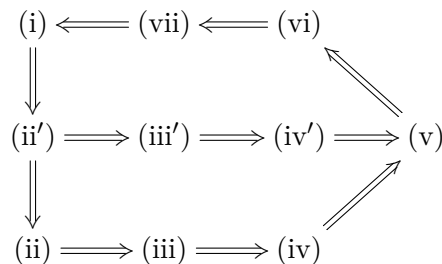
$$\dim(P_{\bar{s}} \cap Q_{\bar{s}}) = \dim P_{\bar{s}} + \dim Q_{\bar{s}} - \dim G_{\bar{s}}.$$

- (vi) *There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a Borel subgroup B_i of P_{S_i} and a Borel subgroup B'_i of Q_{S_i} such that $B_i \cap B'_i$ is a maximal torus of G_{S_i} .*
- (vii) *There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a splitting $(T_i, \dots, R_i)$ of G_{S_i} and a system of positive roots R_i^+ of R_i such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by a subset $R_i^{(1)}$ (resp. $R_i^{(2)}$) of R containing R_i^+ (resp. $-R_i^+$) (see Exp. XXII, 5.4.2 and 5.2.1 for the definitions).*

⁽¹⁶⁾**N.D.E.** Part (ii), which will be useful in 4.5.1, has been added.

⁽¹⁷⁾**N.D.E.** In view of the details added in 4.5.1, the original (which said “we shall not use this fact”) has been modified here.

PROOF. We shall prove the theorem according to the logical diagram



One trivially has $(ii) \Rightarrow (iii)$ and $(ii') \Rightarrow (iii')$. If (iii) holds, the set-theoretic image of the morphism $P \times_S Q \Rightarrow G$ is an open subset of G which contains the identity section; since the fibers of G are connected, this image is dense on each fiber, which proves (iv) . Similarly, one has $(iii') \Rightarrow (iv')$.

One has $(ii') \Rightarrow (ii)$, by the cartesian diagram

$$\begin{array}{ccc}
 P \times_S Q & \longrightarrow & G \\
 \text{pr}_1 \downarrow & & \downarrow \\
 P & \longrightarrow & G/Q.
 \end{array}$$

Moreover, (iv) or (iv') implies (v) , by dimension theory (cf. EGA IV₂, 5.6.6). One notes that one can indeed suppose $S = \text{Spec}(k)$, k an algebraically closed field, and that every nonempty fiber of the morphism (iv) , resp. (iv') , at a point of $G(k)$, resp. of $(G/Q)(k)$, is isomorphic to $P \cap Q$ (as an immediate computation shows).

One has $(vi) \Rightarrow (vii)$, by Exp. XXII, 5.5.1 (iv) and 5.9.2.

One has $(vii) \Rightarrow (i)$, since, to verify that $\text{Lie}(P) + \text{Lie}(Q) = \text{Lie}(G)$, one can reason locally for (fpqc); hence, if (vii) holds, one can suppose G split, $P \supset B_{R_+}$ and $Q \supset B_{-R_+}$ (usual notations), in which case one already has

$$\text{Lie}(B_{R_+}) + \text{Lie}(B_{-R_+}) = \text{Lie}(G).$$

Let us prove that (i) implies (ii') .

Let $u : P \rightarrow G/Q$ be the canonical morphism; to prove that u is smooth, one can content oneself with doing so for the geometric fibers of u , since P and G/Q are smooth over S , and one can therefore suppose that S is the spectrum of an algebraically closed field. Since the morphism u is compatible with the evident action of P (one has $u(pp') = pu(p')$), it suffices, by a translation argument (cf. the proof of VIB 1.3), to verify that u is smooth at $e \in P(k)$, i.e. (SGA 1, II 4.7) that the tangent map to u at e is surjective; but this is naturally identified with the canonical map $\text{Lie}(P) \rightarrow \text{Lie}(G)/\text{Lie}(Q)$, which is surjective if (i) holds.

It therefore remains only to verify the last assertion, that is, $(v) \Rightarrow (vi)$. First suppose that S is the spectrum of an algebraically closed field. By 4.1.1, there exists a maximal torus T contained in P and Q ; let R (resp. R_1 , resp. R_2) be the set of roots of G (resp. P , resp. Q) relative to T . One has:

$$\begin{aligned}
 \dim(G) &= \dim(T) + \text{Card}(R), & \dim(P) &= \dim(T) + \text{Card}(R_1), \\
 \dim(Q) &= \dim(T) + \text{Card}(R_2), & \dim(P \cap Q) &= \dim(T) + \text{Card}(R_1 \cap R_2),
 \end{aligned}$$

by Exp. XXII, 5.4.4 and 5.4.5, for example. Condition (v) is therefore equivalent to

$$\text{Card}(R_1 \cap R_2) = \text{Card}(R_1) + \text{Card}(R_2) - \text{Card}(R),$$

that is, $R_1 \cup R_2 = R$. To prove (vi) , it suffices, by Exp. XXII, 5.9.2 and 5.4.5, to prove that $R_1 \cap -R_2$ contains a system of positive roots of R . One is therefore reduced to proving:

Lemma 4.2.2. *Let R be a “root system” (for example the set of roots of a root datum in the sense of Exposé XXI). Let R_1 and R_2 be two closed subsets of R , each containing a system of positive roots. If $R_1 \cup R_2 = R$, then $R_1 \cap -R_2$ contains a system of positive roots.*

Indeed, since $R_1 \cap -R_2 = R_3$ is evidently closed, and by Exp. XXI, 3.3.6, it suffices to show that $R_3 \cup -R_3 = R$. Now one knows that $R_1 \cup -R_1 = R = R_2 \cup -R_2$, and one concludes by the following elementary fact: if A, A', B, B' are four subsets of a set E , and if $A \cup A' = B \cup B' = A \cup B = A' \cup B' = E$, one has $(A \cap B') \cup (A' \cap B) = E$.

This completes the proof of (v) $\Rightarrow$ (vi) in the case where the base is the spectrum of an algebraically closed field. Let us now return to the general case and suppose (v) satisfied. Let $s \in S$; by the foregoing, one can find a Borel subgroup $\overline{B}$ (resp. $\overline{B}'$) of $P_{\overline{s}}$ (resp. $Q_{\overline{s}}$) such that $\overline{B} \cap \overline{B}'$ is a maximal torus of $G_{\overline{s}}$.

Since the S -scheme $\underline{\text{Bor}}(P) \simeq \underline{\text{Bor}}(P/\text{rad}^u(P))$ of Borel subgroups of P is smooth, one can, applying “Hensel’s lemma” (cf. Exp. XI 1.10) and reasoning locally for the étale topology (i.e. replacing S by an étale $S' \rightarrow S$ covering s , and s by a point of its fiber in S'), suppose that there exists a Borel subgroup B of P projecting onto $\overline{B}$; one can similarly suppose that there exists a Borel subgroup B' of Q projecting onto $\overline{B}'$. Since $\overline{B} \cap \overline{B}'$ is a maximal torus of G , there exists an open subset U of S containing s and such that $B_U \cap B'_U$ is a maximal torus of G_U (Exp. XXII, 5.9.4), which proves (vi). $\square$

Definition 4.2.3. A pair (P, Q) satisfying the equivalent conditions (i) to (vii) of theorem 4.2.1 is said to be *in transversal position*. One also says that P is *in transversal position relative to Q* , or, by abuse of language, that P and Q are in (mutual) transversal position.

By (vi), this definition coincides, in the case of Borel subgroups, with that of Exp. XXII, 5.9.1.

Corollary 4.2.4. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G .*

- (i) *In order that (P, Q) be in transversal position, it is necessary and sufficient that, for each point s of S , the pair $(P_{\overline{s}}, Q_{\overline{s}})$ be in transversal position; if $S' \rightarrow S$ is a surjective morphism, and if $(P_{S'}, Q_{S'})$ is in transversal position, then (P, Q) is in transversal position.*
- (ii) *There exists an open subscheme U of S satisfying the following property: in order that a morphism $S' \rightarrow S$ factor through U , it is necessary and sufficient that $(P_{S'}, Q_{S'})$ be in transversal position.*
- (iii) *Consider the subfunctors*

$$\begin{aligned}\underline{\text{Gen}}(G) &\subset \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G), \\ \underline{\text{Gen}}(/Q) &\subset \underline{\text{Par}}(G), \\ \underline{\text{Gen}}(P/Q) &\subset G,\end{aligned}$$

defined as follows: for $S' \rightarrow S$, $\underline{\text{Gen}}(G)(S')$ is the set of pairs of parabolic subgroups of $G_{S'}$ in transversal position, $\underline{\text{Gen}}(/Q)(S')$ is the set of parabolic subgroups of $G_{S'}$ in transversal position relative to $Q_{S'}$, and $\underline{\text{Gen}}(P/Q)(S')$ is the set of $g \in G(S')$ such that $\text{int}(g)P_{S'}$ is in transversal position relative to $Q_{S'}$.

Each of these functors is representable by an open subscheme universally schematically dense over $S^{(18)}$ (cf. Exp. XVIII § 1) of the corresponding S -scheme $\underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G)$, resp. $\underline{\text{Par}}(G)$, resp. G .

Assertions (i) follow immediately from the description 4.2.1 (i) of the term “transversal position”. To prove (ii), one takes $U = S - \text{Supp}(\text{Coker } u)$, where u is the canonical morphism $\text{Lie}(P) \oplus \text{Lie}(G) \rightarrow \text{Lie}(G)$.

⁽¹⁸⁾**N.D.E.** “Relatively dense” has been replaced by “universally schematically dense over S ”, cf. EGA IV₃, Def. 11.10.8.

Since one has cartesian diagrams

$$\begin{array}{ccc} G & \xrightarrow{f} & \underline{\text{Par}}(G) \\ \uparrow & & \uparrow \\ \underline{\text{Gen}}(P/Q) & \longrightarrow & \underline{\text{Gen}}(/Q) \end{array} \quad \begin{array}{ccc} \underline{\text{Par}}(G) & \xrightarrow{f'} & \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G) \\ \uparrow & & \uparrow \\ \underline{\text{Gen}}(/Q) & \longrightarrow & \underline{\text{Gen}}(G) \end{array}$$

(where $f(g) = \text{int}(g)P$ and $f'(R) = (R, Q)$), it suffices to verify (iii) in the case of $\underline{\text{Gen}}(G)$. Let then P_0 be the canonical parabolic subgroup of $G_{\underline{\text{Par}}(G)}$; put

$$X = \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G), \\ P = \text{pr}_1^*(P_0), \quad Q = \text{pr}_2^*(P_0);$$

applying assertion (ii) to the parabolic subgroups P and Q of G_X , one constructs an open subscheme U of X which, as one immediately verifies, is indeed identified with $\underline{\text{Gen}}(G)$. It remains to verify the density assertion, which can be done on the geometric fibers;⁽¹⁹⁾ one can therefore suppose $S = \text{Spec}(k)$, k an algebraically closed field; since $\underline{\text{Par}}(G)$ is smooth, it suffices to verify that $\underline{\text{Gen}}(G)$ meets every irreducible component of $\underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G)$; in other words, by 3.3, it suffices to see that, if $t, t' \in \text{Of}(\underline{\text{Dyn}}(G))(S)$, there exists a pair (P, P') in transversal position, with $t(P) = t$, $t(P') = t'$.

Now this is immediate: one chooses a pair (B, B') of Borel subgroups of G such that $B \cap B'$ is a maximal torus (one splits G and applies Exp. XXII, 5.9.2), then applies 3.8 to construct $P \supset B$ and $P' \supset B'$, with $t(P) = t$, $t(P') = t'$; P and P' have the desired types and are in transversal position by 4.2.1 (vi).

Corollary 4.2.5. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G , the pair (P, Q) being in transversal position.*

- (i) *Let P' and Q' be two parabolic subgroups of G of the same types as P and Q respectively. In order that the pair (P', Q') be in transversal position, it is necessary and sufficient that it be conjugate to the pair (P, Q) , locally for the étale topology. (N.B. We shall see in § 5 that one can replace the étale topology by the Zariski topology.)*
- (ii) *The canonical morphism $P \times_S Q \rightarrow G$ induces a smooth surjective morphism $P \times_S Q \rightarrow \underline{\text{Gen}}(Q/P)$, and an isomorphism*

$$(P \times_S Q)/(P \cap Q) \simeq \underline{\text{Gen}}(Q/P)$$

(where $P \cap Q = R$ acts on $P \times_S Q$ by $(p, q)r = (pr, r^{-1}q)$).

- (iii) *The canonical morphism $P \rightarrow \underline{\text{Par}}_{t(Q)}(G)$ (defined set-theoretically by $p \mapsto \text{int}(p)Q$) induces a smooth surjective morphism $P \rightarrow \underline{\text{Gen}}(/P) \cap \underline{\text{Par}}_{t(Q)}(G)$, and an isomorphism*

$$P/(P \cap Q) \simeq \underline{\text{Gen}}(/P) \cap \underline{\text{Par}}_{t(Q)}(G).$$

- (iv) *The canonical morphism $G \rightarrow \underline{\text{Par}}_{t(P)}(G) \times_S \underline{\text{Par}}_{t(Q)}(G)$ (defined set-theoretically by $g \mapsto (\text{int}(g)P, \text{int}(g)Q)$) induces a smooth surjective morphism*

$$G \rightarrow \underline{\text{Gen}}(G) \cap (\underline{\text{Par}}_{t(P)}(G) \times_S \underline{\text{Par}}_{t(Q)}(G))$$

and an isomorphism

$$G/(P \cap Q) \simeq \underline{\text{Gen}}(G) \cap (\underline{\text{Par}}_{t(P)}(G) \times_S \underline{\text{Par}}_{t(Q)}(G)).$$

Let us prove (i). It is clear that the condition is sufficient; let us prove that it is necessary. Let therefore (P', Q') be in transversal position. Since P and P' are locally conjugate for the étale topology, one can suppose $P = P'$, and it suffices for us to prove that, if Q and Q' are two parabolic subgroups of G in transversal position relative to P and of the same type, they

⁽¹⁹⁾**N.D.E.** Cf. EGA IV₃, 11.10.10.

are then conjugate, locally for the étale topology, by a section of P . Using 4.2.1 (vi), one can suppose that there exist Borel subgroups B, B', B_1, B'_1 of P, P, Q, Q' respectively such that $B \cap B_1 = T$ and $B' \cap B'_1 = T'$ are maximal tori of G . Now the Killing pairs (B, T) and (B', T') of P are locally conjugate in P for the étale topology (1.16), and one can suppose $B = B', T = T'$, in which case one has $B_1 = B'_1$ by Exp. XXII, 5.9.2, hence $Q = Q'$ by 3.8.

Assertions (ii), (iii) and (iv) are proved in parallel fashion. Let us prove (ii), for example; let $g \in \text{Gen}(Q/P)(S)$, i.e. let $g \in G(S)$ be such that $\text{int}(g)Q$ is in transversal position relative to P . By the preceding proof, Q and $\text{int}(g)Q$ are locally conjugate for the étale topology by a section of P .

Reasoning locally for this topology, one can suppose that there exists $p \in P(S)$ such that $\text{int}(g)Q = \text{int}(p)Q$, hence $p^{-1}g \in \text{Norm}_G(Q)(S) = Q(S)$; this proves the existence of a $q \in Q(S)$ such that $g = pq$. We have therefore proved that the morphism considered in (ii) is covering for the étale topology. Comparing with 4.2.1 (ii), one deduces that it is smooth and surjective. Moreover, an immediate argument shows that the equivalence relation defined on $P \times_S Q$ by the morphism $P \times_S Q \rightarrow G$ is the equivalence relation associated with the action of the group $R = P \cap Q$ (acting by $(p, q)r = (pr, r^{-1}q)$), which proves the last assertion of (ii) (since a smooth surjective morphism is an effective epimorphism, for example).

Remark 4.2.6. If P and Q are in transversal position, one will often denote by $P \cdot Q$ the open subset $\text{Gen}(Q/P)$ of G , a notation justified by 4.2.5 (ii).

Proposition 4.2.7. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G , the pair (P, Q) being in transversal position.*

- (i) *The group $P \cap Q$ is smooth over S (and in fact has connected fibers by 4.1.2); let us then introduce $(P \cap Q)^0$ (cf. Exp. VIB 3.10); this is a subgroup of type (RC) of G (Exp. XXII, 5.11.1), whose unipotent radical (loc. cit., 5.11.4) decomposes as a direct product*

$$\text{rad}^u((P \cap Q)^0) = (\text{rad}^u(P) \cap Q) \times_S (P \cap \text{rad}^u(Q)).$$

- (ii) *If S is affine, $H^1(S, \text{rad}^u(P \cap Q)^0) = 0$. If S is semilocal, $P \cap Q$ contains a maximal torus of G .*

Indeed, $P \cap Q$ is smooth by 4.2.1 (ii) and the cartesian diagram

$$\begin{array}{ccc} P \times_S Q & \longrightarrow & G \\ \uparrow & & \uparrow \\ P \cap Q & \longrightarrow & S. \end{array}$$

To verify the stated assertions about $(P \cap Q)^0$, one can reason locally for the étale topology, hence by 4.2.1 (vii) suppose that one has chosen a splitting (G, T, M, R) of G such that P and Q contain T and are defined respectively by subsets R_1 and R_2 of R , R_1 containing a system of positive roots R_+ , and R_2 containing the opposite system R_- . Let Δ be the set of simple roots of R_+ ; put

$$A_1 = \Delta \cap -R_1, \quad A_2 = \Delta \cap R_2, \quad A = A_1 \cap A_2.$$

By 1.4 (v) and 1.12, one has

$$\begin{aligned} R_1 &= R_+ \cup (R_- \cap -\mathbf{N}A_1), & R_2 &= (R_+ \cap \mathbf{N}A_2) \cup R_-, \\ \text{rad}^u(P) &= \prod_{\substack{\alpha \in R_1 \\ \alpha \notin -R_1}} U_\alpha, & \text{rad}^u(Q) &= \prod_{\substack{\alpha \in R_2 \\ \alpha \notin R_2}} U_\alpha. \end{aligned}$$

By Exp. XXII, 5.6.7, one therefore has

$$\begin{aligned}\mathrm{rad}^u(P) \cap Q &= \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_1}} U_\alpha = \prod_{\alpha \in K_2} U_\alpha, \\ \mathrm{rad}^u(Q) \cap P &= \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_2}} U_\alpha = \prod_{\alpha \in K_1} U_\alpha,\end{aligned}$$

where K_2 is the set of positive roots which are linear combinations of the elements of A_2 , but not linear combinations of the elements of A , and K_1 the set of negative roots which are linear combinations of the elements of A_1 , but not linear combinations of the elements of A . It is clear that, if $\alpha \in K_2$, $\beta \in K_1$, $\alpha + \beta$ is never a root, nor zero, which implies that the two groups above commute.

Moreover, one knows by Exp. XXII, 5.4.5 that $H = (P \cap Q)^0$ is defined by the set of roots $R_1 \cap R_2$, namely

$$R_1 \cap R_2 = (R_+ \cap \mathbf{N}A_2) \cup (R_- \cap -\mathbf{N}A_1).$$

Since $R_1 \cap R_2$ is closed, H is of type (RC) by definition (Exp. XXII 5.11.1), and by *loc. cit.*, 5.11.3 and 5.11.4, one has

$$\mathrm{rad}^u(H) = \prod_{\alpha \in K} U_\alpha,$$

where K is the set of $\alpha \in R_1 \cap R_2$ such that $\alpha \notin -(R_1 \cap R_2)$. Since the symmetric part of $R_1 \cap R_2$ is evidently $R \cap \mathbf{Z}A$, one sees immediately that $K = K_1 \cup K_2$, which ends the proof of (i).

The first assertion of (ii) then follows from (i) and 2.10; let us prove the second. Since $(P \cap Q)^0 / \mathrm{rad}^u((P \cap Q)^0)$ is reductive, it possesses a maximal torus $\overline{T}$ if the base is semilocal. The inverse image of $\overline{T}$ in $(P \cap Q)^0$ is a subgroup N of type (R) of G with solvable fibers, and one has $N^u = \mathrm{rad}^u((P \cap Q)^0)$ (Exp. XXII, 5.6.9). The scheme of maximal tori of N is a principal homogeneous bundle under N^u (*loc. cit.*, 5.6.13), and so possesses a section, since $H^1(S, N^u) = 0$.

4.3. Opposite parabolic subgroups

4.3.1. — If G is a reductive S -group, one defined in Exp. XXIV, 3.16.6 a canonical “outer automorphism” s_G of order $\leq 2^{(20)}$ of G , hence a canonical automorphism of order ≤ 2 of $\mathrm{Dyn}(G)$, hence also an automorphism of order ≤ 2 of $\mathrm{Of}(\mathrm{Dyn}(G))$, which we shall also denote by s_G , or simply s . Two types of parabolic subgroups $t, t' \in \mathrm{Of}(\mathrm{Dyn}(G))(S)$ will be called *opposite* when $t = s_G(t')$.

Theorem 4.3.2. *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G .*

- (a) *If L is a Levi subgroup of P , there exists a unique parabolic subgroup P' of G such that $P \cap P' = L$.*
- (b) *For every parabolic subgroup Q of G , the following conditions are equivalent:*
 - (i) *For every $s \in S$, $((P \cap Q)_{\overline{s}})^0$ (which is smooth by 4.1.1) is reductive.*
 - (ii) *$P \cap Q$ is a Levi subgroup of P and of Q .*
 - (iii) *P and Q have opposite types, and the pair (P, Q) is in transversal position (cf. 4.2.3).*
 - (iv) *P and Q have opposite types, and $\mathrm{rad}^u(P) \cap Q = e$.*
 - (v) *$\mathrm{rad}^u(P) \cap Q = \mathrm{rad}^u(Q) \cap P = e$.*
 - (vi) *The canonical morphism $\mathrm{rad}^u(P) \times_S Q \rightarrow G$ is an open immersion.*
 - (vi') *The canonical morphism $\mathrm{rad}^u(P) \rightarrow G/Q$ is an open immersion.*

⁽²⁰⁾**N.D.E.** “Of order 2” has been replaced by “of order ≤ 2 ”, since s_G can be trivial (for example if G is of type $A_1, B_n, C_n, \dots$).

(vii) *There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a splitting (T_i, M_i, R_i) of G_{S_i} and a subset $R_i^{(1)}$ of R_i such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by $R_i^{(1)}$ (resp. by $R_i^{(2)} = -R_i^{(1)}$).*

PROOF. Let us first prove the second part of the theorem; one first sees that (iii) $\Leftrightarrow$ (vii), by 4.2.1 (vii) and the definition of s_G in the split case (Exp. XXII, 3.16.2 (iv)); one evidently has (ii) $\Rightarrow$ (i); one has (vi') $\Rightarrow$ (vi) by the base change $G \rightarrow G/Q$.

Now suppose (vii) satisfied, and let us prove all the other conditions; since they are local for the étale topology, one can suppose $G = (G, T, M, R)$ split, P defined by the subset R' of R , and Q by the subset $-R'$. If L is the subgroup of type (R) of G containing T defined by $R' \cap -R'$, it is clear by Exp. XXII, 5.11.3 that L is a common Levi subgroup of P and Q . But $P = L \cdot \text{rad}^u(P)$, $Q = L \cdot \text{rad}^u(Q)$, and by Exp. XXII, 5.6.7, $\text{rad}^u(P) \cap Q = Q \cap \text{rad}^u(P) = e$; hence $P \cap Q = L$, and one has proved (ii) and (v). Since P and Q are in transversal position, the canonical morphism $P \rightarrow G/Q$ induces an open immersion $P/P \cap Q \rightarrow G/Q$ (4.2.1); but the canonical morphism $\text{rad}^u(P) \rightarrow P/P \cap Q = P/L$ is an isomorphism, so that one has proved (vi'). In view of what one has already seen, all the assertions are therefore consequences of (vii).

It now suffices for us to prove that any one of assertions (i), (iv), (v), (vi) implies (vii); since we have already proved the equivalence of (ii) and (iii), it suffices to give the proof on the geometric fibers, and one can suppose that S is the spectrum of an algebraically closed field. By 4.1.1, there exists a maximal torus T of G contained in $P \cap Q$. Let R (resp. R_1 , resp. R_2) be the set of roots of G (resp. P , resp. Q) relative to T .

Let R_1^a be the asymmetric part of R_1 (i.e. $R_1^a = \{\alpha \in R_1 \mid -\alpha \notin R_1\}$). Similarly introduce R_2^a . One must prove that $R_1 = -R_2$.

- Condition (i) implies that $R_1 \cap R_2$ is symmetric; let $\alpha \in R_1$; if $\alpha \notin R_2$, then $\alpha \in -R_2$, and if $\alpha \in R_2$, then $\alpha \in R_1 \cap R_2 = -(R_1 \cap R_2) \subset -R_2$; one therefore has $R_1 \subset -R_2$, hence, by symmetry, $R_1 = -R_2$.
- Condition (iv) implies $\text{Card}(R_1) = \text{Card}(R_2)$ and $R_1^a \cap R_2 = \emptyset$; the second condition is equivalent to $R_2 \subset -R_1$; the first then gives $R_2 = -R_1$.
- Condition (v) implies $R_1^a \cap R_2 = R_2^a \cap R_1 = \emptyset$, hence $R_2 \subset -R_1$ and $R_1 \subset -R_2$, which again gives $R_2 = -R_1$.
- Condition (vi) implies $\text{Lie}(\text{rad}^u(P)) \oplus \text{Lie}(Q) = \text{Lie}(G)$, which implies that R is the disjoint union of R_2 and R_1^a , hence that $R_2 = -R_1$.

This completes the proof of the second part of the theorem. Let us prove the first; first note that, by (vii) $\Rightarrow$ (ii), one has already proved the existence, locally for the étale topology, of the group P' sought; it therefore remains to prove its uniqueness, and this can likewise be done locally for the étale topology. One can therefore suppose G split relative to a maximal torus T of L , and P (resp. P') defined by a subset R_1 (resp. R_1') of the root system R .

By hypothesis $R_1 \cap R_1'$ is symmetric; reasoning as above, one obtains $R_1' = -R_1$, which proves that P' is determined by P and L , and completes the proof. $\square$

Definition 4.3.3. Two parabolic subgroups of G satisfying the equivalent conditions (i) to (vii) of 4.3.2 are called *opposite*. If P is a parabolic subgroup of G , and if L is a Levi subgroup of P (resp. and if T is a maximal torus of P), one calls the unique parabolic subgroup Q of G such that $P \cap Q = L$ (resp. such that $P \cap Q$ is the unique Levi subgroup of P containing T , cf. 1.6, or again such that $P \cap Q$ contains T and P and Q are opposite) the *parabolic subgroup opposite to P relative to L* (resp. T).

By 4.3.2 (iii), one immediately deduces from 4.2.4 and 4.2.5 parallel results; let us give a sample of them.

Corollary 4.3.4. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G .*

- (i) *In order that P and Q be opposite, it is necessary and sufficient that, for every point $s \in S$, $P_{\bar{s}}$ and $Q_{\bar{s}}$ be opposite. If $S' \rightarrow S$ is a surjective morphism, and if $P_{S'}$ and $Q_{S'}$ are opposite, then P and Q are opposite.*
- (ii) *The functor $\underline{\text{Opp}}(G)$, such that for $S' \rightarrow S$, $\underline{\text{Opp}}(G)(S')$ is the set of pairs of opposite parabolic subgroups of $G_{S'}$, is representable by an open subscheme of $\underline{\text{Par}}(G)^2$. The functor $\underline{\text{Opp}}(/P)$, such that for $S' \rightarrow S$, $\underline{\text{Opp}}(/P)(S')$ is the set of parabolic subgroups of $G_{S'}$ opposite to $P_{S'}$, is representable by an open subscheme universally schematically dense over $S^{(21)}$ of $\underline{\text{Par}}_{s(t(P))}(G)$.*
- (iii) *Suppose P and Q opposite; let P' and Q' be two parabolic subgroups of G , P' having the same type as P . In order that P' and Q' be opposite, it is necessary and sufficient that, locally for the étale topology, the pair (P', Q') be conjugate to the pair (P, Q) . (N.B. We shall see in § 5 that one can replace the étale topology by the Zariski topology.)*

Corollary 4.3.5. *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G .*

- (i) *The morphism $\underline{\text{Opp}}(/P) \rightarrow \underline{\text{Lev}}(P)$ (cf. 1.9), defined set-theoretically by $Q \mapsto P \cap Q$, is an isomorphism; $\underline{\text{Opp}}(/P)$ is a principal homogeneous bundle under $\text{rad}^u(P)$ ($\text{rad}^u(P)$ acting by inner automorphisms). If S is affine, there exists a parabolic subgroup of G opposite to P .*
- (ii) *Suppose S semilocal; let $\{s_i\}$ be the set of its closed points; for each i , let Q_i be a parabolic subgroup of G_{s_i} opposite to P_{s_i} . There exists a parabolic subgroup Q of G opposite to P and such that $Q_{s_i} = Q_i$ for each i .*
- (iii) *The morphism $\underline{\text{Opp}}(G) \rightarrow \underline{\text{PL}}$ (cf. 3.15), defined set-theoretically by $(P, Q) \mapsto (P, P \cap Q)$, is an isomorphism.*

All this follows from the first part of the theorem and from 1.9, 2.3 and 2.8.

Remark 4.3.6. Let P and Q be two opposite parabolic subgroups of G , and let $P \cdot Q$ be the open subscheme of G , the sheaf-theoretic image of $P \times_S Q$, introduced in 4.2.6. The “product morphism” $G \times_S G \rightarrow G$ induces isomorphisms:

$$\text{rad}^u(P) \times_S Q \xrightarrow{\sim} P \cdot Q \xleftarrow{\sim} P \times_S \text{rad}^u(Q).$$

Indeed, this follows from 4.3.2 (or from 4.2.5 (ii)) and from the fact that $P \cap Q$ is a Levi subgroup of P and of Q , hence that $P = \text{rad}^u(P) \cdot (P \cap Q)$ and $Q = \text{rad}^u(Q) \cdot (P \cap Q)$.

One similarly has a commutative diagram

$$\begin{array}{ccc} \text{rad}^u(P) & \hookrightarrow & G/Q \\ \sim \downarrow & & \downarrow \sim \\ \underline{\text{Opp}}(/P) & \hookrightarrow & \underline{\text{Par}}_{t(Q)}(G), \end{array}$$

where the vertical arrows are induced by $g \mapsto \text{int}(g)Q$.

4.4. Osculatory position

Proposition 4.4.1. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G . The following conditions are equivalent:*

- (i) *$P \cap Q$ is a parabolic subgroup of G .*

⁽²¹⁾**N.D.E.** “Relatively dense” has been replaced by “universally schematically dense over S ”, cf. EGA IV₃, Def. 11.10.8.

- (ii) $P \cap Q$ locally contains a Borel subgroup of G for the étale topology.
- (iii) $P \cap Q$ locally contains a maximal torus of G for the étale topology, and, for every $S' \rightarrow S$ and every maximal torus T of $G_{S'}$ contained in $P_{S'}$ and $Q_{S'}$, the opposite of $P_{S'}$ relative to T is in transversal position relative to $Q_{S'}$.
- (iv) There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a maximal torus T_i of G_{S_i} contained in P_{S_i} and Q_{S_i} , and such that the opposite of P_{S_i} relative to T_i is in transversal position relative to Q_{S_i} .
- (v) There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a splitting (T_i, M_i, R_i) of G_{S_i} such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by a set of roots $R_i^{(1)}$ (resp. $R_i^{(2)}$), $R_i^{(1)} \cap R_i^{(2)}$ containing a system of positive roots of R .

Moreover, if these conditions hold, one has $t(P \cap Q) = t(P) \cap t(Q)$ (with the notations of 3.2).

One has (v) $\Rightarrow$ (ii) and (iii) $\Rightarrow$ (iv) trivially. Moreover, (ii) $\Rightarrow$ (i) by 1.18. One has (iv) $\Rightarrow$ (v): indeed, one can suppose G split, P (resp. Q) defined by the set of roots R_1 (resp. R_2); the opposite of P is then defined by $-R_1$, and one is reduced to lemma 4.2.2. One proves (i) $\Rightarrow$ (iii) by splitting, in the same way. Finally, the last assertion of the theorem can be proved locally for the étale topology; one can suppose that $P \cap Q$ contains a Borel subgroup B of G , and one is reduced to 3.7.

Definition 4.4.2. Two parabolic subgroups of G satisfying conditions (i) to (v) of 4.4.1 are said to be *in osculatory position*.

Corollary 4.4.3. Let P and Q be two parabolic subgroups in osculatory position, and let P' and Q' be two parabolic subgroups of G of the same types as P and Q respectively; in order that P' and Q' be in osculatory position, it is necessary and sufficient that the pair (P', Q') be conjugate to the pair (P, Q) , locally for the étale topology.

It suffices to prove that, if P and P' are in osculatory position relative to Q , they are conjugate, locally for (ét), by a section of Q . Now $P \cap Q$ and $P' \cap Q$ are two parabolic subgroups of the same type contained in Q , and so are conjugate, locally for (ét), by a section of Q , by part (ii) of the lemma below. One can therefore suppose $P \cap Q = P' \cap Q$; one then has $P = P'$, by part (i) of the same lemma:

Lemma 4.4.4. Let P, P' and Q be three parabolic subgroups of the reductive S -group G .

- (i) In order that $P = P'$, it is necessary and sufficient that P and P' be in osculatory position and of the same type.
- (ii) If $P \subset Q$, $P' \subset Q$, and if $g \in G(S)$ is such that $\text{int}(g)P$ and P' are in osculatory position, then $g \in Q(S)$.

Part (i) follows trivially from the last assertion of 4.4.1. Let us prove (ii): Q and $\text{int}(g)Q$ contain $P' \cap \text{int}(g)P$, and so are in osculatory position; they coincide by (i), hence $g \in \underline{\text{Norm}}_G(Q)(S) = Q(S)$.

Note that assertions (iii) and (iv) of the theorem immediately give:

Corollary 4.4.5. Let P, P' and Q be three parabolic subgroups of the reductive S -group G containing the same maximal torus T of G . Suppose P and P' opposite relative to T . In order that Q be in osculatory position relative to P , it is necessary and sufficient that it be in transversal position relative to P' . Under these conditions, $P \cap Q$ is also in transversal position relative to P' .

Corollary 4.4.6. Let P and Q be two parabolic subgroups of G containing the same maximal torus T . In order that P and Q be in transversal position, it is necessary and sufficient that

there exist two parabolic subgroups $P' \subset P$ and $Q' \subset Q$ of G opposite relative to T ; one can even choose $t(P') = t(P) \cap s(t(Q))$.⁽²²⁾

The condition is evidently sufficient (4.2.1 (i) and 4.3.2 (iii)). Let us show that it is necessary; let P^- (resp. Q^-) be the opposite of P (resp. Q) relative to T . By 4.4.5, $P^- \cap Q$ is in transversal position relative to P and Q^- , hence also relative to $P \cap Q^-$ by a new application of 4.4.5; moreover

$$\begin{aligned} t(P^- \cap Q) &= t(P^-) \cap t(Q) = s(t(P)) \cap s(t(Q^-)) \\ &= s(t(P) \cap t(Q^-)) = s(t(P \cap Q^-)), \end{aligned}$$

hence $P^- \cap Q = P'$ and $P \cap Q^- = Q'$ are opposite (4.3.2 (iii)); but $P' \cap Q' \supset T$, so they are indeed opposite relative to T .

4.5. Standard position

In this section, we briefly indicate how some of the preceding results generalize.

Proposition 4.5.1. *If P_1 and P_2 are two parabolic subgroups of the reductive S -group G , the following conditions are equivalent.*⁽²³⁾

- (i) $P_1 \cap P_2$ is smooth.
- (ii) $P_1 \cap P_2$ is a subgroup of type (R) (or of type (RC)) of G .
- (iii) $P_1 \cap P_2$ locally contains a maximal torus of G for the (fpqc) topology.
- (iv) $P_1 \cap P_2$ locally contains a maximal torus of G for the Zariski topology.

When S is semilocal, these conditions are moreover equivalent to:

- (v) $P_1 \cap P_2$ contains a maximal torus of G .

PROOF. ⁽²³⁾ Evidently, (iv) $\Rightarrow$ (iii) (and (v) $\Rightarrow$ (iv) when S is semilocal), and (ii) $\Rightarrow$ (i) by Exp. XXII, Def. 5.2.1. We shall show (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), then that (iii) implies (i) and (iv) (and also (v) when S is semilocal). Put $K = P_1 \cap P_2$. By 4.1.1 and [BT65], 4.5, each geometric fiber $K_{\bar{s}}$ contains a maximal torus $T_{\bar{s}}$ of $G_{\bar{s}}$ and is connected, and the set of roots of $K_{\bar{s}}$ relative to $T_{\bar{s}}$ is a closed subset of the set of roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$. Thus, if K is smooth over S , it is a subgroup of type (RC) (hence a fortiori of type (R)), cf. Exp. XXII 5.2.1 and 5.11.1. One therefore has (i) $\Leftrightarrow$ (ii).

If K is a subgroup of type (R), it locally contains a maximal torus of G for the étale topology, by Exp. XXII 2.2 and Exp. XIX 6.1. Thus (ii) $\Rightarrow$ (iii).

Suppose (iii) satisfied, and let us show that K is smooth. By (fpqc) descent, one can suppose that $G = (G, T, M, R)$ is split, where T is a maximal torus contained in $P \cap Q$, and that there exist two closed subsets R_1 and R_2 of R such that $P = H_{R_1}$ and $Q = H_{R_2}$.

Since K has connected fibers, it follows from Exp. XXII 5.4.5 that K equals $H_{R_1 \cap R_2}$, and so is smooth over S and of type (RC).

Let R_1^s be the symmetric part of R_1 , and $R_1^a = R_1 - R_1^s$; then $\text{rad}^u(P_1) = H_{R_1^a}$. As noted in the proof of [BT65], 4.4, $R' = (R_1^s \cap R_2) \cup R_1^a$ is a closed subset of R such that $R' \cup (-R') = R$; thus, by Exp. XXI 3.3.6, R' contains a system of positive roots of R . Moreover, the symmetric part of R' is $R_1^s \cap R_2^s$, which is contained in $R_1 \cap R_2$.

One deduces from the foregoing that $P' = K \cdot \text{rad}^u(P)$ is a parabolic subgroup of G , and that K is a subgroup of type (RC) of P' such that $\text{rad}^u(K) = K \cap \text{rad}^u(P')$. Thus, by 2.11, K possesses a Levi subgroup L . It then follows from Exp. XIV 3.20 and 3.21 that K satisfies assertion (iv), as well as assertion (v) when S is semilocal. This proves 4.5.1. $\square$

⁽²²⁾N.D.E. Recall that the involution s was defined in 4.3.1.

⁽²³⁾N.D.E. The proof of the equivalence of these conditions has been added (as has condition (v), used implicitly in 6.17 of the original); consequently no. 4.5.1 has been changed into proposition 4.5.1 plus definition 4.5.1.1.

Definition 4.5.1.1. ⁽²⁴⁾ When the preceding conditions hold, one says that P and Q are in *mutual standard position*; this is the case, for example, if P and Q are in transversal position, or in osculatory position, or if the base is the spectrum of a field. This notion is stable under extension of the base and local for the (fpqc) topology.

4.5.2. — Let (P, Q) and (P', Q') be two pairs of parabolic subgroups of G in standard position, and let $\underline{H}$ be the subfunctor of G defined as follows: $\underline{H}(S')$ is the set of $g \in G(S')$ such that $\text{int}(g)P = P'$ and $\text{int}(g)Q = Q'$. This is a closed subscheme of G , smooth over S , and formally principal homogeneous under $P \cap Q$. One deduces that the following conditions are equivalent:

- (i) (P, Q) and (P', Q') are locally conjugate for the (fpqc) topology,
- (ii) (P, Q) and (P', Q') are locally conjugate for the étale topology.
- (iii) (P, Q) and (P', Q') are conjugate on every geometric fiber.

One then says that the pairs (P, Q) and (P', Q') have the *same type of mutual position*. This notion is stable under base change and local for the (fpqc) topology.

4.5.3. — Let $\underline{\text{Stand}}(G)$ be the subfunctor of $\underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G)$ “consisting of pairs in mutual standard position”. Then $\underline{\text{Stand}}(G)$ is representable; there exist a finite étale S -scheme $\underline{\text{TypeStand}}$ (“scheme of types of mutual standard position”) and a smooth morphism of finite presentation, with irreducible geometric fibers (and hence in particular faithfully flat),

$$t_2 : \underline{\text{Stand}}(G) \longrightarrow \underline{\text{TypeStand}},$$

which is a quotient of $\underline{\text{Stand}}(G)$ by the action of G : two sections of $\underline{\text{Stand}}(G)$ (over any $S' \rightarrow S$) have the same type of mutual position if and only if they have the same image under t_2 .

One has a commutative diagram

$$\begin{array}{ccc} \underline{\text{Stand}}(G) & \xrightarrow{t_2} & \underline{\text{TypeStand}} \\ \downarrow & & \downarrow q \\ \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G) & \xrightarrow{t \times t} & \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \times_S \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \end{array}$$

where the morphism q can be described by descent as follows: if (P, Q) is a pair of parabolic subgroups of G in standard relative position, and if T is a maximal torus of $P \cap Q$, then the morphism $\underline{\text{Norm}}_G(T) \rightarrow \underline{\text{Stand}}(G)$ defined set-theoretically by $n \mapsto (P, \text{int}(n)Q)$ induces an isomorphism

$$W_P(T) \backslash W_G(T) / W_Q(T) \simeq q^{-1}(t(P), t(Q)).$$

(The first side denotes the sheaf of double cosets. . . .) These assertions are proved without difficulty (note in particular that $t_2^{-1}(t_2(P, Q)) \simeq G/(P \cap Q)$).

4.5.4. — Now let P be a fixed parabolic subgroup of G , and let $\underline{\text{Par}}(G; P)$ be the functor of parabolic subgroups of G in standard position relative to P . For each $t \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$, similarly put $\underline{\text{Par}}_t(G; P) = \underline{\text{Par}}(G; P) \cap \underline{\text{Par}}_t(G)$. One sees immediately that the preceding two functors are obtained from $\underline{\text{Stand}}(G)$ by fibered products, and so are representable by S -schemes smooth and of finite presentation over S , with nonempty fibers. One has a canonical morphism t_P induced by t_2 (i.e. $t_P(Q) = t_2(P, Q)$)

$$t_P : \underline{\text{Par}}_t(G; P) \longrightarrow q^{-1}(t(P), t)$$

which is smooth and of finite presentation, with irreducible geometric fibers. The canonical morphism $\underline{\text{Par}}_t(G; P) \rightarrow \underline{\text{Par}}_t(G)$ is a surjective monomorphism, and can therefore be

⁽²⁴⁾**N.D.E.** See N.D.E. (23). Moreover, for an example of parabolic subgroups P, Q which are not in standard position, see 7.11 below.

considered as a *cellular decomposition* of $\underline{\text{Par}}_t(G)$ (indexed by the set of connected components of $q^{-1}(t(P), t)$).

4.5.5. — Now suppose that the type t is of the form $t(Q)$, where Q is a parabolic subgroup of G in standard position relative to P , and that $P \cap Q$ contains a maximal torus T .

Then $\underline{\text{Par}}_t(G) \simeq G/Q$ and $q^{-1}(t(P), t) \simeq W_P(T) \backslash W_G(T) / W_Q(T)$, which gives a diagram

$$\begin{array}{ccc} \underline{\text{Par}}_{t(Q)}(G; P) & \xrightarrow{f} & W_P(T) \backslash W_G(T) / W_Q(T) \\ \downarrow i & & \\ \underline{\text{Par}}_{t(Q)}(G) & \xrightarrow{\sim} & G/Q \end{array}$$

where i is a surjective monomorphism, and where f is smooth and of finite presentation, with irreducible geometric fibers. Moreover, if Q_1 and Q_2 are two sections of $\underline{\text{Par}}_{t(Q)}(G; P)$ (over an $S' \rightarrow S$), that is, two parabolic subgroups of $G_{S'}$ conjugate (locally for (fpqc)) to Q and in standard position relative to $P_{S'}$, then Q_1 and Q_2 are conjugate by a section of P (locally for (fpqc)) if and only if $f(Q_1) = f(Q_2)$. If S is the spectrum of an algebraically closed field k , one thus finds the relation

$$P(k) \backslash G(k) / Q(k) \simeq W_P(T)(k) \backslash W_G(T)(k) / W_Q(T)(k).$$

In general, if one assumes that the scheme $W_P(T) \backslash W_G(T) / W_Q(T)$ is constant and of the form E_S (which is the case, for example, when G is split relative to T and S is connected), the $f^{-1}(e)$, $e \in E$, form a decomposition of $\underline{\text{Par}}_{t(Q)}(G; P)$ into open and closed subschemes, which are homogeneous spaces under P , smooth and of finite presentation over S , with irreducible geometric fibers.

4.5.6. — Let us return to the general situation of 4.5.4. The scheme $q^{-1}(t(P), t)^{(25)}$ always has two particular sections, corresponding respectively to the types “transversal position” and “osculatory position”. The inverse image of the first section is a relatively dense open subscheme of $\underline{\text{Par}}_t(G; P)$, as one has seen above; it is the cell of maximum relative dimension of the decomposition.

The inverse image of the second section is presumably a closed subscheme of $\underline{\text{Par}}_t(G; P)$; it is the cell of minimum relative dimension of the decomposition.

5. Conjugacy theorem

Theorem 5.1. *Let S be a semi-local scheme, G a reductive S -group, P and P' two opposite parabolic subgroups (4.3.3) of G . Then*

$$\text{rad}^u(P)(S) P' \cdot P = G,$$

i.e. the union of the open subschemes $uP' \cdot P$ (4.3.6), as u ranges over $\text{rad}^u(P)(S)$, is the whole of G .

The proof proceeds in several steps:

5.1.1. — It suffices to give the proof in the case where S is the spectrum of a field k ; this follows at once from 2.6.

5.1.2. — Let $L = P \cap P'$. Suppose that L has a Borel subgroup B_L ; let T be a maximal torus of B_L (Exp. XXII, 5.9.7); one checks at once that $B = B_L \cdot \text{rad}^u(P)$ is a Borel subgroup of P ; let B' be the Borel subgroup of G opposite to B relative to T (i.e. such that $B \cap B' = T$). One has $B' \subset P'$, as one checks at once by splitting G relative to T . Let us prove that

$$(x) \quad B^u(S) B' \cdot B \subset \text{rad}^u(P)(S) P' \cdot P.$$

⁽²⁵⁾**N.D.E.** One has corrected $t^{-1}(t(P), t)$ to $q^{-1}(t(P), t)$ and, below, $\underline{\text{Par}}_t(G)$ to $\underline{\text{Par}}_t(G; P)$ (twice).

Since one has

$$B^u(S) \subset P(S) = \text{rad}^u(P)(S) \cdot L(S) \subset \text{rad}^u(P)(S)P'(S),$$

it suffices to prove that $B' \cdot B \subset P' \cdot P$, which is obvious. It follows from (x) that it suffices to prove 5.1 for the pair (B, B') .

5.1.3. — The theorem is true if k is algebraically closed; indeed, the condition of 5.1.2 is satisfied, and one concludes by Exp. XXII, 5.7.10.

5.1.4. — The theorem is true when k is an infinite field. Indeed, $\text{rad}^u(P)(k)$ is dense in $\text{rad}^u(P)(\bar{k})$ by 2.7, and the theorem is true for $\bar{k}$.

5.1.5. — One is therefore reduced to the case where k is a finite field. Now $\text{Bor}(L)$ is a *smooth* homogeneous space of L ; it therefore follows from Lang's theorem (Am. J. of Maths., 78, 1956⁽²⁶⁾) that L has a Borel subgroup B_L . By 5.1.2, one can therefore assume that $P = B$ and $P' = B'$ are Borel subgroups. One writes $T = B \cap B'$.

5.1.6. — Let K be the algebraic closure of k ; choose a pinning of the triple (G_K, B_K, T_K) ; let R_+ (resp. Δ) be the set of positive (resp. simple) roots. By Exp. XXII, 5.7.2, it suffices to prove that for every $\alpha \in \Delta$, one has

$$(1) \quad u_\alpha B'^u(K) \subset B^u(k) B'^u(K) B(K).$$

Let α_i be the different roots conjugate to α over k (they are elements of Δ , since B is “defined over k ”), and let R' be the set of roots that are linear combinations of the α_i . Write $R'_- = R_- \cap R'$. Since “ R' is defined over k ”, there is a subtorus Q of T such that Q_K is the maximal torus of the common kernel of the α_i .

Write $Z = \text{Centr}_G(Q)$, $B_Z = B \cap Z$, $B'_Z = B' \cap Z$ (cf. Exp. XXII, 5.10.2). Let us show that it suffices to verify the desired assertion in Z , that is,

$$(2) \quad u_\alpha \cdot B'^u_Z(K) \subset B^u_Z(k) B'^u_Z(K) B_Z(K).$$

One has $(B'^u_Z)_K = \prod_{\alpha \in R'_-} U_\alpha$; let R'' be the complement of R' in R , and put

$$V = \prod_{\alpha \in R'' \cap R_-} U_\alpha.$$

One has at once $(B'^u)_K = (B'^u_Z)_K \cdot V$, and $(B_Z)_K$ normalizes V (Exp. XXII, 5.6.7). One therefore deduces successively from (2)

$$\begin{aligned} u_\alpha \cdot B'^u(K) &= u_\alpha \cdot B'^u_Z(K) V(K) \\ &\subset B^u_Z(k) B'^u_Z(K) B_Z(K) V(K) \\ &\subset B^u_Z(k) B'^u_Z(K) V(K) B_Z(K), \end{aligned}$$

which implies (1) at once.

We are therefore reduced to the case where $G = Z$, that is, where the Galois group of K over k acts transitively on the simple roots.

5.1.7. — The assertion to be proved is equivalent to the fact that G/B is the union of the translates by $B^u(k)$ of the open image of B'^u , an assertion that does not change if one replaces G by its adjoint group (or, for that matter, by any group giving the same adjoint group). One can therefore assume that G is adjoint.

5.1.8. — Consider then the Dynkin diagram of G_K . The Galois group acts *transitively* on this Dynkin diagram. But this Galois group has only cyclic quotients, and the Dynkin

⁽²⁶⁾N.D.E. See also [DG70], § III.5, 7.4.

diagram has no cycles. It follows at once that this diagram is of type nA_1 , $n \geq 0$, or mA_2 , $m \geq 0$. Using the canonical decomposition of Exp. XXIV, 5.9, one can write

$$G = \prod_{D/K} G_0,$$

where D is a *finite* K -scheme and G_0 is either a torus, or of type A_1 , or of type A_2 .

By Exp. XXIV, 5.12, B comes from a Borel subgroup B_0 of G_0 , T from a maximal torus T_0 of G_0 ; B' comes from the Borel subgroup B'_0 of G_0 opposite to B_0 relative to T_0 . One has

$$B'^u(k) = B_0^u(D), \quad B'^u \cdot T \cdot B^u = \prod_{D/k} B_0'^u \cdot T_0 \cdot B_0^u,$$

and it suffices for us to prove the desired assertion for the triple (G_0, B_0, T_0) .

5.1.9. — One can therefore assume that G is of type $\emptyset$, A_1 or A_2 . Since G has a Borel subgroup B , G is quasi-splittable relative to B (Exp. XXIV, 3.9.1), hence splittable if it is of type $\emptyset$ or A_1 . Since the theorem has already been proved in the split case (Exp. XXII 5.7.10), only the case A_2 remains to be treated. By Exp. XXIV, 3.11 there is a morphism $E \rightarrow \text{Spec}(k)$, a Galois principal bundle under the group $\mathbf{Z}/2\mathbf{Z}$ of automorphisms of the Dynkin diagram of type A_2 , such that $G = \text{Q-Pin}_{E/\text{Spec}(k)}(A_2)$. If E has a section, G is splittable and the theorem is proved. Otherwise, one necessarily has $E = \text{Spec}(k')$, where k' is a quadratic extension of k . Finally, as one has seen in 5.1.7, one can assume that G is simply connected (i.e. that G is a form of $\text{SL}_{3,k}$).

5.1.10. — One is therefore in the following situation: one has a finite field k , a quadratic extension k' of k . The group $\text{SL}_{3,k'}$ of 3×3 matrices of determinant 1 is pinned as follows: the maximal torus is the group of diagonal matrices, the Borel subgroup is the group of upper triangular matrices, the “pins” are the elements:

$$u_\alpha = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad u_\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

One sees at once that the big cell Ω is defined by

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in \Omega(S) \iff a \text{ and } ae - bd \text{ are invertible,}$$

and that

$$B^u(k) = \left\{ \begin{pmatrix} 1 & \bar{x} & z \\ 0 & 1 & x \\ 0 & 0 & 1 \end{pmatrix} \middle| x, z \in k', \quad z + \bar{z} = x\bar{x} \right\}.$$

We must prove inclusion (1) of 5.1.6, that is, show that for all $a, b, c \in K$ (algebraic closure of k), there exist $x, z \in k'$ such that $z + \bar{z} = x\bar{x}$ and

$$(1) \quad \begin{pmatrix} 1 & \bar{x} & z \\ 0 & 1 & x \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{pmatrix} \in \Omega(K).$$

- If $a \neq -1$, one takes $x = z = 0$.
- If $a = -1$, the conditions to be satisfied read

$$\begin{cases} z + \bar{z} = x\bar{x}, \\ bz - \bar{x} \neq 0, \\ (b + c)\bar{z} + bx - 1 \neq 0. \end{cases}$$

Let q be the number of elements of k ($q \geq 2$). One knows that for every $m \in k$, the equation $z + \bar{z} = m$, with $z \in k'$, has q solutions.

- If $b = 0$, take $x = 1$; one must solve $z + \bar{z} = 1$, $c\bar{z} \neq 1$, which is always possible by the preceding remark.
- If $b \neq 0$, take $x = 0$; one must solve

$$z + \bar{z} = 0, \quad z \neq 0, \quad (b + c)z \neq 1.$$

This is always possible if $q \geq 3$. If $k = \mathbf{F}_2$, it is possible if $b + c \neq 1$; one can take $z = 1$.

- Only the case $k = \mathbf{F}_2$, $b + c = 1$, $b \neq 0$ therefore remains to be treated. The system then reads

$$z + \bar{z} = x\bar{x}, \quad bz \neq \bar{x}, \quad \bar{z} + bx \neq 1.$$

If $b = 1$ (resp. $b \notin k'$), put $x = 1$; then the last two conditions read $z \neq b^{-1}, 1 - b$, and they follow from $z + \bar{z} = 1$, which has solutions. Finally, if $b \in k' - k$, one can take $x = \bar{b}$, $z = b$. Q.E.D.

Corollary 5.2. *Let S be a semi-local scheme, G a reductive S -group, P and P' two opposite parabolic subgroups of G . The canonical map*

$$\mathrm{rad}^u(P)(S) \cdot \mathrm{rad}^u(P')(S) \longrightarrow (G/P)(S)$$

is surjective (in particular, one has $(G/P)(S) = G(S)/P(S)$). Every parabolic subgroup Q of G of the same type as P is of the form $\mathrm{int}(uu')P$, with $u \in \mathrm{rad}^u(P)(S)$ and $u' \in \mathrm{rad}^u(P')(S)$.

The second assertion is obviously equivalent to the first; let us prove the latter. Let s_i be the closed points of S , let V be the open subscheme of G/P that is the image of P' (and is isomorphic to $\mathrm{rad}^u(P')$, cf. 4.3.6), and let $x \in G/P(S)$. By 5.1, for each i there is a section $u_i \in \mathrm{rad}^u(P)(\kappa(s_i))$ such that $u_i x_{s_i}$ is a section of V_{s_i} . If $u \in \mathrm{rad}^u(P)(S)$ lifts the u_i (2.6), ux is a section of V , since such an assertion is checked on the closed fibers. But $\mathrm{rad}^u(P')(S) \xrightarrow{\sim} V(S)$ is bijective, and one concludes at once.

Corollary 5.3. *Let S be a semi-local scheme, G a reductive S -group, P , P' and Q three parabolic subgroups of G . There is $g \in G(S)$ such that $\mathrm{int}(g)Q$ is in transversal position relative to P and P' .*

With the notation of 4.2.4(ii), one must verify that the open subscheme of G that is universally schematically dense over $S^{(27)}$, $\underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P')$, has a section over S . In fact, choose a parabolic subgroup of G opposite to Q (4.3.5(i)), say Q_1 , and put $U = \mathrm{rad}^u(Q)$, $U' = \mathrm{rad}^u(Q_1)$. We shall show that there is $g \in U(S)U'(S)$ answering the question; in this form it follows from 4.2.4(i) and 2.6 that it suffices to verify the assertion on the fibers at the closed points of S , and one can therefore assume that S is the spectrum of a field k .

If k is algebraically closed, there is $g \in G(k)$ answering the question; now g is written $uu'q$, with $u \in U(k)$, $u' \in U'(k)$, $q \in Q(k)$ (5.2), and one has $\mathrm{int}(uu')Q = \mathrm{int}(g)Q$.

If k is infinite, consider the open subscheme V of $U \times_k U'$ defined by the cartesian diagram

$$\begin{array}{ccc} U \times_k U' & \longrightarrow & G \\ \uparrow & & \downarrow \\ V & \longrightarrow & \underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P') \end{array}$$

since $V(\bar{k}) \neq \emptyset$ by what one has just seen, V is dense in $U \times_k U'$, hence has a section by 2.7.

If k is finite, P (resp. P') has a Borel subgroup B (resp. B'), by Lang's theorem (cf. 5.1.5), the schemes $\underline{\mathrm{Bor}}(P) \simeq \underline{\mathrm{Bor}}(P/\mathrm{rad}^u(P))$ and $\underline{\mathrm{Bor}}(Q) \simeq \underline{\mathrm{Bor}}(Q/\mathrm{rad}^u(Q))$ being smooth. If B_1 is a Borel subgroup opposite to B (4.3.5(i)), there are $a \in B^u(k)$ and $a_1 \in B_1^u(k)$ such that $\mathrm{int}(aa_1)B = B'$ (5.2); then $B_0 = \mathrm{int}(aa_1)B_1 = \mathrm{int}(a)B_1$ is opposite to B' and to B ; if Q_0 is the unique parabolic subgroup of G containing B_0 and of the same type as Q (3.8), Q_0

⁽²⁷⁾**N.D.E.** One has replaced “relatively dense” by “universally schematically dense over S ”, cf. EGA IV₃, Def. 11.10.8.

is in transversal position relative to P and P' (4.2.1(vi)). Moreover, by 5.2, Q_0 is written $\text{int}(uu')Q$, with $u' \in U'(k)$, $u \in U(k)$, which was to be proved.

Corollary 5.4. *Let S be a semi-local scheme, G a reductive S -group, P and Q two parabolic subgroups of G . There is $g \in G(S)$ such that $\text{int}(g)P$ is in osculatory position relative to Q , i.e. (4.4.2) such that $\text{int}(g)P \cap Q$ is a parabolic subgroup of G .*

Indeed, by 4.3.5(i), there is a parabolic subgroup P' of G opposite to P . By 5.3, there is a parabolic subgroup P'_1 of G of the same type as P' , in transversal position relative to P and Q . If T is a maximal torus of $P'_1 \cap Q$ (4.2.7(ii)), and if P_1 is the opposite of P'_1 relative to T , then P_1 and Q are in osculatory position, by 4.4.5. Moreover, since P and P_1 are opposite to P'_1 , there is $g \in \text{rad}^u(P'_1)(S)$ such that $\text{int}(g)P = P_1$ (4.3.5(i)). Q.E.D.

Observe, moreover, that for the same reason, there is $u \in \text{rad}^u(P)(S)$ such that $\text{int}(u)P' = P'_1$, hence g is written $\text{int}(u)u'$, with $u' \in \text{rad}^u(P')(S)$, which gives $P_1 = \text{int}(uu'u^{-1})P = \text{int}(uu')P$ and, in passing, proves 5.2 again.

Statements 5.3 and 5.4 are the essential results of this paragraph. Let us first state a few consequences of 5.4.

Corollary 5.5. *Let S be a semi-local scheme, G a reductive S -group.*

- (i) *If P and Q are two parabolic subgroups of G and if $t(P) \subset t(Q)$ (cf. 3.3), there is $g \in G(S)$ such that $\text{int}(g)P \subset Q$.*
- (ii) *Let*

$$P_1 \supset P_2 \supset \cdots \supset P_n \quad \text{and} \quad P'_1 \supset P'_2 \supset \cdots \supset P'_n$$

be two chains of parabolic subgroups of G such that $t(P_i) = t(P'_i)$. There is $g \in G(S)$ such that $\text{int}(g)P_i = P'_i$ for every i .

- (iii) *Let P, Q, P', Q' be four parabolic subgroups of G such that $t(P) = t(P')$ and $t(Q) = t(Q')$. If the pairs (P, P') and (Q, Q') are in transversal (resp. osculatory) position, there is $g \in G(S)$ such that $\text{int}(g)P = P'$ and $\text{int}(g)Q = Q'$.*
- (iv) *Let P and P' be two parabolic subgroups of the same type, L (resp. L') a Levi subgroup of P (resp. P'). There is $g \in G(S)$ such that $\text{int}(g)P = P'$ and $\text{int}(g)L = L'$.*

PROOF. (i) follows at once from 5.4; (ii) is proved by induction on n , the case $n = 0$ being trivial; one can therefore assume that $P_i = P'_i$ for $i = 1, \dots, n-1$; by 5.2 there is $g \in G(S)$ such that $\text{int}(g)P_n = P'_n$; but then P_n and $\text{int}(g)P_n$ are contained in $P_{n-1} = P'_{n-1}$, hence $g \in P_{n-1}(S)$ (4.4.4(ii)) and $\text{int}(g)P_i = P'_i$ for $i = 1, \dots, n$.

Moreover, (iv) follows at once from 5.2 and 1.8. Let us prove (iii) in the case “transversal position”; the assertion follows from (iv) when the types of P and Q are opposite (4.3.3(iii)); in the general case, by 4.2.7(iii) and 4.4.6 one can find parabolic subgroups P_1, P'_1, Q_1, Q'_1 of P, P', Q, Q' respectively, such that P_1 and P'_1 are opposite, as are Q_1 and Q'_1 , and such that $t(P_1) = t(P'_1)$; there is therefore $g \in G(S)$ such that $\text{int}(g)P_1 = P'_1$, $\text{int}(g)Q_1 = Q'_1$, and one can assume that $P_1 = P'_1$ and $Q_1 = Q'_1$; but then P and P' are in osculatory position and of the same type, hence $P = P'$ (4.4.4(i)); for the same reason $Q = Q'$.

It remains for us to prove assertion (iii) in the case “osculatory position”. By the conjugacy theorem (5.2), one can assume that $P = P'$; by the same theorem, one can find $g \in G(S)$ such that $\text{int}(g)Q = Q'$; but then $g \in P(S)$ by 4.4.4(ii), and one has $\text{int}(g)P = P = P'$. □

Definition 5.6. Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G . One says that P is *minimal* if whenever Q is a parabolic subgroup of G contained in P , one has $Q = P$.

One will note that this notion is not, in general, stable under passage to fibers.

Corollary 5.7. *Let S be a semi-local scheme, G a reductive S -group.*

- (i) *Let $t, t' \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$. If there exist in G a parabolic subgroup of type t and a parabolic subgroup of type t' , there exists a parabolic subgroup of type $t \cap t'$. In*

particular, there is a least element $t_{\min}$ in the set of the $t(P)$, as P ranges over the set of parabolic subgroups of G .

- (ii) Every parabolic subgroup of G contains a minimal parabolic subgroup. For a parabolic subgroup of G to be minimal, it is necessary and sufficient that it be of type $t_{\min}$. Two minimal parabolic subgroups of G are conjugate by an element of $G(S)$.

This follows at once from 5.4 and 5.5(i).

Remark 5.8. A parabolic subgroup opposite to a minimal parabolic subgroup is also minimal; this implies $s(t_{\min}) = t_{\min}$.

Corollary 5.9. Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G . The canonical morphism

$$G \longrightarrow G/P = X$$

makes G a locally trivial X -bundle (in the Zariski sense) with group P_X . If L is a Levi subgroup of P , the canonical morphism (cf. 3.12)

$$G \longrightarrow G/L = Y$$

makes G a locally trivial Y -bundle (in the Zariski sense) with group L_Y .

It suffices to prove that if one has a morphism $S' \longrightarrow S$, where S' is local, and a morphism $S' \longrightarrow X$ (resp. $S' \longrightarrow Y$), it lifts to a morphism $S' \longrightarrow G$. In other words, one can assume that S is local and must show that the map $G(S) \longrightarrow X(S)$ (resp. $G(S) \longrightarrow Y(S)$) is surjective.

The first assertion was proved in 5.2; let us prove the second. Let $y \in Y(S)$; its canonical image in $X(S)$ comes from a $g \in G(S)$; the projection y' of g in $Y(S)$ therefore has the same projection as y in $X(S)$. There is therefore a unique $u \in \text{rad}^u(P)(S)$ such that $y'u = y$, and the projection of gu in $Y(S)$ is indeed y .

Corollary 5.10. Let S be a semi-local scheme, G a reductive S -group.

- (i) Let P be a parabolic subgroup of G , L a Levi subgroup of P . The canonical maps (cf. 3.21) induce bijections

$$H^1(S, L) \xrightarrow{\sim} H^1(S, P) \xrightarrow{\sim} H^1_{t(P)}(S, G).$$

- (ii) Let $t, t' \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$; one has (cf. 3.21)

$$H^1_t(S, G) \cap H^1_{t'}(S, G) = H^1_{t \cap t'}(S, G).$$

- (iii) If P and Q are two parabolic subgroups of G in osculatory position, the following canonical diagram is cartesian and consists of injections:

$$\begin{array}{ccc} H^1(S, P) & \longrightarrow & H^1(S, G) \\ \uparrow & & \uparrow \\ H^1(S, P \cap Q) & \longrightarrow & H^1(S, Q) \end{array}$$

Let us prove (i). The map $H^1(S, L) \longrightarrow H^1(S, P)$ is bijective by 2.3; the map $H^1(S, P) \longrightarrow H^1_{t(P)}(S, G)$ is surjective (3.21); let us show that it is injective, i.e. that the canonical map $H^1(S, P) \longrightarrow H^1(S, G)$ is injective. Let Q be a principal bundle under P , Q_1 the associated principal bundle under G , P' and G' the corresponding twisted forms of P and G . It is clear that G' is a reductive S -group and that P' is a parabolic subgroup of it. The set of elements of $H^1(S, P)$ that have the same image as the class of Q in $H^1(S, G)$ is naturally identified with the kernel of the canonical map $H^1(S, P') \longrightarrow H^1(S, G')$, and the latter, by the cohomology exact sequence, with the set of orbits of $G'(S)$ in $(G'/P')(S)$ (for these

arguments in nonabelian cohomology, see Giraud's thesis⁽²⁸⁾). But $G'(S)$ acts transitively on $(G'/P')(S)$ by 5.2.

Let us prove (ii): let Q be a principal homogeneous bundle under G and G^Q the corresponding twisted form of G . By definition (3.21), we must prove that G^Q has a parabolic subgroup of type $t \cap t'$ if and only if it has parabolic subgroups of type t and t' , which is simply the conjunction of 3.8 and 5.7(i). Finally, (iii) follows at once from (i) and (ii).

Let us now state a consequence of 5.3.

Corollary 5.11. *Let S be a semi-local scheme, G a reductive S -group, P a parabolic subgroup of G . If $P \neq G$, there are at least three distinct parabolic subgroups of G of the same type as P ; in other words, $P \neq G$ implies $(G(S) : P(S)) \geq 3$.*

Indeed, let P' be a parabolic subgroup of G opposite to P (4.3.5(i)). Since $P \neq G$, one has $\text{rad}^u(P') \neq e$ (by 4.3.2, for example). By 2.1, $\text{rad}^u(P')(S) \neq e$; let therefore $u \in \text{rad}^u(P')(S)$, $u \neq e$. Then $\text{int}(u)P \neq P$, and by 5.3 there is a P_1 of the same type as P and opposite to P and $\text{int}(u)P$; then P_1, P and $\text{int}(u)P$ are three distinct parabolic subgroups of G of the same type as P .

6. Parabolic subgroups and split tori

Proposition 6.1. *Let S be a scheme, G a reductive S -group, $\mathfrak{g} = \text{Lie}(G/S)$, and Q a split subtorus of G . Write $Q = D_S(M)$ and let*

$$\mathfrak{g} = \coprod_{\alpha \in M} \mathfrak{g}^\alpha$$

be the decomposition of $\mathfrak{g}$ under the action of Q . Let M_1 be a subset of M such that $0 \in M_1$ and $\alpha, \beta \in M_1 \Rightarrow \alpha + \beta \in M_1$.

- (i) *There is a unique smooth subgroup H_{M_1} of G , with connected fibers, containing $\underline{\text{Centr}}_G(Q)$ and whose Lie algebra is $\coprod_{\alpha \in M_1} \mathfrak{g}^\alpha$.*
- (ii) *One has the following implications:*

$$\begin{aligned} M_1 = \{0\} &\implies H_{M_1} = \underline{\text{Centr}}_G(Q), \\ M_1 = -M_1 &\implies H_{M_1} \text{ is reductive,} \\ M_1 \cup (-M_1) = M &\implies H_{M_1} \text{ and } H_{-M_1} \text{ are opposite} \\ &\quad \text{parabolic subgroups of } G, \text{ with common} \\ &\quad \text{Levi subgroup } H_{M_1 \cap -M_1}. \end{aligned}$$

To prove (i) and (ii), which are local for the (fpqc) topology, one can assume that Q is contained in a maximal torus T of G ; one can moreover split G relative to T . Assertion (i) then follows at once from Exp. XXII, 5.3.5, 5.4.5 and 5.4.7; the assertions of (ii) follow from Exp. XXII, 5.3.5, 5.10.1, 5.11.3 and from this exposé, 1.4 and 4.3.2.

Corollary 6.2. *Let S be a scheme, G a reductive S -group, Q a split subtorus of G . There is a parabolic subgroup of G of which $\underline{\text{Centr}}_G(Q)$ is a Levi subgroup.*

Indeed, writing $Q = D_S(M)$, one chooses a total ordering on the group M and calls M_1 the set of positive elements of M ; the group H_{M_1} answers the question.

Corollary 6.3. *If the reductive S -group G has a noncentral split subtorus, it has a proper parabolic subgroup (i.e. $\neq G$).*

By 5.9 and 5.10, one deduces from 6.2:

⁽²⁸⁾N.D.E. See [Gi71], § III.3.

Corollary 6.4. *Let S be a scheme, G a reductive S -group, Q a split subtorus of G . The canonical morphism $G \rightarrow G/\underline{\text{Centr}}_G(Q)$ is a locally trivial fibration.*

If S is semi-local, the map $G(S) \rightarrow (G/\underline{\text{Centr}}_G(Q))(S)$ is surjective, and the map $H^1(S, \underline{\text{Centr}}_G(Q)) \rightarrow H^1(S, G)$ is injective.

6.5. — Suppose that S is connected. If T is an S -torus and if T' and T'' are two split subtori of T , their product $T' \cdot T''^{(29)}$ is also a split subtorus of T . Indeed, it is identified with the quotient of $T' \times_S T''$ by $T' \cap T''$, a quotient that is split by Exp. IX, 2.11. It follows that T has a largest split subtorus; one denotes it by T_{spl} .

Lemma 6.6. *Let S be a connected scheme, T an isotrivial S -torus, T_{spl} its largest split subtorus. The following conditions are equivalent:*

- (i) *There is a homomorphism $T \rightarrow \mathbf{G}_{m,S}$ distinct from e .*
- (ii) *$T_{\text{spl}} \neq e$.*

Since T is assumed to be isotrivial, there are a finite group Γ , a connected Galois principal covering $S' \rightarrow S$ with group Γ , and an isomorphism $T_{S'} \simeq D_{S'}(M)$; M is then endowed with a Γ -module structure, and one has a natural isomorphism $\text{Hom}_S(T, \mathbf{G}_{m,S}) = H^0(\Gamma, M)$.

Moreover, let V be the vector subspace of $M \otimes \mathbf{Q}$ generated by the elements of the form $g(m) - m$, $g \in \Gamma$, $m \in M$. One checks at once that $(T_{\text{spl}})_{S'}$ is identified with $D_{S'}(M/M \cap V)$. Assertion (i) is therefore equivalent to $H^0(\Gamma, M) \neq 0$, or again to $H^0(\Gamma, M \otimes \mathbf{Q}) \neq 0$, whereas assertion (ii) is equivalent to $M \neq M \cap V$, or again to $M \otimes \mathbf{Q} \neq V$. Now one has $M \otimes \mathbf{Q} = H^0(\Gamma, M \otimes \mathbf{Q}) \oplus V$, as one checks at once (consider the projector $M \otimes \mathbf{Q} \rightarrow H^0(\Gamma, M \otimes \mathbf{Q})$ that sends x to the average of its transforms by Γ).

Lemma 6.7. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G such that $P \neq G$, L a Levi subgroup of P , Q its radical. There is a homomorphism $Q \rightarrow \mathbf{G}_{m,S}$ distinct from e .*

Consider the unipotent radical U of P ; it is invariant under $\text{int}(P)$, hence under $\text{int}(Q)$. Consider the invertible $\mathcal{O}_S$ -module $\det(\text{Lie}(U))$, the “maximum exterior power” of the locally free $\mathcal{O}_S$ -module $\text{Lie}(U)$. The adjoint representation defines a group homomorphism

$$f: Q \rightarrow \underline{\text{Aut}}(\det(\text{Lie}(U))) = \mathbf{G}_{m,S}.$$

If $P \neq G$, then $U \neq e$. Choose an $s \in S$ such that $U_{\bar{s}} \neq e$. Splitting $G_{\bar{s}}$ relative to a maximal torus containing $Q_{\bar{s}}$, one sees at once that $f_{\bar{s}} \neq e$.

Proposition 6.8. *Let S be a connected semi-local scheme, G a reductive S -group, P a parabolic subgroup of G , L a Levi subgroup of P , Q its radical, Q_{spl} the largest split subtorus of Q (i.e. the largest split central subtorus of L). Then*

$$L = \underline{\text{Centr}}_G(Q_{\text{spl}}).$$

Put $L' = \underline{\text{Centr}}_G(Q_{\text{spl}})$; it is a reductive subgroup of G containing L ; moreover, $P' = P \cap L'$ is a parabolic subgroup of L' , with Levi subgroup L (1.20). If $L' \neq L$, then $L' \not\subset P$ (since L is a maximal reductive subgroup of P , cf. 1.7), hence $P' \neq L'$.

Let G_1 be the derived group of G , and $P_1 = P' \cap G_1$. By 1.19, P_1 is a parabolic subgroup of the semisimple group G_1 , $L_1 = L \cap G_1$ is a Levi subgroup of it, and

$$Q_1 = \text{rad}(L_1) = (\text{rad}(L) \cap G_1)^0 = (Q \cap G_1)^0.$$

Since L_1 has a maximal torus T_1 (Exp. XIV, 3.20), and the latter is isotrivial (Exp. XXIV, 4.1.5), Q_1 , which is a subtorus of T_1 , is also isotrivial (Exp. IX, 2.11); since $P_1 \neq G_1$, one can apply 6.7 and 6.6, and therefore has $(Q_1)_{\text{spl}} \neq e$, whence $(Q_{\text{spl}} \cap G_1)^0 \neq e$, hence $Q_{\text{spl}} \not\subset \text{rad}(L')$ (since $\text{rad}(L') \cap G_1$ is finite), which contradicts the definition of L' .

⁽²⁹⁾**N.D.E.** One has adopted multiplicative notation, i.e. replaced “their sum $T' + T''$ ” by “their product $T' \cdot T''$ ”.

Corollary 6.9. *Let S be a connected semi-local scheme, G a reductive S -group, Q a critical subtorus of G (i.e. such that $\text{rad}(\text{Centr}_G(Q)) = Q$). For $\text{Centr}_G(Q)$ to be a Levi subgroup of a parabolic subgroup of G , it is necessary and sufficient that $\text{Centr}_G(Q) = \text{Centr}_G(Q_{\text{spl}})$, that is, that $\text{Lie}(G)^Q = \text{Lie}(G)^{Q_{\text{spl}}}$.*

This follows from 6.2 and 6.8.

Corollary 6.10. *Let S be a connected semi-local scheme, G a reductive S -group, L a subgroup of G . The following conditions are equivalent:*

- (i) *There is a parabolic subgroup of G of which L is a Levi subgroup.*
- (ii) *There is a split subtorus of G of which L is the centralizer.*
- (iii) *There is a homomorphism $\mathbf{G}_{m,S} \rightarrow G$ of which L is the centralizer.*

Indeed, one has (i) $\Rightarrow$ (ii) by 6.8, and (iii) $\Rightarrow$ (i) by 6.1; it remains to prove (ii) $\Rightarrow$ (iii). Suppose therefore that $L = \text{Centr}_G(Q)$, with $Q = D_S(M)$; write $\mathfrak{g} = \text{Lie}(G/S)$ and

$$\mathfrak{g} = \coprod_{\alpha \in M} \mathfrak{g}^\alpha,$$

and let R be the set of the $\alpha \in M - \{0\}$ such that $\mathfrak{g}^\alpha \neq 0$.

Since R is finite and does not contain 0, there is a homomorphism $u : M \rightarrow \mathbf{Z}$ such that $u(\alpha) \neq 0$ for every $\alpha \in R$. By duality, u gives a homomorphism $\mathbf{G}_{m,S} \rightarrow Q$, hence a homomorphism $f : \mathbf{G}_{m,S} \rightarrow G$. One has $\text{Centr}_G(f) \supset \text{Centr}_G(Q)$; these are two smooth subgroups of G , with connected fibers; their Lie algebras coincide (since both are equal to $\mathfrak{g}^0$); they therefore coincide, by a usual argument.

Corollary 6.11. *Let S be a connected semi-local scheme, G a reductive S -group. The maps*

$$L \mapsto \text{rad}(L)_{\text{spl}}, \quad Q \mapsto \text{Centr}_G(Q)$$

are mutually inverse bijections, reversing the natural order structures, between the set of subgroups L of G that are Levi subgroups of parabolic subgroups of G and the set of split subtori Q of G such that $\text{rad}(\text{Centr}_G(Q))_{\text{spl}} = Q$.

Corollary 6.12. *Let S be a connected semi-local scheme, G a reductive S -group. Consider the following assertions:*

- (i) *There is a parabolic subgroup of G distinct from G .*
- (ii) *G has a noncentral split subtorus.*
- (ii bis) *G has a noncentral split subtorus of relative dimension 1.*
- (iii) *There is a group homomorphism $\mathbf{G}_{a,S} \rightarrow G$ that is a closed immersion.*

Then one has (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (ii bis) $\Rightarrow$ (iii).

The only new assertion is (i) $\Rightarrow$ (iii). Let therefore P be a parabolic subgroup of G distinct from G . Then $U = \text{rad}^u(P) \neq e$. Consider the last nontrivial subgroup U_n of the composition series of U (2.1). One has an isomorphism $U_n \simeq W(\mathcal{E}_n)$, where $\mathcal{E}_n$ is a locally free $\mathcal{O}_S$ -module, hence free⁽³⁰⁾. Since $\mathcal{E}_n \neq 0$, there is a monomorphism $\mathcal{O}_S \rightarrow \mathcal{E}_n$ that is locally a direct summand, hence a closed immersion $\mathbf{G}_{a,S} = W(\mathcal{O}_S) \hookrightarrow W(\mathcal{E}_n) \simeq U_n$, which implies (iii) at once.

Remark 6.12.1. When S is the spectrum of a field of characteristic 0, it follows from the Jacobson–Morozov theorem that (iii) $\Rightarrow$ (ii bis). The four preceding conditions are then equivalent (“Godement’s criterion”, cf. [BT65], 8.5).*

⁽³⁰⁾**N.D.E.** Since S is assumed to be semi-local and connected.

*This is more generally true when S is the spectrum of a perfect field (Tits).³¹

⁽³¹⁾**N.D.E.** This is Corollary 3.7 of the article [BT71].

Definition 6.13. Let S be a connected semi-local scheme, G a reductive S -group. One says that G is *anisotropic* if G contains no split subtorus not reduced to e .

Corollary 6.14. *Let S be a connected semi-local scheme. For the reductive S -group G to be anisotropic, it is necessary and sufficient that it have no parabolic subgroup $P \neq G$ and that its radical be anisotropic.*

Using now 6.6, Exp. XXIV, 4.1.5, and Exp. XXII, 6.2, one deduces:

Corollary 6.15. *Let S be a connected semi-local scheme, G an isotrivial reductive S -group (for example, G semisimple, or S normal (Exp. X 5.16)). For G to be anisotropic, it is necessary and sufficient that G have no parabolic subgroup $P \neq G$ and that $\text{Hom}_{S\text{-gr.}}(G, \mathbf{G}_{m,S}) = e$.*

Proposition 6.16. *Let S be a connected semi-local scheme, G a reductive S -group. The maximal split subtori of G are the largest split central subtori of the Levi groups of the minimal parabolic subgroups of G . Two such tori are conjugate by an element of $G(S)$.*

Let Q be a maximal split subtorus of G .⁽³²⁾ Then, by 6.2, $L = \underline{\text{Centr}}_G(Q)$ is a Levi subgroup of a parabolic subgroup P of G , and since $Q \subset \text{rad}(\underline{\text{Centr}}_G(Q))_{\text{spl}}$, the maximality of Q implies $Q = \text{rad}(\underline{\text{Centr}}_G(Q))_{\text{spl}}$. By 6.11, L is a minimal element of the set of Levi subgroups of parabolic subgroups of G , hence P is a minimal parabolic subgroup of G by 1.20. It then follows from 5.7 and 5.5(iv) that two subtori such as Q are conjugate by a section of $G(S)$. The conjugacy of the Q and of the pairs (P, L) then implies the first assertion of 6.16.

Corollary 6.17. *Let S be a connected semi-local scheme, P and P' two minimal parabolic subgroups in standard position (4.5.1.1). Then $P \cap P'$ contains a common Levi subgroup of P and P' .*

Indeed, $P \cap P'$ contains a maximal torus T of G , by 4.5.1(v); let L be the unique Levi subgroup of P containing T . One has

$$\text{rad}(P) \cap T = \text{rad}(P) \cap L = \text{rad}(L)$$

by 1.21, hence $\text{rad}(P) \cap T$ contains $\text{rad}(L)_{\text{spl}}$, which is a maximal split subtorus of G , and is therefore necessarily equal to T_{spl} . One therefore has $L = \underline{\text{Centr}}_G(T_{\text{spl}})$, and by symmetry L is also a Levi subgroup of P' .

Remark 6.18. It follows from 1.21 that the parabolic subgroup P of G is minimal if and only if $\text{rad}(P)$ contains a maximal split subtorus of G ; then, by the proof of 6.17, if T is a maximal torus of P , T_{spl} is a maximal split torus of G and of $\text{rad}(P)$, and $\underline{\text{Centr}}_G(T_{\text{spl}})$ is a Levi subgroup of P . Moreover, every Levi subgroup of P is obtained in this way.

7. Relative root datum

In this paragraph, S will denote a *nonempty connected semi-local* scheme, G a reductive S -group, Q a maximal split subtorus of G , and L the centralizer of Q in G , i.e. $L = \underline{\text{Centr}}_G(Q)$.

7.1. — Since Q is the largest split central subtorus of L , every section of $G(S)$ that normalizes L also normalizes Q . One therefore has (cf. 7.1.1)

$$\underline{\text{Norm}}_G(L)(S) = \underline{\text{Norm}}_G(Q)(S).$$

Moreover, one has seen in 6.4 that the map $G(S) \rightarrow (G/L)(S)$ is surjective. It follows that one has a canonical identification

$$W_G(Q)(S) = (\underline{\text{Norm}}_G(Q)/\underline{\text{Centr}}_G(Q))(S) \simeq \underline{\text{Norm}}_G(L)(S)/L(S).$$

One will denote by M the group $\text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S})$, so that one has a canonical isomorphism $Q \simeq D_S(M)$. One will write W for the group of automorphisms of M defined by $W_G(Q)(S)$.

⁽³²⁾**N.D.E.** One has expanded the sentence that follows.

One therefore has isomorphisms

$$W \simeq W_G(Q)(S) \simeq \underline{\text{Norm}}_G(L)(S)/L(S).$$

7.1.1. — In general, one does not have $\underline{\text{Norm}}_G(L) = \underline{\text{Norm}}_G(Q)$. Take, for example, for S the spectrum of a field k having a quadratic extension k' , and for G the unitary group $\text{Q-Pin}_{k'/k}(A_2)$ (cf. Exp. XXIV, 3.11.2).

Since the minimal parabolic subgroups of G are its Borel subgroups, their Levi subgroups are maximal tori, and one has $(\underline{\text{Norm}}_G(L)/L)(\bar{k}) = \mathfrak{S}_3$.

Moreover, since G is not split, the maximal split tori of G are of dimension ≤ 1 , hence isomorphic to $\mathbf{G}_{m,k}$. Since $\underline{\text{Norm}}_G(Q)/L$ acts faithfully on Q , one has $(\underline{\text{Norm}}_G(Q)/Q)(\bar{k}) = \mathbf{Z}/2\mathbf{Z}$.

7.2. — If P is a parabolic subgroup of G with Levi group L (such a subgroup exists by 6.2), P is necessarily minimal (cf. 6.18). By the conjugacy of minimal parabolic subgroups of G (5.7), that of Levi subgroups of a parabolic subgroup (1.8), and the equalities $P = \underline{\text{Norm}}_G(P)$ and $\underline{\text{Norm}}_G(L) \cap P = L$ (1.6), one obtains: *the set of (minimal) parabolic subgroups of G with Levi group L is principal homogeneous under the group W .*

7.3. — The Lie algebra of G decomposes under the action of Q as

$$\text{Lie}(G) = \text{Lie}(L) \oplus \coprod_{\alpha \in R} \text{Lie}(G)^\alpha,$$

where R is the set of nonzero characters of Q such that $\text{Lie}(G)^\alpha \neq 0$ (roots of G relative to Q).

Denote by M^* the group $\text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, Q)$, which is in duality with M and on which W acts naturally by transport of structure.

Theorem 7.4. *With the notation of 7.3, there is a unique map $\alpha \mapsto \alpha^*$ from R to M^* defining in (M, M^*) a root datum (Exp. XXI, 1.1) whose Weyl group is W .*

Moreover, the parabolic subgroups P of G with Levi group L and the systems of positive roots R_+ of R correspond bijectively by the relation

$$\text{Lie}(P) = \text{Lie}(L) \oplus \coprod_{\alpha \in R_+} \text{Lie}(G)^\alpha.$$

7.4.1. — Suppose first that the existence of the desired map $\alpha \mapsto \alpha^*$ has been proved. By Exp. XXI, 3.4.10, s_α is the unique element of W such that for every $m \in M$, $s_\alpha(m) - m$ is a rational multiple of α , which shows that s_α is determined by α ; since one then has⁽³³⁾ $\alpha^*(m)\alpha = m - s_\alpha(m)$, one sees that α^* is determined by α , which proves the uniqueness of the map $\alpha \mapsto \alpha^*$.

7.4.2. — Let $\alpha \in R$, and let L_α (resp. H_α , resp. $H_{-\alpha}$) be the unique smooth subgroup with connected fibers of G containing L and such that (cf. 6.1(i))

$$\text{Lie}(L_\alpha) = \text{Lie}(L) \oplus \coprod_{\gamma \in \mathbf{Z}\alpha \cap R} \text{Lie}(G)^\gamma,$$

resp.

$$\text{Lie}(H_\alpha) = \text{Lie}(L) \oplus \coprod_{\gamma \in \mathbf{N}\alpha \cap R} \text{Lie}(G)^\gamma,$$

resp.

$$\text{Lie}(H_{-\alpha}) = \text{Lie}(L) \oplus \coprod_{\gamma \in -\mathbf{N}\alpha \cap R} \text{Lie}(G)^\gamma;$$

L_α is a reductive subgroup of G , H_α and $H_{-\alpha}$ are parabolic subgroups of it with Levi subgroup L , and H_α and $H_{-\alpha}$ are opposite relative to L (cf. 6.1(ii)).

⁽³³⁾**N.D.E.** One has corrected $\alpha^*(m)$ to $\alpha^*(m)\alpha$.

By 7.2, there is therefore $s_\alpha \in \underline{\text{Norm}}_{L_\alpha}(Q)(S)/L(S) \subset W$ such that $s_\alpha(H_\alpha) = H_{-\alpha}$. One has $s_\alpha(\alpha) = -\alpha$ (since α (resp. $-\alpha$) is the common divisor of the elements of R occurring in H_α (resp. $H_{-\alpha}$)), and one has $s_\alpha^2 = \text{id}$ (since $s_\alpha^2(H_\alpha)$ and H_α are both opposite to $H_{-\alpha}$ relative to L). One has therefore constructed an $s_\alpha \in W$ satisfying the following properties:

$$(x) \quad s_\alpha(\alpha) = -\alpha, \quad s_\alpha^2 = \text{id},$$

$$(xx) \quad s_\alpha \text{ can be represented by an element of } L_\alpha(S).$$

Observe, moreover, that s_α is constructed canonically from α , and in particular that

$$(xxx) \quad \text{for every } w \in W, \text{ one has } ws_\alpha w^{-1} = s_{w(\alpha)}.$$

7.4.3. — We now propose to prove the assertion:

$$(xxxx) \quad \text{for every } m \in M, \quad s_\alpha(m) - m \in \mathbf{Z}\alpha.$$

Since S is connected, this assertion is local for the (fpqc) topology. One can therefore assume that $L_\alpha = G_1$ is splittable relative to a maximal torus T_1 of L . Let therefore (G_1, T_1, M_1, R_1) be such a splitting. The monomorphism $Q \rightarrow T_1$ identifies M with a quotient of M_1 ; let $p: M_1 \rightarrow M$ be the canonical map.

The image of R_1 under p consists possibly of zero and of the roots of G_1 relative to Q (hence of the elements of R that are integral multiples of α); one therefore has $p(R_1) \subset \mathbf{Z}\alpha$. By (xx), there is an element of $\underline{\text{Norm}}_{G_1}(L)(S)$ that induces s_α on Q . By Exp. XXII, 5.10.10, there is therefore a section $w \in (W_1)_S(S)$ that induces s_α on Q (one denotes by W_1 the Weyl group of the root datum $(M_1, R_1, \dots)$). After restricting S , one can therefore assume that there is $w \in W_1$ inducing s_α on Q , hence satisfying $p(w(m_1)) = s_\alpha(p(m_1))$ for every $m_1 \in M_1$. But, by definition of W_1 , w is a product of reflections with respect to elements of R_1 , hence $w(m_1) - m_1$ is a linear combination with integer coefficients of the elements of R_1 . It follows that $s_\alpha(p(m_1)) - p(m_1)$ is a linear combination with integer coefficients of the elements of $p(R_1) \subset \mathbf{Z}\alpha$, hence an integral multiple of α , which was to be proved.

7.4.4. — One can therefore define an element $\alpha^* \in M^*$ by⁽³⁴⁾

$$\alpha^*(m)\alpha = m - s_\alpha(m).$$

By (x), one has $(\alpha^*, \alpha) = 2$; moreover, it follows from (xxx) that for every pair $(\alpha, \alpha') \in R \times R$, one has $s_\alpha(\alpha') \in R$ and $s_\alpha(\alpha^*) = s_\alpha(\alpha')^*$, which proves (cf. Exp. XXI, 1.1) that the map $\alpha \mapsto \alpha^*$ constructed does indeed define a root datum in (M, M^*) .

7.4.5. — Let W' be the Weyl group of this root datum (the group of transformations of M generated by the s_α); one has $W' \subset W$.

Let, moreover, $\geq$ be a total ordering on the free abelian group M ; put $R_+ = \{\alpha \in R \mid \alpha \geq 0\}$. One knows that R_+ is a system of positive roots of R . Let $w \in W$, represented by an $n \in \underline{\text{Norm}}_G(L)(S) = \underline{\text{Norm}}_G(Q)(S)$. Put $P = H_{R_+}$ (notation of 6.1); by *loc. cit.*, P is a parabolic subgroup of G , with Levi subgroup L . One obviously has $\text{int}(n)P = H_{w(R_+)}$. It then follows from 7.3 that $w(R_+) = R_+$ implies $w = e$. Since the group W' acts transitively on the systems of positive roots of R (Exp. XXI, 3.3.7), and since the stabilizer of R_+ in W is the identity, one concludes at once that $W = W'$. One also concludes that $W = W'$ acts simply transitively both on the set of systems of positive roots of R and on the set of parabolic subgroups of G with Levi group L , which implies the last assertion of 7.4. Q.E.D.

7.5. — If P and P_1 are two minimal parabolic subgroups of G , with Levi subgroups L and L_1 , and if one denotes by Q and Q_1 the maximal split central tori of L and L_1 , then the pairs (P, Q) and (P_1, Q_1) are conjugate: there is $g \in G(S)$ such that $\text{int}(g)P = P_1$ and $\text{int}(g)Q = Q_1$.

⁽³⁴⁾**N.D.E.** One has corrected $\alpha^*(m)$ to $\alpha^*(m)\alpha$.

Indeed, P and P_1 are conjugate (5.7), and one can therefore assume that $P = P_1$; then L and L_1 are conjugate by a section of $P(S)$ (1.8). Moreover, if g and g' are two sections of G conjugating the pairs (P, Q) and (P_1, Q_1) , then $g'^{-1}g$ normalizes P and Q , hence P and L ; but

$$\underline{\text{Norm}}_G(P) \cap \underline{\text{Norm}}_G(L) = P \cap \underline{\text{Norm}}_G(L) = L = \underline{\text{Centr}}_G(Q).$$

The isomorphism $Q \xrightarrow{\sim} Q_1$ induced by $\text{int}(g)$ is therefore independent of g .

Let $\mathcal{R}$ and $\mathcal{R}_1$ be the root data defined by means of 7.4 in $\text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S})$ and $\text{Hom}_{S\text{-gr.}}(Q_1, \mathbf{G}_{m,S})$, and let R_+ and R_{1+} be the systems of positive roots corresponding to P and P_1 . The canonical isomorphism $Q \xrightarrow{\sim} Q_1$ defined above transforms $(\mathcal{R}, R_+)$ into $(\mathcal{R}_1, R_{1+})$. One deduces at once that one can define the *pinned relative root datum*⁽³⁵⁾ of G over S , by identifying the different $(\mathcal{R}, R_+)$ by means of the transitive system of isomorphisms described above.

From now on, we shall denote this pinned root datum by

$$(\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+) = \mathcal{R}(G/S);$$

for every pair $P \supset Q$ as above, one therefore has a canonical isomorphism

$$\underline{M} \xrightarrow{\sim} \text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S})$$

transforming $\underline{R}$ (resp. $\underline{R}_+$) into the set of roots of G (resp. of P) relative to Q , and $W(\underline{R})$ into $W_G(Q)(S)$.

7.6. — Let Q still be a maximal split torus of G , P a (minimal) parabolic subgroup of G with Levi group $\underline{\text{Centr}}_G(Q)$, $(\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+)$ the corresponding pinned root datum (7.4), and Δ the set of simple roots of R_+ . For every $A \subset \Delta$, let $R_A \subset R$ be the set

$$R_A = R_+ \cup (\mathbf{Z}A \cap R_-)$$

consisting of the positive roots and the negative roots that are linear combinations of the elements of A . It is a closed set (Exp. XXI, 3.1.4) of roots, and every closed set containing R_+ can be written uniquely in this form (Exp. XXI, 3.3.10). By 6.1, there is a unique subgroup P_A of G , smooth and with connected fibers, containing $\underline{\text{Centr}}_G(Q)$ and such that

$$\text{Lie}(P_A) = \text{Lie}(G)^0 \oplus \coprod_{\alpha \in R_A} \text{Lie}(G)^\alpha.$$

It then follows at once from 6.1, from the conjugacy of minimal parabolic subgroups, and from the fact that the set of roots of a parabolic subgroup of G containing Q is closed (which follows at once from 1.4 by splitting), that:

Proposition 7.7.

- (i) *The map $A \mapsto P_A$ is a bijection from the set of subsets of Δ onto the set of parabolic subgroups of G containing P . This bijection preserves the natural inclusion order relations.*
- (ii) *Every parabolic subgroup of G is conjugate by a section of $G(S)$ to a unique P_A .*

7.8. — Let $P \supset Q$ be as above. Consider the relative root datum (7.5) of G over S and the canonical isomorphism

$$f : (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+) \xrightarrow{\sim} (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+).$$

The set Δ of simple roots of R_+ is transformed into the set $\underline{\Delta}$ of simple roots of $\underline{R}_+$, hence every subset A of Δ into a subset $f(A) \subset \underline{\Delta}$.

⁽³⁵⁾**N.D.E.** Recall (Exp. XXIII 1.5) that a pinned root datum is a root datum endowed with a choice of a system of positive roots (or of simple roots).

Definition 7.9.0. ⁽³⁶⁾Let H be any parabolic subgroup of G . By 7.7(ii), it is conjugate to a unique P_A . Write $t_r(H) = f(A) \subset \underline{\Delta}$. One checks at once, by means of the conjugacy theorems, that $t_r(H)$ is independent of the choice of the pair $P \supset Q$. One says that it is the *relative type* of H .

Proposition 7.9.

- (i) The map $H \mapsto t_r(H)$ induces a bijection between the set of conjugacy classes (by $G(S)$) of parabolic subgroups of G and the set of subsets of $\underline{\Delta}$.
- (ii) Let H be a parabolic subgroup of G , P a minimal parabolic subgroup contained in H , Q the maximal split central torus of a Levi subgroup of P , Δ the set of simple roots of P relative to Q , and $f : \Delta \xrightarrow{\sim} \underline{\Delta}$ the canonical isomorphism. Then, for every $\alpha \in \Delta$, one has the equivalence:

$$f(\alpha) \in t_r(H) \iff \text{Lie}(H)^{-\alpha} \neq 0,$$

and one has $H = P_A$, where $A = f^{-1}(t_r(H))$.

- (iii) If H and H' are two parabolic subgroups of G containing P , then (see 3.8.1(ii) and 5.5(i) for other equivalent conditions):

$$t_r(H) \subset t_r(H') \iff t(H) \subset t(H').$$

7.10. — One can study the relative positions of two minimal parabolic subgroups; the results are as follows (one refers to 4.5.2 for the notation $t_2(P, P_1)$):

- (1) If P, P_1, P', P'_1 are four minimal parabolic subgroups of G , then $t_2(P, P_1) = t_2(P', P'_1)$ (i.e. (P, P_1) and (P', P'_1) are conjugate locally for (fpqc)) if and only if there is $g \in G(S)$ such that $\text{int}(g)P = P'$ and $\text{int}(g)P_1 = P'_1$.
- (2) Fix, in particular, a minimal parabolic subgroup P with Levi subgroup L , and let T be a maximal torus of L . Consider the scheme $\underline{\text{Par}}_{t_{\min}}(G; P)$ of minimal parabolic subgroups of G in standard position relative to P . One has a morphism (cf. 4.5.5)

$$f : \underline{\text{Par}}_{t_{\min}}(G; P) \longrightarrow W_P(T) \backslash W_G(T) / W_P(T),$$

whose fibers are “the orbits of P in $\underline{\text{Par}}_{t_{\min}}(G; P)$ ”. By (1), f therefore induces a monomorphism

$$P(S) \backslash \underline{\text{Par}}_{t_{\min}}(G; P)(S) \hookrightarrow (W_P(T) \backslash W_G(T) / W_P(T))(S).$$

The image of this morphism is identified with W ; this is the *Bruhat theorem*: every orbit of $P(S)$ in $\underline{\text{Par}}_{t_{\min}}(G; P)(S)$ contains one and only one parabolic subgroup of G with Levi group L (that is, of the form $\text{int}(n)P$, where $n \in \underline{\text{Norm}}_G(L)$).

- (3) In other words, let $E(S)$ be the set of $g \in G(S)$ such that $\text{int}(g)P$ and P are in mutual standard position. Then $E(S)$ has a partition into double cosets modulo $P(S)$ indexed by W :

$$E(S) = P(S)W P(S)$$

(obvious notation). Putting $U = \text{rad}^u(P)(S)$, one can also write

$$E(S) = U(S) \cdot \underline{\text{Norm}}_G(L)(S) \cdot U(S),$$

thus exhibiting a partition of $E(S)$ into double cosets modulo $U(S)$, indexed by $\underline{\text{Norm}}_G(L)(S)$.

- (4) If S is the spectrum of a field, then $E(S) = G(S)$, and one recovers [BT65], 5.15.

Counterexamples 7.11. Let $S = \text{Spec}(\mathbf{Z}/4\mathbf{Z})$, $G = \text{SL}_{2,S}$. Let B be the usual Borel subgroup consisting of the matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $c = 0$. Let $g = \begin{pmatrix} -1 & 1 \\ 2 & 1 \end{pmatrix} \in G(S)$, and put $B' = \text{int}(g)B$. Then $B(S) = B'(S)$, and $B \cap B'$ contains no maximal torus⁽³⁷⁾. This shows,

⁽³⁶⁾**N.D.E.** One has added the numbering 7.9.0 to bring out the definition of “relative type”.

⁽³⁷⁾**N.D.E.** Indeed, B' is the subgroup of G defined by the equation $c = 2(a + b)$; then $B \cap B'$ is not flat over S , hence contains no maximal torus, by 4.5.1.

on the one hand, that two distinct minimal parabolic subgroups can have the same group of sections; on the other hand, that in general there is no criterion allowing one to recognize whether two minimal parabolic subgroups P and P' are in standard position solely by means of the groups $P(S)$ and $P'(S)$. In particular, the subset $E(S)$ of $G(S)$ does not seem to be definable solely by means of the situation $\{G(S), P(S), \underline{\text{Norm}}_G(L)(S)\}$ (in the preceding case, this subset is defined by $c \neq 2^{(38)}$).

7.12. — One now proposes to study the variation of $\mathcal{R}(G/S)$ with S . Let therefore S' be an S -scheme, also semi-local, connected and nonempty. Let Q be a maximal split torus of G ; then $Q_{S'}$ is a split torus of $G_{S'}$; let Q' be a maximal split torus of $G_{S'}$ containing $Q_{S'}$. Put

$$M = \text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S}) \simeq \text{Hom}_{S'\text{-gr.}}(Q_{S'}, \mathbf{G}_{m,S'}), \\ M' = \text{Hom}_{S'\text{-gr.}}(Q', \mathbf{G}_{m,S'}).$$

The monomorphism $Q_{S'} \rightarrow Q'$ induces an epimorphism $u : M' \rightarrow M$. Write

$$L = \underline{\text{Centr}}_G(Q), \quad L' = \underline{\text{Centr}}_{G_{S'}}(Q');$$

one has $L' \subset L_{S'}$.

If H is a subgroup of G containing L , then $H_{S'}$ contains L' , and one has

$$\text{Lie}(H) = \text{Lie}(L) \oplus \coprod_{\alpha \in R_H} \text{Lie}(H)^\alpha, \\ \text{Lie}(H_{S'}) = \text{Lie}(L') \oplus \coprod_{\alpha' \in R'_{H_{S'}}} \text{Lie}(H_{S'})^{\alpha'},$$

where R_H (resp. $R'_{H_{S'}}$) denotes the set of roots of H (resp. $H_{S'}$) relative to Q (resp. Q'). One immediately deduces that

$$R_H \subset u(R'_{H_{S'}}) \subset R_H \cup \{0\}.$$

Taking $H = G$, one first sees that $R \subset u(R') \subset R \cup \{0\}$; then taking for H a minimal parabolic subgroup P with Levi subgroup L , one sees that $R'_{H_{S'}}$ contains a system of positive roots of R' , hence (7.4) that there is a minimal parabolic subgroup P' of $G_{S'}$ with Levi subgroup L' contained in $P_{S'}$. One has therefore constructed a diagram

$$\begin{array}{ccccc} Q_{S'} & \subset & L_{S'} & \subset & P_{S'} \\ \cap & & \cup & & \cup \\ Q' & \subset & L' & \subset & P'. \end{array}$$

If R_+ (resp. Δ) is the system of positive (resp. simple) roots of R defined by P , and if one defines R'_+ and Δ' similarly, one readily checks that

$$R_+ \subset u(R'_+) \subset R_+ \cup \{0\}, \quad \Delta \subset u(\Delta') \subset \Delta \cup \{0\}.$$

Let now $w \in W \simeq \underline{\text{Norm}}_G(Q)(S)/\underline{\text{Centr}}_G(Q)(S)$, and let $n \in \underline{\text{Norm}}_G(Q)(S)$ be a representative of w . One has $\text{int}(n)Q = Q$, hence $\text{int}(n)L = L$, hence $\text{int}(n)L_{S'} = L_{S'}$. Then Q' and $\text{int}(n)Q'$ are two maximal split tori of $L_{S'}$, hence are conjugate by a section $x \in L(S')$, and one has $\text{int}(nx)Q' = Q'$, hence $nx \in \underline{\text{Norm}}_{G_{S'}}(Q')(S')$. Let w' be the image of $n' = nx$ in $W' \simeq \underline{\text{Norm}}_{G_{S'}}(Q')(S')/\underline{\text{Centr}}_{G_{S'}}(Q')(S')$. It is clear that the action of w' on M' is compatible with the projection $u : M' \rightarrow M$, and that the induced action on M coincides with the one defined by w .

Using now the definition of relative root data and the conjugacy theorems, one proves without difficulty:

⁽³⁸⁾**N.D.E.** More generally, for every S -scheme S' , $E(S')$ is defined by the condition: “ c is zero or invertible”.

Theorem 7.13. *Let S and S' be two nonempty connected semi-local schemes, $S' \rightarrow S$ a morphism of S -schemes, G a reductive S -group,*

$$\mathcal{R}(G/S) = (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+),$$

$$\mathcal{R}(G_{S'}/S') = (\underline{M}', \underline{M}'^*, \underline{R}', \underline{R}'^*, \underline{R}'_+)$$

the relative pinned root data. There is a canonical homomorphism

$$\underline{u} : \underline{M}' \rightarrow \underline{M}$$

satisfying the following conditions:

- (i) $\underline{u}$ is surjective.
- (ii) For every $w \in \underline{W}$, there is an element w' of $\underline{W}'$ compatible with $\underline{u}$ and inducing w on $\underline{M}$.
- (iii) For every subset X of $\underline{M}$, write $X^\wedge = X \cap (\underline{M} - \{0\})$. Then

$$\underline{u}(\underline{R}'_+)^\wedge = \underline{R}_+, \quad \underline{u}(\underline{\Delta}')^\wedge = \underline{\Delta}.$$

- (iv) For every parabolic subgroup H , consider $t_r(H) \subset \underline{\Delta}$ and $t_r(H_{S'}) \subset \underline{\Delta}'$. Then

$$\begin{aligned} t_r(H_{S'}) &= \underline{u}^{-1}(t_r(H) \cup \{0\}) \cap \underline{\Delta}' \\ &= \{\alpha' \in \underline{\Delta}' \mid \underline{u}(\alpha') \in t_r(H) \text{ or } \underline{u}(\alpha') = 0\}. \end{aligned}$$

Remark 7.14. If G is splittable over S , its maximal split tori are maximal tori, and the relative notions introduced here then coincide with the absolute notations already introduced. The preceding theorem therefore gives a description of the relative root datum $\mathcal{R}(G/S)$ and of the relative type t_r by means of the absolute root datum and the absolute type of the group $G_{S'}$, where S' is chosen in such a way that $G_{S'}$ is splittable (cf. Exp. XXIV, 4.4.1). Let us refer to [BT65], 6.12 and following, for this description.

7.15. — Let S be a *henselian local* scheme, s_0 its closed point, S_0 the spectrum of the residue field of s_0 , identified with a closed subscheme of S ; for every object X over S , denote by X_0 the object over S_0 obtained from X by base change. Let, finally, G be a reductive S -group. For every parabolic subgroup P of G , P_0 is a parabolic subgroup of G_0 ; conversely, for every parabolic subgroup $\overline{P}$ of G_0 , there is a parabolic subgroup P of G such that $P_0 = \overline{P}$ (this follows from Hensel's lemma and the fact that $\underline{\text{Par}}(G)$ is a smooth S -scheme); in particular (cf. 5.7), a parabolic subgroup P of G is minimal if and only if P_0 is minimal. Once such a subgroup P of G has been chosen, an analogous argument shows that the maximal split subtori of P_0 are of the form T_0 , where T is a maximal split subtorus of P . It follows without difficulty that *the relative root data of G over S and of G_0 over S_0 are canonically isomorphic*, so that the theory of parabolic subgroups of G reduces to that of parabolic subgroups of G_0 .

Observe, moreover, that every reductive S_0 -group is of the form G_0 (Exp. XXIV, Prop. 1.21), which conversely allows one to reduce the study of parabolic subgroups of a reductive S_0 -group to the corresponding study over S .

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⁽³⁹⁾**N.D.E.** To this reference, which appeared in the original, one has added the references that follow.